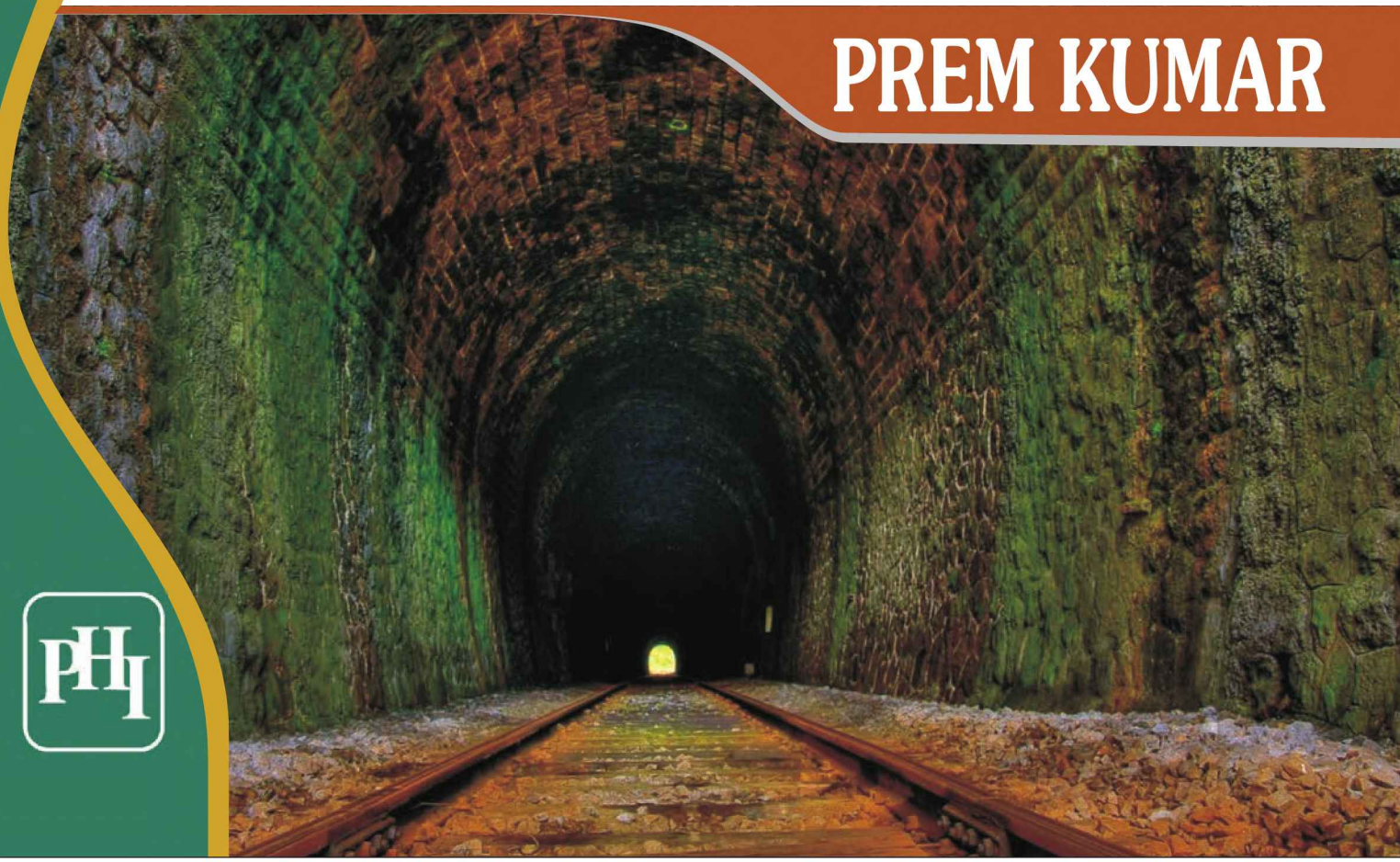
Eastern Economy Edition

The Complete PHI Learning Guide to MATHEMATICS for JEE (Main)

PREM KUMAR



PHI

The Complete **PHI Learning** *Guide to*

MATHEMATICS

for JEE (Main)

PREM KUMAR
Head of MATHS POINT
An Educational Institution
Delhi

PHI Learning Private Limited
New Delhi-110001
2013

THE COMPLETE PHI LEARNING GUIDE TO MATHEMATICS FOR JEE (MAIN)
Prem Kumar

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Syllabus

UNIT 1: SETS, RELATIONS AND FUNCTIONS: Sets and their representation; Union, intersection and complement of sets and their algebraic properties; Power set; Relation, Types of relations, equivalence relations, functions; one-one, into and onto functions, composition of functions.

UNIT 2: COMPLEX NUMBERS AND QUADRATIC EQUATIONS: Complex numbers as ordered pairs of reals, Representation of complex numbers in the form $a + ib$ and their representation in a plane, Argand diagram, algebra of complex numbers, modulus and argument (or amplitude) of a complex number, square root of a complex number, triangle inequality, Quadratic equations in real and complex number system and their solutions. Relation between roots and coefficients, nature of roots, formation of quadratic equations with given roots.

UNIT 3: MATRICES AND DETERMINANTS: Matrices, algebra of matrices, types of matrices, determinants and matrices of order two and three. Properties of determinants, evaluation of determinants, area of triangles using determinants. Adjoint and evaluation of inverse of a square matrix using determinants and elementary transformations, Test of consistency and solution of simultaneous linear equations in two or three variables using determinants and matrices.

UNIT 4: PERMUTATIONS AND COMBINATIONS: Fundamental principle of counting, permutation as an arrangement and combination as selection, Meaning of $P(n, r)$ and $C(n, r)$, simple applications.

UNIT 5: MATHEMATICAL INDUCTION: Principle of Mathematical Induction and its simple applications.

UNIT 6: BINOMIAL THEOREM AND ITS SIMPLE APPLICATIONS: Binomial theorem for a positive integral index, general term and middle term, properties of Binomial coefficients and simple applications.

UNIT 7: SEQUENCES AND SERIES: Arithmetic and Geometric progressions, insertion of arithmetic, geometric means between two given numbers. Relation between A.M. and G.M. Sum up to n terms of special series: S_n, S_{n^2}, S_{n^3}. Arithmetico-Geometric progression.

UNIT 8: LIMIT, CONTINUITY AND DIFFERENTIABILITY: Real-valued functions, algebra of functions, polynomials, rational, trigonometric, logarithmic and exponential functions, inverse functions. Graphs of simple functions. Limits, continuity and differentiability. Differentiation of the sum, difference, product and quotient of two functions. Differentiation of trigonometric, inverse trigonometric, logarithmic, exponential, composite and implicit functions; derivatives of order up to two. Rolle's and Lagrange's Mean Value Theorems. Applications of derivatives: Rate of change of quantities, monotonic—increasing and decreasing functions, Maxima and minima of functions of one variable, tangents and normals.

UNIT 9: INTEGRAL CALCULUS: Integral as an anti-derivative. Fundamental integrals involving algebraic, trigonometric, exponential and logarithmic functions. Integration by substitution, by parts and by partial fractions. Integration using trigonometric identities. Evaluation of simple integrals of the type. Integral as limit of a sum. Fundamental Theorem of Calculus. Properties of definite integrals. Evaluation of definite integrals, determining areas of the regions bounded by simple curves in standard form.

UNIT 10: DIFFERENTIAL EQUATIONS: Ordinary differential equations, their order and degree. Formation of differential equations. Solution of differential equations by the method of separation of variables, solution of homogeneous and linear differential equations of the type: $dy/dy + p(x)y = q(x)$.

UNIT 11: COORDINATE GEOMETRY: Cartesian system of rectangular coordinates 10 in a plane, distance formula, section formula, locus and its equation, translation of axes, slope of a line, parallel and perpendicular lines, intercepts of a line on the coordinate axes. Straight lines. Various forms of equations of a line, intersection of lines, angles between two lines, conditions for concurrence of three lines, distance of a point from a line, equations of internal and external bisectors of angles between two lines, coordinates of centroid, orthocentre and circumcentre of a triangle, equation of family of lines passing through the point of intersection of two lines. Circles, conic sections. Standard form of equation of a circle, general form of the equation of a circle, its radius and centre, equation of a circle when the end points of a diameter are given, points of intersection of a line and a circle with the centre at the origin and condition for a line to be tangent to a circle, equation of the tangent. Sections of cones, equations of conic sections (parabola, ellipse and hyperbola) in standard forms, condition for $y = mx + c$ to be a tangent and point(s) of tangency.

UNIT 12: THREE DIMENSIONAL GEOMETRY: Coordinates of a point in space, distance between two points, section formula, direction ratios and direction cosines, angle between two intersecting lines. Skew lines, the shortest distance between them and its equation. Equations of a line and a plane in different forms, intersection of a line and a plane, coplanar lines.

UNIT 13: VECTOR ALGEBRA: Vectors and scalars, addition of vectors, components of a vector in two dimensions and three dimensional space, scalar and vector products, scalar and vector triple product.

UNIT 14: STATISTICS AND PROBABILITY: Measures of Dispersion: Calculation of mean, median, mode of grouped and ungrouped data calculation of standard deviation, variance and mean deviation for grouped and ungrouped data. Probability: Probability of an event, addition and multiplication theorems of probability, Baye's theorem, probability distribution of a random variate, Bernoulli trials and Binomial distribution.

UNIT 15: TRIGONOMETRY: Trigonometrical identities and equations. Trigonometrical functions. Inverse trigonometrical functions and their properties. Heights and Distances.

UNIT 16: MATHEMATICAL REASONING: Statements, logical operations and, or, implies, implied by, if and only if. Understanding of tautology, contradiction, converse and contrapositive.

Preface

In pursuance of the recent directives and changes promulgated by the Ministry of Human Resource Development (MHRD), Government of India, with regard to the merger of AIEEE and IIT–JEE, a combined Joint Entrance Examination (JEE) will be held hereinafter in two phases, with effect from April 2013:

1. JEE (Main)
2. JEE (Advanced)

The JEE (Main), a de facto replacement of AIEEE, will be conducted by the Central Board of Secondary Education (CBSE). The JEE (Main) will have the same syllabus and pattern of examination as those of AIEEE. Its scores (with 40% weightage to CBSE scores) will be used for admissions to BE/BTech courses at NIITs, IIITs, Delhi Technological University (DTU) and other Centrally Funded Technical Institutions (CFTIs), as per the hitherto practice followed by CBSE in respect of AIEEE.

The JEE (Advanced), hitherto known as IIT–JEE, will be conducted by the JEE Apex Board for only the top 1,50,000 candidates in JEE (Main), for admission to IITs, IIT BHU and ISM Dhanbad.

The JEE (Main) will have one objective type question paper as per Paper I of the erstwhile AIEEE. It will consist of Physics, Chemistry and Mathematics. The duration of the paper will be 3 hours.

The first JEE (Main) will be held in April 2013. It has now been proposed to conduct the JEE test twice in a year. The second JEE (Main) will be held in November/December 2013, the third in April 2014, and so on.

This mathematics book has been specifically designed keeping in mind the standard of the examination for the **JEE (Main)**. Having taught mathematics to a large number of students of Classes XI and XII for more than a decade now, I am well versed with the areas which require special treatment. Accordingly, I have given utmost importance to the concepts and methodologies required for solving a wide variety of objective type questions. The real inspiration behind writing this book has been the feedback of my students over the years.

I wish to thank all my students for their suggestions without which it would not have been possible to bring out this book. I am also indebted to the editorial and production teams of PHI Learning for many of their helpful suggestions, and also care and devotion shown during the production of this book.

I am grateful to my father Sh. Brham Dev and my family members for providing me with moral support during my busy periods of writing this book.

Although every effort has been made to keep this book free from errors, I shall feel highly obliged to students and teachers for pointing out errors and for making suggestions towards the improvement of the book.

Prem Kumar
mathspoint9818159014@gmail.com

A Suggestion for Students

"··· any idea or problem or body of knowledge can be presented in a form simple enough so that any particular learner can understand it in a recognizable form."

—BRUNER

I hope you have read the Preface carefully. The JEE (Main) paper will be objective type in nature and hence quite different from the CBSE Board exams. The purpose of this book is to provide students with a firm understanding of the concepts along with rigorous practice of solving problems with speed and accuracy.

Mathematics is rich, elegant and beautiful—but not necessarily easy. There are three steps for achieving proficiency in mathematics—concept building, proper application of concepts and last but not least, practice and more practice. Do every exercise, even when you think you can do it easily. There is always a reason for every exercise, so do not leave any out. Make all-out efforts to solve every problem yourself sitting on a study table with paper and pencil at hand. Try hard to crack and remember the trick. Develop speed and accuracy through regular practice.

It is not luck, but good hard work, patience and perseverance that you should bank upon to win through in the examination.

You must have realized by now that the JEE (Main) is the **gateway** to JEE (Advanced). You have to be one of the 1,50,000 top rankers in JEE (Main) to become eligible for taking the test of the JEE (Advanced).

1

GEORG CANTOR

Sets, Relations and Functions

SETS
- Definition of Set
- Types of Sets
- Venn Diagrams
- Operations on Sets
- Laws of Algebra of Sets
- Cardinal Number

RELATIONS
- Ordered Pair
- Cartesian Product
- Relations
- Equivalence Relation
- Different Types of Relations
- Composition of Relations
- Inverse Relation

FUNCTIONS
- Functions
- Types of Functions
- Inverse Functions
- Composition of Functions

SETS

DEFINITION OF SET

The German mathematician, Georg Cantor (1845–1918), developed the theory of sets. In our mathematical language, all living and non-living things in the universe are known as objects. Any well-defined collection of distinct objects is known as a **set**.

What does a "well-defined" collection of objects mean? It means that it must be possible to decide whether a given object belongs to the given set or not. Any such object in the set is called its **member** or **element**.

Generally, sets are denoted by capital letters such as A, B, C, … and their elements are denoted by lowercase letters such as a, b, c, …, x, y, z.

Let A be a nonempty set. If x is an element of A, then we write "$x \in A$" which is read as "x is an element of A" or "x belongs to A". If x is not an element of A, then we write "$x \notin A$" and read as "x is not an element of A" or "x does not belong to A".

For example, if set $A = \{2, 4, 6, 8\}$, then 2, 4, 6 and 8 are called the elements or members of the set A. Thus, we can write, $2 \in A$, $4 \in A$, but $3 \notin A$.

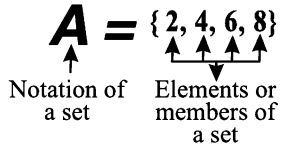


Fig. 1.1 An example of a set of all positive even integers less than 10.

Illustration 1. Which of the following collections is a set?
(a) The collection of all boys in your class.
(b) The collection of intelligent boys in your class.
(c) The collection of smart boys in your class.
(d) The collection of tall boys in your class.

Ans. (a)

Solution: Clearly, collections in sets (b), (c) and (d) are not well-defined collections. So, they do not form a set.

☞ REPRESENTATION OF SETS

Usually, sets are represented in the following two ways:

1. Roster form or tabular form
2. Set builder form or rule method

1. Roster Form

In this method, all the elements of a set are listed, separated by commas, and put within braces { } as already seen by us. This method is also known as roster method or tabular method or listed method. Another example is:

A = set of all prime numbers less than 11 = {2, 3, 5, 7}.

2. Set Builder Method

In this method, instead of listing all the elements of a set, we list the property or properties satisfied by the elements of the set and write it as $A = \{x : P(x)\}$. It is read "A is the set of all elements x such that x has the property $P(x)$". The symbol ':' stands for *such that*. For example:

$$A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$
$$= \{x : x \in N \text{ and } x \leq 10\}$$

This method is also known as rule method or property method

Note

- The order of elements in a set has no importance. For example, $\{a, b, c\}$ and $\{c, a, b\}$ are the same sets.
- The repetition of elements in a set does not affect the set. For example, $\{a, b, c\}$ and $\{a, a, b, c\}$ are the same sets.

Illustration 2. If A is the set whose elements are obtained by adding 1 to each of the even numbers, then the set builder notation of A is
(a) $A = \{x : x \text{ is even}\}$
(b) $A = \{x : x \text{ is odd } x > 1\}$
(c) $A = \{x : x \text{ is odd and } x \in N\}$
(d) $A = \{x : x \text{ is an integer}\}$

Ans. (c)

Solution: When we add 1 to even integers, we obtain odd integers.

$\therefore$ $A = \{x : x \text{ is an odd integer}\}$. That is,
$A = \{x : x \text{ is odd and } x \in N\}$

☞ TYPES OF SETS

1. The Empty Set

A set having no element in it is called the *null set* or the *empty set* or the *void set*. It is denoted by { } or ϕ. The symbol ϕ is pronounced 'phi'.

Let A be a set of natural numbers less than 1. Then A is an empty set, i.e. $A = \{\}$ since natural numbers consist of all positive integers: 1, 2, 3, ...

Note

- The set $\{0\}$ is not an empty set as it contains one element 0 (zero).
- If ϕ represents the null set, then ϕ is never written within braces, i.e., $\{\phi\}$ is not the null set.
- It is wrong to call 'an empty set' or 'a null set' as there is one and only one empty set though it may have many descriptions. Therefore, it is always called 'the empty set' or 'the null set'.

Illustration 3. Which of the following is an empty set?
(a) $\{x : x \text{ is a real number and } x^2 - 1 = 0\}$
(b) $\{x : x \text{ is a real number and } x^2 + 1 = 0\}$
(c) $\{x : x \text{ is a real number and } x^2 - 9 = 0\}$
(d) $\{x : x \text{ is a real number and } x^2 = x + 2\}$

Ans. (b)

Solution: We observe that there is no real number x such that $x^2 + 1 = 0$, i.e. $x^2 = -1$ because the square of a real number cannot be negative.

$\therefore$ $\{x : x \text{ is a real number and } x^2 + 1 = 0\} = \phi$

2. Singleton Set

A set which has only one element in it is called a singleton set. For example,
(i) A = the set of present president of India.
(ii) $\{2\}$, $\{0\}$, $\{\phi\}$
are singleton sets.

Illustration 4. Which of the following is a singleton set?
(a) $\{x : |x| < 1, \text{ and } x \in Z\}$
(b) $\{x : |x| = 5, \text{ and } x \in Z\}$
(c) $\{x : x^2 = 1, \text{ and } x \in Z\}$
(d) $\{x : x^2 + x + 1 = 0, \text{ and } x \in \mathbb{R}\}$

where Z consists of all the integers and $\mathbb{R}$ consists of all the real numbers.

Ans. (a)

Solution: If $|x| < 1 \Rightarrow -1 < x < 1$ $\therefore$ $x = 0$, since 0 is the only integer between -1 and 1. This set, therefore, has only one element in it which is 0.

3. Pair Set

A set having two elements is called a pair set. For example, $A = \{1, 2\}$, $B = \{0, 3\}$ and $C = \{a, b\}$ are pair sets.

4. Set of Sets

A set S having all its elements as sets is called a set of sets or a family of sets or a class of sets. For example, the set $\{1, 2\}$, $\{2, 5\}$, $\{3, 6, 8\}$ has three elements in it, and each element is a set by itself.

On the other hand, the set

$$S = \{\{1, 2\}, 3, \{4\}\}$$

is not a set of sets as the element 3 is not a set.

5. Finite Set

A set whose elements can be counted, that is in which the process of counting its elements comes to an end, is called a finite set. In other words, a set A is called finite if it has n distinct elements, where n is a positive integer or zero. For example, $A = \{1, 2, 3, 4\}$, A is a finite set as it has 4 elements. The number of elements in a set is called the cardinality of the set.

Illustration 5. Which of the following sets is not finite?

(a) $\{(x, y): x^2 + y^2 \le 1 \le x + y, \text{ and } x, y \in \mathbb{R}\}$

(b) $\{(x, y): x^2 + y^2 \le 1 \le x + y, \text{ and } x, y \in Z\}$

(c) $\{(x, y): x^2 \le y \le |x|, \text{ and } x, y \in Z\}$

(d) $\{(x, y): x^2 + y^2 = 1, \text{ and } x, y \in Z\}$

where $\mathbb{R}$ is the set of real numbers and Z consists of all the integers.

Ans. (a)

Solution: (a) The set $\{(x, y): x^2 + y^2 \le 1 \le x + y, \text{ and } x, y \in \mathbb{R}\}$ consists of all points in the first quadrant which lie inside the circle $x^2 + y^2 = 1$ and above the line $x + y = 1$. So, it is not a finite set.

(b) The set $\{(x, y): x^2 + y^2 \le 1 \le x + y, \text{ and } x, y \in Z\}$
$= \{(1, 0), (0, 1)\}$. So it is a finite set.

(c) The set $\{(x, y): x^2 \le y \le |x|, \text{ and } x, y \in Z\}$
$= \{(0, 0), (1, 1), (-1, 1)\}$. So it is a finite set.

(d) The set $\{(x, y): x^2 + y^2 = 1, \text{ and } x, y \in Z\}$
$= \{(1, 0), (0, 1), (-1, 0), (0, -1)\}$. So it is a finite set.

Hence, the option (a) is correct.

6. Infinite Set

A set is said to be infinite, if it has an unlimited (uncountable) number of elements. In other words, a set which is not finite is called infinite. For example, the set Z of all the integers $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$ is an infinite set.

7. Equal Sets

The sets A and B are said to be identical or equal, written as $A = B$, when they have exactly the same elements. To be more precise, A and B are equal if every element of A is an element of B and conversely every element of B is also an element of A. For example,

$$X = \{1, 2, 3, 4\} \text{ and } Y = \{2, 2, 1, 3, 4\}$$

are equal sets. Here, Y has a repeated element 2 and we know that a set is known by its distinct elements.

8. Equivalent Sets

Two finite sets A and B are equivalent if their cardinal numbers are the same, i.e. if $n(A) = n(B)$. This equivalence relation is written as $A \leftrightarrow B$ or $A \sim B$ (and read as "A is equivalent to B"). For example, if

A = {letters of FOLLOW} and

B = {letters of FLOW}, then

$A = B$ because each set can be written as {F, O, L, W}. Also $n(A) = 4 = n(B) = 4$. So $A \leftrightarrow B$.

Note

- Equal sets are always equivalent. But the converse is not always true.
- Two empty sets are always equal.
- Two infinite sets are always equivalent.

9. Overlapping Sets

The sets are called overlapping (or joint) if they have at least one element in common. For example,

$$A = \{1, 2, 3\} \text{ and } B = \{2, 4, 6\}$$

are overlapping (joint) sets because they have the element 2 as common.

10. Subset

If every element of one set is also an element of a second set, then the first set is a subset of the second set. We use the symbol '$\subseteq$' to mean "is a subset of". Thus, A is a subset of B will be written as $A \subseteq B$. In other words, A is contained in B, as shown diagrammatically in Fig. 1.2.

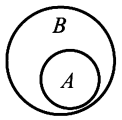


Fig. 1.2 $A \subseteq B$.

Algebra is a part of maths. Hence "algebra is a subset of maths". Thus, algebra $\subseteq$ math.

If $A = \{3, 5, 7\}$ and $B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, then $A \subseteq B$. $A \not\subseteq B$ means that A is not a subset of B.

Note

- Every set is a subset of itself, i.e. $A \subseteq A$, $B \subseteq B$, and so on.
- The empty set is a subset of every set, i.e. $\phi \subseteq A$, $\phi \subseteq B$, and so on.
- If A has n elements, then the number of subsets of set A is 2^n.

Illustration 6. If $A = \{x : x = 4n + 1, \text{and } 2 \le n \le 5\}$, then the number of subsets of A is

(a) 16 (b) 15

(c) 4 (d) none of these

Ans. (a)

Solution: Given that $A = \{x : x = 4n + 1, \text{and } 2 \le n \le 5\}$

The number of elements in $A = 4$.

So, the number of subsets = $2^4 = 16$.

11. Superset

If a set A is a subset of a set B, then the set B is called the superset of A. Symbolically, we write $B \supseteq A$, read "B is a superset of A", and diagrammatically represented as in Fig. 1.3.

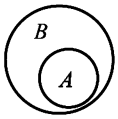


Fig. 1.3 $B \supseteq A$.

$B \not\supseteq A$ is read "B is not a superset of A".

If $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4, 5, 6\}$, then since all the elements of set A are contained in set B, therefore, we have $A \subseteq B$, but $B \not\subseteq A$. Also, set B is a superset of set A.

12. Proper Subset

Set A is a proper subset of $B (A \neq B)$ if every element of set A is an element of set B and at least one element of B is not an element of A. The symbol for proper subset is '$\subset$', i.e. if A is a proper subset of B, we write $A \subset B$. For example, if $A = \{4, 6\}$ and $B = \{2, 4, 6, 8, 10\}$, then A is a proper subset of B.

The empty set is always a proper subset of every set.

Illustration 7. Which of the following does not have a proper subset?

(a) $\{x : x \in Q\}$

(b) $\{x : x \in N, \text{and } 3 < x < 4\}$

(c) $\{x : x \in Q, \text{and } 3 < x < 4\}$

(d) none of these

Ans. (b)

Solution: The set $\{x : x \in N, \text{and } 3 < x < 4\}$ is empty because there is no natural number between 3 and 4. The empty set cannot have a proper subset.

12.1 *Number of subsets and number of proper subsets*

If a given set has n elements in it, then it will have

(a) 2^n subsets

(b) $2^n - 1$ proper subsets

Number of elements in $\{x, y\} = 2$, $\therefore n = 2$. Hence, the number of subsets in $\{x, y\} = 2^n = 2^2 = 4$ and the number of proper subsets in $\{x, y\} = 2^n - 1 = 2^2 - 1 = 3$.

13. Power Set

The collection of all subsets of a set A is called the power set of A. It is denoted by $P(A)$. In $P(A)$, every element is a set, i.e. $P(A) = \{s : s \subseteq A\}$. Since the empty set and the set A itself are subsets of A, they are therefore the elements of $P(A)$. Thus, the power set of a given set is always nonempty.

If $A = \{1, 2, 3\}$

$P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

Note

- If A is the empty set ϕ, then $P(A)$ has just one element ϕ, i.e. $P\{\phi\} = \{\phi\}$.
- Each element of a power set is a set.
- Power set of any set is always nonempty.
- If set A has n elements, then $P(A)$ has 2^n elements.

14. Comparable Sets

Two sets A and B are said to be *comparable* if one of them is a subset of the other, i.e. if either $A \subseteq B$ or $B \subseteq A$.

The sets $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4, 5, 6, 7\}$ are comparable sets, because $A \subseteq B$.

15. Universal or Fundamental Set

A set that contains all the elements under consideration in a given problem is called a universal set. It is denoted by U or ξ (pxi). For example, if we are considering days of the week, then

$A = \{\text{Sunday, Monday}\}$

is a subset of the universal set U or $\xi = \{\text{days of the week}\}$.

Note

- A universal set for the sets under consideration is not unique, i.e. we may have more than one universal set for the same sets under consideration.
- Every set under discussion is a subset of the universal set.
- $U \subseteq U$ because every set is a subset of itself.

☞ VENN DIAGRAMS

A Swiss mathematician, Euler, gave an idea to represent a set by the points in a closed curve. Later a British mathematician John Venn (1834–1923) put this idea to practice. So, the diagrams drawn to represent sets are called Venn–Euler diagrams or simply Venn diagrams.

In Venn diagrams, a universal set U is represented by a rectangular region, while sets within U are represented by circles. Also, an element of a set A is represented by a point within the circle of set A.

If $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $A = \{2, 4, 6, 8\}$, then the Venn diagram is drawn as shown in Fig. 1.4.

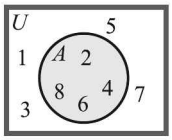


Fig. 1.4 Venn diagram.

OPERATIONS ON SETS

Now, we introduce some operations on sets to construct new sets from the given ones.

1. Union of Sets

The union of two sets A and B is an operation that yields a new set whose elements are either in set A or in set B or in both the sets A and B. The union of A and B is denoted by $A \cup B$ which is read as "A union B".

Symbolically, $A \cup B = \{x : x \in A \text{ or } x \in B\}$

Clearly, $x \in A \cup B \Rightarrow x \in A$ or $x \in B$

and $x \notin A \cup B \Rightarrow x \notin A$ and $x \in B$

The Venn diagram of $A \cup B$ is shown in Fig. 1.5 where the shaded portion represents the set $A \cup B$.

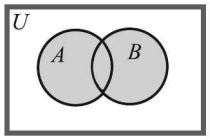


Fig. 1.5 Venn diagram of $A \cup B$.

If $A = \{a, b, c, d\}$ and $B = \{x, y, z\}$, then

$$A \cup B = \{a, b, c, d, x, y, z\}$$

1.1 *General form*

The union of a finite number of sets $A_1, A_2, \ldots, A_n$ is represented by

$$A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_n \quad \text{or} \quad \bigcup_{i=1}^{n} A_i$$

Symbolically,

$$\bigcup_{i=1}^{n} A_i = \{x : x \in A_i \text{ for at least one } A_i, \text{ i.e. one or more of the sets } A_i\}$$

Note

- Include the common elements in A and B only once in $A \cup B$.
- Union of sets is commutative, i.e. $A \cup B = B \cup A$; $C \cup D = D \cup C$, and so on.

Illustration 9. Suppose $A_1, A_2, \ldots, A_{30}$ are thirty sets each having 5 elements and $B_1, B_2, \ldots, B_n$ are n sets each having 3 elements. Let $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^{n} B_j = S$ and let each element of S belong to exactly 10 of A_i's and exactly 9 of B_j's, then the value of n is equal to

(a) 15 (b) 3

(c) 45 (d) none of these

Ans. (c)

Solution: If the elements in $A_1, A_2, \ldots, A_{30}$ were not repeated, then the number of elements in $A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_{30}$, i.e. S would be 30×5. But since each element of S belongs to exactly ten of A_i's, so

$$S = \frac{30 \times 5}{10} = 15 \qquad \text{(i)}$$

Similarly, if the elements in $B_1, B_2, \ldots, B_n$ were not repeated, then the total number of elements in S would be $3n$, but since each element of S belongs to exactly nine of B_j's,

so $$S = \frac{3n}{9} \Rightarrow 15 = \frac{3n}{9} \Rightarrow n = 45$$

2. Intersection of Sets

If A and B are two sets, then the intersection of A and B is a set of all those elements which are in both A and B. The intersection of A and B is denoted by $A \cap B$ and read as "A intersection B".

Symbolically, $A \cap B = \{x : x \in A \text{ and } x \in B\}$

If $x \in A \cap B \Rightarrow x \in A$ and $x \in B$

and if $x \notin A \cap B \Rightarrow x \notin A$ and $x \notin B$

The Venn diagram of $A \cap B$ is shown in Fig. 1.6 as the shaded region which represents the set $A \cap B$.

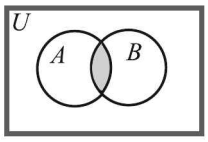


Fig. 1.6 Venn diagram of $A \cap B$.

If $A = \{f, l, o, r\}$ and $B = \{f, l, o, w, e, r\}$, then

$$A \cap B = \{f, l, o, r\}$$

Sets which have at least one element in common are called overlapping or intersection sets.

2.1 *General form*

The intersection of a finite number of sets $A_1, A_2, A_3, \ldots, A_n$ is represented by

$$A_1 \cap A_2 \cap A_3 \cap \ldots \cap A_n \quad \text{or} \quad \bigcap_{i=1}^{n} A_i$$

Symbolically, $\bigcap_{i=1}^{n} A_i = \{x : x \in A_i \text{ for all } A_i\}$

Illustration 10. If $A = \{x : f(x) = 0\}$ and $B = \{x : g(x) = 0\}$, then $A \cap B$ will be

(a) $[f(x)]^2 + [g(x)]^2 = 0$ (b) $f(x)/g(x)$

(c) $g(x)/f(x)$ (d) none of these

Ans. (a)

Solution: Given that $A = \{x : f(x) = 0\}$ and $B = \{x : g(x) = 0\}$, then

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$
$$= \{x : f(x) = 0 \text{ and } g(x) = 0\}$$
$$= [f(x)]^2 + [g(x)]^2 = 0$$

3. Disjoint Sets

Two sets are called disjoint (non-overlapping) if they have no elements in common.

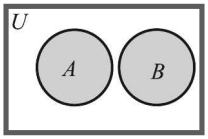


Fig. 1.7 Venn diagram of two disjoint sets.

Suppose

A = {Students of class X} and

B = {Students of class XII}

Here the sets A and B are disjoint as the students of class X and those of class XII will not be common.

4. Difference of Two Sets

Let A and B be two sets. The difference of A and B, written as $A - B$, is the set of all those elements of A which do not belong to B.

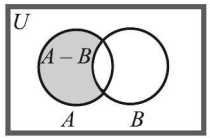


Fig. 1.8 Venn diagram of $A - B$.

Thus, $A - B = \{x : x \in A \text{ and } x \notin B\}$.

Clearly, $x \in A - B \Leftrightarrow x \in A$ and $x \notin B$

The difference $B - A$ is the set of all those elements of B that do not belong to A, i.e. $B - A = \{x : x \in B \text{ and } x \notin A\}$.

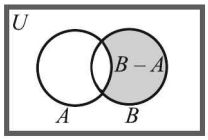


Fig. 1.9 Venn diagram of $B - A$.

If $A = \{1, 2, 3, 4\}$ and $B = \{4, 5, 6, 7, 8\}$, then

$A - B = \{1, 2, 3\}$ and $B - A = \{5, 6, 7, 8\}$

4.1 *Symmetric difference*

If A and B are two sets, then the set $(A - B) \cup (B - A)$ is known as symmetric difference of sets A and B and is denoted by $A \Delta B$.

The Venn diagram of $A \Delta B$ is as shown in Fig. 1.10 where the shaded region represents $A \Delta B$.

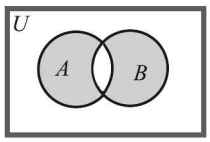


Fig. 1.10 Venn diagram of $A \Delta B$.

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, then

$A \Delta B = (A - B) \cup (B - A)$

$= \{1, 2, 3\} \cup \{4, 5, 6\} = \{1, 2, 3, 4, 5, 6\}$

4.2 *Complement set*

The complement of the set A is the set of all elements of a universal set which are not the elements of A. The complement of a set A is written as A' and read as "complement of set A".

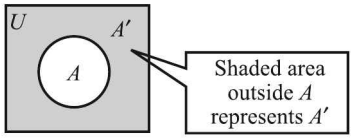


Fig. 1.11 Venn diagram of complement set.

If the universal set $U = \{1, 2, 3, 4, 5, 6, 7\}$

$A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$, then

complement of set $A = A'$

$= \{\text{elements which are in } U \text{ but not in set } A\}$

$= \{2, 4, 6, 7\}$

complement of set $B = B'$

$= \{\text{elements which are in } U \text{ but not in set } B\}$

$= \{1, 3, 5, 7\}$

Note

- $\phi' = U$
- $U' = \phi$
- $(A')' = A$
- $A \cup A' = U$
- $A \cap A' = \phi$

☞ LAWS OF ALGEBRA OF SETS

If A, B and C are three sets, then the following algebraic laws are satisfied:

(i) Idempotent law

(a) $A \cup A = A$ (b) $A \cap A = A$

(ii) Identity law

(a) $A \cup \phi = A$ (b) $A \cap U = A$

(iii) Commutative law

(a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$

(iv) Associative law

(a) $(A \cup B) \cup C = A \cup (B \cup C)$

(b) $A \cap (B \cap C) = (A \cap B) \cap C$

(v) Distributive law

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(vi) De Morgan's laws

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

(c) $A - (B \cup C) = (A - B) \cap (A - C)$

(d) $A - (B \cap C) = (A - B) \cup (A - C)$

☞ CARDINAL NUMBER OF A FINITE SET

The number of distinct elements in a finite set A is called its cardinal number and it is denoted by $n(A)$.

If $A = \{-3, -1, 8, 10, 13, 17\}$, then $n(A) = 6$

1. Cardinal Number of Some Sets

If A, B and C are finite sets and U is the finite universal set, then

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cup B) = n(A) + n(B)$; if A and B are disjoint sets

(iii) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

(iv) $n(A - B) = n(A) - n(A \cap B)$

(v) $n(A \Delta B) = n(A) + n(B) - 2n(A \cap B)$

(vi) $n(A') = n(U) - n(A)$

(vii) $n(A' \cup B') = n(U) - n(A \cap B)$

(viii) $n(A' \cap B') = n(U) - n(A \cup B)$

(ix) $n(A \cap B') = n(A) - n(A \cup B)$

(x) $n(A' \cap B) = n(B) - n(A \cup B)$

RELATIONS

☞ ORDERED PAIR

Two elements a and b listed in a specific order form an ordered pair, denoted by (a, b). In an ordered pair (a, b), a is regarded as the first element and b the second element. It is evident from the definition that

(i) $(a, b) \neq (b, a)$ (ii) $(a, b) = (c, d)$ iff $a = c$, $b = d$

☞ CARTESIAN PRODUCT OF SETS

If A and B are two nonempty sets, then the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$ is called the Cartesian product of the set A with set B and is denoted by $A \times B$. Thus,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

The Cartesian product of two sets

$$A = \{a_1, a_2\} \text{ and } B = \{b_1, b_2\}$$

is therefore $A \times B = \{(a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2)\}$

Illustration 11. If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $(A \cap B) \times A$ is

(a) $\{(1, 3), (2, 3), (3, 3)\}$ (b) $\{(3, 1), (3, 2), (3, 3)\}$

(c) $\{(1, 3), (3, 1), (3, 2)\}$ (c) none of these

Ans. (b)

Solution: We have $(A \cap B) = \{3\}$ and $A = \{1, 2, 3\}$

$\therefore$ $(A \cap B) \times A = \{(3, 1), (3, 2), (3, 3)\}$.

Note

- If $A \neq B$, then $A \times B \neq B \times A$
- If A has p elements and B has q elements, then $A \times B$ has pq elements.
- If $A = \phi$ and $B = \phi$, then $A \times B = \phi$.
- The Cartesian product of n sets $A_1, A_2, A_3, \ldots, A_n$ is the set of all n-tuples $(a_1, a_2, a_3, \ldots, a_n)$, $a_i \in A_i$, $i = 1, 2, 3, \ldots, n$.

  Thus, $A_1 \times A_2 \times A_3 \times \ldots \times A_n$ or $\prod_{i=1}^{n} A_i = \{(a_1, a_2, a_3, \ldots, a_n) : a_i \in A_i, i = 1, 2, 3, \ldots, n\}$.

1. Graph of $A \times B$

Let A and B be two nonempty subsets of R. Then the graph of $A \times B$ is the set of all points in the plane represented by the ordered pairs of $A \times B$. For example, if $A = \{1, 2, 3\}$, and $B = \{3, 4, 5\}$, then

$A \times B = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5)\}$

The graph of $A \times B$ will thus be as shown in Fig. 1.12.

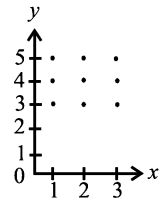


Fig. 1.12 Graph of $A \times B$.

2. Properties of Cartesian Product

If A, B and C are three sets, then

(i) (a) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(b) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

(ii) $A \times (B - C) = (A \times B) - (A \times C)$

(iii) $A \times B = B \times A \Leftrightarrow A = B$

(iv) If $A \subseteq B \Rightarrow A \times A \subseteq (A \times B) \cap (B \times A)$ and $A \times C \subseteq B \times C$ for any set C.

(v) $A \times (B' \cup C')' = (A \times B) \cap (A \times C)$

(vi) $A \times (B' \cap C')' = (A \times B) \cup (A \times C)$

If A, B, C, D are four sets, then

(i) $A \subseteq B$ and $C \subseteq D \Rightarrow A \times C \subseteq B \times D$

(ii) $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$

Note If A and B have n common elements, then $A \times B$ and $B \times A$ will have n^2 common elements.

☞ RELATIONS

Let A and B be two nonempty sets. Then, a **relation R from A to B** is a subset of $A \times B$.

Thus, a relation R from A to $B \Rightarrow R \subseteq A \times B$ and $(a, b) \in R$, which we write as aRb and read as "a is related to b by R".

Let R be a relation between two sets $A = \{1, 2, 3\}$ and $B = \{1, 4, 9\}$ defined as "square root of", which we then write as

$$1R1, 2R4, 3R9$$

$\therefore$ $$R = \{(1, 1), (2, 4), (3, 9)\}$$

☞ DOMAIN AND RANGE OF RELATIONS

If R is a relation from a set A to a set B, then the set of all the first components or coordinates of the ordered pairs belonging to R is called the domain of R, while the set of

all the second components or coordinates of the ordered pairs in R is called the range of R. Thus,

Domain $(R) = \{a: (a, b) \in R\}$

Range $(R) = \{b: (a, b) \in R\}$.

Note Let A and B be two nonempty finite sets having p and q elements, respectively. Then the total number of relations from A to $B = 2^{pq}$.

CO-DOMAIN OF A RELATION

If R is a relation from A to B, then B is called the co-domain of relation R.

Let $A = \{2, 3, 5\}$ and $B = \{4, 7, 10, 8\}$.

Let $aRb \Leftrightarrow a$ divides b.

Then, $R = \{(2, 4), (2, 10), (2, 8), (5, 10)\}$.

Here, domain of $R = \{2, 5\}$ and range of $R = \{4, 10, 8\}$ and co-domain of $R = B = \{4, 7, 10, 8\}$.

INVERSE RELATION

Let A and B be two sets and let R be a relation from A to B. Then, the inverse of R, denoted by R^{-1}, is a relation from B to A obtained from R by interchanging the components of ordered pairs in R.

Thus, $R^{-1} = \{(b, a); (a, b) \in R\}$

Clearly, domain (R) = range (R^{-1})

and range (R) = domain (R^{-1}).

RELATION ON A SET

Let A be a nonempty set. Then, a relation from A to itself, i.e. a subset of $A \times A$, is called a relation on A, instead of being called a relation from A to A.

Illustration 12. Let $A = \{1, 2, 3, \ldots, 10\}$ and $R = \{(x, y): x + 2y = 10$, and $x, y \in A\}$ be a relation on A. Then, $R^{-1} =$

(a) $\{(2, 4), (4, 3), (6, 2), (8, 1)\}$

(b) $\{(4, 2), (3, 4), (2, 6), (1, 8)\}$

(c) $\{(4, 2), (3, 4), (2, 6)\}$

(d) $\{(x, y): x + 2y = 10,\ x, y \in A\}$

Ans. (b)

Solution: We have $x, y \in A \Leftrightarrow x + 2y = 10 \Leftrightarrow y = \dfrac{10 - x}{2}$.

We observe that

$x = 2 \Rightarrow y = 4$, $x = 4 \Rightarrow y = 3$,

$x = 6 \Rightarrow y = 2$, $x = 8 \Rightarrow y = 1$

Also, $y \notin A$ for $x = 1, 3, 5, 7, 9, 10$.

$\therefore$ $R = \{(2, 4), (4, 3), (2, 6), (8, 1)\}$

Hence, $R^{-1} = \{(4, 2), (3, 4), (2, 6), (1, 8)\}$

Also, $R^{-1} = \{(y, x): x + 2y = 10$ and $x, y \in A\}$

or $R = \{(x, y): 2x + y = 10$ and $x, y \in A\}$

TYPES OF RELATIONS

1. Void Relation

Let A be a nonempty set and $\phi \subseteq A \times A$. It follows that ϕ is a relation on A which is called the empty or void relation.

2. Universal Relation

Let A be a nonempty set and $A \times A \subseteq A \times A$. It follows that $A \times A$ is a relation on A which is called the universal relation.

3. Identity Relation

The relation I_A on set A is called the identity relation if every element of A is related to itself only. Symbolically, the relation $I_A = \{(a, a): a \in A\}$ is called the identity relation on A.

If $A = \{1, 2, 3, 4\}$, then the identity relation on A is given by $I_A = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$.

4. Reflexive Relation

R is a reflexive relation if $(a, a) \in R$, for all $a \in A$. It should be noted if there exists any $a \in A$ such that $(a, a) \notin R$, then R is not reflexive.

Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2)\}$

Then R is not reflexive since $3 \in A$ but $(3, 3) \notin R$.

Note

- The identity relation is always a reflexive relation but its converse need not be true.
- The universal relation on a non-void set A is reflexive.

5. Symmetric Relation

A relation R is called symmetric on a set A iff $(x, y) \in R \Rightarrow (y, x) \in R$. That is, if x is related to y, then y is also related to x.

Let $A = \{2, 3, 4\}$, $R_1 = \{(2, 3), (3, 2)\}$

and $R_2 = \{(2, 3), (3, 2), (3, 4)\}$

Here R_1 is a symmetric relation on A but R_2 is not a symmetric relation on A because $(3, 4) \in R_2$ but $(4, 3) \notin R_2$.

6. Anti-symmetric Relation

A relation R on a set A is called anti-symmetric if $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$. Thus, if $a \neq b$, then a may be related to b or b may be related to a, but never both.

Let N be the set of natural numbers. A relation $R \subseteq N \times N$ is defined by xRy, iff x divides y, i.e. (x/y).

Then, $xRy, yRx \Rightarrow x$ divides y, y divides x.

Note

- A relation which is not symmetric is not necessarily anti-symmetric.
- The identity relation on a set A is an anti-symmetric relation.

7. Transitive Relation

A relation R on a set A is called transitive, if $(a, b) \in R$, $(b, c) \in R \Rightarrow (a, c) \in R$.

In other words, if a is related to b and b is related to c, then a is related to c.

Let $A = \{4, 5, 6\}$, $R = \{(4, 5), (5, 4), (4, 4)\}$ and

$R_1 = \{(5, 5), (5, 6), (6, 5), (6, 6)\}$

Then, R is not transitive since $(5, 4) \in R$, $(4, 5) \in R$ but $(5, 5) \in R$.

But R_1 is transitive. Since $(5, 6) \in R_1$, $(6, 5) \in R_1$ then $(5, 5) \in R_1$. Also $(6, 5) \in R_1$, $(5, 6) \in R_1$ then $(6, 6) \in R_1$.

☞ EQUIVALENCE RELATION

A relation R on a set A is called an equivalence relation, if

(i) R is reflexive, i.e., $(a, a) \in R$, for all $a \in A$

(ii) R is symmetric, i.e., $(a, b) \in R \Rightarrow (b, a) \in R$

(iii) R is transitive, i.e., $(a, b) \in R$, $(b, c) \in R \Rightarrow (a, c) \in R$

Let $A = \{1, 2, 3\}$, and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$.

Here, it is easy to verify that R is reflexive, symmetric and transitive. So, R is an equivalence relation.

Note

- The intersection of two equivalence relations on a set is an equivalence relation on the set.
- The union of two equivalence relations on a set is not necessarily an equivalence relation on the set.
- The inverse of an equivalence relation on a set is also an equivalence relation on the set.
- A relation R on a set A is symmetric iff $R = R^{-1}$.

☞ PARTIAL ORDER RELATION

A relation R on a set A is said to be a partial order if R is simultaneously reflexive, transitive and anti-symmetric on A.

Let N be the set of all natural numbers. Let a relation R be defined on N by $R = \{(a, b): a, b \in N \text{ and } a \le b\}$.

Then R is reflexive because $a \le a$ for all $a, b, c \in N$. R is transitive because $a \le b$ and $b \le c \Rightarrow a \le c$, for all $a, b, c \in N$.

R is anti-symmetric because $a \le b$ and $b \le a \Rightarrow a = b \in N$. Thus, R is a partial order relation.

Note

- A partial order relation is not an equivalence relation on A because $a \le b$ does not imply $b \le a$.
- A nonempty set A together with a partial order relation R (also called partial ordering) is said to be a **partially ordered set**.

☞ TOTAL ORDER RELATION

A relation R on a set A is said to be a total order if R is a partial order relation on A such that given any $x, y \in A$, we must have either xRy or yRx.

Let S be a given nonempty set. We define a relation R on power set $P(S)$ by $XRY \Rightarrow X \in Y$ for all $X, Y \in P(S)$.

Clearly, R is reflexive, anti-symmetric and transitive. So R is a partial order relation. But it is not a total order relation because if $S = \{1, 2, 3\}$, then $\{2\}, \{3\} \in P(S)$ but neither $\{2\} \in \{3\}$ nor $\{3\} \in \{2\}$.

Note A nonempty set A together with a total order relation R is called a **totally ordered** set or chain.

☞ EQUIVALENCE CLASSES OF AN EQUIVALENCE RELATION

Let R be an equivalence relation on the nonempty set A. Let $a \in A$. Then the equivalence class of each element a of A denoted by $[a]$ or $[\bar{a}]$, is the set of all those elements of A which are related to a under the relation R. Thus $[a] = \{x \in A: xRa\}$. It is easy to see that

(i) $b \in [a] \Rightarrow a \in [b]$

(ii) $b \in [a] \Rightarrow a = [b]$

Two equivalence classes are therefore either symmetric or identical. As an example, we consider a very important relation $x = y \pmod n$. If and only if n divides $(x - y)$, n is a fixed positive integer.

Consider $n = 5$, then

$[0] = \{x : x \equiv (\text{mod } 5)\} = \{5p : p \in I\} = \{\pm 5, \pm 10, \pm 15, \ldots\}$

$[1] = \{x : x \equiv (\text{mod } 5)\} = \{x : x - 1 = 5k \text{ and } k \in I\}$

$= \{5k + 1 : k \in I\} = \{1, 6, 11, \ldots, -4, -9, \ldots\}$

One can easily see that there are only 5 distinct equivalence classes, viz. [0], [1], [2], [3] and [4] when $n = 5$.

☞ COMPOSITION OF RELATIONS

Let $R \subseteq A \times B$ and $S \subseteq B \times C$ be two relations. Then, we can define a new relation, the composition of R and S, denoted by $S \circ R \subseteq A \times C$, and defined by $(a, c) \in S \circ R$, iff for some $b \in B$, we have $(a, b) \in R$, $(b, c) \in S$.

Let $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$, $C = \{\alpha, \beta, \gamma\}$

$R \subseteq (A \times B) = \{(1, a), (1, c), (2, d)\}$

$S \subseteq (B \times C) = \{(a, \alpha), (c, \gamma), (d, \beta)\}$

Then, $S \circ R \subseteq A \times C = \{(1, \alpha), (1, \gamma), (2, \beta)\}$

Note

- One should be careful in computing the relation $R \circ S$. Actually $S \circ R$ starts with R and $R \circ S$ starts with S. In general, $S \circ R \ne R \circ S$.
- $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$ is known as reversal rule.

FUNCTIONS

FUNCTIONS OR MAPPINGS

Let A and B be nonempty sets. Then, a function f from A to B is a rule which associates the elements of set A with the elements of set B such that all the elements of set A are associated with the elements of set B in a unique way.

If f associates $x \in A$ to $y \in B$, then we say that y is the image of the element x and denote it by $f(x)$ and write as $y = f(x)$. The element x is called the pre-image or inverse image of y in B. A function f from A to B is denoted by $f: A \to B$.

REAL FUNCTION

If $f: A \to B$ be a function from a nonempty set A to another nonempty set B, where $A, B \subseteq \mathbb{R}$ (set of all real numbers), then we say that f is a real-valued function or in short a real function.

FEATURES OF A MAPPING $f: A \to B$

1. For each element $x \in A$, there exists a unique element $y \in B$.
2. The element $y \in B$ is called the image of $x \in A$ under the mapping f.
3. If there is an element in A which has more than one image in B, then $f: A \to B$ is not a function. But distinct elements of A may be associated with the same element of B.
4. If there is an element in A which does not have an image in B, then $f: A \to B$ is not a function.

Note Throughout this chapter, a "function" will mean a "real function".

Illustration 13. If $A = \{1, 2, 3\}$, and $B = \{a, b, c, d, e\}$, then $f(1) = a, f(2) = b, f(3) = c$ gives a map $f: A \to B$ (Fig. 1.13).

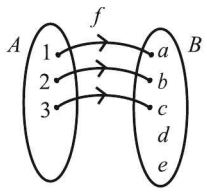


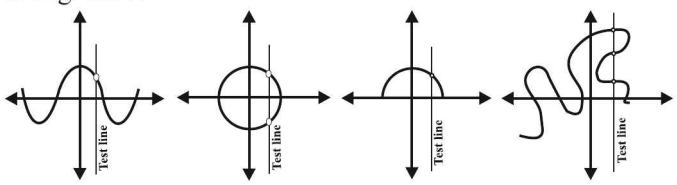
Fig. 1.13 Map $f: A \to B$.

DIFFERENCE BETWEEN A RELATION AND A FUNCTION

	Relation R	Function f
Domain	Domain of R may be a subset of A.	Domain of f is equal to set A.
Association	An element of A may be related to more than one element of B.	One or more elements of A are associated to one and only one element of B.

Geometrical Test for a Relation to be a Function

Geometrically, if we draw a vertical parallel line (VPL), i.e. any line which is parallel to the y-axis ($x = a$), then if this line intersects the graph of the expression in more than one point, then the expression is a relation, else if it intersects at only one point, the expression is a function, This is shown in Fig. 1.14.



(a) Function (b) Relation (c) Relation (d) Function

Fig. 1.14 Geometrical test for a relation to be a function.

Reasoning: We know that only one-to-one and many-to-one correspondences represent a function. Also many-to-many and one-to-many correspondences do not represent a function. This is what VPL test checks for: By drawing a vertical parallel line, we are actually keeping the x part the same throughout for different values of y and if this line cuts or intersects the graph in more than one point, then for all different points we get all those different values of y for the same x, i.e. same values of x give different values of y (images) which is one-to-many.

VALUE OF A FUNCTION

The value of a function $y = f(x)$ at $x = a$ is denoted by $f(a)$. It is obtained by putting $x = a$ in $f(x)$.

Illustration 14. If $f(x) = \log\left(\frac{1-x}{1+x}\right)$, $x \in (-1,1)$, then

$$f(0) = [f(x)]_{x=0} = \log\left(\frac{1-0}{1+0}\right) = \log 1 = 0$$

$$f\left(\frac{1}{2}\right) = \log\left[\frac{1-(1/2)}{1+(1/2)}\right] = \log\frac{1}{3}, \text{ etc.}$$

Note If for some value of x, say $x = a$, the denominator vanishes, we say $f(a)$ is undefined (or does not exist).

DOMAIN AND RANGE OF A FUNCTION

If $f: X \to Y$ be a function, then the set X is said to be the domain of f. The range of f

= set of all image points in Y under the map f.

$= f(X) = \{f(x) : f(x) \in Y \text{ and } x \in X\}$

The set Y is also called the co-domain of f.

Clearly, $f(X) \subseteq Y$.

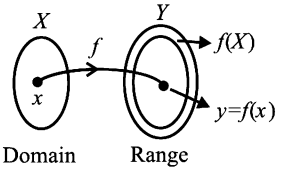


Fig. 1.15 Domain and range of a function.

Illustration 15. If $X = \{0, \pm1, \pm2\}$, $Y = \{0, 1, 2, 4\}$, then the rule $f: X \to Y$ given by $f(x) = x^2$ is a map from X to Y.
$\therefore$ Domain of $f = X$, range of $f = \{0, 1, 4\}$.

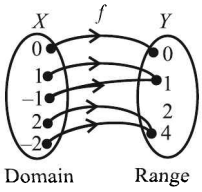


Fig. 1.16 Domain and range of f.

NUMBERS AND THEIR SETS

- Natural numbers: $N = \{1, 2, 3, 4, 5, \ldots\}$
- Whole numbers: $W = \{0, 1, 2, 3, \ldots\}$
- Integers: $Z = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$
  $= \{0, \pm1, \pm2, \pm3, \ldots\}$
- Rational numbers: $Q = \left\{\frac{p}{q} : p, q \in Z, \text{ and } q \neq 0\right\}$
- Irrational numbers: The numbers which are not rational or which cannot be written in the form of p/q are called irrational numbers, i.e. $\{\sqrt{3}, 2^{1/3}, 5^{1/4}, \pi, e, \ldots\}$
- Real numbers: $\{x$, where x is a rational or irrational number$\}$ $\mathbb{R} = \{1, 1000, 20/6, \pi, \sqrt{2}, -10, -20/3, \ldots\}$, i.e. set of real number is $(-\infty, \infty)$
- Positive real numbers: $\mathbb{R}^+ = (0, \infty)$
- Negative real numbers: $\mathbb{R}^- = (-\infty, 0)$
- $\mathbb{R}_0 = (0, \infty)$: all real numbers except 0 (zero) $= \mathbb{R} - \{0\}$
- Imaginary numbers: $a \pm \sqrt{-k}$, where $a \in \mathbb{R}$ and $k \in \mathbb{R}$.

TYPES OF FUNCTIONS

1. One-one function (injective function)
2. Many-one function
3. Onto (surjective) function
4. Into function
5. One-one and onto (bijective) function

1. One-one Function (Injective Function)

A function $f: A \to B$ is called a one-one function (mapping), if distinct elements of A have distinct image in B. Such a function is also known as an injective function or an injection or monomorphism.

Methods to test one-one function

(a) **Analytically:** If $x_1, x_2 \in A$, then $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ or equivalently, $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.

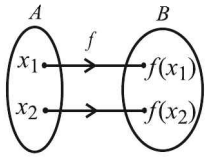


Fig. 1.17 One-one function.

(b) **Graphically:** If any line parallel to the x-axis cuts the graph of the function at most at one point, then the function is one-one.

(c) **Monotonically:** If any function is entirely increasing or decreasing in whole domain, then $f(x)$ is one-one.

Note To find one-one or injective mapping, we mainly use the graphic or monotonic method.

Illustration 16. Prove that the map $f: A \to B$ given by $f(x) = 2x + 5$ is one-one.

Solution: Let us consider set $A = \{-1, 2, 4\}$, and then $B = \{3, 9, 13)$, i.e. two different values of x do not have the same value of $f(x)$. Hence, $f(x)$ is one-one.

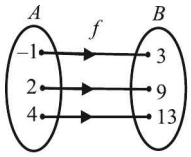


Fig. 1.18 An example of a one-one function.

I. ***Alternative Method (Analytically)***

Let $x_1, x_2 \in A$, then $f(x_1) = f(x_2)$
$\Rightarrow \quad 2x_1 + 5 = 2x_2 + 5 \quad \Rightarrow \quad 2x_1 = 2x_2$
$\Rightarrow \quad x_1 = x_2$ or if $x_1 \neq x_2$
$\Rightarrow \quad 2x_1 \neq 2x_2 \quad \Rightarrow \quad 2x_1 + 5 \neq 2x_2 + 5$
$\Rightarrow \quad f(x_1) \neq f(x_2)$
Hence $f(x)$ is one-one.

II. ***Alternative Method (Graphically)***

First draw the graph of $y = 2x + 5$ as shown in Figure 1.19.

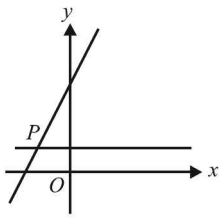


Fig. 1.19 One-one function (graphically).

Since any line parallel to the x-axis cuts the graph at one point only, say P, hence $f(x)$ is one-one.

III. ***Alternative Method (Monotonically)***

Since $f(x) = 2x + 5 \quad \therefore \quad f'(x) = 2 > 0$
$\Rightarrow$ $f(x)$ is strictly an increasing function.
Hence $f(x)$ is one-one.

Geometrical test for injectivity: HPL test

Geometrically, if we draw a horizontal parallel line (HPL), i.e. any line which is parallel to the x-axis ($y = a$), then if this line intersects the graph of the function in at least two points then the function is not a one-to-one function, else if it intersects at only one point, the function is one-to-one, i.e. injective.

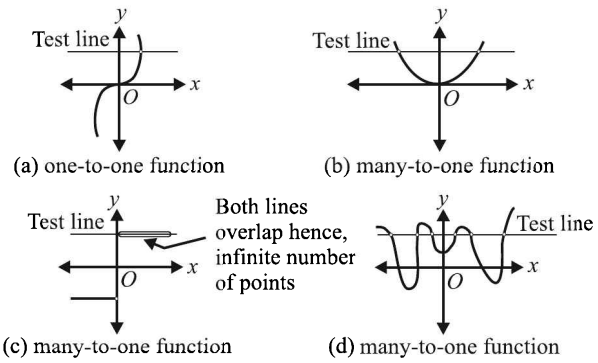


Fig. 1.20 One-to-one and many-to-many functions.

Reasoning: By drawing the horizontal parallel line, we are actually keeping the y part the same throughout for different values of x and if this line cuts or intersects the graph of the curve in more than one point, then for all different points we get all those different x at which this same value of y is obtained, i.e. different values of x and same value of y, which is not one-to-one, i.e. injective.

2. Many-one Function

A function f: $A \to B$ is called a many-one function if there exists at least two distinct elements in A whose image is same in B.

Methods to test many-one function

(a) **Analytically:** If $x_1, x_2 \in A$, then $f(x_1) = f(x_2) \Rightarrow x_1 \neq x_2$

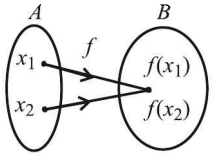


Fig. 1.21 Many-one function.

(b) **Graphically:** If any line parallel to the x-axis cuts the graph of the function at least at two points, then the function is many-one.

(c) **Monotonically:** If any function is neither increasing nor decreasing in whole domain, then $f(x)$ is many-one. Or any continuous function $f(x)$ which has at least one local maxima or local minima, is many-one.

Illustration 17. Prove that the map f: $A \to B$ given by $f(x) = x^2 + x + 1$ for all $x \in \mathbb{R}$ is many-one.

Solution: Let us consider set $A = \{-1, 0, 2\}$, and then $B = \{1, 7\}$. Here for two different values of x, $f(x)$ has the same value. Hence $f(x)$ is many-one.

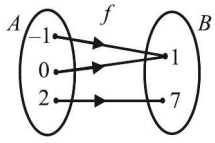


Fig. 1.22 Many-one function.

I. ***Alternative Method (Analytically)***

Let $x_1, x_2 \in A$

Then $f(x_1) = f(x_2) \Rightarrow x_1^2 + x_1 + 1 = x_2^2 + x_2 + 1$

$\Rightarrow x_1^2 - x_2^2 + x_1 - x_2 = 0 \Rightarrow (x_1 - x_2)(x_1 + x_2 + 1) = 0$

$\Rightarrow x_1 \neq x_2 \quad [\because x_1 = -1, x_2 = 0]$

Hence $f(x)$ is many-one.

II. ***Alternative Method (Graphically)***

Let $y = f(x) = x^2 + x + 1$

$$\Rightarrow \quad y = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \quad \text{or} \quad \left(x + \frac{1}{2}\right)^2 = \left(y - \frac{3}{4}\right)$$

which is a parabola whose vertex is at $\left(-\frac{1}{2}, \frac{3}{4}\right)$.

It cuts the y-axis at (0, 1) and neither cuts nor touches the x-axis. Graph of $y = x^2 + x + 1$ is as shown in Fig. 1.23.

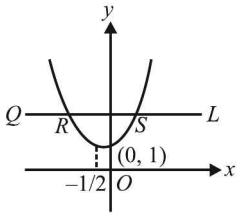


Fig. 1.23 Graph of $y = x^2 + x + 1$.

Since any line QL parallel to the x-axis cuts the graph at two points R and S, hence $f(x)$ is many-one.

III. ***Alternative Method (Monotonically)***

Since $f(x) = x^2 + x + 1 \quad \therefore \quad f'(x) = 2x + 1$

$\Rightarrow \quad f'(x) > 0$ if $x > -1/2$ and $f'(x) < 0$ if $x < -1/2$

Therefore, $f(x)$ is neither increasing nor decreasing and continuous everywhere which has the minimum value 3/4 at $x = (-1/2)$. Hence, $f(x)$ is many-one.

3. Onto Function (Surjective Function)

A function f: $A \to B$ is said to be onto or surjective if every element of B is an image of some element of A under f, i.e. for every $b \in B$, there exists an element a in A such that $f(a) = b$ and $B = f(A)$, where the range of f is the co-domain of f, i.e. set B. Thus, for f: $A \to B$, every element of B is the image of some element of A, i.e. if $B = f(A)$, then the range of f is the co-domain of f.

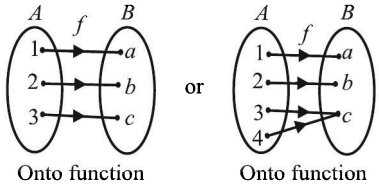


Fig. 1.24 Onto functions.

Methods to test onto functions

Let $A \xrightarrow{f} B$.

(i) Take a point (element) y of the domain B.

(ii) Put down $f(x) = y$

(iii) Solve $f(x) = y$ for x

(iv) Let $x = g(y)$. If for all $y \in B$, $x = g(y)$, then the function is said to be onto.

Note In onto functions every element in B has pre-image(s) in A.

Let $A = \{-1, 1, 2\}$, $B = \{1, 4\}$ and $f: A \to B$ be a function defined by $f(x) = x^2$. Then,

(i) the pre-image of 1 is -1 and 1.

(ii) the pre-image of 4 is 2 and there is not any other element in B, which does not have a pre-image in A. So, the function is onto.

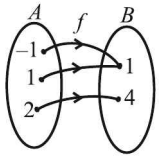


Fig. 1.25 Onto function.

Note In case A and B are the sets of real numbers, then f is many-one and onto.

4. Into Function (Not Onto Function)

$f: A \to B$ is called an into function if there exists any element in B having no pre-image in A.

Thus, $f: A \to B$ is an into function, as shown in Fig. 1.26.

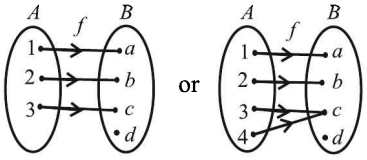


Fig. 1.26 Into functions.

5. One-one and Onto (Bijective) Function

A one-one and onto function is said to be bijective if all the elements of A are joined to all the elements of B distinctly and no element is left vacant. In other words, a function $f: A \to B$ is bijective, if

(a) it is one-one, i.e. $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for all x_1, $x_2 \in A$.

(b) it is onto, i.e. for all $y \in B$, there exists $x \in A$ such that $f(x) = y$.

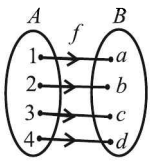


Fig. 1.27 One-one and onto function.

IDENTITY FUNCTION

Let A be a nonempty set. Then, the function, defined by $I_A: A \to A$, $I_A(x) = x$ for all $x \in A$, is called the identity function on A. This is clearly a one-one and onto function with domain A and range A.

EQUAL FUNCTIONS

Two functions f and g are said to be equal, written as $f = g$, if they have the same domain and they satisfy the condition $f(x) = g(x)$ for all x.

Illustration 18. Let $f: [0, \infty) \to [0, 2]$ be defined by $f(x) = 2x/(1 + x)$, then f is

(a) one-one but not onto (b) onto but not one-one

(c) both one-one and onto (d) neither one-one nor onto

Ans. (a)

Solution: Range $\subset$ Co-domain and $f(x_1) = f(x_2)$

$$\Rightarrow \quad \frac{2x_1}{1+x_1} = \frac{2x_2}{1+x_2} \quad \Rightarrow \quad 2x_1 + 2x_1x_2 = 2x_2 + 2x_1x_2$$

$\Rightarrow \quad x_1 = x_2 \qquad \therefore \quad f$ is one-one.

Now, let $y = f(x) = [2x/(1 + x)]$

$\Rightarrow \quad y + xy = 2x \qquad \Rightarrow \quad y = 2x - xy = x(2 - y)$

$\Rightarrow \quad x = y/(2 - y) \qquad y \in [0, 2]$

at $y = 2$, no real value of x exists. So, $f(x) = [2x/(1 + x)]$ is not onto.

INVERSE OF AN ELEMENT

Let f be a function A to B and let $b \in B$. Then, the inverse image of the element b under f is denoted by $f^{-1}(b)$. It consists of those elements in A which have b as their f-image. In symbolic form, we can write

If $f: A \to B$, then $f^{-1}(b) = \{x : x \in A \text{ and } f(x) = b\}$

f^{-1} is read as "f inverse" and $f^{-1}(b)$ is always a subset of A.

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{a, b\}$

Let $f: A \to B$ be defined by

$f = \{(1, a), (2, a), (3, b), (4, a), (5, b)\}$

$f^{-1}(b) = \{3, 5\}$, since 3 and 5 have b in their f-image.

Also, $f^{-1}(a) = \{1, 2, 4\}$.

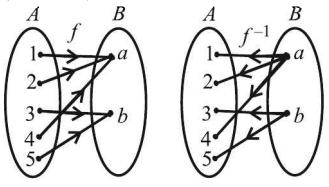


Fig. 1.28 Inverse of elements a and b.

INVERSE OF A FUNCTION

If f is a mapping from set A to set B, then the set A is called the domain and the set B is called the co-domain of mapping f.

Now, if $f: A \to B$ is bijective, i.e. both one-to-one and onto then the mapping $f^{-1}: B \to A$ is called the inverse mapping of f.

Reasoning: Let A and B be two sets and let $f: A \to B$ be a function. If we follow a rule in which the elements of B are associated to their pre-images, then we find that under such a rule there may be some elements in B which are not associated to elements in A. This happens when f is not an onto map. Therefore, all elements in B will be associated to some elements in A if f is an onto map. Also, if it is a many-one function then under the said rule an element in B may be associated to more than one element in A. Therefore, an element in B will be associated to a unique element in A if f is an injective map. If follows from the above discussion that if $f: A \to B$ is a bijection, we can define a new function from B to A which associates each element $y \in B$ to pre-image $f^{-1}(y) \in A$. Such a function is known as the inverse of function f and is denoted by f^{-1}. See Fig. 1.29.

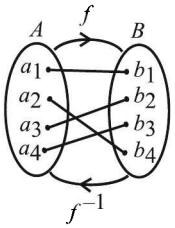


Fig. 1.29 Inverse function.

Note The graph of an inverse function is symmetric about the line $y = x$.

METHOD TO FIND THE INVERSE OF A FUNCTION

(i) Take $f(x_1) = f(x_2)$ and show that $x_1 = x_2$, i.e. to show that f is a one-one function.

(ii) Define f^{-1} by assuming $f^{-1}(x) = c$, where x is a subset of co-domain.

(iii) As $f^{-1}(x) = c \Rightarrow x = f(c)$, solve this equation for c in terms of the variable term (i.e. x).

If $f(x) = 3x + 4$ for all $x \in \mathbb{R}$, then for $f^{-1}(x)$ we take three steps.

Step 1. Let $x_1, x_2 \in \mathbb{R}$, $f(x_1) = f(x_2)$

$\Rightarrow \quad 3x_1 + 4 = 3x_2 + 4 \quad \Rightarrow \quad x_1 = x_2$

$\therefore$ $f(x)$ is one-one.

Step 2. Let $c \in \mathbb{R}$

$\therefore$ $f(x) = c$

$\Rightarrow \quad 3x + 4 = c \quad \therefore \quad x = \dfrac{c-4}{3} \in \mathbb{R} \quad \therefore$ $f(x)$ is onto.

Step 3. Now for $x \in \mathbb{R}$, let $f^{-1}(x) = c$

$\therefore \quad f(c) = x \quad \Rightarrow \quad c = \dfrac{x-4}{3} \quad \Rightarrow \quad f^{-1}(x) = \dfrac{x-4}{3}$

PROPERTIES OF INVERSE FUNCTION

(i) The inverse of a bijective function is unique and bijective.

(ii) Let $f: A \to B$ be a function such that f is bijective and $g: B \to A$ is inverse of f, then
$f \circ g = I_B$ = Identity function of set B
$g \circ f = I_A$ = Identity function of set A

(iii) If $f \circ g = g \circ f$ then either $f^{-1} = g$ or $g^{-1} = f$ and $(f \circ g)(x) = (g \circ f)(x) = x$.

(iv) If f and g are two bijective functions such that $f: A \to B$ and $g: B \to C$ then $g \circ f: A \to C$ is bijective. Also, $(g \circ f)^{-1} = (f^{-1} \circ g^{-1})$.

COMPOSITION OF FUNCTIONS

Let A, B and C be three non-empty sets. Let $f: A \to B$ and $g: B \to C$ be two mappings (or functions) then $g \circ f: A \to C$. This function is called the product or composite of f and g given by

$$g \circ f(x) = g[f(x)] \text{ for all } x \in A$$

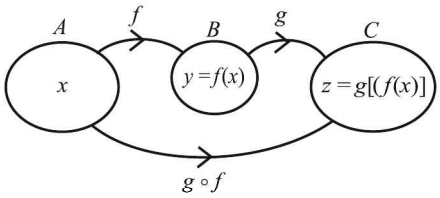


Fig. 1.30 Composite of f and g.

$g \circ f$ exists iff the range of f is a subset of the domain of g. Similarly, $f \circ g$ exists if the range of g is a subset of the domain of f.

Note

- The $g \circ f$ is defined only if for all $x \in A$, $f(x)$ is an element of the domain of g so that we can take its g-image.
- The range of f must be a subset of the domain of g in $g \circ f$.
- (i) $(f \circ g)(x) = f(g(x))$
  (ii) $(f \circ f)(x) = f(f(x))$
  (iii) $(g \circ g)(x) = g(g(x))$
  (iv) $(f \pm g)(x) = f(x) \pm g(x)$
  (v) $\left(\dfrac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}$; $g(x) \neq 0$

PROPERTIES OF COMPOSITE FUNCTIONS

(i) The composition of a function is not commutative, i.e.

$$f \circ g \neq g \circ f$$

(ii) The composition of a function is associative, i.e.

$$f \circ (g \circ h) = (f \circ g) \circ h$$

(iii) The composition of any function with the identity function is the function itself, i.e. if $f: A \to B$, then

$$f \circ I_A = I_B \circ f = f$$

(vi) Let $f: A \to B$, $g: B \to C$ be two functions, then
(a) f and g are injective $\Rightarrow g \circ f$ is injective
(b) f and g are surjective $\Rightarrow g \circ f$ is surjective
(c) f and g are bijective $\Rightarrow g \circ f$ is bijective

(v) An injective mapping from a finite set to itself is bijective.

Symbol	Meaning
N	set of natural numbers (all the +ve integers)
Z	set of all the integers
Q	set of rational numbers
$\mathbb{R}$	set of real numbers
$\mathbb{C}$	set of complex numbers
$\in$	is an element of (or belongs to)
$\notin$	is not an element of (or does not belong to)

Symbol	Meaning
U or ξ	universal set
: or /	such that
ϕ	empty set or null set
$\subseteq$	is a subset of
$\supseteq$	is the superset of
$\subset$	is the proper subset of
$\supset$	is the proper superset of
$\cup$	union
$\cap$	intersection
$-$	difference
Δ	symmetric difference
$\wedge$	and
$\vee$	or
$\forall$	for all (or for every)
$\exists$	there exists
$\sim$	not
$\Rightarrow$	implies
$\Leftrightarrow$	if and only if implies and is implied by

SOLVED PROBLEMS

1. If $A = \{2, 3, 4, 8, 10\}$, $B = \{3, 4, 5, 10, 12\}$ and $C = \{4, 5, 6, 12, 14\}$, then $(A \cap B) \cup (A \cap C)$ is equal to
(a) $\{3, 4, 10\}$ (b) $\{2, 8, 10\}$
(c) $\{4, 5, 6\}$ (d) $\{3, 5, 14\}$

Ans. (a)

Solution:
$A \cap B = \{2, 3, 4, 8, 10\} \cap \{3, 4, 5, 10, 12\} = \{3, 4, 10\}$
$A \cap C = \{4\}$
$\therefore \quad (A \cap B) \cup (A \cap C) = \{3, 4, 10\}$

2. If $aN = \{ax : x \in N\}$, then the set $3N \cap 7N$ is
(a) $21N$ (b) $10N$
(c) $4N$ (d) none of these

Ans. (a)

Solution: $3N = \{x \in N : x \text{ is a multiple of } 3\}$
$7N = \{x \in N : x \text{ is a multiple of } 7\}$
$\therefore \quad 3N \cap 7N = \{x \in N : x \text{ is a multiple of 3 and 7}\}$
$= \{x \in N : x \text{ is a multiple of } 21\} = 21N$

3. Suppose $n(A) = 3$, $n(B) = 6$ and $A \subseteq B$. Then, the number of elements in $A \cup B$ is equal to
(a) 3 (b) 9
(c) 6 (d) none of these

Ans. (c)

Solution: As $A \subseteq B$, $\quad \therefore \quad A \cup B = B$
So, $n(A \cup B) = n(B) = 6$.

4. A class has 175 students. The following data shows the number of students opting for one or more subjects: Mathematics 100, physics 70, chemistry 40, mathematics and physics 30, mathematics and chemistry 28, physics and chemistry 23, mathematics, physics and chemistry 18. How many students opted for mathematics alone?
(a) 35 (b) 48
(c) 60 (d) 22

Ans. (c)

Solution: $n(M \text{ alone}) = n(M) - n(M \cap C) - n(M \cap P) + n(M \cap P \cap C)$
$= 100 - 28 - 30 + 18 = 60$

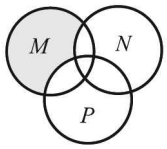


Fig. 1.31

5. Given $n(U) = 20$, $n(A) = 12$, $n(B) = 9$, $n(A \cap B) = 4$, where U is the universal set, A and B are subsets of U, then $n((A \cup B)')$ is equal to
(a) 17 (b) 9
(c) 11 (d) 3

Ans. (d)

Solution: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 12 + 9 - 4 = 17$
Now, $n((A \cup B)') = n(U) - n(A \cup B) = 20 - 17 = 3$

6. If $A_1 = \{1, 2, 3, \ldots, 20\}$, $A_2 = \{a, b, c, d\}$, $A_3 = \{b, d, e, f\}$, then the number of elements of $(A_1 \times A_2) \cup (A_1 \times A_3)$ is
(a) 100 (b) 120
(c) 140 (d) 40

Ans. (b)

Solution: $A_1 \times A_2$ has $20 \times 4 = 80$ elements
$A_1 \times A_3$ has $20 \times 4 = 80$ elements
Number of common elements $= 20 \times 2 = 40$
[$\because$ b and d are common elements]
$\therefore$ Number of elements of $(A_1 \times A_2) \cup (A_1 \times A_3)$
$= 80 + 80 - 40 = 160 - 40 = 120$

7. Total number of equivalence relations defined in the set $A = \{a, b, c\}$ is
(a) 5 (b) 3!
(c) 2^3 (d) 3^3

Ans. (a)

Solution: The smallest equivalence relation is the identity relation $R_1 = \{(a, a), (b, b), (c, c)\}$
Then, two ordered pairs of two distinct elements can be added to give three more equivalence relations
$R_2 = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$
Similarly, R_3 and R_4.
Finally, the largest equivalence relation, that is, the universal relation

$$R_5 = \begin{Bmatrix} (a, a), (b, b), (c, c), (a, b), (b, a), \\ (a, c), (c, a), (b, c), (c, b) \end{Bmatrix}$$

8. Let n be a fixed positive integer. Let a relation R be defined in Z (the set of all integers) as follows: aRb iff $n/(a - b)$, that is, iff $a - b$ is divisible by n. Then, the relation R is
(a) reflexive only (b) symmetric only
(c) transitive only (d) an equivalence relation

Ans. (d)

Solution: R is reflexive since for any integer a we have $a - a = 0$ divisible by n. Hence, aRa for all $a \in Z$. If R is symmetric, let aRb. Then, by definition of R, $a - b = nk$, where $k \in Z$. $b - a = (-k)n$, where $-k \in Z$ and so bRa. Thus, we have $a - b = k_1 n$ and $b - c = k_2 n$, where $k_1 \in k_2 Z$. It then follows that $a - c = (a - b) + (b - c) = k_1 n + k_2 n = (k_1 + k_2)n$, where $k_1 + k_2 \in Z$.

9. Let $A = \{1, 2, 3, 4\}$ and R be a relation in A given by $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 1), (1, 3)\}$. Then, R is
(a) reflexive and symmetric (b) transitive
(c) an equivalence relation (d) none of these

Ans. (a)

Solution: $(1, 1), (2, 2), (3, 3), (4, 4) \in R$; so it is reflexive. $(1, 2), (2, 1) \in R$, so R is symmetric.

10. Let A be the set of all real numbers. Then, the relation $R = \{(a, b) : 1 + ab > 0\}$ on A is
(a) reflexive and symmetric but not transitive
(b) reflexive and transitive but not symmetric
(c) symmetric, transitive but not reflexive
(d) none of these

Ans. (a)

Solution: As $1 + a \cdot a = 1 + a^2 > 0$ for all $a \in A$,
$\therefore$ $(a, a) \in R$ is reflexive
Also $(a, b) \in R \Rightarrow 1 + ab > 0 \Rightarrow 1 + ba > 0$
$\Rightarrow$ $(b, a) \in R$
$\therefore$ $(a, b) \in R$ and $(b, c) \in R$ need not imply $(a, c) \in R$. Hence, R is not transitive.

11. Let R be a relation defined as aRb if $|a| \le b$. Then, the relation R is
(a) reflexive (b) symmetric
(c) transitive (d) none of these

Ans. (c)

Solution: R is not reflexive, if $-a$ is any negative real number, then $|-a| > -a$ so that $-a \not R -a$.
R is not symmetric. Consider the real numbers $a = -2$ and $b = 3$. Then, aRb as $|-2| < 3$. But $b \not R a$ as $|3| \le -2$.
R is transitive: let a, b, c be three real numbers such that $|a| \le b$ and $|b| \le c$. But $|a| \le b \Rightarrow b \ge 0$, and so $|b| \le c \Rightarrow b \le c$. So, it follows $|a| \le c$.

12. If $f = \{(1, 1), (2, 3), (3, 5), (4, 7)\}$ is a function described by the formula, $f(1) = k$, then $f(n)$, $n \in N$, is equal to
(a) $\alpha = 1, \beta = 1$ (b) $\alpha = 2, \beta = -1$
(c) $\alpha = 1, \beta = -2$ (d) $\alpha = -2, \beta = -1$

Ans. (b)

Solution: $(1, 1)$ satisfies $g(x) = \alpha x = \beta \Rightarrow \alpha + \beta = 1$
$(2, 3)$ satisfies $g(x) = \alpha x = \beta \Rightarrow 2\alpha + \beta = 3$
Solving the two equations, we get $\alpha = 2$, $\beta = -1$. It can be checked that ordered pairs satisfy $g(x) = 2x - 1$.

13. In a certain city, only two newspapers A and B are published. It is known that 25% of the city population reads A and 20% reads B, while 8% reads both A and B. It is also known that 30% of those who read A but not B, look into advertisements and 40% of those who read B but not A, look into advertisements, while 50% of those who read both A and B, look into advertisements. What % of the population reads advertisements?
(a) 12.9% (b) 13.9%
(c) 14.8% (d) 15.2%

Ans. (b)

Solution: Let L = set of people who read the paper A. Let M = set of people who read the paper B. Given,
$n(L) = 25$, $n(M) = 20$, $n(L \cap M) = 8$
Now, $n(L - M) = n(L) - n(L \cap M) = 25 - 8 = 17$
$n(M - L) = n(M) - n(L \cap M) = 20 - 8 = 12$
$\therefore$ % of people reading an advertisement
= (30% of 17) + (40% of 12) + (50% of 8)

$$= \frac{51}{10} + \frac{24}{5} + 4 = 13.9\%$$

14. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. How many persons like both coffee and tea?
(a) 19 (b) 20
(c) 21 (d) 22

Ans. (a)

Solution: Let A = set of people who like coffee. Let B = set of people who like tea.

Given, $n(A \cup B) = 70$, $n(A) = 37$, $n(B) = 52$.

To find $n(A \cup B)$:

$$n(A \cap B) = n(A) + n(B) - n(A \cup B)$$
$$= 37 + 52 - 70 = 19$$

15. In a statistical investigation of 1003 families of Kolkata, it was found that 63 families had neither a radio nor a TV, 794 families had a radio and 187 had a TV. The number of families in that group having both a radio and a TV is

(a) 36 (b) 41 (c) 32 (d) none of these

Ans. (b)

Solution: Let R be the set of families having a radio and T be the set of families having a TV, then $n(R \cup T)$ = the number of families having either a radio or a TV = 1003 – 63 = 940.

$n(R) = 794$ and $n(T) = 187$.

Suppose x families had both a radio and a TV, i.e. $n(R \cap T) = x$. The number of families who had only radio = $794 - x$ and the number of families who had only TV = $187 - x$.

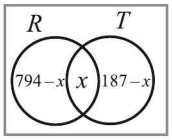


Fig. 1.32

From the Venn diagram, $(794 - x) + x + (187 - x) = 940$ or $x = 981 - 940 = 41$.

Hence, the required number of families having both a radio and a TV = 41.

16. In a group of 1000 people, there are 750 people who can speak Hindi and 400 who can speak English. Then the number of persons who can speak Hindi only is

(a) 300 (b) 400 (c) 600 (d) none of these

Ans. (c)

Solution: Here $n(H \cup E) = 1000$, $n(H) = 750$, $n(E) = 400$.

Using $n(H \cup E) = n(H) + n(E) - n(H \cap E)$

$$1000 = 750 + 400 - n(H \cap E)$$
$$\Rightarrow \quad n(H \cap E) = 1150 - 1000 = 150$$

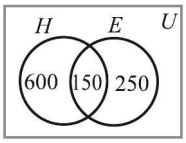


Fig. 1.33

Number of people who can speak Hindi only:

$$n(H \cap E') = n(H) - n(H \cap E)$$
$$= 750 - 150 = 600.$$

17. If $A = \{1, 2, 5\}$ and $B = \{3, 4, 5, 9\}$, then $A \Delta B$ is equal to

(a) $\{1, 2, 5, 9\}$ (b) $\{1, 2, 3, 4, 9\}$
(c) $\{1, 2, 3, 4, 5, 9\}$ (d) none of these

Ans. (b)

Solution: $A \Delta B = (A - B) \cup (B - A)$
$= \{1, 2\} \cup \{3, 4, 9\} = \{1, 2, 3, 4, 9\}$

18. Which one of the following functions is not invertible?

(a) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 1$

(b) $f : \mathbb{R} \to [0, \infty)$, $f(x) = x^2$

(c) $f : \mathbb{R}^+ \to \mathbb{R}^+$, $f(x) = 1/x^3$

(d) none of these

Ans. (b)

Solution: The function $f(x) = x^2$, $x \in \mathbb{R}$ is not one-one because $f(-4) = -f(4) = 16$

$\therefore$ It is not invertible.

19. Let $f(x) = x^2 - x + 1$, $x \geq (1/2)$, then the solution of the equation $f(x) = f^{-1}(x)$ is

(a) $x = 1$ (b) $x = 2$
(c) $x = 1/2$ (d) none of these

Ans. (a)

Solution: $y = x^2 - x + 1 \Rightarrow x^2 - x + (1 - y) = 0$

Here, $a = 1$, $b = -1$ and $c = 1 - y$, therefore

$$x = \frac{1 \pm \sqrt{1 - 4(1 - y)}}{2} \qquad \left[\because x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\right]$$

$$\because \quad x > \frac{1}{2}, \quad \therefore \quad x = \frac{1}{2} + \sqrt{y - \frac{3}{4}}$$

$$\Rightarrow \quad f^{-1}(x) = \frac{1}{2} + \sqrt{x - \frac{3}{4}}$$

Now, $\quad x^2 - x + 1 = \frac{1}{2} + \sqrt{x - \frac{3}{4}}$

Since the graphs of the original and inverse functions can intersect only on the straight line $y = x$, therefore

$$x = f(x) \qquad \Rightarrow \quad x = x^2 - x + 1$$
$$\Rightarrow \quad x^2 - 2x + 1 = 0 \qquad \Rightarrow \quad (x - 1)^2 = 0$$
$$\Rightarrow \quad x = 1.$$

20. Let the function $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + \sin x$ for $x \in \mathbb{R}$. Then f is

(a) one-one and onto (b) one-one but not onto
(c) onto but not one-one (d) neither one-one nor onto

Ans. (a)

Solution: $f(x) = 2x + \sin x \Rightarrow f'(x) = 2 + \cos x \geq 1 > 0$
$[\because \ -1 \leq \cos x \leq 1]$

$\therefore \quad 1 \leq 2 + \cos x \leq 3$

$\Rightarrow$ f is strictly increasing on $\mathbb{R}$

$\Rightarrow$ f is one-one on $\mathbb{R}$.

Again as every vertical line does not meet the graph of $f(x)$, so f is onto. Hence, f is one-one and onto.

21. If $f:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+\frac{1}{x}$, then $f^{-1}(x)$ equals

(a) $\frac{x+\sqrt{x^2-4}}{2}$ (b) $\frac{x}{1+x^2}$

(c) $\frac{x-\sqrt{x^2-4}}{2}$ (d) $1+\sqrt{x^2-4}$

Ans. (a)

Solution: (a) Let $y=x+\frac{1}{x}$

$$\Rightarrow \quad x^2-xy+1=0 \quad \Rightarrow \quad x=\frac{y\pm\sqrt{y^2-4}}{2}$$

$$\because \qquad x\in(2,\infty)$$

$$\therefore \qquad x=\frac{y+\sqrt{y^2-4}}{2}.$$

Hence, $f^{-1}(x)=\frac{x+\sqrt{x^2-4}}{2}$

EXERCISE SET 1

1. The void relation on a set A is
(a) reflexive
(b) symmetric and transitive
(c) reflexive and symmetric
(d) reflexive and transitive

2. If A, B and C are nonempty subsets of a set, then $(A-B)\cup(B-A)$ equals
(a) $(A\cap B)\cup(A\cup B)$ (b) $(A\cup B)-(A\cap B)$
(c) $A-(A\cap B)$ (d) $(A\cup B)-B$

3. Two finite sets have m and n elements. If the total number of subsets of the first set is 56 more than that of the total number of subsets of the second, then the values of m and n are
(a) 7, 6 (b) 6, 3 (c) 5, 1 (d) 8, 7

4. If $X\cup\{1,2\}=\{1,2,3,5,9\}$, then which of the following statements are true?
(a) the smallest set of X is $\{3,5,9\}$
(b) the smallest set of X is $\{2,3,5,9\}$
(c) the largest set of X is $\{1,2,3,5\}$
(d) the largest set of X is $\{2,3,5,9\}$

5. If $A=\{1,3,5,7,9,11,13,15,17\}$, $B=\{2,4,6,...,18\}$ and U is the universal set, then $A'\cup((A\cup B)\cap B')$ is
(a) A (b) U (c) B (d) none of these

6. In $n(U)=700$, $n(A)=200$, $n(B)=300$, and $n(A\cap B)=100$, then $n(A'\cap B')$ is equal to
(a) 400 (b) 240 (c) 300 (d) none of these

7. Let A and B be two sets. Then $(A\cup B)'\cup(A'\cap B)$ is equal to
(a) A' (b) A (c) B' (d) none of these

8. Let U be the universal set and $A\cup B\cup C=U$. Then $\{(A-B)\cup(B-C)\cup(C-A)\}'$ is equal to
(a) $A\cup B\cup C$ (b) $A\cup(B\cap C)$
(c) $A\cap B\cap C$ (d) $A\cap(B\cup C)$

9. Let $A=\{1,2,3\}$. The total number of distinct relations that can be defined over A is
(a) 2^9 (b) 6 (c) 8 (d) none of these

10. For any three sets A, B and C, the set $(A\cup B\cup C)\cap(A\cap B'\cap C')'\cap C'$ is equal to
(a) $B\cap C'$ (b) $B'\cap C'$
(c) $B\cap C$ (d) $A\cap B\cap C$

11. If the set A has p elements and B has q elements, then the number of elements in $A\times B$ is
(a) $p+q$ (b) $p+q+1$
(c) pq (d) p^2

12. If A and B are two given sets, then $A\cap(A\cap B)'$ is equal to
(a) A (b) B (c) ϕ (d) $A\cap B'$

13. If $N_a=\{an:n\in N\}$, then $N_5\cap N_7=$
(a) N_7 (b) N (c) N_{35} (d) N_5

14. If the function

$$f(x)=\cos^2 x+\cos^2\left(\frac{\pi}{3}+x\right)-\cos x\cdot\cos\left(\frac{\pi}{3}+x\right)$$

is constant (independent of x), then the value of this constant is
(a) 0 (b) 3/4 (c) 1 (d) 4/3

15. The shaded region in the given figure is

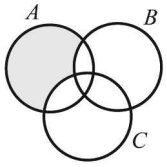


Fig. 1.34

(a) $A\cap(B\cup C)$ (b) $A\cup(B\cap C)$
(c) $A\cap(B-C)$ (d) $A-(B\cup C)$

16. If P, Q and R are the subsets of a set A, then $R\times(P'\cup Q')'=?$

(a) $(R\times P)\cap(R\times Q)$ (b) $(R\times Q)\cap(R\times P)$

(c) $(R\times P)\cup(R\times Q)$ (d) none of these

17. If $A=\{x\in\mathbb{C}:x^2=1\}$ and $B=\{x\in\mathbb{C}:x^4=1\}$, then $A\Delta B$ is equal to

(a) $\{-1,1\}$ (b) $\{-1,1,i,-i\}$

(c) $\{-i,i\}$ (d) none of these

18. Which of the following relations is/are correct?

(1) $A-B=A-(A\cap B)$

(2) $A=(A\cap B)\cup(A-B)$

(3) $A-(B\cup C)=(A-B)\cup(A-C)$

(a) (1) and (3) (b) (2) only

(c) (2) and (3) (d) (1) and (2)

19. If $n(A)=4$, $n(B)=3$ and $n(A\times B\times C)=24$, then $n(C)=?$

(a) 288 (b) 1 (c) 17 (d) 2

20. If $f(x)=2^x$, then $f(0), f(1), f(2),\ldots$ are in

(a) A.P (b) G.P.

(c) H.P. (d) arbitrary

21. Let X be the universal set for sets A and B. If $n(A)=200$, $n(B)=300$ and $n(A\cap B)=100$, then $n(A'\cap B')$ is equal to 300 provided $n(X)$ is equal to

(a) 600 (b) 700 (c) 800 (d) 900

22. The number of elements in the set $\{(a,b):2a^2+3b^2=35$ and $a,b\in Z\}$, where Z is the set of all integers, is

(a) 2 (b) 4 (c) 8 (d) 12

23. If $f(x)=\cos(\log x)$, then

$$f(x)f(y)-\frac{1}{2}\left[f\left(\frac{x}{y}\right)+f(xy)\right]$$ has the value

(a) 1 (b) 1/2

(c) –2 (d) 0

24. Let A and B be two sets defined as follows:

$A=\{(x,y):|x-3|<1$ and $|y-3|<1\}$

$B=\{(x,y):4x^2+9y^2-32x-54y+109\le 0\}$. Then,

(a) $A\subseteq B$ (b) $B\subseteq A$

(c) $A=B$ (d) none of these

25. In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is

(a) at least 30 (b) at most 20

(c) exactly 25 (d) none of these

26. The relation 'less than' on the set of natural numbers is

(a) only symmetric (b) only transitive

(c) only reflexive (d) an equivalence relation

27. The inverse of the function $y=\dfrac{10^x-10^{-x}}{10^x+10^{-x}}+1$ is

(a) $y=\log_{10}\left(\dfrac{x}{2-x}\right)$ (b) $y=\dfrac{1}{2}\log_{10}\left(\dfrac{x}{2-x}\right)$

(c) $y=\dfrac{1}{2}\log_{10}\left(\dfrac{x}{1-x}\right)$ (d) none of these

28. If $A=\{(x,y):x^2+y^2\le 1$ and $x,y\in\mathbb{R}\}$ and $B=\{(x,y):x^2+y^2\le 4$ and $x,y\in\mathbb{R}\}$, then

(a) $A-B=A$ (b) $B-A=B$

(c) $A-B=\phi$ (d) $B-A\neq\phi$

29. If $A=\{1,2,3\}$ and $B=\{3,8\}$, then $(A\cup B)\times(A\cap B)$ is equal to

(a) $\{(8,3),(8,2),(8,1),(8,8)\}$

(b) $\{(1,2),(2,2),(3,3),(8,8)\}$

(c) $\{(3,1),(3,2),(3,3),(3,8)\}$

(d) $\{(1,3),(2,3),(3,3),(8,3)\}$

30. If R be a relation $<$ (less than) from $A=\{1,2,3,4\}$ to $B=\{1,3,5\}$, that is, $(a,b)\in R\Leftrightarrow a<b$, then $R\circ R^{-1}$ is

(a) $\{(1,3),(1,5),(2,3),(2,5),(3,5),(4,5)\}$

(b) $\{(3,1),(5,1),(3,2),(5,2),(5,3),(5,4)\}$

(c) $\{(3,3),(3,5),(5,3),(5,5)\}$

(d) $\{(3,3),(3,4),(4,5)\}$

31. If $f(x)=\cos[\pi^2]x+\cos[-\pi^2]x$, where $[x]$ stands for the greatest integer function, then

(a) $f\left(\dfrac{\pi}{2}\right)=1$ (b) $f(\pi)=1$

(c) $f(-\pi)=0$ (d) $f\left(\dfrac{\pi}{4}\right)=2$

32. If $A=\{\theta:2\cos^2\theta+\sin\theta\le 2\}$ and $B=\left\{\theta:\dfrac{\pi}{2}\le\theta\le\dfrac{3\pi}{2}\right\}$, then $A\cap B$ is equal to

(a) $\{\theta:\pi/2\le\theta\le 5\pi/6\}$

(b) $\{\theta:\pi\le\theta\le 3\pi/2\}$

(c) $\{\theta:\pi/2\le\theta\le 5\pi/6\}\cup\{\theta:\pi\le\theta\le 3\pi/2\}$

(d) none of these

33. An integer m is said to be related to another integer n if m is a multiple of n. Then, the relation is

(a) reflexive and symmetric

(b) reflexive and transitive

(c) symmetric and transitive

(d) an equivalence relation

34. Let A be the set of all determinants of order 3 with entries 0 or 1 only. B is the subset of A consisting of all determinants with value 1 and C is the subset of A consisting of all determinants with value –1. If $n(B)$ and $n(C)$ denote the number of elements in B and C respectively, then

(a) $C = \phi$ (b) $n(B) = n(C)$
(c) $A = B \cup C$ (d) $n(B) = 2n(C)$

35. A market research group conducted a survey of 2000 consumers and reported that 1720 consumers liked the product P_1 and 1450 consumers liked the product P_2. The least number of consumers who must have liked both the products is
(a) 1170 (b) 3170
(c) 270 (d) none of these

36. Let R be a relation over the set $N \times N$, defined by (a, b) $R(c, d) \Rightarrow a + d = b + c$. Then, R is
(a) reflexive only (b) symmetric only
(c) transitive only (d) an equivalence relation

37. If $X = \{8^n - 7n - 1 : n \in N\}$ and $Y = \{49(n-1) : n \in N\}$, then
(a) $X \subseteq Y$ (b) $Y \subseteq X$
(c) $X = Y$ (d) none of these

38. A college awarded 38 medals in football, 15 in basket ball and 20 in cricket. If these medals went to a total of 58 men and only three men got medals in all the three sports, the number of students who received medals in exactly two of the three sports is
(a) 18 (b) 15 (c) 9 (d) 6

39. The function $f(x) = \log\left(\frac{1+x}{1-x}\right)$ satisfies the equation
(a) $f(x+2) - 2f(x+1) + f(x) = 0$
(b) $f(x+1) + f(x) = f(x)(x+1)$
(c) $f(x_1) f(x_2) = f(x_1 + x_2)$
(d) $f(x_1) + f(x_2) = f\left(\frac{x_1 + x_2}{1 + x_1 x_2}\right)$

40. If $n(A) = 3$, and $n(B) = 4$, then $n(A \times A \times B)$ is equal to
(a) 36 (b) 12
(c) 108 (d) none of these

41. In a class of 55 students, the number of students studying different subjects is 23 in mathematics, 24 in physics, 19 in chemistry, 12 in mathematics and physics, 9 in mathematics and chemistry, 7 in physics and chemistry and 4 in all the three subjects. The number of students who have taken exactly one subject is
(a) 6 (b) 9
(c) 7 (d) none of these

42. If $f(x + 2y, x - 2y) = xy$, then $f(x, y)$ equals
(a) $\frac{x^2 - y^2}{8}$ (b) $\frac{x^2 - y^2}{4}$
(c) $\frac{x^2 + y^2}{4}$ (d) $\frac{x^2 - y^2}{2}$

43. If $aN = \{ax : x \in N\}$ and $bN \cap cN = dN$, where $b, c \in N$ are relatively prime, then
(a) $d = bc$ (b) $c = bd$
(c) $b = cd$ (d) none of these

44. If $A = \{(x, y) : x^2 + y^2 = 25\}$ and $B = \{(x, y) : x^2 + 9y^2 = 144\}$, then $A \cap B$ contains
(a) one point (b) three points
(c) two points (d) four points

45. Which of the following functions is the inverse of itself?
(a) $f(x) = \frac{1-x}{1+x}$ (b) $g(x) = 5^{\log x}$
(c) $h(x) = 2^{x(x-1)}$ (d) none of these

46. If $f_1(x) = \frac{x}{4} + 6$ for all $x \in \mathbb{R}$ and $f_n(x) = f_1(f_{n-1}(x))$ for $n \geq 2$, then $f_n(x) = ?$
(a) $\frac{x-8}{4^n} + 8$ (b) $\frac{x+8}{4^n} + 8$
(c) $\frac{x-8}{4^n} - 8$ (d) none of these

47. If $f(x+1) + f(x-1) = 2f(x)$ and $f(0) = 0$, then $f(n)$, $n \in N$, is
(a) $nf(1)$ (b) $\{f(1)\}^n$
(c) 0 (d) none of these

48. The function $f(x) = \log(x + \sqrt{x^2 + 1})$ is
(a) neither an even nor an odd function
(b) an even function
(c) an odd function
(d) a periodic function

49. If $A = \{1, 2, 3, 4\}$, then which of the following is a function from A to itself?
(a) $f_1 = \{(x, y) : y = x + 1\}$ (b) $f_2 = \{(x, y) : x + y > 4\}$
(c) $f_3 = \{(x, y) : y < x\}$ (d) $f_4 = \{(x, y) : x + y = 5\}$

50. Let $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ be two functions given by $f(x) = 2x - 3$ and $g(x) = x^3 + 5$. Then, $(f \circ g)^{-1}(x)$ is equal to
(a) $\left(\frac{x+7}{2}\right)^{1/3}$ (b) $\left(x - \frac{7}{2}\right)^{1/3}$
(c) $\left(\frac{x-2}{7}\right)^{1/3}$ (d) $\left(\frac{x-7}{2}\right)^{1/3}$

51. If A, B, C be three sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$, then
(a) $A = B$ (b) $B = C$ (c) $A = C$ (d) $A = B = C$

52. A set contains $2n + 1$ elements. The number of subsets of this set containing more than n elements is equal to
(a) 2^{n-1} (b) 2^n (c) 2^{n+1} (d) 2^{2n}

53. If two sets A and B have 99 elements in common, then the number of elements common to each of the sets $A \times B$ and $B \times A$ are
(a) 2^{99} (b) 99^2 (c) 100 (d) 18

54. Let R be a relation on the set N defined by $\{(x, y): x, y \in N \text{ and } 2x + y = 41\}$. Then R is
(a) reflexive (b) symmetric
(c) transitive (d) none of these

55. If $f(x) = ax^2 + bx + c$, then the values of a and b for which the identity $f(x+1) - f(x) = 8x + 3$ is satisfied are:
(a) $a = 1$ and $b = 4$ (b) $a = 1$ and $b = -4$
(c) $a = 4$ and $b = 1$ (d) $a = 4$ and $b = -1$

56. Let R be the relation on the set of all real numbers defined by aRb iff $|a - b| \le 1$. Then, R is
(a) reflexive and symmetric
(b) symmetric only
(c) transitive only
(d) anti-symmetric only

57. If the function f: $[1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is
(a) $\left(\frac{1}{2}\right)^{x(x-1)}$ (b) $\left(\frac{1}{2}\right)(1 + \sqrt{1 + 4\log_2 x})$
(c) $\left(\frac{1}{2}\right)(1 - \sqrt{1 + 4\log_2 x})$ (d) not defined

EXERCISE SET 2

1. Let $R = \{(x, y): x^2 + y^2 = 1 \text{ and } x, y \in R\}$ be a relation in R. The relation R is
(a) reflexive (b) symmetric
(c) transitive (d) anti-symmetric

2. If $A = \{x: x^2 - 5x + 6 = 0\}$, $B = \{2, 4\}$ and $C = \{4, 5\}$, then $A \times (B \cap C)$ is
(a) $\{(2, 4), (3, 4)\}$ (b) $\{(4, 2), (4, 3)\}$
(c) $\{(2, 4), (3, 4), (4, 4)\}$
(d) $\{(2, 2), (3, 3), (4, 4), (5, 5)\}$

3. Let R be a relation in the set N of natural numbers defined by the relation nRm which implies n is a factor of m. The relation R is
(a) reflexive and symmetric only
(b) symmetric and transitive only
(c) reflexive and transitive relation
(d) an equivalence relation

4. If f: $R \to R$ and g: $R \to R$ defined by $f(x) = 2x + 3$ and $g(x) = x^2 + 7$, then the value of x for which $f(g(x)) = 25$ is
(a) ± 1 (b) ± 2 (c) ± 3 (d) ± 4

5. Let R and S be two non-void relations on a set A. Which of the following statements is false?
(a) R and S are transitive implies $R \cap S$ is transitive.
(b) R and S are transitive implies $R \cup S$ is transitive.
(c) R and S are symmetric implies $R \cup S$ is symmetric.
(d) R and S are reflexive implies $R \cup S$ is reflexive.

6. If $A = \{1, 2, 3\}$, then the relation $R = \{(1, 2), (2, 3), (1, 3)\}$ in A is
(a) symmetric and transitive only
(b) symmetric only
(c) transitive only
(d) none of these

7. Let f and g be two functions from R to R defined as

$$f(x) = \begin{cases} 7x^2 + x - 8, & x \le 1 \\ 4x + 5, & 1 < x \le 7 \\ 8x + 3, & x > 7 \end{cases}, \quad g(x) \begin{cases} |x|, & x < -3 \\ 0, & -3 \le x < 2 \\ x^2 + 4, & x \ge 2 \end{cases}$$

then
(a) $(f \circ g)(-3) = 8$ (b) $(f \circ g)(9) = 683$
(c) $(g \circ f)(0) = -8$ (d) $(g \circ f)(6) = 427$

8. Which of the following functions is an even function?
(a) $f(x) = \frac{a^x + a^{-x}}{a^x - a^{-x}}$ (b) $f(x) = \frac{a^x + 1}{a^x - 1}$
(c) $f(x) = x\frac{a^x - 1}{a^x + 1}$ (d) $f(x) = \log_2(x + \sqrt{x^2 + 1})$

9. A relation R from $\mathbb{C}$ to $\mathbb{R}$ is defined by xRy iff $|x| = y$. Which of the following is correct?
(a) $(2 + 3i)R13$ (b) $3R(-3)$
(c) $(1 + i)R2$ (d) $iR1$

10. Let $f:[0,1] \to [0,1]$ and $g:[0,1] \to [0,1]$ be two functions defined by $f(x) = \frac{1-x}{1+x}$ and $g(x) = 4x(1-x)$, then $(f \circ g)(x)$ equals
(a) $\frac{8x(1-x)}{(1+x)^2}$ (b) $\frac{4(1-x)}{1+x}$
(c) $\frac{1 - 4x + 4x^2}{1 + 4x - 4x^2}$ (d) none of these

11. If R is a relation from a set A to a set B and S is a relation from B to C, then the relation $S \circ R$
(a) is from C to A (b) is from A to C
(c) does not exist (d) none of these

12. Let $f(x)=\dfrac{\alpha x}{x+1}$, $x\neq 1$. Then for what value of α, is $f(f(x))=x$?

(a) $\sqrt{2}$ (b) $-\sqrt{2}$ (c) 1 (d) -1

13. The function $f: N \to N$ (N is the set of natural numbers) defined by $f(n)=2n+3$, is

(a) surjective (b) injective
(c) bijective (d) none of these

14. Let R be a relation defined by $R = \{(a, b): a \geq b$ and $a, b \in R\}$, then R is

(a) an equivalence relation in R
(b) reflexive, transitive but not symmetric
(c) symmetric, transitive but not reflexive
(d) only reflexive

15. Let $p(x)=a^2+bx$, $q(x)=lx^2+mx+n$. If $p(1) - q(1) = 0, p(2) - q(2) = 1$ and $p(3) - q(3) = 4$, then $p(4) - q(4)$ equals

(a) 0 (b) 5 (c) 6 (d) 9

16. R is a relation from $\{11, 12, 13\}$ to $\{8, 10, 12\}$ defined by $y = x - 3$. The relation R^{-1} is

(a) $\{(11, 8), (13, 10)\}$ (b) $\{(8, 11), (10, 13)\}$
(c) $\{(8, 11), (9, 12), (10, 13)\}$
(d) none of these

17. If $f(x)=\dfrac{3x+2}{5x-3}$, then

(a) $f^{-1}(x)=f(x)$ (b) $f^{-1}(x)=-f(x)$

(c) $(f\circ f)(x)=-x$ (d) $f^{-1}(x)=-\dfrac{1}{19}f(x)$

18. For real numbers x and y, we define xRy iff $x-y+\sqrt{5}$ is an irrational number. The relation R is

(a) reflexive (b) symmetric
(c) transitive (d) none of these

19. If the mappings $f: A \to B$ and $g: B \to C$ are both bijective, then the mapping $g \circ f: A \to C$ is

(a) neither one-one nor onto
(b) one-one and onto
(c) one-one but not onto
(d) onto but not one-one

20. Given the relation $R = \{(1, 2), (2, 3)\}$ on the set $\{1, 2, 3\}$, the minimum number of ordered pairs which when added to R make it an equivalence relation is

(a) 5 (b) 6 (c) 7 (d) 8

21. Let $f(x)$ be defined for all $x > 0$ and be continuous. Let $f(x)$ satisfy $f(x/y) = f(x) - f(y)$ for all x, y and $f(e) = 1$. Then,

(a) $f(x)$ is bounded (b) $f(1/x) \to 0$ as $x \to 0$
(c) $xf(x) \to 1$ as $x \to 0$ (d) $f(x) = \log x$

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct option.

(a) Statement 1 is false, Statement 2 is true.
(b) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1.
(d) Statement 1 is true, Statement 2 is false.

22. Statement 1: Two finite sets have m and n elements. The total number of subsets of the first set is 56 more than the total number of subsets of the second set. The sets have 6 and 3 elements.

Statement 2: A set of n elements has 2^n subsets.

23. Statement 1: If X and Y are two sets, then $X \cap (X \cup Y)'$ is equal to ϕ.

Statement 2: If $A = \{(x, y): y = (1/x)$ and $x \neq 0, x \in \mathbb{R}\}$ and $B = \{(x, y): y = -x$ and $x \in \mathbb{R}\}$, then $A \cap B$ has only one element.

24. Statement 1: If $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$ and $C = \{1, 2\}$, then $(A - B) \times (A \cap C) = \{(1, 2), (1, 3)\}$.

Statement 2: If $A = \{x: x \geq 4$ and $x \in N\}$ and $B = \{x : x \leq 5$, and $x \in N\}$, then $A \cap B = \{4, 5\}$.

25. Statement 1: Let $P = \{(x, y): x^2 + y^2 = 1$ and $x, y \in \mathbb{R}\}$. The relation P is not reflexive.

Statement 2: The relation P is symmetric.

26. Statement 1: Let X be a family of sets and R be a relation defined on X by 'A is disjoint from B'. The relation R is reflexive.

Statement 2: The relation R is not transitive.

Previous Years' Questions

27. Let $R = \{(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)\}$ be a relation on the set $A = \{1, 2, 3, 4\}$. The relation R is

(a) not symmetric (b) transitive
(c) a function (d) reflexive

[AIEEE 2004]

28. Let $f: [4, \infty[\to [4, \infty[$ be defined by $f(x)=5^{x(x-4)}$, then $f^{-1}(x)$ is

(a) $2-\sqrt{4+\log_5 x}$ (b) $2+\sqrt{4+\log_5 x}$

(c) $(1/5)^{x(x-4)}$ (d) none of these

[DCE 2009]

29. If f: $R \to S$, defined by $f(x) = \sin x - \sqrt{3}\cos x + 1$, is onto, then the interval of S is
(a) [0, 1] (b) [–1, 1] (c) [0, 3] (d) [–1, 3]
[AIEEE 2004]

30. If the graph of the function $y = f(x)$ is symmetrical about the line $x = 2$, then
(a) $f(x) = f(-x)$ (b) $f(2+x) = f(2-x)$
(c) $f(x+2) = f(x-2)$ (d) $f(x) = -f(-x)$
[AIEEE 2004]

31. The domain of the function $f(x) = \dfrac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is
(a) [1, 2] (b) [2, 3) (c) [2, 3] (d) [1, 2)
[AIEEE 2004]

32. Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on the set A = {3, 6, 9, 12}. The relation is
(a) reflexive and symmetric only
(b) an equivalence relation
(c) reflexive only
(d) reflexive and transitive only.
[AIEEE 2005]

33. Let $f:(-1,1) \to B$ be a function defined by $f(x) = \tan^{-1}[2x/(1-x^2)]$, then f is both one-one and onto when B is in the interval
(a) $\left[0, \frac{\pi}{2}\right)$ (b) $\left(0, \frac{\pi}{2}\right)$ (c) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (d) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
[AIEEE 2005]

34. A function is matched below against an interval where it is supposed to be increasing. Which of the following pairs is incorrectly matched?

Interval	*Function*
(a) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(b) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(c) $(-\infty, -4]$	$x^3 + 6x^2 + 6$
(d) $\left(-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$

[AIEEE 2005]

35. A real-valued function $f(x)$ satisfies the functional equation $f(x-y) = f(x)f(y) - f(a-x)f(a+y)$, where a is a given constant and $f(0) = 1$. Then $f(2a-x)$ is equal to
(a) $f(x)$ (b) $-f(x)$
(c) $f(-x)$ (d) $f(a) + f(a-x)$
[AIEEE 2005]

36. Let W denote the words in English dictionary. If the relation R is defined as $R = \{(x, y) \in W \times W$: the words x and y have at least one letter in common$\}$, then R is
(a) not reflexive, symmetric and transitive
(b) reflexive, symmetric and not transitive
(c) reflexive, symmetric and transitive
(d) reflexive, not symmetric and transitive
[AIEEE 2006]

37. The set $S = \{1, 2, 3, \ldots, 12\}$ is to be partitioned into three sets A, B and C of equal size. Thus, $A \cup B \cup C = S$, and $A \cap B = B \cap C = A \cap C = \phi$. The number of ways to partition S is
(a) $\dfrac{12!}{(4!)^3}$ (b) $\dfrac{12!}{(4!)^4}$ (c) $\dfrac{12!}{3!(4!)^3}$ (d) $\dfrac{12!}{3!(4!)^4}$
[AIEEE 2007]

38. Let R be the real line. Consider the following subsets of the plane $R \times R$:
$S = \{(x, y)$: $y = x + 1$ and $0 < x < 2\}$
$T = \{(x, y)$: $x - y$ is an integer$\}$
Which one of the following is true?
(a) T is an equivalence relation on R but S is not.
(b) Neither S nor T is an equivalence relation on R.
(c) Both S and T are equivalence relations on R.
(d) S is an equivalence relation on R but T is not.
[AIEEE 2008]

39. Let f: $N \to Y$ be a function defined as $f(x) = 4x + 3$, where, $Y = \{y \in N: y = 4x + 3$ for some $x \in N\}$.
Show that f is invertible and its inverse is
(a) $g(y) = \dfrac{y-3}{4}$ (b) $g(y) = \dfrac{3y+4}{3}$
(c) $g(y) = 4 + \dfrac{y+3}{4}$ (d) $g(y) = \dfrac{y+3}{4}$
[AIEEE 2008]

40. If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then
(a) $A = C$ (b) $B = C$ (c) $A \cap B = \phi$ (d) $A = B$
[AIEEE 2009]

41. For real x, let $f(x) = x^3 + 5x + 1$, then
(a) f is onto R but not one-one
(b) f is one-one and onto R
(c) f is neither one-one nor onto R
(d) f is one-one but not onto R
[AIEEE 2009]

42. Let $f(x) = (x+1)^2 - 1, x \geq -1$.
Statement 1: The set $\{x: f(x) = f^{-1}(x)\} = \{0, -1\}$.
Statement 2: f is a bijection.
Select the correct choice:
(a) Statement 1 is true, statement 2 is true, statement 2 is not a correct explanation for statement 1.
(b) Statement 1 is true, statement 2 is false.

(c) Statement 1 is false, statement 2 is true.
(d) Statement 1 is true, statement 2 is true; Statement 2 is the correct explanation for Statement 1.

[AIEEE 2009]

43. The number of relations of a set with four elements is equal to
(a) 2^{16} (b) 2^{12} (c) 2^8 (d) 2^4

[UP SEE 2004]

44. Which of the following is a true statement?
(a) $\{a\} \in \{a, b, c\}$ (b) $\{a\} \subseteq \{a, b, c\}$
(c) $\phi \in \{a, b, c\}$ (d) none of these

[UPSEE 2005]

45. If $f(x) = \dfrac{2x-1}{x+5} (x \neq -5)$, then $f^{-1}(x)$ is equal to
(a) $\dfrac{x-5}{2x-1}, x \neq \dfrac{1}{2}$ (b) $\dfrac{5x+1}{2-x}, x \neq 2$
(c) $\dfrac{x-5}{2x+1}, x \neq \dfrac{1}{2}$ (d) $\dfrac{5x-1}{2-x}, x \neq 2$

[UPSEE 2006]

46. If $f(x) = \dfrac{x}{x-1}$, then $\dfrac{(f \circ f \circ f \circ \ldots \circ f)(x)}{19 \text{ times}}$ is equal to
(a) $\dfrac{x}{x-1}$ (b) $\left(\dfrac{x}{x-1}\right)^{19}$ (c) $\dfrac{19x}{x-1}$ (d) x

[UPSEE 2006, WB JEE 2006]

47. The domain of the function $f(x) = \exp\left(\sqrt{5x-3-2x^2}\right)$ is
(a) $\left[\dfrac{3}{2}, \infty\right)$ (b) $\left[1, \dfrac{3}{2}\right]$ (c) $[-\infty, 1]$ (d) $\left(1, \dfrac{3}{2}\right)$

[UPSEE 2006, WBJEE 2006]

48. The range of $f(x) = \cos x - \sin x$ is
(a) $[-1, 1]$ (b) $(-1, 2)$
(c) $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$ (d) $\left[-\sqrt{2}, \sqrt{2}\right]$

[UPSEE 2007]

49. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is
(a) $\{1, 2, 3, 4, 5\}$ (b) $\{1, 2, 3, 4, 5, 6\}$
(c) $\{1, 2, 3, 4\}$ (d) $\{1, 2, 3\}$

[UPSEE 2007]

50. A survey shows that 63% of the Americans like cheese whereas 76% like apples. If x% of the Americans like both cheese and apples, then
(a) $x = 39$ (b) $x = 63$
(c) $39 \leq x \leq 63$ (d) none of these

[UPSEE 2008]

51. Let $f(x) = \dfrac{1}{2} - \tan\left(\dfrac{\pi x}{2}\right)$, $-1 < x < 1$ and $g(x) = \sqrt{3 + 4x - 4x^2}$, then dom $(f + g)$ is given by
(a) $\left[\dfrac{1}{2}, 1\right]$ (b) $\left[\dfrac{1}{2}, -1\right)$
(c) $\left[-\dfrac{1}{2}, 1\right)$ (d) $\left[-\dfrac{1}{2}, -1\right]$

[UPSEE 2008]

52. Let $E = \{1, 2, 3, 4\}$ and $F = \{1, 2\}$. Then the number of onto functions from E to F is
(a) 14 (b) 16 (c) 12 (d) 8

[IIT-JEE 2001]

53. Suppose $f(x) = (x+1)^2$ for $x \geq -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals
(a) $-\sqrt{x} - 1, x \geq 0$ (b) $\dfrac{1}{(x+1)^2}, x > -1$
(c) $\sqrt{x+1}, x \geq -1$ (d) $\sqrt{x} - 1, x \geq 0$

[IIT-JEE 2002]

54. Range of the function $f(x) = \dfrac{x^2 + x + 2}{x^2 + x + 1}, x \in \mathbb{R}$ is
(a) $(1, \infty)$ (b) $\left(1, \dfrac{11}{7}\right)$ (c) $\left(1, \dfrac{7}{3}\right]$ (d) $\left(1, \dfrac{7}{5}\right)$

[IIT-JEE 2003]

55. The natural domain of the function $\sqrt{\sin^{-1}(2x) + \dfrac{\pi}{6}}$, $x \in \mathbb{R}$ is
(a) $\left[-\dfrac{1}{4}, \dfrac{1}{2}\right]$ (b) $\left[-\dfrac{1}{4}, \dfrac{1}{4}\right]$ (c) $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$ (d) $\left[-\dfrac{1}{2}, \dfrac{1}{4}\right]$

[IIT-JEE 2004, Pb. CET 2006]

56. If $f(x) = \begin{cases} x; x \in Q \\ 0; x \notin Q \end{cases}$ and $g(x) = \begin{cases} x; x \notin Q \\ 0; x \in Q \end{cases}$, then $(f - g)$ will be
(a) one-one onto (b) one-one into
(c) many-one onto (d) many-one into

[IIT-JEE 2005]

57. If $(\circ)$ is a binary operation such that $a \circ b = ab + 1$, where $a, b \in Z$, then $(\circ)$ is
(a) commutative and associative
(b) commutative but not associative
(c) associative but not commutative
(d) neither associative nor commutative

[DCE 2003]

58. Let $g(x)=1+x-[x]$ and $f(x)=\begin{cases}-1, & x<0\\ 0, & x=0\\ 1, & x>0\end{cases}$, then for all $x, f[g(x)]$ is equal to

(a) x (b) 1 (c) $f(x)$ (d) $g(x)$

[DCE 2006]

59. If $g(x)$ is the inverse of the function $f(x)$ and $f'(x)=\frac{1}{1+x^5}$, then $\frac{d}{dx}g(x)$ is

(a) $\frac{1}{1+[g(x)]^5}$ (b) $\frac{1}{1+[f(x)]^5}$

(c) $1+[g(x)]^5$ (d) $1+f(x)$

[DCE 2007]

60. Let f be a function defined by $f(x)=(x-1)^2+1, (x\geq 1)$.
Statement 1: The set $\{x: f(x)=f^{-1}(x)\}=\{1, 2\}$.
Statement 2: f is a bijection function and $f^{-1}(x)=1+\sqrt{x-1}, x\geq 1$.

(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is false.
(d) Statement 1 is false, Statement 2 is true.

[AIEEE 2011]

61. Let A, B, C be the pairwise independent events with $P(C)>0$ and $P(A\cap B\cap C)=0$. Then $P(A'\cap B'/C)$ is equal to

(a) $P(1)-P(B')$ (b) $P(A')+P(B')$
(c) $P(A')-P(B')$ (d) $P(A')-P(B)$

[AIEEE 2011]

62. Consider the following relation R on the set of real square matrices of order 3.

$R=\{(A,B): A=P^{-1}BP$ for some invertible matrix $P\}$
Statement 1: R is equivalence relation.
Statement 2: For any two invertible 3×3 matrices M and N, $(MN)^{-1}=N^{-1}M^{-1}$.

(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is false.
(d) Statement 1 is false, Statement 2 is true.

[AIEEE 2011]

63. Let $X=\{1, 2, 3, 4, 5\}$. The number of different ordered pairs (Y, Z) that can be formed such that $Y\subseteq X, Z\subseteq X$ and $Y\cap Z$ is empty, is

(a) 5^2 (b) 3^5 (c) 2^5 (d) 5^3

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (b) **2.** (b) **3.** (b) **4.** (a) and (c) **5.** (b) **6.** (c) **7.** (a) **8.** (c) **9.** (a)
10. (a) **11.** (c) **12.** (d) **13.** (c) **14.** (b) **15.** (d) **16.** (a) and (b) **17.** (c) **18.** (d)
19. (d) **20.** (b) **21.** (b) **22.** (c) **23.** (d) **24.** (a) **25.** (c) **26.** (b) **27.** (b)
28. (c) and (d) **29.** (d) **30.** (c) **31.** (c) **32.** (c) **33.** (b) **34.** (b) **35.** (a) **36.** (d)
37. (a) **38.** (c) **39.** (d) **40.** (a) **41.** (d) **42.** (a) **43.** (a) **44.** (d) **45.** (a) **46.** (a)
47. (a) **48.** (c) **49.** (d) **50.** (d) **51.** (b) **52.** (d) **53.** (b) **54.** (d) **55.** (d) **56.** (a)
57. (b)

Exercise Set 2

1. (b) **2.** (a) **3.** (c) **4.** (b) **5.** (b) **6.** (c) **7.** (b) **8.** (c) **9.** (d) **10.** (c)
11. (b) **12.** (d) **13.** (b) **14.** (b) **15.** (d) **16.** (b) **17.** (a) **18.** (a) **19.** (b) **20.** (c)
21. (d) **22.** (b) **23.** (d) **24.** (a) **25.** (c) **26.** (a) **27.** (c) **28.** (d) **29.** (d) **30.** (c)
31. (b) **32.** (d) **33.** (c) **34.** (d) **35.** (b) **36.** (b) **37.** (a) **38.** (a) **39.** (a) **40.** (b)
41. (b) **42.** (b) **43.** (b) **44.** (b) **45.** (b) **46.** (a) **47.** (b) **48.** (d) **49.** (d) **50.** (c)
51. (c) **52.** (a) **53.** (d) **54.** (c) **55.** (a) **56.** (a) **57.** (b) **58.** (b) **59.** (c) **60.** (a)
61. (d) **62.** (d) **63.** (b)

HINTS AND SOLUTIONS—EXERCISE SET 1

1. The void relation ϕ on A is not reflexive as $(a,a) \notin \phi$ for any $a \in A$. The void relation is symmetric and transitive.

2. $(A-B)\cup(B-A) = (A\cup B)-(A\cap B)$

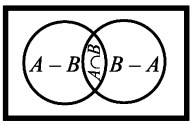


Fig. 1.35

3. Since the two finite sets have m and n elements, so the number of subsets of these sets will be 2^m and 2^n respectively. According to the equation $2^m - 2^n = 56$. Putting $m = 6$, $n = 3$ we get

$$2^6 - 2^3 = 56 \text{ or } 64 - 8 = 56.$$

4. The smallest set of X must have three elements different from the elements 1 and 2 and the largest set of X must have five elements.
$\therefore$ The smallest set of X is $\{3, 5, 9\}$ and the largest set of X is $\{1, 2, 3, 5, 9\}$.

5. We have $(A\cup B)\cap B' = A$

$$(A\cup B)\cap B')\cup A' = A\cup A' = U$$

Hence, (b) is the correct answer.

6.
$$\begin{aligned} n(A'\cap B') &= n(U) - n(A\cup B) \\ &= 700 - n(A\cup B) \end{aligned}$$

Now
$$\begin{aligned} n(A\cup B) &= n(A) + n(B) - n(A\cap B) \\ &= 200 + 300 - 100 = 400 \end{aligned}$$

Hence, $n(A'\cup B') = 700 - 400 = 300$

7.
$$\begin{aligned} A\cup B)'\cup(A'\cap B) &= (A'\cap B')\cup(A'\cap B) \\ &= (A'\cup A')\cap(A'\cup B)\cap(B'\cap A')\cap(B'\cap B) \\ &= A'\cap\{A'\cup(B\cap B')\}\cup U = A'\cap(A'\cup\phi)\cap U \\ &= A'\cap A'\cap U = A'\cap U = A'. \end{aligned}$$

8. $(A-B)\cup(B-C)\cup(C-A)$ is represented by the unshaded portion in Fig. 1.36. The complement of the unshaded portion is the shaded portion $A\cap B\cap C$.

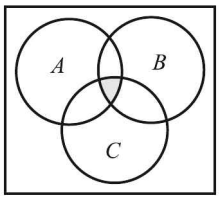


Fig. 1.36

9. $n(\mathrm{A}\times\mathrm{A}) = n(\mathrm{A})\cdot n(\mathrm{A}) = 3^2 = 9$. So, the total number of subsets of $A\times A$ is 2^9 and a subset of $A\times A$ is a relation over the set A.

10. We have

$$(A\cap B'\cap C')' = A'\cup B\cup C$$

$$\begin{aligned} \therefore\quad & (A\cup B\cup C)\cap(A\cap' B\cap C')' \\ &= (A\cup B\cup C)\cap(A'\cup B\cup C) \\ &= (A\cap A')\cup(B\cup C) \quad \text{[By distributivity of } \cup \text{ over } \cap] \\ &= \phi\cup(B\cup C) = (B\cup C) \end{aligned}$$

Hence,

$$\begin{aligned} & (A\cup B\cup C)\cap(A\cap B'\cap C')'\cap C' \\ &= (B\cup C)\cap C' \\ &= (B\cap C')\cup(C\cap C') \\ &= B\cap C' \end{aligned}$$

11. $n(A\times B) = pq$

12.
$$\begin{aligned} A\cap(A\cap B)' &= A\cap(A'\cup B') \\ &= (A\cap A')\cup(A\cap B') \\ &= \phi\cup(A\cap B') = A\cap B' \end{aligned}$$

13. $N_5\cap N_7 = N_{35}$
$\because$ 5 and 7 are relatively prime numbers.

14. Since the function is constant $\forall x \in \mathbb{R}$,
$\therefore$ the value of the constant is obtained by assigning any suitable value to x. Choosing $x = 0$, the desired value of the constant

$$\begin{aligned} &= \cos^2 0 + \cos^2\frac{\pi}{3} - \cos 0\cos\frac{\pi}{3} \\ &= 1 + \frac{1}{4} - \frac{1}{2} = \frac{5}{4} - \frac{1}{2} = \frac{3}{4} \end{aligned}$$

15. The shaded portion in the figure is $A-(B\cup C)$.

16.
$$\begin{aligned} R\times(P'\cup Q')' &= R\times[(P')'\cap(Q')'] \\ &= R\times(P\cap Q) = (R\times P)\cap(R\times Q) \\ &= (R\times Q)\cap(R\times P) \end{aligned}$$

17.
$$\begin{aligned} x^2 = 1 &\Rightarrow x = -1, 1 \qquad \therefore A = \{-1, 1\} \\ x^4 = 1 &\Rightarrow x^2 = -1, 1 \\ &\Rightarrow x = -i, i, -1, 1 \qquad \therefore B = \{-i, i, -1, 1\} \end{aligned}$$

$$\therefore A\Delta B = (A-B)\cup(B-A) = \phi\cup\{-i, i\} = \{-i, i\}$$

18. Statements (1) and (2) are correct. Hence, option (d) is correct.

19. $n(A) = 4$, $n(B) = 3$

$$n(A)\times n(B)\times n(C) = n(A\times B\times C)$$

$$= 4\times 3\times n(C) = 24 \quad\Rightarrow\quad n(C) = \frac{24}{12} = 2$$

20. $f(0)=2^0, f(1)=2^1, f(2)=2^2$.
Clearly, they are in G.P.

21. We know that $n(A\cup B)=n(A)+n(B)-n(A\cap B)$
$\therefore \quad n(A\cup B)=200+300-100=400$
Also, $n(A'\cap B')=n((A\cup B)')=n(X)-n(A\cup B)$
$\Rightarrow \quad 300=n(X)-400 \quad \Rightarrow \quad n(X)=700$

22. The given set is $\{(a,b): 2a^2+3b^2=35 \text{ and } a,b\in Z\}$
We can see that, $2(\pm 2)^2+3(\pm 3)^2=35$
and $2(\pm 4)^2+3(\pm 1)^2=35$
$\therefore$ (2, 3), (2, −3), (−2, −3), (−2, 3), (4, 1), (4, −1), (−4, −1), (−4, 1) are the 8 elements of the set.
$\therefore n=8$

23. $f(x)f(y)-\frac{1}{2}\left[f\left(\frac{x}{y}\right)+f(xy)\right]$
$=\cos(\log x)\cos(\log y)-\frac{1}{2}\times$
$\left[\cos\left(\log\left(\frac{x}{y}\right)\right)+\cos(\log(xy))\right]$
$=\cos(\log x)\cos(\log y)-\frac{1}{2}[\cos(\log x-\log y)$
$+\cos(\log x+\log y)]=0$

24. Graph of A:
$|x-3|<1 \quad \Leftrightarrow \quad -1<x-3<1 \quad \Leftrightarrow \quad 2<x<4$
and $|y-3|<1 \quad \Leftrightarrow \quad -1<y-3<1 \quad \Leftrightarrow \quad 2<y<4$
$\therefore A$ consists of all points inside (not on) the square bounded by the lines $x=2$, $x=4$, $y=2$ and $y=4$. This region has been shown by the shaded area in the Fig. 1.37.

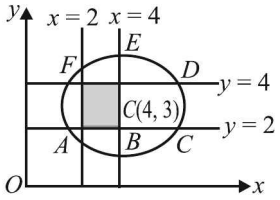


Fig. 1.37

We have $4x^2+9y^2-32x-54y+109\le 0$
$\Rightarrow \quad 4(x^2-8x)+9(y^2-6y)+109\le 0$
$\Rightarrow \quad 4(x-4)^2+9(y-3)^2\le 36$
$\Rightarrow \quad \frac{(x-4)^2}{3^2}+\frac{(y-3)^2}{2^2}\le 1$
Thus, B is the set of all points lying inside the ellipse having its centre at (4, 3) and major and minor axes of lengths 3 and 2 units. It can be easily seen by drawing the graphs of two regions that $A\subseteq B$.

25. Let the number of newspapers be x. If every student reads one newspaper, the number of students would be $60x$. Since every student reads 5 newspapers,
$\therefore$ Number of students $=\frac{x\times 60}{5}=300$
$\Rightarrow x=25$.

26. As $x<y, y<z \quad \Rightarrow \quad x<z$ for all $x,y,z\in N$
$\therefore \quad xRy, yRz \Rightarrow xRz$ $\therefore$ Relation is transitive
As $x<y$ does not give $y<x$.
$\therefore$ Relation is not symmetric.
As $x<x$ does not hold, hence the relation is not reflexive.

27. $y=\frac{10^x-10^{-x}}{10^x+10^{-x}}+1 \quad \Rightarrow \quad y-1=\frac{10^{2x}-1}{10^{2x}+1}$
$\Rightarrow \quad \frac{y-1+1}{y-1-1}=\frac{10^{2x}-1+10^{2x}+1}{10^{2x}-1-10^{2x}-1}$
[By componedo and dividendo]
$\Rightarrow \quad \frac{y}{y-2}=\frac{2(10^{2x})}{-2}$
$\Rightarrow \quad \frac{y}{2-y}=10^{2x} \quad \Rightarrow \quad 2x=\log_{10}\left(\frac{y}{2-y}\right)$
$\Rightarrow \quad x=\frac{1}{2}\log_{10}\left(\frac{y}{2-y}\right)=f^{-1}(y)$
Hence, the inverse of the given function is
$y=\frac{1}{2}\log_{10}\left(\frac{x}{2-x}\right)$

28. Clearly, A is the set of all points lying inside or on the circle with centre at the origin and radius 1 and B is the set of all points lying inside or on the circle with centre at the origin and radius 2 units. Clearly, $A\subseteq B$. Therefore, $A-B=\phi$ and $B-A\neq\phi$.

29. $A\cup B=\{1,2,3\}\cup\{3,8\}=\{1,2,3,8\}$
and $A\cap B=\{1,2,3\}\cap\{3,8\}=\{3\}$
$\therefore \quad (A\cup B)\times(A\cap B)=\{1,2,3,8\}\times\{3\}$
$=\{(1,3),(2,3),(3,3),(8,3)\}$

30. We have $R=\{(1,3),(1,5),(2,3),(2,5),(3,5),(4,5)\}$
$R^{-1}=\{(3,1),(5,1),(3,2),(5,2),(5,3),(5,4)\}$
Hence, $R\circ R^{-1}=\{(3,3),(3,5),(5,3),(5,5)\}$

31. $f(-\pi)=\cos(-9\pi)+\cos(10\pi)$
$=\cos 9\pi+\cos 10\pi=-1+1=0$

32. We have $2\cos^2\theta+\sin\theta\le 2$
$\Rightarrow \quad 2(1-\sin^2\theta)+\sin\theta\le 2$
$\Rightarrow \quad 2\sin^2\theta-\sin\theta\ge 0 \quad \Rightarrow \quad \sin\theta(2\sin\theta-1)\ge 0$
$\Rightarrow \quad \sin\theta\ge 0$ and $2\sin\theta-1\ge 0$ or
$\Rightarrow \quad \sin\theta\le 0$ and $2\sin\theta-1\le 0$

Case I: We have $\sin\theta \geq 0$ and $2\sin\theta - 1 \geq 0$

$\Rightarrow \quad \sin\theta \geq 0$ and $\sin\theta \geq \frac{1}{2}$

$\Rightarrow \quad \frac{\pi}{6} \leq \theta \leq \frac{5\pi}{6}$

$\therefore \quad A \cap B = \left\{\theta : \frac{\pi}{2} \leq \theta \leq \frac{5\pi}{6}\right\}$

$$\left[\because B = \left\{\theta : \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}\right\}\right]$$

Case II: We have $\sin\theta \leq 0$ and $2\sin\theta - 1 \leq 0$

$\Rightarrow \quad \sin\theta \leq 0$ and $\sin\theta \leq \frac{1}{2} \quad \Rightarrow \quad \sin\theta \leq 0$

$\Rightarrow \quad \pi \leq \theta \leq 2\pi$

$\therefore A \cap B = \left\{\theta : \pi \leq \theta \leq \frac{3\pi}{2}\right\}$

$$\left[\because B = \left\{\theta : \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}\right\}\right]$$

Thus, $A \cap B = \left\{\theta : \frac{\pi}{2} \leq \theta \leq \frac{5\pi}{6}\right\} \cup \left\{\theta : \pi \leq \theta \leq \frac{3\pi}{2}\right\}$

33. For any integer n, we have $n|n \Rightarrow nRn$. So, nRn for all $n \in Z \Rightarrow R$ is reflexive. Now 2|6 but 6|2
$\Rightarrow$ (2, 6) . So R is not symmetric.
Let $(m, n) \in R$ and $(n, p) \in R$

Then, $\left.\begin{matrix}(m, n) \in R \Rightarrow m|n \\ (n, p) \in R \Rightarrow n|p\end{matrix}\right] \Rightarrow m|p \Rightarrow (m, p) \in R$.

So, R is transitive. Hence, R is reflexive and transitive but it is not symmetric.

34. C cannot be the empty set because, for instance,

$-1 = \begin{vmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{vmatrix} \in C$. We also have

$$\begin{vmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ 0 & 1 & 1\end{vmatrix} = 2$$

So, $A \neq B \cup C$. In general, the determinant

$$\begin{vmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{13} & a_{32} & a_{33}\end{vmatrix}$$

$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{31}a_{22}$

with the a's being 0 or 1, equals 1 only if $a_{11}a_{22}a_{33} = 1$ and the remaining terms are zero, if $a_{12}a_{23}a_{31} = 1$ and the remaining terms are zero. Since there are three similar relations for determinants that equal – 1, we must have $n(B) = n(C)$.

35. Let U be the set of all consumers who were questioned, A be the set of consumers who liked product P_1 and B be the set of consumers who liked product P_2. It is given that

$$n(U) = 2000,\ n(A) = 1720 \text{ and } n(B) = 1450$$

$\therefore \quad n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow \quad n(A \cup B) = 1720 + 1450 - n(A \cap B)$

$\Rightarrow \quad n(A \cup B) = 3170 - n(A \cap B)$

Now, $A \cup B \subset U$

$\Rightarrow n(A \cup B) \leq n(U) \Rightarrow 3170 - n(A \cap B) \leq 2000$

$\Rightarrow n(A \cap B) \geq 1170$

Hence, the least number of consumers who liked both the products is 1170.

36. We have $(a, b)R(c, d)$ for all $(a, b) \in N \times N$. As $a + b = b + a$, hence, R is reflexive. R is symmetric, for we have $(a, b)R(c, d) \quad \Rightarrow \quad a + d = b + c$

$\Rightarrow \quad d + a = c + b \quad \Rightarrow \quad c + b = d + a$

$\Rightarrow \quad (a, b)R(e, f)$

Then, by definition of R, we have $a + d = b + c$ and $c + f = d + e$
Hence by addition, we get

$$a + d + c + f = b + c + d + e$$

or $$a + f = b + e$$

Hence, $(a, b)R(e, f)$. Thus, $(a, b)R(c, d)$ and $(c,d)R(e,f) \quad \Rightarrow \quad (a,b)R(e,f)$

37. Since $8^n - 7n - 1 = (1+7)^n - 7n - 1$

$= 1 + {}^nC_1 \cdot 7 + {}^nC_2 \cdot 7^2 + {}^nC_3 \cdot 7^3 + \cdots + {}^nC_n \cdot 7^n - 7n - 1$

$= 1 + 7n + {}^nC_2 \cdot 7^2 + {}^nC_3 \cdot 7^3 + \cdots + {}^nC_n \cdot 7^n - 7n - 1$

$= {}^nC_2 7^2 + {}^nC_3 7^3 + \cdots + {}^nC_n 7^n$

$= 49({}^nC_2 + {}^nC_3 7 + \cdots + {}^nC_n 7^{n-2})$

$\therefore \quad 8^n - 7n - 1$ is a multiple of 49 for all $n \in N$.

$\therefore$ X contains elements which are multiples of 49 and clearly Y contains all multiples of 49.

$\therefore \quad X \subseteq Y$.

38. Let F, B and C denote the sets of students who received medals in football, basketball and cricket, respectively. Then, we have

$$n(F) = 37,\ n(B) = 15,\ n(C) = 20,$$
$$n(F \cup B \cup C) = 58 \text{ and } n(F \cap B \cap C) = 3$$

Now,

$$n(F \cup B \cup C) = n(F)+n(B)+n(C)-n(F \cap B) - n(B \cap C)-n(C \cap F) + n(F \cap B \cap C)$$

$$\Rightarrow \quad 58 = 38 + 15 + 20 - \{n(F \cap B)+n(B \cap C) + n(C \cap F)\}+3$$

$$\Rightarrow n(F \cap B)+n(B \cap C)+n(C \cap F)\} = 18$$

Hence, the number of students who received medals in exactly two of the three sports

$$= n(F \cap B)+n(B \cap C)+n(C \cap F)-3n(F \cap B \cap C)$$
$$= 18-3\times 3 = 9$$

39. We have $f(x_1)+f(x_2) = \log\left(\frac{1+x_1}{1-x_1}\right)+\log\left(\frac{1+x_2}{1-x_2}\right)$

$$= \log\left[\left(\frac{1+x_1}{1-x_1}\right)\left(\frac{1+x_2}{1-x_2}\right)\right] = \log\left(\frac{1+x_1+x_2+x_1x_2}{1-x_1-x_2+x_1x_2}\right)$$

$$= \log\left(\frac{1+\frac{x+x_2}{1+x_1x_2}}{1-\frac{x_1+x_2}{1+x_1x_2}}\right) = f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$$

Also, (a), (b) and (c) cannot be true.

40. $n(A \times A \times B) = n(A) \times n(A) \times n(B)$

$$= 3\times 3\times 4 = 36$$

41. We have $n(M) = 23$, $n(P) = 24$, $n(C) = 19$, $n(M \cap P) = 12$, $n(M \cap C) = 9$, $n(P \cap C) = 7$ and $n(M \cap P \cap C) = 4$

$\therefore$ Number of students studying exactly one subject

$$= n(M)+n(P)+n(C)-2[n(M \cap P)+n(P \cap C) + n(M \cap C)]+3n(M \cap P \cap C)$$
$$= 23+24+19-2\{12+9+7\}+3\times 4$$
$$= 66-2\times 28+12 = 22$$

42. We have $f(x+2y, x-2y) = xy$ (i)

Let $x+2y = u$ and $x-2y = v$

Then, $x = \frac{u+v}{2}$ and $y = \frac{u-v}{4}$

Substituting the values of x and y in (i), we obtain

$$f(u, v) = \frac{u^2-v^2}{8} \quad \Rightarrow \quad f(x, y) = \frac{x^2-y^2}{8}$$

43. bN = the set of +ve integral multiples of b

cN = the set of +ve integral multiples of c

$\therefore$ $bN \cap cN$ = the set of +ve integral multiples of bc.

$= b\ c\text{N}$ [$\because$ b and c are relatively prime]

or $d = bc$ $\therefore$ LCM of b and c = bc.

44. Clearly, A is the set of all points on the circle $x^2+y^2 = 25$ and B is the set of all points on the ellipse $x^2+9y^2 = 144$. These two intersect at four points P, Q, R and S. Hence, $A \cap B$ contains four points.

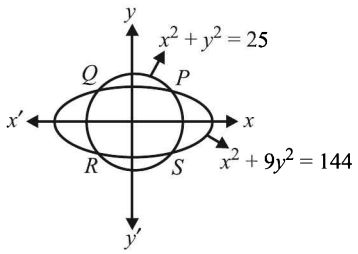


Fig. 1.41

45. As $f \circ f(x) = f(f(x)) = f\left(\frac{1-x}{1+x}\right)$

$$= \frac{1-[(1-x)/(1+x)]}{1+[(1-x)/(1+x)]} = x, \text{ for all } x$$

So, the inverse of f is f itself. It can be easily seen that $g \circ g(x) \neq x$ and $h \circ h(x) \neq x$.

46. Since $f_n(x) = f_1(f_{n-1}(x))$ for $n \geq 2$

$$\therefore \quad f_2(x) = f_1(f_1(x)) = \frac{1}{4}f_1(x)+6$$

$$= \frac{1}{4}\left(\frac{x}{4}+6\right)+6 = \frac{x}{4^2}+\frac{6}{4}+6$$

$$f_3(x) = f_1(f_2(x)) = f_1\left(\frac{x}{4^2}+\frac{6}{4}+6\right)$$

$$= \frac{x}{4^3}+\frac{6}{4^2}+\frac{6}{4}+6$$

$$f_4(x) = f_2(f_3(x)) = f_1\left(\frac{x}{4^3}+\frac{6}{4^2}+\frac{6}{4}+6\right)$$

$$= \frac{x}{4^4}+\frac{6}{4^3}+\frac{6}{4^2}+\frac{6}{4}+6$$

and so on.

$$\therefore \quad f_n(x) = \frac{x}{4^n}+\frac{6}{4^{n-1}}+\frac{6}{4^{n-2}}+\cdots+\frac{6}{4}+6$$

$$= \frac{x}{4^n}+6\left[1+\frac{1}{4}+\frac{1}{4^2}+\cdots+\frac{1}{4^{n-1}}\right]$$

$$= \frac{x}{4^n}+6\frac{1-\frac{1}{4^n}}{1-\frac{1}{4}} = \frac{x}{4^n}+8\left(1-\frac{1}{4^n}\right) = \frac{x-8}{4^n}+8$$

47. Putting $x = 1$, $f(2)+f(0) = 2f(1) \quad \Rightarrow \quad f(2) = 2f(1)$

Putting $x = 2$, $f(3)+f(1) = 2f(2)$

$\Rightarrow \quad f(3) = 2 \times 2f(1) - f(1) = 3f(1)$, and so on

$\therefore \quad f(n) = nf(1), \quad$ for $n = 1, 2, \ldots, n$

Also, $f(n+1) + f(n-1) = 2f(n)$

$\Rightarrow \quad f(n+1) + (n-1)f(1) = 2nf(1)$

$\Rightarrow \quad f(n+1) = (n+1)f(1)$

48. $f(-x) = \log(-x + \sqrt{x^2+1})$

$$= \log\left(\frac{(\sqrt{x^2+1} - x)(\sqrt{x^2+1} + x)}{\sqrt{x^2+1} + x}\right)$$

$$= \log\left(\frac{x^2 + 1 - x^2}{\sqrt{x^2+1} + x}\right)$$

$$= -\log(\sqrt{x^2+1} + x) = -f(x)$$

$\therefore f(x)$ is an odd function.

49. We have $f_1 = \{(1, 2), (2, 3), (3, 4)\}$

$f_2 = \{(1, 4), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4), (4, 1), (4, 2), (4, 3)\}$

$f_3 = \{(2, 1), (3, 1), (3, 1), \ldots\}$

$f_4 = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$

Obviously, f_4 is a function from A to itself and f_1, f_2, f_3 are not functions from A to themselves.

50. As $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$, given by $f(x) = 2x - 3$ and $g(x) = x^3 + 5$, respectively, are bijections, f^{-1} and g^{-1} exist. We have $f(x) = 2x - 3$

$$\therefore \quad f(x) = y \Rightarrow 2x - 3 = y \Rightarrow x = \frac{y+3}{2}$$

$$\Rightarrow \quad f^{-1}(y) = \frac{y+3}{2}$$

Thus, f^{-1} is given by $f^{-1}(x) = \frac{x+3}{2}$ for all $x \in \mathbb{R}$

Similarly, $g^{-1}(x) = (x-5)^{1/3}$ for all $x \in \mathbb{R}$

Now, $(fog)^{-1}(x) = (g^{-1}of^{-1})(x) = g^{-1}(f^{-1}(x))$

$$= g^{-1}\left(\frac{x+3}{2}\right) = \left(\frac{x+3}{2} - 5\right)^{1/3} = \left(\frac{x-7}{2}\right)^{1/3}$$

51. $B = C$

52. Let the original set contain $(2n + 1)$ elements, then the number of subsets containing more than n elements, that is, subsets containing $(n + 1)$ elements, $(n + 2)$ elements, …, $(2n + 1)$ elements will be required to be computed.

$\therefore$ Required number of subsets

$$= {}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + \cdots + {}^{2n+1}C_{2n} + {}^{2n+1}C_{2n+1}$$

$$= {}^{2n+1}C_n + {}^{2n+1}C_{n+1} + \cdots + {}^{2n+1}C_1 + {}^{2n+1}C_0$$

$$= {}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \cdots + {}^{2n+1}C_{n-1} + {}^{2n+1}C_n$$

$$= \frac{1}{2}\left[(1+1)^{2n+1}\right] = \frac{1}{2}\left[2^{2n+1}\right] = 2^{2n}$$

53. $n((A \times B) \cap (B \times A))$

$= n((A \cap B) \times (B \cap A)) = n(A \cap B) \cdot n(B \cap A)$

$= n(A \cap B) \cdot n(A \cap B) = (99)(99) = 99^2$

54. On the set N of natural numbers, $R = \{(x, y) : x, y \in N$ and $2x + y = 41\}$

Here $(1, 1) \notin R$ as $2 \cdot 1 + 1 = 3 \neq 41$. So, R is not reflexive. $(1, 39) \in R$ but $(39, 1) \notin R$. So, R is not symmetric. $(20, 1), (1, 39) \in R$. But $(20, 39) \notin R$. So, R is not transitive.

55. $f(x+1) - f(x) = 8x + 3$ gives

$$a(x+1)^2 + b(x+1) + c - ax^2 - bx - c = 8x + 3$$

or $\qquad 2ax + a + b = 8x + 3$

$\Rightarrow \quad 2a = 8$ and $a + b = 3$

$\Rightarrow \quad a = 4 \ \therefore \ 4 + b = 3 \Rightarrow b = -1$

[By equating coefficients of x and constant terms]

Hence, $a = 4$ and $b = -1$

56. $|a - a| = 0 < 1 \quad \therefore \ aRa$ for all $a \in R$

$\therefore R$ is reflexive.

Again $aRb \ \Rightarrow \ |a - b| \leq 1$ and $|b - a| \leq 1 \ \Rightarrow \ bRa$

$\therefore R$ is symmetric.

Again, $1R\frac{1}{2}$ and $\frac{1}{2}R1$ but $\frac{1}{2} \neq 1$.

$\therefore R$ is not anti-symmetric.

Further, $1R2$ and $2R3$ but not $1R3$. $\quad [\because |1 - 3| = 2 > 1]$

$\therefore R$ is not transitive.

57. $f(x) = 2^{x(x-1)}$, i.e. $y = 2^{x(x-1)}$

$\Rightarrow \log_2 y = x(x-1) = x^2 - x$

$\Rightarrow x^2 - x - \log_2 y = 0$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 + 4\log_2 y}}{2}$$

Since $x \in [1, \infty) \quad \therefore -$ve sign is ruled out.

$$\therefore \quad x = \frac{1}{2}(1 + \sqrt{1 + 4\log_2 y})$$

$$\Rightarrow \quad f^{-1}(y) = \frac{1}{2}(1 + \sqrt{1 + 4\log_2 y})$$

$$\Rightarrow \quad f^{-1}(x) = \frac{1}{2}(1 + \sqrt{1 + 4\log_2 x})$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. We have $R = \{(x, y) : x^2 + y^2 = 1 \text{ and } x, y \in R\}$

If $4 \in R$, then $(4)^2 + (4)^2 = 32 \neq 1 \quad \therefore (4, 4) \notin R$

Hence R is not reflexive.

Let $(x, y) \in R \quad \therefore x^2 + y^2 = 1$

$\Rightarrow y^2 + x^2 = 1$

$\Rightarrow (y, x) \in R$

$\therefore R$ is symmetric.

Now, $(0, 1), (1, 0) \in R$, because

$(0)^2 + (1)^2 = 1$ and $(1)^2 + (0)^2 = 1$

But $(0)^2 + (0)^2 = 0 \neq 1 \quad \therefore (0, 0) \notin R$

$\therefore$ Hence R is not transitive

2. Clearly, $A = \{2, 3\}, B = \{2, 4\}, C = \{4, 5\}$ and $B \cap C = \{4\}$

$\therefore A \times (B \cap C) = \{(2, 4), (3, 4)\}$

3. Let $n \in N \quad \therefore n$ is a factor of n.

$\therefore nRn \quad \therefore R$ is reflexive.

$3R6$ because 3 is a factor of 6,

Also 6 is not a factor of 3.

$\therefore (6, 3) \notin R, \quad \therefore R$ is not symmetric.

Let aRb and bRc

$\therefore \quad b = ak_1$ and $c = bk_2$ for $k_1, k_2 \notin N$

$\therefore \quad c = (ak_1)k_2 = a(k_1k_2)$ or a is a factor of c

$\therefore \quad aRc, \quad \therefore R$ is transitive.

4. $f(g(x)) = f(x^2 + 7) = 2(x^2 + 7) + 3 = 25$

$\Rightarrow \quad 2x^2 = 8$

$\Rightarrow \quad x = \pm 2$

5. Let $A = \{1, 2, 3\}$

Let the two transitive relations on set A be

$R = \{(1, 1), (1, 2)\}$

and $S = \{(2, 2), (2, 3)\}$

Now, $R \cup S = \{(1, 1), (1, 2), (2, 2), (2, 3)\}$

Here, $(1, 2), (2, 3) \in R \cup S$

$\Rightarrow (1, 3) \notin R \cup S$

$\therefore R \cup S$ is not transitive.

6. We have $R = \{(1, 2), (2, 3), (1, 3)\}$

Here $(2, 3) \in R$ but $(3, 2) \notin R$.

$\therefore R$ is not symmetric.

The relation R is transitive only, because if $(1, 2) \in R$ and $(2, 3) \in R$, then $(1, 3) \in R$

7. We have, $g(-3) = 0$

$\Rightarrow \quad f(g(-3)) = f(0) = 7(0)^2 + 0 - 8 = -8$

$\Rightarrow \quad f \circ g(-3) = -8$

$g(9) = 9^2 + 4 = 85$

$\Rightarrow \quad f(g(9)) = f(85) = 8 \cdot 85 + 3 = 683$

$\Rightarrow \quad f \circ g(9) = 683$

$f(0) = 7 \cdot 0^2 + 0 - 8 = -8$

$\Rightarrow \quad g(f(0)) = g(-8) = |-8| = 8$

$f(6) = 4 \cdot 6 + 5 = 29$

$\Rightarrow \quad g(f(6)) = g(29) = (29)^2 + 4 = 845$

$\Rightarrow \quad g \circ f(6) = 845$

8. For (c): $f(-x) = -x\dfrac{a^{-x} - 1}{a^{-x} + 1}$

$= x\dfrac{a^x - 1}{a^x + 1} = f(x)$

$\therefore f(x)$ is even. For others $f(x)$ is odd.

9. We have $|2 + 3i| = \sqrt{2^2 + 3^2} = \sqrt{13} \neq 13$

$\therefore (2 + 3i)R13$ is wrong.

$|3| = \sqrt{3^2 + 0^2} = \sqrt{9} = 3 \neq -3$

$\therefore 3R(-3)$ is wrong

$|1 + i| = \sqrt{1^2 + 1^2} = \sqrt{2} \neq 2$

$\therefore (1 + i)R2$ is wrong

$|i| = \sqrt{0^2 + 1^2} = \sqrt{1} = 1$

$\therefore iR1$ is correct.

10. $(f \circ g)(x) = f(g(x)) = \dfrac{1 - g(x)}{1 + g(x)} \qquad \left[\because f(x) = \dfrac{1 - x}{1 + x}\right]$

$= \dfrac{1 - 4x(1 - x)}{1 + 4x(1 - x)} = \dfrac{1 - 4x + 4x^2}{1 + 4x - 4x^2}$

11. Since $R \subseteq A \times B$ and $S \subseteq B \times C$, we have $S \circ R \subseteq A \times C$

$\therefore S \circ R$ is a relation from A to C.

12. $f(f(x)) = \dfrac{\alpha f(x)}{f(x) + 1} = \dfrac{\alpha \cdot \dfrac{\alpha x}{x + 1}}{\dfrac{\alpha x}{x + 1} + 1}$

$= \dfrac{\alpha^2 x}{\alpha x + x + 1} = \dfrac{\alpha^2 x}{(\alpha + 1)x + 1}$

$f(f(x)) = x \quad \Rightarrow \quad \dfrac{\alpha^2 x}{(\alpha + 1)x + 1} = x$

$\Rightarrow \quad \alpha^2 = (\alpha + 1)x + 1 \Rightarrow \alpha + 1 = 0 \Rightarrow \alpha = -1$

13. For any $x, y \in N$, we have

$f(x) = f(y) \quad \Rightarrow \quad 2x + 3 = 2y + 3$

$\Rightarrow \quad x = y$

So, f is injective. f is not surjective because $1 \in N$ does not have its pre-image.

14. R is reflexive because $a \geq a$ for $a \in R$

We have $(4, 2) \in R$ because $4 \geq 2$ but $(2, 4) \notin R$.

$\therefore$ R is not symmetric.

Let $(a, b), (b, c) \in R \quad \therefore a \geq b, b \geq c.$

$\Rightarrow a \geq c \quad \Rightarrow \quad (a, c) \in R \quad \Rightarrow \quad R$ is transitive.

15. $p(x) = a^2 + bx,\ q(x) = lx^2 + mx + n$

$\therefore \quad p(1) - q(1) = 0$ gives

$$a^2 + b - 1 - m - n = 0 \qquad \text{(i)}$$

$$p(2) - q(2) = 1 \text{ gives}$$

$$a^2 + 2b - 4l - 2m - n = 1 \qquad \text{(ii)}$$

$$p(3) - q(3) = 4 \text{ gives}$$

$$a^2 + 3b - 9l - 3m - n = 4 \qquad \text{(iii)}$$

$$p(4) - q(4) = a^2 + 4b - 16l - 4m - n = ?$$

Eq. (iii) – Eq. (ii) gives: $b - 5l - m = 3$

$\therefore \quad b = 5l + m + 3$

Eq. (ii) – (i) gives $b - 3l - m = 1$

$\therefore \quad b = 3l + m + 1$

$\therefore \quad 5l + m + 3 = 3l + m + 1$

$\Rightarrow \quad 2l = -2 \quad \Rightarrow \quad l = -1$

$\therefore \quad -5 + m + 3 = b \quad \Rightarrow \quad m = 2 + b$

$\therefore \quad a^2 + b = l + m + n$ gives

$$a^2 + b = -1 + 2 + b + n = l + m + n$$

$\Rightarrow \quad a^2 = n + 1$

$$p(4) - q(4) = a^2 + 4b - 16l - 4m - n$$
$$= a^2 + 4b - 16(-1) - 4(2 + b) - n$$
$$= a^2 + 4b + 16 - 8 - 4b - n$$
$$= n + 1 + 8 - n = 9$$

16. Let $A = \{11, 12, 13\}$, and $B = \{8, 10, 12\}$

$\therefore \quad R = \{(11, 8), (13, 10)\}$

$R^{-1} = \{(8, 11), (10, 13)\}$

17. $y = \dfrac{3x + 2}{5x - 3} \quad \Rightarrow \quad x = \dfrac{3y + 2}{5y - 3} = f^{-1}(y)$

$\therefore \quad f^{-1}(x) = \dfrac{3x + 2}{5x - 3} = f(x)$

18. $x \in R \Rightarrow x - x + \sqrt{5} = \sqrt{5}$ is an irrational number.

$\therefore (x, x) \in R$, so R is reflexive.

$(\sqrt{5}, 1) \in R$ because $\sqrt{5} - 1 + \sqrt{5} = 2\sqrt{5} - 1$, which is an irrational number.

Also, $(1, \sqrt{5}) \in R$ if $1 - \sqrt{5} + \sqrt{5} = 1$, which is not an irrational number.

$\therefore (1, \sqrt{5}) \notin R \quad \therefore R$ is not symmetric.

We have $(\sqrt{5}, 1), (1, 2\sqrt{5}) \in R$ because

$\sqrt{5} - 1 + \sqrt{5} = 2\sqrt{5} - 1$ and $1 - 2\sqrt{5} + \sqrt{5} = 1 - \sqrt{5}$ are irrational numbers.

Also $(\sqrt{5}, 2\sqrt{5}) \in R$ if $\sqrt{5} - 2\sqrt{5} + \sqrt{5} = 0$, which is not an irrational number.

$\therefore (\sqrt{5}, 2\sqrt{5}) \notin R$. So, R is not transitive.

19. $g \circ f$ is injective (i.e. one-one). For, we have

$(g \circ f)(x) = (g \circ f)(y) \quad (x, y \in A)$

$\Rightarrow \quad g[f(x)] = g[f(y)]$

$\Rightarrow \quad f(x) = f(y) \qquad [\because g \text{ is one-one}]$

$\Rightarrow \quad x = y \qquad [\because f \text{ is one-one}]$

$g \circ f$ is surjective (i.e. onto). For let z be any arbitrary element of C. Then since g is onto, there exists an element $y \in B$ such that $g(y) = z$. Again since f is onto, there exists an element $x \in A$ such that $f(x) = y$. Thus, to each $z \in C$, there exists an element $x \in A$ such that

$$z = g(y) = g[f(x)] = (g \circ f)(x)$$

Hence, $g \circ f$ is both injective and surjective and so it is one-one and onto (bijective).

20. For reflexive add (1, 1), (2, 2), (3, 3) $\Rightarrow$ 3

For symmetric add (2, 1), (3, 2) $\Rightarrow$ 2

For transitive add (1, 3), (3, 1) $\Rightarrow$ 2

$\therefore$ toal = 3 + 2 + 2 = 7

21. Take $f(x) = \log x$

Then $f\left(\dfrac{x}{y}\right) = f(x) - f(y)$

Clearly $f(x)$ is not bounded and

$f\left(\dfrac{1}{x}\right) = -\log x \to \infty \quad \text{as } x \to 0$

Also $xf(x) = x \log x \to 0$ as $x \to 0$.

Hence (d) is correct.

22. Statement 2 is trivially true. Also, $2^6 - 2^3 = 64 - 8 = 56$

$\therefore$ Statement 1 is true.

Also, Statement 2 is a correct explanation for Statement 1.

23. $X \cap (X \cup Y)' = X \cap (X' \cap Y')$

$= (X \cap X') \cap Y' = \phi \cap Y' = \phi$

$\therefore$ Statement 1 is true.

Now, $\qquad y = \dfrac{1}{x} \Rightarrow xy = 1$

$\therefore$ The set A contains all the points on the hyperbola $xy = 1$.

$y = -x$ is a line passing through the origin and having inclination 135°.

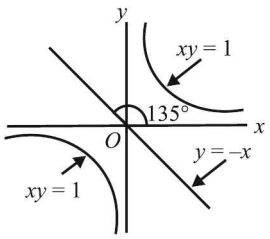


Fig. 1.48

There is no common point.

$\therefore A \cap B = \phi$. $\therefore$ Statement 2 is false.

24. We have $A - B = \{1, 2, 3\}$ and $A \cap C = \{1, 2\}$

$\therefore \quad (A-B)\times(A\cap C) = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$

$\therefore$ Statement 1 is false.

$A = \{4, 5, 6, 7, \ldots\}, B = \{1, 2, 3, 4, 5\}$

$\therefore \quad A \cap B = \{4, 5\}$.

$\therefore$ Statement 2 is true.

25. $2 \in R$ and $2^2 + 2^2 = 8 \neq 1$.

$\therefore$ P is not reflexive.

$\therefore$ Statement 1 is true.

Let $(x, y) \in P \quad \Rightarrow \quad x^2 + y^2 = 1$

$\Rightarrow (y, x) \in P$

$\therefore$ P is symmetric

$\therefore$ Statement 2 is true.

But, Statement 2 is not a correct explanation for Statement 1.

26. Let A be a nonempty set in X.

$\therefore \quad A \cap A = A \neq \phi \quad \therefore \quad A$ is not related to A.

$\therefore$ R is not reflexive

$\therefore$ Statement 1 is false.

Let $A = \{1, 2, 3\}, B = \{4, 5\}, C = \{3, 7\} \in X$.

$\therefore \quad A \cap B = \phi, B \cap C = \phi$

$\therefore$ ARB and BRC

Also, $3 \in A \cap C \quad \therefore A \cap C \neq \phi$

$\therefore$ A is not related to C

$\therefore$ R is not transitive.

$\therefore$ Statement 2 is true.

27. R is a function as $A = \{1, 2, 3, 4\}$ and $(2, 4) \in R$ and $(2, 3) \in R$.

R is not reflexive as $(1, 1) \notin R$

R is not symmetric as $(2, 3) \in R$ but $(3, 2) \notin R$.

R is not transitive as $(1, 3) \in R$ and $(3, 1) \in R$ but $(1, 1) \notin R$.

28. $f(x) = 5^{x(x-4)}$ for f: $[4, \infty[\to [4, \infty[$

At $x = 4$

$f(x) = 5^{4(4-4)} = 1$

which does not lie in the interval $[4, \infty[$.

$\therefore$ Function is not bijective.

Hence, $f^{-1}(x)$ is not defined.

29. Let $f(x) = g(x) + 1$

where $g(x) = 2\left[\frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x\right] = 2\sin(x - 60°)$

$\therefore \quad -2 \le 2\sin(x - 60°) \le 2$

$-1 \le 2\sin(x - 60°) + 1 \le 3$

30. If $y = f(x)$ is symmetrical about the line $x = \alpha$, then $f(x + \alpha) = f(x - \alpha)$

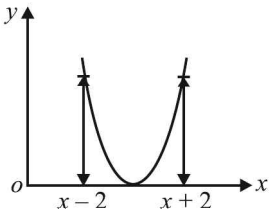


Fig. 1.39

$\therefore \quad f(x+2) = f(x-2)$

31. $f(x) = \frac{p(x)}{q(x)}$ (say), then the domain of $f(x)$ is $D_f p(x) \cap D_f q(x), q(x) \neq 0$. Now the domain of $p(x)$ is

$$-\frac{\pi}{2} \le \sin^{-1}(x-3) \le \frac{\pi}{2}$$

$$\Rightarrow \quad -\sin\frac{\pi}{2} \le x - 3 \le \sin\frac{\pi}{2}$$

$$\Rightarrow \quad 2 \le x \le 4 \qquad \text{(i)}$$

Again $9 - x^2 > 0 \Rightarrow x^2 < 9, |x| < 3$

i.e. $-3 < x < 3$ (ii)

From (i) and (ii), we have

$\therefore$ $2 \le x < 3$ is the correct domain.

32. For $(3, 9) \in R$, $(9, 3) \notin R$

$\therefore$ Relation is not symmetric which means our choices (a) and (b) are out of question. We therefore need to prove reflexivity and transitivity.

For reflexive $a \in A, (a, a) \in R$ which holds, i.e. R is reflexive. Again, for transitivity of $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$ which is also true in $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$.

33. For $x \in (-1, 1)$ we have $f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$$\therefore \quad f(\tan\theta) = \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)$$

$$= \tan^{-1}(\tan 2\theta) = 2\tan^{-1}x$$

$$\Rightarrow \quad -\frac{\pi}{2} < \tan^{-1}\left(\frac{2x}{1-x^2}\right) < \frac{\pi}{2}$$

34. $f(x) = x^3 + 6x^2 + 6$

$f'(x) = 3x^2 + 12x = 3x(x+4)$

$f'(x) > 0 \quad \Rightarrow \quad x < -4$ and $x > 0$

The interval $x < -4$, i.e $(-\infty, -4]$ matches correctly and after checking others we find that $f(x) = 3x^2 - 2x + 1$ $\Rightarrow f'(x) > 0$ for $x > 1/3$ which is not given in the choice.

35. Given $f(x-y) = f(x)f(y) - f(a-x)f(a+y)$

Let $x = 0 = y$. Then

$f(0) = (f(0))^2 - (f(a))^2$

$\Rightarrow \quad 1 = 1 - (f(a))^2 \Rightarrow f(a) = 0$

$\therefore f(2a - x) = f(a - (x - a))$

$= f(a) f(x - a) - f(a + x - a) f(0)$

$= 0 - f(x)\,(1) = -f(x)$

$[\because f(a) = 0, f(0) = 1]$

36. Given relation R such that $R = \{(x, y) \in W \times W$: the words x and y have at least one letter in common$\}$, where W denotes a set of words in English dictionary.

Clearly $(x, x) \in R$ for all $x \in W$

$\therefore$ (x, x) has every letter common. $\therefore$ R is reflexive. Let $(x, y) \in R$, then $(y, x) \in R$ as x and y have at least one letter in common. It implies R is symmetric. But R is not transitive.

$\therefore$ Let x = DON, y = NEST, z = SHE, then $(x, y) \in R$ and $(y, z) \in R$. But $(x, z) \notin R$.

$\therefore$ R is reflexive, symmetric but not transitive.

37. Number of ways $= {}^{12}C_4 \times {}^{8}C_4 \times {}^{4}C_4 = \dfrac{12!}{(4!)^3}$.

38. To be an equivalence relation, the relation must be all—reflexive, symmetric and transitive.

$T = \{(x, y) : x - y \in Z\}$ is

(a) reflexive for $(x, x) \in Z$, i.e. $x - x = 0 \in Z$

(b) symmetric for $(x, y) \in Z \Rightarrow x - y \in Z$

$\Rightarrow \quad y - x \in Z$, i.e. $(y, x) \in Z$

(c) transitive for $(x, y) \in Z$ and $(y, w) \in Z$

$\Rightarrow \quad x - y \in Z$ and $y - w \in Z$, giving

$x - w \in Z$, i.e. $(x, w) \in Z$.

$\therefore$ T is an equivalence relation on R.

$S = \{(x, y) : y = x + 1$ and $0 < x < 2\}$ is not reflexive, for $(x, x) \in S$ would imply $x = x + 1$

$\Rightarrow 0 = 1$ (impossible)

Thus, S is not an equivalence relation.

39. Let $f(x_1) = f(x_2), x_1, x_2 \in N$

$\Rightarrow \quad 4x_1 + 3 = 4x_2 + 3 \quad \Rightarrow \quad x_1 = x_2$

Thus, $f(x_1) = f(x_2) \quad \Rightarrow \quad x_1 = x_2$. Hence, the function is one-one. Let $y \in Y$ be a number of the form $y = 4k + 3$, for some $k \in N$, then $y = f(x)$

$\Rightarrow \quad 4k + 3 = 4x + 3 \quad \Rightarrow \quad x = k \in N$

Thus, corresponding to any $y = Y$, we have $x \in N$. The function then is onto.

The function, being both one-one and onto, is invertible.

$y = 4x + 3 \quad \Rightarrow \quad x = \dfrac{y-3}{4} \quad \therefore \quad f^{-1}(x) = \dfrac{x-3}{4}$

or $g(y) = \dfrac{y-3}{4}$ is the inverse of the function.

40. **First Method**

Let $x \in C$, suppose $x \in A \quad \Rightarrow \quad x \in A \cap C$

$\Rightarrow \quad x \in A \cap B \qquad (\because A \cap C = A \cap B)$

Thus $x \in B$.

Again suppose, $x \notin A \quad \Rightarrow \quad x \in C \cup A$

$\Rightarrow \quad x \in B \cup A \quad \Rightarrow \quad x \in B$

Thus in both cases $x \in C \quad \Rightarrow \quad x \in B$

Hence, $C \subseteq B$ (i)

Similarly, we can show that $B \subseteq C$. (ii)

Combining (i) and (ii), we get $B = C$.

Second Method

$A \cap B = A \cap C$. Forming union with B on both sides,

$B \cup (A \cap B) = B \cup (A \cap C)$

$\Rightarrow \quad B = (B \cup A) \cap (B \cup C)$

[Using $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$]

$= (C \cup A) \cap (C \cup B) = C \cup (A \cap B) = C \cup (A \cap C) = C$

Thus, $B = C$.

41. The function is $f : R \to R, f(x) = x^3 + 5x + 1$

Let $y \in R$ then $y = x^3 + 5x + 1$

$\Rightarrow \quad x^3 + 5x + 1 - y = 0$

As a polynomial of odd degree has always at least one real root, corresponding to any $y \in$ co-domain then there exists some $x \in$ domain such that $f(x) = y$. Hence, f is onto. Also f is continuous on R, because it is a polynomial function $f'(x) = 3x^2 + 5 > 0$ $\therefore$ f is strictly increasing. Hence, f is one-one as well.

42. The solution of $f(x) = f^{-1}(x)$ is given by $f(x) = x$, which gives $(x + 1)^2 - 1 = x$

$\Rightarrow \quad (x+1)^2 - (x+1) = 0 \quad \Rightarrow \quad (x+1)x = 0$

$\therefore$ $x = -1, 0$. But as no co-domain of f is specified, nothing can be said about f being onto or not.

43. Number of reflexive relations of a set of 4 elements $= 2^{4^2 - 4} = 2^{12}$.

44. (a) is incorrect since $\{a\}$ is a set not an element.

(c) is incorrect since ϕ is a set.

(b) is correct.

45. Let $y = \dfrac{2x-1}{x+5}$

$\Rightarrow \quad x = \dfrac{5y+1}{2-y}$

$\therefore f^{-1}(x) = \dfrac{5x+1}{2-x}, x \neq 2$

46. $f(x) = \dfrac{x}{x-1}$

$\Rightarrow \quad f(f(x)) = \dfrac{\frac{x}{x-1}}{\frac{x}{x-1}-1} = x$

$\Rightarrow \quad f(f(f(x))) = \dfrac{x}{x-1}$, and so on

$\Rightarrow (f \circ f \circ f \circ \ldots)$ (19 times) $= \dfrac{x}{x-1}$

47. $5x - 3 - 2x^2 \geq 0$

$\Rightarrow \quad 2x^2 - 5x + 3 \leq 0$

$\Rightarrow \quad 2x^2 - 3x - 2x + 3 \leq 0$

$\Rightarrow \quad x(2x-3) - 1(2x-3) \leq 0$

$\Rightarrow \quad (2x-3)(x-1) \leq 0$

$\Rightarrow x \in \left[1, \dfrac{3}{2}\right]$

48. $f(x) = \cos x - \sin x$

$= \sqrt{2}\left(\dfrac{1}{\sqrt{2}}\cos x - \dfrac{1}{\sqrt{2}}\sin x\right) = \sqrt{2}\cos\left(\dfrac{\pi}{4} + x\right)$

Since $-1 \leq \cos x \leq 1 \quad \Rightarrow \quad -1 \leq \cos\left(\dfrac{\pi}{4}+x\right) \leq 1$

$\Rightarrow \quad -\sqrt{2} \leq \sqrt{2}\cos\left(\dfrac{\pi}{4}+x\right) \leq \sqrt{2}$

$\therefore$ Range is $[-2, \sqrt{2}]$.

49. $7 - x > 0, x - 3 \geq 0, 7 - x \geq x - 3$

$\Rightarrow \quad x < 7, x \geq 3, 2x \leq 10$

$\Rightarrow \quad x = 3, 4, 5 \qquad (\because x \text{ is an integer})$

$\Rightarrow$ range $= \{f(3), f(4), f(5)\}$

$= \{{}^4P_0, {}^3P_1, {}^2P_2\} = \{1, 3, 2\}$

50. $n(C) = 63, n(A) = 76$ and $n(C \cap A) = x$

We know that

$n(C \cup A) = n(C) + n(A) - n(C \cap A)$

$\Rightarrow \quad 100 = 63 + 76 - x \quad \Rightarrow \quad x = 139 - 100 = 39$

and $\quad n(C \cap A) \leq n(C)$

$\Rightarrow \quad x \leq 63 \qquad \therefore 39 \leq x \leq 63$

51. $f(x) = \dfrac{1}{2} - \tan\left(\dfrac{\pi x}{2}\right), -1 < x < 1$

Given, domain of $f(x)$ is $d_1 = (-1, 1)$

For domain of $g(x)$, $3 + 4x - 4x^2 \geq 0$

$\Rightarrow \quad (2x-3)(2x+1) \leq 0$

$\therefore$ Domain of $g(x)$ is $d_2 = \left[-\dfrac{1}{2}, \dfrac{3}{2}\right]$

Hence, domain of $(f+g) = d_1 \cap d_2 = \left[-\dfrac{1}{2}, 1\right)$.

52. The number of onto functions from E to F

$= 2^4 - 2 = 16 - 2 = 14.$

53. $f(x) = (x+1)^2$ for $x \geq -1$

Reflection of the graph of $f(x)$ w.r.t. the line $y = x$ is the inverse of the function.

$\therefore \quad x = (y+1)^2 \quad \Rightarrow \quad \sqrt{x} = y + 1$

$\Rightarrow \quad y = \sqrt{x} - 1$

54. $f(x) = \dfrac{x^2+x+2}{x^2+x+1} = 1 + \dfrac{1}{x^2+x+1}$

$= 1 + \dfrac{1}{\left(x+\frac{1}{2}\right)^2 + \frac{3}{4}}$

$\therefore$ Range $= \left(1, 1 + \dfrac{1}{0+\frac{3}{4}}\right]$ i.e. $\left(1, \dfrac{7}{3}\right]$.

55. For $D_f, -1 \leq 2x \leq 1$ (i)

$(\because \sin^{-1}(t)$ is defined for $-1 \leq t \leq 1$ only)

and $\quad \sin^{-1}(2x) + \dfrac{\pi}{6} \geq 0$

$\therefore \quad \sin^{-1}(2x) \geq -\dfrac{\pi}{6} \quad \Rightarrow \quad 2x \geq -\dfrac{1}{2}$

$\Rightarrow \quad x \geq -\dfrac{1}{4} \quad \Rightarrow \quad -\dfrac{1}{4} \leq x$ (ii)

From (i), $-\dfrac{1}{2} \leq x \leq \dfrac{1}{2}$ (iii)

From (ii) and (iii), $-\dfrac{1}{4} \leq x \leq \dfrac{1}{2}$

$\therefore \quad D_f = \left[-\dfrac{1}{4}, \dfrac{1}{2}\right]$

56. Let $h(x) = f(x) - g(x) = \begin{cases} x-0; x \notin Q \\ 0-x; x \in Q \end{cases}$

i.e. $h(x) = \begin{cases} x; x \notin Q \\ -x; x \in Q \end{cases}$

$\Rightarrow$ $h(x)$ is one-one and onto.

57. o is commutative.

Since, $b \circ a = ba + 1 = ab + 1 = a \circ b$

Now $(a \circ b) \circ c = (ab+1) \circ c = (ab+1)c + 1$

$= abc + c + 1$

and $a \circ (b \circ c) = a \circ (bc+1) = a(bc+1) + 1$

$= abc + a + 1$

$\Rightarrow$ $(a \circ b) \circ c \neq a \circ (b \circ c)$

$\Rightarrow$ $\circ$ is non-associative.

58. $g(x) = 1 + x - [x]$ is greater than 1 since $x - [x] > 0$

$\therefore$ $f[g(x)] = 1$.

59. $g[f(x)] = x = f[g(x)]$

$\Rightarrow$ $g'[f(x)]f'(x) = 1$

$\Rightarrow$ $g'[f(x)]\dfrac{1}{f'(x)} = 1 + x^5$

$\Rightarrow$ $g'[f(x)] = 1 + g[f(x)]^5$

$\Rightarrow$ $g'(x) = 1 + [g(x)]^5$

60. $f(x) = (x-1)^2 + 1, x \geq 1$

$f : [1, \infty) \to [1, \infty)$ is a bijective function

$\Rightarrow$ $y = (x-1)^2 + 1$ $\Rightarrow$ $(x-1)^2 = y - 1$

$\Rightarrow$ $x = 1 \pm \sqrt{y-1}$ $\Rightarrow$ $f^{-1}(y) = 1 \pm \sqrt{y-1}$

$\Rightarrow$ $f^{-1}(x) = 1 + \sqrt{x-1}$ $\{\because x \geq 1\}$

So Statement 2 is correct.

Now $f(x) = f^{-1}(x)$ $\Rightarrow$ $f(x) = x$

$\Rightarrow$ $(x-1)^2 + 1 = x$ $\Rightarrow$ $x^2 - 3x + 2 = 0$

$\Rightarrow$ $x = 1, 2$

So Statement 1 is correct.

61. $P(A' \cap B'/C) = \dfrac{P((A' \cap B') \cap C)}{P(C)}$

$= \dfrac{P(C) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)}{P(C)}$

$= \dfrac{P(C) - P(A) \cdot P(C) - P(B)P(C) + 0}{P(C)}$

$= 1 - P(A) - P(B)$

$= P(A') - P(B)$

62. Statement 1:

$A = P^{-1}BP$

$PA = BP$

$P = \{(A, A) : PA \neq AP$ in general not reflexive$\}$

so, not equivalence relation.

Statement 2: $(MN)^{-1} = N^{-1}M^{-1}$, true from defination.

63. The number of ordered pairs is $3 \times 3 \times 3 \times 3 \times 3 = 3^5$

[$\because$ Every element has 3 options, either it can be in y but not in z, or it can be in z but not in y or it can be neither in y nor z.]

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. Which of the following is the empty set?
(a) $\{x : x$ is a real number and $x^2 + 1 = 0\}$
(b) $\{x : x$ is a real number and $x^2 - 9 = 0\}$
(c) $\{x : x$ is a real number and $x^2 = x + 3\}$
(d) $\{x : x$ is a real number and $x^2 - 7 = 0\}$

2. If A and B are not disjoint, then $n(A \cup B)$ is equal to
(a) $n(A) + n(B) + n(A \cap B)$
(b) $n(A) - n(B)$
(c) $n(A) + n(B) - n(A \cap B)$
(d) none of these

3. If $A \subseteq B$, then $A \cap B$ is equal to
(a) B' (b) A' (c) A (d) B

4. The number of subsets that can be formed from the set $A = \{4, 5, 6\}$ are
(a) 6 (b) 7
(c) 8 (d) none of these

5. If A is the set of the null set, then the number of elements in the power set $P(A)$ is
(a) 0 (b) 2
(c) 1 (d) none of these

6. If $n(A) = 5$ and $n(B) = 8$ and $A \subseteq B$, then the number of elements in $A \cup B$ is equal to
(a) 5 (b) 8 (c) 3 (d) 2

7. If $S = \{1, 3, 5, 7\}$, then the total number of subset of S is
(a) 8 (b) 32 (c) 16 (d) 15

8. If A and B are any two sets, then $A \times B = B \times A$, iff
(a) $A \subseteq B$ (b) $B \subseteq A$
(c) $A \cap B = \phi$ (d) $A = B$

9. The relation defined on the set of natural numbers N, $R=\{(x, y): x<y, x \in N, y \in N\}$, is
(a) reflexive (b) symmetric
(c) transitive (d) none of these

10. $(A \cup B)-(A \cap B)$ is equal to
(a) $A \cap B'$ (b) $A \cup B'$
(c) $(A-B) \cup (B-A)$ (d) $A-B \cap B-A$

11. Let A and B be two sets in the universal set, then $A-B$ equals
(a) $A' \cap B$ (b) $A \cap B'$
(c) $A \cap B$ (d) $A \cup B$

12. Which of the following relations is incorrect?
(a) $(A \cap B)' = A' \cup B'$
(b) $(A \cup B)' = A' \cap B'$
(c) $(A \times (B \cup C)) = (A \times B) \cup (A \times C)$
(d) $A' = \phi$

13. If $A=\{1, 2, 3, 4\}$, $B=\{3, 4, 5, 6\}$ and $C=\{1, 2, 5, 6\}$, then $(A \cap B) \cap C =$
(a) $\{3, 4\}$ (b) $\{5, 6\}$
(c) ϕ (d) none of these

14. Which of the following statements is false?
(a) $A-B=A \cap B'$ (b) $A-B=A-(A \cap B)$
(c) $A-B=A-B'$ (d) $A-B=(A \cup B)-B$

15. Let $A=\{1, 2, 4\}$ and $B=\{2, 6, 10\}$, then $n(A \times B)$ is equal to
(a) 6 (b) 9
(c) 3 (d) none of these

16. If A and B are not disjoint, then $n(A \cup B)$ is equal to
(a) $n(A)$ (b) $n(B)$
(c) $n(A)+n(B)$ (d) $n(A)+n(B)-n(A \cap B)$

17. Let L denote the set of all straight lines in a plane. Let a relation R be defined by $\alpha R \beta$ iff $\alpha \perp \beta, \alpha, \beta \in L$ then R is
(a) reflexive (b) symmetric
(c) transitive (d) none of these

18. Which of the following is the null set?
(a) $A=\{x : x \text{ is} > 1 \text{ and } x \text{ is} < 1\}$
(b) $B=\{x : x+3=3\}$
(c) $C=\{\phi\}$
(d) $D=\{x : x \text{ is} \geq 1 \text{ and } x \text{ is} \leq 1\}$

19. If $A=\{2, 3, 4, 8, 10\}$, $B=\{3, 4, 5, 10, 12\}$ and $C=\{4, 5, 6, 12, 145\}$, then $(A \cup B) \cap (A \cup C)$ is equal to
(a) $\{2, 3, 4, 5, 8, 10, 12\}$ (b) $\{2, 4, 8, 10, 12\}$
(c) $\{3, 8, 10, 12\}$ (d) $\{2, 8, 10\}$

20. Of the 20 teachers of mathematics and physics, 12 teach mathematics and 4 teach both the subjects. The number of teachers teaching physics is
(a) 9 (b) 4 (c) 8 (d) 12

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. In a group of 500 persons, if 350 can speak Hindi and 250 can speak English, then the number of persons speaking both the languages are
(a) 150 (b) 200
(c) 100 (d) 250

2. In a group of 150 persons, if 35 drink milk but not tea and 65 drink milk, then the number of persons who drink both tea and milk are
(a) 25 (b) 30
(c) 35 (d) none of these

3. Cardinal number of $P(P(P(\phi)))$ is
(a) 1 (b) 2
(c) 3 (d) 4

4. A class of 100 students sat in an examination in maths and physics. Every student cleared at least 1 subject, 55 students cleared maths and 67 students cleared physics. Then the number of students who cleared physics only is
(a) 22 (b) 33
(c) 10 (d) 45

5. The smallest set such that $A \cup \{1, 2\} = \{1, 2, 3, 5, 9\}$ is
(a) $\{1, 2\}$ (b) $\{1, 2, 3, 5, 9\}$
(c) $\{3, 5, 9\}$ (d) none of these

6. Consider the set A of all determinants of order 3 with entries 0 or 1 only. Let B be the subset of A consisting of all determinants with value 1. Let C be the subset of the set of all determinants with value –1. Then:
(a) B has as many elements as C
(b) $C=\phi$
(c) B has twice as many elements as C
(d) $A = B \cup C$

7. Let a binary operation '$\forall$' on Q (set of all rational numbers) be defined by $a \circ b = a+2b$ for all $a, b \in Q$, then
(a) Q is closed under the given operation
(b) the given operation is commutative
(c) the given operation is associative
(d) Q is not closed

8. Let S be the set of integers. For $a, b \in S$, aRb if and only if $|a-b|<1$, then
(a) R is not reflexive
(b) R is not symmetric
(c) $R = r = \{(a, a); a \in 1\}$
(d) R is not an equivalence relation

9. Let I be the set of integers, N the set of non-negative integers, N_p the set of non-positive integers, E the set of even integers and P the set of prime numbers. Then
(a) $N \cap N_p = \phi$ (b) $I \sim N = N_p$
(c) $N \Delta N_p = I \sim \{0\}$ (d) $E \cap P = \phi$

10. X and Y are two sets and X has 40 elements, $X \cup Y$ has 60 elements and $X \cap Y$ has 10 elements. Then the number of elements in Y is
(a) 30 (b) 10
(c) 40 (d) 50

11. R is the relation "x is a brother of y" in the set of children in a family. Then R is
(a) reflexive (b) symmetric
(c) transitive (d) none of these

12. Let $f: N \to N$ be defined as

$$f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$$

Then f is
(a) one-to-one (b) one-to-one and onto
(c) onto (d) none of these

13. For the sets of real numbers given by

$$R_1 = \{(x, y): x \in \mathbb{R}, y \in \mathbb{R}\ x^2 + y^2 \le 25\}$$

$$R_2 = \{(x, y): x \in \mathbb{R}, y \in \mathbb{R},\ 9y \ge 4x^2\}$$

$R_1 \cap R_2$ is
(a) one-to-one
(b) onto
(c) one-to-one and onto
(d) none of these

14. A survey showed that 60% of the Indians like oranges whereas 75% like mangoes. If k % of the Indians like both oranges and mangoes, then
(a) $35 \le k \le 60$ (b) $45 \le k \le 60$
(c) $35 \ge k$ and $k \ge 60$ (d) none of these

15. If $A = \{3, 5\}$ and $B = \{5, 6, 7\}$, then $(A \cap B) \times (A \cup B)$ is
(a) $\{(5, 3)\ (5, 4)\ (5, 6)\}$
(b) $\{(5, 3), (5, 5), (5, 6), (5, 7)\}$
(c) $\{(6, 5), (6, 3), (6, 6), (6, 7)\}$
(d) none of these

16. If A and B are two sets and U is the universal set such that $n(U) = 800$, $n(A) = 300$, $n(B) = 400$ and $n(A \cap B) = 150$, then $n(A' \cap B')$ is equal to
(a) 400 (b) 150
(c) 350 (d) 250

17. If $A \subseteq B$, then
(a) $A' \subseteq B'$ (b) $A' = B'$
(c) $B' \subseteq A'$ (d) $B' \not\subseteq A'$

18. If $A = \{x, y, t\}$, $B = \{x, z, u\}$ and $C = \{x, z, v\}$, then
(a) $A - B = \{y, u\}$
(b) $B - C = \{z, v\}$
(c) $C - A = \{x, t\}$
(d) none of these

19. Let $A = Z \cup \{\sqrt{2}\}$. Define a relation R in A by aRb if and only if $a + b \in Z$. The relation R is then
(a) reflexive
(b) symmetric and transitive
(c) only transitive
(d) none of these

20. If $f(x) = 1 - \frac{1}{x}$, then $f\left[f\left(\frac{1}{x}\right)\right]$ is
(a) $\frac{1}{x}$ (b) $\frac{1}{1+x}$
(c) $\frac{x}{x-1}$ (d) $\frac{1}{x-1}$

ANSWERS

Level 1

1. (a)	**2.** (c)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (d)	**9.** (c)	**10.** (c)
11. (b)	**12.** (d)	**13.** (c)	**14.** (c)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (a)	**20.** (d)

Level 2

1. (c)	**2.** (b)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (c)	**7.** (a)	**8.** (c)	**9.** (c)	**10.** (a)
11. (c)	**12.** (c)	**13.** (d)	**14.** (a)	**15.** (b)	**16.** (d)	**17.** (c)	**18.** (d)	**19.** (b)	**20.** (d)

2

WILLAM ROWAN HAMILTON

Complex Numbers

- Introduction
- Definition of a Complex Number
- Integral Powers of Iota
- Algebraic Operations with Complex Numbers
- Geometrical Representation of a Complex Number
- Vector Representation of a Complex Number
- Trigonometrical or Polar Form of a Complex Number
- Eulerian Representation (Exponential Form)
- Logarithm of a Complex Number
- Conjugate of a Complex Number
- Properties of Conjugate
- Argument or Amplitude of a Complex Number
- Properties of Arguments
- De Moivre's Theorem
- Square Roots of a Complex Number
- Cube Roots of Unity
- The *n*th Roots of Unity
- Properties of the *n*th Roots of Unity
- Geometrical Representation of Algebraic Operations on Complex Numbers
- Multiplication of Complex Numbers
- Division of Complex Numbers
- Some Important Geometrical Results
- Recognizing Some Loci by Inspection
- Equation of Tangent to a Given Circle

INTRODUCTION

We know that there is no real number whose square is negative. So, there exists no real number which satisfies the equation $x^2+1=0$. Euler (1707–1783) was the first mathematician to introduce the symbol i, pronounced iota, for the square root of -1 with the explicit property $i^2=-1$. But the use of i for $\sqrt{-1}$ should be made carefully. For example, we cannot write

$$i^2 = i\cdot i = \sqrt{-1}\cdot\sqrt{-1} = \sqrt{(-1)\cdot(-1)} = \sqrt{+1} = +1$$

DEFINITION OF A COMPLEX NUMBER

An expression of the form $x + iy$, where x and y are real numbers and $i=\sqrt{-1}$, is called a complex number. It is denoted by z, i.e.

$$z = x + iy$$

A complex number may also be defined as an ordered pair of real numbers and may be denoted by the symbol (x, y). If we write $z=(x, y)$, then x is called the real part and y the imaginary part of the complex number z and may be denoted by $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ respectively.

Two complex numbers are said to be equal if and only if their real parts and imaginary parts are separately equal to one another

i.e. $x+iy=c+id \Leftrightarrow x=c$ and $y=d$

For example, if $z=8+5i$, then Re(z) = 8 and Im(z) = 5.

Illustration 1. If $(x^2y-2)+i(x+2xy-5)=0$, find the values of x and y.

Solution: We have $(x^2y-2)+i(x+2xy-5)=0$

$\Rightarrow \quad (x^2y-2)+i(x+2xy-5)=0+i0$

Equating the real and imaginary parts of both sides, we get

$$x^2y-2=0 \quad \Rightarrow \quad y=\frac{2}{x^2} \qquad \text{(i)}$$

and $$x+2xy-5=0 \qquad \text{(ii)}$$

Putting the value of y from (i) in (ii), we get

$$x+2x\left(\frac{2}{x^2}\right)-5=0 \quad \Rightarrow \quad x+\frac{4}{x}-5=0$$

$$\Rightarrow \quad x^2-5x+4=0 \quad \Rightarrow \quad (x-1)(x-4)=0$$

When $x=1$, $y=\frac{2}{(1)^2}=2$

When $x=4$, $y=\frac{2}{(4)^2}=\frac{1}{8}$.

INTEGRAL POWERS OF IOTA

$i^0=1, i^1=i, i^2=-1, i^3=i^2\cdot i=(-1)i=-i, i^4=(i^2)^2=(-1)^2=1,$

$i^5=i^4\cdot i=i, i^6=i^4\cdot i^2=-1,$ and so on.

Note that $i^4=1$. If n is any integer, let $n=4m+r$, where $m, r \in Z$ and r = 0, 1, 2 or 3, then we have

$$i^n=i^{4m+r}=(i^4)^m\cdot i^r=1^m\cdot i^r=i^r$$

For example, $i^{100}=(i^4)^{25}=1$; $i^{102}=(i^4)^{25}\cdot i^2=i^2=-1$

$$i^{-3}=i^{4\cdot(-1)+1}=i^1=i;\ i^{-35}=i^{4\cdot(-9)+1}=i^1=i$$

ALGEBRAIC OPERATIONS WITH COMPLEX NUMBERS

Let $z_1=x_1+iy_1$ and $z_2=x_2+iy_2$ be two complex numbers,

1. **Addition**

$$(x_1+iy_1)+(x_2+iy_2)=(x_1+x_2)+i(y_1+y_2)$$

2. **Subtraction**

$$(x_1+iy_1)-(x_2+iy_2)=(x_1-x_2)+i(y_1-y_2)$$

3. **Multiplication**

$$(x_1+iy_1)(x_2+iy_2)=x_1x_2+i(x_1y_2)+i(x_2y_1)+i^2(y_1y_2)$$

$$= x_1x_2+i(x_1y_2+x_2y_1)+(-1)(y_1y_2)$$

$$= x_1x_2+i(x_1y_2+x_2y_1)-(y_1y_2)$$

$$= (x_1x_2-y_1y_2)+i(x_1y_2+x_2y_1)$$

4. **Division**

$\frac{x_1+iy_1}{x_2+iy_2}$ (when at least one of x_2 and y_2 is nonzero)

Multiplying the numerator and denominator by the conjugate of the denominator,

$$\frac{x_1+iy_1}{x_2+iy_2}=\frac{(x_1+iy_1)(x_2-iy_2)}{(x_2+iy_2)(x_2-iy_2)}$$

$$=\frac{(x_1x_2+y_1y_2)+i(-x_1y_2+x_2y_1)}{(x_2x_2+y_2y_2)+i(-x_2y_2+x_2y_2)}$$

$$=\frac{x_1x_2+y_1y_2}{x_2^2+y_2^2}+\frac{i(-x_1y_2+x_2y_1)}{x_2^2+y_2^2}$$

Note

- Addition and subtraction are easier to perform in Cartesian form while multiplication and division are easier to perform in polar form.
- Addition and subtraction of complex numbers follow the closure law, the commutative law and the associative law.
- Multiplication and division of complex numbers follow the closure law, the commutative law, the associative law and the distributive law.

GEOMETRICAL REPRESENTATION OF A COMPLEX NUMBER

The complex number $z=x+iy=(x, y)$ is represented by the point P as shown in Fig. 2.1. xOx' and yOy' are called real axis and imaginary axis, respectively. This plane is called *Argand plane* or *Argand diagram* or *complex plane* or *Gaussian plane*.

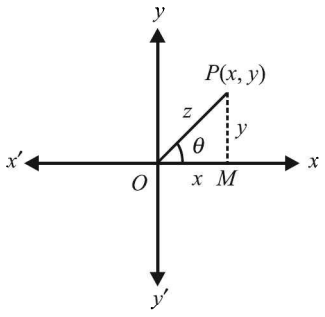


Fig. 2.1

The distance of any complex number from the origin is called the *modulus* of the complex number and is denoted by $|z|$, that is, $|z|=\sqrt{x^2+y^2}$.

The angle of any complex number with the positive direction of x-axis (however, z must lie in first quadrant) is called *amplitude* or *argument* of z, that is,

$$\text{amp}(z)=\arg(z)=\tan^{-1}\left(\frac{y}{x}\right).$$

☞ VECTOR REPRESENTATION OF A COMPLEX NUMBER

If P is a point (x, y) on the Argand plane corresponding to the complex number $z = x + iy$, then

$$\overrightarrow{OP} = x\hat{i} + y\hat{j} \qquad \therefore \quad |\overrightarrow{OP}| = \sqrt{x^2 + y^2} = |z|$$

and arg (z) = direction of the vector $\overrightarrow{OP} = \tan^{-1}(y/x)$.

☞ TRIGONOMETRICAL OR POLAR FORM OF A COMPLEX NUMBER

Let $z = x + iy$, where z is represented by $P(x, y)$ in the Argand plane. By geometrical representation,

$$OP = \sqrt{x^2 + y^2} = |z|; \quad \angle POM = \arg(z)$$

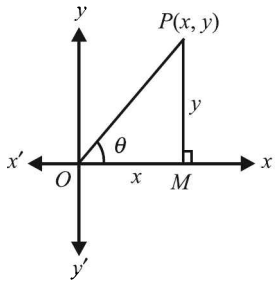


Fig. 2.2

In the triangle OPM,

$$x = OP\cos(\angle POM) = |z|\cos[\arg(z)]$$

$$y = OP\sin(\angle POM) = |z|\sin[\arg(z)]$$

$\because \quad z = x + iy$

$\therefore \quad z = |z|\{\cos[\arg(z)] + i\sin[\arg(z)]\}$

If $|z| = r$ and $\arg(z) = \theta$, then

$z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ (Euler's form). Here we should take the principal value of θ.

Illustration 2. Write the polar form of $-\frac{1}{2} - \frac{i\sqrt{3}}{2}$.

Solution: Here $r\cos\theta = -\frac{1}{2}$, $r\sin\theta = -\frac{\sqrt{3}}{2}$

Squaring and adding, we get

$$r^2\cos^2\theta + r^2\sin^2\theta = \frac{1}{4} + \frac{3}{4}$$

$\therefore \quad r^2(\cos^2\theta + \sin^2\theta) = 1$ or $r^2 = 1$ or $r = 1$

Dividing, we get

$$\tan\theta = \sqrt{3} = \tan\frac{\pi}{3}$$

Since $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ lies in the third quadrant,

$\therefore \quad$ Principal value of θ is $-\pi + \frac{\pi}{3} = \frac{-2\pi}{3}$

$\therefore$ Polar form of $-\frac{1}{2} - i\frac{\sqrt{3}}{2}$ is $\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)$

☞ EULERIAN REPRESENTATION (EXPONENTIAL FORM)

As we have $e^{i\theta} = \cos\theta + i\sin\theta$, z can be expressed as $z = re^{i\theta}$, where $|z| = r$ and $\theta = \arg(z)$.

Logarithm of a Complex Number

$$\log(x + iy) = \log_e(re^{i\theta}) = \log_e r + \log_e e^{i\theta} = \log_e r + i\theta$$

$$= \log_e\sqrt{x^2 + y^2} + i\tan^{-1}\left(\frac{y}{x}\right)$$

Thus, $\log_e(z) = \log_e|z| + i\arg(z)$.

☞ CONJUGATE OF A COMPLEX NUMBER

If there exists a complex number $z = x + iy$ for $(x, y) \in \mathbb{R}$, then its conjugate is defined as $\bar{z} = x - iy$.

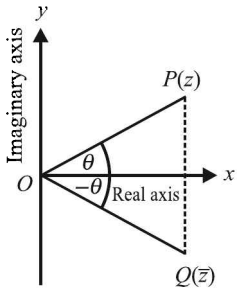


Fig. 2.3

Hence, we have $\text{Re}(z) = \frac{z + \bar{z}}{2}$ and $\text{Im}(z) = \frac{z - \bar{z}}{2i}$.

Geometrically, the conjugate of z is the reflection or the point image of z in the real axis.

☞ PROPERTIES OF CONJUGATE

Let $z = x + iy$, $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be any three complex numbers, then

Property 1: $\bar{\bar{z}} = z$

Reasoning: Since $\bar{\bar{z}} = \overline{(\overline{x + iy})} = \overline{(x - iy)} = x + iy$

$\therefore \quad \bar{\bar{z}} = x + iy = z$

Property 2: $z + \bar{z} = 2\text{Re}(z)$

Reasoning: Since $z = x + iy$,

$\therefore \quad z + \bar{z} = (x + iy) + (x - iy) = 2x = 2\text{Re}(z)$

Property 3: $z - \bar{z} = 2\text{Im}(z)i$

Reasoning: Since $z = x + iy$

$\therefore \quad z - \bar{z} = (x + iy) - (x - iy) = 2yi = 2\text{Im}(z)i$

Property 4: If $z = \bar{z}$, then Im $(z) = 0$ or z is purely real.

Reasoning: Since $z = \bar{z}$,

$\Rightarrow \quad z - \bar{z} = 0 \quad \Rightarrow \quad 2\,\text{Im}(z)i = 0 + 0i$

$\therefore$ Im $(z) = 0$ and hence z is a purely real number.

Property 5: If $z = -\bar{z}$, then Re $(z) = 0$ or z is purely imaginary.

Reasoning: Since $z + \bar{z} = 0$,

$\Rightarrow \quad 2\text{Re}(z) = 0 \quad \Rightarrow \quad z$ is purely imaginary.

Property 6: $z(\bar{z}) = [\text{Re}(z)]^2 + [\text{Im}(z)]^2 = |z|^2$

Reasoning: Now, $z(\bar{z}) = (x + iy)(x - iy)$

$\Rightarrow \quad z(\bar{z}) = (x)^2 - (iy)^2 = x^2 + y^2$

$\therefore \quad z(\bar{z}) = [\text{Re}(z)]^2 + [\text{Im}(z)]^2$

Property 7: $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

Reasoning: Now,

$$\overline{z_1 + z_2} = \overline{(x_1 + iy_1) + (x_2 + iy_2)} = \overline{(x_1 + x_2) + i(y_1 + y_2)}$$

$$\Rightarrow \quad \overline{z_1 + z_2} = (x_1 + x_2) - i(y_1 + y_2) = (x_1 - iy_1) + (x_2 - iy_2)$$

$$\therefore \quad \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

Property 8: $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$

Property 9: $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$

Reasoning: Here, $\overline{z_1 \cdot z_2} = \overline{(x_1 + iy_1) \cdot (x_2 + iy_2)}$

$$\Rightarrow \quad \overline{z_1 \cdot z_2} = \overline{(x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)}$$

$$\Rightarrow \quad \overline{z_1 \cdot z_2} = (x_1x_2 - y_1y_2) - i(x_1y_2 + x_2y_1)$$

$$\Rightarrow \quad \overline{z_1 \cdot z_2} = x_1(x_2 - iy_2) - iy_1(x_2 - iy_2) = (x_1 - iy_1)(x_2 - iy_2)$$

$$\therefore \quad \overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$$

Property 10: $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$, where $z_2 \neq 0$

Property 11: $\overline{(z^n)} = (\bar{z})^n$

Property 12:

$$\overline{\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}_{m \times n}} = \begin{bmatrix} \bar{x}_{11} & \bar{x}_{12} & \cdots & \bar{x}_{1n} \\ \bar{x}_{21} & \bar{x}_{22} & \cdots & \bar{x}_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ \bar{x}_{m1} & \bar{x}_{m2} & \cdots & \bar{x}_{mn} \end{bmatrix}_{m \times n}$$

Property 13:

$$\text{Re}(z_1\bar{z}_2) = \text{Re}(\bar{z}_1 z_2) = \frac{1}{2}(z_1\bar{z}_2 + z_2\bar{z}_1) = (x_1x_2 + y_1y_2)$$

Property 14:

$$\text{Im}(z_1\bar{z}_2) = \text{Im}(\bar{z}_1 z_2) = \frac{1}{2}(z_1\bar{z}_2 - z_2\bar{z}_1) = (x_2y_1 - x_1y_2)$$

ARGUMENT OR AMPLITUDE OF A COMPLEX NUMBER $z = x + iy$ FOR DIFFERENT SIGNS OF x AND y

(i) When $(x, y) \in$ first quadrant, $x > 0, y > 0$. The principal value of arg$(z) = \theta = \tan^{-1}\left(\frac{y}{x}\right)$. Here, θ is an acute angle and positive. See Fig. 2.4.

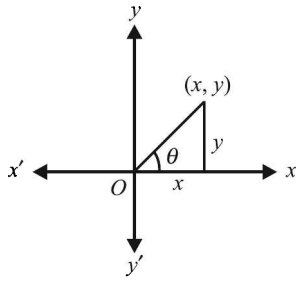


Fig. 2.4

(ii) When $(x, y) \in$ second quadrant, $x < 0$, $y > 0$. The principal value of arg$(z) = \theta = \pi - \tan^{-1}\left(\frac{y}{|x|}\right)$. Here, θ is an obtuse angle and positive. See Fig. 2.5.

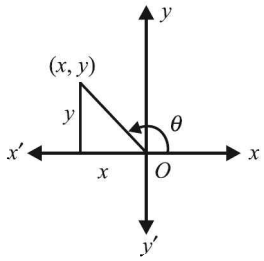


Fig. 2.5

(iii) When $(x, y) \in$ third quadrant, $x < 0, y < 0$. The principal value of arg$(z) = \theta = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$. Here, θ is an obtuse angle and negative. See Fig. 2.6.

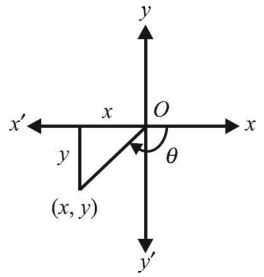


Fig. 2.6

(iv) When $(x, y) \in$ fourth quadrant, $x > 0$, $y < 0$. The principal value of arg$(z) = \theta = -\tan^{-1}\left(\frac{|y|}{x}\right)$. Here, θ is an acute angle and negative. See Fig. 2.7.

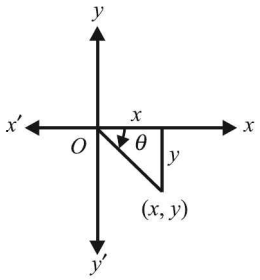


Fig. 2.7

SHORTCUT METHOD FOR FINDING THE ARGUMENT OF A COMPLEX NUMBER

Step 1. Find $\tan\theta = \left|\frac{y}{x}\right|$ and this gives the value of θ in the first quadrant.

Step 2. Find the quadrant in which z lies by the sign of its x and y co-ordinates.

Step 3. Then the argument of z will be $\theta, \pi-\theta, \theta-\pi$ and $-\theta$ according as z lies in the first, second, third or fourth quadrant respectively.

Note

- Argument of the complex number 0 is not defined.
- If $z_1 = z_2 \Leftrightarrow |z_1| = |z_2|$, then $\arg(z_1) = \arg(z_2)$.
- If $\arg(z) = \frac{\pi}{2}$ or $-\frac{\pi}{2}$, z is purely imaginary.
- If $\arg(z) = 0$ or π, z is purely real.

Illustration 3. Find the principal values of arguments of $z_1 = 2+2i$, $z_2 = -2+2i$, $z_3 = -2-2i$ and $z_4 = 2-2i$.

Solution: Since z_1, z_2, z_3 and z_4 lie in the first quadrant, second quadrant, third quadrant and fourth quadrant respectively, the principal values of arguments of the given complex numbers are given by

$$\arg(z_1) = \tan^{-1}(1) = \frac{\pi}{4},\ \arg(z_2) = \pi - \tan^{-1}(1) = \frac{3\pi}{4}$$

$$\arg(z_3) = -\pi + \tan^{-1}(1) = -\frac{3\pi}{4},\ \arg(z_4) = -\tan^{-1}(1) = -\frac{\pi}{4}$$

Hence, the principal values of z_1, z_2, z_3 and z_4 are $\frac{\pi}{4}$, $\frac{3\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{4}$, respectively.

PROPERTIES OF ARGUMENTS

(i) $\arg(z_1z_2) = \arg(z_1) + \arg(z_2)$. In general, $\arg(z_1z_2z_3\ldots z_n) = \arg(z_1) + \arg(z_2) + \arg(z_3) + \cdots + \arg(z_n)$

(ii) $\arg(z_1\bar{z}_2) = \arg(z_1) - \arg(z_2)$ $[\because\ \arg(\bar{z}_2) = -\arg(z_2)]$

(iii) $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

(iv) $\arg(z/\bar{z}) = 2\arg(z)$

(v) $\arg(z^n) = n\arg(z)$

(vi) $\arg(\bar{z}) = -\arg(z) = \arg(1/z)$

(vii) $z_1\bar{z}_2 + \bar{z}_1z_2 = 2|z_1||z_2|\cos(\theta_1 - \theta_2)$, where $\theta_1 = \arg(z_1)$ and $\theta_2 = \arg(z_2)$

(viii) If z is purely imaginary, then $\arg(z) = \pm\frac{\pi}{2}$.

(ix) If z is purely real, then $\arg(z) = 0$ or π.

(x) The angle between two segments joining z_1 and z_2 and joining z_3 and z_4 is $\theta = \pm\arg\left(\frac{z_2 - z_1}{z_3 - z_4}\right)$.

DE MOIVRE'S THEOREM

(i) If n is any rational number, then $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

(ii) If $z = (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)(\cos\theta_3 + i\sin\theta_3) \ldots (\cos\theta_n + i\sin\theta_n)$, then

$$z = \cos(\theta_1+\theta_2+\theta_3+\ldots+\theta_n) + i\sin(\theta_1+\theta_2+\theta_3+\cdots+\theta_n).$$

(iii) If $z = r(\cos\theta + i\sin\theta)$ and n is a positive integer, then

$$z^{1/n} = r^{1/n}\left[\cos\left(\frac{2k\pi+\theta}{n}\right) + i\sin\left(\frac{2k\pi+\theta}{n}\right)\right]$$

where $k = 0, 1, 2, 3, \ldots, (n-1)$.

Illustration 4. If $x_r = \cos\left(\frac{\pi}{3^r}\right) + i\sin\left(\frac{\pi}{3^r}\right)$, prove that $x_1x_2x_3\ldots = i$

Solution: We have $x_r = \cos\left(\frac{\pi}{3^r}\right) + i\sin\left(\frac{\pi}{3^r}\right)$

$$\therefore\ x_1x_2x_3\ldots$$

$$= \cos\left(\frac{\pi}{3}+\frac{\pi}{3^2}+\frac{\pi}{3^3}+\ldots\right) + i\sin\left(\frac{\pi}{3}+\frac{\pi}{3^2}+\frac{\pi}{3^3}+\ldots\right)$$

$$= \cos\left[\frac{\pi/3}{1-(\pi/3)}\right] + i\sin\left[\frac{\pi/3}{1-(\pi/3)}\right]$$

$$= \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = 0 + i = i$$

SQUARE ROOTS OF A COMPLEX NUMBER

Let $x+iy$ be a complex number such that $\sqrt{x+iy} = a+ib$, where x and y are real numbers. Then,

$$\sqrt{x+iy} = a+ib \quad\Rightarrow\quad x+iy = (a+ib)^2$$

$$\Rightarrow \qquad x+iy = (a^2-b^2) + 2iab$$

On equating the real and imaginary parts, we get

$$a^2 - b^2 = x \quad \text{and} \quad 2ab = y \qquad \text{(i)}$$

Now, $$(a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2 \qquad \text{(ii)}$$

$$\Rightarrow \quad (a^2 + b^2)^2 = x^2 + y^2$$

$$\Rightarrow \quad a^2 + b^2 = \sqrt{x^2 + y^2} \qquad \left[\because a^2 + b^2 \geq 0\right] \qquad \text{(iii)}$$

Solving (i) and (iii), we get

$$a^2 = \left(\frac{1}{2}\right)\left(\sqrt{x^2 + y^2} + x\right) \text{ and } b^2 = \left(\frac{1}{2}\right)\left(\sqrt{x^2 + y^2} - x\right) \qquad \text{(iv)}$$

If y is positive, then the signs of a and b from (i) will be the same, i.e.

$$\sqrt{x+iy} = a + ib = \pm\left[\sqrt{\left(\frac{1}{2}\right)\left(\sqrt{x^2 + y^2} + x\right)} + i\sqrt{\left(\frac{1}{2}\right)\left(\sqrt{x^2 + y^2} - x\right)}\right]$$

If y is negative, then the signs of a and b will be opposite, i.e.

$$\sqrt{x+iy} = a + ib = \pm\left[\sqrt{\left(\frac{1}{2}\right)\left(\sqrt{x^2 + y^2} + x\right)} - i\sqrt{\left(\frac{1}{2}\right)\left(\sqrt{x^2 + y^2} - x\right)}\right]$$

Note

- The square roots of i are $\pm\left(\frac{1+i}{\sqrt{2}}\right)$. (Here $y = 1$.)
- The square roots of $-i$ are $\pm\left(\frac{1-i}{\sqrt{2}}\right)$. (Here $y = -1$.)
- The square roots of ω are: $\pm\omega^2$.
- The square roots of ω^2 are: $\pm\omega$.

☞ SHORTCUT METHOD OF FINDING THE SQUARE ROOT OF A COMPLEX NUMBER

The square roots of $z = x + iy$ from (iv) above are:

$$\pm\left[\sqrt{\frac{|z|+x}{2}} + i\sqrt{\frac{|z|-x}{2}}\right] \text{ for } y > 0$$

$$\pm\left[\sqrt{\frac{|z|+x}{2}} - i\sqrt{\frac{|z|-x}{2}}\right] \text{ for } y < 0$$

Illustration 5(a). Find the square roots of $3 - 4i$.

Solution: Let $z = x + iy = 3 - 4i$

$$|z| = \sqrt{9+16} = 5$$

$$\Rightarrow \text{ Square roots are: } \pm\left[\sqrt{\frac{5+3}{2}} - i\sqrt{\frac{5-3}{2}}\right]$$

$$= \pm(2 - i)$$

Illustration 5(b). Find the square roots of $-15 - 8i$.

Solution: Let $\sqrt{-15-8i} = a + ib$. Then,

$$\Rightarrow \quad -15 - 8i = (a + ib)^2$$

$$\Rightarrow \quad -15 - 8i = (a^2 - b^2) + 2iab$$

$$\Rightarrow \quad -15 = a^2 - b^2 \qquad \text{(i)}$$

and $$2ab = -8 \qquad \text{(ii)}$$

Now, $$(a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$$

$$\Rightarrow \quad (a^2 + b^2)^2 = (-15)^2 + 64 = 289$$

$$\Rightarrow \quad a^2 + b^2 = 17 \qquad \text{(iii)}$$

On solving (i) and (iii), we get

$$a^2 = 1 \text{ and } b^2 = 16 \quad \Rightarrow \quad a = \pm 1 \text{ and } b = \pm 4$$

From (ii), $2ab$ is negative, so a and b are of opposite signs.

$\therefore$ $(a = 1 \text{ and } b = -4)$ or $(a = -1 \text{ and } b = 4)$

Hence, $\sqrt{-15-8i} = a + ib = \pm(1 - 4i)$.

☞ CUBE ROOTS OF UNITY

Let $$x = \sqrt[3]{1}$$

$$\Rightarrow \quad x^3 - 1 = 0$$

$$\Rightarrow \quad (x-1)(x^2 + x + 1) = 0$$

$$\therefore \quad x = 1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$$

If the second root is represented by ω, then the third root is ω^2.

$\therefore$ The cube roots of unity are 1, ω, ω^2.

Properties

1. $1 + \omega + \omega^2 = 0$
2. $\omega^3 = 1$
3. $\omega^{3n} = 1$, $\omega^{3n+1} = \omega$, $\omega^{3n+2} = \omega^2$
4. $\bar{\omega} = \omega^2$ and $(\bar{\omega})^2 = \omega$
5. A complex number $x + iy$, for which $|x : y| = 1 : \sqrt{3}$ or $\sqrt{3} : 1$, can always be expressed in terms of i, ω or ω^2. For example:

   (i) $1 + i\sqrt{3} = -(-1 - i\sqrt{3}) = -2\left(\frac{-1-i\sqrt{3}}{2}\right) = -2\omega^2$

   (ii) $\sqrt{3} + i = \frac{i(\sqrt{3}+i)}{i} = \frac{-1+i\sqrt{3}}{i} = \left(\frac{-1+i\sqrt{3}}{2}\right)\frac{2}{i} = \frac{2\omega}{i}$

7. $a+b\omega+c\omega^2=0 \Rightarrow a=b=c$ if a, b, c are real.
8. $\omega\bar{\omega}=\omega^3, \omega=e^{2\pi i/3}, \omega^2=e^{-2\pi i/3}$

Important Relations

1. $x^2+x+1=(x-\omega)(x-\omega^2)$
2. $x^2-x+1=(x+\omega)(x+\omega^2)$
3. $x^2+xy+y^2=(x-y\omega)(x-y\omega^2)$
4. $x^2-xy+y^2=(x+y\omega)(x+y\omega^2)$
5. $x^2+y^2=(x+iy)(x-iy)$
6. $x^3+y^3=(x+y)(x+y\omega)(x+y\omega^2)$
7. $x^3-y^3=(x-y)(x-y\omega)(x-y\omega^2)$
8. $x^2+y^2+z^2-xy-yz-zx$
   $=(x+y\omega+z\omega^2)(x+y\omega^2+z\omega)$
9. $x^3+y^3+z^3-3xyz$
   $=(x+y+z)(x+y\omega+z\omega^2)(x+y\omega^2+z\omega)$

THE *n*th ROOTS OF UNITY

Let x be the nth root of unity. Then

$$x^n=1=1+i(0)=\cos 0°+i\sin 0°$$
$$=\cos(2k\pi+0)+i\sin(2k\pi+0)$$
$$=\cos 2k\pi+i\sin 2k\pi$$

$$\Rightarrow \quad x=\cos\frac{2k\pi}{n}+i\sin\frac{2k\pi}{n},\ k=0, 1, 2, ..., (n-1)$$

Let $\alpha=\cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}$. Then, the nth roots of unity are $1, \alpha, \alpha^2, \ldots, \alpha^{n-1}$

PROPERTIES OF THE *n*th ROOTS OF UNITY

1. The sum of the nth roots of unity is zero.

   i.e. $1+\alpha+\alpha^2+\ldots+\alpha^{n-1}=\frac{1-\alpha^n}{1-\alpha}=0$

   $$\Rightarrow \quad \sum_{k=0}^{n-1}\cos\frac{2k\pi}{n}=0$$

   Thus, the sum of the roots of unity is zero.

2. The product of the nth roots of unity

   $$1\alpha\alpha^2\ldots\alpha^{n-1}=\alpha^{\frac{n(n-1)}{2}}=\left(\cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}\right)^{n\left(\frac{n-1}{2}\right)}$$
   $$=\cos[\pi(n-1)]+i\sin[\pi(n-1)]$$

   If n is even, $\alpha^{\frac{n(n-1)}{2}}=-1$

   If n is odd, $\alpha^{\frac{n(n-1)}{2}}=1$.

3. The points represented by 'n' in nth roots of unity are located at the vertices of a regular polygon of n sides inscribed in a unit circle having centre at the origin, one vertex being on the positive real axis (geometrically represented as shown). See Fig. 2.8.

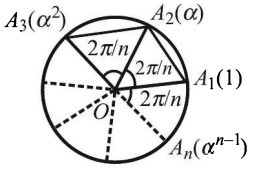


Fig. 2.8

For n = 3, we get the cube roots of unity and they are 1, $\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}$ and $\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}$, that is, 1, $\frac{-1+i\sqrt{3}}{2}$ and $\frac{-1-i\sqrt{3}}{2}$. They are generally denoted by 1, ω and ω^2 and geometrically represented by the vertices of an equilateral triangle whose circumcentre is the origin and circumradius is unity. See Fig. 2.9.

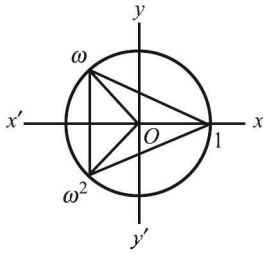


Fig. 2.9

GEOMETRICAL REPRESENTATION OF ALGEBRAIC OPERATIONS ON COMPLEX NUMBERS

1. Addition

Let the complex numbers $z_1=x_1+iy_1=(x_1, y_1)$ and $z_2=x_2+iy_2=(x_2, y_2)$ be represented by the points P and Q on the Argand plane. See Fig. 2.10.

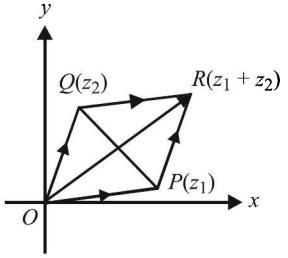


Fig. 2.10

Complete the parallelogram $OPRQ$. Then the mid-points of PQ and OR are the same. The mid-point of $PQ=\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.

Hence,

$$R = (x_1 + x_2,\ y_1 + y_2)$$
$$= (x_1 + x_2) + i(y_1 + y_2) = (x_1 + iy_1) + (x_2 + iy_2)$$
$$= z_1 + z_2 = (x_1, y_1) + (x_2, y_2)$$

In vector notation, we have

$$z_1 + z_2 = \overrightarrow{OP} + \overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PR} = \overrightarrow{OR}$$

2. Subtraction

In Fig. 2.11, $OPRQ'$ is a parallelogram. $OP = Q'R = |z_1|$ and $OQ = OQ' = PR = |z_2|$

Also in the triangle OPR, we have $|OP - PR| \le OR \Rightarrow \big||z_1| - |z_2|\big| \le |z_1 - z_2|$.

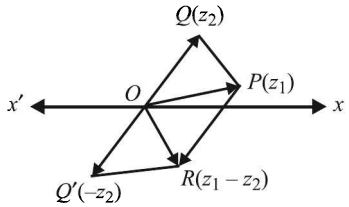


Fig. 2.11

MULTIPLICATION OF COMPLEX NUMBERS

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers, then

$$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2)$$
$$= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$$
$$\Rightarrow \quad z_1 z_2 = [\mathrm{Re}(z_1)\,\mathrm{Re}(z_2) - \mathrm{Im}(z_1)\,\mathrm{Im}(z_2)] + i[\mathrm{Re}(z_1)\,\mathrm{Im}(z_2) + \mathrm{Re}(z_2)\,\mathrm{Im}(z_1)]$$

DIVISION OF COMPLEX NUMBERS

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers, then

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 x_2 + y_1 y_2) + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2}$$

SOME IMPORTANT GEOMETRICAL RESULTS AND EQUATIONS

1. Distance Formula

The distance between $A(z_1)$ and $B(z_2)$ is given by $AB = |z_2 - z_1|$.

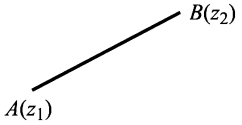


Fig. 2.12

2. Section Formula

The point $P(z)$ which divides the join of the segment AB in the ratio $m : n$ is given by $z = \dfrac{mz_2 + nz_1}{m + n}$.

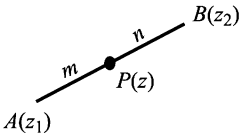


Fig. 2.13

3. Mid-point Formula

The mid-point $M(z)$ of the segment AB is given by $z = \dfrac{1}{2}(z_1 + z_2)$.

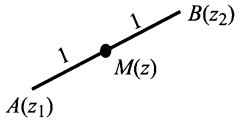


Fig. 2.14

4. Condition(s) for Four Points $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ to Represent the Vertices of a Geometrical Figure

(i) **Parallelogram:** The diagonals AC and BD must bisect each other. See Fig. 2.15.

$$\Leftrightarrow \quad \frac{1}{2}(z_1 + z_3) = \frac{1}{2}(z_2 + z_4) \quad \Leftrightarrow \quad z_1 + z_3 = z_2 + z_4$$

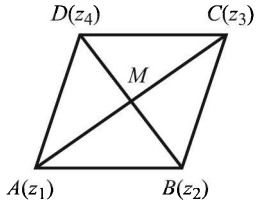


Fig. 2.15

(ii) **Rhombus**

(a) The diagonals AC and BD bisect each other

$$\Leftrightarrow \quad z_1 + z_3 = z_2 + z_4$$

(b) Any two adjacent sides are equal, for instance, $AD = AB$

$$\Leftrightarrow \quad |z_4 - z_1| = |z_2 - z_1|$$

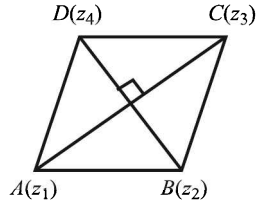


Fig. 2.16

(iii) **Square**

(a) The diagonals AC and BD bisect each other

$$\Leftrightarrow \quad z_1 + z_3 = z_2 + z_4$$

(b) Any two adjacent sides are equal, for instance, $AD = AB$

$$\Leftrightarrow \quad |z_4 - z_1| = |z_2 - z_1|$$

(c) The two diagonals are equal, that is, $AC = BD$

$\Leftrightarrow \quad |z_3 - z_1| = |z_4 - z_2|$

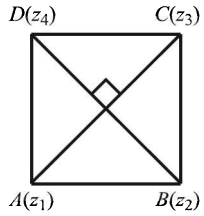


Fig. 2.17

(iv) **Rectangle**

(a) The diagonals AC and BD bisect each other

$\Leftrightarrow \quad z_1 + z_3 = z_2 + z_4$

(b) The diagonals AC and BD are equal

$\Leftrightarrow \quad |z_3 - z_1| = |z_4 - z_2|$

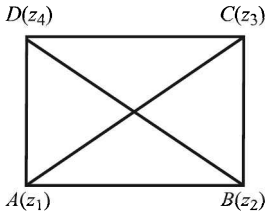


Fig. 2.18

5. Centroid, Incentre, Orthocentre and Circumcentre of a Triangle

Let ABC be a triangle with vertices $A(z_1), B(z_2)$ and $C(z_3)$. See Fig. 2.19.

(i) **Centroid:** $G(z)$ of the ΔABC is the point of concurrence of medians of ΔABC and is given by

$$z = \frac{1}{3}(z_1 + z_2 + z_3)$$

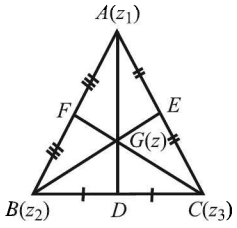


Fig. 2.19

(ii) **Incetre:** $I(z)$ of the ΔABC is the point of concurrence of internal bisectors of angles of ΔABC and is given by $z = \dfrac{az_1 + bz_2 + cz_3}{a+b+c}$. See Fig. 2.20.

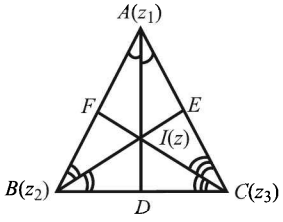


Fig. 2.20

(iii) **Circumcentre:** $S(z)$ of the ΔABC is the point of concurrence of perpendicular bisectors of sides of ΔABC and is given by

$$z = \frac{(z_2 - z_3)|z_1|^2 + (z_3 - z_1)|z_2|^2 + (z_1 - z_2)|z_3|^2}{\overline{z}_1(z_2 - z_3) + \overline{z}_2(z_3 - z_1) + \overline{z}_3(z_1 - z_2)}$$

See Fig. 2.21.

or
$$z = \frac{\begin{vmatrix} |z_1|^2 & z_1 & 1 \\ |z_2|^2 & z_2 & 1 \\ |z_3|^2 & z_3 & 1 \end{vmatrix}}{\begin{vmatrix} \overline{z}_1 & z_1 & 1 \\ \overline{z}_2 & z_2 & 1 \\ \overline{z}_3 & z_3 & 1 \end{vmatrix}}$$

Also, $z = \dfrac{z_1(\sin 2A) + z_2(\sin 2B) + z_3(\sin 2C)}{\sin 2A + \sin 2B + \sin 2C}$

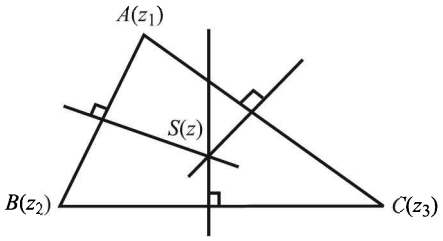


Fig. 2.21

(iv) **Orthocentre:** $H(z)$ of the ΔABC is the point of concurrence of altitudes of ΔABC (see Fig. 2.22) and is given by

$$z = \frac{\begin{vmatrix} z_1^2 & \overline{z}_1 & 1 \\ z_2^2 & \overline{z}_2 & 1 \\ z_3^2 & \overline{z}_3 & 1 \end{vmatrix} + \begin{vmatrix} |z_1|^2 & z_1 & 1 \\ |z_2|^2 & z_2 & 1 \\ |z_3|^2 & z_3 & 1 \end{vmatrix}}{\begin{vmatrix} \overline{z}_1 & z_1 & 1 \\ \overline{z}_2 & z_2 & 1 \\ \overline{z}_3 & z_3 & 1 \end{vmatrix}}$$

Also, $z = \dfrac{z_1(\tan A) + z_2(\tan B) + z_3(\tan C)}{\tan A + \tan B + \tan C}$

or $z = \dfrac{z_1(a \sec A) + z_2(b \sec B) + z_3(c \sec C)}{a \sec A + b \sec B + c \sec C}$

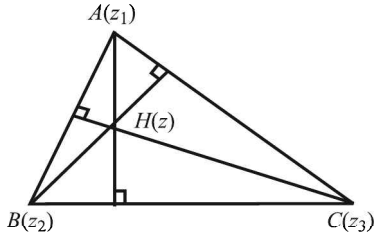


Fig. 2.22

Euler's line: The centroid G of a triangle lies on the segment joining the orthocentre H and the circumcentre S of the triangle. G divides the join of H and S in the ratio 2 : 1.

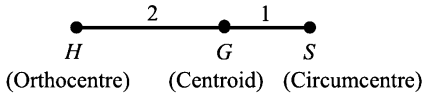


Fig. 2.23

6. Area of a Triangle

The area of ΔABC with vertices $A(z_1), B(z_2)$ and $C(z_3)$ is given by

$$\Delta = \frac{1}{4}\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix}$$

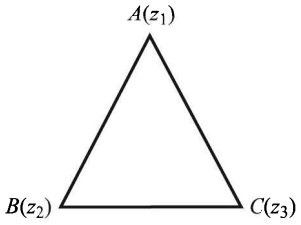


Fig. 2.24

7. Condition for a Triangle to be Equilateral

A triangle ABC with vertices $A(z_1)$, $B(z_2)$ and $C(z_3)$ is equilateral if and only if

$$\frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2} = 0$$

$$\Leftrightarrow \quad z_1^2 + z_2^2 + z_3^2 = z_2 z_3 + z_3 z_1 + z_1 z_2$$

$$\Leftrightarrow \quad \begin{vmatrix} 1 & z_2 & z_3 \\ 1 & z_3 & z_1 \\ 1 & z_1 & z_2 \end{vmatrix} = 0$$

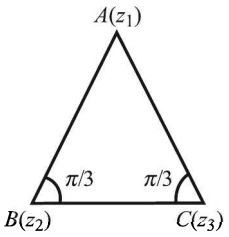


Fig. 2.25

8. Equation of a Straight Line

(i) **Non-parametric form:** The equation of a straight line joining the two points $A(z_1)$ and $B(z_2)$ is given by

$$\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} = 0$$

Also, $$\frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1}$$ (Fig. 2.26)

or $$z(\bar{z}_1 - \bar{z}_2) - \bar{z}(z_1 - z_2) + z_1\bar{z}_2 - z_2\bar{z}_1 = 0$$

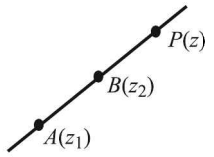


Fig. 2.26

(ii) **Parametric form:** The equation of the line joining the points $A(z_1)$ and $B(z_2)$ is $z = tz_1 + (1-t)z_2$, where t is a real parameter.

(iii) **General equation of a straight line:** The general equation of a straight line is $\bar{a}z + a\bar{z} + b = 0$, where a is a complex number and b is a real number.

9. Complex Slope of a Line

If $A(z_1)$ and $B(z_2)$ are two points in the complex plane, then the complex slope of line AB is defined as

$$\mu = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$$

Two lines with complex slopes μ_1 and μ_2 are

(i) parallel, if $\mu_1 = \mu_2$

(ii) perpendicular, if $\mu_1 + \mu_2 = 0$.

The complex slope of the line $\bar{a}z + a\bar{z} + b = 0$ is given by $-(a/\bar{a})$.

10. Length of the Perpendicular from a Point to a Line

Let the given line be $z\bar{a} + \bar{z}a + b = 0$ and the given point be $z_c = x_c + iy_c$. See Fig. 2.27.

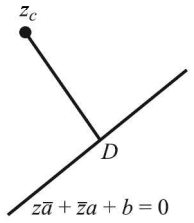


Fig. 2.27

Replacing z by $x+iy$ in the given equation, we get $x(a+\bar{a}) + iy(\bar{a}-a) + b = 0$. The distance of $P(z_c)$ from this line is

$$\frac{|x_c(a+\bar{a}) + iy_c(\bar{a}-a) + b|}{\sqrt{(a+\bar{a})^2 - (a-\bar{a})^2}} = \frac{|z_c\bar{a} + \bar{z}_c + b|}{\sqrt{4(\mathrm{Re}(a))^2 + 4(\mathrm{Im}(a))^2}}$$

$$= \frac{|z_c\bar{a} + \bar{z}_c + b|}{2|a|}$$

11. Equation of a Circle

(i) The equation of the circle with centre at z_0 and radius r is $|z - z_0| = r$.

or $z = z_0 + re^{i\theta}, \quad 0 \le \theta \le 2\pi$ (parametric form)

or $$z\bar{z} - z_0\bar{z} - \bar{z}_0 z + z_0\bar{z}_0 - r^2 = 0$$

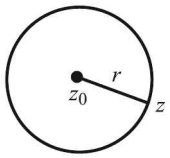


Fig. 2.28

(ii) The general equation of a circle is $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$, where a is a complex number and b is a real number. The centre of $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$ is $-a$ and its radius is $\sqrt{a\bar{a} - b}$.

(iii) **Diameter form of a circle:** The equation of the circle one of whose diameters is the segment joining $A(z_1)$ and $B(z_2)$ is

$$(z - z_1)(\bar{z} - \bar{z}_2) + (\bar{z} - \bar{z}_1)(z - z_2) = 0$$

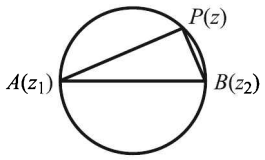


Fig. 2.29

(iv) The equation of the circle passing through two points $A(z_1)$ and $B(z_2)$ is

$$(z - z_1)(\bar{z} - \bar{z}_2) + (\bar{z} - \bar{z}_1)(z - z_2) + k\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

where k is a parameter.

(v) The equation of the circle passing through three non-collinear points: Let the three non-collinear points be $A(z_1)$, $B(z_2)$ and $C(z_3)$. Let $P(z)$ be any point on the circle. Then either $\angle ACB = \angle APB$ (when the angles are in the same segment) or $\angle ACB + \angle APB = \pi$ (when the angles are in the opposite segments). Then

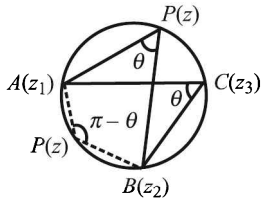


Fig. 2.30

$$\arg\left(\frac{z_3 - z_2}{z_3 - z_1}\right) - \arg\left(\frac{z - z_2}{z - z_1}\right) = 0$$

or

$$\arg\left(\frac{z_3 - z_2}{z_3 - z_1}\right) + \arg\left(\frac{z - z_1}{z - z_2}\right) = \pi$$

$$\Rightarrow \quad \arg\left[\left(\frac{z_3 - z_2}{z_3 - z_1}\right)\left(\frac{z - z_1}{z - z_2}\right)\right] = 0$$

or $$\arg\left[\left(\frac{z_3 - z_2}{z_3 - z_1}\right)\left(\frac{z - z_1}{z - z_2}\right)\right] = \pi$$

$$\left[\begin{array}{l}\text{using } \arg(z_1/z_2) = \arg(z_1) - \arg(z_2) \\ \text{and } \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)\end{array}\right]$$

In any case, we get $\dfrac{(z - z_1)(z_3 - z_2)}{(z - z_2)(z_3 - z_1)}$ as purely real.

$$\Leftrightarrow \quad \frac{(z - z_1)(z_3 - z_2)}{(z - z_2)(z_3 - z_1)} = \frac{(\bar{z} - \bar{z}_1)(\bar{z}_3 - \bar{z}_2)}{(\bar{z} - \bar{z}_2)(\bar{z}_3 - \bar{z}_1)}$$

(vi) Condition for four points to be concyclic. Four points z_1, z_2, z_3 and z_4 will lie on the same circle if and only if $\dfrac{(z_4 - z_1)(z_3 - z_2)}{(z_4 - z_2)(z_3 - z_1)}$ is purely real.

$$\Leftrightarrow \quad \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} \text{ is purely real.}$$

RECOGNIZING SOME LOCI BY INSPECTION

1. $\arg(z) = \alpha$
   If α is a real number, then $\arg(z) = \alpha$ represents a ray starting at the origin (excluding the origin) and making an angle α with the real axis. See Fig. 2.31.

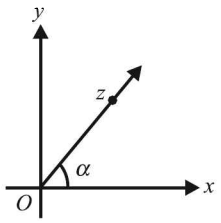


Fig. 2.31

2. $(z - z_0) = \alpha$
   If α is a real number and z_0 is a fixed point, then $\arg(z - z_0) = \alpha$ represents a ray starting at z_0 (excluding the point z_0) and making an angle α with the real axis.

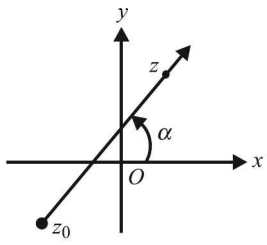


Fig. 2.32

3. If z_1 and z_1 are two fixed points, then $|z - z_1| = |z - z_2|$ represents a perpendicular bisector of the segment joining $A(z_1)$ and $B(z_2)$. See Fig. 2.33.

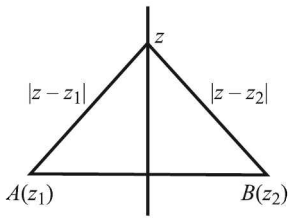


Fig. 2.33

4. If z_1 and z_2 are two fixed points, and $k > 0$, $k \neq 1$ is real number, then $\dfrac{|z-z_1|}{|z-z_2|} = k$ represents a circle. For $k = 1$, it represents the perpendicular bisector of the segment joining $A(z_1)$ and $B(z_2)$.

5. $|z-z_1|+|z-z_2| = k$

   Let z_1 and z_1 be two fixed points and k be a positive real number.

   (a) If $k > |z_1 - z_2|$, then $|z-z_1|+|z-z_2| = k$ represents an ellipse with foci at $A(z_1)$ and $B(z_2)$ and length of major axis = k.

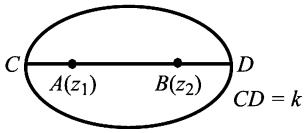


Fig. 2.34

   (b) If $k = |z_1 - z_2|$, then $|z-z_1|+|z-z_2| = k$ represents the segment joining z_1 and z_2.

   (c) If $k < |z_1 - z_2|$, then $|z-z_1|+|z-z_2| = k$ does not represent any curve in the Argand plane.

6. $|z-z_1|-|z-z_2| = k$

   Let z_1 and z_2 be two fixed points, k be a positive real number.

   (a) If $k \neq |z_1 - z_2|$, then $|z-z_1|-|z-z_2| = k$ represents a hyperbola with foci at $A(z_1)$ and $B(z_2)$. See Fig. 2.35.

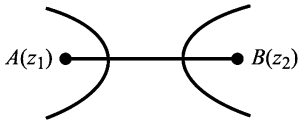


Fig. 2.35

   (b) If $k = |z_1 - z_2|$, then $\big||z-z_1|-|z-z_2|\big| = k$ represents a straight line joining $A(z_1)$ and $B(z_2)$ but excluding the segment AB. See Fig. 2.36.

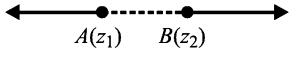


Fig. 2.36

7. $|z-z_1|^2 + |z-z_2|^2 = |z_1 - z_2|^2$

   If z_1 and z_2 are two fixed points, then

$$|z-z_1|^2 + |z-z_2|^2 = |z_1 - z_2|^2$$

   represents a circle with z_1 and z_2 as the extremities of a diameter. See Fig. 2.37.

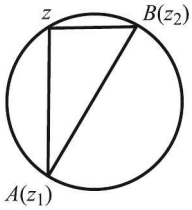


Fig. 2.37

8. $\arg\left(\dfrac{z-z_1}{z-z_2}\right) = \alpha$

   Let z_1 and z_2 be two fixed points, and α be a real number such that $0 \le \alpha \le \pi$.

   (a) If $0 \le \alpha \le \pi$ and $\alpha \neq \dfrac{\pi}{2}$, then the $\arg\left(\dfrac{z-z_1}{z-z_2}\right) = \alpha$ represents a segment of the circle passing through $A(z_1)$ and $B(z_2)$. See Fig. 2.38.

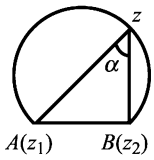


Fig. 2.38

   (b) If $\alpha = \dfrac{\pi}{2}$, then the $\arg\left(\dfrac{z-z_1}{z-z_2}\right) = \alpha\left(=\dfrac{\pi}{2}\right)$ represents a circle with diameter as the segment joining $A(z_1)$ and $B(z_2)$. See Fig. 2.39.

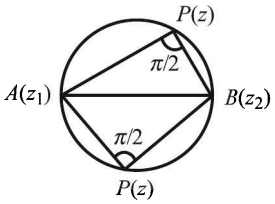


Fig. 2.39

   (c) If $\alpha = \pi$, then the $\arg\left(\dfrac{z-z_1}{z-z_2}\right) = \alpha(=\pi)$ represents the straight line joining $A(z_1)$ and $B(z_2)$ but excluding the segment AB. See Fig. 2.40.

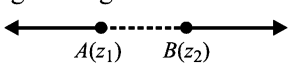


Fig. 2.40

(d) If $\alpha = 0$, then the $\arg\left(\frac{z-z_1}{z-z_2}\right) = \alpha (= 0)$ represents the segment joining $A(z_1)$ and $B(z_2)$. See Fig. 2.41.

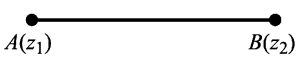


Fig. 2.41

EQUATION OF A TANGENT TO A GIVEN CIRCLE

Let $|z - z_0| = r$ be the given circle and we have to obtain the equation of the tangent at $A(z_1)$. Let us take any point $P(z)$ on the tangent line at $A(z_1)$. Clearly, $\angle PAB = \pi/2$. See Fig. 2.42.

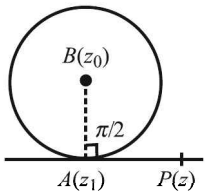


Fig. 2.42

Hence, $\frac{z-z_1}{z_0-z_1}$ is purely imaginary.

$$\Rightarrow \quad \frac{z-z_1}{z_0-z_1} + \frac{\bar{z}-\bar{z}_1}{\bar{z}_0-\bar{z}_1} = 0$$

$$\Rightarrow \quad (z-z_1)(\bar{z}_0-\bar{z}_1) + (\bar{z}-\bar{z}_1)(z_0-z_1) = 0$$

$$\Rightarrow \quad z(\bar{z}_0-\bar{z}_1) + \bar{z}(z_0-z_1) + 2|z_1|^2 - z_1\bar{z}_0 - \bar{z}_1 z_0 = 0$$

In particular, if the given circle is $|z| = r$, then the equation of the tangent at $z = z_1$ would be $z\bar{z}_1 + \bar{z}z_1 = 2|z_1|^2 = 2r^2$.

If $\left|\frac{z-z_1}{z-z_2}\right| = \lambda$ $(\lambda \in \mathbb{R}^+, \lambda \neq 1)$, where z_1 and z_2 are the given complex numbers and z is an arbitrary complex number, then z would lie on the circle.

Note

- If we take C to be the mid-point of AB, it can be easily proved that $CA \cdot CB = (CA)^2$, i.e. $|z_1 - z_0||z_2 - z_0| = r^2$, where the point C is denoted by z_0 and r is the radius of the circle.
- If $\lambda = 1 \Rightarrow |z - z_1| = |z - z_2|$, then $P(z)$ would lie on the right bisector of the line joining the points $A(z_1)$ and $B(z_2)$. Note that in this case, z_1 and z_2 are the mirror images of each other with respect to the right bisector.

SOLVED PROBLEMS

1. For any integer n, the arg of $z = \frac{(\sqrt{3}+i)^{4n+1}}{(1-i\sqrt{3})^{4n}}$ is

(a) $\pi/6$ (b) $\pi/3$
(c) $\pi/2$ (d) $2\pi/3$

Ans. (a)

Solution: $z = \frac{2^{4n+1}}{2^{4n}} \frac{e^{i(4n+1)\pi/6}}{e^{-(i4n)\pi/3}} = 2e^{i(12n+1)\pi/6}$

$= 2e^{2n\pi i} \cdot e^{i\pi/6} = 2e^{i\pi/6} \quad \therefore \arg(z) = \pi/6$

2. $\cos\left(i \log \frac{a-ib}{a+ib}\right)$ is equal to

(a) ab (b) $\frac{a^2-b^2}{a^2+b^2}$
(c) $\frac{a^2-b^2}{2ab}$ (d) $\frac{2ab}{a^2+b^2}$

Ans. (b)

Solution: Let $a + ib = re^{i\theta}$ and $a - ib = re^{-i\theta}$.

$$\therefore \quad i \log \frac{a-ib}{a+ib} = i \log(e^{-2\theta i}) = i(-2\theta i) \log e = 2\theta$$

$$\therefore \quad \cos\left(i \log \frac{a-ib}{a+ib}\right) = \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$$

$$= \frac{1-(b^2/a^2)}{1+(b^2/a^2)} = \frac{a^2-b^2}{a^2+b^2}$$

3. If α and β are two different complex numbers with $|\beta| = 1$, then $\left|\frac{\beta-\alpha}{1-\bar{\alpha}\beta}\right|$ is equal to

(a) 0 (b) 1/2
(c) 1 (d) 2

Ans. (c)

Solution: $|\beta| = 1 \quad \therefore |\beta|^2 = \beta\bar{\beta} = 1$ (see Proporty 6 of conjugate)

$$\therefore \quad \text{L.H.S.} = \left|\frac{\beta-\alpha}{\beta\bar{\beta}-\bar{\alpha}\beta}\right| = \frac{|\beta-\alpha|}{|\beta||\bar{\beta}-\bar{\alpha}|}$$

$$= \frac{|\beta-\alpha|}{1 \cdot |\overline{\beta-\alpha}|} = 1 \qquad \left(\because \ |\bar{z}| = |z|\right)$$

4. If $\left(\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)^{48} = 3^{24}(x+iy)$, then the ordered pair (x, y) is

(a) (0, 2) (b) (0, 1)
(c) (1, 0) (d) (1, 1)

Ans. (c)

Solution:

$$\left(\frac{3}{2}+\frac{i\sqrt{3}}{2}\right)^{48}=\left\{\sqrt{3}\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)\right\}^{48}$$

$$=3^{24}\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)^{48}$$

$$=3^{24}(e^{i\pi/6})^{48}=3^{24}(e^{i8\pi})$$

$$=3^{24}(\cos 8\pi+i\sin 8\pi)=3^{24}(1+i\cdot 0)$$

$$=3^{24}(x+iy)$$

$\therefore \quad (x, y)=(1, 0)$

5. If $(a_1+ib_1)(a_2+ib_2)\cdots(a_n+ib_n)=A+iB$, then $(a_1^2+b_1^2)(a_2^2+b_2^2)\cdots(a_n^2+b_n^2)$ is equal to

(a) 1 (b) A^2+B^2

(c) $A+B$ (d) $\frac{1}{A^2}+\frac{1}{B^2}$

Ans. (b)

Solution: Now, $(a_1^2+b_1^2)(a_2^2+b_2^2)\cdots(a_n^2+b_n^2)$

$=[(a_1+ib_1)(a_2+ib_2)\cdots(a_n+ib_n)]$

$[(a_1-ib_1)(a_2-ib_2)\cdots(a_n-ib_n)]$

$=[A+iB][A-iB]$

$=A^2+B^2$

6. The value of the expression $\left(\frac{i+\sqrt{3}}{2}\right)^8+\left(\frac{i-\sqrt{3}}{2}\right)^8$ is

(a) -1 (b) 0

(c) 1 (d) none of these

Ans. (a)

Solution: $\left(\frac{i+\sqrt{3}}{2}\right)^8+\left(\frac{i-\sqrt{3}}{2}\right)^8$

$$=(-i)^8\left[\frac{-1+\sqrt{3}i}{2}\right]^8+(-i)^8\left[\frac{-1-\sqrt{3}i}{2}\right]^8$$

$$=\omega^8+\omega^{16}=\omega^2+\omega=-1$$

7. Let z_1 and z_2 be two complex numbers such that $|z_1-z_2|=|z_1|-|z_2|$. Then the $\arg(z_1/z_2)$ is

(a) $\pi/4$ (b) $\pi/2$

(c) 0 (d) none of these

Ans. (c)

Solution: If $z_1=r_1e^{i\theta_1}$ and $z_2=r_2e^{i\theta_2}$, then

L.H.S. $=[r_1^2+r_2^2-2r_1r_2\cos(\theta_1-\theta_2)]^{1/2}$

R.H.S. $=r_1-r_2$

$\therefore$ L.H.S. = R.H.S. if $\cos(\theta_1-\theta_2)=1$

or $\theta_1-\theta_2=0$

or $\arg(z_1)-\arg(z_2)=\arg(z_1/z_2)=0$

8. If $i=\sqrt{-1}$, then

$4+5\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{334}+3\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{365}$ is equal to

(a) $1-i\sqrt{3}$ (b) $-1+i\sqrt{3}$

(c) $i\sqrt{3}$ (d) $-i\sqrt{3}$

Ans. (c)

Solution: $-\frac{1}{2}+\frac{i\sqrt{3}}{2}=\omega$ (cube root of unity), $\omega^{3k}=1$

$\therefore \quad 4+5\omega^{334}+3\omega^{365}$

$=4+5\cdot\omega^{333}\cdot\omega+3\cdot\omega^{363}\cdot\omega^2$

$=4+5\omega+3\omega^2=1+3+3\omega+3\omega^2+2\omega=1+2\omega$ $(\because\ 1+\omega+\omega^2=0)$

$=1+(-1+i\sqrt{3})=i\sqrt{3}$

9. The points z_1, z_2, z_3 and z_4 in the complex plane are the vertices of a parallelogram taken in order if and only if

(a) $z_1+z_4=z_2+z_3$ (b) $z_1+z_3=z_2+z_4$

(c) $z_1+z_2=z_3+z_4$ (d) none of these

Ans. (b)

Solution: Equating the complex numbers corresponding to the mid-points of the two diagonals, we have

$$\frac{z_1+z_3}{2}=\frac{z_2+z_4}{2} \text{ or } z_1+z_3=z_2+z_4$$

10. Let $z_1=6+i$ and $z_2=4-3i$. Let z be a complex number such that the $\arg\left(\frac{z-z_1}{z_2-z}\right)=\frac{\pi}{2}$, then z satisfies the relation

(a) $|z-(5-i)|=5$ (b) $|z-(5-i)|=\sqrt{5}$

(c) $|z-(5+i)|=5$ (d) $|z-(5+i)|=\sqrt{5}$

Ans. (b)

Solution: $\arg\left(\frac{z-z_1}{z_2-z}\right)=\frac{\pi}{2}$, $z-z_1=(z_2-z)\pi/2$

$$(z_1-z)=(z_2-z)(-\pi/2)$$

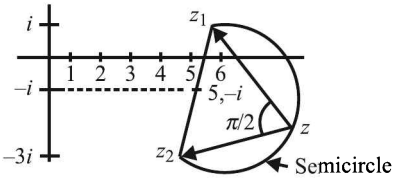


Fig. 2.43

Centre of semicircle $=\frac{z_1+z_2}{2}=5-i \quad \Rightarrow \quad (5, -i)$

$\therefore \quad |z-(5-i)|=r$

where $r=\frac{z_1-z_2}{2}=\frac{(6+i)-(4-3i)}{2}$

$$=\frac{2+4i}{2}=\frac{\sqrt{2^2+4^2}}{2}=\frac{\sqrt{20}}{2}=\sqrt{5}$$

11. If z_1 and z_2 are the conjugate complex numbers and z_3, z_4 are also conjugate, then $\arg(z_3/z_2)$ is equal to

(a) $\arg(z_1/z_4)$ (b) $\arg(z_4/z_1)$

(c) $\arg(z_2/z_4)$ (d) $\arg(z_1/z_3)$

Ans. (a)

Solution: $z_1 = re^{i\theta}, z_2 = re^{-i\theta}$ (conjugate complex numbers)

$z_3 = Re^{i\phi}, z_4 = Re^{-i\phi}$ (conjugate complex numbers)

$$\frac{z_3}{z_2} = \frac{R}{r}e^{i(\theta+\phi)} \quad \therefore \quad \arg\left(\frac{z_3}{z_2}\right) = \theta+\phi$$

$$\frac{z_1}{z_4} = \frac{r}{R}e^{i(\theta+\phi)} \quad \therefore \quad \arg\left(\frac{z_1}{z_4}\right) = \theta+\phi$$

Hence the correct answer is (a).

12. If $z = x+iy$ such that $|z+1| = |z-1|$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$, then

(a) $x = \sqrt{2}+1,\ y = 0$ (b) $x = 0,\ y = \sqrt{2}+1$

(c) $x = 0,\ y = \sqrt{2}-1$ (d) $x = \sqrt{2}-1,\ y = 0$

Ans. (b)

Solution: $|z+1| = |z-1| \Rightarrow (x+1)^2 + y^2 = (x-1)^2 + y^2$

$$\Rightarrow x = 0.$$

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$$

$$\Rightarrow \arg\left(\frac{[(x-1)+iy][(x+1)-iy]}{(x+1)^2+y^2}\right) = \frac{\pi}{4}$$

$$\Rightarrow \arg\left(\frac{(x^2+y^2-1)+2iy}{(x+1)^2+y^2}\right) = \frac{\pi}{4}$$

$$\Rightarrow y > 0,\ x^2+y^2-1 > 0,\ \frac{2y}{x^2+y^2-1} = 1$$

$$\Rightarrow x^2+y^2-2y-1 = 0,\ y > 0,\ x^2+y^2-1 > 0$$

$$\therefore \quad \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$$

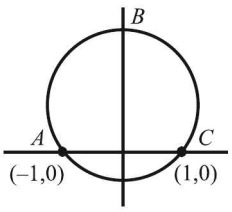


Fig. 2.44

is the arc ABC of the circle $x^2+y^2-2y-1=0$.

Solving with $x = 0$, we get

$$y = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2},\ y > 0$$

$$\therefore \quad y = 1+\sqrt{2}$$

13. The least positive integer n for which $\left(\frac{1+i}{1-i}\right)^n$ is real is

(a) 2 (b) 4

(c) 8 (d) none of these

Ans. (a)

Solution: $\frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = \frac{2i}{2} = i$

So, $\left(\frac{1+i}{1-i}\right)^n = (i)^n$

$\Rightarrow \quad n = 2$ (least positive integer)

14. In Argand diagram, the locus given by

$$\arg[(1+2i)z - 1 + 3i] = \frac{\pi}{4}$$

(a) is a straight line through the origin

(b) is a the straight line through the points $(2 - 2i)$ and $(5 - 3i)$

(c) is a circle with the line joining $(1 + 2i)$ and $(1 - 3i)$ as the diameter

(d) none of these

Ans. (b)

Solution: Let $z = x + iy$

$$\therefore \quad (1+2i)z - 1 + 3i = x - 2y - 1 + i(2x + y + 3)$$

Since $\arg[(1+2i)z - 1 + 3i] = \frac{\pi}{4}$,

$$\Rightarrow \tan^{-1}\left(\frac{2x+y+3}{x-2y-1}\right) = \frac{\pi}{4} \Rightarrow \frac{2x+y+3}{x-2y-1} = \tan\frac{\pi}{4} = 1$$

$\Rightarrow 2x + y + 3 = x - 2y - 1 \Rightarrow x + 3y + 4 = 0$

which is the locus of a straight line passing through $(2 - 2i)$ and $(5 - 3i)$.

15. If we express

$$z = \frac{(\cos 2\theta - i\sin 2\theta)^4(\cos 4\theta + i\sin 4\theta)^{-5}}{(\cos 3\theta + i\sin 3\theta)^{-2}(\cos 3\theta - i\sin 3\theta)^{-9}}$$

in the form of $x + iy$, we get

(a) $\cos 49\theta - i\sin 49\theta$ (b) $\cos 23\theta + i\sin 23\theta$

(c) $\cos 49\theta + i\sin 49\theta$ (d) $\cos 21\theta + i\sin 21\theta$

Ans. (a)

Solution: The given complex number is

$$z = \frac{(\cos 2\theta - i\sin 2\theta)^4(\cos 4\theta + i\sin 4\theta)^{-5}}{(\cos 3\theta + i\sin 3\theta)^{-2}(\cos 3\theta - i\sin 3\theta)^{-9}}$$

The given number may be written as

$$z = \frac{[(\cos\theta + i\sin\theta)^{-2}]^4[(\cos\theta + i\sin\theta)^4]^{-5}}{[(\cos\theta + i\sin\theta)^3]^{-2}[(\cos\theta + i\sin\theta)^{-3}]^{-9}}$$

$$= \frac{(\cos\theta + i\sin\theta)^{-8}(\cos\theta + i\sin\theta)^{-20}}{(\cos\theta + i\sin\theta)^{-6}(\cos\theta + i\sin\theta)^{27}}$$

$$= (\cos\theta + i\sin\theta)^{-8-20+6-27} = (\cos\theta + i\sin\theta)^{-49}$$

$$= \cos 49\theta - i\sin 49\theta$$

16. If $2|z-1|=|z-2|$ and if $x^2+y^2=\lambda x$, then λ is equal to

(a) 1/3 (b) 2/3

(c) 4/3 (d) none of these

Ans. (c)

Solution: Squaring both sides of $2|z-1|=|z-2|$,

$$4[(x-1)^2+y^2]=(x-2)^2+y^2$$

$$\Rightarrow \quad 3(x^2+y^2)=4x$$

$$\therefore \quad x^2+y^2=\frac{4}{3}x=\lambda x \quad \Rightarrow \quad \lambda=\frac{4}{3}$$

17. The point of intersection of the curves $\arg(z-3i)=\frac{3\pi}{4}$ and $\arg(2z+1-2i)=\frac{\pi}{4}$ is

(a) $\frac{1}{4}(3+9i)$ (b) $\frac{1}{4}(9+3i)$

(c) $1+i$ (d) none of these

Ans. (d)

Solution: $\arg(z-3i)=\arg(x+(y-3)i)=\frac{3\pi}{4}$

$$\Rightarrow \quad x<0,\ y-3>0,\ \frac{y-3}{x}=\tan\frac{3\pi}{4}=-1$$

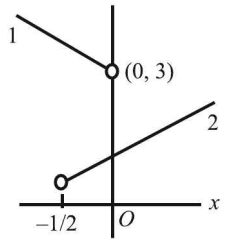


Fig. 2.45

$$\Rightarrow \quad x+y-3=0,\ x<0,\ y>3 \qquad \text{(i)}$$

$$\arg(2z+1-2i)=\arg[(2x+1)+i(2y-2)]=\frac{\pi}{4}$$

$$\Rightarrow \quad x>-\frac{1}{2},\ y>1 \text{ and } \frac{2(y-1)}{2x+1}=\tan\frac{\pi}{4}=1$$

$$\Rightarrow \quad 2x-2y+3=0,\ x>-\frac{1}{2},\ y>1 \qquad \text{(ii)}$$

Hence, (i) and (ii) do not have any point of intersection.

18. For positive integers n_1 and n_2, the value of the expression $(1+i)^{n_1}+(1+i^3)^{n_1}+(1+i^5)^{n_2}+(1+i^7)^{n_2}$, $i=\sqrt{-1}$, is a real number if and only if

(a) $n_1=n_2+1$ (b) $n_1=n_2-1$

(c) $n_1=n_2$ (d) $n_1>0, n_2>0$

Ans. (d)

Solution: Using $i^3=-i, i^5=i$ and $i^7=-i$, we can write the given expression as

$$\begin{aligned}&(1+i)^{n_1}+(1-i)^{n_1}+(1+i)^{n_2}+(1-i)^{n_2}\\ &=2[{}^{n_1}C_0+{}^{n_1}C_2i^2+{}^{n_1}C_4i^4+\cdots]\\ &\quad+2[{}^{n_2}C_0-{}^{n_2}C_2i^2+{}^{n_2}C_4i^4-\cdots]\\ &=2[{}^{n_1}C_0-{}^{n_1}C_2+{}^{n_1}C_4-\cdots]\\ &\quad+2[{}^{n_2}C_0-{}^{n_2}C_2+{}^{n_2}C_4-\cdots]\end{aligned}$$

This is a real number if the values of n_1 and n_2 are greater than zero.

19. If α and β are the roots of $z+\frac{1}{z}=2(\cos\theta+i\sin\theta)$, $(0<\theta<\pi)$, then

(a) $|\alpha-i|>|\beta-i|$ (b) $|\alpha-i|<|\beta-i|$

(c) $|\alpha-1|=|\beta-1|$ (d) $|\alpha-i|=|\beta-i|$

Ans. (d)

Solution: Since $z+\frac{1}{z}=2(\cos\theta+i\sin\theta)=2e^{i\theta}$, we have

$$z^2-2e^{i\theta}z+1=0 \quad \Rightarrow \quad z=e^{i\theta}\pm\sqrt{e^{2i\theta}-1}$$

$$\begin{aligned}e^{2i\theta}-1&=(\cos2\theta+i\sin2\theta)-1\\&=-(1-\cos2\theta)+i\sin2\theta\\&=-2\sin^2\theta+2i(\sin\theta\cos\theta)\\&=2\sin\theta(-\sin\theta+i\cos\theta)\\&=2\sin\theta\left[\cos\left(\frac{\pi}{2}+\theta\right)+i\sin\left(\frac{\pi}{2}+\theta\right)\right]\end{aligned}$$

Now, if $\alpha=e^{i\theta}+\sqrt{e^{2i\theta}-1}$, we have

$$\alpha=(\cos\theta+i\sin\theta)+\sqrt{2\sin\theta}\times\left[\cos\left(\frac{\pi}{2}+\theta\right)+i\sin\left(\frac{\pi}{2}+\theta\right)\right]^{1/2}$$

$$\alpha=(\cos\theta+i\sin\theta)+\sqrt{2\sin\theta}\times\left[\cos\left(\frac{\pi}{4}+\frac{\theta}{2}\right)+i\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right]$$

$$\alpha=\cos\theta+\sqrt{2\sin\theta}\cos\left(\frac{\pi}{4}+\frac{\theta}{2}\right)+i\left[\sin\theta+\sqrt{2\sin\theta}\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right]$$

Now,

$$|\alpha-i|^2=\left[\cos\theta+\sqrt{2\sin\theta}\cos\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right]^2+\left[\sin\theta+\sqrt{2\sin\theta}\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)-1\right]^2$$

$$|\alpha-i|^2 = 2+2\sqrt{2\sin\theta}\cos\left(\frac{\pi}{4}+\frac{\theta}{2}-\theta\right) - 2\sqrt{2\sin\theta}\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)$$

$$|\alpha-i|^2 = 2+2\sqrt{2\sin\theta}\cos\left(\frac{\pi}{4}-\frac{\theta}{2}\right) - 2\sqrt{2\sin\theta}\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)$$

or $|\alpha-i|^2 = 2+2\sqrt{2\sin\theta}\left[\cos\left(\frac{\pi}{4}-\frac{\theta}{2}\right)-\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right]$

$\therefore \quad |\alpha-i|^2 = 2+2\sqrt{2\sin\theta}(0) = 2$

Similarly, $|\beta-i|^2 = 2$

Hence $|\alpha-i| = |\beta-i|$.

20. The value of $\text{Re}\left(\frac{(1+i)^2}{3-i}\right)$ is

(a) –1/5 (b) 1/5
(c) 1/10 (d) –1/10

Ans. (a)

Solution: Given: $z = \frac{(1+i)^2}{3-i}$.

$$= \frac{(1+i)^2}{3-i}\times\frac{3+i}{3+i} = \frac{(1+2i-1)(3+i)}{3^2-i^2}$$

$$= \frac{3+6i-3+i-2-i}{9+1} = \frac{-2+6i}{10}$$

Therefore, $\text{Re}\left(\frac{-2+6i}{10}\right) = -\frac{2}{10} = -\frac{1}{5}$

EXERCISE SET 1

1. The equation $z\bar{z}+a\bar{z}+\bar{a}z+b=0, b\in\mathbb{R}$ represents a circle if

(a) $|a|^2 = b$ (b) $|a^2| > b$
(c) $|a|^2 < b$ (d) none of these

2. If $\frac{c+i}{c-i} = a+ib$, where a, b, c are real, then $a^2 + b^2$ is equal to

(a) 7 (b) 1
(c) c^2 (d) $-c^2$

3. $\frac{3+2i\sin\theta}{1-2i\sin\theta}$ will be real, if θ is

(a) $2n\pi$ (b) $n\pi + (\pi/2)$
(c) $n\pi$ (d) none of these

4. If $1, \alpha_1, \alpha_2, \ldots, \alpha_{3n}$ are the roots of equation $x^{3n+1}-1=0$ and ω, ω^2 are the imaginary cube roots of unity, then $\frac{(\omega^2-\alpha_1)(\omega^2-\alpha_2)\ldots(\omega^2-\alpha_{3n})}{(\omega-\alpha_1)(\omega-\alpha_2)\ldots(\omega-\alpha_{3n})} = ?$

(a) ω (b) $-\omega$
(c) 1 (d) ω^2

5. If $(1+i)(1+2i)(1+3i)\ldots(1+ni) = x+iy$, then the value of $2\times5\times10\times\cdots\times(1+n^2)$ is

(a) x^2+y^2 (b) x^2-y^2
(c) $\frac{\sqrt{x}}{2}+\frac{\sqrt{y}}{3}$ (d) $\frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}}$

6. If $\left(\frac{1+i}{1-i}\right)^x = 1$, then for any +ve integer n, x is equal to

(a) $4n$ (b) $4n + 1$
(c) $2n$ (d) $2n + 1$

7. If $1, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}$ are the n roots of unity, then $(1-\alpha_1)(1-\alpha_2)\ldots(1-\alpha_{n-1})$ is equal to

(a) $n - 1$ (b) n
(c) -1 (d) 1

8. If z is a complex number, then the minimum value of $|z|+|z-1|$ is

(a) 1 (b) 0
(c) 1/2 (d) none of these

9. If $|z^n\sin\theta_0 + z^{n-1}\sin\theta_1 + z^{n-2}\sin\theta_2 + \cdots + z\sin\theta_{n-1} + \sin\theta_n| = 2$, then all the roots of the equation lie

(a) inside the circle $|z| = 1/2$
(b) outside the circle $|z| = 1/2$
(c) on the circle $|z| = 1/2$
(d) none of these

10. If the conjugate and the reciprocal of a complex number $z = x+iy$ are equal, then

(a) $x+y=1$ (b) $x^2+y^2=1$
(c) $x=1$ and $y=0$ (d) $x=0$ and $y=1$

11. The value of the sum $\sum_{n=1}^{13}(i^n+i^{n+1})$, where $i=\sqrt{-1}$, is

(a) i (b) $i - 1$
(c) $-i$ (d) 0

12. If 1, ω and ω^2 are the three cube roots of unity and α, β and γ are the cube roots of p ($p < 0$), then for any x, y and z the expression $\frac{x\alpha+y\beta+z\gamma}{x\beta+y\gamma+z\alpha}$ is equal to

(a) 1 (b) ω
(c) ω^2 (d) none of these

13. The smallest positive integral value of n for which $\left(\frac{1-i}{1+i}\right)^n$ is purely imaginary with positive imaginary part, is
(a) 1 (b) 3
(c) 5 (d) none of these

14. If $\left|\frac{z-2}{z+2}\right| = n$, where n is a constant and $n \neq 1$, then the locus of z is
(a) a straight line (b) a circle
(c) a parabola (d) an ellipse

15. Let $\alpha = z_1 + z_2$, $\beta = \omega z_1 + \omega^2 z_2$ and $\gamma = \omega^2 z_1 + \omega z_2$, where ω is a complex cube root of unity, then
(a) $\alpha^2 + \beta^2 + \gamma^2 = 8 z_1 z_2$
(b) $\alpha\beta + \beta\gamma + \gamma\alpha = 3 z_1 z_2$
(c) $\alpha^3 + \beta^3 + \gamma^3 = 3(z_1^3 + z_2^3)$
(d) $\alpha\beta\gamma = 2(z_1^3 + z_2^3)$

16. If z_1, z_2 and z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$, then the value of $|z_1 + z_2 + z_3|$ is
(a) equal to 1 (b) less than 1
(c) equal to 3 (d) greater than 3

17. If $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \cdots + a_{2n} x^{2n}$, then $a_0 + a_3 + a_6 + \cdots$ is equal to
(a) 3^n (b) 3^{n-1}
(c) 3^{n-2} (d) none of these

18. If $z = x + iy$, then $|3z - 1| = |3(z - 2)|$ represents
(a) x-axis (b) y-axis
(c) circle (d) $x = c$

19. For a complex number z, the minimum value of $|z| + |z - 2|$ is
(a) 1 (b) 2
(c) 3 (d) none of these

20. If $|z_1| = |z_2| = \cdots = |z_n| = 1$, then the value of $|z_1 + z_2 + z_3 + \cdots + z_n|$ is
(a) 1 (b) $|z_1| + |z_2| + \cdots + |z_n|$
(c) $\left|\frac{1}{z_1} + \frac{1}{z_2} + \cdots + \frac{1}{z_n}\right|$ (d) none of these

21. If z_1 and z_2 are the roots of the equation $z^2 + az + b = 0$, then prove that the origin, z_1 and z_2 form an equilateral triangle if and only if
(a) $a^2 = 3b$ (b) $b^2 = 3a$
(c) $a^2 + 3b = 0$ (d) $b^2 + 3a = 0$

22. The value of z satisfying the equation $\log z + \log z^2 + \cdots + \log z^n = 0$ is
(a) $\cos\left[\frac{4m\pi}{n(n+1)}\right] + i \sin\left[\frac{4m\pi}{n(n+1)}\right]$, $m = 0, 1, 2, \ldots$
(b) $\cos\left[\frac{4m\pi}{n(n+1)}\right] - i \sin\left[\frac{4m\pi}{n(n+1)}\right]$, $m = 0, 1, 2, \ldots$
(c) $\sin\left(\frac{4m\pi}{n}\right) + i \cos\left(\frac{4m\pi}{n}\right)$, $m = 0, 1, 2, \ldots$
(d) 0

23. If $a = \cos\alpha + i \sin\alpha$, $b = \cos\beta + i \sin\beta$, $c = \cos\gamma + i \sin\gamma$ and $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 1$, then $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha)$ is equal to
(a) 3/2 (b) –(3/2)
(c) 0 (d) 1

24. The greatest value of the moduli of the complex number z satisfying the equation $\left|z - \frac{4}{z}\right| = 2$ is
(a) $\sqrt{5}$ (b) $\sqrt{5} - 1$
(c) $\sqrt{5} + 1$ (d) none of these

25. If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1 - z^2}$ lie on
(a) a line not passing through the origin
(b) $|z| = \sqrt{2}$
(c) the x-axis
(d) the y-axis

26. If z is a complex number, then $|3z - 1| = 3|z - 2|$ represents
(a) the y-axis (b) a circle
(c) the x-axis (d) a line parallel to the y-axis

27. If a, b, c are integers not all equal and $\omega \neq 1$ is a cube root of unity, then min value of $|a + b\omega + c\omega^2|$ is
(a) 0 (b) 1
(c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

28. The values of x and y satisfying the equation $\frac{(1+i)x - 2i}{3 + i} + \frac{(2 - 3i)y + i}{3 - i} = i$ are
(a) $x = -1$, $y = 3$ (b) $x = 3$, $y = -1$
(c) $x = 0$, $y = 1$ (d) $x = 1$, $y = 0$

29. The locus of the complex number z in an Argand plane satisfying the equation $\arg(z + i) - \arg(z - i) = (\pi/2)$ is
(a) the boundary of a circle
(b) the interior of a circle
(c) the exterior of a circle
(d) none of these

30. Let $3-i$ and $2+i$ be affixes of two points A and B in the Argand plane and P represent the complex number $z=x+iy$. Then the locus of P if $|z-3+i|=|z-2-i|$ is
(a) the circle on AB as diameter
(b) the line AB
(c) the perpendicular bisector of AB
(d) none of these

31. The complex numbers $\sin x+i\cos 2x$ and $\cos x-i\sin 2x$ are conjugate to each other for
(a) $x=n\pi$ (b) $x=\left(n+\frac{1}{2}\right)\pi$
(c) $x=0$ (d) no value of x

32. If the equation $(p+iq)x^2+(m+in)x+r=0$ has real roots, where p, q, m, n and r are real $(r\neq 0)$, then pn^2+rq^2-mnq is equal to
(a) 1 (b) 0
(c) -1 (d) none of these

33. If $|z|=1$ and $\omega=\frac{z-1}{z+1}$ (where $z\neq -1$), then the value of $\text{Re}(\omega)$ is
(a) 0 (b) $-\frac{1}{|z+1|^2}$
(c) $\frac{\sqrt{2}}{|z+1|^2}$ (d) $\left|\frac{z}{z+1}\right|.\frac{1}{|z+1|^2}$

34. If $z=\frac{1+7i}{(2-i)^2}$, then the polar form of z is
(a) $\sqrt{2}\left[\cos\frac{\pi}{4}+i\sin\frac{3\pi}{4}\right]$ (b) $\sqrt{2}\left[\cos\frac{3\pi}{4}-i\sin\frac{3\pi}{4}\right]$
(c) $\sqrt{2}\left[\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right]$ (d) none of these

35. The points representing the complex number z, for which $|z+3i|^2-|z-3|^2=6$, lie on the line given by
(a) $x+y=1$ (b) $x+y=-1$
(c) $x+y=2$ (d) $x+y=0$

36. If $\tan^{-1}(\alpha+i\beta)=x+iy$, then x is equal to
(a) $\frac{1}{2}\tan^{-1}\left(\frac{2\alpha}{1-\alpha^2-\beta^2}\right)$ (b) $\frac{1}{2}\tan^{-1}\left(\frac{2\alpha}{1+\alpha^2+\beta^2}\right)$
(c) $\tan^{-1}\left(\frac{2\alpha}{1-\alpha^2-\beta^2}\right)$ (d) none of these

37. The value of $-2i\sum_{p=1}^{6}\left(\sin\frac{2p\pi}{7}+i\cos\frac{2p\pi}{7}\right)$ is
(a) $2i$ (b) $-2i$
(c) 2 (d) 1

38. If $x+iy=\sqrt{\frac{a+ib}{c+id}}$, then the value of $(x^2+y^2)^2$ is
(a) $\frac{a^2+b^2}{c^2+d^2}$ (b) $\frac{c^2+d^2}{a^2+b^2}$
(c) $\frac{a^2-b^2}{c^2-d^2}$ (d) $\sqrt{\frac{a^2+b^2}{c^2+d^2}}$

39. If for the complex numbers z_1 and z_2, $|z_1+z_2|=|z_1-z_2|$, then $\arg z_1 \sim \arg z_2$ is equal to
(a) π (b) $\pi/2$
(c) $\pi/4$ (d) none of these

40. If n is a positive integer greater than unity and z is a complex number satisfying the equation $z^n=(z+1)^n$, then
(a) $\text{Re}(z)<0$ (b) $\text{Re}(z)>0$
(c) $\text{Re}(z)=0$ (d) none of these

41. The locus of z, if $\arg\left(\frac{z-1}{z+1}\right)=\frac{\pi}{2}$, is
(a) a circle (b) a semicircle
(c) a straight line (d) none of these

42. Suppose z_1, z_2, z_3 are the vertices of an equilateral triangle inscribed in the circle $|z|=2$. If $z_1=(1+i\sqrt{3})$ and z_1, z_2, z_3 are in the clockwise sense, then
(a) $z_2=1-i\sqrt{3}, z_3=-2$ (b) $z_2=2, z_3=1-i\sqrt{3}$
(c) $z_2=-1+i\sqrt{3}, z_3=-2$ (d) none of these

43. If $8iz^3+12z^2-18z+27i=0$, then
(a) $|z|=\frac{3}{2}$ (b) $|z|=\frac{2}{3}$
(c) $|z|=1$ (d) $|z|=\frac{3}{4}$

44. If p, q, r are position integers and ω is an imaginary cube root of unity and $f(x)=x^{3p}+x^{3q+1}+x^{3r+2}$, then $f(\omega)$ is equal to
(a) ω (b) $-\omega^2$
(c) 0 (d) none of these

45. If $\arg\left(\frac{z-1}{z+1}\right)=\frac{\pi}{4}$, then the locus of z is
(a) a straight line (b) a circle
(c) an ellipse (d) a hyperbola

46. Let z be a complex number and a be a real parameter such that $z^2+az+a^2=0$, then
(a) the locus of z is a pair of straight lines
(b) the locus of z is a circle
(c) $\arg(z)=\pm\frac{5\pi}{3}$
(d) $|z|=-2|a|$

47. If z is a point on the tangent at point z_1 of the circle $|z|=1$ and z_2 and z_3 are any two other points on the circle, then $\arg\left(\frac{z-z_1}{z_2-z_1}\right)$ is equal to

(a) $\arg\left(\frac{z_1}{z_2-z_1}\right)$ (b) $\frac{1}{2}\arg\left(\frac{z_1}{z_2}\right)$

(c) $\arg(z_1 z_2)$ (d) none of these

48. A complex number z is such that $\arg\left(\frac{z-2}{z+2}\right)=\frac{\pi}{3}$. The points representing this complex number will lie on

(a) an ellipse (b) a parabola

(c) a circle (d) a straight line

49. The smallest positive integer n for which $(1+i)^{2n} = (1+i)^{2n}$ is

(a) 4 (b) 8

(c) 2 (d) 12

50. Let A, B and C, respectively, represent the complex numbers z_1, z_2, z_3 on the complex plane. If the circumcentre of the triangle ABC lies at the origin, then the orthocentre is represented by the complex number

(a) $z_1+z_2-z_3$ (b) $z_2+z_3-z_1$

(c) $z_3+z_1-z_2$ (d) $z_1+z_2+z_3$

51. If $a \neq 1$ is any nth root of unity, then $S = 1+3\alpha+5\alpha^2+\cdots$ up to n terms, is equal to

(a) $\frac{2n}{1-\alpha}$ (b) $-\frac{2n}{1-\alpha}$

(c) $\frac{n}{1-\alpha}$ (d) $-\frac{n}{1-\alpha}$

52. If $\frac{3}{2+\cos\theta+i\sin\theta}=a+ib$, then $(a-2)^2+b^2$ is equal to

(a) 0 (b) 1

(c) –1 (d) 2

53. Complex numbers z_1, z_2 and z_3 are associated with vertices A, B and C of a triangle. D, E and F are the mid-points of sides AB, BC and CA respectively. If $I(z_4)$ and $I'(z_5)$ are the incentres of triangles ABC and DEF respectively and $G(z_6)$ is the centroid of the triangle ABC, then

(a) $2z_4+z_5=3z_6$ (b) $2z_5+z_4=3z_6$

(c) $z_4+z_5=2z_6$ (d) none of these

54. If $3^{49}(x+iy)=\left(\frac{3}{2}+i\frac{\sqrt{3}}{2}\right)^{100}$ and $x=ky$, then k is

(a) $-\frac{1}{3}$ (b) $\sqrt{3}$

(c) $-\sqrt{3}$ (d) $-\frac{1}{\sqrt{3}}$

55. If z_1, z_2, z_3, z_4 are the four complex numbers represented by the vertices of a quadrilateral taken in order such that $z_1-z_4=z_2-z_3$ and $\arg\left(\frac{z_4-z_1}{z_2-z_1}\right)=\frac{\pi}{2}$, then the quadrilateral is

(a) a square (b) a rhombus

(c) a rectangle (d) none of these

56. Let $A_0A_1A_2A_3A_4A_5$ be a regular hexagon inscribed in a circle of unit radius. Then the product of the lengths of the line segments A_0A_1, A_0A_2 and A_0A_4 is

(a) $\frac{3}{4}$ (b) $3\sqrt{3}$

(c) 3 (d) $\frac{3\sqrt{3}}{2}$

57. If $x=a+b$, $y=a\alpha+b\beta$ and $z=a\beta+b\alpha$, where α and β are complex cube roots of unity, then xyz is equal to

(a) a^2+b^2 (b) a^3+b^3

(c) a^3b^3 (d) a^3-b^3

58. If 1, α, α^2, …, α^{n-1} are the nth roots of unity, then $\sum_{k=1}^{n-1}\frac{1}{2-\alpha^k}$ is equal to

(a) $(n-2)2^n$ (b) $\frac{(n-2)2^{n-1}+1}{2^n-1}$

(c) $\frac{(n-2)2^{n-1}}{2^n-1}$ (d) none of these

59. In Fig. 2.46 if O, P and Q represent the origin and the complex numbers z and $z+iz$, respectively, then the value of $\angle OPQ$ is

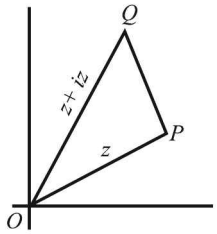


Fig. 2.46

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$

60. $e^{2mi\cot^{-1}p}\cdot\left(\frac{pi+1}{pi-1}\right)^m$ is equal to

(a) 0 (b) 1

(c) –1 (d) none of these

61. If $a = \cos\theta + i\sin\theta$, then $\dfrac{1+a}{1-a}$ is equal to

(a) $\cot\dfrac{\theta}{2}$ (b) $\cot\theta$

(c) $i\cot\dfrac{\theta}{2}$ (d) $i\tan\dfrac{\theta}{2}$

62. If $\dfrac{(a+ib)^2}{a-ib} - \dfrac{(a-ib)^2}{a+ib} = x+iy$, then x is equal to

(a) $\dfrac{-2b^3}{(a^2+b^2)^2}$ (b) $\dfrac{6a^2b}{(a^2+b^2)^2}$

(c) 0 (d) none of these

63. If α, β, γ are the cube roots of $p < 0$, then for any x, y, z $\dfrac{\alpha^2x^2+\beta^2y^2+\gamma^2z^2}{\beta^2x^2+\gamma^2y^2+\alpha^2z^2}$ is

(a) 1 (b) $\dfrac{\alpha}{\gamma}$

(c) $\dfrac{\beta}{\alpha}$ (d) $\dfrac{\gamma}{\beta}$

64. PQ and PR are two infinite rays and QAR is an arc. The point lying in the shaded region excluding the boundary, satisfies.

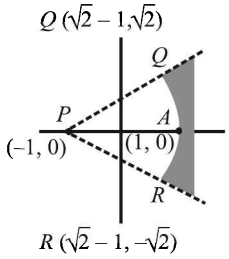


Fig. 2.47

(a) $|z-1|>2 : |\arg(z-1)| < \dfrac{\pi}{4}$

(b) $|z-1|>2 : |\arg(z-1)| < \dfrac{\pi}{2}$

(c) $|z-1|>2 : |\arg(z+1)| < \dfrac{\pi}{4}$

(d) $|z-1|>2 : |\arg(z+1)| < \dfrac{\pi}{2}$

65. If $x+iy = \dfrac{3}{2+\cos\theta+i\sin\theta}$, then x^2+y^2 is equal to

(a) $3x-4$ (b) $4x-3$

(c) $4x+3$ (d) none of these

EXERCISE SET 2

1. $\dfrac{1-ix}{1+ix} = a-ib$ and $a^2+b^2=1$, where a and b are real, then x is equal to

(a) $\dfrac{2a}{(1+a)^2+b^2}$ (b) $\dfrac{2b}{(1+a)^2+b^2}$

(c) $\dfrac{2a}{(1+b)^2+a^2}$ (d) $\dfrac{2b}{(1+b)^2+a^2}$

2. k non-overlapping regular polygons of $p_1, p_2, \ldots, p_k$ sides have one common vertex so that no gap is left at that vertex. Then $\dfrac{1}{p_1}+\dfrac{1}{p_2}+\cdots+\dfrac{1}{p_k}$ is equal to

(a) $\dfrac{k}{2}$ (b) $\dfrac{k}{2}-1$

(c) $1+\dfrac{k}{2}$ (d) $\dfrac{k+1}{2}$

3. The polar form of the complex number $1+\sqrt{3}i$ is

(a) $\cos\dfrac{\pi}{3}$ (b) $\sin\dfrac{\pi}{3}$

(c) $\cos\dfrac{\pi}{3}+i\sin\dfrac{\pi}{3}$ (d) $2\left(\cos\dfrac{\pi}{3}+i\sin\dfrac{\pi}{3}\right)$

4. If ω is a complex cube root of unity, then for positive integral values of n, the product $\omega\cdot\omega^2\cdot\omega^3\cdot\ldots\cdot\omega^n$ will be

(a) $\dfrac{1-i\sqrt{3}}{2}$ (b) $\dfrac{-1+i\sqrt{3}}{2}$

(c) 1 (d) both (b) and (c)

5. If $\dfrac{(a+i)^2}{2a-1} = p+iq$, then p^2+q^2 is equal to

(a) $\dfrac{(a^2+1)^2}{2a^2+1}$ (b) $\dfrac{(a^2+1)^2}{2a^2-1}$

(c) $\dfrac{(a^2+1)^2}{4a^2-1}$ (d) none of these

6. Let z_1 and z_2 be two non-real complex cube roots of unity and $|z-z_1|^2+|z-z_2|^2=\lambda$ be the equation of a circle with z_1, z_2 as the ends of a diameter, then the value of λ is

(a) 4 (b) 3

(c) 2 (d) $\sqrt{2}$

7. The polar form of the complex number $\frac{1+7i}{(2-i)^2}$ is

(a) $\sqrt{2}\left(\cos\frac{2\pi}{4}+i\sin\frac{2\pi}{4}\right)$

(b) $\sqrt{2}\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$

(c) $\sqrt{2}\left(\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}\right)$

(d) $\sqrt{2}\left(\cos\frac{3\pi}{4}+i\sin\frac{\pi}{4}\right)$

8. If ω is a complex root of the equation $z^3 = 1$, then $\omega+\omega^{\left(\frac{1}{2}+\frac{3}{8}+\frac{9}{32}+\frac{27}{128}+\cdots\right)}$ is equal to

(a) -1 (b) 0

(c) 9 (d) i

9. $\left(\frac{1+i\sin\frac{\pi}{8}+\cos\frac{\pi}{8}}{1-i\sin\frac{\pi}{8}+\cos\frac{\pi}{8}}\right)^8$ is equal to

(a) 2^8 (b) 0

(c) -1 (d) 1

10. If z_1, z_2, z_3, z_4 are the vertices of a quadrilateral taken in order such that $z_1 + z_3 = z_2 + z_4$ and $|z_1 - z_3| = |z_2 - z_4|$, then the $\arg\left(\frac{z_1-z_2}{z_3-z_2}\right)$ is equal to

(a) $\frac{\pi}{2}$ (b) $\pm\frac{\pi}{2}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

11. If n_1 and n_2 are positive integers, then

$$(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2} + (1+i^7)^{n_2}$$

is a real number if and only if

(a) $n_1 = n_2 + 1$

(b) $n_1 + 1 = n_2$

(c) $n_1 = n_2$

(d) n_1 and n_2 are any two positive integers

12. If α, β, γ and a, b, c are complex numbers such that $\frac{\alpha}{a}+\frac{\beta}{b}+\frac{\gamma}{c}=1+i$ and $\frac{a}{\alpha}+\frac{b}{\beta}+\frac{c}{\gamma}=0$, then the value of $\frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2}+\frac{\gamma^2}{c^2}$ is equal to

(a) 0 (b) -1

(c) $2i$ (d) $-2i$

13. If $-1+\sqrt{-3} = re^{i\theta}$, then the value of θ is

(a) $2\pi/3$ (b) $-(2\pi/3)$

(c) $\pi/3$ (d) $-(\pi/3)$

14. If $\omega_n = \cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}$, then $(x+y\omega_3+z\omega_3^2)(x+y\omega_3^2+z\omega_3)$ is equal to

(a) 0

(b) $x^2+y^2+z^2$

(c) $x^2+y^2+z^2-yz-zx-xy$

(d) $x^2+y^2+z^2+yz+zx+xy$

15. If $z_r = \cos\frac{r\alpha}{n^2}+i\sin\frac{r\alpha}{n^2}$, where $r = 1, 2, 3, \ldots, n$, then $\lim_{n\to\infty}(z_1 z_2 z_3 \ldots z_n)$ is equal to

(a) $\cos\alpha + i\sin\alpha$ (b) $\cos\left(\frac{\alpha}{2}\right)-i\sin\left(\frac{\alpha}{2}\right)$

(c) $e^{i\alpha/2}$ (d) $\sqrt[3]{e^{i\alpha}}$

16. If the cube roots of unity are $1, \omega, \omega^2$, then the roots of the equation $x^3 - 3x^2 + 3x + 63 = 0$ are

(a) $-3, 1-3\omega, 1+3\omega^2$ (b) $-4, 1-4\omega, 1-4\omega^2$

(c) $-3, 1-4\omega, 1-4\omega^2$ (d) $-3, 1+4\omega, 1+4\omega^2$

17. If $\frac{\tan\alpha - i\left(\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2}\right)}{1+2i\sin\frac{\alpha}{2}}$ is purely imaginary, then α is given by

(a) $n\pi+\frac{\pi}{4}$ (b) $n\pi-\frac{\pi}{4}$

(c) $(2n+1)\pi$ (d) $2n\pi+\frac{\pi}{4}$

18. The inequality $|z-4| < |z-2|$ represents the region given by

(a) $\mathrm{Re}(z) > 0$ (b) $\mathrm{Re}(z) < 0$

(c) $\mathrm{Re}(z) > 3$ (d) none of these

19. If the roots of $(z-1)^n = i(z+1)^n$ are plotted in the Argand plane, they are

(a) on a parabola (b) concyclic

(c) collinear (d) the vertices of a triangle

20. If z_1 and z_2 are two non-zero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$, then the value of $\arg(z_1) - \arg(z_2)$ is

(a) 0 (b) $-\pi$

(c) $\frac{\pi}{2}$ (d) $\frac{-\pi}{2}$

21. The imaginary part of
$(z-1)(\cos\alpha - i\sin\alpha) + (z-1)^{-1}(\cos\alpha + i\sin\alpha)$
is zero, if
(a) $|z-1|=2$ (b) $\arg(z-1)=2\alpha$
(c) $\arg(z-1)=\alpha$ (d) $|z|=1$

22. If $1, \omega, \omega^2, \ldots, \omega^{n-1}$ are the n, nth roots of unity, then $(7-\omega)(7-\omega^2)\cdots(7-\omega^{n-1})$ is equal to
(a) $\frac{7^n+1}{6}$ (b) $\frac{5^n-1}{6}$
(c) $\frac{7^n-1}{6}$ (d) none of these

23. If z is a complex number such that $|2z-5|+|2z+5|=10$, then the locus of z is
(a) a circle (b) a straight line
(c) a parabola (d) none of these

24. The complex numbers $z=x+iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right|=1$ will lie on
(a) the x-axis
(b) the line $y=5$
(c) a circle through the origin
(d) none of these

25. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1-z_3}{z_2-z_3}=\frac{1-i\sqrt{3}}{2}$ are the vertices of a triangle
(a) which is of zero area
(b) which is right-angled isosceles
(c) which is equilateral
(d) which is obtuse-angled isosceles

26. If $x_r = \cos\frac{\pi}{2^r} + i\sin\frac{\pi}{2^r}$, then the value of $x_1 \times x_2 \times x_3 \times \ldots \times \infty$ is
(a) 0 (b) 1
(c) π (d) -1

27. If $a=\cos\alpha + i\sin\alpha$, $b=\cos\beta + i\sin\beta$, $c=\cos\gamma + i\sin\gamma$ and $\frac{b}{c}+\frac{c}{a}+\frac{a}{b}=1$, then
$\cos(\beta-\gamma)+\cos(\gamma-\beta\alpha)+\cos(\alpha-\beta)$ is equal to
(a) 3/2 (b) $-(3/2)$
(c) 0 (d) 1

28. Let y be a set of all complex numbers z such that $|z|=1$ and the defined relation R on y by $z_1 R z_2$ is
$|\arg z_1 - \arg z_2| = \frac{2\pi}{3}$, then R is
(a) symmetric (b) transitive
(c) anti-symmetric (d) reflexive

29. If z_1, z_2, z_3 are the vertices of an equilateral triangle with z_0 as its centroid, then $z_1^2+z_2^2+z_3^2$ is equal to
(a) z_0^2 (b) $9z_0^2$
(c) $3z_0^2$ (d) $2z_0^2$

30. The locus represented by $|z-1|=|z+i|$ is
(a) a circle of radius 1
(b) an ellipse with foci at 1 and $-i$
(c) a line through the origin
(d) a circle on the join of 1 and $-i$ as a diameter

31. For all complex numbers z_1, z_2 satisfying $|z_1|=12$ and $|z_2-3-4i|=5$, the minimum value of $|z_1-z_2|$ is
(a) 0 (b) 2
(c) 7 (d) 17

32. If n and x are two real numbers, then $e^{2ni\cot^{-1}x}\left(\frac{xi+1}{xi-1}\right)^n$ is equal to
(a) $(\cos x + i\sin x)$ (b) $\frac{1}{2}n$
(c) 1 (d) $\frac{1}{2}(n+1)$

33. The maximum value of $|z|$ where z satisfies the condition $|z+(2/z)|=2$ is
(a) $\sqrt{3}-1$ (b) $\sqrt{3}+1$
(c) $\sqrt{3}$ (d) $\sqrt{2}+\sqrt{3}$

34. If a, b, c are the roots $y^3-3y^2+3y+26=0$ and ω is the cube root of unity, then the value of $\frac{a-1}{b-1}+\frac{b-1}{c-1}+\frac{c-1}{a-1}$ is equal to
(a) $-3\omega^2$ (b) $3\omega^2$
(c) ω^2 (d) $2\omega^2$

35. If α, β are two complex numbers, then $|\alpha|^2+|\beta|^2$ is equal to
(a) $\frac{1}{2}\left(|\alpha+\beta|^2-|\alpha-\beta|^2\right)$ (b) $\frac{1}{2}\left(|\alpha+\beta|^2+|\alpha-\beta|^2\right)$
(c) $|\alpha+\beta|^2+|\alpha-\beta|^2$ (d) none of these

36. If the imaginary part of $\frac{2z+1}{iz+1}$ is -2, then the locus of the point representing z in the complex plane is
(a) a circle (b) a straight line
(c) a parabola (d) none of these

37. If $x+(1/x)=2\cos\theta$, then the value of x is
(a) $\cos\theta \pm i\sin\theta$ (b) $\cos 2\theta + i\sin 2\theta$
(c) $(\cos\theta + i\sin\theta)^2$ (d) $\cos 4\theta - i\sin 4\theta$

38. If $x+(1/x)=1$, $p=x^{4000}+(1/x^{4000})$ and q is the digit at the unit's place in the number $(2^{2^n}+1), n$ being a natural number greater than 1, then the value of $(p+q)$ is
(a) 8 (b) 6
(c) 7 (d) none of these

39. Let z be a complex number satisfying $|z-5i|\le 1$ such that arg(z) is minimum. Then z is equal to
(a) $\frac{2\sqrt{6}}{5}+\frac{24i}{5}$ (b) $\frac{24}{5}+\frac{2\sqrt{6}i}{5}$
(c) $\frac{2\sqrt{6}}{5}-\frac{24i}{5}$ (d) none of these

40. Let $A=\frac{2}{\sqrt{3}}e^{-i(\pi/6)}$, $B=\frac{2}{\sqrt{3}}e^{i(\pi/2)}$, $C=\frac{2}{\sqrt{3}}e^{-(5\pi/6)}$, be three points forming a triangle ABC in the Gaussian plane, then the triangle ABC is
(a) equiangular (b) isosceles
(c) scalene (d) none of these

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) or (d) given below.
(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1
(c) Statement 1 is true; Statement 2 is false
(d) Statement 1 is false; Statement 2 is true

41. **Statement 1:** The equation $z\bar{z}+\bar{a}z+a\bar{z}+\lambda=0$, where a is a complex number, represents a circle in the Argand plane, if λ is real.
Statement 2: The radius of the circle $z\bar{z}+\bar{a}z+a\bar{z}+\lambda=0$ is $\sqrt{a\bar{a}-\lambda}$.

42. **Statement 1:** The locus of the centre of a circle which touches the circles $|z-z_1|=a$ and $|z-z_2|=b$ externally (z, z_1 and z_2 are complex numbers) will be a hyperbola.
Statement 2: $|z-z_1|-|z-z_2|<|z_2-z_1|$
$\Rightarrow$ z lies on hyperbola.

43. **Statement 1:** If $|z|<\sqrt{2}-1$, then $|z^2+2z\sin\alpha|<1$.
Statement 2: $|z_1+z_2|<|z_1|+|z_2|$ and $|\sin\alpha|\le 1$.

44. **Statement 1:** If $z=\sqrt{(5+12i)}+\sqrt{(12i-5)}$, then the principal values of arg(z) are $\pm\frac{\pi}{4}, \pm\frac{3\pi}{4}$, where $i=\sqrt{-1}$.
Statement 2: If $z=a+ib$, then

$$\sqrt{z}=\pm\left\{\sqrt{\left(\frac{|z|+a}{2}\right)}+i\sqrt{\left(\frac{|z|-a}{2}\right)}\right\}, \text{ for } b>0$$

and $$\sqrt{z}=\pm\left\{\sqrt{\left(\frac{|z|+a}{2}\right)}-i\sqrt{\left(\frac{|z|-a}{2}\right)}\right\}, \text{ for } b<0.$$

Previous Years' Questions

45. If z and ω are two nonzero complex numbers, such that $|z|=|\omega|$ and arg(z) + arg(ω) = π, then z is equal to
(a) $\bar{\omega}$ (b) $-\bar{\omega}$
(c) ω (d) $-\omega$
[AIEEE 2002]

46. The locus of the centre of a circle which touches the circles $|z-z_1|=a$ and $|z-z_2|=b$ externally (where z_1, z_2, $\in \mathbb{C}$) will be
(a) an ellipse (b) a hyperbola
(c) a circle (d) none of these
[AIEEE 2002]

47. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x+iy$, then
(a) $x=3, y=1$ (b) $x=1, y=3$
(c) $x=0, y=3$ (d) $x=0, y=0$
[AIEEE 2002]

48. If $\omega\neq 1$ is a cube root of unity, then
$\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -1+\omega-i & -1 \end{vmatrix}$ is equal to
(a) 0 (b) 1
(c) i (d) ω
[AIEEE 2002]

49. If z and ω are two nonzero complex numbers such that $|z\omega|=1$ and arg(z) – arg(ω) = $\pi/2$, then $\bar{z}\omega$ is equal to
(a) 1 (b) –1
(c) i (d) $-i$
[AIEEE 2003]

50. If $\left(\frac{1+i}{1-i}\right)^x = i$, then
(a) $x=2n,\ n\in Z^+$ (b) $x=4n+1,\ n\in Z^+$
(c) $x=2n+1,\ n\in Z^+$ (d) $x=4n,\ n\in Z^+$
[AIEEE 2003]

51. Let z_1 and z_2 be the two roots of the equation $z^2+az+b=0$, where z is a complex number, and let us assume that the origin, z_1 and z_2 form an equilateral triangle, then
(a) $a^2=2b$ (b) $a^2=3b$
(c) $a^2=4b$ (d) $a^2=b$
[AIEEE 2003]

52. Let z, ω be complex numbers such that $\bar{z} + i\bar{\omega} = 0$ and $\arg(z\omega) = \pi$. Then $\arg(z)$ is equal to
(a) $\pi/4$ (b) $\pi/2$ (c) $3\pi/4$ (d) $5\pi/4$
[AIEEE 2004]

53. If $z = x - iy$ and $z^{1/3} = p + iq$, then $\left(\frac{x}{p} + \frac{y}{q}\right)\Big/(p^2 + q^2)$ is equal to
(a) 1 (b) -1 (c) 2 (d) -2
[AIEEE 2004]

54. If $|z^2 - 1| = |z|^2 + 1$, then z lies on
(a) a circle (b) the imaginary axis
(c) the real axis (d) an ellipse
[AIEEE 2004]

55. If the cube roots of unity are 1, ω, ω^2, then the roots of the equation $(x-1)^3 + 8 = 0$ are
(a) $-1, -1, -1$ (b) $-1, -1 + 2\omega, -1 - 2\omega^2$
(c) $-1, 1 + 2\omega, 1 + 2\omega^2$ (d) $-1, 1 - 2\omega, 1 - 2\omega^2$
[AIEEE 2005]

56. $\sin^{-1}\left[\frac{1}{i}(z-1)\right]$, where z is non-real, can be the angle of a triangle , if
(a) $\text{Re}(z) = 1$, $\text{Im}(z) = 2$
(b) $\text{Re}(z) = 1$, $-1 \le \text{Im}(z) \le 1$
(c) $\text{Re}(z) + \text{Im}(z) = 0$
(d) none of these
[AIEEE 2005]

57. If $\omega = \frac{z}{z - (1/3)i}$ and $|\omega| = 1$, then z lies on
(a) a parabola (b) an ellipse
(c) a straight line (d) a circle
[AIEEE 2005]

58. The value of $\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$ is
(a) i (b) 1 (c) -1 (d) $-i$
[AIEEE 2006]

59. If $z^2 + z + 1 = 0$, where z is a complex number, then the value of $\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \cdots + \left(z^6 + \frac{1}{z^6}\right)^2$ is
(a) 18 (b) 54 (c) 6 (d) 12
[AIEEE 2006]

60. If $|z + 4| \le 3$, then the maximum value of $|z + 1|$ is
(a) 6 (b) 0 (c) 4 (d) 10
[AIEEE 2007]

61. If the conjugate of a complex number is $1/(i-1)$, then that complex number is
(a) $1/(i+1)$ (b) $-[1/(i+1)]$
(c) $1/(i-1)$ (d) $-[1/(i-1)]$
[AIEEE 2008]

62. If $\left|z - \frac{4}{z}\right| = 2$, then the maximum value of $|z|$ is equal to
(a) $\sqrt{5} + 1$ (b) 2 (c) $2 + \sqrt{2}$ (d) $\sqrt{3} + 1$
[AIEEE 2009]

63. Which of the following is not applicable for a complex number?
(a) addition (b) subtraction
(c) division (d) inequality
[Assam J.E.E. 1998, Delhi C.E.E. 1999]

64. If $z = x + iy$, $z^{1/3} = a - ib$ and $\frac{x}{a} - \frac{y}{b} = k(a^2 - b^2)$, then the value of k is equal to
(a) 2 (b) 4 (c) 6 (d) 1
[DCE 2005]

65. The complex numbers $\sin x + i\cos 2x$ and $\cos x - i\sin 2x$ are conjugate to each other for
(a) $x = n\pi$ (b) $x = 0$
(c) $x = \left(n + \frac{1}{2}\right)\pi$ (d) no value of x
[IIT 1988]

66. If $\log_{1/2}\frac{|z|^2 + 2|z| + 4}{2|z|^2 + 1} < 0$, then the region traced by z is
(a) $|z| < 3$ (b) $1 < |z| < 3$
(c) $|z| > 1$ (d) $|z| < 2$
[DCE 2005]

67. If $e^{i\theta} = \cos\theta + i\sin\theta$, then for the ΔABC, $e^{iA} \cdot e^{iB} \cdot e^{iC}$ is
(a) $-i$ (b) 1
(c) -1 (d) none of these
[IIT 1981]

68. If α, β, γ are the roots of $x^3 - 3x^2 + 3x + 7 = 0$ (ω is the cube root of unity), then $\frac{\alpha - 1}{\beta - 1} + \frac{\beta - 1}{\gamma - 1} + \frac{\gamma - 1}{\alpha - 1}$ is
(a) $3/\omega$ (b) ω^2 (c) $2\omega^2$ (d) $3\omega^2$
[DCE 2002]

69. If $z \ne 1$ and $z^2/(z-1)$ is real, then the point represented by the complex number z lies
(a) either on the real axis or on a circle passing through the origin
(b) on a circle with centre at the origin
(c) either on the real axis or on a circle not passing through the origin
(d) on the imaginary axis
[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (b)	**2.** (b)	**3.** (c)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (b)	**8.** (a)	**9.** (b)	**10.** (b)
11. (b)	**12.** (c)	**13.** (b)	**14.** (b)	**15.** (c)	**16.** (a)	**17.** (b)	**18.** (d)	**19.** (b)	**20.** (c)
21. (a)	**22.** (a)	**23.** (d)	**24.** (c)	**25.** (d)	**26.** (d)	**27.** (b)	**28.** (b)	**29.** (a)	**30.** (c)
31. (d)	**32.** (b)	**33.** (a)	**34.** (c)	**35.** (a)	**36.** (a)	**37.** (a)	**38.** (a)	**39.** (b)	**40.** (a)
41. (b)	**42.** (a)	**43.** (a)	**44.** (c)	**45.** (b)	**46.** (a)	**47.** (b)	**48.** (c)	**49.** (c)	**50.** (d)
51. (b)	**52.** (b)	**53.** (b)	**54.** (d)	**55.** (c)	**56.** (c)	**57.** (b)	**58.** (b)	**59.** (c)	**60.** (b)
61. (c)	**62.** (c)	**63.** (d)	**64.** (c)	**65.** (b)					

Exercise Set 2

1. (b)	**2.** (b)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (b)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (b)
11. (a)	**12.** (c)	**13.** (a)	**14.** (c)	**15.** (c)	**16.** (c)	**17.** (a)	**18.** (c)	**19.** (c)	**20.** (a)
21. (c)	**22.** (c)	**23.** (d)	**24.** (a)	**25.** (c)	**26.** (d)	**27.** (d)	**28.** (a)	**29.** (c)	**30.** (c)
31. (b)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (b)	**36.** (b)	**37.** (a)	**38.** (b)	**39.** (a)	**40.** (a)
41. (a)	**42.** (d)	**43.** (a)	**44.** (b)	**45.** (b)	**46.** (b)	**47.** (d)	**48.** (a)	**49.** (d)	**50.** (d)
51. (b)	**52.** (c)	**53.** (d)	**54.** (b)	**55.** (d)	**56.** (b)	**57.** (c)	**58.** (d)	**59.** (d)	**60.** (a)
61. (b)	**62.** (a)	**63.** (d)	**64.** (b)	**65.** (d)	**66.** (a)	**67.** (c)	**68.** (a)	**69.** (a)	

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Adding and subtracting $a\bar{a}$ on both sides, the given relation becomes

$$(z+a)(\bar{z}+\bar{a})+b=a\bar{a}$$

or $(z+a)\overline{(z+a)}=|a|^2-b$

or $|z+a|^2=|a|^2-b$

The above equation represents a circle centred at $-a$ provided $|a|^2-b>0$ (i.e. $|a^2| > b$) as its radius is $\sqrt{|a|^2-b}$.

Note: If $|a|^2=b$ then it will be a point circle as its radius will be zero.

2. $\left(\frac{c+i}{c-i}\right)\left(\frac{c+i}{c+i}\right)=a+ib$

$$\Rightarrow \quad \frac{(c^2-1)+2ic}{c^2+1}=a+ib$$

$$\therefore \quad a=\frac{c^2-1}{c^2+1},\ b=\frac{2c}{c^2+1}$$

and $a^2+b^2=\frac{1}{(c^2+1)^2}(c^4-2c^2+1+4c^2)$

$$=\frac{(c^2+1)^2}{(c^2+1)^2}=1$$

3. We have

$$\left(\frac{3+2i\sin\theta}{1-2i\sin\theta}\right)=\frac{(3+2i\sin\theta)(1+2i\sin\theta)}{(1-2i\sin\theta)(1+2i\sin\theta)}$$

$$=\left(\frac{3-4\sin^2\theta}{1+4\sin^2\theta}\right)+i\left(\frac{8\sin\theta}{1+4\sin^2\theta}\right)$$

For this expression to be real, Im(z) should be zero.

i.e. $\frac{8\sin\theta}{1+4\sin^2\theta}=0 \quad \Rightarrow \quad \sin\theta=0$

$\therefore \quad \theta=n\pi$, where $n = 0, 1, 2, 3,\ldots$

4. Since 1, α_1, $\alpha_2,\ldots,$ α_{3n} are the roots of the equation $x^{3n+1} - 1 = 0$, according to the factor theorem,

$$x^{3n+1}-1=(x-1)(x-\alpha_1)\ldots(x-\alpha_{3n})$$

$$\Rightarrow \quad (x-\alpha_1)\cdots(x-\alpha_{3n})=\frac{x^{3n+1}-1}{x-1}$$

$\Rightarrow \quad (x-\alpha_1)\ldots(x-\alpha_{3n}) = 1 + x + x^2 + \cdots + x^{3n}$

Now, let $E = \dfrac{(\omega^2-\alpha_1)(\omega^2-\alpha_2)\ldots(\omega^2-\alpha_{3n})}{(\omega-\alpha_1)(\omega-\alpha_2)\ldots(\omega-\alpha_{3n})}$

$\Rightarrow \qquad E = \dfrac{\omega^{6n+2}-1}{\omega^2-1}\cdot\dfrac{\omega-1}{\omega^{3n+1}-1}$

$\therefore \qquad E = \dfrac{(\omega^2-1)(\omega-1)}{(\omega^2-1)(\omega-1)} = 1$

5. The given complex equation is

$$(1+i)(1+2i)(1+3i)\ldots(1+ni) = x+iy$$

Taking modulus of both sides, we get

$$|(1+i)(1+2i)(1+3i)\ldots(1+ni)| = |x+iy|$$

or $\qquad \sqrt{2}\sqrt{5}\sqrt{10}\ldots\sqrt{(1+n^2)} = \sqrt{x^2+y^2}$

or $\qquad 2\times5\times10\times\cdots\times(1+n^2) = x^2+y^2.$

6. $\dfrac{1+i}{1-i} = \dfrac{\sqrt{2}e^{\pi i/4}}{\sqrt{2}e^{-(\pi i/4)}} = e^{\pi i/2} = \cos\dfrac{\pi}{2} + i\sin\dfrac{\pi}{2} = i$

$\therefore \quad \left(\dfrac{1+i}{1-i}\right)^x = (i)^x = 1 \text{ (given)}$

The above relation is possible if $x = 4n$, where n is a +ve integer.

7. Being the roots of unity, 1, α_1, α_2, …, α_{n-1} are the roots of $x^n - 1 = 0$, so that

$$x^n - 1 = (x-1)(x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_{n-1})$$

$\Rightarrow \qquad (x-\alpha_1)(x-\alpha_2)\ldots(x-a_{n-1}) = \dfrac{x^n-1}{x-1}$

$\Rightarrow \quad (x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_{n-1}) = x^{n-1}+x^{n-2}+\cdots+x+1$

Putting $x = 1$ in the above equation, we get

$$(1-\alpha_1)(1-\alpha_2)\ldots(1-\alpha_{n-1}) = \underbrace{1+1+\ldots+1+1}_{n \text{ times}} = n.$$

8. We know that

$$|-z| = |z|$$

and $\qquad |z_1+z_2| \le |z_1| + |z_2|$

Now $\quad |z| + |z-1| = |z| + |1-z| \ge |z+(1-z)| = |1| = 1$

Hence, the minimum value of $|z| + |z-1|$ is 1.

9. $\left|z^n\sin\theta_0 + z^{n-1}\sin\theta_1 + z^{n-2}\sin\theta_2 + \cdots + z\sin\theta_{n-1} + \sin\theta_n\right| = 2$

According to triangle inequality, we have

$$\left|z^n\sin\theta_0\right| + \left|z^{n-1}\sin\theta_1\right| + \left|z^{n-2}\sin\theta_2\right| + \cdots + \left|\sin\theta_n\right| \ge 2$$

$\Rightarrow \qquad 2 \le |\sin\theta_n| + |z||\sin\theta_{n-1}| + \cdots + |z|^n|\sin\theta_0|$

Now, let us assume that the roots lie inside the circle $|z| = 1/2$.

Then, $\quad 2 \le 1 + \dfrac{1}{2} + \left(\dfrac{1}{2}\right)^2 + \cdots + \left(\dfrac{1}{2}\right)^n = 1\left\{\dfrac{1-(1/2)^{n-1}}{1-(1/2)}\right\}$

$\Rightarrow \qquad 2 \le 2\left(1 - \dfrac{1}{2^{n-1}}\right)$

which is not possible. Hence all of the roots of the given equation must lie outside the circle $|z| = 1/2$.

10. Since the reciprocal $(1/z)$ and the conjugate $(\bar{z})$ of the given complex number are equal, we have $\bar{z} = (1/z)$

or $\qquad z\bar{z} = 1 \quad$ or $\quad (x+iy)(x-iy) = 1$

or $\qquad x^2 - i^2y^2 = 1 \quad$ or $\quad x^2 + y^2 = 1$

11. Since the sum of any four consecutive powers of i is zero,

$$\therefore \quad \sum_{n=1}^{13}(i^n + i^{n+1}) = \sum_{n=1}^{13} i^n + \sum_{n=1}^{13} i^{n+1}$$

$$= (i + i^2 + i^3 + \cdots + i^{13}) + (i^2 + i^3 + i^4 + \cdots + i^{14})$$

$$= (0 + i) + (0 + i^2)$$

$$= i - 1$$

12. $p^{1/3} = (-p)^{1/3}(-1)^{1/3}$

$$= -(-p)^{1/3} - (-p)^{1/3}\omega - (-p)^{1/3}\omega^2$$

$$= -q - q\omega - q\omega^2, \text{ where } q = (-p)^{1/3}$$

Let $\alpha = -q, \beta = -q\omega, \gamma = -q\omega^2$

$$\therefore \quad \frac{x\alpha + y\beta + z\gamma}{x\beta + y\gamma + z\alpha} = \frac{-q(x + y\omega + z\omega^2)}{-q(x\omega + y\omega^2 + z)}$$

$$= \frac{x\omega^3 + y\omega^4 + z\omega^2}{x\omega + y\omega^2 + z} = \omega^2$$

13. $\dfrac{1-i}{1+i} = \dfrac{(1-i)^2}{1-i^2} = \dfrac{-2i}{2} = -i$

$\therefore \quad \left(\dfrac{1-i}{1+i}\right)^n = (-i)^n = -i, i, -i$ for $n = 1, 3, 5$ respectively.

Hence $n = 3$ is the correct answer.

14. As we know, the equation of the form $\left|\dfrac{z-2}{z+2}\right| = n$ is a circle, if $n \ne 1$.

15. $\alpha + \beta + \gamma = (1 + \omega + \omega^2)z_1 + (1 + \omega + \omega^2)z_2$

$$= 0 + 0 = 0 \qquad \{\because 1 + \omega + \omega^2 = 0\}$$

$$\therefore \qquad \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

or $\alpha^3 + \beta^3 + \gamma^3 = 3(z_1 + z_2)(\omega z_1 + \omega^2 z_2)(\omega^2 z_1 + \omega z_2)$

or $\qquad \alpha^3 + \beta^3 + \gamma^3 = 3(z_1 + z_2)(z_1^2 - z_1z_2 + z_2^2)$

$$\therefore \qquad \alpha^3 + \beta^3 + \gamma^3 = 3(z_1^3 + z_2^3)$$

and $\qquad \alpha\beta\gamma = z_1^3 + z_2^3$

Also, $\alpha^2 + \beta^2 + \gamma^2 = 6z_1z_2$

$\therefore \quad \alpha\beta + \beta\gamma + \gamma\alpha = -3z_1z_2.$

Hence $\alpha^3 + \beta^3 + \gamma^3 = 3(z_1^3 + z_2^3)$ is the correct choice.

16. Given: Complex numbers z_1, z_2 and z_3, and $|z_1| = |z_2| = |z_3|$

$$= \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1.$$

Since $|z_1| = |z_2| = |z_3| = 1$, therefore, $z_1\bar{z}_1 = z_2\bar{z}_2 = z_3\bar{z}_3 = 1$.

We know that $|z_1| = |\bar{z}_1|$; $|z_2| = |\bar{z}_2|$ and $|z_3| = |\bar{z}_3|$.

$$\therefore \quad |z_1 + z_2 + z_3| = \left|\overline{z_1 + z_2 + z_3}\right|$$
$$= |\bar{z}_1 + \bar{z}_2 + \bar{z}_3|$$
$$= \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$$

17. Putting $x = 1$, ω, ω^2 respectively, we get

$$3^n = a_0 + a_1 + a_2 + \cdots + a_{2n}$$
$$(1 + \omega + \omega^2)^n = a_0 + a_1\omega + a_2\omega^2 + \cdots + a_{2n}\omega^{2n}$$
$$(1 + \omega^2 + \omega^4)^n = a_0 + a_1\omega^2 + a_2\omega^4 + \ldots + a_{2n}\omega^{4n}$$

Adding these, we get

$$3^n + (1 + \omega + \omega^2)^n + (1 + \omega^2 + \omega^4)^n$$
$$= 3a_0 + a_1(1 + \omega + \omega^2) + a_2(1 + \omega^2 + \omega^4)$$
$$+ a_3(1 + \omega^3 + \omega^6) + \cdots$$

$\therefore \quad 3^n + 0^n + 0^n = 3a_0 + 3a_3 + 3a_6 + \cdots$

$\therefore \quad 3^{n-1} = a_0 + a_3 + a_6 + \cdots$

18. Taking mod of both sides, we have

$$(3x - 1)^2 + 9y^2 = 9[(x - 2)^2 + y^2]$$

or $30x = 35$ $\therefore$ $x = 7/6$ which represents a line $x = c$ parallel to the y-axis.

19.

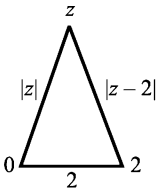


Fig. 2.48

Figure 2.48 shows a triangle with vertices 0, 2 and z. Hence $|z| + |z - 2| \geq 2$

20. We have $|z_k| = 1$, where $k = 1, 2, \ldots, n$

$$\Rightarrow |z_k|^2 = 1 \quad \Rightarrow z_k\bar{z}_k = 1 \quad \Rightarrow \bar{z}_k = \frac{1}{z_k}$$

$$\therefore \quad |z_1 + z_2 + \cdots + z_n| = \left|\overline{z_1 + z_2 + \cdots + z_n}\right| \quad (\because |z| = |\bar{z}|)$$
$$= |\bar{z}_1 + \bar{z}_2 + \cdots + \bar{z}_n|$$
$$= \left|\frac{1}{z_1} + \frac{1}{z_2} + \cdots + \frac{1}{z_n}\right|$$

21. $z_1 + z_2 = -a$, $z_1z_2 = b$

Now, 0, z_1, z_2 form an equilateral triangle, if and only if

$$0^2 + z_1^2 + z_2^2 = 0(z_1 + z_2) + z_1z_2$$
$$\Rightarrow \quad (z_1 + z_2)^2 = 3z_1z_2$$
$$\Rightarrow \quad a^2 = 3b$$

22. We have $\log z + \log z^2 + \log z^3 + \cdots + \log z^n = 0$

$$\Rightarrow \log(zz^2z^3 \ldots z^n) = 0 \quad \Rightarrow \quad \log\left\{z^{[n(n+1)/2]}\right\} = 0$$
$$\Rightarrow \quad z^{[n(n+1)/2]} = 1 \quad \Rightarrow \quad z = 1^{[2/n(n+1)]}$$
$$\Rightarrow \quad z = (\cos 0° + i \sin 0°)^{[2/n(n+1)]}$$
$$\Rightarrow \quad z = (\cos 2m\pi + i \sin 2m\pi)^{[2/n(n+1)]}, \; m \in 0, 1, 2, \ldots$$
$$\Rightarrow \quad z = \cos\left(\frac{4m\pi}{n(n+1)}\right) + i\sin\left(\frac{4m\pi}{n(n+1)}\right), m = 0, 1, 2, \ldots$$

23. $\dfrac{a}{b} + \dfrac{b}{c} + \dfrac{c}{a} = 1$

$$\Rightarrow \quad \frac{\text{cis}\alpha}{\text{cis}\beta} + \frac{\text{cis}\beta}{\text{cis}\gamma} + \frac{\text{cis}\gamma}{\text{cis}\alpha} = 1$$

(where cis represents $\cos\theta + i \sin\theta$)

$$\Rightarrow \quad \text{cis}(\alpha - \beta) + \text{cis}(\beta - \gamma) + \text{cis}(\gamma - \alpha) = 1$$

Equating the real parts of both sides, we get

$$\Rightarrow \quad \cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = 1$$

24. We have $2 = \left|z - \dfrac{4}{z}\right| \geq |z| - \left|\dfrac{4}{z}\right|$

$$\Rightarrow \quad |z| - \left|\frac{4}{z}\right| \leq 2$$
$$\Rightarrow \quad |z|^2 - 2|z| - 4 \leq 0 \quad \text{or} \quad (|z| - 1)^2 - 5 \leq 0$$
$$\Rightarrow \quad (|z| - 1)^2 \leq 5 \quad \text{or} \quad |z| - 1 \leq \sqrt{5}$$
$$\Rightarrow \quad |z| \leq \sqrt{5} + 1.$$

Hence, the greatest value of $|z|$ is $\sqrt{5} + 1$.

25. Since $|z| = 1$, $\therefore$ $z = e^{i\theta}$ as $r = 1$

$$\therefore \quad \frac{z}{1 - z^2} = \frac{1}{z^{-1} - z} = \frac{1}{e^{-i\theta} - e^{i\theta}} = \frac{-1}{2i\sin\theta}$$
$$= 0 + i \cdot \frac{1}{2\sin\theta} \qquad \left(\because \frac{1}{i} = -i\right)$$

$\therefore$ The real part of $z/(1 - z^2)$ is zero. Hence, all its values lie on the y-axis.

26. $|3z-1|=3|z-2|$

$\Rightarrow \quad \left|z-\frac{1}{3}\right|=|z-2|$

$\Rightarrow$ z is a perpendicular bisector of (1/3, 0) and (2, 0)

$\Rightarrow \quad x=\frac{(1/3)+2}{2}=\frac{7}{6}$

which is a line parallel to the y-axis.

27. Let $E=|a+b\omega+c\omega^2|$

$\Rightarrow \quad E=|a+b\omega+c(-1-\omega)|$

$\Rightarrow \quad E=|(a-c)-(c-b)\omega\,|\geq\big||a-c|-|(c-b)\omega|\big|$

$\Rightarrow \quad E\geq\big||a-c|-|b-c|\big|\geq\big||a|-|c|-|b|-|c|\big|$

$\Rightarrow \quad E\geq\big||a|-|b|\big|$

Since it is given that a, b and c are distinct integers, hence the minimum value of their absolute difference is unity, such that

$$E\geq\big||a|-|b|\big|\geq 1$$

Hence, the minimum value of the given expression $E\left(=|a+b\omega+c\omega^2|\right)$ is unity.

28. The given complex equation is

$$\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y+i}{3-i}=i$$

or $\dfrac{(3-i)(1+i)x-(3-i)2i+(3+i)(2-3i)y+(3+i)i}{(3+i)(3-i)}=i$

or $\quad (4+2i)x+(9-7i)y-3i-3=i(3^2-i^2)=10i$

or $\quad (2x-7y-3)i+4x+9y-3=10i$

Equating the real and imaginary parts, we get

$$2x-7y-3=10\,;\,4x+9y-3=0$$

or $\quad 2x-7y=13 \qquad$ (i)

and $\quad 4x+9y=3 \qquad$ (ii)

Multiplying Eq. (i) by 2 and subtracting it from Eq. (ii), we get

$$23y=-23$$

or $\quad y=-1.$

Substituting the value of y in Eq. (ii), we get $x = 3$.

29. We have $\arg(z+i)-\arg(z-i)=\frac{\pi}{2}$

$\Rightarrow \quad \arg\left(\frac{z+i}{z-i}\right)=\frac{\pi}{2}\ \therefore\ \mathrm{Re}\left(\frac{z+i}{z-i}\right)=0$

$\Rightarrow \quad \frac{\left(\frac{z+i}{z-i}\right)+\overline{\left(\frac{z+i}{z-i}\right)}}{2}=0\Rightarrow\left(\frac{z+i}{z-i}\right)+\left(\frac{\bar{z}+\bar{i}}{\bar{z}-\bar{i}}\right)=0$

$\Rightarrow \quad (z+i)(\bar{z}+i)+(z-i)(\bar{z}-i)=0$

$\Rightarrow \quad z\bar{z}+i(z+\bar{z})-1+z\bar{z}-i(z+\bar{z})-1=0$

$\Rightarrow \quad 2(z\bar{z})=2 \quad\Rightarrow\quad z\bar{z}=1$ or $|z|^2=1$

$\Rightarrow \quad |z|=1.$

The equation represents a circle centred at origin and radius 1 unit.

30.

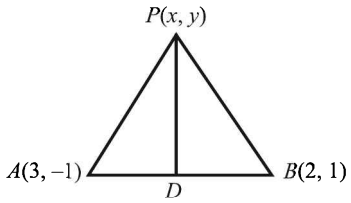


Fig. 2.49

The given equation implies $PA = PB$ for all positions of P. Hence, its locus is the right bisector of AB.

31. Suppose $(\cos x - i \sin 2x)$ is the conjugate of $\sin x + i \cos 2x$

$\Rightarrow \quad \cos x+i\sin 2x=\sin x+i\cos 2x$

$\therefore \quad \cos x=\sin x$ and $\sin 2x=\cos 2x$

$\Rightarrow$ $\tan x=1$ and $\tan 2x=1$; which is impossible.

Hence our supposition is not correct for any value of x.

32. Let α be a real root satisfying the given equation, such that $(p\alpha^2+m\alpha+r)+i(q\alpha^2+n\alpha)=0.$

According to the law of equality of complex numbers,

$\alpha=-\frac{n}{q}$ and $\frac{pn^2}{q^2}-\frac{mn}{q}+r=0.$

$\Rightarrow \quad pn^2+rq^2-mnq=0$

33. Complex numbers $|z|=1$ and $\omega=\frac{z-1}{z+1}$, where $z=-1$.

Substituting $z=\cos\theta+i\sin\theta$, we get

$$\omega=\frac{\cos\theta+i\sin\theta-1}{\cos\theta+i\sin\theta+1}=\frac{\cos\theta-1+i\sin\theta}{\cos\theta+1+i\sin\theta}$$

$$=\frac{\left[1-2\sin^2\left(\frac{\theta}{2}\right)\right]-1+i\sin\theta}{\left[2\cos^2\left(\frac{\theta}{2}\right)-1\right]+1+i\sin\theta}=\frac{-2\sin^2\left(\frac{\theta}{2}\right)+i\sin\theta}{2\cos^2\left(\frac{\theta}{2}\right)+i\sin\theta}$$

$$=\frac{-2\sin^2\left(\frac{\theta}{2}\right)+i\left[2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right]}{2\cos^2\left(\frac{\theta}{2}\right)+i\left[2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right]}$$

$$=\frac{-\sin^2\left(\frac{\theta}{2}\right)+i\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{\cos^2\left(\frac{\theta}{2}\right)+i\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}$$

$$= i\frac{\sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right)}\left[\frac{\cos\left(\frac{\theta}{2}\right)+i\sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right)+i\sin\left(\frac{\theta}{2}\right)}\right] \qquad (\because i^2=-1)$$

$$= i\tan\left(\frac{\theta}{2}\right)=0+i\tan\left(\frac{\theta}{2}\right)$$

Thus, the real part of Re(ω) = 0.

34. We have $z=\dfrac{1+7i}{(2-i)^2}=\dfrac{1+7i}{4+i^2-4i}=\dfrac{1+7i}{3-4i}$

$$=\frac{(1+7i)(3+4i)}{(3-4i)(3+4i)}=\frac{-25+25i}{3^2+4^2}=-1+i$$

$\therefore \quad r=|z|=\sqrt{(-1)^2+1^2}=\sqrt{2}$

Again, $z=-1+i, \quad \therefore \quad x=-1, y=1$

Also, $\tan\theta=\left|\dfrac{y}{x}\right|=\left|\dfrac{1}{-1}\right|=1, \quad \therefore \quad \theta=\dfrac{\pi}{4}.$

$\because \quad x=-1<0$ and $y=1>0$

Hence z lies in the 2nd quadrant.

$\therefore \quad \arg(z)=\pi-\theta=\pi-\dfrac{\pi}{4}=\dfrac{3\pi}{4}.$

$\therefore \quad$ Polar form of $z=\sqrt{2}\left(\cos\dfrac{3\pi}{4}+i\sin\dfrac{3\pi}{4}\right)$

35. $[x^2+(y+3)^2]-[(x-3)^2+y^2]=6$

$\Rightarrow \quad 6x+6y=6 \quad \Rightarrow \quad x+y=1$

36. We have: $\tan^{-1}(\alpha+i\beta)=x+iy$

$\Rightarrow \quad \alpha+i\beta=\tan(x+iy)$

Taking conjugate, we have

$$(\alpha-i\beta)=\tan(x-iy)$$

Now, $\tan 2x=\tan[(x+iy)+(x-iy)]$

$\Rightarrow \quad \tan 2x=\dfrac{\tan(x+iy)+\tan(x-iy)}{1-\tan(x+iy)\tan(x-iy)}$

$$=\frac{(\alpha+i\beta)+(\alpha-i\beta)}{1-(\alpha+i\beta)(\alpha-i\beta)}$$

$$=\frac{2\alpha}{1-(\alpha^2+\beta^2)}$$

$\Rightarrow \quad x=\dfrac{1}{2}\tan^{-1}\left(\dfrac{2\alpha}{1-\alpha^2-\beta^2}\right).$

37. Let $E=-2i\sum_{p=1}^{6}\left(\cos\dfrac{2p\pi}{7}+i\sin\dfrac{2p\pi}{7}\right)$

$\Rightarrow \quad E=-2i\sum_{p=1}^{6}e^{i(2p\pi/7)}$

Let $\alpha=e^{i(2\pi/7)}$

$\therefore \quad E=-2i(\alpha+\alpha^2+\cdots+\alpha^6)$

$\Rightarrow \quad E=-2i[(1+\alpha+\alpha^2+\cdots+\alpha^6)-1]$

$\Rightarrow \quad E=-2i\left[\left(\dfrac{1-\alpha^7}{1-\alpha}\right)-1\right]$

Since $\alpha^7=[e^{i(2\pi/7)}]^7=e^{i2\pi}=1$, we can write

$\Rightarrow \quad E=-2i\left[\dfrac{1-1}{1-e^{i(2\pi/7)}}-1\right]$

$\Rightarrow \quad E=-2i(0-1) \quad \therefore \quad E=2i$

38. Complex equation $x+iy=\sqrt{\dfrac{a+ib}{c+id}}$ (i)

We know that the conjugate of the given equation is

$$x-iy=\sqrt{\frac{a-ib}{c-id}} \qquad \text{(ii)}$$

Multiplying Eqs. (i) and (ii), we get

$$(x+iy)(x-iy)=\sqrt{\frac{a+ib}{c+id}}\times\sqrt{\frac{a-ib}{c-id}}$$

or $\quad x^2+y^2=\sqrt{\dfrac{a^2+b^2}{c^2+d^2}}$

Squaring both sides, we get

$$(x^2+y^2)^2=\frac{a^2+b^2}{c^2+d^2}$$

39. Let $z_1=x_1+iy_1$ and $z_2=x_2+iy_2$

Now $\quad z_1+z_2=(x_1+x_2)+i(y_1+y_2)$

and $z_1-z_2=(x_1-x_2)+i(y_1-y_2)$

As $|z_1+z_2|=|z_1-z_2|$, we get

$$(x_1+x_2)^2+(y_1+y_2)^2=(x_1-x_2)^2+(y_1-y_2)^2 \qquad \text{(i)}$$

Now, $\quad \arg z_1-\arg z_2=\tan^{-1}\dfrac{y_1}{x_1}-\tan^{-1}\dfrac{y_2}{x_2}$

$$=\tan^{-1}\frac{(y_1/x_1)-(y_2/x_2)}{1+[(y_1/x_1)\cdot(y_2/x_2)]}=\tan^{-1}\frac{x_2y_1-y_2x_1}{x_1x_2+y_1y_2}$$

$= \tan^{-1}\infty$, by (i) $x_1x_2+y_1y_2=0$

$\therefore \quad |\arg z_1-\arg z_2|=\dfrac{\pi}{2}.$

40. $z/(1+z)=1^{1/n}$, i.e. nth root of unity

$\therefore \quad \left|\dfrac{z}{1+z}\right|=1$ or $|z|^2=|1+z|^2$

$x^2+y^2=(x+1)^2+y^2 \quad \Rightarrow \quad 2x+1=0$

$\therefore \quad x=-(1/2)$ and y may be any number $\in \mathbb{R}$.

$\therefore \quad \text{Re}(z)=x=-(1/2)<0.$

41. We have: $\arg(z-1)-\arg(z+1)=\pi/2$.

It is clear from Fig. 2.50 that it is a semicircle of equation $x^2+y^2=1$.

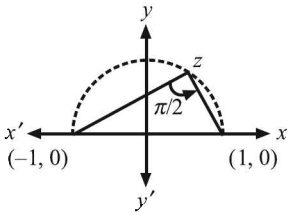


Fig. 2.50

42. The equilateral triangle is inscribed in a circle with centre at origin and radius 2 units.

$\therefore$ Centroid of the equilateral triangle is the origin and the angle subtended by the sides at the centre is $2\pi/3$. See Fig. 2.51.

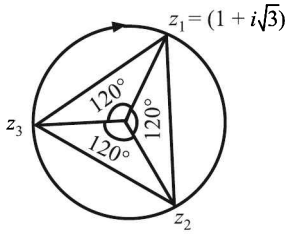


Fig. 2.51

$$\therefore \quad \frac{z_3}{z_1}=\frac{|z_3|}{|z_1|}e^{i2\pi/3} \quad \Rightarrow \quad z_3=z_1e^{i2\pi/3}$$

(in clockwise direction)

$$\Rightarrow \quad z_3=(1+i\sqrt{3})\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$$

$$\Rightarrow \quad z_3=(1+i\sqrt{3})\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)=-2$$

Also, $\frac{z_2}{z_1}=\frac{|z_2|}{|z_1|}e^{-(i2\pi/3)}$ (in clockwise direction)

$$\Rightarrow \quad z_2=z_1e^{-(i2\pi/3)}$$

$$\Rightarrow \quad z_2=(1+i\sqrt{3})\left(\cos\frac{2\pi}{3}-i\sin\frac{2\pi}{3}\right)=1-i\sqrt{3}$$

$\therefore$ In clockwise sense, the other vertices of triangle are $z_2=(1-i\sqrt{3})$ and $z_3==-2$

Shortcut Method:

Since we know that cube roots of unity 1, ω, ω^2 form the vertices of an equilateral triangle inscribed in a circle of radius unity in anticlockwise order, hence, one of the vertices of the equilateral triangle inscribed in a circle of radius 2 units is $1+i\sqrt{3}$, i.e. $-2\omega^2$. Therefore by convention, the other two vertices will be -2, -2ω in anticlockwise order or -2ω, $(-1+i\sqrt{3})$, -2 in clockwise order.

43. We have $8iz^3+12z^2-18z+27i=0$

$$\Rightarrow \quad 4z^2(2iz+3)+9i(2iz+3)=0$$

$$\Rightarrow \quad (2iz+3)(4z^2+9i)=0$$

$$\Rightarrow \quad 2iz+3=0 \text{ or } 4z^2+9i=0$$

$$\therefore \quad |z|=3/2$$

44. We have: $f(\omega)=\omega^{3p}+\omega^{3q+1}+\omega^{3r+2}$

$$=\omega^{3p}+\omega^{3q}\cdot\omega+\omega^{3r}\cdot\omega^2$$

$$=(\omega^3)^p+(\omega^3)^q\cdot\omega+(\omega^3)^r\cdot\omega^2$$

$$=1+\omega+\omega^2=0 \qquad (\because \omega^3=1)$$

45. Putting $z=x+iy$ and $\arg(z_1/z_2)=\arg(z_1)-\arg(z_2)$, we get

$$\tan^{-1}\frac{y}{x-1}-\tan^{-1}\frac{y}{x+1}=\frac{\pi}{4}$$

or $$\frac{[y/(x-1)]-[y/(x+1)]}{1+[y^2/(x^2-1)]}=\tan\frac{\pi}{4}=1$$

or $$2y=x^2+y^2-1$$

or $$x^2+y^2-2y-1=0 \quad \Rightarrow \quad \text{circle}$$

46. $z^2+ax+a^2=0 \quad \Rightarrow \quad z=a\omega, a\omega^2$

(where ω is a non-real root of unity)

$\Rightarrow$ Locus of z is a pair of straight lines

and $$\arg(z)=\arg(a)+\arg(\omega)$$

or $$\arg(z)=\arg(a)+\arg(\omega^2)$$

$$\Rightarrow \quad \arg(z)=\pm\frac{2\pi}{3}$$

Also, $|z|=|a||\omega|$ or $|z|=|a||\omega^2|$

$$\Rightarrow \quad |z|=|a|$$

Hence the option (a) is the correct choice.

47.

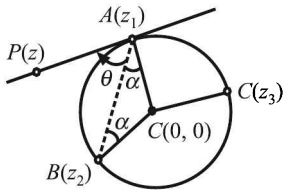


Fig. 2.52

Let $\angle BAP=\theta$

Since we know that the radius meets the point of contact of tangent to the circle at right angles, such that

$$\theta+\alpha=90^\circ \quad \Rightarrow \quad \alpha=90^\circ-\theta$$

Again in isosceles triangle ABC, we have

$$\angle CAB=\angle CAB=90^\circ-\theta$$

Such that $\angle BCA=2\theta$

Given that $\theta=\arg\left(\frac{z-z_1}{z_2-z_1}\right)$ (i)

Now in ΔABC, we have

$$\arg\left(\frac{z_1-0}{z_2-0}\right)=2\theta \quad \Rightarrow \quad \theta=\frac{1}{2}\arg\left(\frac{z_1}{z_2}\right)$$

Substituting this value of θ in (i), we get

$$\arg\left(\frac{z-z_1}{z_2-z_1}\right)=\frac{1}{2}\arg\left(\frac{z_1}{z_2}\right)$$

48. Complex number $= z$ and its $\arg\left(\frac{z-2}{z+2}\right)=\frac{\pi}{3}$.

We know that the standard form of a complex number is $z = x + iy$.

Therefore, $\arg\left[\frac{(x+iy)-2}{x+iy+2}\right]=\frac{\pi}{3}$

or $\arg[(x-2)+iy]-\arg[(x+2)+iy]=\frac{\pi}{3}$

or $\tan^{-1}\left(\frac{y}{x-2}\right)-\tan^{-1}\left(\frac{y}{x+2}\right)=\frac{\pi}{3}$

or $\tan^{-1}\left\{\frac{[y/(x-2)]-[y/(x+2)]}{1+[y^2/(x^2-4)]}\right\}=\frac{\pi}{3}$

$\therefore \quad \frac{(x+2)y-y(x-2)}{x^2-4+y^2}=\tan\left(\frac{\pi}{3}\right)=\sqrt{3}$

or $\frac{xy+2y-xy+2y}{x^2+y^2-4}=\sqrt{3}$ or $4y=\sqrt{3}(x^2+y^2-4)$

or $x^2+y^2-\frac{4y}{\sqrt{3}}-4=0$

Since it is the equation of a circle, therefore, the points representing the given complex number will lie on a circle.

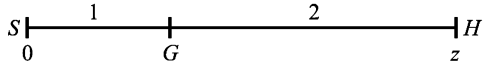
49. We have $(1+i)^{2n}=(1-i)^{2n}$

$\Rightarrow \quad \left(\frac{1+i}{1-i}\right)^{2n}=1 \quad \Rightarrow \quad \left(\frac{(1+i)^2}{2}\right)^{2n}=1$

$\Rightarrow \quad \left(\frac{1-1+2i}{2}\right)^{2n}=1 \quad \Rightarrow \quad (i)^{2n}=1$

$\Rightarrow$ $n = 2$ is the smallest positive integer because $i^4 = 1$.

50. The centroid (G) of the ΔABC divides the orthocentre (H) and the circumcentre (S) in the ratio 2 : 1.



Here $G=\frac{1}{3}z=\frac{1}{3}(z_1+z_2+z_3)$

$\therefore \quad z=z_1+z_2+z_3$

51. $S=1+3\alpha+5\alpha^2+\cdots+(2n-1)\alpha^{n-1}$ (i)

$\Rightarrow \quad \alpha S=\alpha+3\alpha^2+5\alpha^3+\cdots+(2n-1)\alpha^n$ (ii)

Subtracting Eq. (ii) from Eq. (i), we get

$$(1-\alpha)S=1+2\alpha+2\alpha^2+\cdots+2\alpha^{n-1}-(2n-1)\alpha^n$$

$$=2(1+\alpha+\alpha^2+\cdots+\alpha^{n-1})-1-(2n-1)\alpha^n$$

$$=\frac{2(1-\alpha^n)}{1-\alpha}-2n=-2n \qquad (\because\ \alpha^n=1)$$

$$\therefore \qquad S=\frac{-2n}{1-\alpha}$$

52. $(a-2)+ib=-\frac{(1+2\cos\theta)+2i\sin\theta}{(2+\cos\theta)+i\sin\theta}$

$\Rightarrow \quad (a-2)^2+b^2=\frac{(1+2\cos\theta)^2+4\sin^2\theta}{(2+\cos\theta)^2+\sin^2\theta}$

$$=\frac{5+4\cos\theta}{5+4\cos\theta}=1$$

53.

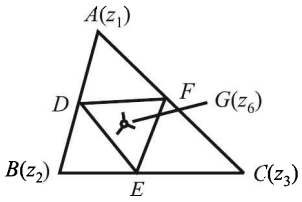


Fig. 2.53

$$I(z_4)=\frac{az_1+bz_2+cz_3}{a+b+c} \qquad \text{(i)}$$

where a, b and c are the sides of the triangle ABC.

Also, $I(z_5)=\dfrac{\frac{a}{2}\left(\frac{z_2+z_3}{2}\right)+\frac{b}{2}\left(\frac{z_1+z_3}{2}\right)+\frac{c}{2}\left(\frac{z_1+z_2}{2}\right)}{\frac{a}{2}+\frac{b}{2}+\frac{c}{2}}$

Again, $G(z_6)=\frac{z_1+z_2+z_3}{3}$

$\Rightarrow \quad 3z_6=z_1+z_2+z_3$ (ii)

$$z_5=\frac{a(z_2+z_3)+b(z_1+z_3)+c(z_1+z_2)}{2(a+b+c)}$$

$\Rightarrow \quad 2z_5=\frac{az_2+az_3+bz_1+bz_3+cz_1+cz_2}{a+b+c}$ (iii)

Adding Eqs. (i) and (iii), we have

$$z_4+2z_5=z_1+z_2+z_3=3z_6$$

Hence option (b) is the correct choice.

54. As $3^{49}(x+iy)=\left(\frac{3}{2}+i\frac{\sqrt{3}}{2}\right)^{100}$

$$=\left[i\sqrt{3}\left(\frac{1-i\sqrt{3}}{2}\right)\right]^{100}$$

$\Rightarrow \quad 3^{49}(x+iy)=i^{100}\cdot 3^{50}\cdot(-\omega)^{100}$

$$\Rightarrow \quad 3^{49}(x+iy) = 3^{50}\cdot\omega = 3^{50}\cdot\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)$$

$$\therefore \quad x+iy = -\frac{3}{2}+i\frac{3\sqrt{3}}{2}$$

or $$x = -\frac{3}{2} = -\frac{1}{\sqrt{3}}\times\frac{3\sqrt{3}}{2}$$

$$\Rightarrow \quad x = -\frac{1}{\sqrt{3}}y, \qquad \therefore \quad k = -\frac{1}{\sqrt{3}}$$

55. We have $z_1 - z_4 = z_2 - z_3$ or $\frac{z_1+z_3}{2} = \frac{z_2+z_4}{2}$

i.e. the diagonals bisect each other.

$\therefore$ It is a parallelogram.

Also, $$\arg\left(\frac{z_4-z_1}{z_2-z_1}\right) = \frac{\pi}{2}$$

$\Rightarrow$ Angle at z_1 is a right angle.

$\therefore$ It is a rectangle.

56. Let the vertices be $z_0, z_1, \ldots, z_5$ at centre O as the origin.

$$OA_1 = OA_0 e^{i\theta}$$

or $$z_1 = z_0 e^{i\theta}$$

$$|z_0| = 1,\ A_0A_1 = |z_1 - z_0| = |z_0 e^{i\theta} - z_0|$$

$$\therefore \quad A_0A_1 = |z_0||\cos\theta + i\sin\theta - 1|$$

$$= 1\cdot\sqrt{(\cos\theta-1)^2+\sin^2\theta} = \sqrt{2(1-\cos\theta)}$$

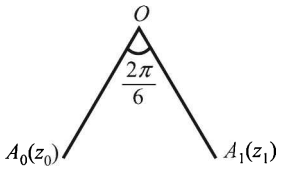


Fig. 2.54

$$\therefore A_0A_1 = \sqrt{2\cdot 2\sin^2\frac{\theta}{2}} = 2\sin\frac{\theta}{2}, \text{ where } \theta = \frac{2\pi}{6} = \frac{\pi}{3}$$

Replacing θ by 2θ and 4θ, we get

$$A_0A_2 = 2\sin\frac{2\theta}{2} = 2\sin\theta$$

$$A_0A_4 = 2\sin\frac{4\theta}{2} = 2\sin 2\theta$$

$$\therefore \quad (A_0A_1)(A_0A_2)(A_0A_4) = 8\sin\frac{\pi}{6}\sin\frac{\pi}{3}\sin\frac{2\pi}{3}$$

$$= 8\cdot\frac{1}{2}\cdot\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2} = 3$$

57. $x = a + b,\ y = a\alpha + b\beta$ and $z = a\beta + b\alpha$

Now, $\quad xyz = (a+b)(a\omega + b\omega^2)(a\omega^2 + b\omega)$

where $\alpha = \omega$ and $\beta = \omega^2$

$$\therefore \quad xyz = (a+b)(a^2 + ab\omega^2 + ab\omega + b^2)$$

$$\Rightarrow \quad (a+b)(a^2 - ab + b^2) = a^3 + b^3$$

58. Given: $x^n - 1 = (x-1)(x-\alpha)(x-\alpha^2)\cdots(x-\alpha^{n-1})$

$$\Rightarrow \log(x^n-1) = \log(x-1) + \log(x-\alpha) + \log(x-\alpha^2) + \cdots + \log(x-\alpha^{n-1})$$

Differentiating w.r.t. x, we have

$$\frac{nx^{n-1}}{x^n-1} = \frac{1}{x-1} + \frac{1}{x-\alpha} + \frac{1}{x-\alpha^2} + \cdots + \frac{1}{x-\alpha^{n-1}}$$

Substituting $x = 2$, we get

$$\frac{n\cdot 2^{n-1}}{2^n-1} = \frac{1}{1} + \frac{1}{2-\alpha} + \frac{1}{2-\alpha^2} + \cdots + \frac{1}{2-\alpha^{n-1}}$$

$$\Rightarrow \quad \frac{n\cdot 2^{n-1}}{2^n-1} - 1 = \sum_{k=1}^{n-1}\frac{1}{2-\alpha^k}$$

$$\Rightarrow \quad \frac{n\cdot 2^{n-1} - 2^n + 1}{2^n-1} = \sum_{k=1}^{n-1}\frac{1}{2-\alpha^k}$$

$$\therefore \quad \sum_{k=1}^{n-1}\frac{1}{2-\alpha^k} = \frac{(n-2)2^{n-1}+1}{2^n-1}$$

59. Given: Points O, P and Q represent the origin and the complex numbers z and $z + iz$, respectively.

We know that $z = r(\cos\theta + i\sin\theta)$.

$\therefore$ $OP = |z| = r$, and

$$z + iz = r(\cos\theta + i\sin\theta) + ir(\cos\theta + i\sin\theta)$$

$$= r[(\cos\theta - \sin\theta) + i(\sin\theta + \cos\theta)]$$

$$= r\cdot\sqrt{2}\left[\left(\frac{1}{\sqrt{2}}\cos\theta - \frac{1}{\sqrt{2}}\sin\theta\right) + i\left(\frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta\right)\right]$$

$$= \sqrt{2}r\left[\left(\cos\theta\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)\sin\theta\right) + i\left(\sin\theta\cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)\cos\theta\right)\right]$$

$$= \sqrt{2}r\left[\cos\left(\theta+\frac{\pi}{4}\right) + i\sin\left(\theta+\frac{\pi}{4}\right)\right].$$

$$\therefore \quad \arg(z+iz) = \left(\theta + \frac{\pi}{4}\right)$$

and $$OQ = |z+iz| = \sqrt{2}r$$

and $$\angle POQ = \arg(z+iz) - \arg(z) = \theta + \frac{\pi}{4} - \theta = \frac{\pi}{4}$$

We also know that in ΔOPQ,

$$(PQ)^2 = (OP)^2 + (OQ)^2 - 2(OP)(OQ)\cos\frac{\pi}{4}$$

$$= r^2 + (\sqrt{2}r)^2 - 2(r)(\sqrt{2}r)\cos\frac{\pi}{4} = r^2 + 2r^2 - 2r^2$$

$\therefore$ $PQ = r$. Hence, $\angle OPQ = 90° = \frac{\pi}{2}$.

60. Let $\cot^{-1} p = \theta$, then $p = \cot\theta$

$$\therefore \quad e^{2mi\cot^{-1}p}\cdot\left(\frac{pi+1}{pi-1}\right)^m = e^{2mi\theta}\cdot\left(\frac{i\cot\theta+1}{i\cot\theta-1}\right)^m$$

$$= e^{2mi\theta}\cdot\left(\frac{i(\cot\theta-i)}{i(\cot\theta+i)}\right)^m = e^{2mi\theta}\cdot\left(\frac{\cot\theta-i}{\cot\theta+i}\right)^m$$

$$= e^{2mi\theta}\cdot\left(\frac{\cos\theta-i\sin\theta}{\cos\theta+i\sin\theta}\right)^m = e^{2mi\theta}\cdot\left(\frac{e^{-i\theta}}{e^{i\theta}}\right)^m$$

$$= e^{2mi\theta}(e^{-2i\theta})^m = e^0 = 1$$

61. $$\frac{1+a}{1-a} = \frac{e^{-i\theta/2}(1+e^{i\theta})}{e^{-i\theta/2}(1-e^{i\theta})} = \frac{e^{-i(\theta/2)}+e^{i(\theta/2)}}{e^{-i(\theta/2)}-e^{i(\theta/2)}}$$

$$= \frac{2\cos(\theta/2)}{-2i\sin(\theta/2)} = i\cot\frac{\theta}{2}$$

62. $$\frac{(a+ib)^2}{a-ib} - \frac{(a-ib)^2}{a+ib}$$

$$\Rightarrow \frac{(a+ib)^3-(a-ib)^3}{a^2+b^2}$$

$$\Rightarrow \frac{(a^3+i^3b^3+3a^2ib-3ab^2)-(a^3-i^3b^3-3ia^2b-3ab^2)}{a^2+b^2}$$

$$\Rightarrow \frac{2i^3b^3+6a^2ib}{a^2+b^2} \Rightarrow \frac{6a^2ib-2ib}{a^2+b^2}$$

$$\therefore \frac{i(6a^2b-2b)}{a^2+b^2} \Rightarrow x+iy \Rightarrow x=0$$

63. α, β, γ are the cube roots of $p < 0$.

$$\therefore \quad \alpha,\beta,\gamma = (p)^{1/3} = p, p\omega, p\omega^2$$

Let $E = \dfrac{\alpha^2x^2+\beta^2y^2+\gamma^2z^2}{\beta^2x^2+\gamma^2y^2+\alpha^2z^2}$

$$= \frac{p^2x^2+p^2\omega^2y^2+p^2\omega z^2}{p^2\omega^2x^2+p^2\omega y^2+p^2z^2} = \frac{x^2+\omega^2y^2+\omega z^2}{\omega^2x^2+\omega y^2+z^2}$$

$$= \frac{\omega(x^2+\omega^2y^2+\omega z^2)}{\omega(\omega^2x^2+\omega y^2+z^2)}$$

$$\Rightarrow E = \omega\left(\frac{x^2+\omega^2y^2+\omega z^2}{\omega^3x^2+\omega^2y^2+\omega z^2}\right) = \omega$$

$$\Rightarrow E = \omega = \frac{\gamma}{\beta}$$

64. Two infinite rays are PQ and PR and arc = QAR. We know that the equation of the line PQ is $\arg(z+1) = \frac{\pi}{4}$.

Similarly, the equation of the line PR is $\arg(z+1) = -\frac{\pi}{4}$.

We also know that for the shaded region,

$-\frac{\pi}{4} < \arg(z+1) < \frac{\pi}{4}$ or $|\arg(z+1)| < \frac{\pi}{4}$.

We also know that $|PQ| = \sqrt{(\sqrt{2})^2+(\sqrt{2})^2} = 2$ and $|PA| = 2$. Similarly, $|PR| = 2$.

Therefore, arc QAR is a circle with radius = 2 and equation $|z-1| = 2$ and all the points in the shaded region excluding the boundary satisfy $|z-1| > 2$ and $|\arg(z+1)| < \frac{\pi}{4}$.

65. $$x+iy = \frac{3}{2+\cos\theta+i\sin\theta}$$

$$= \frac{3(2+\cos\theta-i\sin\theta)}{(2+\cos\theta)^2+\sin^2\theta}$$

$$= \frac{6+3\cos\theta-3i\sin\theta}{4+\cos^2\theta+4\cos\theta+\sin^2\theta}$$

$$= \left(\frac{6+3\cos\theta}{5+4\cos\theta}\right)+i\left(\frac{-3\sin\theta}{5+4\cos\theta}\right)$$

Equating the real and imaginary parts, we get

$$x = \frac{3(2+\cos\theta)}{5+4\cos\theta},\ y = \frac{-3\sin\theta}{5+4\cos\theta}$$

$$\therefore x^2+y^2 = \frac{9}{(5+4\cos\theta)^2}(4+\cos^2\theta+4\cos\theta+\sin^2\theta)$$

$$= \frac{9}{5+4\cos\theta} = 4\left(\frac{6+3\cos\theta}{5+4\cos\theta}\right)-3$$

$$= 4x-3$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $\dfrac{1-ix}{1+ix}=a-ib \quad\Rightarrow\quad \dfrac{(1-ix)(1-ix)}{(1+ix)(1-ix)}=a-ib$

$\Rightarrow \dfrac{1-x^2-2ix}{1+x^2}=a-ib \quad\Rightarrow\quad \dfrac{1-x^2}{1+x^2}=a \text{ and } \dfrac{2x}{1+x^2}=b$

Now, x can be written as

$$x=\frac{2x/(1+x^2)}{2/(1+x^2)}=\frac{2x/(1+x^2)}{[(1-x^2)/(1+x^2)]+1}$$

$$=\frac{b}{1+a}=\frac{2b}{1+1+2a}$$

$$\Rightarrow \quad \frac{2b}{1+(a^2+b^2)+2a}=\frac{2b}{(1+a)^2+b^2}$$

2. Each interior angle of the regular polygon of p sides is

$$\pi-\frac{2\pi}{p_i} \quad (i=1 \text{ to } k)$$

Since no gap is left at the vertex, the sum of the interior angles of k regular polygons meeting at a common vertex must be 2π.

So, $$\left(\pi-\frac{2\pi}{p_1}\right)+\left(\pi-\frac{2\pi}{p_2}\right)+\cdots+\left(\pi-\frac{2\pi}{p_k}\right)=2\pi$$

$$\Rightarrow \quad k-2\left(\frac{1}{p_1}+\frac{1}{p_2}+\cdots+\frac{1}{p_k}\right)=2$$

$$\therefore \quad \frac{1}{p_1}+\frac{1}{p_2}+\cdots+\frac{1}{p_k}=\frac{k}{2}-1$$

3. Complex number, $z=1+\sqrt{3}i$.

Let $x+iy=1+\sqrt{3}i$.

Equating the real and imaginary parts, we get

$x=1$ and $y=\sqrt{3}$.

We know that modulus

$$r=\sqrt{x^2+y^2}=\sqrt{1^2+(\sqrt{3})^2}=\sqrt{1+3}=\sqrt{4}=2$$

We also know that $\tan\theta=\dfrac{y}{x}=\dfrac{\sqrt{3}}{1}=\sqrt{3}$

or $$\theta=\tan^{-1}(\sqrt{3})=\frac{\pi}{3}$$

Therefore, the polar form of the given number

$$=r(\cos\theta+i\sin\theta)=2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right).$$

4. $\omega^1\cdot\omega^2\cdot\omega^3\cdot\ldots\cdot\omega^n=\omega^{1+2+3+\cdots+n}=\omega^{n(n+1)/2}=S_n$

Putting $n=1, 2, 3, \ldots n$, we get

$S_1=\omega^1=\omega, \qquad S_2=\omega^3=1$

$S_3=\omega^6=1,\ldots., \qquad S_7=\omega^{28}=\omega$

$\therefore$ We always get 1 and ω.

5. We have $p+iq=\dfrac{(a+i)^2}{2a-1}$ (i)

Changing i to $-i$, we get $p-iq=\dfrac{(a-i)^2}{2a+1}$ (ii)

Multiplying Eqs. (i) and (ii), we get

$$(p+iq)(p-iq)=\frac{(a+i)^2}{2a-1}\times\frac{(a-i)^2}{2a+1}$$

$$\Rightarrow \quad p^2+q^2=\frac{(a^2+1)^2}{4a^2-1}$$

6. Since z_1 and z_2 are two non-real complex cube roots of unity,

$\therefore\ z_1=\omega$ and $z_2=\omega^2$

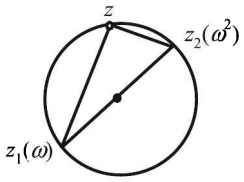


Fig. 2.55

z is any point on the circle such that

$$|z-z_1|^2+|z-z_2|^2=\lambda$$

Now, z_1 and z_2 are the ends of a diameter and we know that the angle subtended by the diameter of the circle on the circle is a right angle.

$\therefore$ In the right-angled triangle represented by the vertices $z, z_1(\omega)$ and $z_2(\omega^2)$,

$$|z-z_1|^2+|z-z_2|^2=|z_1-z_2|^2$$

$$\Rightarrow \quad \lambda=|z_1-z_2|^2=|\omega-\omega^2|^2$$

$$\therefore \quad \lambda=|\sqrt{3}|^2=3$$

7. The given complex number may be written as

$$z=\frac{1+7i}{4+i^2-4i}=\frac{1+7i}{3-4i}$$

$$=\frac{1+7i}{3-4i}\times\frac{3+4i}{3+4i}=\frac{-25+25i}{25}=-1+i$$

Let $x+iy=-1+i$.

Equating the real and imaginary parts, we get

$x=-1$ and $y=1$.

We know that modulus

$$r=\sqrt{x^2+y^2}=\sqrt{(-1)^2+(1)^2}=\sqrt{2}$$

We also know that $\tan\theta=\frac{y}{x}=\frac{1}{-1}=-1$

or $\theta=\tan^{-1}(-1)=\frac{3\pi}{4}$.

Therefore, the polar form of the given number

$$=r(\cos\theta+i\sin\theta)=\sqrt{2}\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$$

8. $\omega+\omega^{\left(\frac{1}{2}+\frac{3}{8}+\frac{9}{32}+\frac{27}{128}+\cdots\right)}=\omega+\omega^{\left(\frac{1/2}{1-3/4}\right)}$

$$=\omega+\omega^2=-1 \quad (\because 1+\omega+\omega^2=0)$$

9. $\left(\frac{1+\cos\frac{\pi}{8}+i\sin\frac{\pi}{8}}{1+\cos\frac{\pi}{8}-i\sin\frac{\pi}{8}}\right)^8=\left(\frac{2\cos^2\frac{\pi}{16}+2i\sin\frac{\pi}{16}\cos\frac{\pi}{16}}{2\cos^2\frac{\pi}{16}-2i\sin\frac{\pi}{16}\cos\frac{\pi}{16}}\right)^8$

$$=\left(\frac{\cos\frac{\pi}{16}+i\sin\frac{\pi}{16}}{\cos\frac{\pi}{16}-i\sin\frac{\pi}{16}}\right)^8$$

$$=\left[\left(\cos\frac{\pi}{16}+i\sin\frac{\pi}{16}\right)\left(\cos\frac{\pi}{16}-i\sin\frac{\pi}{16}\right)^{-1}\right]^8$$

$$=\left[\left(\cos\frac{\pi}{16}+i\sin\frac{\pi}{16}\right)\left(\cos\frac{\pi}{16}+i\sin\frac{\pi}{16}\right)\right]^8$$

$$=\left[\left(\cos\frac{\pi}{16}+i\sin\frac{\pi}{16}\right)^2\right]^8$$

$$=\cos\pi+i\sin\pi \qquad \text{(using De Moivre's theorem)}$$

$$=-1$$

Shortcut Method

Let $z=\cos\frac{\pi}{8}+i\sin\frac{\pi}{8}$

$\therefore$ $\frac{1}{z}=\cos\frac{\pi}{8}-i\sin\frac{\pi}{8}$

Now from the given equation, we have

$$\left(\frac{1+\cos\frac{\pi}{8}+i\sin\frac{\pi}{8}}{1+\cos\frac{\pi}{8}-i\sin\frac{\pi}{8}}\right)^8=\left(\frac{1+z}{1+z^{-1}}\right)^8=\left[\frac{(1+z)z}{1+z}\right]^8$$

$$=z^8$$

$$\Rightarrow \left(\cos\frac{\pi}{8}+i\sin\frac{\pi}{8}\right)^8 \Rightarrow \cos\pi=-1$$

10. $z_1+z_3=z_2+z_4 \Rightarrow \frac{z_1+z_3}{2}=\frac{z_2+z_4}{2}$

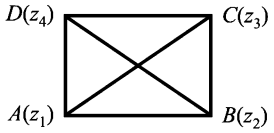


Fig. 2.56

$\Rightarrow$ (mid-point of z_1 and z_3) = (mid-point of z_2 and z_4)

$\Rightarrow$ Quadrilateral $ABCD$ is a parallelogram.

Also, $|z_1-z_3|=|z_2-z_4|$

i.e. $\begin{pmatrix}\text{Distance between}\\ z_1 \text{ and } z_3\end{pmatrix}=\begin{pmatrix}\text{Distance between}\\ z_2 \text{ and } z_4\end{pmatrix}$

$\Rightarrow$ Lengths of two diagonals are equal.

$\therefore$ The parallelogram $ABCD$ is a rectangle.

Now, $\arg\left(\frac{z_1-z_2}{z_3-z_2}\right)$, i.e. angle between z_1, z_2 and z_3 (taken in order) is $\pm\frac{\pi}{2}$ which is the angle between the sides AB and BC.

11. $(1+i)^{n_1}+(1-i^3)^{n_1}+(1+i^5)^{n_2}+(1-i^7)^{n_2}$

$$=(1+i)^{n_1}+(1-i)^{n_1}+(1+i)^{n_2}+(1-i)^{n_2}$$

$$=2^{n_1/2}\left(\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right)^{n_1}+2^{n_1/2}\left(\frac{1}{\sqrt{2}}-i\frac{1}{\sqrt{2}}\right)^{n_1}$$

$$+2^{n_2/2}\left(\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right)^{n_2}+2^{n_2/2}\left(\frac{1}{\sqrt{2}}-i\frac{1}{\sqrt{2}}\right)^{n_2}$$

$$=2^{n_1/2}\left[\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)^{n_1}+\left(\cos\frac{\pi}{4}-i\sin\frac{\pi}{4}\right)^{n_1}\right]$$

$$+2^{n_2/2}\left[\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)^{n_2}+\left(\cos\frac{\pi}{4}-i\sin\frac{\pi}{4}\right)^{n_2}\right]$$

$$=2^{n_1/2}\left[\cos\frac{n_1\pi}{4}+i\sin\frac{n_1\pi}{4}+\cos\frac{n_1\pi}{4}-i\sin\frac{n_1\pi}{4}\right]$$

$$+2^{n_2/2}\left[\cos\frac{n_2\pi}{4}+i\sin\frac{n_2\pi}{4}+\cos\frac{n_2\pi}{4}-i\sin\frac{n_2\pi}{4}\right]$$

$$=2^{n_1/2}\cdot 2\cos\frac{n_1\pi}{4}+2^{n_2/2}\cdot 2\cos\frac{n_2\pi}{4}=\text{real number}$$

i.e. n_1, n_2 are any two positive integers.

12. Let $\frac{\alpha}{a}=P, \frac{\beta}{b}=Q, \frac{\gamma}{c}=R$

$\therefore$ $P+Q+R=1+i$ (i)

and $\frac{1}{P}+\frac{1}{Q}+\frac{1}{R}=0$ [given]

$\Rightarrow$ $PQ+QR+RP=0$ (ii)

Squaring (i), we have

$$P^2+Q^2+R^2+2(PQ+QR+RP)=2i$$

$\Rightarrow$ $P^2+Q^2+R^2+2(0)=2i$ [using (ii)]

$\therefore$ $P^2+Q^2+R^2=2i$

13. The given complex number may be written as

$$-1+\sqrt{-3}=-1+\sqrt{3}i=re^{i\theta}=r(\cos\theta+i\sin\theta)$$

Comparing the real and imaginary parts,

$$r\cos\theta=-1 \quad \text{(i)}$$

$$r\sin\theta=\sqrt{3} \quad \text{(ii)}$$

Dividing (ii) by (i),

$$\frac{r\sin\theta}{r\cos\theta}=\frac{\sqrt{3}}{-1} \quad \text{or} \quad \tan\theta=-\sqrt{3}$$

or $\theta=\tan^{-1}(-\sqrt{3})=\frac{2\pi}{3}$

14. Given $\omega_n=\cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}$

$\Rightarrow$ $\omega_3=\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}=-\frac{1}{2}+\frac{i\sqrt{3}}{2}=\omega$

and $\omega_3^2=\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)^2$

$$=\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}=-\frac{1}{2}-\frac{i\sqrt{3}}{2}=\omega^2$$

$\therefore$ $(x+y\omega_3+z\omega_3^2)(x+y\omega_3^2+z\omega_3)$

$=(x+y\omega+z\omega^2)(x+y\omega^2+z\omega)$

$=x^2+y^2+z^2-xy-yz-zx$

15. We have $z_r=\cos\frac{r\alpha}{n^2}+i\sin\frac{r\alpha}{n^2}$, where $r=1, 2, 3, ..., n$.

$\therefore$ $z_1=\cos\frac{\alpha}{n^2}+i\sin\frac{\alpha}{n^2}$, $z_2=\cos\frac{2\alpha}{n^2}+i\sin\frac{2\alpha}{n^2}, \cdots,$

$z_n=\cos\frac{n\alpha}{n^2}+i\sin\frac{n\alpha}{n^2}$

$\therefore$ $\lim_{n\to\infty}(z_1z_2z_3\cdots z_n)$

$$=\lim_{n\to\infty}\left(\cos\frac{\alpha}{n^2}+i\sin\frac{\alpha}{n^2}\right)\left(\cos\frac{2\alpha}{n^2}+i\sin\frac{2\alpha}{n^2}\right)\cdots\left(\cos\frac{n\alpha}{n^2}+i\sin\frac{n\alpha}{n^2}\right)$$

$$=\lim_{n\to\infty}\left\{\cos\left[\frac{\alpha}{n^2}(1+2+3+\cdots+n)\right]+i\sin\left[\frac{\alpha}{n^2}(1+2+3+\cdots+n)\right]\right\}$$

$$=\lim_{n\to\infty}\left[\cos\frac{\alpha n(n+1)}{2n^2}+i\sin\frac{\alpha n(n+1)}{2n^2}\right]$$

$$=\cos\frac{\alpha}{2}+i\sin\frac{\alpha}{2}=e^{i\alpha/2}$$

16. Given equation $(x^3-3x^2+3x-1)+64=0$

$\Rightarrow$ $(x-1)^3=(-4)^3$ $\Rightarrow$ $x-1=-4, -4\omega, -4\omega^2$

$\therefore$ $x=-3, 1-4\omega, 1-4\omega^2$

17. $\text{Re}\left\{\frac{\tan\alpha-i[\sin(\alpha/2)+\cos(\alpha/2)]}{1+2i\sin(\alpha/2)}\right\}=0$

$$\text{Re}\left\{\frac{\left[\tan\alpha-i\left(\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2}\right)\right]\left(1-2i\sin\frac{\alpha}{2}\right)}{1+4\sin^2\frac{\alpha}{2}}\right\}=0$$

$$\Rightarrow \text{Re}\left\{\frac{\tan\alpha-2\sin\frac{\alpha}{2}\left(\cos\frac{\alpha}{2}+\sin\frac{\alpha}{2}\right)-i\left(\tan\alpha\cdot2\sin\frac{\alpha}{2}+\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2}\right)}{1+4\sin^2\frac{\alpha}{2}}\right\}=0$$

$\Rightarrow$ $\tan\alpha=2\sin^2\frac{\alpha}{2}+\sin\alpha$

$\Rightarrow$ $\frac{\sin\alpha}{\cos\alpha}=\sin\alpha+1-\cos\alpha$

$\Rightarrow$ $\sin\alpha=\sin\alpha\cdot\cos\alpha+\cos\alpha-\cos^2\alpha$

$\Rightarrow$ $\sin\alpha(1-\cos\alpha)=\cos\alpha(1-\cos\alpha)$

$\Rightarrow$ $\sin\alpha=\cos\alpha, \cos\alpha=1$

$\Rightarrow$ $\alpha=n\pi+\frac{\pi}{4}, \alpha=2n\pi\ (n\in Z)$

18. Given $|z-4|^2<|z-2|^2$

$\Rightarrow$ $|(x-4)+iy|^2<|(x-2)+iy|^2$

$\Rightarrow$ $(x-4)^2+y^2<(x-2)^2+y^2$

$\Rightarrow$ $-4x<-12$ $\Rightarrow$ $4x>12$, $x>3$

$\Rightarrow$ $\text{Re}(z)>3$.

19. Since, $\left(\frac{z-1}{z+1}\right)^n=i \Rightarrow \left(\frac{z-1}{z+1}\right)^n=\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}$

$\Rightarrow$ $\frac{z-1}{z+1}=\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)^{1/n}$

$\Rightarrow$ $\frac{z-1}{z+1}=\left[\cos\left(2k\pi+\frac{\pi}{2}\right)+i\sin\left(2k\pi+\frac{\pi}{2}\right)\right]^{1/n}$

$$\Rightarrow \quad \frac{z-1}{z+1} = \cos\frac{(4k+1)\pi}{2n} + i\sin\frac{(4k+1)\pi}{2n}$$

$k = 0, 1, 2, 3, \ldots, (n-1)$

Let $\quad \frac{(4k+1)\pi}{2n} = 2\alpha$

$$\therefore \quad \frac{z-1}{z+1} = \cos 2\alpha + i\sin 2\alpha$$

Using componendo and dividendo, we have

$$\frac{z-1+z+1}{z-1-z-1} = \frac{\cos 2\alpha + i\sin 2\alpha + 1}{\cos 2\alpha + i\sin 2\alpha - 1}$$

$$\Rightarrow \quad \frac{2z}{-2} = \frac{2\cos^2\alpha + i2\sin\alpha\cos\alpha}{-2\sin^2\alpha + i2\sin\alpha\cos\alpha}$$

$$\Rightarrow \quad -z = \cot\alpha \frac{\cos\alpha + i\sin\alpha}{i(\cos\alpha + i\sin\alpha)} \quad \Rightarrow \quad -z = -i\cot\alpha$$

$$\Rightarrow \quad z = i\cot\frac{(4k+1)\pi}{4n}$$

i.e. z is the locus of all those points which lie on the x-axis

$\therefore$ All these points are collinear.

20. Nonzero complex numbers are z_1 and z_2 and the equation is $|z_1+z_2| = |z_1| + |z_2|$.

Squaring both sides,

$$|z_1+z_2|^2 = \left(|z_1| + |z_2|\right)^2$$

$$\Rightarrow \quad |z_1|^2 + |z_2|^2 + 2\,\mathrm{Re}\,|z_1\bar{z}_2| = |z_1|^2 + |z_2|^2 + 2|z_1||z_2|$$

$$\Rightarrow \quad \mathrm{Re}\,|z_1\bar{z}_2| = |z_1||z_2| = |z_1\bar{z}_2|$$

$$\Rightarrow \quad z_1\bar{z}_2 \in \mathbb{R}$$

Therefore, $\arg(z_1\bar{z}_2) = 0 \Rightarrow \arg(z_1) + \arg(\bar{z}_2) = 0$

$\Rightarrow \arg(z_1) - \arg(z_2) = 0$

21. Let $z - 1 = r(\cos\theta + i\sin\theta) = re^{i\theta}$

$$\therefore \quad \text{Given expression} = re^{i\theta}\cdot e^{-i\alpha} + \frac{1}{re^{i\theta}}\cdot e^{i\alpha}$$

$$= re^{i(\theta-\alpha)} + \frac{1}{r}e^{-i(\theta-\alpha)}$$

Since the imaginary part of the given expression is zero, we have

$$r\sin(\theta-\alpha) - \frac{1}{r}\sin(\theta-\alpha) = 0$$

$$r^2 - 1 = 0 \quad \Rightarrow \quad r^2 = 1$$

$$\Rightarrow \quad r = 1 \quad \Rightarrow \quad |z-1| = 1$$

$$\sin(\theta-\alpha) = 0 \quad \Rightarrow \quad \theta - \alpha = 0$$

$$\Rightarrow \quad \theta = \alpha \quad \Rightarrow \quad \arg(z-1) = \alpha$$

22. Let $x = (1)^{1/n}$

$\Rightarrow \quad x^n - 1 = 0$ has n roots

$$\therefore \quad (x^n - 1) = (x-1)(x-\omega)(x-\omega^2)\cdots(x-\omega^{n-1})$$

$$\frac{x^n-1}{x-1} = (x-\omega)(x-\omega^2)\cdots(x-\omega^{n-1})$$

Putting $x = 7$ on both sides,

$$\therefore \quad (7-\omega)(7-\omega^2)(7-\omega^3)\cdots(7-\omega^{n-1}) = \frac{7^n-1}{6}$$

23. $|2z-5| + |2z+5| = 10$

$$\Rightarrow \quad \left|z - \frac{5}{2}\right| + \left|z + \frac{5}{2}\right| = 2$$

which represents the locus of a point $z(x, y)$ such that the sum of its distances from two fixed points $\left(\frac{5}{2}, 0\right)$ and $\left(\frac{-5}{2}, 0\right)$ is a constant.

$\therefore$ The locus of z is an ellipse by definition.

24. $\left|\frac{z-5i}{z+5i}\right| = 1 \quad \Rightarrow \quad \left|\frac{x+i(y-5)}{x+i(y+5)}\right| = 1$

$$\Rightarrow \quad x^2 + (y-5)^2 = x^2 + (y+5)^2$$

$\Rightarrow \quad 20y = 0;\ y = 0$, which is the x-axis.

25. Since $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$

$$\Rightarrow \quad \frac{2(z_1-z_3)}{z_2-z_3} - 1 = -i\sqrt{3} \quad \Rightarrow \quad \frac{2z_1 - z_2 - z_3}{z_2 - z_3} = -i\sqrt{3}$$

Squaring both sides,

$$(2z_1 - z_2 - z_3)^2 = -3(z_2 - z_3)^2$$

$$\Rightarrow \quad 4z_1^2 + z_2^2 + z_3^2 - 4z_1z_2 + 2z_2z_3 - 4z_3z_1$$

$$= -3(z_2^2 + z_3^2 - 2z_2z_3)$$

$$\Rightarrow \quad 4z_1^2 + 4z_2^2 + 4z_3^2 - 4z_1z_2 - 4z_2z_3 - 4z_3z_1 = 0$$

$$\Rightarrow \quad z_1^2 + z_2^2 + z_3^2 - z_1z_2 - z_2z_3 - z_3z_1 = 0$$

which is the condition for a triangle having vertices z_1, z_2 and z_3 to be an equilateral triangle.

26. Complex number $x_r = \cos\frac{\pi}{2^r} + i\sin\frac{\pi}{2^r}$.

We know that $x_1 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$.

Similarly, $x_2 = \cos\frac{\pi}{2^2} + i\sin\frac{\pi}{2^2}$

and $x_3 = \cos\frac{\pi}{2^3} + i\sin\frac{\pi}{2^3}$, and so on.

$\therefore \quad x_1 \times x_2 \times x_3 \times x_4 \times \cdots \times \infty$

$= \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\left(\cos\frac{\pi}{2^2} + i\sin\frac{\pi}{2^2}\right) \times \cdots \times \infty$

$= \cos\left(\frac{\pi}{2} + \frac{\pi}{2^2} + \frac{\pi}{2^3} + \cdots\right) + i\sin\left(\frac{\pi}{2} + \frac{\pi}{2^2} + \frac{\pi}{2^3} \cdots\right)$

$= \cos\left(\frac{\pi/2}{1-1/2}\right) + i\sin\left(\frac{\pi/2}{1-1/2}\right)$

$= \cos\pi + i\sin\pi = -1$

27. We have $a = \cos\alpha + i\sin\alpha$, $b = \cos\beta + i\sin\beta$ and $c = \cos\gamma + i\sin\gamma$

Now, $\frac{b}{c} = \frac{\cos\beta + i\sin\beta}{\cos\gamma + i\sin\gamma} \times \frac{\cos\gamma - i\sin\gamma}{\cos\gamma - i\sin\gamma}$

$= \cos\beta\cos\gamma + \sin\beta\sin\gamma + i(\sin\beta\cos\gamma - \sin\gamma\cos\beta)$

$\Rightarrow \quad \frac{b}{c} = \cos(\beta-\gamma) + i\sin(\beta-\gamma)$ (i)

Similarly, $\frac{c}{a} = \cos(\gamma-\alpha) + i\sin(\gamma-\alpha)$ (ii)

and $\frac{a}{b} = \cos(\alpha-\beta) + i\sin(\alpha-\beta)$ (iii)

Adding (i), (ii) and (iii), we get

$\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) + i[\sin(\beta-\gamma) + \sin(\gamma-\alpha) + \sin(\alpha-\beta)] = 1$

Equating real parts, we get

$\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) = 1$

28. Let $z = e^{i\theta} \quad \therefore \quad \arg(z) = \theta$

Let $\arg(z_1) = \theta_1$ and $\arg(z_2) = \theta_2$

$\therefore \quad z_1 R z_2 \quad \Leftrightarrow \quad |\arg(z_1) - \arg(z_2)| = \frac{2\pi}{3}$

$\therefore \quad z_1 R z_2 \quad \Leftrightarrow \quad |\theta_1 - \theta_2| = \frac{2\pi}{3} \quad \Leftrightarrow \quad |\theta_2 - \theta_1| = \frac{2\pi}{3}$

$\Leftrightarrow \quad z_2 R z_1$

but $z_1 \neq z_2$ (If $z_1 = z_2$, then $2\pi/3 = 0$ which is not true)

$\Rightarrow$ R is symmetric.

29. Since z_1, z_2, z_3 are the vertices of an equilateral triangle,

$\therefore \quad z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

Hence, $(z_1 + z_2 + z_3)^2 = 3(z_1^2 + z_2^2 + z_3^2)$

Since z_0 is the centroid of the triangle,

$\therefore \quad \frac{z_1 + z_2 + z_3}{3} = z_0 \quad \Rightarrow \quad (3z_0)^2 = 3(z_1^2 + z_2^2 + z_3^2)$

$\therefore \quad z_1^2 + z_2^2 + z_3^2 = 3z_0^2$

30. We have $|z-1| = |z+i|$

$\Rightarrow \quad |(x-1) + iy| = |x + i(y+1)|$

$\Rightarrow \quad (x-1)^2 + y^2 = x^2 + (y+1)^2$

$\Rightarrow$ $x + y = 0$, which is a line through the origin.

31. Complex numbers are z_1 and z_2, where $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$.

We know that $|z_1 - z_2| \geq ||z_1| - |z_2||$

and $\quad 5 = |z_2 - (3+4i)| \geq ||z_2| - 5|$

$\Rightarrow \quad 0 \leq |z_2| \leq 10 \quad \Rightarrow \quad |z_2| \leq 10$

Therefore, $||z_1| - |z_2|| = |12 - 10| = 2$

$\Rightarrow \quad |z_1 - z_2| \geq 2$.

32. Assuming $\cot^{-1}x = \theta$

$\Rightarrow \quad \cot\theta = x \quad \Rightarrow \quad \tan\theta = \frac{1}{x}$

Now, $e^{2i\cot^{-1}x} = e^{2i\theta} = \cos 2\theta + i\sin 2\theta$

$\Rightarrow \quad e^{2i\cot^{-1}x} = \left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right) + i\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$

$\Rightarrow \quad e^{2i\cot^{-1}x} = \left[\frac{1-(1/x^2)}{1+(1/x^2)}\right] + i\left[\frac{2(1/x)}{1+(1/x^2)}\right]$

$\Rightarrow \quad e^{2i\cot^{-1}x} = \left(\frac{x^2-1}{x^2+1}\right) + i\left(\frac{2x}{x^2+1}\right)$

$\Rightarrow \quad e^{2i\cot^{-1}x} = \frac{x^2-1+2ix}{x^2+1} = \frac{(x+i)^2}{(x+i)(x-i)}$

$\Rightarrow \quad e^{2i\cot^{-1}x} = \frac{x+i}{x-i} = \frac{ix+i^2}{ix-i^2} = \frac{ix-1}{ix+1}$

$\Rightarrow \quad (e^{2i\cot^{-1}x})^n = \left(\frac{ix-1}{ix+1}\right)^n$

$\therefore \quad e^{2ni\cot^{-1}x}\left(\frac{ix+1}{ix-1}\right)^n = 1$

33. $\left|z + \frac{2}{z}\right| = 2 \quad \Rightarrow \quad |z| - \frac{2}{|z|} \leq 2$

$\Rightarrow \quad |z|^2 - 2|z| - 2 \leq 0$

This is a quadratic equation in $|z|$.

$\therefore \quad |z| \leq \frac{2 \pm \sqrt{4+8}}{2} \leq 1 \pm \sqrt{3}$

Hence, the maximum value of $|z|$ is $1 + \sqrt{3}$.

34. From the given equation, we have

$$(y-1)^3 = -27 \quad \Rightarrow \quad \left(\frac{y-1}{-3}\right)^3 = 1$$

$$\therefore \quad \frac{y-1}{-3} = (1)^{1/3}$$

$$\therefore \quad \frac{y-1}{-3} = 1,\ \omega,\ \omega^2$$

As a, b, c are the roots of the given equation,

$$\therefore \quad a-1 = -3,\ b-1 = -3\omega,\ c-1 = -3\omega^2$$

$$\therefore \quad \frac{a-1}{b-1} + \frac{b-1}{c-1} + \frac{c-1}{a-1} = \frac{1}{\omega} + \frac{1}{\omega} + \frac{\omega^2}{\omega^3}$$

$$= \omega^2 + \omega^2 + \omega^2$$

$$= 3\omega^2$$

35. $|\alpha+\beta|^2 = (\alpha+\beta)(\overline{\alpha+\beta}) = (\alpha+\beta)(\bar{\alpha}+\bar{\beta})$

$$\therefore \quad |\alpha+\beta|^2 = \alpha\bar{\alpha} + \beta\bar{\beta} + \alpha\bar{\beta} + \bar{\alpha}\beta$$

$$\Rightarrow \quad |\alpha+\beta|^2 = |\alpha|^2 + |\beta|^2 + \alpha\bar{\beta} + \bar{\alpha}\beta \qquad \text{(i)}$$

Also, $|\alpha-\beta|^2 = (\alpha-\beta)(\bar{\alpha}-\bar{\beta})$

$$\Rightarrow \quad |\alpha-\beta|^2 = \alpha\bar{\alpha} + \beta\bar{\beta} - \alpha\bar{\beta} - \bar{\alpha}\beta$$

$$\therefore \quad |\alpha-\beta|^2 = |\alpha|^2 + |\beta|^2 - \alpha\bar{\beta} - \bar{\alpha}\beta \qquad \text{(ii)}$$

Adding (i) and (ii), we have

$$|\alpha|^2 + |\beta|^2 = \frac{1}{2}\left(|\alpha+\beta|^2 + |\alpha-\beta|^2\right)$$

36. Let $z = x+iy$

$$\therefore \quad \frac{2z+1}{iz+1} = \frac{2(x+iy)+1}{i(x+iy)+1} = \frac{(2x+1)+2iy}{(1-y)+ix} \cdot \frac{(1-y)-ix}{(1-y)-ix}$$

$$= \frac{(2x+1)(1-y)+2xy+i[-x(2x+1)+2y(1-y)]}{(1-y)^2+x^2}$$

Because the Imaginary part of $\left(\frac{2z+1}{iz+1}\right) = -2$, we have

$$\frac{-x(2x+1)+2y(1-y)}{(1-y)^2+x^2} = -2$$

or $\quad -2x^2 - 2y^2 - x + 2y = -2(1+y^2-2y) - 2x^2$

i.e. $x+2y-2=0$, which is a straight line.

37. The equation $x + (1/x) = 2\cos\theta$ may be written as

$$\frac{x^2+1}{x} = 2\cos\theta$$

or $x^2 - 2x\cos\theta + 1 = 0$

Factorizing, we get

$$x = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2} = \frac{2\cos\theta \pm 2\sqrt{\cos^2\theta - 1}}{2}$$

$$= \cos\theta \pm \sqrt{-\sin^2\theta} = \cos\theta \pm i\sin\theta.$$

38. Since $x + \frac{1}{x} = 1$

$$\Rightarrow \quad x^2 - x + 1 = 0 \quad \Rightarrow \quad x = \frac{1 \pm i\sqrt{3}}{2}$$

$$\Rightarrow \quad x = -\omega \text{ or } -\omega^2$$

Now, $x^{4000} = \omega^{4000}$ or $\omega^{8000} = \omega$ or ω^2

$$\therefore \quad p = x^{4000} + \frac{1}{x^{4000}} = \omega + \frac{1}{\omega} \text{ or } \omega^2 + \frac{1}{\omega^2}$$

$$\Rightarrow \quad p = -1 \qquad (\because \quad 1 + \omega + \omega^2 = 0)$$

Secondly, let $q = 2^{2^n} + 1$, where $n > 1$

Now, for $n > 1$, we have

$$2^n = 4\lambda, \text{ where } \lambda \in N$$

$$\therefore \quad q = 2^{4\lambda} + 1 = (16)^{\lambda} + 1$$

$$\Rightarrow \quad q = (10k+6) + 1 = 10k + 7$$

$\therefore$ q = the digit at unit's place in $2^{2^n} + 1 = 7$

Hence, $p + q = 6$.

39. We have $|z - 5i| \le 1$

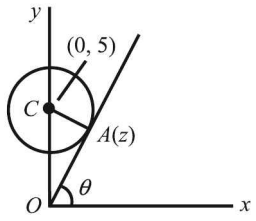


Fig. 2.57

Let $\theta = \angle AOX$ = min. arg(z),

$$\therefore \quad \angle AOC = 90° - \theta \quad \Rightarrow \quad \sin(90° - \theta) = \frac{1}{5}$$

$$\Rightarrow \quad \cos\theta = \frac{1}{5}$$

$$\therefore \quad z = OA\cos\theta + iOA\sin\theta$$

$$\Rightarrow \quad z = \sqrt{5^2-1}\left(\frac{1}{5}\right) + i\sqrt{5^2-1}\sqrt{1-\frac{1}{5^2}}$$

$$= \frac{2\sqrt{6}}{5}(1 + i2\sqrt{6}) = \frac{2\sqrt{6}}{5} + \frac{i24}{5}$$

40. $A = \frac{2}{\sqrt{3}}e^{-i(\pi/6)} = \frac{\sqrt{3}-1i}{\sqrt{3}}$, $B = \frac{2i}{\sqrt{3}}$, $C = \frac{-\sqrt{3}-i}{\sqrt{3}}$

Now $|A-B| = 2 = |B-C| = |C-A|$

$\Rightarrow$ The triangle ABC is equiangular (equilateral).

41. Given: $z\bar{z} + \bar{a}z + a\bar{z} + a\bar{a} = a\bar{a} - \lambda$

$$\Rightarrow \quad (z+a)(\bar{z}+\bar{a}) = a\bar{a} - \lambda$$

$$\Rightarrow \quad |z+a| = \sqrt{a\bar{a} - \lambda} \qquad (\because \quad z\bar{z} = |z|^2)$$

Since $a\bar{a}$ is real, λ should be real. Hence, $\sqrt{a\bar{a} - \lambda}$ represents the radius of the circle. Hence, option (a) is correct.

42. Let $A(z_1)$ and $B(z_2)$ be the centres of the given circles and P be the centre of the variable circle which touches the given circles externally, then $|AP| = a + r$ and $|BP| = b + r$, where r is the radius of the variable circle.

Now, $|AP| - |BP| = |a - b|$,

$\Rightarrow \quad \big||AP| - |BP|\big| = |a - b|$ is a constant.

Hence, the locus of P can possibly be

1. a right bisector of AB, if $a = b$
2. a hyperbola, if $|a - b| < |AB| = |z_2 - z_1|$
3. an empty set, if $|a - b| > |AB| = |z_2 - z_1|$
4. the set of all points on line AB except those which lie between A and B, if $|a - b| = |AB| \neq 0$.

Statement 2 is true. Statement 1 is false.
Hence option (d) is correct.

43. Now, $|z^2 + 2z\sin\alpha| \le |z^2| + |2z\sin\alpha|$

$\le |z^2| + 2|z||\sin\alpha| < (\sqrt{2} - 1)^2 + 2(\sqrt{2} - 1) \cdot 1 \le 1$

$\therefore \quad |z^2 + 2z\sin\alpha| < 1$

Hence option (a) is correct.

44. Now, $\sqrt{(5 + 12i)} = \pm\left[\sqrt{\frac{13+5}{2}} + i\sqrt{\frac{13-5}{2}}\right]$

$= \pm(3 + 2i) \qquad (\because \ b > 0)$

and $\sqrt{(-5 + 12i)} = \pm\left\{\sqrt{\left(\frac{13-5}{2}\right)} + i\sqrt{\left(\frac{13+5}{2}\right)}\right\}$

$= \pm(2 + 3i) \qquad (\because \ b > 0)$

$\therefore \quad z = \pm(3 + 2i) \pm (2 + 3i)$

$\Rightarrow \quad z = 5 + 5i, 1 - i, -1 + i, -5 - 5i$

Hence, the principal values of z are $\pm\frac{\pi}{4}, \pm\frac{3\pi}{4}$. Statement 2 is a true statement.

45. Let $\arg(\omega) = \theta$, then $\arg(z) = \pi - \theta$

$\therefore \quad \omega = |\omega|(\cos\theta + i\sin\theta)$

and $z = |\omega|[\cos(\pi - \theta) + i\sin(\pi - \theta)]$

$= |\omega|(-\cos\theta + i\sin\theta)$

$= -|\omega|(\cos\theta - i\sin\theta) = -\bar{\omega}$

46.

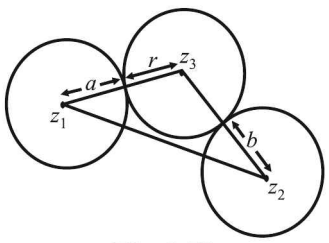


Fig. 2.58

A hyperbola is the locus of a point which moves in such a way that the difference of its distances from two fixed points (foci) is always constant.

Here, $|z_3 - z_1| - |z_3 - z_2| = (a + r) - (b + r)$

$= a - b =$ a constant.

Hence, option (b) is the correct answer.

47. Applying $R_2 \Rightarrow R_2 + R_1$, we get

$$\begin{vmatrix} 6i & -3i & 1 \\ 4 + 6i & 0 & 0 \\ 20 & 3 & i \end{vmatrix} = x + iy$$

$\Rightarrow \quad -(4 + 6i)(-3i^2 - 3) = x + iy$

$\Rightarrow \quad 0 = x + iy \quad \Rightarrow \quad x = 0, \ y = 0$

$\therefore$ Option (d) holds.

48. Applying $R_1 \Rightarrow R_1 + R_3 - R_2$ on the given determinant

$$\begin{vmatrix} (1-i) - (1-i) & \omega^2 + \omega - (-1) & (\omega^2 - 1) - (\omega^2 - 1) \\ 1 - i & -1 & \omega^2 - 1 \\ -i & -1 + \omega - i & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 0 \\ 1 - i & -1 & \omega^2 - 1 \\ -i & -1 + \omega - i & -1 \end{vmatrix} = 0$$

$\therefore$ Option (a) holds.

49. Let $z = r_1 e^{i\theta}$ and $\omega = r_2 e^{i\phi}$

$\Rightarrow \quad \bar{z} = r_1 e^{-i\theta}$

Given $|z\omega| = 1 \quad \Rightarrow \quad |r_1 e^{i\theta} \cdot r_2 e^{i\phi}| = 1 \qquad$ (i)

$\therefore \quad r_1 r_2 = 1$

And also given $\arg(z) - \arg(\omega) = \frac{\pi}{2}$

$\therefore \quad \theta - \phi = \frac{\pi}{2}$

Now, $\bar{z}\omega = r_1 e^{-i\theta} \cdot r_2 e^{i\phi} \qquad$ (ii)

$= r_1 r_2 e^{-i(\theta - \phi)}$

$\therefore$ From (i) and (ii), we get

$$\bar{z}\omega = 1.e^{-i\pi/2} = \cos\frac{\pi}{2} - i\sin\frac{\pi}{2} = -i$$

$\therefore \quad \bar{z}\omega = -i$

50. Given $\left(\frac{1+i}{1-i}\right)^x = 1$

$\Rightarrow \quad \left(\frac{1+i}{1-i} \times \frac{1+i}{1+i}\right)^x = 1$

$$\Rightarrow \quad \left(\frac{2i}{2}\right)^x = 1 \quad \Rightarrow \quad i^x = 1$$

$$\Rightarrow \quad i^x = (i)^{4n} \quad \Rightarrow \quad x = 4n,\ n \in Z^+$$

51. As z_1, z_2 are the roots of $z^2 + az + b = 0$,

$$\therefore \quad z_1 + z_2 = -a,\ z_1 z_2 = b$$

Again O, z_1, z_2 are the vertices of an equilateral triangle.

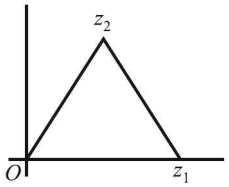


Fig. 2.59

$$\therefore \quad 0^2 + z_1^2 + z_2^2 = 0 z_1 + z_1 z_2 + z_2 0$$

or $$z_1^2 + z_2^2 = z_1 z_2$$

$$\Rightarrow \quad (z_1 + z_2)^2 = 3 z_1 z_2$$

$$\Rightarrow \quad a^2 = 3b^2$$

52. $\bar{z} + i\bar{\omega} = 0 \quad \Rightarrow \quad \overline{\bar{z} + i\bar{\omega}} = 0 \quad \Rightarrow \quad z - i\omega = 0$

Now, $\arg(z\omega) = \pi$

$$\Rightarrow \arg\left(\frac{z^2}{i}\right) = \pi \quad \Rightarrow \quad \arg(z^2) - \arg(i) = \pi$$

$$\Rightarrow \ 2\arg(z) - \frac{\pi}{2} = \pi \quad \Rightarrow \quad \arg(z) = \frac{3\pi}{4}$$

53. $z^{1/3} = p + iq$

$$\therefore \quad x - iy = (p + iq)^3 = p^3 + 3p^2(iq) + 3p(iq)^2 + (iq)^3$$

$$\Rightarrow \quad x = p^3 - 3pq^2 \quad \Rightarrow \quad -y = 3p^2 q - q^3$$

$$\Rightarrow \quad \frac{x}{p} = p^2 - 3q^2 \quad \Rightarrow \quad \frac{-y}{q} = 3p^2 - q^2$$

$$\Rightarrow \quad \frac{x}{p} + \frac{y}{q} = -2(p^2 + q^2)$$

$$\therefore \quad \frac{(x/p) + (y/q)}{p^2 + q^2} = -2$$

$\therefore$ Option (d) holds.

54. $\left|z^2 - 1\right| = |z|^2 + 1$

Let $z = x + iy \quad \Rightarrow \quad (x-1)^2 + y^2 = (x^2 + y^2) + 1$

$$\Rightarrow \quad 2x = 0 \quad \Rightarrow \quad x = 0$$

$\Rightarrow$ z lies on the imaginary axis.

55. $(x-1)^3 + 8 = 0 \Rightarrow \left(\frac{x-1}{-2}\right)^3 = 1 \quad \Rightarrow \quad \frac{x-1}{-2} = (1)^{1/3}$

$$\Rightarrow \quad \frac{x-1}{-2} = 1,\ \omega,\ \omega^2 \quad \Rightarrow \quad x = -1,\ 1 - 2\omega,\ 1 - 2\omega^2$$

56. $\frac{z-1}{i}$ must be real.

Now, $\frac{z-1}{i} = \frac{x - 1 + iy}{i} = y - (x-1)i$ is real

$\therefore \quad x - 1 = 0$ or $x = 1 \quad \Rightarrow \quad \text{Re}(z) = 1$

Then, $\sin^{-1}\left[\frac{1}{i}(z-1)\right] = \sin^{-1} y$ So, $-1 \le \text{Im}(z) \le 1$.

57. $\left|\frac{z}{z - (i/3)}\right| = 1 \quad \Rightarrow \quad |z| = \left|z - \frac{i}{3}\right|$

Clearly, the locus of z is a perpendicular bisector of the line joining points having complex numbers $0 + i0$ and $0 + (i/3)$. Hence, z lies on a straight line.

58. Using $\sum_{k=1}^{n}\left[\sin\left(\frac{2k\pi}{n+1}\right) \mp i\cos\left(\frac{2k\pi}{n+1}\right)\right] = \pm i$, we get

$$\sum_{k=1}^{10}\left[\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right] = -i$$

59. $z^2 + z + 1 = 0 \quad \Rightarrow \quad z = \frac{-1 \pm \sqrt{-3}}{2} = \omega,\ \omega^2$

Putting $z = \omega$,

$$\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \cdots + \left(z^6 + \frac{1}{z^6}\right)^2$$

$$= (-1)^2 + (-1)^2 + (1+1)^2 + (-1)^2 + (-1)^2 + (1+1)^2$$

$$= 12$$

The same result is obtained by putting $z = \omega^2$.

60. z lies on or inside the circle with centre $(-4, 0)$ and radius 3 units.

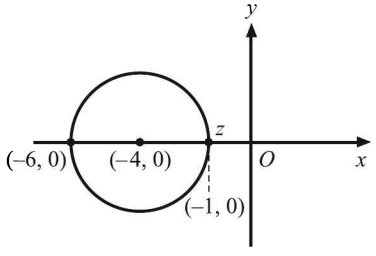


Fig. 2.60

Hence, the maximum distance of z from $(-1, 0)$ is 6 units. See Fig. 2.60.

61. $\bar{z} = \dfrac{1}{i-1} = x - iy$ and $z\bar{z} = |z|^2 = x^2 + y^2$

$\therefore \quad (x + iy)\dfrac{1}{i-1} = x^2 + y^2$ (i)

$\Rightarrow \quad x + iy = i(x^2 + y^2) - (x^2 + y^2)$

$\Rightarrow \quad x = -(x^2 + y^2)$ and $y = x^2 + y^2$

$\Rightarrow \quad x = -y \quad (\because\ x^2 + y^2 = y)$

Now, from (i),

$$\frac{x(1-i)}{i-1} = 2x^2 \quad \Rightarrow \quad 2x^2 + x = 0$$

$\Rightarrow x = 0, x = -(1/2)$

When $x = 0$, $y = 0$; when $x = -(1/2)$, $y = 1/2$

$x = 0, y = 0$ is not admissible.

For $x = -(1/2), y = 1/2$

$$z = x + iy = \frac{1}{2}(-1 + i)$$

$$= \frac{1}{2}\frac{(-1+i)(-1-i)}{-1-i} = -\frac{1}{2}\left(\frac{2}{1+i}\right) = -\frac{1}{1+i}$$

62. We have for any two complex numbers α and β

$\big||\alpha| - |\beta|\big| \le |\alpha - \beta|$

Now, $\left||z| - \left|\dfrac{4}{|z|}\right|\right| \le \left|z - \dfrac{4}{z}\right| \quad \Rightarrow \quad \left||z| - \dfrac{4}{|z|}\right| \le 2$

Set $|z| = r > 0$, then $\left|r - \dfrac{4}{r}\right| \le 2$

$$\Rightarrow \quad -2 \le r - \frac{4}{r} \le 2$$

The left inequality gives $r^2 + 2r - 4 \ge 0$

The corresponding roots are $r = \dfrac{-2 \pm \sqrt{20}}{2} = -1 \pm \sqrt{5}$

Thus, $r \ge \sqrt{5} - 1$ or $r \le -1 - \sqrt{5}$

$\Rightarrow \quad r \ge \sqrt{5} - 1$ (as $r > 0$) (i)

Again consider the right inequality,

$$r - \frac{4}{r} \le 2 \quad \Rightarrow \quad r^2 - 2r - 4 \le 0$$

The corresponding roots are $r = \dfrac{2 \pm \sqrt{20}}{2} = 1 \pm \sqrt{5}$

Thus, $1 - \sqrt{5} \le r \le 1 + \sqrt{5}$

But $r > 0$, hence $r \le 1 + \sqrt{5}$ (ii)

Equations (i) and (ii) give, $\sqrt{5} - 1 \le r \le \sqrt{5} + 1$

So, the maximum value is $\sqrt{5} + 1$.

63. Inequality is not applicable.

64. $z = x + iy$ (i)

$$z^{1/3} = a - ib \quad \Rightarrow \quad z = (a - ib)^3$$

$$z = a^3 - i^3 b^3 - 3ab^2 - 3a^2 bi \quad \text{(ii)}$$

Comparing (i) and (ii), we get

$$a^3 - 3ab^2 = x$$

$$-(-b^3 + 3a^2 b) = y$$

$$\Rightarrow \quad \frac{x}{a} = a^2 - 3b^2 \quad \Rightarrow \quad -\frac{y}{b} = -b^2 + 3a^2$$

$$\Rightarrow \quad \frac{x}{a} - \frac{y}{b} = 4(a^2 - b^2).$$ So, $k = 4$.

65. Let $z_1 = \sin x + i\cos 2x$, $z_2 = \cos x - i\sin 2x$

Suppose $\bar{z}_1 = z_2$

$\Rightarrow \quad \sin x - i\cos 2x = \cos x - i\sin 2x$

$\Rightarrow \quad \sin x = \cos x$ and $\cos 2x = \sin 2x$

$\Rightarrow \quad \tan x = 1$ and $\tan 2x = 1$

$$\Rightarrow \quad x = \frac{\pi}{4} \quad \text{and} \quad x = \frac{\pi}{8}$$

which is not possible at the same time.

Thus, there is no value of x for which z_1 and z_2 are conjugate to each other.

66. $\log_{1/2}\dfrac{|z|^2 + 2|z| + 4}{2|z|^2 + 1} < 0 = \log_{1/2} 1$

$$\Rightarrow \quad \frac{|z|^2 + 2|z| + 4}{2|z|^2 + 1} > 1$$

or $\quad |z|^2 + 2|z| + 4 > 2|z|^2 + 1$

or $\quad |z|^2 - 2|z| - 3 < 0$

or $\quad (|z| + 1)(|z| - 3) < 0$

$\therefore \quad |z| - 3 < 0$ or $|z| < 3$.

67. $e^{iA} \cdot e^{iB} \cdot e^{iC}$

$= (\cos A + i\sin A)(\cos B + i\sin B)(\cos C + i\sin C)$

$= \cos(A + B + C) + i\sin(A + B + C)$

$= \cos\pi + i\sin\pi = -1$

68. We have $x^3 - 3x^2 + 3x + 7 = 0$

$$\Rightarrow \quad (x-1)^3 = (-2)^3$$

$$\Rightarrow \quad \left(\frac{x-1}{-2}\right)^3 = (1)^3$$

$$\Rightarrow \quad x - 1 = -2, -2\omega, -2\omega^2$$

Since the roots are α, β, γ, we have

$$\alpha - 1 = -2, \beta - 1 = -2\omega, \gamma - 1 = -2\omega^2$$

$$\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1} = \frac{1}{\omega} + \frac{1}{\omega} + \omega^2 = \frac{3}{\omega}$$

69. $\dfrac{z^2}{z-1}=\dfrac{\bar{z}^2}{\bar{z}-1}$

$\Rightarrow \quad z\bar{z}\,z-z^2=z\bar{z}\,\bar{z}-\bar{z}^2$

$\Rightarrow \quad |z|^2(z-\bar{z})-(z-\bar{z})(z+\bar{z})=0$

Either $z=\bar{z}$ real axis

or $\quad |z|^2=z+\bar{z} \quad \Rightarrow \quad z\bar{z}-z-\bar{z}=0$

represents a circle passing through the origin.

Alternative method

Let $\quad z=x+iy \quad (\because \quad x\neq 1 \text{ as } z\neq 1)$

$z^2=(x^2-y^2)+i(2xy)$

$z^2/(z-1)$ is real $\Rightarrow$ its imaginary part = 0

$\Rightarrow \quad 2xy(x-1)-y(x^2-y^2)=0$

$\Rightarrow \quad y(x^2+y^2-2x)=0$

$\Rightarrow \quad y=0;\ x^2+y^2-2x=0$

$\therefore$ z lies either on the real axis or on a circle through the origin.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. z_1, z_2, z_3 are the vertices of an equilateral triangle inscribed in a circle $|z|=2$. If $z_1=1+i\sqrt{3}$, then the values of z_2 and z_3 are

(a) $-2, -1-i\sqrt{3}$ (b) $-2, i\sqrt{3}$

(c) $-2, 1-i\sqrt{3}$ (d) none of these

2. For any two complex numbers z_1, z_2 and real numbers a and b, $|az_1-bz_2|^2+|bz_1+az_2|^2$ is equal to

(a) $(a^2+b^2)(|z_1|^2+|z_2|^2)$ (b) $(a^2-b^2)(|z_1|^2+|z_2|^2)$

(c) $(a^2+b^2)(|z_1|+|z_2|)^2$ (d) none of these

3. If z is a complex number such that $|z|=4$ and arg(z) = $5\pi/6$, then z is equal to

(a) $-2\sqrt{3}+2i$ (b) $2\sqrt{3}+i$

(c) $2\sqrt{3}-2i$ (d) $-\sqrt{3}+i$

4. If $\text{Re}\left(\dfrac{z-8i}{z+6}\right)=0$, then z lies on the curve

(a) $x^2+y^2+6x-8y=0$ (b) $4x-3y+24=0$

(c) $4ab$ (d) none of these

5. If $(2+i)(2+2i)(2+3i)\cdots(2+ni)=x+iy$, then $5\cdot 8\cdot 13\ldots(4+n^2)$ is equal to

(a) x^2-y^2 (b) x^2+y^2

(c) x^4-y^4 (d) x^4+y^4

6. If a and b are real numbers between 0 and 1 such that the points $z_1=a+i, z_2=b+i, z_3=0$ form an equilateral triangle, then a and b are equal to

(a) $a=b=2+\sqrt{3}$ (b) $a=b=2-\sqrt{3}$

(c) $a=b=-2+\sqrt{3}$ (d) $a=b=-2-\sqrt{3}$

7. If p, q, r are positive integers, ω is an imaginary cube root of unity and $f(x)=x^{3p}+x^{3q+1}+x^{3r+2}$, then $f(\omega)$ is equal to

(a) 1 (b) 0

(c) -1 (d) none of these

8. The common roots of the equations $z^3+2z^2+2z+1=0$ and $z^{1985}+z^{100}+1=0$ are

(a) ω (b) -1

(c) 1 (d) none of these

9. If $(\cos\theta+i\sin\theta)(\cos 2\theta+i\sin 2\theta)\ldots(\cos n\theta+i\sin n\theta)=1$, then the value of θ is

(a) $4m\pi,\ m\in Z$ (b) $\dfrac{2m\pi}{n(n+1)},\ m\in Z$

(c) $\dfrac{4m\pi}{n(n+1)},\ m\in Z$ (d) $\dfrac{m\pi}{n(n+1)},\ m\in Z$

10. The equation $|z-1|^2+|z+1|^2=2$ represents

(a) a circle of radius 1 (b) a straight line

(c) the ordered pair (0, 0) (d) none of these

11. The maximum and minimum values of $|z+1|$ when $|z+3|\le 3$ are

(a) (5, 0) (b) (6, 0) (c) (7, 1) (d) (5, 1)

12. If $z=x+iy, z^{1/3}=a-ib$ and $\dfrac{x}{a}-\dfrac{y}{b}=\lambda(a^2-b^2)$, then λ is equal to

(a) 1 (b) 2 (c) 3 (d) 4

13. The three vertices of a triangle are represented by the complex numbers 0, z_1 and z_2. If the triangle is equilateral, then

(a) $z_1^2-z_2^2=z_1z_2$ (b) $z_2^2-z_1^2=z_1z_2$

(c) $z_1^2+z_2^2=z_1z_2$ (d) $z_1^2+z_2^2+z_1z_2=0$

14. If the area of the triangle on the Argand plane formed by the complex numbers z, iz, $z+iz$ is 200 square units, then $|z|$ is equal to

(a) 10 (b) 20

(c) 30 (d) none of these

15. If $x=a+b+c, y=a\alpha+b\beta+c$ and $a\beta+b\alpha+c$ where α and β are complex cube roots of unity, then xyz is equal to

(a) $2(a^3+b^3+c^3)$ (b) $2(a^3-b^3-c^3)$

(c) $a^3+b^3+c^3-3abc$ (d) $a^3-b^3-c^3$

16. If $x + (1/x) = 1$, then $x^{2000} + (1/x^{2000})$ is equal to
(a) 1 (b) -1
(c) 0 (d) none of these

17. If ω is the imaginary cube root of unity, then the value of $\tan\left[(\omega^{100} + \omega^{101})\pi + \frac{\pi}{4}\right]$ is
(a) -1 (b) 1
(c) $1/\sqrt{2}$ (d) none of these

18. If $z_r = \cos\frac{2r\pi}{5} + i\sin\frac{2r\pi}{5}$, r = 0, 1, 2, 3, 4, ... then $z_1 \cdot z_2 \cdot z_3 \cdot z_4 \cdot z_5$ is equal to
(a) -1 (b) 0
(c) 1 (d) none of these

19. The curve represented by $\mathrm{Re}(z^2) = 4$ is
(a) a parabola
(b) an ellipse
(c) a circle
(d) a rectangular hyperbola

20. If $(x - 1)^4 - 16 = 0$, then the sum of non-real complex values of x is
(a) 2 (b) 0
(c) 4 (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If $z + \sqrt{2}|z+1| + i = 0$ and $z = x + iy$, then
(a) $x = -2$ (b) $x = 2$
(c) $y = -2$ (d) $y = 1$

2. If $\arg(z) \le 0$, then $\arg(-z) - \arg(z)$ is equal to
(a) -1 (b) π
(c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{2}$

3. If α is the nth root of unity, then $1 + 2\alpha + 3\alpha^2 + \cdots$ to n terms is equal to
(a) $\frac{-n}{(1-\alpha)^2}$ (b) $\frac{-n}{1-\alpha}$
(c) $\frac{-2n}{1-\alpha}$ (d) $\frac{-2n}{(1-\alpha)^2}$

4. In G.P., if the first term and the common ratio are both $\frac{1}{2}(\sqrt{3} + i)$, then the absolute value of its nth term is
(a) 1 (b) 2^n
(c) 4^n (d) none of these

5. Find the least value of n ($n \in N$) for which $\left(\frac{1+i}{1-i}\right)^n$ is real.
(a) 1 (b) 2
(c) 3 (d) 4

6. If $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \cdots + a_{2n}x^{2n}$, then $a_0 + a_3 + a_6 + a_9 + \cdots$ is equal to
(a) 3^n (b) 3^{n-1}
(c) 3^{n+1} (d) none of these

7. If $|z| = \max\left[|z-1|, |z+1|\right]$, then
(a) $|z + \bar{z}| = \frac{1}{2}$ (b) $z + \bar{z} = 1$
(c) $|z + \bar{z}| = 1$ (d) none of these

8. If $z_1 = -3 + 5i$, $z_2 = -5 - 3i$ and z is a complex number lying on the line segment joining z_1 and z_2, then arg(z) can be
(a) $-\frac{3\pi}{4}$ (b) $-\frac{\pi}{4}$
(c) $\frac{\pi}{6}$ (d) $\frac{5\pi}{6}$

9. The expression $\left[\frac{1+i\tan\alpha}{1-i\tan\alpha}\right]^n - \frac{1+i\tan n\alpha}{1-i\tan n\alpha}$ when simplified reduces to
(a) zero (b) $2\sin n\alpha$
(c) $2\cos n\alpha$ (d) none of these

10. If $z = x + iy$ satisfies $\arg(z-1) = \arg(z+3i)$, then the value of $(x-1) : y$ is equal to
(a) 2 : 1 (b) 1 : 3
(c) -1 : 3 (d) none of these

11. The points z_1, z_2, z_3, z_4 in the complex plane taken in order are the vertices of a parallelogram if and only if,
(a) $z_1 + z_2 = z_3 + z_4$ (b) $z_1 + z_3 = z_2 + z_4$
(c) $z_1z_2 = z_3z_4$ (d) none of these

12. If P is the affix of z in the Argand diagram and P moves so that $(z - i)/(z + 1)$ is always purily imaginary, then the locus of z is
(a) circle centre [(1/2), (1/2)], radius $1/\sqrt{2}$
(b) circle centre [$-$(1/2), $-$(1/2)], radius $1/\sqrt{2}$
(c) circle centre (2, 2) and radius (1/2)
(d) none of these

13. Let A, B, C represent the complex numbers z_1, z_2, z_3, respectively, on the complex plane. If the circumcentre of the triangle ABC lies at the origin, then the orthocentre is represented by the complex number
(a) $z_1 + z_2 - z_3$ (b) $z_2 + z_3 - z_1$
(c) $z_3 + z_1 - z_2$ (d) $z_1 + z_2 + z_3$

14. If a, b, c are integers not all equal and ω is a cube root of unity ($\omega \neq 1$), then the minimum value of $|a+b\omega+c\omega^2|$ is

(a) 1 (b) 0

(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$

15. If z_1 and z_2 are two complex numbers and if $\arg\left(\frac{z_1+z_2}{z_1-z_2}\right)=\frac{\pi}{2}$ but $|z_1+z_2|\neq|z_1-z_2|$, then the figure formed by the points represented by 0, z_1, z_2 and z_1+z_2 is

(a) a parallelogram but not a rectangle or a rhombus

(b) a rectangle but not a square

(c) a rhombus but not a square

(d) a square

16. If $\prod_{p=1}^{r} e^{ip\theta}=1$, where Π denotes the continued product, then the most general value of θ is

(a) $\frac{2n\pi}{r(r-1)}$ (b) $\frac{2n\pi}{r(r+1)}$

(c) $\frac{4n\pi}{r(r-1)}$ (d) $\frac{4n\pi}{r(r+1)}$

17. $\sin^{-1}\left[\frac{1}{i}(z-1)\right]$, where z is non-real, can be the angle of a triangle if

(a) $\text{Re}(z)=1, \text{Im}(z)=2$

(b) $\text{Re}(z)=1, 0<\text{Im}(z)\leq 1$

(c) $\text{Re}(z)+\text{Im}(z)=0$

(d) none of these

18. If $|z-i|\leq 2$ and $z_0=5+3i$, then the maximum value of $|iz+z_0|$ is

(a) $2+\sqrt{31}$ (b) 7

(c) $\sqrt{31}-2$ (d) none of these

19. Multiplying a complex number by i rotates the vector representing the complex number through an angle of

(a) 180° (b) 90°

(c) 60° (d) 360°

20. The area of the triangle whose vertices are represented by the complex numbers, $0, z, ze^{i\alpha}$ $(0<\alpha<\pi)$ is equal to

(a) $\frac{1}{2}|z|^2\cos\alpha$ (b) $\frac{1}{2}|z|^2\sin\alpha$

(c) $\frac{1}{2}|z|^2\sin\alpha\cos\alpha$ (d) $\frac{1}{2}|z|^2$

ANSWERS

Level 1

1. (c)	**2.** (a)	**3.** (a)	**4.** (a)	**5.** (b)	**6.** (b)	**7.** (d)	**8.** (a)	**9.** (c)	**10.** (c)
11. (a)	**12.** (d)	**13.** (c)	**14.** (b)	**15.** (c)	**16.** (b)	**17.** (b)	**18.** (c)	**19.** (d)	**20.** (a)

Level 2

1. (a)	**2.** (b)	**3.** (b)	**4.** (a)	**5.** (b)	**6.** (b)	**7.** (d)	**8.** (d)	**9.** (a)	**10.** (b)
11. (b)	**12.** (a)	**13.** (d)	**14.** (a)	**15.** (c)	**16.** (d)	**17.** (b)	**18.** (b)	**19.** (b)	**20.** (b)

3

WILLAM ROWAN HAMILTON

Quadratic Equations

- General Equation in One Variable
- Real Polynomial
- Complex Polynomial
- Degree of Polynomial
- Identity and Equation
- Polynomial Equation
- Roots of an Equation
- Quadratic Equation
- Roots of the Quadratic Equation
- Nature of Roots of Quadratic Equation
- Special Cases
- Sum and Product of Roots of Quadratic Equations
- Sum of Roots
- Product of Roots
- Formation of Quadratic Equation If the Roots are Given
- Elementary Applications
- Symmetric Functions
- Graph of a Quadratic Polynomial
- Sign of Quadratic Expression
- Position of a Point w.r.t. Roots of an Equation

☞ GENERAL EQUATION IN ONE VARIABLE

Let $$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n$$

The above equation in a single variable x where parameters $a_0, a_1, a_2, a_3, ..., a_{n-1}, a_n$ are coefficients of x^n, x^{n-1}, x^{n-2}, …, x and x^0, respectively, is called a *general equation* in one independent variable x or a polynomial in x of the nth degree.

☞ REAL POLYNOMIAL

Let $$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n$$

where a_0, a_1, a_2, …, a_n are real numbers and x is a real independent variable. Then $f(x)$ is called a *real polynominal* of real independent variable x with real coefficients.

☞ COMPLEX POLYNOMIAL

Let $$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

where a_0, a_1, a_2, …, a_n are complex numbers and x is a varying complex independent variable. Then $f(x)$ is called a *complex polynomial*.

DEGREE OF POLYNOMIAL

A polynomial $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, real or complex, is a polynomial of degree n if $a_n \neq 0$ or we can say that the value of the highest power (index) of the variable is known as the degree of the polynomial.

Note Polynomials of degrees 2, 3 and 4 are known as quadratic, cubic and biquadratic polynomials, respectively.

EQUATION

An equation is a statement of equality between two expressions which is not true for all values of the variables involved.

IDENTITY AND EQUATION

A statement of equality of two expressions which is satisfied for each value of a variable is called an *identity*.
For example: $(x - 4)^2 + 8x = x^2 + 16$ is an identity.
A statement of equality between two expressions which is satisfied for definite values of a variable is known as an equation.
For example: $x^2 - 5x + 6 = 0$ is an equation, which is not satisfied for any value of x except 2 and 3.

POLYNOMIAL EQUATION

If $f(x)$ is a polynomial, real or complex, then $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n = 0$ is called the polynomial equation.

ROOTS OF AN EQUATION

The value of the variable satisfying the given equation is called its root. In other words, $x = \alpha$ is the root of $f(x) = 0$, if and only if $f(\alpha) = 0$, and simltaneously if $f(\alpha) = 0$ then α is the root of the equation $f(x) = 0$.

For example: If $f(x) = x^2 + 3x + 2 = 0$
and $f(-1) = (-1)^2 + 3(-1) + 2 = 1 + 2 - 3 = 0$
$\Rightarrow$ $f(-1) = 0$
$\therefore$ -1 is a root of $f(x) = 0$

Note

- There are n real or imaginary roots of an equation of degree n.
- An odd degree equation has at least one real root, whose sign is opposite to that of its last term, provided that the coefficient of the highest degree term is positive.
- Every equation of an even degree, whose constant term is negative and the coefficient of the highest degree term is positive, has at least one negative root.

QUADRATIC EQUATION

Any polynomial equation of degree TWO is called a quadratic equation.
Examples: $ax^2 + bx + c = 0$
$ax^2 + 2hxy + by^2\ 2gx + 2fy + c = 0$

ROOTS OF THE QUADRATIC EQUATION

Let the quadratic equation be $ax^2 + bx + c = 0$
$[a, b, c \in \mathbb{R}$ and $(a \neq 0)]$

Step 1 Take out a such that the coefficient of x^2 is unity. That is,

$$ax^2 + bx + c = 0$$

$$\Rightarrow \quad a\left(x^2 + \frac{bx}{a} + \frac{c}{a}\right) = 0$$

Step 2 Add and subtract the square of half the coefficient of x,

$$a\left\{x^2 + \left(\frac{b}{a}\right)x + \frac{c}{a} + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2\right\} = 0$$

The coefficient of x is (b/a); half the coefficient of x is $(b/2a)$. The square of half the coefficient of x is $(b/2a)^2$.

Step 3 $$a\left[\left(x^2 + \frac{bx}{a} + \left(\frac{b}{2a}\right)^2\right) - \left(\left(\frac{b}{2a}\right)^2 - \frac{c}{a}\right)\right] = 0$$

$$\Rightarrow \quad a\left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b^2 - 4ac}{4a^2}\right)\right] = 0$$

$$\Rightarrow \quad a\left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{\sqrt{b^2 - 4ac}}{2a}\right)^2\right] = 0$$

Step 4 Solve for x now.

$$a\left(x + \frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a}\right)\left(x + \frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}\right) = 0$$

$a = 0$ is not possible $\because\ a \neq 0$

$$x + \frac{b + \sqrt{b^2 - 4ac}}{2a} = 0 \quad \Big| \quad x + \frac{b - \sqrt{b^2 - 4ac}}{2a} = 0$$

$$x = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \quad \Big| \quad x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hence, the two roots of the equation $ax^2 + bx + c = 0$ are $\alpha = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \dfrac{-b - \sqrt{b^2 - 4ac}}{2a}$, where the quantity $b^2 - 4ac$, denoted by D, is called the discriminant of the equation.

GENERAL QUADRATIC EXPRESSION

The general equation of two degrees in two variables, i.e. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ can be resolved into two linear factors if

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc - 2fgh - bg^2 - af^2 - ch^2 = 0$$

NATURE OF ROOTS OF QUADRATIC EQUATIONS

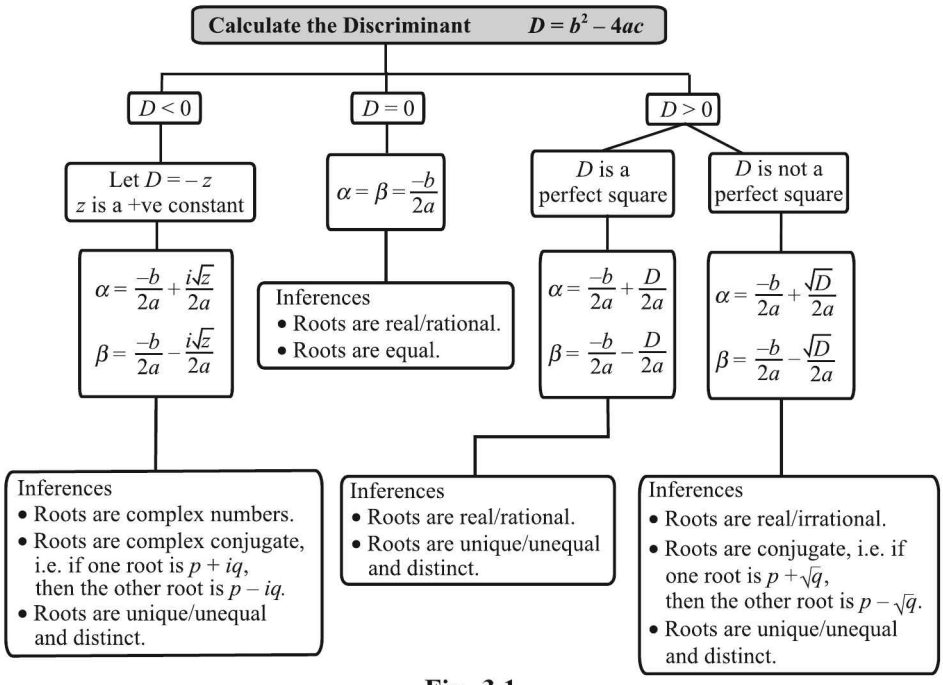


Fig. 3.1

We can examine the nature of roots of quadratic equation $ax^2 + bx + c = 0$ from another angle. From step 3 the quadratic equation $ax^2 + bx + c = 0$ can be written as:

$$\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a^2} = 0, \text{ where } D = b^2 - 4ac \tag{i}$$

If $D < 0$, and since $[x + (b/2a)]^2$ is positive for any x, Eq. (i) will have no real roots. If $D = 0$, then $[x + (b/2a)]^2 = 0$, whence we find $x = -(b/2a)$ which is the only root of the equation.

If $D > 0$, Eq. (i) can be rewritten as follows:

$$\left(x + \frac{b}{2a}\right)^2 - \left(\frac{\sqrt{D}}{2a}\right)^2 = 0$$

Factorizing, we get

$$\left(x + \frac{b}{2a} - \frac{\sqrt{D}}{2a}\right)\left(x + \frac{b}{2a} + \frac{\sqrt{D}}{2a}\right) = 0$$

which yields

$$x = \frac{-b + \sqrt{D}}{2a} \quad \text{and} \quad x = \frac{-b - \sqrt{D}}{2a}$$

or

$$x = \frac{-b \pm \sqrt{D}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{ii}$$

Putting $D = 0$ in Eq. (ii), we get $x = -(b/2a)$. Hence the formula (ii) can be used not only when $D > 0$, but also when $D = 0$.

SUM AND PRODUCT OF ROOTS OF QUADRATIC EQUATION

Let $ax^2 + bx + c = 0$ [$(a, b, c \in \mathbb{R})$ and $(a \neq 0)$] be a quadratic equation having roots α and β. Then it implies that

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

$$\Rightarrow \quad a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) = a(x^2 - (\alpha + \beta)x + \alpha\beta) \quad [\because a \neq 0]$$

SUM OF ROOTS

Sum of roots $(S) = \alpha + \beta$

$$\therefore \qquad S = -\frac{b}{a} = -\left\{\frac{\text{coefficient of } x}{\text{coefficient of } x^2}\right\}$$

PRODUCT OF ROOTS

Product of roots $(P) = \alpha\beta$

$$\therefore \qquad P = \frac{c}{a} = \left\{\frac{\text{constant term}}{\text{coefficient of } x^2}\right\}$$

FORMATION OF QUADRATIC EQUATION IF THE ROOTS ARE GIVEN

If α and β are the roots of a quadratic equation, then according to Factor Theorem, we have

$$(x - \alpha)(x - \beta) = 0$$

$$\Rightarrow \qquad x^2 - (a + b)x + \alpha\beta = 0$$

$\Rightarrow \quad x^2 - Sx + P = 0$, where S = sum of roots and P = product of roots

$\therefore \quad x^2 -$ (sum of roots)x + (product of roots) = 0

ELEMENTARY APPLICATIONS

Let $ax^2 + bx + c = 0$ [$(a, b, c \in \mathbb{R})$ and $(a \neq 0)$] be a quadratic equation having roots α and β.

When one root of the equation is zero,

$\Rightarrow$ Product of roots $\frac{c}{a} = 0$

$\Rightarrow \quad c = 0$

SYMMETRIC FUNCTIONS

A function in α and β is said to be a *symmetric function* of α and β if it remains unaltered/unchanged when α and β are interchanged.

In order to find the value of a symmetric function in α and β, express the given function in terms of sum $(\alpha + \beta)$ or product $(\alpha\beta)$ or both. Some useful results are:

(i) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

(ii) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

(iii) $\alpha^4 + \beta^4 = (\alpha + \beta)(\alpha^3 + \beta^3) - \alpha\beta(\alpha^2 + \beta^2)$

(iv) $\alpha^5 + \beta^5 = (\alpha^2 + \beta^2)(\alpha^3 + \beta^3) - (\alpha^2\beta^3 + a^3\beta^2)$

$= (\alpha^2 + \beta^2)(\alpha^3 + \beta^3) - \alpha^2\beta^2(\alpha + \beta)$

(v) $\alpha - \beta = \sqrt{(\alpha - \beta)^2 - 4\alpha\beta}$

(vi) $\alpha^2 - \beta^2 = (\alpha - \beta)(\alpha + \beta)$

(vii) $\alpha^3 - \beta^3 = (\alpha + \beta)\left\{(\alpha + \beta)^2 - \alpha\beta\right\}$

(viii) $\alpha^4 - \beta^4 = (\alpha^2 + \beta^2)(\alpha^2 - \beta^2)$

$= (\alpha^2 + \beta^2)(\alpha + \beta)(\alpha - \beta)$

QUADRATIC EXPRESSIONS

Graph of a Quadratic Polynomial

Let $y = f(x) = ax^2 + bx + c$ be a quadratic expression or a quadratic polynomial in independent variable x, where a, b, and c are real numbers and $a \neq 0$.

Now, $y = ax^2 + bx + c$

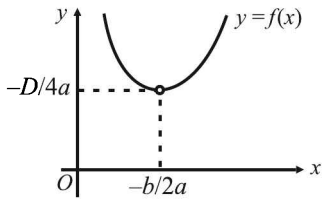


Fig. 3.2

$$\Rightarrow \qquad y = a\left\{x^2 + \frac{b}{a}x + \frac{c}{a}\right\}$$

$$\Rightarrow \qquad y = a\left\{x^2 + \frac{b}{a}x + \frac{c}{a} + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2\right\}$$

$$\Rightarrow \qquad y = a\left\{\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2}\right\}$$

$$\Rightarrow \qquad y = a\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a^2}$$

$$\Rightarrow \qquad y + \frac{D}{4a} = a\left(x + \frac{b}{2a}\right)^2$$

$$\therefore \qquad Y = aX^2$$

which is the equation of the parabola.

We know that the vertex of the parabola $y^2 = 4ax$ is $(0, 0)$. Therefore, in these conditions, we have $(X, Y) = (0, 0)$, which implies that

$$\left(x + \frac{b}{2a}, y + \frac{D}{4a}\right) = (0, 0)$$

$$\Rightarrow \qquad (x, y) = \left(-\frac{b}{2a}, -\frac{D}{4a}\right)$$

Hence, the shape of the curve

$$y = f(x) = ax^2 + bx + c$$

is a parabola, having the vertex at $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$.

☞ SIGN OF QUADRATIC EXPRESSION

Case I: The parabola intersects the x-axis in two real and distinct points.

The parabola of the quadratic equation will intersect the x-axis in two distinct points if $D > 0$.

In this case, the parabola meets the x-axis at

$$\alpha = \frac{-b - \sqrt{D}}{2a} \text{ and } \beta = \frac{-b + \sqrt{D}}{2a}$$

(A) For $a > 0$, we have

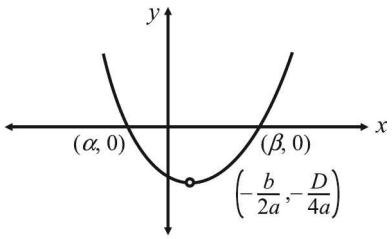


Fig. 3.3

$$y_{\min} = \frac{-D}{4a} \text{ at } x = \frac{-b}{2a},\ y_{\max} \to \infty$$

and $$y = f(x)\begin{cases} < 0 & \text{for} \quad \alpha < x < \beta \\ = 0 & \text{for} \quad x = \alpha, \beta \\ > 0 & \text{for} \quad x < \alpha \text{ or } x > \beta \end{cases}$$

(B) For $a < 0$, we have

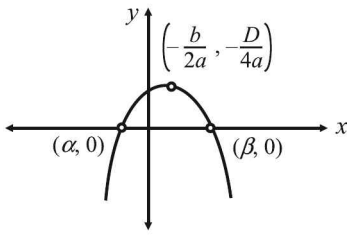


Fig. 3.4

$$y_{\max} = \frac{-D}{4a} \text{ at } x = \frac{-b}{2a},\ y_{\min} \to \infty$$

and $$y = f(x)\begin{cases} < 0 & \text{for} \quad x < \alpha \text{ or } x < \beta \\ = 0 & \text{for} \quad x = \alpha, \beta \\ > 0 & \text{for} \quad \alpha < x < \beta \end{cases}$$

Case II: The parabola touches the x-axis.

The parabola will just touch the x-axis at one point if $D = 0$. In this case, the parabola touches the x-axis at

$$\alpha = \beta = \frac{-b}{2a}$$

(A) For $a > 0$, we have

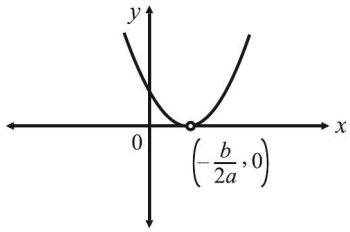


Fig. 3.5

$$y_{\min} = 0 \text{ at } x = \frac{-b}{2a},\ y_{\max} \to \infty$$

and $$y = f(x)\begin{cases} > 0 & \text{for} \quad x \neq \alpha \\ = 0 & \text{for} \quad x = \alpha \end{cases}$$

(B) For $a < 0$, we have

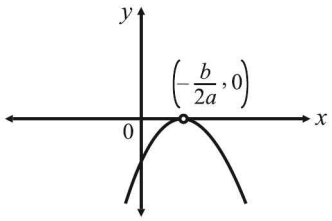


Fig. 3.6

$$y_{\max} = 0 \text{ at } x = \frac{-b}{2a},\ y_{\min} \to \infty$$

and $$y = f(x)\begin{cases} < 0 & \text{for} \quad x \neq \alpha \\ = 0 & \text{for} \quad x = \alpha \end{cases}$$

Case III: The parabola neither touches nor intersects the x-axis.

The parabola will not intersect the x-axis if $D < 0$.

In this case, the parabola remains either completely above the x-axis or completely below the x-axis according as $a > 0$ or $a < 0$.

(A) For $a > 0$, we have

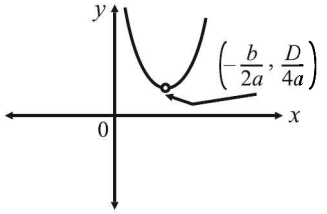
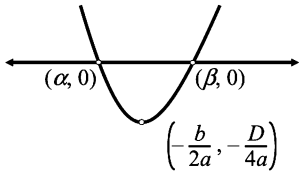
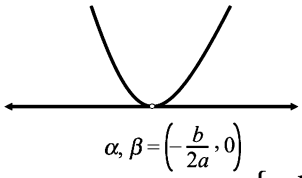
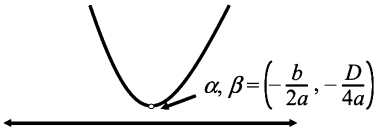
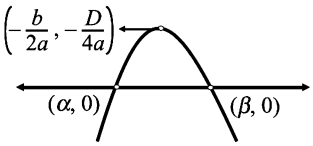
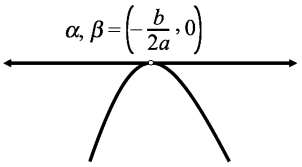
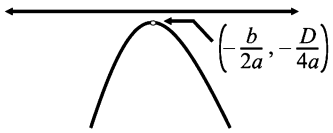
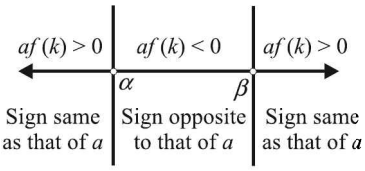
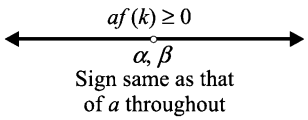
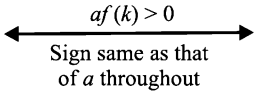


Fig. 3.7

$$y_{\min} = \frac{-D}{4a} \text{ at } x = \frac{-b}{2a},\ y_{\max} \to \infty$$

and $$y = f(x) > 0 \text{ for all } x$$

	$D > 0$	$D = 0$	$D < 0$
$a > 0$			
	$f(x) > 0, x \in (-\infty, \alpha) \cup (\beta, \infty)$	$f(x) > 0, x \in (-\infty, \infty) - \left\{-\frac{b}{2a}\right\}$	$f(x) > 0$, for all $x \in \mathbb{R}$
	$f(x) < 0, x \in (\alpha, \beta)$	$f(x) = 0, x - \left\{-\frac{b}{2a}\right\}$	
		$\therefore \quad f(x) \geq 0$, for all $x \in \mathbb{R}$	
$a < 0$			
	$f(x) > 0, x \in (\alpha, \beta)$	$f(x) < 0, x \in (-\infty, \infty) - \left\{-\frac{b}{2a}\right\}$	$f(x) < 0$, for all $x \in \mathbb{R}$
	$f(x) < 0, x \in (-\infty, \alpha) \cup (\beta, \infty)$	$f(x) = 0, x = \left\{-\frac{b}{2a}\right\}$	
		$\therefore f(x) \leq 0$, for all $x \in \mathbb{R}$	
Inferences			

(B) For $a < 0$, we have

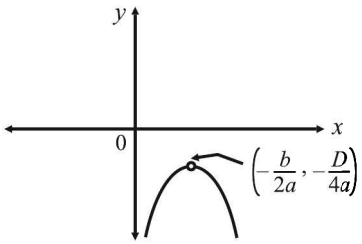


Fig. 3.8

$$y_{max} = \frac{-D}{4a} \text{ at } x = \frac{-b}{2a}, \; y_{min} \to -\infty$$

and $\quad y = f(x) < 0$ for all x

POSITION OF A POINT w.r.t. ROOTS OF AN EQUATION

Let k be any point such that $k \in \mathbb{R}$ and α and β be the roots of the quadratic equation $ax^2 + bx + c = 0$, then

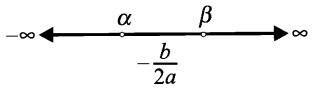


Fig. 3.9

Let (k_1, k_2) be an interval bounded by the points k_1 and k_2 such that $k_1, k_2 \in \mathbb{R}$ and $k_1 < k_2$. Also, α and β are the roots of the quadratic equations, then

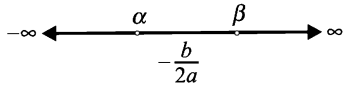


Fig. 3.10

(a)	k	$af(k) > 0$	$k < -\frac{b}{2a}$	k lies outside and is less than both the roots
(b)	k	$af(k) > 0$		k lies between both the roots
(c)	k		$k > -\frac{b}{2a}$	k lies outside and is greater than both the roots

MAXIMUM AND MINIMUM VALUE OF $ax^2 + bx + c$

$\because$ $ax^2 + bx + c$, we have

$$a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right]$$ making the perfect square

$$a\left[x^2 + \frac{b}{a}x + \frac{c}{a} + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2\right]$$

$$\Rightarrow\ a\left[\left(x + \frac{b}{2a}\right)^2 + \left(\frac{c}{a} - \frac{b^2}{4a^2}\right)\right]$$

$$\Rightarrow\ a\left[\left(x + \frac{b}{2a}\right)^2 + \left(\frac{4ac - b^2}{4a^2}\right)\right]$$

$$\Rightarrow\ a\left(x + \frac{b}{2a}\right)^2 + \left(\frac{4ac - b^2}{4a}\right)$$

Case I: When $a > 0$

$\because$ $ax^2 + bx + c \geq \frac{4ac - b^2}{4a}$

$\therefore$ Minimum value of $ax^2 + bx + c$ is $\frac{4ac - b^2}{4a}$ at $x = -\frac{b}{2a}$.

Case II: When $a < 0$

$\because$ $ax^2 + bx + c \leq \frac{4ac - b^2}{4a}$

$\therefore$ Maximum value of $ax^2 + bx + c$ is $\frac{4ac - b^2}{4a}$ at $x = -\frac{b}{2a}$.

SOLVED PROBLEMS

1. The greatest negative integer satisfying $x^2 - 4x - 77 < 0$ and $x^2 > 4$ is

(a) 6 (b) -7

(c) -8 (d) -17

Ans. (a)

Solution: Since $x^2 - 4x - 77 < 0$

$\Rightarrow$ $x^2 - 11x + 7x - 77 < 0$

$\Rightarrow$ $(x - 11)(x + 7) < 0$

$\therefore$ $-7 < x < 11$ (i)

Also, $x^2 > 4$

$\Rightarrow$ $x^2 - 4 > 0$

$\therefore$ $x < -2$ or $x > 2$ (ii)

Using (i) and (ii), we have

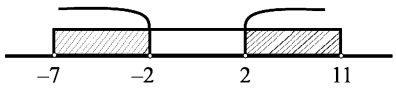


Fig. 3.11

$\therefore$ The gratest negative integer satisfying the above inequality is -6.

2. The product of the real roots of the equation $|2x + 3|^2 - 3|2x + 3| + 2 = 0$ is

(a) 5/4 (b) 5/2

(c) 5 (d) 2

Ans. (b)

Solution: Let $|2x + 3| = y$. The equation becomes

$$y^2 - 3y + 2 = 0$$

$\Rightarrow$ $(y - 1)(y - 2) = 0$

$\Rightarrow$ $y = 1, 2$

$\therefore$ $|2x + 3| = 1, 2$

$\Rightarrow$ $2x + 3 = \pm 1, \pm 2$

$\Rightarrow$ $2x = -3 \pm 1, -3 \pm 2$

$\Rightarrow$ $x = -1, -2, -\frac{1}{2}, -\frac{5}{2}$

Thus, the product of the roots is 5/2.

3. If α and β are the roots of the equation $2x^2 + 6x + b = 0$ $(b < 0)$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is less than

(a) 2 (b) -2

(c) 18 (d) none of these

Ans. (b)

Solution: We have $\alpha + \beta = -3$ and $\alpha\beta = \frac{b}{2}$. Since $b < 0$, the discriminant $D = 36 - 8b > 0$. So, α and β are real. Now,

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2}{\alpha\beta} - 2 = \frac{18}{b} - 2$$

$\Rightarrow$ $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} < -2$ $[\because b < 0]$

4. If α and β are the roots of $ax^2 + bx + c = 0$, then the equation $ax^2 - bx(x - 1) + c(x - 1)^2 = 0$ has roots

(a) $\left(\frac{\alpha}{1-\alpha}\right), \left(\frac{\beta}{1-\beta}\right)$ (b) $\left(\frac{1-\alpha}{\alpha}\right), \left(\frac{1-\beta}{\beta}\right)$

(c) $\left(\frac{\alpha}{1+\alpha}\right), \left(\frac{\beta}{1+\beta}\right)$ (d) $\left(\frac{1+\alpha}{\alpha}\right), \left(\frac{1+\beta}{\beta}\right)$

Ans. (c)

Solution: Since α and β are the roots of $ax^2 + bx + c = 0$, therefore

$$\alpha + \beta = -\frac{b}{a},\ \alpha\beta = \frac{c}{a}$$

The equation $ax^2 - bx(x-1) + c(x-1)^2 = 0$ can be written as

$$x^2(a-b+c) + (b-2c)x + c = 0$$

Sum of the roots of this equation is

$$S = -\frac{b-2c}{a-b+c} = \frac{-b+2c}{a-b+c} = \frac{-\frac{b}{a}+\frac{2c}{a}}{1-\frac{b}{a}+\frac{c}{a}}$$

$$\Rightarrow \quad S = \frac{\alpha+\beta+2\alpha\beta}{1+\alpha+\beta+\alpha\beta} = \frac{\alpha}{\alpha+1} + \frac{\beta}{\beta+1}$$

Product of the roots $= \dfrac{c}{a-b+c}$

$$\Rightarrow \quad P = \frac{\frac{c}{a}}{1-\frac{b}{a}+\frac{c}{a}}$$

$$\Rightarrow \quad P = \frac{\alpha\beta}{1+\alpha+\beta+\alpha\beta} = \frac{\alpha}{\alpha+1}\cdot\frac{\beta}{\beta+1}$$

Thus, $ax^2 - bx(x-1) + c(x-1)^2 = 0$ has $\dfrac{\alpha}{\alpha+1}, \dfrac{\beta}{\beta+1}$ as its two roots.

5. If α and β are the roots of the equation $x^2 + px + p^2 + q = 0$, then the value of $\alpha^2 + \alpha\beta + \beta^2 + q$ = ?

(a) 0 (b) 1

(c) q (d) $2q$

Ans. (a)

Solution: $E = (\alpha+\beta)^2 - \alpha\beta + q = p^2 - (p^2+q) + q = 0$

6. If $5^x + (2\sqrt{3})^{2x} \geq 13^x$, then the solution set for x is

(a) $[2, +\infty]$ (b) $\{2\}$

(c) $(-\infty, 2]$ (d) $[0, 2]$

Ans. (c)

Solution: $\left(\dfrac{5}{13}\right)^x + \left(\dfrac{12}{13}\right)^x \geq 1$

$\therefore \quad \cos^x\alpha + \sin^x\alpha \geq 1$, where $\cos\alpha = \dfrac{5}{13}$

Equality holds for $x = 2$. If $x < 2$, both $\cos\alpha$ and $\sin\alpha$ increase (being positive fractions). So,

$\cos^x\alpha + \sin^x\alpha \geq 1$ if $x < 2$.

Thus, $x \leq 2$.

7. If x, y, z are in G.P. and $x \neq y \neq z$, when x, y, z satisfy the condition $x^n + y^n = z^n$, then

(a) $\left(\dfrac{y}{z}\right)^n = \left(\dfrac{x}{z}\right)^{n/2}$ (b) $\left(\dfrac{y}{z}\right)^n = \dfrac{\sqrt{5}-1}{2}$

(c) $\left(\sqrt{\dfrac{x}{z}}\right)^n = \dfrac{\sqrt{5}-1}{2}$ (d) all of these

Ans. (d)

Solution: Given x, y, z are in G.P.

$$\therefore \quad \frac{y}{x} = \frac{z}{x} \quad \Rightarrow \quad \frac{y^2}{z} = x$$

$$\Rightarrow \quad \frac{y^2}{z^2} = \frac{x}{z} \quad \Rightarrow \quad \left(\frac{y}{z}\right)^2 = \frac{x}{z}$$

$$\Rightarrow \quad \left(\frac{y}{z}\right) = \left(\frac{x}{z}\right)^{1/2} \Rightarrow \left(\frac{y}{z}\right)^n = \left(\frac{x}{z}\right)^{n/2}$$

[Hence (a) is correct.]

Again $x^n + y^n = z^n$

$$\Rightarrow \quad \left(\frac{x}{z}\right)^n + \left(\frac{y}{z}\right)^n = 1 \Rightarrow \left(\frac{y}{z}\right)^{2n} + \left(\frac{y}{z}\right)^n = 1$$

$$\Rightarrow \quad R^2 + R - 1 = 0$$

Let $R = \left(\dfrac{y}{z}\right)^n \quad \Rightarrow \quad R = \dfrac{-1 \pm \sqrt{5}}{2}$

Now let $R = \dfrac{\sqrt{5}-1}{2}$

$$\Rightarrow \quad \left(\frac{y}{z}\right)^n = \frac{\sqrt{5}-1}{2}$$ [(b) is correct.]

Again $\left(\dfrac{y}{z}\right)^n = \left(\dfrac{x}{z}\right)^{n/2} = \dfrac{\sqrt{5}-1}{2}$ [$\therefore$ (c) is also correct)]

8. If $\log_{2x+3}(6x^2 + 23x + 21) = 4 - \log_{3x+7}(4x^2 + 12x + 9)$, then x equals

(a) -4 (b) -2

(c) $-1/4$ (d) none of these

Ans. (c)

Solution: Definition: $\log_a x = t \Rightarrow x = a^t$ such that $a > 0$, $a \neq 1$ Now $\log_{2x+3}(2x+3)(3x+7)$

$$= 4 - \log_{3x+7}(2x+3)^2 \qquad \text{(i)}$$

Now $2x + 3 > 0$ and $2x + 3 \neq 1$

$3x + 7 > 0$ and $3x + 7 \neq 1$

$\therefore \quad 1 + \log_{2x+3}(3x+7) = 4 - \log_{3x+7}(2x+3)^2$

Definition: $\because \log_a(ab) = 1 + \log_a b$

$$\Rightarrow \quad \log_{2x+3}(3x+7) + 2\log_{3x+7}(2x+3) - 3 = 0$$

$$t + \frac{2}{t} - 3 = 0$$

$t^2 - 3t + 2 = 0, \qquad t = \log_{2x+3}(3x+7)$

$\therefore \quad t = 1, t = 2$

$\therefore \quad 2x+3 = 3x+7$ or $(2x+3)^2 = 3x+7$

$x = -4$ or $4x^2 + 9x + 2 = 0$

$x = -4$ or $x = -2, -\dfrac{1}{4}$

$\therefore \quad x = -4, -2, -\frac{1}{4}$

From these three values, only $x = -1/4$ satisfies Eq. (i).

9. If α, β are the roots of the equation $(x - a)(x - b) + c = 0$ $(c \neq 0)$, then the roots of the equation $(x - c - \alpha)(x - c - \beta) = c$ are

(a) a and $b + c$ (b) $a^2 + c^2$ and $b^2 + c^2$

(c) $a + c$ and $a - c$ (d) $a + c$ and $b + c$

Ans. (d)

Solution: Given $(x - a)(x - b) + c = 0$ has roots α and β.

$\therefore \quad x^2 - x(a + b) + (ab + c) = 0$

$\therefore\ S = \alpha + \beta = a + b,\ P = \alpha\beta = ab + c$

Now to find the roots of the equation

$$(x - c - \alpha)(x - c - \beta) = c$$

i.e. $(x^2 + c^2 - 2xc) - (x - c)(a + b) + ab + c = c$

i.e. $(x - c)^2 - (x - c)(a + b) + ab + c = c$

(quadratic in $x - c$)

i.e. $(x - c - a)(x - c - b) = 0$

$\therefore\ x = a + c,\ b + c$ are the required roots.

10. The roots of the equation $(x-\alpha)^3 + (x-\beta)^3 + (x-\gamma)^3 = 0$ are

(a) one real and two imaginary

(b) α, β, γ

(c) all real

(d) all imaginary

Ans. (a)

Solution: 1. Maximum roots of the equation that may exist are 3 and complex roots always exist in pair so choice (d) cannot be possible.

2. If $f(x)$ be a polynomial and $f'(x) > 0$, $\forall x \in \mathbb{R}$, then $f'(x) = 0$ has no real roots; so α, β, γ cannot be the roots of $f(x) = 0$. So choice (b) and (c) cannot be possible so we are left with only one choice which is (a).

11. If α and β are roots of the equation $x^2 - 15x + 1 = 0$, then the value of $\left(\frac{1}{\alpha} - 15\right)^{-2} + \left(\frac{1}{\beta} - 15\right)^{-2}$ is

(a) 225 (b) 900

(c) 223 (d) none of these

Ans. (c)

Solution: α and β are the roots of $x^2 - 15x + 1 = 0$

$\therefore\ \alpha + \beta = 15,\ \alpha\beta = 1$

Now $\left(\frac{1}{\alpha} - 15\right)^{-2} + \left(\frac{1}{\beta} - 15\right)^{-2}$

$= (-\alpha)^{-2} + (-\beta)^{-2}$

$\left(\because\ \frac{1}{\alpha} = \beta,\ \frac{1}{\beta} = \alpha,\ \alpha + \beta = 15\right)$

$= \left(\frac{1}{-\alpha}\right)^2 + \left(\frac{1}{-\beta}\right)^2 = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$

$= (\alpha + \beta)^2 - 2\alpha\beta \qquad (\because\ \alpha\beta = 1)$

$= 225 - 2 = 223$

12. In a triangle PQR, $\angle R = \frac{\pi}{2}$. If $\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of the equation $ax^2 + bx + c = 0$ $(a \neq 0)$, then

(a) $a + b = c$ (b) $b + c = 0$

(c) $a + c = b$ (d) $b = c$

Ans. (a)

Solution: We have, $P + Q = \frac{\pi}{2}$

$\therefore \quad \tan\left(\frac{P+Q}{2}\right) = \tan\frac{\pi}{4}$

$\Rightarrow \quad \frac{\tan(P/2) + \tan(Q/2)}{1 - \tan(P/2) \cdot \tan(Q/2)} = 1 \quad \Rightarrow \quad \frac{-b/a}{1 - (c/a)} = 1$

$\Rightarrow \quad a + b = c$

13. The number of real roots of the equation $|x^2| - 3|x| + 2 = 0$ is

(a) 1 (b) 2

(c) 3 (d) 4

Ans. (d)

Solution: Given equation is $(|x| - 1)(|x| - 2) = 0$

$\Rightarrow \quad |x| = 1,\ |x| = 2$

$\Rightarrow \quad x \pm 1, \pm 2$

14. If α and β are the roots of $4x^2 + 3x + 7 = 0$, then the value of $\frac{1}{\alpha^3} + \frac{1}{\beta^3}$ is

(a) $-(27/64)$ (b) 63/16

(c) 225/343 (d) none of these

Ans. (c)

Solution: $\frac{\alpha^3 + \beta^3}{\alpha^3\beta^3} = \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha^3\beta^3} = \frac{225}{343}$

15. If $a + b + c > 0$ and $a < 0 < b < c$, then the roots of the equation $a(x-b)(x-c) + b(x-c)(x-a) + c(x-a)(x-b) = 0$ are

(a) both positive

(b) both negative

(c) one positive and one negative

(d) complex

Ans. (c)

Solution: The equation is

$$(a + b + c)x^2 - 2(ab + bc + ca)x + 3abc = 0$$

Discriminant $= 4(ab+bc+ca)^2 - 12(a+b+c)abc \geq 0$
$(\because a+b+c>0$ and $abc<0)$
Also, product of roots = abc = negative
Thus, both roots are of opposite nature.

16. If α and β are roots of $ax^2 - 2bx + c = 0$, then $\alpha^3\beta^3 + \alpha^2\beta^3 + \alpha^3\beta^2$ is

(a) $\dfrac{c^2(c+2b)}{a^3}$ (b) $\dfrac{bc^3}{a^3}$

(c) $\dfrac{c^2}{a^3}$ (d) none of these

Ans. (a)

Solution: $\alpha^3\beta^3 + \alpha^2\beta^2(\alpha+\beta) = \alpha^2\beta^2(\alpha\beta+\alpha+\beta)$
$= \dfrac{c^2(c+2b)}{a^3}$

17. If $a > b > c$ and $a^3 + b^3 + c^3 = 3abc$, then the quadratic equation $ax^2 + bx + c = 0$ has roots which are

(a) both real (b) both +ve
(c) both –ve (d) one +ve and one –ve

Ans. (a)

Solution: $a^3+b^3+c^3-3abc$
$= \dfrac{1}{2}(a+b+c)\,[(a-b)^2+(b-c)^2+(c-a)^2] = 0$
$[\because \quad a>b>c$, so $(a-b)^2 + (b-c)^2 + (a-c)^2 > 0]$
$\therefore \qquad a+b+c = 0 \qquad$ (i)
Comparing (i) with $ax^2 + bx + c = 0$, we find $x = 1$.
Also, $ax^2 + bx + c = 0$ for $x \leq c/a$
Therefore, the roots of $ax^2 + bx + c = 0$ are 1 and c/a.

18. If $x + 1$ is a factor of $x^4 + (p - 3)x^3 - (3p - 5)x^2 + (2p - 9)x + 6$, then the value of p is

(a) 2 (b) 4
(c) –4 (d) 0

Ans. (b)

Solution: Let
$f(x) = x^4 + (p-3)x^3 - (3p-5)x^2 + (2p-9)x - 6$
Since $x + 1$ is a factor of $f(x)$, -1 must be a root of $f(x) = 0$.
$\therefore \quad f(-1) = 1 - (p-3) - (3p-5) - (2p-9) + 6 = 0$
$\Rightarrow \qquad -6p + 24 = 0$
$\Rightarrow \qquad p = 4$

EXERCISE SET 1

1. If p and q are distinct rational numbers, then the roots of equation $(p+q)^2x^2 - 5(p^2-q^2)x - 2(p-q)^2 = 0$ are

(a) rational and unequal (b) real and equal
(c) complex (d) none of these

2. If $x = \sqrt{7+4\sqrt{3}}$, then $x + \dfrac{1}{x}$ is equal to

(a) 2 (b) 3
(c) 4 (d) 6

3. If α and β are the roots of the equation $ax^2 + 3x + 2 = 0$ $(a < 0)$, then $\dfrac{\alpha^2}{\beta} + \dfrac{\beta^2}{\alpha}$ is greater than

(a) 0 (b) 1
(c) 2 (d) none of these

4. Let a, b and c be real numbers $(a \neq 0)$. If α is a root of $a^2x^2 + bx + c = 0$, β is a root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies

(a) $2\gamma = \alpha + \beta$ (b) $\gamma - \alpha = \dfrac{\beta}{2}$
(c) $\gamma = \alpha$ (d) $\alpha < \gamma < \beta$

5. If α and β are the roots of the equation $8x^2 - 3x + 27 = 0$, then $\left(\dfrac{\alpha^2}{\beta}\right)^{1/3} + \left(\dfrac{\beta^2}{\alpha}\right)^{1/3}$ is equal to

(a) 1/4 (b) 1/3
(c) 7/2 (d) 1/16

6. If α and β are the roots of $ax^2 + bx + c = 0$ and $\alpha + \beta$, $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$ are in G.P., then

(a) $\Delta \neq 0$ (b) $b\Delta = 0$
(c) $cb \neq 0$ (d) $c\Delta = 0$

where $\Delta = b^2 - 4ac$.

7. The values of p and q $(p \neq 0, q \neq 0)$ for which p and q are the roots of the equation $x^2 + px + q = 0$ are

(a) $p = 1, q = -2$ (b) $p = 1, q = 2$
(c) $p = -1, q = 2$ (d) $p = -1, q = -2$

8. If $a(p + q)^2 + 2bpq + c = 0$ and $a(p + r)^2 + 2bpr + c = 0$, then the value of qr is

(a) $p^2 + \dfrac{a}{c}$ (b) $p^2 + \dfrac{c}{a}$

(c) $p^2 + \dfrac{a}{b}$ (d) $p^2 + \dfrac{b}{a}$

9. The set of all p for which $(p - 2)x^2 + 2(p + 1)x + p + 4 < 3$, for all real x, is given by

(a) $(-\infty, 2)$ (b) $(-1, 2)$
(c) $(-\infty, -1)$ (d) $(-\infty, 2)$

10. If α and β are the roots of $ax^2 + bx + c = 0$; $\alpha + h$ and $\beta + h$ are the roots of $px^2 + qx + r = 0$ and D_1 and D_2 are the respective discriminants of these equations, then $D_1 : D_2 =$

(a) a^2/p^2 (b) b^2/q^2
(c) c^2/r^2 (d) none of these

11. If the equation $\frac{x^2-bx}{ax-c}=\frac{m-1}{m+1}$ has roots equal in magnitude but opposite in sign, then m is equal to

(a) $\frac{a+b}{a-b}$ (b) $\frac{a-b}{a+b}$

(c) $\frac{b-a}{b+a}$ (d) none of these

12. The values of m for which the equation $mx^2 + (2m-1)x + m - 1 = 0$ has roots of opposite sign are given by

(a) no values of m (b) for all values of m

(c) $-1 < m < 0$ (d) $0 < m < 1$

13. The number of real solutions of the equation $2^{x/2} + (\sqrt{2}+1)^x = (5+2\sqrt{2})^{x/2}$ is

(a) infinite (b) 4

(c) 2 (d) 1

14. If the equation $\frac{a}{x-a}+\frac{b}{x-b}=1$ has two roots equal in magnitude and opposite in sign, then the value of $a+b$ is

(a) 0 (b) 1

(c) -1 (d) none of these

15. The values of a for which both the roots of the equation $(a-6)x^2 = a(x-3)$ are positive are given by

(a) $(0, 6)$ (b) $\left(0, \frac{72}{11}\right)$

(c) $\left(6, \frac{72}{11}\right)$ (d) $\left(\frac{72}{13}, 6\right)$

16. If all the real values of x obtained from the equation $4^x - (a-3)2^x + a - 4 = 0$ are non-positive, then

(a) $a \in (4, +\infty)$ (b) $a \in (4,5]$

(c) $a \in (0, 4)$ (d) none of these

17. If the sum of the roots of $ax^2 + bx + c = 0$ is equal to the sum of the squares, then

(a) $2ac = ab + b^2$ (b) $2ab = bc + c^2$

(c) $2bc = ac + c^2$ (d) none of these

18. The equations $kx^2 + x + k = 0$ and $kx^2 + kx + 1 = 0$ have exactly one root in common for

(a) $k = \frac{-1}{2}, 1$ (b) $k = 1$

(c) $k = \frac{-1}{2}$ (d) $k = \frac{1}{2}$

19. The product of the roots of $\sqrt[3]{8+x}+\sqrt[3]{8-x}=1$ is

(a) -189 (b) -5

(c) -21 (d) -9

20. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\frac{1}{\alpha+\beta}, \frac{1}{\alpha}+\frac{1}{\beta}$ is

(a) $acx^2 + (a^2 + bc)x + bc = 0$

(b) $bcx^2 + (b^2 + ac)x + ab = 0$

(c) $abx^2 + (c^2 + ab)x + ca = 0$

(d) none of these

21. Let α and β be the roots of the equation $x^2 + x + 1 = 0$. The equation whose roots are α^{19} and β^7 is

(a) $x^2 - x - 1 = 0$ (b) $x^2 - x + 1 = 0$

(c) $x^2 + x - 1 = 0$ (d) $x^2 + x + 1 = 0$

22. If $x^2 - x - 2$ is a factor of $x^4 - px^2 - q = 0$, then $(p + q)$ equals

(a) 0 (b) 1

(c) -1 (d) 9

23. If x is real, then $\frac{x^2-2x+1}{x+1}$ does not lie

(a) between 0 and 8 (b) between -8 and 8

(c) between -8 and 0 (d) between 16 and 16

24. If α, β and γ are the roots of the equation $x^3 + 3x - 7 = 0$, then the equation whose roots are $2\alpha + 1$, $2\beta + 1$ and $2\gamma + 1$ is

(a) $x^3 - 3x^2 + 15x - 69 = 0$

(b) $x^3 - 3x^2 + 15x + 69 = 0$

(c) $x^3 - 7x + 3 = 0$

(d) none of these

25. If the equation $(\log_{16} x)^2 - (\log_{16} x) + \log_{16} k = 0$ has a unique solution (where k is constant), then the value of k is

(a) 1 (b) 2

(c) 0 (d) 4

26. The equation $(x-a)^3 + (x-b)^3 + (x-c)^3 = 0$ has

(a) all roots real (b) one real, two complex

(c) three real roots a, b, c (d) none of these

27. The solution set of the inequality $(x^2 - x - 2)(x^2 + 5x + 4) \geq 0$ is given by

(a) $[-\infty, -4)$ (b) $[-4, 2)$

(c) $[-\infty, 2) \cup [4, \infty)$ (d) none of these

28. The roots of the equation $(b-c)x^2 + (c-a)x + (a-b) = 0$ are

(a) $\frac{c-a}{a-b}, 1$ (b) $\frac{b-c}{a-b}, 1$

(c) $\frac{a-b}{b-c}, 1$ (d) $\frac{c-a}{b-c}, 1$

29. If one root of the equation is $ix^2 - 2(i+1)x + (2-i) = 0$ is $2 - i$, then the other root is

(a) $-i$ (b) $2 + i$

(c) i (d) $2 - i$

30. If $-3 < \frac{x^2 - ax - 2}{x^2 + x + 1} < 2$ for all $x \neq \mathbb{R}$, then a satisfies
(a) $-6 < a < 7$ (b) $-1 < a < 2$
(c) $-2 < a < 1$ (d) $-7 < a < 6$

31. If α and β are the roots of $ax^2 + bx + c = 0$ and $p_n = \alpha^n + \beta^n$, then $ap_{n+1} + bp_n + cp_{n-1}$ is equal to
(a) 1 (b) 0
(c) $a + b + c$ (d) $-(a + b + c)$

32. If α and β are the roots of $ax^2 + bx + c = 0$, then those of $ax^2 + 2bx + 4c = 0$ are
(a) $\alpha/2, \beta/2$ (b) $2\alpha, 2\beta$
(c) $2\alpha, -2\beta$ (d) $-\alpha, -\beta$

33. $\frac{x+3}{x^2 - x - 2} \geq \frac{1}{x-4}$ for all x satisfying
(a) $-2 < x < 1$ or $x > 4$ (b) $-1 < x < 2$ or $x > 4$
(c) $x < -1$ or $2 < x < 4$ (d) none of these

34. The equation $\sqrt{x + 3 - 4\sqrt{x-1}} + \sqrt{x + 8 - 6\sqrt{x-1}} = 1$ has
(a) only two solutions
(b) only one solution
(c) no solution
(d) more than two solutions

35. If $(2x+1)^2 - |2x+1| - 6 < 0$, then
(a) $-1 < x < 2$ (b) $-2 < x < 1$
(c) $-2 < x < -1$ (d) $1 < x < 2$

36. If $\tan A$ and $\tan B$ are the roots of the equation $x^2 - px + q = 0$, then the value of $\sin^2(A + B)$ is
(a) $\frac{p^2}{p^2 + (1-q)^2}$ (b) $\frac{p^2}{p^2 - (1-q)^2}$
(c) $\frac{p^2}{p^2 - (1-q)^2}$ (d) none of these

37. If $x^2 - 2x\cos\theta + 1 = 0$, then the value of $x^{2n} - 2x^n \cos n\theta + 1$ is equal to
(a) $\cos 2n\theta$
(b) $\sin 2n\theta$
(c) 0
(d) some real number other than 0

38. If $(x^2 - 4|x| + 4)(x^2 - 4|x| + 3) < 0$, then x lies in the interval
(a) $(-3, -1)$ (b) $(1, 3)$
(c) $(2, -1)$ (d) either (a) or (b) above

39. The value of x for which $\log_3(2^{1-x} + 3), \log_9 4$ and $\log_{27}(2^x - 1)^3$ form an A.P. is
(a) 11/6 (b) 6/11
(c) $\log_2(11/6)$ (d) 1

40. The solution set of $\left|\frac{x}{x-1}\right| + |x| = \frac{x^2}{x-1}$ is
(a) $[1, \infty)$ (b) $(-1, \infty)$
(c) $(1, \infty)$ (d) none of these

41. The number of integral solutions of $\frac{x+2}{x^2+1} > \frac{1}{2}$ is
(a) 5 (b) 3
(c) 4 (d) none of these

42. The inequality $|2x - 3| < 1$ is valid when x lies in the interval
(a) (3, 4) (b) (1, 2)
(c) (−1, 2) (d) (−4, 3)

43. If $x^2 + y^2 + z^2 = a$, then the value of $xy + zy + zx$ lies in the interval
(a) $[-a, a]$ (b) $\left[\frac{-a}{2}, \frac{a}{2}\right]$
(c) $\left[\frac{-a}{2}, a\right]$ (d) $\left[-a, \frac{a}{2}\right]$

44. For the equation $\frac{1}{x+a} - \frac{1}{x+b} = \frac{1}{x+c}$, if the product of the roots is zero, then the sum of the roots is
(a) $-\frac{2bc}{b+c}$ (b) $\frac{2bc}{b+c}$
(c) 0 (d) $\frac{2ab}{b+c}$

45. If $a + b + c = 0$, then $x^{a^2/bc} \cdot x^{b^2/ca} \cdot x^{c^2/ab}$ is equal to
(a) 1 (b) x
(c) x^2 (d) x^3

46. If $a + 3b + 9c = 0, ac < 0$ and one root of the equation $ax^2 + bx + c = 0$ is the square of the other, then
(a) a and b have the same sign
(b) b and c have the opposite sign
(c) both the roots are rational
(d) a, b and c are irrational

47. If α and β are the roots of $x^2 + px + 1 = 0$ and γ and δ are the roots of $x^2 + qx + 1 = 0$, then the value of $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$ is equal to
(a) $q^2 - p^2$ (b) q^2
(c) $p^2 - q^2$ (d) p^2

48. If α, β and γ are the roots of the equation $x^3 + ax + b = 0$, then $\frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha^2 + \beta^2 + \gamma^2} = ?$
(a) $3b/2a$ (b) $-3b/2a$
(c) $3b$ (d) $2a$

49. If the roots of the equation $ax^2 + bx + c = 0$ are real and distinct, then
(a) both the roots are greater than $-b/2a$
(b) both the roots are less than $-b/2a$
(c) one of the roots exceeds $-b/2a$
(d) none of these

50. If a, b and c are real numbers such that $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ lies in the interval
(a) $\left[-1, \frac{1}{2}\right]$ (b) $\left[-\frac{1}{2}, 1\right]$
(c) $\left[\frac{1}{2}, 2\right]$ (d) $[-1, 2]$

51. If $x = (\beta - \gamma)(\alpha - \delta)$, $y = (\gamma - \alpha)(\beta - \delta)$ and $z = (\alpha - \beta)(\gamma - \delta)$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is
(a) 0 (b) $\alpha^6 + \beta^6 + \gamma^6 + \delta^6$
(c) $\alpha^6\beta^6\gamma^6\delta^6$ (d) none of these

52. If the equation $ax^2 - 2bx + c = 0$ has real roots which are reciprocal of each other, then one has
(a) $|b| \geq |a|$ (b) $|b| \geq |c|$
(c) $a = c$ (d) all of these

53. If $\log_2 x + \log_x 2 = \frac{10}{3} = \log_2 y + \log_y 2$ and $x \neq y$, then $(x + y) = ?$
(a) 2 (b) 65/8
(c) 37/6 (d) none of these

54. The expression $\frac{x^2 - 2x + p^2}{x^2 + 2x + p^2}$, where $p > 1$, lies between
(a) $\frac{p-1}{p+1}$ and 1 (b) $p^2 - 1$ and $p^2 + 1$
(c) p and p^2 (d) $\frac{p-1}{p+1}$ and $\frac{p+1}{p-1}$

55. a, b and c are positive real numbers forming a G.P. If $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then d/a, e/b, f/c are in
(a) H.P. (b) G.P.
(c) A.P. (d) none of these

56. If a, b and c are distinct negative numbers that are in H.P., then
(a) the equation $ax^2 - 2bx + c = 0$ has distinct real roots
(b) the equation $ax^2 - 2bx + c = 0$ has at least one –ve root
(c) $ax^2 - 2bx + c < 0$ for all real x
(d) $ax^2 - 2bx + c > 0$ for all real x

57. Suppose a and b are rational and α and β are the roots of $x^2 + 2ax + b = 0$, then the equation with rational coefficients with one root $\alpha + \beta\sqrt{\alpha^2 + \beta^2}$ is
(a) $x^2 + 4ax + 2b = 0$ (b) $x^2 + 4ax - 2b = 0$
(c) $x^2 - 4ax + 2b = 0$ (d) $x^2 - 4ax - 2b = 0$

58. If x is real, the expression $\frac{x^2 + 34x - 71}{x^2 + 2x - 7}$ takes all values which do not lie between a and b $(a < b)$, then a and b are
(a) $a < 5$, $b > 9$ (b) 3, 5
(c) 5, 9 (d) none of these

59. If the roots of the equation $x^2 + 2ax + b = 0$ are real and distinct and differ by at most $2m$, then b lies in the interval
(a) $(a^2, a^2 + m^2)$ (b) $(a^2 - m^2, a^2)$
(c) $[a^2 - m^2, a^2)$ (d) none of these

60. If the roots of $ax^2 + 2bx + c = 0$ $(a \neq 0)$ are non-real complex and $a + c < 2b$, then
(a) $c > 0$ (b) $c < 0$
(c) $4a + c < 4b$ (d) $4a + c > 4b$

61. Equation $\frac{a^2}{x - \alpha} + \frac{b^2}{x - \beta} + \frac{c^2}{x - \gamma} = m - n^2x$ (a, b, c, m, $n \in \mathbb{R}$) has necessarily
(a) all the roots real
(b) all the roots imaginary
(c) two real and two imaginary roots
(d) two rational and two irrational roots

EXERCISE SET 2

1. If $a > 0$, $a \neq 1$, then the equation $\log_x a + \log_{ax} a + 3\log_{a^2x} a = 0$ has
(a) infinite number of real roots
(b) two real roots
(c) exactly one real root
(d) no real root

2. If α and β are the roots of $ax^2 + c = bx$, then the equation $(a + cy)^2 = b^2y$ in y has the roots
(a) $\alpha\beta^{-1}, \alpha^{-1}\beta$ (b) α^{-2}, β^{-2}
(c) α^{-1}, β^{-1} (d) α^2, β^2

3. If $x^2 - 3x + 2$ is one of the factors of the expression $x^4 - px^2 + q$, then
(a) $p = 4$ and $q = 5$ (b) $p = 5$ and $q = 4$
(c) $p = -5$ and $q = -4$ (d) $p = -4$ and $q = -5$

4. The solution set of $\left|\frac{x+1}{x}\right| + |x+1| = \frac{|x+1|^2}{|x|}$ is
(a) $\{x | x \geq 1 \text{ or } x \leq -1\}$ (b) $\{-1, 1\}$
(c) $\{x | x \geq 0\}$ (d) $\{x | x > 0\} \cup \{-1\}$

5. If a and b are non-zero distinct roots of $x^2 + ax + b = 0$, then the least value of $x^2 + ax + b$ is
(a) $-9/4$ (b) 1
(c) 2/3 (d) 9/4

6. If one root of the equation $z^2 + (p + iq)z + m + in = 0$ is real, where $p, q, m, n \in \mathbb{R}$, then $n^2 + mq^2 - pqn$ will be equal to
(a) 0 (b) -1
(c) 1 (d) none of these

7. If $a + b + c = 0$ and a, b, c are rational, then the roots of the quadratic equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$ are
(a) rational (b) irrational
(c) imaginary (d) none of these

8. If α and β are the roots of the equation $2x^2 - 35x + 2 = 0$ then the value of $(2a - 35)^3 (2\beta - 35)^3$ is equal to
(a) 8 (b) 1
(c) 64 (d) none of these

9. If S is the set of all real x such that $\dfrac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains
(a) $\left(-\infty, -\dfrac{3}{2}\right)$ (b) $\left(-\dfrac{3}{2}, \dfrac{1}{4}\right)$
(c) $\dfrac{1}{4}, \dfrac{1}{2}$ (d) $\left(\dfrac{1}{2}, 3\right)$

10. If x is real, then the values of $\dfrac{x+2}{2x^2+3x+6}$ are
(a) $-1/3$ and $-1/3$ (b) 1/13 and 1/3
(c) $-1/3$ and 1/13 (d) $-1/13$ and 1/3

11. If α and β are the roots of $ax^2 + bx + c = 0$ and $\alpha + h$ and $\beta + h$ are the roots of $px^2 + qx + r = 0$, then $h = ?$
(a) $-\dfrac{1}{2}\left(\dfrac{a}{b} - \dfrac{p}{q}\right)$ (b) $\left(\dfrac{b}{a} - \dfrac{q}{p}\right)$
(c) $\dfrac{1}{2}\left(\dfrac{b}{a} - \dfrac{q}{p}\right)$ (d) none of these

12. Let $x, y \in [0, 10]$, then the number of solutions (x, y) of the inequation $3^{\sec^2 x - 1}\sqrt{9y^2 - 6y + 2} \le 1$ is
(a) 4 (b) 2
(c) 1 (d) infinite

13. Let $p, q \in \{1, 2, 3, 4\}$. The number of equations of the form $px^2 + qx + 1 = 0$ having real roots is
(a) 7 (b) 8
(c) 9 (d) 15

14. If $\left(m_r, \dfrac{1}{m_r}\right)$; $r = 1, 2, 3, 4$ is four pairs of values of x and y that satisfy the equation $x^2 + y^2 + 2gx + 2fy + c = 0$, then the value of $m_1 \cdot m_2 \cdot m_3 \cdot m_4$ is
(a) 0 (b) 1
(c) -1 (d) none of these

15. For all real values of x, the expression $\dfrac{(x-b)(x-c)}{(x-a)}$ will assume all real values provided
(a) $a \le c \le b$ (b) $b \ge a \ge c$
(c) $b \le c \le a$ (d) $a \ge b \ge c$

16. The number of real roots of the equation $x^2 - x + |5x + 3| = 0$ is
(a) none (b) 1
(c) 2 (d) more than 2

17. For all real values of x, the function $\dfrac{(x-a)(x-b)}{(x-c)}$ will assume all real values provided
(a) $a > b > c$ (b) $a < b < c$
(c) $a > b < c$ (d) $a < c < b$ and $a > c > b$

18. If $x^3 + 3x^2 - 9x + c$ is of the form $(x - \alpha)^2 (x - \beta)$, then $c = ?$
(a) -5 (b) 27
(c) -27 (d) 0

19. If α, β, γ and σ are the roots of the equation $x^4 + 4x^3 - 6x^2 + 7x - 9 = 0$, then the value of $(1 + \alpha^2)(1 + \beta^2)(1 + \gamma^2)(1 + \sigma^2)$ is
(a) 9 (b) 11
(c) 13 (d) 5

20. The set of all real numbers x for which $x^2 - |x + 2| + x > 0$ is
(a) $(-\infty, -2) \cup (2, \infty)$ (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(c) $(-\infty, -1) \cup (1, \infty)$ (d) $(\sqrt{2}, \infty)$

21. If a, and b are the roots of the equation $(x + c)(x + d) = p$, then the roots of the equation $(x - a)(x - b) + p = 0$ are
(a) c, d (b) $-c, -d$
(c) $p + c, p + d$ (d) $p - c, p - d$

22. If the roots of the equation $(a^2 + b^2)t^2 - 2(ac + bd)t + (c^2 + d^2) = 0$ are equal, then
(a) $ab = cd$ (b) $ac = bd$
(c) $ad + bc = 0$ (d) $a/b = c/d$

23. The number of real roots of $(x-1)^2 + (x-2)^2 + (x-3)^2 = 0$ is
(a) 2 (b) 1
(c) 0 (d) 3

24. If a is a positive integer, the number of values of a satisfying $\int_0^{\pi/2}\left[a^2\left(\dfrac{\cos 3x}{4} + \dfrac{3}{4}\cos x\right) + a\sin x - 20\cos x\right]dx \le -\dfrac{a^2}{3}$ is
(a) only one (d) two
(c) three (d) four

25. The greatest intergral value of m for which the equation $(2m-1)x^2-4x+(2m-1)$ has real roots is given by
(a) 0 (b) 1
(c) 2 (d) none of these

26. If the roots of equation $x^2+2cx+ab=0$ are real and un-equal, then the roots of $x^2-2(a+b)x+a^2+b^2+2c^2=0$
(a) real (b) complex
(c) real complex (d) non-real complex

27. The real roots of $|x|^3-3x^2+3|x|-2=0$ are
(a) 0, 2 (b) ± 1
(c) ± 2 (d) 1, 2

28. The number of solutions of the equation $x^2-2-2[x]=0$ (where [ ] denotes the greatest integer function) is
(a) one (b) two
(c) zero (d) infinity

29. The quadratic equation $(1-\sin\theta)x^2+2(1-\sin\theta)x-3\sin\theta=0$ has both roots complex for all θ lying in the interval
(a) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ (b) $\left(0,\frac{3\pi}{2}\right)$
(c) $\left(\frac{\pi}{6},\frac{7\pi}{6}\right)$ (d) $\left(\frac{7\pi}{6},\frac{11\pi}{6}\right)$

30. For what values of k, the equation $x^2-2(1+3k)x+7(3+2k)=0$ has equal roots?
(a) 1 and –10/9 (b) 2 and –10/9
(c) 3 and –10/9 (d) 4 and –10/9

31. Both the roots of the equation $(x-b)(x-c)+(x-c)(x-a)+(x-a)(x-b)=0$ are always
(a) positive (b) negative
(c) real (d) none of these

32. If α and $\beta(\alpha<\beta)$ are the roots of the equation $x^2+bx+c=0$, where $c<0<b$, then
(a) $0<\alpha<\beta$ (b) $\alpha<0<\beta<|\alpha|$
(c) $\alpha<\beta<0$ (d) $\alpha<0<|\alpha|<\beta$

33. If the roots of the equations $x^2+bx+c=0$ and $x^2-cx+b=0$ differ by the same quantity, then the value of $b+c$ is
(a) 4 (b) 1
(c) 0 (d) –4

34. For the equation $3x^2+px+3=0$, $p>0$, if one of the roots is the square of the other then p is equal to
(a) 1/3 (b) 1
(c) 3 (d) 2/3

35. The solution set for $(\sqrt{2+\sqrt{2}})^x+(\sqrt{2-\sqrt{2}})^x=2\cdot 2^{x/4}$ is
(a) {2} (b) {0}
(c) [0, 2] (d) none of these

36. If $x^2+\alpha x+\beta=0$ and $x^2+px+q=0$ have a common root, then the common root is
(a) $\frac{q+\beta}{\alpha+p}$ (b) $\frac{q-\beta}{\alpha+p}$
(c) $\frac{q-\beta}{\alpha-p}$ (d) $-\frac{q+\beta}{\alpha+p}$

37. If the roots of $ax^2+bx+c=0$ are of the forms $\frac{\alpha}{\alpha-1}$ and $\frac{\alpha+1}{\alpha}$, then the value of $(a+b+c)^2$ is
(a) b^2-4ac (b) b^2-2ac
(c) $2b^2-ac$ (d) $4b^2-2ac$

38. The number of distinct real roots of $\begin{vmatrix}\sin x & \cos x & \cos x\\ \cos x & \sin x & \cos x\\ \cos x & \cos x & \sin x\end{vmatrix}=0$ in the interval $-\frac{\pi}{4}\le x\le\frac{\pi}{4}$ is
(a) 0 (b) 2
(c) 1 (d) 3

39. If $2^{x^2}:2^{2x}=8^k:1$, then the equation has only one solution if
(a) $k=1/3$ (b) $k=-1/3$
(c) $k>1/3$ (d) $k<-1/3$

40. If the roots of a quadratic equation $\frac{x-m}{mx+1}=\frac{x+n}{nx+1}$ are reciprocal to each other, then
(a) $n=0$ (b) $m=n$
(c) $m+n=1$ (d) $m^2+n^2=1$

41. If $y=2[x]+3$ and $y=3[x-2]$, then $[x+y]=?$
(a) 20 (b) 25
(c) 30 (d) 40

42. If the equation $\frac{a}{x-a}+\frac{b}{x-b}=1$, has roots equal in magnitude but opposite in sign, then the value of $a+b$ is
(a) –1 (b) 0
(c) 1 (d) none of these

43. If α and $\beta(\alpha<\beta)$ are the roots of the equation $x^2+bx+c=0$, where $c<0<b$, then
(a) $0<\alpha<\beta$ (b) $\alpha<0<\beta<|\alpha|$
(c) $\alpha<\beta<0$ (d) $\alpha<0, |\alpha|<\beta$

44. The maximum value of the quadratic equation $1-x-3x^2=0$, $x\in\mathbb{R}$ is
(a) 1 (b) 1/3
(c) 12/13 (d) 13/12

45. The number of real solutions of the equation $(x+4)^3+(x+3)^3+(x+2)^3+(x+1)^3+(x-5)^3+180=0$ is
(a) none of these (b) 1
(c) 2 (d) 3

46. The number of real solutions of
$\frac{(x+b)(x+c)}{(b-a)(c-a)}+\frac{(x+c)(x+a)}{(c-b)(a-b)}+\frac{(x+a)(x+b)}{(a-c)(b-c)}=1$ is
$(a, b, c \in \mathbb{R})$
(a) 0 (b) 1
(c) 2 (d) infinite

47. The minimum value of the quadratic equation $x^2+2bx+c=0$ is
(a) cb^2 (b) c^2b
(c) $c+b^2$ (d) $c-b^2$

48. If the sum of the roots of the equation $x^2+px+q=0$ is equal to the sum of their squares, then
(a) $p^2-q^2=0$ (b) $p^2+q^2=2q$
(c) $p^2+p=2q$ (d) none of these

49. The equation $(\sqrt{5}+2)^{|x|}+(\sqrt{5}-2)^{|x|}=18$ has
(a) only one solution
(b) two solutions
(c) no solution
(d) any number of solutions

50. If $a \in \mathbb{R}$, the number of distinct real solutions of $x^2-|x|+a=0$ cannot be
(a) 3 (b) 1
(c) 2 (d) 4

51. The greatest value of the expression $\frac{1}{4x^2+2x+1}$, $x \in \mathbb{R}$ is
(a) 4/3 (b) 5/2
(c) 13/14 (d) 14/17

52. If α and β are the roots of the equation $x^2-p(x+1)-q=0$, then the value of $\frac{\alpha^2+2\alpha+1}{\alpha^2+2\alpha+q}+\frac{\beta^2+2\beta+1}{\beta^2+2\beta+q}$ is
(a) 2 (b) 3
(c) 0 (d) 1

53. The roots of the equation $(3-x)^4+(2-x)^4=(5-2x)^4$ are
(a) all real
(b) all imaginary
(c) two real and two imaginary
(d) none of these

54. The greatest integral value of m for which the equation $(2m-3)x^2-4x+(2m-3)=0$ has both negative roots is given by
(a) 1 (b) 2
(c) 3 (d) none of these

55. The minimum value of the quadratic equation $2x^2+x+1=0$, $x \in \mathbb{R}$ is
(a) 7/8 (b) 1/2
(c) 1 (d) 8/7

56. If x is real, then the least value of the expression $(ax^2+bx+c), a>0$ is
(a) $-\frac{b}{2a}$ (b) b^2-4ac
(c) $\frac{4ac-b^2}{4a}$ (d) $\frac{4ac-b^2}{4a^2}$

57. The number of positive integral n for which the equation $\frac{x-8}{n-10}=\frac{n}{x}$ has no real solution is
(a) 5 (b) 6
(c) 7 (d) 8

58. If α and β are the roots of the equation $2x^2-3x-6=0$, then the equation whose roots are α^2+2 and β^2+2 is
(a) $x^2-4x+59=0$ (b) $x^2-59x+118=0$
(c) $4x^2-49x+118=0$ (d) $x^2-51x+123=0$

59. Let α and β be the roots of the equation $(x-a)(x-b)=c$, $c \neq 0$. Then the roots of the equation $(x-\alpha)(x-\beta)+c=0$ are
(a) a, c (b) b, c
(c) a, b (d) $a+c, b+c$

60. If p, q and r are +ve and are in A.P., then the roots of the quadratic equation $px^2+qx+r=0$ are all real for
(a) $\left|\frac{r}{p}-7\right| \geq 4\sqrt{3}$ (b) $\left|\frac{p}{r}-7\right| \geq 4\sqrt{3}$
(c) all p and r (d) no p and r

61. If the roots of the equation $\frac{x^2-bx}{ax-c}=\frac{\lambda-1}{\lambda+1}$ are such that $\alpha+\beta=0$, then the value of λ is
(a) $\frac{a-b}{a+b}$ (b) c
(c) $\frac{1}{c}$ (d) $\frac{a+b}{a-b}$

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c), (d) given below.

(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1
(c) Statement 1 is true; Statement 2 is false
(d) Statement 1 is false; Statement 2 is true

62. Let a, b, c, p and q be real numbers. Suppose α and β are the roots of the equation $x^2 + 2px + q = 0$ and α and $1/\beta$ are the roots of the equation $x^2 + 2bx + c = 0$, where $\beta^2 \notin (-1, 0, 1)$.

Statement 1: $(p^2 - q)(b^2 - ac) \geq 0$

Statement 2: $b \neq pa$ or $c \neq qa$

63. **Statement 1:** If $a + b + c = 0$ and a, b and c are rational, then the roots of the equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$ are rational.

Statement 2: Discriminant of the equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$ is a perfect square.

64. **Statement 1:** If both the roots of the equation $4x^2 - 2x + a = 0, a \in \mathbb{R}$ lie in the interval $(-1, 1)$, then $-2 < a \leq \frac{1}{4}$.

Statement 2: If $f(x) = 4x^2 - 2x + a$ then $D \geq 0$, $f(-1) > 0$ and $f(1) > 0 \Rightarrow -2 < a \leq \frac{1}{4}$.

65. **Statement 1:** The greatest integral value of λ for which $(2\lambda - 1)x^2 - 4x + (2\lambda - 1) = 0$ has real roots is 2.

Statement 2: For real roots of $ax^2 + bx + c = 0, D \geq 0$.

66. **Statement 1:** Let $f(x)$ be a quadratic expression such that $f(0) + f(1) = 0$. If -2 is one of the roots of $f(x) = 0$, then the other root is 3/5.

Statement 2: If α and β are the zeros of $f(x) = ax^2 + bx + c$, then the sum of zeros $= -b/a$ and the product of zeros $= c/a$.

67. **Statement 1:** The equation $-x^2 + x - 1 = \sin^4 x$ has only one solution.

Statement 2: If the curves $y = f(x)$ and $y = g(x)$ cut at one point, the number of solutions is 1.

68. **Statement 1:** If both the roots of the equation $2x^2 - x + a = 0$ $(a \in \mathbb{R})$ lie in $(1, 2)$, then $-1 < a \leq 1/8$.

Statement 2: If $f(x) = 2x^2 - x + a$, then $D \geq 0$, $f(1) > 0, f(2) > 0$ yield $-1 < a \leq 1/8$.

Previous Years' Questions

69. The number of real roots of $3^{2x^2 - 7x + 7} = 9$ is

(a) 0 (b) 2
(c) 1 (d) 4

[AIEEE 2002]

70. If the difference between the corresponding roots of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is same and $a \neq b$, then

(a) $a + b - 4 = 0$ (b) $a + b + 4 = 0$
(c) $a - b - 4 = 0$ (d) $a - b + 4 = 0$

[AIEEE 2002]

71. If p and q are the roots of the equation $x^2 + px + q = 0$, then

(a) $p = 1, q = -2$ (b) $p = 0, q = 1$
(c) $p = -2, q = 0$ (d) $p = -2, q = 1$

[AIEEE 2002]

72. The value of a for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is

(a) 2/3 (b) –2/3
(c) 1/3 (d) –1/3

[AIEEE 2003]

73. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in

(a) G.P. (b) H.P.
(c) A.G.P. (d) A.P.

[AIEEE 2003]

74. The number of real solutions of the equation $x^2 - 3|x| + 2$ is

(a) 3 (b) 2
(c) 4 (d) 1

[AIEEE 2003]

75. If $(1 - p)$ is a root of the quadratic equation $x^2 + px + (1 - p) = 0$, then its roots are

(a) 0, 1 (b) –1, 1
(c) 0, –1 (d) –1, 2

[AIEEE 2004]

76. Let two numbers have A.M. as 9 and G.M. as 4. Then these numbers are the roots of the equation

(a) $x^2 + 18x - 16 = 0$ (b) $x^2 - 18x + 16 = 0$
(c) $x^2 + 18x + 16 = 0$ (d) $x^2 - 18x - 16 = 0$

[AIEEE 2004]

77. If one root of the equation $x^2 + px + 12 = 0$ is 4 while the equation $x^2 + px + q = 0$ has equal roots, then one value of q is

(a) 3 (b) 12
(c) 49/4 (d) 4

[AIEEE 2004, UP SEE 2009]

78. The value of a for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assumes the least value is

(a) 0 (b) 1
(c) 2 (d) 3

[AIEEE 2005]

79. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval

(a) [4, 5] (b) $(-\infty, 4)$
(c) $(6, \infty)$ (d) (5, 6]

[AIEEE 2005]

80. If the roots of the equation $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c$ equals
(a) 3 (b) –2 (c) 1 (d) 2
[AIEEE 2005]

81. If the roots of the quadratic equation $x^2 - px + q = 0$ are tan 30° and tan 15° respectively, then the value of $2 + q - p$ is
(a) 2 (b) 3 (c) 0 (d) 1
[AIEEE 2006]

82. If x is real, the maximum value of $\frac{3x^2+9x+17}{3x^2+9x+7}$ is
(a) 1/4 (b) 41 (c) 1 (d) 17/7
[AIEEE 2006]

83. All the values of m for which both the roots of the equation $x^2-2mx+m^2-1=0$ are greater than -2 but less than 4, lie in the interval
(a) $-2<m<0$ (b) $m>3$
(c) $-1<m<3$ (d) $1<m<4$
[AIEEE 2006]

84. If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is
(a) $(-3, \infty)$ (b) $(3, \infty)$ (c) $(-3, -\infty)$ (d) $(-3, 3)$
[AIEEE 2007]

85. How many real solutions does the equation $x^7 + 14x^5 + 16x^3 + 30x - 560 = 0$ have?
(a) 1 (b) 3 (c) 5 (d) 7
[AIEEE 2008]

86. The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The other roots of the first and second equations are integers in the ratio 4 : 3. Then, the common root is
(a) 1 (b) 4 (c) 3 (d) 2
[AIEEE 2008]

87. If the roots of the equation $bx^2 + cx + a = 0$ are imaginary, then for all real values of x the expression $3b^2x^2+6bcx+2c^2$ is
(a) greater than $4ab$ (b) less than $4ab$
(c) greater than $-4ab$ (d) less than $-4ab$
[AIEEE 2009]

88. If $x=8+3\sqrt{7}$ and $xy=1$, then the value of $\frac{1}{x^2}+\frac{1}{y^2}$ is
(a) 254 (b) 192 (c) 292 (d) 66
[BVP 2007]

89. If one root of the equation $(x - 1)(7 - x) = m$ is three times the other, then m equals
(a) –5 (b) 0 (c) 2 (d) 5
[AMU 1989]

90. If a, b and c are the sides of a ΔABC such that $x^2 - 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ has real roots, then
(a) $\lambda<\frac{4}{3}$ (b) $\lambda>\frac{5}{3}$
(c) $\lambda\in\left(\frac{4}{3},\frac{5}{3}\right)$ (d) $\lambda\in\left(\frac{1}{3},\frac{5}{3}\right)$
[IIT JEE 2006]

91. If a, b and c are real numbers such that $a^2+b^2+c^2=1$, then $ab + bc + ca$ lies in the interval
(a) $\left[\frac{1}{2},2\right]$ (b) $[-1, 2]$
(c) $\left[-\frac{1}{2},1\right]$ (d) $\left[-1,\frac{1}{2}\right]$
[Roorke 1989, IIT 1984]

92. The real roots of the equation $x^{2/3}+x^{1/3}-2=0$ are
(a) 1, 8 (b) –1, –8 (c) –1, 8 (d) 1, –8
[UP SEE 2004]

93. Let α and β be the roots of the equation $x^2 - px + r = 0$ and $\alpha/2$ and 2β be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is
(a) $\frac{2}{9}(p-q)(2q-p)$ (b) $\frac{2}{9}(q-p)(2p-q)$
(c) $\frac{2}{9}(q-2p)(2q-p)$ (d) $\frac{2}{9}(2p-q)(2q-p)$
[IIT 2007]

94. If x is real, then the expression $\frac{x^2+34x-71}{x^2+2x-7}$ can have no value between
(a) 3 and 7 (b) 4 and 8
(c) 5 and 9 (d) 6 and 10
[Roorkee 1989]

95. If the roots of the equation $x^n - 1 = 0$ are $1, a_1, a_2, \ldots, a_{n-1}$ then the value of $(1-a_1)(1-a_2)(1-a_3)\ldots(1-a_{n-1})$ will be
(a) n (b) n^2 (c) n^n (d) 0
[UP SEAT 1990]

96. The equation formed by decreasing each root of $ax^2+bx+c=0$ by 1 is $2x^2+8x+2=0$ when
(a) $a = -b$ (b) $b = -c$
(c) $c = -a$ (d) $b = a + c$
[EAM CET 2000]

97. Let α and β be the roots of $x^2-x+p=0$ and γ and δ be the roots of $x^2-4x+q=0$. If α, β, γ and δ are in G.P. then the integral values of p and q respectively are
(a) –2, –32 (b) –2, 3 (c) –6, 3 (d) –6, –32
[IIT Screeing 2001]

98. The equation $x-\frac{2}{x-1}=1-\frac{2}{x-1}$ has
(a) no root (b) one root
(c) two equal roots (d) infinitely many roots
[IIT 1984]

99. Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down the coefficient of x to get the roots (3, 2). The correct roots of the equation are
(a) 6, 1 (b) 4, 3 (c) $-6,-1$ (d) $-4,-3$
[AIEEE 2011]

100. Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is
(a) 3 (b) 9 (c) 6 (d) 18
[AIEEE 2011]

101. The equation $e^{\sin x} - e^{\sin x} - 4 = 0$ has
(a) infinite number of real roots
(b) no real roots
(c) exactly one real root
(d) exactly four real roots
[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (a)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (a)	**6.** (d)	**7.** (a)	**8.** (b)	**9.** (c)	**10.** (a)
11. (b)	**12.** (d)	**13.** (d)	**14.** (a)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (c)	**19.** (a)	**20.** (b)
21. (d)	**22.** (b)	**23.** (c)	**24.** (a)	**25.** (b)	**26.** (b)	**27.** (d)	**28.** (c)	**29.** (a)	**30.** (b)
31. (b)	**32.** (b)	**33.** (d)	**34.** (d)	**35.** (b)	**36.** (a)	**37.** (c)	**38.** (d)	**39.** (d)	**40.** (c)
41. (b)	**42.** (b)	**43.** (c)	**44.** (a)	**45.** (d)	**46.** (a)	**47.** (a)	**48.** (a)	**49.** (c)	**50.** (b)
51. (a)	**52.** (d)	**53.** (d)	**54.** (d)	**55.** (c)	**56.** (c)	**57.** (a)	**58.** (c)	**59.** (c)	
60. (b) and (c)		**61.** (a)							

Exercise Set 2

1. (b)	**2.** (b)	**3.** (b)	**4.** (d)	**5.** (a)	**6.** (a)	**7.** (a)	**8.** (c)	**9.** (a) and (d)	
10. (d)	**11.** (c)	**12.** (a)	**13.** (a)	**14.** (b)	**15.** (c) and (d)		**16.** (a)	**17.** (d)	**18.** (c)
19. (c)	**20.** (b)	**21.** (b)	**22.** (d)	**23.** (c)	**24.** (d)	**25.** (b)	**26.** (d)	**27.** (c)	**28.** (a)
29. (d)	**30.** (b)	**31.** (c)	**32.** (b)	**33.** (d)	**34.** (c)	**35.** (b)	**36.** (c)	**37.** (a)	**38.** (c)
39. (b)	**40.** (a)	**41.** (c)	**42.** (b)	**43.** (b)	**44.** (d)	**45.** (b)	**46.** (d)	**47.** (d)	**48.** (c)
49. (b)	**50.** (b)	**51.** (a)	**52.** (d)	**53.** (c)	**54.** (a)	**55.** (a)	**56.** (c)	**57.** (b)	**58.** (c)
59. (c)	**60.** (b)	**61.** (a)	**62.** (b)	**63.** (a)	**64.** (a)	**65.** (d)	**66.** (a)	**67.** (d)	**68.** (a)
69. (b)	**70.** (b)	**71.** (a)	**72.** (a)	**73.** (b)	**74.** (c)	**75.** (c)	**76.** (b)	**77.** (c)	**78.** (b)
79. (b)	**80.** (c)	**81.** (b)	**82.** (b)	**83.** (c)	**84.** (d)	**85.** (a)	**86.** (d)	**87.** (c)	**88.** (a)
89. (d)	**90.** (a)	**91.** (c)	**92.** (d)	**93.** (d)	**94.** (c)	**95.** (a)	**96.** (b)	**97.** (a)	**98.** (a)
99. (a)	**100.** (d)	**101.** (b)							

HINTS AND SOLUTIONS—EXERCISE SET 1

1. $D = 25(p^2-q^2)^2 + 8(p+q)^2(p-q)^2$
$= 25(p-q)^2(p+q)^2 + 8(p+q)^2(p-q)^2$
$= 33(p-q)^2(p+q)^2 > 0 \quad (\because p \neq q)$

2. $x = \sqrt{7+4\sqrt{3}} = \sqrt{4+3+2.2\sqrt{3}}$
$= \sqrt{(2+\sqrt{3})^2} = 2+\sqrt{3}$

$\therefore \quad \frac{1}{x} = \frac{1}{2+\sqrt{3}} = \frac{2-\sqrt{3}}{4-3} = 2-\sqrt{3}$

$\Rightarrow \quad x+\frac{1}{x} = (2+\sqrt{3})+(2-\sqrt{3}) = 4$

3. Since $a < 0$, therefore discriminant $D = 9 - 8a > 0$. So, α and β are real.

We have: $\alpha+\beta = (-3/a)$ and $\alpha\beta = (2/a)$

$\therefore \quad \frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha} = \frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}$

$= \frac{(\alpha+\beta)^3}{\alpha\beta} - 3(\alpha+\beta) = -\frac{27}{2a^2}+\frac{9}{a} < 0 \quad [\because a < 0]$

4. We have: $a^2\alpha^2 + b\alpha + c = 0$ (i)

$a^2\beta^2 - b\beta - c$ (ii)

Let $f(x) = a^2x^2 + 2bx + 2c$

$\therefore \quad f(\alpha) = a^2\alpha^2 + 2(b\alpha + c)$
$= a^2\alpha^2 + 2(-a^2\alpha^2)$ [from (i)]
$= -a^2\alpha^2 < 0$

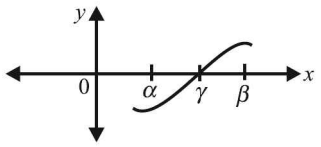


Fig. 3.12

and $f(\beta) = a^2\beta^2 + 2b\beta + 2c = a^2\beta^2 + 2a^2\beta^2 > 0$

$\because$ $f(\alpha) < 0$ and $f(\beta) > 0$, also $f(x)$ is continuous

$\therefore$ $f(\gamma) = 0$, if $\alpha < \gamma < \beta$

5. $\alpha + \beta = \dfrac{3}{8}$

$\alpha\beta = \dfrac{27}{8}$

Expression $= \left(\dfrac{\alpha^2}{\beta}\right)^{1/3} + \left(\dfrac{\beta^2}{\alpha}\right)^{1/3}$

$\Rightarrow \dfrac{\alpha+\beta}{(\alpha\beta)^{1/3}} = \dfrac{\frac{3}{8}}{\left(\frac{27}{8}\right)^{1/3}} = \dfrac{1}{4}$

6. $(\alpha^2 + \beta^2)^2 = (\alpha+\beta)(\alpha^3+\beta^3)$ ($\because$ in G.P.)

or $\left(\dfrac{b^2-2ac}{a^2}\right)^2 = \left(-\dfrac{b}{a}\right)\left(\dfrac{-b^3+3abc}{a^3}\right)$

$\therefore$ $4a^2c^2 = acb^2$ or $ac(b^2 - 4ac) = 0$

or $c\Delta = 0$ as $a \neq 0$, for otherwise the given quadratic equation will reduce to the first degree.

7. Sum of the roots $= p + q = -\dfrac{p}{1}$ (i)

Product of the roots $= pq = \dfrac{q}{1}$ (ii)

From (ii), we have $p = 1$

$\therefore$ From (i), we have $1 + q = -1 \Rightarrow q = -2$

8. The given relations show that q and r are the roots of equation

$$a(p+x)^2 + 2bpx + c = 0$$

or $ax^2 + 2p(a+b)x + ap^2 + c = 0$

$\therefore$ qr = product of roots $= \dfrac{ap^2+c}{a} = p^2 + \dfrac{c}{a}$

9. Let $f(x) = (p-2)x^2 + 2(p+1)x + (p+1)$

$f(x) < 0$ for all p, if

$p - 2 < 0$ and discriminant, $D < 0$

$\Rightarrow$ $p < 2$ and $4(p+1)^2 - 4(p-2)(p+1) < 0$

$\Rightarrow$ $p < 2$ and $p < -1$

Combining, we get $p < -1$

10. Let $A = \alpha + h$ and $B = \beta + h$, then

$A - B = (\alpha + h) - (\beta + h) = \alpha - \beta$

$\Rightarrow$ $(A-B)^2 = (\alpha-\beta)^2$

$\Rightarrow$ $(A+B)^2 - 4AB = (\alpha+\beta)^2 - 4\alpha\beta$

$\Rightarrow$ $\dfrac{b^2}{a^2} - \dfrac{4c}{a} = \dfrac{q^2}{p^2} - \dfrac{4r}{p}$

$\Rightarrow$ $\dfrac{b^2-4ac}{a^2} = \dfrac{q^2-4rp}{p^2}$

$\Rightarrow$ $\dfrac{D_1}{a^2} = \dfrac{D_2}{p^2} \Rightarrow \dfrac{D_1}{D_2} = \dfrac{a^2}{p^2}$

11. $S = 0$

$\therefore$ Coefficient of $x = 0$

or $-b(m+1) - a(m-1) = 0$

or $m = \dfrac{a-b}{a+b}$

12. $D = (2m-1)^2 - 4m(m-1) = 1 > 0$

Product of roots $= \dfrac{m-1}{m} < 0$

$\Rightarrow$ $0 < m < 1$

13. $\left(\dfrac{\sqrt{2}}{\sqrt{5+2\sqrt{2}}}\right)^x + \left(\dfrac{\sqrt{2}+1}{\sqrt{5+2\sqrt{2}}}\right)^x = 1$

which is of the form

$\cos^x\alpha + \sin^x\alpha = 1$

$\therefore$ $x = 2$; hence one real solution.

14. The given equation is $x^2 - 2x(a+b) + 3ab = 0$

$\alpha = -\beta \Rightarrow a + \beta = 0 \Rightarrow a + b = 0$

15. The given equation is $(a-6)x^2 - ax + 3a = 0$

$D = a^2 - 4(a-6)\cdot 3a > 0 \Rightarrow 0 < a < \dfrac{72}{11}$ (i)

Also, $\alpha + \beta = \dfrac{a}{a-6} > 0 \Rightarrow a < 0$ or $a > 6$ (ii)

and $\alpha\beta = \dfrac{3a}{a-6} > 0 \Rightarrow a < 0$ or $a > 6$

$\therefore$ $6 < a < \dfrac{72}{11}$ [From Eqs. (i) and (ii)]

16. $(2^x)^2 - (a-4)2^x - 2^x + (a-4) = 0$

$\Rightarrow$ $(2^x - a + 4)(2^x - 1) = 0$

$\therefore$ $2^x = 1$ or $2^x = a - 4$

As $x \leq 0, 0 < a - 4 \leq 1$

17. $\alpha + \beta = a^2 + \beta^2 = (a+\beta)^2 - 2\alpha\beta$

$\therefore$ $-\dfrac{b}{a} = \left(-\dfrac{b}{a}\right)^2 - \dfrac{2c}{a}$

or $2ac = ab + b^2$

18. Let α be the common root. Then

$$k\alpha^2+\alpha+k=0$$
$$k\alpha^2+k\alpha+1=0$$

Solving, we get $\dfrac{\alpha^2}{1-k^2}=\dfrac{\alpha}{k^2-k}=\dfrac{1}{k^2-k}$

$$\Rightarrow \quad \frac{1-k^2}{k^2-k}=\frac{k^2-k}{k^2-k}=1$$
$$\Rightarrow \quad 2k^2-k-1=0$$
$$\Rightarrow \quad k=\frac{-1}{2},\ 1$$

For $k=1$, equations are identical, thus not possible. Hence, $k=-1/2$.

19. Cubing the given equation both sides, we have $8+x+8-x+3(64-x^2)^{1/3}=1$

$$\Rightarrow \quad (64-x^2)^{1/3}=-5$$
$$\Rightarrow \quad 64-x^2=-125$$
$$\Rightarrow \quad x^2=189$$
$$\Rightarrow \quad x=\pm 3\sqrt{21}$$

20. $S=\dfrac{1}{\alpha+\beta}+\dfrac{\alpha+\beta}{\alpha\beta}=-\dfrac{a}{b}-\dfrac{b}{c}=-\dfrac{ac+b^2}{bc}$

$$P=\frac{1}{\alpha+\beta}\cdot\frac{\alpha+\beta}{\alpha\beta}=\frac{1}{\alpha\beta}=\frac{a}{c}$$

$[\because P=$ product of roots, $S=$ sum of roots$]$

$\therefore$ Equation is $x^2-Sx+P=0$

$$\Rightarrow \quad x^2+\frac{ac+b^2}{bc}x+\frac{a}{c}=0$$
$$\Rightarrow \quad bcx^2+(b^2+ac)x+ab=0$$

21. The roots of $x^2+x+1=0$ are ω and ω^2. Therefore,

$\alpha=\omega, \beta=\omega^2, \quad \therefore \quad \alpha^{19}=\omega^{19}=\omega$

$\beta^7=\omega^{14}=\omega^2$

Hence α^{19} and β^7 are the roots of $x^2+x+1=0$.

22. We have $x^2-x+2=(x-2)(x+1)$

i.e. $x=2$ and $x=-1$ are the roots of the equation $x^4-px^2-q=0$

At $x=-1$, $1-p-q=0$ or $p+q=1$

23. Let $y=\dfrac{x^2-2x+1}{x+1}$

or $\quad x^2-x(y+2)+(1-y)=0$

If $\quad x\in\mathbb{R}$, then $D\geq 0$

$$\Rightarrow \quad (y+2)^2-4(1-y)\geq 0$$
$$\Rightarrow \quad y^2+8y\geq 0$$
$$\Rightarrow \quad y(y+8)\geq 0$$
$$\Rightarrow \quad y\notin[-8,\ 0]$$

24. Let $z=2\alpha+1$

$$\therefore \quad \frac{z-1}{2}=\alpha$$

Also, α is the root of $x^3+3x-7=0$

$$\therefore \quad \left(\frac{z-1}{2}\right)^3+3\left(\frac{z-1}{2}\right)-7=0$$

$\Rightarrow \quad z^3-3z^2+15z-69=0$, having a root $2\alpha+1=z$.

The other two roots being $2\beta+1$ and $2\gamma+1$.

25. $D=0$ because the equation has a unique solution.

So, $\quad 1-4\log_{16}k=0$

then $\quad k=(16)^{1/4}=2$

26. If the given equation is $f(x)=0$, then

$$f'(x)=3[(x-a)^2+(x-b)^2+(x-c)^2]$$

Clearly $f'(x)$ is always +ve for real values of x and hence $f'(x)=0$ has no real roots. Therefore, $f(x)=0$ has one real root and consequently two imaginary roots.

Note $f'(x)=0$ when $x=a, b, c$ all together but this is not possible as a quadratic equation cannot have more than two roots.

27. $(x-2)(x+1)^2(x+4)\geq 0$

$\Rightarrow \quad x\leq -4$

$\Rightarrow \quad x\geq 2$ and $x=-1$. Hence none of the given options (a) to (c) apply.

28. 1 is a root because $(b-c)+(c-a)+(a-b)=0$

Let the other root be α.

$$\therefore \quad \alpha\cdot 1=\frac{a-b}{b-c} \quad \text{or} \quad \alpha=\frac{a-b}{b-c}$$

29. One is tempted to say that if one root is $2-i$, the other root will be $2+i$ as complex roots occur in conjugate pairs. This is so only when the coefficients are real. Here the coefficients are not real. Hence, if the other root is α, then

$$\alpha(2-i)=\text{Product}=\frac{2-i}{i}$$
$$\therefore \quad \alpha=\frac{1}{i}=-i \quad \Rightarrow \quad (a) \text{ is correct}$$

30. Since $x^2+x+1>0$ for all $x\in\mathbb{R}$

$(\because a=1>0$ and $D<0)$

we have $-3x^2-3x-3<x^2-ax-2<2x^2+2x+21$

Solving the inequality $-3x^2-3x-3$, we get $-1<a<7$

Solving the inequality x^2-ax-2, we get $-6<a<2$

Combining, we get $-1<a<2$

31. Expression $=a\left[p_{n+1}+\left(\dfrac{b}{a}\right)p_n+\left(\dfrac{c}{a}\right)p_{n-1}\right]$

$$=a[(\alpha^{n+1}+\beta^{n+1})-(\alpha+\beta)(\alpha^n+\beta^n)+\alpha\beta(\alpha^{n-1}+\beta^{n-1})]$$
$$=0$$

32. $x^2+2\frac{b}{a}x+4\frac{c}{a}=0$

or $x^2-2(\alpha+\beta)x+4\alpha\beta=0$

$\Rightarrow (x-2\alpha)(x-2\beta)=0$

$\therefore$ Roots are $2\alpha, 2\beta$.

33. $\frac{x+3}{(x+1)(x-2)}-\frac{1}{x-4}\geq 0$

$\Rightarrow \frac{(x+3)(x-4)-(x+1)(x-2)}{(x+1)(x-2)(x-4)}\geq 0$

$\Rightarrow \frac{-10}{(x+1)(x-2)(x-4)}\geq 0$

$\Rightarrow x<-1,\ 2<x<4$

34. Put $\sqrt{x-1}=t$

or $x=t^2+1$

so that the given equation becomes

$$\sqrt{t^2+4-4t}+\sqrt{t^2+9-6t}=1$$

or $$\sqrt{(t-2)^2}+\sqrt{(t-3)^2}=1$$

or $$|t-2|+|t-3|=1$$

This equation is satisfied for all values of t lying between 2 and 3, i.e. $2\leq t\leq 3$.

Thus, the given equation is satisfied for all values of x lying between 5 and 10. Hence, there are more than two solutions.

35. If $y=|2x+1|$, then we have

$y^2-y-6<0$

or $(y-3)(y+2)<0$

$\Rightarrow -2<y<3$

$\Rightarrow |2x+1|<3 \quad (\because |2x+1|>-2$ is always true)

$\Rightarrow -2<x<1$

36. We have

$\tan A+\tan B=p$

$\tan A\cdot\tan B=q$

Now, $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}=\frac{p}{1-q}$

$\therefore \sin^2(A+B)=\frac{1-\cos 2(A+B)}{2}$

$$=\frac{1}{2}\left[1-\frac{1-\tan^2(A+B)}{1+\tan^2(A+B)}\right]$$

$$=\frac{\tan^2(A+B)}{1+\tan^2(A+B)}$$

$$=\frac{[p/(1-q)]^2}{1+[p/(1-q)]^2}=\frac{p^2}{(1-q)^2+p^2}$$

37. From the given relation,

$x=\cos\theta\pm i\sin\theta$

Take $x=\cos\theta+i\sin\theta$

then $x^n=\cos n\theta+i\sin n\theta$

and $1/x^n=\cos n\theta-i\sin n\theta$

$\therefore x^n+(1/x^n)=2\cos n\theta$

Hence, $x^{2n}-2x^n\cos n\theta+1=0$

The same result is obtained if we choose $x=\cos\theta-i\sin\theta$

38. $(|x|-2)^2(|x|-1)(|x|-3)<0$

$\Rightarrow 1<|x|<3, x\neq\pm 2$

$\Rightarrow 1<x<3$

or $-3<x<-1, \quad x\neq\pm 2$

39. Condition for A.P. is: $2b=a+c$

$\Rightarrow 2\log_9 4=\log_3(2^{1-x}+3)+\log_{27}(2^x-1)^3$

Make the base 3.

$$\frac{2}{2}\log 4=\log(2^{1-x}+3)+\frac{3}{3}\log(2^x-1)$$

$\therefore 4=\left(\frac{2}{y}+3\right)(y-1)$, where $y=2^x$

$\therefore 4y=(3y+2)(y-1)$ or $3y^2-5y-2=0$

$\therefore y=2,-1/3=2^x$

Since an exponential function cannot be –ve,

$\therefore 2^x=-1/3$ is rejected

$\therefore 2^x=2 \Rightarrow x=1$

40. $\left|\frac{x}{x-1}\right|+|x|=\frac{x^2}{x-1}=x+\frac{x}{x-1}$

It is possible, if $|x|=x$ and $\left|\frac{x}{x-1}\right|=\frac{x}{x-1}$

$\therefore x\geq 0$ and $\frac{x}{x-1}\geq 0; \quad x\neq 1$

$\therefore x\geq 0$ and $x>1 \Rightarrow x>1$

41. $\frac{x+2}{x^2+1}>\frac{1}{2}$

$\Rightarrow -x^2+2x+3>0$

$[\because x^2+1$ is always positive$]$

By sign scheme, $-1<x<3$

As x is an integer, $x=0, 1, 2$

42. $-1<2x-3<1$ or $1<x<2$

43. $(x+y+z)^2=\Sigma x^2+2\Sigma xy\geq 0$

$\Rightarrow a+2\Sigma xy\geq 0$

$\Rightarrow xy+yz+zx\geq -\frac{a}{2}$

Also $x^2+y^2+z^2-(xy+yz+zx)=$

$$\frac{1}{2}\{(x-y)^2+(y-z)^2+(z-x)^2\}$$

$\Rightarrow \quad x^2 + y^2 + z^2 - (xy + yz + zx) \geq 0$

$\Rightarrow \quad xy + yz + zx \leq a$

$\therefore \quad \left[\frac{-a}{2}, a\right]$ is the required value.

44. We have

$$\frac{(x+b)-(x+a)}{(x+a)(x+b)} = \frac{1}{(x+c)}$$

$\Rightarrow \quad (b-a)x + c(b-a) = x^2 + (a+b)x + ab$

$\Rightarrow \quad x^2 + 2ax + ab + ac - bc = 0$

Product of roots $= \frac{ab+ac-bc}{1} = 0$ (Given)

$\therefore \quad ab + ac - bc = 0$

$\Rightarrow \quad a = \frac{bc}{b+c}$

Now, sum of roots $= -\frac{2a}{1} = -\frac{2bc}{b+c}$

45. L.H.S. is $x^{\frac{a^3+b^3+c^3}{abc}} = x^{\frac{3abc}{abc}} = x^3$

$\because \quad a^3 + b^3 + c^3 = 3abc$ when $a + b + c = 0$

46. Let $f(x) = ax^2 + bx + c$

$$f\left(\frac{1}{3}\right) = \frac{a}{9} + \frac{b}{3} + c = \frac{1}{9}(a + 3b + 9c) = 0$$

$\therefore \quad \frac{1}{3}$ is the root of the given equation

Also, product of the roots $= \frac{c}{a} = \frac{ac}{a^2} < 0$. Therefore, the other root must be negative. Hence, it will be $-\frac{1}{\sqrt{3}}$ and the required equation is

$$\left(x - \frac{1}{3}\right)\left(x + \frac{1}{\sqrt{3}}\right) = 0$$

$\Rightarrow \quad 3\sqrt{3}x^2 + (3 - \sqrt{3})x - 1 = 0$

47. We have

$$\alpha + \beta = -p, \quad \alpha\beta = 1$$
$$\gamma + \delta = -q, \quad \gamma\delta = 1$$

Now, $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$

$= (\alpha\beta - \gamma(\alpha + \beta) + \gamma^2)(\alpha\beta + \delta(\alpha + \beta) + \delta^2)$

$= (1 + p\gamma + \gamma^2)(1 - p\delta + \delta^2)$

$= (\gamma^2 + 1 + p\gamma)(\delta^2 + 1 - p\delta)$

$= (-q\gamma + p\gamma)(-q\delta - p\delta)$

$(\because \quad \gamma^2 + q\gamma + 1 = 0, \delta^2 + q\delta + 1 = 0)$

$= (q - p)(q + p)\gamma\delta$

$= (q^2 - p^2) \cdot 1$

$= q^2 - p^2$

48. $\alpha + \beta + \gamma = 0$ as the coefficient of $x^2 = 0$

$\therefore \quad \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma = -3b$

$\alpha^2 + \beta^2 + \gamma^2 = (\Sigma\alpha)^2 - 2\Sigma\alpha\beta = 0 - 2a = -2a$

$\therefore \quad \frac{\Sigma\alpha^3}{\Sigma\alpha^2} = \frac{-3b}{-2a} = \frac{3b}{2a}$

49. The roots of the given equation are

$$\alpha = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

Since α and β are real and distinct, therefore $b^2 - 4ac > 0$. Now if $\alpha > 0$, then $\beta > \frac{-b}{2a}$. And if $\beta > 0$, then $\alpha > -\frac{b}{2a}$

Thus, one of the roots exceeds $-b/2a$.

50. Let $x = ab + bc + ca$

We have $a^2 + b^2 + c^2 = 1$

$\Rightarrow \quad (a + b + c)^2 - 2(ab + bc + ca) = 1$

$\Rightarrow \quad (a + b + c)^2 = 1 + 2x$

$\Rightarrow \quad 1 + 2x \geq 0$

$\Rightarrow \quad x \geq -\frac{1}{2}$

Also $(a - b)^2 + (b - c)^2 + (c - a)^2$

$= 2(a^2 + b^2 + c^2) - 2(ab + bc + ca)$

$\Rightarrow \quad 0 \leq 2(1) - 2x$

$\Rightarrow \quad x \leq 1$

$\therefore \quad x \in \left[-\frac{1}{2}, 1\right]$

51. $x + y + z = (\beta - \gamma)(\alpha - \delta) + (\gamma - \alpha)(\beta - \delta) + (\alpha - \beta)(\gamma - \delta) = 0$

$\therefore \quad x^3 + y^3 + z^3 = 3xyz$

or $\quad x^3 + y^3 + z^3 - 3xyz = 0$

52. $a \cdot \frac{1}{\alpha} = \frac{c}{\alpha}$

$\Rightarrow \quad c = a$

$D = 4b^2 - 4ac > 0$

$\Rightarrow \quad b^2 - ac > 0$

$\Rightarrow \quad b^2 > c^2 \quad (\because a = c)$

$\Rightarrow \quad |b| > |c|$

$\Rightarrow \quad |b| > |a| \quad (\because a = c)$

53. $\log_2 x + \dfrac{1}{\log_2 x} = 3 + \dfrac{1}{3} = \log_2 y + \dfrac{1}{\log_2 y}$

either $\log_2 x = 3$ or $\log_2 y = \dfrac{1}{3}$ [$\because x \neq y$]

$\Rightarrow$ $x = 2^3$ and $y = 2^{1/3}$

$\therefore$ $x + y = 8 + 2^{1/3}$

54. Let $y = \dfrac{x^2 - 2x + p^2}{x^2 + 2x + p^2}$

$\therefore$ $yx^2 + 2yx + p^2 y - x^2 + 2x - p^2 = 0$

$\Rightarrow$ $(y-1)x^2 + (2y+2)x + p^2 y - p^2 = 0$

Since x is real, we have

$$(2y+2)^2 - 4(y-1)(p^2 y - p^2) \geq 0$$

$\Rightarrow$ $$(y+1)^2 - p^2 (y-1)^2 \geq 0$$

$\Rightarrow$ $$(1-p^2)y^2 + (2+2p^2)y + (1-p^2) \geq 0$$

$\Rightarrow$ $$y^2 + \frac{2+2p^2}{1-p^2} y + 1 \leq 0$$

$\Rightarrow$ $$y^2 - \frac{2p^2+2}{p^2-1} y + 1 \leq 0$$

$\Rightarrow$ $$y^2 - \left(\frac{p-1}{p+1} + \frac{p+1}{p-1}\right) y + \left(\frac{p-1}{p+1}\right)\left(\frac{p+1}{p-1}\right) \leq 0$$

$\Rightarrow$ $$\left(y - \frac{p-1}{p+1}\right)\left(y - \frac{p+1}{p-1}\right) \leq 0$$

$\Rightarrow$ $$\frac{p-1}{p+1} \leq y \leq \frac{p+1}{p-1}$$

55. For the first equation $D = 4b^2 - 4ac = 0$ (as given a, b, c, are in G.P.).

$\Rightarrow$ The equation has equal roots which are equal to $-b/a$ each.

Thus, it should also be the root of the second equation.

Thus, $$d\left(\frac{-b}{a}\right)^2 + 2e\left(\frac{-b}{a}\right) + f = 0$$

$\Rightarrow$ $$d\frac{b^2}{a^2} - 2\frac{be}{a} + f = 0$$

$\Rightarrow$ $$d\frac{ac}{a^2} - 2\frac{be}{a} + f = 0 \qquad (\text{as } b^2 = ac)$$

$\Rightarrow$ $$\frac{d}{a} + \frac{f}{c} = 2\frac{be}{ac} = 2\frac{e}{b}$$

Hence d/a, e/b and f/c are in A.P.

56. Since a, b, c are in H.P.,

$\therefore$ $$b = \frac{2ac}{a+c}$$

Also, $D = 4b^2 - 4ac$

Putting $b = \dfrac{2ac}{a+c}$ and solving, we get

$$D = 4\left(\frac{2ac}{a+c}\right)^2 - 4ac = -4ac\left(\frac{a-c}{a+c}\right)^2 = -\text{ve}$$

$\therefore$ $ax^2 - 2bx + c < 0$ for all $x \in \mathbb{R}$

57. Since α, β are the roots of $x^2 + 2ax + b = 0$ [given]

$\therefore$ $\alpha + \beta = -2a$ and $\alpha\beta = b$

Let $\gamma = \alpha + \beta + \sqrt{\alpha^2 + \beta^2}$

$\Rightarrow$ $(\gamma + 2a)^2 = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\Rightarrow$ $(\gamma + 2a)^2 = 4a^2 - 2b$

$\Rightarrow$ $\gamma^2 + 4a\gamma + 2b = 0$

Hence, the required equation is $x^2 + 4ax + 2b = 0$

58. $\dfrac{x^2 + 34x - 71}{x^2 + 2x - 7} = \alpha$

$\Rightarrow$ $(\alpha - 1)x^2 + 2(\alpha - 17)x - 7\alpha + 71 = 0$

Since x is real, we have

$$4(\alpha - 17)^2 - 4(\alpha - 1)(-7\alpha + 71) \geq 0$$

$\Rightarrow$ $$\alpha^2 - 14\alpha + 45 \geq 0$$

$\Rightarrow$ $$(\alpha - 5)(\alpha - 9) \geq 0$$

$\Rightarrow$ $\alpha \leq 5$ or $\alpha \geq 9$

$\therefore$ $a = 5, b = 9$

59. Let the roots be α, β

$\therefore$ $\alpha + \beta = -2a$ and $\alpha\beta = b$

Given, $|\alpha - \beta| \leq 2m$

$\Rightarrow$ $|\alpha - \beta|^2 \leq (2m)^2$

$\Rightarrow$ $(\alpha + \beta)^2 - 4\alpha\beta \leq 4m^2$

$\Rightarrow$ $4a^2 - 4b \leq 4m^2$

$\Rightarrow$ $a^2 - m^2 \leq b$ and $D > 0$ or $4a^2 - 4b > 0$

$\Rightarrow$ $a^2 - m^2 \leq b$ and $b < a^2$

Hence, $b \in [a^2 - m^2, a^2)$

60. Since roots of $ax^2 + 2bx + c = 0$ are non-real complex,

$\therefore$ $f(x) = ax^2 + 2bx + c > 0$ or < 0 for all x

But $f(-1) = a - 2b + c < 0$

$\therefore$ $f(0)$ and $f(-2)$ must be less than zero

$f(0) < 0 \Rightarrow c < 0$

and $f(-2) < 0$

$\Rightarrow$ $4a + c < 4b$

61. Let $p + iq$ be a root of the given equation.

$$\therefore \quad \frac{a^2}{p-\alpha+iq}+\frac{b^2}{p-\beta+iq}+\frac{c^2}{p-\gamma+iq} = m-n^2(p+iq)$$

$$\Rightarrow \quad \frac{a^2(p-\alpha-iq)}{(p-\alpha)^2+q^2}+\frac{b^2[(p-\beta)-iq]}{(p-\beta)^2+q^2}+\frac{c^2[(p-\gamma)-iq]}{(p-\gamma)^2+q^2} = m-n^2p-in^2q$$

According to the law of equality of complex numbers, we have

$$\left\{\left[\frac{a^2}{(p-\alpha)^2+q^2}+\frac{b^2}{(p-\beta)^2+q^2}+\frac{c^2}{(p-\gamma)^2+q^2}\right]+n^2\right\}q=0$$

$\therefore \quad q = 0 \quad [\because \quad a^2 > 0, (p-a)^2 > 0, q^2 > 0$, etc.]

Hence, $x = p + iq = p$ is a real number.

$\therefore$ x is a real root.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. The given equation can be written as

$$\frac{2\log a}{\log x}+\frac{\log a}{\log(ax)}+\frac{3\log a}{\log(a^2x)}=0 \qquad \text{(i)}$$

As $a > 0$ and $a \neq 1$, $\log a \neq 0$, Eq. (i) can be written as

$$\frac{2}{y}+\frac{1}{b+y}+\frac{3}{2b+y}=0$$

where $b = \log a$ and $y = \log x$.

$\Rightarrow \quad 2(b+y)(2b+y)+y(2b+y)+3y(b+y)=0$

$\Rightarrow \quad 4b^2+11by+6y^2=0$

$\Rightarrow \quad y=\frac{-11b\pm\sqrt{121b^2-96b^2}}{12}=-\frac{4b}{3},-\frac{b}{2}$

$\Rightarrow \quad \log x=-\frac{4}{3}\log a \quad \text{or} \quad -\frac{1}{2}\log a$

$\Rightarrow \quad x=a^{-4/3}, a^{-1/2}$

2. $ax^2-bx+c=0$

$\alpha+\beta=\frac{b}{a}, \alpha\beta=\frac{c}{a}$

Also, $(a+cy)^2=b^2y$

$\Rightarrow \quad c^2y^2-(b^2-2ac)y+a^2=0$

$\Rightarrow \quad \left(\frac{c}{a}\right)^2y^2-\left[\left(\frac{b}{a}\right)^2-2\left(\frac{c}{a}\right)\right]y+1=0$

$\Rightarrow \quad (\alpha\beta)^2y^2-(\alpha^2+\beta^2)y+1=0$

$\Rightarrow \quad y^2-(\alpha^{-2}+\beta^{-2})y+a^{-2}\beta^{-2}=0$

$\Rightarrow \quad (y-\alpha^{-2})(y-\beta^{-2})=0$

Hence, the roots are α^{-2}, β^{-2}.

3. Given: One of the factors of the expression $(x^4 - px^2 + q)$ is $x^2 - 3x + 2$. Dividing $x^4 - px^2 + q$ by $x^2 - 3x + 2$, we get remainder $(15 - 3p)x + (2p + q - 14)$. Since $x^2 - 3x + 2$ is one of the factors of $x^4 - px^2 + q$, therefore after dividing $x^4 - px^2 + q$ by $x^2 - 3x + 2$, we get remainder equal to 0.

or $\quad (15-3p)x+(2p+q-14)=0$

Comparing both sides, we get $15 - 3p = 0$ or $p = 5$

and $2p+q-14=0$ or $2(5)+q-14=0$ or $q-4=0$

or $q = 4$

Thus $p=5$ and $q=4$.

4. $\frac{|x+1|}{|x|}+|x+1|=\frac{|x+1|^2}{|x|}$

$\Rightarrow \quad |x+1|\left\{\frac{1}{|x|}+1-\frac{|x+1|}{|x|}\right\}=0$

$\therefore \quad |x+1|=0$

or $\quad 1+|x|-|x+1|=0$

$\Rightarrow \quad x=-1 \qquad |x+1|=0$

If $x<-1$, $\quad 1+|x|-|x+1|=0$

$\Rightarrow \quad 1-x+x+1=0$

$\Rightarrow \quad 2=0$ (absurd)

If $-1 \le x < 0$

$$1+|x|-|x+1|=0$$

$\Rightarrow \quad 1-x-(x+1)=0$

$\Rightarrow \quad x=0$ (not possible)

If $x \ge 0$, $\quad 1+x-(x+1)=0$

$\Rightarrow \quad 0=0$

$\Rightarrow$ x can have any value in the interval

$\therefore \quad x=-1, x>0 \qquad [\because x \neq 0]$

5. $x^2+ax+b=\left(x+\frac{a}{2}\right)^2+b-\frac{a^2}{4} \ge b-\frac{a^2}{4}$

$\Rightarrow$ Least value is $b-\frac{a^2}{4}$ when $x+\frac{a}{2}=0$

As a, b, are roots of the given equation,

$a+b=-a, \quad ab=b$

$\Rightarrow \quad b=-2a \quad \text{or} \quad b(1-a)=0$

$\Rightarrow \quad a=1$ as $b \neq 0$ and hence, $b = -2 \cdot 1 = -2$

Hence, the least value of $b-\frac{a^2}{4}=-2-\frac{1}{4}=-\frac{9}{4}$

6. Since $z^2+(p+iq)z+(m+in)=0$,
let α be a real root of the given equation

$\therefore \quad \alpha^2+(p+iq)\alpha+(m+in)=0$

$\Rightarrow \quad (\alpha^2+p\alpha+m)+i(q\alpha+n)=0$

$\Rightarrow \quad \alpha^2+p\alpha+m=0 \quad$ (i)

and $\quad q\alpha+n=0$

$\Rightarrow \quad \alpha=-(n/q) \quad$ (ii)

Substituting the value of $\alpha=-(n/q)$ from (ii) in (i), we have

$$\frac{n^2}{q^2}-p.\frac{n}{q}+m=0$$

$\therefore \quad n^2+mq^2-pqn=0$

7. Given: Quadratic equation

$$(b+c-a)x^2+(c+a-b)x+(a+b-c)=0 \quad \text{(i)}$$

and $\quad a+b+c=0 \quad$ (ii)

We know that the standard quadratic equation is:

$$Ax^2+Bx+C=0$$

Comparing the given equation (i) with the standard equation, we get

$A=(b+c-a)$, $B=(c+a-b)$ and $C=(a+b-c)$

We also know that

$$B^2-4AC=(c+a-b)^2-4(b+c-a)(a+b-c)$$
$$=(-2b)^2-4(-2a)(-2c)$$
$$=4(b^2-4ac)=4[(a+c)^2-4ac]=(a-c)^2$$

which is a perfect rational number.

Since the coefficients of x^2, x and constant terms are rational, therefore the roots of the equation are also rational.

8. As α and β are the roots of the equation

$2x^2-35x+2=0$, hence $\alpha\beta=1$

$\therefore \quad 2\alpha^2-35\alpha=-2 \quad$ or $\quad 2\alpha-35=-2/\alpha$

$2\beta^2-35\beta=-2 \quad$ or $\quad 2\beta-35=-2/\beta$

Now $\quad (2\alpha-35)^3(2\beta-35)^3=\left(\frac{-2}{\alpha}\right)^3\left(\frac{-2}{\beta}\right)^3$

$$=\frac{8\cdot 8}{\alpha^3\beta^3}=\frac{64}{1}=64$$

9. $\dfrac{2x-1}{2x^3+3x+x}>0 \quad \Rightarrow \quad \dfrac{2x-1}{x(2x+1)(x+1)}>0$

Case I: If $x<-1 \Rightarrow \dfrac{2x-1}{x(2x+1)(x+1)}>0$,

so, $x\in(-\infty,-1)$

Case II: If $x>-1$ but $x<0$. At $x=-(1/2)$

$\Rightarrow \dfrac{2x-1}{x(2x+1)(x+1)}\notin\mathbb{R}$, so, $x\neq-\dfrac{1}{2}$

If $-1<x<-\dfrac{1}{2} \Rightarrow \dfrac{2x-1}{x(2x+1)(x+1)}<0$, so, $x\in\left(-1,-\dfrac{1}{2}\right)$

If $-\dfrac{1}{2}<x<0 \Rightarrow \dfrac{2x-1}{x(2x+1)(x+1)}>0$, so, $x\in\left(-\dfrac{1}{2},0\right)$

Case III:

If $0<x<\dfrac{1}{2} \Rightarrow \dfrac{2x-1}{x(2x+1)(x+1)}<0$, so, $x\notin\left(0,\dfrac{1}{2}\right)$

If $x=\dfrac{1}{2} \Rightarrow \dfrac{2x-1}{x(2x+1)(x+1)}=0$, so, $x\neq\dfrac{1}{2}$

If $x>\dfrac{1}{2} \Rightarrow \dfrac{2x-1}{x(2x+1)(x+1)}>0$, so, $x\in\left(\dfrac{1}{2},\infty\right)$

$\therefore \quad x\in(-\infty,-1)\cup\left(-\dfrac{1}{2},0\right)\cup\left(\dfrac{1}{2},\infty\right)$

10. Given expression $=\dfrac{x+2}{2x^2+3x+6}$

Let $\dfrac{x+2}{2x^2+3x+6}=k$

Therefore, $x+2=2kx^2+3kx+6k$

or $\quad 2kx^2+(3k-1)x+6k-2=0 \quad$ (1)

We know that the standard quadratic equation is

$$ax^2+bx+c=0$$

Comparing Eq. (1) with the standard equation, we get

$a=2k$, $b=(3k-1)$ and $c=(6k-2)$

We also know that for real roots, $b^2-4ac\geq 0$

or $\quad (3k-1)^2-4(2k)(6k-2)\geq 0$

or $\quad 9k^2-6k+1-48k^2+16k\geq 0$

or $\quad -39k^2+10k+1\geq 0$

or $\quad 39k^2-10k-1\leq 0$

or $\quad 39k^2-13k+3k-1\leq 0$

or $\quad 13k(3k-1)+(3k-1)\leq 0$

or $\quad (13k+1)(3k-1)\leq 0$

or $13k+1\leq 0$ and $3k-1\geq 0$ or $k\leq-\dfrac{1}{13}$ and $k\geq\dfrac{1}{3}$

Therefore, $k=-1/13$ and $1/3$

11. We have: $\alpha+\beta=-\dfrac{b}{a}$, $\alpha\beta=\dfrac{c}{a}$,

$$\alpha+h+\beta+h=-\frac{q}{p} \text{ and } (\alpha+h)(\beta+h)=\frac{r}{p}$$

$\Rightarrow \quad \alpha+\beta+2h=-\dfrac{q}{p}$

$\Rightarrow \quad -\dfrac{b}{a}+2h=\dfrac{-q}{p} \qquad \left[\because \alpha+\beta=-\dfrac{b}{a}\right]$

$\Rightarrow \quad h=\dfrac{1}{2}\left(\dfrac{b}{a}-\dfrac{q}{p}\right)$

12. Since $3^{\sec^2 x-1}\sqrt{9y^2-6y+2}\le 1$

$$\Rightarrow \quad 3^{\sec^2 x}\frac{1}{3}\sqrt{9y^2-6y+2}\le 1$$

$$\Rightarrow \quad 3^{\sec^2 x}\sqrt{y^2-\frac{2}{3}y+\frac{2}{9}}\le 1$$

$$\Rightarrow \quad 3^{\sec^2 x}\sqrt{\left(y-\frac{1}{3}\right)^2+\frac{1}{9}}\le 1 \qquad \text{(i)}$$

Now, $3^{\sec^2 x}\ge 3$ and $\sqrt{\left(y-\frac{1}{3}\right)^2+\frac{1}{9}}\ge\frac{1}{3}$

$$\therefore \quad 3^{\sec^2 x}\sqrt{\left(y-\frac{1}{3}\right)^2+\frac{1}{9}}\ge 1 \qquad \text{(ii)}$$

Hence the inequations (i) and (ii) are satisfied if and only if

$$3^{\sec^2 x}\sqrt{\left(y-\frac{1}{3}\right)^2+\frac{1}{9}}=1$$

i.e. $\sec^2 x=1$ and $y=1/3$

$\therefore$ $x=0,\ \pi,\ 2\pi,\ 3\pi$ and $y=1/3$

13. Given: $p, q\in\{1,2,3,4\}$ and quadratic equation $px^2+qx+1=0$. We know that the standard quadratic equation is $ax^2+bx+c=0$.

Comparing the given equation with the standard equation, we get $a=p$, $b=q$ and $c=1$.

We also know that for real roots $b^2-4ac\ge 0$ or $q^2-4p\ge 0$ or $q^2\ge 4p$

Therefore, when $p=1$, then $q^2\ge 4$ or $q=2, 3, 4$. And when $p=2$, then $q^2\ge 8$ or $q=3, 4$ and when $p=3$, then $q^2\ge 12$ or $q=4$. When $p=4$, $q^2\ge 16$ or $q=4$.

Thus, the total order pairs of

$(p,q)=\{(1,2)\ (1,3)\ (1,4)\ (2,3)\ (2,4)\ (3,4)\ (4,4)\}=7$

Therefore, the number of equations = 7

14. $[m_r,\ (1/m_r)]$ satisfies the given equation $x^2+y^2+2gx+2fy+c=0$

Then, $$m_r^2+\frac{1}{m_r^2}+2gm_r+\frac{2f}{m_r}+c=0$$

$$\Rightarrow \quad m_r^4+2gm_r^3+cm_r^2+2fm_r+1=0$$

Now, the roots of the given eqation are m_1, m_2, m_3, m_4.

$$\text{Product of root}=m_1m_2m_3m_4=\frac{\text{Constant term}}{\text{Coefficeint of } m_r^4}=\frac{1}{1}=1$$

15. Let $m=\dfrac{(x-b)(x-c)}{x-a}$

$$\Rightarrow \quad x^2-(b+c+m)x+(bc+am)=0$$

Since x is real, we must have $D\ge 0$

$$\Rightarrow \quad (b+c+m)^2-4(bc+am)\ge 0$$

$$\Rightarrow \quad m^2+2(b+c-2a)m+(b-c)^2\ge 0$$

Since this inequality is always positive, $D<0$

$$\therefore \quad 4(b+c-2a)^2-4(b-c)^2<0$$

$$\Rightarrow \quad (b+c-2a)^2-(b-c)^2<0$$

$$\Rightarrow \quad (b+c-2a+b-c)\ (b+c-2a-b+c)<0$$

$$\Rightarrow \quad 2(b-a)2(c-a)<0$$

$$\Rightarrow \quad (a-b)\ (a-c)<0$$

$$\therefore \quad b<a<c \text{ or } c<a<b$$

according as $b<c$ or $c<b$ respectively.

16. Use $|5x+3|=5x+3$, when $x\ge\dfrac{-3}{5}$

$=-(5x+3)$, when $x<\dfrac{-3}{5}$

When $x\ge\dfrac{-3}{5}$, the equation is $x^2+4x+3=0$

$\Rightarrow$ $x=-1, -3$. Both solutions are not possible.

When $x\le\dfrac{-3}{5}$, the equation is $x^2-6x-3=0$

$\Rightarrow$ $x=3\pm 2\sqrt{3}$ (both not possible)

17. Given: Function $y=\dfrac{(x-a)(x-b)}{(x-c)}$

The given function may be written as

$$(x-c)y=(x-a)(x-b)$$

or $$x^2-x(y+a+b)+(ab+cy)=0 \qquad \text{(i)}$$

We know that the standard quadratic equation is

$$Ax^2+Bx+C=0$$

Comparing Eq. (i) with the standard equation, we get $A=1, B=-(y+a+b)$ and $C=(ab+cy)$

We also know that for real roots, $B^2-4AC\ge 0$

or $$(y+a+b)^2-4(ab+cy)\ge 0$$

or $$y^2+2y(a+b-2c)+(a-b)^2\ge 0$$

We also know that this inequation is always positive if its discriminant is negative. Thus,

$$4(a+b-2c)^2-4(a-b)^2<0$$

or $c^2-c(a+b)+ab<0$ or $(c-a)(c-b)<0$

Therefore, either $c-a<0$ and $c-b>0$ or $c<a$ and $c>b$ or $a>c>b$ or $(c-a)>0$ and $(c-b)<0$ or $c>a$ and $c<b$ or $a<c<b$. Thus, $a<c<b$ and $a>c>b$

18. If $f(x)=x^3+3x^2-9x+c=(x-\alpha)^2(x-b)$, then $(x-\alpha)$ is a factor of order 2. So, $(x-\alpha)$ is a factor of order 1 of $f'(x)$, i.e. $3x^2+6x-9$

Now,

$$f'(x)=3x^2+6x-9=3(x^2+2x-3)$$
$$=3(x+3)\ (x-1)$$

$\therefore \quad f'(x)=0 \Rightarrow x=1 \quad \text{or} \quad x=-3$

This shows that either $\alpha = 1$ or $\alpha = -3$

If $\alpha = 1$, then as α is a root of $x^3+3x^2-9x+c=0$, Therefore,

$$1+3-9+c=0 \Rightarrow c=5$$

If $\alpha = -3$, then as α is a root of $x^3+3x^2-9x+c=0$, therefore, $-27+27+27+c=0 \Rightarrow c=-27$

19. α, β, γ and σ are the roots of the given equation $x^4 + 4x^3 - 6x^2 + 7x - 9 = (x-\alpha)(x-\beta)(x-\gamma)(x-\sigma)$. Putting $x = i$ and then $x = -i$, we get

$$1-4i+6+7i-9=(i-\alpha)(i-\beta)(i-\gamma)(i-\sigma) \text{ and}$$
$$1+4i+6-7i-9=(i-\alpha)(-i-\beta)(-i-\gamma)(-i-\sigma)$$

Multiplying these two, we get

$$(-2+3i)(-2-3i)=(1+\alpha^2)(1+\beta^2)(1+\gamma^2)(i+\sigma^2)$$
$$\Rightarrow \qquad 13=(1+\alpha^2)(1+\beta^2)(1+\gamma^2)(i+\sigma^2)$$

20. $x^2-|x+2|+x>0$

Case 1: $x<-2$

$$\Rightarrow \qquad x^2+x+2+x>0$$
$$\Rightarrow \qquad x^2+2x+1+1>0$$
$$\Rightarrow \qquad (x+1)^2+1>0$$

Always true $\forall x \in \mathbb{R}$ and $x < -2$

Hence, $x < -2$

Case 2: $x \geq -2$

$$\Rightarrow \qquad x^2-x-2+x>0$$
$$\Rightarrow \qquad x^2>2$$
$$\Rightarrow \qquad |x|>\sqrt{2}$$
$$\Rightarrow \quad |x|<-\sqrt{2} \quad \text{or} \quad x>\sqrt{2}$$

But $x \geq -2$

Hence, $x \in [-2,-\sqrt{2}) \cup (\sqrt{2},\infty)$

$\therefore$ From Case 1 and Case 2, we get

$$x \in (-\infty,-\sqrt{2}) \cup (\sqrt{2},\infty)$$

21. $\because$ a and b are the roots of $(x+c)(x+d)-p=0$

$\therefore \quad (x+c)(x+d)-p=(x-a)(x-b)$ (note this step)

$$\Rightarrow \qquad (x-a)(x-b)+p=(x+c)(x+d)$$

Clearly, $-c$ and $-d$ are the roots of the equation,

$$(x-a)(x-b)+p=0$$

22. Given: Quadratic equation

$$(a^2+b^2)t^2-2(ac+bd)t+(c^2+d^2)=0$$

We know that the standard quadratic equation is

$$At^2+Bt+C=0$$

Comparing the given equation with the standrad quadratic equation, we get

$$A=(a^2+b^2),\ B=-2(ac+bd) \text{ and } C=(c^2+d^2)$$

We also know that for equal roots, $B^2 = 4AC$

$$\text{or} \qquad [-2(ac+bd)]^2=4(a^2+b^2)(c^2+d^2)$$
$$\text{or} \qquad 4(ac+bd)^2=4(a^2+b^2)(c^2+d^2)$$
$$\text{or} \quad a^2c^2+b^2d^2+2abcd=a^2c^2+a^2d^2+b^2c^2+b^2d^2$$
$$\text{or} \quad a^2d^2+b^2c^2-2abcd=0 \quad \text{or} \quad (ad-bc)^2=0$$
$$\text{or} \quad ad=bc \quad \text{or} \quad \frac{a}{b}=\frac{c}{d}$$

23. Since $(x-1)^2, (x-2)^2, (x-3)^2$ cannot vanish simultaneously for any real x,

$\therefore$ The number of real roots is 0.

24. Given:

$$\int_0^{\pi/2}\left[\left\{a^2\left(\frac{\cos 3x}{4}+\frac{3}{4}\cos x\right)+a\sin x-20\cos x\right]dx \leq -\frac{a^2}{3}\right.$$

$$\Rightarrow \quad \left[\left[a^2\left(\frac{\sin 3x}{12}+\frac{3}{4}\sin x\right)-a\cos x-20\sin x\right]\right]_0^{\pi/2} \leq -\frac{a^2}{3}$$

$$\Rightarrow \quad a^2\left(-\frac{1}{12}+\frac{3}{4}-0\right)-20+a \leq -\frac{a^2}{3}$$

$$\Rightarrow \quad \frac{2}{3}a^2+a-20+\frac{a^2}{3} \leq 0 \quad \Rightarrow \quad a^2+a-20 \leq 0$$

$$\Rightarrow \quad (a+5)(a-4) \leq 0$$

$\therefore \quad -5 \leq a \leq 4$

Thus, $a = 1, 2, 3, 4$ [$\because \ a \in I$]

25. $D=4^2-4(2m-1)^2>0$

$$\Rightarrow \quad 4[2^2-(2m-1)^2]>0$$

$$\Rightarrow \quad 4[(2-2m+1)(2+2m-1)]>0 \text{ or } -\frac{1}{2}<m<\frac{3}{2}$$

$\therefore$ The greatest integer value of $m = 1$

26. Given: Quadratic equations $x^2 + 2cx + ab = 0$ (i)

and $x^2-2(a+b)x+(a^2+b^2+2c^2)=0$ (ii)

We know that the standard quadratic equation is

$$Ax^2+Bx+C=0$$

Comparing Eqs. (i) and (ii) with the standard equation, we get

$A_1 = 1, B_1 = 2c, C_1 = ab, A_2 = 1, B_2 = -2(a + b)$ and $C_2 = a^2 + b^2 + 2c^2$

We also know that for real and unequal roots,

$$B_1^2-4A_1C_1>0 \quad \text{or} \quad (2c)^2-4\times1\times ab>0$$
$$\text{or} \quad 4c^2-4ab>0 \quad \text{or} \quad 4(c^2-ab)>0$$
$$\text{or} \quad c^2-ab>0 \quad \text{and} \quad B_2^2-4A_2C_2>0$$

or $[-2(a+b)]^2 - 4[a^2+b^2+2c^2] > 0$

or $-8(c^2-ab) > 0$ or $8(c^2-ab) < 0$

Therefore, the roots of the quadratic equation (ii) are non-real complex.

27. $|x|^3 - 3x^2 + 3|x| - 2 = 0$

$\Rightarrow \quad (|x|-2)(|x|^2 - |x| + 1) = 0$

$\Rightarrow \quad |x| = 2$ or $|x|^2 - |x| + 1 = 0$

$\Rightarrow \quad x = \pm 2 \qquad [\because |x|^2 - |x| + 1$ has imaginary roots$]$

28. Since $x^2 - 2 - 2[x] = 0$

$\Rightarrow \quad x^2 - 2 = 2[x]$

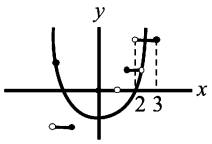


Fig. 3.13

From the above graph, it is clear that there is a solution of the equation between 2 and 3. Hence, the correct answer is (a).

29. $D = 4(1-\sin\theta)^2 + 4(1-\sin\theta)3\sin\theta < 0$

$\Rightarrow \quad 4(1-\sin\theta)(1+2\sin\theta) < 0$

$\Rightarrow \quad \sin\theta < \dfrac{-1}{2}$ or $\sin\theta > 1$ (impossible)

$\Rightarrow \quad \dfrac{7\pi}{6} < \theta < \dfrac{11\pi}{6}$

30. Quadratic equation: $x^2 - 2(1+3k)x + 7(3+2k) = 0$ and its roots are equal.

We know that the standard quadratic equation is

$$ax^2 + bx + c = 0$$

Comparing the given equation with the standard quadratic equation, we get

$a = 1$, $b = -2(1 + 3k)$ and $c = 7(3 + 2k)$

We also know that for equal roots

$$b^2 = 4ac$$

or $[-2(1+3k)]^2 = 4 \times 1 \times 7(3+2k)$

or $4(1+9k^2+6k) = 28(3+2k)$

or $1+9k^2+6k = 7(3+2k)$

or $1+9k^2+6k = 21+14k$

or $9k^2 - 8k - 20 = 0$

$9k^2 - 18k + 10k - 20 = 0$

or $9k(k-2) + 10(k-2) = 0$

or $k(9k-10) + 2(k-10)$

or $(k-2)(9k+10) = 0$

or $k = 2$ and $k = -\dfrac{10}{9}$

31. $(x-b)(x-c) + (x-c)(x-a) + (x-a)(x-b) = 0$

$\Rightarrow \quad 3x^2 - 2(a+b+c)x + (ab+bc+ca) = 0$

$\Rightarrow \quad D = 4(a+b+c)^2 - 12(ab+bc+ca)$

$= 4[a^2+b^2+c^2-ab-bc-ca]$

$= 2[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$

$\therefore$ Both the roots are real.

32. Since $a, b \in x^2 + bx + c = 0$,

such that $c < 0 < b$

i.e. c is –ve and b is +ve

$$\alpha + \beta = -b \quad \alpha\beta = c$$

$\Rightarrow \quad \alpha + \beta < 0 \quad \alpha\beta < 0$

$\Rightarrow$ α and β are of opposite signs.

Since $\alpha < \beta$,

$\Rightarrow$ α is < 0 and β is > 0

Again $\alpha + \beta < 0$

$\Rightarrow$ α is more in magnitude than β.

Hence, $\alpha < 0 < \beta < |\alpha|$

33. Given: Quadratic equations:

$$x^2 + bx + c = 0 \qquad \text{(i)}$$

and $$x^2 - cx + b = 0 \qquad \text{(ii)}$$

and their roots differ by the same quantity.

Let α and β be the roots of the given equations.

We know that the standard quadratic equation is $Ax^2 + Bx + C = 0$

Comparing Eqs. (i) and (ii) with the standard equation, we get

$A_1 = 1, B_1 = b, C_1 = c; A_2 = 1, B_2 = -c$ and $C_2 = b$

We also know that the difference between the roots

$$(\alpha - \beta) = \frac{\sqrt{B^2 - 4AC}}{A} \qquad \text{(iii)}$$

Since the roots of both the equations differ by the same quantity, therefore

$$\frac{\sqrt{B_1^2 - 4A_1C_1}}{A_1} = \frac{\sqrt{B_2^2 - 4A_2C_2}}{A_2}$$

or $$\frac{\sqrt{(b)^2 - 4(1)(c)}}{(1)} = \frac{\sqrt{(-c)^2 - 4(1)b}}{(1)}$$

or $b^2 - 4c = c^2 - 4b$

or $-4(b-c) = b^2 - c^2 = (b+c)(b-c)$

or $b + c = -4$

34. Let α and α^2 be the roots.

$$\therefore \quad \alpha+\alpha^2=\frac{-p}{3} \qquad \text{(i)}$$

and $$\alpha^3=1 \qquad \text{(ii)}$$

Cubing (i), we get

$$\alpha^3+\alpha^6+3\alpha^3(\alpha+\alpha^2)=\frac{-p^3}{27}$$

$$\Rightarrow \quad 1+1+3(1)\left(\frac{-p}{3}\right)=\frac{-p^3}{27}$$

$$\therefore \quad 2-p=\frac{-p^3}{27} \quad \Rightarrow \quad p^3-27p+54=0 \quad \Rightarrow \quad p=3$$

35. Since $(\sqrt{2+\sqrt{2}})^x+(\sqrt{2-\sqrt{2}})^x=2\cdot 2^{x/4}$ (i)

Now, A.M. of two numbers $(\sqrt{2+\sqrt{2}})^x$ and $(\sqrt{2-\sqrt{2}})^x$ is given by

$$\text{A.M.}=\frac{(\sqrt{2+\sqrt{2}})^x+(\sqrt{2-\sqrt{2}})^x}{2}$$

$$\Rightarrow \quad (\sqrt{2+\sqrt{2}})^x+(\sqrt{2-\sqrt{2}})^x=2\text{A.M.}$$

Now G.M. of $(\sqrt{2+\sqrt{2}})^x$ and $(\sqrt{2-\sqrt{2}})^x$ is

$$\sqrt{(\sqrt{2+\sqrt{2}})^x(\sqrt{2-\sqrt{2}})^x}=2^{\frac{x}{4}}$$

Substituting the respective values in Eq. (i), we get 2A.M. = 2G.M.

$$\Rightarrow \quad \text{A.M. = G.M.}$$

Now, A.M. = G.M. if and only if

$$(\sqrt{2+\sqrt{2}})^x=(\sqrt{2-\sqrt{2}})^x$$

$$\therefore \quad x=0$$

36. Given: Quadratic equations

$$x^2+ax+\beta=0$$

and $$x^2+px+q=0$$

Let m be the common root of the equations.

Therefore, $$m^2+\alpha m+\beta=0 \qquad \text{(i)}$$

and $$m^2+pm+q=0 \qquad \text{(ii)}$$

Solving Eqs. (i) and (ii) by the cross multiplication method, we get

$$\frac{m^2}{\alpha q-\beta p}=-\frac{m}{q-\beta}=\frac{1}{p-\alpha} \quad \text{or} \quad m=\frac{q-\beta}{\alpha-p}$$

37. $\dfrac{\alpha}{\alpha-1}+\dfrac{\alpha+1}{\alpha}=\dfrac{-b}{a}$ and $\dfrac{\alpha}{\alpha-1}\cdot\dfrac{\alpha+1}{\alpha}=\dfrac{c}{a}$

$$\Rightarrow \quad \frac{2\alpha^2-1}{\alpha^2-\alpha}=-\frac{b}{a} \qquad \text{(i)}$$

and $$\frac{\alpha+1}{\alpha-1}=\frac{c}{a} \qquad \text{(ii)}$$

From (ii), we get $a\alpha+a=c\alpha-c$

$$\Rightarrow \quad \alpha(c-a)=a+c \quad \Rightarrow \quad \alpha=\frac{c+a}{c-a}$$

Putting in (i), $$\frac{2\left(\frac{c+a}{c-a}\right)^2-1}{\left(\frac{c+a}{c-a}\right)^2-\left(\frac{c+a}{c-a}\right)}=-\frac{b}{a}$$

$$\Rightarrow \quad \frac{2(c+a)^2-(c-a)^2}{(c+a)^2-(c^2-a^2)}=-\frac{b}{a}$$

$$\Rightarrow \quad \frac{(c+a)^2+4ac}{2a^2+2ac}=-\frac{b}{a}$$

$$\Rightarrow \quad (c+a)^2+4ac=-2b(a+c)$$

$$\Rightarrow \quad (c+a)^2+2b(a+c)+4ac=0$$

$$\Rightarrow \quad (c+a)^2+2b(a+c)+b^2=b^2-4ac$$

$$\Rightarrow \quad (c+a+b)^2=b^2-4ac$$

Hence $$(a+b+c)^2=b^2-4ac$$

38. Let

$$\Delta=\begin{vmatrix}\sin x & \cos x & \cos x\\ \cos x & \sin x & \cos x\\ \cos x & \cos x & \sin x\end{vmatrix}=0$$

$$\Rightarrow \quad \Delta=\sin^3 x+2\cos^3 x-3\sin x\cos^2 x=0$$

Since $-\dfrac{\pi}{4}\le x\le\dfrac{\pi}{4} \quad \Rightarrow \quad \cos x\neq 0$ in this range

So dividing the whole expression by $\cos^3 x$, we have

$$\tan^3 x+2-3\tan x=0$$

$$\Rightarrow \quad \tan^3 x-3\tan x+2=0$$

$$\Rightarrow \quad (\tan x-1)^2(\tan x+2)=0$$

Either $\tan x=1$ or $\tan x=-2$

Now, $$-\frac{\pi}{4}\le x\le\frac{\pi}{4}$$

$$\Rightarrow \quad -1\le\tan x\le 1$$

So $\tan x=-2$ is not defined and hence the only solution is $\tan x = 1$

39. $2^{x^2}=2^{2x}\cdot 2^{3k}=2^{2x+3k}$

$$\Rightarrow \quad x^2=2x+3k \quad \Rightarrow \quad x^2-2x-3k=0$$

$$D=4+12k=0 \quad \Rightarrow \quad k=\frac{-1}{3}$$

40. Given: Quadratic equation $\dfrac{x-m}{mx+1}=\dfrac{x+n}{nx+1}$

The given equation may be written as

$$(x-m)(nx+1)=(x+n)(mx+1)$$

or $$nx^2+x-mnx-m=mx^2+x+mnx+n$$

or $$(m-n)x^2+2mnx+(m+n)=0 \qquad \text{(i)}$$

We know that the standard quadratic equation is

$$ax^2 + bx + c = 0$$

Comparing Eq. (i) with the standard quadratic equation, we get

$$a = (m-n),\ b = 2mn \text{ and } c = (m+n)$$

Since the roots of Eq. (i) are reciprocal to each other, therefore $c = a$

or $m - n = m + n$ or $2n = 0$ or $n = 0$

41. $y = 2[x] + 3 = 3[x-2] = 3[x] - 6$

$\Rightarrow [x] = 9$

$\therefore \quad y = 2(9) + 3 = 18 + 3 = 21$

$\therefore \quad [x + y] = [x + 21] = [x] + 21 = 9 + 21 = 30$

42. Rearranging, we have

$$x^2 - x(a+b) + ab = ax + bx - 2ab$$

$\Rightarrow \quad x^2 - 2x(a+b) + 3ab = 0$

Since the roots are equal in magnitude but opposite in sign,

$\therefore$ sum of the roots = 0

$\Rightarrow \quad 2(a+b) = 0 \quad \therefore \quad a + b = 0$

43. $\alpha + \beta = -b < 0 \quad (\because b > 0)$

$\alpha\beta = c < 0$ (it is given that $c < 0$)

Since the product of the roots is –ve, the roots must be of opposite sign, i.e. $\alpha < 0$ and $\beta > 0$

But the sum of the roots is also –ve, thus the numerical value of α, i.e. $|\alpha| > \beta$. $\therefore \quad \alpha < 0 < \beta < |\alpha|$.

44. Given: Quadratic equation $1 - x - 3x^2 = 0$

The given equation may be written as $-3x^2 - x + 1 = 0$

We know that the standard quadratic equation is $Ax^2 + Bx + C = 0$

Comparing the given equation with the standard equation, we get

$$A = -3,\ B = -1 \text{ and } C = 1$$

We also know that the maximum value of the given quadratic equation (for $A < 0$)

$$= \frac{4AC - B^2}{4A} = \frac{4 \times (-3) \times (1) - (-1)^2}{4(-3)} = \frac{-12-1}{-12} = \frac{13}{12}$$

45. A.M. of $(x+4), (x+3), (x+2), (x+1), (x-5)$ is given as $\frac{5x+5}{5} = x + 1$

Let $y = x + 1$, then the given equation reduces to

$$(y+3)^3 + (y+2)^3 + (y+1)^3 + y^3 + (y-6)^3 + 180 = 0$$

$\Rightarrow \quad (y^3 + 9y^2 + 27y + 27) + (y^3 + 6y^2 + 12y + 8)$

$$+ (y^3 + 3y^2 + 3y + 1) + y^3$$

$$+ (y^3 - 18y^2 + 108y - 216) + 180 = 0$$

$\Rightarrow \quad 5y^3 + 150y = 0 \quad$ or $\quad y^3 + 30y = 0$

$\Rightarrow \quad y(y^2 + 30) = 0$

$\Rightarrow \quad y = 0 \qquad [\because y^2 + 30 = 0$ is not possible]

$\therefore \quad x + 1 = 0 \quad$ or $\quad x = -1$

$\therefore$ The number of real solutions = 1

46. The given equation is satisfied by $x = -a, -b$ and $-c$. Hence, it is an identity. Thus, infinite solutions are possible.

47. Given: Quadratic equation $x^2 + 2bx + c = 0$

We know that the standard quadratic equation is

$$Ax^2 + Bx + C = 0$$

Comparing the given equation with the standard equation, we get $A = 1$, $B = 2b$ and $C = c$

We also know that the minimum value of the given quadratic equation (for $A > 0$)

$$= \frac{4AC - B^2}{4A} = \frac{4c - 4b^2}{4} = c - b^2$$

48. Here $\alpha + \beta = \alpha^2 + \beta^2$

$\Rightarrow \quad \alpha + \beta = (\alpha + \beta)^2 - 2\alpha\beta$

$\Rightarrow \quad -p = (-p)^2 - 2q \Rightarrow p^2 + p = 2q$

49. Let $t = (\sqrt{5}+2)^{|x|} = (\sqrt{5}-2)^{-|x|}$

$$\left[\because (\sqrt{5}+2)(\sqrt{5}-2) = 1\right]$$

Now, the equation reduces to $t^2 - 18t + 1 = 0$

$\Rightarrow \quad t = \frac{18 \pm \sqrt{324-4}}{2} = 9 \pm 4\sqrt{5} = (\sqrt{5} \pm 2)^2$

Thus, $|x| = 2 \quad \therefore \quad x = \pm 2$

50. $|x|^2 - |x| + a = 0 \qquad$ (i)

or $\quad y^2 - y + a = 0 \quad$ (Let $y = |x|$) $\qquad$ (ii)

If both the roots of (i) are +ve, four solutions of (i) are possible. If only one root of (ii) is +ve, two solutions of (i) are possible.

If $a = 0$, three distinct solutions, viz. $x = 0, 1, -1$ are there. Thus, in any case, the number of real solutions cannot be one.

51. Given: Expression $\frac{1}{4x^2 + 2x + 1}$, $x \in \mathbb{R}$

Let $\quad y = \frac{1}{4x^2 + 2x + 1} \quad$ or $\quad 4yx^2 + 2yx + y = 1$

or $\quad 4yx^2 + 2yx + (y-1) = 0 \qquad$ (i)

We know that the standard equation is:

$$Ax^2 + Bx + C = 0$$

Comparing the given quadratic equation with the standard equation, we get $A = 4y$, $B = 2y$ and $C = y - 1$.

Since $x \in \mathbb{R}$, therefore the roots of Eq. (i) are real.

Thus, $B^2 - 4AC \geq 0$ or $(2y)^2 - 4(4y)(y-1) \geq 0$

or $4y^2 - 16y(y-1) \geq 0$ or $4y[y - 4(y-1)] \geq 0$

or $4y[-3y + 4] \geq 0$ or $4y[3y - 4] \leq 0$

Therefore, the greatest value of the expression = 4/3.

52. Here $\alpha + \beta = p$ and $\alpha\beta = -(p+q)$. Thus,

$$\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + q} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + q}$$

$$= \frac{(\alpha+1)^2}{(\alpha+1)^2 + (q-1)} + \frac{(\beta+1)^2}{(\beta+1)^2 + (q-1)}$$

$$= \frac{2(\alpha+1)^2(\beta+1)^2 + (q-1)[(\alpha+1)^2 + (\beta+1)^2]}{(\alpha+1)^2(\beta+1)^2 + (q-1)^2 + (q-1)[(\alpha+1)^2 + (\beta+1)^2]} \quad \text{(i)}$$

Now $(\alpha + 1)(\beta + 1) = \alpha\beta + (\alpha + \beta) + 1$

$= -(p + q) + p + 1 = 1 - q$

Putting in (i), we get

$$\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + q} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + q}$$

$$= \frac{2(1-q)^2 + (q-1)[(\alpha+1)^2 + (\beta+1)^2]}{2(1-q)^2 + (q-1)[(\alpha+1)^2 + (\beta+1)^2]} = 1$$

53. Since, $(3-x)^4 + (2-x)^4 = (5-2x)^4$

$\Rightarrow (3-x)^4 + (2-x)^4 = (3-x+2-x)^4$

$\Rightarrow (3-x)^4 + (2-x)^4 = (3-x)^4 + 4(3-x)^3(2-x)$
$+ 6(3-x)^2(2-x)^2 + 4(3-x)(2-x)^3 + (2-x)^4$

$\Rightarrow 2(3-x)(2-x)[2(3-x)^2 + 3(3-x)(2-x)$
$+ 2(2-x)^2] = 0$

$\Rightarrow x = 2, 3$, or $2(3-x)^2 + 3(3-x)(2-x) + 2(2-x)^2 = 0$

$\Rightarrow x = 2, 3$ or $(7x^2 - 35x + 44) = 0$

Since, the equation $7x^2 - 35x + 44 = 0$ has imaginary roots, therefore, the given equation has two real and two imaginary roots.

54. $\because$ Both roots are –ve,

$\therefore$ $D > 0$, sum of roots = –ve and product of roots = +ve

Here $D > 0 \Rightarrow \frac{1}{2} < m < \frac{5}{2}, \alpha + \beta < 0 \Rightarrow m < \frac{3}{2}$

$\alpha \cdot \beta = 1 > 0$

$\therefore$ The greatest integral value of $m = 1$.

55. Given: Quadratic equation $2x^2 + x + 1 = 0, x \in \mathbb{R}$

We know that the standard quadratic equation is

$$Ax^2 + Bx + C = 0$$

Comparing the given quadratic equation with the standard equation, we get $A = 2, B = 1$ and $C = 1$

We also know that the minimum value of the given quadratic equation (for $A > 0$)

$$= \frac{4AC - B^2}{4A} = \frac{4 \times 2 \times 1 - (1)^2}{4 \times 2} = \frac{8-1}{8} = \frac{7}{8}$$

56. $ax^2 + bx + c = a\left[\left(x^2 + \frac{b}{a}x\right) + \frac{c}{a}\right]$

$$= a\left[\left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2}\right]$$

$$= a\left[\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a^2}\right]$$

Least value $= \frac{4ac - b^2}{4a}$ $\left[\because \text{Least of } \left(x + \frac{b}{2a}\right)^2 = 0\right]$

57. The given equation is $x^2 - 8x - (n^2 - 10n) = 0$

$D = 64 + 4(n^2 - 10n) < 0$

$\Rightarrow$ $(n-2)(n-8) < 0 \Rightarrow 2 < n < 8$

$\therefore$ $n = 3, 4, 5, 6, 7$

Also $n = 10$ makes the equation undefined.

58. Given: The first quadratic equation $2x^2 - 3x - 6 = 0$ and its roots are α and β and the roots of the second quadratic equation are $(\alpha^2 + 2)$ and $(\beta^2 + 2)$.

We know that the standard quadratic equation is

$$ax^2 + bx + c = 0$$

Comparing the given equation with the standard equation, we get $a = 2$, $b = -3$ and $c = -6$

We also know that the sum of the first roots

$$(\alpha + \beta) = -\frac{b}{a} = -\frac{(-3)}{2} = \frac{3}{2}$$

and the product of the first roots $(\alpha\beta) = \frac{c}{a} = \frac{-6}{2} = -3$

Therefore, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \left(\frac{3}{2}\right)^2 - 2 \times (-3) = \frac{9}{4} + 6 = \frac{33}{4}$$

We also know that the sum of the second roots

$$(\alpha^2 + 2) + (\beta^2 + 2) = \alpha^2 + \beta^2 + 4 = \frac{33}{4} + 4 = \frac{49}{4}$$

and the product of the second roots $= (\alpha^2 + 2)(\beta^2 + 2) =$

$$\alpha^2\beta^2 + 2(\alpha^2 + \beta^2) + 4 = (-3)^2 + 2\left(\frac{33}{4}\right) + 4 = \frac{59}{2}$$

We also know that the standard quadratic equation is

$x^2 -$ (Sum of roots) $x +$ (Product of roots) $= 0$

or $x^2 - \frac{49}{4}x + \frac{59}{2} = 0$ or $4x^2 - 49x + 118 = 0$

59. Since α and β are the roots of $(x-a)(x-b)=c$,

i.e. $x^2-(a+b)x+ab-c=0$,

$\therefore\ \alpha+\beta=a+b$ and $\alpha\beta=ab-c$

$\Rightarrow\ a+b=\alpha+\beta$ and $ab=\alpha\beta+c$

$\therefore\ a$ and b are the roots of $x^2-(a+b)x+ab=0$

$\Rightarrow\quad x^2-(\alpha+\beta)x+\alpha\beta+c=0$

$\Rightarrow\quad (x-\alpha)(x-\beta)+c=0$

60. For real roots $q^2-4pr\geq 0$

$$\Rightarrow\quad \left(\frac{p+r}{2}\right)^2-4pr\geq 0 \qquad [\because p,q,r \text{ are in A.P.}]$$

$$\Rightarrow\quad p^2+r^2-14pr\geq 0$$

$$\Rightarrow\quad \frac{p^2}{r^2}-14\frac{p}{r}+1\geq 0$$

$$\Rightarrow\quad \left(\frac{p}{r}-7\right)^2-48\geq 0 \quad\Rightarrow\quad \left|\frac{p}{r}-7\right|\geq 4\sqrt{3}$$

61. $\dfrac{x^2-bx}{ax-c}=\dfrac{\lambda-1}{\lambda+1}$

$\Rightarrow\ (\lambda+1)x^2-(\lambda+1)bx=(\lambda-1)ax-(\lambda-1)c$

$\Rightarrow\ (\lambda+1)x^2-x(\lambda b+b+\lambda a-a)+(\lambda-1)c=0$

Since $\alpha+\beta=0$,

$\therefore\quad \lambda b+b+\lambda a-a=0$

$\Rightarrow\quad \lambda(a+b)=a-b$

Hence, $\lambda=\dfrac{a-b}{a+b}$

62. Given, $x^2+2px+q=0$

$\therefore\quad \alpha+\beta=-2p$

$\alpha\beta=q$

and $ax^2+2bx+c=0$

$\therefore\quad \alpha+\dfrac{1}{\beta}=-\dfrac{2b}{a}$

and $\dfrac{\alpha}{\beta}=\dfrac{c}{a}$

Now, $(p^2-q)(b^2-ac)$

$$=\left[\left(\frac{\alpha+\beta}{-2}\right)^2-\alpha\beta\right]\left[\left(\frac{\alpha+\frac{1}{\beta}}{2}\right)^2-\frac{\alpha}{\beta}\right]a^2$$

$$=\frac{(\alpha+\beta)^2}{16}\left(\alpha+\frac{1}{\beta}\right)^2\cdot a^2\geq 0$$

$\therefore$ Statement I is true.

Again, now $pa=-\left(\dfrac{\alpha+\beta}{2}\right)a=-\dfrac{a}{2}(\alpha+\beta)$

and $b=-\dfrac{a}{2}\left(\alpha+\dfrac{1}{\beta}\right)$

Since, $pa\neq b\ \Rightarrow\ \alpha+\dfrac{1}{\beta}\neq\alpha+\beta$

$\Rightarrow\ \beta^2\neq 1,\ \beta\neq\{-1,0,1\}$, which is correct. Similarly, if

$a\dfrac{\alpha}{\beta}\neq a\alpha\beta$

$\Rightarrow\ \alpha\left(\beta-\dfrac{1}{\beta}\right)\neq 0\quad\Rightarrow\quad \alpha\neq 0$ and $\beta-\dfrac{1}{\beta}\neq 0$

$\Rightarrow\ \beta\neq\{-1,0,1\}$

Statement II is true.

Both Statement I and Statement II are true. But Statement II does not explain Statement I.

63. Let $A=b+c-a$, $B=c+a-b$, $C=a+b-c$

$\therefore\quad A+B+C=0$

And the given equation becomes

$$Ax^2+Bx+C=0$$

$\therefore\quad D=B^2-4AC$

$=(A+C)^2-4AC=(A-C)^2$

$=4[c-a]^2$ = perfect square

Hence, the roots of

$$(b+c-a)x^2+(c+a-b)x+(a+b-c)=0$$

are rational.

64. Let $f(x)=4x^2-2x+a$

Also, both roots of $f(x)=0$ lie in the interval $(-1, 1)$

$\therefore\quad D\geq 0, f(-1)>0$ and $f(1)>0$

Now, $D\geq 0$

$\Rightarrow\quad (-2)^2-4.4.a\geq 0\quad\Rightarrow\quad a\leq 1/4$ (i)

and $f(-1)>0$

$\Rightarrow\quad 4(-1)^2-2(-1)+a>0\quad\Rightarrow\quad a>-6$ (ii)

Also, $f(1)>0\quad\Rightarrow\quad 4(1)^2-2(1)+a>0$

$\Rightarrow\quad a>-2$ (iii)

From Eqs. (i), (ii) and (iii), we get $-2<a\leq 1/4$.

65. For real roots,

$\Rightarrow\quad (-4)^2-4(2\lambda-1)(2\lambda-1)\geq 0$

$\Rightarrow\quad (2\lambda-1)^2\leq 4$

$\Rightarrow\quad -2\leq 2\lambda-1\leq 2$

$\Rightarrow\quad -\dfrac{1}{2}\leq\lambda\leq\dfrac{3}{2}$

$\therefore$ Integral values of λ are 0 and 1.

Hence, the greatest integer value of $\lambda=1$

66. Since, $x=-2$ is a root of $f(x)$

$\therefore \quad f(x)=(x+2)(ax+b)$

But $f(0)+f(1)=0$

$\therefore \quad 2b+3a+3b=0 \quad \Rightarrow \quad -\frac{b}{a}=\frac{3}{5}.$

67. Let $f(x)=-x^2+x-1$

Here, $a<0$ and $D=(1)^2-4(1)<0$, then $f(x)<0$

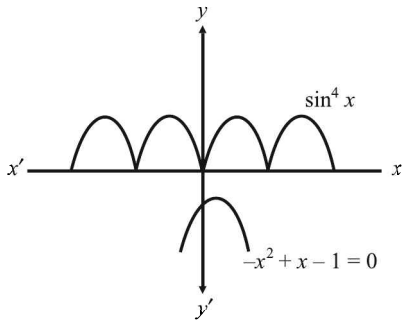


Fig. 3.14

But $\sin^4 x \geq 0$

$\therefore \quad -x^2+x-1 \neq \sin^4 x$

Hence, the number of solutions is 0.

68. Here, the coefficient of $x^2=2>0$, then the conditions are $D\geq 0$, $f(1)>0$ and $f(2)>0$

$\Rightarrow \quad 1-8a\geq 0,\ 2-1+a>0$ and $8-2+a>0$

$\Rightarrow \quad a\leq \frac{1}{8}, a>-1$ and $a>-6$

$\Rightarrow \quad -1<a\leq \frac{1}{8}.$

69. $3^{2x^2-7x+7}=9$

$\Rightarrow \quad 3^{2x^2-7x+7}=3^2$

$\Rightarrow \quad 2x^2-7x+7=2$

$\Rightarrow \quad 2x^2-7x+5=0$

Here, $D=(-7)^2-4(2)(5)=49-40=9>0$

Hence, the given equation has two real roots.

70. Let α and β be the roots of $x^2+bx+a=0$

$\therefore \quad \alpha+\beta=-b$ and $\alpha\beta=a$

Again let γ and δ be the roots of $x^2+ax+b=0$

$\therefore \quad \gamma+\delta=-a$ and $\gamma\delta=b$

Now given

$\alpha-\beta=\gamma-\delta$

$\Rightarrow \quad (\alpha-\beta)^2=(\gamma-\delta)^2$

$\Rightarrow \quad (\alpha+\beta)^2-4\alpha\beta=(\gamma+\delta)^2-4\gamma\delta$

$\Rightarrow \quad b^2-4a=a^2-4b$

$\Rightarrow \quad b^2-a^2=-4(b-a)$

$\Rightarrow \quad b+a+4=0 \qquad [\because a\neq b]$

71. Given $S=p+q=-p$ and product $pq=q$

$\Rightarrow \quad q(p-1)=0 \quad \Rightarrow \quad q=0, p=1$

Now if $q=0$, then $p=0 \Rightarrow p=q$

When $p=1$

$\Rightarrow \quad p+q=-p \quad \Rightarrow \quad q=-2p$

$\Rightarrow \quad q=-2(1) \quad \Rightarrow \quad q=-2$

$\Rightarrow \quad p=1$ and $q=-2$

72. Since one root of the quadratic equation $(a^2-5a+3)x^2+(3a-1)x+2=0$ is twice as large as the other, then let their roots be α and 2α.

$\therefore \quad \alpha+2\alpha=-\frac{3\alpha-1}{a^2-5a+3}$

$\Rightarrow \quad 3\alpha=-\frac{3\alpha-1}{a^2-5a+3}$ and $\alpha\cdot 2\alpha=\frac{2}{a^2-5a+3}$

$\Rightarrow \quad 2\alpha^2=\frac{2}{a^2-5a+3}$

$\Rightarrow \quad \frac{(3a-1)^2}{9(a^2-5a+3)^2}=\frac{1}{(a^2-5a+3)}$

$\Rightarrow \quad (3a-1)^2=9(a^2-5a+3)$

$\Rightarrow \quad 9a^2-6a+1=9a^2-45a+27$

$\Rightarrow \quad 45a-6a=27-1$

$\Rightarrow \quad a=\frac{26}{39}=\frac{2}{3}$

73. Given $\alpha+\beta=\frac{1}{\alpha^2}+\frac{1}{\beta^2}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha^2\beta^2}$

$\Rightarrow \quad 2a^2c=bc^2+ab^2$

$\Rightarrow \quad \frac{2a}{b}=\frac{c}{a}+\frac{b}{c} \Rightarrow \frac{c}{a},\frac{a}{b},\frac{b}{c}$ are in A.P.

$\Rightarrow$ Reciprocals are in H.P.

74. $|x|^2-3|x|+2=0$

$\Rightarrow \quad (|x|-1)(|x|-2)=0$

$\Rightarrow \quad |x|=1,2 \quad \Rightarrow \quad =\pm 1, \pm 2$

Hence, there are 4 real roots.

75. Here $p=1$

$\therefore$ The given equation becomes

$x^2+x=0$

$\Rightarrow \quad x(x+1)=0$

$\Rightarrow \quad x=0,-1$

76. Let the two numbers be α, β

$\therefore \quad \frac{\alpha+\beta}{2}=9$ and $\sqrt{\alpha\beta}=4$

$\therefore$ The required equation is

$x^2 - 2$ (average value of $\alpha, \beta) x + (\sqrt{\text{G.M.}})^2 = 0$

$x^2 - 2(9)x + 16 = 0 \Rightarrow x^2 - 18x + 16 = 0$

77. As $x^2 + px + q = 0$ has equal roots

$\therefore p^2 = 4q$

One root of $x^2 + px + 12 = 0$ is 4.

$\therefore 16 + 4p + 12 = 0 \quad \therefore p = -7$

$\therefore p^2 = 4q \Rightarrow q = \frac{49}{4}$

78. As α, β are the roots of $x^2 - (a-2)x - a - 1 = 0$,

$\therefore \alpha + \beta = a - 2$ and $\alpha\beta = -a - 1$

Now $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (a-2)^2 - 2(-a-1) = a^2 - 2a + 6$

$= (a-1)^2 + 5 > 0$

Hence, the least value occurs when $a = 1$.

79. Let $f(x) = x^2 - 2kx + k^2 + k - 5$

Since both the roots are less than 5, $D \geq 0$, $-\frac{b}{2a} < 5$ and $f(5) > 0$

Now, $D = 4k^2 - 4(k^2 + k - 5)$

$= -4k + 20 \geq 0$

$\Rightarrow k \leq 5$ (i)

$-\frac{b}{2a} < 5 \Rightarrow k < 5$ (ii)

and $f(5) > 0$

$\Rightarrow 25 - 10k + k^2 + k - 5 > 0$

$\Rightarrow k^2 - 9k + 20 > 0$

$\Rightarrow (k-5)(k-4) > 0$

$\Rightarrow k < 4$ and $k > 5$ (iii)

From Eqs. (i), (ii) and (iii), we get $k < 4$

80. Let α and $\alpha + 1$ be the consective integers.

$\therefore (x + \alpha)(x + \alpha + 1) = x^2 - bx + c$

Comparing both sides, we get

$-b = 2\alpha + 1 \qquad c = \alpha^2 + \alpha$

$\therefore b^2 - 4c = (2\alpha + 1)^2 - 4(\alpha^2 + \alpha) = 1$

81. The given roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30°$ and $\tan 15°$

$\Rightarrow \tan 30° + \tan 15° = -p$ and $\tan 30° \tan 15° = q$

Now, $\tan(30° + 15°) = \frac{\tan 30° + \tan 15°}{1 - \tan 30° \cdot \tan 15°}$

$\Rightarrow 1 = \frac{-p}{1-q} \Rightarrow 1 - q = -p$

$\Rightarrow 1 = q - p \Rightarrow 2 + q - p = 2 + 1 = 3$

$\therefore$ The required value = 3

82. Let $y = \frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$

$\Rightarrow 3x^2(y-1) + 9x(y-1) + 7y - 17 = 0$

As x is real, $D \geq 0$

$\Rightarrow 81(y-1)^2 - 4 \times 3(y-1)(7y-17) \geq 0$

$\Rightarrow (y-1)(y-41) \leq 0 \Rightarrow 1 \leq y \leq 41$

$\therefore$ The maximum value of $y = 41$

83. The given equation is $x^2 - 2mx + m^2 - 1 = 0$

$\Rightarrow (x-m)^2 - 1 = 0$

$\Rightarrow (x - m + 1)(x - m - 1) = 0$

$\Rightarrow x = m - 1, m + 1$

From the given condition $m - 1 > -2$ and $m + 1 < 4$, we have

$\Rightarrow m > -1$ and $m < 3$

Hence, $-1 < m < 3$

84. If α, β are the roots of $x^2 + ax + 1 = 0$, then $|\alpha - \beta| < \sqrt{5}$

$\Rightarrow \left|\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}\right| < \sqrt{5}$

$\Rightarrow \left|\sqrt{a^2 - 4}\right| < \sqrt{5}$

$\Rightarrow \sqrt{a^2 - 4} < \sqrt{5}$

$\Rightarrow a^2 - 4 < 5$

$\Rightarrow a^2 < 9$

$\Rightarrow -3 < a < 3$

Hence $a \in (-3, 3)$.

85. $f(x) = x^7 + 14x^5 + 16x^3 + 30x - 560$

$\therefore f'(x) = 7x^6 + 70x^4 + 48x^2 + 30$

$\Rightarrow f'(x)$ is always > 0, $\forall x \in \mathbb{R}$

$\Rightarrow f(x)$ is always an increasing function, so $f(x) = 0$ has at most one real solution.

86. Let the roots of $x^2 - 6x + a = 0$ be α and 4β and that of $x^2 - cx + 6 = 0$ be α and 3β.

$\therefore \alpha + 4\beta = 6$ and $4\alpha\beta = a$

Also, $\alpha + 3\beta = c$ and $3\alpha\beta = 6$

$\Rightarrow a = 8$

$\therefore x^2 - 6x + 8 = 0$

$\Rightarrow x = 2, 4$

and $x^2 - cx + 6 = 0$

$\Rightarrow$ $2^2 - 2c + 6 = 0$

$\Rightarrow$ $c = 5$

$\therefore$ $x^2 - 5x + 6 = 0$

$\Rightarrow$ $x = 2, 3$

Hence the common root is 2.

87. $bx^2 + cx + a = 0$

Roots are imaginary $\Rightarrow$ $c^2 - 4ab < 0$ $\Rightarrow$ $c^2 < 4ab$

$\Rightarrow$ $-c^2 > -4ab$

$$3b^2x^2 + 6bcx + 2c^2$$

Since $3b^2 > 0$, the given expression has the minimum value $= \dfrac{4(3b^2)(2c^2) - 36b^2c^2}{4(3b^2)} = -\dfrac{12b^2c^2}{12b^2}$

$= -c^2 > -4ab$

88. Given, $x = 8 + 3\sqrt{7}$

$\therefore$ $y = \dfrac{1}{8 + 3\sqrt{7}} = 8 - 3\sqrt{7}$

Now, $\dfrac{1}{x^2} + \dfrac{1}{y^2} = \dfrac{x^2 + y^2}{(xy)^2} = \dfrac{(x+y)^2 - 2xy}{(xy)^2}$

$= (x+y)^2 - 2$

$= (8 + 3\sqrt{7} + 8 - 3\sqrt{7})^2 - 2$

$= 16^2 - 2 = 254$

89. The given equation is $(x - 1)(7 - x) = m$

$\Rightarrow$ $x^2 - 8x + 7 + m = 0$

Let the roots be α and 3α.

$\Rightarrow$ $\alpha + 3\alpha = 8 \Rightarrow \alpha = 2$

$\Rightarrow$ $4 - 16 + 7 + m = 0$ (putting $x = 2$ in the equation)

$\Rightarrow$ $m = 5$

90. Since the roots are real, therefore $D \geq 0$

$\Rightarrow$ $4(a+b+c)^2 - 12\lambda(ab + bc + ca) \geq 0$

$\Rightarrow$ $(a+b+c)^2 \geq 3\lambda(ab + bc + ca)$

$\Rightarrow$ $a^2 + b^2 + c^2 \geq (ab + bc + ca)(3\lambda - 2)$

$\Rightarrow$ $3\lambda - 2 \leq \dfrac{a^2 + b^2 + c^2}{ab + bc + ca}$ (i)

Also, $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc} < 1$

$\Rightarrow$ $b^2 + c^2 - a^2 < 2bc$

Similarly, $c^2 + a^2 - b^2 < 2ca$

and $a^2 + b^2 - c^2 < 2ab$

$\Rightarrow$ $a^2 + b^2 + c^2 < 2(ab + bc + ca)$

$\Rightarrow$ $\dfrac{a^2 + b^2 + c^2}{ab + bc + ac} < 2$ (ii)

$\therefore$ From relations (i) and (ii), we get

$3\lambda - 2 < 2$

$\Rightarrow$ $\lambda < \dfrac{4}{3}$

91. Putting $ab + bc + ca = t$, we get

$(a+b+c)^2 = a^2 + b^2 + c^2 + 2t$

$\Rightarrow$ $(a+b+c)^2 = 1 + 2t$

$\Rightarrow$ $1 + 2t \geq 0$

$\Rightarrow$ $-\dfrac{1}{2} \leq t$

Again $(a-b)^2 + (b-c)^2 + (c-a)^2 = 2 - 2t$

$\Rightarrow$ $2 - 2t \geq 0$

$\Rightarrow$ $t \leq 1$

$\Rightarrow$ $-\dfrac{1}{2} \leq t \leq 1$

92. The given equation is

$x^{2/3} + x^{1/3} - 2 = 0$

$\Rightarrow$ $y^2 + y - 2 = 0$ (Putting $x^{1/3} = y$)

$\Rightarrow$ $(y+2)(y-1) = 0$

$\Rightarrow$ $y = -2, 1$

$\therefore$ $x^{1/3} = -2, x^{1/3} = 1 \Rightarrow x = -8, 1$

93. As α and β are the roots of $x^2 - px + r = 0$

$\therefore$ $\alpha + \beta = p$ (i)

and $\alpha\beta = r$ (ii)

Also $\dfrac{\alpha}{2}$ and 2β are the roots of $x^2 - qx + r = 0$

$\therefore$ $\dfrac{\alpha}{2} + 2\beta = q$ or $\alpha + 4\beta = 2q$ (iii)

Solving (i) and (iii) for α and β, we get

$$\beta = \frac{1}{3}(2q - p) \text{ and } \alpha = \frac{2}{3}(2p - q)$$

Substituting the values of α and β in Eq. (ii), we get

$$r = \frac{2}{9}(2p - q)(2q - p)$$

94. Let $\dfrac{x^2 + 34x - 71}{x^2 + 2x - 7} = y$, then $(y - 1)x^2 + 2(y - 17)x + (71 - 7y) = 0$

As x is real, $D \geq 0$

$\Rightarrow$ $4(y-17)^2 \geq 4(y-1)(71 - 7y)$

$\Rightarrow$ $y^2 - 14y + 45 \geq 0$

$\Rightarrow$ $(y-9)(y-5) \geq 0$

$\Rightarrow$ $y - 9 \geq 0, y - 5 \leq 0$

$\Rightarrow$ $y \geq 9$ or $y \leq 5$

$\Rightarrow$ y cannot lie between 5 and 9

95. $x^n - 1 = (x-1)(x-a_1)(x-a_2)\ldots(x-a_{n-1})$

$\Rightarrow \dfrac{x^n-1}{x-1} = (x-a_1)(x-a_2)\ldots(x-a_{n-1})$

$\Rightarrow 1+x+x^2+\cdots+x^n = (x-a_1)(x-a_2)\ldots(x-a_{n-1})$

$\Rightarrow n = (1-a_1)(1-a_2)\ldots(1-a_{n-1})$ [Putting $x = 1$]

96. Let α and β be the roots of the given equation $ax^2 + bx + c = 0$.

$\therefore \quad \alpha+\beta = -\dfrac{b}{a}; \quad \alpha\beta = \dfrac{c}{a}$

The roots of the equation $2x^2 + 8x + 2 = 0$ are $\alpha - 1$ and $\beta - 1$.

Their sum $= \alpha+\beta-2 = -\dfrac{b}{a} - 2 = -\dfrac{8}{2} = -4$

$\Rightarrow \dfrac{b}{a} = 2$

Their product $= (\alpha - 1)(\beta - 1) = \alpha\beta - (\alpha+\beta) + 1$

$= \dfrac{c}{a} + \dfrac{b}{a} + 1 = 1$ [$\because$ New equation is $2x^2 + 8x + 2 = 0$]

$\Rightarrow \quad c + b = 0$

$\Rightarrow \quad b = -c$

97. Clearly, $\alpha+\beta = 1, \alpha\beta = p, \gamma+\delta = 4, \gamma\delta = q\ (p, q \in I)$.

As $\alpha, \beta, \gamma, \delta$ are in G.P. (with common ratio r)

$\therefore \quad \alpha + \alpha r = 1,\ \alpha(r^2 + r^3) = 4$

$\Rightarrow \quad \alpha(1+r) = 1,\ \alpha r^2(1+r) = 4$

$\Rightarrow \quad r^2 \cdot 1 = 4 \Rightarrow r^2 = 4 \Rightarrow r = 2, -2$

If $\quad r = 2, \alpha + 2\alpha = 1 \Rightarrow \alpha = \dfrac{1}{3}$

If $\quad r = -2, \alpha - 2\alpha = 1 \Rightarrow \alpha = -1$

But $\quad p = \alpha\beta - 2$ [$\because$ $r = -2$ and $\alpha = -1$]

For $\quad p = -2, q = \alpha^2 r^5 = 1(-2)^5 = -32$

98. The given equation is $x - \dfrac{2}{x-1} = 1 - \dfrac{2}{x-1}$

Clearly, $x \neq 1$ for the given equation to be defined.

If $x - 1 \neq 0$, we can cancel the common term $-2/(x-1)$ on both sides to get $x = 1$, which is not possible. So the given equation has no root.

99. Let the correct equation be $ax^2 + bx + c = 0$.

Now Sachin's equation

$$\left.\begin{matrix} ax^2 + bx + c' = 0 \\ ax^2 + b'x + c = 0 \end{matrix}\right\} (4, 3)$$

$$-\frac{b}{a} = 7 \qquad \text{(i)}$$

$$\frac{c}{a} = 6 \qquad \text{(ii)}$$

From (i) and (ii), the correct equation is $x^2 - 7x + 6 = 0$, roots are 6 and 1.

100. $P(x) = f(x) - g(x) = ax^2 + bx + c$

Since $(p - 1)$ is the only one root

$\therefore \quad p(x) = k(x+1)^2$

Given $p(-2) = 2 \Rightarrow k = 2$

$\therefore \quad p(2) = 18$

101. $e^{\sin x} - e^{-\sin x} = 4$

Let $e^{\sin x} = t$

$$t - \frac{1}{t} = 4$$

$$t^2 - 4t - 1 = 0 \Rightarrow t = \frac{4 \pm \sqrt{16+4}}{2}$$

$$\Rightarrow t = \frac{4 \pm 2\sqrt{5}}{2} \Rightarrow t = 2 \pm \sqrt{5}$$

$$e^{\sin x} = 2 \pm \sqrt{5}$$

$$-1 \le \sin x \le 1 \qquad \frac{1}{e} \le e^{\sin x} \le e$$

$e^{\sin x} = 2 + \sqrt{5}$ not possible [$\because$ $2.7 < e < 2.8$]

$e^{\sin x} = 2 - \sqrt{5}$ not possible [Never Negative]

$\therefore$ Hence no solution.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. If α and β are the roots of the quadratic equation $x^2 - 3x + 5 = 0$, then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is

(a) $x^2 + 4x + 1 = 0$ (b) $x^2 - 4x - 1 = 0$

(c) $x^2 - 4x + 4 = 0$ (d) $x^2 + 2x + 3 = 0$

2. If α and β are the roots of $Ax^2 + Bx + C = 0$ and α^2 and β^2 are the roots of $x^2 + px + q = 0$, then p is equal to

(a) $\dfrac{B^2 - 2AC}{A^2}$ (b) $\dfrac{2AC - B^2}{A^2}$

(c) $\dfrac{B^2 - 4AC}{A^2}$ (d) $(4AC - B^2)A^2$

3. If $(x + 1)^2$ is greater than $5x - 1$ and less than $7x - 3$, then the integral value of x is equal to
(a) 1 (b) 2
(c) 3 (d) 4

4. The set of values of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is
(a) $(-\infty, 0)$ (b) $(0, 1)$
(c) $(1, \infty)$ (d) $(0, \infty)$

5. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$ is
(a) $\frac{b^2-2ac}{ac}$ (b) $\frac{b^2-2ac}{a^2c^2}$
(c) $\frac{2ac-b^2}{ac}$ (d) $\frac{b^2}{a^2c}$

6. If $y = -2x^2 - 6x + 9$, then the
(a) maximum value of y is -11 and it occurs at $x = 2$
(b) minimum value of y is -11 and it occurs at $x = 2$
(c) maximum value of y is 13.5 and it occurs at $x = -1.5$
(d) minimum value of y is 13.5 and it occurs at $x = -1.5$

7. The real values of a for which the quadratic equation $2x^2 - (a^2 + 8x - 1)x + a^2 - 4a = 0$ possesses roots of opposite signs are given by
(a) $a > 5$ (b) $0 < a < 4$
(c) $a > 0$ (d) $a > 7$

8. The range of the values of $\frac{x}{x^2+4}$ for all real values of x is
(a) $\frac{-1}{2} \le y \le \frac{1}{2}$ (b) $\frac{-1}{4} \le y \le \frac{1}{4}$
(c) $\frac{-1}{6} \le y \le \frac{1}{6}$ (d) none of these

9. The value of the expression $x^2 + 2bx + c$ will be positive if
(a) $b^2 - 4c > 0$ (b) $b^2 - 4c < 0$
(c) $c^2 < b$ (d) $b^2 < c$

10. The number of real roots of the equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ is
(a) 2 (b) 1
(c) infinite (d) none of these

11. If the roots of the equation $x^2 + ax + 25 = 0$ are in the ratio of 2 : 3, then the value of a is
(a) $\frac{\pm 5}{\sqrt{6}}$ (b) $\frac{\pm 25}{\sqrt{6}}$
(c) $\frac{\pm 5}{6}$ (d) none of these

12. If $x^2 + ax + b$ is an integer for every integer x, then
(a) a is always an integer but b need not be an integer
(b) b is always an integer but a need not be an integer
(c) $a + b$ is always an integer
(d) none of these

13. Let $p(x) = 0$ be a polynomial equation of least possible degree, with rational coefficients, having $\sqrt[3]{7} + \sqrt[3]{49}$ as one of its roots. Then the product of all the roots of $p(x) = 0$ is
(a) 7 (b) 49
(c) 56 (d) 63

14. If the expression $\left(mx - 1 + \frac{1}{x}\right)$ is non-negative for all real x, then the minimum value of m must be
(a) $-1/2$ (b) 0
(c) 1/4 (d) 1/2

15. If a, b, c are in G.P., then the equationsa $x^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root if $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

16. If both the roots of the equation
$$x^2 - (p - 4)x + 2e^{2in}p - 4$$
are negative, then p belongs to
(a) $(-\sqrt{2}, 4)$ (b) $(\sqrt{2}, 4)$
(c) $(-4, \sqrt{2})$ (d) $(-\infty, -\sqrt{2})$

17. $\frac{8x^2+16x-51}{(2x-3)(x-4)} > 3$ if x is such that
(a) $x < -4$ (b) $-3 < x < \frac{3}{2}$
(c) $x > \frac{5}{2}$ (d) all of these

18. If p, q, r are in H.P. and p and r are different having same sign, then the roots of the equation $px^2 + 2qx + r = 0$ will be
(a) real (b) equal
(c) imaginary (d) none of these

19. If all the real values of x obtained from the equation $4^x - (a-3)2^x + a - 4 = 0$ are non-positive, then
(a) $a \in (4, 5]$ (b) $a \in (0, 4)$
(c) $a \in (4, +\infty)$ (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. The equation $(a^2 - a - 2)x^2 + (a^2 - 4)x + a^2 - 3a + 2 = 0$ will have more than two solutions if a equals
(a) 2 (b) 1
(c) –2 (d) not possible

2. If the difference between the roots of the equation $x^2 + px + q = 0$ is 1, then
(a) $p^2 = 4q$ (b) $p^2 = 4q + 1$
(c) $p^2 = 4q - 1$ (d) none of these

3. If $0 < a < b < c$ and the roots α and β of the equation $ax^2 + bx + c = 0$ are imaginary, then
(a) $|\alpha| \neq |\beta|$ (b) $|\alpha| > 1$
(c) $|\beta| < 1$ (d) none of these

4. Ramesh and Mahesh solve a quadratic equation. Ramesh reads its constant term wrongly and finds its roots as 8 and 2 whereas Mahesh reads the coefficient of x wrongly and finds its roots as 11 and –1. The correct roots of the equation are
(a) 11, 1 (b) –11, 1
(c) 11, –1 (d) none of these

5. If the roots of $ax^2 + bx + c = 0$ are α and β and the roots of $Ax^2 + Bx + C = 0$ are $\alpha - K$ and $\beta - K$, then $(B^2 - 4AC)/(b^2 - 4ac)$ is equal to
(a) 0 (b) 1 (c) $(A/a)^2$ (d) $(a/A)^2$

6. If α and β are the roots of $x^2 + px + q = 0$ and α^4 and β^4 are the roots of $x^2 - rx + 5 = 0$, then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always
(a) two real roots
(b) two negative roots
(c) two positive roots
(d) one positive and one negative root

7. If 1, ω, ω^2, …, ω^{n-1} are the nth roots of unity, then $(1-\omega)(1-\omega^2)\ldots(1-\omega^{n-1})$ equals
(a) 0 (b) 1 (c) n (d) n^2

8. If the equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root, then it must be equal to
(a) $\dfrac{pq' - p'q}{q - q'}$ (b) $\dfrac{p'q - pq'}{q - q'}$
(c) $\dfrac{q + q'}{p' - p}$ (d) none of these

9. If the equation $\dfrac{x^2 - bx}{ax - c} = \dfrac{m - 1}{m + 1}$ has roots equal in magnitude but opposite in sign, then $m = ?$
(a) $\dfrac{a + b}{a - b}$ (b) $\dfrac{a - b}{a + b}$
(c) $\dfrac{b - c}{b + a}$ (d) none of these

10. If one root of the equations $x^2 + px + q = 0$ and $x^2 + p'x + q^2 = 0$ ($p \neq p'$ and $q \neq q'$) is common, then the root is
(a) $\dfrac{q - q'}{p - p'}$ (b) $\dfrac{pq' - p'q}{q - q'}$
(c) $\dfrac{q - q'}{p' - p}$ or $\dfrac{pq' - p'q}{q - q'}$ (d) $\dfrac{q - q'}{p' - p}$ or $\dfrac{pq - p'q'}{q - q'}$

11. If one root of the equation $x^2 - 30x + p = 0$ is the square of the other, then p is equal to
(a) only 125 (b) 125, –216
(c) 125, 215 (d) only 216

12. The value of $\sqrt{8 + 2\sqrt{8 + 2\sqrt{8 + 2\sqrt{8 + \cdots}}}}$
(a) 10 (b) 6
(c) 8 (d) 4

13. If $\tan\theta$ and $\sec\theta$ are the roots of the equation $ax^2 + bx + c = 0$, then
(a) $a^3 + b^3 + c^3 - 3abc = 0$ (b) $a^2 + b^2 + 2ac = 0$
(c) $a^4 + 4ab^2c - b^4 = 0$ (d) none of these

14. Let $\alpha + \beta$, $\alpha, \beta \in \mathbb{R}$, $\beta \neq 0$ be a root of the equation $x^3 + ax + b = 0$, where $a, b \in \mathbb{R}$. Then the cubic equation with real coefficients, one of whose roots is α, is
(a) $x^3 - ax + b = 0$ (b) $x^3 - 2ax + b = 0$
(c) $8x^3 + 2ax - b = 0$ (d) $8x^3 + 2ax + b = 0$

15. If the roots of the equation $bx^2 + cx + a = 0$ are imaginary, then for all real values of x the expression $3b^2x^2 + 6bcx + 2c^2$ is
(a) greater than $4ab$ (b) less than $4ab$
(c) greater than $-4ab$ (d) less than $-4ab$

16. If $a \in Z$ and the equation $(x - a)(x - 10) + 1 = 0$ has integral roots, then the values of a are
(a) 10, 8 (b) 12, 10
(c) 12, 8 (d) none of these

17. The values of the parameter a for which the quadratic equations $(1 - 2a)x^2 - 6ax - 1 = 0$ and $ax^2 - x + 1 = 0$ have at least one root in common are
(a) $0, \dfrac{1}{2}$ (b) $\dfrac{1}{2}, \dfrac{2}{9}$
(c) $\dfrac{2}{9}$ (d) $0, \dfrac{1}{2}, \dfrac{2}{9}$

18. Let $a = \cos\dfrac{2\pi}{7} + i\sin\dfrac{2\pi}{7}$, $\alpha = a + a^2 + a^4$ and $\beta = a^3 + a^5 + a^6$. Then the equation whose roots are α and β is
(a) $x^2 - x + 2 = 0$ (b) $x^2 + x - 2 = 0$
(c) $x^2 - x - 2 = 0$ (d) $x^2 + x + 2 = 0$

ANSWERS

Level 1

1. (c)	**2.** (b)	**3.** (c)	**4.** (b)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (b)	**9.** (d)	**10.** (d)
11. (b)	**12.** (c)	**13.** (c)	**14.** (b)	**15.** (a)	**16.** (b)	**17.** (d)	**18.** (c)	**19.** (a)	

Level 2

1. (a)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (c)	**6.** (a)	**7.** (c)	**8.** (a)	**9.** (b)	**10.** (c)
11. (a)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (c)	**16.** (c)	**17.** (c)	**18.** (d)		

4 Matrices

JAMES JOSEPH SYLVESTER

- Definition of Matrix
- Types of Matrices
- Addition of Matrices
- Properties of Matrix Addition
- Subtraction of Matrices
- Properties of Matrix Subtraction
- Multiplication of Matrices
- Properties of Multiplication
- Adjoint of a Square Matrix
- Properties of adj A
- Inverse of a Matrix
- Two Theorems Related to Invertible Matrix
- A Method for Finding the Inverse of a Matrix
- Solution of a System of Non-homogeneous Equations
- Solution of a System of Homogeneous and Linear Equations

☞ DEFINITION OF MATRIX

A *matrix* is any rectangular array of numbers written within brackets. A matrix is usually represented by a capital letter and classified by its dimensions. The dimensions of a matrix are its number of rows and columns. An $m \times n$ matrix is usually written as

$$A_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

where a_{ij} represents any number which lies on the ith row (from top) and jth column (from left).

Note The matrix is not a number. It has got no numerical value.

☞ TYPES OF MATRICES

Row Matrix

A matrix having only one row is called a *row matrix*.

Examples:

(a) [2 3 5] (b) [4 5 9]

(c) $[a_1 \quad a_2 \quad \dots \quad a_n]$

Column Matrix

A matrix having only one column is called a column matrix.
Examples:

(a) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ (b) $\begin{bmatrix} 6 \\ 1 \\ 4 \end{bmatrix}$

(c) $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$

Null Matrix or Zero Matrix

A matrix which has all its elements zero is called a *null matrix*.
Examples:

(a) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2}$ (b) $[0]_{1\times 1}$

(c) $\begin{bmatrix} m\left\{\begin{matrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ \underbrace{0 & 0 & \dots & 0}_{n} \end{matrix}\right. \end{bmatrix}_{m\times n}$

Square Matrix

A matrix in which the number of rows is equal to the number of columns is called a *square matrix*. It is denoted by I_nA (where n is the size of the matrix).
Examples:

(a) $[\,2\,]$ (b) $\begin{bmatrix} \alpha & \beta & \gamma \\ x & y & z \\ a & b & c \end{bmatrix}_{3\times 3}$

(c) $\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}_{n\times n}$

Note

- $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ are called the elements of the principal diagonal.
- A square matrix is a singular matrix if its determinant is zero, otherwise it is a non-singular matrix.

Diagonal Matrix

A square matrix whose all the non-diagonal elements are zero, i.e. $a_{ij} = 0,\ \forall i \neq j$, is called a *diagonal matrix*. It can be written as dia $(a_{11}a_{12}\dots a_{nn})$.

Examples:

(a) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$

Scalar Matrix

A square matrix $A = [a_{ij}]_{m\times n}$ is called a *scalar matrix* if

(i) $a_{ij} = 0,\ \forall i \neq j$ and

(ii) $a_{ii} = c$ for all i, where $c \neq 0$

i.e. a diagonal matrix in which all the diagonal elements are equal is called a scalar matrix.

Example: $\begin{bmatrix} k & 0 & \cdots & 0 \\ 0 & k & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & k \end{bmatrix}$

It can be written as dia($kk...k$).

Unit or Identity Matrix

A square matrix each of whose diagonal elements is unity and each of whose non-diagonal elements is equal to zero is called a *unit matrix*.

Example: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Note The identity matrices of order n are denoted by I_n.

Transpose of Matrices

If $A = [a_{ij}]_{m\times n}$, then the transpose of A, i.e. A^T or $A' = [a_{ij}]_{n\times m}$.

Example: If $A = \begin{bmatrix} 4 & 7 & 5 \\ 3 & 6 & 8 \end{bmatrix}$, then $A^T = \begin{bmatrix} 4 & 3 \\ 7 & 6 \\ 5 & 8 \end{bmatrix}$.

Properties of transpose of matrices

1. $(A^T)^T = A$, i.e. the transpose of the transpose of a matrix is the matrix itself.
2. $(A+B)^T = A^T + B^T$, i.e. the transpose of the sum of two matrices is the sum of their transposes.
3. $(kA)^T = kA^T$ (k is a scalar).
4. $(AB)^T = B^TA^T$, i.e. the transpose of the product of two matrices is the product in reverse order of their transposes.

Equal Matrices

Two matrices are said to be *equal* if

(i) they are of the same order

(ii) the elements at the corresponding positions of the two matrices are equal.

Illustration 1. If $\begin{bmatrix} k+5 & 2m+k \\ n-2 & 4p-6 \end{bmatrix} = \begin{bmatrix} -k-3 & 0 \\ 5 & 3p \end{bmatrix}$, then find k, m, n, p.

Solution: $\because$ Two matrices are equal,

$\therefore \quad k+5=-k-3 \Rightarrow 2k=-8 \Rightarrow k=-4$

$n-2=5 \Rightarrow n=7$

$4p-6=3p \Rightarrow p=6$

$2m+k=0 \Rightarrow 2m-4=0 \Rightarrow m=2$

$\therefore \quad (k,m,n,p) \equiv (-4,2,7,6)$

Symmetric Matrices

They are of two types.

Symmetric matrix

A square matrix $A=[a_{ij}]$ is called symmetric if $[a_{ij}]=[a_{ji}]$ for all i, j, that is, if $A^T=A$.

Example: $\begin{bmatrix} 2 & 1 & 7 \\ 1 & 3 & 9 \\ 7 & 9 & 5 \end{bmatrix}$

Skew-symmetric matrix

A square matrix $A=[a_{ij}]$ is skew symmetric if $a_{ij}=-a_{ji}$ for all i, j, that is, if $A^T=A$.

Example: $\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$

Illustration 2. Any square matrix A is skew symmetric

$\Leftrightarrow \quad a_{ij}=-a_{ij} \forall i,j \quad \Leftrightarrow \quad (A_{ij})=-(A^T)_{ij} \forall i,j$

$\Leftrightarrow \quad A=-A^T \quad \Leftrightarrow \quad A^T=-A$

Note Thus, a square matrix A is a skew-symmetric matrix iff $A^T=-A$.

Upper Triangular Matrix

A square matrix whose all the elements below the principal diagonal are zero, i.e. $a_{ij}=0, \forall i>j$, is called an *upper triangular matrix.*

Examples: (a) $\begin{bmatrix} 5 & 4 & 3 \\ 0 & 6 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 5 & 6 & 8 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

Lower Triangular Matrix

A square matrix whose all the elements above the principal diagonal are zero, i.e. $a_{ij}=0, \forall i<j$, is known as a *lower triangular matrix.*

Example: $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ 5 & 1 & 7 & 0 \\ 6 & 2 & 3 & 1 \end{bmatrix}$

Orthogonal Matrix

A square matirx A is called orthogonal if $AA^T=A^TA=I$

$\Rightarrow \quad |AA^T|=|I|$

$\Rightarrow \quad |A||A^T|=1$

$\Rightarrow \quad |A|^2=1 \qquad (\because \quad A^T=-A)$

$\Rightarrow \quad |A|=\pm 1$

Example: $A=\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$

Note If $|A|$, i.e. $\det(A)=1$, then the matrix is called proper.

Conjugate Matrices

If $A=[a_{ij}]$, then $\bar{A}=[\overline{a_{ij}}]$.

Example: $A=\begin{bmatrix} 2+3i & 1+i \\ 2 & 5i \end{bmatrix} \Rightarrow \bar{A}=\begin{bmatrix} 2-3i & 1-i \\ 2 & -5i \end{bmatrix}$

Unitary Matrix

If $\overline{A^T}A=I$, then A is called a *unitary matrix.*

Illustration 3. Show that $A=\begin{bmatrix} 1+i & 1+2i \\ 2+i & 1-i \end{bmatrix}$ is a unitary matrix.

Solution: $A^T=\begin{bmatrix} 1+i & 2+i \\ 1+2i & 1-i \end{bmatrix} \Rightarrow \overline{A^T}=\begin{bmatrix} 1-i & 2-i \\ 1-2i & 1+i \end{bmatrix}$

Now $\overline{A^T}A=\begin{bmatrix} 1-i & 2-i \\ 1-2i & 1+i \end{bmatrix}\begin{bmatrix} 1+i & 1+2i \\ 2+i & 1-i \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}=I$

So A is a unitary matrix.

Hermitian Matrix

If a square matrix is A such that $\overline{A^T}=A$, then A is called a *Hermitian matrix.*

Illustration 4. If $A=\begin{bmatrix} l & x+iy & a+ib \\ x-iy & m & p+iq \\ a-ib & p-iq & n \end{bmatrix}$

show that it is a Hermitian matrix.

Solution: $(A^T)=\begin{bmatrix} l & x-iy & a-ib \\ x+iy & m & p-iq \\ a+ib & p+iq & n \end{bmatrix}$

$$\Rightarrow \quad (\overline{A^T})=\begin{bmatrix} l & x+iy & a+ib \\ x-iy & m & p+iq \\ a-ib & p-iq & n \end{bmatrix}=A$$

Hence, A is Hermitian.

Note The diagonal elements of a hermitian matrix are necessarily real.

Skew-Hermitian Matrix

A square matrix $A=[a_{ij}]$ is said to be a skew-Hermitian matrix if $a_{ij}=-\overline{a}_{ij}, \forall i, j$, i.e. $A^T=-A$.

Illustration 5. If $A=\begin{bmatrix} 2i & -2-3i \\ 2-3i & -i \end{bmatrix}$ show that it is a skew-Hermitian matrix.

Solution: $A=\begin{bmatrix} 2i & -2-3i \\ 2-3i & -i \end{bmatrix}$

$$A^T=\begin{bmatrix} 2i & 2-3i \\ -2-3i & -i \end{bmatrix}$$

$$\Rightarrow \quad \overline{A^T}=\begin{bmatrix} -2i & 2+3i \\ -2+3i & i \end{bmatrix}$$

$$=-\begin{bmatrix} 2i & -2-3i \\ 2-3i & -i \end{bmatrix}=-A$$

Note

- The diagonal elements of a skew-Hermitian matrix are either purely imaginery or zero.
- Every square matrix (with complex elements) can be uniquely expreseed as the sum of Hermitian and skew-Hermitian matrices, i.e. $A=\frac{1}{2}\left(A+\overline{A^T}\right)+\frac{1}{2}\left(A-\overline{A^T}\right)$.

Submatrix

A submatrix is obtained by deleting some rows and columns.

Example: $A=\begin{bmatrix} 2 & 3 & 5 & 6 \\ 5 & 7 & 1 & 0 \\ 2 & 1 & 1 & 7 \end{bmatrix}=\begin{bmatrix} 2 & 3 & 5 & 6 \\ 5 & 7 & 1 & 0 \\ 2 & 1 & 1 & 7 \end{bmatrix}$

$\Rightarrow$ Submatrix of $A=\begin{bmatrix} 5 & 7 & 0 \\ 2 & 1 & 7 \end{bmatrix}$.

Idempotent Matrix

A square matrix A is called *idempotent* if it satisfies the relation $A^2=A$.

Illustration 6. Show that the matrix $A=\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent.

Solution: $A^2=A\cdot A=\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}\times\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$

$$=\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}=A$$

So, A is idempotent.

Periodic Matrix

A square matrix A is called *periodic* if $A^{k+1}=A$, where k is a positive integer. If k is the least positive integer for which $A^{k+1}=A$, then k is said to be the period of A. For $k=1$, we get $A^2=A$ and we call it to be an idempotent matrix.

Nilpotent Matrix

A square matrix A is called *nilpotent matrix* of order m provided it satisfies the relation $A^k=O$ and $A^{k-1}\neq O$, where k is a positive integer and O is a null matrix and k is the order of the nilpotent matrix A.

Illustration 7. Show that $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is a nilpotent matrix of order 3.

Solution: Let $A=\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

$$A^2=A\cdot A=\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}\times\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$=\begin{bmatrix} 1+5+6 & 1+2-3 & 3+6-9 \\ 5+10-12 & 5+4-6 & 15+12-18 \\ -2-5+6 & -2-2+3 & -6-6+9 \end{bmatrix}=\begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$$

$$A^3=A^2\cdot A=\begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}\times\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}=0$$

$\therefore \quad A^3=0 \quad$ i.e. $A^k=0$

Hence $k=3$. So, A is nilpotent of order 3.

Involutory Matrix

A square matrix A is called *involutory* provided it satisfies the relation $A^2 = I$, where I is the identity matrix.

Illustration 8. Show that the matrix $\begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is involutory.

Solution: $A^2 = A \cdot A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \times \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Hence the given matrix A is involutory.

ADDITION OF MATRICES

If $A[a_{ij}]$ and $B[b_{ij}]$ are two matrices of the same order, then $A + B$ is a matrix, each of whose element is the sum of the corresponding elements of A and B. For example,

$$A + B = [a_{ij} + b_{ij}]$$

Illustration 9. If $A = \begin{bmatrix} a & b & c \\ p & q & r \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 4 & 3 \\ 7 & 5 & 1 \end{bmatrix}$, find $A + B$.

Solution: If $A = \begin{bmatrix} a & b & c \\ p & q & r \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 4 & 3 \\ 7 & 5 & 1 \end{bmatrix}$

$$A + B = \begin{bmatrix} a+2 & b+4 & c+3 \\ p+7 & q+5 & r+1 \end{bmatrix}$$

PROPERTIES OF MATRIX ADDITION

If A, B and C are three matrices of the same order $m \times n$ and $A = [a_{ij}], B = [b_{ij}], C = [c_{ij}]$, then the following properties hold in matrix addition:

1. Commutative law i.e. $A + B = B + A$
2. Associative law i.e. $A + (B + C) = (A + B) + C$
3. Distributive law i.e. $k(A + B) = kA + kB$ (where k is a scalar)
4. Additive identity i.e. $A + O = A = O + A$
5. Additive inverse i.e. $A + B = O = B + A$
   Here B is the additive inverse of the matrix A or the negative of A.
6. (i) Left cancellation law, i.e.
   $A + B = A + C \Rightarrow B = C$
   (ii) Right cancellation law, i.e.
   $B + A = C + A \Rightarrow B = C$
7. The equation $A + X = 0$ has a unique solution $X = -A$, where X has the order $m \times n$.

Illustration 10. If $A = \begin{bmatrix} 2 & 3 \\ 5 & 2 \\ 8 & 9 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 5 & 2 \\ 8 & 9 \end{bmatrix}$ and $A + B - 2D = 0$, then find D.

Solution: Given $A + B - 2D = 0 \Rightarrow D = \frac{1}{2}(A + B)$

$$D = \frac{1}{2}\begin{bmatrix} 2 & 3 \\ 5 & 2 \\ 8 & 9 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 2 & 1 \\ 5 & 2 \\ 8 & 9 \end{bmatrix}$$

$$\Rightarrow \quad D = \frac{1}{2}\begin{bmatrix} 4 & 4 \\ 10 & 4 \\ 16 & 18 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 5 & 2 \\ 8 & 9 \end{bmatrix}$$

SUBTRACTION OF MATRICES

If $A[a_{ij}]$ and $B[b_{ij}]$ are two matrices of the same order, then $A - B = [a_{ij} - b_{ij}]$.

PROPERTIES OF MATRIX SUBTRACTION

The properties of matrix subtraction are the same as those of addition.

Illustration 11. Solve the following equation for x and y:

$$2x - y = \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & 2 \end{bmatrix} \text{ and } 2y + x = \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix}$$

Solution: $2x - y = \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & 2 \end{bmatrix}$

Multiplying both sides by 2, we get

$$4x - 2y = \begin{bmatrix} 6 & -6 & 0 \\ 6 & 6 & 4 \end{bmatrix} \quad \text{(i)}$$

Given $$x + 2y = \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix} \quad \text{(ii)}$$

From (i) + (ii), we get

$$5x = \begin{bmatrix} 6 & -6 & 0 \\ 6 & 6 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix} = \begin{bmatrix} 10 & -5 & 5 \\ 5 & 10 & 0 \end{bmatrix}$$

$$\Rightarrow \quad x = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

Putting the value of x, in Eq. (ii), we get

$$\begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix} + 2y = \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix}$$

$$\Rightarrow \quad 2y = \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix} - \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 4 \\ -2 & 2 & -4 \end{bmatrix}$$

$$\Rightarrow \quad y = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 1 & -2 \end{bmatrix}$$

$$\Rightarrow \quad x = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \text{ and } y = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 1 & -2 \end{bmatrix}$$

Note Two matrices are said to be conformable for addition iff they are of the same order.

MULTIPLICATION OF MATRICES

For multiplication unlike addition, A and B may have different sizes. However, the number of columns of A must be the number of rows of B. Thus, if A is $m \times n$, then B must be $n \times p$ for some p. As we shall see, the size of AB is $m \times p$.

Let $A = [a_{ij}]$ be an $m \times n$ matrix and let $B = [b_{ij}]$ be an $n \times p$ matrix. The product AB is the $m \times p$ matrix $C = [c_{ij}]$ such that $c_{ij} = a_{i1}b_{1i} + a_{i2}b_{2j} + a_{i3}b_{2j} + \cdots + a_{in}b_{nj}$ for $i = 1, 2, 3, \ldots, m$ and $j = 1, 2, \ldots, p$.

Size of A	Size of B	Size of AB
2×3	3×5	2×5
4×2	2×3	4×3
3×1	1×3	3×3
1×3	3×1	1×1
5×3	3×5	5×5
5×3	5×3	AB is not defined

Note The following diagram may help you remember the relationship between sizes of matrices when working with a product AB.

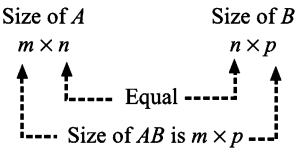


Fig. 4.1

PROPERTIES OF MULTIPLICATION

1. Multiplication of matrices is not commutative, i.e. $AB \neq BA$.

Note If $AB = -BA$, then A and B are said to be anti-commutative.

2. Matrix multiplication is associative if conformability is assured, i.e. $A(BC) = (AB)C$.
3. Matrix multiplication is distributive with respect to addition, i.e. $A(B + C) = AB + AC$.
4. If A and B are two matrices and O is a null matrix such that $AB = 0$, then $A = 0$ or $B = 0$ or both are 0.
5. Multiplication of matrix A by unit matrix I.
$$AI = A = IA$$
6. If A is an $n \times p$ matrix and O is a null matrix, then
   (i) $A_{n \times p} O_{p \times m} = O_{n \times m}$
   (ii) $O_{m \times n} A_{n \times p} = O_{m \times p}$
7. The product of $\underbrace{AA \ldots A}_{m \text{ times}} = A^m$ and $(A^m)^n = A^{mn}$.

ADJOINT OF A SQUARE MATRIX

Let $A = [a_{ij}]$ be a square matrix of order n and let C_{ij} be a cofactor of a_{ij} in A. Then the transpose of the matrix of cofactors of elements of A is called the *adjoint* of A which is denoted by adj A. That is, $\text{adj}\, A = [C_{ij}]^T \Rightarrow (\text{adj}\, A)_{ij} = C_{ij}$ = cofactors of a_{ij} in A.

Illustration 12. Find the adjoint of matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ -1 & 2 & 3 \end{bmatrix}$.

Solution: Let C_{ij} be a cofactor of a_{ij} in A. Then cofactors of elements of A are given by

$$C_{11} = \begin{vmatrix} 1 & -3 \\ 2 & 3 \end{vmatrix} = 9, C_{12} = -\begin{vmatrix} 2 & -3 \\ -1 & 3 \end{vmatrix} = -3, C_{13} = \begin{vmatrix} 2 & 1 \\ -1 & 2 \end{vmatrix} = 5$$

$$C_{21} = -\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = -1, C_{22} = \begin{vmatrix} 1 & 1 \\ -1 & 3 \end{vmatrix} = 4, C_{23} = -\begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix} = -3$$

$$C_{31} = \begin{vmatrix} 1 & 1 \\ 1 & -3 \end{vmatrix} = -4, C_{32} = -\begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} = 5, C_{33} = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1$$

$$\therefore \quad \text{adj}\, A = \begin{bmatrix} 9 & -3 & 5 \\ -1 & 4 & -3 \\ -4 & 5 & -1 \end{bmatrix}^T = \begin{bmatrix} 9 & -1 & -4 \\ -3 & 4 & 5 \\ -5 & -3 & -1 \end{bmatrix}$$

PROPERTIES OF adj A

1. If A is an n-rowed square matrix, then (adj A)A = A(adj A) = $|A| I_n$.

Case 1. If A is a square singular matrix, then (adj A)A = A(adj A) = O (null matrix).

Case 2. If A is non-singular matrix, then $|\text{adj}\, A| = |A|^{n-1}$.

2. adj (AB) = (adj B) · (adj A)
3. $(\text{adj}\, A)^T = \text{adj}\, A^T$
4. $\text{adj}(\text{adj}\, A) = |A|^{n-2} A$ (A is a non-singular matrix)

5. $|\text{adj}(\text{adj A})| = |\text{A}|^{(n-1)^2}$ (A is a non-singular matrix)
6. Adjoint of a diagonal matrix is a diagonal matrix.

☞ INVERSE OF A MATRIX

Let A be a square matrix of order n. If there exists a matrix B such that $AB = I_n = BA$, then B is called the *inverse* of A and is denoted by A^{-1} (read "A inverse").

☞ TWO THEOREMS RELATED TO INVERTIBLE MATRIX

1. Every invertible matrix possesses a unique inverse.
Proof: Let A be an invertible matrix of order $n \times m$. Let B and C be two inverse matrices of A

Then $AB = BA = I_n$ (i)

and $AC = CA = I_n$ (ii)

Now, $AB = I_n$

$\Rightarrow \quad C(AB) = CI_n \quad \Rightarrow \quad (CA)B = CI_n$ (by associativity)

$\Rightarrow \quad I_nB = CI_n \quad [\because\ CA = I_n \text{ by (ii)}]$

$\Rightarrow \quad B = C \quad [\because\ I_nB = B,\ CI_n = C]$

Hence, an invertible matrix possesses a unique inverse.

2. A square matrix is invertible if and only if it is non-singular.
Proof: Let A be an invertible matrix. Then there exists a matrix B such that

$AB = I_n = BA \quad \Rightarrow \quad |AB| = |I_n|$

$\Rightarrow \quad |A||B| = 1 \qquad [\because\ |AB| = |A||B|]$

$\Rightarrow\ |A| \neq 0 \quad \Rightarrow \quad A$ is a non-singular matrix.

☞ A METHOD FOR FINDING THE INVERSE OF A MATRIX

Let A be a non-singular square matrix of order n. Then

$$A(\text{adj } A) = |A| I_n = (\text{adj } A)A$$

$$\Rightarrow \quad A\left(\frac{1}{|A|}\text{adj}\,A\right) = I_n = \left(\frac{1}{|A|}\text{adj}\,A\right)A$$

$$\Rightarrow \quad A^{-1} = \frac{1}{|A|}\text{adj}\,A \quad \text{[by definition of inverse]}$$

☞ MORE RESULTS TO INVERSE OF A MATRIX

If A, B and C are non-singular square matrices, then

1. (i) $AB = AC \ \Rightarrow \ B = C$
   (ii) $BA = CA \ \Rightarrow \ B = C$
2. $(AB)^{-1} = B^{-1}A^{-1}$ in general
   $(ABC\ldots Z)^{-1} = Z^{-1}Y^{-1}\ldots B^{-1}A^{-1}$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $\left(\overline{A^T}\right)^{-1} = \left(\overline{A^{-1}}\right)^T$
5. Inverse of a non-singular diagonal matrix

$$\text{If } A \equiv \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix} \text{ (and } |A| \neq 0\text{), then } A^{-1} \equiv \begin{bmatrix} \frac{1}{x} & 0 & 0 \\ 0 & \frac{1}{y} & 0 \\ 0 & 0 & \frac{1}{z} \end{bmatrix}$$

☞ FINDING THE INVERSE OF A SQUARE MATRIX BY USING ELEMENTARY ROW TRANSFORMATIONS

For a 2 × 2 Matrix

Step I. Make $a_{11} = 1$ by operating either $R_1 \to \frac{1}{a_{11}}R_1$ (if $a_{11} \neq 0$) or by $R_1 \leftrightarrow R_2$. In case of 0, then first operate $R_1 \leftrightarrow R_2$ and then

$$R_1 \to \frac{1}{a_{11}}R_1 \text{ (if need be).}$$

Step II. Make $a_{12} = 0$ by the operation $R_1 \to R_2 - a_{21}R_1$.

Step III. Make $a_{22} = 1$ by the operation $R_2 \to \frac{1}{a_{22}}R_2$.

Step IV. Make $a_{12} = 0$ by the operation $R_1 \to R_1 - a_{12}R_2$.

For a 3 × 3 Matrix

Step I. Make $a_{11} = 1$ either by operating $R_1 \to \frac{1}{a_{11}}$ (if $a_{11} \neq 0$) or by operating $R_1 \leftrightarrow R_2$ or $R_1 \leftrightarrow R_3$ and then $R_1 \to \frac{1}{a_{11}}R_1$ (if need be).

Step II. Operate $R_2 \to R_2 - a_{21}R_1$ and $R_3 \to R_3 - a_{31}R_1$ to obtain $a_{21} = a_{31} = 0$.

Step III. Operate $R_2 \to \frac{1}{a_{22}}R_2$ (if $a_{22} \neq 0$) or $R_2 \leftrightarrow R_3$ following by $R_2 \to \frac{1}{a_{22}}R_2$ (if need be) to make $a_{22} = 1$.

Step IV. Operate $R_3 \to R_3 - a_{32}R_2$ to make $a_{32} = 0$.

Step V. Operate $R_3 \to \frac{1}{a_{33}}R_3$ to make $a_{33} = 1$.

Step VI. Operate $R_2 \to R_2 - a_{23}R_3$ and $R_1 \to R_1 - a_{13}R_3$ to make $a_{23} = a_{13} = 0$.

Step VII. Operate $R_1 \to R_1 - a_{12}R_2$ to make $a_{12} = 0$.

SOLUTION OF A SYSTEM OF NON-HOMOGENEOUS EQUATIONS

Consider the following system of m linear equations (non-homogenous) in n unknowns $x_1, x_2, \ldots, x_n$:

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots.. + a_{1n}x_m &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots.. + a_{2n}x_m &= b_2 \\ \vdots \quad\quad \vdots \quad\quad\quad\quad \vdots \quad\quad \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + \ldots.. + a_{mn}x_n &= b_n \end{aligned}\right\}$$

This system of equations can be written in a matrix form as follows:

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Thus, $AX = B$. Here, A is a square matrix.

$\therefore \quad X = A^{-1}B$

System of non-homogeneous linear equations

S. No.	Condition	Sub-condition	Nature of solution	Number of solutions	Solution
1.	$\lvert A\rvert \neq 0$		Consistent	Unique solution	$X = A^{-1}B$
2.	$\lvert A\rvert = 0$	(i) (Adj A)$B \neq 0$ (ii) (Adj A)$B = 0$	Inconsistent Consistent	No solution Infinite solutions	

SOLUTION OF A SYSTEM OF HOMOGENEOUS AND LINEAR EQUATIONS

Consider the following system of homogeneous linear equations in n unkowns $x_1, x_2, \ldots, x_n$.

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots. + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \ldots. + a_{2n}x_n &= 0 \\ \vdots \quad\quad \vdots \quad\quad\quad\quad \vdots \quad\quad \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + \ldots. + a_{mn}x_n &= 0 \end{aligned}\right\}$$

This system can be written in a matrix form as follows:

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Thus, $AX = 0$

System of homogeneous linear equations

S. No.	Condition	Sub condition	Nature of solution	Solution
1.	$\lvert A\rvert \neq 0$		Trivial solution	$X = A^{-1}0$
2.	$\lvert A\rvert = 0$	(i) (Adj A)$B = 0$ or $B = 0$	Non-trivial solution Infinite solutions	

TRACE OF A MATRIX

If $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, then the *trace* of the matrix A is represented as $Tr(A) = a + e + i$ = the sum of the main diagonal entries.

RANK OF A MATRIX

The rank of a matrix A is the order of the highest-order non-zero minors of A.

$\therefore$ Rank of a matrix is r if

1. Every minor of order $(r+1)$ is zero.
2. There is at least one non-zero minor of order r.

SOLVED PROBLEMS

1. If $\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then

(a) $a = 1, b = 1$ (b) $a = \cos 2\theta, b = \sin 2\theta$

(c) $a = \sin 2\theta, b = \cos 2\theta$ (d) none of these

Ans. (b)

Solution: Since

$$\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\frac{1}{1+\tan^2\theta}\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \frac{1}{1+\tan^2\theta}\begin{bmatrix} 1-\tan^2\theta & -2\tan\theta \\ 2\tan\theta & 1-\tan^2\theta \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{1-\tan^2\theta}{1+\tan^2\theta} & \frac{-2\tan\theta}{1+\tan^2\theta} \\ \frac{2\tan\theta}{1+\tan^2\theta} & \frac{1-\tan^2\theta}{1+\tan^2\theta} \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$\therefore \quad a = \cos 2\theta, b = \sin 2\theta$

2. The inverse of a skew-symmetric matrix of an odd order is

(a) a symmetric matrix (b) a skew-symmetric matrix

(c) diagonal matrix (d) does not exist

Ans. (d)

Solution: Let A be a skew-symmetric matrix of order n.

By definition $A^T = -A$

$$\Rightarrow \quad |A^T| = |-A| \quad \Rightarrow \quad |A| = (-1)^n |A|$$

$$\Rightarrow \quad |A| = -|A| \qquad [\because \ n \text{ is odd}]$$

$$\Rightarrow \quad 2|A| = 0 \quad \Rightarrow \quad |A| = 0$$

$\therefore \quad A^{-1}$ does not exist.

3. Let A be a non-singular square matrix. Then $|\text{adj } A|$ is equal to

(a) $|A|^n$ (b) $|A|^{n-1}$

(c) $|A|^{n-2}$ (d) none of these

Ans. (b)

Solution: Since A is non-singular

$\therefore \quad A^{-1}$ exists

Now, $A(\text{adj}A) = |A|I = (\text{adj}A)A$

$$\Rightarrow \quad |A||\text{adj } A| = |A|^n = |\text{adj } A||A|$$

$$\therefore \quad |\text{adj}A| = |A|^{n-1}$$

4. If A and B are two non-zero square matrices of the same order such that the product $AB = 0$, then

(a) both A and B must be singular

(b) exactly one of them must be singular

(c) both of them are singular

(d) none of these

Ans. (c)

Solution: If possible assume that A is non-singular, then A^{-1} exists. Thus,

$$AB = 0 \quad \Rightarrow \quad A^{-1}(AB) = (A^{-1}A)B = 0$$

$\Rightarrow \quad IB = 0$ or $B = 0$. A contradiction.

5. If $x + y + z = 6, x - y + z = 2, 2x + y - z = 1$, then x, y, z are respectively

(a) 3, 2, 1 (b) 1, 2, 3

(c) 2, 1, 3 (d) none of these

Ans. (b)

Solution: $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$

Let $AX = B \quad \Rightarrow \quad X = A^{-1}B$ (i)

Now, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

$$\therefore \quad |A| = D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - 2R_1$

$$\Rightarrow \quad |A| = D = \begin{vmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & -1 & -3 \end{vmatrix} = 6 \neq 0$$

i.e. matrix A is non-singular.

$\therefore$ We will have a unique solution and the equations are consistent.

Now, $A^{-1} = \text{adj}A \div |A| = \frac{1}{6}\begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$

$$\Rightarrow A^{-1} = \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix} \quad \therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix}\begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6\cdot 0 + \frac{2}{3} + \frac{1}{3} \\ \frac{6}{2} - 1 + 0 \\ \frac{6}{2} + \frac{1}{3} - \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\therefore \quad x = 1, y = 2, z = 3$

6. The value of x for which the matrix

$A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ is the inverse of $B = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ is

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{1}{4}$ (d) $\frac{1}{5}$

Ans. (d)

Solution: We have $AB = \begin{bmatrix} 5x & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10x-2 & 5x \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\Rightarrow \quad x = \frac{1}{5}$$

7. Let $F(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Then $[F(\alpha)]^{-1}$ is equal to

(a) $F(-\alpha)$ (b) $F(\alpha^{-1})$

(c) $F(2\alpha)$ (d) none of these

Ans. (a)

Solution: Now,

$$F(\alpha)F(-\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \quad F(\alpha)F(-\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$\therefore \quad F(-\alpha) = [F(\alpha)]^{-1}$$

8. If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then

(a) $\alpha = a^2 + b^2, \beta = 2ab$ (b) $\alpha = a^2 + b^2, \beta = a^2 - b^2$

(c) $\alpha = 2ab, \beta = a^2 + b^2$ (d) $\alpha = a^2 + b^2, \beta = ab$

Ans. (a)

Solution: We have

$$A^2 = \begin{bmatrix} a & b \\ b & a \end{bmatrix} \begin{bmatrix} a & b \\ b & a \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & 2ab \\ 2ab & b^2 + a^2 \end{bmatrix} = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$$

$$\Rightarrow \quad \alpha = a^2 + b^2, \beta = 2ab$$

9. If $E(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then $E(\alpha)E(\beta)$ is equal to

(a) $E(\alpha^0)$ (b) $E(\alpha\beta)$

(c) $E(\alpha + \beta)$ (d) $E(\alpha - \beta)$

Ans. (c)

Solution: Since $E(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

$$\Rightarrow \quad E(\alpha)E(\beta) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{bmatrix}$$

$$\Rightarrow \quad E(\alpha)E(\beta) = \begin{bmatrix} \cos(\alpha+\beta) & \sin(\alpha+\beta) \\ -\sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix}$$

$$\therefore \quad E(\alpha)E(\beta) = E(\alpha + \beta)$$

10. If $A = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$, then the system of equations

$$a_1x + b_1y + c_1z = 0,\ a_2x + b_2y + c_2z = 0$$

and $a_3x + b_3y + c_3z = 0$ has

(a) only one solution

(b) infinite number of solutions

(c) no solution

(d) more than one but finite number of solutions

Ans. (a)

Solution: Note that A is invertible as $|A| \neq 0$.

Now, $AX = 0 \quad \Rightarrow \quad A^{-1}(AX) = 0$

$\Rightarrow \quad (A^{-1}A)X = 0$ or $IX = 0$ or $X = 0$

$\therefore \quad x = y = z = 0$ is the only solution of the system of equations.

11. If $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then $A^2 + 3A$ equals

(a) $18I_3$ (b) $8A$

(c) $12A$ (d) none of these

Ans. (a)

Solution: $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3I_3$

$$\therefore \quad A^2 = 9I_3 = 3A$$

$$\therefore \quad A^2 + 3A = 3A + 3A = 6A = 6 \times 3I_3 = 18I_3$$

12. If $A=\begin{bmatrix}1 & 0\\ (1/2) & 1\end{bmatrix}$, then A^{100} is equal to

(a) $\begin{bmatrix}1 & 0\\ 50 & 1\end{bmatrix}$ (b) $\begin{bmatrix}1 & 0\\ (1/2)^{100} & 1\end{bmatrix}$

(c) $\begin{bmatrix}1 & 0\\ 25 & 0\end{bmatrix}$ (d) none of these

Ans. (a)

Solution: $A^2=\begin{bmatrix}1 & 0\\ (1/2) & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ (1/2) & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 2(1/2) & 1\end{bmatrix}$

$$A^3=A^2A=\begin{bmatrix}1 & 0\\ 2(1/2) & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ (1/2) & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 3(1/2) & 1\end{bmatrix}$$

Continuing in this way, we get

$$A^{100}=\begin{bmatrix}1 & 0\\ 100(1/2) & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 50 & 1\end{bmatrix}$$

13. If $A=\begin{bmatrix}1 & -5 & 7\\ 0 & 7 & 9\\ 11 & 8 & 9\end{bmatrix}$, then the trace of matrix A is

(a) 17 (b) 25

(c) 3 (d) 12

Ans. (a)

Solution: We know that if $A=[a_{ij}]$ is a square matrix of order $n\times n$, then trace of A is the sum of all diagonal elements.

$$\Rightarrow \qquad \text{tr}(A)=\sum_{i=j}^{n}a_{ij}$$

$\therefore$ If $A=\begin{bmatrix}1 & -5 & 7\\ 0 & 7 & 9\\ 11 & 8 & 9\end{bmatrix}$, then tr$(A)$ = 1 + 7 + 9 = 17.

14. If A and B are square matrices of order 3 such that $|A|=-1,\ |B|=3$, then the determinant of $3AB$ is equal to

(a) –9 (b) –27

(c) –81 (d) 81

Ans. (c)

Solution: det$(3AB)$ = 27 det(AB)

[$\because$ order of A = order of B = 3]

$$= 27|A||B|=27(-1)(3)=-81$$

15. If $X=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$, then the value of X^n is

(a) $\begin{bmatrix}3n & -4n\\ n & -n\end{bmatrix}$ (b) $\begin{bmatrix}2+n & 5-n\\ n & -n\end{bmatrix}$

(c) $\begin{bmatrix}3^n & (-4)^n\\ 1^n & (-1)^n\end{bmatrix}$ (d) none of these

Ans. (d)

Solution: For n = 2

$$X^2=X\cdot X=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}=\begin{bmatrix}5 & -8\\ 2 & -3\end{bmatrix}=\begin{bmatrix}1+(2\times2) & (-2)^{2+1}\\ 2 & 1-(2\times2)\end{bmatrix}$$

For n = 3

$$X^3=X^2\cdot X=\begin{bmatrix}5 & -8\\ 2 & -3\end{bmatrix}\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}=\begin{bmatrix}7 & -16\\ 3 & -5\end{bmatrix}=\begin{bmatrix}1+(2\times3) & (-2)^{3+1}\\ 3 & 1-(2\times3)\end{bmatrix}$$

$$\Rightarrow \quad X^n=\begin{bmatrix}1+2n & (-2)^{n+1}\\ n & 1-2n\end{bmatrix}$$

So, option (d) is correct.

16. $A=\begin{bmatrix}P & 0 & 0\\ 0 & Q & 0\\ 0 & 0 & R\end{bmatrix}_{3\times3}$, where P, Q and R are non-zero real numbers, then A^{-1} equals

(a) $\frac{1}{PQR}I_3$ (b) $\frac{1}{PQR}A$

(c) $\frac{1}{PQR}\begin{vmatrix}1/P & 0 & 0\\ 0 & 1/Q & 0\\ 0 & 0 & 1/R\end{vmatrix}$ (d) $\begin{vmatrix}1/P & 0 & 0\\ 0 & 1/Q & 0\\ 0 & 0 & 1/R\end{vmatrix}$

Ans. (d)

Solution: For the given matrix A, $|A| = PQR$ and

$$\text{adj}A=\begin{bmatrix}QR & 0 & 0\\ 0 & PR & 0\\ 0 & 0 & PQ\end{bmatrix}$$

$$\therefore \quad A^{-1}=\frac{1}{|A|}(\text{adj}A)$$

$$=\frac{1}{PQR}\begin{bmatrix}QR & 0 & 0\\ 0 & PR & 0\\ 0 & 0 & PQ\end{bmatrix}=\begin{bmatrix}1/P & 0 & 0\\ 0 & 1/Q & 0\\ 0 & 0 & 1/R\end{bmatrix}$$

Shortcut method:

In a matrix $A=[a_{ij}]_{m\times m}$ (i.e. A is a square matrix), if $a_{ij}\neq0,\forall i=j$ and $a_{ij}=0,\forall i\neq j$, then to get A^{-1} only write the reciprocal entry of a_{ij} at that place where it exists.

$$\therefore \qquad A^{-1}=\begin{bmatrix}1/P & 0 & 0\\ 0 & 1/Q & 0\\ 0 & 0 & 1/R\end{bmatrix}$$

17. If A and B are two square matrices such that $B=-A^{-1}BA$, then $(A+B)^2$ is equal to

(a) A^2+B^2 (b) 0

(c) $A^2+2AB+B^2$ (d) $A+B$

Ans. (a)

Solution: As $B=-A^{-1}BA$, we get $AB=-BA$ or $AB+BA=0$

Now, $(A+B)^2=(A+B)(A+B)=A^2+AB+BA+B^2$

$$=A^2+0+B^2=A^2+B^2$$

18. If $A=\begin{bmatrix}1 & \tan(\theta/2)\\ -\tan(\theta/2) & 1\end{bmatrix}$ and $AB=1$, then $B=$

(a) $\left[\cos^2(\theta/2)\right]A$ (b) $\left[\cos^2(\theta/2)\right]A^T$

(c) $\left[\cos^2(\theta/2)\right]I$ (d) $\left[\sin^2(\theta/2)\right]A$

Ans. (b)

Solution: Since $AB = 1$

$$\Rightarrow \quad B=A^{-1}$$

$$\Rightarrow \quad B=\begin{bmatrix}1 & \tan(\theta/2)\\ -\tan(\theta/2) & 1\end{bmatrix}^{-1}$$

$$\Rightarrow \quad B=\frac{1}{1+\tan^2(\theta/2)}\begin{bmatrix}1 & -\tan(\theta/2)\\ \tan(\theta/2) & 1\end{bmatrix}$$

$$\Rightarrow \quad B=\frac{1}{\sec^2(\theta/2)}\begin{bmatrix}1 & -\tan(\theta/2)\\ \tan(\theta/2) & 1\end{bmatrix}=\cos^2\frac{\theta}{2}A^T$$

19. If $\begin{bmatrix}1 & 0 & 1\\ -1 & 1 & 0\\ 0 & -1 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}1\\ 1\\ 2\end{bmatrix}$, then $(x, y, z) =$

(a) (1, 1, 1) (b) (1, 2, 3)

(c) (1, –1, 0) (d) (–1, 0 2)

Ans. (d)

Solution: $AX = B \quad \Rightarrow \quad X = A^{-1}B$

Now, $|A|=2$

Also, $A^{-1}=\frac{1}{|A|}\text{adj}A=\frac{1}{2}\begin{bmatrix}1 & -1 & -1\\ 1 & 1 & -1\\ 1 & 1 & 1\end{bmatrix}$

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\frac{1}{2}\begin{bmatrix}1 & -1 & -1\\ 1 & 1 & -1\\ 1 & 1 & 1\end{bmatrix}\begin{bmatrix}1\\ 1\\ 2\end{bmatrix}=\begin{bmatrix}-1\\ 0\\ 2\end{bmatrix}$$

$\therefore \quad x=-1, y=0, z=2$

EXERCISE SET 1

1. If $A(\alpha,\beta)=\begin{bmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & e^{\beta}\end{bmatrix}$, then $A(\alpha,\beta)^{-1}$ is equal to

(a) $A(-\alpha,-\beta)$ (b) $A(-\alpha,\beta)$

(c) $A(\alpha,-\beta)$ (d) $A(\alpha,\beta)$

2. If $\begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}$ is to be the square root of a two-rowed unit matrix, then α, β and γ should satisfy the relation

(a) $1+\alpha^2+\beta\gamma=0$ (b) $1-\alpha^2-\beta\gamma=0$

(c) $1-\alpha^2+\beta\gamma=0$ (d) $\alpha^2+\beta\gamma-1=0$

3. If $A=\begin{bmatrix}2 & 14 & 7\\ 0 & \sin 2x & \cos 2x\\ 0 & \cos 2x & \sin 2x\end{bmatrix}$, then $|A|$ equals

(a) cos 2x (b) –2

(c) –2 cos 4x (d) sin 4x

4. If A is a matrix such that $A^2+A+2I=0$, then which of the following is **not** true?

(a) A is symmetric (b) A is non-singular

(c) $A\neq 0$ (d) $A^{-1}=-\frac{1}{2}(A+I)$

5. If $E(\theta)=\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta\end{bmatrix}$ and θ and ϕ differ by an odd multiple of $\pi/2$, then $E(\theta)E(\phi)$ is

(a) a null matrix (b) a unit matrix

(c) a diagonal matrix (d) none of these

6. If $A = [a_{ij}]$ is a 4 × 4 matrix and C_{ij} is the cofactor of the element a_{ij} in $|A|$, then the expression $a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14}$ is equal to

(a) 0 (b) –1

(c) 1 (d) $|A|$

7. If $A=\begin{bmatrix}0 & c & -b\\ -c & 0 & a\\ b & -a & 0\end{bmatrix}$ and $B=\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}$, then $|AB|$ equals

(a) A^3 (b) B^2

(c) 1 (d) none of these

8. If A and B are square matrices of the same order such that $A^2=A, B^2=B, AB=BA=O$, then

(a) $(A-B)^2=B-A$ (b) $(A-B)^2=A-B$

(c) $(A+B)^2=A+B$ (d) none of these

9. If $A=\begin{bmatrix}2x & 0\\ x & x\end{bmatrix}$ and $A^{-1}=\begin{bmatrix}1 & 0\\ -1 & 2\end{bmatrix}$, then x equals

(a) 2 (b) –(1/2)

(c) 1 (d) 1/2

10. The equations $2x+y=4$, $3x+2y=2$ and $x+y=-2$ have

(a) no solution

(b) one solution

(c) two solutions

(d) infinitely many solutions

11. If $A=\begin{bmatrix}5 & 3\\ 14 & 11\end{bmatrix}$, then $13A^{-1}$ equals

(a) A (b) $2A$

(c) adj A (d) $(1/13)A$

12. If $P=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}$, $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$, $Q=PAP^T$ and $x=P^TQ^{2005}P$, then x is equal to

(a) $\begin{bmatrix}1 & 2005\\ 0 & 1\end{bmatrix}$

(b) $\begin{bmatrix}4+2005\sqrt{3} & 6015\\ 2005 & 4-2005\sqrt{3}\end{bmatrix}$

(c) $\frac{1}{4}\begin{bmatrix}2+\sqrt{3} & 1\\ -1 & 2-\sqrt{3}\end{bmatrix}$

(d) $\frac{1}{4}\begin{bmatrix}2005 & 2-\sqrt{3}\\ 2+\sqrt{3} & 2005\end{bmatrix}$

13. If $f(x)=\begin{bmatrix}\sin x & \text{cosec } x & \tan x\\ \sec x & x\sin x & x\tan x\\ x^2-1 & \cos x & x^2+1\end{bmatrix}$, then $\int_{-a}^{a}|f(x)|dx$ equals

(a) 1 (b) -1

(c) $2a$ (d) 0

14. If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$ are two matrices, then $A^2=B$ is true for

(a) $\alpha=-1$ (b) $\alpha=1$

(c) $\alpha=4$ (d) no real value of α

15. Let $A=\begin{bmatrix}4 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 4\end{bmatrix}$, then A^n equals

(a) $\begin{bmatrix}4^n & 0 & 0\\ 0 & 4^{n-1} & 0\\ 0 & 0 & 4^{n-2}\end{bmatrix}$ (b) $\begin{bmatrix}2^{2n} & 0 & 0\\ 0 & 2^{2n} & 0\\ 0 & 0 & 2^{2n}\end{bmatrix}$

(c) $\begin{bmatrix}4^{n-1} & 0 & 0\\ 0 & 4^{n-1} & 0\\ 0 & 0 & 4^{n-1}\end{bmatrix}$ (d) $\begin{bmatrix}|A|^n & 0 & 0\\ 0 & |A|^n & 0\\ 0 & 0 & |A|^n\end{bmatrix}$

16. For any 2×2 matrix A, if $A(\text{adj } A)=\begin{bmatrix}10 & 0\\ 0 & 10\end{bmatrix}$, then $|A|$ is equal to

(a) 20 (b) 100

(c) 10 (d) 0

17. If $\begin{bmatrix}\cos\frac{2\pi}{3} & -\sin\frac{2\pi}{3}\\ \sin\frac{2\pi}{3} & \cos\frac{2\pi}{3}\end{bmatrix}^k=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then the least value of k equals $(k\neq 0)$

(a) 1 (b) 2

(c) -1 (d) 3

18. Which of the following is correct?

(a) Every identity matrix is a scalar matrix.

(b) Every scalar matrix is an identity matrix.

(c) Every diagonal matrix is an identity matrix.

(d) A square matrix whose each element is 1 is an identity matrix.

19. The system of homogeneous equations $tx+(t+1)y+(t-1)z=0$, $(t+1)x+ty+(t+2)z=0$ and $(t-1)x+(t+2)y+tz=0$ has non-trivial solutions for

(a) exactly three real values of t

(b) exactly two real values of t

(c) exactly one real value of t

(d) infinite number of values of t

20. If $A=\begin{bmatrix}\cos x & \sin x\\ -\sin x & \cos x\end{bmatrix}$ and $A(\text{adj}A)=k\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then the value of k is

(a) $\sin x\cos x$ (b) 1

(c) 2 (d) 3

21. If $A=\begin{bmatrix}1 & 2 & -1\\ -1 & 1 & 2\\ 2 & -1 & 1\end{bmatrix}$, then $\det[\text{adj}(\text{adj}A)]$ is

(a) $(14)^4$ (b) $(14)^3$

(c) $(14)^2(14)^4$ (d) $(14)^1$

22. If A and B are square matrices and A^{-1} and B^{-1} of the same order exist, then the value of $(AB)^{-1}$ is

(a) AB^{-1} (b) $A^{-1}B$

(c) $A^{-1}B^{-1}$ (d) $B^{-1}A^{-1}$

23. The system of equations $ax+by+cz=q-r$, $bx+cy+az=r-p$ and $cx+ay+bz=p-q$ is

(a) consistent if $p=q=r$

(b) inconsistent if $a=b=c$, and p, q, r are distinct

(c) consistent if a, b, c are distinct and $a+b+c=0$

(d) none of these

24. If a matrix A is such that $3A^3+2A^2+5A+I=0$, then A^{-1} is equal to
(a) $-(3A^2+2A+5I)$ (b) $3A^2+2A+5$
(c) $3A^2-2A-5$ (d) none of these

25. Given $2x-y+2z=2$, $x-2y+z=-4$ and $x+y+\lambda z=4$, then the value of λ such that the given system of equations has no solution is
(a) 3 (b) 1
(c) 0 (d) –3

26. Assuming that the sums and products given below are defined, then which of the following is not true for matrices?
(a) $(AB)'=B'A'$
(b) $A+B=B+A$
(c) $AB=AC$ does not imply $B=C$
(d) $AB=0$ implies $A=0$ or $B=0$

27. Which of the following is **not** true?
(a) $A^2-B^2=(A+B)(A-B)$
(b) $(A^T)^T=A$
(c) $(AB)^n=A^nB^n$, where A and B commute
(d) $(A-I)(I+A)=0 \Leftrightarrow A^2=1$

28. If in a square matrix $A=[a_{ij}]$, we find that $a_{ij}=a_{ji}\forall i,j$ then A is a
(a) symmetric matrix (b) diagonal matrix
(c) transpose matrix (d) skew-symmetric matrix

29. If A is a non-singular square matrix of order n, then the rank of A is
(a) equal to n (b) less than n
(c) greater than n (d) none of these

30. If A and B are square matrices of the order 3, such that $|A|=-1$ and $|B|=3$, then the value of $|3AB|$ is
(a) –9 (b) –81
(c) –27 (d) 81

31. If $A=\begin{bmatrix}1&2&x\\0&1&0\\0&0&1\end{bmatrix}$ and $B=\begin{bmatrix}1&-2&y\\0&1&0\\0&0&1\end{bmatrix}$ and $AB=I_3$, then $x+y$ equals
(a) 0 (b) –1
(c) 2 (d) none of these

32. If $A=\begin{bmatrix}1&2\\2&3\end{bmatrix}$ and $A^2-kA-I_2=0$, then the value of k is
(a) 4 (b) 2
(c) 1 (d) –4

33. If A satisfies the equation $x^3-5x^2+4x+k=0$, then A^{-1} exists if
(a) $k\neq 1$ (b) $k\neq 2$
(c) $k\neq -1$ (d) none of these

34. If matrix $A=\begin{bmatrix}\alpha&2\\2&\alpha\end{bmatrix}$ and $|A^3|=125$, then the value of α is
(a) ±3 (b) ±2
(c) ±5 (d) 0

35. If A is a matrix of order 2×3 and AB is a matrix of order 2×5, then B may be a
(a) 3×5 matrix (b) 5×3 matrix
(c) 3×2 matrix (d) 5×2 matrix

36. Let A and B be two square matrices such that $AB=A$ and $BA=B$, then $A^2=$
(a) B (b) A
(c) I (d) 0

37. If $AB=A$ and $BA=A$, where A and B are square matrices, then
(a) $B^2=B$ and $A^2=A$ (b) $B^2\neq B$ and $A^2=A$
(c) $A^2\neq A$ and $B^2=B$ (d) $A^2\neq A$ and $B^2\neq B$

38. If A is an $m\times n$ matrix such that AB and BA are both defined, then B is an
(a) $m\times n$ matrix (b) $n\times m$ matrix
(c) $n\times n$ matrix (d) $m\times m$ matrix

39. If the matrix $\begin{bmatrix}(x-a)^2&(x-b)^2&(x-c)^2\\(y-a)^2&(y-b)^2&(y-c)^2\\(z-a)^2&(z-b)^2&(z-c)^2\end{bmatrix}$ is a zero matrix, then a, b, c, x, y, z are connected by
(a) $a+b+c=0,\ x+y+z=0$
(b) $a+b+c=0,\ x=y=z$
(c) $a=b=c,\ x+y+z=0$
(d) none of these

40. The value of x for which the matrix $A=\begin{bmatrix}6&x-2\\3&x\end{bmatrix}$ has no inverse is
(a) –2 (b) 2
(c) 0 (d) 3

41. If $\begin{bmatrix}x+y&8\\0&x-y\end{bmatrix}=\begin{bmatrix}2&3\\-1&5\end{bmatrix}+\begin{bmatrix}3&5\\1&-2\end{bmatrix}$, then (x,y) equals
(a) (4, 1) (b) (1, 4)
(c) (–4,–1) (d) (1,–4)

42. If $A=\begin{bmatrix}1&0\\1&1\end{bmatrix}$ and $I=\begin{bmatrix}1&0\\0&1\end{bmatrix}$, then which one of the following holds for all $n\geq 1$, by the principle of mathematical induction?
(a) $A^n=2^{n-1}A-(n-1)I$ (b) $A^n=nA-(n-1)I$
(c) $A^n=2^{n-1}A+(n-1)I$ (d) $A^n=nA+(n-1)I$

43. If $A=\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$, $n \in N$, then A^{4n} equals

(a) $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$

44. If A and B are symmetric matrices, then ABA is

(a) symmetric (b) skew-symmetric

(c) diagonal (d) triangular

45. If a, b, c are distinct and $\begin{vmatrix} a & a^2 & a^3-1 \\ b & b^2 & b^3-1 \\ c & c^2 & c^3-1 \end{vmatrix} = 0$, then

(a) $abc=1$ (b) $a+b+c=0$

(c) $a+b+c=1$ (d) $ab+bc+ca=0$

46. If $A=\begin{bmatrix} 2 & 3 & 1 \\ 4 & -2 & 6 \end{bmatrix}$, $B=\begin{bmatrix} 0 & 3 & 2 \\ -2 & 1 & 4 \end{bmatrix}$ and $C=\begin{bmatrix} 7 & 8 \\ 9 & 0 \\ 6 & 5 \end{bmatrix}$, then $2A+3B-C^T$ equals

(a) $\begin{bmatrix} -3 & 6 & 2 \\ -6 & -1 & 19 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & 6 & 2 \\ 6 & 1 & 19 \end{bmatrix}$

(c) $\begin{bmatrix} 11 & 24 & 14 \\ 10 & 7 & 31 \end{bmatrix}$ (d) none of these

47. If $A^2-A+I=0$, then the inverse of A is

(a) A (b) $A+I$

(c) $I-A$ (d) $A-I$

48. If ω is a complex cube root of unity, and $A=\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then A^{100} is equal to

(a) A (b) $-A$

(c) 0 (d) none of these

49. If A = diagonal $(d_1, d_2, \ldots, d_n)$, then A^n is

(a) diag. $(d_1^{n-1}, d_2^{n-1}, \ldots, d_n^{n-1})$

(b) A

(c) diag. $(d_1^n, d_2^n, \ldots, d_n^n)$

(d) none of these

50. If $\begin{bmatrix} 1 & x & 1 \end{bmatrix}\begin{bmatrix} 2 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix}=0$, then the values of x are

(a) 1, 8 (b) –1, 8

(c) –1, –8 (d) 1, –8

51. Let $A=\begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $(10)B=\begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of A, then α is

(a) –2 (b) –1

(c) 2 (d) 5

52. If $A^3=0$, then $I+A+A^2$ equals

(a) $I-A$ (b) $(I-A)^{-1}$

(c) $(I+A)^{-1}$ (d) none of these

53. The inverse of a symmetric matrix is

(a) symmetric (b) skew-symmetric

(c) diagonal (d) none of these

54. If $\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix}A=\begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$, then the matrix A is

(a) $\begin{bmatrix} 1 & -2 & -5 \\ 3 & 4 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3 \\ -2 & 4 \\ -5 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 2 & 5 \\ 3 & 4 & 0 \end{bmatrix}$ (d) none of these

55. If $A=\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2=\begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then

(a) $\alpha=2ab, \beta=a^2+b^2$ (b) $\alpha=a^2+b^2, \beta=ab$

(c) $\alpha=a^2+b^2, \beta=2ab$ (d) $\alpha=a^2+b^2, \beta=a^2-b^2$

56. If A is a non-singular matrix of order 3×3, then adj (adj A) is equal to

(a) $|A|A$ (b) $|A|^2 A$

(c) $|A|^{-1} A$ (d) none of these

57. If A and B are two square matrices such that $AB = A$ and $BA = B$ then

(a) A, B are idempotent (b) only A is idempotent

(c) only B is idempotent (d) none of these

58. The matrices

$$P=\begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix} \text{ and } Q=\frac{1}{9}\begin{bmatrix} 2 & 2 & 1 \\ 12 & -5 & m \\ -8 & 1 & 5 \end{bmatrix}$$

are such that $PQ = I$ (an identity matrix). Solving the equations $\begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$ the value of y comes out to be –3. Then the value of m is equal to

(a) 27 (b) 7

(c) –27 (d) –7

59. If A is a square matrix such that $A^2 = A$ and $(I + A)^n = I + \lambda A$, then $\lambda = ?$

(a) $(2n - 1)$ (b) $2^n - 1$

(c) $2n+1$ (d) none of these

60. If A is skew-symmetric, then A^n for $n \in N$ is

(a) symmetric (b) skew-symmetric

(c) diagonal (d) none of these

61. If ω is the cube root of unity, then

$$\begin{bmatrix} \omega & \omega^2 \\ 1 & \omega \\ \omega^2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix} \text{ equals}$$

(a) $\begin{bmatrix} \omega-\omega^2 \\ \omega-\omega^2 \\ \omega-2\omega^2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}$

(c) $(\omega-\omega^2)\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ (d) none of these

62. Consider the system of equations in x, y, z as $x \sin 3\theta - y + z = 0$, $x\cos 2\theta + 4y + 3z = 0$ and $2x + 7y + 7z = 0$. If this system has a non-trivial solution, then for any integer n, the values of θ are given by

(a) $\left[n + \frac{(-1)^n}{3}\right]\pi$ (b) $\left[n + \frac{(-1)^n}{4}\right]\pi$

(c) $\left[n + \frac{(-1)^n}{6}\right]\pi$ (d) $\frac{n\pi}{2}$

63. If a matrix A is symmetric as well as skew-symmtric, then

(a) A is a diagonal matrix (b) A is a null matrix

(c) A is a unit matrix (d) A is a triangular matrix

64. If $A = \begin{bmatrix} 3 & 4 \\ 4 & 4 \end{bmatrix}$, and $B = \begin{bmatrix} -2 & -2 \\ 0 & -2 \end{bmatrix}$, then $(A+B)^{-1}$ is

(a) is a skew-symmetric matrix

(b) $A^{-1} + B^{-1}$

(c) does not exist

(d) none of these

65. Let p be an odd prime number and T_p be the following set of matrices

$$T_p = \left\{A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, \ldots, p-1\}\right\}$$

The number of A in T_p such that the trace of A is not divisible by p but $\det(A)$ is divisible by p is

(a) $(p-1)(p^2 - p + 1)$ (b) $p^3 - (p-1)^2$

(c) $(p-1)^2$ (d) $(p-1)(p^2 - 2)$

EXERCISE SET 2

1. A and B are two non-zero square matrices such that $AB = 0$. Then,

(a) both A and B are singular

(b) either of them is singular

(c) neither matrix is singular

(d) none of these

2. If $A\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}$, then A equals

(a) $\begin{bmatrix} \frac{-11}{2} & 4 \\ -19 & 12 \end{bmatrix}$ (b) $\begin{bmatrix} \frac{-11}{2} & \frac{11}{2} \\ -19 & -12 \end{bmatrix}$

(c) $\begin{bmatrix} -8 & 4 \\ -19 & 12 \end{bmatrix}$ (d) $\begin{bmatrix} -16 & 8 \\ -38 & 24 \end{bmatrix}$

3. If A is an n-rowed non-singular square matrix, then $|\text{adj}A|$ is equal to

(a) $|A|^n$ (b) $|A|^{n-1}$

(c) $|A|^{n-2}$ (d) $|A|$

4. The inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ is equal to

(a) adj A (b) 2 adj A

(c) A (d) A^T

5. The values of x for which the matrix

$$A = \begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix}$$

is singular are

(a) 0, 3 (b) 1, 3

(c) 2, 3 (d) 3, 3

6. If $A = \begin{bmatrix} 1 & \log_b a \\ \log_a b & 1 \end{bmatrix}$, then $|A|$ is equal to

(a) 1 (b) 0

(c) $\log_a b$ (d) $\log_b a$

7. If A is an invertible matrix, then det (A^{-1}) is equal to

(a) 1 (b) $|A|$

(c) $1/|A|$ (d) none of these

8. If $|A|$ denotes the determinant of matrix $A=\begin{bmatrix}1 & 1 & 1\\ -1 & 1 & 1\\ -1 & -1 & 1\end{bmatrix}$, then $|5A|$ equals

(a) 4 (b) 0
(c) 20 (d) 500

9. If $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$, then A^n is equal to

(a) $2^{n-1}A-(n-1)I$ (b) $nA-(n-1)I$
(c) $2^{n-1}A+(n-1)I$ (d) $nA+(n-1)I$

10. If $A=\begin{bmatrix}1 & -2 & 1\\ 2 & 1 & 3\end{bmatrix}$ and $B=\begin{bmatrix}2 & 1\\ 3 & 2\\ 1 & 1\end{bmatrix}$, then $(AB)^T$ is equal to

(a) $\begin{bmatrix}-3 & -2\\ 10 & 7\end{bmatrix}$ (b) $\begin{bmatrix}-3 & 10\\ -2 & 7\end{bmatrix}$
(c) $\begin{bmatrix}-3 & 7\\ 10 & 2\end{bmatrix}$ (d) none of these

11. If A is an orthogonal matrix, then

(a) $|A|=0$ (b) $|A|=\pm 1$
(c) $|A|=\pm 2$ (d) none of these

12. If A is skew-symmetric and $B=(I-A)^{-1}(I+A)$, then B is

(a) singular (b) symmetric
(c) skew-symmetric (d) orthogonal

13. A square matrix P satisfies $P^2=I-P$, where I is the identity matrix. If $P^n=5I-8P$, then n is equal to

(a) 4 (b) 5
(c) 6 (d) 7

14. If A is invertible, then which of the following is not true?

(a) $A^{-1}=|A|^{-1}$ (b) $(A^2)^{-1}=(A^{-1})^2$
(c) $(A^T)^{-1}=(A^{-1})^T$ (d) none of these

15. If $A^2-3A+2I=0$, then

(a) A is singular (b) $A^{-1}=\dfrac{3I+A}{2}$
(c) $A^{-1}=\dfrac{I-3A}{2}$ (d) $A^{-1}=\dfrac{3I-A}{2}$

16. If a square matrix A is such that $AA^T=I=A^TA$, then $|A|$ is equal to

(a) 0 (b) ± 1
(c) ± 2 (d) none of these

17. If $A=\begin{bmatrix}a & 0 & 0\\ 0 & a & 0\\ 0 & 0 & a\end{bmatrix}$, then the value of $|\text{adj}A|$ is

(a) a^3 (b) a^6
(c) a^9 (d) a^{27}

18. The value of λ such that the system of equations $x-2y+z=-4$, $2x-y+2z=2$, $x+y+\lambda z=4$ has no solution, is

(a) 0 (b) 1
(c) -1 (d) 3

19. Let $M=[a_{uv}]_{n\times n}$ be a matrix, where $a_{uv}=\sin(\theta_u-\theta_v)+i\cos(\theta_u-\theta_v)$, then M is equal to

(a) $\bar{M}$ (b) $-\bar{M}$
(c) $(\bar{M})^T$ (d) $(-\bar{M})^T$

20. If $\begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}$ is to be the square root of a two-rowed unit matrix, then α, β and γ should satisfy the relation

(a) $1+\alpha^2+\beta\gamma=0$ (b) $1-\alpha^2-\beta\gamma=0$
(c) $1-\alpha^2+\beta\gamma=0$ (d) $1+\alpha^2-\beta\gamma=0$

21. If the system of equations $x-ky-z=0$, $kx-y-z=0$ and $x+y-z=0$ has a non-zero solution, then k is equal to

(a) 0, 1 (b) 1, -1
(c) -1, 2 (d) 2, -2

22. For any square matrix A, AA^T is a

(a) unit matrix
(b) symmetric matrix
(c) skew-symmetric matrix
(d) diagonal matrix

23. If A is a square matrix such that $|A|=2$, then for any positive integer n, $|A^n|$ is equal to

(a) 0 (b) $2n$
(c) 2^n (d) n^2

24. If $A=\begin{bmatrix}\dfrac{-1+i\sqrt{3}}{2i} & \dfrac{-1-i\sqrt{3}}{2i}\\ \dfrac{1+i\sqrt{3}}{2i} & \dfrac{1-i\sqrt{3}}{2i}\end{bmatrix}$, $i=\sqrt{-1}$ and $f(x)=x^2+2$, then $f(A)$ is equal to

(a) $\left(\dfrac{5-i\sqrt{3}}{2}\right)\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (b) $\left(\dfrac{3-i\sqrt{3}}{2}\right)\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$
(c) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (d) $(2+i\sqrt{3})\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

25. Which of the following matrices is not invertible?

(a) $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ (b) $\begin{bmatrix}3 & 3\\ 5 & 5\end{bmatrix}$
(c) $\begin{bmatrix}-1 & -1\\ -1 & 2\end{bmatrix}$ (d) $\begin{bmatrix}2 & -2\\ 1 & 1\end{bmatrix}$

26. $A=\begin{bmatrix}0 & 3\\ 2 & 0\end{bmatrix}$ and $A^{-1}=\lambda(\text{adj}\,A)$, then λ is equal to

(a) $-\frac{1}{6}$ (b) $\frac{1}{3}$

(c) $-\frac{1}{3}$ (d) $\frac{1}{6}$

27. If A and B are two matrices such that $A+B$ and AB are both defined, then

(a) A and B can be any matrices

(b) A and B are square matrices not necessarily of same order

(c) A and B are square matrices of same order

(d) number of columns of A = number of rows of B

28. If $\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta\end{bmatrix}\begin{bmatrix}\cos^2\phi & \cos\phi\sin\phi\\ \cos\phi\sin\phi & \sin^2\phi\end{bmatrix}$ is a zero matrix, then θ and ϕ differ by an

(a) even multiple of $\pi/2$ (b) odd multiple of $\pi/2$

(c) even multiple of π (d) odd multiple of π

29. If $A=\begin{bmatrix}i & 0\\ 0 & i\end{bmatrix}$, then $A^{4n}(n\in N)$ equals

(a) $\begin{bmatrix}0 & i\\ i & 0\end{bmatrix}$ (b) $\begin{bmatrix}i & 0\\ 0 & i\end{bmatrix}$

(c) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (d) $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

30. A is a matrix such that $A^2=2A-I$, where I is the identity matrix. Then, for $n\geq 2, A^n$ is equal to

(a) $nA-(n-1)I$ (b) $nA-I$

(c) $2^{n-1}A-(n-1)I$ (d) $2^{n-1}A-I$

31. If A and B are any 2×2 matrices, then $|A+B|=0$ implies

(a) $|A|+|B|=0$ (b) $|A|=0$ or $|B|=0$

(c) $|A|=0$ and $|B|=0$ (d) none of these

32. If A and P are 3×3 matrices with integral entries such that $P^TAP=A$, then det. P is

(a) -1

(b) 1

(c) ± 1

(d) ± 1 provided A is non-singular

33. If the matrix $A=\begin{bmatrix}8 & -6 & 2\\ -6 & 7 & -4\\ 2 & -4 & \lambda\end{bmatrix}$ is singular, then λ is equal to

(a) 3 (b) 4

(c) 2 (d) 5

34. If 1, ω, ω^2 are the cube roots of unity, then

$$\Delta=\begin{vmatrix}1 & \omega^n & \omega^{2n}\\ \omega^{2n} & 1 & \omega^n\\ \omega^n & \omega^{2n} & 1\end{vmatrix}$$

has the value

(a) 0 (b) ω

(c) ω^2 (d) 1

35. If each element of a 3×3 matrix A is multiplied by 3, then the determinant of the newly formed matrix is

(a) 3 det A (b) 9 det A

(c) (det A)^3 (d) 27 det A

36. The inverse matrix of $\begin{bmatrix}0 & 1 & 2\\ 1 & 2 & 3\\ 3 & 1 & 1\end{bmatrix}$ is

(a) $\begin{bmatrix}\frac{1}{2} & -\frac{1}{2} & \frac{1}{2}\\ -4 & 3 & -1\\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2}\end{bmatrix}$ (b) $\begin{bmatrix}\frac{1}{2} & -4 & \frac{5}{2}\\ 1 & -6 & 3\\ 1 & 2 & -1\end{bmatrix}$

(c) $\frac{1}{2}\begin{bmatrix}1 & 2 & 3\\ 3 & 2 & 1\\ 4 & 2 & 3\end{bmatrix}$ (d) $\frac{1}{2}\begin{bmatrix}1 & -1 & -1\\ -8 & 6 & -2\\ 5 & -3 & 1\end{bmatrix}$

37. If A is a 3×3 non-singular matrix, then det [adj A] is equal to

(a) (det A)^2 (b) (det A)^3

(c) det A (d) (det A)$^{-1}$

38. The inverse of the matrix $\begin{bmatrix}\cos 2\theta & -\sin 2\theta\\ \sin 2\theta & \cos 2\theta\end{bmatrix}$ is

(a) $\begin{bmatrix}\cos 2\theta & -\sin 2\theta\\ \sin 2\theta & \cos 2\theta\end{bmatrix}$ (b) $\begin{bmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{bmatrix}$

(c) $\begin{bmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & \cos 2\theta\end{bmatrix}$ (d) $\begin{bmatrix}\cos 2\theta & \sin 2\theta\\ -\sin 2\theta & \cos 2\theta\end{bmatrix}$

39. If α, β and γ are the roots of the equation $x^3+px+q=0$, then the value of det $\begin{bmatrix}\alpha & \beta & \gamma\\ \beta & \gamma & \alpha\\ \gamma & \alpha & \beta\end{bmatrix}$ is

(a) p (b) q

(c) p^2-2q (d) 0

40. Let $F(\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$, then $[F(\alpha)]^{-1}$ is equal to

(a) $F(-\alpha)$ (b) $F(\alpha^{-1})$

(c) $F(2\alpha)$ (d) none of these

41. Let p be a non-singular matrix such that $1 + p + p^2 + \ldots + p^n = O$ (O denotes the null matrix), then p^{-1} is
(a) p^n (b) $-p^n$
(c) $-(1+p+\cdots+p^n)$ (d) none of these

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c), or (d) given below:

(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1
(c) Statement 1 is true; Statement 2 is false
(d) Statement 1 is false; Statement 2 is true

42. Statement 1: The matrix $A=\frac{1}{3}\begin{bmatrix}1 & -2 & 2\\ -2 & 1 & 2\\ -2 & -2 & -1\end{bmatrix}$ is an orthogonal matrix.
Statement 2: If A and B are orthogonal, then AB is also orthogonal.

43. Statement 1: If A is a skew-symmetric matrix of order 3×3, then $\det(A) = 0$ or $|A| = 0$.
Statement 2: If A is a square matrix, then $\det(A) = \det(A^T) = \det(-A^T)$.

44. Statement 1: The inverse of $A=\begin{bmatrix}3 & 4\\ 3 & 5\end{bmatrix}$ does not exist.
Statement 2: The matrix A is non-singular.

45. Statement 1: If a matrix of order 2×2 commutes with every matrix of order 2×2, then it is a scalar matrix.
Statement 2: A scalar matrix of order 2×2 commutes with every 2×2 matrix.

46. Statement 1: $\begin{bmatrix}3 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 7\end{bmatrix}$ is a diagonal matrix.
Statement 2: If $A=[a_{ij}]$ is a square matrix such that $a_{ij} = 0, \forall i \neq j$, then A is called a diagonal matrix.

Previous Years' Questions

47. If $a > 0$ and discriminant of $ax^2 + 2bx + c < 0$ is greater than zero, then $\begin{vmatrix} a & b & ax+b\\ b & c & bx+c\\ ax+b & bx+c & 0\end{vmatrix}$ is
(a) positive
(b) $(ac-b^2)(ax^2+2bx+c)$
(c) negative
(d) 0 **[AIEEE 2002]**

48. If l, m, n are the pth, qth, rth terms of a G.P., all positive, then $\begin{bmatrix}\log l & p & 1\\ \log m & q & 1\\ \log n & r & 1\end{bmatrix}$ equals
(a) -1 (b) 2
(c) 1 (d) 0 **[AIEEE 2002]**

49. Let $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$, $a, b \in N$. Then
(a) there cannot exist any B such that $AB = BA$
(b) there exists more than one but a finite number of B's such that $AB = BA$
(c) there exists exactly one B such that $AB = BA$
(d) there exist infinitely many B's such that $AB = BA$
[AIEEE 2003, 2006]

50. If $A=\begin{bmatrix}a & b\\ b & a\end{bmatrix}$ and $A^2=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix}$, then
(a) $\alpha = 2ab, \beta = a^2 + b^2$
(b) $\alpha = a^2 + b^2, \beta = ab$
(c) $\alpha = a^2 + b^2, \beta = 2ab$
(d) $\alpha = a^2 = b^2, \beta = a^2 - b^2$ **[AIEEE 2003]**

51. If the system of equations $x + 2ay + az = 0$, $x + 3by + bz = 0$ and $x + 4cz + cz = 0$ has a non-zero solution, then a, b, c
(a) are in G.P.
(b) are in H.P.
(c) satisfy $a + 2b + 3c = 0$
(d) are in A.P. **[AIEEE 2003]**

52. If $\begin{vmatrix} a & a^2 & 1+a^3\\ b & b^2 & 1+b^3\\ c & c^2 & 1+c^3\end{vmatrix} = 0$ and vectors $(1, a, a^2), (1, b, b^2)$ and $(1, c, c^2)$ are non-coplanar, then abc equals
(a) -1 (b) 1
(c) 0 (d) 2 **[AIEEE 2003]**

53. Let $A=\begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix}$. The only correct statement about the matrix A is
(a) A^{-1} does not exist (b) $A = (-1)I$
(c) A is a zero matrix (d) $A^2 = I$
[AIEEE 2004]

54. Let $A\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}$ and (10) $B=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & \alpha\\ 1 & -2 & 3\end{bmatrix}$. If B is the inverse of A, then α is
(a) –2 (b) –1 (c) 2 (d) 5
[AIEEE 2004]

55. If $A^2 - A + I = 0$, then the inverse of A is
(a) A (b) $A + I$ (c) $I - A$ (d) $A - I$
[AIEEE 2005]

56. If $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$ and $I=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction?
(a) $A^n = 2^{n-1}A-(n-1)I$ (b) $A^n = nA-(n-1)I$
(c) $A^n = 2^{n-1}A+(n-1)I$ (d) $A^n = nA+(n-1)I$
[AIEEE 2005]

57. If $a^2+b^2+c^2=-2$ and

$$f(x)=\begin{vmatrix}1+a^2x & (1+b^2)x & (1+c^2)x\\ (1+a^2)x & 1+b^2x & (1+c^2)x\\ (1+a^2)x & (1+b^2)x & 1+c^2x\end{vmatrix}$$

then $f(x)$ is a polynomial of degree
(a) 0 (b) 1 (c) 2 (d) 3
[AIEEE 2005]

58. The system of equations

$$\alpha x+y+z=\alpha-1$$
$$x+\alpha y+z=\alpha-1$$
$$x+y+\alpha z=\alpha-1$$

has no solution if α is
(a) –2 or 1 (b) –2 (c) 1 (d) –1
[AIEEE 2005]

59. If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will always be true?
(a) $A = B$
(b) $AB = BA$
(c) either A or B is a zero matrix
(d) either A or B is an identity matrix
[AIEEE 2006]

60. If $A=\begin{vmatrix}5 & 5\alpha & \alpha\\ 0 & \alpha & 5\alpha\\ 0 & 0 & 5\end{vmatrix}$ and $|A^2| = 25$, then $|\alpha|$ equals
(a) 1 (b) 1/5 (c) 5 (d) 5^2
[AIEEE 2007]

61. If $D=\begin{bmatrix}1 & 1 & 1\\ 1 & 1+x & 1\\ 1 & 1 & 1+y\end{bmatrix}$ for $x \neq 0, y \neq 0$, then D is
(a) divisible by x but not by y
(b) divisible by y but not by x
(c) divisible by neither x nor y
(d) divisible by both x and y **[AIEEE 2007]**

62. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz$, $y = az + cx, z = bx + ay$, then $a^2 + b^2 + c^2 + 2abc$ equals to
(a) –1 (b) 1 (c) 0 (d) 2
[AIEEE 2008]

63. Let A be a square matrix all of whose entries are integers, then which of the following is true?
(a) If $|A| \neq \pm 1$, then A^{-1} exist, and all its entries are non-integers
(b) If $|A| = \pm 1$, then A^{-1} exists and all its entries are integers
(c) If $|A| = \pm 1$, then A^{-1} need not exist
(d) If $|A| = \pm 1$, then A^{-1} exists but all its entries are not necessarily integers **[AIEEE 2008]**

64. If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$, then the value of α for which $A^2 = B$ is
(a) 1 (b) –1 (c) 4
(d) no real values **[IIT JEE]**

65. If A is a 3×4 matrix and B is a matrix such that A^TB and B^TA are both defined, then the order of B is
(a) 3×4 (b) 3×3 (c) 4×4 (d) 4×3
[DCE 2004]

66. If $A=\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}$, then A is
(a) idempotent (b) involutary
(c) nilpotent (d) scalar
[DCE 2004]

67. For 2×2 matrices A, B and I, if $A + B = I$ and $2A - 2B = I$, then A equals
(a) $\begin{bmatrix}1/4 & 0\\ 0 & 1/4\end{bmatrix}$ (b) $\begin{bmatrix}1/2 & 0\\ 0 & 1/2\end{bmatrix}$
(c) $\begin{bmatrix}3/4 & 0\\ 0 & 3/4\end{bmatrix}$ (d) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$
[AMU 1992]

68. The value of x for which the matrix product

$$\begin{bmatrix}2 & 0 & 7\\ 0 & 1 & 0\\ 1 & -2 & 1\end{bmatrix}\begin{bmatrix}-x & 14x & 7x\\ 0 & 1 & 0\\ x & -4x & -2x\end{bmatrix}$$

equals an identity matrix is
(a) 1/2 (b) 1/3 (c) 1/4 (d) 1/5
[AMU 1991]

69. If A and B are square matrices of the same order and A is non-singular, then for a positive integer n, $(A^{-1}BA)^n$ is equal to

(a) $A^{-n}B^nA^n$ (b) $A^nB^nA^{-n}$ (c) $A^{-1}B^nA$ (d) $n(A^{-1}BA)$

[AMU 1990]

70. If A is a singular matrix, then adj A is

(a) singular (b) non-singular

(c) symmetric (d) not defined

[CET, 1993, 1997]

71. If $\omega \neq 1$ is the complex cube root of unity and matrix $H = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then H^{70} is equal to

(a) 0 (b) $-H$ (c) H^2 (d) H

[AIEEE 2011]

72. If the trivial solution is the only solution of the system of equations $x - ky + z = 0$, $kx + 3y - kz = 0$ and $3x + y - z = 0$, then the set of all values of k is

(a) $\mathbb{R} - \{2, -3\}$ (b) $\mathbb{R} - \{2\}$

(c) $\mathbb{R} - \{-3\}$ (d) $\{2, -3\}$ **[AIEEE 2011]**

73. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $Au_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, then $u_1 + u_2$ is equal to

(a) $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ (b) $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ (c) $\begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

[AIEEE 2012]

74. Let P and Q be 3×3 matrices $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$, then determinant of $(P^2 + Q^2)$ is equal to

(a) -2 (b) 2 (c) 0 (d) -1

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (a)	**2.** (b)	**3.** (c)	**4.** (a)	**5.** (a)	**6.** (d)	**7.** (d)	**8.** (c)	**9.** (d)	**10.** (b)
11. (c)	**12.** (a)	**13.** (d)	**14.** (d)	**15.** (b)	**16.** (c)	**17.** (d)	**18.** (a)	**19.** (c)	**20.** (b)
21. (a)	**22.** (d)	**23.** (a)	**24.** (a)	**25.** (b)	**26.** (c)	**27.** (a)	**28.** (a)	**29.** (a)	**30.** (b)
31. (a)	**32.** (a)	**33.** (d)	**34.** (a)	**35.** (a)	**36.** (b)	**37.** (a)	**38.** (b)	**39.** (d)	**40.** (a)
41. (a)	**42.** (b)	**43.** (c)	**44.** (a)	**45.** (a)	**46.** (a)	**47.** (c)	**48.** (a)	**49.** (c)	**50.** (c)
51. (d)	**52.** (b)	**53.** (a)	**54.** (a)	**55.** (c)	**56.** (a)	**57.** (a)	**58.** (d)	**59.** (b)	**60.** (d)
61. (d)	**62.** (c)	**63.** (b)	**64.** (d)	**65.** (c)					

Exercise Set 2

1. (b)	**2.** (c)	**3.** (b)	**4.** (a)	**5.** (a)	**6.** (b)	**7.** (c)	**8.** (d)	**9.** (b)	**10.** (b)
11. (b)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (d)	**16.** (b)	**17.** (b)	**18.** (b)	**19.** (d)	**20.** (b)
21. (b)	**22.** (b)	**23.** (c)	**24.** (d)	**25.** (b)	**26.** (a)	**27.** (c)	**28.** (b)	**29.** (c)	**30.** (a)
31. (d)	**32.** (d)	**33.** (a)	**34.** (a)	**35.** (d)	**36.** (a)	**37.** (a)	**38.** (d)	**39.** (d)	**40.** (a)
41. (a)	**42.** (b)	**43.** (c)	**44.** (d)	**45.** (a)	**46.** (b)	**47.** (c)	**48.** (d)	**49.** (d)	**50.** (c)
51. (b)	**52.** (a)	**53.** (d)	**54.** (d)	**55.** (c)	**56.** (b)	**57.** (c)	**58.** (b)	**59.** (b)	**60.** (b)
61. (d)	**62.** (b)	**63.** (b)	**64.** (d)	**65.** (a)	**66.** (c)	**67.** (c)	**68.** (d)	**69.** (c)	**70.** (a)
71. (d)	**72.** (a)	**73.** (d)	**74.** (c)						

HINTS AND SOLUTIONS—EXERCISE SET 1

1. We have

$$A(\alpha, \beta)^{-1} = \frac{1}{e^{\beta}} \begin{bmatrix} e^{\beta}\cos\alpha & -e^{\beta}\sin\alpha & 0 \\ e^{\beta}\sin\alpha & e^{\beta}\cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = A(-\alpha, -\beta)$$

2. Since $\begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ is a square root of I_2, i.e. two-rowed unit matrix

$$\therefore \quad \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}\begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 + \beta\gamma & \alpha\beta - \alpha\beta \\ \gamma\alpha - \gamma\alpha & \gamma\beta + \alpha^2 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \gamma\beta + \alpha^2 \end{bmatrix}$$

$$\therefore \quad \alpha^2 + \beta\gamma = 1 \quad \text{[by law of equality]}$$

$$\Rightarrow \quad 1 - \alpha^2 - \beta\gamma = 0$$

3. $|A| = 2(\sin^2 2x - \cos^2 2x) = -2\cos 4x$

4. Given $A^2 + A + 2I = 0$

$\Rightarrow \quad A^2 + A = -2I \quad \Rightarrow \quad |A^2 + A| = |-2I|$

$\Rightarrow \quad |A||A + I| = (-2)^n \quad \Rightarrow \quad |A| \neq 0$

$\Rightarrow A$ is non-singular; hence, its inverse exists. Also multiplying both sides of the given equation with A^{-1}, we get

$$A^{-1} = -\frac{1}{2}(A + I)$$

5. Let $S = E(\theta)E(\phi)$

$$\Rightarrow S = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$$

$$\Rightarrow S = \begin{bmatrix} \cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi\cos(\theta-\phi) \\ \cos\phi\sin\theta\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi) \end{bmatrix}$$

Since θ and ϕ differ by an odd multiple of $\pi/2$,

$$S = \begin{bmatrix} \cos\theta\cos\phi\cos(2n+1)\frac{\pi}{2} & \cos\theta\sin\phi\cos(2n+1)\frac{\pi}{2} \\ \cos\phi\sin\theta\cos(2n+1)\frac{\pi}{2} & \sin\theta\sin\phi\cos(2n+1)\frac{\pi}{2} \end{bmatrix}$$

$$\therefore \quad S = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

6. By definition, any row of a matrix/determinant A when multiplied by its corresponding cofactors results in the value of $|A|$.

7. $$AB = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}\begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O_3$$

Short cut method: $|AB| = |A||B| = 0$

As A is skew-symmetric of order 3 so $|A| = 0$, whatever be the value of $|B|$.

$$\therefore \quad |AB| = 0$$

8. $(A+B)^2 = A^2 + AB + BA + B^2 = A + 0 + 0 + B = A + B$

$(A-B)^2 = A + B \neq A - B$

9. We know that $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Given $A = \begin{bmatrix} 2x & 0 \\ x & x \end{bmatrix}$

$$\therefore \quad A^{-1} = \frac{1}{2x^2}\begin{bmatrix} x & 0 \\ -x & 2x \end{bmatrix} = \begin{bmatrix} 1/2x & 0 \\ -(1/2x) & 1/x \end{bmatrix}$$

But $A^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$

$$\therefore \quad \begin{bmatrix} 1/2x & 0 \\ -(1/2x) & 1/x \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

$$\Rightarrow \quad \frac{1}{2x} = 1 \quad \Rightarrow \quad x = \frac{1}{2}$$

10. $2x + y = 4$ and $3x + 2y = 2$ gives

$$x = \frac{\begin{vmatrix} 4 & 1 \\ 2 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix}} = \frac{6}{1} = 6; \quad y = \frac{\begin{vmatrix} 2 & 4 \\ 3 & 2 \end{vmatrix}}{1} = \frac{-8}{1} = -8$$

Clearly, $x + y = -2$ is also satisfied.

Hence, the given system has one solution: $x = 6$, $y = -8$.

11. $A^{-1} = \frac{1}{|A|}(\text{adj}\,A)$ using the fact $|A|A^{-1} = \text{adj}\,A$

$$\therefore \quad 13A^{-1} = \text{adj}\,A$$

12. $$P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, Q = PAP^T$$

Here, $$P \cdot P^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \quad P \cdot P^T = I$$

Now, $Q = PAT^T$. Also, $x = P^T Q^{2005} P$

$$\Rightarrow \quad x = P^T(PAP^T)^{2005}P$$

$$\Rightarrow \quad x = P^T(PAP^T)(PAP^T)^{2004}P$$

$$\Rightarrow \quad x = (IA)P^T(PAP^T)^{2004}P$$

$$\Rightarrow \quad x = AP^T(PAP^T)(PAP^T)^{2003}P \ldots$$

$$\Rightarrow \quad x = A^{2004}P^T(PAP^T)P$$

$$\Rightarrow \quad x = A^{2004}(P^TP)A(P^TP) = A^{2004}IAI$$

$$\therefore \quad x = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

13. $$f(x) = \begin{bmatrix} \sin x & \text{cosec}\,x & \tan x \\ \sec x & x\sin x & x\tan x \\ x^2 - 1 & \cos x & x^2 + 1 \end{bmatrix}, \text{ then}$$

$$|f(x)| = \begin{vmatrix} \sin x & \text{cosec}\,x & \tan x \\ \sec x & x\sin x & x\tan x \\ x^2 - 1 & \cos x & x^2 + 1 \end{vmatrix} = g(x) \quad \text{(assume)}$$

so $g(-x)=\begin{vmatrix} -\sin x & -\text{cosec } x & -\tan x \\ \sec x & x\sin x & x\tan x \\ x^2-1 & \cos x & x^2+1 \end{vmatrix} = -g(x)$

$\therefore \quad g(-x)=-g(x)$

$\Rightarrow \quad g(x)=|f(x)|$ is an odd function.

$\therefore \quad \int_{-a}^{a} |f(x)|\,dx = 0.$

14. Since $A=\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$

Also, $A^2 = B$

$\Rightarrow \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \alpha^2+1 & 0 \\ \alpha+1 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$

$\therefore \quad \alpha^2 + 1 = 1 \quad \Rightarrow \quad \alpha = 0$

$\Rightarrow \quad \alpha+1=5$ or $\alpha=4$

Both the values of α are not possible simultaneously.

$\therefore$ There is no real value of α for which $A^2 = B$.

15. $A=\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}=4I_3$

$\therefore \quad A^n = 4^n(I_3)^n = 2^{2n} I_3$

$\Rightarrow \quad \begin{bmatrix} 2^{2n} & 0 & 0 \\ 0 & 2^{2n} & 0 \\ 0 & 0 & 2^{2n} \end{bmatrix}$

16. Since $A(\text{adj } A) = |A| I$

$\Rightarrow \quad \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}=|A|\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} |A| & 0 \\ 0 & |A| \end{bmatrix}$

$\therefore \quad |A|=10$

17. $A=\begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$, where $t=\dfrac{2\pi}{3}$

$\therefore \quad 3t = 2\pi$

$\therefore \quad A^2=\begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}\begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$

$A^2=\begin{bmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{bmatrix}$

$\Rightarrow \quad A^2 \cdot A = A^3 = \begin{bmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{bmatrix}\begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} \cos 3t & -\sin 3t \\ \sin 3t & \cos 3t \end{bmatrix}=\begin{bmatrix} \cos 2\pi & -\sin 2\pi \\ \sin 2\pi & \cos 2\pi \end{bmatrix}$

$\Rightarrow \quad A^3=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\therefore$ The least value of $k = 3$.

18. We know that every identity matrix is a scalar matrix. Therefore, option (a) is correct.

19. For the system of equations to have non-trivial solution,

$$\Delta=\begin{vmatrix} t & t+1 & t-1 \\ t+1 & t & t+2 \\ t-1 & t+2 & t \end{vmatrix}=0$$

Applying $C_2 \to C_2 - C_1, C_3 \to C_3 - C_1$, we get

$$\Delta=\begin{vmatrix} t & 1 & -1 \\ t+1 & -1 & 1 \\ t-1 & 3 & 1 \end{vmatrix}=\begin{vmatrix} 2t+1 & 0 & 0 \\ t+1 & -1 & 1 \\ t-1 & 3 & 1 \end{vmatrix}=0$$

[using $R_1 \to R_1 + R_2$]

$= (2t+1)(-4) = 0$

$\Rightarrow \quad t=-\dfrac{1}{2}.$

20. We know that $A(\text{adj}A)=|A|I$.

If $A=\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$, then $|A| = 1$

$\therefore \quad A(\text{adj}A) = I.$

It is given that $A(\text{adj}A) = kI$. Therefore, $k = 1$.

21. We know that $\text{adj}(\text{adj}A)=|A|^{n-2}A$ if $A \neq 0$, provided the order of A is n.

$\Rightarrow \quad \text{adj}(\text{adj}A)=|A|A \qquad (\text{as } n = 3)$

$\therefore \quad \det(\text{adj}(\text{adj}A))=|A|^3 \det A=|A|^4$

But $|A|=\begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}=14$

$\therefore \quad \det[\text{adj}(\text{adj}A)] = (14)^4.$

22. Given square matrices A and B are of the same order. We know that if A and B are square matrices then $(AB)^{-1} = B^{-1}A^{-1}$.

23. If $p = q = r$, then $x = 0, y = 0, z = 0$ is a solution of the system of equations.

If a, b, c are distinct, then the determinant of the coefficient matrix is

$$\frac{1}{2}(a+b+c)[(b-c)^2+(c-a)^2+(a-b)^2] \neq 0$$

and hence the system of equations has a unique solution.

24. We have $3A^3 + 2A^2 + 5A + I = 0$

$\Rightarrow \quad I = -3A^3 - 2A^2 - 5A$

$\Rightarrow \quad IA^{-1} = (-3A^3 - 2A^2 - 5A)A^{-1}$

$\Rightarrow \quad A^{-1} = -3A^2 - 2A - 5I$

25. Since the system possesses no solution

$$\therefore \quad \Delta = \begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$\Rightarrow \quad \lambda = 1$

26. We know that $AB = AC$ which implies that $B = C$. Therefore, the option (c) is not true for the matrices.

27. We have $(A + B)(A - B) = A^2 - AB + BA - B^2$.
So, (a) is not true.

28. Given square matrix $A = [a_{ij}]$ and $a_{ij} = a_{ji}$.
We know that in a square matrix if $a_{ij} = a_{ji}$ $\forall i, j$, then ith, jth elements are equal to jth, ith elements in matrix A. Therefore, A is a symmetric matrix.

29. Since A is a non-singular matrix, therefore $|A| \neq 0$
$\Rightarrow$ rank $(A) = n$.

30. Given order of the square matrices A and B, $n = 3$, $|A| = -1$ and $|B| = 3$. We know that if A and B are two square matrices of order n, then $|rAB| = r^n |A||B|$. Comparing $|rAB|$ to the given matrix $|3AB|$, we get $r = 3$. Therefore, $|3AB| = (3)^3 |A||B| = 27(-1)(3) = -81$.

31. We have $I_3 = AB = \begin{bmatrix} 1 & 2 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & x+y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \quad x + y = 0$$

32. Given matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and the matrix equation

$$A^2 - kA - I_2 = 0 \Rightarrow kA = A^2 - I_2 \qquad \text{(i)}$$

We know that $A^2 = A \cdot A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 8 & 13 \end{bmatrix}$

and $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Substituting the values of A^2, A and I_2 in Eq. (i), we get

$$k \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 8 & 12 \end{bmatrix}$$

$$\Rightarrow \quad k \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = 4 \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}.$$

Comparing both sides, we get $k = 4$.

33. Since A satisfies the equation $x^3 - 5x^2 + 4x + k = 0$, $A^3 - 5A^2 + 4A + kI = 0$.
$\Rightarrow$ A^{-1} exists if $k \neq 0$.

34. Matrix $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and matrix equation $|A^3| = 125$.
Since $|A|^3 = 125$, therefore $|A| = 5$. We know that

$$|A| = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix} = \alpha^2 - 4$$

Substituting the value of $|A|$ in the above equation, we get

$\Rightarrow \quad 5 = \alpha^2 - 4 \quad \Rightarrow \quad \alpha^2 = 9 \quad \Rightarrow \quad \alpha = \pm 3$.

35. Number of columns in A = Number of rows in B.
Hence B is a 3×5 matrix.

36. $AB = A$

$\Rightarrow \quad ABA = A.A = A^2$

$\Rightarrow \quad A(BA) = A^2$

$\Rightarrow \quad AB = A^2 \qquad [\because \quad BA = B]$

$\Rightarrow \quad A = A^2 \qquad [\because \quad AB = A]$

Thus, $A^2 = A$

37. We have $A^2 = AA$

$\Rightarrow \quad A^2 = (AB)A \qquad [\because \quad AB = A]$

$\Rightarrow \quad A^2 = A(BA)$

$\Rightarrow \quad A^2 = AB \qquad [\because \quad BA = B]$

$\Rightarrow \quad A^2 = A \qquad [\because \quad AB = A]$

and $B^2 = BB$

$\Rightarrow \quad B^2 = (BA)B \qquad [\because \quad BA = B]$

$\Rightarrow \quad B^2 = B(AB)$

$\Rightarrow \quad B^2 = BA \qquad [\because \quad AB = A]$

$\Rightarrow \quad B^2 = B \qquad [\because \quad BA = B]$

38. Since AB exists, therefore the number of columns in A = the number of rows in B. So, B has n rows. Since BA exists, the number of columns in B = the number of rows in A. So, B has m columns.
$\therefore$ B is an $n \times m$ matrix.

39. For the given matrix to be a zero matrix, each entry in the matrix must be zero, which is possible when $x = y = z = a = b = c$.

40. Matrix $A = \begin{bmatrix} 6 & x-2 \\ 3 & x \end{bmatrix}$.
We know that if $|A| = 0$, then A has no inverse.
Therefore, $|A| = 6x - 3x + 6 = 0$ or $x = -2$.

41. $x+y=5, x-y=3$

$\Rightarrow \qquad x=4, y=1$

42. We have $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

$\therefore \qquad A^2=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$

$$A^3=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}$$

By induction, $A^n=\begin{bmatrix}1 & 0\\ n & 1\end{bmatrix}$ (i)

$$\therefore \quad nA-(n-1)I=n\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}-(n-1)\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}n & 0\\ n & n\end{bmatrix}-\begin{bmatrix}(n-1) & 0\\ 0 & (n-1)\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0\\ n & 1\end{bmatrix}=A^n. \qquad \text{[using (i)]}$$

43. We have $A=\begin{bmatrix}i & 0\\ 0 & i\end{bmatrix}$

Clearly, A is a diagonal matrix. Therefore, A^{4n} is also a diagonal matrix such that

$$A^{4n}=\begin{bmatrix}i^{4n} & 0\\ 0 & i^{4n}\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

44. $(ABA)^T=A^TB^TA^T=ABA$

$\Rightarrow$ ABA is symmetric.

45. Matrix $\begin{bmatrix}a & a^2 & a^3-1\\ b & b^2 & b^3-1\\ c & c^2 & c^3-1\end{bmatrix}=0$

The given matrix may be written as

$$\begin{bmatrix}a & a^2 & a^3\\ b & b^2 & b^3\\ c & c^2 & c^3\end{bmatrix}+\begin{bmatrix}a & a^2 & -1\\ b & b^2 & -1\\ c & c^2 & -1\end{bmatrix}=0$$

$$\Rightarrow \qquad abc\begin{bmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{bmatrix}-\begin{bmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{bmatrix}=0$$

$$\Rightarrow \qquad (abc-1)\begin{bmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{bmatrix}=0$$

$\Rightarrow$ $abc=1$

46. $2A+3B-C^T$

$$=\begin{bmatrix}4 & 6 & 2\\ 8 & -4 & 12\end{bmatrix}+\begin{bmatrix}0 & 9 & 6\\ -6 & 3 & 12\end{bmatrix}-\begin{bmatrix}7 & 9 & 6\\ 8 & 0 & 5\end{bmatrix}$$

$$=\begin{bmatrix}-3 & 6 & 2\\ -6 & -1 & 19\end{bmatrix}$$

47. $A^2-A+I=0 \quad \Rightarrow \quad A^2-A=-I$

$\Rightarrow$ $A(A-I)=-I \quad \Rightarrow \quad A(I-A)=I$.

Hence, $A^{-1}=I-A$.

48. We have

$$A=\omega\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=\omega I_2$$

$\therefore \quad A^{100}=\omega^{100}(I_2)^{100}=\omega^{100}I_2=\omega I_2=A$

49. If $A=\begin{bmatrix}d_1 & 0\\ 0 & d_2\end{bmatrix}$, then

$$A^2=\begin{bmatrix}d_1 & 0\\ 0 & d_2\end{bmatrix}\begin{bmatrix}d_1 & 0\\ 0 & d_2\end{bmatrix}=\begin{bmatrix}d_1^2 & 0\\ 0 & d_2^2\end{bmatrix}.$$

Hence the option (c) is correct.

50. $\begin{bmatrix}1 & x & 1\end{bmatrix}\begin{bmatrix}2 & 3 & 2\\ 0 & 5 & 1\\ 0 & 3 & 2\end{bmatrix}\begin{bmatrix}1\\ 1\\ x\end{bmatrix}=0$

$$\Rightarrow \qquad \begin{bmatrix}2 & 6+5x & 4+x\end{bmatrix}\begin{bmatrix}1\\ 1\\ x\end{bmatrix}=0$$

$\Rightarrow$ $2+(6+5x)+(4+x)x=0 \quad \Rightarrow \quad x^2+9x+8=0$

$\Rightarrow$ $(x+1)(x+8)=0 \quad \Rightarrow \quad x=-1,-8$

51. Here $A=\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}$

$$\therefore \quad |A|=\begin{vmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{vmatrix}=1\cdot(1+3)+1\cdot(2+3)+1\cdot(2-1)$$

$$=4+5+1=10$$

$$\therefore \quad \text{adj}A=\begin{bmatrix}4 & -5 & 1\\ 2 & 0 & -2\\ 2 & 5 & 3\end{bmatrix}^T=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}$$

$$\therefore \quad B=A^{-1}=\frac{1}{10}\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}\Rightarrow (10)B=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}.$$

Hence $\alpha=5$.

52. We have

$$(I-A)(I+A+A^2)=I+A+A^2-A-A^2-A^3$$
$$=I-0=I$$
$$\therefore \quad (I-A)^{-1}=I+A+A^2$$

53. $AA^{-1}=I \quad \Rightarrow \quad (AA^{-1})^T=I^T$

$\Rightarrow \quad (A^{-1})^T A^T=I \quad \Rightarrow \quad (A^{-1})^T=(A^T)^{-1}$

$\Rightarrow \quad (A^{-1})^T=A^{-1} \quad (\because \ A=A^T)$

Hence the inverse of a symmetrix matrix is symmetric.

54. Let $A=\begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}$, then the given equation is

$$\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix}\begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}=\begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} 2x-u & 2y-v & 2z-w \\ x & y & z \\ -3x+4u & -3y+4v & -3z+4w \end{bmatrix}$$
$$=\begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$$

Equating the corresponding elements, we get the values of x, y, z, u, v, and w as 1, –2, –5, 3, 4, and 0.

Hence $\quad A=\begin{bmatrix} 1 & -2 & -5 \\ 3 & 4 & 0 \end{bmatrix}$

55. $A^2=\begin{bmatrix} a & b \\ b & a \end{bmatrix}\begin{bmatrix} a & b \\ b & a \end{bmatrix}=\begin{bmatrix} a^2+b^2 & 2ab \\ 2ab & a^2+b^2 \end{bmatrix}$

But $A^2=\begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$

$\therefore \quad \alpha=a^2+b^2, \ \beta=2ab$

56. If A is a non-singular matrix of order n, then

$$\text{adj}\,(\text{adj}\,A)=|A|^{n-2}A$$

Here, $n = 3$.

$\therefore \quad \text{adj}(\text{adj}\,A)=|A|A$.

57. $AB = A$ and $BA = B$

Now, $AB=A \quad \Rightarrow \quad (AB)A=AA=A^2$

$\Rightarrow \ A(BA)=A^2 \Rightarrow \ AB=A^2 \quad (\because BA=B)$

$\Rightarrow \ A=A^2 \quad (\because AB=A)$

$\Rightarrow A$ is idempotent. Similarly, B is idempotent.

58. $\because \ PQ=I \ \Rightarrow \ P^{-1}=Q$

Now, the system in matrix notation is $PX = B$.

$\therefore \quad X=P^{-1}B=QB$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix}=\frac{1}{9}\begin{bmatrix} 2 & 2 & 1 \\ 13 & -5 & m \\ -8 & 1 & 5 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

$$\Rightarrow \quad y=\frac{1}{9}(13-5+5m)$$

$$\Rightarrow \quad -27=8+5m \qquad (\text{given } y=-3)$$

$\therefore \quad m=-7$

59. We have, $A^2=A$

$\therefore \quad (I+A)^2=(I+A)(I+A)=I+2A+A^2=I+3A$

and $(I+A)^3=(I+A)^2(I+A)$

$=(I+3A)(I+A) \quad [\because \ (I+A)^2=I+3A]$

$= I+4A+3A^2=I+7A \quad [\because \ A^2=A]$

Thus, we have

$(I+A)^2=I+3A$ and $(I+A)^3=I+7A$

$\Rightarrow \quad (I+A)^2=I+(2^2-1)A$

and $\quad (I+A)^3=I+(2^3-1)A$

Hence, $(I+A)^n=I+(2^n-1)A$

$\therefore \quad (I+A)^n=I+\lambda A \quad \Rightarrow \quad \lambda=2^n-1$

60. A is skew symmetric.

$$\Rightarrow \quad A^T=-A \quad \Rightarrow \quad (A^T)^n=(-A)n$$

$$\Rightarrow \quad (A^n)^T=(-1)^nA^n=\begin{bmatrix} A^n \text{ if } n \text{ is even} \\ -A^n \text{ if } n \text{ is odd} \end{bmatrix}$$

Hence the option (d).

61. $\begin{bmatrix} \omega & \omega^2 \\ 1 & \omega \\ \omega^2 & 1 \end{bmatrix}\times\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \end{bmatrix}$

$$=\begin{bmatrix} \omega+\omega^3 & \omega^2+\omega^4 & \omega^3+\omega^2 \\ 1+\omega^2 & \omega+\omega^3 & \omega^2+\omega \\ \omega^2+\omega & \omega^3+\omega^2 & \omega^4+1 \end{bmatrix}$$
$$=\begin{bmatrix} -\omega^2 & -1 & -\omega \\ -\omega & -\omega^2 & -1 \\ -1 & -\omega & -\omega^2 \end{bmatrix}$$

Now, $-\begin{bmatrix} \omega^2 & 1 & \omega \\ \omega & \omega^2 & 1 \\ 1 & \omega & \omega^2 \end{bmatrix}\begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

62. The given system of equations has a non-trivial solution.

$$\therefore \quad \begin{vmatrix} \sin 3\theta & -1 & 1 \\ \cos 2\theta & 4 & 3 \\ 2 & 7 & 7 \end{vmatrix}=0$$

$\Rightarrow 7\sin 3\theta + 14\cos 2\theta - 14 = 0$

$\Rightarrow 3\sin\theta - 4\sin^3\theta + 2 - 4\sin^2\theta - 2 = 0$

$\Rightarrow \sin\theta(4\sin^2\theta + 4\sin\theta - 3) = 0$

$\Rightarrow \sin\theta(2\sin\theta + 3)(2\sin\theta - 1) = 0$

$\Rightarrow \sin\theta = 0$ or, $\sin\theta = (1/2)$

$\Rightarrow \theta = n\pi$ or $\theta = n\pi + (-1)^n \frac{\pi}{6}$

63. As A is symmetric $\Rightarrow A^T = A$

Also, as A is skew-symmetric

$\Rightarrow \quad A^T = -A$

$\therefore \quad A = -A \quad \Rightarrow \quad 2A = 0$

$\Rightarrow A = 0$

$\Rightarrow A$ is null matrix.

64. Now, $A + B = \begin{bmatrix} 3 & 4 \\ 4 & 4 \end{bmatrix} + \begin{bmatrix} -2 & -2 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$

$\Rightarrow \quad |A+B| = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} = 2 - 4 = -2$

Also, adj $(A+B) = \begin{bmatrix} 2 & -2 \\ -2 & 1 \end{bmatrix}$

$\therefore \quad (A+B)^{-1} = -\frac{1}{2}\begin{bmatrix} 2 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1/2 \end{bmatrix}$

Hence $(A + B)^{-1}$ exists. It is not a skew-symmetric matrix.

Now, $A^{-1} = \frac{1}{4}\begin{bmatrix} 4 & -4 \\ -2 & 3 \end{bmatrix}$ and $B^{-1} = \frac{1}{4}\begin{bmatrix} -2 & 2 \\ 0 & -2 \end{bmatrix}$

$\therefore \quad A^{-1} + B^{-1} = \begin{bmatrix} 1 & -1 \\ -1/2 & 3/4 \end{bmatrix} + \begin{bmatrix} -1/2 & 1/2 \\ 0 & -1/2 \end{bmatrix}$

$= \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/4 \end{bmatrix}$

$\therefore \quad (A+B)^{-1} \neq A^{-1} + B^{-1}$

$\therefore$ Option (d) is correct.

65. We have trace of $A = 2a$.

Clearly, trace of A will be divisible by p iff $a = 0$.

$\therefore \quad a \in \{1, 2, ..., (p-1)\}$

We have $\det(A) = a^2 - bc$

$\therefore \quad p \mid \det(A) \quad \Rightarrow \quad p \mid a^2 - bc$

$\Rightarrow \quad a^2 - bc = \lambda p$ for some integer λ

For each value of $a \in \{1, 2, \ldots, (p-1)\}$, there are $(p - 1)$ ordered pairs (b, c) for bc such that $a^2 - bc$ is divisible by p.

Hence, there are $(p - 1)^2$ matrices in T_p such that the trace of A is not divisible by p but the determinant of A is divisible by p.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. As $|AB| = |A||B|$

$\because \quad AB = 0 \quad \Rightarrow \quad |AB| = 0$

$\Rightarrow |A| \cdot |B| = 0$

i.e. either $|A| = 0$ or $|B| = 0$.

2. $A\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}$

Multiplying both sides by $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}^{-1}$, we get

$$A = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}^{-1},$$

but $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}^{-1} = \frac{\begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix}}{10 - 12} = \frac{1}{2}\begin{bmatrix} -5 & 3 \\ 4 & -2 \end{bmatrix}$

$\therefore \quad A = \frac{1}{2}\begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}\begin{bmatrix} -5 & 3 \\ 4 & -2 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -16 & 8 \\ -38 & 24 \end{bmatrix}$

$= \begin{bmatrix} -8 & 4 \\ -19 & 12 \end{bmatrix}$

3. Since $A(\text{adj}A) = |A| I_n$

$\therefore \quad |A||\text{adj}A| = |A|^n$

$\Rightarrow \quad |\text{adj}A| = |A|^{n-1} \qquad [\because |A| \neq 0]$

4. $|A| = 1(16 - 9) - 3(4 - 3) + 3(3 - 4) = 1$

$\because \quad A^{-1} = \frac{1}{|A|}\text{adj}A$, hence $A^{-1} = \text{adj}A$

5. $\det(A) = \begin{vmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{vmatrix} = 0$

$\Rightarrow \quad -(x-3)^2 x = 0 \quad \Rightarrow \quad x = 0, 3$

6. We have $A = \begin{bmatrix} 1 & \log_b a \\ \log_a b & 1 \end{bmatrix}$

$\therefore \quad |A| = 1 - \log_a b \log_a a = 1 - 1 = 0$

7. $AA^{-1} = I \quad \Rightarrow \quad |AA^{-1}| = |I|$

$\Rightarrow \quad |A||A^{-1}| = 1 \quad \Rightarrow \quad |A^{-1}| = \frac{1}{|A|} \qquad [\because |A| \neq 0]$

8. Here $|A| = 4$

$\therefore \quad |5A| = 5^3 \times 4 = 500$

(Note that 5 is taken common from all the three columns or rows)

9. $A^2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, \ldots, A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$

$= \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$

$= nA - (n-1)I$

10. Given that $A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 1 \end{bmatrix}$

$\therefore \quad AB = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 2-6+1 & 1-4+1 \\ 4+3+3 & 2+2+3 \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ 10 & 7 \end{bmatrix}$

$\Rightarrow (AB)^T = \begin{bmatrix} -3 & 10 \\ -2 & 7 \end{bmatrix}$

11. A is orthogonal $\Rightarrow \quad AA^T = A^T A = I$

$\Rightarrow |AA^T| = |I| \quad \Rightarrow \quad |A||A^T| = 1$

$\Rightarrow |A||A| = 1 \quad [\because |A| = |A^T|] \quad \Rightarrow \quad |A| = \pm 1$

12. Now, $BB^T = (I-A)^{-1}(I+A)(I+A)^T[(I-A)^{-1}]^T$

$= (I-A)^{-1}(I+A)(I-A)(I+A)^{-1}$

$= (I-A)^{-1}(I-A)(I+A)(I+A)^{-1}$

$= I \cdot I$

$= I$

$\Rightarrow$ B is an orthogonal matrix.

13. As $P^3 = P(I-P) \quad [\because P^2 = I - P]$

$= PI - P^2 = PI - (I-P)$

$= P - I + P = 2P - I$

Now, $P^4 = P.P^3$

$\Rightarrow \quad P^4 = P(2P-I) \quad \Rightarrow \quad P^4 = 2P^2 - P$

$\Rightarrow \quad P^4 = 2I - 2P - P \quad \Rightarrow \quad P^4 = 2I - 3P$

and $P^5 = P(2I - 3P)$

$\Rightarrow \quad P^5 = 2P - 3(I-P) \quad \Rightarrow \quad P^5 = 5P - 3I$

Also, $P^6 = P(5P - 3I)$

$\Rightarrow \quad P^6 = 5P^2 - 3P \quad \Rightarrow \quad P^6 = 5(I-P) - 3P$

$\Rightarrow \quad P^6 = 5I - 8P$

So, $n = 6$

Alternate solution

As $P^n = 5I - 8P \quad \Rightarrow \quad P^n = 5(I-P) - 3P$

$\Rightarrow \quad P^n = P(5P - 3I) \quad (\because P^2 = I - P)$

$\Rightarrow \quad P^n = P(2P - 3P^2) \quad \Rightarrow \quad P^n = P^2(2I - 3P)$

$\Rightarrow \quad P^n = P^2[2(I-P) - P]$

$\Rightarrow \quad P^n = P^2[2P^2 - P] \quad \Rightarrow \quad P^n = P^3[2P - I]$

$\Rightarrow \quad P^n = P^4[I - P] \quad \Rightarrow \quad P^n = P^4 \cdot P^2 = P^6$

$\Rightarrow n = 6$

14. L.H.S. of option (a) is a matrix while R.H.S. is a determinant which is a number. Hence (a) is not true.

15. $A^{-1}(A^2 - 3A + 2I) = A^{-1} \cdot 0 = 0$

$\Rightarrow \quad 2A^{-1} = 3I - A \quad \Rightarrow \quad A^{-1} = \dfrac{3I - A}{2}$

16. Given matrix A is a square matrix and $AA^T = I = A^T A$

$\Rightarrow |AA^T| = |I| = |A^T A| \quad \Rightarrow \quad |A||A^T| = 1 = |A^T||A|$

$\Rightarrow |A|^2 = 1 \quad [\because |A||A^T| = |A|^2]$

$\Rightarrow |A| = \pm 1$

17. $|A| = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix} = a^3$

$\Rightarrow \quad |A| \cdot |\text{adj}A| = ||A| \cdot I||A|^3$

$\Rightarrow \quad |\text{adj}A| = |A|^2 = (a^3)^2 = a^6$

18. $x - 2y + z = -4 \quad$ (i)

$2x - y + 2z = 2 \quad$ (ii)

$x + y + \lambda z = 4 \quad$ (iii)

$\Rightarrow \quad x - 2y + z = -4$

From Eqs. (i) and (ii), $3y = 10$

From Eqs. (iii) and (i), $3y + (\lambda - 1)z = 8$

$\Rightarrow \quad (\lambda - 1)z = -2$

If $\lambda = 1, 0 \cdot z = -2$, no solution.

If $\lambda \neq 1$, we have unique solution.

19. Given: $M = [a_{uv}]_{n \times n} = [\sin(\theta_u - \theta_v) + i\cos(\theta_u - \theta_v)]$

$\Rightarrow \quad \overline{M} = [\sin(\theta_u - \theta_v) - i\cos(\theta_u - \theta_v)]$

$$\Rightarrow \quad (\overline{M})^T = [\sin(\theta_v - \theta_u) - i\cos(\theta_v - \theta_u)]$$
$$= [-\sin(\theta_u - \theta_v) - i\cos(\theta_u - \theta_v)]$$
$$= -[\sin(\theta_u - \theta_v) + i\cos(\theta_u - \theta_v)]$$
$$= -M$$
$$\Rightarrow \quad M = -(\overline{M})^T$$

20. The two-rowed unit matrix is $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$.

$$\therefore \quad \begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}^2 = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$
$$\Rightarrow \quad \begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}\begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$
$$\Rightarrow \quad \begin{bmatrix}\alpha^2 + \beta\gamma & 0\\ 0 & \beta\gamma + \alpha^2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$
$$\Rightarrow \quad \alpha^2 + \beta\gamma = 1 \quad \Rightarrow \quad 1 - \alpha^2 - \beta\gamma = 0$$

21. $\begin{vmatrix}1 & -k & -1\\ k & -1 & -1\\ 1 & 1 & -1\end{vmatrix} = 0$

$$\Rightarrow \quad \begin{vmatrix}1 & -k & -1\\ k-1 & k-1 & 0\\ 0 & k+1 & 0\end{vmatrix} = 0$$

[using $R_2 - R_1 \to R_2, R_3 - R_1 \to R_3$]

$$\Rightarrow \quad k^2 - 1 = 0 \quad \Rightarrow \quad k = 1, -1$$

22. We have $(AA^T)^T = (A^T)^T A^T = AA^T$

$\therefore$ AA^T is a symmetric matrix.

23. $|A^n| = |A \cdot A \ldots A|$

$$= |A| \cdot |A| \ldots |A| = |A|^n = 2^n$$

24. $A = \begin{bmatrix}\frac{\omega}{i} & \frac{\omega^2}{i}\\ -\frac{\omega^2}{i} & \frac{-\omega}{i}\end{bmatrix} = \frac{\omega}{i}\begin{bmatrix}1 & \omega\\ -\omega & -1\end{bmatrix}$

$$\therefore \quad A^2 = -\omega^2\begin{bmatrix}1-\omega^2 & 0\\ 0 & 1-\omega^2\end{bmatrix}$$
$$= -\begin{bmatrix}-\omega^2 + \omega^4 & 0\\ 0 & -\omega^2 + \omega^4\end{bmatrix}$$
$$= \begin{bmatrix}-\omega^2 + \omega & 0\\ 0 & -\omega^2 + \omega\end{bmatrix}$$

$\because$ $f(x) = x^2 + 2$

$$\therefore \quad f(A) = A^2 + 2I$$
$$= \begin{bmatrix}-\omega^2 + \omega & 0\\ 0 & -\omega^2 + \omega\end{bmatrix} + \begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$$
$$= [-\omega^2 + \omega + 2]\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = (3 + 2\omega)\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$
$$= (2 + i\sqrt{3})\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

25. Since $\begin{bmatrix}3 & 3\\ 5 & 5\end{bmatrix} = 0$,

$\therefore$ $\begin{bmatrix}3 & 3\\ 5 & 5\end{bmatrix}$ is not invertible.

26. Given that $A^{-1} = \lambda(\text{adj}\,A)$

On comparing with $A^{-1} = \frac{1}{|A|}\,\text{adj}\,A$, we get

$$\lambda = \frac{1}{|A|}$$

Now, $|A| = \begin{vmatrix}0 & 3\\ 2 & 0\end{vmatrix} = 0 - 6 = -6$

$$\Rightarrow \quad \lambda = -\frac{1}{6}$$

27. $A + B$ is defined.

$\therefore$ A and B are of same order, i.e. AB is defined.

$\therefore$ The number of colums in A equals the number of rows in B. Hence A and B are square matrices of same order.

28. $\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta\end{bmatrix}\begin{bmatrix}\cos^2\phi & \cos\phi\sin\phi\\ \cos\phi\sin\phi & \sin^2\phi\end{bmatrix}$

$$= \begin{bmatrix}\cos\theta\cos\phi\cos(\theta \sim \phi) & \cos\theta\sin\phi\cos(\theta \sim \phi)\\ \cos\phi\sin\theta\cos(\theta \sim \phi) & \sin\theta\sin\phi\cos(\theta \sim \phi)\end{bmatrix}$$
$$= \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix} = O \quad \left[\text{If } \theta - \phi = (2n+1)\frac{\pi}{2},\ n \in Z\right]$$

29. $A^2 = AA = \begin{bmatrix}i & 0\\ 0 & i\end{bmatrix}\begin{bmatrix}i & 0\\ 0 & i\end{bmatrix}$

$$= \begin{bmatrix}i^2 + 0 & 0 + 0\\ 0 + 0 & 0 + i^2\end{bmatrix} = \begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$$
$$A^4 = A^2A^2 = \begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$$
$$= \begin{bmatrix}1+0 & 0-0\\ -0-0 & 0+1\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = I$$

Hence, $A^{4n} = (A^4)^n = (I)^n = I$

30. Given $A^2 = 2A - I$

Now, $A^3 = A^2 \cdot A = 2A^2 - IA$

$= 2A^2 - A = 2(2A - I) - A$

$= 3A - 2I = 3A - (3-1)I \ldots$

$A^n = nA - (n-1)I$ for $n \geq 2$

31. $|A+B|$ cannot be expressed in terms of $|A|$ and $|B|$. Hence the given equation gives no inference.

32. $|P^T AP| = |A|$

$\Rightarrow \quad |P^T||A||P| - |A| = 0$

$\Rightarrow \quad |A|(|P|^2 - 1) = 0, \ |A| \neq 0 \ [\because \ |P^T| = |P|]$

$\Rightarrow \quad |P|^2 - 1 = 0 \quad \Rightarrow \quad |P| = \pm 1$

33. The given matrix A is singular if

$$|A| = \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & \lambda \end{vmatrix} = 0$$

$\Rightarrow \ 8(7\lambda - 16) + 6(-6\lambda + 8) + 2(24 - 14) = 0$

$\Rightarrow \ 56\lambda - 128 - 36\lambda + 48 + 20 = 0$

$\Rightarrow \ 20\lambda = 60$

$\Rightarrow \ \lambda = 3$

34. By expanding the determinant, we get

$= 1(1 - \omega^{3n}) - \omega^n(\omega^{2n} - \omega^{2n}) + \omega^{2n}(\omega^{4n} - \omega^n)$

$= (1-1) - 0 + \omega^{2n}(\omega^n - \omega^n) = 0$

35. $\because \ |kA| = k^n |A|$, where k is a scalar and A is a matrix of the order $n \times n$. Here, A is of the order 3×3 and $k = 3$.

$\therefore \ |3A| = 3^3|A| = 27|A|$

36. Let $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

$$\therefore \quad |A| = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix} = 0\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$$

$$= -(1-9) + 2(1-6) = 8 - 10 = -2$$

$$\text{and adj } A = \begin{bmatrix} -1 & 8 & -5 \\ 1 & -6 & 3 \\ -1 & 2 & -1 \end{bmatrix}^T = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$$

Hence, $A^{-1} = \dfrac{1}{|A|}(\text{adj } A)$

$$= -\frac{1}{2}\begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

37. Since $A(\text{adj } A) = |A| I$,

$\therefore \ |A||\text{adj } A| = |A||I| = |A|^3 \quad [\because \ A \text{ is a } 3 \times 3 \text{ matrix}]$

$\Rightarrow \ |\text{adj } A| = |A|^2$

38. $A = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$

$\therefore \quad |A| = \cos^2 2\theta + \sin^2 2\theta = 1$

and $\quad \text{adj } A = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$

$$\therefore \quad A^{-1} = \frac{1}{1}\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

39. Since α, β, γ are the roots of $x^3 + px + q = 0$,

$\therefore \quad \alpha + \beta + \gamma = 0$

Using $R_1 \to R_1 + R_2 + R_3$, we get

$$\begin{bmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{bmatrix} = \begin{bmatrix} \alpha+\beta+\gamma & \alpha+\beta+\gamma & \alpha+\beta+\gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{bmatrix}$$

$$= 0$$

$[\because \quad \alpha + \beta + \gamma = 0]$

40. Given that $F(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\Rightarrow \quad F(-\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore \ F(\alpha)F(-\alpha)$

$$= \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\alpha + \sin^2\alpha & \cos\alpha\sin\alpha - \sin\alpha\cos\alpha & 0 \\ \sin\alpha\cos\alpha - \sin\alpha\cos\alpha & \sin^2\alpha + \cos^2\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$\Rightarrow \ [F(\alpha)]^{-1} = F(-\alpha)$

41. Given that p is a non-singular matrix such that

$$1+p+p^2+\cdots+p^n=O$$

$\Rightarrow (1+p)(1+p+p^2+\cdots+p^n)=O$

$\Rightarrow 1-p^{n+1}=O$

$\Rightarrow p^{n+1}=1$

$\Rightarrow p^n\times p^1=1$

$\Rightarrow p^n=1/p$

$\therefore\ p^{-1}=p^n$

42. $\because\ A^TA=\dfrac{1}{3}\begin{bmatrix}1&-2&2\\-2&1&2\\-2&-2&-1\end{bmatrix}\cdot\dfrac{1}{3}\begin{bmatrix}1&-2&-2\\-2&1&-2\\2&2&-1\end{bmatrix}$

$$=\frac{1}{9}\begin{bmatrix}9&0&0\\0&9&0\\0&0&9\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=I$$

$\therefore$ A is orthogonal.

Also, if A and B are orthogonal, then AB is orthogonal.

43. Let $A=\begin{bmatrix}0&-c&b\\c&0&a\\-b&-a&0\end{bmatrix}$, then $A=-A^T$

$\therefore\ \det(A)=\det(-A^T)=-\det(A^T)=-\det A$

$\therefore\ \det A=0$

$\det A^T=\det(-A^T)$ is not true

$\because\ \det(-A^T)=(-1)^3\det(A^T)=-\det A^T$

44. $\because\ |A|=\begin{bmatrix}3&4\\3&5\end{bmatrix}=15-12=3\neq 0$

$\therefore$ A is a non-singular matrix.

$\therefore\ A^{-1}$ exists.

45. Let $A=\begin{bmatrix}a&b\\c&d\end{bmatrix}$, $B=\begin{bmatrix}x&y\\z&u\end{bmatrix}$

Also, $AB=BA$ (given)

$$\Rightarrow \begin{bmatrix}ax+bz&ay+bu\\cx+dz&cy+du\end{bmatrix}=\begin{bmatrix}ax+cy&bx+dy\\az+cu&bz+du\end{bmatrix}$$

On comparing, we get

$$ax+bz=ax+cy$$

$$\Rightarrow bz=cy$$

$$\Rightarrow \frac{z}{c}=\frac{y}{b}=\lambda \quad \text{(say)}$$

$$\Rightarrow y=b\lambda,\ z=c\lambda \qquad \text{(i)}$$

and $ay+bu=bx+dy$

$\Rightarrow ab\lambda+bu=bx+bd\lambda$ [From Eq. (i)]

$\Rightarrow a\lambda+u=x+d\lambda=k$ (say)

For $\lambda=0$; $y=0$, $z=0$, $u=k$, $x=k$

Then, $B=\begin{bmatrix}k&0\\0&k\end{bmatrix}$ = scalar matrix

Also, if $A=\begin{bmatrix}a&b\\c&d\end{bmatrix}$ and $B=\begin{bmatrix}k&0\\0&k\end{bmatrix}$

Then, $AB=BA=\begin{bmatrix}ak&bk\\ck&dk\end{bmatrix}=kA$

46. If $A=[a_{ij}]_{n\times n}$ is a square matrix such that $a_{ij}=0$ for $i\neq j$; then A is called a diagonal matrix. Thus, the given statement is true and $A=\begin{bmatrix}3&0&0\\0&4&0\\0&0&7\end{bmatrix}$ is a diagonal matrix.

47. Appling $C_1\to xC_1$ and $C_1\to C_1+(C_2-C_3)$, we get the given matrix as

$$\frac{1}{x}\begin{vmatrix}0&b&ax+b\\0&c&bx+c\\ax^2+2bx+c&bx+c&0\end{vmatrix}$$

$$=\frac{(ax^2+2bx+c)}{x}[b^2x+bc-acx-bc]$$

$$=(b^2-ac)(ax^2+2bx+c)$$

$$=(+\text{ve})\,(-\text{ve})$$

$$<0$$

48. Let A be the first term and R be the common ratio of G.P.

$\therefore\ l=t_p=AR^{p-1}$

$\Rightarrow \log l=\log A+(p-1)\log R$

Similarly, $\log m=\log A+(q-1)\log R$

and $\log n=\log A+(r-1)\log R$

$$\therefore\ \begin{vmatrix}\log l&p&1\\\log m&q&1\\\log n&r&1\end{vmatrix}=\begin{vmatrix}\log A+(p-1)\log R&p&1\\\log A+(q-1)\log R&q&1\\\log A+(r-1)\log R&r&1\end{vmatrix}$$

$$=\underset{C_1\propto C_3}{\begin{vmatrix}\log A-\log R&p&1\\\log A-\log R&q&1\\\log A-\log R&r&1\end{vmatrix}}+\underset{C_1\propto C_2}{\begin{vmatrix}p\log R&p&1\\q\log R&q&1\\r\log R&r&1\end{vmatrix}}$$

$$=0+0=0$$

49. Here $AB=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}a & 0\\ 0 & b\end{bmatrix}=\begin{bmatrix}a & 2b\\ 3a & 4b\end{bmatrix}$

$$BA=\begin{bmatrix}a & 0\\ 0 & b\end{bmatrix}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}=\begin{bmatrix}a & 2a\\ 3b & 4a\end{bmatrix}$$

Hence, $AB = BA$ only when $a = b$. It means infinitely B's such that $AB = BA$.

50. $A^2=\begin{bmatrix}a & b\\ b & a\end{bmatrix}\begin{bmatrix}a & b\\ b & a\end{bmatrix}=\begin{bmatrix}a^2+b^2 & 2ab\\ 2ab & a^2+b^2\end{bmatrix}$

But $A^2=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix}$

$\Rightarrow \alpha=a^2+b^2$ and $\beta=2ab$

51. For a non-trivial solution, the determinant of the coefficients of various terms vanishes

i.e. $$\begin{vmatrix}1 & 2a & a\\ 1 & 3b & b\\ 1 & 4c & c\end{vmatrix}=0$$

$\Rightarrow \quad (3bc-4bc)-2a(c-b)+a(4c-3b)$

$\Rightarrow \quad \dfrac{2ac}{a+c}=b$

$\Rightarrow$ a, b, c are in H.P.

52. As vectors $(1,a,a^2)$, $(1,b,b^2)$ and $(1,c,c^2)$ are non-coplanar.

$\therefore \quad \begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix}\neq 0$

Now $$\begin{vmatrix}a & a^2 & a^3+1\\ b & b^2 & b^3+1\\ c & c^2 & c^3+1\end{vmatrix}=0$$

$\Rightarrow \quad (1+abc)\begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix}=0$

$\Rightarrow (1+abc)=0 \Rightarrow abc=-1$

53. (i) $|A|=1$, $\therefore$ A^{-1} does not exist is a wrong statement.

(ii) $(-1)I=\begin{pmatrix}-1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -1\end{pmatrix}\neq A \Rightarrow (b)$ is false

(iii) A is clearly a non-zero matrix.

$\therefore$ (c) is false

Now, we are left with option (d) only.

54. Here, $A=\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}$

$\therefore \quad |A|=\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}$

$=1\cdot(1+3)+1\cdot(2+3)+1\cdot(2-1)$

$=4+5+1=10$

$\Rightarrow \text{adj } A=\begin{bmatrix}4 & -5 & 1\\ 2 & 0 & -2\\ 2 & 5 & 3\end{bmatrix}^T=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}$

$\Rightarrow B=A^{-1}=\dfrac{1}{10}\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}$

$\Rightarrow (10)B=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & \alpha\\ 1 & -2 & 3\end{bmatrix}$

Hence, $\alpha=5$

55. $A^2-A+I=0\Rightarrow I=A-A\cdot A$

$IA^{-1}=AA^{-1}-A(AA^{-1})$ or $A^{-1}=I-A$

56. $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

$\therefore \quad A^2=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$, $A^3=\begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}$, so $A^n=\begin{bmatrix}1 & 0\\ n & 1\end{bmatrix}$

and

$$nA-(n-1)I=\begin{pmatrix}n & 0\\ n & n\end{pmatrix}-\begin{pmatrix}n-1 & 0\\ 0 & n-1\end{pmatrix}=\begin{pmatrix}1 & 0\\ n & 1\end{pmatrix}=A^n$$

57. Applying $C_2\to C_2+C_3+C_1$, we get

$$f(x)=1+2x+x(a^2+b^2+c^2)\begin{vmatrix}1+a^2x & 1 & (1+c^2)x\\ (1+a^2)x & 1 & (1+c^2)x\\ (1+a^2)x & 1 & 1+c^2x\end{vmatrix}$$

Applying $R_1\to R_1-R_2, R_2\to R_2-R_3$ and using $a^2+b^2+c^2=-2$, we have

$$f(x)=(1+2x-2x)\begin{vmatrix}1-x & 0 & 0\\ 0 & 0 & x-1\\ (1+a^2)x & 1 & 1+c^2x\end{vmatrix}=(1-x)^2$$

$=x^2-2x+1$

$\therefore$ Degree of $f(x)$ is 2.

58. Let $\Delta=\begin{vmatrix}\alpha & 1 & 1\\ 1 & \alpha & 1\\ 1 & 1 & \alpha\end{vmatrix}=\alpha^3-3\alpha+2=0$

$\Rightarrow \quad (\alpha-1)(\alpha^2+\alpha-2)=0$

$\Rightarrow \quad (\alpha-1)^2(\alpha+2)=0$

$\Rightarrow \alpha=1 \quad \text{or} \quad -2$

$\alpha=1 \Rightarrow x+y+z=0$ gives an infinite number of solutions. Hence, $\alpha=-2$ has no solution.

59. Given: $A^2-B^2=(A+B)(A-B)$

$\Rightarrow \quad 0=BA-AB \Rightarrow BA=AB$

60. $|A^2|=25 \Rightarrow |A|=\pm5$

$\Rightarrow 625\alpha^2=\pm5$

$\Rightarrow \alpha=\frac{1}{5}$

61. $D=\begin{bmatrix}1 & 1 & 1\\ 1 & 1+x & 1\\ 1 & 1 & 1+y\end{bmatrix}$

(Appling $C_2\to C_2-C_1, C_3\to C_3-C_1$)

$$D=\begin{bmatrix}1 & 0 & 0\\ 1 & x & 0\\ 1 & 0 & y\end{bmatrix}=1(xy-0)=xy$$

Hence, D is divisible by both x and y.

62. Given equations have a non-trivial solution if the determinant of the coefficient matrix is zero.

$$\therefore \quad \begin{bmatrix}1 & -c & -b\\ -c & 1 & -a\\ -b & -a & 1\end{bmatrix}=0$$

$\Rightarrow \quad 1-a^2-b^2-c^2-2abc=0$

$\Rightarrow \quad a^2+b^2+c^2+2abc=1$

63. Given: A is a square matrix with all entries as integers.

(a) Now $|A|\neq1,-1$ means it may be zero. Also if $|A|=0$ then A^{-1} does not exist.

$\therefore$ Choice (a) is false.

(b) If $|A|=1,-1$, then A^{-1} certainly exists but A is a square matrix with all integral entries so all cofactors are integers. So adj A matrix has all integral entries.

$$\therefore \quad A^{-1}=\frac{1}{|A|}(\text{adj}A)=\pm(\text{adj}A)$$

$\therefore$ Choice (b) is correct.

(c) $|A|=1,-1$, $\therefore A^{-1}$ must exist but given that A^{-1} does no texist which is false. $\therefore$ Choice (c) is incorrect.

(d) $|A|=1,-1$, it is true that A^{-1} exists but we have to follow that A has all integral entries but choice (d) says that it is not necessary that entries are integers which means adj A may or may not be with integral entries. Hence, the choice (d) is false.

64. Given that $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$ and $A^2=B$

$$\Rightarrow \quad \begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix}\alpha^2 & 0\\ \alpha+1 & 0\end{bmatrix}=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$$

$\Rightarrow \quad \alpha^2=1, \alpha+1=5$

$\Rightarrow \quad \alpha=\pm1, \alpha=4$

$\therefore$ There is no common value. Hence there is no real value of α for which $A^2=B$.

65. Since A is a 3×4 matrix,

$\Rightarrow A^T$ is a 4×3 matrix

Also A^TB is defined

$\Rightarrow (B)_{4\times3}$

Also, BA^T is defined $\Rightarrow B_{3\times4}$

$\therefore$ Order of B is 3×4.

66. Given: $A=\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}$

$$A\cdot A=\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}$$

$$=\begin{bmatrix}a^2b^2-a^2b^2 & ab^3-ab^3\\ -a^3b+a^3b & -a^2b^2+a^2b^2\end{bmatrix}$$

$=0$

$\therefore \quad A^2=0$

$\therefore$ A is a nilpotent matrix of order 2.

67. $A+B=I \Rightarrow 2A+2B=2I$ (i)

and $2A-2B=I$ (ii)

Adding (i) and (ii),

$$\Rightarrow \quad A=\frac{3}{4}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

68. $\begin{bmatrix}2 & 0 & 7\\ 0 & 1 & 0\\ 1 & -2 & 1\end{bmatrix}\begin{bmatrix}-x & 14x & 7x\\ 0 & 1 & 0\\ x & -4x & -2x\end{bmatrix}$

$$=\begin{bmatrix}5x & 0 & 0\\ 0 & 1 & 0\\ 0 & 10x-2 & 5x\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \text{ (given)}$$

$\therefore \quad 5x=1, 10x-2=0, \quad \therefore \quad x=(1/5)$

69. $(A^{-1}BA)^2 = (A^{-1}BA)(A^{-1}BA)$

$= A^{-1}B(AA^{-1})BA = A^{-1}BIBA = A^{-1}B^2A$

$\Rightarrow (A^{-1}BA)^3 = (A^{-1}B^2A)(A^{-1}BA)$

$= A^{-1}B^2(AA^{-1})BA$

$= A^{-1}B^2IBA = A^{-1}B^3A$, and so on

$\Rightarrow (A^{-1}BA)^n = A^{-1}B^nA$

70. As $A\,\text{adj}\,A = |A|I$

$\Rightarrow |A\,\text{adj}\,A| = |A|^n$ [If A is of order $n \times n$]

$\Rightarrow |A||\text{adj}\,A| = |A|^n \Rightarrow |\text{adj}\,A| = |A|^{n-1}$

As A is singular, $|A| = 0$

$\Rightarrow |\text{adj}\,A| = 0$

Hence, adj A is singular.

71. $H^2 = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = \begin{bmatrix} \omega^2 & 0 \\ 0 & \omega^2 \end{bmatrix}$

If $H^k = \begin{bmatrix} \omega^k & 0 \\ 0 & \omega^k \end{bmatrix}$, then $H^{k+1} = \begin{bmatrix} \omega^{k+1} & 0 \\ 0 & \omega^{k+1} \end{bmatrix}$

So by mathematical induction,

$H^{70} = \begin{bmatrix} \omega^{70} & 0 \\ 0 & \omega^{70} \end{bmatrix} = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = H$

72. $x - ky + z = 0$

$kx + 3y - kz = 0$

$3x + y - z = 0$

These equations will have non-trivial solution if

$$\begin{vmatrix} 1 & -k & 1 \\ k & 3 & -k \\ 3 & 1 & -1 \end{vmatrix} = 0$$

$1(-3+k) + k(-k+3k) + 1(k-9) = 0$

$\Rightarrow k - 3 + 2k^2 + k - 9 = 0 \Rightarrow 2k^2 + 2k - 12 = 0$

$\Rightarrow k^2 + k - 6 = 0 \Rightarrow k = -2, k = 2$

So the equations will have only trivial solution when $k \in \mathbb{R} - \{2, -3\}$.

73. $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$. Let $u_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$; $u_2 = \begin{bmatrix} d \\ e \\ f \end{bmatrix}$

$Au_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} a \\ 2a+b \\ 3a+2b+c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow a = 1;\ b = -2;\ c = 1$

$\Rightarrow u_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $Au_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} x \\ 2x+y \\ 3x+2y+z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow x = 0;\ y = 1;\ z = -2$

$\Rightarrow u_2 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$

$\therefore\ u_1 + u_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

74. $P^3 = Q^3$ (i)

$Q^2P = P^2Q$ (ii)

Adding Eq. (i) and (ii)

$P^3 + Q^2P = Q^3 + P^2Q$

$P(P^2 + Q^2) = Q(Q^2 + P^2)$

$(P - Q)(P^2 + Q^2) = 0$

$\Rightarrow$ det $(P^2 + Q^2) = 0$ $(\because\ P - Q \neq 0)$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The number of matrices having 12 elements is

(a) 3 (b) 1

(c) 6 (d) none of these

2. If a matrix A is symmetric as well as skew-symmetric, then A is

(a) a diagonal matrix (b) a null matrix

(c) a unit matrix (d) none of these

3. If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & 4 & 0 \\ 0 & 2 & -1 \\ 1 & -3 & 2 \end{bmatrix}$, then

(a) $AB = \begin{bmatrix} -5 & 8 & 0 \\ 0 & 4 & -2 \\ 3 & -9 & 6 \end{bmatrix}$ (b) $AB = \begin{bmatrix} -2 & -1 & 4 \end{bmatrix}$

(c) $AB = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ (d) AB does not exist

4. If A and B are two square matrices such that $AB = 0$, then

(a) det $A = 0$ or det $B = 0$ (b) det $B = 0$

(c) $B = A^{-1}$ (d) det $A = 0$

5. If $A=\begin{bmatrix}3&0&0\\0&3&0\\0&0&3\end{bmatrix}$ and $B=\begin{bmatrix}a_1&a_2&a_3\\b_1&b_2&b_3\\c_1&c_2&c_3\end{bmatrix}$, then AB is equal to

(a) B (b) $3B$

(c) B^3 (d) $A+B$

6. If $A=\begin{bmatrix}1&2\\2&1\end{bmatrix}$, then adj A = ?

(a) $\begin{bmatrix}1&-2\\-2&1\end{bmatrix}$ (b) $\begin{bmatrix}2&1\\1&1\end{bmatrix}$

(c) $\begin{bmatrix}1&-2\\-2&-1\end{bmatrix}$ (d) $\begin{bmatrix}-1&2\\2&-1\end{bmatrix}$

7. If $I=\begin{bmatrix}1&0\\0&1\end{bmatrix}$, $J=\begin{bmatrix}0&1\\-1&0\end{bmatrix}$ and $B=\begin{bmatrix}\cos\theta&\sin\theta\\-\sin\theta&\cos\theta\end{bmatrix}$, then B = ?

(a) $I\cos\theta+J\sin\theta$ (b) $I\cos\theta-J\sin\theta$

(c) $I\sin\theta+J\cos\theta$ (d) $-I\cos\theta+J\sin\theta$

8. If $A=\begin{bmatrix}0&0&0\\1&0&0\\0&1&0\end{bmatrix}$, then

(a) $A^2=A$ (b) $A^2=0$

(c) $A^2=I$ (d) $A^3=0$

9. If A is a skew-symmetric matrix, then the trace of A is

(a) 1 (b) -1

(c) 0 (d) none of these

10. If $A=\begin{bmatrix}\cos\theta&-\sin\theta&0\\\sin\theta&\cos\theta&0\\0&0&1\end{bmatrix}$, then adj A = ?

(a) A^T (b) I

(c) 0 (d) A^2

11. If $A=\begin{vmatrix}0&0&0\\0&0&0\\0&1&0\end{vmatrix}$, then A is

(a) an invertible matrix (b) an idempotent matrix

(c) a nilpotent matrix (d) none of these

12. If A is any square matrix, then

(a) $A+A^T$ is skew-symmetric

(b) $A-A^T$ is symmetric

(c) AA^T is symmetric

(d) none of these

13. For any 2×2 matrix A, if $A(\text{adj } A)=\begin{bmatrix}10&0\\0&10\end{bmatrix}$, then $|A|$ is equal to

(a) 20 (b) 100

(c) 10 (d) 0

14. If A and B are square matrices of order 3 such that $|A|=-1$ and $|B|=3$, then $|3AB|$ is equal to

(a) -9 (b) -81

(c) -27 (d) 81

15. If A and B are square matrices of order 2, then $(A+B)^2$ is equal to

(a) $A^2+2AB+B^2$ (b) $A^2+AB+BA+B^2$

(c) $A^2+2BA+B^2$ (d) none of these

16. If A is a square matrix such that $A^3=I$ then A^{-1} is equal to

(a) I (b) A

(c) A^2 (d) none of these

17. If B is a non-singular matrix and A is a square matrix, then $\det(B^{-1}AB)$ is equal to

(a) $\det(A^{-1})$ (b) $\det(B^{-1})$

(c) $\det(A)$ (d) $\det(B)$

18. Matrix $\begin{bmatrix}a&b&(a\alpha-b)\\b&c&(b\alpha-c)\\2&1&0\end{bmatrix}$ is non-invertible if

(a) $\alpha=\frac{1}{2}$ (b) a, b, c are in A.P.

(c) a, b, c are in G.P. (d) none of these

19. Let A be a skew-symmetric matrix of order n. Then

(a) $|A|=0$ if n is even (b) $|A|=0$ if n is odd

(c) $|A|=0$ for all $n\in N$ (d) none of these

20. Let A and B be 3×3 matrices. Then $AB=0$ implies

(a) $A=0$ and $B=0$

(b) $|A|=0$ and $|B|=0$

(c) either $|A|=0$ or $|B|=0$

(d) $A=0$ or $B=0$

21. If A, B are two $n\times n$ non-singular matrices, then

(a) AB is non-singular (b) AB is singular

(c) $(AB)^{-1}=A^{-1}B^{-1}$ (d) $(AB)^{-1}$ does not exist

22. If $A=\begin{bmatrix}1&2\\3&-5\end{bmatrix}$ and $B=\begin{bmatrix}1&0\\0&2\end{bmatrix}$ and X be matrices such that $A=BX$, then X is equal to

(a) $\frac{1}{2}\begin{bmatrix}2&4\\3&-5\end{bmatrix}$ (b) $\frac{1}{2}\begin{bmatrix}-2&4\\3&5\end{bmatrix}$

(c) $\begin{bmatrix}2&4\\3&-5\end{bmatrix}$ (d) none of these

23. Each diagonal element of a skew-symmetric matrix is

(a) zero (b) positive

(c) non-real (d) negative

24. If the matrix $A=\begin{bmatrix} y+a & b & c \\ a & y+b & c \\ a & b & y+c \end{bmatrix}$ has rank 3, then

(a) $y \neq (a+b+c)$
(b) $y \neq 1$
(c) $y = 0$
(d) $y \neq -(a+b+c)$ and $y \neq 0$

25. If $A=\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$, $n \in N$ then A^{4n} is

(a) $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$

26. If A is a square matrix such that $A^2 = I$, then $A^{-1} = ?$
(a) $2A$ (b) A
(c) 0 (d) $A + I$

27. Let a, b, c, d, u, v be integers. If the system of equations $ax + by = u$, and $cx + dy = v$ has a unique solution in integers, then
(a) $ad - bc = 1$
(b) $ad - bc = -1$
(c) $ad - bc \neq 0$
(d) $ad - bc$ need not be equal to ± 1

28. If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ then A^{-1} is equal to

(a) $\begin{bmatrix} 7 & -3 & -3 \\ 0 & 1 & 0 \\ -1 & 0 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 7 & -3 & -3 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ (d) none of these

29. If A is a non-zero column matrix of order $m \times 1$ and B is a non-zero row matrix of order $1 \times n$, then the rank of AB is equal to
(a) n (b) m
(c) 1 (d) none of these

30. The matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is
(a) an idempotent matrix
(b) an involutory matrix
(c) a nilpotent matrix
(d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If $A = \text{dig}(2, -1, 3)$ and $B = \text{dig}(-1, 3, 2)$, then $A^2B =$
(a) dig (5, 4, 11) (b) dig (−4, 3, 18)
(c) dig (3, 1, 8) (d) B

2. If $A=\begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ and $B=\begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$, then B^TA^T is
(a) a null matrix
(b) an identity matrix
(c) a scalar matrix, but not an identity matrix
(d) such that $Tr(B^TA^T) = 4$

3. Which of the following relations is true for $A=\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, $B=\begin{bmatrix} 1 & 4 \\ -1 & 1 \end{bmatrix}$?
(a) $(A+B)^2 = A^2 + 2AB + B^2$
(b) $(A-B)^2 = A^2 - 2AB + B^2$
(c) $AB = BA$
(d) none of these

4. The trace of a skew-symmetric matrix is always equal to
(a) $\sum a_{ij}$ (b) $\sum a_{ii}$
(c) zero (d) none of these

5. Which of the following is incorrect?
(a) $A^2 - B^2 = (A+B)(A-B)$
(b) $(A^T)^T = A$
(c) $(AB)^n = A^nB^n$, where A and B commute
(d) $(A-I)(I+A) = 0 \Leftrightarrow A^2 = I$

6. If $X=\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, then value of X^n is

(a) $\begin{bmatrix} 3n & -4n \\ n & -n \end{bmatrix}$ (b) $\begin{bmatrix} 2+n & 5-n \\ n & -n \end{bmatrix}$
(c) $\begin{bmatrix} 3^n & (-4)^n \\ 1^n & (-1)^n \end{bmatrix}$ (d) none of these

7. If the matrix AB is a zero matrix, then
(a) $A = 0$ or $B = 0$
(b) $A = 0$ and $B = 0$
(c) it is not necessary that either $A = 0$ or $B = 0$
(d) all the above statements are wrong

8. If A^T is the transpose of a square matrix A, then

(a) $|A| \neq |A^T|$

(b) $|A| = |A^T|$

(c) $|A| + |A^T| = 0$

(d) $|A| = |A^T|$ only when A is symmetric

9. If $AB = 0$ for the matrices $A = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$ then $\phi - \theta$ is

(a) an odd multiple of $\pi/2$

(b) an odd multiple of π

(c) an even multiple of $\pi/2$

(d) 0

10. If A is a skew-symmetric matrix and n is an even positive integer, then A^n is

(a) a symmetric matrix

(b) a skew-symmetric matrix

(c) a diagonal matrix

(d) none of these

11. If I_n is the identity matrix of order n, then $(I_n)^{-1}$

(a) does not exist (b) equals I_n

(c) equals O (d) none of these

12. If A and B are symmetric matrices, then ABA is a

(a) symmetric matrix (b) skew-symmetric matrix

(c) diagonal matrix (d) scalar matrix

13. If A is a non-singular matrix and A^T denotes the transpose of A, then

(a) $|A| \neq |A^T|$ (b) $|A \cdot A^T| \neq |A|^2$

(c) $|A^T \cdot A| \neq |A^T|^2$ (d) $|A| + |A^T| \neq 0$

14. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}$; $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. If $A - \lambda I$ is a singular matrix, then

(a) $\lambda \in \phi$ (b) $\lambda^2 - 3\lambda - 4 = 0$

(c) $\lambda^2 + 3\lambda + 4 = 0$ (d) $\lambda^2 - 3\lambda - 6 = 0$

15. If for a matrix A of order 2, $A^2 + I = 0$, where I is the identity matrix, then A equals

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

16. From the matrix equation $AB = AC$, we conclude that $B = C$ provided

(a) A is singular (b) A is non-singular

(c) A is symmetric (d) A is square

17. If A is a square matrix of order 3, then the true statement is (where I is the unit matrix)

(a) $\det(-A) = -\det A$

(b) $\det A = 0$

(c) $\det(A + I) = 1 + \det A$

(d) $\det 2A = 2\det A$

18. If A is a square matrix of order $n \times n$ and k is a scalar, then adj (kA) is equal to

(a) k adj A (b) k^n adj A

(c) k^{n-1} adj A (d) k^{n+1} adj A

19. If A and B are symmetric matrices of order $n (A \neq B)$, then

(a) $A + B$ is a skew-symmetric matrix

(b) $A + B$ is a symmetric matrix

(c) $A + B$ is a diagonal matrix

(d) $A + B$ is a zero matrix

20. If $F(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $G(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$, then $[F(\alpha)G(\beta)]^{-1} =$

(a) $F(\alpha) - G(\beta)$ (b) $-F(\alpha) - G(\beta)$

(c) $[F(\alpha)]^{-1}[G(\beta)]^{-1}$ (d) $[G(\beta)]^{-1}[F(\alpha)]^{-1}$

21. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfies the equation $x^2 - (a + b)x + k = 0$, then

(a) $k = bc$ (b) $k = ad$

(c) $k = a^2 + b^2 + c^2 + d^2$ (d) $k = ad - bc$

22. The singularity of matrix

$$\begin{bmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{bmatrix}$$

depends on which of the following parameters?

(a) a (b) p

(c) x (d) none of these

23. If $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ and $B = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$, then $AB =$

(a) A^3 (b) B^2

(c) 0 (d) I

24. Which of the following is true for matrix $A=\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}$

(a) $A + 4I$ is a symmetric matrix

(b) $A - B$ is a diagonal matrix for any value of α if

$$B=\begin{bmatrix}\alpha & -1\\ 2 & 5\end{bmatrix}$$

(c) $A - 4I$ is a skew-symmetric matrix

(d) none of these

25. The value of a for which the system of equations

$$a^3x+(a+1)^3y+(a+2)^3z=0$$
$$ax+(a+1)y+(a+2)z=0$$
$$x+y+z=0$$

has a non-zero solution is

(a) -1 (b) 0

(c) 1 (d) none of these

26. The value of k for which the set of equations

$$3x+ky-2z=0,\ x+ky+3z=0$$

and

$$2x+3y-4z=0$$

has a non-trivial solution over the set of rationals is

(a) $\frac{33}{2}$ (b) $\frac{31}{2}$

(c) 16 (d) 15

27. If $AB = A$ and $BA = B$, then B^2 is equal to

(a) B (b) A

(c) 1 (d) 0

28. The values of x for which the matrix

$$\begin{bmatrix}x+a & b & c\\ a & x+b & c\\ a & b & x+c\end{bmatrix}$$

is non-singular are

(a) $R-\{0\}$ (b) $R-\{-a(a+b+c)\}$

(c) $R-\{0,-(a+b+c)\}$ (d) none of these

29. Which of the following is a nilpotent matrix?

(a) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (b) $\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$

(c) $\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}$ (d) $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$

30. If $A=\begin{bmatrix}2 & 3\\ 5 & -2\end{bmatrix}$, then $19A^{-1}$ is equal to

(a) A^T (b) $2A$

(c) $\frac{1}{2}A$ (d) A

ANSWERS

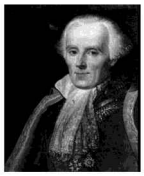
Level 1

1. (c)	**2.** (b)	**3.** (d)	**4.** (a)	**5.** (b)	**6.** (a)	**7.** (a)	**8.** (d)	**9.** (c)	**10.** (a)
11. (c)	**12.** (c)	**13.** (c)	**14.** (b)	**15.** (b)	**16.** (c)	**17.** (c)	**18.** (a)	**19.** (b)	**20.** (c)
21. (a)	**22.** (a)	**23.** (a)	**24.** (d)	**25.** (c)	**26.** (b)	**27.** (c)	**28.** (b)	**29.** (c)	**30.** (c)

Level 2

1. (b)	**2.** (b)	**3.** (d)	**4.** (c)	**5.** (a)	**6.** (d)	**7.** (c)	**8.** (b)	**9.** (a)	**10.** (a)
11. (b)	**12.** (a)	**13.** (d)	**14.** (b)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (c)	**19.** (b)	**20.** (d)
21. (d)	**22.** (c)	**23.** (c)	**24.** (b)	**25.** (a)	**26.** (a)	**27.** (a)	**28.** (c)	**29.** (c)	**30.** (d)

5



P.S. LAPLACE

Determinants

- Definition of Determinant
- Determinant of a Square Matrix of Order 1
- Determinant of a Square Matrix of Order 2
- Determinant of a Square Matrix of Order 3
- Minors and Cofactors
- Sarrus Rule for Expansion
- Properties of Determinants
- Product of Determinants
- Differentiation of a Determinant
- Solving System of Linear Equations
- Some Important Terms Defined
- Crammer's Rule

☞ DEFINITION OF DETERMINANT

A determinant is a pure number associated with a square matrix. Corresponding to each square matrix A, there is associated an

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

expression, called the determinant of A, denoted by det A or $|A|$ and written as

$$|A| = \det A = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

A matrix is an arrangement of numbers and it has no fixed value, but a determinant is a number and it has a fixed value. A determinant having n rows and n columns is called a **determinant of order n**.

Note

- An arrangement of numbers is known as matrix and a matrix having equal number of rows and columns is known as a square matrix.
- The determinant of a matrix, which is not a square matrix, cannot be found out.
- A matrix has no definite value while each determinant has a defined value.

DETERMINANT OF A SQUARE MATRIX OF ORDER 1

Let $A = [a_{11}]$ be a 1×1 matrix, then the determinant of A is the number a_{11} itself, i.e. $|a_{11}| = a_{11}$.

DETERMINANT OF A SQUARE MATRIX OF ORDER 2

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ be a 2×2 matrix, then

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

i.e. the determinant of a 2×2 matrix is obtained by taking the product of the entries on the main diagonal and subtracting from it the product of the entries on the other diagonal.

DETERMINANT OF A SQUARE MATRIX OF ORDER 3

The arrangement of 9 elements in 3 rows and 3 columns represents a determinant of order 3, such that

$$\begin{array}{c} \\ R_1 \rightarrow \\ R_2 \rightarrow \\ R_3 \rightarrow \end{array} \begin{array}{c} \text{Columns} \\ \begin{array}{ccc} C_1 \downarrow & C_2 \downarrow & C_3 \downarrow \end{array} \\ \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \end{array} = \Delta$$

(Rows)

The representation is usually shown as Δ and the value of the determinant is

$$\Delta = a_{11}\underbrace{\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}} - a_{12}\underbrace{\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}} + a_{13}\underbrace{\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}}$$

$\begin{vmatrix} \cdot & \cdot \\ \cdot & \cdot \end{vmatrix}$ determinant left after removing 1st row and 1st column

$\begin{vmatrix} \cdot & \cdot \\ \cdot & \cdot \end{vmatrix}$ determinant left after removing 1st row and 2nd column

$\begin{vmatrix} \cdot & \cdot \\ \cdot & \cdot \end{vmatrix}$ determinant left after removing 1st row and 3rd column

The above is the expansion of the determinant along the first row.

Example: If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, then

$$= |A| = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 1\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} + 3\begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= 1(2-1) - 2(3-1) + 3(3-2)$$

$$= 1 - 2(2) + 3 = 4 - 4 = 0.$$

MINORS AND COFACTORS

1. Minors

Let $A = [a_{ij}]$ be a square matrix of order n. Then the minor M_{ij} of a_{ij} in A is the determinant of a square sub-matrix of order $(n-1)$ obtained by removing the ith row and jth column of A.

Example: If $A = \begin{vmatrix} 1 & 2 & 3 \\ -3 & 2 & -1 \\ 2 & -4 & 3 \end{vmatrix}$, then

$$M_{11} = \begin{vmatrix} 2 & -1 \\ -4 & 3 \end{vmatrix} = 6 - 4 = 2,$$

$$M_{12} = \begin{vmatrix} -3 & -1 \\ 2 & 3 \end{vmatrix} = -9 + 2 = -7$$

$$M_{13} = \begin{vmatrix} -3 & 2 \\ 2 & -4 \end{vmatrix} = 12 - 4 = 8, \; M_{21} = \begin{vmatrix} 2 & 3 \\ -4 & 3 \end{vmatrix} = 6 + 12 = 18$$

$$M_{22} = \begin{vmatrix} 1 & 3 \\ 2 & 3 \end{vmatrix} = 3 - 6 = -3, \; M_{23} = \begin{vmatrix} 1 & 2 \\ 2 & -4 \end{vmatrix} = -4 - 4 = -8$$

$$M_{31} = \begin{vmatrix} 2 & 3 \\ 2 & -1 \end{vmatrix} = -2 - 6 = -8,$$

$$M_{32} = \begin{vmatrix} 1 & 3 \\ -3 & -1 \end{vmatrix} = -1 + 9 = 8$$

and $M_{33} = \begin{vmatrix} 1 & 2 \\ -3 & 2 \end{vmatrix} = 2 + 6 = 8.$

2. Cofactors

Let $A = [a_{ij}]$ be a square matrix of order n. Then the cofactor C_{ij} of a_{ij} in A is $(-1)^{i+j}$ times M_{ij}, where M_{ij} is the minor of a_{ij} in A.

$$\therefore \qquad C_{ij} = (-1)^{i+j} M_{ij}$$

Example: If $A = \begin{vmatrix} 4 & -7 \\ -3 & 2 \end{vmatrix}$, then the cofactors are

$$C_{11} = (-1)^{1+1} M_{11} = M_{11} = 2$$

$$C_{12} = (-1)^{1+2} M_{12} = -M_{12} = -(-3) = 3$$

$$C_{21} = (-1)^{2+1} M_{21} = -M_{21} = 7$$

and $$C_{22} = (-1)^{2+2} M_{22} = M_{22} = 4.$$

SARRUS RULE FOR EXPANSION

Method I: *Using the Sarrus rule*

Sarrus gave a rule for a determinant of order 3. The following diagram, called Sarrus diagram, enables us to write the value of a determinant of order 3 in a very convenient manner.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1b_2c_3 + b_1c_2a_3 + c_1a_2b_3 - a_3b_2c_1 - c_3a_2b_1 - b_3c_2a_1$$

Let $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ be a determinant of order 3. Write the elements as shown in Fig. 5.1. Multiply the elements joined by arrows. Assign the positive sign to an expression if it is formed by a downward arrow and the negative sign to an expression if it is formed by an upward arrow. Note that the first two columns written to the right of the determinant complete the process. The value of the determinant is given by $a_1b_2c_3 + b_1c_2a_3 + c_1a_2b_3 - a_3b_2c_1 - c_3a_2b_1 - b_3c_2a_1$.

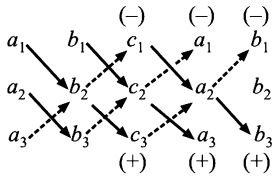
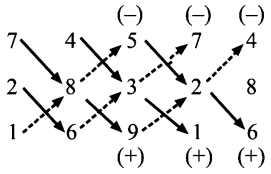


Fig. 5.1

Illustration 1. Find the value of the determinant $A = \begin{vmatrix} 7 & 4 & 5 \\ 2 & 8 & 3 \\ 1 & 6 & 9 \end{vmatrix}$, using the Sarrus rule.

Solution: We write



$$= (7\times8\times9) + (4\times3\times1) + (5\times2\times6) - (1\times8\times5) - (6\times3\times7) - (9\times2\times4)$$

$$= 504 + 12 + 60 - 40 - 126 - 72 = 576 - 238 = 338.$$

Method II: *Using cofactors*

The value of a determinant can also be evaluated by multiplying the elements of any one row (or column) by its corresponding cofactors and then summing the resulting products.

Example: Let $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$, then according to definition,

$$\Delta = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

$$\Rightarrow \quad \Delta = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Also, $\Delta = a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23}$

$$\Delta = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$$

and $\Delta = a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31}$ and similarly the other results can be obtained.

Note Evaluation of a determinant using the cofactor rule is much more efficient than using the Sarrus rule for higher-order determinants.

PROPERTIES OF DETERMINANTS

Determinants have some properties that are useful as they permit to generate equal determinants with different and simpler configurations of entries. This, in turn, helps to find values of determinants with efficiency. The properties of determinants are as follows:

Property 1: The value of a determinant is not changed when its rows are changed into corresponding columns. Naturally when rows are changed into corresponding columns, then columns will be changed into corresponding rows. This operation is called taking the transpose of a determinant.

Example: For any determinant, we have

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{vmatrix}$$

Property 2: If two adjacent rows or columns of a determinant are interchanged, the value of the determinant so obtained is the negative of the value of the original determinant.

Example: For any determinant, we have

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -\begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Note If n interchanges of rows or columns take place, then the resulting determinant is $(-1)^n$ times the given determinant.

Property 3: The value of a determinant is zero if any two rows or columns are identical.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0 \quad (\text{here } R_1 = R_2)$$

Property 4: The value of a diagonal matrix, upper triangular matrix and lower triangular matrix is equal to the product of the principal diagonal elements.

Example: (i) Diagonal matrix

Let $\Delta = \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix}$. Expanding along R_1 (row 1),

we have, $\Delta = a_{11}(a_{22}a_{33} - 0) - 0 + 0$

$\Rightarrow \quad \Delta = a_{11}a_{22}a_{33}$ = Product of principal diagonal elements

(ii) Upper triangular matrix

Let $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{vmatrix}$. Expanding along C_1 (column 1),

we have, $\Delta = a_{11}(a_{22}a_{33} - a_{23}\cdot 0) - 0 + 0$

$\Rightarrow \Delta = a_{11}a_{22}a_{33}$ = Product of principal diagonal elements

(iii) Lower triangular matrix

Let $\Delta = \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$. Expanding along R_1, we have

$\Delta = a_{11}(a_{22}a_{33} - a_{32}\cdot 0) - 0 + 0$

$\Rightarrow \Delta = a_{11}a_{22}a_{33}$ = Product of principal diagonal elements

Property 5: A common factor of all elements of any row (or any column) may be taken outside the sign of the determinant. In other words, if all the elements of the same row (or the same column) are multiplied by a certain number, then the determinant becomes multiplied by that number.

Example: For any determinant, we have

$$\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Property 6: If every element of some column (or row) is the sum of two terms, then the given determinant is equal to the addition of two determinants—the first one with the first term of the sum and the second one with the second term of the sum. The remaining elements of both the determinants are the same as in the given determinant.

$$\begin{vmatrix} a_{11}+\alpha_1 & a_{12} & a_{13} \\ a_{21}+\alpha_2 & a_{22} & a_{23} \\ a_{31}+\alpha_3 & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} \alpha_1 & a_{12} & a_{13} \\ \alpha_2 & a_{22} & a_{23} \\ \alpha_3 & a_{32} & a_{33} \end{vmatrix}$$

Property 7: If the elements of a row (or column) of a determinant are added k times to the corresponding elements of another row (or column), then the value of the determinant thus obtained equals the value of the original determinant.

Example: Let $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ be any determinant.

Then $$\begin{vmatrix} a_{11}+ka_{12} & a_{12} & a_{13} \\ a_{21}+ka_{22} & a_{22} & a_{23} \\ a_{31}+ka_{32} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

[Applying $C_1 \to C_1 + kC_2$]

or $$\begin{vmatrix} a_{11}+ka_{12}+ma_{13} & a_{12} & a_{13} \\ a_{21}+ka_{22}+ma_{23} & a_{22} & a_{23} \\ a_{31}+ka_{32}+ma_{33} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

[Applying $C_1 \to C_1 + kC_2 + mC_3$]

Property 8: If each element of a row (or column) of a determinant is zero, then the value of the determinant is zero.

Property 9: If each element of a determinant is a function of x, such that

$$\Delta(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$$

and if at $x = \alpha$, $\Delta(x) = 0$, i.e. $\Delta(\alpha) = 0$,

then $x = \alpha$ is called the root of the determinant function and conversely $(x - \alpha)$ is the factor of the determinant.

Note If r rows (or columns) become identical when a is substituted for x, then $(x-a)^{r-1}$ is the factor of the given determinant.

☞ PRODUCT OF DETERMINANTS

Let $\Delta_1 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix}$ be any two determinants of third order, then

$$\Delta_1 \cdot \Delta_2 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix}$$

$$\Rightarrow \Delta_1 \cdot \Delta_2 = \begin{vmatrix} a_{11}b_{11}+a_{12}b_{21}+a_{13}b_{31} & a_{11}b_{12}+a_{12}b_{22}+a_{13}b_{32} & a_{11}b_{13}+a_{12}b_{23}+a_{13}b_{33} \\ a_{21}b_{11}+a_{22}b_{21}+a_{23}b_{31} & a_{21}b_{12}+a_{22}b_{22}+a_{23}b_{32} & a_{21}b_{13}+a_{22}b_{23}+a_{23}b_{33} \\ a_{31}b_{11}+a_{32}b_{21}+a_{33}b_{31} & a_{31}b_{12}+a_{32}b_{22}+a_{33}b_{32} & a_{31}b_{13}+a_{32}b_{23}+a_{33}b_{33} \end{vmatrix}$$

i.e. the (i, j)th element in $(\Delta_1 \cdot \Delta_2)$ is obtained by adding the products of the corresponding elements of the ith row of Δ_1 and the jth column of Δ_2.

Note Since we know that the value of the determinant does not change by interchanging the rows and columns, therefore, while writing down the product of two determinants of the same order, we can also follow the row by row multiplication rule, rather than only the row by column rule.

Also column by column and column by row methods of operation on product of determinants will yield the same result.

☞ DIFFERENTIATION OF A DETERMINANT

Differentiation of a determinant is equal to the sum of differentials distributed on each row (or column) one at a time.

Thus, if Δ is a determinant of order 3, such that

$$\Delta(x)=\begin{vmatrix} f(x) & g(x) & h(x) \\ p(x) & q(x) & r(x) \\ u(x) & v(x) & w(x) \end{vmatrix}$$

where $f, g, h, p, q, r, u, v, w$ are functions of x alone. Then

$$\frac{d\Delta(x)}{dx}=\begin{vmatrix} f'(x) & g'(x) & h'(x) \\ p(x) & q(x) & r(x) \\ u(x) & v(x) & w(x) \end{vmatrix}+\begin{vmatrix} f(x) & g(x) & h(x) \\ p'(x) & q'(x) & r'(x) \\ u(x) & v(x) & w(x) \end{vmatrix}+\begin{vmatrix} f(x) & g(x) & h(x) \\ p(x) & q(x) & r(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}.$$

Illustration 2. If a, b, c are in A.P. and

$$f(x)=\begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}, \text{ then find } f'(x).$$

Solution: $f(x)=\begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}$

Differentiating column by column,

$$f'(x)=\begin{vmatrix} 1 & x^2+1 & 1 \\ 1 & 2x^2-1 & 1 \\ 1 & 3x^2-2 & 1 \end{vmatrix}+\begin{vmatrix} x+a & 2x & 1 \\ x+b & 4x & 1 \\ x+c & 6x & 1 \end{vmatrix}+\begin{vmatrix} x+a & x^2+1 & 0 \\ x+b & 2x^2-1 & 0 \\ x+c & 3x^2-2 & 0 \end{vmatrix}$$

0 ($\because$ C_1 and C_3 are identical)

$$+2x\begin{vmatrix} x+a & 1 & 1 \\ x+b & 2 & 1 \\ c-b & 1 & 0 \end{vmatrix}(R_3 \to R_3-R_2)$$

$$=2x\begin{vmatrix} x+a & 1 & 1 \\ b-a & 1 & 0 \\ c-b & 1 & 0 \end{vmatrix}(R_2 \to R_2-R_1)$$

$$=2x(2b-a-c)$$

$$=2\times 0=0 \quad \left(\because a,b,c \text{ are in A.P.} \Rightarrow b=\frac{a+c}{2}\right)$$

☞ INTEGRATION OF A DETERMINANT

Let $\Delta(x)=\begin{vmatrix} a_{11}(x) & a_{12}(x) & a_{13}(x) \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

$$\therefore \quad \int\Delta(x)dx=\begin{vmatrix} \int a_{11}(x)dx & \int a_{12}(x)dx & \int a_{13}(x)dx \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Illustration 3. If $f(x)=\begin{vmatrix} x^3 & \sin x \\ 5 & 7 \end{vmatrix}$, then find the value of $\int_{-a}^{a} f(x)dx$.

Solution: $\int_{-a}^{a} f(x)dx=\begin{vmatrix} \int_{-a}^{a} x^3dx & \int_{-a}^{a}\sin x dx \\ 5 & 7 \end{vmatrix}=\begin{vmatrix} 0 & 0 \\ 5 & 7 \end{vmatrix}=0$

Note The above formula is applicable only if there is a variable only in one row or column, otherwise expand the determinant and then integrate.

☞ APPLICATIONS OF DETERMINANTS IN GEOMETRY

1. Area of a triangle: If $(x_1\, y_1)$, $(x_2\, y_2)$ and $(x_3 y_3)$ are the vertices of a triangle, then

$$\text{the area of the triangle} = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$=\frac{1}{2}[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)]$$

Illustration 4. Using determinants, find the area of ΔABC whose vertices A, B and C are the points (2, 3), (5, 4) and (–1, 6) respectively.

Solution: Area of $\Delta ABC=\frac{1}{2}\begin{vmatrix} 2 & 3 & 1 \\ 5 & 4 & 1 \\ -1 & 6 & 1 \end{vmatrix}$

$$=\frac{1}{2}[2(4-6)-5(3-6)-1(3-4)]$$

$$=\frac{1}{2}[2(-2)-5(-3)-1(-1)]$$

$$=\frac{1}{2}[-4+15+1]=6 \text{ sq. units.}$$

2. Condition of collinearity of three points: Let the three points be $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then these points will be collinear if

the area of $\Delta ABC = 0$

$$\Rightarrow \quad \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}=0$$

Illustration 5. Show that the points $A(1, -1)$ $B(2, 1)$ and $C(x_3, y_3)$, are collinear.

Solution: We know that three points are collinear if the area of the triangle formed by them is zero.

Now the area of ΔABC with vertices $A(1, -1)$, $B(2, 1)$ and $C(4, 5)$ is given by

$$\frac{1}{2}\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & 1 \\ 4 & 5 & 1 \end{vmatrix}$$

$$= \frac{1}{2}[1(1-5)+1(2-4)+1(10-4)]$$

$$= \frac{1}{2}[-4-2+6] = 0$$

Hence, the points $(1, -1)$, $(2, 1)$ and $(4, 5)$ are collinear.

3. Equation of a straight line passing through two points: Let the the two points be $A(x_1, y_1)$ and $B(x_2, y_2)$ and $P(x, y)$be a point on the line joining points A and B, then the equation of the line is given by

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Illustration 6. Using determinants, find the equation of the line joining the points (1, 2) and (3, 6).

Solution: Let $P(x, y)$ be any point on the line AB.

Then the area $(\Delta ABP) = \frac{1}{2}\begin{vmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ x & y & 1 \end{vmatrix}$

$$= \frac{1}{2}\begin{vmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \\ x-1 & y-2 & 0 \end{vmatrix} \quad \left[\text{Applying } \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix}\right]$$

$$= \frac{1}{2}[2(y-2)-4(x-1)]$$

$$= y-2-2x+2 = y-2x$$

As the points are collinear, the area of $\Delta ABP = 0$.

$\therefore\ y - 2x = 0$, which is the required equation of the line AB.

SOLVING SYSTEM OF LINEAR EQUATIONS

Let n simultaneous equations in n unknowns $x_1, x_2, x_3, \ldots, x_n$ be

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= d_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{1n}x_n &= d_2 \\ \vdots \quad \vdots \quad \vdots \quad \vdots \\ a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 + \cdots + a_{in}x_n &= d_i \\ \vdots \quad \vdots \quad \vdots \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n &= d_n \end{aligned}$$

The above system of equations is said to be a system of homogeneous equation if

$d_1 = d_2 = d_3 = \cdots = d_i = \cdots = d_n = 0$. But if at least one of d_1, $d_2, d_3 \ldots, d_i, \ldots, d_n$ is non-zero, the system of equations is said to form a non-homogeneous system.

SOME IMPORTANT TERMS DEFINED

Consistent system: If the system of equations has an infinite number of solutions, the system is said to be a *consistent system* or *indeterminant system* of equations.

Inconsistent system: If the system of equations has no solution, the system of equations is said to be an inconsistent system of equations.

CRAMMER'S RULE

Consider the following three linear simultaneous equations in x, y, z:

$$a_1x + b_1y + c_1z = d_1 \quad \text{(i)}$$

$$a_2x + b_2y + c_2z = d_2 \quad \text{(ii)}$$

$$a_3x + b_3y + c_3z = d_3 \quad \text{(iii)}$$

If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, $\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

then using the Crammer's rule of determinants,

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}, \text{ i.e. } x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$$

The above can be shown diagrammatically as in Figs. 5.2 and 5.3.

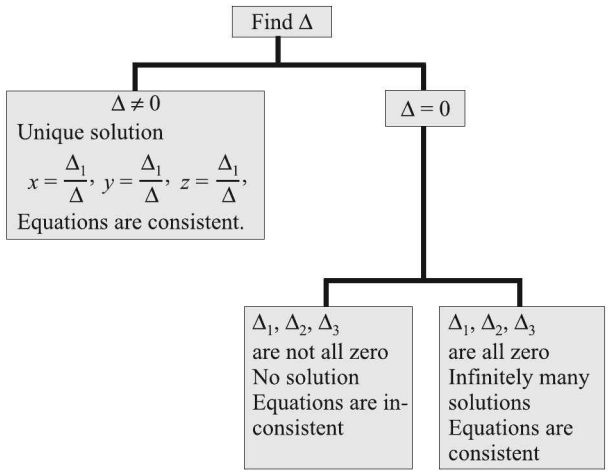


Fig. 5.2 Solution of a non-homogeneous system of linear equations.

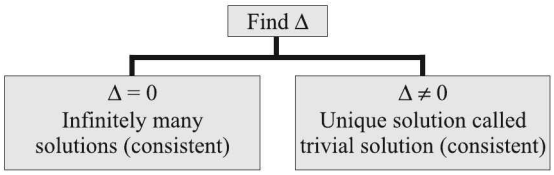


Fig. 5.3 Solution of a homogeneous system of linear equations.

Illustration 7. The system of equations $x + y + z = 2$, $2x + y - z = 3$ and $3x + 2y + kz = 4$ has a unique solution if

(a) $k = 0$ (b) $k \neq 0$

(c) $-1 < k < 1$ (d) $-2 < k < 2$

Solution: The given system will have a unique solution if

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & k \end{vmatrix} \neq 0$$

$\Rightarrow k \neq 0.$

SOLVED PROBLEMS

1. Let λ and α be real. The set of all values of λ for which the system of linear equations

$$\lambda x + (\sin\alpha)y + (\cos\alpha)z = 0$$
$$x + (\sin\alpha)y - (\cos\alpha)z = 0$$
$$-x + (\sin\alpha)y + (\cos\alpha)z = 0$$

has a non-trivial solution is

(a) $[0, \sqrt{2}]$ (b) $[-\sqrt{2}, 0]$

(c) $[-\sqrt{2}, \sqrt{2}]$ (d) none of these

Ans. (c)

Solution: Since the system has a non-trivial solution, therefore

$$\begin{vmatrix} \lambda & \sin\alpha & \cos\alpha \\ 1 & \cos\alpha & \sin\alpha \\ -1 & \sin\alpha & -\cos\alpha \end{vmatrix} = 0$$

$$\Rightarrow \lambda(-\cos^2\alpha - \sin^2\alpha) - (-\sin\alpha\cos\alpha - \sin\alpha\cos\alpha) - (\sin^2\alpha - \cos^2\alpha) = 0$$

$$\Rightarrow -\lambda + \sin 2\alpha + \cos 2\alpha = 0 \Rightarrow \lambda = \sin 2\alpha + \cos 2\alpha$$

$$\Rightarrow \lambda = \sqrt{2}\cos\left(2\alpha - \frac{\pi}{4}\right)$$

Since $-1 \le \cos\left(2\alpha - \frac{\pi}{4}\right) \le 1 \; \forall \alpha \in R$

$\therefore \; -\sqrt{2} \le \lambda \le \sqrt{2}$ i.e. $\lambda \in [-\sqrt{2}, \sqrt{2}]$

2. The value of the determinant $\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$ is

(a) $abc + pqr + xyz$ (b) $(a-x)(y-z)(r-p)$

(c) 0 (d) none of these

Ans. (c)

Solution: Applying $C_1 \to C_1 + C_2 + C_3$, we get the value of the determinant = 0.

3. Which of the following is not the root of the equation $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$?

(a) -9 (b) 2

(c) 3 (d) 7

Ans. (c)

Solution: $\Delta = \begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix}$

$$\Delta = \begin{vmatrix} x+9 & x+9 & x+9 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} (R_1 \to R_1 + R_2 + R_3)$$

$$= \begin{vmatrix} x+9 & 0 & 0 \\ 2 & x-2 & 0 \\ 7 & -1 & x-7 \end{vmatrix} (C_2 \to C_2 - C_1, C_3 \to C_3 - C_1)$$

$= (x + 9)(x - 2)(x - 7)$

$\therefore \quad \Delta = 0$

$\Rightarrow \quad (x + 9)(x - 2)(x - 7) = 0$

$\Rightarrow \quad x = -9, 2, 7.$

4. If $a + b + c = 0$, one root of $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$ is

(a) $x = 0$ (b) $x = 1$

(c) $x = 2$ (d) $x = a^2 + b^2 + c^2$

Ans. (a)

Solution: Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$(a+b+c-x)\begin{vmatrix} 1 & c & b \\ 1 & b-x & a \\ 1 & a & c-x \end{vmatrix} = 0$$

i.e.

$$x\begin{vmatrix} 1 & c & b \\ 1 & b-x & a \\ 1 & a & c-x \end{vmatrix} = 0$$

$\therefore \; x = 0$ is one of the roots of $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0.$

5. If $\Delta = \begin{vmatrix} a & c & c-a & a+c \\ c & b & b-c & b+c \\ a-b & b-c & 0 & a-c \\ x & y & z & 1+x+y \end{vmatrix} = 0$ and a, b, c are not in A.P., then

(a) a, c, b are in H.P. (b) a, c, b are in G.P.
(c) b, a, c in A.P. (d) a, b, c are in G.P.

Ans. (b)

Solution: Applying $C_3 \to C_3 - (C_2 - C_1)$ and $C_4 \to C_4 - (C_1 - C_2)$, we get

$$\Delta = \begin{vmatrix} a & c & 0 & 0 \\ c & b & 0 & 0 \\ a-b & b-c & a+c-2b & 0 \\ x & y & z+x-y & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a & c & 0 \\ c & b & 0 \\ a-b & b-c & a+c-2b \end{vmatrix} \quad \text{(Expanding along } C_4\text{)}$$

$$= (a+c-2b)(ab-c^2) \quad \text{(Expanding along } C_3\text{)}$$

$\therefore \quad \Delta = 0$

$\Rightarrow \quad a+c-2b = 0$ or $ab - c^2 = 0$.

$\Rightarrow$ a, c, b are in G.P. (given that a, b, c are in A.P.)

6. The value of $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$ is

(a) abc (b) 0
(c) $a + b + c$ (d) $4abc$

Ans. (d)

Solution: We have $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = \begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$

(Applying $R_1 \to R_1 - R_2 - R_3$)

$$= \frac{1}{c}\begin{vmatrix} 0 & -2c & -2b \\ 0 & c(c+a-b) & b(c-a-b) \\ c & c & a+b \end{vmatrix}$$

(Applying $R_2 \to cR_2 - bR_3$)

$$= \frac{1}{c} \cdot c(2bc)\left[-(c-a-b)+(c+a-b)\right]$$

$= (2bc)(2a) = 4abc$ (Expanding along C_1)

7. $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$ is equal to

(a) 1 (b) 0
(c) $(1-a)(1-b)(1-c)$ (d) $(a+b+c)$

Ans. (b)

Solution: Let $\Delta = \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$

$$= \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & a+b+c \\ 1 & c & a+b+c \end{vmatrix} (C_3 \to C_2 + C_3)$$

$$= (a+b+c)\begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} = 0 \quad [\because \text{ two columns are identical}]$$

8. The value of $\begin{vmatrix} yz & zx & xy \\ p & 2q & 3r \\ 1 & 1 & 1 \end{vmatrix}$, where x, y, z are, respectively, the pth, $(2q)$th and $(3r)$th terms of an H.P., is

(a) -1 (b) 0
(c) 1 (d) none of these

Ans. (b)

Solution: Let a be the first term and d be the common difference of corresponding A.P.

Then, $\Delta = xyz\begin{vmatrix} 1/x & 1/y & 1/z \\ p & 2q & 3r \\ 1 & 1 & 1 \end{vmatrix}$

$$= xyz\begin{vmatrix} a+(p-1)d & a+(2q-1)d & a+(3r-1)d \\ p & 2q & 3r \\ 1 & 1 & 1 \end{vmatrix}$$

Applying $R_1 \to R_1 - aR_3, R_2 \to R_2 - R_3$ and then taking d common from R_1, we get

$$\Delta = xyzd\begin{vmatrix} (p-1) & (2q-1) & (3r-1) \\ (p-1) & (2q-1) & (3r-1) \\ 1 & 1 & 1 \end{vmatrix} = 0$$

9. The value of the determinant $\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix}$ is

(a) -2 (b) $x^2 + 2$
(c) 2 (d) none of these

Ans. (a)

Solution: We have

$$\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix} = \begin{vmatrix} x+1 & x+2 & 2 \\ x+3 & x+5 & 3 \\ x+7 & x+10 & 4 \end{vmatrix}$$

(Applying $C_3 \to C_3 - C_2$)

$$= \begin{vmatrix} x+1 & 1 & 2 \\ x+3 & 2 & 3 \\ x+7 & 3 & 4 \end{vmatrix} \quad (\text{Applying } C_2 \to C_2 - C_1)$$

$$= \begin{vmatrix} x+1 & 1 & 2 \\ 2 & 1 & 1 \\ 4 & 1 & 1 \end{vmatrix}$$

(Applying $R_2 \to R_2 - R_1, R_3 \to R_3 - R_2$)

$$= \begin{vmatrix} x+1 & 1 & 1 \\ 2 & 1 & 0 \\ 2 & 0 & 0 \end{vmatrix} \quad (\text{Applying, } R_3 \to R_3 - R_2)$$

$= 2(0-1) = -2$ (Expanding along R_3)

10. $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix}$ is equal to

(a) $abc(a+b+c)^2$ (b) $2abc(a+b+c)^2$

(c) $2abc(a+b+c)^3$ (d) $2abc(a+b+c)$

Ans. (c)

Solution: Let $\Delta = \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix}$

(Applying $C_1 \to C_1 - C_3$ and $C_2 \to C_2 - C_3$)

$$= \begin{vmatrix} (b+c)^2 - a^2 & 0 & a^2 \\ 0 & (c+a)^2 - b^2 & b^2 \\ c^2 - (a+b)^2 & c^2 - (a+b)^2 & (a+b)^2 \end{vmatrix}$$

$$= \begin{vmatrix} (b+c+a)(b+c-a) & 0 & a^2 \\ 0 & (c+a+b)(c+a-b) & b^2 \\ (c+a+b)(c-a-b) & (c+a+b)(c-a-b) & (a+b)^2 \end{vmatrix}$$

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ c-a-b & c-a-b & (a+b)^2 \end{vmatrix}$$

[Applying $R_3 \to R_3 - (R_1 + R_2)$]

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ -2b & -2a & 2ab \end{vmatrix}$$

$$= \frac{(a+b+c)^2}{ab} \begin{vmatrix} ab+ac & a^2 & a^2 \\ b^2 & bc+ba & b^2 \\ 0 & 0 & 2ab \end{vmatrix}$$

$$= \frac{(a+b+c)^2}{ab} \cdot ab \cdot 2ab \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ 0 & 0 & 1 \end{vmatrix}$$

$$= 2ab(a+b+c)^2 \begin{vmatrix} b+c & a \\ b & c+a \end{vmatrix}$$

$$= 2ab(a+b+c)^2 [(b+c)(c+a) - ab]$$

$$= 2abc(a+b+c)^2 (a+b+c)$$

$$= 2abc(a+b+c)^3$$

11. If $g(x) = \begin{vmatrix} a^{-x} & e^{x\log_e a} & x^2 \\ a^{-3x} & e^{3x\log_e a} & x^4 \\ a^{-5x} & e^{5x\log_e a} & 1 \end{vmatrix}$, then

(a) $g(x) + g(-x) = 0$ (b) $g(x) - g(-x) = 0$

(c) $g(x) \times g(-x) = 0$ (d) none of these

Ans. (a)

Solution: $g(x) = \begin{vmatrix} a^{-x} & e^{\log_e a^x} & x^2 \\ a^{-3x} & e^{\log_e a^{3x}} & x^4 \\ a^{-5x} & e^{\log_e a^{5x}} & 1 \end{vmatrix}$

$$= \begin{vmatrix} a^{-x} & a^x & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix} \quad [\because e^{\log a^x} = a^x]$$

$$\Rightarrow g(-x) = \begin{vmatrix} a^x & a^{-x} & x^2 \\ a^{3x} & a^{-3x} & x^4 \\ a^{5x} & a^{-5x} & 1 \end{vmatrix} = -\begin{vmatrix} a^{-x} & a^x & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix}$$

(Interchanging 1st and 2nd columns)

$= -g(x)$

$\Rightarrow g(x) + g(-x) = 0$

12. If $f(x) = \begin{vmatrix} \cos^2 x & \cos x \cdot \sin x & -\sin x \\ \cos x \cdot \sin x & \sin^2 x & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$, then for all x

(a) $f(x) = 0$ (b) $f(x) = 1$

(c) $f(x) = 2$ (d) none of these

Ans. (b)

Solution: We have $f(x) = \begin{vmatrix} \cos^2 x & \cos x \cdot \sin x & -\sin x \\ \cos x \cdot \sin x & \sin^2 x & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$

$$= \begin{vmatrix} 1 & 0 & -\sin x \\ 0 & 1 & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$$

(Applying $C_1 \to C_1 - \sin x \cdot C_3$ and $C_2 \to C_2 + \cos x \cdot C_3$)

$$= \begin{vmatrix} 1 & 0 & -\sin x \\ 0 & 1 & \cos x \\ 0 & -\cos x & \sin^2 x \end{vmatrix}$$

(Applying $R_3 \to R_3 - \sin x \cdot R_1$)

$= \sin^2 x + \cos^2 x = 1$ for all x (Expanding along C_1)

13. The value of the determinant $\begin{vmatrix} b+c & a-b & a \\ c+a & b-c & b \\ a+b & c-a & c \end{vmatrix}$ is

(a) $a^3 + b^3 + c^3 - 3abc$ (b) $-(a^3 + b^3 + c^3 - 3abc)$

(c) $-(a^3 + b^3 + c^3)$ (d) none of these

Ans. (b)

Solution: $\begin{vmatrix} b+c & a-b & a \\ c+a & b-c & b \\ a+b & c-a & c \end{vmatrix} = \begin{vmatrix} a+b+c & -b & a \\ a+b+c & -c & b \\ a+b+c & -a & c \end{vmatrix}$

[$C_1 \to C_1 + C_3$ and $C_2 \to C_2 - C_3$]

$$= -(a+b+c)\begin{vmatrix} 1 & b & a \\ 1 & c & b \\ 1 & a & c \end{vmatrix}$$

$$= -(a+b+c)\begin{vmatrix} 1 & b & a \\ 0 & c-b & b-a \\ 0 & a-b & c-a \end{vmatrix}$$

[$R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$]

$= -(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= -(a^3 + b^3 + c^3 - 3abc)$

14. If $a+b+c \neq 0$ and $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$, then the total number of different values of x is equal to

(a) 1 (b) 2

(c) 3 (d) none of these

Ans. (c)

Solution: Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\Delta = \begin{vmatrix} a+b+c-x & c & b \\ a+b+c-x & b-x & a \\ a+b+c-x & a & c-x \end{vmatrix}$$

$$= (a+b+c-x)\begin{vmatrix} 1 & c & b \\ 1 & b-x & a \\ 1 & a & c-x \end{vmatrix}$$

Applying $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$, we get

$$\Delta = (a+b+c-x)\begin{vmatrix} 0 & c-b+x & b-a \\ 0 & b-a-x & a-c+x \\ 1 & a & c-x \end{vmatrix}$$

$= (a+b+c-x)\,[(x+c-b)(x+a-c)$
$+ (x+a-b)(b-a)]$

$= (a+b+c-x)\,[x^2 + x(a-b) + (c-b)(a-c)$
$+ x(b-a) - (a-b)^2$

$= (a+b+c-x)(x^2 + ac - c^2 - ab + bc - a^2 - b^2 - 2ab)$

$= (a+b+c-x)(x^2 - a^2 - b^2 - c^2 + ab + bc + ca)$

$$= (a+b+c-x)\left\{x^2 - \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]\right\}$$

As $\Delta = 0$

$$\Rightarrow \quad x = a+b+c, \pm\sqrt{\frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2)}$$

15. If A, B and C are the angles of a triangle, and

$$\Delta = \begin{vmatrix} e^{-2iA} & e^{iC} & e^{iB} \\ e^{iC} & e^{-2iB} & e^{iA} \\ e^{iB} & e^{iA} & e^{-2iC} \end{vmatrix}$$

find $|\Delta|$.

(a) 2 (b) 3

(c) -1 (d) -4

Ans. (d)

Solution: Let $\alpha = e^{iA}$, $\beta = e^{iB}$ and $\gamma = e^{iC}$, then $\alpha\beta\gamma = e^{i(A+B+C)}$ $= e^{i\pi} = \cos\pi + i\sin\pi = -1$ and

$$\Delta = \begin{vmatrix} 1/\alpha^2 & \gamma & \beta \\ \gamma & 1/\beta^2 & \alpha \\ \beta & \alpha & 1/\gamma^2 \end{vmatrix}$$

$$= \frac{1}{\alpha^2\beta^2\gamma^2}\begin{vmatrix} 1 & \alpha^2\gamma & \alpha^2\beta \\ \beta^2\gamma & 1 & \beta^2\alpha \\ \gamma^2\beta & \gamma^2\alpha & 1 \end{vmatrix} = \begin{vmatrix} 1 & -\alpha/\beta & -\alpha/\gamma \\ -\beta/\alpha & 1 & -\beta/\gamma \\ -\gamma/\alpha & -\gamma/\beta & 1 \end{vmatrix}$$

$$= \frac{1}{\alpha\beta\gamma}\begin{vmatrix} \alpha & -\alpha & -\alpha \\ -\beta & \beta & -\beta \\ -\gamma & -\gamma & \gamma \end{vmatrix}$$

[Multiplying C_1, C_2, C_3 by α, β and γ respectively]

$$= \frac{\alpha\beta\gamma}{\alpha\beta\gamma}\begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & -1 & -1 \\ 0 & 1 & -1 \\ -2 & -1 & 1 \end{vmatrix} = -2(1+1) = -4$$

(Applying $C_1 \to C_1 + C_2$)

16. $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$ is equal to

(a) abc (b) $\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

(c) $abc\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$ (d) $abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

Ans. (d)

Solution: Let $\Delta = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$

$$= abc\begin{vmatrix} \frac{1}{a}+1 & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$$= abc\begin{vmatrix} 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$(R_1 \to R_1 + R_2 + R_3)$

$$= abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$$= abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$$

$(C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1)$

$$= abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$$

17. If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$, then the value of $\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}$ is equal to

(a) -1 (b) 1

(c) -2 (d) 2

Ans. (d)

Solution: Given $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$

Applying $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$ reduces the determinant to

$$\begin{vmatrix} p-a & b-q & 0 \\ 0 & q-b & c-r \\ a & b & r \end{vmatrix} = 0$$

$\Rightarrow (p-a)(q-b)r + a(b-q)(c-r) - b(p-a)(c-r) = 0$

$\Rightarrow$ Dividing throughout by $(p-a)(q-b)(r-c)$, we get

$$\frac{r}{r-c}+\frac{a}{p-a}+\frac{b}{q-b} = 0$$

$$\Rightarrow \quad \frac{r}{r-c}+1+\frac{a}{p-a}+1+\frac{b}{q-b} = 2$$

$$\Rightarrow \quad \frac{r}{r-c}+\frac{p}{p-a}+\frac{q}{q-b} = 2$$

18. If $U_n = \begin{vmatrix} 1 & k & k \\ 2n & k^2+k+1 & k^2+k \\ 2n-1 & k^2 & k^2+k+1 \end{vmatrix}$ and $\sum_{n=1}^{k} U_n = 72$

then $k =$

(a) 8 (b) 9

(c) 6 (d) none of these

Ans. (a)

Solution: $\sum_{n=1}^{k} U_n = \begin{vmatrix} \sum_{n=1}^{k} 1 & k & k \\ 2\sum_{n=1}^{k} n & k^2+k+1 & k^2+k \\ 2\sum_{n=1}^{k} n - \sum_{n=1}^{k} 1 & k^2 & k^2+k+1 \end{vmatrix}$

$$= \begin{vmatrix} k & k & k \\ k(k+1) & k^2+k+1 & k^2+k \\ k^2 & k^2 & k^2+k+1 \end{vmatrix}$$

$$= \begin{vmatrix} k & 0 & k \\ k^2+1 & 1 & k^2+k \\ k^2 & 0 & k^2+k+1 \end{vmatrix}$$

[Applying $C_2 \to C_2 - C_1$]

$$= k(k^2+k+1) - k^3 = k(k+1) = 72$$

$$\Rightarrow \quad k = 8.$$

19. If $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$ and a, b, c are all distinct, then $abc(ab+bc+ca)$ is equal to

(a) $a+b+c$ (b) abc
(c) 0 (d) none of these

Ans. (a)

Solution: Given $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$

$$\Rightarrow \quad \begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} + \begin{vmatrix} a & a^3 & -1 \\ b & b^3 & -1 \\ c & c^3 & -1 \end{vmatrix} = 0$$

$$\Rightarrow \quad abc \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} - 1 \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \quad abc(a-b)(b-c)(c-a)(ab+bc+ca)$$
$$-(a-b)(b-c)(c-a)(a+b+c) = 0$$

$$\Rightarrow \quad abc(ab+bc+ca) - (a+b+c) = 0$$

$$\Rightarrow \quad abc(ab+bc+ca) = a+b+c$$

EXERCISE SET 1

1. If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$, then $f(500)$ is equal to

(a) 0 (b) 1
(c) 500 (d) -500

2. Let $f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$, then $f\left(\frac{\pi}{6}\right) = ?$

(a) 1 (b) 0
(c) -1 (d) 2

3. Let α_1, α_2 and β_1, β_2 be the roots of $ax^2+bx+c=0$ and $px^2+qx+r=0$, respectively. If the system of equations $a_1y + a_2z = 0$ and $\beta_1y + \beta_2z = 0$ has a non-trivial solution, then

(a) $\frac{b^2}{q^2} = \frac{ac}{pr}$ (b) $\frac{c^2}{r^2} = \frac{ab}{pq}$
(c) $\frac{a^2}{p^2} = \frac{bc}{qr}$ (d) none of these

4. A root of the equation $\begin{vmatrix} 3-x & -6 & 3 \\ -6 & 3-x & 3 \\ 3 & 3 & -6-x \end{vmatrix} = 0$ is

(a) 6 (b) 3
(c) 0 (d) none of these

5. If $l_1^2 + m_1^2 + n_1^2 = 1$, etc., and $l_1l_2 + m_1m_2 + n_1n_2 = 0$, etc., and $\Delta = \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}$, then

(a) $|\Delta| = 3$ (b) $|\Delta| = 2$
(c) $|\Delta| = 1$ (d) $\Delta = 0$

6. The value of $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$, ω being a cube root of unity, is

(a) 0 (b) 1
(c) ω^2 (d) ω

7. The value of the determinant $\begin{vmatrix} 1 & \sin 3\theta & \sin^3\theta \\ 2\cos\theta & \sin 6\theta & \sin^3 2\theta \\ 4\cos^2\theta - 1 & \sin 9\theta & \sin^3 3\theta \end{vmatrix}$ is

(a) -1 (b) 0
(c) 1 (d) 2

8. $\begin{vmatrix} b^2-ab & b-c & bc-ac \\ ab-a^2 & a-b & b^2-ab \\ bc-ac & c-a & ab-a^2 \end{vmatrix} = ?$

(a) abc (b) $a+b+c$
(c) 0 (d) $a^2+b^2+c^2$

9. If $a \neq b \neq c$, one value of x which satisfies the equation $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$ is given by

(a) $x=a$ (b) $x=b$
(c) $x=c$ (d) $x=0$

10. If a, b, c are non-zero real numbers, then $\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix}$ vanishes for

(a) $\frac{1}{a}-\frac{1}{b}-\frac{1}{c}=0$ (b) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$
(c) $\frac{1}{b}+\frac{1}{c}-\frac{1}{a}=0$ (d) $\frac{1}{b}-\frac{1}{c}-\frac{1}{a}=0$

11. If $A+B+C=\pi$, then the value of $\begin{vmatrix} \sin(A+B+C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$ is equal to

(a) 0 (b) 1
(c) $2 \sin B \tan A \cos C$ (d) none of these

12. $\begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix} = ?$

(a) $(x+p)(x+q)(x-p-q)$
(b) $(x-p)(x-q)(x+p+q)$
(c) $(x-p)(x-q)(x-p-q)$
(d) $(x+p)(x+q)(x+p+q)$

13. The determinant $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix} = 0$ if

(a) a, b, c are in A.P.
(b) a, b, c are in G.P. or $(x-\alpha)$ is a factor of $ax^2+2bx+c=0$
(c) a, b, c are in H.P.
(d) α is a root of the equation $ax^2+2bx+c=0$

14. $f(x)=\begin{vmatrix} 2\cos x & 1 & 0 \\ 1 & 2\cos x & 1 \\ 0 & 1 & 2\cos x \end{vmatrix}$, then $f\left(\frac{\pi}{3}\right)$ is equal to

(a) -5 (b) -4
(c) -3 (d) -1

15. For a +ve value of x, y, z, the numerical value of the determinant $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ is equal to

(a) 1 (b) 0
(c) $\log x+\log y+\log z$ (d) -1

16. The value of the determinant $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ is

(a) $kabc(a^2+b^2+c^2)$
(b) $k(a+b)(b+c)(c+a)$
(c) $k(a-b)(b-c)(c-a)$
(d) $k(a+b-c)(b+c-a)(c+a-b)$

17. If $D_k = \begin{vmatrix} 1 & n & n \\ 2k & n^2+n+1 & n^2+n \\ 2k-1 & n^2 & n^2+n+1 \end{vmatrix}$ and $\sum_{k=1}^{n} D_k = 56$, then n equals

(a) 4 (b) 6
(c) 7 (d) none of these

18. $\begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}^2$ is equal to

(a) $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

(b) $\begin{vmatrix} b^2+c^2 & ab & ac \\ ba & c^2+a^2 & bc \\ ca & bc & a^2+b^2 \end{vmatrix}$

(c) $\begin{vmatrix} a+b & 0 & c \\ b+c & b & 0 \\ c+a & a & b \end{vmatrix}$

(d) $\begin{vmatrix} b+c & ab & ac \\ c+a & ca & cb \\ a+b & ba & bc \end{vmatrix}$

19. If $a+b+c=0$, then $\Delta = \begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$ when

(a) $x=-1$ (b) $x=\pm\sqrt{\frac{3}{2}(a^2+b^2+c^2)}$
(c) $x=abc$ (d) none of these

20. The number of distinct real solutions of

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0 \text{ in the interval } -\frac{\pi}{4} \le x \le \frac{\pi}{4} \text{ is}$$

(a) 0 (b) 2
(c) 1 (d) 3

21. If $\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$, then $x = ?$

(a) 1, 9 (b) –1, 9
(c) –1, –9 (d) 1, –9

22. The determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$ if

(a) x, y, z are in A.P. (b) x, y, z are in G.P.
(c) x, y, z are in H.P. (d) xy, yz, zx are in A.P.

23. The value of $\Delta = \begin{vmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix}$ is equal to zero, where m is

(a) 6 (b) 4
(c) 5 (d) none of these

24. Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix}$, then the maximum value of $f(x)$ is

(a) 0 (b) 1
(c) 6 (d) –1

25. If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ 1 & 1 & 1 \\ \beta & \gamma & \alpha \end{vmatrix}$ is

(a) $p^2 + q - 2r$ (b) $p^2 - 3q$
(c) $p^2 + q + 2r$ (d) $p^2 + 3q$

26. The value of the determinant $\begin{vmatrix} {}^5C_0 & {}^5C_3 & 1 \\ {}^5C_1 & {}^5C_4 & 1 \\ {}^5C_2 & {}^5C_5 & 1 \end{vmatrix}$ is

(a) 0 (b) $-(6!)$
(c) 80 (d) none of these

27. The value of the determinant $\begin{vmatrix} a+pd & a+qd & a+rd \\ p & q & r \\ d & d & d \end{vmatrix}$ is equal to

(a) 0 (b) –1
(c) 1 (d) $p + q + r$

28. The value of $\Delta = \begin{vmatrix} {}^5C_0 & {}^5C_3 & 14 \\ {}^5C_1 & {}^5C_4 & 1 \\ {}^5C_2 & {}^5C_5 & 1 \end{vmatrix}$ is

(a) 0 (b) -576
(c) 80 (d) none of these

29. If $f(x)$, $g(x)$ and $h(x)$ are three polynomials of degree 2 and $\Delta(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$ then $\Delta(x)$ is a polynomial of degree

(a) 2 (b) 3
(c) 0 (d) at most 3

30. The value of $\begin{vmatrix} 2 & \omega & \omega^2 \\ 1 & 1+\omega^2 & \omega \\ \omega^2 & 1 & 1+\omega \end{vmatrix}$, where ω is the complex cube root of unity, is

(a) 1 (b) ω
(c) ω^2 (d) none of these

31. Let $f(x) = \begin{vmatrix} \cos x & -x & 1 \\ 2\sin x & -x^2 & 2x \\ \tan x & -x & 1 \end{vmatrix}$, find $\lim_{x \to 0} \frac{f'(x)}{x}$.

(a) 3 (b) 4
(c) 2 (d) 3

32. If a, b, c are in A.P., then the determinant $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$ in its simplified form is

(a) $x^3 + 3ax + 7c$ (b) 0
(c) 15 (d) $10x^2 + 5x + 2c$

33. If $\Delta(x) = \begin{vmatrix} x & 1+x^2 & x^3 \\ \log(1+x^2) & e^x & \sin x \\ \cos x & \tan x & \sin^2 x \end{vmatrix}$, then

(a) $\Delta(x)$ is divisible by x (b) $\Delta(x) = 0$
(c) $\Delta'(x) = 0$ (d) none of these

34. If $\Delta(n)=\begin{vmatrix} x^n & \sin x & \cos x \\ n! & \sin\frac{n\pi}{2} & \cos\frac{n\pi}{2} \\ \alpha & \alpha^2 & \alpha^3 \end{vmatrix}$, then the value of $\frac{d^n}{dx^n}[\Delta(x)]$ at $x=0$ is

(a) -1 (b) 0
(c) 1 (d) 2

35. If $\begin{vmatrix} 3x^2-2x+3 & x^2-5 & 3x-2 \\ x & 2-3x & x+1 \\ x^2+2x & 7+2x^2 & 3x-x^2+1 \end{vmatrix} = ax^5+bx^4+cx^3+dx^3+ex+f$, then f is equal to

(a) -12 (b) -15
(c) 15 (d) 12

36. A root of the equation $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix}=0$ is

(a) a (b) b
(c) 0 (d) 1

37. If $2s=a+b+c$, then the value of $\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix}$ is

(a) $\sqrt{s(s-a)(s-b)(s-c)}$
(b) $2s^2(s-a)(s-b)(s-c)$
(c) $2s^3(s-a)(s-b)(s-c)$
(d) none of these

38. If α, β, γ are the roots of $px^3+qx^2+r=0$, then the value of the determinant $\begin{vmatrix} \alpha\beta & \beta\gamma & \gamma\alpha \\ \beta\gamma & \gamma\alpha & \alpha\beta \\ \gamma\alpha & \alpha\beta & \beta\gamma \end{vmatrix}$ is

(a) p (b) q
(c) 0 (d) r

39. If $f(x)=\begin{vmatrix} 1 & 2x & 3x^2 \\ 3 & a & 27 \\ 1 & 3 & 9 \end{vmatrix}$ and $\int_0^3 f(x)dx=0$, then a is equal to

(a) 3 (b) 6
(c) 9 (d) any real number

40. Let $\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = ax^5+bx^4+cx^3+dx^2+\lambda x+\mu$ be an identity in x, where a, b, c, d, λ, μ are independent of x. Then the value of λ is

(a) 3 (b) 2
(c) 4 (d) none of these

41. If $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = \lambda a^2b^2c^2$, then the value of λ is

(a) 1 (b) 2
(c) 4 (d) 3

42. The value of the determinant $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+a & 1 \\ 1 & 1 & 1+a \end{vmatrix}$ is

(a) $a^3\left(1-\frac{2}{a}\right)$ (b) $a^3\left(1+\frac{3}{a}\right)$
(c) $a^3\left(1-\frac{3}{a}\right)$ (d) $a^3\left(1+\frac{2}{a}\right)$

43. If $\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}=64$, then $a=?$

(a) 2 (b) 3
(c) 4 (d) 6

44. For positive numbers x, y, z, the determinant $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_z x & 2 & \log_y z \\ \log_z x & \log_z y & 4 \end{vmatrix}$ equals

(a) 0 (b) 3
(c) $\log x \log y \log z$ (d) none of these

45. If a, b, c are the sides of a triangle and $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}=0$, then

(a) ΔABC is an equilateral triangle
(b) ΔABC is a right-angled isosceles triangle
(c) ΔABC is an isosceles triangle
(d) none of these

46. If $D_r=\begin{vmatrix} 2^{r-1} & 2\cdot 3^{r-1} & 4\cdot 5^{r-1} \\ \alpha & \beta & \gamma \\ 2^{n-1} & 3^{n-1} & 5^{n-1} \end{vmatrix}$, then the value of $\sum_{r=1}^{n} D_r$ is

(a) 0 (b) $\alpha\beta\gamma$
(c) $\alpha+\beta+\gamma$ (d) $\alpha\cdot 2^n+\beta\cdot 3^n+\gamma\cdot 4^n$

47. If a, b, c are non-zero distinct numbers, then the system of equations $ax + by + cz = 0$, $cx + ay + bz = 0$ and $bx + cy + ax = 0$ has a non-trivial solution provided
(a) $a+b+c=0$ (b) $a+b+c\neq 0$
(c) $a+b+c=1$ (d) $a+b+c\ -1$

48. If $f(x)=\begin{vmatrix} x^3 & \cos^2 x & 2^{x^4} \\ \tan^5 x & 1 & \sec 2x \\ \sin^3 x & x^4 & 5 \end{vmatrix}$, then $\int_{-\pi/2}^{\pi/2} f(x)dx = ?$
(a) 2 (b) -2
(c) 0 (d) none of these

49. If $f(x)=\begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin(2x) & 0 & \sin(2x^2) \end{vmatrix}$,
find $f'(0)$.
(a) 2 (b) 4
(c) 7 (d) 1

50. The equations $ax+y=4$, $4x+ay=8$ and $2x+y=2a$ are consistent for
(a) no value of a (b) one value of a
(c) two values of a (d) three values of a

51. If $f(x)=\begin{vmatrix} \sec^2 x & 1 & 1 \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cot^2 x \end{vmatrix}$, then $f(x) = ?$
(a) $\cos^4 x$ (b) $\sin^4 x$
(c) 0 (d) $\sin^2 x \cos^2 x$

52. If $S_r=\begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$, then the value of $\sum_{r=1}^{n} S_r$ is independent of
(a) x only (b) y only
(c) x, y, z, n (d) n only

53. If the system of equations $ax + y + z = 0$, $x + by + z = 0$ and $x + y + cz = 0$ $(a, b, c \neq 1)$ has a non-trivial solution, then the value of $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$ is
(a) -1 (b) 0
(c) 1 (d) none of these

54. The system of equations $3x+ky=2, kx+ky=4$ is inconsistent for
(a) no value of k (b) one value of k
(c) two values of k (d) none of these

55. If x, y, z are in A.P. , then the value of the det A, where
$A=\begin{bmatrix} 4 & 5 & 6 & x \\ 5 & 6 & 7 & y \\ 6 & 7 & 8 & z \\ x & y & z & 0 \end{bmatrix}$ is
(a) 0 (b) 1
(c) 2 (d) none of these

56. If $\sin 2x + 1 = 0$, then $\begin{vmatrix} 0 & \cos x & -\sin x \\ \sin x & 0 & \cos x \\ \cos x & \sin x & 0 \end{vmatrix}^2$ equals
(a) 1 (b) 2
(c) 3 (d) none of these

57. Let λ and α be real. Let S denote the set of all values of λ for which the system of linear equations
$$\lambda x+(\sin\alpha)y+(\cos\alpha)z=0$$
$$x+(\cos\alpha)y+(\sin\alpha)z=0$$
$$-x+(\sin\alpha)y-(\cos\alpha)z=0$$
has a non-trivial solution, then S contains
(a) $(-1, 1)$ (b) $[-\sqrt{2}, \sqrt{2}]$
(c) $[1, \sqrt{2}]$ (d) $(-2, 2)$

58. If the system of equations $2x + ay + 6z = 8$, $x + 2y + bz = 5$ and $x + y + 3z = 4$ has a unique solution, then
(a) $a\neq 2, b\neq 3$ (b) $a\neq 3, b\neq 2$
(c) $a\neq 2, b=3$ (d) $a=2, b\neq 2$

59. The value of $\begin{bmatrix} x & x^2-yz & 1 \\ y & y^2-zx & 1 \\ z & z^2-xy & 1 \end{bmatrix}$ is
(a) 1 (b) -1
(c) 0 (d) $-xyz$

60. The repeated factor of the determinant
$\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$ is
(a) $x-y$ (b) $y-z$
(c) $z-x$ (d) none of these

61. The system of equations $x\sin\theta-\sqrt{3}y=0$ and $x\cos\theta + y = 0$ has an infinite number of solutions, for θ equal to
(a) $\pi/3$ (b) $2\pi/3$
(c) $5\pi/3$ (d) $4\pi/3$

62. If $p+q+r=0=a+b+c$, then the value of the determinant $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$ is

(a) 0 (b) $pa+qb+rc$

(c) 1 (d) none of these

63. If $\alpha = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$ and $= \beta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$, then

(a) $\alpha \neq \beta$ (b) $\alpha = \beta$

(c) $\alpha = 2\beta$ (d) $\alpha = -\beta$

64. One factor of $\begin{vmatrix} a^2+x & ab & ac \\ ab & b^2+x & cb \\ ca & cb & c^2+x \end{vmatrix}$ is

(a) $1/x$

(b) $(a^2+x)(b^2+x)(c^2+x)$

(c) x^2

(d) $a^2+b^2+c^2+x$

EXERCISE SET 2

1. $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = ?$

(a) abc (b) Σa

(c) 0 (d) $(\Sigma a)^2$

2. The determinant

$$\Delta = \begin{bmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) & \cos 2\beta \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{bmatrix}$$

is independent of

(a) α (b) β

(c) α and β (d) neither α nor β

3. The maximum and minimum values of (3×3) determinant whose elements belong to $\{0, 1\}$ are

(a) $1, -1$ (b) $2, -2$

(c) $4, -4$ (d) none of these

4. Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 3\cos 3x \end{vmatrix}$, then $f'\left(\frac{\pi}{2}\right) = ?$

(a) 8 (b) 6

(c) 4 (d) 2

5. The solution of equations $x+2y+3z=1$, $2x+y+3z=2$ and $5x+5y+9z=4$ is

(a) unique (b) infinity

(c) inconsistent (d) none of these

6. The value of $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ (x, y, z being +ve) is

(a) $\log_y x$ (b) $\log_z y$

(c) $\log_x z$ (d) 0

7. If $p+q+r=0$, then $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$ is

(a) 0 (b) pqr

(c) abc (d) $pqr\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$

8. The system of equations $ax+4y+z=0, bx+3y+z=0$, and $cx+2y+z=0$ has a non-trivial solution if a, b, c are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

9. $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix} = ?$

(a) abc (b) $a^2b^2c^2$

(c) 0 (d) none of these

10. The value of the determinant $\Delta = \begin{vmatrix} 1! & 2! & 3! \\ 2! & 3! & 4! \\ 3! & 4! & 5! \end{vmatrix}$ is

(a) 2! (b) 3!

(c) 4! (d) 5!

11. The value of k for which the system of equations $3x+ky-2z=0$, $x+ky+3z=0$ and $2x+3y-4z=0$ has a non-trivial solution is

(a) 15 (b) 31/2

(c) 16 (d) 33/2

12. The value of the determinant (without expansion)

$$\begin{vmatrix} b^2-ab & b-c & bc-ac \\ ab-a^2 & a-b & b^2-ab \\ bc-ac & c-a & ab-a^2 \end{vmatrix} = ?$$

(a) abc (b) $a+b+c$

(c) 0 (d) $a^2+b^2+c^2$

13. If $a^2+b^2+c^2=0$ and $\begin{vmatrix} b^2+c^2 & ab & ac \\ ab & c^2+a^2 & bc \\ ac & bc & a^2+b^2 \end{vmatrix} = ka^2b^2c^2$, then the value of k is
(a) 1 (b) 2
(c) 3 (d) 4

14. The system of equations $2x - y + z = 0$, $x - 2y + z = 0$ and $\lambda x - y + 2z = 0$ has infinite number of non-trivial solutions for
(a) $\lambda = 1$ (b) $\lambda = 5$
(c) $\lambda = -5$ (d) no real value of λ

15. If a, b, c are respectively the pth, qth and rth terms of an H.P., then $\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix} = ?$
(a) abc (b) $p+q+r$
(c) 0 (d) none of these

16. If $a \neq b \neq c$ such that $\begin{vmatrix} a^3-1 & b^3-1 & c^3-1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = 0$, then
(a) $ab+bc+ca=0$ (b) $a+b+c=0$
(c) $abc=1$ (d) $a+b+c=1$

17. The system of equations $2x+3y=8$, $7x-5y+3=0$ and $4x-6y+\lambda=0$ is solvable if λ is
(a) 6 (b) 8
(c) -8 (d) -6

18. If $U_n = \begin{vmatrix} n & 15 & 8 \\ n^2 & 35 & 9 \\ n^3 & 25 & 10 \end{vmatrix}$, then $\sum_{n=1}^{5} U_n = ?$
(a) 0 (b) 25
(c) 625 (d) none of these

19. Let $\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = ax^5+bx^4+cx^3+dx^2+\lambda x+\mu$ be an identity in x, where a, b, c, d, λ, μ are independent of x. Then the value of λ is
(a) 3 (b) 2
(c) 4 (d) none of these

20. If the solutions of the system of equations $3x-y+4z-3=0$, $x+2y-3z+2=0$ and $6x+5y+\lambda z+3=0$ are infinite, then $\lambda = ?$
(a) 7 (b) -7
(c) 5 (d) -5

21. If $\Delta = \begin{vmatrix} 4 & 5 & 6 & x \\ 5 & 6 & 7 & y \\ 6 & 7 & 8 & z \\ x & y & z & 0 \end{vmatrix}$, then Δ is equal to
(a) $(x+z-2y)^2$ (b) $(y+z-2x)^2$
(c) $(z+x-2y)^2$ (d) none of these

22. The value of θ lying between $\theta = 0$ and $\theta = \pi/2$ and satisfying the equation $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$ is
(a) $3\pi/24$ (b) $5\pi/24$
(c) $11\pi/24$ (d) $\pi/24$

23. The value of the determinant $\Delta = \begin{vmatrix} 2a_1b_1 & a_1b_2+a_2b_1 & a_1b_3+a_3b_1 \\ a_1b_2+a_2b_1 & 2a_2b_2 & a_2b_3+a_3b_2 \\ a_1b_3+a_3b_1 & a_3b_2+a_2b_3 & 2a_3b_3 \end{vmatrix}$ is
(a) 1 (b) -1
(c) 0 (d) $a_1a_2a_3b_1b_2b_3$

24. If ω is an imaginary cube root of unity, then $\begin{vmatrix} \lambda+1 & \omega & \omega^2 \\ \omega & \lambda+\omega^2 & 1 \\ \omega^2 & 1 & \lambda+\omega \end{vmatrix}$ is equal to
(a) 0 (b) λ^3+1
(c) λ^3 (d) none of these

25. If $\begin{vmatrix} x^n & x^{n+2} & x^{n+4} \\ y^n & y^{n+2} & y^{n+4} \\ z^n & z^{n+2} & z^{n+4} \end{vmatrix} = \left(\frac{1}{y^2}-\frac{1}{x^2}\right)\left(\frac{1}{z^2}-\frac{1}{y^2}\right)\left(\frac{1}{x^2}-\frac{1}{z^2}\right)$, then $n = ?$
(a) 4 (b) -4
(c) 2 (d) -2

26. If x, y, z are all distinct and $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$, then the value of xyz is
(a) -2 (b) -1
(c) -3 (d) none of these

27. If $f(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$, then $f(\theta) - 2f(\theta) + f(\theta) = ?$

(a) 1 (b) -1
(c) 0 (d) none of these

28. If $\begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix} = \lambda$, then $\begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} = ?$

(a) λ (b) 2λ
(c) $-\lambda$ (d) -2λ

29. The value of the determinant

$$\begin{vmatrix} \log_a\left(\frac{x}{y}\right) & \log_a\left(\frac{y}{z}\right) & \log_a\left(\frac{z}{x}\right) \\ \log_{a^2}\left(\frac{y}{z}\right) & \log_{a^2}\left(\frac{z}{x}\right) & \log_{a^2}\left(\frac{x}{y}\right) \\ \log_{a^3}\left(\frac{z}{x}\right) & \log_{a^3}\left(\frac{x}{y}\right) & \log_{a^3}\left(\frac{y}{z}\right) \end{vmatrix} \text{ is}$$

(a) 1 (b) -1
(c) 0 (d) $1/6 \log_a xyz$

30. The value of $\begin{vmatrix} 1 & 1 & 1 \\ (2^x+2^{-x})^2 & (3^x+3^{-x})^2 & (5^x+5^{-x})^2 \\ (2^x-2^{-x})^2 & (3^x-3^{-x})^2 & (5^x-5^{-x})^2 \end{vmatrix}$ is

(a) 0 (b) 30^x
(c) 30^{-x} (d) none of these

31. The sum of the two non-integral roots of

$\Delta = \begin{vmatrix} x & 3 & 4 \\ 5 & x & 5 \\ 4 & 2 & x \end{vmatrix} = 0$ is

(a) 4 (b) -4
(c) 16 (d) none of these

32. If $\theta \in (0, \pi/2)$, then the value of

$$\begin{vmatrix} (\sin\theta+\operatorname{cosec}\theta)^2 & (\sin\theta-\operatorname{cosec}\theta)^2 & 1 \\ (\cos\theta+\sec\theta)^2 & (\cos\theta-\sec\theta)^2 & 1 \\ (\tan\theta+\cot\theta)^2 & (\tan\theta-\cot\theta)^2 & 1 \end{vmatrix} = ?$$

(a) $\sin\theta+\cos\theta+\tan\theta$ (b) 1
(c) 0 (d) 4

33. If $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = \alpha a^2b^2c^2$, then the value of α is

(a) 4 (b) 3
(c) 2 (d) 1

34. If $\begin{vmatrix} b^2+c^2 & ab & ac \\ ab & c^2+a^2 & bc \\ ca & cb & a^2+b^2 \end{vmatrix}$

$= \begin{vmatrix} b^2+c^2 & a^2 & a^2 \\ b^2 & c^2+a^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix} = \lambda a^2b^2c^2$, then $\lambda = ?$

(a) 2 (b) 1
(c) 4 (d) 3

35. The only integral root of the equation

$\begin{vmatrix} 2-y & 2 & 3 \\ 2 & 5-y & 6 \\ 3 & 4 & 10-y \end{vmatrix} = 0$ is

(a) $y = 2$ (b) $y = 3$
(c) $y = 1$ (d) none of these

36. If x, y, z are in A.P., then the value of the determinant

$\begin{bmatrix} a+2 & a+3 & a+2x \\ a+3 & a+4 & a+2y \\ a+4 & a+5 & a+2z \end{bmatrix}$ is

(a) $2a$ (b) a
(c) 1 (d) 0

37. $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = ?$

(a) $1+\Sigma a$ (b) $1+\Sigma\frac{1}{a}$
(c) $abc\left[1+\Sigma\frac{1}{a}\right]$ (d) none of these

38. The determinant

$\Delta = \begin{vmatrix} a^2 & a & 1 \\ \cos(nx) & \cos(n+1)x & \cos(n+2)x \\ \sin(nx) & \sin(n+1)x & \sin(n+2)x \end{vmatrix}$ is independent of

(a) n (b) a
(c) x (d) none of these

39. If a, b and c are non-zero real numbers, then

$\Delta = \begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$ is equal to

(a) $a^2\,b^2c^2$ (b) $bc + ca + ab$
(c) abc (d) 0

40. If $\begin{vmatrix} p & q-y & r-z \\ p-x & q & r-z \\ p-x & q-y & r \end{vmatrix} = 0$, then the value of $\frac{p}{x}+\frac{q}{y}+\frac{r}{z}$ is

(a) 0 (b) 1
(c) 2 (d) 4 pqr

41. Let m be a positive integer and

$$\Delta_r = \begin{vmatrix} 2r-1 & {}^{m}C_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix} \quad (0 \le r \le m)$$

Then the value of $\sum_{r=0}^{m} \Delta_r$ is given by

(a) 0 (b) m^2-1
(c) 2^m (d) $2^m \sin^2(2^m)$

42. If a, b and c are the roots of the equation $x^3+px+q=0$, $(p \neq 0, q \neq 0)$ the value of the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is

(a) p^2-2q (b) p
(c) 0 (d) q

43. $\begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix} = ?$

(a) x^4 (b) 4
(c) 0 (d) -8

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b) (c) or (d) given below:

(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1
(c) Statement 1 is true; Statement 2 is false
(d) Statement 1 is false; Statement 2 is true

44. Statement 1: The value of the determinant

$A = \begin{vmatrix} 0 & 2 & 1 \\ -2 & 0 & 3 \\ -1 & -3 & 0 \end{vmatrix}$ is equal to 0.

Statement 2: If A is a skew-symmetric determinant of odd order, then $|A|$ is a perfect square.

45. Statement 1: The value of $\begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^{2n} & 1 & \omega^n \\ \omega^n & \omega^{2n} & 1 \end{vmatrix}$ is zero.

where $n \in N$.

Statement 2: $1+\omega+\omega^2=0$

46. Statement 1: The minor of 5 in the determinant

$\begin{vmatrix} 2 & 5 & 8 \\ 0 & 3 & 7 \\ -1 & -2 & 6 \end{vmatrix}$ is -7.

Statement 2: The determinant that is left by cancelling the row and column intersecting at a particular element is called the minor.

47. Statement 1: If a, b, c are different and $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$, then $abc\,(ab+bc+ac) = a+b+c$.

Statement 2: $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (ab+bc+ca)\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$

48. Statement 1:

If $f(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ a & b & c \\ d & e & f \end{vmatrix}$, where f_1, f_2, f_3 are differentiable functions and a, b, c, d, e, f are constants, then

$$\int f(x)dx = \begin{vmatrix} \int f_1(x)dx & \int f_2(x)dx & \int f_3(x)dx \\ a & b & c \\ d & e & f \end{vmatrix} + k$$

Statement 2: Integration of the sum of several functions is equal to the sum of integrations of individual function.

49. Statement 1: If $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$, then $\begin{vmatrix} ka & kb & kc \\ kx & ky & kz \\ kp & kq & kr \end{vmatrix}$ is equal to $k^3\,\Delta$.

Statement 2: If any row of a determinant is multiplied by r, then the value of the determinant is also multiplied by r.

Previous Years' Questions

50. If $\omega(\neq 1)$ is a cubic root of unity, then $\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -1+\omega-i & -1 \end{vmatrix}$ equals

(a) 0 (b) 1
(c) i (d) ω **[AIEEE 2002]**

51. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x+iy$, then

(a) $x=0, y=3$ (b) $x=0, y=0$
(c) $x=3, y=1$ (d) $x=1, y=3$
[AIEEE 2002]

52. If 1, ω, ω^2 are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$ is equal to

(a) 0 (b) 1
(c) ω (d) ω^2 **[AIEEE 2003]**

53. If the system of linear equations

$$x+2ay+az=0$$
$$x+3by+bz=0$$
$$x+4cy+cz=0$$

has a non-zero solution, then a, b, c

(a) are in A.P. (b) satisfy $a+2b+3c=0$
(c) are in H.P. (d) are in G.P.
[AIEEE 2003]

54. If $a_1, a_2, a_3 \ldots$ are in G.P., then $\Delta = \begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$ is equal to

(a) 0 (b) 1
(c) 2 (d) 4 **[AIEEE 2004]**

55. If $a^2+b^2+c^2=-2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$,

then $f(x)$ is a polynomial of degree

(a) 2 (b) 0
(c) 3 (d) 1 **[AIEEE 2005]**

56. The system of equations $\alpha x+y+z=\alpha-1$, $x+\alpha y+z=\alpha-1$ and $x+y+\alpha z=\alpha-1$ has no solution, if α is

(a) either -2 or 1 (b) -2
(c) not -2 (d) 1 **[AIEEE 2005]**

57. If $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x\neq 0, y\neq 0$, then Δ is

(a) divisible by neither x nor y
(b) divisible by y but not by x
(c) divisible by x but not by y
(d) divisible by both x and y **[AIEEE 2007]**

58. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x=cy+bz$, $y=az+cx$, and $z=bx+ay$. Then $a^2+b^2+c^2+2abc$ is equal to

(a) -1 (b) 0
(c) 1 (d) 2 **[AIEEE 2008]**

59. Let A be a square matrix all of whose entries are integers. Then, which one of the following is true?

(a) If $\det(A)=\pm 1$, then A^{-1} need not exist
(b) If $\det(A)=\pm 1$, then A^{-1} exists but all its entries are not necessarily integers
(c) If $\det(A)=\pm 1$, then A^{-1} exists and all its entries are non-integers
(d) If $\det(A)=\pm 1$, then A^{-1} exists and all its entries are integers **[AIEEE 2008]**

60. Let a, b, c be such that $(b+c)\neq 0$. If $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$,

then the value of n is

(a) zero (b) any even integer
(c) any odd integer (d) any integer
[AIEEE 2009]

61. If x is a positive integer, then $\begin{vmatrix} x! & (x+1)! & (x+2)! \\ (x+1)! & (x+2)! & (x+3)! \\ (x+2)! & (x+3)! & (x+4)! \end{vmatrix}$ is equal to

(a) $2x!(x+1)!$ (b) $2x!(x+1)!(x+2)!$
(c) $2x!(x+3)!$ (d) $2(x+1)!(x+2)!(x+3)!$
[DCE 2009]

62. Let $\omega = \frac{-1}{2}+i\frac{\sqrt{3}}{2}$, then the value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$ is

(a) 3ω (b) $3\omega(\omega-1)$
(c) $3\omega^2$ (d) $3\omega(1-\omega)$ **[DCE 2007]**

63. If $C = 2\cos\theta$, then the value of the determinant

$$\Delta = \begin{vmatrix} C & 1 & 0 \\ 1 & C & 1 \\ 6 & 1 & C \end{vmatrix}$$ is

(a) $\dfrac{\sin 4\theta}{\sin\theta}$ (b) $4\cos^2\theta(\cos\theta - 1)$

(c) $\dfrac{2\sin^2 2\theta}{\sin\theta}$ (d) none of these

[DCE 2007]

64. If $D_r = \begin{vmatrix} 2^{r-1} & 2\cdot 3^{r-1} & 4\cdot 5^{r-1} \\ \alpha & \beta & \gamma \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix}$, then the vlaue of $\sum_{r=1}^{n} D_r$ is equal to

(a) $\alpha + \beta + \gamma$ (b) $\alpha\cdot 2^n + \beta\cdot 3^n + \gamma\cdot 4^n$

(c) 0 (d) $\alpha\beta\gamma$ **[DCE 2006]**

65. If $\begin{vmatrix} -12 & 0 & \lambda \\ 0 & 2 & -1 \\ 2 & 1 & 15 \end{vmatrix} = -360$, then the value of λ is

(a) –1 (b) –2 (c) –3 (d) 4

[DCE 2006]

66. If $\Delta = \begin{vmatrix} 1 & \sin\theta & 1 \\ \sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{vmatrix}$, then Δ lies in the interval

(a) [2, 4] (b) [3, 4] (c) [1, 4] (d) none of these

[DCE 2006]

67. The value of the determinant $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$ is equal to

(a) $6xyz$ (b) xyz

(c) $4xyz$ (d) $xy + yz + zx$

[DCE 2004]

68. Statement 1: The determinant of a skew-symmetric matrix of order 3 is zero.

Statement 2: For any matrix A, $\det(A)^T = \det(A)$ and $\det(-A) = -\det(A)$, where det(2) denotes the determinant of matrix B. Then

(a) both statements are true

(b) both statements are false

(c) Statement 1 is false and Statement 2 is true

(d) Statement 1 is true and Statement 2 is false

[AIEEE 2011]

ANSWERS

Exercise Set 1

1. (a)	**2.** (a)	**3.** (a)	**4.** (c)	**5.** (c)	**6.** (a)	**7.** (b)	**8.** (c)	**9.** (d)	**10.** (b)
11. (a)	**12.** (b)	**13.** (b)	**14.** (d)	**15.** (b)	**16.** (c)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (c)
21. (d)	**22.** (b)	**23.** (c)	**24.** (c)	**25.** (b)	**26.** (d)	**27.** (a)	**28.** (b)	**29.** (c)	**30.** (d)
31. (c)	**32.** (b)	**33.** (a)	**34.** (b)	**35.** (b)	**36.** (c)	**37.** (c)	**38.** (c)	**39.** (d)	**40.** (a)
41. (c)	**42.** (b)	**43.** (c)	**44.** (b)	**45.** (c)	**46.** (a)	**47.** (a)	**48.** (c)	**49.** (a)	**50.** (c)
51. (a)	**52.** (c)	**53.** (c)	**54.** (b)	**55.** (a)	**56.** (d)	**57.** (b)	**58.** (a)	**59.** (c)	**60.** (c)
61. (d)	**62.** (a)	**63.** (b)	**64.** (c) and (d)						

Exercise Set 2

1. (c)	**2.** (a)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (d)	**7.** (d)	**8.** (a)	**9.** (c)	**10.** (c)
11. (d)	**12.** (c)	**13.** (d)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (a)	**20.** (d)
21. (a)	**22.** (c)	**23.** (c)	**24.** (c)	**25.** (b)	**26.** (b)	**27.** (c)	**28.** (c)	**29.** (c)	**30.** (a)
31. (b)	**32.** (c)	**33.** (a)	**34.** (c)	**35.** (c)	**36.** (d)	**37.** (c)	**38.** (a)	**39.** (d)	**40.** (c)
41. (a)	**42.** (c)	**43.** (d)	**44.** (c)	**45.** (a)	**46.** (d)	**47.** (a)	**48.** (a)	**49.** (a)	**50.** (a)
51. (b)	**52.** (a)	**53.** (c)	**54.** (a)	**55.** (a)	**56.** (b)	**57.** (d)	**58.** (c)	**59.** (d)	**60.** (c)
61. (b)	**62.** (b)	**63.** (d)	**64.** (c)	**65.** (c)	**66.** (a)	**67.** (c)	**68.** (d)		

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Taking x common from R_2 and $x(x-1)$ common from R_3, we get

$$f(x) = x^2(x-1)\begin{vmatrix} 1 & x & x+1 \\ 2 & x-1 & x+1 \\ 3 & x-2 & x+1 \end{vmatrix}$$

Applying $C_3 \to C_3 - C_2$, we get $f(x) = x^2(x-1)$

$$\begin{vmatrix} 1 & x & 1 \\ 2 & x-1 & 2 \\ 3 & x-2 & 3 \end{vmatrix} = 0$$

Thus, $f(500) = 0$

2. Since $f(\theta)=\begin{vmatrix}\cos^2\theta & \cos\theta\sin\theta & -\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta\\ \sin\theta & -\cos\theta & 0\end{vmatrix}$,

applying $C_1 \to C_1 - \sin\theta C_3$; $C_2 \to C_2 + \cos\theta C_3$

$$f(x)=\begin{vmatrix}1 & 0 & -\sin\theta\\ 0 & 1 & \cos\theta\\ \sin\theta & -\cos\theta & 0\end{vmatrix}$$

Applying $R_3 \to R_3 - \sin\theta R_1 + \cos\theta R_2$

$$f(\theta)=\begin{vmatrix}1 & 0 & -\sin\theta\\ 0 & 1 & \cos\theta\\ 0 & 0 & 1\end{vmatrix}$$

$\Rightarrow$ $f(\theta) = 1(1) = 1$ = constant function

$\therefore$ $f\left(\frac{\pi}{6}\right)=1$

3. Since α_1, α_2 and β_1, β_2 are the roots of $ax^2 + bx + c = 0$ and $px^2 + qx + r = 0$, respectively, therefore,

$$\alpha_1+\alpha_2=\frac{-b}{a},\ \alpha_1\alpha_2=\frac{c}{a}$$

and $\beta_1+\beta_2=\frac{-q}{p},\ \beta_1\beta_2=\frac{r}{p}$

Since the given system of equations has a non-trivial solution,

$\therefore$ $\begin{vmatrix}\alpha_1 & \alpha_2\\ \beta_1 & \beta_2\end{vmatrix}=0$ i.e. $\alpha_1\beta_2 - \alpha_2\beta_1 = 0$

or $$\frac{\alpha_1}{\beta_1}=\frac{\alpha_2}{\beta_2}=\frac{\alpha_1+\alpha_2}{\beta_1+\beta_2}=\sqrt{\frac{\alpha_1\alpha_2}{\beta_1\beta_2}}$$

$\Rightarrow$ $$\frac{pb}{qa}=\sqrt{\frac{pc}{ra}}\Rightarrow\frac{b^2}{q^2}=\frac{ac}{pr}$$

4. Since $\begin{vmatrix}3-x & -6 & 3\\ -6 & 3-x & 3\\ 3 & 3 & -6-x\end{vmatrix}=0,$

applying $C_1 \to C_1 + C_2 + C_3$, we obtain

$$\Rightarrow\quad -x\begin{vmatrix}1 & -6 & 3\\ 1 & 3-x & 3\\ 1 & 3 & -6-x\end{vmatrix}=0$$

Applying $R_2 \to R_2 - R_1; R_3 \to R_3 - R_1$, we get

$$-x\begin{vmatrix}1 & -6 & 3\\ 0 & 9-x & 0\\ 0 & 9 & -9-x\end{vmatrix}=0$$

$\Rightarrow$ $-x(9-x)(-9-x)=0$

$\therefore$ $x = 0, 9, -9$

5. We have $\Delta^2=\Delta\cdot\Delta=\begin{vmatrix}l_1 & m_1 & n_1\\ l_2 & m_2 & n_2\\ l_3 & m_3 & n_3\end{vmatrix}\cdot\begin{vmatrix}l_1 & m_1 & n_1\\ l_2 & m_2 & n_2\\ l_3 & m_3 & n_3\end{vmatrix}$

$$=\begin{vmatrix}l_1^2+m_1^2+n_1^2 & l_1l_2+m_1m_2+n_1n_2 & l_1l_3+m_1m_3+n_1n_3\\ l_1l_2+m_1m_2+n_1n_2 & l_2^2+m_2^2+n_2^2 & l_2l_3+m_2m_3+n_2n_3\\ l_1l_3+m_1m_3+n_1n_3 & l_2l_3+m_2m_3+n_2n_3 & l_3^2+m_3^2+n_3^2\end{vmatrix}$$

$$=\begin{vmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{vmatrix}=1\quad\Rightarrow\quad\Delta=\pm1$$

$\Rightarrow$ $|\Delta| = 1$

6. Operating $C_1 \to C_1 + C_2 + C_3$, the given determinant

$$=\begin{vmatrix}1+\omega+\omega^2 & \omega & \omega^2\\ \omega+\omega^2+1 & \omega^2 & 1\\ \omega^2+1+\omega & 1 & \omega\end{vmatrix}=\begin{vmatrix}0 & \omega & \omega^2\\ 0 & \omega^2 & 1\\ 0 & 1 & \omega\end{vmatrix}=0$$

7. Let $\Delta=\begin{vmatrix}1 & \sin3\theta & \sin^3\theta\\ 2\cos\theta & \sin6\theta & \sin^3 2\theta\\ 4\cos^2\theta-1 & \sin9\theta & \sin^3 3\theta\end{vmatrix}$

Multiplying C_1 by $\sin\theta$, the determinant can be written as

$$\Delta=\frac{1}{\sin\theta}\begin{vmatrix}\sin\theta & \sin3\theta & \sin^3\theta\\ 2\cos\theta\sin\theta & \sin6\theta & \sin^3 2\theta\\ \sin\theta(4\cos^2\theta-1) & \sin9\theta & \sin^3 3\theta\end{vmatrix}$$

$$=\frac{1}{\sin\theta}\begin{vmatrix}\sin\theta & \sin3\theta & \sin^3\theta\\ \sin2\theta & \sin6\theta & \sin^3 2\theta\\ \sin3\theta & \sin9\theta & \sin^3 3\theta\end{vmatrix}$$

Now applying $C_1 \to 3C_1 - 4C_3$, we get

$$\Delta=\frac{1}{\sin\theta}\begin{vmatrix}3\sin\theta-\sin^3\theta & \sin3\theta & \sin^3\theta\\ 3\sin2\theta-4\sin^3 2\theta & \sin6\theta & \sin^3 2\theta\\ 3\sin3\theta-4\sin^3 3\theta & \sin9\theta & \sin^3 3\theta\end{vmatrix}$$

$$\Delta=\frac{1}{\sin\theta}\begin{vmatrix}\sin3\theta & \sin3\theta & \sin^3\theta\\ \sin6\theta & \sin6\theta & \sin^3 2\theta\\ \sin9\theta & \sin9\theta & \sin^3 3\theta\end{vmatrix}=0$$

8. Let $\Delta=\begin{vmatrix}b(b-a) & b-c & c(b-a)\\ a(b-a) & a-b & b(b-a)\\ c(b-a) & c-a & a(b-a)\end{vmatrix}$

$$\Rightarrow\ \Delta=(b-a)^2\begin{vmatrix}b & b-c & c\\ a & a-b & b\\ c & c-a & a\end{vmatrix}$$

Applying $C_1 \to C_1 - C_2 + C_3$, we get

$$\therefore \quad \Delta = (b-a)^2 \begin{vmatrix} 0 & b-c & c \\ 0 & a-b & b \\ 0 & c-a & a \end{vmatrix} = 0$$

9. Here $-(x-a)[-(x+b)(x-c)]$
$+(x-b)[(x+a)(x-c)] = 0$

$\Rightarrow (x-a)(x+b)(x-c) + (x+a)(x-b)(x+c) = 0$

$\Rightarrow x = 0$ [$\because$ there will be no constant term]

10. $\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix} = 0$

$\Rightarrow 3a^2b^2c^2 - [(ab)^3 + (bc)^3 + (ca)^3] = 0$

$\Rightarrow (ab)^3 + (bc)^3 + (ca)^3 - 3a^2b^2c^2 = 0$

We know that

if $x^3 + y^3 + z^3 = 3xyz$, then $x + y + z = 0$

$\therefore ab + bc + ca = 0$

$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

11. Here the given determinant is

$$= \begin{vmatrix} 0 & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(\pi - C) & -\tan A & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ -\cos C & -\tan A & 0 \end{vmatrix} = 0.$$

[$\because$ Determinant is skew-symmetric of odd order]

12. Let $\Delta = \begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we obtain

$$\Rightarrow \quad \Delta = \begin{vmatrix} x+p+q & p & q \\ x+p+q & x & q \\ x+p+q & q & x \end{vmatrix}$$

$$\Rightarrow \quad \Delta = (x+p+q)\begin{vmatrix} 1 & p & q \\ 1 & x & q \\ 1 & q & x \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$, we get

$$\Rightarrow \quad \Delta = (x+p+q)\begin{vmatrix} 1 & p & q \\ 0 & x-p & 0 \\ 0 & q-p & x-q \end{vmatrix}$$

Expanding along C_1, we have

$$\Delta = (x+p+q)(x-p)(x-q)$$

13. Operating $R_3 \to R_3 - \alpha R_1 - R_2$,

$$\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ 0 & 0 & -(a\alpha^2 + 2b\alpha + c) \end{vmatrix} = 0$$

$\Rightarrow -(a\alpha^2 + 2b\alpha + c)(ac - b^2) = 0$

$\Rightarrow b^2 = ac$ or $a\alpha^2 + 2b\alpha + c = 0$

$\Rightarrow a, b, c$ are in G.P. or α is a root of $ax^2 + 2bx + c = 0$.

14. $f\left(\frac{\pi}{3}\right) = \begin{vmatrix} 2\cos\frac{\pi}{3} & 1 & 0 \\ 1 & 2\cos\frac{\pi}{3} & 1 \\ 0 & 1 & 2\cos\frac{\pi}{3} \end{vmatrix}$

$$= \begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix}, \text{ Applying } R_2 \to R_2 - R_1$$

$$f\left(\frac{\pi}{3}\right) = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = -1$$

15. $\Delta = \begin{vmatrix} 1 & \frac{\log y}{\log x} & \frac{\log z}{\log x} \\ \frac{\log x}{\log y} & 1 & \frac{\log z}{\log y} \\ \frac{\log x}{\log z} & \frac{\log y}{\log z} & 1 \end{vmatrix}$

$$= \frac{1}{\log x \log y \log z}\begin{vmatrix} \log x & \log y & \log z \\ \log x & \log y & \log z \\ \log x & \log y & \log z \end{vmatrix} = 0$$

16. Let $\Delta = \begin{vmatrix} ka & k^2 + a^2 & 1 \\ kb & k^2 + b^2 & 1 \\ kc & k^2 + c^2 & 1 \end{vmatrix}$

$$\Rightarrow \Delta = \begin{vmatrix} ka & k^2 & 1 \\ kb & k^2 & 1 \\ kc & k^2 & 1 \end{vmatrix} + \begin{vmatrix} ka & a^2 & 1 \\ kb & b^2 & 1 \\ kc & c^2 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = 0 + k\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$$

$\therefore \Delta = k(a-b)(b-c)(c-a)$

17. $\sum_{k=1}^{n} D_k = \begin{vmatrix} \sum_{k=1}^{n} 1 & n & n \\ 2\sum_{k=1}^{n} k & n^2+n+1 & n^2+n \\ 2\sum_{k=1}^{n} k - \sum_{k=1}^{n} 1 & n^2 & n^2+n+1 \end{vmatrix}$

$$= \begin{vmatrix} n & n & n \\ n^2+n & n^2+n+1 & n^2+n \\ n^2 & n^2 & n^2+n+1 \end{vmatrix}$$

Operating $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_2$

$$= \begin{vmatrix} n & 0 & 0 \\ n^2+n & 1 & -1 \\ n^2 & 0 & n+1 \end{vmatrix}$$

$= n(n+1)$.

By the equation, $n(n+1) = 56 = 7 \cdot 8 \Rightarrow n = 7$.

18. $\begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}^2 = \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix} \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}$

$$= \begin{vmatrix} b^2+c^2 & ab & ca \\ ab & c^2+a^2 & bc \\ ac & bc & a^2+b^2 \end{vmatrix}$$

19. Since $\Delta = \begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$

applying $R_1 \to R_1 + R_2 + R_3$, we get

$$\Rightarrow \Delta = (a+b+c-x)\begin{vmatrix} 1 & 1 & 1 \\ c & b-x & a \\ b & a & c-x \end{vmatrix}$$

Applying $C_2 \to C_2 - C_1$; $C_3 \to C_3 - C_1$

$$\Rightarrow \Delta = (-x)\begin{vmatrix} 1 & 0 & 0 \\ c & b-x-c & a-c \\ b & a-b & c-x-b \end{vmatrix} \quad [\because (a+b+c=0)]$$

$$\Rightarrow \Delta = -x\{(b-x-c)(c-x-b)-(a-b)(a-c)\}$$

$$\Rightarrow \Delta = x(a^2+b^2+c^2-ab-bc-ca-x^2)$$

If $\Delta = 0 \Rightarrow x = 0$

or $x^2 = a^2+b^2+c^2-ab-bc-ca$

$$\Rightarrow x^2 = a^2+b^2+c^2-\frac{1}{2}\{(a+b+c)^2-a^2-b^2-c^2\}$$

$$\Rightarrow x^2 = \frac{3}{2}(a^2+b^2+c^2) \quad [\because a+b+c=0]$$

$$\therefore x = \pm\sqrt{\frac{3}{2}(a^2+b^2+c^2)}$$

20. $\Delta = (2\cos x + \sin x)\begin{vmatrix} 1 & \cos x & \cos x \\ 1 & \sin x & \cos x \\ 1 & \cos x & \sin x \end{vmatrix}$

$(C_1 \to C_1 + C_2 + C_3)$

$$= (2\cos x + \sin x)\begin{vmatrix} 0 & \cos x - \sin x & 0 \\ 1 & \sin x & \cos x \\ 1 & \cos x & \sin x \end{vmatrix}$$

$(R_1 \to R_1 - R_2)$

$= (2\cos x + \sin x)(\cos x - \sin x)^2 = 0$

$\Rightarrow \tan x = -2$ or 1, hence only one solution in $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$

21. Since $\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0,$

applying $C_1 \to C_1 + (C_2 + C_3)$, we get

$$\Rightarrow \begin{vmatrix} x+9 & 3 & 5 \\ x+9 & x+2 & 5 \\ x+9 & 3 & x+4 \end{vmatrix} = 0$$

$$\Rightarrow (x+9)\begin{vmatrix} 1 & 3 & 5 \\ 1 & x+2 & 5 \\ 1 & 3 & x+4 \end{vmatrix} = 0$$

Applying $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$, we get

$$\Rightarrow (x+9)\begin{vmatrix} 1 & 3 & 5 \\ 0 & x-1 & 0 \\ 0 & 0 & x-1 \end{vmatrix} = 0$$

$$\Rightarrow (x+9)(x-1)(x-1) = 0$$

$\therefore x = -9, 1, 1$

22. $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$

$\Rightarrow y^2 = xz \Rightarrow x, y, z$ are in G.P.

23. $\Delta = \begin{vmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix} = 0,$ Applying $C_2 \to C_1 + C_2$

$$\Delta = \begin{vmatrix} {}^{10}C_4 & {}^{10}C_4 + {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_6 + {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_8 + {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix} = 0$$

$$= \begin{vmatrix} {}^{10}C_4 & {}^{11}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{12}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{13}C_9 & {}^{13}C_{m+4} \end{vmatrix} = 0$$

Clearly, $m = 5$ satisfies the above result.

($\because$ C_2, C_3 will be identical)

24. Applying $C_1 \to C_1 + C_2$

$$f(x) = \begin{vmatrix} 2 & \cos^2 x & 4\sin 2x \\ 2 & 1+\cos^2 x & 4\sin 2x \\ 1 & \cos^2 x & 1+4\sin 2x \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$, we have

$$f(x) = \begin{vmatrix} 2 & \cos^2 x & 4\sin 2x \\ 0 & 1 & 0 \\ 1 & \cos^2 x & 1+4\sin 2x \end{vmatrix}$$

$\Rightarrow f(x) = 2 + 4 \sin 2x$

The value of $f(x)$ is maximum when $\sin 2x = 1$.

$\therefore$ Maximum value of $f(x) = 6$

25. Applying $C_1 \to C_1 - C_2$ and $C_2 \to C_2 - C_3$ in succession, we get

$$\Delta = \begin{vmatrix} \alpha-\beta & \beta-\gamma & \gamma \\ 0 & 0 & 1 \\ \beta-\gamma & \gamma-\alpha & \alpha \end{vmatrix}$$

$$\therefore \quad \Delta = \alpha^2 + \beta^2 + \gamma^2 - (\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= (\alpha+\beta+\gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha) = p^2 - 3q$$

26. Let $\Delta = \begin{vmatrix} {}^5C_0 & {}^5C_3 & 1 \\ {}^5C_1 & {}^5C_4 & 1 \\ {}^5C_2 & {}^5C_5 & 1 \end{vmatrix}$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & 10 & 1 \\ 5 & 5 & 1 \\ 10 & 1 & 1 \end{vmatrix}$$

Applying $R_3 \to R_3 - R_1$ and $R_2 \to R_2 - R_1$, we get

$$\Delta = \begin{vmatrix} 1 & 10 & 1 \\ 4 & -5 & 0 \\ 9 & -9 & 0 \end{vmatrix}$$

$$\therefore \quad \Delta = -36 + 45 = 9$$

27. $\begin{vmatrix} a+pd & a+qd & a+rd \\ p & q & r \\ d & d & d \end{vmatrix}$

$$= \begin{vmatrix} a & a & a \\ p & q & r \\ d & d & d \end{vmatrix} + d \begin{vmatrix} p & q & r \\ p & q & r \\ d & d & d \end{vmatrix} = 0 + d(0) = 0$$

28. $\Delta = \begin{vmatrix} {}^5C_0 & {}^5C_3 & 14 \\ {}^5C_1 & {}^5C_4 & 1 \\ {}^5C_2 & {}^5C_5 & 1 \end{vmatrix}$ (Operating $R_1 \to R_1 + R_2 + R_3$)

$$= \begin{vmatrix} {}^5C_0 + {}^5C_1 + {}^5C_2 & {}^5C_3 + {}^5C_4 + {}^5C_5 & 14+1+1 \\ 5 & 5 & 1 \\ 10 & 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 16 & 16 & 16 \\ 5 & 5 & 1 \\ 10 & 1 & 1 \end{vmatrix} = 16 \begin{vmatrix} 1 & 1 & 1 \\ 5 & 5 & 1 \\ 10 & 1 & 1 \end{vmatrix}$$

[$\because$ ${}^5C_0 + {}^5C_1 + {}^5C_2 = {}^5C_3 + {}^5C_4 + {}^5C_5 = 16$]

$$= 16 \begin{vmatrix} 1 & 1 & 1 \\ 5 & 5 & 1 \\ 9 & 0 & 0 \end{vmatrix} = 16 \times 9(-4) = -576$$

29. $\Delta'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$

$$+ \begin{vmatrix} f(x) & g(x) & h(x) \\ f''(x) & g''(x) & h''(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ 0 & 0 & 0 \end{vmatrix}$$

$\Rightarrow \Delta'(x) = 0$

$\Rightarrow$ $\Delta(x)$ is constant, i.e. polynomial of degree 0.

30. $C_1 \to C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} 1+(1+\omega+\omega^2) & \omega & \omega^2 \\ 1+(1+\omega+\omega^2) & -\omega & \omega \\ 1+(1+\omega+\omega^2) & 1 & -\omega^2 \end{vmatrix}$$

$R_1 \to R_1 - R_2$; $R_2 \to R_2 - R_3$

$$\Delta = \begin{vmatrix} 0 & 2\omega & \omega^2-\omega \\ 0 & -\omega-1 & \omega+\omega^2 \\ 1 & 1 & -\omega^2 \end{vmatrix}$$

$$= 2\omega(\omega+\omega^2) - (\omega^2-\omega)(-\omega-1)$$

$$= -2\omega - \omega + 1 = 1 - 3\omega$$

31. Differentiating w.r.t. x, we get

$$-f'(x) = \begin{vmatrix} -\sin x & x & 1 \\ 2\cos x & x^2 & 2x \\ \sec^2 x & x & 1 \end{vmatrix} + \begin{vmatrix} \cos x & 1 & 1 \\ 2\sin x & 2x & 2x \\ \tan x & 1 & 1 \end{vmatrix}$$

$$+ \begin{vmatrix} \cos x & x & 0 \\ 2\sin x & x^2 & 2 \\ \tan x & x & 0 \end{vmatrix}$$

$$\Rightarrow \quad -\frac{f'(x)}{x}=\begin{vmatrix}-\sin x & 1 & 1\\ 2\cos x & x & 2x\\ \sec^2 x & 1 & 1\end{vmatrix}+\begin{vmatrix}\cos x & 1 & 0\\ 2\sin x & x & 2\\ \tan x & 1 & 0\end{vmatrix}$$

[$\because$ the second determinant is zero]

$$\Rightarrow \quad -\lim_{x\to 0}\frac{f'(x)}{x}=\begin{vmatrix}0 & 1 & 1\\ 2 & 0 & 0\\ 1 & 1 & 1\end{vmatrix}+\begin{vmatrix}1 & 1 & 0\\ 0 & 0 & 2\\ 0 & 1 & 0\end{vmatrix}=-2$$

$$\Rightarrow \quad \lim_{x\to 0}\frac{f'(x)}{x}=2$$

32. a, b, c are in A.P. $\Rightarrow \quad 2b=a+c$ (i)

The given determinant is

$$=\frac{1}{2}\begin{vmatrix}x+1 & x+2 & x+a\\ 2x+4 & 2x+6 & 2x+2b\\ x+3 & x+4 & x+c\end{vmatrix}$$

$$=\frac{1}{2}\begin{vmatrix}x+1 & x+2 & x+a\\ 0 & 0 & 2b-(a+c)\\ x+3 & x+4 & x+6\end{vmatrix}$$

[Applying $R_2 \to R_2-(R_1+R_3)$]

$$=\frac{1}{2}\begin{vmatrix}x+1 & x+2 & x+a\\ 0 & 0 & 0\\ x+3 & x+4 & x+6\end{vmatrix}=0$$ [Using (i)]

33. $\Delta(x)=\begin{vmatrix}x & 1+x^2 & x^3\\ \log(1+x^2) & e^x & \sin x\\ \cos x & \tan x & \sin^2 x\end{vmatrix}$

Let $\Delta(x)=A+Bx+Cx^2+Dx^3+\cdots$

$$\Delta(0)=\begin{vmatrix}0 & 1 & 0\\ 0 & 1 & 0\\ 1 & 0 & 0\end{vmatrix}=0 \quad \Rightarrow \quad A=0$$

$$\therefore \quad \Delta'(x)=\begin{vmatrix}1 & 1+x^2 & x^3\\ \frac{2x}{1+x^2} & e^x & \sin x\\ -\sin x & \tan x & \sin^2 x\end{vmatrix}$$

$$+\begin{vmatrix}1 & 2x & x^3\\ \log(1+x^2) & e^x & \sin x\\ \cos x & \sec^2 x & \sin^2 x\end{vmatrix}$$

$$+\begin{vmatrix}x & 1+x^2 & 3x^2\\ \log(1+x^2) & e^x & \cos x\\ \cos x & \tan x & 2\sin x\cos x\end{vmatrix}$$

$$\therefore \quad \Delta'(0)=\begin{vmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{vmatrix}+\begin{vmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 1 & 1 & 0\end{vmatrix}+\begin{vmatrix}0 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 0\end{vmatrix}$$

$$=0+0+1=1$$

Also, $\Delta'(x)=B+2xC+3x^2D \quad \therefore \quad \Delta'(0)=B$

$\therefore \quad B=1 \quad \therefore \quad \Delta(x)=x+Cx^2+Dx^3+\cdots$

$\therefore \quad \Delta(x)$ is divisible by x.

34. $\dfrac{d^n}{dx^n}\Delta(x)=\begin{vmatrix}\frac{d^n}{dx^n}x^n & \frac{d^n}{dx^n}(\sin x) & \frac{d^n}{dx^n}(\cos x)\\ n! & \sin\frac{n\pi}{2} & \cos\frac{n\pi}{2}\\ \alpha & \alpha^2 & \alpha^3\end{vmatrix}$

$$\frac{d^n}{dx^n}\Delta(x)=\begin{vmatrix}n! & \sin\left(x+\frac{n\pi}{2}\right) & \cos\left(x+\frac{n\pi}{2}\right)\\ n! & \sin\frac{n\pi}{2} & \cos\frac{n\pi}{2}\\ \alpha & \alpha^2 & \alpha^3\end{vmatrix}$$

$$\therefore \quad \left(\frac{d^n}{dx^n}\Delta(x)\right)_{x=0}=\begin{vmatrix}n! & \sin\frac{n\pi}{2} & \cos\frac{n\pi}{2}\\ n! & \sin\frac{n\pi}{2} & \cos\frac{n\pi}{2}\\ \alpha & \alpha^2 & \alpha^3\end{vmatrix}=0$$

35. Putting $x=0$, we have

$$f=\begin{vmatrix}3 & -5 & -2\\ 0 & 2 & 1\\ 0 & 7 & 1\end{vmatrix}=3(2-7)=-15$$

36. Since $\begin{vmatrix}0 & x-a & x-b\\ x+a & 0 & x-c\\ x+b & x+c & 0\end{vmatrix}=0,$

Expanding the given determinant, we have

$$2x^3+2x(ac-ab-bc)=0$$

$\therefore \quad x=0$

37. Let $s-a=A, s-b=B, s-c=C.$

$\therefore \quad A+B+C=3s-(a+b+c)=3s-2s=s$

$B+C=2s-(b+c)=a+b+c-(b+c)=a.$

Similarly, $C+A=b, A+B=c$

$$\therefore \quad \Delta = \begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix}$$

$$= \begin{vmatrix} (B+C)^2 & A^2 & A^2 \\ B^2 & (C+A)^2 & B^2 \\ C^2 & C^2 & (A+B)^2 \end{vmatrix}$$

Operating $C_1 \to C_1 - C_3, C_2 \to C_2 - C_3$

$$\Delta = \begin{vmatrix} (B+C)^2 - A^2 & 0 & A^2 \\ 0 & (C+A)^2 - B^2 & B^2 \\ C^2 - (A+B)^2 & C^2 - (A+B)^2 & (A+B)^2 \end{vmatrix}$$

$$= (A+B+C)^2 \begin{vmatrix} B+C-A & 0 & A^2 \\ 0 & C+A-B & B^2 \\ C-A-B & C-A-B & (A+B)^2 \end{vmatrix}$$

Operating $R_3 \to R_3 - (R_1 + R_2)$,

$$\Delta = (A+B+C)^2 \begin{vmatrix} B+C-A & 0 & A^2 \\ 0 & C+A-B & B^2 \\ -2B & -2A & 2AB \end{vmatrix}$$

$$= 2(A+B+C)^2 \begin{vmatrix} B+C-A & 0 & A^2 \\ 0 & C+A-B & B^2 \\ -B & -A & AB \end{vmatrix}$$

$$= 2(A+B+C)^2\{[(B+C-A)]$$
$$[(AB)(C+A-B)+AB^2]+A^2[0+B(C+A-B)]\}$$
$$= 2AB(A+B+C)^2[(B+C-A)(C+A-B+B)$$
$$+A(C+A-B)]$$
$$= 2AB(A+B+C)^2[BC+C^2-CA+AB$$
$$+AC-A^2+AC+A^2-AB]$$
$$= 2AB(A+B+C)^2[BC+C^2+AC]$$
$$= 2ABC(A+B+C)^3$$
$$= 2s^3(s-a)(s-b)(s-c)$$

38. Applying $C_1 \to C_1 + C_2 + C_3$, we have

$$\Delta = (\alpha\beta+\beta\gamma+\gamma\alpha)\begin{vmatrix} 1 & \beta\gamma & \gamma\alpha \\ 1 & \gamma\alpha & \alpha\beta \\ 1 & \alpha\beta & \beta\gamma \end{vmatrix}$$

Also, since $\alpha, \beta, \gamma \in px^3 + qx^2 + r = 0$,

$$\therefore \quad S_2 = \alpha\beta+\beta\gamma+\gamma\alpha = (-1)^2\frac{0}{p} = 0$$

Hence $\Delta = 0$

39. $\int f(x)dx = \begin{vmatrix} x & x^2 & x^3 \\ 3 & a & 27 \\ 1 & 3 & 9 \end{vmatrix}$

$$\therefore \int_0^3 f(x)dx = \begin{vmatrix} 3 & 9 & 27 \\ 3 & a & 27 \\ 1 & 3 & 9 \end{vmatrix} - \begin{vmatrix} 0 & 0 & 0 \\ 3 & a & 27 \\ 1 & 3 & 9 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 9 & 27 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

Hence, a can be any real number.

40. Differentiating both sides w.r.t. x, we have

$$\begin{vmatrix} 1 & 1 & 2x \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} + \begin{vmatrix} 1+x & x & x^2 \\ 1 & 1 & 2x \\ x^2 & x & 1+x \end{vmatrix} + \begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ 2x & 1 & 1 \end{vmatrix}$$
$$= 5ax^4 + 4bx^3 + 3cx^2 + 2dx + \lambda$$

Putting $x = 0$, we have

$$\lambda = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

$\Rightarrow \quad \lambda = 1 + 1 + 1 = 3$

41. Taking a common from R_1 and C_1, b from R_2 and C_2, c from R_3 and C_3, we have

$$\text{L.H.S.} = a^2b^2c^2\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

Applying $R_2 \to R_2 + R_1, R_3 \to R_3 + R_1$,

$$= a^2b^2c^2\begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix} = 4a^2b^2c^2$$

$\therefore \quad \lambda = 4$

42. Applying $R_1 \to R_1 + R_2 + R_3$, we get the given determinant

$$\begin{vmatrix} 3+a & 3+a & 3+a \\ 1 & 1+a & 1 \\ 1 & 1 & 1+a \end{vmatrix}$$

$$= (3+a)\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+a & 1 \\ 1 & 1 & 1+a \end{vmatrix}$$

Applying $C_3 \to C_3 - C_1, C_2 \to C_2 - C_1$

$$= (3+a)\begin{vmatrix}1 & 0 & 0\\ 0 & a & 0\\ 0 & 0 & a\end{vmatrix} = (3+a)a^2$$

$$= a^3\left(1+\frac{3}{a}\right)$$

43. Since $\begin{vmatrix}a & a+b & a+b+c\\ 2a & 3a+2b & 4a+3b+2c\\ 3a & 6a+3b & 10a+6b+3c\end{vmatrix} = 64,$

applying $R_2 \to R_2 - 2R_1; R_3 \to R_3 - 3R_1$, we get

$$\begin{vmatrix}a & a+b & a+b+c\\ 0 & a & 2a+b\\ 0 & 3a & 7a+3b\end{vmatrix} = 64$$

Expanding along C_1, we have

$$a(7a^2+3ab-6a^2-3ab) = 64$$

$\Rightarrow \quad a(a^2) = 64$

$\therefore \quad a = 4$

44. $\Delta = \begin{vmatrix}1 & \frac{\log y}{\log x} & \frac{\log z}{\log x}\\ \frac{\log x}{\log y} & 2 & \frac{\log z}{\log y}\\ \frac{\log x}{\log z} & \frac{\log y}{\log z} & 4\end{vmatrix}$

$$\Rightarrow \quad \Delta = \frac{1}{\log x \log y \log z}\begin{vmatrix}\log x & \log y & \log z\\ \log x & 2\log y & \log z\\ \log x & \log y & 4\log z\end{vmatrix}$$

$$\Rightarrow \quad \Delta = \frac{1}{\log x \log y \log z}\begin{vmatrix}\log x & \log y & \log z\\ 0 & \log y & 0\\ 0 & 0 & 3\log z\end{vmatrix}$$

$$= \frac{3\log x \log y \log z}{\log x \log y \log z} = 3 \text{ (diagonal property)}$$

45. $\Delta = \begin{vmatrix}a^2 & b^2 & c^2\\ (a+1)^2 & (b+1)^2 & (c+1)^2\\ (a-1)^2 & (b-1)^2 & (c-1)^2\end{vmatrix}$

$$= \begin{vmatrix}a^2 & b^2 & c^2\\ 4a & 4b & 4c\\ (a-1)^2 & (b-1)^2 & (c-1)^2\end{vmatrix}$$

[Applying $R_2 \to R_2 - R_3$]

$$= 4\begin{vmatrix}a^2 & b^2 & c^2\\ a & b & c\\ (a-1)^2 & (b-1)^2 & (c-1)^2\end{vmatrix}$$

$$= 4\begin{vmatrix}a^2 & b^2 & c^2\\ a & b & c\\ 1-2a & 1-2b & 1-2c\end{vmatrix}$$ [Applying $R_3 \to R_3 - R_1$]

Applying $R_3 \to R_3 + 2R_2$

$$= 4\begin{vmatrix}a^2 & b^2-a^2 & c^2-a^2\\ a & b-a & c-a\\ 1 & 0 & 0\end{vmatrix}$$

$$= 4\begin{vmatrix}a^2 & b^2 & c^2\\ a & b & c\\ 1 & 1 & 1\end{vmatrix}$$

$= 4(b-a)(b-c)(c-a) = 0$

Either $b-a=0$ or $b-c=0$ or $c-a=0$

Either $a=b$ or $b=c$ or $c=a$

i.e. ΔABC is isosceles.

46. $\Sigma 2^{r-1} = 1+2+2^2+\cdots+2^{n-1} = 1\cdot\frac{2^{n-1}}{2-1} = 2^{n-1}$

$$\Sigma 2\cdot 3^{r-1} = 2(1+3+3^2+\cdots+3^{n-1}) = \frac{2(3^{n-1})}{3-1} = 3^{n-1}$$

$$\Sigma 4\cdot 5^{r-1} = 4(1+5+5^2+\cdots+5^{n-1}) = \frac{4(5^{n-1})}{5-1} = 5^{n-1}$$

$$\therefore \quad \Sigma D_r = \begin{vmatrix}2^{n-1} & 3^{n-1} & 5^{n-1}\\ \alpha & \beta & \gamma\\ 2^{n-1} & 3^{n-1} & 5^{n-1}\end{vmatrix} = 0$$

[$\because R_1$ and R_3 are identical]

47. $\Delta = \begin{vmatrix}a & b & c\\ c & a & b\\ b & c & a\end{vmatrix} = 0$

$$\Rightarrow \quad (a+b+c)\begin{vmatrix}1 & b & c\\ 1 & a & b\\ 1 & c & a\end{vmatrix} = 0 \ (C_1 \to C_1 + C_2 + C_3)$$

48. We have

$$f(-x) = \begin{vmatrix}-x^3 & \cos^2 x & 2^{x^4}\\ -\tan^5 x & 1 & \sec 2x\\ -\sin^3 x & x^4 & 5\end{vmatrix} = -f(x)$$

$$\therefore \quad \int_{\pi/2}^{\pi/2} f(x)dx = 0$$

49. Differentiating w.r.t. x, we get

$$f'(x) = (1+2x)\begin{vmatrix} -\sin(x+x^2) & \cos(x+x^2) & \sin(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin(2x^2) \end{vmatrix}$$

$$+(1-2x)\begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \cos(x-x^2) & -\sin(x-x^2) & \cos(x-x^2) \\ \sin 2x & 0 & \sin(2x^2) \end{vmatrix}$$

$$+2\begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \cos(2x) & 0 & 2x\cos(2x^2) \end{vmatrix}$$

$$\Rightarrow f'(0) = \begin{vmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{vmatrix} + 2\begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$$

$$= 0+0+2(1) = 2$$

50. $\begin{vmatrix} a & 1 & 4 \\ 4 & a & 8 \\ 2 & 1 & 2a \end{vmatrix} = 0$

$\Rightarrow 2a^3 - 24a + 32 = 0$

$\Rightarrow 2(a-2)(a^2+2a-8) = 0$

$\Rightarrow a = 2, -4$

51. Since $f(x) = \begin{vmatrix} \sec^2 x & 1 & 1 \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cot^2 x \end{vmatrix}$,

applying $C_2 \to C_2 - \cos^2 x C_1$, we get

$$\Rightarrow f(x) = \begin{vmatrix} \sec^2 x & 0 & 1 \\ \cos^2 x & \cos^2 x - \cos^4 x & \operatorname{cosec}^2 x \\ 1 & 0 & \cot^2 x \end{vmatrix}$$

Expanding along C_2, we have

$$f(x) = (\cos^2 x - \cos^4 x)(\sec^2 x \cot^2 x - 1)$$

$\Rightarrow f(x) = \cos^2 x \sin^2 x (\operatorname{cosec}^2 x - 1)$

$\therefore f(x) = \cos^2 x \sin^2 x \cot^2 x = \cos^4 x$

52. $\sum_{r=1}^{n} S_r = \begin{vmatrix} \sum_{r=1}^{n} 2r & x & n(n+1) \\ \sum_{r=1}^{n} (6r^2-1) & y & n^2(2n+3) \\ \sum_{r=1}^{n} (4r^3 - 2nr) & z & n^3(n+1) \end{vmatrix}$

$$= \begin{vmatrix} n(n+1) & x & n(n+1) \\ n(n+1)(2n+1) - n & y & n^2(2n+3) \\ n^2(n+1)^2 - n^2(n+1) & z & n^3(n+1) \end{vmatrix}$$

$$= \begin{vmatrix} n(n+1) & x & n(n+1) \\ n^2(2n+3) & y & n^2(2n+3) \\ n^3(n+1) & z & n^3(n+1) \end{vmatrix} = 0$$

[$\because$ C_1 and C_3 are identical]

which is independent of x, y, z, n.

53. Since the the given system of equations has a non-trivial solution

$$\therefore \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$, we get

$$\begin{vmatrix} a & 1 & 1 \\ 1-a & b-1 & 0 \\ 1-a & 0 & c-1 \end{vmatrix} = 0$$

$\Rightarrow a(b-1)(c-1) - (1-a)(c-1) - (1-a)(b-1) = 0$

Dividing by $(1-a)(1-b)(1-c)$, we get

$$\frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0$$

$$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

54. $\begin{vmatrix} 3 & k \\ k & k \end{vmatrix} = 0 \Rightarrow k = 0, 3$ ($k = 0$ is not possible)

$$\Delta_1 = \begin{vmatrix} k & 2 \\ k & 4 \end{vmatrix} \neq 0 \text{ if } k = 3$$

i.e. at $k = 3$, the system is inconsistent.

55. Since x, y, z are in A.P.,

$\Rightarrow 2y = x + z$ (i)

Now, $|A| = \begin{bmatrix} 4 & 5 & 6 & x \\ 5 & 6 & 7 & y \\ 6 & 7 & 8 & z \\ x & y & z & 0 \end{bmatrix}$

Applying $R_2 \to 2R_2 - (R_1 + R_3)$, we get

$$\Rightarrow |A| = \begin{bmatrix} 4 & 5 & 6 & x \\ 0 & 0 & 0 & 0 \\ 6 & 7 & 8 & 0 \\ x & y & z & 0 \end{bmatrix}$$ [using (i)]

$\therefore |A| = 0$

56. $\sin 2x + 1 = 0 \Rightarrow 2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4}$

$$\therefore \begin{vmatrix} 0 & \cos x & -\sin x \\ \sin x & 0 & \cos x \\ \cos x & \sin x & 0 \end{vmatrix}^2 = \begin{vmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{vmatrix}^2$$

$$= \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right)^2 \begin{vmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}^2$$

$$= \frac{1}{8}[1(0-1)-1(1-0)]^2 = \frac{1}{8}(-2)^2 = \frac{1}{2}$$

57. The given system of linear equations will have a non-trivial solution if

$$\begin{vmatrix} \lambda & \sin\alpha & \cos\alpha \\ 1 & \cos\alpha & \sin\alpha \\ -1 & \sin\alpha & -\cos\alpha \end{vmatrix} = 0$$

Expanding the determinant along C_1, we get

$\lambda(-\cos^2\alpha - \sin^2\alpha) - (-\sin\alpha\cos\alpha - \sin\alpha\cos\alpha)$

$-(\sin^2\alpha - \cos^2\alpha) = 0$

$\Rightarrow \quad -\lambda + \sin 2\alpha + \cos 2\alpha = 0$

$\Rightarrow \quad \lambda = \sin 2\alpha + \cos 2\alpha = \sqrt{2}\sin(\pi/4 + 2\alpha)$

$\Rightarrow \quad -\sqrt{2} \le \lambda \le \sqrt{2}$

58. For a unique solution, $\Delta \neq 0$

$$\Rightarrow \begin{vmatrix} 2 & a & 6 \\ 1 & 2 & b \\ 1 & 1 & 3 \end{vmatrix} \neq 0 \quad \Rightarrow \begin{vmatrix} 0 & a-2 & 0 \\ 0 & 1 & b-3 \\ 1 & 1 & 3 \end{vmatrix} \neq 0$$

$(R_1 \to R_1 - 2R_3, R_2 \to R_2 - R_3)$

$\Rightarrow \quad (a-2)(b-3) \neq 0 \quad \Rightarrow a \neq 2, b \neq 3$

59. Let $\Delta = \begin{bmatrix} x & x^2 - yz & 1 \\ y & y^2 - zx & 1 \\ z & z^2 - xy & 1 \end{bmatrix}$

$$\Rightarrow \quad \Delta = \begin{bmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{bmatrix} + \begin{bmatrix} x & -yz & 1 \\ y & -zx & 1 \\ z & -xy & 1 \end{bmatrix}$$

$$\Rightarrow \quad \Delta = \begin{bmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{bmatrix} - \frac{1}{xyz}\begin{bmatrix} x^2 & xyz & x \\ y^2 & zxy & y \\ z^2 & xyz & z \end{bmatrix}$$

$$\Rightarrow \quad \Delta = \begin{bmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{bmatrix} - \begin{bmatrix} x^2 & 1 & x \\ y^2 & 1 & y \\ z^2 & 1 & z \end{bmatrix}$$

In determinant II, interchanging the first $C_3 \leftrightarrow C_2$ and then $C_2 \leftrightarrow C_1$, we get

$$\Delta = \begin{bmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{bmatrix} - \begin{bmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{bmatrix} = 0$$

60. Applying $R_1 \to R_1 + R_2 + R_3$, we get

$$\begin{vmatrix} 2(x+y+z) & x+y+z & x+y+z \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$$

$$= (x+y+z)\begin{vmatrix} 2 & 1 & 1 \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$$

$$= (x+y+z)\begin{vmatrix} 0 & 1 & 1 \\ 0 & z & x \\ x-z & y & z \end{vmatrix}$$

[Operating $C_1 \to C_1 - C_2 - C_3$]

$$= (x+y+z)(x-z)^2$$

Hence, the repeated factor $= z - x$

61. A homogeneous system having infinite solutions means that it is non-trivial, i.e. $\Delta = 0$

$$\begin{vmatrix} \sin\theta & -\sqrt{3} \\ \cos\theta & 1 \end{vmatrix} = 0 \Rightarrow \tan\theta = -\sqrt{3} \Rightarrow \theta = \frac{4\pi}{3}$$

62. Since $p+q+r = 0 = a+b+c$ (given)

$\Rightarrow \quad p^3+q^3+r^3 = 3pqr$ or $a^3+b^3+c^3 = 3abc$

Let $\Delta = \begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$

$\Rightarrow \quad \Delta = pqr(a^3+b^3+c^3) - abc(p^3+q^3+r^3)$

$\therefore \quad \Delta = pqr(3abc) - abc(3pqr) = 0$

63. $$\alpha = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = \frac{1}{xyz}\begin{vmatrix} x & x^2 & xyz \\ y & y^2 & xyz \\ z & z^2 & xyz \end{vmatrix}$$

$$= \frac{xyz}{xyz}\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = \beta$$

64. Let $\Delta = \begin{vmatrix} a^2+x & ab & ac \\ ab & b^2+x & bc \\ ac & bc & c^2+x \end{vmatrix}$

Applying $C_1 \to aC_1, C_2 \to bC_2, C_3 \to cC_3$, we get

$$\Delta = \frac{1}{abc}\begin{vmatrix} a^3+ax & a^2b & a^2c \\ ab^2 & b^3+bx & b^2c \\ ac^2 & bc^2 & c^3+cx \end{vmatrix}$$

$$\Rightarrow \quad \Delta = \begin{vmatrix} a^2+x & a^2 & a^2 \\ b^2 & b^2+x & b^2 \\ c^2 & c^2 & c^2+x \end{vmatrix}$$

Applying $R_1 \to R_1 + R_2 + R_3$, we get

$$\Rightarrow \quad \Delta = (a^2+b^2+c^2+x)\begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2+x & b^2 \\ c^2 & c^2 & c^2+x \end{vmatrix}$$

$$\Rightarrow \quad \Delta = (a^2+b^2+c^2+x)\{(b^2x+c^2x+x^2) - (b^2x)+(-c^2x)\}$$

$$\Rightarrow \quad \Delta = x^2(a^2+b^2+c^2+x)$$

$\therefore$ x^2 is a factor of Δ as well as $a^2 + b^2 + c^2 + x$ is a factor of Δ.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. Apply $C_3 \to C_3 + C_2$ and take $a + b + c$ common from C_3. $\therefore$ $\Delta = 0$ as two columns C_1 and C_3 are identical.

2. Let $\Delta = \begin{bmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) & \cos 2\beta \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{bmatrix}$

Applying $R_1 \to R_1 + R_2 \sin\beta + R_3 \cos\beta$, we get

$$\Delta = \begin{bmatrix} 0 & 0 & \cos 2\beta + 1 \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{bmatrix}$$

$$\Rightarrow \quad \Delta = (\cos 2\beta + 1)(\sin^2\alpha + \cos^2\alpha)$$

$$\therefore \quad \Delta = (\cos 2\beta + 1) \text{ which is independent of } \alpha$$

3. $|A|_{\max} = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 0 - (-1) + 1(1) = 2$

$$\therefore \quad |A|_{\min} = -2$$

4. $f'(x) = \begin{vmatrix} -\sin x & \sin x & \cos x \\ -2\sin 2x & \sin 2x & 2\cos 2x \\ -3\sin 3x & \sin 3x & 3\cos 3x \end{vmatrix} +$

$$+ \begin{vmatrix} \cos x & \cos x & \cos x \\ \cos 2x & 2\cos 2x & 2\cos 2x \\ \cos 3x & 3\cos 3x & 3\cos 3x \end{vmatrix}$$

$$+ \begin{vmatrix} \cos x & \sin x & -\sin x \\ \cos 2x & \sin 2x & -4\sin 2x \\ \cos 3x & \sin 3x & -9\sin 3x \end{vmatrix}$$

$$\therefore \quad f'\left(\frac{\pi}{2}\right) = \begin{vmatrix} -1 & 1 & 0 \\ 0 & 0 & -2 \\ 3 & -1 & 0 \end{vmatrix} + 0 + \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 9 \end{vmatrix}$$

$$= 2(1-3) + 0 + 1 \cdot (9-1)$$

$$= -4 + 8 = 4$$

5. The determinant of coefficients of x, y, z is

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 5 & 5 & 9 \end{vmatrix}$$

$$\Rightarrow \quad \Delta = 1(9-15) - 2(18-15) + 3(10-5)$$

$$\Rightarrow \quad \Delta = -6 - 6 + 15 = 3 \neq 0$$

$\therefore$ The system of equations represents a unique solution.

6. $\log_x y = \dfrac{\log y}{\log x}$

Multiply R_1, R_2, R_3 by $\log x$, $\log y$ and $\log z$, respectively and divide Δ by $\log x \log y \log z$.

$\Delta = 0$ as all rows become identical.

7. $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$

$$= pa(a^2qr - p^2bc) - qb(q^2ca - b^2pr) + rc(pqc^2 - r^2ab)$$

$$= pqr(a^3+b^3+c^3) - abc(p^3+q^3+r^3)$$

$$= pqr(a^3+b^3+c^3) - abc(3pqr)$$

$$[\because p+q+r=0, p^3+q^3+r^3 = 3pqr]$$

$$= pqr(a^3+b^3+c^3-3abc)$$

$$= pqr\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

8. Since the given system of equations has a non-trivial solution, we have

$$\Delta = \begin{vmatrix} a & 4 & 1 \\ b & 3 & 1 \\ c & 2 & 1 \end{vmatrix} = 0$$

[Applying $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$]

$$\Rightarrow \quad \Delta = \begin{vmatrix} a-b & 1 & 0 \\ b-c & 1 & 0 \\ c & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \quad (a-b)-(b-c) = 0$$

$$\Rightarrow \quad a - b = b - c$$

or, $b = \dfrac{a+c}{2}$

$\therefore$ a, b, c are in A.P.

9. $\Delta = \Delta_1 - \Delta_2 = 0$ as $\Delta_2 = \Delta_1$

$$\Delta_2 = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \frac{1}{abc}\begin{vmatrix} a & a^2 & abc \\ b & b^2 & abc \\ c & c^2 & abc \end{vmatrix}$$

$$= \frac{abc}{abc}\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} = \Delta_1$$

10. $\Delta = \begin{vmatrix} 1 & 2 & 6 \\ 2 & 6 & 24 \\ 6 & 24 & 120 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 6 \\ 0 & 2 & 12 \\ 0 & 12 & 84 \end{vmatrix}$

[Operating $R_3 \to R_3 - 6R_1, R_2 \to R_2 - 2R_1$]

$= 168 - 144 = 24 = 4!$

11. The value of x, y, z are non-trivial if $\begin{vmatrix} 3 & k & -2 \\ 1 & k & 3 \\ 2 & 3 & -4 \end{vmatrix} = 0$

$$\Rightarrow \quad 3(-4k-9) - 1(-4k+6) + 2(3k+2k) = 0$$

$$\Rightarrow \quad -12k - 27 + 4k - 6 + 10k = 0$$

$$\Rightarrow \quad 2k = 33 \quad \therefore \quad k = \frac{33}{2}$$

12. Taking $(b-a)$ common from each of C_1 and C_3, we have

$$\Delta = (b-a)^2 \begin{vmatrix} b & b-c & c \\ a & a-b & b \\ c & c-a & a \end{vmatrix}$$

Now applying $C_1 \to C_1 - C_3$, the new C_1 and C_2 become identical

$\therefore \qquad \Delta = 0$

13. $\begin{vmatrix} b^2+c^2 & ab & ac \\ ab & c^2+a^2 & bc \\ ac & bc & a^2+b^2 \end{vmatrix}$

$$= \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} \qquad [\because\ a^2+b^2+c^2 = 0]$$

$$= abc\begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix} = a^2b^2c^2\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= a^2b^2c^2\begin{vmatrix} 0 & 0 & 2 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 4a^2b^2c^2$$

Given: $4a^2b^2c^2 = ka^2b^2c^2 \quad \Rightarrow \quad k = 4$

14. For the system of equations to have non-trivial solutions,

$$\Delta = \begin{vmatrix} 2 & -1 & 1 \\ 1 & -2 & 1 \\ \lambda & -1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow \quad 2(-4+1) + 1(2-\lambda) + 1(-1+2\lambda) = 0$$

$$\Rightarrow \quad -6 + 2 - \lambda - 1 + 2\lambda = 0$$

$$\therefore \quad \lambda = 5$$

15. $a = p$th term of H.P.

$$\therefore \qquad \frac{1}{a} = p\text{th term of A.P.} = A + (p-1)D$$

$$\Delta = abc\begin{vmatrix} 1/a & 1/b & 1/c \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$$

[We have divided R_1 by abc and multiplied Δ by abc.]

$$= abc\begin{vmatrix} A+(p-1)D & A+(q-1)D & A+(r-1)D \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$$

$= \Delta_1 + \Delta_2$

Each of them is zero because of identical rows.

16. $\Delta = \begin{vmatrix} a^3 & b^3 & c^3 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} + \begin{vmatrix} -1 & -1 & -1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$$= abc\begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix} - 1\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= (abc-1)\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= (abc-1)(a-b)(b-c)(c-a)$$

$\therefore \quad \Delta = 0 \Rightarrow abc - 1 = 0 \qquad [\because a \neq b \neq c]$

$\Rightarrow \quad abc = 1$

17. The linear system of equations will possess a solution if

$$\begin{vmatrix} 2 & 3 & -8 \\ 7 & -5 & 3 \\ 4 & -6 & \lambda \end{vmatrix} = 0$$

$\Rightarrow \quad 2(-5\lambda + 18) - 3(7\lambda - 12) - 8(-42 + 20) = 0$

$\Rightarrow \quad -10\lambda + 36 - 21\lambda + 36 + 176 = 0$

$\Rightarrow \quad 31\lambda = 248$

$\therefore \quad \lambda = 8$

18. $\sum_{n=1}^{5} U_n = U_1 + U_2 + U_3 + U_4 + U_5$

Putting $n = 5$ in the formula for $\sum n, \sum n^2, \sum n^3$, we get

$$U_n = \begin{vmatrix} 15 & 15 & 8 \\ 55 & 35 & 9 \\ 225 & 25 & 10 \end{vmatrix} = \begin{vmatrix} 0 & 15 & 8 \\ 20 & 35 & 9 \\ 200 & 25 & 10 \end{vmatrix}$$

by applying $C_1 \to C_1 - C_2$

$$= 20 \times 5 \begin{vmatrix} 0 & 3 & 8 \\ 1 & 7 & 9 \\ 10 & 5 & 10 \end{vmatrix}$$

$$= 100[-1(70-45) + 10(27-56)]$$

$$= 100(-25 - 290) = 100 \times -315$$

$$= -31500$$

19. Differentiating both sides w.r.t. x, we get

$$\begin{vmatrix} 1 & 1 & 2x \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} + \begin{vmatrix} 1+x & x & x^2 \\ 1 & 1 & 2x \\ x^2 & x & 1+x \end{vmatrix} + \begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ 2x & 1 & 1 \end{vmatrix}$$

$$= 5ax^4 + 4bx^3 + 3cx^2 + 2dx + \lambda$$

Putting $x = 0$, we get

$$\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \lambda$$

$\Rightarrow \quad 1 + 1 + 1 = \lambda$

$\Rightarrow \quad \lambda = 3$

20. The system of equations has infinite solutions if

$$\Delta = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix} = 0$$

$\Rightarrow \quad 3(2\lambda + 15) - 1(-\lambda - 20) + 6(3-8) = 0$

$\Rightarrow \quad 6\lambda + 45 + \lambda + 20 - 30 = 0$

$\Rightarrow \quad 7\lambda = -35 \quad \therefore \quad \lambda = -5$

21. Applying $R_1 \to R_1 + R_3 - 2R_2$

$$\Delta = \begin{vmatrix} 0 & 0 & 0 & x+z-2y \\ 5 & 6 & 7 & y \\ 6 & 7 & 8 & z \\ x & y & z & 0 \end{vmatrix} = -(x+z-2y)\begin{vmatrix} 5 & 6 & 7 \\ 6 & 7 & 8 \\ x & y & z \end{vmatrix}$$

Again applying $C_1 + C_3 - 2C_2$, we get

$$\Delta = (x + z - 2y)^2$$

22. Here $\Delta = 0$

$$\Rightarrow \begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$$

[Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$]

$$\Rightarrow \quad (1+\sin^2\theta) + \cos^2\theta + 4\sin 4\theta = 0$$

$$\Rightarrow \quad 2 + 4\sin 4\theta = 0 \Rightarrow \sin 4\theta = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right)$$

$$\therefore \quad 4\theta = n\pi + (-1)^n\left(-\frac{\pi}{6}\right)$$

$$\Rightarrow \quad \theta = \frac{n\pi}{4} + (-1)^{n+1}\frac{\pi}{24}$$

When $n = 2, \theta = \dfrac{11\pi}{24}$

23. $$\Delta = \begin{vmatrix} 2a_1b_1 & a_1b_2 + a_2b_1 & a_1b_3 + a_3b_1 \\ a_1b_2 + a_2b_1 & 2a_2b_2 & a_2b_3 + a_3b_2 \\ a_1b_3 + a_3b_1 & a_3b_2 + a_2b_3 & 2a_3b_3 \end{vmatrix}$$

$$\therefore \quad \Delta = \begin{vmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ a_3 & b_3 & 0 \end{vmatrix}\begin{vmatrix} b_1 & a_1 & 0 \\ b_2 & a_2 & 0 \\ b_3 & a_3 & 0 \end{vmatrix} = 0$$

24. Applying $C_1 \to C_1 + C_2 + C_3$ and putting $1 + \omega + \omega^2 = 0$, we have

$$\Delta = \lambda\begin{vmatrix} 1 & \omega & \omega^2 \\ 1 & \lambda + \omega^2 & 1 \\ 1 & 1 & \lambda + \omega \end{vmatrix}$$

[Expanding and not making two zeros]

$$= \lambda\{(\lambda^2 - \lambda) - 1(\lambda\omega) + 1(\omega - \lambda\omega^2 - \omega)\}$$

$$= \lambda[\lambda^2 - \lambda(1 + \omega + \omega^2)] = \lambda^3$$

25. $\Delta = x^n y^n z^n \begin{vmatrix} 1 & x^2 & x^4 \\ 1 & y^2 & y^4 \\ 1 & z^2 & z^4 \end{vmatrix}$

$= (xyz)^n (x^2 - y^2)(y^2 - z^2)(z^2 - x^2)$

$= \left(\frac{1}{y^2} - \frac{1}{x^2}\right)\left(\frac{1}{z^2} - \frac{1}{y^2}\right)\left(\frac{1}{x^2} - \frac{1}{z^2}\right)$ when $n = -4$

26. Since $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0,$

$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} (1 + xyz) = 0$

$\Rightarrow (x - y)(y - z)(z - x)(1 + xyz) = 0$

$\Rightarrow xyz + 1 = 0 \qquad [\because x \neq y \neq z \text{ (given)}]$

$\therefore \quad xyz = -1$

27. $f'(x) = -\begin{vmatrix} \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\alpha) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$

$+ \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(\beta-\gamma) & \sin(1-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$

$= \quad -0 + 0 = 0$

$\Rightarrow$ $f(x)$ is a constant function = c (say)

$\therefore \quad f(\theta) - 2f(\theta) + f(\theta) = c - 2c + c = 0$

28. $\begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix} = \lambda$, then

interchanging $C_3 \leftrightarrow C_2$ and according to properties of determinants, we have

$\therefore \quad \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} = -\begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix} = -\lambda$

29. Operating $C_1 \to C_1 + C_2 + C_3$, we get $\Delta = 0$

30. Let $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ (2^x + 2^{-x})^2 & (3^x + 3^{-x})^2 & (5^x + 5^{-x})^2 \\ (2^x - 2^{-x})^2 & (3^x - 3^{-x})^2 & (5^x - 5^{-x})^2 \end{vmatrix}$

Applying $R_2 \to R_2 - R_3$, we get

$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 4 \cdot 2^x \cdot 2^{-x} & 4 \cdot 3^x \cdot 3^{-x} & 4 \cdot 5^x \cdot 5^x \\ (2^x - 2^{-x})^2 & (3^x - 3^{-x})^2 & (5^x - 5^{-x})^2 \end{vmatrix}$

$\Rightarrow \quad \Delta = 4 \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ (2^x - 2^{-x})^2 & (3^x - 3^{-x})^2 & (5^x - 5^{-x})^2 \end{vmatrix}$

$\therefore \quad \Delta = 0$

31. Applying $C_1 \to C_1 - C_3$, and taking out $x - 4$ and expanding, we have

$$\Delta = (x-4)(x^2 + 4x - 25) = 0$$

The non-integral roots are given by the second factor and their sum S is -4.

32. Applying $C_1 \to C_1 - C_2$, we get

$$\Delta = \begin{vmatrix} 4 & (\sin\theta - \operatorname{cosec}\theta)^2 & 1 \\ 4 & (\cos\theta - \sec\theta)^2 & 1 \\ 4 & (\tan\theta - \cot\theta)^2 & 1 \end{vmatrix} = 0$$

33. L.H.S. $= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$

$= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix}$

(Applying $R_2 \to R_2 + R_1$ and $R_3 \to R_3 + R_1$)

L.H.S. $= 4a^2b^2c^2 = \alpha a^2b^2c^2$

$\therefore \quad \alpha = 4$

34. Multiplying R_1, R_2, R_3 by a, b, c respectively and hence dividing Δ_1 by abc, and then taking out a, b, c common from C_1, C_2, C_3 respectively, we get $\Delta_1 = \Delta_2$.

Now applying $R_1 \to R_1 - (R_2 + R_3)$ on Δ_2, we get

$$\Delta_2 = \begin{vmatrix} 0 & -2c^2 & -2b^2 \\ b^2 & c^2 + a^2 & b^2 \\ c^2 & c^2 & a^2 + b^2 \end{vmatrix}$$

Expanding with R_1

$\Delta_2 = 2c^2b^2(a^2 + b^2 - c^2) - 2b^2c^2(b^2 - c^2 - a^2)$

$= 2b^2c^2(2a^2) = 4a^2b^2c^2$

$\therefore \quad \lambda = 4$

35. Expanding the determinant, we have

$$(2-y)(26-15y+y^2)-2(20-2y-18)$$
$$+3(8-15+3y)=0$$

$\Rightarrow$ $-y^3+17y^2-43y+27=0$ or $y^3-17y^2-43y-27=0$

$\therefore$ We have $(y-1)(y^2-16y+27)=0$

$\therefore$ Either $y=1$ or $y^2-16y+27=0$

$$\Rightarrow \quad y=\frac{16\pm\sqrt{256-108}}{2}=\frac{8\pm\sqrt{148}}{2}$$

which are not integers

Hence, the only integral root is $y=1$.

36. Since x, y, z are in A.P., therefore, $x+z-2y=0$.

Now operating $R_1 \to R_1+R_3-2R_2$, we get

$$\Delta=\begin{bmatrix} 0 & 0 & 2(x+z-2y) \\ a+3 & a+4 & a+2y \\ a+4 & a+5 & a+2z \end{bmatrix}$$

$$=\begin{bmatrix} 0 & 0 & 0 \\ a+3 & a+4 & a+2y \\ a+4 & a+5 & a+2z \end{bmatrix}=0$$

37. Dividing C_1, C_2, C_3 by a, b, c respectively and then multiplying Δ by abc, we get

$$\Delta=abc\begin{vmatrix} \frac{1}{a}+1 & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b}+1 & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c}+1 \end{vmatrix}$$

Now applying $C_1+C_2+C_3$ and taking $1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ common and then making two zeros, we have

$$\therefore \quad \Delta=abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)=abc\left(1+\sum\frac{1}{a}\right)$$

38. Applying $C_1 \to C_1 \to C_1+C_3-2\cos x\, C_2$, we get

$$\Delta=\begin{vmatrix} a^2-2a\cos x+1 & a & 1 \\ 0 & \cos(n+1)x & \cos(n+2)x \\ 0 & \sin(n+1)x & \sin(n+2)x \end{vmatrix}$$

$[\because \cos nx+\cos(n+2)x=2\cos(n+1)x\cos x$ and $\sin nx+\sin(n+2)x=2\sin(n+1)x\cos x]$

Expanding along C_1, we get

$$\Delta=(a^2-2a\cos x+1)[\cos(n+1)x\sin(n+2)x$$
$$-\sin(n+1)x\cos(n+2)x]$$
$$=(a^2-2a\cos x+1)\sin[(n+2)x-(n+1)x]$$
$$=(a^2-2a\cos x+1)\sin x$$

which is independent of n.

39. $$\Delta=\frac{1}{abc}\begin{vmatrix} ab^2c^2 & abc & ab+ac \\ a^2bc^2 & abc & bc+ab \\ a^2b^2c & abc & ac+bc \end{vmatrix}$$

$$=\frac{a^2b^2c^2}{abc}\begin{vmatrix} bc & 1 & ab+ac \\ ac & 1 & bc+ab \\ ab & 1 & ac+bc \end{vmatrix}$$

[Applying $C_1 \to C_1+C_3$]

$$=abc\begin{vmatrix} ab+bc+ca & 1 & ab+ac \\ ab+bc+ca & 1 & bc+ab \\ ab+bc+ca & 1 & ac+bc \end{vmatrix}$$

$$=abc(ab+bc+ca)\begin{vmatrix} 1 & 1 & ab+ac \\ 1 & 1 & bc+ab \\ 1 & 1 & ac+bc \end{vmatrix}$$

$$=0$$

40. Dividing C_1, C_2, C_3 by x, y, z respectively, we get

$$\Delta=xyz\begin{vmatrix} \frac{p}{x} & \frac{q}{y}-1 & \frac{r}{z}-1 \\ \frac{p}{x}-1 & \frac{q}{y} & \frac{r}{z}-1 \\ \frac{p}{x}-1 & \frac{q}{y}-1 & \frac{r}{z} \end{vmatrix}=0$$

Applying $C_1 \to C_1+C_2+C_3$, taking $\left(\Sigma\frac{p}{x}-2\right)$ common, then making two zeros and expanding, we get

$$\therefore \quad \Delta=xyz\left(\Sigma\frac{p}{x}-2\right)=0 \ \therefore \Sigma\frac{p}{x}=2$$

41. Using the sum property, we get

$$\sum_{r=0}^{m}\Delta_r=\begin{vmatrix} \sum_{r=0}^{m}(2r-1) & \sum_{r=0}^{m}{}^mC_r & \sum_{r=0}^{m}1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix}$$

But $\sum_{r=0}^{m}(2r-1)=\frac{1}{2}(m+1)(2m-1-1)=m^2-1$

$\sum_{r=0}^{m}{}^mC_r=2^m$ and $\sum_{r=0}^{m}1=m+1$. Therefore

$$\sum_{r=0}^{m}\Delta_r=\begin{vmatrix} m^2-1 & 2^m & m+1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix}=0$$

42. Since a, b and c are the roots of $x^3 + px + q = 0$,

$\therefore \quad a+b+c=0$

$\therefore \quad \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Applying $R_1 \to R_1 + R_2 + R_3$, we get

$$= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$= 0$

43. Applying $R_1 \to R_1 + R_3 - 2R_2$ and taking 2 common from R_1, we get

$$\Delta = 2\begin{vmatrix} 1 & 1 & 1 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix}$$

Then making two zeros by $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$ and expanding, we get

$$2\begin{vmatrix} 1 & 0 & 0 \\ (x-1)^2 & 2x-1 & 4x \\ x^2 & 2x+1 & 4x+4 \end{vmatrix} = 2[(2x-1)(4x+4) - 4x(2x+1)]$$

$$= 2(-4) = -8$$

44. Explanation: $A = \begin{vmatrix} 0 & 2 & 1 \\ -2 & 0 & 3 \\ -1 & -3 & 0 \end{vmatrix}$

$A = 0(0+9) - 2(0+3) + 1(6-0)$

$A = 0$

The odd order skew-symmetric determinant is always equal to zero. The even order skew-symmetric determinant is always equal to a perfect square.

So, Assertion is true but Reason is not true.

45. Applying $C_1 \to C_1 + C_2 + C_3$ and taking common from C_1, we get

$$1 + \omega^n + \omega^{2n}\begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ 1 & 1 & \omega^n \\ 1 & \omega^{2n} & 1 \end{vmatrix}$$

$1 + \omega^n + \omega^{2n} = 0 \qquad [\because n \in N]$

Both Assertion and Reason are true and Assertion follows from Reason.

46. The minor of 5 in the determinant is

$$5 = M_{12} = \begin{vmatrix} 0 & 7 \\ -1 & 6 \end{vmatrix} = 0 + 7 = 7$$

So, Assertion is not true but Reason is true.

47. $D = \begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix}$

$$= \begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} + \begin{vmatrix} a & a^3 & -1 \\ b & b^3 & -1 \\ c & c^3 & -1 \end{vmatrix}$$

$$= abc\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} - \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = 0$$

$D \Rightarrow abc\,(D_1) - (D_2) = 0 \qquad$ (i)

Now, $D_1 = abc\begin{vmatrix} \frac{1}{a} & a & a^2 \\ \frac{1}{b} & b & b^2 \\ \frac{1}{c} & c & c^2 \end{vmatrix} = \begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix}$

Applying operations, we get

$$D_1 = (ab+bc+ca)\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

Applying operations $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$

$$D_1 = (ab+bc+ca)(a-b)(b-c)(c-a) \qquad \text{(ii)}$$

$$D_2 = \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$$

Applying operations $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$

$$D_2 = (a-b)(b-c)(c-a)(a+b+c) \qquad \text{(iii)}$$

From (i), (ii) and (iii), we get

$abc(ab+bc+ca) = (a+b+c)$ [$\because$ a, b, c are different]

So, both Assertion and Reason are true.

48. $D(x) = \lambda_1 f_1(x) - \lambda_2 f_2(x) + \lambda_3 f(x)$

where $\lambda_1 = (bf - ce); \lambda_2 = (af - cd); \lambda_3 = (ae - bd)$

Then,

$$\int D dx = \int \lambda_1 f_1(x)dx + \int \lambda_2 f_2(x)dx + \int \lambda_3 f_3(x)dx + k$$

$$= \begin{vmatrix} \int f_1(x)dx & \int f_2(x)dx & \int f_3(x)dx \\ a & b & c \\ d & e & f \end{vmatrix}$$

Assertion is true and follows from Statement 2.

49. If $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$,

then $\begin{vmatrix} ka & kb & kc \\ kx & ky & kz \\ kp & kq & kr \end{vmatrix} = k^3 \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = k^3\Delta$

Thus, both Assertion and Reason are true and Assertion follows from Reason.

50. Applying $R_1 \to R_1 + R_3$, we get

$$\Delta = \begin{vmatrix} 1-i & -1 & \omega^2 - 1 \\ 1-i & -1 & \omega^2 - 1 \\ -i & -1+\omega-i & -1 \end{vmatrix} = 0$$

51. $x + iy = \begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix}$

$$= -3i \begin{vmatrix} 6i & 1 & 1 \\ 4 & -1 & -1 \\ 20 & i & i \end{vmatrix} = 0$$

$\Rightarrow x = 0, y = 0$

52. If n is a multiple of 3, then all the rows are identical and $\Delta = 0$.

If n is not a multiple of 3, then $1 + \omega^n + \omega^{2n} = 0$.

Since $1 + \omega + \omega^2 = 0$,

applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\Delta = \begin{vmatrix} 0 & \omega^n & \omega^{2n} \\ 0 & \omega^{2n} & 1 \\ 0 & 1 & \omega^n \end{vmatrix}$$

$\Rightarrow \Delta = 0$ for all integers n.

53. Operating $C_2 \to C_2 - 2C_3$, we get

for non-zero solution $\begin{vmatrix} 1 & 0 & a \\ 1 & b & b \\ 1 & 2c & c \end{vmatrix} = 0$

$$\Rightarrow \quad 2ac = ab + bc \quad \Rightarrow \quad b = \frac{2ab}{a+c}$$

$\Rightarrow$ a, b, c are in H.P.

54. $a_n = a_1 r^{n-1}$, r is the common ratio.

$\log a_n = \log a_1 + (n-1)\log r$

Applying $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$ in the given determinant, we get

$$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ 3\log r & 3\log r & 3\log r \\ 6\log r & 6\log r & 6\log r \end{vmatrix} = 0$$

Since R_2 and R_3 are proportional.

55. Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$f(x) = \begin{vmatrix} 1 & (1+b^2)x & (1+c^2)x \\ 1 & 1+b^2x & (1+c^2)x \\ 1 & (1+b^2)x & 1+c^2x \end{vmatrix} \quad [\because\ a^2+b^2+c^2+2=0]$$

Applying $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$

$$= \begin{vmatrix} 1 & (1+b^2)x & (1+c^2)x \\ 0 & 1-x & 0 \\ 0 & 0 & 1-x \end{vmatrix} = 1 - x^2$$

Hence the degree of $f(x) = 2$.

56. For no solution or infinitely many solutions,

$$\begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = 0 \Rightarrow \alpha = 1, \alpha = -2.$$

But for $\alpha = 1$, clearly there are infinitely many solutions and when we put $\alpha = -2$ in the given system of equations and adding them together, L.H.S. $\neq$ R.H.S., i.e. no solution.

57. Applying $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$, we get

$$\begin{vmatrix} 0 & -x & 0 \\ 0 & x & -y \\ 1 & 1 & 1+y \end{vmatrix}$$

Expanding along R_1, $x(y) = xy$

$\therefore$ Divisible by both x and y.

58. For non-zero solution, $\begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$

$$1(1-a^2) + c(-c-ab) - b(ac+b) = 0$$

$$1 - a^2 - b^2 - c^2 - 2abc = 0$$

$$a^2 + b^2 + c^2 + 2abc = 1$$

59. As $\det(A) = \pm 1$, A^{-1} exists

and $A^{-1} = \frac{1}{\det(A)}(\text{adj}\,A) = \pm(\text{adj}\,A)$

All entries in adj(A) are integers.

$\therefore$ A^{-1} has integer entries.

60. $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ a & -b & c \end{vmatrix}$

$= \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & a-1 & a \\ b+1 & b-1 & -b \\ c-1 & c+1 & c \end{vmatrix}$

$= \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^{n+1} \begin{vmatrix} a+1 & a & a-1 \\ b+1 & -b & b-1 \\ c-1 & c & c+1 \end{vmatrix}$

$(C_2 \leftrightarrow C_3)$

$= (1+(-1)^{n+2}) \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix}$

From above it is clear that n can be any odd integer.

61. Let

$$\Delta = \begin{vmatrix} x! & (x+1)x! & (x+2)(x+1)x! \\ (x+1)! & (x+2)(x+1)! & (x+3)(x+2)(x+1)! \\ (x+2)! & (x+3)(x+2)! & (x+4)(x+3)(x+2)! \end{vmatrix}$$

Taking common $x!$, $(x+1)!$ and $(x+2)!$ from R_1, R_2 and R_3 respectively

$$= x!(x+1)!(x+2)! \begin{vmatrix} 1 & (x+1) & (x+2)(x+1) \\ 1 & (x+2) & (x+3)(x+2) \\ 1 & (x+3) & (x+4)(x+3) \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_2$,

$$= x!(x+1)!(x+2)! \begin{vmatrix} 1 & (x+1) & (x+2)(x+1) \\ 0 & 1 & 2(x+2) \\ 0 & 1 & 2(x+3) \end{vmatrix}$$

$$= x!(x+1)!(x+2)![1(2x+6-2x-4)]$$

$$= 2(x!)(x+1)!(x+2)!$$

62. $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$,

$$= \begin{vmatrix} 3 & 1 & 1 \\ 0 & -1-\omega^2 & \omega^2 \\ 0 & \omega^2 & \omega^4 \end{vmatrix} = 3(\omega^2 - \omega) = 3\omega(\omega - 1)$$

63. If $C = 2\cos\theta$, expanding along R_1,

$$\Delta = C(C^2 - 1) - 1(C - 6) + 0(1 - 6C)$$

$$= C^3 - 2C + 6$$

$$= 8\cos^3\theta - 4\cos\theta + 6$$

64. $\sum_{r=1}^{n} D_r = \begin{vmatrix} \sum_{r=1}^{n} 2^{r-1} & \sum_{r=1}^{n} 2\cdot 3^{r-1} & \sum_{r=1}^{n} 4\cdot 5^{r-1} \\ \alpha & \beta & \gamma \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix}$

Now, $\sum_{r=1}^{n} 2^{r-1} = 1 + 2 + 2^2 + \cdots + 2^{n-1} = \frac{1\cdot(2^n - 1)}{(2-1)}$

$$= 2^n - 1$$

$$\sum_{r=1}^{n} 2\cdot 3^{r-1} = 2(1 + 3 + 3^2 + \cdots + 3^{n-1}) = \frac{2(3^n - 1)}{(3-1)}$$

$$= 3^n - 1$$

$$\sum_{r=1}^{n} 4\cdot 5^{r-1} = 4[1 + 5 + 5^2 + \cdots + 5^{n-1}] = \frac{4(5^n - 1)}{(5-1)}$$

$$= 5^n - 1$$

$$\therefore \quad \sum_{r=1}^{n} D_r = \begin{vmatrix} 2^n - 1 & 3^n - 1 & 5^n - 1 \\ \alpha & \beta & \gamma \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix} = 0$$

65. Expanding along R_1, we get

$$-12(30+1) - 0(0+2) + \lambda(0-4) = -360$$

$$\Rightarrow \quad -372 - 4\lambda = -360 \quad \Rightarrow \quad -4\lambda = 12 \quad \Rightarrow \quad \lambda = -3$$

66. $\Delta = \begin{vmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{vmatrix}$

$$= \begin{vmatrix} 0 & 0 & 2 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{vmatrix} \quad (R_1 \to R_1 + R_3)$$

$$\Rightarrow \quad \Delta = 2(\sin^2\theta + 1) \qquad \text{(i)}$$

Now, $0 \le \sin^2\theta \le 1$

$$\Rightarrow \quad 1 \le 1 + \sin^2\theta \le 2 \quad \Rightarrow \quad 2 \le 2(1 + \sin^2\theta) \le 4$$

$$\Rightarrow \quad 2 \le \Delta \le 4 \qquad \text{[From Eq. (i)]}$$

So, $\Delta \in [2, 4]$

67. Operating $R_1 \to R_1 - R_2 - R_3$, we get

$$\Delta = \begin{vmatrix} 0 & -2z & -2y \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$$

Expanding, we get $\Delta = 4xyz$.

68. Statement 1: Determinant of a skew-symmetric matrix of odd order is zero.

Statement 2: $\det(A^T) = \det(A)$

$\det(-A) = (-1)^n \det(A)$, where A is an $n \times n$ order matrix.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. If $a, b, c > 0$ and $x, y, z \in \mathbb{R}$, then the determinant

$$\begin{vmatrix} (a^x + a^{-x})^2 & (a^x + a^{-x})^2 & 1 \\ (b^y + a^{-y})^2 & (b^y - a^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix} = ?$$

(a) $a^x b^y c^z$ (b) $a^{-x} b^{-y} c^{-z}$
(c) $a^{2x} b^{2y} c^{2z}$ (d) zero

2. If A, B and C are the angles of a triangle ABC, then the value of the determinant

$$\begin{vmatrix} \sin\frac{A}{2} & \sin\frac{B}{2} & \sin\frac{C}{2} \\ \sin(A+B+C) & \sin\frac{B}{2} & \cos\frac{A}{2} \\ \cos\frac{(A+B+C)}{2} & \tan(A+B+C) & \sin\frac{C}{2} \end{vmatrix}$$

is less than or equal to
(a) 1/2 (b) 1/4
(c) 1/8 (d) none of these

3. If $f(x) = \begin{vmatrix} x & \cos x & e^{x^2} \\ \sin x & x^2 & \sec x \\ \tan x & 1 & 2 \end{vmatrix}$, then the value of

$\int_{-\pi/2}^{\pi/2} f(x)dx$ is equal to
(a) 0 (b) 1
(c) 2 (d) none of these

4. The value of the $\Delta = \begin{vmatrix} a^3 - x & a^4 - x & a^5 - x \\ a^5 - x & a^6 - x & a^7 - x \\ a^7 - x & a^8 - x & a^9 - x \end{vmatrix}$ is

(a) 0 (b) $(a^3 - 1)(a^6 - 1)(a^9 - 1)$
(c) $(a^3 + 1)(a^6 + 1)(a^9 + 1)$ (d) $a^{15} - 1$

5. The determinant $\Delta = \begin{vmatrix} a^2(1+x) & ab & ac \\ ab & b^2(1+x) & bc \\ ac & bc & c^2(1+x) \end{vmatrix}$ is

divisible by
(a) $1 + x$ (b) $(1 + x)^2$
(c) x^2 (d) none of these

6. If $U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N+1 \end{vmatrix}$, then $\sum_{n=1}^{N} U_n$ is equal to

(a) $2\sum_{n=1}^{N} n$ (b) $2\sum_{n=1}^{N} n^2$
(c) $\frac{1}{2}\sum_{n=1}^{N} n^2$ (d) 0

7. If $\begin{vmatrix} 4x & 6x+2 & 8x+1 \\ 6x+2 & 9x+3 & 12x \\ 8x+1 & 12x & 16x+2 \end{vmatrix} = 0$, then x is equal to

(a) $x = 0$ (b) $x = -11$
(c) $x = 97$ (d) $x = -\frac{11}{97}$

8. The determinant $\Delta = \begin{vmatrix} b & c & b\alpha + c \\ c & d & c\alpha + d \\ b\alpha + c & c\alpha + d & a\alpha^3 - c\alpha \end{vmatrix}$ is

equal to zero if
(a) b, c, d are in A.P.
(b) b, c, d are in G.P.
(c) b, c, d are in H.P
(d) α is a root of $ax^3 - bx^2 - 3cx - d = 0$

9. $\begin{vmatrix} 103 & 115 & 114 \\ 111 & 108 & 106 \\ 104 & 113 & 116 \end{vmatrix} + \begin{vmatrix} 113 & 116 & 104 \\ 108 & 106 & 111 \\ 115 & 114 & 103 \end{vmatrix}$ is equal to

(a) a positive number (b) a negative number
(c) zero (d) none of these

10. $\Delta = \begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$ is equal to

(a) abc (b) $a^2b^2c^2$
(c) $ab + bc + ca$ (d) none of these

11. $\Delta = \begin{vmatrix} x^2 + y^2 & ax + by & px + qy \\ ax + by & a^2 + b^2 & ap + bq \\ px + qy & ap + bq & p^2 + q^2 \end{vmatrix}$ is equal to

(a) $p + q$ (b) $a + b + c$
(c) $x + y + z$ (d) 0

12. $\Delta = \begin{vmatrix} 1 + a^2 + a^4 & 1 + ab + a^2b^2 & 1 + ac + a^2c^2 \\ 1 + ab + a^2b^2 & 1 + b^2 + b^4 & 1 + bc + b^2c^2 \\ 1 + ac + a^2c^2 & 1 + bc + b^2c^2 & 1 + c^2 + c^4 \end{vmatrix}$ is

equal to

(a) $(a+b+c)^6$ (b) $(a-b)^2(b-c)^2(c-a)^2$
(c) $4(a-b)(b-c)(c-a)$ (d) none of these

13. The determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$ for all $p \in \mathbb{R}$ if
(a) x, y, z are in A.P. (b) x, y, z are in G.P.
(c) x, y, z are in H.P. (d) xy, yz, zx are in A.P.

14. If the determinant $\begin{vmatrix} a & a+d & a+2d \\ a^2 & (a+d)^2 & (a+2d)^2 \\ 2a+3d & 2(a+d) & 2a+d \end{vmatrix} = 0,$ then
(a) $d=0$ (b) $a+d=0$
(c) $d=0$ if $a+d=0$ (d) none of these

15. If x, y, z are integers in A.P. lying between 1 and 9 and $x51, y41$ and $z31$ are three-digit numbers, then the value of $\begin{vmatrix} 5 & 4 & 3 \\ x51 & y41 & z31 \\ x & y & z \end{vmatrix}$ is
(a) $x+y+z$ (b) $x-y+z$
(c) 0 (d) none of these

16. The value of $\begin{vmatrix} a_1x+b_1y & a_2x+b_2y & a_3x+b_3y \\ b_1x+a_1y & b_2x+a_2y & b_3x+a_3y \\ b_1x+a_1 & b_2x+a_2 & b_3x+a_3 \end{vmatrix}$ is equal to
(a) x^2+y^2 (b) 0
(c) $a_1a_2a_3x^2+b_1b_2b_3y^2$ (d) none of these

17. If α and β are non-real numbers satisfying $x^3 - 1 = 0$, then the value of $\begin{vmatrix} \lambda+1 & \alpha & \beta \\ \alpha & \lambda+\beta & 1 \\ \beta & 1 & \lambda+\alpha \end{vmatrix}$ is equal to
(a) 0 (b) λ^3
(c) λ^3+1 (d) none of these

18. The value of $\begin{vmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix}$ is equal to zero when m is
(a) 6 (b) 4
(c) 5 (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. $\begin{vmatrix} \cos C & \tan A & 0 \\ \sin B & 0 & -\tan A \\ 0 & \sin B & \cos C \end{vmatrix}$ has the value
(a) 0 (b) 1
(c) $\sin A \sin B \sin C$ (d) none of these

2. The value of $\begin{vmatrix} x & x^2-yz & 1 \\ y & y^2-zx & 1 \\ z & z^2-xy & 1 \end{vmatrix}$ is
(a) 1 (b) -1
(c) 0 (d) $-xyz$

3. If $\begin{vmatrix} x^k & x^{k+2} & x^{k+3} \\ y^k & y^{k+2} & y^{k+3} \\ z^k & z^{k+2} & z^{k+3} \end{vmatrix} = (x-y)(y-z)(z-x)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right),$ then
(a) $k=-3$ (b) $k=-1$
(c) $k=1$ (d) $k=3$

4. If $\sqrt{-1} = i$ and ω is a non-real cube root of unity, then the value of $\begin{vmatrix} 1 & \omega^2 & 1+i+\omega^2 \\ -i & -1 & -1-i+\omega \\ 1-i & \omega^2-1 & -1 \end{vmatrix}$ is equal to
(a) 1 (b) i
(c) ω (d) 0

5. If the determinant $\begin{vmatrix} a+p & l+x & u+f \\ b+q & m+y & v+g \\ c+r & n+z & w+h \end{vmatrix}$ splits into exactly K determinants of order 3, each element of which contains only one term, then the value of K is
(a) 6 (b) 7
(c) 8 (d) 9

6. The value of $\begin{vmatrix} i^m & i^{m+1} & i^{m+2} \\ i^{m+5} & i^{m+4} & i^{m+3} \\ i^{m+6} & i^{m+7} & i^{m+8} \end{vmatrix}$, where $i = \sqrt{-1}$, is
(a) 1 if m is a multiple of 4
(d) 0 for all real m
(c) $-i$ if m is a multiple of 3
(d) none of these

7. Let $\begin{vmatrix} \lambda^2+3\lambda & \lambda-1 & \lambda+3 \\ \lambda+1 & -2\lambda & \lambda-4 \\ \lambda-1 & \lambda+4 & 3\lambda \end{vmatrix} = p\lambda^4+q\lambda^3+r\lambda^2+s\lambda+t$ be an identity in λ, where p, q, r, s, t, are independent of λ, then the value of t is

(a) 4 (b) 0
(c) 1 (d) none of these

8. If n is not a multiple of 3 and 1, ω, ω^2 are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^{2n} & 1 & \omega^n \\ \omega^n & \omega^{2n} & 1 \end{vmatrix}$ is equal to

(a) 0 (b) ω
(c) ω^2 (d) 1

9. If $\Delta(x) = \begin{vmatrix} 1 & \cos x & 1-\cos x \\ 1+\sin x & \cos x & 1+\sin x-\cos x \\ \sin x & \sin x & 1 \end{vmatrix}$, then $\int_0^{\pi/2} \Delta(x)dx$ is equal to

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) 0 (d) $-\frac{1}{2}$

10. If a, b, c are real numbers, and $D = \begin{vmatrix} a & 1+2i & 3-5i \\ 1-2i & b & -7-3i \\ 3+5i & -7+3i & c \end{vmatrix}$, then D is

(a) purely real (b) purely imaginary
(c) non-real (d) an integer

11. If $i = \sqrt{-1}, \sqrt[4]{1} = \alpha, \beta, \gamma, \delta$ then $\begin{vmatrix} \alpha & \beta & \gamma & \delta \\ \beta & \gamma & \delta & \alpha \\ \gamma & \delta & \alpha & \beta \\ \delta & \alpha & \beta & \gamma \end{vmatrix}$ is equal to

(a) i (b) $-i$
(c) 1 (d) 0

12. Let $\Delta = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, then

(a) $x + a$ is a factor of Δ
(b) $(x + a)^2$ is a factor of Δ
(c) $(x + a)^3$ is a factor of Δ
(d) $(x + a)^4$ is not a factor of Δ

13. The system of equations $ax + 4y + z = 0$, $bx + 3y + z = 0$ and $cx + 2y + z = 0$ has non-trivial solutions if a, b, c are in

(a) A.P. (b) G.P.
(c) H.P. (d) none of these

14. The system of equations

$$2x - y + z = 0$$
$$x - 2y + z = 0$$
$$\lambda x - y + 2z = 0$$

has an infinite number of non-trivial solutions for

(a) $\lambda = 1$ (b) $\lambda = 5$
(c) 5 (d) no real value of λ

ANSWERS

Level 1

1. (d) **2.** (c) **3.** (a) **4.** (a) **5.** (c) **6.** (b) **7.** (d) **8.** (b) and (d) **9.** (c)
10. (d) **11.** (d) **12.** (b) **13.** (b) **14.** (c) **15.** (c) **16.** (b) **17.** (b) **18.** (c)

Level 2

1. (a) **2.** (c) **3.** (b) **4.** (d) **5.** (c) **6.** (b) **7.** (b) **8.** (a) **9.** (d) **10.** (a)
11. (d) **12.** (a), (b) and (d) **13.** (a) **14.** (c)

6

JACOB BERNOULLI

Permutations and Combinations

- Permutation
- Fundamental Principles of Counting
- Factorial *n*
- Some Important Properties
- Some Important Results
- Circular Permutation
- Arrangment of Objects around a Circle
- Clockwise and Anticlockwise Arrangements
- Permutations of Distinct Objects into Groups
- Combination
- Selection of Objects
- Objects Always Included or Excluded
- Selection of One or More Things
- Division and Distribution of Objects
- Conbinations of Distinct Objects into Groups
- Division of Identical Objects into Groups
- Some Important Results

PERMUTATION

Permutation of things means the *arrangement* of things. The word *arrangment* is used if the order of things is taken into account. Thus, if order of different things changes, then their arrangement also changes.

FUNDAMENTAL PRINCIPLES OF COUNTING

Addition Principle

If one operation can be performed in m different ways and another operation, which is independent of the first operation, can be performed in n different ways, then either of the two operations can be performed in $(m + n)$ ways.

Multiplication Principle

If one operation can be performed in m ways and another can be performed (after the first one has been performed in any one way) in n ways for each of the m ways, then the total number of ways in which both of these operations can be performed is $m \times n$. If there are more than two operations to be performed, then the total number of ways to perform all of them in sequence will be $m \times n \times p \times \ldots$.

FACTORIAL *n*

The product of the first n natural numbers is denoted by $\underline{|n}$ or by $n!$ which is read '**factorial *n***'. For example, $n! = n(n-1)(n-2) \ldots 3 \cdot 2 \cdot 1$.

SOME IMPORTANT PROPERTIES

(i) $n! = n(n-1)!$

(ii) $(2n)! = 2^n \cdot n! \cdot [1 \cdot 3 \cdot 5 \cdot 7 \ldots (2n-1)]$

(iii) $0! = 1! = 1$

(iv) Factorials of negative integers are not defined.

(v) $\dfrac{n!}{r!} = n(n-1)(n-2) \ldots (r+1)$

SOME IMPORTANT RESULTS

1. Without repetition: The number of permutations of n different things taken r at a time is denoted by ${}^nP_r$ or $P(n, r)$ or $A(n, r)$. Thus,

$$ {}^nP_r = \frac{n!}{(n-r)!} \quad (1 \le r \le n) \quad n \in N \text{ and } r \in W $$

***Reasoning*:** Arranging the n things taking r at a time means that the arrangement is equivalent to filling r places with n things.

r places:	1	2	3	4	...	r
Number of choices:	n	$n-1$	$n-2$	$n-3$		$n-(r-1)$

2. With repetition: The number of permutations (or arrangements) of n different things taken r at a time when each thing may occur once, twice, thrice etc. up to n times (i.e., repetition is allowed) in any arrangement is n^r.

Reasoning: Arranging the n things taking r at a time, means that the arrangement is equivalent to filling r places with n things.

r places:	1	2	3	4	...	r
Number of choices:	n	n	n	n		n

Number of ways to arrange = Number of ways to fill all r places = n^r.

(a) The number of permutations (or arrangements) of n different things taken all at a time is given by

$$ {}^nP_n = \frac{n!}{(n-n)!} = n! $$

(b) The number of arrangements that can be formed using n things out of which p are identical and of one kind, q are identical and of another kind, r of them are alike and of third kind and the rest are all different, is given by

$$ \frac{n!}{p!q!r!} $$

Note

- The number of permutations of n distinct object taken all at a time is ${}^nP_n = n!$
- ${}^nP_0 = 1$, ${}^nP_1 = n$ and ${}^nP_{n-1} = n!$
- ${}^nP_r = n \cdot {}^{n-1}P_{r-1} = r \cdot {}^{n-1}P_{r-1} + {}^{n-1}P_r$
- ${}^{n-1}P_r = (n-r)^{n-1}P_{r-1}$

Illustration 1. How many three-letter words can be formed using a, b, c, d, e if

(i) repetition is not allowed (ii) repetition is allowed?

Solution: (i) The number of words that can be formed is equal to the number of ways to fill the three places.

No. of choices: 5 4 3

$\Rightarrow 5 \times 4 \times 3 = 60$ words can be formed.

(ii) The number of words that can be formed is equal to the number of ways to fill the three places.

No. of choices: 5 5 5

The first place can be filled in five ways (a, b, c, d, e). If repetition is allowed, all the remaining places can also be filled in five ways using a, b, c, d, e.

$\therefore$ No. of words which can be formed = $5 \times 5 \times 5 = 125$.

CIRCULAR PERMUTATION

So far, we have considered the permutation of things in a row, called *linear permutation*. Instead of arranging in a row, the things may also be arranged around a closed curve.

ARRANGEMENT OF OBJECTS AROUND A CIRCLE (CLOSED PATH)

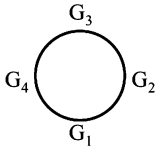
The permutation in a row has a beginning and an end, but there is nothing like beginning or end in a circular permutation. Thus, in the circular permutation, we consider one thing as fixed and the remaining things are arranged as in case of arrangement in a row.

Thus, the number of permutations of 4 persons in a row is 4!, whereas the number of permutations of 4 persons in a circle is (4 – 1)!.

G_1 G_2 G_3 G_4

Arrangement of 4 persons in a row

Fig. 6.1



Arrangement of 4 persons in a circle

Fig. 6.2

Clearly, the following arrangments of G_1, G_2, G_3 and G_4 in a circle will be considered as one and the same arrangement:

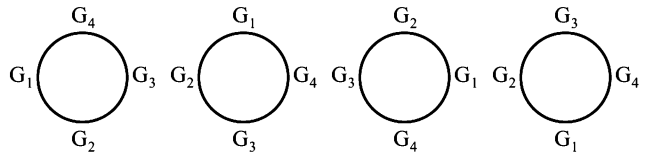


Fig. 6.3

Note carefully that when arranged in a row, G_1, G_2, G_3, G_4, G_4, G_1, G_2, G_3, G_3, G_4, G_1, G_2, G_2, G_3, G_4, G_1 are different permutations but when in arranged in a circle, all these four are considered as one permutation.

DISTINCTION BETWEEN CLOCKWISE AND ANTICLOCKWISE ARRANGEMENTS

Again consider the following circular permutations. In both, the order of the circular arrangement is G_1, G_2, G_3, G_4; in Fig. 6.4, the order is anticlockwise, whereas in Fig. 6.5, the order is clockwise.

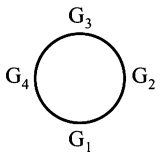


Fig. 6.4

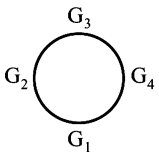


Fig. 6.5

Therefore, when a distinction is made between the anticlokwise and clockwise arrangement of n different things around a circle, then the number of permutations is $(n-1)!$.

But if no distinction is made between the anticlockwise and clockwise arrangements of n different things around a circle, then the number of permutations is $\frac{1}{2}(n-1)!$.

PERMUTATIONS (ARRANGEMENTS) OF DISTINCT OBJECTS INTO GROUPS

1. The number of ways to divide and arrange (permutate) n distinct objects in r different groups is $n!^{n-1}C_{r-1}$ if blank (empty) groups are not allowed and it is $n!^{n+r-1}C_{r-1} = r(r+1)(r+2)\ldots(r+n+1)$ if blank (empty) groups are allowed.
2. The number of ways to divide and arrange (permutate) N distinct objects taken n at a time in r different groups is obtained by multiplying the above result by ${}^nC_n$.

Illustration 2. In how many ways can 5 persons be arranged around a circular table?

Solution: Let the 5 persons be A_1, A_2, A_3, A_4 and A_5.
Let us imagine A_1 as fixed in its position. The remaining 4 persons can be arranged among themselves in 4! ways.

Hence, the number of different arrangements = (5 – 1)! = 4! = 24.

Illustration 3. In how many of the above arrangements, will 2 particular persons be next to each other?

Solution: Let us assume that A_1 and A_2 be the 2 particular persons next to each other.

Treating (A_1, A_2) as one person, we have 4 persons in all to be arranged in a circle: $(A_1A_2), A_3, A_4, A_5$.

These can be arranged in a circle in (4 – 1)! = 3! = 6 ways.

Now A_1 and A_2 can be arranged among themselves in 2! ways.

Hence, the total number of arrangements = $6 \times 2! = 12$.

COMBINATION

Each of the different groups or selections which can be made by some or all of a number of given thigs without reference to the order of the things in each group is called a *combination*.

SELECTION OF OBJECTS

1. Without repetition: The number of combinations (or groups) that can be formed from n different things taken r $(0 \le r \le n)$ at a time is given by

$$^nC_r = \frac{n!}{r!(n-r!)}$$

Reasoning: Let x be the number of groups. Each group contains r things. Hence, each group can be arranged in $r!$ ways. Thus, the number of arrangements is $x(r!)$. But the number of arrangements is ${}^nP_r$. Hence,

$$x(r!) = {}^nP_r$$

$$\Rightarrow \quad x = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!}$$

2. With repetition: The number of combinations of n things taken r at a time when each may occur once, twice, thrice etc. up to r times (i.e. repetition is allowed) in any combination is given by

$$^{n+r-1}C_r$$

PROPERTIES OF ${}^nC_r$

1. ${}^nC_0 = {}^nC_n = 1$
2. ${}^nC_r = {}^nC_{n-r}$
3. If ${}^nC_r = {}^nC_k$, then $r = k$ or $n - r = k$
4. ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
5. $r \cdot {}^nC_r = n\,{}^{n-1}C_{r-1}$
6. $\frac{1}{r+1}\,{}^nC_r = \frac{1}{n+1}\,{}^{n-1}C_{r+1}$

7. $\frac{{}^{n}C_{r}}{{}^{n}C_{r-1}} = \frac{n-r+1}{r}$

8. (i) If n is even, ${}^{n}C_{r}$ is greatest for $r = \frac{n}{2}$.

   (ii) If n is odd, ${}^{n}C_{r}$ is greatest for $r = \frac{n-1}{2}, \frac{n+1}{2}$.

SELECTION OF OBJECTS WHEN PARTICULAR OBJECTS ARE ALWAYS INCLUDED OR EXCLUDED (RESTRICTED PERMUTATION)

1. The number of ways in which r objects can be selected from n different objects if p particular objects are
   (i) always included = ${}^{n-p}C_{r-p}$
   (ii) never included = ${}^{n-p}C_{r}$
2. The number of arrangements of n different objects taking r at a time so that p particular objects are
   (i) always included = ${}^{n-p}C_{r-p}r!$
   (ii) never included = ${}^{n-p}C_{r}r!$

SELECTION OF ONE OR MORE THINGS

1. **Selection from different things:** The number of selections from n different things taking at least one is given by
   $${}^{n}C_{1} + {}^{n}C_{2} + {}^{n}C_{3} + \cdots + {}^{n}C_{n} = 2^{n} - 1$$
2. **Selection from identical things:**
   (i) The number of selections of r things out of n identical things is 1.
   (ii) The total number of selections of zero or more things from p identical things is $p + 1$.
   (iii) The total number of selection of some or all out of $p + q + r$ things, where p are alike of one kind, q are alike of the second kind and the rest are alike of the third kind is $[(p + 1)(q + 1)(r + 1)] - 1$.
3. **Selection of things when both identical and different things are present:** The number of selections taking at least one from $p + q + r + \cdots + n$ objects, where p are exactly alike of one kind, q are alike of the second kind, r are alike of the third kind, and n are different, is $[(p + 1)(q + 1)(r + 1) \ldots 2^{n} - 1]$.

DIVISION AND DISTRIBUTION OF OBJECTS

1. Into groups of unequal size:

(i) The number of ways in which $m + n + p$ things can be divided into unequal groups containing m, n and p things is

$${}^{m+n+p}C_{m} \; {}^{n+p}C_{n} \; {}^{p}C_{p} = \frac{(m+n+p)!}{m!n!p!}$$

(ii) The number of ways to distribute $m + n + p$ things among 3 persons in the groups containing m, n and p things
= (No. of ways to divide) × (No. of groups)!

$$= \frac{(m+n+p)!}{m!n!p!} \times 3!$$

2. Into groups of equal size:

(i) The number of ways in which mn different things can be divided equally in n groups is $\left(\frac{(mn)!}{(m!)!}\right)\frac{1}{n!}$.

(ii) The number of ways in which mn different things can be distributed equally among n different persons
= (No. of ways to divide) × (No. of groups)!

$$= \frac{(mn)!}{(m!)^{n}}$$

SELECTIONS (COMBINATIONS) OF DISTINCT OBJECTS INTO GROUPS

The number of ways of distributing n distinct objects into r groups is r^{n}, if blank groups are allowed and the number is

$$r^{n} - {}^{r}C_{1}(r-1)^{n} + {}^{r}C_{2}(r-2)^{n} + \cdots + (-1)^{r-1}\,{}^{r}C_{r+1}$$

if blank groups are not allowed.

DIVISION OF IDENTICAL OBJECTS INTO GROUPS

1. The number of ways of dividing n identical objects into r groups is ${}^{n+r-1}C_{r-1}$, if blank groups are allowed and the number is ${}^{n-1}C_{r-1}$, if blank groups are not allowed.
2. The number of ways in which n identical objects can be divided into r groups so that no group contains less than p objects and more than $q (p < q)$ is equal to the coefficient of x^{n} in the expansion of $(x^{p} + x^{p-1} + \cdots + x^{q})^{r}$.

SOME IMPORTANT RESULTS

1. The number of combinations of n objects of which p are identical taken r at a time is
   $${}^{n-p}C_{r} + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \cdots + {}^{n-p}C_{0} \text{ if } r \le p$$
   and the number is
   $${}^{n-p}C_{r} + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \cdots + {}^{n-p}C_{r-p} \text{ if } r > p$$
2. The coefficient of x^{r} in the expansion of $(1 - x)^{-n}$ is ${}^{n+r-1}C_{r}$.

3. (i) If there are k objects of one kind, l objects of another kind and so on, then the number of ways of choosing r objects out of these objects is the coefficient of x^r in

$$(1 + x + x^2 + \cdots + x^k)\,(1 + x + x^2 + \cdots + x^l) \ldots$$

(ii) If one object of each kind is to be included in the selection of (i), then the number of ways of choosing r objects is the coefficient of x^r in

$$(x + x^2 + \cdots + x^k)(x + x^2 + \cdots + x^l)\cdots$$

4. If $(a_0 + a_1x + a_2x^2 + \cdots)(b_0 + b_1x + b_2x^2 + \cdots) = (c_0 + c_1x + c_2x^2 + \cdots + c_rx^r + \cdots)$, then on comparing the coefficient of x^r on both sides, we can obtain $c_r = a_0b_r + a_1b_r + \cdots + a_rb_0$ for all $r \geq 0$.

5. If n things are arranged in a row, then the number of ways in which they can be rearranged so that no one of them occupies its original place is

$$n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right]$$

Illustration 4. How many triangles can be formed by joining the vertices of a hexagon?

Solution: Let $A_1, A_2, A_3, \ldots, A_6$ be the vertices of a hexagon. One triangle is formed by selecting a group of 3 points form the 6 given vertices.

No. of triangles = No. of groups of 3 points each from the given 6 points

$$= {}^6C_3 = \frac{6!}{3!3!} = 20$$

Illustration 5. How many diagonals are there in a polygon with n sides?

Solution: No. of lines that can be formed by using the given vertices of a polygon

= No. of groups of 2 points each selected from the n points.

$$= {}^nC_2 = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}$$

Out of ${}^nC_2$ lines, n are the sides of the polygon and the remaining ${}^nC_2 - n$ are the diagonals.

So the number of diagonals $= \frac{n(n-1)}{2} - n$

$$= \frac{n(n-3)}{2}$$

Illustration 6. In how many ways can a cricket team be selected from a group of 25 players containing 10 batsmen, 8 bowlers, 5 all-rounders and 2 wicketkeepers? Assume that the team of 11 plyers requires 5 batsmen, 3 all-rounders, 2 bowlers and 1 wicketkeeper.

Solution: Divide the selection of team into the following four operations:

(i) Selection of batsman can be done (5 from 10) in ${}^{10}C_5$ ways.

(ii) Selection of bowlers can be done (2 from 8) in ${}^8C_2$ ways.

(iii) Selection of all-rounders can be done (3 from 5) in ${}^5C_3$ ways.

(iv) Selection of wicketkeeper can be done (1 from 2) in ${}^2C_1$ ways.

$\Rightarrow$ the team can be selected in

$${}^{10}C_5 \times {}^8C_2 \times {}^5C_3 \times {}^2C_1 \text{ ways}$$

$$= \frac{10! \times 8 \times 7 \times 10 \times 2}{5!5!2!} = 141{,}120 \text{ ways}$$

SOLVED PROBLEMS

1. The value of r for which ${}^{15}C_{r+3} = {}^{15}C_{2r-6}$ is

(a) 2 (b) 4

(c) 6 (d) −9

Ans. (c)

Solution: ${}^{15}C_{r+3} = {}^{15}C_{2r-6}$

$\Rightarrow r + 3 = 2r - 6$ or $15 = (r + 3) + (2r - 6)$

$\Rightarrow r = 9$ or $3r = 18$

$\Rightarrow r = 9$ or $r = 6$

Hence, $r = 6$.

2. If ${}^{2n}C_3 : {}^nC_2 = 44 : 3$, then for which of the following values of r, the value of ${}^nC_r$ will be 15?

(a) $r = 3$ (b) $r = 4$

(c) $r = 6$ (d) $r = 5$

Ans. (b)

Solution: ${}^{2n}C_3 : {}^nC_2 = 44 : 3$

$$\Rightarrow \frac{(2n)!}{3!(2n-3)!} \times \frac{2!(n-2)!}{n!} = \frac{44}{3}$$

$$\Rightarrow \frac{(2n)(2n-1)(2n-2)}{3n(n-1)} = \frac{44}{3}$$

$\Rightarrow 4(2n - 1) = 44$ $\Rightarrow 2n - 1 = 11$

$\Rightarrow 2n = 12$ $\Rightarrow n = 6$

Now, ${}^nC_r = 15$ $\Rightarrow {}^6C_r = 15$

$\Rightarrow {}^6C_r = {}^6C_2$ or ${}^6C_4$ $\Rightarrow r = 2, 4$

Hence, $r = 4$.

3. All the letters of the word AGAIN are permutated in all possible ways and the words so formed (with or without meaning) are written as in the dictionary, then the 50th word is

(a) NAAGI (b) NAAIG
(c) IAANG (d) INAGA

Ans. (b)

Solution: Number of words which begin with A = 4! = 24.

Number of words which begin with G = $\frac{4!}{2!}=12$.

Number of words which begin with I = $\frac{4!}{2!}=12$.

So the 49th word is NAAGI and the 50th word is NAAIG.

4. How many words can be formed by taking four different letters of the word MATHEMATICS?

(a) 756 (b) 1680
(c) 2454 (d) 18

Ans. (c)

Solution: The word MATHEMATICS has 2M's, 2T's, 2A's, H, E, I, C, S.

$\therefore$ Four letters can be chosen in four ways:

(i) 2 alike of one kind and 2 alike of second kind.

$\therefore$ Number of words $= {}^3C_2 \times \frac{4!}{2!2!}=18.$

(ii) 2 alike of one kind and 2 different.

Number of words $= {}^3C_2 \times {}^7C_2 \times \frac{4!}{2!}=756$

(iii) All are different.

Number of words = ${}^8C_4 \times 4! = 1680$.

Hence, the total number of words = 2454.

5. The value of ${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$ is equal to

(a) ${}^{47}C_6$ (b) ${}^{52}C_5$
(c) ${}^{52}C_4$ (d) none of these

Ans. (c)

Solution:

$${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$$
$$= {}^{47}C_4 + {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3$$
$$= {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + ({}^{47}C_3 + {}^{47}C_4)$$
$$= {}^{52}C_4$$

6. If $a_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$, then $\sum_{r=0}^{n} \frac{r}{{}^nC_r}$ euqals

(a) $(n-1)a_n$ (b) na_n
(c) $\frac{1}{2}na_n$ (d) none of these

Ans. (c)

Solution: Given, $a_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$

Let $b_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$

Then, $b_n = \frac{0}{{}^nC_0} + \frac{1}{{}^nC_1} + \frac{2}{{}^nC_2} + \cdots + \frac{n}{{}^nC_n}$ (i)

$\Rightarrow$ $b_n = \frac{n}{{}^nC_0} + \frac{n-1}{{}^nC_1} + \frac{n-2}{{}^nC_2} + \cdots + \frac{0}{{}^nC_n}$ (ii)

$\left[\because {}^nC_0 = {}^nC_n,\ {}^nC_1 = {}^nC_{n-1} \ldots \text{as } {}^nC_r = {}^nC_{n-r}\right]$

Adding Eqs. (i) and (ii), we get

$$2b_n = \frac{n}{{}^nC_0} + \frac{n}{{}^nC_1} + \frac{n}{{}^nC_2} + \cdots + \frac{n}{{}^nC_n}$$
$$= n\left[\frac{1}{{}^nC_0} + \frac{1}{{}^nC_1} + \frac{1}{{}^nC_2} + \cdots + \frac{1}{{}^nC_n}\right]$$

$\Rightarrow$ $2b_n = na_n$ $\therefore$ $b_n = \frac{1}{2}na_n$

7. There are n numbered seats around a round table. The total number of ways in which $n_1 (n_1 < n)$ persons can sit around the round table is equal to

(a) ${}^nC_{n_1}$ (b) ${}^nP_{n_1}$
(c) ${}^nC_{n_1-1}$ (d) ${}^nP_{n-1}$

Ans. (b)

Solution: When seats are numbered, circular permutation is the same as linear permutation as the seat number is also important. Thus, the total number of sitting arrangements $= {}^nP_{n_1}$.

8. The number of five-digit numbers that can be formed without any restriction, is

(a) 990,000 (b) 100,000
(c) 90,000 (d) none of these

Ans. (c)

Solution: The required number of ways = 9 × 10 × 10 × 10 × 10 = 90,000.

9. In how many ways can 15 members of a council sit along a circular table, when the secretary is to sit on one side of the chairman and the deputy secretary on the other side?

(a) 2×12! (b) 24
(c) 2×15! (d) none of these

Ans. (a)

Solution: Total members are 15, but one is left because of circular condition. Therefore, the remaining members are 14, but 3 special members constitute a member. Therefore, the required number of arrangements is 12! × 2, because, the chairman remains between the two specified persons and these persons can sit in two ways.

10. How many different nine-digit numbers can be formed from the digit of the number 223355888 by rearrangement of the digits so that the odd digits occupy even places?

(a) 16 (b) 36

(c) 60 (d) 180

Ans. (c)

Solution: In a nine digits number, there are four even places for the four odd digits 3, 3, 5, 5.

$\therefore$ The required number of ways $= \frac{4!}{2!2!}.\frac{5!}{2!3!} = 60$

11. There are 12 volleyball players in all in a college, out of which a team of 9 players is to be formed. If the captain always remains the same, then in how many ways can the team be formed?

(a) 36 (b) 108

(c) 99 (d) 165

Ans. (d)

Solution: Required number of ways $= {}^{11}C_8 = 165$ (As the captain has already been chosen, now from 11 players, 8 are to be chosen.)

12. How many even numbers of 3 different digits can be formed from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 (repetition of digits is not allowed)?

(a) 224 (b) 280

(c) 324 (d) none of these

Ans. (a)

Solution: The number will be even if the last digit is either 2, 4, 6 or 8 i.e., the last digit can be filled in 4 ways and the remaining two digits can be filled in ${}^8P_2$ ways. Hence, the required number of numbers of three different digits $= {}^8P_2 \times 4 = 224$.

13. Two packs of 52 cards are shuffled together. The number of ways in which a man can be dealt 26 cards so that he does not get two cards of the same suit and same denomination is

(a) ${}^{52}C_{26} \cdot 2^{26}$ (b) ${}^{104}C_{26}$

(c) $2 \cdot {}^{52}C_{26}$ (d) none of these

Ans. (a)

Solution: 26 cards can be chosen out of 52 cards in ${}^{52}C_{26}$ ways. There are two ways in which each card can be dealt, because a card can be either from the first pack or from the second. Hence, the total number of ways $= {}^{52}C_{26} \cdot 2^{26}$.

14. The digits 4, 5, 6, 7 and 8 are written in every possible order. The number of numbers greater than 56,000 is

(a) 72 (b) 96

(c) 90 (d) 98

Ans. (c)

Solution: The number formed by the digits 4, 5, 6, 7 and 8 which is greater than 56,000 is possible in two cases.

Case I. Let the ten thousand digit place number be greater than 5. The number of numbers $= 3\times4\times3\times2\times1 = 72$

Case II. Let the ten thousand digit number be 5 and the thousand digit number be either 6 or greater than 6. Then, the number of numbers $= 3\times3\times2\times1 = 18$

$\therefore$ The required number of ways = 72 + 18 = 90

15. A student is to answer 10 out of 13 questions in an examination so that he must choose at least 4 from the first 5 questions. The number of choices available to him is

(a) 140 (b) 196

(c) 280 (d) 346

Ans. (b)

Solution: The number of total ways

$= {}^5C_4 \times {}^8C_6 + {}^5C_5{}^8C_5 = 196.$

16. How many different nine-digit numbers can be formed from the number 22, 33, 55, 888 by rearranging its digits so that the odd digits occupy even positions?

(a) 16 (b) 36

(c) 60 (d) 180

Ans. (b)

Solution:

1	2	3	4	5	6	7	8	9	O = Odd, E = Even
O	E	O	E	O	E	O	E	O	

Fig. 6.6

$\underbrace{3\ 3\ 5\ 5}_{\text{Odd Digits}}\ \ \underbrace{2\ 2\ 8\ 8\ 8}_{\text{Even Digits}}$

Out of 4 odd digits, 4 even places can be occupied in ${}^4C_4$ ways.

$\therefore$ Total number of ways $= \left(\frac{{}^4C_4 4!}{2!2!}\right)\left(\frac{{}^5C_5 5!}{3!2!}\right) = 60$

17. There are 5 roads leading to a town from a village. The number of ways in which a villager can go to the town and return is

(a) 25 (b) 20

(c) 10 (d) 5

Ans. (a)

Solution: The number of roads is 5.

We know that the number of ways to go to the town is 5 and the number of ways to return is 5. Therefore, the total number of ways in which the villager can go to the town and return is $5\times5 = 25$.

18. Four buses runs between Bhopal and Gwalior. If a man goes from Gwalior to Bhopal by a bus and comes back to Gwalior by another bus, then the total possible ways are

(a) 12 (b) 16
(c) 4 (d) 8

Ans. (a)

Solution: Since the man can go in 4 ways, he can come back in 3 ways.

$\therefore$ Total number of ways = $4 \times 3 = 12$.

19. If ${}^{12}P_r = {}^{11}P_6 + 6 \cdot {}^{11}P_5$, then the value of r is

(a) 3 (b) 6
(c) 9 (d) 12

Ans. (b)

Solution: Equation is ${}^{12}P_r = {}^{11}P_6 + 6 \cdot {}^{11}P_5$.

We know that ${}^{n}P_r = {}^{(n-1)}P_r + r.\ {}^{(n-1)}P_{(r-1)}$.

Therefore, ${}^{12}P_6 = {}^{(12-1)}P_6 + 6 \times {}^{(12-1)}P_{(6-1)}$

$= {}^{11}P_6 + 6 \cdot {}^{11}P_5$

or ${}^{12}P_r = {}^{12}P_6$ or $r = 6$.

20. Ten different letters of an alphabet are given. Words with five letters are formed from these given letters. Then the number of words which have at least one letter repeated is

(a) 69,760 (b) 30,240
(c) 99,748 (d) none of these

Ans. (a)

Solution: The number of words in which all the 5 letters are repeated $= 10^5 = 100{,}000$ and the number of words in which no letter is repeated $= {}^{10}P_5 = \dfrac{10!}{5!} = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30240$.

Hence, the required number of ways = $100000 - 30240 = 69760$.

21. If $n = {}^{m}C_2$, the value of ${}^{n}C_2$ is given by

(a) ${}^{m+1}C_4$ (b) ${}^{m-1}C_4$
(c) ${}^{m+2}C_4$ (d) none of these

Ans. (d)

Solution: We have $n = {}^{m}C_2 = \dfrac{m(m-1)}{2}$

$$\therefore \quad {}^{n}C_2 = \frac{n(n-1)}{2} = \frac{m(m-1)}{4}\left[\frac{m(m-1)}{2} - 1\right]$$

$$= \frac{1}{8}m(m-1)(m^2 - m - 2)$$

$$= \frac{1}{8}m(m-1)(m-2)(m+1)$$

$$= 3\left[\frac{1}{24}(m+1)m(m-1)(m-2)\right] = 3({}^{m+3}C_4)$$

EXERCISE SET 1

1. How many numbers consisting of 5 digits can be formed in which the digits 3, 4 and 7 are used only once and the digit 5 is used twice?

(a) 30 (b) 60
(c) 45 (d) 90

2. Let $x \cdot y \cdot z = 150$, where $x, y, z \in N$. Then number of ordered triplets (x, y, z) satisfying the given equation is

(a) 15 (b) 27
(c) 6 (d) none of these

3. Sixty-six games were played in a tournament where each player is placed against the rest. The number of players is

(a) 33 (b) 12
(c) 13 (d) 11

4. In how many ways can the letters of the word DELHI be arranged, so that the vowels occupy only the even places?

(a) 2 (b) 6
(c) 12 (d) 36

5. A car can accommodate 2 persons in the front and 2 persons in the rear seat. If among 6 persons, 2 can drive, then the number of ways in which the car can be filled is

(a) 10 (b) 20
(c) 30 (d) none of these

6. A person gives a dinner party to 6 guests. The number of ways in which they may be selected from among 10 friends if 2 of the friends will not attend the party together is:

(a) 112 (b) 140
(c) 164 (d) none of these

7. Among the 36 teachers in a school, one principal and one vice-principal are to be appointed. In how many ways this can be done?

(a) 1060 (b) 1160
(c) 1206 (d) 1260

8. The total number of permutations of n different things taken not more then r at a time, where each thing may be repeated any number of times, is

(a) $\dfrac{n(n^n - 1)}{n-1}$ (b) $\dfrac{n^r - 1}{n-1}$
(c) $\dfrac{n(n^r - 1)}{n-1}$ (d) $\dfrac{n(n^r - 1)}{r-1}$

9. In a test a_i students gave wrong answers to at least i questions, where $i = 1, 2, 3, 4, \ldots, k$. No one gave wrong answers to greater than k, then the total number of wrong answers given is

(a) $a_1 + 2a_2 + 3a_3 + \cdots + ka_k$

(b) $a_1 + a_2 + a_3 + \cdots + a_k$

(c) 0

(d) $a_1a_2 + a_2a_3 + \cdots + a_ka_k$

10. The number of diagonals in a polygon of m sides is

(a) $\frac{1}{2!}m(m-5)$ (b) $\frac{1}{2!}m(m-1)$

(c) $\frac{1}{2!}m(m-3)$ (d) $\frac{1}{2!}m(m-2)$

11. The number of five-digit numbers that can be formed with the digits 0, 1, 2, 3 and 4 if the digits can be repeated in the same number is

(a) 25 (b) 250

(c) 2500 (d) 12,500

12. The number of divisors of the number 56 excluding 1 and 56 is

(a) 8 (b) 7

(c) 6 (d) none of these

13. A father with 8 children takes 3 at a time to a zoological garden. If the father does not take the same 3 children more than once, then the number of times he will go to the garden is

(a) 336 (b) 112

(c) 56 (d) 28

14. The exponent of 3 in (100)! is

(a) 33 (b) 44

(c) 48 (d) 52

15. There are n points in a plane in which p points are collinear. How many lines can be formed from these points?

(a) $(n-p)C_2$ (b) ${}^nC_2 - {}^pC_2$

(c) ${}^nC_2 - {}^pC_2 + 1$ (d) ${}^nC_2 + {}^pC_2 - 1$

16. The sum of the digits in the unit place of all numbers formed with the help of 3, 4, 5, 6, taken all at a time, is

(a) 18 (b) 108

(c) 432 (d) 144

17. Out of 5 nurses and 2 doctors, a commitee of 3 members is to be formed. In how many ways can it be done if at least one doctor is to be included?

(a) 5 (b) 15

(c) 25 (d) 50

18. Let y be an element of the set $A = \{1,2,3,5,6,10,15,30\}$ and x_1, x_2, x_3 be integers such that $x_1x_2x_3 = y$, then the number of positive integral solutions of $x_1x_2x_3 = y$ is

(a) 64 (b) 27

(c) 81 (d) none of these

19. In how many ways can 6 persons be chosen, among 4 officers and 8 jawans, so as to include at least one officer?

(a) 896 (b) 780

(c) 668 (d) 448

20. m men and n women are to be seated in a row so that no two women sit together. If $m > n$, then the number of ways in which they can be seated is

(a) $\frac{m!(m+1)!}{(m-n+1)!}$ (b) $\frac{m!(m-1)!}{(m-n+1)!}$

(c) $\frac{(m-1)!(m+1)!}{(m-n+1)!}$ (d) $\frac{(m-1)!(m+1)!}{(m+n+1)!}$

21. There are four postcards and five letter boxes. In how many ways can postcards be dropped if all of them are not dropped in the same letter box.

(a) 620 (b) 3120

(c) 252 (d) 1020

22. The number of ways in which 8 different flowers can be strung to form a garland so that 4 particular flowers are never separated is

(a) $4! \cdot 4!$ (b) 288

(c) 8!/4! (d) $5! \cdot 4!$

23. The number of arrangements that can be made out of the letters of the word "SUCCESS" so that all S's do not come together is

(a) 60 (b) 120

(c) 360 (d) 420

24. A five-digit number divisible by 3 is to be formed using the numerals 0, 1, 2, 3, 4 and 5 without repetition. The total number of ways in which this can be done is

(a) 216 (b) 600

(c) 240 (d) 3125

25. In how many ways can 6 prizes be distributed among 10 students?

(a) 10^6 (b) 6^{10}

(c) 60 (d) $\frac{10!}{4!}$

26. The number of ways of arranging the letters AAAAA BBB CCC D EE F in a row when no two C's are together is

(a) $\frac{15!}{5!3!3!2!} - 3!$ (b) $\frac{15!}{5!3!3!2!} - \frac{13!}{5!3!2!}$

(c) $\frac{12!}{5!3!2!} \times \frac{{}^{13}P_3}{3!}$ (d) $\frac{12!}{5!3!2!} \times {}^{13}P_3$

27. The number of arrangements of the letters of the word BANANA in which the two N's do not appear adjacently is
(a) 40 (b) 60
(c) 80 (d) 100

28. The number of triangles formed by joining 12 points, 7 of which are in the same straight line, is
(a) 185 (b) 220
(c) 92 (d) none of these

29. How many numbers between 1000 and 5000 can be formed with the digits 0, 1, 2, 3, 4, 5 and 6 when repetition is not allowed?
(a) 1050 (b) 720
(c) 210 (d) 480

30. There are n different books and m copies of each in a college library. The number of ways in which a student can make a selection of one or more books is
(a) $(m+1)^n$ (b) $(m+1)^n - 1$
(c) $\dfrac{mn!}{(m!)^m}$ (d) ${}^{mn}C_n \times {}^{n}C_n$

31. The number of ways in which the letters of the word 'vowel' can be arranged so that the letters O and E occupy even places is
(a) 12 (b) 18
(c) 24 (d) none of these

32. A box contains two white balls, three black balls and four red balls. The number of ways in which three balls can be drawn from the box having at least one balls is
(a) 74 (b) 84
(c) 64 (d) 20

33. The number of words that can be made by arranging the letters of the word ARRANGE so that the words begin with A but do not end with E is
(a) 300 (b) 360
(c) 420 (d) none of these

34. Eight chairs are numbered 1 to 8. Two women and three men wish to occupy one chair each. First the women choose the chairs from amongst the chairs marked 1 to 4 and then the men select the chairs from amongst the remaining. The number of possible arrangements is
(a) ${}^6C_3 \times {}^4C_2$ (b) ${}^4C_2 \times {}^4P_3$
(c) ${}^4P_2 \times {}^6P_3$ (d) none of these

35. The number of ways in which the letters of the word STRANGE can be arranged so that the vowels may appear in the odd places is
(a) 1370 (b) 1440
(c) 1470 (d) none of these

36. In a steamer, there are stalls for 12 animals and there are horses, cows and calves (not less then 12 each) ready to be shipped. In how many ways can the ship load be made ?
(a) $3^{12} - 1$ (b) 3^{12}
(c) $(12)^3 - 1$ (d) $(12)^3$

37. The number of words that can be formed with the letters of the word PARALLEL so that all L's do not come together but both A's come together is
(a) 420 (b) 3000
(c) 720 (d) none of these

38. The number of triangles whose vertices are the vertices of an octagon but none of whose sides happen to be the sides of the octagon is
(a) 24 (b) 52
(c) 48 (d) 16

39. Ten different letters of an alphabet are given. Words with five letters are formed from these given letters. Then the number of words which have at least one letter repeated is
(a) 69,760 (b) 30,240
(c) 99,748 (d) none of these

40. There are n books having p copies of each. The number of ways in which a selection can be made from them is
(a) n^p (b) p^n
(c) $(p+1)^n - 1$ (d) $(n+1)^p - 1$

41. The number of words that can be formed with the letters of the word EQUATION so that consonants occur only at even places is
(a) 2880 (b) 3600
(c) 3240 (d) none of these

42. The number of permutations that can be formed by arranging all the letters of word NINETEEN in which no two E's occur together is
(a) $\dfrac{8}{3!3!}$ (b) $\dfrac{5!}{3! \times {}^6C_2}$
(c) $\dfrac{5!}{3!} \times {}^6C_3$ (d) $\dfrac{8!}{5!} \times {}^6C_3$

43. The total number of words formed by 2 vowels and 3 consonants taken from 4 vowels and 5 consonants is equal to
(a) 60 (b) 120
(c) 7200 (d) none of these

44. On the occasion of Deewali, each student of a class sends greeting cards to others. If there are 20 students in the class, then the total number of greeting cards exchanged by the students is
(a) ${}^{20}C_2$ (b) $2 \cdot {}^{20}C_2$
(c) $2 \cdot {}^{20}P_2$ (d) none of these

45. The letters of the word PATNA are arranged in all possible ways as in a dictionary, then the rank of the word PATNA is
(a) 41 (b) 42
(c) 43 (d) 44

46. The sum of all the divisors of $2^5 3^4 5^2$ is
(a) $3^2 \cdot 7^2 \cdot 11^2$ (b) $3^2 \cdot 7 \cdot 11^2 \cdot 31$
(c) $3 \cdot 7 \cdot 11 \cdot 31$ (d) none of these

47. The total number of arrangements which can be made out of the letters of the word ALGEBRA without altering the relative positions of vowels and consonants is
(a) $\frac{7!}{2!}$ (b) $\frac{7!}{2!5!}$
(c) 4!3! (d) $\frac{4!3!}{2}$

48. The number of ways in which we can select four numbers from 1 to 30 so as to exclude every selection of four consecutive numbers is
(a) 27,378 (b) 27,405
(c) 27,399 (d) none of these

49. The number of integers which lie between 1 and 10^6 and which have the sum of the digits equal to 12 is
(a) 8550 (b) 5382
(c) 6062 (d) 8055

50. In how many ways can 7 toys be given away to 3 children when each child is eligible for all the toys?
(a) 3570 (b) 4350
(c) 2187 (d) 2426

51. Four-letter words are to be formed out of the letters of the word PASSPORT. Their number is
(a) 666 (b) 626
(c) 686 (d) 606

52. There are 10 lamps in a hall. Each one of them can be switched on independently. The number of ways is which the hall can be illuminated is
(a) 10^2 (b) 1023
(c) 2^{10} (d) 10!

53. In how many ways can 6 boys and 5 girls can be seated in a row such that two girls are never together?
(a) 6!5! (b) $2 \times 5!5!$
(c) $6! \times {}^7P_5$ (d) $6! \times 6!$

54. In a college examination, a candidate is required to attempt 6 out of 10 questions which are divided into two sections each containing 5 questions. Further, the candidate is not permitted to attempt more than 4 questions from either of the sections. The number of ways in which he can make up a choice of 6 questions is
(a) 15 (b) 200
(c) 100 (d) 50

55. How many 10-digit numbers can be written by using the digits 1 and 2?
(a) ${}^{10}C_1 + {}^9C_2$ (b) 2^{10}
(c) ${}^{10}C_2$ (d) 10!

56. How many numbers between 5000 and 10,000 can be formed using the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 each digit appearing not more than once in each number?
(a) $5 \times {}^8P_3$ (b) $5 \times {}^8P_8$
(c) $5! \times 7!$ (d) 7!

57. There are $2n$ things out of which n are alike and n different. The number of ways of selecting n things is
(a) 2^n (b) ${}^{2n}C_n$
(c) $2^n - 1$ (d) $2^n({}^{2^n}C_n)$

58. The number of arrangements which can be made using all the letters of the word LAUGH keeping the vowels adjacent is
(a) 10 (b) 24
(c) 120 (d) 48

59. The number of all three-element subsets of the set-$\{a_1, a_2, a_3, \ldots, a_n\}$ which contain a_3 is
(a) ${}^nC_3$ (b) ${}^{n-1}C_3$
(c) ${}^{n-1}C_2$ (d) none of these

60. Ten different letters of an alphabet are given. If words with 5 letters are formed from these given letters, then the number of words which have at least one letter repeated is
(a) 69,760 (b) 30,240
(c) 99,748 (d) none of these

EXERCISE SET 2

1. The number of distinct permutations of the letters of the word STATISTICS that begin and end with the letter S is
(a) 5040 (b) 4280
(c) 3360 (d) 1680

2. In a party, 12 persons are to be seated in a round table. If two particular persons are not to sit side by side, then the total number of arrangements is
(a) $9 \times 10!$ (b) $2 \times 10!$
(c) $45 \times 8!$ (d) 10!

3. Everybody in a room shakes hands with everybody else. The total number of handshakes is 66. The total number of persons in the room is
(a) 11 (b) 12
(c) 13 (d) 14

4. $ABCD$ is a convex quadrilateral. The points 3, 4, 5 and 6 are marked on the sides AB, BC, CD and DA respectively. The number of triangles with vertices on different sides is
   (a) 270 (b) 220
   (c) 282 (d) none of these

5. The number of ways in which a committee of 6 persons can be formed out of 10 men and 5 women when at least one women is necessarily selected is
   (a) ${}^{15}P_6 - {}^{10}P_6$ (b) ${}^{15}C_9 - {}^{10}C_4$
   (c) ${}^{15}C_5 - {}^{10}C_5$ (d) none of these

6. The number of ways in which the four faces of a regular tetrahedron can be painted with four different colours is
   (a) 4! (b) 4
   (c) 1 (d) none of these

7. The total number of 9-digit numbers which have all different digits is
   (a) 10! (b) 9!
   (c) $9 \cdot 9!$ (d) $10 \cdot 10!$

8. A guard of 12 men is formed from a group of n soldiers in all possible ways. If the number of times two particular soldiers A and B are together on guard is thrice the number of times three particular soldiers C, D and E are together on guard, then n is equal to
   (a) 18 (b) 24
   (c) 32 (d) 36

9. In a group of 10 boys 6 are to be selected for a training camp. The number of ways in which a particular boy is always selected is x and the number of ways in which particular boy is never selected is y. Then $x - y$ is
   (a) 0 (b) 42
   (c) 84 (d) none of these

10. The number of rectangles in the following figure is

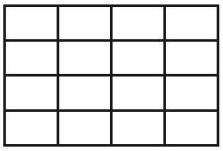


Fig. 6.7

   (a) 5×5 (b) ${}^5P_2 \times {}^5P_2$
   (c) ${}^5C_2 \times {}^5C_2$ (d) none of these

11. The number of numbers divisible by 5 and lying between 3000 and 4000 that can be formed from the digits 3, 4, 5, 6, 7 and 8, no digit being repeated in any number, is
   (a) 6 (b) 12
   (c) 18 (d) none of these

12. The number of ways in which the letters of the word ARRANGE can be arranged such that both R's do not come together is
   (a) 360 (b) 900
   (c) 1260 (d) 1620

13. Eleven animals in a circus have to be caged in 11 cages, one in each cage. If 4 of the cages are too small for 6 of the animals, the number of ways of caging the animals is
   (a) 724,800 (b) 604,800
   (c) 506,420 (d) 648,500

14. Five-Math books and 3 Physics books are placed on a shelf so that the books on the same subject always remain together. The possible arrangements are
   (a) 1440 (b) 1956
   (c) 720 (d) none of these

15. Five-digit numbers are formed with 0, 1, 2, 3, 4. The number of numbers in which at least one digit is repeated, is
   (a) 96 (b) 120
   (c) 2500 (d) 2404

16. The number of ways of selecting 10 balls out of an unlimited number of white, red, blue and green balls is
   (a) 270 (b) 84
   (c) 286 (d) 86

17. The number of ways in which an examiner can assign 20 marks to 5 questions giving not less than 2 marks to any question, is (marks in fraction cannot be given)
   (a) ${}^{14}C_4$ (b) ${}^{14}C_5$
   (c) ${}^{14}C_6$ (d) none of these

18. There are three coplanar parallel ends. If any p points are taken on each of the lines, the maximum number of triangles with vertices at these points is
   (a) $3p^2(p-1)+1$ (b) $3p^2(p-1)$
   (c) $p^2(4p-3)$ (d) none of these

19. Four-digit numbers are formed from digits 0, 1, 2, ..., 9. The number of such numbers when at least one digit including zero is repeated at least once is
   (a) 4644 (b) 4356
   (c) 4464 (d) 4536

20. A shopkeeper sells three varieties of perfumes and he has a large number of bottles of the same size of each variety in his stock. There are 5 places in a row in his showcase. The number of different ways of displaying the three varieties of perfumes in the showcase is
   (a) 6 (b) 50
   (c) 150 (d) none of these

21. A student is allowed to select at most n books from a collection of $(2n + 1)$ books. If the total number of ways in which he can select at least one book is 255, then n is equal to
(a) 3 (b) 4
(c) 5 (d) 8

22. The total number of ways in which a beggar can be given at least one rupee from four 25 paise coins, three 50 paise coins and 2 one-rupee coins is
(a) 54 (b) 53
(c) 51 (d) none of these

23. A student is allowed to select at most n books from a collection of $(2n + 1)$ books. If the total number of ways in which he can select at least one books is 63, then n is equal to
(a) 3 (b) 4
(c) 6 (d) 5

24. The number of zeros at the end of 70! is
(a) 16 (b) 5
(c) 7 (d) 70

25. The total number of ways in which six '+' signs and four '$-$' signs can be arranged is a line such that no two negative signs occur togather is
(a) 210 (b) 232
(c) 200 (d) none of these

26. A person opts for an examination in which there are four papers with a maximum of m marks from each paper. The number of ways in which one can get $2m$ marks is
(a) ${}^{(2m+3)}C_3$
(b) $\frac{1}{3}(m+1)(2m^2+4m+1)$
(c) $\frac{1}{3}(m+1)(2m^2+4m+3)$
(d) none of these

27. Ten players are to be seated in a row for a photograph, so that two particular players sit on the two middle seats. The number of arrangements is
(a) 9! (b) (9!) (2!)
(c) $2 \cdot (8!)$ (d) none of these

28. On a railway route, there are 20 stations. The number of different tickets required in order that it may be possible to travel from every station to every station is
(a) 196 (b) 105
(c) 210 (d) 225

29. There are four balls of different colours and four boxes of same colours in which the balls, one each in a box, could be placed such that a ball does not go to a box of its own colour. The number of ways this can be done is
(a) 9 (b) 18
(c) 12 (d) none of these

30. The number of ways to fill each of the four cells of the table with a distinct natural number such that the sum of the numbers is 10 and the sums of the numbers placed diagonally are equal is

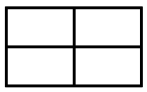


Fig. 6.8

(a) $2! \times 2!$ (b) 8
(c) 2(4!) (d) 4!

31. The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines is
(a) 6 (b) 18
(c) 12 (d) 9

32. The number of divisors of the form $4n + 1$, $n \geq 0$, of the number $10^{10}11^{11}13^{13}$ is
(a) 750 (b) 840
(c) 924 (d) 1024

33. If $500! = 2^m \times$ an integer, then
(a) $m = 494$
(b) $m = 496$
(c) it is equivalent to number of n is $400! = 2^n \times$ an integer
(d) $m = {}^{500}C_2$

34. The number of ways in which a mixed double game can be arranged from amongst 9 married couples, if no husband and wife play in the same game, is
(a) 756 (b) 1512
(c) 3024 (d) none of these

35. The number of diagonals that can be drawn by joining the vertices of an octagon is
(a) 28 (b) 48
(c) 20 (d) none of these

36. Let p be a prime number such that $p \geq 111$.
Let $n = p! + 1$. The number of primes in the list $n + 1$, $n + 2$, $n + 3$, ..., $n + p - 1$ is
(a) $p-1$ (b) 2
(c) 1 (d) none of these

37. There are 6 balls of different colours and 3 boxes of different sizes. Each box can hold all the six balls. The balls are put in the boxes so that no box remains empty. The number of ways this can be done is
(a) 534 (b) 543
(c) 540 (d) 28

38. If x, y, z, ... are $(m + 1)$ distinct prime numbers, the numbers of factors of $x^n yz$ ... is
(a) $m(n + 1)$ (b) $2^n m$
(c) $(n + 1)2^m$ (d) $n \cdot 2m$

39. The total number of arrangements of the letters in the expansion $a^3b^2c^4$ when written at full length is
(a) 1260 (b) 2520
(c) 610 (d) none of these

40. In world cup of cricket there are 12 teams, divided equally in two groups. Teams of each group will play a match against each other. From each group 3 top teams will qualify for the next round. In this round, each team will play against others once. Four top teams of this round will qualify for the semifinal round, where each teams of this round will go to the final round, where each team will play against the others once. Two top teams of this round will go to the final round, where they will play the best of three matches. The minimum number of matches in the World Cup will be
(a) 54 (b) 53
(c) 38 (d) 40

41. The number of signals that can be generated by using 6 differently coloured flags, when any number of them may be hoisted at a time, is
(a) 1956 (b) 1957
(c) 1958 (d) 1959

Assertion-Reason Type Questions

These questions are Assertion and Reason type questions. Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions has four alternative choices. Only one of them is the correct answer. Select the correct answer.
(a) Both Statement 1 and Statement 2 are true and Statement 2 is a correct explanation of Statement 1.
(b) Both Statement 1 and Statement 2 are true but Statement 2 is not correct explanation of Statement 1.
(c) Statement 1 is true, Statement 2 is false.
(d) Statement 1 is false, Statement 2 is true.

42. **Statement 1:** If N is the number of three-digit numbers of the form a, b, c such that $a < b < c$, then N is an even number.
Statement 2: N is divisible by 7.

43. **Statement 1:** 11 sides of the polygon has 44 diagonals.
Statement 2: The number of diagonals in the polygon by $\frac{n(n-3)}{2}$, where n is the number of sides of the polygon.

44. **Statement 1:** 1, 2, 3 and 4 are four numbers. Then 72 matrices can be made using all four numbers.
Statement 2: If the numbers of rows is equal to the number of columns, then it is called a matrix.

45. **Statement 1:** 4320 words can be formed from the letters of the word DAUGHTER so that the vowels always come together.
Statement 2: 36000 words can be formed from the letters of that word so that the vowels never come together.

46. **Statement 1:** If ${}^nC_{r-1} = 36$, ${}^nC_r = 84$ and ${}^nC_{r+1} = 126$, then ${}^rC_2 = 8$.
Statement 2: ${}^nC_r = {}^nC_{n-r}$

47. **Statement 1:** 60 different permutation can be made by the letter of the word BANANA.
Statement 2: The number of permutations of n things of which p are alike of one kind, q are alike of the second kind and the remaining all are distinct, is $\frac{n!}{p!q!}$.

Previous Years' Questions

48. Total number of four-digit odd numbers that can be formed using 0, 1, 2, 3, 5, 7 are
(a) 400 (b) 720
(c) 375 (d) 216
[AIEEE 2002]

49. Numbers greater than 1000 but not greater than 4000 whcih can be formed with the digits 0, 1, 2, 3, 4, (repetition of digits is allowed) are
(a) 576 (b) 450
(c) 375 (d) 350 **[AIEEE 2002]**

50. If ${}^nC_r$ denotes the number of combinations of n things taken r at a time, then the expression ${}^nC_{r+1} + {}^nC_{r-1} + 2 \times {}^nC_r$ equals
(a) ${}^{n+2}C_r$ (b) ${}^{n+2}C_{r+1}$
(c) ${}^{n+1}C_r$ (d) ${}^{n+1}C_{r+1}$ **[AIEEE 2003]**

51. The number of ways in which 6 men and 5 women can arrange at a round table if no two women sit together is given by
(a) $7! \times 5!$ (b) $5! \times 4!$
(c) $6! \times 5!$ (d) 30 **[AIEEE 2003]**

52. A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first 5 questions. The number of choices available to him is
(a) 196 (b) 280
(c) 346 (d) 140 **[AIEEE 2003]**

53. The number of ways of distributing 8 identical balls in 3 distinct boxes so that none of the boxes is empty is
(a) 5 (b) 21
(c) 3^8 (d) ${}^8C_3$ **[AIEEE 2004]**

54. How many ways are there to arrange the letters in the word GARDEN with the vowels in alphabetical order?
(a) 120 (b) 240
(c) 360 (d) 480 **[AIEEE 2004]**

55. If the letters of the word SACHIN are arranged in all possible ways and these words are written out as in a dictionary, then the word SACHIN appears at serial number
(a) 600 (b) 603
(c) 602 (d) 601 **[AIEEE 2005]**

56. In an election, a voter may vote for any number of cnadidates not greater than the number to be elected. There are 10 candidates and 4 are to be elected. If a voter votes for at least one candidate, then the number of ways in which he can vote, is
(a) 6210 (b) 385
(c) 1110 (d) 5040 **[AIEEE 2006]**

57. The set $S = \{1, 2, 3, \ldots, 12\}$ is to partitioned into three sets A, B and C of equal size. Thus, $A \cup B \cup C = S$, and $A \cup B = B \cap C = A \cap C = \phi$. Then the number of ways to partition S is
(a) $\frac{12!}{3!(4!)^3}$ (b) $\frac{12!}{(3!)^4}$
(c) $\frac{12!}{(4!)^3}$ (d) $\frac{12!}{3!(3!)^4}$ **[AIEEE 2007]**

58. How many different words can be formed by jumbling the letters in the word MISSISSIPPI in which no two S's are adjacent?
(a) $8 \cdot {}^6C_4 \cdot {}^7C_4$ (b) $6 \cdot 7 \cdot {}^8C_4$
(c) $6 \cdot 8 \cdot {}^7C_4$ (d) $7 \cdot {}^6C_4 \cdot {}^8C_4$
[AIEEE 2008]

59. In a shop, there are five types of ice-creams available. A child wants to buy six ice-creams.
Statement 1: The number of different ways the child can buy the six ice-creams is ${}^{10}C_5$.
Statement 2: The number of different ways the child can buy the six ice-creams is equal to the number of different ways of arranging 6 A's and 4 B's in row.
(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(c) Statement 1 is true; Statement 2 is false.
(d) Statement 1 is false; Statement 2 is true.
[AIEEE 2008]

60. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on the shelf so that the dictionary is always in the middle. Then the number of such arrangement is
(a) less than 500
(b) all least 500 but less than 750
(c) at least 750 but less than 1000
(d) at least 1000 **[AIEEE 2009]**

61. There is a set of m parallel lines intersecting a set of another n parallel lines in a plane. The number of parallelograms formed, is
(a) ${}^{m-1}C_2 \cdot {}^{n-1}C_2$ (b) ${}^mC_2 \cdot {}^nC_2$
(c) ${}^{m-1}C_2 \cdot {}^nC_2$ (d) ${}^mC_2 \cdot {}^{n-1}C_2$
[DCE 2007]

62. In an examination, a students has to answer 4 questions out of 6. Questions 1 and 2 are, however, compulsory. The number of ways in which the student can make the choice is
(a) ${}^6C_2$ (b) 6
(c) ${}^6C_4$ (d) 4 **[DCE 2006]**

63. If $\frac{1}{{}^5C_r} + \frac{1}{{}^6C_r} = \frac{1}{{}^4C_r}$, then r equals
(a) 4 (b) 2
(c) 5 (d) none of these
[DCE 2006]

64. The sum of the digits in the unit place of all numbers formed with the help of 3, 4, 5 and 6 taken all at a time is
(a) 144 (b) 432
(c) 108 (d) 18 **[DCE 2006]**

65. There are 10 true-false questions in an examination. Then these questions can be answered in
(a) 240 ways (b) 20 ways
(c) 1024 ways (d) 100 ways
[DCE 2005]

66. If there are n numbers of seats and m number of people have to be seated, then how many ways are possible to do this ($m < n$)?
(a) ${}^nP_m$ (b) ${}^nC_m$
(c) ${}^nC_n \times (m-1)!$ (d) ${}^{n-1}P_{m-1}$
[DCE 2004]

67. The number of different words that can be formed from the letters of the word PENCIL so that no two vowels are together is
(a) 120 (b) 260
(c) 240 (d) 480
[DCE 2004]

68. 1, 2, 3, 4 are four numbers. How many numbers can be made using all four numbers?
(a) 1 (b) 3
(c) 2 (d) 64
[DCE 2003]

69. The number of committees of 5 persons consisting of at least one female member that can be formed from 6 males and 4 females is
(a) 246 (b) 252
(c) 6 (d) none of these
[DCE 2003]

70. How many 10-digit numbers can be written by using digits 9 and 2?
(a) ${}^{10}C_1 + {}^{9}C_2$ (b) 2^{10}
(c) ${}^{10}C_2$ (d) 10! **[UP SEE 2008]**

71. If ${}^{8}C_r - {}^{7}C_3 = {}^{7}C_2$, then r is equal to
(a) 3 (b) 4
(c) 8 (d) 6 **[UP SEE 2008]**

72. Out of 6 boys and 4 girls, a group of 7 is to be formed. In how many ways can this be done if the group is to have a majority of boys?
(a) 120 (b) 576
(c) 90 (d) 100 **[UP SEE 2007]**

73. 'X' speaks truth in 60% and 'Y' in 50% of the cases. The probability that they contradict each other narrating the same incident is
(a) $\frac{1}{4}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{2}{3}$ **[UP SEE 2004]**

74. The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN is
(a) 360 (b) 192
(c) 96 (d) 48
[IIT-JEE 2007]

75. If r, s and t are prime numbers and p and q are positive integers such that LCM of p and q is $r^2s^4t^2$, then the number of ordered pairs (p, q) is
(a) 252 (b) 254
(c) 225 (d) 224
[IIT-JEE 2006]

76. There are 10 points in a plane. Out of these, 6 are collinear. If N is the number of triangles formed by joining these points, then
(a) $N \leq 100$ (b) $100 < N \leq 140$
(c) $140 < N \leq 190$ (d) $N > 190$
[AIEEE 2011]

77. Assuming the balls to be identical except for difference in colours, the number of ways in which one or more balls can be selected from 10 white, 9 green and 7 black balls is
(a) 880 (b) 629
(c) 630 (d) 879
[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (b)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (d)	**8.** (c)	**9.** (b)	**10.** (c)
11. (c)	**12.** (c)	**13.** (c)	**14.** (c)	**15.** (c)	**16.** (b)	**17.** (c)	**18.** (a)	**19.** (a)	**20.** (a)
21. (a)	**22.** (b)	**23.** (c)	**24.** (a)	**25.** (a)	**26.** (c)	**27.** (a)	**28.** (a)	**29.** (d)	**30.** (b)
31. (a)	**32.** (c)	**33.** (a)	**34.** (c)	**35.** (b)	**36.** (b)	**37.** (c)	**38.** (d)	**39.** (a)	**40.** (c)
41. (a)	**42.** (c)	**43.** (c)	**44.** (b)	**45.** (b)	**46.** (b)	**47.** (d)	**48.** (a)	**49.** (c)	**50.** (c)
51. (d)	**52.** (b)	**53.** (c)	**54.** (b)	**55.** (b)	**56.** (a)	**57.** (a)	**58.** (d)	**59.** (c)	**60.** (b)

Exercise Set 2

1. (c)	**2.** (a)	**3.** (b)	**4.** (d)	**5.** (b)	**6.** (c)	**7.** (c)	**8.** (c)	**9.** (b)	**10.** (c)
11. (b)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (d)	**16.** (c)	**17.** (a)	**18.** (c)	**19.** (c)	**20.** (c)
21. (b)	**22.** (a)	**23.** (a)	**24.** (a)	**25.** (d)	**26.** (c)	**27.** (c)	**28.** (c)	**29.** (d)	**30.** (b)
31. (b)	**32.** (c)	**33.** (a)	**34.** (b)	**35.** (c)	**36.** (d)	**37.** (c)	**38.** (c)	**39.** (a)	**40.** (b)
41. (a)	**42.** (b)	**43.** (a)	**44.** (c)	**45.** (b)	**46.** (d)	**47.** (a)	**48.** (b)	**49.** (c)	**50.** (b)
51. (c)	**52.** (a)	**53.** (b)	**54.** (a)	**55.** (d)	**56.** (b)	**57.** (c)	**58.** (d)	**59.** (d)	**60.** (d)
61. (b)	**62.** (b)	**63.** (b)	**64.** (c)	**65.** (c)	**66.** (a)	**67.** (d)	**68.** (d)	**69.** (a)	**70.** (b)
71. (a)	**72.** (d)	**73.** (c)	**74.** (c)	**75.** (c)	**76.** (a)	**77.** (d)			

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Five-digit numbers formed by using the digits 3, 4 and 7 only once and 5 using twice = 5!/2! = 60.

2. $xyz = 3 \times 5 \times 7$ (product of primes)
3 will be a factor of x or y or z in 3 ways
5 will be a factor of x or y or z in 3 ways
7 will be a factor of x or y or z in 3 ways
$\therefore$ Total number of ways $= 3 \times 3 \times 3 = 27$

3. If n is the number of players, then ${}^nC_2 = 66$

$$\Rightarrow \frac{n(n-1)}{2} = 66 \qquad \Rightarrow n(n-1) = 132$$
$$\Rightarrow n(n-1) = 12 \times 11 \qquad \Rightarrow n = 12$$

4. Given: The number of letters in the word "DELHI" = 5. We know that the number of vowels in the given word (i.e., E, I) = 2.
And the number of even places which these vowels can occupy (second and fourth) = 2.
Therefore, the number of ways to arrange vowels at even places = ${}^2P_2$. And the number of ways to arrange the remaining 3 letters in 3 places = ${}^3P_3$. Therefore, the total number of ways

$$= {}^2P_2 \times {}^3P_3 = \frac{2!}{(2-2)!} \times \frac{3!}{(3-3)!} = 2! \times 3! = 12$$

5. Since 2 persons can drive the car, therefore, we have to select 1 from these two. This can be done in ${}^2C_1$ ways. Now, from the remaining 5 persons, we have to select 3 which can be done in ${}^5C_3$ ways.
Therefore, the required number of ways in which the car can be filled $= {}^5C_3 \times {}^2C_1 = 10 \times 2 = 20$.

6. The required number of ways
$= {}^{10}C_6 - {}^8C_4 = 210 - 70 = 140$.

7. Number of teachers = 36.
We know that the number of ways to appoint a principal among 36 teachers = 36.
And the number of ways to appoint a vice-principal from the remaining 35 teachers = 35.
Therefore, the total number of ways to appoint a principal and a vice-principal $= 36 \times 35 = 1260$.

8. One place can be filled in n ways by n things; the 1st and 2nd palces can be filled in $n \times n = n^2$ ways.
The 1st , 2nd and 3rd palces can be filled in $n \times n \times n$ ways.
Similarly, the 1st, 2nd, ..., rth places can be filled in $n \times n \times \ldots \times n$ (r times) $= n^r$.
$\Rightarrow$ Required total number $= n + n^2 + n^3 \cdots + n^r$
$\therefore$ Required number of ways $= \dfrac{n(n^r - 1)}{n-1}$

9. Total number of wrong answers given
$= 1(a_1 - a_2) + 2(a_2 - a_3) + \cdots + (k-1)(a_{k-1} - a_k) + ka_k$
$= a_1 + a_2 + \cdots + a_k$

10. Number of diagonals $= {}^mC_2 - m$

$$= \frac{m(m-1)}{2!} - m = \frac{m}{2!}(m-3)$$

11. Given: Total number of digits (0, 1, 2, 3, 4) = 5.
We know that as 0 cannot be placed in ten thousand's place, therefore, the number of ways to fill up the ten thousand's place from the remaining 4 digits = 4.
We also know that as the numbers can be repeated, therefore the number of ways to fill up each of the other 4 places (thousand's, hundred's, ten's and one's) is 5.
Therefore, the total number of ways to form 5-digit numbers $= 4 \times 5 \times 5 \times 5 \times 5 = 2500$.

12. $56 = 2^3 \times 7^1$
$\therefore$ Number of divisors $= (3 + 1)(1 + 1) = 8$
$\therefore$ Number of divisors excluding 1 and 56 $= 8 - 2 = 6$.

13. Number of children = 8 and number of children taken at a time = 3.
We know that number of times the father will go to the garden by taking 3 children out of 8 children

$$= {}^8C_3 = \frac{8!}{3!(8-3)!} = \frac{8!}{3! \times 5!} = \frac{8 \times 7 \times 6 \times 5!}{(3 \times 2 \times 1) \times 5!} = 56.$$

14. Let $E(n)$ denote the exponent of 3 in n.
Here $E(100!) = E(1 \cdot 2 \cdot 3 \ldots 99 \cdot 100)$
$= E[(3 \cdot 1)(3 \cdot 2) \ldots ((3 \cdot 33)]$
$= 33 + E(1 \cdot 2 \cdot 3 \ldots 33)$
Now $E(1 \cdot 2.3 \ldots\ldots 33)$
$= E(3 \cdot 6 \cdot 9 \ldots 33)$
$= E[(3 \cdot 1)(3 \cdot 2) \ldots (3 \cdot 11)]$
$= 11 + E(1 \cdot 2 \cdot 3) \ldots (3 \cdot 11)$
And $E(1 \cdot 2 \cdot 3 \ldots 11) = E(3 \cdot 6 \cdot 9)$
$= E[(3 \cdot 1)(3 \cdot 2)(3 \cdot 3)]$
$= 3 + E(1 \cdot 2 \cdot 3) = 3 + 1 = 4$
Hence, $E(100!) = 33 + 11 + 4 = 48$

15. Given: Number of points in a plane = n and number of collinear points = p.
We know that as one line has two end points, therefore, the total number of lines = ${}^nC_2$.
Since p points are collinear, therefore the total number of lines drawn = ${}^pC_2$.
We also know that as there are p points on line, therefore the number of lines which contain these points = 1. Therefore, the total number of lines = ${}^nC_2 - {}^pC_2 + 1$.

16. When 3 is in the unit place, then the number of numbers = 3! = 6.

$\therefore$ Sum of digits in the unit place = $3\times6=18$

Similary, 4×6, 5×6, 6×6

Hence, sum of digits in the unit place

$= 18+24+30+36=108$.

17. Number of nurses = 5, number of doctors = 2, and number of members in committee = 3.

We know that the number of ways to form a committee of 3 members in such a way to have at least one doctor = 2 nurses out of 5 and 1 doctor out of 2 ($^5C_2$ and $^2C_1$) or 1 nurse out of 5 and 2 doctors out of 2($^5C_1$ and $^2C_2$). Therefore, total number of ways to form the committee

$$= \left[{}^5C_2\times {}^2C_1\right]+\left[{}^5C_1\times {}^2C_2\right]$$

$$= \left(\frac{5!}{2!(5-2)!}\times\frac{2!}{1!(2-1)!}\right)+\left(\frac{5!}{1!(5-1)!}\times\frac{2!}{2!(2-2)!}\right)$$

$$= \left(\frac{5!}{2!\times3!}\times\frac{2!}{1!\times1!}\right)+\left(\frac{5!}{1!\times4!}\times\frac{2!}{2!\times0!}\right)$$

$$= \left(\frac{5\times4\times3!}{(2\times1)\times3!}\times\frac{2\times1}{1\times1}\right)+\left(\frac{5\times4!}{1\times4!}\times\frac{2!}{2!\times1}\right)$$

$= (10\times2)+(5\times1)=20+5=25$.

18. Number of solutions of the given equation is the same as the number of solutions of the equation,

$$x_1x_2x_3x_4 = 30 = 2\times3\times5$$

Here x_4 is there because if $x_1x_2x_3 = 15$, then $x_4 = 2$ and if $x_1x_2x_3 = 5$, then $x_4 = 6$, etc.

x_4 is in fact a dummy variable.

Thus $x_1x_2x_3x_4 = 2\times3\times5$

Each of 2, 3 and 5 will be a factor of exactly one of x_1, x_2, x_3, x_4 ways.

$\therefore$ Required number = $4^3 = 64$.

19. Number of persons to be chosen = 6, number of officers = 4 and number of jawans = 8.

We know that total number of ways to select 6 persons such a way so as to include at least 1 officer = 1 officer out of 4 and 5 jawans out of 8 = ($^4C_1$ and $^8C_4$) or 2 officers out of 4 and 4 jawans out of 8 ($^4C_2$ and $^8C_4$) or 3 officers out of 4 and 3 jawans out of 8 ($^4C_3$ and $^8C_3$) or 4 officers out of 4 and 2 jawans out of 8($^4C_4$ and $^8C_2$).

Therefore, the total number of ways to choose the persons

$$= \left[{}^4C_1\times {}^8C_5\right]+\left[{}^4C_2\times {}^8C_4\right]+\left[{}^4C_3\times {}^8C_3\right]+\left[{}^4C_4\times {}^8C_2\right]$$

$$= \left(\frac{4!}{1!(4-1)!}\times\frac{8!}{5!(8-5)!}\right)+\left(\frac{4!}{2!(4-2)!}\times\frac{8!}{4!(8-4)!}\right)+\left(\frac{4!}{3!(4-3)!}\times\frac{8!}{3!(8-3)!}\right)+\left(\frac{4!}{4!(4-4)!}\times\frac{8!}{2!(8-2)!}\right)$$

$$= \left(\frac{4!}{1!\times3!}\times\frac{8!}{5!\times3!}\right)+\left(\frac{4!}{2!\times2!}\times\frac{8!}{4!\times4!}\right)+\left(\frac{4!}{3!\times1!}\times\frac{8!}{3!\times5!}\right)+\left(\frac{4!}{4!\times0!}\times\frac{8!}{2!\times6!}\right)$$

$$= \left(\frac{4\times3!}{1\times3!}\times\frac{8\times7\times6\times5!}{5!\times(3\times2\times1)}\right)+\left(\frac{4\times3\times2!}{(2\times1)\times2!}\times\frac{8\times7\times6\times5\times4!}{(4\times3\times2\times1)\times4!}\right)+\left(\frac{4\times3!}{3!\times1}\times\frac{8\times7\times6\times5!}{(3\times2\times1)\times5!}\right)+\left(\frac{4!}{4!\times1}\times\frac{8\times7\times6!}{(2\times1)\times6!}\right)$$

$= (4\times56)+(6\times70)+(4\times56)+(1\times28)$

$= 224 + 420 + 224 + 28 = 896$.

20. ×🧍×🧍×🧍×-----×🧍×

First let m men be seated is a row. Now, there are $(m + 1)$ places in which women can be seated. Thus,

$$^{m+1}C_n\times n!=\frac{(m-1)!\times n!}{n!(m+1-n)!}=\frac{(m+1)!}{(m-n+1)!}$$

$\therefore$ Required number of ways of sitting men and women

$$W=\frac{m!(m+1)!}{(m-n+1)!}$$

21. When there is no restriction, each letter can be dropped in any five of the letter boxes in 5 ways. Thus, the total number of ways without restriction = $5\times5\times5\times5$.

However, in 5 ways all the postcards can be dropped in the same letter box.

$\therefore$ Required number of ways = $5^4 - 5 = 620$

22. Number of ways $=\frac{1}{2}(5-1)!\times4!=\frac{24\times24}{2}=288$.

23. Total together of 7(3S's, 2C's, 2 diff.) letters = 7!/3!2!

Total together with 2C's in which 3S's come together = 5!/2!

$\therefore$ Total in which 3S's do not come together =

$$\frac{7!}{3!2!}-\frac{5!}{2!}=\frac{5!}{2!}\left(\frac{42}{6}-1\right)=360$$

For together we have used the string method.

24. Since the numbers to be formed are divisible by 3,

$\therefore$ we take the digits from the sets {1, 2, 3, 4, 5} or {0, 1, 2, 4, 5} [$\because$ sum is divisible by 3]

$\therefore$ Required number of numbers = 5! – 4! = 240 – 24 = 216.

25. A prize can be given to any of the 10 students in 10 ways.
$\therefore$ Required number of ways = 10^6.

26. Total number of letters = 15, number of C's = 3.
First, place 12 letters other than C's at dot places.

$\times\cdot\times\cdot\times\cdot\times\cdot\times\cdot\times\cdot\times\cdot\times\cdot\times\cdot\times\cdot\times\cdot\times$

The number of ways = $\frac{12!}{5!3!2!}$.
Since no two C's are together,
$\therefore$ Place C's at cross places whose number = 13.
Their arrangements are in $\frac{^{13}P_3}{3!}$ ways.

Total number of ways $= \frac{12!}{5!3!2!} \times \frac{^{13}P_3}{3!}$.

27. Total together (string method) of 6 letters (3A's, 2N's, 1B) = 6!/3!2!
Total together with 3A's in which 2N's come together = 5!/3!
Total in which 2N's do not come together

$$\frac{6!}{3!2!} - \frac{5!}{3!} = 60 - 20 = 40$$

28. Number of triangles = $^{12}C_3 - {}^7C_3$

$$= \frac{12\times11\times10}{3\times2\times1} - \frac{7\times6\times5}{3\times2\times1}$$

$$= 2\times11\times10 - 7\times5 = 220 - 35 = 185.$$

29.

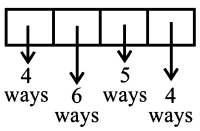


Fig. 6.9

Total ways = $4\times6\times5\times4 = 480$
(the first place can be filled by only 1, 2, 3 or 4)

30. The number of ways in which a student can make a selection of one or more books
$= (m+1)(m+1)\cdots(m+1) - 1$
$= (m + 1)^n - 1$
[$\because$ student can make selection of 0, 1, 2, 3, …, m books out of each type]

31. There are 2 even and 3 odd places.
$\therefore$ $^2P_2 \cdot {}^3P_3 = 2!\cdot3! = 2\cdot6 = 12$

32. (i) 1 black and 2 others
$= {}^3C_1 \times {}^6C_2 = 3\times15 = 45$
(ii) 2 black and 1 more
$= {}^3C_2 \times {}^6C_1 = 3 \times 6 = 18$
(iii) all black = $^3C_3 = 1$
$\therefore$ Total number of ways = 45 + 18 + 1 = 64.

33. Required number = Number of words that begin with A – Number of words that begin with A and end with E = $\frac{6!}{2!} - \frac{5!}{2!} = 300$.

34. Number of ways = $^4P_2 \times {}^6P_3$
[$\because$ First we choose 2 chairs out of 4 and then the number of chairs left is 6 and we choose 3 at a time.]

35. Two vowels, 5 consonants. 3 even and 4 odd places.
Two vowels a, e can be arranged in 4 odd places in $^4P_2$ = 12 ways and 5 consonants can be placed in 5 blank places in $^5P_5$ = 5! = 120 ways.
Hence, by fundamental theorem, $12 \times 120 = 1440$ ways.

36. The first stall can be filled in 3 ways, the second stall in 3 ways and so on.
Similarly, the 2th stall can be filled in 3 ways.
$\therefore$ Number of ways of loading the steamer is $3 \times 3 \times 3 \times \cdots \times 3$ (12 times) = 3^{12}.

37. When both A's are together, taking them as single identity, number of ways = 7!/3! ($\because$ there are 3L's)
When both A's and all three L's are together, taking them as two seperate identity, number of ways = 5!.

Required number of ways $\frac{7!}{3!} - 5! = 840 - 120 = 720.$

38. Required number of triangles = Total number of triangles
– Number of triangles having one side common with the octagon
– Number of tirangles having two sides common with the octagon
$= {}^8C_3 - {}^8C_1 \times {}^4C_1 - 8$
= 16.

39. At least one repeated
= Total with repetition – No repetition
$= 10^5 - {}^{10}P_5 = 100{,}000 - 30{,}240 = 69{,}760.$
when repetition is allowed, each of the five places can be filled in 10 ways. Hence by fundamental theorem, it is 10^5.

40. For each book, we can have 0, 1, 2, …, p copies. Thus, each book can be dealt in $(p + 1)$ ways.
$\therefore$ n books can be dealt in $(p + 1)^n$ ways.
This includes the case when all the books are rejected.
$\therefore$ Required number of ways = $(p + 1)^n - 1$.

41. Vowels are E, U, A, I, O
Consonants are Q, T, N

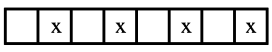


Fig. 6.10

According to the question, for the 3 consonants we have 4 places (shown in the figure).
Thus, the consonants can be arranged in ${}^4P_3$ ways and in the remaining places, vowels can be arranged in ${}^5P_5$ ways.

$\therefore$ Total ways $= {}^4P_3 \times {}^5P_5 = 24 \times 120 = 2880.$

42. First, place N, I, N, T, N at cross places

.x . x . x . x . x .

Then place E's at dot places. In that case, no two E's will be together.

$\therefore$ Total number of ways $= \frac{5!}{3!} \cdot {}^6C_3$

[$\because$ number of N's = 3 out of 5 (N, I, N, T, N)]

43. Total number of words $= {}^4_{V2}C \times {}^5_{C3}C \times 5! = 6 \times 10 \times 120$

$= 7200.$

44. Total number of greeting cards exchanged $= 2 \cdot {}^{20}C_2.$

45. Number of words starting with A,

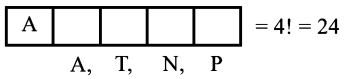


Fig. 6.11(i)

Number of words starting with N,

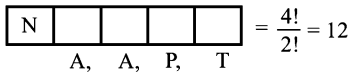


Fig. 6.11(ii)

Number of words starting with PAA

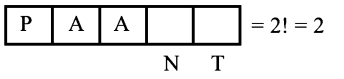


Fig. 6.11(iii)

Number of words starting with PAN

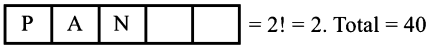


Fig. 6.11(iv)

The next words is PATAN, then PATNA is the 42nd word.

46. There will be $(5+1)(4+1)(2+1) = 90$ divisors. The sum of all the divisors is

$$(2^0 + 2^1 + 2^2 + \cdots + 2^5)(3^0 + 3^1 + 3^2 + \cdots + 3^5)$$
$$(5^0 + 5^1 + 5^2)$$

$$= \frac{2^6 - 1}{2 - 1} \cdot \frac{3^5 - 1}{3 - 1} \cdot \frac{5^3 - 1}{5 - 1} \text{ (sum of G.P.)}$$

$$= 63 \times 121 \times 31 = 3^2 \cdot 7 \cdot 11^2 \cdot 31$$

47. The four consonants all different can be arranged in 2nd, 3rd, 5th and 6th (four) places in 4! = 24 ways and three vowels A, E, A out of which two are alike can be arranged in 1st, 4th and 7th positions in (3!/2!) = 3 ways. Hence, the total number of ways is $24 \times 3 = 72$.

48. The number of ways of selecting any four numbers from 1 to 30 is ${}^{30}C_4$. Four consecutive numbers can be chosen in the following ways: (1, 2, 3); (2, 3, 4, 5), ..., (27, 28, 29, 30). Thus, required number of ways $= {}^{30}C_4 - 27 = 27{,}378.$

49. Consider the product $(x^0 + x^1 + x^2 + \cdots + x^9)(x^0 + x^1 + x^2 + \cdots + x^6)$ ... 6 factors. The number of ways in which the sum of the digits will be equal to 12 is equal to the coefficient of x^{12} in the above product.
So, the required number of ways
= coefficient of x^{12} in $(x^0 + x^1 + x^2 + \cdots + x^9)$ ... 6 factors

$$= \text{coefficient of } x^{12} \text{ in } \left(\frac{1 - x^{10}}{1 - x}\right)^6$$

$$= \text{coefficient of } x^{12} \text{ in } (1 - x^{10})^6 (1 - x)^{-6}$$

$$= \text{coefficient of } x^{12} \text{ in } (1 - x)^{-6}(1 - {}^6C_1 x^{10} + \cdots)$$

$$= \text{coefficient of } x^{12} \text{ in } (1 - x)^{-6} - {}^6C_1.$$

$$= \text{coefficient of } x^{12} \text{ in } (1 - x)^{-6}$$

$$= {}^{12+6-1}C_{6-1} - {}^6C_1 \times {}^{2+6-1}C_{6-1} = {}^{17}C_5 - 6 \times {}^7C_5 = 6062$$

50. Required number of ways $= 3^7 = 2187$
[$\because$ The first toy can be given in 3 ways, the second toy can be given in 3 ways and so on.]

51. 2P's, 2S's, 4 different letters in the word. All different 360, 2 alike, 2 different 240, 2 alike, 2 alike = 6.
$\therefore$ Total = 360 + 240 + 6 = 606.

52. The hall can be illuminated by switching on at least one of 10 bulbs.
$\therefore$ Required number of ways $= 2^{10} - 1 = 1023.$

53. Since there is restriction on girls, let us seat the boys first, which can be done in 6! ways.

$$\times B_1 \times B_2 \times B_3 \times B_4 \times B_5 \times B_6 \times$$

For 5 girls, there are 7 places shown by "×" above (note that here 2 or more boys can sit together) in which they can be seated in ${}^7P_5$ ways.
$\therefore$ Total ways $= {}^7P_5 \times 6!$

54. Required number of ways

$$= {}^{15}C_4 \cdot {}^5C_2 + {}^5C_3 \cdot {}^5C_3 + {}^5C_2 \cdot {}^5C_4$$

$$= 50 + 100 + 50 = 200$$

55. Each of the 10 places can be filled in 2 ways, i.e. by a 1 or a 2. Hence, by fundamental theorem $2 \times 2 \times 2 \ldots$ 10 times $= 2^{10}$.

56. A number between 5000 and 10,000 can have any of the digits 5, 6, 7, 8, 9 at the thousand's place.
Thus, the thousand's place can be filled up in 5 ways. The remaining 3 places can be filled up by the remaining 8 digits in ${}^8P_3$ ways.
$\therefore$ Required number of numbers $= 5 \times {}^8P_3$.

57. Suppose we select r things out of n alike things in one way and then we choose $n - r$ things out of n different things in ${}^nC_{n-r}$ ways, where r may vary from 0 to n. By fundamental theorem, the number of selections will be

$$1 \cdot {}^nC_{n-r} = 1 \cdot {}^nC_r, \text{ where } 0 \le r \le n$$

$$\therefore \quad \text{Total} = \sum_{r=0}^{n} {}^nC_r = C_0 + C_1 + C_2 + \cdots + C_n = 2^n.$$

58. Consider two vowels as one letter.
$\therefore$ Required number of ways $= 4! \cdot 2! = 48$.

59. The number of three-element subsets containing a_3 is equal to the number of ways of selecting 2 elements out of the $n - 1$ elements. So, the required number of subsets is ${}^{n-1}C_2$.

60. Number of five-letter words when one or more of its letters are repeated $= 10^5 = 100{,}000$.
Number of five-letter words when none of these letters is repeated $= {}^{10}P_5 = 30{,}240$.
Number of five-letter words when at least one of these letters is repeated $= 100{,}000 - 30{,}240 = 69{,}760$.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. Number of letters in the word STATISTICS = 10. We know that after fixing two S's (one in the begining and the other in the end), the number of remaining letters $= 10 - 2 = 8$.
Since the remaining latters have three T's and two I's therefore, the number of distinct permutations

$$= \frac{8!}{3! \times 2!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3}{3 \times 2} = 3360$$

2. Number of persons = 12
We know that the number of ways to arrange 12 persons in the round table $= (12 - 1)! = 11!$.
And the number of ways in which 2 particular persons sit side by side $= 10! \times 2!$.
Therefore, the total number of arrangements in which two particular persons are not to sit side by side
$= 11! - (10! \times 2!) = (11 \times 10!) - (10! \times 2 \times 1)$
$= (11 - 2) \times 10! = 9 \times 10!$.

3. ${}^nC_2 = 66$

$$\Rightarrow \frac{n(n-1)}{2} = 66 \quad \Rightarrow \quad n = 12, -11.$$

Hence, $n = 12$ (the –ve value being inadmissible).

4. Number of triangle with vertices on AB, BC, CD $= {}^3C_1 \times {}^4C_1 \times {}^5C_1$.
Similarly, other cases can be found.
$\therefore$ Required number of triangles
$= {}^3C_1 \times {}^4C_1 \times {}^5C_1 + {}^3C_1 \times {}^4C_1 \times {}^6C_1$
$+ {}^3C_1 \times {}^5C_1 \times {}^6C_1 + {}^4C_1 \times {}^5C_1 \times {}^6C_1 = 342$.

5. Number of ways without restriction $= {}^{15}C_6$
Number of ways when no woman is selected $= {}^{10}C_6$
$\therefore$ Required ways $= {}^{15}C_6 - {}^{10}C_6 = {}^{15}C_9 - {}^{10}C_4$
$[\because {}^nC_{n-k} = {}^nC_k]$

6. $\frac{4!}{4!} = 1$, $[\because$ all faces are alike$]$

7. ${}^{10}P_9 - {}^9P_8$ (0 in 1st place)
$= 10! - 9! = 9!(10 - 1) = 9 \times 9!$

Alternative. The first place can be filled in 9 ways (0 cannot be in the first place). The remaining 9 places can be filled by 9 digits (0 can be there) in 9! ways. Hence, by fundamental theorem, $9 \times 9!$ ways.

8. The number of times A and B are together on guard is

$$\binom{n-2}{10}$$

The number of times C, D and E are together on guard is

$$\binom{n-3}{9}$$

According to the question, $\binom{n-2}{10} = 3\binom{n-3}{9}$

$$\Rightarrow n - 2 = 30$$
$$\Rightarrow n = 32$$

9. $x = {}^9C_5$, $y = {}^9C_6$
$\therefore$ $x - y = {}^9C_5 - {}^9C_6 = 42$

10. Since each choice of a pair of horizontal lines and a pair of vertical lines gives us a rectangle, hence the number of rectangles $= {}^5C_2 \times {}^5C_2$.

11. It will be a 4-digit number and have 3 in the 1st place and 5 in the last place because of the given condition. Out of the remaining 4 given digits, we have to choose 2 digits in ${}^4P_2 = 12$ ways.

12. The word ARRANGE has 7 letters in which 2 are A's, 2 are R's and 3 different are N, G and E.

$\therefore$ Total number of arrangements $= \frac{7!}{2!2!} = 1260.$

Number of arrangements in which both RR are together as one $= \frac{6!}{2!} = 360.$

$\therefore$ Required number of ways = 1260 – 360 = 900.

13. There are 5 animals which can be kept in the 4 small cages. The small cages can be filled in ${}^5P_4$ ways and the rest in 7! ways.

$\therefore$ $120 \times 7! = 604{,}800$

14. Required number of ways = 5! 3! 2! = $120 \times 6 \times 2 = 1440$.

(Regarding 5 Math books as one and 3 Physics books as one)

15. Repetition allowed: The first place can be filled in 4 ways only ($\neq 0$) and each remaining in 5 ways. (repetition allowed). Hence the number of numbers = $4 \times 5 \times 5 \times 5 \times 5 = 625 \times 4 = 2500$

Repetition not allowed: The number of numbers = 5!– 4! (0 in 1st place) = 120 – 24 = 96

$\therefore$ The number of numbers with at least one digit repeated

= Repetition – No repetition

= 2500 – 96 = 2404.

16. Required number of ways

= Coefficient of y^{10} in $(1 + y + y^2 + \cdots)^4$

= Coefficient of y^{10} in $(1 - y)^{-4}$

= Coefficient of y^{10} in

$$(1 + {}^4C_1 y + {}^5C_2 y^2 + {}^6C_3 y^3 + \cdots)$$

$$= {}^{13}C_3 = \frac{13 \times 12 \times 11}{2 \times 3} = 286.$$

17. Required number of ways = Coefficient of x^{20} in $(x^2 + x^3 + x^4 \ldots.)^5$

= Coefficient of x^{20} in $x^{10}(1 + x + x^2 + \cdots)^5$

= Coefficient of x^{10} in $(1 - x)^{-5}$

$= {}^{10+5-1}C_{10} = {}^{14}C_{10}$

$= {}^{14}C_4 \quad (\because {}^nC_r = {}^nC_k \Rightarrow r + k = n)$

18. The number of triangles with vertices on different lines

$${}^pC_1 \times {}^pC_1 \times {}^pC_1 = p^3$$

The number of triangles with 2 vertices on one line and the third vertex on any one of the other two lines

$$= {}^3C_1 ({}^pC_2 \times {}^{2p}C_1) = 6p \cdot \frac{p(p-1)}{2}$$

$= 3p^2(p - 1)$

$\therefore$ The required number of triangles

$= p^3 + 3p^2(p-1) = 4p^3 - 3p^2 = p^2(4p - 3)$

(The word 'maximum' shows that no selection of points from each of the three lines is collinear).

19. Required number = Repetitions allowed – No repetition

Repetition allowed = $9 \times 10 \times 10 \times 10 = 9000$

($\because$ The first place can be filled in 9 ways only.)

Repetition not allowed = ${}^{10}P_4 - {}^{10}P_3$

i.e. Total – (Number beginning with zero)

$\Rightarrow 10 \cdot 9 \cdot 8 \cdot 7 - 9 \cdot 8 \cdot 7 = 9 \cdot 9 \cdot 8 \cdot 7 = 4536$

$\therefore$ Required number = 9000 – 4536 = 4464.

20. We have

Possibility	Selections	Arrangements
One triplet and 2 different	${}^3C_1 \times {}^2C_2$	${}^3C_1 \times {}^2C_2 \times \frac{5!}{3!} = 60$
Two pairs and one different	${}^3C_1 \times {}^1C_1$	${}^3C_2 \times {}^2C_1 \times \frac{5!}{2!2!} = 90$

21. ${}^{2n+1}C_1 + {}^{2n+1}C_2 + \cdots + {}^{2n+1}C_n = 255$

$\Rightarrow {}^{2n+1}C_0 + {}^{2n+1}C_1 + \cdots + {}^{2n+1}C_n = 256$

$\Rightarrow \frac{2^{2n+1}}{2} = 256 \quad \Rightarrow \quad n = 4$

22. There are four possibilities:

(i) At least one rupee coin, any number of 50-paise coins, any number of 25-paise coins. This can be done in $2 \times 4 \times 5 = 40$ ways

(ii) At least two 50-paise coins, any number of 25-paise coins. This can be done in $2 \times 5 = 10$ ways.

(iii) One 50-paise coin, at least two 25-paise coins in $1 \times 3 = 3$ ways

(iv) Four 25-paise coins in 1 way.

Total number of ways = 40 + 10 + 3 + 1 = 54 ways.

23. He can select either 1, 2, … or n books. The number of ways is

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + \cdots + {}^{2n+1}C_n$$

$$= \frac{1}{2}\left[{}^{2n+1}C_1 + {}^{2n+1}C_2 + \cdots + {}^{2n+1}C_n + {}^{2n+1}C_{n+1} + \cdots + {}^{2n+1}C_{2n} \right]$$

$$= \frac{1}{2}\left[2^{2n+1} - {}^{2n+1}C_0 - {}^{2n+1}C_{2n+1} \right]$$

$\Rightarrow 2^{2n}-1=63$ (given)

$\Rightarrow 2^{2n}=64=2^6$

$\Rightarrow n=3$

24. We have

$$70!=2^{\alpha}\times 3^{b}\times 5^{c}\times 7^{d}\times\cdots$$

Now,

$a = E_2(70!)$

$$=\left[\frac{70}{2}\right]+\left[\frac{70}{2^2}\right]+\left[\frac{70}{2^3}\right]+\left[\frac{70}{2^4}\right]+\left[\frac{70}{2^5}\right]+\left[\frac{70}{2^6}\right]$$

$= 35 + 17 + 8 + 4 + 2 + 1 = 67$

and $c=E_5(70!)=\left[\frac{70}{5}\right]+\left[\frac{70}{5^2}\right]=14+2=16$

$\therefore\quad 70!=2^{67}\times 5^{16}\times 3^{b}\times 7^{d}\times\cdots$

$\Rightarrow\quad 70!=(2\times 5)^{16}\times 2^{51}\times 3^{b}\times 7^{d}\times\cdots$

$\Rightarrow\quad 70!=10^{16}\times 2^{51}\times 3^{b}\times 7^{d}\times\cdots$

Thus, the number of zeros at end of 70! is 16.

25. O + O + O + O + O + O + O

We can put – sign in any of the 7 circle in the above arrangement i.e. We have to select 4 circles out of 7 which can be done in ${}^7C_4$ ways = 35 ways.

26. The required number

$$= \text{Coefficient of } x^{2m} \text{ in } (x^0+x^1+\cdots+x^m)^4$$

$$= \text{Coefficient of } x^{2m} \text{ in } \left(\frac{1-x^{m+1}}{1-x}\right)^4$$

$$= \text{Coefficient of } x^{2m} \text{ in } (1-x^{m+1})^4\,(1-x)^{-4}$$

$$= \text{Coefficient of } x^{2m} \text{ in } (1-4x^{m+1}+6x^{2m+2}+\cdots)$$

$$\left(1+4x^{m+1}\cdots\frac{(r+1)(r+2)(r+3)}{3!}x^r+\cdots\right)$$

$$=\frac{(2m+1)(2m+2)(2m+3)}{6}-4m\frac{(m+1)(m+2)}{6}$$

$$=\frac{(m+1)(2m^2+4m+3)}{3}$$

27. There are two middle seats out of 10 i.e. 5th and 6th on which two particular players can be seated in 2! ways = 2 ways. Remaining eight players can be seated in the remaining eight seats in ${}^8P_8 = 8!$ ways.

$\therefore$ Total number of arrangements $= 2\cdot(8!)$ by fundamental theorem.

28. Number of different tickets

$= 20 + 19 + 18 + 17 + \cdots + 3 + 2 + 1$

$$=\frac{20}{2}(20+1)=10(21)=210.$$

29. This question is based on the derangement theory which states that if n things are arranged in a row, the number of ways in which they can be deranged so that none of them occupies its original place is

$$n!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\cdots+(-1)+\frac{n!}{n!}\right]$$

Here, $n = 4$, thus the required ways

$$=4!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right)$$

$$=24\left(\frac{1}{2}-\frac{1}{6}+\frac{1}{24}\right)=12-4+1=9$$

Hence, the correct answer option is (a).

30. The natural numbers are 1, 2, 3, 4.

Clearly, in one diagonal we have to place 1, 4 and in the other 2, 3.

The number of ways for (i) = 2! × 2! = 4

The number of ways for (ii) = 2! × 2! = 4

Total number of ways = 4 + 4 = 8.

31. ${}^4C_2\times{}^3C_2 = 6\times 3 = 18$

32. $2^{10}5^{10}11^{11}13^{13}$ has a divisor of the form $2^{\alpha}\cdot 5^{\beta}\cdot 11^{\gamma}\cdot 13^{\delta}$, where $\alpha=0, 1, 2, \ldots, 10$; $\beta=0, 1, 2, \ldots, 10$; $\gamma=1, 2, \ldots 11$; $\delta = 1, 2, \ldots, 13$.

It is of the form $4n + 1$, if

$\alpha=0$; $\beta=0, 1, 2, \ldots. 10$; $\gamma=0, 2, 4, \ldots, 10$; $\delta=0, 1, 2, \ldots, 13$

$\therefore$ Number of divisors $11\times 6\times 14 = 924$.

33. Here we have to count '2' in 500!, To get this, count only one '2' from

(i) multiple of 2 i.e., 2, 4, 6, …, 500 ≡ 250

(ii) multiple of 4 i.e., 4, 8, 12, …, 500 ≡ 125

(iii) multiple of 8 i.e., 8, 16, 24, …, 496 ≡ 62

(iv) multiple of 16 i.e., 16, 32, 48, …, 496 ≡ 31

(v) multiple of 32 i.e., 32, 64, 96, …, 480 ≡ 15

(vi) multiple of 64 i.e., 64, 128, …, 448 ≡ 7

(vii) multiple of 128 i.e., 128, 256, 384 ≡ 3

(viii) multiple of 256 i.e., 256 ≡ 1

Thus, total = 494.

34. We can choose 2 men out of 9 in ${}^9C_2$ ways. Since no husband and wife are to play in the same game, 2 women out of the remaining 7 can be chosen in ${}^7C_2$ ways. If M_1, M_2, W_1 and W_2 are chosen, then a team may consists of M_1 and W_1 or M_2 and W_2. Thus, the number of ways of arranging the game is

$$\left({}^9C_2\right)\left({}^7C_2\right)2=\frac{9\times 8}{2}\times\frac{7\times 6}{2}\times 2=1512.$$

35. ${}^8C_2 - 8 = 28 - 8 = 20$.

36. For $1\le i\le p-1$, $p!$ is divisible by $(i-1)$.

Thus, $n + i = p! + (i + 1)$ is divisible by $(i + 1)$ for $1\le i\le p-1$.

Hence, none of $n + 1, n + 2, \ldots, n + p - 1$ is prime.

37. Each ball has 3 choices. Hence the total number of ways of distribution = $3\times3\times3\times3\times3\times3=3^6$
But this includes tow cases:
(i) One box is empty.
(ii) Two boxes are empty.
The number of ways in which this happens = $3(2^6-1)$
Hence, the required number = $3^6-3(2^6-1)$
$= 729-3\times63=729-189=540.$

38. The total numbers of factors of $x^n yz \ldots$ is equal to the number of ways of selecting one or more out of n identical quantities of one type and the remaining m distinct quantities.
Hence, the required numbers of factors = $(n+1)^{2m}$

39. There are 9 letters viz. 3a's, 2b's and 4c's.
These can be arranged in $\frac{9!}{3!2!4!}$ i.e. 1260 ways.

40. The number of matches in the first round = ${}^6C_2 + {}^6C_2$ = 15 + 15 = 30.
The number of matches in the next round = ${}^6C_2 = 15$.
The number of matches in the semi-final round = ${}^4C_2$ = 6.
$\therefore$ The required number of matches
= 30 + 15 + 6 + 2 = 53
($\because$ for best of three at least two matches are played)

41. The signals can be made by using at a time one or more flags.
The total number of signals when r flages are used at a time from 6 flags is equal to the number of arrangements of 6, taking r at a time i.e. ${}^6C_r \times r!$. Since r can take values 1, 2, 3, 4, 5, 6. Hence, the number of different signals that can be generated is

$$\sum_{r=1}^{6} {}^6C_r \times r! = 6+30+120+360+720+720=1956$$

42. 0 cannot be used. Out of remaining 9 digits, choose 3.
Therefore, $N = {}^9C_3 = 84$
and it is obvious 84 is divisible by 7.
So, both Statement 1 and Statement 2 are true but Statement 1 does not follow from Statement 2.

43. Number of ways of selecting 2 out of $n = {}^nC_2$.
Out of these lines, n lines are the sides of the polygon.
Thus,

$$^nC_2 - n = \frac{n(n-3)}{2}$$

If n = 11, then the number of diagonals is

$$\frac{11\times(11-3)}{2} = 44.$$

So, both Statement 1 and Statement 2 are true and Statement 1 follows from Statement 2.

44. Number of matrices of the form 1×4 is 4!, 4×1 is 4! and 2×2 is 4!
So, the total number of matrices = 4! + 4! + 4! = 72.
Only Statement 1 is true. Statement 2 is false.

45. Considering three vowels as one letter, and 6 letters can be arranged in ${}^6P_6 = 6!$ ways. The vowels A, U, E can be but together in 3! ways.
Hence, the required number of words
$= 6!\times3! = 720\times6 = 4320$
The total number of words in which vowels are never together
= Total number of words – Number of words in which vowels are always together.
= 40320 – 4320 = 36000
So, both Statement 1 and Statement 2 are true. But Statement 1 does not follow from Statement 2.

46. $\frac{{}^nC_{n-r}}{{}^nC_{r+1}} = \frac{r+1}{n-r} \Rightarrow 2n-5r=3$ (i)

$\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{r}{n-(r-1)} \Rightarrow 3n-10r=-3$ (ii)

By solving (i) and (ii), we get $r = 3$
$\Rightarrow {}^rC_2 = {}^3C_2 = 3$
So, Statement 1 is false but Statement 2 is true.

47. Clearly, there are six letters in the word BANANA of which three are alike of one kind (3 A's), two are alike of second kind (2N's) and one of its own kind. Total number of permutations = 6!/3!2! = 60, which follows from Statement 2.

48. Digits are 0, 1, 2, 3, 5, 7.
The last place can be filled by (1, 3, 5, 7) 4 ways as the number is to be odd and the first place can be filled by 5 ways (excluding 0), so the total number will be $5\times6\times6\times4=720$.
($\because$ The second and third places can be filled by 6 ways.)

49. Numbers greater than 1000 and less than or equal to 4000 will be of 4 digits and will have either 1 (except 1000) or 2 or 3 in the first place with 0 in each of the remaining places.
After fixing the 1st place, the second place can be filled by any of the 5 numbers. Similarly, the third place can be filled by 5 ways and 4th place can be filled by 5 ways. Thus, there will be $5\times5\times5=125$ ways in which 1 will be in the first place but this include 1000 also; hence, there will be 124 numbers having 1 in the first place. Similarly, 125 for each 2 and 3 in the first place. One number will be in which 4 is in the first place, i.e. 4000. Hence, the required ways are
124 + 125 + 125 + 1 = 375 ways.

50. ${}^nC_{r+1} + {}^nC_{r-1} + 2{}^nC_r$

$= {}^nC_{r+1} + {}^nC_r + {}^nC_{r-1} + {}^nC_r$

$= {}^{n+1}C_{r+1} + {}^{n+1}C_r$

$= {}^{n+2}C_{r+1}$

51. Number of ways in which 6 men can be arranged at a round table = $(6-1)!$.
Now, women can be arranged in 6! ways.
Total number of ways = $6! \times 5!$.

52. The possibilities are

From first five	Remaining questions
4	6
5	5

Therefore, the required number of choices

$= {}^5C_4 \times {}^8C_6 + {}^5C_5 \times {}^8C_5$

$= 5 \times {}^8C_2 + 1 \times {}^8C_3$

$= 5 \times \frac{8\times 7}{2\times 1} + 1 \times \frac{8\times 7\times 6}{3\times 2\times 1}$

$= 140 + 56 = 196$

53. Required number of ways $= {}^{8-1}C_{3-1}$

$= {}^7C_2 = \frac{7!}{2!5!} = \frac{7\cdot 6}{2\cdot 1} = 21$

54. Taking AE as one letter, then, 5 letters can be arranged in 5! ways, that is, 120 ways.

55. Words starting with A, C, H, I, N are equals to 5!.
$\therefore$ Total words = $5 \times 5! = 600$.
The first word starting with S is SACHIN.
$\therefore$ SACHIN appears in the dictionary at serial number 601.

56. Total number of ways

$= {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4$

$= 10 + 45 + 120 + 210 = 385$

57. $S = \{1, 2, 3, \ldots, 12\}$ is to be partitioned into the sets A, B and C of equal size, i.e. each set of 4 elements. It is given that $A \cap B = B \cap C = A \cap C = \phi$, so in each set all the elements are different. Hence, the number of ways to partitions is $\frac{12!}{(4!)^3 3!} \times 3! = \frac{12!}{(4!)^3}$.

58. Other than S, seven letters M, I, I, I, P, P, I can be arranged in $\frac{7!}{2!4!} = 7\cdot 5\cdot 3$ ways.

Now four S's can be placed in 8 spaces in ${}^8C_4$ ways.
$\therefore$ Desired number of ways $= 7\cdot 5\cdot 3\cdot {}^8C_4$

$= 7\cdot {}^6C_4 \cdot {}^8C_4$.

59. The number of ways the child can buy the six ice-creams is equal to the number of different ways of arranging 6A's and 4B's in a row.
$\therefore$ Number of ways to arrange 6A's and 4B's in a row.

$\frac{10!}{6!4!} = {}^{10}C_4$

and the number of integral solutions of the equation

$x_1 + x_2 + x_3 + x_4 + x_5 = 6$

$= {}^{6+5-1}C_{5-1} = {}^{10}C_4 \neq {}^{10}C_5$

Statement 1 is false and Statements 2 is true.

60. Four novels can be selected from 6 novels in ${}^6C_4$ ways. One dictionary can be selected from 3 dictionaries in ${}^3C_1$ ways. As the dictionary selected is fixed in the middle, the remaining 4 novels can be arranged in 4! ways. Threrefore, the required number of ways of arrangement $= {}^6C_4 \times {}^3C_1 \times 4! = 1080$.

61. Since a set of m parallel lines intersects a set of another n parallel lines in a plane, then the number of parallelograms formed is ${}^mC_2 \cdot {}^nC_2$.

62. Since questions 1 and 2 are compulsory, 4 questions are left from 6 questions and 2 are to be chosen from them.

$\therefore$ Required number of ways $= {}^4C_2 = \frac{4\times 3}{2} = 6$

63. $\frac{1}{{}^5C_r} + \frac{1}{{}^6C_r} = \frac{1}{{}^4C_r}$

$\Rightarrow \frac{(5-r)!r!}{5!} + \frac{(6-1)!r!}{6!} = \frac{(4-r)!r!}{4!}$

$\left[\because {}^nC_r = \frac{n!}{(n-r)!r!}\right]$

$\Rightarrow \frac{(5-r)!}{5!} + \frac{(6-r)!}{6!} = \frac{(4-r)!}{4!}$

$\Rightarrow \frac{(5-r)(4-r)!}{5(4!)} + \frac{(6-r)(5-r)(4-r)!}{6\cdot 5(4!)} = \frac{(4-r)!}{4!}$

$\Rightarrow \frac{5-r}{5} + \frac{(6-r)(5-r)}{30} = 1$

$\Rightarrow 30 - 6r + 30 - 6r - 5r + r^2 = 30$

$\Rightarrow 30 - 6r + 30 - 11r + r^2 = 30$

$\Rightarrow r^2 - 17r + 30 = 0$

$\Rightarrow r^2 - 15r - 2r + 30 = 0$

$\Rightarrow (r-15)(r-2) = 0 \quad \Rightarrow \quad r = 15, 2$

$\Rightarrow r = 2 \qquad (\because r \neq 15)$

64. When the number at the unit place is 3, then the other 3 numbers can be arranged in 3! ways.
$\therefore$ The sum of the digits in the unit place when 3 is at the unit place = $3! \times 3$
Similarly, the sum of the digits when
4 is at the unit place = $3! \times 4$
5 is at the unit place = $3! \times 5$
6 is at the unit place = $3! \times 6$
$\therefore$ The sum of the digits in the unit place of all numbers formed with the help of 3, 4, 5, 6 taken all at a time is

$$(3!\times 3)+(3!\times 4)+(3!\times 5)+(3!\times 6)$$
$$=3!(3+4+5+6)=6\times 18=108$$

65. 10 questions can be answered in 2^{10} ways.
($\because$ Each true-false questions can be answered in 2 ways).

66. Total number of seats = n
and number of people = m
1st person can be seated in n ways.
2nd person can be seated in $(n - 1)$ ways.
$\vdots$
mth person can be seated in $(n - m + 1)$ ways.
$\therefore$ Total number of ways
$= n(n-1)(n-2)\ldots(n-m+1)$
$= {}^nP_m$ ways.

67. Given word is PENCIL.
Number of total letters in the given word = 6
Number of vowels = 2
Number of consonants = 4
Firstly, the 4 consonants in alternate positions can be arranged in 4! ways.
$\therefore$ Remaining two places can be filled by two vowels in ${}^5P_2 = 5$ ways.
$\therefore$ Total number of ways = $4! \, {}^5P_2$
$= 24 \times 20 = 480.$

68. Given four numbers 1, 2, 3 and 4.
One digit number that can be formed = ${}^4P_1 = 4$
Two digit numbers = ${}^4P_2 = 12$
Three digit numbers = ${}^4P_3 = 24$
Four digit numbers = ${}^4P_4 = 24$
Total number of numbers that can be formed = 4 + 12 + 24 + 24 = 64.

69. Total number of persons = 10
4 females and 1 male can be selected in ${}^4C_1 \cdot {}^6C_1$ ways
3 females and 2 males can be selected in ${}^4C_3 \cdot {}^6C_2$ ways
2 females and 3 males can be selected in ${}^4C_2 \cdot {}^6C_3$ ways
1 female and 4 males can be selected in ${}^4C_1 \cdot {}^6C_4$ ways
Total number of ways = 6 + 60 + 120 + 60 = 246.

70. Total number of numbers = $2 \times 2 \times \cdots \times 10$ times
$= 2^{10}$.

71. $\because \; {}^8C_r = {}^7C_3 + {}^7C_2$
$\Rightarrow \; {}^8C_r = {}^8C_3 \qquad (\because \; {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r)$
$\Rightarrow \; r = 3.$

72. The boys are in majority, if the groups are 4B 3G, 5B 2G, 6B, 1G.
Total number of combinations

$$= {}^6C_4 \times {}^4C_3 + {}^6C_5 \times {}^4C_2 + {}^6C_6 \times {}^4C_1$$
$$= 15\times 4+6\times 6+1\times 4$$
$$= 60 + 36 + 4 = 100.$$

73. $P(A)=\dfrac{60}{100}=\dfrac{3}{5}, P(B)=\dfrac{50}{100}=\dfrac{1}{2},$

$$\text{Required probability} = P(A\bar{B}+\bar{A}B)$$
$$= P(A\bar{B})+P(\bar{A}B)$$
$$= P(A)P(\bar{B})+P(\bar{A})P(B)$$
$$= \left(\frac{3}{5}\right)\left(\frac{1}{2}\right)+\left(\frac{2}{5}\right)\left(\frac{1}{2}\right)$$
$$= \frac{5}{10}=\frac{1}{2}$$

74. Arrange the letters of the word COCHIN as in the order of dictionary CCHINO.
Consider the words starting from C.
There are 5! such words. Number of words with the two C's occupying first and second place = 4!.
Number of words starting with CH, CI, CN is 4! each.
Similarly, the number of words before the first word starting with CO = 4! + 4! + 4! + 4! = 96.
The word starting with CO found first in the dictionary is COCHIN.
There are 96 words before COCHIN.

75. LCM of p and q will be r^2, if exponent of r is 2 in at least one of p and q i.e., (0, 2), (2, 0), (1, 2), (2, 1) and (2, 2), so the exponent for r can be selected in $2 \times 2 + 1 = 5$ ways.
Similarly, the exponent for s and t can be selected in $2\times 4+1=9$ and $2\times 2+1=5$ ways, respectively.
Hence, the total number of selections $=5\times 9\times 5=225$ ways.

76. ${}^{10}C_3 - {}^6C_3$

$$=\frac{10\times 9\times 8}{6}-\frac{6\times 5\times 4}{6}=120-20=100$$

77. The number of ways of selecting one or more balls from 10 white, 9 green, and 7 black balls is
$(10 + 1)(9 + 1)(7 + 1) - 1 = 11 \cdot 10 \cdot 8 - 1 = 879$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. Let P_m stand for ${}^mP_m$. Then the expression

$$1 \cdot P_1 + 2 \cdot P_2 + 3 \cdot P_3 + \cdots + n \cdot P_n =$$

(a) $(n + 1)! - 1$ (b) $(n + 1)! + 1$
(c) $(n + 1)!$ (d) none of these

2. How many words can be made with the letters of the word GENIUS if each word neither begins with G nor ends with S?
(a) 24 (b) 240
(c) 480 (d) 504

3. The maximum number of points of intersection of 8 circles is
(a) 56 (b) 28
(c) 24 (d) 16

4. If ${}^nP_r = 5040\left({}^{n-1}C_{r-1} + {}^{n-1}C_r\right)$, then
(a) $r = 5$ (b) $r = 6$
(c) $r = 7$ (d) $r = 8$

5. The number of integer solutions for the equation $x + y + z + t = 20$, where, x, y, z, t are all ≥ -1, is
(a) ${}^{20}C_4$ (b) ${}^{23}C_3$
(c) ${}^{27}C_4$ (d) ${}^{27}C_3$

6. Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, then n is equal to
(a) 5 (b) 7
(c) 6 (d) 4

7. There are m points on a straight line AB and n points on the line AC, none of them being the point A. Triangles are formed with these points as vertices, when,
(i) A is excluded
(ii) A may be included. The ratio of the number of triangles in the two cases is

(a) $\dfrac{m+n-2}{m+n}$ (b) $\dfrac{m+n-2}{m+n-1}$

(c) $\dfrac{m+n-2}{m+n+1}$ (d) $\dfrac{m(n-1)}{(m+1)(n+1)}$

8. Passengers are to travel by a double-decked bus which can accommodate 13 in the upper deck and 7 in the lower deck. The number of ways that they can be distributed if 5 refuse to sit in the upper deck and 8 refuse to sit in the lower deck is
(a) 25 (b) 21
(c) 18 (d) 15

9. How many divisors of 21600 are divisible by 10 but not by 15?
(a) 10 (b) 30
(c) 40 (d) none of these

10. The number of ways in which the number 94864 can be resolved as a product of two factors is
(a) 30 (b) 23
(c) 45 (d) 46

11. If chocolates of a particular brand are all identical then the number of ways in which we can choose 6 chocolates out of 8 different brands available in the market is
(a) ${}^{13}C_6$ (b) ${}^{13}C_8$
(c) 8^6 (d) none of these

12. The number of ways in which 20 different pearls of two colours can be set alternately on a necklace, there being 10 pearls of each colour, is
(a) $9! \times 10!$ (b) $5(9!)^2$
(c) $(9!)^2$ (d) $6!(9!)^2$

13. The exponent of 12 in 100! is
(a) 48 (b) 49
(c) 96 (d) 97

14. The number 24! is divisible by
(a) 6^{24} (b) 24^6
(c) 12^{12} (d) 48^5

15. In an election, the number of candidates is 1 greater than the persons to be elected. If a voter can vote in 254 ways, then the number of candidates is
(a) 7 (b) 10
(c) 8 (d) 6

16. Between two junction stations A and B, there are 12 intermediate stations. The number of ways in which a train can be made to stop at 4 of these stations so that no two of these halting stations are consecutive is
(a) ${}^8C_4$ (b) ${}^9C_4$
(c) ${}^{12}C_4 - 4$ (d) ${}^{10}C_4$

17. Let $A = \{x | x$ is a prime number and $x < 30\}$. The number of different rational numbers whose numerators and denominators belong to A is
(a) 90 (b) 180
(c) 91 (d) 181

18. The greatest integer less than $(\sqrt{2}+1)^6$ is
(a) 198 (b) 197
(c) 196 (d) none of these

19. In the expansion of $(1+x)^{70}$, the sum of the coefficients of odd powers of x is
(a) 0 (b) 2^{69}
(c) 2^{70} (d) 2^{71}

20. Numerically the longest term in the expansion of $(3+2x)^{50}$, when $x = 1/5$ is
(a) 5th (b) 8th
(c) 7th (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. A train going from London to Cambridge stops at 12 intermediate stations. Seventy-five persons enter the train during the journey with 75 different tickets of the same class. The number of different sets of tickets they may be holding is
(a) ${}^{78}C_3$ (b) ${}^{91}C_{75}$
(c) ${}^{84}C_{75}$ (d) none of these

2. Seven different coins are to be divided amongst three persons. If no two of the persons receive the same number of coins but each receives at least one coin and none is left over, then the number of ways in which the division may be made is
(a) 420 (b) 630
(c) 710 (d) none of these

3. The streets of a city are arranged like the lines of a chess board. There are m streets running north to south and n streets running east to west. The number of ways in which a man can travel from NW to SE corner going the shortest possible distance is
(a) $\sqrt{m^2+n^2}$ (b) $\sqrt{(m-1)^2\cdot(n-1)^2}$
(c) $\dfrac{(m+n)!}{m!\cdot n!}$ (d) $\dfrac{(m+n-2)!}{(m-1)!\cdot(n-1)!}$

4. The number of ordered triplets of positive integers which are solutions of the equation $x + y + z = 100$ is
(a) 5081 (b) 6005
(c) 4851 (d) none of these

5. The number of ways in which a batsman can score 14 runs in 6 balls not scoring more than 4 runs in any ball is
(a) ${}^{19}C_5 - 6\cdot{}^{14}C_5 + 15\cdot{}^{9}C_5$
(b) ${}^{19}C_5 + 6\cdot{}^{14}C_5 - 15\cdot{}^{9}C_5$
(c) 386
(d) 286

6. There are n numbered seats around a round table. The number of ways in which $m(m < n)$ persons can sit around the round table is
(a) ${}^{n}C_m \cdot m!$ (b) ${}^{n}C_m - (m-1)!$
(c) $\dfrac{{}^{n}C_m - (m-1)!}{2}$ (d) none of these

7. The number of ways in which 5X's cab be placed in the squares of Fig. 6.12 so that no row remains empty is

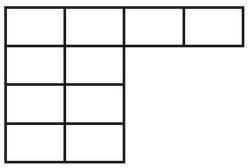


Fig. 6.12

(a) 97 (b) 44
(c) 100 (d) 126

8. Two variants of a test paper are distributed among 12 students. The number of ways of seating of the students in two rows so that the students sitting side by side do not have identical papers and those sitting in the same column have the same paper is
(a) $\dfrac{12!}{6!6!}$ (b) $\dfrac{(12)!}{2^5\cdot 6!}$
(c) $(6!)^2\cdot 2$ (d) $12!\times 2$

9. Let P_n denotes the number of ways of selecting 3 people out of n sitting in a row if no two of them are consecutive and Q_n is the corresponding figure when they are in a circle. If $P_n - Q_n = 6$, then n is equal to
(a) 8 (b) 9
(c) 10 (d) 12

10. The number of times the digits 3 will be written when listing the integers from 1 to 1000 is
(a) 269 (b) 300
(c) 271 (d) 302

11. If $x \in N$ and ${}^{x-1}C_4 - {}^{x-1}C_3 - \dfrac{5}{4}\,{}^{x-2}P_2 < 0$, then x is equal to
(a) 4, 5, 6, 7, 8, 9, 10 (b) $3 < x < 11$
(c) $3 \le x \le 10$ (d) none of these

12. The number of non-negative integral solutions of the equation $x + y + 3z = 33$ is
(a) 120 (b) 135
(c) 210 (d) 520

13. A is a set containing n different elements. A subset P of A is chosen. The set A is reconstructed by replacing the

elements of P. A subset Q of A is again chosen. The number of ways of choosing P and Q so that $P \cap Q$ contains exactly two elements is

(a) ${}^nC_3 \cdot 2^n$ (b) ${}^nC_2 \cdot 3^{n-2}$

(c) 3^{n-2} (d) none of these

14. The number of ways in which 2 Indians, 3 Americans, 3 Italians and 4 French can be seated on a circle, if the people of the same nationality sit together is

(a) $2 \cdot (4!)^2\,(3!)^2$ (b) $2 \cdot (3!)^3 \cdot 4!$

(c) $2 \cdot (3!)\,(4!)^3$ (d) none of these

15. The number of ways in which 3 numbers in A.P. can be selected from 1, 2, 3, …, n is

(a) $\left(\frac{n-1}{2}\right)^2$ if n is even (b) $\frac{n(n-2)}{4}$ if n is odd

(c) $\frac{(n-2)^2}{4}$ if n is odd (d) $\frac{n(n-3)}{4}$ if n is even

16. If $(1+x)^n = \sum_{r=0}^{n} C_r x^r$, then the value of $C_1 + 2C_2 + 3C_3 + \cdots + nC_n$ equals

(a) $n2^n$ (b) $(n+1)2^n$

(c) $n2^{n-1}$ (d) $n \cdot 2^{n+1}$

17. If n is even and ${}^nC_0 < {}^nC_1 < {}^nC_2 < \ldots < {}^nC_r > {}^nC_{r+1} > {}^nC_{r+2} > \cdots > {}^nC_n$, then $r =$

(a) $\frac{n}{2}$ (b) $\frac{n-1}{2}$

(c) $\frac{n-2}{2}$ (d) $\frac{n+2}{2}$

18. If ${}^{13}C_r$ is denoted by C_r, then the value of $C_1 + C_5 + C_7 + C_9 + C_{11}$ is equal to

(a) $2^{12} - 287$ (b) $2^{12} - 165$

(c) $2^{12} - C_3$ (d) $2^{12} - C_2 - C_{13}$

ANSWERS

Level 1

1. (a)	**2.** (d)	**3.** (a)	**4.** (c)	**5.** (d)	**6.** (b)	**7.** (a)	**8.** (b)	**9.** (a)	**10.** (b)
11. (a)	**12.** (b)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (d)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (c)

Level 2

1. (a)	**2.** (b)	**3.** (d)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (b)
11. (a)	**12.** (c)	**13.** (b)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (a)	**18.** (a)		

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G. PEANO

Mathematical Induction

- Introduction
- First Principle of Mathematical Induction
- Problem-solving Trick(s)
- Second Principle of Mathematical Induction

INTRODUCTION

In algebra, there are certain results that are formulated with n number of terms in them, where n is a natural number (i.e. a positive integer). Those results can be proved by a specific technique, known as the *principle of mathematical induction*. We use the symbol $P(n)$ (read "P of n") to denote some proposition which depends on the positive integer n. For example, $P(n)$ might denote the sum of the first n odd positive numbers, that is

$$1 + 3 + 5 + 7 + 9 + \cdots + (2n - 1) = n^2$$

where $n = 1, 2, 3, \ldots, n$.

FIRST PRINCIPLE OF MATHEMATICAL INDUCTION

The proof of the proposition $P(n)$ by mathematical induction for all $n \in N$ consists of the following three steps:

Step 1. Verification step

Verify that the proposition $P(n)$ is true for $n = 1$, i.e. the first natural number or the smallest positive integer. This is also called the **basis** step of the induction and is generally very easy.

Step 2. Induction step

Assume that the proposition will also be true for some $n = k \geq 1$, i.e. we assume $P(k)$ to be true. This is called the **induction step.**

Step 3. Generalization step

If $P(k)$ is true, then prove that the proposition is also true for $n = (k + 1)$, which is the next positive integer (i.e. the next natural number), i.e. we have to prove that $P(k + 1)$ must also be true. In this step we prove that the implication $P(k) \Rightarrow P(k + 1)$ is true.

Next we generalize the result by saying that, since the proposition is proved to be true for $n = k + 1$, then it must also be true for $n = k$, and hence the proposition will be true for all n belonging to the set of natural numbers.

PROBLEM-SOLVING TRICK(S)

- If the proposition is not true for even any one natural number, then it is FALSE.
- If the proposition is true at each step, then it must be TRUE.

- It may be noted that Step 1 may or may not be true even if Step 3 is true and Step 3 may not be true even if Step 1 is true. In that case, the proposition remains FALSE.
- In order to prove a certain statement (proposition) for all of the natural numbers, we have to establish validation at each and every step.

☞ SECOND PRINCIPLE OF MATHEMATICAL INDUCTION

Step 1. Verification step

Verify that the proposition $P(n)$ is true for $n = r$, where r is some fixed integer.

Step 2. Induction step

Assume that the proposition $P(n)$ is true for $n = r, r + 1, r + 2, \ldots, m$.

Step 3. Generalization step

Prove that the proposition $P(n)$ is true for $n = m + 1$. Thus, if true, we generalize the result by saying that since the propositon is true for $n = m + 1$, then it must also be true for $n = r, r + 1, r + 2, \ldots, m$ as assumed in Step 2. Thus, the proposition is true for all $n \geq r$ belonging to the set of natural numbers.

Illustration 1. Show, by mathematical induction, that for all integers $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Solution: Let

$$P(n): 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Basis step: We must first show that $P(1)$ is true. $P(1)$ is the statement

$1 = \frac{1(1+1)}{2}$, which is clearly true.

Induction step: We assume that for some fixed $n = k \geq 1$,

$$P(k): 1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$$

Verification step: We have to show the truth of $P(k + 1)$.

Now, $P(k+1) = 1 + 2 + 3 + \cdots + (k+1) = \dfrac{(k+1)[(k+1)+1]}{2}$

The left-hand side of $P(k + 1)$ can be written as

$1 + 2 + 3 + \cdots + k + (k + 1)$

$$= \frac{k(k+1)}{2} + k + 1$$

$\{\because 1 + 2 + 3 + \cdots + k = [k(k+1)/2]$ as assumed in the induction step$\}$

$$= (k+1)\left(\frac{k}{2} + 1\right)$$

$$= \frac{(k+1)(k+2)}{2} = \frac{(k+1)[(k+1)+1]}{2}$$

= the right-hand side of $P(k + 1)$.

Thus, we have shown that the left-hand side of $P(k + 1)$ is equal to the right-hand side of $P(k + 1)$.

Hence by the principle of mathematical induction, it follows that the given $P(n)$ is true for all integers $n \geq 1$. That is,

$$1 + 2 + 3 + \cdots + n = \sum n = \frac{n(n+1)}{2} \text{ for all } n \in N.$$

Illustration 2. Prove that the sum of the cubes of the first n natural numbers is $\left[\frac{n(n+1)}{2}\right]^2$.

Solution:

$$P(n): 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2, \; n \in N$$

$$P(1): 1^3 = \left[\frac{1(1+1)}{2}\right]^2 \text{ or } 1 = \left[\frac{1(2)}{2}\right]^2 \text{ or } 1 = 1, \text{ which is true.}$$

$$P(k): 1^3 + 2^3 + 3^3 + \cdots + k^3 = \left[\frac{k(k+1)}{2}\right]^2$$

$P(k + 1)$:

$$1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3 = \left\{\frac{(k+1)[(k+1)+1]}{2}\right\}^2$$

$$\text{L.H.S. of } P(k+1) = \left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3$$

[Assuming $P(k)$ to be true]

$$= (k+1)^2\left(\frac{k^2}{4} + k + 1\right)$$

$$= \frac{(k+1)^2(k^2 + 4k + 4)}{4}$$

$$= \frac{(k+1)^2(k+2)^2}{4}$$

$$= \left[\frac{(k+1)(k+2)}{2}\right]^2$$

$$= \left\{\frac{(k+1)[(k+1)+1]}{2}\right\}^2$$

= R.H.S. of $P(k + 1)$

Hence $P(n)$ is true for all $n \in N$, i.e.

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \sum n^3 = \left[\frac{n(n+1)}{2}\right]^2 \text{ for all } n \in N$$

Illustration 3. Prove that for every positive integer n,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Solution:

$P(n)$: $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$, $n \in N$

$P(1)$: $1^2 = \frac{1(1+1)(2+1)}{6} = \frac{(2)(3)}{6} = 1$ or $1 = 1$, which is true.

$P(k)$: $1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$

$P(k+1)$:

$$1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

L.H.S. of $P(k+1) = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

[Assuming $P(k)$ to be true]

$$= (k+1)\left[\frac{k(2k+1)}{6} + (k+1)\right]$$

$$= \frac{(k+1)(2k^2+7k+6)}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

= R.H.S. of $P(k+1)$

Hence $P(n)$ is true for all $n \in N$, i.e.

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

for all $n \in N$

SOLVED PROBLEMS

1. Prove that $1\cdot2\cdot3 + 2\cdot3\cdot4 + 3\cdot4\cdot5 + \cdots$ up to n terms is equal to

(a) $\frac{(n+1)(n+2)(n+3)}{4}$

(b) $\frac{(n+2)(n-2)(n-3)(n+3)}{4}$

(c) $\frac{n(n+1)(n+2)(n+3)}{4}$

(d) none of these

Ans. (c)

Solution: Let $P(n)$: $1\cdot2\cdot3 + 2\cdot3\cdot4 + 3\cdot4\cdot5 + \cdots$ (n terms)

$\Rightarrow P(n): \sum n(n+1)(n+2)$

$\Rightarrow P(n): \sum (n^2+n)(n+2)$

$\Rightarrow P(n): \sum (n^3 + 2n^2 + n^2 + 2n)$

$\Rightarrow P(n): \sum (n^3 + 3n^2 + 2n)$

$\Rightarrow P(n): \sum n^3 + 3\sum n^2 + 2\sum n$

$\Rightarrow P(n): \frac{n^2(n+1)^2}{4} + 3\frac{1}{6}n(n+1)(2n+1) + 2\frac{1}{2}n(n+1)$

$\Rightarrow P(n): \frac{n(n+1)}{4}[n(n+1) + 2(2n+1) + 4]$

$\therefore \quad P(n): \frac{n(n+1)\ (n+2)\ (n+3)}{4}$

2. Prove, by mathematical induction, that

$\frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \cdots + \frac{1}{n(n+1)(n+2)}$ is equal to

(a) $\frac{n(n+1)}{4(n+2)(n+3)}$

(b) $\frac{n(n+3)}{4(n+1)(n+2)}$

(c) $\frac{n(n+2)}{4(n+1)(n+3)}$

(d) none of these

Ans. (b)

Solution: Let

$$P(n): \frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \cdots + \frac{1}{n(n+1)(n+2)}$$

$$= \frac{n(n+3)}{4(n+1)(n+2)}$$

(i) For $n = 1$

L.H.S. $= \frac{1}{1\cdot2\cdot3} = \frac{1}{6}$ and R.H.S. $= \frac{1(1+3)}{4(1+1)(1+2)} = \frac{1}{6}$

$\therefore \quad P(1)$ is true.

(ii) Let $P(k)$ be true, then

$$P(k): \frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \cdots + \frac{1}{k(k+1)(k+2)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} \qquad \text{(i)}$$

(iii) For $n = k+1$,

$$P(k+1): \frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \cdots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} = \frac{(k+1)(k+4)}{4(k+2)(k+3)}$$

$$\text{L.H.S.} = \frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \cdots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

[from Eq. (i)]

$$= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)}$$

$$= \frac{k^3 + 6k^2 + 9k + 4}{4(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)(k+4)}{4(k+2)(k+3)} = \text{R.H.S.}$$

Hence, $P(k+1)$ is true.

Hence, by the principle of mathematical induction for all $n \in N$, $P(n)$ is true.

3. Prove that $P(n): 1+3+5+\cdots+(2n-1)=n^2$ is

(a) true for all $n \in N$ (b) true for $n > 1$

(c) true for no n (d) none of these

Ans. (a)

Solution: Given $P(n): 1+3+5+\cdots+(2n-1)=n^2$

$$P(1): 1=1 \text{ (true)}$$

Let $P(k)=1+3+5+\cdots+(2k-1)=k^2$

$\therefore$ L.H.S. of $P(k+1)=1+3+5+\cdots+(2k-1)+2k+1$

$$= k^2+2k+1=(k+1)^2$$

$$= \text{R.H.S. of } P(k+1)$$

So, $P(n)$ holds for all $n \in N$.

4. For all $n \in N$, $(3)(5^{2n+1})+2^{3n+1}$ is divisible by

(a) 17 (b) 19

(c) 21 (d) 23

Ans. (a)

Solution:

$P(n)$: $3(5^{2n+1}) + 2^{3n+1}$

$P(1)$: $3(5^3) + 2^4 = 3(125) + 16$

$= 375 + 16 = 391 = 17(23)$

So, $P(1)$ is divisible by 17.

Let $P(k)$: $3(5^{2k+1}) + 2^{3k+1} = 17m$

$\therefore$ $P(k+1)$: $3(5^{2k+3}) + 2^{3k+4} = 17\lambda$

L.H.S. of $P(k+1) = 3(5^{2k+3}) + 2^{3k+4}$

$= 3(5^{2k+1})(5^2) + (2^{3k+1})(2^3)$

$= (17m - 2^{3k+1})(25) + 8(2^{3k+1})$

[Assuming $P(k)$ to be true]

$= 17(25m) - 25(2^{3k+1}) + 8(2^{3k+1})$

$= 17(25m) - 17(2^{3k+1})$

$= 17(25m - 2^{3k+1}) = 17\lambda$

Thus, $P(k+1)$ is divisible by 17 whenever $P(k)$ is divisible by 17. Hence $P(n)$ is divisible by 17 for all $n \in N$.

5. For all $n \in N$, $n(n+1)(2n+1)$ is divisible by

(a) 6 (b) 7

(c) 8 (d) 15

Ans. (a)

Solution:

$P(n)$: $n(n + 1)(2n + 1)$

$P(1)$: $1(1 + 1)(2 + 1) = (2)(3) = 6$

So, $P(1)$ is divisible by 6.

Let $P(k)$: $k(k + 1)(2k + 1) = 6m$

$\therefore$ $P(k + 1)$: $(k + 1)(k + 2)(2k + 3) = 6\lambda$

L.H.S. of $P(k + 1) = (k + 1)(k + 2)(2k + 3)$

$$= \frac{(k+2)(2k+3)(6m)}{k(2k+3)}$$ [Assuming $P(k)$ to be true]

$$= 6\left[m \times \frac{(k+2)(2k+3)}{k(2k+3)}\right] = 6\lambda$$

Thus, $P(k + 1)$ is divisible by 6 whenever $P(k)$ is divisible by 6. Hence $P(n)$ is divisible by 6 for all $n \in N$.

6. The smallest positive integer n for which $n! < \left(\frac{n+1}{2}\right)^n$ holds, is

(a) 1 (b) 2

(c) 3 (d) 4

Ans. (b)

Solution: Given $n! < \left(\frac{n+1}{2}\right)^n$

At $n = 1$, $1! \not< 1$

At $n = 2$, $2! < \left(\frac{3}{2}\right)^2$

$\Rightarrow$ $2 < 2.25$, which is true.

7. Prove that $x^{2n-1} + y^{2n-1}$ is divisible by $x + y$, if n is

(a) a positive integer

(b) an even positive integer

(c) an odd positive integer

(d) none of these

Ans. (a)

Solution:

$P(n)$: $x^{2n-1} + y^{2n-1}$

$P(1)$: $x^1 + y^1 = x + y$, which is divisible by $x + y$

Let $P(k)$: $x^{2k-1} + y^{2k-1} = (x + y)m$

$\therefore$ $P(k + 1)$: $x^{2k+1} + y^{2k+1} = (x + y)\lambda$

L.H.S. of $P(k + 1) = x^{2k+1} + y^{2k+1}$

$= x^2(x^{2k-1}) + y^2(y^{2k-1})$

$= x^2[m(x+y) - y^{2k-1}] + y^2(y^{2k-1})$

[Assuming $P(k)$ to be true]

$= (x+y)(mx^2) - y^{2k-1}(x^2 - y^2)$

$= (x + y)[mx^2 - y^{2k-1}(x - y)] = (x + y)\lambda$

Thus, $P(k + 1)$ is divisible by $(x + y)$ whenever $P(k)$ is divisible by $(x + y)$. Hence $P(n)$ is divisible by $x + y$ for all $n \in N$, i.e. for all positive integers.

8. Prove that if $n \in N$, then $11^{n+2} + 12^{2n+1}$ is divisible by
(a) 113 (b) 123
(c) 133 (d) none of these

Ans. (c)

Solution:

$P(n)$: $11^{n+2} + 12^{2n+1}$

$P(1)$: $11^3 + 12^3 = 3059$, which is divisible by 133.

Let $P(k)$: $11^{k+2} + 12^{2k+1} = 133m$

$\therefore$ $P(k + 1)$: $11^{k+3} + 12^{2k+3} = 133\lambda$

L.H.S. of $P(k + 1) = 11^{k+3} + 12^{2k+3}$

$= 11(11^{k+2}) + (12)^2(12^{2k+1})$

$= 11(133m - 12^{2k+1}) + 144(12^{2k+1})$

$= 133(11m) + 133(12^{2k+1})$

$= 133(11m + 12^{2k+1}) = 133\lambda$

Thus, $P(k + 1)$ is divisible by 133 whenever $P(k)$ is divisible by 133. Hence $P(n)$ is divisible by 133 for all $n \in N$.

9. Let the statement be $m^2 > 100$, then the statement $P(k + 1)$ will be true if
(a) $P(1)$ is true (b) $P(2)$ is true
(c) $P(k)$ is true (d) none of these

Ans. (c)

Solution: Let $P(r)$ be true

$\Rightarrow$ $r^2 > 100$

$\Rightarrow$ $r^2 + 2r + 1 > 100 + 2r + 1$

$\Rightarrow$ $(r+1)^2 > 100$

$\Rightarrow$ $P(r+1)$ is true as $r^2 + (2r+1) > r^2 > 100$

$\Rightarrow$ $P(k+1)$ is true (say $r = k$)

$P(k+1)$ is true when every $P(k)$ is so. In order to prove that $P(k+1)$ is true, it is sufficient to consider $P(k)$ is true.

10. The statement $P(n) = 9^n - 8^n$ when divided by 8 always leaves the remainder
(a) 2 (b) 3
(c) 1 (d) 7

Ans. (c)

Solution: $P(n) = 9^n - 8^n$

$\therefore$ $P(1) = 9 - 8 = 1$

$\therefore$ $P(1) - 1 = 0$ which is divisible by 8

$\therefore$ 1 is the remainder when $P(n)$ is divided by 8

Now, $P(2) = 9^2 - 8^2 = 17 = 16 + 1$

$\Rightarrow$ remainder is 1, when divided by 8.

11. Let $P(n) = 2^{3n} - 7n - 1$, then $P(n)$ is divisible by
(a) 63 (b) 36
(c) 49 (d) 25

Ans. (c)

Solution:

$P(n)$: $2^{3n} - 7n - 1$

$P(1)$: $2^3 - 7 - 1 = 8 - 7 - 1 = 0$

$P(2)$: $2^6 - 7(2) - 1 = 64 - 14 - 1 = 49$,

which is divisible by 49

Let $P(k)$: $2^{3k} - 7k - 1 = 49m$

$\therefore$ $P(k + 1) = 2^{3k+3} - 7(k + 1) - 1 = 49\lambda$

L.H.S. of $P(k + 1) = 2^{3k+3} - 7(k+1) - 1$

$= (2^{3k})(2^3) - 7k - 7 - 1$

$= (49m + 7k + 1)(8) - 7k - 8$

[Assuming $P(k)$ to be true]

$= 49(8m) + 56k + 8 - 7k - 8$

$= 49(8m + k) = 49\lambda$

Thus, $P(k + 1)$ is divisible by 49 whenever $P(k)$ is divisible by 49. Hence $P(n)$ is divisible by 49 for all $n \in N$.

12. If a and b are natural numbers such that $a^2 - b^2$ is a prime number, then $a^2 - b^2$ is equal to
(a) $a + b$ (b) $a - b$
(c) ab (d) 1

Ans. (a)

Solution: Consider $a^2 - b^2 = (a+b)(a-b)$ as prime (given).

$\therefore$ $a^2 - b^2$ is divisible by either 1 or by itself.

$\therefore$ $a - b = 1$ and $a^2 - b^2 = a + b$

$\therefore$ $a^2 - b^2 = a + b$ ($\because$ $a + b \neq 1$ as $a, b \in N$)

13. Let $P(n) = 12^n + 125^{n-1}$, then the least divisor of $P(n)$ is
(a) 0 (b) 13
(c) 13λ (d) none of these

Ans. (b)

Solution: $P(n) = 12^n + 125^{n-1}$.

$\Rightarrow$ $P(1) = 12^1 + (125)^{1-1} = 12 + 1 = 13$.

So, the least divisor = 13.

EXERCISE SET

1. Prove that $3 + 13 + 29 + 51 + 79 + \cdots$ up to n terms is equal to
(a) $2n^2 + 7n^3$ (b) $n^2 + 5n^3$
(c) $n^3 + 2n^2$ (d) none of these

2. For all $n \in N$, prove that $a^{2n-1} + b^{2n-1}$ is divisible by
(a) $a + b$ (b) $(a + b)^2$
(c) $a^3 + b^3$ (d) none of these

3. The greatest positive integer which divides $(n + 1)(n + 2)(n + 3)\ldots(n + r)$ for all $n \in N$ is
(a) r (b) $r!$ (c) $n + r$ (d) $(r + 1)!$

4. The inequality $n! > 2^{n-1}$, $n \in N$ is true for
(a) all n (b) no n
(c) all $n > 1$ (d) all $n > 2$

5. For all $n \in N$, $\frac{n^5}{5} + \frac{n^3}{3} + \frac{7}{15n}$ is
(a) an integer (b) a natural number
(c) a positive fraction (d) a rational number

6. Prove that $49^n + 16n - 1$ is divisible by
(a) 3 (b) 64 (c) 19 (d) 29

7. If $a_n = 2^{2^n} + 1$, $n > 1$, then the last digit of a_n is
(a) 3 (b) 5 (c) 7 (d) 8

8. The statement $P(n)$: $1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n + 1)! - 1$ is
(a) true for no n (b) true for all $n > 1$
(c) true for all $n \in N$ (d) none of these

9. $3^{3n} - 26n - 1$ is divisible by
(a) 239 (b) 547 (c) 627 (d) 676

10. For all $n \in N$, prove that $\cos\alpha \cos 2\alpha \cos 4\alpha \ldots \cos(2^{n-1}\alpha)$ is equal to
(a) $\frac{\sin(2^n\alpha)}{2\sin\alpha}$ (b) $\frac{\sin(2^n\alpha)}{2^n\sin\alpha}$
(c) $\frac{\cos(2^n\alpha)}{2^n\cos\alpha}$ (d) none of these

11. If $n \in N$, prove that $7^{2n} - 48n - 1$ is divisible by
(a) 25 (b) 26 (c) 1234 (d) 2304

12. Prove that $2^{3n} - 7n - 1$ is divisible by
(a) 64 (b) 36 (c) 49 (d) 25

13. If $49^n + 16n + k$ is divisible by 64 for $n \in N$, then the numerically least –ve integral value of k is
(a) –2 (b) –1 (c) –3 (d) –4

14. If $10^n + 3 \cdot 4^{n+2} + k$ is divisible by 9 for all $n \in N$, then the least +ve integral value of k is
(a) 5 (b) 3 (c) 7 (d) 1

15. If $P(n)$: $3^n < n!$, $n \in N$, then $P(n)$ is true
(a) for $n \geq 6$ (b) for $n \geq 7$
(c) for $n \geq 3$ (d) for all n

16. For all $n \in N$, $10^n + 3 \cdot 4^{n+2} + 5$ is divisible by
(a) 23 (b) 3 (c) 7 (d) 1

17. If $x^n - 1$ is divisible by $x - k$, then the least +ve integral value of k is
(a) 1 (b) 2 (c) 3 (d) 4

18. If $1 + 5 + 12 + 22 + 35 + \cdots$ to n terms $= \frac{n^2(n+1)}{2}$, then the nth term of the series is
(a) $\frac{n(4n-1)}{3}$ (b) $\frac{n(3n-1)}{2}$
(c) $\frac{n(3n+1)}{2}$ (d) $\frac{n(4n+1)}{3}$

19. The sum of n terms of the series
$$\frac{1^3}{1} + \frac{1^3 + 2^3}{1 + 3} + \frac{1^3 + 2^3 + 3^3}{1 + 3 + 5} + \cdots =$$
(a) $\frac{n(n+1)^3(n+2)}{24}$ (b) $\frac{n(2n^2 + 9n + 13)}{24}$
(c) $\frac{4n^2 + 1}{5}$ (d) none of these

20. If $1^3 + 2^3 + 3^3 + \cdots + 100^3 = k^2$, then k is equal to
(a) 10,100 (b) 5000 (c) 5050 (d) 1010

21. $\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \cdots$ up to $(n + 1)$ terms is equal to
(a) $\frac{n}{n+1}$ (b) $\frac{2n}{n+1}$
(c) $\frac{2}{n(n+1)}$ (d) $\frac{2}{n+2}$

22. For all $n \in N$, $\int_0^{\pi/2} \frac{\sin^2 nx}{\sin x}\, dx$ is equal to
(a) $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ (b) $1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1}$
(c) 0 (d) none of these

Assertion-Reason Type Questions

Read the following questions and choose the correct choice for two statements: one is Assertion (I) and the other is Reason (II). You are to examine these two statements carefully and decide if the Assertion (I) and the Reason (II) are separately true and if so whether the Reason (II) is a correct explanation for the given Assertion (I). Select the correct code from the following:

(a) Both (I) and (II) are individually true and (II) is a correct explanation of (I).

(b) Both (I) and (II) are individually true but (II) is not a correct (proper) explanation of (I).
(c) (I) is true but (II) is false.
(d) (I) is false but (II) is true.

23. Assertion I: $3^{2n}\ \forall n \in N$ leaves the remainder 1 when divided by 8.
Reason II: $9^n = 1 + 8\lambda$.

24. Assertion I: Let $n \in N$, then $p(n) = n(n+1)$ is an even number.
Reason II: The product of two consecutive natural numbers is even.

25. Assertion I: $P(n) = n^2 + n + 1$ is an odd natural number $\forall n \in N$.
Reason II: If 1 is added to an even number, then it becomes an odd number.

26. Assertion I: $P(n) = 62 \times 63 \times 64$ leaves the remainder 59 when divided by 65.
Reason II: $P(n) = 59 \pmod{65}$.

27. Assertion I: $P(n) = n^2 + n + 41$ is prime $\forall n \in N$.
Reason II: If a number is prime, then it contains only two factors, i.e. 1 and the number itself.

28. Assertion I: $S_n = n^3 + 3n^2 + 5n + 3$ is divisible by 3.
Reason II: $t_n = 3\lambda$.

29. Assertion I: Let $P(n) = (n+1)(n+2)(n+3)\ldots(n+r)$, then the greatest integer which divides $P(n)$ is $r!$.
Reason II: The product of r consecutive numbers is divisible by $r!$.

30. Assertion I: The greatest positive integer which divides $(n+11)(n+12)(n+13)(n+14)\ \forall n \in N$ is 24.
Reason II: The product of any r consecutive integers is divisible by $r!$.

31. Assertion I: If $P(n) = 111\ldots1$ (91 times), then $P(n)$ is a prime number.
Reason II: Every prime number has at most two factors.

32. Assertion I: The digit in the unit place of $183! + 3^{183}$ is 7.
Reason II: 183! has unit place 0 and 3^{4k+3} ends with 7.

Previous Years' Questions

33. If $a_n = \sqrt{7 + \sqrt{7 + \sqrt{7 + \cdots}}}$ having n radical signs, then by the method of mathematical induction which is true?
(a) $a_n > 7, \forall n \geq 1$ (b) $a_n > 3, \forall n \geq 1$
(c) $a_n < 4, \forall n \geq 1$ (d) $a_n < 3, \forall n \geq 1$
[AIEEE 2002]

34. Let $S(K) = 1 + 3 + 5 + \cdots + (2K-1) = 3 + K^2$. Then which of the following is true?
(a) $S(1)$ is correct (b) $S(K) = S(K+1)$
(c) $S(K) \neq S(K-1)$
(d) Principle of mathematical induction can be used to prove the formula. **[AIEEE 2004]**

35. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds, for all $n \geq 1$, the principle of mathematical induction?
(a) $A^n = 2^{n-1}A + (n-1)I$ (b) $A^n = nA + (n-1)I$
(c) $A^n = 2^{n-1}A - (n-1)I$ (d) $A^n = nA - (n-1)I$
[AIEEE 2005]

36. The remainder when $8^{2n} - (62)^{2n+1}$ is divided by 9 is
(a) 2 (b) 7
(c) 8 (d) 0
[AIEEE 2009]

37. If $x > -1$, then the statement $(1+x)^n > 1 + nx$ is true for
(a) all $n \in N$ (b) all $n > 1$
(c) all $n > 1$ provided $n \neq 0$
(d) none of these **[Pb. CET 2005]**

38. The statement $P(n)$: $1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n+1)! - 1$ is
(a) true for all $n > 1$ (b) not true for any n
(c) true for all $n \in N$ (d) none of these
[Him. CET 2005]

39. Statement 1: For each natural number n, $(n+1)^7 - n^7 - 1$ is divisible by 7.
Statement 2: For each natural number n, $n^7 - n$ is divisible by 7.
(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is false.
(d) Statement 1 is false, Statement 2 is true.
[AIEEE 2011]

ANSWERS

1. (c)	**2.** (a)	**3.** (b)	**4.** (c)	**5.** (d)	**6.** (b)	**7.** (c)	**8.** (c)	**9.** (d)	**10.** (b)
11. (d)	**12.** (c)	**13.** (b)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (b)	**19.** (b)	**20.** (c)
21. (b)	**22.** (b)	**23.** (a)	**24.** (a)	**25.** (a)	**26.** (a)	**27.** (d)	**28.** (a)	**29.** (a)	**30.** (a)
31. (d)	**32.** (a)	**33.** (b)	**34.** (b)	**35.** (d)	**36.** (a)	**37.** (c)	**38.** (c)	**39.** (a)	

HINTS AND SOLUTIONS—EXERCISE SET

1. $P(n)$: $3+13+29+51+79+\cdots$ up to n terms

$P(1)=3,\ P(1)=1^3+2\cdot1^2$

Again $P(2) = 16 = 2^3+2\cdot2^2$

So, $P(n)=n^3+2\cdot n^2$

2. $P(n)$: $a^{2n-1}+b^{2n-1}$

$P(1)$: $a^1 + b^1 = a + b$, which is divisible by itself, i.e. by $(a + b)$.

$\therefore$ $P(n)$: $a^{2n-1}+b^{2n-1}$ is divisible by $(a + b)$, and is true for $n = 1$.

Let $P(k)$ be true, i.e. $P(k)$: $a^{2k-1} + b^{2k-1}$ is divisible by $(a + b)$

i.e. $a^{2k-1} + b^{2k-1} = m(a + b)$ (i)

Now, $P(k + 1) = a^{2k+1} + b^{2k+1} = a^{2k-1}\cdot a^2 + b^{2k+1}$

$= a^2[m(a+b) - b^{2k-1}] + b^{2k+1}$

$= m(a+b)a^2 - a^2b^{2k-1} + b^{2k+1}$

$= m(a+b)a^2 - b^{2k-1}(a^2 - b^2)$

$= m(a + b)a^2 - (a + b)(a - b)b^{2k-1}$

$= (a + b)[ma^2 - (a - b)b^{2k-1}]$

$\therefore$ $P(k + 1)$ is divisible by $(a + b)$ whenever $P(k)$ is divisible by $(a + b)$.

Hence $P(n)$ is divisible by $(a + b)$ for all $n \in N$.

3. $P(n)$: $(n + 1)(n + 2) \ldots (n + r)$

$P(1)$: $(2)(3) \ldots (r + 1) = r!(r + 1)$, which is divisible by $r!$

Let $P(k)$: $(k + 1)(k + 2) \ldots (k + r) = r!(m)$

$\therefore$ $P(k + 1)$: $(k + 2)(k + 3) \ldots (k + 1 + r) = r!(\lambda)$

L.H.S. of $P(k + 1)$

$= (k+2)(k+3)\cdots(k + r + 1)$

$= \dfrac{(k + 1)(k+2)(k+3)\cdots(k + r + 1)}{k+1}$

$= \dfrac{r!(m)(k + r + 1)}{k+1}$

$= r!(\lambda)$

Thus, $P(k + 1)$ is divisible by $r!$ whenever $P(k)$ is divisible by $r!$.

Hence $P(n)$ is divisible by $r!$ for all $n \in N$.

4. For $n = 1$, $1! > 2^0$, i.e. $1 > 1$, is not true. By induction, $n! > 2^{n-1}\ \forall n \in N$ for $n > 1$.

5. $P(n)$: $\dfrac{n^5}{5}+\dfrac{n^3}{3}+\dfrac{7}{15n}$

$P(1)$: $\dfrac{1}{5}+\dfrac{1}{3}+\dfrac{7}{15} = 1$, which is an integer

Assume $P(k)$: $\dfrac{k^5}{5}+\dfrac{k^3}{3}+\dfrac{7}{15k}$ is an integer

$P(k+1)$: $\dfrac{(k+1)^5}{5}+\dfrac{(k+1)^3}{3}+\dfrac{7}{15(k+1)}$

$= \dfrac{3(k+1)^6+5(k+1)^4+7}{15(k+1)}$

For all $k \in N$,

$3(k + 1)^6 + 5(k + 1)^4 + 7 \in N$

and also $15(k + 1) \in N$, and $15(k + 1) \neq 0$

$\therefore$ $\dfrac{3(k+1)^6+5(k+1)^4+7}{15(k+1)}$ is a rational number

6. $P(n)$: $49^n + 16n - 1$

$P(1)$: $49 + 16 - 1 = 64$, which is divisible by 64

Let $P(k)$: $49^k + 16k - 1 = 64m$

$\therefore$ $P(k + 1)$: $49^{k+1} + 16(k + 1) - 1 = 64\lambda$

L.H.S. of $P(k + 1) = 49^{k+1} + 16(k + 1) - 1$

$= 49(64m - 16k + 1) + 16k + 16 - 1$

[Assuming $P(k)$ to be true]

$= 64(49m) - 48(16k) + 64$

$= 64(49m - 12k + 1) = 64\lambda$

Thus, $P(k + 1)$ is divisible by 64 whenever $P(k)$ is divisible by 64.

Hence $P(n)$ is divisible by 64.

7. For $n = 2$, $a_n = 2^{2^2}+1 = 16+1 = 17$. Assume that $a_k = 2^{2^k}+1 = 10m+7$, where $k > 1$ and m is some positive integer.

Now, $a_{k+1} = 2^{2^{k+1}}+1 = \left(2^{2^k}\right)^2+1$

$= (10m+6)^2+1$

$= 10(10m^2+12m+3)+7$

Hence, the last digit of a_n is 7, $\forall n > 1$.

8. By induction, $P(n)$ is true for all $n \in N$.

9. $P(n)$: $3^{3n} - 26n - 1$

$P(1)$: $3^3 - 26 - 1 = 27 - 27 = 0$

$P(2)$: $3^6 - 52 - 1 = 729 - 53 = 676$, which is divisible by 676

Let $P(k)$: $3^{3k} - 26k - 1 = 676m$

$\therefore$ $P(k + 1)$: $3^{3(k+1)} - 26(k + 1) - 1 = 676\lambda$

L.H.S. of $P(k + 1) = 3^{3k+3} - 26(k + 1) - 1$

$= (3^{3k})(27) - 26k - 27$

$= (676m + 26k + 1)(27) - 26k - 27$

[Assuming $P(k)$ is true]

$= 676(27m) + (26k)(26) + 27 - 27$

$= 676(27m) + 676k$

$= 676(27m + k) = 676\lambda$

Thus, $P(k + 1)$ is divisible by 676 whenever $P(k)$ is divisible by 676.
Hence $P(n)$ is divisible by 676.

10. $P(n)$: $\cos\alpha \cos 2\alpha \cos 4\alpha \ldots \cos 2^{n-1}\alpha$

$P(1)$: $\cos\alpha = \dfrac{\sin 2\alpha}{2\sin\alpha}$

$P(2)$: $\cos\alpha\cos 2\alpha = \dfrac{\sin 4\alpha}{4\sin\alpha}$

Let $P(k)$: $\cos\alpha\cos 2\alpha\cos 4\alpha \ldots \cos 2^{k-1}\alpha$

$= \dfrac{\sin 2^k\alpha}{2^k\sin\alpha}$

$\therefore$ $P(k + 1)$: $\cos\alpha\cos 2\alpha\cos 4\alpha \ldots \cos 2^k\alpha$

$= \dfrac{\sin 2^{k+1}\alpha}{2^{k+1}\sin\alpha}$

L.H.S. of $P(k + 1)$

$= \cos\alpha\cos 2\alpha\cos 4\alpha \ldots \cos 2^k\alpha$

$= \dfrac{\sin 2^k\alpha}{2^k\sin\alpha} \times \cos 2^k\alpha$

[Assuming $P(k)$ to be true]

$= \dfrac{2\sin 2^k\alpha\cos 2^k\alpha}{2^{k+1}\sin\alpha} = \dfrac{\sin 2^{k+1}\alpha}{2^{k+1}\sin\alpha}$

$=$ R.H.S. of $P(k+1)$

Hence $P(n)$ holds true for all $n \in N$. That is,

$$\cos\alpha\cos 2\alpha\cos 4\alpha \ldots \cos 2^{n-1}\alpha = \frac{\sin 2^n\alpha}{2^n\sin\alpha}$$

11. Let $f(n) = 7^{2n} - 48n - 1$

$f(1) = 49 - 49 = 0, f(2) = 2304$

which is divisible by 2304 only.

12. For $n = 1$, $2^{3n} - 7n - 1 = 2^3 - 7 - 1 = 0$

For $n = 2$, $2^{3n} - 7n - 1 = 2^6 - 14 - 1 = 64 - 15 = 49$
which is divisible by 49.

13. $f(n) = 49^n + 16n + k$

$f(1) = 49 + 16 + k = 65 + k = 64 + 1 + k$

If $f(1)$ is to be divisible by 64, then the least –ve integral value of k has to be –1.

14. $f(n) = 10^n + 3\cdot 4^{n+2} + k$

$f(1) = 10 + 3\cdot 4^2 + k = 10 + 48 + k = 58 + k$

$= 9 \times 7 - 5 + k$

If $f(1)$ is to be divisible by 9, then the least +ve integral value of k has to be 5.

15. Given that $P(n)$: $3^n < n!$

Now, $P(7): 3^7 < 7!$ is true.

Let $P(k): 3^k < k!$

$\Rightarrow$ $P(k+1): 3^{k+1} = 3\cdot 3^k < 3\cdot k! < (k+1)!$ ($\because$ $k+1 > 3$)

Hence $P(n)$ is true for $n \geq 7$.

16. $f(n) = 10^n + 3\cdot 4^{n+2} + 5$

$f(1) = 10 + 48 + 5 = 63$, which is divisible by 7 and 3

$f(2) = 100 + 3(256) + 5 = 105 + 768 = 873$, which is divisible by 3

So, $f(n) = 10^n + 3\cdot 4^{n+2} + 5$ is divisible by 3.

17. Let $P(n)$: $x^n - 1 = \lambda(x - k)$

Now $P(1)$: $x - 1 = \lambda_1(x - k)$

Also, $P(2)$: $x^2 - 1 = \lambda_2(x - k)$

$\Rightarrow$ $P(2)$: $(x-1)(x+1) = \lambda_2(x - k)$

$\therefore$ The least value of k for which the proposition $P(n)$ is true is $k = 1$.

18. Given $P(n)$: $1 + 5 + 12 + 22 + 35 + \cdots$ (n terms) $= \dfrac{n^2(n+1)}{2}$

$\therefore$ nth term of L.H.S. $= P(n) - P(n-1)$

$\Rightarrow$ $P(n) - P(n-1) = \dfrac{n^2(n+1)}{2} - \dfrac{(n-1)^2 n}{2}$

$\Rightarrow$ $P(n) - P(n-1) = \dfrac{n}{2}(n^2 + n - n^2 + 2n - 1)$

$\therefore$ $t_n = P(n) - P(n-1) = \dfrac{n}{2}(3n - 1)$

19. Let $P(n)$: $\dfrac{1^3}{1} + \dfrac{1^3 + 2^3}{1+3} + \dfrac{1^3 + 2^3 + 3^3}{1+3+5} + \cdots$ (n terms)

$\Rightarrow$ $P(n)$: $\sum \dfrac{1^3 + 2^3 + \cdots + n^3}{1 + 3 + 5 + \cdots + (2n - 1)}$

$\Rightarrow$ $P(n)$: $\sum\left(\dfrac{\sum n^3}{n^2}\right)$

$\Rightarrow$ $P(n)$: $\sum\left[\dfrac{1}{4}\dfrac{n^2(n+1)^2}{n^2}\right]$

$\Rightarrow$ $P(n)$: $\dfrac{1}{4}\sum(n^2 + 2n + 1)$

$\Rightarrow$ $P(n)$: $\dfrac{1}{4}\left[\sum n^2 + 2\sum n + \sum(1)\right]$

$\Rightarrow$ $P(n)$: $\dfrac{1}{4}\left[\dfrac{n(n+1)}{2} + \dfrac{1}{3}n(n+1)(2n+1) + n\right]$

$\Rightarrow \quad P(n)\text{: } \frac{1}{24}n[3(n+1)+2(n+1)(2n+1)+6]$

$\therefore \quad P(n)\text{: } \frac{1}{24}n\left(2n^2+9n+13\right)$

20. $1^3+2^3+3^3+\cdots+100^3=k^3$

$\Rightarrow \quad \sum_{n=1}^{100} n^3=k^2$

$\Rightarrow \quad \left[\frac{100(100+1)}{2}\right]^2=k^2$

$\Rightarrow \quad k=\frac{100\cdot 101}{2}=5050$

Note that k here will not be negative as k is the sum of cubes of +ve integers.

21. Let $P(n)\text{: } \frac{1}{1}+\frac{1}{1+2}+\frac{1}{1+2+3}+\cdots(n+1)$ terms

$$t_n=\frac{1}{1+2+3+\cdots+n}=\frac{2}{n(n+1)}$$

$\Rightarrow \quad t_n=2\left(\frac{1}{n}-\frac{1}{n+1}\right)$

$\Rightarrow \quad P(n)\text{: } \left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right]$

$\therefore \quad P(n)=2\left(1-\frac{1}{n+1}\right)=\frac{2n}{n+1}$

22. Let $P(n)=\int_0^{\pi/2}\frac{\sin^2 nx}{\sin x}dx$

$$P(1)=\int_0^{\pi/2}\frac{\sin^2 x}{\sin x}dx=\int_0^{\pi/2}\sin x dx=[-\cos x]_0^{\pi/2}=1$$

$$P(2)=\int_0^{\pi/2}\frac{\sin^2 2x}{\sin x}dx=\int_0^{\pi/2}\frac{(2\sin x\cos x)^2}{\sin x}dx$$

$\Rightarrow \quad P(2)=\int_0^{\pi/2}4\sin x\cos^2 x dx$

$\Rightarrow \quad P(2)=4\left[\frac{-\cos^2 x}{3}\right]_0^{\pi/2}=\frac{4}{3}=1+\frac{1}{3}$

$\therefore$ For any $n \in N$,

$$P(n)=\int_0^{\pi/2}\frac{\sin^2 nx}{\sin x}dx=1+\frac{1}{3}+\frac{1}{5}+\cdots+\frac{1}{2n-1}$$

23. Let $P(n)=3^{2n}=9^n$

$\therefore \quad 9^n=(1+8)^n=1+8\lambda$

$\Rightarrow$ Choice (a) is the correct answer.

24. The numbers n and $n+1$ are consecutive numbers, so $n(n+1)$ is an even number.
Hence the option (a) is correct.

25. $P(n)$: n^2+n+1

$P(1)$: $1+1+1=3$, which is an odd number
$\quad = 2+1$

Let $P(k)$: $k^2+k+1=2m+1$ (an odd number)

$\therefore \quad P(k+1)$: $(k+1)^2+(k+1)+1=2\lambda+1$

L.H.S. of $P(k+1)=(k+1)^2+(k+1)+1$
$\quad = k^2+1+2k+k+1+1$
$\quad = (k^2+k+1)+2k+2$
$\quad = (2m+1)+2k+2$
$\quad$ [Assuming $P(k)$ to be true]
$\quad = 2(m+k+1)+1$
$\quad = 2\lambda+1$ (an odd number)

Hence $P(n)=n^2+n+1$ is an odd natural number for $n \in N$.

26. $62=-3(\text{mod } 65)$, $63=-2\text{mod}(65)$

$64=-1(\text{mod } 65)$

$\therefore \quad 62\times 63\times 64=(-3)(-2)(-1)(\text{mod } 65)$
$\quad = -6(\text{mod } 65)=59(\text{mod } 65)$

$\therefore$ Remainder is 59.

27. $P(n)=n^2+n+41$

Putting $n = 41$,

$\therefore \quad P(41)=41^2+41+41=41\times 43$

which is not prime.

$\therefore$ Assertion is false but the reason is obvious.

28. $S_n=n^3+3n^2+5n+3$

$\therefore \quad t_n=S_n-S_{n-1}$
$\quad =(n^3+3n^2+5n+3)$
$\quad -[(n-1)^3+3(n-1)^2+5(n-1)+3]$
$\quad =3n^2+3n+3=3(n^2+n+1)=3\lambda$

29. $P(n)=(n+1)(n+2)(n+3)\ldots(n+r)$, here the number of factors in $P(n)$ is r. So, $P(n)$ is divisible by $r!$.

30. Since the reason II is obvious, so the greatest positive integer which divides the product
$(n+11)(n+12)(n+13)(n+14)$ is $4! = 24$.

31. $P(n)=1+10+10^2+\cdots+10^{90}$

$$=\frac{10^{91}-1}{9}=\frac{(10^7)^{13}-1}{9}=\frac{t^{13}-1}{9}, \text{ where } t=10^7$$

$$=\frac{(t-1)(t^{12}+t^{11}+\cdots+t+1)}{9}$$

$$=\frac{10^7-1}{10-1}(t^{12}+t^{11}+\cdots+t+1)$$

$$=(1+10+10^2+\cdots+10^6)(1+t+t^2+\cdots+t^{12})$$

$\therefore$ $P(n) = 111\ldots1$ (91 times) is not a prime number as $P(n)$ has more than two factors. So assertion (I) is false.

32. For every n!, $\forall n \geq 5$, ends with 0 and 3^{4n} ends with 1,
$\therefore$ $3^{4n+3} = 3^{4n}3^3$ whose unit place is 7.

33. $P(n)$: $a_n = \sqrt{7+\sqrt{7+\sqrt{7+\cdots}}}$

$P(1)$: $a_1 = \sqrt{7} < 7$

$= \sqrt{7} > 3$

Let $P(k)$: $a_k = \sqrt{7+\sqrt{7+\cdots}} < 7$ and > 3

$\therefore$ $P(k+1)$: $a_{k+1} = \sqrt{7+\sqrt{7+\cdots}}$ is also < 7 and > 3

Now, $a_{k+1} = \sqrt{7+a_k}$ [Assuming $P(k)$ to be true]

$\Rightarrow$ $a_{k+1}^2 = 7 + a_k < 7 + 7 < 14$

$\Rightarrow$ $a_{k+1} < \sqrt{14} < 7$ but $\sqrt{14} > 3$

$\Rightarrow$ $a_{k+1} > 3$

So, $a_n < 7$ and $a_n > 3$; $\forall n \geq 1$

34. $S(K) = 1+3+5+\cdots+(2K-1) = 3+K^2$

Putting $K = 1$ on both sides, we get

L.H.S. = 1, R.H.S. = 3 + 1 = 4

$\therefore$ L.H.S. $\neq$ R.H.S.

Putting $(K + 1)$ on both sides in place of K, we have

$\text{L.H.S.} = 1+3+5+\cdots+(2K-1)+(2K+1)$

$\text{R.H.S.} = 3+(K+1)^2 = 3+K^2+2K+1 = K^2+2K+4$

$1+3+5+\cdots+(2K-1)+(2K+1) = 3+K^2+2K+1$

$\Rightarrow$ $1+3+5+\cdots+(2K-1) = 3+K^2$

If $S(K)$ is true, then $S(K + 1)$ is also true. Hence, $S(K) = S(K+1)$.

35. $A^1 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $A^2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

$= 2\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= 2A - (2-1)I$

$A^3 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$

$= 3A - (3-1)I$

Let $A^k = kA - (k-1)I$ (i)

$\therefore$ $A^{k+1} = (k+1)A - kI$

L.H.S. of $A^{k+1} = A^k \cdot A = [kA-(k-1)I]A$ [From (i)]

$= kA^2 - (k-1)IA = kA^2 - (k-1)A$

$= k\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} - (k-1)\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} k & 0 \\ 2k & k \end{bmatrix} - \begin{bmatrix} k-1 & 0 \\ k-1 & k-1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ k+1 & 1 \end{bmatrix}$

$= (k+1)\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} - (k)\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

$= (k+1)A - kI$

36. Using the modulo arithmetic, we have

$8 = -1$ (modulo 9)

Also $62 = -1$ (modulo 9)

$\Rightarrow$ $8^{2n} - (62)^{2n+1} = [(-1)^{2n} - (-1)^{2n+1}] \bmod 9$

$= (1+1) \bmod 9 = 2 \bmod 9$

$\Rightarrow$ Remainder = 2

37. For $n = 1$, $(1+x)^n = 1+x$ and $1+nx = 1+x$

$\therefore$ $P(1)$ is not true.

For $n = 2$, $(1+x)^n = (1+x)^2 = 1+2x+x^2 > 1+2x$

[$\because x^2 \geq 0$] (for $n = 2$)

$\therefore$ $P(n)$ is true for $n = 2$

Let $P(m)$ be true where m is a some +ve integer ≥ 2

$\therefore$ $(1+x)^m > 1+mx$

$\Rightarrow$ $(1+x)^{m+1} > (1+mx)(1+x)$

$= 1+(m+1)x+mx^2 > 1+(m+1)x$

$\therefore$ $P(m+1)$ is true if $P(m)$ is true.

$\therefore$ By induction, $P(n)$ is true for all $n > 1$ and $x \neq 0$.

[$\because$ for $n = 0$, $(1+x)^n = 1$ and $1+nx = 1$]

38. For $n = 1$

L.H.S. = 1; R.H.S. = $2! - 1 = 2 - 1 = 1$

$\therefore$ $P(1)$ is true.

Let $P(m)$ be true.

$\therefore$ $1\times1! + 2\times2! + \cdots + m\times m! = (m+1)! - 1$

Consider $P(m+1)$

$= 1\times1! + 2\times2! + \cdots + m\times m! + (m+1)(m+1)!$

$= (m+1)! - 1(m+1)(m+1)!$

$= (m+1)!(m+1+1) - 1$

$= (m+1)1(m+2) - 1 = (m+2)! - 1$

$= (\overline{m+1}+1)! - 1$

$\therefore$ $P(m+1)$ is true.

$\therefore$ Result is true for all $n \in N$.

39. Statement 2: $P(n)=n^7-n$

Put $n=1$ $\quad 1-1=0$ is divisible by 7

Let $n=k$ $\quad P(k)=k^7-k$ is divisible by 7

Put $n=k+1$

$\therefore\ P(k+1)=(k+1)^7-(k+1)$

$= k^7 + {}^7C_1k^6 + {}^7C_2k^5 + \cdots + {}^7C_6k + 1 - k - 1$

$P(k+1)=(k^2-k)$ + multiple of 7

Statement 1: n^7-n is divisible by 7

$\Rightarrow (n+1)^7-(n+1)$ is divisible by 7

$\Rightarrow (n+1)^7-n^7-1+(n^7-n)$ is divisible by 7

$\Rightarrow (n+1)^7-n^7-1$ is divisible by 7

MULTIPLE CHOICE TYPE QUESTIONS

1. $3+3^2+3^3+\cdots+n$ terms is equal to

(a) $\frac{3}{2}(3^n+1)$ (b) $\frac{3}{2}(3^n-1)$

(c) $3(3^n-1)$ (d) $3(3^n+1)$

2. $1+4+13+40+\cdots+n$ terms is equal to

(a) $\frac{3^{n+1}-2^n}{2n}$ (b) $\frac{3^{n+1}-2n-3}{4}$

(c) $\frac{3n-1+3n^n}{9}$ (d) $\frac{3^{n+1}+2n^2}{8}$

3. $1+3+7+15+\cdots+n$ terms is equal to

(a) $2^{n+1}-n-2$ (b) n^2+n-2

(c) 2^n+n^2-n (d) none of these

4. Sum of the series $S=1^2-2^2+3^2-4^2+\cdots-2002^2+2003^2$ is equal to

(a) 2,007,006 (b) 1,005,004

(c) 2,000,506 (d) none of these

5. The positive interger n for which $2\times2^2+3\times2^3+4\times2^4+\cdots+n\times2^n=2^{n+10}$ is equal to

(a) 510 (b) 511

(c) 512 (d) 513

6. $1\lfloor\underline{1}+2\lfloor\underline{2}+3\lfloor\underline{3}+\cdots+n\lfloor\underline{n}$ is equal to

(a) $\lfloor\underline{n+1}-1$ (b) $\lfloor\underline{n-1}+1$

(c) $\lfloor\underline{n+1}+1$ (d) $\lfloor\underline{n-1}-1$

7. $2\cdot4+4\cdot7+6\cdot10+\cdots+(n-1)$ terms is equal to

(a) $2n^3-2n^2$ (b) $\frac{n^3+3n^2+1}{6}$

(c) $2n^3+2n$ (d) none of these

8. $3\cdot6+4\cdot7+5\cdot8+\cdots+(n-2)$ terms is equal to

(a) n^3+n^2+n+2 (b) $\frac{2n^3+12n^2+10n-84}{6}$

(c) $\alpha n^3+\beta n^2+rn$ (d) $\frac{n^3+6n^2+5n-42}{3}$

9. $1\cdot2\cdot3+2\cdot3\cdot4+3\cdot4\cdot5+n$ terms is equal to

(a) $\frac{n(n+1)(n+2)(3n+5)}{12}$ (b) $\frac{n(n+1)(n+2)(n+3)}{4}$

(c) $2n(n+1)(n+2)(n+3)$ (d) $\frac{n(n+1)(n+2)(n+3)}{12}$

10. $(1)+(1+3)+(1+3+5)+\cdots+n$ brackets is equal to

(a) $\frac{n(n+1)(n+2)}{6}$

(b) $\frac{n(n+1)(3n^2+23n+46)}{12}$

(c) $\frac{2(27n^3+90n^2+450-50)}{4}$

(d) $\frac{n(n+1)(2n+1)}{6}$

11. The sum of $(1^2)+(1^2+2^2)+(1^2+2^2+3^3)+\cdots+n$ brackets is equal to

(a) $\frac{n(n+1)(n+2)}{6}$

(b) $\frac{n(n+1)^2(n+2)}{12}$

(c) $\frac{2(27n^3+90n^2+450-50)}{4}$

(d) $\frac{n(n+1)(2n+1)}{6}$

12. The sum of n brackets of $(1)+\left(\frac{1}{3}+\frac{1}{3^2}\right)+\left(\frac{1}{3^3}+\frac{1}{3^4}+\frac{1}{3^5}\right)+\cdots$ is equal to

(a) $\frac{(3^n-1)^3}{2\cdot4^{n-1}}$ (b) $\frac{3^n-1}{2\cdot3^{(n-1)(n+2)/2}}$

(c) $\frac{3^n+1}{3\cdot7^{n-1}}$ (d) none of these

13. $(1)+(2+3+4)+(5+6+7+8+9)+\cdots+n$ brackets is equal to

(a) $(n-1)^3+n^3$ (b) $(n+1)^3+8n^2$

(c) $\dfrac{(n+1)(n+2)}{6n}$ (d) $(n+1)^3+n^3$

14. $(1)+(2+3)+(4+5+6)+\cdots+n$ brackets is equal to

(a) $\dfrac{n(n+1)(n^2+n+2)}{8}$ (b) $\dfrac{n(n+1)(n^2-n+2)}{8}$

(c) $\dfrac{n(n-1)(n^2+n+2)}{8}$ (d) $\dfrac{n(n-1)(n^2-n+2)}{8}$

15. The value of the sum in the nth bracket of $(1)+(2+3)+(4+5+6+7)+(8+9+10+\cdots+15)+\cdots$ is equal to

(a) $2^n(2^n+2^{n-1}-1)$ (b) $2^{n-1}(2^n+2^{n-1}-1)$

(c) $2^{n-2}(2^n+2^{n-1}-1)$ (d) none of these

16. If the natural numbers are divided into groups of $\{1\}$ $\{2, 3\}$ $\{4, 5, 6\}$ $\{7, 8, 9, 10\}$ ..., then the sum of 50 groups is equal to

(a) 62,525 (b) 65,225

(c) 56,255 (d) 55,625

17. $(1)+(2+3+4)+(5+6+7+6+9)+\cdots+n$ brackets is equal to

(a) $\dfrac{n(1+2)(n+2)}{6}$ (b) $\dfrac{n^2(n^2+1)}{2}$

(c) $n(n+1)(2n+1)$ (d) $\dfrac{n(n+1)(2n+1)}{6}$

ANSWERS

1. (b)	**2.** (b)	**3.** (a)	**4.** (a)	**5.** (d)	**6.** (a)	**7.** (a)	**8.** (d)	**9.** (b)	**10.** (d)
11. (b)	**12.** (b)	**13.** (a)	**14.** (a)	**15.** (c)	**16.** (a)	**17.** (b)			

8

ISAAC NEWTON

Binomial Theorem

- Binomial Theorem
- Types of Binomial Theorems
- Binomial Theorem for Positive Integral Index
- Pascal's Triangle
- General Term in the Expansion of $(x + y)^n$
- Method of Finding the Independent Term
- Properties of Binomial Theorem
- The Numerically Greatest Term in the Expansion of $(1 + x)^n$
- Binomial Theorem for Negative/Rational Index
- Important Expansions
- Multinomial Theorem

BINOMIAL THEOREM

Binomial theorem is the formula by which the expansion of a binomial expression raised to the power of n, where n can be positive, negative or a fraction, can be found.

TYPES OF BINOMIAL THEOREMS

There are two types of binomial theorems:

1. Binomial theorem for positive integral index
2. Binomial theorem for negative/rational index.

BINOMIAL THEOREM FOR POSITIVE INTEGRAL INDEX

It states that if n is any positive integer, then the expansion of $(x + y)^n$ can be written as

$$(x+y)^n = {}^nC_0x^ny^0 + {}^nC_1x^{n-1}y^1 + {}^nC_2x^{n-2}y^2 + \cdots + {}^nC_rx^{n-r}y^r + \cdots + {}^nC_nx^0y^n$$

PASCAL'S TRIANGLE

The coefficients ${}^nC_0, {}^nC_1, {}^nC_2, \ldots, {}^nC_r, \ldots, {}^nC_n$ in the binomial expansion follow a particular pattern represented in the form of a triangle shown in Fig. 8.1. In the figure, each row is bounded to 1 on both sides. Any entry, except the first and the last, in the row is the sum of two entries above it in the preceding row. This pattern is known as **Pascal's triangle**.

Exponent of binomial	Coefficients of successive terms in binomial theorem
$n = 0$	1
$n = 1$	1 1
$n = 2$	1 2 1
$n = 3$	1 3 3 1
$n = 4$	1 4 6 4 1
$n = 5$	1 5 10 10 5 1
$n = 6$	1 6 15 20 15 6 1

Fig. 8.1

☞ GENERAL TERM IN THE EXPANSION OF $(x + y)^n$

Since

$$(x+y)^n = {}^nC_0x^n y^0 + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + \cdots + {}^nC_rx^{n-r}y^r + \cdots + {}^nC_nx^0y^n$$

Term→ (1) (2) (3) $(r+1)$ $(n+1)$

So, if the $(r + 1)$th term is the general term in the expansion of $(x + y)^n$, then

$$T_{r+1} = {}^nC_r x^{n-r} y^r$$

where $r = 0, 1, 2, 3, ..., n$

Note the odd-numbered and even-numbered terms in the above expansion.

Binomial expansion	General term
$(x+y)^n$	${}^nC_r x^{n-r} y^r$
$(x-y)^n$	$(-1)^r\ {}^nC_r x^{n-r} y^r$
$(1+x)^n$	${}^nC_r x^r$
$(1-x)^n$	$(-1)^r\ {}^nC_r x^r$

☞ METHOD OF FINDING THE INDEPENDENT TERM

The term independent of x in the expansion of $(x+y)^n$ is found as follows:

Step 1. Write down the general term in the expansion of $(x+y)^n$, i.e. $(r+1)$th term $\Rightarrow$ $T_{r+1} = {}^nC_r x^{n-r} y^r$

Step 2. Separate the constants and variables and group them separately.

Step 3. Since we need to find the term independent of x in the given binomial expansion, equate to zero the index of x and accordingly we will get the value of r for which there exists a term independent of x in the expansion.

Illustration 1. Find the term independent of x in the expansion of $\left(\sqrt[3]{x} + \frac{1}{2\sqrt[3]{x}}\right)^{18}$, $x > 0$.

Solution: We have

$$T_{r+1} = {}^{18}C_r \left(\sqrt[3]{x}\right)^{18-r} \left(\frac{1}{2\sqrt[3]{x}}\right)^r$$

$$= {}^{18}C_r x^{\frac{18-r}{3}} \cdot \frac{1}{2^r . x^{\frac{r}{3}}} = {}^{18}C_r \frac{1}{2^r} \cdot x^{\frac{18-2r}{3}}$$

Since we have to find a term independent of x, i.e. the term not having x, so take $\frac{18-2r}{3} = 0$. We get $r = 9$. The required term is ${}^{18}C_9 \frac{1}{2^9}$.

☞ PROPERTIES OF BINOMIAL THEOREM

Property 1. The number of terms in the expansion of the binomial expression $(x+y)^n$ is one more than the value of its integral index n, i.e. $(n+1)$ is the number of terms for n as the integral index.

Property 2. In every term of the binomial expansion, the sum of indices of x and y is n.

Property 3. Binomial coefficients equidistant from beginning of the expansion and end of the expansion are equal, i.e. ${}^nC_0 = {}^nC_n$, ${}^nC_1 = {}^nC_{n-1}$ and so on since ${}^nC_r = {}^nC_{r-1}$.

Property 4. The rth term from the end in the expansion of $(x + y)^n$ is the $\{[(n+1)-r]+1\}$th term from the beginning of the expansion.

Property 5. $(x + y)^n + (x - y)^n$

$= 2({}^nC_0x^n + {}^nC_2x^{n-2}y^2 + {}^nC_4x^{n-4}y^4 + \cdots)$

= 2(Sum of the terms with even binomial coefficients)

= 2(Sum of the odd-numbered terms)

Property 6. $(x + y)^n - (x - y)^n$

$= 2({}^nC_1x^{n-1}y^1 + {}^nC_3x^{n-3}y^3 + {}^nC_5x^{n-5}y^5 + \cdots)$

= 2(Sum of the terms with odd binomial coefficients)

= 2(Sum of the even-numbered terms)

Property 7. Middle term(s) in a binomial expansion: Since the number of terms in the expansion of $(x+y)^n$ is $(n + 1)$, so, two cases arise:

Case I: If *n* is even

If n is even, then the number of terms $(n + 1)$ in the expansion will be an odd number.

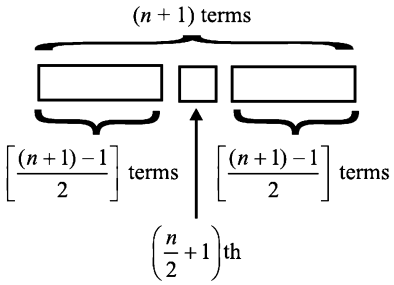

Fig. 8.2

The middle term in the expansion will then be the [(*n*/2) + 1]th term.

Case II: If *n* is odd

If *n* is odd, then the number of terms (*n* + 1) in the expansion will be an even number. Hence, there will be two middle terms, as shown in Fig. 8.3.

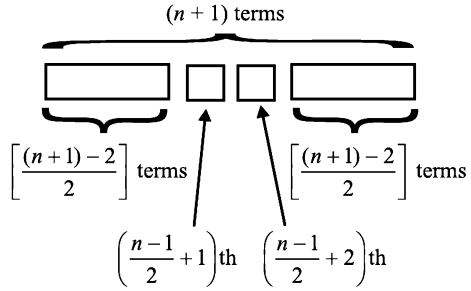

Fig. 8.3

The two middle terms in the expansion will be $\left(\frac{n-1}{2}+1\right)$th and $\left(\frac{n-1}{2}+2\right)$th terms

$\Rightarrow$ the two middle terms in the expansion will be $\left(\frac{n+1}{2}\right)$th and $\left(\frac{n+3}{2}\right)$th terms

Illustration 2. Find the middle terms in the expansion of $[3 - (x^3/6)]^7$.

Solution: The general expression $= \left(3-\frac{x^3}{6}\right)^7$. Here $n = 7$, which is an odd number.

So, $\left(\frac{7+1}{2}\right)$th and $\left(\frac{7+3}{2}\right)$th, i.e. the 4th and 5th terms are the two middle terms.

Now, $T_4 = T_{3+1} = {}^7C_3(3)^{7-3}\left(-\frac{x^3}{6}\right)^3$

$= (-1)^3\,{}^7C_3(3)^4\left(\frac{x^3}{6}\right)^3$

$= -35\times81\times\frac{x^9}{216} = -\frac{105x^9}{8}$

and $T_5 = T_{4+1} = {}^7C_4(3)^{7-4}\left(-\frac{x^3}{6}\right)^4 = {}^7C_4(3)^3\left(-\frac{x^3}{6}\right)^4$

$= 35\times27\times\frac{x^{12}}{1296} = \frac{35x^{12}}{48}$

Hence, the middle terms are $\frac{-105x^9}{8}$ and $\frac{35x^{12}}{48}$.

Illustration 3. Show that the middle term in the expansion of $(1 + x)^{2n}$ is $\frac{1\cdot3\cdot5\ldots(2n-1)}{n!}2^n\cdot x^n$.

Solution: The given expression $= (1 + x)^{2n}$. Here, the index $2n$ is even. So, the $\left(\frac{2n}{2}+1\right)$th, i.e. the $(n + 1)$th term, is the middle term.

Hence, the middle term $= T_{n+1} = {}^{2n}C_n(1)^{2n-n}x^n$

$= {}^{2n}C_n x^n$

$= \frac{(2n)!}{n!(2n-n)!}x^n$

$= \frac{1\cdot2\cdot3\cdot4\ldots(2n-3)(2n-2)(2n-1)(2n)}{n!n!}$

$= \frac{[1\cdot3\cdot5\ldots(2n-3)(2n-1)][2\cdot4\cdot6\ldots(2n-2)(2n)]}{n!n!}x^n$

$= \frac{2^n[1\cdot3\cdot5\ldots(2n-3)(2n-1)][1\cdot2\cdot3\cdot(n-1)(n)]}{n!n!}x^n$

$= \frac{[1\cdot3\cdot5\ldots(2n-3)(2n-1)]\,n!\,2^n x^n}{n!n!}$

$= \frac{1\cdot3\cdot5\ldots(2n-1)}{n!}2^n x^n$

THE NUMERICALLY GREATEST TERM IN THE EXPANSION OF $(1 + x)^n$

Method 1

If T_r and T_{r+1} are the rth and $(r + 1)$th terms in the expansion of $(1 + x)^n$, then

$$\frac{T_{r+1}}{T_r} = \frac{{}^nC_r x^r}{{}^nC_{r-1}x^{r-1}} = \frac{n-r+1}{r}x$$

Step 1. Let T_r (the rth term) be the greatest term.

Step 2. Find T_{r-1}, T_r, T_{r+1} from the given expansion.

Step 3. Put $\frac{T_r}{T_{r+1}} \geq 1$ and $\frac{T_r}{T_{r-1}} \geq 1$. This will give an inequality from where the value or values of r can be obtained.

Step 4. Then, find the rth term T_r, which is the greatest term.

Method 2 (Shortcut)

Step 1. Calculate $m = \frac{x(n+1)}{x+1}$.

Step 2. If m is an integer, then T_m and T_{m+1} are equal and both are the greatest terms.

Step 3. If m is not an integer and $[m]$ is its integral part, then the greatest term is $T_{[m]+1}$.

Note

- Numerically, the greatest term in the expansion of $(1-x)^n, x > 0, n \in N$ is the same as the greatest term in the expansion of $(1+x)^n$.
- To find the greatest term in the expansion of $(x+y)^n$, write it as $x^n\left(1+\frac{y}{x}\right)^n$ and then find the greatest term in the expansion $\left(1+\frac{y}{x}\right)^n$.

Illustration 4. Find the numerically greatest term in the expansion of $(3-5x)^{11}$ when $x = \frac{1}{5}$.

Solution: Since $(3-5x)^{11} = 3^{11}\left(1-\frac{5x}{3}\right)^{11}$, now in the expansion of $\left(1-\frac{5x}{3}\right)^{11}$, we have

$$\frac{T_{r+1}}{T_r} = \frac{(11-r+1)}{r}\left|-\frac{5x}{3}\right|$$

$$= \left(\frac{12-r}{r}\right)\left|-\frac{5}{3}\times\frac{1}{5}\right| \qquad \left(\because x = \frac{1}{5}\right)$$

$$= \left(\frac{12-r}{r}\right)\left(\frac{1}{3}\right) = \left(\frac{12-r}{3r}\right)$$

$\therefore \quad \frac{T_{r+1}}{T_r} \geq 1 \Rightarrow \frac{12-r}{3r} \geq 1$

$\Rightarrow \quad 4r \leq 12 \quad \Rightarrow \quad r \leq 3$

$\therefore \quad r = 2, 3$

So, the greatest terms are T_{2+1} and T_{3+1}.

$\therefore$ Greatest term (when $r = 2$) $= 3^{11}|T_{2+1}|$

$$= 3^{11}\left|{}^{11}C_2\left(-\frac{5}{3}x\right)^2\right|$$

$$= 3^{11}\left|{}^{11}C_2\left(-\frac{5}{3}\times\frac{1}{5}\right)^2\right| \qquad \left(\because x = \frac{1}{5}\right)$$

$$= 3^{11}\left|\frac{11\cdot 10}{1\cdot 2}\times\frac{1}{9}\right| = 55\times 3^9$$

and the greatest term (when $r = 3$) $= 3^{11}|T_{3+1}|$

$$= 3^{11}\left|{}^{11}C_3\left(-\frac{5}{3}x\right)^3\right|$$

$$= 3^{11}\left|{}^{11}C_3\left(-\frac{5}{3}\times\frac{1}{5}\right)^3\right|$$

$$= 3^{11}\left|\frac{11\cdot 109}{1\cdot 2\cdot 3}\times\frac{-1}{27}\right| = 55\times 3^9$$

From the above, we see that the values of both the greatest terms are equal.

Illustration 5. Find the numerically greatest term in the expansion of $(8-5x)^{18}$, where $x = \frac{2}{5}$.

Solution: Shortcut method

The numerically greatest term in the expansion of $(8-5x)^{18}$ = the numerically greatest term in the expansion of $(8+5x)^{18}$

Now, $\quad m = \frac{(18+1)\times 5x}{8+5x} = \frac{19\times 5\times\frac{2}{5}}{8+5\times\frac{2}{5}} = \frac{38}{10} = 3.8$

$\therefore$ The greatest term is

$$T_{[3]+1} = {}^{18}C_3 8^3 \times\left(5\times\frac{2}{5}\right)^{15}$$

$$= \frac{18\times 17\times 16}{3\times 2}\times 2^{9+15} = 816\times 2^{24} = 1632\times 2^{23}.$$

BINOMIAL THEOREM FOR NEGATIVE/ RATIONAL INDEX

Let n be a rational number and x be a real number such that $|x| < 1$, then

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

$$+\frac{n(n-1)(n-2)\ldots(n-(r-1)}{r!}x^r + \cdots + \infty$$

In the above binomial, the first term must be unity. If the first term is not unity and the index of the binomial is either a negative or a fraction, then we expand the binomial as follows:

$$(x+y)^n = \left[y\left(1+\frac{x}{y}\right)\right]^n = y^n\left(1+\frac{x}{y}\right)^n$$

$$\Rightarrow (x+y)^n = y^n\left[1+n\left(\frac{x}{y}\right)+\frac{n(n-1)}{2!}\left(\frac{x}{y}\right)^2+\frac{n(n-1)(n-2)}{3!}\left(\frac{x}{y}\right)^3+\cdots+\frac{n(n-1)\cdots(n-(r-1))}{r!}\left(\frac{x}{y}\right)^r+\cdots\infty\right]$$

The above expansion is valid when $\left|\frac{x}{y}\right| < 1$ or equivalently $|x|<|y|$.

GENERAL TERM(S)

The general term in the expansion of $(1+x)^n$ is given by

$$T_{r+1} = \frac{n(n-1)(n-2)\ldots[n-(r-1)]}{r!}x^r$$

The general term in the expansion of $(1-x)^n$ is given by

$$T_{r+1} = (-1)^r\frac{n(n-1)(n-2)\ldots[n-(r-1)]}{r!}x^r$$

The general term in the expansion of $(1+x)^{-n}$ is given by

$$T_{r+1} = (-1)^r\frac{n(n+1)(n+2)\ldots[n+(r-1)]}{r!}x^r$$

The general term in the expansion of $(1-x)^{-n}$ is given by

$$T_{r+1} = \frac{n(n+1)(n+2)\ldots.[n+(r-1)]}{r!}x^r$$

IMPORTANT EXPANSIONS

Putting $n = 1, 2, 3, \ldots$ in the above expansions of $(1+x)^{-n}$ and $(1-x)^{-n}$, we have

(i) $(1+x)^{-1} = 1-x+x^2-x^3+x^4-\cdots+(-1)^r x^r+\cdots+\infty$

(ii) $(1-x)^{-1} = 1+x+x^2+x^3+x^4+\cdots+x^r+\cdots+\infty$

(iii) $(1+x)^{-2} = 1-2x+3x^2-4x^3+\cdots+(-1)^r(r+1)x^r+\cdots+\infty$

(iv) $(1-x)^{-2} = 1+2x+3x^2+4x^3+\cdots+(r+1)x^r+\cdots+\infty$

(v) $(1+x)^{-3} = 1-3x+6x^2-10x^3+\cdots+(-1)^r \times\frac{(r+1)(r+2)}{2!}x^r+\cdots+\infty$

(vi) $(1-x)^{-3} = 1+3x+6x^2+10x^3+\cdots+\frac{(r+1)(r+2)}{2!}x^r+\cdots+\infty$

(vii) If x is small compared with 1, we find that the values of x^2, x^3, x^4,... become smaller and smaller. Consequently, the terms in the above expansion become smaller and smaller.
If x is so small that its square and higher powers of x can be neglected, we write

$$(1+x)^n = 1+nx$$

$$(1+ax)(1+bx) = 1+(a+b)x$$

[neglecting the higher powers of x^2]

MULTINOMIAL THEOREM

If we wish to expand $(x+y+z)^n$, where n is a positive integer and x, y, z are complex numbers, we follow the steps given below:

Step 1. We substitute $(y + z) = a$.

Step 2. We expand the binomial $(x + a)^n$.

Step 3. Expand $a \equiv (y+z)^r (0 \le r \le n)$ in each term of the binomial expansion of $(x+y)^n$.

Hence, in general, if the multinomial expression is of the form $(a_1 + a_2 + \cdots + a_m)^n$, then its expansion can be written as

$$(a_1+a_2+\cdots+a_m)^n = \sum\frac{n!}{n_1!n_2!\ldots n_m!}a_1^{n_1}a_2^{n_2}\ldots a_m^{n_m}$$

such that $n_1, n_2, n_3, \ldots, n_m$ are all +ve integers, and $n_1 + n_2 + n_3 + \cdots + n_m = n$.

Illustration 6. Find the coefficient of x^3 in the expansion of $(1 + x + x^2)^n$.

Solution: $(1+x+x^2)^n = [1+x(1+x)]^n$

$$= {}^nC_0 + {}^nC_1x(1+x) + {}^nC_2x^2(1+x)^2+\cdots$$

Coefficient of $x^3 = {}^nC_2$ [coefficient of x in $(1+x)^2$]
$+ {}^nC_3$ [coefficient of x^0 in $(1 + x)^3$]

$$= {}^nC_2(2) + {}^nC_3(1)$$

$$= \frac{2n(n-1)}{2!}+\frac{n(n-1)(n-2)}{3!}$$

$$= \frac{n(n-1)}{6}(6+n-2)$$

$$= \frac{n(n-1)(n+4)}{6}.$$

SOLVED PROBLEMS

1. The number of terms which are free from radical signs in the expansion of $(y^{1/5} + x^{1/10})^{55}$ is

(a) 5 (b) 6
(c) 7 (d) none of these

Ans. (b)

Solution: The general term in the expansion of $(y^{1/5} + x^{1/10})^{55}$ is

$$T_{r+1} = {}^{55}C_r (y^{1/5})^{55-r} (x^{1/10})^r = {}^{55}C_r y^{11-r/5} x^{r/10}$$

Clearly, T_{r+1} will be independent of radical signs if $r/5$ and $r/10$ are integers, where $0 \le r \le 55$.

$\therefore$ $r = 0, 10, 20, 30, 40, 50$

Hence, there are 6 terms in the expansion of the given binomial which are independent of radicals.

2. In the expansion of $\left(2x^2 - \frac{1}{x}\right)^{12}$, the term independent of x is

(a) 10th (b) 9th
(c) 8th (d) 7 th

Ans. (b)

Solution: Given expression $= \left(2x^2 - \frac{1}{x}\right)^{12}$.

Let the $(r + 1)$th term of the given expression be independent of x.

We know that in the binomial expansion of $(a + b)^n$, the $(r + 1)$th term $T_{r+1} = {}^nC_r a^{n-r} b^r$

Therefore, the $(r + 1)$th term in the expansion of

$$\left(2x^2 - \frac{1}{x}\right)^{12} = {}^{12}C_r (2x^2)^{12-r} \left(-\frac{1}{x}\right)^r$$

$$= {}^{12}C_r 2^{12-r} x^{24-2r} x^{-r} (-1)^r$$

$$= {}^{12}C_r 2^{12-r} x^{24-3r} (-1)^r$$

Equation to zero the index of x in T_{r+1}, we get $24 - 3r = 0$

or $r = \frac{24}{3} = 8$

Thus, the term independent of x in the expansion of the given expression is the T_{r+1} i.e. $(r + 1)$th $= (8 + 1)$th $=$ 9th term.

3. The middle term(s) in the expression of $\left(3x - \frac{x^3}{6}\right)^7$ is (are)

(a) only the 4th term
(b) only the 5th term
(c) only the 6th term
(d) both the 4th and 5th term

Ans. (d)

Solution: Given expression $= [3x - (x^3/6)]^7$.

We know that the standard expression of the given form is $(a + b)^n$.

Comparing the given expression with the standard form, we get $n = 7$.

Since the value of n is 7, therefore, the number of terms $m = n + 1 = 7 + 1 = 8$ (even)

Thus, the middle terms are $\left(\frac{m}{2}\right)$ and $\left(\frac{m}{2} + 1\right)$, i.e. $\left(\frac{8}{2}\right)$ and $\left(\frac{8}{2} + 1\right)$ or the 4th and 5th terms.

4. The value of the middle term in the expansion of $\left(x + \frac{1}{2x}\right)^{2n}$ is

(a) $\frac{1 \times 3 \times 5 \times \cdots \times (2n-3)}{n!}$ (b) $\frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{n!}$
(c) $\frac{1 \times 3 \times 5 \times \cdots \times (2n+1)}{n!}$ (d) $\frac{1 \times 3 \times 5 \times \cdots \times (2n+3)}{n!}$

Ans. (b)

Solution: Given expression $= \left(x + \frac{1}{2x}\right)^{2n}$.

We know that the standard expression of the given form is $(a + b)^N$.

Comparing the given expression with the standard form, we get $N = 2n$ (even).

Since the value of N is $2n$, therefore, the number of terms $m = N + 1 = 2n + 1$ (odd)

Thus, the middle term is the $\left(\frac{m+1}{2}\right)$th term, i.e. the $\left[\frac{(2n+1)+1}{2}\right]$th term or the $(n + 1)$th term.

We also know that in the binomial expansion of $(a + b)^N$, the $(r + 1)$th term $T_{r+1} = {}^NC_r a^{N-r} b^r$.

Therefore, the middle term, $T_{n+1} = {}^{2n}C_n x^{2n-n} \left(\frac{1}{2x}\right)^n$

$$\frac{{}^{2n}C_n}{2^n} = \frac{(2n)!}{(n!)(n!)(2^n)} = \frac{1 \times 2 \times 3 \times 4 \times 5 \times \cdots \times (2n-1)(2n)}{(n!)(n!)(2^n)}$$

$$= \frac{(2 \times 4 \times 6 \times \cdots \times 2n)[1 \times 3 \times 5 \times \cdots \times (2n-1)]}{(n!)(n!)(2^n)}$$

$$= \frac{2^n (1 \times 2 \times 3 \times \cdots \times n)[1 \times 3 \times 5 \times \cdots \times (2n-1)]}{(n!)(n!)(2^n)}$$

$$= \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{n!}$$

5. If the ninth term in the expansion of $[3^{\log_3\sqrt{25^{x-1}+7}} + 3^{(-1/8)\log_3(5^{x-1}+1)}]^{10}$ is equal to 180 and $x > 1$, then x equals

(a) $\log_{10} 15$ (b) $\log_5 15$

(c) $\log_e 15$ (d) none of these

Ans. (b)

Solution: We know that $a^{\log_a N} = N$

$\therefore \quad 3^{\log_3\sqrt{25^{x-1}+7}} = \sqrt{25^{x-1}+7} = \sqrt{(5^{x-1})^2+7} = \sqrt{y^2+7}$

where $y = 5^{x-1}$. And

$$3^{(-1/8)\log_3(5^{x-1}+1)} = 3^{\log_3(5^{x-1}+1)^{-1/8}} = (5^{x-1}+1)^{-1/8} = (y+1)^{-1/8}$$

$$\therefore \quad [3^{\log_3\sqrt{25^{x-1}+7}} + 3^{(-1/8)\log_3(5^{x-1}+1)}]^{10}$$

$$= [\sqrt{y^2+7} + (y+1)^{-1/8}]^{10}$$

Now, $T_9 = 180$

$$\Rightarrow \quad {}^{10}C_8(\sqrt{y^2+7})^{10-8}[(y+1)^{-1/8}]^8 = 180$$

$$\Rightarrow \quad {}^{10}C_8(y^2+7)(y+1)^{-1} = 180$$

$$\Rightarrow \quad 45\left(\frac{y^2+7}{y+1}\right) = 180$$

$$\Rightarrow \quad y^2+7 = 4y+4 \Rightarrow y^2-4y+3 = 0$$

$$\Rightarrow \quad y = 1, y = 3$$

$$\Rightarrow \quad 5^{x-1} = 1 \text{ or } 5^{x-1} = 3$$

$$\Rightarrow \quad 5^x = 5 \text{ or } 5^x = 15$$

$$\Rightarrow \quad x = 1 \text{ or } x = \log_5 15$$

But $x > 1$, therefore, $x = \log_5 15$.

6. The coefficient of x^{32} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is

(a) $-{}^{15}C_3$ (b) ${}^{15}C_4$

(c) $-{}^{15}C_5$ (d) ${}^{15}C_2$

Ans. (b)

Solution: Given expression $= \left(x^4 - \frac{1}{x^3}\right)^{15}$.

Let the $(r + 1)$th term be the term containing x^{32} in the given expression.

We know that in the binomial expansion of $(a + b)^n$, the $(r + 1)$th term is $T_{r+1} = {}^nC_r a^{n-r} b^r$.

Therefore, the $(r + 1)$th term in the expansion of

$$\left(x^4 - \frac{1}{x^3}\right)^{15} = {}^{15}C_r x^{4(15-r)}\left(-\frac{1}{x^3}\right)^r$$

$$= {}^{15}C_r x^{(60-4r)} x^{(-3r)} (-1)^r = (-1)^r {}^{15}C_r x^{(60-7r)}$$

Since x^{32} occurs in T_{r+1}, therefore, $60 - 7r = 32$ or $7r = 60 - 32 = 28$ or $r = 4$

Thus, the coefficient of $x^{32} = (-1)^4 \times {}^{15}C_4 = {}^{15}C_4$.

7. If the coefficient of x^7 in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$ is equal to the coefficient of x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^{11}$, then the value of ab is

(a) 1 (b) 2

(c) 3 (d) 4

Ans. (a)

Solution: Given that the coefficient of x^7 in $\left(ax^2 + \frac{1}{bx}\right)^{11}$ = the coefficient of x^{-7} in $\left(ax - \frac{1}{bx^2}\right)^{11}$.

We know that in binomial expansion of $(A + B)^n$, the $(r + 1)$th term $T_{r+1} = {}^nC_r A^{n-r} B^r$

Therefore, in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$, the $(r + 1)$th term $T_{r+1} = {}^{11}C_r (ax^2)^{(11-r)}\left(\frac{1}{bx}\right)^r = {}^{11}C_r \frac{a^{(11-r)}}{b^r} x^{(22-3r)}$

Similarly, in the binomial expansion of $\left(ax - \frac{1}{bx^2}\right)^{11}$, the $(s + 1)$th term

$$(T_{s+1}) = {}^{11}C_s (ax)^{(11-s)}\left(-\frac{1}{bx^2}\right)^s = (-1)^s {}^{11}C_s \frac{a^{(11-s)}}{b^s} x^{(11-3s)}$$

Since x^7 occurs in T_{r+1}, therefore, $22 - 3r = 7$ or $3r = 15$ or $r = 5$ and x^{-7} occurs in T_{s+1}, therefore, $11 - 3s = -7$ or $3s = 18$ or $s = 6$

Thus, the coefficient of x^7 in the expansion of $\left(ax + \frac{1}{bx}\right)^{11}$

$$= {}^{11}C_5\left(\frac{a^{(11-5)}}{b^5}\right) = {}^{11}C_5\left(\frac{a^6}{b^5}\right) \quad \text{(i)}$$

Similarly, the coefficient of x^{-7} in the expansion of

$$\left(ax - \frac{1}{bx^2}\right)^{11} = (-1)^6 {}^{11}C_6\left(\frac{a^5}{b^6}\right) = {}^{11}C_6\left(\frac{a^5}{b^6}\right) \quad \text{(ii)}$$

Since the coefficient of x^7 is equal to the coefficient of x^{-7}, therefore ${}^{11}C_5\left(\frac{a^6}{b^5}\right) = {}^{11}C_6\left(\frac{a^5}{b^6}\right)$ or $ab = 1$.

8. The value of ${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$ is

(a) ${}^{47}C_6$ (b) ${}^{47}C_5$

(c) ${}^{52}C_4$ (d) ${}^{52}C_5$

Ans. (c)

Solution: Given expression = ${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$. We know that

$$\begin{aligned}
&{}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3 \\
&= {}^{47}C_4 + {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3 \\
&= ({}^{47}C_4 + {}^{47}C_3) + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3 \\
&= ({}^{48}C_4 + {}^{48}C_3) + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3 \\
&= ({}^{49}C_4 + {}^{49}C_3) + {}^{50}C_3 + {}^{51}C_3 \\
&= ({}^{50}C_4 + {}^{50}C_3) + {}^{51}C_3 = {}^{51}C_4 + {}^{51}C_3 = {}^{52}C_4 \\
&\qquad \text{(where } {}^{n}C_r + {}^{n}C_{r+1} = {}^{n+1}C_{r+1}\text{)}
\end{aligned}$$

9. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \cdots + C_nx^n$, then the value of $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \cdots + \frac{nC_n}{C_{n-1}}$ is

(a) $\frac{n}{2}$ (b) $n(n+1)$

(c) $\frac{n(n+1)}{12}$ (d) $\frac{n(n+1)}{2}$

Ans. (a)

Solution: The given binomial expansion is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \cdots + C_nx^n$$

We know that

$$\begin{aligned}
&\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \cdots + \frac{nC_n}{C_{n-1}} \\
&= \frac{n}{1} + \frac{2\cdot\frac{n(n-1)}{2!}}{n} + \frac{3\cdot\frac{n(n-1)(n-2)}{3!}}{\frac{n(n-1)}{2!}} + \cdots + \frac{n\cdot 1}{n} \\
&= n + (n-1) + (n-2) + \cdots + 1 = \frac{n(n+1)}{2}
\end{aligned}$$

10. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \cdots + C_nx^n$, then the value of $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \cdots + \frac{C_n}{n+1}$ is

(a) $\frac{1}{n+1}$ (b) $2^{n+1} - 1$

(c) $\frac{2^{n-1}+1}{2(n-1)}$ (d) $\frac{2^{n+1}-1}{n+1}$

Ans. (d)

Solution: The given binomial expansion is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \cdots + C_nx^n$$

Substituting $x = 1$ and $n = n + 1$ in the given expression,

$$C_0 + C_1 + C_2 + \cdots + C_{n+1} = 2^{n+1}$$

We know that

$$\begin{aligned}
&C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \cdots + \frac{C_n}{n+1} \\
&= \sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1} = \sum_{r=0}^{n} \frac{1}{n+1}\cdot\frac{n+1}{r+1}\cdot{}^{n}C_r = \frac{1}{n+1}\sum_{r=0}^{n}\frac{n+1}{r+1}{}^{n}C_r \\
&= \frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1} \qquad \left(\because\ {}^{n+1}C_{r+1} = \frac{n+1}{r+1}\,{}^{n}C_r\right) \\
&= \frac{1}{n+1}\left[{}^{n+1}C_1 + {}^{n+1}C_2 + {}^{n+1}C_3 + \cdots + {}^{n+1}C_{n+1}\right] \\
&= \frac{1}{n+1}\left[\left({}^{n+1}C_0 + {}^{n+1}C_1 + \cdots + {}^{n+1}C_{n+1}\right) - \left({}^{n+1}C_0\right)\right]
\end{aligned}$$

Therefore, $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \cdots + \frac{C_n}{n+1} = \frac{1}{n+1}\left[2^{n+1} - 1\right]$

$$= \frac{2^{n+1}-1}{n+1}$$

$$\left(\because\ {}^{n+1}C_0 = 1\right)$$

11. The value of $\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + \cdots$ is

(a) $\frac{2^n}{n+1}$ (b) $\frac{2^n-1}{n+1}$

(c) $\frac{2^n+1}{n-1}$ (d) $\frac{2^n+1}{n+1}$

Ans. (b)

Solution: Given expression = $\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + \cdots$

We know from the binomial expansion that

$$(1+x)^n = 1 + C_1x + C_2x^2 + \cdots + C_rx^r + \cdots + C_nx^n \qquad \text{(i)}$$

Similarly, $(1-x)^n = 1 - C_1x + C_2x^2 + \cdots + C_nx^n$ (ii)

Subtracting Eq. (ii) from Eq. (i), we get

$$(1+x)^n - (1-x)^n = 2(C_1x + C_3x^3 + C_5x^5 + \cdots)$$

Integrating both sides of the above equation within limits 0 and 1, we get

$$\int_0^1 [(1+x)^n - (1-x)^n]\,dx = 2\int_0^1 (C_1x + C_3x^3 + C_5x^5 + \cdots)\,dx$$

or $$\left[\frac{(1+x)^{n+1}}{(n+1)} + \frac{(1-x)^{n+1}}{(n+1)}\right]_0^1 = 2\left[C_1\frac{x^2}{2} + C_3\frac{x^4}{4} + C_5\frac{x^6}{6} + \cdots\right]_0^1$$

$$\frac{1}{2(n+1)}[(2^{n+1}-1)+(0-1)]$$

$$=\left[C_1\frac{(1)^2}{2}+C_3\frac{(1)^4}{4}+C_5\frac{(1)^6}{6}+\cdots\right]$$

or $$\frac{1}{2(n+1)}(2^{n+1}-2)=\frac{C_1}{2}+\frac{C_3}{4}+\frac{C_5}{6}+\cdots$$

Therefore, $$\frac{C_1}{2}+\frac{C_3}{4}+\frac{C_5}{6}+\cdots=\frac{2^n-1}{n+1}$$

12. The value of $(1.1)^5$ is

(a) 1.10651 (b) 1.61051

(c) 3.51051 (d) 3.61051

Ans. (b)

Solution: Given number = $(1.1)^5$.

The given number may be written as $(1 + 0.1)^5$.

We know from the binomial expansion that if a and b are real numbers $\forall n \in N$, then

$$(a+b)^n = {}^nC_0a^nb^0 + {}^nC_1a^{n-1}b^1 + {}^nC_2a^{(n-2)}b^2+\cdots$$
$$+ {}^nC_ra^{n-r}b^r+\cdots+{}^nC_{n-1}a^1b^{n-1} + {}^nC_na^0b^n \quad \text{(i)}$$

Substituting $n = 5$, $a = 1$ and $b = 0.1$ in Eq. (i), we get

$$(1+0.1)^5 = {}^5C_0(1)^5(0.1)^0 + {}^5C_1(1)^4(0.1)^1$$
$$+ {}^5C_2(1)^3(0.1)^2 + {}^5C_3(1)^2(0.1)^3$$
$$+ {}^5C_4(1)^1(0.1)^4 + {}^5C_5(1)^0(0.1)^5$$
$$=[1+(5\times0.1)+(10\times0.01)+(10\times0.001)$$
$$+(5\times0.0001)+(1\times0.00001)]$$
$$=1+0.5+0.1+0.01+0.0005+0.00001$$
$$=1.61051$$

13. If x is very small compared with a such that

$$\left(\frac{a}{a+x}\right)^{1/2}+\left(\frac{a}{a-x}\right)^{1/2}=2+k\frac{x^3}{a^2},$$

then the value of k is

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{3}{4}$ (d) 1

Ans. (c)

Solution: Given expression:

$$\left(\frac{a}{a+x}\right)^{1/2}+\left(\frac{a}{a-x}\right)^{1/2}=2+k\frac{x^3}{a^2}$$

The L.H.S. of the given expression may be written as

$$\frac{1}{\left(1+\frac{x}{a}\right)^{1/2}}+\frac{1}{\left(1-\frac{x}{a}\right)^{1/2}}=\left(1+\frac{x}{a}\right)^{-1/2}+\left(1-\frac{x}{a}\right)^{-1/2}$$

We know that $(1+x)^n = 1+nx+\frac{n(n+1)}{2!}x^2+\cdots$

Therefore,

$$2+k\frac{x^3}{a^2}=\left(1+\frac{x}{a}\right)^{-1/2}+\left(1-\frac{x}{a}\right)^{-1/2}$$
$$=\left(1-\frac{1}{2}\frac{x}{a}+\frac{3}{8}\frac{x^2}{a^2}+\cdots\right)+\left(1+\frac{1}{2}\frac{x}{a}+\frac{3}{8}\frac{x^2}{a^2}+\cdots\right).$$

Since x is very small, therefore neglecting the third and higher powers of x in this expansion, we get

$$2+k\frac{x^3}{a^2}=2+\frac{3}{4}\frac{x^2}{a^2} \text{ or } k=\frac{3}{4}$$

14. The coefficient of x^9 in $(1 + 3x + 3x^2 + x^3)^{15}$ is

(a) ${}^{45}C_9$ (b) ${}^{15}C_9$

(c) 0 (d) none of these

Ans. (a)

Solution: Since $(1+3x+3x^2+x^3)^{15}=[(1+x)^3]^{15}=(1+x)^{45}$, hence, the coefficient of x^9 in $(1 + x)^{45}$ is ${}^{45}C_9$.

15. The total number of terms in the expansion of $(x + a)^{100} + (x - a)^{100}$ after simplification will be

(a) 202 (b) 51

(c) 50 (d) none of these

Ans. (b)

Solution: In $(x + a)^{100} + (x - a)^{100}$, n is even.

$\therefore$ Total number of terms = $\frac{n}{2}+1=\frac{100}{2}+1=51$

16. The number of terms in the expansion of $(1 + 3x + 3x^2 + x^3)^6$ is

(a) 8 (b) 9

(c) 19 (d) 24

Ans. (c)

Solution: $(1+3x+3x^2+x^3)^6=[(1+x)^3]^6=(1+x)^{18}$

$\therefore$ Number of terms = 18 + 1 = 19

17. In $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n$ if the ratio of the 7th term from the beginning to the 7th term from the end is $\frac{1}{6}$, then n =

(a) 7 (b) 8

(c) 9 (d) none of these

Ans. (c)

Solution: $$\frac{1}{6}=\frac{{}^nC_6(2^{1/3})^{n-6}(3^{-1/3})^6}{{}^nC_{n-6}(2^{1/3})^6(3^{-1/3})^{n-6}}$$

$$\Rightarrow \quad 6^{-1}=6^{-4}\cdot6^{n/3}=6^{(n/3)-4}$$

$$\Rightarrow \quad \frac{n}{3}-4=-1$$

$$\Rightarrow \quad n=9$$

18. The number of irrational terms in the expansion of $(5^{1/6} + 2^{1/8})^{100}$ is

(a) 96 (b) 97
(c) 98 (d) 99

Ans. (b)

Solution: The general term in the expansion of $(5^{1/6} + 2^{1/8})^{100}$ is given by

$$T_{r+1} = {}^{100}C_r (5^{1/6})^{100-r} (2^{1/8})^r$$

As 5 and 2 are relatively prime, T_{r+1} will be rational if $\frac{100-r}{6}$ and $\frac{r}{8}$ are both integers, i.e. if $100-r$ is a multiple of 6 and r is a multiple of 8. As $0 \le r \le 100$, multiples of 8 up to 100 and the corresponding values of $100 - r$ are

$$r = 0,\ 8,\ 16,\ 24, \ldots, 88,\ 96$$
$$100 - r = 100, 92, 84, 76, \ldots, 12, 4$$

Out of $100 - r$, multiples of 6 are $84, 60, 36, 12$.

$\therefore$ There are four rational terms.

Hence, the number of irrational terms is $101 - 4 = 97$.

19. The number of terms in $(1+x)^{101}(1+x^2-x)^{100}$ is

(a) 302 (b) 301
(c) 202 (d) 101

Ans. (c)

Solution: Given expression

$$= (1+x)^{101}(1+x^2-x)^{100}$$
$$= (1+x)(1+x^3)^{100}$$
$$= (1+x)\left({}^{100}C_0 + {}^{100}C_1 x^3 + {}^{100}C_2 x^6 + \cdots + {}^{100}C_{100} x^{300}\right)$$

$\therefore$ No. of terms = 101 + 101 = 202

20. The fractional part of $\frac{2^{4n}}{15}$ is

(a) $\frac{1}{15}$ (b) $\frac{2}{15}$
(c) $\frac{4}{15}$ (d) none of these

Ans. (a)

Solution: $\frac{2^{4n}}{15} = \frac{(15+1)^n}{15}$

$$= \frac{{}^nC_0 15^n + {}^nC_1 15^{n-1} + \cdots + {}^nC_{n-1}15 + {}^nC_n}{15}$$
$$= \text{integer} + \frac{1}{15}$$

Hence, the fractional part of $\frac{2^{4n}}{15}$ is $\frac{1}{15}$.

EXERCISE SET 1

1. If in the expansion of $(a-2b)^n$ the sum of the 4th and 5th terms is zero, then the value of $\frac{a}{b}$ is

(a) $\frac{n-4}{5}$ (b) $\frac{n-3}{2}$
(c) $\frac{5}{n-4}$ (d) $\frac{5}{2(n-4)}$

2. The number of terms in the expansion of $(1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5)^{20}$ is

(a) 100 (b) 101
(c) 120 (d) none of these

3. The $(r + 1)$th term in the expansion of $(1 - x)^{-4}$ will be

(a) $\frac{x^r}{r!}$ (b) $\frac{(r+1)(r+2)(r+3)}{6}x^r$
(c) $\frac{(r+2)(r+3)}{2}x^r$ (d) none of these

4. If 7 divides $32^{32^{32}}$, the remainder is

(a) 1 (b) 0
(c) 4 (d) 6

5. The value of x, for which the 6th term in the expansion of $\left\{2^{\log_2\sqrt{9^{x-1}+7}} + \frac{1}{2^{(1/5)\log_2(3^{x-1}+1)}}\right\}^7$ is 84, is equal to

(a) 4 (b) 3
(c) 2 (d) 5

6. The number of irrational terms in the expansion of $(\sqrt[8]{5} + \sqrt[6]{2})^{100}$ is

(a) 97 (b) 98
(c) 96 (d) 99

7. If the rth term in the expansion of $(x/3 - 2/x^2)^{10}$ contains x^4, then r is equal to

(a) 2 (b) 3
(c) 4 (d) 5

8. If a and b are the coefficients of x^n and x^{n+1} respectively in the expansion of $(1 + x)^{2n+1}$, then

(a) $a = 2b$ (b) $b = 2a$
(c) $a = b$ (d) $a + b = (2n+1)!$

9. If in the expansion of $\left(2^x + \frac{1}{4^x}\right)^n$, $\frac{T_3}{T_2} = 7$ and the sum of the coefficients of the 2nd and 3rd terms is 36, then the value of x is

(a) $-\frac{1}{3}$ (b) $-\frac{1}{2}$

(c) $\frac{1}{3}$ (d) $\frac{1}{2}$

10. The fractional part of $\frac{5^{2n}}{24}$ is

(a) $\frac{5}{24}$ (b) $\frac{9}{24}$

(c) $\frac{1}{24}$ (d) none of these

11. If the last term of $\left(2^{1/3}-\frac{1}{\sqrt{2}}\right)^n$ is $\left(\frac{1}{3\cdot 9^{1/3}}\right)^{\log_3 8}$, then the 5th term from the beginning is

(a) ${}^{10}C_6$ (b) $\frac{1}{2}{}^{10}C_4$

(c) $2^{10}C_4$ (d) none of these

12. If the 21st and 22nd terms in the expansion of $(1+x)^{44}$ are equal, then x is

(a) $\frac{7}{8}$ (b) $\frac{8}{7}$

(c) $-\frac{7}{8}$ (d) $-\frac{8}{7}$

13. The number of terms in the expansion of $(2x + 3y - 4z)^n$ is

(a) $n+1$ (b) $n+3$

(c) $\frac{(n+1)(n+2)}{2}$ (d) none of these

14. The sum of the coefficients of the terms independent of x in the expansion of $(1+x+2x^3)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is

(a) 1/3 (b) 19/54

(c) 17/54 (d) 1/4

15. The coefficient of x^{10} in the expansion of $(1 + x^2 - x^3)^8$ is

(a) 528 (b) 506

(c) 496 (d) 476

16. If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n$, then the sum of the series ${}^nC_0 + 3{}^nC_1 + 5{}^nC_2 + \cdots + (2n+1){}^nC_n$ will be

(a) $n\cdot 2^n$ (b) $(n+1)2^n$

(c) $n\cdot 2^n + 1$ (d) 0

17. The sum to $(n + 1)$ terms of the series $\frac{{}^nC_0}{2}-\frac{{}^nC_1}{3}+\frac{{}^nC_2}{4}-\frac{{}^nC_3}{5}+\cdots$ is

(a) $\frac{1}{n+1}$ (b) $\frac{1}{n+2}$

(c) $\frac{1}{n(n+1)}$ (d) none of these

18. If the second, third and fourth terms in the expansion of $(x+y)^n$ are 135, 30 and 10/3 respectively, then

(a) $n=7$ (b) $n=5$

(c) $n=6$ (d) none of these

19. If $(1+x)^n=\sum_{r=0}^{n}{}^nC_rx^r$, then

$$\left(1+\frac{{}^nC_1}{{}^nC_0}\right)\left(1+\frac{{}^nC_2}{{}^nC_1}\right)\cdots\left(1+\frac{{}^nC_n}{{}^nC_{n-1}}\right)=$$

(a) $\frac{n^{n-1}}{(n-1)!}$ (b) $\frac{(n+1)^{n-1}}{(n-1)!}$

(c) $\frac{(n+1)^n}{n!}$ (d) $\frac{(n+1)^{n+1}}{n!}$

20. The sum of the series $\sum_{r=0}^{10}{}^{20}C_r$ is

(a) 2^{20} (b) 2^{19}

(c) $2^{19}+\frac{1}{2}{}^{20}C_{10}$ (d) $2^{19}-\frac{1}{2}{}^{20}C_{10}$

21. The coefficient of x^{17} in $(x-1)(x-2)(x-3)...(x-18)$ is

(a) 342 (b) $\frac{171}{2}$

(c) −171 (d) 684

22. $\frac{1}{n!}+\frac{1}{2!(n-2)!}+\frac{1}{4!(n-4)!}+\cdots$ is equal to

(a) $\frac{2^{n-1}}{n!}$ (b) $\frac{2^n}{(n+1)!}$

(c) $\frac{2^n}{n!}$ (d) $\frac{2^{n-2}}{(n-1)!}$

23. The coefficient of x^8y^{10} in $(x+y)^{18}$ is

(a) 2^{18} (b) ${}^{18}P_{10}$

(c) ${}^{18}C_8$ (d) none of these

24. The greatest integer less than or equal to $(\sqrt{2}+1)^6$ is
(a) 197 (b) 198
(c) 196 (d) 199

25. The fifth term from the end in the expansion of $\left(\frac{x^3}{2}-\frac{2}{x^2}\right)^{12}$ is
(a) $\frac{7920}{x^4}$ (b) $-\frac{7920}{x^4}$
(c) $7220x^{-4}$ (d) $-7520x^4$

26. In the expansion of $(1+x^2)^{40}\left(x^2+2+\frac{1}{x^2}\right)^{-5}$, the coefficient of x^{20} is
(a) ${}^{30}C_{10}$ (b) 1
(c) ${}^{30}C_{25}$ (d) none of these

27. If the coefficients of $(2r + 4)$th and $(r - 2)$th terms in $(1 + x)^{18}$ are equal, then r equals
(a) 12 (b) 10
(c) 8 (d) 6

28. The number of terms in the expansion of $[(a + 4b)^3 (a - 4b)^3]^2$ is
(a) 6 (b) 7
(c) 8 (d) 32

29. If $0 \le r \le n$, then the coefficient of x^r in the expansion of $P=1+(1+x)+(1+x)^2+\cdots+(1+x)^n$ is
(a) ${}^nC_r$ (b) ${}^{n+1}C_{r+1}$
(c) ${}^nC_{r+1}$ (d) none of these

30. The term independent of x in $\left[\sqrt{\left(\frac{x}{3}\right)}+\sqrt{\left(\frac{3}{2x^2}\right)}\right]^{10}$ is
(a) 1 (b) ${}^{10}C_1$
(c) $\frac{5}{12}$ (d) none of these

31. If ${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ..., ${}^nC_n$ (n is a multiple of 3) are binomial coefficients, then ${}^nC_0 + {}^nC_3 + {}^nC_6 + {}^nC_9 + \cdots$ is equal to
(a) 0 (b) 3
(c) $\frac{2^n-2}{3}$ (d) $\frac{1}{3}[2^n+(-1)^n 2]$

32. The coefficient of x^{53} in the expansion of $\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ is
(a) ${}^{100}C_{47}$ (b) ${}^{100}C_{53}$
(c) $-{}^{100}C_{53}$ (d) $-{}^{100}C_{100}$

33. In the expansion of $(1+x+x^3+x^4)^4$, the coefficient of x^4 is
(a) ${}^{40}C_4$ (b) ${}^{10}C_4$
(c) 210 (d) 310

34. The value of $\sum_{r=1}^{12} {}^{24}C_r$ is equal to
(a) $2^{24} - {}^{24}C_{12} - 1$ (b) $2^{23}+\frac{1}{2}{}^{24}C_{12}+1$
(c) 2^{24} (d) $2^{23}+\frac{1}{2}{}^{24}C_{12}-1$

35. If $(1+2x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then $a_r =$
(a) $({}^nC_r)^2$ (b) ${}^nC_r\,{}^nC_{r+1}$
(c) ${}^{2n}C_r$ (d) ${}^{2n}C_{r+1}$

36. If $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$, then the value of ${}^nC_1 + 2\,{}^nC_2 + 3\,{}^nC_3 + \cdots + n\,{}^nC_n$ is
(a) $n \cdot 2^{n-1}$ (b) $(n+2)2^n$
(c) $(n+1)2^{n-1}$ (d) $(n+2)\cdot 2^{n-1}$

37. The coefficient of x^{12} in the expansion of $\sum_{r=0}^{50} {}^{50}C_r (x-1)^{50-r} 2^r$ is
(a) ${}^{50}C_{12}$ (b) 1
(c) 0 (d) none of these

38. If $R=(6\sqrt{6}+14)^{2n+1}$ and $f=R-[R]$, where [ ] denotes the greatest integer function, then Rf equals
(a) 20^n (b) 20^{2n}
(c) 20^{2n+1} (d) none of these

39. If $n > 1$, then $(1+x)^n - nx - 1$ is divisible by
(a) x (b) x^2
(c) x^3 (d) x^4

40. If $\sum_{r=1}^{n} {}^{2n+1}C_r = 63$, then the value of n is equal to
(a) 4 (b) 3
(c) 6 (d) 5

41. If $(2r + 3)$th and $(r - 1)$th terms in the expansion of $(1 + x)^{15}$ have equal coefficients, then r =
(a) 3 (b) 4
(c) 5 (d) 6

42. The coefficient of the term independent of x in $\left(2x-\frac{3}{x}\right)^6$ is
(a) 4320 (b) 216
(c) –216 (d) –4320

43. The number of terms in the expansion of $(1 + 2x + x^2)^{20}$ is
(a) 20 (b) 21
(c) 40 (d) 41

44. If the coefficients of the rth and $(r + 1)$th terms in the expansion of $(3 + 7x)^{29}$ are equal, then r equals
(a) 15 (b) 21
(c) 14 (d) none of these

45. If in the expansion of $\left(2a - \frac{a^2}{4}\right)^9$, the sum of middle terms is S, then which of the following is true?
(a) $S = \left(\frac{63}{32}\right)a^{14}(a+8)$ (b) $S = \left(\frac{63}{32}\right)a^{14}(a-8)$
(c) $S = \left(\frac{63}{32}\right)a^{13}(a-8)$ (d) $S = \left(\frac{63}{32}\right)a^{13}(8-a)$

46. The greatest coefficient in the expansion of $(1 + x)^{2n+2}$ is
(a) $\frac{2n!}{(n!)^2}$ (b) $\frac{(2n+2)!}{n!(n+1)!}$
(c) $\frac{(2n)!}{[n(n+1)]!}$ (d) $\frac{(2n+2)!}{[(n+1)!]^2}$

47. The 14th term from the end in the expansion of $(\sqrt{x} - \sqrt{y})^{17}$ is
(a) ${}^{17}C_5 x^6(-\sqrt{y})^5$ (b) ${}^{17}C_6(\sqrt{x})^{11}y^3$
(c) ${}^{17}C_4 x^{13/2}y^2$ (d) none of these

48. If the expansion of $\left(x^2 + \frac{2}{x}\right)^n$ for positive integer n has the 13th term independent of x, then the sum of the divisors of n is
(a) 36 (b) 38
(c) 39 (d) 32

49. If the 6th term in the expansion of $\left(\frac{1}{x^{8/3}} + x^2 \log_{10} x\right)^8$ is 5600, then x equals
(a) 0 (b) $\log_e 10$
(c) 10 (d) x does not exist

50. If the third term in the expansion of $(x + x^{\log_{10} x})^5$ is 10^6, then x may be
(a) 1 (b) 11
(c) $10^{-5/2}$ (d) 10^2

51. If $x = \frac{1}{3}$, then the greatest term in the expansion of $(1 + 4x)^8$ is the
(a) 4th term (b) 5th term
(c) 6th term (d) 3rd term

52. If $[x]$ denotes the greatest integer less than or equal to x, then $(6\sqrt{6} + 14)^{2n+1}$
(a) is an even integer (b) is an odd integer
(c) depends on n (d) none of these

53. If the middle term in the expansion of $\left(\frac{x}{2} + 2\right)^8$ is 1120, $x \in N$, then x is equal to
(a) 2 (b) 3
(c) –2 (d) –3

54. The coefficient of $\frac{1}{y^2}$ in $\left(y + \frac{c^3}{y^2}\right)^{10}$ is
(a) $187c^{12}$ (b) $\frac{210}{c^{12}}$
(c) $\frac{c^{12}}{210}$ (d) $210c^{12}$

55. There are two bags each of which contains h balls. A man has to select an equal number of balls from both the bags. The number of ways in which a man can choose at least one ball from each bag is
(a) ${}^{2n}C_n$ (b) $({}^nC_n)^2$
(c) ${}^{2n}C_1$ (d) ${}^{2n}C_n - 1$

56. If the largest interval to which x belongs so that the greatest term in $(1+x)^{2n}$ has the greatest coefficient is $\left(\frac{10}{11}, \frac{11}{10}\right)$, then n equals
(a) 9 (b) 10
(c) 11 (d) none of these

57. The middle term in the expansion of $(1+\sqrt{x})^{20}$ is
(a) ${}^{20}C_9 x^9$ (b) ${}^{20}C_{10}x^{10}$
(c) ${}^{20}C_{10}x^5$ (d) none of these

58. Let $R = (5\sqrt{5} + 11)^{2n+1}$ and $f = R - [R]$, where [ ] denotes the greatest integer function, then Rf is equal to
(a) 4^{2n+1} (b) 4^{2n}
(c) 4^{2n-1} (d) none of these

59. If the 4th term in the expansion of $\left(ax + \frac{1}{x}\right)^n$ is $\frac{5}{2}$, then the values of a and n are
(a) (1/2), 6 (b) 1, 3
(c) (1/2), 3 (d) cannot be found

60. If $x+y=1$, then $\sum_{r=0}^{n} r^2\,{}^nC_r x^r y^{y-r}$ equals

(a) nxy (b) $nx(x+yn)$

(c) $nx(nx+y)$ (d) none of these

61. The middle term in the expansion of $(1+x)^{2n}$ is

(a) ${}^{2n}C_n x^n$ (b) ${}^{2n}C_{n+1} x^{n+1}$

(c) ${}^{2n}C_{n-1} x^{n-1}$ (d) $\dfrac{1\cdot 3\cdot 5\cdots(2n-1)}{n!}2^n x^n$

62. The term independent of x in the expansion of $[(t^{-1}-1)x+(t^{-1}+1)^{-1}x^{-1}]^8$ is

(a) $56\left(\dfrac{1-t}{1+t}\right)^3$ (b) $56\left(\dfrac{1+t}{1-t}\right)^3$

(c) $70\left(\dfrac{1-t}{1+t}\right)^4$ (d) $70\left(\dfrac{1+t}{1-t}\right)^4$

63. $2^{3n}-7n-1$ is divisible by

(a) 36 (b) 49

(c) 69 (d) none of these

64. The number of terms in the expansion of $(x+y+z)^{10}$ is

(a) 66 (b) 55

(c) 50 (d) none of these

65. If the sum of the coefficients in the expansion of $(\lambda x^2-2x+1)^{37}$ is equal to the sum of the coefficients in the expansion of $(x-\lambda y)^{37}$, then

(a) λ is equal to 0

(b) λ is equal to 1

(c) λ may be any real number

(d) no such value of λ exists

EXERCISE SET 2

1. If the seventh terms from the beginning and from the end in the expansion of $\left(\sqrt[3]{2}+\dfrac{1}{\sqrt[3]{3}}\right)^n$ are equal, then n equals

(a) 9 (b) 12

(c) 15 (d) 18

2. Prove that the value of $2\left(\dfrac{C_1}{C_0}+2\dfrac{C_2}{C_1}+3\dfrac{C_3}{C_2}+\cdots+n\cdot\dfrac{C_n}{C_{n-1}}\right)$ is

(a) $n(n-1)$ (b) $n(n+1)$

(c) n^2-1 (d) n^2+1

3. If the rth term in the expansion of $\left(\dfrac{x}{3}-\dfrac{2}{x^2}\right)^{10}$ contains x^4, then r is equal to

(a) 2 (b) 3

(c) 14 (d) 5

4. In the expansion of $(1+x)^{50}$, the sum of the coefficients of the odd powers of x is

(a) 0 (b) 2^{49}

(c) 2^{50} (d) 2^{51}

5. If the rth term is the middle term in the expansion of $\left(x^2-\dfrac{1}{2x}\right)^{20}$, then the $(r+3)$th term is

(a) ${}^{20}C_{14}\cdot\dfrac{1}{2^{14}}\cdot x$ (b) ${}^{20}C_{12}\cdot\dfrac{1}{2^{12}}\cdot x^2$

(c) $-\dfrac{1}{2^{13}}\cdot{}^{20}C_7\cdot x$ (d) none of these

6. In the expansion of $(1+x)^n(1+y)^n(1+z)^n$, the sum of the coefficients of the terms of degree r is

(a) $({}^nC_r)^3$ (b) $3\cdot{}^nC_r$

(c) ${}^{3n}C_r$ (d) ${}^nC_{3r}$

7. If the coefficient of the middle term in the expansion of $(1+x)^{2n+2}$ is p and the coefficients of the middle terms in the expansion of $(1+x)^{2n+1}$ are θ and r, then

(a) $p+q=r$ (b) $p+r=q$

(c) $p=q+r$ (d) $p+q+r=0$

8. The 7th term in $\left(\dfrac{1}{y}+y^2\right)^{10}$ when expanded in descending power of y is

(a) $210/y^2$ (b) $y^2/210$

(c) $210y^2$ (d) none of these

9. In the expansion of $\left(3x-\dfrac{1}{x^2}\right)^{10}$, the 5th term from the end is

(a) $\dfrac{16486}{x^8}$ (b) $\dfrac{17010}{x^8}$

(c) $\dfrac{13486}{x^8}$ (d) none of these

10. The $(p+2)$th term from the end in $\left(x-\frac{1}{x}\right)^{2n+1}$ is

(a) $(-1)^p \frac{(2n+1)!}{(2n-p)!(p+1)!}x^{2p-2n+1}$

(b) $(-1)^p \frac{(2n+1)!}{(2n-p)!(p+1)!}x^{2n-2p+1}$

(c) $(-1)^p \frac{(2n+1)!}{(2n-p)!(p+1)!}x^{2p-2n-1}$

(d) none of these

11. The number of terms in the expansion of $(x+y+z)^n$ is

(a) $n+1$ (b) $n+3$

(c) $\frac{1}{2}(n+1)(n+2)$ (d) none of these

12. In the expansion of $\left(x^4-\frac{1}{x^3}\right)^{15}$, the coefficient of x^{39} is

(a) 1365 (b) –1365

(c) 455 (d) –455

13. Which term in the expansion of the binomial $\left[\sqrt[3]{\frac{a}{\sqrt{b}}}+\frac{\sqrt{b}}{\sqrt[6]{a}}\right]^{21}$ contains a and b to one and the same power?

(a) 9th (b) 10th

(c) 11th (d) 12th

14. The coefficient of x^6 in the expansion of $(1+x)^{21}+(1+x)^{22}+\cdots+(1+x)^{30}$ is

(a) ${}^{51}C_6$ (b) ${}^{9}C_6$

(c) ${}^{31}C_7-{}^{21}C_7$ (d) ${}^{30}C_6-{}^{20}C_6$

15. If the sum of all the coefficients in the expansion of $(1-3x+10x^2)^n$ is A and if the sum of all the coefficients in the expansion of $(1+x^2)^n$ is B, then

(a) $A=3B$ (b) $A=B^3$

(c) $B=A^3$ (d) none of these

16. The remainder when $(5)^{99}$ is divided by 13 is equal to

(a) 6 (b) 8

(c) 9 (d) 10

17. If $(5+2\sqrt{6})^n=I+f$, where $I\in N$, $n\in N$ and $0\le f<1$ then I equals

(a) $\frac{1}{f}-f$ (b) $\frac{1}{1+f}$

(c) $\frac{1}{1-f}-f$ (d) $\frac{1}{1-f}+f$

18. The largest term in the expansion of $(3+2x)^{50}$ when $x=\frac{1}{5}$ is

(a) 5th (b) 51th

(c) 7th (d) 8th

19. If $a_n=\sum_{r=0}^{n}\frac{1}{{}^nC_r}$, then $\sum_{r=0}^{n}\frac{r}{{}^nC_r}$ is equal to

(a) $(n-1)a_n$ (b) na_n

(c) $\frac{n}{2}a_n$ (d) none of these

20. The coefficients of x^n in the expansion of $(1+x)^{2n}$ and $(1+x)^{2n-1}$ are in the ratio

(a) 1:2 (b) 1:3

(c) 3:1 (d) 2:1

21. If the second term in the expansion of $\left(\sqrt[13]{a}+\frac{a}{\sqrt{a^{-1}}}\right)^n$ is $14a^{5/2}$, then the value of ${}^nC_3/{}^nC_2$ is

(a) 4 (b) 3

(c) 12 (d) 6

22. In the expansion of $\left(x-\frac{1}{y}\right)^{23}$,

(a) $T_{12}=T_{13}$ (b) $xyT_{13}=T_{12}$

(c) $T_{12}=-xyT_{13}$ (d) $T_{12}+T_{13}=25$

23. The number of non-zero terms in the expansion of $(1+3\sqrt{2}x)^9+(1-3\sqrt{2}x)^9$ is

(a) 9 (b) 0

(c) 5 (d) 10

24. If n is an odd natural number, then $\sum_{r=0}^{n}\frac{(-1)^r}{{}^nC_r}$ equals

(a) 0 (b) $\frac{1}{n}$

(c) $\frac{n}{2^n}$ (d) none of these

25. If the coefficients of T_8 and T_9 in $\left(2+\frac{x}{3}\right)^n$ are equal, then n is

(a) 56 (b) 55

(c) 45 (d) 15

26. If the binomial coefficients of $(m+1)$th and $(m+3)$th terms in the expansion of $(1-3x)^{2n}$ are equal, then the total number of terms is

(a) $2m-1$ (b) $2m+1$

(c) $2m+4$ (d) $2m+3$

27. If the coefficients of three consecutive terms in the expansion of $(1 + x)^n$ are in the ratio 1:7:42, then the value on n is
(a) 60 (b) 70
(c) 55 (d) none of these

28. If $(1+x)^n = \sum_{r=0}^{n} C_r x^r$ then the value of $C_1 + 2C_2 + 3C_3 + \cdots + nC_n$ is equal to
(a) $n2^n$ (b) $(n+1)2^n$
(c) $n2^{n-1}$ (d) $n \cdot 2^{n+1}$

29. The number of distinct terms in the expansion of $(x_1 + x_2 + x_3 + \cdots + x_n)^m$ is
(a) ${}^{n+1}C_m$ (b) ${}^{n+m-1}C_m$
(c) ${}^{n+m}C_m$ (d) ${}^{n+m-1}C_{n-1}$

30. The term independent of x in $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is
(a) 5/27 (b) 7/18
(c) 8/27 (d) 1/24

31. If $T_0, T_1, T_2, \ldots, T_n$ represent the terms in the expansion of $(x + a)^n$, then the value of $(T_0 - T_2 + T_4 - T_6 + \cdots)^2 + (T_1 - T_3 + T_5 - T_7 + \cdots)^2$ is equal to
(a) $(x^2 - a^2)^n$ (b) $(x^2 + a^2)^n$
(c) $(x+a)^n - (x-a)^n$ (d) $(x+a)^n + (x-a)^n$

32. If the $(r + 1)$th term in the expansion of
$$\left(\sqrt[3]{\frac{a}{\sqrt{b}}} + \sqrt{\frac{b}{\sqrt[3]{a}}}\right)^{21}$$
contains the same powers of a and b, then the value of r is
(a) 9 (b) 10
(c) 8 (d) 6

33. The sum of $(n + 1)$ terms of the series $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \frac{C_3}{5} + \cdots$ is
(a) $\frac{1}{(n+1)(n+2)}$ (b) $\frac{1}{n+2}$
(c) $\frac{1}{n(n+1)}$ (d) none of these

34. The coefficient of x^n in the expansion of $(1-2x)^{-1/2}$ is
(a) $\frac{2n!}{n!n!}$ (b) $\frac{2n!}{n!2^n}$
(c) $\frac{2n!}{(n!)^2 2^n}$ (d) none of these

35. The coefficient of $a^8b^6c^4$ in the expansion of $(a + b + c)^{18}$ is
(a) ${}^{18}C_{14} \cdot {}^{14}C_8$ (b) ${}^{18}C_{14}$
(c) ${}^{12}C_8$ (d) none of these

36. The coefficient of $x^k (0 \le k \le n)$ in the expansion of $E = 1 + (1 + x) + (1 + x)^2 + \cdots + (1 + x)^n$ is
(a) ${}^{n+1}C_{k+1}$ (b) ${}^{n}C_k$
(c) ${}^{n+1}C_{n-k}$ (d) ${}^{n}C_{n-k+1}$

37. The coefficient of x^8 in the expansion of $(1 + 2x + 3x^2 + 4x^3 + \cdots)^{1/2}$ is
(a) 1 (b) -1
(c) 0 (d) none of these

38. If x is very small in magnitude compared with a such that $\sqrt{\frac{a}{a+x}} + \sqrt{\frac{a}{a-x}} = 2 + k\frac{x^2}{a^2}$, then the value of k is
(a) 1/4 (b) 1/2
(c) 3/4 (d) 1

39. The expression
$$\left(\sqrt{2x^2+1} + \sqrt{2x^2-1}\right)^6 + \left(\frac{2}{\sqrt{2x^2+1} + \sqrt{2x^2-1}}\right)^6$$
is a polynomial of degree
(a) 6 (b) 8
(c) 10 (d) 12

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement 1 is false, Statement 2 is true.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(d) Statement 1 is true, Statement 2 is false.

40. **Statement 1:** $2^{4n} - 2^n(7n + 1)$ is divisible by the square of 14, where n is a natural number greater than 1.
Statement 2: $(1+x)^n = 1 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n, \forall n \in N$

41. **Statement 1:** The greatest coefficient in the expansion of $(1 + 5x)^8$ is ${}^8C_4 5^4$.
Statement 2: The greatest coefficient in the expansion of $(1 + x)^{2n}$ is the coefficient of the middle term.

42. **Statement 1:** $(\sqrt{2}-1)^n$ can be expressed as $\sqrt{N'}=\sqrt{N'-1}$ for $N' \in N$ and $n \in N$.

Statement 2: $(\sqrt{2}-1)^n$ can be expressed as $A+B\sqrt{2}$, where A and B are integers and $n \in N$.

43. **Statement 1:** The number of terms in the expansion of $\left(x+\frac{1}{x}+1\right)^n$ is $2n+1$.

Statement 2: The number of terms in the expansion of $(x_1+x_2+\cdots+x_m)^n$ is ${}^{m+n-1}C_{m-1}$.

Previous Years' Questions

44. If $|x|<1$, then the coefficient of x^n in the expansion of $(1+x+x^2+x^3+\cdots)^2$ is

(a) n (b) $n-1$
(c) $n+2$ (d) $n+1$ **[AIEEE 2002]**

45. The positive integer just greater than $(1+0.0001)^{10^3}$ is

(a) 4 (b) 5
(c) 2 (d) 3 **[AIEEE 2002]**

46. If the sum of the coefficients of various terms in the expansion of $(a+b)^n$ is 4096, then the greatest coefficient in the expansion is

(a) 1594 (b) 792
(c) 924 (d) 2924 **[AIEEE 2002]**

47. The coefficient of x^5 in $(1+2x+3x^2+\cdots)^{-3/2}$ is

(a) 21 (b) 25
(c) 26 (d) none of these **[AIEEE 2002]**

48. If x is positive, the first negative term in the expansion of $(1+x)^{27/5}$ is the

(a) 7th term (b) 5th term
(c) 8th term (d) 6th term **[AIEEE 2003]**

49. The number of integral terms in $(\sqrt{3}+\sqrt[8]{5})^{256}$ is

(a) 33 (b) 34
(c) 35 (d) 32 **[AIEEE 2003]**

50. If ${}^nC_r$ denotes the number of combinations of n things taken r at a time, then the expression ${}^nC_{r+1}+{}^nC_{r-1}+2\times{}^nC_r$ is equal to

(a) ${}^{n+2}C_{r+1}$ (b) ${}^{n+1}C_r$
(c) ${}^{n+1}C_{r+1}$ (d) ${}^{n+2}C_r$ **[AIEEE 2003]**

51. The coefficient of x^n in the expansion of $(1-x)(1-x)^n$ is

(a) $n-1$ (b) $(-1)^n(1+n)$
(c) $(-1)^{n-1}(1-n)^2$ (d) $(-1)^{n-1}n$ **[AIEEE 2004]**

52. If the coefficients of the middle terms in the expansion of $(1+\alpha x)^4$ and $(1-\alpha x)^6$ are equal, then α equals

(a) $-\frac{3}{10}$ (b) $\frac{10}{3}$
(c) $-\frac{5}{3}$ (d) $\frac{3}{5}$ **[AIEEE 2004]**

53. If $\lim\limits_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b are

(a) $b=2,\ a\in\mathbb{R}$ (b) $a=1,\ b\in\mathbb{R}$
(c) $a,\ b\in\mathbb{R}$ (d) $a=1, b=2$ **[AIEEE 2004]**

54. If the coefficients of the rth, $(r+1)$th and $(r+2)$th terms in the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation

(a) $m^2-m(4r-1)+4r^2+2=0$
(b) $m^2-m(4r+1)+4r^2-2=0$
(c) $m^2-m(4r+1)+4r^2+2=0$
(d) $m^2-m(4r-1)+4r^2-2=0$ **[AIEEE 2005]**

55. The value of ${}^{50}C_4+\sum\limits_{r=1}^{6}{}^{56-r}C_3$ is

(a) ${}^{56}C_4$ (b) ${}^{56}C_3$
(c) ${}^{55}C_3$ (d) ${}^{55}C_4$ **[AIEEE 2005]**

56. If x is so small that x^3 and higher powers of x may be neglected, then $\frac{(1+x)^{3/2}-\left[1+(1/2)x\right]^3}{(1-x)^{1/2}}$ may be approximated to

(a) $\frac{x}{2}-\frac{3}{8}x^2$ (b) $-\frac{3}{8}x^2$
(c) $3x+\frac{3}{8}x^2$ (d) $1-\frac{3}{8}x^2$ **[AIEEE 2005]**

57. If the coefficient of x^7 in $\left(ax^2+\frac{1}{bx}\right)^{11}$ equals the coefficient of x^{-7} in $\left(ax-\frac{1}{bx^2}\right)^{11}$, then a and b satisfy the relation

(a) $a+b=1$ (b) $a-b=1$
(c) $ab=1$ (d) $\frac{a}{b}=1$ **[AIEEE 2005]**

58. For natural numbers m, n, if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y_2 + \cdots$, and $a_1 = a_2 = 10$, then (m, n) is
(a) (20, 45) (b) (35, 20)
(c) (45, 35) (d) (35, 45)
[AIEEE 2006]

59. If the expansion in powers of x of the function $\frac{1}{(1-ax)(1-bx)}$ is $a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots$, then a_n is
(a) $\frac{a^n - b^n}{b-a}$ (b) $\frac{a^{n+1} - b^{n+1}}{b-a}$
(c) $\frac{b^{n+1} - a^{n+1}}{b-a}$ (d) $\frac{b^n - a^n}{b-a}$
[AIEEE 2006]

60. The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \cdots + {}^{20}C_{10}$ is
(a) $-{}^{20}C_{10}$ (b) $\frac{1}{2}{}^{20}C_{10}$
(c) 0 (d) ${}^{20}C_{10}$
[AIEEE 2007]

61. In the binomial expansion of $(a-b)^n, n \geq 5$, if the sum of the 5th and 6th terms is zero, then a/b equals
(a) $\frac{n-5}{6}$ (b) $\frac{n-4}{5}$
(c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$ **[AIEEE 2007]**

62. **Statement I:** $\sum_{r=0}^{n}(r+1)\cdot {}^nC_r = (n+2)2^{n-1}$

Statement II: $\sum_{r=0}^{n}(r+1){}^nC_r \cdot x^r = (1+x)^n + nx(1+x)^{n-1}$
(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true; Statement II is false.
(d) Statement I is false; Statement II is true.
[AIEEE 2008]

63. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is
(a) 0 (b) 2
(c) 7 (d) 8 **[AIEEE 2009]**

64. Given positive integers $r > 1$, $n > 2$ and that the coefficients of $(3r)$th and $(r+2)$th terms in the binomial expansion of $(1+x)^{2n}$ are equal. Then
(a) $n = 2r$ (b) $n = 2r+1$
(c) $n = 3r$ (d) none of these
[IIT-JEE 1983]

65. The coefficient of x^4 in $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$ is
(a) $\frac{405}{256}$ (b) $\frac{504}{259}$
(c) $\frac{450}{263}$ (d) none of these
[IIT-JEE 1983]

66. The expression $\left(x + (x^3-1)^{1/2}\right)^5 + \left(x - (x^3-1)^{1/2}\right)^5$ is a polynomial of degree
(a) 5 (b) 6
(c) 7 (d) 8 **[IIT-JEE 1992]**

67. If in the expansion of $(1+x)^m(1-x)^n$, the coefficient of x and x^2 are 3 and –6, respectively, then m is
(a) 6 (b) 9
(c) 12 (d) 24 **[IIT 1999]**

68. The sum $\sum_{i=0}^{m}\binom{10}{i}\binom{20}{m-i}$ (where $\binom{p}{q} = 0$ if $p < q$) is maximum when m is
(a) 5 (b) 10
(c) 15 (d) 20 **[IIT-JEE 2002]**

69. The coefficient of t^{24} in $(1+t^2)^{12}(1+t^{12})(1+t^{24})$ is
(a) ${}^{12}C_6 + 3$ (b) ${}^{12}C_6 + 1$
(c) ${}^{12}C_6$ (d) ${}^{12}C_6 + 2$
[IIT-JEE 2003]

70. If ${}^{n-1}C_r = (k^2-3){}^nC_{r+1}$, then $k \in$
(a) $(-\infty, -2]$ (b) $[2, \infty)$
(c) $[-\sqrt{3}, \sqrt{3}]$ (d) $(\sqrt{3}, 2]$
[IIT-JEE 2004]

71. If $|x| < 1$, then $1 + n\left(\frac{2x}{1+x}\right) + \frac{n(n+1)}{2!}\left(\frac{2x}{1+x}\right)^2 + \cdots$ is equal to
(a) $\left(\frac{2x}{1+x}\right)^n$ (b) $\left(\frac{1+x}{2x}\right)^n$
(c) $\left(\frac{1-x}{1+x}\right)^n$ (d) $\left(\frac{1+x}{1-x}\right)^n$
[Delhi CEE 1992, Karnataka CET 1995]

72. The coefficient of the term independent of x in the expansion of $\left[\frac{(x+1)}{x^{2/3} - x^{1/3} + 1} - \frac{(x-1)}{x - x^{1/2}}\right]^{10}$ is
(a) 210 (b) 105
(c) 70 (d) 112 **[DEC 2009]**

73. $7^9 + 9^7$ is divisible by
(a) 128 (b) 24
(c) 64 (d) 72 **[DEC 2009]**

74. The value of

$$\frac{1}{81^n}-\frac{10}{81^n}{}^{2n}C_1+\frac{10^2}{81^n}{}^{2n}C_2-\frac{10^3}{81^n}{}^{2n}C_3+\cdots+\frac{10^{2n}}{81^n} \text{ is}$$

(a) 2 (b) 0
(c) 1/2 (d) 1 **[UP SEE 2008]**

75. The greatest term in the expansion of $\sqrt{3}\left(1+\frac{1}{\sqrt{3}}\right)^{20}$ is

(a) $\frac{26,840}{9}$ (b) $\frac{24,840}{9}$
(c) $\frac{25,840}{9}$ (d) none of these

[DCE 2008]

76. The sum of the last eight coefficients in the expansion of $(1+x)^{15}$ is

(a) 2^{16} (b) 2^{15}
(c) 2^{14} (d) none of these

[DCE 2007]

77. The coefficient of x^{20} in the expansion of $(1 + 3x + 3x^2 + x^3)^{20}$ is

(a) ${}^{60}C_{40}$ (b) ${}^{30}C_{20}$
(c) ${}^{15}C_2$ (d) none of these

[DCE 2007]

78. The digit in the unit place of the number $183!+3^{183}$ is

(a) 7 (b) 6
(c) 3 (d) 0

[Karnataka CET 2002]

79. If n is a positive integer, then $(\sqrt{3}+1)^{2n}-(\sqrt{3}-1)^{2n}$ is

(a) an irrational number
(b) an odd positive integer
(c) an even positive integer
(d) a rational number other than positive integers

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (b)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (c)	**6.** (a)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (c)
11. (a)	**12.** (c)	**13.** (c)	**14.** (c)	**15.** (d)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (c)	**20.** (d)
21. (c)	**22.** (a)	**23.** (c)	**24.** (a)	**25.** (a)	**26.** (c)	**27.** (d)	**28.** (b)	**29.** (b)	**30.** (d)
31. (d)	**32.** (c)	**33.** (d)	**34.** (d)	**35.** (c)	**36.** (a)	**37.** (a)	**38.** (c)	**39.** (a) and (b)	
40. (b)	**41.** (c)	**42.** (d)	**43.** (d)	**44.** (b)	**45.** (d)	**46.** (d)	**47.** (c)	**48.** (c)	**49.** (c)
50. (c)	**51.** (c)	**52.** (a)	**53.** (a)	**54.** (d)	**55.** (d)	**56.** (b)	**57.** (c)	**58.** (a)	**59.** (a)
60. (c)	**61.** (d)	**62.** (c)	**63.** (b)	**64.** (a)	**65.** (b)				

Exercise Set 2

1. (b)	**2.** (b)	**3.** (b)	**4.** (b)	**5.** (c)	**6.** (c)	**7.** (c)	**8.** (c)	**9.** (b)	**10.** (a)
11. (c)	**12.** (d)	**13.** (b)	**14.** (c)	**15.** (b)	**16.** (b)	**17.** (c)	**18.** (c)	**19.** (c)	**20.** (d)
21. (a)	**22.** (c)	**23.** (c)	**24.** (a)	**25.** (b)	**26.** (d)	**27.** (c)	**28.** (c)	**29.** (b) and (d)	
30. (b)	**31.** (b)	**32.** (a)	**33.** (a)	**34.** (c)	**35.** (a)	**36.** (c)	**37.** (a)	**38.** (c)	**39.** (a)
40. (b)	**41.** (c)	**42.** (c)	**43.** (c)	**44.** (d)	**45.** (c)	**46.** (c)	**47.** (d)	**48.** (c)	**49.** (a)
50. (a)	**51.** (b)	**52.** (a)	**53.** (b)	**54.** (b)	**55.** (a)	**56.** (b)	**57.** (c)	**58.** (d)	**59.** (c)
60. (b)	**61.** (b)	**62.** (a)	**63.** (b)	**64.** (a)	**65.** (a)	**66.** (c)	**67.** (c)	**68.** (c)	**69.** (d)
70. (d)	**71.** (d)	**72.** (a)	**73.** (c)	**74.** (d)	**75.** (c)	**76.** (c)	**77.** (a)	**78.** (a)	**79.** (a)

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Here, $T_4 = {}^nC_3(a)^{n-3}(-2b)^3$

and $T_5 = {}^nC_4(a)^{n-4}(-2b)^4$

Given $T_4 + T_5 = 0$

$\Rightarrow \quad {}^nC_3(a)^{n-3}(-2b)^3 + {}^nC_4(a)^{n-4}(-2b)^4 = 0$

$\Rightarrow \quad (a)^{n-4}(-2b)^3[a^nC_3 + {}^nC_4(-2b)] = 0$

$\Rightarrow \quad \frac{a}{b} = \frac{2{}^nC_4}{{}^nC_3}$

$$= \frac{2\frac{n!}{4!(n-4)!}}{\frac{n!}{3!(n-3)!}} = \frac{2\times 3!(n-3)!}{4!(n-4)!} = \frac{n-3}{2}$$

2. Given expression $=[(1+x)^5]^{20}=(1+x)^{100}$

$\therefore$ Number of terms = 100 + 1 = 101

3. $(1-x)^{-4}=1\cdot x^0+4x+\dfrac{4\cdot 5}{2}x^2+\cdots$

$$=\left[\frac{1\cdot 2\cdot 3}{6}x^0+\frac{2\cdot 3\cdot 4}{6}x+\frac{3\cdot 4\cdot 5}{6}x^2+\frac{4\cdot 5\cdot 6}{6}x^3+\cdots+\frac{(r+1)(r+2)(r+3)}{6}x^r+\cdots\right]$$

Therefore, $T_{r+1}=\dfrac{(r+1)(r+2)(r+3)}{6}x^r$

4. Here, $32=2^5$

$$(32)^{32}=(2^5)^{32}=2^{160}=(3-1)^{160}$$
$$=3m+1;\ m\in Z$$

Now $32^{32^{32}}=32^{3m+1}=2^{5(3m+1)}=2^{15m+5}$
$$=2^{3(5m+1)}\cdot 2^2$$
$$=4(8)^{5m+1}=4(7+1)^{5m+1}$$
$$=4(7n+1);\ n\in Z$$
$$=28n+4$$

$\therefore$ Remainder = 4

5. We have

$$\left\{2^{\log_2\sqrt{9^{x-1}+7}}+\frac{1}{2^{(1/5)\log_2(3^{x-1}+1)}}\right\}^7$$

$$=\left[\sqrt{9^{x-1}+7}+\frac{1}{(3^{x-1}+1)^{1/5}}\right]^7$$

$\therefore$ $T_6={}^7C_5(\sqrt{9^{x-1}+7})^{7-5}\left[\dfrac{1}{(3^{x-1}+1)^{1/5}}\right]^5$

$$={}^7C_5(9^{x-1}+7)\frac{1}{(3^{x-1}+1)}$$

$\Rightarrow$ $84={}^7C_5\dfrac{(9^{x-1}+7)}{(3^{x-1}+1)}$

$\Rightarrow$ $9^{x-1}+7=4(3^{x-1}+1)$

$\Rightarrow$ $\dfrac{3^{2x}}{9}+7=4\left(\dfrac{3^x}{3}+1\right)$

$\Rightarrow$ $3^{2x}-12(3^x)+27=0$

$\Rightarrow$ $y^2-12y+27=0$ (putting $y=3^x$)

$\Rightarrow$ $(y-3)(y-9)=0$

$\Rightarrow$ $y=3,9 \Rightarrow 3^x=3,9 \Rightarrow x=1,2$

6. Here, $t_{r+1}={}^{100}C_r(\sqrt[8]{5})^{100-r}(\sqrt[6]{2})^r$

Since 2 and 5 are coprime,

$\therefore$ t_{r+1} is rational if $100-r$ is a multiple of 8 and r is a multiple of 6. Also $0\le r\le 100$

$\therefore$ $r=0,6,12,...,96$

$\Rightarrow$ $100-r=4,10,16,...,100$ (i)

But $100-r$ is a multiple of 8

$\Rightarrow$ $100-r=0,8,16,24,...,96$ (ii)

Common terms of (i) and (ii) are 16, 40, 64 and 88.

Thus, $r=84,60,36,12$ give the rational terms.

Hence, the number of irrational terms = $101-4=97$

7. $T_r={}^{10}C_{r-1}\left(\dfrac{x}{3}\right)^{11-r}\left(-\dfrac{2}{x^2}\right)^{r-1}$

$\therefore$ $(11-r)-2(r-1)=4$

$\therefore$ $9=3r$

$\therefore$ $r=3$

8. $a={}^{2n+1}C_n=\dfrac{(2n+1)!}{n!(n+1)!}$

$$b={}^{2n+1}C_{n+1}=\frac{(2n+1)!}{(n+1)!n!}$$

$\therefore$ $a=b$

9. Since $\dfrac{T_3}{T_2}=7$, [given]

$\Rightarrow$ $\dfrac{{}^nC_2(2^x)^{n-2}(4^{-x})^2}{{}^nC_1(2^x)^{n-1}.(4^{-x})}=7$

$\Rightarrow$ $\left(\dfrac{n-1}{2}\right)\cdot\dfrac{1}{(2^x)^3}=7$ (i)

Also, ${}^nC_2+{}^nC_1=36$ [given]

$\Rightarrow$ $\dfrac{n(n-1)}{2}+n=36$

$\Rightarrow$ $n^2+n-72=0$

$\Rightarrow$ $n=8,-9$

$n=-9$ is not possible as in Eq. (i), $n-1$ should be positive.

Substituting $n=8$ in Eq. (i), we get

$$2^{3x}=2^{-1}$$

$\Rightarrow$ $3x=-1$

$\Rightarrow$ $x=-\dfrac{1}{3}$

10. We have $\frac{5^{2n}}{24}=\frac{1}{24}(25)^n=\frac{1}{24}(1+24)^n$

$=\frac{1}{24}({}^nC_0+{}^nC_1\cdot 24+{}^nC_2\cdot 24^2+{}^nC_3\cdot 24^3+\cdots+{}^nC_n\cdot 24^n)$

$=\frac{1}{24}+({}^nC_1+{}^nC_2\cdot 24+\cdots+{}^nC_n\cdot 24^{n-1})$

$=\frac{1}{24}+\text{an integer}$

$\therefore$ The fractional part of $\frac{5^{2n}}{24}=\frac{1}{24}$

11. $T_{n+1}={}^nC_n(2^{1/3})^0\left(-\frac{1}{\sqrt{2}}\right)^n=$ Last term

$=(-1)^n\frac{1}{2^{n/2}}=\left(\frac{1}{3^{5/3}}\right)^{3\log_3 2}=(3)^{-5\log_3 2}$

$=3^{\log_3 2^{-5}}=2^{-5}$

$\therefore\ (-1)^n 2^{-n/2}=2^{-5}$

$\therefore\ n=10,(-1)^n=1$

$\therefore\ T_5={}^{10}C_4(2^{1/3})^6\left(-\frac{1}{\sqrt{2}}\right)^4={}^{10}C_4 2^2\cdot\frac{1}{2^2}={}^{10}C_4={}^{10}C_6$

$(\because\ {}^nC_r={}^nC_{n-r})$

12. $T_{21}=T_{20+1}={}^{44}C_{20}x^{20}$

$T_{22}=T_{21+1}={}^{44}C_{21}(-x)^{21}=-{}^{44}C_{21}x^{21}$

$\therefore\ {}^{44}C_{20}x^{20}={}^{44}C_{21}x^{21}$

$\therefore\ x=-\frac{{}^{44}C_{20}}{{}^{44}C_{21}}=-\frac{44!}{24!20!}\times\frac{21!23!}{44!}$

$=-\frac{21\cdot 20!23!}{20!24\cdot 23!}=-\frac{21}{24}=-\frac{7}{8}$

13. Total number of terms in the expansion of $(2x+3y-4z)^n$ is ${}^{n+k-1}C_{k-1}$

where $k=$ number of algebraic terms in the multinomial.

$\therefore\ {}^{n+3-1}C_{3-1}={}^{n+2}C_2=\frac{(n+2)(n+1)}{2}$

14. $(1+x+2x^3)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$

$=(1+x+2x^3)\left[\sum_{r=0}^{9}{}^9C_r\left(\frac{3}{2}x^2\right)^{9-r}\left(-\frac{1}{3x}\right)^r\right]$

$=(1+x+2x^3)\left[\sum_{r=0}^{9}{}^9C_r\left(\frac{3}{2}\right)^{9-r}\left(-\frac{1}{3}\right)^r x^{18-3r}\right]$

$=\left[\sum_{r=0}^{9}{}^9C_r\left(\frac{3}{2}\right)^{9-r}\left(-\frac{1}{3}\right)^r x^{18-3r}\right]$

$+\left[\sum_{r=0}^{9}{}^9C_r\left(\frac{3}{2}\right)^{9-r}\left(-\frac{1}{3}\right)^r x^{19-3r}\right]$

$+2\left[\sum_{r=0}^{9}{}^9C_r\left(\frac{3}{2}\right)^{9-r}\left(-\frac{1}{3}\right)^r x^{21-3r}\right]$

Clearly, the first and third expansions contain terms independent of x and are obtained by the equations $18-3r=0$ and $21-3r=0$ respectively. So, the sum of the coefficients of the terms independent of x

$={}^9C_6\left(\frac{3}{2}\right)^{9-6}\left(-\frac{1}{3}\right)^6+2\left[{}^9C_7\left(\frac{3}{2}\right)^{9-7}-\left(\frac{1}{3}\right)^7\right]$

$=\frac{7}{18}-\frac{2}{27}=\frac{17}{54}$

15. The given expression can be written as $[1+x^2(1-x)]^8$.

$\therefore\ T_{r+1}={}^8C_r[x^2(1-x)]^r={}^8C_r x^{2r}(1-x)^r$

where r varies from 0 to 8.

Now for $r=0$, 1, 3, we shall not get the term of x^{10} as $x^0(1-x)^0$, $x^2(1-x)$, $x^6(1-x)^3$ will not have any item of x^{10}.

For $r=4$ and 5, we will have the terms $x^8(1-x)^4$ and $x^{10}(1-x)^5$ which will contain the term of x^{10}. For the value of $r=6,7,8$, the power of x will be greater than 10. For $r=4$,

$\therefore$ the coefficient of x^{10} in ${}^8C_4x^8(1-x)^4$ is

${}^8C_4\cdot{}^4C_2=(70)(6)=420$

For $r=5$,

$\therefore$ the coefficient of x^{10} in ${}^8C_5(x^2)^5(1-x)^5$ is

${}^8C_5={}^8C_3=\frac{8\cdot 7\cdot 6}{1\cdot 2\cdot 3}=56$

Hence, the total coefficient of x^{10} is $420+56=476$

16. ${}^nC_0+3{}^nC_1+5{}^nC_2+\cdots+(2n+1){}^nC_n$

$=({}^nC_0+{}^nC_1+\cdots+{}^nC_n)+2{}^nC_1+4{}^nC_2+\cdots+2n{}^nC_n$

$=2^n+2({}^nC_1+2{}^nC_2+\cdots+n{}^nC_n)$

$=2^n+2\left(n+2\cdot\frac{n(n-1)}{2!}+3\cdot\frac{n(n-1)(n-2)}{3!}+\cdots+n\cdot 1\right)$

$=2^n+2\left(n+n(n-1)+\frac{n(n-1)(n-2)}{2!}+\cdots+n\right)$

$=2^n+2n\left(1+(n-1)+\frac{(n-1)(n-2)}{2!}+\cdots+1\right)$

$=2^n+2n\left({}^{n-1}C_0+{}^{n-1}C_1+{}^{n-1}C_2+\cdots+{}^{n-1}C_{n-1}\right)$

$=2^n+2n\cdot 2^{n-1}=2^n+n\cdot 2^n=2^n(n+1)$

17. Since $(1+x)^n={}^nC_0+{}^nC_1x+{}^nC_2x^2+\cdots+{}^nC_nx^n$

$\Rightarrow\ {}^nC_0x+{}^nC_1x^2+{}^nC_2x^3+\cdots+{}^nC_nx^{n+1}=x(1+x)^n$

Integrating w.r.t. x, we have

$$\frac{{}^nC_0x^2}{2}+\frac{{}^nC_1x^3}{3}+\frac{{}^nC_2x^4}{4}+\cdots+\frac{{}^nC_nx^{n+2}}{n+2}$$
$$=\int(1+x-1)(1+x)^n\,dx$$

$$\Rightarrow \quad \frac{{}^nC_0x^2}{2}+\frac{{}^nC_1x^3}{3}+\frac{{}^nC_2x^4}{4}+\cdots$$
$$=\int(1+x)^{n+1}\,dx-\int(1+x)^n\,dx$$

$$\Rightarrow \quad \frac{{}^nC_0x^2}{2}+\frac{{}^nC_1x^3}{3}+\frac{{}^nC_2x^4}{4}+\cdots$$
$$=\frac{(1+x)^{n+2}}{n+2}-\frac{(1+x)^{n+1}}{n+1}+C$$

Putting $x = 0$, we have

$$0=\frac{1}{n+2}-\frac{1}{n+1}+C$$
$$0=\frac{(n+1)-(n+2)}{(n+1)(n+2)}+C$$

$$\Rightarrow \quad C=\frac{1}{(n+1)(n+2)}$$

Putting $x = -1$, we have

$$\frac{{}^nC_0}{2}-\frac{{}^nC_1}{3}+\frac{{}^nC_2}{4}+\cdots=0-0+\frac{1}{(n+1)(n+2)}$$

$$\Rightarrow \quad \frac{{}^nC_0}{2}-\frac{{}^nC_1}{3}+\frac{{}^nC_2}{4}+\cdots=\frac{1}{(n+1)(n+3)}$$

18. We have

$$T_2=135\Rightarrow {}^nC_1x^{n-1}y^1=135$$
$$T_3=30\Rightarrow {}^nC_2x^{n-2}y^2=30$$
$$T_4=\frac{10}{3}\Rightarrow {}^nC_3x^{n-3}y^3=\frac{10}{3}$$

Now, $$\frac{T_2\times T_4}{T_3^2}=\frac{135\times\frac{10}{3}}{(30)^2}$$

$$\Rightarrow \quad \frac{({}^nC_1x^{n-1}y^{-1})\times({}^nC_3x^{n-3}y^3)}{({}^nC_2x^{n-2}y^2)^2}=\frac{135\times\frac{10}{3}}{(30)^2}$$

$$\Rightarrow \quad \frac{{}^nC_1\times{}^nC_3}{({}^nC_2)^2}=\frac{135\times10}{2700} \Rightarrow \frac{n[n(n-1)(n-2)/3!]}{[n(n-1)/2!]^2}=\frac{1}{2}$$

$$\Rightarrow \quad \frac{n-2}{n-1}\times\frac{2}{3}=\frac{1}{2} \Rightarrow 4n-8=3n-3 \Rightarrow n=5$$

19. $$\left(1+\frac{{}^nC_1}{{}^nC_0}\right)\left(1+\frac{{}^nC_2}{{}^nC_1}\right)\cdots\left(1+\frac{{}^nC_n}{{}^nC_{n-1}}\right)$$
$$=\left(1+\frac{n}{1}\right)\left(1+\frac{\frac{n(n-1)}{2}}{n}\right)\cdots\left(1+\frac{1}{n}\right)$$
$$=\left(\frac{n+1}{1}\right)\left(\frac{n+1}{2}\right)\cdots\left(\frac{n+1}{n}\right)=\frac{(n+1)^n}{n!}$$

20. Since ${}^{20}C_0+{}^{20}C_1+{}^{20}C_2+{}^{20}C_3+\cdots+{}^{20}C_{10}+{}^{20}C_{11}+\cdots+{}^{20}C_{20}=2^{20}$

$$\Rightarrow \left({}^{20}C_0+{}^{20}C_{20}\right)+\left({}^{20}C_1+{}^{20}C_{19}\right)+\cdots+\left({}^{20}C_9+{}^{20}C_{11}\right)+{}^{20}C_{10}=2^{20}$$

$$\Rightarrow \quad 2\left({}^{20}C_0+{}^{20}C_1+{}^{20}C_2+\cdots+{}^{20}C_9\right)+{}^{20}C_{10}=2^{20}$$

$$\Rightarrow \quad 2\left({}^{20}C_0+{}^{20}C_1+\cdots+{}^{20}C_{10}\right)=2^{20}-{}^{20}C_{10}$$

$$\Rightarrow \quad {}^{20}C_0+{}^{20}C_1+\cdots+{}^{20}C_{10}=2^{19}-\frac{1}{2}{}^{20}C_{10}$$

21. Coefficient of $x^{17}=-(1+2+3+\cdots+18)=\frac{-18}{2}(1+18)$
$=-9\times19=-171$

22. The given expression $=\frac{1}{n!}\left[{}^nC_0+{}^nC_2+{}^nC_4+\cdots\right]=\frac{2^{n-1}}{n!}$

23. $T_{r+1}={}^{18}C_rx^ry^{18-r}$. Compare this with x^8y^{10}.

$\therefore \quad r=8$. Hence, the coefficient is ${}^{18}C_8$.

24. Let $(\sqrt{2}+1)^6=I+F$, where $I\in N$ and $0<F<1$. Clearly, I is the greatest integer less than or equal to $(\sqrt{2}+1)^6$.

Let $(\sqrt{2}-1)^6 = G$. Then

$$I+F+G=(\sqrt{2}+1)^6+(\sqrt{2}-1)^6$$
$$=2[{}^6C_0(\sqrt{2})^6+{}^6C_2(\sqrt{2})^{6-2}+\cdots]$$
$$=\text{an even integer}$$

$\therefore \quad F+G=1$

Now, $I+F+G=2[{}^6C_0(\sqrt{2})^6+{}^6C_2(\sqrt{2})^{6-2}+{}^6C_4(\sqrt{2})^{6-4}+{}^6C_6(\sqrt{2})^{6-6}]$

$$\Rightarrow \quad I+1=2[8+15\times4+15\times2+1]$$
$$\Rightarrow \quad I=197$$

25. The fifth term from the end has 8 terms before it

$$\therefore \quad T_{8+1}={}^{12}C_8\left(\frac{x^3}{2}\right)^4\left(-\frac{2}{x^2}\right)^8=\frac{12\times11\times10\times9}{1\times2\times3\times4}\cdot\frac{2^8}{2^4}\frac{1}{x^4}$$

$$=\frac{12\times11\times10\times9}{24}\times\frac{16}{x^4}$$
$$=\frac{11\times5\times9\times16}{x^4}=\frac{99\times80}{x^4}=\frac{7920}{x^4}$$

26. $E=(1+x^2)^{40}\left(x+\frac{1}{x}\right)^{-10}=x^{10}(1+x^2)^{30}$

Hence, we have to choose the term of x^{20} in $x^{10}(1+x^2)^{30}$ or the term of $(x^2)^5$ in $(1+x^2)^{30}$ which will be ${}^{30}C_5$ or ${}^{30}C_{25}$ as ${}^nC_r={}^nC_{n-r}$

27. ${}^{18}C_{2r+3}={}^{18}C_{r-3}$ (given)

$\Rightarrow \quad 18=2r+3+r-3$

$\Rightarrow \quad 3r=18\Rightarrow r=6$

28. $[(a+4b)^3(a-4b)^3]^2=(a^2-16b^2)^6$

Number of terms = 6 + 1 = 7

29. Coefficient of x^r in $\left[\frac{(1+x)^{n+1}-1}{(1+x)-1}\right]$

Coefficient of x^{r+1} in $(1+x)^{n+1}={}^{n+1}C_{r+1}$

30. The general term is

$$T_{r+1}={}^{10}C_r\left(\sqrt{\frac{x}{3}}\right)^{10-r}\left(\sqrt{\frac{3}{2x^2}}\right)^r$$
$$={}^{10}C_r\left(\frac{1}{3}\right)^{5-(r/2)}\left(\frac{3}{2}\right)^{r/2}\cdot x^{5-(3r/2)}$$

For the term independent of x, $5-\frac{3r}{2}=0\Rightarrow r=\frac{10}{3}$, which is not a positive integer. Hence, there is no term independent of x.

31. $(1+x)^n={}^nC_0+{}^nC_1x+{}^nC_2x^2+\cdots+{}^nC_nx^n$

Putting $x=1, x=\omega$ and $x=\omega^2$, we get

$(1+1)^n={}^nC_0+{}^nC_1+{}^nC_2+\cdots+{}^nC_n$

$(1+\omega)^n={}^nC_0+{}^nC_1\omega+{}^nC_2\omega^2+{}^nC_3\omega^2+\cdots+{}^nC_n\omega^n$

$(1+\omega^2)^n={}^nC_0+{}^nC_1\omega^2+{}^nC_2\omega^4+{}^nC_3\omega^6+\cdots+{}^nC_n\omega^{2n}$

Adding, we get

$2^n+(-\omega^2)^n+(-\omega)^n=3({}^nC_0+{}^nC_3+{}^nC_6+\cdots)$

$\Rightarrow \frac{1}{3}[2^n+(-1)^n(\omega^{2n}+\omega^n)]={}^nC_0+{}^nC_3+{}^nC_6+\cdots$

Since n is a multiple of 3, $\therefore \omega^{2n}=1, \omega^n=1$

$\therefore \quad {}^nC_0+{}^nC_3+{}^nC_6+\cdots=\frac{1}{3}[2^n+(-1)^n2]$

$\therefore$ Option (d) holds.

32. The given sigma is the expansion of $[(x-3)+2)]^{100}=(x-1)^{100}=(1-x)^{100}$

$\therefore \quad x^{53}$ will occur in T_{54}.

$$T_{54}={}^{100}C_{53}(-x)^{53}$$

$\therefore$ Coefficient is $-{}^{100}C_{53}$.

33. $(1+x+x^3+x^4)^{10}$

$=(1+x)^{10}(1+x^3)^{10}$

$=(1+{}^{10}C_1x+{}^{10}C_2x^2+{}^{10}C_3x^3+{}^{10}C_4x^4+\cdots)$
$(1+{}^{10}C_1x^3+\cdots)$

$\therefore$ Coefficient of $x^4={}^{10}C_4+{}^{10}C_1\cdot{}^{10}C_1=210+(10)(10)=310$

34. $\sum_{r=1}^{12}{}^{24}C_r={}^{24}C_1+{}^{24}C_2+\cdots+{}^{24}C_{12}$

$={}^{24}C_0+{}^{24}C_1+{}^{24}C_2+\cdots+{}^{24}C_{12}+{}^{24}C_{13}+{}^{24}C_{14}+\cdots$
$+{}^{24}C_{24}-({}^{24}C_{13}+{}^{24}C_{14}+\cdots+{}^{24}C_{24})-{}^{24}C_0$

$=2^{24}-1-({}^{24}C_{11}+{}^{24}C_{10}+\cdots+{}^{24}C_0)$

$=2^{24}-1-({}^{24}C_0+{}^{24}C_1+\cdots+{}^{24}C_{11}+{}^{24}C_{12})+{}^{24}C_{12}$

$=2^{24}-2+{}^{24}C_{12}-({}^{24}C_1+{}^{24}C_2+\cdots+{}^{24}C_{12})$

$\therefore \quad 2\left(\sum_{r=1}^{24}{}^{24}C_r\right)=2^{24}-2+{}^{24}C_{12}$

$\Rightarrow \quad \sum_{r=1}^{24}{}^{24}C_r=2^{23}-1+\frac{1}{2}{}^{24}C_{12}$

35. L.H.S. $=[(1+x)^2]^n=(1+x)^{2n}$

$\therefore \quad a_r=$ coefficient of $x^r={}^{2n}C_r$

36. ${}^nC_1+2^nC_2+3^nC_3+\cdots+n^nC_n$

$=n+2\frac{n(n-1)}{2!}+3\frac{n(n-1)(n-2)}{3!}+\cdots$

$=n+n(n-1)+\frac{n(n-1)(n-2)}{2!}+\cdots$

$=n({}^{n-1}C_0+{}^{n-1}C_1+{}^{n-1}C_2+\cdots)$

$=n\cdot2^{n-1}$

37. $\sum_{r=0}^{50}{}^{50}C_r(x-1)^{50-r}\cdot2^r=(x-1+2)^{50}=(x+1)^{50}=(1+x)^{50}$

$=1+{}^{50}C_1x+{}^{50}C_2x^2+\cdots+{}^{50}C_{12}x^{12}+\cdots$

Coefficient of $x^{12}={}^{50}C_{12}$

38. Let $G=(6\sqrt{6}-14)^{2n+1}$. Then

$R-G=(6\sqrt{6}+14)^{2n+1}-(6\sqrt{6}-14)^{2n+1}$

$=2[{}^{2n+1}C_1(6\sqrt{6})^{2n}(14)+\cdots]=$ an even integer

$\Rightarrow [R]+f-G=$ an even integer $\Rightarrow f=G$

So, $Rf=RG=(6\sqrt{6}+14)^{2n+1}+(6\sqrt{6}-14)^{2n+1}=20^{2n+1}$

39. For $n > 1$, we have

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \cdots + {}^nC_nx^n$$

$$\Rightarrow (1+x)^n = 1 + nx + ({}^nC_2x^2 + {}^nC_3x^3 + \cdots + {}^nC_nx^n)$$

$$\Rightarrow (1+x)^n - 1 - nx = x^2({}^nC_2 + {}^nC_3x + {}^nC_4x^2 + \cdots + {}^nC_nx^{n-2})$$

Clearly, R.H.S. is divisible by x^2 and x. So, L.H.S. is also divisible by x as well as x^2.

40. ${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \cdots + {}^{2n+1}C_n$

$= 63 + 1 = A = 2^6$ $[\because {}^{2n+1}C_0 = 1]$

${}^{2n+1}C_0 + {}^{2n+1}C_1 + \cdots + {}^{2n+1}C_n$

$= {}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + \cdots + {}^{2n+1}C_{2n+1} = A$

Also, ${}^{2n+1}C_0 + {}^{2n+1}C_2 + \cdots + {}^{2n+1}C_{2n+1} = 2^{2n+1}$

$\Rightarrow 2A = 2^{2n+1}$

$\Rightarrow 2^6 = 2^{2n} \Rightarrow n = 3$

41. ${}^{15}C_{2r+2} = {}^{15}C_{r-2}$

$\therefore 2r + 2 + r - 2 = 15$

$\therefore r = 5$

42. Here, $T_{r+1} = {}^6C_r(2x)^{6-r}\left(\frac{-3}{x}\right)^r$

$= {}^6C_r 2^{6-r}x^{6-r}(-1)^r\frac{3^r}{x^r}$

$= (-1)^r 2^{6-r}3^r\,{}^6C_r x^{6-2r}$

Now, $6 - 2r = 0 \Rightarrow r = 3$

$\therefore$ Constant term $= (-1)^3 2^3 3^3\,{}^6C_3$

$= -(8)(27)(20)$

$= -4320$

43. We have $(1 + 2x + x^2)^{20} = [(1+x)^2]^{20} = (1+x)^{40}$

$\therefore$ Total number of terms in the expansion = 40 + 1 = 41.

44. $\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r}\frac{a}{x}$ for $(x+a)^n = \frac{30-r}{r}\frac{7x}{3}$ for $(3 + 7x)^{29}$, $(\because n = 29)$

Since the coefficients of the rth and $(r + 1)$th terms are equal,

$\therefore \frac{(30-r)}{r}\cdot\frac{7}{3} = 1 \Rightarrow r = 21$

45. Sum of the middle terms

$= {}^9C_4(2a)^5\left(\frac{-a^2}{4}\right)^4 + {}^9C_5(2a)^4\left(\frac{-a^2}{4}\right)^5$

$= \frac{9!}{4!5!}(2a)^4\left(\frac{-a^2}{4}\right)^4\left(2a - \frac{a^2}{4}\right)$

$= \frac{63}{32}a^4 \cdot a^8 \cdot a(8-a) = \frac{63}{32}a^{13}(8-a)$

46. The greatest coefficient in $(1+x)^{2n+2} = {}^{2n+2}C_{n+1}$

$= \frac{(2n+2)!}{(n+1)!(n+1)!}$

$[\because {}^nC_r$ is greatest and $r = (n/2)$ if n is even$]$

$= \frac{(2n+2)!}{[(n+1)!]^2}$

47. T_{14} from the end of $(\sqrt{x} - \sqrt{y})^{17}$ is T_{14} from the beginning of $(-\sqrt{y} + \sqrt{x})^{17}$

$\therefore T_{14} = {}^{17}C_{13}(-\sqrt{y})^4(\sqrt{x})^{13} = {}^{17}C_4x^{13/2}y^2$

48. $T_{13} = {}^nC_{12}(x^2)^{n-12}\left(\frac{2}{x}\right)^{12} = {}^nC_{12}\cdot 2^{12}\cdot x^{2n-36}$

By the question, $2n - 36 = 0 \Rightarrow n = 18$

But $18 = 2 \times 3^2$

$\therefore$ Sum of divisors of $18 = (1+2)(1+3+3^2) = 39$

49. $T_6 = T_{5+1} = {}^8C_5\left(\frac{1}{x^{8/3}}\right)^3(x^2\log_{10}x)^5 = 5600$

$\Rightarrow \frac{8\times7\times6}{1\times2\times3}\cdot\frac{1}{x^8}x^{10}(\log_{10}x)^5 = 5600$

$\Rightarrow x^2(\log_{10}x)^5 = 100$

$\Rightarrow x^2(\log_{10}x)^5 = (10)^2(\log_{10}10)^5 \Rightarrow x = 10$

50. Let us put $\log_{10}x = z$. Given expression is $(x + x^z)^5$.

$T_3 = {}^5C_2x^3(x^z)^2 = 10x^{3+2z} = 10^6$

$\therefore x^{3+2z} = 10^5$. Taking log, we get

$(3 + 2z)\log_{10}x = 5\log_{10}10$ or $(3 + 2z)z = 5$ or

$2z^2 + 3z - 5 = 0$

$\therefore (z-1)(2z+5) = 0$

$\therefore z = 1, -5/2$

$\therefore \log_{10}x = 1$ or $-5/2$

$\therefore x = 10$ or $10^{-5/2}$

51. $(1 + 4x)^8$

Here, $\frac{T_{r+1}}{T_r} \geq 1$

$\Rightarrow \frac{8-r+1}{r}\left(\frac{4}{3}\right) \geq 1$

$\Rightarrow (9-r)4 \geq 3r \Rightarrow 36 \geq 7r$

$\Rightarrow r \leq 5\frac{1}{7}$

$\therefore$ 6th term is the greatest term.

52. Let $(6\sqrt{6}+14)^{2n+1} = I + F$, where $I \in N$ and $0 < F < 1$

Also let $G = (6\sqrt{6}-14)^{2n+1}$

Then $0 < G < 1$

$\therefore \quad (6\sqrt{6}+14)^{2n+1} - (6\sqrt{6}-14)^{2n+1} = I + F - G$

or $2[{}^{2n+1}C_1(6\sqrt{6})^{2n}(14) + {}^{2n+1}C_3(6\sqrt{6})^{2n-2}(14)^3 + \cdots]$

$= I + F - G$

$\Rightarrow \quad I + F - G$ is an even integer.

$\Rightarrow \quad F - G$ is an integer. [$\because$ I is an integer.]

But $0 < F < 1$ and $0 < G < 1$

$\therefore \quad F - G = 0 \Rightarrow F = G \quad \therefore I$ is an even integer.

Hence, $(6\sqrt{6}+14)^{2n+1}$ is an even integer.

53. The middle term is $\left(\frac{n}{2}+1\right)$th, i.e. $(4+1)$th, i.e. T_5

$\therefore \quad T_5 = {}^8C_4\left(\frac{x}{2}\right)^4 \cdot 2^4 = 1120$

$\frac{8\cdot 7\cdot 6\cdot 5}{1\cdot 2\cdot 3\cdot 4}x^4 = 1120$

$\Rightarrow \quad x^4 = \frac{1120}{70} = 16$

$\Rightarrow \quad (x^2+4)(x^2-4) = 0$

$\therefore \quad x = 2$ only as $x \in N$

54. In the expansion of $\left(y + \frac{c^3}{y^2}\right)^{10}$, the rth term is given by

$$T_r = {}^{10}C_{r-1}y^{10-r+1}\left(\frac{c^3}{y^2}\right)^{r-1} = {}^{10}C_{r-1}y^{11-r}\frac{c^{3r-3}}{y^{2r-2}}$$

$$= {}^{10}C_{r-1}y^{11-r-2r+2}c^{3r-3}$$

Thus, $T_r = {}^{10}C_{r-1}y^{13-3r}c^{3r-3}$ (i)

Let the rth term contain y^{-2}.

$\therefore \quad 13 - 3r = -2 \Rightarrow r = \frac{15}{3} = 5$

Therefore, the 5th term will contain y^{-2}.

From (i), $T_5 = {}^{10}C_{5-1}c^{15-3}y^{-2} = {}^{10}C_4c^{12}y^{-2}$

$\therefore$ Coefficient of $y^{-2} = {}^{10}C_4c^{12}$

$$= \frac{10!c^{12}}{4!(10-4)!} = \frac{10!c^{12}}{4!6!} = \frac{10\cdot 9\cdot 8\cdot 7}{4\cdot 3\cdot 2\cdot 1}\cdot c^{12} = 210c^{12}$$

55. The number of ways of selecting r balls from each bag is equal to ${}^nC_r \times {}^nC_r$.

So, the required number of means $= \sum_{r=1}^{n} {}^nC_r \times {}^nC_r$

$= {}^nC_1^2 + {}^nC_2^2 + \cdots + {}^nC_n^2$

$= ({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \cdots + ({}^nC_n)^2 - ({}^nC_0)^2$

$= {}^{2n}C_n - 1 \qquad [\because \ {}^nC_0^2 + {}^nC_1^2 + \cdots + {}^nC_n^2 = {}^{2n}C_n]$

56. Since the greatest term has the greatest coefficient,

$\therefore$ middle term, i.e. $(n+1)$th term, is the greatest.

$\therefore \quad \frac{T_{n+1}}{T_{n+2}} > 1 \quad \Rightarrow \quad \frac{n+1}{n}x > 1$

$\Rightarrow \quad x > \frac{n}{n+1}$ (i)

Also $\frac{T_{n+1}}{T_{n+2}} > 1 \quad \Rightarrow \quad x < \frac{n+1}{n}$ (ii)

From (i) and (ii), we get

$$\frac{n}{n+1} < x < \frac{n+1}{n}$$

$\therefore$ The largest interval is $\left(\frac{n}{n+1}, \frac{n+1}{n}\right)$,

which is given as $\left(\frac{10}{11}, \frac{11}{10}\right)$

$\therefore \quad n = 10$

57. We have $(1 + x^{1/2})^{20}$. It has 21 terms in its expansion. So the middle term is the 11th term, i.e.

$$T_{11} = {}^{20}C_{10}(x^{1/2})^{10} = {}^{20}C_{10}x^5$$

58. Let $G = (5\sqrt{5}-11)^{2n+1}$, where $0 < G < 1$

[$\because \ 0 < 5\sqrt{5} - 11 < 1$]

Now $R - G = (5\sqrt{5}+11)^{2n+1} - (5\sqrt{5}-11)^{2n+1}$

$= 2[{}^{2n+1}C_1(5\sqrt{5})^{2n}(11) + \cdots + {}^{2n+1}C_{2n+1}(11)^{2n+1}]$

$\Rightarrow \quad R - G$ is an even integer

$\Rightarrow \quad [R] \times f - G$ is an even integer [$\because \ R = [R] + f$]

$\Rightarrow \quad f - G$ is an integer

Now $0 < f < 1$ and $0 < G < 1$

$\Rightarrow \quad 0 \le f - G < 1$

$\therefore \quad f - G = 0 \quad \Rightarrow \quad f = G$

$\therefore \quad Rf = RG$

$= (5\sqrt{5}+11)^{2n+1}(5\sqrt{5}-11)^{2n+11}$

$= (125-121)^{2n+1} = (4)^{2n+1}$

59. It is given that the fourth term in the expansion of $\left(ax + \frac{1}{x}\right)^n$ is $\frac{5}{2}$.

$\therefore \quad {}^nC_3(ax)^{n-3}\left(\frac{1}{x}\right)^3 = \frac{5}{2}$

$\Rightarrow \quad {}^nC_3 a^{n-3} x^{n-6} = \dfrac{5}{2}$ (i)

$\Rightarrow \quad n-6=0$ [$\because$ R.H.S. is independent of x]

Putting $n=6$ in (i), we get

$${}^6C_3 a^3 = \frac{5}{2}$$

$$\Rightarrow \quad a^3 = \frac{1}{8}$$

$$\Rightarrow \quad a = \frac{1}{2}$$

Thus, $n=6, a=\dfrac{1}{2}$

60. $\displaystyle\sum_{r=0}^{n} r^2\, {}^nC_r x^r y^{n-r} = \sum_{r=0}^{n} [r(r-1)+r]\, {}^nC_r x^r y^{n-r}$

$$= \sum_{r=0}^{n} r(r-1)\, {}^nC_r x^r y^{n-r} + \sum_{r=0}^{n} r\, {}^nC_r x^r y^{n-r}$$

$$= \sum_{r=0}^{n-2} r(r-1)\frac{n}{r}\frac{n-1}{r-1}\, {}^{n-2}C_{r-2} \cdot x^2 x^r - 2y^n - r +$$

$$\sum_{r=1}^{n-1} r \cdot \frac{n}{r}\, {}^{n-1}C_{r-1} x.x^{r-1} y^{n-r}$$

$$= n(n-1)x^2 \sum_{r=1}^{n} -2\, {}^{n-2}C_{r-2} \cdot x^{r-2} y^{(n-2)-(r-2)}$$

$$+nx \sum_{r=2}^{n-1} {}^{n-1}C_{r-1} x^{r-1} y^{(n-1)(r-1)}$$

$$= n(n-1)x^2 (x+y)^{n-2} + nx(x+y)^{n-1}$$

$$= n(n-1)x^2 + nx \qquad [\because\ x+y=1]$$

$$= nx[(n-1)x+1] = nx[nx-x+1]$$

$$= nx(nx+y) \qquad [\because\ x+y=1,\ \therefore\ 1-x=y]$$

61. There will be $2n+1$ terms (odd) and hence the middle term will be $(2n/2)+1$, i.e. $(n+1)$th

$$T_{n+1} = {}^{2n}C_n x^n = \frac{(2n)!}{n!n!} x^n$$

$$= \frac{1\cdot 2\cdot 3\cdot 4\cdots(2n-1)(2n)}{n!n!} x^n$$

$$= \frac{[1\cdot 3\cdot 5\cdots(2n-1)]\,[2\cdot 4\cdot 6\cdots 2n]}{n!n!} x^n$$

$$= \frac{[1\cdot 3\cdot 5\cdots(2n-1)]\cdot 2^n n!}{n!n!} x^n$$

$$= \frac{1\cdot 3\cdot 5\cdots(2n-1)}{n!} \cdot 2^n x^n$$

62. Here, $T_{r+1} = {}^8C_r [(t^{-1}-1)x]^{8-r} [(t^{-1}+1)^{-1} x^{-1}]^r$

$$= {}^8C_r (t^{-1}-1)^{8-r} (t^{-1}+1)^{-r} x^{8-2r}$$

Now, $8-2r=0 \ \Rightarrow\ 2r=8 \ \Rightarrow\ r=4$

$\therefore \quad T_5 = {}^8C_4 (t^{-1}-1)^4 (t^{-1}+1)^{-4}$

$$= 70\left(\frac{1}{t}-1\right)^4 \left(\frac{1}{t}+1\right)^{-4}$$

$$= 70(1-t)^4 (1+t)^{-4} = 70\left(\frac{1-t}{1+t}\right)^4$$

63. $2^{3n} = 8^n = (1+7)^n$

Now $E = (1+7)^n - 7n - 1 = {}^nC_2 7^2 + {}^nC_3 7^3 + \cdots$

$\therefore \quad E = 7^2$ (integer). Hence, it is divisible by 49.

64. The number of terms in the expansion of $(x+y+z)^{10}$

$= {}^{10+2}C_2 = {}^{12}C_2 = \dfrac{12\times 11}{1\times 2} = 6\times 11 = 66$

65. Sum of the coefficients is obtained by putting $x = 1$ in L.H.S. or $x = 1 = y$ in R.H.S.

If they are equal, then $(\lambda-1)^{37} = (1-\lambda)^{37} = -(\lambda-1)^{37}$

or $2(\lambda-1)^{37} = 0 \Rightarrow \lambda - 1 = 0 \ \Rightarrow\ \lambda = 1$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. Here, the 7th term from the end is the $(n-5)$th term from the beginning.

$\therefore \quad T_7 = T_{n-5}$

$$\Rightarrow \quad {}^nC_6 (2^{1/3})^{n-6} \left(\frac{1}{3^{1/3}}\right)^6 = {}^nC_{n-6} (2^{1/3})^6 \left(\frac{1}{3^{1/3}}\right)^{n-6}$$

$$\Rightarrow \quad (2^{1/3})^{n-12} = \left(\frac{1}{3^{1/3}}\right)^{n-12}$$

$$\Rightarrow \quad n-12=0 \ \Rightarrow\ n=12$$

2. We know that $\dfrac{{}^nC_{r+1}}{{}^nC_r} = \dfrac{n-r}{r+1}$

Let $E = 2\left(\dfrac{C_1}{C_0} + 2\dfrac{C_2}{C_1} + 3\dfrac{C_3}{C_2} + \cdots + n\dfrac{C_n}{C_{n-1}}\right)$

$$E = 2\left(\frac{n}{1} + 2\cdot\frac{n-1}{2} + 3\frac{n-2}{3} + \cdots + n\frac{n-(n-1)}{n}\right)$$

$$\Rightarrow \quad E = 2(n + (n-1) + (n-2) + \cdots + 1)$$

$$\Rightarrow \quad E = 2\Sigma n = \frac{2n(n+1)}{2}$$

$$\therefore \quad E = n(n+1)$$

3. $T_r = {}^{10}C_{r-1} \left(\dfrac{x}{3}\right)^{11-r} \left(-\dfrac{2}{x^2}\right)^{r-1}$

$\therefore \quad 11-r-2r+2=4$

$\Rightarrow \quad 3r=9 \qquad \therefore \quad r=3$

4. $(1+x)^{50} = C_0 + C_1x + C_2x^2 + C_3x^3 + \cdots + C_{50}x^{50}$

Putting $x = 1$ and -1 and subtracting, we get

$$2^{50} = 2(C_1 + C_3 + C_5 + \cdots)$$

$$\therefore \quad C_1 + C_3 + C_5 + \cdots = 2^{49}$$

or the sum of the coefficients of the odd powers of x is 2^{49}.

5. Here, the 11th term is the middle term.

$$T_{r+3} = T_{14} = {}^{20}C_{13}(x^2)^7\left(-\frac{1}{2x}\right)^{13}$$

$$= -\frac{1}{2^{13}}{}^{20}C_7 x$$

6. The given expansion can be written as

$$\underset{n\text{-factors}}{[(1+x)(1+x)(1+x)\cdots(1+x)]}$$

$$\times \underset{n\text{-factors}}{[(1+y)(1+y)(1+y)\cdots(1+y)]}$$

$$\times \underset{n\text{-factors}}{[(1+z)(1+z)(1+z)\cdots(1+z)]}$$

There are $3n$ factors in this product.

To get a term of degree r, we choose r brackets out of these $3n$ brackets and then multiply the second term in each bracket. There are ${}^{3n}C_r$ such terms each having coefficient 1.

Hence, the sum of the coefficients $= {}^{3n}C_r$.

7. Since the $(n + 2)$th term is the middle term in the expansion of $(1 + x)^{2n+2}$, we have

$$p = {}^{2n+2}C_{n+1}$$

Since the $(n + 1)$th and $(n + 2)$th terms are the middle terms in the expansion of $(1+x)^{2n+1}$, we have

$$q = {}^{2n+1}C_n \text{ and } r = {}^{2n+1}C_{n+1}$$

But $\quad {}^{2n+1}C_n + {}^{2n+1}C_{n+1} = {}^{2n+2}C_{n+1} \quad \therefore\ q + r = p$

8. When $\left(\frac{1}{y} + y^2\right)^{10}$ is expanded, the powers of y go on increasing as the terms proceed. Hence, it is expanded in ascending powers of y. So, $\left(y^2 + \frac{1}{y}\right)^{10}$, when expanded, will be in descending powers of y.

Hence, $T_7 = {}^{10}C_6(y^2)^4\left(\frac{1}{y}\right)^6 = \frac{10\times9\times8\times7}{4\times3\times2\times1}y^2 = 210y^2$

9. There are 11 terms in the expansion of $[3x - (1/x^2)]^{10}$.

$\therefore$ The 5th term from the end = (10 – 5 + 2)th term from the beginning

$$= T_7 = {}^{10}C_6(3x)^{10-6}\cdot\left(-\frac{1}{x^2}\right)^6$$

$$= \frac{10!}{6!4!}\cdot 3^4x^4\cdot\frac{1}{x^{12}} = \frac{17010}{x^8}$$

10. The $(p + 2)$th term from the end

= the $[(2n + 2) - (p + 2 - 1)]$th term from the beginning

= the $(2n - p + 1)$th term from the beginning

$$= {}^{2n+1}C_{2n-p}x^{(2n+1)-(2n-p)}\left(-\frac{1}{x}\right)^{2n-p}$$

$$= (-1)^p\frac{(2n+1)!}{(2n-p)!(p+1)!}x^{2p-2n+1}$$

11. Number of terms in the expansion of $(x + y + z)^n$ where n is a +ve integer

$$(x+y+z)^n = x^n + {}^nC_1x^{n-1}(y+z) + {}^nC_2x^{n-2}(y+z)^2 + \cdots + {}^nC_n(y+z)^n$$

Now we know that the number of terms in the above expansion is $1 + 2 + 3 + 4 + \cdots + (n + 1) = \frac{(n+1)(n+2)}{2}$

12. The general term in $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is

$$T_{r+1} = {}^{15}C_r(x^4)^{15-r}\left(-\frac{1}{x^3}\right)^r = {}^{15}C_rx^{60-7r}(-1)^r$$

If x^{39} occurs in T_{r+1}, then $60 - 7r = 39$

$\Rightarrow\ r = 3$

So, the coefficient of $x^{39} = {}^{15}C_3(-1)^3 = -455$

13. $T_{r+1} = {}^{21}C_r\left[\sqrt[3]{\frac{a}{\sqrt{b}}}\right]^{21-r}\left(\frac{\sqrt{b}}{\sqrt[6]{a}}\right)^r$

$$= {}^{21}C_r\frac{a^{7-\frac{r}{3}}}{b^{\frac{21-r}{6}}}\cdot\frac{b^{r/2}}{a^{r/6}}$$

$$= {}^{21}C_ra^{\frac{42-3r}{6}}b^{\frac{4r-21}{6}}$$

By the given condition, $42 - 3r = 4r - 21$ or $7r = 63$

$\therefore\ r = 9$

Hence, the required term $= T_{10}$

14. The given sequence is a G.P. of 10 terms whose common ratio is $(1 + x)$.

$$\therefore \quad S = (1+x)^{21}\frac{[(1+x)^{10}]-1}{1+x-1}$$

$$\therefore \quad S = \frac{1}{x}[(1+x)^{31} - (1+x)^{21}]$$

$\therefore$ The coefficient of x^6 in S is the coefficient of x^7 in the numerator $= {}^{31}C_7 - {}^{21}C_7 \Rightarrow$ Option (c)

15. Putting $x = 1$ in $(1 - 3x + 10x^2)^n$ and in $(1 + x^2)^n$, the required sum of coefficients

$A = (1 - 3 + 10)^n = 8^n$ and $B = (1 + 1)^n = 2^n$

$\therefore \quad B^3 = 8^n = A$

16. $5^{99} = 5^3 \cdot 5^{96} = 125 \cdot (625)^{24}$

$= (13 \times 9 + 8)(1 + 48 \times 13)^{24}$

$= (13 \times 9 + 8)[1 + {}^{24}C_1(48 \times 13) + {}^{24}C_2(48 \times 13)^2 + \cdots + {}^{24}C_{24}(48 \times 13)^{24}]$

$= (8)$ + terms including powers of 13.

Hence, remainder = 8. Answer option (b).

17. Let $G = (5 - 2\sqrt{6})^n$. Then

$I + f + G = (5 + 2\sqrt{6})^n + (5 - 2\sqrt{6})^n$

$= 2[{}^nC_0 5^n (2\sqrt{6})^0 + {}^nC_2 5^{n-2}(2\sqrt{6})^2 + \cdots]$

= an even integer

$\Rightarrow \quad f + G = 1$

Now, $I + f = (5 + 2\sqrt{6})^n$

$\Rightarrow \quad I = (5 + 2\sqrt{6})^n - f \quad \Rightarrow \quad I = \dfrac{1}{(5 - 2\sqrt{6})^n} - f$

$\Rightarrow \quad I = \dfrac{1}{1 - f} - f \qquad \left[\because \ (5 - 2\sqrt{6})^n = G \text{ and } f + G = 1 \right.$
$\left. \therefore \ (5 - 2\sqrt{6})^n = 1 - f \right]$

18. Now $(3 + 2x)^{50} = 3^{50}\left(1 + \dfrac{2x}{3}\right)^{50}$

$\Rightarrow \quad \dfrac{(n+1)|x|}{|x|+1} = \dfrac{(50+1)(2x/3)}{(2x/3)+1}$

$\Rightarrow \quad \dfrac{(n+1)|x|}{|x|+1} = \dfrac{51 \times (2/15)}{(2/15)+1} \qquad \left[\because \ x = \dfrac{1}{5}\right]$

$\Rightarrow \quad \dfrac{(n+1)|x|}{|x|+1} = \dfrac{102}{17} = 6$

$\therefore$ The (6 + 1)th term, i.e. 7th term is numerically the greatest term.

19. $a_n = \sum_{r=0}^{n} \dfrac{1}{{}^nC_r}$

Let $b = \sum_{r=0}^{n} \dfrac{r}{{}^nC_r} = \sum_{r=0}^{n} \dfrac{n-r}{{}^nC_{n-r}}$ (reverse order)

$= \sum_{r=0}^{n} \dfrac{n-r}{{}^nC_r} \qquad [\because \ {}^nC_r = {}^nC_{n-r}]$

$= n\sum_{r=0}^{n} \dfrac{1}{{}^nC_r} - \sum_{r=0}^{n} \dfrac{r}{{}^nC_r}$

or $b = na_n - b$

$\therefore \quad 2b = na_n$. Hence, $b = \dfrac{na_n}{2}$

20. x^n will occur in T_{n+1}.

$\therefore \quad {}^{2n}C_n \cdot {}^{2n-1}C_n$

$= \dfrac{(2n)!}{n!n!} \cdot \dfrac{n!(n-1)!}{(2n-1)!} = \dfrac{2n}{n} = \dfrac{2}{1}$

21. We have $T_2 = 14a^{5/2}$

$\Rightarrow \quad {}^nC_1 (a^{1/13})^{n-1} (a^{3/2})^1 = 14a^{5/2}$

$\Rightarrow \quad na^{\frac{n-1}{13}} = 14a$

$\Rightarrow \quad n = 14$

$\Rightarrow \quad \dfrac{{}^nC_3}{{}^nC_2} = \dfrac{{}^{14}C_3}{{}^{14}C_2} = \dfrac{n-r+1}{r} = \dfrac{12}{3} = 4$

$(\because \quad n = 14; \ r = 3)$

22. $T_{12} = {}^{23}C_{11} x^{12}\left(-\dfrac{1}{y}\right)^{11}, T_{13} = {}^{23}C_{12} x^{11}\left(-\dfrac{1}{y}\right)^{12}$

$\therefore \quad -xyT_{13} = {}^{23}C_{12} x^{12}\left(-\dfrac{1}{y}\right)^{11} = {}^{23}C_{11} x^{12}\left(-\dfrac{1}{y}\right)^{11}$

$= T_{12}$

Then $T_{12} = -xyT_{13}$

23. The given expression is

$2[{}^9C_0 + {}^9C_2(3\sqrt{2}x)^2 + {}^9C_4(3\sqrt{2}x)^4 + {}^9C_6(3\sqrt{2}x)^6 + {}^9C_8(3\sqrt{2}x)^8]$

$\therefore$ The number of non-zero terms is 5.

24. $\sum_{r=0}^{n} \dfrac{(-1)^r}{{}^nC_r} = \sum_{r=0}^{\frac{n+1}{2}} \left[\dfrac{(-1)^r}{{}^nC_r} + \dfrac{(-1)^{n-r}}{{}^nC_{n-r}}\right]$

$= \sum_{r=0}^{\frac{n+1}{2}} (-1)^r \left[\dfrac{1}{{}^nC_r} + \dfrac{(-1)^n}{{}^nC_{n-r}}\right]$

$= \sum_{r=0}^{\frac{n+1}{2}} (-1)^r \left[\dfrac{1}{{}^nC_r} - \dfrac{1}{{}^nC_r}\right] \qquad \left[\because n \text{ is odd and } {}^nC_r = {}^nC_{n-r}\right]$

$= 0$

25. Since the coefficients of T_8 and T_9 are equal, therefore,

${}^nC_7 2^{n-7}\left(\dfrac{1}{3}\right)^7 = {}^nC_8 2^{n-8}\left(\dfrac{1}{3}\right)^8$

$\Rightarrow \quad \dfrac{n!}{7!(n-7)!} \cdot 2 = \dfrac{n!}{8!(n-8)!} \cdot \dfrac{1}{3}$

$\Rightarrow \quad \dfrac{2}{n-7} = \dfrac{1}{8 \cdot 3}$

$\Rightarrow \quad 48 = n - 7$

$\Rightarrow \quad n = 55$

26. $T_{m+1} = {}^{2n}C_m(-3x)^m$, $T_{m+3} = {}^{2n}C_{m+2}(-3x)^{m+2}$

By the given condition,

$$^{2n}C_m = {}^{2n}C_{m+2}$$

$$\Rightarrow \quad m+m+2=2n \quad \Rightarrow \quad 2m+2=2n$$

Total number of terms $= 2n+1 = 2m+2+1$

$$= 2m+3$$

27. Let $(r+1), (r+2)$ and $(r+3)$ be the three consecutive terms in $(1+x)^n$.

$$\therefore \quad T_{r+1} : T_{r+2} : T_{r+3} = 1:7:42$$

$$\frac{T_{r+1}}{T_{r+2}} = \frac{1}{7} \qquad \frac{T_{r+2}}{T_{r+3}} = \frac{7}{42} = \frac{1}{6}$$

$$\Rightarrow \quad \frac{{}^nC_r}{{}^nC_{r+1}} = \frac{1}{7} \qquad \Rightarrow \quad \frac{{}^nC_{r+1}}{{}^nC_{r+2}} = \frac{1}{6}$$

$$\Rightarrow \quad \frac{{}^nC_{r+1}}{{}^nC_r} = 7 \qquad \Rightarrow \quad \frac{{}^nC_{r+2}}{{}^nC_{r+1}} = 6$$

$$\Rightarrow \quad \frac{n-r}{r+1} = 7 \qquad \Rightarrow \quad \frac{n-(r+1)}{r+2} = 6$$

$$\Rightarrow \quad n-r = 7r+7 \qquad \Rightarrow \quad n-r-1 = 6r+12$$

$$\Rightarrow \quad n-8r = 7 \qquad \Rightarrow \quad n-7r = 13$$

Solving the above two equations, we have $n = 55$.

28. $(1+x)^n = \sum_{r=0}^{n} C_r x^r$

$$\Rightarrow \quad (1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \cdots + C_nx^n \quad \text{(i)}$$

Differentiating (i) with respect to x, we get

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \cdots + nC_nx^{n-1}$$

Putting $x = 1$,

$$\Rightarrow \quad n2^{n-1} = C_1 + 2C_2 + 3C_3 + \cdots + nC_n$$

29. According to Multinomial Theorem, we have

$$(x_1 + x_2 + \cdots + x_n)^m = \sum \frac{m!}{m_1!m_2!\cdots m_n!} \cdot x_1^{m_1} x_2^{m_2} \cdots x_n^{m_n}$$

where $m_1 + m_2 + \cdots + m_n = m$

Thus, the number of distinct terms in the expansion of $(x_1 + x_2 + \cdots + x_n)^m$ equals the number of non-negative integral solutions of $m_1 + m_2 + \cdots + m_n = m$

$\Rightarrow N =$ coefficient of t^m in $(1+t+t^2+\cdots+t^m)^n$

$\Rightarrow N =$ coefficient of t^m in $(1-t^{m+1})^n(1-t)^{-n}$

$\Rightarrow N =$ coefficient of t^m in $(1-t)^{-n}$

$$\therefore \quad N = {}^{m+n-1}C_m = {}^{m+n-1}C_{n-1} \qquad (\because \quad {}^nC_r = {}^nC_{n-r})$$

30. The general term in $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is

$$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r$$

The term containing x^0 is when $2(9-r) - r = 0$

$$\Rightarrow \quad r = 6$$

$\therefore$ T_7 is the term independent of x

$$\Rightarrow \quad T_7 = {}^9C_6 \left(\frac{3}{2}x^2\right)^3 \left(-\frac{1}{3x}\right)^6$$

$$\therefore \quad T_7 = \frac{9\cdot8\cdot7}{1\cdot2\cdot3} \cdot \frac{3^3}{2^3} \cdot \frac{1}{3^6} = \frac{7}{18}$$

31. $(x+a)^n = {}^nC_0x^na^0 + {}^nC_1x^{n-1}a + {}^nC_2x^{n-2}a^2 + \cdots + C_nx^0a^n \quad \text{(i)}$

$$= ({}^nC_0x^n + {}^nC_2x^{n-2}a^2 + {}^nC_4x^{n-4}a^4 + \cdots) + ({}^nC_1x^{n-1}a + {}^nC_3x^{n-3}a^3 + \cdots)$$

$$= (T_0 + T_2 + T_4 + T_6 + \cdots) + (T_1 + T_3 + T_5 + \cdots)$$

Putting $x = ai$ and $-ai$ respectively in (i), we get

$$(x+ai)^n = (T_0 + T_2i^2 + T_4i^4 + \cdots) + (T_1i + T_3i^3 + T_5i^5 + \cdots)$$

$$\therefore \quad (x+ai)^n = (T_0 - T_2 + T_4 - T_6 + \cdots) + i(T_1 - T_3 + T_5 + \cdots) \quad \text{(ii)}$$

and $(x-ai)^n = (T_0 - T_2 + T_4 - T_6 + \cdots) - i(T_1 - T_3 + T_5 + \cdots) \quad \text{(iii)}$

Multiplying (ii) and (iii), we get

$$(x^2+a^2)^n = (T_0 - T_2 + T_4 - T_6 + \cdots)^2 + (T_1 - T_3 + T_5 - \cdots)^2$$

32. The general term in the expansion of the given binomial is

$$T_{r+1} = {}^{21}C_r \left(\sqrt[3]{\sqrt{a}}\right)^{21-r} \left(\sqrt{\frac{b}{\sqrt[3]{a}}}\right)^r$$

$$\Rightarrow \quad T_{r+1} = {}^{21}C_r a^{7-\frac{r}{2}} b^{\frac{2}{3}r-\frac{7}{2}}$$

Since the powers of a and b are the same,

$$\therefore \quad 7 - \frac{r}{2} = \frac{2}{3}r - \frac{7}{2}$$

$$\Rightarrow \quad r = 9$$

33. We have $(1-x)^n = C_0 - C_1x + C_2x^2 - C_3x^3 + \cdots$

$$\Rightarrow \quad x(1-x)^n = C_0x - C_1x^2 + C_2x^3 - C_3x^4 + \cdots$$

$$\Rightarrow \quad \int_0^1 x(1-x)^n\,dx = \int_0^1 (C_0x - C_1x^2 + C_2x^3 - C_3x^4 + \cdots)\,dx$$

Now, L.H.S. $= \int_0^1 x(1-x)^n\,dx$

$$= \int_1^0 (1-t)t^n(-1)dt$$

$$= \int_0^1 (t^n - t^{n+1})dt$$

$$= \left| \frac{t^{n+1}}{n+1} - \frac{t^{n+2}}{n+2} \right|_0^1$$

$$= \frac{1}{n+1} - \frac{1}{n+2} = \frac{1}{(n+1)(n+2)}$$

and R.H.S. $= \left(\frac{C_0 x^2}{2} - \frac{C_1 x^3}{3} + \frac{C_2 x^4}{4} - \cdots \right)_0^1$

$$= \frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \cdots$$

Thus, $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \cdots = \frac{1}{(n+1)(n+2)}$

34. Let x^n occur in the $(r+1)$th term in the expansion of $(1-2x)^{-1/2}$. Let T_{r+1} be the $(r+1)$th term in the given expansion.

$$\therefore T_{r+1} = \frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)\cdots\left(-\frac{1}{2}-(r-1)\right)}{r!}(-2x)^r$$

$$= \frac{(-1)^r\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\cdots\left(\frac{2r-1}{2}\right)}{r!}(-1)^r 2^r x^r$$

$$= \frac{(-1)^{2r}1\cdot 3\cdot 5\cdots(2r-1)2^r}{2^r r!}x^r = \frac{1\cdot 3\cdot 5\cdots(2n-1)}{r!} \quad \text{(i)}$$

For the coefficient of x^n, putting $r = n$ in (i), we have

coefficient of $x^n = \frac{1\cdot 3\cdot 5\cdots(2n-1)}{n!}$

$$= \frac{1\cdot 2\cdot 3\cdot 4\cdot 5\cdot 6\cdots(2n-1)(2n)}{2\cdot 4\cdot 6\cdots 2n\, n!} = \frac{2n!}{2^n n! n!} = \frac{2n!}{2^n (n!)^2}$$

35. In $(a+b+c)^{18}$, the general term is given by

$$\frac{(18)!}{(r_1)!(r_2)!(r_3)!}(a)^{r_1}(b)^{r_2}(c)^{r_3} \quad \text{(i)}$$

$\therefore$ The coefficient of $a^8 b^6 c^4$ is obtained by putting $r_1 = 8,\ r_2 = 6,\ r_3 = 4$ in (i) which gives the general term as

$$\frac{(18)!}{8!\,6!\,4!}a^8 b^6 c^4$$

Thus, the required coefficient $= \frac{(18)!}{8!\,6!\,4!} \times \frac{14!}{14!}$

$$= {}^{18}C_{14} \times {}^{14}C_8$$

36. $E = 1 + (1+x) + (1+x)^2 + \cdots + (1+x)^n$

$$= \frac{(1+x)^{n+1} - 1}{1+x-1} = \frac{(1+x)^{n+1} - 1}{x} \quad \text{[sum of a G.P.]}$$

Coefficient of x^k in E = coefficient of x^{k+1} in $[(1+x)^{n+1} - 1]$

$$= {}^{n+1}C_{k+1}$$

$$= {}^{n+1}C_{n+1-k-1}$$

$$= {}^{n+1}C_{n-k}$$

37. Let $S = (S_1)^{1/2}$

where $S_1 = 1 + 2x + 3x^2 + 4x^3 + \cdots$ which is in G.P.

$$= \frac{a}{1-r} + \frac{dr}{(1-r)^2}, \text{ where } a = 1,\ r = x,\ d = 1$$

$$= \frac{1}{1-x} + \frac{x}{(1-x)^2} = \frac{1-x+x}{(1-x)^2}$$

$$= \frac{1}{(1-x)^2} = (1-x)^{-2}$$

$\therefore\ [(1-x)^{-2}]^{1/2} = (1-x)^{-1}$

$$= 1 + x + x^2 + x^3 + x^4 + \cdots + x^n + \cdots$$

$\therefore$ Coefficient of $x^8 = 1$

38. Since x is small in magnitude compared with a,

$\therefore\quad |x| < |a|$

$\Rightarrow\quad \left|\frac{x}{a}\right| < 1$

Let $\quad E = \left(\frac{a}{a+x}\right)^{1/2} + \left(\frac{a}{a-x}\right)^{1/2}$

$\Rightarrow\quad E = \frac{1}{\left(1+\frac{x}{a}\right)^{1/2}} + \frac{1}{\left(1-\frac{x}{a}\right)^{1/2}}$

$\Rightarrow\quad E = \left(1+\frac{x}{a}\right)^{-1/2} + \left(1-\frac{x}{a}\right)^{-1/2}$

$\Rightarrow\quad E = \left(1 - \frac{1x}{2a} + \frac{3}{8}\frac{x^2}{a^2}\right) + \left(1 + \frac{1x}{2a} + \frac{3}{8}\frac{x^2}{a^2}\right)$

$\Rightarrow\quad E = 2 + \frac{3}{4}\frac{x^2}{a^2}$

$\therefore\quad k = \frac{3}{4}$

39. $\frac{2}{\sqrt{2x^2+1} + \sqrt{2x^2-1}} = \frac{2(\sqrt{2x^2+1} - \sqrt{2x^2-1})}{(2x^2+1) - (2x^2-1)}$

$$= \sqrt{2x^2+1} - \sqrt{2x^2-1}$$

Thus, the given expression can be written as

$$\left(\sqrt{2x^2+1}+\sqrt{2x^2-1}\right)^6+\left(\sqrt{2x^2+1}-\sqrt{2x^2-1}\right)^6$$

But

$$(a+b)^6+(a-b)^6=2(a^6+{}^6C_2a^4b^6+{}^6C_4a^2b^4+b^6)$$

$$\therefore\quad \left(\sqrt{2x^2+1}+\sqrt{2x^2-1}\right)^6+\left(\sqrt{2x^2+1}-\sqrt{2x^2-1}\right)^6$$

$$=2[(2x^2+1)^3+15(2x^2+1)^2(2x^2-1)+15(2x^2-1)+(2x^2-1)^2+(2x^2-1)^3]$$

which is a polynomial of degree 6.

40. $2^{4n}-2^n(7n+1)=2^n[2^{3n}-(7n+1)]$

$=2^n[(1+7)^n-(7n+1)]$

$=2^n[(1+{}^nC_1 7+{}^nC_2 7^2+{}^nC_3 7^3+\cdots+7^n-(7n+1)]$

$=2^n[{}^nC_2 7^2+{}^nC_3 7^3+\cdots+7^n]$

$=2^2\cdot 7^2\cdot 2^{n-2}[{}^nC_2+{}^nC_3 7+\cdots+7^{n-2}]$ [$\because\ n\geq 2$]

$=(14)^2\cdot 2^{n-2}[{}^nC_2+{}^nC_3 7+\cdots+7^{n-2}]$

$\therefore$ Statement 1 is true.

Statement 2 is also true and is also a correct explanation for Statement 1.

41. Greatest coefficient in $(1+x)^{2n}={}^{2n}C_n$ [$\because\ 2n$ is even]

Middle term $=T_{\frac{2n+2}{2}}=T_{n+1}={}^{2n}C_n x^n$

$\therefore$ Coefficient of the middle term $={}^{2n}C_n$

$\therefore$ Statement 2 is true.

In $(1+5x)^8$,

$$\frac{\text{Coefficient of } T_{r+1}}{\text{Coefficient of } T_r}=\frac{{}^8C_r 5^r}{{}^8C_{r-1}5^{r-1}}$$

$$=\frac{8!}{r!(8-r)!}\times\frac{(r-1)!(9-r)!}{8!}\times 5=\frac{5(9-r)}{r}$$

$\therefore\quad \dfrac{5(9-r)}{r}\geq 1$ if $45-5r\geq r$

i.e. if $6r\leq 45$ if $r\leq 7.5$

$\therefore$ Coefficient of T_{r+1} is maximum when $r=4$

$\therefore\quad T_{r+1}=T_{4+1}={}^8C_4(5x)^4$

$\therefore$ Greatest coefficient $={}^8C_4 5^4$

$\therefore$ Statement 1 is true

Also, Statement 2 is not a correct explanation for Statetment 1.

42. $n=1\ \Rightarrow\ (\sqrt{2}-1)^n=\sqrt{2}-1=\sqrt{2}-\sqrt{2-1}$

$n=2\ \Rightarrow\ (\sqrt{2}-1)^2=3-2\sqrt{2}=\sqrt{9}-\sqrt{9-1}$

$n=3\ \Rightarrow\ (\sqrt{2}-1)^3=(3-2\sqrt{2})(\sqrt{2}-1)$

$=3\sqrt{2}-3-4+2\sqrt{2}$

$=5\sqrt{2}-7=\sqrt{50}-\sqrt{50-1}$

$\therefore$ Statement 1 is true.

By expanding $(\sqrt{2}-1)^n$ by binomial theorem, we get

$(\sqrt{2}-1)^n=A+B\sqrt{2}$, where $A, B\in Z$

$\therefore$ Statement 2 is true.

Also, Statement 2 is not a correct explanation for Statement 1.

43. $\left(x+\dfrac{1}{x}+1\right)^n=\left[1+\left(x+\dfrac{1}{x}\right)\right]^n$

$=1+{}^nC_1\left(x+\dfrac{1}{x}\right)+{}^nC_2\left(x+\dfrac{1}{x}\right)^2+{}^nC_3\left(x+\dfrac{1}{x}\right)^3+\cdots+\left(x+\dfrac{1}{x}\right)^n$

$=\dfrac{a_1}{x^n}+\dfrac{a_2}{x^{n-1}}+\cdots+a_{n+1}+a_{n+2}x+a_{n+3}x^2+\cdots+a_{2n+1}x^n$, say

$\therefore$ Total number of terms $=2n+1$

$\therefore$ Statement 1 is true.

Statement 2 is trivially true.

Also, Statement 2 is not a correct explanation for Statement 1.

44. $(1+x+x^2+x^3+\cdots)^2=[(1-x)^{-1}]^2=(1-x)^{-2}$

Coefficient of x^n in $(1+x+x^2+\cdots)^2$

= coefficient of x^n in $(1-x)^{-2}$

$={}^{n+2-1}C_{2-1}={}^{n+1}C_1=n+1$

45. Let $R=\left(1+\dfrac{1}{10^4}\right)^{1000}$

$=1+1000\left(\dfrac{1}{10^4}\right)^1+1000\dfrac{999}{2}\left(\dfrac{1}{10^4}\right)^2+\cdots+\infty$

$<1+\dfrac{1}{10}+\dfrac{1}{10^2}+\dfrac{1}{10^3}+\cdots+\infty<\dfrac{10}{9}$

$\therefore\quad R<\dfrac{10}{9}$

$\therefore$ The number just greater than $\dfrac{10}{9}$ is 2.

46. Consider,

$(a+b)^n={}^nC_0a^n+{}^nC_1a^{n-1}b+{}^nC_2a^{n-2}b^2+\cdots+{}^nC_nb^n$

Putting $a=b=1$, we get

$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n$

$2^n = 4096 = 2^{12}$

$\Rightarrow \quad n = 12$ (even)

Now $(a+b)^n = (a+b)^{12}$

As $n = 12$ is even, hence the coefficient of the greatest term is ${}^nC_{n/2} = {}^{12}C_{12/2} = {}^{12}C_6$

$$= \frac{12!}{6!6!} = \frac{12\times11\times10\times9\times8\times7}{6\times5\times4\times3\times2}$$

$$= 11\times3\times4\times7 = 924$$

47. $(1+2x+3x^2+\cdots)^{-3/2} = [(1-x)^{-2}]^{-3/2} = (1-x)^3$

So, the coefficient of x^5 in $(1+2x+3x^2+\cdots)^{-3/2}$
= the coefficient of x^5 in $(1-x)^3$
There is no x^5 term in $(1-x)^3$. Hence, option (d).

48. The $(r+1)$th term in the expansion of $(1+x)^{27/5}$

$$= \frac{(27/5)\,[(27/5)-1]\cdots[(27/5)-r+1]}{r!}x^r$$

This term will be negative if the last factor in its numerator is negative.

$$\Rightarrow \quad \frac{27}{5} - r + 1 < 0 \quad \Rightarrow \quad \frac{32}{5} < r$$

$\Rightarrow 6.4 < r \Rightarrow$ least value of r is 7
Thus, the first negative term will be the 8th term.

49. $(\sqrt{3}+\sqrt[8]{5})^{256}$

$$T_{r+1} = {}^{256}C_r(\sqrt{3})^{256-r}\left(\sqrt[8]{5}\right)^r = {}^{256}C_r(3)^{(256-r)/2}(5)^{r/8}$$

For integral terms, $\frac{256-r}{2}$ and $\frac{r}{8}$ should both be positive integers.

$\therefore \quad r = 0, 8, 16, ..., 256$

Now, $256 = 0 + (n-1)\,8$ [using $t_n = a + (n-1)d$]

$\therefore \quad \frac{256}{8} = n - 1$

$\therefore \quad n = \frac{256}{8} + 1$

$n = 32 + 1 = 33$

50. Consider ${}^nC_{r-1} + 2{}^nC_r + {}^nC_{r+1}$

$= ({}^nC_{r-1} + {}^nC_r) + ({}^nC_r + {}^nC_{r+1})$

$= {}^{n+1}C_r + {}^{n+1}C_{r+1} = {}^{n+2}C_{r+1}$

51. $(1-x)(1-x)^n = (1-x)[(1+{}^nC_1(-x)+\cdots+{}^nC_{n-1}(-x)^{n-1} + {}^nC_n(-x)^n]$

$\therefore$ Coefficients of $x^n = {}^nC_n(-1)^n - {}^nC_{n-1}(-1)^{n-1}$

$= (-1)^n + (-1)^{n-1}n = (-1)^{n-1}(-1+n)$

$= (-1)^n(n+1)$

52. The coefficient of the middle term in $(1+\alpha x)^4$ = the coefficient of the middle term in $(1-\alpha x)^6$

$\therefore \quad {}^4C_2\alpha^2 = {}^6C_3(-\alpha)^3$

$\Rightarrow \quad \alpha = -\frac{3}{10}$

53. $\lim_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x} = e^2$

$\lim_{x\to\infty} 2x \log\left(1+\frac{a}{x}+\frac{b}{x^2}\right) = 2\log_e e$

$\lim_{x\to\infty} 2x \log\left(1+\frac{a}{x}+\frac{b}{x^2}\right) = 2$

$\Rightarrow \lim_{x\to\infty} 2x \log\left[1+\left(\frac{a}{x}+\frac{b}{x^2}\right)\right] = 2$

$\Rightarrow \lim_{x\to\infty} 2x \left[\left(\frac{a}{x}+\frac{b}{x^2}\right) - \frac{1}{2}\left(\frac{a}{x}+\frac{b}{x^2}\right)^2 + \frac{1}{3}\left(\frac{a}{x}+\frac{b}{x^2}\right)^3 - \cdots\right] = 2$

$\left[\because \log(1+x) = \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots\right)\right]$

$\Rightarrow \lim_{x\to\infty} 2x \left[\left(\frac{a}{x}+\frac{b}{x^2}\right) - \frac{1}{2}\left(\frac{a^2}{x^2}+\frac{b^2}{x^4}+\frac{2ab}{x^3}\right) + \frac{1}{3}\left(\frac{a^3}{x^2}+\frac{b^3}{x^5}+\frac{3a^2b}{x^3}+\frac{3ab^2}{x^4}\right) - \cdots\right] = 2$

$\Rightarrow \lim_{x\to\infty} 2x \left\{\frac{1}{x}\left[\left(a+\frac{b}{x}\right) - \frac{1}{2}\left(\frac{a^2}{x^2}+\frac{b^2}{x^3}+\frac{2ab}{x^2}\right) + \frac{1}{3}\left(\frac{a^3}{x^2}+\frac{b^3}{x^5}+\frac{3a^2b}{x^3}+\frac{3ab^2}{x^4}\right) - \cdots\right]\right\} = 2$

$\Rightarrow \lim_{x\to\infty} 2\cancel{x} \times \frac{1}{\cancel{x}}\left[\left(a+\frac{b}{x}\right) - \frac{1}{2}\left(\frac{a^2}{x}+\frac{b^2}{x^3}+\frac{2ab}{x^2}\right) + \frac{1}{3}\left(\frac{a^3}{x^2}+\frac{b^3}{x^5}+\frac{3a^2b}{x^3}+\frac{3ab^2}{x^4}\right) - \cdots\right] = 2$

$\Rightarrow \lim_{x\to\infty}\left[2a + \text{terms of the form } \frac{A}{x}, \frac{B}{x^2}, \frac{C}{x^3}\cdots\right] = 2$

$\Rightarrow \quad 2a + 0 = 2$

$\Rightarrow$ $a = 1$ and b can assume any of the values in $\mathbb{R}$, i.e. $a = 1$, $b \in \mathbb{R}$. Hence, the correct option is (b).

54. Since the coefficient of the given terms are ${}^mC_{r-1}$, ${}^mC_r$, ${}^mC_{r+1}$ respectively and they are in A.P.,

$\therefore \quad {}^mC_{r-1} + {}^mC_{r+1} = 2{}^mC_r$

$$\Rightarrow \frac{m!}{(r-1)!(m-r+1)!}+\frac{m!}{(r+1)!(m-r-1)!}=2\frac{m!}{r!(m-r)!}$$

$$\Rightarrow \frac{1}{(m-r+1)(m-r)}+\frac{1}{(r+1)r}=\frac{2}{r(m-r)}$$

$$\Rightarrow \frac{r(r+1)+(m-r+1)(m-r)}{r(r+1)(m-r+1)(m-r)}=\frac{2}{r(m-r)}$$

$$\Rightarrow r^2+r+m^2+r^2-2mr+m-r = 2(mr-r^2+r+m-r+1)$$

$$\Rightarrow 4r^2-4mr-m-2+m^2=0$$

$$\Rightarrow m^2-m(4r+1)+4r^2-2=0$$

55. $^{50}C_4+\sum_{r=1}^{6}{}^{56-r}C_3$

Putting $r=6,5,4,3,2,1$, we get

$= {}^{50}C_4+{}^{50}C_3+{}^{51}C_3+{}^{52}C_3+{}^{53}C_3+{}^{54}C_3+{}^{55}C_3$

$[\because {}^nC_r+{}^nC_{r+1}={}^{n+1}C_{r+1}]$

$[\because {}^{50}C_4+{}^{50}C_3={}^{51}C_4]$

$= {}^{51}C_4+{}^{51}C_3+{}^{52}C_3+{}^{53}C_3+{}^{54}C_3+{}^{55}C_3$

$= {}^{52}C_4+{}^{52}C_3+{}^{53}C_3+{}^{54}C_3+{}^{55}C_3$

$= {}^{53}C_4+{}^{53}C_3+{}^{54}C_3+{}^{55}C_3={}^{54}C_4+{}^{54}C_3+{}^{55}C_3$

$= {}^{55}C_4+{}^{55}C_3={}^{56}C_4$

56. $\dfrac{(1+x)^{3/2}-\left(1+\frac{1}{2}x\right)^3}{(1-x)^{1/2}}$

$$\approx \frac{\left(1+\frac{3}{2}x+\frac{(3/2)\cdot(1/2)}{2}x^2\right)-\left(1+\frac{3x}{2}+\frac{3\cdot 2}{2}\cdot\frac{x^2}{4}\right)}{(1-x)^{1/2}}$$

[neglecting higher powers of x^3]

$$\approx -\frac{3x^2}{8}(1-x)^{-1/2}$$

$$\approx -\frac{3x^2}{8}\left(1+\frac{1}{2}x+\frac{\frac{1}{2}\cdot\frac{3}{2}}{2}.x^2\right)$$

$$\approx -\frac{3x^2}{8}$$ [$\because$ higher powers of x^3 can be neglected]

57. T_{r+1} of $\left(ax^2+\frac{1}{bx}\right)^{11}={}^{11}C_r(ax^2)^r\left(\frac{1}{bx}\right)^{11-r}$

T_{r+1} of $\left(ax-\frac{1}{bx^2}\right)^{11}={}^{11}C_r(ax)^r\left(-\frac{1}{bx^2}\right)^{11-r}$

$\therefore$ Coefficient of x^7 in $\left(ax^2+\frac{1}{bx}\right)^{11}={}^{11}C_6\frac{a^6}{b^5}$ and coefficient of x^{-7} in $\left(ax-\frac{1}{bx^2}\right)^{11}={}^{11}C_5\frac{a^5}{b^6}$

Now ${}^{11}C_6\frac{a^6}{b^5}={}^{11}C_5\frac{a^5}{b^6}$

$\therefore$ $ab=1$

58. $(1-y)^m(1+y)^n=(1-{}^mC_1y+{}^mC_2y^2-\cdots)$

$(1+{}^nC_1y+{}^nC_2y^2+\cdots)$

$=1+(n-m)+\left(\frac{m(m-1)}{2}+\frac{n(n-1)}{2}-mn\right)y^2+\cdots$

Given $a_1=a_2=10 \Rightarrow a_1=n-m=10$

$$a_2=\frac{m^2+n^2-m-n-2mn}{2}=10$$

$\Rightarrow n-m=10$ (i)

and $(m-n)^2-(m+n)=20$

$\Rightarrow m+n=80$ (ii)

Solving (i) and (ii), we get $m=35, n=45$

59. $\because (1-ax)^{-1}(1-bx)^{-1}$

$=(1+ax+a^2x^2+\cdots)(1+bx+b^2x^2+\cdots)$

$\therefore$ $a_n=$ coefficient of x^n in $(1-ax)^{-1}(1-bx)^{-1}$

$=a^0b^n+ab^{n-1}+\cdots+a^nb^0$

$$=a^0b^n\left[1+\frac{a}{b}+\left(\frac{a}{b}\right)^2+\cdots\right]=a^0b^n\left[\frac{(a/b)^{n+1}-1}{(a/b)-1}\right]$$

$$=\frac{b^n(a^{n+1}-b^{n+1})}{a-b}\cdot\frac{b}{b^{n+1}}=\frac{a^{n+1}-b^{n+1}}{a-b}=\frac{b^{n+1}-a^{n+1}}{b-a}$$

60. We know that

$(1+x)^{20}={}^{20}C_0+{}^{20}C_1x+\cdots+{}^{20}C_{10}x^{10}+\cdots+{}^{20}C_{20}x^{20}$

Putting $x=-1$ in the above expansion, we get

$0={}^{20}C_0-{}^{20}C_1+\cdots-{}^{20}C_9+{}^{20}C_{10}-{}^{20}C_{11}+\cdots+{}^{20}C_{20}$

$\Rightarrow 0={}^{20}C_0-{}^{20}C_1+\cdots-{}^{20}C_9+{}^{20}C_{10}-{}^{20}C_9+\cdots+{}^{20}C_0$

$\Rightarrow 0=2({}^{20}C_0-{}^{20}C_1+\cdots-{}^{20}C_9)+{}^{20}C_{10}$

$\Rightarrow {}^{20}C_{10}=2({}^{20}C_0-{}^{20}C_1+\cdots+{}^{20}C_{10})$

$\Rightarrow {}^{20}C_0-{}^{20}C_1+\cdots+{}^{20}C_{10}=\frac{1}{2}{}^{20}C_{10}$

61. Given ${}^nC_4a^{n-4}(-b)^4=-[{}^nC_5a^{n-5}(-b)^5]$

$\Rightarrow \frac{a}{b}=\frac{n-4}{5}$

62. Since $\sum_{r=0}^{n} {}^nC_r \cdot x^r = (1+x)^n$, multiplying by x, we get

$$\sum_{r=0}^{n} {}^nC_r \cdot x^{r+1} = x(1+x)^n$$

Differentiating w.r.t. x, we get

$$\sum_{r=0}^{n} (r+1) \cdot {}^nC_r \cdot x^r = (1+x)^n + nx(1+x)^{n-1}$$

$\therefore$ Statement II is true

If $x = 1$, then

$$\sum_{r=0}^{n} (r+1) \cdot {}^nC_r \cdot = 2^n + n(2)^{n-1} = (n+2)2^{n-1}$$

$\therefore$ Statement I is true and Statement II is a correct explanation of Statement I.

63. $8^{2n} - (62)^{2n+1} = (1+63)^n - (63-1)^{2n+1}$

$= (1+63)^n + (1-63)^{2n+1}$

$= [1 + {}^nC_1 63 + {}^nC_2 (63)^2 + \cdots + (63)^n]$

$+ [1 - {}^{(2n+1)}C_1 63 + {}^{(2n+1)}C_2 (63)^2 + \cdots + (-1)(63)^{(2n+1)}]$

$= 2 + 63[{}^nC_1 + {}^nC_2 (63) + ... + (63)^{n-1}$

$- {}^{(2n+1)}C_1 + {}^{(2n+1)}C_2 (63) - \cdots - (63)^{(2n)}]$

$\therefore$ Remainder is 2.

64. Given that r and n are +ve integers such that $r > 1$, $n > 2$. Also in the expansion of $(1+x)^{2n}$, the coefficient of the $(3r)$th term = the coefficient of the $(r+2)$th term

$\Rightarrow {}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$

$\Rightarrow 3r-1 = r+1$ or $3r-1+r+1 = 2n$

[using ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$]

$\Rightarrow r = 1$ or $2r = n$

But $r > 1$ $\therefore n = 2r$

65. $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$

The general term in this expansion is

$$T_{r+1} = {}^{10}C_r \left(\frac{x}{2}\right)^{10-r} \left(\frac{-3}{x^2}\right)^r = {}^{10}C_r x^{10-3r} \frac{(-1)^r 3^r}{2^{10-r}}$$

For the coefficient of x^4, we should have $r = 2$.

$\therefore$ Coefficient of $x^4 = {}^{10}C_2 \frac{(-1)^2 3^2}{2^8} = \frac{405}{256}$

66. The given expression is $(x+\sqrt{x^3-1})^5 + (x-\sqrt{x^3-1})^5$

We know that

$(x+a)^n + (x-a)^n = 2[{}^nC_0 x^n + {}^nC_2 x^{n-2} a^2$

$+ {}^nC_4 x^{n-4} a^4 + \cdots]$

Therefore, the given expression is equal to $2[{}^5C_0 x^5 + {}^5C_2 x^3 (x^3-1) + {}^5C_4 x (x^3-1)^2]$

Maximum power of x involved here is 7. Also only +ve integral powers of x are involved. Therefore, the given expression is a polynomial of degree 7.

67. $(1+x)^m (1-x)^n$

$$= \left(1 + mx + \frac{m(m-1)x^2}{2!} + \cdots\right)\left(1 - nx + \frac{n(n-1)}{2!}x^2 - \cdots\right)$$

$$= 1 + (m-n)x + \left(\frac{n^2-n}{2} - mn + \frac{m^2-m}{2}\right)x^2 + \cdots$$

Given, $m - n = 3$ or $n = m - 3$

Also $\frac{n^2-n}{2} - mn + \frac{m^2-m}{2} = -6$

$\Rightarrow \frac{(m-3)(m-4)}{2} - m(m-3) + \frac{m^2-m}{2} = -6$

$\Rightarrow m^2 - 7m + 12 - 2m^2 + 6m + m^2 - m + 12 = 0$

$\Rightarrow -2m + 24 = 0$

$\Rightarrow m = 12$

68. $\sum_{i=0}^{m} {}^{10}C_i \, {}^{20}C_{m-i} = {}^{10}C_0 \, {}^{20}C_m + {}^{10}C_1 \, {}^{20}C_{m-1} + {}^{10}C_2 \, {}^{20}C_{m-2}$

$+ \cdots + {}^{10}C_m \, {}^{20}C_0$

= Coefficient of x^m in the expansion of the product $(1+x)^{10}(x+1)^{20}$

= Coefficient of x^m in the expansion of $(1+x)^{30} = {}^{30}C_m$

Hence, the maximum value of ${}^{30}C_m$ is ${}^{30}C_{15}$.

69. $(1+t^2)^{12}(1+t^{12})(1+t^{24})$

$= (1+t^{12}+t^{24}+t^{36})(1+t^2)^{12}$

$= (1+t^{12}+t^{24}+t^{36})({}^{12}C_0 + {}^{12}C_1 t^2 + {}^{12}C_2 t^4 + \cdots + {}^{12}C_{12} t^{24})$

Therefore, the coefficient of t^{24} = 1 × coefficient of t^{24} in $(1 + t^2)^{12}$ + 1 × coefficient of t^{12} in $(1 + t^2)^{12}$ + 1 × constant term in $(1 + t^2)^{12}$

$= {}^{12}C_{12} + {}^{12}C_6 + {}^{12}C_0 = 1 + {}^{12}C_6 + 1 = {}^{12}C_6 + 2$

70. ${}^{n-1}C_r = {}^nC_{r+1}(k^2-3)$

$\Rightarrow k^2 - 3 = \frac{{}^{n-1}C_r}{{}^nC_{r+1}} = \frac{r+1}{n}$

As $0 \le r \le n-1$

$\Rightarrow 1 \le r+1 \le n$

$\Rightarrow \frac{1}{n} \le \frac{r+1}{n} \le 1 \quad \Rightarrow \quad \frac{1}{n} \le (k^3-3) \le 1$

$\Rightarrow 3 + \frac{1}{n} \le k^2 \le 4 \quad \Rightarrow \quad \sqrt{3+\frac{1}{n}} \le k \le 2$

As $n \to \infty$

$\Rightarrow \quad \sqrt{3} < k \le 2$

$\Rightarrow \quad k \in (\sqrt{3}, 2]$

71. Required value $= \left(1 - \frac{2x}{1+x}\right)^{-n} = \left(\frac{1+x-2x}{1+x}\right)^{-n}$

$= \left(\frac{1-x}{1+x}\right)^{-n} = \left(\frac{1+x}{1-x}\right)^{n}$

72. $\because \quad \left[\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right]^{10}$

$= \left[\frac{(x^{1/3})^3+1^3}{x^{2/3}-x^{1/3}+1} - \frac{(\sqrt{x})^2-1)}{\sqrt{x}(\sqrt{x}-1)}\right]^{10}$

$= [x^{1/3}+1-(x^{-1/2}+1)]^{10} = (x^{1/3}-x^{-1/2})^{10}$

$\therefore \quad T_{r+1} = {}^{10}C_r (x)^{\frac{10-r}{3}} (-x^{-1/2})^r = {}^{10}C_r (-1)^r x^{\frac{20-5r}{6}}$

For the term independent of x, put $\frac{20-5r}{6} = 0 \Rightarrow r = 4$

$\therefore$ Required coefficient $= {}^{10}C_4 = 210$

73. $\because \quad 7^9 = (8-1)^9 = -1(1-8)^9$

$= -1 + {}^9C_1 8^1 - {}^9C_2 8^2 + \cdots + {}^9C_9 8^9$

and $9^7 = (1+8)^7$

$= 1 + {}^7C_1 8^1 + {}^7C_2 8^2 + {}^7C_3 8^3 + \cdots + {}^7C_7 8^7$

$\therefore \quad 7^9 + 9^7 = 8({}^9C_1 + {}^7C_1) + 8^2({}^7C_2 - {}^9C_2) + \cdots$

$= 8(9+7) + 8^2(21-36) + \cdots$

$= 64 \times 2 + 64(-15) + \cdots$

Hence, it is divisible by 64.

74. $\frac{1}{81^n}(1 - 10 \cdot {}^{2n}C_1 + 10^2 \cdot {}^{2n}C_2 - 10^3 \cdot {}^{2n}C_3 + \cdots + 10^{2n})$

$= \frac{1}{(81)^n}(1-10)^{2n} = 1$

75. The given expression is $\sqrt{3}\left(1 + \frac{1}{\sqrt{3}}\right)^{20}$.

$\therefore \quad T_{r+1} = \sqrt{3} \cdot {}^{20}C_r \left(\frac{1}{\sqrt{3}}\right)^r$ and $T_r = \sqrt{3} \cdot {}^{20}C_{r-1} \left(\frac{1}{\sqrt{3}}\right)^{r-1}$

Now, $\frac{T_{r+1}}{T_r} = \frac{20-r+1}{r}\left(\frac{1}{\sqrt{3}}\right)$

Since $T_{r+1} \ge T_r$

$\Rightarrow \quad 20 - r + 1 \ge \sqrt{3}r$

$\Rightarrow \quad 21 \ge r(\sqrt{3}+1)$

$\Rightarrow \quad r \le \frac{21}{\sqrt{3}+1} = \frac{21}{2.73}$

$\Rightarrow \quad r \le 7 \cdot 692$

$\Rightarrow \quad r = 7$

$\therefore$ The greatest term is

$T_8 = \sqrt{3} \cdot {}^{20}C_7 \left(\frac{1}{\sqrt{3}}\right)^7 = \frac{25840}{9}$

76. We have ${}^{15}C_0 + {}^{15}C_1 + \cdots + {}^{15}C_{15} = 2^{15}$

$\Rightarrow \quad 2({}^{15}C_8 + {}^{15}C_9 + \cdots + {}^{15}C_{15}) = 2^{15} \quad [\because {}^nC_r = {}^nC_{n-r}]$

$\Rightarrow \quad {}^{15}C_8 + {}^{15}C_9 + \cdots + {}^{15}C_{15} = 2^{14}$

77. $\because \quad (1 + 3x + 3x^2 + x^3)^{20} = (1+x)^{60}$

$\therefore$ Coefficient of x^{20} in $(1+x)^{60}$ is ${}^{60}C_{20}$, i.e. ${}^{60}C_{40}$.

78. We know that $n!$ terminates in 0 for $n \ge 5$ and 3^{4n} terminates in 1 $[\because \quad 3^4 = 81]$

$\therefore \quad 3^{180} = (3^4)^{45}$ terminates in 1.

Also, $3^3 = 27$ terminates in 7.

Hence, $183! + 3^{183} = 183! + 3^{180} \times 3^3$ terminates in $0 + 1 \times 7 = 7$.

That is, the digit in the unit place is 7.

79. $(\sqrt{3}+1)^{2n} - (\sqrt{3}-1)^{2n}$

$= \left[(\sqrt{3}+1)^2\right]^n - \left[(\sqrt{3}-1)^2\right]^n$

$= \left[3+1+2\sqrt{3}\right]^n - \left[3+1-2\sqrt{3}\right]^n$

$= (4+2\sqrt{3})^n - (4-2\sqrt{3})^n$

$= \left[2(2+\sqrt{3})\right]^n - \left[2(2-\sqrt{3})\right]^n$

$= 2^n\left[(2+\sqrt{3})^n - (2-\sqrt{3})^n\right]$

$= 2^n\left\{\left[{}^nC_0 2^n + {}^nC_1 2^{n-1}\sqrt{3} + {}^nC_2 2^{n-2} 3 + \cdots\right]\right.$

$\left. -\left[{}^nC_0 2^n - {}^nC_1 2^{n-1}\sqrt{3} + {}^nC_2 2^{n-2} 3 - \cdots\right]\right\}$

$= 2^{n+1}\left[{}^nC_1 2^{n-1}\sqrt{3} + {}^nC_3 2^{n-3} 3\sqrt{3} + \cdots\right] = 2^{n+1}\sqrt{3}$

which is irrational.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. If the second, third and fourth terms in the expansion of $(a + b)^n$ are 135, 30 and 10/3 respectively, then the value of a is

(a) 3 (b) 4 (c) 5 (d) 7

2. If the binomial coefficients of the rth, $(r + 1)$th and $(r + 2)$th terms in the expansion of $(1 + x)^{14}$ are in A.P., then the value of r is

(a) 10 (b) 6 (c) 7 (d) 9

3. If $(1+ax)^n = 1+8x+24x^2+\cdots$, then
(a) $a = 3$ (b) $a = 4$ (c) $a = 2$ (d) $a = 5$

4. The $(m + 1)$th term of $[(x/y) + (y/x)]^{2m+1}$ is
(a) independent of x (b) a constant
(c) dependent on the ratio x/y and m
(d) none of these

5. The sum of the coefficients in the expansion of $(1 + 5x - 7x^3)^{3165}$ is
(a) 1 (b) 2^{3165} (c) 2^{3164} (d) 1

6. In the expansion of $\left(3-\sqrt{\frac{17}{4}+3\sqrt{2}}\right)^{15}$, the 11th term is a
(a) positive integer
(b) positive irrational number
(c) negative integer
(d) negative irrational number

7. The $(r + 1)$th term in the expansion of $(x + a)^n$ will be
(a) ${}^nC_r x^n a^{n-r}$ (b) ${}^nC_r x^{n-r} a^r$
(c) ${}^nC_r x^{n-r} a^n$ (d) ${}^nC_r x^r a^{n-r}$

8. The ratio of the coefficient of terms $x^{n-r}a^r$ and $x^r a^{n-r}$ in the binomial expansion of $(x+a)^n$ will be
(a) $x : a$ (b) $n : r$
(c) $x : n$ (d) none of these

9. The middle term in the expansion of $\left(\frac{x}{a}+\frac{a}{x}\right)^{20}$ is
(a) ${}^{20}C_{11}\frac{x}{a}$ (b) ${}^{20}C_{11}\frac{a}{x}$
(c) ${}^{20}C_{10}$ (d) none of these

10. The integral part of $(7+4\sqrt{3})^n$ $(n \in N)$ is
(a) an even number (b) an odd number
(c) an even or an odd number depending upon the value of n
(d) none of these

11. The sum ${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n$ is equal to
(a) $\frac{2\cdot4\cdot6\cdots2n}{n!}$ (b) n^n
(c) $n!$ (d) 3^n

12. The two successive terms in the expansion of $(1 + x)^{24}$ with coefficients in the ratio 1:4 are
(a) 3rd and 4th (b) 4th and 5th
(c) 5th and 6th (d) 6th and 7th

13. If the second term of the expansion $\left[a^{1/13}+\frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$, then the value of $\frac{{}^nC_3}{{}^nC_2}$ is
(a) 4 (b) 3
(c) 12 (d) 6

14. The value of $\frac{18^3+7^3+3\cdot18\cdot7\cdot25}{\left[\begin{array}{r}3^6+6\cdot243\cdot2+15\cdot81\cdot4+20\cdot27\cdot8\\+15\cdot9\cdot16+6\cdot3\cdot32+64\end{array}\right]}$ is
(a) 1 (b) 2
(c) 3 (d) none of these

15. Given that the term of the expansion $(x^{1/3} - x^{-1/2})^{15}$ which does not contain x is $5m$, where $m \in N$, then $m =$
(a) 1100 (b) 1010
(c) 1001 (d) none of these

16. If the coefficients of 7th and 13th terms in the expansion of $(1 + x)^n$ are equal, then $n =$
(a) 10 (b) 15
(c) 18 (d) 20

17. Which of the following expressions is divisible by 1225?
(a) $6^{2n}-35n-1$ (b) $6^{2n}-35n+1$
(c) $6^{2n}-35n$ (d) $6^{2n}-35n+2$

18. The coefficients of two consecutive terms in the expansion of $(1 + x)^n$ will be equal, if
(a) n is any integer
(b) n is an odd integer
(c) n is an even integer
(d) none of these

19. The coefficient of $x^k (0 \le k \le n)$ in the expansion of $E = 1+(1+x)+(1+x)^2+\cdots+(1+x)^n$ is
(a) ${}^{n+1}C_{k+1}$ (b) ${}^nC_k$
(c) ${}^{n+1}C_{n-k-1}$ (d) none of these

20. The ratio of the coefficient of x^{15} to the term independent of x in $(x^2 + 2/x)^{15}$ is
(a) 12:32 (b) 1:32
(c) 32:12 (d) 32:1

21. If the binomial coefficients of the 2nd, 3rd and 4th terms in the expansion of $\left[\sqrt{2^{\log_{10}(10-3^x)}}+\sqrt[5]{2^{(x-2)\log_{10}3}}\right]^m$ are in A.P. and the 6th term is 21, then the value(s) of x is (are)
(a) 1, 3 (b) 0, 2
(c) 4 (d) -1

22. If $T_0, T_1, T_2, ..., T_n$ represent the terms in the expansion of $(x + a)^n$, then the value of $(T_0 - T_2 + T_4 - T_6 \cdots)^2 + (T_1 - T_3 + T_5 \cdots)^2$ is
(a) $(x^2 - a^2)^n$ (b) $(x^2 + a^2)^n$
(c) $(a^2 - x^2)^n$ (d) none of these

23. If $k \in \mathbb{R}$ and the middle term of $\left(\frac{k}{2}+2\right)^8$ is 1120, then the value of k is

(a) 3 (b) 2
(c) –3 (d) –4

24. The term independent of x in the expansion of $\left(x-\frac{1}{x}\right)^4\left(x+\frac{1}{x}\right)^3$ is

(a) –3 (b) 0
(c) 1 (d) 3

25. ${}^{10}C_1 + {}^{10}C_3 + {}^{10}C_5 + {}^{10}C_7 + {}^{10}C_9 =$

(a) 2^9 (b) 2^{10}
(c) $2^{10} - 1$ (d) none of these

26. The 6th term in the expansion of $\left(2x^2 - \frac{1}{3x^2}\right)$ is

(a) $\frac{4580}{17}$ (b) $-\frac{896}{27}$
(c) $\frac{5580}{17}$ (d) none of these

27. In the expansion of $\left(\frac{a}{x}+bx\right)^{12}$, the coefficient of x^{-10} will be

(a) $12a^{11}$ (b) $12b^{11}a$
(c) $12a^{11}b$ (d) $12a^{11}b^{11}$

28. The remainder when 2^{2003} is divided by 17 is

(a) 1 (b) 2
(c) 8 (d) none of these

29. The coefficient of x in the expansion of $(1-2x^3+3x^5)[1+(1/x)]^8$ is

(a) 56 (b) 65
(c) 154 (d) 62

30. The greatest value of the term independent of x in the expansion of $(x\sin p + x^{-1}\cos p)^{10}$, $p \in \mathbb{R}$, is

(a) 2^5 (b) $\frac{10!}{(5!)^2}$
(c) $\frac{1}{2^5}\cdot\frac{10!}{(5!)^2}$ (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. The 14th term from the end in the expansion of $(\sqrt{x}-\sqrt{y})^{17}$ is

(a) ${}^{17}C_5 x^6(-\sqrt{y})^5$ (b) ${}^{17}C_6(\sqrt{x})^{11}y^3$
(c) ${}^{17}C_4 x^{13/2}y^2$ (d) none of these

2. If $n \in N$, then the sum of the coefficients in the expansion of the binomial $(5x-4y)^n$ is

(a) 1 (b) –1
(c) 1 (d) 0

3. When 5^{99} is divided by 13, the remainder is

(a) 8 (b) 9
(c) 10 (d) none of these

4. The number of values of r satisfying the equation ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$ is

(a) 1 (b) 2
(c) 3 (d) 4

5. The number of elements in the set of values of r for which ${}^{18}C_{r-2} + 2\cdot{}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ is satisfied is

(a) 4 (b) 5
(c) 7 (d) 10

6. If x is nearly equal to one, then the approximate value of $\frac{mx^m - nx^n}{m-n}$ is

(a) x^m (b) x^n
(c) x^{m+n} (d) none of these

7. If the coefficients of the pth, $(p+1)$th and $(p+2)$th terms in the expansion of $(1+x)^n$ are in A.P., then

(a) $n^2 - 2np + 4p^2 = 0$
(b) $n^2 - n(4p+1) + 4p^2 - 2 = 0$
(c) $n^2 - n(4p+1) + 4p^2 = 0$
(d) none of these

8. ${}^nC_0 - \frac{1}{2}{}^nC_1 + \frac{1}{3}{}^nC_2 - \cdots + (-1)^n \frac{{}^nC_n}{n+1} =$

(a) n (b) $\frac{1}{n}$
(c) $\frac{1}{n+1}$ (d) $\frac{1}{n-1}$

9. The expansion of $\frac{1}{(4-3x)^{1/2}}$ by binomial theorem will be valid, if

(a) $x<1$ (b) $|x|<1$

(c) $-\frac{2}{\sqrt{3}}<x<\frac{2}{\sqrt{3}}$ (d) none of these

10. The two consecutive terms in the expansion of $(3+2x)^{74}$ with equal coefficients are

(a) 30th and 31st terms

(b) 29th and 30th terms

(c) 31st and 32nd terms

(d) 28th and 29th terms

11. If $\sum_{r=m}^{n} {}^{r}C_m = {}^{n+1}C_{m+1}$, then $\sum_{r=m}^{n}(n-r+1)\,{}^{r}C_m$ is equal to

(a) ${}^{n+3}C_{m+2}$ (b) ${}^{n+2}C_{m+2}$

(c) ${}^{n+2}C_{m+1}$ (d) none of these

12. In the expansion of $(7^{1/3}+11^{1/9})^{65/61}$, the number of terms free from radicals is

(a) 730 (b) 715

(c) 725 (d) 750

13. $(1+x+x^2+x^3)^5 = a_0+a_1x+a_2x^2+\cdots+a_{15}x^{15}$, then a_{10} is equal to

(a) 99 (b) 101

(c) 100 (d) 110

14. If $C_0, C_1, C_2, ..., C_n$ are the binomial coefficients in the expansion of $(1+x)^n$, n being even, then $C_0+(C_0+C_1)+(C_0+C_1+C_2)+\cdots+(C_0+C_1+C_2+\cdots+C_{n-1})$ is equal to

(a) $n\cdot 2^n$ (b) $n\cdot 2^{n-1}$

(c) $n\cdot 2^{n-2}$ (d) $n\cdot 2^{n-3}$

15. If $(1+x+2x^2)^{20} = a_0+a_1x+a_2x^2+\cdots+a_{40}x^{40}$, then $a_0+a_2+a_4+\cdots+a_{38}$ is equal to

(a) $2^{19}(2^{30}+1)$ (b) $2^{19}(2^{20}-1)$

(c) $2^{19}(2^{19}-1)$ (d) none of these

16. $C_1+2C_2+3C_3+4C_4+\cdots+nC_n =$

(a) 2^n (b) $n\cdot 2^n$

(c) $n\cdot 2^{n-1}$ (d) $n\cdot 2^{n+1}$

17. $\frac{1}{\sqrt[3]{6-3x}} =$

(a) $6^{\frac{1}{3}}\left[1-\frac{x}{6}-\frac{2x^2}{6^2}-\cdots\right]$

(b) $6^{-\frac{1}{3}}\left[1+\frac{x}{6}+\frac{2x^2}{6^2}+\cdots\right]$

(c) $6^{\frac{1}{3}}\left[1-\frac{x}{6}+\frac{2x^2}{6^2}-\cdots\right]$

(d) $6^{-\frac{1}{3}}\left[1-\frac{x}{6}+\frac{2x^2}{6^2}-\cdots\right]$

18. If the coefficients of 5th, 6th and 7th terms in the expansion of $(1+x)^n$ are in A.P., then $n =$

(a) 7 only (b) 14 only

(c) 7 or 14 (d) none of these

19. The integer next above to $(\sqrt{3}+1)^{2m}$ contains

(a) 2^{m+1} as a factor

(b) 2^{m+2} as a factor

(c) 2^{m+3} as a factor

(d) 2^m as a factor

20. If ${}^{13}C_r$ is denoted by C_r, then the value of $C_1+C_5+C_7+C_9+C_{11}$ is equal to

(a) $2^{12}-287$ (b) $2^{12}-165$

(c) $2^{12}-C_3$ (d) $2^{12}-C_2-C_{13}$

21. Let n be an odd natural number greater than 1. Then the number of zeros at the end of the sum $99n+1$ is

(a) 3 (b) 4

(c) 2 (d) none of these

22. The number of terms in the expansion of $\left(x^2+1+\frac{1}{x^2}\right)^n$, $n \in N$, is

(a) $2n$ (b) $3n$

(c) $2n+1$ (d) $3n+1$

23. In the expansion of $\left(x^3-\frac{1}{x^2}\right)^n$, $n\in N$, if the sum of the coefficients of x^5 and x^{10} is 0, then n is

(a) 25 (b) 20

(c) 15 (d) none of these

24. If $(1+x)^n = \sum_{r=0}^{n} a_r x^r$, $b_r = 1+\frac{a_r}{a_{r-1}}$ and $\prod_{r=1}^{n} b_r = \frac{(101)^{100}}{100!}$ then n is equal to

(a) 99 (b) 100

(c) 101 (d) none of these

25. If a and d are two complex numbers, then the sum to $(n + 1)$ terms of the following series

$aC_0 - (a+d)C_1 + (a+2d)C_2 - \cdots + \cdots$ is

(a) $\frac{a}{2^n}$ (b) na

(c) 0 (d) none of these

26. The digit at units place in the number $17^{1995} + 11^{1995} - 7^{1995}$ is

(a) 0 (b) 1

(c) 2 (d) 3

27. The interval in which x must lie so that the greatest term in the expansion of $(1 + x)^{2n}$ has the greatest coefficient is

(a) $\left(\frac{n-1}{n}, \frac{n}{n-1}\right)$ (b) $\left(\frac{n}{n+1}, \frac{n+1}{n}\right)$

(c) $\left(\frac{n}{n+2}, \frac{n+2}{n}\right)$ (d) none of these

28. The integral part of $(4+2\sqrt{5})^{2n+1}$ $(n \in N)$ is

(a) an even number

(b) an odd number

(c) an even or an odd number depending upon the value of n

(d) none of these

29. If the term independent of x in $\left(\sqrt{x} + \frac{k}{x^2}\right)^{10}$ is 405, then the value of k is

(a) 2 (b) –2

(c) 3 (d) none of these

30. The coefficient of x^{11} in the expansion of $(2x^2 + x - 3)^6$ is

(a) 384 (b) 192

(c) 572 (d) 64

ANSWERS

Level 1

1. (a)	**2.** (d)	**3.** (c)	**4.** (c)	**5.** (d)	**6.** (b)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (b)
11. (a)	**12.** (c)	**13.** (a)	**14.** (a)	**15.** (c)	**16.** (c)	**17.** (a)	**18.** (b)	**19.** (a)	**20.** (b)
21. (b)	**22.** (b)	**23.** (b)	**24.** (b)	**25.** (a)	**26.** (b)	**27.** (c)	**28.** (c)	**29.** (c)	**30.** (c)

Level 2

1. (c)	**2.** (a)	**3.** (a)	**4.** (b)	**5.** (c)	**6.** (c)	**7.** (b)	**8.** (c)	**9.** (d)	**10.** (a)
11. (b)	**12.** (a)	**13.** (b)	**14.** (b)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (c)	**19.** (a)	**20.** (a)
21. (c)	**22.** (c)	**23.** (c)	**24.** (b)	**25.** (c)	**26.** (a)	**27.** (b)	**28.** (a)	**29.** (c)	**30.** (b)

9

ISAAC NEWTON

Sequences and Series

- Sequence
- Series
- Arithmetic Progression (A.P.)
- Sum of the First n Terms of an A.P.
- Properties of A.P.
- Arithmetic Mean (A.M.)
- Geometric Progression (G.P.)
- General Term of a G.P.
- Sum of n Terms of a G.P.
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- Properties of G.P.
- Geometric Mean (G.M.)
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- Harmonic Mean (H.M.)
- Relation between A.M., G.M. and H.M.
- Arithmetico-Geometric Series
- Sum of the First n Terms of an Arithmetico Geometric Series
- Sigma Notation
- To Find the nth Term by the Difference Method

SEQUENCE

A *sequence* is a succession of numbers formed according to some rule, e.g. the sequence 1, 4, 7, 10, 13, 16, … whose first element is 1 and its subsequent elements are formed by adding the number 3 to the preceding element. Hence, this is the rule which makes it possible to determine any element of this given sequence using the previous known element.

A sequence is thus a function defined on the domain N (the set of natural numbers) with codomain $\mathbb{R}$ (the set of real numbers) or $\mathbb{C}$ (the set of complex numbers). For example, suppose $f(n)$ is any function on the domain N, given by

$f(n) = n^2 + n + 1$, where $n \in N$, then the values of the function for natural numbers 1, 2, 3,…, n, i.e. the values

$f(1), f(2), f(3), f(4), \ldots, f(n)$ or $3, 7, 13, 21, \ldots$

are called the terms of the sequence. The terms of a sequence are also symbolized as $t_1, t_2, t_3, \ldots, t_n$, where $t_n = f(n)$ is called the general term. The number of terms in a sequence can also be infinite, such as $t_1, t_2, t_3, \ldots, t_n \ldots$.

SERIES

If $t_1, t_2, t_3, \ldots, t_n$ is a sequence, then $S = t_1 + t_2 + t_3 + \cdots + t_n$ is the corrsponding series. A series is finite or infinite according as the number of terms in the sequence is finite or infinite.

☞ ARITHMETIC PROGRESSION (A.P.)

Definition of A.P.

An arithmetic progression (A.P.) is a sequence in which each term increases or decreases regularly by the same constant.

The constant quantity is called the common difference of the A.P.

For example, observe the following sequences.

Sequence	Common difference
2, 4, 6, 8,…	2
9, 5, 1, –3, –7, …	–4
a, $(a + d)$, $(a + 2d)$, $(a + 3d)$, …	d

Note We usually denote

- the first term of an A.P. by a
- the last term of an A.P. by l
- the common difference by d
- the number of terms by n
- the sum of the first n terms by S_n

General Term of an A.P.

Let a be the first term, l be the last term and d be the common difference of a certain sequence in A.P. Then, the nth term of an A.P. from the beginning and the first term of an A.P. from the end would be as given in the box below.

nth term of an A.P. from the beginning	First term of an A.P. from the end
$t_1 = a = a + (1-1)d$	$t_n = l = l - (n-n)d$
$t_2 = a + d$ $= a + (2-1)d$	$t_{n-1} = l - d$ $= l - [n-(n-1)]d$
$t_3 = a + 2d$ $= a + (3-1)d$	$t_{n-2} = l - 2d$ $= l - [n-(n-2)]d$
…	…
$t_r = a + (r-1)d$	$t_r = l - (n-r)d$
…	…
$t_n = a + (n-1)d$	$t_1 = l - (n-1)d$

$\therefore$ nth term or last term of an A.P. $\boxed{t_n = l = a + (n-1)d}$

Note In an A.P., the sum of the terms equidistant from the beginning and end is a constant and equal to the sum of the first and last terms.

Illustration 1. If m times the mth term of an A.P. is equal to n times its nth term, then show that the $(m + n)$th term of the A.P. is zero.

Solution: Let a be the first term and d the common difference, then $m t_m = n t_n$ (given)

$\Rightarrow$ $m[a + (m-1)d] = n[a + (n-1)d]$

$\Rightarrow$ $\left[(m^2 - n^2) - (m-n)\right]d = (n-m)a$

$\Rightarrow$ $[(m-n)(m+n) - (m-n)]d = (n-m)a$

$\Rightarrow$ $(m-n)(m+n-1)d = (n-m)a$

$\Rightarrow$ $(m+n-1)d = -a \quad \Rightarrow \quad a + (m+n-1)d = 0$

$\Rightarrow$ $t_{m+n} = 0$.

☞ SUM OF THE FIRST n TERMS OF AN A.P.

Case I: When the first term and the last term are known

Let a be the first, n be the number of terms and l be the last term of an A.P., then the sum of the first n terms of an A.P. is S_n such that

$$\boxed{S_n = \frac{n}{2}(a + l)}$$

Case II: When the first term and the common difference are known

Let a be the first term, n be the number of terms and d be the common difference of an A.P., then the sum of the first n terms of an A.P. is S_n such that

$$\boxed{S_n = \frac{n}{2}[2a + (n-1)d]}$$

Case III: When the last term and the common difference are known

Let l be the last term, n be the number of terms and d be the common difference of an A.P., then the sum of the first n terms of an A.P. is S_n such that

$$\boxed{S_n = \frac{n}{2}[2l - (n-1)d]}$$

Illustration 2. In an A.P. if the pth term is $1/q$ and the qth term is $1/p$, then prove that the sum of the first pq terms must be $\frac{1}{2}(pq + 1)$.

Solution: Let the first term of an A.P. be a and the common difference be d. Then,

$$t_p = a + (p-1)d = \frac{1}{q} \quad \Rightarrow \quad \frac{1}{q} = a + (p-1)d \qquad \text{(i)}$$

and $$t_q = a + (q-1)d = \frac{1}{p} \quad \Rightarrow \quad \frac{1}{p} = a + (q-1)d \qquad \text{(ii)}$$

Subtracting (ii) from (i), we get

$$(p-q)d = \frac{1}{q} - \frac{1}{p} = \frac{p-q}{pq} \quad \Rightarrow \quad d = \frac{1}{pq} \qquad \text{(iii)}$$

Putting $d = \frac{1}{pq}$ in (i), we get $a + (p-1)\frac{1}{pq} = \frac{1}{q}$

$$\Rightarrow \qquad a=\frac{1}{q}-\frac{1}{q}+\frac{1}{pq} \quad \Rightarrow \quad a=\frac{1}{pq}$$

Now, $S_{pq}=\frac{pq}{2}[2a+(pq-1)d]=\frac{pq}{2}\left[2\frac{1}{pq}+(pq-1)\frac{1}{pq}\right]$

$$\Rightarrow \qquad S_{pq}=\frac{1}{2}(pq+1)$$

PROPERTIES OF A.P.

Property 1: If each term of an A.P. is increased, decreased, multiplied and divided by the same quantity, then the resulting series will also be an A.P.

	Increased by k	Decreased by k	Multiplied by k	Divided by $k\,(\neq 0)$
First term	$a+k$	$a-k$	ak	$\frac{a}{k}$
Common difference	d	d	dk	$\frac{d}{k}$

Here, a is the first term and d is the common difference of a, $(a+d)$, $(a+2d)$,..., $a+(n-1)d$,... which is an A.P.

Property 2: If both $a_1, a_2, a_3, ..., a_n$ and $b_1, b_2, b_3, ..., b_n$ are in A.P., then the sequence(s) $a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3, \ldots, a_n \pm b_n$ will also be an A.P. with the first term as $(a_1 \pm b_1)$ and the common difference as $(d_1 \pm d_2)$.

Property 3: If both $a_1, a_2, a_3, \ldots, a_n$ and $b_1, b_2, b_3, \ldots, b_n$ are in A.P., then the sequence(s) $a_1b_1, a_2b_2, a_3b_3, \ldots, a_nb_n$ and $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \ldots, \frac{a_n}{b_n}$ will **not** be in A.P.

Property 4: The common difference of an A.P. can be **zero, positive** or **negative**.

Property 5: Equal numbers, zero or non-zero, are in A.P.

Property 6: *Selection of terms in an A.P.* For a sequence to be in A.P. with odd number of terms, take the common difference as d, otherwise for an A.P. with even number of terms, take the common difference as $2d$.

Number of terms	Sequence
2	$a-d, a+d$
3	$a-d, a, a+d$
4	$a-3d, a-d, a+d, a+3d$
5	$a-2d, a-d, a, a+d, a+2d$
6	$a-5d, a-3d, a-d, a+d, a+3d, a+5d$
7	$a-3d, a-2d, a-d, a, a+d, a+2d, a+3d$

Property 7: If $a_1, a_2, a_3, \ldots, a_{n-2}, a_{n-1}, a_n$ are in A.P., then

$$a_1+a_n=(a_2+a_{n-1})=(a_3+a_{n-2})=\ldots=a_r+a_{n-(r-1)}$$

In an A.P., the sum of two terms equidistant from the beginning and the end is a constant and is equal to the sum of the first term and last term.

Property 8: Any term of an A.P. (except the first and the last term) is half the sum of two terms equidistant from it.

Property 9: If x is the pth term, y the qth term and z the rth term of an A.P., then $\begin{vmatrix} x & y & z \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}=0$

Property 10: A sequence is an A.P. if its nth term is a linear expression in n such that $t_n=an+b$, where a and b are constants.

Note If the general term of the series is a linear function of type $t_n = an + b$, then the sequence is an A.P. and the coefficient of n is the common difference of the A.P. Also for $n = 1$, we obtain the first term of the A.P.

Property 11: Any two degree quadratic equation having zero constant can be the sum of an A.P. such that $S_n=an^2+bn$, where a and b are constants.

Note The sum of the first n terms of the series is a quadratic equation of the type $S_n=an^2+bn$.

In this case, the common difference of the A.P. is half the coefficient of n^2.

Illustration 3. If $a+b+c\neq 0$ and $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in A.P., prove that $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are also in A.P.

Solution: We have $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ in A.P.

Adding 1 to each term, we get

$$\left(\frac{b+c}{a}+1\right), \left(\frac{c+a}{b}+1\right), \left(\frac{a+b}{c}+1\right)$$

in A.P., i.e. $\frac{a+b+c}{a}, \frac{c+a+b}{b}, \frac{a+b+c}{c}$ are in A.P.

Dividing each term by $a+b+c$, we get $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ in A.P.

Illustration 4. If the sum of three numbers in A.P. is –3 and their product is 8, then find the sum of the squares of the numbers.

Solution: Let the three numbers in A.P. be $a-d, a, a+d$.

$$\therefore \qquad (a-d)+a+(a+d)=-3$$

$$\Rightarrow \qquad 3a=-3 \quad \Rightarrow \quad a=-1$$

and $$(a-d)(a)(a+d)=8$$

$$\Rightarrow \qquad (-1)[(-1)^2-d^2]=8 \quad \Rightarrow \quad d^2=9$$

$\Rightarrow \quad d = \pm 3$

If $d = 3$, then the numbers are $-4, -1$ and 2.

If $d = -3$, then the numbers are $2, -1$ and -4.

Hence, the sum of squares of numbers for $d = \pm 3$

$= (-4)^2 + (-1)^2 + (2)^2 = (2)^2 + (-1)^2 + (-4)^2$

$= 16 + 1 + 4 = 21$

☞ ARITHMETIC MEAN (A.M.)

If three numbers are in A.P., then the middle one is called the arithmetic mean between the other two, i.e. if a, b, c are in A.P., then b is called the arithmetic mean between a and b.

Single A.M. between Two Numbers *a* and *b*

Let A be the A.M. between a and b, then a, A, b are in A.P.

$\Rightarrow \quad A - a = b - A \quad \Rightarrow \quad 2A = a + b$

$\Rightarrow \quad A = \frac{1}{2}(a + b)$

To Insert *n* Arithmetic Means between Two Numbers *a* and *b*

Let $x_1, x_2, x_3, \ldots, x_n$ be n A.Ms. between a and b.

Then $a, x_1, x_2, x_3, \ldots, x_n, b$ are in A.P.

Total number of terms $= (n + 2)$. If d is the common difference of an A.P., then

$t_{n+2} = t_1 + (n+1)d \quad \Rightarrow \quad b = a + (n+1)d$

$\therefore \quad d = \frac{b-a}{n+1}$

Now, $t_2 = x_1 = a + d = a + \left(\frac{b-a}{n+1}\right)$

$t_3 = x_2 = a + 2d = a + 2\left(\frac{b-a}{n+1}\right) \ldots$

$t_{n+1} = x_n = a + nd = a + n\left(\frac{b-a}{n+1}\right)$

Note Between any two given numbers, an infinite set of arithmetic means can be inserted.

Illustration 5. Find the A.M. between 11 and 89.

Solution: A.M. $= \frac{11+89}{2} = \frac{100}{2} = 50$

Illustration 6. Insert 6 arithmetic means between 3 and 24.

Solution: Let $A_1, A_2, A_3, A_4, A_5, A_6$ be six A.M.'s between 3 and 24.

$\therefore \quad 3, A_1, A_2, A_3, A_4, A_5, A_6, 24$ are in A.P.

We have $t_8 = 24 \quad \Rightarrow \quad 3 + (8-1)d = 24$

$[\because \ t_n = a + (n-1)d]$

$\Rightarrow \quad 3 + 7d = 24 \quad \Rightarrow \quad d = 3 \quad (\because \ a = 3)$

$\therefore \quad A_1 = a + d = 3 + 3 = 6$

$A_2 = A_1 + d = 6 + 3 = 9$

$A_3 = A_2 + d = 9 + 3 = 12$

$A_4 = A_3 + d = 12 + 3 = 15$

$A_5 = A_4 + d = 15 + 3 = 18$

$A_6 = A_5 + d = 18 + 3 = 21$

☞ GEOMETRIC PROGRESSION (G.P.)

A *geometirc progression* (G.P.) is the sequence of numbers, whose first term is non-zero and every other term is obtained by multiplying its preceding term by a constant quantity.

This constant quantity is called the *common ratio* of the G.P. For example, if a is the first and r is the common ratio of the G.P. then the G.P. can be written as

$$a, ar, ar^2, ar^3, \ldots, ar^{n-1}, \ldots (a \neq 0)$$

Note We usually denote

- the first term of a G.P. by a
- the last term of a G.P. by l
- the common ratio of a G.P. by r
- the number of rerms in a G.P. by n
- the sum of the first n terms of a G.P. by S_n

☞ GENERAL TERM OF A G.P.

If $t_n = n$th term of a G.P., then $t_n = ar^{n-1}$, where r is the common ratio such that $r = (a_k / a_{k-1})$.

Let a be the first term, l the last and r the common ratio of a G.P. The nth term of a G.P. from the beginning and the first term of the G.P. from the end are shown in the box below.

nth term of a G.P. from the beginning	First term of a G.P. from the end
$t_1 = a = ar^{1-1}$	$t_n = l$
$t_2 = ar = ar^{2-1}$	$t_{n-1} = \frac{1}{r} = \frac{1}{r^{n-(n-1)}}$
$t_3 = ar^2 = ar^{3-1}$	$t_{n-2} = \frac{1}{r^2} = \frac{1}{r^{n-(n-2)}}$
... ...	... ...
$t_{n-1} = ar^{n-2} = ar^{(n-1)-1}$	$t_2 = \frac{1}{r^{n-2}}$
$t_n = ar^{n-1}$	$t_1 = \frac{1}{r^{n-1}}$

Illustration 7. The first term of a G.P. is 1. The sum of the third and fifth terms is 90. Find the common ratio of the G.P.

Solution: Let r be the common ratio of the G.P. Then,

$t_3 = ar^2 \quad \Rightarrow \quad t_3 = (1)r^2 = r^2 \quad [\because a = 1]$

and $t_5 = ar^4 \quad \Rightarrow \quad t_5 = (1)(r)^4 = r^4 \quad [\because a = 1]$

Given, $t_3 + t_5 = 90 \quad \Rightarrow \quad r^2 + r^4 = 90$

$\Rightarrow \quad r^4 + r^2 - 90 = 0 \quad \Rightarrow \quad (r^2 - 9)(r^2 + 10) = 0$

$\Rightarrow \quad r^2 = 9 \quad \Rightarrow \quad r = \pm 3 \qquad [\because r^2 + 10 \neq 0]$

Hence, the common ratio is ± 3.

SUM OF *n* TERMS OF A G.P.

Sum of n terms of a G.P., $S_n = \begin{cases} \dfrac{a(r^n - 1)}{r - 1}, & \text{when } r > 1 \\ \dfrac{a(1 - r^n)}{1 - r}, & \text{when } r < 1 \end{cases}$

Illustration 8. How many terms of the G.P. $3, \frac{3}{2}, \frac{3}{4}, \ldots$ are needed to give the sum $\frac{3069}{512}$?

Solution: Let n be the number of terms needed. Given that $a = 3$, $r = \frac{1}{2}$ and $S_n = \frac{3069}{512}$.

Since $S_n = \dfrac{a(1 - r^n)}{1 - r}$

Therefore, $\dfrac{3069}{512} = \dfrac{3\left(1 - \frac{1}{2^n}\right)}{1 - \frac{1}{2}} = 6\left(1 - \frac{1}{2^n}\right)$

$\Rightarrow \quad \dfrac{3069}{3072} = 1 - \dfrac{1}{2^n}$

$\Rightarrow \quad \dfrac{1}{2^n} = 1 - \dfrac{3069}{3072} = \dfrac{3}{3072} = \dfrac{1}{1024}$

$\Rightarrow \quad 2^n = 1024 = 2^{10}$, which gives $n = 10$.

SUM OF INFINITE TERMS OF A G.P.

Case I: When $|r| < 1$

As $n \to \infty$ and $|r| < 1 \quad \Rightarrow \quad r^n \to 0$

$\therefore \qquad S_\infty = \dfrac{a}{1 - r}$

Case II: When $|r| > 1$. As $n \to \infty$ and $|r| > 1$

$\Rightarrow \qquad r^n \to \infty \quad \therefore \quad S_\infty = \infty$

Illustration 9. Prove that $9^{\frac{1}{3}} \cdot 9^{\frac{1}{9}} \cdot 9^{\frac{1}{27}} \ldots$ to $\infty = 3$.

Solution: We have $9^{\frac{1}{3}} \cdot 9^{\frac{1}{9}} \cdot 9^{\frac{1}{27}} \ldots \infty = 9^{\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots \infty} = 9^S$ (i)

where $S = \dfrac{1}{3} + \dfrac{1}{9} + \dfrac{1}{27} + \cdots + \infty$

This is an infinite G.S. with $a = \frac{1}{3}$, $r = \dfrac{t_2}{t_1} = \dfrac{\frac{1}{9}}{\frac{1}{3}} = \frac{1}{3}$

Since $|r| = \left|\frac{1}{3}\right| = \frac{1}{3} < 1$, therefore the sum up to infinity is as follows:

$$S_\infty = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2} \qquad \left[\because \; S_\infty = \frac{a}{1 - r}\right] \qquad \text{(ii)}$$

Putting this value of S in (i), we get

$$9^{\frac{1}{3}} \cdot 9^{\frac{1}{9}} \cdot 9^{\frac{1}{27}} \ldots \infty = 9^{\frac{1}{2}} = (3^2)^{\frac{1}{2}} = 3$$

PROPERTIES OF G.P.

Property 1: If each term of a G.P. is multiplied and divided by the same quantity which is non-zero, then the resulting series will also be a G.P., but if each term of the series is increased or decreased by the same quantity, the resulting series is not a G.P.

	Increased by k	Decreased by k	Multiplied by k	Divided by k ($\neq 0$)
First term	$a + k$	$a - k$	ak	$\frac{a}{k}$
Common ratio	Not possible	Not possible	r	r

Here, a ($\neq 0$) is the first term and r is the common ratio of $a, ar, ar^2, \ldots, ar^{n-1}, \ldots$ which is a G.P.

Property 2: If $a_1, a_2, a_3, \ldots, a_n$ and $b_1, b_2, b_3, \ldots, b_n$ are in G.P., then the sequence(s) $a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3, \ldots, a_n \pm b_n$ will not be a G.P.

Property 3: If $a_1, a_2, a_3, \ldots, a_n$ and $b_1, b_2, b_3, \ldots, b_n$ are in G.P., then the sequences

$$a_1 b_1, a_2 b_2, a_3 b_3, \ldots, a_n b_n \text{ and } \frac{a_1}{a_1}, \frac{a_2}{a_2}, \frac{a_3}{a_3}, \ldots, \frac{a_n}{a_n}$$

will also be in G.P.

Property 4: If each term of a G.P. is raised to one and same power, the resulting sequence also forms a G.P.

Property 5: If $a_1, a_2, a_3, \ldots$ are in G.P., then $\log a_1$, $\log a_2$, $\log a_3 \ldots$ are in A.P.

Property 6: Equal non-zero numbers are in G.P.

Property 7: *Selection of terms in a G.P.* For a sequence to be in G.P. with odd number of terms, take the common ratio as r, otherwise for a G.P. with even number of terms, take the common ratio as r^2.

Number of terms	Sequence
2	$\frac{a}{r}, ar$
3	$\frac{a}{r}, a, ar$
4	$\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$
5	$\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$
6	$\frac{a}{r^5}, \frac{a}{r^3}, \frac{a}{r}, ar, ar^2, ar^5$
7	$\frac{a}{r^3}, \frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2, ar^3$

Property 8: If $a_1, a_2, a_3, \ldots, a_{n-2}, a_{n-1}, a_n$ are in G.P., then

$$a_1 a_n = (a_2 a_{n-1}) = (a_3 a_{n-2}) = \cdots = [a_r, a_{n-(r-1)}]$$

In a G.P., the product of two equidistant terms from the beginning and end is a constant and equal to the product of the first term and the last term.

Illustration 10. The sum of the first three terms of a G.P. is 39/10 and their product is 1. Find the common ratio and the terms.

Solution: Let the first three terms of a G.P. be a/r, a, ar. Then,

$$\frac{a}{r} + a + ar = \frac{39}{10} \qquad \text{(i)}$$

and $$\frac{a}{r} \cdot a \cdot ar = 1 \quad \Rightarrow \quad a^3 = 1$$

$$\Rightarrow \qquad a = 1 \qquad \text{(ii)}$$

Putting the value of a from (ii) in (i), we get

$$\frac{1}{r} + 1 + r = \frac{39}{10} \quad \Rightarrow \quad 1 + r + r^2 = \frac{39r}{10}$$

$$\Rightarrow \quad r^2 - \frac{29}{10}r + 1 = 0 \quad \Rightarrow \quad 10r^2 - 29r + 10 = 0$$

$$\Rightarrow \quad 10r^2 - 25r - 4r + 10 = 0$$

$$\Rightarrow \quad 5r(2r-5) - 2(2r-5) = 0$$

$$\Rightarrow \quad (2r-5)(5r-2) = 0 \quad \Rightarrow \quad r = (5/2) \text{ and } (2/5)$$

$\Rightarrow$ G.P.: 2/5, 1, 5/2 and 5/2, 1, 2/5.

Illustration 11. The third term of a G.P. is 4. Find the product of its first five terms.

Solution: Let a be the first term and r be the common ratio of the given G.P., then

$$t_3 = 4 \quad \Rightarrow \quad ar^2 = 4 \qquad \text{(i)}$$

Now the product of the first five terms $= a \cdot ar \cdot ar^2 \cdot ar^3 \cdot ar^4$

$$= a^5 r^{10} = (ar^2)^5 = 4^5 \qquad \text{[using (i)]}$$

$$= 1024.$$

GEOMETRIC MEAN (G.M.)

If three numbers are in G.P., the middle one is called the *geometric mean* between the other two.

Inserting Single Geometric Mean between Two Numbers *a* and *b*

Let G be the geometric mean between a and b, then a, G, b are in G.P.

$$\therefore \qquad \frac{G}{a} = \frac{b}{G} \quad \text{or} \quad G^2 = ab \quad \Rightarrow \quad G = \sqrt{ab}$$

Inserting *n* Geometric Means between Two Given Numbers

Let a and b be the given numbers and $x_1, x_2, x_3, \ldots, x_n$ be n geometric means inserted between them. Then, by definition, $a, x_1, x_2, \ldots, x_n, b$ are in G.P.

$$\Rightarrow \qquad b = t_{n+2} = ar^{n+1}$$

$$\therefore \qquad r^{n+1}\left(\frac{b}{a}\right) \quad \Rightarrow \quad r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$$

Hence, $x_1 = a\left(\frac{b}{a}\right)^{\frac{1}{n+1}}$, $x_2 = a\left(\frac{b}{a}\right)^{\frac{2}{n+1}}, \ldots,$ $x_n = a\left(\frac{b}{a}\right)^{\frac{n}{n+1}}$

Note

- The product of n geometric means between a and b is equal to the nth power of geometric mean of a and b,

  i.e. $\qquad G_1 G_2 \ldots G_n = (ab)^{n/2} = G^n$
- The geometric mean of n positive numbers $a_1, a_2, a_3, \ldots, a_n$ is $(a_1 a_2 a_3 \ldots a_n)^{1/n}$.
- If A and G are the A.M. and G.M. between two positive numbers, then the numbers are $A \pm \sqrt{A^2 - G^2}$.
- If A and G are respectively A.M. and G.M. between two positive numbers a and b, then the quadratic equation having a, b as its roots is $x^2 - 2Ax + G^2 = 0$.

Illustration 12. If x, y, z are three positive numbers forming a geometric sequence, then show that $\log_a x, \log_a y, \log_a z$ form an arithmetic sequence, where a is positive not equal to 1.

Solution: We are given that x, y, z are in G.P.

$$\Rightarrow \qquad \frac{y}{x} = \frac{z}{y} \quad \Rightarrow \quad y^2 = zx$$

Taking log on both sides (with respect to the base a), we get

$$\log_a (y^2) = \log_a (zx)$$

$$\Rightarrow \qquad 2\log_a y = \log_a z + \log_a x$$

$\Rightarrow$ $\log_a x$, $\log_a y$, $\log_a z$ are in A.P.

☞ HARMONIC PROGRESSION (H.P.)

A sequence of non-zero numbers is said to form a harmonic progression (H.P.) if the reciprocals of its terms are in arithmetic progression. For example, the sequences

(i) $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \cdots$ and

(ii) $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \ldots$

are harmonic progressions as the sequences formed by their reciprocals $a, a+d, a+2d, \ldots$ and 2, 4, 6,... are arithmetic progressions.

Illustration 13. If $a>1, b>1, c>1$ are in G.P., then prove that $\frac{1}{1+\log a}, \frac{1}{1+\log b}, \frac{1}{1+\log c}$ are in H.P.

Solution: $\because$ a, b, c are in G.P.,

$\therefore \quad b^2 = ac \quad \Rightarrow \quad 2\log b = \log a + \log c$

$\Rightarrow \quad 2(1+\log b) = (\log a + 1) + (1 + \log c)$

$\Rightarrow$ $(1+\log a), (1+\log b), (1+\log c)$ are in A.P.

$[\because 2B = A + C]$

$\Rightarrow \frac{1}{1+\log a}, \frac{1}{1+\log b}, \frac{1}{1+\log c}$ are in H.P.

Note We usually denote

- the first term of an H.P. by $1/a$
- the corresponding term of an A.P. by a
- the common difference of the corresponding terms of an A.P. by d
- the nth term of an H.P. is the reciprocal of the nth term of the corresponding A.P.

*n*th Term in an H.P.

Consider the H.P. $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \cdots$

The corresponding A.P. is $a, a+d, a+2d, \ldots$

The nth term of A.P. $= a + (n-1)d$

$\therefore$ The nth term of H.P. $= \frac{1}{a+(n-1)d}$

☞ HARMONIC MEAN (H.M.)

If three numbers are in H.P., then the middle one is called the *harmonic mean* between the other two.

Inserting a Single Harmonic Mean between Two Numbers *a* and *b*

If H is the harmonic mean between a and b, then a, H, b are in H.P., i.e. $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ are in A.P.

$\Rightarrow \quad \frac{1}{H} = \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a+b}{2ab} \quad \therefore \quad H = \frac{2ab}{a+b}$

Inserting *n* Harmonic Means between *a* and *b*

If $x_1, x_2, x_3, \ldots, x_n$ are n harmonic means between a and b, then $a, x_1, x_2, \ldots, x_n, b$ are in H.P.

$\therefore \frac{1}{a}, \frac{1}{x_1}, \frac{1}{x_2}, \ldots, \frac{1}{x_n}, \frac{1}{b}$ are in A.P. containing $(n+2)$ terms

$\Rightarrow \quad \frac{1}{b} = \frac{1}{a} + (n+1)d \quad \Rightarrow \quad d = \frac{1}{n+1}\left(\frac{1}{b} - \frac{1}{a}\right) = \frac{a-b}{ab(n+1)}.$

$\therefore \quad \frac{1}{x_1} = \frac{1}{a} + d = \frac{1}{a} + \frac{a-b}{ab(n+1)},$

$\frac{1}{x_2} = \frac{1}{a} + \frac{2(a-b)}{ab(n+1)}, \ldots, \frac{1}{x_n} = \frac{1}{a} + \frac{n(a-b)}{ab(n+1)}$

Thus, the required n harmonic means are

$$\frac{(n+1)ab}{a+bn}, \frac{(n+1)ab}{2a+b(n-1)}, \ldots, \frac{(n+1)ab}{na+b}$$

Illustration 14. Find the sum of the following series:

$$10^3 + 11^3 + 12^3 + \cdots + 20^3.$$

Solution: Let S be the required sum, then

$S = 10^3 + 11^3 + 12^3 + \cdots + 20^3$

$= (1^3 + 2^3 + 3^3 + \cdots + 20^3) - (1^3 + 2^3 + 3^3 + \cdots + 9^3)$

$= \sum_{n=1}^{20} n^3 - \sum_{n=1}^{9} n^3 \quad \because \sum n^3 = \left[\frac{n(n+1)}{2}\right]^2$, we get

$= \left(\frac{20(20+1)}{2}\right)^2 - \left(\frac{9(9+1)}{2}\right)^2 = \left(\frac{20\times 21}{2}\right)^2 - \left(\frac{9\times 10}{2}\right)^2$

$= (210)^2 - (45)^2 = (210+45)(210-45)$

$= 255 \times 165 = 42{,}075$

☞ RELATION BETWEEN A.M., G.M. AND H.M.

Let a and b be two real positive and unequal numbers, then their arithmetic mean, geometric mean and harmonic mean, between them are respectively,

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}, \quad H = \frac{2ab}{a+b}$$

Now, $AH = \left(\frac{a+b}{2}\right)\left(\frac{2ab}{a+b}\right) = ab = G^2 \Rightarrow G^2 = AH$ (i)

i.e. G is the geometric mean between A and H. Again

$$A - G = \frac{a+b}{2} - \sqrt{ab} = \frac{a+b-2\sqrt{ab}}{2} = \left(\frac{\sqrt{a}-\sqrt{b}}{\sqrt{2}}\right)^2 > 0$$

$\therefore \quad A - G > 0 \quad (\because a \text{ and } b \text{ are positive}) \quad \Rightarrow \quad A > G \quad$ (ii)

From (i), $\frac{G}{H} = \frac{A}{G} > 1 \quad \Rightarrow \quad \frac{G}{H} > 1 \quad \Rightarrow \quad G > H \quad$ (iii)

From (ii) and (iii), $A > G > H$. Hence, G lies between A and H.

Note If the two numbers are equal, i.e. $a = b$, then

$$A - G = \left(\frac{\sqrt{a} - \sqrt{b}}{\sqrt{2}}\right)^2 = 0$$

$\therefore \quad A = G$, hence from (i), $\frac{G}{H} = \frac{A}{G} = 1$

$\therefore \quad G = H$. Hence, $A = G = H$.

Thus, $A \geq G \geq H$ inequality holds only when $a = b$.

Illustration 15. If the A.M. and H.M. between two numbers are 27 and 12, respectively, then find their G.M.

Solution: Let A, G and H denote respectively the A.M., G.M. and H.M. between the two numbers.

Then, $A = 27$ and $H = 12$.

Since A, G, H are in G.P., therefore, $G^2 = AH$

$\Rightarrow \qquad G^2 = 27 \times 12 \quad \Rightarrow \quad G = 18.$

☞ ARITHMETICO-GEOMETRIC SERIES

A series in which each term is formed by the product of the corresponding terms of an A.P. and a G.P. is called an *arithmetico-geometric series*. For example, the series

(i) $a + (a+d)r + (a+2d)r^2 + \cdots + [a + (n-1)d]\, r^{n-1} + \cdots$

(ii) $1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots$ to n terms

are A.G. (arithetico-geometric) series, where a, $(a + d)$, $(a + 2d)$, …, $[a + (n-1)d]$… and 1, 2, 3,… are in A.P. and $1, r, r^2, \ldots$ and $2, 2^2, 2^3, \ldots$ are in G.P.

☞ SUM OF THE FIRST *n* TERMS OF AN ARITHMETICO-GEOMETRIC SERIES

Let $a, (a+d)r, (a+2d)r^2, \ldots, [a+(n-1)d]r^{n-1}$ be the n terms in an A.G., then the sum of the series given by S is

$$S_n = a + (a+d)r + (a+2d)r^2 + \cdots + [a + (n-1)d]r^{n-1}$$

$$rS_n = ar + (a+d)\, r^2 + \cdots + [a + (n-2)d]r^{n-1} + [a + (n-1)d]r^n$$

$$\Rightarrow \quad S_n(1-r) = a + d(r + r^2 + \cdots + r^{n-1}) - [a + (n-1)d]r^n$$

$$\Rightarrow \qquad S_n(1-r) = a + dr\frac{(1 - r^{n-1})}{1-r} - [a + (n-1)d]r^n$$

$$\therefore \qquad S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{1-r}$$

$$S_\infty = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{1-r}$$

If $|r| < 1$ and $n \to \infty \quad \Rightarrow \quad r^n \to 0$, i.e. $\lim_{n\to\infty} r^n = 0$

$\therefore \qquad S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

☞ SIGMA NOTATION

(a) $\sum n = 1 + 2 + 3 + \cdots + n = \frac{n}{2}(n+1)$

(b) $\sum n^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

(c) $\sum n^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n}{2}(n+1)\right]^2$

(d) $\sum n^4 = 1^4 + 2^4 + \cdots + n^4 = \frac{n(n+1)(2n+1)}{6}\left(\frac{3n^2 + 3n - 1}{5}\right)$

(e) Sum of the product of n numbers taken two at a time

$$= \frac{n(n^2-1)(3n+2)}{24}$$

Note

- $\frac{1}{a_1 a_2 \cdots a_r} + \frac{1}{a_2 a_3 \cdots a_{r+1}} + \cdots + \frac{1}{a_n a_{n+1} \cdots a_{n+r-1}}$

  $$= \frac{1}{(r-1)(a_2 - a_1)}\left[\frac{1}{a_1 a_2 \ldots a_{r-1}} - \frac{1}{a_{n+1} a_{n+2} \ldots a_{n+r-1}}\right]$$

- $(a_1 a_2 \ldots a_r) + (a_2 a_3 \ldots a_{r+1}) + (a_n a_{n+1} \ldots a_{n+r-1})$

  $$= \frac{1}{(r+1)(a_2 - a_1)}[a_n a_{n+1} \ldots a_{n+r} - a_0 a_1 a_2 \ldots a_n]$$

  where $a_1, a_2 \ldots a_n$ are in A.P. and $a_0 = a_1 - d$

- $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

- $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)}$

  $$= \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$$

- $1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}$

- $1 \cdot 2 \cdot 3 \cdot 4 + 2 \cdot 3 \cdot 4 \cdot 5 + \cdots + n(n+1)(n+2)(n+3)$

  $$= \frac{1}{5} n(n+1)(n+2)(n+3)(n+4)$$

☞ TO FIND THE *n*th TERM BY THE DIFFERENCE METHOD

If $t_1, t_2, \ldots, t_n$ are the terms of any series and their differences $(t_2 - t_1), (t_3 - t_2), (t_4 - t_3), \ldots, (t_n - t_{n-1})$ are either in A.P. or in G.P., then t_n can be easily found out. Thus, for finding t_n, the sum of the series can be calculated by the $\sum$ notation.

This method of finding t_n is known as the *difference method*. Sometimes this method is used several times.

Illustration 16. The sum of the infinite series

$1+\frac{4}{3}+\frac{9}{3^2}+\frac{16}{3^3}+\cdots+\infty$ is

(a) 4/2 (b) 9/2
(c) 4/9 (d) 1

Ans. (b)

Solution: This is clearly not an arithmetico-geometric series, since 1, 4, 9, 16,... are not in A.P. However, their successive differences (4 – 1), (9 – 4), (16 – 9),... are in A.P.

Let $S_\infty = 1+\frac{4}{3}+\frac{9}{3^2}+\frac{16}{3^3}+\cdots+\infty$

$\frac{1}{3}S_\infty = \frac{1}{3}+\frac{4}{3^2}+\frac{9}{3^3}+\cdots+\infty$

On subtraction, we get

$\frac{2}{3}S_\infty = 1+\frac{3}{3}+\frac{5}{3^2}+\frac{7}{3^3}+\cdots+\infty$

$\frac{1}{3}\cdot\frac{2}{3}S_\infty = \frac{1}{3}+\frac{3}{3^2}+\frac{5}{3^3}+\cdots+\infty$

[multiplying both sides by 1/3]

On subtracting the two series, we get

$$\left(\frac{4}{9}\right)\cdot S_\infty = 1+\frac{2}{3}+\frac{2}{3^2}+\frac{2}{3^3}+\cdots+\infty$$

$$=1+\frac{2}{3}\left(1+\frac{1}{3}+\frac{1}{3^2}+\cdots+\infty\right)=1+\frac{2}{3}\times\frac{1}{\left(1-\frac{1}{3}\right)}=2$$

$\therefore \quad S_\infty = \left(2\times\frac{9}{4}\right)=\frac{9}{2}$

SOLVED PROBLEMS

1. In an A.P., the pth term is q and the $(p + q)$th term is 0. Then the qth term is

(a) $-p$ (b) p
(c) $p + q$ (d) $p - q$

Ans. (b)

Solution: $t_p = a+(p-1)d = q$ (i)

$t_{p+q} = a+(p+q-1)d = 0$ (ii)

(ii)–(i) gives $qd = -q \Rightarrow d = -1$

$\therefore$ Now $a+(q-1)d+pd = 0$ [by (ii)]

$\Rightarrow \quad t_q - p = 0 \Rightarrow t_q = p$

2. If a, b, c are in A.P., then $10^{ax+10}, 10^{bx+10}, 10^{cx+10}$ are in

(a) A.P. (b) G.P. when $x > 0$
(c) G.P. for all x (d) G.P. when $x < 0$

Ans. (c)

Solution: Here, $2b = a + c$

$\Rightarrow \quad 2bx = ax + cx$ for all x

$\Rightarrow \quad 2(bx+10) = (ax+10)+(cx+10)$

$\Rightarrow \quad 10^{2bx+20} = 10^{ax+10}\cdot 10^{cx+10}$

$\Rightarrow \quad (10^{bx+10})^2 = 10^{ax+10}\cdot 10^{cx+10}$

$\therefore \quad 10^{ax+10}, 10^{bx+10}, 10^{cx+10}$ are in G.P. for all x.

3. Let S_n denote the sum to n terms of an A.P.

Let $S_n = n^2p$, $S_m = m^2p$, where m, n, p are +ve integers and $m \neq n$. Then, $S_p =$

(a) $\frac{(m+n)p^2}{mn}$ (b) $\frac{mnp^2}{m+n}$
(c) p^3 (d) $\frac{(p+m+n)^2}{pmn}$

Ans. (c)

Solution: $S_n = \frac{n}{2}[2a+(n-1)d] = n^2p$

$\Rightarrow \quad 2a+(n-1)d = 2np$ (i)

$S_m = \frac{m}{2}[2a+(m-1)d] = m^2p$

$\Rightarrow \quad 2a+(m-1)d = 2mp$ (ii)

(i)–(ii) gives $(n-m)d = 2(n-m)p$

$\therefore \quad d = 2p$

$\therefore$ (i) gives $2a + (n-1)2p = 2np$

$\Rightarrow \quad 2a = 2p \Rightarrow a = p$

Now, $S_p = \frac{p}{2}[2a+(p-1)d]$

$=\frac{p}{2}[2p+(p-1)2p]$

$=\frac{p}{2}\cdot 2p\cdot p = p^3$

4. If x, y, z are the pth, qth and rth terms, respectively, of an A.P. and also of a G.P., then $x^{y-z}y^{z+x}z^{x+y}$ is equal to

(a) xyz (b) 0
(c) 1 (d) none of these

Ans. (c)

Solution: Let a be the first term and d be the common difference of the A.P. Then

$x = a+(p-1)d$, $y = a+(q-1)d$ and $z = a+(r-1)d$

Let a be the first term and r the common ratio of the G.P., then $x = ar^{p-1}$, $y = ar^{q-1}$, $z = ar^{z-1}$.

Let $E = (x^{y-z} y^{z-x} z^{x-y})$

$\Rightarrow \quad E = (ar^{p-1})^{(q-r)d} (ar^{q-1})^{(r-p)d} (ar^{r-1})^{(p-q)d}$

$\therefore \quad E = a^0 r^0 = 1$

5. If the roots of the equation $(b-c)x^2 + (c-a)x + (a-b) = 0$ are equal, then a, b, c are in

(a) A.P. (b) G.P.

(c) H.P (d) none of these

Ans. (a)

Solution: For equal roots, the discriminant D is

$D = (c-a)^2 - 4(b-c)(a-b) = 0$

$\Rightarrow \quad c^2 + a^2 - 2ca - 4(ab - b^2 - ca + bc) = 0$

$\Rightarrow \quad c^2 + a^2 + 2ca + 4b^2 - 4ab - 4bc = 0$

$\Rightarrow \quad (c + a - 2b)^2 = 0 \quad \Rightarrow \quad c + a - 2b = 0$

$\Rightarrow \quad 2b = c + a \quad \Rightarrow \quad a, b, c$ are in A.P.

6. If $a_1, a_2, a_3, \ldots$ is an A.P. such that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + \cdots + a_{23} + a_{24}$ is equal to

(a) 909 (b) 75

(c) 750 (d) 900

Ans. (d)

Solution: Since $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$

We know that the sum of the terms equidistant from the beginning and end are equal.

$\Rightarrow \quad (a_1 + a_{24}) + (a_5 + a_{20}) + (a_{10} + a_{15}) = 225$

$\Rightarrow \quad 3(a_1 + a_{24}) = 225 \quad \Rightarrow \quad a_1 + a_{24} = \dfrac{225}{3}$

Now, the sum of the first 24 terms in A.P. is given by

$$a_1 + a_2 + a_3 + \cdots + a_{24} = \frac{24}{2}(a_1 + a_{24})$$

$$\therefore \quad a_1 + a_2 + a_3 + \cdots + a_{24} = 12\left(\frac{225}{3}\right) = 900$$

7. If $\log_{10} 2$, $\log_{10}(2^x - 1)$ and $\log_{10}(2^x + 3)$ are three consecutive terms of an A.P., then

(a) $x = 0$ (b) $x = 1$

(c) $x = \log_2 5$ (d) none of these

Ans. (c)

Solution: By the given condition,

$$2\log_{10}(2^x - 1) = \log_{10} 2 + \log_{10}(2^x + 3)$$

$\Rightarrow \quad (2^x - 1)^2 = 2(2^x + 3) \quad \Rightarrow \quad 2^{2x} - 2.2^x + 1 = 2.2^x + 6$

$\Rightarrow \quad 2^{2x} - 4 \cdot 2^x - 5 = 0 \quad \Rightarrow \quad (2^x - 5)(2^x + 1) = 0$

$\Rightarrow \quad 2^x = 5$ or $2^x = -1$

But $2^x = -1$ is not possible.

$\therefore \quad 2^x = 5 \quad \Rightarrow \quad \log_2 2^x = \log_2 5$

$\Rightarrow \quad x \log_2 2 = \log_2 5 \quad \Rightarrow \quad x = \log_2 5 \quad [\because \log_2 2 = 1]$

8. The first and last terms of an A.P. are a and l respectively. If S be the sum of all the terms of the A.P., then the common difference is

(a) $\dfrac{l^2 - a^2}{2S - (l + a)}$ (b) $\dfrac{l^2 - a^2}{2S - (l - a)}$

(c) $\dfrac{l^2 + a^2}{2S + (l + a)}$ (d) $\dfrac{l^2 + a^2}{2S - (l + a)}$

Ans. (a)

Solution: Let n be the number of terms and d be the common difference of the A.P. Then

$$l = a + (n-1)d \quad \text{and} \quad S = \frac{n}{2}(a + l)$$

$\Rightarrow \quad \dfrac{l - a}{n - 1} = d$, where $n = \dfrac{2S}{a + l}$

$$\therefore \quad d = \frac{l - a}{\dfrac{2S}{l + a} - 1} = \frac{l^2 - a^2}{2S - (l + a)}$$

9. $a^x = b^y = c^z = d^t$ and a, b, c, d are in G.P., then x, y, z, t are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

Ans. (c)

Solution: Let r be the common ratio of the G.P. $a, b, c, d, \ldots$

$\therefore \quad b = ar; \; c = ar^2; \; d = ar^3$

Again $a^x = b^y = c^z = d^t$

$\Rightarrow \quad x \log a = y \log b = z \log c = t \log d$

$$\Rightarrow \quad \frac{x}{y} = \frac{\log b}{\log a} = \frac{\log ar}{\log a} = 1 + \frac{\log r}{\log a} \qquad \text{(i)}$$

Similarly $$\frac{x}{z} = 1 + 2\frac{\log r}{\log a} \qquad \text{(ii)}$$

and $$\frac{x}{t} = 1 + 3\frac{\log r}{\log a} \qquad \text{(iii)}$$

By (i) and (ii), $\dfrac{x}{z} = 1 + 2\left(\dfrac{x}{y} - 1\right)$

$$\Rightarrow \quad \frac{1}{z} = \frac{1}{x} + 2\left(\frac{1}{y} - \frac{1}{x}\right) = \frac{2}{y} - \frac{1}{x}$$

$$\Rightarrow \quad \frac{1}{x} + \frac{1}{z} = \frac{2}{y} \qquad \text{(iv)}$$

By (i), (ii) and (iii), we have

$$\frac{x}{y} + \frac{x}{t} = 2 + 4\frac{\log r}{\log a} = 2\left(1 + 2\frac{\log r}{\log a}\right) = 2\frac{x}{z}$$

$$\Rightarrow \quad \frac{1}{y} + \frac{1}{t} = \frac{2}{z} \qquad \text{(v)}$$

From (iv) and (v), we can say that $\frac{1}{x},\frac{1}{y},\frac{1}{z},\frac{1}{t}$ are in A.P.

$\Rightarrow$ x, y, z, t are in H.P.

10. If $\frac{3+5+7+\cdots+n \text{ terms}}{5+8+11+\cdots+10 \text{ terms}}=7$, then the value of n is

(a) 35 (b) 26
(c) 37 (d) 40

Ans. (a)

Solution: $$\frac{3+5+7+\cdots+n \text{ terms}}{5+8+11+\cdots+10 \text{ terms}}=7$$

$$\Rightarrow \frac{\frac{n}{2}[6+(n-1)2]}{\frac{10}{2}[10+(10-1)3]}=7 \Rightarrow \frac{n(n+2)}{5(37)}=7$$

$$\Rightarrow n^2+2n=35\times 37 \Rightarrow (n+37)(n-35)=0$$

$\therefore$ $n=35$ as $n=-37$ is not admissible.

11. Let S_1, S_2, S_3 be the sum of n terms of three series in A.P., the first term of each being 1 and the common difference being 1, 2, 3 respectively. If $S_1+S_3=\lambda S_2$, then the value of λ is

(a) 1 (b) 2
(c) 3 (d) none of these

Ans. (b)

Solution: $S_1=\frac{n}{2}[2+(n-1)1]=\frac{n}{2}(n+1)$

$$S_2=\frac{n}{2}[2+(n-1)2]=n(n)=n^2$$

$$S_3=\frac{n}{2}[2+(n-1)3]=\frac{n}{2}(3n-1)$$

Since $S_1+S_3=\lambda S_2$

$\therefore$ $$\frac{n}{2}(n+1+3n-1)=\lambda n^2$$

$\Rightarrow$ $2n^2=\lambda n^2 \Rightarrow \lambda=2$

12. If $a_1, a_2, a_3, \ldots, a_n$ are in A.P., where $a_i > 0$ for all i, then the value of

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\cdots+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}} \text{ is}$$

(a) $\frac{1}{\sqrt{a_1}+\sqrt{a_n}}$ (b) $\frac{1}{\sqrt{a_1}-\sqrt{a_n}}$
(c) $\frac{n}{\sqrt{a_1}-\sqrt{a_n}}$ (d) $\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$

Ans. (d)

Solution: $a_1, a_2, a_3, \ldots, a_n$ are in A.P.

Let the common difference be d. Also, let

$$E=\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\cdots+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$$

Rationalizing, we have

$$\Rightarrow E=\frac{\sqrt{a_1}-\sqrt{a_2}}{a_1-a_2}+\frac{\sqrt{a_2}-\sqrt{a_3}}{a_2-a_3}+\cdots+\frac{\sqrt{a_{n-1}}-\sqrt{a_n}}{a_{n-1}-a_n}$$

$$\Rightarrow E=\frac{\sqrt{a_1}-\sqrt{a_2}}{-d}+\frac{\sqrt{a_2}-\sqrt{a_3}}{-d}+\cdots+\frac{\sqrt{a_{n-1}}-\sqrt{a_n}}{-d}$$

$$\Rightarrow E=-\frac{1}{d}(\sqrt{a_1}-\sqrt{a_2}+\sqrt{a_2}-\sqrt{a_3}+\cdots+\sqrt{a_{n-1}}-\sqrt{a_n})$$

$$\Rightarrow E=-\frac{1}{d}(\sqrt{a_1}-\sqrt{a_n})=-\frac{1}{d}\left(\frac{a_1-a_n}{\sqrt{a_1}+\sqrt{a_n}}\right)$$

$$\Rightarrow E=-\frac{1}{d}\left\{\frac{a_1-[a_1+(n-1)d]}{\sqrt{a_1}+\sqrt{a_n}}\right\}$$

$\therefore$ $$E=\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$$

13. If every term of a G.P. is positive and also if every term is the sum of the two preceding terms, then the common ratio of the G.P. is

(a) 1 (b) $\frac{\sqrt{5}-1}{2}$
(c) $\frac{\sqrt{5}+1}{2}$ (d) $\frac{1-\sqrt{5}}{2}$

Ans. (c)

Solution: $t_3=t_1+t_2 \Rightarrow ar^2=a+ar$

$$\Rightarrow r^2=1+r \Rightarrow r^2-r-1=0$$

$$\Rightarrow r=\frac{1\pm\sqrt{1+4}}{2}=\frac{1\pm\sqrt{5}}{2} \quad \therefore \quad r=\frac{1+\sqrt{5}}{2}$$

$$\left[\because \text{ terms are +ve, and } \frac{1-\sqrt{5}}{2}<0\right]$$

14. If the sums of the first p terms, first q terms and first r terms of an A.P. are x, y and z respectively, then

$$\frac{x}{p}(q-r)+\frac{y}{q}(r-p)+\frac{z}{r}(p-q)=$$

(a) 0 (b) 2
(c) pqr (d) $\frac{8xyz}{pqr}$

Ans. (a)

Solution: Let a be the first term and d be the common difference of the A.P. Since it is given that

$$x=\frac{p}{2}[2a+(p-1)d]$$

$$y=\frac{q}{2}[2a+(q-1)d]$$

and $$z=\frac{r}{2}[2a+(r-1)d]$$

$\therefore$ $$\frac{2x}{p}=2a+(p-1)d \quad \text{(i)}$$

$$\frac{2y}{q}=2a+(q-1)d \quad \text{(ii)}$$

$$\frac{2z}{r}=2a+(r-1)d \quad \text{(iii)}$$

Multiplying (i), (ii) and (iii) by $(q-r)$, $(r-p)$ and $(p-q)$ respectively and adding, we get

$$\frac{2x}{p}(q-r)+\frac{2y}{q}(r-p)+\frac{2z}{r}(p-q)=0$$

$\therefore$ $$\frac{x}{p}(q-r)+\frac{y}{q}(r-p)+\frac{z}{r}(p-q)=0$$

15. Let a, b, c be in A.P. and $|a|<1$, $|b|<1$, $|c|<1$.

If $x=1+a+a^2+\cdots+\infty$, $y=1+b+b^2+\cdots+\infty$

$z=1+c+c^2+\cdots+\infty$, then x, y, z are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

Ans. (c)

Solution: $x=\frac{1}{1-a}$, $y=\frac{1}{1-b}$, $z=\frac{1}{1-c}$

$$\left[\because \text{ sum of G.P. to } \infty=\frac{a}{1-r}\right]$$

Now, a, b, c are in A.P. $\Rightarrow$ $1-a, 1-b, 1-c$ are in A.P.

$=\frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ are in H.P. $\Rightarrow$ x, y, z are in H.P.

16. If $x^{18}=y^{21}=z^{28}$, then

$3, 3\log_y x, 3\log_z y, 7\log_x z$ are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

Ans. (a)

Solution: Let $x^{18}=y^{21}=z^{28}=k$

Taking log, we have

$$18\log x=21\log y=28\log z=\log k$$

$$\frac{\log x}{\log y}=\frac{21}{18}, \frac{\log y}{\log z}=\frac{28}{21}, \frac{\log z}{\log x}=\frac{18}{28}$$

$$\Rightarrow \quad \log_y x=\frac{21}{18}, \log_z y=\frac{28}{21}, \log_x z=\frac{18}{28}$$

$$\Rightarrow \quad 3\log_y x=\frac{7}{2}, 3\log_z y=4, 7\log_x z=\frac{9}{2}$$

$\therefore$ $3, 3\log_y x, 3\log_z y, 7\log_x z$ are in A.P.

17. The 10th common term between the series $3 + 7 + 11 + \cdots$ and $1 + 6 + 11 + \cdots$ is

(a) 191 (b) 193

(c) 211 (d) none of these

Ans. (a)

Solution: t_n of $3 + 7 + 11 + \cdots = 3 + (n-1)4 = 4n - 1$

t_m of $1 + 6 + 11 + \cdots = 1 + (m-1)5 = 5m - 4$

For a common term, $4n-1=5m-4$, i.e. $4n=5m-3$

For $n = 3$, $m = 3$ gives the first common term, i.e. 11.

$n = 8$, $m = 7$ gives the second common term, i.e. 31.

$n = 13$, $m = 11$ gives the third common term, i.e. 51.

$n = 18$, $m = 15$ gives the fourth common term, i.e. 71.

Now, the 10th term of 3, 8, 13, 18,... $= 3 + (10-1)5 = 48$

For the 10th common term, $n = 48$.

$\therefore$ Reqd. common term $= 4(48) - 1$

$= 192 - 1 = 191$.

18. If $\frac{x+y}{1-xy}$, y, $\frac{y+z}{1-yz}$ are in A.P., then

$x, \frac{1}{y}, z$ will be in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

Ans. (c)

Solution: Since $\frac{x+y}{1-xy}$, y, $\frac{y+z}{1-yz}$ are in A.P.

$$\Rightarrow \quad y-\frac{x+y}{1-xy}=\frac{y+z}{1-yz}-y$$

$$\Rightarrow \quad \frac{y-xy^2-x-y}{1-xy}=\frac{y+z-y+y^2z}{1-yz}$$

$$\Rightarrow \quad \frac{-x}{1-xy}=\frac{z}{1-yz}$$

$$\Rightarrow \quad -x+xyz=z-xyz$$

$$\Rightarrow \quad 2xyz=x+z \quad \Rightarrow \quad y=\frac{x+z}{2xz}$$

$$\Rightarrow \quad \frac{1}{y}=\frac{2xz}{x+z} \quad \therefore \quad x, \frac{1}{y}, z \text{ are in H.P.}$$

19. If the third term of a G.P. is 4, then the product of the first five terms is

(a) 4^3 (b) 4^5

(c) 4^4 (d) none of these

Ans. (b)

Solution: $t_3 = ar^2 = 4$

$$t_1 \cdot t_2 \cdot t_3 \cdot t_4 \cdot t_5 = a \cdot ar \cdot ar^2 \cdot ar^3 \cdot ar^4$$

$$= a^5r^{10} = (ar^2)^5 = 4^5$$

20. If $a > 0, b > 0, c > 0$ and the minimum value of $a(b^2+c^2)+b(c^2+a^2)+c(a^2+b^2)$ is λabc, then λ is

(a) 2 (b) 1

(c) 6 (d) 3

Ans. (c)

Solution: Consider the numbers ab^2, ac^2, bc^2, ba^2, cb^2, ca^2.

Since A.M. ≥ G.M.,

$$\therefore \quad \frac{ab^2+ac^2+bc^2+ba^2+ca^2+cb^2}{6} \geq (ab^2.ac^2.bc^2.ba^2.ca^2.cb^2)^{1/6}$$

$$\Rightarrow \quad a(b^2+c^2)+b(c^2+a^2)+c(a^2+b^2) \geq 6(a^6b^6c^6)^{1/6} = 6abc$$

$\therefore$ Min. value of $a(b^2+c^2)+b(c^2+a^2)+c(a^2+b^2) = 6abc$

$\therefore \quad \lambda = 6.$

EXERCISE SET 1

1. If the interior angles of a polygon are in A.P. with common difference 5° and the smallest angle is 120°, then the number of sides of the polygon is

(a) 9 or 16 (b) 9

(c) 16 (d) 13

2. If a, b, c, d are in G.P., then $(a^3+b^3)^{-1}$, $(b^3+c^3)^{-1}$ and $(c^3+d^3)^{-1}$ are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

3. If $a_1, a_2, a_3, \ldots$ are in A.P., and t_p, t_q, t_r are also in A.P., then p, q, r are in

(a) H.P. (b) G.P.

(c) A.P. (d) none of these

4. In an A.P., if $t_m = \frac{1}{n}$, $t_n = \frac{1}{m}$, then $a - d$ is equal to

(a) 0 (b) 1

(c) $\frac{1}{mn}$ (d) $\frac{1}{m+n}$

5. If a, b, c are in A.P., p, q, r are in H.P. and ap, bq, cr are in G.P., then $\frac{p}{r}+\frac{r}{p}$ is equal to

(a) $\frac{a}{c}-\frac{c}{a}$ (b) $\frac{a}{c}+\frac{c}{a}$

(c) $\frac{b}{q}+\frac{q}{b}$ (d) $\frac{b}{q}-\frac{q}{b}$

6. In a geometric progression, if the $(p + q)$th term is m and the $(p - q)$th term is n, then its pth term is

(a) 0 (b) mn

(c) $\sqrt{mn}$ (d) $\frac{1}{2}(m+n)$

7. If $A = 1 + r^a + r^{2a} + r^{3a} + \cdots + \infty$ and $B = 1 + r^b + r^{2b} + r^{3b} + \cdots + \infty$, then $\frac{a}{b}$ is equal to

(a) $\log_{1-B}(1-A)$ (b) $\log_{\frac{B-1}{B}}\left(\frac{A-1}{A}\right)$

(c) $\log_B A$ (d) none of these

8. If the sum of n terms of an A.P. is $3n^2+5n$ and $t_m = 164$, then $m =$

(a) 26 (b) 27

(c) 28 (d) none of these

9. If the A.M. and G.M. between two numbers are in the ratio $m : n$, then the numbers are in the ratio

(a) $m+\sqrt{n^2-m^2} : m-\sqrt{n^2-m^2}$

(b) $m+\sqrt{m^2+n^2} : m-\sqrt{m^2+n^2}$

(c) $m+\sqrt{m^2-n^2} : m-\sqrt{m^2-n^2}$

(d) none of these

10. Let $f(x) = 2x+1$. Then the number of real values of x for which the three unequal numbers $f(x)$, $f(2x)$, $f(4x)$ are in G.P. is

(a) 1 (b) 2

(c) 0 (d) none of these

11. The number of terms common between the series $1+2+4+8+\ldots$ to 100 terms and $1+4+7+10+\ldots$ to 100 terms is

(a) 6 (b) 4

(c) 5 (d) none of these

12. If the sides of a triangle are in G.P., and its larger angle is twice the smallest, then the common ratio r satisfies the inequality

(a) $0 < r < \sqrt{2}$ (b) $1 < r < \sqrt{2}$

(c) $1 < r < 2$ (d) none of these

13. In an A.P. of which a is the first term, if the sum of the first p terms is zero, then the sum of the next q terms is

(a) $-\frac{a(p+q)p}{q+1}$ (b) $\frac{a(p+q)p}{p+1}$

(c) $-\frac{a(p+q)q}{p-1}$ (d) none of these

14. The harmonic mean of $\frac{a}{1-ab}$ and $\frac{a}{1+ab}$ is

(a) a (b) $\frac{a}{1-a^2b^2}$

(c) $\frac{1}{1-a^2b^2}$ (d) $\frac{a}{\sqrt{1-a^2b^2}}$

15. Let $\{a_n\}$ be a G.P. such that $\frac{a_4}{a_6}=\frac{1}{4}$ and $a_2+a_5=216$. Then a_1 equals

(a) 10 (b) 12

(c) 7 or 54/7 (d) none of these

16. If ax^3+bx^2+cx+d is divisible by ax^2+c, then a, b, c, d are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

17. If S_n denotes the sum of the first n terms of an A.P., then $\frac{S_{3n}-S_{n-1}}{S_{2n}-S_{2n-1}}$ is always equal to

(a) $2n$ (b) $2n + 1$

(c) $2n - 1$ (d) none of these

18. If a, b and c are in H.P., then the value of $\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)\left(\frac{1}{c}+\frac{1}{a}-\frac{1}{b}\right)$ is

(a) $\frac{2}{bc}+\frac{2}{b^2}$ (b) $\frac{3}{c^2}+\frac{2}{ca}$

(c) $\frac{3}{b^2}-\frac{2}{ab}$ (d) $\frac{1}{a^2}-\frac{2}{3abc}$

19. If $a_1, a_2, a_3, \ldots, a_{2n+1}$ are in A.P., then $\frac{a_{2n+1}-a_1}{a_{2n+1}+a_1}+\frac{a_{2n}-a_2}{a_{2n}+a_2}+\cdots+\frac{a_{n+2}-a_n}{a_{n+2}+a_n}$ is equal to

(a) $\frac{n(n+1)}{2}\cdot\frac{a_2-a_1}{a_{n+1}}$ (b) $\frac{n(n+1)}{2}$

(c) $(n+1)(a_2-a_1)$ (d) none of these

20. If the sum of the series 2, 5, 8, 11,... is 60,100, then n is

(a) 100 (b) 200

(c) 150 (d) 250

21. If x, y, z are in G.P. and $a^x = b^y = c^z$, then

(a) $\log_b a=\log_a c$ (b) $\log_c b=\log_a c$

(c) $\log_b a=\log_c b$ (d) none of these

22. The soultion of the system of equations $1 + x + x^2 + x^3+\cdots+x^{23}=0$, $1+x+x^2+x^3+\cdots+x^{19}=0$ is

(a) $x=-2$ (b) $x=2$

(c) $x=-1$ (d) $x=1$

23. If a, b, c are three unequal numbers such that a, b, c are in A.P. and $b-a$, $c-b$, a are in G.P., then $a : b : c$ is

(a) 1 : 2 : 3 (b) 1 : 3 : 5

(c) 2 : 3 : 4 (d) 1 : 2 : 4

24. If a, b, c, d are non-zero real numbers such that $(a^2+b^2+c^2)(b^2+c^2+d^2)\le(ab+bc+cd)^2$, then a,b,c,d are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

25. If $(1-p)(1+3x+9x^2+27x^3+81x^4+243x^5)=1-p^6$, $p\neq 1$, then the value of p/x is

(a) 1/3 (b) 3

(c) 1/2 (d) 2

26. The value of $2^{1/4}\cdot 4^{1/8}\cdot 8^{1/16}\cdot 16^{1/32},\ldots$ is

(a) 1 (b) 2

(c) 3 (d) 5

27. If $\log\left(\frac{5c}{a}\right)$, $\log\left(\frac{3b}{5c}\right)$, $\log\left(\frac{a}{3b}\right)$ are in A.P., where a, b, c are in G.P., then a, b, c are the lengths of the sides of

(a) an isosceles triangle (b) an equilateral triangle

(c) a right-angled triangle (d) none of these

28. If repetition of digits is allowed, then the number of even natural numbers having three digits is

(a) 250 (b) 350

(c) 450 (d) 550

29. If $a^2+b^2, ab+bc$ and b^2+c^2 are in G.P., then a, b, c are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

30. If the series $1+\frac{2}{x}+\frac{3}{x^2}+\cdots+=\frac{1}{3}$, then the value of x is

(a) 1/3 (b) 1/6

(c) $\frac{\sqrt{3}-1}{2}$ (d) $\frac{\sqrt{3}+1}{2}$

31. If a, b, c are in G.P. and $\log a-\log 2b, \log 2b-\log 3c$ and $\log 3c-\log a$ are in A.P., then a, b, c are the lengths of the sides of a triangle which is

(a) acute-angled (b) obtuse-angled

(c) right-angled (d) equilateral

32. If the sum of an infinitely decreasing G.P. is 3, and the sum of the squares of its terms is 9/2, then the sum of the cubes of the terms is

(a) 105/13 (b) 108/13

(c) 729/8 (d) none of these

33. If $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz}$ and p, q, r be in A.P., then x, y, z are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

34. The value of $\frac{1^3+2^3+3^3+4^3+\cdots+12^3}{1^2+2^2+3^2+4^2+\cdots+12^2}$ is
(a) $\frac{234}{25}$ (b) $\frac{243}{35}$
(c) $\frac{263}{27}$ (d) $\frac{253}{31}$

35. If $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots+\infty=\frac{\pi^2}{6}$, then $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots$ equals
(a) $\frac{\pi^2}{8}$ (b) $\frac{\pi^2}{12}$
(c) $\frac{\pi^2}{3}$ (d) $\frac{\pi^2}{2}$

36. If the sum of the first $2n$ terms of the A.P. 2, 5, 8,... is equal to the sum of the first n terms of the A.P. 57, 59, 61,..., then n equals
(a) 10 (b) 12
(c) 11 (d) 13

37. For what values of the parameter a are there real values of x such that $5^{1+x}+5^{1-x}$, $\frac{a}{2}$, 25^x+25^{-x} are three consecutive terms of an A.P.?
(a) 12 (b) 14
(c) 16 (d) 10

38. The sum of the series $\frac{2}{3}+\frac{8}{9}+\frac{26}{27}+\frac{80}{81}+\cdots+n$ terms is
(a) $n-\frac{1}{2}(3^n-1)$ (b) $n+\frac{1}{2}(3^n-1)$
(c) $n+\frac{1}{2}(1-3^{-n})$ (d) $n+\frac{1}{2}(3^{-n}-1)$

39. The sum of r terms of an A.P. is denoted by S_r and $\frac{S_m}{S_n}=\frac{m^2}{n^2}$. Find the ratio of the mth term and the nth term of the A.P.
(a) $\frac{2m-1}{2n-1}$ (b) $\frac{2m+1}{2n-1}$
(c) $\frac{2m-1}{2n+1}$ (d) $\frac{2m+1}{2n+1}$

40. If the sum of the first p terms of an A.P. is q and the sum of the first q terms is p, then the sum of the first $(p+q)$ terms is
(a) $p+q$ (b) 0
(c) $-(p+q)$ (d) $-2(p+q)$

41. The sum of $\frac{1^3}{1}+\frac{1^3+2^3}{1+3}+\frac{1^3+2^3+3^3}{1+3+5}+\cdots$ to 16 terms is
(a) 246 (b) 646
(c) 446 (d) 746

42. The sum of the series $1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots$ is
(a) $\frac{e^x-e^{-x}}{2}$ (b) $\frac{e^x+e^{-x}}{2}$
(c) $\frac{e^{-x}-e^x}{2}$ (d) $\frac{-(e^x+e^{-x})}{2}$

43. The real numbers x_1, x_2, x_3 satisfying the equation $x^3-x^2+\beta x+\gamma=0$ are in A.P. The intervals in which β and γ lie are
(a) $\left(-\infty, \frac{1}{3}\right], \left[-\frac{1}{27}, \infty\right)$ (b) $\left(-8, \frac{1}{3}\right), \left[-\frac{1}{27}, \infty\right)$
(c) $\left(-\infty, \frac{1}{3}\right], \left(-\frac{1}{27}, \infty\right)$ (d) none of these

44. If $n!$, $3\times n!$ and $(n+1)!$ are in G.P., then $n!$, $5\times n!$ and $(n+1)!$ are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

45. Let the sequence $a_1, a_2, a_3, \ldots, a_n, \ldots$ form an A.P. Then $a_1^2-a_2^2+a_3^2-a_4^2+\cdots+a_{2n-1}^2-a_{2n}^2$ is equal to
(a) $\frac{n}{2n-1}(a_1^2-a_{2n}^2)$ (b) $\frac{2n}{n-1}(a_{2n}^2-a_1^2)$
(c) $\frac{n}{n+1}(a_1^2+a_{2n}^2)$ (d) none of these

46. The sum of the series $\frac{1}{2.3}+\frac{1}{4.5}+\frac{1}{6.7}+\cdots$ is
(a) $\log\left(\frac{e}{2}\right)$ (b) $\log\left(\frac{2}{3}\right)$
(c) $\frac{e}{2}$ (d) $\frac{2}{e}$

47. If 1, a and p are in A.P. and l, g and p are in G.P., then
(a) $1+2a+g^2=0$ (b) $1+2a-g^2=0$
(c) $1-2a-g^2=0$ (d) $1-2a+g^2=0$

48. In an A.P. if $S_1=t_1+t_2+t_3+\cdots+t_n$ (n odd), $S_2=t_1+t_3+t_5+\cdots+t_n$, then S_1/S_2 =
(a) $\frac{2n}{n+1}$ (b) $\frac{n}{n+1}$
(c) $\frac{n+1}{2n}$ (d) $\frac{n+1}{n}$

49. If x, y, z are real and $4x^2+9y^2+16z^2-6xy-12yz-8zx=0$, then x, y, z are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

50. The sum of the series $\log_4 2-\log_8 2+\log_{16} 2+\cdots+\infty$ is
(a) e^2 (b) $\log 2+1$
(c) $\log 2-1$ (d) $1-\log 2$

51. If the pth term of an A.P. is q and the qth term is p, then the rth term is
(a) $q-p+r$ (b) $p-q+r$
(c) $p+q+r$ (d) $p+q-r$

52. If nine harmonic means are inserted between 2 and 3, then the value of $A+\frac{6}{H}$ (where A is any of the A.M.'s and H the corresponding H.M.'s) is
(a) 8 (b) 5
(c) 10 (d) none of these

53. If $\sum_{n=1}^{n} n, \frac{\sqrt{10}}{3}\cdot\sum_{n=1}^{n} n^2, \sum_{n=1}^{n} n^3$ are in G.P., then the value of n is
(a) 2 (b) 3
(c) 4 (d) none of these

54. If $S=\frac{1}{1\cdot 2}-\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}-\frac{1}{4\cdot 5}+\cdots+\infty$, then the value of e^S is
(a) $\frac{e}{4}$ (b) $\frac{4}{e}$
(c) $\log\left(\frac{e}{4}\right)$ (d) $\log\left(\frac{4}{e}\right)$

55. If the arithmetic and geometric means of two distinct, positive numbers are A and G respectively, then their harmonic mean is
(a) A/G (b) G/A
(c) G^2/A (d) A^2/G

56. If $a_1, a_2, a_3, \ldots, a_n$ is an A.P. of non-zero terms, then $\frac{1}{a_1a_2}+\frac{1}{a_2a_3}+\cdots+\frac{1}{a_{n-1}a_n}=$
(a) $\frac{1}{a_1a_n}$ (b) $\frac{n}{a_1a_n}$
(c) $\frac{n-1}{a_1a_n}$ (d) none of these

57. If $H_1, H_2, \ldots, H_n$ are the n harmonic means between a and b, then $\frac{H_1+a}{H_1-a}+\frac{H_n+b}{H_n-b}=$
(a) n (b) $n+1$
(c) $2n$ (d) $2n-2$

58. If the first, second and middle terms of an A.P. are a, b, c, respectively, then their sum is
(a) $\frac{2(c-a)}{b-a}$ (b) $\frac{2c(c-a)}{b-a}+c$
(c) $\frac{2c(b-a)}{c-a}$ (d) $\frac{2b(c-a)}{b-a}$

59. If a, b, c, d are non-zero real numbers such that $(a^2+b^2+c^2)(b^2+c^2+d^2)\le(ab+bc+cd)^2$, then a, b, c, d are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

60. The sum to infinity of the series $1+\frac{4}{5}+\frac{7}{5^2}+\frac{10}{5^3}+\cdots$ is
(a) $\frac{16}{35}$ (b) $\frac{11}{8}$
(c) $\frac{35}{16}$ (d) $\frac{8}{11}$

61. If the sum of n terms of the series $\frac{2n+1}{2n-1}+3\left(\frac{2n+1}{2n-1}\right)^2+5\left(\frac{2n+1}{2n-1}\right)^3+\cdots$ is 36, then the value of n is
(a) 2 (b) 3
(c) 4 (d) 1

62. If 1, $\log_y x$, $\log_z y$, $-15\log_x z$ are in A.P., then
(a) $z^3=x$ (b) $x=y^{-1}$
(c) $z^{-3}=y$ (d) all of these

63. Let $a_1, a_2, a_3, \ldots$ be in A.P. and a_p, a_q, a_r be in G.P. Then $a_q : a_p$ is equal to
(a) $\frac{r-p}{q-p}$ (b) $\frac{q-p}{r-p}$
(c) $\frac{r-q}{q-p}$ (d) none of these

64. The A.M. of two given positive numbers is 2. If the larger number is increased by 1, the G.M. of the numbers becomes equal to the A.M. of the given numbers. Then the H.M. of the given numbers is
(a) $\frac{3}{2}$ (b) $\frac{2}{3}$
(c) $\frac{1}{2}$ (d) none of these

65. Three numbers are in G.P. If we double the middle term, we get an A.P. Then the common ratio of the G.P. equals
(a) $2\pm\sqrt{3}$ (b) $3\pm\sqrt{2}$
(c) $3\pm\sqrt{5}$ (d) $5\pm\sqrt{3}$

EXERCISE SET 2

1. If $3+\frac{1}{4}(3+d)+\frac{1}{4^2}(3+2d)+\cdots+\infty=8$, then the value of d is
(a) 9 (b) 5 (c) 1 (d) none of these

2. If $a_n>1$ for all $n\in N$, then $\log_{a_2}a_1+\log_{a_3}a_2+\cdots+\log_{a_n}a_{n-1}+\log_{a_{n+1}}a_n$ has the minimum value
(a) 1 (b) 2 (c) 0 (d) none of these

3. $\cos x = b$. For what b do the roots of the equation form an A.P.?
(a) -1 (b) 1/2 (c) $\sqrt{3}/2$ (d) none of these

4. Find the sum of n terms of the series
$\cot^{-1}3+\cot^{-1}7+\cot^{-1}13+\cot^{-1}21+\cdots$
(a) $\tan^{-1}\frac{n}{n+1}$ (b) $\tan^{-1}\frac{n}{n-1}$
(c) $\tan^{-1}\frac{n}{n+2}$ (d) none of these

5. In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of this progression equals
(a) $\frac{1}{2}(1-\sqrt{5})$ (b) $\frac{1}{2}\sqrt{5}$
(c) $\sqrt{5}$ (d) $\frac{1}{2}(\sqrt{5}-1)$

6. The number of terms common between the series 1 + 2 + 4 + 8 +⋯ to 100 terms and 1 + 4 + 7 + 10 +⋯ to 100 terms is
(a) 6 (b) 4 (c) 5 (d) none of these

7. If the sum of the first n positive integers is 1/5 times the sum of their squares, then n equals
(a) 5 (b) 6 (c) 7 (d) 8

8. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a<b<c$ and $a+b+c=3/2$, then the value of a is
(a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2\sqrt{3}}$
(c) $\frac{1}{2}-\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}-\frac{1}{\sqrt{2}}$

9. The value of $\sum_{n=1}^{m}\log\frac{a^{2n-1}}{b^{m-1}}$ $(a\neq 0, 1;\ b\neq 0, 1)$ is
(a) $m\log\frac{a^{2m}}{b^{m-1}}$ (b) $\log\frac{a^{2m}}{b^{m-1}}$
(c) $\frac{m}{2}\log\frac{a^{2m}}{b^{2m-2}}$ (d) $\frac{m}{2}\log\frac{a^{2m}}{b^{m+1}}$

10. If $a, a_1, a_2, a_3, \ldots, a_{2n}, b$ are in A.P. and $g, g_1, g_2, g_3, \ldots, g_{2n}, b$ are in G.P. and H is the H.M. of a and b, then
$\frac{a_1+a_{2n}}{g_1g_{2n}}+\frac{a_2+a_{2n-1}}{g_2g_{2n-1}}+\cdots+\frac{a_n+a_{n+1}}{g_ng_{n+1}}$ is equal to
(a) $2n/H$ (b) $2nH$ (c) nH (d) n/H

11. The sum of the first n terms of the series $\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}+\cdots$ is
(a) 2^n-1 (b) $1-2^{-n}$
(c) $2^{-n}-n+1$ (d) $2^{-n}+n-1$

12. If $t_r=2^{r/2}+2^{-r/2}$, then $\sum_{r=1}^{10}t_r^2$ is equal to
(a) $\frac{2^{21}-1}{2^{10}}+20$ (b) $\frac{2^{21}-1}{2^{10}}+19$
(c) $\frac{2^{21}-1}{2^{20}}-1$ (d) none of these

13. If the fourth, seventh and tenth terms of a G.P. are f, g, h respectively, then
(a) $f^2=g^2+h^2$ (b) $g^2=fh$
(c) $f^2=gh$ (d) $fgh+fg+1=0$

14. If the 7th term of an H.P. is 8 and the 8th term is 7, then its 15th term is
(a) 16 (b) 14 (c) 27/14 (d) 56/15

15. The sum to n terms of the series
$\frac{1}{\sqrt{1}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{7}}+\cdots$ is
(a) $\sqrt{2n+1}$ (b) $(1/2)\sqrt{2n+1}$
(c) $\sqrt{2n+1}-1$ (d) $(1/2)(\sqrt{2n+1}-1)$

16. If x_1, x_2, x_3 and y_1, y_2, y_3 are both in G.P. with the same common ratio, then the points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$
(a) lie on a straight line (b) lie on a circle
(c) lie on an ellipse
(d) are the vertices of a triangle

17. If $0<x<(\pi/2)$, then the minimum value of $(\sin x+\cos x+\operatorname{cosec}2x)^3$ is
(a) 27 (b) 13.5 (c) 6.75 (d) none of these

18. The product of n positive numbers is 1, then their sum is a positive integer, which is
(a) equal to 1 (b) equal to $n+n^2$
(c) divisible by n (d) never less than n

19. The harmonic mean of the roots of the equation $(5+\sqrt{2})x^2-(4+\sqrt{5})x+8+2\sqrt{5}=0$ is
(a) 2 (b) 4 (c) 6 (d) 8

20. The sum of the series
$S = 1^2 - 2^2 + 3^2 - 4^2 + \cdots + 2001^2 - 2002^2 + 2003^2$ is
(a) 2,007,006 (b) 1,005,004
(c) 2,000,506 (d) none of these

21. If α, β are the roots of $x^2 - 3x + a = 0$ and γ, δ the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in order) form an increasing G.P., then
(a) $a = 3, b = 12$ (b) $a = 12, b = 3$
(c) $a = 2, b = 32$ (d) $a = 4, b = 16$

22. The 99th term of the series $2 + 7 + 14 + 23 + 34 + \cdots$ is
(a) 9999 (b) 9998 (c) 10,000 (d) none of these

23. Let the positive numbers a, b, c, d be in A.P. Then abc, abd, acd, bcd are
(a) not in A.P./G.P./H.P. (b) in A.P.
(c) in G.P. (d) in H.P.

24. The product of n positive numbers is 1. Their sum is
(a) a positive integer (b) divisible by n
(c) equal to $n + 1/n$ (d) greater than or equal to n

25. The value of $(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\cdots+\infty\right)}$ is
(a) 2 (b) 3 (c) 4 (d) none of these

26. A G.P. consists of an even number of terms. If the sum of all the terms is 5 times the sum of the terms occupying odd places, the common ratio will be
(a) 2 (b) 3 (c) 4 (d) 5

27. If $(m + 1)$th, $(n + 1)$th and $(r + 1)$th terms of an A.P. are in G.P. and m, n, r are in H.P., then the ratio of the first term of the A.P. to its common difference in terms of n is
(a) $n/2$ (b) $-(n/2)$ (c) $n/3$ (d) $-(n/3)$

28. If $\ln(a + c)$, $\ln(c - a)$, $\ln(a - 2b + c)$ are in A.P., then
(a) a, b, c are in A.P. (b) $a^2b^2c^2$ are in A.P
(c) a, b, c are in G.P. (d) a, b, c are in H.P.

29. The sum of $0.2 + 0.004 + 0.00006 + 0.0000008 + \cdots$ to ∞ is
(a) $\frac{200}{891}$ (b) $\frac{2000}{9801}$ (c) $\frac{1000}{9801}$ (d) none of these

30. $\tan 80°, \tan 70° + \tan 10°, \tan 10°$ are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

31. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a < b < c$ and $a + b + c = 3/2$, then the value of a is
(a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2\sqrt{3}}$
(c) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (d) $\frac{1}{2} - \frac{1}{\sqrt{2}}$

32. The minimum value of n such that $1 + 3 + 3^2 + \cdots + 3^n > 1000$ is
(a) 7 (b) 8 (c) 9 (d) none of these

33. Let x, y, z be three positive prime numbers. The progression in which $\sqrt{x}, \sqrt{y}, \sqrt{z}$ can be three terms (not necessarily consecutive) is
(a) A.P. (b) G.P. (c) H.P. (d) none of these

34. If the first, second and the last terms of an A.P. are a, b, c respectively, then the sum of the A.P. is
(a) $\frac{(a+b)(a+c-2b)}{2(b-a)}$ (b) $\frac{(b+c)(a+b-2c)}{2(b-a)}$
(c) $\frac{(a+c)(b+c-2a)}{2(b-a)}$ (d) none of these

35. It is known that $\sum_{r=1}^{\infty} \frac{1}{(2r-1)^2} = \frac{\pi^2}{8}$. Then $\sum_{r=1}^{\infty} \frac{1}{r^2}$ is equal to
(a) $\frac{\pi^2}{24}$ (b) $\frac{\pi^2}{3}$ (c) $\frac{\pi^2}{6}$ (d) none of these

36. If $x > 1, y > 1, z > 1$ are in G.P., then $\frac{1}{1+\ln x}, \frac{1}{1+\ln y}, \frac{1}{1+\ln z}$ are in
(a) A.P. (b) H.P. (c) G.P. (d) none of these

37. If three positive real numbers a, b, c $(c > a)$ are in H.P., then $\log(a + c) + \log(a - 2b + c)$ is equal to
(a) $2\log(c - b)$ (b) $2\log(a + c)$
(c) $2\log(c - a)$ (d) $\log a + \log b + \log c$

38. Let $S_1, S_2, \ldots$ be squares such that for each $n \geq 1$, the length of a side of S_n equals the length of a diagonal of S_{n+1}. If the length of a side of S_1 is 10 cm, then for which of the following values of n is the area of S_n less than 1 cm^2?
(a) 6 (b) 5 (c) 2 (d) 10

39. If the mth, nth and pth terms of an A.P. and G.P. are equal and are respectively x, y, z, then
(a) $x^y y^z x^x = x^z y^x x^y$
(b) $(x - y)^x (y - z)^y = (z - x)^z$
(c) $(x - y)^z (y - z)^x = (z - x)^y$
(d) none of these

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions: Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice as given below:

(a) Statement 1 is false, Statement 2 is true.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(c) Statement 1 true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(d) Statement 1 is true, Statement 2 is false.

40. **Statement 1:** If $a + b + c = 12$ $(a, b, c > 0)$, then the maximum value of abc is 64.
Statement 2: The maximum value of abc occurs when $a = b = c$.

41. **Statement 1:** If positive numbers a^{-1}, b^{-1}, c^{-1} are in A.P., then the product of the roots of the equation $x^2 - \lambda x + 2b^{101}a^{101} - c^{101} = 0\,(\lambda \in \mathbb{R})$ has the –ve sign.
Statement 2: If a, b, c are in H.P., then $\frac{a^n + c^n}{2} > b^n$.

42. **Statement 1:** If $x^2 + y^2 = 1$, then $|x + y| \le \sqrt{2}$.
Statement 2: A.M. ≥ G.M.

43. **Statement 1:** The sums of n terms of two A.Ps. are in the ratio $7n+1 : 4n+17$. The ratio of the nth terms is $14n + 6 : 8n + 13$.
Statement 2: If $S_n = an^2 + bn + c$, then $t_n = S_n - S_{n-1}$.

Previous Years' Questions

44. $1^3 - 2^3 + 3^3 - 4^3 + \cdots + 9^3 =$
(a) –475 (b) 425 (c) 475 (d) –425
[AIEEE 2002]

45. The value of $2^{1/4} \cdot 4^{1/8} \cdot 8^{1/16} \ldots \infty$ is
(a) 1 (b) 2 (c) 3/2 (d) 4
[AIEEE 2002]

46. If the fifth term of a G.P. is 2, then the product of its first 9 terms is
(a) 256 (b) 512 (c) 1024 (d) none of these
[AIEEE 2002]

47. If a, b, c are in A.P., then (a/bc), $(1/c)$, $(2/b)$ will be in
(a) A.P. (b) G.P. (c) H.P. (d) none of these
[MPCET 2002]

48. The sum of infinite number of terms of a G.P. is 20 and the sum of their squares is 100. The common ratio of the G.P. is
(a) 1/5 (b) 8/5 (c) 3/5 (d) 5
[AIEEE 2002]

49. If $1, \frac{1}{2}\log_3(3^{1-x} + 2), \log_3(4 \cdot 3^x - 1)$ are in A.P., then x equals
(a) $\log_3 4$ (b) $1 - \log_3 4$
(c) $1 - \log_4 3$ (d) $\log_4 3$
[AIEEE 2002]

50. If a, b, c are respectively the pth, qth and rth terms of a G.P., then $\Delta = \begin{vmatrix} \log a & \log b & \log c \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$ is equal to
(a) 1 (b) 0 (c) –1 (d) none of these
[AIEEE 2002]

51. Let $f(x)$ be a polynominal of degree two. If $f(1) = f(-1)$ and a, b, c are in A.P., then $f'(a), f'(b), f'(c)$ are in
(a) A.G.P. (b) A.P. (c) G.P. (d) H.P.
[AIEEE 2003]

52. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then (a/c), (b/a) and (c/b) are in
(a) geometric progression
(b) harmonic progression
(c) arithmetico-geometric progression
(d) arithmetic progression
[AIEEE 2003]

53. If $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$ and $f(1) = 7$, then $\sum_{r=1}^{n} f(r)$ is
(a) $\frac{7n(n+1)}{2}$ (b) $\frac{7n}{2}$
(c) $\frac{7(n+1)}{2}$ (d) $7n(n+1)$
[AIEEE 2003]

54. If a, b, c are in A.P., $|a|, |b|, |c| < 1$ and $x = 1 + a + a^2 + \cdots + \infty$, $y = 1 + b + b^2 + \cdots + \infty$, $z = 1 + c + c^2 + \cdots + \infty$, then x, y, z shall be in
(a) H.P. (b) G.P. (c) A.P. (d) none of these
[AIEEE 2004]

55. Let t_r be the rth term of an A.P. whose first term is a and common difference d. If for some positive intergers m, n, $m \ne n$, $t_m = 1/n$ and $t_n = 1/m$, then $a - d$ is equal to
(a) $\frac{1}{m} + \frac{1}{n}$ (b) 1 (c) $\frac{1}{mn}$ (d) 0
[AIEEE 2004]

56. The sum of the first n terms of the series $12 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \cdots$ is $\frac{n(n+1)^2}{2}$ when n is even. When n is odd, the sum is
(a) $\frac{3n(n+1)}{2}$ (b) $\frac{n^2(n+1)}{2}$
(c) $\frac{n(n+1)^2}{4}$ (d) $\left(\frac{n(n+1)}{2}\right)^2$
[AIEEE 2004]

57. If two numbers have arithmetic mean 9 and geometric mean 4, then these numbers are the roots of the equation

(a) $x^2+18x+16=0$ (b) $x^2-18x+16=0$

(c) $x^2+18x-16=0$ (d) $x^2-18x-16=0$

[AIEEE 2004]

58. If p, q, r are in G.P. and $\tan^{-1}p$, $\tan^{-1}q$, $\tan^{-1}r$ in A.P., then p, q, r satisfy the relation

(a) $p=q=r$ (b) $p\neq q\neq r$

(c) $p+q=r$ (d) none of these

[AIEEE 2005]

59. If $x=\sum_{n=0}^{\infty}a^n$, $y=\sum_{n=0}^{\infty}b^n$, $z=\sum_{n=0}^{\infty}c^n$, where a, b, c are in A.P. and $|a|<1$, $|b|<1$, $|c|<1$, then x, y, z are in

(a) H.P. (b) A.G.P. (c) A.P. (d) G.P.

[AIEEE 2005]

60. Let $a_1,a_2,a_3,\ldots$ be the terms of an A.P. If $\frac{a_1+a_2+\ldots+a_p}{a_1+a_2+\ldots+a_q}=\frac{p^2}{q^2}, p\neq q$, then $\frac{a_6}{a_{21}}$ equals

(a) 41/11 (b) 11/41

(c) 2/7 (d) 7/2

[AIEEE 2006]

61. If $a_1,a_2,\cdots,a_n$ are in H.P., then the expression $a_1a_2+a_2a_3+\cdots+a_{n-1}a_n$ is equal to

(a) $(n-1)a_1a_n$ (b) $n(a_1-a_n)$

(c) $(n-1)(a_1-a_n)$ (d) na_1a_n

[AIEEE 2006]

62. In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of this progression equals

(a) $\frac{1}{2}(\sqrt{5}-1)$ (b) $\frac{1}{2}\sqrt{5}$

(c) $\sqrt{5}$ (d) $\frac{1}{\sqrt{2}}(1-\sqrt{5})$

[AIEEE 2007]

63. In a G.P. of positive terms, any term is equal to the sum of the next two terms. Then, the common ratio of the G.P. is

(a) $\frac{\sqrt{5}-1}{2}$ (b) $\frac{\sqrt{5}+1}{2}$

(c) $-\frac{\sqrt{5}+1}{2}$ (d) $\frac{1-\sqrt{5}}{2}$

[AIEEE 2007]

64. The first two terms of a geometric progression add up to 12. The sum of the third and fourth terms is 48. If the terms of the geometric progression are alternatively positive and negative, then the first term is

(a) –12 (b) 12 (c) 4 (d) –4

[AIEEE 2008]

65. The sum to infinity of the series $1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\frac{14}{3^4}+\cdots$ is

(a) 2 (b) 3 (c) 4 (d) 6

[AIEEE 2009]

66. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the nth minute. If $a_1=a_2=\ldots=a_{10}=150$ and $a_{10},a_{11},\ldots$ are in A.P., with common difference –12, then the time taken by him to count all the notes is

(a) 125 minutes (b) 135 minutes

(c) 24 minutes (d) 34 minutes

[AIEEE 2010]

67. If the A.M. of two numbers is twice their G.M., then the ratio of the greatest number to the smallest number is

(a) $7-4\sqrt{3}$ (b) $7+4\sqrt{3}$

(c) 21 (d) 5 **[DCE 2007]**

68. If b^2, a^2, c^2 are in A.P., then $a+b$, $b+c$, $c+a$ will be in

(a) A.P. (b) G.P. (c) H.P. (d) none of these

[DCE 2007]

69. If the sum of the first n terms of a series be $5n^2+2n$, then its second term is

(a) 16 (b) 17 (c) 27/14 (d) 56/15

[M.P.C.E.T. 1996]

70. Let a_n be the nth term of an A.P. If $\sum_{r=1}^{100}a_{2r}=\alpha$ and $\sum_{r=1}^{100}a_{2r-1}=\beta$, then the common difference of the A.P. is

(a) $\alpha-\beta$ (b) $\frac{\alpha-\beta}{100}$ (c) $\beta-\alpha$ (d) $\frac{\alpha-\beta}{200}$

[AIEEE 2011]

71. **Statement 1:** The sum of the series 1 + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + ⋯ + (361 + 380 + 400) is 8000.

Statement 2: $\sum_{k=1}^{N}[k^3-(k-1)^3]=n^3$ for any natural number n.

(a) Statement 1 is false, statement 2 is true

(b) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1

(c) Satement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1

(d) Satement 1 is true, statement 2 is false

[AIEEE 2012]

72. If 100 times the 100th term of an AP with non-zero common difference equals the 50 times its 50th term, then the 150th term of this AP is

(a) –150 (b) 150 times its 50th term
(c) 150 (d) zero

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (b)	**2.** (b)	**3.** (c)	**4.** (a)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (b)	**9.** (c)	**10.** (d)
11. (c)	**12.** (b)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (b)	**18.** (c)	**19.** (a)	**20.** (b)
21. (c)	**22.** (c)	**23.** (a)	**24.** (b)	**25.** (b)	**26.** (b)	**27.** (d)	**28.** (c)	**29.** (b)	**30.** (c)
31. (b)	**32.** (b)	**33.** (c)	**34.** (a)	**35.** (a)	**36.** (c)	**37.** (a)	**38.** (d)	**39.** (a)	**40.** (c)
41. (c)	**42.** (b)	**43.** (a)	**44.** (a)	**45.** (a)	**46.** (a)	**47.** (d)	**48.** (a)	**49.** (c)	**50.** (d)
51. (d)	**52.** (b)	**53.** (c)	**54.** (b)	**55.** (c)	**56.** (c)	**57.** (c)	**58.** (b)	**59.** (b)	**60.** (c)
61. (c)	**62.** (d)	**63.** (c)	**64.** (a)	**65.** (a)					

Exercise Set 2

1. (a)	**2.** (d)	**3.** (a)	**4.** (c)	**5.** (d)	**6.** (c)	**7.** (c)	**8.** (d)	**9.** (c)	**10.** (a)
11. (d)	**12.** (b)	**13.** (b)	**14.** (d)	**15.** (d)	**16.** (a)	**17.** (b)	**18.** (d)	**19.** (b)	**20.** (a)
21. (c)	**22.** (b)	**23.** (d)	**24.** (d)	**25.** (c)	**26.** (c)	**27.** (b)	**28.** (d)	**29.** (b)	**30.** (a)
31. (d)	**32.** (a)	**33.** (d)	**34.** (c)	**35.** (c)	**36.** (b)	**37.** (c)	**38.** (d)	**39.** (a)	**40.** (b)
41. (b)	**42.** (b)	**43.** (a)	**44.** (b)	**45.** (b)	**46.** (b)	**47.** (d)	**48.** (c)	**49.** (b)	**50.** (b)
51. (b)	**52.** (b)	**53.** (a)	**54.** (a)	**55.** (d)	**56.** (b)	**57.** (b)	**58.** (a)	**59.** (a)	**60.** (b)
61. (a)	**62.** (a)	**63.** (a)	**64.** (a)	**65.** (b)	**66.** (d)	**67.** (b)	**68.** (c)	**69.** (b)	**70.** (b)
71. (b)	**72.** (d)								

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Let there be n sides of the polygon. Then the sum of all the interior angles is $(2n - 4)$ right angles.

$\therefore \quad \frac{n}{2}[240+(n-1)5]=(2n-4)\times 90$

$\Rightarrow \quad n^2-25n+144=0$

$\Rightarrow \quad n=9,16$

But for n = 16, the minimum angle is 120° and the largest angle is 120 + (16 – 1)5 = 195, which is impossible because there will be one angle 180° in this series.

Hence, n = 9

2. Let $b=ar, c=ar^2$ and $d=ar^3$. Then

$$\frac{1}{a^3+b^3}=\frac{1}{a^3(1+r^3)}$$

$$\frac{1}{b^3+c^3}=\frac{1}{a^3r^3(1+r^3)}$$

and $$\frac{1}{c^3+d^3}=\frac{1}{a^3r^6(1+r^3)}.$$

Clearly $(a^3+b^3)^{-1},(b^3+c^3)^{-1}$ and $(c^3+d^3)^{-1}$ are in GP with common ratio $1/r^3$.

3. $t_p=a_1+(p-1)d;\ t_q=a_1+(q-1)d;\ t_r=a_1+(r-1)d$ where d is the common difference of the A.P. Since t_p, t_q, t_r are in A.P., we have

$a_1+(p-1)d,\ a_1+(q-1)d,\ a_1+(r-1)d$ in A.P.

$\Rightarrow \quad (p-1)d,\ (q-1)d,\ (r-1)d$ are in A.P.

$\Rightarrow \quad p-1,\ q-1,\ r-1$ are in A.P.

$\Rightarrow \quad p, q, r$ are in A.P.

4. $t_m = a + (m - 1)d$

$$\Rightarrow \qquad \frac{1}{n}=a+(m-1)d \qquad \text{(i)}$$

and $\qquad t_n = a + (n - 1)d$

$$\Rightarrow \quad \frac{1}{m} = a + (n-1)d \qquad \text{(ii)}$$

Subtracting (ii) from (i),

$$\frac{1}{n} - \frac{1}{m} = (m - 1 - n + 1)d$$

$$\Rightarrow \quad \frac{m-n}{mn} = (m-n)\,d$$

$$\Rightarrow \quad d = \frac{1}{mn}$$

Putting $d = 1/mn$ in (i),

$$\Rightarrow \quad \frac{1}{n} = a + \frac{m-1}{mn} = a + \frac{1}{n} - \frac{1}{mn}$$

$$\Rightarrow \quad a = \frac{1}{mn}$$

$$\Rightarrow \quad a - d = \frac{1}{mn} - \frac{1}{mn} = 0$$

5. Let $E = \frac{p}{r} + \frac{r}{p} = \frac{p^2+r^2}{pr} = \frac{(p+r)^2 - 2pr}{pr}$

Since p, q, r are in H.P.

$$\therefore \quad q = \frac{2pr}{p+r}$$

$$\Rightarrow \quad E = \frac{(4p^2r^2/q^2) - 2pr}{pr}$$

As ap, bq, cr are in G.P., i.e. $b^2q^2 = acpr$

$$\Rightarrow \quad E = \frac{4pr}{q^2} - 2 = \frac{4b^2}{ac} - 2$$

Since a, b, c are in A.P., i.e. $2b = a + c$

$$\Rightarrow \quad E = \frac{(a+c)^2}{ac} - 2$$

$$\therefore \quad \frac{p}{r} + \frac{r}{p} = \frac{a}{c} + \frac{c}{a}$$

6. Given: $(p + q)$th term of G.P. is $T_{p+q} = m$ and its $(p - q)$th term $T_{p-q} = n$.

Let a be the first term and r be the common ratio of the given G.P.

We know that the nth term of the G.P. $t_n = ar^{n-1}$

Therefore, its $(p + q)$th term $t_{p+q} = ar^{(p+q-1)} = m$ (i)

Similarly, its $(p-q)th$ term $t_{p-q} = ar^{(p-q-1)} = n$ (ii)

Multiplying Eq. (i) by Eq. (ii), we get

$$a^2r^{2p-2} = mn$$

or $$ar^{p-1} = \sqrt{mn}$$

where ar^{p-1} = pth term of the G.P.

7. $A = \frac{1}{1-r^a} \Rightarrow 1 - r^a = \frac{1}{A} \Rightarrow r^a = 1 - \frac{1}{A} = \frac{A-1}{A}$

$B = \frac{1}{1-r^b} \Rightarrow 1 - r^b = \frac{1}{B} \Rightarrow r^b = 1 - \frac{1}{B} = \frac{B-1}{B}$

$\therefore \quad a\log r = \log\left(\frac{A-1}{A}\right),\ b\log r = \log\left(\frac{B-1}{B}\right)$

$\therefore \quad \frac{a}{b} = \frac{\log[(A-1)/A]}{\log[(B-1)/B]} = \log_{\frac{B-1}{B}}\left(\frac{A-1}{A}\right)$

8. $t_m = S_m - S_{m-1}$

$\therefore \quad 164 = 3m^2 + 5m - 3(m-1)^2 - 5(m-1) = 6m + 2$

$\therefore \quad 6m = 162$ or $m = 27$

9. Let the two numbers be a and b. Then, $\frac{(a+b)/2}{\sqrt{ab}} = \frac{m}{n}$

$$\Rightarrow \quad \sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}} = \frac{2m}{n}$$

$$\Rightarrow \quad \frac{a}{b} - 2\sqrt{\frac{a}{b}}\frac{m}{n} + 1 = 0 \quad \Rightarrow \quad x^2 - \frac{2m}{n}x + 1 = 0$$

where $x = a/b$.

$$\Rightarrow \quad x = \frac{m \pm \sqrt{m^2 - n^2}}{n}$$

$$\therefore \quad a : b = m + \sqrt{m^2 - n^2} : m - \sqrt{m^2 - n^2}$$

10. As $f(x)$, $f(2x)$, $f(4x)$ are in G.P.

$$\therefore \quad (4x+1)^2 = (2x+1)(8x+1)$$

$$\Rightarrow \quad 16x^2 + 8x + 1 = 16x^2 + 10x + 1 \Rightarrow x = 0 \qquad \text{(i)}$$

Here $f(x)$, $f(2x)$, $f(4x)$ are three unequal numbers but if $x = 0$, we have

$$f(x) = f(2x) = f(4x) = f(0) = 1$$

which is not possible

$\therefore$ There is no solution.

11. For G.P., $t_n = 2^{n-1}$; for A.P. $t_m = 1 + (m-1)3$. They would be common if $2^{n-1} = 3m - 2$

or $\quad 2^{n-2} + 1 = \frac{3m}{2} \le 150 \quad \Rightarrow \quad n \le 9,\ m \le 100$

As $2^{n-1} = 3m - 2$;

$\Rightarrow$ $(n = 1, m = 1)$; $(n = 3, m = 2)$; $(n = 5, m = 6)$; $(n = 7, m = 22)$; $(n = 9, m = 86)$

And for $n = 2, 4, 6, 8$; $m \in$ fraction, which is not possible.

12. Let the sides of the triangle be a/r, a and ar, with $a > 0$ and $r > 1$. Let α be the smallest angle, so that the largest angle is 2α. Then α is opposite to the side a/r, and 2α is opposite to the side ar. Applying sine rule, we get

$$\frac{a/r}{\sin\alpha} = \frac{ar}{\sin 2\alpha} \quad \Rightarrow \quad \frac{\sin 2\alpha}{\sin\alpha} = r^2$$

$\Rightarrow \quad 2\cos\alpha = r^2$

$\Rightarrow \quad r^2 < 2 \qquad [\because \ \alpha \neq 0 \qquad \therefore \ 2\cos\alpha < 2]$

$\Rightarrow \quad r < \sqrt{2}$

Hence, $1 < r < \sqrt{2}$

13. $S_p = 0 \quad \therefore \ \frac{p}{2}[2a + (p-1)d] = 0 \ \Rightarrow \ d = \frac{-2a}{p-1}$ (i)

The sum of the next q terms = sum of an A.P. whose first term will be $t_{p+1} = a + pd$.

$$\Rightarrow S = \frac{q}{2}[2(a+pd) + (q-1)d]$$

$$= \frac{q}{2}[2a + (p-1)d + (p+q)d]$$

$$= \frac{q}{2}\left[0 - (p+q)\frac{2a}{p-1}\right]$$

$$= -a\frac{(p+q)q}{p-1}$$

14. Given terms are $\frac{a}{1-ab}$ and $\frac{a}{1+ab}$

We know that as the terms $\frac{a}{1-ab}$ and $\frac{a}{1+ab}$ are in H.P., therefore $\frac{1-ab}{a}$ and $\frac{1+ab}{a}$ are in A.P.

We know that the arithmetic mean of a and $b = \frac{a+b}{2}$

Therefore, the arithmetic mean of $\frac{1-ab}{a}$ and $\frac{1+ab}{a}$ = $\frac{[(1-ab)/a] + [(1+ab)/a]}{2} = \frac{1}{a}$.

Thus, the harmonic mean of the given terms = a.

15. $\frac{a_4}{a_6} = \frac{1}{4} \ \Rightarrow \ \frac{a_1 r^3}{a_1 r^5} = \frac{1}{4} \ \Rightarrow \ \frac{1}{r^2} = \frac{1}{4}$

$r^2 = 4 \ \Rightarrow \ r = \pm 2$

(a = first term, r = common ratio of G.P.)

$r \neq -2$ ($\because$ If $r = -2$, then $a_2 + a_5 \neq 216$)

$\Rightarrow \quad a_1 r + a_1 r^4 = 216$

$\Rightarrow \quad a_1(2) + a_1(16) = 216$

or $18a_1 = 216 \ \Rightarrow \ a_1 = \frac{216}{18} = 12$

16. If $ax^3 + bx^2 + cx + d$ is divisible by $ax^2 + c$, the remainder should be zero. If $ax^3 + bx^2 + cx + d$ is divided by $ax^2 + c$, the remainder obtained is $\frac{bc}{a} - d$.

$\therefore \quad \frac{bc}{a} - d = 0 \ \Rightarrow \ bc = ad \ \Rightarrow \ \frac{b}{a} = \frac{d}{c}$

$\Rightarrow \ a, b, c, d$ are in G.P.

17. $S_{3n} = \frac{3n}{2}[2a + (3n-1)d]$

$S_{n-1} = \frac{n-1}{2}[2a + (n-2)d]$

$\Rightarrow \quad S_{3n} - S_{n-1}$

$$= \frac{1}{2}\{2a(3n-n+1) + d[3n(3n-1) - (n-1)(n-2)]\}$$

$$= \frac{1}{2}[2a(2n+1) + d(8n^2 - 2)]$$

$$= a(2n+1) + d(4n^2 - 1) = (2n+1)[a + (2n-1)d]$$

$$S_{2n} - S_{2n-1} = t_{2n} = a + (2n-1)d$$

$$\Rightarrow \quad \frac{S_{3n} - S_{n-1}}{S_{2n} - S_{2n-1}} = 2n + 1$$

18. Given a, b and c are in H.P.

We know that as a, b and c are in H.P., therefore,

$$\frac{1}{a} + \frac{1}{c} = \frac{2}{b} \text{ or } \frac{1}{c} = \frac{2}{b} - \frac{1}{a}$$

Therefore, $\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a}\right)\left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b}\right)$

$$\Rightarrow \quad \left(\frac{1}{b} + \frac{2}{b} - \frac{1}{a} - \frac{1}{a}\right)\left(\frac{2}{b} - \frac{1}{b}\right)$$

$$\Rightarrow \quad \left(\frac{3}{b} - \frac{2}{a}\right)\left(\frac{1}{b}\right) = \frac{3}{b^2} - \frac{2}{ab}$$

19. $\frac{a_{2n+1} - a_1}{a_{2n+1} + a_1} = \frac{a_1 + (2n+1-1)d - a_1}{a_1 + (2n+1-1)d + a_1}$

$$= \frac{a_1 + 2nd - a_1}{a_1 + 2nd + a_1} = \frac{nd}{a_1 + nd}$$

$$\frac{a_{n+2} - a_n}{a_{n+2} + a_n} = \frac{a_1 + (n+2-1)d - (a_1 + (n-1)d)}{a_1 + (n+2-1)d + (a_1 + (n-1)d)}$$

$$= \frac{(n+1)d - (n-1)d}{2a_1 + (n+1)d + (n-1)d)}$$

$$= \frac{2d}{2a_1 + 2nd} = \frac{d}{a_1 + nd}$$

$\therefore$ Given expression

$$= \frac{d(1+2+3+\cdots+n)}{a_1 + nd} = \frac{d \cdot n[(n+1)/2]}{a_1 + nd}$$

$$= \frac{n(n+1)}{2} \cdot \frac{a_2 - a_1}{a_{n+1}}$$

20. $3n^2 + n - 120{,}200 = 0$

$3 \times 120{,}200 = 3 \times 200 \times 601 = 600 \times 601$

or $(n - 200)(3n + 601) = 0$

$\therefore \ n = 200$ as the other value is not admissible.

21. x, y, z are in G.P. $\Rightarrow \ y^2 = xz$ (i)

We have $a^x = b^y = c^z = \lambda$ (say)

$\Rightarrow \ x\log a = y\log b = z\log c = \log\lambda$

$\Rightarrow \quad x=\dfrac{\log\lambda}{\log a},\ y=\dfrac{\log\lambda}{\log b},\ z=\dfrac{\log\lambda}{\log c}$

Putting the values of x, y, and z in (i), we get

$$\left(\frac{\log\lambda}{\log b}\right)^2=\frac{\log\lambda}{\log a}\frac{\log\lambda}{\log c}$$

$\Rightarrow (\log b)^2=\log a\log c \Rightarrow \log_b a=\log_c b$

22. Given: System of equations

$$1+x+x^2+x^3+\cdots+x^{23}=0 \qquad \text{(i)}$$

and $$1+x+x^2+x^3+\cdots+x^{19}=0 \qquad \text{(ii)}$$

where $x\neq 0$.

We know that the given series is a geometric series. Substituting Eq. (ii) in Eq. (i), we get

$$x^{20}+x^{21}+x^{22}+x^{23}=0$$

or $$x^{20}(1+x+x^2+x^3)=0$$

Since $x^{20}\neq 0$, therefore, $1+x+x^2+x^3=0$

or $1(1+x)+x^2(1+x)=0$ or $(1+x)(1+x^2)=0$

Thus, either $x=-1$ or $x^2=-1$ or $x=\sqrt{-1}$

Since x equal to $\sqrt{-1}$ is not possible, therefore, $x=-1$.

23. Since a, b, c are in A.P.,

$\therefore \quad b-a=c-b$

Also $b-a$, $c-b$, a are in G.P.

$\therefore \quad (c-b)^2=(b-a)a$

$\Rightarrow \quad (b-a)^2=(b-a)a \Rightarrow b-a=a \Rightarrow b=2a$

Also $c=2b-a=2(2a)-a=3a$

$\therefore \quad a:b:c=a:2a:3a=1:2:3$

24. On simplification, $(b^2-ac)^2+(c^2-bd)^2+(ad-bc)^2 \leq 0$ which is possible iff each of

$$(b^2-ac)=(c^2-bd)=(ad-bc)=0$$

$\Rightarrow b^2=ac,\ c^2=bd,\ ad=bc$

$\Rightarrow \dfrac{b}{a}=\dfrac{c}{b}=\dfrac{d}{c} \qquad \therefore \quad a, b, c, d$ are in G.P.

25. $(1-p)(1+3x+9x^2+27x^3+81x^4+243x^5)=1-p^6$

$$\Rightarrow (1+3x+9x^2+27x^3+81x^4+243x^5)=\frac{1-p^6}{1-p}$$

$$\Rightarrow (1+3x+9x^2+27x^3+81x^4+243x^5)$$
$$=1+p^2+p^3+p^4+p^5$$

Comparing, we get $p=3x$ or $p/x=3$.

26. The given expression may be written as

$$2^{1/4}\cdot 2^{2/8}\cdot 2^{3/16}\cdot 2^{4/32}\cdots=2^{\left(\frac{1}{4}+\frac{2}{8}+\frac{3}{16}+\frac{4}{32}+\cdots\right)}$$
$$=2^{\frac{1}{4}\left(1+2\cdot\frac{1}{2}+3\cdot\frac{1}{4}+4\cdot\frac{1}{8}+\cdots\right)} \qquad \text{(i)}$$

We know that the arithmetico-geometric series in the brackets is

$$S_{-\infty}=1+2\cdot\frac{1}{2}+3\cdot\frac{1}{4}+4\cdot\frac{1}{8}+\cdots$$
$$=1+2\left(\frac{1}{2}\right)+3\left(\frac{1}{2}\right)^2+4\left(\frac{1}{2}\right)^3+\cdots$$

We also know that the index of $2^{1/4}$ is of an infinite arithmetico-geometric series with the first term $a=1$, common difference $d=1$ and common ratio $r=1/2$. We know that the sum of the infinite arithmetico-geometric series

$$S_\infty=\frac{a(1-r)+dr}{(1-r)^2}=\frac{1}{[1-(1/2)]^2}=4$$

Substituting the value of S_∞ in Eq. (i), we get $=2^{(1/4)\times 4}=2^1=2$.

27. Since the given numbers are in A.P.,

$$\therefore \quad 2\log\frac{3b}{5c}=\log\frac{5c}{a}+\log\frac{a}{3b}$$

$$\Rightarrow \left(\frac{3b}{5c}\right)^2=\frac{5c}{a}\cdot\frac{a}{3b} \Rightarrow 3b=5c$$

$$\Rightarrow b=\frac{5c}{3}$$

Also, $b^2=ac \quad \therefore \quad \dfrac{25c^2}{9}=ac \Rightarrow 9a=25c$

$$\therefore \quad \frac{9a}{5}=5c=3b \Rightarrow \frac{a}{5}=\frac{b}{3}=\frac{c}{9/5} \Rightarrow b+c<a$$

$\therefore$ a, b, c cannot be the sides of a triangle.

28. $S=100+102+104+\cdots+998$

$998=t_n=100+(n-1)2 \Rightarrow n=450$

29. $a^2+b^2,\ ab+bc,\ b^2+c^2$ are in G.P.

$\Rightarrow (ab+bc)^2=(a^2+b^2)(b^2+c^2)$

$\Rightarrow a^2b^2+b^2c^2+2ab^2c=a^2b^2+a^2c^2+b^2c^2+b^4$

$\Rightarrow b^4+a^2c^2-2ab^2c=0 \Rightarrow (b^2-ac)^2=0$

$\Rightarrow b^2=ac \Rightarrow$ a, b, c are in G.P.

30. Given: Series $1+\frac{2}{x}+\frac{3}{x^2}+\cdots+=\frac{1}{3}$

We know that the given series is an infinite arithmetico-geometric series with the first term $a = 1$, common difference $d = 1$ and common ratio $r = 1/x$.
Therefore, the sum of the given series

$$S_\infty = \frac{a}{1-r}+\frac{dr}{(1-r)^2}=\frac{1}{1-\frac{1}{x}}+\frac{1\times\frac{1}{x}}{\left(1-\frac{1}{x}\right)^2}$$

or $\frac{1}{3}=\frac{x}{x-1}+\frac{x}{(x-1)^2}$ or $2x^2+2x-1=0$ (i)

$$\therefore\quad x=\frac{-2\pm\sqrt{(2)^2-4\times2\times(-1)}}{2\times2}$$

$$=\frac{-2\pm\sqrt{4+8}}{4}=\frac{-2\pm2\sqrt{3}}{4}=\frac{\pm\sqrt{3}-1}{2}$$

or $x=\frac{\sqrt{3}-1}{2}$. The other value of x is not admissible.

31. By the given condition,

$$2(\log 2b-\log 3c)$$
$$=\log a-\log 2b+\log 3c-\log a$$
$$=\log 3c-\log 2b$$
$$\Rightarrow\quad 3(\log 2b-\log 3c)=0\quad\Rightarrow\quad 2b=3c$$
$$\Rightarrow\quad b=\frac{3c}{2}$$

Also, $b^2=ac\ \therefore\ \frac{9c^2}{4}=ac\ \Rightarrow\ c=\frac{4a}{9}$

and also, $b=\frac{3c}{2}=\frac{3}{2}\cdot\frac{4a}{9}=\frac{2a}{3}$

Since $a+b=a+\frac{2a}{3}=\frac{5a}{3}>c$

$$b+c=\frac{2a}{3}+\frac{4a}{9}=\frac{10a}{9}>a$$

$$c+a=\frac{13a}{9}>b$$

$\therefore$ a, b, c are the sides of a triangle. Also as a is the greatest side, let us find angle A of ΔABC.

$$\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{\frac{4a^2}{9}+\frac{16a^2}{81}-a^2}{2\cdot\frac{2a}{3}\cdot\frac{4a}{9}}=-\frac{29}{48}<0$$

Hence, ΔABC is an obtuse-angled triangle.

32. Let the GP be $a, ar, ar^2, \ldots$, where $0 < r < 1$. Then,

$a+ar+ar^2+\cdots=3$ and $a^2+a^2r^2+a^2r^4+\cdots=9/2$

$$\Rightarrow\quad \frac{a}{1-r}=3\quad\text{and}\quad\frac{a^2}{1-r^2}=\frac{9}{2}$$

$$\Rightarrow\quad \frac{9(1-r)^2}{1-r^2}=\frac{9}{2}\ \Rightarrow\ \frac{1-r}{1+r}=\frac{1}{2}\ \Rightarrow\ r=\frac{1}{3}$$

Putting $r=\frac{1}{3}$ in $\frac{a}{1-r}=3$, we get $a = 2$

Now, the required sum of the cubes is

$$a^3+a^3r^3+a^3r^6+\cdots=\frac{a^3}{1-r^3}=\frac{8}{1-(1/27)}=\frac{108}{13}$$

33. We have $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz}=\lambda$ (say)

$$\Rightarrow\quad p=\frac{a-x}{\lambda x},\ q=\frac{a-y}{\lambda y},\ r=\frac{a-z}{\lambda z}$$

Now p, q, r are in A.P.

$$\Rightarrow\quad \frac{a-x}{\lambda x},\frac{a-y}{\lambda y},\frac{a-z}{\lambda z}\ \text{are in A.P.}$$

$$\Rightarrow\quad \frac{a-x}{x},\frac{a-y}{y},\frac{a-z}{z}\ \text{are in A.P.}$$

$$\Rightarrow\quad \frac{a}{x}-1,\frac{a}{y}-1,\frac{a}{z}-1\ \text{are in A.P.}$$

$$\Rightarrow\quad \frac{a}{x},\frac{a}{y},\frac{a}{z}\ \text{are in A.P.}$$

$$\Rightarrow\quad \frac{1}{x},\frac{1}{y},\frac{1}{z}\ \text{are in A.P.}$$

$$\Rightarrow\quad x,\ y,\ z\ \text{are in H.P.}$$

34. Given: Series $\frac{1^3+2^3+3^3+4^3+\cdots+12^3}{1^2+2^2+3^2+4^2+\cdots+12^2}$

We know that the sum of the squares of the series

$$1^2+2^2+3^2+\cdots+n^2=\sum n^2=\frac{n(n+1)(2n+1)}{6}$$

And the sum of the cubes of the series

$$1^3+2^3+3^3+\cdots+n^3=\sum n^3=\left[\frac{n(n+1)}{2}\right]^2$$

Therefore, $\frac{1^3+2^3+3^3+\cdots+n^3}{1^2+2^2+3^2+\cdots+n^2}=\frac{\sum_{n=1}^{n=12}n^3}{\sum_{n=1}^{n=12}n^2}$

$$=\frac{\{[n(n+1)]/2\}^2}{[n(n+1)(2n+1)]/6}=\frac{3n(n+1)}{2(2n+1)}$$

$$=\frac{3\times12(12+1)}{2\times(24+1)}=\frac{36\times13}{2\times25}=\frac{234}{25}$$

35. $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\frac{1}{7^2}\cdots$

$$=\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\frac{1}{5^2}+\frac{1}{6^2}+\cdots\right)$$

$$-\left(\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\cdots\right)$$

$$=\frac{\pi^2}{6}-\frac{1}{4}\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots\right)$$

$$=\frac{\pi^2}{6}-\frac{1}{4}\cdot\frac{\pi^2}{6}=\frac{\pi^2}{6}\cdot\frac{3}{4}=\frac{\pi^2}{8}$$

36. $S_{2n}=S'_n$

$$\therefore\quad \frac{2n}{2}[2\cdot 2+(2n-1)3]=\frac{n}{2}[2\cdot 57+(n-1)2]$$

$$\therefore\quad 6n+1=n+56$$

$\Rightarrow\ 5n=55$ or $n=11\ \Rightarrow$ option (c)

37. We should have

$$2\cdot\frac{a}{2}=(5^{1+x}+5^{1-x})+(25^x+25^{-x})$$

or $a=5(5^x+5^{-x})+(5^{2x}+5^{-2x})$

$$=5[(5^{x/2}-5^{-x/2})^2+2\cdot 5^{x/2}\cdot 5^{-x/2}]$$

$$+[(5^x-5^{-x})^2+2\cdot 5^x\cdot 5^{-x}]$$

$$=5[(5^{x/2}-5^{-x/2})^2+2]+(5^x-5^{-x})^2+2$$

$\therefore\quad a\geq 5\cdot 2+2=12$

38. Given: Series $\frac{2}{3}+\frac{8}{9}+\frac{26}{27}+\frac{80}{81}+\cdots n$ terms.

We know that the nth term of the given series

$$t_n=\frac{3^n-1}{3^n}=1-\left(\frac{1}{3}\right)^n$$

Therefore, the sum of the n terms of the series

$$\sum t_n=n-\sum_{n=1}^{n}\left(\frac{1}{3}\right)^n$$

$$=n-\frac{\frac{1}{3}[1-(1/3)^n]}{1-(1/3)}=n-\frac{1}{2}(1-3^{-n})$$

$$=n+\frac{1}{2}(3^{-n}-1)$$

39. Let the first term be a and the common difference be d.

$$\therefore\quad S_m=\frac{m}{2}[2a+(m-1)d]\ \text{and}\ S_n=\frac{n}{2}[2a+(n-1)d]$$

$$\therefore\quad \frac{S_m}{S_n}=\frac{\frac{m}{2}[2a+(m-1)d]}{\frac{n}{2}[2a+(n-1)d]}\ \text{or}\ \frac{m^2}{n^2}=\frac{m}{n}\cdot\frac{2a+(m-1)d}{2a+(n-1)d}$$

or $\frac{2a+(m-1)d}{2a+(n-1)d}=\frac{m}{n}$

This is an identity in m and n.

So, putting $2m-1$ for m and $2n-1$ for n, we get

$$\frac{2a+(2m-1-1)d}{2a+(2n-1-1)d}=\frac{2m-1}{2n-1}$$

or $$\frac{a+(m-1)d}{a+(n-1)d}=\frac{2m-1}{2n-1}$$

or $$\frac{m\text{th term}}{n\text{th term}}=\frac{2m-1}{2n-1}$$

40. $$S_p=\frac{p}{2}(a+l)=q \qquad \text{(i)}$$

$$S_q=\frac{q}{2}(a+l)=p \qquad \text{(ii)}$$

Subtracting (ii) from (i), we get

$$\frac{1}{2}(a+l)(p-q)=q-p\quad\text{or}\quad\frac{1}{2}(a+l)=-1$$

$$S_{p+q}=\frac{(p+q)}{2}(a+l)=-(p+q)$$

41. $$t_n=\frac{1^3+2^3+3^3+\cdots+n^3}{1+3+5+\cdots+n\ \text{terms}}=\frac{\sum n^3}{\frac{n}{2}[2\cdot 1+(n-1)2]}$$

$$\Rightarrow\qquad t_n=\frac{1}{4}\cdot\frac{n^2(n+1)^2}{n^2}=\frac{1}{4}(n^2+2n+1) \qquad \text{(i)}$$

Now, $S_n=\frac{1}{4}(\sum n^2+2\sum n+1)$

$$=\frac{1}{4}\left[\frac{n(n+1)(2n+1)}{6}+2\cdot\frac{n(n+1)}{2}+n\right]$$

$$=\frac{n}{24}(2n^2+3n+1+6n+6+6)$$

$$=\frac{n}{24}(2n^2+9n+13)$$

Putting $n=16$, we get

$$S_n=\frac{16}{24}[2(256)+144+13]=\frac{2}{3}(669)=446$$

42. Given: Exponential series $1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots$

We know that $e^x=1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}\cdots$ (i)

And $e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} \cdots$ (ii)

Additing Eqs. (i) and (ii), we get

$$e^x + e^{-x} = \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}\right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \cdots\right)$$

$$= 2 + \frac{2x^2}{2!} + \frac{2x^4}{4!} + \cdots$$

$$= 2\left[1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right]$$

or $$1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \frac{e^x + e^{-x}}{2}$$

43. Since x_1, x_2, x_3 are in A.P., we may take $x_1 = \alpha - d$, $x_2 = \alpha$, $x_3 = \alpha + d$.

Since x_1, x_2, x_3 are the roots of $x^3 - x^2 + \beta x + \gamma = 0$,

$\therefore\ x_1 + x_2 + x_3 = 1 \Rightarrow 3\alpha = 1 \Rightarrow \alpha = \frac{1}{3}$

$$x_1x_2 + x_2x_3 + x_3x_1 = \beta$$

$\Rightarrow (\alpha - d)\alpha + \alpha(\alpha + d) + (\alpha - d)(\alpha + d) = \beta$

$\Rightarrow \alpha^2 - \alpha d + \alpha^2 + \alpha d + \alpha^2 - d^2 = \beta$

$\Rightarrow 3\left(\frac{1}{3}\right)^2 - d^2 = \beta \Rightarrow d^2 = \frac{1}{3} - \beta$

i.e. $\beta = \frac{1}{3} - d^2 \le \frac{1}{3} \quad \therefore\ \beta \in \left(-\infty, \frac{1}{3}\right)$

$x_1x_2x_3 = -\gamma \Rightarrow (\alpha - d)\alpha(\alpha + d) = -\gamma$

$\Rightarrow \alpha^3 - \alpha d^2 = -\gamma \Rightarrow \frac{1}{27} - \frac{1}{3}d^2 = -\gamma$

$\Rightarrow \gamma = \frac{1}{3}d^2 - \frac{1}{27} \ge -\frac{1}{27}$

Thus, $\gamma \in \left[-\frac{1}{27}, \infty\right)$

Hence, the required intervals are $\left(-\infty, \frac{1}{3}\right], \left[-\frac{1}{27}, \infty\right)$.

44. Since $n!$, $3(n!)$ and $(n+1)!$ are in G.P.,

$\therefore\ \frac{3(n!)}{n!} = \frac{(n+1)!}{3(n)!}$

$\Rightarrow 3 = \frac{1}{3}(n+1) \Rightarrow n + 1 = 9$ or $n = 8$

$\therefore\ n!, 5(n)!$ and $(n+1)!$ are $8!, 5(8)!, 9!$

$\Rightarrow 5(8)! - 8! = 9! - 5(8)!$

or $4(8)! = 4 \cdot (8)!$

$\therefore\ n!, 5(n)!$ and $(n+1)!$ are in A.P.

45. Let d be the common difference of the A.P. Then

$d = a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \cdots = a_{2n} - a_{2n-1}$

Now, $a_1^2 - a_2^2 + a_3^2 - a_4^2 + \cdots + a_{2n-1}^2 - a_{2n}^2$

$= (a_1 + a_2)(a_1 - a_2) + (a_3 + a_4)(a_3 - a_4) + \cdots + (a_{2n-1} + a_{2n})(a_{2n-1} - a_{2n})$

$= -d(a_1 + a_2 + a_3 + \cdots + a_{2n})$

$= -d\frac{2n}{2}(a_1 + a_{2n})$

$= -dn\frac{a_1^2 - a_{2n}^2}{a_1 - a_{2n}} = \frac{dn(a_1^2 - a_{2n}^2)}{a_{2n} - a_1}$

$= \frac{n}{2n-1}(a_1^2 - a_{2n}^2)$ [using $a_{2n} = a_1 + (2n-1)d$]

46. Given: Series $\frac{1}{2.3} + \frac{1}{4.5} + \frac{1}{6.7} + \cdots$

The given series may be written as

$$\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \cdots$$

$$= 1 - 1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \frac{1}{6} - \frac{1}{7} + \cdots$$

$$= 1 - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \cdots\right) = 1 - \log 2$$

$$= \log e - \log 2 = \log\left(\frac{e}{2}\right)$$

$\left(\text{where } 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} = \log 2\right)$

47. Since 1, a and p are in A.P.,

$\therefore\ a = \frac{p+1}{2} \Rightarrow p = 2a - 1$

Since 1, g, p are in A.P.,

$\therefore\ g^2 = p = 2a - 1 \quad \therefore\ 1 - 2a + g^2 = 0$

48. If there are 9 odd terms, then t_2, t_4, t_6, t_8 will be $[(9-1)/2] = 4$ in number and $t_1 + t_3 + t_5 + t_7 + t_9$ will be $[(9+1)/2] = 5$ in number. Hence, S_1 is an A.P. of n terms, but S_2 is an A.P. of $(n+1)/2$ terms with common difference $2d$.

$$S_1 = \frac{n}{2}[2a + (n-1)d] \quad \text{(i)}$$

$$S_2 = \frac{1}{2}\left(\frac{n+1}{2}\right)\left[2a + \left(\frac{n+1}{2} - 1\right)2d\right]$$

$$= \frac{n+1}{4}[2a + (n-1)d] \quad \text{(ii)}$$

$\therefore\ \frac{S_1}{S_2} = \frac{2n}{n+1}$

49. Multiplying the given expression by 2 and rewriting it as $(2x-3y)^2+(3y-4z)^2+(4z-2x)^2=0$

$\Rightarrow\ 2x=3y=4z\ \Rightarrow\ \frac{1}{x},\frac{1}{y},\frac{1}{z}$ are in A.P.

$\Rightarrow\ x, y, z$ are in H.P.

50. Given: Series $\log_4 2-\log_8 2+\log_{16} 2-\log_{32} 2+\ldots+\infty$

The given series may be written as

$$\frac{\log 2}{\log 4}-\frac{\log 2}{\log 8}+\frac{\log 2}{\log 16}-\frac{\log 2}{\log 32}+\cdots+\infty$$

$$=\frac{\log 2}{2\log 2}-\frac{\log 2}{3\log 2}+\frac{\log 2}{4\log 2}-\frac{\log 2}{5\log 2}+\cdots+\infty$$

$$=\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\cdots+\infty=1-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}+\cdots+\infty$$

$$=1-\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\infty\right)=1-\log 2$$

$$\left(\text{where } 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}+\cdots+=\log 2\right)$$

51. $t_p=a+(p-1)d=q$ and $t_q=a+(q-1)d=p$

$t_p-t_q=q-p=(p-q)d\ \Rightarrow\ d=-1$

$\therefore\ t_p=q=a+(p-1)(-1)=a-p+1$

or $a=q+p-1$

Now, $t_r=a+(r-1)d=q+p-1+(r-1)(-1)$

$=q+p-1-r+1=p+q-r$

52. Let $H_1, H_2, \ldots, H_9$ be the harmonic means between 2 and 3, then, 2, H_1, $H_2, \ldots$, H_9, 3 are in H.P.

$\therefore\ \frac{1}{2},\frac{1}{H_1},\frac{1}{H_2},\cdots,\frac{1}{H_9},\frac{1}{3}$ are in A.P. with common difference $D=\frac{2-3}{(9+1)\times 2\times 3}=-\frac{1}{60}\left[\because D=\frac{a-b}{(n+1)ab}\right]$

$\therefore\ \frac{1}{H_i}=\frac{1}{2}+iD;\ i=1, 2, 3, \ldots, 9\ \Rightarrow\ \frac{1}{H_i}=\frac{1}{2}-\frac{i}{60}$

$\Rightarrow\ \frac{6}{H_i}=3-\frac{i}{10}$ (i)

Let $A_1, A_2, \ldots, A_9$ be the nine A.M.'s between 2 and 3. Then, 2, $A_1, A_2, \ldots, A_9$, 3 are in A.P. with common difference $d=\frac{3-2}{9+1}=\frac{1}{10}\quad\left[\because d=\frac{b-a}{n+1}\right]$

$\therefore\ A_i=2+id;\ i=1, 2, \ldots, 9\ \Rightarrow\ A_i=2+\frac{i}{10}$ (ii)

From (i) and (ii), we have

$A_i+\frac{6}{H_i}=2+\frac{i}{10}+3-\frac{i}{10}=5$ for $i=1, 2, \ldots, 9$

53. Since $\sum_{n=1}^{n} n,\ \frac{\sqrt{10}}{3}\cdot\sum_{n=1}^{n} n^2,\ \sum_{n=1}^{n} n^3$ are in G.P., we have

$\frac{n(n+1)}{2},\frac{\sqrt{10}}{3}\cdot\frac{n(n+1)(2n+1)}{6},\frac{n^2(n+1)^2}{4}$ in G.P.

$\Rightarrow\ \frac{10\cdot n^2(n+1)^2(2n+1)^2}{9\cdot 36}=\frac{n(n+1)}{2}\cdot\frac{n^2(n+1)^2}{4}$

$\Rightarrow\ 20(2n+1)^2=81n(n+1)$

$\Rightarrow\ n^2+n-20=0$

$\therefore\ n=4\quad(\because n=-5$ is not admissible)

54. Given: $S=\frac{1}{1\cdot 2}-\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}-\frac{1}{4\cdot 5}+\cdots+\infty$

The given series may be written as

$$S=\left(\frac{1}{1\cdot 2}+\frac{1}{3\cdot 4}+\cdots\right)-\left(\frac{1}{2\cdot 3}+\frac{1}{4\cdot 5}+\cdots\right)$$

$$=\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots\right)-\left(\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\cdots\right)$$

$$=\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots\right)+\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots\right)-1 \quad \text{(i)}$$

We know that $\log(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots$ (ii)

Substituting $x=1$ in Eq. (ii), we get

$$\log 2=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots \quad \text{(iii)}$$

Comparing Eqs. (i) and (iii), we get

$$S=\log 2+\log 2-1=2\log 2-\log e$$

$$=\log 4-\log e=\log\left(\frac{4}{e}\right)$$

Therefore, $e^S=e^{\log\left(\frac{4}{e}\right)}=\frac{4}{e}$

55. Since $G^2=AH,\ \therefore\ H=\frac{G^2}{A}$

56. $\frac{1}{a_1}-\frac{1}{a_2}=\frac{a_2-a_1}{a_1a_2}=\frac{d}{a_1a_2}$

$\therefore\ \frac{1}{a_1a_2}=\frac{1}{d}\left[\frac{1}{a_1}-\frac{1}{a_2}\right],\ \frac{1}{a_2a_3}=\frac{1}{d}\left[\frac{1}{a_2}-\frac{1}{a_3}\right]\cdots$

$$\frac{1}{a_{n-1}a_n}=\frac{1}{d}\left[\frac{1}{a_{n-1}}-\frac{1}{a_n}\right]$$

Hence by addition, we get

$$S=\frac{1}{d}\left[\frac{1}{a_1}-\frac{1}{a_n}\right]=\frac{1}{d}\left[\frac{a_n-a_1}{a_1a_n}\right]=\frac{1}{d}\frac{(n-1)d}{a_1a_n}=\frac{n-1}{a_1a_n}$$

57. $$\frac{H_1+a}{H_1-a}+\frac{H_n+b}{H_n-b}=\frac{\frac{1}{a}+\frac{1}{H_1}}{\frac{1}{a}-\frac{1}{H_1}}+\frac{\frac{1}{b}+\frac{1}{H_n}}{\frac{1}{b}-\frac{1}{H_n}}$$

$$=\frac{\frac{1}{a}+\frac{1}{a}+d}{\frac{1}{a}-d-\frac{1}{a}}+\frac{\frac{1}{b}+\frac{1}{b}-d}{\frac{1}{b}+d-\frac{1}{b}}$$

$$=\frac{\frac{2}{a}+d}{-d}+\frac{\frac{2}{b}-d}{d}=\frac{\frac{2}{b}-\frac{2}{a}}{d}-2$$

$$=2(n+1)-2$$

$$[\because \quad \frac{1}{b}=\frac{1}{a}+(n+2-1)d\ ]$$

$$=2n$$

58. We have: first term = a, second term = b

$\therefore$ d = common difference = $b-a$

It is given that the middle term is c. It means that there is an odd number of terms in the A.P. Let there be $(2n+1)$ terms in the A.P. Then, the $(n+1)$th term is the middle term.

$\therefore$ Middle term $=c \Rightarrow a+nd=c$

$$\therefore \quad a+n(b-a)=c \Rightarrow n=\frac{c-a}{b-a}$$

$$\therefore \text{ Sum } =\frac{2n+1}{2}[2a+(2n+1-1)d]$$

$$=\frac{1}{2}\left[2\left(\frac{c-a}{b-a}\right)+1\right]\left[2a+2\left(\frac{c-a}{b-a}\right)(b-a)\right]$$

$$=\frac{1}{2}\left[\frac{2(c-a)}{b-a}+1\right](2c)=\frac{2c(c-a)}{b-a}+c$$

59. On simplification, we get

$$(b^2-ac)^2+(c^2-bd)^2+(ad-bc)^2\le 0$$

$$\Rightarrow b^2=ac,\ c^2=bd,\ ad=bc \Rightarrow \frac{b}{a}=\frac{c}{b}=\frac{d}{c}$$

$\therefore$ a, b, c, d are in G.P.

60. Let $S=1+\frac{4}{5}+\frac{7}{5^2}+\frac{10}{5^3}+\cdots$

$$\therefore \quad \frac{1}{5}S=\frac{1}{5}+\frac{4}{5^2}+\frac{7}{5^3}+\cdots$$

Subtracting, we get

$$\frac{4}{5}S=1+\frac{3}{5}+\frac{3}{5^2}+\frac{3}{5^3}+\cdots+\infty$$

$$=1+\frac{3/5}{1-(1/5)}=1+\frac{3/5}{4/5}=1+\frac{3}{4}=\frac{7}{4}$$

$$\therefore \quad S=\frac{7}{4}\times\frac{5}{4}=\frac{35}{16}$$

61. Let $\frac{2n+1}{2n-1}=r$

Then the given series is

$$S=r+3r^2+5r^3+7r^4+\cdots+(2n-1)r^n$$

$$rS=r^2+3r^3+5r^4+\cdots+(2n-3)r^n+(2n-1)r^{n+1}$$

$$(1-r)S=r+2r^2+2r^3+\cdots+2r^n-(2n-1)r^{n+1}$$

$$\Rightarrow \quad 36(1-r)=r+\frac{2r^2(1-r^{n-1})}{1-r}-(2n-1)r^{n+1}$$

$$\Rightarrow \quad 36(1-r)^2=r-r^2+2r^2-2r^{n+1}-(2n-1)r^{n+1}$$

$$+(2n-1)r^{n+2}$$

$$\Rightarrow \quad 36(1-r)^2=r+r^2-(2n+1)r^{n+1}+(2n-1)r^{n+2}$$

$$\Rightarrow \quad 36(1-r)^2=r+r^2-(2n-1)\left[\frac{2n+1}{2n-1}r^{n+1}-r^{n+2}\right]$$

$$\Rightarrow \quad 36(1-r)^2=r+r^2-(2n-1)\left[r.r^{n+1}-r^{n+2}\right]$$

$$\Rightarrow \quad 36(1-r)^2=r(1+r)$$

$$\Rightarrow \quad 36\left[1-\frac{2n+1}{2n-1}\right]^2=\frac{2n+1}{2n-1}\left[1+\frac{2n+1}{2n-1}\right]$$

$$\Rightarrow \quad 36\left[\frac{-2}{2n-1}\right]^2=\frac{2n+1}{2n-1}\left[\frac{4n}{2n-1}\right] \Rightarrow 36=n(2n+1)$$

$$\Rightarrow \quad n=4$$

62. If d is the common difference of the given A.P., then

$\log_y x=1+d$, $\log_z y=1+2d$, $-15\log_x z=1+3d$

$$\therefore \quad x=y^{1+d},\ y=z^{1+2d},\ z=x^{-\frac{1+3d}{15}}$$

Eliminating y and z, we get

$$x=z^{(1+d)(1+2d)}=x^{-\frac{(1+3d)(1+d)(1+2d)}{15}}$$

$$\therefore \quad (1+d)(1+2d)(1+3d)=-15$$

or $\quad 6d^3+11d^2+6d+16=0$

$$(d+2)(6d^2-d+8)=0 \qquad \therefore \quad d=-2$$

The other factor gives the imaginary values of d.

$\therefore$ $x=y^{-1}, y=z^{-3}, z=x^{1/3}$ or $x=y^{-1}, x=z^3, y=z^{-3}$

63. $a_1+(p-1)d, a_1+(q-1)d, a_1+(r-1)d$ are in G.P.

$$\therefore \quad \frac{a_q}{a_p}=\frac{a_1+(q-1)d}{a_1+(p-1)d}=\frac{a_1+(r-1)d}{a_1+(q-1)d}=\frac{r-q}{q-p}$$

64. Let a and b be two positive numbers.

$$\therefore \quad \frac{a+b}{2}=2 \text{ and } \sqrt{a(b+1)}=2$$

$$\Rightarrow \quad a+b=4 \text{ and } (ab+a)=4$$

On solving, we get $a = 1$ and $b = 3$

$\therefore$ H.M. of a, $b = \dfrac{2ab}{a+b} = \dfrac{6}{4} = \dfrac{3}{2}$

65. Let the three numbers in G.P. be a, ar, ar^2.
By the given condition, $a, 2ar, ar^2$ are in A.P.

$\Rightarrow\ 4ar = a + ar^2 \ \Rightarrow\ 4r = 1 + r^2$

$\Rightarrow\ r^2 - 4r + 1 = 0 \ \Rightarrow\ r = \dfrac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. The given series is an arithmetico-geometric series. The sum of the series is given by

$$\frac{3}{1-1/4} + \frac{d \times 1/4}{(1-1/4)^2} \quad \left[\text{Using } S = \frac{a}{1-r} + \frac{dr}{(1-r)^2}\right]$$

$$\Rightarrow \qquad 4 + \frac{4d}{9} = 8 \ \Rightarrow\ d = 9$$

2. $A \geq G \ \Rightarrow\ \dfrac{\log_{a_2} a_1 + \log_{a_3} a_2 + \cdots + \log_{a_{n+1}} a_n}{n}$

$$\geq \sqrt[n]{\log_{a_2} a_1 \cdot \log_{a_3} a_2 \cdot \cdots \cdot \log_{a_{n+1}} a_n}$$

$\Rightarrow\ \log_{a_2} a_1 + \log_{a_3} a_2 + \cdots + \log_{a_{n+1}} a_n \geq \sqrt[n]{1}$

$\therefore\ \log_{a_2} a_1 + \log_{a_3} a_2 + \cdots + \log_{a_{n+1}} a_n \geq n$

Hence option (d).

3. For $b = -1$, $\cos x = b \ \Rightarrow\ \cos x = -1$ is satisfied for $x = \pi, 3\pi, 5\pi$, etc. which forms an A.P. with common difference 2π. For no other value of x, out of 1/2 and $\sqrt{3}/2$, the roots will form an A.P.

4. Consider the sequesnce 3, 7, 13, 21,..., i.e. 3 + (3 + 4) + (3 + 4 + 6) + (3 + 4 + 6 + 8) + $\cdots$

$\therefore$ nth term of the sequence

$= 3 + 4 + 6 + 8 + \cdots$ to n terms

$= 1 + (2 + 4 + 6 + 8 + \cdots$ to n terms)

$= 1 + 2(1 + 2 + 3 + 4 + \cdots$ to n terms)

$= 1 + 2 \cdot \dfrac{n(n+1)}{2} = n^2 + n + 1$

For the given series,

$$t_n = \cot^{-1}(n^2 + n + 1) = \tan^{-1}\frac{1}{n^2 + n + 1}$$

$$= \tan^{-1}\frac{1}{1 + n(n+1)} = \tan^{-1}\frac{(n+1) - n}{1 + n(n+1)}$$

$$= \tan^{-1}(n+1) - \tan^{-1} n$$

$\therefore\ t_1 = \tan^{-1} 2 - \tan^{-1} 1$

$t_2 = \tan^{-1} 3 - \tan^{-1} 2$

$t_3 = \tan^{-1} 4 - \tan^{-1} 3$

$t_n = \tan^{-1}(n+1) - \tan^{-1} n$

Adding, L.H.S. $= \tan^{-1}(n+1) - \tan^{-1} 1$

$$= \tan^{-1}\frac{(n+1) - 1}{1 + (n+1) \cdot 1} = \tan^{-1}\frac{n}{n+2}$$

5. $t_n = t_{n+1} + t_{n+2} \ \Rightarrow\ ar^{n-1} = ar^n + ar^{n+1}$

$\Rightarrow\ r^2 + r - 1 = 0 \quad \therefore\ r = \dfrac{-1 \pm \sqrt{5}}{2}$

We choose only the +ve value of r as G.P. consists of only positive terms.

$\therefore\ r = \dfrac{\sqrt{5} - 1}{2}$

6. For G.P., $t_n = 2^{n-1}$; for AP, $t_m = 1 + (m-1)3$. They are common if

$$2^{n-1} = 3m - 2 \text{ or } 2^{n-2} + 1 = \frac{3m}{2} \leq 150$$

$\Rightarrow\ n \leq 9,\ m \leq 100$

By trial, n = 1, m = 1; n = 3, m = 2; n = 5, m = 6; n = 7, m = 22; n = 9, m = 86.

7. $\Sigma n = \dfrac{1}{5}\Sigma n^2$ (given)

$\therefore\ \dfrac{n(n+1)}{2} = \dfrac{1}{5}\dfrac{n(n+1)(2n+1)}{6}$

$\Rightarrow\ 1 = \dfrac{2n+1}{15} \Rightarrow 2n + 1 = 15 \Rightarrow n = 7$

8. Since a, b, c are in A.P., $\therefore\ a + c = 2b$ (i)

Also a^2, b^2, c^2 are in G.P., $\therefore\ a^2c^2 = b^4$ (ii)

Again, $a + b + c = \dfrac{3}{2}$ (given)

$\therefore\ b = \dfrac{3}{2} - (a + c)$ (iii)

From (i), we get $b = \dfrac{a+c}{2}$

Putting b in (iii), we get $\dfrac{a+c}{2} = \dfrac{3}{2} - (a + c)$

$\Rightarrow\ \dfrac{3}{2}(a + c) = \dfrac{3}{2} \ \Rightarrow\ a + c = 1$

Therefore, (iii) gives $b = \dfrac{3}{2} - 1 = \dfrac{1}{2}$

Also $c = 1 - a$

Therefore, (ii) gives $a^2(1-a)^2 = \dfrac{1}{16}$

$\Rightarrow\ a(1-a) = \pm\dfrac{1}{4}$

Case I. If $a < 0$, then $a(1-a) = -\frac{1}{4} \Rightarrow a^2 - a - \frac{1}{4} = 0$

$\Rightarrow a = \frac{1 \pm \sqrt{1+1}}{2} = \frac{1 \pm \sqrt{2}}{2}$

But $a < 0 \quad \therefore \quad a = \frac{1-\sqrt{2}}{2} = \frac{1}{2} - \frac{1}{\sqrt{2}}$

Case II. When $a > 0$, then

$a(1-a) = \frac{1}{4} \Rightarrow a^2 - a + \frac{1}{4} = 0 \Rightarrow \left(a - \frac{1}{2}\right)^2 = 0 \Rightarrow a = \frac{1}{2}$

But $a < b$ and $b = (1/2)$.

Therefore, this solution is rejected.

Hence, $a = \frac{1}{2} - \frac{1}{\sqrt{2}}$

9. $\sum_{n=1}^{m} \log \frac{a^{2n-1}}{b^{m-1}} = \sum_{n=1}^{m} \left[(2n-1)\log a - \log(b^{m-1})\right]$

$= \left[2\sum_{n=1}^{m} n - \sum_{n=1}^{m} 1\right] \log a - \sum_{n=1}^{m} \log b^{m-1}$

$= \left[\frac{2(m)(m+1)}{2} - m\right] \log a - m \log b^{m-1}$

$= m^2 \log a - m \log(b^{m-1})$

$= \frac{m}{2}\left(2m \log a - 2 \log b^{m-1}\right) = \frac{m}{2}\left(\log \frac{a^{2m}}{b^{2m-2}}\right)$

10. Here $a + b = a_1 + a_{2n} = a_2 + a_{2n-1} = \ldots = a_n + a_{n+1}$

$ab = g_1 \cdot g_{2n} = g_2 \cdot g_{2n-1} = \ldots = g_n \cdot g_{n+1}$

and $H = \frac{2ab}{a+b}$ or $\frac{ab}{a+b} = \frac{H}{2}$

$\Rightarrow \frac{g_1 \cdot g_{2n}}{a_1 + a_{2n}} = \frac{g_2 \cdot g_{2n-1}}{a_2 + a_{2n-1}} = \cdots = \frac{g_n \cdot g_{n+1}}{a_n + a_{n+1}} = \frac{H}{2}$

$\Rightarrow \frac{a_1 + a_{2n}}{g_1 \cdot g_{2n}} = \frac{a_2 + a_{2n-1}}{g_2 \cdot g_{2n-1}} = \cdots = \frac{a_n + a_{n+1}}{g_n \cdot g_{n+1}} = \frac{2}{H}$

$\Rightarrow \frac{a_1 + a_{2n}}{g_1 \cdot g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 \cdot g_{2n-1}} + \cdots + \frac{a_n + a_{n+1}}{g_n \cdot g_{n+1}} = \frac{2n}{H}$

Hence the correct option is (a).

11. $S = \frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \cdots$ to n terms

$= \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{4}\right) + \left(1 - \frac{1}{8}\right) + \left(1 - \frac{1}{16}\right) + \cdots$ to n terms

$= (1+1+1+\cdots+1) - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots \text{to } n \text{ terms}\right)$

$= n - \frac{1/2\,[1-(1/2)^n]}{1-(1/2)} = n - \left(1 - \frac{1}{2^n}\right)$

$= n - 1 + \frac{1}{2^n} = n - 1 + 2^{-n}$

12. $t_r = 2^{r/2} + 2^{-r/2} \Rightarrow t_r^2 = 2^r + 2^{-r} + 2$

$\Rightarrow \sum_{r=1}^{10} t_r^2 = \left(2 + 2^2 + \cdots + 2^{10}\right) + \left(\frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{10}}\right) + 20$

$= 2(2^{10} - 1) + \left(1 - \frac{1}{2^{10}}\right) + 20$

$= \frac{2^{21} - 1}{2^{10}} + 19$

13. t_4, t_7 and t_{10} of a G.P. are also in G.P. of common ratio r^3. ($\because \quad t_4 = ar^3 = f$, $t_7 = ar^6 = g$, $t_{10} = ar^9 = h$),

$\therefore \quad g^2 = fh$

14. t_7 of an H.P. $= \frac{1}{a+6d} = 8 \quad \Rightarrow \quad a + 6d = \frac{1}{8}$

t_8 of an H.P. $= \frac{1}{a+7d} = 7$

$\Rightarrow \quad a + 7d = \frac{1}{7} \quad \therefore \quad d = \frac{1}{7} - \frac{1}{8} = \frac{1}{56}$

Now, $t_{15} = \frac{1}{a+14d} = \frac{1}{a+7d+7d} = \frac{1}{\frac{1}{7} + \frac{1}{8}} = \frac{56}{15}$

15. The required sum of n terms is

$\frac{1}{\sqrt{1}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{7}} + \cdots + \frac{1}{\sqrt{2n-1}+\sqrt{2n+1}}$

$= \frac{1}{2}[(\sqrt{3} - \sqrt{1}) + (\sqrt{5} - \sqrt{3}) + (\sqrt{7} - \sqrt{5})$

$+ (\sqrt{2n+1} - \sqrt{2n-1})]$

$= \frac{1}{2}(\sqrt{2n+1} - 1)$

16. x_1, x_2, x_3 are a, ar, ar^2 respectively

y_1, y_2, y_3 are b, br, br^2 respectively

$\therefore$ A, B, C are (a, b), (ar, br), (ar^2, br^2) respectively.

Slope of AB = b/a = slope of BC.

Hence, the points are collinear.

17. $\frac{\sin x + \cos x + \text{cosec } 2x}{3} \geq \sqrt[3]{\sin x \cdot \cos x \cdot \text{cosec} 2x} = \sqrt[3]{\frac{1}{2}}$

$\therefore \quad (\sin x + \cos x + \text{cosec} 2x)^3 \geq \frac{27}{2} = 13.5$

18. Given $x_1 x_2 \cdots x_n = 1$

Since A.M. $\geq$ G.M., we have

$\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right) \geq (x_1 x_2 \ldots x_n)^{1/2} = (1)^{1/2} = 1$

$\therefore \quad x_1 + x_2 + \cdots + x_n \geq n$

$\therefore \quad x_1 + x_2 + \cdots + x_n$ can never be less than n.

19. $\frac{1}{H}=\frac{1}{2}\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)$

$$=\frac{\alpha+\beta}{2\alpha\beta}=\frac{4+\sqrt{5}}{(5+\sqrt{2})}\cdot\frac{(5+\sqrt{2})}{8+2\sqrt{5}}\cdot\frac{1}{2}=\frac{1}{4}$$

$\therefore \quad H=4$

20. We can write S as

$$S=(1-2)(1+2)+(3-4)(3+4)+\cdots+(2001-2002)(2001+2002)+2003^2$$

$$=-(1+2+3+4+\cdots+2001+2002)+2003^3$$

$$=-\frac{1}{2}(2002)(2003)+2003^2=2,007,006$$

21. $\alpha,\beta,\gamma,\delta$ being in increasing G.P., they may be taken as k, kr, kr^2, kr^3, where $r>1$.

The sum of the roots of the two equations,

$S_1=\alpha+\beta=k(1+r)=3,\ S_2=\gamma+\delta=kr^2(1+r)=12$

$\therefore \quad 3r^2=12 \quad$ or $\quad r=2 \quad \therefore \quad k=1$

$P_1=\alpha\beta=k^2r=a,\ P_2=\gamma\beta=k^2r^5=b$

Substituting for r and k, $a=2$, $b=32$.

22. Let $S_{99}=2+7+14+23+34+\cdots+t_{98}+t_{99}$ (i)

Also, $S_{99}=2+7+14+23+\cdots+t_{98}+t_{99}$ (ii)

(i)$-$(ii) $\Rightarrow$ $0=(2+5+7+9+11+\cdots$ up to 99 terms$)-t_{99}$

$\therefore \quad t_{99}=2+(5+7+9+11+\cdots$ up to 98 terms$)$

$$=2+\frac{98}{2}[2(5)+(98-1)2]=2+49(10+194)$$

$$=2+49\times204=9998$$

23. a, b, c, d are in A.P.

$\therefore\ d, c, b, a$ are also in A.P.

$\Rightarrow \quad \frac{d}{abcd},\frac{c}{abcd},\frac{b}{abcd},\frac{a}{abcd}$ are also in A.P.

$\Rightarrow \quad \frac{1}{abc},\frac{1}{abd},\frac{1}{acd},\frac{1}{bcd}$ are in A.P.

$\Rightarrow \quad abc, abd, acd, bcd$ are in H.P.

24. $A=\frac{a_1+a_2+\cdots+a_n}{n},\ G=\sqrt[n]{a_1a_2\ldots a_n}=1$

$A\geq G \quad \Rightarrow \quad a_1+a_2+\cdots+a_n\geq n$

25. Let $S=\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\cdots$, then $S=\frac{1/3}{1-1/3}=\frac{1}{2}$

Let $y=(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\cdots+\infty\right)}$

Then $y=(0.16)^{\log_{2.5}(1/2)}$

$\Rightarrow \quad \log y=\log_{2.5}(1/2)\log(0.16)$

$\Rightarrow \quad \log y=-\frac{\log 2}{\log 2.5}\times 2\log 0.4$

$\Rightarrow \quad \log y=-\frac{\log 2}{\log(1/0.4)}\times 2\log 0.4$

$\Rightarrow \quad \log y=-\frac{\log 2}{-\log 0.4}\times 2\log 0.4$

$\Rightarrow \log y=2\log 2=\log 4 \Rightarrow y=4$

26. $S_{2n}=5(t_1+t_3+t_5+\cdots+n)$

$\therefore \quad \frac{a(1-r^{2n})}{1-r}=5\cdot\frac{a[1-(r^2)^n]}{1-r^2}$

$\therefore \quad 1+r=5 \quad \Rightarrow \quad r=4$

27. $t_{m+1}=a+md,\ t_{n+1}=a+nd,\ t_{r+1}=a+rd$

$\therefore \quad (a+nd)^2=(a+md)(a+rd)$ (i)

[as per the given condition]

Also m, n, r are in H.P.

$\therefore \quad n=\frac{2mr}{m+r} \quad \Rightarrow \quad mr=\frac{(m+r)n}{2}$ (ii)

By (i), we have

$$\left(\frac{a}{d}+n\right)^2=\left(\frac{a}{d}+m\right)\left(\frac{a}{d}+r\right)$$

$$\Rightarrow \quad \left(\frac{a}{d}\right)^2+n^2+\frac{2an}{d}=\left(\frac{a}{d}\right)^2+\frac{a}{d}(m+r)+mr$$

$$\Rightarrow \quad n^2=\frac{a}{d}\left[(m+r)-2n\right]+mr$$

$$\Rightarrow \quad \frac{a}{d}=\frac{n^2-mr}{m+r-2n} \qquad \text{(iii)}$$

By (ii) and (iii), we get

$$\frac{a}{d}=\frac{n^2-\frac{(m+r)n}{2}}{m+r-2n}=\frac{\frac{n}{2}[2n-(m+r)]}{-[2n-(m+r)]}=-\frac{n}{2}$$

Hence, $\frac{a}{d}=-\frac{n}{2}$

28. $\ln(a+c),\ \ln(c-a),\ \ln(a-2b+c)$ are in A.P.

$\Rightarrow \quad a+c,\ c-a,\ a-2b+c$ are in G.P.

$\Rightarrow \quad (c-a)^2=(a+c)(a-2b+c)$

$\Rightarrow \quad (c-a)^2=(a+c)^2-2b(a+c)$

$\Rightarrow \quad 2b(a+c)=(a+c)^2-(c-a)^2$

$\Rightarrow \quad 2b(a+c)=4ac$

$\Rightarrow \quad b=\frac{2ac}{a+c} \quad \Rightarrow \quad a,b,c$ are in H.P.

29. Sum $(S) = \frac{2}{10}+\frac{4}{10^3}+\frac{6}{10^5}+\frac{8}{10^7}+\cdots$ (i)

This is an arithmetico-geometric series, where $a = 2$, $r = 1/100$ and $d = 2$.

$\therefore \quad \frac{1}{10^2}S = \frac{2}{10^3}+\frac{4}{10^5}+\cdots+\infty$ (ii)

Subtracting (ii) from (i),

$$\left(S-\frac{1}{10^2}S\right)=\frac{2}{10}+\frac{2}{10^3}+\frac{2}{10^5}+\cdots+\infty \quad \text{(G.P.)}$$

$$S\left(\frac{99}{100}\right)=\frac{2/10}{[1-(1/10)]} \Rightarrow \frac{20}{99} \quad \left(\because S_\infty=\frac{a}{1-r}\right)$$

$\therefore \quad S\left(\frac{99}{100}\right)=\frac{20}{99}$ or $S=\frac{2000}{9801}$

30. $\tan 80^\circ = \tan(70^\circ+10^\circ)=\frac{\tan 70^\circ+\tan 10^\circ}{1-\tan 70^\circ\tan 10^\circ}$

$\Rightarrow \tan 80^\circ-\tan 80^\circ\cdot\tan 70^\circ\cdot\tan 10^\circ=\tan 70^\circ+\tan 10^\circ$

$\Rightarrow \tan 80^\circ-\tan 70^\circ=\tan 70^\circ+\tan 10^\circ$

$(\because \tan 80^\circ\cdot\tan 10^\circ=1)$

$\Rightarrow \tan 80^\circ=\tan 10^\circ+2\tan 70^\circ$

$\Rightarrow \tan 80^\circ+\tan 10^\circ=2(\tan 10^\circ+\tan 70^\circ)$

Hence the given sequence is in A.P.

31. We have $a+c=2b \Rightarrow 3b=a+b+c=3/2 \Rightarrow b=1/2$

Thus, $a + c = 1$ and $a^2c^2=b^4=1/16$

$\Rightarrow a^2(1-a)^2=1/16 \Rightarrow a(1-a)=\pm 1/4$

$\Rightarrow a^2-a\pm 1/4=0$

$\Rightarrow 4a^2-4a\pm 1=0 \Rightarrow (2a-1)^2=0$ or $(2a-1)^2=2$

$\Rightarrow a=\frac{1}{2}$ or $a=\frac{1}{2}\pm\frac{1}{\sqrt{2}}$

As $a<b<c$ and $b=\frac{1}{2}$, we get $a=\frac{1}{2}-\frac{1}{\sqrt{2}}$

32. The sum of G.P. of $(n + 1)$ terms $=\frac{1\cdot 3^{n+1}-1}{3-1}>1000$

or $3^{n+1}>2001$ (i)

Now $3^7=3^4\cdot 3^3=2187$, $3^6=729$

Hence we must have $n+1>7 \quad \therefore n>6$

Since $n > 6$ and n is an integer, the minimum value of n must be 7.

33. If in A.P., $\sqrt{y}=\sqrt{x}+(n-1)d$ and $\sqrt{z}=\sqrt{x}+(m-1)d$

$\therefore \quad \frac{\sqrt{y}-\sqrt{x}}{\sqrt{z}-\sqrt{x}}=\frac{n-1}{m-1}$, a rational number

As, x, y, z are prime, $\frac{\sqrt{y}-\sqrt{x}}{\sqrt{z}-\sqrt{x}}$ is irrational.

$\therefore$ irrational = rational (which is absurd)

So $\sqrt{x},\sqrt{y},\sqrt{z}$ are not in A.P.

Similarly, they are not in G.P. or H.P.

34. Let there be n terms in the A.P. Then,

$c=a+(n-1)(b-a) \qquad [\because d=b-a]$

$\Rightarrow n=\frac{b+c-2a}{b-a}$

$\because$ Sum of n terms $=\frac{n}{2}(a+c)=\frac{(a+c)(b+c-2a)}{2(b-a)}$

35. Here $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots$ to $\infty=\frac{\pi^2}{8}$

Let $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots$ to $\infty = x$. Then,

$$x=\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots \text{ to } \infty$$

$$=\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots \text{ to } \infty\right)+\left(\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\cdots \text{ to } \infty\right)$$

$$=\frac{\pi^2}{8}+\frac{1}{4}\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots \text{ to } \infty\right)=\frac{\pi^2}{8}+\frac{1}{4}x$$

or $x = \pi^2/6$.

36. $y^2=xz \Rightarrow 2\ln y=\ln x+\ln z$

$\Rightarrow$ $\ln x$, $\ln y$, $\ln z$ are in A.P.

$\Rightarrow$ $1+\ln x$, $1+\ln y$, $1+\ln z$ are in A.P.

$\Rightarrow \frac{1}{1+\ln x}, \frac{1}{1+\ln y}, \frac{1}{1+\ln z}$ are in H.P.

37. If a, b, c are in H.P., $b=\frac{2ac}{a+c}$

We have

$\log(a + c) + \log(a - 2b + c)$

$$=\log(a+c)+\log\left[(a+c)-\frac{4ac}{a+c}\right]$$

$$=\log[(a+c)^2-4ac]=\log[(c-a)^2]=2\log(c-a)$$

38. If a is the side of a square, then $d=a\sqrt{2}$.

By the given condition, $a_n=\sqrt{2}a_{n+1}$

or $a_{n+1}=\frac{a_n}{\sqrt{2}}=\frac{a_{n-1}}{(\sqrt{2})^2}=\frac{a_{n-2}}{(\sqrt{2})^3}=\ldots=\frac{a_1}{(\sqrt{2})^n}$

Replacing n by $n - 1$, we get

$$a_n=\frac{a_1}{(\sqrt{2})^{n-1}}=\frac{10}{(2)^{(n-1)/2}}$$

Area of $S_n<1 \Rightarrow a_n^2<1 \Rightarrow \frac{100}{2^{n-1}}<1$ or $200<2^n$

or $2^n>200$

Now $2^7=128<200$, $2^8=256>200$

$\therefore n=10 \Rightarrow$ Option (d)

39. Let a be the first term and d the common difference of the A.P. Let A be the first term and R the common ratio of the G.P.

$\therefore \quad x = a + (m-1)d$ (i)

$x = AR^{m-1}$ (ii)

$y = a + (n-1)d$ (iii)

$y = AR^{n-1}$ (iv)

$z = a + (p-1)d$ (v)

$z = AR^{p-1}$ (vi)

From (iii) and (v),

$y - z = (n-p)d$ (vii)

By (ii) and (vii),

$x^{y-z} = (AR^{m-1})^{(n-p)d} = A^{(n-p)d} R^{(m-1)(n-p)d}$

Similarly, $y^{z-x} = A^{(p-m)d} R^{(n-1)(p-m)d}$

$z^{x-y} = A^{(m-n)d} R^{(p-1)(m-n)d}$

$\therefore \quad x^{y-z} y^{z-x} z^{x-y} = A^0 R^0 = 1$

$\therefore \quad x^y y^z z^x = x^z y^x z^y$

40. We know that A.M. ≥ G.M.

$\therefore \quad \dfrac{a+b+c}{3} \geq (abc)^{1/3}$

$\Rightarrow \quad \dfrac{12}{3} \geq (abc)^{1/3} \Rightarrow abc \leq 64$

$\therefore$ Maximum value of abc is 64 and this occurs when $a = b = c$.

$\therefore$ Statement 1 true.

Also, Statement 2 is the correct explanation for Statement 1.

41. a, b, c are in H.P.

$\therefore \quad \sqrt{ac} > b \qquad (\because$ G.M. > H.M.)

$\Rightarrow \quad (\sqrt{ac})^n > b^n \Rightarrow \sqrt{a^n c^n} > b^n$ (i)

Also $\dfrac{a^n + c^n}{2} > \sqrt{a^n c^n} \qquad (\because$ A.M. > G.M.)

$\therefore \quad \dfrac{a^n + c^n}{2} > b^n$ (ii)

$\therefore$ Statement 2 is true.

$\Rightarrow \quad 2b^n - a^n - c^n < 0 \Rightarrow 2b^{101} - a^{101} - c^{101} < 0$

$\therefore$ Product of roots of the given equation

$$= \frac{2b^{101} - a^{101} - c^{101}}{1} < 0$$

$\therefore$ Statement 1 is true.

Also, Statement 2 is the correct explanation for Statement 1.

42. Statement 2 is trivially true.

$(x+y)^2 = x^2 + y^2 + 2xy = 1 + 2xy$ (i)

A.M. ≥ G.M. $\Rightarrow \dfrac{x+y}{2} \geq \sqrt{xy}$

$\Rightarrow \quad (x+y)^2 \geq 4xy$

$\Rightarrow \quad (x+y)^2 \geq 2[(x+y)^2 - 1]$ [using (i)]

$\Rightarrow \quad (x+y)^2 \leq 2 \Rightarrow |x+y| \leq \sqrt{2}$

$\therefore$ Statement 1 is true.

Also, Statement 2 is the correct explanation for Statement 1.

43. Statement 2 is trivially true.

We have $\dfrac{S_n}{S_n'} = \dfrac{7n+1}{4n+17} = \dfrac{(7n+1)n}{(4n+17)n} = \dfrac{7n^2+n}{4n^2+17n}$

Let $S_n = (7n^2+n)\lambda$ and $S_n' = (4n^2+17n)\lambda$

$\therefore \quad \dfrac{t_n}{t_n'} = \dfrac{S_n - S_{n-1}}{S_n' - S_{n-1}'}$

$$= \frac{\lambda\{(7n^2+n) - [7(n-1)^2 + (n-1)]\}}{\lambda\{(4n^2+17n) - [4(n-1)^2 + 17(n-1)]\}}$$

$$= \frac{7n^2 + n - (7n^2 - 14n + 7 + n - 1)}{4n^2 + 17n - (4n^2 - 8n + 4 + 17n - 17)}$$

$$= \frac{14n - 6}{8n + 13}$$

$\therefore$ Statement 1 is false.

44. $1^3 - 2^3 + 3^3 - 4^3 + \cdots + 9^3 = \sum_{n=1}^{9} n^3 - 2(2^3 + 4^3 + 6^3 + 8^3)$

$$= \sum_{n=1}^{9} n^3 - 2 \cdot 2^3 \left(\sum_{n=1}^{4} n^3 \right) = \left(\frac{9 \cdot 10}{2} \right)^2 - 16 \left(\frac{4 \cdot 5}{2} \right)^2$$

$= 2025 - 1600 = 425$

45. $2^{1/4} \cdot 4^{1/8} \cdot 8^{1/16} \ldots \infty$

$= 2^{1/4} \cdot (2^2)^{1/8} \cdot (2^3)^{1/16} \ldots \infty$

$= 2^{1/4} \cdot 2^{2/8} \cdot 2^{3/16} \ldots \infty$

$= 2^{1/4 + 2/8 + 3/16 + \ldots \infty} = 2^S$

where $S = \dfrac{1}{4} + \dfrac{2}{8} + \dfrac{3}{16} + \cdots$ (i)

$\therefore \quad \dfrac{S}{2} = \dfrac{1}{8} + \dfrac{2}{16} + \cdots$ (ii)

Subtracting (ii) from (i),

$\therefore \quad S\left(1 - \dfrac{1}{2}\right) = \dfrac{1}{4} + \dfrac{1}{8} + \dfrac{1}{16} + \cdots = \dfrac{1/4}{1 - (1/2)} = \dfrac{1}{2}$

$\Rightarrow \quad \dfrac{S}{2} = \dfrac{1}{2} \Rightarrow S = 1$

$\therefore$ The given expression $= 2^1 = 2$.

46. $t_5 = ar^4 = 2$

Product of the first 9 terms of the G.P.

$= a(ar)(ar^2)\ldots(ar^8)$

$= a^9 r^{1+2+\cdots+8}$

$= a^9 r^{\frac{8}{2}(1+8)} = a^9 r^{36}$

$= (ar^4)^9 = 2^9 = 512$

47. Since a, b, c are in A.P.,

$\therefore \quad 2b = a + c$

Since $\frac{a}{bc} + \frac{2}{b} = \frac{a+2c}{bc} \neq \frac{2}{c}$, (i)

$\therefore \quad \frac{a}{bc}, \frac{1}{c}, \frac{2}{b}$ are not in A.P.

Since $\frac{bc}{a} + \frac{b}{2} = \frac{2bc+ab}{2a} \neq c$,

therefore, the given numbers are not in H.P.

Again $\frac{a}{bc} \cdot \frac{2}{b} = \frac{2a}{b^2c} \neq \frac{1}{c^2}$

Therefore, the given numbers are not in G.P.

48. Let the G.P. be $a, ar, \ldots$

$\therefore \quad \frac{a}{1-r} = 20$ (i)

and $\frac{a^2}{1-r^2} = 100$ (ii)

From (i), $a^2 = 400(1-r)^2$

From (ii), $a^2 = 100(1-r^2)$

$\therefore \quad 400(1-r)^2 = 100(1-r^2)$

$\Rightarrow \quad 4(1-r) = 1+r \quad \Rightarrow \quad 5r = 3 \quad \Rightarrow \quad r = \frac{3}{5}$

49. Here, $1, \frac{1}{2}\log_3(3^{1-x}+2), \log_3(4\cdot 3^x - 1)$ are in A.P.

$\Rightarrow \quad 2 \cdot \frac{1}{2}\log_3(3^{1-x}+2) = 1 + \log_3(4\cdot 3^x - 1)$

$\Rightarrow \quad \log_3(3^{1-x}+2) = \log_3 3(4\cdot 3^x - 1) \quad (\because \quad \log_3 3 = 1)$

$\Rightarrow \quad 3^{1-x} + 2 = 3(4\cdot 3^x - 1)$

$\Rightarrow \quad 3\cdot 3^{-x} + 2 = 12\cdot 3^x - 3$

Putting $3^x = y$, we get $\frac{3}{y} + 2 = 12y - 3$

$\Rightarrow \quad 12y^2 - 5y - 3 = 0$

$\Rightarrow \quad (3y+1)(4y-3) = 0 \quad \Rightarrow \quad y = -\frac{1}{3}, \frac{3}{4}$

Rejecting the –ve value, $y = \frac{1}{3} \quad \Rightarrow \quad 3^x = \frac{3}{4}$

$\Rightarrow \quad x = \log_3\left(\frac{3}{4}\right) = \log_3 3 - \log_3 4 = 1 - \log_3 4$

50. Let A be the first term and R the common ratio of the G.P., then,

$a = AR^{p-1} \Rightarrow \log a = \log A + (p-1)\log R$ (i)

$b = AR^{q-1} \Rightarrow \log b = \log A + (q-1)\log R$ (ii)

$c = AR^{r-1} \Rightarrow \log c = \log A + (r-1)\log R$ (iii)

Multiplying (i), (ii) and (iii) by $(q-r)$, $(r-p)$ and $(p-q)$ respectively and adding, we get

$(q-r)\log a + (r-p)\log b + (p-q)\log c = 0$

$\Rightarrow \quad \Delta = 0$

51. Let $f(x) = px^2 + qx + r$

Then, $f(1) = f(-1) = p + q + r = p - q + r$

$\Rightarrow \quad 2q = 0 \quad \Rightarrow \quad q = 0$

$\therefore \quad f(x) = px^2 + r$

$\therefore \quad f'(a) = 2pa + r, f'(b) = 2pb + r$

$f'(c) = 2pc + r$

$\because \quad a, b, c$ are in A.P.

$\Rightarrow \quad 2pa, 2pb, 2pc$ are in A.P.

$\Rightarrow \quad 2pa + r, 2pb + r, 2pc + r$ are in A.P.

$\therefore \quad f'(a), f'(b), f'(c)$ are in A.P.

52. Let α, β be the roots of $ax^2 + bx + c = 0$

$\Rightarrow \quad \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\beta^2 + \alpha^2}{\alpha^2\beta^2}$

$\Rightarrow \quad \alpha + \beta = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$

$\Rightarrow \quad \frac{-b}{a} = \frac{(-b/a)^2 - 2c/a}{(c/a)^2} \quad \Rightarrow \quad \frac{-b}{a} = \frac{b^2 - 2ca}{c^2}$

$\Rightarrow \quad -bc^2 = ab^2 - 2a^2c \quad \Rightarrow \quad 2a^2c = ab^2 + bc^2$

$\Rightarrow \quad \frac{2a}{b} = \frac{b}{c} + \frac{c}{a} \quad \Rightarrow \quad \frac{c}{a}, \frac{a}{b}, \frac{b}{c}$ are in A.P.

$\Rightarrow \quad \frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in H.P.

53. $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$

$\Rightarrow \quad f(x) = xf(1)$ for all $x, y \in \mathbb{R}$

$\Rightarrow \quad f(x) = 7x$, for all $x, y \in \mathbb{R}$

$\therefore \quad \sum_{r=1}^{n} f(r) = \sum_{r=1}^{n} 7r = 7\sum_{r=1}^{n} r = \frac{7n(n+1)}{2}$

54. Clearly, $x = \frac{1}{1-a}, \quad y = \frac{1}{1-b}, \quad z = \frac{1}{1-c}$

Since a, b, c are in A.P.

$\Rightarrow \quad 1-a, 1-b, 1-c$ are also in A.P.

$\Rightarrow \quad \frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ are in H.P.

$\therefore \quad x, y, z$ are in H.P.

55. We have

$$t_m = \frac{1}{n} \text{ and } t_n = \frac{1}{m}$$

$$\Rightarrow \quad a+(m-1)d = \frac{1}{n} \text{ and } a+(n-1)d = \frac{1}{m}$$

$$\Rightarrow \quad a = d = \frac{1}{mn} \quad \Rightarrow \quad a-d = 0$$

56. When n is odd, the last term will be n^2.

$\therefore$ The sum is

$$1^2+2\cdot 2^2+3^2+2\cdot 4^2+5^2+2\cdot 6^2+\cdots+2\cdot(n-1)^2+n^2$$

$$= \frac{(n-1)n^2}{2}+n^2$$

$$\left[\text{Replacing } n \text{ by } n-1 \text{ in } \frac{n(n+1)^2}{2}\right]$$

$$= \frac{n^3-n^2+2n^2}{2} = \frac{n^3+n^2}{2} = \frac{n^2(n+1)}{2}$$

57. Let a and b be two numbers.

By the question, $A = 9$, $G = 4$

$$\Rightarrow \quad \frac{a+b}{2} = 9 \text{ and } \sqrt{ab} = 4$$

$$\Rightarrow \quad a+b = 18 \text{ and } ab = 16$$

Therefore, the required equation is

$$x^2-(a+b)x+ab = 0$$

$$x^2-18x+16 = 0$$

58. $q^2 = pr$

$$2\tan^{-1}q = \tan^{-1}p+\tan^{-1}r$$

$$2\tan^{-1}q = \tan^{-1}\left(\frac{p+r}{1-pr}\right)$$

$$\tan^{-1}\left(\frac{2q}{1-q^2}\right) = \tan^{-1}\left(\frac{p+r}{1-pr}\right)$$

$$\frac{2q}{1-pr} = \frac{p+r}{1-pr}$$

It means p, q, r are in both A.P. and G.P., i.e. $p = q = r$.

59. $x = \sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \qquad [\because \ |a|<1]$

$$y = \sum_{n=0}^{\infty} b^n = \frac{1}{1-b} \qquad [\because \ |b|<1]$$

$$z = \sum_{n=0}^{\infty} c^n = \frac{1}{1-c} \qquad [\because \ |c|<1]$$

$$\therefore \quad 1-a = \frac{1}{x} \quad \Rightarrow \quad a = 1-\frac{1}{x}$$

Similarly, $b = 1-\frac{1}{y}$, $c = 1-\frac{1}{z}$

Since a, b, c are in A.P.

$$\therefore \quad 1-\frac{1}{x}, 1-\frac{1}{y}, 1-\frac{1}{z} \text{ are in A.P.}$$

$$\Rightarrow \quad -\frac{1}{x}, -\frac{1}{y}, -\frac{1}{z} \text{ are in A.P.}$$

$$\Rightarrow \quad \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in A.P.}$$

$\Rightarrow$ x, y, z are in H.P.

60. $$\frac{\frac{p}{2}[2a_1+(p-1)d]}{\frac{q}{2}[2a_1+(q-1)d]} = \frac{p^2}{q^2}$$

$$\Rightarrow \quad \frac{(2a_1-d)+pd}{(2a_1-d)+qd} = \frac{p}{q}$$

After solving, $a_1 = \frac{d}{2}$ or $d = 2a_1$

$$\frac{a_6}{a_{21}} = \frac{a_1+5d}{a_1+20d} = \frac{11a_1}{41a_1} = \frac{11}{41}$$

61. Since $a_1, a_2, a_3, \ldots, a_n$ are in H.P.

$$\therefore \quad \frac{1}{a_1}, \frac{1}{a_2}, \ldots, \frac{1}{a_n} \text{ are in A.P.}$$

Let d be the common difference of the A.P.

Now $a_1a_2+a_2a_3+\cdots+a_{n-1}a_n$

$$= \frac{1}{d}[a_1-a_2+a_2-a_3+\ldots a_{n-1}-a_n]$$

$$\left[\frac{1}{a_2}-\frac{1}{a_1} = d \ \Rightarrow \ \frac{a_1-a_2}{a_2a_1} = d \ \Rightarrow \ \frac{a_1-a_2}{d} = a_1a_2, \text{ etc.}\right]$$

$$= \frac{1}{d}(a_1-a_n)$$

$$= \frac{1}{d}\left[\frac{a_1-a_n}{(n-1)}(n-1)\right]$$

$$= \frac{(n-1)(a_1-a_n)}{\frac{1}{a_n}-\frac{1}{a_1}} \qquad \left[\because \quad \frac{1}{a_n} = \frac{1}{a_1}+(n-1)d\right]$$

$$= (n-1)a_1a_n$$

62. We have $a = ar+ar^2$

$$\Rightarrow \quad r^2+r-1 = 0 \quad \Rightarrow \quad r = \frac{-1\pm\sqrt{1+4}}{2}$$

For terms to be positive

$$\Rightarrow \quad r = \frac{\sqrt{5}-1}{2}$$

63. Let $a_1, a_2, a_3, \ldots, a_n$ be a G.P. with common ratio r.

Then, $a_n = a_{n+1} + a_{n+2}$ [given]

$\Rightarrow \quad a_1 r^{n-1} = a_1 r^n + a_1 r^{n+1}$

$\Rightarrow \quad r^{-1} = 1 + r$

$\Rightarrow \quad r^2 + r - 1 = 0 \quad \Rightarrow \quad r = \dfrac{-1 \pm \sqrt{5}}{2} \quad \Rightarrow r = \dfrac{\sqrt{5}-1}{2}$

$[\because \quad r > 0$ (given)$]$

64. $a, -ar, ar^2, -ar^3$ (when r is positive) are the four terms of the G.P. series

$$a - ar = 12 \text{ and } ar^2(1-r) = 48$$

$$\therefore \qquad \frac{ar^2(1-r)}{a(1-r)} = \frac{48}{12}$$

$\Rightarrow \quad r^2 = 4 \quad \therefore \quad r = 2$

$\Rightarrow \quad a(1-2) = 12 \quad \Rightarrow \quad a = -12$

65. Let $S = 1 + \dfrac{2}{3} + \dfrac{6}{3^2} + \dfrac{10}{3^2} + \dfrac{14}{3^4} + \cdots$ (i)

$$\frac{1}{3}S = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} + \cdots \qquad \text{(ii)}$$

Subtracting (ii) from (i), we get

$$S\left(1 - \frac{1}{3}\right) = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \cdots$$

$$\Rightarrow \quad \frac{2}{3}S = \frac{4}{3} + \frac{4}{3^2}\left(1 + \frac{1}{3} + \frac{1}{3^2} + \cdots\right)$$

$$\Rightarrow \quad \frac{2}{3}S = \frac{4}{3} + \frac{4}{3^2}\left[\frac{1}{1-(1/3)}\right]$$

$$= \frac{4}{3} + \frac{4}{3^2}\frac{3}{2} = \frac{4}{3} + \frac{2}{3} = \frac{6}{3}$$

$$\Rightarrow \quad \frac{2}{3}S = \frac{6}{3} \quad \Rightarrow S = 3$$

66. The number of notes counted in the first 10 minutes $= 150 \times 10 = 1500$.

Suppose the person counts the remaining 3000 currency notes in n minutes. Then,

3000 = sum of n terms of an A.P. with the first term 148 and common difference –2.

$$\Rightarrow \quad 3000 = \frac{n}{2}[2 \times 148 + (n-1) \times (-2)]$$

$$\Rightarrow \quad 3000 = n(149 - n)$$

$$\Rightarrow \quad n^2 - 149n + 3000 = 0$$

$$\Rightarrow \quad (n-125)(n-24) = 0 \Rightarrow \quad n = 125, \ 24$$

Clearly, n = 125 is not possible

$\therefore$ Total time taken = (10 + 24) minutes = 34 minutes

67. $A = 2G$

$$\frac{A}{G} = \frac{2}{1}$$

$$\frac{a+b}{2\sqrt{ab}} = 2$$

$$a + b = 4\sqrt{ab}$$

$$\frac{a}{\sqrt{ab}} + \frac{b}{\sqrt{ab}} = 4$$

$$\sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}} = 4$$

Let $\sqrt{\dfrac{a}{b}} = x$

$$\Rightarrow \quad x + \frac{1}{x} = 4 \qquad \Rightarrow \quad x^2 - 4x + 1 = 0$$

$$\Rightarrow \quad x = \frac{4 \pm \sqrt{16-4}}{2} \qquad \Rightarrow \quad x = \frac{4 \pm 2\sqrt{3}}{2}$$

$$\Rightarrow \quad x = 2 \pm \sqrt{3} \qquad \Rightarrow \quad x = 2 + \sqrt{3}$$

then $\sqrt{\dfrac{a}{b}} = (2 + \sqrt{3})$

$$\Rightarrow \quad \frac{a}{b} = 4 + 3 + 4\sqrt{3} \qquad \Rightarrow \quad \frac{a}{b} = 7 + 4\sqrt{3}$$

68. b^2, a^2, c^2 are in A.P. (given)

$$a^2 - b^2 = c^2 - a^2$$

$$(a-b)(a+b) = (a-c)(a+c)$$

$$\frac{1}{b+c} - \frac{1}{a+b} = \frac{1}{c+a} - \frac{1}{b+c}$$

$\dfrac{1}{a+b}, \dfrac{1}{b+c}, \dfrac{1}{c+a}$ are in A.P.

So, $(a + b)$, $(b + c)$, $(c + a)$ are in H.P.

69. $S_n = 5n^2 + 2n$

$\therefore \quad S_1 = 5 + 2 = 7 \quad \therefore \quad t_1 = 7$

$\therefore \quad S_2 = 5(4) + 4 = 24$

$\therefore \quad t_1 + t_2 = 24 \quad \therefore \quad t_2 = 24 - 7 = 17$

70. $\displaystyle\sum_{r=1}^{100} a_{2r} = \alpha$

$$\Rightarrow \quad \alpha - \beta = \sum_{r=1}^{100}(a_{2r} - a_{2r-1})$$

$$= \sum_{r=1}^{100} d = 100d \Rightarrow d = \frac{\alpha - \beta}{100}$$

71. Statement 2: $\sum_{K=1}^{n} K^3 - (K-1)^3$

$$\Rightarrow \quad \left(\frac{n(n+1)}{2}\right)^2 - \left(\frac{n(n-1)}{2}\right)^3 = n^3$$

Statement '2' is correct.

Statement 1: $(1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \cdots + (20^3 - 19^3) = 8000$.

Statement '1' is correct and statement '2' explains statement '1'.

72. $100(T_{100}) = 50(T_{50})$

$$\Rightarrow \quad 2(a+99d) = a+49d$$

$$\Rightarrow \quad 2a + 198d = a + 49d$$

$$\Rightarrow \quad a+149d = 0$$

$$\Rightarrow \quad T_{150} = 0$$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The first term of an A.P. of consecutive integers is $p^2 + 1$. The sum of $(2p + 1)$ terms of this series can be expressed as

(a) $(p+1)^2$ (b) $(2p+1)(p+1)^2$

(c) $(p+1)^3$ (d) $p^3+(p+1)^3$

2. Consider an A.P. with the first term a and the common difference d. Let S_k denote the sum of its first k terms. If S_{kx}/S_x is independent of x, then

(a) $a = d/2$ (b) $a = d$

(c) $a = 2d$ (d) none of these

3. If $x \in \mathbb{R}$, the numbers $5^{1+x} + 5^{1-x}$, $a/2$, $25^x + 25^{-x}$ form an A.P. then a must lie in the interval

(a) [1, 5] (b) [2, 5] (c) [5, 12] (d) [12, ∞)

4. The third term of a G.P. is 4. The product of the first five terms is

(a) 4^3 (b) 4^5 (c) 4^4 (d) none of these

5. The first, second and middle terms of an A.P. are a, b, c respectively. Their sum is

(a) $\frac{2(c-a)}{b-a}$ (b) $\frac{2c(c-a)}{b-a}+c$

(c) $\frac{2c(b-a)}{c-a}$ (d) $\frac{2b(c-a)}{b-a}$

6. For a sequence $\{a_n\}$, $a_1 = 2$ and $\frac{a_{n+1}}{a_n} = \frac{1}{3}$. The $\sum_{r=1}^{20} a_r$ is

(a) $\frac{20}{2}[4+19\times 3]$ (b) $3\left(1-\frac{1}{3^{20}}\right)$

(c) $2(1 - 3^{20})$ (d) none of these

7. The rational number, which equals the number $2.\overline{357}$ with recurring decimal, is

(a) $\frac{2355}{1001}$ (b) $\frac{2379}{997}$

(c) $\frac{2355}{999}$ (d) none of these

8. If $x > 1$ if $\left(\frac{1}{x}\right)^a, \left(\frac{1}{x}\right)^b, \left(\frac{1}{x}\right)^c$ are in G.P., then a, b, c are in

(a) A.P. (b) G.P. (c) H.P (d) none of these

9. If $3+\frac{1}{4}(3+d)+\frac{1}{4^2}(3+2d)+\cdots+$ up to $\infty = 8$, then the value of d is

(a) 9 (b) 5 (c) 1 (d) none of these

10. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{a}{b}, \frac{b}{a}, \frac{c}{b}$ are in

(a) A.P. (b) G.P. (c) H.P. (d) none of these

11. If $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, then the minimum value of $\cos^3\theta + \sec^3\theta$ is

(a) 1 (b) 2 (c) 0 (d) none of these

12. For the A.P. given by $a_1, a_2, \ldots, a_n \ldots$, the equation satisfied is

(a) $a_1 + 2a_2 + a_3 = 0$

(b) $a_1 - 2a_2 + a_3 = 0$

(c) $a_1 + 3a_2 - 3a_3 - a_4 = 0$

(d) $a_1 - 4a_3 + 5a_3 - 3a_4 + a_5 = 0$

13. The harmonic mean of the roots of the equation $\left(5+\sqrt{2}\right)x^2 - \left(4+\sqrt{5}\right)x + 8 + 2\sqrt{5} = 0$ is

(a) 2 (b) 4

(c) 6 (d) 8

14. For a positive integer n, let $a(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots \frac{1}{(2^n)-1}$. Then

(a) $a(100) \le 100$ (b) $a(100) > 100$

(c) $a(200) \le 100$ (d) $a(200) \ge 100$

15. Let the positive numbers a, b, c, d be in A.P. Then abc, abd, bcd are
(a) not in A.P/G.P./H.P. (b) in A.P.
(c) in G.P. (d) in H.P.

16. An infinite G.P. has the first term as x and the sum to infinity as 5. Then the range of values of x is
(a) $x \le -10$ (b) $x \ge 10$
(c) $0 < x < 10$ (d) $-10 \le x \le 10$

17. If $a > 1$, $b > 1$, then the minimum value of $\log_b a + \log_a b$ is
(a) 0 (b) 1
(c) 2 (d) none of these

18. If the sum of the first n terms of a sequence is of the form $An^2 + Bn$, where A and B are constants, then the sequence is
(a) an A.P. (b) a G.P.
(c) an H.P. (d) none of these

19. The minimum value of $4^x + 4^{1-x}$, $x \in \mathbb{R}$, is
(a) 2 (b) 4
(c) 1 (d) none of these

20. In the sequence 1, 2, 2, 3, 3, 3, 4, 4, 4, 4,..., where n consecutive terms have the value n, the 150th term is
(a) 17 (b) 16
(c) 18 (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. Three numbers form an increasing G.P. If the middle number is doubled, then the new numbers are in A.P. Then the common ratio of the GP is
(a) $2-\sqrt{3}$ (b) $2+\sqrt{3}$ (c) $\sqrt{3}-2$ (d) $3+\sqrt{2}$

2. If G_1 and G_1 are two geometric means and A is the arithmetic mean inserted between two positive numbers, then the value of $\frac{G_1^2}{G_2}+\frac{G_2^2}{G_1}$ is
(a) $A/2$ (b) A
(c) $2A$ (d) none of these

3. If x, y, z are three real numbers of the same sign, then the value of $\frac{x}{y}+\frac{y}{z}+\frac{z}{x}$ lies in the interval
(a) $[2,+\infty)$ (b) $[3,+\infty)$ (c) $(3,+\infty)$ (d) $(-\infty,3)$

4. If $\{a_n\}$ and $\{b_n\}$ are two sequences given by
$$a_n = (x)^{1/2^n} + (y)^{1/2^n}$$
and $$b_n = (x)^{1/2^n} - (y)^{1/2^n}$$
for all $n \in N$, then the value of $a_1 a_2 a_3 \ldots a_n$ is equal to
(a) $x - y$ (b) $\frac{x+y}{b_n}$
(c) $\frac{x-y}{b_n}$ (d) $\frac{xy}{b_n}$

5. If $a^2 + 9b^2 + 16c^2 = 2(3ab + 6bc + 4ca)$, where a, b, c are non-zero numbers, then a, b, c are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

6. Let $a_1, a_2, \ldots, a_{10}$ be in A.P. and $h_1, h_2, \ldots, h_{10}$ be in H.P. If $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$, then $a_4 h_2$ is
(a) 2 (b) 3
(c) 5 (d) 6

7. If the sum of n terms of an A.P. is $nA + n^2B$, where A and B are constants, then its common difference will be
(a) $A - B$ (b) $A + B$
(c) $2A$ (d) $2B$

8. The positive integer n for which $2\times2^2 + 3\times2^3 + 4\times2^4 + \cdots + n\times2^n = 2^{n+10}$ is
(a) 510 (b) 511
(c) 512 (d) 513

9. Let $a, b, c > 0$ and $4a^2 + 9b^2 + 16c^2 - 6ab - 12bc - 8ac = 0$, then b is
(a) $\le \sqrt{ac}$ (b) $\ge \sqrt{ab}$
(c) $\ge \frac{a+c}{2}$ (d) $\ge \sqrt{ac}$

10. If $\sum_{r=1}^{n} t_r = \frac{1}{12} n(n+1)(n+2)$, the value of $\sum_{r=1}^{n} \frac{1}{t_r}$ is
(a) $\frac{2n}{n+1}$ (b) $\frac{2n}{(n+1)}$
(c) $\frac{4n}{n+1}$ (d) $\frac{3n}{n+2}$

11. The sum of all two-digit numbers which, when divided by 4, yields unity as a remainder is
(a) 1190 (b) 1197
(c) 1210 (d) none of these

12. If a, b, c are in A.P., p, q, r are in H.P. and ap, bq, cr are in G.P., then $\frac{p}{r}+\frac{r}{p}$ is equal to
(a) $\frac{a}{c}+\frac{c}{a}$ (b) $\frac{a}{c}-\frac{c}{a}$
(c) $\frac{b}{q}-\frac{q}{b}$ (d) $\frac{b}{q}-\frac{a}{p}$

13. The sides of a triangle ABC (a, b, c) are in G.P. If r is the common ratio of this G.P. then

(a) $r \in \left(\frac{\sqrt{5}-1}{2}, \infty\right)$ (b) $r \in \left(\frac{\sqrt{5}+1}{2}, \infty\right)$

(c) $r \in \left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2}\right)$ (d) $r \in \left(\frac{\sqrt{5}+1}{2}, \frac{\sqrt{5}+3}{2}\right)$

14. 150 workers were engaged to finish a piece of work in a certain number of days. Four workers left the second day, 4 more workers left the third day and so on. It then took eight more days to finish the work. The number of days in which the work was completed is

(a) 15 (b) 20
(c) 25 (d) 30

15. The common difference of the A.P. in which $t_7 = 9$ and $t_1 t_2 t_7$ is least is

(a) $\frac{33}{2}$ (b) $\frac{5}{4}$
(c) $\frac{33}{20}$ (d) none of these

16. The H.M. between two numbers is 16/5, their A.M. is A and G.M. is G. If $2A + G^2 = 26$, then the numbers are

(a) 6, 8 (b) 4, 8
(c) 2, 8 (d) 1, 8

17. If $1^2 + 2^2 + \cdots + n^2 = 1015$, then the value of n is

(a) 15 (b) 14
(c) 13 (d) none of these

18. If $a, b, c \in \mathbb{R}^+$, then the minimum value of $a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) =$

(a) abc (b) $2abc$
(c) $3abc$ (d) $6abc$

19. The total number of positive real values of x such that x, $[x]$, $\{x\}$ are in H.P., where [.] denotes the greatest integer function and {.} the fraction part, is equal to

(a) 0 (b) 1
(c) 2 (d) none of these

20. The sum of those integers from 1 to 100 which are not divisible by 3 or 5 is

(a) 2489 (b) 4735
(c) 2317 (d) 2632

ANSWERS

Level 1

1. (d)	**2.** (a)	**3.** (d)	**4.** (b)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (a)	**9.** (a)	**10.** (c)
11. (b)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (d)	**16.** (c)	**17.** (c)	**18.** (a)	**19.** (b)	**20.** (a)

Level 2

1. (b)	**2.** (c)	**3.** (d)	**4.** (c)	**5.** (c)	**6.** (d)	**7.** (b)	**8.** (d)	**9.** (a)	**10.** (c)
11. (c)	**12.** (a)	**13.** (b)	**14.** (d)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (b)	**20.** (d)

10 Functions

LEJEUNE DIRICHLET

- Definition of Function
- Working Rules for Finding Domain of a Function
- Working Rules for Finding Range of a Function
- Types of Functions
- Even Odd Functions
- Periodic Functions
- Properties of Periodic Functions
- Composition of Functions
- Properties of Composite Functions

DEFINITION OF FUNCTION

Function: Let X and Y be two nonempty sets. A function f, defined from X to Y, is a rule or a collection of rules, which associates to each element x in X a unique element y in Y.

Symbolically, we write it as $f: X \to Y$, which is read as "function defined from the set X to the set Y".

(i) The unique element y of Y is called the value of f at x (the image of x under f). It is written as $f(x)$. Thus, $y = f(x)$.

(ii) The element x of X is called pre-image (or inverse image) of y.

(iii) The set X is called the domain of f.

(iv) The set Y is called the co-domain of f.

(v) The set consisting of all images of the elements of X under f is called the range of f. This is denoted by $f(x)$.
Thus, the range of $f = f(x) = \{f(x)$: for all $x \in X\}$.
This is a subset of Y, which may or may not be equal to Y.

Note A function is also termed mapping or correspondence or transformation.

Examples:

(i) Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8, 10\}$.
Here the rule, i.e. f, which associates to each element x in A, the element $2x$ in B is a function from A to B. The rule written as $f(x) = 2x$ is depicted in Fig. 10.1.

(ii) Let $A = \{a, b, c, d, e\}$ and $B = \{p, q, r\}$.
Here the rule, which is depicted in Fig. 10.2, is not a function from A to B because the elements a and c in A have been associated with two elements p and q each of B.

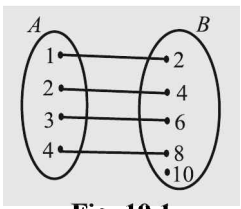


Fig. 10.1

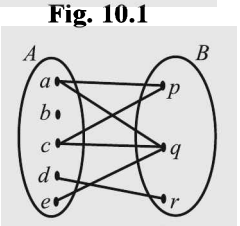


Fig. 10.2

☞ WORKING RULES FOR FINDING DOMAIN OF A FUNCTION

1. Expression under even root (i.e. square root, fourth root) is greater than or equal to zero.
2. Denominator should not be zero.
3. If the domains of $y = f(x)$ and $y = g(x)$ are D_1 and D_2 respectively, then the domain of $f(x) \pm g(x)$ or $f(x) \cdot g(x)$ *is* $D_1 \cap D_2$, while domain of $\frac{f(x)}{g(x)}$ is $D_1 \cap D_2 - \{x : g(x) = 0\}$.

Illustration 1. Find the domain of $f(x) = \sqrt{\cos(\sin x)}$.

Solution: $f(x) = \sqrt{\cos(\sin x)}$ is defined if

$$\cos(\sin x) \geq 0 \qquad \text{(i)}$$

As we know $-1 \leq \sin x \leq 1$ for all x.

$$\cos\theta \geq 0$$

[Here, $\theta = \sin x$ lies in the third and fourth quadrants.]

i.e. $\cos(\sin x) \geq 0$ for all x, i.e. $x \in \mathbb{R}$

Thus, the domain of $f(x) \in \mathbb{R}$.

☞ WORKING RULES FOR FINDING RANGE OF A FUNCTION

Step 1: Find the domain for the function $y = f(x)$.

Step 2: Change the function $y = f(x)$ as $x = f(y)$.

Step 3: Solve $x = \phi(y)$.

Step 4: Find the values of y for x in the domain of f.

Step 5: The set of values of y in step 4 is the range of $f(x)$.

Illustration 2. Find the range of $f(x) = \sin^2 x - \sin x + 1$.

Solution: $f(x) = \sin^2 x - \sin x + 1 = \left(\sin x - \frac{1}{2}\right)^2 + \frac{3}{4}$

Now, $-1 \leq \sin x \leq 1 \quad \Rightarrow \quad -\frac{3}{2} \leq \sin x - \frac{1}{2} \leq \frac{1}{2}$

$$\Rightarrow \quad 0 \leq \left(\sin x - \frac{1}{2}\right)^2 \leq \frac{9}{4}$$

$$\Rightarrow \quad \frac{3}{4} \leq \left(\sin x - \frac{1}{2}\right)^2 + \frac{3}{4} \leq 3$$

Hence, the range is $\left[\frac{3}{4}, 3\right]$.

☞ TYPES OF FUNCTIONS

1. Constant Function

A function which does not change as its parameters vary, i.e. the function whose rate of change is zero, is called a *constant function*. In short, we can say that a constant function is a function that always gives or returns the same value.

Domain: $x \in (-\infty, \infty)$, i.e. $x \in \mathbb{R}$

Range: $y \in \{c\}$

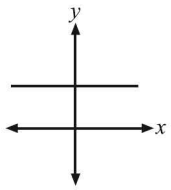


Fig. 10.3

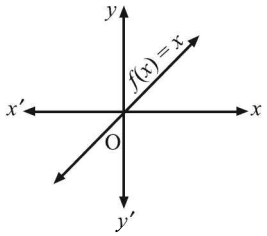


Fig. 10.4

2. Identity Function

An identity function is a function of the form $y = f(x) = x$, which is a straight line passing through the origin and having slope unity.

Domain: $x \in (-\infty, \infty)$, i.e. $x \in \mathbb{R}$

Range: $y \in (-\infty, \infty)$, i.e. $y \in \mathbb{R}$

3. Polynomial Function

A polynomial function is a function of the form

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n$$

Here the value of n must be a non-negative integer, i.e. it must be a whole number and a_a, a_1, a_2, …, $a_n \in \mathbb{R}$. The degree of the polynomial is the highest value for n, where $a_n \neq 0$, i.e. n.

Domain: The entire number line, i.e. $x \in \mathbb{R}$

Range: Varies from function to function.

4. Rational Function

If $h(x)$ and $g(x)$ are polynomial functions, $g(x) \neq 0$, then the function $f(x) = \frac{h(x)}{g(x)}$ is known as a *rational function*.

Domain: $x \in \mathbb{R} - \{x : g(x) = 0\}$

Range: Varies from function to function.

5. Irrational Function

An *irrational function* is a function of the form

$$y = \{f(x)\}^{p/q}$$

where p and q are integers, have no common factor and also $q \neq 0$, $q > 0$. For example,

$$y = f(x) = \frac{4x^{5/2} - 9x^{3/2}}{x^{1/2} - 7}$$

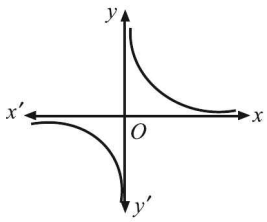


Fig. 10.5

6. Reciprocal Function

A *reciprocal function* is the inverse of an identity function, such that $y = f(x) = \frac{1}{x}$, which is an equation of a rectangular hyperbola whose asymptotes (here x-axis and y-axis) intersect at right angles.

Domain: $x \in \mathbb{R} - \{0\}$
Range: $y \in \mathbb{R} - \{0\}$

7. Exponential Function

A function of the form $f(x) = a^x$ is an *exponential function*. It is the inverse of a logarithmic function. The value of the function depends upon the value of a.

For $0 < a < 1$, the function decreases, and for $a > 1$, the function increases.

Domain: $x \in \mathbb{R}$
Range: $y \in \mathbb{R}^+$

8. Logarithmic Function

A function of the form $f(x) = \log_a x$ is a *logarithmic function*. The value of the function depends upon the value of a. For $0 < a < 1$, the function decreases, and for $a > 1$, the function increases.

Domain: $x > 0$, i.e. $x \in (0, \infty)$
Range: $y \in \mathbb{R}$

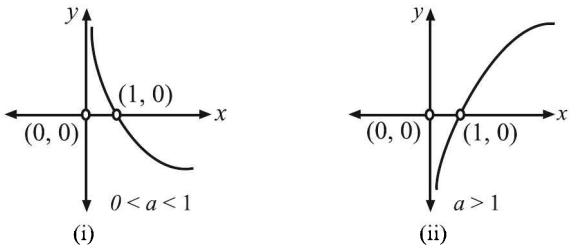


Fig. 10.6

8.1 *Properties of a logarithmic function*

1. $\log_e(xy) = \log_e x + \log_e y \qquad x, y > 0$
2. $\log_e\left(\frac{x}{y}\right) = \log_e x - \log_e y \qquad x, y > 0$
3. $\log_e x^a = a \log_e x \qquad x > 0$
4. $\log_b a = \frac{\log_c a}{\log_c b}$, where c is any constant such that $c \in (0, \infty) - \{1\}$. Also, $a, b > 0$

   (i) $\log_a a = 1$ (ii) $\log_b a = \frac{1}{\log_a b}$
5. $a^{\log_a b} = b$
6. If $\log_b a = c$, then $a = b^c$, where $a > 0$ and $b \in (0, \infty) - \{1\}$
7. $a^{\log_c b} = b^{\log_c a}$
8. If $\log_a x > \log_a y$, and
   if $a > 1$ then $x > y$
   i.e. for $a > 1$, the inequality remains the same,
   else if $0 < a < 1$ then $x < y$, and for
   $0 < a < 1$, the inequality gets reversed.

Illustration 3. Find the domain of the function

$$f(x) = \left[\log_{10}\left(\frac{5x - x^2}{4}\right)\right]^{1/2}.$$

Solution: $f(x) = \left[\log_{10}\left(\frac{5x - x^2}{4}\right)\right]^{1/2}$ (i)

From (i), clearly $f(x)$ is defined for those values of x for which

$$\log_{10}\left[\frac{5x - x^2}{4}\right] \geq 0.$$

$$\Rightarrow \quad \left(\frac{5x - x^2}{4}\right) \geq 10^0 \quad \Rightarrow \quad \left(\frac{5x - x^2}{4}\right) \geq 1$$

$$\Rightarrow \quad x^2 - 5x + 4 \leq 0 \quad \Rightarrow \quad (x-1)(x-4) \leq 0$$

Hence, the domain of the function is [1, 4].

9. Trigonometric Function

A function of trigonometric ratios is known as *trigonometric function*.

Function	Domain	Range
$\sin x$	$\mathbb{R}$	$[-1, 1]$
$\cos x$	$\mathbb{R}$	$[-1, 1]$
$\tan x$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2} : n \in I\right\}$	$\mathbb{R}$
$\cot x$	$\mathbb{R} - \{n\pi : n \in I\}$	$\mathbb{R}$
$\sec x$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2} : n \in I\right\}$	$(-\infty, -1] \cup [1, \infty)$
$\operatorname{cosec} x$	$\mathbb{R} - \{n\pi : n \in I\}$	$(-\infty, -1] \cup [1, \infty)$

10. Inverse Trigonometric Function

A function involving inverse trigonometric ratios is known as *inverse trigonometric function*.

Function	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\operatorname{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$

11. Modulus Function

A function $y = f(x) = |x|$ is known as *modulus function*.

$$y = f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Domain: $f(x) = x \in \mathbb{R}$

Range: $f(x) = [0, \infty)$

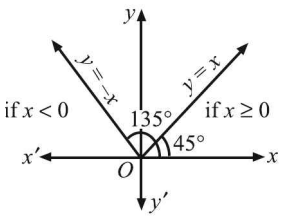


Fig. 10.7

11.1 *Properties of a modulus function*

1. For any real number x, $\sqrt{x^2} = |x|$.
2. If a, b are positive real numbers, then
   (i) $x^2 \leq a^2 \Leftrightarrow |x| \leq a \Leftrightarrow -a \leq x \leq a$
   (ii) $x^2 \geq a^2 \Leftrightarrow |x| \geq a \Leftrightarrow x \leq -a$ or $x \geq a$
   (iii) $a^2 \leq x^2 \leq b^2 \Leftrightarrow a \leq |x| \leq b \Leftrightarrow x \in [-b, -a] \cup [a, b]$
3. $|x + y| \geq \big||x| - |y|\big|$
4. $|x \pm y| \leq |x| + |y|$

12. Greatest Integer Function

The greatest integer function is denoted as $y = f(x) = [x]$, where $[x]$ means the greatest integer n, which is less than or equal to x, i.e. $n \leq x$.

$[x]$ = [Integer + Proper positive fraction] = Integer

For example,

$[4.2] = [4 + 0.2] = 4$

$[-4.2] = [-5 + 0.8] = -5$

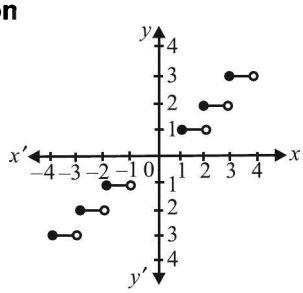


Fig. 10.8

12.1 *Properties of a greatest integer function*

1. If $[x] = x$, then x must belong to the set of integers
2. $[x + I] = [x] + [I] = [x] + I$
3. $[x + y] \geq [x] + [y]$
4. If $[f(x)] \geq I$, then $f(x) \geq I$
   If $[f(x)] \leq I$, then $f(x) < I + 1$
5. $[-x] = \begin{cases} -[x] & x \in I \\ -[x] - 1 & x \notin I \end{cases}$ i.e. $[x] + [-x] = 0$ / i.e. $[x] + [-x] = -1$

13. Smallest Integer Function (Ceiling Function)

For any real number x, we use the symbol $\lceil x \rceil$ to denote the smallest integer greater than or equal to x.

The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \lceil x \rceil$ for all $x \in \mathbb{R}$ is called the *smallest integer function* or the *ceiling function*. It is also a step function.

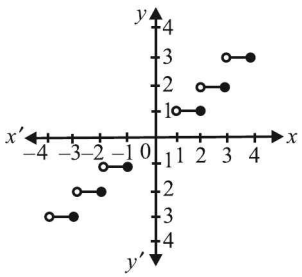


Fig. 10.9

Domain: $x \in \mathbb{R}$

Range: $y \in I$

14. Fractional Part Function

$y = f(x) = \{x\}$, by definition.

Mathematically,

$$\{x\} = x - [x]$$

It always returns the fractional part of the variable and is always positive.

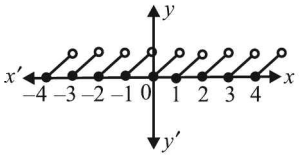


Fig. 10.10

Domain: $x \in \mathbb{R}$

Range: $y \in [0, 1)$

14.1 *Properties of a fractional part function*

1. If $0 \leq x < 1$ then $\{x\} = x$
2. If $x \in I$ then $\{x\} = 0$
3. If $x \notin I$ and $x > 0$ then $-\{x\} = 1 - \{x\}$

15. Signum Function

This function is defined by

$$f(x) = \begin{cases} \dfrac{|x|}{x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

Thus, we have

$$f(x) = \begin{cases} 1, & \text{when } x > 0 \\ 0, & \text{when } x = 0 \\ -1, & \text{when } x < 0 \end{cases}$$

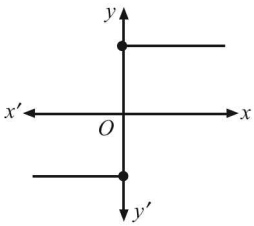


Fig. 10.11

Domain: $x \in \mathbb{R}$

Range: $y \in \{-1, 0, 1\}$

EVEN ODD FUNCTIONS

1. **Even function:** A function f is said to be an even function of x if $f(-x) = f(x)$. For example, $f(x) = x^2 + 4$, $f(x) = 3\cos x - 5$, etc. are all even functions of x.
2. **Odd function:** A function f is said to be an odd function of x if $f(-x) = -f(x)$. For example, $f(x) = x^3 + 7x$, $f(x) = 7\sin^3 x + \tan x$, etc. are all odd functions of x.
3. **Extension of a function:** If a function $f(x)$ is defined in $[0, a]$, then $0 \leq x \leq a \Rightarrow -a \leq -x \leq 0$
   $\Rightarrow x \in [-a, 0]$.
   We define $f(x)$ in $[-a, 0]$ such that $f(x) = f(-x)$.
   If g is the even extension, then

$$g(x) = \begin{cases} f(x), & x \in [0, a] \\ f(-x), & x \in [-a, 0] \end{cases}$$

4. **Odd extension:** If $f(x)$ is defined in $[0, a]$, then
$0 \le x \le a \Rightarrow -a \le -x \le 0$
$\Rightarrow -x \in [-a, 0]$.
We define $f(x)$ in $[-a, 0]$ such that $f(x) = -f(-x)$.
If g is the odd extension, then
$$g(x) = \begin{cases} f(x); & x \in [0, a] \\ -f(-x); & x \in [-a, 0] \end{cases}$$

PERIODIC FUNCTION

A function f is said to be periodic with period α if
$$f(x + \alpha) = f(x) \text{ for all } x,$$
where α is the least positive real number. For example, sin x, cos x, sec x and cosec x are all periodic functions with period 2π and tan x and cot x are periodic functions with period π.

Standard results on some periodic functions

S. No.	Function	Period
1.	$\sin^n x, \cos^n x, \text{cosec}^n x$	π (if n is an even number) 2π (if n is an odd number)
2.	$\tan^n x, \cot^n x$	π
3.	$\lvert\sin x\rvert, \lvert\cos x\rvert, \lvert\tan x\rvert, \lvert\sec x\rvert$ $\lvert\text{cosec } x\rvert, \lvert\cot x\rvert$	π
4.	$x - [x]$	1
5.	$\sin^{-1}(\sin x), \cos^{-1}(\cos x)$, $\text{cosec}^{-1}(\text{cosec } x), \sec^{-1}(\sec x)$	2π
6.	$\tan^{-1}(\tan x), \cot^{-1}(\cot x)$	π

PROPERTIES OF A PERIODIC FUNCTION

1. If $f(x)$ is periodic with period p, then $af(x) + b$, where $a, b \in \mathbb{R}$ $(a \neq 0)$, is also periodic with period p.
2. If $f(x)$ is periodic with period p, then $f(ax) + b$, where $a, b \in \mathbb{R}$ $(a \neq 0)$, is also periodic with period $\frac{p}{|a|}$.
3. Suppose $f(x)$ has period $p = \frac{m}{n}$ $(m, n \in N$ and co-prime) and $g(x)$ has period $q = \frac{r}{s}$ $(r, n \in N$ and co-prime) and let t be the LCM of p and q, i.e. $\frac{\text{LCM of } (m, r)}{\text{HCF of } (r, s)}$.
Then t shall be the period of $f + g$ provided there does not exist a positive number k $(< t)$ for which $f(x + k) + g(x + k) = f(x) + g(x)$, else k will be the period.
The same rule is applicable for any other algebraic combination of $f(x)$ and $g(x)$.
The LCM of p and q always exists if p/q is a rational quantity. If p/q is irrational, then the algebraic combination of f and g is non-periodic.

Illustration 4. Find the periods (if periodic) of $f(x) = \tan \pi/2$ where $[\cdot]$ denotes the greatest integer function.

Solution: $f(x) = \tan\frac{\pi}{2}[x], \tan\frac{\pi}{2}[x + T]$
$= \tan\frac{\pi}{2}[x] \Rightarrow \tan\frac{\pi}{2}[x + T]$
$= n\pi + \frac{\pi}{2}[x] \Rightarrow$ Period $= 2$.

COMPOSITION OF FUNCTION

Let A, B and C be three nonempty sets. Let $f: A \to B$ and $g: B \to C$ be two mappings (or functions), then $g \circ f: A \to C$. This function is called the product or composite of f and g given by the function
$$g \circ f(x) = g[f(x)], \text{ for all } x \in A$$

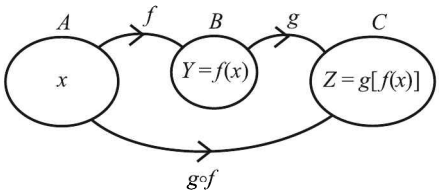


Fig. 10.12

$g \circ f$ exists if the range of f is a subset of the domain of g. Similarly, $f \circ g$ exists if the range of g is a subset of the domain of f.

Note

- The $g \circ f$ is defined only if for all $x \in A$, $f(x)$ is an element of the domain of g so that we can take its g-image.
- The range of f must be a subset of the domain of g in $g \circ f$.
- (i) $(f \circ g)(x) = f[g(x)]$
(ii) $(f \circ f)(x) = f[f(x)]$
(iii) $(g \circ g)(x) = g[g(x)]$
(iv) $(f \pm g)(x) = f(x) \pm g(x)$
(v) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}; g(x) \neq 0$

Properties of a Composite Function

1. The composition of a function is not commutative, i.e. $f \circ g \neq g \circ f$.
2. The composition of a function is associative, i.e. $f \circ (g \circ h) = (f \circ g) \circ h$.

3. The composition of any function with the identity function is the function itself, i.e. if $f: A \to B$, then $f \circ I_A = I_B \circ f = f$
4. Let $f: A \to B$, $g: B \to C$ be two functions, then
   (a) f and g are injective $\Rightarrow g \circ f$ is injective
   (b) f and g are surjective $\Rightarrow g \circ f$ is surjective
   (c) f and g are bijective $\Rightarrow g \circ f$ is bijective
5. An injective mapping from a finite set to itself is bijective.

SOLVED PROBLEMS

1. Let $f(x) = \dfrac{9^x}{9^x + 3}$. Then $f(x) + f(1-x)$ is

(a) 2 (b) 1
(c) -1 (d) none of these

Ans. (b)

Solution: $f(x) = \dfrac{9^x}{9^x + 3}$ (i)

and $f(1-x) = \dfrac{9^{1-x}}{9^{1-x} + 3}$

$\Rightarrow \quad f(1-x) = \dfrac{\frac{9}{9^x}}{\frac{9}{9^x} + 3} = \dfrac{9}{9 + 3.9^x}$

$\Rightarrow \quad f(1-x) = \dfrac{3}{(3 + 9^x)}$ (ii)

Adding (i) and (ii), we get

$$f(x) + f(1-x) = \frac{9^x}{9^x + 3} + \frac{3}{3 + 9^x} = 1$$

$\Rightarrow \quad f(x) + f(1-x) = 1$

2. The expression $\left[x + \sqrt{x^2 - 1}\right]^5 + \left[x - \sqrt{x^2 - 1}\right]^5$ is a polynomial of degree

(a) 5 (b) 6
(c) 10 (d) 20

Ans. (a)

Solution: $\left[x + \sqrt{x^2 - 1}\right]^5 + \left[x - \sqrt{x^2 - 1}\right]^5$

$= x^5 + {}^5C_1 x^4 \sqrt{x^2 - 1} + {}^5C_2 x^3 (x^2 - 1)$
$+ {}^5C_3 x^2 (x^2 - 1)^{3/2} + {}^5C_4 x (x^2 - 1)^2 + {}^5C_5 (x^2 - 1)^{5/2}$
$+ x^5 - {}^5C_1 x^4 \sqrt{x^2 - 1} + {}^5C_2 x^3 (x^2 - 1)$
$- {}^5C_3 x^2 (x^2 - 1)^{3/2} + {}^5C_4 x (x^2 - 1) - {}^5C_5 (x^2 - 1)^{5/2}$
$= 2x^5 + 2\,{}^5C_2 x^3 (x^2 - 1) + 2\,{}^5C_4 x (x^2 - 1)^2,$

which is a polynomial of degree 5.

3. Let $f(x) = 4\cos\sqrt{x^2 - \dfrac{\pi^2}{9}}$. Then

(a) the domain of f is $\left[\dfrac{\pi}{3}, +\infty\right)$
(b) the range of f is $[-1, 1]$
(c) the domain of f is $\left(-\infty, -\dfrac{\pi}{3}\right] \cup \left[\dfrac{\pi}{3}, +\infty\right)$
(d) none of these

Ans. (c)

Solution: Since $f(x) = 4\cos\sqrt{x^2 - \dfrac{\pi^2}{9}}$

By definition, the domain of f can be defined if

$$x^2 - \frac{\pi^2}{9} \geq 0 \quad \Rightarrow \quad x^2 \geq \left(\frac{\pi}{3}\right)^2$$

$\Rightarrow \quad |x| \geq \dfrac{\pi}{3}$

either $x \leq -\dfrac{\pi}{3}$ or $x \geq \dfrac{\pi}{3}$

i.e. $D_f = x \in \left(-\infty, -\dfrac{\pi}{3}\right] \cup \left[\dfrac{\pi}{3}, \infty\right)$.

4. The domain of the function $f(x) = \dfrac{1}{\sqrt{x^2 - 3x + 2}}$ is

(a) $(-\infty, 1)$ (b) $(-\infty, 1) \cap (2, \infty)$
(c) $(-\infty, 1] \cup [2, \infty)$ (d) $(2, \infty)$

Ans. (b)

Solution: For $f(x)$ to be defined, we must have

$x^2 - 3x + 2 = (x-1)(x-2) > 0$
$\Rightarrow \quad x < 1$ or > 2
Domain of $f = (-\infty, 1) \cup (2, \infty)$

5. The function $f(x) = \sin[\log(x + \sqrt{x^2 + 1})]$ is

(a) even (b) odd
(c) neither even nor odd (d) periodic

Ans. (b)

Solution: $f(x) = \sin[\log(x + \sqrt{1 + x^2})]$

$\Rightarrow \quad f(-x) = \sin[\log(-x + \sqrt{1 + x^2})]$

$\Rightarrow \quad f(-x) = \sin\log\left[(\sqrt{1 + x^2} - x)\dfrac{(\sqrt{1 + x^2} + x)}{(\sqrt{1 + x^2} + x)}\right]$

$\Rightarrow \quad f(-x) = \sin\log\left[\dfrac{1}{(x + \sqrt{1 + x^2})}\right]$

$\Rightarrow \quad f(-x)=\sin\left[-\log(x+\sqrt{1+x^2})\right]$

$\Rightarrow \quad f(-x)=-\sin\left[\log(x+\sqrt{1+x^2})\right]$

$\Rightarrow \quad f(-x)=-f(x)$

$\Rightarrow$ $f(x)$ is an odd function.

6. If the function $f(x)=\cos^2 x+\cos^2\left(\frac{\pi}{3}+x\right)$ $-\cos x\cdot\cos\left(\frac{\pi}{3}+x\right)$ is constant (independent of x), then the value of this constant is

(a) 0 (b) 3/4

(c) 1 (d) 4/3

Ans. (b)

Solution: Since the function is constant for $x\in\mathbb{R}$, the value of the constant is obtained by assigning any suitable value of x. Choose $x=0$, the desired value of the constant

$$=\cos^2 0+\cos^2\frac{\pi}{3}-\cos 0\cos\frac{\pi}{3}$$

$$=1+\frac{1}{4}-\frac{1}{2}=\frac{3}{4}.$$

7. If $f(x)=|x|+2$, then the range of $f[f(x)]$ is

(a) $(-\infty, 4)$ (b) $[4, \infty)$

(c) $[-4, 4]$ (d) none of these

Ans. (b)

Solution: $f(x)=|x|+2$

Now, $f[f(x)]=\big||x|+2\big|+2$

$\Rightarrow \quad f[f(x)]=|x|+2+2=|x|+4$

Since $|x|\geq 0$ for all $x\in\mathbb{R}$

$\Rightarrow$ $4+|x|\geq 4$ for all $x\in\mathbb{R}$

$\therefore$ Range of $f=R_f=[4,\infty)$

8. The domain of the function $f(x)=\log_{0.2}\log_{0.5}\log_{0.25}x$ is equal to

(a) $(1, \infty)$ (b) $(1/4, \infty)$

(c) $(1/4, 1)$ (d) none of these

Ans. (c)

Solution: We have, $f(x)=\log_{0.2}\log_{0.5}\log_{0.25}x$

$\log_{(1/2)}\log_{(1/4)}x>0 \quad\Rightarrow\quad 0<\log_{1/4}x<1$

$\Rightarrow \quad 1>x>\frac{1}{4} \qquad\Rightarrow\quad \frac{1}{4}<x<1$

9. The domain of the function $f(x)=\log(\sqrt{x-4}+\sqrt{6-x})$ is

(a) $[4, \infty)$ (b) $(-\infty, 6]$

(c) $[4, 6]$ (d) none of these

Ans. (c)

Solution. $f(x)=\log(\sqrt{x-4}+\sqrt{6-x})$

$\Rightarrow \quad x-4\geq 0 \quad$ and $\quad 6-x\geq 0$

$\Rightarrow \quad x\geq 4 \quad$ and $\quad x\leq 6$

$\Rightarrow$ Domain of $f(x)=[4,6]$.

10. If $f(x)=\log\left(\frac{1+x}{1-x}\right)$ and $g(x)=\frac{3x+x^3}{1+3x^2}$, then $f[g(x)]$ is equal to

(a) $f(3x)$ (b) $[f(x)]^3$

(c) $-f(x)$ (d) $3f(x)$

Ans. (d)

Solution: $f[g(x)]=\log\left[\frac{1+g(x)}{1-g(x)}\right]$

$$=\log\left(\frac{1+\frac{3x+x^2}{1+3x^2}}{1-\frac{3x+x^3}{1+3x^2}}\right)=\log\left(\frac{1+3x^2+3x+x^3}{1+3x^2-3x-x^3}\right)$$

$$=\log\left(\frac{1+x}{1-x}\right)^3=3\log\left(\frac{1+x}{1-x}\right)=3f(x).$$

11. The range of the function $f(x)=x^2+\frac{1}{x^2+1}$ is

(a) $[1, +\infty)$ (b) $[2, +\infty)$

(c) $\left[\frac{3}{2}, +\infty\right)$ (d) none of these

Ans. (a)

Solution: Since $f(x)=x^2+\frac{1}{x^2+1}$

$\Rightarrow \quad f(x)=x^2+\left(\frac{1}{x^2+1}-1\right)+1$

$\Rightarrow \quad f(x)=(x^2+1)-\left(\frac{x^2}{x^2+1}\right)$

$\Rightarrow \quad f(x)=(1+x^2)-\left(1-\frac{1}{1+x^2}\right)\geq 1$ for all $x\in\mathbb{R}$

$\therefore$ Range of $f=[1,\infty)$.

12. The domain of $f(x)=\sin^{-1}[2-4x^2]$ (where [ ] denotes the greatest integer function) is

(a) $[-1,1]$ (b) $(-2,2)$

(c) $\left[-\frac{\sqrt{3}}{2},0\right)\cup\left(0,\frac{\sqrt{3}}{2}\right]$ (d) none of these

Ans. (c)

Solution: For function to be defined, we must have

$-1 \le 2-4x^2 < 2$

$\therefore \quad 2-4x^2 < 2 \quad \Rightarrow \quad x \in \mathbb{R} - \{0\}$

and $2-4x^2 \ge -1 \quad \Rightarrow \quad x^2 \le \frac{3}{4}$

$\Rightarrow \quad x \in \left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right] - \{0\}.$

13. The domain of the function $\log|x^2-9|$ is

(a) $\mathbb{R}$ (b) $\mathbb{R} - [-3, 3]$

(c) $\mathbb{R} - \{-3, 3\}$ (d) none of these

Ans. (c)

Solution: For $x = -3, 3, |x^2 - 9| = 0$

Therefore, $\log|x^2 - 9|$ does not exist at $x = -3, 3$.

Hence, the domain of the function is $\mathbb{R} - \{-3, 3\}$.

14. Which of the following is a rational function?

(a) $\frac{1}{3}\sqrt{4x^3+4x+7}$

(b) $\frac{3x^3-7x+1}{x-2}, x \ne 2$

(c) $\frac{3x^5+5x^3+2x+7}{x^{3/2}}, x > 0$

(d) $\frac{\sqrt{1+x}}{2+5x}, x \ne -\frac{2}{5}$

Ans. (b)

Solution: $\frac{3x^3-7x+1}{x-2}, x \ne 2$ is a rational function.

15. The domain of $f(x) = \sqrt{1-\sqrt{1-\sqrt{1-x^2}}}$ is

(a) [0, 1] (b) [–1, 1]

(c) $(-\infty, \infty)$ (d) none of these

Ans. (b)

Solution: $f(x) = \sqrt{1-\sqrt{1-\sqrt{1-x^2}}}$

For $f(x)$ to be defined, we must have

Case I: $1-x^2 \ge 0 \quad \therefore \quad -1 \le x \le 1$ (i)

Case II: $1-\sqrt{1-x^2} \ge 0 \quad \Rightarrow \quad \sqrt{1-x^2} \le 1$

$\Rightarrow \quad 1-x^2 \le 1 \quad \Rightarrow \quad x^2 \ge 0$

$\therefore \quad -\infty < x < \infty$ (ii)

Case III: $1-\sqrt{1-\sqrt{1-x^2}} \ge 0$

$\Rightarrow \quad 1 \ge \sqrt{1-\sqrt{1-x^2}} \quad \Rightarrow \quad \sqrt{1-x^2} \ge 0$

$\Rightarrow \quad 1 \ge 1-\sqrt{1-x^2} \quad \Rightarrow \quad \sqrt{1-x^2} \ge 0$

$\Rightarrow \quad 1-x^2 \ge 0 \quad \therefore \quad -1 \le x \le 1$ (iii)

From the above cases, we have $-1 \le x \le 1$

$\therefore$ The domain of $f = D_f = [-1, 1]$.

EXERCISE SET 1

1. The range of $f(x) = \sin^{-1}(\sqrt{x^2+x+1})$ is

(a) $\left(0, \frac{\pi}{2}\right]$ (b) $\left(0, \frac{\pi}{3}\right]$

(c) $\left[\frac{\pi}{3}, \frac{\pi}{2}\right]$ (d) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$

2. The domain of the function $f(x) = \sqrt{\log_{0.5} x}$ is

(a) (0, 1] (b) $(0, \infty)$

(c) $(0.5, \infty)$ (d) $[1, \infty)$

3. The domain of the function

$f(x) = \frac{1}{1-x} + 2^{\sin^{-1}x} + \frac{1}{\sqrt{x-2}}$ is

(a) $(-\infty, \infty) - \{1\}$ (b) $(2, \infty)$

(c) $[-1, 1)$ (d) ϕ

4. The period of the function $\left|\sin^3\frac{x}{2}\right| + \left|\cos^5\frac{x}{5}\right|$ is

(a) 2π (b) 10π

(c) 8π (d) 5π

5. The function $f(x) = \sin\left(\frac{\pi x}{n!}\right) - \cos\left(\frac{\pi x}{(n+1)!}\right)$ is

(a) not periodic

(b) periodic, with period $2n!$

(c) periodic, with period $(n + 1)$

(d) none of these

6. The domain of $f(x) = \sqrt{\frac{1-|x|}{2-|x|}}$ is

(a) $(-\infty, \infty) - [-2, 2]$

(b) $(-\infty, \infty) - [-1, 1]$

(c) $[-1, 1] \cup (-\infty, -2) \cup (2, \infty)$

(d) none of these

7. The range of $f(x) = 3\tan\sqrt{\frac{\pi^2}{9} - x^2}$ is

(a) $[-3\sqrt{3}, 3\sqrt{3}]$ (b) $[0, \sqrt{3}]$

(c) $[0, 3\sqrt{3}]$ (d) none of these

8. The domain of the function $f(x) = \cos^{-1}\left(\frac{3}{4+2\sin x}\right)$ is

(a) $\left[-\frac{\pi}{6}+2n\pi, \frac{7\pi}{6}+2n\pi\right]$

(b) $\left(-\frac{\pi}{6}+2n\pi, \frac{7\pi}{6}+2n\pi\right)$

(c) $\left(-\frac{\pi}{6}+2n\pi, \frac{7\pi}{6}+2n\pi\right]$

(d) $\left[-\frac{\pi}{6}+2n\pi, \frac{7\pi}{6}+2n\pi\right)$

9. The domain of the function

$f(x)=\sqrt[3]{1-3x}+3\cos^{-1}\left(\frac{2x-1}{3}\right)+e^{3\tan x}$ is

(a) $[-1, 2]-\left\{\frac{\pi}{2}\right\}$ (b) $(-1, 2)$

(c) $(-\infty, \infty)$ (d) none of these

10. The range of the function $y=\sin[x]$ for all $x\in\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$, when $[x]$ denotes the greatest integer less than and equal to x, is

(a) $\{0\}$ (b) $\{0, -1\}$

(c) $\{0, \pm \sin 1\}$ (d) $\{0, -\sin 1\}$

11. The inverse of the function $f(x)=\frac{e^x-e^{-x}}{e^x+e^{-x}}+2$ is given by

(a) $\log_e\left(\frac{x+2}{x-1}\right)^{1/2}$ (b) $\log_e\left(\frac{x-1}{3-x}\right)^{1/2}$

(c) $\log_e\left(\frac{x}{2-x}\right)^{1/2}$ (d) $\log_e\left(\frac{x-1}{x+1}\right)^{-2}$

12. If $A=\left\{x:\frac{\pi}{6}\le x\le\frac{\pi}{3}\right\}$ and $f(x)=\cos x-x(1+x)$, then $f(A)$ is equal to

(a) $\left[\frac{1}{2}-\frac{\pi}{3}-\frac{\pi^2}{9}, \frac{\sqrt{3}}{2}-\frac{\pi}{6}-\frac{\pi^2}{36}\right]$

(b) $\left[\frac{1}{2}+\frac{\pi}{3}-\frac{\pi^2}{9}, \frac{\sqrt{3}}{2}+\frac{\pi}{6}-\frac{\pi^2}{36}\right]$

(c) $\left(\frac{1}{2}-\frac{\pi}{3}-\frac{\pi^2}{9}, \frac{\sqrt{3}}{2}-\frac{\pi}{6}-\frac{\pi^2}{36}\right)$

(d) none of these

13. Which one of the following is periodic?

(a) $f(x)=x-[x]$, where $[x]\le x$

(b) $f(x)=x\sin(1/x)$ for $x\ne 0, f(0)=0$

(c) $f(x)=x\cos x$

(d) none of these

14. The range of the function $f(x)=\sqrt{\frac{x}{1+x}}$ is

(a) $(0, \infty)$ (b) $[0, \infty)$

(c) $[0, \infty)-\{1\}$ (d) none of these

15. Let $f(x)=\sin^2\frac{x}{2}+\cos^2\frac{x}{2}$ and $g(x)=\sec^2 x-\tan^2 x$. The two functions are equal over the set

(a) ϕ

(b) $\mathbb{R}$

(c) $\mathbb{R}-\left\{x: x=(2n+1)\frac{\pi}{2}; n\in I\right\}$

(d) none of these

16. The domain of the function $f(x)=\frac{1}{1+2\sin x}$ is

(a) $\mathbb{R}$

(b) $\mathbb{R}-\left\{n\pi+(-1)^n\frac{5\pi}{6}, n\in Z\right\}$

(c) $\mathbb{R}-\left\{n\pi+(-1)^n\frac{11}{6}, n\in Z\right\}$

(d) none of these

17. If $f(x+1)+f(x-1)=2f(x)$ and $f(0)=0$, then $f(n), n\in N$, is

(a) $nf(1)$ (b) $[f(1)]^n$

(c) 0 (d) none of these

18. If $f(x)=\frac{x}{x-1}=\frac{1}{y}$, then the value of $f(y)$ is

(a) x (b) $x-1$

(c) $1-x$ (d) $1+x$

19. The domain of the function

$y=\cos^{-1}\left(\frac{1+x^4}{2x^2}\right)+\sqrt{x(x+2)}$ is

(a) $[-1, 1]$ (b) $(-1, 1)$

(c) $\{-1, 1\}$ (d) $\{1\}$

20. The range of the function $f(x)=3|\sin x|-2|\cos x|$ is

(a) $[-2, \sqrt{13}]$ (b) $[-2, 3]$

(c) $[3, \sqrt{13}]$ (d) none of these

21. Let f be a function satisfying $f(x+y)=f(x)+f(y)$ for all $x, y\in\mathbb{R}$. If $f(1)=k$, then $f(n), n\in N$, is equal to

(a) k^n (b) nk

(c) $-k^n$ (d) none of these

22. If $f(x)=\frac{1-x}{1+x}$, then the value of $f[f(\cos 2\theta)]$ is

(a) $\tan 2\theta$ (b) $\sec 2\theta$

(c) $\cos 2\theta$ (d) $\cot 2\theta$

23. The domain of $f(x)=\dfrac{\log_2(x+3)}{x^2+3x+2}$ is

(a) $\mathbb{R}-\{-1,-2\}$ (b) $(-2,\infty)$

(c) $\mathbb{R}-\{-1,-2,-3\}$ (d) $(-3,\infty)-\{-1,-2\}$

24. The domain of the definition of $f(x)=\log_2[1-\log_2(x^2-5x+8)]$ is

(a) (1, 3) (b) (2, 3)

(c) [2, 3] (d) none of these

25. The largest set of real values of x for which

$$f(x)=\sqrt{(x+2)(5-x)}-\frac{1}{\sqrt{x^2-4}}$$

is a real function is

(a) $[1,2)\cup(2,5]$ (b) (2, 5]

(c) [3, 4] (d) none of these

26. If $g(x)=1+\sqrt{x}$ and $f[g(x)]=3+2\sqrt{x}+x$, then the value of $f(x)$ is

(a) $1+2x^2$ (b) $2+x^2$

(c) $1+x$ (d) $2+x$

27. If $0<a<1$ and $f(x)=\sqrt{\dfrac{\log_a|x-2|}{|x|}}$, then the domain of $f(x)$ is

(a) [1, 2) (b) (2, 3]

(c) $[1,2)\cup(2,3]$ (d) $\mathbb{R}-(1,3)$

28. The interval for which $\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}=\dfrac{\pi}{2}$ holds is

(a) $[0,\infty)$ (b) [0, 3]

(c) [0, 1] (d) [0, 2]

29. The domain of the function $f(x)=\sqrt{x-\sqrt{1-x^2}}$ is

(a) $\left[-1,-\dfrac{1}{\sqrt{2}}\right]\cup\left[\dfrac{1}{\sqrt{2}},1\right]$

(b) [−1, 1]

(c) $\left(-\infty,-\dfrac{1}{\sqrt{2}}\right]\cup\left[\dfrac{1}{\sqrt{2}},+\infty\right)$

(d) $\left[\dfrac{1}{\sqrt{2}},1\right]$

30. The range of values of

$f(x)=1+\sin x+\sin^3 x+\sin^5 x,\ldots,x\in\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$ is

(a) (0, 1) (b) $(-\infty,\infty)$

(c) (−2, 2) (d) none of these

31. The range of the function $y=\sin^{-1}\left(\dfrac{x}{1+x^2}\right)$ is

(a) $\left[-\dfrac{\pi}{6},\dfrac{\pi}{6}\right]$ (b) $\left[-\dfrac{\pi}{4},\dfrac{\pi}{4}\right]$

(c) $\left[-\dfrac{\pi}{3},\dfrac{\pi}{3}\right]$ (d) $\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]$

32. If $f(x)=\lim\limits_{t\to\infty}\dfrac{(1+\sin\pi x)^t-1}{(1+\sin\pi x)^t+1}$, then the range of $f(x)$ is

(a) {−1, 1} (b) {0, 1}

(c) [−1, 1] (d) {−1, 0, 1}

33. The domain of the function $f(x)=\sqrt{x^2-[x]^2}$, where $[x]$ = the greatest integer less than or equal to x, is

(a) $\mathbb{R}$ (b) $[0,+\infty)$

(c) $(-\infty,0]$ (d) none of these

34. Which of the following functions has period π?

(a) $2\cos\left(\dfrac{2\pi x}{3}\right)+3\sin\left(\dfrac{\pi x}{3}\right)$

(b) $|\tan x|+\cos 2x$

(c) $4\cos\left(2\pi x+\dfrac{\pi}{2}\right)+2\sin\left(\pi x+\dfrac{\pi}{4}\right)$

(d) none of these

35. The period of $f(x)=\sin\sqrt{2x}+\cos\sqrt{3x}$ is

(a) π (b) $\dfrac{\pi}{\sqrt{6}}$

(c) 2π (d) none of these

36. If $f(x)=\sqrt{3|x|-x-2}$ and $g(x)=\sin x$, then the domain of the definition of $f\circ g(x)$ is

(a) $\left\{2n\pi+\dfrac{\pi}{2}\right\}$

(b) $\bigcup\limits_{n\in I}\left(2n\pi+\dfrac{7\pi}{6},2n\pi+\dfrac{11\pi}{6}\right)$

(c) $\left\{2n\pi+\dfrac{7\pi}{6}\right\}_{n\in 1}$

(d) $\left(2n\pi+\dfrac{7\pi}{6},2n\pi+\dfrac{11\pi}{6}\right)\bigcup\limits_{n,m\in I}\left(2n\pi+\dfrac{\pi}{2}\right)$

37. The domain of $f(x)=\sin^{-1}\left(\dfrac{1+x^2}{2x}\right)+\sqrt{1-x^2}$ is

(a) {1} (b) (−1, 1)

(c) {−1, 1} (d) none of these

38. Period of the function $f(x)=e^{[\sin 3x]-\sin 3x}$, where [ ] is the greatest integer function, is

(a) 2π (b) 6π

(c) $2\pi/3$ (d) none of these

39. If $f(x)=\sin x+\tan\frac{1}{2}x+\sin\frac{1}{2^2}x+\tan\frac{1}{2^3}x+\cdots$
$+\sin\frac{1}{2^{n-1}}x+\tan\frac{1}{2^n}x$
is a periodic function with period $k\pi$, then $k=$
(a) 1 (b) 2
(c) 2^n (d) $1/2^n$

40. The domain of the function
$f(x)=\frac{1}{\sqrt{x^{12}-x^9+x^4-x+1}}$ is
(a) $(-\infty,-1)$ (b) $(1,\infty)$
(c) $(-1,1)$ (d) $(-\infty,\infty)$

41. The range of the function $f(x)=\sqrt{x^2+x+1}$ is equal to
(a) $[0,\infty)$ (b) $\left[\frac{\sqrt{3}}{2},\infty\right)$
(c) $\left(\frac{\sqrt{3}}{2},\infty\right)$ (d) none of these

42. Let $f(x)=\cos^{-1}\left(\frac{x^2}{1+x^2}\right)$. The range of f is
(a) $\left[0,\frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$
(c) $\left[-\frac{\pi}{2},0\right]$ (d) none of these

43. The domain of the function $f(x)=\sqrt{1-\sqrt{1-\sqrt{1-x^2}}}$ is
(a) $(-\infty,1)$ (b) $(-1,\infty)$
(c) $[0,1]$ (d) $[-1,1]$

44. The value of $\log_2\log_2\sqrt{\sqrt{\cdots\sqrt{2}}}$, there being n square root signs, is equal to
(a) n (b) 1
(c) $-n$ (d) none of these

45. If $f(x)=\sin(\log x)$, then the value of
$f(xy)+f\left(\frac{x}{y}\right)-2f(x)\cos(\log y)$ is
(a) -1 (b) 0
(c) 1 (d) none of these

46. The function $f:(-\infty,-1]\to(0,e^5)$, defined by
$f(x)=e^{x^3-3x+2}$, is
(a) many-one and onto (b) many-one and into
(c) one-one and onto (d) one-one and into

47. The period of the function $f(x)=\sin^4 x+\cos^4 x$ is
(a) π (b) $\pi/2$
(c) 2π (d) none of these

48. If for a real number x, $[x]$ denotes the greatest integer less than or equal to x, then for any $n\in N$
$\left[\frac{n+1}{2}\right]+\left[\frac{n+2}{4}\right]+\left[\frac{n+4}{8}\right]+\left[\frac{n+8}{16}\right]+\cdots=$
(a) n (b) $n-1$
(c) $n+1$ (d) $n+2$

49. The domain of the function
$y=\cos^{-1}\left[\frac{1+x^4}{2x^2}\right]+\sqrt{x(x+2)}$ is
(a) $[-1,1]$ (b) $(-1,1)$
(c) $\{-1,1\}$ (d) $\{1\}$

50. The domain of $f(x)=\sqrt{\sin(\cos x)}+\cos^{-1}(2+2x)$ is
(a) $\left[\frac{-3}{2},\frac{-1}{2}\right]$ (b) $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$
(c) $\left[\frac{-\pi}{2},\frac{-3}{2}\right]\cup\left[\frac{-1}{2},\frac{\pi}{2}\right]$ (d) none of these

51. The domain of the function
$f(x)=\log_{10}\frac{x-5}{x^2-10x+24}-\sqrt[3]{x+5}$ is
(a) $(4,5)$ (b) $(6,\infty)$
(c) $(4,5)\cup(6,\infty)$ (d) $(4,5]\cup(6,\infty)$

52. Which of the following functions is non-periodic?
(a) $\frac{2^x}{2^{[x]}}$ (b) $\sin^{-1}[(x)]$
(c) $\sin^{-1}(\sqrt{\cos x})$ (d) $\sin^{-1}[\cos(x^2)]$

53. The range of the function
$f(x)=\sin^4 x(1+\sin^2 x)+\cos^4 x(1+\cos^2 x)$ is
(a) $[0,2]$ (b) $\left[0,\frac{3}{4}\right]$
(c) $\left[\frac{3}{4},2\right]$ (d) $\mathbb{R}$

54. The range of the function $y=\sin\left(x+\frac{\pi}{6}\right)+\cos\left(x-\frac{\pi}{6}\right)$ is
(a) $[-\sqrt{2},\sqrt{2}]$
(b) $[-\sqrt{2}(\sqrt{3}+1),\sqrt{2}(\sqrt{3}+1)]$
(c) $\left[-\frac{\sqrt{3}+1}{\sqrt{2}},\frac{\sqrt{3}+1}{\sqrt{2}}\right]$
(d) none of these

55. The range of the function $f(x)=\frac{x-1}{x+3}$ is
(a) $\mathbb{R}$ (b) $\mathbb{R}-\{-1\}$
(c) $\mathbb{R}-\{1\}$ (d) $\mathbb{R}-\{-3\}$

EXERCISE SET 2

1. The domain of $f(x)=\sin^{-1}\left(\frac{x^2}{2}\right)$ is

(a) $[-2, 2]$ (b) $\left[-\frac{1}{2},\frac{1}{2}\right]$

(c) $[-\sqrt{2},\sqrt{2}]$ (d) none of these

2. The domain of the function $f(x)=\log_2\log_3\log_4 x$ is

(a) $[4, \infty)$ (b) $(4, \infty)$

(c) $(-\infty, 4)$ (d) none of these

3. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ given by $f(x)=\frac{x^2-4}{x^2+1}$. Then $f(x)$ is

(a) one-one and into

(b) one-one and onto

(c) many-one and into

(d) many-one and onto

4. The function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=6^x+6^{|x|}$ is

(a) one-one and onto

(b) many-one and onto

(c) one-one and into

(d) many-one and into

5. Which of the following functions from Z to itself is bijective?

(a) $f(x)=x^3$ (b) $f(x)=x+2$

(c) $f(x)=2x+1$ (d) $f(x)=x^2+x$

6. The range of the function $f(x)=x^2-6x+7$ is

(a) $(-\infty, 0)$ (b) $[-2, \infty)$

(c) $(-\infty, \infty)$ (d) $(-\infty, -2)$

7. The range of the function $f(x)=\frac{x^2+x+2}{x^2+x+1}, x\in\mathbb{R}$ is

(a) $(1, \infty)$ (b) $\left(1,\frac{11}{7}\right)$

(c) $\left(1,\frac{7}{3}\right]$ (d) $\left(1,\frac{7}{5}\right)$

8. The domain of the function

$f(x)=\log_{1/2}\left(x-\frac{1}{2}\right)+\log_2\sqrt{4x^2-4x+5}$ is

(a) $\left[\frac{1}{2},\infty\right)$ (b) $\left(\frac{1}{2},\infty\right)$

(c) $(-\infty, \infty)$ (d) none of these

9. The domain of the function $f(x)=\sin^{-1}\left[\log\left(\frac{x^2}{5}\right)\right]$ is

(a) $[-5,-1]\cup[1,5]$ (b) $[-5,-5]$

(c) $(-5,-1)\cup(1,5)$ (d) none of these

10. The domain of the function $f(x)=\sqrt{\log\left(\frac{1}{|\sin x|}\right)}$ is

(a) $\mathbb{R}-[-\pi, \pi]$ (b) $\mathbb{R}-\{n\pi, n\in Z\}$

(c) $\mathbb{R}-\{2n\pi \mid n\in Z\}$ (d) $\mathbb{R}$

11. The range of the function for real x of $y=\frac{1}{2-\sin 3x}$ is

(a) $\frac{1}{3}\le y\le 1$ (b) $-\frac{1}{3}\le y<1$

(c) $-\frac{1}{3}>y>1$ (d) $\frac{1}{3}>y>1$

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement I is false, Statement II is true.

(b) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

(c) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.

(d) Statement I is true, Statement II is false.

12. **Statement I:** If $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be two mappings such that $f(x)=\sin x$ and $g(x)=x^2$, then $(f\circ g)(x)\ne(g\circ f)(x)$

Statement II: $(f\circ g)(x)=g(x)f(x)=g[f(x)]$

13. **Statement II:** The domains of $f(x)=\sqrt{\cos(\sin x)}$ and $g(x)=\sqrt{\sin(\cos x)}$ are the same.

Statement II: $-1\le\cos(\sin x)\le 1$ and $-1\le\sin(\cos x)\le 1$

14. **Statement I:** If $f(x)=x^2-x+1, x\ge\frac{1}{2}$ and $g(x)=\frac{1}{2}+\sqrt{x-\frac{3}{4}}$, then the number of solutions of the equation $f(x)=g(x)$ is two.

Statement II: $f(x)$ and $g(x)$ are mutually inverse.

15. **Statement I:** If $f(x+y)+f(x-y)=2f(x)\times f(y)$ for all $x, y\in\mathbb{R}$ and $f(0)\ne 0$, then $f(x)$ is an even function.

Statement II: If $f(-x)=f(x)$ then $f(x)$ is an even function.

16. **Statement I:** The domain of the function

$f(x)=\sin^{-1}x+\cos^{-1}x+\tan^{-1}x$ is $[-1, 1]$

Statement II: $\sin^{-1} x$ and $\cos^{-1} x$ are defined in $|x| \le 1$ and $\tan^{-1} x$ is defined for all x.

17. Statement I: A function $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = x - [x]$ (where $[x]$ is the greatest integer $\le x$) for all $x \in \mathbb{R}$. f is not invertible.
Statement II: $f(x)$ is a periodic function.

Previous Years' Questions

18. The domain of the definition of the function

$$f(x) = \sqrt{\log_{10}\left(\frac{5x - x^2}{4}\right)} \text{ is}$$

(a) [1, 4] (b) [1, 0]
(c) [0, 5] (d) [5, 0] **[AIEEE 2002]**

19. Which of the following functions is not periodic?
(a) $|\sin 3x| + \sin^2 x$ (b) $\cos\sqrt{x} + \cos^2 x$
(c) $\cos 4x + \tan^2 x$ (d) $\cos 2x + \sin x$
[AIEEE 2002]

20. The domain of $\sin^{-1}\left[\log_3\left(\frac{x}{3}\right)\right]$ is
(a) [1, 9] (b) [–1, 9]
(c) [–9, 1] (d) [–9, –1] **[AIEEE 2002]**

21. If f is a periodic function, then
(a) f has to be bounded (b) f has to be onto
(c) f has to be into (d) f may not be bounded
[AIEEE 2003]

22. If $f: \mathbb{R} \to \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$, for all x, $y \in \mathbb{R}$ and $f(1) = 7$, then $\sum_{r=1}^{n} f(r)$ is
(a) $\frac{7n}{2}$ (b) $\frac{7(n+1)}{2}$
(c) $7n(n+1)$ (d) $\frac{7n(n+1)}{2}$
[AIEEE 2003]

23. A function f from the set of natural numbers to integers defined by

$$f(n) = \begin{cases} \frac{n-1}{2}, & \text{when } n \text{ is odd} \\ -\frac{n}{2}, & \text{when } n \text{ is even} \end{cases} \text{ is}$$

(a) onto but not one-one
(b) both one-one and onto
(c) neither one-one nor onto
(d) one-one but not onto **[AIEEE 2003)]**

24. The domain of the definition of the function

$$f(x) = \frac{3}{4 - x^2} + \log_{10}(x^3 - x) \text{ is}$$

(a) $(-1, 0) \cup (1, 2)$
(b) $(1, 2) \cup (2, \infty)$
(c) $(-1, 0) \cup (1, 2) \cup (2, \infty)$
(d) $(1, 2)$ **[AIEEE 2003]**

25. The domain of the function $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9 - x^2}}$ is
(a) [2, 3] (b) [2, 3)
(c) [1, 2] (d) [1, 2) **[AIEEE 2004]**

26. If $f: R \to S$ defined by $f(x) = \sin x - \sqrt{3}\cos x + 1$ is onto, then the interval of S is
(a) [0, 3] (b) [–1, 1]
(c) [0, 1] (d) [–1, 3] **[AIEEE 2004]**

27. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is
(a) {1, 2, 3} (b) {1, 2, 3, 4, 5, 6}
(c) {1, 2, 3, 4} (d) {1, 2, 3, 4, 5}
[AIEEE 2004]

28. The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$, then
(a) $f(x+2) = f(x-2)$ (b) $f(2+x) = f(2-x)$
(c) $f(x) = f(-x)$ (d) $f(x) = -f(-x)$
[AIEEE 2004]

29. Let $f: (-1, 1) \to B$ be a function defined by $f(x) = \tan^{-1}\frac{2x}{1-x^2}$ then f is both one-one and onto when B is in the interval
(a) $\left[0, \frac{\pi}{2}\right)$ (b) $\left(0, \frac{\pi}{2}\right)$
(c) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (d) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
[AIEEE 2005]

30. A real-valued function $f(x)$ satisfies the functional equation $f(x-y) = f(x)f(y) - f(a-x)f(a+y)$, where a is a constant and $f(0) = 1$. $f(2a - x)$ is equal to
(a) $f(x)$ (b) $-f(x)$
(c) $f(-x)$ (d) $f(a) + f(a-x)$
[AIEEE 2005]

31. The largest interval lying in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which the function $f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)$ is defined as

(a) $[0,\pi]$ (b) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

(c) $\left[-\frac{\pi}{4},\frac{\pi}{2}\right)$ (d) $\left[0,\frac{\pi}{2}\right)$ **[AIEEE 2007]**

32. Let $f: N \to Y$ be a function defined as $f(x) = 4x + 3$, where $\{y \in N : y = 4x+3 \text{ for some } x \in N\}$. Then the inverse is

(a) $g(y)=\frac{3y+4}{3}$ (b) $g(y)=4+\frac{y+3}{4}$

(c) $g(y)=\frac{y+3}{4}$ (d) $g(y)=\frac{y-3}{4}$

[AIEEE 2008]

33. For real x let $f(x) = x^3 + 5x + 1$, then

(a) f is one-one but not onto $\mathbb{R}$

(b) f is onto $\mathbb{R}$ but not one-one

(c) f is one-one and onto $\mathbb{R}$

(d) f is neither one-one nor onto $\mathbb{R}$ **[AIEEE 2009]**

34. Let $f(x)=(x+1)^2-1, x \geq -1$

Statement I: The set $\{x : f(x) = f^{-1}(x)\} = \{0, -1\}$

Statement II: f is a bijection.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

[AIEEE 2009]

35. The domain of the definition of the function $f(x)$ given by the equation $2^x + 2^y = 2$ is

(a) $0 < x \leq 1$ (b) $0 \leq x \leq 1$

(c) $-\infty < x \leq 0$ (d) $-\infty < x < 1$

[IIT-JEE 2000]

36. If $f:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+\frac{1}{x}$, then $f^{-1}(x)$ equals

(a) $\frac{(x+\sqrt{x^2-4})}{2}$ (b) $\frac{x}{1+x^2}$

(c) $\frac{(x-\sqrt{x^2-4})}{2}$ (d) $1+\sqrt{x^2-4}$

[IIT-JEE 2001]

37. For a real number x, $[x]$ denotes the integral part of x. The value of

$$\left[\frac{1}{2}\right]+\left[\frac{1}{2}+\frac{1}{100}\right]+\left[\frac{1}{2}+\frac{2}{100}\right]+\cdots+\left[\frac{1}{2}+\frac{99}{100}\right] \text{ is}$$

(a) 49 (b) 50

(c) 48 (d) 51 **[IIT-JEE 2001]**

38. Suppose $f(x)=(x+1)^2$ for $x \geq -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals

(a) $-\sqrt{x}-1, x \geq 0$ (b) $\frac{1}{(x+1)^2}, x > -1$

(c) $\sqrt{x+1}, x \geq -1$ (d) $\sqrt{x}-1, x \geq 0$

[IIT-JEE 2002]

39. The domain of the definition of the function

$$f(x)=\sqrt{\sin^{-1}(2x)+\frac{\pi}{6}} \text{ for real-valued } x \text{ is}$$

(a) $\left[-\frac{1}{4},\frac{1}{2}\right]$ (b) $\left[-\frac{1}{2},\frac{1}{2}\right]$

(c) $\left(-\frac{1}{2},\frac{1}{9}\right)$ (d) $\left[-\frac{1}{4},\frac{1}{4}\right]$

[IIT-JEE 2003]

40. If $f(x)=x^2+2bx+2c^2$ and $g(x)=-x^2-2cx+b^2$ such that $f(x) > \max g(x)$, then the relation between b and c is

(a) no real value of b and c (b) $0 < c < b\sqrt{2}$

(c) $|c| < |b|\sqrt{2}$ (d) $|c| > |b|\sqrt{2}$

[IIT-JEE 2003]

41. If $f(x)=\sin x+\cos x, g(x)=x^2-1$, then $g[f(x)]$ is invertible in the domain

(a) $\left[0,\frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{4},\frac{\pi}{4}\right]$

(c) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (d) $[0, \pi]$

[IIT-JEE 2004]

42. If the functions $f(x)$ and $g(x)$ are defined on $\mathbb{R} \to \mathbb{R}$ such that $f(x)=\begin{cases}0, & x \in \text{irrational}\\ x, & x \in \text{rational}\end{cases}$ and

$g(x)=\begin{cases}0, & x \in \text{irrational}\\ x, & x \in \text{rational}\end{cases}$ then $(f-g)(x)$ is

(a) one-one and onto

(b) neither one-one nor onto

(c) one-one but not onto

(d) onto but not one-one **[IIT-JEE 2005]**

43. Suppose $f:[-2,2]\to\mathbb{R}$ is defined by

$f(x)=\begin{cases}-1, & \text{for } -2 \leq x \leq 0\\ x-1, & \text{for } 0 \leq x \leq 2\end{cases}$ then $\{x \in [-2, 2] : x \leq 0$ and $f(|x|) = x\}$ is equal to

(a) $\{-1\}$ (b) $\{0\}$

(c) $\left\{-\frac{1}{2}\right\}$ (d) ϕ **[EAMCET 2003]**

44. If $e^{f(x)} = \frac{10+x}{10-x}$, $x \in (-10, 10)$ and $f(x) = kf\left(\frac{200x}{100+x^2}\right)$, then k is equal to

(a) 0.5 (b) 0.6

(c) 0.7 (d) 0.8 **[EAMCET 2003]**

45. The function $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = 3^{-x}$. Observe the following statements

I. f is one-one II. f is onto

III. f is a decreasing function

Out of these, true statements are

(a) only I, II (b) only II, III

(c) only I, III (d) I, II, III

[EAMCET 2004]

46. The function $f: \mathbb{C} \to \mathbb{C}$ defined by $f(x) = \frac{ax+b}{cx+d}$ for $x \in \mathbb{C}$, where $bd \neq 0$, reduces to a constant function, if

(a) $a = c$ (b) $b = d$

(c) $ad = bc$ (d) $ab = cd$

[EAMCET 2005]

47. If $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = x - [x] - \frac{1}{2}$ for $x \in \mathbb{R}$, where $[x]$ is the greatest integer not exceeding x, then $\left\{x \in \mathbb{R} : f(x) = \frac{1}{2}\right\}$ is equal to

(a) ϕ, the empty set

(b) N, the set of all natural numbers

(c) Z, the set of all integers

(d) $\mathbb{R}$, the set of all real numbers **[EAMCET 2006]**

48. If $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = \frac{1}{2-\cos 3x}$ for each $x \in \mathbb{R}$, then the range of f is

(a) (1, 2) (b) $\left(\frac{1}{3}, 3\right)$

(c) $\left[\frac{1}{4}, \frac{1}{2}\right]$ (d) $\left[\frac{1}{3}, 1\right]$

[EAMCET 2007]

49. If $f: [-6, 6] \to 6$ is defined by $f(x) = x^2 - 3$ for $x \in \mathbb{R}$, then $(f \circ f \circ f)(-1) + (f \circ f \circ f)(0) + (f \circ f \circ f)(1)$ is equal to

(a) $f(4\sqrt{2})$ (b) $f(3\sqrt{2})$

(c) $f(2\sqrt{2})$ (d) $f(\sqrt{2})$

[EAMCET 2008]

50. If $f: [2, 3] \to \mathbb{R}$ is defined by $f(x) = x^3 + 3x - 2$, then the range $f(x)$ is contained in the interval

(a) [1, 12] (b) [12, 34]

(c) [35, 50] (d) [12, 12]

[EAMCET 2009]

51. The domain of $f(x) = \log|\log x|$ is

(a) $(-\infty, 1)$ (b) $(0, \infty)$

(c) $(1, \infty)$ (d) $(0, 1) \cup (1, \infty)$

[DCE 2002]

52. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$, then for all x, $f[g(x)]$ is equal to

(a) x (b) 1

(c) $f(x)$ (d) $g(x)$ **[DCE 2006]**

53. The range of the function $f(x) = \frac{x}{1+x^2}$ is

(a) $(-\infty, \infty)$ (b) $[-1, 1]$

(c) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (d) $[-\sqrt{2}, \sqrt{2}]$

[DCE 2009]

54. The function $f(x) = \log\left(\frac{1+x}{1-x}\right)$ satisfies the equation

(a) $f(x_1) \cdot f(x_2) = f(x_1 + x_2)$

(b) $f(x_1) + f(x_2) = f\left(\frac{x_1 + x_2}{1 + x_1 x_2}\right)$

(c) $f(x) + f(x+1) = f(x^2 + x)$

(d) $f(x+2) - 2f(x+1) + f(x) = 0$

[WBJEE 2008, DCE 1997]

55. If $f(x) = \frac{x}{x-1}$, then the value of $\frac{f(a)}{f(a+1)}$ is

(a) $f\left(\frac{-a}{a-1}\right)$ (b) $f(-a)$

(c) $f\left(\frac{1}{a}\right)$ (d) $f(a^2)$

[AMU 2002, Punjab CEET 1997]

56. For which of the following functions, is $\frac{f(m) - f(n)}{m-n}$ constant for all numbers m and n, $m \neq n$?

(a) $f(x) = \cos x$ (b) $f(x) = \log x$

(c) $f(x) = x^2 + 1$ (d) $f(x) = 4x + 7$

[AMU 2002]

57. If $f(x)=\dfrac{4^x}{4^x+2}$, then

$f\left(\dfrac{1}{1997}\right)+f\left(\dfrac{2}{1997}\right)+\cdots+f\left(\dfrac{1996}{1997}\right)$ is equal to

(a) 499/501 (b) 249
(c) 348/435 (d) 998

[Orissa JEE 2000, DCE 2006, WBJEE 2005]

58. If $f(x)=\log\left(\dfrac{1+x}{1-x}\right)$, then the value of $f\left(\dfrac{2x}{1+x^2}\right)$ is

(a) $2f(x)$ (b) $3f(x)$
(c) $[f(x)]^2$ (d) $[f(x)]^3$

[UPSET 2003, Rajasthan PET 1999, Punjab CEET 1998]

59. If $f(x)=\dfrac{1-x}{1+x}$, then $f[f(\cos 2\theta)]$ is equal to

(a) $\cos 2\theta$ (b) $\tan 2\theta$ (c) $\sec 2\theta$ (d) $\cot 2\theta$

[KCET 1998, Punjab CEET 2002]

60. If $f(x)=\dfrac{x+1}{x-1}, x\neq 1$, then $(f\circ f\circ f\circ f)x$ is equal to

(a) $1/x$ (b) x
(c) x^2 (d) indeterminate

[Rajasthan PET 2004, Punjab CEET 1996]

61. If the function $f(x)$ is differentiable at $x=a$, then

$\lim\limits_{x\to a}\dfrac{x^2f(a)-a^2f(x)}{x-a}$ is

(a) $-a^2f'(a)$ (b) $af(a)-a^2f'(a)$
(c) $2af(a)-a^2f'(a)$ (d) $2af(a)+a^2f'(a)$

[AIEEE 2011]

ANSWERS

Exercise Set 1

1. (c)	**2.** (a)	**3.** (d)	**4.** (b)	**5.** (d)	**6.** (c)	**7.** (c)	**8.** (a)	**9.** (a)	**10.** (d)
11. (b)	**12.** (a)	**13.** (a)	**14.** (c)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (c)	**19.** (d)	**20.** (d)
21. (b)	**22.** (c)	**23.** (d)	**24.** (b)	**25.** (b)	**26.** (b)	**27.** (c)	**28.** (c)	**29.** (d)	**30.** (b)
31. (a)	**32.** (d)	**33.** (d)	**34.** (b)	**35.** (d)	**36.** (d)	**37.** (c)	**38.** (c)	**39.** (c)	**40.** (d)
41. (b)	**42.** (d)	**43.** (d)	**44.** (c)	**45.** (b)	**46.** (d)	**47.** (b)	**48.** (a)	**49.** (d)	**50.** (a)
51. (c)	**52.** (d)	**53.** (c)	**54.** (c)	**55.** (c)					

Exercise Set 2

1. (c)	**2.** (b)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (b)	**9.** (a)	**10.** (b)
11. (a)	**12.** (d)	**13.** (a)	**14.** (a)	**15.** (c)	**16.** (b)	**17.** (b)	**18.** (a)	**19.** (b)	**20.** (a)
21. (d)	**22.** (d)	**23.** (b)	**24.** (c)	**25.** (b)	**26.** (d)	**27.** (a)	**28.** (b)	**29.** (c)	**30.** (b)
31. (d)	**32.** (d)	**33.** (c)	**34.** (c)	**35.** (d)	**36.** (a)	**37.** (b)	**38.** (d)	**39.** (a)	**40.** (d)
41. (b)	**42.** (a)	**43.** (c)	**44.** (a)	**45.** (c)	**46.** (c)	**47.** (a)	**48.** (d)	**49.** (a)	**50.** (b)
51. (d)	**52.** (b)	**53.** (c)	**54.** (b)	**55.** (d)	**56.** (d)	**57.** (d)	**58.** (a)	**59.** (a)	**60.** (b)
61. (c)									

HINTS AND SOLUTIONS—EXERCISE SET 1

1. For domain point of view, $0\le x^2+x+1\le 1$, but

$x^2+x+1\ge\dfrac{3}{4}\quad\Rightarrow\quad\dfrac{\sqrt{3}}{2}\le\sqrt{x^2+x+1}\le 1$

$\Rightarrow\quad\dfrac{\pi}{3}\le\sin^{-1}(\sqrt{x^2+x+1})\le\dfrac{\pi}{2}$.

2. $f(x)$ is defined if $\log_{0.5}x\ge 0$, i.e. if $\log_{1/2}x\ge 0$

i.e. if $\log_{1/2}x=t$, where $t\ge 0$

i.e. if $x=\left(\dfrac{1}{2}\right)^t, t\neq 0$

i.e. if $x>0$ and $x\le 1$. $\left[\because\dfrac{1}{2^t}=\dfrac{1}{2^0}=1, \text{for } t=0\right]$

3. $1/1-x$ is defined, if $1-x\neq 0$, i.e. $x\neq 1$ (i)

and $2^{\sin^{-1}x}$ is defined, if $-1\le x\le 1$ (ii)

Also, $\dfrac{1}{\sqrt{x-2}}$ is defined, if $x-2>0\Rightarrow x>2$ (iii)

From (i), (ii) and (iii), the domain of $f=\phi$ (empty set)

4. $f(x)=\left|\sin^3\dfrac{x}{2}\right|+\left|\cos^5\dfrac{x}{5}\right|$

period of $\sin^3 x$ is 2π

$\Rightarrow$ period of $\sin^3\dfrac{x}{2}$ is $\dfrac{2\pi}{1/2}=4\pi$

$\Rightarrow$ period of $\left|\sin^3\dfrac{x}{2}\right|$ is 2π

period of $\cos^5 x$ is 2π

$\Rightarrow$ period of $\cos^5\frac{x}{2}$ is $\frac{2\pi}{\left(\frac{1}{5}\right)}=10\pi$

$\Rightarrow$ period of $\left|\cos^5\frac{x}{2}\right|$ is 5π

Now period of $f(x)$ = LCM of $\{2\pi, 10\pi\} = 10\pi$.

5. $f(x)=\sin\left(\frac{\pi x}{n!}\right)-\cos\left[\frac{\pi x}{(n+1)!}\right]=f_1-f_2$

period of $f_1=\frac{2\pi}{\left(\frac{\pi}{n!}\right)}=2(n!)=A$ (say)

period of $f_2=2[(n+1)!]=B$ (say)

Now, LCM of A and B is $2[(n+1)!]$

Hence, the function $f(x)$ is periodic with period $2[(n+1)!]$.

6. $\frac{1-|x|}{2-|x|}\geq 0 \quad\Rightarrow\quad \frac{|x|-1}{|x|-2}\geq 0$

$\Rightarrow \quad |x|\leq 1 \quad$ and $\quad |x|>2$

$\Rightarrow \quad -1\leq x\leq 1 \quad$ and $\quad x<-2$ or $x>2$

$\Rightarrow \quad D_f=[-1,1]\cup(-\infty,-2)\cup(2,\infty)$.

7. $f(x)=3\tan\sqrt{\frac{\pi^2}{9}-x^2}$

Since domain of $f=D_f=\left[-\frac{\pi}{3},\frac{\pi}{3}\right]$

Now, for all $x\in\left[-\frac{\pi}{3},\frac{\pi}{3}\right]$,

$\theta=\sqrt{\frac{\pi^2}{9}-x^2}$ lies between $\left[0,\frac{\pi}{3}\right]$

As far as the tangent (tan) function is concerned, we know that tan is an increasing function throughout its domain. So in this case, tan θ will attain its minimum $\theta = 0$ and maximum at $\theta = \pi/3$

i.e. $\quad 0\leq\sqrt{\frac{\pi^2}{9}-x^2}\leq\frac{\pi}{3}$

$\Rightarrow \quad \tan 0\leq\tan\sqrt{\frac{\pi^2}{9}-x^2}\leq\tan\frac{\pi}{3}$

$\Rightarrow \quad 3(0)\leq 3\tan\sqrt{\frac{\pi^2}{9}-x^2}\leq 3\sqrt{3}$

$\Rightarrow \quad 0\leq f(x)\leq 3\sqrt{3}$ for all $x\in\left[-\frac{\pi}{3},\frac{\pi}{3}\right]$

$\therefore$ Range of $f=R_f=[0,3\sqrt{3}]$

8. $f(x)$ is defined, if $-1\leq\frac{3}{4+2\sin x}\leq 1$

Since $4+2\sin x>0$ for all real x, therefore

$$\frac{3}{4+2\sin x}\leq 1\Rightarrow 3\leq 4+2\sin x\Rightarrow\sin x\geq-\frac{1}{2}$$

$\Rightarrow \quad -\frac{\pi}{6}+2n\pi\leq x\leq\frac{7\pi}{6}+2n\pi, n\in I$

$\therefore$ Domain of $f=\left[-\frac{\pi}{6}+2n\pi,\frac{7\pi}{6}+2n\pi\right]$

9. $\sqrt[3]{1-3x}$ and $e^{3\tan x}$ is defined for all real $x\in\left\{(2n+1)\frac{\pi}{2}\right\}$

$3\cos^{-1}\left(\frac{2x-1}{3}\right)$ is defined, if $-1\leq\frac{2x-1}{3}\leq 1$

$\Rightarrow \quad -3\leq 2x-1\leq 3\Rightarrow -1\leq x\leq 2$

Domain of $f=[-1,2]-\left\{\frac{\pi}{2}\right\}$

10. Since $y=\sin[x]$ for all $x\in\left(-\frac{\pi}{4},\frac{\pi}{4}\right)$

i.e. $x\in(-0.785,0.785)$

Case I: $-0.785<x<0$

If $x\in(-0.785,0)$, then $[x]=-1$

$\therefore \quad y=\sin[x]=\sin(-1)=-\sin 1$

Case II: $0\leq x<0.785$

If $x\in[0,0.785)$, then $[x]=0$

$\therefore \quad y=\sin(0)=0$

Hence, range of the function = $\{0, -\sin 1\}$

11. $y=\frac{e^x-e^{-x}}{e^x+e^{-x}}+2 \quad\Rightarrow\quad \frac{y-2}{1}=\frac{e^x-e^{-x}}{e^x+e^{-x}}$.

By componendo and dividendo, $\frac{y-1}{y-3}=\frac{2e^x}{-2e^{-x}}$

$\Rightarrow \quad e^{2x}=\frac{y-1}{3-y} \quad\Rightarrow\quad x=\frac{1}{2}\log\left(\frac{y-1}{3-y}\right)$.

Hence, $f^{-1}(x)=\log_e\left(\frac{x-1}{3-x}\right)^{1/2}$.

12. We have $f(x)=\cos x-x(1+x)$

Since in the interval $\left[\frac{\pi}{6},\frac{\pi}{3}\right]$, $\cos x$ decreases and $x(1+x)$ increases,

$\therefore \quad f(x)$ decreases in $\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

$\therefore \quad f\left(\frac{\pi}{3}\right)\leq f(x)\leq f\left(\frac{\pi}{6}\right), x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

Hence, $f(A)=\left[\frac{1}{2}-\frac{\pi}{3}\left(1+\frac{\pi}{3}\right),\frac{\sqrt{3}}{2}-\frac{\pi}{6}\left(1+\frac{\pi}{6}\right)\right]$

$=\left[\frac{1}{2}-\frac{\pi}{3}-\frac{\pi^2}{9},\frac{\sqrt{3}}{2}-\frac{\pi}{6}-\frac{\pi^2}{36}\right]$

13. $x-[x]$ is periodic with period 1.

14. Let $y=f(x)=\sqrt{\frac{x}{1+x}}$, where $y\geq 0$ always

$\Rightarrow\quad y^2=\frac{x}{1+x}\quad\Rightarrow\quad y^2(1+x)=x$

$\Rightarrow\quad y^2+y^2x=x\quad\Rightarrow\quad (1-y^2)x=y^2$

$\Rightarrow\quad x=\frac{y^2}{1-y^2}=f^{-1}(y)$

$\Rightarrow\quad f^{-1}(x)=\frac{y^2}{1-y^2}$

which is valid if $y\neq -1,1$

$\therefore$ Range of $f=R_f=[0,\infty)-\{1\}$

15. $f(x)=g(x)$

over $\mathbb{R}-\left\{x:x=(2n+1)\frac{\pi}{2};n\in I\right\}$.

16. For $f(x)$ to get defined, we must have $1+2\sin x\neq 0$

$\Rightarrow\quad \sin x\neq -\frac{1}{2}\quad\Rightarrow\quad x\neq n\pi+(-1)^n\frac{5\pi}{6},n\in Z$

Hence, domain of the function is

$$\mathbb{R}-\left\{n\pi+(-1)^n\frac{5\pi}{6},n\in Z\right\}.$$

17. Putting $x=1, f(2)+f(0)=2f(1)$

$\Rightarrow\quad f(2)=2f(1)$.

Putting $x=2, f(3)+f(1)=2f(2)$

$\Rightarrow\quad f(3)=2\times 2f(1)-f(1)=3f(1)$, etc.

$\therefore\quad f(n)=nf(1)$, for $n=1, 2, \ldots, n$.

$f(n+1)+f(n-1)=2f(n)$

$\Rightarrow\quad f(n+1)+f(n-1)f(1)=2nf(1)$

$\Rightarrow\quad f(n+1)=(n+1)f(1)$.

18. Function $f(x)=\frac{x}{x-1}=\frac{1}{y}$ (i)

The given function may be written as $y=\frac{x-1}{x}$.

And $f(y)=\frac{y}{y-1}$.

Substituting the value of y in Eq. (i), we get

$$f(y)=\frac{\frac{x-1}{x}}{\frac{x-1}{x}-1}=\frac{x-1}{-1}=1-x.$$

19. $-1\leq\frac{1+x^4}{2x^2}\leq 1\quad\Rightarrow\quad\frac{1+x^4}{2x^2}\leq 1$

$\left(\because\frac{1+x^4}{2x^2}\geq -1 \text{ is always true}\right)$

$\Rightarrow\quad (1-x^2)^2\leq 0\quad\Rightarrow\quad x=\pm 1$

but $x=-1$ does not define $\sqrt{x(x+2)}$

20. Since $f(x)=3|\sin x|-2|\cos x|=y$ (say)

Squaring both sides, we have

$y^2=(3|\sin x|-2|\cos x|)^2$

$y=9\sin^2 x+4\cos^2 x-6|\sin x||\cos x|$

$\Rightarrow\quad y^2(9-9\cos^2 x)+(4-4\sin^2 x)-6|\sin x||\cos x|$

$\Rightarrow\quad y^2=13-(9\cos^2 x+4\sin^2 x+6|\sin x||\cos x|)$

$\Rightarrow\quad y^2=13-(3|\cos x|+2|\sin x|^2)\leq 13$

$\Rightarrow\quad y^2\leq 13\quad\Rightarrow\quad |y|\leq\sqrt{13}$

$\Rightarrow\quad -\sqrt{13}\leq y\leq\sqrt{13}$.

21. Putting $x=1, y=1$, we get

$f(2)=f(1)+f(1)=2f(1)=2k$

Putting $x=2, y=1$, we get

$f(3)=f(2)+f(1)=3f(1)=3k$, etc.

22. Function is $f(x)=\frac{1-x}{1+x}$.

We know that

$f(\cos 2\theta)=\frac{1-\cos 2\theta}{1+\cos 2\theta}=\frac{2\sin^2\theta}{2\cos^2\theta}=\tan^2\theta$.

Therefore,

$f[f(\cos 2\theta)]=f(\tan^2\theta)=\frac{1-\tan^2\theta}{1+\tan^2\theta}=\cos 2\theta$

23. $x+3>0$ and $x^2+3x+2\neq 0$

24. Since $f(x)=\log_2[1-\log_2(x^2-5x+8)]$

$\Rightarrow\quad 1-\log_2(x^2-5x+8)>0$

$\Rightarrow\quad \log_2(x^2-5x+6)<1=\log_2 2$

$\Rightarrow\quad x^2-5x+8<2$

$\Rightarrow\quad x^2-5x+6<0$

$\therefore\quad x\in(2,3)$

25. The domain of $\sqrt{(x+2)(5-x)}$ is the set of values of x for which $(x+2)(5-x)\geq 0$, i.e. $x\in[-2, 5]$.

The domain of $\frac{1}{\sqrt{x^2-4}}$ is the set of values of x for which $x^2-4>0$, i.e. $x\in(-\infty,-2)\cup(2,+\infty)$.

The common values are in (2, 5].

26. Equations are $g(x)=1+\sqrt{x}$ (i)

and $f[g(x)]=3+2\sqrt{x}+x$ (ii)

Substituting the value of $g(x)$ in Eq. (ii), we get

$$f(1+\sqrt{x})=3+2\sqrt{x}+x.$$

Now substituting $y=1+\sqrt{x}$,

$\Rightarrow (y-1)=\sqrt{x} \Rightarrow (y-1)^2=x$

we get

$(y)=3+2(y-1)+(y-1)^2=3+2y-2+y^2-2y+1$

$\Rightarrow f(y)=2+y^2 \Rightarrow f(x)=2+x^2$.

27. Required conditions are

$$\frac{\log_a|x-2|}{|x|}\geq 0;\ |x-2|\neq 0,|x|\neq 0$$

$\because \quad 0<a<1;\ \frac{\log_a|x-2|}{|x|}\geq 0$

$\Rightarrow \quad |x-2|\leq 1$

$\Rightarrow \quad 1\leq x\leq 3$ and as $|x-2|\neq 0, x\neq 2$

$\therefore$ Domain is $[1, 2)\cup(2, 3]$.

28. We know that

$\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$ if $x\in[-1,1]$

Here, $\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}=\frac{\pi}{2}$ if

$x\geq 0$ and $-1\leq\sqrt{x}\leq 1$

$\Rightarrow 0\leq x\leq 1$

$\therefore \quad x\in[0,1]$.

29. Here, $x-\sqrt{1-x^2}\geq 0 \quad \therefore \quad x\geq\sqrt{1-x^2}>0$

$\therefore \quad x^2\geq 1-x^2 \quad \Rightarrow \quad x^2\geq\frac{1}{2}$

Also $1-x^2\geq 0$ or $x^2\leq 1$

$\Rightarrow \quad x^2\geq\frac{1}{2} \quad \Rightarrow \quad x<-\frac{1}{\sqrt{2}}$ or $x\geq\frac{1}{\sqrt{2}}$

$x^2\leq 1 \quad \Rightarrow \quad -1\leq x\leq 1$.

Thus, $x>0, x^2\geq\frac{1}{2}$ and $x^2\leq 1 \quad \Rightarrow \quad x\in\left[\frac{1}{\sqrt{2}},1\right]$.

30. We have $f(x)=1+\frac{\sin x}{\cos^2 x}$

$$\Rightarrow f'(x)=\frac{\cos^2 x(\cos x)+\sin x(2\cos x\sin x)}{\cos^4 x}$$

$$=\frac{\cos x(\cos^2 x+2\sin^2 x)}{\cos^4 x}=\frac{1+\sin^2 x}{\cos^3 x}$$

$\Rightarrow f'(x)>0$

$\therefore \quad f(x)$ is an increasing function.

$$\lim_{x\to-\frac{\pi}{2}}\left(1+\frac{\sin x}{\cos^2 x}\right)=-\infty \quad \text{and} \quad \lim_{x\to\frac{\pi}{2}}\left(1+\frac{\sin x}{\cos^2 x}\right)=\infty$$

$\therefore$ Range of $f=(-\infty,\infty)$

31. Note the fact that $-1\leq\frac{2x}{1+x^2}\leq 1$,

therefore, $\frac{-1}{2}\leq\frac{x}{1+x^2}\leq\frac{1}{2}$

$$\therefore \quad \sin^{-1}\left(-\frac{1}{2}\right)\leq\sin^{-1}\frac{x}{1+x^2}\leq\sin^{-1}\left(\frac{1}{2}\right).$$

32. Since $f(x)=\lim_{t\to\infty}\frac{(1+\sin\pi x)^t-1}{(1+\sin\pi x)^t+1}$

Case I: $0<\sin\pi x\leq 1$

Now, $f(x)=\lim_{t\to\infty}\frac{1-\frac{1}{(1+\sin\pi x)^t}}{1+\frac{1}{(1+\sin\pi x)^t}}$

As $0<\sin\pi x\leq 1 \quad \Rightarrow \quad 1\leq 1+\sin\pi x<2$

$$\Rightarrow \quad \frac{1}{2}<\frac{1}{1+\sin\pi x}\leq 1$$

As $t\to\infty \quad \Rightarrow \quad \left(\frac{1}{1+\sin x}\right)^t\to 0$

$$\therefore \quad f(x)=\frac{1-0}{1+0}=1$$

Case II: $\sin\pi x=0$

$$f(x)=\lim_{t\to\infty}\frac{1^t-1}{1^t+1}=0$$

Case III: $-1\leq\sin\pi x<0$

$$f(x)=\lim_{t\to\infty}\frac{(1+\sin\pi x)^t-1}{(1+\sin\pi x)^t+1}$$

As $-1\leq\sin\pi x<0 \quad \Rightarrow \quad 0\leq 1+\sin\pi x<1$

As $t\to\infty \quad \Rightarrow \quad (1+\sin\pi x)^t\to 0$

$$\therefore \quad f(x)=\frac{0-1}{0+1}=-1$$

Hence, $f(x)=\begin{cases}-1 & -1\leq\sin\pi x<0\\ 0 & \sin\pi x=0\\ 1 & 0<\sin\pi x\leq 1\end{cases}$

$\therefore$ Range of $f=R_f=\{-1,0,1\}$

33. $x^2-[x]^2\geq 0 \Rightarrow x^2\geq [x]^2$

This is true for all positive x and all negative integer x.

34. Period of $\left[2\cos\left(\frac{2\pi x}{3}\right)+3\sin\left(\frac{\pi x}{3}\right)\right]$

$= \text{LCM of}\left(\frac{2\pi}{2\pi/3},\frac{2\pi}{\pi/3}\right)=6$

Period of $\left[4\cos\left(2\pi x+\frac{\pi}{2}\right)+2\sin\left(\pi x+\frac{\pi}{4}\right)\right]$

$= \text{LCM of}\left(\frac{2\pi}{2\pi},\frac{2\pi}{\pi}\right)=2$

Period of $\{|\tan x|+\cos 2x\} = \text{LCM of}\left(\pi,\frac{2\pi}{2}\right)=\pi$

35. Required period = LCM of $\frac{2\pi}{\sqrt{2}}$ and $\frac{2\pi}{\sqrt{3}}$, which does not exist (note that LCM of a rational and an irrational number or two irrational numbers of different kind does not exist).

36. We have $f(x)=\sqrt{3|x|-x-2}$ and $g(x)=\sin x$

$\therefore\ f\circ g(x)=\sqrt{3|\sin x|-\sin x-2}$ which is defined, if $3|\sin x|-\sin x-2\geq 0$

If $\sin x>0$, then $2\sin x-2\geq 0 \Rightarrow \sin x\geq 1$

$\Rightarrow \sin x=1\Rightarrow x=\frac{\pi}{2}$

If $\sin x<0$, then $-4\sin x-2\geq 0\Rightarrow -1\leq \sin x\leq -\frac{1}{2}$

$\therefore\ x\in\left(2n\pi+\frac{7\pi}{6},2n\pi+\frac{11\pi}{6}\right)\bigcup_{n,m\in I}\left(2n\pi+\frac{\pi}{2}\right) n,m\in I$

37. Here $\left|\frac{1+x^2}{2x}\right|\leq 1$ and $1-x^2\geq 0$. The first inequality implies $x = 1, -1$ which satisfies the second inequation.

38. Required period = period of $\sin 3x=\frac{2\pi}{3}$.

39. The functions $\sin x, \sin\left(\frac{x}{2^2}\right), \sin\left(\frac{x}{2^4}\right),\ldots,\sin\left(\frac{x}{2^{n-1}}\right)$ are periodic with periods 2π, $2^3\pi$, $2^5\pi$, ..., $2^n\pi$, respectively.

Also, the functions

$$\tan\left(\frac{x}{2}\right),\tan\left(\frac{x}{2^3}\right),\tan\left(\frac{x}{2^5}\right),\ldots,\tan\left(\frac{x}{2^n}\right)$$

are periodic with periods 2π, $2^3\pi$, $2^5\pi$, ..., $2^n\pi$, respectively.

Hence, the period of whole function $f(x)$ is actually the LCM of 2π, $2^3\pi$, $2^5\pi$, ..., $2^n\pi$, such that the period of $f(x)$ is $2^n\pi$.

Given that $f(x)$ is periodic with period $k\pi$.

$\Rightarrow 2^n\pi=k\pi$

$\therefore\ k=2^n$

40. $f(x)$ is defined for $x^{12}-x^9+x^4-x+1>0$

$\Rightarrow x^4(x^8+1)-x(x^8+1)+1>0$

$\Rightarrow (x^8+1)x(x^3-1)+1>0$

If $x\geq 1$ or $x\leq -1$, then the above expression is positive.

If $-1<x\leq 0$, the above inequality still holds.

If $0<x<1$, then $x^{12}-x(x^8+1)+(x^4+1)>0$.

$[\because x^4+1>x^8+1$ and so $x^4+1>x(x^8+1)]$

The domain of $f=(-\infty,\infty)$.

41. $f(x)=\sqrt{x^2+x+1}$

$\Rightarrow f(x)=\sqrt{x^2+x+\left(\frac{1}{4}\right)^2+1-\left(\frac{1}{4}\right)^2}$

$\Rightarrow f(x)=\sqrt{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}\geq\sqrt{\frac{3}{4}}$

$\Rightarrow f(x)\geq\frac{\sqrt{3}}{2}$

Hence, the range of $f=R_f=\left[\frac{\sqrt{3}}{2},\infty\right)$.

42. $\left|\frac{x^2}{1+x^2}\right|\leq 1$. This is true for all $x\in\mathbb{R}$.

So, the domain = $\mathbb{R}$.

Now $\frac{x^2}{1+x^2}=1-\frac{1}{1+x^2}$.

$\therefore\ 0\leq\frac{x^2}{1+x^2}<1$.

$\therefore\ \cos^{-1}0\geq\cos^{-1}\frac{x^2}{1+x^2}>\cos^{-1}1$

$\Rightarrow \frac{\pi}{2}\geq\cos^{-1}\frac{x^2}{1+x^2}>\cos^{-1}1$

$\Rightarrow \frac{\pi}{2}\geq\cos^{-1}\frac{x^2}{1+x^2}>0$

$\therefore$ The range $=\left(0,\frac{\pi}{2}\right]$.

43. $f(x)$ is defined, if

$1-\sqrt{1-\sqrt{1-x^2}}\geq 0, 1-\sqrt{1-x^2}\geq 0$ and

$1-x^2\geq 0$

$1-x^2 \geq 0 \Rightarrow (x+1)(x-1) \leq 0 \Rightarrow -1 \leq x \leq 1$

Clearly for these values, the other two inequalities hold.

Thus, the domain of $f = [-1, 1]$.

44. Let $E = \log_2 \log_2 \left(\underset{1\ 2\ 3 \quad\quad n}{\sqrt{\sqrt{\sqrt{\ldots\sqrt{2}}}}} \right)$

$\Rightarrow E = \log_2 \log_2 \left[\left((2)^{\frac{1}{2}} \right)^{\frac{1}{2}} \right]^{\frac{1}{2} \cdots (n \text{ times})}$

$\Rightarrow E = \log_2 \log_2 \left(2^{\frac{1}{2^n}} \right) = \log_2 \left(\frac{1}{2^n} \log_2 2 \right)$

$\Rightarrow E = \log_2 (2^{-n})$

$\therefore \quad E = -n$

45. Let $E = f(xy) + f\left(\frac{x}{y}\right) - 2f(x)\cos(\log y)$

$\Rightarrow E = \sin(\log xy) + \sin\left[\log\left(\frac{x}{y}\right)\right] - 2f(x)\cos(\log y)$

$\Rightarrow E = \sin(\log x + \log y) + \sin(\log x - \log y) - 2f(x)\cos(\log y)$

$\Rightarrow E = 2\sin(\log x)\cos(\log y) - 2\sin(\log x)\cos(\log y)$

$\therefore \quad E = 0$

46. We have $f(x) = e^{x^3 - 3x + 2}$

Let $g(x) = x^3 - 3x + 2$

$\Rightarrow g'(x) = 3x^2 - 3 = 3(x^2 - 1)$

$\Rightarrow g'(x) \geq 0$ for $x \in (-\infty, -1]$

$\therefore$ $g(x)$ is an increasing function. $\therefore$ $f(x)$ is one-one.

Now, the range of $f(x)$ is $(0, e^4]$, but the co-domain is $(0, e^5)$.

$\therefore$ $f(x)$ is an into function.

47. We have $f(x) = \sin^4 x + \cos^4 x$

$= (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x$

$= 1 - \frac{1}{2}(\sin 2x)^2 = 1 - \frac{1}{2}\left[\frac{1 - \cos 4x}{2}\right]$

$= \frac{3}{4} + \frac{1}{4}\cos 4x$.

Since $\cos x$ is periodic with period 2π, therefore $\cos 4x$ is periodic with period $2\pi/4$ and hence $f(x)$ is periodic with period $\pi/2$.

48. For any $x \in \mathbb{R}$, we have

$$[x] = \left[\frac{x}{2}\right] + \left[\frac{x+1}{2}\right] \qquad \text{(i)}$$

$\Rightarrow [n] = \left[\frac{n+1}{2}\right] + \left[\frac{n}{2}\right]$

$\Rightarrow n = \left[\frac{n+1}{2}\right] + \left[\frac{n}{2}\right]$

$\Rightarrow n = \left[\frac{n+1}{2}\right] + \left[\frac{n}{4}\right] + \left[\frac{\frac{n}{2}+1}{2}\right]$ [using Eq. (i)]

$\Rightarrow n = \left[\frac{n+1}{2}\right] + \left[\frac{n+2}{4}\right] + \left[\frac{n}{4}\right]$

$\Rightarrow n = \left[\frac{n+1}{2}\right] + \left[\frac{n+2}{4}\right] + \left[\frac{n}{8}\right] + \left[\frac{\frac{n}{4}+1}{2}\right]$ [using Eq. (i)]

$\Rightarrow n = \left[\frac{n+1}{2}\right] + \left[\frac{n+2}{4}\right] + \left[\frac{n+4}{8}\right] + \left[\frac{n}{8}\right]$

Continuing in this manner, we have

$$\left[\frac{n+1}{2}\right] + \left[\frac{n+2}{4}\right] + \left[\frac{n+4}{8}\right] + \left[\frac{n+8}{16}\right] + \cdots = n$$

49. $-1 \leq \frac{1+x^4}{2x^2} \leq 1$

$\Rightarrow \frac{1+x^4}{2x^2} \leq 1 \qquad \left(\because \frac{1+x^4}{2x^2} \geq -1 \text{ is always true}\right)$

$\Rightarrow (1-x^2)^2 \leq 0 \quad \Rightarrow \quad x = \pm 1$

But $x = -1$ does not define $\sqrt{x(x+2)}$.

50. $\sqrt{\sin(\cos x)}$ is defined, if $0 \leq \cos x \leq 1$

i.e. $2n\pi - \frac{\pi}{2} \leq x \leq 2n\pi + \frac{\pi}{2}$

and $\cos^{-1}(2 + 2x)$ is defined, if $-1 \leq 2 + 2x \leq 1$

$\Rightarrow -\frac{3}{2} \leq x \leq \frac{-1}{2}$

Combining both the cases, we get the domain as $\left[-\frac{3}{2}, \frac{-1}{2}\right]$.

51. $\log_{10} \frac{x-5}{x^2 - 10x + 24}$ is defined, if

$\frac{x-5}{x^2 - 10x + 24} > 0 \Rightarrow \frac{(x-4)(x-5)(x-6)}{(x-4)^2(x-6)^2} > 0$

$\Rightarrow (x-4)(x-5)(x-6) > 0$

$\Rightarrow 4 < x < 5$ or $x > 6$ (i)

$\sqrt[3]{x+5}$ is defined for all real x. (ii)

From (i) and (ii), we get the domain of $f = (4, 5) \cup (6, \infty)$.

52. Period $\frac{2^x}{2^{[x]}}$ is 1. Period of $\sin^{-1}\{x\}$ is 1. Period of $\sin^{-1}(\sqrt{\cos x})$ is 2π, whereas $\sin^{-1}(\cos x^2)$ is non-periodic.

53. $f(x) = \sin^4 x + \cos^4 x + \sin^6 x + \cos^6 x$

$= (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x + (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x \cos^2 x (\sin^2 x + \cos^2 x)$

$= 1 - \frac{\sin^2 2x}{2} + 1 - \frac{3\sin^2 2x}{4}$

$= 2 - \frac{5}{4}\sin^2 2x$

Since $0 \le \sin^2 2x \le 1$,

$\therefore \quad \frac{3}{4} \le f(x) \le 2$.

54. Given expression

$= \left(\sin x \cos\frac{\pi}{6} + \cos x \sin\frac{\pi}{6}\right) + \left(\cos x \cos\frac{\pi}{6} + \sin x \sin\frac{\pi}{6}\right)$

$= \sin x\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right) + \cos x\left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)$

$= \frac{\sqrt{3}+1}{2}(\sin x + \cos x)$

Since $-\sqrt{a^2+b^2} \le a\sin x + b\cos x \le \sqrt{a^2+b^2}$

and $-\sqrt{2} \le \sin x + \cos x \le \sqrt{2}$

Hence, y lies between $\frac{\sqrt{3}+1}{-\sqrt{2}}$ and $\frac{\sqrt{3}+1}{\sqrt{2}}$.

55. $y = \frac{x-1}{x+3} \quad \Rightarrow \quad xy + 3y = x - 1$

$\Rightarrow \quad x = \frac{-3y-1}{y-1}$

$\therefore$ Range $= \mathbb{R} - \{1\}$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $f(x) = \sin^{-1}\left(\frac{x^2}{2}\right)$ is defined if $-1 \le \frac{x^2}{2} \le 1$ or $-2 \ge x^2 \ge 2$

{as x^2 cannot be negative}

$\Rightarrow \quad -\sqrt{2} \le x \le \sqrt{2}$

Therefore, the domain of $f(x) \in [-\sqrt{2}, \sqrt{2}]$.

2. $f(x)$ is defiend if $\log_3 \log_4 x > 0, \log_4 x > 0$ and $x > 0$

$\Rightarrow \quad \log_4 x > 3^0 = 1, x > 4^0$ and $x > 0$

$\Rightarrow \quad x > 4^1, x > 1$ and $x > 0 \Rightarrow x > 4$

$\therefore$ Domain of $f = (4, \infty)$

3. $f(x) = f(-x)$. So, f is many-one. Also

$f(x) = 1 - \frac{5}{x^2+1} > 1 - 5 = -4$. So, f is into.

4. $6^x > 0, 6^{|x|} > 0$ for all $x \in \mathbb{R}$. So f is into. For different x, 6^x and $6^{|x|}$ are different positive numbers. Clearly, f is one-one.

5. Here $f: Z \to Z$ should be a bijective function. Clearly, $f(x)$ must be $x + 2$ as for this function each image has its pre-image and each image has one and only one pre-image.

6. Function $f(x) = (x^2 - 6x + 7)$

Let $f(x) = y = x^2 - 6x + 9 - 2 = (x-3)^2 - 2$

or $y + 2 = (x-3)^2$ or $x = \sqrt{y+2} + 3$

We know that for range $y + 2 \ge 0$ or $y \ge -2$

Therefore, the range of the function is $[-2, \infty)$.

7. Function $f(x) = \frac{x^2+x+2}{x^2+x+1}$

Let $y = f(x)$ or $y = \frac{x^2+x+2}{x^2+x+1}$

or $x^2y + xy + y = x^2 + x + 2$

or $(y-1)x^2 + (y-1)x + (y-2) = 0$ (i)

We know that this equation is a quadratic equation if $y \ne 1$ and x is real $b^2 - 4ac \ge 0$

or $(y-1)^2 - 4(y-1)(y-2) \ge 0$

or $(y-1)[(y-1) - 4(y-2)] \ge 0$

or $(y-1)(-3y+7) \ge 0$ or $(y-1)$ and $(-3y+7) \ge 0$

or $y \ge 1$ and $3y \le 7$ or $y \ge 1$ and $y \le \frac{7}{3}$

Since $y \ne 1$, therefore the range of $f(x) = \left(1, \frac{7}{3}\right]$.

8. $\log_{1/2}\left(x - \frac{1}{2}\right)$ is defined if $x - \frac{1}{2} > 0$, i.e. $x > \frac{1}{2}$ and $\log_2 \sqrt{4x^2 - 4x + 5}$ is defined if $4x^2 - 4x + 5 > 0$

$\Rightarrow \quad 4\left[\left(x - \frac{1}{2}\right)^2 + 1\right] > 0$ which is true for all real x

$\therefore$ Domain of $f = \left(\frac{1}{2}, \infty\right)$.

9. Domain of $\sin^{-1} x$ is $[-1, 1]$.

$\therefore \quad -1 \le \log_5\left(\frac{x^2}{5}\right) \le 1 \quad \Rightarrow \quad 5^{-1} \le \frac{x^2}{5} \le 5^1$

$\Rightarrow \quad 1 \le x^2 \le 25 \quad \Rightarrow \quad 1 \le |x| \le 5$

$\Rightarrow \quad x \in [-5, -1] \cup [1, 5]$

10. $f(x)$ is defined if $\log\left(\frac{1}{|\sin x|}\right) \geq 0$

i.e. if $\frac{1}{|\sin x|} \geq 1$ and $|\sin x| \neq 0$

i.e. if $|\sin x| \neq 0$ $\left(\because \frac{1}{|\sin x|} \geq 1, \text{ for all } x\right)$

i.e. if $x \neq n\pi, n \in Z$

$\therefore$ $D_f = \mathbb{R} - \{n\pi | n \in Z\}$

11. $y = \frac{1}{2 - \sin 3x}$

Since $-1 \leq \sin 3x \leq 1$

$\Rightarrow$ $1 \geq -\sin 3x \geq -1$

$\Rightarrow$ $3 \geq 2 - \sin 3x \geq 1$

$\Rightarrow$ $\frac{1}{3} \leq \frac{1}{2 - \sin 3x} \leq 1$

$\Rightarrow$ $\frac{1}{3} \leq y \leq 1$ $\Rightarrow$ Range is $\left[\frac{1}{3}, 1\right]$.

12. Statement II is false because $(f \circ g)(x) = f[g(x)]$ and $(g \circ f)(x) = g[f(x)]$.

We have $f(x) = \sin x$ and $g(x) = x^2$

$\therefore$ $(f \circ g)(x) = f[g(x)] = f(x^2) = \sin x^2$ and

$(g \circ f)(x) = g[f(x)] = g(\sin x) = \sin^2 x$

$\therefore$ $(f \circ g)(x) \neq (g \circ f)(x)$

$\therefore$ Statement I is true.

13. $x \in \mathbb{R} \Rightarrow -1 \leq \sin x \leq 1$

$\Rightarrow$ $-\frac{\pi}{2} \leq \sin x \leq \frac{\pi}{2}$

$\Rightarrow$ $\cos(\sin x) \geq 0$

$\Rightarrow$ $f(x)$ is defined

$\therefore$ $D(f) = \mathbb{R}$

$x \in \mathbb{R} \Rightarrow -1 \leq \cos x \leq 1$

$\Rightarrow$ $-\frac{\pi}{2} \leq \cos x \leq \frac{\pi}{2}$

$\Rightarrow$ $-1 \leq \sin(\cos x) \leq 1$

$\Rightarrow$ $g(x)$ may not be defined

$\therefore$ $D(g) \neq \mathbb{R}$

$\therefore$ Statement I is false.

Statement II is trivially true.

14. Let $y = f(x) = x^2 - x + 1$

$\therefore$ $x^2 - x + (1 - y) = 0$

$\therefore$ $x = \frac{1 \pm \sqrt{1 - 4(1-y)}}{2} = \frac{1 \pm \sqrt{4y - 3}}{2}$

$= \frac{1}{2} \pm \sqrt{y - \frac{3}{4}} = \frac{1}{2} + \sqrt{y - \frac{3}{4}}$ $\left(\because x \geq \frac{1}{2}\right)$

$\therefore$ $f^{-1}(x) = \frac{1}{2} + \sqrt{x - \frac{3}{4}}$

$\therefore$ $g(x) = f^{-1}(x)$

$\therefore$ $f(x)$ and $g(x)$ are mutually inverse.

$\therefore$ Statement II is true.

Since $f(x)$ and $g(x)$ are mutually inverse, they will intersect along the line $y = x$ and at exactly one point.

$f(x) = x \Rightarrow x^2 - x + 1 = \infty \Rightarrow (x-1)^2 = 0 \Rightarrow x = 1$

$\therefore$ Statement I is false.

15. We have

$$f(x+y) + f(x-y) = 2f(x)f(y) \quad \text{(i)}$$

Replacing x by y and y by x, we get

$$f(y+x) + f(y-x) = 2f(y)f(x) \quad \text{(ii)}$$

$\Rightarrow$ $f(x+y) + f(x-y) = f(y+x) + f(y-x)$

$\Rightarrow$ $f(x-y) = f(y-x)$

Putting $y = 2x$, we get $f(x-2x) = f(2x-x)$

$\Rightarrow$ $f(-x) = f(x)$

$\therefore$ $f(x)$ is an even function.

Statement I is true.

Statement II is trivially true and it is not a correct explanation for Statement I.

16. $D(\sin^{-1} x) = [-1, 1]$, $D(\cos^{-1} x) = [-1, 1]$ and $D(\tan^{-1} x) = \mathbb{R}$

$\therefore$ $D(\sin^{-1} x + \cos^{-1} x + \tan^{-1} x)$

$= [-1, 1] \cap [-1, 1] \cap \mathbb{R} = [-1, 1]$

$\therefore$ Statement I is true.

$\therefore$ Statement II is trivially true and it is not a correct explanation for Statement I.

17. $f(x) = x - [x]$ is a periodic function with period I.

$\therefore$ Statement II is true.

We have $f(7.4) = 7.4 - [7.4] = 7.4 - 7 = 0.4$

and $f(8.4) = 8.4 - [8.4] = 8.4 - 8 = 0.4$

$\therefore$ $f(x)$ is not one-one and hence not invertible

$\therefore$ Statetment I is true.

Also, Statement II is a correct explanation for Statement I.

18. Since $f(x) = \sqrt{\log_{10}\left(\frac{5x - x^2}{4}\right)}$

$\therefore$ $f(x)$ exists only for

$$\log_{10}\left(\frac{5x - x^2}{4}\right) \geq 0$$

$$\Rightarrow \quad \frac{5x-x^2}{4} \geq 1$$

$$\Rightarrow \quad x^2-5x+4 \leq 0$$

$$\Rightarrow \quad (x-1)(x-4) \leq 0$$

$\Rightarrow \quad x \in [1, 4]$

19. Since $\cos\sqrt{x}$ is not periodic,

$\therefore$ $\cos\sqrt{x}+\cos^2 x$ is not periodic although $\cos^2 x$ is periodic.

20. Here $f(x)=\sin^{-1}\left[\log_3\left(\frac{x}{3}\right)\right]$

Here $x > 0$ and $-1 \leq \log_3\left(\frac{x}{3}\right) \leq 1$

$$\Rightarrow \quad 3^{-1} \leq \frac{x}{3} \leq 3^1 \quad \Rightarrow \quad \frac{1}{3} \leq \frac{x}{3} \leq 3$$

$\Rightarrow \quad 1 \leq x \leq 9$

Hence $D_f = [1, 9]$

21. Periodic functions may not be bounded, e.g. $\tan x$, $\cot x$.

22. $\sum_{r=1}^{n} f(r) = f(1)+f(2)+f(3)+\cdots+f(n)$

$$= f(1)+2f(1)+3f(1)+\cdots+nf(1)$$

[since $f(x+y)=f(x)+f(y)$]

$$= (1+2+3+\cdots+n)\, f(1) = f(1)\Sigma n$$

$$= \frac{7n(n+1)}{2} \qquad [\because f(1) = 7 \text{ (given)}]$$

23. When n is even

Let $f(2m_1)=f(2m_2) \Rightarrow -\frac{2m_1}{2}=-\frac{2m_2}{2} \Rightarrow m_1=m_2$

When n is odd

Let $f(2m_1+1)=f(2m_2+1)$

$$\Rightarrow \quad \frac{2m_1+1-1}{2}=\frac{2m_2+1-1}{2} \Rightarrow m_1=m_2$$

$\therefore$ $f(x)$ is one-one

Since $f: I \to I$ for all $m \in I$, m is an even number, there exists $n \in N$ such that $= n = 2m + 1$

Hence, f is both one-one and onto.

24. $f(x)=\frac{3}{4-x^2}+\log_{10}\left(x^3-x\right)$

f is defined when $x \neq \pm 2$ and $x^3-x>0$

$\Rightarrow \quad x \neq \pm 2$ and $(x^2-1)>0$

$\Rightarrow \quad x \neq \pm 2, x \in (-1,0)\cup(1,\infty)$

$\Rightarrow \quad x \in (-1,0)\cup(1,2)\cup(2,\infty)$

25. We have $f(x)=\frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}=\frac{g(x)}{h(x)}$ (say)

$\sin^{-1}(x-3)$ is defined if $-1 \leq x-3 \leq 1 \Rightarrow D_g = [2, 4]$

For $h(x), 9-x^2>0 \Rightarrow x^2<9 \Rightarrow D_h = (-3,3)$

Hence,

$$D_f = D_g \cap D_h, h(x) \neq 0 = [2,4] \cap (-3,3) = [2,3)$$

26. We have $f(x)=\sin x-\sqrt{3}\cos x+1$

$$= 2\left(\sin x \cdot \frac{1}{2}-\cos x \cdot \frac{\sqrt{3}}{2}\right)+1$$

$$= 2\left(\sin x\cos\frac{\pi}{2}-\cos x.\sin\frac{\pi}{3}\right)+1 = 2\sin\left(x-\frac{\pi}{3}\right)+1$$

Clearly f is onto, when the interval of S is [–1, 3]

[$\because$ Max. value = 2(1) + 1 = 3 and Min. value = 2(–1) + 1 = –1]

27. We have $f(x) = {}^{7-x}P_{x-3} = \frac{(7-x)!}{(10-2x)!}$

We must have $7-x>0, x \geq 3$ and $7-x \geq x-3$

$\Rightarrow \quad x<7, x \geq 3$ and $x \leq 5$

$\Rightarrow \quad 3 \leq x \leq 5$

$\Rightarrow \quad x = 3, 4, 5$

Now, $f(3)=\frac{4!}{4!}=1, f(4)=\frac{3!}{2!}=3, f(5)=\frac{2!}{0!}=2$

Hence, $R_f = \{1, 2, 3\}$

28. Graph is symmetrical about the line $x = 2$

$\Rightarrow \quad f(2-x)=f(2+x)$

[$\because$ Graph is symmetrical about x = 0, i.e. y-axis if $f(-x)=f(x)$ if $f(0-x)=f(0+x)$]

29. For $-1<x<1$, $\tan^{-1}\frac{2x}{1-x^2}=2\tan^{-1}x$

$\therefore$ Range of $f(x)=\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$.

30. We have $f(x-y)=f(x)f(y)-f(a-x)f(a+y)$

Putting $x = a$ and $y = a - x$, we get

$$f(a-(x-a))=f(a)f(x-a)-f(0)f(x) \qquad \text{(i)}$$

Putting $x = 0$, $y = 0$, we get

$$f(0)=f(0)f(0)-f(a)f(a)$$

$$\Rightarrow \quad f(0)=[f(0)]^2-[f(a)]^2$$

$$\Rightarrow \quad 1=(1)^2-[f(a)]^2$$

$$\Rightarrow \quad f(a)=0$$

$$\Rightarrow \quad f(2a-x)=-f(x)$$

31. $f(x)=4^{-x^2}+\cos^{-1}\left(\frac{x}{2}-1\right)+\log(\cos x)$

Here, 4^{-x^2} is defined for $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, $\cos^{-1}\left(\frac{x}{2}-1\right)$ is defined if $-1\le\frac{x}{2}-1\le 1\Rightarrow 0\le x\le 4$ and $\log(\cos x)$ is defined if $\cos x>0$.

$\Rightarrow\quad -\frac{\pi}{2}<x<\frac{\pi}{2}$

Hence, $f(x)$ is defined for $x\in\left[0,\frac{\pi}{2}\right)$.

32. Function is increasing $\Rightarrow$ Function is one-one $\Rightarrow$ Function is invertible

$$x=\frac{y-3}{4}=g(y)$$

33. $f(x)=x^3+5x+1$

Now, $f'(x)=3x^2+5>0$, for all $x\in\mathbb{R}$

$\therefore$ $f(x)$ is strictly an increasing function.

$\therefore$ $f(x)$ is a one-one function.

Clearly, $f(x)$ is a continous function and also increasing on $\mathbb{R}$.

$\therefore\quad \lim\limits_{x\to-\infty} f(x)=-\infty$ and $\lim\limits_{x\to\infty}=\infty$

$f(x)$ takes every value between $-\infty$ and ∞.

Thus, $f(x)$ is an onto function.

34. $f(x)=(x+1)^2-1, x\ge -1$

$\Rightarrow\quad f'(x)=2(x+1)\ge 0$ for $x\ge -1$

$\Rightarrow$ $f(x)$ is one-one.

Since co-domain of the given function is not given, hence it can be considered as $\mathbb{R}$, the set of real numbers and consequently $\mathbb{R}$ is not onto.

Hence, f is not bijective. Statement II is false.

Also, $f(x)=(x+1)^2-1\ge -1$ for $x\ge -1$

$\Rightarrow\quad R_f=[-1,\infty)$

$$f^{-1}(x)=\sqrt{x+1}-1$$

Clearly, $f(x)=f^{-1}(x)$ at $x=0$ and $x=-1$

$\therefore$ Statement I is true.

35. It is given that $2^x+2^y=2$ for all $x, y\in\mathbb{R}$

$\Rightarrow\quad 2^y=2-2^x$

$\Rightarrow\quad y=\log_2(2-2^x)$

We must have $2-2^x>0$

$\Rightarrow\quad 2^x<2$

$\Rightarrow\quad x<1$

36. $f:[1,\infty)\to[2,\infty)$

$$f(x)=x+\frac{1}{x}=y\Rightarrow x^2-yx+1=0$$

$$\Rightarrow\quad x=\frac{y\pm\sqrt{y^2-4}}{2}$$

But for $x>1$, we have $y>2$

$$\Rightarrow\quad f^{-1}(x)=\frac{x+\sqrt{x^2-4}}{2}$$

37. We have $\left[\frac{1}{2}+\frac{x}{100}\right]=\begin{cases}0, & \text{if} \quad 0<x<49\\ 1, & \text{if} \quad 50\le x\le 99\end{cases}$

Thus,

$$\left[\frac{1}{2}\right]+\left[\frac{1}{2}+\frac{1}{100}\right]+\left[\frac{1}{2}+\frac{2}{100}\right]+\cdots+\left[\frac{1}{2}+\frac{99}{100}\right]=50$$

38. Given that $f(x)=(x+1)^2 x\ge -1$

Now if $g(x)$ is the reflection of $f(x)$ in the line $y=x$ then it can be obtained by interchanging x and y in $f(x)$,

i.e. $y=(x+1)^2$ changes to $x=(y+1)^2$

$\Rightarrow\quad y+1=\sqrt{x}$

$\Rightarrow\quad y=\sqrt{x}-1$ is defined for all $x\ge 0$

$\therefore\quad g(x)=\sqrt{x}-1$, for all $x\ge 0$

39. For $f(x)=\sqrt{\sin^{-1}(2x)+\frac{\pi}{6}}$ to be defined and real,

$\sin^{-1}2x+\pi/6\ge 0$

$$\Rightarrow\quad \sin^{-1}2x\ge -\frac{\pi}{6} \qquad \text{(i)}$$

But we know that $-\pi/2\le\sin^{-1}2x\le\pi/2$ (ii)

Combining (i) and (ii), we get $-\pi/6\le\sin^{-1}2x\le\pi/2$

$\Rightarrow\quad -1/2\le 2x\le 1\quad\Rightarrow\quad D_f=\left[-\frac{1}{4},\frac{1}{2}\right]$

40. We have

$$f(x)=x^2+2bx+2c^2; g(x)=-x^2-2cx+b^2$$

$\Rightarrow\quad f(x)=(x+b)^2+2c^2-b^2$

and $\quad g(x)=-(x+c)^2+b^2+c^2$

$\Rightarrow\quad f_{min}=2c^2-b^2$ and $g_{\max}=b^2+c^2$

for $f_{\min}>g_{\max}, 2c^2-b^2>b^2+c^2$

$\Rightarrow\quad c^2>2b^2$

$\Rightarrow\quad |c|>|b|\sqrt{2}$

41. $f(x)=\sin x+\cos x, g(x)=x^2-1$

$\Rightarrow\quad g[f(x)]=(\sin x+\cos x)^2-1=\sin 2x$

Clearly, $g[f(x)]$ is invertible in $-\frac{\pi}{2}\le 2x\le\frac{\pi}{2}$

[$\because$ $\sin\theta$ is invertible when $-\pi/2\le\theta\le\pi/2$]

$\Rightarrow\quad -\frac{\pi}{4}\le x\le\frac{\pi}{4}$

42. $(f-g):\mathbb{R}\to\mathbb{R}$ such that

$$(f-g)(x)=\begin{cases}-x & \text{if } x\in \text{rational}\\ x & \text{if } x\in \text{irrational}\end{cases}$$

Since $f\cdot g:\mathbb{R}\to\mathbb{R}$ for any x, there is only one value of $[f(x)-g(x)]$ whether x is rational or irrational. Moreover, as $x\in\mathbb{R}, f(x)-g(x)$ also belongs to $\mathbb{R}$. Therefore, $(f-g)$ is one-one onto.

43. Since $x\in[-2,2], x\le 0$ and $f(|x|)=x$

For $-2\le x\le 0$

$$f(-x)=x\Rightarrow(-x)-1=x\Rightarrow x-\frac{1}{2}$$

44. Here, $f(x)=\log\dfrac{10+x}{10-x}$

Given that $f(x)=kf\left(\dfrac{200x}{100+x^2}\right)$

$$\Rightarrow \log\frac{10+x}{10-x}=k\cdot\log\left\{\frac{10+\frac{200x}{100+x^2}}{10-\frac{200x}{100+x^2}}\right\}$$

$$=k\log\left(\frac{10+x}{10-x}\right)^2$$

$$\Rightarrow \log\frac{10+x}{10-x}=2k\cdot\log\frac{10+x}{10-x}$$

$$\Rightarrow k=0.5$$

45. Since $f:\mathbb{R}\to\mathbb{R}$ such that $f(x)=3^{-x}$

Let y_1 and y_2 be two elements of $f(x)$ such that $y_1=y_2$

$\Rightarrow 3^{-x_1}=3^{-x_2}\Rightarrow x_1=x_2$

Since if two images are equal, then their elements are equal, therefore it is a one-one function.

Since $f(x)$ is positive for every value of x, therefore $f(x)$ is into.

On differentiating w.r.t. x, we get $\dfrac{dy}{dx}=-3^{-x}\log 3<0$ for every value of x.

$\therefore$ It is a decreasing function.

$\therefore$ Statements I and III are true.

46. $f(x)=\dfrac{ax+b}{cx+d}$

It reduces to a constant function if $\dfrac{a}{c}=\dfrac{b}{d}\Rightarrow ad=bc$

47. $f(x)=\dfrac{1}{2}$

$$\Rightarrow x-[x]-\frac{1}{2}=\frac{1}{2}$$

$$\Rightarrow x-[x]=1$$

$\Rightarrow \{x\}=1$, where $\{x\}$ is the fractional part of x. This is not possible since $0\le\{x\}<1$ for not all $x\in\mathbb{R}$.

Hence $\left\{x\in\mathbb{R}: f(x)=\dfrac{1}{2}\right\}=\phi$

48. We have

$$-1\le\cos 3x\le 1$$

$$\Rightarrow -1\le-\cos 3x\le 1$$

$$\Rightarrow 1\le 2-\cos 3x\le 3$$

$$\Rightarrow \frac{1}{3}\le\frac{1}{2-\cos 3x}\le 1$$

$\therefore$ Range $(f)=\left[\dfrac{1}{3},1\right]$

49. $f(x)=x^2-3$

Now, $f(1)=(1)^2-3=-2$

$\Rightarrow f\circ f(-1)=f(-2)=(-2)^3-3=1$

$\Rightarrow f\circ f\circ f(-1)=f(1)=1^2-3=-2$

Now, $f(0)=0^2-3=-3$

$\Rightarrow f\circ f(0)=f(-3)=(-3)^2-3=6$

$\Rightarrow f\circ f\circ f(0)=f(6)=6^2-3=33$

Again, $f(1)=1^2-3=-2$

$\Rightarrow f\circ f(1)=f(-2)=(-2)^2-3=1$

$\Rightarrow f\circ f\circ f(1)=(1)^2-3=-2$

$\therefore f\circ f\circ f(-1)+f\circ f\circ f(0)+f\circ f\circ f(1)=-2+33-2=29$

Now, $f(4\sqrt{2})=(4\sqrt{2})^2-3=32-3=29$

50. $f(x)=x^3+3x-2$

On differentiating w.r.t. x, we get

$$f'(x)=3x^2+3$$

Put $f'(x)=0\Rightarrow 3x^2+3=0$

$$\Rightarrow x^2=-1$$

$\therefore$ $f(x)$ is either increasing or decreasing.

At $x=2, f(2)=2^3+3(2)-2=12$

At $x=3, f(3)=3^3+3(3)-2=34$

$\therefore$ $f(x)\in[12,34]$

51. $f(x)=\log|\log x|$ is defined when $|\log x|>0$ and $x>0$

i.e. when $x\ne 1$ and $0<x<\infty$

i.e. when $x\in(0,1)\cup(1,\infty)$

$\therefore$ Dom $(f)=(0,1)\cup(1,\infty)$

52. $g(x)=1+x-[x]$ is greater than 1 since $x-[x]>0$ $f[g(x)]=1$.

53. Let $y=\dfrac{x}{1+x^2}$

$\Rightarrow \quad x^2y-x+y=0$

For x to be real,

$1-4y^2\geq 0$

$\Rightarrow \quad (1-2y)(1+2y)\geq 0$

$\Rightarrow \quad \left(\dfrac{1}{2}-y\right)\left(\dfrac{1}{2}+y\right)\geq 0$

$\Rightarrow \quad -\dfrac{1}{2}\leq y\leq \dfrac{1}{2}$

$\therefore \quad y=f(x)\in\left[-\dfrac{1}{2},\dfrac{1}{2}\right]$

54. $f\left(\dfrac{x_1+x_2}{1+x_1x_2}\right)=\log\left\{\dfrac{1+\dfrac{x_1+x_2}{1+x_1x_2}}{1-\dfrac{x_1+x_2}{1+x_1x_2}}\right\}$

$=\log\left\{\dfrac{1+x_1x_2+x_1+x_2}{1+x_1x_2-x_1-x_2}\right\}$

$=\log\left\{\dfrac{(1+x_1)(1+x_2)}{(1-x_1)(1-x_2)}\right\}=\log\left\{\left(\dfrac{1+x_1}{1-x_1}\right)\cdot\left(\dfrac{1+x_2}{1-x_2}\right)\right\}$

$=\log\left(\dfrac{1+x_1}{1-x_1}\right)+\log\left(\dfrac{1+x_2}{1-x_2}\right)=f(x_1)+f(x_2)$

55. $\dfrac{f(a)}{f(a+1)}=\dfrac{\left\{\dfrac{a}{a-1}\right\}}{\left\{\dfrac{a+1}{(a+1)-1}\right\}}$

$=\dfrac{a}{a-1}\cdot\dfrac{a}{a+1}=\dfrac{a^2}{a^2-1}=f(a^2)$

56. For $f(x)=\cos x$, we have

$\dfrac{f(m)-f(n)}{m-n}=\dfrac{\cos m-\cos n}{m-n}$, which is not constant.

For $f(x)=\log x$, we have

$\dfrac{f(m)-f(n)}{m-n}=\dfrac{\log m-\log n}{m-n}=\dfrac{\log\left(\dfrac{m}{n}\right)}{m-n}$, which is not constant.

For $f(x)=x^2+1$, we have

$\dfrac{f(m)-f(n)}{m-n}=\dfrac{(m^2+1)-(n^2+1)}{m-n}=\dfrac{m^2-n^2}{m+n}=m+n,$

which is not constant.

For $f(x)=4x+7$, we have;

$\dfrac{f(m)-f(n)}{m-n}=\dfrac{(4m+7)-(4n+7)}{m-n}=\dfrac{4(m-n)}{m-n}=4$

which is constant.

57. $f(x)=\dfrac{4^x}{4^x+2}$

$\Rightarrow \quad f(1-x)=\dfrac{4^{1-x}}{4^{1-x}+2}=\dfrac{\dfrac{4}{4^x}}{\dfrac{4}{4^x}+2}=\dfrac{4}{4+2\cdot 4^x}$

$=\dfrac{2}{2+4^x}=\dfrac{2}{4^x+2}$

$\therefore \quad f(x)+f(1-x)=\dfrac{4^x}{4^x+2}+\dfrac{2}{4^x+2}=\dfrac{4^x+2}{4^x+2}=1$

Now, $f\left(\dfrac{1}{1997}\right)+f\left(\dfrac{2}{1997}\right)+\cdots+f\left(\dfrac{1996}{1997}\right)$

$=\left[f\left(\dfrac{1}{1997}\right)+f\left(\dfrac{1996}{1997}\right)\right]+\left[f\left(\dfrac{2}{1997}\right)+f\left(\dfrac{1995}{1997}\right)\right]+$

$+\left[f\left(\dfrac{998}{1997}\right)+f\left(\dfrac{999}{1997}\right)\right]$

$\left[\because \text{Total}=\dfrac{1996}{2}=998 \text{ pairs will be formed}\right]$

$=\left[f\left(\dfrac{1}{1997}\right)+f\left(1-\dfrac{1}{1997}\right)\right]$

$+\left[f\left(\dfrac{2}{1997}\right)+f\left(1-\dfrac{2}{1997}\right)\right]+\cdots$

$+\left[f\left(\dfrac{998}{1997}\right)+f\left(1-\dfrac{998}{1997}\right)\right]$

= 1 + 1 + … 998 terms = 998.

58. $f\left(\dfrac{2x}{1+x^2}\right)=\log\left\{\dfrac{1+\dfrac{2x}{1+x^2}}{1-\dfrac{2x}{1+x^2}}\right\}$

$=\log\left\{\dfrac{1+x^2+2x}{1+x^2-2x}\right\}=\log\left\{\dfrac{(1+x)^2}{(1-x)^2}\right\}$

$=2\log\left\{\dfrac{1+x}{1-x}\right\}=2f(x)$

59. $f[f(\cos 2\theta)]=f\left(\dfrac{1-\cos 2\theta}{1+\cos 2\theta}\right)=f\left(\dfrac{2\sin^2\theta}{2\cos^2\theta}\right)$

$=f(\tan^2\theta)$

$=\dfrac{1-\tan^2\theta}{1+\tan^2\theta}=\cos 2\theta$

60. $(f\circ f\circ f\circ f)(x) = f\{f\{f\{f(c)\}\}\}$

$$= f\left\{f\left\{f\left(\frac{x+1}{x-1}\right)\right\}\right\}$$

$$= f\left\{f\left(\frac{\frac{x+1}{x-1}+1}{\frac{x+1}{x-1}-1}\right)\right\}$$

$$= f\left\{f\left(\frac{2x}{2}\right)\right\} = f\{f(x)\} = f\left(\frac{x+1}{x-1}\right)$$

$$\left[\frac{\frac{x+1}{x-1}+1}{\frac{x+1}{x-1}-1}\right] = \frac{2x}{2} = x$$

61. $\lim\limits_{x\to a}\dfrac{x^2f(a)-a^2f(x)}{x-a}$

$$= \lim_{x\to a}\frac{x^2f(a)-a^2f(a)+a^2f(a)-a^2f(x)}{x-a}$$

$$= \lim_{x\to a}\frac{(x^2-a^2)f(a)-a^2(f(x)-f(a))}{x-a}$$

$$= \lim_{x\to a}(x+a)f(a)-a^2\left[\frac{f(x)-f(a)}{(x-a)}\right]$$

$$= 2af(a)-a^2f'(a).$$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The domain of the function $f(x)=\dfrac{\sqrt{-\log_{o.3}(x-1)}}{\sqrt{x^2+2x+8}}$ is

(a) (1, 4) (b) (–2, 4)
(c) (2, 4) (d) [2, ∞)

2. If $f(x+2y, x-2y) = xy$, then $f(x, y)$ equals

(a) $\dfrac{x^2-y^2}{8}$ (b) $\dfrac{x^2-y^2}{4}$
(c) $\dfrac{x^2+y^2}{4}$ (d) $\dfrac{x^2-y^2}{2}$

3. Range of $f(x) = \log(3x^2-4x+5)$ is

(a) $\left[\log\frac{11}{3}, \infty\right)$ (b) $[\log 10, \infty)$
(c) $\left[\log\frac{11}{6}, \infty\right)$ (d) $\left[\log\frac{11}{12}, \infty\right)$

4. The domain of the function
$f(x) = \log_{10}[1-\log_{10}(x^2-5x+16)]$ is

(a) (2, 3) (b) [2, 3]
(c) (2, 3] (d) [2, 3)

5. The range of $f(x) = 4^x + 2^x + 1$ is

(a) (0, ∞) (b) (1, ∞)
(c) (2, ∞) (d) (3, ∞)

6. If $f(x) = \cos^{-1}\left(\dfrac{2-|x|}{4}\right) + [\log(3-x)]^{-1}$, then its domain is

(a) [–2, 6] (b) [–6, 2) ∪ (2, 3)]
(c) [–6, 2] (d) [–2, 2) ∪ (2, 3]

7. If x and y satisfy the equation $y = 2[x] + 3$ and $y = 3[x-2]$ simultaneously, the $[x + y]$ is

(a) 21 (b) 9
(c) 30 (d) 12

8. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = (x-1)(x-2)(x-3)$ is

(a) one-one but not onto (b) onto but not one-one
(c) both one-one and onto (d) neither one-one nor onto

9. The function $f(x) = \cot^{-1}\sqrt{(x+3)x} + \cos^{-1}\sqrt{x^2+3x+1}$ is defined on the set S, where S is equal to

(a) {0, 3} (b) (0, 3)
(c) {0, –3} (d) [–3, 0]

10. If $f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$, where $[x]$ stands for the greatest integer function, then

(a) $f\left(\dfrac{\pi}{2}\right) = 1$ (b) $f(\pi) = 1$
(c) $f(-\pi) = 0$ (d) none of these

11. If $f(1) = 1$ and $f(n+1) = 2f(n)+1$ if $n \geq 1$, then $f(n)$ is equal to

(a) $2^n + 1$ (b) 2^n
(c) $2^n - 1$ (d) $2^{n-1} - 1$

12. The domain of the function $f(x) = \sqrt{\dfrac{1-|x|}{|x|-2}}$ is

(a) $(-\infty, -1) \cup (1, \infty)$ (b) $(-\infty, -2) \cup (2, \infty)$
(c) $(-2, -1] \cup [1, 2)$ (d) none of these

13. If $f(x)=(ax^2+b)^3$, then $f(x)f(y)-\frac{1}{2}\left[f\left(\frac{x}{y}\right)+f(xy)\right]$ has the value

(a) -1 (b) $1/2$
(c) -2 (d) none of these

14. If $f(x)=(ax^2+b)^3$, then the function g such that $f[g(x)]=g[f(x)]$ is

(a) $g(x)=\left(\frac{b-x^{1/3}}{a}\right)^{1/2}$ (b) $g(x)=\frac{1}{(ax^2+b)^3}$

(c) $g(x)=(ax^2+b)^{1/3}$ (d) $g(x)=\left(\frac{x^{1/3}-b}{a}\right)^{1/2}$

15. The domain of the function $f(x)=\log_x \cos x$ is

(a) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)-\{1\}$ (b) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]-\{1\}$
(c) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ (d) none of these

16. If $f(x)=\frac{x}{\sqrt{1+x^2}}$, then $(f\circ f\circ f)(x)$ is

(a) $\frac{3x}{\sqrt{1+x^2}}$ (b) $\frac{x}{\sqrt{1+3x^2}}$
(c) $\frac{3x}{\sqrt{1-x^2}}$ (d) none of these

17. If $f(x)=\sin^2 x+\sin^2(x+\pi/3)+\cos x\cos(x+\pi/3)$ and $g(5/4)=1$, then $(g\circ f)(x)$ is equal to

(a) 1 (b) 0
(c) $\sin x$ (d) none of these

18. If $3f(x)-f\left(\frac{f}{x}\right)=\log x^4$, then $\log x^4 f(e^{-x})$ is

(a) $1+x$ (b) $1/x$
(c) x (d) $-x$

19. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=\frac{2x^2-x+5}{7x^2+2x+10}$, then f is

(a) one-one bot not onto
(b) onto but not one-one
(c) onto as well as one-one
(d) neither onto nor one-one

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. The total number of solutions of $\big||x|-5\big|=2$ is

(a) 0 (b) 1
(c) 2 (d) 4

2. The function $f(x)=\log\left(\frac{1+\sin x}{1-\sin x}\right)$ is

(a) even (b) odd
(c) neither even nor odd (d) both even and odd

3. The range of the function $f(x)=\frac{1}{x^2-2x+2}$ is

(a) (0, 1] (b) (0, 1)
(c) [0, 1) (d) [0, 1]

4. If $f(x)=\sin\sqrt{[a]}x$ (where [.] denotes the greatest integer function) has π as its fundamental period, then

(a) $a=1$ (b) $a=9$
(c) $a\in[1,2)$ (d) $a\in[4,5)$

5. The complete solution set of the inequality $\frac{1}{|x-1|(x-2)^2}\geq 0$ is

(a) $\mathbb{R}$ (b) $\mathbb{R}-\{1\}$
(c) $\mathbb{R}-\{1,2\}$ (d) none of these

6. The inverse of the function $y=\frac{e^x-e^{-x}}{e^x+e^{-x}}$ is

(a) $\frac{1}{2}\log\frac{1+x}{1-x}$ (b) $\frac{1}{2}\log\frac{2+x}{2-x}$
(c) $\frac{1}{2}\log\frac{1-x}{1+x}$ (d) $2\log(1+x)$

7. The period of $f(x)=\tan^{-1}(\tan x)+\tan 2x$ is

(a) $\pi/2$ (b) 2π
(c) π (d) none of these

8. If $f(0,\infty)\to\mathbb{R}$, where $f(x)=\log x$, then $f^{-1}(x)$ is

(a) e^x (b) e^{-x}
(c) x (d) none of these

9. Given the function $f(x)=\frac{a^x+a^{-x}}{2}(a>0)$. If $f(x+y)+f(x-y)=kf(x)\cdot f(y)$, then k has the value equal to

(a) 1 (b) 2 (c) 4 (d) 1/2

10. If $f(x)=\cos\left[\frac{\pi^2}{2}\right]x+\sin\left[-\frac{\pi^2}{2}\right]x$, $[x]$ denoting the greatest integer function, then

(a) $f(0)=1$ (b) $f\left(\frac{\pi}{3}\right)=\frac{1}{\sqrt{3}+1}$
(c) $f(\pi/2)=0$ (d) $f(\pi)=0$

11. The domain of the function

$$f(x)=\log_{1/2}\left[-\log_2\left(1+\frac{1}{\sqrt[4]{x}}\right)-1\right] \text{ is}$$

(a) $0<x<1$ (b) $0<x\le 1$

(c) $x\ge 1$ (d) a null set

12. $f:[2,\infty)\to(-\infty,4]$ where $f(x)=x(4-x)$, then $f^{-1}(x)$ is

(a) $2-\sqrt{4-x}$ (b) $2+\sqrt{4-x}$

(c) $-2+\sqrt{4-x}$ (d) $-2-\sqrt{4-x}$

13. The image of the interval $\mathbb{R}$ when the mapping $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=\cot^{-1}(x^2-4x+3)$ is

(a) $\left[\frac{\pi}{4},\frac{3\pi}{4}\right]$ (b) $\left[\frac{\pi}{4},\pi\right)$

(c) $(0,\pi)$ (d) $\left(0,\frac{3\pi}{4}\right]$

14. If $f(x)+2f(1-x)=x^2+1$ for all $x\in\mathbb{R}$, then $f(x)$ is

(a) $\frac{1}{3}(x^2+4x-3)$ (b) $\frac{2}{3}(x^2+4x-3)$

(c) $\frac{1}{3}(x^2-4x+3)$ (d) $\frac{2}{3}(x^2-4x+3)$

15. If $f(x)=\frac{4a-7}{3}x^3+(a-3)x^2+x+5$ is a one-one function, then

(a) $2\le a\le 8$ (b) $1\le a\le 2$

(c) $0\le a\le 1$ (d) none of these

16. The range of $f(x)=\frac{\tan(\pi[x^2-x])}{1+\sin(\cos x)}$ (where [.] denotes the greatest integer function) is

(a) $(-\infty,\infty)\sim[0,\tan 1]$ (b) $(-\infty,\infty)\sim[\tan 2,0]$

(c) $[\tan 2,\tan 1]$ (d) none of these

17. The function $f(x)=\frac{x}{e^x-1}+\frac{x}{2}+1$ is

(a) an odd function

(b) an even function

(c) neither an odd nor an even function

(d) a periodic function

18. Let $f:N\to\mathbb{R}; f(x)2x-1, g:\mathbb{R}\to\mathbb{R}; g(x)=\frac{x^2}{2}$, then $(g\circ f)(0)$ is

(a) 1 (b) 1/2

(c) −1/2 (d) none of these

19. If the graph of the function $f(x)=\frac{a^x-1}{x^n(a^x+1)}$ is symmetric about the y-axis, then n is equal to

(a) 2 (b) 2/3

(c) 1/4 (d) −1/3

ANSWERS

Level 1

1. (d) **2.** (a) **3.** (a) **4.** (a) **5.** (b) **6.** (b) **7.** (c) **8.** (b) **9.** (c) **10.** (c)
11. (c) **12.** (c) **13.** (d) **14.** (d) **15.** (d) **16.** (b) **17.** (a) **18.** (d) **19.** (d)

Level 2

1. (d) **2.** (b) **3.** (a) **4.** (d) **5.** (c) **6.** (a) **7.** (c) **8.** (a) **9.** (b)
10. (a), (b) and (c) **11.** (d) **12.** (b) **13.** (d) **14.** (c) **15.** (a) **16.** (d) **17.** (b)
18. (d) **19.** (d)

11

ISAAC NEWTON

Limit, Continuity and Differentiability

- Limit
- Indeterminate Forms
- Algebra of Limits
- One-sided Limits
- Right-hand Limit
- Left-hand Limit
- Limit of a Function Derived from One-sided Limits
- Some Important Limits
- L'Hôpital's Rule
- Some Important Expansions
- Logarithmic Limits
- Exponential Limits
- Useful Trigonometric Results
- Continuity at a Point
- Continuity in an Open and Closed Interval
- Properties of Continuous and Discontinuous Functions
- Types of Discontinuity
- Types of Derivatives
- Relation between Continuity and Differentiability
- Properties of Differentiable Functions

☞ MEANING OF $x \to a$

$x \to a$ is read as 'x tends to a' or 'x approaches a', where x is a variable. It can be changed so that its value comes nearer and nearer to a, $0 < |x-a|$, where (i) $x \neq a$ and (ii) $|x-a|$ becomes smaller and smaller as we please.

☞ LIMIT

We say that $\lim_{x \to a} f(x) = l$, if for each $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - l| < \varepsilon$, whenever $|x-a| < \delta$. This means, smaller is the difference between x and a, smaller will be the difference between $f(x)$ and l.

☞ INDETERMINATE FORMS

If a function $f(x)$ takes any of the following forms when $x = a$, then we say that $f(x)$ is indeterminate at $x = a$.

1. $0/0$
2. ∞/∞
3. $\infty - \infty$
4. $0 \times \infty$
5. 1^{∞}
6. 0^0
7. ∞^0

☞ ALGEBRA OF LIMITS

1. $\lim_{x\to a}[f(x)+g(x)]=\lim_{x\to a}f(x)+\lim_{x\to a}g(x)$
2. $\lim_{x\to a}[f(x)-g(x)]=\lim_{x\to a}f(x)-\lim_{x\to a}g(x)$
3. $\lim_{x\to a}[kf(x)]=k\lim_{x\to a}f(x)$
4. $\lim_{x\to a}[f(x)\cdot g(x)]=k\left[\lim_{x\to a}f(x)\right].\left[\lim_{x\to a}g(x)\right]$
5. $\lim_{x\to a}\left[\frac{f(x)}{g(x)}\right]=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}$, when $\lim_{x\to a}g(x)\neq 0$
6. $\lim_{x\to a}[f(x)]^n=\left[\lim_{x\to a}f(x)\right]^n$
7. $\lim_{x\to a}e^{f(x)}=e^{\lim_{x\to a}f(x)}$
8. $\lim_{x\to a}[f(x)]^{g(x)}=e^{\left[\lim_{x\to a}g(x)\log_e f(x)\right]}$
9. $\log\left[\lim_{x\to a}f(x)\right]=\lim_{x\to a}[\log f(x)]$, when $\lim_{x\to a}f(x)>0$

☞ ONE-SIDED LIMITS

This method is applied to find the limit at $x = a$ when the function is defined differently for $x > a$, $x = a$ and $x < a$.

☞ RIGHT-HAND LIMIT

We say that the right-hand limit of $f(x)$ at $x = a$ is A if $f(x) \to A$ when $x \to a$ through values greater than a, and we write

$$\lim_{x\to a^+}f(x)=A \quad \text{or} \quad \lim_{x\to a+0}f(x)=A \quad \text{or} \quad f(a+0)=A$$

☞ WORKING RULE FOR FINDING $\lim_{x\to a^+}f(x)$

Replace x by $(a + h)$ and take the limit as $h \to 0$, i.e. $\lim_{x\to a^+}f(x)=\lim_{h\to 0}f(a+h)$.

☞ LEFT-HAND LIMIT

We say that the left-hand limit of $f(x)$ at $x = a$ is B if $f(x) \to B$ when $x \to a$ through values less than a, and we write

$$\lim_{x\to a^-}f(x)=B \text{ or } \lim_{x\to a-0}f(x)=B \text{ or } f(a-0)=B$$

☞ WORKING RULE FOR FINDING $\lim_{x\to a^-}f(x)$

Replace x by $(a - h)$ and take the limit as $h \to 0$, i.e. $\lim_{x\to a^-}f(x)=\lim_{h\to 0}f(a-h)$.

☞ LIMIT OF A FUNCTION DERIVED FROM ONE-SIDED LIMITS

We say that $\lim_{x\to a}f(x)=l$ if $\lim_{x\to a^+}f(x)=\lim_{x\to a^-}f(x)=l$. However, if $\lim_{x\to a^+}f(x)\neq\lim_{x\to a^-}f(x)$ or if any of the limits $\lim_{x\to a^+}f(x)$ or $\lim_{x\to a^-}f(x)$ does not exist, then we say that $\lim_{x\to a}f(x)$ does not exist.

Illustration 1. Evaluate $\lim_{x\to 1}\frac{x^3-x^2\log x+\log x-1}{x^2-1}$.

Solution: The given limit

$$=\lim_{x\to 1}\frac{(x^3-1)-(x^2-1)\log x}{x^2-1}$$

$$=\lim_{x\to 1}\frac{(x-1)(x^2+x+1)-(x-1)(x+1)\log x}{(x-1)(x+1)}$$

$$=\lim_{x\to 1}\frac{(x-1)[x^2+x+1-(x+1)\log x]}{(x-1)(x+1)}$$

$$=\frac{1^2+1+1-(1+1)\log 1}{(1+1)}=\frac{3}{2}.$$

☞ SOME IMPORTANT LIMITS

1. If $f(x)$ is a polynomial, then $\lim_{x\to a}f(x)=f(a)$
2. If $a\neq 0$ and $n\in Q$, then $\lim_{x\to a}\left(\frac{x^n-a^n}{x-a}\right)=na^{n-1}$
3. $\lim_{x\to a}\left(\frac{x^m-a^m}{x^n-a^n}\right)=\frac{m}{n}a^{m-n}$
4. $\lim_{x\to 0}\frac{\sin x}{x}=1=\lim_{x\to 0}\frac{x}{\sin x}$
5. $\lim_{x\to 0}\frac{\tan x}{x}=1=\lim_{x\to 0}\frac{x}{\tan x}$
6. $\lim_{x\to 0}\frac{\sin^{-1}x}{x}=1=\lim_{x\to 0}\frac{x}{\sin^{-1}x}$
7. $\lim_{x\to 0}\frac{\tan^{-1}x}{x}=1=\lim_{x\to 0}\frac{x}{\tan^{-1}x}$
8. $\lim_{x\to 0}\frac{\sin x^0}{x}=\frac{\pi}{180}$
9. $\lim_{x\to 0}\cos x=1$
10. $\lim_{x\to a}\frac{\sin(x-a)}{x-a}=1$
11. $\lim_{x\to a}\frac{\tan(x-a)}{x-a}=1$
12. $\lim_{x\to a}\sin^{-1}x=\sin^{-1}a, |a|\leq 1$

13. $\lim_{x\to a}\cos^{-1}x=\cos^{-1}a, |a|\le 1$
14. $\lim_{x\to a}\tan^{-1}x=\tan^{-1}a, -\infty<a<\infty$

L'HÔPITAL'S RULE

If $\frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ when $x=a$, then

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

This is known as **L'Hôpital's rule.**

SOME IMPORTANT EXPANSIONS THAT ARE USEFUL FOR FINDING LIMITS

1. For $|y|<1$, $(1+x)^n$
$$=\left[1+nx+\frac{n(n-1)}{\underline{|2}}x^2+\frac{n(n-1)(n-2)}{\underline{|3}}x^3\ldots\right]$$
2. $\left(\frac{x^n-a^n}{x-a}\right)=(x^{n-1}+ax^{n-2}+a^2x^{n-3}+\cdots+a^{n-1})$
3. $e^x=\left[1+x+\frac{x^2}{\underline{|2}}+\frac{x^3}{\underline{|3}}+\cdots+\frac{x^n}{\underline{|n}}+\cdots\right]$
4. $a^x=\left[1+x(\log x)+\frac{x^2}{\underline{|2}}(\log a)^2+\cdots\right]$
5. $\log(1+x)=\left[x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots\right]$
6. $\sin x=\left[x-\frac{x^3}{\underline{|3}}+\frac{x^5}{\underline{|5}}-\frac{x^7}{\underline{|7}}+\cdots\right]$
7. $\cos x=\left[1-\frac{x^2}{\underline{|2}}+\frac{x^4}{\underline{|4}}-\frac{x^6}{\underline{|6}}+\cdots\right]$
8. $\tan x=\left[x-\frac{x^3}{3}+\frac{2x^5}{15}+\cdots\right]$
9. $\tan^{-1}x=\left[x-\frac{x^3}{3}+\frac{x^5}{5}-\cdots\right]$
10. $e^{-x}=\left[1-x+\frac{x^2}{\underline{|2}}-\frac{x^3}{\underline{|3}}+\cdots\right]$
11. $\log(1-x)=-\left[x+\frac{x^2}{2}+\frac{x^3}{3}+\cdots\right]$
12. $\sec x=\left[1+\frac{x^2}{\underline{|2}}+\frac{5x^4}{\underline{|4}}+\cdots\right]$
13. $\sin^{-1}x=\left[x-\frac{1}{2}\cdot\frac{x^3}{3}+\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{x^5}{5}+\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot\frac{x^7}{7}+\cdots\right]$

LOGARITHMIC LIMITS

We use the series $\log(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots\infty$, where $-1<x\le 1$ and expansion is true only if base is e.

1. $\lim_{x\to 0}\frac{\log(1+x)}{x}=1$
2. $\lim_{x\to e}\log_e x=1$
3. $\lim_{x\to 0}\frac{\log(1-x)}{x}=-1$
4. $\lim_{x\to 0}\frac{\log_a(1+x)}{x}=\log_a e; a>0,\neq 1$

EXPONENTIAL LIMITS

Based on series expansion, we use the series

$$e^x=1+x+\frac{x^2}{\underline{|2}}+\frac{x^3}{\underline{|3}}+\cdots\infty$$

1. $\lim_{x\to 0}\frac{e^x-1}{x}=1$
2. $\lim_{x\to 0}\frac{a^x-1}{x}=\log_e a$
3. $\lim_{x\to 0}\frac{e^{\lambda x}-1}{x}=\lambda\ (\lambda\neq 0)$

BASED ON THE FORM 1^∞

To evaluate the exponential form 1^∞, we use the following results:

If $\lim_{x\to a}f(x)=\lim_{x\to a}g(x)=0$, then

$$\lim_{x\to a}[1+f(x)]^{1/g(x)}=\lim_{e^{x\to a}}\frac{f(x)}{g(x)}\ \text{or}$$

when $\lim_{x\to a}f(x)=1$ and $\lim_{x\to a}g(x)=\infty$, then

$$\lim_{x\to a}[f(x)]^{g(x)}=\left[\lim_{x\to a}f(x)\right]^{\lim_{x\to a}g(x)}$$

1. $\lim_{x\to 0}(1+x)^{1/x}=e$
2. $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x=e$

3. $\lim_{x\to 0}(1+\lambda x)^{1/x} = e^{\lambda}$
4. $\lim_{x\to\infty}\left(1+\frac{\lambda}{x}\right)^{x} = e^{\lambda}$
5. $\lim_{x\to\infty} a^{x} = \begin{cases}\infty, & \text{if } a > 1\\ 0, & \text{if } a < 1\end{cases}$

Illustration 2. Evaluate $\lim_{x\to\pm\infty} x(\sqrt{x^2+k}-x),\ k>0$.

Solution: $\lim_{x\to\pm\infty} x(\sqrt{x^2+k}-x)\frac{(\sqrt{x^2+k}+x)}{(\sqrt{x^2+k}+x)}$

$$= \lim_{x\to\pm\infty}\frac{x(x^2+k-x^2)}{(\sqrt{x^2+k}+x)}$$

$$= \lim_{x\to\pm\infty}\frac{xk}{\left[|x|\sqrt{\left(1+\frac{k}{x^2}\right)}\right]+x}$$

Here we have to consider two cases:

(i) When $x\to\infty;\ |x| = -x$, then we have

$$\lim_{x\to\infty}\frac{xk}{x\sqrt{\left(1+\frac{k}{x^2}\right)}+x} = \lim_{x\to\infty}\frac{xk}{x\left[\sqrt{\left(1+\frac{k}{x^2}\right)}+1\right]} = \frac{k}{2}$$

(ii) When $x\to-\infty;\ |x| = -x$, then we have

$$\lim_{x\to-\infty}\frac{xk}{-x\sqrt{\left(1+\frac{k}{x^2}\right)}+x} = \lim_{x\to-\infty}\frac{xk}{x\left[-\sqrt{\left(1+\frac{k}{x^2}\right)}+1\right]}$$

$$= \frac{k}{-1^{-}+1} = \frac{k}{0^{-}} \to -\infty.$$

☞ TO FIND $\lim_{x\to\infty} f(x)$

Replace x by $1/y$ and take the limit as $y\to 0$.

Some Limits

1. If $|x|<1$, then $\lim_{n\to\infty} x^n = \infty$
2. If $x > 1$, then $\lim_{n\to\infty} x^n = \infty$
3. $\lim_{x\to\infty} e^{x} = \infty$
4. $\lim_{x\to\infty} e^{-x} = 0$
5. $\lim_{x\to\infty} \log x = \infty$
6. $\lim_{n\to\infty}\frac{1}{n} = 0$ and $\lim_{n\to\infty}\frac{1}{n^2} = 0$
7. $\lim_{n\to\infty} x^{1/n} = 1$
8. $\lim_{x\to\infty}\frac{\sin x}{x} = \lim_{x\to\infty}\frac{\cos x}{x}$
9. $\lim_{x\to\infty}\frac{\sin 1/x}{1/x} = 1$

☞ USEFUL TRIGONOMETRIC RESULTS

1. $\sin(A+B) = \sin A\cos B + \cos A\sin B$
2. $\sin(A-B) = \sin A\cos B - \cos A\sin B$
3. $\cos(A+B) = \cos A\cos B - \sin A\sin B$
4. $\cos(A-B) = \cos A\cos B + \sin A\sin B$
5. $\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B}$
6. $\tan(A-B) = \frac{\tan A-\tan B}{1+\tan A\tan B}$
7. $\sin C+\sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$
8. $\sin C-\sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$
9. $\cos C+\cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$
10. $\cos C-\cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$

☞ CONTINUITY AT A POINT

A function $f(x)$ is said to be continuous at $x = a$ if $\lim_{x\to a^-} f(a) = \lim_{x\to a^+} f(x) = f(a)$, i.e. LHL = RHL = value of the function at a, i.e. $\lim_{x\to a} f(x) = f(a)$. If $f(x)$ is not continuous at $x = a$, we say that $f(x)$ is discontinuous at $x = a$.

Illustration 3. Discuss the continuity of the function $[\cos x]$ at $x = (\pi/2)$, where [ ] denotes the greatest integer function.

Solution: LHL $= \lim_{x\to(\pi/2)^-}[\cos x] = 0$

RHL $= \lim_{x\to(\pi/2)^+}[\cos x] = -1$

$$f\left(\frac{\pi}{2}\right) = \left[\cos\frac{\pi}{2}\right] = 0$$

Clearly, LHL ≠ RHL.
So, the function is discontinuous at $x = (\pi/2)$.

☞ CONTINUITY IN AN OPEN INTERVAL

A function $f(x)$ is said to be continuous in an open interval (a, b) if it is continuous at each and every point of (a, b), i.e. $y = [x]$ is continuous in (1, 2).

☞ CONTINUITY IN A CLOSED INTERVAL

A function $f(x)$ is said to be continuous in a closed interval $[a, b]$ if

(i) it is continuous in (a, b).

(ii) the value of the function at b is equal to the left-hand limit at b, i.e. $f(b) = \lim_{x \to b^+} f(x)$.

(iii) the value of the function at a is equal to the right-hand limit at a, i.e. $f(a) = \lim_{x \to a^+} f(x)$.

Illustration 4. Check the continuity of the function $f(x) = [x^2] - [x]^2$ for all $x \in \mathbb{R}$ at the end points of the interval $[-1, 0]$, where [ ] denotes the greatest integer function.

Solution: Continuity at $x = -1$

$$f(-1) = [(-1)^2] - [-1]^2 = [1] - (-1)^2 = 1 - 1 = 0$$

$$\text{RHL} = \lim_{x \to -1^+} \{[x^2] - [x]^2\} = 0 - 1 = -1$$

So, $f(-1) \neq \text{RHL}$

Continuity at $x = 0$

$$f(0) = [(0)^2] - [0]^2 = 0 - 0 = 0$$

$$\text{LHL} = \lim_{x \to 0^+} \{[x^2] - [x]^2\} = 0 - 1 = -1$$

So, $f(0) \neq \text{LHL}$

Hence, the function is not continuous at the end points of the interval $[-1, 0]$.

☞ PROPERTIES OF CONTINUOUS FUNCTIONS

Let $f(x)$ and $g(x)$ be continuous functions at $x = a$. Then,

(i) $cf(x)$ is continuous at $x = a$, where c is any constant.

(ii) $f(x) \pm g(x)$ is continuous at $x = a$.

(iii) $f(x) \cdot g(x)$ is continuous at $x = a$.

(iv) $f(x)/g(x)$ is continuous at $x = a$, provided $g(a) \neq 0$.

(v) If $f(x)$ is continuous on $[a, b]$, such that $f(a)$ and $f(b)$ are of opposite signs, then there exists at least one solution of the equation $f(x) = 0$ in the open interval (a, b).

☞ DISCONTINUOUS FUNCTION

If $f(x)$ is not continuous at $x = a$, then $f(x)$ is said to be discontinuous at $x = a$ and this point is called a point of discontinuity.

☞ TYPES OF DISCONTINUITY

Removable Discontinuity

A function f is said to possess removable discontinuity if at $x = a$,

$$L = R \neq V$$

$$\Rightarrow \quad \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) \neq f(a)$$

i.e. the left-hand limit and the right-hand limit at $x = a$ exist and are equal, but they are not equal to the value of the function at $x = a$. This is shown in Fig. 11.1.

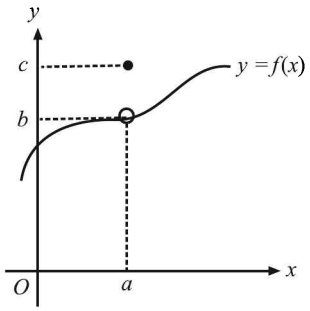


Fig. 11.1

Irremovable Discontinuity

A function f is said to possess irremovable discontinuity if at $x = a$, the left-hand limit is not equal to the right-hand limit, i.e. $L \neq R$.

$$\Rightarrow \quad \lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$$

Discontinuity of First Kind

A function f is said to possess discontinuity of first kind at $x = a$ if at $x = a$, both the left-hand limit and the right-hand limit exist finitely but are unequal.

This can be illustrated with the help of Fig. 11.2.

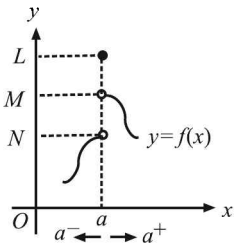


Fig. 11.2

Here, $\lim_{x \to a^-} f(x) = N$ and $\lim_{x \to a^+} f(x) = M$. Also, $f(a) = L$.

Clearly, from the figure, $N \neq M$.

Discontinuity of Second Kind

A function f is said to possess discontinuity of second kind at $x = a$ if at $x = a$, both the left-hand limit and the right-hand limit do not exist and are infinite. This can be illustrated with the help of Fig. 11.3.

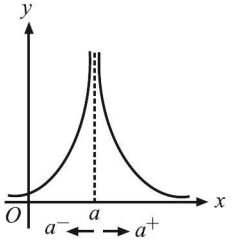


Fig. 11.3

☞ DIFFERENTIABILITY

Let $f(x)$ be a real-valued function defined on an interval (a, b) and $x_1 \in (a, b)$. Then the function $f(x)$ is said to be *differentiable* (or *derivable*) at x_1 if

$$\lim_{h\to 0}\frac{f(x_1+h)-f(x_1)}{(x_1+h)-(x_1)}=\frac{\text{small change in } y}{\text{small change in } x}$$

or equivalently, $\lim\limits_{x\to x_1}\dfrac{f(x)-f(x_1)}{x-x_1}$.

The value of the limit is denoted by $f'(x_1)$ or by $Df(x_1)$ and is usually called *derivative* of $f(x)$.

☞ TYPES OF DERIVATIVES

Left-hand Derivatives

Regressive derivative or left-hand derivative of $f(x)$ at $x = x_1$ is given by

$$\text{LHD} = Lf'(x_1)=\lim_{h\to 0}\frac{f(x_1-h)-f(x_1)}{(x_1-h)-(x_1)}$$

Right-hand Derivatives

Progressive derivative or right-hand derivative of $f(x)$ at $x = x_1$ is given by

$$\text{RHD} = Rf'(x_1)=\lim_{h\to 0}\frac{f(x_1+h)-f(x_1)}{(x_1+h)-(x_1)}$$

Illustration 5. Prove that the function $f(x)=|x|+|x-1|$ is not differentiable at $x = 1$.

Solution: $f(x)=|x|+|x-1|=\begin{cases}-2x+1, & x<0\\ 1, & 0\le x<1\\ 2x-1, & 1\le x\end{cases}$

$$\text{LHL} = \lim_{x\to 1^-} f(x)=\lim_{x\to 1^-} 1=1$$

$$\text{RHL} = \lim_{x\to 1^+} f(x)=\lim_{x\to 1^+}(2x-1)=1$$

and $$f(1)=2\times 1-1=1$$

$\because \lim\limits_{x\to 1^-} f(x)=\lim\limits_{x\to 1^+} f(x)=f(1)$

$\therefore$ $f(x)$ is continuous at $x = 1$,

Now, $$Rf'(1)=\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$$

$$=\lim_{h\to 0}\frac{2(1+h)-1-1}{h}=2$$

$$Lf'(1)=\lim_{h\to 0}\frac{f(1+h)-f(1)}{-h}$$

$$=\lim_{h\to 0}\frac{1-1}{-h}=0$$

$\because$ $Rf'(1)\neq Lf'(1)$

$\therefore$ $f(x)$ is not differentiable at $x = 1$.

☞ RELATION BETWEEN CONTINUITY AND DIFFERENTIABILITY

If a function is differentiable at a point, then it is necessarily continuous at that point. But the converse is not necessarily true, i.e. every continuous function need not be differentiable.

☞ PROPERTIES OF DIFFERENTIABLE FUNCTIONS

1. Every polynomial function is differentiable at each $x \in \mathbb{R}$.
2. The exponential function a^x, $a > 0$, is differentiable at each $x \in \mathbb{R}$.
3. Every constant function is differentiable at each $x \in \mathbb{R}$.
4. The logarithmic function is differentiable at each point in its domain.
5. Trigonometric and inverse trigonometric functions are differentiable in their respective domains.
6. The sum, difference, product and quotient of two differentiable functions are differentiable.
7. The composition of a differentiable function is a differentiable function.
8. Absolute functions are always continuous throughout but not differentiable at their critical points.

SOLVED PROBLEMS

1. The value of $\lim\limits_{x\to 0^+}\dfrac{1}{3x}$ is

(a) $-\infty$ (b) -1

(c) 0 (d) $+\infty$

Ans. (d)

Solution: Expression is $\lim\limits_{x\to 0^+}\dfrac{1}{3x}$.

We know that $\lim\limits_{x\to 0^+}\dfrac{1}{3x}=\dfrac{1}{3\times 0}=\dfrac{1}{0}=+\infty$

2. The right-hand limit of the function sec x at $x = -(\pi/2)$ is

(a) $-\infty$ (b) -1

(c) 0 (d) ∞

Ans. (d)

Solution: Function $f(x)$ = sec x and point $x = -(\pi/2)$. We know that right-hand limit of the function $f(x)$ at $x = -(\pi/2)$ is

$$\lim_{x\to[-(\pi/2)]^+} f(x)=\lim_{x\to[-(\pi/2)]}\sec x.$$

Substituting $x = h + \left(-\frac{\pi}{2}\right)$ and $h \to 0$, we get

$$\lim_{h\to 0} \sec\left[h + \left(-\frac{\pi}{2}\right)\right] = \lim_{h\to 0} \sec\left(\frac{\pi}{2} - h\right)$$

$$= \lim_{h\to 0} \operatorname{cosec} h = \operatorname{cosec} 0 = \infty$$

$[\because \sec(-\theta) = \sec\theta]$

3. The value of $\lim_{x\to -1} \frac{x^9+1}{x^{14}+1}$ is

(a) $\frac{9}{14}$ (b) $-\frac{9}{14}$

(c) $\frac{3}{78}$ (d) $-\frac{3}{18}$

Ans. (b)

Solution: Expression is $\lim_{x\to -1} \frac{x^9+1}{x^{14}+1}$.

We know that the given expression is in an indeterminate form at $x = -1$.

Applying L'Hôpital's rule, we get

$$\lim_{x\to -1} \frac{x^9+1}{x^{14}+1} = \lim_{x\to -1} \frac{9x^8}{14x^{13}} = -\frac{9}{14}.$$

4. The value of $\lim_{x\to -1} \frac{x^2+3x+2}{x^2+4x+3}$ is

(a) 0 (b) 1

(c) 2 (d) 1/2

Ans. (d)

Solution: Expression is $\lim_{x\to -1} \frac{x^2+3x+2}{x^2+4x+3}$.

The given expression may be written as

$$\lim_{x\to -1} \frac{x^2+2x+x+2}{x^2+3x+x+3}$$

$$= \lim_{x\to -1} \frac{x(x+2)+1(x+2)}{x(x+3)+1(x+3)}$$

$$= \lim_{x\to -1} \frac{(x+1)(x+2)}{(x+1)(x+3)} = \lim_{x\to -1} \frac{x+2}{x+3} = \frac{1}{2}.$$

5. $\lim_{x\to 0} \frac{x\sqrt[3]{z^2-(z-x)^2}}{\left(\sqrt[3]{8xz-4x^2}+\sqrt[3]{8xz}\right)^4}$ is equal to

(a) $\frac{z}{2^{11/3}}$ (b) $\frac{1}{2^{23/3}\cdot z}$

(c) $2^{21/3} z$ (d) none of these

Ans. (b)

Solution: $\lim_{x\to 0} \frac{x\sqrt[3]{z^2-(z-x)^2}}{\left(\sqrt[3]{8xz-4x^2}+\sqrt[3]{8xz}\right)^4}$

$$= \lim_{x\to 0} \frac{x\sqrt[3]{2xz-x^2}}{\left(\sqrt[3]{8xz-4x^2}+\sqrt[3]{8z}\sqrt[3]{x}\right)^4}$$

$$= \lim_{x\to 0} \frac{x^{4/3}\sqrt[3]{2z-x}}{x^{4/3}\left(\sqrt[3]{8z-4x}+\sqrt[3]{8z}\right)^4} = \frac{\sqrt[3]{2z}}{\left[2\sqrt[3]{8z}\right]^4} = \frac{1}{2^{23/3} z}$$

6. The value of $\lim_{x\to 1} \frac{1+\log x - x}{1-2x+x^2}$ is

(a) –1 (b) –(1/2)

(c) 0 (d) 1

Ans. (b)

Solution: Expression is $\lim_{x\to 1} \frac{1+\log x - x}{1-2x+x^2}$.

We know that the given expression is in an indeterminate form at $x = 1$. Applying L'Hôpital's rule, we get

$$\lim_{x\to 1} \frac{1+\log x - x}{1-2x+x^2} = \lim_{x\to 1} \frac{(1/x)-1}{-2+2x}$$

$$= \lim_{x\to 1} \frac{1-x}{2x(x-1)} = \lim_{x\to 1} \frac{-1}{2x} = -\frac{1}{2}.$$

7. The value of $\lim_{x\to 0} \frac{\sin(\pi\cos^2 x)}{x^2}$ is

(a) $-\pi$ (b) $\pi/2$

(c) π (d) $3\pi/2$

Ans. (c)

Solution: Expression is $\lim_{x\to 0} \frac{\sin(\pi\cos^2 x)}{x^2}$.

We know that

$$\lim_{x\to 0} \frac{\sin(\pi\cos^2 x)}{x^2} = \lim_{x\to 0} \frac{\sin[\pi(1-\sin^2 x)]}{x^2}$$

$$= \lim_{x\to 0} \frac{\sin(\pi - \pi\sin^2 x)}{x^2}$$

$$= \lim_{x\to 0} \frac{\sin\pi\cos\left(\pi\sin^2 x\right) - \cos\pi\sin\left(\pi\sin^2 x\right)}{x^2}$$

$$= \lim_{x\to 0} \frac{0+\sin(\pi\sin^2 x)}{x^2} = \lim_{x\to 0} \frac{\sin\left(\pi\sin^2 x\right)}{x^2}$$

$$= \lim_{x\to 0} \frac{\sin\left(\pi\sin^2 x\right)}{\pi\sin^2 x} \times \pi \times \left(\frac{\sin x}{x}\right)^2$$

$$= (1)(\pi)(1)^2 = \pi \quad \left(\text{where } \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1\right)$$

8. The value of $\lim_{\alpha\to\beta}\left[\frac{\sin^2\alpha-\sin^2\beta}{\alpha^2-\beta^2}\right]$ is

(a) 0 (b) 1

(c) $\frac{\sin\beta}{\beta}$ (d) $\frac{\sin 2\beta}{2\beta}$

Ans. (d)

Solution: Expression is $\lim_{\alpha\to\beta}\left[\frac{\sin^2\alpha-\sin^2\beta}{\alpha^2-\beta^2}\right]$.

We know that

$$\lim_{\alpha\to\beta}\left[\frac{\sin^2\alpha-\sin^2\beta}{\alpha^2-\beta^2}\right]$$

$$=\lim_{\alpha\to\beta}\left[\frac{\sin(\alpha-\beta)\sin(\alpha+\beta)}{(\alpha-\beta)(\alpha+\beta)}\right]$$

$$=\lim_{\alpha\to\beta}\left[\frac{\sin(\alpha-\beta)}{(\alpha-\beta)}\right]\lim_{\alpha\to\beta}\left[\frac{\sin(\alpha+\beta)}{(\alpha+\beta)}\right]$$

$$=\lim_{\alpha-\beta\to 0}\left[\frac{\sin(\alpha-\beta)}{(\alpha-\beta)}\right]\cdot\left(\frac{\sin 2\beta}{2\beta}\right)$$

$$=1\cdot\frac{\sin 2\beta}{2\beta}=\frac{\sin 2\beta}{2\beta}\cdot\left(\text{where }\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1\right)$$

9. The value of $\lim_{x\to(\pi/2)}\frac{[1-\tan(x/2)][1-\sin x]}{[1+\tan(x/2)][\pi-2x]^3}$ is

(a) 0 (b) ∞

(c) 1/32 (d) 1/8

Ans. (c)

Solution: Expression is $\lim_{x\to(\pi/2)}\frac{[1-\tan(x/2)][1-\sin x]}{[1+\tan(x/2)][\pi-2x]^3}$.

The given expression may be written as

$$\lim_{x\to(\pi/2)}\frac{\tan(\pi/4)-\tan(x/2)}{1+\tan(x/4)\times\tan(x/2)}\times\frac{(1-\sin x)}{(\pi-2x)^3}$$

$$=\lim_{x\to(\pi/2)}\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\times\frac{(1-\sin x)}{(\pi-2x)^3}$$

Substituting $x=\left(\frac{\pi}{2}-h\right)$ in this expression, we get

$$=\lim_{h\to 0}\tan\left[\frac{\pi}{4}-\frac{1}{2}\left(\frac{\pi}{2}-h\right)\right]\cdot\frac{1-\sin\left(\frac{\pi}{2}-h\right)}{\left[\pi-2\left(\frac{\pi}{2}-h\right)\right]^3}$$

$$=\lim_{h\to 0}\tan\left(\frac{\pi}{4}-\frac{\pi}{4}+\frac{h}{4}\right)\cdot\frac{1-\sin\left(\frac{\pi}{2}-h\right)}{(\pi-\pi+2h)^3}$$

$$=\lim_{h\to 0}\tan\left(\frac{h}{2}\right)\cdot\frac{1-\cosh}{(2h)^3}$$

$$=\lim_{h\to 0}\tan\frac{h}{2}\frac{\left[1-1+2\sin^2\left(\frac{h}{2}\right)\right]}{8h^3}$$

$$=\lim_{h\to 0}\frac{\tan\left(\frac{h}{2}\right)\times 2\times\sin^2\left(\frac{h}{2}\right)}{2\times\left(\frac{h}{2}\right)\times 8\times 4\left(\frac{h}{2}\right)^2}=\frac{1}{32}$$

$$=\lim_{h\to 0}\frac{\tan\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)}\times\left(\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}\right)^2=\frac{1}{32}\times 1\times 1=\frac{1}{32}$$

10. The value of $\lim_{n\to\infty}\left[\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+\ldots+\frac{1}{3n}\right]$ is

(a) 0 (b) $\log_e 4$

(c) $\log_e 3$ (d) $\log_e 2$

Ans. (c)

Solution: Expression is $\lim_{n\to\infty}\left[\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{3n}\right]$.

We know that

$$\lim_{n\to\infty}\left[\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{3n}\right]$$

$$=\lim_{n\to\infty}\left\{\frac{1}{n}\left[\frac{1}{1}+\frac{1}{1+\frac{1}{n}}+\frac{1}{1+\frac{2}{n}}+\cdots+\frac{1}{1+\frac{2n}{n}}\right]\right\}$$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{2n}\frac{1}{\left(1+\frac{r}{n}\right)}.$$

Substituting $(r/n)=x$ and $(1/n)=dx$.

We also know that when $r=0$, then $x=0$. And when $r=2n$, then $x=2$.

Therefore, $I=\int_0^2\frac{1}{1+x}dx=[\log_e(1+x)]_0^2=\log_e 3-\log_e 1$

$\Rightarrow\quad I-\log_e 3.$

11. The value of $\lim_{n\to\infty}\left[\frac{1}{1-n^2}+\frac{2}{1-n^2}+\frac{3}{1-n^2}+\cdots+\frac{n}{1-n^2}\right]$ is

(a) ∞ (b) 1

(c) 0 (d) −(1/2)

Ans. (d)

Solution: Expression is

$$\lim_{n\to\infty}\left[\frac{1}{1-n^2}+\frac{2}{1-n^2}+\frac{3}{1-n^2}+\cdots+\frac{n}{1-n^2}\right]$$

We know that

$$\lim_{n\to\infty}\left[\frac{1}{1-n^2}+\frac{2}{1-n^2}+\frac{3}{1-n^2}+\cdots+\frac{n}{1-n^2}\right]$$

$$=\lim_{n\to\infty}\frac{n(n+1)}{2(1-n^2)}$$

$$=\lim_{n\to\infty}\frac{n}{2(1-n)}=\frac{1}{2}\lim_{n\to\infty}\frac{n}{1-n}$$

$$=\frac{1}{2}\lim_{n\to\infty}\left[\frac{1}{(1/n)-1}\right]=-\frac{1}{2}.$$

$$\left[\text{where } 1+2+3+\cdots+n=\frac{n(n+1)}{2}\right]$$

12. If $f(x)=\begin{cases}\dfrac{x^5-32}{x-2}, & \text{where } x\neq 2\\ x, & \text{where } x=2\end{cases}$ is continuous at $x = 2$, then the value of k is

(a) 10 (b) 15
(c) 35 (d) 80

Ans. (d)

Solution: Function $f(x)=\begin{cases}\dfrac{x^5-32}{x-2}, & \text{where } x\neq 2\\ x, & \text{where } x=2\end{cases}$ is continuous at $x = 2$.

We know that

$$\lim_{x\to 2}f(x)=\lim_{x\to 2}\left(\frac{x^5-32}{x-2}\right)=\lim_{x\to 2}\left(\frac{x^5-2^5}{x-2}\right).$$

We also know that $\lim_{x\to a}\left(\dfrac{x^n-a^n}{x-a}\right)=na^{n-1}$.

Therefore, $\lim_{x\to 2}f(x)=5\times 2^4=80$.

Since $f(2) = k$ and $f(x)$ is continuous at $x = 2$, therefore

$\lim_{x\to 2}f(x)=f(2) \Rightarrow k=80$.

13. If $f(x)=\begin{cases}x+\lambda, & -1>x>3\\ 4, & x=3\\ 3x-5, & x>3\end{cases}$ is continuous at $x = 3$, then the value of λ is

(a) -1 (b) 0
(c) 1 (d) does not exist

Ans. (c)

Solution: Function $f(x)=\begin{cases}x+\lambda, & -1>x>3\\ 4, & x=3\\ 3x-5, & x>3\end{cases}$ is continuous at $x = 3$. We know that as $f(x)$ is continuous at $x = a$, therefore $\lim_{x\to a}f(x)=f(a)$.

Since the given function is continuous at $x = 3$, therefore

$\lim_{x\to 3-}(x+\lambda)=f(3) \Rightarrow 3+\lambda=4 \Rightarrow \lambda=4-3=1$.

14. If $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ are continuous functions, then the value of $\int_{-\pi/2}^{\pi/2}[f(x)+f(-x)][g(x)-g(-x)]dx$ is

(a) 0 (b) -1
(c) 1 (d) π

Ans. (a)

Solution: Two continuous functions are $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$. Let

$$F(x)=\left|f(x)+f(-x)\right|\left|g(x)-g(-x)\right| \quad \text{(i)}$$

and $$F(-x)=\left|f(-x)+f(x)\right|\left|g(-x)-g(x)\right| \quad \text{(ii)}$$

Adding Eqs. (i) and (ii), we get

$$F(x)+F(-x)=\left|2f(x)+2f(-x)\right|$$

$$\left|g(x)-g(-x)+g(-x)-g(x)\right|$$

$\Rightarrow F(x)+F(-x)=0 \Rightarrow F(x)=-F(-x)$

$\Rightarrow F(x)$ is an odd function.

Therefore, $\int_{-\pi/2}^{\pi/2}[f(x)+f(-x)][g(x)-g(-x)]dx=0$.

15. If the function $f(x)=\begin{cases}1+\sin\dfrac{\pi}{2}x, & \text{for } -\infty<x\le 1\\ ax+b & \text{for } 1<x<3\\ 6\tan\dfrac{\pi x}{12} & \text{for } 3\le x<6\end{cases}$ is continuous in the interval $(-\infty, 6)$, then the values of a and b are

(a) (0, 2) (b) (1, 1)
(c) (2, 0) (d) (2, 1)

Ans. (c)

Solution: Function $f(x)=\begin{cases}1+\sin\dfrac{\pi}{2}x, & \text{for } -\infty<x\le 1\\ ax+b & \text{for } 1<x<3\\ 6\tan\dfrac{\pi x}{12} & \text{for } 3\le x<6\end{cases}$ is continuous in the interval $(-\infty, 6)$. We know that $f(x)$ will also be continuous at $x = 1$ and $x = 3$. Therefore, for continuity at $x = 1$, $f(1)=\lim_{x\to 1}f(x)$

$\Rightarrow \quad 1+\sin\frac{\pi}{2}=\lim_{x\to 1}(ax+b) \quad \Rightarrow \quad a+b=2$ (i)

Similarly, for continuity at $x = 3$, $f(3)=\lim_{x\to 3} f(x)$

$\Rightarrow \quad 6\tan\frac{3\pi}{12}=\lim_{x\to 3}(ax+b) \quad \Rightarrow \quad 3a+b=6$ (ii)

Solving Eqs. (i) and (ii), we get $a = 2$ and $b = 0$.

16. The function $f(x) = [x]$ is
(a) derivable for all x
(b) continuous for all x
(c) a constant function
(d) discontinuous only for integral x

Ans. (d)

Solution: Function $f(x) = [x]$. Let $x \in I$.

We know that when $x \in I$ then the right-hand limit of the function at $x = k \in I$ is

$$\text{RHL} = \lim_{x\to k^+} f(x) = \lim_{x\to k^+}[x] = k$$

And the left-hand limit at $x = k \in I$ is

$$\text{LHL} = \lim_{x\to k^-} f(x) = \lim_{x\to k^-}[x] = (k-1)$$

Since RHL ≠ LHL, therefore the function is discontinuous only for integral x.

17. If $f(x)=\frac{1}{2}x-1$, then on the interval $[0, \pi]$
(a) $\tan[f(x)]$ and $1/f(x)$ are continuous.
(b) $\tan[f(x)]$ and $1/f(x)$ are discontinuous.
(c) $\tan[f(x)]$ is continuous but $1/f(x)$ is discontinuous.
(d) $\tan f(x)$ is discontinuous but $1/f(x)$ is continuous.

Ans. (c)

Solution: Function is $f(x)=\frac{1}{2}x-1$ and interval is $[0, \pi]$.

We know that $\tan[f(x)]=\tan\left(\frac{1}{2}x-1\right)$ is continuous when $-\frac{\pi}{2}<\frac{1}{2}x-1<\frac{\pi}{2}$

$\Rightarrow \quad -\pi+2<x<\pi+2$ and $[0,\pi]\subset(-\pi+2,\pi+2)$.

We also know that as $\frac{1}{f(x)}=\frac{2}{x-2}$ is not defined when $x = 2$, therefore $1/f(x)$ is not continuous on the interval $[0, \pi]$.

18. The point of discontinuity of $f(x)=\tan\left(\frac{\pi x}{x+1}\right)$ other than $x = -1$ is
(a) $x = 0$ (b) $x = \pi$
(c) $x=\frac{2m+1}{1-2m}$ (d) $x=\frac{2m-1}{2m+1}$

Ans. (c)

Solution: Function is $f(x)=\tan\left(\frac{\pi x}{x+1}\right)$.

We know that $f(x)$ is discontinuous at those points, when

$$\tan\left(\frac{\pi x}{x+1}\right)=\tan\frac{\pi}{2} \quad \Rightarrow \quad \frac{\pi x}{x+1}=m\pi+\frac{\pi}{2}$$

$$\Rightarrow \quad \frac{x}{x+1}=m+\frac{1}{2}$$

Therefore, $x=\frac{2m+1}{1-2m}$.

EXERCISE SET 1

1. $\lim_{\theta\to 0}\frac{4\theta(\tan\theta-2\theta\tan\theta)}{(1-\cos 2\theta)^2}$ is equal to
(a) $-\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

2. $\lim_{x\to 0}\frac{4^x-1}{3^x-1}$ equals
(a) 1 (b) $\log_4 3$
(c) $\log 4$ (d) $\log_3 4$

3. If α is a repeated root of $ax^2+bx+c=0$, then $\lim_{x\to\alpha}\frac{\sin(ax^2+bx+c)}{(x-\alpha)^2}$ is
(a) 0 (b) a
(c) b (d) c

4. The value of $\lim_{x\to 0}\frac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x}$ is
(a) −1 (b) 0
(c) 1 (d) none of these

5. $\lim_{x\to -2}\frac{\sin^{-1}(x+2)}{x^2+2x}$ is equal to
(a) $-\frac{1}{2}$ (b) 0
(c) ∞ (d) none of these

6. If $f: IR \to IR$ is defined by

$$f(x)=\begin{cases}\dfrac{x-2}{x^2-3x+2}, & \text{if } x\in IR-(1,2)\\ 2, & \text{if } x=1\\ 1, & \text{if } x=2\end{cases}$$

then $\lim\limits_{x\to 2}\dfrac{f(x)-f(2)}{x-2}$ is equal to

(a) –1 (b) –(1/2)
(c) 1 (d) does not exist

7. The value of $\lim\limits_{x\to 0}\left\{\dfrac{3x-\sin^{-1}x}{4x-\tan^{-1}x}\right\}$ is

(a) $\dfrac{3}{4}$ (b) $\dfrac{4}{3}$
(c) $\dfrac{2}{3}$ (d) $\dfrac{3^2}{4^2}$

8. The value of $\lim\limits_{x\to 0}\dfrac{\sin\sqrt[3]{x}\ln(1-5x)}{(\tan^{-1}\sqrt{x})^2(e^{\sqrt[3]{x}}-1)}$ is

(a) $-\dfrac{5}{3}$ (b) –5
(c) $-\dfrac{2}{3}$ (d) $-\dfrac{3}{5}$

9. The value of $\lim\limits_{x\to 0}\dfrac{(1-\cos x)^2}{\tan^3 x-\sin^3 x}$ is

(a) $\dfrac{1}{12}$ (b) 0
(c) ∞ (d) dose not exist

10. The value of $\lim\limits_{x\to-\infty} x+\sqrt{x^2+x^2\sin\left(\dfrac{1}{x}\right)}$ is

(a) –2 (b) 0
(c) 2 (d) none of these

11. $\lim\limits_{n\to\infty}\left\{\dfrac{1+\frac{1}{3}+\frac{1}{9}+\cdots \text{ to } (n+1) \text{ terms}}{1+\frac{1}{2}+\frac{1}{4}+\cdots \text{ to } (n+1) \text{ terms}}\right\}=$

(a) $\dfrac{3}{4}$ (b) $\dfrac{4}{3}$
(c) $\dfrac{3}{2}$ (d) $\dfrac{2}{3}$

12. $\lim\limits_{n\to\infty}\left(\dfrac{1+a+a^2+\cdots+a^{n-1}}{1+b+b^2+\cdots+b^{n-1}}\right)=0$, if

(a) $a=b$ (b) $a<b$
(c) $a>b$ (d) $1<a<b$

13. $\lim\limits_{n\to\infty}\dfrac{1}{1\cdot 3}+\dfrac{1}{3\cdot 5}+\dfrac{1}{5\cdot 7}+\cdots+\dfrac{1}{(2n-1)\cdot(2n+1)}$ is equal to

(a) 1/4 (b) 1/3
(c) 1/2 (d) none of these

14. $\lim\limits_{x\to 0}\left(\dfrac{1^x+2^x+3^x+\cdots+n^x}{n}\right)^{a/x}$ is equal to

(a) $(\underline{|n})^a$ (b) $\dfrac{1}{a\underline{|n}}$
(c) $(\underline{|n})^{a/n}$ (d) $\dfrac{n}{(\underline{|n})^a}$

15. The value of $\lim\limits_{x\to\frac{\pi}{2}}\left\{\dfrac{1^{\cos^2 x}+2^{\cos^2 x}+\cdots+n^{\cos^2 x}}{n}\right\}^{\frac{1}{\cos^2 x}}$ is

(a) $\dfrac{1}{(\underline{|n})^n}$ (b) $(\underline{|n})^{\frac{1}{n}}$
(c) $\dfrac{n^n}{\underline{|n}}$ (d) $\left(\dfrac{\underline{|n}}{1-n}\right)^{\frac{n}{n+1}}$

16. $\lim\limits_{x\to\infty} x^2\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\ldots\infty}}}}$ is equal to

(a) 0 (b) 1/2
(c) 1 (d) 2/3

17. Let $f(x)=\begin{cases}\dfrac{(x^3+x^2-16x+20)}{(x-2)^2} & \text{if } x\neq 2\\ k & \text{if } x=2\end{cases}$. If $f(x)$ is continuous for all x, then k =

(a) 7 (b) 2
(c) 0 (d) –1

18. The function $f(x)=\sin^{-1}(\cos x)$ is

(a) discontinuous at $x = 0$
(b) continuous at $x = 0$
(c) differentiable at $x = 0$
(d) none of these

19. The function $f(x)=\begin{cases}x+2, & 1\le x\le 2\\ 4, & x=2\\ 3x-2, & x>2\end{cases}$ is continuous at

(a) $x = 2$ only (b) $x \le 1$
(c) $x \ge 1$ (d) none of these

20. Let the function $f(x)=\dfrac{x}{1+e^{1/x}}$, if $x\neq 0; f(x)=0$. If $x = 0$, the derivative from the right $f'(0^+)$ and the derivative from the left $f'(0^-)$ are
(a) 0 and 1 (b) –1 and 1
(c) 1 and –1 (d) 1 and 0

21. The function $f(x)=1+|\sin x|$ is
(a) continuous nowhere
(b) continuous everywhere and not differentiable at $x = 0$
(c) differentiable nowhere
(d) differentiable at $x = 0$

22. Let $f(x)=\begin{cases}0, & x<0\\ x^2, & x\geq 0\end{cases}$.
Then for all x,
(a) f'' is differentiable (b) f is differentiable
(c) f' is not continuous (d) f is not continuous

23. The function $f(x)=\begin{cases}\dfrac{1}{4x^2-1}, & x\neq 0\\ 0, & x=0\end{cases}$ is continuous
(a) everywhere except at $x = 0$ and $x = \infty$
(b) nowhere
(c) everywhere
(d) everywhere except at $x = 0$

24. At point $x = 1$, the function $f(x)=\begin{cases}x^3-1; & 1<x<\infty\\ x-1; & -\infty<x\leq 1\end{cases}$
is
(a) continuous and differentiable
(b) continuous but not differentiable
(c) discontinuous and differentiable
(d) discontinuous and not differentiable

25. A function is defined as $f(x)=\begin{cases}e^x; & x\leq 0\\ |x-1|; & x>0\end{cases}$,
then $f(x)$ is
(a) continuous at $x = 0, 1$
(b) differentiable at $x = 0$
(c) differentiable at $x = 1$
(d) none of these

26. For the function $f(x)=\begin{cases}\dfrac{1-x}{|x-1|}, & x<1\\ 1, & x=1\\ x^2, & x>1\end{cases}$ which of the following is true?
(a) It is continuous at all points
(b) It is continuous at all points except at $x = 1$
(c) It is differentiable at all points
(d) none of these

27. If $f(x)=\begin{cases}\dfrac{1-\cos x}{x^2}, & x<0\\ \dfrac{1}{2}e^x, & x\geq 0\end{cases}$,
then at $x = 0$, f is
(a) continuous (b) not continuous
(c) differentiable (d) none of these

28. The function $f(x)=\begin{cases}|2x-3|\cdot[x], & x\geq 1\\ \sin\left(\dfrac{\pi x}{2}\right), & x<1\end{cases}$
(where $[x]$ denotes the greatest integer $\leq x$) is
(a) continuous at $x = 2$
(b) differentiable at $x = 1$
(c) continuous but not differentiable at $x = 1$
(d) none of these

29. Let $f(x)=\begin{cases}\dfrac{x^4-5x^2+4}{(x-1)(x-2)}, & x\neq 1,2\\ 6, & x=1\\ 12, & x=2\end{cases}$
then $f(x)$ is continuous on the set
(a) $\mathbb{R}$ (b) $\mathbb{R} - \{1\}$
(c) $\mathbb{R} - \{2\}$ (d) $\mathbb{R} - \{1, 2\}$

30. The set of all points, where the function $f(x)=\dfrac{x}{1+|x|}$ is differentiable, is
(a) $(-\infty,\infty)$ (b) $(0,\infty)$
(c) $(-\infty,0)\cup(0,\infty)$ (d) none of these

31. The value of
$\lim\limits_{n\to\infty}\dfrac{1+2^4+3^4+\cdots+n^4}{n^5}-\lim\limits_{n\to\infty}\dfrac{1+2^3+3^3+\cdots+n^3}{n^5}$ is
(a) 0 (b) 1/4
(c) 1/5 (d) 1/30

32. The function $f(x)=\dfrac{1+\sin x-\cos x}{1-\sin x-\cos x}$ $x\neq 0$ is continuous at $x = 0$ if $f(0)$ is
(a) 1 (b) –1
(c) $\sqrt{2}$ (d) $-\sqrt{2}$

33. The value of $\lim_{x\to 1}\frac{x+x^2+\cdots+x^n-n}{x-1}$ is

(a) n (b) $\frac{n+1}{2}$

(c) $\frac{n(n+1)}{2}$ (d) $\frac{n(n-1)}{2}$

34. If the function $f(x)=\begin{cases}(1+|\sin x|)^{\frac{a}{|\sin x|}}, & -\frac{\pi}{6}<x<0\\ b, & x=0\\ e^{\frac{\tan 2x}{\tan 3x}}, & 0<x<\frac{\pi}{6}\end{cases}$ is

continuous at $x = 0$, then

(a) $a=\log_e b,\ a=\frac{2}{3}$ (b) $b=\log_e a,\ a=\frac{2}{3}$

(c) $a=\log_e b,\ b=2$ (d) none of these

35. If $f(x)=\sin|x|-e^{|x|}$ then at $x = 0$, $f(x)$ is

(a) continuous but not differentiable
(b) neither continuous nor differentiable
(c) both continuous and differentiable
(d) none of these

36. The value of $\lim_{x\to 1}\frac{x^n+x^{n-1}+x^{n-2}+\cdots+x^2+x-n}{x-1}$ is

(a) $\frac{n(n+1)}{2}$ (b) 0

(c) 1 (d) n

37. If $f(x)=\begin{cases}\frac{x\log\cos x}{\log(1+x^2)}, & x\neq 0\\ 0, & x=0\end{cases}$

then

(a) $f(x)$ is not continuous at $x = 0$
(b) $f(x)$ is continuous at $x = 0$
(c) $f(x)$ is continuous at $x = 0$ but not differentiable at $x = 0$
(d) none of these

38. The value of $\lim_{x\to\infty}\left[\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right]$ is

(a) 1/2 (b) 1

(c) 0 (d) none of these

39. The set of all points where $f(x)=\sqrt[3]{x^2|x|}-|x|-1$ is not differentiable is

(a) $\{0\}$ (b) $\{-0,0,1\}$

(c) $\{0,1\}$ (d) none of these

40. $\lim_{x\to 0}\frac{x\sqrt{y^2-(y-x)^2}}{\left(\sqrt{8xy-4x^2}+\sqrt{8xy}\right)^3}$ equals

(a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2}$

(c) $\frac{1}{4}$ (d) none of these

41. Let $f(x)=(x+|x|)|x|$. Then for all x,

(a) f is continuous
(b) f' is differentiable for some x
(c) f' is continuous
(d) f'' is continuous

42. The value of $\lim_{x\to 0}\frac{l_n(1+2h)-2l_n(1+h)}{h^2}$ is

(a) 1 (b) –1

(c) 0 (d) none of these

43. If $f(x)=\begin{cases}ax^2+b, & b\neq 0, c\leq 1\\ bx^2+ax+c, & x>1\end{cases}$

then $f(x)$ is continuous and differentiable at $x = 1$ if

(a) $c=0, a=2b$ (b) $a=b, c\in\mathbb{R}$

(c) $a=b, c=0$ (d) $a=b, c\neq 0$

44. $\lim_{n\to\infty}\frac{1-(2\sin x)^{2n}}{1+(2\sin x)^{2n}}$ is not continuous at $x =$

(a) $\pi/6$ (b) $\pi/4$

(c) $\pi/2$ (d) none of these

45. Let $f(x)=\sin\frac{1}{x}, x\neq 0$. Then $f(x)$ can be continuous at $x = 0$

(a) if $f(0) = 1$
(b) if $f(0) = 0$
(c) if $f(0) = -1$
(d) for no definite value of $f(0)$

46. The set of all points where the function $f(x)=\frac{x}{1+|x|}$ is differentiable is

(a) $(-\infty,\infty)$ (b) $(\infty, 0)$

(c) $(-\infty, 0)\cup(0,\infty)$ (d) none of these

47. If $f(x)=[x]$, where [] is the greatest integer function, and $g(x)=x(1-x^2)(4-x^2)$, then $g[f(x)]$ is

(a) discontinuous at all integers
(b) continuous at all integers
(c) continuous at $x = 0, \pm 1, \pm 2$
(d) discontinuous at $x = 0, \pm 1, \pm 2$

48. Let $f(x)=[\cos x+\sin x], 0<x<2\pi$, where $[x]$ denotes the greatest integer less than or equal to x. The number of points of discontinuity of $f(x)$ is
(a) 6 (b) 5
(c) 4 (d) 3

49. If $[x]$ denotes the greatest integer $\le x$, then $f(x) = \dfrac{\tan(\pi[x-\pi])}{1+[x]^2}$ is
(a) discontinuous at some x
(b) continuous at all x, but the derivative $f'(x)$ does not exist for some x
(c) $f'(x)$ exists for all x but $f''(x)$ does not exist
(d) $f'(x)$ exists for all x

50. The number of points where $f(x)=\dfrac{1}{\log|x^2-3x+3|}$ is discontinuous is
(a) 2 (b) 4
(c) 6 (d) 0

51. Let $f(x)=\sin x$ if x is rational and $f(x)=1-2\cos x$ if x is irrational, then
(a) $f(x)$ is nowhere continuous
(b) $f(x)$ is continuous at one point only
(c) $f(x)$ is injective
(d) $f(x)$ is continuous at infinite number of points

52. The value of k for which the function
$$f(x)=\begin{cases}\sin\dfrac{1}{x}, & x\neq 0\\ k, & x=0\end{cases}$$
is continuous is
(a) 8 (b) 1
(c) -1 (d) none of these

53. If $f(x)=\dfrac{x}{1+e^{1/x}}, x\neq 0$ and $f(0)=0$, then at $x=0$
(a) f is continuous
(b) f is discontinuous
(c) f is left continuous only
(d) f is right continuous only

EXERCISE SET 2

1. If $a=\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{1}{(r+2)r!}$ and $b=\lim\limits_{x\to 0}\dfrac{e^{\sin x}-e^{x}}{\sin x-x}$, then
(a) $a=b$ (b) $a=2b$
(c) $2a=b$ (d) $a+b=0$

2. The equivalent definition of the function given by
$$f(x)=\begin{cases}2x, & x\ge 0\\ 0, & x<0\end{cases}$$ is
(a) $f(x)=|x|$ (b) $f(x)=2x$
(c) $f(x)=|x|+x$ (d) $f(x)=2|x|$

3. The value of $\lim\limits_{x\to 2a}\dfrac{\sqrt{x-2a}+\sqrt{x}-\sqrt{2a}}{\sqrt{x^2-4a^2}}$ is
(a) $\dfrac{1}{\sqrt{a}}$ (b) $\dfrac{1}{2\sqrt{a}}$ (c) $\dfrac{\sqrt{a}}{2}$ (d) $2\sqrt{a}$

4. Let f be the function defined by
$$f(x)=\begin{cases}\dfrac{x^2-1}{x^2-2|x-1|-1}, & x\neq 1\\ 1/2, & x=1\end{cases}$$
then
(a) the function is continuous for all values of x
(b) the function is continuous only for $x>1$
(c) the function is continuous at $x=1$
(d) the function is not continuous at $x=1$

5. The value of
$$\lim_{x\to(\pi/2)}\tan^{-1}\left[\frac{\sin(a\tan^3 x+b\tan^2 x+c\tan x)}{a\tan^3 x+b\tan^2 x+c\tan x}\right]$$ is
(a) 0 (b) 1
(c) $\pi/4$ (d) $\pi/2$

6. $\lim\limits_{n\to\infty}\left(\tan\theta+\dfrac{1}{2}\tan\dfrac{\theta}{2}+\dfrac{1}{2^2}\tan\dfrac{\theta}{2^n}+\cdots+\dfrac{1}{2^n}\tan\dfrac{\theta}{2^n}\right)=$
(a) $\dfrac{1}{\theta}$ (b) $\dfrac{1}{\theta}-2\cot 2\theta$
(c) $2\cot 2\theta$ (d) none of these

7. $\lim\limits_{x\to 0}\dfrac{1-\cos^3 x}{x\sin x\cos x}=$
(a) 2/5 (b) 3/5
(c) 3/2 (d) 3/4

8. Let $f(x)=\begin{cases}x^2, & x \text{ an integer}\\ \dfrac{k(x^2-4)}{2-x}, & \text{otherwise}\end{cases}$ then $\lim\limits_{x\to 2}f(x)$
(a) exists only when $k=1$
(b) exists for every real k
(c) exists for every real k except $k=1$
(d) does not exist

9. Let $f(x)=\dfrac{\sin\{\pi[x-\pi]\}}{1+[x]^2}$ where $[x]$ stands for the greatest integer function. Then $f(x)$ is
(a) discontinuous at integral points
(b) continuous everywhere but not differentiable
(c) differentiable once but $f''(x), f'''(x),\ldots$ do not exist
(d) differentiable for all x

10. If $f(x)=\begin{cases}\dfrac{1-\sin x}{\pi-2x}, & x\neq\dfrac{\pi}{2}\\ \lambda, & x=\dfrac{\pi}{2}\end{cases}$
is continuous at $x = \pi/2$, then the value of λ is
(a) -1 (b) 1 (c) 0 (d) 2

11. $\lim\limits_{x\to0}\left(1^{\operatorname{cosec}^2x}+2^{\operatorname{cosec}^2x}+\cdots+n^{\operatorname{cosec}^2x}\right)^{\sin^2x}=$
(a) 0 (b) ∞ (c) n (d) none of these

12. Fundamental period of the function
$f(x)=\sin^4x+\cos^4x=$
(a) π (b) 2π (c) $\pi/2$ (d) none of these

13. The value of $\lim\limits_{x\to2}\dfrac{\sqrt{1+\sqrt{2+x}}-\sqrt{3}}{x-2}$ is
(a) $1/8\sqrt{3}$ (b) $1/4\sqrt{3}$ (c) 0 (d) none of these

14. If $f(x)$ is differentiable everywhere, then
(a) $|f(x)|$ is differentiable everywhere
(b) $|f|^2$ is differentiable everywhere
(c) $f|f|$ is not differentiable at some point
(d) none of these

15. If $f(x)=\begin{cases}\dfrac{\sqrt{1+kx}-\sqrt{1-kx}}{x}, & \text{for } -1\le x<0\\ 2x^2+3x-2, & \text{for } 0\le x\le1\end{cases}$
is continuous at $x = 0$, then k is equal to
(a) -4 (b) -3 (c) -2 (d) -1

16. $\lim\limits_{x\to0}\dfrac{\sqrt{x-\sin x}}{x+\sin^2x}$ is equal to
(a) 1 (b) 0 (c) ∞ (d) none of these

17. The function $f(x)=(x^2-1)|x^2-3x+2|+\cos|x|$ is not differentiable at
(a) -1 (b) 0 (c) 1 (d) 2

18. If $f(x)=\begin{cases}\dfrac{(4x-1)^3}{\sin(x/a)\log[1+(x^2/3)]}, & x\neq0\\ 9(\log4)^3, & x=0\end{cases}$ is a continuous function at $x = 0$, then the value of a is equal to
(a) 3 (b) 1 (c) 2 (d) 0

19. If $f(x)$ and $g(x)$ are differentiable functions and $f(1) = g(1) = 2$, then $\lim\limits_{x\to1}\dfrac{f(1)g(x)-f(x)g(1)-f(1)+g(1)}{g(x)-f(x)}$ is equal to
(a) 0 (b) 1 (c) 2 (d) none of these

20. Let $f(x)=\begin{cases}\dfrac{\cos^2x-\sin^2x-1}{\sqrt{x^2+4}-2}, & x\neq0\\ a, & x=0\end{cases}$ then the value of a in order that $f(x)$ may be continuous at $x = 0$ is
(a) -8 (b) 8 (c) -4 (d) 4

21. If $f(x)=\begin{cases}\dfrac{x-1}{2x^2-7x+5}, & \text{for } x\neq1\\ -\dfrac{1}{3}, & \text{for } x=1\end{cases}$,
then $f'(1)$ is equal to
(a) $-(1/9)$ (b) $-(2/9)$ (c) -13 (d) $1/3$

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:
(a) Statement I is false, Statement II is true.
(b) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(c) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(d) Statement I is true, Statement II is false.

22. Statement I: $\lim\limits_{x\to\infty}\dfrac{5x^3-4x^2+2x-1}{7x^3+x-1}=\dfrac{5}{7}$

Statement II: If $f(x)$ and $g(x)$ are polynomials of the same degree, then $\lim\limits_{x\to\infty}\dfrac{f(x)}{g(x)}$ is a finite non-zero number.

23. Statement I: $\lim\limits_{x\to0}\dfrac{\sqrt{1-\cos2x}}{x}$ does not exist.

Statement II: $|\sin x|=\begin{cases}\sin x, & 0<x<\pi/2\\ -\sin x, & -\pi/2<x<0\end{cases}$

24. Statement I: $\lim\limits_{x\to\infty}\left(\dfrac{a_1^{1/x}+a_2^{1/x}+\cdots+a_n^{1/x}}{n}\right)^{nx}=\prod\limits_{i=1}^{n}a_i$,
where $a_i > 0$ for $i = 1, 2,\ldots, n$.

Statement II: If $\lim\limits_{x\to a}f(x)\to1$ and $\lim\limits_{x\to a}g(x)\to\infty$, then $\lim\limits_{x\to a}[f(x)]^{g(x)}=\lim\limits_{x\to a}e^{[f(x)-1]g(x)}$

Previous Years' Questions

25. $\lim_{x\to 0}\dfrac{\sqrt{1-\cos 2x}}{\sqrt{2x}}$ is

(a) 1 (b) –1
(c) 0 (d) does not exist

[AIEEE 2002]

26. $\lim_{x\to\infty}\left(\dfrac{x^2+5x++3}{x^2+x+2}\right)^x$ is equal to

(a) e^4 (b) e^2
(c) e^3 (d) e **[AIEEE 2002]**

27. For $x\in\mathbb{R}$, $\lim_{x\to\infty}\left(\dfrac{x-3}{x+2}\right)^x$ is

(a) e (b) e^{-1}
(c) e^{-5} (d) e^5 **[AIEEE 2002]**

28. If $f(a) = g(a) = k$ and their nth derivatives $f''(a)$ and $g^n(a)$ exist and are not equal for some n, and further if

$$\lim_{x\to a}\frac{f(a)(g)(x)-f(a)-g(a)f(x)+g(a)}{g(x)-f(x)}=4,$$

then the value of k is

(a) 0 (b) 4
(c) 2 (d) 1 **[AIEEE 2003]**

29. $\lim_{x\to\frac{\pi}{2}}\dfrac{[1-\tan(x/2)][1-\sin x]}{[1+\tan(x/2)][\pi-2x]^3}$ is

(a) ∞ (b) 1/8
(c) 0 (d) 1/32

[AIEEE 2003]

30. If $\lim_{x\to 0}\dfrac{\log(3+x)-\log(3-x)}{x}=k$, the value of k is

(a) –1/3 (b) 2/3
(c) –2/3 (d) 0

[AIEEE 2003]

31. The value of $\lim_{x\to 0}\dfrac{\int_0^{x^2}\sec^2 t\,dt}{x\sin x}$ is

(a) 2 (b) 1
(c) 0 (d) 3

[AIEEE 2003]

32. Let $f(x)=\dfrac{1-\tan x}{4x-\pi}, x\neq\dfrac{\pi}{4}, x\in\left[0,\dfrac{\pi}{2}\right]$, $f(x)$ is continuous in $\left[0,\dfrac{\pi}{2}\right]$, then $f\left(\dfrac{\pi}{4}\right)$ is

(a) –(1/2) (b) 1/2
(c) 1 (d) –1

[AIEEE 2004]

33. If $\lim_{x\to\infty}\left(1+\dfrac{a}{x}+\dfrac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b are

(a) $a\in\mathbb{R}, b=2$ (b) $a=1, b\in\mathbb{R}$
(c) $a\in\mathbb{R}, b\in\mathbb{R}$ (d) $a=1$ and $b=2$

[Jamia Millia Islamia 2007, AIEEE 2004]

34. Let α and β be the distinct roots of $ax^2+bx+c=0$, then $\lim_{x\to\alpha}\dfrac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$ is equal to

(a) 0 (b) $\dfrac{a^2}{2}(\alpha-\beta)^2$
(c) $\dfrac{1}{2}(\alpha-\beta)^2$ (d) $-\dfrac{a}{2}(\alpha-\beta)^2$

[AIEEE 2005]

35. The function $f:\mathbb{R}/\{0\}\to\mathbb{R}$ given by

$f(x)=\dfrac{1}{x}-\dfrac{2}{e^{2x}-1}$ can be made continuous at $x = 0$ by defining $f(0)$ as

(a) –1 (b) 0
(c) 1 (d) 2 **[AIEEE 2007]**

36. $\lim_{x\to 1}(1-x)\tan\left(\dfrac{\pi x}{2}\right)$ is equal to

(a) $\pi/2$ (b) π
(c) $2/\pi$ (d) 0

[Jamia Millia Islamia 2004]

37. If $f(x)=\begin{cases}\dfrac{[x]^2+\sin[x]}{[x]} & \text{for } [x]\neq 0\\ 0 & \text{for } [x]=0\end{cases}$ where $[x]$ denotes the greatest integer less than or equal to x, then $\lim_{x\to 0} f(x)$ equals

(a) 1 (b) 0
(c) –1 (d) none of these

[Jamia Millia Islamia 2005]

38. The value of $\lim_{x\to 0}\dfrac{e^x+\log(1+x)-(1-x)^{-2}}{x^2}$ is equal to

(a) 0 (b) –3
(c) –1 (d) infinity

[Jamia Millia Islamia 2006]

39. $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}e^{r/n}$ is

(a) e (b) $e-1$ (c) $1-e$ (d) $e+1$

[Jamia Millia Islamia 2007]

40. $\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}$ is

(a) 0 (b) 1/2

(c) 2 (d) none of these

[Jamia Millia Islamia 2007]

41. The value of $\lim_{x\to 0}\frac{5^x-5^{-x}}{2x}$ is

(a) log 5 (b) 0 (c) 1 (d) 2 log 5

[Jamia Millia Islamia 2008]

42. $\lim_{x\to\infty}\frac{\sqrt{x^2+1}-\sqrt[3]{x^3+1}}{\sqrt[4]{x^4+1}-\sqrt[5]{x^4+1}}$ equals

(a) 1 (b) 0

(c) −1 (d) none of these

[Jamia Millia Islamia 2009]

43. A function $f(x)=\begin{cases}1+x, & x\le 2\\ 5-x, & x>2\end{cases}$ is

(a) not continuous at $x=2$

(b) differentiable at $x=2$

(c) continuous but not differentiable at $x=2$

(d) none of these

[Jamia Millia Islamia 2009]

44. If $f(x)=\left|\log_e|x|\right|$, then $f'(x)$ equals

(a) $\frac{1}{|x|}$, $x\neq 0$

(b) $\frac{1}{x}$ for $|x|>1$ and $\frac{-1}{x}$ for $|x|<1$

(c) $\frac{-1}{x}$ for $|x|>1$ and $\frac{1}{x}$ for $|x|<1$

(d) $\frac{1}{x}$ for $x>0$ and $-\frac{1}{x}$ for $x<0$

[Jamia Millia Islamia 2009]

45. The set of points where the function $f(x)=x|x|$ is differentiable is

(a) $(-\infty,\infty)$ (b) $(-\infty,0)\cup(0,\infty)$

(c) $(0,\infty)$ (d) $[0,\infty)$

[Jamia Millia Islamia 2009]

46. The value of $\lim_{h\to 0}\frac{\log(1+2h)-2\log(1+h)}{h^2}$ is

(a) 1 (b) −1

(c) 0 (d) none of these

[IIT 1997]

47. The value of $\lim_{x\to 0}\frac{\int_0^{x^2}\cos t^2\,dt}{x\sin x}$ is

(a) 3/2 (b) 1

(c) −1 (d) none of these

[IIT 1997]

48. The value of $\lim_{x\to 0}\left(\frac{1+5x^2}{1+3x^2}\right)^{1/x^2}$ is

(a) e^2 (b) e

(c) e^{-1} (d) none of these

[IIT 1996, CCE (Delhi) 2001, 2002]

49. $\lim_{x\to 1}\frac{\sqrt{1-\cos 2(x-1)}}{x-1}$

(a) exists and it equals $\sqrt{2}$

(b) exists and it equals $-\sqrt{2}$

(c) does not exist because $(x-1)\to 0$

(d) does not exist because the left-hand limit is not equal to the right-hand limit

[IIT 1998]

50. $\lim_{x\to\pi/4}\frac{\int_2^{\sec^2 x} f(t)dt}{x^2-\pi^2/16}$ equals

(a) $\frac{8}{\pi}f(2)$ (b) $\frac{2}{\pi}f(2)$

(c) $\frac{2}{\pi}f\left(\frac{1}{2}\right)$ (d) $4f(2)$

[IIT 2007]

51. Let $L=\lim_{x\to 0}\frac{a-\sqrt{a^2-x^2}-x^2/4}{x^4}$, $a>0$. If L is finite, then

(a) $a=2, L=\frac{1}{64}$ (b) $a=1, L=\frac{1}{64}$

(c) $a=3, L=\frac{1}{32}$ (d) $a=1, L=\frac{1}{32}$

[IIT 2009]

52. The value of $\lim_{x\to 7}\frac{2-\sqrt{x-3}}{x^2-49}$ is

(a) 2/9 (b) −2/49 (c) 1/64 (d) −1/56

[MP PET 2003]

53. If $f(x)=(x+1)^{\cot x}$ is continuous at $x = 0$, then $f(0)$ is equal to

(a) 0 (b) $-e$
(c) e (d) none of these

[MP PET 2004]

54. If $\lim_{x\to a}\frac{a^x-x^a}{x^x-a^a}=-1$, then a is equal to

(a) 1 (b) 0
(c) e (d) $(1/e)$

[MP PET 2005]

55. Let $f(x)=\begin{cases}1, & \text{when } x \text{ is rational}\\ 0, & \text{when } x \text{ irrational}\end{cases}$, then $\lim_{x\to 0} f(x)$ is

(a) 0 (b) 1
(c) 1/2 (d) none of these

[MP PET 2006]

56. Define $F(x)$ as the product of two real functions

$f_1(x)=x, x\in\mathbb{R}$ $f_2(x)=\begin{cases}\sin(1/x), & \text{If } x\neq 0\\ 0, & \text{If } x=0\end{cases}$ as follows:

Statement 1: $F(x)$ is continuous on $\mathbb{R}$.
Statement 2: $f_1(x)$ and $f_2(x)$ are continuous on $\mathbb{R}$.

(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is false.
(d) Statement 1 is false, Statement 2 is true.

[AIEEE 2011]

57. Let $f:\mathbb{R}\to[0,\infty)$ be such that $\lim_{x\to 5} f(x)$ exists and $\lim_{x\to 5}\frac{[f(x)]^2-9}{\sqrt{|x-5|}}=0$. Then $\lim_{x\to 5} f(x)$ equals

(a) 0 (b) 1
(c) 2 (d) 3

[AIEEE 2011]

58. If $f:\mathbb{R}\to\mathbb{R}$ is a function defined by

$$f(x)=[x]\cos\frac{2x-1}{2}\pi$$

where $[x]$ denotes the greatest integer function, then f is

(a) continuous for every real x
(b) discontinuous only at $x = 0$
(c) discontinuous only at non-zero integral values of x
(d) continuous only at $x = 0$

[AIEEE 2012]

59. Consider the function, $f(x) = |x - 2| + |x - 5|$, $x \in R$.
Statement 1: $f'(4) = 0$
Statement 2: f is continuous in [2, 5], differentiable in (2, 5) and $f(2) = f(5)$.

(a) Statement 1 is false, statement 2 is true
(b) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1
(c) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1
(d) Statement 1 is true, statement 2 is false

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (b)	**2.** (d)	**3.** (a)	**4.** (d)	**5.** (a)	**6.** (a)	**7.** (c)	**8.** (b)	**9.** (d)	**10.** (d)
11. (a)	**12.** (d)	**13.** (c)	**14.** (c)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (b)	**19.** (c)	**20.** (a)
21. (b)	**22.** (b)	**23.** (d)	**24.** (b)	**25.** (a)	**26.** (a)	**27.** (a)	**28.** (c)	**29.** (a)	**30.** (a)
31. (c)	**32.** (b)	**33.** (c)	**34.** (a)	**35.** (c)	**36.** (a)	**37.** (b)	**38.** (a)	**39.** (d)	**40.** (d)
41. (c)	**42.** (b)	**43.** (a)	**44.** (a)	**45.** (d)	**46.** (a)	**47.** (c)	**48.** (c)	**49.** (d)	**50.** (d)
51. (a)	**52.** (d)	**53.** (a)							

Exercise Set 2

1. (c)	**2.** (c)	**3.** (b)	**4.** (d)	**5.** (a)	**6.** (b)	**7.** (c)	**8.** (b)	**9.** (d)	**10.** (c)
11. (c)	**12.** (c)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (a)	**19.** (c)	**20.** (c)
21. (b)	**22.** (c)	**23.** (b)	**24.** (b)	**25.** (d)	**26.** (a)	**27.** (c)	**28.** (b)	**29.** (d)	**30.** (b)
31. (b)	**32.** (a)	**33.** (b)	**34.** (b)	**35.** (c)	**36.** (c)	**37.** (c)	**38.** (b)	**39.** (b)	**40.** (b)
41. (a)	**42.** (b)	**43.** (c)	**44.** (b)	**45.** (a)	**46.** (b)	**47.** (b)	**48.** (a)	**49.** (d)	**50.** (a)
51. (a)	**52.** (d)	**53.** (c)	**54.** (a)	**55.** (b)	**56.** (c)	**57.** (d)	**58.** (a)	**59.** (c)	

HINTS AND SOLUTIONS—EXERCISE SET 1

1. $\lim_{\theta\to 0}\dfrac{4\theta(\tan\theta-\sin\theta)}{(1-\cos 2\theta)^2}=\lim_{\theta\to 0}\dfrac{4\theta\sin\theta(1-\cos\theta)}{(2\sin^2\theta)^2\cdot\cos\theta}$

$=\lim_{\theta\to 0}\dfrac{\theta\cdot(1-\cos\theta)}{\sin\theta\cdot\sin^2\theta\cdot\cos\theta}$

$=\lim_{\theta\to 0}\dfrac{\theta}{\sin\theta}\cdot\lim_{\theta\to 0}\dfrac{(1-\cos\theta)(1+\cos\theta)}{\sin^2\theta.\cos\theta(1+\cos\theta)}$

$=\dfrac{1}{\lim_{\theta\to 0}\left(\dfrac{\sin\theta}{\theta}\right)}\cdot\lim_{\theta\to 0}\dfrac{(1-\cos^2\theta)}{\sin^2\theta\cos\theta(1+\cos\theta)}$

$=\dfrac{1}{1}\cdot\lim_{\theta\to 0}\dfrac{\sin^2\theta}{\sin^2\theta.\cos\theta(1+\cos\theta)}$

$=\lim_{\theta\to 0}\dfrac{1}{\cos\theta(1+\cos\theta)}=\dfrac{1}{2}$

2. $\lim_{x\to 0}\dfrac{4^x-1}{3^x-1}=\lim_{x\to 0}\dfrac{4^x-1}{x}\cdot\dfrac{x}{3^x-1}$

$=\left[\lim_{x\to 0}\left(\dfrac{4^x-1}{x}\right)\right]\cdot\dfrac{1}{\left[\lim_{x\to 0}\left(\dfrac{3^x-1}{x}\right)\right]}$

$=\dfrac{\log_e 4}{\log_e 3}=\log_3 4.$

3. Since α is a repeated root of $ax^2+bx+c=0$, so we have $ax^2+bx+c=a(x-\alpha)^2$

Now, $\lim_{x\to\alpha}\dfrac{\sin(ax^2+bx+c)}{(x-\alpha)^2}=\lim_{x\to\alpha}\dfrac{a\sin\left[a(x-\alpha)^2\right]}{a(x-\alpha)^2}$

$=a\lim_{x\to\alpha}\dfrac{\sin\left[a(x-\alpha)^2\right]}{a(x-\alpha)^2}=a\times 1=a.$

$\left[\because as\ x\to\alpha, a(x-\alpha)^2\to 0\right]$

4. Let $f(x)=\dfrac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x}=\dfrac{\sqrt{\frac{1}{2}(2\sin^2 x)}}{x}=\dfrac{|\sin x|}{x}.$

Then, $\lim_{x\to 0-}f(x)=\lim_{h\to 0}f(0-h)=\lim_{h\to 0}f(-h)$

$=\lim_{h\to 0}\dfrac{|\sin(-h)|}{(-h)}=\lim_{h\to 0}\dfrac{|-\sin h|}{-h}$

$=-\lim_{h\to 0}\dfrac{\sin h}{h}=-1,$

and $\lim_{x\to 0+}f(x)=\lim_{h\to 0}f(0+h)=\lim_{h\to 0}f(h)=\lim_{h\to 0}\dfrac{|\sin h|}{h}$

$=\lim_{h\to 0}\dfrac{\sin h}{h}=1$

$\therefore\ \lim_{x\to 0-}f(x)\neq\lim_{x\to 0+}f(x)$

Hence, $\lim_{x\to 0}f(x)$, i.e. $\lim_{x\to 0}\dfrac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x}$ does not exist.

5. Let $L=\lim_{x\to -2}\dfrac{\sin^{-1}(x+2)}{x^2+2x}=\lim_{x\to -2}\dfrac{\sin^{-1}(x+2)}{x(x+2)}.$

Put $t=x+2$. Then, as $x\to -2, t\to 0$. Also, $x=t-2$.

$\therefore\ L=\lim_{t\to 0}\dfrac{\sin^{-1}t}{(t-2)t}=\lim_{t\to 0}\dfrac{\sin^{-1}t}{t}\cdot\lim_{t\to 0}\dfrac{1}{(t-2)}$

$=1\times\dfrac{1}{(0-2)}=1\times\left(-\dfrac{1}{2}\right)=-\dfrac{1}{2}\quad\left[\because\lim_{t\to 0}\dfrac{\sin^{-t}t}{t}=1\right]$

6. LHL $=\lim_{x\to 2-}\dfrac{f(x)-f(2)}{x-2}=\lim_{h\to 0}\dfrac{f(2-h)-f(2)}{(2-h)-2}$

$=\lim_{h\to 0}\dfrac{f(2-h)-f(2)}{-h}\qquad\left[\dfrac{0}{0}\text{ form}\right]$

$=\lim_{h\to 0}\dfrac{-f'(2-h)}{-1}=\lim_{h\to 0}f'(2-h)$

$=\lim_{h\to 0}\dfrac{-1}{[(2-h)-1]^2}$

$\left[\because f(x)=\dfrac{x-2}{x^2-3x+2}\quad\text{when } x\neq 1,2\right.$

$\left.=\dfrac{1}{x-1}\Rightarrow f'(x)=\dfrac{-1}{(x-1)^2}\quad\text{when } x\neq 1,2\right]$

$=\lim_{h\to 0}\dfrac{-1}{(1-h)^2}=\dfrac{-1}{(1-0)^2}=-1.$

RHL $=\lim_{x\to 2+}\dfrac{f(x)-f(2)}{x-2}=\lim_{h\to 0}\dfrac{f(2+h)-f(2)}{(2+h)-2}$

$=\lim_{h\to 0}\dfrac{f(2+h)-f(2)}{h}\quad\left[\dfrac{0}{0}\text{ form}\right]$

$$= \lim_{h\to 0}\frac{f'(2+h)}{1} = \lim_{h\to 0}\frac{-1}{\{(2+h)-1\}^2}$$

$$\left[\because\ f'(x) = \frac{-1}{(x-1)^2} \text{ when } x \neq 1, 2\right]$$

$$= \lim_{h\to 0}\frac{-1}{(h+1)^2} = \frac{-1}{(0+1)^2} = -1$$

$\therefore$ LHL = RHL $= -1$.

Hence, $\lim\limits_{x\to 2}\dfrac{f(x)-f(2)}{x-2} = -1$.

7. $\lim\limits_{x\to 0}\left\{\dfrac{3x-\sin^{-1}x}{4x-\tan^{-1}x}\right\} = \lim\limits_{x\to 0}\left\{\dfrac{3-\dfrac{\sin^{-1}x}{x}}{4-\dfrac{\tan^{-1}x}{x}}\right\} = \dfrac{3-1}{4-1} = \dfrac{2}{3}$.

$$\left[\because\ \lim_{x\to 0}\frac{\sin^{-1}x}{x} = 1 \text{ and } \lim_{x\to 0}\frac{\tan^{-1}x}{x} = 1\right]$$

8. $\lim\limits_{x\to 0}\dfrac{\sin\sqrt[3]{x}\log(1-5x)}{(\tan^{-1}\sqrt{x})^2(e^{\sqrt[3]{x}}-1)}$

$$= \lim_{x\to 0}\frac{\sin x^{1/3}}{x^{1/3}}\cdot\frac{x^{1/3}}{(e^{x^{1/3}}-1)}\cdot\frac{\log(1-5x)}{(-5x)}\cdot\frac{x}{(\tan^{-1}\sqrt{x})^2}\cdot(-5)$$

$$= (-5)\left\{\lim_{x\to 0}\frac{\sin x^{1/3}}{x^{1/3}}\right\}\cdot\left\{\lim_{x\to 0}\frac{x^{1/3}}{e^{x^{1/3}}-1}\right\}\cdot\left\{\lim_{x\to 0}\frac{\log(1-5x)}{(-5x)}\right\}\cdot\left\{\lim_{x\to 0}\left(\frac{\sqrt{x}}{\tan\sqrt{x}}\right)^2\right\}$$

$$= (-5)\times 1\times 1\times 1\times 1^2 = -5.$$

9. Let $f(x) = \dfrac{(1-\cos x)^2}{\tan^3 x - \sin^3 x}$

Then, $\lim\limits_{x\to 0+} f(x) = \lim\limits_{h\to 0} f(0+h) = \lim\limits_{h\to 0} f(h)$

$$= \lim_{h\to 0}\frac{(1-\cos h)^2}{\tan^3 h - \sin^3 h}$$

$$= \lim_{h\to 0}\frac{\cos^3 h(1-\cos h)^2}{\sin^3 h(1-\cos^3 h)}$$

$$= \lim_{h\to 0}\frac{\cos^3 h(1-\cos h)}{\sin^3 h(1+\cos h+\cos^2 h)}$$

$$= \lim_{h\to 0}\frac{\cos^3 h\cdot 2\sin^2\dfrac{h}{2}}{8\sin^3\dfrac{h}{2}\cos^3\dfrac{h}{2}\cdot(1+\cos h+\cos^2 h)}$$

$$= \lim_{h\to 0}\frac{\cos^3 h}{4\left(\sin\dfrac{h}{2}\right)\cos^3\dfrac{h}{2}(1+\cos h+\cos^2 h)}$$

$$= \frac{1}{12}\cdot\frac{1}{0} = \infty,$$

$$= \lim_{h\to 0-} f(x) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} f(-h)$$

$$= \lim_{h\to 0}\frac{(1-\cos(-h)^2}{\tan^3(-h)-\sin^3(-h)}$$

$$= \lim_{h\to 0}\frac{\cos^3 h(1-\cos h)^2}{\sin^3 h(\cos^3 h-1)}$$

$$= -\left\{\lim_{h\to 0}\frac{\cos^3 h(1-\cos h)^2}{\sin^3 h(1-\cos^3 h)}\right\} = -\infty.$$

[solving as above]

$\therefore$ $= \lim\limits_{x\to 0+} f(x) \neq \lim\limits_{x\to 0-} f(x)$.

Hence, $\lim\limits_{x\to 0} f(x)$, i.e. $\lim\limits_{x\to 0}\dfrac{(1-\cos x)^2}{\tan^3 x - \sin^3 x}$ does not exist.

10. $\lim\limits_{x\to -\infty} x + \sqrt{x^2 + x^2\sin\left(\dfrac{1}{x}\right)}$

$$= \lim_{x\to -\infty}\left\{\frac{x^2\left[x^2+x^2\sin(1/x)\right]}{x-\sqrt{x^2+x^2\sin(1/x)}}\right\}$$

$$= \lim_{x\to -\infty}\frac{-x^2\sin(1/x)}{x-\sqrt{x^2+x^2\sin(1/x)}}$$

$$= \lim_{x\to -\infty}\frac{-x^2\sin(1/x)}{x\left[1+\sqrt{1+\sin(1/x)}\right]}$$

[Note: As $x\to -\infty \Rightarrow \sqrt{x^2} = -x$]

$$= \frac{\lim\limits_{x\to -\infty}\left\{\dfrac{-\sin(1/x)}{(1/x)}\right\}}{\lim\limits_{x\to -\infty}\left\{1+\sqrt{1+\sin(1/x)}\right\}} = \frac{-1}{1+\sqrt{1+0}} = \frac{-1}{2}.$$

11. $\lim\limits_{n\to\infty}\left\{\dfrac{1+\dfrac{1}{3}+\dfrac{1}{9}+\cdots \text{ to } (n+1) \text{ terms}}{1+\dfrac{1}{2}+\dfrac{1}{4}+\cdots \text{ to } (n+1) \text{ terms}}\right\}$

$$= \lim_{n\to\infty}\frac{\left(1-\dfrac{1}{3^n+1}\right)}{\left(1-\dfrac{1}{3}\right)}\cdot\frac{\left(1-\dfrac{1}{2}\right)}{\left(1-\dfrac{1}{2^n+1}\right)}$$

$$= \frac{3}{4}\cdot\lim_{n\to\infty}\frac{\left(1-\dfrac{1}{3^n+1}\right)}{\left(1-\dfrac{1}{2^n+1}\right)} = \frac{3}{4}\cdot\frac{(1-0)}{(1-0)} = \frac{3}{4}.$$

12. $\lim_{n\to\infty}\left(\frac{1+a+a^2+\cdots+a^{n-1}}{1+b+b^2+\cdots+b^{n-1}}\right)$

$=\lim_{n\to\infty}\left\{\frac{(a^n-1)\cdot(b-1)}{(a-1)\cdot b^n-1}\right\}$

$=\frac{(b-1)}{(a-1)}\cdot\lim_{n\to\infty}\frac{(a^n-1)}{(b^n-1)}$

$=\frac{(b-1)}{(a-1)}\cdot\lim_{n\to\infty}\left\{\frac{\left(\frac{a}{b}\right)^n-\frac{1}{b^n}}{\left(1-\frac{1}{b^n}\right)}\right\}.$

Now, the given limit = 0

$\Leftrightarrow \lim_{n\to\infty}\left(\frac{a}{b}\right)^n=0$ and $\lim_{n\to\infty}\frac{1}{b^n}=0$

$\Leftrightarrow 1<a<b.$

13. The given series is

$$S_n=\frac{1}{1\cdot3}+\frac{1}{3\cdot5}+\frac{1}{5\cdot7}+\cdots+\frac{1}{(2n-1)\cdot(2n+1)}.$$

Its nth term $t_n=\frac{1}{(2n-1)\cdot(2n+1)}=\frac{1}{2}\left[\frac{1}{2n-1}-\frac{1}{2n+1}\right].$

$\therefore\quad S_n=\frac{1}{2}\left[\left(1-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{7}\right)+\cdots\right.$

$\left.+\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)\right]$

$=\frac{1}{2}\left(1-\frac{1}{2n+1}\right).$

$\therefore$ Given limit $=\lim_{n\to\infty}S_n=\lim_{n\to\infty}\frac{1}{2}\left(1-\frac{1}{2n+1}\right)$

$=\frac{1}{2}\lim_{n\to\infty}\left(1-\frac{1}{2n+1}\right)=\frac{1}{2}(1-0)=\frac{1}{2}.$

14. Let $f(x)=\left(\frac{1^x+2^x+3^x+\cdots+n^x}{n}\right).$

Then $\lim_{x\to0}f(x)=\frac{1^0+2^0+3^0+\cdots+n^0}{n}=\frac{n}{n}=1.$

$\therefore\quad \lim_{x\to0}\left[f(x)\right]^{g(x)}=e^{\lim_{x\to0}g(x)\{f(x)-1\}}$

Now, $\lim_{x\to0}\left(\frac{1^x+2^x+3^x+\cdots+n^x}{n}\right)^{a/x}$

$=e^{\lim_{x\to0}\frac{a}{x}\left\{\frac{1^x+2^x+3^x+\cdots+n^x}{n}-1\right\}}=e^{\lim_{x\to0}\left\{\frac{(1^x+2^x+3^x+\cdots+n^x)-n}{x}\right\}\frac{a}{n}}$

$=e^{\frac{a}{n}\lim_{x\to0}\left\{\frac{(1^x-1)}{x}+\frac{(2^x-1)}{x}+\frac{(3^x-1)}{x}+\cdots+\frac{(n^x-1)}{x}\right\}}$

$=e^{\frac{a}{n}(\log1+\log2+\log3+\cdots+\log n)}=e^{\frac{a}{n}\log(\underline{|n})}=e^{\log(\underline{|n})^{a/n}}$

$=(\underline{|n})^{a/n}$

15. Let $f(x)=\frac{1^{\cos^2x}+2^{\cos^2x}+\cdots+n^{\cos^2x}}{n}.$

Then, $\lim_{x\to\frac{\pi}{2}}f(x)=\frac{1^0+2^0+\cdots+n^0}{n}$

$=\frac{1+1+\cdots n\text{ terms}}{n}=\frac{n}{n}=1.$

$\therefore\quad \lim_{x\to\frac{\pi}{2}}[f(x)]^{g(x)}=e^{\lim_{x\to\frac{\pi}{2}}[f(x)-1]g(x)}$

$\therefore\quad \lim_{x\to\frac{\pi}{2}}\left[\frac{1^{\cos^2x}+2^{\cos^2x}+\cdots+n^{\cos^2x}}{n}\right]^{\frac{1}{\cos^2x}}$

$=e^{\lim_{x\to\frac{\pi}{2}}\left[\frac{1^{\cos^2x}+2^{\cos^2x}+\cdots+n^{\cos^2x}}{n}-1\right]\frac{1}{\cos^2x}}$

$=e^{\lim_{x\to\frac{\pi}{2}}\left[\frac{\left(1^{\cos^2x}+2^{\cos^2x}+\cdots+n^{\cos^2x}\right)-n}{\cos^2x}\right]\frac{1}{n}}$

$=e^{\lim_{x\to\frac{\pi}{2}}\frac{1}{n}\left[\frac{1^{\cos^2x}-1}{\cos^2x}+\frac{2^{\cos^2x}-1}{\cos^2x}+\cdots+\frac{n^{\cos^2x}-1}{\cos^2x}\right]}$

$=e^{\frac{1}{n}\left[\lim_{\cos^2x\to0}\left(\frac{1^{\cos^2x}-1}{\cos^2x}\right)+\lim_{\cos^2x\to0}\left(\frac{2^{\cos^2x}-1}{\cos^2x}\right)+\cdots+\lim_{\cos^2x\to0}\left(\frac{n^{\cos^2x}-1}{\cos^2x}\right)\right]}$

$=e^{\frac{1}{n}(\log1+\log2+\cdots+\log n)}=e^{\frac{1}{n}\log(1.2.3\ldots n)}=e^{\log(\underline{|n})^{1/n}}$

$=(\underline{|n})^{1/n}$

16. Let $y=\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\ldots\infty}}}}$

$\Rightarrow\quad y=\sqrt{\left(1-\cos\frac{1}{x}\right)y}\Rightarrow y^2-\left(1-\cos\frac{1}{x}\right)y=0$

$\Rightarrow\quad y=0$ or $y=\left(1-\cos\frac{1}{x}\right)$

$$L=\lim_{x\to\infty}x^2\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\ldots\infty}}}}$$

$$=\lim_{x\to\infty}x^2y \;=\lim_{x\to\infty}x^2\left(1-\cos\frac{1}{x}\right)$$

Put $\frac{1}{x}=t$. Then as $x\to\infty, t\to 0$.

$$\therefore\quad L=\lim_{t\to0}\frac{1}{t^2}(1-\cos t)=\lim_{t\to0}\frac{2\sin^2(t/2)}{(t/2)^2\cdot4}$$

$$=\frac{1}{2}\left[\lim_{t\to0}\frac{\sin(t/2)}{(t/2)}\right]^2$$

$$=\frac{1}{2}\times1^2=\frac{1}{2}.$$

17. $f(2)=k$

$$\lim_{x\to2}f(x)=\lim_{x\to2}\frac{x^3+x^2-16x+20}{(x-2)^2}$$

$$=\lim_{x\to2}\frac{(x-2)^2(x+5)}{(x-2)^2}$$

$$=\lim_{x\to2}(x+5)=7$$

$\Rightarrow\ k=7$

18. $f(x)=\sin^{-1}(\cos x)=\sin^{-1}\left[\sin\left(\frac{\pi}{2}-x\right)\right]$

$=\frac{\pi}{2}-x$ in the vicinity of 0

$\Rightarrow\ \sin^{-1}(\cos x)$ is continuous at zero.

19. At $x=2,\ f(x)=4$

$$\lim_{x\to2^+}f(x)\lim_{x\to2^+}(3x-2)=4$$

$$\lim_{x\to2^-}f(x)=\lim_{x\to2^-}(x+2)=4$$

Therefore, the function is continuous for $x\geq1$.

20. $f'(0^+)\lim_{h\to0^+}=\frac{f(0+h)-f(0)}{h}=\lim_{h\to0^+}\frac{h}{h\left(1+e^{1/h}\right)}$

$$=\lim_{h\to0^+}\frac{1}{1+e^{1/h}}=0$$

$$f'(0^-)\lim_{h\to0^-}=\frac{f(h)-f(0)}{h}=\lim_{h\to0^-}\frac{h}{h\left(1+e^{-1/h}\right)}$$

$$=\lim_{h\to0^-}\frac{1}{1+e^{-1/h}}=+1$$

21. $f(x)=1+|x|$ and $g(x)=\sin x$ are continuous everywhere. Therefore $f\circ g$ is continuous everywhere.

$\Rightarrow\ 1+|\sin x|$ is continuous everywhere.

At $x=0$

$$\lim_{h\to0^+}\frac{1+|\sin h|-1}{h}=\lim_{h\to0^+}\frac{\sin h}{h}=1$$

$$\lim_{h\to0^-}\frac{1-\sin h-1}{h}=-1$$

$\Rightarrow$ It is not differentiable at x = 0.

22. For all $x, f(x)$ is differentiable, but the only point where differentiability is to be examined is $x=0$.

$$\lim_{h\to0^+}\frac{f(h)-f(0)}{h}=\lim_{h\to0^+}\frac{h^2}{h}=0$$

$$\lim_{h\to0^-}\frac{f(h)-f(0)}{h}=\lim_{h\to0^-}\frac{0}{h}=0$$

$\Rightarrow f$ is differentiable.

23. To examine $f(x)$ at $x=0$ (as at rest of the points, it is continuous).

$$f(0)=0$$

$$\lim_{x\to0^+}f(x)=\lim_{x\to0^+}\frac{1}{4^x-1}=+\infty$$

$$\lim_{x\to0^-}f(x)=\lim_{x\to0^-}\frac{1}{4^x-1}=-\infty$$

24. $f(1)=0,\ f(x)$ is continuous.

$$\lim_{h\to0^+}\frac{f(1+h)-f(1)}{h}=\lim_{h\to0^+}\frac{(1+h)^3-1}{h}$$

$$=\lim_{h\to0^+}\frac{h^3+3h^3+3h}{h}$$

$$=3$$

$$\lim_{h\to0^-}\frac{f(1+h)-f(1)}{h}=\lim_{h\to0^-}\frac{1+h-1}{h}=1$$

$\Rightarrow\ f(x)$ is not differentiable.

25. $\lim_{x\to0^+}f(x)=\lim_{x\to0^+}|x-1|=1$

[$\because\ f(0)=1$]

$$\lim_{x\to0^-}f(x)=\lim_{x\to0^-}e^x=1$$

$\Rightarrow$ $f(x)$ is continuous at $x = 0$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} x - 1 = 0$$

$[\because \quad f(1) = 0]$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} 1 - x = 0$$

$\Rightarrow$ $f(x)$ is continuous at $x = 0, 1$

At $x = 0$,

$$\lim_{h\to 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0^+} \frac{1-h-1}{h} = -1$$

$$\lim_{h\to 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0^-} \frac{e^h - 1}{h} = 1$$

Hence, it is not differentiable.

26. To examine this function at $x = 1$,

$$f(1) = 1$$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} x^2 = 1$$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \lim_{x\to 1^-} \frac{1-x}{|x-1|} = 1$$

Hence, it is continuous at all points.

27. To examine the continuity at $x = 0, f(0) = \frac{1}{2}$

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{1}{2} e^x = \frac{1}{2}$$

$$\lim_{x\to 0^-} \frac{1-\cos x}{x^2} = \frac{1}{2}$$

$\Rightarrow$ f is continuous.

To examine the differentiability at $x = 0$,

$$\lim_{h\to 0^+} \frac{f(h) - \frac{1}{2}}{h} = \lim_{h\to 0^+} \frac{1}{2}\left(\frac{e^h - 1}{h}\right) = 1$$

$$\lim_{h\to 0^-} \frac{f(h) - \frac{1}{2}}{h} = \lim_{h\to 0^-} \frac{1}{h}\left(\frac{1-\cos h}{h^2} - \frac{1}{2}\right) = 0$$

$\Rightarrow$ f is not differentiable.

28. At $x = 1$, $f(x) = 1$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} |2x-3|[x] = 1$$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \sin\frac{\pi x}{2} = 1$$

$\Rightarrow$ $f(x)$ is continuous at $x = 1$.

$$\lim_{h\to 0^+} \frac{f(1+h) - 1}{h} = \lim_{h\to 0^+} \frac{|2h-1| - 1}{h}$$

$$= \lim_{h\to 0^+} \frac{1 - 2h - 1}{h} = -2$$

$$\lim_{h\to 0^-} \frac{f(1+h) - 1}{h} = \lim_{h\to 0^-} \frac{\sin\left(\frac{\pi}{2} + \frac{\pi h}{2}\right) - 1}{h}$$

$$= \lim_{h\to 0^-} \frac{\cos\frac{\pi h}{2} - 1}{h} = 0$$

$\Rightarrow$ $f(x)$ is not differentiable at $x = 1$.

29. At $x = 1$, $f(x) = 6$

$$\lim_{x\to 1} f(x) = \lim_{x\to 1} \frac{(x-2)(x+2)(x-1)(x+1)}{(x-1)(x-2)}$$

$$= \lim_{x\to 1} (x+1)(x+2) = 6$$

$\Rightarrow$ $f(x)$ is continuous at $x = 1$.

Similarly, it can be shown that it is continuous at $x = 2$.

$\Rightarrow$ $f(x)$ is continuous for all on the set $\mathbb{R}$.

30. $f(x) = \frac{x}{1+|x|}$

$\Rightarrow$
$$f(x) = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$$

Let us check the differentiability of $f(x)$ at $x = 0$.

$\Rightarrow$ $f(0^-) = \lim_{h\to 0} \frac{f(0-h) - f(0)}{-h}$

$\Rightarrow$ $f'(0^-) = \lim_{h\to 0} \frac{\frac{0-h}{1-(0-h)} - 0}{-h}$

$\Rightarrow$ $f'(0^-) = \lim_{h\to 0} \frac{-h}{(1+h)(-h)}$

Now, $f'(0^+) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$

$\Rightarrow$ $f'(0^+) = \lim_{h\to 0} \frac{\frac{0+h}{1+(0+h)} - 0}{h}$

$\Rightarrow$ $f'(0^+) = \lim_{h\to 0} \frac{1}{1+h}$ $[\because \quad h \to 0 \text{ and } h \neq 0]$

$\therefore \quad f'(0^+)=1$

Since $f'(0^-)=f(0^+)$

$\Rightarrow$ $f(x)$ is differentiable at $x = 0$

$\therefore$ $f(x)$ is differentiable for all $x \in \mathbb{R}$.

31. $\lim\limits_{n\to\infty}\dfrac{1^3+2^3+\cdots+n^3}{n^5}$

$$=\lim_{n\to\infty}\frac{\left(\frac{n(n+1)}{2}\right)^2}{n^5}=\frac{1}{4}\lim_{n\to\infty}\frac{(n+1)^2}{n^3}$$

$$=\frac{1}{4}\lim_{n\to\infty}\frac{1}{n}\left(1+\frac{1}{n}\right)^2=\frac{1}{4}\cdot 0(1+0)^2=0.$$

$\therefore$ Required limit $=\lim\limits_{n\to\infty}\dfrac{1^4+2^4+\cdots+n^4}{n^5}-0$

$$=\lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{n^4}+\frac{2^4}{n^4}+\cdots+\frac{n^4}{n^4}\right]$$

$$=\int_0^1 x^4dx=\left[\frac{x^5}{5}\right]_0^1=\frac{1}{5}-0=\frac{1}{5}.$$

32. (Ans. –1) use $1-\cos x=2\sin^2\dfrac{x}{2}$

and $\sin x=2\sin\dfrac{x}{2}\cos\dfrac{x}{2}$.

33. $\lim\limits_{x\to1}\dfrac{x+x^2\cdots+x^n-n}{x-1}$ $\left(\dfrac{0}{0}\text{ form}\right)$ (L'Hôpital's rule)

$$=\lim_{x\to1}\frac{1+2x+\cdots+nx^{n-1}}{1}=1+2+\cdots+n$$

$$=\frac{n(n+1)}{2}$$

34. Since $f(x)$ is continuous at $x = 0$,

$\therefore$ $\lim\limits_{x\to0^-}f(x)=f(0)=\lim\limits_{x\to0^+}f(x)$

$\Rightarrow$ $\lim\limits_{x\to0^-}(1+|\sin x|)^{\frac{a}{|\sin x|}}=b=\lim\limits_{x\to0^+}e^{\frac{\tan 2x}{\tan 3x}}$

$\Rightarrow$ $e^a=b=e^{2/3}$

$\therefore$ $a=\dfrac{2}{3}$ and $a=\log_e b$

35. Both $\sin|x|$ and $-e^{|x|}$ are continuous functions and the sum of the two continuous functions is continuous also.

$$f(x)=\sin x-e^x, x\le x<\pi$$

$$=-\sin x-e^{-x}, -\pi<x\le 0$$

Since the function is continuous,

$$f'(x)=\cos x-e^x, 0<x<\pi -\cos x+e^{-x}, -\pi<x<0$$

Clearly, $f'(0-)=f'(0+)=0$

36. $\lim\limits_{x\to1}\dfrac{x^n+x^{n-1}+x^{n-2}+\cdots+x^2+x-n}{x-1}$ $\left(\dfrac{0}{0}\text{ form}\right)$

$$=\lim_{x\to1}\left\{\frac{(x^n-1)}{(x-1)}+\frac{(x^{n-1}-1)}{(x-1)}+\cdots+\frac{(x-1)}{(x-1)}\right\}$$

$$=n+(n-1)+\cdots+2+1=\frac{n(n+1)}{2}$$

37. Let $E=\lim\limits_{x\to0}\dfrac{f(x)-f(0)}{x-0}$

$$\Rightarrow E=\lim_{x\to0}\frac{\log\cos x}{\log(1+x^2)}$$

$$\Rightarrow E=\lim_{x\to0}\frac{\log(1-1+\cos x)}{\log(1+x^2)}\cdot\frac{1-\cos x}{1-\cos x}$$

$$\Rightarrow E=\lim_{x\to0}\frac{\log[1-(1-\cos x)]}{1-\cos x}\cdot\frac{1-\cos x}{\log(1+x^2)}$$

$$\Rightarrow E=-\lim_{x\to0}\log\frac{[1-(1-\cos x)]}{-(1-\cos x)}\cdot\frac{2\sin^2\frac{x}{2}}{4\left(\frac{x}{2}\right)^2}\cdot\frac{x^2}{\log(1+x^2)}$$

$$\Rightarrow E=-\frac{1}{2}$$

Hence, $f(x)$ is differentiable at $x = 0$ and so continuous at $x = 0$.

38. $\lim\limits_{x\to\infty}\left[\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right]$

$$=\lim_{x\to\infty}\frac{x+\sqrt{x+\sqrt{x}}-x}{\sqrt{x+\sqrt{x+\sqrt{x}}}+\sqrt{x}}=\lim_{x\to\infty}=\frac{\sqrt{x+\sqrt{x}}}{\sqrt{x+\sqrt{x+\sqrt{x}}}+\sqrt{x}}$$

$$=\lim_{x\to\infty}\frac{\sqrt{x}\left(1+\frac{1}{\sqrt{x}}\right)^{1/2}}{\sqrt{x}\left[\left(1+\frac{1}{\sqrt{x}}\sqrt{1+\frac{1}{\sqrt{x}}}\right)^{1/2}+1\right]}=\frac{1}{1+1}=\frac{1}{2}$$

39. $f(x) = -x + x - 1, x < 0$

$= x - x - 1, x \geq 0$

$f(x) = -1 =$ a constant function, hence differential for all x.

40. $\lim_{x\to 0} \dfrac{x\sqrt{y^2-(y-x)^2}}{\left(\sqrt{8xy-4x^2}+\sqrt{8xy}\right)^3}$

$= \lim_{x\to 0} \dfrac{x\sqrt{2xy-x^2}}{\left(\sqrt{8xy-4x^2}+\sqrt{8xy}\right)^3}$

$= \lim_{x\to 0} \dfrac{x^{3/2}\sqrt{2y-x}}{x^{3/2}\left(\sqrt{8y-4x}+\sqrt{8y}\right)^3}$

$= \lim_{x\to c} \dfrac{\sqrt{2y-x}}{x^{3/2}\left(\sqrt{8y-4x}+\sqrt{8y}\right)^3} = \dfrac{\sqrt{2y}}{\left(2\sqrt{8y}\right)^3}$

$= \dfrac{\sqrt{2y}}{32\sqrt{2}y^{3/2}} = \dfrac{1}{32y}.$

41. $f(x) = \begin{cases} (x-x)(-x) = 0, & x < 0 \\ (x+x)x = 2x^2, & x \geq 0 \end{cases}$

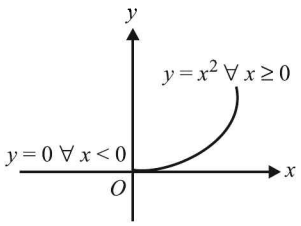


Fig. 11.4

It is evident from the graph of $f(x)$ that it is continuous and differentiable for all x. Also

$$f'(x) = \begin{cases} 0, & x < 0 \\ 4x, & x \geq 0 \end{cases}$$

$\therefore$ $f'(x)$ is continuous for all x but it is not differentiable at $x = 0$.

42. $\lim_{h\to 0} \dfrac{l_n(1+2h) - 2l_n(1+h)}{h^2}$

$= \lim_{h\to 0} \dfrac{\log(1+2h) - 2\log(1+h)}{h^2}$

$= \lim_{h\to 0} \dfrac{\log\left(\dfrac{1+2h}{(1+h)^2}\right)}{h^2} = -\lim_{h\to 0} \dfrac{\log\left(\dfrac{(1+h)^2}{1+2h}\right)}{h^2}$

$= \lim_{h\to 0} \dfrac{\log\{1+[h^2/(1+2h)]\}}{[h^2/(1+2h)](1+2h)}$

$= -\lim_{h\to 0} \dfrac{1}{1+2h} \cdot \lim_{h\to 0} \log\left(1+\dfrac{h^2}{1+2h}\right)^{(1+2h)/h^2}$

$= -\dfrac{1}{1+0}.\log e = -\log e = -1.$

43. Since, $f(x)$ is continuous at $x = 1$,

$\therefore$ $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)$

$\Rightarrow$ $a + b = b + a + c$

$\Rightarrow$ $c = 0$

As $f(x)$ is differentiable at $x = 1$

$\therefore$ (LHD at $x = 1$) = (RHD at $x = 1$)

$\Rightarrow$ $2a + 0 = 2b + a + 0$

$\Rightarrow$ $a = 2b$

$\therefore$ $a = 2b,\ c = 0$

44. Note that $(2\sin x)^{2n} = \infty$ if $|2\sin x| > 1$

$= 1$ if $|2\sin x| = 1$

$= 0$ if $|2\sin x| < 1$

Thus, doubtful points are those where

$$|\sin x| = \frac{1}{2} \quad \text{or} \quad x = n\pi + \frac{\pi}{6}$$

45. For continuity at $x = 0$

$$\lim_{h\to 0} \sin\frac{1}{0+h} = \lim_{h\to 0} \sin\frac{1}{0-h} = f(0)$$

But $\lim_{h\to 0} \sin\dfrac{1}{0+h}$ and $\lim_{h\to 0} \sin\left(\dfrac{1}{0-h}\right)$ are any number oscillating between 1 and -1 which is not equal to a defined number.

$\therefore$ $f(0)$ cannot be a definite number.

46. $f(x) = \dfrac{x}{1+x}, x \geq 0 = \dfrac{x}{1-x}, x < 0$

$$f'(x) = \frac{(1+x)-x}{(1+x)^2} = \frac{1}{(1+x)^2}, x > 0$$

$$= \frac{(1-x)-x(-1)}{(1-x)^2} = \frac{1}{(1-x)^2}, x < 0$$

$f'(0+) = f'(0-)$

47. Clearly for all integers between –3 and 3, $g(x) = 0$

$\therefore$ for $x = \pm 1, \pm 2, 0 \cdot g[f(x)] = 0$

i.e. function is continuous at x = 0, ±1, ±2 and discontinuous at other integers.

48. $\cos x + \sin x = \sqrt{2}\cos\left(x - \frac{\pi}{4}\right)$

So, $f(x) = \left[\sqrt{2}\cos\left(x - \frac{\pi}{4}\right)\right]$

Since $[x]$ is discontinuous at integral values of x, now,

$\sqrt{2}\cos\left(x - \frac{\pi}{4}\right)$ is an integer at

$$x = \frac{\pi}{2}, \frac{\pi}{2} + \frac{\pi}{4}, \pi + \frac{\pi}{4}, \frac{3\pi}{2} + \frac{\pi}{4}$$

49. Here $[x - \pi]$ = an integer for all x.

$\therefore$ $[x - \pi]\pi$ = integral multiple of π.

$\therefore$ $f(x) = 0$

50. $x^2 - 3x + 3 > 0$ for all $x\ (a > 0, D < 0)$

$x^2 - 3x + 3 = 1 \Rightarrow x = 1, 2$

$x^2 - 3x + 3 = -1 \Rightarrow x =$ imaginary

Thus, function is discontinuous at x = 1, 2 (only when denominator becomes 0).

51. $f(x)$ is continuous if $\sin x = 1 - 2\cos x$, which has infinite solutions

$$f\left(\frac{\pi}{6}\right) = f\left(\frac{-\pi}{6}\right) = 1 - 2\cos\frac{\pi}{6}$$

Thus, f is not injective.

52. Since $f(x) = \begin{cases} \sin\frac{1}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$

If $f(x)$ is continuous at x = 0,

$$\lim_{x \to 0} f(x) = f(0) = k$$

$$\Rightarrow \lim_{x \to 0} \sin\frac{1}{x} = k$$

$\Rightarrow k = \sin\infty$ can take any finite value between –1 and 1.

$\therefore$ $f(x)$ is discontinuous at x = 0

53. For $f(x)$ to be continuous at x = 0, we must have

$$L = R = V$$

$$\therefore \quad L = f(0^-) = \lim_{x \to 0^-} f(x) = \lim_{h \to 0} f(0 - h)$$

$$\Rightarrow \qquad L = \lim_{h \to 0} \frac{0 - h}{1 + e^{-1/h}}$$

As $h \to 0$

$$\Rightarrow \frac{1}{h} \to \infty \Rightarrow -\frac{1}{h} \to -\infty$$

$$\Rightarrow e^{-1/h} \to 0$$

$$\therefore \quad L = \frac{0}{1} = 0$$

Also, $R = f(0^+) = \lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0 + h)$

$$\Rightarrow R = \lim_{h \to 0} \frac{0 + h}{1 + e^{1/h}} = \lim_{h \to 0} \frac{he^{-1/h}}{e^{-1/h} + 1}$$

$$\Rightarrow R = \frac{0 \cdot 0}{0 + 1} = 0$$

Also, $V = f(0) = 0$ {given}

Since $L = R = V$

$\therefore$ The function is continuous at x = 0.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. Let $t_r = \frac{1}{(r+2)r!} = \frac{r+1}{(r+2)!} = \frac{(r+2)-1}{(r+2)!}$

$$\Rightarrow t_r = \frac{1}{(r+1)!} - \frac{1}{(r+2)!}$$

Now, $\sum_{r=1}^{n} t_r = \left(\frac{1}{2!} - \frac{1}{3!}\right) + \left(\frac{1}{3!} - \frac{1}{4!}\right)$

$$t_r = + \left(\frac{1}{4!} - \frac{1}{5!}\right) + \cdots + \left(\frac{1}{(n+1)!} - \frac{1}{(n+2)!}\right)$$

$$\Rightarrow \sum_{r=1}^{n} t_r = \frac{1}{2!} - \frac{1}{(n+2)!}$$

Hence, $a = \lim_{n \to \infty}\left[\frac{1}{2} - \frac{1}{(n+2)!}\right] = \frac{1}{2}$

Now, $b=\lim_{x\to 0}\dfrac{e^{\sin x}-e^{x}}{\sin x-x}$

$\Rightarrow\ b=\lim_{x\to 0}e^{x}\cdot\left(\dfrac{e^{\sin x-x}-1}{\sin x-x}\right)=e^{0}\cdot 1=1$

$\therefore\quad b=2a$

2. $2x=x+x$

$$|x|=x,x>0,|x|=-x,x<0$$

Both of these are satisfied by (c).

3. $\lim_{x\to 2a}\dfrac{\sqrt{x-2a}}{\sqrt{x^2-4a^2}}+\dfrac{\sqrt{x}-\sqrt{2a}}{\sqrt{x^2-4a^2}}$

$$=\lim_{x\to 2a}\frac{1}{\sqrt{x+2a}}+\frac{\sqrt{x}-\sqrt{2a}}{\sqrt{(x-2a)(x+2a)}}$$

$$=\frac{1}{2\sqrt{a}}+\lim_{x\to 2a}\frac{\sqrt{x}-\sqrt{2a}}{\sqrt{\left(\sqrt{x}-\sqrt{2a}\right)\left(\sqrt{x}+\sqrt{2a}\right)(x+2a)}}$$

$$=\frac{1}{2\sqrt{a}}+\lim_{x\to 2a}\frac{\sqrt{\sqrt{x}-\sqrt{2a}}}{\sqrt{\left(\sqrt{x}+\sqrt{2a}\right)(x+2a)}}=\frac{1}{2\sqrt{a}}+0$$

4. For $x<1,\quad f(x)=\dfrac{x^2-1}{x^2+2x+-3}=\dfrac{x+1}{x+3}$

$\therefore\quad \lim_{x\to 1^-}f(x)=\dfrac{1}{2}$

For $x>1,\quad f(x)=\dfrac{x^2-1}{x^2-2x+1}=\dfrac{x+1}{x-1}$

$\therefore\quad \lim_{x\to 1^+}f(x)=\infty$

$\therefore$ The function is not continuous at $x=1$.

5. Now, $\lim_{x\to(\pi/2)}\tan^{-1}\left[\dfrac{\sin(a\tan^3 x+b\tan^2 x+c\tan x)}{a\tan^3 x+b\tan^2 x+c\tan x}\right]$

$\therefore\quad \lim_{x\to\frac{\pi}{2}}\tan x$ is ∞ or $-\infty$ as x approaches $\pi/2$ from left or right respectively.

$$-1\ge\sin a(\tan^3 x+b\tan^2 x+c\tan x)\le 1$$

But $a\tan^3 x+b\tan^2 x+c\tan x$ either approaches ∞ or approaches $-\infty$.

So, $\dfrac{\sin(a\tan^3 x+b\tan^2 x+c\tan x)}{a\tan^3 x+b\tan^2 x+c\tan x}$ approaches 0 as x approaches $\pi/2$. Hence, limit is 0.

6. $\tan 2\theta=\dfrac{\sin 2\theta}{\cos 2\theta}$

$$\Rightarrow\ \frac{1}{\cot 2\theta}=\frac{2\sin\theta\cos\theta}{\cos^2\theta-\sin^2\theta}$$

$$\Rightarrow\ \cot 2\theta=\frac{\cos^2\theta-\sin^2\theta}{2\sin\theta\cos\theta}$$

$$\Rightarrow\ 2\cot 2\theta=\frac{\cos^2\theta}{\sin\theta\cos\theta}-\frac{\sin^2\theta}{\sin\theta\cos\theta}$$

$$\Rightarrow\ 2\cot 2\theta=\cot\theta-\tan\theta$$

$$\Rightarrow\ \tan\theta=\cot\theta-2\cot 2\theta \qquad \text{(i)}$$

Now, $\tan\theta=\cot\theta-2\cot 2\theta$

$$\frac{1}{2}\tan\frac{\theta}{2}=\frac{1}{2}\cot\frac{\theta}{2}-\cot\theta$$

$$\Rightarrow\ \frac{1}{2^2}\tan\frac{\theta}{2^2}=\frac{1}{2^2}\cot\frac{\theta}{2^2}-\frac{1}{2}\cot\frac{\theta}{2}$$

$$\Rightarrow\ \frac{1}{2^n}\tan\frac{\theta}{2^n}=\frac{1}{2^n}\cot\frac{\theta}{2^n}-\frac{1}{2^{n-1}}\cot\frac{\theta}{2^{n-1}}$$

$$\Rightarrow\ S=-2\cot 2\theta+\frac{1}{2^n}\cot\frac{\theta}{2^n}$$

$$\therefore\quad E=\lim_{n\to\infty}\left(\tan\theta+\frac{1}{2}\tan\frac{\theta}{2}+\frac{1}{2^2}\tan\frac{\theta}{2^2}+\cdots+\frac{1}{2^n}\tan\frac{\theta}{2^n}\right)$$

$$\Rightarrow\ E=\lim_{n\to\infty}S=\lim_{n\to\infty}\left(-2\cot 2\theta+\frac{1}{2^n}\cot\frac{\theta}{2^n}\right)$$

$$\Rightarrow\ E=-2\cot 2\theta+\lim_{n\to\infty}\frac{1}{\theta}\left[\frac{\theta/2^n}{\tan(\theta/2^n)}\right]$$

$$\therefore\quad E=-2\cot 2\theta+\frac{1}{\theta}$$

7. $\lim_{x\to 0}\dfrac{1-\cos^3 x}{x\sin x\cos x}=\lim_{x\to 0}\dfrac{(1-\cos x)(1+\cos^2 x+\cos x)}{x\sin x\cos x}$

$$=\lim_{x\to 0}\frac{2\sin^2(x/2)+(1+\cos^2 x+\cos x)}{2x\sin(x/2)\cos(x/2)\cos(x/2)\cos x}$$

$$=\lim_{x\to 0}\frac{1}{2}\cdot\frac{\sin(x/2)}{x/2}\cdot\frac{1+\cos^2 x+\cos x}{\cos(x/2)\cos x}$$

$$=\frac{1}{2}\cdot 1\cdot\frac{3}{1}=\frac{3}{2}$$

8. $\lim_{x\to 2^+}f(x)=\lim_{x\to 2^+}\dfrac{k(x^2-4)}{2-x}=\lim_{x\to 2^+}k(-2-x)=-4k$

$$\lim_{x\to 2^-}f(x)=\lim_{x\to 2^-}\frac{k(x^2-4)}{2-x}=\lim_{x\to 2^-}k(-2-x)=-4k$$

9. $[x]^2+1\neq 0$. Also $[x-\pi]$ is always an integer.

$\therefore$ $p[x-\pi]$ is of the form $k\pi$ (k is any integer) and hence $f(x)$ is identically the zero function for all x. So it is a constant function.

10. Since $f(x)$ is continuous at $x=\dfrac{\pi}{2}$,

$$\therefore \quad f\left(\frac{\pi}{2}\right)=\lim_{x\to\pi/2} f(x)$$

$$\Rightarrow \qquad \lambda=\lim_{x\to\pi/2}\frac{1-\sin x}{\pi-2x} \qquad \left(\frac{0}{0}\text{ form}\right)$$

Applying L'Hôpital's rule

$$\Rightarrow \qquad \lambda=\lim_{x\to\pi/2}\frac{-\cos x}{-2}$$

$$\Rightarrow \lambda=0$$

11. Let $y=\operatorname{cosec}^2 x$

$$\therefore \qquad E=\lim_{y\to\infty}\left(1^y+2^y+\cdots+n^y\right)^{\frac{1}{y}}$$

$$\Rightarrow E=\lim_{y\to\infty}(n^y)^{\frac{1}{y}}\left[\left(\frac{1}{n}\right)^y+\left(\frac{2}{n}\right)^y+\cdots+\left(\frac{n-1}{n}\right)^y+1\right]^{\frac{1}{y}}$$

$$\Rightarrow E=n\lim_{y\to\infty}(0+0+\cdots+0+1)^{\frac{1}{y}}$$

$$\therefore \quad E=n$$

12. $\sin^4 x+\cos^4 x=\left(\sin^2 x+\cos^2 x\right)^2-2\sin^2 x\cos^2 x$

$$=1-\frac{1}{2}\sin^2 2x$$

$$=1-\frac{1}{4}(1-\cos 4x)=\frac{3}{4}+\frac{1}{4}\cos 4x$$

Since the function cos 4x has period $2\pi/4=\pi/2$, the given function also has period $\pi/2$.

13. The required limit

$$=\lim_{x\to 2}\frac{\left[1+\sqrt{2+x}-3\right]}{(x-2)\left[\sqrt{1+\sqrt{2+x}}+\sqrt{3}\right]}$$

$$=\lim_{x\to 2}\frac{\left(\sqrt{x+2}-2\right)\left(\sqrt{x+2}+2\right)}{(x-2)\left(\sqrt{1+\sqrt{2+x}}+\sqrt{3}\right)\left(\sqrt{x+2}+2\right)}$$

$$=\lim_{x\to 2}\frac{(x+2)-4}{(x-2)\left(\sqrt{1+\sqrt{2+x}}+\sqrt{3}\right)\left(\sqrt{x+2}+2\right)}$$

$$=\lim_{x\to 2}\frac{1}{\left(\sqrt{1+\sqrt{2+x}}+\sqrt{3}\right)\left(\sqrt{x+2}+2\right)}$$

$$=\frac{1}{2\sqrt{3}}\cdot\frac{1}{4}=\frac{1}{8\sqrt{3}}$$

14. If $f(x)$ is differentiable, $[f(x)]^2$, i.e. $|f(x)|^2$, is also differentiable.

Taking $f(x)=x$, we have $|f|(x)=|f(x)|=|x|$, which is not differentiable at $x=0$.

Thus, f is differentiable in $\mathbb{R} \not\Rightarrow |f|$ is differentiable in $\mathbb{R}$. Also

$$(f|f|)(x)=f(x)|f(x)|=\begin{cases}-[f(x)]^2, & \text{if } f(x)<0\\ [f(x)]^2, & \text{if } f(x)\geq 0\end{cases}$$

which is differentiable in $\mathbb{R}$ if f is differentiable in $\mathbb{R}$.

15. $$\text{LHL}=\lim_{x\to 0^-}\frac{\sqrt{1+kx}-\sqrt{1-kx}}{x}$$

$$=\lim_{x\to 0^-}\frac{2kx}{x\left(\sqrt{1+kx}+\sqrt{1-kx}\right)}=k$$

$$\text{RHL}=\lim_{x\to 0^+}\left(2x^2+3x-2\right)=-2$$

Since it is given that $f(x)$ is continuous,

$\therefore$ LHL = RHL $\Rightarrow$ $k=-2$

16. $$\text{Limit}=\sqrt{\lim_{x\to 0}\frac{\sqrt{x-\sin x}}{x+\sin^2 x}}=\sqrt{\lim_{x\to 0}\frac{1-\cos x}{1+\sin 2x}}=0.$$

17. $(x^2-3x+2)=(x-1)(x-2)=+\text{ve}$

when $x<1$ or $x>2$, $-$ve, when $1\leq x\leq 2$

Also $\cos|x|=\cos x$, $\because \cos(-x)=\cos x$

$$\therefore \quad f(x)=-(x^2-1)(x^2-3x+2)+\cos x, \quad 1\leq x\leq 2$$

$$=(x^2-1)(x^2-3x+2)+\cos x, \quad x>2$$

Evidently $f(x)$ is not differentiable at $x=2$ as $L'\neq R'$.

18. We have $$f(x)=\begin{cases}\dfrac{(4^x-1)^3}{\sin(x/a)\log\left[1+(x^2/3)\right]}, & x\neq 0\\ 9(\log 4)^3, & x=0\end{cases}$$

For $f(x)$ to be continuous at $x=0$, we should have

$$\lim_{x\to 0} f(x)=f(0)=9(\log 4)^3$$

$$\therefore \lim_{x\to 0} f(x) = \lim_{x\to 0}\left(\frac{4^x-1}{x}\right)^3 \frac{x/a}{\sin\left(\frac{x}{a}\right)}\cdot\frac{ax^2}{\log\left(1+\frac{x^2}{3}\right)}$$

$$= (\log 4)^3 a \lim_{x\to 0}\left(\frac{x^2}{\frac{1}{3}x^2-\frac{1}{18}x^4+\cdots}\right)$$

$$\Rightarrow 9(\log 4)^3 = 3a(\log 4)^3$$

$$\Rightarrow a = 3$$

19. Limit $= \lim_{x\to 1}\frac{f(1)g'(x)-f'(x)g(1)}{g'(x)-f'(x)}$

$$= \lim_{x\to 1}\frac{2[g'(x)-f'(x)]}{g'(x)-f'(x)} = \lim_{x\to 1} 2 = 2.$$

20. Given: $f(x) = \begin{cases}\dfrac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+4}-2}, & (x\neq 0)\\ a, & (x=0)\end{cases}$

is continuous at $x = 0$, then

$$\lim_{x\to 0} f(x) = \lim_{x\to 0}\frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+4}-2}$$

$$= \lim_{x\to 0}\frac{\cos 2x-1}{\sqrt{x^2+4}-2}$$

$$= \lim_{x\to 0}\left[\frac{\frac{-2\sin 2x}{2x}}{\sqrt{x^2+4}}\right] \text{ (by L'Hôpital's rule)}$$

$$= \lim_{x\to 0}\frac{-\sin 2x\sqrt{x^2+4}}{x}$$

$$= -\lim_{x\to 0}\left[2\cos 2x\sqrt{x^2+4}+\sin 2x\frac{x}{\sqrt{x^2+4}}\right]$$

(by L'Hôpital's rule)

$$= -\left[2\sqrt{4}\right] = -4$$

Thus, $\lim_{x\to 0} f(x) = -4$

For function to be continuous at $x = 0$.

$$f(0) = \lim_{x\to 0} f(x)$$

$$\Rightarrow \quad a = -4$$

21. $f'(1^+) = \dfrac{d}{dx}\left(\dfrac{x-1}{2x^2-7x+5}\right)$

$$= \frac{d}{dx}\left[\frac{x-1}{(x-1)(2x-5)}\right]$$

$$= \frac{d}{dx}\left[\frac{1}{2x-5}\right]$$

$$\Rightarrow f'(1^+) = -\frac{2}{(2x-5)^2}$$

$$= -\frac{2}{(2-5)^2} = -\frac{2}{9}$$

22. $\lim_{x\to a}\dfrac{5x^3-4x^2+2x-1}{7x^3+x-1}$

$$= \lim_{x\to a}\frac{5-\frac{4}{x}+\frac{2}{x^2}-\frac{1}{x^3}}{7+\frac{1}{x^2}-\frac{1}{x^3}} = \frac{5}{7}$$

23. $\lim_{x\to 0}\dfrac{\sqrt{1-\cos 2x}}{x} = \sqrt{2}\lim_{x\to 0}\dfrac{|\sin x|}{x}$

$$= \sqrt{2}\begin{cases}\lim\limits_{x\to 0}\dfrac{\sin x}{x}, & 0<x<\dfrac{\pi}{2}\\ \lim\limits_{x\to 0}\dfrac{-\sin x}{x}, & -\dfrac{\pi}{2}<x<0\end{cases}$$

$$= \sqrt{2}\begin{cases}1, & 0<x<\dfrac{\pi}{2}\\ -1, & -\dfrac{\pi}{2}<x<0\end{cases}$$

$\therefore$ LHL $= -\sqrt{2}$ and RHL $= \sqrt{2}$

$\therefore$ Given limit does not exist.

24. $\lim_{x\to\infty}\left(\dfrac{a_1^{1/k}+a_2^{1/k}+\cdots+a_n^{1/x}}{n}\right)^{nx}$

$$= e^{\lim\limits_{x\to\infty}\left(\frac{a_1^{1/k}+a_2^{1/k}+\cdots+a_n^{1/x}}{n}-1\right)(nx)}$$

$$= e^{\lim\limits_{x\to\infty}\left(\frac{a_1^{1/k}+a_2^{1/k}+\cdots+a_n^{1/x}-n}{n}\right)(nx)}$$

$$= e^{\lim\limits_{x\to\infty}\left\{\frac{(a_1^{1/x}-1)}{\frac{1}{x}}+\frac{(a_2^{1/x}-1)}{\frac{1}{x}}+\cdots+\frac{(a_n^{1/x}-1)}{\frac{1}{x}}\right\}}$$

25. $\lim_{x\to 0}\dfrac{\sqrt{1-\cos 2x}}{\sqrt{2x}}=\lim_{x\to 0}\dfrac{\sqrt{1-\left(1-2\sin^2 x\right)}}{\sqrt{2x}}$

$$=\lim_{x\to 0}\frac{\sqrt{2\sin^2 x}}{\sqrt{2x}}=\lim_{x\to 0}\frac{|\sin x|}{x},$$

which does not exist.

$[\because$ LHL $=-1$ and RHL $=1]$

26. $\lim_{x\to\infty}\left(\dfrac{x^2+5x+3}{x^2+x+2}\right)^x=\lim_{x\to\infty}\left(1+\dfrac{4x+1}{x^2+x+2}\right)^x$

$$=\lim_{x\to\infty}\left[(1+\alpha)^{1/\alpha}\right]^{\alpha x}$$

where $\alpha=\dfrac{4x+1}{x^2+x+2}$

when $x\to\infty,\alpha\to 0$ and $\lim_{\alpha\to 0}(1+\alpha)^{1/\alpha}=e$

Also $\alpha x=\dfrac{4x^2+1}{x^2+x+2}\to 4$ as $x\to\infty$

$\therefore$ Given limit $=e^4$.

27. $\lim_{x\to\infty}\left(\dfrac{x-3}{x+2}\right)^x=\lim_{x\to\infty}\left[\dfrac{1-(3/x)}{1+(2/x)}\right]^x$

$$=\lim_{x\to\infty}\frac{\{[1-(3/x)]^{-(x/3)}\}^{-3}}{\{[1+(2/x)]^{x/2}\}^2}=\frac{e^{-3}}{e^2}=e^{-5}.$$

28. By L'Hôpital's rule,

$$4=\frac{f(a)g'(a)-g(a)f'(a)}{g'(a)-f'(a)}$$

$$=\frac{k\left[g'(a)-f'(a)\right]}{g'(a)-f'(a)}\therefore k=4$$

$\therefore$ Option (b) holds.

29. $\lim_{x\to\frac{\pi}{2}}\dfrac{[1-\tan(x/2)][1-\sin x]}{[1+\tan(x/2)][\pi-2x]^3}$

Let $x=\dfrac{\pi}{2}-h$. Thus, $x\to\dfrac{\pi}{2}\Rightarrow h\to 0$.

$\therefore$ Given limit $=\lim_{h\to 0}\dfrac{\left[1-\tan\left(\dfrac{\pi}{4}-\dfrac{h}{2}\right)\right]}{\left[1+\tan\left(\dfrac{\pi}{4}-\dfrac{h}{2}\right)\right]}\cdot\dfrac{[1-\cos h]}{(2h)^3}$

$$=\lim_{h\to 0}\frac{\tan\left[\dfrac{\pi}{4}-\left(\dfrac{\pi}{4}-\dfrac{h}{2}\right)\right]}{8h^3}\cdot 2\sin^2\frac{h}{2}$$

$$=\lim_{h\to 0}\frac{\tan(h/2)}{2\cdot(h/2)}\cdot\frac{2\sin^2(h/2)}{4(h/2)^2}\cdot\frac{1}{8}=\frac{1}{32}$$

30. $\lim_{x\to 0}\dfrac{\log(3+x)-\log(3-x)}{x}$

$$=\lim_{x\to 0}\frac{\log 3+\log\left(1+\dfrac{x}{3}\right)-\log 3-\log\left(1-\dfrac{x}{3}\right)}{x}$$

$$=\lim_{x\to 0}\frac{\log\left(1+\dfrac{x}{3}\right)-\log\left(1-\dfrac{x}{3}\right)}{x}$$

$$=\lim_{x\to 0}\frac{\left(\dfrac{x}{3}-\dfrac{x^2}{2\cdot 3^2}+\cdots\right)-\left(-\dfrac{x}{3}-\dfrac{x^2}{2\cdot 3^2}\cdots\right)}{x}$$

$$=\lim_{x\to 0}\frac{\dfrac{2x}{3}+\dfrac{2x^3}{3^3\cdot 3}}{x}=\frac{2}{3}.$$

31. $\lim_{x\to 0}\dfrac{\int_0^{x^2}\sec^2 t\,dt}{x\sin x}$

$$=\lim_{x\to 0}\frac{2x\sec^2 x^2}{\sin x+x\cos x}\qquad\text{[by L'Hôpital's rule]}$$

$$=\lim_{x\to 0}\frac{2\sec^2 x^2}{(\sin x/x)+\cos x}=\frac{2(1)}{1+1}=1.$$

32. Since $f(x)$ is continuous in $[0, (\pi/2)]$

$\therefore$ it is continuous at $x=(\pi/4)$

$\therefore$ $f\left(\dfrac{\pi}{4}\right)=\lim_{x\to(\pi/4)}f(x)=\lim_{x\to(\pi/4)}\dfrac{1-\tan x}{4x-\pi}$ $\left(\dfrac{0}{0}\text{ form}\right)$

$$=\lim_{x\to(\pi/4)}\frac{-\sec^2 x}{4}=-\frac{2}{4}=-\frac{1}{2}$$

$\therefore$ Option (a) holds.

33. $\lim_{x\to\infty}\left(1+\dfrac{a}{x}+\dfrac{b}{x^2}\right)^{2x}=e^2$

$\Rightarrow\lim_{x\to\infty}\left(1+\dfrac{ax+b}{x^2}\right)^{2x}=e^2$

$\Rightarrow\lim_{x\to\infty}\left[\left(1+\dfrac{ax+b}{x^2}\right)^{\frac{x^2}{ax+b}}\right]^{\frac{2x\cdot(ax+b)}{x^2}}=e^2$

$\Rightarrow e^{2a}=c^2\Rightarrow\quad a=1$

Thus, $a=1$ and $b\in\mathbb{R}$.

34. $\lim\limits_{x\to\alpha}\dfrac{1-\cos a\cdot(x-\alpha)(x-\beta)}{(x-\alpha)^2}$

$$\left[\because ax^2+bx+c=1(x-\alpha)(x-\beta)\right]$$

$$=\lim_{x\to\alpha}\frac{2\sin^2\dfrac{a(x-\alpha)(x-\beta)}{2}}{\dfrac{a^2(x-\alpha)^2(x-\beta)^2}{4}}\times\frac{a^2(x-\beta)^2}{4}$$

$$=\frac{2a^2(x-\beta)^2}{4}(1)^2=\frac{a^2}{2}(\alpha-\beta)^2$$

35. For continuity,

$$f(0)=\lim_{x\to0}\left(\frac{1}{x}-\frac{2}{e^{2x}-1}\right)=\lim_{x\to0}\frac{e^{2x}-1-2x}{x\left(e^{2x}-1\right)}$$

$$=\lim_{x\to0}\frac{\left[1+(2x)+\dfrac{(2x)^2}{2!}+\dfrac{(2x)^3}{3!}+\cdots\right]-1-2x}{x\left[1+(2x)+\dfrac{(2x)^2}{2!}+\cdots-1\right]}$$

$$=\lim_{x\to0}\frac{\dfrac{(2x)^2}{2!}+\dfrac{(2x)^3}{3!}+\cdots}{x\left[(2x)+\dfrac{(2x)^2}{2!}+\cdots\right]}$$

$$=\lim_{x\to0}\frac{\dfrac{2^2}{2!}+\dfrac{2^3}{3!}x+\cdots}{2+\dfrac{(2)^2}{2!}x+\cdots}=\frac{2+0+\cdots}{2+0+\cdots}=\frac{2}{2}=1.$$

36. Put $1-x=y$ as $x\to1, y\to0$

$$\therefore\quad\lim_{y\to0}y\tan\frac{\pi(1-y)}{2}=\lim_{y\to0}\frac{2}{\pi}\frac{(\pi y/2)}{\tan(\pi y/2)}=\frac{2}{\pi}$$

37. As $x\to0-$ (i.e. approaches 0 from the left) $[x]=-1$

$$\therefore\quad\lim_{x\to0-}f(x)=\lim_{x\to0-}\frac{1+\sin(-1)}{-1}=-1+\sin1$$

whereas if $x\to0+$ we get $[x]=0$

$$\therefore\quad f(x)=0\ \Rightarrow\ \lim_{x\to0+}f(x)=0$$

Thus, $\lim\limits_{x\to0}f(x)$ does not exist.

38. $\lim\limits_{x\to0}\dfrac{e^x+\log(1+x)-(1-x)^{-2}}{x^2}$ $\left[\dfrac{0}{0}\text{ form}\right]$

$$=\lim_{x\to0}\frac{e^x+(1+x)^{-1}-2(1-x)^{-3}}{2x}$$

[by L'Hôpital's rule]

$$=\lim_{x\to0}\frac{e^x-(1+x)^{-2}-6(1-x)^{-4}}{2}$$

[by L'Hôpital's rule]

$$=\frac{e^0-1-6}{2}=-3$$

39. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{1}{n}e^{r/n}=\int_0^1e^x\,dx=\left[e^x\right]_0^1=e-1.$

40. $\lim\limits_{x\to0}\dfrac{\tan x-\sin x}{x^3}=\lim\limits_{x\to0}\dfrac{\sin x-\sin x\cos x}{\cos x.x^3}$

$$=\lim_{x\to0}\frac{\sin x}{\cos x}.\frac{1-\cos x}{x^3}$$

$$=\lim_{x\to0}\frac{\tan x}{x}.\frac{2\sin^2(x/2)^2}{x^2}$$

$$=\lim_{x\to0}\frac{\tan x}{x}.\frac{2}{4}\left(\frac{\sin(x/2)}{x/2}\right)^2=\frac{1}{2}$$

41. $\lim\limits_{x\to0}\dfrac{5^x-5^{-x}}{2x}=\lim\limits_{x\to0}\dfrac{5^x\log5+5^{-x}\log5}{2}$

[by L'Hôpital's rule]

$$=\frac{\log5+\log5}{2}$$

$$=\log5.$$

42. $\lim\limits_{x\to\infty}\dfrac{\sqrt{x^2+1}-\sqrt[3]{x^3+1}}{\sqrt[4]{x^4+1}-\sqrt[5]{x^4+1}}$

$$=\lim_{x\to\infty}\frac{\sqrt{1+\dfrac{1}{x^2}}-\sqrt[3]{1+\dfrac{1}{x^3}}}{\sqrt[4]{1+\dfrac{1}{x^4}}-\sqrt[5]{\dfrac{1}{x}+\dfrac{1}{x^5}}}=\frac{1-1}{1-0}=0$$

43. $\text{LHL}=\lim\limits_{x\to2^-}f(x)=\lim\limits_{h\to0}1+(2-h)=3$

$\text{RHL}=\lim\limits_{x\to2^+}f(x)=\lim\limits_{h\to0}5-(2+h)=3, f(2)=3$

Hence, f is continuous at $x=2$.

Now, $Rf'(2)=\lim_{h\to 0}\dfrac{f(2+h)-f(2)}{h}$

$=\lim_{h\to 0}\dfrac{5-(2+h)-3}{h}=-1$

$Lf'(2)=\lim_{h\to 0}\dfrac{f(2-h)-f(2)}{-h}$

$=\lim_{h\to 0}\dfrac{1+(2-h)-3}{-h}=1$

$\therefore\ Rf'(2)\neq Lf'(2)$

$\therefore$ f is not differentiable at $x = 2$.

44. For $x > 1$, we have

$$f(x)=|\log|x||=\log x \ \Rightarrow\ f'(x)=\frac{1}{x}$$

For $x<-1$, we have

$$f(x)=|\log|x||=\log(-x) \ \Rightarrow\ f'(x)=\frac{1}{x}$$

For $0<x<1$, we have

$$f(x)=|\log|x||=-\log x \ \Rightarrow\ f'(x)=\frac{-1}{x}$$

For $-1<x<0$, we have

$$f(x)=-\log(-x) \ \Rightarrow\ f'(x)=-\frac{1}{x}$$

Hence, $f'(x)=\begin{cases}(1/x), & |x|>1\\ -(1/x), & |x|<1\end{cases}$

45. We have $f(x)=\begin{cases}x^2, & x\geq 0\\ -x^2, & x<0\end{cases}$

Clearly, $f(x)$ is differentiable for all $x > 0$ and for all $x < 0$. So, we check the differentiability at $x = 0$.
Now, (RHD at $x = 0$)

$$\left[\frac{d}{dx}(x)^2\right]_{x=0}=(2x)_{x=0}=0$$

and (LHD at $x = 0$)

$$\left[\frac{d}{dx}(-x)^2\right]_{x=0}=(-2x)_{x=0}=0$$

$\therefore$ (LHD at $x = 0$) = (RHD at $x = 0$)

So, $f(x)$ is differentiable for all x, i.e. the set of all points where $f(x)$ is differentiable is $(-\infty, \infty)$.

46. We have

$$\lim_{h\to 0}\frac{\log(1+2h)-2\log(1+h)}{h^2}$$

$$=\lim_{h\to 0}\frac{\log\left[\dfrac{1+2h}{(1+h)^2}\right]}{h^2}=-\lim_{h\to 0}\frac{\log\left[\dfrac{(1+h)^2}{1+2h}\right]}{h^2}$$

$$=-\lim_{h\to 0}\frac{\log\left(1+\dfrac{h^2}{1+2h}\right)}{\left(\dfrac{h^2}{1+2h}\right)(1+2h)}=-1=-1$$

47. We have

$$\lim_{x\to 0}\frac{\int_0^{x^2}\cos t^2\,dt}{x\sin x}$$

$$=\lim_{x\to 0}\frac{2x\cos x^4}{x\cos x+\sin x}\qquad\text{[using L'Hôpital's rule]}$$

$$=\lim_{x\to 0}\frac{2\cos x^4-8x^4\sin x^4}{2\cos x-x\sin x}=\frac{2-0}{2-0}=1$$

48. We have

$$\lim_{x\to 0}\left(\frac{1+5x^2}{1+3x^2}\right)^{\frac{1}{x^2}}=e^{\lim_{x\to 0}\frac{2x^2}{1+3x^2}\cdot\frac{1}{x^2}}=e^{\lim_{x\to 0}\frac{2}{1+3x^2}}=e^2$$

49. We have

$$\lim_{x\to 1}\frac{\sqrt{1-\cos 2(x-1)}}{x-1}$$

$$=\lim_{x\to 1}\frac{\sqrt{2\sin^2(x-1)}}{x-1}=\lim_{x\to 1}\sqrt{2}\frac{|\sin(x-1)|}{x-1}$$

$\therefore$ $(\text{LHL at } x=1)$

$$=\lim_{x\to 1^-}\sqrt{2}\frac{|\sin(x-1)|}{x-1}=\lim_{x\to 1}-\sqrt{2}\frac{\sin(x-1)}{x-1}=-\sqrt{2}$$

and $(\text{RHL at } x=1)$

$$=\lim_{x\to 1^+}\sqrt{2}\frac{\sin(x-1)|}{x-1}=\lim_{x\to 1}-\sqrt{2}\frac{\sin(x-1)}{x-1}=\sqrt{2}$$

Clearly, $(\text{LHL at } x=1)\neq(\text{RHL at } x=1)$

So, $\lim_{x\to 1}\dfrac{\sqrt{1-\cos 2(x-1)}}{x-1}$ does not exist.

50. $\lim\limits_{x\to\pi/4}\dfrac{\int_{2}^{\sec^2 x} f(t)dt}{x^2-\pi^2/16}$

$= \lim\limits_{x\to\pi/4}\dfrac{2\sec^2 x\tan x f\left(\sec^2 x\right)}{2x}$

[using Leibniz and L'Hôpital's rules]

$= \dfrac{\sec^2\frac{\pi}{4} f\left(\sec^2\frac{\pi}{4}\right)\tan\frac{\pi}{4}}{\pi/4} = \dfrac{8}{\pi}f(2)$

51. We have

$$L = \lim_{x\to 0}\frac{a-\sqrt{a^2-x^2}-\frac{x^2}{4}}{x^4}$$

$$\Rightarrow L = \lim_{x\to 0}\frac{a-a[1-(x^2/a^2)]^{1/2}-(x^2/4)}{x^4}$$

$$\Rightarrow L = \lim_{x\to 0}\frac{a-a\left(1-\frac{1}{2}\cdot\frac{x^2}{a^2}-\frac{1}{8}\cdot\frac{x^4}{a^4}+\frac{1}{16}\cdot\frac{x^6}{a^6}\cdots\right)-\frac{x^2}{4}}{x^4}$$

$$\Rightarrow L = \lim_{x\to 0}\frac{\left\{\frac{1}{2}\cdot\frac{x^2}{a}+\frac{1}{8}\cdot\frac{x^4}{a^3}-\frac{1}{16}\cdot\frac{x^6}{a^5}\cdots\right\}-\frac{x^2}{4}}{x^4}$$

$$\Rightarrow L = \lim_{x\to 0}\frac{x^2}{2}\left(\frac{1}{a}-\frac{1}{2}\right)+\frac{1}{8a^3}-\frac{1}{16}\cdot\frac{x^2}{a^5}+\cdots$$

$\Rightarrow \dfrac{1}{a}-\dfrac{1}{2}=0$ and in that case $L=\dfrac{1}{8a^3}$

[$\because$ L is finite]

$\Rightarrow a = 2$ and $L = \dfrac{1}{64}$

52. $\lim\limits_{x\to 7}\dfrac{2-\sqrt{x-3}}{x^2-49} = \lim\limits_{x\to 7}\dfrac{-\frac{1}{2\sqrt{x-3}}}{2x}$

$$= -\frac{1}{4\cdot 7\cdot 2} = -\frac{1}{56}$$

53. $\lim\limits_{x\to 0}(1+x)^{\cot x} = \lim\limits_{x\to 0}\left[(1+x)^{1/x}\right]^{x\cot x}$

$$= \lim_{x\to 0} e^{x\cot x} = e$$

Since $f(x)$ is continuous at $x = 0$,

$\therefore\ f(0) = \lim\limits_{x\to 0} f(x) = e$

54. $\lim\limits_{x\to a}\dfrac{a^x-x^a}{x^x-a^a} = -1$

$$\Rightarrow \lim_{x\to a}\frac{a^x\log_e a - ax^{a-1}}{x^x(1+\log_e x)-0} = -1$$

[by L'Hôpital's rule]

$$\Rightarrow \frac{a^a\log_e a - a^a}{a^a(1+\log_e a)} = -1$$

$\Rightarrow 2\log_e a = 0 \Rightarrow a = 1$

55. $\text{LHL} = \lim\limits_{x\to 0^-} f(x) = \lim\limits_{h\to 0} f(0-h) = 1$

[$\because$ $(0-h)$ is rational]

$\text{RHL} = \lim\limits_{x\to 0^+} f(x) = \lim\limits_{h\to 0} f(0+h) = 1$

[$\because$ $(0+h)$ is rational]

Hence, LHL = RHL = 1

56. $f_1(x) = x$ is continuous on $\mathbb{R}$

and $f_2(x) = \begin{cases}\sin\frac{1}{x}, & x\neq 0\\ 0, & x=0\end{cases}$ is discontinuous at $x = 0$

$\Rightarrow$ product of $f_1(x)$ and $f_2(x)$ is continuous

$\therefore$ Statement 1 is correct and Statement 2 is false.

57. $\lim\limits_{x\to 5}\dfrac{(f(x)^2)-9}{\sqrt{|x-5|}} = 0$

$\lim\limits_{x\to 5}[f(x))^2 - 9] = 0$

$\lim\limits_{x\to 5} f(x) = 3$

58. Doubtful points are $x = n$, $n \in I$

$$\text{LHL} = \lim_{x\to n^-}[x]\cos\left(\frac{2x-1}{2}\right)\pi = (n-1)\cos\left(\frac{2n-1}{2}\right)\pi = 0$$

$$\text{RHL} = \lim_{x\to n^+}[x]\cos\left(\frac{2n-1}{2}\right)\pi = n\cos\left(\frac{2n-1}{2}\right)\pi = 0$$

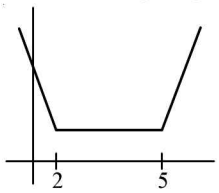
$$f(n) = 0$$

Hence continuous.

59. $f(x) = 7 - 2x;\ x < 2 = 3;\ 2 \le x \le 5 = 2x - 7;\ x > 5$

$f(x)$ is a constant function in [2, 5].

f is a continuous function in [2, 5].



MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. If $\lim_{x\to 0}(1+a\sin x)^{\operatorname{cosec} x}=3$, then a is
(a) ln 2 (b) ln 3
(c) ln 4 (d) e^3

2. $\lim_{x\to 5}\frac{x-5}{|x-5|}$ equals
(a) 2 (b) 0
(c) –3 (d) none of these

3. Let $f(x)=\lim_{x\to\infty}\sin^{2n}x$, then the number of point(s) where $f(x)$ is discontinuous is
(a) 0 (b) 1
(c) 2 (d) infinitely many

4. The number of points at which the function $f(x)=\frac{1}{\log|x|}$ is discontinuous is
(a) 1 (b) 2
(c) 3 (d) 4

5. Let $f(x)=\begin{cases}x^p\sin\frac{1}{x}, & x\neq 0\\ 0, & x=0\end{cases}$
Then $f(x)$ is continuous but differentiable at $x = 0$ if
(a) $p < 0$ (b) $p = 0$
(c) $0 < p \le 1$ (d) $p \ge 1$

6. If $f(x)=|x-25|$ and $g(x)=f[f(x)]$ then for $x > 50$, $g'(x)$ is equal to
(a) 0 (b) 1
(c) 25 (d) none of these

7. In order that the function $f(x)=(x+1)^{\cot x}$ is continuous at $x = 0$, $f(0)$ must be defined as
(a) 0 (b) e
(c) $1/e$ (d) none of these

8. $\lim_{x\to 0}\frac{x(1-\sqrt{1-x^2})}{\sqrt{1-x}\cdot(\sin^{-1}x)^3}$ is equal to
(a) 1/2 (b) 1/3
(c) 1/8 (d) none of these

9. For $m, n\in I^+$, $\lim_{x\to 0}\frac{\sin x^n}{(\sin x)^m}$ is equal to
(a) 1, if $n < m$ (b) 0, if $n > m$
(c) n/m (d) 0, if $n = m$

10. The integer n for which $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is
(a) 1 (b) 2
(c) 3 (d) 4

11. If $f(x)$ is continuous and $f(0) = 2$, then $\lim_{x\to 0}\frac{\int_0^x f(u)du}{x}$ is
(a) 0 (b) 2
(c) $f(2)$ (d) $f(1)$

12. If $\lim_{x\to 0}(1+ax+bx^2)^{2/x}=e^3$, then
(a) $a = 3, b = 0$ (b) $a = 3/2, b \neq 1$
(c) $a = 3/2, b = 4$ (d) $a = 2, b = 3$

13. Let $g(x)$ be the inverse of the function $f(x)$ and $f'(x) = 1/(1 + x^3)$, then $g'(x) =$
(a) $\frac{1}{1+[g(x)]^3}$ (b) $\frac{1}{1+[f(x)]^3}$
(c) $1+[g(x)]^3$ (d) $1+[f(x)]^3$

14. For all real values of a, the maximum of real roots of the equation $x^5+5x+a=0$ is
(a) 3 (b) 5
(c) 1 (d) 2

15. $\lim_{n\to\infty}\frac{n^p\sin^2(n!)}{n+1}, 0<p<1$, is equal to
(a) 0 (b) ∞
(c) 1 (d) none of these

16. If $f(x)=\frac{16^x-2^x}{10^x-5^x}$, then $\lim_{x\to\infty}f(x)$ is
(a) 2 (b) 8
(c) 3 (d) none of these

17. Let $S_n=\frac{1}{1\cdot 4}+\frac{1}{4\cdot 7}+\frac{1}{7\cdot 10}+\cdots$ to n terms. Then $\lim_{n\to\infty}s_n$ is equal to
(a) 1/3 (b) 3
(c) 1/4 (d) ∞

18. $\lim_{n\to 0}\frac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}$ is equal to
(a) 10/3 (b) 3/10
(c) 6/5 (d) 5/6

19. Let the rth term t_r of a series be given by $t_r=\frac{r}{1+r^2+r^4}$.
Then $\lim_{n\to\infty}\sum_{r=1}^{n}t_r$ is
(a) 1/4 (b) 1
(c) 1/2 (d) none of these

20. If $\lim_{n\to 0}\frac{[(a-x)-\tan x]\sin(nx)}{x^2}=0$, where n is a non-zero real number, then a is equal to
(a) 0 (b) $1 + (1/n)$
(c) n (d) $n + (1/n)$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If $\lim_{x\to\infty}\left(1+\frac{\lambda}{x}+\frac{\mu}{x^2}\right)^{2x}=e^2$, then

(a) $\lambda=1,\ \mu=2$

(b) $\lambda=2,\ \mu=1$

(c) $\lambda=1,\ \mu=$ any real constant

(d) $\lambda=\mu=1$

2. If α is a repeated root of $ax^2+bx+c=0$, then $\lim_{x\to\alpha}\frac{\sin(ax^2+bx+c)}{(x-\alpha)^2}$ is

(a) 0 (b) a

(c) b (d) c

3. The function $f(x) = [x]^2 - [x^2]$ (where [.] denotes the greatest integer function) is discontinuous at

(a) all integers

(b) all integers except 0 and 1

(c) all integers except 1

(d) all integers except 0

4. If $y=(\sin x)^{\sin x}$, then dy/dx =

(a) $(\sin x)^{\sin x}\cos x\,(1+\log\sin x)$

(b) $(\sin x)^{\sin x}(1+\log\sin x)$

(c) $(\sin x)^{\sin x}\,(1-\log\sin x)$

(d) none of these

5. Let $f(x+y)=f(x)\cdot f(y)$ for all x and y. If $f(5)=2$ and $f'(0) = 3$ then $f'(5)$ is equal to

(a) 5 (b) 6

(c) 0 (d) none of these

6. $\lim_{x\to0}\frac{e^{px}-e^{qx}}{x}=$

(a) $\ln p/q$ (b) $\ln pq$

(c) $p-q$ (d) none of these

7. $\lim_{x\to0}\frac{\sin^{-1}x-\tan^{-1}x}{x^3}$ is equal to

(a) 2 (b) 1

(c) –1 (d) 1/2

8. $\lim_{x\to0}\frac{x^4\left(\cot^4x-\cot^2x+1\right)}{\left(\tan^4x-\tan^2x+1\right)}$ is equal to

(a) 1 (b) 0

(c) 2 (d) none of these

9. $\lim_{x\to-1+}\frac{\sqrt{\pi}-\sqrt{\cos^{-1}x}}{\sqrt{x+1}}$ is equal to

(a) $1/\sqrt{\pi}$ (b) $1/\sqrt{(2\pi)}$

(c) 1 (d) 0

10. Let $f:\mathbb{R}^+\to\mathbb{R}$ be a continuous function satisfying $f\left(\frac{x}{y}\right)=f(x)-f(y)$ for all $x,y\in\mathbb{R}^+$. If $f'(1)=1$, then

(a) f' is bounded (b) $\lim_{x\to0}f\left(\frac{1}{x}\right)=0$

(c) $\lim_{x\to0}\frac{f(x)}{x}=1$ (d) $\lim_{x\to0}x\cdot f(x)=0$

11. $f(x)$ and $g(x)$ are continuous in $[a, b]$ differentiable in (a, b) with $f(a) = 4, f(b) = 10, g(a) = 1, g(b) = 3$. Then for $a < c < b, f'(c) = kg'(c)$, where k is equal to

(a) 1/2 (b) 2

(c) 3 (d) none of these

12. $\lim_{x\to0}\left(\frac{4^x+9^x}{2}\right)^{1/x}=$

(a) 2 (b) 6

(c) 16 (d) none of these

13. Let $h(x) = \min\{x, x^2\}, x\in\mathbb{R}$. Then $h(x)$ is

(a) differentiable everywhere

(b) non-differentiable at three values of x

(c) non-differentiable at two values of x

(d) none of these

14. $\lim_{n\to\infty}\frac{2\cdot3^n-3\cdot5^n}{3\cdot3^n+4\cdot5^n}=$

(a) 2/3 (b) –3/4

(c) 1 (d) none of these

15. If $f(x)=x, x\le1$ and $f(x)=x^2+bx+c\ (x>1)$ and $f'(x)$ exists finitely for all $x\in\mathbb{R}$, then

(a) $b = -1, c\in\mathbb{R}$ (b) $c = 1, b\in\mathbb{R}$

(c) $b = 1, c = -1$ (d) $b = -1, c = 1$

16. $\lim_{x\to\infty}\frac{(2+x)^{40}(4+x)^5}{(2-x)^{45}}=$

(a) –1 (b) 1

(c) 16 (d) 32

17. If the function $f(x)=\begin{cases}Ax-B, & x\le1\\ 3x, & 1<x<2\\ Bx^2-A, & x\ge2\end{cases}$

is continuous at $x = 1$ and discontinuous at $x = 2$, then

(a) $A = 3 + B, B\neq3$

(b) $A = 3 + B, B = 3$

(c) $A = 3 + B$

(d) none of these

18. $\lim_{x\to\infty}\frac{x^n}{e^x}=0 \quad (n\in I)$ for

(a) no value of n (b) all values of n
(c) negative values of n (d) positive values of n

19. Let $f''(x)$ be continuous at $x = 0$ and $f''(0) = 4$. Then $\lim_{x\to 0}\frac{2f(x)-3f(2x)+f(4x)}{x^2}$ is equal to

(a) 11 (b) 2
(c) 12 (d) none of these

20. $\lim_{x\to\infty}\frac{8}{x^8}\left[1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right]$ is equal to

(a) $\frac{1}{16}$ (b) $-\frac{1}{16}$
(c) $\frac{1}{32}$ (d) $-\frac{1}{32}$

ANSWERS

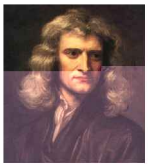
Level 1

1. (b)	**2.** (d)	**3.** (d)	**4.** (c)	**5.** (c)	**6.** (b)	**7.** (b)	**8.** (a)	**9.** (b)	**10.** (c)
11. (b)	**12.** (c)	**13.** (c)	**14.** (c)	**15.** (a)	**16.** (c)	**17.** (a)	**18.** (a)	**19.** (a)	**20.** (d)

Level 2

1. (c)	**2.** (b)	**3.** (c)	**4.** (a)	**5.** (b)	**6.** (c)	**7.** (d)	**8.** (a)	**9.** (b)	**10.** (d)
11. (c)	**12.** (b)	**13.** (c)	**14.** (b)	**15.** (d)	**16.** (a)	**17.** (a)	**18.** (c)	**19.** (c)	**20.** (c)

12



ISAAC NEWTON

Differentiation

- Definition of Differentiation
- Geometrical Meaning of Derivative
- Laws of Derivatives
- Some Standard Derivatives
- Derivative of an Implicit Function
- Derivative of a Function with Respect to Another Function
- Logarithmic Differentiation
- Properties of Logarithmic Functions
- Derivative of a Parametric Function
- Differentiation by Substitution
- Derivative by Trigonometrical Transformations
- Differentiation of Determinants
- Successive Differentiations

☞ DEFINITION OF DIFFERENTIATION

Differentiation is a method to compute the rate at which a dependent variable y changes with respect to the change in the independent variable x. This rate of change is called the *derivative* of y with respect to x.

Consider a function $y = f(x)$. Let us change x by a small quantity, say h. Then the corresponding change in y can be obtained from $f(x + h) - f(x)$.

From the definition of dy/dx, we get

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{\text{change in } y}{\text{change in } x} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{x+h-x} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

So, for finding the derivatives by first principles, we have to use the following formula:

$$\frac{dy}{dx} = f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

☞ GEOMETRICAL MEANING OF DERIVATIVE

The tangent to the curve $y = f(x)$ at the point (x, y) makes an angle θ with the x-axis.

Then $\dfrac{dy}{dx} = \tan\theta$.

Thus the derivative (dy/dx) or $f'(x)$ represents the slope of the tangent to the curve at the point (x, y) (Fig. 12.1).

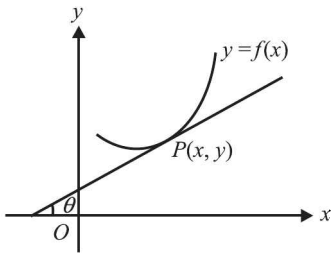


Fig. 12.1

☞ LAWS OF DERIVATIVES

Let u and v be the differentiable functions of x and k be a constant. Then,

1. $\frac{d}{dx}(ku) = k \cdot \frac{du}{dx}$
2. $\frac{d}{dx}(u+v) = \frac{du}{dx} + \frac{dv}{dx}$ [Sum Rule]
3. $\frac{d}{dx}(u-v) = \frac{du}{dx} - \frac{dv}{dx}$ [Difference Rule]
4. $\frac{d}{dx}(u \cdot v) = u \cdot \frac{d}{dx}v + v\frac{d}{dx}u$ [Product Rule]
5. $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{d}{dx}u - u\frac{d}{dx}v}{v^2}$ [Quotient Rule]
6. Chain Rule: If y is a function of t and t is a function of x, then $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$
7. If $\frac{d}{dx}F(x) = f(x)$, then $\frac{d}{dx}F(ax+b) = af(ax+b)$
8. If x and y are separately given as functions of a single variable t (called a parameter), i.e. $x = f(t)$ and $y = g(t)$, then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}$

Note Differentiation of an even function is an odd function and differentiation of an odd function is an even function.

☞ SOME STANDARD DERIVATIVES

1. $\frac{d}{dx}(c) = 0$, where c is independent of x.
2. $\frac{d}{dx}(x^n) = nx^{n-1}$
3. $\frac{d}{dx}(\log_e x) = \frac{1}{x}$
4. $\frac{d}{dx}(\log_a x) = \frac{1}{x \log_e a}$
5. $\frac{d}{dx}(e^x) = e^x$
6. $\frac{d}{dx}(a^x) = a^x \log a$
7. $\frac{d}{dx}(|x|) = \frac{|x|}{x}$
8. $\frac{d}{dx}(\log_e |x|) = \frac{1}{x}$
9. $\frac{d}{dx}(x^x) = x^x(1+\log x)$
10. $\frac{d}{dx}(\sin x) = \cos x$
11. $\frac{d}{dx}(\cos x) = -\sin x$
12. $\frac{d}{dx}(\tan x) = \sec^2 x, \left\{x \neq n\pi + \frac{\pi}{2}\right\}$
13. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x, \{x \neq n\pi\}$
14. $\frac{d}{dx}(\sec x) = \sec x \tan x, \left\{x \neq n\pi + \frac{\pi}{2}\right\}$
15. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x, \{x \neq n\pi\}$
16. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$
17. $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, -1 < x < 1$
18. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, x \in \mathbb{R}$
19. $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}, x \in \mathbb{R}$
20. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, |x| > 1$
21. $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}, |x| > 1$

Illustration 1. Find $\frac{dy}{dx}$ for $y = \sin(x^2 + a)$.

Solution: *1st method:* Let $y = \sin(x^2 + a)$

Putting $t = x^2 + a$, we get

$y = \sin t$ and $t = x^2 + a$

$$\frac{dy}{dt}=\cos t \qquad \frac{dt}{dx}=2x$$

then $\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}=2x\cos(x^2+a)$

2nd method: $y=\sin(x^2+a)$

$$\frac{dy}{dx}=\cos(x^2+a)\frac{d}{dx}(x^2+a) \quad \text{[Differentiation w.r.t. } x\text{]}$$

$$\frac{dy}{dx}=2x\cos(x^2+a)$$

Illustration 2. If $y=e^{ax}\cdot\cos(bx+c)$, find $\frac{dy}{dx}$.

Solution: $y=e^{ax}\cos(bx+c)$

On differentiating, we get

$$\frac{dy}{dx}=\left[\frac{d}{dx}(e^{ax})\right]\cdot\cos(bx+c)+e^{ax}\cdot\left\{\frac{d}{dx}[\cos(bx+c)]\right\}$$

$$=\frac{d(e^{ax})}{d(ax)}\cdot\frac{d(ax)}{dx}\cdot\cos(bx+c)+e^{ax}\cdot\frac{d\cos(bx+c)}{d(bx+c)}\cdot\frac{d(bx+c)}{dx}$$

$$=e^{ax}\cdot a\cdot\cos(bx+c)+e^{ax}\cdot[-\sin(bx+c)]\cdot b$$

$$=ae^{ax}\cos(bx+c)-be^{ax}\sin(bx+c)$$

Illustration 3. Differentiate $\tan^{-1}\left(\frac{a\cos x-b\sin x}{b\cos x+a\sin x}\right)$,

$-\frac{\pi}{2}<x<\frac{\pi}{2}$ w.r.t. x.

Solution: Let $y=\tan^{-1}\left(\frac{a\cos x-b\sin x}{b\cos x+a\sin x}\right)$,

Dividing the numerator and denominator by $b\cos x$, we get

$$y=\tan^{-1}\left(\frac{\frac{a\cos x}{b\cos x}-\frac{b\sin x}{b\cos x}}{\frac{b\cos x}{b\cos x}+\frac{a\sin x}{b\cos x}}\right)=\tan^{-1}\left(\frac{\frac{a}{b}-\tan x}{1+\frac{a}{b}\tan x}\right)$$

$$=y=\tan^{-1}\left(\frac{a}{b}\right)-\tan^{-1}(\tan x)$$

$$\Rightarrow\ y=\tan^{-1}\left(\frac{a}{b}\right)-x \qquad \left[\because -\frac{\pi}{2}<x<\frac{\pi}{2}\right]$$

$$\Rightarrow\ \frac{dy}{dx}=0-1=-1$$

Illustration 4. Find $\frac{dy}{dx}$ for $y=\log(x-\sqrt{a^2+x^2})$.

Solution: Let $y=\log(x-\sqrt{a^2+x^2})$.

Then $\frac{dy}{dx}=\frac{d}{dx}\left[\log(x-\sqrt{a^2+x^2})\right]$ $\qquad\left[\because \frac{d}{dx}\log x=\frac{1}{x}\right]$

$$=\frac{1}{x-\sqrt{a^2+x^2}}\frac{d}{dx}(x-\sqrt{a^2+x^2})$$

$$=\frac{1}{x-\sqrt{a^2+x^2}}\left[1-\frac{1}{2}(a^2+x^2)^{-1/2}\cdot\frac{d}{dx}(a^2+x^2)\right]$$

$$=\frac{1}{x-\sqrt{a^2+x^2}}\left[1-\frac{1}{2\sqrt{a^2+x^2}}\cdot 2x\right]$$

$$=\frac{1}{x-\sqrt{a^2+x^2}}\frac{\sqrt{a^2+x^2}-x}{\sqrt{a^2+x^2}}$$

$$=\frac{1}{x-\sqrt{a^2+x^2}}\times\frac{-(x-\sqrt{a^2+x^2})}{\sqrt{a^2+x^2}}=-\frac{1}{\sqrt{a^2+x^2}}$$

DERIVATIVE OF AN IMPLICIT FUNCTION

Consider an equation of the form $f(x, y) = c$. The function $f(x, y)$ is called an *implicit function* of x and y.

To find the derivative $\frac{dy}{dx}$, we differentiate each term w.r.t. x, keeping in mind that $\frac{d}{dx}[\phi(y)]=\frac{d}{dy}[\phi(y)]\cdot\frac{dy}{dx}$.

Illustration 5. If $x^3+y^3=3axy$, find $\frac{dy}{dx}$.

Solution: $x^3+y^3=3axy$

On differentiating w.r.t. x on both sides, we get

$$3x^2+3y^2\left(\frac{dy}{dx}\right)=3a\left(x\frac{dy}{dx}+y\right)$$

$$\Rightarrow\ (y^2-ay)\left(\frac{dy}{dx}\right)=(ay-x^2)\Rightarrow\frac{dy}{dx}=\frac{ay-x^2}{y^2-ax}.$$

Illustration 6. If $x^y+y^x=a^b$ (where a and b are constants), then find $\frac{dy}{dx}$.

Solution: We have $x^y+y^x=a^b$

$$\Rightarrow\ e^{\log x^y}+e^{\log y^x}=a^b \qquad \left[\begin{array}{l}\because e^{\log x}=x\\ \therefore e^{\log x^y}=x^y\ \&\ e^{\log y^x}=y^x\end{array}\right]$$

$$\Rightarrow \quad e^{y\log x} + e^{x\log y} = a^b$$

Differentiating both sides w.r.t., x we get

$$\frac{d}{dx}(e^{y\log x}) + \frac{d}{dx}(e^{x\log y}) = \frac{d}{dx}(a^b)$$

$$\Rightarrow \quad e^{y\log x}\frac{d}{dx}(y\log x) + e^{x\log y}\frac{d}{dx}(x\log y) = 0$$

$$\Rightarrow \quad x^y\left[y\cdot\frac{1}{x} + \log x\cdot\frac{dy}{dx}\right] + y^x\left[x\cdot\frac{1}{y}\frac{dy}{dx} + \log y\cdot 1\right] = 0$$

$$\Rightarrow \quad (x^y\log x + x\cdot y^{x-1})\frac{dy}{dx} + (y\cdot x^{y-1} + y^x\log y) = 0$$

Hence, $\dfrac{dy}{dx} = -\left\{\dfrac{yx^{y-1} + y^x\cdot\log y}{x^y\log x + x\cdot y^{x-1}}\right\}$

Alternative method:

We have $x^y + y^x = a^b$

Let $u = x^y$ (i)

and $v = y^x$ (ii)

$$u + v = a^b$$

Differentiating w.r.t. x, we get

$$\frac{du}{dx} + \frac{dv}{dx} = 0 \qquad \text{(iii)}$$

Taking log on both sides of Eq. (i), we have

$$\log u = y\log x$$

$$\frac{1}{u}\cdot\frac{du}{dx} = y\cdot\frac{1}{x} + \log x\frac{dy}{dx}$$

$$\frac{du}{dx} = 4\left[\frac{y}{x} + \log x\frac{dy}{dx}\right] = x^y\left[\frac{y}{x} + \log x\frac{dy}{dx}\right]$$

$$\frac{du}{dx} = x^{y-1}y + x^y\log x\frac{dy}{dx}$$

Taking log on both sides of Eq. (ii), we have

$$\log v = x\log y$$

$$\Rightarrow \quad \frac{1}{v}\frac{dv}{dx} = x\frac{1}{y}\frac{dy}{dx} + \log y\cdot 1$$

$$\Rightarrow \quad \frac{dv}{dx} = v\left[\frac{x}{y}\frac{dy}{dx} + \log y\right]$$

$$\Rightarrow \quad \frac{dv}{dx} = y^x\left[\frac{x}{y}\frac{dy}{dx} + \log y\right]$$

$$\Rightarrow \quad \frac{dv}{dx} = \left[y^{x-1}x\frac{dy}{dx} + y^x\log y\right]$$

Now putting the values of $\dfrac{du}{dx}$ and $\dfrac{dv}{dx}$ in Eq. (iii), we get

$$x^{y-1} + x^y\log x\frac{dy}{dx} + y^{x-1}x\frac{dy}{dx} + y^x\log y = 0$$

$$\frac{dy}{dx}\left[x^y\log x + y^{x-1}x\right] = -\left[x^{y-1} + y^x\log y\right]$$

$$\frac{dy}{dx} = \frac{-\left[x^{y-1} + y^x\log y\right]}{\left[x^y\log x + y^{x-1}x\right]}$$

DERIVATIVE OF A FUNCTION w.r.t. ANOTHER FUNCTION

Let $u = f(x)$ and $v = g(x)$. Then the derivative of $f(x)$ w.r.t. $g(x)$ is $\dfrac{du}{dv} = \dfrac{du/dx}{dv/dx}$.

Illustration 7. Find the derivative of x^2 w.r.t. $\sqrt{x}$.

Solution: Let $u = x^2$ and $v = \sqrt{x}$.

Then $\dfrac{du}{dx} = 2x$ and $\dfrac{dv}{dx} = \dfrac{1}{2\sqrt{x}}$

$$\therefore \quad \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{2x}{\left(\dfrac{1}{2\sqrt{x}}\right)} = 4x^{3/2}.$$

Illustration 8. Find the derivative of $a\sin^3\theta$ w.r.t. $a\cos^3\theta$.

Solution: Let $u = a\sin^3\theta$ and $v = a\cos^3\theta$

Now, $\dfrac{du}{d\theta} = a.3\sin^2\theta.\cos\theta$ and $\dfrac{dv}{d\theta} = a.3\cos^2\theta(-\sin\theta)$

Now, $\dfrac{du}{dv} = \dfrac{du}{d\theta}\times\dfrac{d\theta}{dv} = -\tan\theta$

Illustration 9. Find the derivative of $\tan^{-1}\dfrac{\sqrt{1+x^2}-1}{x}$ w.r.t. $\tan^{-1}x$.

Solution: Let $u = \tan^{-1}\dfrac{\sqrt{1+x^2}-1}{x}$ and $v = \tan^{-1}x$

Now, putting $x = \tan\theta$, we get

$$u = \tan^{-1}\frac{\sec\theta - 1}{\tan\theta} = \tan^{-1}\frac{1-\cos\theta}{\sin\theta} = \tan^{-1}\frac{2\sin^2\dfrac{\theta}{2}}{2\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}}$$

$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{\theta}{2} = \frac{1}{2}$$

Also, $v = \tan^{-1}(\tan\theta) = \theta \quad \therefore \quad \frac{dv}{d\theta} = 1$

Now, $\frac{du}{dv} = \frac{du}{d\theta} \times \frac{d\theta}{dv} = \frac{1}{2}$

Note

- If y and t are functions of x, then
$$\frac{d}{dx}(y') = ty^{t-1}\frac{dy}{dx} + (y^t \log y)\frac{dt}{dx}$$
- If $y = t^n$ and $t = f(x)$, then $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = nt^{n-1}\frac{dt}{dx}$

☞ LOGARITHMIC DIFFERENTIATION

The functions which can be evaluated by using logarithmic differentiation are of two types.

1. If $y = [f(x)]^{g(x)}$,
then $\frac{dy}{dx} = [f(x)]^{g(x)} \cdot \frac{d}{dx}[g(x), \log f(x)]$
2. $y = f_1(x) \cdot f_2(x) \ldots, f_n(x)$
$$\Rightarrow \log y = \log f_1(x) + \log f_2(x) + \cdots + f_n(x)$$
$$\Rightarrow \frac{1}{y}\cdot\frac{dy}{dx} = \frac{f_1'(x)}{f_1(x)} + \frac{f_2'(x)}{f_2(x)} + \cdots + \frac{f_n'(x)}{f_n(x)}$$
$$\Rightarrow \frac{dy}{dx} = y \cdot \left[\frac{f_1'(x)}{f_1(x)} + \frac{f_2'(x)}{f_2(x)} + \cdots + \frac{f_n'(x)}{f_n(x)}\right]$$

☞ PROPERTIES OF LOGARITHMIC FUNCTION

1. $\log_e(ab) = \log_e a + \log_e b \quad (a, b > 0)$
2. $\log_e\left(\frac{a}{b}\right) = \log_e a - \log_e b \quad (a, b > 0)$
3. $\log_e a^m = m\log_e a \quad (a > 0, m \in \mathbb{R})$
4. $\log_a a = 1 \quad (a > 0, a \neq 1)$
5. $\log_{b^m} a = \frac{1}{m}\log_b a \quad (a, b > 0, b \neq 1 \text{ and } m \in \mathbb{R} - \{0\})$
6. $\log_b a = \frac{1}{\log_a b} \quad (a, b > 0 \text{ and } a, b \neq \{0\})$
7. $\log_b a = \frac{\log_m a}{\log_m b} \quad (a, b, m > 0 \neq \{1\})$
8. $a^{\log_a m} = m \quad (a > 0 \neq \{1\}, m > 0$
9. $a^{\log_c b} = b^{\log_c a} \quad a, b, c, > 0 \text{ and } c \neq 1$
10. If $\log_m x > \log_m y \Rightarrow \begin{cases} x > y, & \text{if } m > 1 \\ x < y, & \text{if } 0 < m < 1 \end{cases}$
$(m, x, y > 0, m \neq 1)$
11. $\log_m a = b \Rightarrow a = m^b \quad (m, a > 0; m \neq 1; b \in \mathbb{R})$
12. $\log_m a > b \Rightarrow \begin{cases} a > m^b, & \text{if } m > 1 \\ a < m^b, & \text{if } 0 < m < 1 \end{cases}$
13. $\log_m a < b \Rightarrow \begin{cases} a < m^b, & \text{if } m > 1 \\ a > m^b, & \text{if } 0 < m < 1 \end{cases}$

Illustration 10. If $x^y \cdot y^x = 1$, find $\frac{dy}{dx}$.

Solution: Taking log on both sides, we get
$$y\log x + x\log y = \log 1$$
Differentiating both sides, we get
$$y\frac{d}{dx}(\log x) + \left(\frac{d}{dx}y\right)\cdot\log x + \left(\frac{d}{dx}x\right)\cdot\log y + x\left(\frac{d}{dx}\log y\right) = 0$$
$$\Rightarrow \quad y\cdot\frac{1}{x} + \log x\cdot\frac{dy}{dx} + 1\cdot\log y + x\frac{1}{y}\frac{dy}{dx} = 0$$
$$\Rightarrow \quad \left(\log x + \frac{x}{y}\right)\frac{dy}{dx} = -\left(\frac{y}{x} + \log y\right)$$
$$\Rightarrow \quad \frac{dy}{dx} = -\frac{y + x\log y}{x + y\log x}\frac{y}{x}$$

Illustration 11. Find $\frac{dy}{dx}$ of $y = x^{x^x}$.

Solution: Since $y = x^{x^x}$
$$= (x)^{x^x} \qquad \left[(x)^{x^x} \neq (x^x)^x \because (x^x)^x = x^{x^2}\right]$$
$$= e^{x^x \log x}$$
$$\therefore \quad \frac{dy}{dx} = e^{x^x \log x}\frac{d}{dx}(x^x \log x)$$
$$= x^{x^x}\left[x^x\cdot\frac{1}{x} + \log x\cdot\frac{d}{dx}(x^x)\right] = x^{x^x}\left[x^{x-1} + \log x\frac{d}{dx}e^{x\log x}\right]$$
$$= x^{x^x}\left[x^{x-1} + \log x\cdot e^{x\log x}\frac{d}{dx}(x\log x)\right]$$

$$= x^{x^x}\left[x^{x-1} + \log x \cdot x^x \cdot \left(x \cdot \frac{1}{x} + \log x \cdot 1\right)\right]$$

$$= x^{x^x} \cdot x^{x-1}(1 + x\log x(1 + \log x \cdot 1))$$

Alternative method:

$$y = x^{x^x}$$

$$\therefore \quad \frac{dy}{dx} = x^{x^x} \log x \frac{d}{dx}(x^x) + x^x \cdot (x)^{x^x - 1} \cdot 1$$

$$= x^{x^x} \log x[x^x \log x \cdot 1 + x \cdot x^{x-1} \cdot 1] + x^{x-1} \cdot x^{x^x}$$

$$= x^{x^x} . x^{x-1}[1 + x\log x(1 + \log x)]$$

Illustration 12. If $y = \sqrt{x + \sqrt{y + \sqrt{x + \sqrt{y + \cdots \infty}}}}$ then find $\frac{dy}{dx}$.

Solution: We have $y = \sqrt{x + \sqrt{y + \sqrt{x + \sqrt{y + \cdots \infty}}}}$

$$\Rightarrow \quad y = \sqrt{x + \sqrt{y + y}}$$

$$\Rightarrow \quad y^2 - x = \sqrt{2y}$$

$$\Rightarrow \quad (y^2 - x)^2 = 2y$$

Differentiating both sides w.r.t. x, we get

$$2(y^2 - x)\left(2y\frac{dy}{dx} - 1\right) = 2\frac{dy}{dx}$$

$$\Rightarrow \quad (2y(y^2 - x) - 1)\frac{dy}{dx} = (y^2 - x)$$

Hence, $\frac{dy}{dx} = \frac{(y^2 - x)}{(2y^3 - 2xy - 1)}$.

☞ DERIVATIVE OF A PARAMETRIC FUNCTION

Let $x = f(t)$ and $y = g(t)$. Then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

If $\frac{dy}{dx} = F(t)$, then

$$\frac{d^2y}{dx^2} = \frac{d}{dt}[F(t)] \cdot \frac{dt}{dx} = \frac{\frac{d}{dt}[F(t)]}{dx/dt}$$

Illustration 13. If $x = a\left(\cos t + \log \tan \frac{t}{2}\right)$ and $y = a \sin t$, find $\frac{dy}{dx}$.

Solution: We have $x = a\left(\cos t + \log \tan \frac{t}{2}\right)$

$$\Rightarrow \quad \frac{dy}{dx} = a\left(-\sin t + \frac{1}{\tan t/2}\sec^2 \frac{t}{2}\frac{1}{2}\right)$$

$$= a\left(-\sin t + \frac{1}{2\sin \frac{t}{2}\cos \frac{t}{2}}\right)$$

$$= a\left(-\sin t + \frac{1}{\sin t}\right) = a\left(1 - \frac{\sin^2 t}{\sin t}\right)$$

$$= \frac{a\cos^2 t}{\sin t}$$

Also, $y = a \sin t$

$$\Rightarrow \quad dy = a \cos t.$$

$$\therefore \quad \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = a\cos t \times \frac{\sin t}{a\cos^2 t} = \tan t .$$

Illustration 14. If $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$, find $\frac{dy}{dx}$.

Solution: Here, $x = a(\theta - \sin\theta)$

On differentiating both sides, we get

$$\frac{dx}{d\theta} = a(1 - \cos\theta) \quad \text{and} \quad y = a(1 - \cos\theta)$$

On differentiating both sides, we get

$$\frac{dy}{d\theta} = a(\sin\theta)$$

So, $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx} = \frac{a\sin\theta}{a(1 - \cos\theta)}$

$$= \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = \cot\frac{\theta}{2}$$

☞ DIFFERENTIATION BY SUBSTITUTION

S. No.	Function	Substitution
1.	$\sqrt{a^2-x^2}$	$x=a\sin\theta$ or $a\cos\theta$
2.	$\sqrt{x^2-a^2}$	$x=a\sec\theta$ or $a\operatorname{cosec}\theta$
3.	$\sqrt{(x-a)(b-x)}$	$x=a\cos^2\theta+b\sin^2\theta$
4.	$\sqrt{(x-a)(x-b)}$	$x=a\sec^2\theta-b\tan^2\theta$
5.	$\sqrt{x^2+a^2}$	$x=\alpha\tan\theta$ or $\alpha\cot\theta$
6.	$\sqrt{\frac{x}{a+x}}$	$x=a\tan^2\theta$
7.	$\sqrt{\frac{a-x}{a+x}}$	$x=\cos 2\theta$
8.	$\sqrt{ax-x^2}$	$x=a\sin^2\theta$
9.	$\sqrt{\frac{x}{a-x}}$	$x=a\sin^2\theta$
10.	$\sqrt{a^2-x^2}$	$x^2=a^2\cos 2\theta$

☞ DERIVATIVE BY TRIGONOMETRICAL TRANSFORMATIONS

By using trigonometrical transformations before differentiation, the work is shorten considerably. Some important results are given below:

1. $\sin 2x=2\sin x\cos x=2\sin x\sqrt{(1-\sin^2 x)}=\dfrac{2\tan x}{1+\tan^2 x}$
2. $\cos 2x=2\cos^2 x-1=1-2\sin^2 x=\dfrac{1-\tan^2 x}{1+\tan^2 x}$
3. $\tan 2x=\dfrac{2\tan x}{1-\tan^2 x}$
4. $\tan^2 x=\dfrac{1-\cos 2x}{1+\cos 2x}$
5. $\sin 3x=3\sin x-4\sin^3 x$
6. $\cos 3x=4\cos^3 x-3\cos x$
7. $\tan 3x=\dfrac{3\tan x-\tan^3 x}{1-3\tan^2 x}$
8. $\tan\left(\dfrac{\pi}{4}+x\right)=\dfrac{1+\tan x}{1-\tan x}$
9. $\tan\left(\dfrac{\pi}{4}-x\right)=\dfrac{1-\tan x}{1+\tan x}$
10. $\sqrt{(1\pm\sin x)}=\cos\dfrac{x}{2}\pm\sin\dfrac{x}{2}$
11. $\tan^{-1}x\pm\tan^{-1}y=\tan^{-1}\left(\dfrac{x\pm y}{1\mp xy}\right)$
12. $\sin^{-1}x\pm\sin^{-1}y=\sin^{-1}(x\sqrt{1-y^2}\pm y\sqrt{1-x^2})$
13. $\cos^{-1}x\pm\cos^{-1}y=\cos^{-1}(xy\mp\sqrt{1-x^2}\sqrt{1-y^2})$
14. $\sin^{-1}x+\cos^{-1}x=\dfrac{\pi}{2}$
15. $\tan^{-1}x+\cot^{-1}x=\dfrac{\pi}{2}$
16. $\sec^{-1}x+\operatorname{cosec}^{-1}x=\dfrac{\pi}{2}$

☞ DIFFERENTIATION OF DETERMINANTS

If $y=\begin{vmatrix} f(x) & g(x) & h(x)\\ p(x) & q(x) & r(x)\\ u(x) & v(x) & w(x)\end{vmatrix}$,

Then $\dfrac{dy}{dx}=\begin{vmatrix} f'(x) & g'(x) & h'(x)\\ p(x) & q(x) & r(x)\\ u(x) & v(x) & w(x)\end{vmatrix}$

$$+\begin{vmatrix} f(x) & g(x) & h(x)\\ p'(x) & q'(x) & r'(x)\\ u(x) & v(x) & w(x)\end{vmatrix}+\begin{vmatrix} f(x) & g(x) & h(x)\\ p(x) & q(x) & r(x)\\ u'(x) & v'(x) & w'(x)\end{vmatrix}$$

i.e. to differentiate a determinant, differentiate one row (or column) at a time, keeping the other unchanged.

Illustration 15. If $f(x)=\begin{vmatrix} x & x^2 & x^3\\ 1 & 2x & 3x^2\\ 0 & 2 & 6x\end{vmatrix}$, find $f'(x)$.

Solution: Here $f(x)=\begin{vmatrix} x & x^2 & x^3\\ 1 & 2x & 3x^2\\ 0 & 2 & 6x\end{vmatrix}$

On differentiating, we get

$$\Rightarrow \qquad f'(x)=\begin{vmatrix} \dfrac{d}{dx}(x) & \dfrac{d}{dx}(x^2) & \dfrac{d}{dx}(x^3)\\ 1 & 2x & 3x^2\\ 0 & 2 & 6x\end{vmatrix}$$

$$+\begin{vmatrix} x & x^2 & x^3 \\ \frac{d}{dx}(1) & \frac{d}{dx}(2x) & \frac{d}{dx}(3x^2) \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ \frac{d}{dx}(0) & \frac{d}{dx}(2) & \frac{d}{dx}(6x) \end{vmatrix}$$

$$f'(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 0 & 2 & 6x \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix}$$

We know that if any two rows or columns of a determinant are equal, then the value of the determinant is zero.

$$\Rightarrow \quad 0+0+\begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix} = 6(2x^2 - x^2) = 6x^2$$

Illustration 16. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$, where p is constant, then find $\frac{d^3}{dx^3}[f(x)]$ at $x = 0$.

Solution: Given, $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

On differentiating it, we get

$$\frac{d}{dx}f(x) = \begin{vmatrix} \frac{d}{dx}x^3 & \frac{d}{dx}\sin x & \frac{d}{dx}\cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$$+\begin{vmatrix} x^3 & \sin x & \cos x \\ 0 & 0 & 0 \\ p & p^2 & p^3 \end{vmatrix} + \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

So, $\frac{d}{dx}f(x) = \begin{vmatrix} 3x^2 & \cos x & -\sin x \\ 0 & 0 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

☞ SUCCESSIVE DIFFERENTIATIONS

Let $y = f(x)$ be a differentiable function of x. Then the differential coefficient of $f(x)$, denoted by $\frac{dy}{dx}$ or y' or f', is called the *first differential coefficient* of $f(x)$. If $\frac{dy}{dx}$ is also differentiable, then the differential coefficient of $\frac{dy}{dx}$, denoted by $\frac{d^2y}{dx^2}$ or y'' or f'', is called the *second differential coefficient* of $f(x)$. Similarly, we get successive differential coefficients $\frac{d^3y}{dx^3}, \frac{d^4y}{dx^4}, \cdots, \frac{d^ny}{dx^n}$.

☞ IMPORTANT FORMULAE USED IN SUCCESSIVE DIFFERENTIATIONS

1. $\frac{d^n}{dx^n}(x^n) = n!$
2. $\frac{d^n}{dx^n}a^x = a^x(\log a)^n$
3. $\frac{d^n}{dx^n}(\sin x) = \sin\left(x + \frac{n\pi}{2}\right)$
4. $\frac{d^n}{dx^n}(\cos x) = \cos\left(x + \frac{n\pi}{2}\right)$
5. $\frac{d^n}{dx^n}e^{mx} = m^n e^{mx}$
6. $\frac{d^n}{dx^n}(\log x) = (-1)^{n-1}(n-1)!x^{-n}$
7. $\frac{d^n}{dx^n}[e^{ax}\sin(bx+c)] = r^n e^{ax}\sin(bx + c + n\phi)$, where $r^2 = a^2 + b^2$ and $\phi = \tan^{-1}\left(\frac{b}{a}\right)$
8. $\frac{d^n}{dx^n}[e^{ax}\cos(bx+c)] = r^n e^{ax}\cos(bx + c + n\phi)$ where $r^2 = a^2 + b^2$ and $\phi = \tan^{-1}\left(\frac{b}{a}\right)$

SOLVED PROBLEMS

1. If $\sin^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right)=\log a$ then $\frac{dy}{dx}=$

(a) $\frac{-y}{x}$ (b) $\frac{y}{x}$

(c) $\frac{x}{y}$ (d) $-\frac{x}{y}$

Ans. (b)

Solution: $\frac{x^2-y^2}{x^2+y^2}=\sin(\log a)=\lambda$, [$\because$ sin (log a) is constant]

$$x^2-y^2=\lambda(x^2+y^2)$$

$$x^2-y^2=\lambda x^2+\lambda y^2$$

$$\therefore \quad x^2(1-\lambda)=y^2(1+\lambda) \qquad \text{(i)}$$

$$\Rightarrow \quad 2x(1-\lambda)=2y\frac{dy}{dx}(1+\lambda) \qquad \text{(ii)}$$

Dividing (ii) by (i), we have

$$\frac{2}{x}=\frac{2}{y}\frac{dy}{dx} \qquad \therefore \quad \frac{dy}{dx}=\frac{y}{x}$$

2. If $y = a\cos(\log x) - b\sin(\log x)$, then the value of $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+y$ is

(a) 0 (b) 1

(c) 2 (d) 3

Ans. (a)

Solution: $y = a\cos(\log x) - b\sin(\log x)$

On differentiating w.r.t. x, we get

$$\therefore \quad \frac{dy}{dx}=\frac{a[-\sin(\log x)]}{x}-\frac{b\cos(\log x)}{x}$$

$$=-\frac{[a\sin(\log x)+b\cos(\log x)]}{x}$$

$$\Rightarrow \quad x\frac{dy}{dx}=-[a\sin(\log x)+b\cos(\log x)]$$

Again, differentiating w.r.t. x, we get

$$x\frac{d^2y}{dx^2}+\frac{dy}{dx}=-\left[\frac{a\cos(\log x)}{x}-\frac{b\sin(\log x)}{x}\right]=-\frac{y}{x}$$

$$\Rightarrow \quad x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+y=0$$

3. If $y=\log_{x^2+4}(7x^2-5x+1)$, then $\frac{dy}{dx}=$

(a) $\frac{1}{\log_e(x^2+4)}\left(\frac{14x-5}{7x^2-5x+1}-\frac{2xy}{x^2+4}\right)$

(b) $\frac{1}{\log_e(x^2+4)}\left(\frac{14x-5}{7x^2-5x+1}+\frac{2xy}{x^2+4}\right)$

(c) $-\frac{1}{\log_e(x^2+4)}\left(\frac{14x-5}{7x^2-5x+1}-\frac{2xy}{x^2+4}\right)$

(d) none of these

Ans. (a)

Solution: $y=\log_{x^2+4}(7x^2-5x+1)=\frac{\log_e(7x^2-5x+1)}{\log_e(x^2+4)}$

$$=\frac{\log_e f(x)}{\log_e g(x)}$$

$$\therefore \quad \frac{dy}{dx}=\frac{\log_e g(x)\frac{f'(x)}{f(x)}-\log_e f(x)\frac{g'(x)}{g(x)}}{[\log_e g(x)]^2}$$

$$=\frac{1}{\log_e g(x)}\left[\frac{f'(x)}{f(x)}-y\frac{g'(x)}{g(x)}\right]$$

$$=\frac{1}{\log_e(x^2+4)}\left(\frac{14x-5}{7x^2-5x+1}-\frac{2xy}{x^2+4}\right)$$

4. If $x=\sqrt{a^{\sin^{-1}t}}$ and $y=\sqrt{a^{\cos^{-1}t}}$ then $\frac{dy}{dx}=$

(a) $\frac{y}{x}$ (b) $\frac{x}{y}$

(c) $-\frac{x}{y}$ (d) $-\frac{y}{x}$

Ans. (d)

Solution: Let $x=\sqrt{a^{\sin^{-1}t}}$ (i)

$$y=\sqrt{a^{\cos^{-1}t}} \qquad \text{(ii)}$$

Multiplying (i) and (ii), we get

$$xy=\sqrt{a^{\sin^{-1}t}}\times\sqrt{a^{\cos^{-1}t}}$$

$$\Rightarrow \quad xy=\sqrt{a^{\sin^{-1}t}\times a^{\cos^{-1}t}} \quad \Rightarrow \quad xy=\sqrt{a^{\sin^{-1}t+\cos^{-1}t}}$$

$$\Rightarrow \quad xy=\sqrt{a^{\frac{\pi}{2}}} \qquad \left[\because \sin^{-1}t+\cos^{-1}t=\frac{\pi}{2}\right]$$

Differentiating w.r.t. x, we get

$$x\frac{dy}{dx}+y=0 \qquad \Rightarrow \quad \frac{dy}{dx}=-\frac{y}{x}$$

5. If $\sqrt{x+y}+\sqrt{y-x}=c$, then $\frac{d^2y}{dx^2}$ is equal to

(a) $\frac{2}{c}$ (b) $-\frac{2}{c^2}$

(c) $-\frac{2}{c^2}$ (d) none of these

Ans. (c)

Solution: $\sqrt{x+y}+\sqrt{y-x}$.

We know that $(y+x)-(y-x)=2x$

Dividing, we get

$$\sqrt{y+x}-\sqrt{y-x}=\frac{2x}{c} \qquad \therefore \quad 2\sqrt{y+x}=c+\frac{2x}{c}$$

$$\Rightarrow \quad 4(y+x)=c^2+\frac{4x^2}{c^2}+4x$$

$$\Rightarrow \quad 4y=c^2+\frac{4x^2}{c^2} \qquad \Rightarrow \quad 4\frac{dy}{dx}=4.\frac{2x}{c^2}$$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{2x}{c^2} \qquad \Rightarrow \quad \frac{d^2y}{dx^2}=\frac{2}{c^2}$$

6. The derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t. $\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is

(a) -1 (b) 1

(c) $\frac{1}{2}$ (d) 2

Ans. (a)

Solution: Let $y=\sin^{-1}\frac{2x}{1+x^2}, z=\sin^{-1}\frac{1-x^2}{1+x^2}$

Put $x = \tan\theta$ $\qquad \therefore \quad y=\sin^{-1}(\sin 2\theta)=2\theta$

$z=\sin^{-1}(\cos 2\theta)$

$$= \sin^{-1}\left[\sin\left(\frac{\pi}{2}-2\theta\right)\right]=\frac{\pi}{2}-2\theta$$

$$\therefore \quad \frac{dy}{d\theta}=2, \quad \frac{dz}{d\theta}=-2 \quad \therefore \quad \frac{dy}{dz}=\frac{dy}{d\theta}\times\frac{d\theta}{dz}=\frac{2}{-2}=-1$$

7. If $f(9) = 9$ and $f'(9) = 4$, then $\lim_{x\to 9}\frac{\sqrt{f(x)}-3}{\sqrt{x}-3}$ equals

(a) 9 (b) 4

(c) 36 (d) none of these

Ans. (b)

Solution: $\lim_{x\to 9}\frac{\sqrt{f(x)}-3}{\sqrt{x}-3}=\lim_{x\to 9}\frac{f'(x)-9}{x-9}\times\frac{\sqrt{x}+3}{\sqrt{f(x)}+3}$

$$= f'(9)\times\left(\frac{\sqrt{9}+3}{\sqrt{f(9)}+3}\right)$$

$$= f'(9)\times\frac{3+3}{3+3}=f'(9)=4$$

8. If $y=f\left(\frac{2x-1}{x^2+1}\right)$ and $f'(x)=\sin x^2$, then $\frac{dy}{dx}$ is

(a) $2\left(\frac{x^2+x+1}{(x^2+1)^2}\right)\cdot\sin\left(\frac{2x-1}{x^2+1}\right)^2$

(b) $-2\left(\frac{x^2-x-1}{(x^2+1)^2}\right)\cdot\sin\left(\frac{2x+1}{x^2-1}\right)^2$

(c) $2\left(\frac{x^2-x-1}{(x^2+1)^2}\right)\cdot\sin\left(\frac{2x-1}{x^2+1}\right)^2$

(d) $-2\left(\frac{x^2-x-1}{(x^2+1)^2}\right)\cdot\sin\left(\frac{2x-1}{x^2+1}\right)^2$

Ans. (d)

Solution: $y=f\left(\frac{2x-1}{x^2+1}\right)$

$$\Rightarrow \quad \frac{dy}{dx}=f'\left(\frac{2x-1}{x^2+1}\right)\cdot\frac{(x^2+1)\cdot 2-(2x-1)\cdot 2x}{(x^2+1)^2}$$

$$= \frac{-2x^2+2x+2}{(x^2+1)^2}\cdot f'\left(\frac{2x-1}{x^2+1}\right)$$

$$= -2\left(\frac{x^2-x-1}{(x^2+1)^2}\right)\cdot\sin\left(\frac{2x-1}{x^2+1}\right)^2$$

$$\left[\because f'(x)=\sin^2 x \Rightarrow f'\left(\frac{2x-1}{x^2+1}\right)=\sin\left(\frac{2x-1}{x^2+1}\right)^2\right]$$

9. If $y=\sqrt{\log x+\sqrt{\log x+\sqrt{\log x+\cdots+\infty}}}$, then $\frac{dy}{dx}$ =

(a) $\frac{x}{2y-1}$ (b) $\frac{\log x}{2y-1}$

(c) $\frac{1}{x(2y-1)}$ (d) none of these

Ans. (c)

Solution: Here, $y=\sqrt{\log x+y} \Rightarrow y^2 = \log x + y$

$\Rightarrow \quad y^2 - y = \log x,$

Differentiating w.r.t. x, we get

$$2y\frac{dy}{dx}-\frac{dy}{dx}=\frac{1}{x} \qquad \Rightarrow \quad \frac{dy}{dx}=\frac{1}{x(2y-1)}$$

10. If $y=\sqrt{\cos x^2+\sqrt{\cos x^2+\sqrt{\cos x^2+\cdots+\infty}}}$, then $\frac{dy}{dx}$ =

(a) $\frac{-\sin x}{2y-1}$ (b) $\frac{-2x\sin x^2}{2y-1}$

(c) $\frac{-\sin x^2}{x(2y-1)}$ (d) none of these

Ans. (b)

Solution: Given $y=\sqrt{\cos x^2+y} \Rightarrow y^2 - y = \cos x^2,$

Differentiating w.r.t. x, we get

$$2y\frac{dy}{dx}-\frac{dy}{dx}=-(\sin x^2)\cdot 2x$$

$$= \frac{dy}{dx}=\frac{-2x\sin x^2}{2y-1}$$

11. If $y=ae^{mx}+be^{-mx}$, then y_2 is equal to

(a) $-m^2y$ (b) m^2y

(c) my_1 (d) none of these

Ans. (b)

Solution: $y_1 = ae^{mx}m + be^{-mx}(-m)$

$\Rightarrow \quad y_2 = mae^{mx}m - mbe^{-mx}(-m)$

$\Rightarrow \quad y_2 = m^2(ae^{mx}+be^{-mx}) = m^2y$

12. If $y = a \sin mx + b \cos mx$, then $\frac{d^2y}{dx^2}$ is equal to

(a) m^2y (b) $-m^2y$

(c) my_1 (d) none of these

Ans. (b)

Solution: $y_1 = a \sin mx \cdot m + b(-\sin mx)\, m$

$\Rightarrow \quad y_2 = ma(-\sin mx)\, m - mb \cos mx \cdot m$

$= -m^2y$

13. If $x = a\cos^3\theta$, $y = a\sin^3\theta$, then $\sqrt{1+\left(\frac{dy}{dx}\right)^2}$ =

(a) $\tan^2\theta$ (b) $\sec^2\theta$

(c) $-\sec\theta$ (d) $|\sec\theta|$

Ans. (d)

Solution: $x = a\cos^3\theta \quad \Rightarrow \frac{dx}{d\theta}=3a\cos^2\theta(-\sin\theta)$

$y = a\sin^3\theta \quad \Rightarrow \frac{dy}{d\theta}=3a\sin^2\theta(\cos\theta)$

$$\frac{dy}{dx}=\frac{dy}{d\theta}\times\frac{d\theta}{dx}=\frac{3a\sin^2\theta(\cos\theta)}{3a\cos^2\theta(-\sin\theta)}$$

$$\Rightarrow \frac{dy}{dx}=-\tan\theta$$

$$\Rightarrow \sqrt{1+\left(\frac{dy}{dx}\right)^2}=\sqrt{1+(-\tan\theta)^2}=\sqrt{1+\tan^2\theta}=|\sec\theta|$$

14. If $e^{x-y} = x^y$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{x}{(1+\log x)^2}$ (b) $\frac{x\log x}{(1+\log x)^2}$

(c) $\frac{\log x}{(1+\log x)^2}$ (d) none of these

Ans. (c)

Solution: $e^{x-y} = x^y$, taking log on both sides, we get

$(x-y)\log_e = y\log x \quad [\because \log_e e = 1]$

$\Rightarrow \quad x = y + y\log x \quad \Rightarrow x = y(1+\log x)$

$$\Rightarrow \quad y=\frac{x}{1+\log x}$$

Now differentiating w.r.t. x, we get

$$\frac{dy}{dx}=\frac{(1+\log x)\cdot 1 - x\cdot\frac{1}{x}}{(1+\log x)^2}$$

$$\Rightarrow \frac{dy}{dx}=\frac{\log x}{(1+\log x)^2}$$

15. If $\sqrt{x^2+y^2}=ae^{\tan^{-1}(y/x)}$, $a > 0$, $(y(0) > 0)$, then $y''(0)$ equals

(a) $\frac{a}{2}e^{\pi/2}$ (b) $ae^{\pi/2}$

(c) $-\frac{2}{a}e^{-\pi/2}$ (d) $\frac{a}{2}e^{-\pi/2}$

Ans. (c)

Solution: Given $\sqrt{x^2+y^2}=ae^{\tan^{-1}(y/x)}$ (i)

Differentiating w.r.t. x, we get

$$\frac{2x+2y\frac{dy}{dx}}{2\sqrt{x^2+y^2}} = ae^{\tan^{-1}(y/x)} \times \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{x\frac{dy}{dx}-y}{x^2}$$

$$\Rightarrow \sqrt{x^2+y^2}\left(x+y\frac{dy}{dx}\right) = ae^{\tan^{-1}(y/x)}\left(x\frac{dy}{dx}-y\right)$$

$$\Rightarrow \sqrt{x^2+y^2}\left(x+y\frac{dy}{dx}\right) = \sqrt{x^2+y^2}\times\left(x\frac{dy}{dx}-y\right)$$

[using (i)]

$$\Rightarrow \quad \frac{dy}{dx} = \frac{x+y}{x-y} \qquad \text{(ii)}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(x-y)\left(1+\frac{dy}{dx}\right)-(x+y)\left(1-\frac{dy}{dx}\right)}{(x-y)^2}$$

$$= \frac{-2y+2x\frac{dy}{dx}}{(x-y)^2} = \frac{-2y+2x\left(\frac{x+y}{x-y}\right)}{(x-y)^2} \quad \text{[using (ii)]}$$

$$= \frac{2(x^2+y^2)}{(x-y)^3}$$

$\therefore\ y''(0) =$ value of $\frac{d^2y}{dx^2}$ at $x = 0$

$$= \frac{-2[y(0)]^2}{[y(0)]^3} = -\frac{2}{y(0)} \qquad \text{(iii)}$$

Also, from (i) when $x = 0$,

$$|y(0)| = ae^{\pi/2}$$

$$\Rightarrow \quad y(0) = ae^{\pi/2} \qquad \text{(iv)}$$

From (iii) and (iv), $y''(0) = -\frac{2}{a}e^{-\pi/2}$.

Here, the result has been obtained assuming that $y(0) > 0$.

EXERCISE SET 1

1. $\frac{d}{dx}[\log(\tan x)]$ is equal to

(a) 2 sec 2x (b) sec 2x
(c) cosec 2x (d) 2 cosec 2x

2. If $y = 2^{\log x}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{2^{\log x}}{x}$ (b) $\frac{2^{\log x}}{\log 2}$

(c) $2^{\log x}\cdot\log 2$ (d) $\frac{2^{\log x}\cdot\log 2}{x}$

3. If $y = x\log x - x$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{1}{\log x}$ (b) $\frac{1}{x\log x}$

(c) $\log x$ (d) $\frac{1}{\log x - x}$

4. If $y = \tan^{-1}\left(\frac{a\cos x - b\sin x}{b\cos x + a\sin x}\right)$, $\frac{dy}{dx} =$

(a) $\frac{1}{1+x^2}$ (b) $\frac{1}{\sqrt{1-x^2}}$
(c) -1 (d) none of these

5. If $f(x) = \left(\frac{x^a}{x^b}\right)^{a+b}\cdot\left(\frac{x^b}{x^c}\right)^{b+c}\cdot\left(\frac{x^c}{x^a}\right)^{c+a}$ then $f'(x)$ is equal to

(a) 1 (b) 0
(c) x^{a+b+c} (d) none of these

6. The value of the derivative of $|x-1|+|x-3|$ at $x = 2$ is

(a) -2 (b) 0
(c) 2 (d) not defined

7. If $y = \log_e(\log_e x)$, then $e^y\frac{dy}{dx}$ is equal to

(a) e^y (b) $\frac{1}{x}$

(c) $\frac{1}{\log x}$ (d) $\frac{1}{x\log x}$

8. If

$$y = \frac{1}{1+x^{\beta-\alpha}+x^{\gamma-\alpha}} + \frac{1}{1+x^{\alpha-\beta}+x^{\gamma-\beta}} + \frac{1}{1+x^{\alpha-\gamma}+x^{\beta-\gamma}},$$

then $\frac{dy}{dx}$ is equal to

(a) 0 (b) 1
(c) $(\alpha+\beta+\gamma)x^{\alpha+\beta+\gamma-1}$ (d) none of these

9. If $y = ce^{\frac{x}{x-a}}$, then $\frac{dy}{dx}$ equals

(a) $a^2(x-a)^2$ (b) $a(x-a)^2$

(c) $-\frac{ay}{(x-a)^2}$ (d) none of these

10. If $y = \tan^{-1}(\text{cosec } x + \cot x)$, then $\frac{dy}{dx}$ equals

(a) $-\frac{1}{2}$ (b) -1

(c) 0 (d) 2

11. If $y = 1 + \frac{x}{\lfloor 1} + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \ldots$, $\frac{dy}{dx}$ is equal to

(a) 0 (b) 1

(c) x (d) y

12. If $y = 1 - x + \frac{x^2}{\lfloor 2} - \frac{x^3}{\lfloor 3} + \frac{x^4}{\lfloor 4} - \ldots$, then $\frac{d^2y}{dx^2}$ is equal to

(a) $-x$ (b) $-y$

(c) x (d) y

13. If $y = x^{x^{x^{x^{\ldots}}}}$, then $\frac{dy}{dx}$ is equal to

(a) yx^{y-1} (b) $\frac{y^2}{x(1-y\log x)}$

(c) $\frac{y}{x(1+y\log x)}$ (d) none of these

14. If $f(a) = 2, f'(a) = 1$, $g(a) = 1, g'(a) = 2$, then $\lim_{x\to a}\frac{g(x)f(a) - g(a)f(x)}{x-a}$ is equal to

(a) $-\frac{1}{3}$ (b) $\frac{1}{3}$

(c) -3 (d) 3

15. The derivative of $\log_{10} x$ with respect to x^2 is

(a) $\frac{1}{2x^2}\log_e 10$ (b) $\frac{2}{x^2}\log_{10} e$

(c) $\frac{1}{2x^2}\log_{10} e$ (d) none of these

16. If $\tan^{-1}\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}$, then $\frac{dy}{dx} =$

(a) $\frac{1}{\sqrt{1-x^2}}$ (b) $\frac{1}{2\sqrt{1-x^2}}$

(c) $\frac{x}{\sqrt{1-x^2}}$ (d) none of these

17. If $f(x) = |x|^{|\sec x|}$, then $f'\left(-\frac{\pi}{3}\right)$ is

(a) $\frac{2\pi}{3} - \frac{\pi^2}{3\sqrt{3}}\log\frac{\pi}{3}$ (b) $\frac{\pi}{3} - \frac{\pi^2\sqrt{3}}{2}\log\frac{\pi}{3}$

(c) $-\frac{2\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}\log\frac{\pi}{3}$ (d) none of these

18. If $f(x) = \begin{vmatrix} \sec\theta & \tan^2\theta & 1 \\ \theta\sec x & \tan x & x \\ 1 & \tan x - \tan\theta & 0 \end{vmatrix}$, then $f'(\theta)$ is

(a) 0 (b) -1

(c) independent of θ (d) none of these

19. If $e^x = \frac{\sqrt{1+t} - \sqrt{1-t}}{\sqrt{1+t} + \sqrt{1-t}}$ and $\tan\frac{y}{2} = \sqrt{\frac{1-t}{1+t}}$, then $\frac{dy}{dx}$ at $t = \frac{1}{2}$ is

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$

(c) 0 (d) none of these

20. If $f(x) = (\log_{\cot x}\tan x)(\log_{\tan x}\cot x)^{-1} + \tan^{-1}\left(\frac{x}{\sqrt{4-x^2}}\right)$, then $f(x)$ at $x = 0$ is

(a) 7/2 (b) 1/4

(c) 1/3 (d) none of these

21. If $f(x) = \sqrt{x^2+9}$, then $\lim_{x\to 4}\frac{f(x) - f(4)}{x-4}$ has the value

(a) 5/4 (b) $-4/5$

(c) 4/5 (d) none of these

22. The derivative of $\tan^{-1}\left(\frac{1+x}{1-x}\right)$ is

(a) $\frac{2}{1+x^2}$ (b) $\frac{1}{1+x^2}$

(c) $\frac{1+x^2}{1-x^2}$ (d) none of these

23. If $y = \log_a x + \log_x a + \log_x x + \log_a a$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{1}{x} + x\log a$ (b) $\frac{\log a}{x} + \frac{x}{\log a}$

(c) $\frac{1}{x\log a} + x\log a$ (d) $\frac{1}{x\log a} - \frac{\log a}{x(\log x)^2}$

24. If $f(x) = x^n$, then the value of

$$f(1)+\frac{f'(1)}{\lfloor 1}+\frac{f''(1)}{\lfloor 2}+\frac{f'''(1)}{\lfloor 3}+\cdots+\frac{f^{(n)}(1)}{\lfloor n} \text{ is}$$

(a) n (b) 2^n

(c) 2^{n-1} (d) $\dfrac{n(n+1)}{2}$

25. If $y=\sin^{-1}2x\cdot\sqrt{1-x^2}+\sec^{-1}\dfrac{1}{\sqrt{(1-x^2)}}$, then $\dfrac{dy}{dx}$ =

(a) $\dfrac{1}{\sqrt{1-x^2}}$ (b) $\dfrac{2}{\sqrt{1-x^2}}$

(c) $\dfrac{3}{\sqrt{1-x^2}}$ (d) none of these

26. Derivative of the function $f(x) = \log_5(\log_{7x})$, $x > 7$ is

(a) $\dfrac{1}{x(\log 5)(\log 7)(\log_7 x)}$ (b) $\dfrac{1}{x(\log 5)(\log 7)}$

(c) $\dfrac{1}{x(\log x)}$ (d) none of these

27. $\dfrac{d}{dx}\left(\sin^2\cot^{-1}\dfrac{1}{\sqrt{\dfrac{1+x}{1-x}}}\right)=$

(a) 0 (b) $-\dfrac{1}{2}$

(c) $\dfrac{1}{2}$ (d) –1

28. If $y=\cot^{-1}\left[\dfrac{\sqrt{1+\sin x}+\sqrt{1+\sin x}}{\sqrt{1+\sin x}-\sqrt{1+\sin x}}\right]$, then $\dfrac{dy}{dx}=$

(a) $\dfrac{1}{2}$ (b) $\dfrac{2}{3}$

(c) 3 (d) 1

29. The derivative of $\cos^{-1}\dfrac{1-x^2}{1+x^2}$ w.r.t. $\cot^{-1}\dfrac{1-3x^2}{3x+x^3}$ is

(a) $\dfrac{1}{2}$ (b) $\dfrac{2}{3}$

(c) 1 (d) $\dfrac{3}{2}$

30. If $y^{1/m}=\left[x+\sqrt{1+x^2}\right]$, then $(1+x^2)y_2+xy_1$ is equal to

(a) m^2y (b) my^2

(c) m^2y^2 (d) none of these

31. If $y = (1 + x)^{1/x}$, then $\dfrac{dy}{dx}=$

(a) $\dfrac{x}{1+x}(1+x)^{1/x}-\log(1+x)$

(b) $(1+x)^{1/x}\dfrac{\left[\dfrac{x}{1+x}-\log(1+x)\right]}{x^2}$

(c) $(1+x)^{1/x}\cdot x\log(1+x)$

(d) none of these

32. If $f(x)=\log_{x^2}(\log x)$, then $f'(x)$ at $x = e$ is

(a) 0 (b) 1

(c) $\dfrac{1}{e}$ (d) $\dfrac{1}{2e}$

33. If $y=\dfrac{2}{\sqrt{a^2-b^2}}\tan^{-1}\left(\sqrt{\dfrac{a-b}{a+b}}\tan\dfrac{x}{2}\right), a>b>0$, then

(a) $\dfrac{dy}{dx}=\dfrac{1}{a+b\cos x}$ (b) $\dfrac{d^2y}{dx^2}=\dfrac{b\sin x}{(a+b\cos x)^2}$

(c) $\dfrac{dy}{dx}=\dfrac{1}{a-b\cos x}$ (d) $\dfrac{d^2y}{dx^2}=\dfrac{-b\sin x}{(a-b\cos x)^2}$

34. If $y=\cos^{-1}\sqrt{\dfrac{\sqrt{1+x^2}+1}{2\sqrt{1+x^2}}}$ w.r.t. x, then $\dfrac{dy}{dx}\left(\cot^{-1}\dfrac{1-3x^2}{3x-x^3}\right)$ is

(a) $\dfrac{1}{2(1+x^2)}$ (b) $\dfrac{1}{1+x^2}$

(c) $-\dfrac{1}{2(1+x^2)}$ (d) none of these

35. If $f(x)=|x-2|$ and $g(x)=f[f(x)]$, then for $x > 20$, $g'(x)$ is equal to

(a) –1 (b) 0

(c) 1 (d) 2

36. If $y=(\log x)^x+x^{\log x}$, then $\dfrac{dy}{dx}=$

(a) $(\log x)^x\left[\log(\log x)+\dfrac{1}{\log x}\right]+x^{\log x}\cdot\left(\dfrac{2\log x}{x}\right)$

(b) $(\log x)^x\left[\log(\log x)-\dfrac{1}{\log x}\right]-x^{\log x}\cdot\left(\dfrac{2\log x}{x}\right)$

(c) $(\log x)^x\left[\log(\log x)+\dfrac{1}{\log x}\right]-x^{\log x}\cdot\left(\dfrac{2\log x}{x}\right)$

(d) none of these

37. If $f(x)=\cot^{-1}\left(\frac{x^x-x^{-x}}{2}\right)$, then $f'(1)$ is equal to

(a) –1 (b) 1
(c) log 2 (d) – log 2

38. If $u = f(x^3)$, $v = g(x^2)$, $f'(x) = \cos x$ and $g'(x) = \sin x$, then $\frac{du}{dv}$ is

(a) $\frac{3}{2}x\cdot\cos x^3\cdot\operatorname{cosec} x^2$ (b) $\frac{2}{3}\sin x^3\cdot\sec x^2$
(c) $\tan x$ (d) none of these

39. $\frac{d}{dx}\left(\cos^{-1}\frac{x-x^{-1}}{x+x^{-1}}\right)=$

(a) $\frac{1}{1+x^2}$ (b) $\frac{-2}{1+x^2}$
(c) $\frac{2}{1-x^2}$ (d) none of these

40. The differential coefficient of the function $|x-1|+|x-3|$ at the point $x = 2$ is

(a) –2 (b) 0
(c) 2 (d) undefined

41. If $y=\frac{ax^2}{(x-a)(x-b)(x-c)}+\frac{bx}{(x-b)(x-c)}+\frac{c}{x-c}+1$, then $\frac{y'}{y}=$

(a) $\frac{1}{x}\left(\frac{a}{a-x}+\frac{b}{b-x}+\frac{c}{c-x}\right)$

(b) $\frac{a}{a-x}+\frac{b}{b-x}+\frac{c}{c-x}$

(c) $\frac{1}{x}\left(\frac{a}{x-a}+\frac{b}{x-b}+\frac{c}{x-c}\right)$

(d) none of these

42. Let $f(x) = x^n$, n being a non-negative integer. The value of n for which the equality $f'(x + y) = f'(x) + f'(y)$ is valid for all $x, y > 0$ is

(a) 0, 1 (b) 1, 2
(c) 2, 4 (d) none of these

43. If $f(x) = x + \tan x$ and f is inverse of g, then $g'(x)$ is equal to

(a) $\frac{1}{1+[g(x)-x]^2}$ (b) $\frac{1}{2-[g(x)-x]^2}$

(c) $\frac{1}{2+[g(x)-x]^2}$ (d) none of these

44. If S_n denotes the sum of mn terms of a G.P. whose common ratio is r, then $(r-1)\frac{dS_n}{dr}$ is equal to

(a) $(n-1)S_n + nS_{n-1}$ (b) $(n-1)S_n - nS_{n-1}$
(c) $(n-1)S_n$ (d) none of these

45. If $xe^{xy} = y+\sin^2 x$, then at $x=0$, $\frac{dy}{dx}$ is equal to

(a) –1 (b) 1
(c) 0 (d) none of these

46. If $\sqrt{1-x^6}+\sqrt{1-y^6} = a^3(x^3-y^3)$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$ (b) $\frac{y^2}{x^2}\sqrt{\frac{1-y^6}{1-x^6}}$

(c) $\frac{x^2}{y^2}\sqrt{\frac{1-x^6}{1-y^6}}$ (d) none of these

47. If g is inverse of f and $f'(x)=\frac{1}{1+x^n}$, then $g'(x)$ equals

(a) $1 + x^n$ (b) $1 + [f(x)]^n$
(c) $1 + [g(x)]^n$ (d) none of these

48. If $f'(1) = 2$ and $g'(\sqrt{2})=4$, then the derivative of $f(\tan x)$ w.r.t. $g(\sec x)$ at $x=\frac{\pi}{4}$ is

(a) 1 (b) $\sqrt{2}$
(c) $\sqrt{2}/2$ (d) 2

49. If $y=f(x)=\left(\frac{\tan^m x}{\tan^n x}\right)^{m+n}\left(\frac{\tan^n x}{\tan^p x}\right)^{n+p}\left(\frac{\tan^p x}{\tan^m x}\right)^{p+m}$, then $\frac{dy}{dx}$ =

(a) 0 (b) 1
(c) $\tan^{m+n+p}x$ (d) none of these

50. The value of $\frac{d}{dx}(|x-1|+|x-5|)$ at $x = 3$ is

(a) –2 (b) 0
(c) 2 (d) 4

51. Let $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}$ for all real x and y for $f'(0) = -1$, $f(0) = 1$, then $f'(2) =$

(a) $\frac{1}{2}$ (b) 1

(c) –1 (d) $-\frac{1}{2}$

52. If $y=(1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^n})$, then $\frac{dy}{dx}$ at $x = 0$ is

(a) 1 (b) –1
(c) 0 (d) none of these

53. If $y = \sin px$ and y_n is the nth derivative of y, then $\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix}$ is equal to

(a) 1 (b) 0
(c) –1 (d) none of these

54. If $y=\sin^{-1}\frac{\sqrt{(1+x)}+\sqrt{(1-x)}}{2}$, then $\frac{dy}{dx}=$

(a) $\frac{1}{\sqrt{(1-x^2)}}$ (b) $-\frac{1}{\sqrt{(1-x^2)}}$
(c) $-\frac{1}{2\sqrt{(1-x^2)}}$ (d) none of these

55. If $f(x) = \cos x\cdot\cos 2x\cdot\cos 4x\cdot\cos 8x\cdot\cos 16x$, then the value of $f'\left(\frac{\pi}{4}\right)$ is

(a) 1 (b) $\sqrt{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) 0

56. If $f(x)=x\tan^{-1}x$, then $f'(1)$ equals

(a) $\frac{1}{2}+\frac{\pi}{4}$ (b) $-\frac{1}{2}+\frac{\pi}{4}$
(c) $-\frac{1}{2}-\frac{\pi}{4}$ (d) $\frac{1}{2}-\frac{\pi}{4}$

57. If $f(x)=\frac{x^2-x}{x^2+2x}$, then $\frac{df^{-1}(x)}{dx}$ is equal to

(a) $-\frac{3}{(1-x)^2}$ (b) $\frac{3}{(1-x)^2}$
(c) $\frac{1}{(1-x)^2}$ (d) none of these

58. If $2x^3-3x^2y^2+4x-y+7=0$ and $y(1) = 1$, then the value of $y^n(1)$ is equal to

(a) $-\frac{343}{474}$ (b) $-\frac{474}{343}$
(c) $\frac{474}{343}$ (d) $\frac{343}{474}$

59. If $\frac{\cos^4\alpha}{x}+\frac{\sin^4\alpha}{y}=\frac{1}{x+y}$, then $\frac{dy}{dx}=$

(a) $\sqrt{\left(\frac{1-y^2}{1-x^2}\right)}$ (b) $\sqrt{\left(\frac{1-x^2}{1-y^2}\right)}$
(c) $\sqrt{(1-x^2)(1-y^2)}$ (d) none of these

60. If $y=\sqrt{\sin x+\sqrt{\sin x+\sqrt{\sin x+\ldots+\infty}}}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{\cos x}{2y-1}$ (b) $\tan^2\alpha$
(c) 0 (d) $(x^2 + y^2)\sec^2\alpha$

61. If $y = ax^{n+1} + bx^{-n}$, then $x^2\frac{d^2y}{dx^2}=$

(a) $n(n - 1)y$ (b) $n(n + 1)y$
(c) ny (d) n^2y

62. If $x^2 + y^2 = 1$, then

(a) $yy''-2(y')^2+1=0$ (b) $yy''+(y')^2+1=0$
(c) $yy''+(y')^2-1=0$ (d) $yy''+2(y')^2+1=0$

63. If $x = \text{cosec}\,\theta - \sin\theta$, $y = \text{cosec}^n\theta - \sin^n\theta$, then $(x^2+4)\left(\frac{dy}{dx}\right)^2-n^2y^2=$

(a) n^2 (b) $2n^2$
(c) $3n^2$ (d) $4n^2$

64. If $\sin^{-1}[\sqrt{x-ax}-\sqrt{a-ax}]$, then $\frac{dy}{dx}=$

(a) $\frac{1}{\sin\sqrt{a-ax}}$ (b) $\sin\sqrt{x}\sin\sqrt{a}$
(c) $\frac{1}{2\sqrt{x}\sqrt{1-x}}$ (d) 0

65. If variables x and y are related by the equation $x=\int_0^y\frac{1}{\sqrt{1+9u^2}}\,du$, then $\frac{d^2y}{dx^2}$ is equal to

(a) $\sqrt{1+9y^2}$ (b) $\frac{1}{1+9y^2}$
(c) $9y$ (d) $\frac{1}{9}y$

66. If $f(0) = 0 = g(0)$ and $f'(0) = 6 = g'(0)$, then $\lim_{x\to 0}\frac{f(x)}{g(x)}$ is given by

(a) 1 (b) 0
(c) 12 (d) –1

67. The derivative of $f(\tan x)$ with respect to $g(\sec x)$ at $x=\frac{\pi}{4}$, where $f'(1)=2$ and $g'(\sqrt{2})=4$, is

(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$

(c) 1 (d) 0

68. If $y=x^2+\cfrac{1}{x^2+\cfrac{1}{x^2+\cfrac{1}{x^2+\ldots+\infty}}}$, then $\frac{dy}{dx}=$

(a) $\frac{2xy}{2y-x^2}$ (b) $\frac{xy}{y+x^2}$

(c) $\frac{xy}{y-x^2}$ (d) $\frac{2xy}{2+(x^2/y)}$

69. If $y=\sum_{n=1}^{x}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)$, then $\frac{dy}{dx}=$

(a) $\frac{1}{1+x^2}$ (b) $\frac{1}{1+(1+x)^2}$

(c) 0 (d) none of these

70. $\frac{d^2}{dx^2}(2\cos x\cos 3x)$ is equal to

(a) $2^2(\cos 2x + 2^2\cos 4x)$
(b) $2^2(\cos 2x - 2^2\cos 4x)$
(c) $2^2(-\cos 2x + 2^2\cos 4x)$
(d) $-2^2(\cos 2x + 2^2\cos 4x)$

71. If $f(x) = e^x g(x)$, $g(0)=2, g'(0)=1$, then $f'(0)=$

(a) 1 (b) 3

(c) 2 (d) 0

EXERCISE SET 2

1. If $y=\left(1+\frac{1}{x}\right)^x$, then $\frac{dy}{dx}$ equals

(a) $\left(1+\frac{1}{x}\right)^x\left[\log\left(1+\frac{1}{x}\right)-\frac{1}{1+x}\right]$

(b) $\left(1+\frac{1}{x}\right)^x\left[\log\left(1+\frac{1}{x}\right)\right]$

(c) $\left(x+\frac{1}{x}\right)^x\left[\log(x+1)-\frac{x}{x+1}\right]$

(d) $\left(x+\frac{1}{x}\right)^x\left[\log\left(1+\frac{1}{x}\right)+\frac{x}{1+x}\right]$

2. If $\sin(x+y)=\log(x+y)$, then $\frac{dy}{dx}$ is equal to

(a) 2 (b) –2

(c) 1 (d) –1

3. If $f(x)=|\cos x-\sin x|$, then $f'\left(\frac{\pi}{4}\right)$ is equal to

(a) $\sqrt{2}$ (b) $-\sqrt{2}$

(c) 0 (d) none of these

4. If $f(a)=a^2, \phi(a)=b^2$ and $f'(a)=3\phi'(a)$, then $\lim_{x\to a}\frac{\sqrt{f(x)}-a}{\sqrt{\phi(x)}-b}$ is

(a) $\frac{b^2}{a^2}$ (b) $\frac{b}{a}$

(c) $\frac{2b}{a}$ (d) none of these

5. If $y=\cos^{-1}\left(\frac{2\cos x-3\sin x}{\sqrt{13}}\right)$, then $\frac{dy}{dx}$ is equal to

(a) 0 (b) 1

(c) constant $\neq 1$ (d) none of these

6. If $x^2+y^2=t-\frac{1}{t}$ and $x^4+y^4=t^2+\frac{1}{t^2}$ then $x^3y\frac{dy}{dx}$ is equal to

(a) 0 (b) 1

(c) –1 (d) none of these

7. If $y=f(x)=\tan^{-1}\left[\frac{\log(e/x^2)}{\log(ex^2)}\right]+\tan^{-1}\left(\frac{3+2\log x}{1-6\log x}\right)$, then $\frac{d^ny}{dx^n}$ is equal to

(a) $\tan^{-1}[(\log x)^n]$ (b) 0

(c) $\frac{1}{2}$ (d) none of these

8. If $f(x)$ is an odd differentiable function defined on $(-\infty,\infty)$ such that $f'(3)=2$, then $f'(-3)$ equals

(a) 0 (b) 1

(c) 2 (d) 4

9. If $y = \log \tan \frac{x}{2} + \sin^{-1}(\cos x)$, then $\frac{dy}{dx}$ is

(a) cosec $x - 1$ (b) cosec x
(c) cosec $x + 1$ (d) x

10. Let $y = e^{2x}$, then $\frac{d^2y}{dx^2} \cdot \frac{d^2x}{dy^2}$ is

(a) 1 (b) $-2e^{-2x}$
(c) $2e^{-2x}$ (d) xe^{-2x}

11. If $f(x) = \sin\left\{\frac{\pi}{3}[x] - x^2\right\}$ for $2 < x < 3$ and $[x]$ denotes the greatest integer less than or equal to x, then $f'[\sqrt{\pi/3}]$ is equal to

(a) $-\sqrt{\frac{\pi}{3}}$ (b) $\sqrt{\frac{\pi}{3}}$

(c) $-\sqrt{\frac{\pi}{2}}$ (d) $-\sqrt{\pi}$

12. $y = \tan^{-1}\frac{1}{1+x+x^2} + \tan^{-1}\frac{1}{x^2+3x+3}$

$+\tan^{-1}\frac{1}{x^2+5x+7} + \cdots + n$ terms, then $y'(0) =$

(a) $\frac{n^2}{n^2+1}$ (b) $\frac{-n^2}{n^2+1}$

(c) $\frac{-1}{n^2+1}$ (d) $\frac{1}{n^2+1}$

13. Let $f(x) = \alpha(x)\beta(x)\gamma(x)$ for all real x, where $\alpha(x)$, $\beta(x)$ and $\gamma(x)$ are differentiable functions of x.
If $f'(2) = 18f(2)$, $\alpha'(2) = 3\alpha(2)$, $\beta'(2) = -4\beta(2)$ and $\gamma'(2) = k\gamma(2)$, then the value of k is
(a) 14 (b) 16
(c) 19 (d) none of these

14. If $y = \log|x|$, then $\frac{dy}{dx} =$

(a) $\frac{1}{x}$ (b) $-\frac{1}{x}$

(c) $\frac{1}{|x|}$ (d) none of these

15. If $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$, then $\frac{dy}{dx}$ equals

(a) 0 (b) 1
(c) -1 (d) 2

16. If $y = (x + \sqrt{x^2+a^2})^n$, then $\frac{dy}{dx} =$

(a) $\frac{nx}{\sqrt{x^2+a^2}}$ (b) $\frac{n}{\sqrt{x^2+a^2}}$

(c) $\frac{ny}{\sqrt{x^2+a^2}}$ (d) none of these

17. Differential coefficient of $\log_{10} x$ w.r.t. $\log_x 10$ is

(a) $-\frac{(\log x)^2}{(\log 10)^2}$ (b) $\frac{(\log_{10} x)^2}{(\log 10)^2}$

(c) $\left(\frac{\log_x 10}{\log 10}\right)$ (d) $-\frac{(\log 10)^2}{(\log x)^2}$

18. If $x = \sqrt{\frac{1-t^2}{1+t^2}}$ and $y = \frac{\sqrt{1+t^2} - \sqrt{1-t^2}}{\sqrt{1+t^2} + \sqrt{1-t^2}}$, then the value of $\frac{d^2y}{dx^2}$ at $t = 0$ is given by

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) -1

19. If $y = x^y$, then $x(1 - y\log x) \cdot \frac{dy}{dx} =$

(a) x^2 (b) y^2
(c) xy^2 (d) none of these

20. If $y = |x|^{|\sin x|}$, then the value of $\frac{dy}{dx}$ at $x = -\frac{\pi}{4}$ is equal to

(a) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}}\left[\frac{1}{\sqrt{2}}\log\frac{4}{\pi} - \frac{2\sqrt{2}}{\pi}\right]$

(b) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}}\left[\frac{1}{\sqrt{2}}\log\frac{4}{\pi} + \frac{2\sqrt{2}}{\pi}\right]$

(c) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}}\left[\frac{1}{\sqrt{2}}\log\frac{\pi}{4} - \frac{2\sqrt{2}}{\pi}\right]$

(d) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}}\left[\frac{1}{\sqrt{2}}\log\frac{\pi}{4} + \frac{2\sqrt{2}}{\pi}\right]$

21. If $y = \log_{10} x + \log_e y$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{y}{y-1}$ (b) $\frac{y}{x}$

(c) $\frac{\log_{10} e}{x}\left(\frac{y}{y-1}\right)$ (d) none of these

22. If $f(x)=\frac{\sin^{-1}x}{\sqrt{1-x^2}}$, then the value of $(1-x^2)f'(x)-xf(x)$ is

(a) 1 (b) 2
(c) 3 (d) 4

23. If $y=\log\left(\frac{x}{a+bx}\right)^x$, then $x^3\frac{d^2y}{dx^2}$ is equal to

(a) $\left(\frac{dy}{dx}+x\right)^2$ (b) $\left(\frac{dy}{dx}-y\right)^2$

(c) $\left(x\frac{dy}{dx}+y\right)^2$ (d) $\left(x\frac{dy}{dx}-y\right)^2$

24. If $y=\cos^{-1}\left(\frac{5\cos x-12\sin x}{13}\right)$, $x\in\left(0,\frac{\pi}{2}\right)$, then $\frac{dy}{dx}$ is equal to

(a) 1 (b) -1
(c) 0 (d) none of these

25. If $y=\sqrt{(a-x)(x-b)}-(a-b)\tan^{-1}\sqrt{\frac{a-x}{x-b}}$, then $\frac{dy}{dx}=$

(a) 1 (b) $\sqrt{\frac{a-x}{x-b}}$

(c) $\sqrt{(a-x)(x-b)}$ (d) $\frac{1}{\sqrt{(a-x)(b-x)}}$

26. If $y=x^{(x^x)}$, then $\frac{dy}{dx}=$

(a) $y[x^x(\log ex)\cdot\log x+x^x]$

(b) $y[x^x(\log ex)\cdot\log x+x]$

(c) $y[x^x(\log ex)\cdot\log x+x^{x-1}]$

(d) $y[x^x(\log_e x)\cdot\log x+x^{x-1}]$

27. If $y=\sin^{-1}(x\sqrt{1-x}+\sqrt{x}\sqrt{1-x^2})$, then $\frac{dy}{dx}=$

(a) $\frac{-2x}{\sqrt{1-x^2}}+\frac{1}{2\sqrt{x-x^2}}$

(b) $\frac{-1}{\sqrt{1-x^2}}-\frac{1}{2\sqrt{x-x^2}}$

(c) $\frac{1}{\sqrt{1-x^2}}+\frac{1}{2\sqrt{x-x^2}}$

(d) none of these

28. If $y=\frac{2}{\sqrt{a^2-b^2}}\tan^{-1}\left(\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}\right)$, $a>b>0$, then

(a) $\frac{dy}{dx}=\frac{1}{a+b\cos x}$ (b) $\frac{dy}{dx}=\frac{1}{a-b\cos x}$

(c) $\frac{d^2y}{dx^2}=\frac{-b\sin x}{(a-b\cos x)^2}$ (d) none of these

29. If $y=\tan^{-1}\frac{x-\sqrt{1-x^2}}{x+\sqrt{1-x^2}}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{-1}{\sqrt{1-x^2}}$ (b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{-x}{\sqrt{1-x^2}}$ (d) none of these

30. If $x=t^t$ and $y=t^{t^t}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{t^{t^t}\cdot\left[(1+\log t)\log t+\frac{1}{t}\right]}{(1+\log t)}$

(b) $\frac{t^{t^t}\cdot\left[1+\log t+\frac{1}{t}\right]}{(1+\log t)}$

(c) $\frac{t^{t^t}\cdot\left[(1+\log t)\log t+\frac{1}{t}\right]}{(1+\log t)^2}$

(d) none of these

31. If $f(x)=\left(\frac{a+x}{b+x}\right)^{a+b+2x}$, then $f'(x)=(0)$ is equal to

(a) $\left(2\log\frac{a}{b}+\frac{a^2-b^2}{ab}\right)\left(\frac{a}{b}\right)^{a+b}$

(b) $\left(2\log\frac{a}{b}+\frac{b^2-a^2}{ab}\right)\left(\frac{a}{b}\right)^{a+b}$

(c) $\left(2\log\frac{a}{b}+\frac{a^2+b^2}{ab}\right)\left(\frac{a}{b}\right)^{a+b}$

(d) none of these

32. If $e^x+e^y=e^{x+y}$, then the value of $\frac{dy}{dx}$ at (1, 1) is

(a) 0 (b) -1
(c) 1 (d) none of these

33. If $y=\tan^{-1}\left(\frac{4x}{1+5x^2}\right)+\tan^{-1}\left(\frac{2+3x}{3-2x}\right)$, then $\frac{dy}{dx}$ =

(a) $\frac{1}{1+x^2}$ (b) $\frac{5}{1+25x^2}$

(c) 1 (d) none of these

34. The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ w.r.t.

$\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at $x = 0$ is

(a) $\frac{1}{8}$ (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) 1

35. If $f(x) = \sin^{-1}(\cos x)$, then the value of $f(10) + f'(10)$ is equal to

(a) $11-\frac{7\pi}{2}$ (b) $\frac{7\pi}{2}-11$

(c) $\frac{5\pi}{2}-11$ (d) none of these

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement 1 is false, Statement 2 is true.

(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

(d) Statement 1 is true, Statement 2 is false.

36. Statement 1:

Derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t.

$\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is 1, for $0 < x < 1$.

Statement 2:

$\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ for $-1\le x\le 1$.

37. Statement 1:

If $y = f(x) = x^2$, $x > 0$, then $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right)=1$.

Statement 2:

$\frac{dy}{dx}\cdot\frac{dx}{dy}=1$

38. Statement 1:

$\frac{d}{dx}(\sin^{-1}(2x^2+1)+\cos^{-1}(2x^2+1))$ is zero.

Statement 2:

Derivative of a constant is zero.

39. Let $y = \sin(3 \sin^{-1} x)$

Statement 1: $y''(0) = 0$

Statement 2: $y'(0) = 3$

Previous Years' Questions

40. If $\sin y = x \sin(x + y)$, then $\frac{dy}{dx}$ is

(a) $\frac{\sin a}{\sin^2(a+y)}$ (b) $\frac{\sin^2(a+y)}{\sin a}$

(c) $\sin a \sin^2(a+y)$ (d) $\frac{\sin^2(a+y)}{\sin a}$

[AIEEE 2002]

41. If $y=(x+\sqrt{1+x^2})^n$, then $(1+x^2)\frac{d^2y}{dx^2}+x\frac{dy}{dx}$ is

(a) n^2y (b) $-n^2y$

(c) $-y$ (d) $2n^2y$

[AIEEE 2002]

42. If $x^y = e^{x-y}$, then $\frac{dy}{dx}$ is

(a) $\frac{1+x}{1+\log x}$ (b) $\frac{1-\log x}{1+\log x}$

(c) not defined (d) $\frac{\log x}{(1+\log x)^2}$

[AIEEE 2002]

43. Let $f(2) = 4$ and $f'(2) = 4$. Then $\lim_{x\to 2}\frac{xf(2)-2f(x)}{x-2}$ is given by

(a) 2 (b) –2

(c) – 4 (d) 3

[AIEEE 2003]

44. If $f(x+y)=f(x)\cdot f(y)\ \forall\ x, y\in\ \mathbb{R}$ and $f(5)=2, f(0)=2$, then $f'(5)$ is

(a) 0 (b) 1
(c) 6 (d) 2

[AIEEE 2003]

45. If $x=e^{y+e^{y+e^{y+\cdots\infty}}}$, $x>0$, then $\frac{dy}{dx}$ is

(a) $\frac{1-x}{x}$ (b) $\frac{1}{x}$
(c) $\frac{x}{1+x}$ (d) $\frac{1+x}{x}$

[AIEEE 2004]

46. Let f be differentiable for all x.
If $f(1)=-2$ and $f'(x)\geq 2$ for $x\in[1,6]$, then

(a) $f(6)<5$ (b) $f(6)=5$
(c) $f(6)\geq 8$ (d) $f(6)<8$

[AIEEE 2005]

47. If f is a real-valued differentiable function satisfying $|f(x)-f(y)|\leq(x-y)^2, x, y\in\mathbb{R}$ and $f(0)=0$, then $f(1)$ equals

(a) 2 (b) 1
(c) –1 (d) 0

[AIEEE 2005]

48. Suppose $f(x)$ is differentiable at $x=1$ and $\lim_{h\to 0}\frac{1}{h}f(1+h)=5$, then $f'(1)$ equals

(a) 5 (b) 6
(c) 3 (d) 4

[AIEEE 2005]

49. If $x^m y^n=(x+y)^{m+n}$, then $\frac{dy}{dx}$ is

(a) $\frac{x}{y}$ (b) $\frac{y}{x}$
(c) $\frac{x+y}{xy}$ (d) xy

[AIEEE 2006]

50. The set of all points where the function $f(x)=\frac{x}{1+|x|}$ is differentiable is

(a) $(-\infty,\infty)$ (b) $(0,\infty)$
(c) $(-\infty,0)\cup(0,\infty)$ (d) none of these

[AIEEE 2006]

51. Let y be an implicit function of x defined by $x^{2x}-2x^x\cot y-1=0$. Then, $y'(1)$ is

(a) –1 (b) 1
(c) log 2 (d) – log 2

[AIEEE 2007]

52. The first derivative of the function $\left[\cos^{-1}\left(\sin\sqrt{\frac{1+x}{2}}\right)+x^x\right]$ with respect to x at $x=1$ is

(a) $-\frac{1}{2}$ (b) 0
(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

[UPSEAT 1998, MP PET 1998, AMU 2003]

53. $f(x)=\log_x(\log x)$, then $f'(x)$ at $x=e$ is

(a) 0 (b) 1
(c) $\frac{1}{e}$ (d) $\frac{1}{2}e$

[Punjab CEET 2002, DCE 2000, Kerala PET 2003, Rajasthan PET 2000, MP PET 2000]

54. The differential coefficient of $f(\log x)$ w.r.t. x, where $f(x)=\log x$, is

(a) $(x\log x)^{-1}$ (b) $\frac{x}{\log x}$
(c) $\frac{\log x}{x}$ (d) none of these

[DCE 2000, Punjab CEET 2007, AMU 1995]

55. If $y=e^{1+\log_e x}$, then the value of $\frac{dy}{dx}$ is equal to

(a) 0 (b) e
(c) 1 (d) $\log_e xe^{\log_e ex}$

[AMU 2002, DCE 2002, MP PET 1996, Punjab CEET 2001]

56. If $y=\sec^{-1}\left(\frac{x+1}{x-1}\right)+\sin^{-1}\left(\frac{x-1}{x+1}\right)$, then $\frac{dy}{dx}$ is equal to

(a) 0 (b) 1
(c) $\left(\frac{x+1}{x-1}\right)$ (d) $\left(\frac{x-1}{x+1}\right)$

[Kerala PET 2005, DCE 2000, 2003, KET 1998, UPSEAT 2002]

57. The derivative of $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$ with respect to $\sqrt{1-x^2}$ at $x=\frac{1}{2}$ is

(a) –2 (b) 1
(c) 2 (d) 4

[Punjab CEET 2003, DCE 1994, Haryana CEET 2002]

58. The derivative of $f(x)=|x|^3$ at $x = 0$ is

(a) -1 (b) 0

(c) 1 (d) none of these

[Rajasthan PET 2001, Kerala PET 2003, Haryana CEET 2002]

59. If $x^p y^q = (x+y)^{p+q}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{px-qy}{(x+y)}$ (b) $\frac{qx-py}{px+qy}$

(c) $\frac{(p+q)(x+y)}{px-qx}$ (d) $\frac{y}{x}$

[J&KCET 2005, AIEEE 2006, Rajasthan PET 1999]

60. If $\sin y = x(\sin(a+y)$, then the value of $\frac{dy}{dx}$ is equal to

(a) $\sin(a+y)$ (b) $\frac{\sin\sqrt{a}}{\sin(a+y)}$

(c) $\frac{\sin^2(a+y)}{\sin a}$ (d) none of these

[Punjab CEET 2001, KCET 2000, UPSEAT 2002, Kerala PET 2005, EAMCET 1995]

61. If $f'(x)=\sin(\log x)$ and $y=f\left(\frac{2x+3}{3-2x}\right)$, then $\frac{dy}{dx}$ equals

(a) $\sin(\log x)\cdot\frac{1}{x\log x}$

(b) $\frac{12}{(3-2x)^2}\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

(c) $\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

(d) none of these

[Punjab CEET 2007, AMU 1996]

62. The value of $\frac{d}{dx}\left[\log\left(x+\frac{1}{x}\right)\right]$ is

(a) $\left(x+\frac{1}{x}\right)$ (b) $\left(1+\frac{1}{x}\right)$

(c) $\frac{\left(1-\frac{1}{x^2}\right)}{\left(\frac{1}{x}+x\right)}$ (d) $\frac{\left(1+\frac{1}{x^2}\right)}{\left(x+\frac{1}{x}\right)}$

[MP PET 1995, Rajasthan PET 1996]

63. The derivative of $\tan^{-1}\left[\frac{2x}{1-x^2}\right]$ w.r.t. $\sin^{-1}\left[\frac{2x}{1+x^2}\right]$ is

(a) 1 (b) 2

(c) 3 (d) 4

[UPSEAT 2001, AMU 1996]

64. Differential coefficient of $\sec(\tan^{-1}x)$ is

(a) $\frac{x}{1+x^2}$ (b) $\frac{x}{\sqrt{1+x^2}}$

(c) $\frac{1}{\sqrt{1+x^2}}$ (d) $x\sqrt{1+x^2}$

[Haryana CEET 2001, REE 1993, DCE 2002]

65. If $y=\log_{\cos x}\sin x$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{(\cot x\log\cos x+\tan x\log\sin x)}{(\log\cos x)^2}$

(b) $\frac{\sec x\log\cos x+\operatorname{cosec} x\log\sin x}{(\log\cos x)^2}$

(c) $\frac{\tan x\log\cos x+\cot x\log\sin x}{(\log\cos x)^2}$

(d) $\frac{\operatorname{cosec} x\log\cos x+\sec x\log\sin x}{(\log\cos x)^2}$

[Jamia Millia Islamia 2003]

ANSWERS

Exercise Set 1

1. (d)	**2.** (d)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (a)
11. (d)	**12.** (d)	**13.** (b)	**14.** (d)	**15.** (b)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (d)
21. (c)	**22.** (b)	**23.** (d)	**24.** (b)	**25.** (c)	**26.** (a)	**27.** (c)	**28.** (a)	**29.** (b)	**30.** (a)
31. (b)	**32.** (d)	**33.** (a) and (b)		**34.** (a)	**35.** (c)	**36.** (a)	**37.** (a)	**38.** (a)	**39.** (b)

40. (b)	**41.** (a)	**42.** (d)	**43.** (c)	**44.** (b)	**45.** (b)	**46.** (a)	**47.** (c)	**48.** (c)	**49.** (a)
50. (b)	**51.** (c)	**52.** (a)	**53.** (b)	**54.** (c)	**55.** (b)	**56.** (a)	**57.** (b)	**58.** (b)	**59.** (b)
60. (a)	**61.** (b)	**62.** (b)	**63.** (d)	**64.** (c)	**65.** (c)	**66.** (a)	**67.** (a)	**68.** (a)	**69.** (b)
70. (d)	**71.** (b)								

Exercise Set 2

1. (a)	**2.** (d)	**3.** (d)	**4.** (d)	**5.** (b)	**6.** (b)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (b)
11. (a)	**12.** (b)	**13.** (c)	**14.** (a)	**15.** (c)	**16.** (c)	**17.** (a)	**18.** (b)	**19.** (b)	**20.** (a)
21. (c)	**22.** (a)	**23.** (d)	**24.** (a)	**25.** (b)	**26.** (c)	**27.** (c)	**28.** (a)	**29.** (b)	**30.** (a)
31. (b)	**32.** (b)	**33.** (b)	**34.** (b)	**35.** (a)	**36.** (d)	**37.** (a)	**38.** (b)	**39.** (c)	**40.** (b)
41. (a)	**42.** (d)	**43.** (c)	**44.** (c)	**45.** (a)	**46.** (c)	**47.** (d)	**48.** (a)	**49.** (b)	**50.** (a)
51. (a)	**52.** (d)	**53.** (c)	**54.** (a)	**55.** (b)	**56.** (a)	**57.** (d)	**58.** (b)	**59.** (d)	**60.** (c)
61. (b)	**62.** (c)	**63.** (a)	**64.** (b)	**65.** (a)					

HINTS AND SOLUTIONS—EXERCISE SET 1

1. $\frac{d}{dx}[\log(\tan x)] = \frac{1}{\tan x}\cdot\frac{d}{dx}(\tan x) = \frac{\sec^2 x}{\tan x}$

$= \frac{1}{\sin x \cos x} = \frac{2}{\sin 2x} = 2\operatorname{cosec} 2x$

2. $y = 2^{\log x}$

$\Rightarrow \frac{dy}{dx} = 2^{\log x}\cdot\log 2\cdot\frac{1}{x} = \frac{2^{\log x}\cdot\log 2}{x}$

3. $y = x\log x - x$

$\Rightarrow \frac{dy}{dx} = \left(x\cdot\frac{1}{x} + \log x\cdot 1\right) - 1 = 1 + \log x - 1 = \log x$

4. Put $a = r\cos\alpha$, $b = r\sin\alpha$

$\therefore\ y = \tan^{-1}\cot(x+\alpha)$

$= \tan^{-1}\tan\left[\frac{\pi}{2} - (x+\alpha)\right]$ or $y = \frac{\pi}{2} - x - \alpha$

$\therefore \frac{dy}{dx} = -1.$

5. We have

$f(x) = (x^{a-b})^{a+b}\cdot(x^{b-c})^{b+c}\cdot(x^{c-a})^{c+a}$

$= x^{a^2-b^2}\cdot x^{b^2-c^2}\cdot x^{c^2-a^2}$

$= x^{a^2-b^2+b^2-c^2+c^2-a^2} = x^0 = 1$

$\therefore\ f'(x) = 0$

6. Let $f(x) = |x-1| + |x-3|$

$$= \begin{cases} -(x-1)-(x-3), & \text{when } x < 1 \\ (x-1)-(x-3), & \text{when } 1 \le x < 3 \\ (x-1)+(x-3), & \text{when } x \ge 3 \end{cases}$$

$$= \begin{cases} -2x+4, & \text{when } x < 1 \\ 2, & \text{when } 1 \le x < 3 \\ 2x-4, & \text{when } x \ge 3 \end{cases}$$

$$\therefore\ f'(x) = \begin{cases} -2, & \text{when } x < 1 \\ 0, & \text{when } 1 \le x < 3 \\ 2, & \text{when } x \ge 3 \end{cases}$$

At $x = 2$, we have $f'(x) = 0$.

7. $y = \log_e(\log_e x)$

$\Rightarrow \frac{dy}{dx} = \frac{1}{\log_e x}\cdot\frac{1}{x}$

$\Rightarrow e^y\frac{dy}{dx} = \frac{e^y}{x\log_e x}$

$= \frac{\log_e x}{x\log_e x}$ $\quad[\because e^y = e^{\log_e(\log_e x)} = \log_e x]$

$= \frac{1}{x}.$

8. We have

$$y=\frac{1}{1+\frac{x^\beta}{x^\alpha}+\frac{x^\gamma}{x^\alpha}}+\frac{1}{1+\frac{x^\alpha}{x^\beta}+\frac{x^\gamma}{x^\beta}}+\frac{1}{1+\frac{x^\alpha}{x^\gamma}+\frac{x^\beta}{x^\gamma}}$$

$$=\frac{x^\alpha}{x^\alpha+x^\beta+x^\gamma}+\frac{x^\beta}{x^\alpha+x^\beta+x^\gamma}+\frac{x^\gamma}{x^\alpha+x^\beta+x^\gamma}$$

$$=\frac{x^\alpha+x^\beta+x^\gamma}{x^\alpha+x^\beta+x^\gamma}=1$$

$$\therefore \quad \frac{dy}{dx}=0$$

9. $y=ce^{\frac{x}{x-a}}$

$$\Rightarrow \frac{dy}{dx}=c\cdot e^{\frac{x}{x-a}}\cdot\frac{d}{dx}\left[\frac{x}{x-a}\right]$$

$$= c.e^{\frac{x}{x-a}}\left[\frac{(x-a).1-x.1}{(x-a)^2}\right]=ce^{\frac{x}{x-a}}\left[\frac{-a}{x-a^2}\right]$$

$$= y\left[\frac{-a}{(x-a)^2}\right]=-\frac{ay}{(x-a)^2}.$$

10. $y = \tan^{-1}(\text{cosec } x + \cot x)$

$$= \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)=\tan^{-1}\left(\frac{2\cos^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}}\right)$$

$$= \tan^{-1}\left(\cot\frac{x}{2}\right)=\frac{\pi}{2}-\cot^{-1}\left(\cot\frac{\pi}{2}\right)$$

$$\left[\because \quad \tan^{-1}A+\cot^{-1}A=\frac{\pi}{2}\right]$$

$$= \frac{\pi}{2}-\frac{x}{2} \quad \Rightarrow \quad \frac{dy}{dx}=-\frac{1}{2}.$$

11. $y=1+\frac{x}{\lfloor 1}+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+\frac{x^4}{\lfloor 4}+\cdots$

$$\Rightarrow \frac{dy}{dx} = 0 + 1+\frac{2x}{\lfloor 2}+\frac{3x^2}{\lfloor 3}+\frac{4x^3}{\lfloor 4}+\cdots$$

$$= 1+\frac{x}{\lfloor 1}+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+\cdots=y$$

12. $y=1-x+\frac{x^2}{\lfloor 2}-\frac{x^3}{\lfloor 3}+\frac{x^4}{\lfloor 4}-\cdots$

$$\Rightarrow \frac{dy}{dx}=(-1)+\frac{2x}{\lfloor 2}-\frac{3x^2}{\lfloor 3}+\frac{4x^3}{\lfloor 4}-\cdots$$

$$=(-1)\left(1-x+\frac{x^2}{\lfloor 2}-\frac{x^3}{\lfloor 3}+\cdots\right)=(1)\cdot y=-y$$

$$\Rightarrow \frac{d^2y}{dx^2}=-\frac{dy}{dx}=-(-y)=y$$

or

$$y=1-x+\frac{x^2}{\lfloor 2}-\frac{x^3}{\lfloor 3}+\frac{x^4}{\lfloor 4}-\cdots=e^{-x}$$

$$\Rightarrow \frac{dy}{dx}=-e^{-x} \Rightarrow \frac{d^2y}{dx^2}=e^{-x}=y$$

13. We have $y=x^{x^{x^{x\cdots}}}$

$$\Rightarrow y = x^y \qquad \Rightarrow \quad y=e^{y\log x}$$

$$\Rightarrow \frac{dy}{dx}=e^{y\log x}\frac{d}{dx}(y\log x)$$

$$\Rightarrow \frac{dy}{dx}=x^y\left(\frac{dy}{dx}\log x+\frac{y}{x}\right)$$

$$\Rightarrow \frac{dy}{dx}=y\left(\frac{dy}{dx}\log x+\frac{y}{x}\right) \qquad [\because y=x^y]$$

$$\Rightarrow \frac{dy}{dx}=\frac{y^2}{x(1-y\log x)}$$

14. $\lim\limits_{x\to a}\dfrac{g(x)f(a)-g(a)f(x)}{(x-a)}$ $\left[\dfrac{0}{0}\text{ form}\right]$

$$= \lim_{x\to a}\frac{f(a)\cdot g'(x)-g(a)\cdot f'(x)}{1}$$

$$= f(a)\cdot[g'(x)]_{x=a}$$

$$=-g(a)\cdot f'(a)=(2\times 2)-(1\times 1)=4-1=3.$$

15. Let $u = \log_{10} x$ and $v = x^2$. Then,

$$\frac{du}{dv}=\frac{\frac{du}{dx}}{\frac{dv}{dx}}=\frac{\frac{1}{x\log_e 10}}{2x}=\frac{1}{2x^2}\log_{10}e$$

16. Put $x = \cos 2\theta$

$$\therefore \quad y=\tan^{-1}\left\{\frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}\right\}$$

$$= \tan^{-1}\tan\left(\frac{\pi}{4}-\theta\right)=\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x.$$

17. Around $x=-\frac{\pi}{3}$, we have $x<0$ and $\sec x>0$.

$\therefore\ |x|=-x$ and $|\sec x|=\sec x$

$\therefore\ f(x)=(-x)^{\sec x}$

$\Rightarrow\ \log f(x)=\sec x\log(-x)$

$\Rightarrow \frac{1}{f(x)}\cdot f'(x)=\sec x\cdot\frac{1}{-x}\cdot(-1)+\log(-x)\sec x\tan x$

$\Rightarrow\ f'(x)=(-x)^{\sec x}\left[\frac{\sec x}{x}+\log(-x)\sec x\tan x\right].$

$\Rightarrow\ f'\left(-\frac{\pi}{3}\right)=\left(\frac{\pi}{3}\right)^2\left[\frac{-6}{\pi}+2\sqrt{3}\log\frac{\pi}{3}\right]$

$$=\left(-\frac{2\pi}{3}+\frac{2\pi^2}{3\sqrt{3}}\log\frac{\pi}{3}\right)$$

18. We have $f(x)=\begin{vmatrix}\sec\theta & \tan^2\theta & 1\\ \theta\sec x & \tan x & x\\ 1 & \tan x-\tan\theta & 0\end{vmatrix}$

$$f'(x)=\begin{vmatrix}0 & 0 & 0\\ \theta\sec x & \tan x & x\\ 1 & \tan x-\tan\theta & 0\end{vmatrix}$$

$$+\begin{vmatrix}\sec\theta & \tan^2\theta & 1\\ \theta\sec x\tan x & \sec^2 x & 1\\ 1 & \tan x-\tan\theta & 0\end{vmatrix}$$

$$+\begin{vmatrix}\sec\theta & \tan^2\theta & 1\\ \theta\sec x & \tan x & x\\ 0 & \sec^2 x & 0\end{vmatrix}$$

$$\therefore f'(\theta)=0+\begin{vmatrix}\sec\theta & \tan^2\theta & 1\\ \theta\sec\theta\tan\theta & \sec^2\theta & 1\\ 1 & 0 & 0\end{vmatrix}$$

$$+\begin{vmatrix}\sec\theta & \tan^2\theta & 1\\ \theta\sec\theta & \tan\theta & \theta\\ 0 & \sec^2\theta & 0\end{vmatrix}$$

$\Rightarrow f'(\theta)=\tan^2\theta-\sec^2\theta+\sec^2\theta(\theta\sec\theta-\theta\sec\theta)=-1$

19. Put $t=\cos\theta$

At $t=\frac{1}{2}$, $\cos\theta=\frac{1}{2}\ \Rightarrow\ \theta=\frac{\pi}{3}$

$\therefore\ e^x=\frac{\cos(\theta/2)-\sin(\theta/2)}{\cos(\theta/2)+\sin(\theta/2)}=\frac{1-\tan(\theta/2)}{1+\tan(\theta/2)}$

$$=\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)$$

$\tan\frac{y}{2}=\frac{\sin(\theta/2)}{\cos(\theta/2)}=\tan\frac{\theta}{2}\ \therefore\ y=\theta.$

$\therefore\quad x=\log\tan\left(\frac{\pi}{4}-\frac{y}{2}\right)$

$$\frac{dy}{dx}=\frac{1}{\tan\left(\frac{\pi}{4}-\frac{y}{2}\right)}\cdot\sec^2\left(\frac{\pi}{4}-\frac{y}{2}\right)\times\left(-\frac{1}{2}\right)$$

$$=-\frac{1}{2\sin\left(\frac{\pi}{4}-\frac{y}{2}\right)\cos\left(\frac{\pi}{4}-\frac{y}{2}\right)}$$

$$=-\frac{1}{\sin\left(\frac{\pi}{2}-y\right)}=-\frac{1}{\cos y}$$

or $\frac{dx}{dy}=-\frac{1}{\cos\theta}=-\frac{1}{\cos(\pi/3)}=-2$

$\therefore\ \frac{dy}{dx}=-\frac{1}{2}.$

20. Since $f(x)=(\log_{\cot x}\tan x)(\log_{\tan x}\cot x)^{-1}+\tan^{-1}\left(\frac{x}{\sqrt{4-x^2}}\right)$

$$=(\log_{(\tan x)^{-1}}\tan x)[\log_{\tan x}(\tan x)^{-1}]^{-1}+\tan^{-1}\left(\frac{x}{\sqrt{4-x^2}}\right)$$

$$=(-1)\times(-1)^{-1}+\tan^{-1}\frac{x}{\sqrt{4-x^2}}$$

$$=1+\tan^{-1}\frac{x}{\sqrt{4-x^2}}$$

$$f'(x)=0+\frac{1}{1+\frac{x^2}{4-x^2}}\times\frac{\sqrt{4-x^2}\cdot 1-x\frac{1}{2\sqrt{4-x^2}}\times(-2x)}{(\sqrt{4-x^2})^2}$$

$$= \frac{1}{4} \times \frac{4 - x^2 + x^2}{\sqrt{4 - x^2}} = \frac{1}{\sqrt{4 - x^2}}$$

$$\therefore \quad f'(0) = \frac{1}{\sqrt{4}} = \frac{1}{2}.$$

21. We have $f(x) = \sqrt{x^2 + 9} \quad \Rightarrow \quad f'(x) = \dfrac{x}{\sqrt{x^2 + 9}}$

Now, $\lim\limits_{x \to 4} \dfrac{f(x) - f(4)}{x - 4} = f'(4)$

[By defferentiation of derivative]

$$\Rightarrow \lim_{x \to 4} \frac{f(x) - f(4)}{x - 4} = \frac{4}{\sqrt{4^2 + 9}} = \frac{4}{5}$$

22. $y = \tan^{-1} \tan\left(\dfrac{\pi}{4} + \theta\right) = \dfrac{\pi}{4} + \theta$

$$= \frac{\pi}{4} + \tan^{-1} x \qquad \therefore \quad \frac{dy}{dx} = \frac{1}{1 + x^2}$$

23. $\because y = \log_a x + \dfrac{\log a}{\log x} + 1 + 1$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x} \log_a e - \log a \left(\frac{1}{\log x}\right)^2 \frac{1}{x}$$

$$= \frac{1}{x \log a} - \frac{\log a}{x (\log x)^2}$$

24. We have $f(x) = x^n$

$$\Rightarrow \quad f^r(x) = \frac{\lfloor n}{\lfloor (n - r)} x^{n-r}, \; r = 1, 2, 3, \ldots, n$$

$$\Rightarrow \quad f^r(1) = \frac{\lfloor n}{\lfloor (n - r)} = {}^n\mathrm{C}_r \cdot \lfloor r \quad \Rightarrow \quad \frac{f^r(1)}{\lfloor r} = {}^n\mathrm{C}_r$$

$$\therefore \quad f(1) + \frac{f'(1)}{\lfloor 1} + \frac{f''(1)}{\lfloor 2} + \frac{f'''(1)}{\lfloor 3} + \ldots + \frac{f^{(n)}(1)}{\lfloor n}$$

$$= \sum_{r=0}^{n} {}^n C_r = 2^n$$

25. Put $x = \sin\theta \qquad \therefore \quad y = 2\theta + \theta = 3\theta$

$$y = 3\sin^{-1} x \qquad \therefore \quad \frac{dy}{dx} = \frac{3}{\sqrt{(1 - x^2)}}$$

26. $f(x) = \log_5(\log_7 x) \quad \Rightarrow \quad f(x) = \log_5\left(\dfrac{\log_e x}{\log_e 7}\right)$

$$\Rightarrow \quad f(x) = \log_5 \log_e x - \log_5 \log_e 7$$

$$\Rightarrow \quad f(x) = \frac{\log_e \log_e x}{\log_e 5} - \log_5 \log_e 7$$

On differentiating w.r.t. x, we get

$$f'(x) = \frac{1}{x \log_e x \log_e 5} - 0$$

$$= \frac{1}{\dfrac{x \log_e x}{\log_e 7} \log_e 7 \cdot \log_e 5}$$

$$= \frac{1}{x \log_7 x \log 7 \cdot \log 5}$$

27. Put $x = \cos\theta \Rightarrow \theta = \cos^{-1} x$

$$\therefore \quad y = \sin^2 \cot^{-1} \frac{1}{\sqrt{\dfrac{1 + \cos\theta}{1 - \cos\theta}}}$$

$$= \sin^2 \cot^{-1} \frac{1}{\cot\dfrac{\theta}{2}} = \sin^2 \cot^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$= \sin^2 \cot^{-1} \cot\left(\frac{\pi}{2} - \frac{\theta}{2}\right)$$

$$= \sin^2\left(\frac{\pi}{2} - \frac{\theta}{2}\right) = \cos^2\frac{\theta}{2}$$

$$= \frac{1 + \cos\theta}{2} = \frac{1 + x}{2}$$

28. $y = \cot^{-1}\left[\dfrac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}}\right]$

$$= \cot^{-1}\left[\frac{2 + 2\cos x}{2\sin x}\right] = \cot^{-1}\left[\frac{1 + \cos x}{\sin x}\right]$$

$$= \cot^{-1}\left[\cot\frac{x}{2}\right] = \frac{x}{2} \qquad \therefore \quad \frac{dy}{dx} = \frac{1}{2}$$

29. $x = \tan\theta, \quad y = 2\theta, \; z = \cot^{-1}\left(\dfrac{1}{\tan 3\theta}\right) = 3\theta$

$$\therefore \quad y = \frac{2}{3} z \quad \text{or} \quad \frac{dy}{dz} = \frac{2}{3}$$

30. We have

$$y^{1/m} = \left[x + \sqrt{1 + x^2}\right] \quad \Rightarrow y = [x + \sqrt{1 + x^2}]^m$$

$$\Rightarrow \quad \frac{dy}{dx} = m[x + \sqrt{1 + x^2}]^{m-1}\left(1 + \frac{x}{\sqrt{x^2 + 1}}\right)$$

$$= m\frac{[x\sqrt{1+x^2}]m}{\sqrt{1+x^2}}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{my}{\sqrt{1+x^2}} \quad \Rightarrow y_1^2(1+x^2) = m^2y^2$$

$$\Rightarrow \quad 2y_1y_2(1+x^2) + 2xy_1^2 = 2m^2yy_1$$

$$\Rightarrow \quad y_2(1+x^2) + xy = m^2y$$

31. Since $y = (1+x)^{1/x}$

$$\therefore \quad \log y = \frac{1}{x}\log(1+x)$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = \frac{x\cdot\frac{1}{1+x} - \log(1+x)\cdot 1}{x^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = y\left[\frac{\frac{x}{1+x} - \log(1+x)}{x^2}\right]$$

$$= (1+x)^{1/x}\left[\frac{\frac{x}{1+x} - \log(1+x)}{x^2}\right]$$

32. $y = f(x) = \dfrac{\log(\log x)}{\log x^2}$

Put $\log x = t \quad \Rightarrow \quad \dfrac{dt}{dx} = \dfrac{1}{x}$

$$\Rightarrow \quad \left(\frac{dt}{dx}\right)_{x=e} = \frac{1}{e}$$

Also, when $x = e$, $t = 1$

$$\therefore \quad y = \frac{\log t}{2t}$$

$$\frac{dy}{dt} = \frac{1}{2}\frac{t\cdot\frac{1}{t} - 1\cdot\log t}{t^2} = \frac{1}{2}\,\frac{1-\log t}{t^2}$$

$$\left(\frac{dy}{dt}\right)_{t=1} = \frac{1}{2}\frac{(1-\log t)}{1^2} = \frac{1}{2}$$

$$\therefore \quad \frac{dy}{dx} = \frac{dy}{dt}\times\frac{dt}{dx} = \frac{1}{2}\times\frac{1}{e} = \frac{1}{2e}$$

33. $\dfrac{dy}{dx} = \dfrac{2}{\sqrt{a^2-b^2}}\cdot\dfrac{1}{1+\left(\sqrt{\dfrac{a-b}{a+b}}\tan\dfrac{x}{2}\right)^2}\times\sqrt{\dfrac{a-b}{a+b}}\sec^2\dfrac{x}{2}\cdot\dfrac{1}{2}$

$$= \frac{1}{\sqrt{a-b}\sqrt{a+b}}\cdot\frac{1}{1+\frac{a-b}{a+b}\cdot\frac{\sin^2\frac{x}{2}}{\cos^2\frac{x}{2}}}\sqrt{\frac{a-b}{a+b}}\sec^2\frac{x}{2}$$

$$= \frac{1}{a+b}\,\frac{(a+b)\cos^2\frac{x}{2}}{(a+b)\cos^2\frac{x}{2} + (a-b)\sin^2\frac{x}{2}}\sec^2\frac{x}{2}$$

$$= \frac{1}{a\left(\cos^2\frac{x}{2} + \sin^2\frac{x}{2}\right) + b\left(\cos^2\frac{x}{2} - \sin^2\frac{x}{2}\right)}$$

$$= \frac{1}{a+b\cos x}$$

$$\therefore \quad \frac{d^2y}{dx^2} = \frac{-1}{(a+b\cos x)^2}(-b\sin x) = \frac{b\sin x}{(a+b\cos x)^2}$$

34. $y = \cos^{-1}\sqrt{T}$. Put $x = \tan\theta$, then

$$T = \frac{1+\sec\theta}{2\sec\theta} = \frac{1+\cos\theta}{2} = \cos^2\frac{\theta}{2}$$

$$\therefore \quad y = \cos^{-1}\sqrt{\cos^2\frac{\theta}{2}} = \frac{\theta}{2} = \frac{1}{2}\tan^{-1}x$$

$$\therefore \quad \frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

35. $f[f(x)] = f(|x-2|) = \big||x-2|-2\big|$

$$= x - 4$$

$\Rightarrow g(x) = x - 4 \qquad \therefore g'(x) = 1$

36. Let $y = (\log)^x + x^{\log x}$

Put $u = (\log x)^x$, $v = x^{\log x} \quad \therefore y = u + v$

$$\therefore \quad \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \qquad \text{(i)}$$

Now, $u = (\log x)^x$

$\Rightarrow \log u = x\log(\log x)$

$$\Rightarrow \frac{1}{u}\frac{du}{dx} = x\cdot\frac{1}{\log x}\cdot\frac{1}{x} + \log(\log x)\cdot 1$$

$$= \frac{1}{\log x} + \log(\log x)$$

$$\Rightarrow \frac{du}{dx} = u\left[\log(\log x) + \frac{1}{\log x}\right]$$

$$= (\log x)^x\left[\log(\log x) + \frac{1}{\log x}\right]$$

$v = x^{\log x}$

$\Rightarrow \log c = \log x \cdot \log x = (\log x)^2$

$$\Rightarrow \frac{1}{v}\frac{dv}{dx} = 2\log x\cdot\frac{1}{x} \quad \Rightarrow \frac{dv}{dx} = 2\frac{\log x}{x}\cdot x^{\log x}$$

$\therefore$ (i) gives

$$\frac{dy}{dx} = (\log x)^x\left[\log(\log x) + \frac{1}{\log x}\right] + x^{\log x}\cdot\left(\frac{2\log x}{x}\right)$$

37. Since $f(x) = \cot^{-1}\left(\frac{x^x - x^{-x}}{2}\right) = \cot^{-1}\left(\frac{x^{2x}-1}{2x^x}\right)$

Put $x^x = \tan\theta$

$$\therefore \quad f(x) = \cot^{-1}\left(\frac{\tan^2\theta - 1}{2\tan\theta}\right)$$

$$= \cot^{-1}(-\cot 2\theta) = \pi - \cot^{-1}(\cot 2\theta)$$

$\Rightarrow y = \pi - 2\theta = \pi - 2\tan^{-1}(x^x)$

On differentiating w.r.t. x, we get

$$\frac{dy}{dx} = -\frac{2}{1+x^2}\cdot x^2(1+\log x)$$

At $x = 1$, $\frac{dy}{dx} = \frac{-2}{1+1}(1+0) = -1$

38. $$\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{f'(x^3)3x^2}{g'(x^2)2x} = \frac{\cos x^3 \times 3x^2}{\sin x^2 \times 2x}$$

$$= \frac{3}{2}x\cos x^3 \operatorname{cosec} x^2$$

39. $$y = \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) = \cos^{-1}\left(-\frac{1-x^2}{1+x^2}\right)$$

$$= \pi - \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \pi - 2\tan^{-1}x \text{ etc.}$$

40. Let $f(x) = |x-1| + |x-3|$

$$f(x) = \begin{cases} -(x-1)-(x-3), & x<1 \\ (x-1)-(x-3), & 1\le x<3 \\ (x-1)+(x-3), & x\ge 3 \end{cases}$$

$$f(x) = \begin{cases} 4-2x, & x<1 \\ 2, & 1\le x<3 \\ 2x-4, & x\ge 3 \end{cases}$$

At $x = 2$,

$f(x) = 2 \quad \Rightarrow \quad f'(x) = 0$

41. $$y = \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \frac{c+x-c}{x-c}$$

$$= y = \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \frac{x}{x-c}$$

$$= y = \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{x}{x-c}\left(\frac{b}{x-b}+1\right)$$

$$= y = \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{x}{x-c}\left(\frac{b+x-b}{x-b}\right)$$

$$= \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{x^2}{(x-b)(x-c)}$$

$$= \frac{ax^2 + x^2(x-a)}{(x-a)(x-b)(x-c)} = \frac{x^3}{(x-a)(x-b)(x-c)}$$

$\therefore \log y = 3\log x - \log(x-a) - \log(x-b) - \log(x-c)$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{3}{x} - \frac{1}{x-a} - \frac{1}{x-b} - \frac{1}{x-c}$$

$$= \frac{y'}{y} = \frac{3}{x} - \left(\frac{1}{x-a} + \frac{1}{x-b} + \frac{1}{x-c}\right)$$

$$= \left(\frac{1}{x} - \frac{1}{x-a}\right) + \left(\frac{1}{x} - \frac{1}{x-b}\right) + \left(\frac{1}{x} - \frac{1}{x-c}\right)$$

$$= -\frac{1}{x}\left(\frac{a}{x-a} + \frac{b}{x-b} + \frac{c}{x-c}\right)$$

$$= \frac{1}{x}\left(\frac{a}{a-x} + \frac{b}{b-x} + \frac{c}{c-x}\right)$$

42. We have $f(x) = x^n \Rightarrow f(x+y) = (x+y)^n$

$\Rightarrow \quad f'(x+y) = n(x+y)^{n-1}$

Also, $\quad f'(x) = nx^{n-1}$ and $f'(y) = ny^{n-1}$

$\therefore \quad f'(x+y) = f'(x) + f'(y)$

$\Rightarrow \quad n(x+y)^{n-1} = n.x^{n-1} + n.y^{n-1}$

$(x+y)^{n-1} = x^{n-1} + y^{n-1}$

Clearly, this is true for $n = 2$ only.

43. $f(x) = x + \tan x$

$f[f^{-1}(y)] = f^{-1}(y) + \tan f^{-1}(y)$

$y = g(y) + \tan g(y)$ and $x = g(x) + \tan g(x)$

Differentiating, we get $1 = g'(x) + \sec^2 g(x) \cdot g'(x)$

$$\Rightarrow \quad g'(x) = \frac{1}{1+\sec^2 g(x)} = \frac{1}{2+[g(x)-x]^2}$$

44. We have $S_n = \dfrac{a(r^n - 1)}{r-1}$

$\Rightarrow \quad (r-1)S_n = ar^n - a$

On differentiating both sides w.r.t. r, we get

$$(r-1)\frac{dS_n}{dr} + S_n = nar^{n-1} - 0$$

$$\Rightarrow \quad (r-1)\frac{dS_n}{dr} = nar^{n-1} - S_n$$

$= n[n\text{th term of G.P.}] - S_n$

$= n(S_n - S_{n-1}) - S_n$

$= (n-1)S_n - nS_{n-1}$

45. When $x = 0$, we get $y = 0$

On differentiating both sides of the given equation, w.r.t. x, we get

$$e^{xy} + xe^{xy}\left(x\frac{dy}{dx} + y\right) = \frac{dy}{dx} + 2\sin x\cos x$$

On putting $x = 0$, $y = 0$, we get

$$e^0 + 0e^0\left(0\cdot\left[\frac{dy}{dx}\right]_{(0,0)} + 0\right) = \left[\frac{dy}{dx}\right]_{(0,0)} + 2\sin 0\cos 0$$

$$\Rightarrow \quad \left[\frac{dy}{dx}\right]_{(0,0)} = 1$$

46. Put $x^3 = \sin\theta, y^3 = \sin\phi$

$\therefore \quad \cos\theta + \cos\phi = a^3(\sin\theta - \sin\phi)$

$$\Rightarrow \quad 2\cos\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2} = a^3\cdot 2\cos\frac{\theta+\phi}{2}\sin\frac{\theta-\phi}{2}$$

$$\Rightarrow \quad \cot\left(\frac{\theta-\phi}{2}\right) = a^3$$

$$\Rightarrow \quad \frac{\theta-\phi}{2} = \cot^{-1}a^3 = \text{constant}$$

$\Rightarrow \quad \theta - \phi = \text{constant}$

$\Rightarrow \quad \sin^{-1}x^3 - \sin^{-1}y^3 = \text{constant}$

$$\Rightarrow \quad \frac{3x^2}{\sqrt{1-x^6}} - \frac{1}{\sqrt{1-y^6}}\cdot 3y^2\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}.$$

47. Since g is the inverse of f,

$\therefore \quad f\circ g(x) = x$ for all x

$$\Rightarrow \quad \frac{d}{dx}[f\circ g(x)] = 1 \text{ for all } x$$

$$\Rightarrow \quad f'[g(x)]\cdot g'(x) = 1$$

$$\Rightarrow \quad f'[g(x)] = \frac{1}{g(x)}$$

$$\Rightarrow \quad \frac{1}{1+[g(x)]^n} = \frac{1}{g'(x)} \qquad \left[\because \quad f'(x) = \frac{1}{1+x^n}\right]$$

$\Rightarrow \quad g'(x) = 1 + [g(x)]^n$

48. Now, $\dfrac{df(\tan x)}{dg(\sec x)} = \dfrac{f'(\tan x)\sec^2 x}{g'(\sec x)\sec x\tan x}$

$$= \frac{f'(\tan x)\sec x}{g'(\sec x)\tan x} = \frac{f'\left(\tan\frac{\pi}{4}\right)\sec\frac{\pi}{4}}{g'\left(\sec\frac{\pi}{4}\right)\tan\frac{\pi}{4}} \qquad \left(\text{putting } x = \frac{\pi}{4}\right)$$

$$= \frac{f'(1)\sqrt{2}}{g'(\sqrt{2})(1)} = \frac{2\sqrt{2}}{4(1)} = \frac{1}{\sqrt{2}}$$

49. $y = f(x) = (\tan x)^{(m^2-n^2)+(n^2-p^2)+(p^2-m^2)} = \tan^0 x = 1$

$\therefore \quad \dfrac{dy}{dx} = 0$

50. Let $f(x) = |x-1| + |x-5|$

$$f(x) = \begin{cases} -(x-1)-(x-5), & x<1 \\ (x-1)-(x-5), & 1\le x\le 5 \\ x-1+x-5, & x>5 \end{cases}$$

$$f(x) = \begin{cases} 6-2x, & x<1 \\ 4, & x\le 5 \\ 2x-6, & x>5 \end{cases}$$

$\because \quad x = 3 \in (1,5)$ for $x = 3$

$f(x) = 4 \quad \Rightarrow \quad f'(x) = 0$

51. $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$ (i)

$\therefore \; f(x) = f\left(\frac{2x+0}{2}\right) = \frac{f(2x)+f(0)}{2}$ (ii)

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{f\left(\frac{2x+2h}{2}\right)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{\frac{f(2x)+f(2h)}{2} - \frac{f(2x)+f(0)}{2}}{h}$$

[by (i) and (ii)]

$$= \lim_{2h\to 0}\frac{f(2h)-f(0)}{2h} = f'(0) = -1$$

$\therefore \; f'(x) = -1 \Rightarrow f(x) = -x + c$

When $x = 0, f(0) = 1$

$\therefore \quad 1 = 0 + c, \qquad \therefore \; c = 1$

$\Rightarrow f(x) = -x + 1 \qquad \Rightarrow f(2) = -2 + 1 = -1$

52. We have

$y = (1+x)(1+x^2)(1+x^4) \ldots (1+x^{2^n})$

$$\Rightarrow \quad y = \frac{(1-x)(1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^n})}{(1-x)}$$

$$\Rightarrow \quad y = \frac{1-x^{2^{n+1}}}{1-x}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-2^{n+1}x^{2^{n+1}-1}(1-x) + (1-x^{2^{n+1}})}{(1-x)^2}$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{x=0} = 1$$

53. $$D = \begin{vmatrix} \sin px & p\cos px & -p^2\sin px \\ -p^3\cos px & p^4\sin px & p^5\cos px \\ -p^6\sin px & -p^7\cos px & p^8\sin px \end{vmatrix}$$

$$= p^9\begin{vmatrix} \sin px & p\cos px & -p^2\sin px \\ -\cos px & p\sin px & p^2\cos px \\ -\sin px & -p\cos px & p^2\sin px \end{vmatrix}$$

$$= -p^9\begin{vmatrix} \sin px & p\cos px & -p^2\sin px \\ \cos px & p\sin px & p^2\cos px \\ \sin px & p\cos px & -p^2\sin px \end{vmatrix} = 0$$

54. Put $x = \cos\theta$,

$1 + \cos\theta = 2\cos^2(\theta/2)$, $1 - \cos\theta = 2\sin^2(\theta/2)$

$y = \sin^{-1}\{(1/\sqrt{2})[\cos(\theta/2) + \sin(\theta/2)]\}$

$= \sin^{-1}\sin(\theta/2 + \pi/4) = \theta/2 + \pi/4$

$y = \frac{1}{2}\cos^{-1}x + \frac{\pi}{4} \qquad \therefore \quad \frac{dy}{dx} = \frac{-1}{2\sqrt{(1-x^2)}}$

55. We know that $\cos A \cos 2A \cos 2^2A \cdots \cos 2^{n-1}A$

$$= \frac{\sin(2^n A)}{2^n \sin A}$$

$$\therefore \quad \cos x \cos 2x \cos 4x \cos 8x \cos 16 = \frac{\sin 32x}{32\sin x}$$

$$\Rightarrow \quad f(x) = \frac{1}{32}\cdot\frac{\sin 32x}{\sin x}$$

$$\therefore \quad f'(x) = \frac{1}{32}\times\frac{\sin x(32\cos 32x) - \sin 32x\cos x}{\sin^2 x}$$

$$\Rightarrow \quad f'\left(\frac{\pi}{4}\right) = \frac{1}{\sin\frac{\pi}{4}} = \sqrt{2}.$$

56. $f'(x) = \frac{x}{1+x^2} + \tan^{-1}x$

$$\Rightarrow \quad f'(1) = \frac{1}{2} + \tan^{-1}(1) = \frac{1}{2} + \frac{\pi}{4}.$$

57. We have $f(x) = \frac{x^2 - x}{x^2 + 2x}$

Clearly, $f(0) = f(-2)$ is not defined.

So domain of $f = \mathbb{R} - \{0, -2\}$

Then, in this domain, we have

$$y = f(x) = \frac{x-1}{x+2} \quad \Rightarrow \quad yx + 2y = x - 1$$

$$\Rightarrow \quad x = \frac{2y+1}{1-y} \Rightarrow f^{-1}(x) = \frac{2x+1}{1+x}$$

$$\therefore \quad \frac{df^{-1}(x)}{dx} = \frac{2(1-x)+2x+1}{(1-x)^2} = \frac{3}{(1-x)^2}$$

58. Differentiating the given equation w.r.t. x, we get

$$6x^2 - 6xy^2 - 6x^2yy' + 4 - y' = 0 \qquad \text{(i)}$$

On putting $x = 1$ and $y = 1$, we get

$$6 - 6 - 6y'(1) + 4 - y'(1) = 0$$

$$\Rightarrow \quad y'(1) = \frac{4}{7}$$

Differentiating Eq. (i) w.r.t. x, we have

$$12x - 6y^2 - 12xyy' - 12xyy' - 6x^2y'^2 - 6x^2yy'' - y'' = 0$$

$$\Rightarrow \quad y''(1) = -\frac{474}{343}$$

59. $(x+y)\left(\frac{\cos^4\alpha}{x} + \frac{\sin^4\alpha}{y}\right) = 1 = (\cos^2\alpha + \sin^2\alpha)^4$

$$\therefore \quad \frac{y}{x}\cos^4\alpha + \frac{x}{y}\sin^4\alpha - 2\sin^2\alpha\cos^2\alpha = 0$$

or $$\left(\sqrt{\frac{y}{x}}\cos^2\alpha - \sqrt{\frac{x}{y}}\sin^2\alpha\right)^2 = 0$$

$$\therefore \quad \tan^2\alpha = \frac{y}{x} \text{ or } y = x\tan^2\alpha \quad \therefore \quad \frac{dy}{dx} = \tan^2\alpha$$

60. $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots \infty}}}$

$$\Rightarrow \quad y = \sqrt{\sin x + y} \qquad \Rightarrow \quad y^2 = \sin x + y$$

On differentiating w.r.t. x, we get

$$2y\frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\cos x}{2y - 1}$$

61. $y = ax^{n+1} + bx^{-n}$

$$\frac{dy}{dx} = (n+1)ax^n - nbx^{-n-1}$$

$$\frac{d^2y}{dx^2} = (n+1)nax^{n-1} + n(n+1)bx^{-n-2}$$

$$\therefore \quad x^2\frac{d^2y}{dx^2} = (n+1)na\cdot x^{n+1} + n(n+1)bx^{-n}$$

$$= n(n+1)[a\cdot x^{n+1} + bx^{-n}] = n(n+1)y$$

62. We have $x^2 + y^2 = 1$

Differentiating w.r.t. x, we get

$$2x + 2yy' = 0 \quad \Rightarrow \quad x + yy' = 0$$

Again differentiating w.r.t. x, we get

$$1 + (y')^2 + yy'' = 0$$

63. $\because\ x = \text{cosec}\,\theta - \sin\theta$

$$\Rightarrow \quad x^2 + 4 = (\text{cosec}\,\theta - \sin\theta)^2 + 4 = (\text{cosec}\,\theta + \sin\theta)^2$$

and

$$y^2 + 4 = (\text{cosec}^n\theta - \sin^n\theta)^2 + 4 = (\text{cosec}^n\theta + \sin^n\theta)^2$$

Now, $\frac{dy}{dx} = \frac{dy}{d\theta}\times\frac{d\theta}{dx}$

$$= \frac{n(\text{cosec}^{n-1}\theta)(-\text{cosec}\,\theta\cot\theta) - n\sin^{n-1}\theta\cos\theta}{-\text{cosec}\,\theta\cot\theta - \cos\theta}$$

$$= \frac{n(\text{cosec}^n\theta\cot\theta + \sin^{n-1}\theta\cos\theta)}{(\text{cosec}\,\theta\cot\theta + \cos\theta)}$$

$$= \frac{n\cot\theta(\text{cosec}^n\theta + \sin^n\theta)}{\cot\theta(\text{cosec}\,\theta + \sin\theta)}$$

$$= \frac{n(\text{cosec}^n\theta + \sin^n\theta)}{(\text{cosec}\,\theta + \sin\theta)} = \frac{n\sqrt{y^2+4}}{\sqrt{x^2+4}}$$

Squaring both sides, we get $\left(\frac{dy}{dx}\right)^2 = \frac{n^2(y^2+4)}{x^2+4}$

or $$(x^2+4)\left(\frac{dy}{dx}\right)^2 = n^2(y^2+4)$$

64. $y = \sin^{-1}\left[\sqrt{x}\sqrt{1-a} - \sqrt{a}\sqrt{1-x}\right]$

Put $\sqrt{x} = \sin\theta$ and $\sqrt{a} = \sin\phi$

$$\therefore \quad y = \sin^{-1}\sin(\theta - \phi) = \theta - \phi$$

$$= \sin^{-1}\sqrt{x} - \sin^{-1}\sqrt{a}$$

$$\therefore \quad \frac{dy}{dx} = \frac{1}{2\sqrt{x}\sqrt{1-x}} - 0, \text{ etc.}$$

65. We know that by Newton–Leibniz formula,

if $I = \int_u^v f(t)\,dx$, then $\frac{dI}{dx} = f(v)\frac{dv}{dx} - f(u)\frac{du}{dx}$

where u and v are functions of x.

$$\therefore \quad \frac{dx}{dy} = \frac{1}{\sqrt{1+9y^2}} \quad \Rightarrow \quad \frac{dy}{dx} = \sqrt{1+9y^2}$$

$$\therefore \quad \frac{d^2y}{dx^2} = \frac{9y}{\sqrt{1+9y^2}}\cdot\frac{dy}{dx} = \frac{9y}{\sqrt{1+9y^2}}\cdot\sqrt{1+9y^2} = 9y$$

66. $\lim_{x\to 0}\frac{f(x)}{g(x)} = \lim_{x\to 0}\frac{f'(0)}{g'(0)}$ [by L'Hôpital's rule]

$$= \frac{6}{6} = 1$$

67. Let $u = f(\tan x)$ and $v = g(\sec x)$. Then

$$\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{f'(\tan x)\sec^2 x}{g'(\sec x)\sec x \tan x}$$

$$\Rightarrow \frac{du}{dv} = \frac{f'(\tan x)}{g'(\sec x)} \operatorname{cosec} x$$

$$\Rightarrow \left(\frac{du}{dv}\right)_{x=\frac{\pi}{4}} = \sqrt{2}\frac{f'(1)}{g'(\sqrt{2})} = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$$

68. $y = x^2 + \frac{1}{y} \quad \Rightarrow \quad y^2 = x^2 y + 1$

$$\Rightarrow 2y\frac{dy}{dx} = y \times 2x + x^2\frac{dy}{dx} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{2xy}{2y - x^2}$$

69. $T_r = \tan^{-1}\frac{1}{1 + r + r^2} = \tan^{-1}\frac{(r+1) - r}{1 + (r+1)r}$

or $T_r = [\tan^{-1}(r+1) - \tan^{-1} r]$

Put $r = 1, 2, 3, \ldots, x$ and add, the terms cancel diagonally.

$\therefore \quad y = \tan^{-1}(x+1) - \tan^{-1}(1)$

$\therefore \quad \frac{dy}{dx} = \frac{1}{1 + (x+1)^2}$

70. Let $y = 2\cos x \cos 3x$

On differentiating w.r.t. x, we get

$$\frac{dy}{dx} = 2\cos x(-3\sin 3x) + 2\cos 3x(-\sin x)$$

$$= -3(\sin 4x + \sin 2x) + (-1)[\sin 4x + \sin(-2x)]$$

Again differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2} = -3(4\cos 4x + 2\cos 2x) - 1(4\cos 4x - 2\cos 2x)$$

$$= -16\cos 4x - 4\cos 2x = -4(\cos 2x + 4\cos 4x)$$

$$= -2^2(\cos 2x + 2^2\cos 4x)$$

71. $f'(x) = e^x g'(x) + e^x g(x)$

$\therefore \quad f'(0) = g'(0) + g(0) = 1 + 2 = 3$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $y = \left(1 + \frac{1}{x}\right)^x$

$$\Rightarrow \log y = x\log\left(1 + \frac{1}{x}\right) = x\log\left(\frac{x+1}{x}\right)$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = x \cdot \frac{1}{\frac{x+1}{x}} \cdot \frac{x\cdot 1 - (x+1)\cdot 1}{x^2} + \log\left(\frac{x+1}{x}\right)\cdot 1$$

$$= \frac{x^2}{x+1}\cdot\frac{-1}{x^2} + \log\left(1 + \frac{1}{x}\right)$$

$$= \log\left(1 + \frac{1}{x}\right) - \frac{1}{x+1}$$

$$\Rightarrow \frac{dy}{dx} = y\left[\log\left(1 + \frac{1}{x}\right) - \frac{1}{x+1}\right]$$

$$= \left(1 + \frac{1}{x}\right)^x\left[\log\left(1 + \frac{1}{x}\right) - \frac{1}{x+1}\right]$$

2. $\sin(x + y) = \log(x + y)$

$$\Rightarrow \cos(x+y)\left(1 + \frac{dy}{dx}\right) = \frac{1}{x+y}\cdot\left[1 + \frac{dy}{dx}\right]$$

$$\Rightarrow 1 + \frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -1$$

3. We have $f(x) = |\cos x - \sin x|$

$$\Rightarrow f(x) = \begin{cases} \cos x - \sin x, & \text{for } 0 < x \le \frac{\pi}{x} \\ \sin x - \cos x, & \text{for } \frac{\pi}{4} < x < \frac{\pi}{2} \end{cases}$$

Clearly,

$$(\text{LHD at } x = \frac{\pi}{4}) = \left[\frac{d}{dx}(\cos x - \sin x)\right]_{\text{at } x=\frac{\pi}{4}}$$

$$= (-\sin x - \cos x)_{\text{at } x=\frac{\pi}{4}} = -\sqrt{2}$$

and

$$(\text{RHD at } x = \frac{\pi}{4}) = \left[\frac{d}{dx}(\sin x - \cos x)\right]_{\text{at } x=\frac{\pi}{4}}$$

$$= (\cos x + \sin x)_{\text{at } x=\frac{\pi}{4}} = \sqrt{2}$$

$\therefore$ (LHD at $x=\frac{\pi}{4}$) $\neq$ (RHD at $x=\frac{\pi}{4}$)

Thus, $f'\left(\frac{\pi}{4}\right)$ does not exist.

4. $\lim\limits_{x\to a}\dfrac{\sqrt{f(x)}-a}{\sqrt{\phi(x)}-b}$

$= \lim\limits_{x\to a}\dfrac{f(x)-a^2}{\phi(x)-b^2}\cdot\dfrac{\sqrt{\phi(x)}+b}{\sqrt{f(x)}+a}$

$= \lim\limits_{x\to a}\dfrac{f(x)-f(a)}{\phi(x)-\phi(a)}\cdot\dfrac{\sqrt{\phi(x)}+b}{\sqrt{f(x)}+a}$

$= \dfrac{f'(a)}{\phi'(a)}\cdot\dfrac{\sqrt{\phi(a)}+b}{\sqrt{f(a)}+a} = 3\cdot\dfrac{b+b}{a+a}$

$= 3\cdot\dfrac{2b}{2a}=\dfrac{3b}{a}$

5. Put $2 = r\cos\theta, 3 = r\sin\theta \ \therefore\ r=\sqrt{13}, \tan\theta=\dfrac{3}{2}$

$\therefore\quad y=\cos^{-1}\left(\dfrac{r\cos x\cos\theta - r\sin x\sin\theta}{\sqrt{13}}\right)$

$= \cos^{-1}[\cos(x+\theta)] = x+\theta$

$= x+\tan^{-1}\left(\dfrac{3}{2}\right) \qquad \therefore\ \dfrac{dy}{dx}=1$

6. $x^2+y^2=t-\dfrac{1}{t}$ and $x^4+y^4=t^2+\dfrac{1}{t^2}$

Now $(x^2+y^2)^2=t^2+\dfrac{1}{t^2}-2=x^4+y^4-2$

$\therefore\quad x^4+y^4+2x^2y^2=x^4+y^4-2$

$\therefore\quad x^2y^2=-1$

$\therefore\quad y^2=-\dfrac{1}{x^2} \qquad \therefore\ 2y\dfrac{dy}{dx}=\dfrac{2}{x^3}\Rightarrow x^3y\dfrac{dy}{dx}=1$

7. $y=f(x)$

$y=\tan^{-1}\left(\dfrac{\log(e/x^2)}{\log(ex^2)}\right)+\tan^{-1}\left(\dfrac{3+2\log x}{1-6\log x}\right)$

$y=\tan^{-1}\left(\dfrac{\log e-\log x^2}{\log e+\log x^2}\right)+\tan^{-1}\left(\dfrac{3+2\log x}{1-6\log x}\right)$

$y=\tan^{-1}\left(\dfrac{1-2\log x}{1+2\log x}\right)+\tan^{-1}\left(\dfrac{3+2\log x}{1-3\times 2\log x}\right)$

$[\because \log_e e=1]$

$= \tan^{-1}(1)-\tan^{-1}(2\log x)+\tan^{-1}(3)+\tan^{-1}(2\log x)$

$= \tan^{-1}(1)+\tan^{-1}(3)$

$\therefore\ y=$ constant $\quad\therefore\ \dfrac{dy}{dx}=0$ and $\dfrac{d^2y}{dx^2}=0$

8. Since $f(x)$ is odd $\ \therefore\ f(-x)=-f(x)$

$\therefore\quad f'(-x)(-1)=-f'(x)$

$\Rightarrow\quad f'(-x)=f'(x)$

$\therefore\quad f'(-3)=f'(3)=2$

9. $\dfrac{dy}{dx}=\dfrac{1}{\tan\frac{x}{2}}\cdot\sec^2\dfrac{x}{2}\cdot\dfrac{1}{2}+\dfrac{1}{\sqrt{1-\cos^2 x}}(-\sin x)$

$= \operatorname{cosec} x - 1$

10. $y=e^{2x} \ \Rightarrow\ \dfrac{dy}{dx}=2e^{2x}\ \Rightarrow\ \dfrac{d^2y}{dx^2}=4e^{2x}$

Again, $y=e^{2x} \ \Rightarrow\ 2x=\log y$

$\Rightarrow\ 2\dfrac{dx}{dy}=\dfrac{1}{y} \ \Rightarrow\ \dfrac{dx}{dy}=\dfrac{1}{2y}$

$\Rightarrow\ \dfrac{d^2x}{dy^2}=-\dfrac{1}{2y^2}$

$\therefore\quad \dfrac{d^2y}{dx^2}\cdot\dfrac{d^2x}{dy^2}=4e^{2x}\cdot\dfrac{1}{-2y^2}$

$= -\dfrac{2e^{2x}}{e^{4x}}=-2e^{-2x}$

11. When $2<x<3$, $[x]=2$

$\therefore\quad y=\sin\left[\dfrac{2\pi}{3}-x^2\right]$

$\therefore\quad \dfrac{dy}{dx}=-2x\cos\left(\dfrac{2\pi}{3}-x^2\right)$

Put $x=\sqrt{\dfrac{\pi}{3}}$ or $x^2=\dfrac{\pi}{3}$

$\therefore\quad \dfrac{dy}{dx}=-2\sqrt{\dfrac{\pi}{3}}\cos\left(\dfrac{2\pi}{3}-\dfrac{\pi}{3}\right)=-2\cdot\dfrac{1}{2}\sqrt{\dfrac{\pi}{3}}=-\sqrt{\dfrac{\pi}{3}}$

12. $T_1 = \tan^{-1}\dfrac{(x+1)-x}{1+(x+1)x} = \tan^{-1}(x+1) - \tan^{-1}x$

$T_2 = \tan^{-1}\dfrac{(x+2)-(x+1)}{1+(x+2).(x+1)}$

$= \tan^{-1}(x+2) - \tan^{-1}(x+1)$

$\vdots$

$T_n = \tan^{-1}(x+n) - \tan^{-1}[x+(n-1)]$

$\therefore\ y = \tan^{-1}(x+n) - \tan^{-1}x$ as the terms cancel diagonally

$\therefore\ \dfrac{dy}{dx} = y' = \dfrac{1}{1+(x+n)^2} - \dfrac{1}{1+x^2}$

$\Rightarrow\ y'(0) = \dfrac{1}{1+n^2} - 1 = \dfrac{-n^2}{n^2+1}$

13. We have $f(x) = \alpha(x)\beta(x)\gamma(x)$ for all real x

$\Rightarrow\ f'(x) = \alpha'(x)\beta(x)\gamma(x) + \alpha(x)\beta'(x)\gamma(x)$
$+\alpha(x)\beta(x)\gamma'(x)$

$\Rightarrow\ f'(2) = \alpha'(2)\beta(2)\gamma(2) + \alpha(2)\beta'(2)\gamma(2)$
$+\alpha(2)\beta(2)\gamma'(2)$

$\Rightarrow\ 18f(2) = 3\alpha(2)\beta(2)\gamma(2) - 4\alpha(2)\beta(2)\gamma(2)$
$+k\alpha(2)\beta(2)\gamma(2)$

[$Qf'(2) = 18f(2), \alpha'(2) = 3\alpha(2), \beta'(2) = -4\beta(2)$ and
$\gamma'(2) = k\gamma(2)$

$\Rightarrow\ 18f(2) = (-1+k)\alpha(2)\beta(2)\gamma(2) = (k-1)f(2)$

[$\because f(2) = \alpha(2)\beta(2)\gamma(2)$]

$\Rightarrow\ k - 1 = 18$

$\therefore\ k = 19$

14. The function $y = \log|x|$ is defined for all real x except $x = 0$ and $\log|x| = \begin{cases}\log x & \text{if } x>0\\ \log(-x) & \text{if } x<0\end{cases}$

Hence,

$\dfrac{d}{dx}\log|x| = \left\{\dfrac{1}{x}, x>0; = \left(\dfrac{1}{-x}\right)(-1) = \dfrac{1}{x} \text{ if } x<0\right\}$

$\dfrac{dy}{dx} = \dfrac{1}{x},\ x \neq 0$

Thus, $\dfrac{d}{dx}\log|x| = \dfrac{1}{x}, x \neq 0$

15. $y = (1-x^{1/2})(1+x^{1/2}) = 1-x$

$\therefore\ \dfrac{dy}{dx} = -1$

16. $\dfrac{dy}{dx} = n[x+\sqrt{x^2+a^2}\ ]^{n-1}\cdot\left[1+\dfrac{2x}{2\sqrt{x^2+a^2}}\right]$

$= \dfrac{n[x+\sqrt{x^2+a^2}\ ]^n}{\sqrt{x^2+a^2}} = \dfrac{ny}{\sqrt{x^2+a^2}}.$

17. $yz = \log_{10}x \cdot \log_x 10 = \log_{10}10 = 1$

$\therefore\ y = \dfrac{1}{z}$

$\therefore\ \dfrac{dy}{dz} = -\dfrac{1}{z^2}$

$= -(\log_{10}x)^2 = -\left(\dfrac{\log x}{\log 10}\right)^2$

18. Dividing both the numerator and denominator by $\sqrt{1+t^2}$, we get

$y = \dfrac{1-x}{1+x} = \dfrac{-1-x+2}{1+x} = -1 + \dfrac{2}{1+x}$

$\therefore\ \dfrac{dy}{dx} = \dfrac{-2}{(1+x)^2}$ and $\dfrac{d^2y}{dx^2} = \dfrac{4}{(1+x)^3} = \dfrac{4}{8} = \dfrac{1}{2}.$

Because when $t = 0$, $x = 1$.

19. $y = x^y$ $\qquad\therefore\ \log y = y \log x$

$\dfrac{1}{y}\cdot\dfrac{dy}{dx} = \dfrac{dy}{dx}\log x + y\cdot\dfrac{1}{x}$ etc.

20. In the neighbourhood of $x = -\dfrac{\pi}{4}$, both x and $\sin x$ are negative so that $|x| = -x$ and $|\sin x| = -\sin x$

$\therefore\ y = (-x)^{-\sin x}$ or $\log y = -\sin x \log(-x)$

or $\dfrac{1}{y}\dfrac{dy}{dx} = -\left[\cos x\log(-x) + \sin x\cdot\left(-\dfrac{1}{x}\right)(-1)\right]$

$\therefore\ \dfrac{dy}{dx} = -y\left[\cos x\log(-x) + \dfrac{1}{x}\sin x\right]$

Now putting $x = -\dfrac{\pi}{4}$ we get the result.

21. We have $y = \log_{10}x + \log_e y$

$\therefore\ \dfrac{dy}{dx} = \dfrac{1}{x}\log_{10}e + \dfrac{1}{y}\dfrac{dy}{dx}$

$\Rightarrow\ \dfrac{dy}{dx}\left(\dfrac{y-1}{y}\right) = \dfrac{\log_{10}e}{x} \Rightarrow \dfrac{dy}{dx} = \dfrac{\log_{10}e}{x}\left(\dfrac{y}{y-1}\right)$

22. $f(x)=\dfrac{\sin^{-1}x}{\sqrt{1-x^2}} \Rightarrow (1-x^2)[f(x)]^2=(\sin^{-1}x)^2$

On differentiating w.r.t. x, we get

$$(1-x^2)2f(x)f'(x)+[f(x)]^2(-2x)=\frac{2\sin^{-1}x}{\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2)2f(x)f'(x)+[f(x)]^2(-2x)=2f(x)$$

$$\Rightarrow (1-x^2)f'(x)-xf(x)=1$$

23. $\because\ y=x\log\left(\dfrac{x}{a+bx}\right)=x[\log x-\log(a+bx)]$

or $\left(\dfrac{y}{x}\right)=\log x-\log(a+bx)$

On differentiating both sides w.r.t. x, we get

$$\left(\frac{x\dfrac{dy}{dx}-y\cdot 1}{x^2}\right)=\frac{1}{x}-\frac{b}{a+bx}=\frac{a}{x(a+bx)} \qquad \text{(i)}$$

or $\left(x\dfrac{dy}{dx}-y\right)=\left(\dfrac{ax}{a+bx}\right)$

On taking log on both sides, we get

$$\log\left(x\frac{dy}{dx}-y\right)=\log(ax)-\log(a+bx)$$

On differentiating both sides w.r.t. x, we get

$$\frac{x\dfrac{d^2y}{dx^2}+\dfrac{dy}{dx}-\dfrac{dy}{dx}}{\left(x\dfrac{dy}{dx}-y\right)}=\frac{1}{x}-\frac{b}{a+bx}=\frac{a}{x(a+bx)}$$

$$=\frac{\left(x\dfrac{dy}{dx}-y\right)}{x^2}$$

or $x^3\dfrac{d^2y}{dx^2}=\left(x\dfrac{dy}{dx}-y\right)^2$

24. Let $\cos\alpha=\dfrac{5}{13}$. Then $\sin\alpha=\dfrac{12}{13}$.

So, $y=\cos^{-1}(\cos\alpha\cdot\cos x-\sin\alpha\cdot\sin x)$

$\Rightarrow y=\cos^{-1}[\cos(x+\alpha)]=x+\alpha$

[$\because\ x+\alpha$ is in the first or the second quadrant]

$\Rightarrow \dfrac{dy}{dx}=1$

25. Let $x=a\cos^2\theta+b\sin^2\theta$

$\therefore\ a-x=a-a\cos^2\theta-b\sin^2\theta=(a-b)\sin^2\theta$

and $x-b=a\cos^2\theta+b\sin^2\theta-b=(a-b)\cos^2\theta$

$\therefore\ y=(a-b)\sin\theta\cos\theta-(a-b)\tan^{-1}\tan\theta$

$$=\frac{a-b}{2}\sin 2\theta-(a-b)\theta$$

$$\therefore\ \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{(a-b)\cos 2\theta-(a-b)}{(b-a)\sin 2\theta}=\frac{1-\cos 2\theta}{\sin 2\theta}$$

$$=\tan\theta=\sqrt{\frac{a-x}{x-b}}$$

26. $y=x^{(x^x)}$

$\Rightarrow \log y=x^x\log x$

$\Rightarrow \dfrac{1}{y}\dfrac{dy}{dx}=\dfrac{dz}{dx}\cdot\log x+\dfrac{1}{x}z,$ (where $x^x=z$)

$\Rightarrow \dfrac{dy}{dx}=x^{(x^x)}[x^x(\log ex)\cdot\log x+x^{x-1}]$

$\left[\because \dfrac{dz}{dx}=x^x\log ex\right]$

27. Putting $x=\sin A$ and $\sqrt{x}=\sin B$,

$\Rightarrow y=\sin^{-1}(\sin A\sqrt{1-\sin^2 B}+\sin B\sqrt{1-\sin^2 A})$

$y=\sin^{-1}(\sin(A+B)=A+B=\sin^{-1}x+\sin^{-1}\sqrt{x}$

$\Rightarrow \dfrac{dy}{dx}=\dfrac{1}{\sqrt{1-x^2}}+\dfrac{1}{2\sqrt{x-x^2}}$

28. $\dfrac{dy}{dx}=\dfrac{2}{\sqrt{a^2-b^2}}\cdot\dfrac{1}{1+\left(\sqrt{\dfrac{a-b}{a+b}}\tan\dfrac{x}{2}\right)}\times\sqrt{\dfrac{a-b}{a+b}}\sec^2\dfrac{x}{2}\cdot\dfrac{1}{2}$

$$=\frac{1}{\sqrt{a-b}\sqrt{a+b}}\cdot\frac{1}{1+\dfrac{a-b}{a+b}\cdot\dfrac{\sin^2\dfrac{x}{2}}{\cos^2\dfrac{x}{2}}}\cdot\frac{\sqrt{a-b}}{\sqrt{a+b}}\sec^2\frac{x}{2}$$

$$=\frac{1}{a+b}\cdot\frac{(a+b)\cos^2\dfrac{x}{2}}{(a+b)\cos^2\dfrac{x}{2}+(a-b)\sin^2\dfrac{x}{2}}\sec^2\frac{x}{2}$$

$$=\frac{1}{a\left(\cos^2\dfrac{x}{2}+\sin^2\dfrac{x}{2}\right)+b\left(\cos^2\dfrac{x}{2}-\sin^2\dfrac{x}{2}\right)}$$

$$=\frac{1}{a+b\cos x}.$$

29. Put $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$

$$\therefore\ y = \tan^{-1}\left(\frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}\right) = \tan^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right)$$

$$= \tan^{-1}\tan\left(\frac{\pi}{4}-\theta\right) = \frac{\pi}{4}-\theta = \frac{\pi}{4}-\cos^{-1}x$$

$$\therefore\ \frac{dy}{dx} = 0-\left(-\frac{1}{\sqrt{1-x^2}}\right) = \frac{1}{\sqrt{1-x^2}}$$

30. We have $x = t^t = e^{t\log t}$

$$\Rightarrow \frac{dx}{dt} = e^{t\log t}(1+\log t) = t^t(1+\log t)$$

Also, $y = t^{t^t} \Rightarrow \log y = t^t\log t = e^{t\log t}\cdot\log t$

$$\Rightarrow \frac{1}{y}\frac{dy}{dt} = e^{t\log t}(1+\log t)\log t + e^{t\log t}\cdot\frac{1}{t}$$

$$\Rightarrow \frac{dy}{dt} = t^{t^t}\cdot t^t\left[(1+\log t)\log t+\frac{1}{t}\right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt}\times\frac{dt}{dx} = \frac{t^{t^t}\left[(1+\log t)\log t+\frac{1}{t}\right]}{(1+\log t)}.$$

31. We have

$$\log f(x) = (a+b+2x)[\log(a+b)-\log(b+x)]$$

Differentiating both sides with respect to x, we get

$$\frac{f'(x)}{f(x)} = 2[\log(a+x)-\log(b+x)] + (a+b+2x)\left[\frac{1}{a+x}-\frac{1}{b+x}\right]$$

$$\Rightarrow f'(0) = f(0)\left[2(\log a-\log b)+(a+b)\left(\frac{1}{a}-\frac{1}{b}\right)\right]$$

$$= \left(\frac{a}{b}\right)^{a+b}\left[2\log\frac{a}{b}+\frac{b^2-a^2}{ab}\right]$$

32. We have $e^x+e^y = e^{x+y} \Rightarrow e^{-y}+e^{-x} = 1$

Differentiating with respect to x, we get

$$-e^{-y}\frac{dy}{dx}-e^{-x} = 0 \Rightarrow \frac{dy}{dx} = -e^{y-x}$$

$$\therefore\ \left(\frac{dy}{dx}\right)_{(1,1)} = -e^{1-1} = -1$$

33. $\tan^{-1}\dfrac{4x}{1+5x^2} = \tan^{-1}\left(\dfrac{5x-x}{1+5x\cdot x}\right) = \tan^{-1}5x-\tan^{-1}x$

and $\tan^{-1}\left(\dfrac{2+3x}{3-2x}\right) = \tan^{-1}\left(\dfrac{\frac{2}{3}+x}{1-\frac{2}{3}x}\right) = \tan^{-1}\dfrac{2}{3}+\tan^{-1}x$

$$\therefore\ y = \tan^{-1}5x+\tan^{-1}\frac{2}{3}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1+25x^2}\cdot 5 = \frac{5}{1+25x^2}$$

34. Let $y = \tan^{-1}\left(\dfrac{\sqrt{1-x^2}-1}{x}\right)$

Put $x = \tan\theta$

$$\therefore\ y = \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right) = \tan^{-1}\left(\frac{\sec\theta-1}{\frac{\sin\theta}{\cos\theta}}\right)$$

$$= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right) = \tan^{-1}\left(\frac{2\sin^2\theta/2}{2\sin(\theta/2)\cos(\theta/2)}\right)$$

$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{\theta}{2} = \frac{1}{2}\tan^{-1}x$$

$$\therefore\ \frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

Let $z = \tan^{-1}\left(\dfrac{2x\sqrt{1-x^2}}{1-2x^2}\right)$. Put $x = \sin\phi$

$$\therefore\ z = \tan^{-1}\left(\frac{2\sin\phi\cos\phi}{1-2\sin^2\phi}\right) = \tan^{-1}\left(\frac{\sin 2\phi}{\cos 2\phi}\right) = 2\phi$$

$$\therefore\ z = 2\sin^{-1}x \qquad \therefore\ \frac{dz}{dx} = \frac{2}{\sqrt{1-x^2}}$$

$$\therefore\ \frac{dy}{dz} = \frac{\frac{1}{2(1+x^2)}}{\frac{2}{\sqrt{1-x^2}}} = \frac{\sqrt{1-x^2}}{4(1-x^2)}$$

At $x = 0$, $\dfrac{dy}{dz} = \dfrac{\sqrt{1-0}}{4(1+0)} = \dfrac{1}{4}$

35. $f(10)=\sin^{-1}\cos 10=\sin^{-1}\sin\left(\frac{\pi}{2}-10\right)$

$$= -\sin^{-1}\sin\left(10-\frac{\pi}{2}\right)$$

$$= -\sin^{-1}\sin\left(3\pi-10+\frac{\pi}{2}\right)=-\left(3\pi+\frac{\pi}{2}-10\right)$$

$$= 10-\frac{7\pi}{2}$$

$$f'(x)=\frac{-\sin x}{\sqrt{1-\cos^2 x}}=\frac{-\sin x}{|\sin x|} \Rightarrow f'(10)=\frac{-\sin 10}{|\sin 10|}=1$$

So, $f(10)+f'(10)=11-\frac{7\pi}{2}$

36. $\sin^{-1}\left(\frac{2x}{1+x^2}\right)=2\tan^{-1}x$ for $-1\le x\le 1$

and $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)=\begin{cases}2\tan^{-1}x, & x\ge 0\\ -2\tan^{-1}x, & x<0\end{cases}$

$\therefore$ For $0<x<1$, $\sin^{-1}\frac{2x}{1+x^2}=\cos^{-1}\frac{1-x^2}{1+x^2}$

$\therefore$ Derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t. $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is 1.

$\therefore$ Statement 1 is true but Statement 2 is false.

37. $y=x^2 \Rightarrow \frac{dy}{dx}=2x$ and $\frac{d^2y}{dx^2}=2$

Also, $1=2x\frac{dy}{dx} \Rightarrow \frac{dx}{dy}=\frac{1}{2x}$ (i)

$\Rightarrow \frac{d^2x}{dy^2}=-\frac{1}{2x^2}\frac{dx}{dy}=-\frac{1}{4x^3}$ [by (i)]

$\therefore \frac{d^2y}{dx^2}\cdot\frac{d^2x}{dy^2}\ne 1$

But $\frac{dy}{dx}\cdot\frac{dx}{dy}=1$ always.

38. Since $\sin^{-1}(2x^2+1)+\cos^{-1}(2x^2+1)=\frac{\pi}{2}$

$\therefore \frac{d}{dx}(\sin^{-1}(2x^2+1)+\cos^{-1}(2x^2+1)=0$

39. $y'=\cos(3\sin^{-1}x)\cdot\frac{3}{\sqrt{1-x^2}}$

$\Rightarrow y'(0)=3$. Also, $\sqrt{1-x^2}\,y'=\cos(3\sin^{-1}x)$

$\Rightarrow (1-x^2)y'^2=\cos^2(3\sin^{-1}x)=9(1-y^2)$

$\Rightarrow 2(1-x^2)y'.y''-2xy'^2=-18yy'$

$\Rightarrow (1-x^2)y''-xy'+9y=0$

Putting $x=0$, we get

$y''(0)=-9y(0)=0$.

40. $x=\frac{\sin y}{\sin(a+y)}$

$$\therefore \frac{dy}{dx}=\frac{\sin(a+y)\cos y-\sin y\cos(a+y)}{\sin^2(a+y)}$$

$$=\frac{\sin(a+y-y)}{\sin^2(a+y)}=\frac{\sin a}{\sin^2(a+y)}$$

$$\therefore \frac{dy}{dx}=\frac{\sin^2(a+y)}{\sin a}$$

41. $y=(x+\sqrt{1+x^2})^n$

$$\Rightarrow \frac{dy}{dx}=n(x+\sqrt{1+x^2})^{n-1}\left(1+\frac{1}{2\sqrt{1+x^2}}\cdot 2x\right)$$

$$=n(x+\sqrt{1+x^2})^{n-1}\left(1+\frac{x}{\sqrt{1+x^2}}\right)$$

$$=\frac{n(x+\sqrt{1+x^2})^{n-1}(x+\sqrt{1+x^2})}{\sqrt{1+x^2}}$$

$$=\frac{n(x+\sqrt{1+x^2})^n}{\sqrt{1+x^2}}=\frac{ny}{\sqrt{1+x^2}}$$

$$\therefore (1+x^2)\left(\frac{dy}{dx}\right)^2=n^2y^2$$

$$\Rightarrow (1+x^2)2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2}+2x\left(\frac{dy}{dx}\right)^2=n^2\cdot dy\frac{dy}{dx}$$

$$\Rightarrow (1+x^2)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=n^2y$$

42. Since $x^y=e^{x-y}$ $\therefore \log x^y=\log^{x-y}$

$\Rightarrow y\log x=x-y$

$\Rightarrow y(1+\log x)=x$

$\Rightarrow \quad y = \dfrac{x}{1+\log x}$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{(1+\log x)\cdot 1 - x\cdot\frac{1}{x}}{(1+\log)^2} = \dfrac{\log x}{(1+\log x)^2}$

43. $\lim\limits_{x\to 2}\dfrac{xf(2)-2f(x)}{x-2}$

$= \lim\limits_{x\to 2}\dfrac{xf(2)-2f(x)+2f(2)-2f(2)}{x-2}$

$= \lim\limits_{x\to 0}\dfrac{(x-2)f(2)-2[f(x)-f(2)]}{x-2}$

$= \lim\limits_{x\to 2}\dfrac{(x-2)f(2)}{x-2} - 2\lim\limits_{x\to 2}\dfrac{f(x)-f(2)}{x-2}$

$= f(2) - 2f'(2) = 4 - 2(4) - 2(4)$

$= -4$

44. $f(x + y) = f(x)\, f(y)$

$\Rightarrow \quad f(x+5) = f(x)f(5) = 2f(x)$

$\Rightarrow \quad f'(x+5) = 2f'(x)$

$\Rightarrow \quad f'(0+5) = 2f'(0) \;= 2(3) = 6$

$\Rightarrow \quad f'(5) = 6$

45. $x = e^{y+x}$

$\Rightarrow \quad \log x = (y+x)\log e = y + x$

$\Rightarrow \quad y = \log x - x$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{1}{x} - 1 = \dfrac{1-x}{x}$

46. Given that $f(1) = -2$ and $f'(x) \geq 2$

$\Rightarrow \quad \dfrac{dy}{dx} \geq 2$

$\Rightarrow \quad dy \geq 2dx$

$\Rightarrow \quad \int_{f(1)}^{f(6)} dy \;\geq\; 2\int_{1}^{6} dx \quad \Rightarrow \quad f(6) - f(1) \geq 10$

$\Rightarrow \quad f(6) \geq 10 + f(1) \quad \Rightarrow \quad f(6) \geq 8$

47. $\lim\limits_{x\to y}\dfrac{|f(x)-f(y)|}{|x-y|} \;\leq\; \lim\limits_{x\to y}|x-y|$

$\Rightarrow \quad |f'(y)| \leq 0 \qquad\qquad \Rightarrow \quad f'(y) = 0$

$\Rightarrow$ $f(y)$ = constant as $f(0) = 0$

$\Rightarrow \quad f(y) = 0 \qquad\qquad \Rightarrow f(1) = 0$

48. $f'(1) = \lim\limits_{h\to 0}\dfrac{f(1+h)-f(h)}{h} \;=\; \lim\limits_{h\to 0}\dfrac{f(1+h)}{h} - \lim\limits_{h\to 0}\dfrac{f(1)}{h}$

Since $\lim\limits_{h\to 0}\dfrac{f(1+h)}{h} = 5$

$\therefore \quad \lim\limits_{h\to 0}\dfrac{f(1)}{h}$ must be finite as $f'(1)$ exists and $\lim\limits_{h\to 0}\dfrac{f(1)}{h}$

can be finite only if $f(1) = 0$ and $\lim\limits_{h\to 0}\dfrac{f(1)}{h} = 0$

$\therefore \quad f'(1) = \lim\limits_{h\to 0}\dfrac{f(1+h)}{h} = 5$

49. We have $x^m y^n = (x+y)^{m+n}$

$\Rightarrow \quad m\log x + n\log y = (m+n)\log(x+y)$

Differentiating w.r.t. x, we get

$\dfrac{m}{x} + \dfrac{n}{y}\cdot\dfrac{dy}{dx}$

$= (m+n)\dfrac{1}{x+y}\left[1+\dfrac{dy}{dx}\right]$

$\Rightarrow \quad \dfrac{dy}{dx}\left(\dfrac{n}{y} - \dfrac{m+n}{x+y}\right) = \dfrac{m+n}{x+y} - \dfrac{m}{x}$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{y}{x}$

50. $f(x) = \dfrac{x}{1-x}$ if $x \leq 0$ and $f(x) = \dfrac{x}{1+x}$ if $x \geq 0$

At $x = 0$, the left-handed derivative

$= \lim\limits_{h\to 0}\dfrac{f(0-h)-f(0)}{-h}$

$= \lim\limits_{h\to 0}\dfrac{f(-h)-f(0)}{-h} = \lim\limits_{h\to 0}\dfrac{\frac{-h}{1+h}-0}{-h}$

$= \lim\limits_{h\to 0}\dfrac{1}{1+h} = 1$ and the right-handed derivative

$= \lim\limits_{h\to 0}\dfrac{f(0+h)-f(0)}{h}$

$= \lim\limits_{h\to 0}\dfrac{f(h)-f(0)}{h} = \lim\limits_{h\to 0}\dfrac{\frac{h}{1+h}-0}{h}$

$= \lim\limits_{h\to 0}\dfrac{1}{1+h} = 1$

$\therefore$ LHD = RHD at $x = 0$

Hence, $f(x)$ is differentiable at $x = 0$

Further, $f'(x) = \dfrac{1}{(1-x)^2}$, where $x \le 0$

and $f'(x) = \dfrac{1}{(1+x)^2}$, where $x \ge 0$

Thus, $f'(x)$ exists for all values of x in the interval $(-\infty, \infty)$.

51. We have

$$x^{2x} - 2x^x \cot y - 1 = 0 \qquad \text{(i)}$$

When $x = 1$, we get

$1 - 2\cot y - 1 = 0 \Rightarrow \cot y = 0 \Rightarrow y = \dfrac{\pi}{2}$

Differentiating (i) w.r.t. x, we get

$$2x^{2x}(1+\log x) - 2x^x(1+\log x)\cot y + 2x^x \operatorname{cosec}^2 y \frac{dy}{dx} = 0$$

Putting $x = 1$, $y = \dfrac{\pi}{2}$, we get

$$= 2 - 2\times 0 + 2\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -1$$

52. Let $y = \left[\cos^{-1}\left(\sin\sqrt{\dfrac{1+x}{2}}\right) + x^x\right]$

$$= \left\{\cos^{-1}\left[\cos\left(\frac{\pi}{2} - \sqrt{\frac{1+x}{2}}\right)\right] + x^x\right\}$$

$$= \frac{\pi}{2} - \sqrt{\frac{1+x}{2}} + x^x$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{2} - \sqrt{\frac{1+x}{2}}\right) + \frac{d}{dx}(x^x)$$

$$\left[\text{Let } Y = x^x \Rightarrow \log Y = x\log x \Rightarrow \frac{1}{4}\cdot\frac{dY}{dx} = x\cdot\frac{1}{x} + \log x \Rightarrow \frac{dY}{dx} = x^x(1+\log x)\right]$$

$$\Rightarrow \left[-\frac{1}{2}\left(\frac{1-x}{2}\right)^{-\frac{1}{2}}\cdot\frac{1}{2}\right] + x^x(1+\log x)$$

$$\Rightarrow -\frac{1}{4}\frac{\sqrt{2}}{\sqrt{1+x}} + x^x(1+\log x)$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{x=1} = -\frac{1}{4}\cdot\frac{\sqrt{2}}{\sqrt{1+1}} + 1^1(1+\log 1) = -\frac{1}{4} + 1 = \frac{3}{4}.$$

53. $f(x) = \log_x(\log x)$

$$\Rightarrow f(x) = \frac{\log_e(\log_e x)}{\log_e x}$$

$$\Rightarrow f'(x) = \left[\frac{\log_e x \times \dfrac{1}{x\log_e x} - \dfrac{\log_e(\log_e x)}{x}}{(\log_e x)^2}\right]$$

$$= \frac{1 - \log_e(\log_e x)}{x(\log_e x)^2}$$

$$\Rightarrow f'(e) = \frac{1-\log_e(\log_e e)}{e(\log_e x)^2} = \frac{1-\log_e 1}{e} = \frac{1-0}{e} = \frac{1}{e}$$

54. $f(x) = \log \Rightarrow f(\log x) + \log_e(\log_e x)$

$$\Rightarrow \frac{d}{dx}[f(\log x)] = \left(\frac{1}{\log_e x}\right)\cdot\left(\frac{1}{x}\right)$$

$$= \frac{1}{(x\log x)} = (x\log x)^{-1}$$

55. $y = e^{1+\log_e x} = e\cdot e^{\log_e x} = ex$

$$\Rightarrow \frac{dy}{dx} = e$$

56. $y = \sec^{-1}\left(\dfrac{x+1}{x-1}\right) + \sin^{-1}\left(\dfrac{x-1}{x+1}\right)$

$$= \cos^{-1}\left(\frac{x-1}{x+1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right)$$

$$\left[\because \sec^{-1}(t) = \cos^{-1}\left(\frac{1}{t}\right)\right]$$

$$= \frac{\pi}{2} \qquad \left[\because \cos^{-1}(4) + \sin^{-1}(4) = \frac{\pi}{2}\right]$$

$$\Rightarrow \frac{dy}{dx} = 0.$$

57. Let $u = \sec^{-1}\left(\dfrac{1}{2x^2-1}\right)$ and $v = \sqrt{1-x^2}$.

We shall first evaluate $\dfrac{du}{dx}$.

Put $x = \cos\theta$ so that

$$u = \sec^{-1}\left(\frac{1}{2\cos^2\theta - 1}\right) = \sec^{-1}\left(\frac{1}{\cos 2\theta}\right)$$

$$= \sec^{-1}(\sec 2\theta) = 2\theta = 2\cos^{-1}x$$

$$\Rightarrow \frac{du}{dx} = \frac{-2}{\sqrt{1-x^2}} \qquad \text{(i)}$$

Also, $v = \sqrt{1-x^2}$

$$\Rightarrow \frac{dv}{dx} = \frac{1}{2\sqrt{1-x^2}} \cdot (-2x) = \frac{-x}{\sqrt{1-x^2}} \qquad \text{(ii)}$$

Now, $\dfrac{du}{dv} = \dfrac{\left(\dfrac{du}{dx}\right)}{\left(\dfrac{dv}{dx}\right)} = \dfrac{-2}{\sqrt{1-x^2}} \times \dfrac{\sqrt{1-x^2}}{-x}$

$$\Rightarrow \frac{du}{dv} = \frac{2}{x} \qquad \text{(iii)}$$

Putting $x = \dfrac{1}{2}$ in (iii), we get

$$\left(\frac{du}{dv}\right)_{x=\frac{1}{2}} = 4$$

58. $f(x) = |x|^3 = \begin{cases} -x^3, & \text{when } x < 0 \\ x^3, & \text{when } x \geq 0 \end{cases}$

$$\therefore \quad Rf'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h\to 0} \frac{h^3 - 0}{h} = \lim_{h\to 0} h^2 = 0$$

and $Lf'(0) = \lim\limits_{h\to 0} \dfrac{f(0-h) - f(0)}{-h} = \lim\limits_{h\to 0} \dfrac{-(-h)^3 - 0}{h}$

$$= \lim_{h\to 0} \frac{h^3}{-h} = \lim_{h\to 0} (-h^2) = 0$$

$$\therefore \quad f'(0) = Rf'(0) = Lf'(0) = 0.$$

59. $x^p y^q = (x+y)^{p+q}$

$$\Rightarrow p\log x + q\log y = (p+q)\log(x+y)$$

$$\Rightarrow \frac{p}{x} + \frac{q}{y}\frac{dy}{dx} = \frac{(p+q)}{(x+y)} \cdot \left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \left[\frac{q}{y} - \frac{(p+q)}{(x+y)}\right]\frac{dy}{dx} = \frac{(p+q)}{(x+y)} - \frac{p}{x}$$

$$\Rightarrow \left[\frac{qx - py}{y(x+y)}\right]\frac{dy}{dx} = \frac{qx - px}{x(x-y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}.$$

60. $\sin y = x\sin(a+y) \quad \Rightarrow \quad x = \dfrac{\sin y}{\sin(a+y)} \qquad \text{(i)}$

On differentiating (i) w.r.t. x, we get

$$1 = \frac{\sin(a+y)\cos y\frac{dy}{dx} - \sin y\cos(a+y)\frac{dy}{dx}}{\sin^2(a+y)}$$

$$\Rightarrow 1 = \frac{[\sin(a+y)\cos y - \sin y\cos(a+y)]\frac{dy}{dx}}{\sin^2(a+y)}$$

$$\Rightarrow 1 = \frac{\sin(a+y-y)}{\sin^2(a+y)} \cdot \frac{dy}{dx}$$

$$\Rightarrow 1 = \frac{\sin a}{\sin^2(a+y)} \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}.$$

61. $y = f\left(\dfrac{2x+3}{3-2x}\right)$

$$\Rightarrow \frac{dy}{dx} = f'\left(\frac{2x+3}{3-2x}\right) \cdot \frac{d}{dx}\left(\frac{2x+3}{3-2x}\right)$$

$$= \left\{\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]\right\} \cdot \left[\frac{(3-2x)\cdot 2 - (2x+3)\cdot(-2)}{(3-2x)^2}\right]$$

$$= \frac{12}{(3-2x)^2} \cdot \sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$$

62. $\dfrac{d}{dx}\left[\log\left(x + \dfrac{1}{x}\right)\right] = \dfrac{1}{\left(x + \dfrac{1}{x}\right)} \cdot \dfrac{d}{dx}\left(x + \dfrac{1}{x}\right)$

$$= \frac{1}{\left(x + \frac{1}{x}\right)} \cdot \left(1 - \frac{1}{x^2}\right) = \frac{\left(1 - \frac{1}{x^2}\right)}{\left(x + \frac{1}{x}\right)}.$$

63. Let $u = \tan^{-1}\left[\dfrac{2x}{1-x^2}\right]$.

Put $x = \tan\theta$. Then,

$$u = \tan^{-1}\left[\frac{2\tan\theta}{1-\tan\theta}\right] = \tan^{-1}(\tan 2\theta)$$

$$= 2\theta = 2\tan^{-1} x$$

$$\therefore \quad \frac{du}{dx} = \frac{2}{1+x^2}.$$

Let $v = \sin^{-1}\left[\frac{2x}{1-x^2}\right]$.

Put $x = \tan\phi$. Then

$$v = \sin^{-1}\left[\frac{2\tan\phi}{1+\tan^2\phi}\right] = \sin^{-1}(\sin 2\phi)$$

$$= 2\phi = 2\tan^{-1}x$$

$$\therefore \quad \frac{du}{dx} = \frac{2}{1+x^2}.$$

Thus, $\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{\frac{2}{1+x^2}}{\frac{2}{1+x^2}} = 1$

64. Let $y = \sec(\tan^{-1}x)$.

Put $x = \tan\theta$. Then

$$y = \sec[\tan^{-1}(\tan\theta)] = \sec\theta = \sqrt{1+\tan^2\theta}$$

$$= \sqrt{1+x^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{2}(1+x^2)^{-\frac{1}{2}}\cdot 2x \quad \Rightarrow \quad \frac{dy}{dx} = \frac{x}{\sqrt{1+x^2}}.$$

65. $y = \log_{\cos x}\sin x = \frac{\log\sin x}{\log\cos x}$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\log(\cos x)\cdot\left(\frac{\cos x}{\sin x}\right) - \log(\sin x)\cdot\left(-\frac{\sin x}{\cos x}\right)}{(\log\cos x)^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{(\cot x\log\cos x + \tan x\log\sin x)}{(\log\cos x)^2}.$$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. If $y = f(x)$ is an odd differentiable function defined on $(-\infty, \infty)$ such that $f'(3) = -2$, then $f'(-3)$ equals

(a) 4 (b) 2

(c) –2 (d) 0

2. If $x = \frac{1+t}{t^3}$ and $y = \frac{3}{2t^2} + \frac{2}{t}$, then $x\left(\frac{dy}{dx}\right)^3 - \frac{dy}{dx} =$

(a) 0 (b) –1

(c) 1 (d) 2

3. If $f(x) = (\cos x + i\sin x)(\cos 3x + i\sin 3x)$ ... $\cos(2n-1)x + i\sin(2n-1)x$, then $f''(x)$ is equal to

(a) $n^2 f(x)$ (b) $-n^4 f(x)$

(c) $-n^2 f(x)$ (d) $n^4 f(x)$

4. If $x = \cos t$ and $y = \log_e^t$, then

(a) $\frac{dy}{dx} = -\frac{2}{\pi}$ at $t = \frac{\pi}{2}$ (b) $\frac{dy}{dx} = \frac{4}{\pi^2}$ at $t = \frac{\pi}{2}$

(c) $\frac{dy}{dx} = \frac{144}{\pi^2}$ at $t = \frac{\pi}{6}$ (d) $\frac{dy}{dx} = -\frac{12}{\pi}$ at $t = \frac{\pi}{6}$

5. If $y = \sin^{-1}\left(\frac{\sin\alpha\sin x}{1-\cos\alpha\sin x}\right)$, then $y'(0)$ is

(a) 1 (b) 2 tan α

(c) (1/2) tan α (d) sin α

6. If $F(x) = \frac{1}{x^2}\int_4^x [4t^2F'(t)]dt$, then $F'(4)$ equals

(a) $\frac{32}{9}$ (b) $\frac{64}{3}$

(c) $\frac{64}{9}$ (d) none of these

7. A triangle has two of its vertices at $P(a, 0)$ and $Q(0, b)$, and the third vertex $R(x, y)$ is moving along the straight line $y = x$. If A is the area of the triangle, then $\frac{dA}{dx}$ is equal to

(a) $\frac{a-b}{2}$ (b) $\frac{a-b}{4}$

(c) $-\left(\frac{a-b}{2}\right)$ (d) $\frac{a+b}{4}$

8. If $y = \int_0^x f(t)\sin[k(x-t)]dt$, then $\frac{d^2y}{dx^2} + k^2y$ equals

(a) 0 (b) y

(c) $kf(x)$ (d) $k^2f(x)$

9. If $y^2 = ax^2 + bx + c$, where a, b and c are constants, then $y^3\frac{d^2y}{dx^2}$ is equal to

(a) a constant

(b) a function of x

(c) a function of y

(d) a function of both x and y

10. The differential coefficient of $a^{\log_{10}\operatorname{cosec}^{-1}x}$ is

(a) $\dfrac{a^{\log_{10}(\operatorname{cosec}^{-1}x)}}{\operatorname{cosec}^{-1}x}\dfrac{1}{x\sqrt{x^2-1}}\log_{10}a$

(b) $-\dfrac{a^{\log_{10}(\operatorname{cosec}^{-1}x)}}{\operatorname{cosec}^{-1}x}\cdot\dfrac{1}{|x|\sqrt{x^2-1}}\log 10a$

(c) $\dfrac{-a^{\log_{10}(\operatorname{cosec}^{-1}x)}}{\operatorname{cosec}^{-1}x}\cdot\dfrac{1}{|x|\sqrt{x^2-1}}\log_a 10$

(d) $\dfrac{a^{\log_{10}\operatorname{cosec}^{-1}x}}{\operatorname{cosec}^{-1}x}\cdot\dfrac{1}{x\sqrt{x^2-1}}\log_a 10$

11. If $f(x)=\tan^{-1}\left[\dfrac{\log(e/x^2)}{\log(ex^2)}\right]+\tan^{-1}\left(\dfrac{3+2\log x}{1-6\log x}\right)$, then $\dfrac{d^n y}{dx^n}$ is

(a) $\tan^{-1}[(\log x)^n]$ (b) 0

(c) $\dfrac{1}{2}$ (d) none of these

12. If $y=\sin^2\alpha+\cos^2(\alpha+\beta)+2\sin\alpha\sin\beta\cos(a+\beta)$, then $\dfrac{d^3y}{d\alpha^3}$ is

(a) $\dfrac{\sin^3(\alpha+\beta)}{\cos\alpha}$ (b) $\cos(\alpha+3\beta)$

(c) 0 (d) none of these

13. If $y=f(x^3), z=g(x^5), f'(x)=\tan x$ and $g'(x)=\sec x$, then the value of $\dfrac{dy}{dz}$ is

(a) $\dfrac{3}{5x^2}\cdot\dfrac{\tan x^3}{\sec x^5}$ (b) $\dfrac{5x^2}{3}\cdot\dfrac{\sec x^5}{\tan x^3}$

(c) $\dfrac{3x^2}{5}\cdot\dfrac{\tan x^3}{\sec x^5}$ (d) none of these

14. If $ax^2+2hxy+by^2=1$, then $\dfrac{d^2y}{dx^2}$ equals

(a) $\dfrac{h^2+ab}{(hx+by)^3}$ (b) $\dfrac{h^2-ab}{(hx+by)^2}$

(c) $\dfrac{h^2+ab}{(hx+by)^3}$ (d) $\dfrac{h^2-ab}{(hx+by)^3}$

15. If $\cos\dfrac{x}{2}\cos\dfrac{x}{2^2}\cos\dfrac{x}{2^3}\cdots\infty=\dfrac{\sin x}{x}$, then

(a) $\dfrac{1}{2}\tan\dfrac{x}{2}+\dfrac{1}{2^2}\tan\dfrac{x}{2^2}+\dfrac{1}{2^3}\tan\dfrac{x}{2^3}+\cdots\infty=-\cot x+\dfrac{1}{x}$

(b) $\dfrac{1}{2}\tan\dfrac{x}{2}+\dfrac{1}{2^2}\tan\dfrac{x}{2^2}+\dfrac{1}{2^3}\tan\dfrac{x}{2^3}+\cdots\infty=\cot x-\dfrac{1}{x}$

(c) $\dfrac{1}{2^2}\sec^2\dfrac{x}{2}+\dfrac{1}{2^4}\sec^2\dfrac{x}{2^2}+\cdots+\infty=\operatorname{cosec}^2x-\dfrac{1}{x^2}$

(d) $\dfrac{1}{2^2}\sec^2\dfrac{x}{2}+\dfrac{1}{2^4}\sec^2\dfrac{x}{2^2}+\cdots+\infty=-\operatorname{cosec}^2x+\dfrac{1}{x^2}$

16. If $x=t+\dfrac{1}{t}, y=t-\dfrac{1}{t}$, then $\dfrac{d^2y}{dx^2}$ is equal to

(a) $-4t(t^2-1)^{-2}$

(b) $-4t^3(t^2-1)^{-3}$

(c) $(t^2+1)(t^2-1)^{-1}$

(d) $-4t^2(t^2-1)^{-2}$

17. If $y=\tan^{-1}\left(\dfrac{x}{\sqrt{1-x^2}}\right)$, then $\dfrac{dy}{dx}=$

(a) $-\dfrac{1}{\sqrt{1-x^2}}$ (b) $\dfrac{x}{\sqrt{1-x^2}}$

(c) $\dfrac{1}{\sqrt{1-x^2}}$ (d) $\dfrac{\sqrt{1-x^2}}{x}$

18. If $y=\sec x^0$, then $\dfrac{dy}{dx}=$

(a) $\sec x\tan x$

(b) $\sec x^0\tan x^0$

(c) $\dfrac{\pi}{180}\sec x^0\tan x^0$

(d) $\dfrac{180}{\pi}\sec x^0\tan x^0$

19. If $y=\log\tan\left(\dfrac{\pi}{4}+\dfrac{x}{2}\right)$, then $\dfrac{dy}{dx}=$

(a) $-\tan x$ (b) $-\sec x$

(c) $\tan x$ (d) $\sec x$

20. If $y=\tan^{-1}(\sec x+\tan x)$, then $\dfrac{dy}{dx}=$

(a) 1 (b) $-1/2$

(c) 2 (d) $1/2$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If $f(x + y) = f(x)\, f(y)$ for all $x, y \in \mathbb{R}$ and $f(4) = 3$, $f'(0) = 4$, then $f'(4) =$

(a) 12 (b) 4
(c) 3 (d) 6

2. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$ (b) $\frac{y^2}{x^2}\sqrt{\frac{1-y^6}{1-x^6}}$

(c) $\frac{x^2}{y^2}\sqrt{\frac{1-x^6}{1-y^6}}$ (d) none of these

3. If $y = \log(\sqrt{x-a} + \sqrt{x-b})$, then $\frac{dy}{dx}$ is

(a) $\frac{1}{2\sqrt{x-a}\sqrt{x-b}}$ (b) $\frac{1}{2\sqrt{a-x}\sqrt{x-b}}$

(c) $\frac{-1}{2\sqrt{x-a}\sqrt{x-b}}$ (d) $-\frac{1}{2\sqrt{a-x}\sqrt{x-b}}$

4. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant.

Then $\frac{d^3}{dx^3}[(f(x)]$ at $x = 0$ is

(a) p (b) $p + p^2$
(c) $p + p^3$ (d) independent of p

5. If $\sin y = x\sin(a + y)$ and $\frac{dy}{dx} = \frac{A}{1 + x^2 - 2x\cos a}$, then the value of A is

(a) $\sin a$ (b) $\cos a$
(c) $-\cos a$ (d) $-\sin a$

6. If $y = \sqrt{(x-1)(x-2)(x-3)(x-4)}$, then $\frac{dy}{dx}$ is

(a) $\frac{1}{2}y\left[\frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3} + \frac{1}{x-4}\right]$

(b) $\frac{y}{2}\left[\frac{1}{x-1} - \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4}\right]$

(c) $y\left[\frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3} + \frac{1}{x-4}\right]$

(d) $\frac{y}{4}\left[\frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3} + \frac{1}{x-4}\right]$

7. If $y = \sqrt{x + \sqrt{y + \sqrt{x + \sqrt{y + \cdots \infty}}}}$, then $\frac{dy}{dx}$ is

(a) $\frac{(y^2 - x)}{2y^3 - 2xy - 1}$ (b) $\frac{y^2 - x}{2y^3 - 2xy + 1}$

(c) $\frac{y^2 - x}{2y^3 - xy - 1}$ (d) $\frac{y^2 - x}{2x^3 - xy - 1}$

8. If $f(x) = \cos^{-1}\left[\frac{1 - (\log_e x)^2}{1 + (\log_e x)^2}\right]$, then $f'(e)$ is equal to

(a) $\frac{2}{e}$ (b) $\frac{1}{e}$

(c) $\frac{3}{e}$ (d) $\frac{4}{e}$

9. If $y = x^{(\log x)^{\log^{\log x}}}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{y\log y}{x\log x}(2\log\log x + 1)$

(b) $\frac{x\log x}{y\log y}(2\log\log x + 1)$

(c) $\frac{2y\log y}{x\log x}(\log\log x + 1)$

(d) none of these

10. If $f(9) = 9$ and $f'(9) = 4$, then $\lim_{x\to 9}\frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3}$ equals

(a) 9 (b) 4
(c) 36 (d) none of these

11. If g is the inverse of f and $f'(x) = \frac{1}{1 + x^{n'}}$, then $g'(x)$ equals

(a) $1 + x^n$ (b) $1 + [f(x)]^n$
(c) $1 + [g(x)]^n$ (d) none of these

12. $\frac{d}{dx}\cos^{-1}\left(\frac{4x^3}{27} - x\right) =$

(a) $\frac{3}{\sqrt{9 - x^2}}$ (b) $\frac{-1}{\sqrt{9 - x^2}}$

(c) $\frac{-3}{\sqrt{9 - x^2}}$ (d) $\frac{1}{\sqrt{9 - x^2}}$

13. If $ax^2 + 2hxy = by^2 = 1$, then $\frac{d^2y}{dx^2}$ is equal to

(a) $\frac{ab - h^2}{(hx + by)^2}$ (b) $\frac{h^2 - ab}{(hx + by)^3}$

(c) $\frac{h^2 + ab}{(hx + by)^3}$ (d) none of these

14. Let $f(t) - \ln(t)$. Then $\frac{d}{dx}\left[\int_{x^2}^{3} f(t)dt\right]$

(a) has a value 0 when $x = 0$
(b) has a value 0 when $x = 1$ and $x = 4/9$
(c) has a value $9e^2 - 4e$ when $x = e$
(d) has a differential coefficient $27e - 8$ for $x = e$

15. If $y = \cos^{-1}\left(\frac{2\cos x - 3\sin x}{\sqrt{13}}\right)$, then $\frac{dy}{dx}$ is

(a) zero (b) constant = 1
(c) constant ≠ 1 (d) none of these

16. If $\sqrt{\frac{v}{\mu}} + \sqrt{\frac{\mu}{v}} = 6$ then $\frac{dv}{d\mu} =$

(a) $\frac{17\mu - v}{\mu - 17v}$ (b) $\frac{\mu - 17v}{17\mu - v}$

(c) $\frac{17\mu + v}{\mu - 17v}$ (d) $\frac{\mu + 17v}{17\mu - v}$

17. If $y = \cot^{-1}\sqrt{x^2 - 1} + \sec^{-1} x$, then $x\frac{dy}{dx}$ equals

(a) x (b) 1
(c) 0 (d) -1

18. If $f(x) = \left(\frac{a + x}{b + x}\right)^{a+b+2x}$, then $f'(0)$ is

(a) $\left(2\log\frac{a}{b} + \frac{a^2 - b^2}{ab}\right)\left(\frac{a}{b}\right)^{a+b}$

(b) $\left(2\log\frac{a}{b} + \frac{b^2 - a^2}{ab}\right)\left(\frac{a}{b}\right)^{a+b}$

(c) $\left(2\log\frac{a}{b} + \frac{a^2 + b^2}{ab}\right)\left(\frac{a}{b}\right)^{a+b}$

(d) none of these

19. If $x^p \cdot y^q = (x + y)^{p+q}$, then $\frac{dy}{dx}$ is

(a) independent of p
(b) independent of q
(c) dependent on both p and q
(d) $\frac{y}{x}$

20. Let g be the inverse function of f and $f'(x) = \frac{x^{10}}{(1 + x^2)}$. If $g(2) = a$, then $g'(2)$ is equal to

(a) $\frac{a}{2^{10}}$ (b) $\frac{1 + a^2}{a^{10}}$

(c) $\frac{a^{10}}{1 + a^2}$ (d) $\frac{1 + a^{10}}{a^2}$

ANSWERS

Level 1

1. (c) **2.** (c) **3.** (b) **4.** (a) and (d) **5.** (d) **6.** (a) **7.** (c) **8.** (c) **9.** (a)
10. (b) **11.** (b) **12.** (c) **13.** (a) **14.** (d) **15.** (a) and (c) **16.** (b) **17.** (c) **18.** (c)
19. (d) **20.** (d)

Level 2

1. (a) **2.** (a) **3.** (a) **4.** (d) **5.** (a) **6.** (a) **7.** (a) **8.** (b) **9.** (a) **10.** (b)
11. (c) **12.** (b) **13.** (b) **14.** (b) (c) and (d) **15.** (b) **16.** (b) **17.** (c) **18.** (b)
19. (a) (b) and (d) **20.** (b)

13

ISAAC NEWTON

Applications of Derivatives

- Rate of Change of Quantities
- Tangents and Normals
- Slope of a Line in Terms of Coordinates
- Slope of a Line When Its Equation Is Given
- Slopes of the Tangent and the Normal
- Equations of the Tangent and the Normal
- Angle between Two Lines
- Angle of Intersection of Two Curves
- Orthogonal Curves
- Length of Tangent, Normal, Subtangent and Subnormal
- Increasing and Decreasing Functions
- Maxima and Minima Functions
- Using Derivative (First Derivative Test)
- Rolle's Theorem
- Lagrange's Mean Value Theorem

RATE OF CHANGE OF QUANTITIES

1. Average rate of change of y w.r.t. $x = \frac{\delta y}{\delta x}$.
2. Instantaneous rate of change of y w.r.t. $x = \frac{dy}{dx}$.
3. Related rates of change $= \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ if $\frac{dx}{dt} \neq 0$.
4. Marginal cost is the rate of change of total cost with respect to the output.
5. Marginal revenue is the rate of change of total revenue with respect to the number of units sold.

Illustration 1. A balloon, which always remains spherical on inflation, is being inflated by pumping in 900 cm^3 gas per second. Find the rate at which the radius of the balloon increases when the radius is 15 cm.

Solution: Let V be the volume and r the radius of the balloon.

$\therefore$ $$V = \frac{4}{3}\pi r^3$$

or $$\frac{dV}{dr} = 4\pi r^2 = 4\pi(15)^2 \qquad (\because\ r = 15 \text{ cm})$$

$$= 900\ \pi \qquad \text{(i)}$$

We are given that $$\frac{dV}{dt} = 900\ \text{cm}^3/\text{s} \qquad \text{(ii)}$$

Using (i) and (ii), we get

$$\frac{dr}{dt}=\frac{dr}{dV}\cdot\frac{dV}{dt}=\frac{1}{900\pi}\times 900=\frac{1}{\pi}=\frac{7}{22}$$

$\therefore$ Radius is changing at $\frac{7}{22}$ cm/s.

☞ TANGENTS AND NORMALS

Slope of a Line

Slope = tan θ, where θ is $\angle XCA$, angle made by the line with the positive direction of the x-axis in anticlockwise direction.

Generally, the slope is denoted by m. Thus, m = tan θ.

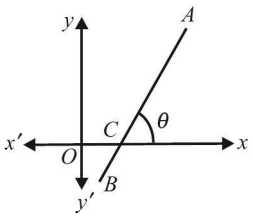


Fig. 13.1

Slope of a Line in Terms of Coordinates of Any Two Points on It

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points on a line. Then,

$$m=\tan\theta=\frac{QR}{PR}=\frac{y_2-y_1}{x_2-x_1}$$

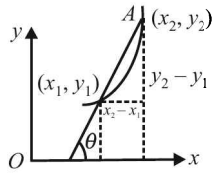


Fig. 13.2

Slope of a Line When Its Equation Is Given

If $ax + by + c = 0$ is the equation of a line, then the slope is given by

$$m=-\frac{a}{b}=-\frac{\text{Coefficient of } x}{\text{Coefficient of } y}$$

Illustration 2. Find the slope of the tangent to the curve $y = x^3 - x$ at $x = 2$.

Solution: Here $y = x^3 - x$

$\therefore$ $$\frac{dy}{dx}=3x^2-1$$

or $$\left(\frac{dy}{dx}\right)_{x=2}=3(2)^2-1=11$$

Thus, the slope of the tangent = 11.

Slopes of the Tangent and the Normal

The slope of the tangent is given by $\frac{dy}{dx}$, at P.

The slope of the normal is given by $-\frac{dx}{dy}$, at P.

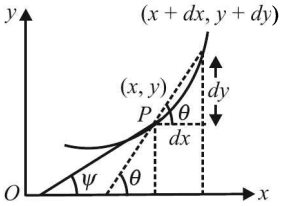


Fig. 13.3

☞ EQUATIONS OF THE TANGENT AND THE NORMAL

1. Equation of a Tangent

The equation of the tangent at (x_1, y_1) is

$$y-y_1=m(x-x_1) \quad\Rightarrow\quad y-y_1=\frac{dy}{dx}(x-x_1)$$

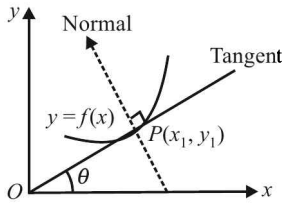


Fig. 13.4

Illustration 3. Find the equation of the tangent to the curve $\sqrt{x}+\sqrt{y}=a$ at the point $\left(\frac{a^2}{4},\frac{a^2}{4}\right)$.

Solution: The given equation is $\sqrt{x}+\sqrt{y}=5$.

Differentiating w.r.t. x, we get

$$\frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{y}}\cdot\frac{dy}{dx}=0 \quad\text{or}\quad \frac{dy}{dx}=-\frac{\sqrt{y}}{\sqrt{x}}$$

$\therefore$ $$\frac{dy}{dx}\text{ at }\left(\frac{a^2}{4},\frac{a^2}{4}\right)=-\frac{\sqrt{\frac{a^2}{4}}}{\sqrt{\frac{a^2}{4}}}=-1$$

The equation of the tangent at $\left(\frac{a^2}{4},\frac{a^2}{4}\right)$ is

$$\left(y-\frac{a^2}{4}\right)=-1\left(x-\frac{a^2}{4}\right)$$

or $$4y-a^2=-4x+a^2$$

or $$4x+4y=2a^2 \quad\text{or}\quad 2x+2y=a^2$$

2. Equation of a Normal

The equation of the tangent at (x_1, y_1) is given by

$$y-y_1=-\frac{dx}{dy}(x-x_1)$$

Illustration 4. Find the equation of the normal on the curve $2y = 3 - x^2$ at the point (1, 1).

Solution: Given: Equation of the curve $2y = 3 - x^2$.

Differentiating the given equation, we get

$$2\frac{dy}{dx}=0-2x \quad\text{or}\quad \frac{dy}{dx}=-x$$

We know that at (1, 1), $\left(\frac{dy}{dx}\right)_{(1,1)}=-1$

And the slope of the normal $= -\dfrac{1}{\left(\dfrac{dy}{dx}\right)_{(1,1)}} = 1$

We also know that the equation of the normal at (x_1, y_1) is

$$y - y_1 = -\frac{1}{\left(\dfrac{dy}{dx}\right)_{(x_1, y_1)}} (x - x_1)$$

or $(y-1) = 1(x-1)$ or $y - 1 = x - 1$ or $x - y = 0$.

Note

(i) If $\dfrac{dy}{dx} = 0$, then the tangent at (x_1, y_1) is parallel to the x-axis and its equation is $y = y_1$.

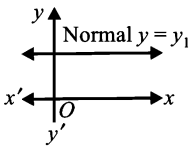


Fig. 13.5

(ii) If $\dfrac{dy}{dx} = \infty$, then the tangent at (x_1, y_1) is perpendicular to the x-axis and its equation is $x = x_1$.

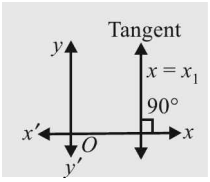


Fig. 13.6

Angle between Two Lines

The angle between two lines having slopes m_1 and m_2 is given by $\tan\theta = \pm\left(\dfrac{m_1 - m_2}{1 + m_1 m_2}\right)$.

Note

- If the lines are parallel, then $m_1 = m_2$.
- If the lines are perpendicular, then $m_1 m_2 = -1$.
- If $\dfrac{dy}{dx} > 0$, the tangent makes an acute angle with the x-axis.
- If $\dfrac{dy}{dx} < 0$, the tangent makes an obtuse angle with the x-axis.
- If the tangent is equally inclined to the axes, then $\dfrac{dy}{dx} = \tan 45°$ or $\tan 135 = \pm 1$.

ANGLE OF INTERSECTION OF TWO CURVES

If $y_1 = m_1 x + c_1$ and $y_2 = m_2 x + c_2$ are two curves, the angle between these curves is $\tan\theta = \dfrac{m_1 - m_2}{1 + m_1 m_2}$.

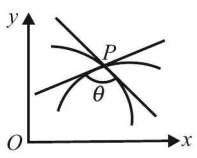


Fig. 13.7

ORTHOGONAL CURVES

If the angle of intersection of two curves is a right angle, then the two curves are said to be *orthogonal curves*.

If the curves are orthogonal, then $\theta = \pi/2$

$$\Rightarrow \quad 1 + \left(\frac{dy}{dx}\right)_{c_1}\left(\frac{dy}{dx}\right)_{c_2} = 0$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{c_1}\left(\frac{dy}{dx}\right)_{c_2} = -1$$

$$\Rightarrow \quad m_1 m_2 = -1$$

If the two curves touch each other, then $m_1 = m_2$ at the point of intersection.

Illustration 5. Prove that the curves $x = y^2$ and $xy = k$ cut at right angles if $8k^2 = 1$.

Solution: We are given that

$$x = y^2 \quad \text{(i)}$$

and $$xy = k \quad \text{(ii)}$$

Differentiating (i) w.r.t. x, we get

$$1 = 2y\frac{dy}{dx}$$

$$\therefore \quad \frac{dy}{dx} = \frac{1}{2y} \quad \text{(iii)}$$

Similarly, from (ii), we get

$$x \cdot \frac{dy}{dx} + y = 0 \quad \text{or} \quad \frac{dy}{dx} = -\frac{y}{x} \quad \text{(iv)}$$

When the curves cut each other, the tangents at the point of intersection are perpendicular to each other.

$\therefore$ From (iii) and (iv), we get

$$\frac{1}{2y} \times \left(-\frac{y}{x}\right) = -1 \quad \text{or} \quad \frac{1}{2x} = 1$$

$$\Rightarrow \quad x = \frac{1}{2}$$

Putting $x = \dfrac{1}{2}$ in (ii), we get

$$y = 2k$$

Putting $y = 2k$ and $x = \dfrac{1}{2}$ in (i), we get

$$\frac{1}{2} = (2k)^2$$

$$\therefore \quad 8k^2 = 1$$

LENGTH OF TANGENT, NORMAL, SUBTANGENT AND SUBNORMAL

The tangent and normal at the point $P(x, y)$ on the curve meet the x-axis at the points T and N, respectively. Let M be the foot of the ordinate at P. Then,

1. Length of the tangent $= PT = \left|y \operatorname{cosec}\theta\right|$

$$= \left|y\sqrt{1+\cot^2\theta}\right| = \left|\frac{x\sqrt{1+\left(\frac{dy}{dx}\right)^2}}{\frac{dy}{dx}}\right|$$

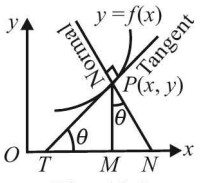


Fig. 13.8

2. Length of the normal $= PN = \left|y\sec\theta\right|$

$$= \left|y\sqrt{1+\tan^2\theta}\right| = \left|y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\right|$$

3. Length of the subtangent $= TM = \left|y\cot\theta\right| = \left|\frac{y}{\left(\frac{dy}{dx}\right)}\right|$

4. Length of the subnormal $= MN = \left|y\tan\theta\right| = \left|y\left(\frac{dy}{dx}\right)\right|$

☞ INCREASING AND DECREASING FUNCTIONS (MONOTONICITY)

1. Increasing Function

A function $f(x)$ is said to be an increasing function on an interval I, if $x_1 < x_2$ $\Rightarrow f(x_1) \le f(x_2), \forall x_1, x_2 \in I$

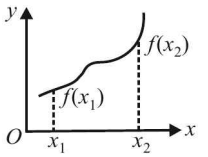


Fig. 13.9

2. Strictly Increasing Function

A function $f(x)$ is said to be a strictly increasing function on an interval I, if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2), \forall x_1, x_2 \in I$

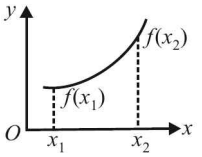


Fig. 13.10

3. Decreasing Function

A function $f(x)$ is said to be a decreasing function on an interval I, if $x_1 < x_2 \Rightarrow f(x_1) \ge f(x_2), \forall x_1, x_2 \in I$

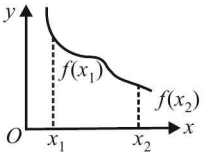


Fig. 13.11

4. Strictly Decreasing Function

A function $f(x)$ is said to be a strictly decreasing function on an interval I, if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2), \forall x_1, x_2 \in I$

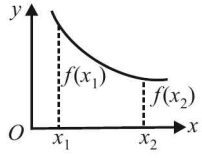


Fig. 13.12

Illustration 6. Find the intervals in which the function $f(x) = x^3 - 12x^2 + 36x + 17$ is

(a) increasing (b) decreasing.

Solution: We have $f(x) = x^3 - 12x^2 + 36x + 17$

$$f'(x) = 3x^2 - 24x + 36$$
$$= 3(x^2 - 8x + 12) = 3(x-6)(x-2)$$

(a) For $f(x)$ to be increasing, $f'(x) \ge 0$

$\therefore \quad 3(x-6)(x-2) \ge 0$

$\Rightarrow \quad x-6 \ge 0$ and $x-2 \ge 0$ or $x-6 \le 0$ and $x-2 \le 0$

$\therefore \quad x \ge 6$ and $x \ge 2$ or $x \le 6$ and $x \le 2$

$\Rightarrow \quad x \ge 6$ or $x \le 2$

$\therefore \quad f(x)$ is increasing in $(-\infty, 2) \cup (6, \infty)$.

(b) For $f(x)$ to be decreasing, $f'(x) \le 0$

$\therefore \quad 3(x-6)(x-2) \le 0$

$\Rightarrow \quad x-6 \ge 0$ and $x-2 \le 0$ or $x-6 \le 0$ and $x-2 \ge 0$

$\Rightarrow \quad x \ge 6$ and $x \le 2$ or $x \le 6$ and $x \ge 2$

$\Rightarrow \quad x \ge 6$ or $x \ge 2$

$\therefore \quad f(x)$ is decreasing in (2, 6).

5. Monotonic Function

A function $f(x)$ is said to be monotonic on an interval I if it is either increasing or decreasing on I.

6. Test of Monotonic Function

(i) $f(x)$ is said to be increasing in $[a, b]$ if $f'(x) \ge 0$ in $[a, b]$.
(ii) $f(x)$ is said to be strictly increasing in $[a, b]$ if $f'(x) > 0$ in (a, b).
(iii) $f(x)$ is said to be decreasing in $[a, b]$ if $f'(x) \le 0$ in $[a, b]$.
(iv) $f(x)$ is said to be strictly decreasing in $[a, b]$ if $f'(x) < 0$ in (a, b).

Note

- If a function $f(x)$ is said to be increasing in $[a, b]$, then f^{-1} exists and is also increasing in $[a, b]$.
- If $f(x)$ and $g(x)$ are both increasing functions, then $(g \circ f)(x)$ is an increasing function.
- If the function $f(x)$ is increasing and $g(x)$ is decreasing, then the composite $(g \circ f)(x)$ is also decreasing.
- A function may be increasing in some interval I_1 and decreasing in some other interval I_2.

☞ MAXIMA AND MINIMA OF FUNCTIONS

1. Local Maximum

A function $y = f(x)$ is said to have a local maximum value at a point $x = a$, if $f(x) \le f(a)$, $\forall x \in (a - h, a + h)$, for small $h > 0$, i.e. $f(a)$ is the greatest of all the values of $f(x)$ in the interval $(a - h, a + h)$. The point $x = a$ is called a point of local maximum of the function $f(x)$.

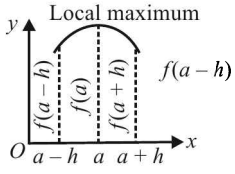


Fig. 13.13

2. Local Minimum

A function $y = f(x)$ is said to have a local minimum value at a point $x = a$, if $f(x) \geq f(a)$, $\forall x \in (a - h, a + h)$, for small $h > 0$, i.e. $f(a)$ is the smallest of all the values of $f(x)$ in the interval $(a - h, a + h)$.

The point $x = a$ is called a point of local minimum of the function $f(x)$.

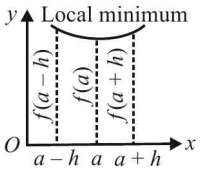


Fig. 13.14

☞ USING DERIVATIVE (FIRST DERIVATIVE TEST) TO FIND LOCAL MAXIMA AND MINIMA OF $f(x)$

Condition I: $f'(x) = 0$

Condition II:

$x \to a$	$f'(x)$ slightly $< a$	$f'(x)$ slightly $> a$	Nature of $f(x)$ at the critical point a
$f'(x)$	+ve	–ve	Maxima
$f'(x)$	–ve	+ve	Minima
$f'(x)$	+ve	+ve	Neither maxima nor minima
$f'(x)$	–ve	–ve	Neither minima nor maxima

☞ TO FIND LOCAL MAXIMA AND LOCAL MINIMA USING FIRST- AND SECOND-ORDER DERIVATIVES

Condition I: $\frac{dy}{dx} = 0$ (for maxima and minima)

Condition II:

(i) For maxima, $\frac{d^2y}{dx^2} < 0$

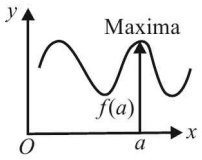


Fig. 13.15(i)

(ii) For minima, $\frac{d^2y}{dx^2} > 0$

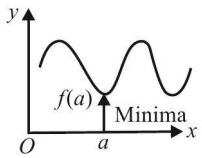


Fig. 13.15(ii)

(iii) For point of inflexion, $\frac{d^2y}{dx^2} = 0$

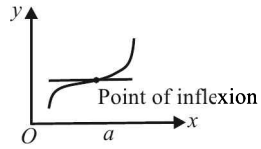


Fig. 13.15(iii)

Illustration 7. Find the minimum value of $f(x) = x^4$.

Solution: $\because f(x) = x^4$

$\therefore \quad f'(x) = 4x^3$

Now $f'(x) = 0 \Rightarrow x = 0$

and $f''(x) = 12x^2$

At $x = 0$, $f''(x) = 0$

$\therefore$ The test fails.

Now we use the 1st derivative. By that test, $f'(x) < 0$ when $x < 0$ and $f'(x) > 0$ for $x > 0$.

$\therefore$ $f'(x)$ changes sign from –ve to +ve.

$\therefore$ $f(x)$ has a minimum value at $x = 0$ and the minimum value = 0.

Illustration 8. Find the local maximum and local minimum values of the function $f(x) = x^3 - 9x^2 + 15x + 11$.

Solution: $f(x) = x^3 - 9x^2 + 15x + 11$

$\therefore \quad f'(x) = 3x^2 - 18x + 15$

At stationary points, $f'(x) = 0$

$\therefore \quad 3x^2 - 18x + 15 = 0$

or $x^2 - 6x + 5 = 0$ or $(x - 5)(x - 1) = 0$

or $x = 5, x = 1$

Now $f''(x) = 6x - 18$

For $x = 5, f''(x) = 30 - 18 = 12 > 0$

$\therefore$ $f(x)$ has a minimum value at $x = 5$.

Minimum value = $f(5) = 5^3 - 9(5)^2 + 15 \times 5 + 11$

$= 125 - 225 + 75 + 11 = -14$

For $x = 1$, $f''(x) = 6 - 18 = -12 < 0$

$\therefore$ $f(x)$ has a maximum value at $x = 1$.

Maximum value = $f(1) = 1^3 - 9(1)^2 + 15 \times 1 + 11$

$= 1 - 9 + 15 + 11 = 18$.

Hence, the minimum value is -14 at $x = 5$ and the maximum value is 18 at $x = 1$.

☞ ROLLE'S THEOREM

If a function $y = f(x)$ is defined in $[a, b]$ and

(i) $f(x)$ is continuous in $[a, b]$,

(ii) $f(x)$ is differentiable in (a, b) and
(iii) $f(a) = f(b)$,
then there will be at least one value of $c \in (a, b)$ such that $f'(c) = 0$.

Illustration 9. Verify Rolle's theorem for the function
$$f(x) = x^2 + 2x - 8, x \in [-4, 2].$$

Solution: (i) $f(x) = x^2 + 2x - 8$
$f(x)$ being a polynomial function is continuous in $[-4, 2]$.

(ii) $f'(x) = 2x + 2$ which exists in $]-4, 2[$
Thus, $f(x)$ is differentiable in $]-4, 2[$.

(iii) $f(2) = 2^2 + 2 \cdot 2 - 8 = 4 + 4 - 8 = 0$
and $f(-4) = (-4)^2 + 2(-4) - 8 = 16 - 8 - 8 = 0$
$\therefore \quad f(-4) = f(2)$
Thus, all the conditions of Rolle's theorem are satisfied. So there must exist at least one point c in $]-4, 2[$ such that $f'(c) = 0$.
$\therefore \quad f'(c) = 2c + 2 = 0$
$\Rightarrow \quad c = -1 \in -]4, 2[$.
Hence, Rolle's theorem is verified.

LAGRANGE'S MEAN VALUE THEOREM

If a function $f(x)$ is said to be defined on $[a, b]$ and
(i) continuous in $[a, b]$ and
(ii) differentiable in (a, b),
then there will be at least one value of $c \in (a, b)$ such that
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Illustration 10. Verify Lagrange's mean value theorem for the function $f(x) = x^2 + 2x + 3, [4, 6]$.

Solution: $\because f(x) = x^2 + 2x + 3$ is an algebraic polynomial in x,
$\therefore$ $f(x)$ is continuous in [4, 6] and derivative in]4, 6[.
$\therefore$ All the conditions of the mean value theorem are satisfied by $f(x)$ in [4, 6].
$\therefore$ By Lagrange's mean value theorem, there exists some $c \in]4, 6[$ such that
$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(6) - f(4)}{6 - 4} \qquad \text{(i)}$$
Now $f(4) = 4^2 + 2.4 + 3 = 16 + 8 + 3 = 27$ (ii)
and $f(6) = 6^2 + 2.6 + 3 = 36 + 12 + 3 = 51$ (iii)
Also, $f'(x) = 2x + 2$
$\therefore \quad f'(c) = 2c + 2$ (iv)
From (i) to (iv), we get
$$2c + 2 = \frac{51 - 27}{2} = 12 \quad \Rightarrow \quad 2c = 12 - 2 = 10$$
$\therefore \quad c = 5$
Clearly, $5 \in]4, 6[$
Hence, Lagrange's mean value theorem is verified.

SOLVED PROBLEMS

1. The rate of change of the surface area of a sphere of radius r, when the radius is increasing at the rate of 2 cm/s, is proportional to
(a) $1/r$ (b) $1/r^2$
(c) r (d) r^2

Ans. (c)

Solution: Surface area of the sphere $s = 4\pi r^2$ and $\frac{dr}{dt} = 2$
$$\therefore \quad \frac{ds}{dt} = 4\pi \times 2r \frac{dr}{dt} = 8\pi r \times 2 = 16\pi r$$
$$\Rightarrow \quad \frac{ds}{dt} \propto r$$

2. The length of the normal to the curves, $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, $a + \theta = \frac{\pi}{2}$ is
(a) $a/\sqrt{2}$ (b) $2a$
(c) $2\sqrt{2}a$ (d) $\sqrt{2}a$

Ans. (d)

Solution: Here $\frac{dx}{d\theta} = a(1 + \cos\theta)$ and $\frac{dy}{d\theta} = a(\sin\theta)$
$$\Rightarrow \quad \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a\sin\theta}{a(1 + \cos\theta)}$$
$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{x=\pi/2} = \frac{\sin\pi/2}{1 + \cos\pi/2} = 1$$
and length of the normal is $\left[y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\right]_{\theta=\pi/2}$
$$= a\left(1 - \cos\frac{\pi}{2}\right)\sqrt{1 + 1^2} = \sqrt{2a}$$

3. The number of critical points of the function $f(x) = |x|e^{-x}$ is
(a) 1 (b) 2
(c) 3 (d) none of these

Ans. (b)

Solution: $f(x) = xe^{-x}, x \geq 0$
$= -xe^{-x}, x < 0$

$$f'(x) = e^{-x}(1-x); x > 0$$
$$= -e^{-x}(1-x); x < 0$$

Clearly $f'(x)$ is not defined at $x = 0$ and $f'(x) = 0$ at $x = 1$ so $x = 0, 1$ are critical points.

4. If the distance s meters traveresed by a particle in t seconds is given by $s = t^3 - 3t^2$, then the velocity of the particle when the acceleration is zero in m/s is

(a) 3 (b) -2
(c) -3 (d) 2

Ans. (c)

Solution: We have $s = t^3 - 3t^2$

On differentiating with respect to t, we get

$$\frac{ds}{dt} = 3t^2 - 6t \qquad \text{(i)}$$

Again differentiating Eq. (i), we get

$$\frac{d^2s}{dt^2} = 6t - 6 = 0 \Rightarrow t = 1$$

On putting the value of $t = 1$ in Eq. (i), we get

$$\frac{ds}{dt} = 3\times 1 - 6\times 1 = 3 - 6 = -3 \text{ m/s}$$

5. The slope of the tangent to the locus $y = \cos^{-1}(\cos x)$ at $x = -\frac{\pi}{4}$ is

(a) 1 (b) 0
(c) 2 (d) -1

Ans. (d)

Solution: If $-\frac{\pi}{2} < x < 0$

then $y = \cos^{-1}(\cos x) = \cos^{-1}[\cos(-x)] = -x$

$$\therefore \quad \frac{dy}{dx} = -1 \Rightarrow \left(\frac{dy}{dx}\right)_{x=-\pi/4} = -1$$

6. If 20 is divided into two parts such that the product of one part and the cube of the other is maximum, then the ratio of the two parts is

(a) 1:1 (b) 2:3
(c) 1:3 (d) 1:4

Ans. (c)

Solution: Let one part be x, then the other will be $(20 - x)$.

Let $y = x(20-x)^3$

$$\frac{dy}{dx} = (20-x)^3 + x.3(20-x)^2(-1)$$
$$= (20-x)^2(20-4x)$$

$$\therefore \quad \frac{dy}{dx} = 0 \Rightarrow x = 5, \text{ so the other part is 15.}$$

7. If a particle is moving such that the velocity acquired is proportional to the square root of the distance covered, then its acceleration is

(a) a constant (b) $\propto s^2$
(c) $\propto \frac{1}{s^2}$ (d) $\propto s$

Ans. (a)

Solution: If the velocity acquired by the particle is proportional to the square root of the distance covered, then its acceleration is a constant.

8. The point at which the tangent to the curve $y = 2x^2 - x + 1$ is parallel to $y = 3x + 9$ will be

(a) (2, 1) (b) 1, 2)
(c) (3, 9) (d) (–2, 1)

Ans. (b)

Solution: Given: Equation of the curve $y = 2x^2 - x + 1$ and equation of the line $y = 3x + 9$. Differentiating the given equation, we get

$$\frac{dy}{dx} = 4x - 1.$$

We know that the slope of the tangent to the curve $\left(\frac{dy}{dx}\right) = 4x - 1$. Since this tangent is parallel to $y = 3x + 9$, therefore slope of the tangent = slope of the line or $4x - 1 = 3$ or $x = 1$

Substituting $x = 1$ in the given equation, we get $y = 2x^2 - x + 1 = 2\times(1)^2 - 1 + 1 = 2$. Thus, the point is $(x, y) = (1, 2)$.

9. If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on the x-axis and y-axis, respectively, then the value of a^2b is

(a) $27c^3$ (b) $\frac{4}{27}c^3$
(c) $\frac{27}{4}c^3$ (d) $\frac{4}{9}c^3$

Ans. (c)

Solution: $x^2y = c^3$

Differentiating with respect to x, we have

$$x^2\frac{dy}{dx} + 2xy = 0 \Rightarrow \frac{dy}{dx} = -\frac{2y}{x}$$

The equation of the tangent at (h, k), $y - k = -\frac{2k}{h}(x - h)$

$y = 0$ gives $x = \frac{3h}{2} = a$ and $x = 0$ gives $y = 3k = b$

Now, $a^2b = \frac{9h^2}{4}\,3k = \frac{27}{4}h^2k = \frac{27}{4}c^3$

10. Equation of the tangents to the curve $f(x) = x^2$ at the point where the slope of the tangent is equal to the functional value of the curve is

(a) $4x - y - 6 = 0, x = 0$ (b) $4x - y + 6 = 0, x = 0$
(c) $4x - y - 4 = 0, y = 0$ (d) $4x - y + 6 = 0, y = 0$

Ans. (c)

Solution: According to the question, $f'(x) = f(x)$

$\Rightarrow 2x = x^2 \Rightarrow x = 0, 2$

at $x = 0$, $y = 0$ and at $x = 2$, $y = 4$

so we have to find the equation of the tangent at (0, 0) (2, 4) at (0, 0) $f'(x) = 2x = 0$.

Thus, the equation of the tangent is $y = 0$

at (2, 4); $f'(x) = 2 \times 2 = 4$, the equation of the tangent is

$y - 4 = 4(x - 2) \Rightarrow 4x - y - 4 = 0$

11. A wire of length a is cut into two parts which are bent, respectively, in the form of a square and a circle. The least value of the sum of the areas so formed is

(a) $\dfrac{a^2}{\pi + 4}$ (b) $\dfrac{a}{\pi + 4}$

(c) $\dfrac{a}{4(\pi + 4)}$ (d) $\dfrac{a^2}{4(\pi + 4)}$

Ans. (d)

Solution: Given $4x + 2\pi r = a$

$$A = x^2 + \pi r^2 = \frac{1}{16}(a - 2\pi r)^2 + \pi r^2$$

$\dfrac{dA}{dr} = 0$ gives $r = \dfrac{a}{2(\pi + 4)}$ for which

$\dfrac{d^2A}{dr}$ is +ve and hence minimum.

Now $4x = a - 2\pi r = a - \dfrac{a\pi}{\pi + 4}$

$$\frac{4a}{\pi + 4} \Rightarrow x = \frac{a}{\pi + 4}$$

$$\Rightarrow A = x^2 + r^2\pi = \frac{a^2}{4(\pi + 4)}$$

12. If the line $x\cos\alpha + y\sin\alpha = p\ (p > 0)$ is a normal to the curve $(x + a)y = p$, then α lies in

(a) $[n\pi, (n+1)\pi]$ (b) $[n\pi, (n+1/2)\pi]$

(c) $[(n+1/2)\pi, (n+1)\pi]$ (d) $[2n\pi, (2n+1)\pi]$

Ans. (c)

Solution: Given curve $y = \dfrac{p}{x + a}(p > 0)$

Differentiating $\dfrac{dy}{dx} = \dfrac{-p}{(x + a)^2}$

The slope of the normal $\dfrac{-1}{\left(\dfrac{dy}{dx}\right)} = \dfrac{(x + a)^2}{p}$

Also the slope of the normal $x\cos\alpha + y\sin\alpha = p$ is $-\cot\alpha$

$$\therefore \quad -\cot\alpha = \frac{(x + a)^2}{p} > 0 \quad \text{or} \quad \cot\alpha < 0$$

or $n\pi + \dfrac{\pi}{2} < \alpha < n\pi + \pi$

13. If $f(x) = 2x + \cot^{-1}x + \log(\sqrt{1 + x^2} - x)$, then $f(x)$

(a) decreases in $[0, \infty)$

(b) increases in $[0, \infty)$

(c) neither increases nor decreases in $[0, \infty)$

(d) increases in $(-\infty, \infty)$

Ans. (d)

Solution: We have $f(x) = 2x + \cot^{-1}x + \log(\sqrt{1 + x^2} - x)$

$$\therefore \quad f'(x) = 2 - \frac{1}{1 + x^2} + \frac{1}{\sqrt{1 + x^2} - x}\left(\frac{x}{\sqrt{1 + x^2}} - 1\right)$$

$$= \frac{1 + 2x^2}{1 + x^2} - \frac{1}{\sqrt{1 + x^2}} = \frac{1 + 2x^2}{1 + x^2} - \frac{\sqrt{(1 + x^2)}}{1 + x^2}$$

$$= \frac{x^2 + \sqrt{1 + x^2}\{\sqrt{1 + x^2} - 1\}}{1 + x^2} > 0 \text{ for all } x$$

Hence, $f(x)$ is an increasing function on $(-\infty, \infty)$.

14. Which of the following does not hold Rolle's theorem in $[-1, 1]$?

(a) $f(x) = |x|$ (b) $f(x) = x^2 - 1$

(c) $f(x) = x^2 + x - 1$ (d) none of these

Ans. (a)

Solution: $|x|$ is not differentiable at $x = 0$.

15. Which of the following does not follow the mean value theorem in $[0,1]$?

(a) $f(x) = \left(\dfrac{1}{2} - x\right)^2, x > 1/2$

$$= \left(\frac{1}{2} - x\right), x \le 1/2$$

(b) $f(x) = \dfrac{\sin x}{x}, x \neq 0$

$= 1, x = 0$

(c) $f(x) = x|x|$

(d) $f(x) = |x|$

Ans. (a)

Solution: $f(x)$ is not differentiable at $x = 1/2$.

16. If the angle between the curves $y^2 = 16x$ and $9x^2 + by^2 = 16$ is $\pi/2$, then the value of b is

(a) 2 (b) 4

(c) 9/2 (d) 13/4

Ans. (c)

Solution: Given: Equation of the curves $y^2 = 16x$ (i)

and $9x^2 + by^2 = 16$ (ii)

Angle between the curves (i) and (ii) = $\pi/2$.

Differentiating Eq. (i), we get

$$2y\frac{dy}{dx} = 16 \quad \text{or} \quad \frac{dy}{dx} = \frac{8}{y} = m_1$$

and differentiating Eq. (ii), we get

$$18x + 2by\frac{dy}{dx} = 0 \quad \text{or} \quad by\frac{dy}{dx} = -9x$$

or $$\frac{dy}{dx} = \frac{-9x}{by} = m_2$$

We know that $m_1 \times m_2 = -1$

or $$\left(\frac{8}{y}\right) \times \left(\frac{-9x}{by}\right) = -1 \quad \text{or} \quad by^2 = 72x$$

or $$16bx = 72x$$

or $$b = \frac{9}{2}.$$

17. The slope of the normal to the curve $y = \cos 2x$ at $x = \frac{\pi}{6}$ is

(a) $\sqrt{3}$ (b) $\frac{2}{\sqrt{3}}$

(c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{3}}$

Ans. (d)

Solution: Given: Equation of the curve $y = \cos 2x$.

Differentiating the given equation, we get

$\frac{dy}{dx} = -2\sin 2x$. We know that at

$$x = \frac{\pi}{6}, \left(\frac{dy}{dx}\right)_{\pi/6} = -2\sin\frac{\pi}{3} = -2 \times \frac{\sqrt{3}}{2} = -\sqrt{3}.$$

Since $-\sqrt{3}$ is the slope of the tangent, therefore the slope of the normal $= \frac{1}{\sqrt{3}}$.

EXERCISE SET 1

1. If a particle is moving such that the velocity acquired is proportional to the square root of the distance covered, then its acceleration is

(a) a constant (b) $\propto s^2$

(c) $\propto \frac{1}{s^2}$ (d) $\propto \frac{1}{s}$

2. The curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ touches the straight line $\frac{x}{a} + \frac{y}{b} = 2$ at the point (a, b)

(a) for $x = 3$ (b) for $x = 2$

(c) for any value of n (d) for no value of n

3. For the curve $x^{m+n} = a^{m-n}y^{2n}$, $(\text{subtangent})^m$ is proportional to

(a) $(\text{subnormal})^m$ (b) $(\text{subnormal})^n$

(c) y^{2n} (d) none of these

4. If $2a + 3b + 6c = 0$, then $f(x) = ax^2 + bx + c = 0$ has

(a) at least one solution in [1, 2]

(b) at least one solution in [0, 1]

(c) at least one solution in [–1, 0]

(d) two solutions in [0, 1]

5. The length of the subtangent to the curve $\sqrt{x} + \sqrt{y} = 3$ at the point (4, 1) is

(a) –2 (b) 1/2

(c) –3 (d) 4

6. Tangents are drawn to the curve $x^2y = 1 - y$ at the points where it is met by the curve $xy = 1 - y$. The point of intersection of these tangents is

(a) (0, –1) (b) (0, 1)

(c) (1, 1) (d) none of these

7. In a ΔABC, the sides b and c are given. If there is an error ΔA in measuring angle A, then the error Δa in side a is given by

(a) $\frac{S}{2a}\Delta A$ (b) $\frac{2S}{a}\Delta A$

(c) $bc\sin A\Delta A$ (d) none of these

8. If m is the slope of a tangent to the curve $e^{2y} = 1 + 4x^2$, then

(a) $m < 1$ (b) $|m| \le 1$

(c) $|m| > 1$ (d) none of these

9. If the function $f(x) = ax^3 + bx^2 + 11x - 6$ satisfies the conditions of Rolle's theorem in [1, 3] and $f'\left(2 + \frac{1}{\sqrt{3}}\right) = 0$, then the values of a and b are respectively

(a) 1, –6 (b) –2, 1

(c) –1, 1/2 (d) –1, 6

10. If $a+b+c=0$, then the equation $3ax^2+2bx+c=0$ has
(a) at least one real root in [0, 1]
(b) one root in [–1, 0] and the other in [1, 2]
(c) two imaginary roots
(d) two coincident roots

11. If the two curves $y=a^x$ and $y=b^x$ intersect at an angle α, then $\tan\alpha$ equals
(a) $\dfrac{\log a-\log b}{1+\log a\log b}$ (b) $\dfrac{\log a+\log b}{1-\log a\log b}$
(c) $\dfrac{\log a-\log b}{1-\log a\log b}$ (d) none of these

12. The number of points on the curve $x^{3/2}+y^{3/2}=a^{3/2}$ where the tangents are equally inclined to the axes is
(a) 1 (b) 2
(c) 4 (d) none of these

13. Tangents to the curve $y=x^3+3x$ at $x=-1$ and $x=1$ are
(a) parallel
(b) intersecting obliquely but not at an angle of 45°
(c) intersecting at right angles
(d) intersecting at an angle of 45°

14. The function f defined by $f(x)=4x^4-2x+1$ is increasing for
(a) $x<1$ (b) $x>0$
(c) $x<\dfrac{1}{2}$ (d) $x>\dfrac{1}{2}$

15. The length of the subnormal of the curve $y^2=x^3$ at the point (4, 8) is
(a) 3/8 (b) 8/3
(c) 24 (d) 34

16. A particle moves in a straight line so that $s=\sqrt{t}$, then its acceleration is proportional to
(a) (velocity)^3 (b) velocity
(c) (velocity)^2 (d) (velocity)$^{3/2}$

17. A stone is thrown vertically upwards from the top of a tower 64 m high according to the law $s=48t-16t^2$. The greatest height above the ground is
(a) 64 m (b) 36 m
(c) 100 m (d) none of these

18. If the tangent to the curve $x+y=e^{xy}$ is parallel to the y-axis, then the point of contact is
(a) (1, 0) (b) (0, 1)
(c) (1, 1) (d) none of these

19. The slope of the tangent to the curves $x=a\sin t, y=a\left(\cos t+\log\tan\dfrac{t}{2}\right)$ at the point t is
(a) $\tan t/2$ (b) $\cot t$
(c) $\tan t$ (d) none of these

20. The area of the triangle formed by the tangent to the curve $y=\dfrac{8}{(4+x^2)}$ at $x=2$ and the coordinate axes is
(a) 2 sq. units (b) 4 sq. units
(c) 8 sq. units (d) 7/2 sq. units

21. The equation of the tangent at (–4, 4) on the curve $x^2=-4y$ is
(a) $2x+y+4=0$ (b) $2x-y-12=0$
(c) $2x+y-4=0$ (d) $2x-y+4=0$

22. The slope of the tangent to the curves $x=t^2+3t-8$, $y=2t^2-2t-5$ at the point (2, –1) is
(a) 22/7 (b) 6/7
(c) 6 (d) none of these

23. Rolle's theorem is not applicable to the function $f(x)=|x|$ in the interval [–3, 3] because
(a) f is not derivable in (–3, 3)
(b) $f(x)\geq 0\ \forall x$ in (–3, 3)
(c) $f(3)\neq f(-3)$
(d) f is continuous in (–3, 3)

24. The point on the curve $y=x^2$, where the slope of the tangent is equal to the x-coordinate of the point is
(a) (2, 0) (b) (0, 2)
(c) (0, 0) (d) $\left(-\dfrac{1}{2},\dfrac{1}{2}\right)$

25. If the function $f:\mathbb{R}\to\mathbb{R}$ is defined by $f(x)=\tan x-x$, then $f(x)$
(a) increases (b) decreases
(c) remains constant (d) becomes zero

26. The distance travelled s (in metres) by a particle in t second is given by $s=t^3+2t^2+t$. The speed of the particle after 18 will be
(a) 8 cm/s (b) 6 cm/s
(c) 2 cm/s (d) none of these

27. The curve $y-e^{xy}+x=0$ has a vertical tangent at the point
(a) (1, 1) (b) at no point
(c) (0, 1) (d) (1, 0)

28. For the curve $xy=c^2$, the subnormal at any point varies as
(a) x^2 (b) x^3 (c) y^2 (d) y^3

29. Rolle's theorem is applicable in case of $\phi(x)=a^{\sin x}$ in
(a) any interval (b) the interval $[0, \pi]$
(c) the interval $(0, \pi/2)$ (d) none of these

30. If Rolle's theorem holds for the function $f(x)=x^3+bx^2+cx, 1\leq x\leq 2$ at the point 4/3, the values of b and c are
(a) $b=8, c=-5$ (b) $b=-5, c=8$
(c) $b=5, c=-8$ (d) $b=-5, c=-8$

31. The normals to the curves $x = a(\theta + \sin\theta), y = a(1 - \cos\theta)$ at the point $\theta = (2n+1)\pi, n \in I$ are all
(a) parallel to the x-axis
(b) parallel to the y-axis
(c) parallel to the line $y = x$
(d) none of these

32. If the tangent to the curves $x = at^2, y = 2at$ is perpendicular to the x-axis, then its point of contact is
(a) (a, a) (b) $(0, a)$ (c) $(0, 0)$ (d) $(a, 0)$

33. If $a + b + c = 0$, then the quadratic equation $3ax^2 + 2bx + c = 0$ has
(a) imaginary roots
(b) at least one real root in $[0, 1]$
(c) one root in $[2, 3]$ and the other in $[3, 6]$
(d) none of these

34. Let $f(x) = x^3 + \frac{3}{2}x^2 + 3x + 3$, then $f(x)$ is
(a) a decreasing function (b) an increasing function
(c) an odd function (d) an even function

35. An object is moving in the clockwise direction around the unit circle $x^2 + y^2 = 1$. As it passes through the point $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, its y-coordinate is decreasing at the rate of 3 unit per second. The rate at which the x-coordinate changes at this point is (in unit per second)
(a) 2 (b) $3\sqrt{3}$ (c) $\sqrt{3}$ (d) $2\sqrt{3}$

36. Let $f(x) = x^3 - 6x^2 + 9x + 8$, then $f(x)$ is strictly decreasing in
(a) $(-\infty, 1)$ (b) $(1, 3)$
(c) $(3, \infty)$ (d) $(-\infty, 1) \cup (3, \infty)$

37. The function $y = a(1 - \cos x)$ is maximum when x is equal to
(a) π (b) $\pi/2$ (c) $-\pi/2$ (d) $-\pi/6$

38. If the curve $y = ax^3 + bx^2 + cx$ is inclined at 45° to the x-axis at $(0, 0)$ but it touches the x-axis at $(1, 0)$, then the values of a, b and c are given by
(a) $a = -1, b = 2, c = 1$ (b) $a = 1, b = -2, c = 1$
(c) $a = 1, b = 1, c = -2$ (d) $a = -2, b = 1, c = 1$

39. The function $f(x) = x^2 - 2x$ is increasing in the interval
(a) $(-2, -1)$ (b) $(-1, 0)$ (c) $(0, 1)$ (d) $(1, 2)$

40. The position of a point in time t is given by $x = a + bt - ct^2$, $y = at + bt^2$. Its acceleration at time t is
(a) $b - c$ (b) $b + c$ (c) $2b - 2c$ (d) $2\sqrt{b^2 + c^2}$

41. The point on the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at which the normal is parallel to the x-axis is
(a) $(0, 0)$ (b) $(0, a)$ (c) $(a, 0)$ (d) (a, a)

42. If $f'(x) = g(x)(x - a)^2$, where $g(a) \neq 0$ and is continuous at $x = a$, then
(a) f is increasing near a if $g(a) > 0$
(b) f is increasing near a if $g(a) < 0$
(c) f is increasing near a if $g(a) = 0$
(d) none of these

43. The function $f(x) = \frac{x}{1 + |x|}$ is
(a) non-differentiable at $x = 0$
(b) strictly increasing
(c) strictly decreasing
(d) neither increasing nor decreasing

44. The coordinates of a point on the curve $y = x \log x$ at which the normal is parallel to the line $2x - 2y = 3$ are
(a) $(0, 0)$ (b) (e, e)
(c) $(e^2, 2e^2)$ (d) $(e^{-2}, -2e^{-2})$

45. Two towns A and B are 60 km apart. A school is to be built to serve 150 students in town A and 50 students in town B. If the total distance to be travelled by all the 200 students is to be as small as possible, then the school should be built at
(a) town B (b) 45 km from town A
(c) town A (d) 45 km from town B

46. The curves $ax^2 + by^2 = 1$ and $a'x^2 + b'y^2 = 1$ intersect orthogonally if
(a) $\frac{1}{a} + \frac{1}{a'} = \frac{1}{b} + \frac{1}{b'}$ (b) $\frac{1}{a} - \frac{1}{b} = \frac{1}{a'} - \frac{1}{b'}$
(c) $\frac{1}{a} + \frac{1}{b} = \frac{1}{a'} + \frac{1}{b'}$ (d) none of these

47. The critical points of the function $f(x) = (x - 2)^{2/3}(2x + 1)$ are
(a) 1 and 2 (b) 1 and $-1/2$
(c) -1 and 2 (d) 1

48. A ladder 10-m long rests against a vertical wall with the lower end on the horizontal ground. The lower end of the ladder is pulled along the ground away from the wall at the rate of 3 cm/s. The height of the upper end while it is descending at the rate of 4 cm/s is
(a) $4\sqrt{3}$ m (b) $5\sqrt{3}$ m (c) 6 m (d) 8 m

49. If the tangent to the curve $y = 2x^2 - x + 1$ at a point p is parallel to $y = 3x + 4$, the coordinates of p are
(a) $(2, 1)$ (b) $(1, 2)$ (c) $(-1, 2)$ (d) $(2, -1)$

50. Let $f(x) = (x^2 - 4)^{1/3}$, then f has a
(a) local maxima at $x = 0$
(b) local minima at $x = 0$
(c) point of inflexion at $x = 0$
(d) none of these

EXERCISE SET 2

1. If $f(x)=\dfrac{1}{x+1}-\log(1+x), x>0$, then f is
(a) an increasing function
(b) a decreasing function
(c) both an increasing and a decreasing function
(d) none of these

2. If $f(x)=x+\dfrac{1}{x'}$, then
(a) relative minimum > relative maximum
(b) relative maximum > relative minimum
(c) relative maximum does not exist
(d) relative minimum does not exist

3. If $f(x)=\dfrac{1}{4x^2+2x+1}$, then its maximum value is
(a) 4/3 (b) 2/3 (c) 1 (d) 3/4

4. The function $f(x)=\cos x-2px$ is monotonically decreasing for
(a) $p\le\dfrac{1}{2}$ (b) $p\ge\dfrac{-1}{2}$ (c) $p\le 2$ (d) $p\ge 2$

5. The function $f(x)=x+\cos x$ is
(a) always increasing (b) always decreasing
(c) increasing for certain range of x
(d) none of these

6. A circle of radius unity is inscribed in an isosceles triangle. The least perimeter of the triangle is
(a) $6\sqrt{3}$ (b) 9 (c) $2\sqrt{3}$ (d) $3\sqrt{3}$

7. Gas is being pumped into a spherical balloon at the rate of 30 ft^3/min. Then the rate at which the radius increases when it reaches the value 15 ft is
(a) $1/30\pi$ ft/min (b) $1/15\pi$ ft/min
(c) 1/20 ft/min (d) 1/25 ft/min

8. The difference between the greatest and least values of the function $f(x)=\sin 2x-x$ on $\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]$ is
(a) $\dfrac{\sqrt{3}+\sqrt{2}}{2}$ (b) $\dfrac{\sqrt{3}+\sqrt{2}}{2}+\dfrac{\pi}{6}$
(c) $\dfrac{\sqrt{3}}{2}+\dfrac{\pi}{3}$ (d) $\dfrac{\sqrt{3}+\sqrt{2}}{2}-\dfrac{\pi}{3}$

9. The function $f(x) = x + \sin x$ has
(a) a minimum but no maximum
(b) a maximum but no minimum
(c) neither maximum nor minimum
(d) both maximum and minimum

10. Let $f(x)=\int e^x(x-1)(x-2)dx$. Then, f decreases in the interval
(a) $(-\infty,-2)$ (b) $(-2,-1)$ (c) $(1, 2)$ (d) $(2, \infty)$

11. If the line $ax+by+c=0$ is a tangent to the curve $xy = 4$, then
(a) $a>0, b>0$ (b) $a>0, b<0$
(c) $a<0, b>0$ (d) none of these

12. The angle of intersection of the curves $y=x^2$ and $x=y^2$ at (1, 1) is
(a) $\tan^{-1}(4/3)$ (b) $\tan^{-1}(1)$
(c) 90° (d) $\tan^{-1}(3/4)$

13. If $f(x)=xe^{x(1-x)}$, then $f(x)$ is
(a) increasing on $[-(1/2), 1]$
(b) decreasing on $\mathbb{R}$
(c) increasing on $\mathbb{R}$
(d) decreasing on $[-(1/2), 1]$

14. An isosceles triangle of a vertical angle 2θ is inscribed in a circle of radius a. Then the area of the triangle is maximum when $\theta=$
(a) $\pi/6$ (b) $\pi/4$ (c) $\pi/3$ (d) $\pi/2$

15. The greatest value of $f(x)=\cos\left[xe^{|x|}+7x^2-3x\right], x\in[-1,\infty)$ is
(a) 0 (b) -1
(c) 1 (d) none of these

16. The volume of a ball is increasing at the rate of 4π cm^3/s. The rate of increase of the radius when the volume is 288π cm^3 is
(a) 1/6 cm/s (b) 1/36 cm/s
(c) 1/9 cm/s (d) 1/24 cm/s

17. A particle moves along the x-axis so that its position is given by $x=2t^3-3t^2$ at times t seconds. What is the time interval during which the particle will be on the negative half of the axis?
(a) $0<t<2/3$ (b) $0<t<1$
(c) $0<t<3/2$ (d) $1/2<t<1$

18. The angle between the tangents to the curve $y^2 = 2ax$ at the points, where $x = (a/2)$, is
(a) $\pi/6$ (b) $\pi/4$ (c) $\pi/3$ (d) $\pi/2$

19. The velocity v m/s of a particle is proportional to the cube of the time. If the velocity after 2 s is 4 m/s, then v is equal to
(a) t^3 (b) $t^3/2$ (c) $t^3/3$ (d) $t^3/4$

20. The function $f(x)=2x^3-3x^2-12x+4$ has
(a) no maximum and minimum
(b) one maximum and one minimum
(c) two maxima
(d) two minima

Assertion-Reason Type Questions

The following questions consist of two statements, one labelled as Statement I (Assertion) and the other labelled as Statement II (Reason). You are to examine these two statements carefully and decide if the Statement I and the Statement II are individually true and if so, whether the Statement II is a correct explanation for the given Statement I. Select your answers to these items using the codes given below and mark your answer sheet accordingly.

(a) Statement I is false, Statement II is true.
(b) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(c) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(d) Statement I is true, Statement II is false.

21. **Statement I:** At the point $(-1, 3)$, the tangent to the curve with parametric equations $x = 2 - 3\sin\theta$, $y = 3 + 2\cos\theta$ is vertical.
Statement II: For vertical tangent, $dy/dx = 0$.

22. **Statement I:** The equation $x^2 = x\sin x + \cos x$ has only one solution.
Statement II: The derivative of $x^2 - x\sin x - \cos x$ is $x(2 - \cos x)$.

23. **Statement I:** The function $f(x) = (x-5)^3$ has neither maximum nor minimum at $x = 5$.
Statement II: $f'(x) = 0, f''(x) = 0$ at $x = 5$.

24. **Statement I:** The function $f(x) = x^2 + (128/x)$ has minimum value 48 at $x = 4$.
Statement II: As x increases through 4, $f'(x)$ changes sign from +ve to –ve.

25. **Statement I:** The points on the curve $y^2 = x + \sin x$ at which the tangent is perpendicular to the y-axis lie on a parabola.
Statement II: The tangent is perpendicular to the y-axis if $dy/dx = 0$.

Previous Years' Questions

26. The greatest value of $f(x) = (x + 1)^{1/3} - (x - 1)^{1/3}$ on $[0, 1]$ is
(a) 1 (b) 2
(c) 3 (d) 1/3 **[AIEEE 2002]**

27. The function $f(x) = \cot^{-1}x + x$ increases in the interval
(a) $(1, \infty)$ (b) $(-1, \infty)$
(c) $(-\infty, \infty)$ (d) $(0, \infty)$ **[AIEEE 2002]**

28. The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 - 2 = 0$
(a) cut at right angle (b) touch each other
(c) cut at an angle $\pi/3$ (d) cut at an angle $\pi/4$
[AIEEE 2002]

29. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals
(a) 3 (b) 1
(c) 2 (d) 1/2 **[AIEEE 2003]**

30. The real number x when added to its inverse gives the minimum value of the sum at x equal to
(a) 2 (b) 1
(c) –1 (d) –2 **[AIEEE 2003]**

31. The normal to the curves $x = a(1 + \cos\theta), y = a\sin\theta$ at θ always passes through the fixed point
(a) $(a, 0)$ (b) $(0, a)$
(c) $(0, 0)$ (d) (a, a) **[AIEEE 2004]**

32. A function $y = f(x)$ has a second-order derivative $f'' = 6(x - 1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function is
(a) $(x - 1)^2$ (b) $(x - 1)^3$
(c) $(x + 1)^3$ (d) $(x + 1)^2$ **[AIEEE 2004]**

33. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^2/min. When the thickness of the ice is 15 cm, then the rate at which the thickness of the ice decreases is
(a) $5/6\pi$ cm/min (b) $1/54\pi$ cm/min
(c) $1/18\pi$ cm/min (d) $1/36\pi$ cm/min
[AIEEE 2005]

34. The normal to the curves $x = a(\cos\theta + \theta\sin\theta), y = a(\sin\theta - \theta\cos\theta)$ at any point θ is such that
(a) it is at a constant distance from the origin
(b) it passes through $\left(\frac{a\pi}{2}, -a\right)$
(c) it makes an angle $\frac{\pi}{2} - \theta$ with the x-axis
(d) it passes through the origin **[AIEEE 2005]**

35. The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minimum at
(a) $x = -2$ (b) $x = 0$
(c) $x = 1$ (d) $x = 2$ **[AIEEE 2006]**

36. The angle between the tangent to the curve $y = x^2 - 5x + 6$ at the points $(2, 0)$ and $(3, 0)$ is
(a) $\pi/2$ (b) $\pi/6$
(c) $\pi/4$ (d) $\pi/3$ **[AIEEE 2006]**

37. The normal to a curve at $p(x, y)$ meets the x-axis at G. If the distance of G from the origin is twice the abscissa of p, then the curve is
(a) an ellipse (b) a parabola
(c) a circle (d) a hyperbola
[AIEEE 2007]

38. A value of c for which the conclusion of the mean value theorem holds for the function $f(x) = \log_e x$ on the interval [1, 3] is
(a) $2 \log_3 e$ (b) $1/2 \log_e 3$
(c) $\log_3 e$ (d) $\log_e 3$ **[AIEEE 2007]**

39. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in
(a) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (b) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$
(c) $\left(0, \frac{\pi}{2}\right)$ (d) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
[AIEEE 2007]

40. Suppose the cubic $x^3 - px + q$ has three distinct real roots where $p > 0$ and $q > 0$. Then, which one of the following holds?
(a) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$
(b) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$
(c) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$
(d) The cubic has minima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$
[AIEEE 2008]

41. How many real solutions does the equation $x^7 + 14x^5 + 16x^3 + 30x - 560 = 0$ have?
(a) 5 (b) 7
(c) 1 (d) 3 **[AIEEE 2008]**

42. Given $p(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $p'(x) = 0$. If $p(-1) < p(1)$, then in the interval $[-1, 1]$.
(a) $p(-1)$ is the minimum and $p(1)$ is the maximum of p
(b) $p(-1)$ is not the minimum but $p(1)$ is the maximum of p
(c) $p(-1)$ is the minimum and $p(1)$ is not the maximum of p
(d) neither $p(-1)$ is the minimum nor $p(1)$ is the maximum of p **[AIEEE 2009]**

43. Let $f: \mathbb{R} \to \mathbb{R}$ be a continuous function defined by $f(x) = \frac{1}{e^x + 2e^{-x}}$.

Statement 1: $f(c) = \frac{1}{3}$, for some $c \in \mathbb{R}$.

Statement 2: $0 < f(x) \le \frac{1}{2\sqrt{2}}$, for all $x \in \mathbb{R}$.
(a) Statement 1 is true, Statement 2 is also true; Statement 2 is the correct explanation of Statement 1.
(b) Statement 1 is true, Statement 2 is also true; Statement 2 is not the correct explanation of Statement 1.
(c) Statement 1 is true, Statement 2 is false.
(d) Statement 1 is false, Statement 2 is true.
[AIEEE 2010]

44. The equation of the tangent to the curve $y = x + \frac{4}{x^2}$ that is parallel to the x-axis is
(a) $y = 0$ (b) $y = 1$
(c) $y = 2$ (d) $y = 3$ **[AIEEE 2010]**

45. The point in the interval $[0, 2\pi]$, where $f(x) = e^x \sin x$ has the maximum slope, is
(a) $\pi/4$ (b) $\pi/2$
(c) π (d) $3\pi/2$ **[AIEEE 2010]**

46. The maximum value of $x^3 - 3x$ in the interval [0, 2] is
(a) -2 (b) 0
(c) 2 (d) 1 **[DCE 2003]**

47. For a given integer k, in the interval $\left[2\pi k + \frac{\pi}{2}, 2\pi k - \frac{\pi}{2}\right]$, the graph of sin x is
(a) increasing from -1 to 1
(b) decreasing from -1 to 0
(c) decreasing from 0 to 1
(d) none of these **[DCE 2003]**

48. If $f(x)$ satisfies the condition for Rolle's theorem in [3, 5], then $\int_3^5 f(x)dx$ equals
(a) 2 (b) -1
(c) 0 (d) $-4/3$ **[DCE 2004]**

49. The equation of the tangent to the curve $\frac{x^2}{3} - \frac{y^2}{2} = 1$, which is parallel to $y = x$, is
(a) $y = x \pm 1$ (b) $y = x - 1/2$
(c) $y = x + 1/2$ (d) $y = 1 - x$ **[DCE 2007]**

50. For what value of a, $f(x) = -x^3 + 4ax^2 + 2x - 5$ is decreasing for all x?
(a) (1, 2) (b) (3, 4)
(c) $\mathbb{R}$ (d) no value of a
[DCE 2007]

51. The point(s) on the curve $y^3+3x^2=12y$, where the tangent is vertical, is (are)

(a) $\left(\pm\frac{4}{\sqrt{3}},-2\right)$ (b) $\left(\pm\sqrt{\frac{11}{3}},1\right)$

(c) (0, 0) (d) $\left(\pm\frac{\sqrt{4}}{3},2\right)$ **[IIT 2002]**

52. Find which function does not obey the mean value theorem in [0, 1]

(a) $f(x)=\begin{cases}\frac{1}{2}-x, x<\frac{1}{2}\\ \left(\frac{1}{2}-x\right)^2, x\geq\frac{1}{2}\end{cases}$ (b) $f(x)=\begin{cases}\frac{\sin x}{x}, x\neq 0\\ 1, x=0\end{cases}$

(c) $f(x)=x|x|$ (d) $f(x)=|x|$ **[IIT 2003]**

53. If the minimum value of $f(x)=x^2+2bx+2c^2$ is greater than the maximum value of $g(x)=-x^2-2cx+b^2$, then for x is real

(a) $|c|>|b|\sqrt{2}$ (b) $|c|\sqrt{2}>b$

(c) $0<c<\sqrt{2}b$ (d) no real value of a

[IIT 2003]

54. A tangent is drawn at the point $(3\sqrt{3}\cos\theta,\sin\theta)$, $0<\theta<\pi/2$ of the ellipse $\frac{x^2}{27}+\frac{y^2}{1}=1$. The least value of the sum of the intercepts on the coordinate axes by this tangent is attained at $\theta=$

(a) $\pi/6$ (b) $\pi/3$

(c) $\pi/8$ (d) $\pi/4$ **[IIT 2003]**

55. If $f(x)=x^3+bx^2+cx+d$ and $0<b^2<c$, then in $(-\infty,\infty)$, $f(x)$

(a) is increasing (b) has real maximum

(c) is decreasing (d) is bounded **[IIT 2004]**

56. If $f(x)=x^\alpha\log x$ and $f(0)=0$, then the value of α for which Rolle's theorem can be applied in [0, 1] is

(a) -2 (b) -1

(c) 0 (d) 1/2 **[IIT 2004]**

57. If $f(x)$ is a differentiable increasing function, then $\lim\limits_{x\to 0}\frac{f(x^2)-f(x)}{f(x)-f(0)}$ equals

(a) 1 (b) 0

(c) -1 (d) 2 **[IIT 2004]**

58. The line $2x+\sqrt{6}y=2$ is a tangent to the curve $x^2-2y^2=4$. The point of contact is

(a) $(4,-\sqrt{6})$ (b) $(7,-2\sqrt{6})$

(c) (2, 3) (d) $(\sqrt{6},1)$ **[IIT 2004]**

59. If $f:\mathbb{R}\to\mathbb{R}$ and $f(1)=1, f(2)=4, f(3)=9$, where $f(x)$ is a twice differentiable function, then

(a) $f''(x)=2\ \forall x\in[1,3]$

(b) $f'(x)=2$ for some $x\in(1,3)$

(c) $f''(x)=f'(x)=5$ for some $x\in(1,3)$

(d) none of these

[IIT 2005]

60. $f(x)$ is cubic polynomial which has local maxima at $x=-1$. If $f(2)=18, f(1)=-1$ and $f'(x)$ has local minima at $x=0$, then

(a) the distance between $(-1, 2)$ and $[a, f(a)]$ where $x=a$ is the point of local minima, is $2\sqrt{5}$

(b) $f(x)$ is increasing for $x\in[1,2\sqrt{5}]$

(c) $f(x)$ has local minima at $x\neq 1$

(d) the value of $f(0)$ is 5

[IIT 2006]

61. Let f be a function defined by $f(x)=\begin{cases}\frac{\tan x}{x}, x\neq 0\\ 1, x=0\end{cases}$

Statement 1: $x=0$ is a point of minima of f.

Statement 2: $f'(0)=0$.

(a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

(b) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

(c) Statement 1 is true, Statement 2 is false.

(d) Statement 1 is false, Statement 2 is true.

[AIEEE 2011]

62. A spherical balloon is filled with 4500π cubic metres of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic metres per minute, then the rate (in metres per minute) at which the radius of the balloon decreases 49 minutes after the leakage bagan is

(a) 9/7 (b) 7/9 (c) 2/9 (d) 9/2

[AIEEE 2012]

63. Let $a, b\in R$ be such that the function f given by

$$f(x)=\ln|x|+bx^2+ax,\quad x\neq 0$$

has extreme values at $x=-1$ and $x=2$.

Statement 1: f has local maximum at $x=-1$ and at $x=2$.

Statement 2: $a=1/2$ and $b=-1/4$.

(a) Statement 1 is false, statement 2 is true

(b) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1

(c) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1

(d) Statement 1 is true, statement 2 is false

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (a)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (a)	**6.** (b)	**7.** (b)	**8.** (b)	**9.** (a)	**10.** (a)
11. (a)	**12.** (a)	**13.** (a)	**14.** (d)	**15.** (c)	**16.** (a)	**17.** (c)	**18.** (a)	**19.** (b)	**20.** (b)
21. (d)	**22.** (b)	**23.** (a)	**24.** (c)	**25.** (a)	**26.** (a)	**27.** (d)	**28.** (d)	**29.** (b)	**30.** (b)
31. (a)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (b)	**36.** (b)	**37.** (a)	**38.** (b)	**39.** (d)	**40.** (d)
41. (b)	**42.** (a)	**43.** (b)	**44.** (d)	**45.** (c)	**46.** (b)	**47.** (a)	**48.** (c)	**49.** (b)	**50.** (b)

Exercise Set 2

1. (b)	**2.** (a)	**3.** (a)	**4.** (b)	**5.** (a)	**6.** (a)	**7.** (a)	**8.** (c)	**9.** (a)	**10.** (c)
11. (a)	**12.** (d)	**13.** (a)	**14.** (a)	**15.** (c)	**16.** (b)	**17.** (c)	**18.** (d)	**19.** (b)	**20.** (b)
21. (b)	**22.** (a)	**23.** (b)	**24.** (d)	**25.** (a)	**26.** (b)	**27.** (c)	**28.** (a)	**29.** (c)	**30.** (b)
31. (a)	**32.** (b)	**33.** (c)	**34.** (a)	**35.** (d)	**36.** (a)	**37.** (d)	**38.** (a)	**39.** (b)	**40.** (b)
41. (c)	**42.** (b)	**43.** (c)	**44.** (d)	**45.** (b)	**46.** (c)	**47.** (a)	**48.** (d)	**49.** (a)	**50.** (d)
51. (d)	**52.** (a)	**53.** (d)	**54.** (a)	**55.** (a)	**56.** (d)	**57.** (c)	**58.** (c)	**59.** (b)	**60.** (b)
61. (b)	**62.** (c)	**63.** (b)							

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Let v be the velocity of the particle when the distance covered is s. Then

$$v \propto \sqrt{s} \qquad \text{[Given]}$$

$$\Rightarrow \quad v = \lambda\sqrt{s}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{\lambda}{2\sqrt{s}}$$

$$\Rightarrow \quad v\frac{dv}{dx} = \frac{\lambda v}{2\sqrt{s}} = \frac{\lambda^2}{2} = \text{constant}$$

$\therefore$ The acceleration is a constant.

2. Here $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$. $\therefore \frac{n}{a}\left(\frac{x}{a}\right)^{n-1} + \frac{n}{b}\left(\frac{y}{b}\right)^{n-1}\cdot\frac{dy}{dx} = 0$

$$\therefore \quad \frac{dy}{dx} = \frac{-\frac{1}{a}\left(\frac{x}{a}\right)^{n-1}}{\frac{1}{b}\left(\frac{y}{b}\right)^{n-1}} = \frac{b^n x^{n-1}}{a^n y^{n-1}}$$

At (a,b), $\frac{dy}{dx} = -\frac{b^n}{a^n}\cdot\frac{a^{n-1}}{b^{n-1}} = -\frac{b}{a}$

Tangent at (a, b) is $y - b = -\frac{b}{a}(x-a)$

$$\Rightarrow \quad \frac{y}{b} - 1 = -\frac{x}{a} + 1, \quad \text{i.e. } \frac{x}{a} + \frac{y}{b} = 2.$$

This shows that $\frac{x}{a} + \frac{y}{b} = 2$ touches the given curve for all n.

3. Since $x^{m+n} = a^{m-n}y^{2n}$, differentiating w.r.t. x, we have

$$(m+n)x^{m+n-1} = a^{m-n}(2n)y^{(2n-1)}y'$$

$$\Rightarrow \quad y' = \frac{(m+n)x^{m+n-1}}{2na^{m-1}y^{2n-1}}$$

Length of subtangent $= |ST| = \frac{Y}{Y'}$

$$\Rightarrow \quad |\mathrm{ST}| = \frac{Y\,2na^{m-n}y^{2n-1}}{(m+n)x^{m+n-1}}$$

$$\Rightarrow \quad |\mathrm{ST}| = \frac{2na^{m-n}y^{2n}}{(m+n)x^{m+n-1}} = \frac{2na^{m-n}x^{m+n}}{(m+n)a^{m-n}x^{m+n-1}}$$

$$\Rightarrow \quad |\mathrm{ST}| = \left(\frac{2n}{m+n}\right)x$$

Now, subnormal $= |\mathrm{SN}| = yy'$

$$\Rightarrow \quad |\mathrm{ST}| = \frac{y(m+n)x^{m+n-1}}{2na^{m-n}y^{2n}y^{-1}} = \frac{(m+n)y^2}{2na^{m-n}}\frac{x^{m+n-1}}{y^{2n}}$$

$$\Rightarrow \quad |\mathrm{SN}| = \frac{(m+n)y^2}{2na^{m-n}}\frac{x^{m+n-1}a^{m-n}}{x^{m+n}} = \frac{(m+n)y^2}{2nx}$$

Finding the ratio $\frac{|\mathrm{ST}|^m}{|\mathrm{SN}|^n}$, we see that it is equal to some constant.

4. Apply Rolle's theorem for $\frac{ax^3}{3} + \frac{bx^2}{2} + cx$ [0, 1].

5. Equation of the curve is $\sqrt{x} + \sqrt{y} = 3$ and coordinates of the point $(x, y) = (4, 1)$. Differentiating the given equation, we get

$$-\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0 \quad \text{or} \quad \frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}}\frac{dy}{dx} = 0$$

or $\frac{dy}{dx}=-\sqrt{\left(\frac{y}{x}\right)}$. We know that at (4, 1)

$\left(\frac{dy}{dx}\right)_{(4,1)}=\sqrt{\left(\frac{1}{4}\right)}=-\frac{1}{2}$. We also know that the length of the subtangent $=\frac{1}{\frac{dy}{dx}}=\frac{1}{\left(-\frac{1}{2}\right)}=-2$.

6. Solving, we have $x^2y=1-y=xy$

$xy(x-1)=0\therefore x=0, y=0, x=1$

$x=0, y=1\therefore (0,1), x=1, y=\frac{1}{2}\therefore\left(1,\frac{1}{2}\right)$

$y=0\Rightarrow 1=0$ (not possible)

Also $x^2=\frac{1-y}{y}=\frac{1}{y}-1$

$\therefore\quad 2x\frac{dx}{dy}=-\frac{1}{y^2}\therefore\frac{dy}{dx}=-2xy^2$

$$m_1=\left(\frac{dy}{dx}\right)_{(0,1)}=0, m_2=\left(\frac{dy}{dx}\right)_{(1,1/2)}=-\frac{1}{2}$$

Hence, two tangents are $y=1$ and $x+2y-2=0$. They intersect at (0, 1).

7. In ΔABC, we have

$2bc\cos A=b^2+c^2-a^2$

$\Rightarrow\quad d(2bc\cos A)=d(b^2+c^2-a^2)$

$\Rightarrow\quad -2bc\sin A\,dA=-2a\,da$

$\Rightarrow\quad bc\sin A\,dA=a\,da$

$\Rightarrow\quad \frac{2}{a}\left(\frac{1}{2}bc\sin A\right)dA=da$

$\Rightarrow\quad da=\frac{2S}{a}dA$

$\Rightarrow\quad \Delta a=\frac{2S}{a}dA$ [$\because dx\cong\Delta a$ and $dA\cong\Delta A$]

8. We have

$$e^{2y}=1+4x^2\Rightarrow e^{2y}\cdot 2\frac{dy}{dx}=8x$$

$$\Rightarrow\quad \frac{dy}{dx}=\frac{4x}{e^{2y}}=\frac{4x}{1+4x^2}$$

$\therefore$ Slope of the tangent $= m=\frac{4x}{1+4x^2}$

$$\Rightarrow\quad |m|=\frac{4|x|}{1+4|x|^2}\le 1$$

$$\left[\because (1-2|x|)^2\ge 0\Rightarrow\ 1+4|x|^2-4|x|\ge 0\right.$$
$$\left.\Rightarrow\quad \frac{4|x|}{1+4|x|^2}\le 1\right]$$

9. $f(1)=f(3)$

$\Rightarrow\quad a+b+11-6=27a+9b+33-6$

$\Rightarrow\quad 26a+8b-22\quad\Rightarrow\quad 13a+4b=-11$ (i)

$\Rightarrow\quad f'(x)=3ax^2+bx+11\quad\Rightarrow\quad f'\left(2+\frac{1}{3}\right)$

$=3a\left(4+\frac{1}{3}+\frac{4}{\sqrt{3}}\right)+2b\left(2+\frac{1}{\sqrt{3}}\right)+11-6$ (ii)

Solving (i) and (ii), we get the values of a and b.

10. Let $f(x)=ax^3+bx^2+cx$ in [0, 1]

$f(x)$ is continuous in [0, 1] and derivable in (0, 1).

Also $f(0)=0, f(1)=a+b+c=0$

$\therefore\quad f(0)=f(1)$

$\therefore$ $f(x)$ satisfies all the conditions of Rolle's theorem in (0, 1).

By Rolle's theorem, $f(x)$ has at least one real root in (0, 1).

11. The angle between the two curves is the angle between their tangents at their point of contact.

$\therefore$ The point of intersection of the two curves is $a^x=b^x$.

For distinct values of a and b, the equality is true, if $x=0$.

Now, if $x=0$, then $y=a^0=1$

$\therefore$ The two curves intersect at (0, 1).

Let m_1 and m_2 be the slopes of the tangent to the curve $y=a^x$ and $y=b^x$, respectively at P (0, 1).

$$\therefore\quad m_1=\left.\frac{dy}{dx}\right|_p=a^x\log a|_{(0,1)}=\log a$$

$$\text{Also,}\quad m_2=\left.\frac{dy}{dx}\right|_p=b^x\log b|_{(0,1)}=\log b$$

If α is the angle of intersection of the tangents, then

$$\tan\alpha=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$$

$$\Rightarrow\quad \tan\alpha=\left|\frac{\log a-\log b}{1+\log a\log b}\right|$$

12. $\frac{dy}{dx}=-\frac{f_x}{f_y}=-\frac{x^{1/2}}{y^{1/2}}=\pm 1$

$\therefore\quad y^{1/2}=x^{1/2}\quad$ or $\quad y^{1/2}=-x^{1/2}$

Also $x^{3/2}+y^{3/2}=a^{3/2}$

$\therefore\quad 2x^{3/2}=a^{3/2}\quad$ or $\quad 0=a^{3/2}\quad$ (not possible)

$\therefore\quad x=\frac{a}{2^{2/3}}=y$. Hence only one point.

13. $\frac{dy}{dx}=3x^2+3$.

At $x=-1, \frac{dy}{dx}=3+3=6=m_1$

At $x=-1, \frac{dy}{dx}=3+3=6=m_2$

Since $m_1 = m_2$, therefore, tangents are parallel.

14. We have $f(x)=4x^4-2x+1$. On differentiating, we get

$f'(x)=16x^3-2$. Put $f'(x)=0$ for maxima

$\Rightarrow \quad 16x^3-2=0 \quad \Rightarrow \quad x=\frac{1}{2}$

Taking interval $\left(-\infty, \frac{1}{2}\right)$ and $\left(\frac{1}{2}, \infty\right)$

In interval $\left(\frac{1}{2}, \infty\right)$, put $x = 1$

$f'(1)=16-2=14$ (increasing)

$\therefore$ Function is increasing for $x>\frac{1}{2}$.

15. Equation of the curve $y^2 = x^3$.

Differentiating the given equation, we get

$$2y\frac{dy}{dx}=3x^2 \quad \text{or} \quad \frac{dy}{dx}=\frac{3x^2}{2y}.$$

We know that at (4, 8),

$$\left(\frac{dy}{dx}\right)_{(4,8)}=\frac{3\times(4)^2}{(2\times 8)}=\frac{3\times 16}{16}=3.$$

We also know that the equation of the subnormal is $y\frac{dy}{dx}=(8\times 3)=24$. Therefore, the length of the subnormal $=24$.

16. That $s=\sqrt{t}$

$\therefore \quad \frac{ds}{dt}=\frac{1}{2\sqrt{t}}$

$\Rightarrow \quad v=\frac{1}{2\sqrt{t}} \quad \Rightarrow \quad \frac{dv}{dt}=-\frac{1}{2\cdot 2t^{3/2}}$

$\Rightarrow \quad a=-\frac{2}{(2\sqrt{t})^3}$

$\Rightarrow \quad a=-2v^3 \quad \Rightarrow \quad a \propto v^3$

17. $v=\frac{ds}{dt}=48-32t$

For maximum height $v = 0$

$\therefore \quad 48-32t=0 \ \therefore \ t=\frac{3}{2}$

Height above the ground

$$=64+48\left(\frac{3}{2}\right)-16\left(\frac{9}{4}\right)$$

$$=64+72-36=64+36=100.$$

18. $f(x,y)=x+y-e^{xy}=0$

$\therefore \quad \frac{dy}{dx}=-\frac{f_x}{f_y}$.

If the tangent is parallel to the y-axis, then $\frac{dy}{dx}=\infty$

or $\quad f_y=0$, i.e. $1-e^{xy}\cdot x=0 \quad$ or $\quad 1-x(x+y)=0$

The above relation is satisfied by $x = 1, y = 0$.

19. The slope of the tangent $=\frac{dy}{dx}=\frac{dy}{dt}\Big/\frac{dx}{dt}$

$$=\frac{a\left(-\sin t+\frac{1}{\tan t/2}\sec^2 t/2.\frac{1}{2}\right)}{a\cos t}$$

$$=\frac{\left(-\sin t+\frac{1}{2\sin t/2\cos t/2}\right)}{\cos t}$$

$$=\frac{-\sin t+\frac{1}{\sin t}}{\cos t}=\frac{1-\sin^2 t}{\sin t\cos t}$$

$$=\frac{\cos^2 t}{\sin t\cos t}=\cot t.$$

20. At $x=2, y=\frac{8}{4+2^2}=1$

$\therefore$ The slope of the tangent to the curve $y=\frac{8}{4+x^2}$ (i)

at (2, 1) is $m_T=\left.\frac{dy}{dx}\right|_{(2,1)}=-\left.\frac{16x}{(4+x^2)^2}\right|_{(2,1)}$

$\Rightarrow \quad m_T=-\frac{1}{2}$

Now the equation of the tangent is

$$y-1=-\frac{1}{2}(x-2)$$

$\Rightarrow \quad x+2y=4$

Intercepts of the tangent with the x-axis $(y=0)$ and y-axis $(x = 0)$ are 4 and 2, respectively.

$\therefore$ Area of the triangle formed by the tangent with the intersection of the coordinate axis is

$$\frac{1}{2}(4)(2)=4 \text{ sq. units.}$$

21. Equation is $x^2=-4y$. On differentiating, we get

$$2x=-4\frac{dy}{dx} \Rightarrow \frac{dy}{dx}=-\frac{x}{2}.$$

The slope of the tangent at (–4, 4) is $\left(\frac{dy}{dx}\right)_{(-4,-4)}=\frac{4}{2}=2$

We know that the equation of the tangent at the point $(x_1\ y_1)$ is

$$(y-y_1)=\left(\frac{dy}{dx}\right)_{(x_1y_1)}(x-x_1)$$

If the point is $(-4, -4)$ then $y+4=2(x+4)$

$\Rightarrow\ 2x-y+4=0.$

22. $\dfrac{dy}{dx}=\dfrac{\dot{y}}{\dot{x}}=\dfrac{4t-2}{2t+3}$ (i)

For $(2,-1)$, $t^2-3t-8=2$ or $t^2+3t-10=0$

$\therefore\ (t+5)(t-2)$ and $2t^2-2t-5=-1$ or $t^2-t-2=0$.

$\therefore\ (t+1)(t-2)$

Hence for $(2, -1)$, $t=2$ (common value)

At $t=2, \dfrac{dy}{dx}=\dfrac{8-2}{7}=\dfrac{6}{7}$ by (i)

23. L $f'(0)=\lim\limits_{h\to0^-}\dfrac{f(h)-f(0)}{h}$

$$=\lim_{h\to0^-}\frac{|h|}{h}=-1$$

$$Rf'(0)=\lim_{h\to0^+}\frac{f(h)-f(0)}{h}=\lim_{h\to0^+}\frac{|h|}{h}=1$$

$\therefore\ f(x)$ is not derivable at $x=0$ $[\because Lf'(0)\neq Rf'(0)]$

$\therefore\ f$ is not derivable in $(-3, 3)$.

24. $\dfrac{dy}{dx}=2x=x$ (given)

$\Rightarrow x=0\Rightarrow y=0$

$\therefore$ The required point is $(0, 0)$.

25. $f(x)=\tan x-x$

On differentiating, we get

$$f'(x)=\sec^2x-1=\frac{1}{\cos^2x}-1=\frac{1-\cos^2x}{\cos^2x}$$

Since $0\le\cos^2x\le1$ for all values of x,

$\therefore\ f'(x)>0$ for all values of x. Thus, $f(x)$ always increases.

26. $s=t^3+2t^2+t$

$\Rightarrow\ v=\dfrac{ds}{dt}=3t^2+4t+1$

Speed of the particle after 1 s

$$v_{(t=1)}\left(\frac{ds}{dt}\right)_{(t=1)}$$

$$=3\times1^2+4\times1+1=3+5=8\text{ cm/s}$$

27. $y+x=e^{xy}$, $\dfrac{dy}{dx}=-\dfrac{f_x}{f_y}=\infty\Rightarrow f_y=0$

$\therefore\ 1-xe^{xy}=0$ or $1-x(x+y)=0$

It is satisfied by $(1, 0)$.

28. Given curve is $xy=c^2$. On differentiating, we get

$$\frac{dy}{dx}=\frac{-c^2}{x^2}\qquad\text{(i)}$$

Subnormal $=y\dfrac{dy}{dx}$

$$=\frac{y\times(-c^2)}{x^2}=\frac{-yc^2}{\left(\frac{c^2}{y}\right)^2}\qquad\left[\text{from (i), } x=\frac{c^2}{y}\right]$$

$$=\frac{-yc^2y^2}{c^4}=\frac{-y^3}{c^2}$$

$\therefore$ Subnormal varies as y^3.

29. Rolle's theorem is applicable in $[0, \pi]$.

30. Here $f(1)=f(2)$

$\Rightarrow\ 1+b+c=8+4b+2c$

$\Rightarrow\ 3b+c+7=0$

Also $f'(x)=3x^2+2bx+c$ (i)

$\therefore\ f'\left(\dfrac{4}{3}\right)=0\Rightarrow3\left(\dfrac{16}{9}\right)+2b\left(\dfrac{4}{3}\right)+x=0$

$\Rightarrow\ 16+8b+3c=0$ (ii)

Solving (i) and (ii), we get $b=-5$, $c=8$.

31. We have $x=a(\theta+\sin\theta)$ and $y=a(1-\cos\theta)$

$\Rightarrow\ \dfrac{dx}{d\theta}=a(1+\cos\theta)$ and $\dfrac{dy}{d\theta}=a\sin\theta$

$\therefore\ \dfrac{dy}{dx}=\dfrac{dy/d\theta}{dx/d\theta}=\dfrac{\sin\theta}{1+\cos\theta}=\dfrac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}=\tan\dfrac{\theta}{2}$

$\therefore\ \left[\dfrac{dy}{dx}\right]_{\theta=(2n+1)\pi}=\tan(2n+1)\dfrac{\pi}{2}=\infty$

$\therefore$ Slope of the normal is 0.

Hence, the normal is parallel to the x-axis at $\theta=(2n+1)\pi$.

32. We have $x=at^2$

$$\frac{dx}{dt}=2at\text{ and }y=2at\Rightarrow\frac{dx}{dt}=2a$$

$\therefore$ Tangent $\dfrac{dy}{dx}=\dfrac{2a}{2at}=\dfrac{1}{t}$

$\Rightarrow\ \dfrac{1}{t}=\infty\ \Rightarrow\ t=0\ \Rightarrow$ point of contact is $(0, 0)$.

33. Let $f(x)=ax^3+bx^2+cx$

$\therefore\ f'(x)=3ax^2+2bx+c$

Clearly $f(x)$ is continuous in $[0, 1]$, derivable in $(0, 1)$ and $f(0)=0=f(1)$

$\because\ f(1)=a+b+c=0$ (given)

$\therefore$ By Rolle's theorem, there exists at least one real $x \in (0, 1)$ such that $f'(x) = 0$, i.e. $3ax^2 + 2bx + c = 0$

Hence, there exists at least one real root of

$$3ax^2 + 2bx + c = 0 \text{ in } (0, 1)$$

34. $f'(x) = 3x^2 + 3x + 3$

$$= 3(x^2 + x + 1) = 3\left[\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}\right] \geq \frac{9}{4} > 0$$

$\Rightarrow$ $f(x)$ is an increasing function.

35. The equation of the given circle is $x^2 + y^2 = 1$.

On differentiating w.r.t. t, we get

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

But we have $x = \frac{1}{2}, y = \frac{\sqrt{3}}{2}$ and $\frac{dy}{dt} = -3$, then

$$\frac{1}{2}\frac{dx}{dt} + \frac{\sqrt{3}}{2}(-3) = 0 \Rightarrow \frac{dx}{dt} = 3\sqrt{3}.$$

36. $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3)$

$= 3(x-3)(x-1)$. If $f(x)$ is decreasing, then

$f'(x) < 0 \Rightarrow 3(x-3)(x-1) < 0$

$\Rightarrow (x-3)(x-1) < 0 \Rightarrow$ either $x - 3 > 0, x - 1 < 0$

or $x - 3 < 0, x - 1 > 0 \Rightarrow$ either $x > 3, x < 1$

or $x < 3, x > 1 \Rightarrow 3 < x < 1$ or $1 < x < 3$

$\Rightarrow x \in (1, 3)$ [$\because 3 < x < 1$ is not possible]

37. $y = a(1 - \cos x)$. On differentiating, we get $y' = a \sin x$

Put $y' = 0$ for maxima or minima

$\Rightarrow \sin x = 0 \Rightarrow x = 0, \pi$

Again differentiating, we get

$y'' = a\cos x \Rightarrow y''(0) = a$ and $y''(\pi) = -a$

Hence, y is maximum when $x = \pi$.

38. $\frac{dy}{dx} = 3ax^2 + 2bx + c$ (given)

At $(0, 0)$, $\frac{dy}{dx} = c = \tan 45° = 1 \Rightarrow c = 1$.

At $(1, 0)$, $\frac{dy}{dx} = 3a + 2b + c$.

Tangent at $(1, 0)$ is $y - 0 = (3a + 2b + c) \cdot (x - 1)$

This is x-axis if $3a + 2b + c = 0$,

which is true if $a = 1$, $b = -2$

Hence, $a = 1$, $b = -2$, $c = 1$.

39. $f'(x) = 2x - 2 = 2(x-1) \geq 0$

If $f(x)$ is increasing $\Rightarrow x \geq 1$.

40. Point is $x = a + bt - ct^2$

Acceleration in the x direction $= \frac{d^2x}{dt^2} = -2c$ and

acceleration in the y direction $= \frac{d^2y}{dt^2} = 2b$

$\therefore$ Resultant acceleration

$$= \sqrt{\left(\frac{d^2x}{dt^2}\right)^2 + \left(\frac{d^2y}{dt^2}\right)^2} = \sqrt{(-2c)^2 + (2b)^2} = 2\sqrt{b^2 + c^2}$$

41. The equation of the given curve is $\sqrt{x} + \sqrt{y} = \sqrt{a}$

$$\therefore \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$$

The normal is parallel to the x-axis, if

$$\left(\frac{dx}{dy}\right)_{(x_1, y_1)} = 0 \Rightarrow x_1 = 0$$

$\therefore$ From the equation of the curve, $y_1 = a$

$\therefore$ Required point is $(0, a)$.

42. Since g is continuous at $x = a$,

$\therefore$ if $g(a) > 0$, then there exists an open interval I containing a so that $g(x) > 0$ for all $x \in I$

$\Rightarrow f'(x) > 0$ for all $x \in I \Rightarrow f$ is increasing near a.

43. $f'(x) = \frac{(1 + |x|) \cdot 1 - x(\pm 1)}{(1 + |x|)^2}$

$$\frac{1 + |x| - |x|}{(1 + |x|)^2} = \frac{1}{(1 + |x|)^2} > 0, \forall x \in \mathbb{R}$$

$\therefore$ f is strictly increasing.

44. Curve is $y = x \log x$

On differentiating with respect to x, we get

$$\frac{dy}{dx} = 1 + \log x$$

The slope of the normal $= -\frac{1}{(dy/dx)} = \frac{-1}{1 + \log x}$

The slope of the given line $2x - 2y = 3$ is 1.

Since these given lines are parallel,

$$\therefore \frac{-1}{1 + \log x} = 1 \Rightarrow \log x = -2 \Rightarrow x = e^{-2} \text{ and } y = -2e^{-2}$$

$\therefore$ Coordinates of the point are $(e^{-2}, -2e^{-2})$.

45. Let the distance of the school from A $= x$

Let the distance of the school from B $= 60 - x$

Total distance covered by 200 students

$$= 2[150x + 50(60 - x)]$$

$$= 2[100x + 3000]$$

This is minimum when $x = 0$

$\therefore$ The school should be at town A.

46. Here $ax^2+by^2=1 \Rightarrow 2ax+2by\frac{dy}{dx}=0.$

$\therefore \quad m_1=\frac{dy}{dx}=-\frac{ax}{by}.$

From $a'x^2+b'y^2=1, m_2=\frac{dy}{dx}=-\frac{a'x}{b'y}.$

Since the given curves intersect orthogonally,

$\therefore \quad m_1m_2=-1 \Rightarrow \frac{aa'x^2}{bb'y^2}=-1 \qquad$ (i)

Also $ax^2+by^2=a'x^2+b'y^2$

$\Rightarrow (a-a')x^2=(b'-b)y^2 \Rightarrow \frac{x^2}{y^2}=\frac{b'-b}{a-a'} \qquad$ (ii)

From (i) and (ii), we get $\frac{aa'(b'-b)}{bb'(a-a')}=-1$

$\Rightarrow \quad \frac{a-a'}{aa'}=\frac{b-b'}{bb} \Rightarrow \frac{1}{a'}-\frac{1}{a}=\frac{1}{b'}-\frac{1}{b}$

$\Rightarrow \quad \frac{1}{a}-\frac{1}{b}=\frac{1}{a'}-\frac{1}{b'}$

47. $f(x)=(x-2)^{2/3}(2x+1)$

$f'(x)=\frac{2}{3}(x-2)^{-1/3}(2x+1)+(x-2)^{2/3}\cdot 2$

Clearly $f'(x)$ is not defined at $x=2$.

$\therefore \quad x=2$ is a critical point. Another critical point is given by $f'(x)=0$,

i.e. $\frac{2}{3}\frac{2x+1}{(x-2)^{1/3}}+2(x-2)^{2/3}=0$

$\Rightarrow \quad \frac{2}{3}(2x+1)+2(x-2)=0$

$\Rightarrow \quad 4x+2+6x-12=0$

$\Rightarrow \quad 10x-10=0 \Rightarrow x=0$

Hence, critical points are 1 and 2.

48. Let $AB = x$ m, $BC = y$ m and $AC = 10$ m

$\therefore \quad x^2+y^2=100 \qquad$ (i)

On differentiating w.r.t. x, we get

$2x\frac{dx}{dt}+2y\frac{dy}{dt}=0$

$\Rightarrow \quad 2x(3)-2y(4)=0$

$\Rightarrow \quad x=\frac{4y}{3}$

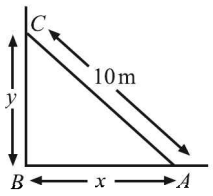


Fig. 13.16

On putting this value in Eq. (i), we get

$\frac{16}{9}y^2+y^2=100$

$\Rightarrow \quad y^2=\frac{100\times 9}{25}=36 \Rightarrow y=6$ m

49. Given curve is $y = 2x^2 - x + 1$. Let the coordinates of p be (h, k). On differentiating, we get

$\frac{dy}{dx}=4x-1$

At the point (h, k), the slope is $\left(\frac{dy}{dx}\right)_{(h,k)}=4h-1.$

Since the tangent is parallel to the given line,

$y=3x+4 \quad \Rightarrow \quad 4h-1=3 \Rightarrow h=1, k=2$

$\therefore$ The point p is (1, 2).

50. $f'(x)=\frac{2x}{3}(x^2-4)^{-2/3} \Rightarrow f'(0)=0.$

Also for $x<0$ (slightly), $f'(x)<0$ and for $x>0$ (slightly), $f'(x)>0$

$\therefore$ f has a local minima at $x = 0$.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. Curve is $f(x)=\frac{1}{x+1}-\log(1+x)$.

On differentiating w.r.t. x, we get

$f'(x)=-\frac{1}{(x+1)^2}-\frac{1}{1+x}$

$\Rightarrow \quad f'(x)=-\left[\frac{1}{x+1}+\frac{1}{(1+x)^2}\right]$

$\Rightarrow \quad f'(x)=-\text{ve}$, when $x>0$

$\therefore$ $f(x)$ is a decreasing function.

2. $f(x)=x+\frac{1}{x}, f'(x)=1-\frac{1}{x^2}, f''(x)=\frac{2}{x^3}$

For max. and mini. $f'(x)=0$

$\Rightarrow x^2-1=0 \quad \Rightarrow \quad x=\pm 1.$

At $x=1, f''(1)=2>0 \therefore x=1$ is a point of minima.

At $x=-1, f''(-1)=-2<0$

$\therefore \quad x=-1$ is a point of maxima.

Max. $f(x)=-1-1=-2.$

Min. $f(x)=1+1=2.$

$\therefore$ Relative min. > relative max.

3. Curve $f(x)=\frac{1}{4x^2+2x+1}$. On differentiating, we get

$\Rightarrow \quad f(x)=\frac{-(8x+2)}{(4x^2+2x+1)^2}$

For maxima or minima, put $f'(x) = 0$

$\Rightarrow \quad 8x+2=0 \Rightarrow x=-\frac{1}{4}$

Again differentiating, we get

$$f''(x) = -\frac{\left[\begin{array}{l}(4x^2+2x+1)^2(8)-(8x+2)2\\ \quad\times(4x^2+2x+1)(8x+2)\end{array}\right]}{(4x^2+2x+1)^4}$$

At $x = -\frac{1}{4}$, $f''\left(-\frac{1}{4}\right) = -\text{ve}$

$f(x)$ is maximum at $x = -1/4$

$\therefore$ Maximum value is

$$f\left(-\frac{1}{4}\right)_{\max} = \frac{1}{4\times\frac{1}{16}-2\times\frac{1}{4}+1} = \frac{1}{\frac{1}{4}-\frac{2}{4}+1}$$

$$= \frac{4}{1-2+4} = \frac{4}{3}.$$

4. $f(x)$ will be monotonically decreasing if

$f'(x) \le 0$. Now $f'(x) = -\sin x - 2p$

$\therefore$ $f'(x) \le 0 \Rightarrow -\sin x - 2p \le 0$

$$\Rightarrow\ p \ge -\frac{1}{2} \quad \left[\begin{array}{l}\because -1 \le \sin x \le 1\\ \therefore 1 \ge -\sin x \ge -1\\ \frac{1}{2} \ge -\frac{\sin x}{2} \ge -\frac{1}{2}\end{array}\right]$$

5. $\because f(x) = x + \cos x$

On differentiating, we get

$f'(x) = 1 - \sin x$, $f'(x) > 0$ for all values of x

($\because \sin x$ is lying between -1 and $+1$)

$\therefore$ $f(x)$ is always increasing.

6. The least perimeter of an isosceles triangle in which a circle of radius r can be inscribed is $6r\sqrt{3}$. In this case $r = 1$, $\therefore$ Required perimeter $= 6\sqrt{3}$.

7. $\frac{dv}{dt} = 30\ \text{ft}^3/\text{min}$ and $r = 15$ ft

Volume of spherical balloon $v = \frac{4}{3}\pi r^3$; $\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt}$

$$\Rightarrow \qquad 30 = 4\pi r^2 \frac{dr}{dt}$$

$$\Rightarrow \qquad \frac{dr}{dt} = \frac{30}{4\times\pi\times15\times15} = \frac{1}{30\pi}\ \text{ft/min}$$

8. $f'(x) = 2\cos 2x - 1$ $\therefore$ $f'(x) = 0 \Rightarrow 2\cos 2x - 1 = 0$

$$\Rightarrow \cos 2x = \frac{1}{2} \quad \Rightarrow \quad 2x = \frac{\pi}{3}, -\frac{\pi}{3}$$

$$\Rightarrow\ x = \frac{\pi}{6}, -\frac{\pi}{6} \quad \text{Now, } f\left(\frac{-\pi}{2}\right) = \frac{\pi}{2}, f'\left(\frac{\pi}{2}\right) = -\frac{\pi}{2}$$

$$f'\left(-\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2} + \frac{\pi}{6} \text{ and } f'\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

Clearly $\frac{\sqrt{3}}{2} - \frac{\pi}{6}$ is the greatest value of $f(x)$ and its least value $= -\frac{\pi}{2}$. Hence the required difference

$$= \frac{\sqrt{3}}{2} - \frac{\pi}{6} - \left(-\frac{\pi}{2}\right) = \frac{\sqrt{3}}{2} + \frac{\pi}{3}$$

9. Function $f(x) = x + \sin x$

On differentiating, we get

$$f'(x) = 1 + \cos x$$

For maxima or minima, put

$f'(x) = 0 \ \Rightarrow\ 1 + \cos x = 0 \Rightarrow \cos x = -1 \Rightarrow x = \pi$

Again differentiating, we get

$f''(x) = -\sin x$, at $x = \pi$, $f''(\pi) = 0$. Again differentiating, we get

$f'''(x) = -\cos x$, $f'''(\pi) = 1$ At $x = \pi$, $f(x)$ is minimum.

10. $f(x) = \int e^x (x-1)(x-2)dx$.

On differentiating w.r.t. x, we get

$$f'(x) = e^e (x-1)(x-2)$$

Since $f(x)$ is decreasing,

$\therefore$ $f'(x) < 0 \Rightarrow e^x (x-1)(x-2) < 0$

$\Rightarrow$ $(x-1)(x-2) < 0 \Rightarrow 1 < x < 2$

11. We have

$$xy = 4 \quad \Rightarrow \quad x\cdot\frac{dy}{dx} + y\cdot 1 = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{y}{x} = -\frac{4}{x^2} \qquad (\because xy = 4)$$

$\therefore$ Slope of the tangent $= -\frac{4}{x^2}$

Slope of the line $ax + by + c = 0$ is $= -\frac{a}{b}$

Since the given line is a tangent to the curve,

$\therefore$ $-\frac{4}{x^2} = -\frac{a}{b} \Rightarrow \frac{a}{b} > 0$

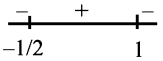


Fig. 13.17

If is possible only when $a > 0$, $b > 0$ or $a < 0$, $b < 0$.

12. The slope of the curve $y = x^2 \Rightarrow \frac{dy}{dx} = 2x$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(1,1)} = 2 = m_1 \text{ and } x = y^2 \Rightarrow 1 = 2y\frac{dy}{dx}$$

$\Rightarrow \quad \frac{dy}{dx}=\frac{1}{2y}\Rightarrow\left(\frac{dy}{dx}\right)_{(1,1)}=\frac{1}{2}=m_2$

$\therefore$ Angle of intersection at the point of (1, 1)

$$\tan\theta=\frac{m_1-m_2}{1+m_1m_2}=\frac{2-\frac{1}{2}}{1+2\times\frac{1}{2}}=\frac{3}{4}$$

$\Rightarrow \quad \theta=\tan^{-1}\left(\frac{3}{4}\right)$

13. $\because f(x)=xe^{x(1-x)}$

On differentiating w.r.t. x, we get

$$f'(x)=e^{x(1-x)}+x\cdot e^{x(1-x)}\cdot(1-2x)$$
$$=e^{x(1-x)}[1+x(1-2x)]$$
$$=e^{x(1-x)}\cdot(-2x^2+x+1)$$

It is clear that $e^{x(1-x)}>0$ for all x.

Now, by the sign rule for $-2x^2+x+1$

$f'(x)>0$, if $x\in\left[-\frac{1}{2},1\right]$. So, $f(x)$ is increasing on $\left[-\frac{1}{2},1\right]$

14. We have $AD=a+a\cos 2\theta$

$BC=2BD=2a\sin 2\theta$

$\therefore$ area of ΔABC

$$\Delta=\frac{1}{2}(BC)(AD)$$
$$=\frac{1}{2}(2a\sin 2\theta)a(1+\cos 2\theta)$$
$$=a^2(\sin 2\theta+\sin 2\theta\cos 2\theta)$$
$$=a^2\sin 2\theta+\frac{a^2}{2}\sin 4\theta$$

$\therefore \quad \frac{d\Delta}{d\theta}=2a^2\cos 2\theta+2a^2\cos 4\theta$

For max. or min. of Δ, $\frac{d\Delta}{d\theta}=0$

$\therefore \quad \cos 2\theta+\cos 4\theta=0$

$\Rightarrow \quad 2\theta=\pi-4\theta\Rightarrow 6\theta=\pi\Rightarrow\theta=\frac{\pi}{6}$

$\frac{d^2\Delta}{d\theta^2}=-4a^2\sin 2\theta-8a^2\sin 4\theta<0$

For $\theta=\pi/6$

15. Since $\cos\theta\le 1$ for all θ,

$\therefore$ $f(x)\le 1$ for all x

$\therefore$ Greatest value $=1$.

16. Let v be the volume and r be the radius at time t.

$\therefore \quad v=\frac{4}{3}\pi r^3$

When $v=288\pi$ cm^3, we have

$$288\pi\frac{4}{3}\pi r^3\Rightarrow r=6\text{ cm}$$

Now $v=\frac{4}{3}\pi r^3$

$\therefore \quad \frac{dv}{dt}=\frac{4}{3}\pi\cdot 3r^2\frac{dr}{dt}=4\pi r^2\frac{dr}{dt}$

$$=4\pi(6)^2\frac{dr}{dt}=144\pi\frac{dr}{dt}$$

But $\frac{dv}{dt}=4\pi$ (given)

$\therefore \quad 4\pi=144\pi\frac{dr}{dt}$ $\therefore \quad \frac{dr}{dt}=\frac{1}{36}$ cm/s

17. Here $x=2t^3-3t^2$. If $x<0$ then $2t^3-3t^2<0$

$\Rightarrow \quad t^2(2t-3)<0\Rightarrow 2t-3<0\Rightarrow t<3/2$.

$\therefore$ Required interval is $0<t<3/2$.

18. We have $y^2=2ax$ (i)

Put $x=\frac{a}{2}; y^2=2a\left(\frac{a}{2}\right) \quad \Rightarrow \quad y=\pm a$

$\therefore$ The points are $\left(\frac{a}{2},a\right)$ and $\left(\frac{a}{2},-a\right)$

On differentiating Eq. (i) w.r.t. x, we get

$$2y\frac{dy}{dx}=2a \quad \Rightarrow \quad \frac{dy}{dx}=\frac{a}{y}$$

At $\left(\frac{a}{2},a\right)$, $\frac{dy}{dx}=\frac{a}{y}=\frac{a}{a}=1=m_1$ (say)

At $\left(\frac{a}{2},-a\right)$, $\frac{dy}{dx}=\frac{a}{y}=\frac{a}{-a}=-1=m_2$

Since $m_1m_2=-1$, the two tangents are at right angles.

19. $V=kt^3$

When $t=2, V=4$ $\therefore$ $4=k(8)\Rightarrow k=\frac{1}{2}$.

$\therefore \quad V=\frac{1}{2}t^3$.

20. $f(x)=2x^3-3x^2-12x+4$

On differentiating w.r.t. x, we get

$$f'(x)=6x^2-6x-12$$

For maxima or minima, put $f'(x)=0$

$\Rightarrow \quad x^2-x-2=0\Rightarrow x=2,-1$

Again differentiating, we get $f''(x)=12x-6$

$\Rightarrow \quad f''(2)=+\text{ve}, \quad f''(-1)=-\text{ve}$

$\therefore$ Given function has one maximum and one minimum.

21. At the point (–1, 3), $\sin\theta = 1$, $\cos\theta = 0$

$\therefore \quad \theta = \dfrac{\pi}{2}$

$\therefore \quad \dfrac{dx}{d\theta} = -3\cos\theta, \dfrac{dy}{d\theta} = -2\sin\theta$

$\because \quad \dfrac{dx}{dy} = \dfrac{-3\cos\theta}{-2\sin\theta}$

$\therefore \quad \left.\dfrac{dx}{dy}\right]_{(-1,3)} = \left[\dfrac{3}{2}\cot\theta\right]_{\theta=\frac{\pi}{2}} = 0$

22. $f(x) = x^2 - x\sin x - \cos x$

$\Rightarrow \quad f'(x) = x(2 - \cos x)$

$\therefore$ Statement II is true.

But $f(-\pi) = \pi^2 + 1 > 0, f(0) < 0, f(\pi) > 0$

$\therefore \quad f(x) = 0$ for at least one x in $(-\pi, 0)$ and for at least one x in $(0, \pi)$

$\therefore$ Statement I is false.

23. $f'(x) = 3(x-5)^2, f''(x) = 6(x-5), f'''(x) = 5$

$\therefore \quad f'(5) = 0, f''(5) = 0, f'''(x) \neq 0$

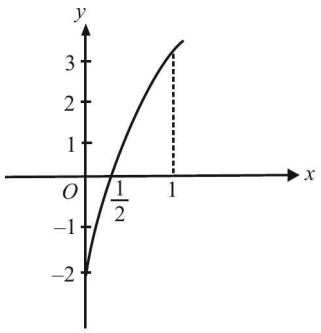


Fig. 13.18

24. $f'(x) = 2x - \dfrac{128}{x^2}$

$f'(x) = 0 \quad \Rightarrow \quad 2x^3 - 128 = 0$

$\Rightarrow \quad x = 4.$

Also, $f'(4^-) < 0$ and $f'(4^+) > 0$.

25. $y^2 = x + \sin x$

$\Rightarrow \quad 2y\dfrac{dy}{dx} = 1 + \cos x$

$\therefore \quad \dfrac{dy}{dx} = 0 \quad \Rightarrow \quad 1 + \cos x = 0$

$\Rightarrow \cos x = -1 \quad \Rightarrow \quad \sin x = 0.$

$\therefore \quad y^2 = x,$ which is a parabola.

26. We have $f(x) = (x+1)^{1/3} - (x-1)^{1/3}$

$\therefore \quad f'(x) = \dfrac{1}{3}\left[\dfrac{1}{(x+1)^{2/3}} - \dfrac{1}{(x-1)^{2/3}}\right]$

$= \dfrac{(x-1)^{2/3} - (x+1)^{2/3}}{3(x^2-1)^{2/3}}$

Clearly, $f'(x)$ does not exist at $x = \pm 1$.

Now, $f'(x) = 0$, then $(x-1)^{2/3} = (x+1)^{2/3}$

$\Rightarrow \quad x = 0$

Clearly, $f'(x) \neq 0$ for any other value of $x \in [0,1]$.

The value of $f(x)$ at $x = 0$ is 2.

Hence, the greatest value of $f(x)$ is 2.

27. Since $f(x) = \cot^{-1} x + x$

$\therefore \quad f'(x) = -\dfrac{1}{1+x^2} + 1 = \dfrac{x^2}{1+x^2}$

Hence, $f(x)$ is an increasing function, since $f'(x) \geq 0$ for all x.

28. Equations of two curves are

$$x^3 - 3xy^2 + 2 = 0 \qquad \text{(i)}$$

and $$3x^2y - y^3 - 2 = 0 \qquad \text{(ii)}$$

On differentiating Eqs. (i) and (ii) w.r.t. x, we get

$\left(\dfrac{dy}{dx}\right)_{C_1} = \dfrac{x^2 - y^2}{2xy}$ and $\left(\dfrac{dy}{dx}\right)_{C_2} = \dfrac{-2xy}{x^2 - y^2}$

Now, $\left(\dfrac{dy}{dx}\right)_{C_1} \times \left(\dfrac{dy}{dx}\right)_{C_2} = \left(\dfrac{x^2 - y^2}{2xy}\right)\left(\dfrac{-2xy}{x^2 - y^2}\right) = -1$

Hence, the two curves cut at the right angle.

29. $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$

$\Rightarrow \quad f'(x) = 6x^2 - 18ax + 12a^2$

For maxima or minima, put $f'(x) = 0$

$\Rightarrow \quad 6(x^2 - 3ax + 2a^2) = 0$

$\Rightarrow \quad x^2 - 3ax + 2a^2 = 0$

$\Rightarrow \quad x^2 - 2ax - ax + 2a^2 = 0$

$\Rightarrow \quad x(x-2a) - a(x-2a) = 0$

$\Rightarrow \quad (x-a)(x-2a) = 0$

$\Rightarrow \quad x = a, x = 2a$

Now, $f''(x) = 12x - 18a$

At $x = a$, $f''(x) = 12a - 18a = -6a$

$\therefore \quad f(x)$ will be maximum at $x = a$, i.e. $p = a$

At $x = 2a$, $f''(x) = 24a - 18a = -6a$

$\therefore$ $f(x)$ will be minimum at $x = 2a$, i.e. $q = 2a$

Given, $p^2 = q$

$\Rightarrow$ $a^2 = 2a \Rightarrow a = 2$

30. Let $f(x) = x + \frac{1}{x} \Rightarrow f'(x) = 1 - \frac{1}{x^2}$

For maxima and minima, put $f'(x) = 0$

$\Rightarrow$ $1 - \frac{1}{x^2} = 0 \Rightarrow x = \pm 1$

Now, $f''(x) = \frac{2}{x^3}$

At $x = 1$, $f''(x) = +ve$, minima.

and at $x = -1$, $f''(x) = -ve$, maxima.

Thus, $f(x)$ attains minimum value at $x = 1$.

31. $x = a(1 + \cos\theta), y = a\sin\theta$

$\Rightarrow$ $\frac{dx}{d\theta} = a(-\sin\theta), \frac{dy}{d\theta} = a\cos\theta$

$\therefore$ $\frac{dy}{dx} = \frac{-\cos\theta}{\sin\theta}$

Equation of the normal at the given point is

$$y - a\sin\theta = \frac{\sin\theta}{\cos\theta}[x - a(1 + \cos\theta)]$$

It is clear that in the given options, the normal passes through the point $(a, 0)$.

32. $f''(x) = 6(x - 1)$

$\Rightarrow$ $f'(x) = 3(x-1)^2 + c$ (i)

At the point (2, 1), the tangent to the graph is $y = 3x - 5$

Slope of the tangent $= 3 \Rightarrow f'(2) = 3$

$\therefore$ $f'(2) = 3(2-1)^2 + c = 3$

$\Rightarrow$ $3 + c = 3 \Rightarrow c = 0$

From Eq. (i), we have

$f'(x) = 3(x-1)^2$

$\Rightarrow$ $f(x) = (x-1)^3 + k$ (ii)

Since the graph passes through (2, 1),

$\therefore$ $1 = (2-1)^2 + k$

$\Rightarrow$ $k = 0$

$\therefore$ Equation of the function is $f(x) = (x-1)^3$.

33. $\frac{d}{dt}\left(\frac{4}{3}\pi r^3\right) = -50$

$\Rightarrow$ $\frac{dr}{dt} = -\frac{50}{4\pi r^2}$

$\Rightarrow$ $\left(\frac{dr}{dt}\right)_{r=15} = -\frac{50}{4\pi \times 255} = -\frac{1}{18\pi}$ cm/min

Hence, the thickness of the ice decreases by $1/18\pi$ cm/min.

34. $x = a(\cos\theta + \theta\sin\theta)$ and $y = a(\sin\theta - \theta\cos\theta)$

$\Rightarrow$ $\frac{dx}{d\theta} = a(-\sin\theta + \sin\theta + \theta\cos\theta)$

$\Rightarrow$ $\frac{dx}{d\theta} = a\theta\cos\theta$ and

$\Rightarrow$ $\frac{dy}{d\theta} = a(\cos\theta - \cos\theta + \theta\sin\theta)$

$\Rightarrow$ $\frac{dy}{d\theta} = a\theta\sin\theta$

$\therefore$ $\frac{dy}{dx} = \tan\theta$

So, the equation of the normal is

$$y - a\sin\theta + a\theta\cos\theta = -\frac{\cos\theta}{\sin\theta}(x - a\cos\theta - a\theta\sin\theta)$$

$\Rightarrow$ $y\sin\theta - a\sin^2\theta + a\theta\cos\theta\sin\theta$

$= -x\cos\theta + a\cos^2\theta + a\theta\sin\theta\cos\theta$

$\Rightarrow$ $x\cos\theta + y\sin\theta = a$

It is always at a constant distance a from the origin.

35. Since $f(x) = \frac{x}{2} + \frac{2}{x}$

$\therefore$ $f'(x) = \frac{1}{2} - \frac{2}{x^2}$

For maxima or minima, put $f'(x) = 0$

$\therefore$ $\frac{1}{2} - \frac{2}{x^2} = 0 \Rightarrow x = \pm 2$

Now, $f''(x) = \frac{4}{x^3}$

$\Rightarrow$ $f''(2) = \frac{4}{8} = \frac{1}{2} > 0$, minima and

$f''(-2) = -\frac{4}{8} = -\frac{1}{2} < 0$, maxima

Hence, $f(x)$ has local minimum at $x = 2$.

36. Since $y = x^2 - 5x + 6$

$\therefore$ $\frac{dy}{dx} = 2x - 5$

Now, $m_1 = \left(\frac{dy}{dx}\right)_{(2,0)} = 4 - 5 = -1$

and $m_2 = \left(\frac{dy}{dx}\right)_{(3,0)} = 6 - 5 = 1$

Now, $m_1 m_2 = -1 \times 1 = -1$

Hence, angle between the tangent and the curve is $\pi/2$.

37. Let the equation of the normal be

$$Y - y = -\frac{dx}{dy}(X - x)$$

It meets the x-axis at G. Therefore, the coordinates of G are $\left(x+y\frac{dy}{dx},0\right)$.

According to the given condition,

$$x+y\frac{dy}{dx}=2x \Rightarrow y\,dy=x\,dx$$

On integrating, we get

$$\frac{y^2}{2}=\frac{x^2}{2}+c \Rightarrow x^2-y^2=-2c.$$

38. Using the mean value theorem, we get

$$f'(c)=\frac{f(3)-f(1)}{3-1}$$

$$\Rightarrow \quad \frac{1}{c}=\frac{\log_e 3-\log_e 1}{2}$$

$$\Rightarrow \quad c=\frac{2}{\log_e 3}=2\log_3 e$$

39. Since $f(x)=\tan^{-1}(\sin x+\cos x)$

$$\Rightarrow \quad f'(x)=\frac{1}{1+(\sin x+\cos x)^2}(\cos x-\sin x)$$

$$=\frac{\sqrt{2}\cos\left(x+\frac{\pi}{4}\right)}{1+(\sin x+\cos x)^2}$$

For $f(x)$ to be increasing,

$$-\frac{\pi}{2}<x+\frac{\pi}{4}<\frac{\pi}{2}$$

$$\Rightarrow \quad -\frac{3\pi}{4}<x<\frac{\pi}{4}$$

Hence, option (b) is correct, which lies in the above interval.

40. Let $f(x)=x^3-px+q$

then $f'(x)=3x^2-p$

put $f'(x)=0 \Rightarrow x=\sqrt{\frac{p}{3}},-\sqrt{\frac{p}{3}}$

Now, $f''(x)=6x$

$\therefore$ At $x=\sqrt{\frac{p}{3}}, f''(x)=6\sqrt{\frac{p}{3}}>0$, minima and at

$x=-\sqrt{\frac{p}{3}}, f''(x)<0$, maxima

41. Let $f(x)=x^7+14x^5+16x^3+30x-560$

$f'(x)=7x^6+70x^4+48x^2+30>0$, for all $x\in\mathbb{R}$

$\therefore$ $f(x)$ is increasing

$\therefore$ $f(x)=0$ has only one solution.

42. $p(x)=x^4+ax^3+bx^2+cx+d$

$\Rightarrow$ $p'(x)=4x^3+3ax^2+2bx+c$

$\because$ $x=0$ is a solution for $p'(x)=0, \Rightarrow$ $c=0$

$\therefore$ $p(x)=x^4+ax^3+bx^2+d$

Also, we have $p(-1)<p(1)$

$\Rightarrow$ $1-a+b+d<1+a+b+d$

$\Rightarrow$ $a>0$

$\because$ $p'(x)=0$, only when $x=0$ and $p(x)$ is differentiable in $(-1, 1)$, we should have the maximum and minimum at the points $x=-1$, 0 and 1 only.

Also, we have $p(-1)<p(1)$

$\therefore$ maximum of $p(x)=\max\{p(0),p(1)\}$ and minimum of $p(x)=\min\{p(-1),p(0)\}$. In the interval [0, 1],

$$p'(x)=4x^3+3ax^2+2bx$$

$$=x(4x^2+3ax+2b)$$

$\because$ $p'(x)$ has only one root $x=0$, then $4x^2+3ax+2b=0$ has no real roots.

$\therefore$ $(3a)^2-32b<0$

$\Rightarrow$ $\frac{3a^2}{32}<b$

$\therefore$ $b>0$

Thus, we have $a>0$ and $b>0$

$\therefore$ $p'(x)=4x^3+4ax^2+2bx>0, \forall x\in(0,1)$

Hence, $p(x)$ is increasing in [0, 1].

$\therefore$ Maximum of $p(x)=p(1)$.

Similarly, $p(x)$ is decreasing in $[-1, 0]$.

Therefore, minimum of $p(x)$ does not occur at $x=-1$.

43. $\because$ $f(x)=\frac{1}{e^x+\frac{2}{e^x}}$

Using A.M. $\geq$ G.M., $\frac{e^x+\frac{2}{e^x}}{2}\geq\left(e^x\cdot\frac{2}{e^x}\right)^{\frac{1}{2}}$, as $e^x>0$

$\Rightarrow$ $e^x+\frac{2}{e^x}\geq 2\sqrt{2} \Rightarrow 0<\frac{1}{e^x+\frac{2}{e^x}}\leq\frac{1}{2\sqrt{2}}$

$\therefore$ $0<f(x)\leq\frac{1}{2\sqrt{2}}$, for all $x\in\mathbb{R}$

$\Rightarrow$ Statement 2 is true and Statement 1 as for some c

$\Rightarrow$ $f(c)=\frac{1}{3}$, which is not true.

Alternate method

$$f(x)=\frac{1}{e^x+2e^{-x}}=\frac{e^x}{e^{2x}+2}$$

$$\Rightarrow \quad f'(x)=\frac{(e^{2x}+2)e^x-2e^{2x}\cdot e^x}{(e^{2x}+2)^2}$$

$$\Rightarrow \quad f'(x)=0\Rightarrow e^{2x}+2=2e^{2x}$$

$$\Rightarrow \quad e^{2x}=2\Rightarrow e^x=\sqrt{2}$$

Maximum value of $f(x)=\frac{\sqrt{2}}{4}=\frac{1}{2\sqrt{2}}$

44. We have $y=x+\frac{4}{x^2}$

On differentiating with respect to x, we get

$$\frac{dy}{dx}=1-\frac{8}{x^3}$$

Since the tangent is parallel to the x-axis, therefore

$$\frac{dy}{dx}=0$$

$\Rightarrow x^3=8\Rightarrow x=2$ and $y=3$

45. $f(x)=e^x\sin x$

$f'(x)=e^x\cos x+\sin xe^x$

$f''(x)=-e^x\sin x+\cos xe^x+e^x\cos x+e^x\sin x$

$=2e^x\cos x$

For maximum slope, $f''(x)=0$

$\Rightarrow \quad 2e^x\cos x=0$

$\Rightarrow \quad \cos x=0$

$\Rightarrow \quad x=\frac{\pi}{2},\frac{3\pi}{2}$, for all $x\in[0,2\pi]$

$f'''(x)=2 \qquad [-e^x\sin x+e^x\cos x]$

at $x=\frac{\pi}{2}, f'''(x)<0$

and at $x=\frac{3\pi}{2}, f'''(x)>0$

$\therefore$ Slope is maximum at $x=\pi/2$

46. $f(x)=x^3-3x$

$\therefore \quad f'(x)=3x^2-3$

For maxima, $f'(x)=6x$

$\Rightarrow \quad 3x^2-3=0\Rightarrow x=\pm1$

$\therefore \quad x=1\in[0,2]$

At $x=1$, $f''(x)>0$, minima

$f(0)=0, f(1)=-2$ and $f(2)=2$

Hence, the maximum value is 2.

47. Since $2\pi k-\frac{\pi}{2}\le\sin x\le 2\pi k+\frac{\pi}{2}$

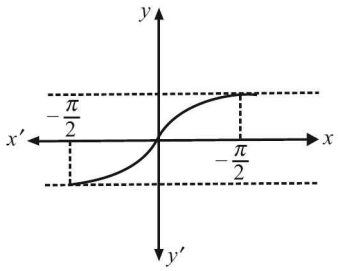


Fig. 13.19

For $k=0$,

$$-\frac{\pi}{2}<\sin x<\frac{\pi}{2}$$

which increases from -1 to 1.

Similarly, for other values of k, it increases from -1 to 1.

48. $f(x)$ satisfies Rolle's theorem in [3, 5].

$\therefore$ $f(x)$ is continuous in [3, 5] and $f(x)$ is differentiable in]3, 5[and

$f(a)=f(b)=0$, i.e. $f(3)=f(5)=0$

Let $\quad f(x)=(x-3)(x-5)$

$\therefore \quad f(x)=x^2-8x+15$

$$\therefore \quad \int_3^5(x^2-8x+15)dx=\left[\frac{x^3}{3}-\frac{8x^2}{2}+15x\right]_3^5$$

$$=\left(\frac{50}{3}-18\right)=-\frac{4}{3}$$

49. $\frac{x^2}{3}-\frac{y^2}{2}=1$ (i)

$$\Rightarrow \quad \frac{2x}{3}-y\frac{dy}{dx}=0$$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{2x}{3y}$$

Now, slope of the line $y=x$ is 1.

Since the tangent is parallel to the given line, then

$$\frac{2x}{3y}=1\Rightarrow x=\frac{3y}{2} \qquad \text{(ii)}$$

From Eqs. (i) and (ii), we get y = ± 2

$\therefore$ Points of contact are (3, 2) and (–3, –2).

$\therefore$ Equations of tangent are

$y-2=(x-3)\Rightarrow x-y-1=0$

and $\quad y+2=x+3\Rightarrow x-y+1=0$

50. $\because \quad f(x)=-x^3+4ax^2+2x-5$

$\because \quad f'(x)=-3x^2+8ax+2$

Since $f(x)$ is decreasing, for all x, therefore $f'(x)=<0$

$\Rightarrow \quad -3x^2+8ax+2<0$

$\Rightarrow \quad 3x^2-8ax-2>0$

Now, $D=(-8a)^2-4(3)(-2)<0$

$\Rightarrow \quad 64a^2+24<0$

$\Rightarrow \quad a^2<-\frac{3}{8}$

Hence, no value of a exists.

51. We have $y^3+3x^2=12y$

Differentiating w.r.t. x, we get

$$3y^2\frac{dy}{dx}+6x=12\frac{dy}{dx}$$

$\Rightarrow \quad 3(y^2-4)\frac{dy}{dx}=-6x \Rightarrow \frac{dy}{dx}=-\frac{2x}{y^2-4}$

$\Rightarrow \quad \frac{dx}{dy}=\frac{y^2-4}{2x}.$

Since the tangent is vertical,

$\therefore \quad \frac{dx}{dy}=0 \Rightarrow -\frac{y^2-4}{2x}=0 \Rightarrow y^2-4=0$

$\Rightarrow \quad y=\pm 2.$

But $y=-2$ gives $-8+3x^2=-24 \Rightarrow 3x^2=-16$

$\Rightarrow$ x is not real.

When $y=2$, then $8+3x^2=24 \Rightarrow 3x^2=16$

$\Rightarrow \quad x^2=\frac{16}{3} \Rightarrow x=\pm\frac{4}{\sqrt{3}}.$

Hence, the required points are $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$.

52. The function is not differentiable at $x=\frac{1}{2}$.

53. $f(x)=x^2+2bx+2c^2$ and $g(x)=-x^2-2cx+b^2$

$\therefore \quad f'(x)=2x+2b$ and $g'(x)=-2x-2c$ and

$f''(x)=2>0 \quad g''(x)=-2<0$

$\therefore$ $f(x)$ is minimum when $f'(x)=0$, i.e. when $x=-b$ and $g(x)$ is maximum when $g'(x)=0$, i.e. when $x=-c$.

$\therefore \quad \left|b^2-2b^2+2c^2\right|>\left|-c^2+2c^2+b^2\right|$

$\Rightarrow \quad \left|2c^2-b^2\right|>\left|c^2+b^2\right|$

$\therefore$ No real values of a and b are possible.

54. Point $P(3\sqrt{3}\cos\theta, \sin\theta)$ lies on $\frac{x^2}{27}+\frac{y^2}{1}=1.$

$\therefore$ The tangent at P is

$$\frac{x(3\sqrt{3}\cos\theta)}{27}+\frac{y(\sin\theta)}{1}=1$$

$\Rightarrow \quad \frac{x\cos\theta}{3\sqrt{3}}+\frac{y\sin\theta}{1}=1$

$\therefore$ Sum of intercepts, $l=\frac{3\sqrt{3}}{\cos\theta}+\frac{1}{\sin\theta}.$

Now $\frac{dl}{d\theta}3\sqrt{3}\sec\theta\tan\theta-\operatorname{cosec}\theta\cot\theta.$

l is least $\frac{dl}{d\theta}=0 \Rightarrow \frac{3\sqrt{3}\sin\theta}{\cos^2\theta}-\frac{\cos\theta}{\sin^2\theta}=0$

$\Rightarrow \quad 3\sqrt{3}\sin^3\theta-\cos^3\theta=0$

$\Rightarrow \quad 3\sqrt{3}\tan^3\theta-1=0$

$\Rightarrow \quad \tan^3\theta=\frac{1}{3\sqrt{3}}$

$\Rightarrow \quad \tan^3\theta=\frac{1}{3^{3/2}}$

$\Rightarrow \quad \tan\theta=\frac{1}{\sqrt{3}} \Rightarrow \theta=\frac{\pi}{6}$

55. We have $f(x)=x^3+bx^2+cx+d.$

$\therefore \quad f'(x)=3x^2+2bx+c$ and $D=4b^2-12c<0$

$$\left[\because b^2<c \Rightarrow 4b^2<4c \Rightarrow 4b^2<12c \atop \Rightarrow b^2-12c<0\right]$$

$\therefore \quad f'(x)>0$

$\Rightarrow$ $f(x)$ is an increasing function.

56. $\lim\limits_{x\to 0^+} f(x)=f(0)$

$\Rightarrow \quad \lim\limits_{x\to 0^+}\frac{\log x}{x^{-\alpha}}=0$

$\Rightarrow \quad \lim\limits_{x\to 0^+}\frac{1/x}{-\alpha n^{-\alpha-1}}=0 \Rightarrow \alpha>0$

57. $\lim\limits_{x\to 0}\frac{f(x^2)-f(x)}{f(x)-f(0)}$

$$=\lim_{x\to 0}\frac{2xf'(x^2)-f'(x)}{f'(x)-(0)} \qquad \text{[L'Hôpital's rule]}$$

$$=\frac{0-f'(0)}{f'(0)}=-1.$$

58. We have $2x+\sqrt{6}y=2$ (i)

and $x^2-2y^2=4$ (ii)

From (i), we get $x=\frac{2-\sqrt{6}y}{2}$ (iii)

Putting in (ii), we get $\left(\frac{2-\sqrt{6}y}{2}\right)^2-2y^2=4$

$\Rightarrow \quad 4+6y^2-4\sqrt{6}y-8y^2=16$

$\Rightarrow \quad 2y^2+4\sqrt{6}y+12=0 \quad \Rightarrow \quad y^2+2\sqrt{6}y+6=0$

$\Rightarrow \quad (y+\sqrt{6})^2=0 \quad \Rightarrow \quad y=-\sqrt{6},-\sqrt{6}.$

Putting in (iii), we get $x=\dfrac{2-\sqrt{6}(-\sqrt{6})}{2}=\dfrac{2+6}{2}=4, 4.$

Hence, the point of contact is $(4,-\sqrt{6})$.

59. Let $g(x)=f(x)-x^2$. We have

$g(1)=f(1)-1=1-1=0.$

$g(2)=f(2)-4=4-4=0.$ and

$g(3)=f(3)-9=9-9=0$

Thus, by Rolle's theorem,

$g'(x)=0$ for $c\in(1,2)$ and

$g'(x)=0$ for $d\in(2,3)$.

Again by Rolle's theorem,

$g''(x)=0$ for $x\in(1,3)$.

$\therefore \quad f''(x)=2$ for $x\in(c,d)$.

60. The reqd. equation which satisfies the condition is

$$f(x)=\frac{1}{4}(19x^3-57x+34)$$

$f(x)$ has local maxima at $x=-1$ and local minima at $x=1$

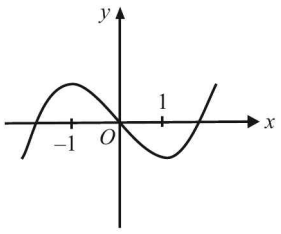


Fig. 13.20

Hence, $f(x)$ is increasing for $x\in[1, 2\sqrt{5}]$.

61. $f(x)=\begin{cases}\tan x/x, & x\neq 0\\ 1, & x=0\end{cases}$

In the right neighbourhood of '0', $\tan x > x$

In the left neighbourhood of '0', $\tan x < x$

$\dfrac{\tan x}{x}>1$ as $(x<0)$

at $x=0 \quad f(x)=1$

$\Rightarrow$ $x=0$ is the point of minima

So Statement 1 is true and Statement 2 is obviously true.

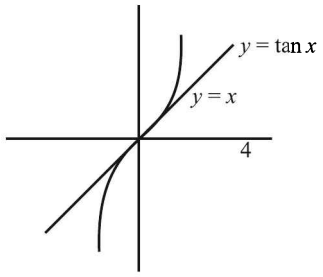


Fig. 13.21

62. $v=\dfrac{4}{3}\pi r^3$

After 49 min volume $=4500\pi-49(72\pi)=972\pi$

$\dfrac{4}{3}\pi r^3=972\pi \quad \Rightarrow \quad r^3=729 \quad \Rightarrow \quad r=9$ m

$v=\dfrac{4}{3}\pi r^3 \quad \Rightarrow \quad \dfrac{dv}{dt}=\dfrac{4}{3}\times 3\pi r^2\dfrac{dr}{dt}$

$\Rightarrow \quad 72\pi=4\pi r^2\dfrac{dr}{dt} \quad \Rightarrow \quad \dfrac{dr}{dt}=\dfrac{72}{4\cdot 9\cdot 9}=\dfrac{2}{9}$ [$\because$ $r=9$ m]

63. $f'(x)=\dfrac{1}{x}+2bx+a$

at $x=-1, \quad -1-2b+a=0$

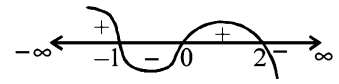
$$a-2b=1 \qquad \text{(i)}$$

at $x=2, \quad \dfrac{1}{2}+4b+a=0$

$$a+4b=-\frac{1}{2} \qquad \text{(ii)}$$

On solving Eqs. (i) and (ii), $a=(1/2)$, $b=-(1/4)$

$$f'(x)=\frac{1}{x}-\frac{x}{2}+\frac{1}{2}=\frac{2-x^2+x}{2x}=\frac{-(x+1)(x-2)}{2x}$$



So maxima is at -1, 2.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The values of p for which the function

$f(x)=\left(\dfrac{\sqrt{p+4}}{1-p}-1\right)x^5-3x+\log 5$ decreases for all real x are

(a) $(-\infty,\infty)$ (b) $\left[-4,\dfrac{3-\sqrt{21}}{2}\right]\cup(1,\infty)$

(c) $\left[-3,\dfrac{5-\sqrt{27}}{2}\right]\cup(2,\infty)$ (d) $[1,\infty)$

2. The number of points of maxima, minima and non-differentiability of $f(x)=\big|x^2-4|x|\big|$ is respectively

(a) 2, 3, 2 (b) 3, 3, 2

(c) 2, 3, 3 (d) 3, 2, 3

3. The number of critical points of $f(x)=\max(\sin x,\cos x)$ for $x\in(0,2\pi)$ is

(a) 2 (b) 5

(c) 3 (d) none of these

4. If the function $f(x)$ increases in the interval (a, b), then the function $\phi(x) = [f(x)]^2$
(a) increases in (a, b)
(b) decreases in (a, b)
(c) we cannot say whether $\phi(x)$ increases or decreases in (a, b)
(d) none of these

5. If the line $ax + by + c = 0$ is normal to the curve $xy + 5 = 0$, then
(a) $a > 0, b > 0$ (b) $b > 0, a < 0$
(c) $b < 0, a > 0$ (d) none of these

6. If $f(x) = |4x - x^2 - 3|$ when $x \in [0, 4]$, then
(a) $x = 1$ is a global maximum
(b) $x = 3$ is a global maximum
(c) $x = 2$ is a local maximum
(d) none of these

7. The function $f(x) = x^3 - 6x^2 + ax + b$ satisfies the conditions of Rolle's theorem in $[1, 3]$. The values of a and b are
(a) $11, -6$ (b) $-6, 11$
(c) $-11, 6$ (d) $6, -11$

8. The maximum value of $f(x) = |x \ln x|$ in $x \in (0, 1)$ is
(a) $1/e$ (b) e
(c) 1 (d) none of these

9. If the function $f(x) = \cos|x| - 2ax + b$ increases for all $x \in \mathbb{R}$, then
(a) $a \le b$ (b) $a = \frac{b}{2}$
(c) $a \le -\frac{1}{2}$ (d) $a \ge -\frac{3}{2}$

10. The number of solutions of the equation $a^{f(x)} + g(x) = 0$, where $a > 0$, $g(x) = 0$, where $a < 0$, $g(x) \neq 0$ and has minimum value $\frac{1}{2}$ is
(a) 1 (b) 2
(c) 0 (d) infinity

11. All points on the curve $y^2 = 4a\left(x + a\sin\frac{x}{a}\right)$ at which the tangents are parallel to the x-axis lie on a
(a) circle (b) parabola
(c) line (d) none of these

12. The product of the minimum value of x^x and the maximum value of $\left(\frac{1}{x}\right)^x$ is
(a) e (b) e^{-1}
(c) 1 (d) e^2

13. Let $f(x) = |x - 1| + a$ if $x \le 1$ and $f(x) = 2x + 3$ if $x > 1$. If $f(x)$ has a local minimum at $x = 1$, then
(a) $a = 7$ (b) $a = 8$
(c) $a = 9$ (d) $a \le 5$

14. The curves $x^3 + pxy^2 = -2$ and $3x^2y - y^3 = 2$ are orthogonal for
(a) $p = 3$ (b) $p = -3$
(c) no value of p (d) $p = \pm 3$

15. A function f is such that $f'(2) = f''(2) = 0$ and $f(x)$ has a local maximum of -17 at $x = 2$, then $f(x)$ may be
(a) $f(x) = -17 - (x-2)^n n \ge 4$, where n is odd
(b) $f(x) = -17 - (x-2)^n n \ge 3$
(c) $f(x) = -17 + (x-2)^n n \ge 3$
(d) $f(x) = 17 + (x-2)^n n \le 4$, where n is odd

16. If $f(x) = (\sin^2 x - 1)^n (2 + \cos^2 x)$ then $x = \pi/2$ is a point of
(a) local maximum, if n is odd
(b) local minimum, if n is odd
(c) local maximum, if n is even
(d) none of these

17. If a curve with equation of the form $y = ax^4 + bx^3 + cx + d$ has zero gradient at the point $(0,1)$ and also touches the x-axis at the point $(-1, 0)$, then the values of x for which the curve has a negative gradient are
(a) $x > -1$ (b) $x < 1$
(c) $x < -1$ (d) $-1 \le x \le 1$

18. The radius of a right circular cylinder of the greatest curved surface which can be inscribed in a given right circular cone is
(a) one third that of the cone
(b) $1/\sqrt{2}$ times that of the cone
(c) 2/3 that of the cone
(d) 1/2 that of the cone

19. The number that exceeds its square by the greatest amount is
(a) -1 (b) 0
(c) 1/2 (d) 1

20. Three normals are drawn to the parabola $y^2 = 4x$ from the point $(c, 0)$. These normals are real and distinct when
(a) $c = 4$ (b) $c = 1$
(c) $c = 2$ (d) $c = 3$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. The dimensions of the rectangle of maximum area that can be inscribed in the ellipse $(x/4)^2 + (y/3)^2 = 1$ are
(a) $\sqrt{8}, \sqrt{2}$ (b) 4, 3
(c) $2\sqrt{8}, 3\sqrt{2}$ (d) $\sqrt{2}, \sqrt{6}$

2. The maximum value of $x^{1/x}$ is
(a) $1/e$ (b) $e^{1/e}$
(c) e (d) $1/e^e$

3. If the curves $ax^2 + by^2 = 1$ and $Ax^2 + By^2 = 1$ intersect orthogonally, then
(a) $\frac{1}{a}+\frac{1}{A}=\frac{1}{b}+\frac{1}{B}$ (b) $\frac{1}{a}-\frac{1}{A}=\frac{1}{b}-\frac{1}{B}$
(c) $\frac{1}{a}+\frac{1}{b}=\frac{1}{B}-\frac{1}{A}$ (d) $\frac{1}{a}-\frac{1}{b}=\frac{1}{A}-\frac{1}{B}$

4. Tangents are drawn to $x^2 + y^2 = 16$ from the point $P(0, h)$. These tangents meet the x-axis at A and B. If the area of the triangle PAB is minimum, then
(a) $h = 12\sqrt{2}$ (b) $h = 6\sqrt{2}$
(c) $h = 8\sqrt{2}$ (d) $h = 4\sqrt{2}$

5. A population $p(t)$ of 1000 bacteria introduced into the nutrient medium grows according to the relation $p(t) = 1000 + \frac{1000t}{100 + t^2}$. The maximum size of this bacterial population is
(a) 1100 (b) 1250
(c) 1050 (d) 5250

6. In a regular triangular prism, the distance from the centre of one base to one of the vertices of the other base is l. The altitude of the prism for which the volume is greatest is
(a) $l/2$ (b) $l/\sqrt{3}$
(c) $l/3$ (d) $l/4$

7. The angle at which the curve $y = ke^{kx}$ intersects the y-axis is
(a) $\tan^{-1}(k^2)$ (b) $\cot^{-1}(k^2)$
(c) $\sin^{-1}\left(\frac{1}{\sqrt{1+k^4}}\right)$ (d) $\frac{\pi}{2} - \tan^{-1}(k^2)$

8. Two vertices of a rectangle are on the positive x-axis. The other two vertices lie on the circle $x^2 + y^2 = 1$. Then the maximum area of the rectangle is
(a) 4/ 3 (b) 3/5
(c) 4/5 (d) 3/4

9. The lower corner of a leaf in a book is folded over so as to just reach the inner edge of the page. The fraction of width folded over if the area of the folded part is minimum is
(a) 5/8 (b) 2/3
(c) 3/4 (d) 4/5

10. The maximum value of $\frac{\log x}{x}$ is equal to
(a) $2/e$ (b) $1/e$
(c) e (d) 1

11. The minimum value of $\left(1+\frac{1}{\sin^n \alpha}\right)\left(1+\frac{1}{\cos^n \alpha}\right)$ is
(a) 1 (b) 2
(c) $(1+2^{n/2})^2$ (d) none of these

12. The length of the subtangent to the curve $x^2 + xy + y^2 = 7$ at $(1, -3)$ is
(a) 3 (b) 5
(c) 3/5 (d) 15

13. The point(s) on the curve $y^3 + 3x^2 = 12y$ where the tangent is vertical is (are)
(a) $(\pm 4/\sqrt{3} - 2)$ (b) $(\pm\sqrt{11/3}, 1)$
(c) (0, 0) (d) $(\pm 4/\sqrt{3}, 2)$

14. If PQ and PR are the two sides of a triangle, then the angle between them which gives maximum area of the triangle is
(a) π (b) $\pi/3$
(c) $\pi/4$ (d) $\pi/2$

15. The points of contact of the vertical tangents to $x = 2 - 3 \sin\theta$ and $y = 3 + 2 \cos\theta$ are
(a) (2, 5), (2, 1) (b) (–1, 3), (5, 3)
(c) (2, 5) (5, 3) (d) (–1, 3), (2, 1)

16. Let $f : R \to R$, where $f(x) = x^3 - ax, a \in \mathbb{R}$. Then the set of values of a so that $f(x)$ is increasing in its entire domain is
(a) $(-\infty, 0)$ (b) $(0, \infty)$
(c) $(-\infty, \infty)$ (d) none of these

17. Let $f(x) = x - \sin x$ and $g(x) = x - \tan x$, where $x \in (0, \pi/2)$. Then for these values of x,
(a) $f(x)\, g(x) > 0$ (b) $f(x)\, g(x) > 0$
(c) $\frac{f(x)}{g(x)} > 0$ (d) none of these

18. The function $f(x) = \sqrt{3}\sin x - \cos x$ will increase monotonically in the interval
(a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{3}, \frac{2\pi}{3}\right]$
(c) $[0, \pi]$ (d) none of these

19. Area of the triangle formed by the positive x-axis and the normal and the tangent to $x^2 + y^2 = 4$ at $(1, \sqrt{3})$ is

(a) $2\sqrt{3}$ sq. units (b) $\sqrt{3}$ sq. units

(c) $4\sqrt{3}$ sq. units (d) none of these

20. The greatest value of the function $f(x) = \dfrac{\sin 2x}{\sin [2x + (\pi/4)]}$ in the interval $[0, \pi/4]$ is

(a) $1/\sqrt{2}$ (b) $\sqrt{2}$

(c) 1 (d) $-\sqrt{2}$

ANSWERS

Level 1

1. (b)	**2.** c)	**3.** (c)	**4.** (c)	**5.** (a)	**6.** (c)	**7.** (a)	**8.** (a)	**9.** (c)	**10.** (c)
11. (b)	**12.** (c)	**13.** (d)	**14.** (b)	**15.** (a)	**16.** (a)	**17.** (c)	**18.** (d)	**19.** (c)	
20. (a) and (d)									

Level 2

1. (c)	**2.** (b)	**3.** (b, d)	**4.** (d)	**5.** (c)	**6.** (b)	**7.** (b, d)	**8.** (c)	**9.** (b)	**10.** (b)
11. (c)	**12.** (d)	**13.** (d)	**14.** (d)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (a)	**20.** (b)

14

G.W. LEIBNIZ

Indefinite Integrals

- Indefinite Integrals
- Standard Formulae
- Integration by Substitution
- Forms of Partial Fractions
- Integration by Parts
- Different Types of Integrals

INDEFINITE INTEGRALS

Integration

Integration is called the inverse process of differentiation. This helps to find the function whose differential coefficient is known. Thus, a function $F(x)$ is called a *primitive* or an *antiderivative* of a function $f(x)$ if

$$\frac{d}{dx}[F(x)] = f(x) \quad \text{or} \quad \int f(x)\,dx = F(x) + c$$

where the symbol '$\int$' is the notation of integration, $f(x)$ is called the integrand and x is called the variable of integration.

STANDARD FORMULAE

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + c$
2. $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{(n+1)a} + c$
3. $\int 0\,dx = c$
4. $\int 1\,dx = x + c$
5. $\int k\,dx = kx + c$
6. $\int \sin x\,dx = -\cos x + c, \ \int \sin(ax)dx = -\frac{\cos(ax)}{a} + c$
7. $\int \cos x\,dx = \sin x + c, \ \int \cos(ax)dx = \frac{\sin(ax)}{a} + c$
8. $\int \sec^2 x\,dx = \tan x + c, \ \int \sec^2(ax)dx = \frac{\tan(ax)}{a} + c$
9. $\int \text{cosec}^2 x\,dx = -\cot x + c, \ \int \text{cosec}^2(ax)dx = -\frac{\cot(ax)}{a} + c$
10. $\int \sec x \tan x\,dx = \sec x + c$
11. $\int \text{cosec}\, x \cot x\,dx = -\text{cosec}\, x + c$
12. $\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1} x + c$ or $-\cos^{-1} x + c$
13. $\int \frac{1}{1+x^2}\,dx = \tan^{-1} x + c$ or $-\cot^{-1} x + c$
14. $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + c$ or $-\text{cosec}^{-1} x + c$

15. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c$

16. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$

17. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$

18. $\int \frac{1}{x}dx = \log|x| + c$

19. $\int e^x dx = e^x + c$

20. $\int e^{ax}dx = \frac{e^{ax}}{a} + c$

21. $\int a^x dx = \frac{a^x}{\log_e a} + c,\ \int a^{mx}dx = \frac{a^{mx}}{m\log_e a} + c$

22. $\int \frac{f'(x)}{f(x)} = \log|f(x)| + c$

23. $\int [f(x)]^n f'(x)\,dx$

24. $\int \cot x dx = \log|\sin x| + c,\ \int \cot ax dx = \frac{1}{a}\log\sin|ax| + c$

25. $\int \tan x\,dx = -\log|\cos x| + c$ or $\log|\sec x| + c$,

$\int \tan ax\,dx = -\frac{1}{a}\log|\cos ax| + c$ or $\frac{1}{a}\log|\sec ax| + c$

26. $\int \sec x\,dx = \log|(\sec x + \tan x)| + c$ or $\log\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + c$

$\int \sec ax\,dx = \frac{1}{a}\log|(\sec ax + \tan ax)| + c$

or $\frac{1}{a}\log\left|\tan\left(\frac{\pi}{4} + \frac{ax}{2}\right)\right| + c$

27. $\int \text{cosec}\, x\,dx = \log|\text{cosec}\, x - \cot x| + c$ or $\log\left|\tan\frac{x}{2}\right| + c$

$\int \text{cosec}\, ax dx = \frac{1}{a}\log|\text{cosec}\, ax - \cot ax| + c$

or $\frac{1}{a}\log\left|\tan\frac{ax}{2}\right| + c$

28. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c$

29. $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$

30. $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log\left|x + \sqrt{x^2 + a^2}\right| + c$

31. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log\left|x + \sqrt{x^2 - a^2}\right| + c$

INTEGRATING BY SUBSTITUTION

The given integral $\int f(x)\,dx$ is transformed into a standard form by changing the independent variable x into t by putting $x = g(t)$.

When $x = g(t), \frac{dx}{dt} = g'(t)$ or $dx = g'(t)dt$

$\therefore \quad \int f(x)dx = \int f[g(t)]g'(t)dt$

After integration, we transform the variable t to the original variable x.

Note We make the substitution for a function whose derivative generally occurs in the integral.

Illustration 1. Evaluate $\int (a + b\cos x)^n \sin x dx$.

Solution: Let $I = \int (a + b\cos x)^n \sin x dx$

Put $t = a + b\cos x \ \Rightarrow\ dt = -b\sin dx$

$\Rightarrow \quad I = \int t^n\left(-\frac{dt}{b}\right) = -\frac{1}{b}\int t^n dt = -\frac{t^{n+1}}{b(n+1)} + c$

$\therefore \quad I = -\frac{(a + b\cos x)^{n+1}}{b(n+1)} + c$

Illustration 2. Evaluate $\int \frac{\log x \sin[1 + (\log x)^2]}{x} dx.$

Solution: Let $I = \int \frac{\log x \sin[1 + (\log x)^2]}{x} dx$

Put $1 + (\log x)^2 = t$

$\Rightarrow \quad dt = 2(\log x).\frac{1}{x}dx$

$\Rightarrow \quad I = \int \frac{\sin t}{2} dt = -\frac{1}{2}\cos t + c$

$\therefore \quad I = -\frac{1}{2}\cos[1 + (\log x)^2] + c$

Function	Substitution
$f(\sqrt{a^2 - x^2})$	$x = a\sin\theta$ or $x = a\cos\theta$
$f(\sqrt{x^2 - a^2})$	$x = a\sec\theta$ or $x = a\,\text{cosec}\,\theta$
$f(x^2 + a^2), f(\sqrt{x^2 + a^2})$	$x = a\tan\theta$ or $x = a\cot\theta$
$f\left(\sqrt{\frac{a-x}{a+x}}\right), f\left(\sqrt{\frac{a+x}{a-x}}\right)$	$x = a\cos 2\theta$
$f\left(\sqrt{\frac{x-a}{b-x}}\right), f(\sqrt{(x-a)(x-b)})$	$x = a\cos^2\theta + b\sin^2\theta$

FORMS OF PARTIAL FRACTIONS

1. $\dfrac{px+q}{(x-a)(x-b)}=\dfrac{A}{x-a}+\dfrac{B}{x-b};\ a\neq b$
2. $\dfrac{px+q}{(x-a)^2}=\dfrac{A}{x-a}+\dfrac{B}{(x-a)^2}$
3. $\dfrac{px^2+qx+r}{(x-a)(x-b)(x-c)}=\dfrac{A}{x-a}+\dfrac{B}{x-b}+\dfrac{C}{x-c}$
4. $\dfrac{px^2+qx+r}{(x-a)^2(x-b)}=\dfrac{A}{x-a}+\dfrac{B}{(x-a)^2}+\dfrac{C}{x-b}$
5. $\dfrac{px^2+qx+r}{(x-a)^3(x-b)}=\dfrac{A}{x-a}+\dfrac{B}{(x-a)^2}+\dfrac{C}{(x-a)^3}+\dfrac{D}{x-b}$
6. $\dfrac{px^2+qx+r}{(x-a)(x^2+bx+c)}=\dfrac{A}{x-a}+\dfrac{Bx+C}{x^2+bx+c}$, where x^2+bx+c cannot be factorized.
7. $\dfrac{px^3}{(qx-a)(x-b)(x-c)}=\dfrac{p}{q}+\dfrac{A}{x-a}+\dfrac{B}{x-b}+\dfrac{C}{x-c}$

Illustration 3. Evaluate $\int\dfrac{x+5}{(x+1)(x-2)^2}dx$.

Solution: Let $I=\int\dfrac{x+5}{(x+1)(x-2)^2}dx$

Since the integrand is a proper fraction, then factorizing the denominator into partial fractions, we get

$$\frac{x+5}{(x+1)(x-2)^2}=\frac{A}{x+1}+\frac{B}{x-2}+\frac{C}{(x-2)^2}$$

$$\Rightarrow\quad x+5=A(x-2)^2+B(x+1)(x-2)+C(x+1)$$

Comparing the respective coefficients on both sides of the equality, we have

$$A=\frac{4}{9},\ B=-\frac{4}{9}\text{ and }C=\frac{7}{3}$$

$$\Rightarrow\quad I=\int\frac{x+5}{(x+1)(x-2)^2}dx=\frac{4}{9}\int\frac{1}{x+1}dx$$
$$-\frac{4}{9}\int\frac{1}{x-2}dx+\frac{7}{3}\int\frac{1}{(x-2)^2}dx$$

$$\therefore\quad I=\frac{4}{3}\log|x+1|-\frac{4}{9}\log|x-2|-\frac{7}{3}\left(\frac{1}{x-2}\right)+c$$

Illustration 4. Evaluate $\int\dfrac{1}{(x^2+1)(x^2+4)}dx$.

Solution: Let $I=\int\dfrac{1}{(x^2+1)(x^2+4)}dx$

Since the integrand is a proper fraction, we have to factorize the denominator into partial fractions. Moreover, the denominator has two pairs of conjugate complex roots.

$$\therefore\quad\frac{1}{(x^2+1)(x^2+4)}=\frac{Ax+B}{x^2+1}+\frac{Cx+D}{x^2+4}$$

$$\Rightarrow\quad 1=(Ax+B)(x^2+4)+(Cx+D)(x^2+1)$$

Comparing the respective coefficients on both sides of the equation, we have

$$A=0,\ B=\frac{1}{3},\ C=0\text{ and }D=-\frac{1}{3}$$

$$\Rightarrow\quad I=\int\frac{0(x)+\left(\frac{1}{3}\right)}{x^2+1}dx+\int\frac{(0)x+\left(-\frac{1}{3}\right)}{x^2+4}dx$$

$$\Rightarrow\quad I=\frac{1}{3}\int\frac{1}{x^2+1}dx-\frac{1}{3}\int\frac{1}{x^2+4}dx$$

$$\therefore\quad I=\frac{1}{3}\left[\tan^{-1}(x)-\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)\right]+c$$

INTEGRATION BY PARTS

1. If u and v are two functions of x, then

$$\int uv\,dx=u\left(\int v dx\right)=\int\left\{\frac{du}{dx}\int v dx\right\}dx$$

(1st function) × (Integral of 2nd function) – Integral of (Diff. of 1st function) × (Integral of 2nd function)

Note

(i) We can also choose the first function as the function which comes first in the word **ILATE**, where

I-Stands for the inverse trigonometric functions

$(\sin^{-1}x,\cos^{-1}x,\tan^{-1}x,\text{etc.})$

L-Stands for the logarithmic functions

$[\log x,\log 2x,(\log x)^2,\text{etc.}]$

A-Stands for the algebraic functions

$[x,x+1,(2x+1)^3,\text{etc.}]$

T-Stands for the trigonometric functions

$[\sin x,\cos 5x,(\tan x)^2,\text{etc.}]$

E-Stands for the exponential functions

$(e^x,e^{5x},e^{x^2},\text{etc.})$

(ii) If both the functions are trigonometrical, take that function as v whose integral is simpler.

(iii) If both the functions are algebraic, take that function as u whose differential is simpler.

2. $\int e^x[f(x)+f'(x)]dx=e^x f(x)+c$
3. $\int\sqrt{x^2-a^2}\,dx=\dfrac{x\sqrt{x^2-a^2}}{2}-\dfrac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+c$

4. $\int\sqrt{x^2+a^2}\,dx=\frac{x\sqrt{x^2+a^2}}{2}+\frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+c$

5. $\int\sqrt{a^2-x^2}\,dx=\frac{x\sqrt{a^2-x^2}}{2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+c$

6. In rational algebraic functions, if the degree of the numerator is greater than or equal to the degree of the denominator, then always divide the numerator by the denominator and use the result.

$$\frac{\text{Numerator}}{\text{Denominator}}=\text{Quotient}+\frac{\text{Remainder}}{\text{Denominator}}$$

e.g. $\int\frac{x^3}{x+2}dx=\int\left(x^2-2x+4-\frac{8}{x+2}\right)dx$

$$=\frac{x^3}{3}-x^2+4x-8\log|x+2|+c$$

7. For integrals of the form $\int\sin^n x\,dx$ and $\int\cos^n x\,dx$, where $n\le 4$, we express $\sin^n x$ and $\cos^n x$ in terms of sines and cosines of multiples of x by using trigonometrical identities given below:

(i) $\sin^2 x=\frac{1-\cos 2x}{2}$

(ii) $\cos^2 x=\frac{1+\cos 2x}{2}$

(iii) $\sin 3x=3\sin x-4\sin^3 x$

(iv) $\cos 3x=4\cos^3 x-3\cos x$

8. For integrals of the form

$\int\sin mx\cos nx\,dx$, $\int\sin mx\sin nx\,dx$,

$\int\cos mx\cos nx\,dx$ and $\int\cos mx\sin nx\,dx$,

we use the following trigonometrical identities:

(i) $2\sin A\cos B=\sin(A+B)+\sin(A-B)$

(ii) $2\cos A\sin B=\sin(A+B)-\sin(A-B)$

(iii) $2\cos A\cos B=\cos(A+B)+\cos(A-B)$

(iv) $2\sin A\sin B=\cos(A-B)-\cos(A+B)$

9. Integrals of the form $\int\sin^m x\cos^n x\,dx$, where m, n are positive integers. In this form, the following substitutions are useful:

(i) If m is odd, i.e. power of sin x is odd, put cos $x = t$.

(ii) If n is odd, i.e. power of cos x is odd, put sin $x = t$.

(iii) If both m and n are even, then we use De'Moivre's theorem.

Illustration 5. Evaluate $\int\sin^2 x\cos^3 x\,dx$.

Solution: Let $I=\int\sin^2 x\cos^3 x\,dx$

Put $\sin x=t\Rightarrow dt=\cos x\,dx$

$$\Rightarrow\quad I=\int t^2\cos^3 x\left(\frac{dt}{\cos x}\right)$$

$$\Rightarrow\quad I=\int t^2\cos^2 x\,dt=\int t^2(1-t^2)\,dt$$

$$\Rightarrow\quad I=\int t^2dt-\int t^4dt=\frac{t^3}{3}-\frac{t^5}{5}+c$$

$$\therefore\quad I=\frac{\sin^3 x}{3}-\frac{\sin^5 x}{5}+c$$

10. Integrals of the form $\int\frac{dx}{ax^2+bx+c}$. We can express $\int\frac{dx}{a[(x+\alpha)^2+\beta^2]}$ (by perfect square). Then we can use the formula $\int\frac{dx}{a^2+x^2}$.

11. Integrals of the form $\int\frac{dx}{\sqrt{ax^2+bx+a}}$. We can express $\int\frac{dx}{a[(x+\alpha)^2+\beta^2]}$ (by perfect square). Then we can use the formula $\int\frac{dx}{\sqrt{a^2+x^2}}$.

12. Integrals of the form $\int(px+q)\sqrt{ax^2+bx+c}\,dx$.

Let $px+q=A\frac{d}{dx}(ax^2+bx+c)+B$. Then, find A and B by comparing coefficients of like powers. Now the integrals can be easily integrated.

Illustration 6. Evaluate $\int(3x-2)\sqrt{x^2+2x+3}dx$.

Solution: Let $I=\int(3x-2)\sqrt{x^2+2x+3}dx$.

The given integral is of the form

$$\int(px+q)\sqrt{ax^2+bx+c}\,dx$$

Let $px+q=A\frac{d}{dx}(ax^2+bx+c)+B$. The fixed A and B by comparing the coefficients of the like powers.

$$3x-2=A\frac{d}{dx}(x^2+2x+3)+B$$

$$\Rightarrow\quad 3x-2=A(2x+2)+B$$

$$\Rightarrow\quad 3x-2=2Ax+(2A+B)$$

Comparing the respective values in the above equality, we have

$$A=\frac{3}{2}\text{ and }2A+B=-2$$

$$\Rightarrow\quad A=\frac{3}{2}\text{ and }B=-5$$

Now, $$I=\int\left[\frac{3}{2}\sqrt{x^2+2x+3}\,\frac{d}{dx}(x^2+2x+3)+(-5)\sqrt{x^2+2x+3}\right]dx$$

$$\Rightarrow \quad I=\frac{3}{2}\int\sqrt{x^2+2x+3}\,\frac{d}{dx}(x^2+2x+3)dx \qquad -5\int\sqrt{x^2+2x+3}dx$$

$$\Rightarrow \quad I=2\left(\frac{3}{2}\frac{(x^2+2x+3)^{3/2}}{3}\right)-5\int\sqrt{(x+1)^2+2}dx$$

$$\Rightarrow \quad I=(x^2+2x+3)^{\frac{3}{2}}-5\left[\frac{x+1}{2}\sqrt{(x+1)^2+2}+\frac{2}{2}\log\left|(x+1)+\sqrt{(x+1)^2+2}\right|\right]+c$$

13. Integrals of the form $\int\frac{(px^2+qx+r)}{\sqrt{ax^2+bx+c}}dx$

Let

$$px^2+qx+r=A(ax^2+bx+c)+B\frac{d}{dx}(ax^2+bx+c)+C$$

Then, find A, B and C by comparing the coefficients of like powers. The integrals can now be easily integrated.

14. Integrals of the form $\int\frac{dx}{L\sqrt{Q}}$, where L and Q are linear or quadratic expression in x.

(i) When Q is linear, put $Q = t^2$.

(ii) When Q is quadratic and L is linear, put $L = 1/t$.

(iii) When Q and L are both quadratic, put $x = 1/t$.

15. Integrals of the form $\int\frac{dx}{a\sin^2x+b\cos^2x}$, $\int\frac{dx}{a+b\sin^2x}$, $\int\frac{dx}{a+b\cos^2x}$, $\int\frac{dx}{(a\sin x+b\cos x)^2}$ and $\int\frac{dx}{a+b\sin^2x+c\cos^2x}$.

Step I: Divide both the numerator and the denominator by $\cos^2 x$.

Step II: Replace $\sec^2 x$, if any, in the denominator by $1 + \tan^2 x$.

Step III: Put $\tan x = t$, so that $\sec^2 x dx = dt$.

Step IV: This reduces the integral to the form $\int\frac{dt}{at^2+bt+c}$ and then integrate.

16. Integrals of the form $\int\frac{dx}{a+b\cos x}$ or $\int\frac{dx}{a+b\sin x}$ or $\int\frac{dx}{a\cos x+b\sin x}$.

Use $\cos x=\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$ or $\sin x=\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}$, then put $\tan\frac{x}{2}=t$. Now we can integrate.

17. Integrals of the form $\int\frac{a\sin x+b\cos x}{c\sin x+d\cos x}dx$. Let numerator = A (denominator) + $B\left[\frac{d}{dx}(\text{numerator})\right]$.

Find A and B by comparing the coefficients of $\sin x$ and $\cos x$. Then it can be easily integrated.

Illustration 7. Evaluate $\int\frac{\sin x+2\cos x}{2\sin x+\cos x}dx$.

Solution: Let $I=\int\left(\frac{\sin x+2\cos x}{2\sin x+\cos x}\right)dx$

Assuming

$$\sin x+2\cos x=A(2\sin x+\cos x)+B(2\cos x-\sin x)$$

and comparing the respective coefficients, we have

$$2A-B=1 \quad \text{(i)}$$

$$A+2B=2 \quad \text{(ii)}$$

Solving Eqs. (i) and (ii), we have

$$A=\frac{4}{5} \text{ and } B=\frac{3}{5}$$

Now, $I=\int\frac{\frac{4}{5}(2\sin x+\cos x)}{(2\sin x+\cos x)}dx+\int\frac{\frac{3}{5}(2\cos x-\sin x)}{(2\sin x+\cos x)}dx$

$$\prec \quad I=\frac{4x}{5}+\frac{3}{5}\log|2\sin x+\cos x|+c$$

18. Integrals of the form $\int\frac{x^2+1}{x^4+kx^2+1}dx$, where k is any constant. Divide the numerator (Nr) and denominator (Dr) by x^2 and put $x\mp\frac{1}{x}=t$.

Illustration 8. Evaluate $\int\frac{x^2+1}{x^4-x^2+1}dx$.

Solution: Let $I=\int\frac{x^2+1}{x^4-x^2+1}dx$

$$\Rightarrow \quad I=\int\frac{1+\frac{1}{x^2}}{x^2-1+\frac{1}{x^2}}dx \quad \Rightarrow \quad I=\int\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}-2+1}dx$$

$$\Rightarrow \quad I=\int\frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+1}dx$$

Put $t=x-\frac{1}{x}\Rightarrow dt=\left(1+\frac{1}{x^2}\right)dx$

$\therefore \quad I = \int \frac{dt}{t^2+1}$

$\Rightarrow \quad I = \tan^{-1}(t) + c$, where $t = x - \frac{1}{x}$

☞ INEXPRESSIBLE INTEGRALS (WHICH INTEGRALS CANNOT BE FOUND)

Any function continuous on an interval (a, b) has an antiderivative in that interval. In other words, there exists a function $F(x)$ such that $F'(x) = f(x)$.

But not every antiderivative $F(x)$, even when it exists, is expressible in closed form in terms of elementary functions such as polynomial, logarithmic, exponential, trigonometric, etc. Then, we say that such antiderivatives or integrals "cannot be found" or are "inexpressible integrals". Some of them are listed as follows:

1. $\int \sqrt{\sin x}\,dx$
2. $\int \sqrt{(1+x^3)}\,dx$
3. $\int \sqrt{\cos x}\,dx$
4. $\int \sqrt[3]{(1+x^2)}\,dx$
5. $\int \sin x^2\,dx$
6. $\int \cos x^2\,dx$
7. $\int x \tan x\,dx$
8. $\int \sqrt{(1-A^2\sin^2 x)}\,dx$
9. $\int \frac{dx}{\log x}$
10. $\int \frac{\sin x}{x}\,dx$
11. $\int \frac{\cos x}{x}\,dx$
12. $\int \frac{x^2}{1+x^5}\,dx$
13. $\int e^{x^2}\,dx$
14. $\int e^{-x^2}\,dx$
15. $\int \frac{dx}{(1-x^2)(1-A^2x)} (A^2 < 1)$

☞ DIFFERENT TYPES OF INTEGRALS

Integrals Involving *ax + b*

1. $\int \frac{dx}{ax+b} = \frac{1}{a}\log(ax+b) + c$
2. $\int \frac{x}{ax+b}\,dx = \frac{x}{a} - \frac{b}{a^2}\log(ax+b) + c$
3. $\int \frac{x^2}{ax+b}\,dx = \frac{(ax+b)^2}{2a^3} - \frac{2b(ax+b)}{a^3} + \frac{b^2}{a^3}\log(ax+b) + c$
4. $\int \frac{x^3}{ax+b}\,dx = \frac{(ax+b)^3}{3a^4} - \frac{3b(ax+b)^2}{2a^4} + \frac{3b^2(ax+b)}{a^4} - \frac{b^3}{a^4}\log(ax+b) + c$
5. $\int \frac{dx}{x(ax+b)} = \frac{1}{b}\log\left(\frac{x}{ax+b}\right) + c$
6. $\int \frac{dx}{x^2(ax+b)} = -\frac{1}{bx} + \frac{a}{b^2}\log\left(\frac{ax+b}{x}\right) + c$
7. $\int \frac{dx}{x^3(ax+b)} = \frac{2ax-b}{2b^2x^2} + \frac{a^2}{b^3}\log\left(\frac{x}{ax+b}\right) + c$
8. $\int \frac{dx}{(ax+b)^2} = \frac{-1}{a(ax+b)} + c$
9. $\int \frac{xdx}{(ax+b)^2} = \frac{b}{a^2(ax+b)} + \frac{1}{b^2}\log(ax+b) + c$
10. $\int \frac{x^2dx}{(ax+b)^2} = \frac{ax+b}{a^3} - \frac{b^2}{a^3(ax+b)} - \frac{2b}{a^3}\log(ax+b) + c$

Integrals Involving $\sqrt{ax + b}$

1. $\int \frac{dx}{\sqrt{ax+b}} = \frac{2\sqrt{ax+b}}{a} + c$
2. $\int \frac{xdx}{\sqrt{ax+b}} = \frac{2(ax-2b)}{3a^2}\sqrt{ax+b} + c$
3. $\int \frac{x^2dx}{\sqrt{ax+b}} = \frac{2(3a^2x^2-4abx+8b^2)}{15a^3}\sqrt{ax+b} + c$
4. $\int \frac{dx}{x(\sqrt{ax+b})} = \frac{1}{\sqrt{b}}\log\left(\frac{\sqrt{ax+b}-\sqrt{b}}{\sqrt{ax+b}+\sqrt{b}}\right)$ or $\frac{2}{\sqrt{-b}}\tan^{-1}\sqrt{\frac{ax+b}{-b}} + c$
5. $\int \frac{dx}{x^2(\sqrt{ax+b})} = -\frac{\sqrt{ax+b}}{bx} - \frac{a}{2b}\int \frac{dx}{x(\sqrt{ax+b})} + c$
6. $\int \sqrt{ax+b}\,dx = \frac{2\sqrt{(ax+b)^3}}{3a} + c$
7. $\int x(\sqrt{ax+b})\,dx = \frac{2(3ax-2b)}{15a^2}\sqrt{(ax+b)^3} + c$
8. $\int x^2(\sqrt{ax+b})\,dx = \frac{2(15a^2x^2-12abx+8b^2)}{105a^3}\sqrt{(ax+b)} + c$
9. $\int \frac{\sqrt{ax+b}}{x}dx = 2\sqrt{ax+b} + b\int \frac{dx}{x(\sqrt{ax+b})} + c$
10. $\int \frac{\sqrt{ax+b}}{x^2}dx = -\frac{\sqrt{ax+b}}{x} + \frac{a}{2}\int \frac{dx}{x(\sqrt{ax+b})} + c$

Integrals Involving *ax + b* and *px + q*

1. $\int \frac{dx}{(ax+b)(px+q)} = \frac{1}{bp-aq}\log\left(\frac{px+q}{ax+b}\right) + c$
2. $\int \frac{xdx}{(ax+b)(px+q)} = \frac{1}{bp-aq} \times \left[\frac{b}{a}\log(ax+b) - \frac{q}{p}\log(px+q)\right] + c$

3. $\int \frac{dx}{(ax+b)^2(px+q)} = \frac{1}{bp-aq} \times \left[\frac{1}{ax+b} + \frac{p}{bp-aq}\log\left(\frac{px+q}{ax+b}\right)\right] + c$

4. $\int \frac{xdx}{(ax+b)^2(px+q)} = \frac{1}{bp-aq} \times \left[\frac{q}{bp-aq}\log\left(\frac{ax+b}{px+q}\right) - \frac{b}{a(ax+b)}\right] + c$

5. $\int \frac{x^2dx}{(ax+b)^2(px+q)} = \frac{b^2}{(bp-aq)a^2(ax+b)} + \frac{1}{(bp-aq)^2} \left[\frac{q^2}{p}\log(px+q) + \frac{b(bp-2aq)}{a^2}\log(ax+b)\right] + c$

Integrals Involving $\sqrt{ax+b}$ and $px+q$

1. $\int \frac{px+q}{\sqrt{ax+b}}dx = \frac{2(apx+3aq-2bp)}{3a^2}\sqrt{ax+b} + c$

2. $\int \frac{dx}{(px+q)(\sqrt{ax+b})} = \begin{cases} \frac{1}{\sqrt{bp-ap}}\log\left(\frac{\sqrt{p(ax+b)}-\sqrt{bp-aq}}{\sqrt{p(ax+b)}+\sqrt{bp-aq}}\right) \\ \frac{2}{\sqrt{aq-bp\sqrt{p}}}\tan^{-1}\sqrt{\frac{p(ax+b)}{ab-bp}} \end{cases} + c$

3. $\int \frac{\sqrt{ax+b}}{px+q}dx =$

$$\begin{cases} \frac{2\sqrt{ax+b}}{p} + \frac{\sqrt{bp-aq}}{p\sqrt{p}}\log\left(\frac{\sqrt{p(ax+b)}-\sqrt{bp-aq}}{\sqrt{p(ax+b)}+\sqrt{bp-aq}}\right) \\ \frac{2\sqrt{ax+b}}{p} - \frac{2\sqrt{aq-bp}}{p\sqrt{p}}\tan^{-1}\sqrt{\frac{p(ax+b)}{ab-bp}} \end{cases} + c$$

Integrals Involving $\sqrt{ax+b}$ and $\sqrt{px+q}$

1. $\int \frac{dx}{\sqrt{(ax+b)(px+q)}} =$

$$\begin{cases} \frac{2}{\sqrt{ab}}\log\left(\sqrt{a(px+q)}+\sqrt{p(ax+b)}\right) \\ \frac{2}{\sqrt{-ap}}\tan^{-1}\sqrt{\frac{-p(ax+b)}{a(px+q)}} \end{cases} + c$$

2. $\int \frac{xdx}{\sqrt{(ax+b)(px+q)}} = \frac{\sqrt{(ax+b)(px+q)}}{ap} - \frac{bp+aq}{2ap}\int \frac{dx}{(ax+b)(px+q)} + c$

3. $\int \sqrt{(ax+b)(px+q)}dx =$

$$\begin{cases} \frac{2apx+bp+aq}{4ap}\sqrt{(ax+b)(px+q)} \\ -\frac{(bp-aq)^2}{8ap}\int \frac{dx}{\sqrt{(ax+b)(px+q)}} \end{cases} + c$$

4. $\int \sqrt{\frac{px+q}{ax+b}}dx =$

$$\frac{\sqrt{(ax+b)(px+q)}}{a} + \frac{aq+bp}{2a}\int \frac{dx}{\sqrt{(ax+b)(px+q)}} + c$$

5. $\int \frac{dx}{(px+q)\sqrt{(ax+b)(px+q)}} = \frac{2\sqrt{ax+b}}{(aq+bp)\sqrt{px+q}} + c$

Integrals Involving $x^2 + a^2$

1. $\int \frac{xdx}{x^2+a^2} = \frac{1}{2}\log(x^2+a^2) + c$

2. $\int \frac{x^2dx}{x^2+a^2} = x - a\tan^{-1}\frac{x}{a} + c$

3. $\int \frac{x^3dx}{x^2+a^2} = \frac{x^2}{2} - \frac{a^2}{2}\log(x^2+a^2) + c$

4. $\int \frac{dx}{x(x^2+a^2)} = \frac{1}{2a^2}\log\left(\frac{x^2}{x^2+a^2}\right) + c$

5. $\int \frac{dx}{x^2(x^2+a^2)} = -\frac{1}{a^2x} - \frac{1}{a^3}\tan^{-1}\frac{x}{a} + c$

Integrals Involving $x^2 - a^2$

1. $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left(\frac{x-a}{x+a}\right) + c$

2. $\int \frac{xdx}{x^2-a^2} = \frac{1}{2}\log(x^2-a^2) + c$

3. $\int \frac{x^2dx}{x^2-a^2} = x + \frac{a}{2}\log\left(\frac{x-a}{x+a}\right) + c$

4. $\int \frac{dx}{x(x^2-a^2)} = \frac{1}{2a^2}\log\left(\frac{x^2-a^2}{x^2}\right) + c$

5. $\int \frac{dx}{x^2(x^2-a^2)} = \frac{1}{a^2x} + \frac{1}{2a^3}\log\left(\frac{x-a}{x+a}\right) + c$

Integrals Involving $a^2 - x^2$

1. $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log\left(\frac{a+x}{a-x}\right) + c$
2. $\int \frac{xdx}{a^2 - x^2} = -\frac{1}{2} \log(a^2 - x^2) + c$
3. $\int \frac{x^2 dx}{a^2 - x^2} = -x + \frac{a}{2} \log\left(\frac{a+x}{a-x}\right) + c$
4. $\int \frac{x^3 dx}{a^2 - x^2} = -\frac{x^2}{2} - \frac{a^2}{2} \log(a^2 - x^2) + c$
5. $\int \frac{dx}{x(a^2 - x^2)} = \frac{1}{2a^2} \log\left(\frac{x^2}{a^2 - x^2}\right) + c$
6. $\int \frac{dx}{x^2(a^2 - x^2)} = -\frac{1}{a^2 x} + \frac{1}{a^3} \log\left(\frac{a+x}{a-x}\right) + c$

Integrals Involving $\sqrt{x^2 + a^2}$

1. $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log(x + \sqrt{x^2 + a^2}) + c$
2. $\int \frac{xdx}{\sqrt{x^2 + a^2}} = \sqrt{x^2 + a^2} + c$
3. $\int \frac{x^2 dx}{\sqrt{x^2 + a^2}} = \frac{x\sqrt{x^2 + a^2}}{2} - \frac{a^2}{2} \log(x + \sqrt{x^2 + a^2}) + c$
4. $\int \frac{x^3 dx}{\sqrt{x^2 + a^2}} = \frac{(x^2 + a^2)^{3/2}}{3} - a^2\sqrt{x^2 + a^2} + c$
5. $\int \frac{dx}{x\sqrt{x^2 + a^2}} = -\frac{1}{a} \log\left(\frac{a + \sqrt{x^2 + a^2}}{x}\right) + c$
6. $\int \frac{dx}{x^2\sqrt{x^2 + a^2}} = -\frac{\sqrt{x^2 + a^2}}{a^2 x} + c$

Integrals Involving $\sqrt{x^2 - a^2}$

1. $\int \frac{xdx}{\sqrt{x^2 - a^2}} = \sqrt{x^2 - a^2} + c$
2. $\int \frac{x^2 dx}{\sqrt{x^2 - a^2}} = \frac{x\sqrt{x^2 - a^2}}{2} + \frac{a^2}{2} \log(x + \sqrt{x^2 - a^2}) + c$
3. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1}\left|\frac{x}{a}\right| + c$
4. $\int \frac{dx}{x^2\sqrt{x^2 - a^2}} = \frac{\sqrt{x^2 - a^2}}{a^2 x} + c$
5. $\int \sqrt{x^2 - a^2}\, dx = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \log(x + \sqrt{x^2 - a^2}) + c$

Integrals Involving $\sqrt{a^2 - x^2}$

1. $\int \frac{xdx}{\sqrt{a^2 - x^2}} = -\sqrt{a^2 - x^2} + c$
2. $\int \frac{x^2 dx}{\sqrt{a^2 - x^2}} = -\frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1}\frac{x}{a} + c$
3. $\int \frac{dx}{x\sqrt{a^2 - x^2}} = -\frac{1}{a} \log\left(\frac{a + \sqrt{a^2 - x^2}}{x}\right) + c$
4. $\int \frac{dx}{x^2\sqrt{a^2 - x^2}} = -\frac{\sqrt{a^2 - x^2}}{a^2 x} + c$
5. $\int x\sqrt{a^2 - x^2}\, dx = -\frac{(a^2 - x^2)^{3/2}}{3} + c$
6. $\int x^2\sqrt{a^2 - x^2}\, dx = -\frac{x(a^2 - x^2)^{3/2}}{4} + \frac{a^2 x\sqrt{a^2 - x^2}}{8} + \frac{a^4}{8} \sin^{-1}\frac{x}{a} + c$
7. $\int \frac{\sqrt{a^2 - x^2}}{x} dx = \sqrt{a^2 - z^2} - a \log\left(\frac{a + \sqrt{a^2 - x^2}}{x}\right) + c$
8. $\int \frac{\sqrt{a^2 - x^2}}{x^2} dx = -\frac{\sqrt{a^2 - x^2}}{x} - \sin^{-1}\frac{x}{a} + c$
9. $\int \frac{dx}{(a^2 - x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2 - x^2}} + c$
10. $\int \frac{xdx}{(a^2 - x^2)^{3/2}} = \frac{1}{\sqrt{a^2 - x^2}} + c$
11. $\int \frac{x^2 dx}{(a^2 - x^2)^{3/2}} = \frac{x}{\sqrt{a^2 - x^2}} - \sin^{-1}\frac{x}{a} + c$
12. $\int \frac{dx}{x(a^2 - x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2 - x^2}} - \frac{1}{a^3} \log\left(\frac{a + \sqrt{a^2 - x^2}}{x}\right) + c$
13. $\int \frac{dx}{x^2(a^2 - x^2)^{3/2}} = -\frac{\sqrt{a^2 - x^2}}{a^{4x}} + \frac{x}{a^4\sqrt{a^2 - x^2}} + c$

Integrals Involving $ax^2 + bx + c$

1. $\int \frac{dx}{ax^2 + bx + c} =$

$$\begin{cases} \frac{2}{\sqrt{4ac - b^2}} \tan^{-1}\frac{2ax + b}{\sqrt{4ac - b^2}} \\ \frac{1}{\sqrt{b^2 - 4ac}} \log\left(\frac{2ax + b - \sqrt{b^2 - 4ac}}{2ax + b + \sqrt{b^2 - 4ac}}\right) + c \end{cases}$$

2. $\int \frac{xdx}{ax^2+bx+c} =$

$$\frac{1}{2a}\log(ax^2+bx+c) - \frac{b}{2x}\int \frac{dx}{ax^2+bx+c} + c$$

3. $\int \frac{x^2dx}{ax^2+bx+c} =$

$$\frac{x}{a} - \frac{b}{2a^2}\log(ax^2+bx+c) + \frac{b^2-2ac}{2a^2}\int \frac{dx}{ax^2+bx+c} + c$$

Integrals Involving $\sqrt{ax^2+bx+c}$

1. $\int \frac{dx}{\sqrt{ax^2+bx+c}} =$

$$\int \frac{dx}{\sqrt{ax^2+bx+c}} = \begin{cases} \frac{1}{\sqrt{a}}\log(2\sqrt{a}\sqrt{ax^2+bx+c} + 2ax + b \\ -\frac{1}{\sqrt{a}}\sin^{-1}\left(\frac{2ax+b}{\sqrt{b^2-4ac}}\right) + c \end{cases}$$

2. $\int \frac{xdx}{\sqrt{ax^2+bx+c}} = \frac{\sqrt{ax^2+bx+c}}{a} - \frac{b}{2a}\int \frac{dx}{\sqrt{ax^2+bx+c}} + c$

3. $\int \frac{x^2dx}{\sqrt{ax^2+bx+c}} =$

$$\frac{2ax-3b}{4a^2}\sqrt{ax^2+bx+c} + \frac{3b^2-4ac}{8a^2}\int \frac{dx}{\sqrt{ax^2+bx+c}} + c$$

Integrals Involving $x^3 + a^3$

1. $\int \frac{dx}{x^3+a^3} =$

$$\frac{1}{6a^2}\log\frac{(x+a)^2}{x^2-ax+a^2} + \frac{1}{a^2\sqrt{3}}\tan^{-1}\frac{2x-a}{a\sqrt{3}} + c$$

2. $\int \frac{xdx}{x^3+a^3} =$

$$\frac{1}{6a^2}\log\frac{x^2-ax+a^2}{(x+a)^2} + \frac{1}{a^2\sqrt{3}}\tan^{-1}\frac{2x-a}{a\sqrt{3}} + c$$

3. $\int \frac{x^2dx}{x^3+a^3} = \frac{1}{3}\log(x^3+a^3) + c$

4. $\int \frac{dx}{x(x^3+a^3)} = \frac{1}{3a^3}\log\left(\frac{x^3}{x^3+a^3}\right) + c$

5. $\int \frac{dx}{x^2(x^3+a^3)} =$

$$-\frac{1}{a^3x} - \frac{1}{6a^4}\log\frac{x^2-ax+a^2}{(x+a)^2} - \frac{1}{a^4\sqrt{3}}\tan^{-1}\frac{2x-a}{a\sqrt{3}} + c$$

SOLVED PROBLEMS

1. $\int \frac{\cos 2x}{(\sin x+\cos x)^2}dx$ is equal to

(a) $\frac{-1}{\sin x+\cos x}+c$ (b) $\log(\sin x+\cos x)+c$

(c) $\log(\sin x-\cos x)+c$ (d) $\log(\sin x+\cos x)^2+c$

Ans. (b)

Solution: Write $\cos 2x = \cos^2 x - \sin^2 x$

$$\therefore \quad I = \int \frac{\cos x-\sin x}{\sin x+\cos x}dx = \int \frac{f'(x)}{f(x)}dx$$

Alt. $D' = 1+\sin 2x$

$$\therefore \quad I = \frac{1}{2}\log(1+\sin 2x)$$

$$= \frac{1}{2}\log(\sin x+\cos x)^2 \text{ etc.}$$

2. $\int \frac{\cos 2x}{\cos x}dx$ is equal to

(a) $2\sin x + \log(\sec x - \tan x) + c$

(b) $2\sin x - \log(\sec x - \tan x) + c$

(c) $2\sin x + \log(\sec x + \tan x) + c$

(d) $2\sin x - \log(\sec x + \tan x) + c$

Ans. (d)

Solution: $I = \int \frac{2\cos^2 x - 1}{\cos x}dx$

$$I = \int (2\cos x - \sec x)dx = 2\int \cos x dx - \int \sec x dx$$

3. $\int \frac{dx}{x^{1/5}(1+x^{4/5})^{1/2}}$ equals

(a) $\sqrt{1+x^{4/5}}+c$ (b) $x^{4/5}(1+x^{4/5})^{1/2}+c$

(c) $\frac{5}{2}\sqrt{1+x^{4/5}}+c$ (d) none of these

Ans. (c)

Solution: Putting $1+x^{4/5}=t$ so that $\frac{4}{5}x^{-1/5}dx=dt$

i.e. $\frac{1}{x^{1/5}}dx=\frac{5}{4}dt$

$\therefore \quad I=\int\frac{5/4dt}{t^{1/2}}=\frac{5}{4}\int t^{-1/2}dt$

$=\frac{5}{4}\frac{\sqrt{t}}{1/2}+c=\frac{5}{2}\sqrt{1+x^{4/5}}+c$

4. $\int\frac{3\sin x+2\cos x}{3\cos x+2\sin x}dx=$

(a) $\frac{12}{13}+\frac{5}{13}\log(3\cos x+2\sin x)$

(b) $\frac{12}{13}x-\frac{5}{13}\log(3\cos x+2\sin x)$

(c) $\frac{5}{13}x+\frac{12}{3}\log(3\cos x+2\sin x)$

(d) none of these

Ans. (b)

Solution: Write $N^r=l(D^r)+m(\text{d.c. of } D^r)$

Let $3\sin x+2\cos x=l(3\cos x+2\sin x)$

$+m(-3\sin x+2\cos x)$

Comparing the coefficients of sin x and cos x on both sides, we get $3=2l-3m, 2=3l+2m$

Solving, we get $l=12/13, m=-5/13$

$\therefore \quad I=l\int dx+m\int\frac{-3\sin x+2\cos x}{3\cos x+2\sin x}dx$

$=lx+m\log(3\cos x+2\sin x)$

5. $\int 7^{7^{7^x}}\cdot 7^{7^x}\cdot 7^x dx$ equals

(a) $\frac{7^{7^x}}{(\log 7)^3}+c$ (b) $7^{7^{7^x}}(\log 7)^3+c$

(c) $\frac{7^{7^{7^x}}}{(\log 7)^3}+c$ (d) none of these

Ans. (c)

Solution: Putting $7^{7^{7^x}}=t$ so that $7^{7^{7^x}}\cdot 7^{7^x}\cdot 7^x(\log 7)^3dx=dt$

$\therefore \quad I=\int\frac{dt}{(\log 7)^3}=\frac{1}{(\log 7)^3}t+c=\frac{7^{7^{7^x}}}{(\log 7)^3}+c$

6. $\int\frac{\sec x dx}{\sqrt{\sin(2x+\alpha)+\sin\alpha}}=$

(a) $\sqrt{2(\sin x+\tan\alpha)\sec\alpha}$

(b) $\sqrt{2(\tan x+\tan\alpha)\sec\alpha}$

(c) $\sqrt{(\tan x+\tan\alpha)\sec\alpha}$

(d) none of these

Ans. (b)

Solution: $\int\frac{\sec x dx}{\sqrt{2\sin(x+\alpha)\cos x}}$

Dividing the numerator and the denominator by cos x, we get

$I=\frac{1}{\sqrt{2}}\int\frac{\sec^2 x dx}{\sqrt{\frac{\sin x\cos\alpha+\cos x\sin\alpha)}{\cos x}}}$

$I=\frac{1}{\sqrt{2}}\int\frac{\sec^2 x dx}{\sqrt{\tan x\cos\alpha+\sin\alpha}}$

Now put $\tan x\cdot\cos\alpha+\sin\alpha=t$

$\therefore \quad \sec^2 x\cdot\cos\alpha dx=dt$

$I=\frac{1}{\sqrt{2}\cos\alpha}\int\frac{dt}{\sqrt{t}}=\frac{1}{\sqrt{2}\cos\alpha}2\sqrt{t}$

$I=\sqrt{2\frac{1}{\cos^2\alpha}}=\frac{\sqrt{2(\tan x\cdot\cos\alpha+\sin\alpha)}}{\cos^2\alpha}$

$I=\sqrt{2\sec\alpha(\tan x+\tan\alpha)}$

7. $\int\frac{2^x+3^x}{5^x}dx$ is equal to

(a) $\frac{(2/5)^x}{\log_3(3/5)}+\frac{(3/5)^x}{\log_3(2/5)}+c$

(b) $\frac{(2/5)^x}{\log_3(2/5)}+\frac{(3/5)^x}{\log_3(3/5)}+c$

(c) $\frac{(3/5)^x}{\log_3(2/5)}+\frac{(2/5)^x}{\log_3(3/5)}+c$

(d) none of these

Ans. (d)

Solution: We have

$\int\frac{2^x+3^x}{5^x}dx$

$=\int\left(\frac{2^x}{5^x}+\frac{3^x}{5^x}\right)dx=\int\left[\left(\frac{2}{5}\right)^x+\left(\frac{3}{5}\right)^x\right]dx$

$=\frac{(2/5)^x}{\log_e(2/5)}+\frac{(3/5)^x}{\log_e(3/5)}+c$.

8. $\int \frac{1}{\sqrt{(\sin^3 x \cos x)}}dx$ is equal to

(a) $\frac{-2}{\sqrt{(\tan x)}}+c$ (b) $2\sqrt{\tan x}+c$

(c) $\frac{2}{\sqrt{(\tan x)}}+c$ (d) $-2\sqrt{\tan x}+c$

Ans. (a)

Solution: $I=\int \frac{1}{\sqrt{(\tan^3 x \cos^4 x)}}dx=\int \frac{\sec^2 x}{(\tan x)^{3/2}}dx$

Now put $\tan x = t$

$\therefore \quad I=\int t^{-3/2}dt$

$$= \frac{t^{-3/2+1}}{-(3/2)+1}=\frac{-2}{\sqrt{t}}=\frac{-2}{\sqrt{\tan x}}+c$$

9. $\int \sqrt{\frac{1+x}{x}}dx =$

(a) $\sqrt{x^2+x}+\frac{1}{2}\log\left|\left(x+\frac{1}{2}\right)+\sqrt{x^2+x}\right|+c$

(b) $\sqrt{x^2+x}-\frac{1}{2}\log\left|\left(x+\frac{1}{2}\right)+\sqrt{x^2+x}\right|+c$

(c) $\sqrt{x^2-x}-\frac{1}{2}\log\left|\left(x+\frac{1}{2}\right)+\sqrt{x^2+x}\right|+c$

(d) none of these

Ans. (a)

Solution: $\int \sqrt{\frac{1+x}{x}}dx=\int \sqrt{\frac{1+x}{x}}\cdot\sqrt{\frac{1+x}{1+x}}dx=\int \frac{1+x}{\sqrt{x(1+x)}}dx$

$$= \int \frac{1+x}{\sqrt{x^2+x}}dx$$

$$\therefore \quad \int \sqrt{\frac{1+x}{x}}dx=\int \frac{1+x}{\sqrt{x^2+x}}dx=\int \frac{\frac{1}{2}(2x+1)+\frac{1}{2}}{\sqrt{x^2+x}}dx$$

$$=\frac{1}{2}\int \frac{2x+1}{\sqrt{x^2+x}}dx+\frac{1}{2}\int \frac{1}{\sqrt{x^2+x}}dx$$

$$=\frac{1}{2}\int \frac{1}{\sqrt{t}}dt+\frac{1}{2}\int \frac{1}{\sqrt{\left(x+\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2}}dx$$

where $t=x^2+x$

$$=\sqrt{t}+\frac{1}{2}\log\left|\left(x+\frac{1}{2}\right)+\sqrt{x^2+x}\right|+c$$

$$=\sqrt{x^2+x}+\frac{1}{2}\log\left|\left(x+\frac{1}{2}\right)+\sqrt{x^2+x}\right|+c$$

10. $\int \frac{(x^4-x)^{1/4}}{x^5}dx$ is equal to

(a) $\frac{4}{15}\left(1-\frac{1}{x^3}\right)^{5/4}+c$ (b) $\frac{4}{5}\left(1-\frac{1}{x^3}\right)^{5/4}+c$

(c) $\frac{4}{15}\left(1+\frac{1}{x^3}\right)^{5/4}+c$ (d) none of these

Ans. (a)

Solution: $I=\int \frac{x[1-(1/x^3)]^{1/4}}{x^5}dx$

Put $1-\frac{1}{x^3}=t$

$\therefore \quad \frac{3}{x^4}dx=dt \qquad \therefore \quad I=\frac{1}{3}\int t^{1/4}dt$ etc.

11. $\int \frac{x^2+1}{x^4+1}dx =$

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^3-1}{\sqrt{2}x}\right)-c$ (b) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^3-1}{\sqrt{2}x}\right)+c$

(c) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^3+1}{\sqrt{2}x}\right)-c$ (d) none of these

Ans. (b)

Solution: $I=\int \frac{x^2+1}{x^4+1}dx=\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx=\int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}dx$

Let $x-\frac{1}{x}=t \Rightarrow d\left(x-\frac{1}{x}\right)=dt \Rightarrow \left(1+\frac{1}{x^2}\right)dx=dt$

$$\therefore \quad I=\int \frac{dt}{t^2+(\sqrt{2})^2}=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)+c$$

$$=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x-1/x}{\sqrt{2}}\right)+c$$

$$=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)+c$$

12. $\int \frac{xdx}{(x^2+1)^{4/5}(x^2+2)^{6/5}}=$

(a) $\frac{2}{5}\left(\frac{x^2+2}{x^2+1}\right)^{1/5}+c$ (b) $\frac{5}{2}\left(\frac{x^2+2}{x^2+1}\right)^{-1/5}+c$

(c) $\frac{5}{2}\left(\frac{x^2+1}{x^2+2}\right)^{1/5}+c$ (d) none of these

Ans. (b) and (c)

Solution: $\frac{4}{5}+\frac{6}{5}=2$

$$I=\int\frac{xdx}{(x^2+1)^2\left(\frac{x^2+2}{x^2+1}\right)^{6/5}}=\int\frac{xdx}{(x^2+1)^2\left[1+\frac{1}{x^2+1}\right]^{6/5}}$$

Put $1+\frac{1}{x^2+1}=t$

$$\therefore\quad \frac{-2x}{(x^2+1)^2}dx=dt$$

$$\therefore\quad I=-\frac{1}{2}\int t^{-6/5}dt$$

$$=-\frac{1}{2}\frac{t^{-1/5}}{-1/5}=\frac{5}{2}t^{-1/5}$$

$$I=\frac{5}{2}\left(1+\frac{1}{x^2+1}\right)^{-1/5}=\frac{5}{2}\left(\frac{x^2+2}{x^2+1}\right)^{-1/5}$$

or $$I=\frac{5}{2}\left(\frac{x^2+1}{x^2+2}\right)^{1/5}+c$$

13. $\int\frac{x^2-8x+7}{(x^2-3x-10)^2}dx=P\log|x-5|+Q\cdot\frac{1}{x-5}$

$-R\cdot\log|x+2|-S\cdot\frac{1}{x+2}+c$. Then

(a) $P=-\frac{45}{98}$ (b) $Q=\frac{8}{49}$

(c) $R=\frac{15}{49}$ (d) all of these

Ans. (d)

Solution: We have $\frac{x^2-8x+7}{(x^2-3x-10)^2}=\frac{(x-7)(x-1)}{(x-5)^2(x+2)^2}$

Let $\frac{(x-7)(x-1)}{(x-5)^2(x+2)^2}=\frac{A}{x-5}+\frac{B}{(x-5)^2}+\frac{C}{x+2}+\frac{D}{(x+2)^2}$

Then, $(x-7)(x-1)=A(x-5)(x+2)^2+B(x+5)^2$

$$+C(x+2)(x-5)^2+D(x-5)^2 \qquad \text{(i)}$$

On putting $x = 5$ and $x = -2$ respectively, we get

$$-8=B(7)^2\quad\Rightarrow\quad B=\frac{-8}{49}$$

and $$27=D(-7)^2\quad\Rightarrow\quad D=\frac{27}{49}$$

Now, on putting $x = 0$, in Eq. (i), we get

$$7=-20A+4B+50C+25D$$

$$\Rightarrow\qquad 7=-20A-\frac{32}{49}+50C+27\times\frac{25}{49}$$

$$\Rightarrow\qquad 20A-50C=\frac{300}{49}\quad\Rightarrow\quad 2A-5C=\frac{30}{49} \qquad \text{(ii)}$$

On putting $x = 1$ in Eq. (i), we get

$$0=-16A+9B+48C+16D$$

$$\Rightarrow\qquad 0=-16A+48C-\frac{72}{49}+\frac{442}{49}$$

$$\Rightarrow\qquad 16A-48C=\frac{360}{49}$$

$$\Rightarrow\qquad 2A-6C=\frac{45}{49} \qquad \text{(iii)}$$

On solving Eqs. (ii) and (iii), we get

$$A=-\frac{45}{98},C=-\frac{15}{49}$$

$$\therefore\quad \frac{(x-7)(x-1)}{(x-5)^2(x+2)^2}$$

$$=-\frac{45}{98}\cdot\frac{1}{x-5}-\frac{8}{49}\cdot\frac{1}{(x-5)^2}-\frac{15}{49}\frac{1}{x+2}+\frac{27}{49}\frac{1}{(x+2)^2}$$

Integrating both sides, we get

$$\int\frac{(x-7)(x-1)}{(x-5)^2(x+2)}dx$$

$$=-\frac{45}{98}\int\frac{1}{x-5}dx-\frac{8}{49}\int\frac{1}{(x-5)^2}dx-\frac{15}{49}\int\frac{1}{x+2}dx$$

$$+\frac{27}{49}\int\frac{1}{(x+2)^2}dx$$

$$=-\frac{45}{98}\log|x-5|+\frac{8}{49}\frac{1}{x-5}-\frac{15}{49}\log|x+2|$$

$$-\frac{27}{49}\frac{1}{(x+2)}+c.$$

$\therefore\quad P=\frac{-45}{98},\ Q=\frac{8}{49},\ R=\frac{15}{49}$ and $S=\frac{27}{49}$.

14. $\int\frac{e^{\log(1+1/x^2)}}{x^2+1/x^2}dx$ is

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left(x-\frac{1}{x}\right)$ (b) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(x-\frac{1}{x}\right)$

(c) $\frac{1}{\sqrt{2}}\log\left(\frac{x+1}{x\sqrt{2}}\right)$ (d) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{x\sqrt{2}}\right)$

Ans. (d)

Solution: $I=\int\frac{[1+(1/x^2)]}{[x-(1/x)]^2+2}\,dx$

Put $x-\frac{1}{x}=t$

15. $\int\frac{1}{\sin x-\sin 2x}dx$ equals

(a) $-\frac{1}{2}\log|1-\cos x|-\frac{1}{6}\log|1+\cos x|$
$+\frac{2}{3}\log|1+2\cos x|+c$

(b) $\frac{1}{2}\log|1-\cos x|+\frac{1}{6}\log|1+\cos x|$
$+\frac{2}{3}\log|1+2\cos x|+c$

(c) $\frac{1}{2}\log|1+\cos x|-\frac{1}{6}\log|1+\cos x|$
$-\frac{2}{3}\log|1+2\cos x|+c$

(d) none of these

Ans. (a)

Solution:

$$I=\int\frac{1}{\sin x-\sin 2x}dx$$
$$=\int\frac{1}{(\sin x-2\sin x\cos x)}dx=\int\frac{1}{\sin x(1-2\cos x)}dx$$
$$=\int\frac{\sin x}{\sin^2 x(1-2\cos x)}dx=\int\frac{\sin x}{(1-\cos^2 x)(1-2\cos x)}dx$$

Putting $\cos x=t$, and $-\sin x dx=dt$ or $\sin x dx=-dt$, we get

$$I=\int\frac{-dt}{(1-t^2)(1-2t)}$$
$$=\int\frac{1}{(t-1)(1-t)(1-2t)}dt$$
$$=\int\frac{1}{(t-1)(2)(-1)}+\frac{1}{(-2)(1+t)(3)}$$
$$+\frac{1}{(-1/2)(3/2)(1-2t)}dt$$
$$=\frac{1}{2}\log|1-t|-\frac{1}{6}\log|1+t|-\frac{4}{3}\times\frac{1}{2}\log|1-2t|+c$$
$$=-\frac{1}{2}\log|1-\cos x|-\frac{1}{6}\log|1+\cos x|$$
$$+\frac{2}{3}\log|1-2\cos x|+c$$

16. $\int\sqrt{\frac{\cos x-\cos^3 x}{1-\cos^3 x}}dx$ is equal to

(a) $\frac{2}{3}\sin^{-1}(\cos^{3/2}x)+c$ (b) $\frac{3}{2}\sin^{-1}(\cos^{3/2}x)+c$

(c) $\frac{2}{3}\cos^{-1}(\cos^{3/2}x)+c$ (d) none of these

Ans. (c)

Solution: $I=\int\frac{\sqrt{\cos x}\sin x}{\sqrt{[1-(\cos^{3/2}x)^2]}}dx$

Put $\cos^{3/2}x=t$, etc.

$\therefore\quad I=-\frac{2}{3}\int\frac{dt}{\sqrt{(1-t^2)}}$

17. $\int e^x\frac{(x^2+1)}{(x+1)^2}dx=$

(a) $\left(\frac{x-1}{x+1}\right)e^x+c$ (b) $e^x\left(\frac{x+1}{x-1}\right)+c$

(c) $e^x(x+1)(x-1)+c$ (d) none of these

Ans. (a)

Solution:

$$\int\frac{e^x(x^2+1)}{(x+1)^2}dx$$
$$=\int\frac{e^x(x^2-1+2)}{(x+1)^2}dx=\int e^x\left[\frac{x-1}{x+1}+\frac{2}{(x+1)^2}\right]dx$$
$$=\int e^x[f(x)+f'(x)]dx,$$

where $f(x)=\frac{x-1}{x+1}$

and $f'(x)=\frac{2}{(x+1)^2}$

$$=e^x\left(\frac{x-1}{x+1}\right)+c$$

18. $\int\frac{x}{\sqrt{x+a}+\sqrt{x+b}}dx$

$$=\frac{1}{a-b}\left[A-\frac{2}{5}(x+b)^{5/2}+\frac{2b}{3}(x+b)^{3/2}\right]+c.$$

Then A is equal to

(a) $\frac{2}{5}(x-a)^{5/2}+\frac{2a}{3}(x-a)^{3/2}$

(b) $\frac{2}{5}(x+a)^{5/2}-\frac{2a}{3}(x+a)^{3/2}$

(c) $\frac{2}{5}(x-a)^{5/2}-\frac{2a}{3}(x-a)^{3/2}$

(d) none of these

Ans. (b)

Solution: We have

$$\int \frac{x}{\sqrt{x+a}+\sqrt{x+b}}\,dx$$

$$= \int \frac{x(\sqrt{x+a}-\sqrt{x+b})}{(\sqrt{x+a}+\sqrt{x+b})(\sqrt{x+a}-\sqrt{x+b})}\,dx$$

$$= \int \frac{x(\sqrt{x+a}-\sqrt{x+b})}{a-b}\,dx$$

$$= \frac{1}{a-b}\int (x\sqrt{x+a}-x\sqrt{x+b})\,dx$$

$$= \frac{1}{a-b}\int [(x+a-a)\sqrt{x+a}-(x+b-b)\sqrt{x+b})]dx$$

$$= \frac{1}{a-b}\left[\frac{2}{5}(x+a)^{5/2}-\frac{2a}{3}(x+a)^{3/2}-\frac{2}{5}(x+b)^{5/2}+\frac{2b}{3}(x+b)^{3/2}\right]+c$$

19. $\int \frac{1}{\sqrt{\sin^3 x\sin(x+\alpha)}}dx, \alpha \neq n\pi, n\in Z$ is equal to

(a) $-2\text{cosec}\alpha(\cos\alpha-\tan x\sin\alpha)^{1/2}+c$

(b) $-2(\cos\alpha+\cot x\sin\alpha)^{1/2}+c$

(c) $-2\text{cosec}\alpha(\cos\alpha+\cot x\sin\alpha)^{1/2}+c$

(d) $-2\text{cosec}\alpha(\sin\alpha+\cot x\cos\alpha)^{1/2}+c$

Ans. (c)

Solution: $\sin^3 x\sin(x+\alpha)=\sin^3 x(\sin x\cos\alpha+\cos x\sin\alpha)$

$$=\sin^4 x(\cos\alpha+\cot x\sin\alpha)$$

$$I=\int \frac{1}{\sqrt{\sin^3 x\sin(x+\alpha)}}dx$$

$$=\int \frac{1}{\sin^2 x\sqrt{\cos\alpha+\cot x\sin\alpha}}dx$$

$$=\int \frac{\text{cosec}^2 x}{\sqrt{\cos\alpha+\cot x\sin\alpha}}dx$$

Putting $\cos\alpha+\cot x\sin\alpha=t$ and $-\text{cosec}^2 x\sin\alpha dx=dt$, we have

$$I=\int -\frac{1}{\sin\alpha\sqrt{t}}dt=-\frac{1}{\sin\alpha}\int t^{-1/2}dt$$

$$=-\frac{1}{\sin\alpha}\left(\frac{t^{1/2}}{1/2}\right)+c$$

$$\Rightarrow I=-2\text{cosec}\alpha\sqrt{t}+c=-2\text{cosec}\alpha(\cos\alpha+\cot x\sin\alpha)^{1/2}+c$$

20. $\int\sqrt{1+\text{cosec}\,x}dx=$

(a) $\sin^{-1}(\tan x-\sec x)$

(b) $\sin^{-1}(\cos x)$

(c) $\sin^{-1}\left(\sin\frac{x}{2}\cdot\cos\frac{x}{2}\right)$

(d) $\pm 2\sin^{-1}\left(\sin\frac{x}{2}-\cos\frac{x}{2}\right)$

Ans. (d)

Solution: $I=\sqrt{\frac{1+\sin x}{\sin x}}=\pm\frac{\sin\frac{x}{2}+\cos\frac{x}{2}}{\sqrt{\sin x}}$

Now put $\sin\frac{x}{2}-\cos\frac{x}{2}=t$ as the direction cosine is present in the numerator.

$$\therefore \quad \frac{1}{2}\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)dx=dt$$

Also, squaring, $1-\sin x=t^2 \quad \therefore \quad 1-t^2=\sin x$

$$\therefore \quad I=\pm\int\frac{2dt}{\sqrt{1-t^2}}=\pm 2\sin^{-1}\left(\sin\frac{x}{2}-\cos\frac{x}{2}\right)$$

EXERCISE SET 1

1. If $\int\frac{x\tan^{-1}x}{\sqrt{1+x^2}}dx=\sqrt{1+x^3}f(x)+A\log(x+\sqrt{x^2+1})+c$, then

(a) $f(x)=\tan^{-1}x, A=-1$

(b) $f(x)=\tan^{-1}x, A=1$

(c) $f(x)=2\tan^{-1}x, A=-1$

(d) $f(x)=2\tan^{-1}x, A=1$

2. $\int\frac{a^{x/2}}{\sqrt{a^{-x}-a^x}}dx$ is equal to

(a) $\frac{1}{\log a}\sin^{-1}(a^x)+c$ (b) $\frac{1}{\log a}\tan^{-1}(a^x)+c$

(c) $2\sqrt{a^{-x}-a^x}+c$ (d) $\log(a^x-1)+c$

3. $\int\frac{a+b\sin x}{(b+a\sin x)^2}dx$ equals

(a) $\frac{-\cos x}{b+a\sin x}$ (b) $\frac{\cos x}{b+a\sin x}$

(c) $\frac{\sin x}{b+a\sin x}$ (d) none of these

4. The value of $\int e^{\sin^2 x}\sin 2x\,dx$ is

(a) $e^{\sin 2x}+c$ (b) $e^{\sin^2 x}+c$

(c) $\sin^2 x+c$ (d) $\sin 2x+c$

5. $\int\left(x+\frac{1}{x}\right)^2 dx=$

(a) $\frac{1}{3}\left(x+\frac{1}{x}\right)^3+c$ (b) $\frac{x^3}{3}+2x-\frac{1}{x}+c$

(c) $\frac{x^3}{3}+2x+\frac{1}{x}+c$ (d) none of these

6. $\int\frac{1}{(1-x^8)}\left(\cos^{-1}\frac{2x}{1+x^2}+\tan^{-1}\frac{2x}{1-x^2}\right)dx$ is equal to

(a) $\frac{\pi}{16}\log\left|\frac{1+x}{1-x}\right|+\frac{\pi}{8}\tan^{-1}x+\frac{\pi}{8\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)$
$-\frac{\pi}{16\sqrt{2}}\log\left|\frac{x^2+\sqrt{2}x+1}{x^2+\sqrt{x}+1}\right|+c$

(b) $\frac{\pi}{16}\log\left|\frac{1-x}{1+x}\right|+\frac{\pi}{8}\tan^{-1}x+\frac{\pi}{8\sqrt{2}}\tan^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)$
$-\frac{\pi}{16\sqrt{2}}\log\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{x}+1}\right|+c$

(c) $\frac{\pi}{16}\log\left|\frac{1-x}{1+x}\right|+\frac{\pi}{8}\tan^{-1}x+\frac{\pi}{8\sqrt{2}}\tan^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)$
$-\frac{\pi}{16\sqrt{2}}\log\left|\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{x}+1}\right|+c$

(d) none of these

7. $\int\frac{dx}{\sin x-\cos x+\sqrt{2}}$ equals

(a) $-\frac{1}{\sqrt{2}}\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)+c$ (b) $\frac{1}{\sqrt{2}}\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)+c$

(c) $\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)+c$ (d) $-\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)+c$

8. $\int\left(x^x\right)^2(1+\log x)dx$

(a) $\frac{\left(x^x\right)^2}{2}+k$ (b) x^x+k

(c) $\frac{x^x}{2}+k$ (d) none of these

9. The value of $\int\frac{\sin x}{3+4\cos^2 x}dx$ is

(a) $\log(3+4\cos^2 x)+c$

(b) $\frac{-1}{2\sqrt{3}}\tan^{-1}\left(\frac{\cos x}{\sqrt{3}}\right)+c$

(c) $\frac{-1}{2\sqrt{3}}\tan^{-1}\left(\frac{2\cos x}{\sqrt{3}}\right)+c$

(d) $\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{2\cos x}{\sqrt{3}}\right)+c$

10. $\int\frac{x^5}{x^2+1}dx=$

(a) $\frac{x^4}{4}+\frac{x^2}{2}+\tan^{-1}x+c$

(b) $\frac{x^4}{4}-\frac{x^2}{2}+\frac{1}{2}\log(x^2+1)+c$

(c) $\frac{x^4}{4}+\frac{x^3}{3}+\tan^{-1}x+c$

(d) $\frac{x^4}{4}+\frac{x^3}{3}-\tan^{-1}x+c$

11. If $\int x\log\left(1+\frac{1}{x}\right)dx=f(x)\log(x+1)+g(x)x^2+Lx+c$, then

(a) $f(x)=\frac{x^2}{2}$ (b) $g(x)=\log x$

(c) $L=1$ (d) none of these

12. $\int\frac{dx}{1-\cos x-\sin x}$ is equal to

(a) $\log\left|1+\cot\frac{x}{2}\right|+c$ (b) $\log\left|1-\tan\frac{x}{2}\right|+c$

(c) $\log\left|1-\cot\frac{x}{2}\right|+c$ (d) $\log\left|1+\tan\frac{x}{2}\right|+c$

13. $\int\frac{(x-1)e^x}{(x+1)^3}dx$ is

(a) $\frac{e^x}{x+1}$ (b) $e^x\left(\frac{x}{x+1}\right)$

(c) $\frac{e^x(x-1)}{(x+1)^2}$ (d) $\frac{e^x}{(x+1)^2}$

14. The value of $\int\frac{\sin x}{\sqrt{4-\cos^2 x}}dx$ is

(a) $\sin^{-1}\left(\frac{\cos x}{2}\right)+c$ (b) $\frac{1}{2}\sin^{-1}\left(\frac{\cos x}{2}\right)+c$

(c) $-\frac{1}{2}\sin^{-1}\left(\frac{\cos x}{2}\right)+c$ (d) $-\sin^{-1}\left(\frac{\cos x}{2}\right)+c$

15. $\int \frac{\sin 2x dx}{a^2 \sin^2 x + b^2 \cos^2 x} dx =$

(a) $\log(a^2 \sin^2 x + b^2 \cos^2 x) + c$

(b) $\frac{1}{a^2 - b^2} \log(a^2 \sin^2 x + b^2 \cos^2 x) + c$

(c) $a^2 \sin^2 x + b^2 \cos^2 x$

(d) none of these

16. $\int \frac{\sin^8 x - \cos^8 x}{1 - 2\sin^2 x \cos^2 x} dx =$

(a) $\frac{1}{2}\sin 2x + c$ (b) $-\frac{1}{2}\sin 2x + c$

(c) $-\frac{1}{2}\sin x + c$ (d) $-\sin^2 x + c$

17. $\int \frac{dx}{7 + 5\cos x}$ is equal to

(a) $\frac{1}{\sqrt{6}} \tan^{-1}\left(\frac{1}{\sqrt{6}} \tan \frac{x}{2}\right) + c$

(b) $\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{1}{\sqrt{3}} \tan \frac{x}{2}\right) + c$

(c) $\frac{1}{4} \tan^{-1}\left(\tan \frac{x}{2}\right) + c$

(d) $\frac{1}{7} \tan^{-1}\left(\tan \frac{x}{2}\right) + c$

18. $\int \frac{\log(x/e)}{(\log x)^2} dx$ is

(a) $\frac{\log x}{x}$ (b) $\frac{x}{\log x}$

(c) $\frac{x}{(\log x)^2}$ (d) none of these

19. $\int \frac{\sin(x-a)}{\sin(x+a)} dx =$

(a) $x\cos 2a + \sin 2a \log|\sin(x+a)| + c$

(b) $x\cos 2a - \sin 2a \log|\sin(x+a)| + c$

(c) $x\cos 2a + \cos 2a \log|\sin(x+a)| + c$

(d) $x\sin 2a - \cos 2a \log|\sin(x+a)| + c$

20. If $\int \cot^{-1}(1 - x + x^2) dx = x\tan^{-1} x - \frac{1}{2}\log(1 + x^2) + A$, then A is equal to

(a) $-(1-x)\tan^{-1}(1-x^2) + \frac{1}{2}\log[1 + (1-x)^2] + c$

(b) $(1-x)\tan^{-1}(1-x^2) + \frac{1}{2}\log[1 + (1-x)^2] + c$

(c) $-(1-x)\tan^{-1}(1-x) + \frac{1}{2}\log[1 + (1-x)^2] + c$

(d) none of these

21. $\int \frac{1 + x + \sqrt{x + x^2}}{\sqrt{x} + \sqrt{1+x}} dx$ is equal to

(a) $\frac{1}{2}\sqrt{1+x} + c$ (b) $\frac{2}{3}(1+x)^{3/2} + c$

(c) $\sqrt{1+x} + c$ (d) $2(1+x)^{3/2} + c$

22. $\int \left[\frac{1}{(x+a)^3 (x+b)^5}\right]^{1/4} dx$ is equal to

(a) $\frac{1}{(b-a)}\left(\frac{x+b}{x+a}\right)^{1/4} + k$ (b) $-\frac{1}{(b-a)}\left(\frac{x+a}{x+b}\right)^{1/4} + k$

(c) $\frac{4}{b-a}\left(\frac{x+a}{x+b}\right)^{3/4} + k$ (d) $\frac{4}{b-a}\left(\frac{x+a}{x+b}\right)^{1/4} + k$

23. $\int \frac{dx}{\cos x + \sqrt{3}\sin x}$ equals

(a) $\frac{1}{2}\log\tan\left(\frac{x}{2} - \frac{\pi}{12}\right) + c$ (b) $\log\tan\left(\frac{x}{2} + \frac{\pi}{12}\right) + c$

(c) $\log\tan\left(\frac{x}{2} - \frac{\pi}{12}\right) + c$ (d) $\frac{1}{2}\log\tan\left(\frac{x}{2} + \frac{\pi}{12}\right) + c$

24. $\int \frac{xe^x}{(1+x)^2} dx$ is equal to

(a) $\frac{e^x}{x+1} + c$ (b) $e^x(x+1) + c$

(c) $-\frac{e^x}{(x+1)^2} + c$ (d) $\frac{e^x}{1+x^2} + c$

25. $\int \frac{dx}{\cos^3 x \sqrt{2\sin 2x}}$ is equal to

(a) $\sqrt{\tan x} + \frac{\tan^{5/2} x}{5} + c$

(b) $\sqrt{\tan x} + \frac{2}{5}\tan^{5/2} x + c$

(c) $2\sqrt{\tan x} + \frac{2}{5}\tan^{5/2} x + c$

(d) none of these

26. $\int \frac{x^2 - 1}{\left(x^4 + 3x^2 + 1\right)\tan^{-1}\left(x + 1/x\right)} dx =$

(a) $\tan^{-1}(x + 1/x) + c$ (b) $\cot^{-1}(x + 1/x) + c$

(c) $\log(x + 1/x) + c$ (d) $\log[\tan^{-1}(x + 1/x)] + c$

27. $\int \frac{\sin\sqrt{x}}{\sqrt{x}} dx =$

(a) $\cos\sqrt{x} + c$ (b) $-2\cos\sqrt{x} + c$

(c) $2\cos\sqrt{x} + c$ (d) $2\sin\sqrt{x} + c$

28. $\int x^{51}\left(\tan^{-1}x+\cot^{-1}x\right)dx=$

(a) $\frac{x^{52}}{52}(\tan^{-1}x+\cot^{-1}x)+c$

(b) $\frac{x^{52}}{52}(\tan^{-1}x-\cot^{-1}x)+c$

(c) $\frac{\pi x^{52}}{104}+\frac{\pi}{2}+c$

(d) $\frac{x^{52}}{52}\cdot\frac{\pi}{2}+c$

29. $\int\frac{\sqrt{5+x^{10}}}{x^{16}}dx$ is equal to

(a) $-\frac{1}{75}\left(1+\frac{5}{x^{10}}\right)+c$ (b) $-\frac{1}{50}\left(1+\frac{5}{x^{10}}\right)^{3/2}+c$

(c) $-\frac{1}{50}\left(1+\frac{5}{x^{10}}\right)^{1/2}+c$ (d) $-\frac{1}{75}\left(1+\frac{5}{x^{10}}\right)^{3/2}+c$

30. $\int e^{\tan^{-1}x}\left(1+\frac{x}{1+x^2}\right)dx$ is equal to

(a) $xe^{\tan^{-1}x}$ (b) $\frac{1}{2}xe^{\tan^{-1}x}$

(c) $e^{\tan^{-1}x}$ (d) $\frac{1}{2}e^{\tan^{-1}x}$

31. $\int\frac{dx}{5+4\sin x}=$

(a) $\tan^{-1}\frac{5\tan(x/2)+4}{3}+c$

(b) $\frac{1}{3}\tan^{-1}\left(\frac{5x+4}{3}\right)+c$

(c) $\frac{2}{3}\tan^{-1}\left(\frac{5\tan(x/2)+4}{3}\right)$

(d) none of these

32. $\int\frac{dx}{x(\log x)(\log\log x)...\underbrace{(\log\log...x)}_{8\text{ times}}}$ is equal to

(a) $\underbrace{(\log\log...x)}_{8\text{ times}}+c$ (b) $\underbrace{(\log\log...x)}_{7\text{ times}}+c$

(c) $\underbrace{(\log\log...x)}_{9\text{ times}}+c$ (d) none of these

33. $\int\frac{\sqrt{\tan x}}{2\sin x\cos x}dx$ is equal to

(a) $\sqrt{\tan x}$ (b) $2\sqrt{\tan x}$

(c) $\frac{1}{2}\sqrt{\tan x}$ (d) $\frac{1}{2}\tan x$

34. $\int\frac{1}{\sin^2 x\cos^2 x}dx=$

(a) $\tan x+\cot x+c$ (b) $\cot x-\tan x+c$

(c) $\tan x-\cot x+c$ (d) none of these

35. If $\int\frac{dx}{\cos^3 x\sqrt{\sin 2x}}=a(\tan^2 x+b)\sqrt{\tan x}+c$, then

(a) $a=\frac{\sqrt{2}}{5}, b=\frac{1}{\sqrt{5}}$ (b) $a=\frac{\sqrt{2}}{5}, b=5$

(c) $a=\frac{\sqrt{2}}{5}, b=-\frac{1}{\sqrt{5}}$ (d) $a=\frac{\sqrt{2}}{5}, b=\sqrt{5}$

36. $\int e^{e^{e^x}}\cdot e^{e^x}\cdot e^x dx$ is equal to

(a) $\frac{1}{2}2e^{e^x}$ (b) $\left(e^{e^{e^x}}\right)^2$

(c) $e^{e^{e^x}}$ (d) $\frac{1}{2}e^{e^{e^x}}$

37. If $\int\frac{1}{(x^2+1)(x^2+4)}dx=A\tan^{-1}x+B\tan^{-1}\frac{x}{2}+c$, then

(a) $A=\frac{1}{3}$ (b) $B=\frac{2}{3}$

(c) $A=-\frac{1}{3}$ (d) $B=-\frac{1}{6}$

38. $\int\frac{x^2-1}{(x^4+3x^2+1)\tan^{-1}[x+(1/x)]}dx$ is equal to

(a) $\tan^{-1}\left(x+\frac{1}{x}\right)+c$ (b) $\cot^{-1}\left(x+\frac{1}{x}\right)+c$

(c) $\log\left(x+\frac{1}{x}\right)+c$ (d) $\log\left[\tan^{-1}\left(x+\frac{1}{x}\right)\right]+c$

39. $\int\frac{dx}{\sin(x-a)\sin(x-b)}=$

(a) $\frac{1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|$

(b) $\frac{1}{\sin(b-a)}\log\left|\frac{\sin(x-b)}{\sin(x-a)}\right|+c$

(c) $\frac{1}{\sin(b-a)}\log\left|\sin(x-a)\sin(x-b)\right|+c$

(d) none of these

40. $\int\frac{\sin 2x}{\sin^4+\cos^4 x}dx=$

(a) $\tan^{-1}(\tan^2 x)+c$ (b) $\tan^{-1}(\cot^2 x)+c$

(c) $\cot^{-1}(\tan^2 x)+c$ (d) $\cot^{-1}(\cot^2 x)+c$

41. $\int \frac{dx}{(\sin x - 2\cos x)(2\sin x + \cos x)}$ is equal to

(a) $\log_e \sqrt[5]{\frac{\tan x - 2}{2\tan x + 1}} + c$

(b) $\frac{1}{4}\log\left(\frac{\tan x - 2}{\tan x + 2}\right) + c$

(c) $-\frac{1}{4}\log\left(\frac{2\sin x - \cos x}{\sin x + 2\cos x}\right) + c$

(d) none of these

42. $\int \frac{dx}{1 + \sin x + \cos x} =$

(a) $\log\left|1 + \cot\frac{x}{2}\right|$ (b) $\log|1 + \tan x| + c$

(c) $\log\left|1 + \tan\frac{x}{2}\right| + c$ (d) none of these

43. $\int 5^{5^{5^x}} \cdot 5^{5^x} \cdot 5^x \, dx$ is equal to

(a) $\frac{5^{5^x}}{(\log 5)^3} + c$ (b) $5^{5^{5^x}} (\log 5)^3 + c$

(c) $\frac{5^{5^{5^x}}}{(\log 5)^3} + c$ (d) none of these

44. If $f(x) = \lim_{n\to\infty} n^2 (x^{1/n} - x^{1/(n+1)}), x > 0$, then $\int x f(x) dx$ is equal to

(a) $x^2/2$ (b) 0

(c) $x^2 \log x - \frac{1}{2}x^2 + c$ (d) none of these

45. $\int \frac{2\cos x}{\sin x + \cos x} dx =$

(a) $\frac{x}{2} + \frac{1}{2}\log|\sin x + \cos x| + c$

(b) $x - \log|\sin x + \cos x| + c$

(c) $x + \log|\sin x + \cos x| + c$

(d) none of these

46. $\int \frac{\sin x}{\sin x - \cos x} dx =$

(a) $x + \log(\sin x - \cos x) + c$

(b) $\frac{1}{2}[x + \log(\sin x - \cos x)] + c$

(c) $\frac{1}{2}[x + \log(\sin x + \cos x)] + c$

(d) $\frac{1}{2}[x - \log(\sin x + \cos x)] + c$

47. The value of $\int \frac{dx}{(2\sin x + \cos x)^2}$ is

(a) $\frac{1}{2 + \cot x} + c$ (b) $-\frac{1}{2}\left(\frac{1}{2\tan x - 1}\right) + c$

(c) $\frac{1}{2}\left(\frac{1}{2\tan x + 1}\right) + c$ (d) $\frac{1}{2}\log(2\tan x + 1) + c$

48. $\int \frac{\cos 5x + \cos 4x}{1 - 2\cos 3x} dx$ is equal to

(a) $\frac{\sin 2x}{2} + \sin x + c$ (b) $-\frac{\sin 2x}{2} - \sin x + c$

(c) $\frac{\sin 2x}{2} - \sin x + c$ (d) $-\frac{\sin 2x}{2} + \sin x + c$

49. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx =$

(a) $\cos x + c$ (b) $x + c$

(c) $\sin x + c$ (d) $\tan x + c$

50. The value of $\int \frac{e^{\tan^{-1} x}}{(1 + x^2)} dx$ is

(a) $\tan^{-1} x + c$ (b) $e^{\tan^{-1} x} + c$

(c) $\frac{1}{1 + x^2} + c$ (d) $\frac{2xe^{\tan^{-1} x}}{(1 + x^2)^2} + c$

51. $\int (\sqrt{\tan x} + \sqrt{\cot x}) dx =$

(a) $\sqrt{2}\sin^{-1}(\sin x - \cos x) + c$

(b) $\sqrt{2}\sin^{-1}(\sin x + \cos x) + c$

(c) $\sqrt{2}\tan^{-1}(\sin x - \cos x) + c$

(d) none of these

52. If $I = \int \left(\sqrt{\cot x} - \sqrt{\tan x}\right) dx$, then I equals

(a) $\sqrt{2}\log\left(\sqrt{\tan x} - \sqrt{\cot x}\right) + c$

(b) $\sqrt{2}\log\left|\sin x + \cos x + \sqrt{\sin 2x}\right| + c$

(c) $\sqrt{2}\log\left|\sin x - \cos x + \sqrt{2}\sin x \cos x\right| + c$

(d) $\sqrt{2}\log\left|\sin(x + \pi/4) + \sqrt{2}\sin x \cos x\right| + c$

53. $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx$ equals

(a) $\sin x - 2(\sin x)^{-1} + c$

(b) $\sin x - 6\tan^{-1}(\sin x) + c$

(c) $\sin x - 2(\sin x)^{-1} - 6\tan^{-1}(\sin x) + c$

(d) $\sin x - 2(\sin x)^{-1} + 5\tan^{-1}(\sin x) + c$

54. The value of $\int e^{\log(\tan x)}dx$ is

(a) $\tan x + c$ (b) $\log(\sec x) + c$

(c) $\log(\tan x) + c$ (d) $e^{\tan x} + c$

55. If $I = \int \frac{x^5}{\sqrt{1+x^3}}dx$, then I is equal to

(a) $\frac{2}{9}(1+x^3)^{5/2} + \frac{2}{3}(1+x^3)^{3/2} + c$

(b) $\frac{2}{9}(1+x^3)^{5/2} - \frac{2}{3}(1+x^3)^{1/2} + c$

(c) $\log\left|\sqrt{x} + \sqrt{1+x^3}\right| + c$

(d) $x^2\log(1+x^3) + c$

56. $\int \frac{\sin\theta + \cos\theta}{\sqrt{\sin 2\theta}}d\theta$ equals

(a) $\log\left|\cos\theta - \sin\theta + \sqrt{\sin 2\theta}\right|$

(b) $\log\left|\sin\theta - \cos\theta + \sqrt{\sin 2\theta}\right|$

(c) $\sin^{-1}(\sin\theta - \cos\theta) + c$

(d) $\sin^{-1}(\sin\theta + \cos\theta) + c$

57. The value of $\int \frac{1-\tan x}{1+\tan x}dx$ is

(a) $\log\left[\sec\left(\frac{\pi}{4} - x\right)\right] + c$ (b) $\log\left[\cos\left(\frac{\pi}{4} + x\right)\right] + c$

(c) $\log\left[\sin\left(\frac{\pi}{4} + x\right)\right] + c$ (d) $\log\left[\tan\left(\frac{\pi}{2} + x\right)\right] + c$

58. If $I = \int \frac{dx}{\sin(x-a)\cos(x-b)}$, then I is equal to

(a) $\frac{1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\cos(x-b)}\right| + c$

(b) $\frac{1}{\cos(a-b)}\log\left|\frac{\sin(x-a)}{\cos(x-b)}\right| + c$

(c) $\frac{1}{\sin(a+b)}\log\left|\frac{\sin(x-a)}{\cos(x-b)}\right| + c$

(d) $\frac{1}{\cos(a+b)}\log\left|\frac{\sin(x-a)}{\cos(x-b)}\right| + c$

59. $\int \sqrt[3]{\frac{\sin^n x}{\cos^{n+6} x}}dx$, $n \in N$, is equal to

(a) $\frac{3}{n}(\tan x)^{(n/3)+1} + c$ (b) $\frac{3}{3+n}(\tan x)^{(n/3)+1} + c$

(c) $\frac{3}{n}(\cos x)^{n+1} + c$ (d) none of these

60. $\int \sqrt{\tan\theta}d\theta$ is equal to

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right) + \frac{1}{2\sqrt{2}}\log\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right| + c$

(b) $\frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right) + \frac{1}{2\sqrt{2}}\log\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right| + c$

(c) $\frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right) + \frac{1}{\sqrt{2}}\log\left|\frac{x^2+\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right| + c$

(d) none of these

EXERCISE SET 2

1. If $I = \int \sqrt{1+\sin x}dx$, then I is equal to

(a) $-2\sqrt{1-\sin x} + c$ (b) $\sin(x/2) + \cos(x/2) + c$

(c) $\cos(x/2) - \sin(x/2) + c$ (d) $2\sqrt{1-\sin x} + c$

2. $\int \frac{\sin x}{\sin 4x}dx$ equals

(a) $\frac{1}{8}\log\left|\frac{\cos x - 1}{\cos x + 1}\right| - \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}\cos x - 1}{\sqrt{2}\cos x + 1}\right| + c$

(b) $\frac{1}{8}\log\left|\frac{\sin x - 1}{\sin x + 1}\right| - \frac{1}{4\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x - 1}{\sqrt{2}\sin x + 1}\right| + c$

(c) $\frac{1}{4}\log\left|\frac{\sin x - 1}{\sin x + 1}\right| - \frac{1}{\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x - 1}{\sqrt{2}\sin x + 1}\right| + c$

(d) none of these

3. The value of $\int e^x \sin x dx$ is

(a) $e^x(\sin x - \cos x) + c$ (b) $e^x(\sin x + \cos x) + c$

(c) $\frac{1}{2}e^x(\sin x - \cos x) + c$ (d) $\frac{1}{2}e^x(\sin x + \cos x) + c$

4. If $I = \int \frac{dx}{\sec x + \operatorname{cosec} x}$, then I equals

(a) $\frac{1}{2}\left(\cos x + \sin x - \frac{1}{\sqrt{2}}\log(\operatorname{cosec} x - \cos x)\right) + c$

(b) $\frac{1}{2}\left(\sin x - \cos x - \frac{1}{\sqrt{2}}\log\left|\operatorname{cosec} x + \cot x\right|\right) + c$

(c) $\frac{1}{\sqrt{2}}\left(\sin x + \cos x + \frac{1}{2}\log\left|\operatorname{cosec} x - \cos x\right|\right) + c$

(d) none of these

5. $\int \frac{(x+1)e^x}{\cos^2(xe^x)}dx$ equals

(a) $\tan(xe^x)$ (b) $\sqrt{\tan(xe^x)}$

(c) $\tan^{-1}(xe^x)$ (d) none of these

6. If $I = \int \frac{\cos x}{\sin(x-a)}dx$, then I equals

(a) $\cos a \log|\sin(x-a)| - x\sin a + c$

(b) $\cos a \log|\sin(x-a)| + x\sin a + c$

(c) $-\cos a \log|\sin(x-a)| - x\sin a + c$

(d) $x\sin a - \cos a \log|\sin(x-a)| + c$

7. $\int \frac{\cot x}{\sqrt{\sin x}}dx$ is

(a) $\frac{-2}{\sqrt{\sin x}}$ (b) $\frac{2}{\sqrt{\sin x}} + c$

(c) $2\sqrt{\sin x} + c$ (d) $\frac{1}{2\sqrt{\sin x}} + c$

8. If $I = \int \frac{dx}{\sin^4 x + \cos^4 x}$, then I is equal to

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan 2x}{\sqrt{2}}\right) + c$

(b) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1+\cos 2x}{\sqrt{2}}\right) + c$

(c) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan x + \cot x}{\sqrt{2}}\right) + c$

(d) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{2}}\right) + c$

9. The value of $\int \frac{x-1}{(x+1)(x-2)}dx$ is

(a) $\log|x+1| + \frac{1}{3}\log|x-2| + c$

(b) $\frac{1}{3}\log|x+1| + \frac{2}{3}\log|x-2| + c$

(c) $\frac{2}{3}\log|x+1| + \frac{1}{3}\log|x-2| + c$

(d) $\frac{1}{6}\log|x^2+2| + \frac{1}{7}\log|x+2| + c$

10. If $I = \int \frac{\sin^4 x}{\sin^4 x + \cos^4 x}dx$, then I equals

(a) $\frac{1}{2}\left[x + \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}+\sin 2x}{\sqrt{2}-\sin 2x}\right|\right] + c$

(b) $\frac{1}{2}\left[x - \frac{1}{2\sqrt{2}}\log\left|\frac{1+\sin 2x}{1-\sin 2x}\right|\right] + c$

(c) $\frac{1}{2}\left[x - \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}+\sin 2x}{\sqrt{2}-\sin 2x}\right|\right] + c$

(d) $\frac{1}{2}\left[x + \frac{1}{2\sqrt{2}}\log\left|\frac{1+\sqrt{2}\sin 2x}{1-\sqrt{2}\sin 2x}\right|\right] + c$

11. $\int \frac{dx}{9+16\sin^2 x}$ is equal to

(a) $\frac{1}{3}\tan^{-1}\left(\frac{3\tan x}{5}\right)$ (b) $\frac{1}{5}\tan^{-1}\left(\frac{\tan x}{15}\right)$

(c) $\frac{1}{15}\tan^{-1}\left(\frac{\tan x}{5}\right)$ (d) $\frac{1}{15}\tan^{-1}\left(\frac{5\tan x}{3}\right)$

12. The value of $\int \frac{1-\cos x}{\cos x(1+\cos x)}dx$ is

(a) $\log\tan\frac{x}{2} + c$

(b) $\log|\sec x + \tan x| + \frac{3}{2}\tan x + c$

(c) $\log|\sec x + \tan x| - 2\tan\frac{x}{2} + c$

(d) $2\log|\sec x + \tan x| - \frac{3}{2}\tan x + c$

13. If $I = \int \frac{dx}{x^{1/2} + x^{1/3}}$, then I equals

(a) $2\sqrt{x} - 3x^{1/3} + 6x^{1/6} - \log(1+x^{1/6}) + c$

(b) $3x^{1/3} - 6x^{1/6} - \log(1+x^{1/6}) + c$

(c) $\sqrt{x} + 3x^{1/3} - 6x^{1/6} + \log(1+x^{1/6}) + c$

(d) $3x^{1/3} - \sqrt{x} + 6x^{1/6} - \log(1+x^{1/6}) + c$

14. If $\int \frac{2^x}{\sqrt{1-4^x}}dx = k\sin^{-1}(2^x)$, then k is equal to

(a) $\log 2$ (b) $\frac{1}{2}$

(c) $\frac{1}{2}\log 2$ (d) $\frac{1}{\log 2}$

15. If $I=\int\frac{(3\sin\theta-2)\cos\theta}{5-\cos^2\theta-4\sin\theta}d\theta$, then I equals

(a) $3\log(2-\sin\theta)+\frac{4}{2-\sin\theta}+c$

(b) $3\log|2-\sin\theta|+\frac{4}{2+\sin\theta}+c$

(c) $3\log(2+\sin\theta)-\frac{2}{2+\cos\theta}+c$

(d) $3\log(2+\sin\theta)-\frac{2}{2-\cos\theta}+c$

16. The equation of a curve is $y=f(x)$. The tangents at $[1, f(1)], [2, f(2)]$ and $[3, f(3)]$ make angles 30°, 60° and 45°, respectively, with +ve direction of the x-axis, then the value of $\int_2^3 f'(x)f''(x)dx+\int_1^3 f''(x)dx$ is

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$

(c) $-\sqrt{3}$ (d) $-\frac{1}{\sqrt{3}}$

17. $\int\cos^3 x e^{\log(\sin x)}dx$ is equal to

(a) $-\frac{\sin^4 x}{4}+c$ (b) $-\frac{\cos^4 x}{4}+c$

(c) $\frac{e^{\sin}x}{4}+c$ (d) none of these

18. $\int\frac{xdx}{1+x^4}=$

(a) $\tan^{-1}x^2+c$ (b) $\frac{1}{2}\tan^{-1}x^2+c$

(c) $\log(1+x^4)+c$ (d) none of these

19. If $I=\int\frac{dx}{4+5\sin x}$, then I equals

(a) $\frac{1}{3}\log\left|\frac{\tan(x/2)+5}{3}\right|+c$

(b) $\frac{1}{3}\log\left|\frac{2\tan(x/2)+1}{\tan(x/2)+2}\right|+c$

(c) $\frac{1}{3}\log\left|\frac{2\tan(x/2)+1}{\tan(x/2)-2}\right|+c$

(d) $\frac{1}{\sqrt{3}}\log\left|\frac{2\tan(x/2)-1}{\tan(x/2)+1}\right|+c$

20. $\int\cos^{\frac{-3}{7}}x\sin^{\frac{-11}{7}}xdx=$

(a) $\log\left|\sin^{\frac{4}{7}}x\right|+c$ (b) $\frac{4}{7}\tan^{\frac{4}{7}}x+c$

(c) $-\frac{7}{4}\tan^{-\frac{4}{7}}x+c$ (d) $\log\left|\cos^{\frac{3}{7}}x\right|+c$

21. The integral $\int\frac{2x-3}{(x^2+x+1)^2}dx$ is equal to

(a) $-\frac{8x+7}{3(x^2+x+1)}-\frac{16}{3\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+c$

(b) $-\frac{1}{x^2+x+1}-\frac{4}{3}\tan^{-1}(4x+3)+c$

(c) $\frac{1}{2(x^2+x+1)}-\frac{(2x+1)^2}{(x^2+x+1)^2}+c$

(d) $\frac{1}{4(x^2+x+1)}+\frac{2}{3}\tan^{-1}(2x+1)+c$

22. $\int|x|^3dx$ is equal to

(a) $\frac{x^4}{4}$ (b) $-\frac{x^4}{4}$

(c) $\frac{|x|^4}{4}$ (d) none of these

23. If $I=\int\frac{\tan x}{\sqrt{a+b\tan^2 x}}dx\ (a>b)$, then I equals

(a) $\frac{1}{\sqrt{b-a}}\sin^{-1}\left(\sqrt{\frac{a+b\tan^2 x}{b-a}}\right)+c$

(b) $\frac{1}{\sqrt{b-a}}\cos^{-1}\left(\sqrt{\frac{a+b\tan^2 x}{b-a}}\right)+c$

(c) $\frac{1}{\sqrt{b-a}}\tan^{-1}\left(\sqrt{\frac{a+b\tan^2 x}{b-a}}\right)+c$

(d) $\frac{1}{\sqrt{b-a}}\tan^{-1}\left(\sqrt{\frac{b-a}{a+b\tan^2 x}}\right)+c$

24. $\int\sec^{2/3}x\,\mathrm{cosec}^{4/3}xdx$ is equal to

(a) $3(\tan x)^{1/3}$ (b) $3(\cot x)^{-1/3}$

(c) $-3(\tan x)^{-1/3}$ (d) $-3(\cot x)^{-1/3}$

25. $I=\int\frac{1}{1-\cos^4 x}dx$ is equal to

(a) $2\sqrt{2}\left(\cot x+\sqrt{2}\tan^{-1}\sqrt{2}\cot x\right)$

(b) $-\frac{1}{2\sqrt{2}}\left[\sqrt{2}\cot x+\tan^{-1}\left(\sqrt{2}\cot x\right)\right]+c$

(c) $2\sqrt{2}\left[\cot x+\tan^{-1}\left(\cot x\right)\right]$

(d) none of these

26. $\int\frac{\cos 2x}{(\sin x+\cos x)^2}dx=$

(a) $\frac{-1}{\sin x+\cos x}+c$

(b) $\log(\sin x+\cos x)+c$

(c) $\log(\sin x-\cos x)+c$

(d) $\log(\sin x+\cos x)^2+c$

27. If $I=\int\frac{\sin^2 x}{(3+4\cos x)^3}dx$, then I equals

(a) $\frac{3\cos x+8}{(3+4\cos x)^2}+c$ (b) $\frac{3+8\cos x}{16(3+4\cos x)^2}+c$

(c) $\frac{3+\cos x}{(3+4\cos x)^2}+c$ (d) $\frac{3-8\cos x}{16(3+4\cos x)^2}+c$

28. $\int(1-\cos x)\text{cosec}^2 xdx$ equals

(a) $\tan\frac{x}{2}+c$ (b) $\frac{1}{2}\tan\frac{x}{2}+c$

(c) $\cot\frac{x}{2}+c$ (d) $2\tan\frac{x}{2}+c$

29. $\int\frac{\sin x\cos x}{a\cos^2 x+b\sin^2 x}dx=$

(a) $\frac{1}{2(b-a)}\log\left|a\cos^2 x+b\sin^2 x\right|+c$

(b) $\frac{1}{(b-a)}\log\left|a\cos^2 x+b\sin^2 x\right|+c$

(c) $\frac{1}{2}\log\left|a\cos^2 x+b\sin^2 x\right|+c$

(d) none of these

30. $\int\frac{\cos x+x\sin x}{x(x+\cos x)}dx=$

(a) $\log[x(x+\cos x)]+c$ (b) $\log\left(\frac{x}{x+\cos x}\right)+c$

(c) $\log\left(\frac{x+\cos x}{x}\right)+c$ (d) none of these

31. If $I=\int\frac{dx}{\sin^3 x+\cos^3 x}$, then I equals

(a) $\frac{1}{3\sqrt{2}}\log\left|\frac{\sqrt{2}+t}{\sqrt{2}-t}\right|+\tan^{-1}(t)+c$

(b) $\frac{1}{3\sqrt{2}}\log\left|\frac{\sqrt{2}-t}{\sqrt{2}+t}\right|+\frac{1}{3}\tan^{-1}(t)+c$

(c) $\frac{1}{3\sqrt{2}}\log\left|\frac{\sqrt{2}+t}{\sqrt{2}-t}\right|+\frac{2}{3}\tan^{-1}(t)+c$

(d) $\frac{1}{3\sqrt{2}}\log\left|\frac{\sqrt{2}+t}{\sqrt{2}-t}\right|-\frac{2}{3}\tan^{-1}(t)+c$

32. The value of $\int\frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}dx$ is

(a) $\sin x-6\tan^{-1}(\sin x)+c$

(b) $\sin x-2(\sin x)^{-1}+c$

(c) $\sin x-2(\sin x)^{-1}-6\tan^{-1}(\sin x)+c$

(d) $\sin x-2(\sin x)^{-1}+5\tan^{-1}(\sin x)+c$

33. $\int\frac{(1+x)^2}{x+x^3}dx=$

(a) $\log_e x+\log_e(1+x^2)+c$

(b) $\log_e x+\tan^{-1}x+c$

(c) $\log_e x+2\tan^{-1}x+c$

(d) none of these

34. $\int\frac{x^{7/2}}{\sqrt{x^9+1}}dx$ is equal to

(a) $\sin^{-1}\sqrt{x^2+9}+c$ (b) $\frac{1}{2}\ln\left|\frac{x^9+1}{x^9-1}\right|+c$

(c) $\frac{2}{9}\ln\left|x^{\frac{9}{2}}\sqrt{x^9+1}\right|+c$ (d) none of these

35. $\int\frac{dx}{\cos x+\sqrt{3}\sin x}=$

(a) $\log\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)+c$ (b) $\log\tan\left(\frac{x}{2}-\frac{\pi}{3}\right)+c$

(c) $\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)+c$ (d) none of these

Assertion-Reason Type Questions

Each of the following questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) or (d) given as follows:

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true; Statement II is false.
(d) Statement I is false; Statement II is true.

36. Statement I:

$$\int \frac{(2-2x)}{\sqrt{(4+2x-x^2)}}dx = 2\sqrt{(4+2x-x^2)} + \sin^{-1}\left(\frac{x-1}{\sqrt{5}}\right) + c$$

Statement II: $\int \frac{dx}{\sqrt{a^2-x^2}} = \frac{x}{2}\sqrt{(a^2-x^2)} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{2}\right)$

37. Statement I: If $I_n = \int \cot^n x dx$, then

$$5(I_6 + I_4) = -\cot^5 x$$

Statement II: If $I_n = \int \cot^n x dx$, then

$$I_n = -\frac{\cot^{n-1}}{n} - I_{n-2}, \text{ where } n \geq 2$$

38. Statement I: $\int e^{x^2} dx = e^{x^2} + c$

Statement II: $\int e^x dx = e^x + c$

39. Statement I: The function $f(x) = \int \sin^2 x dx$ satisfies $f(x+\pi) = f(x)$, for all $x \in \mathbb{R}$

Statement II: $= \sin^2(x+\pi) = \sin^2 x$

Previous Years' Questions

40. $\int \frac{dx}{x(x^n+1)}$ is equal to

(a) $\frac{1}{n}\log\left(\frac{x^n}{x^n+1}\right) + c$ (b) $\frac{1}{n}\log\left(\frac{x^n+1}{x^n}\right) + c$

(c) $\log\left(\frac{x^n}{x^n+1}\right) + c$ (d) none of these

[AIEEE 2002]

41. If $\int \frac{\sin x}{\sin(x-\alpha)}dx = Ax + B\log\sin(x-\alpha) + c$, then the value of (A, B) is

(a) $(-\sin\alpha, \cos\alpha)$ (b) $(\cos\alpha, \sin\alpha)$

(c) $(\sin\alpha, \cos\alpha)$ (d) $(-\cos\alpha, \sin\alpha)$

[AIEEE 2004]

42. $\int \frac{dx}{\cos x - \sin x}$ is equal to

(a) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2} - \frac{\pi}{8}\right)\right| + c$

(b) $\frac{1}{\sqrt{2}}\log\left|\cot\left(\frac{x}{2}\right)\right| + c$

(c) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2} - \frac{3\pi}{8}\right)\right| + c$

(d) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2} + \frac{3\pi}{8}\right)\right| + c$ **[AIEEE 2004]**

43. $\int \left[\frac{(\log x - 1)}{1+(\log x)^2}\right]^2 dx$ is equal to

(a) $\frac{x}{(\log x)^2+1} + c$ (b) $\frac{xe^x}{1+x^2} + c$

(c) $\frac{x}{x^2+1} + c$ (d) $\frac{\log x}{(\log x)^2+1} + c$

[AIEEE 2005]

44. $\int \frac{dx}{\cos x + \sqrt{3}\sin x}$ equals

(a) $\frac{1}{2}\log\tan\left(\frac{x}{2} + \frac{\pi}{12}\right) + c$

(b) $\frac{1}{2}\log\tan\left(\frac{x}{2} - \frac{\pi}{12}\right) + c$

(c) $\log\tan\left(\frac{x}{2} + \frac{\pi}{12}\right) + c$

(d) $\log\tan\left(\frac{x}{2} - \frac{\pi}{12}\right) + c$ **[AIEEE 2007]**

45. The value of $\sqrt{2}\int \frac{\sin x dx}{\sin[x-(\pi/4)]}$ is

(a) $x + \log\left|\cos\left(x - \frac{\pi}{4}\right)\right| + c$

(b) $x - \log\left|\sin\left(x - \frac{\pi}{4}\right)\right| + c$

(c) $x + \log\left|\sin\left(x - \frac{\pi}{4}\right)\right| + c$

(d) $x - \log\left|\cos\left(x - \frac{\pi}{4}\right)\right| + c$ **[AIEEE 2008]**

46. $\int \cos x \log\left(\tan\frac{x}{2}\right)dx$ is equal to

(a) $\sin x \log\left(\tan\frac{x}{2}\right)+c$

(b) $\sin x \log\tan\frac{x}{2}-x+c$

(c) $\sin x \log\left(\tan\frac{x}{2}\right)+x+c$

(d) none of these **[IIT 1998]**

47. $\int\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{1/2}\frac{dx}{x}$ is equal to

(a) $2\left[\log\left|\frac{1+\sqrt{1-x}}{\sqrt{x}}\right|+\cos^{-1}\sqrt{x}\right]+c$

(b) $-2\left[\log\left|\frac{1+\sqrt{1-x}}{\sqrt{x}}\right|-\cos^{-1}\sqrt{x}\right]+c$

(c) $-2\left[\log\left|\frac{\sqrt{x}}{1+\sqrt{1-x}}\right|-\cos^{-1}\sqrt{x}\right]+c$

(d) none of these **[IIT 1999]**

48. $\int\frac{dx}{(2x-7)\sqrt{x^2-7x+12}}$ is equal to

(a) $2\sec^{-1}(2x-7)+c$ (b) $\sec^{-1}(2x-7)+c$

(c) $\frac{1}{2}\sec^{-1}(2x-7)+2$ (d) none of these

[IIT 1997]

49. $\int\frac{dx}{1+3\sin^2 x}$ is equal to

(a) $\frac{1}{3}\tan^{-1}(3\tan^2 x)$ (b) $\frac{1}{2}\tan^{-1}(2\tan x)$

(c) $\tan^{-1}(\tan x)$ (d) none of these

[Roorkee 1989]

50. $\int\frac{a^{\sqrt{x}}}{\sqrt{x}}dx$ is equal to

(a) $2a^{\sqrt{x}}\log_e a$ (b) $2a^{\sqrt{x}}\log_a e$

(c) $2a^{\sqrt{x}}\log_{10} a$ (d) $2a^{\sqrt{x}}\log_a 10$

[Roorkee 1991]

51. $\int\frac{[\sqrt{1+x^2}+x]^n}{\sqrt{1+x^2}}dx$ is equal to

(a) $(\sqrt{1+x^2}+x)^n+c$ (b) $\frac{1}{n}(\sqrt{1+x^2}+x)^n+c$

(c) $\frac{(\sqrt{1+x^2}+x)^{n+1}}{n+1}+c$ (d) none of these

[Roorkee 1995]

52. $\int\frac{x^2-1}{x\sqrt{(x^2+\alpha x+1)(x^2+\beta x+1)}}dx$ is equal to

(a) $\log\left[\frac{\sqrt{x^2+\alpha x+1}+\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right]+c$

(b) $2\log\left[\frac{\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right]+c$

(c) $\log\left[\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}\right]+c$

(d) none of these **[Roorkee 1997]**

53. The integral $\int\frac{dx}{(1+\sin x)^{1/2}}$ is

(a) $\sqrt{2}\log\left|\cot\frac{3\pi}{8}-\frac{x}{4}+c\right|$

(b) $\sqrt{2}\log\left|\operatorname{cosec}\left(\frac{\pi}{4}+\frac{x}{2}\right)-\cot\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+c$

(c) $\sqrt{2}\log\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+c$

(d) $\sqrt{2}\log\left|\sec\left(\frac{\pi}{4}-\frac{x}{2}\right)+\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|+c$

[IIT Hyderabad 2002, Roorkee 2000]

54. $\int\frac{2x}{(1+x^2)\sqrt{x^4-1}}dx$ is equal to

(a) $\sqrt{\frac{x^2+1}{x^2-1}}+c$ (b) $\sqrt{\frac{x^2-1}{x^2+1}}+c$

(c) $\sqrt{x^4-1}+c$ (d) none of these

[Roorkee 1998, 2002]

55. $\int\frac{x+\sin x}{1+\cos x}dx$ is equal to

(a) $x\tan\frac{x}{2}+c$ (b) $x\sec^2\frac{x}{2}+c$

(c) $\log\cos\frac{x}{2}$ (d) none of these

[DCE 2004]

56. If $\int\frac{1}{x+x^5}dx=f(x)+c$, then the value of $\int\frac{x^4}{x+x^5}dx$ equals

(a) $\log x-f(x)+c$ (b) $f(x)+\log x+c$

(c) $f(x)-\log x+c$ (d) none of these

[DCE 2005]

57. If $\int \frac{\cos 4x+1}{\cot x-\tan x}dx = k\cos 4x + c$, then

(a) $k=-1/2$ (b) $k=-1/8$

(c) $k=-1/4$ (d) none of these

[DCE 2005]

58. If $\int f(x)dx = g(x)+c$, then $\int f^{-1}(x)dx$ is equal to

(a) $xf^{-1}(x)+c$ (b) $f[g^{-1}(x)]+c$

(c) $xf^{-1}(x)-g[f^{-1}(x)]+c$ (d) $g^{-1}(x)+c$

[DCE 2006]

59. $\int \frac{dx}{x(x^7+1)}$ is equal to

(a) $\log\left(\frac{x^7}{x^7+1}\right)+c$ (b) $\frac{1}{7}\log\left(\frac{x^7}{x^7+1}\right)+c$

(c) $\log\left(\frac{x^7+1}{x^7}\right)+c$ (d) $\frac{1}{7}\log\left(\frac{x^7+1}{x^7}\right)+c$

[DCE 2007]

60. Let $g(x)$ be a differentiable function satisfying $\frac{d}{dx}[g(x)]=g(x)$ and $g(0)=1$, then $\int g(x)\left(\frac{2-\sin 2x}{1-\cos 2x}\right)dx$ is equal to

(a) $g(x)\cot x+c$ (b) $-g(x)\cot x+c$

(c) $\frac{g(x)}{1-\cos 2x}+c$ (d) none of these

[Kerala PET 2002]

61. $\int \frac{dx}{(2x-7)\sqrt{x^2-7x+12}}$ is equal to

(a) $2\sec^{-1}(2x-7)+c$ (b) $\sec^{-1}(2x-7)+c$

(c) $\frac{1}{2}\sec^{-1}(2x-7)+2$ (d) none of these

[Punjab CET 2002, IIT 1997]

62. $\int e^{3\log x}\left(x^4+1\right)^{-1}dx =$

(a) $\log(x^4+1)+c$ (b) $(1/4)\log(x^4+1)+c$

(c) $-\log(x^4+1)+c$ (d) none of these

[KUKCEET 1999, MPCET 2001]

63. The value of $\int \sec^3 x dx$ will be

(a) $(1/2)[\sec x\tan x+\log(\sec x+\tan x)]+c$

(b) $(1/3)[\sec x\tan x+\log(\sec x+\tan x)]+c$

(c) $(1/4)[\sec x\tan x+\log(\sec x+\tan x)]+c$

(d) $(1/8)[\sec x\tan x+\log(\sec x+\tan x)]+c$

[KUKCEET 1999, UPSEAT 2000]

64. $\int \frac{dx}{\sqrt{2ax-x^2}}$ is equal to

(a) $\sin^{-1}[(x-a)/a]$ (b) $\tan^{-1}[(x-a)/a]$

(c) $\cos^{-1}[(x-a)/a]$ (d) $(1/a)\tan^{-1}[(x-a)/a]$

65. Let [.] denote the greatest integer function, then the value of $\int_0^{1.5} x[x^2]\,dx$ is

(a) 0 (b) 3/2

(c) 3/4 (d) 5/4 **[AIEEE 2011]**

66. If the integral $\int \frac{5\sin x}{\tan x-2}dx = x+a\ln\left|\sin x-2\cos x\right|+k$, then a is equal to

(a) -1 (b) -2

(c) 1 (d) 4 **[AIEEE 2012]**

67. The population $p(t)$ at time t of a certain mouse species satisfies the differential equation $dp(t)/dt = 0.5\,p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is

(a) 2 ln 18 (b) ln 9

(c) (1/2) ln 18 (d) ln 18 **[AIEEE 2012]**

ANSWERS

Exercise Set 1

1. (a)	**2.** (a)	**3.** (a)	**4.** (b)	**5.** (b)	**6.** (a)	**7.** (c)	**8.** (a)	**9.** (c)	**10.** (b)
11. (d)	**12.** (c)	**13.** (d)	**14.** (d)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (b)	**19.** (b)	**20.** (c)
21. (b)	**22.** (d)	**23.** (d)	**24.** (a)	**25.** (a)	**26.** (d)	**27.** (b)	**28.** (a) and (d)		**29.** (d)
30. (a)	**31.** (c)	**32.** (c)	**33.** (a)	**34.** (c)	**35.** (b)	**36.** (c)	**37.** (a) and (d)		**38.** (d)
39. (b)	**40.** (a)	**41.** (a)	**42.** (c)	**43.** (c)	**44.** (d)	**45.** (c)	**46.** (b)	**47.** (a)	**48.** (b)
49. (b)	**50.** (b)	**51.** (a)	**52.** (b)	**53.** (c)	**54.** (b)	**55.** (b)	**56.** (c)	**57.** (c)	**58.** (b)
59. (b)	**60.** (a)								

Exercise Set 2

1. (a)	**2.** (b)	**3.** (c)	**4.** (d)	**5.** (a)	**6.** (a)	**7.** (a)	**8.** (a)	**9.** (c)	**10.** (c)
11. (d)	**12.** (c)	**13.** (a)	**14.** (d)	**15.** (a)	**16.** (d)	**17.** (b)	**18.** (b)	**19.** (b)	**20.** (c)
21. (a)	**22.** (d)	**23.** (c)	**24.** (c)	**25.** (b)	**26.** (b)	**27.** (b)	**28.** (a)	**29.** (a)	**30.** (b)
31. (c)	**32.** (c)	**33.** (c)	**34.** (c)	**35.** (c)	**36.** (c)	**37.** (c)	**38.** (d)	**39.** (a)	**40.** (a)
41. (b)	**42.** (d)	**43.** (a)	**44.** (a)	**45.** (c)	**46.** (b)	**47.** (b)	**48.** (b)	**49.** (b)	**50.** (b)
51. (b)	**52.** (b)	**53.** (c)	**54.** (a)	**55.** (a)	**56.** (a)	**57.** (b)	**58.** (c)	**59.** (b)	**60.** (b)
61. (b)	**62.** (b)	**63.** (a)	**64.** (a)	**65.** (c)	**66.** (d)	**67.** (a)			

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Let $I = \int \frac{x \tan^{-1} x}{\sqrt{1+x^2}} dx$

$$I = \frac{1}{2}\int \tan^{-1} x \frac{2x}{\sqrt{1+x}} dx$$

Using ILATE, we have

$$I = \frac{1}{2}\left[\tan^{-1} x \cdot 2\sqrt{1+x^2} - \int \frac{1}{1+x^2} 2\sqrt{1+x^2}\, dx\right]$$

$$\Rightarrow I = \frac{1}{2}\left[2\sqrt{1+x^2}\tan^{-1} x - 2\int \frac{1}{\sqrt{1+x^2}} dx\right]$$

$$\therefore \quad I = \sqrt{1+x^2}\tan^{-1} x - \log(x+\sqrt{1+x^2}) + c$$

Hence, $f(x) = \tan^{-1} x$ and $A = -1$

2. Let $I = \int \frac{a^{x/2}}{\sqrt{a^{-x} - a^x}} dx = \int \frac{a^x}{\sqrt{1-a^{2x}}} dx$

Put $a^x = t \Rightarrow a^x dx = \frac{1}{\log a} dt$

$$\therefore \quad I = \frac{1}{\log a}\int \frac{dt}{\sqrt{1-t^2}}$$

$$= \frac{1}{\log a}\sin^{-1}(t) + c = \frac{1}{\log a}\sin^{-1}(a^x) + c$$

3. Let $t = \frac{d(\sin x)}{b + a\sin x}$

$$\therefore \quad dt = \frac{(b+a\sin x)(-\sin x) - (\cos x)(a\cos x)}{(b+a\sin x)^2} dx$$

$$\therefore \quad -\frac{(a+b\sin x)}{(b+a\sin x)^2} dx = dt$$

$$\Rightarrow \frac{a+b\sin x}{(b+a\sin x)^2} dx = -dt$$

Integrating both sides, we get

$$\Rightarrow \int \frac{a+b\sin x}{(b+a\sin x)^2} dx = -1 + k = -\frac{\cos x}{b+a\sin x} + k$$

4. Integration $I = \int e^{\sin^2 x} \sin 2x dx$

Substituting $e^{\sin^2 x} = t$ and $e^{\sin^2 x} \times 2\sin x \times \cos x dx = dt$

or $e^{\sin^2 x} \sin 2x dx = dt$ in the given integration, we get

$$I = \int dt = t + c = e^{\sin^2 x} + c$$

5. $$\int \left(x + \frac{1}{x}\right)^2 dx = \int \left(x^2 + \frac{1}{x^2} + 2\right) dx$$

$$= \int x^2 dx + \int x^{-2} dx + 2\int dx$$

$$= \frac{x^3}{3} - \frac{1}{x} + 2x + c$$

6. We have

$$\int \frac{1}{(1-x^8)}\left(\cos^{-1}\frac{2x}{1+x^2} + \tan^{-1}\frac{2x}{1-x^2}\right) dx$$

$$= \int \frac{1}{(1-x^8)}\left(\frac{\pi}{2} - \sin^{-1}\frac{2x}{1+x^2} + \tan^{-1}\frac{2x}{1-x^2}\right) dx$$

$$= \int \frac{1}{(1-x^8)}\left(\frac{\pi}{2} - 2\tan^{-1} x + 2\tan^{-1} x\right) dx$$

$$= \frac{\pi}{2}\int \frac{1}{1-x^8} dx \quad = \frac{\pi}{4}\int \left(\frac{1}{1-x^4} + \frac{1}{1+x^4}\right) dx$$

$$= \frac{\pi}{4}\int \frac{1}{1-x^4} dx + \frac{\pi}{4}\int \frac{1}{1+x^4} dx$$

$$= \frac{\pi}{8}\int \left(\frac{1}{1-x^2} + \frac{1}{1+x^2}\right) dx + \frac{\pi}{8}\int \left(\frac{x^2+1}{x^4+1} - \frac{x^2-1}{x^4+1}\right) dx$$

$$= \frac{\pi}{8}\int \frac{1}{1-x^2} dx + \frac{\pi}{8}\int \frac{1}{1+x^2} dx$$

$$+ \frac{\pi}{8}\int \frac{1+(1/x^2)}{x^2+(1/x^2)} dx - \frac{\pi}{8}\int \frac{1-(1/x^2)}{x^2+(1/x^2)} dx$$

$$= \frac{\pi}{8}\int \frac{1}{1-x^2} dx + \frac{\pi}{8}\int \frac{1}{1+x^2} dx$$

$$+ \frac{\pi}{8}\int \frac{1}{[x-(1/x)]^2 + (\sqrt{2})^2} d\left(x - \frac{1}{x}\right)$$

$$- \frac{\pi}{8}\int \frac{d[x+(1/x)]}{[x+(1/x)]^2 - (\sqrt{2})^2} dx$$

$$= \frac{\pi}{16}\log\left|\frac{1+x}{1-x}\right| + \frac{\pi}{8}\tan^{-1}x + \frac{\pi}{8\sqrt{2}}\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right) - \frac{\pi}{16\sqrt{2}}\log\left|\frac{x+\frac{1}{x}-\sqrt{2}}{x+\frac{1}{x}+\sqrt{2}}\right| + c$$

$$= \frac{\pi}{16}\log\left|\frac{1+x}{1-x}\right| + \frac{\pi}{8}\tan^{-1}x + \frac{\pi}{8\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right) - \frac{\pi}{16\sqrt{2}}\log\left|\frac{x^2+\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right| + c$$

7. Let $I = \int \frac{dx}{\sin x - \cos x + \sqrt{2}}$

$$= \int \frac{dx}{\sin x\frac{\sqrt{2}}{\sqrt{2}} - \cos x\frac{\sqrt{2}}{\sqrt{2}} + \sqrt{2}}$$

$$= \int \frac{dx}{\sqrt{2}\left(\sin x\sin\frac{\pi}{4} - \cos x\cos\frac{\pi}{4} + 1\right)}$$

$$= \frac{1}{\sqrt{2}}\int \frac{dx}{1-\cos\left(x+\frac{\pi}{4}\right)} = \frac{1}{\sqrt{2}}\int \frac{dx}{1-\cos 2\left(\frac{x}{2}+\frac{\pi}{8}\right)}$$

$$= \frac{1}{\sqrt{2}}\int \frac{dx}{2\sin^2\left(\frac{x}{2}+\frac{\pi}{8}\right)} = \frac{1}{2\sqrt{2}}\int \operatorname{cosec}^2\left(\frac{x}{2}+\frac{\pi}{8}\right)dx$$

$$= -\frac{1}{2\sqrt{2}}\cdot\frac{-\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)}{\frac{1}{2}} + c = \frac{1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right) + c$$

8. $x^x = t$

$\therefore \quad x\log x = \log t$

$$(1+\log x)dx = \frac{1}{t}dt$$

$$x^x(1+\log x)dx = dt$$

Now $\int (x^x)^2(1+\log x)dx$

$$= \int x^x x^x(1+\log x)dx = \frac{(x^x)^2}{2} + k$$

9. Integration $I = \int \frac{\sin x}{3+4\cos^2 x}dx$

Substituting cos $x = t$ and $-\sin x dx = dt$ or $\sin x dx = -dt$ in the given integration, we get

$$I = \int \frac{-dt}{3+4t^2} = -\frac{1}{4}\int \frac{dt}{(\sqrt{3}/2)^2 + t^2}$$

$$= -\frac{1}{4}\times\frac{1}{\sqrt{3}/2}\tan^{-1}\left(\frac{t}{\sqrt{3}/2}\right) + c$$

$$= -\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{2\cos x}{\sqrt{3}}\right) + c$$

10. $\frac{x^5}{x^2+1} = x^3 - x + \frac{x}{x^2+1}\int \frac{x^5}{x^2+1}dx$

$$= \int x^3dx - \int xdx + \frac{1}{2}\int \frac{2x}{x^2+1}dx$$

$$= \frac{x^4}{4} - \frac{x^2}{2} + \frac{1}{2}\log(x^2+1) + c$$

11. Let $I = \int x\log\left(1+\frac{1}{x}\right)dx$

Using ILATE, we have

$$\Rightarrow \quad I = \frac{x^2}{2}\log\left(1+\frac{1}{x}\right) - \int \frac{1}{1+(1/x)}\left(-\frac{1}{x^2}\right)\frac{x^2}{2}dx$$

$$\Rightarrow \quad I = \frac{x^2}{2}\log\left(\frac{x+1}{x}\right) + \frac{1}{2}\int \frac{x}{x+1}dx$$

$$\Rightarrow \quad I = \frac{x^2}{2}\log\left(\frac{x+1}{x}\right) + \frac{1}{2}\int \frac{x+1-1}{x+1}dx$$

$$\Rightarrow \quad I = \frac{x^2}{2}\log\left(1+\frac{1}{x}\right) + \frac{1}{2}x - \frac{1}{2}\log(x+1) + c$$

$$\therefore \quad I = \left(\frac{x^2-1}{2}\right)\log(1+x) - \frac{x^2}{2}\log x + \frac{1}{2}x + c$$

12. Let $I = \int \frac{dx}{1-\cos x - \sin x}$

Put $\cos x = \frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}$ and $\sin x = \frac{2\tan(x/2)}{1+\tan^2(x/2)}$

$$\therefore \quad I\int \frac{dx}{1-\frac{[1-\tan^2(x/2)]}{[1+\tan^2(x/2)]} - \frac{2\tan(x/2)}{1+\tan^2(x/2)}}$$

$$= \int \frac{\sec^2(x/2)dx}{\left[1+\tan^2(x/2)-1+\tan^2(x/2)-2\tan(x/2)\right]}$$

$$= \int \frac{\sec^2(x/2)\,dx}{2\tan^2(x/2) - 2\tan(x/2)}$$

$$= \int \frac{(1/2)\sec^2(x/2)dx}{\tan^2(x/2) - \tan(x/2)}$$

Put $\tan\frac{x}{2}=t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx=dt$

$\therefore \quad I=\int\frac{dt}{t^2-t}=\int\frac{dt}{t(t-1)}$

$=\int\left[\frac{1}{t-t}-\frac{1}{t}\right]dt=\int\frac{dt}{t-1}-\int\frac{dt}{t}$

$=\log(t-1)-\log t+c=\log\left|\frac{t-1}{t}\right|+c$

$=\log\left|\frac{\tan(x/2)-1}{\tan(x/2)}\right|+c=\log\left|1-\cot\frac{x}{2}\right|+c$

13. $\int\frac{x-1}{(x+1)^3}e^x dx$

$=\int\frac{x+1-2}{(x+1)^3}e^x dx$

$=\int e^x\left[\frac{1}{(x+1)^2}-\frac{2}{(x+1)^3}\right]dx$

(It is of $\int e^x(f(x)+f'(x)dx$ form)

$=e^x f(x)+c$, where $f(x)=\frac{1}{(x+1)^2}$

14. Integration $I=\int\frac{\sin x}{\sqrt{4-\cos^2 x}}dx$

Substituting $\cos x=t$ and $-\sin x dx=dt$ in the given integration, we get

$$I=-\int\frac{dt}{\sqrt{(2)^2-t^2}}=-\sin^{-1}\left(\frac{t}{2}\right)+c=-\sin^{-1}\left(\frac{\cos x}{2}\right)+c$$

15. $\frac{d}{dx}(a^2\sin^2 x+b^2\cos^2 x)$

$=(a^2\cdot 2\sin x\cos x+b^2-2\cos x\sin x)$

$=(a^2-b^2)\sin 2x$

$\therefore$ Given integral

$=\frac{1}{a^2-b^2}\int\frac{(a^2-b^2)\sin 2x}{a^2\sin^2 x+b^2\cos^2 x}dx$

$=\frac{1}{a^2-b^2}\log(a^2\sin^2 x+b^2\cos^2 x)+c$

$$\left[\int\frac{f'(x)}{f(x)}dx=\log|f(x)|+c\right]$$

16. Let $I=\int\frac{\sin^8 x-\cos^8 x}{1-2\sin^2 x\cos^2 x}dx$

$\Rightarrow I=\int\frac{(\sin^2 x-\cos^2 x)(\sin^4 x+\cos^4 x)}{1-2\sin^2 x\cos^2 x}$

$\Rightarrow I=\int\frac{(-\cos 2x)(1-\sin^2 x\cos^2 x)}{(1-2\sin^2 x\cos^2 x)}dx$

$\therefore \quad I=\int-\cos 2x dx=-\frac{1}{2}\sin 2x+c$

17. Let $I=\int\frac{dx}{7+5\cos x}=\int\frac{dx}{7+5\frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}}$

$=\int\frac{\sec^2(x/2)dx}{7+7\tan^2(x/2)+5-5\tan^2(x/2)}$

$=\int\frac{\sec^2(x/2)\,dx}{12+2\tan^2(x/2)}=\int\frac{\frac{1}{2}\sec^2(x/2)\,dx}{6+\tan^2(x/2)}$

Put $\tan\frac{x}{2}=t \Rightarrow \frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx=dt$

$\therefore \quad I=\int\frac{dt}{t^2+(\sqrt{6})^2}=\frac{1}{\sqrt{6}}\tan^{-1}\left(\frac{t}{\sqrt{6}}\right)+c$

$=\frac{1}{\sqrt{6}}\tan^{-1}\left|\frac{\tan(x/2)}{\sqrt{6}}\right|+c$

18. $\int\frac{\log(x/e)}{(\log x)^2}dx$

$=\int\frac{\log x-1}{(\log x)^2}dx=\int e^t\left(\frac{1}{t}-\frac{1}{t^2}\right)dt,\ t=\log x$

$=\frac{e^t}{t}=\frac{x}{\log x}$

19. Let $\sin(x-a)$

i.e. $\sin x\cos a-\cos x\sin a$

$=l(\sin x\cos a-\cos x\sin a)$

$+m\frac{d}{dx}(\sin x\cos a+\cos x\sin a)$

$=l(\sin x\cos a+\cos x\sin a)$

$+m(\cos x\cos a-\sin x\sin a)$

$\therefore \quad \cos a=l\cos a-m\sin a \quad -\sin a=l\sin a+m\cos a$

$\therefore \quad l\cos a-m\sin a-\cos a=0$

$l\sin a+m\cos a+\sin a=0$

$\therefore \quad \frac{l}{\cos 2x}=\frac{m}{-\sin 2a}=\frac{1}{1}$

$\therefore \quad l=\cos 2a, m=-\sin 2a$

$\therefore \quad \int \frac{\sin(x-a)}{\sin(x+a)}dx$

$$= \int \left[l + \frac{m(d/dx)\sin(x+a)}{\sin(x+a)} \right] dx$$

$$= lx + m\log\left|\sin(x+a)\right| + c$$

$$= (\cos 2a)x - \sin 2a \log\left|\sin(x+a)\right| + c$$

20. We have

$\int \cot^{-1}(1-x+x^2)dx$

$$= \int \cot^{-1}[1-x(1+x)]\,dx = \int \tan^{-1}\left[\frac{1}{1-x(1-x)}\right]dx$$

$$= \int \tan^{-1}\left[\frac{x+(1-x)}{1-x(1-x)}\right]dx = \int [\tan^{-1} + \tan^{-1}(1-x)]dx$$

$$= \int \tan^{-1} x dx + \int \tan^{-1}(1-x)\,dx$$

$$= I_1 - I_2, \qquad \text{(i)}$$

where $I_1 = \int \tan^{-1} x dx$ and $I_2 = \int \tan^{-1}(1-x)dx$.

Now, $I_1 = \int \tan^{-1} x dx = \int \tan^{-1}_{I} x \underset{II}{1} dx$

$$= x\tan^{-1} x - \int \frac{x}{1+x^2}dx$$

$$= x\tan^{-1} x - \frac{1}{2}\int \frac{1}{1+x^2} d(1+x^2)$$

$$= x\tan^{-1} x - \frac{1}{2}\log(1+x^2) \qquad \text{(ii)}$$

and $I_2 = \int \tan^{-1}(1-x)dx$

$$= -\int \tan^{-1}(1-x)d(1-x)$$

$$= -\left\{(1-x)\tan^{-1}(1-x) - \frac{1}{2}\log[1+(1-x)^2]\right\}$$

Substituting the values of I_1 and I_2 in (i), we get

$$\int \cot^{-1}(1-x+x^2)dx = x\tan^{-1} x - \frac{1}{2}\log(1+x^2)$$

$$-(1-x)\tan^{-1}(1-x) + \frac{1}{2}\log[(1+(1-x^2)] + c\,.$$

21. Let $I = \int \frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}}dx$

$$= \int \frac{\sqrt{(1+x)^2} + \sqrt{x}\sqrt{1+x}}{(\sqrt{x}+\sqrt{1+x})}dx$$

$$= \int \frac{\sqrt{1+x}(\sqrt{1+x}+\sqrt{x})}{(\sqrt{x}+\sqrt{1+x})}dx$$

$$= \int \sqrt{1+x}dx = \frac{2}{3}(1+x)^{3/2} + c$$

22. $\int \frac{dx}{\left[(x+a)^3(x+b)^5\right]^{1/4}} = \int \left(\frac{x+b}{x+a}\right)^{-1/4} \frac{1}{(x+a)(x+b)}dx$

Put $\frac{x+b}{x+a} = t^4 = \frac{4}{b-a}\left(\frac{x+a}{x+b}\right)^{1/4} + k$

23. Put $1 = r\sin a, \sqrt{3} = r\cos a$ so that $1+3 = r^2 \Rightarrow r = 2$

$\tan\alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$

$\therefore$ Given integral

$$= \int \frac{dx}{r\sin(x+\alpha)} = \frac{1}{2}\int \operatorname{cosec}\left(\frac{x+\alpha}{2}\right)dx$$

$$= \frac{1}{2}\log\tan\left(\frac{x+\alpha}{2}\right) + c = \frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right) + c$$

24. Let $I = \int e^x \frac{x}{(x+1)^2}dx$

$$\Rightarrow I = \int e^x \left(\frac{1}{x+1} + \frac{-1}{(x+1)^2}\right)dx$$

$$\therefore \quad I = e^x \frac{1}{x+1} + c$$

$$\left\{\because \int e^x[f(x)+f'(x)]\,dx = e^x f(x)\right\}$$

25. Let $I = \int \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$

$$= \int \frac{dx}{\cos^3 x\sqrt{4\sin x\cos x}} = \frac{1}{2}\int \frac{dx}{\cos^{7/2} x \sin^{1/2} x}$$

$$= \frac{1}{2}\int \frac{\sec^4 x}{\sqrt{\tan x}}dx = \frac{1}{2}\int \frac{(1+\tan^2 x)\sec^2 x}{\sqrt{\tan x}}dx$$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$

$$\therefore \quad I = \frac{1}{2}\int \frac{1+t^2}{\sqrt{t}}dt$$

$$= \frac{1}{2}\int t^{-1/2}dt + \frac{1}{2}\int t^{3/2}dt = t^{1/2} + \frac{t^{5/2}}{5} + c$$

$$= \sqrt{\tan x} + \frac{1}{5}\tan^{5/2} x + c$$

26. $\int \frac{x^2\left(1-1/x^2\right)}{x^2\left(x^2+1/x^2+3\right)\tan^{-1}\left(x+1/x\right)}dx$

$$= \int \frac{1}{(1+t^2)(\tan^{-1} t)}\; t = x + \frac{1}{x}$$

$$= \log[\tan^{-1}(x+1/x)] + c$$

27. Put $\sqrt{x}=z \quad \therefore \quad \frac{1}{2\sqrt{x}}dx=dz$

$\therefore$ Given integral $=\int \sin z\cdot 2dz$

$=-2\cos z+c=-2\cos\sqrt{x}+c$

28. Let $I=\int x^{51}\left(\tan^{-1}x+\cot^{-1}x\right)dx$

Since $\tan^{-1}+\cot^{-1}x=\frac{\pi}{2}$

$\Rightarrow \quad I=\int x^{51}\frac{\pi}{2}dx=\frac{\pi}{2}\frac{x^{52}}{52}+c$

$\therefore \quad I=\frac{x^{52}}{52}(\tan^{-1}x+\cot^{-1}x)+c$

29. $\int\frac{\sqrt{5+x^{10}}}{x^{16}}dx$

$=\int\sqrt{\frac{5+x^{10}}{x^{10}}}\cdot\frac{1}{x^{11}}dx=\int\sqrt{\frac{5}{x^{10}}+1}\cdot\frac{1}{x^{11}}dx$

$=-\frac{1}{50}\int\sqrt{1+\frac{5}{x^{10}}}\cdot\left(\frac{-50}{x^{11}}\right)dx$

$=-\frac{1}{50}\int\sqrt{1+\frac{5}{x^{10}}}d\left(1+\frac{1}{x^{10}}\right)dx$

$=-\frac{1}{50}\left\{\frac{[1+(5/x^{10})]^{3/2}}{3/2}\right\}+c=-\frac{1}{75}\left(1+\frac{5}{x^{10}}\right)^{3/2}+c$

30. $\int e^{\tan^{-1}x}\left(1+\frac{x}{1+x^2}\right)dx$

$=\int e^{\tan^{-1}x}dx+\int e^{\tan^{-1}x}\frac{x}{1+x^2}dx$

$=e^{\tan^{-1}x}.x-\int e^{\tan^{-1}x}\frac{1}{1+x^2}xdx$

$+\int e^{\tan^{-1}x}\frac{x}{1+x^2}dx=xe^{\tan^{-1}x}$

31. $I=\int\frac{dx}{5+8\sin(x/2)\cos(x/2)}$

$\Rightarrow \quad I=\int\frac{\sec^2(x/2)}{5\sec^2(x/2)+8\tan(x/2)}dx$

$\Rightarrow \quad I=\int\frac{\sec^2(x/2)}{5+5\tan^2(x/2)+8\tan(x/2)}dx$

Put $\tan\frac{x}{2}=t \quad \Rightarrow \quad \sec^2\frac{x}{2}\cdot\frac{1}{2}dx=dt$

$\therefore \quad I=\int\frac{2dt}{5t^2+8t+5}$

$\Rightarrow \quad I=\frac{2}{5}\int\frac{2dt}{t^2+\frac{8}{5}t+1}=\frac{2}{5}\int\frac{dt}{t^2+\frac{8t}{5}+\frac{16}{25}+\frac{9}{25}}$

$\Rightarrow \quad I=\frac{2}{5}\int\frac{dt}{\left(t+\frac{4}{5}\right)^2+\left(\frac{3}{5}\right)^2}\Rightarrow I=\frac{2}{5}\frac{5}{3}\tan^{-1}\frac{t+(4/5)}{3/5}+c$

$\therefore \quad I=\frac{2}{5}\tan^{-1}\frac{5t+4}{3}=\frac{2}{3}\tan^{-1}\left(\frac{5\tan(x/2)+4}{3}\right)+c$

32. Let $I=\int\frac{dx}{x(\log x)(\log\log x)...\underbrace{(\log\log...x)}_{8\text{ times}}}$

Put $\underbrace{(\log\log...x)}_{8\text{ times}}=t$

$\Rightarrow \quad I=\frac{1}{\underbrace{(\log\log...x)}_{7\text{ times}}...(\log\log x)(\log x)x}dx=dt$

$\therefore \quad I=\int\frac{dt}{t}=\log t+c=\underbrace{(\log\log\log...x)}_{9\text{ times}}+c$

33. $\int\frac{\sqrt{\tan x}}{2\sin x\cos x}dx$

$=\frac{1}{2}\int\frac{\sqrt{\tan x}}{\tan x}\sec^2 xdx=\frac{1}{2}\int\frac{\sec^2 xdx}{\sqrt{\tan x}}$

$=\frac{1}{2}\cdot\frac{(\tan x)^{1/2}}{1/2}=\sqrt{\tan x}$

34. $I=\int\frac{(\sin^2 x+\cos^2 x)}{\sin^2 x\cos^2 x}dx$

$\Rightarrow \quad I=\int(\sec^2 x+\text{cosec}^2)dx$

$\therefore \quad I=\tan x-\cot x+c$

35. We have $\int\frac{dx}{\cos^3 x\sqrt{\sin 2x}}=a(\tan^2 x+b)\sqrt{\tan x}+c$

Let $I=\frac{1}{\sqrt{2}}\int\frac{dx}{\cos^3 x\sqrt{\sin x\cos x}}$

$=\frac{1}{\sqrt{2}}\int\sec^4 x\frac{dx}{\sqrt{\tan x}}$

$=\frac{1}{\sqrt{2}}\int\frac{(1+\tan^2 x)\sec^2 x}{\sqrt{\tan x}}dx$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$

$$I = \frac{1}{\sqrt{2}}\int (1+t^2)\frac{dt}{\sqrt{t}}$$

$$= \frac{1}{\sqrt{2}}\int (t^{-1/2} + t^{3/2})dt = \frac{1}{\sqrt{2}}\left(2t^{1/2} + \frac{2}{5}t^{5/2}\right) + c$$

$$= \frac{1}{\sqrt{2}}\left[2\sqrt{\tan x} + \frac{2}{5}(\tan x)^{5/2}\right] + c$$

$$= \left(\sqrt{2} + \frac{\sqrt{2}}{5}\tan^2 x\right)\sqrt{\tan x}$$

$$= a(\tan^2 x + b)\sqrt{\tan x}$$

$$\Rightarrow \quad a = \frac{\sqrt{2}}{5}, ab = \sqrt{2} \Rightarrow b = 5$$

36. Put $e^{e^x} = z \quad \therefore \quad e^{e^x} e^x dx = dz$

$\therefore$ Given integral $= \int e^z dz = e^z = e^{e^{e^x}}$

37. Let $I = \int \frac{1}{(x^2+1)(x^2+4)} dx$

$$\Rightarrow \quad I = \frac{1}{3}\int\left(\frac{1}{(x^2+1)} - \frac{1}{(x^2+4)}\right)dx$$

$$\Rightarrow \quad I = \frac{1}{3}\tan^{-1} x - \frac{1}{6}\tan^{-1}\frac{x}{2} + c$$

$$\therefore \quad A = \frac{1}{3} \quad \text{and} \quad B = -\frac{1}{6}$$

38. Let $I = \int \frac{x^2 - 1}{(x^4 + 3x^2 + 1)\tan^{-1}\left(x + \frac{1}{x}\right)} dx$

$$= \int \frac{\left(1 - \frac{1}{x^2}\right)}{\left(x^2 + \frac{1}{x^2} + 3\right)\tan^{-1}\left(x + \frac{1}{x}\right)} dx$$

Let $x + \frac{1}{x} = t$

$$\Rightarrow \quad \left(1 - \frac{1}{x^2}\right) = dx = dt \text{ and } \left(x + \frac{1}{x}\right)^2 = t^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = t^2 - 2$$

$$\therefore \quad I = \int \frac{1}{(t^2+1)\tan^{-1} t} dt$$

$$= \log(\tan^{-1} t) + c = \log\left[\tan^{-1}\left(x + \frac{1}{x}\right)\right] + c$$

39. Integral

$$= \frac{1}{\sin(b-a)}\int \frac{\sin[(x-a)-(x-b)]}{\sin(x-a)\sin(x-b)} dx$$

$$= \frac{1}{\sin(b-a)}$$

$$\int \frac{\sin(x-a)\cos(x-b) - \cos(x-a)\sin(x-b)}{\sin(x-a)\sin(x-b)} dx$$

$$= \frac{1}{\sin(b-a)}\int\left[\frac{\cos(x-b)}{\sin(x-b)} - \frac{\cos(x-a)}{\sin(x-a)}\right]dx$$

$$= \frac{1}{\sin(b-a)}[\log|\sin(x-b)| - \log|\sin(x-a)|] + c$$

$$= \frac{1}{\sin(b-a)}\log\frac{|\sin(x-b)|}{|\sin(x-a)|} + c$$

40. $I = \int \frac{\sin 2x}{\sin^4 + \cos^4 x} dx$

Dividing the numerator and denominator by $\cos^4 x$, we get

$$I = \int \frac{2\sin x \cos x \sec^4 x}{1 + \tan^4 x} dx$$

$$\Rightarrow \quad I = \int \frac{2\tan x \sec^2 x}{1 + \tan^4 x} dx$$

Let $\tan^2 x = t \Rightarrow \tan x \sec^2 x dx = dt$

$$\therefore \quad I = \int \frac{dt}{1+t^2} = \tan^{-1} t = \tan^{-1}(\tan^2 x) + c$$

41. Let $I = \int \frac{dx}{(\sin x - 2\cos x)(2\sin x + \cos x)}$

$$= \int \frac{\sec^2 x dx}{(\tan x - 2)(2\tan x + 1)}$$

Let $\tan x = t$

$$\Rightarrow \quad \sec^2 x dx = dt$$

$$\therefore \quad I = \int \frac{dt}{(t-2)(2t+1)} = \frac{1}{5}\int \frac{1}{t-2} dt - \frac{2}{5}\int \frac{1}{2t+1} dt$$

$$= \frac{1}{5}\log(t-2) - \frac{1}{5}\log(2t+1) + c$$

$$= \log\sqrt[5]{\frac{\tan x - 2}{2\tan x + 1}} + c$$

42. Put $\tan\frac{x}{2}=z \;\therefore \frac{1}{2}\sec^2\frac{x}{2}dx=dz \Rightarrow dx=\frac{2dz}{1+z^2}$

$\therefore$ Given integral

$$=\int\frac{\frac{2dz}{1+z^2}}{1+\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}+\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}=\int\frac{\frac{2dz}{1+z^2}}{1+\frac{2z}{1+z^2}+\frac{1-z^2}{1+z^2}}$$

$$=\int\frac{2dz}{1+z^2+2z+1-z^2}=\int\frac{dz}{z+1}$$

$$=\log|z+1|+c=\log\left|\tan\frac{x}{2}+1\right|+c$$

43. Let $I=\int 5^{5^{5^x}}\cdot 5^{5^x}\cdot 5^x\,dx$

Put $5^{5^{5^x}}=t \Rightarrow 5^{5^{5^x}}5^{5^x}5^x(\log 5)^3\,dx=dt$

$\therefore \quad I=\frac{1}{(\log 5)^3}\int 1dt=\frac{t}{(\log 5)^3}+c$

$\Rightarrow I=\frac{5^{5^{5^x}}}{(\log 5)^3}+c$

44. $f(x)=\lim_{n\to\infty}n^2x^{1/(n+1)}\left[x^{\frac{1}{n}-\frac{1}{n+1}}-1\right]$

$$=\lim_{n\to\infty}\frac{x^{1(n+1)}\left(x^{1/[n(n+1)]}-1\right)}{\frac{1}{n(n+1)}\times\frac{n(n+1)}{n^2}}\log x$$

Hence, $\int xf(x)dx=\int x\log x\,dx=\frac{x^2}{2}\log x-\frac{1}{4}x^2+c$

45. Let $2\cos x=l(\sin x+\cos x)+m\frac{d}{dx}(\sin x+\cos x)$

$$=l(\sin x+\cos x)+m(\cos x-\sin x)$$

$$=(l-m)\sin x+(l+m)\cos x$$

$\therefore \quad \left.\begin{matrix}l-m=0\\ l+m=2\end{matrix}\right] \Rightarrow 2l=2 \Rightarrow l=1 \qquad \therefore \quad m=1$

$\therefore$ Given integral

$$=\int\left[l+m\frac{\frac{d}{dx}(\sin x+\cos x)}{\sin x+\cos x}\right]dx$$

$$=lx+m\log|\sin x+\cos x|+c$$

$$=x+\log|\sin x+\cos x|+c$$

46. Let $I=\int\frac{\sin x}{\sin x-\cos x}dx$

$$\Rightarrow I=\frac{1}{2}\int\frac{(\sin x-\cos x)+(\cos x+\sin x)}{\sin x-\cos x}dx$$

$$\Rightarrow I=\frac{1}{2}\int 1+\frac{\cos x+\sin x}{\sin x-\cos x}dx$$

$$\therefore \quad I=\frac{1}{2}[x+\log(\sin x-\cos x)]+c$$

47. Integration $I=\int\frac{dx}{(2\sin x+\cos x)^2}$

Multiplying both the numerator and denominator by cosec^2x in the given integration, we get

$$I=\int\frac{\text{cosec}^2x}{\text{cosec}^2x(2\sin x+\cos x)}dx=\int\frac{\text{cosec}^2x}{(2+\cot x)^2}dx$$

Substituting $2+\cot x=t$ and $-\text{cosec}^2x\,dx=dt$, we get

$$I=\int\frac{-dt}{t^2}=-\int\frac{1}{t^2}dt=\frac{1}{t}+c=\frac{1}{2+\cot x}+c$$

48. Integral

$$=\int\frac{\sin 3x(\cos 5x+\cos 4x)}{\sin 3x-\sin 6x}dx$$

$$=\int\frac{2\sin\frac{3x}{2}\cos\frac{3x}{2}\cdot 2\cos\frac{9x}{2}\cos\frac{x}{2}}{-2\cos\frac{9x}{2}\sin\frac{3x}{2}}$$

$$=-\int 2\cos\frac{3x}{2}\cos\frac{x}{2}dx=-\int(\cos 2x+\cos x)dx$$

$$=-\left(\frac{\sin 2x}{2}+\sin x\right)+c$$

49. Let $I=\int\frac{\sin x+\cos x}{\sqrt{1+\sin 2x}}dx$

Since $1+\sin 2x=\cos^2x+\sin^2x+2\sin x\cos x$

$\Rightarrow 1+\sin 2x=(\sin x+\cos x)^2$

Now, $I=\int\frac{(\sin x+\cos x)}{\sqrt{1+\sin 2x}}dx$

$\Rightarrow I=\int\frac{\cos x+\sin x}{\cos x+\sin x}dx=\int 1dx=x+c$

50. Integration $I=\int\frac{e^{\tan^{-1}x}}{(1+x^2)}dx$

Substituting $\tan^{-1}(x)=t$ and $\frac{1}{1+x^2}dx=dt$ in the given integration, we get

$$I=\int e^t dt=e^t+c=e^{\tan^{-1}x}+c$$

51. $\int(\sqrt{\tan x}+\sqrt{\cot x})dx$

$$=\int\left(\sqrt{\frac{\sin x}{\cos x}}+\sqrt{\frac{\cos x}{\sin x}}\right)dx$$

$$=\int\frac{\sin x+\cos x}{\sqrt{\sin x\cos x}}dx$$

Put $(\sin x-\cos x)=z \quad \therefore \quad (\cos x+\sin x)dx=dz$

Also $(\sin^2 x+\cos^2 x-2\sin x\cos x)=z^2$

$1-2\sin x\cos x=z^2 \Rightarrow \sin x\cos x=\dfrac{1-z^2}{2}$

$\therefore$ Given integral

$$=\int\frac{dz}{\sqrt{(1-z^2)/2}}=\sqrt{2}\int\frac{dz}{\sqrt{1-z^2}}$$

$$=\sqrt{2}\sin^{-1}z+c$$

$$=\sqrt{2}\sin^{-1}(\sin x-\cos x)+c$$

52. $I=\int\dfrac{\cos x-\sin x}{\sqrt{\cos x\sin x}}dx$

Put $\sin x+\cos x=t$, so that $2\sin x\cos x=t^2-1$

$$\therefore \quad I=\sqrt{2}\int\frac{dt}{\sqrt{t^2-1}}=\sqrt{2}\log\left|t+\sqrt{t^2-1}\right|+c$$

$$=\sqrt{2}\log\left|\sin x+\cos x+\sqrt{\sin 2x}\right|+c$$

53. Put $\sin x=t$, so that $\cos x dx=dt$

$$\therefore \quad I=\int\frac{\cos^2 x+\cos^4 x}{t^2+t^4}dt=\int\frac{(1-t^2)+(1-t^2)^2}{t^2+t^4}dt$$

$$=\int\frac{(1-t^2)(2-t^2)}{t^2+t^4}dt=\int\left(1+\frac{2}{t^2}-\frac{6}{1+t^2}\right)dt$$

$$=t-\frac{2}{t}-6\tan^{-1}(t)+c$$

$$=\sin x-2(\sin x)^{-1}-6\tan^{-1}(\sin x)+c$$

54. Integration $I=\int e^{\log(\tan x)}dx$

We know that $e^{\log x}=x$

Therefore, $I=\int e^{\log(\tan x)}dx=\int\tan x dx=\log(\sec x)+c$

(where $e^{\log x}=x$)

55. Put $1+x^3=t^2, 3x^2dx=2t\,dt$, so that

$$I=\int\frac{(t^2-1)}{t}\cdot\frac{2}{3}tdt=\frac{2}{3}\left(\frac{t^3}{3}-t\right)+c$$

$$=\frac{2}{9}(1+x^3)^{5/2}-\frac{2}{3}(1+x^3)^{1/2}+c$$

56. Put $\sin\theta-\cos\theta=t$, so that

$$(\cos\theta+\sin\theta)d\theta=dt$$

Also $1-\sin 2\theta=t^2 \Rightarrow \sin 2\theta=1-t^2$

$$\therefore \quad I=\int\frac{dt}{\sqrt{1-t^2}}=\sin^{-1}t+c$$

$$=\sin^{-1}(\sin\theta-\cos\theta)+c$$

57. Integration $I=\int\dfrac{1-\tan x}{1+\tan x}dx$

We know that

$$\frac{1-\tan x}{1+\tan x}=\frac{\tan\frac{\pi}{4}-\tan x}{1+\tan\frac{\pi}{4}\times\tan x}=\tan\left(\frac{\pi}{4}-x\right)$$

Therefore,

$$I=\int\frac{1-\tan x}{1+\tan x}dx$$

$$=\int\tan\left(\frac{\pi}{4}-x\right)dx=\int\cot\left[\frac{\pi}{2}-\left(\frac{\pi}{4}-x\right)\right]dx$$

$$=\int\cot\left(\frac{\pi}{4}+x\right)dx=\log\left[\sin\left(\frac{\pi}{4}+x\right)\right]+c$$

58. We multiply and divide the integrand by $\cos(x-b)$, so that

$$\cos(a-b)I=\int\frac{\cos[(x-b)-(x-a)]}{\sin(x-a)\cos(x-b)}dx$$

$$=\int\frac{\cos(x-b)\cos(x-a)+\sin(x-b)\sin(x-a)}{\sin(x-a)\cos(x-b)}dx$$

$$=\int[\cot(x-a)+\tan(x-b)]dx$$

$$=[\log(\sin(x-a)-\log|\cos(x-b)|]+c$$

$$=\log\left|\frac{\sin(x-a)}{\cos(x-b)}\right|+c$$

59. $\int\sqrt[3]{\dfrac{\sin^n x}{\cos^{n+6}x}}dx=\int\sqrt[3]{\dfrac{\sin^n x}{\cos^n x\cdot\cos^6 x}}dx$

$$=\int\sqrt[3]{\tan^n}\sec^2 x dx=\int(\tan x)^{n/3}d(\tan x)$$

$$=\frac{(\tan x)^{(n/3)+1}}{(n/3)+1}+c$$

$$=\frac{3}{3+n}(\tan x)^{(n/3)+1}+c$$

60. On putting $\tan\theta = x^2$ and $d(\tan\theta) = dx^2$

$\Rightarrow \sec^2\theta d\theta = 2xdx, \Rightarrow d\theta = \dfrac{2x}{1+x^2}dx,$ we get

$$\int\sqrt{\tan\theta}d\theta = \int\sqrt{x^2}\frac{2x}{1+x^4}dx$$

$$= \int\frac{2x^2}{x^4+1}dx = \int\frac{2}{x^2+\frac{1}{x^2}}dx$$

$$= \int\frac{\left(1+\frac{1}{x^2}\right)+\left(1-\frac{1}{x^2}\right)}{x^2+\frac{1}{x^2}}dx$$

$$= \int\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx + \int\frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$$

$$= \int\frac{1}{\left(x+\frac{1}{x}\right)^2+2}d\left(x-\frac{1}{x}\right) + \int\frac{1}{\left(x+\frac{1}{x}\right)^2-2}d\left(x+\frac{1}{x}\right)$$

$$= \int\frac{du}{u^2+(\sqrt{2})^2} + \int\frac{1}{v^2-(\sqrt{2})}dv$$

where $u = x-\dfrac{1}{x}$ and $v = x+\dfrac{1}{x}$.

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{u}{\sqrt{2}}\right) + \frac{1}{2\sqrt{2}}\log\left|\frac{v-\sqrt{2}}{u+\sqrt{2}}\right| + c$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{x\sqrt{2}}\right) + \frac{1}{2\sqrt{2}}\log\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right| + c.$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $I = \int\frac{\sqrt{1+\sin x}\sqrt{1-\sin x}}{\sqrt{1-\sin x}}dx$

$$= \int\frac{\cos x}{\sqrt{1-\sin x}}dx = -2\sqrt{1-\sin x} + c$$

2. $I = \int\frac{\sin x}{\sin 4x}dx = \int\frac{\sin x}{2\sin 2x\cos 2x}dx$

$$= \int\frac{\sin x}{4\sin x\cos x\cos 2x}dx$$

$$= \frac{1}{4}\int\frac{dx}{\cos x\cos 2x} = \frac{1}{4}\int\frac{\cos x}{\cos^2 x\cos 2x}dx$$

$$= \frac{1}{4}\int\frac{\cos x}{(1-\sin^2 x)(1-2\sin^2 x)}dx$$

Putting $\sin x = t$ so that $\cos x dx = dt$

$\therefore \quad I = \dfrac{1}{4}\int\dfrac{dt}{(1-t^2)(1-2t^2)}$

$$= \frac{1}{4}\int\left[\frac{1}{t^2-1} - \frac{1}{t^2-1/2}\right]dt$$

$$= \frac{1}{4}\left[\frac{1}{2}\log\left|\frac{t-1}{t+1}\right| - \frac{1}{\sqrt{2}}\log\left|\frac{t-1/\sqrt{2}}{t+1/\sqrt{2}}\right|\right] + c$$

$$= \frac{1}{8}\log\left|\frac{\sin x-1}{\sin x+1}\right| - \frac{1}{4\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x-1}{\sqrt{2}\sin x+1}\right| + c$$

3. Integration $I = \int e^x\sin x dx$

Integrating the given integration by parts, we get

$$I = -e^x\cos x - \int e^x(-\cos x)dx$$

$$I = -e^x\cos x + \int e^x\cos x dx = -e^x\cos x + e^x\sin x -$$

$$\int e^x\sin x dx = -e^x\cos x + e^x\sin x - 1$$

or $2I = e^x\sin x - e^x\cos x$

or $I = \dfrac{1}{2}e^x(\sin x - \cos x) + c$

4. $I = \int\frac{\sin x\cos x}{\sin x+\cos x}dx$

$$= \frac{1}{2}\int\frac{(\sin x+\cos x)^2-1}{\sin x+\cos x}dx$$

$$= \frac{1}{2}\int\left[\sin x+\cos x - \frac{1}{\sqrt{2}\sin(x+\pi/4)}\right]dx$$

$$= \frac{1}{2}\left[\sin x+\cos x - \frac{1}{\sqrt{2}}\log\left|\operatorname{cosec}\left(x+\frac{\pi}{4}\right) - \cot\left(x+\frac{\pi}{4}\right)\right|\right] + c$$

5. Put $xe^x = t$, so that $(xe^x+e^x)dx = dt$

i.e. $(x+1)e^x dx = dt$

$\therefore \quad I = \int\dfrac{dt}{\cos^2 t} = \int\sec^2 t dt = \tan t = \tan(xe^x)$

6. Put $x-a=\theta$, so that

$$I=\int\frac{\cos x(a+\theta)}{\sin\theta}\,d\theta=\int\frac{\cos a\cos\theta-\sin a\sin\theta}{\sin\theta}\,d\theta$$

$$=(\cos a)\int\cot\theta\,d\theta-(\sin a)\int d\theta$$

$$=(\cos a)\log|\sin\theta|-(\sin a)\theta+c$$

$$=(\cos a)\log|\sin(x-a)|-(\sin a)x+c$$

where $c'=c-a\sin a$

7. $\int\frac{\cos x}{(\sin x)^{3/2}}dx=\int\cos x(\sin x)^{-3/2}dx$

$$=\frac{(\sin x)^{-1/2}}{-1/2}=\frac{-2}{\sqrt{\sin x}}$$

8. $\sin^4 x+\cos^4 x=\frac{1}{4}[(1-\cos 2x)^2+(1+\cos 2x)^2]$

$$=\frac{1}{2}(1+\cos^2 2x)$$

$$\therefore\quad I=2\int\frac{dx}{1+\cos^2 2x}=2\int\frac{\sec^2 2x}{\sec^2 2x+1}dx$$

$$=2\int\frac{\sec^2 2x}{2+\tan^2 2x}dx$$

$$=\int\frac{dt}{2+t^2},\text{ where } t=\tan 2x$$

$$=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)+c$$

$$=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan 2x}{\sqrt{2}}\right)+c$$

9. Integration $I=\int\frac{x-1}{(x+1)(x-2)}dx$

We know that $\frac{x-1}{(x+1)(x-2)}=\frac{A}{x+1}+\frac{B}{x-2}$ (i)

or $x-1=A(x-2)+B(x+1)$ (ii)

Substituting $x-2=0$ or $x=2$ in Eq. (ii), we get

$$1=3B\quad\text{or}\quad B=\frac{1}{3}$$

Substituting $x+1=0$ or $x=-1$ in Eq. (ii), we get

$$-2=-3A$$

Substituting the values of A and B in Eq. (i), we get

$$\frac{x-1}{(x+1)(x-2)}=\frac{2}{3}\times\frac{1}{x+1}+\frac{1}{3}\times\frac{1}{x-2}$$

Integrating this equation, we get

$$\int\frac{x-1}{(x+1)(x-2)}dx=\frac{2}{3}\int\frac{1}{x+1}dx+\frac{1}{3}\int\frac{1}{x-2}dx$$

$$=\frac{2}{3}\log|x+1|+\frac{1}{3}\log|x-2|+c$$

10. Using $\sin^2 x=\frac{1}{2}(1-\cos 2x)$ and

$$\cos^2 x=\frac{1}{2}(1+\cos 2x),$$

we can write

$$I=\int\frac{(1-\cos 2x)^2}{2(1+\cos^2 2x)}dx$$

$$=\frac{1}{2}\int\left[1-\frac{2\cos 2x}{2-\sin^2 2x}\right]dx$$

$$=\frac{1}{2}\left[x-\frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}+\sin 2x}{\sqrt{2}-\sin 2x}\right|\right]+c$$

11. $I=\int\frac{dx}{9+16\sin^2 x}$

$$=\int\frac{dx}{9\cos^2 x+25\sin^2 x}$$

$$=\int\frac{\sec^2 x}{9+25\tan^2 x}dx$$

Put $\tan x=t$ so that $\sec^2 x\,dx=dt$

$$=\int\frac{dt}{9+25t^2}=\frac{1}{25}\int\frac{1}{t^2+(3/5)^2}dt$$

$$=\frac{1}{25}\cdot\frac{1}{3/5}\tan^{-1}\frac{t}{3/5}=\frac{1}{15}\tan^{-1}\left(\frac{5t}{3}\right)$$

$$=\frac{1}{15}\tan^{-1}\left(\frac{5\tan x}{3}\right)$$

12. Integration $I=\int\frac{1-\cos x}{\cos x(1+\cos x)}dx$

Let $\cos x=y$

Therefore, $\frac{(1-\cos x)}{\cos x(1+\cos x)}=\frac{1-y}{y(1+y)}$

or $\frac{1-y}{y(1+y)}=\frac{A}{y}+\frac{B}{1+y}$ (i)

or $(1-y)=A(1+y)+By$ (ii)

Substituting $y = 0$ in Eq. (ii), we get $A = 1$
Substituting $y = -1$ in Eq. (ii), we get $B = -2$
Substituting the values of A and B in Eq. (i), we get

$$\frac{1-y}{y(1+y)} = \frac{1}{y} + \frac{1}{1+y}$$

or $$\frac{1-\cos x}{\cos x(1+\cos x)} = \frac{1}{\cos x} - \frac{2}{1+\cos x}$$

Integrating this equation, we get

$$I = \int \frac{1-\cos x}{\cos x(1+\cos x)}dx = \int \frac{1}{\cos x}dx \int \frac{2}{1+\cos x}dx$$

$$= \int \sec dx - \int \frac{2}{2\cos^2(x/2)}dx$$

$$= \int \sec x dx - \int \sec^2 \frac{x}{2}dx$$

$$= \log|\sec x + \tan x| - 2\tan\frac{x}{2} + c$$

(where $y = \cos x$)

13. Write $I = \int \frac{dx}{x^{1/3}(1+x^{1/6})}$

Put $x^{1/6} = t$ or $x = t^6$, so that

$$I = 6\int \frac{t^5 dt}{t^2(1+t)} = 6\int \frac{t^3}{t+1}dt$$

$$= 6\int \frac{t^3+1-1}{t+1}dt$$

$$= 6\int \left[\left(t^3 - 1 + 1\right) - \frac{1}{t+1}\right]dt$$

$$= 6\left[\frac{t^3}{3} - \frac{t^2}{2} + t - \log|t+1|\right] + c$$

$$= 2\sqrt{x} - 3x^{1/3} + 6x^{1/6} - \log(1+x^{1/6}) + c$$

14. Put $2^x = t$ so that $2^x \log 2 dx = dt$

$$\therefore \quad I = \frac{1}{\log 2}\int \frac{dt}{\sqrt{1-t^2}} = \frac{1}{\log 2}\cdot \sin^{-1} t$$

$$= \frac{\sin^{-1}(2^x)}{\log 2} = k\sin^{-1}(2^x)$$

$$\Rightarrow \quad k = \frac{1}{\log 2}$$

15. Put $\sin\theta = x$, so that

$$I = \int \frac{(3x-2)}{5-\left(1-x^2\right)-4x}dx = \int \frac{3x-2}{(x-2)^2}dx$$

Put $x - 2 = t$, so that

$$I = \int \frac{3(t+2)-2}{t^2}dt = \int\left(\frac{3}{t} + \frac{4}{t^2}\right)dt$$

$$= 3\log|t| - \frac{4}{t} + c$$

$$= 3\log|\sin\theta - 2| - \frac{4}{\sin\theta - 2} + c$$

$$= 3\log(2-\sin\theta) + \frac{4}{2-\sin\theta} + c$$

16. $y = f(x) \quad \therefore \quad \frac{dy}{dx} = f'(x) = \tan\theta$

$\therefore \quad f(1) = \tan 30° = \frac{1}{\sqrt{3}}, \; f'(2) = \tan 60° = \sqrt{3}$

$f(3) = \tan 45° = 1$

$$I = \frac{1}{2}\{[f'(x)]^2\}_2^3 + [f'(x)^2]_1^3$$

$$= \frac{1}{2}\{[f'(3)]^2 - [f'(2)]^2\} + [f'(3) - f'(1)]$$

$$= \frac{1}{2}[1^2 - (\sqrt{3})^2] + \left[1 - \frac{1}{\sqrt{3}}\right]$$

$$= \frac{1}{2}[-2] + 1 - \frac{1}{\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

17. Let $I = \int \cos^3 x e^{\log\sin x}dx$

$$\Rightarrow \quad I = \int \cos^3 x \sin x dx$$

Put $t = \cos x \quad \Rightarrow \quad dt = -\sin x dx$

$$\therefore \quad I = -\int t^3 dt$$

$$\Rightarrow \quad I = -\frac{t^4}{4} + c = -\frac{\cos^4 x}{4} + c$$

18. $I = \int \frac{(1/2)d(x^2)}{1+(x^2)^2} = \frac{1}{2}\tan^{-1}(x^2) + c$

19. Put $\tan(x/2) = t$, so that

$$4 + 5\sin x = 4 + 5\left(\frac{2t}{1+t^2}\right) = \frac{2}{1+t^2}(2 + 2t^2 + 5t) \text{ and } dx$$

$$= 2dt/(1+t^2)$$

$$\therefore \quad I = \int\left(\frac{dt}{2t^2+5t+2}\right) = \int \frac{dt}{(2t+1)(t+2)}$$

$$= \int\left[\frac{2}{3}\frac{1}{1+2t} - \frac{1}{3}\frac{1}{t+2}\right]dt$$

$$= \frac{1}{3}[\log|1+2t| - \log|t+2|] + c$$

$$= \frac{1}{3}\log\left|\frac{1+2\tan(x/2)}{\tan(x/2)+2}\right| + c$$

20. $\int \cos^{\frac{-3}{7}} x \sin^{\frac{-11}{7}} x dx$

$$= \int \frac{1}{\sin^{\frac{11}{7}} x} \frac{1}{\cos^{\frac{3}{7}} x} dx$$

$$= \int \frac{1}{\frac{\sin^{\frac{11}{7}} x}{\cos^{\frac{11}{7}} x} . \cos^{\frac{11}{7}} x \cos^{\frac{3}{7}} x} dx$$

$$= \int \frac{\sec^2 x}{\tan^{\frac{11}{7}} x} dx$$

Put $\tan x = t$ so that $\sec^2 x ds = dt$

$$= \int \frac{dt}{t^{\frac{11}{7}}} = \int t^{\frac{-11}{7}} dt$$

$$= \frac{t^{-\frac{4}{7}}}{-4/7} + c = -\frac{7}{4} \tan^{-\frac{4}{7}} x + c$$

21. Let $I = \int \frac{2x-3}{(x^2+x+1)^2} dx = \int \frac{(2x+1)-4}{(x^2+x+1)^2} dx$

$$\Rightarrow I = \int \frac{2x+1}{(x^2+x+1)^2} dx - 4I_1$$

$$\Rightarrow I = -\frac{1}{x^2+x+1} - 4I_1$$

Now, $I_1 = \int \frac{1}{\left[\left(\frac{x+1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2\right]^2} dx$

Putting $\frac{x+1}{2} = \frac{\sqrt{3}}{2} \tan\theta$ in I_1, we have

$$I_1 = \int \frac{\frac{\sqrt{3}}{2} \sec^2\theta}{\left[\left(\frac{\sqrt{3}}{2} \tan\theta\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2\right]^2} d\theta$$

$$\Rightarrow I_1 = \int \frac{8}{3\sqrt{3}} \int \cos^2\theta d\theta = \frac{8}{3\sqrt{3}} \int \frac{1+\cos 2\theta}{2} d\theta$$

$$\Rightarrow I_1 = \frac{4}{3\sqrt{3}} (\theta + \sin 2\theta) + c$$

$$\Rightarrow I_1 = \frac{4}{3\sqrt{3}} \left[\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + 2\frac{\tan\theta}{1+\tan^2\theta}\right] + c$$

$$\Rightarrow I_1 = \frac{4}{3\sqrt{3}} \left[\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{\sqrt{3}}{4}\left(\frac{2x+1}{x^2+x+1}\right)\right] + c$$

Using (i), we have

$$I = -\frac{1}{x^2+x+1} - \frac{16}{3\sqrt{3}} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) - \frac{4}{3}\left(\frac{2x+1}{x^2+x+1}\right) + c$$

22. $|x| = \pm x$

$\therefore \quad I = \pm \frac{x^4}{4}$ for all x, +ve or –ve

(a) is correct only for +ve x

(b) is correct only for –ve x

Note $|x|^4 = x^4$ only and so it cannot be option (c).

23. Put $a + b\tan^2 x = t^2$

$$\Rightarrow 2b \tan x \sec^2 x dx = 2t dt$$

$$\Rightarrow \tan x dx = \frac{t}{b} \frac{b}{b+(t^2-a)} dt$$

$$\therefore \quad I = \int \frac{t}{t} \cdot \frac{dt}{t^2+(b-a)}$$

$$= \frac{1}{\sqrt{b-a}} \tan^{-1}\left(\frac{1}{\sqrt{b-a}}\right) + c$$

$$= \frac{1}{\sqrt{b-a}} \tan^{-1}\left(\sqrt{\frac{a+b\tan^2 x}{b-a}}\right) + c$$

Alternative Solution

(i)

$$I = \int \frac{\sin x}{\sqrt{a\cos^2 x + b\sin^2 x}} dx$$

Put $\cos x = t$ so that

$$I = \int \frac{-dt}{\sqrt{b-(b-a)t^2}}$$

$$= \frac{1}{\sqrt{b-a}} \cos^{-1}\left(\sqrt{\frac{b-a}{b}} \cos x\right) + c$$

But $\cos^{-1} x = \tan^{-1}\left(\sqrt{\frac{1-x^2}{x}}\right)$, therefore

$$I = \frac{1}{\sqrt{b-a}} \tan^{-1}\left(\sqrt{\frac{b}{(b-a)\cos^2 x} - 1}\right) + c$$

$$= \frac{1}{\sqrt{b-a}} \tan^{-1}\left(\sqrt{\frac{b(1+\tan^2 x) - b + a}{b-a}}\right) + c$$

$$= \frac{1}{\sqrt{b-a}} \tan^{-1}\left(\sqrt{\frac{a+b\tan^2 x}{b-a}}\right) + c$$

24. $I=\int\frac{1}{\cos^{2/3}x\sin^{4/3}x}dx=\int\frac{1}{\cos^{2/3+4/3}x\tan^{4/3}x}dx$

$I=\int\frac{\sec^2 x}{\sin^{4/3}x}dx$.

Put $\tan x=t$

$\therefore\quad \sec^2 x dx=dt$

$I=\int\frac{dt}{t^{4/3}}=\int t^{-4/3}dt=\frac{t^{-1/3}}{-1/3}=-3t^{-1/3}$

$\therefore\quad I=-3(\tan x)^{-1/3}$

25. $I=\int\frac{1}{(1-\cos^2 x)\sin^2 x}dx$

$\Rightarrow\quad I=\int\frac{(1+\cot^2 x)\text{cosec}^2 x dx}{1+2\cot^2 x}$

Put $\cot x=$ t

$\Rightarrow\quad -\text{cosec}^2 x dx=dt$

$\therefore\quad I=-\int\frac{(1+t^2)dt}{1+2t^2}=-\frac{1}{2}\int\frac{2+2t^2}{1+2t^2}dt$

$\Rightarrow\quad I=-\frac{1}{2}\int dt-\frac{1}{2}\int\frac{dt}{1+2t^2}$

$\Rightarrow\quad I=-\frac{1}{2}\int dt-\frac{1}{4}\int\frac{dt}{t^2+\frac{1}{2}}$

$\Rightarrow\quad I=-\frac{1}{2}t-\frac{1}{4}\cdot\frac{1}{(1/\sqrt{2})}\tan^{-1}\left(\frac{t}{1/\sqrt{2}}\right)+c$

$\therefore\quad I=-\frac{1}{2}\cot x-\frac{1}{2\sqrt{2}}\tan^{-1}\left(\sqrt{2}\cot x\right)+c$

26. $I=\int\frac{\cos^2 x-\sin^2 x}{(\sin x+\cos x)^2}x$

$=\int\frac{\cos x-\sin x}{\sin x+\cos x}dx=\log(\sin x+\cos x)+c$

27. Write $I=\int\frac{2\sin x\cos x}{(3+4\cos x)^3}dx$ and put $3+4\cos x=t$ so that $-4\sin x dx=dt$ and

$I=\frac{-1}{8}\int\frac{(t-3)}{t^3}dt=\frac{1}{8}\left(\frac{1}{t}-\frac{3}{2}\frac{1}{t^2}\right)+c$

$=\frac{2t-3}{16t^2}=\frac{8\cos x+3}{16(3+4\cos x)^2}+c$

28. $I=\int(1-\cos x)\text{cosec}^2 x dx$

$=\int\text{cosec}^2 x dx-\int\text{cosec}\,x\cot x dx$

$=-\cot x+\text{cos ec}\,x+c$

$=-\frac{\cos x}{\sin x}+\frac{1}{\sin x}+c=\frac{1-\cos x}{\sin x}+c$

$=\frac{2\sin^2 x/2}{2\sin x/2\cos x/2}+c=\tan\frac{x}{2}+c$

29. $I=\int\frac{\sin x\cos x}{a\cos^2 x+b\sin^2 x}dx$

Let $a\ \cos^2 x+b\sin^2 x=t$

$\Rightarrow\quad(-2a\cos x\sin x+2b\sin x\cos x)dx=dt$

$\Rightarrow\quad 2(b-a)\sin x\cos x dx=dt$

$\Rightarrow\quad I=\int\frac{dt}{2(b-a)t}=\frac{1}{2(b-a)}\log|t|+c$

$\therefore\quad I=\frac{1}{2(b-a)}\log\left|a\cos^2 x+b\sin^2 x\right|+c$

30. $I=\int\frac{(x+\cos x)-x(1-\sin x)}{x(x+\cos)}dx$

$=\int\left(\frac{1}{x}-\frac{1-\sin x}{x+\cos x}\right)dx=\log x-\log(x+\cos x)+c$

$=\log\left(\frac{x}{x+\cos x}\right)+c$

31. Using $\sin^3 x+\cos^3 x=$ $(\sin x+\cos x)(\sin^2 x+\cos^2-\sin x\cos x)$, we can write

$I=\int\frac{\sin x+\cos x}{(\sin x+\cos)^2(1-\sin x\cos x)}dx$

Put $\sin x-\cos x=t$ so that

$1-2\sin x\cos x=t^2$ and

$I=\int\frac{2dt}{(2-t^2)(1+t^2)}=\frac{2}{3}\int\left[\frac{1}{2-t^2}+\frac{1}{1+t^2}\right]dt$

$=\frac{2}{3}\left[\frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}+t}{\sqrt{2}-t}\right|+\tan^{-1}(t)\right]+c$

32. Let $I=\int\frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}dx$

Put $\sin x=t\ \Rightarrow\ dt=\cos x dx$

$\therefore\quad I=\int\frac{1-t^2+(1-t^2)^2}{t^2+t^4}dt$

$$\Rightarrow I=\int\frac{(1-t^2)(2-t^2)}{t^2+t^4}dt$$

$$\Rightarrow I=\int\frac{2-3t^2+t^4}{t^2+t^4}dt$$

$$\Rightarrow I=\int\left(1+\frac{2}{t^2}-\frac{6}{1+t^2}\right)dt$$

$$\Rightarrow I=t-\frac{2}{t}-6\tan^{-1}(t)+c$$

$$\therefore \quad I=\sin x-2(\sin x)^{-1}-6\tan^{-1}(\sin x)+c$$

33. $\int\frac{(1+x)^2}{x+x^3}dx=\int\frac{(1+x^2)+2x}{x(1+x^2)}dx$

$$=\int\left(\frac{1}{x}+\frac{2}{1+x^2}\right)dx=\log_e x+2\tan^{-1}x+c$$

34. Let $I=\int\frac{x^{\frac{7}{2}}}{\sqrt{x^9+1}}dx$

Let $x^{\frac{9}{2}}=t$

$$\Rightarrow dt=\frac{9}{2}x^{\frac{7}{2}}dx \Rightarrow \frac{2}{9}dt=x^{\frac{7}{2}}dx$$

$$\therefore \quad I=\frac{2}{9}\int\frac{dt}{\sqrt{t^2+1}}=\frac{2}{9}\log\left|t+\sqrt{t^2+1}\right|+c$$

$$\Rightarrow I=\frac{2}{9}\log\left|x^{\frac{9}{2}}+\sqrt{x^9+1}\right|+c$$

35. $I=\int\frac{dx}{2\cos\left(x-\frac{\pi}{3}\right)}=\frac{1}{2}\int\sec\left(x-\frac{\pi}{3}\right)dx$

$$=\frac{1}{2}\log\left[\sec\left(x-\frac{\pi}{3}\right)+\tan\left(x-\frac{\pi}{3}\right)\right]+c$$

$$=\frac{1}{2}\log\tan\left(\frac{\pi}{2}+\frac{x-\pi/3}{2}\right)+c$$

$$=\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)+c$$

36. Let $I=\int\frac{(2-2x)}{\sqrt{(4+2x-x^2)}}dx+\int\frac{dx}{\sqrt{(4+2x-x^2)}}$

$$=2\sqrt{4+2x-x^2}+\int\frac{dx}{\sqrt{5-(x-1)^2}}$$

$$=2\sqrt{4+2x-x^2}+\sin^{-1}\left(\frac{x-1}{\sqrt{5}}\right)+c$$

37. $I_n=\int\cot^n x\,dx=\int\cot^{n-2}x(\operatorname{cosec}^2x-1)dx$

$$=\int\cot^{n-2}x\operatorname{cosec}^2x\,dx-I_{n-2}$$

$$=-\frac{\cot^{n-1}}{n-1}-I_{n-2}$$

Put $n=6, 5(I_6+I_4)=-\cot^5 x$

38. Since $\int e^{x^2}dx$ cannot be expressed in terms of elementary function, the integral is known as inexpressible or that 'cannot be found'.

39. Now, $f(x+\pi)=\int\sin^2(x+\pi)dx=\int\sin^2 x\,dx=f(x)$

40. Let $I=\int\frac{dx}{x(x^n+1)}$

Put $x^n+1=t$

$$\Rightarrow nx^{n-1}dx=dt$$

$$\therefore \quad I=\frac{1}{n}\int\frac{dt}{t(t-1)} \quad =\frac{1}{n}\int\left(\frac{1}{t-1}-\frac{1}{t}\right)dt$$

$$=\frac{1}{n}\log\left(\frac{t-1}{t}\right)+c=\frac{1}{n}\log\left(\frac{x^n}{x^n+1}\right)+c$$

41. $\int\frac{\sin x}{\sin(x-\alpha)}dx=\int\frac{\sin\overline{(x-\alpha}+\alpha)}{\sin(x-\alpha)}dx$

$$=\int\frac{\sin x(x-\alpha)\cos\alpha+\cos(x-\alpha)\sin a}{\sin(x-\alpha)}dx$$

$$=\cos a\int dx+\sin\alpha\int\frac{\cos(x-\alpha)}{\sin(x-\alpha)}dx$$

$$=x\cos\alpha+\sin\alpha,\log\sin(x-\alpha)+c$$

$$\therefore \quad A=\cos\alpha, B=\sin\alpha$$

$$\therefore \quad (A,B)=(\cos\alpha,\sin\alpha)$$

42. $\int\frac{dx}{\cos x-\sin x}$

$$=\frac{1}{\sqrt{2}}\int\frac{dx}{\left(\frac{1}{\sqrt{2}}\cos x-\frac{1}{\sqrt{2}}\sin x\right)}$$

$$=\frac{1}{\sqrt{2}}\int\frac{dx}{\cos\left(x+\frac{\pi}{4}\right)}=\frac{1}{\sqrt{2}}\int\sec\left(x+\frac{\pi}{4}\right)dx$$

$$=\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}+\frac{\pi}{8}\right)\right|+c$$

$$=\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2}+\frac{3\pi}{8}\right)\right|+c$$

43. As we know, derivative of the function should be equal to the integral.

Let us assume a function $f(x)=\dfrac{x}{(\log x)^2+1}+c$ from option (a).

$$\therefore \quad \frac{d}{dx}\left(\frac{x}{(\log x)^2+1}+c\right)=\frac{(\log x)^2+1-x[2\log x\cdot(1/x)]}{[(\log x)^2+1]^2}$$

$$=\frac{(\log x)^2+1-2\log x}{[(\log x)^2+1]^2}$$

$$=\frac{(\log x-1)^2}{[(\log x)^2+1]^2}$$

$$=\left[\frac{(\log x-1)}{1+(\log x)^2}\right]^2$$

Hence, this result is equal to the given integral.

44. $$\int\frac{dx}{\cos x+\sqrt{3}\sin x}=\frac{1}{2}\int\sec\left(x-\frac{\pi}{3}\right)dx$$

$$=\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{6}+\frac{\pi}{4}\right)+c$$

$$=\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+c$$

45. $$\sqrt{2}\int\frac{\sin x dx}{\sin\left(x-\frac{\pi}{4}\right)}=\sqrt{2}\int\frac{\sin\left(x-\frac{\pi}{4}+\frac{\pi}{4}\right)dx}{\sin\left(x-\frac{\pi}{4}\right)}$$

$$=\sqrt{2}\left(\cos\frac{\pi}{4}+\cot\left(x-\frac{\pi}{4}\right)\sin\frac{\pi}{4}\right)dx$$

$$=\int dx+\int\cot\left(x-\frac{\pi}{4}\right)dx$$

$$=x+\ln\left|\sin\left(x-\frac{\pi}{4}\right)\right|+c$$

46. $$\int\cos x\log\left(\tan\frac{x}{2}\right)dx$$

$$=\log\left(\tan\frac{x}{2}\right).\sin x-\int\frac{1}{\tan\frac{x}{2}}\cdot\sec^2\frac{x}{2}\cdot\frac{1}{2}\sin x dx+c$$

(integrating by parts)

$$=\sin x\log\left(\tan\frac{x}{2}\right)-\int\frac{1}{2\sin\frac{x}{2}\cos\frac{x}{2}}\cdot\sin x dx+c$$

$$=\sin x\cdot\log\left(\tan\frac{x}{2}\right)-\int dx+c$$

$$=\sin x\cdot\log\left(\tan\frac{x}{2}\right)-x+c$$

47. Put $x=\cos^2\theta \Rightarrow x=-2\cos\theta\sin\theta d\theta$

$$=\int\left(\frac{1-\cos\theta}{1+\cos\theta}\right)^{1/2}\left(\frac{-2\cos\theta\sin\theta d\theta}{\cos^2\theta}\right)dx$$

$$=-\int\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\cdot\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\cos\theta}d\theta$$

$$=-\int\frac{2\sin^2\frac{\theta}{2}}{\cos\theta}d\theta=-2\int\frac{1-\cos\theta}{\cos\theta}d\theta$$

$$=-2\int(\sec\theta-1)d\theta$$

$$=-2[\log|\sec\theta+\tan\theta|-\theta]$$

$$=-2\left[\log\left|\frac{1}{\sqrt{x}}+\frac{\sqrt{1-x}}{\sqrt{x}}\right|-\cos^{-1}\sqrt{x}\right]+c$$

$$=-2\left[\log\left|\frac{1+\sqrt{1-x}}{\sqrt{x}}\right|-\cos^{-1}\sqrt{x}\right]+c$$

48. $$\int\frac{dx}{(2x-7)\sqrt{x^2-7x+12}}$$

$$=\int\frac{2dx}{(2x-7)\sqrt{4x^2-28x+48}}$$

$$=\int\frac{2dx}{(2x-7)\sqrt{(2x-7)^2-1}}$$

$$=\sec^{-1}(2x-7)+c \qquad \left[\because\int\frac{dx}{x\sqrt{x^2-1}}=\sec^{-1}x+c\right]$$

49. $\because \quad I=\int\dfrac{dx}{1+3\sin^2 x}$

$$=\int\frac{dx}{\cos^2 x+4\sin^2 x}=\int\frac{\sec^2 x}{1+4\tan^2 x}dx$$

Put $\tan x=z \qquad \therefore \quad \sec^2 x dx=dz$

$$\therefore \quad I=\int\frac{dz}{1+4z^2}=\frac{1}{4}\int\frac{dz}{x^2+(1/2)^2}$$

$$=\frac{1}{4}\cdot\frac{1}{1/2}\tan^{-1}\frac{z}{1/2}$$

$$=\frac{1}{2}\tan^{-1}(2z)=\frac{1}{2}\tan^{-1}(2\tan x)$$

50. Put $\sqrt{x}=z \quad \therefore \frac{1}{2\sqrt{x}}dx=dz\Rightarrow\frac{dx}{\sqrt{x}}=2dz$

$\therefore$ Given integral

$$=2\int a^z dz=\frac{2a^z}{\log_e a}=\frac{2a^{\sqrt{x}}}{\log_e a}=2a^{\sqrt{x}}\log_a e$$

51. Put $(\sqrt{1+x^2}+x)^n+z$

$$\therefore \quad n(\sqrt{1+x^2}+x)^{n-1}\cdot\left(\frac{2x}{2\sqrt{1+x^2}}+1\right)dx=dz$$

$$\Rightarrow \quad \frac{(\sqrt{1+x^2}+x)^n}{\sqrt{1+x^2}}dx=\frac{dz}{n}$$

$$\therefore \quad \text{Given integral}=\int\frac{dz}{n}=\frac{1}{n}z+c$$

$$=\frac{1}{n}(\sqrt{1+x^2}+x)^n+c$$

52. Integral

$$=\int\frac{x^2\left(1-\frac{1}{x^2}\right)dx}{x^2\left[\sqrt{\left(x+\frac{1}{x}+\alpha\right)}\sqrt{x+\frac{1}{x}+\beta}\right]}$$

Put $x+\frac{1}{x}=z \quad \therefore \left(1-\frac{1}{x^2}\right)dx=dz$

$\therefore$ Given integral

$$=\int\frac{dz}{\sqrt{(z+\alpha)(z+\beta)}}$$

$$=\int\frac{dz}{\sqrt{z^2+(\alpha+\beta)z+\alpha\beta}}$$

$$=\int\frac{dz}{\sqrt{\left(z+\frac{\alpha+\beta}{2}\right)^2-\left(\frac{\alpha-\beta}{2}\right)^2}}$$

$$=\log\left|z+\frac{\alpha+\beta}{2}-\sqrt{\left(z+\frac{\alpha+\beta}{2}\right)^2-\left(\frac{\alpha-\beta}{2}\right)^2}\right|+c$$

$$=\log\left|x+\frac{1}{x}+\frac{\alpha+\beta}{2}-\sqrt{(z+\alpha)(z+\beta)}\right|+c$$

$$=\log\left|x+\frac{1}{x}+\frac{\alpha+\beta}{2}-\sqrt{\left(x+\frac{1}{x}+\alpha\right)\left(x+\frac{1}{x}+\beta\right)}\right|+c$$

$$=\log\left[\frac{2x^2+(\alpha+\beta)x+2}{2x}-\frac{\sqrt{(x^2+ax+1)(x^2+\beta x+1)}}{x}\right]+c$$

$$=\log\frac{1}{2}\left(\frac{\sqrt{x^2+ax+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right)^2+c$$

$$=2\log\left(\frac{\sqrt{x^2+ax+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right)+c$$

where $\log(1/2)$ has been merged with c.

53. $\int\frac{dx}{(1+\sin x)^{1/2}}=\int\frac{dx}{\cos\frac{x}{2}+\sin\frac{x}{2}}=\frac{1}{r}\int\frac{dx}{\sin\left(\frac{x}{2}+\alpha\right)}$

where $1=r\sin\alpha, 1=r\cos\alpha$

$$=\frac{1}{\sqrt{2}}\int\operatorname{cosec}\left(\frac{x}{2}+\alpha\right)dx=\frac{1}{\sqrt{2}}\left(\frac{\log\left|\tan\left(\frac{x}{4}+\frac{\alpha}{2}\right)\right|}{\frac{1}{2}}\right)+c$$

$$=\sqrt{2}\log\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+c \qquad [\because \alpha=\pi/4, r=\sqrt{2}]$$

54. $\int\frac{2x}{(1+x^2)\sqrt{x^4-1}}dx=\int\frac{-2x}{(x^2-1)^{3/2}\sqrt{x^2+1}}dx$

Put $\sqrt{\frac{x^2+1}{x^2-1}}=z$

$$\therefore \quad \frac{1}{2}\left(\frac{x^2-1}{x^2+1}\right)^{1/2}\cdot\frac{(x^2-1)\cdot 2x-(x^2+1)2x}{(x^2-1)^2}dx=dz$$

$$\Rightarrow \quad \frac{\sqrt{x^2-1}}{\sqrt{x^2+1}}\cdot\frac{-2x}{(x^2-1)^2}dx=dz$$

$$\Rightarrow \quad \frac{-2x}{(x^2-1)^{3/2}\sqrt{x^2+1}}dx=dz$$

$$\therefore \quad \text{Given integral}=\int dz=z+c=\sqrt{\frac{x^2+1}{x^2-1}}+c$$

55. $\int \frac{x+\sin x}{1+\cos x}dx = \int \frac{x+2\sin(x/2)\cos(x/2)}{2\cos^2(x/2)}dx$

$= \int \frac{x}{2}\sec^2\frac{x}{2}dx + \int \tan\frac{x}{2}dx$

$= \frac{1}{2}[x\cdot 2\tan\frac{x}{2} - \int 2\tan\frac{x}{2}dx] + \int \tan\frac{x}{2}dx$

$= x\tan\frac{x}{2} - \int \tan\frac{x}{2}dx + \int \tan\frac{x}{2}dx$

$= x\tan\frac{x}{2} + c$

56. Given, $\int \frac{1}{x+x^5}dx = f(x)+c$ (i)

Let $I = \int \frac{x^4}{x+x^5}dx \Rightarrow I = \int \frac{x^4+1-1}{x+x^5}dx$

$\Rightarrow I = \int \frac{x^4+1}{x(x^4+1)}dx - \int \frac{1}{x(x^4+1)}dx$

$\Rightarrow I = \int \frac{1}{x}dx - \int \frac{1}{x^5+x}dx$

$\Rightarrow I = \log x - [f(x)+c]$ [from Eq. (i)]

$\Rightarrow I = \log x - f(x) + c_1$ (where $c_1 = c$)

57. $\int \frac{\cos 4x+1}{\cot x - \tan x}dx = k\cos 4x + c$ (i)

Let $I = \int \frac{\cos 4x+1}{\cot x-\tan x}dx = \int \frac{2\cos^2 2x}{\frac{\cos^2 x-\sin^2 x}{\sin x\cos x}}dx$

$= \int \frac{2\cos^2 2x}{\cos 2x}\cdot \sin x\cos x dx = \int \cos 2x\cdot \sin 2x dx$

$\Rightarrow I = \frac{1}{2}\int 2\cos 2x\cdot \sin 2x dx = \frac{1}{2}\int \sin 4x dx$

$\Rightarrow I = \frac{1}{2}\frac{(-\cos 4x)}{4} + c$

$I = \frac{-\cos 4x}{8} + c$ (ii)

On comparing Eqs. (i) and (ii), we get

$$k = \frac{-1}{8}$$

58. Let $I = \int f^{-1}(x)dx$ (i)

and $\int f(x)dx = g(x)+c$ (ii)

$\therefore$ From Eq. (i), let $f^{-1}(x) = u$

$\Rightarrow x = f(u) \Rightarrow dx = f'(u)du$

$\therefore I = \int uf'(u)du$

$= uf(u) - \int f(u)du$

$= uf(u) - g(u) + c$ [from Eq. (ii)]

$\therefore$ On putting $u = f^{-1}(x), f(u) = x$

we get, $I = xf^{-1}(x) - g[f^{-1}(x)] + c$

59. Let $I = \int \frac{dx}{x(x^7+1)}$

Put $x^7 = t$ and $dx = \frac{1}{7x^6}dt$

$\therefore I = \frac{1}{7}\int \frac{dt}{t(t+1)} = \frac{1}{7}\log\left(\frac{t}{t+1}\right) + c$

$= \frac{1}{7}\log\left(\frac{x^7}{x^7+1}\right) + c$

60. We have $\frac{d}{dx}[g(x)] = g(x)$

$\Rightarrow g'(x) = g(x)$

$\Rightarrow \int \frac{g'(x)}{g(x)}dx = \int 1dx$

$\Rightarrow \log_e[g(x)] = x + \log c_1$

$\Rightarrow g(x) = c_1e^x$

Now, $g(0) = 1 \Rightarrow c_1 = 1$

$\therefore g(x) = e^x$

So, $\int g(x)\frac{2-\sin x}{1-\cos x}dx$

$= \int e^x(\text{cosec}^2 x - \cot x)dx$

$= -e^x\cot x + c = -g(x)\cot x + c$

61. $\int \frac{dx}{(2x-7)\sqrt{x^2-7x+12}}$

$= \int \frac{2dx}{(2x-7)\sqrt{4x^2-28x+48}}$

$= \int \frac{2dx}{(2x-7)\sqrt{(2x-7)^2-1}}$

$= \sec^{-1}(2x-7) + c$

62. $\int e^{3\log x}\cdot(x^4+1)^{-1}dx$

$= \int e^{\log x^3}\cdot\dfrac{dx}{x^4+1}$

$= \int\dfrac{x^3dx}{x^4+1}$

$= \dfrac{1}{4}\int\dfrac{4x^3}{x^4+1}dx$

$= \dfrac{1}{4}\log(x^4+1)+c$

63. Let $I = \int\sec^3 x dx = \int\sec x\cdot\sec^2 x dx$

$= \int\sqrt{1+\tan^2 x}\cdot\sec^2 x dx$

Put $\tan x = z \quad \therefore \quad \sec^2 x dx = dz$

$\Rightarrow I = \sqrt{1+z^2}dz$

$= \dfrac{z\sqrt{z^2+1}}{2}+\dfrac{1}{2}\log\left|z+\sqrt{z^2+1}\right|+c$

$= \dfrac{\tan x\cdot\sec x}{2}+\dfrac{1}{2}\log(\tan x+\sec x)+c$

$= \dfrac{1}{2}[\sec x\tan x+\log(\sec x+\tan x)]+c$

64. $\int\dfrac{dx}{\sqrt{2ax-x^2}} = \int\dfrac{dx}{\sqrt{a^2-(x-a)^2}}$

$= \sin^{-1}\left(\dfrac{x-a}{a}\right)$

65. $\int_0^1 x[x^2]dx+\int_1^{\sqrt{2}} x[x^2]dx+\int_{\sqrt{2}}^{1.5} x[x^2]dx$

$= \int_0^1 x\cdot 0dx+\int_1^{\sqrt{2}} xdx+\int_{\sqrt{2}}^{1.5} 2xdx = 0+\left[\dfrac{x^2}{2}\right]_1^{\sqrt{2}}+\left[x^2\right]_{\sqrt{2}}^{1.5}$

$= \dfrac{1}{2}(2-1)+(2.25-2) = \dfrac{1}{2}+\cdot 25 = \dfrac{1}{2}+\dfrac{1}{4}=\dfrac{3}{4}$

66. $\int\dfrac{5\tan x}{\tan x-2}dx = \int\dfrac{5\sin x}{\sin x-2\cos x}dx$

$\Rightarrow \int\left[\dfrac{2(\cos x+2\sin x)+(\sin x-2\cos x)}{\sin x-2\cos x}\right]dx$

$= 2\int\left(\dfrac{\cos x+2\sin x}{\sin x-2\cos x}\right)dx+\int dx+k$

$= 2\log|\sin x-2\cos x|+x+k$

$\therefore \quad a = 2$

67. $\dfrac{d[p(t)]}{dt} = \dfrac{1}{2}p(t)-450$

$$\frac{d[p(t)]}{dt} = \frac{p(t)-900}{2}$$

$$2\int\frac{d[p(t)]}{p(t)-900} = \int dt$$

$$2\ln|p(t)-900| = t+c$$

$t = 0 \quad \Rightarrow \quad 2\ln 50 = 0+c \quad \Rightarrow \quad c = 2\ln 50$

$\therefore \quad 2\ln|p(t)-900| = t+2\ln 50$

$P(t) = 0 \quad \Rightarrow \quad 2\ln 900 = t+2\ln 50$

$t = 2(\ln 900-\ln 50) = 2\ln\left(\dfrac{900}{50}\right) = 2\ln 18$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. Integration of $f(x) = \sqrt{1+x^2}$ with respect to x^2 is

(a) $\dfrac{2}{3}\dfrac{(1+x^2)^{3/2}}{x}+c$ (b) $\dfrac{2}{3}(1+x^2)^{3/2}+c$

(c) $\dfrac{2x}{3}(1+x^2)^{3/2}+c$ (d) none of these

2. If $I = \int\dfrac{1}{2p}\sqrt{\dfrac{p-1}{p+1}}dp = f(p)+c$, then $f(p)$ is equal to

(a) $\dfrac{1}{2}\ln(p-\sqrt{p^2-1})$ (b) $\left(\dfrac{1}{2}\cos^{-1}p+\dfrac{1}{2}\sec^{-1}p\right)$

(c) $\ln\sqrt{p+\sqrt{p^2-1}}-\dfrac{1}{2}\sec^{-1}p$

(d) none of these

3. $\int[f(x)g''(x)-f''(x)g(x)]dx$ is equal to

(a) $f(x)/g'(x)$

(b) $f'(x)g(x)-f(x)g'(x)$

(c) $f(x)g'(x)-f'(x)g(x)$

(d) $f(x)g'(x)+f'(x)g'(x)$

4. The equation of a curve passing through the point (0, 1) be given by $y = \int x^2\cdot e^3 dx$. If the equation of the curve be written in the form $x = f(y)$, then $f(y)$ is equal to

(a) $\sqrt{\log_e(3y-2)}$ (b) $\sqrt[3]{\log_e(3y-2)}$

(c) $\sqrt[3]{\log_e(2-3y)}$ (d) none of these

5. If $\int\frac{4e^{x}+6e^{-x}}{9e^{x}-4e^{-x}}dx = Ax + B\ln(9e^{2x}-4)+c$, then

(a) $A=-\frac{3}{2},\ B=\frac{35}{36},\ c=0$

(b) $A=\frac{35}{36},\ B=-\frac{3}{2},\ c\in\mathbb{R}$

(c) $A=-\frac{3}{2},\ B=\frac{35}{36},\ c\in\mathbb{R}$

(d) none of these

6. $\int\frac{dx}{(x+a)^{\frac{8}{7}}(x-b)^{\frac{6}{7}}}$ is equal to

(a) $\left(\frac{7}{a+b}\right)\left(\frac{x+a}{x-b}\right)^{\frac{1}{7}}+c$

(b) $\left(\frac{7}{a+b}\right)\left(\frac{x-b}{x+a}\right)^{\frac{1}{7}}+c$

(c) $\frac{6}{a+b}\left(\frac{x-b}{x+a}\right)^{\frac{1}{6}}+c$

(d) $\frac{6}{a+b}\left(\frac{x+a}{x-b}\right)^{\frac{1}{6}}+c$

7. If $\int\frac{dx}{x^4+x^3}=\frac{A}{x^2}+\frac{B}{x}+\ln\left|\frac{x}{x+1}\right|+c$, then

(a) $A=\frac{1}{2},\ B=1$ (b) $A=1,\ B=-\frac{1}{2}$

(c) $A=-\frac{1}{2},\ B=1$ (d) none of these

8. $\int\left(x+\frac{1}{x}\right)^{n+5}\left(\frac{x^2-1}{x^2}\right)dx$ is equal to

(a) $\frac{\left(x+\frac{1}{x}\right)^{n+6}}{n+6}+c$

(b) $\left(\frac{x^2+1}{x^2}\right)^{n+6}(n+6)+c$

(c) $\left(\frac{x}{x^2+1}\right)^{n+6}(n+6)+c$

(d) none of these

9. $\int e^{\tan\theta}(\sec\theta-\sin\theta)d\theta$ equals

(a) $-e^{\tan\theta}\sin\theta+c$ (b) $e^{\tan\theta}\sin\theta+c$

(c) $e^{\tan\theta}\sec\theta+c$ (d) $e^{\tan\theta}\cos\theta+c$

10. If $\int xe^{x}\cos x dx = f(x)+c$, then $f(x)$ is equal to

(a) $\frac{e^{x}}{2}[(1-x)\sin x - x\cos x]$

(b) $\frac{e^{x}}{2}[(x-1)\sin x + x\cos x]$

(c) $\frac{e^{x}}{2}[(1+x)\sin x - x\cos x]$

(d) none of these

11. $\int\sqrt{\frac{1-\cos x}{\cos\alpha\cos x}}dx$, where $0<\alpha<x<\pi$, equals

(a) $2\ln\left(\cos\frac{\alpha}{2}-\cos\frac{x}{2}\right)+c$

(b) $\sqrt{2}\ln\left(\cos\frac{\alpha}{2}-\cos\frac{x}{2}\right)+c$

(c) $2\sqrt{2}\ln\left(\cos\frac{\alpha}{2}-\cos\frac{x}{2}\right)+c$

(d) $-2\sin^{-1}\left[\frac{\cos(x/2)}{\cos(\alpha/2)}\right]+c$

12. $\int\frac{1}{x[6(\log x)^2+7\log x+2]}dx$ is equal to

(a) $\log\left|\frac{2\log x+1}{3\log x+2}\right|+c$ (b) $\log\left|\frac{3\log x+1}{2\log x+2}\right|+c$

(c) $\log\left|\frac{\log x+1}{\log x+2}\right|+c$ (d) $\log\left|\frac{\log x+3}{\log x+1}\right|+c$

13. Antiderivative of $\frac{\sin^2 x}{1+\sin^2 x}$ w.r.t. x is

(a) $x-\frac{\sqrt{2}}{2}\arctan\left(\sqrt{2}\tan x\right)+c$

(b) $x-\frac{1}{\sqrt{2}}\arctan\left(\frac{\tan x}{\sqrt{2}}\right)+c$

(c) $x-\sqrt{2}\arctan\left(\sqrt{2}\tan x\right)+c$

(d) $x-\sqrt{2}\arctan\left(\frac{\tan x}{\sqrt{2}}\right)+c$

14. $I=\int e^{ax}\cos bx dx$ is equal to

(a) real part of $\int e^{(a+bi)x}dx$

(b) imaginary part of $\int e^{(a+bi)x}dx$

(c) neither real nor imaginary part of $\int e^{(a+bi)x}dx$

(d) none of these

15. $\int\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$ equals

(a) $\sqrt{x}\sqrt{1-x}-2\sqrt{1-x}+\cos^{-1}\left(\sqrt{x}\right)+c$

(b) $\sqrt{x}\sqrt{1-x}+2\sqrt{1-x}+\cos^{-1}\left(\sqrt{x}\right)+c$

(c) $\sqrt{x}\sqrt{1-x}-2\sqrt{1-x}-\cos^{-1}\left(\sqrt{x}\right)+c$

(d) $\sqrt{x}\sqrt{1-x}+2\sqrt{1-x}-\cos^{-1}\left(\sqrt{x}\right)+c$

16. $\int e^{2ax}\frac{1-\cos 2ax}{1+\sin 2ax}dx$ is equal to

(a) $-\frac{1}{a}e^{2ax}\cos\left(\frac{\pi}{4}+ax\right)+c$

(b) $-\frac{1}{2a}e^{2ax}\cot\left(\frac{\pi}{4}+ax\right)+c$

(c) $-\frac{1}{2a}e^{2ax}\cos\left(\frac{\pi}{4}+ax\right)+c$

(d) $-\frac{1}{a}e^{2ax}\text{cosec}\left(\frac{\pi}{4}+ax\right)+c$

17. $\int\left[\ln(1+\sin x)+x\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right]dx$ is equal to

(a) $x\ln(1+\sin x)+c$ (b) $\ln(1+\sin x)+c$

(c) $-x\ln(1+\sin x)+c$ (d) $\ln(1-\sin x)+c$

18. If $I=\int\frac{dx}{(2x+5)\sqrt{4x^2+20x+16}}=\frac{1}{m}\sec^{-1}\left(\frac{2x+5}{p}\right)+c$, then

(a) $m = 1/6, p = 1/3$ (b) $m=-6, p=3$

(c) $m=6, p=3$ (d) $m=-6, p=-3$

19. For $x > 1$, $\int\sin^{-1}\left(\frac{2x}{1+x^2}\right)dx$ is equal to

(a) $2x\tan^{-1}(x)-\ln(x^2+1)+c$

(b) $2x\cot^{-1}x+\log_e(x^2+1)+c$

(c) $2x\tan^{-1}(x)+\log_e(x^2+1)+c$

(d) none of these

20. $\int\sin^{-1}(\cos x)dx=$

(a) $\frac{\pi x}{2}$ (b) $\frac{\pi x^2}{2}$

(c) $\frac{\pi x-x^2}{2}$ (d) $\frac{\pi x+x^2}{2}$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. $I=\int\frac{(10x^9+10^x\log_e 10)}{(x^{10}+10^x)}dx$ is equal to

(a) $10^x+x^{10}+c$ (b) $10^x-x^{10}+c$

(c) 10^x+x^{10} (d) $\log_e(10^x+x^{10})+c$

2. $\int\left(\frac{2+\sin 2x}{1+\cos 2x}\right)e^x dx=$

(a) $e^x\cot x+c$ (b) $-e^x\cot x+c$

(c) $-e^x\tan x+c$ (d) $e^x\tan x+c$

3. $I=\int\sqrt{1+2\tan x(\sec x+\tan x)}dx$ is equal to

(a) $\log_e\left|\sec^2 x+\tan x\sec x\right|+c$

(b) $\log_e\left|1+\tan x(\sec x+\tan x\right|+c$

(c) $\log_e\left|\sin x(\sec x-\tan x)\right|+c$

(d) none of these

4. $\int\frac{x^2+x-1}{x^2+x-6}dx=$

(a) $x+\log(x+3)+\log(x-2)+c$

(b) $x-\log(x+3)+\log(x-2)+c$

(c) $x-\log(x+3)-\log(x-2)+c$

(d) none of these

5. If $A=\int e^x\sin x$ and $B=\int e^x\cos x dx$, then A^2+B^2 is equal to

(a) $\frac{e^{2x}}{2}$ (b) $\frac{1}{2}e$

(c) $\frac{e^{3x}}{3}$ (d) none of these

6. $\int\frac{e^x}{e^{2x}+5e^x+6}dx$ equals

(a) $\log\left(\frac{e^x+3}{e^x+2}\right)+c$ (b) $\log\left(\frac{e^x+2}{e^x+3}\right)+c$

(c) $\frac{1}{2}\log\left(\frac{e^x+2}{e^x+3}\right)+c$ (d) none of these

7. If $I=\int x\log_e\left(1+\frac{1}{x}\right)dx$

$=p(x)\log_e(x+1)+g(x)x^2+Lx+c$, then

(a) $p(x)=\frac{1}{2}x^2$ (b) $g(x)=\log_e x$

(c) $L = 1$ (d) none of these

8. $\int \frac{dx}{4\sin^2 x + 4\sin x\cos x + 5\cos^2 x}$ is equal to

(a) $\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$ (b) $\frac{1}{4}\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$

(c) $4\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$ (d) none of these

9. If n is an odd positive integer, then $\int |x^n|\, dx$ is equal to

(a) $\left|\frac{x^{n+1}}{n+1}\right| + c$ (b) $\frac{x^{n+1}}{n+1} + c$

(c) $\frac{|x^n|x}{n+1} + c$ (d) none of these

10. $\int \frac{1}{4 + 5\sin x} dx$ is equal to

(a) $\frac{1}{3}\log\left|\frac{\tan x/2 + 5}{3}\right| + c$ (b) $\frac{1}{3}\log\left|\frac{2\tan x/2 + 1}{\tan x/2 + 2}\right| + c$

(c) $\frac{1}{3}\log\left|\frac{2\tan x/2 + 1}{\tan x/2 - 2}\right| + c$

(d) $\frac{1}{\sqrt{3}}\log\left|\frac{2\tan x/2 - 1}{\tan x/2 + 1}\right| + c$

11. $\int [1 + 2\tan x(\tan x + \sec x)]^{1/2} dx$ is equal to

(a) $\log\sec x(\sec x - \tan x) + c$

(b) $\log\operatorname{cosec}(\sec x + \tan x) + c$

(c) $\log\sec x(\sec x + \tan x + c)$

(d) $\log(\sec x + \tan x) + c$

12. If $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx = A\cos 4x + c$, then A is

(a) 1/8 (b) $-(1/8)$ (c) 1/4 (d) $-(1/4)$

13. $\int \frac{e^{x-1} + x^{e-1}}{e^x + x^e} dx =$

(a) $\log(e^x + x^e)$ (b) $e\log(e^x + x^e)$

(c) $\frac{1}{e}\log(e^x + x^e)$ (d) none of these

14. The value of $\int \frac{ax^2 - b}{x\sqrt{c^2x^2 - (ax^2 + b)^2}} dx$ is

(a) $\sin^{-1}\left[\frac{ax + (b/x)}{c}\right] + k$ (b) $\sin^{-1}\left[\frac{ax^2 + (b/x^2)}{c}\right] + k$

(c) $\cos^{-1}\left[\frac{ax + (b/x)}{c}\right] + k$ (d) $\cos^{-1}\left(\frac{ax^2 + (b/x^2)}{c}\right) + k$

15. The value of $\int e^x \frac{1 + nx^{n-1} - x^{2n}}{(1 - x^n)\sqrt{1 - x^{2n}}} dx$ is

(a) $\frac{e^x\sqrt{1 - x^n}}{1 - x^n} + c$ (b) $e^x\frac{\sqrt{1 + x^{2n}}}{1 - x^{2n}} + c$

(c) $\frac{e^x\sqrt{1 - x^{2n}}}{1 - x^{2n}} + c$ (d) $e^x\frac{\sqrt{1 - x^{2n}}}{1 - x^n} + c$

16. $\int \frac{3\sin x + 2\cos x}{3\cos x + 2\sin x} dx =$

(a) $\frac{12}{13} + \frac{5}{13}\log(3\cos x + 2\sin x) + c$

(b) $\frac{12}{13} - \frac{5}{13}\log(3\cos x + 2\sin x) + c$

(c) $\frac{5}{13}x + \frac{12}{3}\log(3\cos x + 2\sin x) + c$

(d) none of these

17. $\int x^x(1 + \log x) dx =$

(a) $x^x\log x + c$ (b) $\frac{x^x}{\log x} + c$

(c) $x^x + c$ (d) $x^x(x + 1) + c$

18. $\int \frac{dx}{x\log x\log(\log x)} =$

(a) $\log(\log x)$ (b) $\log[\log(\log x)]$

(c) $[\log(\log x)]^2$ (d) none of these

19. $\int \frac{1}{[(x-1)^3(x+2)^5]^{1/4}} dx$ is equal to

(a) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4} + c$ (b) $\frac{4}{3}\left(\frac{x+2}{x-1}\right)^{1/4} + c$

(c) $\frac{1}{3}\left(\frac{x-1}{x+2}\right)^{1/4} + c$ (d) $\frac{1}{3}\left(\frac{x+2}{x+1}\right)^{1/4} + c$

20. $\int \frac{1}{x}\left(\log_{ex} e \cdot \log_{e^2x} e \cdot \log_{e^3x} e\right) dx$ is equal to

(a) $\frac{1}{2}\log(\log_e e^2x) - \log(\log_e e^3x) + \log(\log_e e^4x) + c$

(b) $\frac{1}{2}\log(\log_e x) - \log(\log_e x) + \log(\log_e x) + c$

(c) $\frac{1}{2}\log(\log_e ex) - \log(\log_e e^2x^2) + \log(\log_e e^3x^3) + c$

(d) $\frac{1}{2}\log(\log_e ex) - \log(\log_e e^2x) + \log(\log_e e^3x) + c$

ANSWERS

Level 1

1. (b)	**2.** (c)	**3.** (c)	**4.** (a)	**5.** (c)	**6.** (b)	**7.** (c)	**8.** (a)	**9.** (d)	**10.** (b)
11. (d)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (a)	**16.** (b)	**17.** (a)	**18.** (c)	**19.** (b)	**20.** (c)

Level 2

1. (d)	**2.** (d)	**3.** (d)	**4.** (b)	**5.** (a)	**6.** (b)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (b)
11. (c)	**12.** (a)	**13.** (b)	**14.** (a)	**15.** (d)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (a)	**20.** (d)

15

G.W. LEIBNIZ

Definite Integrals

- Definite Integrals
- Properties of Definite Integrals
- Integrals as a Limit of a Sum
- Another Form of Definite Integrals
- Some Useful Results for Direct Applications
- Some Useful Reduction Formulae

DEFINITE INTEGRALS

If $f(x)$ is a continuous function defined on an interval $[a, b]$ and let the integral of $f(x)$ be $F(x)$. Then, the definite integral of $f(x)$ over $[a, b]$ denoted by $\int_a^b f(x)dx$ is defined as $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$.

This means, to evaluate a definite integral, follow the steps given below:

Step 1: Integrate the given function (do not include the constant).

Step 2: Substitute the upper limit (b) into the integral.

Step 3: Substitute the lower limit (a) into the integral.

Step 4: Subtract the second value from the first value, the answer will be a number.

These form a part of the **Fundamental Theorem of Calculus.**

Terminology: We have the following terminology associated with the symbol $\int_a^b f(x)dx$ as shown in Fig. 15.1.

PROPERTIES OF DEFINITE INTEGRALS

Property 1: $\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(k)dk = \int_a^b f(z)dz \ldots$

The point of this property is to notice that as long as the functions and limits are the same, the variable of integration that we use in the definite integral will not affect the answer.

Property 2: $\int_a^a f(x)dx = 0$

If the upper and lower limits are the same, then there is no work to do, the integral is zero.

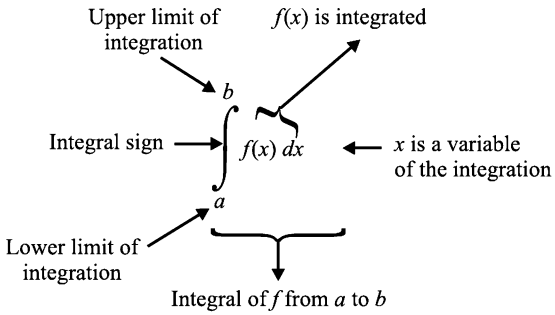


Fig. 15.1

Property 3: $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

We can interchange the limits on any definite integral. All that we need to do is to tack a minus sign onto the integral when we do.

Property 4: $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, when $a < c < b$, where c is any number.

This property is more important than we might realize at first. One of the main uses of this property is that it tells us how we can integrate a function over the adjacent intervals $[a, c]$ and $[c, b]$.

Points to remember: However, c needs to be always between a and b.

Property 5: $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$

We can break up definite integrals across a sum or difference.

Property 6: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Property 7:

$$\int_{-a}^{a} f(x)\,dx = \begin{cases} 0, & \text{when } f(x) \text{ is an odd function} \\ 2\int_0^a f(x)\,dx, & \text{when } f(x) \text{ is an even function} \end{cases}$$

Property 8:

$$\int_0^{2a} f(x)\,dx = \begin{cases} 0, & \text{if } f(2a-x) = -f(x) \\ 2\int_0^a f(x)\,dx, & \text{if } f(2a-x) = f(x) \end{cases}$$

Property 9: $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

Property 10: Leibniz's rule

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)dt = h'(x)f[h(x)] - g'(x)f[g(x)]$$

In particular, $\frac{d}{dx}\int_a^{h(x)} f(t)\,dt = h'(x)f[h(x)]$ [where a is any constant independent of x]

or $$\frac{d}{dx}\int_a^x f(t)dt = f(x)$$

Illustration 1. Evaluate $\frac{d}{dx}\left[\int_{x^3}^{x^4} \frac{1}{\log t}\,dt\right]$.

Solution:

$$\frac{d}{dx}\left[\int_{x^3}^{x^4} \frac{1}{\log t}\,dt\right] = \frac{1}{\log x^4}\frac{d}{dx}x^4 - \frac{1}{\log x^3}\frac{d}{dx}x^3$$

$$= \frac{1}{4\log x}4x^3 - \frac{1}{3\log x}3x^2$$

$$= \frac{4x^3}{4\log x} - \frac{3x^2}{3\log x}$$

$$= \frac{x^3}{\log x} - \frac{x^2}{\log x}$$

$$= \left[\frac{x^3 - x^2}{\log x}\right]$$

INTEGRALS AS A LIMIT OF A SUM

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a) + f(a+h) + f(a+2h) + \cdots + f(a+(n-1)h)]$$

where $nh = b - a$.

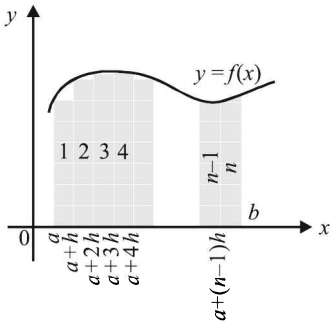


Fig. 15.2

ANOTHER FORM OF DEFINITE INTEGRALS

$$\int_a^b f(x)dx = (b-a)\lim_{n\to\infty}\frac{1}{n}[f(a) + f(a+h) + f(a+2h) + \cdots + f(a+(n-1)h)]$$

where $h = \frac{b-a}{n}$.

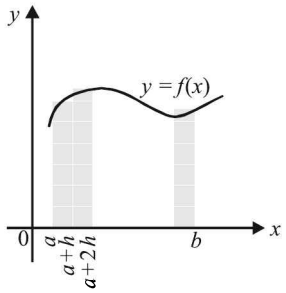


Fig. 15.3

SOME USEFUL RESULTS FOR DIRECT APPLICATIONS

1. $1 + 2 + 3 + \cdots + (n-1) = \frac{n(n-1)}{2}$.
2. $1^2 + 2^2 + 3^2 + \cdots + (n-1)^2 = \frac{n(n-1)(2n-1)}{6}$.

3. $1^3+2^3+3^3+\cdots+(n-1)^3=\left[\frac{n(n-1)}{2}\right]^2$

4. $a+ar+ar^2+\cdots+ar^{n-r}=\frac{a(r^n-1)}{(r-1)}$

5. $\sin a+\sin(a+h)+\sin(a+2h)+\cdots+\sin[a+(n-1)h]$
$=\frac{\sin\{a+[(n-1)/2]h\}\sin(nh/2)}{\sin(h/2)}$

6. $\cos a+\cos(a+h)+\cos(a+2h)+\cdots+\cos[a+(n-1)h]$
$=\frac{\cos\{a+[(n-1)/2]h\}\sin(nh/2)}{\sin(h/2)}$

Illustration 2. Find the value of the integral
$$\int_0^{\pi/2}\frac{\sin^2 x}{\sin x+\cos x}dx.$$

Solution: Let $I=\int_0^{\pi/2}\frac{\sin^2 x}{\sin x+\cos x}dx$ (i)

Now, $I=\int_0^{\pi/2}\frac{\sin^2[(\pi/2)-x]}{\sin[(\pi/2)-x]+\cos[(\pi/2)-x]}dx$

[using property 6]

$$I=\int_0^{\pi/2}\frac{\cos^2 x}{\sin x+\cos x}dx \quad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi/2}\frac{\sin^2 x+\cos^2 x}{\sin x+\cos x}dx=\int_0^{\pi/2}\frac{1}{\sin x+\cos x}dx$$
$$=\frac{1}{\sqrt{2}}\int_0^{\pi/2}\frac{1}{\cos\left(x-\frac{\pi}{4}\right)}dx=\frac{1}{\sqrt{2}}\int_0^{\pi/2}\sec\left(x-\frac{\pi}{4}\right)dx$$
$$=\frac{1}{\sqrt{2}}\left[\log\left\{\sec\left(x-\frac{\pi}{4}\right)+\tan\left(x-\frac{\pi}{4}\right)\right\}\right]_0^{\pi/2}$$
$$=\frac{1}{\sqrt{2}}[\log(\sqrt{2}+1)-\log(\sqrt{2}-1)]$$
$$=\frac{1}{\sqrt{2}}\log\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$$
$$=\frac{1}{\sqrt{2}}\log(\sqrt{2}+1)^2=\sqrt{2}\log(\sqrt{2}+1)$$

Illustration 3. Find the value of the integral $\int_{-\pi/2}^{\pi/2}\frac{\cos x dx}{1+e^x}$.

Solution: Let $I=\int_{-\pi/2}^{\pi/2}\frac{\cos x}{1+e^x}dx$
$$=\int_{-\pi/2}^{0}\frac{\cos x}{1+e^x}dx+\int_0^{\pi/2}\frac{\cos x}{1+e^x}dx$$

Putting $x=-t$ in the first integral
$$=-\int_{\pi/2}^{0}\frac{\cos t}{1+e^{-t}}dt+\int_0^{\pi/2}\frac{\cos x}{1+e^x}dx$$
$$=\int_0^{\pi/2}\frac{e^t\cos t}{e^t+1}dt+\int_0^{\pi/2}\frac{\cos x}{1+e^x}dx$$
$$=\int_0^{\pi/2}\frac{e^x\cos x}{1+e^x}dx+\int_0^{\pi/2}\frac{\cos x}{1+e^x}dx$$
$$=\int_0^{\pi/2}\frac{(1+e^x)\cos x}{1+e^x}dx=\int_0^{\pi/2}\cos x dx$$
$$=|\sin x|_0^{\pi/2}=\sin\frac{\pi}{2}-\sin 0=1-0=1$$

Illustration 4. Find the value of the integral $\int_1^2\frac{dx}{x(1+\log x)^2}$.

Solution: $\int_1^2\frac{dx}{x(1+\log x)^2}=\int_1^2(1+\log x)^{-2}\frac{1}{x}dx$
$$=\left|\frac{(1+\log x)^{-1}}{-1}\right|_1^2=-\frac{1}{1+\log 2}+\frac{1}{1+\log 1}$$
$$=1-\frac{1}{1+\log 2}=\frac{\log 2}{1+\log 2}$$

Illustration 5. Find the value of the integral
$$\int_0^{\pi}\left[\cos\left(x+\frac{\pi}{3}\right)\right]dx.$$

Solution: $\int_0^{\pi}\left[\cos\left(x+\frac{\pi}{3}\right)\right]dx$
$$=\int_0^{\pi/6}\left[\cos\left(x+\frac{\pi}{3}\right)\right]dx+\int_{\pi/6}^{\pi}\left[\cos\left(x+\frac{\pi}{3}\right)\right]dx$$
$$=0-\int_{\pi/6}^{\pi}dx=-\left(\pi-\frac{\pi}{6}\right)=\frac{-5\pi}{6}$$

SOME USEFUL REDUCTION FORMULAE

1. If n is a positive integer, then
$$\int_0^{\pi/2}\sin^n x dx=\int_0^{\pi/2}\cos^n x dx$$
$$=\begin{cases}\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdots, & \text{when } n \text{ is odd}\\ \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdots\frac{1}{2}\cdot\frac{\pi}{2}, & \text{when } n \text{ is even}\end{cases}$$

2. $\int_0^{\pi/2} \sin^m x \cos^n x\,dx$

$$= \begin{cases} \frac{m-1}{m+n}\cdot\frac{m-3}{m+n-2}\cdots\frac{2}{3+n}\cdot\frac{1}{1+n}, \\ \qquad \text{if } m \text{ is odd and } n \text{ may be even or odd} \\ \frac{m-1}{m+n}\cdot\frac{m-3}{m+n-2}\cdots\frac{1}{2+n}\cdot\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdots\frac{2}{3} \\ \qquad \text{if } m \text{ is even and } n \text{ is odd} \\ \frac{m-1}{m+n}\cdot\frac{m-3}{m+n-2}\cdots\frac{1}{2+n}\cdot\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdots\frac{1}{2}\cdot\frac{\pi}{2} \\ \qquad \text{if both } m \text{ and } n \text{ are even} \end{cases}$$

These formulae can be expressed as an angle formula as

$$\int_0^{\pi/2} \sin^m x \cos^n x\,dx = \frac{[(m-1)(m-3)\ldots][(n-1)(n-3)\ldots]}{(m+n)(m+n-2)}$$

To be multiplied by $\pi/2$, when m and n are both even integers.

SOLVED PROBLEMS

1. $\int_0^{\pi} \frac{dx}{1+\sin x} =$

(a) 1 (b) 2
(c) 3 (d) 0

Ans. (b)

Solution: $\int_0^{\pi} \frac{dx}{1+\sin x} = \int_0^{\pi} \frac{1-\sin x}{1-\sin^2 x}dx$

$$= \int_0^{\pi} \frac{1-\sin x}{\cos^2 x}dx = \int_0^{\pi} (\sec^2 - \sec x \tan x)dx$$

$$= |\tan x - \sec x|_0^{\pi}$$

$$= (\tan\pi - \sec\pi) - (\tan 0 - \sec 0)$$

$$= (0-(-1)) - (0-1) = 2$$

2. $\int_{\pi/2}^{\pi/2} \sqrt{\frac{1-\cos 2x}{2}}dx$ equals

(a) 1 (b) 2
(c) 1/2 (d) 0

Ans. (b)

Solution: $I = \int_{-\pi/2}^{\pi/2} |\sin x|dx = 2\int_0^{\pi/2} |\sin x|dx$ [Even function]

$$= 2\int_0^{\pi/2} \sin x dx$$

$$= -2[\cos x]_0^{\pi/2} = 2$$

3. If $I = \int_{-\pi}^{\pi} \frac{\sin^2 x}{1+a^x}dx$, $a > 0$, then I equals

(a) π (b) $\pi/2$
(c) $a\pi$ (d) $a\pi/2$

Ans. (b)

Solution: $I = \int_{-\pi}^{\pi} \frac{\sin^2 x}{1+a^x}dx$ (i)

$$I = \int_{-\pi}^{\pi} \frac{(\sin(-x))^2}{1+a^{-x}}dx$$

$$= \int_{-\pi}^{\pi} \frac{a^x \sin^2 x}{1+a^x}dx \quad \text{(ii)}$$

Adding (i) and (ii), we get

$$2I = \int_{-\pi}^{\pi} \sin^2 x dx$$

$$= 2\int_0^{\pi} \sin^2 x dx$$

$$= \int_0^{\pi} (1-\cos 2x)dx$$

$$= \left(x - \frac{\sin 2x}{2}\right)_0^{\pi} = \pi$$

$\Rightarrow I = \pi/2$

4. $\int_0^{100} (x-[x])dx$ is equal to

(a) 50 (b) 100
(c) 200 (d) none of these

Ans. (a)

Solution: We know that $x-[x]$ is a periodic function of period 1.

$$\therefore \int_0^{100} (x-[x])dx = 100\int_0^1 (x-[x])dx$$

$$= 100\int_0^1 x dx - 100\int_0^1 [x]dx$$

$$= 100\left[\frac{x^2}{2}\right]_0^1 - 100\int_0^1 0dx = 50$$

5. $\int_0^{\pi/2} \frac{d\theta}{5+3\cos\theta} =$

(a) $\tan^{-1}\frac{1}{2}$ (b) $\tan^{-1}\frac{1}{3}$

(c) $\frac{1}{2}\tan^{-1}\frac{1}{2}$ (d) $\frac{1}{3}\tan^{-1}\frac{1}{3}$

Ans. (c)

Solution: $\int_0^{\pi/2} \frac{d\theta}{5+3\cos\theta} = \int_0^{\pi/2} \frac{d\theta}{5+3.\frac{1-\tan^2\theta/2}{1+\tan^2\theta/2}}$

$$= \int_0^{\pi/2} \frac{\sec^2\theta/2 d\theta}{8+2\tan^2\theta/2}$$

$$= \int_0^{\pi/2} \frac{\frac{1}{2}\sec^2(\theta/2)d\theta}{4+\tan^2(\theta/2)} = \int_0^1 \frac{dt}{4+t^2}$$

[Put $\tan\frac{\theta}{2} = t \therefore \frac{1}{2}\sec^2\frac{\theta}{2}d\theta = dt$]

$$= \frac{1}{2}\left[\tan^{-1}\frac{t}{2}\right]_0^1$$

$$= \frac{1}{2}\left(\tan^{-1}\frac{1}{2} - \tan^{-1}0\right) = \frac{1}{2}\tan^{-1}\frac{1}{2}$$

Note: For $\int_0^{\pi} \frac{d\theta}{5+3\cos\theta}$, the value

$$= \frac{1}{2}(\tan^{-1}\infty - \tan^{-1}0) = \frac{\pi}{4}$$

6. $\int_{-1}^{3/2} |x\sin\pi x| dx$ equals

(a) $\frac{3\pi+1}{\pi}$ (b) $\frac{3\pi+1}{\pi^2}$

(c) $\frac{3\pi-1}{\pi}$ (d) none of these

Ans. (b)

Solution: $\int_{-1}^{3/2} |x\sin\pi x| dx = \int_{-1}^{1} |x\sin\pi x| dx + \int_1^{3/2} |x\sin\pi x| dx$

$$= 2\int_0^1 |x\sin\pi x| dx + \int_1^{3/2} |x\sin\pi x| dx$$

($\because |x\sin\pi x|$ is an even function)

$$= 2\int_0^1 x\sin\pi x dx - \int_1^{3/2} x\sin\pi x dx$$

[$\because \sin\theta$ is $-$ve when $\pi \le \theta < \frac{3\pi}{2}$]

But $\int x\sin\pi x dx = \left(\frac{-x}{\pi}\cos\pi x + \frac{\sin\pi x}{\pi^2}\right)$

$$\therefore I = 2\left[\frac{-x}{\pi}\cos\pi x + \frac{\sin\pi x}{\pi^2}\right]_0^1 - \left[\frac{-x}{\pi}\cos\pi x + \frac{\sin\pi x}{\pi^2}\right]_1^{3/2}$$

$$= \frac{3}{\pi} + \frac{1}{\pi^2}$$

7. If $I = \int_0^a \sqrt{\frac{a-x}{a+x}} dx, a > 0$, then I equals

(a) $\frac{1}{2}\left(a - \frac{\pi}{2}\right)$ (b) $\frac{a}{2}(\pi - 1)$

(c) $\frac{1}{\sqrt{2}}a(\pi-1)$ (d) $a\left(\frac{\pi}{2} - 1\right)$

Ans. (d)

Solution: We can write

$$I = \int_0^a \frac{a-x}{\sqrt{a^2-x^2}} dx$$

$$= \left[a\sin^{-1}\left(\frac{x}{a}\right) + \sqrt{a^2-x^2}\right]_0^a$$

$$= a\left(\frac{\pi}{2} - 1\right)$$

8. The value of $\int_0^{\pi/2} \frac{\cos^2\theta}{\cos^2\theta + 4\sin^2\theta} d\theta$ is equal to

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{6}$ (d) none of these

Ans. (c)

Solution: Let $I = \int_0^{\pi/2} \frac{\cos^2\theta}{\cos^2\theta + 4\sin^2\theta} d\theta$

$$= \int_0^{\pi/2} \frac{1}{1+4\tan^2\theta} d\theta$$

$$= \int_0^{\pi/2} \frac{\sec^2\theta}{(1+\tan^2\theta)(1+4\tan^2\theta)} d\theta$$

Put $\tan\theta = t \Rightarrow \sec^2\theta d\theta = dt$

$$\therefore \quad I = \int_0^{\infty} \frac{1}{(1+t^2)(1+4t^2)} dt$$

$$I = \int_0^{\infty} \left[\frac{4}{3(1+4t^2)} - \frac{1}{3(1+t^2)}\right] dt$$

$$= \left[\frac{2}{3}\tan^{-1}(2t) - \frac{1}{3}\tan^{-1}(t)\right]_0^{\infty}$$

$$= \frac{\pi}{6}$$

9. $\int_0^{\pi}\sqrt{|\cos x|-|\cos^3 x|}\,dx$ equals

(a) 4/3 (b) 1
(c) 2/3 (d) none of these

Ans. (a)

Solution: $I=\int_0^{\pi}\sqrt{|\cos x|\left(1-\cos^2 x\right)}dx$

$$(\because |\cos^2 x| = \cos^2 x)$$

$$=\int_0^{\pi}\sqrt{|\cos x|}\,|\sin x|\,dx$$

$$=\int_0^{\pi/2}\sqrt{\cos x}\sin x\,dx+\int_{\pi/2}^{\pi}\sqrt{-\cos x}\sin x\,dx$$

Let $t=\cos x$

$\Rightarrow \quad dt=-\sin x\,dx$

$$I=-\int_1^0\sqrt{t}\,dt-\int_0^{-1}\sqrt{-t}$$

$$=\left[\frac{t^{3/2}}{3/2}\right]_1^0+\left[\frac{(-t)^{3/2}}{3/2}\right]_0^{-1}$$

$$=\frac{2}{3}+\frac{2}{3}=\frac{4}{3}$$

10. If $I=\int_{-3}^{2}(|x+1|+|x+2|+|x-1|)dx$, then I equals

(a) $\frac{31}{2}$ (b) $\frac{35}{2}$
(c) $\frac{37}{2}$ (d) $\frac{39}{2}$

Ans. (a)

Solution: We can write

$$I=I_1+I_2+I_3$$

where $I_1=\int_{-3}^{2}|x+1|dx$, etc.

Put $x+1=t$, so that

$$I_1=\int_{-2}^{3}|t|dt=\int_{-2}^{0}(-t)dt+\int_0^3 tdt$$

$$=-\frac{1}{2}t^2\Big]_{-2}^{0}+\frac{1}{2}t^2\Big]_0^3=\frac{13}{2}$$

Similarly, $I_2=I_3=\frac{9}{2}$

Thus, $I=\frac{31}{2}$

11. $\int_{-2}^{2}|x|dx$ is equal to

(a) 0 (b) 1
(c) 2 (d) 4

Ans. (d)

Solution: $\int_{-2}^{2}|x|dx=-\int_{-2}^{0}xdx+\int_0^2 xdx$

$$=-\left[\frac{x^2}{2}\right]_{-2}^{0}+\left[\frac{x^2}{2}\right]_0^2$$

$$=-(0-2)+(2-0)=4$$

12. $\int_{\pi/6}^{\pi/3}\frac{\sin x+\cos x}{\sqrt{\sin 2x}}dx=$

(a) $2\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$ (b) $\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$

(c) $2\sin^{-1}\left(\frac{\sqrt{3}+1}{2}\right)$ (d) $\sin^{-1}\left(\frac{\sqrt{3}+1}{2}\right)$

Ans. (a)

Solution: Put $\sin x-\cos x=z$

$\therefore \qquad (\cos x+\sin x)dx=dz$

Also $1-\sin 2x=z^2 \Rightarrow \sin 2z=\sqrt{1-z^2}$

When $x=\frac{\pi}{6}, z=\frac{1}{2}-\frac{\sqrt{3}}{2}=\frac{1-\sqrt{3}}{2}$

When $x=\frac{\pi}{3}, z=\frac{\sqrt{3}}{2}-\frac{1}{2}$

$\therefore$ Given integral =

$$\int_{\frac{1-\sqrt{3}}{2}}^{\sqrt{3}/2-1/2}\frac{dz}{\sqrt{1-z^2}}$$

$$=\left|\sin^{-1}z\right|_{\frac{1-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}-\frac{1}{2}}$$

$$=\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)-\sin^{-1}\left(\frac{1-\sqrt{3}}{2}\right)$$

$$=\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)+\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$$

$$=2\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$$

13. If $I=\int_0^{\pi/2}\frac{\sqrt{\sec x}}{\sqrt{\operatorname{cosec} x}+\sqrt{\sec x}}dx$, then I equals

(a) $\pi/4$ (b) $\pi/3$
(c) $\pi/2$ (d) $\pi/6$

Ans. (a)

Solution: $\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$ (i)

We get $I=\int_0^{\pi/2}\frac{\sqrt{\operatorname{cosec} x}}{\sqrt{\sec x}+\sqrt{\operatorname{cosec} x}}dx$ (ii)

Adding (i) and (ii), we get

$$2I=\int_0^{\pi/2}dx=\frac{\pi}{2}$$

$\Rightarrow I=\pi/4$

14. $\int_{1/2}^{2}\frac{1}{x}\operatorname{cosec}^{101}\left(x-\frac{1}{x}\right)dx$ is equal to

(a) 1/4 (b) 1
(c) 0 (d) 101/2

Ans. (c)

Solution: $I=\int_{1/2}^{2}\frac{1}{x}\operatorname{cosec}^{101}\left(x-\frac{1}{x}\right)dx$

Let $\frac{1}{x}=t,-\frac{1}{x^2}dx=dt$

$$\therefore\quad I=-\int_{2}^{1/2}\frac{1}{t}\operatorname{cosec}^{101}\left(t-\frac{1}{t}\right)dt$$

$\Rightarrow I=-I \Rightarrow 2I=0$
$\Rightarrow I=0$

15. $\int_0^{5/2}[x]+[-x]dx$ equals

(a) 5/2 (b) –(5/2)
(c) 7/2 (d) none of these

Ans. (b)

Solution: $[x]+[-x]=0,\ x\in I$

$=-1,\ x\notin I$

$$\therefore\quad I=\int_0^1[x]+[-x]dx+\int_1^2[x]+[-x]dx+\int_2^{5/2}[x]+[-x]$$

$$=-\int_0^1 dx-\int_1^2 dx-\int_2^{5/2}dx$$

$$=-\left[1+(2-1)+\left(\frac{5}{2}-2\right)\right]=\frac{-5}{2}$$

16. If $I=\int_{-2}^{2}\left|1-x^4\right|dx$, then I equals

(a) 6 (b) 8
(c) 12 (d) 21

Ans. (c)

Solution: As $\left|1-x^4\right|$ is an even function, so

$$I=2\int_0^2\left|1-x^4\right|dx$$

$$=2\int_0^1(1-x^4)dx+2\int_1^2(x^4-1)dx$$

$$=2\left(x-\frac{x^5}{5}\right)\Big]_0^1+2\left(\frac{x^5}{5}-x\right)\Big]_1^2$$

$$=2\left(1-\frac{1}{5}\right)+2\left(\frac{32}{5}-2-\frac{1}{5}+1\right)=12$$

EXERCISE SET 1

1. $\int_0^{\pi/2}\frac{\cos\theta}{\sqrt{4-\sin^2\theta}}d\theta$ is equal to

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{5}$

2. $\int_0^{\pi/4}\frac{\sin x+\cos x}{1+\cos^2 x}dx=$

(a) $\sqrt{2}+\log(\sqrt{2}+1)$ (b) $\frac{\sqrt{2}+\log(\sqrt{2}+1)}{2}$
(c) $\log(\sqrt{2}+1)$ (d) $\sqrt{2}$

3. $\int_{\pi/4}^{3\pi/4}\tan^{-1}(\tan x)dx$ equals

(a) $\pi^2/16$ (b) $\pi^2/8$
(c) $\pi^2/32$ (d) none of these

4. If $I=\int_0^{\pi/2}\frac{dx}{5+3\sin x}=\lambda\tan^{-1}\left(\frac{1}{2}\right)$, then the value of λ is

(a) 1 (b) 1/2
(c) 1/3 (d) 1/4

5. $\int_0^{\pi/2} \frac{\sin 2t}{\sin^4 t + \cos^4 t} dt =$

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) none of these

6. $\int_0^{\pi/4} \log(1+\tan x)dx =$

(a) $\pi \log 2$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{8}\log 2$ (d) 0

7. $\int_0^{\pi/2} \log \sin x dx$ is equal to

(a) $-\left(\frac{\pi}{2}\right)\log 2$ (b) $\pi \log \frac{1}{2}$

(c) $-\pi \log \frac{1}{2}$ (d) $\frac{\pi}{2}\log 2$

8. If $I = \int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$, then I equals

(a) $\frac{\pi}{12}$ (b) $\frac{\pi}{6}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

9. The value of $\int_0^{\pi/2} \cos x \cdot e^{\sin x} dx$ is

(a) 0 (b) 1

(c) –1 (d) $e-1$

10. $\int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x}+1}$ is equal to

(a) 0 (b) 1

(c) $-\pi/2$ (d) $\pi/2$

11. If $I = \int_{\pi/3}^{2\pi/3} \frac{x}{1+\sin x} dx$, then I equals

(a) 2 (b) $2+\sqrt{3}$

(c) $2-\sqrt{3}$ (d) 0

12. $\int_0^2 |x-1| dx$ is equal to

(a) –1 (b) 2

(c) 3 (d) 1

13. The value of $\lim_{x\to\infty} \frac{\left(\int_0^x e^x dx\right)^2}{\int_0^x e^{2x^2} dx}$ is

(a) 1 (b) 2

(c) 3 (d) 0

14. If $\int_0^1 \frac{\sin t}{1+t} dt = \alpha$, then the value of $I = \int_{4\pi-2}^{4\pi} \frac{\sin(x/2)}{4\pi+2-x} dt$ is

(a) $\alpha/2$ (b) $-\alpha$

(c) $-\alpha/2$ (d) α

15. $\int_3^6 2[x]dx$ is equal to

(a) 12 (b) 30

(c) 18 (d) 24

16. $\int_{-1}^{1} (\tan^{-1} x + \tan^{-1} \frac{1}{x})dx$ equals

(a) π (b) $\pi/2$

(c) 0 (d) none of these

17. If $I = \int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx$, then I equals

(a) $-\sqrt{2}\pi$ (b) $\sqrt{2}\pi$

(c) $\frac{\pi}{\sqrt{2}}$ (d) $-\frac{\pi}{\sqrt{2}}$

18. $\int_1^3 \frac{\cos(\log x)}{x} dx$ is equal to

(a) $\log(\sin 3)$ (b) $\log 3$

(c) $\sin(\log 3)$ (d) $\sin(\log 2)$

19. $\int_0^{\pi} \frac{dx}{1+2\sin^2 x} dx$ equals

(a) 0 (b) $\frac{\pi}{\sqrt{3}}$

(c) $-\frac{\pi}{\sqrt{3}}$ (d) $\frac{2\pi}{\sqrt{3}}$

20. If $n > 1$, and $I = \int_0^{\infty} \frac{dx}{\left(x+\sqrt{1+x^2}\right)^n}$ then I equals

(a) $\frac{n}{n^2-1}$ (b) $\frac{2n}{n^2-1}$

(c) $\frac{n}{2(n^2-1)}$ (d) $\sqrt{n^2-1}$

21. $\int_{-\pi/3}^{\pi/3} \frac{x\sin x}{\cos^2 x}dx$ is equal to

(a) $\frac{\pi}{3} - \log\tan\frac{3\pi}{2}$ (b) $2\left[\frac{2\pi}{3} - \log\tan\frac{5\pi}{12}\right]$

(c) $3\left[\frac{\pi}{2}\log\sin\frac{\pi}{12}\right]$ (d) none of these

22. $\int_0^{2\pi} (\sin x + |\sin x|)dx$ is equal to

(a) 0 (b) 4

(c) 8 (d) 1

23. $\int_0^{\pi/2} \frac{\sin x}{x}dx$ lies in the interval

(a) $\left(0, \frac{2}{\pi}\right)$ (b) $\left(\frac{2}{\pi}, 1\right)$

(c) $\left(\frac{\pi}{2}, \pi\right)$ (d) $\left(1, \frac{\pi}{2}\right)$

24. If $I = \int_8^{15} \frac{dx}{(x-3)\sqrt{x+1}}$, then I equals

(a) $\frac{1}{2}\log\frac{5}{3}$ (b) $2\log\frac{1}{3}$

(c) $\frac{1}{2}\log\frac{1}{5}$ (d) $2\log\frac{5}{3}$

25. $I = \int_0^{\pi/2} \frac{\sin 2x}{1+4\cos^2 x}dx$ is equal to

(a) $\frac{1}{4}\log 2$ (b) $\frac{1}{4}\log 4$

(c) $\frac{1}{4}\log 3$ (d) $\frac{1}{4}\log 5$

26. $\int_0^1 \tan^{-1}\left(\frac{1}{x^2 - x + 1}\right)dx$ is equal to

(a) $\log 2$ (b) $-\log 2$

(c) $\frac{\pi}{2} + \log 2$ (d) $\frac{\pi}{2} - \log 2$

27. $\int_0^{10} |\sin 2\pi x|dx$ equals

(a) $\frac{5}{\pi}$ (b) $\frac{10}{\pi}$

(c) $\frac{20}{\pi}$ (d) none of these

28. $\int_0^{\pi/2} \frac{dx}{1+\tan x}$ is

(a) a multiple of $\pi/4$ (b) a multiple of $\pi/2$

(c) equal to $\pi/4$ (d) a multiple of π

29. $\int_{\pi/6}^{\pi/4} \frac{1}{\sqrt{\cos x \sin^3 x}}dx$ is equal to

(a) 1 (b) 1/3

(c) –2 (d) none of these

30. $\int_0^{\pi} \frac{xdx}{a^2\cos^2 x + b^2\sin^2 x}$ is equal to

(a) $\frac{\pi}{ab}$ (b) $\frac{\pi}{2ab}$

(c) $\frac{\pi^2}{ab}$ (d) $\frac{\pi^2}{2ab}$

31. $\int_0^1 \frac{\log(1+x)}{1+x^2}dx$ is equal to

(a) $\frac{\pi}{8}$ (b) $\frac{\pi}{8}\log 2$

(c) $\log 2$ (d) none of these

32. $\int_0^{\pi/2} \frac{\cos x - \sin x}{1+\cos x\sin x}dx$ is equal to

(a) 0 (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

33. $I = \int_0^{\pi/2} \frac{\cos x dx}{(1+\sin x)(2+\sin x)}$ is equal to

(a) $\log\frac{3}{4}$ (b) $\log\frac{4}{3}$

(c) $\log 4$ (d) $\log 3$

34. $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\cot x}}$ is equal to

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$

(c) $\frac{\pi}{12}$ (d) $\frac{\pi}{2}$

35. $\int_{-\pi/2}^{\pi/2} \sin^2 x\cos^2 x(\sin x + \cos x)dx$ is equal to

(a) $\frac{2}{15}$ (b) $\frac{4}{15}$

(c) $\frac{6}{15}$ (d) $\frac{8}{15}$

36. The value of $\int_0^{\pi/2} \frac{2^{\sin x}}{2^{\sin x}+2^{\cos x}}dx$ is

(a) 2 (b) π

(c) $\pi/4$ (d) 2π

37. $\int_0^{\pi/2} \frac{\cos 2x dx}{(\sin x+\cos x)^2}$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) none of these

38. The value of $\int_0^1 \frac{dx}{x+\sqrt{1-x^2}}$ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{1}{2}$ (d) $\frac{\pi}{4}$

39. $\int_0^{2\pi} \theta \sin^2\theta\cos\theta d\theta$ is equal to

(a) $\frac{\pi}{16}$ (b) $\frac{3\pi}{16}$

(c) $\frac{16\pi}{3}$ (d) 0

40. The value of $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ is

(a) 0 (b) 1

(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

41. The value of $\int_0^1 \frac{2^{x+1}-5^{x-1}}{10^x}dx$ is

(a) $\frac{8}{5}\log_5 e-\frac{1}{10}\log_2 e$ (b) $\frac{8}{5}\log 5-\frac{\log 2}{10}$

(c) $\frac{8}{5\log 5}+\frac{1}{10\log 2}$ (d) $\frac{1}{10\log 2}-\frac{8}{5\log 2}$

42. The value of $\int_{-\pi/4}^{\pi/4} x^3\sin^4 x dx$ is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{8}$ (d) 0

43. $\int_{-1}^{3}(|x-2|+[x])dx$ is equal to

(a) 7 (b) 5

(c) 4 (d) 3

44. $\lim_{x\to 0}\frac{\int_0^{x^2}\tan\sqrt{t}dt}{x^3}$ is equal to

(a) $\frac{2}{3}$ (b) $\frac{3}{2}$

(c) 1 (d) none of these

45. The value of the integral $\int_{-1}^{3}\left(\tan^{-1}\frac{x}{x^2+1}+\tan^{-1}\frac{x^2+1}{x}\right)dx$ is equal to

(a) π (b) 2π

(c) 4π (d) none of these

46. $\int_0^1 \frac{d}{dx}\left[\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right]dx$ is equal to

(a) 0 (b) π

(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

47. The value of $\int_{\pi/6}^{5\pi/6}\sqrt{4-4\sin^2 t}dt$ is

(a) 0 (b) 2

(c) 1 (d) none of these

48. $\int_0^{\infty}\log\left(x+\frac{1}{x}\right)\frac{dx}{1+x^2}$ is equal to

(a) $\pi\log 2$ (b) $-\pi\log 2$

(c) $\left(\frac{\pi}{2}\right)\log 2$ (d) $-\left(\frac{\pi}{2}\right)\log 2$

49. The value of $\int_0^{\pi/2}\log|\tan x+\cot x|dx$ is

(a) $\pi\log 2$ (b) $-\pi\log 2$

(c) $\frac{\pi}{2}\log 2$ (d) $-\frac{\pi}{2}\log 2$

50. $\int_0^{\infty}\frac{x\log x dx}{(1+x^2)^2}$ is equal to

(a) 0 (b) 1

(c) ∞ (d) none of these

51. If $\int_0^{\pi/2} \frac{d\theta}{9\sin^2\theta + 4\cos^2\theta} = K\pi$, then $K =$

(a) $\frac{1}{10}$ (b) $\frac{1}{12}$

(c) $\frac{1}{8}$ (d) $\frac{1}{3}$

52. $\int_{-1}^{1} [x \sin \pi x] dx$ is equal to

(a) 2 (b) –2

(c) 1 (d) 0

53. Let $f(x) = x - [x]$, for every real number x, where $[x]$ is an integral part of x. Then $\int_{-1}^{1} f(x)dx$ is

(a) 1 (b) 2

(c) 0 (d) 1/2

54. The value of the integral $\int_{-\pi/4}^{\pi/4} \sin^{-4} x dx$ is

(a) –8/3 (b) 3/2

(c) 8/3 (d) none of these

55. If $I_k = \int_{-2k\pi}^{2k\pi} |\sin x|[\sin x]dx$, for all $k \in N$, where [.] denotes the greatest integer function, then $\sum_{k=1}^{10} I_k$ is equal to

(a) –110 (b) –440

(c) –330 (d) –220

56. $\int_{-1}^{1} \left(\frac{x^7 - 3x^5 + 7x^3 - x}{\cos^2 x} + \cos^{-1} x \right) dx$ equals

(a) 0 (b) $\pi/2$

(c) π (d) 2π

57. The value of $\int_{-1}^{1} \left(\sqrt{1 + x + x^2} - \sqrt{1 - x + x^2} \right) dx$ is

(a) 0 (b) 1

(c) –1 (d) none of these

58. The value of $\int_1^5 (\sqrt{x + 2\sqrt{x-1}} + \sqrt{x - 2\sqrt{x-1}}) dx$ is

(a) $\frac{8}{3}$ (b) $\frac{16}{3}$

(c) $\frac{32}{3}$ (d) $\frac{64}{3}$

59. The value of $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)}$ is

(a) 0 (b) 1

(c) e (d) e^2

60. $\int_0^1 x(1-x)^{99} dx$ equals

(a) $\frac{1}{10,010}$ (b) $\frac{1}{10,100}$

(c) $\frac{11}{10,100}$ (d) $\frac{11}{10,010}$

EXERCISE SET 2

1. The value of $\int_0^{\frac{3\pi}{2}} \frac{|\tan^{-1}\tan x| - |\sin^{-1}\sin x|}{|\tan^{-1}\tan x| + |\sin^{-1}\sin x|} dx$ is equal to

(a) $\frac{\pi}{2}$ (b) π

(c) $\frac{3\pi}{2}$ (d) none of these

2. The value of $\int_0^2 [x - x^2] dx$ is

(a) $\frac{\sqrt{3}-3}{2}$ (b) $\frac{\sqrt{5}-5}{2}$

(c) $\frac{\sqrt{7}-7}{2}$ (d) none of these

3. The value of the integral $\int_3^4 \frac{[x^2]}{[x^2 - 14x + 49] + [x^2]} dx$, where [.] denotes the greatest integer function, is

(a) 1 (b) $\frac{3}{2}$

(c) $\frac{1}{2}$ (d) none of these

4. $\int_0^1 \log[\sqrt{1-x} + \sqrt{1+x}] dx$ is equal to

(a) $\frac{1}{2}\left[\log 2 - \frac{\pi}{2} + 1\right]$ (b) $\frac{1}{2}\left[\log 2 + \frac{\pi}{2} + 1\right]$

(c) $\frac{1}{2}\left[\log 2 + \frac{\pi}{2} - 1\right]$ (d) none of these

5. If $I_1 = \int_0^{\pi/4} \sin^2 x dx$ and $I_2 = \int_0^{\pi/4} \cos^2 x dx$, then

(a) $I_1 = I_2$ (b) $I_1 < I_2$

(c) $I_1 > I_2$ (d) $I_2 = I_1 + \frac{\pi}{4}$

6. Let $a_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin x} dx$. Then $a_2 - a_1, a_3 - a_2, a_4 - a_3$ are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

7. $\int_0^{\pi/4} \frac{\sin x + \cos x}{3 + \sin 2x} dx$ is equal to

(a) $-\frac{1}{4}\log 3$ (b) $\frac{1}{4}\log 3$

(c) $-\frac{1}{3}\log 4$ (d) none of these

8. If $\int_{-1}^{4} f(x)dx = 4$ and $\int_2^4 [3 - f(x)]dx = 7$, then the value of $\int_{-1}^{2} f(x)dx$ is

(a) –2 (b) 3

(c) 4 (d) 5

9. $\int_{-\pi/2}^{\pi/2} \sqrt{\frac{1-\cos 2x}{2}} dx$ is

(a) 0 (b) 2

(c) 1/2 (d) none of these

10. If $f(a+b-x) = f(x)$, then $\int_a^b xf(x)dx$ is equal to

(a) $\frac{a+b}{2}\int_a^b f(b-a)dx$ (b) $\frac{a+b}{2}\int_a^b f(x)dx$

(c) $\frac{b-a}{2}\int_a^b f(x)dx$ (d) none of these

11. If $f(x)$ and $g(x)$, $x \in \mathbb{R}$ are continuous functions, then the value of the integral $\int_{-\pi/2}^{\pi/2} \{[f(x) + f(-x)][g(x) - g(-x)]\} dx$ is

(a) π (b) $\pi/2$

(c) 1 (d) 0

12. $\int_{-1}^{1} \frac{|x+2|}{x+2} dx$ is

(a) 1 (b) 2

(c) 0 (d) –1

13. The value of the integral $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x} + \sqrt{\tan x}} dx$ is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) π (d) none of these

14. $\int_{-1}^{1} \sin^3 x \cos^2 x dx$ is equal to

(a) –1 (b) 1

(c) 0 (d) none of these

15. If $n > 1$, then $\int_1^{\infty} \frac{1}{\left(x + \sqrt{1+x^2}\right)^n} dx$ must be equal to

(a) $\frac{1}{n^2-1}$ (b) $\frac{-1}{n^2-1}$

(c) $\frac{2}{n^2-1}$ (d) $\frac{1}{n^2+1}$

16. $\int_0^{\pi/2} |\sin x - \cos x| dx$ is equal to

(a) 0 (b) $2(\sqrt{2}-1)$

(c) $\sqrt{2}-1$ (d) $2(\sqrt{2}+1)$

17. The value of $\int_{-3}^{3} (ax^5 + bx^3 + cx + k)dx$, where a, b, c and k are constants, depends only on

(a) a and k (b) a and b

(c) a, b and c (d) k

18. $\int_{-1}^{1} \frac{\sin x + x^2}{3 - |x|} dx$ is equal to

(a) $\int_0^1 \frac{\sin x + x^2}{3-|x|} dx$ (b) 0

(c) $\int_0^1 \frac{x^2}{3-|x|} dx$ (d) $\int_0^1 \frac{\sin x}{3-|x|} dx$

19. The value of the integral $\int_{-\pi}^{\pi} \sin mx \sin nx dx$, for $m \neq n$ ($m, n \in I$), is

(a) 0 (b) π

(c) $\pi/2$ (d) 2π

20. $3a\int_0^1 \left(\frac{ax-1}{a-1}\right)^2 dx$ is equal to

(a) $a - 1 + (a-1)^2$ (b) $a + a^{-2}$

(c) $a - a^2$ (d) $a^2 + \frac{1}{a^2}$

21. $\int_0^{2\pi} |\sin^3 x| dx$ is equal to

(a) 0 (b) π

(c) $\frac{8}{3}$ (d) $\frac{3}{8}$

22. $\int_0^{\pi} \frac{1}{a^2 - 2a\cos x + 1} dx, a > 1$ is

(a) $\frac{\pi}{a^2 - 1}$ (b) $\frac{2\pi}{a^2 - 1}$

(c) $\frac{\pi}{1 - a^2}$ (d) none of these

23. If $\int_0^1 e^{x^2}(x - \alpha)dx = 0$, then

(a) $1 < \alpha < 2$ (b) $\alpha < 0$

(c) $0 < \alpha < 1$ (d) $\alpha = 0$

24. $\int_1^3 [x^2] dx$ is

(a) 3 (b) 8/3

(c) $5 - \sqrt{3} - \sqrt{2}$ (d) none of these

25. The value of the integral $\int_0^{\pi/2} (\sin^{100} x - \cos^{100} x)dx$ is

(a) $\frac{1}{100}$ (b) $\frac{100!}{(100)^{100}}$

(c) $\frac{\pi}{100}$ (d) 0

26. $\int_0^2 x^2 \left[\frac{x}{2}\right] dx$ is

(a) 0 (b) 1

(c) 1/2 (d) none of these

27. The value of $\int_0^{\infty} \frac{dx}{(a^2 + x^2)}$ is equal to

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{2a}$

(c) $\frac{\pi}{a}$ (d) $\frac{1}{2a}$

28. $\int_{-\pi/2}^{\pi/2} \sin^2 x \log\left(\frac{2 - \sin x}{2 + \sin x}\right) dx$ is

(a) 1 (b) 2

(c) 0 (d) none of these

29. If $a < \int_0^{2\pi} \frac{1}{10 + 3\cos x} dx < b$, then the ordered pair (a, b) is

(a) $\left(\frac{2\pi}{7}, \frac{2\pi}{3}\right)$ (b) $\left(\frac{2\pi}{13}, \frac{2\pi}{7}\right)$

(c) $\left(\frac{\pi}{10}, \frac{2\pi}{13}\right)$ (d) none of these

30. All the values of a for which $\int_1^2 [a^2 + (4 - 4a)x + 4x^3] dx \le 12$ are given by

(a) $a = 3$ (b) $a \le 4$

(c) $0 \le a < 3$ (d) none of these

31. $\int_0^a \sqrt{a^2 - x^2} dx$ is equal to

(a) πa^2 (b) $\frac{1}{2}\pi a^2$

(c) $\frac{1}{3}\pi a^2$ (d) $\frac{1}{4}\pi a^2$

32. The value of $\int_1^{16} \tan^{-1}\sqrt{\sqrt{x} - 1} dx$ is

(a) $\frac{16\pi}{3} + 2\sqrt{3}$ (b) $\frac{4}{3}\pi - 2\sqrt{3}$

(c) $\frac{4}{3}\pi + 2\sqrt{3}$ (d) $\frac{16}{3}\pi - 2\sqrt{3}$

33. $\int_{\pi/4}^{\pi/2} \text{cosec}^2 x dx$ is equal to

(a) -1 (b) 1

(c) 0 (d) $\frac{1}{2}$

34. $\lim_{n \to \infty} \frac{\sqrt{1} + 2\sqrt{2} + 3\sqrt{3} + \cdots + n\sqrt{n}}{n^{5/2}}$ is

(a) $\int_0^1 x\sqrt{x} dx$ (b) 0

(c) 1 (d) none of these

35. If $f(x) = \tan x - \tan^3 x + \tan^5 x - \cdots \infty$ with $0 < x < \frac{\pi}{4}$, then $\int_0^{\pi/4} f(x)dx$ is equal to

(a) 1 (b) 0

(c) $\frac{1}{4}$ (d) $\frac{1}{2}$

36. $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{\pi^2 - 3\pi x + 3x^2} dx$ is equal to

(a) $\frac{\pi^2}{16}$ (b) $\frac{\pi^2}{8}$

(c) $\frac{\pi^2}{4}$ (d) $\frac{\pi^2}{32}$

37. If f is a continuous function, then

(a) $\int_{-2}^{2} f(x)dx = \int_{0}^{2} [f(x) - f(-x)]dx$

(b) $\int_{-3}^{5} 2f(x)dx = \int_{-6}^{10} f(x-1)dx$

(c) $\int_{-3}^{5} f(x)dx = \int_{-4}^{4} f(x-1)dx$

(d) $\int_{-3}^{5} f(x)dx = \int_{-2}^{6} f(x-1)dx$

38. If $f : D \to R$ be such that

$f(x) = \sqrt{\sin x(\cos x)} + \ln(-2\cos^2 x + 3\cos x - 1)$,

then $\int_{x_1}^{x_2} \left[\cos x - \frac{1}{2}\right] dx$, where $x_1, x_2 \in D$ and [.] denotes the greatest integer function, is equal to

(a) 0 (b) $\frac{1}{2}(x_2, x_1)$

(c) $x_1 - x_2$ (d) $\frac{1}{2}(x_1 - x_2)$

39. $\int_{-1}^{1} \frac{\cosh}{1 + e^{2x}} dx$ is equal to

(a) 0 (b) 1

(c) $\frac{e^2 - 1}{2e}$ (d) $\frac{e^2 + 2}{2e}$

40. The value of the integral $\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x} dx$ is

(a) 0 (b) 2/3

(c) 4/3 (d) none of these

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement I is false, Statement II is true.

(b) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

(c) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement II.

(d) Statement I is true, Statement II is false.

41. Statement I: $\int_0^{\pi/2} (\sin^4 x + \cos^4 x)dx$ lies in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{8}\right)$.

Statement II: $\sin^4 x + \cos^4 x$ is periodic with period $(\pi/2)$.

42. Statement I: $\int_{\pi/6}^{\pi/3} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx = \frac{\pi}{12}$

Statement II: $\int_{\pi/6}^{\pi/3} \frac{f(x)}{f(x) + f\left(\frac{\pi}{2} - x\right)} dx = \frac{\pi}{12}$

43. Statement I: $\int_0^1 e^{-x} \cos^2 x dx < \int_0^1 e^{-x^2} \cos^2 x dx$

Statement II: $\int_a^b f(x)dx \geq \int_a^b g(x)dx \forall f(x) \geq g(x)$

44. Statement I: $\int_{-1}^{1} \frac{|x|}{x} dx = 2$

Statement II: $\frac{|x|}{x}$ is not defined at $x = 0$

Previous Years' Questions

45. $\int_0^{10/\pi} |\sin x| dx$ is

(a) 20 (b) 8

(c) 10 (d) 18 **[AIEEE 2002]**

46. If $I_n = \int_0^{\pi/4} \tan^n x dx$, then $\lim_{n\to\infty} n[I_n + I_{n+2}]$ equals

(a) $\frac{1}{2}$ (b) 1

(c) ∞ (d) zero **[AIEEE 2002]**

47. $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ equals

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) zero (d) 1 **[AIEEE 2002]**

48. If $f(a+b-x)=f(x)$, then $\int_a^b xf(x)$ is equal to

(a) $\frac{a+b}{2}\int_a^b f(b-x)dx$ (b) $\frac{a+b}{2}\int_a^b f(x)dx$

(c) $\frac{b-a}{2}\int_a^b f(x)dx$ (d) $\frac{a+b}{2}\int_a^b f(a+b+x)dx$

[AIEEE 2003]

49. $\lim_{n\to\infty}\frac{1+2^4+3^4+\cdots+n^4}{n^5}-\lim_{n\to\infty}\frac{1+2^3+3^3+\cdots+n^3}{n^5}$ is

(a) $\frac{1}{30}$ (b) 0

(c) $\frac{1}{4}$ (d) $\frac{1}{5}$ **[AIEEE 2003]**

50. The value of the integral $I=\int_0^1 x(1-x)^n dx$ is

(a) $\frac{1}{n+2}$ (b) $\frac{1}{n+1}-\frac{1}{n+2}$

(c) $\frac{1}{n+1}+\frac{1}{n+2}$ (d) $\frac{1}{n+1}$

[AIEEE 2003]

51. Let $\frac{d}{dx}F(x)=\left(\frac{e^{\sin x}}{x}\right), x>0$. If $\int_1^4 \frac{3}{x}e^{\sin x^3}dx=F(k)-F(1)$, then one of the possible values of k is

(a) 16 (b) 63

(c) 64 (d) 15 **[AIEEE 2003]**

52. Let $f(x)$ be a function satisfying $f'(x)=f(x)$ with $f(0)=1$ and $g(x)$ be a function that satisfies $f(x)+g(x)=x^2$. Then the value of the integral $\int_0^1 f(x)g(x)dx$ is

(a) $e+\frac{e^2}{2}-\frac{3}{2}$ (b) $e-\frac{e^2}{2}-\frac{3}{2}$

(c) $e+\frac{e^2}{2}+\frac{5}{2}$ (d) $e-\frac{e^2}{2}-\frac{5}{2}$

[AIEEE 2003]

53. If $f(x)=\frac{e^x}{1+e^x}, I_1=\int_{f(-a)}^{f(a)} xg[x(1-x)]dx$ and $I_2=\int_{f(-a)}^{f(a)} g[x(1-x)]dx$, then the value of $\frac{I_2}{I_1}$ is

(a) 2 (b) –3

(c) –1 (d) –1 **[AIEEE 2004]**

54. The value of $\int_{-2}^{3}|1-x^2|dx$ is

(a) $\frac{28}{3}$ (b) $\frac{14}{3}$

(c) $\frac{7}{3}$ (d) $\frac{1}{3}$ **[AIEEE 2004]**

55. The value of $I=\int_0^{\pi/2}\frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}}dx$ is

(a) 0 (b) 1

(c) 2 (d) 3 **[AIEEE 2004]**

56. If the variables x and y are connected by the relation $x=\int_1^y \frac{dz}{\sqrt{1+6z^3}}$, then $\frac{d^2y}{dx^2}$ is proportional to

(a) y (b) y^2

(c) y^3 (d) none of these

[AIEEE 2004]

57. If $I_1=\int_0^1 2^{x^2}dx$, $I_2=\int_0^1 2^{x^3}dx$, $I_3=\int_1^2 2^{x^2}dx$, $I_4=\int_0^2 2^{x^3}dx$, then

(a) $I_1>I_2$ (b) $I_2>I_1$

(c) $I_3>I_4$ (d) $I_3=I_4$

[AIEEE 2005]

58. Let F: $\mathbb{R}\to\mathbb{R}$ be a differentiable function having $f(2)=6f'(2)=\frac{1}{48}$. Then $\lim_{x\to 2}\int_6^{f(x)}\frac{4t^3}{x-2}dt$ equals

(a) 36 (b) 24

(c) 18 (d) 12 **[AIEEE 2005]**

59. The value of $\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}dx, a>0$ is

(a) $\frac{\pi}{2}$ (b) $a\pi$

(c) 2π (d) $\frac{\pi}{a}$ **[AIEEE 2005]**

60. $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}+\cdots+\frac{n}{n^2}\sec^2 1\right]$ equals

(a) $\frac{1}{2}\tan 1$ (b) $\tan 1$

(c) $\frac{1}{2}\text{cosec}\,1$ (d) $\frac{1}{2}\sec 1$

[AIEEE 2005]

61. The value of the integral $\int_3^6\frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}}dx$ is

(a) 1/2 (b) 3/2

(c) 2 (d) 1 **[AIEEE 2006]**

62. $\int_0^{\pi} xf(\sin x)dx$ is equal to

(a) $\pi\int_0^{\pi} xf(\cos x)dx$ (b) $\pi\int_0^{\pi} f(\sin x)dx$

(c) $\frac{\pi}{2}\int_0^{\pi/2} f(\sin x)dx$ (d) $\pi\int_0^{\pi/2} f(\cos x)dx$

[AIEEE 2006]

63. $\int_{-3\pi/2}^{-\pi/2}[(x+\pi)^3+\cos^2(x+3\pi)]dx$ is equal to

(a) $\left(\frac{\pi^4}{32}\right)+\left(\frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2}\right)$

(c) $\left(\frac{\pi}{4}\right)-1$ (d) $\frac{\pi^4}{32}$ **[AIEEE 2006]**

64. The value of $\int_1^a [x]f'(x)dx, a>1$, where $[x]$ denotes the greatest integer not exceeding x, is

(a) $[a]f(a)-\{f(1)+f(2)+\cdots+f([a])\}$

(b) $[a]f([a])-\{f(1)+f(2)+\cdots+f(a)\}$

(c) $af([a])-\{f(1)+f(2)+\cdots+f(a)\}$

(d) $af(a)-\{f(1)+f(2)+\cdots+f([a])\}$

[AIEEE 2006]

65. Let $F[x]=f(x)+f\left(\frac{1}{x}\right)$, where $f(x)=\int_1^x \frac{\log t}{1+t}dt$.

Then $F(e)$ equals

(a) 0 (b) 1

(c) z (d) 1/2 **[AIEEE 2007]**

66. The solution for x of the equation $\int_{\sqrt{2}}^{x} \frac{dt}{t\sqrt{t^2-1}}=\frac{\pi}{2}$ is

(a) π (b) $\sqrt{3}/2$

(c) $2\sqrt{2}$ (d) none of these

[AIEEE 2007]

67. Let $I=\int_0^1 \frac{\sin x}{\sqrt{x}}dx$ and $J=\int_0^1 \frac{\cos x}{\sqrt{x}}dx$. Then which one of the following is true?

(a) $I>(2/3)$ and $J<2$ (b) $I>(2/3)$ and $J>2$

(c) $I<(2/3)$ and $J<2$ (d) $I<(2/3)$ and $J>2$

[AIEEE 2008]

68. $\int_0^{\pi}[\cot x]dx$, [.] denotes the greatest integer function, is equal to

(a) $\pi/2$ (b) 1

(c) -1 (d) $-\pi/2$ **[AIEEE 2009]**

69. Let $p(x)$ be a function defined on $\mathbb{R}$ such that $\lim_{x\to\infty}\frac{f(3x)}{f(x)}=1, p'(x)=p'(1-x)$ for all $x\in[0,1], p(0)=1$ and $p(1) = 41$. Then $\int_0^1 p(x)\,dx$ equals

(a) $\sqrt{41}$ (b) 21

(c) 41 (d) 42 **[AIEEE 2010]**

70. If $g(x)\int_0^x \cos 4t\,dt$, then $g(x+\pi)$ equals

(a) $g(x)/g(\pi)$ (b) $g(x)+g(\pi)$

(c) $g(x)-g(\pi)$ (d) $g(x)\cdot g(\pi)$

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (b)	**2.** (b)	**3.** (d)	**4.** (b)	**5.** (c)	**6.** (c)	**7.** (a)	**8.** (a)	**9.** (d)	**10.** (d)
11. (c)	**12.** (d)	**13.** (d)	**14.** (b)	**15.** (d)	**16.** (c)	**17.** (b)	**18.** (c)	**19.** (b)	**20.** (a)
21. (b)	**22.** (b)	**23.** (d)	**24.** (a)	**25.** (d)	**26.** (d)	**27.** (c)	**28.** (a) and (c)		**29.** (d)
30. (d)	**31.** (b)	**32.** (a)	**33.** (b)	**34.** (c)	**35.** (b)	**36.** (c)	**37.** (a)	**38.** (d)	**39.** (d)
40. (d)	**41.** (a)	**42.** (d)	**43.** (a)	**44.** (d)	**45.** (b)	**46.** (c)	**47.** (b)	**48.** (a)	**49.** (a)
50. (a)	**51.** (b)	**52.** (d)	**53.** (a)	**54.** (a)	**55.** (d)	**56.** (c)	**57.** (a)	**58.** (c)	**59.** (b)
60. (b)									

Exercise Set 2

1. (d)	**2.** (b)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (d)	**9.** (b)	**10.** (b)
11. (d)	**12.** (b)	**13.** (b)	**14.** (c)	**15.** (a)	**16.** (b)	**17.** (d)	**18.** (c)	**19.** (a)	**20.** (a)
21. (c)	**22.** (a)	**23.** (c)	**24.** (c)	**25.** (d)	**26.** (a)	**27.** (b)	**28.** (c)	**29.** (b)	**30.** (a)
31. (d)	**32.** (d)	**33.** (b)	**34.** (a)	**35.** (c)	**36.** (d)	**37.** (b)	**38.** (a)	**39.** (c)	**40.** (c)
41. (c)	**42.** (b)	**43.** (b)	**44.** (a)	**45.** (a)	**46.** (b)	**47.** (a)	**48.** (b)	**49.** (d)	**50.** (b)
51. (c)	**52.** (b)	**53.** (a)	**54.** (a)	**55.** (c)	**56.** (b)	**57.** (a)	**58.** (c)	**59.** (a)	**60.** (a)
61. (b)	**62.** (d)	**63.** (b)	**64.** (a)	**65.** (d)	**66.** (d)	**67.** (c)	**68.** (d)	**69.** (b)	**70.** (b or c)

HINTS AND SOLUTIONS—EXERCISE SET 1

1. $\int_0^{\pi/2} \frac{\cos\theta}{\sqrt{4-\sin^2\theta}} d\theta = \left[\sin^{-1}\left(\frac{\sin\theta}{2}\right)\right]_0^{\pi/2}$

$= \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

2. Integral $= \int_0^{\pi/4} \frac{\sin x + \cos x}{2\cos^2 x} dx$

$= \frac{1}{2}\int_0^{\pi/4} \sec x \tan x dx + \frac{1}{2}\int_0^{\pi/4} \sec x dx$

$= \frac{1}{2}|\sec x|_0^{\pi/4} + \frac{1}{2}|\log \sec x + \tan x|_0^{\pi/4}$

$= \frac{1}{2}\left[\sec\frac{\pi}{4} - \sin 0\right] + \frac{1}{2}$

$\left[\log\left|\sec\frac{\pi}{4} + \tan\frac{\pi}{4}\right| - \log|\sec 0 + \tan 0|\right]$

$= \frac{1}{2}(\sqrt{2}) + \frac{1}{2}[\log(\sqrt{2}+1) - \log(1)]$

$= \frac{\sqrt{2}}{2} + \frac{1}{2}\log(\sqrt{2}+1) \quad = \frac{\sqrt{2} + \log(\sqrt{2}+1)}{2}$

3. $I = \int_{\pi/4}^{\pi/2} \tan^{-1}(\tan x) dx + \int_{\pi/2}^{3\pi/4} \tan^{-1}(\tan x) dx$

$= \int_{\pi/4}^{\pi/2} x dx + \int_{\pi/2}^{3\pi/4} (x - \pi) dx \quad = \frac{3\pi^2}{32} - \frac{3\pi^2}{32} = 0$

4. $I = \int_0^{\pi/2} \frac{dx}{5 + 3\cos x}$

Put $\tan(x/2) = t$, so that

$dx = \frac{2dt}{1+t^2}$ and $5 + 3\cos x = \frac{8 + 2t^2}{1+t^2}$

$\therefore \quad I = \int_0^1 \frac{dt}{4+t^2} = \frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right)\Big]_0^1 \quad = \frac{1}{2}\tan^{-1}\left(\frac{1}{2}\right)$

Thus, $\lambda = \frac{1}{2}$

5. $\int_0^{\pi/2} \frac{\sin 2t}{\sin^4 t + \cos^4 t} dt = \int_0^{\pi/2} \frac{2\sin t \cos t dt}{\sin^4 t + \cos^4 t}$

$= \int_0^{\pi/2} \frac{2 \cdot \frac{\sin t}{\cos t} \cdot \frac{1}{\cos^2 t} dt}{1 + \tan^4 t} = \int_0^{\pi/2} \frac{2\tan t \cdot \sec^2 t dt}{1 + \tan^4 t}$

Put $\tan^2 t = z \quad \therefore \quad 2\tan t \cdot \sec^2 t dt = dz$

When $t = 0, z = 0$, when $t = \frac{\pi}{2}, z = \infty$

$\therefore$ Given integral $= \int_0^\infty \frac{dz}{1+z^2} = |\tan^{-1} z|_0^\infty$

$= \tan^{-1}\infty - \tan^{-1} 0 = \frac{\pi}{2} - 0 = \frac{\pi}{2}$

6. $I = \int_0^{\pi/4} \log(1 + \tan x) dx$

Since $\int_0^a f(x)dx = \int_0^a f(a-x)dx$, therefore

$I = \int_0^{\pi/4} \log\left[1 + \tan\left(\frac{\pi}{4} - x\right)\right] dx$

$= \int_0^{\pi/4} \log\left(1 + \frac{1 - \tan x}{1 + \tan x}\right) dx$

$I = \int_0^{\pi/4} \log\left(\frac{2}{1 + \tan x}\right) dx$

$= \int_0^{\pi/4} \log 2 dx - \int_0^{\pi/4} \log(1 + \tan x) dx$

$\Rightarrow \quad 2I = \log 2 \cdot \frac{\pi}{4} \quad \Rightarrow \quad I = \frac{\pi}{8}\log 2$

7. Let $I = \int_0^{\pi/2} \log\sin x dx$ (i)

$I = \int_0^{\pi/2} \log\sin\left(\frac{\pi}{2} - x\right) dx = \int_0^{\pi/2} \log\cos x dx$ (ii)

On adding Eqs. (i) and (ii), we get

$2I = \int_0^{\pi/2} \log\sin x\cos x dx$

$= \int_0^{\pi/2} \log\sin 2x dx - \int_0^{\pi/2} \log 2 dx$

$= \frac{1}{2}\int_0^{\pi} \log\sin t dt - \frac{\pi}{2}\log 2$

$= \frac{1}{2} \cdot 2\int_0^{\pi/2} \log\sin t dt - \frac{\pi}{2}\log 2$

$\Rightarrow \quad 2I = I - \frac{\pi}{2}\log 2 \Rightarrow I = -\frac{\pi}{2}\log 2$

8. We can write

$$I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx \qquad \text{(i)}$$

Using $\int_a^b f(x)dx=\int_a^b f(a+b-x)\,dx$, we can write

$$I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos x(\pi/2-x)}}{\sqrt{\cos x(\pi/2-x)}+\sqrt{\sin x(\pi/2-x)}}\,dx$$

$$=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx \qquad \text{(ii)}$$

Adding Eqs. (i) and (ii), we get

$$2I=\int_{\pi/6}^{\pi/3}dx=x]_{\pi/6}^{\pi/3}=\frac{\pi}{6}$$

$$\Rightarrow\ I=\frac{\pi}{12}$$

9. $\int_0^{\pi/2}\cos x\cdot e^{\sin x}dx=\left|e^{\sin x}\right|_0^{\pi/2}=e^1-e^0=e-1$

10. $I=\int_{-\pi/2}^{\pi/2}\frac{dx}{e^{\sin x}+1}$ (i)

$$\therefore\qquad I=\int_{-\pi/2}^{\pi/2}\frac{e^{\sin x}dx}{e^{\sin x}+1} \qquad \text{(ii)}$$

$$\left(\text{using }\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right)$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_{-\pi/2}^{\pi/2}1dx=\pi\ \Rightarrow\ I=\frac{\pi}{2}$$

11. $I=\int_{\pi/3}^{2\pi/3}\frac{dx}{1+\sin x}$ (i)

Using $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$, we can write

$$I=\int_{\pi/3}^{2\pi/3}\frac{\pi-x}{1+\sin(\pi-x)}dx \qquad \text{(ii)}$$

Adding Eqs. (i) and (ii), we get

$$2I=\int_{\pi/3}^{2\pi/3}\frac{dx}{1+\sin x}$$

Put $\tan(x/2)=t$, so that

$$2I=\int_{1/\sqrt{3}}^{\sqrt{3}}\frac{2dt/(1+t^2)}{1+2t/(1+t^2)}$$

$$\Rightarrow\ I=\int_{1/\sqrt{3}}^{\sqrt{3}}\frac{dt}{(1+t^2)}=\frac{1}{1+t^2}\Big]_{1/\sqrt{3}}^{\sqrt{3}}$$

$$=-\frac{1}{1+\sqrt{3}}+\frac{\sqrt{3}}{\sqrt{3}+1}=\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)\left(\frac{\sqrt{3}-1}{\sqrt{3}-1}\right)$$

$$=2-\sqrt{3}$$

12. $\int_0^2|x-1|dx=\int_0^1|x-1|dx+\int_1^2|x-1|dx$

$$=\int_0^1(1-x)dx+\int_1^2(x-1)dx$$

$$=\left|x-\frac{x^2}{2}\right|_0^1+\left|\frac{x^2}{2}-x\right|_1^2$$

$$=1-\frac{1}{2}+(2-2)-\left(\frac{1}{2}-1\right)=\frac{1}{2}+\frac{1}{2}=1$$

13. $\lim_{x\to\infty}\frac{\left(\int_0^x e^x dx\right)^2}{\int_0^x e^{2x^2}dx}=\lim_{x\to\infty}\frac{(e^x-1)^2}{\int_0^x e^{2x^2}dx}$

$$=\lim_{x\to\infty}\frac{2(e^x-1)e^x}{e^{2x^2}} \qquad \text{[using L'Hôpital's rule]}$$

$$=2\lim_{x\to\infty}\frac{e^x-1}{e^{2x^2-x}}$$

$$=2\lim_{x\to\infty}\frac{e^x}{e^{2x^2-x}(4x-1)} \qquad \text{[using L'Hôpital's rule]}$$

$$=2\lim_{x\to\infty}\frac{1}{e^{2x^2-2x}(4x-1)}=0$$

14. Put $\frac{x}{2}=\theta$, so that

$$I=\int_{2\pi-1}^{2\pi}\frac{\sin\theta}{4\pi+2-2\theta}(2)d\theta=\int_{2\pi-1}^{2\pi}\frac{\sin\theta}{2\pi+1-\theta}d\theta$$

Now, put $2\pi-\theta=t$, so that

$$I=\int_1^0\frac{\sin(2\pi-t)}{1+t}(-1)dt$$

$$=-\int_0^1\frac{\sin t}{1+t}dt=-\alpha$$

15. $\int_3^6 2[x]dx=2\left[\int_3^4[x]dx+\int_4^5[x]dx+\int_5^6[x]dx\right]$

$$=2\left[\int_3^4 3dx+\int_4^5 4dx+\int_5^6 5dx\right]$$

$$=2[3(1)+4(1)+5(1)]=24$$

16. Since $\tan^{-1}\frac{1}{x}=\cot^{-1}x, x>0=\cot^{-1}x-\pi, x<0$

$$\therefore\quad I=\int_{-1}^{0}(\tan^{-1}x+\cot^{-1}x-\pi)dx+\int_{0}^{1}(\tan^{-1}x+\cot^{-1}x)dx$$

$$=-\frac{\pi}{2}\int_{-1}^{0}dx+\frac{\pi}{2}\int_{0}^{1}dx \quad \left(\because\ \tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}\right)$$

$$=\frac{-\pi}{2}+\frac{\pi}{2}=0$$

17. We can write

$$I=I_1+I_2$$

where $I_1=\int_{-\pi}^{\pi}\frac{2x}{1+\cos^2 x}dx=0$

and $I_2=4\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}$

$$=4\int_0^{\pi}\frac{(\pi-x)\sin(\pi-x)}{1+(\cos(\pi-x))^2}dx$$

$$=4\pi\int_0^{\pi}\frac{dx}{1+\cos^2 x}-I_2$$

$$\Rightarrow\quad I_2=2\pi\int_0^{\pi}\frac{\operatorname{cosec}^2 x}{\operatorname{cosec}^2 x+\cot^2 x}dx$$

$$=2\pi\int_0^{\pi}\frac{\operatorname{cosec}^2 x dx}{1+2\cot^2 x}$$

$$=\frac{-2\pi}{\sqrt{2}}\tan^{-1}\left(\sqrt{2}\cot x\right)\Big]_0^{\pi}$$

$$=-\sqrt{2}\left(-\frac{\pi}{2}-\frac{\pi}{2}\right)=\sqrt{2}\pi$$

18. Put $\log x=z\ \therefore\frac{1}{x}dx=dz$

When $x=1, z=\log 1=0\ \ x=3, z=\log 3$

$\therefore$ Given integral $=\int_0^{\log 3}\cos(z)dz$

$$=\sin z=\sin(\log 3)$$

19. $I=\int_0^{\pi}\frac{dx}{1+2\sin^2 x}$

$$=\int_0^{\pi}\frac{\sec^2 x}{\sec^2 x+2\tan^2 x}dx$$

$$=\int_0^{\pi/2}\frac{\sec^2 x}{1+3\tan^2 x}dx+\int_{\pi/2}^{\pi}\frac{\sec^2 x}{1+3\tan^2 x}dx$$

[$\because \tan x$ and $\sec x$ are discontinuous at $x=(\pi/2)$]

Let $\tan x=t$,

$\Rightarrow \sec^2 x dx=dt$

$$I=\int_0^{\infty}\frac{dt}{1+3t^2}+\int_{-\infty}^{0}\frac{dt}{1+3t^2}$$

$$=\frac{1}{\sqrt{3}}\left[\tan^{-1}\sqrt{3}t\right]_0^{\infty}+\frac{1}{\sqrt{3}}\left[\tan^{-1}\sqrt{3}t\right]_{-\infty}^{0}$$

$$=\frac{1}{\sqrt{3}}\left(\frac{\pi}{2}-0\right)+\frac{1}{\sqrt{3}}\left[0-\left(\frac{-\pi}{2}\right)\right]=\frac{\pi}{\sqrt{3}}$$

20. As x and $\sqrt{1+x^2}$ are involved, put

$$\sqrt{1+x^2}=t-x \quad \text{or} \quad 1+x^2=t^2-2tx+x^2$$

$$\Rightarrow\quad x=\frac{1}{2}\left(t-\frac{1}{t}\right)\quad\Rightarrow\quad dx=\frac{1}{2}\left(t-\frac{1}{t^2}\right)$$

$$\therefore\quad I=\int_1^{\infty}\frac{1}{2}\left(1-\frac{1}{t^2}\right)\frac{dt}{t^n}$$

$$=\frac{1}{2}\left(\frac{t^{-n+1}}{1-n}-\frac{t^{-n-1}}{n+1}\right)\Big]_1^{\infty}=\frac{1}{2}\left(\frac{1}{1-n}\frac{1}{t^{n-1}}+\frac{1}{n+1}\frac{1}{t^{n+1}}\right)\Big]_1^{\infty}$$

$$=0+\frac{1}{2}\left(\frac{1}{n-1}+\frac{1}{n+1}\right)=\frac{n}{n^2-1}$$

21. $\int_{-\pi/3}^{\pi/3}\frac{x\sin x}{\cos^2 x}dx=2\int_0^{\pi/3}x(\sec x\tan x)dx$

[$\because$ function is even]

$$=2\left[|x\cdot\sec x|_0^{\pi/3}-\int_0^{\pi/3}\sec x dx\right]$$

$$=2\left[\frac{\pi}{3}\cdot(2)-|\log(\sec x+\tan x)|_0^{\pi/3}\right.$$

$$=2\left[\frac{2\pi}{3}-\log(2+\sqrt{3})+0\right]$$

$$=2\left[\frac{2\pi}{3}-\log\tan\frac{5\pi}{12}\right]$$

$$\left[\because\ \tan\frac{5\pi}{12}=\tan 75^\circ=2+\sqrt{3}\right]$$

22. $\int_0^{2\pi}(\sin x+|\sin x|)dx$

$$=\int_0^{\pi}(\sin x+\sin x)dx+\int_{\pi}^{2\pi}(\sin x-\sin x)dx$$

$$=\int_0^{\pi}2\sin x dx+\int_{\pi}^{2\pi}0dx=2[-\cos x]_0^{\pi}+0$$

$$=-2(\cos\pi-\cos 0)=-2(-1-1)=4$$

23. If $f(x)=\dfrac{\sin x}{x}$,

then $f'(x)=\dfrac{x\cos x-\sin x}{x^2}=\dfrac{(x-\tan x)-\cos x}{x^2}$

Since $\sin x < x < \tan x$ for $0<x<\dfrac{\pi}{2}$

$$f'(x)<0$$

Thus, $f(x)$ is a decreasing function.

Its least value $f\left(\dfrac{\pi}{2}\right)=\dfrac{1}{\pi/2}=\dfrac{2}{x}$ and greatest value

$$f(0)=\lim_{x\to 0}+\frac{\sin x}{x}=1$$

$$=\left(\frac{\pi}{2}-0\right)\frac{2}{x}\le\int_0^{\pi/2}\frac{\sin x}{x}dx\le\left(\frac{\pi}{2}-0\right)\cdot 1$$

$$=1\le\int_0^{\pi/2}\frac{\sin x}{x}dx\le\frac{\pi}{2}$$

24. Put $\sqrt{x+1}=t$ or $x+1=t^2$

$$\therefore\quad I=\int_3^4\frac{2t}{(t^2-4)t}dt=\frac{2}{(2)(2)}\log\left|\frac{t-2}{t+2}\right|\Bigg]_3^4$$

$$=\frac{1}{2}\left[\log\frac{1}{3}-\log\frac{1}{5}\right]=\frac{1}{2}\log\frac{5}{3}$$

25. Put $\cos^2 x=z\therefore -2\cos x\sin x dx=dz$

$\Rightarrow \sin 2x dx=-dz$

When $x=0,\ z=1, x=\pi/2, z=0$

$$\therefore\quad I=\int_1^0\frac{-dz}{1+4z}=\int_0^1\frac{dz}{4z+1}$$

$$=\frac{1}{4}\left|\log(4z+1)\right|_0^1=\frac{1}{4}\log 5$$

26. $\displaystyle\int_0^1\tan^{-1}\left(\frac{1}{x^2-x+1}\right)dx=\int_0^1\tan^{-1}\left[\frac{x-(x-1)}{1+x(x-1)}\right]$

$$=\int_0^1\tan^{-1}x dx-\int_0^1\tan^{-1}(x-1)dx$$

$$=\int_0^1\tan^{-1}x dx+\int_0^1\tan^{-1}(1-1+x)dx$$

$$=\int_0^1\tan^{-1}x dx+\int_0^1\tan^{-1}x dx=2\int_0^1\tan^{-1}x dx$$

$$=2\left[x\tan^{-1}x-\int\frac{x}{1+x^2}dx\right]_0^1$$

$$=2\left[x\tan^{-1}x-\frac{1}{2}\log(1+x^2)\right]_0^1$$

$$=2\left[\left(1\tan^{-1}1-\frac{1}{2}\log 2\right)-\left(0-\frac{1}{2}\log 1\right)\right]$$

$$=2\left[\frac{\pi}{4}-\frac{1}{2}\log 2-0\right]=\frac{\pi}{2}-\log 2$$

27. Period of $|\sin 2\pi|=\dfrac{\pi}{2\pi}=\dfrac{1}{2}$

$$\therefore\quad I=\int_0^{10}|\sin 2\pi x|dx$$

$$=20\int_0^{1/2}|\sin 2\pi x|dx=20\int_0^{1/2}\sin 2\pi x dx$$

$$=20\left[\frac{-\cos 2\pi x}{2\pi}\right]_0^{1/2}=\frac{-20}{2\pi}[-1-1]=\frac{20}{\pi}$$

28. $I=\displaystyle\int_0^{\pi/2}\frac{dx}{1+\tan x}=\int_0^{\pi/2}\frac{dx}{1+\tan(\pi/2-x)}$ (Property 8)

$$=\int_0^{\pi/2}\frac{dx}{1+\cot x}=\int_0^{\pi/2}\frac{\tan x}{1+\tan x}dx$$

$$=\int_0^{\pi/2}\left(1-\frac{1}{1+\tan x}\right)dx=\frac{\pi}{2}-I$$

$\Rightarrow\ I=\pi/4$

29. Let $I=\displaystyle\int_{\pi/6}^{\pi/4}\frac{dx}{\sqrt{\cos x\sin^3 x}}$

$$=\int_{\pi/6}^{\pi/4}\frac{\sec^2 x}{\sqrt{\tan^3 x}}dx=\left|\frac{(\tan x)^{-1/2}}{-1/2}\right|_{\pi/6}^{\pi/4}$$

$$=-2\left[(1)-\left(\frac{1}{\sqrt{3}}\right)^{-1/2}\right]$$

$$=-2\left[1-(\sqrt{3})^{1/2}\right]=2(3)^{1/4}-2$$

30. Let $I=\displaystyle\int_0^{\pi}\frac{xdx}{a^2\cos^2 x+b^2\sin^2 x}$ (i)

and $I=\displaystyle\int_0^{\pi}\frac{(\pi-x)dx}{a^2\cos^2(\pi-x)+b^2\sin^2(\pi-x)}$

$$\Rightarrow\ I=\int_0^{\pi}\frac{(\pi-x)dx}{a^2\cos^2 x+b^2\sin^2 x}\qquad\text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi}\frac{(x+\pi-x)dx}{a^2\cos^2 x+b^2\sin^2 x}$$

$$\Rightarrow\ 2I=\pi\int_0^{\pi}\frac{dx}{a^2\cos^2 x+b^2\sin^2 x}$$

$$\Rightarrow\quad I=\frac{\pi}{2}\int_0^{\pi}\frac{dx}{a^2\cos^2 x+b^2\sin^2 x}$$

$$I=2\cdot\frac{\pi}{2}\int_0^{\pi/2}\frac{dx}{a^2\cos^2 x+b^2\sin^2 x}$$

On dividing the numerator and the denominator by $\cos^2 x$, we get

$$I=\pi\int_0^{\pi/2}\frac{\sec^2 x dx}{a^2+b^2\tan^2 x}$$

Put $b\tan x=t\Rightarrow b\sec^2 x dx=dt$

$$\therefore\quad I=\frac{\pi}{b}\int_0^{\infty}\frac{dt}{a^2+t^2}=\frac{\pi}{b}\cdot\frac{1}{a}\left[\tan^{-1}\frac{t}{a}\right]_0^{\infty}$$

$$=\frac{\pi}{ab}\left[\frac{\pi}{2}-0\right]=\frac{\pi^2}{2ab}$$

31. Let $I=\int_0^1\frac{1}{1+x^2}\log(1+x)dx$. Put $x=\tan\theta$

$$\therefore\quad I=\int_0^{\pi/4}\frac{\log(1+\tan\theta)\sec^2\theta}{1+\tan^2\theta}d\theta$$

$$=\int_0^{\pi/4}\log(1+\tan\theta)d\theta$$

Also, $I=\int_0^{\pi/4}\log\left[1+\tan\left(\frac{\pi}{4}-\theta\right)\right]d\theta$

$$=\int_0^{\pi/4}\log\left(1+\frac{1-\tan\theta}{1+\tan\theta}\right)d\theta=\int_0^{\pi/4}\log\left(\frac{2}{1+\tan\theta}\right)dx$$

$$=\log 2\int_0^{\pi/4}d\theta-I\Rightarrow 2I=\log 2.\left(\frac{\pi}{4}\right)$$

$$\Rightarrow\quad I=\frac{\pi\log 2}{8}$$

32. Let $I=\int_0^{\pi/2}\frac{\cos x-\sin x}{1+\cos x\sin x}dx$ (i)

On putting $x=\left(\frac{\pi}{2}-x\right)$ in Eq. (i), we get

$$I=\int_0^{\pi/2}\frac{\cos\left(\frac{\pi}{2}-x\right)-\sin\left(\frac{\pi}{2}-x\right)}{1+\cos\left(\frac{\pi}{2}-x\right)\sin\left(\frac{\pi}{2}-x\right)}dx$$

$$=\int_0^{\pi/2}-\left(\frac{\cos x-\sin x}{1+\cos x\sin x}\right)dx \qquad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi/2}0dx=0\Rightarrow I=0$$

33. Put $\sin x=z\ \therefore\ \cos x dx=dz$

When $x=0, z=0$; when $x=\pi/2, z=1$

$$\therefore\quad I=\int_0^1\frac{dz}{(1+z)(2+z)}=\int_0^1\left[\frac{1}{z+1}-\frac{1}{z+2}\right]dz$$

$$=\left|\log\left(\frac{z+1}{z+2}\right)\right|_0^1=\log\left(\frac{2}{3}\right)-\log\left(\frac{1}{2}\right)=\log\frac{4}{3}$$

34. Let $I=\int_{\pi/6}^{\pi/3}\frac{dx}{1+\sqrt{\cot x}}$

$$\Rightarrow\ I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cot x}}dx \qquad \text{(i)}$$

$$\Rightarrow\ I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}+\sqrt{\cot\left(\frac{\pi}{2}-x\right)}}dx$$

$$\left[\because\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$$

$$\Rightarrow\ I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx \qquad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_{\pi/6}^{\pi/3}1dx=[x]_{\pi/6}^{\pi/3}\ \Rightarrow\ I=\frac{1}{2}\left(\frac{\pi}{3}-\frac{\pi}{6}\right)=\frac{\pi}{12}$$

35. $\int_{-\pi/2}^{\pi/2}\sin^2 x\cos^2 x(\sin x+\cos x)dx$

$$=\int_{-\pi/2}^{\pi/2}\sin^3 x\cos^2 x dx+\int_{-\pi/2}^{\pi/2}\sin^2 x\cos^3 x dx$$

$[\because\ \sin^3 x\cos^2 x$ is odd and $\sin^2 x\cos^3 x$ is even$]$

$$=0+2\int_0^{\pi/2}\sin^2 x\cos^3 x dx$$

Put $\sin x=z\quad\therefore\ \cos x dx=dz$

$$\therefore\quad\text{Given integral}=2\int_0^1 z^2(1-z^2)dz$$

$$=2\left[\frac{z^3}{3}-\frac{z^5}{5}\right]_0^1=2\left(\frac{1}{3}-\frac{1}{5}\right)=\frac{4}{15}$$

36. Let $I=\int_0^{\pi/2}\frac{2^{\sin x}}{2^{\sin x}+2^{\cos x}}dx$ (i)

$$\Rightarrow\ I=\int_0^{\pi/2}\frac{2^{\sin(\pi/2-x)}}{2^{\sin(\pi/2-x)}+2^{(\cos\pi/2-x)}}dx$$

$$\Rightarrow\ I=\int_0^{\pi/2}\frac{2^{\cos x}}{2^{\cos x}+2^{\sin x}}dx \quad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi/2}\frac{2^{\sin x}+2^{\cos x}}{2^{\sin x}+2^{\cos x}}dx=\frac{\pi}{2}$$

$$\Rightarrow\quad I=\frac{\pi}{4}$$

37. $\int_0^{\pi/2}\frac{\cos 2x}{(\sin x+\cos x)^2}dx=\int_0^{\pi/2}\frac{\cos^2 x-\sin^2 x}{(\sin x+\cos x)^2}dx$

$$=\int_0^{\pi/2}\frac{\cos x-\sin x}{\cos x+\sin x}dx=\left|\log(\cos x+\sin x)\right|_0^{\pi/2}$$

$$=\log 1-\log 1=0$$

38. Let $I=\int_0^1\frac{dx}{x+\sqrt{1-x^2}}$

Put $x=\sin\theta\Rightarrow dx=\cos\theta d\theta$

$$I=\int_0^{\pi/2}\frac{\cos\theta d\theta}{\sin\theta+\cos\theta} \quad \text{(i)}$$

$$\Rightarrow\ I=\int_0^{\pi/2}\frac{\cos\left(\frac{\pi}{2}-\theta\right)d\theta}{\sin\left(\frac{\pi}{2}-\theta\right)+\cos\left(\frac{\pi}{2}-\theta\right)}$$

$$=\int_0^{\pi/2}\frac{\sin\theta}{\cos\theta+\sin\theta}d\theta \quad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi/2}1d\theta\ \Rightarrow\ I=\frac{\pi}{4}$$

39. Let $I=\int_0^{2\pi}\theta\sin^2\theta\cos\theta d\theta$

$$=\left|\theta\cdot\frac{\sin^3\theta}{3}\right|_0^{2\pi}-\int_0^{2\pi}1\cdot\frac{\sin^3\theta}{3}d\theta$$

$$=-\frac{1}{3}\int_0^{2\pi}\sin^3\theta d\theta$$

But $\sin 3\theta=3\sin\theta-4\sin^3\theta$

$$\therefore\quad \sin^3\theta=\frac{3}{4}\sin\theta-\frac{1}{4}\sin 3\theta$$

$$\therefore\quad I=-\frac{1}{3}\int_0^{2\pi}\left(\frac{3}{4}\sin\theta-\frac{1}{4}\sin 3\theta\right)d\theta$$

$$=-\frac{1}{3}\left[-\frac{3}{4}\cos\theta\frac{1}{12}\cos 3\theta\right]_0^{2\pi}$$

$$=-\frac{1}{3}\left[-\frac{3}{4}+\frac{1}{12}+\frac{3}{4}-\frac{1}{12}\right]=0$$

40. Let $I=\int_0^{\pi/2}\frac{dx}{1+\tan^3 x}dx=\int_0^{\pi/2}\frac{1}{1+\frac{\sin^3 x}{\cos^3 x}}dx$

$$=\int_0^{\pi/2}\frac{\cos^3 x}{\cos^3 x+\sin^3 x}dx \quad \text{(i)}$$

$$=\int_0^{\pi/2}\frac{\cos^3\left(\frac{\pi}{2}-x\right)}{\cos^3\left(\frac{\pi}{2}-x\right)+\sin^3\left(\frac{\pi}{2}-x\right)}dx$$

$$\Rightarrow\ I=\int_0^{\pi/2}\frac{\sin^3 x}{\sin^3 x+\cos^3 x}dx \quad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi/3}\frac{\sin^3 x+\cos^3 x}{\sin^3 x+\cos^3 x}dx$$

$$\Rightarrow\ 2I=\int_0^{\pi/2}1dx$$

$$\Rightarrow\ 2I=[x]_0^{\pi/2}=\frac{\pi}{2}\quad\Rightarrow\quad I=\frac{\pi}{4}$$

41. $\int_0^1\frac{2^{x+1}-5^{x-1}}{10^x}dx=\int_0^1\left[2\left(\frac{2}{10}\right)^x-\frac{1}{5}\left(\frac{5}{10}\right)^x\right]dx$

$$=2\int_0^1\left(\frac{1}{5}\right)^x dx-\frac{1}{5}\int_0^1\left(\frac{1}{2}\right)^x dx$$

$$=2\left|\frac{\left(\frac{1}{5}\right)^x}{\log\frac{1}{5}}\right|_0^1-\frac{1}{5}\left|\frac{\left(\frac{1}{2}\right)^x}{\log\frac{1}{2}}\right|_0^1=\frac{2\left[\frac{1}{5}-1\right]}{\log\frac{1}{5}}-\frac{1}{5}\frac{\left[\frac{1}{2}-1\right]}{\log\frac{1}{2}}$$

$$=2\left[\frac{-\frac{4}{5}}{-\log 5}\right]-\frac{1}{5}\cdot\frac{\left(\frac{1}{2}\right)}{-\log 2}$$

$$=\frac{8}{5\log_e 5}-\frac{1}{10\log_e 2}=\frac{8}{5}\log_5 e-\frac{1}{10}\log_2 e$$

42. Let $I=\int_{-\pi/4}^{\pi/4} x^3\sin^4 x\,dx$

Since $x^3\sin^4 x$ is an odd function,

$\therefore\ I=0$

43. $\int_{-1}^{3}(|x-2|+[x])dx$

$=\int_{-1}^{3}|x-2|dx+\int_{2}^{3}[x]dx=I_1+I_2$

$I_1=\int_{-1}^{2}|x-2|dx+\int_{2}^{3}|x-2|dx=\int_{-1}^{2}(2-x)dx+\int_{2}^{3}(x-2)dx=5$

[verify]

$I_2=\int_{-1}^{0}1dx+\int_{0}^{1}0dx+\int_{1}^{2}1.dx+\int_{2}^{3}2dx=2$

$\therefore$ Given integral = 5 + 2 = 7 [verify]

44. $\lim_{x\to0}\frac{2x\tan|x|}{3x^2}=\frac{2}{3}\lim_{x\to0}\frac{\tan|x|}{x}$

RHL is $\frac{2}{3}$ and LHL is $-\frac{2}{3}$.

So, limit does not exist.

45. $\int_{-1}^{3}\left(\tan^{-1}\frac{x}{x^2+1}+\tan^{-1}\frac{x^2+1}{x}\right)dx$

$=\int_{-1}^{3}\left(\tan^{-1}\frac{x}{x^2+1}+\cot^{-1}\frac{x}{x^2+1}\right)dx$

$=\int_{-1}^{3}\frac{\pi}{2}dx=\frac{\pi}{2}|x|_{-1}^{3}=\frac{\pi}{2}(4)=2\pi$

46. $\int_0^1\frac{d}{dx}\left[\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right]dx$

$=\left[\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right]_0^1=\sin^{-1}(1)-\sin^{-1}(0)=\frac{\pi}{2}$

47. $I=\int_{\pi/6}^{5\pi/6}\sqrt{4-4\sin^2 t}\,dt2=\int_{\pi/6}^{5\pi/6}|\cot t|dt$

$=2\int_{\pi/6}^{\pi/2}|\cos t|dt+2\int_{\pi/2}^{5\pi/6}|\cot t|dt$

$=2\int_{\pi/6}^{\pi/2}\cos t\,dt+2\int_{\pi/2}^{5\pi/6}-\cot t\,dt$

$=2\cdot|\sin t|_{\pi/6}^{\pi/2}-2|\sin t|_{\pi/2}^{5\pi/6}$

$=2\left[\sin\frac{\pi}{2}-\sin\frac{\pi}{6}\right]-\left[\sin\frac{5\pi}{6}-\sin\frac{\pi}{2}\right]$

$=2\left(1-\frac{1}{2}\right)-2\left(\frac{1}{2}-1\right)=2\left(\frac{1}{2}\right)-2\left(-\frac{1}{2}\right)=1+1=2$

48. Let $I=\int_0^\infty\log\left(x+\frac{1}{x}\right)\frac{1}{1+x^2}dx$

Put $x=\tan\theta\ \Rightarrow\ dx=\sec^2\theta d\theta$

$\therefore\ I=\int_0^{\pi/2}\log\left(\tan\theta+\frac{1}{\tan\theta}\right)\frac{\sec^2\theta}{\sec^2\theta}d\theta$

$\Rightarrow\ I=\int_0^{\pi/2}\log\frac{(1+\tan^2\theta)}{\tan\theta}d\theta$

$\Rightarrow\ I=2\int_0^{\pi/2}\log\sec\theta d\theta-\int_0^{\pi/2}\log\tan\theta d\theta$

$\Rightarrow\ I=2\int_0^{\pi/2}\log\sec\theta d\theta\quad\left(\because\int_0^{\pi/2}\log\tan\theta d\theta=0\right)$

$\Rightarrow\ I=-2\int_0^{\pi/2}\log\cos\theta d\theta=-2\times\frac{-\pi}{2}\log 2$

$\left(\because\int_0^{\pi/2}\log\cos\theta d\theta=-\frac{\pi}{2}\log 2\right)$

$\therefore\ I=\pi\log 2$

49. $\int_0^{\pi/2}\log|\tan x+\cot x|dx$

$=\int_0^{\pi/2}\log\left|\frac{\sin^2 x+\cos^2 x}{\cos x\sin x}\right|dx=\int_0^{\pi/2}\log\left(\frac{1}{\sin x\cos x}\right)dx$

[$\because$ sin x and cos x are +ve in the first quadrant]

$=\int_0^{\pi/2}\log 1dx-\int_0^{\pi/2}\log\sin x dx-\int_0^{\pi/2}\log\cos x dx$

$=0-\left(-\frac{\pi}{2}\log 2\right)-\left(-\frac{\pi}{2}\log 2\right)$

$=\frac{\pi}{2}\log 2+\frac{\pi}{2}\log 2=\pi\log 2$

50. $\because\ I=\int_0^\infty\frac{x\log x}{(1+x^2)^2}dx$

Put $x=\tan\theta\ \Rightarrow\ dx=\sec^2\theta d\theta$

$\therefore\ I=\int_0^{\pi/2}\frac{\tan\theta\log(\tan\theta)}{\sec^4\theta}.\sec^2\theta d\theta$

$=\int_0^{\pi/2}\frac{\sin\theta\log(\tan\theta)}{\cos\theta\cdot\frac{1}{\cos^2\theta}}d\theta=\int_0^{\pi/2}\sin\theta\cos\theta\log(\tan\theta)d\theta$

$=\frac{1}{2}\int_0^{\pi/2}\sin 2\theta\log(\tan\theta)d\theta=0$

$\left(\because\int_0^{\pi/2}\sin 2\theta\log\tan\theta d\theta=0\right)$

51. Given integral $= \int_0^{\pi/2} \frac{\sec^2\theta d\theta}{4+9\tan^2\theta}$

$$= \frac{1}{9}\int_0^{\pi/2} \frac{\sec^2\theta d\theta}{(2/3)^2+\tan^2\theta}$$

Put $\tan\theta = z, \sec^2\theta d\theta = dz$

$$= \frac{1}{9}\int_0^{\infty} \frac{dz}{(2/3)^2+z^2}$$

$$= \frac{1}{9}\frac{1}{\frac{2}{3}}\left|\tan^{-1}\frac{z}{\frac{2}{3}}\right|_0^{\infty} = \frac{1}{6}(\tan^{-1}\infty - \tan^{-1}0)$$

$$= \frac{1}{6}\cdot\frac{\pi}{2} = K\pi \quad \text{[given]} \quad \therefore \quad K = \frac{1}{12}$$

52. $\int_{-1}^{1}[x\sin\pi x]dx = 2\int_0^1[x\sin\pi x]dx$ (i)

(as $[x\sin\pi x]$ is an even function)

Now, $0 \le x \le 1$

$\Rightarrow \; 0 \le \pi x \le \pi \;\Rightarrow\; 0 \le \sin\pi x \le 1$ (ii)

$\Rightarrow \; 0 \le x\sin\pi x < 1 \;\Rightarrow\; [x\sin\pi x] = 0$

From Eqs. (i) and (ii), we get

$$\int_0^1[x\sin\pi x]dx = 0$$

53. $\int_{-1}^{1}(x-[x])dx = \int_{-1}^{0}(x-[x])dx + \int_0^1(x-[x])dx$

$$= \int_{-1}^{0}(x+1)dx + \int_0^1(x-0)dx$$

$$= \left[\frac{(x+1)^2}{2}\right]_{-1}^{0} + \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}+\frac{1}{2} = 1$$

54. Let $I = \int_{-\pi/4}^{\pi/4}\sin^{-4}x dx = \int_{-\pi/4}^{\pi/4}\frac{dx}{\sin^4 x}$

$$= \int_{-\pi/4}^{\pi/4}\text{cosec}^{-4}x dx$$

Put $\cot x = t \quad \therefore \; -\text{cosec}^2 x dx = dt$

$$\therefore \quad I = -\int_{-1}^{1}(1+t^2)dt$$

$$= -2\int_0^1(1+t^2)dt = -2\left[t+\frac{t^3}{3}\right]_0^1 = -\frac{8}{3}$$

55. $\therefore \; I_k = \int_{-2k\pi}^{0}|\sin x|[\sin x]dx + \int_0^{2k\pi}|\sin x|[\sin x]dx$

$$= -\int_{2k\pi}^{0}|\sin x|[-\sin x]dx + \int_0^{2k\pi}|\sin x|[\sin x]dx$$

$$= \int_0^{2k\pi}|\sin x|([\sin x]+[-\sin x])dx$$

$$= 2k\int_0^{\pi}|\sin x|([\sin x]+[-\sin x])dx$$

$$= 2k\int_0^{\pi}\sin x(0-1)dx$$

$$= -2k[-\cos]_0^{\pi} \qquad \begin{pmatrix}\because \; \sin x > 0, x \in (0,\pi) \\ \therefore \; -\sin x < 0, x \in (0,\pi)\end{pmatrix}$$

$$= -4k$$

$$\therefore \quad \sum_{k=1}^{10} I_k = -4\sum_{k=1}^{10} k = -4\cdot\frac{10\cdot 11}{2} = -220$$

56. We have $f(x) = \frac{x^7-3x^5+7x^3-x}{\cos^2 x}$, which is an odd function.

$$\therefore \qquad \int_{-1}^{1} f(x)dx = 0$$

$\therefore$ Reqd. integral,

$$I_1 = \int_{-1}^{1}\cos^{-1}(x)dx$$

$$= \int_{-1}^{1}\cos^{-1}(-x)dx = \int_{-1}^{1}(\pi - \cos^{-1}x)dx$$

$$= 2\pi - I$$

$$\therefore \quad 2I_1 = 2\pi \Rightarrow I_1 = \pi$$

57. Since $\sqrt{1+x+x^2} - \sqrt{1-x+x^2}$ is an odd function of x,

$\therefore$ given integral = 0.

58. $\because \; \sqrt{x+2\sqrt{(x-1)}} = \sqrt{(\sqrt{(x-1)})^2+1^2+2\sqrt{(x-1)}}$

$$= \sqrt{(x-1)}+1$$

and $\sqrt{x-2\sqrt{(x-1)}} = \sqrt{(\sqrt{(x-1)})^2+1^2-2\sqrt{(x-1)}}$

$$= \left|\sqrt{(x-1)}-1\right|$$

Then, $\int_1^5\left[\sqrt{x+2\sqrt{(x-1)}}+\sqrt{x-2\sqrt{(x-1)}}\right]dx$

$$= \int_1^5\left[\sqrt{(x-1)}+1\right]dx + \int_1^5\left|\sqrt{(x-1)}-1\right|dx$$

$$= \int_1^5 \left[\sqrt{(x-1)}+1\right]dx + \int_1^2 \left[1-\sqrt{(x-1)}\right]dx + \int_2^5 \left[\sqrt{(x-1)}-1\right]dx$$

$$= \int_0^4 (\sqrt{x}+1)dx + \int_0^1 (1-\sqrt{x})dx + \int_1^4 (\sqrt{x}-1)dx$$

$$= \left[\frac{2}{3}(x^{3/2})+x\right]_0^4 + \left[x-\frac{2}{3}x^{3/2}\right]_0^1 + \left[\frac{2}{3}x^{3/2}-x\right]_1^4$$

$$= \left(\frac{16}{3}+4\right)+\left(1-\frac{2}{3}\right)+\left(\frac{16}{3}-4\right)+\left(\frac{2}{3}-1\right)$$

$$= \frac{32}{3}$$

59. $I_1 = \left|\frac{1}{2}\log(1+t^2)\right|_{1/e}^{\tan s}$

(By taking $1+t^2 = x$ and $2tdt = dx$)

$$I_2 = \int_{1/e}^{\cot x}\left(\frac{1}{t}-\frac{t}{1+t}\right)dt$$

$$= \left|\log t - \frac{1}{2}\log(1+t^2)\right|_{1/e}^{\cot x}$$

$$I_1 + I_2 = \log\sec x - \frac{1}{2}\log\left(1+\frac{1}{e^2}\right) + \log\cot x$$

$$-\frac{1}{2}\log(1+\cot^2 x) - \log\frac{1}{e} + \frac{1}{2}\log\left(1+\frac{1}{e^2}\right)$$

$$= \log\sec x + \log\cot x - \log\operatorname{cosec} x + 1$$

$$= \log\frac{\sec x\cot x}{\operatorname{cosec} x} + 1$$

$$= \log 1 + 1 = 1$$

60. $\int_0^1 x(1-x)^{99}dx = \int_0^1 (1-x)[1-(1-x)]^{99}dx$

$$= \int_0^1 (1-x)^{99}dx = \int_0^1 (x^{99}-x^{100})dx$$

$$= \left[\frac{x^{100}}{100}-\frac{x^{101}}{101}\right]_0^1 = \frac{1}{100}-\frac{1}{101} = \frac{1}{10,100}$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $\because$ Integrand is discontinuous at $\pi/2$, then

$$\int_0^{\pi/2} 0\cdot dx + \int_{\pi/2}^{3\pi/2} 0\cdot dx = 0$$

$\because$ $0 < x < \frac{\pi}{2}$, $\left|\tan^{-1}\tan x\right| = \left|\sin^{-1}\sin x\right|$

and $\frac{\pi}{2} < x < \frac{3\pi}{2}$, $\left|\tan^{-1}\tan x\right| = \left|\sin^{-1}\sin x\right|$

2. The graph of $f(x) = x - x^2$ is $x - x^2 = -1$ at some point becomes

at $x = 1, x - x^2 = 0$

and at $x = 2, x - x^2 = -2$

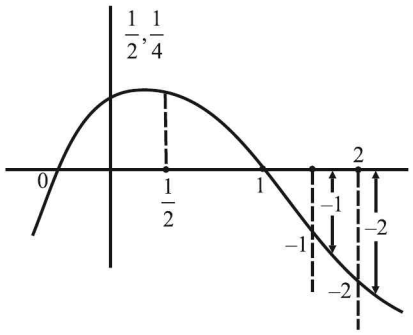


Fig. 15.4

$$\Rightarrow x_0 = \frac{1+\sqrt{5}}{2}$$

$$\Rightarrow x - x^2 = -1$$

$$\int_0^2 [x-x^2]\,dx$$

$$= \int_0^1 [x-x^2]dx + \int_1^{\frac{1+\sqrt{5}}{2}} [x-x^2]dx + \int_{\frac{1+\sqrt{5}}{2}}^2 [x-x^2]dx$$

$$= \int_0^1 0dx + \int_1^{\frac{1+\sqrt{5}}{2}} -1\,dx + \int_{\frac{1+\sqrt{5}}{2}}^2 -2dx$$

$$= -1\left(\frac{1+\sqrt{5}}{2}-1\right) - 2\left(2-\frac{1+\sqrt{5}}{2}\right)$$

$$= -3 + \frac{1+\sqrt{5}}{2} = \frac{\sqrt{5}-5}{2}$$

3. Let $I = \int_3^4 \frac{[x^2]}{[x^2-14x+49]+[x^2]}dx$ (i)

$$= \int_3^4 \frac{[x^2]}{[(7-x)^2]+[x^2]}dx$$

$$= \int_3^4 \frac{[(7-x^2)]}{[(7-(7-x))^2]+[(7-x)^2]}dx$$

$$\left[\text{Using } \int_3^4 f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$\Rightarrow \quad I = \int_3^4 \frac{[(7-x^2)]}{[x^2]+[(7-x)^2]}dx \qquad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I = \int_3^4 1\,dx = 1 \qquad \therefore \quad I = \frac{1}{2}$$

4. The given integral can be written as

$$\int_0^1 \log\left[\sqrt{1-x}+\sqrt{1+x}\right]\cdot 1\,dx$$

Integrating by parts, we get

$$\log(\sqrt{1-x}+\sqrt{1+x})\cdot x\Big|_0^1$$

$$-\int_0^1 \frac{1\left(-\frac{1}{2\sqrt{1-x}}+\frac{1}{2\sqrt{1+x}}\right)}{\sqrt{1-x}+\sqrt{1+x}}\cdot x\cdot dx$$

$$= \frac{1}{2}\log 2 - 0 - \frac{1}{2}\int_0^1 \frac{\left(\sqrt{1-x}-\sqrt{1+x}\right)x}{\left(\sqrt{1-x}+\sqrt{1+x}\right)\left(\sqrt{1-x^2}\right)}dx$$

$$= \frac{1}{2}\log 2 - \frac{1}{2}\int_0^1 \frac{\left(1-x+1+x-2\sqrt{1-x^2}\right)x}{(1-x-1-x)\left(\sqrt{1-x^2}\right)}dx$$

$$= \frac{1}{2}\log 2 - \frac{1}{2}\int_0^1 \frac{\left(2-2\sqrt{1-x^2}\right)x}{-2x\sqrt{1-x^2}}dx$$

$$= \frac{1}{2}\log 2 - \frac{1}{2}\int_0^1 \frac{\sqrt{1-x^2}-1}{\sqrt{1-x^2}}dx$$

$$= \frac{1}{2}\log 2 - \frac{1}{2}\int_0^1 dx + \frac{1}{2}\int_0^1 \frac{1}{\sqrt{1-x^2}}dx$$

$$= \frac{1}{2}\log 2 - \frac{1}{2} + \frac{1}{2}\left[\frac{\pi}{2}\right]$$

$$= \frac{1}{2}\left[\log 2 + \frac{\pi}{2} - 1\right]$$

5. For $x \in \left(0, \frac{\pi}{4}\right)$, $\cos^2 x > \sin^2 x$

$$\Rightarrow \int_0^{\pi/4} \cos^2 x\,dx > \int_0^{\pi/4} \sin^2 x\,dx$$

$$\Rightarrow I_1 < I_2$$

6. $a_n - a_{n-1} = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin x}dx - \int_0^{\pi/2} \frac{\sin^2(n-1)x}{\sin x}dx$

$$= \frac{1}{2}\int_0^{\pi/2} \frac{2\sin^2 nx - 2\sin^2(n-1)x}{\sin x}dx$$

$$= \frac{1}{2}\int_0^{\pi/2} \frac{(1-\cos 2nx)-(1-\cos 2(n-1)x)}{\sin x}dx$$

$$= \frac{1}{2}\int_0^{\pi/2} \frac{\cos 2(n-1)x - \cos 2nx}{\sin x}dx$$

$$= \frac{1}{2}\int_0^{\pi/2} \frac{2\sin(2n-1)x\sin x}{\sin x}dx$$

$$= \int_0^{\pi/2} \sin(2n-1)x\,dx$$

$$= \left[-\frac{\cos(2n-1)x}{2n-1}\right]_0^{\pi/2} = -\frac{1}{2n-1}[0-1]$$

$$= \frac{1}{2n-1}$$

$\therefore \quad a_2 - a_1 = \frac{1}{3}, a_3 - a_2 = \frac{1}{5}, a_4 - a_3 = \frac{1}{7}, \cdots$ which are in H.P. [$\because$ 3, 5, 7,... are in A.P.]

7. $\sin x - \cos x = t \Rightarrow (\cos x + \sin x)dx = dt$

On squaring $1 - \sin 2x = t^2$

$$\Rightarrow I = \int_{-1}^0 \frac{1}{3+1-t^2}dt = \int_{-1}^0 \frac{1}{4-t^2}dt$$

$$= \frac{1}{4}\log\left(\frac{2+t}{2-t}\right)_{-1}^0 = \frac{1}{4}\log 3$$

8. Since $\int_2^4 [3 - f(x)]\,dx = 7$

$$\Rightarrow 3[x]_2^4 - \int_2^4 f(x)dx = 7$$

$$\Rightarrow \int_2^4 f(x)dx = 6 - 7 = -1$$

$$\therefore \quad \int_{-1}^2 f(x)dx = \int_{-1}^4 f(x)dx - \int_2^4 f(x)dx$$

$$= 4 - (-1) = 5$$

9. $I=\int_{-\pi/2}^{\pi/2}\sqrt{\frac{1-\cos 2x}{2}}dx=\int_{-\pi/2}^{\pi/2}|\sin x|dx$

$=2\int_0^{\pi/2}|\sin x|dx=2\int_0^{\pi/2}\sin x dx=2[-\cos x]_0^{\pi/2}$

$=-2(\cos\pi/2-\cos 0)=2$

10. $\int_a^b xf(x)dx=\int_a^b(a+b-x)f(a+b-x)dx$

$\left[\because\ \int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$

$\Rightarrow\ \int_a^b xf(x)dx=\int_a^b(a+b-x)f(x)dx$

$\Rightarrow\ 2\int_a^b xf(x)dx=\int_a^b(a+b)f(x)dx$

$\Rightarrow\ \int_a^b f(x)dx=\frac{a+b}{2}\int_a^b f(x)dx$

11. Let $I=\int_{-\pi/2}^{\pi/2}[f(x)+f(-x)][g(x)-g(-x)]\,dx$

Again let

$h(x)=[f(x)+f(-x)][g(x)-g(-x)]$

$\Rightarrow\ h(-x)=[f(-x)+f(x)][\{g(-x)-g(x)]$

$\Rightarrow\ h(-x)=-h(x)$

Hence, $h(x)$ is an odd function.

$\therefore\ I=0$

12. $-1<x<1$

$\Rightarrow\ -1+2<x+2<1+2$

$\Rightarrow\ 1<x+2<3$, i.e. $x+2>0$

$\therefore\ \int_{-1}^{1}\frac{|x+2|}{x+2}dx=\int_{-1}^{1}\frac{x+2}{x+2}dx=\int_{-1}^{1}1\cdot dx$

$=[x]_{-1}^{1}=1-(-1)=2$

13. The integral is the same as $\int_0^{\pi/2}\frac{\cos x}{\cos x+\sin x}dx$, which can be evaluated by property

$$\int_0^a f(x)dx=\int_0^a f(a-x)dx$$

14. Let $I=\int_{-1}^{1}\sin^3 x\cos^2 x dx$. Again, let $f(x)=\sin^3 x\cos^2 x$

$f(-x)=\sin^3(-x)\cos^2(-x)=-\sin^3 x\cos^2 x$

$\therefore\ f(x)$ is an odd function.

Hence, $\int_{-1}^{1}\sin^3 x\cos^2 x dx=0$

15. Put $x+\sqrt{x^2+1}=t$, then $\sqrt{x^2+1}=t-x$

$\Rightarrow\ x^2+1=t^2+x^2-2tx\ \Rightarrow\ x=\frac{t^2-1}{2t}$

$dx=\frac{t^2-1}{2t^2}dt$

When $x\to\infty$, $t\to\infty$; when $x\to 1, t\to 1$

$\Rightarrow\ I=\int_1^{\infty}\frac{t^2-1}{2t^2}\cdot\frac{1}{t^n}dt$

$\Rightarrow\ \frac{1}{2}\int_1^{\infty}\left(\frac{1}{t^n}+\frac{1}{t^{n+2}}\right)dt=\frac{1}{n^2-1}$

16. $I=\int_0^{\pi/4}-(\sin x-\cos x)dx+\int_{\pi/4}^{\pi/2}(\sin x-\cos x)dx$

$=[\cos x+\sin x]_0^{\pi/4}+[-\cos x-\sin x]_{\pi/4}^{\pi/2}$

$=\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}-1-0\right)+\left(-0-1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right)$

$=\frac{4}{\sqrt{2}}-2=2\sqrt{2}-2=2(\sqrt{2}-1)$

17. $\int_{-3}^{3}(ax^5+bx^3+cx+k)dx$

$=\left[\frac{ax^6}{6}+\frac{bx^4}{4}+\frac{cx^2}{2}+kx\right]_{-3}^{3}$

$=\frac{a\cdot 3^6}{6}+\frac{b\cdot 3^4}{4}+\frac{c\cdot 3^2}{3}+k(3)-\frac{a\cdot 3^6}{6}-\frac{b\cdot 3^4}{4}-\frac{c\cdot 3^2}{3}+k(3)$

$=6k$

i.e. integral depends upon k.

18. Split into two parts. The first one will be zero

$$\int_{-1}^{1}\frac{\sin x+x^2}{3-|x|}dx=\int_{-1}^{1}\frac{\sin x}{3-|x|}dx+\int_0^1\frac{x^2}{3-|x|}dx$$

19. $\int_{-\pi}^{\pi}\sin mx\sin nx dx=\frac{1}{2}\int_{-\pi}^{\pi}2\sin mx\sin nx dx$

$=\frac{2}{2}\int_0^{\pi}[\cos(m-n)x-\cos(m+n)x]dx$

$=\left(\frac{\sin(m-n)x}{m-n}-\frac{\sin(m+n)x}{m+n}\right)_0^{\pi}=0$

20. $3a\int_0^1\left(\frac{ax-1}{a-1}\right)^2 dx=\frac{3a}{(a-1)^2}\left[\frac{(ax-1)^3}{3}\times\frac{1}{a}\right]_0^1$

$=\frac{1}{(a-1)^2}\left[(a-1)^3+1\right]$

$=(a-1)+(a-1)^{-2}$

21. $I = \int_0^{2\pi} |\sin^3 x|\, dx = \int_0^{\pi} \sin^3 x\, dx - \int_{\pi}^{2\pi} \sin^3 x\, dx$

Consider $I_1 = \int_0^{\pi} \sin^3 x\, dx = \int_0^{\pi} \sin x \cdot \sin^2 x\, dx$

$= \int_0^{\pi} \sin x\,(1 - \cos^2 x)\, dx$

Let $\cos x = t$ at $x = 0$, $t = 1$

$dt = -\sin x\, dx$ at $x = \pi$, $t = -1$

$\Rightarrow I_1 = \int_1^{-1} (1 - t^2)\, dt$

$= \int_{-1}^{1} (1 - t^2)\, dt = \left[t - \frac{t^3}{3}\right]_{-1}^{1}$

$= 1 - \frac{1}{3} + 1 - \frac{1}{3} = \frac{4}{3}$

Consider $I_2 = \int_{\pi}^{2\pi} \sin^3 x\, dx = \int_{\pi}^{2\pi} \sin x(\sin^2 x)\, dx$

$= \int_{\pi}^{2\pi} \sin x\,(1 - \cos^2 x)\, dx$

Let $t = \cos x$ at $t = \pi$, $\cos x = -1$

$dt = -\sin x\, dx$ at $t = 2\pi$, $\cos x = 1$

$\Rightarrow I_2 = \int_{-1}^{1} (1 - t^2)\, dt = -\left[t - \frac{t^3}{3}\right]_{-1}^{1} = -\frac{4}{3}$

$\Rightarrow I = I_1 + I_2 = \frac{4}{3} - \left(-\frac{4}{3}\right) = \frac{8}{3}$

So, correct option is (c).

22. Put $\tan\frac{x}{2} = t$ so that $dx = \frac{2dt}{1+t^2}$

$\therefore I = \int_0^{\infty} \frac{2dt}{(a^2+1)(1+t^2) - 2a(1-t^2)}$

$= \int_0^{\infty} \frac{2dt}{(a+1)^2 t^2 + (a-1^2)}$

$= \frac{1}{(a+1)^2} \int_0^{\infty} \frac{2}{t^2 + [(a-1)/(a+1)]^2}\, dt$

$= \frac{2}{(a+1)^2} \cdot \frac{1}{[(a-1)/(a+1)]} \left[\tan^{-1}\left[\left(\frac{a+1}{a-1}\right)t\right]\right]_0^{\infty}$

$= \frac{2}{a^2-1}(\pi/2 - 0) = \frac{\pi}{a^2-1}$

23. $\because \int_0^1 e^{x^2}(x-\alpha)dx = 0$

$\therefore$ $e^{x^2}(x-\alpha)$ must be both +ve and –ve for $x \in (0,1)$,

i.e. $e^x(x-\alpha) = 0$ for one $x \in (0,1)$ $\therefore$ $\alpha \in (0,1)$

24. $I = \int_0^1 [x^2]dx + \int_1^{\sqrt{2}} [x^2]dx + \int_{\sqrt{2}}^{\sqrt{3}} [x^2]dx + \int_{\sqrt{3}}^{2} [x^2]dx$

$= \int_0^1 0dx + \int_1^{\sqrt{2}} 1dx + \int_{\sqrt{2}}^{\sqrt{3}} 2dx + \int_{\sqrt{3}}^{2} 3dx$

$= 0 + (\sqrt{2} - 1) + 2(\sqrt{3} - \sqrt{2}) + 3(2 - \sqrt{3})$

$= 5 - \sqrt{2} - \sqrt{3}$

25. Let $I = \int_0^{\pi/2} \sin^{100} x dx - \int_0^{\pi/2} \cos^{100} x dx$

$= 0 - 0 = 0 \quad = 0$

26. $\int_0^2 x^2 \left[\frac{x}{2}\right] dx = 0$

[$\because$ for $x \in [0,2)$, $(x/2) \in [0,1)$, hence $(x/2) = 0$, for all $x \in [0,2\}$]

27. $\int_0^{\infty} \frac{dx}{(a^2+x^2)} = \left[\frac{1}{a}\tan^{-1}\frac{x}{a}\right]_0^{\infty} = \frac{1}{a}\left(\frac{\pi}{2} - 0\right) = \frac{\pi}{2a}$

28. Let $\phi(x) = \sin^2 x \log\left(\frac{2-\sin x}{2+\sin x}\right)$

$\therefore \quad \phi(-x) = \sin^2 x \log\left(\frac{2+\sin x}{2-\sin x}\right) = -\phi(x)$

i.e. ϕ is an odd function $\therefore \int_{-\pi/2}^{\pi/2} \phi(x)dx = 0$

29. We have $7 \le 10 + 3\cos x \le 13$

$\Rightarrow \frac{1}{13} \le \frac{1}{10+3\cos x} \le \frac{1}{7}$

$\Rightarrow \frac{2\pi}{13} \le \int_0^{2\pi} \frac{1}{10+3\cos x} dx \le \frac{2\pi}{7}$

30. We have $\int_1^2 [a^2 + (4-4a)x + 4x^3]\, dx \le 12$

$\Rightarrow \left[a^2 x(2-2a)x^2 + x^4\right]_1^2 \le 12$

$\Rightarrow a^2 + (2-2a)(4-1) + (16-1) \le 12$

$\Rightarrow a^2 + 6 - 6a + 15 \le 12 \quad \Rightarrow \quad a^2 - 6a + 9 \le 0$

$\Rightarrow (a-3)^2 \le 0 \quad \Rightarrow \quad a = 3$

31. $\int_0^a \sqrt{a^2 - x^2}\, dx = \left[\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_0^a$

$= \left[0 + \frac{a^2}{2}\sin^{-1}(1) - 0 - \frac{a^2}{2}\sin^{-1}(0)\right]$

$= \frac{a^2}{2} \cdot \frac{\pi}{2} - 0 = \frac{a^2\pi}{4}$

32. $\int_1^{16} \tan^{-1}\sqrt{\sqrt{x}-1}dx = \int_1^{16} \tan^{-1}\sqrt{\sqrt{x}-1}\cdot 1\,dx$

$= x\tan^{-1}\sqrt{\sqrt{x}-1}\Big|_1^{16} - \int_1^{16} \frac{1}{1+(\sqrt{x}-1)}\cdot\frac{1}{2\sqrt{\sqrt{x}-1}}\cdot\frac{1}{2\sqrt{x}}\cdot x\,dx$

[Integrating by parts]

$= \frac{16\pi}{3} - \frac{1}{4}\int_1^{16}\frac{dx}{\sqrt{\sqrt{x}-1}}$

Put $\sqrt{x} = 1+t^2 \Rightarrow x = t^4+2t^2+1$

so that $dx = (4t^3+4t)dt$

When $x=1, t=0$. When $x=16, t=\sqrt{3}$

$= \frac{16\pi}{3} - \frac{1}{4}\int_0^{\sqrt{3}}\frac{4t(t^2+1)dt}{t}$

$= \frac{16\pi}{3} - \int_0^{\sqrt{3}}(t^2+1)dt = \frac{16\pi}{3} - \left[\frac{t^3}{3}+t\right]_0^{\sqrt{3}}$

$= \frac{16\pi}{3} - [\sqrt{3}+\sqrt{3}] = \frac{16\pi}{3} - 2\sqrt{3}$

33. $\int_{\pi/4}^{\pi/2}\text{cosec}^2 x\,dx = [-\cot x]_{\pi/4}^{\pi/2} = \left(-\cot\frac{\pi}{2}+\cot\frac{\pi}{4}\right) = 1$

34. Given $= \lim_{n\to\infty}\sum_{r=1}^{n}\frac{r\sqrt{r}}{n^{5/2}}$

$= \lim_{n\to\infty}\sum_{r=1}^{n}\left(\frac{r}{n}\right)^{3/2}\frac{1}{n} = \int_0^1 x^{3/2}dx = \int_0^1 x\sqrt{x}\,dx$

35. $f(x) = \tan x - \tan^3 x + \tan^5 x - \cdots \infty$

$f(x) = \frac{\tan x}{1+\tan^2 x} = \frac{\tan x}{\sec^2 x} = \frac{\sin 2x}{2}$

$\therefore \int_0^{\pi/4} f(x)dx = \int_0^{\pi/4}\frac{\sin 2x}{2}dx$

$= \left[-\frac{\cos 2x}{4}\right]_0^{\pi/4} = \frac{1}{4}$

36. Let $I = \int_0^{\pi}\frac{x^3\cos^4 x\sin^2 x}{\pi^2-3\pi x+3x^2}dx$

$= \int_0^{\pi}\frac{(\pi-x)^3\cos^4(\pi-x)\sin^2(\pi-x)}{\pi^2-3\pi(\pi-x)+3(\pi-x)^2}dx$

$= \int_0^{\pi}\frac{(\pi^3-3\pi^2x+3\pi x^2-x^3)\cos^4 x\sin^2 x}{\pi^2-3\pi x+3x^2}dx$

$= \int_0^{\pi}\frac{(\pi^3-3\pi^2x+3\pi x^2)\cos^4 x\sin^2 x}{\pi^2-3\pi x+3x^2}dx - I$

$\Rightarrow 2I = \pi\int_0^{\pi}\frac{(\pi^2-3\pi x+3x^2)\cos^4 x\sin^2 x}{(\pi^2-3\pi x+3x^2)}dx$

$\therefore I = \pi\int_0^{\pi}\cos^4 x\sin^2 x\,dx = 2\pi\cdot\frac{3\cdot1\cdot1}{6\cdot4\cdot2}\frac{\pi}{2}$

$\therefore I = \frac{\pi^2}{32}$

37. $f(x)$ is a continuous function.

Let us consider $f(x) = x$

$\therefore \int_{-3}^{5} 2x\,dx = [x^2]_{-3}^{5} = 16$

and $\int_{-6}^{10}(x-1)dx = \left[\frac{x^2}{2}-x\right]_{-6}^{10} = 16$

$\therefore \int_{-3}^{5} 2f(x)dx = \int_{-6}^{10} f(x-1)dx$

38. Clearly, $\sqrt{\sin(\cos x)}$ is defined for all $x \in \mathbb{R}$. $\ln(-2\cos^2 x + 3\cos x - 1)$ is defined, if $-2\cos^2 x + 3\cos x - 1 > 0$

$\Rightarrow 2\cos^2 x - 3\cos x + 1 < 0$

$\Rightarrow (2\cos x - 1)(\cos x - 1) < 0$

$\Rightarrow \frac{1}{2} < \cos x < 1$

$\Rightarrow 0 < \cos x - \frac{1}{2} < \frac{1}{2}$

$\Rightarrow \left[\cos x - \frac{1}{2}\right] = 0$

$\therefore \int_{x_1}^{x_2}\left[\cos x - \frac{1}{2}\right]dx = \int_{x_1}^{x_2} 0\,dx = 0$

39. $\int_{-1}^{1}\frac{\cosh}{1+e^{2x}}dx = \int_{-1}^{1}\frac{e^x+e^{-x}}{2(1+e^{2x})}dx \quad \left[\because \cosh = \frac{e^x+e^{-x}}{2}\right]$

$= \frac{1}{2}\int_{-1}^{1}\frac{1+e^{2x}}{(1+e^{2x})e^x}dx = \frac{1}{2}\int_{-1}^{1}e^{-x}dx$

$= -\frac{1}{2}[e^{-x}]_{-1}^{1} = -\frac{1}{2}(e^{-1}-e^{1}) = \frac{e^2-1}{2e}$

40. $\int_{-\pi/2}^{\pi/2}\sqrt{\cos x-\cos^3 x}dx = \int_{-\pi/2}^{\pi/2}\sqrt{\cos x}\,|\sin x|dx$

$= 2\int_0^{\pi/2}\sqrt{\cos x}\,|\sin x|dx$

$= 2\int_0^{\pi/2}\sqrt{\cos x}\sin x\,dx$

$= -2\int_1^0\sqrt{t}\,dt$, where $t = \cos x$

$= -\frac{4}{3}(0-1) = \frac{4}{3}$

41. $\sin^4 x+\cos^4 x=(\sin^2 x+\cos^2 x)^2-2\sin^2 x\cos^2 x$

$$=1-\frac{1}{2}(2\sin x\cos x)^2$$

$$=1-\frac{1}{2}\sin^2 2x$$

$$=1-\frac{1}{4}(1-\cos 4x)$$

$$=\frac{3}{4}+\frac{1}{4}\cos 4x$$

which is periodic with period $\frac{2\pi}{4}=\frac{\pi}{2}$

$\therefore$ Least and greatest values of $\sin^4 x+\cos^4 x$ are $\frac{3}{4}-\frac{1}{4}$ and $\frac{3}{4}+1$, i.e. $\frac{1}{2}$ and $\frac{7}{4}$ $(\because\ -1\le\cos 4x\le 1)$

$$\therefore\quad \left(\frac{\pi}{2}-0\right)\times\frac{1}{2}<\int_0^{\pi/2}(\sin^4 x+\cos^4 x)dx<\left(\frac{\pi}{2}-0\right)\times\frac{7}{4}$$

$$\Rightarrow\quad \frac{\pi}{4}<\int_0^{\pi/2}(\sin^4 x+\cos^4 x)dx<\frac{7\pi}{8}$$

42. $I=\int_{\pi/6}^{\pi/3}\dfrac{f(x)dx}{f(x)+f[(\pi/2)-x]}$ (i)

$$=\int_{\pi/6}^{\pi/3}\frac{f\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)dx}{f\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)+f\left[\frac{\pi}{2}-\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)\right]}$$

(By property)

$$=\int_{\pi/6}^{\pi/3}\frac{f[(\pi/2)-x]}{f[(\pi/2)-x]+f(x)}dx \qquad \text{(ii)}$$

Adding (i) and (ii), we get

$$2I=\int_{\pi/6}^{\pi/3}1dx=\frac{\pi}{3}-\frac{\pi}{6}=\frac{\pi}{6}$$

$$\therefore\quad I=\frac{\pi}{12}$$

Taking $f(x)=\sin^3 xdx$, we get $I=\dfrac{\pi}{12}$

43. For $0<x<1, x>x^2\Rightarrow -x<-x^2$

$\therefore\quad e^{-x}<e^{-x^2}$

$\therefore\quad e^{-x}\cos^2 x<e^{-x^2}\cos^2 x$

$\therefore\quad \int_0^1 e^{-x}\cos^2 xdx<\int_0^1 e^{-x^2}\cos^2 dx$

44. $\int_{-1}^{1}\frac{|x|}{x}dx=\int_{-1}^{0}\frac{|x|}{x}dx+\int_0^1\frac{|x|}{x}dx$

$$=\int_{-1}^{0}\frac{-x}{x}dx+\int_0^1\frac{x}{x}dx=\int_{-1}^{0}dx+\int_0^1 dx=0$$

45. $\int_0^{10/\pi}|\sin x|dx=10\int_0^{\pi}|\sin x|dx$

$$=10\left(\int_0^{\pi/2}\sin xdx+\int_{\pi/2}^{\pi}\sin xdx\right)$$

$$=10\left[[-\cos x]_0^{\pi/2}+[-\cos x]_{\pi/2}^{\pi}\right]$$

$$=10\left[\left(-\cos\frac{\pi}{2}+\cos 0\right)+\left(-\cos\pi+\cos\frac{\pi}{2}\right)\right]$$

$$=10[(-0+1)+(1+0)]=20$$

46. $I_n=\int_0^{\pi/4}\tan^n xdx$

$$\therefore\quad I_{n+2}=\int_0^{\pi/4}\tan^{n+2}xdx$$

$$\therefore\quad I_n+I_{n+2}=\int_0^{\pi/4}\tan^n x(1+\tan^2 x)dx$$

$$=\int_0^{\pi/4}\tan^n x\sec^2 xdx=\int_0^1 t^n dt$$

[Putting $\tan x=t$ so that $\sec^2 xdx=dt$.

When $x=0,t=0$; when $x=\frac{\pi}{4},t=1$]

$$=\left[\frac{t^{n+1}}{n+1}\right]_0^1=\frac{1}{n+1}$$

$$\therefore\quad \lim_{n\to\infty}n[I_n+I_{n+2}]=\lim_{n\to\infty}n\frac{1}{n+1}$$

$$=\lim_{n\to\infty}\frac{1}{1+1/n}=\frac{1}{1+0}=1$$

47. Let $I=\int_0^{\pi/2}\dfrac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ (i)

$$\therefore\quad I=\int_0^{\pi/2}\frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}+\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}dx$$

$$\Rightarrow\ cI=\int_0^{\pi/2}\frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx \qquad \text{(ii)}$$

Adding (i) and (ii), we get

$$2I=\int_0^{\pi/2}I.dx=[x]_0^{\pi/2}=\frac{\pi}{2}-0=\frac{\pi}{2}$$

Hence $I=\frac{\pi}{4}$

48. Let $I=\int_a^b xf(x)dx$ (i)

$$\Rightarrow\ I=\int_a^b(a+b-x)f(x)dx$$

[since $f(a+b-x)=f(x)$, given]

$$\Rightarrow\ I=(a+b)\int_a^b f(x)dx-I \qquad \text{[from Eq. (i)]}$$

$$\Rightarrow\ 2I=(a+b)\int_a^b f(x)dx$$

$$\Rightarrow\ I=\left(\frac{a+b}{2}\right)\int_a^b f(x)dx$$

49. $\lim_{n\to\infty}\frac{1+2^4+3^4+\cdots+n^4}{n^5}-\lim_{n\to\infty}\frac{1+2^3+3^3+\cdots+n^3}{n^5}$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\right)^4-\lim_{n\to\infty}\frac{1}{n}\times\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\right)^3$$

$$=\int_0^1 x^4dx-\lim_{n\to\infty}\frac{1}{n}\times\int_0^1 x^3dx=\left[\frac{x^5}{5}\right]_0^1-0=\frac{1}{5}$$

50. $I=\int_0^1 x(1-x)^n dx$

$$=\int_0^1(1-x)[1-(1-x)]^n dx=\int_0^1(1-x)x^n dx$$

$$=\int_0^1 x^n dx-\int_0^1 x^{x+1}dx=\left[\frac{x^{n+1}}{n+1}\right]_0^1-\left[\frac{x^{n+2}}{n+2}\right]_0^1$$

$$=\left(\frac{1}{n+1}-0\right)-\left(\frac{1}{n+2}-0\right)=\frac{1}{n+1}-\frac{1}{n+2}$$

51. For $\int_1^4\frac{3}{x}e^{\sin x^3}dx$, putting $x^3=t$ so that $3x^2dx=dt$.

When $x=1,t=1$; when $x=4,t=64$

$$\therefore\ \int_1^4\frac{3}{x}e^{\sin x^3}dx=\int_1^{64}\frac{3}{x}e^{\sin t}\frac{dt}{3x^2}=\int_1^{64}\frac{1}{t}e^{\sin t}dt$$

$$=\int_1^{64}\frac{d}{dt}F(t)\cdot dt=F(64)-F(1)$$

Hence, $k=64$.

52. $\int_0^1 f(x)g(x)dx=\int_0^1 e^x(x^2-e^x)dx$

[$\because$ $f(x)=e^x$ satisfies $f'(x)=f(x)$ and $f(0)=1$]

$$=\int_0^1 x^2e^x dx-\int_0^1 e^{2x}dx$$

$$=\left[x^2e^x\right]_0^1-\int_0^1(2x)e^x dx-\left[\frac{e^{2x}}{2}\right]_0^1$$

$$=e-2\int_0^1 xe^x dx-\frac{1}{2}[e^2-1]$$

$$=e-2\left[\left[xe^x\right]_0^1-\int_0^1 e^x dx\right]-\frac{1}{2}e^2+\frac{1}{2}$$

$$=e-2[e-(e-1)]-\frac{1}{2}e^2+\frac{1}{2}$$

$$=e-2-\frac{1}{2}e^2+\frac{1}{2}=e-\frac{e^2}{2}-\frac{3}{2}$$

53. $f(a)=\frac{e^x}{1+e^x}$ (i)

and $f(-a)=\frac{e^{-a}}{1+e^{-a}}=\frac{1}{1+e^a}$ (ii)

On adding Eqs. (i) and (ii), we get

$$f(a)+f(-a)=1$$

$$\Rightarrow\ f(a)=1-f(-a)$$

Let $f(-a)=t$ then $f(a)=1-t$

Now, $I_1=\int_t^{1-t}gx[x(1-x)]dx$ (iii)

$$\Rightarrow\ I_1=\int_t^{1-t}(1-x)g[x(1-x)]dx \qquad \text{(iv)}$$

On adding Eqs. (iii) and (iv), we get

$$2I_1=\int_t^{1-t}g[x(1-x)]dx=I_2\ \Rightarrow\ \frac{I_2}{I_1}=2$$

54. $\int_{-2}^3\left|1-x^2\right|dx=\int_{-2}^{-1}\left|1-x^2\right|dx+\int_{-1}^1\left|1-x^2\right|+\int_1^3\left|1-x^2\right|dx$

$$=\int_{-2}^{-1}(x^2-1)dx+\int_{-1}^1(1-x^2)+\int_1^3-(1-x^2)dx$$

[$\because$ When $x<-1\Rightarrow -x>1\Rightarrow x^2>1\Rightarrow 1-x^2<0$

$\Rightarrow |1-x^2|=x^2-1$;

when $x<1\Rightarrow x^2<1\Rightarrow 1-x^2>0\Rightarrow |1-x^2|=1-x^2$;

when $x>1\Rightarrow x^2>1\Rightarrow 1-x^2<0\Rightarrow |1-x^2|=-(1-x^2)$]

$$=\left[\frac{x^3}{3}-x\right]_{-2}^{-1}+\left[x-\frac{x^3}{3}\right]_{-1}^{1}+\left[-x+\frac{x^3}{3}\right]_{1}^{3}$$

$$=\left[\left(-\frac{1}{3}+1\right)-\left(-\frac{8}{3}+2\right)\right]+\left[\left(1-\frac{1}{3}\right)-\left(-1+\frac{1}{3}\right)\right]+\left[(-3+9)-\left(-1+\frac{1}{3}\right)\right]$$

$$=\frac{2}{3}+\frac{2}{3}+\frac{2}{3}+\frac{2}{3}+6+\frac{2}{3}=6+\frac{10}{3}=\frac{28}{3}$$

55. $I=\int_0^{\pi/2}\frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}}dx$

$$=\int_0^{\pi/2}\frac{(\sin x+\cos x)^2}{\sqrt{\sin^2 x+\cos^2 x+2\sin x\cos x}}dx$$

$$=\int_0^{\pi/2}\frac{(\sin x+\cos x)^2}{\sqrt{(\sin x+\cos x)^2}}dx$$

$$=\int_0^{\pi/2}(\sin x+\cos x)dx=\left[-\cos x+\sin x\right]_0^{\pi/2}$$

$$=\left(-\cos\frac{\pi}{2}+\sin\frac{\pi}{2}\right)-(-\cos 0+\sin 0)$$

$$=(-0+1)-(-1+0)=1+1=2$$

56. We have $\frac{dx}{dy}=\frac{1}{\sqrt{1+6y^3}}\cdot 1$

$$\Rightarrow \frac{dx}{dy}=\sqrt{1+6y^3}$$

$$\therefore \frac{d^2y}{dx^2}=\frac{1}{2\sqrt{1+6y^3}}\cdot 18y^2\cdot\frac{dy}{dx}$$

$$=9y^2\cdot\frac{1}{\sqrt{1+6y^3}}\cdot\sqrt{1+6y^3}=9y^3$$

57. For $0<x<1, x^2>x^3$ and for $1<x<2$

$\therefore\ 2^{x^2}>2^{x^3}$ for $0<x<1$

$\therefore\ \int_0^1 2^{x^2}dx>\int_0^1 2^{x^3}dx.$

Hence $I_1>I_2$.

58. $\lim_{x\to 2}\int_6^{f(x)}\frac{4t^3}{x-2}dt=\lim_{x\to 2}\frac{\int_6^{f(x)}4t^3dt}{x-2}$ $\left[\frac{0}{0}\text{ form}\right]$

$$=\lim_{x\to 2}\frac{4[f(x)]^3\times f'(x)}{1}$$

$$=4[f(2)]^3\times f'(2)$$

$$=4[(6)]^3\times\frac{1}{48}=4\left(\frac{18}{4}\right)=18$$

59. Let $I=\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}dx, a>0$ (i)

$$\therefore\ I=\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^{-x}}dx$$

$$\left[\text{using }\int_a^b f(x)dx=\int_a^b f(a+b-x)\,dx\right]$$

$$\Rightarrow\ I=\int_{-\pi}^{\pi}\frac{a^x\cos^2 x}{1+a^x}dx \quad \text{(ii)}$$

Adding (i) and (ii), we get

$$2I=\int_{-\pi}^{\pi}\cos^2 x dx$$

$$=2\int_0^{\pi}\cos^2 x dx=4\int_0^{\pi/2}\cos^2 x dx$$

$$=4\left(\frac{1}{2}\right)\left(\frac{\pi}{2}\right)=\pi$$

Hence $I=\frac{\pi}{2}$

60. Let

$$\lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{n}\sec^2\left(\frac{1}{n}\right)^2+\frac{2}{n}\sec^2\left(\frac{2}{n}\right)^2+\cdots+\frac{n}{n}\sec^2\left(\frac{n}{n}\right)^2\right]$$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\right)\sec^2\left(\frac{r}{n}\right)^2$$

$$\Rightarrow\ A=\int_0^1\sec^2(x)^2dx$$

Put $x^2=t\ \Rightarrow\ xdx=\frac{dt}{2}$

$$\therefore\ A=\frac{1}{2}\int_0^1\sec^2 tdt=\frac{1}{2}[\tan t]_0^1=\frac{1}{2}\tan 1$$

61. $I=\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}}\,dx$ (i)

$\therefore \quad I=\int_3^6 \frac{\sqrt{9-x}}{\sqrt{x}+\sqrt{9-x}}\,dx$ (ii)

$$\left[\because \int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$$

Adding (i) and (ii), we get

$$2I=\int_3^6 1.dx=[x]_3^6=6-3=3$$

Hence $I=\frac{3}{2}$

62. Let $I=\int_0^{\pi} xf(\sin x)dx$ (i)

$\therefore \quad I=\int_0^{\pi/2}(\pi-x)f[\sin(\pi-x)]dx$

$\Rightarrow \quad I=\int_0^{\pi}(\pi-x)f(\sin x)dx$ (ii)

Adding (i) and (ii), we get

$$2I=\int_0^{\pi}\pi f(\sin x)dx$$

$$\Rightarrow \quad I=\frac{\pi}{2}\int_0^{\pi} f(\sin x)dx \quad \Rightarrow \quad I=\frac{\pi}{2}\cdot 2\int_0^{\pi} f(\sin x)\,dx$$

$$\Rightarrow \quad I=\pi\int_0^{\pi/2} f\left[\sin\left(\frac{\pi}{2}-x\right)\right]dx$$

$$\Rightarrow \quad I=\pi\int_0^{\pi/2} f(\cos x)dx$$

63. Let $I=\int_{-3\pi/2}^{-\pi}[(x+\pi)^3+\cos^2 x]dx$ (i)

$$\Rightarrow \quad I=\int_{-3\pi/2}^{-\pi/2}\left[\left(-\frac{\pi}{2}-\frac{3\pi}{2}-x+\pi\right)^3+\cos^2\left(-\frac{\pi}{2}-\frac{3\pi}{2}-x\right)\right]dx$$

$$\Rightarrow \quad I=\int_{-3\pi/2}^{-\pi/2}[-(x+\pi)^3+\cos^2 x]dx \qquad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_{-3\pi/2}^{-\pi/2} 2\cos^2 x\,dx$$

$$=\int_{-3\pi/2}^{-\pi/2}(1+\cos 2x)dx=\left[x+\frac{\sin 2x}{2}\right]_{-3\pi/2}^{-\pi/2}$$

$$=\left[-\frac{\pi}{2}+\frac{\sin(-\pi)}{2}-\left(-\frac{3\pi}{2}+\frac{\sin(-3\pi}{2}\right)\right]=\pi$$

$$\Rightarrow \quad I=\frac{\pi}{2}$$

64. Since $\int_1^a [x]f'(x)dx=\int_1^2 f'(x)dx+\int_2^3 2f'(x)dx+\cdots$

$$+\int_{[a]}^{a}[a]f'(x)dx$$

$$=[f(x)]_1^2+2[f(x)]_2^3+\cdots+[a][f(x)]_{[a]}^a$$

$$=f(2)-f(1)+2f(3)-2f(2)+\cdots+[a]f(a)-[a]f([a])$$

$$=[a]f(a)-\{f(1)+f(2)+\cdots+f([a])\}$$

65. $F[x]=\int_0^x \frac{\log t}{1+t}at+\int_0^{1/x}\frac{\log t}{1+t}dt$

$$=\int_0^x\left(\frac{\log t}{1+t}+\frac{\log t}{(1+t)t}\right)dt=\int_0^x\frac{\log t}{t}dt=\frac{1}{2}(\log x)^2$$

$\therefore \quad F[e]=\frac{1}{2}$

66. $\int_{\sqrt{2}}^{x}\frac{dt}{t\sqrt{t^2-1}}=\frac{\pi}{2}$

$$\Rightarrow \quad \left[\sec^{-1}t\right]_{\sqrt{2}}^{x}=\frac{\pi}{2} \quad \Rightarrow \quad \sec^{-1}x-\sec^{-1}\sqrt{2}=\frac{\pi}{2}$$

$$\Rightarrow \quad \sec^{-1}x-\frac{\pi}{4}=\frac{\pi}{2} \quad \Rightarrow \quad \sec^{-1}x=\frac{\pi}{2}+\frac{\pi}{4}=\frac{3\pi}{4}$$

$$\Rightarrow \quad x=-\sqrt{2}$$

67. Since $I=\int_0^1\frac{\sin x}{\sqrt{x}}dx<\int_0^1\frac{x}{\sqrt{x}}dx$ because in $x \in (0, 1)$, $x > 1$ sin x

$$I<\int_0^1\sqrt{x}dx=\frac{2}{3}[x^{3/2}]_0^1\Rightarrow I<\frac{2}{3}$$

For $x\in(0,1)$, $\frac{\cos x}{\sqrt{x}}<\frac{1}{\sqrt{x}}$

and $$J=\int_0^1\frac{\cos x}{\sqrt{x}}dx<\int_0^1 x^{-\frac{1}{2}}dx=2$$

$$\Rightarrow \quad J<2$$

68. Let $I = \int_0^{\pi} [\cot x]\,dx$ (i)

$\Rightarrow I = \int_0^{\pi} [\cot(\pi - x)]\,dx = \int_0^{\pi} [-\cot x]\,dx$ (ii)

On adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi} [\cot \pi]\,dx + \int_0^{\pi} [-\cot x]\,dx = \int_0^{\pi} (-1)\,dx$$

$[\because [x] + [-x] = -1, \text{ if } x \notin z \text{ and } [x] + [-x] = 0, \text{ if } x \in z]$

$\therefore \quad I = -\frac{\pi}{2}$

69. We have

$p'(x) = p'(1-x), \text{ for all } x \in [0,1], \; p(0) = 1, \; p(1) = 41$

$\Rightarrow \; p(x) = -p(1-x) + C$

At $x = 0, p(0) = -p(1) + C$

$\Rightarrow 1 = -41 + C \quad \Rightarrow \quad C = 42$

$\therefore \quad p(x) + p(1-x) = 42$

Now, $I = \int_0^1 p(x)\,dx = \int_0^1 p(1-x)\,dx$

$$\Rightarrow 2I = \int_0^1 (p(x) + p(1-x))\,dx = \int_0^1 42\,dx = 42$$

$\Rightarrow I = 21$

70. $g(x + \pi) = \int_0^x \cos 4t\,dt = g(x) + \int_0^{\pi} \cos 4t\,dt = g(x) + g(\pi)$

Here $g(\pi) = \int_0^{\pi} \cos 4t\,dt = 0$

So answers are (b) or (c).

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. If $\int_1^x \frac{dt}{|t|\sqrt{t^2-1}} = \frac{\pi}{6}$, then x can be equal to

(a) $\frac{2}{\sqrt{3}}$ (b) $\sqrt{3}$

(c) 2 (d) none of these

2. The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\tan^5 x}$ is

(a) 1 (b) $\pi/12$

(c) $\pi/6$ (d) none of these

3. $\int_1^{10\pi} ([\sec^{-1} x] + [\cot^{-1} x])\,dx$, where [.] denotes the greatest integer function, is equal to

(a) $10\pi - \sec 1$ (b) $10\pi + \sec 1$

(c) 10π (d) $\sec 1$

4. If $I = \int_{-1}^{1} \left(\frac{x^2 + \sin x}{1+x^2}\right) dx$, then its value is

(a) 0 (b) 2

(c) $\pi/2$ (d) $2 - \pi/2$

5. Let $f: \mathbb{R} \to \mathbb{R}$, $g: \mathbb{R} \to \mathbb{R}$ be continuous functions. Then the value of the integral $\int_{\ln\lambda}^{\ln 1/\lambda} \frac{f(x^2/4)[f(x) - f(-x)]}{g(x^2/4)[g(x) + g(-x)]} dx$

(a) depends on λ (b) is a non-zero constant

(c) is zero (d) none of these

6. The value of $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} \left(\frac{r}{n+r}\right)$ is

(a) ln 2 (b) 1 + ln 2

(c) 1 – ln 2 (d) 0

7. If $f(0) = 1, f(2) = 3, f'(2) = 5$ and $f'(0)$ is finite, then $\int_0^1 x \cdot f''(2x)\,dx$ is equal to

(a) 0 (b) 1

(c) 2 (d) none of these

8. If $I(m,n) = \int_0^1 t^m (1+t)^n\,dt$, then the expression for $I(m, n)$ in terms of $I(m+1, n-1)$ is

(a) $\frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$

(b) $\frac{n}{m+1} I(m+1, n-1)$

(c) $\frac{2^n}{m+1} + \frac{n}{m+1} I(m+1, n-1)$

(d) $\frac{m}{n+1} I(m+1, n-1)$

9. If $u_{10} = \int_0^{\pi/2} x^{10} \sin x\,dx$, then the value of $u_{10} + 90u_8$ is

(a) $9\left(\frac{\pi}{2}\right)^8$ (b) $\left(\frac{\pi}{2}\right)^9$

(c) $10\left(\frac{\pi}{2}\right)^9$ (d) $9\left(\frac{\pi}{2}\right)^9$

10. $\int_0^{\infty}[2e^{-x}]dx$, where [.] denotes the greatest integer function, is equal to
(a) 0 (b) ln 2
(c) e^2 (d) $2e^{-1}$

11. If $I_1=\int_e^{e^2}\frac{dx}{\ln x}$ and $I_2=\int_1^2\frac{e^x}{x}dx$ then
(a) $I_1=I_2$ (b) $2I_1=I_2$
(c) $I_1=2I_2$ (d) none of these

12. If $f(p,q)=\int_0^{\pi/2}\cos^p x\cos qxdx$, then
(a) $f(p,q)=\frac{q}{p+q}f(p-1,q-1)$
(b) $f(p,q)=\frac{p}{p+q}f(p-1,q-1)$
(c) $f(p,q)=-\frac{p}{p+q}f(p-1,q-1)$
(d) $f(p,q)=-\frac{q}{p+q}f(p-1,q-1)$

13. The value of $\int_{2-\log 3}^{3+\log 3}\frac{\log(4+x)}{\log(4+x)+\log(9-x)}dx$
(a) cannot be evaluated (b) is equal to 5/2
(c) is equal to 1 + 2 log 3 (d) is equal to $\frac{1}{2}+\log 3$

14. If $f(a+b-x)=f(x)$, then $\int_a^b xf(x)dx$ is equal to
(a) $\frac{a-b}{2}\int_a^b f(x)dx$ (b) $\left(\frac{a+b}{2}\right)\int_a^b f(x)dx$
(c) 0 (d) none of these

15. Let $I_1=\int_0^{3\pi}f(\cos^2 x)dx, I_2=\int_0^{2\pi}f(\cos^2 x)dx$ and $I_3=\int_0^{\pi}f(\cos^2 x)dx$, then
(a) $I_1+2I_3=3I_2$ (b) $I_1=2I_2+I_3$
(c) $I_2+I_3=I_1$ (d) $I_1=2I_3$

16. For an integer n, the integral $\int_0^{\pi}e^{\cos^2 x}\cos^3(2n+1)xdx$ has the value
(a) π (b) 1
(c) 0 (d) none of these

17. If $f(x)=\int_0^x(2\cos^2 3t+3\sin^2 3t)dt, f(x+\pi)$ is equal to
(a) $f(x)+f(\pi)$ (b) $f(x)+2f(\pi/2)$
(c) $f(x)+4f(\pi/4)$ (d) none of these

18. The value of $\int_{1/e}^{\tan x}\frac{tdt}{1+t^2}+\int_{1/e}^{\cot x}\frac{dt}{t(1+t^2)}$ is equal to
(a) 1 (b) 1/2
(c) $\pi/4$ (d) none of these

19. If $\int_0^{11}\frac{11^x}{11^{[x]}}dx=\frac{k}{\log 11}$ (where [ ] denotes the greatest integer function), then the value of k is
(a) 11 (b) 101
(c) 110 (d) none of these

20. $f(x)=\begin{cases}e^{\cos x}\sin x, & \text{for } |x|\le 2\\ 2, & \text{otherwise}\end{cases}$; then $\int_{-2}^{3}f(x)dx$ is equal to
(a) 0 (b) 1
(c) 2 (d) 3

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. $\lim_{h\to 0}\frac{\int_e^{x+h}\ln^2 tdt-\int_a^x\ln^2 tdt}{h}$ equals
(a) 0 (b) $\ln^2 x$
(c) 2 ln x/x (d) does not exist

2. $\lim_{n\to\infty}\left[\frac{1}{2n}+\frac{1}{2n+1}+\cdots+\frac{1}{6n}\right]$ is equal to
(a) ln 3 (b) ln 6
(c) ln 2 (d) 2 ln 2

3. The value of the function $f(x)=1+x+\int_1^x(\ln^2 t+2\ln t)dt$, where $f'(x)$ vanishes, is
(a) e^{-1} (b) 0
(c) $2e^{-1}$ (d) $1+2e^{-1}$

4. The value of the integral $\int_{e^{-1}}^{e^2}\left|\frac{\log_e x}{x}\right|dx$ is
(a) 3/2 (b) 5/2
(c) 3 (d) 5

5. $\lim_{n\to\infty}\sum_{r=1}^{n}\left(\frac{r^3}{r^4+n^4}\right)$ equals

(a) $\log 2$ (b) $\frac{1}{2}\log 2$

(c) $\frac{1}{3}\log 2$ (d) $\frac{1}{4}\log 2$

6. If $\int_0^x f(t)dt = x + \int_x^1 t, f(t)dt$, then the value of $f(1)$ is

(a) 1/2 (b) 0

(c) 1 (d) $-(1/2)$

7. $\lim_{n\to\infty}\sum_{r=2n+1}^{3n}\frac{n}{r^2-n^2}$ is equal to

(a) $\log\sqrt{\frac{2}{3}}$ (b) $\log\sqrt{\frac{3}{2}}$

(c) $\log\frac{2}{3}$ (d) $\log\frac{3}{2}$

8. Let $f(x) = x - [x]$ for every real number x, where [.] denotes the greatest integer function. Then $\int_{-1}^{1} f(x)dx$ is

(a) 1 (b) 2

(c) 0 (d) 1/2

9. The value of $\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\cdots\left(1+\frac{n^2}{n^2}\right)\right]^{1/n}$ is

(a) $\frac{e^{\pi/2}}{2e^2}$ (b) $2e^2e^{\pi/2}$

(c) $\frac{2}{e^2}e^{\pi/2}$ (d) none of these

10. If f is a continuous function, then $\frac{1}{k}\int_{ak}^{bk} f\left(\frac{x}{k}\right)dx$ is

(a) $\frac{1}{k}\int_a^b f(x)dx$ (b) $\int_a^b f(x)dx$

(c) $k\int_a^b f(x)dx$ (d) none of these

11. If $f(x)$ is a function satisfying $f\left(\frac{1}{x}\right)+x^2f(x)=0$ for all non-zero x, then $\int_{\sin\theta}^{\operatorname{cosec}\theta} f(x)dx$ equals

(a) $\sin\theta+\operatorname{cosec}\theta$ (b) $\sin^2\theta$

(c) $\operatorname{cosec}^2\theta$ (d) none of these

12. If $I_1 = \int_0^{n\pi} f(|\cos x|)dx$ and $I_2 = \int_0^{5\pi} f(|\cos x|)dx$, then $(n \in N)$

(a) $nI_1 = 5I_2$ (b) $I_1 + I_2 = n + 5$

(c) $\frac{I_1}{n} = \frac{I_2}{5}$ (d) none of these

13. $\lim_{n\to\infty}\left[\sin\frac{\pi}{2n}\cdot\sin\frac{2\pi}{2n}\cdot\sin\frac{3\pi}{2n}\cdots\sin\frac{(n-1)\pi}{n}\right]^{1/n}$ is equal to

(a) 1/2 (b) 1/3

(c) 1/4 (d) none of these

14. The value of k if $\int_0^{\pi/3}\frac{\cos x}{3+4\sin x}dx = k\cdot\log\left(\frac{3+2\sqrt{3}}{3}\right)$ is equal to

(a) 1/2 (b) 1/3

(c) 1/4 (d) 1/8

15. If $[x]$ stands for the greatest integer function, the value of $\int_4^{10}\frac{[x^2]}{[x^2-28x+196]+[x^2]}dx$ is

(a) 0 (b) 1

(c) 3 (d) none of these

16. The value of $\int_{-\pi/4}^{\pi/4}\frac{dx}{\sec^2 x(1+\sin x)}$ is

(a) $\pi/4$ (b) π

(c) $\pi/2$ (d) 2π

17. If $f(a+b-x) = f(x)$, then $\int_a^b xf(x)dx$ is equal to

(a) $\frac{a+b}{2}\int_a^b f(b-x)dx$ (b) $\frac{a+b}{2}\int_a^b f(x)dx$

(c) $\frac{b-a}{2}\int_a^b f(x)dx$ (d) none of these

18. Let $f(x) = \min(\tan x, \cot x), 0 \le x \le \frac{\pi}{2}$, then $\int_0^{\pi/2} f(x)dx$ is equal to

(a) ln 2 (b) $\ln\sqrt{2}$

(c) ln 3 (d) none of these

19. If $\int_0^{100} f(x)dx = a$, then $\sum_{r=1}^{100}\left[\int_0^1 f(r-1+x)dx\right] =$

(a) $100a$ (b) a

(c) 0 (d) $10a$

20. If $f(x) = \int_{x^2}^{x^2+1} e^{-t^2}dt$, then $f(x)$ increases in

(a) (2, 2) (b) no value of x

(c) $(0, \infty)$ (d) $(-\infty, 0)$

ANSWERS

Level 1

1. (a)	**2.** (b)	**3.** (a)	**4.** (d)	**5.** (c)	**6.** (c)	**7.** (c)	**8.** (a)	**9.** (c)	**10.** (b)
11. (a)	**12.** (b)	**13.** (d)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (a) and (b)		**18.** (a)	**19.** (c)
20. (c)									

Level 2

1. (b)	**2.** (a)	**3.** (d)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (b)
11. (d)	**12.** (c)	**13.** (c)	**14.** (c)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (b)	**20.** (d)

16

AUGUSTIN-LOUIS CAUCHY

Applications of Integration

- Different Areas Bounded by the Curve
- Area between Two Curves
- Some Well-known Curves

DIFFERENT AREAS BOUNDED BY THE CURVE

- Area bounded by the curve $y = f(x)$, the x-axis and between the ordinates at $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^b y\,dx = \int_a^b f(x)\,dx$$

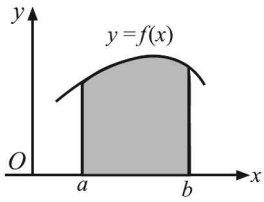


Fig. 16.1

- Area bounded by the curve $y = f(x)$, the y-axis and between the abscissas at $y = c$ and $y = d$ is given by

$$\text{Area} = \int_c^d x\,dy = \int_c^d g(y)\,dy$$

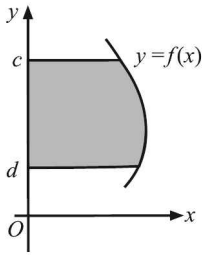


Fig. 16.2

- Area bounded by the two curves $y = f(x)$ and $y = g(x)$, such that $0 \le g(x) \le f(x)$ for all $x \in [a, b]$ and between the ordinates at $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^b [f(x) - g(x)]\,dx$$

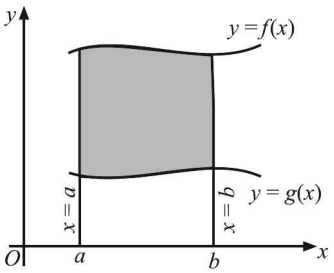


Fig. 16.3

- The area bounded by the curve $y = h(x)$, the x-axis and the two ordinates $x = a$ and $x = b$ is given by

$$\text{Area} = \left|\int_a^c y\,dx\right| + \left|\int_c^b y\,dx\right|$$

where c is a point in between a and b.

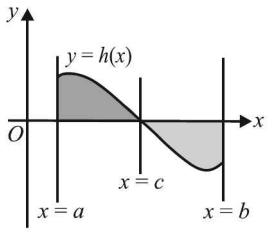


Fig. 16.4

Note For **symmetry**, we remember the following points:

1. **Symmetry about x-axis:** If on changing y to $-y$, there is no change in the equation, then the curve is symmetrical about the x-axis.
   Example: $y^2 = 4ax$ is symmetrical about the x-axis.
2. **Symmetry about y-axis:** If on changing x to $-x$, there is no change in the equation, then the curve is symmetrical about the y-axis.
   Example: $x^2 = 4ay$ is symmetrical about the y-axis.
3. **Symmetry in opposite quadrants:** If on changing x to $-x$ and y to $-y$, there is no change in the equation, then the curve is symmetrical in opposite quadrants.
   Example: $x^2 + y^2 = r^2$ is symmetrical in opposite quadrants.
4. **Symmetry about the line $y = x$:** If on interchanging x and y, there is no change in the equation, then the curve is symmetrical about the line $y = x$.
   Example: $x^2 + y^2 = r^2$ is symmetrical about the line $y = x$.

AREA BETWEEN TWO CURVES AND AXIS

1. Area bounded by the two curves $x = f(y)$ and $x = g(y)$ such that $0 \le g(y) \le f(y)$ for all $y \in [c, d]$ and between the abscissas at $y = c$ and $y = d$ is given by

$$\text{Area} = \int_c^d [f(y) - g(y)]\,dy$$

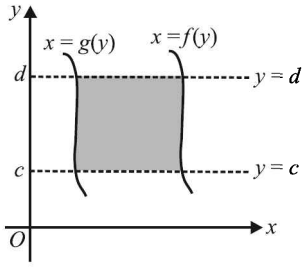


Fig. 16.5

2. The area bounded by the curves $y = f(x)$ and $y = g(x)$ between the ordinates $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^c f(x)dx + \int_c^b g(x)dx$$

where $x = c$ is the point of intersection of the two curves.

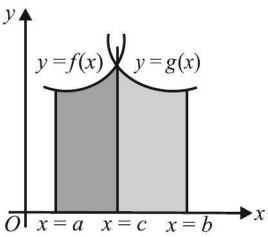


Fig. 16.6

ASYMPTOTES

Asymptotes are the lines which are tangents to the curve at infinity. In other words, the branch of the curve moves almost parallel to asymptot. To find the equation of asymptotes of any equation of the curve, we can use the following steps:

Step I: Find y in terms of x and if there exists real value(s) of x such that the denominator becomes zero, then x equals to that value(s), which is (are) asymptote(s) to that curve.

Step II: Similarly, find x in terms of y and if there exists real value(s) of y such that the denominator becomes zero, then y equals to that value(s), which is (are) asymptote(s) to that curve.

Example: Let the equation of the curve be $x^2y^2 = a^2(y^2 - x^2)$. Since the degree of the equation is 4 and both x^4 and x^3 are missing, hence, equate the coefficient of y^2 to zero, i.e. $x^2 - a^2 = 0$. Hence, $x = \pm a$ are the asymptotes parallel to the y-axis.

SOME WELL-KNOWN CURVES

1. Circle

(i) $x^2 + y^2 = a^2$

This represents a circle with the centre at (0, 0) and the radius as a. It is symmetric about both the axes.

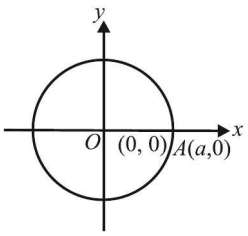


Fig. 16.7

(ii) $(x - h)^2 + (y - k)^2 = a^2$

This is a circle with the centre at (h, k) and the radius as a.

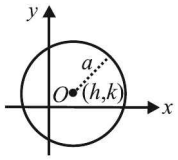


Fig. 16.8

(iii) $x^2 + y^2 + 2gx + 2fy + c = 0$

This is a circle with the centre at $(-g, -f)$ and radius as $\sqrt{g^2 + f^2 - c}$.

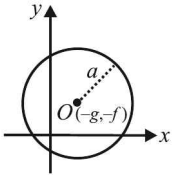


Fig. 16.9

(iv) $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$

End points of the diameter are (x_1, y_1) and (x_2, y_2).

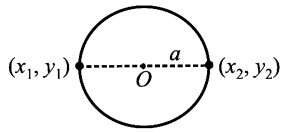


Fig. 16.10

2. Ellipse

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

It is symmetric about both the axes, meeting the x-axis at $(\pm a, 0)$ and the y-axis at $(0, \pm b)$, where $b < a$. Here,

$AA_1 = 2a$ = Length of major axis,

$BB_1 = 2b$ = Length of minor axis.

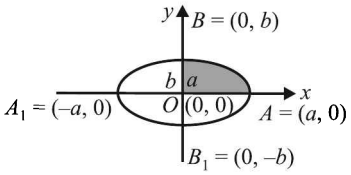


Fig. 16.11

3. Parabola

(i) $y^2 = 4ax$ (standard equation)

It is symmetric about the x-axis.

Vertex : (0, 0)
Focus : $(a, 0)$
Axis : x-axis or $y = 0$
Directrix : $x = -a$
Latus rectum : LM

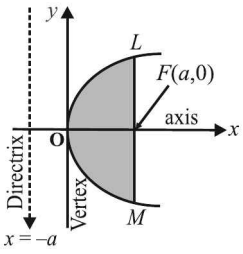


Fig. 16.12

(ii) $y^2 = -4ax$

It is symmetric about the x-axis.

Vertex : (0, 0)
Focus : $(-a, 0)$
Axis : x-axis or $y = 0$
Directrix : $x = a$
Latus rectum : LM

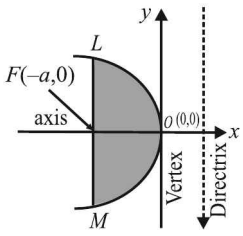


Fig. 16.13

(iii) $x^2 = 4ay$

It is symmetrical about the y-axis.

Vertex : (0, 0)
Focus : $(0, a)$
Axis : y-axis or $x = 0$
Directrix : $y = -a$
Latus rectum : LM

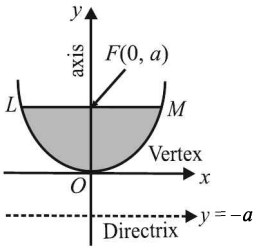


Fig. 16.14

(iv) $x^2 = -4ay$

It is symmetrical about the y-axis.

Vertex : (0, 0)
Focus : $(0, -a)$
Axis : y-axis or $x = 0$
Directrix : $y = a$
Latus rectum : LM

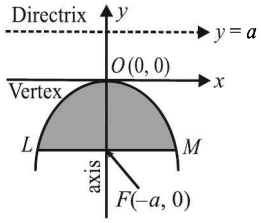


Fig. 16.15

4. Exponential curves

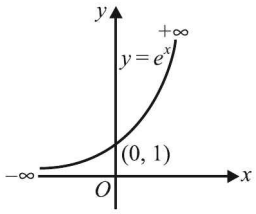


Fig. 16.16

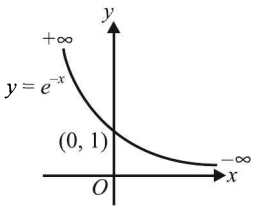


Fig. 16.17

5. Logarithmic curves

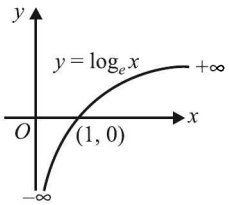


Fig. 16.18

6. Modulus function $y = |x|$

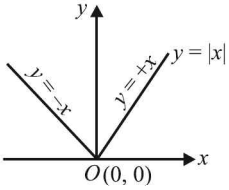
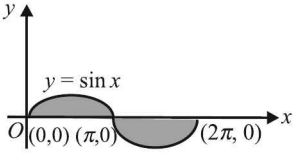
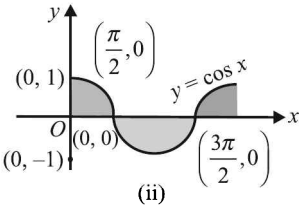


Fig. 16.19

7. Trigonometric curves



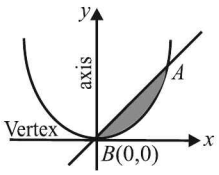
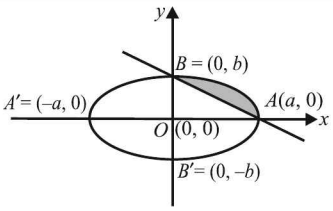
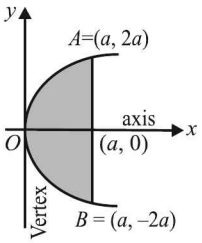
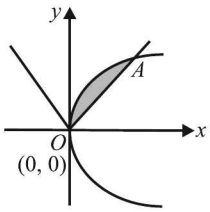
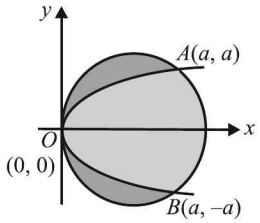
(i)

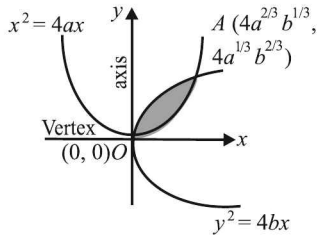
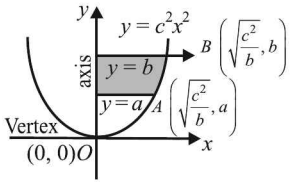
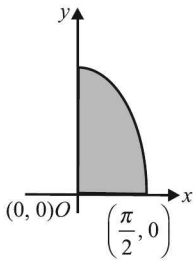
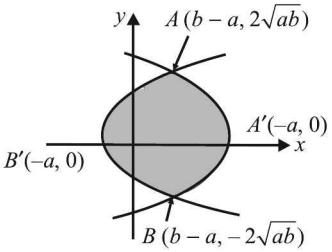
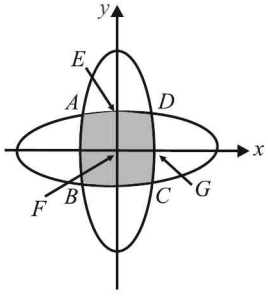


(ii)

Fig. 16.20

☞ AREA BETWEEN TWO CURVES

Curves	Point of Intersection	Area	Shaded Region
$F(x,y)$: $ax^2 \le y \le mx$ $\therefore\ y = ax^2$, $y = mx$	$B(0, 0)$, $A\left(\frac{m}{a}, \frac{m^2}{a}\right)$	$\frac{1}{6}\frac{m^2}{a^2}$ sq. unit	 **Fig. 16.21**
$f(x, y)$: $\frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1$ or $\frac{x}{a}+\frac{y}{b} \ge 1$ or $\frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1 \le \frac{x}{a}+\frac{y}{b}$	$A(a, 0)$, $B(0, b)$	$ab\frac{(\pi-2)}{4}$ sq. unit	 **Fig. 16.22**
Area of parabola $y^2 = 4ax$ and its latus rectum $x = a$	$A(a, 2a)$, $B(a, -2a)$	$\frac{8}{3}a^2$ sq. unit	 **Fig. 16.23**
$f(x, y)$: $y^2 = 4ax$ and $y = \lvert mx \rvert$	$O(0, 0)$, $A\left(\frac{4a}{m^2}, \frac{4a}{m}\right)$	$\frac{8a^2}{3m^3}$ sq. unit	 **Fig. 16.24**
$f(x, y)$: $x^2 + y^2 \le 2ax$ and $y^2 \ge ax$	$O(0, 0)$, $A(a, a)$, $B(a, -a)$	(i) For $x \ge 0$, $y \ge 0$ (area of 1st quadrant) Area = $a^2\left(\frac{\pi}{4}-\frac{2}{3}\right)$ sq. unit (ii) For $x \ge 0$ (for 1st and 4th quadrants) Area $= 2a^2\left(\frac{\pi}{4}-\frac{2}{3}\right)$ $= a^2\left(\frac{\pi}{4}-\frac{2}{3}\right)$	 **Fig. 16.25**

$f(x, y)$: $x^2 = 4ay$ $y^2 = 4bx$	$O(0, 0)$, $A \equiv \left\{ \begin{matrix} 4a^{2/3}b^{1/3}, \\ 4a^{1/3}b^{2/3} \end{matrix} \right\}$	Area $= \frac{16}{3}(ab)$ sq. unit	 **Fig. 16.26**
Area bounded by $y = c^2x^2$ x-axis and the lines $y = a = b$	$O(0, 0)$, A $A = \left(\frac{\sqrt{c^2}}{a}, a \right)$ $B = \left(\frac{\sqrt{c^2}}{b}, b \right)$	Area $= \frac{2(b^{3/2} - a^{3/2})}{3c}$ sq. unit	 **Fig. 16.27**
$y = n \cos 3x$, $\forall 0 \le x \le \frac{\pi}{6}$	$A\ 0 \le x \le \frac{\pi}{6}$ $\therefore\ 0 \le 3x \le \frac{\pi}{2}$	Area $= \frac{n}{3}$ sq. unit	 **Fig. 16.28**
Area bounded by $y^2 = 4a(x + a)$ $y^2 = 4b(b - x)$	$A\ (b - a, 2\sqrt{ab})$ $B\ (b - a, -2\sqrt{ab})$	Area $= \frac{8}{3}\sqrt{ab}\ (a + b)$ sq. unit	 **Fig. 16.29**
Common area bounded by the ellipses $\frac{x^2}{b^2} + \frac{y^2}{a^2} = \frac{1}{a^2b^2}$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{a^2b^2}$ $0 < a < b$	$x = y$ $= \frac{1}{\sqrt{a^2 + b^2}}$ $x = y$ $= -\frac{1}{\sqrt{a^2 + b^2}}$	Area of $ABCD$ $= 4 \times$ area of $EFGD$ $= \frac{4}{ab}\tan^{-1}\left(\frac{a}{b}\right)$ sq. unit	 **Fig. 16.30**

SOLVED PROBLEMS

1. The area of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

(a) πab (b) $\frac{\pi(a^2+b^2)}{4}$

(c) $\pi(a+b)$ (d) none of these

Ans. (a)

Solution:

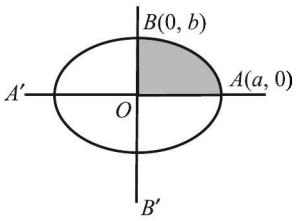


Fig. 16.31

$$A = 4\int_0^a y\,dx$$

The parametric equations of the ellipse are $x = a\cos t$ and $y = b\sin t$.

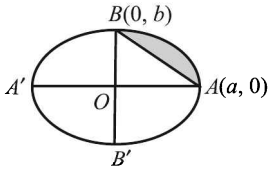


Fig. 16.32

$$\therefore\quad A = 4\int_{\pi/2}^{0}(b\sin t)(-a\sin t)\,dt$$

$$= 4ab\int_0^{\pi/2}\sin^2 t\,dt = 4ab = 4ab\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \pi ab$$

Note You can also do it by

$$a = 4\int_0^a \frac{b}{a}\sqrt{a^2-x^2}\,dx, \text{ etc.}$$

2. Area lying in the first quadrant and bounded by the circle $x^2+y^2=4$ and the line $x=y\sqrt{3}$ is

(a) π (b) $\pi/2$

(c) $\pi/3$ (d) none of these

Ans. (c)

Solution: The line and the curve meet at $P(\sqrt{3},1)$ in the 1st quadrant. Draw perpendicular PM.

$$\therefore\quad A = ar(\Delta OPM) + \int_{\sqrt{3}}^{2} y\,dx$$

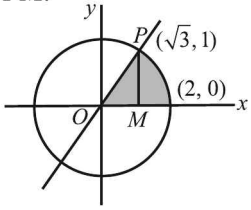


Fig. 16.33

Now $x = 2\cos\theta, y = 2\sin\theta$ and adjust the limits

$$= \frac{1}{2}\sqrt{3}\cdot 1 + \int_{\pi/\theta}^{0}(2\sin\theta)(-2\sin\theta)\,d\theta$$

$$= \frac{\sqrt{3}}{2} + 4\int_0^{\pi/6}\frac{(1-\cos 2\theta)}{2}d\theta$$

$$= \frac{\sqrt{3}}{2} + 2\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/6} = \frac{\sqrt{3}}{2} + 2\left[\frac{\pi}{6} - \frac{1}{2}\frac{\sqrt{3}}{2}\right] = \frac{\pi}{3}$$

Alternative method: $\angle POA = 30° = \frac{\pi}{6}, r = 2$. Area of sector of a circle is $\frac{1}{2}r^2\theta = \frac{1}{2}\cdot 4\cdot\frac{\pi}{6} = \frac{\pi}{3}$. But we have to use integration.

3. The area of the loop between the curve $y = a\sin x$ and the x-axis is

(a) a (b) $2a$

(c) $3a$ (d) none of these

Ans. (b)

Solution: When $y = 0$, we have $a\sin x = 0$, where $x = 0, \pi$. So we have two consecutive values 0 and π of x for which $y = 0$.

Hence, one loop of the curve lies between $x = 0$ and $x = \pi$.

$$\therefore\quad \text{Area of this loop} = \int_0^{\pi} y\,dx = \int_0^{\pi} a\sin x\,dx = 2a$$

4. The area common to the circle $x^2+y^2=16a^2$ and the parabola $y^2 = 6ax$ is

(a) $\frac{4a^2}{3}(4\pi-\sqrt{3})$ (b) $\frac{4a^2}{3}(8\pi-3)$

(c) $\frac{4a^2}{3}(4\pi+\sqrt{3})$ (d) none of these

Ans. (c)

Solution: Solving the two, we get $x^2+6ax-16a^2=0$

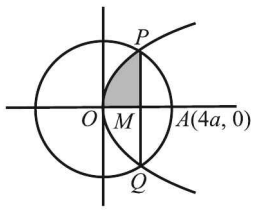


Fig. 16.34

or $(x+8a)(x-2a)=0 \quad \therefore\ x = 2a$

$\therefore\quad y = \pm 2\sqrt{3a}$

The required common area = $2[APOA]$

$$= 2\int y\,dx + 2\int y\,dx$$

$$= 2\int_0^{2a}\sqrt{6a}\sqrt{x}\,dx + 2\int_{2a}^{4a}\sqrt{(4a)^2-x^2}\,dx$$

$$= 2\cdot\sqrt{6a}\,\frac{2}{3}\left[x^{3/2}\right]_0^{2a}$$

$$+ 2\cdot\left[\frac{x}{2}\sqrt{(4a)^2-x^2} + \frac{1}{2}(4a)^2\sin^{-1}\frac{x}{4a}\right]_{2a}^{4a}$$

$$= 2\cdot\sqrt{6a}\frac{2}{3}(2a)\sqrt{2a} + 2\left[(0 - a2\sqrt{3a}) + 8a^2\left(\sin^{-1}1 - \sin^{-1}\frac{1}{2}\right)\right]$$

$$= 2\cdot 2\sqrt{3}\cdot\frac{4}{3}a^2 + 2[-2\sqrt{3}a^2 + 8a^2(\pi/2 - \pi/6)]$$

$$= \frac{16}{3}\sqrt{3}a^2 - 4\sqrt{3}a^2 + 16a^2\frac{\pi}{3}$$

$$= \frac{4\sqrt{3}a^2}{3} + \frac{16\pi a^2}{3} = \frac{4a^2}{3}(4\pi + \sqrt{3})$$

5. The area enclosed between the curves $y = x^3$ and $y = \sqrt{x}$ is [in square units]

(a) $\frac{5}{3}$ (b) $\frac{5}{4}$

(c) $\frac{5}{12}$ (d) $\frac{12}{5}$

Ans. (c)

Solution: $y = \sqrt{x}$ or $y^2 = x$ $(y \geq 0)$ and $y = x^3$.

We get points of intersection (0, 0) and (1, 1)

$\therefore$ Required area $= \int_0^1 (\sqrt{x} - x^3)dx$

$$= \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^4}{4}\right]_0^1 = \frac{5}{12}\text{ sq. unit.}$$

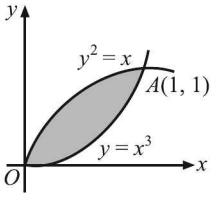


Fig. 16.35

6. Area of the region bounded by the curves $y = 2^x$, $y = 2x - x^2, x = 0$ and $x = 2$ is given by

(a) $\frac{3}{\log 2} - \frac{4}{3}$ (b) $\frac{3}{\log 2} + \frac{4}{3}$

(c) $3\log 2 - \frac{4}{3}$ (d) $3\log^2 - \frac{4}{3}$

Ans. (a)

Solution: Required area

$$= \int_0^2 2^x dx - \int_0^2 (2x - x^2)dx$$

$$= \left[\frac{2^x}{\log 2} - x^2 + \frac{x^3}{3}\right]_0^2$$

$$= \frac{4}{\log 2} - 4 + \frac{8}{3} - \frac{1}{\log 2}$$

$$= \frac{3}{\log 2} - \frac{4}{3}$$

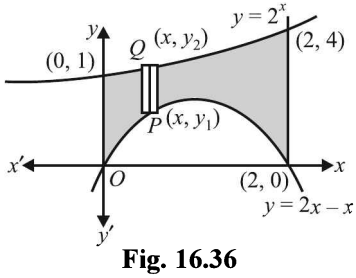


Fig. 16.36

7. Area bounded by the curves satisfying the conditions $\frac{x^2}{25} + \frac{y^2}{36} \leq 1 \leq \frac{x}{5} + \frac{y}{6}$ is given by

(a) $15\left(\frac{\pi}{2} - 1\right)$ sq. units (b) $\frac{15}{4}\left(\frac{\pi}{2} - 1\right)$ sq. units

(c) $30(\pi - 1)$ sq. units (d) $\frac{15}{2}(\pi - 2)$ sq. units

Ans. (d)

Solution: $\frac{x^2}{25} + \frac{y^2}{36} = 1$ and $\frac{x}{5} + \frac{y}{6} = 1$

Solving these two equations, we get (5, 0), (0, 6).

Area of ellipse

$$= 6\int_0^5 \sqrt{1 - \left(\frac{x}{5}\right)^2}dx = \frac{6\times 5\times\pi}{4}$$

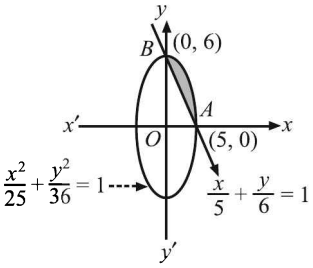


Fig. 16.37

Again, area of the triangle OAB

$$= \frac{1}{2}OA\times OB = \frac{1}{2}\times 5\times 6 = 15$$

$\therefore$ Area bounded by the ellipse and the line

$$= \frac{15}{2}\pi - \frac{2\times 15}{2} = \frac{15}{2}(\pi - 2)\text{ sq. units.}$$

Shortcut method:

Using fact: Area bounded by the curves $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x}{a} + \frac{y}{b} = 1$ is given by

= Area of ellipse $\forall x \geq 0, y \geq 0$ – Area bounded by the line with coordinate axis

$$= \frac{\pi}{4}ab - \frac{1}{2}ab$$

$$= \frac{\pi}{4}(5\cdot 6) - \frac{1}{2}(5\cdot 6)$$

$$= \frac{(5\cdot 6)}{4}(\pi - 2)$$

(If $a > b$)

$$= \frac{15}{4}(\pi - 2)\text{ sq. units (For } a < b \text{ replace } x \leftrightarrow y)$$

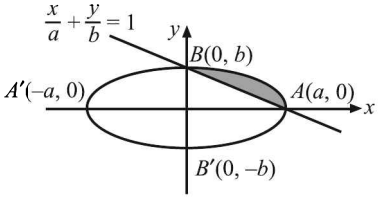


Fig. 16.38

8. The area bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$ is given by

(a) $\frac{45}{7}$ (b) $\frac{25}{4}$

(c) $\frac{\pi}{18}$ (d) $\frac{9}{2}$

Ans. (d)

Solution: The two meet at $3 - x = x^2 + 1$, i.e. $x^2 + x - 2 = 0$

$\Rightarrow (x+2)(x-1) = 0 \quad \Rightarrow \quad x = -2, 1$

$\therefore$ Required area $= \int_{-2}^{1} [(3-x) - (x^2+1)]dx$

$$= \int_{-2}^{1} (2 - x - x^2)dx = \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{1}$$

$$= \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - \frac{4}{2} + \frac{8}{3}\right) = \frac{9}{2}$$

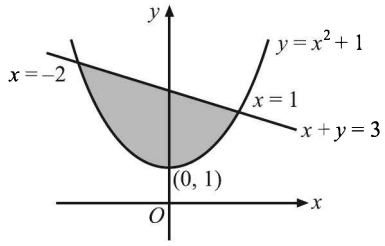


Fig. 16.39

9. The area bounded by the curves $x^2 + y^2 \le 8x$ and $y^2 \ge 4x$ lying in the first quadrant is not equal to

(a) $32\left(\frac{3\pi}{8} - 1\right)$ (b) $\frac{32}{3}\left(\frac{3\pi}{8} - 1\right)$

(c) $4\pi - \frac{32}{3}$ (d) $\frac{1}{3}(12\pi - 32)$

Ans. (a)

Solution: Given $x^2 + y^2 \le 8x$

Let $x^2 + y^2 = 8x$ (i)

$y^2 \ge 4x, y^2 = 4x$ (ii)

$\therefore$ From (i) and (ii), we have

$x^2 = 4x \ \therefore \ x = 0, x = 4$

$\therefore$ Set of points (or points of intersection) are (0, 0) (4, 4). Required area

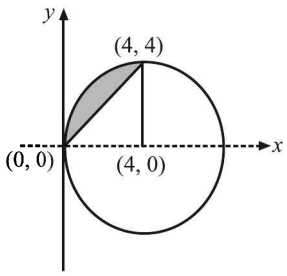


Fig. 16.40

$$= \int_0^4 \sqrt{4^2 - (x-4)^2}\,dx - \int_0^4 2\sqrt{x}\,dx \qquad (\because x^2 + y^2 = 8x)$$

$$\Rightarrow \quad (x-4)^2 + y^2 = 4^2 \quad \Rightarrow \quad y^2 = 4^2 - (x-4)^2$$

$$= \left[\frac{(x-4)}{2}\sqrt{4^2 - (x-4)^2} + \frac{16}{2}\sin^{-1}\left(\frac{x-4}{4}\right)\right]_0^4$$

$$-2 \times \frac{2}{3}\left(x^{\frac{3}{2}}\right)_0^4 = 8 \times \frac{\pi}{2} - \frac{4}{3} \times (4)^{3/2}$$

$$= \left(4\pi - \frac{32}{3}\right) = \frac{32}{3}\left(\frac{3\pi}{8} - 1\right) \text{ sq. units}$$

Shortcut method:

Area bounded by the curves $y^2 + x^2 \le 2ax$ (here $a = 4$) and $y^2 \ge ax$ (here $a = 4$) is given by $\left(\frac{\pi a^2}{4} - \frac{2a^2}{3}\right)$.

Putting $a = 4$, we get the required area.

$\therefore$ Area $= \frac{\pi \times 4^2}{4} - \frac{2 \times 4^2}{3} = 4\pi - \frac{32}{3}$

$$= \frac{32}{3}\left(\frac{3\pi}{8} - 1\right) \text{ sq. units}$$

Here options (b), (c) and (d) represent the area that lies in the first quadrant. So option (a) does not represent the area of the 1st quadrant bounded by the curves.

10. The area bounded by the curves $y^2 = x$ and $y = x^2$ is

(a) $\frac{2}{3}$ sq. unit (b) 1 sq. unit

(c) $\frac{1}{2}$ sq. unit (d) none of these

Ans. (d)

Solution: Required area $= \int_0^1 (\sqrt{x} - x^2)dx$

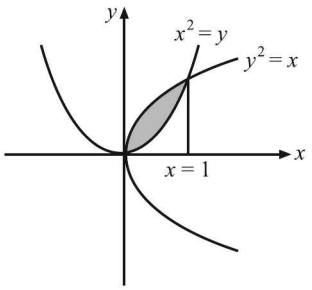


Fig. 16.41

$$= \left[\frac{2x^{3/2}}{3} - \frac{x^3}{3}\right]_0^1 = \left(\frac{2}{3} - \frac{1}{3}\right) = \frac{1}{3} \text{ sq. unit}$$

EXERCISE SET 1

1. The area common to the circle $x^2 + y^2 = 16a^2$ and the parabola $y^2 = 6ax$ is

(a) $\frac{4a^2}{3}(4\pi - \sqrt{3})$ sq. units

(b) $\frac{4a^2}{3}(8\pi - 3)$ sq. units

(c) $\frac{4a^2}{3}(4\pi + \sqrt{3})$ sq. units

(d) none of these

2. The area of the region bounded by the curves $y = |x-1|$ and $y = 3 - |x|$ is

(a) 6 sq. units (b) 2 sq. units

(c) 3 sq. units (d) 4 sq. units

3. AOB is the positive quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in which $OA = a$ and $OB = b$. The area between the arc AB and the chord AB of the ellipse is

(a) $\frac{1}{2}ab(\pi + 2)$ (b) $\frac{1}{4}ab(\pi - 4)$

(c) $\frac{1}{4}ab(\pi - 2)$ (d) none of these

4. The area lying in the first quadrant and bounded by the curve $y = x^3$ and the line $y = 4x$ is

(a) 2 (b) 3

(c) 4 (d) 5

5. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is

(a) 1 (b) 2

(c) $2\sqrt{2}$ (d) 4

6. The area bounded by the parabola $y^2 = x$ and the straight line $2y = x$ is

(a) $\frac{4}{3}$ sq. units (b) 1 sq. unit

(c) $\frac{2}{3}$ sq. unit (d) $\frac{1}{3}$ sq. unit

7. The area bounded by the ellipse $x^2 + 9y^2 = 9$ and the straight line $x + 3y = 3$ is

(a) 3π (b) 4π

(c) 6π (d) 9π

8. The area of a loop of the curve $y^2 = x^2(4 - x^2)$ is

(a) $\frac{8}{3}$ (b) $\frac{16}{3}$

(c) $\frac{32}{3}$ (d) none of these

9. The area bounded by the curve $y = x^3$, the x-axis and two ordinates $x = 1$ and $x = 2$ is

(a) $\frac{15}{2}$ sq. units (b) $\frac{15}{4}$ sq. units

(c) $\frac{17}{2}$ sq. units (d) $\frac{17}{4}$ sq. units

10. If A is the area between the curve $y = \sin x$ and the x-axis in the interval $[0, \pi/4]$, then in the same interval, the area between the curve $y = \cos x$ and the x-axis is

(a) A (b) $\pi/2 - A$

(c) $1 - A$ (d) none of these

11. The area of the smaller portion of the circle $x^2 + y^2 = 4$ cut off by the line $x = 1$ is

(a) $\frac{4\pi + 3\sqrt{3}}{3}$ (b) $\frac{4\pi - 3\sqrt{3}}{3}$

(c) $\frac{2\pi - 3\sqrt{3}}{3}$ (d) $\frac{4\pi - \sqrt{3}}{3}$

12. The area of the smaller segment cut off from the circle $x^2 + y^2 = 9$ by $x = 1$ is

(a) $\frac{1}{2}(9\sec^{-1}(3) - \sqrt{x})$ (b) $9\sec^{-1}(3) - \sqrt{8}$

(c) $\sqrt{8} - 9\sec^{-1}(3)$ (d) none of these

13. The area enclosed by the curve $xy^2 = 4(2 - x)$ and the y-axis is

(a) π (b) 2π

(c) 4π (d) $\pi/2$

14. The area bounded by lines $y = 2 + x, y = 2 - x$ and $x = 2$ is

(a) 3 (b) 4

(c) 8 (d) 16

15. The area bounded by $y = \cos x$, $x = -\frac{\pi}{2}$, $x = 2\pi$ and the axis of x in square units is

(a) 4 (b) 5

(c) 6 (d) 7

16. The area of the figure bounded by the curve $y = \log x$ and $y = (\log x)^2$ is

(a) $3 - e$ (b) $3 + e$

(c) $5 - e$ (d) none of these

17. The area in the first quadrant between $x^2 + y^2 = \pi^2$ and $y = \sin x$ is

(a) $\frac{\pi^3 - 8}{4}$ (b) $\frac{\pi^3}{4}$

(c) $\frac{\pi^3 - 16}{4}$ (d) $\frac{\pi^3 - 8}{2}$

18. The area bounded by the curves $y^2 = x^3$ and $|y| = 2x$ is given by
(a) 2 (b) 32/5
(c) 32/3 (d) none of these

19. The area lying between the curves $y^2 = 4x$ and $y = 2x$ is equal to
(a) 2/3 (b) 1/3
(c) 35/6 (d) none of these

20. The area included between the curves $\sqrt{x} + \sqrt{|y|} = 1$ and $|x| + |y| = 1$ is
(a) 2/3 (b) 1/6
(c) 4/3 (d) none of these

21. The area of the figure bounded by $y = e^x, y = e^{-x}$ and the straight line $x = 1$ is
(a) $e + \frac{1}{e}$ (b) $e - \frac{1}{e}$
(c) $e + \frac{1}{e} - 2$ (d) $e + \frac{1}{e} + 2$

22. The area of the region between the curve $y = 4 - x^2$, $0 \le x \le 3$ and the x-axis is
(a) 16/3 (b) 7/3
(c) 3 (d) 23/3

23. Ratio of the area cut off from a parabola by any double ordinate is that of the corresponding rectangle contained by that double ordinate and its distance from the vertex is
(a) 1 : 2 (b) 1 : 3
(c) 2 : 3 (d) 1

24. The area enclosed between the curves $y = \sqrt{x}$, $x = 2y + 3$ and the x-axis is
(a) 18 (b) 27
(c) 9 (d) none of these

25. The positive value of the parameter a for which the area of the figure bounded by $y = \sin ax$, $y = 0$, $x = \frac{\pi}{a}$ and $x = \frac{\pi}{3a}$ is 3 is equal to
(a) 2 (b) $\frac{1}{2}$
(c) $\frac{2+\sqrt{3}}{3}$ (d) none of these

26. The area bounded by the parabola $y^2 = 4x$ and the line $x + y = 3$ is
(a) 16/3 (b) 32/3
(c) 64/3 (d) none of these

27. The equation of the curve whose slope at any point is thrice its abscissa and which passes through $(-1, -3)$ is
(a) $y = x^2 - 4$ (b) $3y = 2(x^2 - 4)$
(c) $2y = 3(x^2 - 3)$ (d) none of these

28. The area bounded by the curve $y = x^2 + 1$ and the straight line $x + y = 3$ is
(a) $\frac{35}{3}$ (b) $\frac{9}{2}$
(c) $\frac{32}{3}$ (d) $\frac{16}{3}$

29. The area bounded by the curve $|x| + y = 1$ and the x-axis is
(a) 4 (b) 2
(c) 1 (d) 1/2

30. The area enclosed by the curve $y = x^2 e^{x^3}$ between the lines $x = 1$ and $x = 2$ and the x-axis is
(a) $\frac{e^2 - e}{3}$ (b) $\frac{e^8 - e}{3}$
(c) $\frac{e^8 - 1}{3}$ (d) $\frac{e^7}{3}$

31. The area bounded by the curve $y = x(x-1)(x-2)$ and the x-axis is equal to
(a) 1 (b) 1/2
(c) 1/4 (d) none of these

32. The area bounded by the curve $xy^2 = a^2(a - x)$ and the y-axis is
(a) $\pi a^2/2$ (b) πa^2
(c) $3\pi a^2$ (d) none of these

33. The area bounded by the curve $y = \sin^2 x$ and lines $x = \frac{\pi}{2}$, $x = \pi$ and the x-axis is
(a) $\frac{\pi}{2}$ sq. units (b) $\frac{\pi}{4}$ sq. units
(c) $\frac{\pi}{8}$ sq. units (d) none of these

34. The area bounded by the curve $y = x[x]$, the x-axis and the ordinates $x = 1$, $x = -1$ is given by
(a) 0 (b) 1/3
(c) 2/3 (d) none of these

35. The value of c for which the area of the figure bounded by the curve $y = 8x^2 - x^5$, the straight lines $x = 1$ and $x = c$ and the x-axis is equal to 16/3 is
(a) 2 (b) $\sqrt{8 - \sqrt{17}}$
(c) 3 (d) none of these

36. If the area cut off from a parabola by any double ordinate is k times the corresponding rectangle contained by that double ordinate and its distance from the vertex, then k is

(a) 2/3 (b) 1/3
(c) 3/2 (d) 3

37. The area of the region bounded by $x=\frac{1}{2}, x=2, y=\log x$ and $y=2^x$ is

(a) $\frac{3}{2}-\frac{5}{2}\log 2+\frac{3}{\log 2}$ (b) $\frac{3}{2}+\frac{5}{2}\log 2+\frac{3}{\log 2}$

(c) $\frac{3}{2}+\frac{5}{2}\log 2+\frac{3}{\log 2}$ (d) $\frac{3}{2}-\frac{5}{2}\log 2+\frac{3}{\log 2}$

38. The area enclosed by the curve $\frac{x^2}{25}+\frac{y^2}{16}=1$ is

(a) 10π sq. units (b) 20π sq. units
(c) 5π sq. units (d) 4π sq. units

39. For which of the following values of m, is the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equal to 9/2?

(a) -4 (b) -2
(c) 2 (d) none of these

40. The area of the figure bounded by the straight lines $x = 0, x = 2$ and the curves $y=2^x, y=2x-x^2$ is

(a) $\left(\frac{4}{\log 2}-\frac{8}{3}\right)$ sq. units (b) $\left(\frac{4}{\log 2}+\frac{8}{3}\right)$ sq. units

(c) $\left(\frac{8}{\log 3}-\frac{4}{3}\right)$ sq. units (d) $\left(\frac{3}{\log 2}-\frac{4}{3}\right)$ sq. units

41. The area bounded by the curves $y^2=4a(x+a)$ and $y^2=4b(b-x)$, where $a, b > 0$, is

(a) $\frac{8}{3}(a+b)\sqrt{ab}$ sq. units

(b) $\frac{2}{3}(a+b)\sqrt{ab}$ sq. units

(c) $\frac{2}{3}(a+b)2\sqrt{ab}$ sq. units

(d) none of these

42. The area of the region bounded by the curves $y=|x-2|$, $x = 1, x = 3$ and the x-axis is

(a) 3 (b) 2
(c) 1 (d) 4

43. The area bounded by the parabolas $y=4x^2$ and $y=\frac{x^2}{9}$ and the line $y = 2$ is

(a) $\frac{5\sqrt{2}}{3}$ sq. units (b) $\frac{10\sqrt{2}}{3}$ sq. units

(c) $\frac{15\sqrt{2}}{3}$ sq. units (d) $\frac{20\sqrt{2}}{3}$ sq. units

44. If the area bounded by the curve $y^2 = 16x$ and the line $y = mx$ is 2/3, then m is equal to

(a) 3 (b) 4
(c) 1 (d) 2

45. Let $f(x)$ = minimum $f(x)=\left\{e^x, \frac{3}{2}, 1+e^{-x}\right\} 0\le x\le 1$. Then the area bounded by $y = f(x)$, the x-axis, y-axis and the line $x = 1$ is

(a) $\left\{2-\frac{1}{e}+\frac{3}{2}\log\frac{4}{3}-\log 2\right\}$ sq. units

(b) $\left\{\frac{1}{e}+\frac{3}{2}\log\frac{4}{3}-\log 2\right\}$ sq. units

(c) $\left\{3-\frac{1}{e}+\frac{3}{2}\log\frac{4}{3}+\log 2\right\}$ sq. units

(d) $\left\{3-\frac{2}{e}+\frac{3}{2}\log\frac{3}{4}+\log 2\right\}$ sq. units

46. The area bounded by the curves $x^2 = 4y$ and $x = 4y - 2$ is

(a) $\frac{9}{8}$ sq. units (b) $\frac{9}{4}$ sq. units

(c) $\frac{9}{2}$ sq. units (d) 9 sq. units

47. Let $f(x)=\begin{cases}\min[(x-1)^2, 2x(1-x)] & \text{if } 0\le x\le 1/2\\ \min[2x(1-x), x^2] & \text{if } \frac{1}{2}\le x\le 1\end{cases}$

Then the area bounded by the curve and the x-axis is

(a) $\frac{40}{118}$ sq. unit (b) $\frac{21}{108}$ sq. unit

(c) $\frac{40}{128}$ sq. unit (d) $\frac{31}{108}$ sq. unit

48. The area bounded by the curves $y = f(x), y = x$ and the lines $x = 1, x = t$ is $(t+\sqrt{1+t^2})-\sqrt{2}-1$ sq. unit for all $t > 1$. If $f(x)$ satisfies $f(x) > x$ for all $x > 1$, then $f(x)$ is equal to

(a) $x+1+\frac{x}{\sqrt{1+x^2}}$ (b) $x+\frac{x}{\sqrt{1+x^2}}$

(c) $1+\frac{x}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{1+x^2}}$

49. The area bounded by the curve $\sqrt{x}+\sqrt{y}=1$ and the co-ordinate axes is

(a) 1 (b) 1/2
(c) 1/3 (d) 1/6

50. The area of the region bounded by $y=|x-1|$ and $y = 1$ is

(a) 2 (b) 1
(c) 1/2 (d) none of these

EXERCISE SET 2

1. The area of the curve $x^2 + y^2 = 2ax$ is

(a) πa^2 (b) $2\pi a^2$

(c) $4\pi a^2$ (d) $\frac{1}{2}\pi a^2$

2. The area bounded by the curve $y = 2x - x^2$ and the line $x + y = 0$ is

(a) $\frac{35}{6}$ sq. units (b) $\frac{43}{6}$ sq. units

(c) $\frac{19}{6}$ sq. units (d) $\frac{9}{2}$ sq. units

3. The area bounded by the curves $y = \sqrt{x}\, 2y + 3 = x$ and the x-axis in the 1st quadrant is

(a) 9 (b) $\frac{27}{4}$

(c) 36 (d) 18

4. The area bounded by the curve $y = 4x - x^2$ and the x-axis is

(a) $\frac{30}{7}$ sq. units (b) $\frac{31}{7}$ sq. units

(c) $\frac{32}{7}$ sq. units (d) $\frac{34}{3}$ sq. units

5. The area bounded by the curves $y^2 = 20x$ and $x^2 = 16y$ is

(a) 80π sq. units (b) 100 sq. units

(c) $\frac{160}{3}$ sq. units (d) $\frac{320}{3}$ sq. units

6. The area bounded by the parabolas $y = (x+1)^2$ and $y = (x-1)^2$ and the line $y = \frac{1}{4}$ is

(a) 4 (b) $\frac{1}{6}$

(c) $\frac{4}{3}$ (d) $\frac{1}{3}$

7. The area of the region $\{(x, y): x^2 + y^2 \le 1 \le x + y\}$ is

(a) $\frac{1}{2}$ sq. unit (b) $\frac{3\pi - 2}{4}$ sq. units

(c) $\frac{\pi - 2}{4}$ sq. units (d) $\frac{\pi}{3}$ sq. units

8. The area bounded by the curve $y^2 = x$, line $y = 4$ and y-axis is

(a) $\frac{16}{3}$ (b) $\frac{64}{3}$

(c) $7\sqrt{2}$ (d) none of these

9. The area bounded by the parabolas $y = 5x^2$ and $y - 9 = 2x^2$ is

(a) $6\sqrt{2}$ sq. units (b) $12\sqrt{2}$ sq. units

(c) $4\sqrt{3}$ sq. units (d) $12\sqrt{3}$ sq. units

10. The area enclosed by the parabola $y^2 = 2x$ and its tangents through the point (–2, 0) is

(a) 3 (b) 4

(c) 4/3 (d) 8/3

11. The larger area bounded by $y^2 = 4x$ and $x^2 + y^2 - 2x - 3 = 0$ is

(a) $2\pi - \frac{4}{3}$ (b) $2\pi + \frac{4}{3}$

(c) $2\pi + \frac{8}{3}$ (d) $2\pi + \frac{2}{3}$

12. The area bounded by the angle bisectors of the lines $x^2 - y^2 + 2y = 1$ and $x + y = 3$ is

(a) 2 sq. units (b) 3 sq. units

(c) 4 sq. units (d) 6 sq. units

13. The area bounded by the curve $y = x^2$ and $y = \frac{2}{(1+x^2)}$ is

(a) $2\pi - \frac{1}{3}$ (b) $2\pi + \frac{1}{3}$

(c) $\pi - \frac{2}{3}$ (d) $\pi + \frac{2}{3}$

14. The area bounded by $x = \sin t$ and $y = \cos t + 3$, where $-2008 < t < 2008$, is

(a) π (b) 3π

(c) 2π (d) none of these

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice as given below:

(a) Statement 1 is false, Statement 2 is true.

(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

(d) Statement 1 is true, Statement 2 is false.

15. Statement 1: The area under the curve $y = \sin x$ from $x = 0$ to $x = \pi$ above the x-axis is more than the area under the curve $y = \sin^2 x$ from $x = 0$ to $x = \pi$ above the x-axis.
Statement 2: $t \Rightarrow t^2$ for $0 < t < 1$

16. **Statement 1:** The area under the curve $y = e^{x^3}$ between the lines $x = a$ and $x = b$ above the x-axis is $\int_a^b e^{x^3} dx$

Statement 2: e^{x^3} is an increasing function of x.

17. **Statement 1:** If $f(x) = (x-1)(x-3)(x-8)$, then the area under the curve $y = |f(x)|$ above the x-axis and between the lines $x = 4$ and $x = 6$ is $\int_2^6 (x-1)(x-3)(x-8)dx$.

Statement 2: $(x-1)(x-3)(x-8) \le 0, \forall x \in [4, 6]$.

18. **Statement 1:** The area of the ellipse $\frac{x^2}{3} + \frac{y^2}{2} = 1$ is less than the area bounded by $\frac{|x|}{3} + \frac{|y|}{2} \le 1$.

Statement 2: The length of the major axis of the ellipse $\frac{x^2}{3} + \frac{y^2}{2} = 1$ is more than the distance between the points of $\frac{|x|}{3} + \frac{|y|}{2} = 1$ on the x-axis.

Previous Years' Questions

19. The area bounded by the curves $y = \cos x$ and $y = \sin x$ between the ordinates $x = 0$ and $x = \frac{3\pi}{2}$ is

(a) $(4\sqrt{2} - 2)$ sq. units (b) $(4\sqrt{2} + 2)$ sq. units
(c) $(4\sqrt{2} - 1)$ sq. units (d) $(4\sqrt{2} + 1)$ sq. units

[AIEEE 2010]

20. The area of the region bounded by the parabola $(y-2)^2 = x - 1$, the tangent to the parabola at the point (2, 3) and the x-axis is

(a) 6 sq. units (b) 9 sq. units
(c) 12 sq. units (d) 3 sq. units

[AIEEE 2009]

21. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to

(a) $\frac{4}{3}$ sq. units (b) $\frac{5}{3}$ sq. units
(c) $\frac{1}{3}$ sq. unit (d) $\frac{2}{3}$ sq. unit

[AIEEE 2008]

22. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is

(a) 2/3 sq. unit (b) 1 sq. unit
(c) 1/6 sq. unit (d) 1/3 sq. unit

[AIEEE 2007]

23. The area enclosed between the curve $y = \log_e(x + e)$ and the coordinate axes is

(a) 4 sq. units (b) 3 sq. units
(c) 2 sq. units (d) 1 sq. unit

[AIEEE 2005]

24. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the co-ordinate axes. If S_1, S_2 and S_3 are respectively the areas of these parts numbered from top to bottom, then $S_1 : S_2 : S_3$ is

(a) 1:1:1 (b) 2:1:2
(c) 1:2:3 (d) 1:2:1 **[AIEEE 2005]**

25. The area of the region bounded by the curves $y = |x - 2|$, $x = 1$, $x = 3$ and the x-axis is

(a) 1 (b) 2
(c) 3 (d) 4 **[AIEEE 2004]**

26. The area bounded by the curves $y = \sqrt{5 - x^2}$ and $y = |x - 1|$ is

(a) $\left(\frac{5\pi}{4} - 2\right)$ sq. units (b) $\frac{5\pi - 2}{4}$ sq. units
(c) $\frac{5\pi - 2}{2}$ sq. units (d) $\left(\frac{\pi}{2} - 5\right)$ sq. units

[DCE 2009]

27. Let y be the function which passes through (1, 2) having slope $(2x + 1)$. The area bounded between the curve and the x-axis is

(a) 6 sq. units (b) $\frac{5}{6}$ sq. unit
(c) $\frac{1}{6}$ sq. unit (d) none of these

[DCE 2005]

28. The area between the parabola $y = x^2$ and the line $y = x$ is

(a) $\frac{1}{6}$ sq. unit (b) $\frac{1}{3}$ sq. unit
(c) $\frac{1}{2}$ sq. unit (d) none of these

[DCE 2005]

29. The area bounded by $y = \log x$, the x-axis and ordinates $x = 1$, $x = 2$ is

(a) $\frac{1}{2}(\log 2)^2$ (b) $\log\left(\frac{2}{e}\right)$
(c) $\log\left(\frac{4}{e}\right)$ (d) $\log 4$ **[UP SEE 2007]**

30. The area bounded by the curve $y = \log_e$, $x = 0, y \le 0$ and the x-axis is
(a) 1 sq. unit (b) 1/2 sq. unit
(c) 2 sq. units (d) none of these
[UP SEE 2005]

31. The area of the region between the curves $y = \sqrt{\frac{1+\sin x}{\cos x}}$ and $y = \sqrt{\frac{1-\sin x}{\cos x}}$ bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is
(a) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$ (b) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$
(c) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$ (d) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$
[IIT-JEE 2008]

32. The area bounded by the curves $y = |x-1|$ and $y = 1$ is
(a) 1 (b) 2
(c) 1/2 (d) none of these
[IIT-JEE 2005]

33. The area enclosed between the curves $y = ax^2$ and $x = ay^2 (a > 0)$ is 1 sq. unit. Then the value of a is
(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{2}$
(c) 1 (d) $\frac{1}{3}$ **[IIT-JEE 2004]**

34. The area bounded by the curves $y = \sqrt{x}, 2y + 3 = x$ and the x-axis in the first quadrant is
(a) 9 (b) 27/4
(c) 36 (d) 18 **[IIT-JEE 2003]**

35. If $f(x)$ is a continuous function such that the area bounded by the curve $y = f(x)$, the x-axis and the lines $x = a$ and $x = 0$ is $\frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a$, the value of $f\left(\frac{\pi}{2}\right)$ is
(a) 1/2 (b) $a/2$
(c) $a^2/2$ (d) $\pi/2$ **[AMU 2009]**

36. The area lying in the first quadrant and bounded by the circle $x^2 + y^2 = 4$, the line $x = \sqrt{3}y$ and the x-axis is
(a) π sq. units (b) $\pi/2$ sq. units
(c) $\pi/3$ sq. units (d) none of these
[AMU 2007]

37. The area bounded by $y = |\sin x|$, the x-axis and the line $|x| = \pi$ is
(a) 2 sq. units (b) 3 sq. units
(c) 4 sq. units (d) none of these
[OJEE 2009]

38. The area of the region bounded by the curves $y = x \log x$ and $y = 2x - 2x^2$ is
(a) 1/2 sq. unit (b) 3/12 sq. unit
(c) 7/12 sq. unit (d) none of these
[OJEE 2009]

39. The area bounded by the curves $y^2 = 8x$ and $x^2 = 8y$ is
(a) 64 sq. units (b) 64/3 sq. units
(c) 8/3 sq. units (d) none of these
[OJEE 2008]

40. The area bounded by the curve $y = x$, x-axis and ordinates $x = -1$ to $x = 2$ is
(a) 0 sq. unit (b) 1/2sq. unit
(c) 3/2 sq. units (d) 5/2 sq. units
[OJEE 2008]

41. The area bounded by $y^2 = x$, $y = 0$, $x = 1$, $x = 4$ is
(a) 28/3 sq. units (b) 3/28 sq. unit
(c) 8/3 sq. units (d) 4/3 sq. units
[OJEE 2007]

42. If the area is bounded by the curves $y = x - bx^2$ and $y = (1/b)x^2$, where $b > 0$ is maximum, then $b =$
(a) 0 (b) 1
(c) 2 (d) none of these
[BITSAT 2004, OJEE 2005]

43. The area bounded by $y = -x^2 + 2x + 3$ and $y = 0$ is
(a) 32 sq. units (b) 32/3 sq. units
(c) 1/32 sq. unit (d) 1/3 sq. unit
[OJEE 2004]

44. If the area above the x-axis bounded by the curves $y = 2^{kx}$ and $x = 0$ and $x = 2$ is $\frac{3}{\log 2}$, then the value of k is
(a) 1/2 (b) 1
(c) -1 (d) 2 **[OJEE 2003]**

45. The area bounded by the curve $y = x|x|$, the x-axis and the ordinates $x = 1, x = -1$ is given by
(a) 0 (b) 1/3
(c) 2/3 (d) none of these
[Jamia Millia Islamia 2009]

46. The area of the region bounded by the curves $y = |x - 2x = 1, x = 3$ and the x-axis is
(a) 1 (b) 2
(c) 3 (d) 4
[Jamia Millia Islamia 2006, MP PET 2007]

47. The area bounded by the curve $y = \log_e x, x = 0, y \le 0$, and the x-axis is
(a) 1 sq. unit (b) 1/2 sq. unit
(c) 2 sq. units (d) none of these
[Jamia Millia Islamia 2004, WB JEE 2006]

48. The area bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is equal to
(a) 8/9 sq. unit (b) 9/8 sq. units
(c) 4/3 sq. units (d) none of these
[Jamia Millia Islamia 2004]

49. Let $f(x)$ be a non-negative continuous function such that the area bounded by the curve $y = f(x)$, the x-axis and the ordinates $x = (\pi/4)$ and $x = \beta > (\pi/4)$ is $\left(\beta \sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta\right)$. The $f\left(\frac{\pi}{2}\right)$ is
(a) $\left(1 - \frac{\pi}{4} + \sqrt{2}\right)$ (b) $\left(1 - \frac{\pi}{4} - \sqrt{2}\right)$
(c) $\left(\frac{\pi}{4} - \sqrt{2} + 1\right)$ (d) $\left(\frac{\pi}{4} + \sqrt{2} - 1\right)$
[MP PET 2007]

50. The area bounded by the curve $y = x + \sin x$ and its inverse function between the ordinates $x = 0$ and $x = 2\pi$ is
(a) 8π sq. units (b) 4π sq. units
(c) 8 sq. units (d) none of these
[MP PET 2006]

51. The area bounded by the curves $y^2 = 4x$ and $x^2 = 4y$ is
(a) 32/3 (b) 16/3
(c) 8/3 (d) 0 **[AIEEE 2011]**

52. The area bounded between the parabolas $x^2 = x^2 = (y/4)$ and $x^2 = 9y$ and the straight line $y = 2$ is
(a) $20\sqrt{2}$ (b) $\frac{10\sqrt{2}}{3}$
(c) $\frac{20\sqrt{2}}{3}$ (d) $10\sqrt{2}$ **[AIEEE 2012]**

ANSWERS

Exercise Set 1

1. (c)	**2.** (d)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (a)	**7.** (a)	**8.** (b)	**9.** (b)	**10.** (c)
11. (b)	**12.** (b)	**13.** (a)	**14.** (b)	**15.** (b)	**16.** (a)	**17.** (a)	**18.** (a)	**19.** (b)	**20.** (a)
21. (c)	**22.** (d)	**23.** (c)	**24.** (c)	**25.** (b)	**26.** (c)	**27.** (c)	**28.** (b)	**29.** (c)	**30.** (b)
31. (b)	**32.** (b)	**33.** (b)	**34.** (c)	**35.** (d)	**36.** (a)	**37.** (a)	**38.** (b)	**39.** (b)	**40.** (d)
41. (a)	**42.** (c)	**43.** (d)	**44.** (b)	**45.** (a)	**46.** (a)	**47.** (d)	**48.** (a)	**49.** (d)	**50.** (b)

Exercise Set 2

1. (a)	**2.** (d)	**3.** (a)	**4.** (c)	**5.** (d)	**6.** (d)	**7.** (c)	**8.** (a)	**9.** (d)	**10.** (d)
11. (c)	**12.** (a)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (c)	**17.** (a)	**18.** (d)	**19.** (a)	**20.** (b)
21. (a)	**22.** (c)	**23.** (d)	**24.** (a)	**25.** (a)	**26.** (b)	**27.** (c)	**28.** (a)	**29.** (c)	**30.** (a)
31. (b)	**32.** (a)	**33.** (a)	**34.** (a)	**35.** (a)	**36.** (c)	**37.** (c)	**38.** (c)	**39.** (b)	**40.** (a)
41. (a)	**42.** (b)	**43.** (b)	**44.** (b)	**45.** (c)	**46.** (a)	**47.** (a)	**48.** (b)	**49.** (a)	**50.** (c)
51. (b)	**52.** (c)								

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Given equation of the curves are $x^2 + y^2 = 16a^2$ and $y^2 = 6ax$. The points of intersection are $x = 2a$, $y = \pm 2\sqrt{3a}$.

$\therefore$ Required area,

$A = 2$ [area of curve $APOP$]

$= 2$[area of curve $OMPO$ + area of curve $MAPM$]

$$= 2\left[\int_0^{2a} \sqrt{6a}\sqrt{x}\,dx\right] + 2\left[\int_{2a}^{4a} \sqrt{(4a)^2 - x^2}\,dx\right]$$

$$= 2 \cdot \frac{2}{3}\sqrt{6a}\left[x^{3/2}\right]_0^{2a} + 2\left[\frac{x}{2}\sqrt{(4a)^2 - x^2} + \frac{1}{2}(4a)^2 \sin^{-1}\frac{x}{4a}\right]_{2a}^{4a}$$

$$= \frac{4}{3}\sqrt{6a}(2a)^{3/2} + 2\left[(0 - 2a\sqrt{3}a) + 8a^2\left(\sin^{-1}1 - \sin^{-1}\frac{1}{2}\right)\right]$$

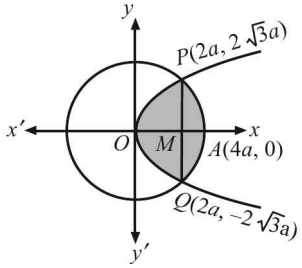


Fig. 16.42

$$= \frac{16}{3}\sqrt{3}a^2 - 4\sqrt{3}a^2 + 16a^2\frac{\pi}{3}$$

$$\Rightarrow \quad A = \frac{4\sqrt{3}a^2}{3} + \frac{16\pi a^2}{3} = \frac{4a^2}{3}(4\pi + \sqrt{3}) \text{ sq. units.}$$

2. Required area is shown shaded in the figure. The two curves meet where

$$3 - |x| = |x-1|$$

$$\Rightarrow \quad |x| + |x-1| = 3$$

$$\Rightarrow \quad x = -1, 2$$

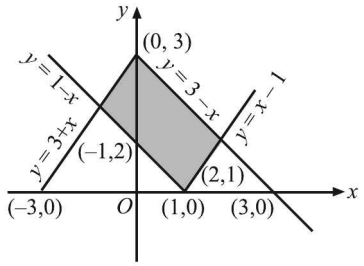


Fig. 16.43

$\therefore$ Required area

$$= \int_{-1}^{3} (3 - |x| - |x-1|)dx$$

$$= \left[3x - \frac{x|x|}{2} - \frac{(x-1)|x-1|}{2}\right]_{-1}^{2}$$

$$= \left(6 - 2 - \frac{1}{2}\right) - \left(-3 + \frac{1}{2} + 2\right) = 4.$$

3. Required area

$$= \int_0^a (y_1 - y_2)dx = \int_0^a \left[\frac{b}{a}\sqrt{a^2 - x^2} - \frac{b}{a}(a-x)\right]dx$$

$$= \frac{b}{a}\left[\frac{1}{2}x\sqrt{a^2 - x^2} + \frac{1}{2}a^2 \sin^{-1}\frac{x}{a}\right]_0^2$$

$$+ \frac{b}{2a}\left[(a-x)^2\right]_0^a$$

$$= \frac{b}{a}\left[\frac{1}{2}a^2 \sin^{-1}(1)\right] + \frac{b}{2a}(0 - a^2)$$

$$= \frac{\pi}{4}ab - \frac{1}{2}ab = \frac{ab}{4}(\pi - 2)$$

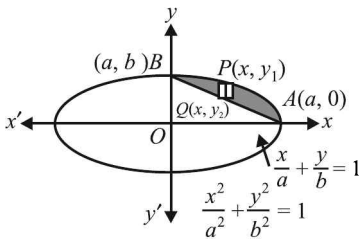


Fig. 16.44

Aliter. Required area = Area of the ellipse in first quadrant – Area of $\Delta OAB = \frac{\pi ab}{4} - \frac{1}{2}ab = \frac{ab}{4}(\pi - 2)$.

4. $y = x^3$ is a curve known as semi-cubical parabola.

If $x \Rightarrow -x$ and $y \Rightarrow -y$, the equation does not change. It is symmetrical in the 1st and 3rd quadrants. The line $y = 4x$ meets it at $4x = x^3$.

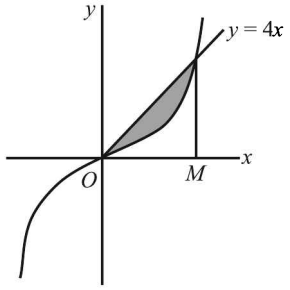


Fig. 16.45

$\therefore \quad x = 0, 2, -2$

$\therefore \quad y = 0, 8, -8$

$\therefore$ Area in the 1st quadrant = $\int_0^2 (y_1 - y_2)dx$

L C

$$A = \int_0^2 (4x - x^3) = \left(2x^2 - \frac{x^4}{4}\right)_0^2 = 4$$

Note $\Delta OPM = \frac{1}{2} \cdot 2 \cdot 8 = 8.$

5. The first curve is $y = x - 1, x \geq 0$ (A)

$y = -x - 1, x < 0$ (B)

The second curve is $y = -x + 1, x \geq 0$ (C)

$y = -x + 1, x < 0$ (D)

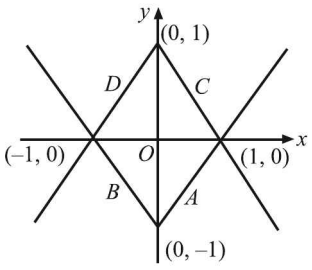


Fig. 16.46

Since each side of the square is $\sqrt{2}$, thus the required area $= (\sqrt{2})^2 = 2.$

6. Given curves are $y^2 = x$ and $2y = x$

$\Rightarrow \quad y^2 = 2y \quad \Rightarrow \quad y = 2$

$$\therefore \quad A=\int_0^2 (y^2-2y)dy=\left[\frac{y^3}{3}-y^2\right]_0^2$$

$$=\frac{4}{3} \text{ sq. units} \qquad \text{[neglect negative sign]}$$

7. The intersection points of $x^2+9y^2=9$ and $x+3y=3$ are (0, 1) and (3, 0).

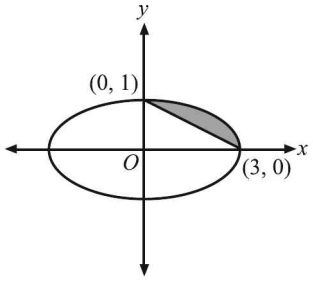


Fig. 16.47

Let $\quad A=\int_0^1 \pi x^2 dy-\int_0^1 \pi x^2 dy$

$$\Rightarrow \quad A=\int_0^1 \pi(9-9y^2)dy-\int_0^1 \pi 9(1-y)^2 dy$$

$$\Rightarrow \quad A=9\pi\left[y-\frac{y^3}{3}\right]_0^1-9\pi\left[\frac{(1-y)^3}{-3}\right]_0^1$$

$$\Rightarrow \quad A=9\pi\left(1-\frac{1}{3}\right)-\frac{9\pi}{3}(0-1)$$

$$\therefore \quad A=6\pi-3\pi=3\pi.$$

8. $y^2=x^2(4-x^2)$ is symmetric about both the x-axis and y-axis. Differentiating, we get

$$2y\frac{dy}{dx}=x^2(-2x)+(4-x^2)2x$$

$$=2x(4-2x^2)$$

Sign scheme of $\frac{dy}{dx}$ is as follows:

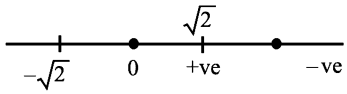


Fig. 16.48(i)

i.e. curve is increasing in $(0,\sqrt{2})$ and decreasing in $(\sqrt{2},\infty)$.

At $x=0$, $y=0$ and at $y=0$, $x=0, \pm 2$.

Thus, the curve will be as shown in Fig. 16.48(ii).

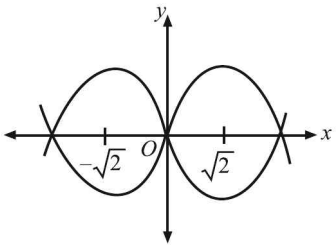


Fig. 16.48(ii)

$$\text{Required area } =4\int_0^2 x\sqrt{4-x^2}dx \;=\; 2\int_0^2 -2x\sqrt{4-x^2}dx$$

$$=2\left[\frac{(4-x^2)^{3/2}}{3/2}\right]_0^2=2\left(0-\frac{8}{3}\right)=\frac{16}{3}.$$

9. Given: Equation of curve $y=x^3$ and ordinates $x=1$ to $x=2$. We know that the area bounded by the curve and the x-axis from $x=a$ to $x=b$.

$$A=\int_a^b y dx=\int_1^2 x^3 dx=\left(\frac{x^4}{4}\right)_1^2=\frac{2^4}{4}-\frac{1^4}{4}=\frac{15}{4} \text{ sq. units}$$

10. $A=\left(1-\frac{1}{\sqrt{2}}\right)$, $A_1=\frac{1}{\sqrt{2}}=1-A$.

11. $\text{Required area}=2\int_1^2 \sqrt{4-x^2}dx=\left[\frac{x}{2}\sqrt{4-x^2}+\frac{4}{2}\sin^{-1}\frac{x}{2}\right]_1^2$

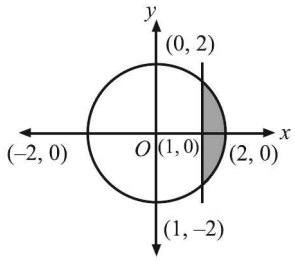


Fig. 16.49

$$=\left[0+2\sin^{-1}(1)-\left(\frac{1}{2}\sqrt{3}+2\sin^{-1}\left(\frac{1}{2}\right)\right)\right]$$

$$=4\left(\frac{\pi}{2}\right)-\sqrt{3}-4\left(\frac{\pi}{6}\right)=\frac{4\pi-3\sqrt{3}}{3}.$$

12. Required area $A=2\int_1^3 \sqrt{9-x^2}dx$

$$=2\cdot\frac{1}{2}\left[x\sqrt{9-x^2}+9\sin^{-1}\frac{x}{3}\right]_1^3$$

$$=\left[9\frac{\pi}{2}-\sqrt{8}-9\sin^{-1}\left(\frac{1}{3}\right)\right]$$

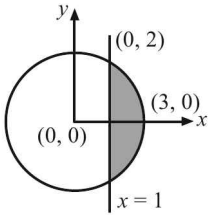


Fig. 16.50

$$= \left\{9\left[\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{3}\right)\right] - \sqrt{8}\right\}$$

$$= \left[9\cos^{-1}\left(\frac{1}{3}\right) - \sqrt{8}\right] = 9\sec^{-1}(3) - \sqrt{8}.$$

13. (i) The curve is symmetric about the x-axis.

(ii) $y^2 = \dfrac{4(2-x)}{x} \geq 0$

$\Rightarrow \quad 0 \leq x \leq 2$

(iii) As $x \to 0, y \to \infty$ and $x \to 2, y \to 0$

(iv) On differentiation,

$\dfrac{dy}{dx} = \dfrac{-8}{x^2} \cdot \dfrac{1}{2y} < 0$, i.e. decreasing

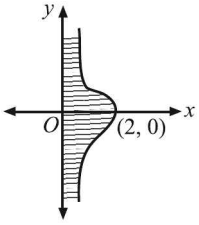


Fig. 16.51

So, the graph will be as given in Fig. 16.51.

Required area

$$= 2\int_0^2 \sqrt{\frac{4(2-x)}{x}}\,dx = 4\int_0^2 \sqrt{\frac{2-x}{x}}\,dx$$

Now, put $x = 2\sin^2\theta$ and solve.

14. Equations of lines $y = 2 + x$ (i)

$y = 2 - x$ (ii)

and $x = 2$ (iii)

Solving Eqs. (i), (ii) and (iii), we get vertices of the triangle $(x_1, y_1) = (2, 4)$, $(x_2, y_2) = (2, 0)$ and $(x_3, y_3) = (0, 2)$. We know that the area of the triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3)

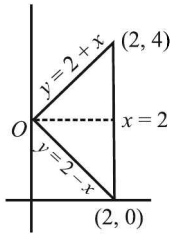


Fig. 16.52

$$A = \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$= \frac{1}{2}[2(0-2) + 2(2-4) + 0(4-0)] = 4 \text{ sq. units}$$

15. Split $\left(-\dfrac{\pi}{2}, 2\pi\right)$ to $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right), \left(\dfrac{\pi}{2}, \dfrac{3\pi}{2}\right), \left(\dfrac{3\pi}{2}, 2\pi\right)$

y is +ve in the 1st and 3rd intervals and it is –ve in the 2nd interval.

$$A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} y\,dx + \left|\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} y\,dx\right| + \int_{\frac{3\pi}{2}}^{2\pi} y\,dx$$

$$A = [\sin x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \left|[\sin x]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}\right| \Rightarrow A = (1+1) + [0 - (-1)].$$

$A = 2 + 2 + 1 = 5$ sq. units

16. Solving the given curves, $\log x = (\log x)^2$

$\Rightarrow \quad \log x(\log x - 1) = 0$

$\Rightarrow \quad x = 1, e$

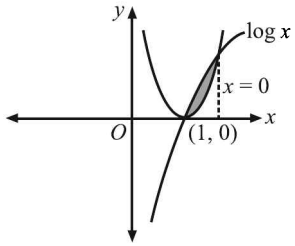


Fig. 16.53

Required area

$$= \int_0^1 [\log x - (\log x)^2\,dx]$$

But $\int \log x\,dx = x\log x - x$

and $\int (\log x)^2\,dx = (\log x)^2 x - \int \dfrac{2(\log x - x)}{x} \cdot x\,dx$

$$= (\log x)^2 x - 2(x\log x - x)$$

$\therefore$ Required area

$$= \left[x\log x - x(\log x)^2 x + 2(x\log x - x)\right]_1^e$$

$$= [e - e - e + 2(e-e)] - [0 - 1 - 0 + 2(0-1)]$$

$$= -e + 3.$$

17. The area of the circle in the 1st quadrant is $\dfrac{\pi(\pi^2)}{4} = \dfrac{\pi^3}{4}$.

Also the area of the curve $y = \sin x$ and the x-axis is 2 sq. units.

$\therefore$ Required area $A = \dfrac{\pi^3}{4} - 2 = \dfrac{\pi^3 - 8}{4}$.

18. Both $y^2 = x^3$ and $|y| = 2x$ are symmetric about the y-axis and their graphs are as shown in Fig. 16.54.

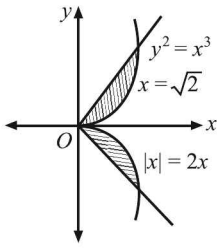


Fig. 16.54

Solving $x^3 = 2x$, we get

$$x = 0, \sqrt{2}$$

Required area $= 2\int_0^{\sqrt{2}} (2x - x^3)\,dx = 2$.

19. $\int_0^1 (y_1 - y_2)\,dx = \int_0^1 (2\sqrt{x} - 2x)\,dx = \dfrac{1}{3}$

20. Consider the curve $\sqrt{x} + \sqrt{|y|} = 1$.

(i) It is symmetric about the x-axis.

(ii) $\sqrt{|y|} = 1 - \sqrt{x}$ and $\sqrt{x} = 1 - \sqrt{|y|}$

$\Rightarrow 0 \leq x, y \leq 1$

(iii) For $x>0$, $y>0$

$$\sqrt{y}=1-\sqrt{x}\,\frac{1}{2\sqrt{y}}\frac{dy}{dx}$$

$$=\frac{-1}{2\sqrt{x}}\frac{dy}{dx}=-\sqrt{\frac{y}{x}}<0.$$

Functions are decreasing.

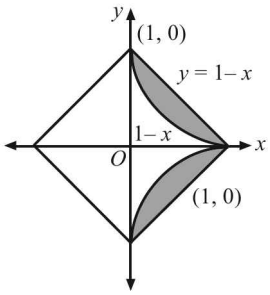


Fig. 16.55

(iv) Curve passes through (1, 0) and (0, 1).

$$\text{Required area} = 2\int_0^1 (2\sqrt{x}-2x)dx$$

$$= 2\int (1-x)-(1-\sqrt{x})^2\,dx=\frac{2}{3}$$

21. $A=\int_0^1 (e^x-e^{-x})dx=\left[e^x+e^{-x}\right]_0^1$

$$=e+\frac{1}{e}-2.$$

22. $y=4-x^2>0 \quad\Rightarrow\quad -2<x<2$

and $y=4-x^2<0 \quad\Rightarrow\quad x<-2$ or $x>2$

$$\text{The required area} = \int_0^2 (4-x^2)dx+\left|\int_2^3 (4-x^2)dx\right|$$

$$=\frac{16}{3}+\left|-\frac{7}{3}\right|=\frac{23}{3}.$$

23. Let $y^2=4ax$ be a parabola and let $x = b$ be a double ordinate. Then,

A_1 = Area enclosed by the parabola $y^2 = 4ax$ and the double ordinate $x = b$

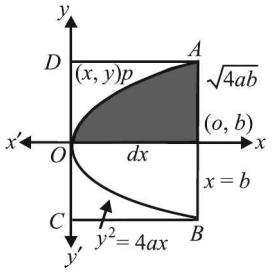


Fig. 16.56

$$=2\int_0^b y\,dx=2\int_0^b \sqrt{4ax}\,dx=4\sqrt{a}\int_0^b \sqrt{x^3}\,dx$$

$$=4\sqrt{a}\left[\frac{2}{3}x^{3/2}\right]_0^b$$

$$=4\sqrt{a}\times\frac{2}{3}b^{3/2}=\frac{8}{3}a^{1/2}b^{3/2}$$

And, A_2 = Area of the rectangle $ABCD$

$$=AB\times AD=2\sqrt{4ab}\times b=4a^{1/2}b^{3/2}$$

$$\therefore\quad A_1:A_2=8/3a^{1/2}b^{3/2}:4a^{1/2}b^{3/2}=2/3:1=2:3$$

24. Solving $\sqrt{x}=\frac{x-3}{2}$, we get $x = 9$

$\therefore$ Required area

$$=\int_0^9\left(\sqrt{x}-\frac{x-3}{2}\right)dx$$

$$=18-9=9$$

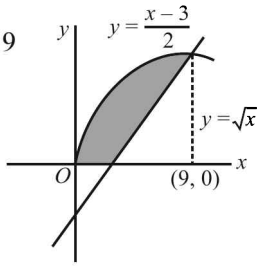


Fig. 16.57

25. We have $\int_{\pi/3a}^{\pi/a}\sin x\,dx=3$

$$\Rightarrow\quad \frac{1}{a}\left[1+\frac{1}{2}\right]=3$$

$$\Rightarrow\quad 2a=1$$

$$\Rightarrow\quad a=\frac{1}{2}$$

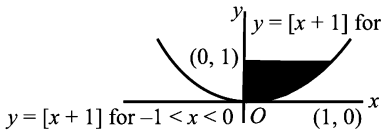


Fig. 16.58

26. $A=\int_{-6}^{2}(x_1-x_2)dy$

The line meets the parabola at (9, –6) and (1, 2).

27. Given $\frac{dy}{dx}=3x$

Integrating, we get

$$y=\frac{3x^2}{2}+c$$

Since it passes through (–1, –3), therefore

$$-3=\frac{3}{2}+c \quad\Rightarrow\quad c=-3-\frac{3}{2}=\frac{-9}{2}$$

$$\therefore\quad y=\frac{3x^2}{2}-\frac{9}{2}\quad\Rightarrow\quad 2y=3(x^2-3)$$

28.

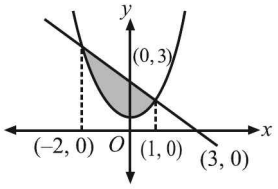


Fig. 16.59

Solving given equations, we get

$$y=x^2+1=3-x$$

or, $x^2+x-2=0, x=1,-2$

$$\text{Required area} = \int_{-2}^{1}\left[(3-x)-(x^2+1)\right]dx$$

$$=\int_{-2}^{1}(2-x-x^2)dx=\left[2x-\frac{x^2}{2}-\frac{x^3}{3}\right]_2^1=\frac{9}{2}$$

29. The given curve consists of two straight lines

$x+y=1 \qquad (x\ge 0)$

$-x+y=1 \qquad (x<0)$

$$A=\Delta ABC=2\Delta OBC=2\cdot\frac{1}{2}(1,1)=1$$

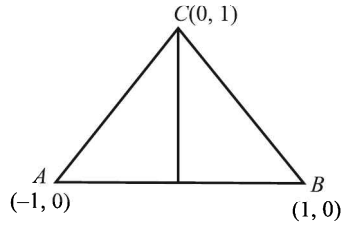


Fig. 16.60

$$2\int_0^1 y\,dx = 2\int_0^1 (1-|x|)\,dx = 2\int_0^1 (1-x)\,dx$$

$$= 2\left[x - \frac{x^2}{2}\right]_0^1 = 2\cdot\frac{1}{2} = 1$$

30. For $1 \le x \le 2$, $x^2 e^{x^3} > 0$

$\therefore$ Required area $= \int_1^2 x^2 e^{x^3}\,dx$

$$= \frac{1}{3}\int_1^2 3x^2 e^{x^3}\,dx = \frac{1}{3}\left[e^{x^3}\right]_1^2 = \frac{e^8 - e}{3}$$

31. $y = f(x) = x(x-1)(x-2)$

is +ve for $x > 2$ –ve for $1 < x < 2$; +ve for $0 < x < 1$, –ve for $x < 0$.

$$\text{Area} = \int_0^1 y\,dx + \left|\int_1^2 y\,dx\right|$$

$$= \left[\frac{x^4}{4} - x^3 + x^2\right]_0^1 + \left|\left[\frac{x^4}{4} - x^3 + x^2\right]_1^2\right|$$

$$= \frac{1}{4} + \left|-\frac{1}{4}\right| = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \text{ sq. unit}$$

32. The given curve is symmetrical about the x-axis and meets it at $(a, 0)$. The line $x = 0$, i.e. the y-axis is asymptote.

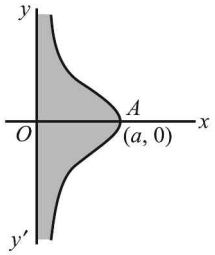


Fig. 16.61

$$\text{Area} = \int_0^\infty x\,dy = 2\int_0^\infty \frac{a^3}{y^2 + a^2}\,dx$$

$$= 2\cdot a^3 \cdot \frac{1}{a}\left[\tan^{-1}\frac{y}{a}\right]_0^\infty = 2a^2 \cdot \frac{\pi}{2} = \pi a^2$$

Note You can also use the formula

$$2\int_0^a y\,dx = 2\int_0^a a\sqrt{\left(\frac{a-x}{x}\right)}\,dx$$

Now, put $x = a\sin^2\theta$, etc.

33. Required area $A = \int_{\pi/2}^{\pi} \sin^2 x\,dx$

$$= \frac{1}{2}\int_{\pi/2}^{\pi} (1 - \cos 2x)\,dx$$

$$= \frac{1}{2}\left[x - \frac{\sin 2x}{2}\right]_{\pi/2}^{\pi}$$

$$= \frac{\pi}{4} \text{ sq. units.}$$

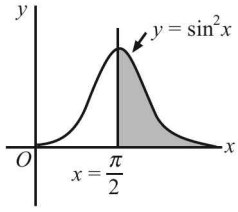


Fig. 16.62

34. Required area $= \left|\int_{-1}^{1} x|x|\,dx\right|$

$$= \left|\int_{-1}^{0} x|x|\,dx\right| + \int_0^1 x|x|\,dx = \left|\int_{-1}^{0} -x^2\,dx\right| + \int_0^1 x^2\,dx$$

$$= \left|\left[-\frac{x^3}{3}\right]_{-1}^{0}\right| - \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

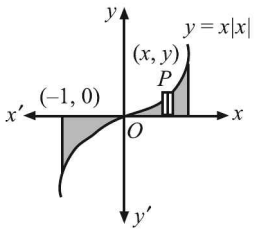


Fig. 16.63

35. $\int_c^1 y\,dx = \frac{16}{3} \Rightarrow \int_c^1 (8x^2 - x^5)\,dx, c < 1$

$\therefore$ $$\frac{16}{3} = \frac{8}{3}(1-c^3) - \frac{1}{6}(1-c^6) \quad \text{(i)}$$

$$\frac{16}{3} - \frac{8}{3} + \frac{1}{6} = c^3 - \left(-\frac{8}{3} + \frac{1}{6}\right)$$

or $$\left(\frac{8}{3} + \frac{1}{6}\right) = c^3\left(-\frac{8}{3} + \frac{c^3}{6}\right)$$

By inspection, we see that $c = -1$ clearly statisfies the above equation. If $c > 1$, then $\frac{16}{3} = \frac{8}{3}(c^3 - 1) - \frac{1}{6}(c^6 - 1)$, from (i)

or $\frac{24}{3}-\frac{1}{6}=c^3\left(\frac{8}{3}-\frac{c^3}{6}\right)$ or $47=16z-z^2$

or $z^2-16z+47=0$, where $z=c^3$

None of the values given satisfies the above equation.

36. Required area $= 2\int_0^a \sqrt{4ax}\,dx$

$= k(\alpha)(2\sqrt{4a\alpha})$

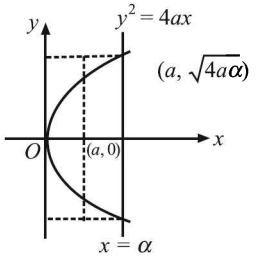


Fig. 16.64

$\frac{8\sqrt{a}}{3}\alpha^{3/2}=4\sqrt{a}k\alpha^{3/2} \Rightarrow k=\frac{2}{3}$

37. $A=\int_{\frac{1}{2}}^{2}(2^x-\log x)dx=\left(\frac{2x}{\log 2}\right)_{\frac{1}{2}}^{2}-(x\log x-x)_{\frac{1}{2}}^{2}$

$=\frac{3}{\log 2}-\left[(2\log 2-2)-\left(\frac{1}{2}\log\frac{1}{2}-\frac{1}{2}\right)\right]$

$=\frac{3}{2}-\frac{5}{2}\log 2+\frac{3}{\log 2}$

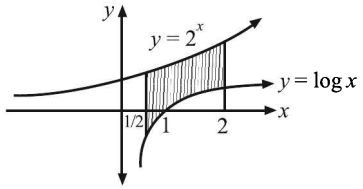


Fig. 16.65

38. Given equation of the ellipse is $\frac{x^2}{25}+\frac{y^2}{16}=1$.

Here, $a=5$, $b=4$

We know that the area of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\pi ab=\pi(5)(4)=20\pi$ sq. units

39. $A=\int_0^{1-m}(x-x^2-mx)dx=\left[(1-m)\frac{x^2}{2}-\frac{x^3}{3}\right]_0^{1-m}$

$\Rightarrow \frac{(1-m)^3}{2}-\frac{(1-m)^3}{3}=\frac{9}{2}$

$\Rightarrow \frac{(1-m)^3}{6}=\frac{9}{2} \Rightarrow (1-m)^3=27$

$\Rightarrow 1-m=3 \Rightarrow m=-2$

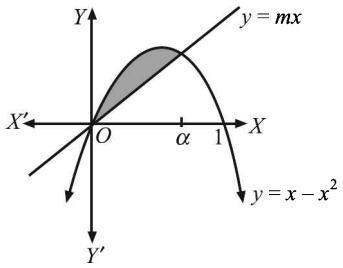


Fig. 16.66

40. The equation $y=2x-x^2$ represents a parabola opening downwards having vertex at (1, 1) and crossing the x-axis at (0, 0) and (2, 0).

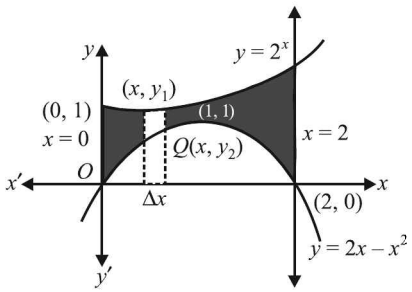


Fig. 16.67

The equation $y=2^x$ represents the exponential curve as shown in Fig. 16.67. Lines $x=0$ and $x=2$ are shown in the figure. The area bounded by these curves is shaded in the figure. We slice the shaded region into vertical strips. For the approxmating rectangle shown in the figure, we have length = (y_1-y_2), width = Δx

$\therefore$ Area = $(y_1-y_2)\Delta x$

The approximating rectangle can move horizontally between $x=0$ and $x=2$. So the required area

$=\int_0^2(y_1-y_2)dx=\int_0^2(2^x-2x+x^2)dx$

$\left[\because P(x, y_1) \text{ and } Q(x, y_2) \text{ lie on } y=2^x \text{ and } y=2x-x^2 \text{ respectively. } \therefore y_1=2^x \text{ and } y_2=2x-x^2\right]$

$=\left[\frac{2x}{\log 2}-x^2+\frac{x^3}{3}\right]^2$

$=\frac{4}{\log 2}-4+\frac{8}{3}-\frac{1}{\log 2}=\left(\frac{3}{\log 2}-\frac{4}{3}\right)$ sq. units

41. The two curves are $y^2=4b(x-a)$ (i)

and $y^2=-4b(x-b)$ (ii)

Clearly, these two curves represent two parabolas having their vertices at $(-a, 0)$ and $(b, 0)$ respectively as shown in Fig. 16.68.

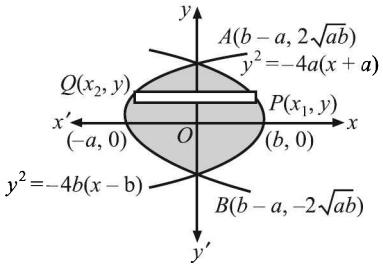


Fig. 16.68

To find the coordinates of their points of intersection, we solve Eqs. (i) and (ii) together. Solving these equations, we find that the two curves intersect at $A(b-a, 2\sqrt{ab})$ and $B(b-a, -2\sqrt{ab})$. We have to find the area of the shaded region in the figure. Let us slice this region into horizontal strips. For the approximating rectangle shown in the figure, we have

Length = $PQ = (x_1 - x_2)$, width = Δy and

area = $(x_1 - x_2)\Delta y$

The approximating rectangle can move vertically between B and A.

$\therefore$ Required area

$$= \int_{-2\sqrt{ab}}^{2\sqrt{ab}} (x_1 - x_2)\,dy$$

$$= \int_{-2\sqrt{ab}}^{2\sqrt{ab}} \left[\left(b - \frac{y^2}{4b}\right)\left(\frac{y^2}{4b} - a\right)\right] dy$$

$$\left[\because P(x_1, y) \text{ and } Q(x_2, y) \text{ lie on (ii) and (i) respectively}\right]$$
$$\left[\because y^2 = -4b(x_1 - b) \text{ and } y^2 = 4a(x_2 + a)\right]$$

$$= \int_{-2\sqrt{ab}}^{2\sqrt{ab}} \left[a + b - \left(\frac{a+b}{4ab}\right) y^2\right] dy$$

$$= 2\int_{0}^{2\sqrt{ab}} \left[a + b - \left(\frac{a+b}{4ab}\right) y^2\right] dy$$

$$= 2\left[(a+b)y - \left(\frac{a+b}{4ab}\right)\frac{y^3}{3}\right]_0^{2\sqrt{ab}}$$

$$= 2\left[(a+b)2\sqrt{ab} - \left(\frac{a+b}{4ab}\right)\frac{8ab\sqrt{ab}}{3}\right]$$

$$= 2\left[(a+b)2\sqrt{ab} - \frac{2}{3}(a+b)\sqrt{ab}\right]$$

$$= \frac{8}{3}(a+b)\sqrt{ab} \text{ sq. units}$$

42. Required area $= \int_1^3 y\,dx = \int_1^3 |x-2|\,dx$

$$= \int_1^2 |x-2|\,dx = \int_2^3 |x-2|dx = \int_1^2 (2-x)dx = \int_2^3 (x-2)dx$$

$$= \left|2x - \frac{x^2}{2}\right|_1^2 + \left|\frac{x^2}{2} - 2x\right|_2^3$$

$$= (4-2) - \left(2 - \frac{1}{2}\right) + \left[\left(\frac{9}{2} - 6\right) - (2-4)\right]$$

$$= 2 - \frac{3}{2} - \frac{3}{2} + 2 = 4 - 3 = 1$$

43. Required area $= 2\int_0^2 \left(3\sqrt{y} - \frac{\sqrt{y}}{2}\right) dy$

$$= 2\int_0^2 \left(\frac{5\sqrt{y}}{2}\right) dy = 5(y^{3/2})_0^2 \frac{2}{3}$$

$$= \frac{10}{3}(\sqrt{8} - 0)$$

$$= \frac{20\sqrt{2}}{3} \text{ sq. units}$$

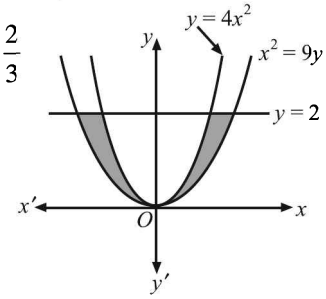


Fig. 16.69

44. Required area $= \int_0^{16/m^2} (\sqrt{16x} - mx)dx = \frac{2}{3}$

$$\Rightarrow \left[4 \times \frac{2}{3}x^{3/2} - m\frac{x^2}{2}\right]_0^{16/m^2} = \frac{2}{3}$$

$$\Rightarrow \frac{8}{3} \times \frac{64}{m^3} - \frac{m}{2}\frac{256}{m^4} = \frac{2}{3}$$

$$\Rightarrow \frac{1}{m^3}\left[\frac{512}{3} - 128\right] = \frac{2}{3}$$

$$\Rightarrow m = 4$$

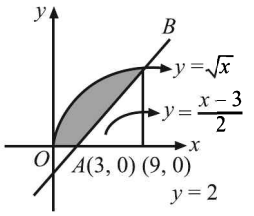


Fig. 16.70

45. It is evident from the graph of $y = e^x$, $y = \frac{3}{2}$ and $y = 1 + e^{-x}$ that

$$f(x) = \begin{cases} e^x, & 0 \le x \le \log(3/2) \\ 3/2, & \log 3/2 \le x \le \log 2 \\ 1 + e^x, & \log 2 \le x \le 1 \end{cases}$$

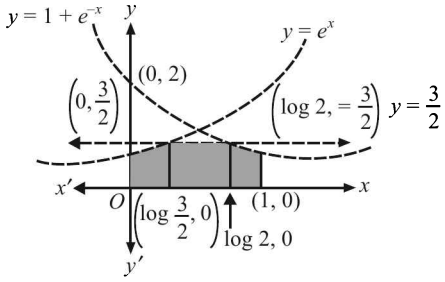


Fig. 16.71

$\therefore$ Required area

$$= \int_0^{\log 3/2} e^x dx + \int_{\log 3/2}^{\log 2} \frac{3}{2} dx + \int_{\log 2}^{1} (1+e^{-x}) dx$$

$$= [e^x]_0^{\log(3/2)} + \frac{3}{2}[x]_{\log 3/2}^{\log 2} + [x - e^{-x}]_{\log 2}^{1}$$

$$= \left(\frac{3}{2} - 1\right) + \frac{3}{2}\left(\log 2 - \log\frac{3}{2}\right) + \left(1 - \frac{1}{e}\right) - \left(\log 2 - \frac{1}{2}\right)$$

$$= \left(2 - \frac{1}{e} + \frac{3}{2}\log\frac{4}{3} - \log 2\right) \text{ sq. units}$$

46. Eliminating y, we get

$x^2 - x - 2 = 0 \Rightarrow x = -1, 2$

$$\text{Area} = \int_{-1}^{2}\left(\frac{x}{4} + \frac{1}{2} - \frac{x^2}{4}\right)dx$$

$$= \frac{1}{8}(4-1) + \frac{3}{2} - \frac{1}{12}(8+1)$$

$$= \frac{3}{8} + \frac{3}{2} - \frac{3}{4} = \frac{9}{8} \text{ sq. units}$$

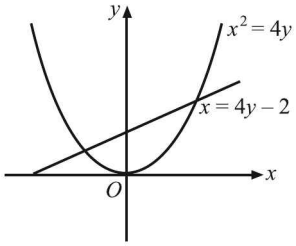


Fig. 16.72

47. The graph of $f(x)$ will be the curve $OBFGNO$ and $f(x)$ will be given by

$$f(x) = \begin{cases} 2x(1-x), & 0 \le x \le \frac{1}{3} \\ (x-1)^2, & \frac{1}{3} \le x \le \frac{1}{2} \\ x^2, & \frac{1}{2} \le x \le \frac{2}{3} \\ 2x(1-x), & \frac{2}{3} \le x \le 1 \end{cases}$$

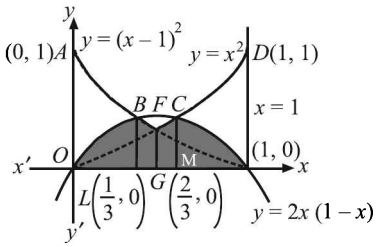


Fig. 16.73

Required area = Area of the region $OBLO$ + Area of the region $BFGL$ + Area of the region $FCMG$ + Area of the region $CMND$

$$= \int_0^{1/3} 2x(1-x)dx + \int_{1/3}^{1/2} (x-1)^2 dx + \int_{1/2}^{2/3} x^2 dx + \int_{2/3}^{1} 2x(1-x)dx$$

$= 31/108$ sq. unit

48. It is given that $f(x) > x$, for all $x > 1$. So, the area bounded by $y = f(x)$, $y = x$ and the lines $x = 1$, $x = t$ is given by

$$\int_1^t [f(x) - x]dx$$

But this area is given equal to $(t + \sqrt{1+t^2} - \sqrt{2} - 1)$ sq. units

Therefore,

$$\int_1^t [f(x) - x]dx = t + \sqrt{1+t^2} - \sqrt{2} - 1, \text{ for all } t > 1$$

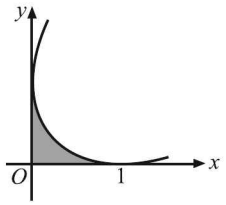


Fig. 16.74

On differentiating both sides w.r.t. t, we get

$$f(t) - t = 1 + \frac{t}{\sqrt{1+t^2}} \text{ for all } t > 1$$

$$\Rightarrow \quad f(t) = t + 1 + \frac{t}{\sqrt{1+t^2}} \text{ for all } t > 1$$

Hence, $f(x) = x + 1 + \dfrac{x}{\sqrt{1+x^2}}$ for all $x > 1$.

49. $A = \int_0^1 y dx$

$$= \int_0^1 (1 - \sqrt{x})dx$$

$$= \int_0^1 (1 - 2\sqrt{x} + x)dx$$

$$= 1 - \frac{4}{3} + \frac{1}{2} = \frac{1}{6} \text{ sq. unit}$$

50. Graph of the curve is shown in Fig. 16.75.

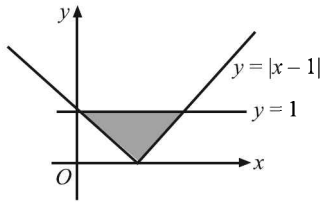


Fig. 16.75

$$\text{Required area} = \left|\int_0^1 [(x-1) - (1-x)]dx\right| = 1 \text{ sq. unit}$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $(x-a)^2 + y^2 = a^2$. Circle with radius a.

2. Eliminating y, we have $2x - x^2 = -x$

$\Rightarrow \quad x(x-3) = 0 \quad \Rightarrow \quad x = 0, x = 3$

$$A = \int_0^3 (2x - x^2 - (-x)dx = \int_0^3 (3x - x^2)dx$$

$$= \frac{27}{2} - 9 = \frac{9}{2} \text{ sq. units}$$

3. Graph of the function is shown in Fig. 16.76.

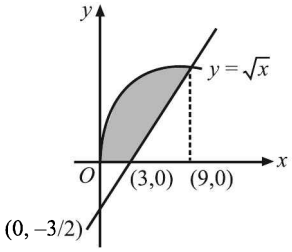


Fig. 16.76

$$\text{Required area} = \int_0^9 \sqrt{x}dx - \int_3^9 \frac{x-3}{2}dx$$

$$= \frac{2}{3}x^{3/2}\Big|_0^9 - \frac{1}{2}\left(\frac{x^2}{2} - 3x\right)\Big|_3^9$$

$$= 18 - \frac{1}{2}(18) = 9 \text{ sq. units}$$

4. $y = 4x - x^2$ meets the x-axis when

$4x - x^2 = 0 \quad \Rightarrow \quad x = 0, 4.$

$$\therefore \quad \text{Required area} = \int_0^4 ydx = \int_0^4 (4x - x^2)dx = \frac{32}{3} \text{ sq. units}$$

5. Eliminating y, we get

$x^4 = 256 \times 20x \quad \Rightarrow \quad x = 0, x = 8(10)^{1/3}$

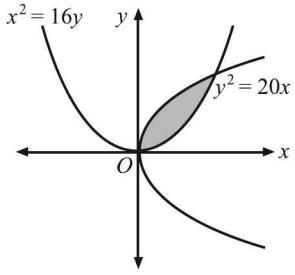


Fig. 16.77

$$A = \int_0^{8(10)^{1/3}} \left(\sqrt{20x} - \frac{x^2}{16}\right)dx = \frac{640}{3} - \frac{320}{3} = \frac{320}{3} \text{ sq. units}$$

6. Graphically, these curves and area can be represented as shown in Fig. 16.78.

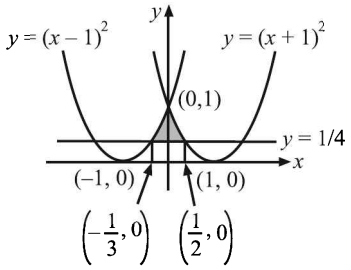


Fig. 16.78

$$\text{Area} = 2\int_0^{\frac{1}{2}} \left((x-1)^2 - \frac{1}{4}\right)dx$$

$$= 2\left[\frac{(x-1)^3}{3} - \frac{1}{4}x\right]_0^{\frac{1}{2}} = 2\left[\frac{\left(-\frac{1}{2}\right)^3}{3} - \frac{1}{8} - \left(-\frac{1}{3} - 0\right)\right]$$

$$= \frac{1}{3} \text{ sq. unit.}$$

7.

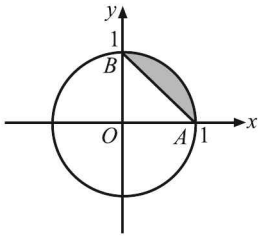


Fig. 16.79

The area is inside the circle and above the chord.

$$\int_0^1 \sqrt{1-x^2}dx - \int_0^1 (1-x)dx = \left(\frac{\pi}{4} - \frac{1}{2}\right) \text{ sq. units}$$

8. Graph of the function is shown in Fig. 16.80.

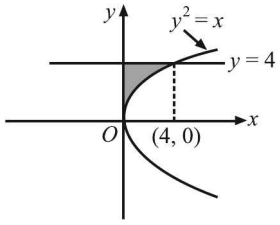


Fig. 16.80

$$\text{Required area} = \int_0^4 (y^2 - 4)dx = \frac{y^3}{3} - 4y\Big|_0^4$$

$$= \frac{64}{3} - 16 = \frac{16}{3} \text{ sq. units}$$

9. Eliminating y, we have

$5x^2 = 9 + 2x^2 \quad \Rightarrow \quad x = \pm\sqrt{3}$

$$A=\int_{-\sqrt{3}}^{\sqrt{3}}(9+2x^2-5x^2)dx=2\int_0^{\sqrt{3}}(9-3x^2)dx$$

$$=2\left[9\sqrt{3}-3\sqrt{3}\right]=12\sqrt{3}\text{ sq. units}$$

10.

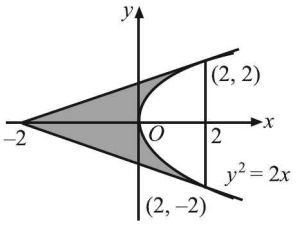


Fig. 16.81

The tangents are $y=mx+\frac{a}{m}, y=mx+\frac{1}{2m}$ since $a=\frac{1}{2}$

It passes through (–2, 0).

$\therefore \quad 4m^2=1 \quad \Rightarrow \quad m=\pm\frac{1}{2}$

The tangents are $y=\frac{x}{2}+1, y=-\frac{x}{2}-1$

$$A=2\int_0^3\left(\frac{y^2}{2}-2y+2\right)dxy$$

$$=2\left[\frac{4}{3}-4+4\right]=\frac{8}{3}\text{ sq. units}$$

11.

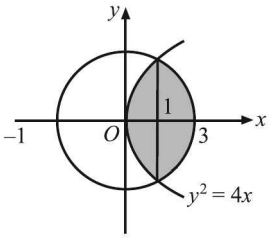


Fig. 16.82

$x^2+y^2-2x-3=0 \quad \Rightarrow \quad (x-1)^2+y^2=4$

It meets $y^2 = 4x$ at $x = 1$.

$$A=2\int_3^1\sqrt{4x}dx+\text{area of semi-circle}$$

$$=\frac{8}{3}+2\pi$$

12. The angle bisectors of the line given by $x^2-y^2+2y=1$ are $x=0, y=1$.

The required area

$=\frac{1}{2}\times2\times2=2$ sq. units

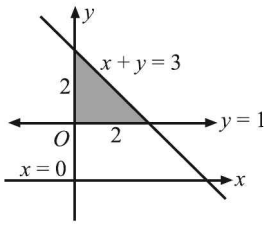


Fig. 16.83

13. Graphically, this can be represented as shown in Fig. 16.84.

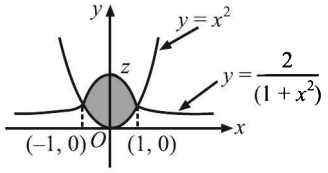


Fig. 16.84

Points of intersection of these two curves are

$$y=\frac{2}{y}-1$$

$\Rightarrow y^2+y-2=0 \quad \Rightarrow \quad (y-1)(y+2)=0$

$\Rightarrow y=-2, 1 \quad \Rightarrow \quad y=1$ as $y\geq0$

At $y = 1, x \pm 1$

Coordinates are $(1,1),(-1,1)$.

$$\text{Required area}=2\left[\int_0^1\frac{2}{1+x^2}dx-\int_0^1x^2dx\right]$$

$$=2\left[\left[2\tan^{-1}x\right]_0^1-\frac{x^3}{3}\Big|_0^1\right]$$

$$=2\left[\frac{2\pi}{4}-\frac{1}{3}\right]=\pi-\frac{2}{3}\text{ sq. units}$$

14. $x^2+(y-3)^2=1$

It is a circle with centre (0, 3) and radius = 1.

$\therefore$ Area $=\pi r^2=\pi\times1=\pi$.

15. For $0<t<1 \quad \Rightarrow \quad t>t^2$

$\therefore \quad \sin x>\sin^2 x$

$\therefore \quad \int_0^{\pi}\sin xdx>\int_0^{\pi}\sin^2 xdx$

16. $y=e^{x^3}$

$\Rightarrow \quad \frac{dy}{dx}=e^{x^3}\cdot3x^2>0$ for each x.

$\therefore$ y is an increasing function of x.

17. For $x\in[4,6], f(x)=(x-1)(x-3)(x-8)\leq0$

$\therefore$ Required area $=\left|\int_4^6f(x)dx\right|$

$$=\left|\int_4^6(x-1)(x-3)(x-8)dx\right|$$

$\therefore$ Statement 1 is false.

18. $\frac{x^2}{(\sqrt{3})^2}+\frac{y^2}{(\sqrt{2})^2}=1$

Area of ellipse $=\pi(\sqrt{3})(\sqrt{2})=3\cdot14\times1\cdot7\times1\cdot41$

$=7.8$ sq. units.

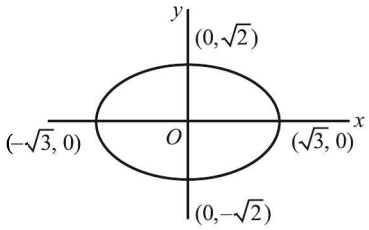


Fig. 16.85

Area bounded by $\frac{|x|}{3}+\frac{|y|}{2}=1$ or $2|x|+3|y|=6$ is the area of the parallelogram as shown.

$$= 4\times\frac{1}{2}\times 3\times 2 = 12 \text{ sq. units}$$

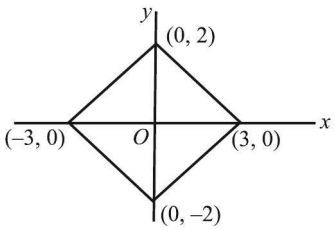


Fig. 16.86

Also, length of the major axis = $2\sqrt{6} < 3+3$.

19.

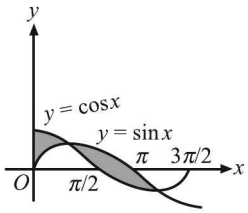


Fig. 16.87

Required area

$$= \int_0^{\pi/4}(\cos x-\sin x)dx + \int_{\pi/4}^{5\pi/4}(\sin x-\cos x)dx + \int_{5\pi/4}^{3\pi/2}(\cos x-\sin x)dx$$

$$= (\sin x+\cos x)_0^{\pi/4} + (-\cos x-\sin x)_{\pi/4}^{5\pi/4} + (\sin x+\cos x)_{5\pi/4}^{3\pi/2}$$

$$= (4\sqrt{2}-2) \text{ sq. units}$$

20. The equation of the tangent at (2, 3) to the given parabola is $x = 2y - 4$.

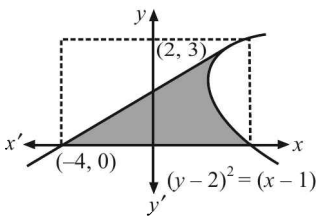


Fig. 16.88

$\therefore$ Required area

$$= \int_0^3 [(y-2)^2+1-2y+4]dy$$

$$= \left[\frac{(y-2)^3}{3}-y^2+5y\right]_0^3$$

$$= \frac{1}{3}-9+15+\frac{8}{3} = 9 \text{ sq. units}$$

21. On solving the given curves, we get $y = \pm 1$ and $x = -2$.

$\therefore$ Required area $= \left|\int_{-1}^{1}(x_1-x_2)dy\right| = \left|\int_{-1}^{1}(1-3y^2+2y^2)dy\right|$

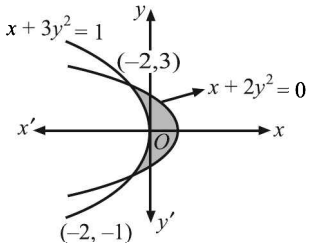


Fig. 16.89

$$= \left|2\int_0^1(1-y^2)dy\right| = \left|2\left[y-\frac{y^3}{3}\right]_0^1\right| = \frac{4}{3} \text{ sq. units}$$

22. Required area $= \int_0^1(\sqrt{x}-x)dx$

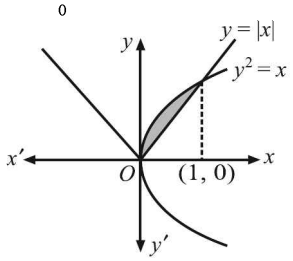


Fig. 16.90

$$= \left[\frac{2}{3}x^{3/2}-\frac{x^2}{2}\right]_0^1 = \frac{1}{6} \text{ sq. unit}$$

23. Required area, $A = \int_{1-e}^{0}\log_e(x+e)dx$

Put $x+e=t \Rightarrow dx=dt$

$\therefore$ $A = \int_1^e \log_e t\,dt$

$$= [t\log t - t]_1^e$$

$$= (e-e-0+1)$$

$$= 1 \text{ sq. unit}$$

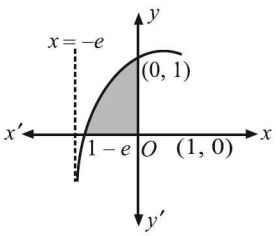


Fig. 16.91

24. Now, $S_1 = S_3 = \int_0^4 \frac{x^2}{4}dx$

$$= \frac{1}{4}\left[\frac{x^3}{3}\right]_0^4 = \frac{1}{12}\times 64 = \frac{16}{3} \text{ sq. units}$$

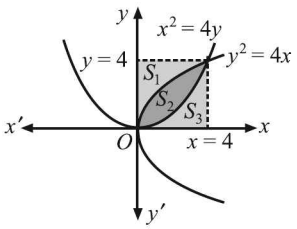


Fig. 16.92

$\therefore \quad S_2 + S_3 = \int_0^4 \sqrt{4x}dx$

$$= 2\left[\frac{x^{3/2}}{3/2}\right]_0^4 = \frac{4}{3}\times 8 = \frac{32}{3} \text{ sq. units}$$

$\Rightarrow \quad S_2 = \frac{16}{3}$ sq. units

$\therefore \quad S_1 : S_2 : S_3 = \frac{16}{3} : \frac{16}{3} : \frac{16}{3} = 1:1:1$

25. Required area $= \int_1^3 y dx = \int_1^3 |x-2|dx$

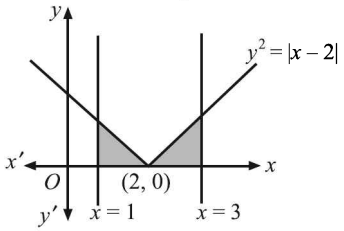


Fig. 16.93

$$= \int_1^2 -(x-2)dx + \int_2^3 (x-2)dx = \int_1^2 (2-x)dx + \int_2^3 (x-2)dx$$

$$= \left[2x - \frac{x^2}{2}\right]_1^2 + \left[\frac{x^2}{2} - 2x\right]_2^3$$

$$= (4-2) - \left(2 - \frac{1}{2}\right) + \left(\frac{9}{2} - 6\right) - (2-4)$$

$$= 2 - \frac{3}{2} - \frac{3}{2} + 2 = 4 - 3 = 1$$

26. Given curves are $y = \sqrt{5-x^2}$ and $y = |x-1|$

or $y^2 + x^2 = 5$ and $y = |x-1|$

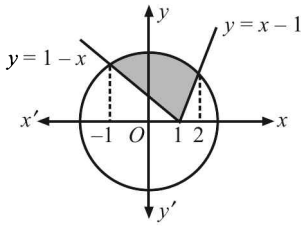


Fig. 16.94

$\therefore$ Required area

$$= \int_{-1}^{2} \sqrt{5-x^2}dx - \int_{-1}^{1}(1-x)dx - \int_{1}^{2}(x-1)dx$$

$$= \left[\frac{x}{2}\sqrt{5-x^2} + \frac{5}{2}\sin^{-1}\frac{x}{\sqrt{5}}\right]_{-1}^{2} - \left[x - \frac{x^2}{2}\right]_{-1}^{1} - \left[\frac{x^2}{2} - x\right]_{1}^{2}$$

$$= \left[1 + \frac{5}{2}\sin^{-1}\frac{2}{\sqrt{5}} + 1 + \frac{5}{2}\sin^{-1}\frac{1}{\sqrt{5}}\right]$$

$$-\left[1 - \frac{1}{2} - \left(-1 - \frac{1}{2}\right)\right] - \left[2 - 2 - \left(\frac{1}{2} - 1\right)\right]$$

$$= 2 + \frac{5}{2}\sin^{-1}\left(\frac{2}{\sqrt{5}}\sqrt{1-\frac{1}{5}} + \frac{1}{\sqrt{5}}\sqrt{1-\frac{4}{5}}\right) - \frac{5}{2}$$

$$= \frac{5}{2}\sin^{-1}(1) - \frac{1}{2} = \frac{5\pi}{4} - \frac{1}{2} = \left(\frac{5\pi - 2}{4}\right) \text{ sq. units}$$

27. Given slope of the curve is $2x + 1$.

$\therefore \quad \frac{dy}{dx} = 2x + 1 \quad \Rightarrow \quad y = \frac{2x^2}{2} + x + c = x^2 + x + c$

Also, it passes through (1, 2).

$\therefore \quad 2 = 1 + 1 + c \quad \Rightarrow \quad c = 0$

Equation of the curve is $y = x^2 + x$.

$\therefore$ Points of intersection of $y = x^2 + x = x(x+1)$ and the x-axis, i.e, $y = 0$ are $x = 0, x = -1$

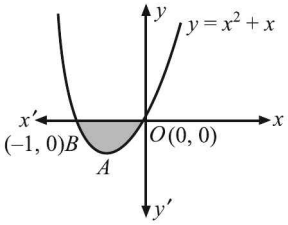


Fig. 16.95

$\therefore$ Required area

$$= \left|\int_{-1}^{0}(x^2 + x)dx\right|$$

$$= \left|\left[\frac{x^3}{3} + \frac{x^2}{2}\right]_{-1}^{0}\right| = \left|\left[0 - \left(-\frac{1}{3} + \frac{1}{2}\right)\right]\right|$$

$$= \left|-\frac{1}{6}\right| = \frac{1}{6} \text{ sq. unit}$$

28. The intersection points of a line and a parabola are $O(0, 0)$ and $A(1, 1)$.

$\therefore$ Required area $= \int_0^1 (x - x^2)dx$

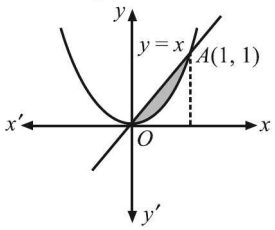


Fig. 16.96

$$= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \text{ sq. unit}$$

29. We have $y = \log x,\ x = 1, x = 2$

$\therefore$ Required area $= \int_1^2 y dx$

$$\Rightarrow A = \int_1^2 \log x dx = [x\log x - x]_1^2$$

$$= 2\log 2 - 2 - \log 1 + 1 = 2\log 2 - 1$$

$$= 2\log 2 - \log e = \log 4 - \log e$$

$$= \log\left(\frac{4}{e}\right) \qquad \left(\because \log m - \log n = \log\frac{m}{n}\right)$$

30. Given curve $y = \log_e x, x = 0, y \le 0$ and x-axis

$\therefore$ Bounded region is shown in Fig. 16.97.

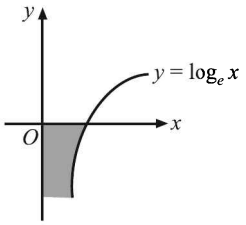


Fig. 16.97

$\therefore$ Required area $= \left|\int_0^1 \log_e dx\right|$

$$= \left[\left|x\log_e x - x\right|\right]_0^1 = 1 \text{ sq. unit.}$$

31. Required area $= \int_0^{\pi/4} \left(\sqrt{\frac{1+\sin x}{\cos x}} - \sqrt{\frac{1-\sin x}{\cos x}}\right) dx$

$$\left[\because \frac{1+\sin x}{\cos x} > \frac{1-\sin x}{\cos x} > 0\right]$$

$$= \int_0^{\pi/4} \left(\sqrt{\frac{1+\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}{\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}} - \sqrt{\frac{1-\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}{\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}}\right) dx$$

$$= \int_0^{\pi/4} \frac{1+\tan\frac{x}{2} - 1 + \tan\frac{x}{2}}{\sqrt{1-\tan^2\frac{x}{2}}} dx = \int_0^{\pi/4} \frac{2\tan\frac{x}{2}}{\sqrt{1-\tan^2\frac{x}{2}}} dx$$

Put $\tan\frac{x}{2} = t \quad \Rightarrow \quad \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

$\therefore$ Required area $= \int_0^{\tan\frac{\pi}{8}} \frac{4tdt}{(1+t^2)\sqrt{1-t^2}}$

$$= \int_0^{\sqrt{2}-1} \frac{4tdt}{(1+t^2)\sqrt{1-t^2}} \qquad \left[\because \tan\frac{\pi}{8} = \sqrt{2} - 1\right]$$

32. The given curves are

(i) $y = x - 1, x > 1$ (ii) $y = -(x-1), x < 1$

(iii) $y = 3$

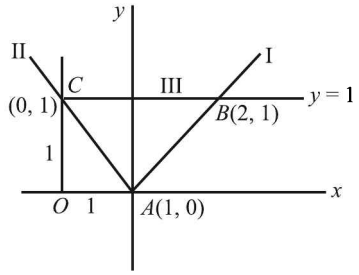


Fig. 16.98

These three lines enclose a triangle whose area is

$$\frac{1}{2} \cdot b \times h = \frac{1}{2} \cdot 2 \cdot 1 = 1$$

33. The points of intersection of the given curves are (0, 0) and $\left(\frac{1}{a}, \frac{1}{a}\right)$.

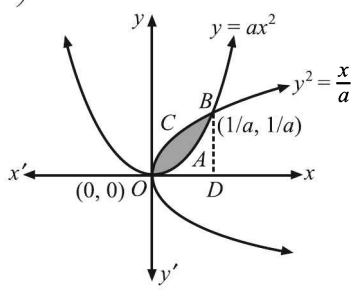


Fig. 16.99

$\therefore$ Required area $OABCO$ = Area of $OCBDO$ – Area of $OABDO$

$$\Rightarrow \int_0^{1/a}\left(\sqrt{\frac{x}{a}}-ax^2\right)dx=1 \Rightarrow \left[\frac{1}{\sqrt{a}}\cdot\frac{x^{3/2}}{3/2}-\frac{ax^3}{3}\right]_0^{1/a}=1$$

$$\Rightarrow \frac{2}{3a^2}-\frac{1}{3a^2}=1 \qquad \Rightarrow \quad a^2=\frac{1}{3}$$

$$\Rightarrow a=\frac{1}{\sqrt{3}} \qquad [\text{as } a>0]$$

34. Required area $= \int_0^9 \sqrt{x}dx-\int_3^9\left(\frac{x-3}{2}\right)dx$

$$=\left(\frac{x^{3/2}}{3/2}\right)_0^9-\frac{1}{2}\left(\frac{x^2}{2}-3x\right)_3^9$$

$$=\left(\frac{2}{3}\cdot 27\right)-\frac{1}{2}\left[\left(\frac{81}{2}-27\right)-\left(\frac{9}{2}-9\right)\right]$$

$= 9$ sq. units

35. According to the given condition, the area of the curve

$$=\int_0^a f(x)dx$$

$$\Rightarrow \frac{a^2}{2}+\frac{a}{2}\sin a+\frac{\pi}{2}\cos a$$

$$=\int_0^a f(x)dx$$

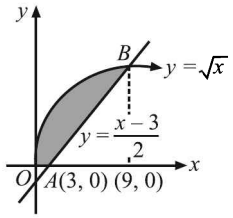


Fig. 16.100

On differentiating both sides w.r.t. a, we get

$$a+\frac{1}{2}\sin a+\frac{a}{2}\cos a-\frac{\pi}{2}\sin a=f(a)$$

$$\Rightarrow f\left(\frac{\pi}{2}\right)=\frac{\pi}{2}+\frac{1}{2}\sin\frac{\pi}{2}+\frac{\pi}{4}\cos\frac{\pi}{2}-\frac{\pi}{2}\sin\frac{\pi}{2}$$

$$\Rightarrow f\left(\frac{\pi}{2}\right)=\frac{\pi}{2}+\frac{1}{2}-\frac{\pi}{2}$$

$$\Rightarrow f\left(\frac{\pi}{2}\right)=\frac{1}{2}$$

36. Required area $= \int_0^1 (x_2-x_1)dy$

$$=\int_0^1(\sqrt{4-y^2}-\sqrt{3})dy$$

$$=\left[\frac{1}{2}y\sqrt{4-y^2}+\frac{1}{2}(4)\sin^{-1}\frac{y}{2}-\frac{\sqrt{3}y^2}{2}\right]_0^1$$

$$=\frac{\sqrt{3}}{2}+2\sin^{-1}\left(\frac{1}{2}\right)-\frac{\sqrt{3}}{2}-2\sin^{-1}0$$

$$=\frac{\pi}{3} \text{ sq. units}$$

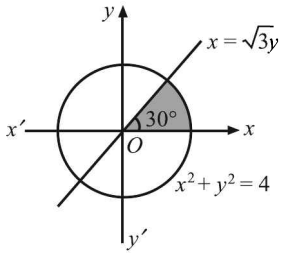


Fig. 16.101

37. Required area $= 2\int_0^{\pi}\sin x dx$

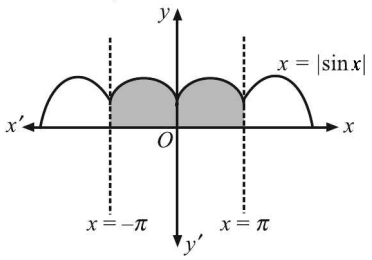


Fig. 16.102

$$= 2\left[-\cos x\right]_0^{\pi}=2[1+1]=4 \text{ sq. units}$$

38. Required area $= \int_0^1\left[(2x-2x^2)-(x\log x)\right]dx$

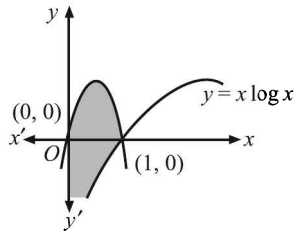


Fig. 16.103

$$=\left[x^2-\frac{2x^3}{3}-\left(\frac{x^2}{2}\log x-\frac{x^2}{4}\right)\right]_0^1$$

$$=\left[1-\frac{2}{3}-\left(0-\frac{1}{4}\right)\right]=\frac{7}{12} \text{ sq. unit}$$

39. Given curves are $y^2=8x \Rightarrow y=\sqrt{8x}$

and $x^2=8y \Rightarrow y=\frac{x^2}{8}$

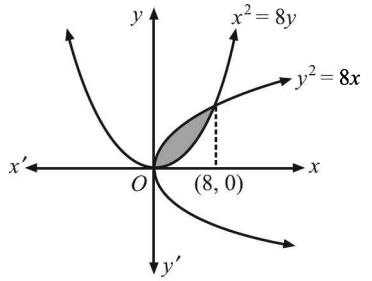


Fig. 16.104

The points of intersection of two curves are (0, 0), (8, 8).

Now, the required area $= \int_0^8 \left(\sqrt{8x} - \frac{x^2}{8}\right) dx$

$$= \left[\frac{\sqrt{8}x^{3/2}}{3/2} - \frac{x^3}{8.3}\right]_0^8 = \frac{64}{3} \text{ sq. units}$$

40. Required area

$$= \left|\int_{-1}^{0} x\,dx\right| + \left|\int_{0}^{2} x\,dx\right|$$

$$= \left|\left[\frac{x^2}{2}\right]_{-1}^{0}\right| + \left|\left[\frac{x^2}{2}\right]_{0}^{2}\right|$$

$$= \left|-\frac{1}{2}\right| + |2| = 2 + \frac{1}{2} = \frac{5}{2} \text{ sq. units}$$

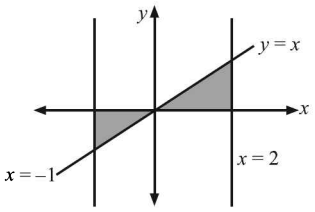


Fig. 16.105

41. Required area $= 2\int_1^4 \sqrt{x}\,dx$

$$= 2\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{4}{3}[8-1] = \frac{28}{3} \text{ sq. units}$$

42. Area bounded by the given curves is

$$A = \int_0^{\frac{b}{b^2+1}} \left(x - bx^2 - \frac{x^2}{b}\right) dx$$

$$= \left|\frac{x^2}{2} - \left(\frac{b^2+1}{b}\right)\frac{x^3}{3}\right|_0^{\frac{b}{b^2+1}} = \frac{1}{6}\left(\frac{b}{b^2+1}\right)^2$$

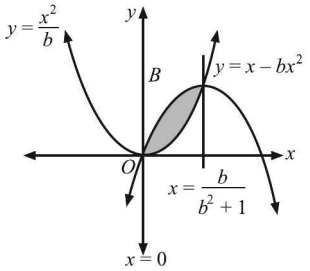


Fig. 16.106

$$\therefore \quad \log A = \frac{1}{6}[2\log b - 2\log(b^2+1)]$$

$$\Rightarrow \quad \frac{1}{A}\frac{dA}{db} = \frac{1}{3}\left(\frac{1}{b} - \frac{2b}{b^2+1}\right)$$

$$\Rightarrow \quad \frac{dA}{db} = \frac{A}{3}\left[\frac{b^2+1-2b^2}{b(b^2+1)}\right]$$

$$= \frac{A}{3}\frac{1-b^2}{b(b^2+1)} = -\frac{(b-1)(b+1)}{b(b^2+1)}\cdot\frac{A}{3}$$

$$\Rightarrow \quad \frac{dA}{db} = 0 \quad \Rightarrow \quad b = 1 \qquad [\because b > 0]$$

Also, $\frac{dA}{db} > 0$ for $0 < b < 1$ and $\frac{dA}{db} < 0$ for $b > 1$

$\therefore$ A is maximum when $b = 1$.

43. Intersection points of the given curves are (–1, 0) and (3, 0).

Required area $= \int_{-1}^{3} (-x^2 + 2x + 3)\,dx$

$$= \left[-\frac{x^3}{3} + \frac{2x^2}{2} + 3x\right]_{-1}^{3}$$

$$= \left[-9 + 9 + 9 - \left(\frac{1}{3} + 1 - 3\right)\right] = \frac{32}{3} \text{ sq. units}$$

44. Given area $= \int_0^2 2^{kx}\,dx = \frac{3}{\log 2}$

$$\Rightarrow \left[\frac{2^{kx}}{\log_e 2}\right]_0^2 = \frac{3}{\log 2} \quad \Rightarrow \quad \frac{2^{kx}}{\log_e 2} - \frac{1}{\log_e 2} = \frac{3}{\log 2}$$

$$\Rightarrow 2^{2k} - 1 = 3 \quad \Rightarrow \quad 2^{2k} = 2^2$$

$$\Rightarrow 2k = 2 \quad \Rightarrow \quad k = 1$$

45. Required area

$$= \left|\int_{-1}^{1} x|x|\,dx\right|$$

$$= \left|\int_{-1}^{0} x|x|\,dx\right| + \int_{0}^{1} x|x|\,dx$$

$$= \left|\int_{-1}^{0} -x^2\,dx\right| + \int_{0}^{1} x^2\,dx$$

$$= \left|\left[\frac{-x^3}{3}\right]_{-1}^{0}\right| + \left[\frac{x^3}{3}\right]_{0}^{1} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

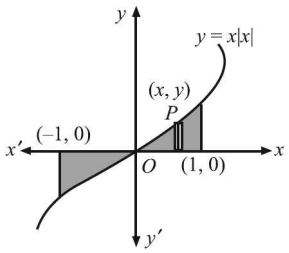


Fig. 16.107

46. Required area

$$= \int_1^3 |x-2|\,dx$$

$$= \int_1^2 (2-x)\,dx + \int_2^3 (x-2)\,dx$$

$$= \left[2x - \frac{x^2}{2}\right]_1^2 + \left[\frac{x^2}{2} - 2x\right]_2^3$$

$$= 2 - \frac{3}{2} - \frac{3}{2} + 2 = 1 \text{ sq. unit}$$

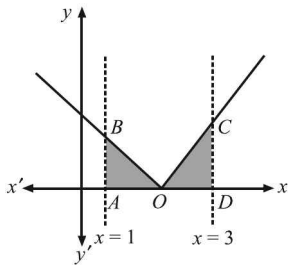


Fig. 16.108

47. Required area $= \left|\int_0^{-\infty} e^y dy\right|$

$= \left|\left[e^y\right]_0^{-\infty}\right| = 1$ sq. unit

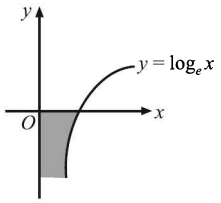


Fig. 16.109

48. The points of intersection of the parabola and the line are $A(2, 1)$ and $B\ [-1, (1/4)]$.

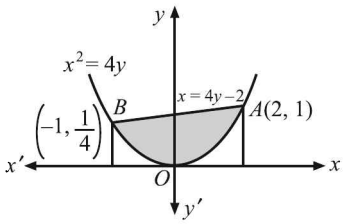


Fig. 16.110

$\therefore$ The required area

$$= \left[\int_{-1}^{2} y dx\right] - \left[\int_{-1}^{2} y dx\right] = \int_{-1}^{2} \frac{1}{4}(x+2)dx - \int_{-1}^{2} \frac{1}{4}x^2 dx$$

$$= \frac{1}{4}\left[\frac{x^2}{2} + 2x\right]_{-1}^{2} - \frac{1}{4}\left[\frac{x^3}{3}\right]_{-1}^{2} = \frac{9}{8} \text{ sq. units}$$

49. Given: $\int_{\pi/4}^{\beta} f(x)dx = \beta \sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta$

On differentiating w.r.t. β on both sides, we get

$$f(\beta) = \sin\beta + \beta\cos\beta - \frac{\pi}{4}\sin\beta + \sqrt{2}$$

Put $\beta = (\pi/2)$

Then $f\left(\frac{\pi}{2}\right) = \sin\frac{\pi}{2} + \frac{\pi}{2}\cos\frac{\pi}{2} - \frac{\pi}{4}\sin\frac{\pi}{2} + \sqrt{2}$

$$= 1 - \frac{\pi}{4} + \sqrt{2}$$

50. Required area

$$= 4\left[\int_0^{\pi}(x + \sin x)dx - \int_0^{\pi} x dx\right]$$

$$= 4\left[\frac{\pi}{2} - \cos\pi + \cos 0° - \frac{\pi^2}{2}\right]$$

$$= 4 \times 2 = 8 \text{ sq. units}$$

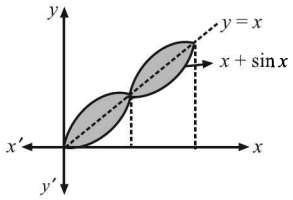


Fig. 16.111

51. Area $= \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4}\right)dx = \left[2\left(\frac{x^{3/2}}{3/2}\right) - \frac{x^3}{12}\right]_0^4$

$$= \frac{4}{3} \times 8 - \frac{64}{12} = \frac{32}{3} - \frac{16}{3} = \frac{16}{3}$$

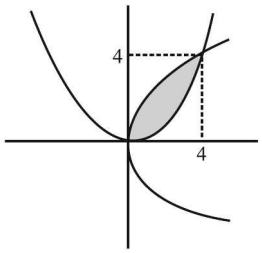


Fig. 16.112

52. $y = 4x^2$

$$y = \frac{1}{9}x^2$$

$$\text{Area} = 2\int_0^2 \left(3\sqrt{y} - \frac{\sqrt{y}}{2}\right)dy$$

$$= 2\left[\frac{5y\sqrt{y}}{2(3/2)}\right]_0^2$$

$$= 2 \cdot \frac{5}{3} 2\sqrt{2} = \frac{20\sqrt{2}}{3}$$

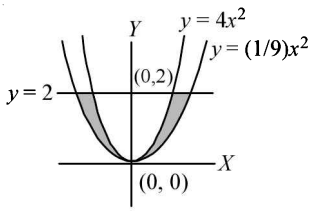


Fig. 16.113

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The area bounded by the curve $y = x^2 + 2x + 1$, the tangent at (1, 4) and the y-axis is

(a) 1 (b) 1/2

(c) 1/3 (d) 1/4

2. The area bounded by the curve $y = \frac{1}{x^2}$ and its asymptote from $x = 1$ to $x = 3$ is

(a) 1/3 (b) 2/3

(c) 1/2 (d) 1/6

3. The area bounded by the curve $y = \sec x$, the x-axis and the lines $x = 0$ and $x = \frac{\pi}{4}$ is

(a) $\sqrt{2}$ (b) 1/2 log 2

(c) $\log(\sqrt{2}+1)$ (d) $\log(\sqrt{2}-1)$

4. The area of the region satisfying $|x|+|y|\le\sqrt{\pi}$ is
(a) 2 (b) 3π
(c) π (d) 4π

5. The area of the smaller portion of the circle $x^2+y^2=a^2$ cut off by the line $x=\frac{a}{2}(a>0)$ is
(a) $\frac{\pi a^2}{3}$ sq. units (b) $a^2\left[\frac{\pi}{3}-\frac{\sqrt{3}}{2}\right]$ sq. units
(c) $a^2\left[\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right]$ sq. units (d) $a^2\left[\frac{\pi}{2}-\sqrt{3}\right]$ sq. units

6. The area of the figure bounded by the right of the line $y=x+1$, $y=\cos x$ and the x-axis is
(a) 1/2 (b) 2/3
(c) 5/6 (d) 3/2

7. The area of the figure bounded by the curves $y=\ln x$, $y=\ln(-x)$, $y=-\ln x$ and $y=-\ln(-x)$ is
(a) 4 (b) 6
(c) 8 (d) none of these

8. The area bounded by $y=\frac{|x|}{x}, x\neq 0$, the y-axis and the curve $y=x^3$ is
(a) $3\sqrt{3}$ (b) $\frac{3}{2}$
(c) $\frac{2}{3}$ (d) $2\sqrt{3}$

9. If the area bounded by the curve $x^2=ky$, $k>0$ and the line $y=3$ is 12 sq. units, then k is
(a) 3 (b) $3\sqrt{3}$
(c) $\frac{3}{4}$ (d) none of these

10. The area of the region bounded by the pairs of lines $y=|x-1|$ and $y=3-|x|$ is
(a) 3 sq. units (b) 4 sq. units
(c) 6 sq. units (d) 2 sq. units

11. Let $f(x)$ be a continuous function such that the area bounded by the curve $y=f(x)$, the x-axis and the two ordinates $x=0$ and $x=a$ is $\frac{a^2}{2}+\frac{a}{2}\sin a+\frac{\pi}{2}\cos a$. Then $f\left(\frac{\pi}{2}\right)$ is
(a) $\frac{1}{2}$ (b) $\frac{\pi^2}{8}+\frac{\pi}{4}$
(c) $\frac{\pi+1}{2}$ (d) $\frac{\pi}{2}$

12. The area bounded by the curve $f(x)=ce^x(c>0)$, the x-axis and the two ordinates $x=p$ and $x=q$ is proportional to
(a) $f(p)\cdot f(q)$ (b) $|f(p)-f(q)|$
(c) $f(p)+f(q)$ (d) $\sqrt{f(p)f(q)}$

13. The area bounded by $y=\tan^{-1}x$, $y\cot^{-1}x$ and the y-axis is
(a) $\log\sqrt{2}$ (b) log 4
(c) log 8 (d) log 2

14. The area of the ellipse $\frac{(x+1)^2}{4}+y^2=1$ falling in the first quadrant is
(a) $\frac{1}{6}(4\pi-3\sqrt{3})$ (b) $\frac{1}{12}(4\pi-3\sqrt{3})$
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{3}(\pi-\sqrt{3})$

15. The area bounded by the curves $y=\sin^{-1}x$, $y=\cos^{-1}x$ and y-axis is
(a) $(2+\sqrt{2})$ (b) $(2-\sqrt{2})$
(c) $(1+\sqrt{2})$ (d) $(\sqrt{2}-1)$

16. The value of the parameter a such that the area bounded by $y=a^2x^2+ax+1$, the coordinate axes and the line $x=1$ attains its least value is equal to
(a) $-\frac{1}{4}$ (b) $-\frac{1}{2}$
(c) $-\frac{1}{3}$ (d) -1

17. The curve $x=4-3y-y^2$ cuts the y-axis at two points P and Q. Then the area enclosed by the y-axis and the portion of the curve which lies between P and Q is
(a) 20 sq. units (b) 18 sq. units
(c) 17 sq. units (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. The area of the curve $y^2=(7-x)(5+x)$ above the x-axis and between the ordinates $x=-5$ and $x=1$ is
(a) 9π (b) 18π
(c) 15π (d) none of these

2. The area bounded by $x^2+y^2-2x=0$ and $y=\sin\frac{\pi x}{2}$ in the upper half of the circle is
(a) $\pi/2-4/\pi$ (b) $\pi/4-2/\pi$
(c) $\pi-8/\pi$ (d) none of these

3. The area common to the circle $x^2 + y^2 = 64$ and the parabola $y^2 = 12x$ is equal to

(a) $\frac{16}{3}(4\pi + \sqrt{3})$ (b) $\frac{16}{3}(8\pi - \sqrt{3})$

(c) $\frac{16}{3}(4\pi - \sqrt{3})$ (d) none of these

4. The area of the region bounded by $y = x^2$ and $y = \frac{2}{1+x^2}$ is

(a) $\pi - \frac{2}{3}$ (b) $\pi - \frac{1}{2}$

(c) $2\pi - \frac{2}{3}$ (d) none of these

5. The area of the region formed by

$x^2 + y^2 - 6x - 4y + 12 \le 0$, $y \le x$ and $x \le \frac{5}{2}$ is

(a) $\frac{\pi}{6} - \frac{\sqrt{3}+1}{8}$ (b) $\frac{\pi}{6} + \frac{\sqrt{3}+1}{8}$

(c) $\frac{\pi}{6} - \frac{\sqrt{3}-1}{8}$ (d) none of these

6. The curve $y = a\sqrt{x} + bx$ passes through the point (1, 2) and the area enclosed by the curve, the x-axis and the line $x = 4$ is 8 sq. units. The values of a and b are

(a) $a = 4, b = -2$ (b) $a = 3, b$ –1

(c) $a = -3, b = 5$ (d) $a = -1, b = 3$

7. The area of the closed figure bounded by $y = x, y = -x$ and the tangent to the curve $y = \sqrt{x^2 - 5}$ at the point (3, 2) is

(a) 5 (b) $\frac{15}{2}$

(c) 10 (d) $\frac{35}{2}$

8. Let $f(x) = \min\left[\tan x, \cot x, \frac{1}{\sqrt{3}}\right], \forall x \in \left(0, \frac{\pi}{2}\right)$. Then the area bounded by $y = f(x)$ and the x-axis is

(a) $\log\left(\frac{4}{3}\right) + \frac{\pi}{6\sqrt{3}}$ (b) $\log\left(\frac{2}{\sqrt{3}}\right) + \frac{\pi}{12\sqrt{3}}$

(c) $\log\left(\frac{4}{3}\right) + \frac{\pi}{12\sqrt{3}}$ (d) none of these

9. If A_m represents the area bounded by the curve $y = \log x^m$, the x-axis and the lines $x = 1$ and $x = e$, then $A_m + mA_{m-1}$ is

(a) m (b) m^2

(c) $m^2/2$ (d) $m^2 - 1$

10. Let $f(x) = ax^2 + bx + c$, where $a \in \mathbb{R}^+$ and $b^2 - 4ac < 0$. The area bounded by $y = f(x)$, the x-axis and the lines $x = 0, x = 1$ is

(a) $\frac{1}{6}[3f(1) + f(-1) + 2f(0)]$

(b) $\frac{1}{12}[5f(1) + f(-1) + 8f(0)]$

(c) $\frac{1}{6}[3f(1) - f(-1) + 2f(0)]$

(d) $\frac{1}{12}[5f(1) - f(-1) + 8f(0)]$

11. Consider a rectangle $ABCD$ formed by the points $A \equiv (0,0), B \equiv (6,0), C \equiv (6,0)$ and $D \equiv (0,4)$. $P(x, y)$ is a moving interior point of the rectangle, moving in such a way that $d(P, AB) \le$ min. $\{(d(P, BC), d(P, CD), d(P, AD)\}$. Here, $d(P, AB)$, $d(P, BC)$, $d(P, CD)$ and $d(P, AD)$ represent the distances of the point P from the sides AB, BC, CD and AD, respectively. The area of the region representing all possible positions of the point P is

(a) 8 (b) 4

(c) 12 (d) 6

12. The area bounded by the curves $x = a\cos^3 t$ and $y = a\sin^3 t$ is

(a) $\frac{3\pi a^2}{8}$ (b) $\frac{3\pi a^2}{16}$

(c) $\frac{3\pi a^2}{32}$ (d) $3\pi a^2$

13. If the area bounded by the curve $y = f(x)$, the lines $x = 1$, $x = b$ and the x-axis is $(b-1)\cos(3b+4), b > 1$, then $f(x)$ is

(a) $(x-5)\sin(3x-4)$

(b) $(x-1)\sin(x+1) + (x+1)\cos(x-1)$

(c) $\cos(3x+4) - 3(x-1)\sin(3x+4)$

(d) $(x-5)\cos(3x+4)$

14. The area common to the region determined by $y \ge \sqrt{x}$ and $x^2 + y^2 < 2$ has the value

(a) $\pi - 2$ (b) $2\pi - 1$

(c) $3\pi - \sqrt{2}/3$ (d) none of these

15. The area bounded by the curve $y = \max.\{\sin x, \cos x\}$, the x-axis and between the lines $x = \frac{\pi}{4}$ and $x = 2\pi$ is

(a) $\frac{(4\sqrt{2}-1)}{\sqrt{2}}$ (b) $(4\sqrt{2}-1)$

(c) $\frac{(4\sqrt{2}-1)}{2}$ (d) none of these

16. Consider a triangle OAB formed by the points $O \equiv (0, 0)$ $A \equiv (2,0), B \equiv (1,\sqrt{3})$. $P(x, y)$ is an arbitrary interior point of the triangle, moving in such a way that $d(P,OA) + d(P,PB) + d(P,OB) = \sqrt{3}$, where $d(P, OA), d(P, AB), d(P, OB)$ represent the distances of 'P' from the sides OA, AB and OB, respectively. The area of the region representing all possible positions of the point P is

(a) $2\sqrt{3}$ (b) $\sqrt{6}$

(c) $\sqrt{3}$ (d) none of these

17. If A_1 is the area enclosed by the curve $xy = 1$, the x-axis and the ordinates $x = 1$, $x = 2$ and A_2 is the area enclosed by the curve $xy = 1$, the x-axis and the ordinates $x = 2$, $x = 4$, then

(a) $A_1 = 2A_2$ (b) $A_2 = 2A_1$

(c) $A_2 = 2A_1$ (d) $A_1 = A_2$

18. The slope to the curve $y = f(x)$ at $(x, f(x))$ is $2x+1$. If the curve passes through the point (1, 2), then the area of the region bounded by the curve $y = f(x)$, the x-axis and the line $x = 1$ is

(a) $\frac{5}{6}$ (b) $\frac{6}{5}$

(c) $\frac{1}{6}$ (d) 6

ANSWERS

Level 1

1. (c)	**2.** (b)	**3.** (c)	**4.** (a)	**5.** (c)	**6.** (d)	**7.** (a)	**8.** (b)	**9.** (a)	**10.** (b)
11. (a)	**12.** (b)	**13.** (d)	**14.** (c)	**15.** (b)	**16.** (c)	**17.** (d)			

Level 2

1. (a)	**2.** (a)	**3.** (a)	**4.** (a)	**5.** (c)	**6.** (b)	**7.** (a)	**8.** (a)	**9.** (b)	**10.** (d)
11. (a)	**12.** (a)	**13.** (c)	**14.** (d)	**15.** (a)	**16.** (c)	**17.** (d)	**18.** (a)		

17

HENRI POINCARE

Differential Equations

- Definition of a Differential Equation
- Order and Degree of a Differential Equation
- Solution of a Differential Equation
- General Solution of a Differential Equation
- Particular Solution
- Differential Equation of First Order and First Degree
- Homogeneous Function
- Homogeneous Differential Equation
- Linear Differential Equations
- Equation Reducible to Homogeneous Form
- Bernoulli's Differential Equations

☞ DEFINITION OF A DIFFERENTIAL EQUATION

A differential equation is an equation which contains a derivative such as dy/dx.

An equation involving an independent variable, a dependent variable and the derivatives of the dependent variable is called a **differential equation**.

Examples: (i) $x\frac{dy}{dx}+y=x^3$ (ii) $\left(\frac{d^2y}{dx^2}\right)+4\left(\frac{dy}{dx}\right)^3+3y=0$

☞ ORDER AND DEGREE OF A DIFFERENTIAL EQUATION

Order

The order of the highest order derivative appearing in a differential equation is called the **order** of the differential equation.

Degree

The power of the highest order derivative appearing in a differential equation, after it is made free from radicals and fraction, is called the **degree** of the differential equation.

Examples:

(i) $\frac{d^2y}{dx^2}-\frac{dy}{dx}+y=0$. This differential equation is of second order as the order of the highest order derivative $\frac{d^2y}{dx^2}$ is two. The degree of this differential equation is one because the power of the highest order derivative $\frac{d^2y}{dx^2}$ is one.

(ii) $\left(\frac{d^3y}{dx^3}\right)^2 - 3\left(\frac{dy}{dx}\right)^3 + 2 = 0$. This differential equation is of third order because the highest order derivative is $\frac{d^3y}{dx^3}$ and its degree is 2 because the power of the highest order derivative $\frac{d^3y}{dx^3}$ is 2.

(iii) $\sqrt{1-x^2}\,dx + \sqrt{1-y^2}\,dy = 0$. This differential equation can be written as $\frac{dy}{dx} = \frac{\sqrt{1-x^2}}{\sqrt{1-y^2}}$. It is of first order and first degree.

SOLUTION OF A DIFFERENTIAL EQUATION

Any relation between the dependent and independent variables (not involving the derivatives) which, when substituted in the differential equation, reduces it to an identity is called a **solution** of the differential equation. Consider the differential equation $dy/dx = \cos x$ (i)

Then, $y = \sin x$ is a solution of Eq. (i).

GENERAL SOLUTION

If the solution of a differential equation of nth order contains n arbitrary constants, the solution is called the **general solution** or the **complete primitive** or the **complete integral** of the differential equation.

Note A differential equation can be derived from its solution by the process of differentiation and elimination of constants. Because of this fact, the solution of a differential equation is also called its primitive or integral.

PARTICULAR SOLUTION

A solution obtained by giving particular values to the arbitrary constants in the general solution is called a **particular solution** or a **particular integral**.

Note The general solution of a differential equation may have several forms but the arbitrary constants (parameters in one form) will be related to the parameters in the other form.

DIFFERENTIAL EQUATIONS OF FIRST ORDER AND FIRST DEGREE

A differential equation of first order and first degree involving x, y and dy/dx can be put in any one of the following forms:

$$\frac{dy}{dx} = f(x, y) \quad \text{or} \quad f\left(x, y, \frac{dy}{dx}\right)$$

or $$f(x, y)dx + g(x, y)dy = 0.$$

METHODS OF SOLVING A FIRST ORDER FIRST DEGREE DIFFERENTIAL EQUATION

Type 1: Differential equation of the type $dy/dx = f(x)$

To solve this type of differential equation, we integrate both sides to obtain the general solution. Thus,

$$\frac{dy}{dx} = f(x) \quad \Rightarrow \quad dy = f(x)dx$$

Integrating both sides, we obtain

$$\int dy = \int f(x)dx + C \quad \text{or} \quad y = \int f(x)dx + C$$

Illustration 1. Find the solution of the differential equation $xy\frac{dy}{dx} = \frac{1+y^2}{1+x^2}(1+x+x^2)$.

Solution: $xy\frac{dy}{dx} = \frac{1+y^2}{1+x^2}(1+x+x^2)$

$$\Rightarrow \quad \frac{y}{1+y^2}dy = \frac{(1+x+x^2)}{x(1+x^2)}dx$$

$$\Rightarrow \quad \int \frac{y}{1+y^2}dy = \int\left(\frac{1}{x} + \frac{1}{1+x^2}\right)dx$$

$$\Rightarrow \quad \frac{1}{2}\log(1+y^2) = \log x + \tan^{-1}x + c$$

$$\therefore \quad \log\frac{\sqrt{1+y^2}}{x} = \tan^{-1}x + c$$

Type 2: Differential equation of the type $dy/dx = f(y)$

To solve this type of equation, we integrate both sides to obtain the general solution. Thus,

$$\frac{dy}{dx} = f(y) \quad \Rightarrow \quad \frac{dx}{dy} = \frac{1}{f(y)} \quad \Rightarrow \quad dx = \frac{1}{f(y)}dy$$

Integrating both sides, we get

$$\int dx = \int \frac{1}{f(y)}dy + C \quad \text{or} \quad x = \int \frac{1}{f(y)}dy + C$$

Type 3: Equations with variables separable

If a given differential equation is capable of being expressed in the form $f_1(x)dx = f_2(y)dy$, then by integrating on both sides, we obtain the required solution.

Type 4: Equations reducing to a variable separable form
Differential equations of the form $dy/dx = f(ax + by + c)$ can be reduced to a variable separable form by the substitution of $ax + by + c = v$.

☞ HOMOGENEOUS FUNCTION

A function $f(x, y)$ in x and y is called a homogeneous function of degree n if the degree of each term is n.

☞ HOMOGENEOUS DIFFERENTIAL EQUATION

A differential equation of the form $\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$, where $f(x, y)$ and $g(x, y)$ are homogeneous functions of the same degree in x and y, is called a homogeneous differential equation.

Examples:

(i) $\frac{dy}{dx} = x \sin\left(\frac{y}{x}\right)$ is a homogeneous differential equation, as $x \sin\left(\frac{y}{x}\right)$, being a function of $\frac{y}{x}$, is a homogeneous function.

(ii) $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ is a homogeneous differential equation, as $(x^2 + y^2)$ and xy are homogeneous functions.

Note A function of x/y is always a homogeneous function.

☞ STEPS OF SOLVING A HOMOGENEOUS EQUATION

Step 1. Put the differential equation in the form $\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$.

Step 2. Put $y = vx$, differentiate with respect to x and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ in the equation in step 1 and cancel out x from the right-hand side. The equation reduces to the form $v + x\frac{dv}{dx} = F(v)$.

Step 3. Shift v on R.H.S. and separate the variables v and x.

Step 4. Integrate both the sides to obtain the solution in terms of v and x.

Step 5. Replace by y/x in the solution contained in Step 4 to obtain the solution in terms of x and y.

Illustration 2. The slope of the tangent at (x, y) to a curve passing through $(1, \pi/4)$ is given by $y/x - \cos^2 y/x$. Find the equation of the curve.

Solution: $\frac{dy}{dx} = \frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$

Put $\frac{y}{x} = v \Rightarrow y = vx$

so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \quad v + x \cdot \frac{dv}{dx} = v - \cos^2 v$

$\Rightarrow \quad x\frac{dv}{dx} = -\cos^2 v \Rightarrow \sec^2 v \cdot dv = \frac{-dx}{x}$

Integrating $\tan v + \log x = c \Rightarrow \tan\frac{y}{x} + \log x = c$

It passes through $(1, \pi/4)$.

$\therefore \quad \tan \pi/4 + \log 1 = c \Rightarrow c = 1$

$\therefore \quad \tan\frac{y}{x} + \log x = 1 = \log e$

$\Rightarrow \quad \tan\frac{y}{x} = \log\left(\frac{e}{x}\right) \Rightarrow y = x\tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$

☞ LINEAR DIFFERENTIAL EQUATIONS

A differential equation in which the dependent variable and all its derivatives occur in the first degree only and are not multiplied together is called a **linear differential equation**.

Examples:

(i) $\frac{dy}{dx} + 6y = x^2$ is a linear equation of order 1.

(ii) $x\frac{d^2y}{dx^2} + 6\frac{dy}{dx} - 9 = 0$ is a linear equation of order 2.

Type 1: $\frac{dy}{dx} + Py = Q$

where P and Q are constants or functions of x only, is called a linear differential equation.

Steps of solving linear equation $\left(\frac{dy}{dx} + Py = Q\right)$

Step 1. Find I.F. $= e^{\int P dx}$
(where I.F. is called integrated factor)

Step 2. The solution (I.F.) is

$$y \times (\text{I.F.}) = \int [Q \times (\text{I.F.})]\, dx + c$$

Step 3. If needed, use the formula $e^{\log f(x)} = f(x)$, $e^{-\log_e f(x)} = \frac{1}{f(x)}$

Example: $e^{\log x^2} = x^2$, $e^{2\log\tan x} = e^{\log(\tan x)^2} = \tan^2 x$

Note The coefficient of dy/dx must be made unity.

Type 2: $\frac{dx}{dy} + Px = Q$

where P and Q are constants or functions of y only, is called a linear differential equation.

Illustration 3. Find the solution of the differential equation $\cos x \frac{dy}{dx} + y \sin x = 1$.

Solution: $\frac{dy}{dx} + y\tan x = \sec x$

$P = \tan x, Q = \sec x$

$\therefore \quad \int Px = \int \tan dx = -\int \frac{-\sin x}{\cos x} dx = -\log\cos x$

$\therefore \quad \text{I.F.} = e^{\int P dx} = e^{-\log\cos x}$

$= e^{\log(\cos x)^{-1}} = (\cos x)^{-1} = \sec x$

Steps of solving linear equation $\left(\frac{dx}{dy} + Px = Q\right)$

Step 1. Find I.F. = $e^{\int p dy}$.

Step 2. The solution (I.F.) is $x \times (\text{I.F.}) = \int [Q \times (\text{I.F.})]\, dy + c$

☞ EQUATION REDUCIBLE TO HOMOGENEOUS FORM

1. A differential equation of the form $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$, where $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$, is solved by putting t equal to either $a_1x + b_1y$ or $a_2x + b_2y$.

2. A differential equation of the form $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$, where $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, is solved by putting $x = X + h$ and $y = Y + k$, where h and k are constants, satisfying the equations $a_1h + b_1k + c_1 = 0$ and $a_2h + b_2k + c_2 = 0$. The given equation reduces to $\frac{dY}{dX} = \frac{a_1X + b_1Y}{a_2X + b_2Y}$. This is a homogeneous differential equation and is solved by putting $y = vx$.

Illustration 4. Find the solution of the differential equation $\frac{dy}{dx} + \frac{3x + 2y - 5}{2x + 3y - 5} = 0$.

Solution: Put $x = X + h, y = Y + k$

$\therefore \quad \left.\begin{matrix} dx = dX \\ dy = dY \end{matrix}\right] \Rightarrow \frac{dy}{dx} = \frac{dY}{dX}$

$$\frac{dY}{dX} + \frac{3X + 2Y + 3h + 2k - 5}{2X + 3Y + 2h + 3k - 5} = 0$$

Put $\quad 3h + 2k - 5 = 0$

$2h + 3k - 5 = 0$

$\therefore \quad \frac{h}{5} = \frac{k}{5} = \frac{1}{5} \therefore h = 1,\ k = 1$

$\therefore \quad \frac{dY}{dX} + \frac{3X + 2Y}{2X + 3Y} = 0$

Put $\quad Y = vX$

$\therefore \quad v + \frac{X dv}{dX} + \frac{3 + 2v}{2 + 3v} = 0$

$\therefore \quad X\frac{dv}{dX} + \frac{3 + 2v + 2v + 3v^2}{2 + 3v} = 0$

$\therefore \quad \frac{3v + 2}{3v^2 + 4v + 3} dv + \frac{dX}{X} = 0$

$\Rightarrow \quad \frac{6v + 4}{3v^2 + 4v + 3} dv + \frac{2dX}{X} = 0$

$\therefore \quad \log(3v^2 + 4v + 3) + 2\log X = \log c$

$\Rightarrow \quad \log\left(3\frac{Y^2}{X^2} + 4\frac{Y}{X} + 3\right) + \log X^2 = \log c$

$\Rightarrow \quad \log(3Y^2 + 4YX + 3X^2) = \log c$

$\Rightarrow \quad 3(y-1)^2 + 4(y-1)(x-1) + 3(x-1)^2 = c_1$

$\Rightarrow \quad 3(x^2 + y^2) + 4xy - 10(x + y) = k$

where $\quad k = c_1 - 7$

☞ BERNOULLI'S DIFFERENTIAL EQUATIONS

The differential equation of the type $\frac{dy}{dx} + P(y) = Qy^n$, where P and Q are constants or functions of x alone and n is a non-

zero constant other than unity is known as **Bernoulli's equation**. This equation can be reduced to a linear differential equation by dividing both the sides by y^n and putting $t = \frac{1}{y^{n-1}}$. Then we can solve it easily.

The Differential Equation of the Type $\frac{d^2y}{dx^2} = f(x)$

Steps for solving differential equation $\left(\frac{d^2y}{dx^2} = f(x)\right)$

Step 1. Obtain the differential equation. Let the differential equation be $\frac{d^2y}{dx^2} = f(x)$.

Step 2. Integrate both the sides of the differential equation in Step 1 with respect to x to obtain a first-order first-degree differential equation.

Step 3. Integrate both the sides of the first-order differential equation obtained in Step 2 with respect to x, to get the desired solution.

The Differential Equation of the Type $\frac{d^2y}{dx^2} = f(y)$

Steps for solving differential equation $\left(\frac{d^2y}{dx^2} = f(y)\right)$

Step 1. Obtain the differential equation. Let the differential equation be $\frac{d^2y}{dx^2} = f(y)$.

Step 2. Multiply both the sides of the differential equation by $2\frac{dy}{dx}$ to get $2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} = 2f(y)\frac{dy}{dx}$ or $\frac{d}{dx}\left(\frac{dy}{dx}\right)^2 = 2f(y)\frac{dy}{dx}$.

Step 3. Integrate both the sides with respect to x.

$$\int \frac{d}{dx}\left(\frac{dy}{dx}\right)^2 dx = \int 2f(y)\frac{dy}{dx}\cdot dx + c_1 \Rightarrow \left(\frac{dy}{dx}\right)^2 = 2\int f(y)dy + c_1$$

Step 4. Obtain the value of $\frac{dy}{dx}$ from the differential equation obtained in Step 3. $\frac{dy}{dx} = \sqrt{F(y)}$.

Step 5. Separate the variables and integrate to get the desired solution. $\frac{dy}{\sqrt{F(y)}} = dx \Rightarrow \int \frac{dy}{\sqrt{F(y)}} = x + c$

SOLVED PROBLEMS

1. The family of curves $y = e^{a\sin x}$, where a is an arbitrary constant, is represented by the differential equation

(a) $\log y = \tan x\frac{dy}{dx}$ (b) $y\log y = \tan x\frac{dy}{dx}$

(c) $y\log y = \sin x\frac{dy}{dx}$ (d) $\log y = \cos x\frac{dy}{dx}$

Ans. (b)

Solution: Curve is $y = e^{a\sin x} \Rightarrow \sin x = \frac{\log y}{a}$

$$\therefore \quad \frac{dy}{dx} = e^{a\sin x}.a\cos x$$

$$\Rightarrow \quad \frac{dy}{dx} = y\cos x.\frac{\log y}{\sin x}$$

$$\Rightarrow \quad y\log y = \tan x\frac{dy}{dx}$$

2. The differential equation of the family of curves $y = ae^x + bxe^x + cx^2e^x$, where a, b and c are aribitrary constants, is

(a) $y''' + 3y'' + 3y' + y = 0$

(b) $y''' + 3y'' - 3y' - y = 0$

(c) $y''' - 3y'' - 3y' + y = 0$

(d) $y''' - 3y'' + 3y' - y = 0$

Ans. (d)

Solution: $y = ae^x + bxe^x + cx^2e^x$ (i)

On differentiating w.r.t. x, we get

$$y' = ae^x + b(xe^x + e^x) + c(x^2e^x + 2xe^x)$$

$$\Rightarrow \quad y' = ae^x + bxe^x + cx^2e^x + be^x + 2cxe^x$$

$$\Rightarrow \quad y' = y + be^x + 2cxe^x \quad \text{(ii)}$$

Again differentiating w.r.t. x, we get

$$y'' = y' + be^x + 2c(xe^x + e^x)$$

$$\Rightarrow \quad y'' = y' + be^x + 2cxe^x + 2ce^x$$

$$\Rightarrow \quad y'' = 2y' - y + 2ce^x \quad \text{(iii)}$$

[from Eq. (ii)]

Again, differentiating w.r.t. x, we get

$$y''' = 2y'' - y' + 2ce^x$$

$$\Rightarrow \quad y''' = 2y'' - y' + (y'' - 2y' + y)$$

[from Eq. (iii)]

$$\Rightarrow \quad y''' - 3y'' + 3y' - y = 0$$

3. The differential equation of all circles which pass through the origin and whose centres lie on the y-axis is

(a) $(x^2 - y^2)\frac{dy}{dx} - 2xy = 0$

(b) $(x^2 - y^2)\frac{dy}{dx} + 2xy = 0$

(c) $(x^2 - y^2)\frac{dy}{dx} - xy = 0$

(d) $(x^2 - y^2)\frac{dy}{dx} + xy = 0$

Ans. (a)

Solution: The equation of the circle through the origin, whose centre lies on the y-axis, can be written as $x^2 + y^2 + 2fy = 0$, f being a parameter.

Differenting w.r.t. x, we get $2x + 2y\frac{dy}{dx} + 2f\frac{dy}{dx} = 0$

$$\Rightarrow \quad 2xy + (2y^2 - x^2 - y^2)\frac{dy}{dx} = 0$$

[Put value of $2f = \frac{-x^2 - y^2}{y}$]

$$\Rightarrow \quad (x^2 - y^2)\frac{dy}{dx} - 2xy = 0$$

4. The differential equation of the family of lines which pass through $(1, -1)$ is

(a) $y = (x+1)\frac{dy}{dx} + 1$ (b) $y = (x+1)\frac{dy}{dx} - 1$

(c) $y = (x-1)\frac{dy}{dx} + 1$ (d) $y = (x-1)\frac{dy}{dx} - 1$

Ans. (d)

Solution: The gradient of a line = dy/dx.

$\therefore$ The equation of the line passing through $(1, -1)$ is

$$y + 1 = \frac{dy}{dx}(x-1)$$

$$\Rightarrow \quad \frac{dx}{x-1} = \frac{dy}{y+1} \quad \Rightarrow \quad \int\frac{dx}{x-1} = \int\frac{dy}{y+1}$$

$$\Rightarrow \quad \log(x-1) = \log(y+1) - \log c$$

$$\Rightarrow \quad \log(y+1) = \log[c(x-1)]$$

$$\therefore \quad y = c(x-1) - 1$$

5. The solution of $(3x + 2y + 1)dx - (3x + 2y - 1)dy = 0$ is

(a) $\log(15x + 10y - 1) = c$

(b) $\log(15x + 10y - 1) + \frac{5}{2}(x - y) = c$

(c) $\frac{5}{2}(x-2) + \log(15x) = c$

(d) none of these

Ans. (b)

Solution: We have $\frac{dy}{dx} = \frac{3x + 2y + 1}{3x + 2y - 1}$

Put $3x + 2y = z$

$$\therefore \quad 3 + 2\frac{dy}{dx} = \frac{dz}{dx}$$

$$\therefore \quad \frac{1}{2}\left(\frac{dz}{dx} - 3\right) = \frac{z+1}{z-1}$$

$$\Rightarrow \quad \frac{dz}{dx} = 3 + \frac{2z+2}{z-1} = \frac{5z-1}{z-1} \quad \Rightarrow \quad \frac{z-1}{5z-1}dz = dx$$

$$\Rightarrow \quad \frac{1}{5}\left(\frac{5z-1-4}{5z-1}\right)dz = dx$$

$$\Rightarrow \quad \frac{1}{5}\int dz - \frac{4}{5}\times\frac{1}{5}\int\frac{5dz}{5z-1} = \int dx + c$$

$$\Rightarrow \quad \frac{z}{5} - \frac{4}{25}\log(5z-1) = x + c$$

$\Rightarrow$ $5(3x + 2y) - 4\log(5(3x + 2y) - 1) = 25x +$ constant

$\Rightarrow$ $4\log(15x + 10y - 1) + 10x - 10y =$ constant

$$\Rightarrow \quad \log(15x + 10y - 1) + \frac{5}{2}(x - y) = c$$

6. The general solution of the equation $\frac{dy}{dx} = \frac{x^2}{y^2}$ is

(a) $x^3 - y^3 = c$ (b) $x^3 + y^3 = c$

(c) $x^2 + y^2 = c$ (d) $x^2 - y^2 = c$

Ans. (a)

Solution: $\frac{dy}{dx} = \frac{x^2}{y^2} \Rightarrow y^2dy = x^2dx \Rightarrow \frac{y^3}{3} = \frac{x^3}{3} + \frac{c}{3}$

$$\Rightarrow \quad x^3 - y^3 = c$$

7. The order and degree of the differential equation $\sqrt{\frac{dy}{dx}} - 4\frac{dy}{dx} - 7x = 0$ are

(a) 1 and $\frac{1}{2}$ (b) 2 and 1

(c) 1 and 1 (d) 1 and 2

Ans. (d)

Solution: $\sqrt{\frac{dy}{dx}} = 4\frac{dy}{dx} + 7x$

$$\Rightarrow \quad \frac{dy}{dx} = \left(4\frac{dy}{dx} + 7x\right)^2$$

The order is 1 and the degree is 2.

8. The degree of the differential equation satisfying $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x - y)$ is

(a) 1 (b) 2
(c) 3 (d) none of these

Ans. (a)

Solution: Putting $x = \sin A$, $y = \sin B$, we get

$$a(\sin A - \sin B) = \cos A + \cos B$$

$$\Rightarrow \quad 2a\cos\frac{A+B}{2}\sin\frac{A-B}{2} = 2a\cos\frac{A+B}{2}\cos\frac{A-B}{2}$$

$$\Rightarrow \quad \cot\frac{A-B}{2} = a \Rightarrow A - B = 2\cot^{-1} a$$

$$\Rightarrow \quad \sin^{-1} x - \sin^{-1} y = 2\cot^{-1} a \ = \text{constant}$$

Differentiating both sides w.r.t. x, we get

$$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0 \quad \therefore \quad \frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

This is of degree 1.

9. The orthogonal trajectories of the family of curves $y = cx^k$ are given by

(a) $x^2 + cy^2 = \text{constant}$ (b) $x^2 + ky^2 = \text{constant}$
(c) $kx^2 + y^2 = \text{constant}$ (d) $x^2 - ky^2 = \text{constant}$

Ans. (b)

Solution: Differentiating, we have

$$\frac{dy}{dx} = ckx^{k-1} \Rightarrow c = \frac{1}{k}x^{1-k}\frac{dy}{dx}$$

Putting this value in the given equation, we have

$$y = \frac{1}{k}x^{1-k}\frac{dy}{dx}x^k = \frac{1}{k}x\frac{dy}{dx}$$

Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$y = -\frac{1}{k}x\frac{dx}{dy} \Rightarrow ky\,dy + x\,dx = 0$$

$$\Rightarrow \quad ky^2 + x^2 = \text{constant}$$

10. The solution of the differential equation $(1 + \cos x)\frac{dy}{dx} = 1 - \cos x$ is

(a) $y = \tan\frac{x}{2} + x + c$ (b) $y = 2\tan\frac{x}{2} - x + c$
(c) $y = \tan\frac{x}{2} - x + c$ (d) $y = x - 2\tan\frac{x}{2} + c$

Ans. (b)

Solution: $(1+\cos x)\frac{dy}{dx} = 1 - \cos x$

$$\Rightarrow \quad \int dy = \int\frac{1-\cos x}{1+\cos x}dx \quad \Rightarrow \quad \int dy = \int\tan^2\frac{x}{2}dx + c$$

$$\Rightarrow \quad y = \int\left(\sec^2\frac{x}{2} - 1\right)dx + c$$

$$\therefore \quad y = 2\tan\frac{x}{2} - x + c$$

11. For solving $\frac{dy}{dx} = 4x + y + 1$, the suitable substitution is

(a) $y = vx$ (b) $y = 4x + v$
(c) $y = 4x$ (d) $y + 4x + 1 = v$

Ans. (d)

Solution: Put $4x + y + 1 = v$. Differentiating w.r.t. x, we get

$$4 + \frac{dy}{dx} = \frac{dv}{dx} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{dv}{dx} - 4$$

$$\Rightarrow \quad \frac{dv}{dx} - 4 = v \quad \Rightarrow \quad \frac{dv}{dx} = 4 + v \quad \Rightarrow \quad \frac{dv}{4+v} = dx$$

Integrating, we have $\log(4 + v) = x + c$

12. The general solution of the differential equation $x(1 + y^2)dx + y(1 + x^2)dy = 0$ is

(a) $(1 + x^2)(1 + y^2) = c$
(b) $(1 + x^2) = c(1 + y^2)$
(c) $(1 + y^4) = c(1 + x^2)$
(d) $(1 + x^2)(1 + y^2) = 0$, c is constant

Ans. (a)

Solution: General solution is given by

$$\int\frac{2x}{1+x^2}dx + \int\frac{2y}{1+y^2}dy = \text{constant}$$

$$\Rightarrow \quad \log(1 + x^2) + \log(1 + y^2) = \log c$$

$$\Rightarrow \quad (1 + x^2)(1 + y^2) = c$$

13. The solution of the differential equation

$\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$ is

(a) $e^y = e^x + \frac{x^3}{3} + c$ (b) $e^y = e^x + c$
(c) $y = e^x + c$ (d) $y = x + x^2 + c$

Ans. (a)

Solution: $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$

$$\Rightarrow \quad e^y\frac{dy}{dx} = e^x + x^2$$

$$\int e^y dy = \int(e^x + x^2)dx$$

$$\therefore \quad e^y = e^x + \frac{x^3}{3} + c$$

14. The family $y = Ax + A^3$ of curves will correspond to a differential equation of order

(a) 3 (b) 2

(c) 1 (d) none of these

Ans. (c)

Solution: $y = Ax + A^3 \Rightarrow \dfrac{dy}{dx} = A$

$\therefore \quad y = x\dfrac{dy}{dx} + \left(\dfrac{dy}{dx}\right)^3$. This is of order 1.

(Or since only one arbitrary constant A is present, therefore, the order of the differential equation is one.)

15. The solution of the differential equation $x^2dy + y(x + y)dx = 0$ is

(a) $x^2y = c^2(2x + y)$ (b) $xy^2 = c^2(2x + y)$

(c) $x^2y = c^2(x + 2y)$ (d) $xy^2 = c^2(x + 2y)$

Ans. (a)

Solution: $x^2\dfrac{dy}{dx} + xy + y^2 = 0$

Dividing by x^2y^2, we get

$$y^{-2}\frac{dy}{dx} + \frac{1}{x}\cdot\frac{1}{y} = -\frac{1}{x^2}$$

Put $\dfrac{1}{y} = v$ so that $-\dfrac{1}{y^2}\dfrac{dy}{dx} = \dfrac{dv}{dx}$

$\therefore \quad \dfrac{-dy}{dx} + \dfrac{1}{x}v = \dfrac{-1}{x^2} \Rightarrow \dfrac{dv}{dx} - \dfrac{1}{x}v = \dfrac{1}{x^2}$ [Linear in v]

Here $P = \dfrac{-1}{x}, Q = \dfrac{1}{x^2}, \int P dx = -\log x = \log x^{-1}$

$\therefore \quad$ I.F. $e^{\log x^{-1}} = x^{-1} = \dfrac{1}{x}$

$\therefore$ The solution is $v\dfrac{1}{x} = \int \dfrac{1}{x}\cdot\dfrac{1}{x^2}dx + c = \dfrac{x^{-2}}{-2} + c'$

$\Rightarrow \quad \dfrac{1}{y}\dfrac{1}{x} = \dfrac{-1}{2x^2} + c'$

$\Rightarrow \quad 2x = -y + 2c'x^2y$

$\Rightarrow \quad 2c'x^2y = 2x + y \quad \Rightarrow \quad x^2y = c^2(2x + y)$

16. The solution of the differential equation $x \log x \dfrac{dy}{dx} + y = 2 \log x$ is

(a) $y = \log x + \dfrac{c}{\log x}$ (b) $y = \log x - \dfrac{c}{\log x}$

(c) $y = \log x - c \log x$ (d) none of these

Ans. (a)

Solution: $x \log x \dfrac{dy}{dx} + y = 2 \log x$

$\Rightarrow \quad \dfrac{dy}{dx} + \dfrac{1}{x \log x}y = \dfrac{2}{x}$

I.F. $= e^{\int \frac{1}{x\log x}dx} = e^{\log(\log x)} = \log x$

Hence, the solution of the differential equation is

$$y \cdot \log x = \int \frac{2}{x}\log x dx + c$$

$\Rightarrow \quad y \cdot \log x = (\log x)^2 + c$

$\therefore \quad y = \log x + \dfrac{c}{\log x}$

EXERCISE SET 1

1. The differential equation of the family of curves $y = e^x(A \cos x + B \sin x)$, where A and B are arbitrary constants, is

(a) $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + 2y = 0$

(b) $\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} - 2y = 0$

(c) $\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2 + y = 0$

(d) $\dfrac{d^2y}{dx^2} - 7\dfrac{dy}{dx} + 2y = 0$

2. The solution of the differential equation $\dfrac{dy}{dx} + \dfrac{1 + \cos 2y}{1 - \cos 2x} = 0$ is

(a) $\tan y + \cot x = c$ (b) $\tan y \cot x = c$

(c) $\tan y - \cot x = c$ (d) none of these

3. The solution of the differential equation $(6x + 2y - 10)\dfrac{dy}{dx} = 2x + 9y - 20$ is

(a) $(y + 2x)^2 = c(x - 2y + 5)$

(b) $(y + 2x)^3 = c(x + 2y - 5)$

(c) $(y - 2x)^2 = c(x + 2y - 5)$

(d) none of these

4. The curve whose subtangents are proportional to the abscissae of the points of tangency (k being the constant of proportionality) is

(a) $y^k = cx^2$ (b) $y^k = cx^3$

(c) $y^k = cx$ (d) none of these

5. The solution of the differential equation $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$ is

(a) $\log[1 + \tan(x + y)] = x + c$

(b) $\log[1 - \tan(x + y)] = x + c$

(c) $\log\left[1 + \tan\frac{1}{2}(x + y)\right] = x + c$

(d) $\log\left[1 - \tan\frac{1}{2}(x + y)\right] = x + c$

6. The solution of $\frac{dy}{dx} = \frac{2x - 6y + 7}{x - 3y + 4}$ is

(a) $2x - y + \frac{1}{5}\log(5x - 15y + 17) = c$

(b) $2x - 6y + \frac{1}{5}\log(5x - 15y + 17) = c$

(c) $2x + y + \frac{1}{5}\log(5x - 15y + 17) = c$

(d) none of these

7. The curves which satisfy $(1 - x^2)\frac{dy}{dx} + xy = ax$ are

(a) straight lines and ellipses
(b) circles and ellipses
(c) parabolas and ellipses
(d) ellipses and hyperbolas

8. The solution of the differential equation $(x^2 - yx^2)dy + (y^2 + x^2y^2)dx = 0$ is

(a) $\frac{1}{x} + y + \log y = c$ (b) $\frac{1}{x} + y + \frac{1}{y^2} = c$

(c) $\frac{1}{y} + \frac{1}{x} + \log y - x = c$ (d) none of these

9. The solution of the differential equation $\sqrt{a + x}\frac{dy}{dx} + xy = 0$ is

(a) $y = Ae^{2/3(2a - x)\sqrt{x+a}}$ (b) $y = Ae^{-2/3(a-x)\sqrt{x+a}}$

(c) $y = Ae^{2/3(a+x)\sqrt{x+a}}$ (d) $y = Ae^{-2/3(2a-x)\sqrt{x+a}}$

10. The curve for which the normal at any point (x, y) and the line joining the origin to that point form an isosceles triangle with the x-axis as base is

(a) an ellipse (b) a rectangular hyperbola
(c) a circle (d) none of these

11. The solution of the differential equation $x\frac{dy}{dx} + y = x^2$ $3x + 2$ is

(a) $xy = \frac{x^3}{3} + \frac{3}{2}x^2 + 2x + c$

(b) $xy = \frac{x^4}{4} + x^3 + x^2 + c$

(c) $xy = \frac{x^4}{4} + \frac{x^3}{3} + x^2 + c$

(d) $xy = \frac{x^4}{4} + x^3 + x^2 + cx$

12. The general solution of $\frac{dy}{dx} = 2xe^{x^2 - y}$ is

(a) $e^{x^2 - y} = c$ (b) $e^{-y} + e^{x^2} = c$

(c) $e^{y} = e^{x^2} + c$ (d) none of these

13. The integral curve satisfying $\frac{dy}{dx} = \frac{x^2 + y^2}{x^2 - y^2}$, $y(1) = 1$ has the slope at the point (1, 0) of the curve, is equal to

(a) -1 (b) 1
(c) $-5/3$ (d) $5/3$

14. If $\frac{dy}{dx} = \frac{xy + y}{xy + x}$, then the solution of this equation is

(a) $y = xe^x + c$ (b) $y = e^x + c$

(c) $y = cxe^{x-y}$ (d) $y = x + c$

15. If $x = A\cos 4t + B\sin 4t$, then $\frac{d^2x}{dt^2}$ is equal to

(a) $-16x$ (b) $16x$
(c) x (d) $-x$

16. The degree and order of the differential equation $y = px + \sqrt[3]{a^2p^2 + b^2}$, where $p = \frac{dy}{dx}$, are respectively

(a) 3, 1 (b) 1, 3
(c) 1, 1 (d) 3, 3

17. The solution of the differential equation $\frac{dy}{dx} = 2xy$ is

(a) $y = ce^{x^2}$ (b) $y^2 = 2x^2 + c$

(c) $y = ce^{-x^2} + c$ (d) $y = x^2 + c$

18. The differential equation of $y = \sec(\tan^{-1} x)$ is

(a) $(1 + x^2)\frac{dy}{dx} = y + x$ (b) $(1 + x^2)\frac{dy}{dx} = y - x$

(c) $(1 + x^2)\frac{dy}{dx} = xy$ (d) $(1 + x^2)\frac{dy}{dx} = \frac{x}{y}$

19. The order and degree of the differential equation $\left(1+\left(\frac{dy}{dx}\right)^2\right)^{3/4}=\left(\frac{d^2y}{dx^2}\right)^{1/3}$ are respectively

(a) 2, 4 (b) 2, 3
(c) 6, 4 (d) 6, 9

20. The solution of the differential equation $\frac{dy}{dx}=\frac{x(2\log x+1)}{\sin y+y\cos y}$ is

(a) $y\sin y=x^2\log x+\frac{x^2}{2}+c$

(b) $y\cos y=x^2(\log x+1)+c$

(c) $y\cos y=x^2\log x+\frac{x^2}{2}+c$

(d) $y\sin y=x^2\log x+c$

21. The differential equation of all the straight lines passing through the origin is

(a) $y=\sqrt{x}\frac{dy}{dx}$ (b) $\frac{dy}{dx}=y+x$

(c) $\frac{dy}{dx}=y-x$ (d) none of these

22. The order of the differential equation $\frac{d^2y}{dx^2}=\sqrt{1+\left(\frac{dy}{dx}\right)^2}$ is

(a) 3 (b) 2
(c) 1 (d) 4

23. The solution of the differential equation $x+y\frac{dy}{dx}=2y$ is

(a) $\log(y-x)=c+\frac{y-x}{x}$

(b) $\log(y-x)=c+\frac{x}{y-x}$

(c) $y-x=c+\log\frac{x}{y-x}$

(d) $y+x=c+\log\frac{x}{y-x}$

24. The solution of the differential equation $\frac{dx}{x}+\frac{dy}{y}=0$ is

(a) $xy=c$ (b) $x+y=c$
(c) $\log x\log y=c$ (d) $x^2+y^2=c$

25. The order of the differential equation of all the tangent lines to the parabola $y=x^2$ is

(a) 1 (b) 2
(c) 3 (d) 4

26. A curve is such that the mid-point of the portion of the tangent intercepted between the point where the tangent is drawn and the point, where the tangent meets the y-axis, lies on the line $y=x$. If the curve passes through (1, 0), then the curve is

(a) $2y=x^2-x$ (b) $y=x^2-x$
(c) $y=x-x^2$ (d) $y=2(x-x^2)$

27. The solution of $\frac{dy}{dx}=2^{y-x}$ is

(a) $2^x+2^y=c$ (b) $2^x-2^y=c$
(c) $\frac{1}{2^x}-\frac{1}{2^y}=c$ (d) $\frac{1}{2^x}+\frac{1}{2^y}=c$

28. The solution of the differential equation $x=1+xy\frac{dy}{dx}+\frac{x^2y^2}{2!}\left(\frac{dy}{dx}\right)^2+\frac{x^3y^3}{3!}\left(\frac{dy}{dx}\right)^3+\cdots$ is

(a) $y=\ln(x)+c$ (b) $y^2=(\ln x^2)+c$
(c) $y=\log x+xy$ (d) $xy=x^y+c$

29. The solution of the differential equation $x\frac{dy}{dx}=y(\log y-\log x+1)$ is

(a) $\log\frac{x}{y}=cy$ (b) $\log\frac{y}{x}=cy$
(c) $\log\frac{x}{y}=cx$ (d) $\log\frac{y}{x}=cx$

30. The differential equation $\cot y\,dx=x\,dy$ has a solution of the form

(a) $y=\cos x$ (b) $x=c\sec y$
(c) $x=\sin y$ (d) $y=\sin x$

31. The form of the differential equation of all family of lines $y=mx+\frac{4}{m}$ by eliminating the arbitrary constant m is

(a) $\frac{d^2y}{dx^2}=0$

(b) $x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}+4=0$

(c) $x\left(\frac{dy}{dx}\right)^2+y\frac{dy}{dx}+4=0$

(d) $\frac{dy}{dx}=0$

32. The family of curves represented by $\frac{dy}{dx}=\frac{x^2+x+1}{y^2+y+1}$ and the family of curves represented by $\frac{dy}{dx}+\frac{y^2+y+1}{x^2+x+1}=0$

(a) touch each other (b) are orthogonal
(c) are one and the same (d) none of these

33. The general solution of $\frac{dy}{dx}=\frac{2x-y}{x+2y}$ is

(a) $x^2-xy+y^2=c$ (b) $x^2-xy-y^2=c$
(c) $x^2+xy-y^2=c$ (d) $x^2+xy^2=c$

34. $y=cx-c^2$ is the general solution of the differential equation

(a) $(y')^2-xy'+y=0$ (b) $y''=0$
(c) $y'=c$ (d) $(y')^2+xy'+y=0$

35. The solution of the differential equation $\frac{dy}{dx}=\frac{xy}{x^2+y^2}$ is

(a) $cy^2=e^{\frac{x^2}{y^2}}$ (b) $cy=e^{\frac{x}{y}}$
(c) $y=e^{x^2}+y+c$ (d) $y=e^x+e^y+c$

36. An object falling from rest in air is subject not only to the gravitational force but also to air resistance. Assume that the air resistance is proportional to the velocity with constant of proportionality as $k > 0$ and acts in a direction opposite to motion ($g = 9.8$ m/s^2). Then the velocity cannot exceed

(a) $9.8/k$ m/s (b) $98/k$ m/s
(c) $k/9.8$ m/s (d) none of these

37. The equation of the curve which is such that the portion of the x-axis cut off between the origin and the tangent at any point is proportional to the ordinate of that point is

(a) $x=y(a-b\log y)$ (b) $\log x=by^2+a$
(c) $x^2=y(a-b\log y)$ (d) none of these

38. If a is an arbitrary constant, then the solution of the differential equation $\frac{dy}{dx}+\sqrt{\frac{1-y^2}{1-x^2}}=0$ is

(a) $x\sqrt{1-y^2}+y\sqrt{1-x^2}=a$
(b) $y\sqrt{1-y^2}+x\sqrt{1-x^2}=a$
(c) $x\sqrt{1-y^2}-y\sqrt{1-x^2}=a$
(d) $y\sqrt{1-y^2}-x\sqrt{1-x^2}=a$

39. The differential equation representing the family of curves $y=xe^{cx}$ (c is a constant) is

(a) $\frac{dy}{dx}=\frac{y}{x}\left(1-\log\frac{y}{x}\right)$ (b) $\frac{dy}{dx}=\frac{y}{x}\log\left(\frac{y}{x}\right)+1$
(c) $\frac{dy}{dx}=\frac{y}{x}\left(1+\log\frac{y}{x}\right)$ (d) $\frac{dy}{dx}+1=\frac{y}{x}\log\left(\frac{y}{x}\right)$

40. If $\phi(x)$ is a differentiable function, then the solution of the differential equation $dy+[y\phi'(x)-\phi(x)\phi'(x)]dx=0$ is

(a) $y=[\phi(x)-1]+ce^{-\phi(x)}$
(b) $y\phi(x)=[\phi(x)^2]+c$
(c) $ye^{\phi(x)}=\phi(x)e^{\phi(x)}+c$
(d) $y-\phi(x)=\phi(x)e^{-\phi(x)}$

41. The equation of a curve that passes through the point (1, 2) and satisfies the differential equation $\frac{dy}{dx}=\frac{2xy}{x^2+1}$ is

(a) $y(x^2+1)=1$ (b) $y(x^2-1)=1$
(c) $y(x^2+1)=4$ (d) none of these

42. Let $f(x)$ be differentiable on the interval $(0,\infty)$ such that $f(1) = 1$ and $\lim_{t\to x}\frac{t^2f(x)-x^2f(t)}{t-x}=1$ for each $x > 0$. Then $f(x)$ is

(a) $\frac{1}{3x}+\frac{2}{3}x^2$ (b) $-\frac{1}{3x}+\frac{4}{3}x^2$
(c) $-\frac{1}{x}+\frac{2}{x^2}$ (d) $\frac{1}{x}$

43. The solution of $(y+x+5)dy=(y-x+1)dx$ is

(a) $\log[(y+3)^2+(x+2)^2]+\tan^{-1}\frac{y+3}{x+2}=c$
(b) $\log[(y+3)^2+(x-2)^2]+\tan^{-1}\frac{y-3}{x-2}=c$
(c) $\log[(y+3)^2+(x+2)^2]+2\tan^{-1}\frac{y+3}{x+2}=c$
(d) $\log[(y+3)^2+(x+2)^2]-2\tan^{-1}\frac{y+3}{x+2}=c$

44. The solution of the differential equation $\log_e\left(\frac{dx}{dy}\right)=3x-5y$ is

(a) $5e^{5y}=3e^{3x}+c$ (b) $3e^{5y}-5e^{3x}=c$
(c) $5e^{5y}=-3e^{-3x}+c$ (d) $3e^{5y}-5e^{-3x}=c$

45. The differential equation $\frac{dy}{dx}=\frac{\sqrt{1-y^2}}{y}$ determines a family of circles with
(a) variable radii and a fixed centre at (0, 1)
(b) variable radii and a fixed centre at (0, –1)
(c) a fixed radius 1 and variable centre along the x-axis
(d) a fixed radius 1 and variable centre along the axis (0, –1)

46. The solution of the differential equation $\frac{d^2y}{dx^2}+\frac{1}{x^2}=0$ is
(a) $(\log_e x)^2$ (b) $\log_e x^2$
(c) $\log_e x$ (d) $2\log_x e$

47. The solution of the equation $ydx+xdy=xy(dy-dx)$ is
(a) $x-y-\log(xy)=c$ (b) $x-y+\log xy=c$
(c) $x-y\log(x/y)=c$ (d) $x+\log xy=c$

48. The differential equation of all 'Simple Harmonic Motions' of the given period $\frac{2\pi}{n}$ is
(a) $\frac{d^2x}{dt^2}=nx=0$ (b) $\frac{d^2x}{dt^2}+n^2x=0$
(c) $\frac{d^2x}{dt^2}-n^2x=0$ (d) $\frac{d^2x}{dt^2}+\frac{1}{n^2}x=0$

49. The solution of the given differential equation $\frac{dy}{dx}+2xy=y$ is
(a) $y=ce^{x-x^2}$ (b) $y=ce^{x^2-x}$
(c) $y=ce^{x}$ (d) $y=ce^{-x^2}$

50. The solution of the equation $\frac{dy}{dx}+\sin\frac{x+y}{2}=\sin\frac{x-y}{2}$ is
(a) $\log\left|\tan\frac{y}{2}\right|+2\sin\frac{x}{2}=c$ (b) $\log\left|\tan\frac{y}{4}\right|+2\sin\frac{x}{2}=c$
(c) $\log\left|\tan\frac{y}{4}\right|-2\sin\frac{x}{2}=c$ (d) none of these

51. The equation of a curve passing through (2, 7/2) and having gradient $1-\frac{1}{x^2}$ at (x, y) is
(a) $y=x^2+x+1$ (b) $xy=x^2+x+1$
(c) $xy=x+1$ (d) none of these

52. The solution of the differential equation $(1+y^2)dx+(1+x^2)dy=0$ is
(a) $1-xy=k$ (b) $\frac{x+y}{1-xy}=k$
(c) $\frac{x+y}{xy}=k$ (d) $\frac{1}{x}+\frac{1}{y}=k$

53. The solution of the equation $\frac{dy}{dx}=1+xy+x+y$ is
(a) $(1+y)(1+x)=c$ (b) $\log_e(1+y)=1+x+c$
(c) $\log|1+y|=x+\frac{x^2}{2}+c$ (d) $\log|1+y|+x+\frac{x^2}{2}=c$

54. A particle moves in a straight line with a velocity given by $\frac{dx}{dt}=x+1$ (x is the distance described). The time taken by the particle to traverse a distance of 99 m is
(a) $\log_{10} e$ (b) $2\log_e 10$
(c) $2\log_{10} e$ (d) $\frac{1}{2}\log_{10} e$

55. The differential equation for the family of curves $x^2+y^2-2ay=0$, where a is an arbitrary constant, is
(a) $(x^2-y^2)y'=2xy$ (b) $2(x^2+y^2)y'=xy$
(c) $2(x^2-y^2)y'=xy$ (d) $(x^2+y^2)y'=2xy$

56. If $\frac{dy}{dx}=\frac{x+y}{x}$, then
(a) $y=x\log|x|+cx$ (b) $y=x\log|x|+c$
(c) $y=\log|x|+cx$ (d) $xy=\log|x|+cx$

57. $y=2e^{2x}-e^{-x}$ is a solution of the differential equation
(a) $y_2+y_1+2y=0$ (b) $y_2-y_1+2y=0$
(c) $y_2+y_1=0$ (d) $y_2-y_1-2y=0$

58. The solution of the differential equation $\tan y\sec^2 xdx+\tan x\sec^2 ydy=0$ is
(a) $\frac{\tan x}{\tan y}=c$ (b) $\tan x\tan y=c$
(c) $\tan x+\tan y=c$ (d) $\tan x-\tan y=c$

59. The solution of $x^4\frac{dy}{dx}+x^3y+\operatorname{cosec} xy=0$ is
(a) $2x^2\cos xy+cx^2+1=0$
(b) $2x^2\cos xy=cx^2$
(c) $2x^2\cos xy=cx^2+1$
(d) $x^2\cos xy+cx^2=1$

60. The differential equation of the family of straight lines whose slope is equal to the y-intercept is
(a) $(x+1)\frac{dy}{dx}-y=0$ (b) $(x+1)\frac{dy}{dx}+y=0$
(c) $\frac{dy}{dx}=\frac{x-1}{y-1}$ (d) $\frac{dy}{dx}=\frac{x+1}{y+1}$

EXERCISE SET 2

1. The solution of $y' = 1 + x + y^2 + xy^2$ and $y(0) = 0$ is

(a) $y^2 = \tan\left(x + \frac{x^2}{x}\right) - 1$ (b) $y^2 = 1 + e^{\left(x + \frac{x^2}{2}\right)}$

(c) $y = \tan(c + x + x^2)$ (d) $y = \tan(x + x^2/2)$

2. The solution of $\cos^2 x \frac{dy}{dx} + y = \tan x$ is

(a) $y = \tan x + 1 + ce^{-\tan x}$

(b) $y = \tan x - 1 + ce^{-\tan x}$

(c) $ye^{\tan x} = \tan e^{\tan x} + e^{\tan x} + c$

(d) none of these

3. If c_1, c_2, c_3, c_4, c_5 and c_6 are constants, then the order of the differential equation whose general solution is given by $y = c_1 \cos(x + c_2) + c_3 \sin(x + c_4) + c_5 e^x + c_6$ is

(a) 6 (b) 5

(c) 4 (d) 3

4. If the slope of the tangent at (x, y) to a curve passing through $(1, \pi/4)$ is given by $\frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$, then the equation of the curve is

(a) $y = \tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$

(b) $y = x\tan^{-1}\left[\log\left(\frac{x}{e}\right)\right]$

(c) $y = x\tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$

(d) none of these

5. The solution of $x\frac{dy}{dx} + y = y^2 \log x$ is

(a) $y(\log x + cx + 1) = 1$

(b) $(\log x + cx + 1) = y$

(c) $y(\log x + c + x) = 1$

(d) $y(\log x + cx + 1) = 0$

6. A differential equation associated to the primitive $y = a + be^{5x} + ce^{-7x}$ is

(a) $y_3 + 2y_2 - y_1 = 0$ (b) $4y_3 + 5y_2 - 20y_1 = 0$

(c) $y_3 + 2y_2 - 35y_1 = 0$ (d) none of these

7. A continuously differentiable function $\phi(x) \in (0, \pi)$ satisfying $y' = 1 + y^2$, $y(0) = 0 = y(\pi)$ is

(a) $\tan x$ (b) $x(x - \pi)$

(c) $(x - \pi)(1 - e^x)$ (d) not possible

8. The solution of the differential equation $\frac{d^2y}{dx^2} = \sin 3x + e^x + x^2$ when $y'(0) = 1$ and $y(0) = 0$ is

(a) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$

(b) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x$

(c) $\frac{-\cos 3x}{3} + e^x + \frac{x^4}{12} + \frac{1}{3}x + 1$

(d) none of these

9. The solution of the differential equation $x(4ydx + 2xdy) + y^3(3ydx + 5xdy) = 0$ is given by

(a) $x^2y^4 + x^5y^3 + c = 0$ (b) $x^4y^2 + x^3y^5 + c = 0$

(c) $x^4y^2 + x^2y^5 + c = 0$ (d) none of these

10. The order of the differential equation whose solution is $y = a\cos x + b\sin x + ce^{-x}$ is

(a) 3 (b) 1

(c) 2 (d) 4

11. The solution of the differential equation $\left(\frac{x + y - 1}{x + y - 2}\right)\frac{dy}{dx} = \left(\frac{x + y + 1}{x + y + 2}\right)$, given that $y = 1$ when $x = 1$, is

(a) $\log\left|\frac{(x - y)^2 - 2}{2}\right| = 2(x + y)$

(b) $\log\left|\frac{(x - y)^2 + 2}{2}\right| = 2(x - y)$

(c) $\log\left|\frac{(x + y)^2 + 2}{2}\right| = 2(x - y)$

(d) none of these

12. The solutions of $y = x\left[\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^3\right]$ are given by

(a) the constant function $y = 0$

(b) $y = p^{-3}e^{\frac{1}{2}p^2}(p + p^3)$

(c) $y = p^3e^{-\frac{1}{2}p^2}(p + p^3)$

(d) $ye^{-\frac{1}{2}p^2} = p^{-2} + 1$

13. The differential equation of the family of parabolas with the focus as the origin and the axis as the x-axis is

(a) $y\left(\frac{dy}{dx}\right)^2 + 4x\frac{dy}{dx} = 4y$

(b) $-y\left(\frac{dy}{dx}\right)^2 = 2x\frac{dy}{dx} - y$

(c) $y\left(\frac{dy}{dx}\right)^2 + y = 2xy\frac{dy}{dx}$

(d) $y\left(\frac{dy}{dx}\right)^2 + 2xy\frac{dy}{dx} + y = 0$

14. The equation of the curve for which the tangent at $P(x, y)$ cuts the y-axis at $(0, y^3)$ is

(a) $xy^2 = x + y$ (b) $x^2(y^2 - 1) = cy^2$

(c) $y(x^2 - 1) = cx^2$ (d) $yx + x^2 = c$

15. The equation of the curve satisfying $xdy - ydx = \sqrt{x^2 - y^2}\,dx$ and $y(1) = 0$ is

(a) $y = x^2 \log(\sin x)$ (b) $y = x \sin(\log x)$

(c) $y^2 = x(x-1)^2$ (d) $y = 2x^2(x-1)$

16. The equation of the curve through the point (1, 1) and whose slope is $\frac{2ay}{x(y-a)}$ is

(a) $y^a \cdot x^{2a} = e^{y-1}$ (b) $y^a \cdot x^{2a} = e^y$

(c) $y^{2a} \cdot x^a = e^{y-1}$ (d) none of these

17. The integrating factor of the differentiable equation $(xy - 1)\frac{dy}{dx} + y^2 = 0$ is

(a) $\frac{1}{x}$ (b) $\frac{1}{y}$

(c) $\frac{1}{xy}$ (d) xy

18. The degree of the differential equation satisfying $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x - y)$ is

(a) 1 (b) 2

(c) 3 (d) none of these

19. The solution of the differential equation $\frac{\sqrt{x}dx + \sqrt{y}dy}{\sqrt{x}dx - \sqrt{y}dy} = \sqrt{\frac{y^3}{x^3}}$ is given by

(a) $\frac{3}{2}\log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1}\left(\frac{y}{x}\right)^{3/2} + c = 0$

(b) $\frac{2}{3}\log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1}\frac{y}{x} + c = 0$

(c) $\frac{2}{3}\log\left(\frac{y}{x}\right) + \log\left(\frac{x+y}{x}\right) + \tan^{-1}\left(\frac{y^{3/2}}{x^{3/2}}\right) + c = 0$

(d) none of these

20. The equation of the curve satisfying the differential equation $y_2(x^2 + 1) = 2xy_1$ passing through the point (0, 1) and having the slope of the tangent at $x = 0$ as 3 is

(a) $y = x^2 + 3x + 2$ (b) $y^2 = x^2 + 3x + 1$

(c) $y = x^3 + 3x + 1$ (d) none of these

21. The differential equation of the curve $y = a\cos(x - b)$, where a and b are constants, is

(a) $\frac{d^2y}{dx^2} - y = 0$ (b) $\frac{d^2y}{dx^2} + ay = 0$

(c) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$ (d) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - y = 0$

22. The equation of the curve satisfying the differential equation $y(x + y^3)dx = x(y^3 - x)dy$ and passing through the point (1, 1) is

(a) $y^3 - 2x + 3x^2y = 0$ (b) $y^3 + 2x + 3x^2y = 0$

(c) $y^3 + 2x - 3x^2y = 0$ (d) none of these

23. If $x = \sin t, y = \cos pt$, then

(a) $(1 - x^2)y_2 + xy_1 + p^2y = 0$

(b) $(1 - x^2)y_2 + xy_1 - p^2y = 0$

(c) $(1 + x^2)y_2 - xy_1 + p^2y = 0$

(d) $(1 - x^2)y_2 - xy_1 + p^2y = 0$

24. The solution of the equation $y\sin x\frac{dy}{dx} = \cos x\left(\sin x - \frac{y^2}{2}\right)$, given $y = 1$ when $x = \frac{\pi}{2}$, is

(a) $y^2 \sin x = 2\sin^2 x + 1$ (b) $y^2 = 2\sin x$

(c) $x^2 = \sin y$ (d) $x^2 = 2\sin y$

25. The differential equation of the family of curves $y = a\cos(x + b)$ is

(a) $\frac{d^2y}{dx^2} - y = 0$ (b) $\frac{d^2y}{dx^2} + y = 0$

(c) $\frac{d^2y}{dx^2} + 2y = 0$ (d) none of these

26. The solution of the differential equation $x^2\frac{dy}{dx}-xy=1+\cos\frac{y}{x}$ is

(a) $\cos\frac{y}{x}=1+\frac{c}{x}$ (b) $x^2=(c+x^2)\tan\frac{y}{x}$

(c) $\tan\frac{y}{2x}=c-\frac{1}{2x^2}$ (d) $\tan\frac{y}{x}=c+\frac{1}{x}$

27. If $y=(x+\sqrt{1+x^2})^n$, then $(1+x^2)\frac{d^2y}{dx^2}+x\frac{dy}{dx}$ is

(a) n^2y (b) $-n^2y$

(c) $-y$ (d) $2x^2y$

28. If $y=a\sin(5x+c)$, then

(a) $\frac{dy}{dx}=5y$ (b) $\frac{dy}{dx}=-5y$

(c) $\frac{d^2y}{dx^2}=-25y$ (d) $\frac{d^2y}{dx^2}=25y$

29. The general solution of the differential equation $y(x^2y+e^x)dx-e^xdy=0$ is

(a) $x^3y-3e^x=cy$ (b) $x^3y+3e^x=cy$

(c) $y^3x-3e^y=cx$ (d) $y^3x+3e^y=cx$

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement I is false, Statement II is true.
(b) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(c) Statement I is true, Statement II is true; StatementII is not a correct explanation for Statement I.
(d) Statement I is true, Statement II is false.

30. Statement I: The differential equation of the family of curves represented by $y=(a+be^c)x$ is $x\frac{dy}{dx}-y=0$, where a, b and c are constants.

Statement II: In general, a differential equation of nth order is obtained on eliminating n arbitrary constants.

31. Statement I: The differenital equation of all circles in a plane is of order 3.

Statement II: Only one circle can be drawn through any three points in a plane.

32. Statement I: The degree of the differenital equation $\frac{d^3y}{dx^3}+3\frac{dy}{dx}=\log\left(\frac{d^2y}{dx^2}\right)$ is one.

Statement II: The degree of a differential equation is the highest power of the highest order derivative involved after the equation has been expressed as a polynomial in derivatives.

Previous Years' Questions

33. The order and degree of the differential equation $\left(1+3\frac{dy}{dx}\right)^{2/3}=4\frac{d^3y}{dx^3}$ are

(a) $\left(1,\frac{2}{3}\right)$ (b) (3, 1)

(c) (3, 3) (d) (1, 2)

[AIEEE 2002]

34. The differential equation of all non-vertical lines in a plane is

(a) $\frac{d^2y}{dx^2}=0$ (b) $\frac{d^2x}{dy^2}=0$

(c) $\frac{dy}{dx}=0$ (d) $\frac{dx}{dy}=0$

[AIEEE 2002]

35. The solution of the equation $\frac{d^2y}{dx^2}=e^{-2x}$ is

(a) $y=\frac{e^{-2x}}{4}$ (b) $y=\frac{e^{-2x}}{4}+cx+d$

(c) $y=\frac{1}{4}e^{-2x}+cx^2+d$ (d) $y=\frac{1}{4}e^{-2x}+c+d$

[AIEEE 2002]

36. The degree and order of the differential equation of the family of all parabolas whose axis is the x-axis, are respectively

(a) 1, 2 (b) 3, 2

(c) 2, 3 (d) 2, 1

[AIEEE 2003]

37. The solution of the differential equation $(1+y^2)+(x-e^{\tan^{-1}y})\frac{dy}{dx}=0$ is

(a) $x^2e^{2\tan^{-1}y}=e^{\tan^{-1}y}+K$

(b) $2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+K$

(c) $xe^{\tan^{-1}y}=\tan^{-1}y+K$

(d) $x-2=Ke^{-\tan^{-1}y}$ **[AIEEE 2003]**

38. Let $(1+t)\frac{dy}{dt} - ty = 1$, $y(0) = -1$, then $y(t)$ at $t = 1$ is

(a) $-(1/2)$ (b) $e + (1/2)$
(c) $e - (1/2)$ (d) $1/2$ **[AIEEE 2003]**

39. The solution of the differential equation $ydx + (x + x^2y)dy = 0$ is

(a) $-(1/xy) = c$ (b) $-(1/xy) + \log y = c$
(c) $(1/xy) + \log y = c$ (d) $\log y = cx$

[AIEEE 2004]

40. The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant, is

(a) $2(x^2 - y^2)y' = xy$ (b) $2(x^2 + y^2)y' = xy$
(c) $(x^2 - y^2)y' = 2xy$ (d) $(x^2 + y^2)y' = 2xy$

[AIEEE 2004]

41. A function $y = f(x)$ has a second-order derivative $f'' = 6(x-1)$. If its graph passes through the point (2, 1) and at that point the tangent to the graph is $y = 3x - 5$, then the function is

(a) $(x-1)^2$ (b) $(x-1)^3$
(c) $(x+1)^3$ (d) $(x+1)^2$ **[AIEEE 2004]**

42. If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is

(a) $\log(y/x) = cx$ (b) $\log(y/x) = cy$
(c) $y \log(y/x) = cx$ (d) $x \log(y/x) = cy$

[AIEEE 2005]

43. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$ is a parameter, is of order and degree as follows:

(a) order 1, degree 3 (b) order 2, degree 3
(c) order 1, degree 2 (d) order 2, degree 1

[AIEEE 2006]

44. The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants, is of

(a) second order and second degree
(b) first order and second degree
(c) first order and first degree
(d) second order and first degree **[AIEEE 2006]**

45. The differential equation of all circles passing through the origin and having their centres on the x-axis is

(a) $x^2 = y^2 + xy\frac{dy}{dx}$ (b) $x^2 = y^2 + 3xy\frac{dy}{dx}$
(c) $y^2 = x^2 + 2xy\frac{dy}{dx}$ (d) $y^2 = x^2 - 2xy\frac{dy}{dx}$

[AIEEE 2007]

46. The normal to a curve at $P(x, y)$ meets the x-axis at G. If the distance of G from the origin is twice the abscissa of P, then the curve is

(a) a parabola (b) a circle
(c) a hyperbola (d) an ellipse

[AIEEE 2007]

47. The differential equation of the family of circles with fixed radius 5 unit and centre on the line $y = 2$ is

(a) $(x-2)^2 y'^2 = 25 - (y-2)^2$
(b) $(x-2)y'^2 = 25 - (y-2)^2$
(c) $(y-2)y'^2 = 25 - (y-2)^2$
(d) $(y-2)^2 y'^2 = 25 - (y-2)^2$

[AIEEE 2008]

48. The orthogonal trajectories of the system of curves $\left(\frac{dy}{dx}\right)^2 = a/x$ are

(a) $9a(y+c)^2 = 4x^3$ (b) $y + c = \frac{-2}{3\sqrt{a}}x^{3/2}$
(c) $y + c = \frac{2}{3\sqrt{a}}x^{3/2}$ (d) none of these

[AIEEE 2008]

49. The solution of the differential equation $\cos x dy = y(\sin x - y)dx$, $0 < x < \frac{\pi}{2}$, is

(a) $\sec x = (\tan x + c)y$ (b) $y \sec x = \tan x + c$
(c) $y \tan x = \sec x + c$ (d) $\tan x = (\sec x + c)y$

[AIEEE 2010]

50. The differential equation whose solution is $(x - h)^2 + (y - k)^2 = a^2$ is (a is a constant)

(a) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2\frac{d^2y}{dx^2}$
(b) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2\left(\frac{d^2y}{dx^2}\right)^2$
(c) $\left[1 + \left(\frac{dy}{dx}\right)\right]^3 = a^2\left(\frac{d^2y}{dx^2}\right)^2$
(d) none of these **[IIT 1992]**

51. The solution of the differential equation $(2x - 10y^3)\frac{dx}{dy} + y = 0$ is

(a) $x + y = ce^{2x}$ (b) $y^2 = 2x^3 + c$
(c) $xy^2 = 2y^5 + c$ (d) $x(y^2 + xy) = 0$

[IIT 1993]

52. A curve passing through the point (1, 1) has the property that the perpendicular distance of the origin from the normal at any point P on the curve is equal to the distance of P from the x-axis. The equation of the curve is

(a) $x^2+y^2+2x=0$ (b) $x^2+y^2-2x=0$

(c) $x^2+y^2-2y=0$ (d) none of these

[IIT 1999]

53. The solution of the differential equation $(2x\cos y + y^2\cos x)dx + (2y\sin x - x^2\sin y)dy = 0$ is

(a) $x^2\cos y + y^2\sin x = c$

(b) $x\cos y - y\sin x = c$

(c) $x^2\cos^2 y + y^2\sin^2 x = c$

(d) none of these **[DCE 2002]**

54. If $\frac{dy}{dx}+y=2e^{2x}$, then y is equal to

(a) $ce^{-x}+\frac{2}{3}e^{2x}$ (b) $(1+x)e^{-x}+\frac{2}{3}e^{2x}+c$

(c) $ce^{-x}+\frac{2}{3}e^{2x}+c$ (d) $e^{-x}+\frac{2}{3}e^{2x}+c$

[DCE 2003]

55. The solution of the differential equation $\frac{dy}{dx}+\frac{y}{x}=x^2$ is

(a) $y=\frac{x^2}{4}+cx^{-2}$ (b) $y=x^{-1}+cx^{-3}$

(c) $y=\frac{x^3}{4}+cx^{-1}$ (d) $xy=x^2+c$

[DCE 2004]

56. The solution of the differential equation $(1+y^2)+(x-e^{\tan^{-1}y})\frac{dy}{dx}=0$ is

(a) $2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+k$

(b) $2xe^{\tan^{-1}y}=e^{\tan^{-1}y}+k$

(c) $xe^{\tan^{-1}y}=e^{\tan^{-1}y}+k$

(d) $xe^{\tan^{-1}y}=e^{\tan^{-1}y}+k$ **[DCE 2005]**

57. The differential equation of the family of the curves $x^2+y^2-2ax=0$ is

(a) $x^2-y^2-2xyy''=0$ (b) $y^2-x^2=2xyy'$

(c) $x^2+y^2+2y''=0$ (d) none of these

[DCE 2007]

58. The solution of $y_1(x^2y^3+xy)=1$ is

(a) $1/x=2-y^2+Ce^{-y^2/2}$

(b) the solution of an equation which is reducible to a linear equation

(c) $2/x=1-y^2+e^{-y/2}$

(d) $e^{y^2/2}\left(\frac{1-2x}{x}+y^2\right)=c$ **[M.P.C.E.T. 2000]**

59. $\tan^{-1}x+\tan^{-1}y=c$ is the general solution of the differential equation

(a) $\frac{dy}{dx}=\frac{1+y^2}{1+x^2}$ (b) $\frac{dy}{dx}=\frac{1+x^2}{1+y^2}$

(c) $(1+x^2)dy+(1+y^2)dx=0$

(d) $\frac{dy}{dx}=\frac{1-y^2}{1-x^2}$ **[Kerala P.E.T. 2001]**

60. If the solution of the differential equation $(x^2+4y^2+4xy)dy = (2x+4y+1)dx$ is $V-2\log|V^2+4V+2|+\frac{3}{\sqrt{2}}\log\left|\frac{V+2-\sqrt{2}}{V+2+\sqrt{2}}\right|=x+6$, then

(a) $V=x+2y$ (b) $V=2x+y$

(c) $V=x+y$ (d) none of these

[Roorkee 1991]

61. Consider the differential equation $y^2dx+\left(x-\frac{1}{y}\right)dy=0$. If $y(1)=1$, then x is given by

(a) $4-\frac{2}{y}-\frac{e^{1/y}}{e}$ (b) $3-\frac{1}{y}+\frac{e^{1/y}}{e}$

(c) $1+\frac{1}{y}-\frac{e^{1/y}}{e}$ (d) $1-\frac{1}{y}+\frac{e^{1/y}}{e}$

[AIEEE 2011]

ANSWERS

Exercise Set 1

1. (a)	**2.** (c)	**3.** (c)	**4.** (c)	**5.** (c)	**6.** (a)	**7.** (d)	**8.** (c)	**9.** (a)	**10.** (b)
11. (a)	**12.** (c)	**13.** (b)	**14.** (c)	**15.** (a)	**16.** (a)	**17.** (a)	**18.** (c)	**19.** (a)	**20.** (d)
21. (d)	**22.** (b)	**23.** (b)	**24.** (a)	**25.** (a)	**26.** (c)	**27.** (c)	**28.** (b)	**29.** (d)	**30.** (b)

31. (b)	**32.** (b)	**33.** (b)	**34.** (a)	**35.** (a)	**36.** (a)	**37.** (a)	**38.** (a)	**39.** (c)	**40.** (a)
41. (d)	**42.** (a)	**43.** (c)	**44.** (b)	**45.** (c)	**46.** (c)	**47.** (b)	**48.** (b)	**49.** (a)	**50.** (b)
51. (b)	**52.** (b)	**53.** (c)	**54.** (b)	**55.** (a)	**56.** (c)	**57.** (d)	**58.** (b)	**59.** (c)	**60.** (a)

Exercise Set 2

1. (d)	**2.** (b)	**3.** (c)	**4.** (c)	**5.** (a)	**6.** (c)	**7.** (d)	**8.** (a)	**9.** (b)	**10.** (a)
11. (d)	**12.** (a) and (b)		**13.** (b)	**14.** (b)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (a)	**19.** (d)
20. (c)	**21.** (b)	**22.** (c)	**23.** (d)	**24.** (a)	**25.** (b)	**26.** (c)	**27.** (a)	**28.** (c)	**29.** (b)
30. (b)	**31.** (c)	**32.** (a)	**33.** (c)	**34.** (a)	**35.** (b)	**36.** (a)	**37.** (b)	**38.** (a)	**39.** (b)
40. (c)	**41.** (b)	**42.** (a)	**43.** (a)	**44.** (d)	**45.** (c)	**46.** (c)	**47.** (d)	**48.** (a), (b) and (c)	
49. (a)	**50.** (c)	**51.** (b)	**52.** (b)	**53.** (a)	**54.** (a)	**55.** (c)	**56.** (a)	**57.** (b)	
58. (a), (b) and (d)		**59.** (c)	**60.** (a)	**61.** (c)					

HINTS AND SOLUTIONS—EXERCISE SET 1

1. $y = e^x(A\cos x + B\sin x)$

$$\frac{dy}{dx} = e^x(-A\sin x + B\cos x) + e^x(A\cos x + B\sin x)$$

$$\frac{dy}{dx} = e^x(-A\sin x + B\cos x) + y \qquad \text{(i)}$$

Again differentiating with respect to x, we get

$$\frac{d^2y}{dx^2} = e^x(-A\sin x + B\cos x) + e^x(-A\cos x - B\sin x) + \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \left(\frac{dy}{dx} - y\right) + \frac{dy}{dx} \qquad \text{[using (i)]}$$

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$$

2. $\frac{dy}{dx} + \frac{1+\cos 2y}{1-\cos 2x} = 0 \Rightarrow \frac{1}{1+\cos 2y}dy + \frac{dx}{1-\cos 2x} = 0$

$$\Rightarrow \quad \frac{1}{2}\int \sec^2 y\,dy + \frac{1}{2}\int \operatorname{cosec}^2 x\,dx = 0$$

$$\Rightarrow \quad \tan y - \cot x = c$$

3. $\frac{dy}{dx} = \frac{2x+9y-20}{6x+2y-10}$

Put $x = X + h$ and $y = Y + k$

$$\frac{dY}{dX} = \frac{2X+9Y}{6X+2Y}$$

Put $Y = 2X$

where $2h + 9k - 20 = 0$

$6h + 2k - 10 = 0$

$$\frac{h}{-50} = \frac{k}{-100} = \frac{1}{-50} \quad \therefore \quad h = 1, k = 2$$

$$v + X\frac{dv}{dX} = \frac{2+9v}{6+2v}$$

$$\therefore \quad X\frac{dv}{dX} = \frac{2+9v-6v-2v^2}{6+2v} = \frac{2+3v-2v^2}{6+2v}$$

$$\frac{6+2v}{2+3v-2v^2}dv = \frac{dX}{X}$$

$$\Rightarrow \quad \int\left[\frac{2}{1+2v} + \frac{2}{2-v}\right]dv = \int\frac{dX}{X} + \log c_1$$

$$\Rightarrow \quad \log(1+2v) - 2\log(2-v) = \log c_1 X$$

$$\Rightarrow \quad \log\frac{1+2v}{(2-v)^2} = \log c_1 X$$

$$\Rightarrow \quad \frac{1+2(Y/X)}{[2-(Y/X)]^2} = c_1 X \Rightarrow \frac{(X+2Y)X}{(2X-Y)^2} = c_1 X$$

$$\Rightarrow \quad (X+2Y) = (2X-Y)^2 c_1$$

$$\Rightarrow \quad c(x+2y-5) = (2x-y)^2$$

$$\Rightarrow \quad (y-2x)^2 = c(x+2y-5)$$

4. We know that the subtangent $= \frac{y}{dy/dx} = y\frac{dx}{dy}$

By the question, $y\frac{dx}{dy} = kx \Rightarrow \frac{dx}{x} = k\frac{dy}{y}$

Integrating, we get

$$\log A + \log|x| = k\log|y|$$

$$\Rightarrow \quad A|x| = |y^k| \Rightarrow y^k = \pm Ax = cx$$

5. $\frac{dy}{dx}=\cos(x+y)+\sin(x+y)$

Let $x+y=v \quad \Rightarrow \quad 1+\frac{dy}{dx}=\frac{dv}{dx}$

$\therefore \quad \frac{dv}{dx}-1=\cos v+\sin v$

$\Rightarrow \quad \frac{dv}{dx}=1+\cos v+\sin v \quad \Rightarrow \quad \int\frac{1}{1+\cos v+\sin v}dv=\int dx$

$\Rightarrow \quad \int\frac{1}{2\cos^2\frac{v}{2}+2\sin\frac{v}{2}\cos\frac{v}{2}}dv=x+c$

$\Rightarrow \quad \int\frac{\sec^2\frac{v}{2}dv}{2+2\tan\frac{v}{2}}=x+c$

Let $\tan\frac{v}{2}=t \quad \Rightarrow \quad dt=\frac{1}{2}\sec^2\frac{v}{2}dv$

$\Rightarrow \int\frac{dt}{1+t}=x+c \Rightarrow \log(1+t)=x+c$

$\Rightarrow \quad \log\left(1+\tan\frac{v}{2}\right)=x+c$

$\therefore \quad \log\left[1+\tan\frac{1}{2}(x+y)\right]=x+c$

6. Put $x-3y=z \quad \therefore \quad 1-3\frac{dy}{dx}=\frac{dz}{dx}$

$\therefore \quad \frac{1}{3}\left[1-\frac{dz}{dx}\right]=\frac{2z+7}{z+4} \quad \Rightarrow \quad 1-\frac{dz}{dx}=\frac{6z+21}{z+4}$

$\Rightarrow \quad \frac{dz}{dx}=1-\frac{6z+21}{z+4}=\frac{z+4-6z-21}{z+4}=\frac{-5z-17}{z+4}$

$\Rightarrow \quad \frac{z+4}{5z+17}dz+dx=0 \quad \Rightarrow \quad \frac{5z+20}{5z+17}dz+5dx=0$

$$\left(1+\frac{3}{5z+17}\right)dx+5dx=0$$

$$z+\frac{3}{5}\log(5z+17)+5x=c$$

$\Rightarrow \quad x-3y+\frac{3}{5}\log(5x-15y+117)+5x=c_1$

$\Rightarrow \quad 6x-3y+\frac{3}{5}\log(5x-15y+17)=c_1$

$\Rightarrow \quad 2x-y+\frac{1}{5}\log(5x-15y+17)=c$

7. We have $(1-x^2)\frac{dy}{dx}+xy=ax$

$\Rightarrow \quad \frac{dy}{dx}+\frac{x}{1-x^2}y=\frac{ax}{1-x^2}$

I.F. $=e^{\int\frac{x}{1-x^2}dx}=e^{-\frac{1}{2}\log(1-x^2)}=\frac{1}{\sqrt{1-x^2}}$

$\therefore$ The solution is

$$y\cdot\frac{1}{\sqrt{1-x^2}}=\int\frac{ax}{1-x^2}\cdot\frac{1}{\sqrt{1-x^2}}dx+c$$

$\Rightarrow \quad y\cdot\frac{1}{\sqrt{1-x^2}}=\frac{a}{\sqrt{1-x^2}}+c \Rightarrow y=a+c\sqrt{1-x^2}$

$\Rightarrow \quad (y-a)^2=c^2(1-x^2) \Rightarrow (y-a)^2+c^2x^2=c^2$

When $-1<x<1$, it is an ellipse.

When $x^2>1$, it is a hyperbola.

8. $(x^2-yx^2)dy+(y^2+x^2y^2)dx=0$

$\Rightarrow \quad x^2(1-y)dy+y^2(1+x^2)dx=0$

Dividing by x^2y^2

$$\left(\frac{1-y}{y^2}\right)dy+\left(\frac{1-x^2}{x^2}\right)dx=0$$

$\Rightarrow \quad \left(\frac{1}{y^2}-\frac{1}{y}\right)dy+\left(\frac{1}{x^2}+1\right)dx=0$

Integrating, we have

$$-\frac{1}{y}-\log y-\frac{1}{x}+x=c$$

$\therefore \quad \frac{1}{y}+\frac{1}{x}+\log y-x=c$

9. $\sqrt{a+x}\frac{dy}{dx}+xy=0 \quad \Rightarrow \quad \frac{dy}{y}+\frac{x}{\sqrt{a+x}}dx=0$

$\Rightarrow \quad \int\frac{dy}{y}+\int\sqrt{a+x}dx-a\int\frac{dy}{\sqrt{a+x}}=c$

$\Rightarrow \quad \log y+\frac{2}{3}(a+x)^{3/2}-2a\cdot(a+x)^{1/2}=\log A$

$\Rightarrow \quad \log\frac{y}{A}=2(a+x)^{1/2}\left[a-\frac{1}{3}(a+x)\right]$

$=\frac{2}{3}(a+x)^{1/2}(2a-x) \quad \Rightarrow \quad y=Ae^{2/3(2a-x)\sqrt{a+x}}$

10. It is given that the triangle OPG is an isosceles triangle. Therefore, $OM = MG$ = subnormal

$\Rightarrow x=y\frac{dy}{dx} \Rightarrow xdx=ydx$

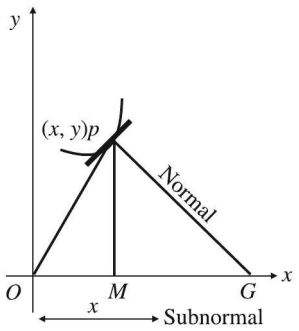


Fig. 17.1

On integration, we get $x^2 - y^2 = c$, which is a rectangular hyperbola.

11. $x\dfrac{dy}{dx} + y = x^2 + 3x + 2 \Rightarrow \dfrac{dy}{dx} + \dfrac{1}{x}y = x + 3 + \dfrac{2}{x}$

$\therefore$ I.F. $= e^{\int \frac{1}{x}dx} = e^{\log_e x} = x$

Hence, the solution is

$\Rightarrow \quad y \cdot x = \int (x^2 + 3x + 2)dx + c$

$\therefore \quad y \cdot x = \dfrac{x^3}{3} + \dfrac{3}{2}x^2 + 2x + c$

12. We have $dy = 2xe^{x^2} \cdot e^{-y}dx$

$\Rightarrow \quad e^y dy = 2xe^{x^2}dx \quad \Rightarrow \quad e^y = e^{x^2} + c$

13. We have $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{x^2 - y^2}$

$\therefore$ Slope at $(1, 0) = \dfrac{1+0}{1-0} = 1$

14. $\dfrac{dy}{dx} = \dfrac{xy + y}{xy + x}$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{y(1+x)}{x(1+y)} \quad \Rightarrow \quad \left(\dfrac{1+y}{y}\right)dy = \left(\dfrac{1+x}{x}\right)dx$

$\Rightarrow \quad \int\left(\dfrac{1}{y} + 1\right)dy = \int\left(\dfrac{1}{x} + 1\right)dx$

$\Rightarrow \quad \log y + y = \log x + x + \log c \quad \Rightarrow \quad \log \dfrac{y}{cx} = x - y$

$\therefore \quad y = cxe^{(x-y)}$

15. Here, $x = A\cos 4t + B\sin 4t$

On differentiating w.r.t. t, we get

$$\frac{dx}{dt} = -4A\sin 4t + 4B\cos 4t$$

Again on differentiating w.r.t. t, we get

$$\frac{d^2x}{dt^2} = -16A\cos 4t - 16B\sin 4t$$

$$= -16(A\cos 4t + B\sin 4t) \quad \Rightarrow \quad \frac{d^2x}{dt^2} = -16x$$

$$\Rightarrow \quad (1 + x^2)\frac{dy}{dx} = xy$$

16. The given differential equation is

$$y = x\frac{dy}{dx} + \left[a^2\left(\frac{dy}{dx}\right)^2 + b^2\right]^{1/3}$$

$$\Rightarrow \quad \left(y - x\frac{dy}{dx}\right)^3 = a^2\left(\frac{dy}{dx}\right)^2 + b^2$$

$\therefore$ The degree and order of the above differential equation are 3 and 1 respectively.

17. $\dfrac{dy}{dx} = 2xy$

$\Rightarrow \quad \dfrac{dy}{y} = 2xdx \quad \Rightarrow \quad \log y = x^2 + \log c$

$\therefore \quad y = ce^{x^2}$

18. The given equation is $y = \sec(\tan^{-1} x)$

On differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \sec(\tan^{-1} x)\tan(\tan^{-1} x) \cdot \frac{1}{1+x^2} = \frac{xy}{1+x^2}$$

$[\because \quad \tan(\tan^{-1} x) = x]$

$$\Rightarrow \quad (1 + x^2)\frac{dy}{dx} = xy$$

19. The given differential equation can be written as

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/4\times 12} = \left(\frac{d^2y}{dx^2}\right)^{1/3\times 12}$$

$$\Rightarrow \quad \left[1 + \left(\frac{dy}{dx}\right)^2\right]^9 = \left(\frac{d^2y}{dx^2}\right)^4$$

Here, we see that the order of the highest derivative is 2 and the degree is 4.

20. $\dfrac{dy}{dx} = \dfrac{x(2\log x + 1)}{\sin y + y\cos y}$

$\Rightarrow \quad (\sin y + y\cos y)dy = x(2\log x + 1)dx$

Integrating both sides, we have

$$\underbrace{\int \sin y dy + \int y\cos y dy}_{I_1} = \underbrace{\int x(2\log x + 1)dx}_{I_2} \qquad \text{(i)}$$

$\therefore \quad I_1 = \int \sin y dy + (y \sin y - \int \sin y dy)$

Also, $I_2 = \int (2x \log x + x) dx$

$\Rightarrow \quad I_2 = x^2 \log x - \int x^2 \frac{1}{x} dx + \frac{x^2}{2} + c$

$\Rightarrow \quad y \sin y = x^2 \log x - \frac{x^2}{2} + \frac{x^2}{2} + c$ [using (i)]

$\therefore \quad y \sin y = x^2 \log x + c$

21. The equation of all the straight lines passing through the origin (0, 0) is

$$y = mx$$

Hence, the required differential equation of all such lines is $y = \left(\frac{dy}{dx}\right)x \qquad \left(\because \ m = \frac{dy}{dx}\right)$

22. The given differential equation is

$$\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \quad \Rightarrow \quad \left(\frac{d^2y}{dx^2}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

Hence, the order is 2.

23. $x + y\frac{dy}{dx} = 2y \quad \Rightarrow \quad \frac{dy}{dx} = \frac{2y - x}{y}$

Let $y = vx \quad \Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \quad v + x\frac{dv}{dx} = \frac{2vx - x}{vx} = \frac{2v - 1}{v}$

$\Rightarrow \quad x\frac{dv}{dx} = \frac{2v - 1}{v} - v = \frac{2v - 1 - v^2}{v}$

$\Rightarrow \quad \int \frac{v}{(v-1)^2} dv = -\frac{dx}{x} \quad \Rightarrow \quad \int \frac{(v-1)+1}{(v-1)^2} dv = -\int \frac{dx}{x}$

$\Rightarrow \quad \int \left[\frac{1}{v-1} + \frac{1}{(v-1)^2}\right] dv = -\int \frac{dx}{x}$

$\Rightarrow \quad \log(v-1) - \frac{1}{(v-1)} = -\log x + c$

$\Rightarrow \quad \log\left(\frac{y-x}{x}\right) - \left(\frac{x}{y-x}\right) = -\log x + c$

$\therefore \quad \log(y-x) = c + \frac{x}{(y-x)}$

24. The given equation is $\frac{dx}{x} + \frac{dy}{y} = 0$.

On integrating, we get

$$\int \frac{dx}{x} + \int \frac{dy}{y} = 0 \quad \Rightarrow \quad \log x + \log y = \log c$$

$\Rightarrow \quad \log(xy) = \log c \quad \Rightarrow xy = c$

25. The given curve is $y = x^2$

For this curve, there is only one tangent line, i.e. x-axis ($y = 0$).

$\therefore \quad \frac{dy}{dx} = 0$

Hence, the order is 1.

26. The point on the y-axis is $\left(0, y - x\frac{dy}{dx}\right)$. According to the given condition,

$$\frac{x}{2} = y - \frac{x}{2}\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = 2\frac{y}{x} - 1$$

Putting $\frac{y}{x} = v$, we get

$$x\frac{dv}{dx} = v - 1 \Rightarrow \ln\left|\frac{y}{x} - 1\right| = \ln|x| + c \Rightarrow 1 - \frac{y}{x} = x$$

[as $y(1) = 0$]

27. The given equation is $\frac{dy}{dx} = 2^y \cdot 2^{-x} \Rightarrow 2^{-y} dy = 2^{-x} dx$

On integrating both sides, we get

$$-\frac{2^{-y}}{\log 2}(-1) = \frac{2^{-x}}{\log 2}(-1) + c_1$$

$\Rightarrow \quad -\frac{2^{-y}}{\log 2} = -\frac{2^{-x}}{\log 2} + c_1 \quad \Rightarrow \quad -2^{-y} = -2^{-x} + c_1 \log 2$

$\therefore \quad \frac{1}{2^x} - \frac{1}{2^y} = c_1 \log 2 = c$

28. The given equation is reduced to $x = e^{xy(dy/dx)}$.

$\Rightarrow \quad \log x = xy\frac{dy}{dx} \quad \Rightarrow \quad \int y dy = \int \frac{1}{x} \log x dx$

$\Rightarrow \quad \frac{y^2}{2} = \frac{(\log x)^2}{2} + c'$

29. $x\frac{dy}{dx} = y(\log y - \log x + 1) \quad \Rightarrow \quad x\frac{dy}{dx} = y\left(\log\frac{y}{x} + 1\right)$

Let $y = vx \quad \Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \quad v + x\frac{dv}{dx} = v(\log v + 1) \quad \Rightarrow \quad \int \frac{dv}{v \log v} = \int \frac{dx}{x}$

$\Rightarrow \quad \log(\log v) = \log x + \log c \quad \Rightarrow \quad \log\frac{y}{x} = cx$

30. Given: $\cot y dx = x dy$

$\Rightarrow \quad \frac{dx}{x} = \frac{dy}{\cot y} \quad \Rightarrow \quad \frac{dx}{x} = \tan y dy$

On integrating both sides, we get

$$\int \frac{1}{x}dx = \int \tan y dy \quad \Rightarrow \quad \log x = \log \sec y + \log c$$

$$\Rightarrow \quad \log x = \log c \sec y \quad \Rightarrow \quad x = c \sec y$$

31. $y = mx + \dfrac{4}{m}$ (i)

$$\therefore \qquad \frac{dy}{dx} = m$$

From Eq. (i), we get

$$y = x\left(\frac{dy}{dx}\right) + \frac{4}{\left(\frac{dy}{dx}\right)} \quad \Rightarrow \quad y\left(\frac{dy}{dx}\right) = x\left(\frac{dy}{dx}\right)^2 + 4$$

$$\Rightarrow \qquad x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} + 4 = 0$$

which is the required differential equation.

32. For the family of curves represented by the first differential equation, the slope of the tangent at any point (x, y) is given by

$$\left(\frac{dy}{dx}\right)_{c_1} = -\frac{x^2 + x + 1}{y^2 + y + 1}$$

For the family of curves represented by the second differential equation, the slope of the tangent at any point is given by

$$\left(\frac{dy}{dx}\right)_{c_2} = -\frac{y^2 + y + 1}{x^2 + x + 1}$$

Since $\left(\dfrac{dy}{dx}\right)_{c_1} \times \left(\dfrac{dy}{dx}\right)_{c_2} = -1$

Hence, the two curves are orthogonal.

33. Given that $\dfrac{dy}{dx} = \dfrac{2x - y}{x + 2y}$

Let $y = vx \Rightarrow y = vx \Rightarrow \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$$\therefore \quad v + x\frac{dv}{dx} = \frac{2 - v}{1 + 2v} \quad \Rightarrow \quad x\frac{dv}{dx} = \frac{2 - v - v(1 + 2v)}{1 + 2v}$$

$$\Rightarrow \quad \int \frac{1 + 2v}{2(1 + v - v^2)}dv = \int \frac{1}{x}dx$$

$$\Rightarrow \quad \log k - \frac{1}{2}\log(1 - v - v^2) = \log x$$

$$\Rightarrow \quad 2\log k - \log(1 - v - v^2) = 2\log x$$

$$\Rightarrow \quad \log c = \log[x^2(1 - v - v^2)] \Rightarrow \quad c = x^2\left(1 - \frac{y}{x} - \frac{y^2}{x^2}\right)$$

$$\Rightarrow \quad x^2 - xy - y^2 = c$$

34. We have $y = cx - c^2$ (i)

On differentiating w.r.t. x, we get

$$y' = c$$

On putting this value in Eq. (i), we get

$$y = x(y') - (y')^2$$

$$\Rightarrow \qquad (y')^2 - xy' + y = 0$$

35. Given that $\dfrac{dy}{dx} = \dfrac{xy}{x^2 + y^2}$

Let $y = vx \Rightarrow \dfrac{dy}{dx} = 1 + x\dfrac{dv}{dx}$

$$\therefore \quad v + x\frac{dv}{dx} = \frac{x^2 v}{x^2 + v^2 x^2} = \frac{v}{1 + v^2}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{v}{1 + v^2} - v = -\frac{v^3}{1 + v^2}$$

$$\Rightarrow \quad \frac{-(1 + v^2)}{v^3}dv = \frac{dx}{x} \quad \Rightarrow \quad \int\left(-\frac{1}{v^3} - \frac{1}{v}\right)dv = \int \frac{dx}{x}$$

$$\Rightarrow \quad \frac{1}{2v^2} - \log v = \log x + \log c$$

$$\Rightarrow \quad \frac{1}{v^2} - 2\log v = 2\log x + \log c \Rightarrow \frac{x^2}{y^2} = \log\left(x^2 \cdot \frac{y^2 c}{x^2}\right)$$

$$\therefore \quad cy^2 = e^{x^2/y^2}$$

36. Let $V(t)$ be the velocity of the object at time t.

Given $\dfrac{dV}{dt} = 9.8 - kV \Rightarrow \dfrac{dV}{9.8 - kV} = dt$

Integrating, we get

$$\log(9.8 - kV) = -kt + C$$

$$\Rightarrow \quad 9.8 - kV = \text{constant } C \cdot e^{-kt}$$

But $V(0) = 0 \Rightarrow C = 9.8.$

Thus $9.8 - kV = 9.8e^{-kt}$

$$\Rightarrow \quad kV = 9.8(1 - e^{-kt}) \Rightarrow V(t) = \frac{9.8}{k}(1 - e^{kt}) < \frac{9.8}{k}$$

for all t. Hence, $V(t)$ cannot exceed $\dfrac{9.8}{k}$.

37. Let the equation of the curve be $y = f(x)$. It is given that $OT \propto y$.

$$\Rightarrow \qquad OT = by$$

$$\Rightarrow \qquad OM - TM = by$$

$$\Rightarrow \qquad x - \frac{y}{dy/dx} = by$$

[$\because$ TM = length of the subtanget]

$$\Rightarrow \qquad x - y\frac{dx}{dy} = b.y$$

$$\Rightarrow \qquad \frac{dx}{dy} - \frac{x}{y} = -b$$

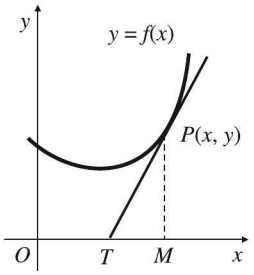


Fig. 17.2

It is a linear differential equation. Its equation is given by $\frac{x}{y} = -b\log y + a \Rightarrow x = y(a - b\log y)$

38. We have $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0 \Rightarrow \sqrt{1-x^2}\,dy + \sqrt{1-y^2}\,dx = 0$

$$\Rightarrow \quad \frac{1}{\sqrt{1-x^2}}dx + \frac{1}{\sqrt{1-y^2}}dy = 0$$

$$\Rightarrow \quad \sin^{-1}x + \sin^{-1}y = \sin^{-1}a \qquad \text{[On integrating]}$$

$$\Rightarrow \quad \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) = \sin^{-1}a$$

$$\Rightarrow \quad x\sqrt{1-y^2} + y\sqrt{1-x^2} = a$$

39. Given $y = xe^{cx}$ (i)

$$\Rightarrow \quad \frac{dy}{dx} = e^{cx} + xe^{cx}\cdot c = \frac{y}{x} + y\cdot c \qquad \text{(ii)}$$

From Eq. (i), we get

$$\log y = \log x + cx$$

$$\Rightarrow \quad c = \frac{1}{x}\log\frac{y}{x}$$

From Eq. (ii), we get

$$\frac{dy}{dx} = \frac{y}{x} + \frac{y}{x}\log\frac{y}{x} = \frac{y}{x}\left(1 + \log\frac{y}{x}\right)$$

40. We have $dy + [y\phi'(x) - \phi(x)\phi'(x)]\,dx = 0$

$$\Rightarrow \quad \frac{dy}{dx} + \phi'(x)\cdot y = \phi(x)\phi'(x) \qquad \text{(i)}$$

This is a linear differential equation with

$$\text{I.F. } e^{\int\phi'(x)dx} = e^{\phi(x)}$$

Multiplying (i) by $f(x)$ and integrating, we get

$$ye^{\phi(x)} = \int\phi(x)\phi'(x)e^{\phi(x)}dx$$

$$\Rightarrow \quad ye^{\phi(x)} = \int e^{\phi(x)}\phi(x)\phi'(x)\,dx$$

$$\Rightarrow \quad ye^{\phi(x)} = \int\phi(x)e^{\phi(x)}\phi'(x)\,dx$$

Let $t = \phi(x) \Rightarrow dt = \phi'(x)\,dx$

$$\Rightarrow \quad ye^{t} = \int te^{t}dt = t(e^{t}) - e^{t} = e^{t}(t-1)$$

$$\Rightarrow \quad ye^{\phi(x)} = e^{\phi(x)}[\phi(x) - 1] + c$$

$$\Rightarrow \quad y = [\phi(x) - 1] + ce^{-\phi(x)}$$

41. $\int\frac{dy}{y} = \int\frac{2x}{x^2+1}dx$

$$\log y = \log(x^2+1) + c$$

It passes through (1, 2).

$$\log 2 = \log 2 + c \quad \text{or} \quad c = 0$$

$$\therefore \quad \log y = \log(x^2+1) \quad \Rightarrow \quad y = x^2 + 1$$

42. $\lim\limits_{t\to x}\frac{t^2f(x) - x^2f(t)}{t-x} = 1$

$$\Rightarrow \quad \lim_{t\to x}\frac{t^2f(x) - x^2f(x) + x^2f(x) - x^2f(t)}{t-x} = 1$$

$$\Rightarrow \quad \lim_{t\to x}\left(\frac{t^2-x^2}{t-x}\right)f(x) - x^2\lim_{t\to x}\frac{f(t)-f(x)}{t-x} = 1$$

$$\Rightarrow \quad f(x)\lim_{t\to x}(t+x) - x^2\lim_{t\to x}\frac{f(t)-f(x)}{t-x} = 1$$

$$\Rightarrow \quad 2xf(x) - x^2f'(x) = 1 \quad \Rightarrow \quad x^2f'(x) - 2xf(x) = -1$$

$$\Rightarrow \quad \frac{x^2f'(x) - 2xf(x)}{(x^2)^2} = -\frac{1}{x^4}$$

$$\Rightarrow \quad \frac{d}{dx}\left(\frac{f(x)}{x^2}\right) = -\frac{1}{x^4} \quad \Rightarrow \quad \frac{f(x)}{x^2} = \frac{1}{3x^3} + c \qquad \text{(i)}$$

It is given that $f(1) = 1$. So, putting $x = 1$ and $f(1) = 1$ in (i), we get $c = \frac{2}{3}$.

Putting $c = 2/3$ in (i), we get

$$\frac{f(x)}{x^2} = \frac{1}{3x^3} + \frac{2}{3} \Rightarrow f(x) = \frac{1}{3x} + \frac{2}{3}x^2$$

43. The intersection point of $y - x + 1 = 0$ and $y + x + 5 = 0$ is $(-2, -3)$. Put $x = X - 2$, $y = Y - 3$. The given equations reduce to $\frac{dY}{dX} = \frac{Y-X}{Y+X}$. This is a homogeneous equation. So putting $Y = vX$, we get

$$X\frac{dv}{dX} = -\frac{v^2+1}{v+1} \quad \Rightarrow \quad \left(-\frac{v}{v^2+1} - \frac{1}{v^2+1}\right)dv = \frac{dX}{X}$$

$$\Rightarrow \quad -\frac{1}{2}\log(v^2+1) - \tan^{-1}v = \log|X| + \text{constant}$$

$$\Rightarrow \quad \log(Y^2 + X^2) + 2\tan^{-1}\frac{Y}{X} = \text{constant}$$

$$\Rightarrow \quad \log[(y+3)^2 + (x+2)^2] + 2\tan^{-1}\frac{y+3}{x+2} = c$$

44. $\dfrac{dx}{dy} = e^{3x-5y} = \dfrac{e^{3x}}{e^{5y}}$

or $\quad \int e^{5y}dy = \int e^{3x}dx \quad$ or $\quad \dfrac{e^{5y}}{5} = \dfrac{e^{3x}}{3} + c_1$

or $\quad 3e^{5y} = 5e^{3x} + c \quad$ (where $c = 15c_1$)

45. The given differential equation is $\dfrac{dy}{dx} = \dfrac{\sqrt{1-y^2}}{y}$

$$\Rightarrow \quad \frac{-y}{\sqrt{1-y^2}}dy = -dx$$

$$\Rightarrow \quad \sqrt{1-y^2} = -x + c$$

$$\Rightarrow \quad 1 - y^2 = (x-c)^2 \Rightarrow (x-c)^2 + y^2 = 1$$

Clearly, it represents a family of circles of fixed radius 1 and variable centre $(c, 0)$ along the x-axis.

46. The differential equation is $\dfrac{d^2y}{dx^2} + \dfrac{1}{x^2} = 0.$

The given equation may be written as $\dfrac{d^2y}{dx^2} = -\dfrac{1}{x^2}.$

Integrating this equation, we get $\dfrac{dy}{dx} = -\left(-\dfrac{1}{x}\right) = \dfrac{1}{x}$

Again integrating, we get $y = \log_e x.$

47. $ydx + xdy = xy(dy - dx)$ or $\int \dfrac{d(xy)}{xy} = \int (dy - dx)$ $\log xy = y - x + c$ or $x + \log xy - y = c$

48. The displacement x for all SHM is given by

$$x = a\cos(nt + b) \Rightarrow \frac{dx}{dt} = -na\sin(nt + b)$$

$$\Rightarrow \quad \frac{d^2x}{dt^2} = -n^2 a\cos(nt + b)$$

$$\Rightarrow \quad \frac{d^2x}{dt^2} = -n^2x \Rightarrow \frac{d^2x}{dt^2} + n^2x = 0$$

49. The differential equation is $\dfrac{dy}{dx} + 2xy = y.$

The given equation may be written as

$$\frac{dy}{dx} = (1-2x)y \quad \text{or} \quad \frac{dx}{dy} = (1-2x)dx$$

Integrating, we get

$$\log y = x - x^2 + \log c \quad \text{or} \quad \log\frac{y}{c} = x - x^2$$

or $\quad y = ce^{x-x^2}$

50. $\dfrac{dy}{dx} = \sin\left(\dfrac{x-y}{2}\right) - \sin\left(\dfrac{x+y}{2}\right)$

$$\frac{dy}{dx} = -2\cos\frac{x}{2}\sin\frac{y}{2}$$

or $\quad \int \text{cosec}\,\dfrac{y}{2}\,dy = -2\int \cos\dfrac{x}{2}\,dx$

or $\quad \log\dfrac{\tan\frac{y}{4}}{1/2} = \dfrac{-2\sin\frac{x}{2}}{1/2} = c_1$

$$\log\tan\frac{y}{4} + 2\sin\frac{x}{2} = c \quad (\text{where } c = 2c_1)$$

51. We have $\dfrac{dy}{dx} = 1 - \dfrac{1}{x^2} \Rightarrow y = x + \dfrac{1}{x} + c$

This passes through (2, 7/2).

$$\therefore \quad \frac{7}{2} = 2 + \frac{1}{2} + c \Rightarrow c = 1$$

Thus, the equation of the curve is

$$y = x + \frac{1}{x} = 1 \Rightarrow xy = x^2 + x + 1$$

52. This differential equation is $(1+y^2)dx + (1+x^2)dy = 0.$

The given equation may be written as

$$\frac{dx}{1+x^2} + \frac{dy}{1+y^2} = 0 \qquad \text{(i)}$$

Integrating Eq. (i), we get

$$\int\frac{1}{1+x^2}dx + \int\frac{1}{1+y^2}dy = c \quad \text{or} \quad \tan^{-1}x + \tan^{-1}y = c$$

We know that $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$

Therefore, $\tan^{-1}\left(\dfrac{x+y}{1-xy}\right) = c$ or $\dfrac{x+y}{1-xy} = \tan c$

or $\quad \dfrac{x+y}{1-xy} = k \qquad (\text{where } k = \tan c)$

53. $\dfrac{dy}{dx} = (1+x)(1+y)$ or $\int\dfrac{dy}{1+y} = \int(1+x)dx$

$$\Rightarrow \quad \log|1+y| = x + \frac{x^2}{2} + c$$

54. We have $\dfrac{dx}{dt} = x + 1$

$$\Rightarrow \quad \frac{1}{x+1}dx = dt \Rightarrow \log(x+1) = t + c$$

Putting $t = 0, x = 0$

$\log 1 = c \Rightarrow c = 0 \qquad \therefore t = \log(x+1)$

Putting $x = 99$, we get

$$t = \log_e 100 = 2\log_e 10$$

55. The equation for the family of curves is $x^2 + y^2 - 2ay = 0$. The given equation may be written as $x^2 + y^2 = 2ay$ or $a = \dfrac{x^2 + y^2}{2y}$.

Differentiating the given equation, we get

$$2x + 2yy' - 2ay' = 0 \qquad \text{(i)}$$

Substituting the value of a in Eq. (i), we get

$$2x + 2yy' - 2\left(\frac{x^2 + y^2}{2y}\right)y' = 0$$

or $$x + yy' - \left(\frac{x^2 + y^2}{2y}\right)y' = 0$$

or $$2xy + 2y^2y' - (x^2 + y^2)y' = 0$$

or $$2xy + y^2y' - x^2y' = 0 \text{ or } 2xy - (x^2 - y^2)y' = 0$$

or $$(x^2 - y^2)y' = 2xy$$

56. $\dfrac{dy}{dx} = \dfrac{x + y}{x}$

$$xdy = xdx + ydx \text{ or } xdy - ydx = xdx$$

or $$\frac{xdy - ydx}{x^2} = \frac{x}{x^2}dx \text{ or } \int d(y/x) = \int \frac{1}{x}dx$$

or $$\frac{y}{x} = \log|x| + c \text{ or, } y = x\log|x| + cx$$

57. $y = 2e^{2x} - e^{-x} \Rightarrow y_1 = 4e^{2x} + e^{-x}$

$$\Rightarrow y_2 = 8e^{2x} - e^{-x} \Rightarrow y_2 = 4e^{2x} + e^{-x} + 4e^{2x} - 2e^{-x}$$

$$\Rightarrow y_2 = y_1 + 2(2e^{2x} - e^{-x})$$

$$\Rightarrow y_2 = y_1 + 2y \Rightarrow y_2 - y_1 - 2y = 0$$

58. The differential equation is

$$\tan y \sec^2 x dx + \tan x \sec^2 y dy = 0$$

The given equation may be written as

$$\frac{\sec^2 x}{\tan x}dx = -\frac{\sec^2 y}{\tan y}dy$$

Integrating this equation, we get

$$\int\frac{\sec^2 x}{\tan x}dx = -\int\frac{\sec^2 y}{\tan y}dy$$

or $$\log\tan x = -\log\tan y + \log c$$

or $$\log\tan x + \log\tan y = \log c$$

or $$\log(\tan x \tan y) = \log c$$

Therefore, $$\tan x \tan y = c$$

59. The given equation is $x^3\left(x\dfrac{dy}{dx} + y\right) = -\operatorname{cosec} xydx$

or $$x^3(xdy + ydy) = -\operatorname{cosec} xydx$$

or $$\int\sin xyd(xy) = -\int x^{-3}dx \text{ or } -\cos xy = -\frac{x^{-2}}{2} - c$$

$$2\cos xy = c + \frac{1}{x^2} \Rightarrow 2x^2\cos xy = cx^2 + 1$$

60. The equation of the line whose slope is equal to the y-intercept is $y = cx + c = c(x + 1)$

$$\Rightarrow \frac{dy}{dx} = c \qquad \therefore \quad \frac{dy}{dx} = \frac{y}{x+1}$$

$$\Rightarrow (x+1)\frac{dy}{dx} - y = 0$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. The differential equation is $y' = 1 + x + y^2 + xy^2$ and $y(0) = 0$. The given equation may be written as

$$\frac{dy}{dx} = 1 + x + y^2 + xy^2 = (1 + x) + y^2(1 + x)$$

$$= (1 + x)(1 + y^2)$$

or $$\frac{dy}{1 + y^2} = (1 + x)dx$$

Integrating this equation, we get

$$\tan^{-1} y = x + \frac{x^2}{2} + c \qquad \text{(i)}$$

Since $y(0) = 0 \Rightarrow$ at $x = 0, y = 0$ therefore $c = 0$

Substituting the value of c in Eq. (i), we get

$$\tan^{-1} y = x + \frac{x^2}{2} \text{ or } y = \tan\left(x + \frac{x^2}{2}\right)$$

2. $\cos^2 x\dfrac{dy}{dx} + y = \tan x$ or $\dfrac{dy}{dx} + \sec^2 xy = \tan x \sec^2 x$

$$\text{I.F.} = e^{\int\sec^2 xdx} = e^{\tan x}$$

Multiplying both sides by I.F. and integrating, we have

$$ye^{\tan x} = \int\tan x \sec^2 x = e^{\tan x}dx$$

or $$ye^{\tan x} = \int ze^z dz \text{ (let } z = \tan x)$$

or $$ye^{\tan x} = ze^z - e^z + c \qquad \text{(use integration by parts)}$$

or $$ye^{\tan x} = \tan xe^{\tan x} - e^{\tan x} + c$$

Divide both sides by $e^{\tan x}$ to get the result.

3. $y = c_1(\cos x + c_2) + c_3\sin(x + c_4) + c_5e^x + c_6$

$$y = c_1(\cos x\cos c_2 - \sin x\sin c_2)$$

$$+c_3(\sin x\cos c_4 + \cos x\sin c_4) + c_5e^x + c_6$$

$$\cos x(c_1\cos c_2 + c_3\sin c_4) + \sin x(-c_1\sin c_2 + c_3\cos c_4) + c_5e^x + c_6$$

$$= A\cos x + B\sin x + Ce^x + D$$

where $A = c_1\cos c_2 + c_3\sin c_4$

$B = -c_1\sin c_2 + c_3\cos c_4$, $C = c_5$, $D = c_6$

Hence, the order is 4.

4. We have $\frac{dy}{dx} = \frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$

Putting $y = v$, x and $\frac{dy}{dx} = v + x\frac{dv}{dx}$, we get

$$v + x\frac{dv}{dx} = v - \cos^2 v$$

$$\Rightarrow \frac{dv}{\cos^2 v} = -\frac{dx}{x} \Rightarrow \sec^2 v\,dv = -\frac{1}{x}dx$$

On integration, we get

$$\tan v = -\log x + \log C \Rightarrow \tan\left(\frac{y}{x}\right) = -\log + \log c$$

This passes through $(1, \pi/4)$. Therefore, $1 = \log c$

$$\therefore \tan\left(\frac{y}{x}\right) = -\log x + 1$$

$$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log e \Rightarrow y = x\tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$$

5. The given equation can be written as

$$\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = \frac{\log x}{x}$$

Let $\frac{1}{y} = t$, then $-\frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$

$\therefore$ Equation becomes $-\frac{dt}{dx} + \frac{t}{x} = \frac{\log x}{x}$

or $$\frac{dt}{dx} - \frac{t}{x} = -\frac{\log x}{x}$$

I.F. $= e^{\int \frac{-1}{x}dx} = e^{-\log_e x} = 1/x$

Multiplying both sides by I.F. and integrating, we get

$$\frac{t}{x} = -\int\frac{\log x}{x^2}dx$$

Solve the R.H.S. using integration by parts and put $t = 1/y$ to get the result.

6. Differentiating the given equation successively, we get

$$y_1 = 5be^{5x} - 7ce^{-7x} \quad \text{(i)}$$

$$y_2 = 25be^{5x} + 49ce^{-7x} \quad \text{(ii)}$$

and $$y_3 = 125be^{5x} - 343ce^{-7x} \quad \text{(iii)}$$

Multiplying (i) by 7 and then adding to (ii), we get

$$y_2 + 7y_1 = 60be^{5x} \quad \text{(iv)}$$

Multiplying (i) by 5 and then subtracting it from (ii), we get

$$y_2 - 5y_1 = 84ce^{-7x} \quad \text{(v)}$$

Putting the values of b and c, obtained from (iv) and (v) respectively, in (i), we get

$$y_3 + 2y_2 - 35y_1 = 0$$

7. The differential equation is $y' = 1 + y^2$, $y(0) = 0 = y(\pi)$.

The given equation may be written as

$$\frac{dy}{dx} = 1 + y^2$$

or $$\frac{dy}{1+y^2} = dx$$

Integrating this equation, we get

$$\tan^{-1} y = x + c \quad \text{(i)}$$

Since $y(0) = 0 \Rightarrow$ at $x = (0)$, $y = 0$,

Therefore, $\tan^{-1} 0 = 0 + c$ or $c = 0$

Substituting the value of c in Eq. (i), we get

$$\tan^{-1}(y) = x \text{ or } y = \tan x$$

Therefore, the solution is

$$y = \phi(x) = \tan x \quad \text{(ii)}$$

We know that $\phi(x)$ or $\tan x$ is not continuous in $(0, \pi)$.

And at $x = \frac{\pi}{2}$, $\tan\left(\frac{\pi}{2}\right) = \infty$. Thus, no solution is possible.

8. Integrating the given differential equation, we have

$$\frac{dy}{dx} = \frac{-\cos 3x}{3} + e^x + \frac{x^3}{3} + c_1$$

but $y'(0) = 1$, so $1 = \left(-\frac{1}{3}\right) + 1 + c_1 \Rightarrow c_1 = 1/3$

Again integrating, we get

$$y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x + c_2$$

but $y(0) = 0$, so $0 = 0 + 1 + c_2 \Rightarrow c_2 = -1$. Thus,

$$y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$$

9. Multiplying by x^2y, we get

$$x^3y(4ydx + 2xdy) + x^2y^4(3ydx + 5xdy) = 0$$

$$4x^3y^2dx + 2x^4ydy + 3x^2y^5dx + 5x^3y^4dy = 0$$

$$\int d(x^4y^2) + \int d(x^3y^5) = 0,\ x^4y^2 + x^3y^5 + c = 0$$

10. In the given equation, there are three parameters. So it is a differential equation of third order.

11. Let $x+y=u$. Then $1+\frac{dy}{dx}=\frac{du}{dx}$

So, the given differential equation becomes

$$\left(\frac{u-1}{u-2}\right)\left(\frac{du}{dx}-1\right)=\frac{u+1}{u+2} \Rightarrow \frac{du}{dx}-1=\left(\frac{u+1}{u+2}\right)\left(\frac{u-2}{u-1}\right)$$

$$\Rightarrow \frac{du}{dx}=\frac{u^2-u-2}{u^2+u-2}+1 \Rightarrow \frac{du}{dx}=\frac{2u^2-4}{u^2+u-2}$$

$$\Rightarrow \frac{u^2+u-2}{u^2-2}du=2dx \Rightarrow \left(1+\frac{u}{u^2-2}\right)du=2dx$$

On integrating, we get

$$u+\frac{1}{2}\log\left|u^2-2\right|=2x+c$$

$$\Rightarrow (x+y)+\frac{1}{2}\log\left|(x+y)^2-2\right|=2x+c$$

$$\Rightarrow 2(y-x)+\log\left|(x+y)^2-2\right|=2c \quad \text{(i)}$$

When $x = 1$, we have $y = 1$

$$\log 2=2c$$

$\therefore$ Substituting the value of c in Eq. (i), we get

$$2(y-x)+\log\left|(x+y)^2-2\right|=\log 2$$

$$\Rightarrow 2(y-x)+\log\left|\frac{(x+y)^2-2}{2}\right|=0$$

Hence, option (d) is the correct answer.

12. Clearly, the constant function $y = 0$ is a solution.

Differentiating the given equation w.r.t. x, we get

$$p=\frac{dy}{dx}=(p+p^3)+x\left(\frac{dp}{dx}+3p^2\frac{dp}{dx}\right)$$

$$\Rightarrow -p^3=x(1+3p^2)\frac{dp}{dx} \Rightarrow -\frac{dx}{x}=\left(\frac{1}{p^3}+\frac{3}{p}\right)dp$$

$$\Rightarrow \log xp^3=(1/2)p^2+C \Rightarrow xp^3=e^{\frac{1}{2}p^2}$$

Putting the value of x in the given equation, we have

$$y=p^{-3}e^{\frac{1}{2}p^2}(p+p^3)$$

13. Focus $S = (0, 0)$. Let $P(x, y)$ be any point on the parabola.

Since $SP^2=PM^2$,

$$\Rightarrow (x-0)^2+(y-0)^2=(x+a)^2$$

$$\Rightarrow y^2=2ax+a^2 \quad \text{(i)}$$

$$\Rightarrow 2y\frac{dy}{dx}=2a \quad \text{(ii)}$$

From Eqs. (i) and (ii), we get

$$y^2=2y\frac{dy}{dx}\cdot x+\left(y\frac{dy}{dx}\right)^2$$

$$\Rightarrow y^2\left(\frac{dy}{dx}\right)^2+2xy\frac{dy}{dx}=y^2$$

$$\Rightarrow -y\left(\frac{dy}{dx}\right)^2=2x\frac{dy}{dx}-y$$

14. The equation of the tangent at $P(x, y)$ to the curve is

$$Y-y=\frac{dy}{dx}(X-x)$$

Since it passes through $(0, y^3)$,

$$\Rightarrow y^3-y=\frac{dy}{dx}(0-x) \Rightarrow \frac{dy}{y^3-y}=-\frac{dx}{x}$$

$$\Rightarrow \frac{y^2-1}{y^2}=\frac{c}{x^2} \Rightarrow x^2(y^2-1)=cy^2$$

15. The equation can be written as

$$x^2\frac{xdy-ydx}{x^2}=x\sqrt{1-(y/x)^2}\,dx$$

$$\Rightarrow \frac{d(y/x)}{\sqrt{1-(y/x)^2}}=\frac{dx}{x} \Rightarrow \sin^{-1}y/x=\log x+\text{constant}$$

Since $y\,(1) = 0$, so the constant $= 0$

Hence, $y=x\sin(\log x)$

16. We have slope $=\frac{dy}{dx}=\frac{2ay}{x(y-a)} \Rightarrow \frac{y-a}{y}dy=\frac{2a}{x}dx$

Integrating both sides, we get

$$a\log|y|-y=-2a\log|x|+\log c \Rightarrow y^a\cdot x^{2a}=ce^y$$

This passes through (1, 1). Therefore, $1=ce \Rightarrow c=\frac{1}{e}$

So, the equation of the curve is $y^a\cdot x^{2a}=e^{y-1}$.

17. The given equation can be written as $xydy+y^2dx-dy=0$.

Multiplying it by $\frac{1}{y}$, we get

$$xdy+ydx-\frac{1}{y}dy=0$$

$$\Rightarrow \int d(xy)=\int\frac{dy}{y}=0 \Rightarrow xy-\log y-c$$

$$\Rightarrow \frac{1}{y} \text{ is the integrating factor}$$

18. We have $a(x-y)=\sqrt{1-x^2}+\sqrt{1-y^2}$

Putting $x=\sin A$, $y=\sin B$, we get

$$\cos A+\cos B=a(\sin A-\sin B)$$

$$\Rightarrow \cot\frac{A-B}{2}=a \Rightarrow A-B=2\cot^{-1}a$$

$$\Rightarrow \sin^{-1}x-\sin^{-1}y=2\cot^{-1}a$$

Differentiating w.r.t. to x, we get

$$\frac{1}{\sqrt{1-x^2}}-\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=0 \Rightarrow \frac{dy}{dx}=\sqrt{\frac{1-y^2}{1-x^2}}$$

Clearly, it is a differential equation of first order and first degree.

19. We have $\dfrac{\sqrt{x}dx+\sqrt{y}dy}{\sqrt{x}dx-\sqrt{y}dy}=\sqrt{\dfrac{y^3}{x^3}}$

$$\Rightarrow \frac{d(x^{3/2})+d(y^{3/2})}{d(x^{3/2})-d(y^{3/2})}=\frac{y^{3/2}}{x^{3/2}}$$

$$\Rightarrow \frac{du+dv}{du-dv}=\frac{v}{u}, \text{ where } u=x^{3/2} \text{ and } v=y^{3/2}$$

$$\Rightarrow udu+udv=vdu-vdv \Rightarrow udu+vdv=vdu-udv$$

$$\Rightarrow \frac{udu+vdv}{u^2+v^2}=\frac{vdu-udv}{u^2+v^2}$$

$$\Rightarrow \frac{d(u^2+v^2)}{u^2+v^2}=-2d\tan^{-1}\left(\frac{v}{u}\right)+c$$

On integrating, we get

$$\log(u^2+v^2)=-2\tan^{-1}\left(\frac{v}{u}\right)+c$$

$$\Rightarrow \frac{1}{2}\log(x^3+y^3)+\tan^{-1}\left(\frac{y}{x}\right)^{3/2}=\frac{c}{2}$$

20. The given differential equation can be written as $y_2/y_1 = 2x/(x^2+1)$. Integrating both the sides, we have $\log y_1 = \log(x^2+1) + c$, which implies $y_1(0) = \log 1 + c$, i.e. $c = \log 3$. Therefore, $\log y_1 = \log(x^2+1) + \log 3$, which implies $y_1 = 3(x^2+1)$ or $y = x^3 + 3x + A$, so $1 = y(0) = 0 + 0 + A$, i.e. $A = 1$. Hence, the required equation of the curve is $y = x^3 + 3x + 1$.

21. $y = a\cos(x-b)$

$$dy/dx = -a\sin(x-b)$$

$$d^2y/dx^2 = -a\cos(x-b) = -ay$$

$$\Rightarrow \quad d^2y/dx^2 + ay = 0$$

22. We have $y(x+y^3)dx = x(y^3-x)dy$

$$\Rightarrow y^3(ydx-xdy)+x(ydx+xdy)=0$$

$$\Rightarrow x^2y^3\cdot\left(\frac{ydx-xdy}{x^2}\right)+xd(xy)=0$$

$$\Rightarrow -\frac{y}{x}d\left(\frac{y}{x}\right)+\frac{d(xy)}{x^2y^2}=0$$

On integrating, we get

$$\frac{-\left(\frac{y}{x}\right)^2}{2}-\frac{1}{xy}=c \Rightarrow \frac{y^2}{2x^2}+\frac{1}{xy}+c=0$$

$$\Rightarrow y^3+2x+2cx^2y=0$$

Since it passes through the point (1, 1), therefore,

$1 + 2 + 2c = 0$ or $c=\dfrac{-3}{2}$

$\therefore$ The curve is $y^3+2x-3x^2y=0$.

23. Given, $x=\sin t$, $y=\cos pt$

$$\frac{dx}{dy}=\cos t, \frac{dy}{dt}=-p\sin pt$$

$$\therefore \frac{dy}{dx}=-\frac{p\sin pt}{\cos t}$$

$$\Rightarrow y_1=\frac{-p\sqrt{1-y^2}}{\sqrt{1-x^2}} \Rightarrow y_1\sqrt{1-x^2}=-p\sqrt{1-y^2}$$

$$\Rightarrow y_1^2(1-x^2)=p^2(1-y^2)$$

$$\Rightarrow 2y_1y_2(1-x^2)-2xy_1^2=-2yy_1p^2 \quad \text{[differentiating]}$$

$$\Rightarrow (1-x^2)y_2-xy_1+p^2y=0$$

24. Put $y^2\sin x = V$

Then $2y\dfrac{dy}{dx}\sin x+y^2\cos x=\dfrac{dV}{dx}$

So, the given equation becomes $\dfrac{1}{2}\dfrac{dV}{dx}=\sin x\cos x$

$$\Rightarrow \quad dV = 2\sin x\cos x dx$$

Integrating, we get

$$V=2\sin^3 x+c \Rightarrow y^3\sin x=\sin^2 x+c$$

On putting $y=1, x=\dfrac{\pi}{2}$, we get

$$1=2+c \Rightarrow c=1$$

So, the solution is $y^2\sin x = 2\sin^2 x + 1$.

25. Given $y=a\cos(x+b) \Rightarrow \dfrac{dy}{dx}=-a\sin(x+b)$

$$\Rightarrow \frac{d^2y}{dx^2}=-a\cos(x+b)=-y \Rightarrow \frac{d^2y}{dx^2}+y=0$$

26. We have $x^2\dfrac{dy}{dx}-xy=1+\cos\dfrac{y}{x}=2\cos^2\dfrac{y}{2x}$

$$\Rightarrow \sec^2\left(\frac{y}{2x}\right)\cdot\left[x^2\frac{dy}{dx}-xy\right]=2$$

$$\Rightarrow \frac{1}{2}\sec^2\left(\frac{y}{2x}\right)\cdot\frac{x\frac{dy}{dx}-y}{x^2}=\frac{1}{x^3} \Rightarrow \frac{d}{dx}\left(\tan\frac{y}{2x}\right)=\frac{1}{x^3}$$

Integrating, we get

$$\tan\frac{y}{2x}=-\frac{1}{2x^2}+c$$

which is the required solution.

27. Given $y=(x+\sqrt{1+x^2})^n$

$$\Rightarrow \frac{dy}{dx}=n[x+\sqrt{1+x^2}]^{n-1}\left(1+\frac{x}{\sqrt{x^2+1}}\right)$$

$$=\frac{n[x+\sqrt{1+x^2}]^n}{\sqrt{1+x^2}}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2(1+x^2)=n^2y^2$$

Again, differentiating, we get

$$2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2}(1+x^2)+2x\left(\frac{dy}{dx}\right)^2=2n^2y\frac{dy}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2}(1+x^2)+x\frac{dy}{dx}=n^2y \qquad \left[\text{divide by } 2\frac{dy}{dx}\right]$$

28. Given, $y=a\sin(5x+c) \Rightarrow \frac{dy}{dx}=5a\cos(5x+c)$

$$\Rightarrow \frac{d^2y}{dx^2}=-25a\sin(5x+c)=-25y$$

29. We have $y(x^2y+e^x)dx-e^xdy=0$

$$\Rightarrow e^x\frac{dy}{dx}=x^2y^2+ye^x$$

Dividing by y^2e^x, we get

$$\frac{1}{y^2}\frac{dy}{dx}-\frac{1}{y}=x^2e^{-x}$$

Put $\frac{1}{y}=V$ so that $\frac{-1}{y^2}\frac{dy}{dx}=\frac{dV}{dx}$

We thus have $\frac{dV}{dx}+V=-x^2e^{-x}$, which is linear

$\therefore$ I.F. $=e^{\int 1dx}=e^x$

Hence, the solution is $V\cdot e^x=-\int x^2e^{-x}\cdot e^xdx+\frac{c}{3}$

$$\Rightarrow \frac{1}{y}e^x=-\frac{x^3}{3}+\frac{c}{3} \Rightarrow x^3y+3e^x=cy$$

30. $y=(a+be^c)x$

$$\Rightarrow \frac{dy}{dx}=(a+be^c)=\frac{y}{x}$$

$$\Rightarrow \frac{xdy}{dx}-y=0$$

Hence, Statement I is true.

Statement II is obviously true.

Hence, option (b) is the correct answer.

31. Statement II is obviously true.

The general equation of all the circles in a plane is

$$x^2+y^2+2gx+2fy+c=0$$

Since it contains 3 arbitrary constants, so the order of the differential equation in 3.

So, Statement I is true, but Statement II is not a correct explanation of Statement I.

32. Since the given differential equation cannot be written as a polynomial in derivatives, so its degree is not defined.

Hence, Statement I is false.

But, Statement II is true.

So, the correct answer is option (a).

33. Clearly, the order is 3.

Again $\left(1+3\frac{dy}{dx}\right)^2=64\left(\frac{d^3y}{dx^3}\right)^3$

$\therefore$ The degree is 3.

34. The general equation of all non-vertical lines in a plane is $ax+hy=1$, where $h\neq 0$.

$$\Rightarrow a+h\frac{dy}{dx}=0 \Rightarrow h\frac{d^2y}{dx^2}=0$$

$$\Rightarrow \frac{d^2y}{dx^2}=0 \qquad (\because\ h\neq 0)$$

35. $\frac{d^2y}{dx^2}=e^{-2x} \Rightarrow \frac{dy}{dx}=\frac{e^{-2x}}{-2}+c$

$$\Rightarrow y=\frac{e^{-2x}}{(-2)(-2)}+cx+d=\frac{1}{4}e^{-2x}+cx+d$$

36. The equation of the family of parabolas whose axis is the x-axis is given by

$$y^2=4a(x-k)$$

$$\therefore\ 2y\frac{dy}{dx}=4a \Rightarrow y\frac{dy}{dx}=2a \Rightarrow y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$$

Hence, the degree is 1 and the order is 2.

37. $(1+y^2)+(x-e^{\tan^{-1}y})\dfrac{dy}{dx}=0$

$$\Rightarrow \quad \frac{dy}{dx}+\frac{x}{1+y^2}=\frac{e^{\tan^{-1}y}}{1+y^2}$$

This is linear in x. $\left[\text{Type } \dfrac{dy}{dx}+Px=Q\right]$

$$\int P\,dy=\int\frac{dy}{1+y^2}=\tan^{-1}y$$

$$\therefore \quad e^{\int P\,dy}=e^{\tan^{-1}y}$$

$\therefore$ The solution is

$$x\cdot e^{\tan^{-1}y}=\int e^{\tan^{-1}y}\cdot\frac{e^{\tan^{-1}y}}{1+y^2}dy+c$$

Put $\tan^{-1}y=z \quad \therefore \quad \dfrac{dy}{1+y^2}=dz$

$$=\int e^{2z}dz+c=\frac{1}{2}e^{2z}+c$$

$$=\frac{1}{2}e^{2\tan^{-1}y}+c$$

$$\Rightarrow \quad 2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+K \qquad [\text{where } 2c=K]$$

38. $\dfrac{dy}{dt}-\dfrac{t}{t+1}y=\dfrac{1}{1+t}$ is a linear differential equation.

$$\therefore \quad \text{I.F.}=e^{-\int\frac{1}{1+t}dt}=e^{(-1+\log|t+1|)}=(t+1)\cdot e^{-t}$$

$$\therefore \quad y(t+1)e^{-t}=\int e^{-t}dt+c=-e^{-t}+c$$

when $y(0)=-1$ then $c=0$

$$\therefore \quad y(1)=-\frac{1}{2}$$

39. Here $ydx+xdy=-x^2ydy$

$$\Rightarrow \quad \frac{ydx+xdy}{x^2y^2}=-\frac{1}{y}dy \quad \Rightarrow \quad d\left(-\frac{1}{xy}\right)=-\frac{1}{y}dy$$

$\therefore$ Integrating both sides, we get

$$-\frac{1}{xy}+\log y=c$$

40. The equation of the family of curves is

$$x^2+y^2-2ay=0 \qquad \text{(i)}$$

Differentiating w.r.t. x, we get

$$2x+2yy'-2ay'=0 \quad \Rightarrow \quad \frac{2(x+yy')}{y'}=2a \qquad \text{(ii)}$$

From (i) and (ii), we get

$$x^2+y^2=\frac{2(x+yy')}{y'}\cdot y \quad \Rightarrow \quad (x^2+y^2)y'=2y(x+yy')$$

$$\Rightarrow \quad (x^2-y^2)y'=2xy$$

41. Since $f''(x)=6(x-1)$

$$\Rightarrow \quad f'(x)=3(x-1)^2+c \quad [\text{integrating}] \qquad \text{(i)}$$

Also, at the point (2, 1), the tangent to the graph is $y=3x-5$ and slope of the tangent = 3.

$$\Rightarrow \quad f'(2)=3$$

$$3(2-1)^2+c=3 \qquad [\text{from Eq. (i)}]$$

$$\Rightarrow \quad 3+c=3 \quad \Rightarrow \quad c=0$$

From Eq. (i), we have

$$f'(x)=3(x-1)^2$$

$$\Rightarrow \quad f(x)=(x-1)^2+k \quad [\text{integrating}] \qquad \text{(ii)}$$

Since it passes through (2, 1),

$$\therefore \quad 1=(2-1)^2+k \Rightarrow k=0$$

Hence, the equation of the function is $f(x)=(x-1)^3$.

42. $\dfrac{dy}{dx}=\dfrac{y}{x}\cdot\left[\log\dfrac{y}{x}+1\right]$ is a homogeneous equation.

Put $y=vx$, then $v+x\dfrac{dv}{dx}=\dfrac{dy}{dx} \quad \Rightarrow \quad \dfrac{dv}{v\log v}=\dfrac{dx}{x}$

Integrating, we get $\log\left(\dfrac{y}{x}\right)=cx$

43. Differentiating, we get

$$2yy'=2c \quad \Rightarrow \quad c=yy'$$

$$\therefore \quad y^2=2yy'(x+\sqrt{yy'}) \quad \Rightarrow \quad (y^2-2yy'x)^2=4(yy')^3$$

$\Rightarrow$ degree = 3 and order = 1

44. $Ax^2+By^2=1 \qquad \text{(i)}$

Differentiating, we get

$$2xA+2By\frac{dy}{dx}=0 \quad \Rightarrow \quad xA+By\frac{dy}{dx}=0 \qquad \text{(ii)}$$

Again differentiating, we get

$$A+By\frac{d^2y}{dx^2}+B\left(\frac{dy}{dx}\right)^2=0 \qquad \text{(iii)}$$

From (ii) and (iii), $x\left[-By\dfrac{d^2y}{dx^2}-B\left(\dfrac{dy}{dx}\right)^2\right]+By\dfrac{dy}{dx}=0$

$$\Rightarrow \quad xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}x=0$$

$\therefore$ order = 2 and degree = 1

45. Let $x^2+y^2-2gx=0$ (i)

$\Rightarrow 2x+2y\frac{dy}{dx}-2g=0 \Rightarrow 2g=\left(2x+2y\frac{dy}{dx}\right)$

On putting the value of $2g$ in Eq. (i), we get

$$x^2+y^2-\left(2x+2y\frac{dy}{dx}\right)x=0 \Rightarrow y^2=x^2+2xy\frac{dy}{dx}$$

46. The slope of the tangent $=\frac{dy}{dx}$

$\therefore$ The slope of the normal $=-\frac{dx}{dy}$

$\therefore$ The equation of the normal is $Y-y=-\frac{dx}{dy}(X-x)$

This meets the x-axis ($Y=0$), where

$$-y=-\frac{dx}{dy}(X-x) \Rightarrow X=x+y\frac{dy}{dx}$$

$\therefore$ G is $\left(x+y\frac{dy}{dx},0\right)$

$\therefore$ $OG=2x \Rightarrow x+y\frac{dy}{dx}=2x$

$\Rightarrow y\frac{dy}{dx}=x \Rightarrow ydy=xdx$ [variables separable]

Integrating, we get

$$\frac{y^2}{2}=\frac{x^2}{2}+\text{constant}$$

$\Rightarrow x^2-y^2=c$, which is a hyperbola.

47. Let the family of circles be $(x-\alpha)^2+(y-2)^2=5^2$

$\Rightarrow x^2+\alpha^2-2\alpha x+y^2-21-4y=0$ (i)

$\Rightarrow 2x-2\alpha+2y\frac{dy}{dx}-4\frac{dy}{dx}=0 \Rightarrow \alpha=x\frac{dy}{dx}(y-2)$

On putting the value of α in Eq. (i), we get

$$\left[x-x-\frac{dy}{dx}(y-2)\right]^2+(y-2)^2=5^2$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2(y-2)^2=25-(y-2)^2$$

48. Replacing $\frac{dy}{dx}$ by $-dx/dy$, we get

$$\left(\frac{dy}{dx}\right)^2=x/a \Rightarrow dy=\pm(x^{1/2}/a^{1/2})dx$$

Integrating, we get $y+c=\pm\frac{2}{3a^{1/2}}x^{3/2}$

$$\Rightarrow 9a(y+c)^2=4x^3$$

49. Since $\cos x dy=y\sin x dx-y^2dx$

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx}-\frac{1}{y}\tan x=-\sec x$$

Put $-\frac{1}{y}=z \Rightarrow \frac{1}{y^2}\frac{dy}{dx}=\frac{dz}{dx}$

$$\Rightarrow \frac{dz}{dx}+(\tan x)z=-\sec x$$

This is a linear differential equation. Therefore,

$$\text{I.F.}=e^{\int\tan x dx}=e^{\log\sec x}=\sec x$$

Hence, the solution is

$$z\cdot(\sec x)=\int-\sec x\cdot\sec x\,dx+c_1$$

$$\Rightarrow -\frac{1}{y}(\sec x)=\int-\tan x+c_1$$

$$\Rightarrow \sec x=y(\tan x+c)$$

50. $(x-h)^2+(y-k)^2=a^2$ (i)

Here h and k are arbitrary constants.

Differentiating (i) on both sides w.r.t. x, we get

$$2(x-h)+2(y-k)\frac{dy}{dx}=0 \Rightarrow (x-h)+(y-k)\frac{dy}{dx}=0 \quad \text{(ii)}$$

Differentiating (ii) w.r.t. x, we get

$$1+(y-k)\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$$

$$\therefore (y-k)=-\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$$

Putting in (ii), we get

$$x-h=-(y-k)\frac{dy}{dx}=\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]\frac{dy}{dx}}{\frac{d^2y}{dx^2}}$$

Putting in (i), we get

$$\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^2\left(\frac{dy}{dx}\right)^2}{\left(\frac{d^2y}{dx^2}\right)^2}+\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^2}{\left(\frac{d^2y}{dx^2}\right)^2}=a^2$$

$$\Rightarrow \left[1+\left(\frac{dy}{dx}\right)^2\right]^2\left[\left(\frac{dy}{dx}\right)^2+1\right]=a^2\left(\frac{d^2y}{dx^2}\right)^2$$

$$\Rightarrow \left[1+\left(\frac{dy}{dx}\right)^2\right]^3=a^2\left(\frac{d^2y}{dx^2}\right)^2$$

51. The given differential equation can be written as

$$y\frac{dx}{dy} = -2x + 10y^3 \Rightarrow \frac{dx}{dy} + \frac{2x}{y} = 10y^2$$

$$\left[\text{Type } \frac{dx}{dy} + P'x = Q\right]$$

$$\int P'dy = \int \frac{2}{y}dy = 2\log y = \log y^2$$

$\therefore$ $e^{\int P'dy} = y^2$

$\therefore$ The solution is $x \cdot y^2 = \int 10y^2 \cdot y^2 dx + c$

$$= \frac{10y^5}{5} + c = 2y^5 + c$$

52. The normal at $P(x, y)$ is $Y - y = -\frac{dx}{dy}(X - x)$ (i)

The length of the perpendicular from the origin to (i) is

$$\frac{\left|y + x\frac{dx}{dy}\right|}{\sqrt{1 + \left(\frac{dx}{dy}\right)^2}}$$

Also, the distance between P and the x-axis is $|y|$.

$$\therefore \quad \frac{\left|y + x\frac{dx}{dy}\right|}{\sqrt{1 + \left(\frac{dx}{dy}\right)^2}} = |y| \Rightarrow y^2 + x^2\left(\frac{dx}{dy}\right)^2 + 2xy\frac{dx}{dy}$$

$$= y^2\left[1 + \left(\frac{dx}{dy}\right)^2\right] \Rightarrow \left(\frac{dx}{dy}\right)^2 (x^2 - y^2) + 2xy\frac{dx}{dy} = 0$$

$$\Rightarrow \quad \frac{dx}{dy}\left[\left(\frac{dx}{dy}\right)(x^2 - y^2) + 2xy\right] = 0$$

$$\frac{dx}{dy} = 0 \quad \text{or} \quad \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

But $\frac{dx}{dy} = 0 \Rightarrow x = K$, where K is a constant

Since the curve passes through (1, 1),

$\therefore$ we get $x = 1$

Again $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$\therefore \quad v + x\frac{dv}{dx} = \frac{v^2x^2 - x^2}{2x \cdot vx} = \frac{v^2 - 1}{2v}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{-1 - v^2}{2v}$$

$$\Rightarrow \quad \frac{2v}{1 + v^2}dv + \frac{dx}{x} = 0 \Rightarrow \log(1 + v^2) + \log x = \log c$$

$$\Rightarrow \log\left(1 + \frac{y^2}{x^2}\right)x = \log c \Rightarrow (x^2 + y^2) = cx$$

It passes through (1, 1) $\therefore$ $1 + 1 = c$

$\therefore$ $x^2 + y^2 = 2x$

Hence, the required curve is

$$x^2 + y^2 - 2x = 0$$

53. The given differential equation is

$$(2x\cos y + y^2\cos x)dx + (2y\sin x - x^2\sin y)dy = 0$$

Let $M = 2x\cos y + y^2\cos x$

and $N = 2y\sin x - x^2\cos y$

$$\frac{dM}{dy} = -2x\sin y + 2y\cos x$$

$$\frac{dM}{dy} = \frac{dN}{dx}$$

whereas $\frac{dN}{dx} = 2y\cos x - 2x\sin y$

$\Rightarrow$ Exact differential equation

$\therefore$ The solution is $\int Mdx + \int Ndy = c$

i.e. $\int (2x\cos y + y^2\cos x)dx$

$$+ \int (2y\sin x - x^2\sin y)dy = c$$

$$\Rightarrow \quad x^2\cos y + y^2\sin x = c$$

54. $\frac{dy}{dx} + y = 2e^{2x}$

$$\text{I.F.} = e^{\int dx} = e^x$$

$$\therefore \quad y \cdot e^x = \int (2e^{2x})e^x dx + c = 2\frac{e^{3x}}{3} + c$$

$$y = \frac{2}{3}e^{2x} + ce^{-x}$$

55. $\frac{dy}{dx} + \frac{1}{x}y = x^2$

$$\text{I.F.} = e^{\int \frac{1}{x}dx} = e^{\log x} = x$$

$$\therefore \quad yx = \int xx^2 dx + c = \frac{x^4}{4} + c$$

$$y = \frac{x^3}{4} + cx^{-1}$$

56. $(1+y^2)+(x-e^{\tan^{-1}y})\dfrac{dy}{dx}=0$

$$(1+y^2)\frac{dx}{dy}=-(x-e^{\tan^{-1}y}) \Rightarrow \frac{dx}{dy}+\frac{1}{1+y^2}x=\frac{e^{\tan^{-1}y}}{1+y^2}$$

Here $\text{I.F.}=e^{\int\frac{1}{1+y^2}x}=e^{\tan^{-1}y}$

$$\therefore\ x(e^{\tan^{-1}y})=\int e^{\tan^{-1}y}\frac{e^{\tan^{-1}y}}{1+y^2}dy+k=\frac{(e^{\tan^{-1}y})^2}{2}+k$$

$$2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+k$$

57. The given differential equation is

$$x^2+y^2-2ax=0$$

Differentiating w.r.t. x, we get

$$2x+2yy'=2a \quad\Rightarrow\quad x+yy'=a$$

$$\therefore\ x^2+y^2-2(x+yy')x=0 \quad\Rightarrow\quad y^2-x^2=2xyy'$$

58. Rewriting the given equation, we have

$$\frac{dx}{dy}=x^2y^3+xy$$

$\Rightarrow x^{-2}\dfrac{dx}{dy}-x^{-1}y=y^3$ which is reducible to a linear form.

Putting $x^{-1}=u$, we have $\dfrac{du}{dy}+yu=-y^3$

The I.F. of this equation is $e^{y^2/2}$. So the solution is

$$ue^{y^2/2}=-\int y^3e^{y^2/2}dy+c=-(y^2e^{y^2/2}-2e^{y^2/2})+c$$

$$\Rightarrow\quad 1/x=(2-y^2)+ce^{-y^2/2}$$

or $\quad(1-2x)/x=-y^2+ce^{-y^2/2}$

59. $\tan^{-1}x+\tan^{-1}y=c \Rightarrow \dfrac{1}{1+x^2}dx+\dfrac{1}{1+y^2}dy=0$

$$\Rightarrow (1+x^2)dy+(1+y^2)dx=0$$

60. $(x^2+4y^2+4xy)dy=(2x+4y+1)dx$

$$\Rightarrow \frac{dy}{dx}=\frac{2x+4y+1}{(x+2y)^2}=\frac{2(x+2y)+1}{(x+2y)^2}$$

Put $x+2y=V \ \therefore\ 1+2\dfrac{dy}{dx}=\dfrac{dV}{dx}$

$$\therefore\ \frac{1}{2}\left(\frac{dV}{dx}-1\right)=\frac{2V+1}{V^2} \quad\Rightarrow\quad \frac{dV}{dx}-1=\frac{4V+2}{V^2}$$

$$\Rightarrow\quad \frac{dV}{dx}=1+\frac{4V+2}{V^2}=\frac{V^2+4V+2}{V^2}$$

$$\Rightarrow\quad \frac{V^2}{V^2+4V+2}dV=dx$$

$$\Rightarrow\quad \left(1-\frac{2(2V+4)}{V^2+4V+2}+\frac{6}{V^2+4V+2}\right)dV=dx$$

$$\Rightarrow\ V-2\log|V^2+4V+2|+6\int\frac{dV}{(V+2)^2-2}=x+c$$

$$\Rightarrow\ V-2\log|V^2+4V+2|+6\cdot\frac{1}{2\sqrt{2}}\log\left|\frac{V+2-\sqrt{2}}{V+2+\sqrt{2}}\right|=x+c$$

$$\Rightarrow\ V-2\log|V^2+4V+2|+\frac{3}{\sqrt{2}}\log\left|\frac{V+2-\sqrt{2}}{V+2+\sqrt{2}}\right|=x+c$$

Hence, $\quad V=x+2y$

61. $\dfrac{dx}{dy}+\dfrac{x}{y^2}=\dfrac{1}{y^3}$

$$\text{I.F.}=e^{\int\frac{1}{y^2}dy}=e^{-\frac{1}{y}}$$

So, $x\cdot e^{-\frac{1}{y}}=\int\dfrac{1}{y^3}e^{-\frac{1}{y}}dy$

Let $\dfrac{-1}{y}=t \quad\Rightarrow\quad \dfrac{1}{y^2}dy=dt$

$$\Rightarrow\ I=-\int te^t dt=e^t-te^t \quad\Rightarrow\quad e^{-\frac{1}{y}}+\frac{1}{y}e^{-\frac{1}{y}}+c$$

$$\Rightarrow\ xe^{-\frac{1}{y}}=e^{-\frac{1}{y}}+\frac{1}{y}e^{-\frac{1}{y}}+c \quad\Rightarrow\quad x=1+\frac{1}{y}+c\cdot e^{1/y}$$

Since $y(1) = 1$,

$$\therefore\ c=-\frac{1}{e} \quad\Rightarrow\quad x=1+\frac{1}{y}-\frac{1}{e}\cdot e^{1/y}$$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The order and degree of the differential equation $r=\dfrac{[1+(dy/dx)^2]^{3/2}}{d^2y/dx^2}$ are respectively

(a) 2, 2 (b) 2, 3 (c) 2, 1 (d) none of these

2. The order of the differential equation whose general solution is given by $y=(C_1+C_2)\sin(X+C_3)-C_4e^{X+C_5}$ is

(a) 5 (b) 4 (c) 2 (d) 3

3. The degree and order of the differential equation of all the parabolas whose axis is the x-axis are
(a) 2, 1 (b) 1, 2 (c) 3, 2 (d) none of these

4. The order and degree of the differential equation of all the tangent lines to the parabola $x^2 = 4y$ is
(a) 1, 2 (b) 2, 2 (c) 3, 1 (d) 4, 1

5. The solution of the differential equation
$\frac{dy}{dx} = \frac{y}{x} + \frac{\phi(y/x)}{\phi'(y/x)}$ is
(a) $x\phi\left(\frac{y}{x}\right) = k$ (b) $\phi\left(\frac{y}{x}\right) = kx$
(c) $y\phi\left(\frac{y}{x}\right) = k$ (d) $\phi\left(\frac{y}{x}\right) = ky$

6. The solution of the equation $\frac{dy}{dx} = \frac{3x-4y-2}{3x-4y-3}$ is
(a) $(x-y)^2 + c = \log(3x-4y+1)$
(b) $x-y+c = \log(3x-4y+4)$
(c) $x-y+c = \log(3x-4y-3)$
(d) $x-y+c = \log(3x-4y+1)$

7. The differential equation of all the conics whose centre lie at the origin is of the order
(a) 2 (b) 3 (c) 4 (d) none of these

8. If p and q are the order and the degree of the differential equation $y^2\left(\frac{d^2y}{dx^2}\right)^2 + 3x\left(\frac{dy}{dx}\right)^{1/3} + x^2y^2 = \sin x$, then
(a) $p > q$ (b) $\frac{p}{q} = \frac{1}{2}$ (c) $p = q$ (d) $p < q$

9. The degree of the differential equation $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x\log\frac{d^2y}{dx^2}$ is
(a) 1 (b) 2
(c) 3 (d) none of these

10. If the x-intercept of any tangent is 3 times the x-coordinate of the point of tangent, then the equation of the curve, given that it passes through (1, 1), is
(a) $y = \frac{1}{x}$ (b) $y = \frac{1}{x^2}$
(c) $y = \frac{1}{\sqrt{x}}$ (d) none of these

11. Which of the following transformations reduces the differential equation $\frac{dz}{dx} + \frac{z}{x}\log z = \frac{z}{x^2}(\log z)^2$ to the form $P(x)u = Q(x)$?
(a) $u = \log z$ (b) $u = e^z$
(c) $u = (\log z)^{-1}$ (d) $u = (\log z)^2$

12. The solution of the differential equation $\left(\frac{dy}{dx}\right)^2 - x\left(\frac{dy}{dx}\right) + y = 0$ is
(a) $y = 3$ (b) $y = 2x$
(c) $y = 2x - 4$ (d) $y = 2x^2 - 4$

13. The solution of the differential equation $ydx - xdy + xy^2dx = 0$ can be
(a) $2x + x^2y = \lambda y$ (b) $2y + y^2x = \lambda y$
(c) $2y - y^2x = \lambda y$ (d) none of these

14. The curve for which the normal at every point passes through a fixed point (h, k) is
(a) $(x-h)^2 + y = c$
(b) $x + y = c$
(c) $(x-y)^2 + (y-k)^2 = c$
(d) none of these

15. The differential equation of all the straight lines which are at a unit distance from the origin is
(a) $\left(y - x\frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$
(b) $\left(y + x\frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$
(c) $\left(y - x\frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$
(d) $\left(y + x\frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$

16. If $\frac{dy}{dx} = e^{-2y}$ and $y = 0$ when $x = 5$, the value of x for $y = 3$ is
(a) e^5 (b) $e^6 + 1$
(c) $\frac{e^6 + 9}{2}$ (d) $\log_e 6$

17. A particle moves in a straight path such that its velocity is always 4 times its acceleration. If its velocity at time $t = 0$ is 2 m/s, its velocity at $t = 2$ s is
(a) $\sqrt{2e}$ m/s (b) $2\sqrt{e}$ m/s
(c) $\frac{\sqrt{e}}{2}$ m/s (d) $\frac{2}{\sqrt{e}}$ m/s

18. The solution of the differential equation $(1 + x^2)dy + (1 + y^2)dx = 0$ is
(a) $x - y = c(1 + xy)$ (b) $x + y = c(1 - xy)$
(c) $xy = c(x + y)$ (d) $xy = c(x - y)$

19. If $\frac{dy}{dx}=1+x+y+xy$ and $y(-1)=0$, then the function y is

(a) $e^{(1-x)^2/2}$ (b) $e^{(1-x)^2/2}-1$

(c) $\log_e(1+x)-1$ (d) $1+x$

20. The solution of $\frac{dv}{dt}+\frac{k}{m}v=-g$ is

(a) $v=ce^{-\frac{k}{m}t}-\frac{mg}{k}$ (b) $v=c-\frac{mg}{k}e^{-\frac{k}{m}t}$

(c) $ve^{-\frac{k}{m}t}=c-\frac{mg}{k}$ (d) $ve^{\frac{k}{m}t}=c-\frac{mg}{k}$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If $y_2(x)$ is a solution of the differential equation $dy/dx + f(x)y = 0$, then a solution of the differential equation $\frac{dy}{dx}+f(x)y=r(x)$ is

(a) $\frac{1}{y(x)}\int y_1(x)\,dx$

(b) $y_1(x)\int\frac{r(x)}{y_1(x)}dx$

(c) $\int r(x)y_1(x)\,dx$

(d) none of these

2. The value of $\lim_{x\to\infty} y(x)$ obtained from the differential equation $\frac{dy}{dx}=y-y^2$, where $y(0) = 2$, is

(a) 0 (b) 1

(c) ∞ (d) none of these

3. The differential equation whose general solution is $y=c_1x+\frac{c_2}{x}$ for all values of c_1 and c_2 is

(a) $\frac{d^2y}{dx^2}+\frac{x^2}{y}+\frac{dy}{dx}=0$ (b) $\frac{d^2y}{dx^2}+\frac{y}{x^2}-\frac{dy}{dx}=0$

(c) $\frac{d^2y}{dx^2}+\frac{1}{2x}\frac{dy}{dx}=0$ (d) $\frac{d^2y}{dx^2}+\frac{1}{x}\frac{dy}{dx}-\frac{y}{x^2}=0$

4. The family of curves passing through (0, 0) and satisfying the differential equation $\frac{y_2}{y_1}=1$ (where $y_n=d^n y/dx^x$) is

(a) $y=k$ (b) $y=kx$

(c) $y=k(e^x+1)$ (d) $y=k(e^x-1)$

5. The differential equation of all the conics whose axes coincide with the axes of coordinates is of the order

(a) 2 (b) 3

(c) 4 (d) 1

6. If $y_1(x)$ and $y_2(x)$ are the two solutions of $\frac{dy}{dx}+f(x)y = r(x)$, then $y_1(x) + y_2(x)$ is the solution of

(a) $\frac{dy}{dx}+f(y)y=0$ (b) $\frac{dy}{dx}+2f(x)y=r(x)$

(c) $\frac{dy}{dx}+f(x)y=2r(x)$ (d) $\frac{dy}{dx}+2f(x)y=2r(x)$

7. The solution of the equation $\frac{dy}{dx}=\cos(x-y)$ is

(a) $y+\cot\left(\frac{x-y}{2}\right)=c$ (b) $x+\cot\left(\frac{x-y}{2}\right)=c$

(c) $x+\tan\left(\frac{x-y}{2}\right)=c$ (d) none of these

8. The equation of the curve, passing through (2, 5) and having the area of the triangle formed by the x-axis, the ordinate of a point on the curve and the tangent at the point as 5 sq. units is

(a) $xy=10$ (b) $x^2=10y$

(c) $y^2=10x$ (d) $xy^{1/2}=10$

9. The degree of the differential equation satisfying the relation $\sqrt{1+x^2}+\sqrt{1+y^2}=\lambda(x\sqrt{1+y^2}-y\sqrt{1+x^2})$ is

(a) 1 (b) 2

(c) 3 (d) none of these

10. The solution of the differential equation $y_1y_3=3y_2^2$ is

(a) $x=A_1y^2+A_2y+A_3$ (b) $x=A_1y+A_2$

(c) $x=A_1y^2+A_2y$ (d) none of these

11. The equation of the curve which is such that the portion of the x-axis cut off between the origin and the tangent at any point is proportional to the ordinate of that point is

(a) $x=y(b-a\log y)$ (b) $\log x=by^2+a$

(c) $x^2=y(a-b\log y)$ (d) none of these

12. The solution of $\frac{dy}{dx}+\sqrt{\frac{1-y^2}{1-x^2}}=0$ is

(a) $\sin^{-1}x, \sin^{-1}y = c$ (b) $\sin^{-1}x = c\sin^{-1}y$

(c) $\sin^{-1}x - \sin^{-1}y = c$ (d) $\sin^{-1}x + \sin^{-1}y = c$

13. The solution of the differential equation $\frac{dy}{dx} + yf'(x) = f(x)f'(x)$ is

(a) $f(x) + \log[1 + y - f(x)] = c$

(b) $f(x) + \log[f(x) + y] = c$

(c) $\log[1 + y - f(x)] = c$

(d) none of these

14. The equation of the curve satisfying the differential equation $y^2(x^2 + 1) = 2xy_1$ passing through the point (0, 1) and having slope of tangent at $x = 0$ as 3 is

(a) $y = x^2 + 3x + 2$ (b) $y^2 = x^2 + 3x + 1$

(c) $y = x^3 + 3x + 1$ (d) none of these

15. The slope of a curve at any point is the reciprocal of twice the ordinate at that point and it passes through the point (4, 3). The equation of the curve is

(a) $x^2 = y + 5$ (b) $y^2 = x - 5$

(c) $y^2 = x + 5$ (d) $x^2 = y + 5$

16. S1: The differential equation of parabolas having their vertices at the origin and foci on the x-axis is an equation whose variables are separable

S2: The differential equation of straight lines which are at a fixed distance p from the origin is an equation of degree 2

S3: The differential equation of all conics whose axes coincide with the axes of coordinates is an equation of order 2

(a) TTT (b) TFT

(c) FFT (d) TTF

17. The solution of the differential equation $\log\left(\frac{dy}{dx}\right) = 4x - 2y - 2$, $y = 1$, when $x = 1$, is

(a) $2e^{2y+2} = e^{4x} + e^2$ (b) $2e^{2y-2} = e^{4x} + e^4$

(c) $2e^{2y+2} = e^{4x} + e^4$ (d) $3e^{2y+2} = e^{3x} + e^4$

18. The general solution of the equation $\frac{dy}{dx} = \frac{x+y}{x-y}$ is

(a) $\tan^{-1}\left(\frac{y}{x}\right) = \log(x^2 + y^2) + c$

(b) $2\tan\left(\frac{y}{x}\right) = \log(x^2 + y^2) + c$

(c) $2\tan^{-1}\left(\frac{y}{x}\right) = \log(x^2 + y^2) + c$

(d) $\tan^{-1}\left(\frac{y}{x}\right) = 2\log(x^2 + y^2) + c$

19. The degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^{2/3} + 4 - 3\frac{d^2y}{dx} + 5\frac{dy}{dx} = 0$ is

(a) 1 (b) 2

(c) 3 (d) none of these

20. The solution of the differential equation $(x^2 \sin^3 - y^2 \cos x)dx + (x^3 \cos y \sin^2 y - 2y \sin x)dy = 0$ is

(a) $x^3 \sin^3 y = 3y^2 \sin x + c$

(b) $x^3 \sin^3 y + 3y^2 \sin x = c$

(c) $x^2 \sin^3 y + y^3 \sin x = c$

(d) $2x^2 \sin y + y^2 \sin x = c$

ANSWERS

Level 1

1. (a)	**2.** (d)	**3.** (b)	**4.** (a)	**5.** (b)	**6.** (d)	**7.** (b)	**8.** (d)	**9.** (d)	**10.** (c)
11. (c)	**12.** (c)	**13.** (a)	**14.** (c)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (b)	**19.** (b)	**20.** (a)

Level 2

1. (a)	**2.** (b)	**3.** (d)	**4.** (d)	**5.** (a)	**6.** (c)	**7.** (b)	**8.** (a)	**9.** (a)	**10.** (a)
11. (a)	**12.** (d)	**13.** (a)	**14.** (c)	**15.** (c)	**16.** (a)	**17.** (c)	**18.** (c)	**19.** (b)	**20.** (a)

18

RENE DESCARTES

Cartesian System of Rectangular Coordinates and Straight Lines

- ❑ Distance Formula
- ❑ Section Formula
- ❑ Coordinates of Centroid
- ❑ Area of a Triangle
- ❑ Incentre of a Triangle
- ❑ Locus and Equation to a Locus
- ❑ Change of Axes
- ❑ Straight Line
- ❑ First-degree Equation
- ❑ Slope of a Line
- ❑ Slope of a Line Passing through Two Given Points
- ❑ Equation of a Line in Various Forms
- ❑ Properties of a Line Whose Equation is $Ax + By + C = 0$
- ❑ Pair of Straight Lines
- ❑ Equation of a Pair of Lines Joining Origin to the Point of Intersection

☞ CARTESIAN OR RECTANGULAR COORDINATE AXES

In Fig. 18.1, xOx' and yOy' are two mutually perpendicular lines through a point O.

***X*-axis:** The horizontal line xOx' is called x-axis or axis of x.

Note: x-coordinate is called the abscissa.

***Y*-axis:** The vertical line yOy' is called y-axis or axis of y.
Note: Y-coordinate is called the ordinate.

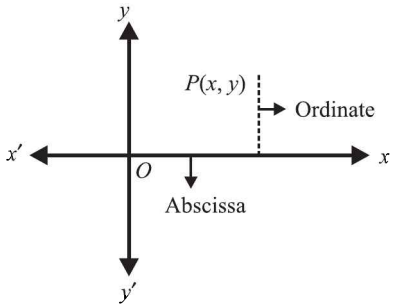


Fig. 18.1

☞ COORDINATE AXES

The two lines xOx' and yOy' together are called coordinate axes or axes. The point O is called the origin.

☞ QUADRANTS

The two axes xOx' and yOy' divide the plane into four regions, called quadrants. The quadrants are named as follows:

xOy' → I Quadrant; yOx' → II Quadrant

$x'Oy'$ → III Q uadrant; $y'Ox$ → IV Quadrant

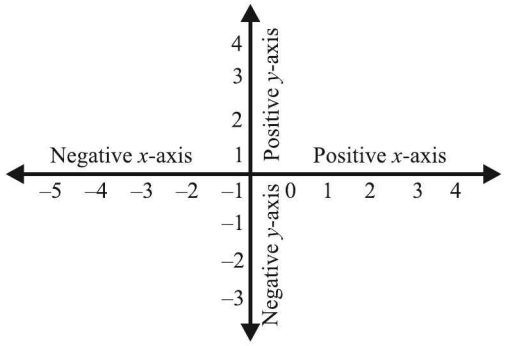


Fig. 18.2

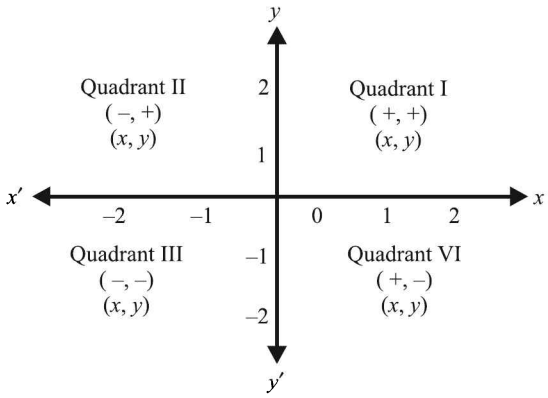


Fig. 18.3

ORIGIN

The point of intersection of coordinate axes is known as origin $O(0, 0)$.

EQUATION

(i) Equation of the x-axis is $y = 0$.

(ii) Equation of the y-axis is $x = 0$.

GRAPH

(i) The line $x = a$ is always parallel to the y-axis.

(ii) The line $y = a$ is always parallel to the x-axis.

DISTANCE FORMULA

Distance between two points in the plane is the length of the line segment joining them. The distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$\therefore$ Distance between two points

$$= \sqrt{\left(\begin{matrix}\text{Difference of}\\ \text{abscissae}\end{matrix}\right)^2 + \left(\begin{matrix}\text{Difference of}\\ \text{ordinates}\end{matrix}\right)^2}$$

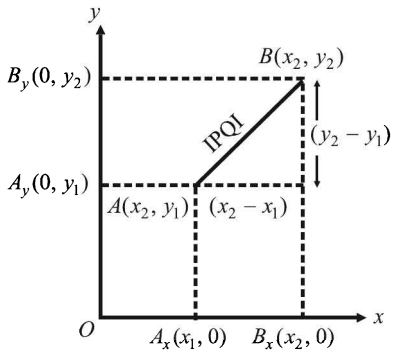
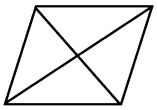


Fig. 18.4

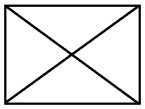
Remember:

Parallelogram



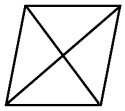
(a) Opposite sides are equal
(b) Diagonals are unequal
(c) Diagonals bisect each other

Rectangle



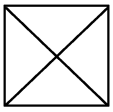
(a) Opposite sides are equal
(b) Diagonals are equal
(c) Diagonals bisect each other

Rhombus



(a) All four sides are equal
(b) Diagonals are unequal
(c) Diagonals bisect each other at right angles

Square



(a) All four sides are equal
(b) Diagonals are equal
(c) Diagonals bisect each other at right angles

Note

- Every parallelogram is not a rectangle, but every rectangle is a parallelogram.
- Every rhombus is not a square, but every square is a rhombus.
- Every parallelogram is not a rhombus, but every rhombus is a parallelogram.
- Every rectangle is not a square, but every square is a rectangle.

SECTION FORMULA

If $P(x, y)$ divides the line joining $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m : n$, then

$$x = \frac{mx_2 + nx_1}{m+n};\ y = \frac{my_2 + ny_1}{m+n}$$

A. Internal Division

The coordinates of the point which divides the line segment joining the points (x_1, y_1) and (x_2, y_2) internally in the ratio $m : n$ are given by

$$(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

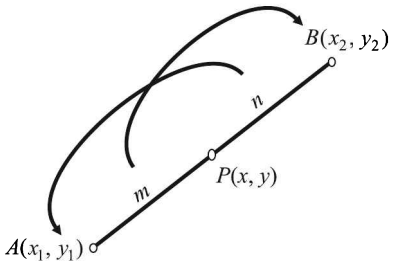


Fig. 18.5

B. External Division

The coordinates of the point which divides the line segment joining the points (x_1, y_1) and (x_2, y_2) externally in the ratio $m : n$ (i.e. internally in the ratio $m : -n$) are given by

$$(x, y) = \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$$

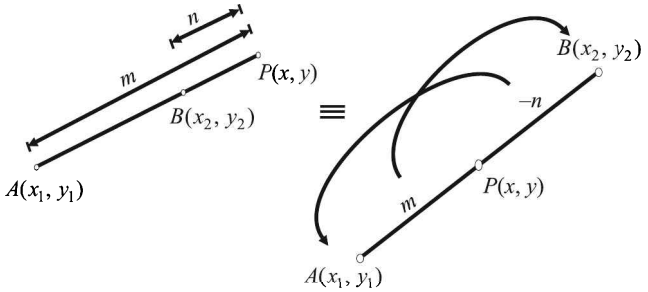


Fig. 18.6

C. Mid-point Formula

If $P(x, y)$ is the mid-point of AB, then $m : n :: 1 : 1$, and the coordinates of point P are given by

$$P(x, y) \equiv P(x, y) \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

COORDINATES OF CENTROID

Centroid is the point at which all the medians in a triangle intersect.

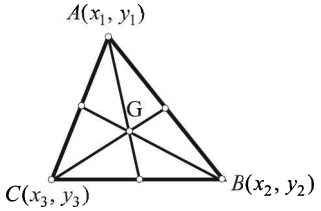


Fig. 18.7

The coordinates of the centroid of a triangle whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are

$$G\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

Note

- If m/n is positive, the division is internal, but if m/n is negative, the division is external.
- If P divides AB internally in the ratio $m : n$ and Q divides AB externally in the ratio $m : n$, then P and Q are said to be harmonic conjugate of each other w.r.t. AB.

  Mathematically, $\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$, i.e. AP, AB and AQ are in H.P.

AREA OF A TRIANGLE

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the coordinates of vertices of the triangle ABC taken in anticlockwise order, then the area of the triangle can be represented as follows:

A. Determinant Method

$$\Delta = ar(\Delta) = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\therefore\quad \Delta = \frac{1}{2}\left[x_1(y_2 - y_3) - y_1(x_2 - x_3) + 1(x_2y_3 - x_2y_3)\right]$$

B. Stair Method

$$\Delta = ar(\Delta) = \frac{1}{2}\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}$$

$$\therefore\quad \Delta = \frac{1}{2}\left[(x_1y_2 + x_2y_3 + x_3y_1) - (y_1x_2 + y_2x_3 + y_3x_1)\right]$$

Note: The area of a triangle can be positive or negative according as the vertices are taken in anticlockwise or clockwise order, respectively.

CONDITIONS OF COLLINEARITY

Three points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are collinear if and only if

(i) $\Delta ABC = 0 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(ii) $m_{AB} = m_{BC} = m_{CA}$, i.e. $\left(\frac{y_1 - y_2}{x_1 - x_2}\right) = \left(\frac{y_2 - y_3}{x_2 - x_3}\right)$

(iii) $AC = AB + BC$ or $AB \sim BC$

(iv) A divides the line segment BC in some ratio.

AREA OF A TRAPEZIUM

If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ and $D(x_4, y_4)$ are the vertices of a trapezium, then the area of the trapezium is given by

$$\frac{1}{2}[(y_2 + y_1)(x_1 - x_2) + (y_1 + y_3)(x_3 - x_1) - (y_2 + y_3)(x_3 - x_2)]$$

$$= \frac{1}{2} (\text{sum of parallel sides}) \times (\text{distance between parallel sides})$$

AREA OF A QUADRILATERAL

If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ and $D(x_4, y_4)$ are the vertices of a quadrilateral $ABCD$, then the area of the quadrilateral is given by

$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \\ x_1 & y_1 \end{vmatrix} = \frac{1}{2}|\{(x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1) - (y_1x_2 + y_2x_3 + y_3x_4 + y_4x_1)\}|$$

INCENTRE OF A TRIANGLE

The point at which all the angle bisectors meet is called **incentre** of a triangle. It is the point which is equidistant from all the sides of a triangle.

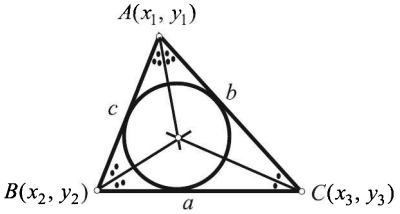


Fig. 18.8

The coordinates of the incentre of a triangle whose vertices are (x_1, y_1), (x_2, y_2) and (x_3, y_3) are given as

$$(x, y) \equiv \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}\right)$$

CIRCUMCENTRE OF A TRIANGLE

The circumcentre of a triangle is the point of intersection of the perpendicular bisectors of the sides of a triangle. Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of ΔABC, then the coordinates of the circumcentre of the triangle is given by

$$\left(\frac{ax_1\cos A + bx_2\cos B + cx_3\cos C}{a\cos A + b\cos B + c\cos C}, \frac{ay_1\cos A + by_2\cos B + cy_3\cos C}{a\cos A + b\cos B + c\cos C}\right)$$

where a, b and c are the sides of the triangle.

ORTHOCENTRE OF A TRIANGLE

The point at which the perpendiculars of the triangle meet is called **orthocentre**.

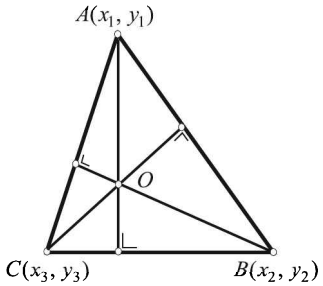


Fig. 18.9

The coordinates of the orthocentre of the triangle whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are given as

$$O\left(\frac{x_1\tan A + x_2\tan B + x_3\tan C}{\tan A + \tan B + \tan C}, \frac{y_1\tan A + y_2\tan B + y_3\tan C}{\tan A + \tan B + \tan C}\right)$$

LOCUS AND EQUATION TO A LOCUS

Geometrical locus: The curve described by a point which moves under a given condition or conditions is called its locus.

Equation to locus of a point: The equation to the locus of a point is the relation which is satisfied by the coordinates of every point on the locus of the point.

HOW TO FIND LOCUS

Step I: Assume the coordinates of the point say (h, k) whose locus is to be found.

Step II: Write the given condition in mathematical form in solving h, k.

Step III: Eliminate the variable(s) if any.

Step IV: Replace h by x and k by y in the result obtained in Step III.

The equation so obtained is the locus of the point which moves under some stated condition(s).

CHANGE OF AXES

A. Translation of Axes

If we shift the origin without rotating the axes, we call it 'translation of axes'.

If we shift the origin to (h, k), then

$$x = X + h \quad \text{and} \quad y = Y + k$$

or equivalently $X = x - h$ and $Y = y - k$

where (X, Y) and (x, y) are current (new) and original coordinates respectively.

B. Rotation of Axes

Let O be the origin. Let $P \equiv (x, y)$ with respect to axes Ox and Oy and let $P \equiv (x', y')$ with respect to axes Ox' and Oy', where $\angle x'Ox = \angle yOy' = \theta$.

In Fig. 18.10,

$$SR = x\tan\theta,\ OS = x\sec\theta,\ PS = y - x\tan\theta$$

Now, in ΔPQS,

$$\sin\theta = \frac{SQ}{PS} = \frac{x' - x\sec\theta}{y - x\tan\theta}$$

$$\Rightarrow\ x' = y\sin\theta - x\frac{\sin^2\theta}{\cos\theta} + \frac{x}{\cos\theta} = y\sin\theta + x\frac{1-\sin^2\theta}{\cos\theta}$$

$$\Rightarrow\ x' = x\cos\theta + y\sin\theta$$

Also $\cos\theta = \dfrac{PQ}{PS} = \dfrac{y'}{y - x\tan\theta}$

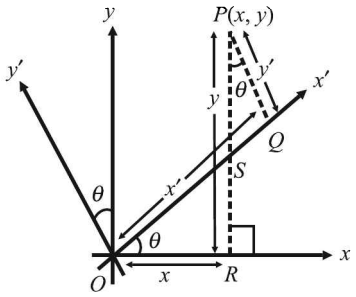


Fig. 18.10

$$\Rightarrow\ y' = -x\sin\theta + y\cos\theta$$

$$\Rightarrow\ x = x'\cos\theta - y'\sin\theta,\ y = x'\sin\theta + y'\cos\theta$$

	x	y
x'	$\cos\theta$	$\sin\theta$
y'	$-\sin\theta$	$\cos\theta$

Note The above relations can also be obtained from the following table:

	$x\downarrow$	$y\downarrow$
$x\rightarrow$	$\cos\theta$	$\sin\theta$
$y\rightarrow$	$\sin\theta$	$\cos\theta$

C. Double Transformation (Origin Shifted and Axes Rotated)

If the origin is changed to $O'(\alpha, \beta)$ and axes are rotated about the new origin O' by an angle θ in the anticlockwise sense such that the new coordinates of $P(x, y)$ become (x', y'), then the equations of transformation will be

$$x = \alpha + x'\cos\theta - y'\sin\theta$$

and $y = \beta + x'\cos\theta + y'\sin\theta$

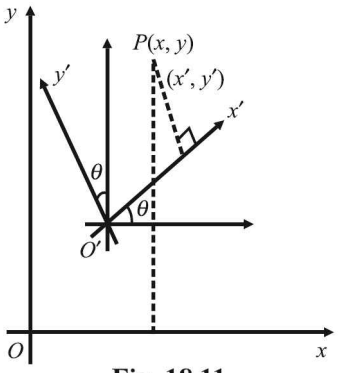


Fig. 18.11

STRAIGHT LINE

A straight line is a curve such that every point on the line segment joining any two points on it lies on it.

FIRST-DEGREE EQUATION

Every first-degree equation like $ax + by + c = 0$ would be the equation of a straight line.

SLOPE OF A LINE

The slope (or gradient) of a line is defined as the tangent of the angle θ which a line makes with the +ve x-axis. It is denoted by m. Thus,

$$m = \tan\theta$$

Examples:

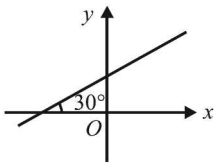


Fig. 18.12

$m = \tan 30° = 1/\sqrt{3}$

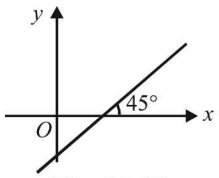


Fig. 18.13

$m = \tan 45° = 1$

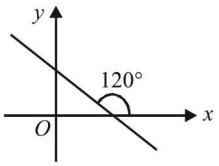


Fig. 18.14

$m = \tan 120° = -\sqrt{3}$

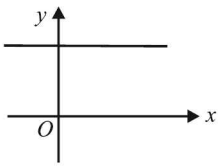


Fig. 18.15

$m = \tan 0° = 0$

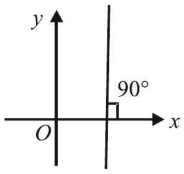


Fig. 18.16

$m = \tan 90° = \infty$

Note

- m can be defined as $\tan\theta$ for $0 \le \theta \le \pi$ and $\theta \ne \pi/2$.
- The slope of a line parallel to the x-axis = 0 and perpendicular to the x-axis is undefined.

☞ SLOPE OF A LINE PASSING THROUGH TWO GIVEN POINTS

The slope of a line joining two points (x_1, y_1) and (x_2, y_2) is equal to $\dfrac{y_2 - y_1}{x_2 - x_1}$.

The angle θ between two lines is given by $\tan\theta = \dfrac{m_1 - m_2}{1 + m_1 m_2}$.

Two lines are parallel '||' if $m_1 = m_2$.

Two lines are perpendicular '⊥' if $m_1 \cdot m_2 = -1$.

☞ EQUATION OF A LINE IN VARIOUS FORMS

1. Equation of the line L of slope m and cutting off an intercept b on y-axis:
   Any point $P(x, y)$ on the line L satisfies the following:

$$\tan\theta = m = \frac{y - b}{x}$$

   $\Rightarrow \quad y = mx + b$ is the equation of the line L.
   This is also known as **slope intercept form**.

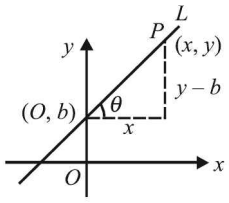


Fig. 18.17

2. Equation of the line L of slope m and passing through a given point (x_1, y_1):

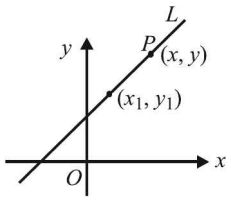


Fig. 18.18

Any point $P(x, y)$ on the line L satisfies the following:

$$\text{slope} = m = \frac{y - y_1}{x - x_1}$$

$\Rightarrow \quad y - y_1 = m(x - x_1)$ is the equation of the line L.
This is also known as **point slope form**.

3. Equation of the line L passing through two given points (x_1, y_1) and (x_2, y_2):

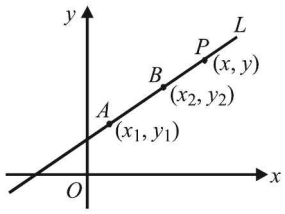


Fig. 18.19

Any point $P(x, y)$ on the line L satisfies the following:

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

The equation of the line in the **determinant form** is given as

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_1 & y_1 & 1 \end{vmatrix} = 0$$ which is the equation of the line.

This is also known as **two point form**.

4. Equation of the line L cutting intercepts a and b on x and y axes:

$$\frac{x}{a} + \frac{y}{b} = 1$$ is the equation of the line.

This is also known as **intercept form**.

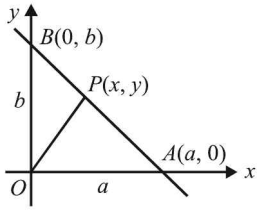


Fig. 18.20

5. Equation of the line L in terms of p, the length of perpendicular from the origin upon it and α the angle which p makes with the +ve x-axis:
   $x \cos\alpha + y \sin\alpha = p$ is the equation of the line L.

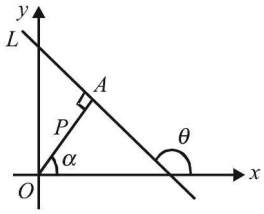


Fig. 18.21

This is also known as **normal form of the equation of the line**.

☞ PROPERTIES OF A LINE WHOSE EQUATION IS $Ax + By + C = 0$

1. Slope: Slope $= m = -\frac{A}{B} = -\frac{\text{coefficient of } x}{\text{coefficient of } y}$
2. Intercepts
   (i) y-intercept: $b =$
   $$-\frac{C}{B} = -\frac{\text{coefficient of constant term}}{\text{coefficient of } y}$$
   (ii) x-intercept: $a =$
   $$-\frac{C}{A} = -\frac{\text{coefficient of constant term}}{\text{coefficient of } x}$$
3. (i) Distance (d) of a point $P(x_1, y_1)$ (not lying on the line) from the line (L) $Ax + By + C = 0$:
   $$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

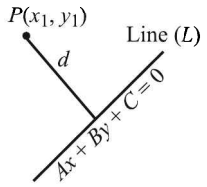


Fig. 18.22

   (ii) Distance of line (L) $Ax + By + C = 0$ from the origin (0, 0):
   $$d = \frac{|C|}{\sqrt{A^2 + B^2}}$$

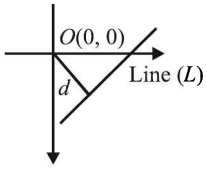


Fig. 18.23

   (iii) Distance between two parallel lines (L) $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$:
   $$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

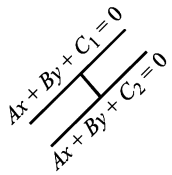


Fig. 18.24

4. The equation of any line parallel to $Ax + By + C = 0$ can be taken as $Ax + By + k = 0$, where k is any number.
5. The equation of any line perpendicular to $Ax + By + C = 0$ can be taken as $Bx - Ay + k = 0$, where k is any number.
6. Concurrency of three lines: The lines $A_1x + B_1y + C_1 = 0$, $A_2x + B_2y + C_2 = 0$ and $A_3x + B_3y + C_3 = 0$ pass through a common point.
   (i) First method: Solve any two equations and find out the point of intersection. If this point also satisfies the third equation, then the three lines are concurrect.
   (ii) Second method: $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = 0$

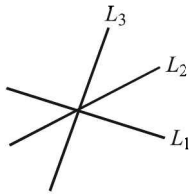


Fig. 18.25

☞ PAIR OF STRAIGHT LINES

Pair of Lines through the Origin

1. *General equation*

$ax^2 + 2hxy + by^2 = 0$ is a homogeneous equation of second degree that represents two straight lines passing through the origin. The individual equations of these straight lines can be obtained by factorizing the expression $ax^2 + 2hxy + by^2$ into two real linear factors. The equation represents two lines in the real xy-plane only if $h^2 - ab \geq 0$.

2. *Slopes of the lines* $ax^2 + 2hxy + by^2 = 0$

Let m_1 and m_2 be the slopes of the lines through the origin. They are expressed by

$$\left[m_1 = \frac{-h + \sqrt{h^2 - ab}}{b} \quad \text{and} \quad m_2 = \frac{-h - \sqrt{h^2 - ab}}{b} \right]$$

Note $h^2 - ab \geq 0$ is the necessary condition for the equation representing real lines.

3. *Angle between the pair of lines* $ax^2 + 2hxy + by^2 = 0$

If the acute angle between the lines $ax^2 + 2hxy + by^2 = 0$ is θ, then

$$\tan\theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2} \right| = \frac{\sqrt{\frac{4h^2}{b^2} - \frac{4a}{b}}}{|1 + (a/b)|}$$

$$\Rightarrow \quad \theta = \tan^{-1} \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right|$$

If the lines are perpendicular, then $\theta = 90°$

$\therefore \quad a + b = 0$

If the lines are parallel, then $\theta = 0°$ or $180°$ and $h^2 = ab$.

If $h^2 - ab \geq 0$, then the lines are real distinct.

If $h^2 - ab = 0$, then the lines are real and distinct.

If $h^2 - ab < 0$, then the lines are imaginary.

Note

- If $ax^2 + 2hxy + by^2 = 0$ represents two lines, then $bx^2 - 2hxy + ay^2 = 0$ represents two lines perpendicular to them.
- If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two parallel lines, then $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$.

4. *Angle bisector of lines $ax^2 + 2hxy + by^2 = 0$*

Let m_1 and m_2 be the slopes of the lines $ax^2 + 2hxy + by^2 = 0$.

$\Rightarrow$ lines are $y - m_1x = 0$ and $y - m_2x = 0$

and $\left[m_1 + m_2 = \frac{-2h}{b} \text{ and } m_1m_2 = \frac{a}{b}\right]$

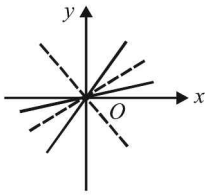


Fig. 18.26

Bisectors of angles are given by

$$\frac{y - m_1x}{\sqrt{1 + m_1^2}} = \pm\left(\frac{y - m_2x}{\sqrt{1 + m_2^2}}\right)$$

The equation of bisectors is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$.

5. *Point of intersection of lines*

The point of intersection of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $\left(\frac{bg - hf}{h^2 - ab}, \frac{af - gh}{h^2 - ab}\right)$.

6. *Distance between parallel lines represented by general equation*

If the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents the general equation of a pair of lines which are parallel to each other, then the distance between them is given by

$$d = \left|2\sqrt{\frac{g^2 - ac}{a(a + b)}}\right| \text{ or } \left|2\sqrt{\frac{f^2 - bc}{b(a + b)}}\right|$$

7. *Pair of lines perpendicular to the lines*

Let L_1 and L_2 be the lines $ax^2 + 2hxy + by^2 = 0$. Let P_1 be the line perpendicular to L_1 and P_2 be the line perpendicular to L_2. We have to find the equation of P_1P_2. $bx^2 - 2hxy + ay^2 = 0$ is the equation of the pair of lines perpendicular to the pair of lines $ax^2 + 2hxy + by^2 = 0$.

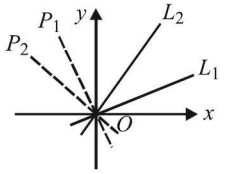


Fig. 18.27

Note By interchanging the coefficients of x^2 and y^2 and reversing the sign of the xy term, we can get the equation of P_1P_2 from L_1L_2.

EQUATION OF A PAIR OF LINES JOINING ORIGIN TO THE POINT OF INTERSECTION

The equation of a pair of lines joining the origin to the point of intersection of $lx + my + n = 0$ and the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is given by

$$ax^2 + 2hxy + by^2 - (2gx + 2fy)\left(\frac{lx + my}{n}\right) + c\left(\frac{lx + my}{n}\right)^2 = 0.$$

SOLVED PROBLEMS

1. If the distance between the points $(a, 2)$ and $(3, 4)$ is 8, then the value of a is

(a) $\left(2 + 3\sqrt{15}\right)$ (b) $\left(2 - 3\sqrt{15}\right)$

(c) $\left(2 \pm 3\sqrt{15}\right)$ (d) $\left(3 \pm 2\sqrt{15}\right)$

Ans. (d)

Solution: Given: Coordinates of point $P(x_1, y_1) = (a, 2)$; coordinates of point $Q(x_2, y_2) = (3, 4)$ and the distance between the points P and $Q = 8$.

We know that distance between the points P and Q

$$(PQ) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

or $8 = \sqrt{(a - 3)^2 + (2 - 4)^2}$ or $64 = (a - 3)^2 + 4$

or $(a - 3)^2 = 60$ or $a - 3 = \pm 2\sqrt{15}$

or $a = \left(3 \pm 2\sqrt{15}\right)$

2. If the sum of the distance of a point from two perpendicular lines in a plane is 1, then its locus is
(a) a straight line (b) a circle
(c) a square (d) an ellipse

Ans. (a)

Solution: $x + y = 1$ (where x and y are distance of the point from the two perpendicular axes)

Since the given equation represents a straight line, therefore the locus of the point is also a straight line.

3. The triangle joining the points $P(2, 7)$, $Q(4 - 1)$ and $R(-2, 6)$ is
(a) an equilateral triangle (b) a right-angled triangle
(c) an isosceles triangle (d) a scalene triangle

Ans. (b)

Solution: Given: Coordinates of point $P(x_1, y_1) = (2, 7)$, coordinates of point $Q(x_2, y_2) = (4, -1)$ and coordinates of point $R(x_3, y_3) = (-2, 6)$.

We know that the distance between points P and Q

$$(PQ) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(4-2)^2 + (-1-7)^2} = \sqrt{4+64} = \sqrt{68}$$

Similarly, the distance between Q and R

$$(QR) = \sqrt{(-2-4)^2 + (6+1)^2} = \sqrt{36+49} = \sqrt{85}$$

And the distance between points P and R

$$(PR) = \sqrt{(-2-2)^2 + (6-7)^2}$$

$$= \sqrt{16+1} = \sqrt{17}$$

Since $(QR)^2 = (PQ)^2 + (PR)^2$, therefore the given triangle is a right-angled triangle.

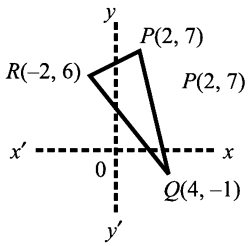


Fig. 18.28

4. The coordinates of the points on the line joining the points $P(3, -4)$ and $Q(-2, 5)$ that is twice as far from P as from Q are
(a) (1, 7) (b) (–1, 7)
(c) (–7, 14) (d) (7, 14)

Ans. (c)

Solution: Given: Coordinates of the first point $P(x_1, y_1) = (3, -4)$, and coordinates of the second point $Q(x_2, y_2) = (-2, 5)$. Let $R(x, y)$ be the point which joins the given points and $PR = 2PQ$ or $PR : RQ = 2 : 1$.

We know that R divides PQ internally or externally in the ratio 2 : 1.

We also know that the coordinates of point R are

$$(x, y) = \left[\frac{(2\times x_2) \pm (1\times x_1)}{2\pm 1}, \frac{(2\times 2) \pm (1\times y_1)}{2\pm 1}\right]$$

$$= \left[\frac{2\times(-2)\pm 1\times(3)}{2+1}, \frac{2\times(5)\pm 1\times(-4)}{2+1}\right]$$

Taking negative sign, we get

$$(x, y) = \left[\frac{2\times(-2)-1\times 3}{(2-1)}, \frac{2\times(5)-1\times(-4)}{(2-1)}\right]$$

$$= (-7, 14)$$

Therefore, coordinates of $R = (-7, 14)$.

5. The slope of the line joining the points on the curve $y = x^2 + 2x$, whose abscissas are 1 and 3, is
(a) 6 (b) 5
(c) 4 (d) 3

Ans. (a)

Solution: Given: Equation of the curve $y = x^2 + 2x$ and abscissas $(x) = 1$ and 3

Substituting $x = 1$ in the given equation of the curve, we get

$$y = (1)^2 + (2\times 1) = 3$$

Thus, the coordinates of the first point $(x_1, y_1) = (1, 3)$. Again substituting $x = 3$ in the given equation of the curve, we get

$$y = (3)^2 + (2\times 3) = 15$$

Thus, the coordinates of the second point $(x_2, y_2) = (3, 15)$. We know that the slope of the line joining the points $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{15-3}{3-1} = 6.$

6. The angle between the lines $y = \left(2-\sqrt{3}\right)x + 6$ and $y = \left(2+\sqrt{3}\right)x - 9$ is

(a) $\tan^{-1}\left(\frac{\pi}{2}\right)$ (b) $\tan^{-1}\left(\frac{\pi}{6}\right)$

(c) $\tan^{-1}\left(\pm\sqrt{3}\right)$ (d) $\tan^{-1}\left(\pm\sqrt{2}\right)$

Ans. (c)

Solution: Given: Equation of lines $y = \left(2-\sqrt{3}\right)x + 6$ and $y = \left(2+\sqrt{3}\right)x + 9$.

Let θ be the angle between the given lines.

We know that the equation of the line is $y = x_1x + c_1$ or $m_1 = (2-\sqrt{3})$ and the line is $y = m_2x + c_2$ or $m_2 = (2+\sqrt{3})$.
We also know that

$$\tan\theta = \pm\left(\frac{m_2 - m_1}{1 + m_1m_2}\right) = \pm\frac{\left(2+\sqrt{3}\right)-\left(2-\sqrt{3}\right)}{1+\left(2-\sqrt{3}\right)\left(2+\sqrt{3}\right)}$$

$$= \pm\frac{2\sqrt{3}}{1+4-3} = \pm\frac{2\sqrt{3}}{2} = \pm\sqrt{3}$$

or $\theta = \tan^{-1}\left(\pm\sqrt{3}\right)$

7. The point $P(a, b)$ lies on the line $3x + 2y = 13$ and the point $Q(b, a)$ lies on the line $4x - y = 5$. The equation of the line PQ is

(a) $x - y = 5$ (b) $x + y = 5$

(c) $x + y = -5$ (d) $x - y = -5$

Ans. (b)

Solution: Given: Point $P(a, b)$ lies on the line $3x + 2y = 13$ and point $Q(b, a)$ lies on the line $4x - y = 5$.
We know that as the point P lies on the line $3x + 2y = 13$, therefore

$$3a + 2b = 13 \qquad \text{(i)}$$

Similarly, as the point Q lies on the line $4x - y = 5$, therefore

$$4b - a = 5 \qquad \text{(ii)}$$

Multiplying Eq. (ii) by 3 and adding the result to Eq. (i), we get

$$14b = 28 \quad \text{or} \quad b = 2 \quad \text{and} \quad a = 3$$

Therefore, the coordinates of point $P(x_1, y_1) = (3, 2)$ and $Q(x_2, y_2) = (2, 3)$.
We also know that the equation of the lines is

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) \quad \text{or} \quad (y - 2) = \frac{3-2}{2-3}(x - 3)$$

or $y - 2 = -x + 3$ or $x + y = 5$

8. The equations of the straight lines passing through the point (4, 3) and making intercepts on the coordinate axes, whose sum is –1, are

(a) $\frac{x}{2} + \frac{y}{3} = 1$ and $\frac{x}{2} + \frac{y}{1} = 1$

(b) $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

(c) $\frac{x}{2} + \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

(d) $\frac{x}{2} - \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

Ans. (b)

Solution: Given coordinates of point $(x_1, y_1) = (4, 3)$ and sum of intercepts = –1.
Let a and b be the intercepts on the x-axis and y-axis respectively.
Therefore, $a + b = -1$ or $b = -1 - a$
We know that the equation of the straight line making intercepts a and b on the coordinate axis is

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{or} \quad \frac{x}{a} + \frac{y}{-(1+a)} = 1 \qquad \text{(i)}$$

Since the straight line (i) passes through the point (x_1, y_1), therefore $\frac{4}{a} + \frac{3}{-(1+a)} = 1$.

or $4 + 4a - 3a = a + a^2$ or $a^2 = 4$ or $a = \pm 2$.

Substituting the value of a in Eq. (i), we get equations of lines $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$.

9. If P and P' are the perpendiculars from the origin, upon the straight lines $x \sec\theta + y \operatorname{cosec}\theta = a$ and $x\cos\theta - y\sin\theta = a\cos 2\theta$, then the value of $4P^2 + P'^2$ is

(a) $-12a^2$ (b) $-5a^2$

(c) a^2 (d) $5a^2$

Ans. (c)

Solution: Given: Equations of lines $x\sec\theta + y \operatorname{cosec}\theta = a$

or $x\sec\theta + y\operatorname{cosec}\theta - a = 0 \qquad \text{(i)}$

and $x\cos\theta - y\sin\theta = a\cos 2\theta$

or $x\cos\theta - y\sin\theta - a\cos 2\theta = 0 \qquad \text{(ii)}$

We know that the coordinates of the origin $(x_1, y_1) = (0, 0)$.
We also know that the standard equation of the line is $ax + by + c = 0$.
Comparing Eqs. (i) and (ii) with the standard equation, we get

$$a_1 = \sec\theta, b_1 = \operatorname{cosec}\theta, c_1 = -a, a_2 = \cos\theta,$$

$$b_2 = -\sin\theta \text{ and } c_2 = -a\cos 2\theta$$

Therefore, the length of the perpendicular from the origin to the line (i) is given by

$$(P) = \frac{|ax_1 + by_1 + c_1|}{\sqrt{a^2 + b^2}} = \frac{|\sec\theta \times (0) + \operatorname{cosec}\theta \times (0) - a|}{\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta}}$$

$$= \frac{a\cos\theta\sin\theta}{\sqrt{\sin^2\theta + \cos^2\theta}} = a\sin\theta\cos\theta$$

Similarly, the length of the perpendicular from the origin to the line (ii) is given by

$$(P') = \frac{|a_2x_1 + b_2y_1 + c_2|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|\cos\theta \times (0) - \sin\theta \times (0) - a\cos 2\theta|}{\sqrt{\cos^2\theta + \sin^2\theta}} = a\cos 2\theta$$

Therefore, $4P^2 + P'^2 = 4a^2 \sin^2\theta\cos^2\theta + a^2\cos^2 2\theta$

$= a^2(2\sin\theta\cos\theta)^2 + a^2\cos^2 2\theta$

$= a^2(\sin^2 2\theta + \cos^2 2\theta) = a^2$

10. What are the coordinates of a point on $x + y + 9 = 0$, whose distance from the line $x + 2y + 2 = 0$ is $\sqrt{5}$?

(a) $(-1,1)$ (b) $(-9,6)$

(c) $(5,6)$ (d) $(9,6)$

Ans. (b)

Solution: Given: Equations of lines

$$x + y + 9 = 0 \quad \text{(i)}$$

and $$x + 2y + 2 = 0 \quad \text{(ii)}$$

Distance between the given lines $(p) = \sqrt{5}$

Since the point $A(x_1, y_1)$ lies on the line (i), therefore

$$x_1 + y_1 + 9 = 0 \quad \text{(iii)}$$

We know that the standard equation of the line is

$$ax + by + c = 0$$

Comparing Eq. (ii) with the standard equation, we get $a_1 - 1$, $b_1 = 2$ and $c_1 = 2$.

We know that the perpendicular distance of the point (x_1, y_1) from the line (ii)

$$(p) = \frac{|x_1 + 2y_1 + 2|}{\sqrt{1^2 + 2^2}} \quad \text{or} \quad \sqrt{5} = \frac{x_1 + 2y_1 + 2}{\sqrt{5}}$$

or $$x_1 + 2y_1 + 2 = 5 \quad \text{or} \quad x_1 + 2y_1 = 3 \quad \text{(iv)}$$

Subtracting Eq. (iii) from Eq. (iv), we get

$$y_1 = 6 \text{ and } x_1 = -9$$

Therefore, coordinates of the point $(x_1, y_1) = (-9,6)$.

11. If four points are $A(6, 3)$, $B(-3, 5)$, $C(4, -2)$ and $P(x, y)$, then the ratio of the areas of ΔPBC and ΔABC is

(a) $\frac{x + y - 2}{7}$ (b) $\frac{x - y - 2}{7}$

(c) $\frac{x - y + 2}{2}$ (d) $\frac{x + y + 2}{2}$

Ans. (a)

Solution: Given: Coordinates of points $A(x_1, y_1) = (6, 3)$, $B(x_2, y_2) = (-3, 5)$, $C(x_3, y_3) = (4, -2)$ and $P(x, y)$.

We know that the area of

$$\Delta PBC = \frac{1}{2}[x(y_2 - y_3) + x_3(y - y_2) + x_2(y_3 - y)]$$

$$= \frac{1}{2}[x(5 + 2) + 4(y - 5) - 3(-2 + y)]$$

$$= \frac{1}{2}[7x + 7y - 14]$$

Similarly, the area of

$$\Delta ABC = \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1)] + x_3(y_1 - y_2)$$

$$= 6(5 + 2) - 3(-2 - 3) + 4(3 - 5) = 49$$

Therefore, the ratio of the areas of ΔPBC and ΔABC

$$= \frac{7x + 7y - 14}{49} = \frac{7(x + y - 2)}{49} = \frac{x + y - 2}{7}$$

12. If the two vertices of a triangle are (3, 5) and (–7, 4) and its centroid is (2, –1), then the coordinates of the third vertex are

(a) (1, –2) (b) (10, –2)

(c) (4, 5) (d) (6, 7)

Ans. (b)

Solution: Given: Coordinates of the first vertex of a triangle $(x_1, y_1) = (3, -5)$, coordinates of the second vertex $(x_2, y_2) = (-7, 4)$ and coordinates of its centroid $(x, y) = (2, -1)$.

Let (x_3, y_3) be the coordinates of the third vertex of the triangle.

We know that the coordinates of the centroid of the triangle are given by

$$(x, y) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

or $$(2, -1) = \left(\frac{3 - 7 + x_3}{3}, \frac{-5 + 4 + y_3}{3}\right) = \left(\frac{x_3 - 4}{3}, \frac{y_3 - 1}{3}\right)$$

or $2 = \frac{x_3 - 4}{3}$ and $\frac{y_3 - 1}{3} = -1$ or $x_3 = 10$ and $y_3 = -2$

Therefore, the coordinates of the third vertex are (10, –2).

13. The orthocentre of the triangle whose sides are given by $4x - 7y + 10 = 0$, $x + y - 5 = 0$ and $7x + 4y - 15 = 0$ is

(a) (–1, –2) (b) (1, –2)

(c) (–1, 2) (d) (1, 2)

Ans. (d)

Solution: Given: Equations of sides of the triangle are

$$4x - 7y + 10 = 0 \quad \text{(i)}$$

$$x + y - 5 = 0 \quad \text{(ii)}$$

and $$7x + 4y - 15 = 0 \quad \text{(iii)}$$

We know that lines (i) and (iii) are perpendicular to each other.

Multiplying Eq. (ii) by 7 and then adding the result with Eq. (i), we get

$x = 1$ and $y = 2$

We also know that the point of intersection of the given lines (i) and (ii), $(x, y) = (1, 2)$.

Therefore, the coordinates of the triangle are (1, 2).

14. If the slope of one of the lines $x^2+kxy+2y^2=0$ is twice that of the other, then the value of k is

(a) ± 3 (b) ± 2

(c) $\pm(3/2)$ (d) ± 1

Ans. (a)

Solution: Given: Equation of the pair of lines

$$x^2+kxy+2y^2=0 \quad \text{or} \quad y^2+\frac{k}{2}xy+\frac{x^2}{2}=0 \qquad \text{(i)}$$

and slope of one line $(m_1) = 2m_2$ (where m_2 is the slope of the other line).

We know that the standard equation of the pair of lines having slopes m_1 and m_2 and passing through the origin is $y^2-(m_1+m_2)xy+(m_1m_2)x^2=0$.

Comparing Eq. (i) with the standard equati on, we get

$$(m_1+m_2)=-\frac{k}{2} \text{ or } (2m_2+m_2)=-\frac{k}{2} \text{ or } 3m_2=-\frac{k}{2}$$

or $$m_2=-\frac{k}{6} \qquad \text{(ii)}$$

and $$m_1\times m_2=\frac{1}{2} \text{ or } (2m_2\times m_2)=\frac{1}{2} \text{ or } 2m_2^2=\frac{1}{2}$$

or $$m_2^2=\frac{1}{4} \qquad \text{(iii)}$$

Substituting the value of m_2 from Eq. (ii) in Eq. (iii), we get

$$\left(-\frac{k}{6}\right)^2=\frac{1}{4} \quad \text{or} \quad k^2=9$$

or $k=\pm 3$.

15. If the equation $x^2+kxy+y^2-5x-7y+6=0$ represents a pair of straight lines, then the value of k is

(a) $\frac{5}{3}$ (b) $\frac{10}{3}$

(c) $\frac{3}{2}$ (d) $\frac{3}{10}$

Ans. (b)

Solution: Given: Equation of the pair of straight lines is

$$x^2+kxy+y^2-5x-7y+6=0$$

We know that the standard equation of the pair of straight lines is $ax^2+2hxy+by^2+2gx+2fy+c=0$. Comparing the given equation with the standard equation, we get

$$a=1, b=1, c=6, f=-\frac{7}{2}, g=-\frac{5}{2} \text{ and } h=\frac{k}{2}$$

We also know that the condition for the pair of straight lines is

$$abc+2fgh-af^2-bg^2-ch^2=0$$

or $$(1\times 1\times 6)+2\times\left(-\frac{7}{2}\right)\times\left(-\frac{5}{2}\right)\times\left(\frac{k}{2}\right)-1\times\left(-\frac{7}{2}\right)^2$$
$$-1\times\left(-\frac{5}{2}\right)^2-6\times\left(\frac{k}{2}\right)^2=0$$

or $$6+(-7)\times\left(-\frac{5}{2}\right)\times\left(\frac{k}{2}\right)-\frac{49}{4}-\frac{25}{4}-\frac{3k^2}{2}=0$$

or $$-\frac{3k^2}{2}+\frac{35k}{4}-\frac{25}{2}=0 \text{ or } 6k^2-35k+50=0$$

or $$(2k-5)(3k-10)=0 \text{ or } k=\frac{10}{3}$$

16. The angle between the pair of straight lines $3x^2+4xy-3y^2=0$ is

(a) $60°$ (b) $90°$

(c) $\tan^{-1}\left(\frac{3}{4}\right)$ (d) $\left(\tan^{-1}\frac{4}{3}\right)$

Ans. (b)

Solution: Given: Equation of the pair of straight lines $3x^2+4xy-3y^2=0$.

We know that the standard equation of the pair of straight lines is $ax^2+2hxy+by^2=0$. Comparing the given equation with the standard equation, we get

$a = 3$, $h = 2$ and $b = -3$

We also know that the relation for the angle between the pair of straight lines (θ) is

$$\tan\theta=\frac{2\sqrt{h^2-ab}}{a+b}=\frac{2\sqrt{(2)^2-(3)\times(-3)}}{3+(-3)}=\infty$$

or $\theta=\tan^{-1}(\infty)=90°$.

17. The equation of lines parallel to $3x-4y-5=0$ at a unit distance from it is

(a) $3x-4y-10=0$ (b) $5x+3y-5=0$

(c) $3x+4y+10=0$ (d) $6x+2y+4=0$

Ans. (a)

Solution: Given: Equation of the line is $3x-4y-5=0$.

We know that the equation of any line parallel to the given line is

$$3x-4y+\lambda=0 \qquad \text{(i)}$$

We also know that the standard equation of the line is

$$ax+by+c=0$$

Comparing Eq. (i) with the standard equat ion, we get

$a_1 = 3$, $b_1 = -4$ and $c_1 = \lambda$

Substituting $x = -1$ in the given equation $3x - 4y - 5 = 0$, we get $y = -2$.

We also know that the length of the perpendicular from the point (–1, –2) to the line $3x-4y+\lambda=0$ is

$$\frac{a_1x_1+b_1y_1+c_1}{\sqrt{a_1^2+b_1^2}}=\frac{3\times(-1)+(-4)\times(-2)+\lambda}{\sqrt{(3)^2+(-4)^2}}=1$$

or $\left|\frac{5+\lambda}{5}\right|=1$ or $|5+\lambda|=5$ or $5+\lambda=\pm5$.

Thus, $\lambda=-10$.

Substituting the value of λ in Eq. (i), we get

$$3x-4y-10=0$$

Therefore, the equation of the parallel line is

$$3x-4y-10=0$$

18. The equation of the straight line passing through the point of intersection of lines $5x - 6y - 1 = 0$ and $3x + 2y + 5 = 0$ and perpendicular to the line $3x - 5y + 11 = 0$ is

(a) $5x+3y+8=0$ (b) $3x-5y+8=0$

(c) $5x+3y+11=0$ (d) $3x-5y+11=0$

Ans. (a)

Solution: Given: Equations of lines

$$5x-6y-1=0 \quad \text{(i)}$$

$$3x+2y+5=0 \quad \text{(ii)}$$

and $3x-5y+11=0$ or $y=\frac{3}{5}x+\frac{11}{5}$ (iii)

Multiplying Eq. (ii) by Eq. (iii) and adding the result with Eq. (i), we get

$$x=-1 \quad \text{and} \quad y=-1$$

Therefore, the coordinates of the point of intersection of the lines (i) and (ii) are $(x_1, y_1)=(-1,1)$.

We know that the standard equation of the line is

$$y=mx+c$$

Comparing Eq. (iii) with the standard equation of the line, we get

$$m=\frac{3}{5}$$

Therefore, the slope of the perpendicular line

$$(m_1)=-\frac{5}{3}$$

We know that the equation of the perpendicular line passing through the point (x_1, y_1) is $(y - y_1) = m_1(x - x_1)$

or $(y+1)=-\frac{5}{3}(x+1)$ or $3y+3=-5x-5$

or $5x+3y+8=0$

EXERCISE SET 1

1. The locus of the point which moves so that the difference of its distance from two straight lines which are at right angles is equal to the distance from another fixed straight line is

(a) a straight line (b) a circle

(c) a parabola (d) an ellipse

2. The third vertex of an equilateral triangle whose vertices are (2, 4) and (2, 6) is

(a) $(2+\sqrt{3},5)$ (b) $(2\sqrt{3},5)$

(c) $(2,5)$ (d) $(\sqrt{3},5)$

3. Two vertices of a triangle are (4, –3) and (–2, 5). If the orthocentre of the triangle is at (1, 2), then the third vertex is

(a) (–33, –26) (b) (33, 26)

(c) (26, 33) (d) none of these

4. If a rod of length l slides with its ends on two perpendicular lines, then the locus of its mid-point is

(a) $2x^2+2y^2=l^2$ (b) $4x^2+4y^2=l^2$

(c) $x^2+y^2=l^2$ (d) $x^2+2y^2=l^2$

5. The circumcentre of the triangle formed by $A(3, 7)$, $B(3 - 2)$, $C(5, 7)$ is

(a) (3, 7) (b) (3, –2)

(c) (5, 7) (d) (4, 5/2)

6. The diagonals AC and BD of a rhombus intersect at (5, 6). If $A \equiv (3, 2)$, then the equation of the diagonal BD is

(a) $y-x=1$ (b) $2y-x=7$

(c) $y-2x+4=0$ (d) $2y+x=17$

7. A straight line through the origin meets the parallel lines $4x + 4y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively. Then the point O divides the segment PQ in the ratio

(a) 1 : 2 (b) 3 : 4

(c) 2 : 1 (d) 4 : 3

8. The fourth vertex of the rhombus formed by the points (–1, 5), (–2, –3), (5, 1) is

(a) (–6, 9) (b) (6, 9)

(c) (6, –9) (d) none of these

9. If the point $P(a^2, a)$ lies in the region corresponding to the acute angle between the lines $2y = x$ and $4y = x$, then

(a) $a \in (2, 6)$ (b) $a \in (4, 6)$

(c) $a \in (2, 4)$ (d) none of these

10. A ray of light passing through the point (1, 2) is reflected on the x-axis at a point P and passes through the point (5, 3). Then the abscissa of the points is

(a) 3 (b) 13/3

(c) 13/5 (d) 13/4

11. The mid-points of AB, BC, CA of a ΔABC are (6, –1) (–4, –3), (2, –5) respectively. The centroid of ΔABC is

(a) (4, 1) (b) $\left(\frac{4}{3},-3\right)$

(c) $\left(\frac{4}{3},3\right)$ (d) $\left(-\frac{4}{3},3\right)$

12. A line is drawn perpendicular to the line $y = 5x$, meeting the coordinate axes at A and B. If the area of the triangle OAB is 10 sq. units, where O is the origin, then the equation of the drawn line is

(a) $x+5y=10$ (b) $x-5y=10$

(c) $x+4y=10$ (d) $x-4y=10$

13. The length of the perpendicular from the origin to a line is 7 and the line makes an angle 150° with the positive direction of the y-axis. The equation of the line is

(a) $\sqrt{3}x+y=14$ (b) $\sqrt{3}x-y=14$

(c) $\sqrt{3}x+y+14=0$ (d) $\sqrt{3}x-y+14=0$

14. The vertices of a triangle are $A(0, 0)$, $B(0, 2)$ and $C(2, 0)$. The distance between the circumcentre and orthocentre is

(a) 0 (b) $\sqrt{2}$

(c) $1/\sqrt{2}$ (d) none of these

15. The coordinates of two consecutive vertices A and B of a regular hexagon $ABCDEF$ are (1, 0) and (2, 0) respectively. The equation of the diagonal CE is

(a) $\sqrt{3}x+y=4$ (b) $x+\sqrt{3}y+4=0$

(c) $x+\sqrt{3}y=4$ (d) none of these

16. The area of the parallelogram formed by the lines $4y-3x=1$, $4y-3x-3=0$, $3y-4x+1=0$ and $3y-4x+2=0$ is

(a) 3/8 (b) 2/7

(c) 1/6 (d) none of these

17. The foot of the perpendicular from (2, 3) upon the line $4x-5y+8=0$ is

(a) (0, 0) (b) (1, 1)

(c) $\left(\frac{41}{78},\frac{128}{75}\right)$ (d) $\left(\frac{78}{41},\frac{128}{41}\right)$

18. The equations of the lines which pass through the origin and are inclined at an angle $\tan^{-1} m$ to the line $y = mx + c$ are

(a) $x=0,\ 2mx+(m^2-1)y=0$

(b) $y=0,\ 2mx+(m^2-1)y=0$

(c) $y=0,\ 2mx+(1-m^2)y=0$

(d) none of these

19. If t_1, t_2 and t_3 are distinct, the points $(t_1, 2at_1+at_1^3)$, $(t_2, 2at_2+at_2^3)$ and $(t_3, 2at_3+at_3^3)$ are collinear if

(a) $t_1t_2t_3=1$ (b) $t_1+t_2+t_3=t_1t_2t_3$

(c) $t_1+t_2+t_3=0$ (d) $t_1+t_2+t_3=-1$

20. The image (or reflection) of the point (4, –13) in the line $5x+y+6=0$ is

(a) (–1, –14) (b) (3, 4)

(c) (1, 2) (d) (–4, 13)

21. The equation of a straight line on which the length of the perpendicular from the origin is four units and the linemakes an angle of 120° with the x-axis is

(a) $x\sqrt{3}+y+8=0$ (b) $x\sqrt{3}-y=8$

(c) $x\sqrt{3}+y=8$ (d) $x-\sqrt{3}y+8=0$

22. The area of the parallelogram formed by the lines $4y-3x-a=0$, $3y-4x+a=0$, $4y-3x-3a=0$ and $3y-4x+2a=0$ is

(a) $\frac{a^2}{5}$ (b) $\frac{a^2}{7}$

(c) $\frac{2a^2}{7}$ (d) $\frac{2a^2}{9}$

23. The equations of the lines on which the perpendiculars from the origin make 30° angle with the x-axis and which form a triangle of area $50/\sqrt{3}$ with axes are

(a) $x+\sqrt{3}y\pm10=0$ (b) $\sqrt{3}x+y\pm10=0$

(c) $x\pm\sqrt{3}y-10=0$ (d) none of these

24. If a vertex of a triangle is (1, 1) and the middle points of the two sides through it are (–2, 3) and (5, 2), then the centroid of the triangle is

(a) $\left(\frac{5}{3},3\right)$ (b) $\left(\frac{5}{3},-3\right)$

(c) $\left(-\frac{5}{3},3\right)$ (d) $\left(-\frac{5}{3},-3\right)$

25. If the distance between the straight lines $y=mx+c_1$ and $y=mx+c_2$ is $|c_1-c_2|$, then

(a) $m = 0$ (b) $m = 1$

(c) $m = 2$ (d) $m = -2$

26. If the lines $ax+y+1=0$, $x+by+1=0$ and $x + y + c = 0$ (a, b and c being distinct and different from 1) are concurrent, then $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$ is equal to

(a) 1 (b) –1

(c) 0 (d) none of these

27. If the lines joining the origin and points of intersection of $kx+hy=2hk$ and the circle $(x-y)^2+(y-k)^2=a^2$ are perpendicular, then $h^2+k^2=$
(a) 0 (b) a
(c) a^2 (d) -1

28. If m_1 and m_2 are the roots of the equation $x^2+(\sqrt{3}+2)x+(\sqrt{3}-1)=0$, then the area of the triangle formed by the lines $y=m_1x$, $y=m_2x$ and $y=2$ is
(a) $\sqrt{33}-\sqrt{11}$ (b) $x\sqrt{33}+\sqrt{11}$
(c) $\sqrt{33}+\sqrt{7}$ (d) none of these

29. Which pair of points lie on the same side of $3x-8y-7=0$?
(a) (0, –1) and (0, 0) (b) (4, –3) and (0, 1)
(c) (–3, –4) and (1, 2) (d) (–1, –1) and (3, 7)

30. If k is a parameter, then the equation of the family of lines parallel to the line $3x+4y+5=0$ is
(a) $4x-3y+k=0$ (b) $3x-4y+k=0$
(c) $3x+4y+k=0$ (d) $4x+3y+k=0$

31. The coordinates of the third vertex of an equilateral triangle whose two vertices are at (3, 4) and (–2, 3) are
(a) (1, 1) or (1, –1)
(b) $\left(\frac{1+\sqrt{3}}{2},\frac{7-5\sqrt{3}}{2}\right)$ or $\left(\frac{1-\sqrt{3}}{2},\frac{7+5\sqrt{3}}{2}\right)$
(c) $(-\sqrt{3},\sqrt{3})$ or $(\sqrt{3},-\sqrt{3})$
(d) none of these

32. The line passing through (1, 1) and parallel to the line $2x-3y+5=0$ is
(a) $3x+2y=5$ (b) $2x-3y+1=0$
(c) $3x-2y=1$ (d) $2x+3y=5$

33. If $x^2+2hxy+y^2=0$ represents the equation of the straight lines through the origin which make an angle α with the straight line $y+x=0$, then which of the following is not true?
(a) $\sec 2\alpha=h$ (b) $\cos\alpha=\sqrt{\frac{1+h}{2h}}$
(c) $2\sin\alpha=\sqrt{\frac{1+h}{h}}$ (d) $\cot\alpha=\sqrt{\frac{h+1}{h-1}}$

34. If the sum of the squares of intercepts on the axes cut off by the tangents to the curve $x^{2/3}+y^{2/3}=a^{2/3}$ $(a>0)$ at $\left(\frac{a}{8},\frac{a}{8}\right)$ is 2, then a has the value
(a) 1 (b) 2
(c) 4 (d) 8

35. If $xy+x+y+1=0$, and $x+ay-3=0$ are concurrent, then $a=$
(a) 3 (b) –4
(c) 2 (d) none of these

36. The lines $x\cos\alpha+y\sin\alpha=p_1$ and $x\cos\beta+y\sin\beta=p_2$ will be perpendicular if
(a) $\alpha=\beta$ (b) $|\alpha-\beta|=\frac{\pi}{2}$
(c) $\alpha=\frac{\pi}{2}$ (d) $\alpha\pm\beta=\frac{\pi}{2}$

37. The area of a parallelogram formed by the lines $ax\pm bx\pm c=0$ is
(a) $\frac{c^2}{ab}$ (b) $\frac{2c^2}{ab}$
(c) $\frac{c^2}{2ab}$ (d) none of these

38. The equation of the straight line passing through the intersection of the lines $x-2y=1$ and $x+3y=2$ and parallel to $3x+4y=0$ is
(a) $3x+4y+5=0$ (b) $3x+4y-10=0$
(c) $3x+4y-5=0$ (d) $3x+4y+6=0$

39. If a and b are real numbers between 0 and 1 such that the points $(a, 1)$, $(1, b)$ and $(0, 0)$ form an equilateral triangle, then a and b are
(a) $2-\sqrt{3},2-\sqrt{3}$ (b) $\sqrt{3}-1,\sqrt{3}-1$
(c) $\sqrt{2}-1,\sqrt{2}-1$ (d) none of these

40. If P and Q are the points (–3, 4) and (2, 1), then the coordinates of the point R on PQ produced such that $PR = 2QR$ are
(a) (2, 4) (b) (3, 7)
(c) (7,–2) (d) $\left(-\frac{1}{2},\frac{5}{2}\right)$

41. If the foot of the perpendicular from the origin to a straight line is at the point (3, –4), then the equation of the line is
(a) $3x-4y=25$ (b) $3x-4y+25=0$
(c) $4x+3y-25=0$ (d) $4x-3y+25=0$

42. The equation of a line through the point (1, 2) whose distance from the point (3, 1) has the greatest possible value is
(a) $y=x$ (b) $y=2x$
(c) $y=-2x$ (d) $y=-x$

43. If the equation $6x^2 - xy - 12y^2 - 8x + 29y - 14 = 0$ represents a pair of lines, then the angle between them is
(a) $\tan^{-1}(17/6)$ (b) $\tan^{-1}(3/4)$
(c) $\pi/4$ (d) $\pi/3$

44. If a, b, c are in A.P., then the straight line $ax + by + c = 0$ will always pass through a fixed point whose coordinates are
(a) (1, –2) (b) (–1, 2) (c) (1, 2) (d) (–1, –2)

45. The angle between the lines represented by $x^2 + 2xy \sec\theta + y^2 = 0$ is
(a) 4θ (b) 2θ (c) θ (d) none of these

46. If the family of lines
$$(2\cos\theta + 3\sin\theta)x + (3\cos\theta - 5\sin\theta)y - (5\cos\theta - 2\sin\theta) = 0$$
is concurrent at (x_1, y_1) and the family of lines $(a + 2b)x + (a - 3b)y + b - a = 0$ for different values of a and b is concurrent at (x_2, y_2), then
(a) $x_2 + y_2 = x_1$ (b) $x_2 + y_2 = y_1$
(c) $6x_2 + y_2 = 2x_1 + y_1$ (d) none of these

47. The sides of a triangle are $3x + 4y$, $4x + 3y$ and $5x + 5y$ units, where $x > 0$, $y > 0$. The triangle is
(a) right angled (b) acute angled
(c) obtuse angled (d) isosceles

48. The acute angle between the lines $x - y = 0$ and $y = 0$ is
(a) $\pi/6$ (b) $\pi/4$ (c) $\pi/3$ (d) $5\pi/6$

49. If the two vertices of a triangle are (–2, 3) and (5, –1), the orthocentre lies at the origin and the centroid on the line is $x + y = 7$, then the third vertex lies at
(a) (7, 4) (b) (8, 14) (c) (12, 21) (d) none of these

50. The straight line which is parallel to the x-axis and crosses the curve $y = \sqrt{x}$ at an angle of 45° is
(a) $x = 1/4$ (b) $y = 1/4$ (c) $x = 1/2$ (d) $y = 1$

51. If the second-degree equation $x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ represents a pair of straight lines, the distance between them is
(a) 4 (b) $4/\sqrt{3}$ (c) 2 (d) $2\sqrt{3}$

52. The area of the triangle with vertices (a, b), (x_1, y_1) and (x_2, y_2), where a, x_1, x_2 are in G.P. with common ratio r and b, y_1, y_2 are in G.P. with common ratio s, is
(a) $ab(r-1)(s-1)(s-r)$
(b) $\frac{1}{2}ab(r+1)(s+1)(s-r)$
(c) $\frac{1}{2}ab(r-1)(s-1)(s-r)$
(d) $ab(r+1)(s+1)(r-s)$

53. The area of the triangle formed by the two rays whose combined equation is $y = |x|$ and the line is $x + 2y = 2$ is
(a) 8/3 sq. unit^2 (b) 4/3 sq. unit^2
(c) 4 sq. unit^2 (d) 16/3 sq. unit^2

54. If the angle between two straight lines represented by $2x^2 + 5xy + 3y^2 + 7y + 4 = 0$ is $\tan^{-1} m$, then m equals
(a) 1 (b) 7 (c) 1/5 (d) 7/5

55. The straight lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is
(a) isosceles (b) equilateral
(c) right-angled (d) none of these

56. The equations of the sides of a triangle are $x = 0$, $y = m_1x + c_1$ and $y = m_2x + c_2$. The area of the triangle is
(a) $\dfrac{c_1 - c_2}{m_1 - m_2}$ (b) $\dfrac{1}{2}\dfrac{(c_1 - c_2)^2}{m_1 - m_2}$
(c) $\dfrac{1}{2}\dfrac{c_1 - c_2}{(m_1 - m_2)^2}$ (d) $\dfrac{(c_1 - c_2)^2}{(m_1 - m_2)^2}$

57. If the straight lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$ form a triangle with the origin as orthocentre, then (a, b) is given by
(a) (6, 4) (b) (–3, 3) (c) (–8, 8) (d) (0, 7)

58. A point equidistant from the lines $4x + 3y + 10 = 0$, $5x - 2y + 26 = 0$ and $7x - 24y - 50 = 0$ is
(a) (1, –1) (b) (1, 1) (c) (0, 0) (d) (0, 1)

59. The point of intersection of two lines given by $2x^2 - 5xy + 2y^2 - 3x + 3y + 1 = 0$ is
(a) $\left(\frac{1}{2}, \frac{1}{3}\right)$ (b) $\left(-\frac{1}{7}, -\frac{1}{7}\right)$
(c) $\left(-\frac{1}{3}, \frac{1}{3}\right)$ (d) $\left(\frac{1}{3}, -\frac{1}{3}\right)$

60. The value of K such that $3x^2 - 11xy + 10y^2 - 7x + 13y + K = 0$ may represent a pair of straight lines is
(a) 3 (b) 4 (c) 6 (d) 8

61. The triangle formed by the lines whose combined equation is $(y^2 - 4xy - x^2)(x + y - 1) = 0$ is
(a) equilateral (b) right-angled
(c) isosceles (d) obtuse-angled

62. The distance between the parallel lines given by $(x+7y)^2+4\sqrt{2}(x+7y)-42=0$ is

(a) 2 (b) 4 (c) $4\sqrt{2}$ (d) $8\sqrt{2}$

63. The number of triangles formed by the curves $x^3-x^2-x-2=0$ and $4x^2+x^2y^2+2y^2-3xy^2-12x+8=0$ is equal to

(a) 0 (b) 1 (c) 2 (d) 3

64. The angle between the lines given by $2x^2-7x+3y^2=0$ is

(a) 60° (b) 45° (c) 30° (d) tan–1(7/6)

65. A straight line passing through $P(3, 1)$ meets the coordinate axes at A and B. It is given that the distance of this straight line from the origin O is maximum. The area of the triangle OAB is equal to

(a) 50/3 sq. units (b) 25/3 sq. units
(c) 20/3 sq. units (d) 100/3 sq. units

66. $A(1, 3)$ and $C(7, 5)$ are two opposite vertices of a square. The equation of a side through A is

(a) $x+2y-7=0$ or $2x-y+1=0$
(b) $x-2y+5=0$
(c) $2x+y-5=0$
(d) none of these

67. If the equation $ax^2+2hxy+by^2+2gx+2fy+c=0$ represents a pair of parallel lines, then

(a) $\frac{a}{h}=\frac{b}{h}=\frac{f}{g}$ (b) $\frac{a}{h}=\frac{h}{b}=\frac{f}{g}$
(c) $\frac{a}{h}=\frac{h}{b}=\frac{g}{f}$ (d) none of these

68. The vertices of a ΔABC are $A(0, 0)$, $B(-3, -1)$ and $C(-1, -3)$. The equation of a line parallel to BC and intersecting sides AB and AC whose distance from the origin is 1/2 is

(a) $x+y+\frac{1}{2}=0$ (b) $x+y-\frac{1}{2}=0$
(c) $x+y-\frac{1}{\sqrt{2}}=0$ (d) $x+y+\frac{1}{\sqrt{2}}=0$

69. The centroid of the triangle whose three sides are given by the combined equation $(x^2+7xy+2y^2)(y-1)=0$ is

(a) $\left(-\frac{7}{3},\frac{2}{3}\right)$ (b) $\left(\frac{7}{3},\frac{2}{3}\right)$
(c) $\left(\frac{2}{3},0\right)$ (d) none of these

70. The angle between the pair of lines represented by $x^2-7xy+12y^2=0$ is

(a) $\sin^{-1}\frac{1}{12}$ (b) $\sin^{-1}\frac{1}{13}$
(c) $\sin^{-1}\frac{1}{\sqrt{170}}$ (d) $\sin^{-1}\frac{1}{\sqrt{85}}$

EXERCISE SET 2

1. Line L is perpendicular to the line $5x-y=1$. The area of the triangle formed by the line and coordinate axes is 5. Its equation is

(a) $x+5y=\sqrt{2}$ (b) $x+5y=5\sqrt{2}$
(c) $x+5y=-5\sqrt{2}$ (d) $x+5y=-\sqrt{2}$

2. Line L has intercepts a and b on the coordinate axes. When the axes are rotated through a given angle keeping the origin fixed, the same line L has intercepts p and q, then

(a) $a^2+b^2=p^2+q^2$ (b) $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{p^2}+\frac{1}{q^2}$
(c) $a^2+p^2=b^2+q^2$ (d) $\frac{1}{a^2}+\frac{1}{p^2}=\frac{1}{b^2}+\frac{1}{q^2}$

3. The equation of the line parallel to the x-axis and bisecting the join of (1, 4) and (–2, 6) is

(a) $y = 5$ (b) $y = 3$
(c) $y + 1 = 0$ (d) $x = 5$

4. If the curve $x^2+y^2+2gx+2fy+c=0$ intercepts on the line $lx+my=1$, a length which subtends a right angle at the origin, then

(a) $c(l^2+m^2)+2(gl+fm+1)=0$
(b) $c(l^2+m^2)-2(gl+fm+1)=0$
(c) $c(l^2+m^2)+2(gl+fm-1)=0$
(d) none of these

5. If one of the lines of the pair $ax^2+2hxy+by^2=0$ bisects the angle between the positive directions of the axes, then a, b, h satisfy the relation

(a) $a+b=2|h|$ (b) $a+b=-2h$
(c) $a-b=2|h|$ (d) $(a-b)^2=4h^2$

6. The points (x_r, y_r), r = 1, 2, 3 are the vertices of an equilateral triangle of side a, then the square of the determinant $D=\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ equals

(a) $4a^4$ (b) $3a^4$
(c) $\frac{3}{4}a^4$ (d) none of these

7. The centroid of an equilateral triangle is (0, 0). If two vertices of the triangle lie on $x+y=2\sqrt{2}$, then one of them will have its coordinates as
   (a) $\left(\sqrt{2}+\sqrt{6},\sqrt{2}-\sqrt{6}\right)$
   (b) $\left(\sqrt{2}+\sqrt{3},\sqrt{2}-\sqrt{3}\right)$
   (c) $\left(\sqrt{2}+\sqrt{5},\sqrt{2}-\sqrt{5}\right)$
   (d) none of these

8. If a variable line intersects the coordinate axes at A and B and passes through a fixed point (a, b), then the locus of the vertex C of the rectangle $OACB$, where O is the origin, is
   (a) $\frac{x}{a}+\frac{y}{b}=1$ (b) $\frac{a}{x}+\frac{b}{y}=1$
   (c) $x+y=\frac{1}{a}+\frac{1}{b}$ (d) $ax+by=1$

9. $A(2,1)$, $B(4,-1)$ and $C(3,2)$ are the vertices of a triangle and if through A and C, lines are drawn parallel to opposite sides to intersect at D, then the area of $ABCD$ in sq. units is
   (a) 12 (b) 8 (c) 6 (d) 4

10. The graph of the function $y=\cos x\cos(x+2)-\cos^2(x+1)$ is
    (a) a straight line passing through $(0, -\sin^2 1)$ with slope 2
    (b) a straight line passing through (0, 0)
    (c) a parabola with vertex $(1, -\sin^2 1)$
    (d) a straight line passing through the point $\left(\frac{\pi}{2}, -\sin^2 1\right)$ and parallel to the x-axis

11. P and Q are points on the line joining $A(-2,5)$ and B (3, 1) such that $AP = PQ = QB$. Then the mid-point of PQ is
    (a) $\left(\frac{1}{2},3\right)$ (b) $\left(-\frac{1}{4},4\right)$
    (c) (2, 3) (d) (1, 4)

12. If the pair of straight lines $ax^2+2hxy-ay^2=0$ and $bx^2+2gxy-by^2=0$ are such that each bisects the angle between the other, then
    (a) $ab+gh=0$ (b) $h^2-ab=0$
    (c) $ah+bg=0$ (d) $ag+bh=0$

13. The equation of the straight line passing through the origin and the middle point of the intercept of the line $ax+by+c=0$ between the axes is
    (a) $ax+by=0$ (b) $ax-by=0$
    (c) $bx+ay=0$ (d) $bx-ay=0$

14. The limiting position of the point of intersection of the lines $3x+4y=1$ and $(1+c)x+3c^2y=2$ as c tends to 1 is
    (a) (4, –5) (b) (–5, 4)
    (c) (5,–4) (d) (–4,5)

15. Let PQR be a right-angled isosceles triangle right-angled at $P(2, 1)$. If the equation of the line QR is $2x+y=3$, then the equation representing PQ and PR is
    (a) $3x^2-3y^2+8xy+20x+10y+25=0$
    (b) $3x^2-3y^2+8xy-20x-10y+25=0$
    (c) $3x^2-3y^2+8xy+10x+15y+20=0$
    (d) $3x^2-3y^2-8xy-10x-15y-20=0$

16. The lines $ax+by+c=0$, $bx+cy+a=0$ and $ax+ay+b=0$ are concurrent if
    (a) $\sum a^3=3abc$ (b) $\sum a=0$
    (c) $\sum a^2=\sum ab$ (d) none of these

17. If the equation of the base of an equilateral triangle is $x+y=2$ and vertex is (2, –1), then the length of the side of the triangle equals
    (a) $\sqrt{1/3}$ (b) $\sqrt{3}$ (c) $\sqrt{2/3}$ (d) $\sqrt{3/2}$

18. The straight line $(A^2-3B^2)x^2+8ABxy+(B^2-3A^2)y^2=0$ forms with the line $Ax+By+C=0$ an equilateral triangle of area
    (a) $\frac{c^2}{\sqrt{2}\cdot(A^2+B^2)}$ (b) $\frac{c^2}{\sqrt{3}\cdot(A^2+B^2)}$
    (c) $\frac{c^2}{A^2+B^2}$ (d) none of these

19. The foot of the perpendicular drawn from (0, 5) on the line $3x-4y-5=0$ is
    (a) (3, 2) (b) (3, 1)
    (c) (1, 3) (d) (2, 3)

Assertion-Reason Type Questions

In this section, each question contains Statement I (Assertion) and Statement II (Reason). Each of these questions has following four choices (a), (b), (c) and (d), only one of which is the correct answer.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

20. Statement I: Reflections of the point (–3, 2) in the line $x+y=0$ is (–2, 3).

Statement II: The reflection of a point $P(\alpha,\beta)$ in the line $ax+by+c=0$ is the point $Q(\alpha',\beta')$ if $\left(\frac{\alpha+\alpha'}{2},\frac{\beta+\beta'}{2}\right)$ lies on the line.

21. Statement I: If the perpendicular bisector of the line segment joining points $A(a, 3)$ and $B(1, 4)$ has the y-intercept –4, then $a = \pm 4$.

Statement II: Locus of a point equidistant from two given points is the perpendicular bisector of the line joining the given points.

22. Statement I: The lines $a_1x+b_1y+c_1=0$, $a_2x+b_2y+c_2=0$ and $a_3x+b_3y+c_3=0$ are concurrent if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$.

Statement II: The area of the triangle formed by three concurrent lines is zero.

23. Statement I: Each point on the line $y-x+12=0$ is equidistant from the lines $3x+4y-12=0$ and $4x+3y-24=0$.

Statement II: The locus of a point equidistant from two given lines is the angle bisector of the two lines.

Previous Years' Questions

24. A triangle with vertices (4, 0) (–1, –1), (3,5) is
(a) isosceles and right-angled
(b) isosceles but not right-angled
(c) right-angled but not isosceles
(d) neither right-angled nor isosceles **[AIEEE 2002]**

25. The points $(-a, -b)$ (0, 0), (a, b) and (a^2, ab) are
(a) collinear
(b) vertices of a rectangle
(c) vertices of a parallelogram
(d) none of these **[AIEEE 2002]**

26. The incentre of the triangle with vertices $(1,\sqrt{3}), (0,0)$ and (2, 0) is

(a) $\left(1,\frac{\sqrt{3}}{2}\right)$ (b) $\left(\frac{2}{3},\frac{1}{\sqrt{3}}\right)$

(c) $\left(\frac{2}{3},\frac{\sqrt{3}}{2}\right)$ (d) $\left(1,\frac{1}{\sqrt{3}}\right)$

[AIEEE 2002]

27. A straight line through the point (2, 2) intersects the lines $\sqrt{3}x+y=0$ and $\sqrt{3}x-y=0$ at the point A and B. The equation to line AB, so that the ΔOAB is equilateral is
(a) $x-2=0$ (b) $y-2=0$
(c) $x+y-4=0$ (d) none of these **[AIEEE 2002]**

28. Three straight lines $2x+11y-5=0, 4x-3y-2=0$ and $24x+7y-20=0$
(a) form a triangle
(b) are only concurrent
(c) are concurrent with one line bisecting the angle between the other two
(d) none of these **[AIEEE 2002]**

29. A square of side a lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha[0<\alpha<(\pi/4)]$ with the positive direction of the x-axis. The equation of its diagonal not passing through the origin is
(a) $y(\cos\alpha+\sin\alpha)+x(\sin\alpha-\cos\alpha)=a$
(b) $y(\cos\alpha+\sin\alpha)+x(\sin\alpha+\cos\alpha)=a$
(c) $y(\cos\alpha+\sin\alpha)+x(\cos\alpha-\sin\alpha)=a$
(d) $y(\cos\alpha-\sin\alpha)-x(\sin\alpha-\cos\alpha)=a$
[AIEEE 2003]

30. If the pair of straight lines $x^2-2pxy-y^2=0$ and $x^2-2qxy-y^2=0$ are such that each pair bisects the angle between the other pair, then
(a) $p=q$ (b) $p=-q$ (c) $pq=1$ (d) $pq=-1$
[AIEEE 2003]

31. If the equation of the locus of a point equidistant from the points (a_1b_1) and (a_2b_2) is $(a_1-a_2)x+(b_1-b_2)y+c=0$, then c =

(a) $a_1^2-a_2^2+b_1^2-b_2^2$ (b) $\frac{1}{2}(a_1^2+a_2^2+b_1^2+b_2^2)$

(c) $\sqrt{a_1^2+b_1^2-a_2^2-b_2^2}$ (d) $\frac{1}{2}(a_2^2+b_2^2-a_1^2-b_1^2)$

[AIEEE 2003]

32. The locus of the centroid of the triangle whose vertices are $(a\cos t, a\sin t), (b\sin t, -b\cos t)$ and (1, 0), where t is a parameter, is
(a) $(3x+1)^2+(3y)^2=a^2-b^2$
(b) $(3x-1)^2+(3y)^2=a^2-b^2$
(c) $(3x-1)^2+(3y)^2=a^2+b^2$
(d) $(3x+1)^2+(3y)^2=a^2+b^2$ **[AIEEE 2003]**

33. The equation of the bisector of the acute angles between the lines $3x-4y+7=0$ and $12x+5y-2=0$ is
(a) $99x-27y-81=0$ (b) $11x-3y+9=0$
(c) $21x+77y-101=0$ (d) $21x+77y+101=0$
[AIEEE 2004]

34. Let $A(2, -3)$ and $B(-2, 1)$ be the vertices of a triangle ABC. If the centroid of the triangle moves on the line $2x+3y=1$, then the locus of the vertex C is the line

(a) $2x+3y=9$ (b) $2x-3y=7$

(c) $3x+2y=5$ (d) $3x-2y=3$

[AIEEE 2004]

35. If the sum of the slopes of the lines given by $x^2-2cxy-7y^2=0$ is four times their product, then c has the value

(a) 1 (b) –1 (c) 2 (d) –2

[AIEEE 2004]

36. The equations of the straight lines passing through the point (4, 3) and making intercepts on the coordinate axes whose sum is –1 are

(a) $\frac{x}{2}+\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(b) $\frac{x}{2}-\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(c) $\frac{x}{2}+\frac{y}{3}=1$ and $\frac{x}{2}+\frac{y}{1}=1$

(d) $\frac{x}{2}-\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$ **[AIEEE 2004]**

37. If one of the lines given by $6x^2-xy+4cy^2=0$ and $3x+4y=0$ form a triangle, then c equals

(a) 3 (b) –1 (c) 1 (d) –3

[AIEEE 2004]

38. If a vertex of a triangle is (1, 1) and the mid-points of two sides through this vertex are (–1, 2) and (3, 2), then the centroid of the triangle is

(a) $\left(\frac{-1}{3},\frac{7}{3}\right)$ (b) $\left(-1,\frac{7}{3}\right)$

(c) $\left(\frac{1}{3},\frac{7}{3}\right)$ (d) $\left(1,\frac{7}{3}\right)$

[AIEEE 2005]

39. The line parallel to the x-axis and passing through the intersection of the lines $ax+2by+3b=0$ and $bx-2ay-3a=0$, where $(a, b)\neq(0, 0)$, is

(a) above the x-axis at a distance of 2/3 from it

(b) above the x-axis at a distance of 3/2 from it

(c) below the x-axis at a distance of 2/3 from it

(d) below the x-axis at a distance of 3/2 from it

[AIEEE 2005]

40. If non-zero numbers a, b, c are in H.P., then the straight line $\frac{x}{a}+\frac{y}{b}+\frac{1}{c}=0$ always passes through a fixed point. That point is

(a) (–1, –2) (b) (–1, 2)

(c) [1, –(1/2)] (d) (1, –2)

[AIEEE 2005]

41. If (a, a^2) falls inside the angle made by the lines $y = (x/2)$, $x > 0$ and $y = 3x$, $x > 0$, then a belongs to

(a) $\left(0,\frac{1}{2}\right)$ (b) $(3, \infty)$

(c) $\left(\frac{1}{2},3\right)$ (d) $\left(-3,-\frac{1}{2}\right)$

[AIEEE 2006]

42. A straight line through the point $A(3, 4)$ is such that its intercept between the axes is bisected at A. Its equation is

(a) $3x+4y=25$ (b) $x+y=7$

(c) $3x-4y+7=0$ (d) $4x+3y=24$

[AIEEE 2006]

43. If one of the lines of $my^2+(1-m^2)xy-mx^2=0$ is a bisector of the angle between the lines $xy = 0$, then m is

(a) 2 (b) 1 (c) 2 (d) –1/2

[AIEEE 2007]

44. Let $P(-1,0)$, $Q(0,0)$ and $R(3,3\sqrt{3})$ be three points. Theequation of the bisector of the angle PQR is

(a) $\frac{\sqrt{3}}{2}x+y=0$ (b) $x+\frac{\sqrt{3}}{2}y=0$

(c) $x+\sqrt{3}y=0$ (d) $\sqrt{3}x+y=0$

[AIEEE 2007]

45. Let $A(h, k)$, $B(1, 1)$ and $C(2, 1)$ be the vertices of a right-angled triangle with AC as its hypotenuse. If the area of the triangle is 1, then the set of values which k can take is given by

(a) {0, 2} (b) {–1,3} (c) {–3,–2} (d) {1, 3}

[AIEEE 2007]

46. The perpendicular bisector of the line segment joining $P(1, 4)$ and $Q(k, 3)$ has y-intercept –4. Then a possible value of k is

(a) 1 (b) 2 (c) –2 (d) –4

[AIEEE 2008]

47. The lines $p(p^2+1)x-y+q=0$ and $(p^2+1)^2x+(p^2+1)y+2q=0$ are perpendicular to a common line for

(a) no value of p

(b) exactly one value of p

(c) exactly two values of p

(d) more than two values of p **[AIEEE 2009]**

48. Three distinct points A, B and C are given in the two-dimensional coordinate plane such that the ratio of the distance of any one of them from the point (1, 0) to the distance from the point (–1, 0) is equal to 1/3. Then the circumcentre of the triangle ABC is at point

(a) $\left(\frac{5}{4},0\right)$ (b) $\left(\frac{5}{2},0\right)$

(c) $\left(\frac{5}{3},0\right)$ (d) $\left(0,\frac{5}{6}\right)$

[AIEEE 2009]

49. Let the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ lie in the plane $x+3y-az+b=0$. Then (a, b) equals

(a) (6, –17) (b) (–6, 7)

(c) (5, –15) (d) (–5, 15)

[AIEEE 2009]

50. The line L given by $\frac{x}{5}+\frac{y}{b}=1$ passes through the point (13, 32). The line K is parallel to L and has the equation $\frac{x}{c}+\frac{y}{3}=1$. Then the distance between L and K is

(a) $\sqrt{17}$ (b) $\frac{17}{\sqrt{15}}$

(c) $\frac{23}{\sqrt{15}}$ (d) $\frac{23}{\sqrt{17}}$ **[AIEEE 2010]**

51. The equation of the pair of straight lines drawn through (1, 1) and perpendicular to the pair of lines $3x^2-7xy+2y^2=0$ is

(a) $2x^2+7xy-11x+6=0$

(b) $2(x-1)^2+7(x-1)(y-1)-3y^2=0$

(c) $2(x-1)^2+7(x-1)(y-1)-3(y-1)^2=0$

(d) none of these **[DCE 2002]**

52. The product of the perpendicular from (α, β) to the lines $ax^2+2hxy+by^2=0$ is

(a) $\frac{a\alpha^2+2h\alpha\beta+b\beta^2}{\sqrt{4h^2+(a-b)^2}}$

(b) $a\alpha^2-2h\alpha\beta+b\beta^2-\sqrt{4h^2-(a-b)^2}$

(c) $\frac{a\alpha^2-2h\alpha\beta+b\beta^2}{\sqrt{4h^2-(a-b)^2}}$

(d) none of the above **[DCE 2002]**

53. The equation of a straight line passing through the point of intersection of $x-y+1=0$ and $3x+y-5=0$ and perpendicular to one of them is

(a) $x-3y-5=0$ (b) $x-3y+5=0$

(c) $x-y+3=0$ (d) $x+y+3=0$

[DCE 2002]

54. The y-intercept of a line passing through (2, 2) and perpendicular to the line $3x+y=3$ is

(a) 1/3 (b) 2/3

(c) 1 (d) 4/3 **[DCE 2003]**

55. If the intercept on the straight line $y = mx$ by the lines $y = 2$ and $y = 6$ is less than 5, then m belongs to

(a) $\left]-\infty,-\frac{4}{3}\right[\cup\left]\frac{4}{3},\infty\right[$ (b) $\left]-\infty,\frac{4}{3}\right[$

(c) $\left]-\frac{4}{3},\frac{4}{3}\right[$ (d) $\left]-\frac{4}{3},\infty\right[$

[UPSEE 2001, DCE 2004]

56. A line passes through the origin and is perpendicular to two given lines $2x+y+6=0$ and $4x+2y-9=0$. The ratio in which the origin divides this line is

(a) 1 : 2 (b) 2 : 1

(c) 4 : 2 (d) 4 : 3 **[DCE 2005]**

57. If (sin θ, cos θ) and (3, 2) lie on the same side of the line $x+y=1$, then θ lies between

(a) $\left(0,\frac{\pi}{4}\right)$ (b) $\left(0,\frac{\pi}{2}\right)$

(c) $(0,\pi)$ (d) $\left(\frac{\pi}{4},\frac{\pi}{2}\right)$

[DCE 2005]

58. If $\frac{x^2}{a}+\frac{y^2}{b}+\frac{2xy}{h}=0$ represents a pair of straight lines such that the slope of one line is twice the other, then $ab^2:h^2$ is

(a) 9 : 8 (b) 8 : 9

(c) 1 : 2 (d) 2 : 1 **[DCE 2005]**

59. The three straight lines $ax+by=c$, $bx+cy=a$ and $cx+ay=b$ are collinear, if

(a) $b + c = a$ (b) $c + a = b$

(c) $a + b + c = 0$ (d) $a + b = c$

[DCE 2006]

60. The equation of the pair of lines joining the origin to the points of intersection of $x^2+y^2=9$ and $x + y = 3$ is

(a) $x^2+(3-x)^2=9$ (b) $xy=0$

(c) $(3+y)^2+y^2=9$ (d) $(x-y)^2=9$

[DCE 2006]

61. If the point (a, a) falls between the lines $|x+y|=2$, then

(a) $|a|=2$ (b) $|a|=1$

(c) $|a|<1$ (d) $|a|<\frac{1}{2}$ **[DCE 2007]**

62. A line passes through the point of intersection of the lines $3x+y+1=0$ and $2x-y+3=0$ and makes equal intercepts with the axes. Then the equation of the line is

(a) $5x+5y-3=0$ (b) $x+5y-3=0$

(c) $5x-y-3=0$ (d) $5x+5y+3=0$

[DCE 2007]

63. The bisector of the acute angle formed between the lines $4x-3y+7=0$ and $3x-4y+14=0$ has the equation

(a) $x-y-3=0$ (b) $x-y+3=0$

(c) $x+y+3=0$ (d) $3x+y-1=0$

[UP SEE 2000, DCE 2008]

64. If the point $P(a, b)$ lies on the straight line $3x+2y=13$ and the point $Q(b, a)$ lies on the straight line $4x-y=5$, then the equation of the line PQ is

(a) $x-y-5=0$ (b) $x-y+5=0$

(c) $x+y-5=0$ (d) $x+y+5=0$

[DCE 2008]

65. The circumcentre of a triangle formed by the lines $xy+2x+2y+4=0$ and $x+y+2=0$ is

(a) $(-1, -1)$ (b) $(0, -1)$

(c) $(1, 1)$ (d) $(-1, 0)$

[DCE 2008]

66. The centroid of the triangle formed by the lines $(x^2 + 7xy + 2y^2)(y - 1) = 0$ is

(a) $\left(-\frac{7}{3},\frac{-2}{3}\right)$ (b) $\left(-\frac{7}{3},\frac{2}{3}\right)$

(c) $\left(\frac{7}{3},\frac{-2}{3}\right)$ (d) $\left(\frac{7}{3},\frac{2}{3}\right)$

[DCE 2008]

67. Point Q is symmetric to $P(4, -1)$ with respect to the bisector of the first quadrant. The length of PQ is

(a) $3\sqrt{2}$ (b) $5\sqrt{2}$

(c) $7\sqrt{2}$ (d) $9\sqrt{2}$

[DCE 2009]

68. The orthocentre of the triangle formed by the lines $xy = 0$ and $x + y = 1$ is

(a) $\left(\frac{1}{2},\frac{1}{2}\right)$ (b) $\left(\frac{1}{3},\frac{1}{3}\right)$

(c) $\left(\frac{1}{4},\frac{1}{4}\right)$ (d) $(0, 0)$

[DCE 2009]

69. The equation of the bisector of the acute angles between the lines $3x-4y+7=0$ and $12x+5y-2=0$ is

(a) $99x-27y+81=0$ (b) $11x-3y-9=0$

(c) $21x+77y-101=0$ (d) $21x+77y+101=0$

[UPSEE 2004]

70. The lines represented by the equation $x^2 - y^2 - x + 3y - 2 = 0$ are

(a) $x-y-2=0,\ x+y+1=0$

(b) $x-y-1=0,\ x+y+2=0$

(c) $x-y+1=0,\ x+y-2=0$

(d) $x-y+2=0,\ x+y-1=0$

[BCECE 2009, EMCET 2006]

71. If $a+b+c=0$, then the line $3ax+4by+c=0$ always passes through a fixed point whose coordinates are

(a) $\left(-\frac{1}{3},-\frac{1}{4}\right)$ (b) $\left(-\frac{1}{3},\frac{1}{4}\right)$

(c) $\left(\frac{1}{3},-\frac{1}{4}\right)$ (d) $\left(\frac{1}{3},\frac{1}{4}\right)$

[Rajasthan PET 2002]

72. The line $(p+2q)x+(p-3q)y=p-q$ for different values of p and q passes through the fixed point

(a) $\left(\frac{2}{5},\frac{2}{5}\right)$ (b) $\left(\frac{2}{5},\frac{3}{5}\right)$

(c) $\left(\frac{3}{5},\frac{2}{5}\right)$ (d) $\left(\frac{3}{5},\frac{3}{5}\right)$

[Rajasthan PET 2006]

73. If $A(3, 5)$, $B(-5, -4)$ and $C(7, 10)$ are the vertices of a parallelogram taken in order, then the coordinates of the fourth vertex are

(a) (10, 12) (b) (12, 17)

(c) (15, 19) (d) (18, 21)

[IIT 1998, Rajasthan PET 2001, 2004, Kerala PET 2002]

74. If the equation of the base of an equilateral triangle is $2x-y=1$ and the vertex is $(-1, 2)$, then the length of the side of the triangle is

(a) $\sqrt{\frac{15}{2}}$ (b) $\sqrt{\frac{16}{5}}$ (c) $\sqrt{\frac{20}{3}}$ (d) $\sqrt{\frac{23}{2}}$

[Rajasthan PET 2003, 2004, AMU 2000, UPSEE 2003, Kerala PET 2000, 2005]

75. If the point P is equidistant from $A(1, 3)$, $B(-3, 5)$ and $C(5, -1)$, then PA is equal to

(a) 5 (b) $5\sqrt{5}$ (c) 25 (d) $5\sqrt{10}$

[EAMCET 2003]

76. Obtain the angle between the lines $\{(x,0): x \in \mathbb{R}\}$ and $\{(0,y): y \in \mathbb{R}\}$

(a) 0 (b) $\pi/4$

(c) $\pi/2$ (d) π

[Gujarat CET 2006]

77. The ratio in which the line $3x+4y=7$ divides the line joining the points (–2, 1) and (1, 2) is

(a) 2 : 3 (b) 4 : 5

(c) 7 : 3 (d) 9 : 4

[Gujarat CET 2008]

78. The curve that passes through the point (2, 3) and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact is given by

(a) $2y-3x=0$ (b) $y = 6/x$

(c) $x^2+y^2=13$ (d) $\left(\frac{x}{2}\right)^2+\left(\frac{y}{3}\right)^2=2$

[AIEEE 2011]

79. The lines $x+y=|a|$ and $ax-y=1$ intersect each other in the first quadrant. Then the set of all possible values of a is in the interval

(a) $(0, \infty)$ (b) $[1, \infty)$

(c) $(-1, \infty)$ (d) $(-1, 1]$

[AIEEE 2011]

80. If $A(2, -3)$ and $B(-2, 1)$ are two vertices of a triangle and the third vertex moves on the line $2x + 3y = 9$, then the locus of the centroid of the triangle is

(a) $x - y = 1$ (b) $2x + 3y = 1$

(c) $2x + 3y = 3$ (d) $2x - 3y = 1$

[AIEEE 2011]

81. If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals

(a) 29/5 (b) 5

(c) 6 (d) 11/5

[AIEEE 2012]

82. A line is drawn through the point (1, 2) to meet the coordinate axes at P and Q such that it forms a triangle OPQ, where O is the origin. If the triangle OPQ is least, then the slope of the line PQ is

(a) –1/4 (b) –2

(c) –2 (d) –1/2

[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (a)	**2.** (a)	**3.** (b)	**4.** (b)	**5.** (d)	**6.** (d)	**7.** (b)	**8.** (b)	**9.** (c)	**10.** (c)
11. (b)	**12.** (a)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (c)	**20.** (a)
21. (a) and (c)		**22.** (c)	**23.** (b)	**24.** (a)	**25.** (a)	**26.** (a)	**27.** (c)	**28.** (b)	**29.** (d)
30. (c)	**31.** (b)	**32.** (b)	**33.** (c)	**34.** (c)	**35.** (b)	**36.** (b)	**37.** (b)	**38.** (c)	**39.** (a)
40. (c)	**41.** (a)	**42.** (b)	**43.** (a)	**44.** (a)	**45.** (c)	**46.** (a), (b) and (c)		**47.** (c)	**48.** (b)
49. (d)	**50.** (c)	**51.** (c)	**52.** (c)	**53.** (b)	**54.** (c)	**55.** (a)	**56.** (b)	**57.** (c)	**58.** (c)
59. (d)	**60.** (b)	**61.** (b)	**62.** (a)	**63.** (a)	**64.** (b)	**65.** (a)	**66.** (a)	**67.** (c)	**68.** (d)
69. (a)	**70.** (c)								

Exercise Set 2

1. (b) and (c)		**2.** (b)	**3.** (a)	**4.** (a)	**5.** (b)	**6.** (c)	**7.** (a)	**8.** (b)	**9.** (d)
10. (d)	**11.** (a)	**12.** (a)	**13.** (b)	**14.** (b)	**15.** (b)	**16.** (a) and (b)		**17.** (c)	**18.** (b)
19. (b)	**20.** (c)	**21.** (a)	**22.** (a)	**23.** (a)	**24.** (a)	**25.** (a)	**26.** (d)	**27.** (b)	**28.** (c)
29. (c)	**30.** (d)	**31.** (d)	**32.** (c)	**33.** (c)	**34.** (a)	**35.** (c)	**36.** (d)	**37.** (d)	**38.** (d)
39. (d)	**40.** (d)	**41.** (c)	**42.** (d)	**43.** (b)	**44.** (d)	**45.** (b)	**46.** (d)	**47.** (b)	**48.** (a)
49. (b)	**50.** (d)	**51.** (d)	**52.** (a)	**53.** (b)	**54.** (d)	**55.** (a)	**56.** (d)	**57.** (b)	**58.** (a)
59. (c)	**60.** (b)	**61.** (c)	**62.** (a)	**63.** (b)	**64.** (a)	**65.** (a)	**66.** (b)	**67.** (b)	**68.** (d)
69. (a)	**70.** (c)	**71.** (d)	**72.** (b)	**73.** (c)	**74.** (c)	**75.** (d)	**76.** (c)	**77.** (d)	**78.** (b)
79. (b)	**80.** (b)	**81.** (c)	**82.** (c)						

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Let the two fixed straight lines be the coordinate axes and let the third line be $x-y=0$. Let $P(h, k)$ be a variable point. Then

$$\left||h|-|k|\right|=\left|\frac{h-k}{\sqrt{2}}\right|$$

$$\Rightarrow \quad |h|-|k|=\pm\frac{h-k}{\sqrt{2}}$$

$\therefore$ Locus of $P(h, k)$ is $|x|-|y|=\pm\left(\frac{x-y}{\sqrt{2}}\right)$

Clearly, it represents a straight line.

2. Third vertex $=\left(\frac{2+2\pm\sqrt{3}(4,6)}{2}, \frac{4+6\mp\sqrt{3}(2-2)}{2}\right)$

$=[2\pm(-\sqrt{3}), 5]=(2+\sqrt{3}, 5)$ or $(2-\sqrt{3}, 5)$

$\therefore$ Required vertex is $(2+\sqrt{3}, 5)$.

3. Let the third vertex be (h, k). Now the slope of AD is $\frac{k-2}{h-1}$. Slope of BC is $\frac{5+3}{-2-4}=\frac{-4}{3}$.

Slope of BE is $\frac{-3-2}{4-1}=\frac{-5}{3}$, and slope of AC is $\frac{k-5}{h+2}$.

Since $AD \perp BC$

$$\Rightarrow \quad \frac{k-2}{h-1}\times\frac{-4}{3}=-1\Rightarrow 3h-4k+5=0 \qquad \text{(i)}$$

Again since $BE \perp AC$

$$\Rightarrow \quad -\frac{5}{3}\times\frac{k-5}{h+2}=-1 \;\Rightarrow\; 3h-5k+31=0 \qquad \text{(ii)}$$

On solving (i) and (ii), we get

$$h = 33,\ k = 26$$

Hence, the third vertex is (33, 26).

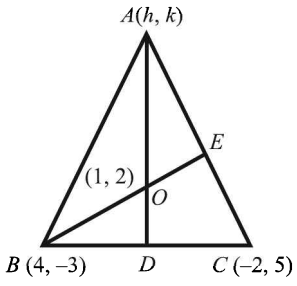


Fig. 18.29

4. Let the two perpendicular lines be the coordinate axes. Let AB be a rod of length l. Let the coordinates of A and B be $(a, 0)$ and $(0, b)$, respectively. As the rod slides, the values of a and b change. So a and b are two variables. Let $P(h, k)$ be the mid-point of the rod AB in one of the infinite positions it attains. Then

$$h=\frac{a+0}{2} \quad \text{and} \quad k=\frac{0+b}{2}$$

$$\Rightarrow \quad h=\frac{a}{2} \quad \text{and} \quad k=\frac{b}{2}$$

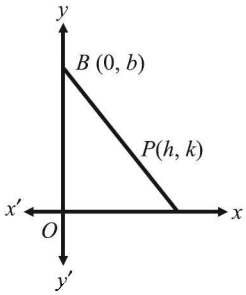


Fig. 18.30

From ΔOAB, we have

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow \quad a^2+b^2=l^2$$

$$\Rightarrow \quad (2h)^2+(2k)^2=l^2$$

$$\Rightarrow \quad 4h^2+4k^2=l^2$$

Therefore, the locus of the point (h, k) is $4x^2+4y^2=l^2$.

5. Here, ABC is a right-angled Δ with $\angle A = 90°$.

$\therefore$ Circumcentre $\left(\frac{3+5}{2}, \frac{-2+7}{2}\right)=\left(4, \frac{5}{2}\right)$

6. $m_{AC}=\frac{6-2}{5-3}=2\Rightarrow m_{BD}=-\frac{1}{2}$. Thus, the equation of BD is

$$(y-6)=-\frac{1}{2}(x+5)$$

i.e. $2y+x-17=0$

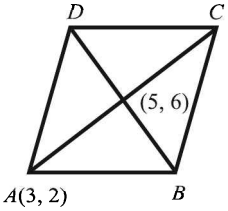


Fig. 18.31

7. Clearly, triangles OPA and OQC are similar.

$$\Rightarrow \quad \frac{OP}{OQ}=\frac{OA}{OC}$$

$$\Rightarrow \quad \frac{OP}{OQ}=\frac{9/4}{3}$$

$$\Rightarrow \quad \frac{OP}{OQ}=\frac{3}{4}$$

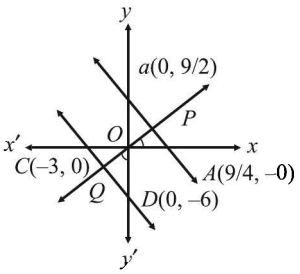


Fig. 18.32

8. Fourth vertex is $(-1+2+5, 5+3+1)$

i.e. (6, 9).

9. Acute angle is formed by lines in the first and third quadrants. But $a^2 > 0$, hence point $P(a^2, a)$ lies in the first quadrant.

We have $a-\frac{a^2}{4}>0$ and $a-\frac{a^2}{2}<0$

$\Rightarrow \quad 0<a<4, a\in(-\infty,0)\cup(2,\infty)$

$\Rightarrow \quad a\in(2,4)$

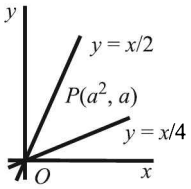


Fig. 18.33

10. Let the coordinates of P be $(a, 0)$. We have

$$\angle APN=\angle BPN=\theta \quad \text{(say)}$$

$\tan(90°-\theta)$ = Slope of PB and $\tan(90°+\theta)$ = Slope of PA

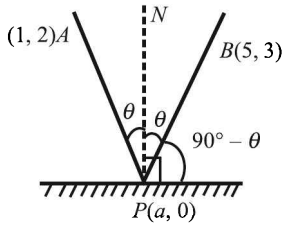


Fig. 18.34

$$\Rightarrow \quad \cot\theta=\frac{3}{5-a} \text{ and } -\cot\theta=\frac{2}{1-a}$$

$$\Rightarrow \quad \frac{3}{5-a}=\frac{2}{1-a} \Rightarrow 3-3a=-10+2a \Rightarrow a=\frac{13}{5}$$

11. Centroid of ΔABC

$$=\left(\frac{6-4+2}{3},\frac{-1-3-5}{3}\right)=\left(\frac{4}{3},-3\right)$$

12. Let the equation of line be $\frac{x}{a}+\frac{y}{b}=1$.

AB is perpendicular to $y = 5x$

$$\Rightarrow \quad -\frac{b}{a}\times 5=-1\Rightarrow 5b=a$$

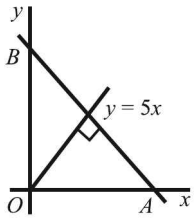


Fig. 18.35

Area of

$$\Delta OAB=\frac{1}{2}|ab| \Rightarrow 10$$

$$=\frac{1}{2}|5b^2| \Rightarrow b^2=4$$

$$\Rightarrow \quad b=\pm 2\Rightarrow a=\pm 10$$

The line can be $\frac{x}{10}+\frac{y}{2}=1$ or $\frac{x}{10}+\frac{y}{2}=-1$.

13. Here $p = 7$ and $\alpha = 30°$

$\therefore$ Equation of the required line is $x \cos 30° + y \sin 30° = 7$.

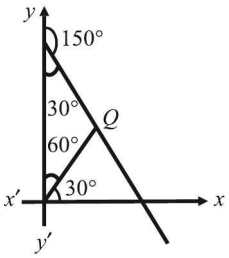


Fig. 18.36

or $x\frac{\sqrt{3}}{2}+y\frac{1}{2}=7$ or $x\sqrt{3}+y=14$

14. Clearly, ABC is a right-angled triangle, with right angle at A.

$\therefore$ Orthocentre is $A(0, 0)$ and circumcentre is $\left(\frac{2+0}{2},\frac{0+2}{2}\right)$, i.e. (1, 1).

$\therefore$ Required distance $=\sqrt{(1-0)^2+(1-0)^2}=\sqrt{2}$

15. $AB = 1$

$$\Rightarrow\ C \equiv (2+1\times\cos 60°, 1\times\sin 60°) = \left(\frac{5}{2}, \frac{\sqrt{3}}{2}\right)$$

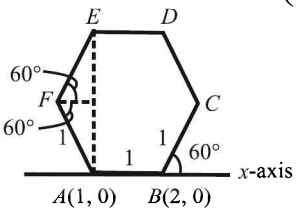


Fig. 18.37

$$E = (1,1\times\sin 60° + 1\times\sin 60°) = (1, \sqrt{3})$$

Therefore, the equation of CE is

$$y - \sqrt{3} = \frac{\sqrt{3} - \frac{\sqrt{3}}{2}}{1 - \frac{5}{2}}(x-1)$$

16. Let the equation of sides AB, BC, CD and DA of the parallelogram $ABCD$ be respectively

$$y = \frac{3}{4}x + \frac{1}{4} \qquad \text{(i)}$$

$$y = \frac{3}{4}x + \frac{3}{4} \qquad \text{(ii)}$$

$$y = \frac{4}{3}x - \frac{1}{3} \qquad \text{(iii)}$$

$$y = \frac{4}{3}x - \frac{2}{3} \qquad \text{(iv)}$$

Here $m = \frac{3}{4}, n = \frac{4}{3}, a = \frac{1}{4}, b = \frac{3}{4}, c = -\frac{1}{3}, d = -\frac{2}{3}$

$\therefore$ Area of the parallelogram $ABCD$

$$= \left|\frac{(a-b)(c-d)}{m-n}\right|$$

$$= \left|\frac{\left(\frac{1}{4}-\frac{3}{4}\right)\left(-\frac{1}{3}+\frac{2}{3}\right)}{\frac{3}{4}-\frac{4}{3}}\right| = \left|\frac{-\frac{1}{2}\times\frac{1}{3}}{-\frac{7}{12}}\right| = \frac{2}{7}$$

17. If foot is (h, k), then

$$\frac{h-2}{4} = \frac{k-3}{-5} = \frac{-(8-15+8)}{16+25} = -\frac{1}{41}$$

$$\therefore\quad h = 2 - \frac{4}{41} = \frac{78}{41};\ k = 3 + \frac{5}{41} = \frac{128}{41}$$

$\therefore$ Required point is $\left(\frac{78}{41}, \frac{128}{41}\right)$.

18. Angle between both the lines is $\tan^{-1} m \pm \tan^{-1} m$

$$= \tan^{-1}\frac{2m}{1-m^2} \text{ or } \tan^{-1} 0$$

Therefore, equations of lines are $y = 0$, $y = \frac{2mx}{1-m^2}$.

19. The given points are collinear if

$$\begin{vmatrix} t_1 & 2at_1 + at_1^3 & 1 \\ t_2 & 2at_2 + at_2^3 & 1 \\ t_3 & 2at_3 + at_3^3 & 1 \end{vmatrix} = 0 \Rightarrow a\begin{vmatrix} t_1 & 2t_1 + t_1^3 & 1 \\ t_2 & 2t_2 + t_2^3 & 1 \\ t_3 & 2t_3 + t_3^3 & 1 \end{vmatrix} = 0$$

Applying $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$, we get

$$\begin{vmatrix} t_1 & 2t_1 + t_1^3 & 1 \\ t_2 - t_1 & 2(t_2 - t_1) + (t_2^3 - t_1^3) & 0 \\ t_3 - t_1 & 2(t_3 - t_1) + (t_3^3 - t_1^3) & 0 \end{vmatrix} = 0$$

$$\Rightarrow (t_2 - t_1)(t_3 - t_1)\begin{vmatrix} t_1 & 2t_1 + t_1^3 & 1 \\ 1 & 2 + t_2^2 + t_1^2 + t_2 t_1 & 0 \\ 1 & 2 + t_3^2 + t_1^2 + t_3 t_1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (t_2 - t_1)(t_3 - t_1)(t_3 - t_2)(t_3 + t_2 + t_1) = 0$$

$$\Rightarrow t_1 + t_2 + t_3 = 0 \quad [\because t_1 \neq t_2 \neq t_3]$$

20. If (h, k) is the image, then

$$\frac{h-4}{5} = \frac{k+13}{1} = \frac{-2(20-13+6)}{25+1} = -1$$

$\therefore$ $h = -5 + 4 = -1$

$k = -1 - 13 = -14$

Point is $(-1, -14)$.

21. Slope $= -\sqrt{3}$

$\therefore$ Line is $y = -\sqrt{3}x + c$

$\Rightarrow \sqrt{3}x + y = c$

Now $\left|\frac{c}{2}\right| = 4$

$\Rightarrow c = \pm 8 \Rightarrow x\sqrt{3} + y = \pm 8$

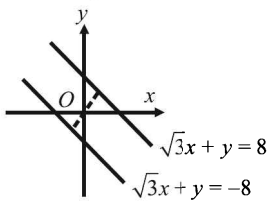


Fig. 18.38

22. Required area $= \left|\frac{(-a+3a)(a-2a)}{-9+16}\right| = \frac{2a^2}{7}$

23. Let p be the length of the perpendicular from the origin on the given line. Then its equation in normal form is

$x\cos 30° + y\sin 30° = p$ or $\sqrt{3}x + y = 2p$

This meets the coordinate axes at $A\left(\frac{2p}{\sqrt{3}}, 0\right)$ and $B(0, 2p)$.

$\therefore$ Area of $\Delta AOB = \frac{1}{2}\left(\frac{2p}{\sqrt{3}}\right)2p = \frac{2p^2}{\sqrt{3}}$

By hypothesis, $\frac{2p^2}{\sqrt{3}} = \frac{50}{\sqrt{3}} \Rightarrow p = \pm 5$

Hence, the lines are $\sqrt{3}x + y \pm 10 = 0$.

24. By joint formula, the points B and C are (9, 3) and (–5, 5). Hence, coordinates of G are given by

$$\left(\frac{\Sigma x_1}{3}, \frac{\Sigma y_1}{3}\right) = \left(\frac{5}{3}, 3\right)$$

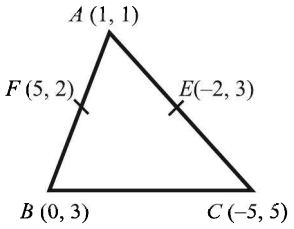


Fig. 18.39

25. By the given condition,

$$\frac{|c_1 - c_2|}{\sqrt{1 + m^2}} = |c_1 - c_2|$$

$\Rightarrow 1 = 1 + m^2 \Rightarrow m = 0$

26. Since the given lines are concurrent,

$$\therefore \quad \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a & 1-a & 1-a \\ 1 & b-1 & 0 \\ 1 & 0 & c-1 \end{vmatrix} = 0$$

[Applying $C_2 \to C_2 - C_1, C_3 \to C_3 - C_1$]

$\Rightarrow a(b-1)(c-1) - (1-a)(c-1) - (1-a)(b-1) = 0$

[Expanding along R_1]

$\Rightarrow a(1-b)(1-c) - (1-a)(1-c) + (1-a)(1-b) = 0$

$\Rightarrow \frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0$

[Dividing by $(1-a)(1-b)(1-c)$]

$\Rightarrow -1 + \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0$

$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$

27. The line is $\frac{x}{2h} + \frac{y}{2k} = 1$ and the circle is

$$x^2 + y^2 - 2(hx + ky) + (h^2 + k^2 - a^2) = 0$$

Making it homogeneous, we get

$$(x^2 + y^2) - 2(hx + ky)(x/2h + y/2k)$$

$$-(h^2 + k^2 - a^2)(x/2h + y/2k)^2 = 0$$

If these lines are perpendicular, then $A + B = 0$

$$\left[1 - 1 - \frac{(h^2 + k^2 - a^2)}{4h^2}\right] + \left[1 - 1 - \frac{(h^2 + k^2 - a^2)}{4k^2}\right] = 0$$

or $$(h^2 + k^2 - a^2)\frac{(h^2 + k^2)}{2h^2k^2} = 0$$

$\therefore \quad h^2 + k^2 - a^2 = 0$ or $h^2 + k^2 = a^2$

28. Since m_1 and m_2 are the roots of the equation

$$x^2 + (\sqrt{3} + 2)x + (\sqrt{3} - 1) = 0,$$

$\therefore \quad m_1 + m_2 = -(\sqrt{3} + 2)$ and $m_1 m_2 = (\sqrt{3} - 1)$

$\therefore \quad m_1 - m_2 = \sqrt{(m_1 + m_2)^2 - 4m_1 m_2}$

$$= \sqrt{3 + 4 + 4\sqrt{3} - 4\sqrt{3} + 4} = \sqrt{11}$$

The coordinates of the vertices of the given triangle are

$$(0,0), \left(\frac{2}{m_1}, 2\right) \text{ and } \left(\frac{2}{m_2}, 2\right)$$

So, the area of the triangle

$$= \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 2/m_1 & 2 & 1 \\ 2/m_2 & 2 & 1 \end{vmatrix} = \frac{1}{2}\left|\frac{4}{m_1} - \frac{4}{m_2}\right|$$

$$= 2\frac{|m_2 - m_1|}{|m_1 m_2|} = 2\frac{\sqrt{11}}{\sqrt{3} - 1}$$

$$= 2 \cdot \sqrt{11}\frac{(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)} = (\sqrt{33} + \sqrt{11})$$

29. $L_1(-1, -1) = 3(-1) - 8(-1) - 7 < 0$

$L_2(3, 7) = 3 \times 3 - 8 \times 7 - 7 < 0$

Hence (–1, –1) and (3, 7) lie on the same side of the line.

30. The family of lines parallel to $3x + 4y + 5 = 0$ is

$$3x + 4y + k = 0$$

31. $PA^2 = PB^2 = 26 = AB^2$

$$PA^2 = PB^2 \Rightarrow 5h + k = 6 \quad \text{(i)}$$

$$PA^2 = 26 \quad \text{(ii)}$$

$$\Rightarrow \quad (h-3)^2 + (k-4)^2 = 26$$

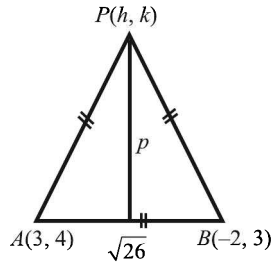


Fig. 18.40

Eliminating k between (i) and (ii), we get

$$2h^2 - 2h - 1 = 0$$

$$\therefore \quad h = \frac{1+\sqrt{3}}{2} \quad \text{or} \quad h = \frac{1-\sqrt{3}}{2}$$

$$\therefore \quad k = \frac{7-5\sqrt{3}}{2} \quad \text{or} \quad \frac{7+5\sqrt{3}}{2} \quad \text{by (i)}$$

32. The line is $2x - 3y + K = 0$.

It passes through (1, 1), so $2 - 3 + K = 0$

$\therefore \quad K = 1$

$\therefore$ The line is $2x - 3y + 1 = 0$.

33. Let the equations of the lines given by $x^2 + 2hxy + y^2 = 0$ be $y = m_1 x$ and $y = m_2 x$. Since these make an angle α with $y + x = 0$ whose slope is -1,

$$\frac{m_1 + 1}{1 - m_1} = \tan\alpha = \frac{-1 - m_2}{1 - m_2}$$

$$\Rightarrow \quad m_1 + m_2 = \frac{(\tan\alpha - 1)^2 + (\tan\alpha + 1)^2}{\tan^2\alpha - 1}$$

$$= -\frac{2\sec^2\alpha \times \cos^2\alpha}{\cos 2\alpha} = -2\sec 2\alpha = -2h$$

$$\Rightarrow \quad \sec 2\alpha = h$$

$$\Rightarrow \quad \cos 2\alpha = \frac{1}{h} \Rightarrow 2\cos^2\alpha - 1 = \frac{1}{h}$$

$$\Rightarrow \quad \cos\alpha = \sqrt{\frac{1+h}{2h}} \text{ and } \cot\alpha = \sqrt{\frac{h+1}{h-1}}$$

34. $x^{2/3} + y^{2/3} = a^{2/3}$

Differentiating, we get

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{y^{1/3}}{x^{1/3}}. \text{ At } (a/8, a/8), \frac{dy}{dx} = -\frac{a^{1/3}}{a^{1/3}} = -1$$

The equation of the tangent at $(a/8, a/8)$ is

$$y - \frac{a}{8} = -(x - a/8) \quad \Rightarrow \quad x + y - \frac{a}{4} = 0$$

Sum of intercepts $= \frac{a}{4} + \frac{a}{4} = \frac{a}{2} = 2$ (given)

$\therefore \quad a = 4$

35. $x + 1 = 0$, $y + 1 = 0$, $x + ay - 3 = 0$ are concurrent. The first two intersect at $(-1, -1)$, which will satisfy the 3rd $\Rightarrow a = -4$.

36. The lines are $\perp$ if $\beta - \alpha = \frac{\pi}{2}$ or $\alpha - \beta = \frac{\pi}{2}$, i.e.

if $$|\alpha - \beta| = \frac{\pi}{2}$$

37. $ax \pm by \pm c = 0$

$$\Rightarrow \quad \frac{x}{\pm c/a} + \frac{y}{\pm c/b} = 1 \text{ which meets the axes at}$$

$$A\left(\frac{c}{a}, 0\right), C\left(-\frac{c}{a}, 0\right), B\left(0, \frac{c}{b}\right), D\left(0, -\frac{c}{b}\right)$$

Therefore, the diagonals AC and BD of the quadrilateral $ABCD$ are perpendicular. Hence, it is a rhombus whose area is given by

$$= \frac{1}{2}AC \times BD = \frac{1}{2} \times \frac{2c}{a} \times \frac{2c}{b} = \frac{2c^2}{ab}$$

38. Solving $x - 2y = 1$ and $x + 3y = 2$, we get $x = 7/5$, $y = 1/5$.

The equation of the line parallel to $3x + 4y = 0$ is

$$3x + 4y + k = 0$$

It passes through (7/5, 1/5).

$$\therefore \quad \frac{21}{5} + \frac{4}{5} + k = 0 \quad \Rightarrow \quad k = -5$$

$\therefore$ The required line is $3x + 4y - 5 = 0$.

39. $(a-1)^2 + (1-b)^2 = a^2 + 1 = b^2 + 1 = (\text{side})^2$

The last two imply $a = b$ or $a = -b$

The first two imply $b^2 - 2b - 2a + 1 = 0$

If $a = -b$, then $b^2 + 1 = 0$ (not possible)

If $a = b$, then $b^2 - 4b + 1 = 0$

$\therefore \quad b = 2 + \sqrt{3}, 2 - \sqrt{3}$

Since $b < 1 \quad \therefore \quad b = 2 - \sqrt{3} = a$

40. R divides PQ externally in the ratio 2 : 1.

$\therefore$ R is $\left(\frac{4+3}{2-1}, \frac{2-4}{2-1}\right)$, i.e. $(7, -2)$

41. Let $P(3, -4)$ be the foot of the perpendicular from the origin O on the required line. Then the slope of OP $= \frac{-4-0}{3-0} = \frac{-4}{3}$, and therefore the slope of the required line is 3/4. Hence its equation is

$$y + 4 = \frac{3}{4}(x - 3) \Rightarrow 3x - 4y = 25$$

42. The line through (1, 2) is

$$y-2=m(x-1) \qquad \text{(i)}$$

$\Rightarrow\ mx-y-m+2=0$

$\therefore$ D distance from (3, 1) $=\dfrac{3m-1-m+2}{\sqrt{m^2+1}}$

$\Rightarrow\ D=\dfrac{2m+1}{\sqrt{m^2+1}}$

$\therefore\ \dfrac{dD}{dm}=\dfrac{2-m}{(m^2+1)^{3/2}}=0$ for max/min

$\Rightarrow\ m=2$

Also $\left.\dfrac{d^2D}{dm}\right]_{m=2}$ is –ve

Putting in (i), $y-2=2(x-1)$

$\Rightarrow\ y=2x.$

43. The given equation can be rewritten as $(2x-3y+2)(3x+4y-7)=0$ and its factors are $(2x-3y+2)$ and $(3x+4y-7)$. Now use the formula for angle between them, i.e.

$$\tan\theta=\frac{m_1\sim m_2}{1+m_1m_2}$$

Alternative method:

You may use $\tan\theta=\dfrac{2\sqrt{h^2-ab}}{a+b}$

$$a=6, b=-12, h=-1/2$$

$\therefore\ \tan\theta=\left|\dfrac{\sqrt{289}}{-6}\right|=\dfrac{17}{6}$

$\therefore\ \theta=\tan^{-1}\dfrac{17}{6}$

44. Since a, b, c are in A.P.,

$\therefore\qquad 2b=a+c$ or $a-2b+c=0$

$\therefore$ The given line passes through (1, –2).

45. Let ϕ be the angle between the lines represented by

$$x^2+2xy\sec\theta+y^2=0 \qquad \text{(i)}$$

Here, $a=1, b=1, h=\sec\theta$

Hence $\tan\phi=\dfrac{2\sqrt{(\sec^2\theta-1\times1)}}{1+1}$

$\Rightarrow\ \tan\phi=\dfrac{2\sqrt{(\sec^2\theta-1)}}{2}=\tan\theta$

$\therefore\ \phi=\theta$

Hence, the angle between the lines represented by (i) is θ.

46. The given family of lines is:

$$\cos\theta(2x+3y-5)+\sin\theta(3x-5y+2)=0 \qquad \text{(i)}$$

and $\qquad a(x+y-1)+b(2x-3y+1)=0 \qquad \text{(ii)}$

$\Rightarrow\quad 2x+3y-5=0$ and $3x-5y+2=0 \qquad \text{(iii)}$

Solving, $x=1, y=1$

And $x+y-1=0$ and $2x-3y+1=0$

Solving we get $x=\dfrac{2}{5}, y=\dfrac{3}{5}$

Hence $x_1=1, y_1=1, x_2=\dfrac{2}{5}, y_2=\dfrac{3}{5}$

47. If the sides are a, b, c respectively, then

$$a^2+b^2=25(x^2+y^2)+48xy$$

$$c^2=25(x^2+y^2)+50xy$$

Clearly $a^2+b^2<c^2$, $\because$ x and y both are +ve

$\therefore\qquad \cos C=\dfrac{a^2+b^2-c^2}{2ab}=-\text{ve}$

Hence $\angle C$ is obtuse.

48. The angle between $y=x$ and $y=0$ is $45°=\dfrac{\pi}{4}$.

49. Let $O(0, 0)$ be the orthocentre; $A(h, k)$ the third vertex; $B(-2,3)$ and $C(5,-1)$ the other two vertices. Then the slope of the line through A and O is k/h, while the line through B and C has the slope –4/7. By the property of the orthocentre, these two lines must be perpendicular, so we have

$$\left(\frac{k}{h}\right)\left(-\frac{4}{7}\right)=-1\ \Rightarrow\ \frac{k}{h}=\frac{7}{4} \qquad \text{(i)}$$

Also $\qquad \dfrac{5-2+h}{3}+\dfrac{-1+3+k}{3}=7$

$\Rightarrow\qquad h+k=16 \qquad \text{(ii)}$

which is not satisfied by the points given in option (a), (b) or (c).

50. The equation of the line parrallel to the axis is $y = k$.

It meets $y=\sqrt{x}$, if $k=\sqrt{x}\ \Rightarrow x=k^2$.

$\therefore$ The point of intersection is (k^2, k).

Now, as $y=\sqrt{x}$, $\dfrac{dy}{dx}=\dfrac{1}{2\sqrt{x}}$

At $(k^2,k)\ \dfrac{dy}{dx}=\dfrac{1}{2k}=\tan45°$ $\qquad\therefore\ k=\dfrac{1}{2}$

[$\because$ line crosses the curve at an angle of 45°]

$\therefore$ The required line is $y=\dfrac{1}{2}$.

51. The two parallel lines are

$$(x+\sqrt{2}y+p)(x+\sqrt{2}y+q)=0$$

where $p + q = 4$ and $pq = 1$

$\therefore \quad p-q=\sqrt{(16-4)}=2\sqrt{3}$

$\therefore$ The required distance $=\frac{p}{\sqrt{3}}-\frac{q}{\sqrt{3}}=\frac{1}{\sqrt{3}}(2\sqrt{3})=2$

52. Since a, x_1, x_2 are in G.P. with common ratio r,

$\therefore \quad x_1=ar, x_2=ar^2$

Similarly, $y_1=bs, y_2=bs^2$

$\therefore$ The required area $=\frac{1}{2}\begin{vmatrix} a & b & 1 \\ ar & bs & 1 \\ ar^2 & bs^2 & 1 \end{vmatrix}$

$$=\frac{1}{2}\begin{vmatrix} a & b & 1 \\ ar-a & bs-b & 0 \\ ar^2-a & bs^2-b & 0 \end{vmatrix}$$

(Operate R_3-R_1, R_2-R_1)

$$=\frac{1}{2}[a(r-1)b(s^2-1)-b(s-1)a(r^2-1)]$$

$$=\frac{1}{2}ab(r-1)(s-1)[(s+1)-(r+1)]$$

$$=\frac{1}{2}ab(r-1)(s-1)(s-r)$$

53. The lines are $y=x$, $y=-x$ and $x+2y=2$ as shown in the figure.

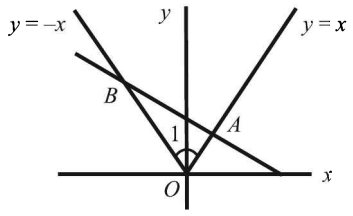


Fig. 18.41

Solving $y=x, x+2y=2$, the point $A=\left(\frac{2}{3},\frac{2}{3}\right)$

$\therefore \quad OA=\sqrt{\frac{4}{9}+\frac{4}{9}}=\frac{2\sqrt{2}}{3}$

Solving $y=-x, x+2y=2$, the point $B=(-2,2)$

$\therefore \quad OB=\sqrt{4+4}=2\sqrt{2}$

$\therefore$ Area of $\Delta OAB=\frac{1}{2}\times OA\times OB=\frac{1}{2}\times\frac{2\sqrt{2}}{3}\times 2\sqrt{2}=\frac{4}{3}$

54. Here $a = 2$, $b = 3$ and $h=\frac{5}{2}$

$\therefore \quad \tan\theta=\frac{2\sqrt{h^2-ab}}{a+b}=\frac{2\sqrt{\frac{25}{4}-(2)(3)}}{2+3}=\frac{1}{5}$

$\Rightarrow \quad \theta=\tan^{-1}\frac{1}{5}$

But $\theta=\tan^{-1}m$

$\therefore \quad m=\frac{1}{5}$

55. Solving the given equations in pairs, we get the vertices as $A(1,1)$, $B(2,-2)$, $C(-2,2)$. Clearly

$$AB=AC=\sqrt{10}, BC=4\sqrt{2}$$

$$AB^2+AC^2\neq BC^2$$

56. Point of intersection of $x = 0$, $y=m_1x+c_1$ is $(0, c_1)$

Point of intersection of $x = 0$, $y=m_2x+c_2$ is $(0,c_2)$

Point of intersection of $y=m_1x+c_1, y=m_2x+c_2$ is

$$\left(\frac{c_1-c_2}{m_2-m_1},\frac{m_2c_1-m_1c_2}{m_2-m_1}\right)$$

$\therefore$ The required area

$$=\pm\frac{1}{2}\begin{vmatrix} 1 & 0 & c_1 \\ 1 & 0 & c_2 \\ 1 & \frac{c_1-c_2}{m_2-m_1} & \frac{m_2c_1-m_1c_2}{m_2-m_1} \end{vmatrix}$$

$$=\pm\frac{1}{2}\frac{c_1-c_2}{m_2-m_1}[-c_2+c_1]$$

$$=\pm\frac{1}{2}\frac{(c_1-c_2)^2}{m_2-m_1}=\frac{1}{2}\cdot\frac{(c_1-c_2)^2}{m_1-m_2}$$

57. The equation of AO is $2x+3y-1+\lambda(x+2y-1)=0$

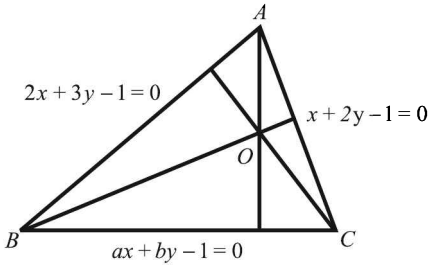


Fig. 18.42

where $\lambda = -1$ since the line passes through the origin, so, $x + y = 0$.

Since AO is perpendicular to BC, $(-1)\left(-\frac{a}{b}\right)=-1$

$\therefore \quad a = -b$

Similarly, $(2x+3y-1)+\mu(ax-ay-1)=0$ will be the equation of BO for $\mu=-1$.

BO is perpendicular to

$$AC \Rightarrow \left[-\frac{(2-a)}{3+a}\right]\left(-\frac{1}{2}\right)=-1$$

$\therefore \quad a=-8,\ b=8$

58. Since the perpendicular distances from (0, 0) are

$$\frac{10}{5}=2,\ \frac{26}{13}=2,\ \frac{50}{25}=2$$

$\therefore$ (0, 0) is equidistant from the given lines.

59. Let the factors be $(x-2y+p)(2x-y+q)$.

Comparing the coefficients of x, y and constant, we get $2p+q=-3,\ p+2q=-3,\ pq=1$. These give $p=q=-1$.

Hence, the two lines are $x-2y-1=0, 2x-y-1=0$ whose point of intersection is clearly $\left(\frac{1}{3},-\frac{1}{3}\right)$.

60. Clearly $\Delta=\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}=\begin{vmatrix} 3 & -\frac{11}{2} & -\frac{7}{2} \\ -\frac{11}{2} & 10 & \frac{13}{2} \\ -\frac{7}{2} & \frac{13}{2} & K \end{vmatrix}=0$

[Here $a=3, b=10, c=K, f=-\frac{13}{2}, g=-\frac{7}{2}, h=-\frac{11}{2}$]

if $3\left(10K-\frac{169}{4}\right)+\frac{11}{2}\left(\frac{11K}{2}+\frac{91}{4}\right)-\frac{7}{2}\left(-\frac{143}{4}+\frac{70}{4}\right)=0$

i.e. if $30K-\frac{507}{4}-\frac{121K}{4}+\frac{1001}{8}+\frac{1001}{8}-\frac{7}{2}\times 35=0$

i.e. if $30K-\frac{121K}{4}=\frac{507}{4}+\frac{245}{2}-\frac{1001}{4}$

$$=\frac{507+490-1001}{4}=-\frac{4}{4}-\frac{K}{4}=-1 \Rightarrow K=4$$

61. The pair $y^2-4xy-x^2=0$ contains lines which are at right angle because $a + b = 0$.

62. $(x+7y)^2+4\sqrt{2}(x+7y)-42=0$

$\Rightarrow \quad (x+7y-3\sqrt{2})(x+7y+7\sqrt{2})=0$

$\therefore$ Distance between these lines

$$=\left|\frac{7\sqrt{2}-(-3\sqrt{2})}{\sqrt{1+49}}\right| =\frac{10\sqrt{2}}{5\sqrt{2}}=2.$$

63. The first equation is satisfied by $x = 2$.

It can be written as $(x-2)(x^2+x+1)=0, \ \therefore \quad x=2$

The second equation can be written as

$$4(x^2-3x+2)+y^2(x^2-3x+2)=0$$

or $(x-1)(x-2)(y^2+4)=0$.

$\therefore \quad x = 1, x = 2$

Hence, there are only two lines $x = 1$, $x = 2$. Therefore, no triangle is formed.

Note: $x^2+x+1=0$ and $y^2+4=0$ do not give real values of x and y.

64. The required angle is given by

$$\tan\theta=\frac{2\sqrt{h^2-ab}}{a+b}.$$

$$=\frac{2\sqrt{\left(-\frac{7}{2}\right)^2-(2)(3)}}{2+3}$$

$$=\frac{2}{5}\sqrt{\left(\frac{49}{4}-6\right)}=\frac{2}{5}\cdot\frac{5}{2}=1 \quad \therefore \ \theta=45°.$$

65. Line AB will be farthest from the origin if OP is right-angled to the line drawn.

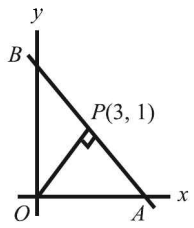


Fig. 18.43

$$m_{OP}=\frac{1}{3} \Rightarrow m_{AB}=-3$$

Thus, the equation of AB is

$$(y-1)=-3(x-3)$$

$$\Rightarrow \ A\equiv\left(\frac{10}{3},0\right),\ B\equiv(0,10)$$

$$\Rightarrow \ \Delta OAB=\frac{1}{2}(OA)(OB)=\frac{1}{2}\times\frac{10}{3}\times 10=\frac{100}{6} \text{ sq. units}$$

66. Let m be the slope of the line through A.

Slope of $AC=\frac{5-3}{7-1}=\frac{2}{6}=\frac{1}{3}$

$$\therefore \quad \tan 45°=\left|\frac{m-\frac{1}{3}}{1+m,\frac{1}{3}}\right| \Rightarrow 1=\left|\frac{3m-1}{m+3}\right|$$

$\Rightarrow\ 3m-1=m+3$ or $3m-1=-m-3$

$\Rightarrow\ 2m=4$ or $4m=-2$

$\Rightarrow\ m=2$ or $m=-\frac{1}{2}$

$\therefore$ The equation of the side AD through (1, 3) with slope 2 is $y-3=2(x-1)\ \Rightarrow\ 2x-y+1=0$.

The equation of the side AB through $A(1, 3)$ with slope $-(1/2)$ is

$$y-3=-\frac{1}{2}(x-1)\quad\Rightarrow\quad 2y-6=-x+1$$

$\Rightarrow\ x+2y-7=0$

67. Since the lines are to be parallel, the two factors will have the same terms of x and y.

$\therefore\quad \left(\sqrt{a}x+\sqrt{b}y+p\right)\left(\sqrt{a}x+\sqrt{b}y+q\right)$

Comparing the coefficients, we have

$$2h=2\sqrt{ab},\sqrt{a}(p+q)=2g$$

$$\sqrt{b}(p+q)=2f, pq=c$$

From the 1st, we have $h^2=ab$ or $\frac{a}{h}=\frac{h}{b}=\lambda$ (say) and on dividing the 2nd and 3rd, we have

$$\sqrt{\frac{a}{b}}=\frac{g}{f}\qquad\therefore\quad bg^2=af^2$$

or $\frac{h}{\lambda}g^2=h\lambda f^2\qquad\therefore\quad \lambda^2=\frac{g^2}{f^2}$

or $\lambda=\frac{g}{f}\qquad\therefore\quad \frac{a}{h}=\frac{h}{b}=\frac{g}{f}=\lambda$

68. The equation of the line BC is $x+y+4=0$.

$\therefore$ The equation of a line parallel to BC is $x + y + k = 0$

This is at a distance 1/2 from the origin.

$$\therefore\qquad \left|\frac{k}{\sqrt{2}}\right|=\frac{1}{2}$$

$$\therefore\qquad k=\pm\frac{1}{\sqrt{2}}$$

Since BC and the required line are on the same side of the origin,

$$\therefore\qquad k=\frac{1}{\sqrt{2}}$$

Hence, the required line is $x+y+\frac{1}{\sqrt{2}}=0$.

69. The sides are $y = 1$ and the pair $x^2+7xy+2y^2=0$. Clearly, one vertex is (0, 0) and the y-coordinate of each of the other two vertices is 1. Putting $y = 1$ in the second equation, we get $x^2+7x+2=0$.

If x_1 and x_2 are the roots, then $x_1 + x_2 = -7$.

$\therefore$ The centroid $=\left(\frac{0+x_1+x_2}{3},\frac{0+1+1}{3}\right)=\left(-\frac{7}{3},\frac{2}{3}\right)$.

70. Here $a = 1$, $b = 12$ and $h = -7/2$

$$\therefore\quad \tan\theta=\frac{2\sqrt{h^2-ab}}{a+b}=\frac{2\sqrt{\frac{49}{4}-12}}{1+12}=\frac{1}{13}$$

$$\Rightarrow\ \theta=\tan^{-1}\frac{1}{13}=\sin^{-1}\frac{1}{\sqrt{170}}$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $AB\perp$ to L is $x+5y=k$

Area of ΔOAB = 5

$$\therefore\quad \frac{1}{2}ab=\pm5$$

or $$k\cdot\frac{k}{5}=\pm10$$

$\therefore\ k^2=50$ or $k^2=-50$ (rejected)

$\therefore\ k=\pm5\sqrt{2}$

2. Suppose we rotate the coordinate axes in the anticlockwise direction through an angle α. The equation of the line L with respect to old axes is $\frac{x}{a}+\frac{y}{b}=1$. In this equation, replacing x by $\cos\alpha - y\sin\alpha$ and $xby\sin\alpha+y\cos\alpha$, the equation of the line with respect to the new axes is

$$\frac{x\cos\alpha-y\sin\alpha}{a}+\frac{x\sin\alpha+y\cos\alpha}{b}=1$$

$$\Rightarrow\quad x\left(\frac{\cos\alpha}{a}+\frac{\sin\alpha}{b}\right)+y\left(\frac{\cos\alpha}{b}-\frac{\sin\alpha}{a}\right)=1\qquad\text{(i)}$$

The intercepts made by (i) on the coordinate axes are given as p and q.

Therefore, $\frac{1}{p}=\frac{\cos\alpha}{a}+\frac{\sin\alpha}{b}$ and $\frac{1}{q}=\frac{\cos\alpha}{b}-\frac{\sin\alpha}{a}$

Squaring and adding, we get

$$\frac{1}{p^2}+\frac{1}{q^2}=\frac{1}{a^2}+\frac{1}{b^2}.$$

3. The equation of the line parallel to the x-axis is $y = k$. It passes through the mid-point of the join of A (1, 4) and $B(-2, 6)$, i.e. through $\left(\frac{1-2}{2}, \frac{4+6}{3}\right)$, i.e. $\left(\frac{-1}{2}, 5\right)$.

$\therefore \quad k = 5$

$\therefore$ The required line is $y = 5$.

4. The equation of the curve is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \qquad \text{(i)}$$

The equation of the line is

$$lx + my = 1 \qquad \text{(ii)}$$

The equation of the lines joining the origin to the points of intersection of (i) and (ii) is obtained by making (i) homogeneous with the help of (ii).

$$\therefore \quad x^2 + y^2 + 2(gx + fy)(lx + my) + c(lx + my)^2 = 0$$

or $$(1 + 2gl + cl^2)x^2 + 2(gm + fl + clm)xy + (1 + 2fm + cm^2)y^2 = 0$$

These lines will be at right angles if the coefficient of x^2 + the coefficient of $y^2 = 0$.

i.e. if $1 + 2gl + cl^2 + 1 + 2fm + cm^2 = 0$

or $c(l^2 + m^2) + 2(gl + fm + 1) = 0$

5. One of the lines will be $y = x \tan 45°$, i.e. $y = x$.

Putting $y = x$ in the given equation, we get

$$a + b = -2h$$

6. $D = 2\Delta$

$$\therefore \quad D^2 = 4\Delta^2 = 4\left(\frac{\sqrt{3}}{4}a^2\right)^2 = \frac{3}{4}a^4$$

$$\because \quad \Delta = \frac{1}{2}a \cdot a \sin 60° = \frac{\sqrt{3}}{4}a^2$$

7. Let the vertices B and C lie on the given line.

$$OD = \frac{2\sqrt{2}}{\sqrt{2}} = 2$$

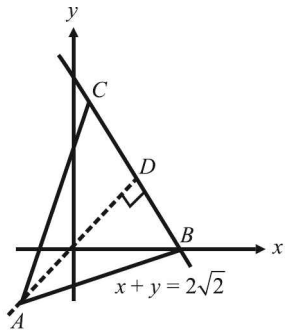


Fig. 18.44

The equation of OD is $y = x \Rightarrow x = y = \sqrt{2}$ (for point D). Also $BD = OD$.

$\tan 60° = 2\sqrt{3}$ for the coordinates of B and C. Using the parametric equation of the line, we get

$$\frac{x - \sqrt{2}}{-\frac{1}{\sqrt{2}}} = \frac{y - \sqrt{2}}{\frac{1}{\sqrt{2}}} = \pm 2\sqrt{3}$$

$$\Rightarrow \quad C \equiv \left(\sqrt{2} + \sqrt{6}, \sqrt{2} - \sqrt{6}\right)$$

$$D \equiv \left(\sqrt{2} - \sqrt{6}, \sqrt{2} + \sqrt{6}\right)$$

8. Let $y = nx + c$ be the variable line.

Since it passes throught (a, b),

$$\therefore \quad b = ma + c \Rightarrow c = b - ma$$

$$\therefore \quad y = mx + b - ma$$

Let C be (h, k).

k = y-coordinate of the point, where the variable line meets the y-axis.

$$\therefore \quad k = b - ma \qquad \text{(i)}$$

and h = coordinate of the point, where the variable line meets the x-axis.

$$\therefore \quad h = a - \frac{b}{m} \qquad \text{(ii)}$$

From (i) and (ii), we get

$$\frac{a}{h} + \frac{b}{m} = 1$$

$\Rightarrow$ The locus of C is $\frac{a}{x} + \frac{b}{y} = 1$

9. Clearly, the figure $ABCD$ is a parallelogram and hence its area is

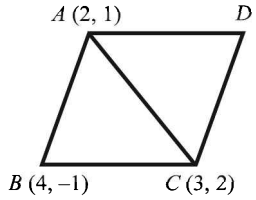


Fig. 18.45

$$2\Delta ABC = 2 \cdot \frac{1}{2}[2(-1-2) + 4(2-1) + 3(1+1)]$$

$$= -6 + 4 + 6 = 4 \text{ sq. units}$$

10. We have

$$y = \cos x \cos(x+2) - \cos^2(x+1)$$

$$y = \frac{1}{2}[2\cos x \cos(x+2) - 2\cos^2(x+1)]$$

$$= \frac{1}{2}[\cos(2x+2) + \cos 2 - 1 - \cos(2x+2)]$$

$$= \frac{1}{2}(\cos 2 - 1) = \frac{1}{2}(1 - 2\sin^2 1 - 1)$$

$$= \sin^2 1$$

which is a straight line passing through $\left(\frac{\pi}{2}, -\sin^2 1\right)$ and parallel to the x-axis.

11. The mid-point of PQ is the same as that of AB in view of the given conditions, i.e. $\left(\frac{1}{2}, 3\right)$

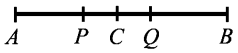


Fig. 18.46

12. The equation of the bisectors of the angles between the lines represented by $ax^2 + 2hxy - ay^2 = 0$ is

$$\frac{x^2 - y^2}{a - (-a)} = \frac{xy}{h} \Rightarrow hx^2 - 2axy - hy^2 = 0$$

Also $bx^2 + 2gxy - by^2 = 0$ represents the bisectors of the angles between the lines given by

$$ax^2 + 2hxy - ay^2 = 0$$

$\therefore$ Comparing the coefficients, we have

$$\frac{h}{b} = \frac{-2a}{ag} = \frac{-h}{-b}$$

$\Rightarrow \quad gh + ab = 0$

13. $A\left(-\frac{c}{a}, 0\right), B\left(0, -\frac{c}{a}\right)$. If P is the mid-point of AB, then P is $\left(-\frac{c}{2a}, -\frac{c}{2b}\right)$. $\therefore$ OP is $y = mx$.

or $\quad y = \frac{a}{b}x \quad$ or $\quad ax - by = 0$

14. We have $\qquad 3x + 4y = 1 \qquad$ (i)

and $\qquad (1 + c)x + 3c^2 y = 2 \qquad$ (ii)

Solving (i) and (ii)

$$x = \frac{3c^2 - 8}{9c^2 - 4c - 4}, y = \frac{5 - c}{9c^2 - 4c - 4}$$

Now, $\lim_{c \to 1} \frac{3c^2 - 8}{9c^2 - 4c - 4} = \frac{3 - 8}{9 - 4 - 4} = \frac{-5}{1} = -5$

and $\lim_{c \to 1} \frac{5 - c}{9c^2 - 4c - 4} = \frac{5 - 1}{9 - 4 - 4} = \frac{4}{1} = 4$

Hence, the limiting position of the point of intersection is (–5, 4).

$\therefore \quad h^2 + k^2 - a^2 = 0 \quad$ or $\quad h^2 + k^2 = a^2$

15. Let m be the slope of PQ.

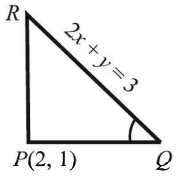


Fig. 18.47

The slope of $PR = -(1/m)$

Also the slope of $QR = -2$

Now ΔPQR is isosceles. [given]

$\Rightarrow \quad \angle PQR = \angle PRQ \Rightarrow \tan\angle PQR = \tan\angle PRQ$

$$\Rightarrow \left|\frac{m+2}{1-2m}\right| = \left|\frac{-(1/m)+2}{1+(2/m)}\right|$$

$$\Rightarrow \left|\frac{m+2}{1-2m}\right| = \left|\frac{-1+2m}{m+2}\right| \Rightarrow (m+2)^2 = (1-2m)^2$$

$\Rightarrow \quad m + 2 = \pm(1 - 2m) \Rightarrow m = 3$ or $1/3$

$\therefore$ The equation of PQ is $y - 1 = 3(x - 2)$

$\Rightarrow \quad 3(x - 2) - (y - 1) = 0$

and the equation of PR is $y - 1 = -\frac{1}{3}(x - 2)$

$\Rightarrow \qquad (x - 2) - 3(y - 1) = 0$

Their joint equation is

$$[3(x-2) - (y-1)][(x-2) + 3(y-1)] = 0$$

$$\Rightarrow \quad 3(x-2)^2 - 3(y-1)^2 + 8(x-2)(y-1) = 0$$

$$\Rightarrow \quad 3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$$

16. The lines are concurrent if $\Delta = 0$ which on expansion gives

$$a^3 + b^3 + c^3 - 3abc = 0$$

or $\quad (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0$

or $\quad (a + b + c) = 0$ because the 2nd factor

$$a^2 + b^2 + c^2 - ab - bc - ca$$

$$= \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

It will be zero only when $a - b = 0, b - c = 0, c - a = 0$ i.e. $a = b = c$.

In this case, the three lines will reduce to a single line $x + y + 1 = 0$.

17. Since the triangle is equilateral, the altitude coincides with the median

$$\angle ABC = 60°$$

Now, $|AD| = \frac{|2 - 1 - 2|}{\sqrt{1+1}} = \frac{1}{\sqrt{2}}$

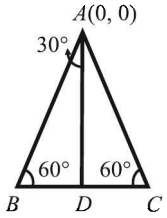


Fig. 18.48

In the right-angled ΔADB, $\frac{AD}{AB} = \sin 60° = \frac{\sqrt{3}}{2}$

$$\Rightarrow \quad AB = AD\left(\frac{2}{\sqrt{3}}\right) = \frac{1}{\sqrt{2}}\left(\frac{2}{\sqrt{3}}\right) = \sqrt{\frac{2}{3}}$$

Hence, the length of the side of the equilateral triangle $= \sqrt{\frac{2}{3}}$.

18. Let $y = mx$ be any line through the origin. It makes an angle of $\pm 60°$ with $Ax + By + C = 0$.

$$\therefore \qquad \tan(\pm 60°) = \frac{m + A/B}{1 - m \cdot A/B}$$

or $\quad \pm\sqrt{3} = \dfrac{mB + A}{B - mA}$

$\therefore \quad 3(B - mA)^2 = (mB + A)^2$

The combined equation of the lines AB and AC is obtained by putting $m = y/x$.

$$\therefore \qquad 3\left(B - \frac{y}{x}A\right)^2 = \left(B \cdot \frac{y}{x} + A\right)^2$$

or $\quad 3(Bx - Ay)^2 = (By + Ax)^2$

which on simplifications reduces to the given equation and hence proved.

Now, if p is perpendicular from A on BC, then

$$p = \frac{C}{\sqrt{(A^2 + B^2)}} \qquad \text{(i)}$$

$$\text{Area of } \Delta = \frac{1}{2} \cdot \frac{BC}{AD} = \frac{1}{2}(2BD) \cdot AD$$

$$= p(\tan 30°)p$$

$$= \frac{1}{\sqrt{3}}p^2 = \frac{C^2}{\sqrt{3} \cdot (A^2 + B^2)} \qquad \text{[By (i)]}$$

19. The given line is $3x - 4y - 5 = 0$ (i)

Any line perpendicular to (i) is $4x + 3y + k = 0$ (ii)

It is passes through (0, 5), then $0 + 15 + k = 0$

$\Rightarrow \quad k = -15$

Putting in (ii), we get

$$4x + 3y - 15 = 0 \qquad \text{(iii)}$$

Solving (i) and (iii), the foot of perpendicular is (3, 1).

20. The reflection of a point $P(\alpha, \beta)$ in the line $ax + by + c = 0$ is the point $Q(\alpha', \beta')$ such that the mid-point of PQ, i.e. $\left(\dfrac{\alpha + \alpha'}{2}, \dfrac{\beta + \beta'}{2}\right)$ lies on the line. PQ is perpendicular to the line $ax + by + c = 0$. Also, the coordinates of Q are given by

$$\frac{\alpha - \alpha'}{a} = \frac{\beta' - \beta}{b} = -2\left(\frac{a\alpha + b\beta + c}{a^2 + b^2}\right)$$

So, Statement II is not true.

The reflection $(-3, 2)$ in the line $x + y = 0$ has the coordinates of (x, y) given by

$$\frac{x + 3}{1} = \frac{y - 2}{1} = -\frac{2(-3 + 2)}{1^2 + 1^2}$$

$\Rightarrow \quad x + 3 = y - 2 = 1 \Rightarrow x = -2$ and $y = 3$

So, Statement I is true.

21. Statement II is true. Using Statement II, the equation of the perpendicular bisector of line joining $A(a, 3)$ and $B(1, 4)$ is given by

$$\sqrt{(x - a)^2 + (y - 3)^2} = \sqrt{(x - 1)^2 + (y - 4)^2}$$

$\Rightarrow \quad 2(a - 1)x - 2y = a^2 - 8$

This has y-intercept -4. So, it passes through $(0, -4)$

$$\therefore \qquad a^2 - 8 = 8 \Rightarrow a = \pm 4$$

Hence, Statement I is true. Also, Statement II is a correct explanation for Statement I.

22. The area of the triangle formed by the lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ is given by

$$\Delta = \frac{1}{2C_1C_2C_3}\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2$$

where C_1, C_2, C_3 are cofactors of c_1, c_2, c_3 respectively in the matrix

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

If the lines are concurrent, then

$$\Delta = 0 \Rightarrow \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

So, Statement II is true. Also, Statement I is true and Statement II is a correct explanation for Statement I.

23. Statement II is true and it is the correct explanation for Statement I as $y-x+12=0$ is one of the angle bisectors of the given lines. So, Statement I also true.

24. Let $A(4, 0)$, $B(-1, -1)$ and $C(3, 5)$ be the given points.

Then $|AB| = \sqrt{(-1-4)^2+(-1-0)^2}$

$= \sqrt{25+1} = \sqrt{26}$

$|BC| = \sqrt{(3+1)^2+(5+1)^2}$

$= \sqrt{16+36} = \sqrt{52}$

and $|CA| = \sqrt{(4-3)^2+(0-5)^2}$

$= \sqrt{1+25} = \sqrt{26}$

Clearly, $|AB| = |CA| \Rightarrow$ triangle is isosceles and $BC^2 = AB^2 + CA^2$ [$\because 52 = 26+26$]
$\Rightarrow$ triangle is right-angled.

25. The slopes of OA, OB and OC are b/a each and hence the points are collinear

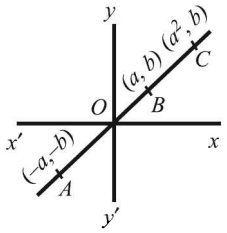


Fig. 18.49

26. Let $A(1,\sqrt{3}), B(0,0)$ and $C(2,0)$ be the given vertices.

Now $|AB| = \sqrt{(0-1)^2+(0-\sqrt{3})^2} = \sqrt{1+3} = 2$

$|BC| = \sqrt{(2-0)^2+(0-0)^2} = \sqrt{4+0} = 2$

and $|CA| = \sqrt{(2-1)^2+(0-\sqrt{3})^2} = \sqrt{1+3} = 2$

Thus, ΔABC is equilateral.
$\therefore$ Incentre and centroid coincide.

But the centroid of ΔABC is $\left(\frac{1+0+2}{3}, \frac{\sqrt{3}+0+0}{3}\right)$

i.e. $\left(1, \frac{1}{\sqrt{3}}\right)$. Hence, the incentre is also $\left(1, \frac{1}{\sqrt{3}}\right)$.

27. $\sqrt{3}x + y = 0$ makes an angle of 120° with OX and $\sqrt{3}x - y = 0$ makes an angle of 60° with OX. So, the required line is $y - 2 = 0$.

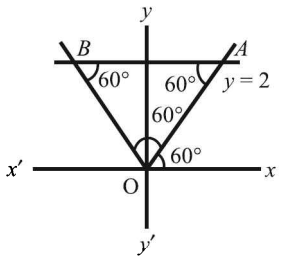


Fig. 18.50

28. The bisectors of $24x+7y-20=0$ and $4x-3y-2=0$ are $\frac{24x+7y-20}{\sqrt{576+49}} = \pm\frac{4x-3y-2}{\sqrt{16+9}}$.

$\Rightarrow \frac{24x+7y-20}{25} = \pm\frac{4x-3y-2}{5}$

Taking +ve sign, $\frac{24x+7y-20}{25} = \frac{4x-3y-2}{5}$

$\Rightarrow \frac{24x+7y-20}{5} = \frac{4x-3y-2}{1}$

$\Rightarrow 24x+7y-20 = 20x-15y-10$

$\Rightarrow 4x+22y-10=0 \Rightarrow 2x+11y-5=0$

29. The coordinates of A and B are as shown:
The equation of the diagonal AB is

$$y - a\sin\alpha = \frac{a\cos\alpha - a\sin\alpha}{-a\sin\alpha - a\cos\alpha}(x - a\cos\alpha)$$

$$\Rightarrow \quad y - a\sin\alpha = -\frac{\cos\alpha - \sin\alpha}{\cos\alpha + \sin\alpha}(x - a\cos\alpha)$$

$$\Rightarrow \quad y(\cos\alpha+\sin\alpha) - a(\sin\alpha\cos\alpha + \sin^2\alpha)$$
$$= -(\cos\alpha - \sin\alpha)x + a\cos\alpha(\cos\alpha - \sin\alpha)$$

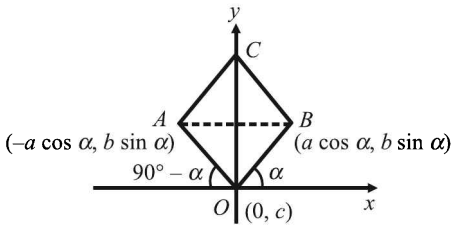


Fig. 18.51

$\Rightarrow y(\cos\alpha+\sin\alpha) + x(\cos\alpha - \sin\alpha) = a$

30. The given equation is $x^2 - 2pxy - y^2 = 0$.
On comparing with $ax^2 + 2hxy + by^2 = 0$, we get

$$a = 1, b = -1, 2h = -2p \Rightarrow h = -p$$

Equation of the bisector of angles is

$$\frac{x^2-y^2}{a-b} = \frac{xy}{h} \quad \Rightarrow \quad \frac{x^2-y^2}{1+1} = \frac{xy}{-p}$$

$$\Rightarrow x^2 - y^2 = -\frac{xy}{p} \quad \Rightarrow \quad x^2 + \frac{2xy}{p} - y^2 = 0 \qquad \text{(i)}$$

But the given equation of the bisector of angle is

$$x^2 - 2qxy - y^2 = 0 \qquad \text{(i)}$$

On comparing Eqs. (i) and (ii), we get

$$\frac{2}{p} = -2q \Rightarrow pq = -1$$

31. Let $P(x, y)$ be any point.

Let $A(a_1, b_1)$ and $B(a_2, b_2)$ be the given points.

By the question, $|PA| = |PB| \Rightarrow PA^2 = PB^2$

$$\Rightarrow (x-a_1)^2 + (y-b_1)^2 = (x-a_2)^2 + (y-b_2)^2$$

$$\Rightarrow a_1^2 + b_1^2 - 2a_1x - 2b_1y = a_2^2 + b_2^2 - 2a_2x - 2b_2y$$

$$\Rightarrow 2(a_1 - a_2)x + 2(b_1 - b_2)y + a_2^2 + b_2^2 - a_1^2 - b_1^2 = 0$$

$$\Rightarrow (a_1 - a_2)x + (b_1 - b_2)y + \frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2) = 0$$

If it is the same as $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then

$$c = \frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$$

32. If (α, β) is the centroid, then

$$\alpha = \frac{a\cos t + b\sin t + 1}{3} \quad \text{and} \quad \beta = \frac{a\sin t - b\cos t}{3}$$

$$a\cos t + b\sin t = 3\alpha - 1 \quad \text{and} \quad a\sin t - b\cos t = 3\beta$$

Squaring and adding, we get

$$a^2 + b^2 = (3\alpha - 1)^2 + (3\beta)^2$$

Hence, locus of (α, β) is $(3x-1)^2 + (3y)^2 = a^2 + b^2$.

33. Given lines are $3x - 4y + 7 = 0$

or $$y = \frac{3}{4}x + \frac{7}{4} \qquad \text{(i)}$$

and $$12x + 5y - 2 = 0$$

or $$y = -\frac{12}{5}x + \frac{2}{5} \qquad \text{(ii)}$$

The equations of the bisectors of the angles between these lines are

$$\frac{3x - 4y + 7}{\sqrt{3^2 + 4^2}} = \pm\frac{12x + 5y - 2}{\sqrt{12^2 + 5^2}}$$

$\therefore$ Required equation of the bisector of the acute angle between these lines is

$$\frac{3x - 4y + 7}{5} = \frac{12x + 5y - 2}{13}$$

$$\Rightarrow 39x - 52y + 91 = 60x + 25y - 10$$

$$\therefore \quad 21x + 77y - 101 = 0$$

34. Let C be (α, β).

The centroid is

$$\left(\frac{2 - 2 + \alpha}{3}, \frac{-3 + 1 + \beta}{3}\right), \text{ i.e. } \left(\frac{\alpha}{3}, \frac{\beta - 2}{3}\right)$$

This lies on $2x + 3y = 1$

$$2\left(\frac{\alpha}{3}\right) + 3\left(\frac{\beta - 2}{3}\right) = 1 \quad \Rightarrow \quad 2\alpha + 3\beta - 6 = 3$$

$$\Rightarrow 2\alpha + 3\beta = 9$$

Hence, locus of (α, β) is $2x + 3y = 9$.

35. The given pair of lines is $x^2 - 2cxy - 7y^2 = 0$

On comparing with $ax^2 - 2hxy + by^2 = 0$,

we get

$$a = 1, 2h = -2c, b = -7$$

$$m_1 + m_2 = -\frac{2h}{b} = -\frac{2c}{7} \quad \text{and} \quad m_1m_2 = \frac{a}{b} = -\frac{1}{7}$$

Given that $m_1 + m_2 = 4m_1m_2$

$$\Rightarrow -\frac{2c}{7} = -\frac{4}{7} \Rightarrow c = \frac{4}{2} = 2$$

36. Let a and b be the intercepts on the axes.

Then the equation of the line is

$$\frac{x}{a} + \frac{y}{b} = 1 \qquad \text{(i)}$$

Since (i) passes through (4, 3),

$$\therefore \quad \frac{4}{a} + \frac{3}{b} = 1 \qquad \text{(ii)}$$

Also $$a + b = -1 \qquad \text{[Given]} \quad \text{(iii)}$$

From (iii), we get $$b = -1 - a \qquad \text{(iv)}$$

Putting in (ii), we get $\frac{4}{a} + \frac{3}{-1-a} = 1$

$$\Rightarrow -4 - 4a + 3a = -a - a^2 \Rightarrow a^2 = 4 \Rightarrow a = \pm 2$$

When $a = 2$, then from (iv), $b = -1 - 2 = -3$

When $a = -2$, then from (iv), $b = -1 + 2 = 1$

$\therefore$ The lines are $\frac{x}{2} + \frac{y}{-3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

i.e. $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

37. Since $3x + 4y = 0 \Rightarrow y = -\frac{3}{4}x = mx$,

where $m = -\frac{3}{4}$

Since $6x^2 - xy + 4cy^2 = 0$,

$\Rightarrow\ 4c\cdot\left(\frac{y}{x}\right)^2-\frac{y}{x}+6=0\ \Rightarrow\ 4cm^2-m+6=0$

$\Rightarrow\ 4c\left(\frac{9}{16}\right)+\frac{3}{4}+6=0\ \Rightarrow\ \frac{9c}{4}+\frac{3}{4}+6=0$

$\Rightarrow\ 9c+3+24=0\ \Rightarrow\ c=-3$

38. Here $\alpha+1=2(-1), \beta+1=2(2)$

$\Rightarrow\ \alpha=-3, \beta=3$

$\therefore$ B is (–3, 3)

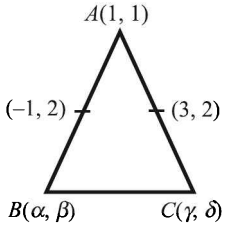


Fig. 18.52

But $\gamma+1=6, \delta+1=4$

$\Rightarrow\ \gamma=5, \delta=3$

$\therefore$ C is (5, 3)

$\therefore$ Centroid is $\left(\frac{1-3+5}{3}, \frac{1+3+3}{3}\right)$, i.e. $\left(1, \frac{7}{3}\right)$

39. The line passing through the intersection of $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$ is

$$(ax+2by+3b)+\lambda(bx-2ay-3a)=0 \qquad \text{(i)}$$

Since this line is parallel to the x-axis, the coefficient of $x = 0$.

$\Rightarrow\ a+\lambda b=0\ \Rightarrow\ \lambda=-a/b$

$\therefore$ (i) becomes

$$b(ax+2by+3b)-a(bx-2ay-3a)=0$$

$\Rightarrow\ 2b^2y+3b^2+2a^2y+3a^2=0$

$\Rightarrow\ 2(b^2+a^2)y+3(b^2+a^2)=0\ \Rightarrow\ y=\frac{-3}{2}$

$\therefore$ The required line is below the x-axis at a distance of 3/2 from the x-axis.

40. a, b, c are in H.P. $\Rightarrow\ \frac{2}{b}=\frac{1}{a}+\frac{1}{c}$

$\Rightarrow\ \frac{x}{a}+\frac{y}{b}+\frac{1}{c}=0$

which passes through (1, –2).

41. Given (a, a^2) falls inside the angle made by

$$y=\frac{x}{2}, x>0 \quad \text{and} \quad y=3x, x>0$$

$\Rightarrow\ a^2-3a<0$ and $a^2-\frac{a}{2}>0$

$\Rightarrow\ \frac{1}{2}<a<3\ \Rightarrow\ a\in\left(\frac{1}{2}, 3\right)$

42. Let the straight line be $\frac{x}{a}+\frac{y}{b}=1$.

It meets the x-axis at $P(a, 0)$ and y-axis at $Q(0, b)$. Mid-point of PQ is $A(3, 4)$ of $\frac{a}{2}=3, \frac{b}{2}=4$.

$\Rightarrow\ a=6, b=8$

$\therefore$ Required line is $\frac{x}{6}+\frac{y}{8}=1\ \Rightarrow\ 4x+3y=24$

43. Sum of slopes $=-\frac{1-m^2}{m}=0$.

$\Rightarrow\ m=\pm1$.

44. The given points are $P(-1,0)$, $Q(0,0)$ and $R(3,3\sqrt{3})$. The equation of the line PQ is

$$(y-0)=\left(\frac{0-0}{0+1}\right)\cdot(x+1)\text{, i.e. } y=0$$

This is of the form $a_1x+b_1y+c_1=0$, where

$$a_1=0,\ b_1=1,\ c_1=0$$

The equation of QR is

$$(y-0)=\left(\frac{3\sqrt{3}-0}{3-0}\right)(x+0),$$

i.e. $y=\sqrt{3}$, i.e. $\sqrt{3}x-y=0$.

This is of the form $a_2x+b_2y+c_2=0$, where

$$a_2=\sqrt{3}, b_2=-1, c_2=0$$

$\therefore\ a_1a_2+b_1b_2=0\cdot(\sqrt{3})+1\cdot(-1)$

$=-1<0$

Now, we have to find the bisector of the obtuse angle $\angle PQR$ between the lines PQ and PR.

The required bisector is given by

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+a_1^2}}=-\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+a_2^2}}$$

i.e. $\frac{y}{\sqrt{0^2+1^2}}=-\frac{\sqrt{3}x-y}{\sqrt{(\sqrt{3})^2+(-1)^2}}$

i.e. $2y=-\sqrt{3}x-y$, i.e. $\sqrt{3}x+y=0$

45. $\frac{1}{2}\begin{vmatrix}h & k & 1\\ 1 & 1 & 1\\ 2 & 1 & 1\end{vmatrix}=1\Rightarrow\frac{1}{2}|h(0)-k(1-2)+(1-2)|=1$

$\Rightarrow\ \frac{1}{2}|k-1|=1\ \Rightarrow\ k=3,-1$

Hence, k is 25 given by {–1, 3}.

46. Slope of the bisector = $k-1$

$$\text{Mid-point} = \left(\frac{k+1}{2}, \frac{7}{2}\right)$$

Equation of the bisector is

$$y - \frac{7}{2} = (k-1)\left(x - \frac{(k+1)}{2}\right)$$

Put $x = 0$ and $y = -4 \Rightarrow k = \pm 4$

47. Lines must be parallel, therefore slopes are equal

$\Rightarrow p(p^2+1) = -(p^2+1) = -1$

48. Let (x, y) denote the coordinates in A, B and C planes.

Then, $$\frac{(x-1)^2 + y^2}{(x+1)^2 + y^2} = \frac{1}{9}$$

$\Rightarrow 9x^2 + 9y^2 - 18x + 9 = x^2 + y^2 + 2x + 1$

$\Rightarrow 8x^2 + 8y^2 - 20x + 8 = 0 \Rightarrow x^2 + y^2 - \frac{5}{2}x + 1 = 0$

$\therefore$ A, B and C lie on a circle with $C\left(\frac{5}{4}, 0\right)$.

49. Direction ratios of line $= (3, -5, 2)$

Direction ratios of normal to the plane $= (1, 3-a)$

The line is perpendicular to the normal. Therefore, $3(1) - 5$.

$(3) + 2(-a) = 0 \Rightarrow 3 - 15 - 2a = 0 \Rightarrow 2a - 12 \Rightarrow a = -6$

Also $(2, 1-2)$ lies on the plane.

$$2 + 3 + 6(-2) + b = 0 \Rightarrow b = 7$$

$\therefore$ $(a, b) = (-6, 7)$.

50. The given lines are

$$L = \frac{x}{5} + \frac{y}{b} - 1 = 0 \qquad \text{(i)}$$

$$K = \frac{x}{c} + \frac{y}{3} - 1 = 0 \qquad \text{(ii)}$$

The line L passes through (13, 32).

$\therefore$ $\frac{13}{5} + \frac{32}{b} - 1 = 0 \quad \Rightarrow \quad b = -20$

So line L is $\frac{x}{5} - \frac{y}{20} - 1 = 0$ i.e. $y = 4x - 20$

Its slope is 4.

$\therefore$ Slope of the line K, i.e. $y = \frac{-3}{c}x + 3$ must also be 4.

[$\because$ Lines K and L are parallel]

$\therefore$ $\frac{-3}{c} = 4 \quad \Rightarrow \quad c = \frac{-3}{4}$

So, line K is $c = \left(-\frac{4}{3}\right)x + \frac{y}{3} - 1 = 0$, i.e. $4x - y + 3 = 0$

Distance between the lines $L \equiv 4x - y - 20 = 0$ and $K \equiv 4x - y + 3 = 0$ is given by

$$d = \frac{|-20-3|}{\sqrt{4^2 + (-1)^2}} = \frac{23}{\sqrt{17}}$$

51. The lines are represented by $2y^2 - 7xy + 3x^2 = 0$

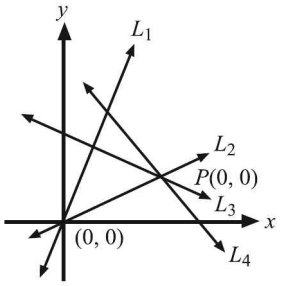


Fig. 18.53

$$\Rightarrow y = \frac{7x \pm \sqrt{4ax^2 - 24x^2}}{4} = \frac{7x \pm 5x}{4}$$

$\Rightarrow y = 3x, y = \frac{1}{2}x$

$\Rightarrow$ Slopes of the perpendicular line are –(1/3) and –2.

$\Rightarrow$ Equation of the pair of lines through (1, 1) and perpendicular to the given pair is

$$\left[y - 1 + \frac{1}{3}(x-1)\right]\left[y - 1 + 2(x-1)\right] = 0$$

$\Rightarrow [3(y-1) + (x-1)][y + 1 + 2(x-1)] = 0$

$\Rightarrow 2(x-1)^2 + 7(x-1)(y-1) + 3(y-1)^2 = 0$

52. $ax^2 + 2hxy + by^2 = (y - m_1 x)(y - m_2 x)$

$\Rightarrow m_1 + m_2 = \frac{-2h}{b}$ and $m_1 + m_2 = \frac{a}{b}$

Let p_1 and p_2 be the perpendiculars drawn from the given point (α, β) to the given pair of lines.

$$\therefore \quad p_1 p_2 = \frac{\beta - m_1\alpha}{\sqrt{1+m_1^2}} \cdot \frac{\beta - m_2\alpha}{\sqrt{1+m_2^2}}$$

$$= \frac{\beta^2 - (m_1 + m_2)\alpha\beta + m_1 m_2 \alpha^2}{\sqrt{1+m_1^2}\sqrt{1+m_2^2}}$$

$$= \frac{\beta^2 + \frac{2h}{b}(\alpha\beta) + \left(\frac{a}{b}\right)\alpha^2}{\sqrt{1 + m_1^2 m_2^2 + m_1^2 m_2^2}}$$

$$= \frac{(b\beta^2 + 2h\alpha\beta + a\alpha^2)\frac{1}{b}}{\sqrt{1+(m_1+m_2)^2 - 2m_1m_2 + m_1^2m_2^2}}$$

$$= \frac{(b\beta^2 + 2h\alpha\beta + a\alpha^2)\frac{1}{b}}{\sqrt{1+\frac{a^2}{b^2}+\frac{4h^2}{b^2}-\frac{2a}{b}}} = \frac{(b\beta^2 + 2h\alpha\beta + a\alpha^2)}{\sqrt{a^2-2ab+b^2+4h^2}}$$

$$\Rightarrow \quad p_1p_2 = \frac{a\alpha^2 + 2h\alpha\beta + b\beta^2}{\sqrt{4h^2+(a-b)^2}}$$

53. The given lines are

$$x - y + 1 = 0 \qquad \text{(i)}$$

and $$3x + y - 5 = 0 \qquad \text{(ii)}$$

Solving (i) and (ii), we get $x = 1, y = 2$

$\therefore$ The point of intersection of the given lines is (1, 2).

Case 1. Equation of a line perpendicular to line (i) and passing through (1, 2)

Slope of the line $x - y + 1 = 0$ is 1

$\therefore$ Slope of the required line is –1.

Let this line be $y = -x + c_1$ (iii)

Since it passes through (1, 2), so we have

$2 = -1 + c_1$ [Putting $x = 1, y = 2$ in (iii)]

$\Rightarrow c_1 = 3$

$\therefore \quad y = -x + 3$, i.e. $x + y - 3 = 0$ is the required line.

Case. 2. Equation of a line perpendicular to line (ii) and passing through (1, 2)

Slope of the line $3x + y - 5 = 0$ is –3.

$\therefore$ Slope of the required line is 1/3.

Let this line be

$$= y = \frac{1}{3}x + c_2 \qquad \text{(iv)}$$

Since it passes through (1, 2), so we have

$$2 = \frac{1}{3} + c_2 \quad \text{[Putting } x = 1, y = 2 \text{ in (iv)]}$$

$$\Rightarrow \quad y = \frac{1}{3}x + \frac{5}{3}, \text{ i.e. } x - 3y + 5 = 0 \text{ is the required line.}$$

From the two cases, we get the lines

$$x + y - 3 = 0 \quad \text{and} \quad x - 3y + 5 = 0$$

Out of these, $x - 3y + 5 = 0$ is given in the alternatives.

54. Because $x - 3y = 2 - 3(2)$ is the line whose y-intercept = 4/3.

55. The given straight line is

$$y = mx \qquad \text{(i)}$$

The line $y = 2$ cuts the line (i) at the point $A\left(\frac{2}{m}, 2\right)$ [Putting $y = 2$ in (i)]

Also, the line $y = 6$ cuts the line (i) at the point $B\left(\frac{6}{m}, 6\right)$.

$\therefore$ The intercept on line (1) by the lines $y = 2$ and $y = 6$ is

$$AB = \sqrt{\left(\frac{6}{m} - \frac{2}{m}\right)^2 + (6-2)^2}$$

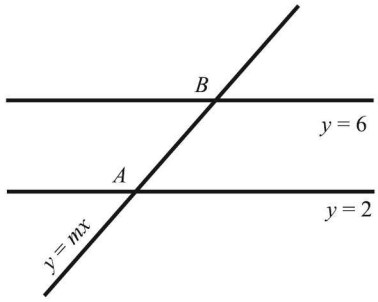


Fig. 18.54

Now, $AB < 5$

$$\Rightarrow \quad \left(\frac{6}{m} - \frac{2}{m}\right)^2 + 16 < 25$$

$$\Rightarrow \quad \left(\frac{4}{m}\right)^2 < 9$$

$$\Rightarrow \quad \left|\frac{4}{m}\right| < 3 \quad \Rightarrow \quad -3 < \frac{4}{m} < 3$$

$$\Rightarrow \quad m < -\frac{4}{3} \quad \text{and} \quad \frac{4}{3} < m$$

$$\therefore \quad m \in \left]-\infty, -\frac{4}{3}\right[\cup \left]\frac{4}{3}, \infty\right[$$

56. As the slope of the two lines is –2, we can take the line $y = (x/2)$ which is perpendicular to both the lines. But the two lines are on the opposite side of the origin.

So $P\left(\frac{9}{5}, \frac{9}{10}\right)$ and $Q\left(\frac{-12}{5}, \frac{-6}{5}\right)$ are feet of the perpendicular of the given lines.

Let $OP : OQ = \lambda : 1$, then

$$0 = \frac{\lambda\left(\frac{-12}{5}\right) + \frac{9}{5}}{\lambda + 1} \quad \text{and} \quad 0 = \frac{\lambda\left(\frac{-6}{5}\right) - \frac{12}{5}}{\lambda + 1}$$

$$\Rightarrow \quad \lambda = \frac{3}{4}, \text{ i.e. } OP : OQ = 3 : 4.$$

Note: Alternative method. Ratio of distance from the origin

$$= \left|\frac{6/\sqrt{5}}{(9/2)/\sqrt{5}}\right| = \frac{4}{3}$$

57. The given line is $x+y=1$, i.e. $L\equiv x+y-1=0$ (i)

Since $(\sin\theta,\cos\theta)$ and (3, 2) lie on the same side of line (i), so we have

$L(\sin\theta,\cos\theta)$ and $L(3, 2)$ are of the same sign.

Now, $L(3,2)\equiv 3+2-1=4>0$

$\therefore\quad L(\sin\theta,\cos\theta)>0 \Rightarrow \sin\theta+\cos\theta-1>0$

$$\Rightarrow \sqrt{2}\left[\sin\left(\theta+\frac{\pi}{4}\right)\right]>1 \Rightarrow \sin\left(\theta+\frac{\pi}{4}\right)>\frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin\left(\theta+\frac{\pi}{4}\right)>\sin\frac{\pi}{4} \text{ and } \sin\left(\theta+\frac{\pi}{4}\right)<\sin\frac{3\pi}{4}$$

$$\Rightarrow \theta+\frac{\pi}{4}>\frac{\pi}{4} \text{ and } \theta+\frac{\pi}{4}<\frac{3\pi}{4}$$

$$\Rightarrow 0<\theta<\frac{\pi}{2} \quad\Rightarrow\quad \theta\in\left(0,\frac{\pi}{2}\right)$$

58. Let m_1 and m_2 be the slopes.

$$\therefore\quad m_1+m_2=-\frac{2b}{h} \text{ and } m_1m_2=\frac{b}{a}$$

and $m_2=2m_1$

$$\therefore\quad 3m_1=-\frac{2b}{h} \text{ and } 2m_1^2=\frac{b}{a}$$

$$\therefore\quad \frac{9m_1^2}{2m_1^2}=\frac{4b^2}{h^2}\times\frac{a}{b} \Rightarrow ab:h^2=9:8$$

59. Given lines are concurrent, if

$$\begin{vmatrix} a & b & -c \\ b & c & -a \\ c & a & -b \end{vmatrix}=0$$

$$\Rightarrow \begin{vmatrix} a+b+c & a+b+c & -(a+b+c) \\ b & c & -a \\ c & a & -b \end{vmatrix}=0$$

$$\Rightarrow (a+b+c)\begin{vmatrix} 1 & 1 & -1 \\ b & c & -a \\ c & a & -b \end{vmatrix}=0$$

$$\Rightarrow (a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ b & c-c & b+a \\ c & a-c & c+b \end{vmatrix}=0$$

$$\Rightarrow (a+b+c)[c^2-b^2-(a-c)(a+b)]=0$$

$$\Rightarrow a+b+c=0$$

60. We have $x^2+y^2=9$ (i)

and $x+y=3$ (ii)

From Eqs. (i) and (ii), we make a homogeneous equation

$$\Rightarrow x^2+y^2=(x+y)^2 \Rightarrow x^2+y^2=x^2+y^2+2xy$$

$$\Rightarrow 2xy=0 \Rightarrow xy=0$$

61. For point (a, a) to fall between $x+y+2=0$ and $x+y-2=0$, satisfy the inequality

$(a+a+2)(a+a-2)<0$, i.e. $-1<a<1$

$\Rightarrow |a|<1$.

62. Solving $3x+y+1=0$ and $2x-y+3=0$, we get the intersection point $\left(-\frac{4}{5},\frac{7}{5}\right)$.

Hence, the required line is $x+y=-\frac{4}{5}+\frac{7}{5}=\frac{3}{5}$,

i.e. $5x+5y=3$.

63. One of the given lines is

$$4x-3y+7=0 \quad \text{(i)}$$

This is of the form $a_1x+b_1y+c_1=0$ where a_1 = 4, b_1 = 3 and c_1 = 7.

The other line is

$$3x-4y+14=0 \quad \text{(ii)}$$

This is of the form $a_2x+b_2y+c_2=0$ where a_2 = 3, b_2 = –4 and c_2 = 14.

Then, $a_1a_2+b_1b_2=(4\times3)+[(-3)\times(-4)]$

$$=12+12=24>0$$

$\therefore$ The equation of the bisector of the acute angle between the lines (i) and (ii) is given by

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=-\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$

$$\Rightarrow \frac{4x-3y+7}{\sqrt{4^2+(-3)^2}}=-\frac{3x-4y+14}{\sqrt{3^2+(-4)^2}}$$

$$\Rightarrow 7x-7y+21=0 \quad\Rightarrow\quad x-y+3=0$$

64. Since $P(a, b)$ lies on the line $3x+2y=13$, so have

$$3a+2b=13 \quad \text{(i)}$$

Since $Q(b, a)$ lies on the line $4x-y=5$, so we have

$$4b-a=5 \quad \text{(ii)}$$

Solving (i) and (ii), we get a = 3 and b = 2

$\therefore$ The points P and Q have coordinates (3, 2) and (2, 3), respectively.

$\therefore$ Equation of the line PQ is given by

$$(y-2)=\left(\frac{3-2}{2-3}\right)(x-3),\ \text{i.e.}\ x-y-5=0$$

65. Given, $xy+2x+2y+4=0$ (i)

and $x+y+2=0$ (ii)

From Eqs. (i) and (ii), we get

$$xy=0\Rightarrow x=y=0$$

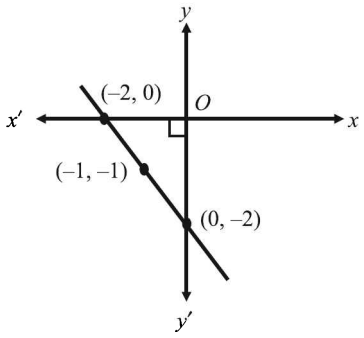


Fig. 18.55

$\therefore$ Vertices of the triangle are $(-2,0),(0,0),(0,-2)$. (In a right-angled triangle, circumcentre is the mid-point of the hypotenuse.)

$\therefore$ $(-1,-1)$ is the circumcentre.

66. Let the lines represented by the equation $x^2+7xy+2y^2=0$ be $x+ay=0$ and $x+by=0$.

[$\because$ These lines pass through (0, 0)]

Then, $(x+ay)(x+by)=x^2+7xy+2y^2$

$\Rightarrow$ $x^2+(a+b)xy+aby^2=x^2+7xy+2y^2$

$\therefore$ $a+b=7$ and $ab=2$.

Now, the three sides of the triangles are represented by the lines $x+ay=0, x+by=0$ and $y-1=0$.

$\therefore$ The vertices of this triangle are $O(0,0)$, $A(-a,1)$ and $B(-b,1)$. [Solving the equations of sides in pairs]

$\therefore$ The centroid of this triangle is given by

$$C\left(\frac{0-a-b}{3},\frac{0+1+1}{3}\right)$$

i.e. $C\left(\frac{-7}{3},\frac{2}{3}\right)$ [$\because 0-a-b=-(a+b)=-7$]

Another method:

The centroid of the triangle formed by the pair of lines $ax^2+2hxy+by^2=0$ and the line $lx+my+n=0$ is

$$C=\left[\frac{2(bl-hm)}{3(am^2-2hm+bl^2)},\frac{2(am-hl)}{3(am^2-2hm+bl^2)}\right]$$

Here $a=1,h=\frac{7}{2},b=2,l=0,m=1,n=-1$

$\therefore$ Centroid of the required triangle is

$$C=\left\{\frac{2[0-(7/2)]}{3(1-0+0)},\frac{2(1-0)}{3(1-0+0)}\right\},\ \text{i.e.}\ C\left(\frac{-7}{3},\frac{2}{3}\right)$$

67. The bisector of the first quadrant is the line $y=x$. The reflection of the point $P(4,-1)$ in the line $y=x$ is the point $(-1,4)$, i.e. the point symmetric to $P(4,-1)$ w.r.t. $y=x$ is $(-1,4)$.

$\therefore$ The point Q is $(-1,4)$.

So, we have

$$PQ=\sqrt{[4-(-1)]^2+(-1-4)^2}=\sqrt{5^2+(-5)^2}$$

$$=\sqrt{50}=5\sqrt{2}$$

68. Given lines are $xy=0$ and $x+y=1$

$\Rightarrow$ $x=0$, $y=0$ and $x+y=1$

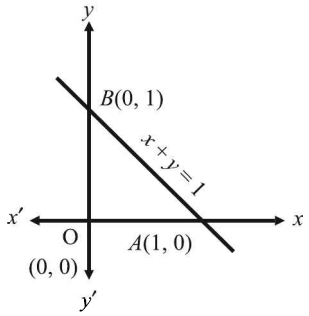


Fig. 18.56

In a ΔOAB, OB and OA are the altitudes which intersect at O.

$\therefore$ The required orthocentre is (0, 0).

69. For $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$, angle bisector of the acute angle is

(i) when $a_1a_2+b_1b_2>0$ keeping c_1 and c_2

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=-\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$

(ii) when $a_1a_2+b_1b_2<0$ keeping c_1 and c_2

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=+\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$

So here $3x-4y+7=0$ and $-12x-5y+2=0$ are two lines (keeping constants positive) $3(-12)+(-4)(-5)<0$

Hence, $$\frac{3x-4y+7}{\sqrt{3^2+(-4)^2}}=\frac{-12x-5y+2}{\sqrt{(-12)^2+(-5)^2}}$$

$\Rightarrow$ $$\frac{3x-4y+7}{5}=\frac{-12x-5y+2}{13}$$

$\Rightarrow (39+60)x+(-52+25y)y+(91-10)=0$

$\Rightarrow 99x-27y+81=0$

is the equation of the bisector of the acute angle.

70. The given equation of the pair of lines is

$$x^2-y^2-x+3y-2=0 \qquad \text{(i)}$$

$$\Rightarrow \left(x^2-x+\frac{1}{4}\right)-\left(y^2-3y+\frac{9}{4}\right)=0$$

$$\Rightarrow \left(x-\frac{1}{2}\right)^2-\left(y-\frac{3}{2}\right)^2=0$$

$$\Rightarrow (x-y+1)(x+y-2)=0$$

$\therefore$ The equations of the lines represented by Eq. (i) are $x-y+1=0$ and $x+y-2=0$.

71. The given line is $3ax+4by+c=0$

$$\Rightarrow 3ax+4by+(-a-b)=0$$

$$[\because a+b+c=0 \Rightarrow c=-a-b]$$

$$\Rightarrow (3x-1)a+(4y-1)b=0 \qquad \text{(i)}$$

Equation (i) clearly represents the family of straight lines passing through the point of intersection of the lines $3x-1=0$ and $4y-1=0$.

Solving $3x-1=0$ and $4y-1=0$, we get

$$x=\frac{1}{3},\ y=\frac{1}{4}$$

$\therefore$ For different values of a and b, the given equation represents a line that definitely passes through the point $\left(\frac{1}{3},\frac{1}{4}\right)$.

72. The given equation is

$$(p+2q)x+(p-3q)y=p-q$$

$$\Rightarrow p(x+y-1)+q(2x-3y+1)=0 \qquad \text{(i)}$$

It clearly represents the family of stright lines passing through the point of intersection of the lines $x+y-1=0$ and $2x-3y+1=0$.

Solving the equations $x+y-1=0$ and $2x-3y+1=0$ we get

$$x=\frac{2}{5},y=\frac{3}{5}$$

$\therefore$ For different values p and q, the given equation represents a line that definitely passes through the point $\left(\frac{2}{5},\frac{3}{5}\right)$.

73. Let $D(h, k)$ be the fourth vertex. The diagonals of a parallelogram bisect each other.

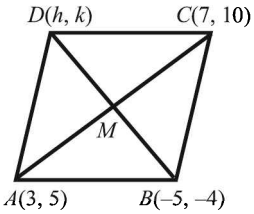


Fig. 18.57

$\therefore$ The diagonals AC and BD have the same midpoint,

i.e. $$\left(\frac{3+7}{2},\frac{5+10}{2}\right)\equiv\left(\frac{-5+h}{2},\frac{-4+k}{2}\right)$$

$\therefore$ $$\frac{3+7}{2}=\frac{-5+h}{2} \text{ and } \frac{5+10}{2}=\frac{-4+k}{2}$$

$\Rightarrow h = 15$ and $k = 19$

Hence, the fourth vertex is (15, 19).

74. Let ABC be the required triangle with vertex $A(-1,2)$ and the equation of the base BC be $2x-y=1$.

Then, the length of the altitude AD = the perpendicular distance of the line BC from the vertex A.

$$=\frac{|2(-1)-2-1|}{\sqrt{2^2+(-1)^2}}=\frac{5}{\sqrt{5}}=\sqrt{5}$$

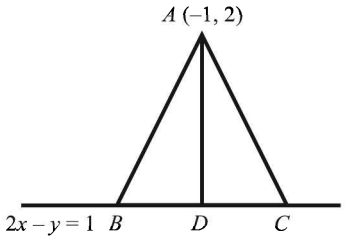


Fig. 18.58

Let the length of each side of the equilateral ΔABC be a. Then $AD=\frac{\sqrt{3}}{2}a$.

$$\Rightarrow a=\frac{2}{\sqrt{3}}\cdot AD=\frac{2}{\sqrt{3}}\times\sqrt{5}=\sqrt{\frac{20}{3}}$$

i.e. the length of each side of ΔABC is $\sqrt{\frac{20}{3}}$.

75. Let (x, y) be the coordinates of the point P.

Then, $PA = AB$

$$\Rightarrow PA^2=PB^2$$

$$\Rightarrow (x-1)^2+(y-3)^2=(x+3)^2+(y-5)^2$$

$$\Rightarrow 2x-y+6=0 \qquad \text{(i)}$$

and $PA = PC$

$$\Rightarrow PA^2=PC^2$$

$\Rightarrow \quad (x-1)^2+(y-3)^2=(x-5)^2+(y+1)^2$

$\Rightarrow \quad x-y-2=0 \qquad$ (ii)

Solving (i) and (ii), we get

$x=-8, y=-10$

$\therefore \quad PA=\sqrt{(x-1)^2+(y-3)^2}=\sqrt{(-9)^2+(-13)^2}$

$=250=5\sqrt{10} \qquad$ [Using $x=-8$, $y=-10$]

76. The line $[(x,0): x\in\mathbb{R}]$ is the locus of all those points for which $y = 0$. It therefore represents the x-axis.

The line $[(0,y): y\in\mathbb{R}]$ is the locus of all points for which $x = 0$. It therefore represents the y-axis.

Clearly, the angle between the two lines is $\pi/2$.

77. Let the line $3x + 4y = 7$ divide the line joining $A(-2, 1)$ and $B(1, 2)$ in the ratio $k : 1$. Then the coordinates of the point of intersection M are given by

$$\left(\frac{1\cdot k+(-2)\cdot 1}{k+1}, \frac{2k+1\cdot 1}{k+1}\right), \text{ i.e. } \left(\frac{k-2}{k+1}, \frac{2\cdot k+1}{k+1}\right)$$

Now, this point $M\left(\frac{k-2}{k+1}, \frac{2k+1}{k+1}\right)$ lies on the line $3x+4y=7$.

$$\therefore \quad 3\left(\frac{k-2}{k+1}\right)+4\left(\frac{2k+1}{k+1}\right)=7$$

$$\Rightarrow \quad 4k=9 \Rightarrow k=\frac{9}{4}$$

Hence, the required ratio is $\frac{9}{4}:1$, i.e. 9 : 4.

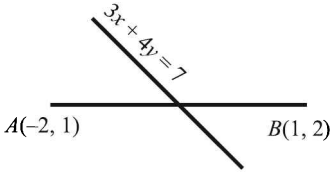


Fig. 18.59

78. $Y-y=\frac{dy}{dx}(X-x)$

x-intercept is $\left(x-\frac{y}{dy/dx}, 0\right)$

y-intercept is $\left(0, y-\frac{xdy}{dx}\right)$

According to the statement,

$$x-\frac{y}{dy/dx}=2x \text{ and } y-\frac{xdy}{dx}=2y$$

$$\frac{-y}{\frac{dy}{dx}}=x \quad \Rightarrow \quad \frac{-xdy}{dx}=y$$

$$\frac{dx}{x}+\frac{dy}{y}=0$$

$\log y=\log x+\log c$

$$y=\frac{c}{x}$$

$\Rightarrow \quad c=6$

Hence $y=\frac{6}{x}$.

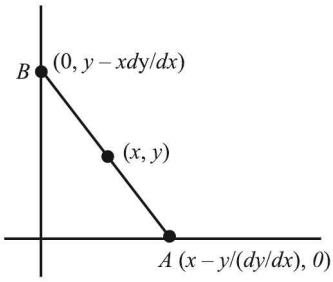


Fig. 18.60

79.

$x+y=\pm a$	$x+y=\lvert a\rvert$
$ax-y=1$	$ax-y=1$
$(1+a)x=\lvert a\rvert+1$	$ax+ay=a\lvert a\rvert$
$x=\frac{\lvert a\rvert+1}{1+a}$	$(a+1)y=a\lvert a\rvert-1$
	$y=\frac{a\lvert a\rvert-1}{1+a}$
$\frac{\lvert a\rvert+1}{1+a}>0$	$\frac{a\lvert a\rvert-1}{1+a}>0$

$\Rightarrow \quad a > 1$

80.

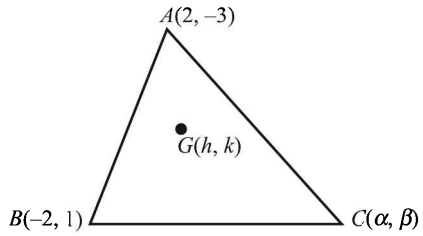


Fig. 18.61

$a=3h, \ \beta-2=3k \quad \Rightarrow \quad \beta=3k+2$

Third vertex on the line

$2x+3y=9$

$2\alpha+3\beta=9$

$2(3h)+3(3k+2)=9$

$2h+3k=1, \ 2x+3y-1=0$

81. Coordinate of O

$$=\frac{3\times 2+2\times 1}{5}, \frac{3\times 4+2\times 1}{5}$$

$$\Rightarrow \quad \left(\frac{8}{5}, \frac{14}{5}\right)$$

Satisfying the equation $2x + y = k$

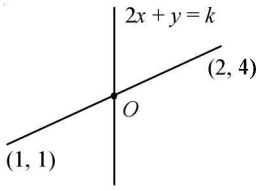


Fig. 18.62

$$2\left(\frac{8}{5}\right)+\left(\frac{14}{5}\right)=k$$

$$\Rightarrow \frac{16}{5}+\frac{14}{5}=k \Rightarrow k=\frac{30}{5}=6$$

82. Let the line passing through (1, 2) be $\frac{x}{a}+\frac{y}{b}=1$.

$$\therefore \frac{1}{a}+\frac{2}{b}=1 \Rightarrow a=\frac{b}{b-2}$$

Area of triangle, $A=\frac{1}{2}ab=\frac{1}{2}\cdot\frac{b}{(b-2)}\cdot b$

$$\frac{dA}{db}=\frac{1}{2}\left[\frac{(b-2)2b-b^2\cdot 1}{(b-2)^2}\right]=0$$

$\Rightarrow b=0, 4$

$\therefore b=4$ and $a=2$

Hence slope = –2.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The area of the quadrilateral formed by the line $|x|+|y|=2$ is

(a) 8 (b) 6

(c) 4 (d) none of these

2. The position of the moving point in the xy-plane at time t is given by $\left(u\cos\alpha t, u\sin\alpha t-\frac{1}{2}gt^2\right)$, where u, α and g are constants. The locus of the moving point is

(a) a circle (b) a parabola

(c) an ellipse (d) none of these

3. If the image of the point A (1, 2) by the line mirror $y = x$ is the point B and the image of B by the line mirror $y = 0$ is the point (α, β), then

(a) $\alpha=1, \beta=-2$ (b) $\alpha=0, \beta=0$

(c) $\alpha=2, \beta=-1$ (d) none of these

4. Consider the equation $y-y_1=m(x-x_1)$. If m and x_1 are fixed and different lines are drawn for different values of y_1, then

(a) the line will pass through a fixed point

(b) there will be a set of parallel lines

(c) all the lines will be parallel to the line $y = y_1$

(d) none of these

5. On the portion of the straight line $x + 2y = 4$ intercepted between the axes a square is constructed on the side of the line away from the origin. Then the point of intersection of its diagonals has coordinates

(a) (2, 3) (b) (3, 2)

(c) (3, 3) (d) none of these

6. The point (4, 1) undergoes the following three transformations successively:

(i) Reflection about the line $y = x$

(ii) Transaction through a distance 2 units along the positive direction of the x-axis.

(iii) Rotation through an angle $\pi/4$ about the origin in the anticlockwise direction.

The final position of the point is given by the coordinates

(a) $\left(\frac{4}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (b) $\left(-\frac{1}{\sqrt{2}},\frac{7}{\sqrt{2}}\right)$

(c) $\left(\frac{1}{\sqrt{2}},\frac{7}{\sqrt{2}}\right)$ (d) $\left(\frac{3}{\sqrt{2}},\frac{4}{\sqrt{2}}\right)$

7. The equations of bisectors of two lines L_1 and L_2 are $2x-16y-5=0$ and $64x+8y+35=0$. If the line L_1 passes through (–11, 4), the equation of the acute angle bisector of L_1 and L_2 is

(a) $2x-16y-5=0$ (b) $64x+8y+35=0$

(c) data insufficient (d) none of these

8. The point of intersection of the lines $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$ lies on the line

(a) $x-y=0$

(b) $(x-y)(a+b)=2ab$

(c) $(lx+my)(a-b)=(l+m)ab$

(d) $(lx-my)(a-b)=(1-m)ab$

9. The vertices of the triangle formed by (0, 0), (5, 12) and (16, 12) is

(a) (7, 9) (b) (9, 7)

(c) (–9, 7) (d) (–7, 9)

10. If the algebraic sum of the perpendicular distances from $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ to a variable line is zero, then the line passes through

(a) the orthocentre of ΔABC

(b) centroid of ΔABC

(c) incentre of ΔABC

(d) circumcentre of ΔABC

11. If one vertex of an equilateral triangle is at (1, 2) and the base is $x+y-2=0$, then the length of each of the side is

(a) $\sqrt{3/2}$ (b) $\sqrt{2/3}$

(c) 2/3 (d) 3/2

12. If the points of intersection of the curves
$C_1 = \lambda x^2 + 4y^2 - 2xy - 9x + 3$
and $C_2 = 2x^2 + 3y^2 - 4xy + 3x - 1$
subtend a right angle at the origin, then the value of λ is
(a) 19 (b) 9
(c) –19 (d) –9

13. If the slope of a line passing through the point $A(3, 2)$ is 3/4, then the points on the line which are 5 units away from A are
(a) (5, 5), (–1, – 1) (b) (7, 5), (–1, – 1)
(c) (5, 7), (–1, – 1) (d) (7, 5), (1, 1)

14. Coordinates of the orthocentre of the triangle whose sides are $x = 3, y = 4$ and $3x + 4y = 6$ will be
(a) (0, 0) (b) (3, 0)
(c) (0, 4) (d) (3, 4)

15. If a line is perpendicular to the line $5x - y = 0$ and forms a triangle, with the coordinate axes, of area 2 sq. units, then its equation is
(a) $x + 5y \pm 5\sqrt{2} = 0$ (b) $x - 5y \pm 5\sqrt{2} = 0$
(c) $5x + y \pm 5\sqrt{2} = 0$ (d) $5x - y \pm 5\sqrt{2} = 0$

16. The equation of the line with slope $-(3/2)$ and which is concurrent with lines $4x + 3y - 7 = 0$ and $8x + 5y - 1 = 0$ is
(a) $2y - 3x - 2 = 0$ (b) $3x + 2y - 2 = 0$
(c) $3x + 2y - 63 = 0$ (d) none of these

17. If the lines represented by $px^2 - qxy - y^2 = 0$ make the angles α and β with the x-axis, then the value of $\tan(\alpha + \beta)$ is
(a) $\frac{-q}{1+p}$ (b) $\frac{q}{1+p}$
(c) $\frac{p}{1+q}$ (d) $\frac{-p}{1+q}$

18. The locus of the point which moves so that the difference of its distances from two fixed straight lines which are at right angles is equal to the the distance from another fixed straight line is
(a) a straight line (b) a circle
(c) a parabola (d) an ellipse

19. If the point (2, –3) lies on $Kx^2 - 3y^2 + 2x + y - 2 = 0$, then K is equal to
(a) 1/7 (b) 16
(c) 7 (d) 12

20. If $A(2, 2)$, $B(-4, -4)$ and $C(5, -8)$ are the vertices of any triangle, the length of the median passing through C will be
(a) $\sqrt{65}$ (b) $\sqrt{117}$
(c) $\sqrt{85}$ (d) $\sqrt{113}$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If the equal sides AB and AC (each equal to a) of right angled isosceles triangle ABC are produced to P and Q so that $BP \cdot CQ = AB^2$, then the line PQ always passes through the fixed point
(a) $(a, 0)$ (b) $(0, a)$
(c) (a, a) (d) none of these

2. A straight line through $P(1, 2)$ is such that its intercept between the axes is bisected at P. Its equation is
(a) $x + 2y = 5$ (b) $x - y + 1 = 0$
(c) $x + y - 3 = 0$ (d) $2x + y - 4 = 0$

3. A straight line through the origin meets the parallel lines $4x + 2y + 6 = 0$ and $2x + y + 6 = 0$ at points P and Q, respectively. Then the point O divides the segment PQ in the ratio
(a) 1 : 2 (b) 3 : 4
(c) 2 : 1 (d) 4 : 3

4. The area of the triangle formed by the lines $x = 0$, $y = 0$ and $\frac{x}{a} + \frac{y}{b} = 1$ is
(a) ab (b) $ab/2$
(c) $2ab$ (d) $ab/3$

5. In a ΔABC, the side AB has the equation $2x + 3y = 29$ and the side AC has the equation $x + 2y = 16$. If the mid-point of BC is (5, 6), then the equation of BC is
(a) $2x + y = 7$ (b) $x + y = 11$
(c) $2x - y = 17$ (d) none of these

6. Angles made with the x-axis by two lines drawn through the point (1, 2) and cutting the line $x + y = 4$ at a distance $\sqrt{6}/3$ from the point (1, 2) are
(a) $\pi/6$ and $\pi/3$ (b) $\pi/8$ and $3\pi/8$
(c) $\pi/12$ and $5\pi/12$ (d) none of these

7. A line L is perpendicular to the line $5x - y = 1$ and the area of the triangle formed by the line L and the coordinate axes is 5. The equation of the line L is
(a) $x + 5y = 5$ (b) $x + 5y = \pm 5\sqrt{2}$
(c) $x - 5y = 5$ (d) $x - 5y = 5\sqrt{2}$

8. Which pair of points lies on the same side of $3x-8y-7=0$?
(a) (0, –1) and (0, 0) (b) (4, –3) and (0, 1)
(c) (–3, –4) and (1, 2) (d) (–1, –1) and (3, 7)

9. If a ray of light passing through the point (1, 2) is reflected on the x-axis at a point P and passes through the point (5, 3), then the abscissa of the point is
(a) 3 (b) 13/3 (c) 13/5 (d) 13/4

10. The reflection of the point (4, –13) in the line $5x+y+6=0$ is
(a) (–1, –14) (b) (3, 4)
(c) (1, 2) (d) (–4, 13)

11. The equations of the line which passes through the point (3, 4) and the sum of its intercept on the axes is 14 are
(a) $4x-3y=24,\ x-y=7$
(b) $4x+3y=24,\ x+y=7$
(c) $4x+3y+24=0,\ x+y+7=0$
(d) $4x-3y+24=0,\ x-y+7=0$

12. The lines represented by $ax+by+c=0$, where $3a+2b+4c=0$, are concurrent at the point
(a) $\left(\frac{1}{2},\frac{3}{4}\right)$ (b) (1, 3)
(c) (3, 1) (d) $\left(\frac{3}{4},\frac{1}{2}\right)$

13. The length of the perpendicular from the origin to a line is 7 and the line makes an angle of 150° with the positive direction of the y-axis. The equations of the line is
(a) $\sqrt{3}x+y=14$ (b) $\sqrt{3}x-y=14$
(c) $\sqrt{3}x+y+14=0$ (d) $\sqrt{3}x-y+14=0$

14. The combined equation of the bisectors of the angle between the lines represented by $(x^2+y^2)\sqrt{3}=4xy$ is
(a) $y^2-x^2=0$ (b) $xy=0$
(c) $x^2+y^2=2xy$ (d) $\frac{x^2-y^2}{\sqrt{3}}=\frac{xy}{2}$

15. The area of the parallelogram formed by the lines $4y-3x=1,\ 4y-3x-3=0,\ 3y-4x+1=0,\ 3y-4x+2$ is
(a) 3/8 (b) 2/7
(c) 1/6 (d) none of these

16. The equation of the bisector of the acute angle between the lines $3x-4y+7=0$ and $12x+5y-2=0$ is
(a) $21x+77y-101=0$ (b) $11x-3y+9=0$
(c) $11x-3y-9=0$ (d) $22x-77y+101=0$

17. The equation of a straight line passing through the origin and making with the x-axis an angle twice the size of the angle made by the line $y=0.2x$ with the x-axis is
(a) $y=0.4x$ (b) $y=(5/12)x$
(c) $6y-5x=0$ (d) none of these

18. A variable straight line passes through a fixed point $(a,\ b)$ intersecting the coordinates axes at A and B. If O is the origin, then the locus of the centroid of the triangle OAB is
(a) $bx+ay-3xy=0$ (b) $bx+ay-2xy=0$
(c) $ax+by-3xy=0$ (d) $ax+by-2xy=0$

19. The set of values of b for which the origin and the point (1, 1) lie on the same side of the straight line $a^2x+aby+1=0$, for all $a\in\mathbb{R}, b>0$ are
(a) $b\in(2,4)$ (b) $b\in(0,2)$
(c) $b\in[0,2]$ (d) $(2,\infty)$

20. Drawn from the origin are two mutually perpendicular straight lines forming an isosceles triangle together with the straight line $2x+y=a$. Then the area of the triangle is
(a) $\frac{a^2}{2}$ (b) $\frac{a^2}{3}$
(c) $\frac{a^2}{5}$ (d) none of these

ANSWERS

Level 1

1. (a)	**2.** (b)	**3.** (c)	**4.** (b)	**5.** (c)	**6.** (b)	**7.** (a)	**8.** (a)	**9.** (a)	**10.** (b)
11. (b)	**12.** (c)	**13.** (b)	**14.** (d)	**15.** (a)	**16.** (b)	**17.** (a)	**18.** (a)	**19.** (c)	**20.** (c)

Level 2

1. (c)	**2.** (d)	**3.** (b)	**4.** (b)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (a)
11. (b)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (b)	**18.** (a)	**19.** (b)	**20.** (c)

19

APOLLONIUS

Circles

- Definition of Circle
- Equation of a Circle
- Different Types of Circles
- Intercepts Made on the Axes by a Circle
- Position of a Point with Respect to a Circle
- Intersection of a Line and a Circle
- Two Circles Touching Each Other
- Equation of Common Chord of Circles
- Tangent to the Circle
- Equation of Normal
- Equation of Chord of Contact
- Equation of Chord with Mid-point
- Power of a Point with Respect to a Circle
- Pole and Polar
- Family of Circles
- Radical Axis
- Orthogonal Circles

DEFINITION OF CIRCLE

A **circle** is the locus of a point which moves in a plane such that its distance from a fixed point always remains constant. The fixed point is called the *centre* and the constant distance is called the *radius* of the circle.

EQUATION OF A CIRCLE

1. The equation of a circle with centre $A = (h, k)$ and radius $AP = r$ is given by
   $(x - h)^2 + (y - k)^2 = r^2$

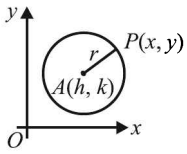


Fig. 19.1

2. The equation of a circle with centre at origin $O(0, 0)$ and radius = r is given by
   $x^2 + y^2 = r^2$
3. The equation of a circle with centre $(-g, -f)$ and radius = r is given by
   $x^2 + y^2 + 2gx + 2fy + c = 0$, where $c = g^2 + f^2 - r^2$
   This is also known as the **general form of equation** of the circle. For this circle,

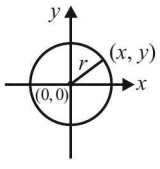


Fig. 19.2

   (a) Centre $\equiv (-g, -f) \equiv \left(\frac{-\text{coff. of } x}{2}, \frac{-\text{coff. of } y}{2}\right)$

   (b) Radius $= r = \sqrt{g^2 + f^2 - c}$

DIFFERENT TYPES OF CIRCLES

1. **Centred at *x*-axis**
   ⇒ *y*-coordinate of centre ≡ 0, i.e. $f = 0$
   ⇒ Centre ≡ $(-g, 0)$
   ⇒ $x^2 + y^2 + 2gx + c = 0$ is the equation of the circle.

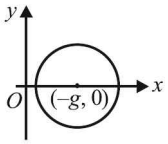


Fig. 19.3

2. **Centred at *y*-axis**
   ⇒ *x*-coordinate of centre ≡ 0, i.e. $g = 0$
   ⇒ Centre ≡ $(0, -f)$
   ⇒ $x^2 + y^2 + 2fy + c = 0$ is the equation of the circle.

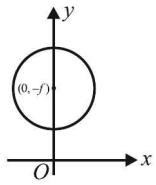


Fig. 19.4

3. **Centred at origin**
   ⇒ Centre ≡ $(-g, -f) \equiv (0, 0)$
   ⇒ $x^2 + y^2 + c = 0$ is the equation of the circle ($c = -r^2$).

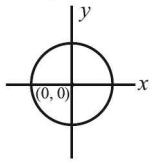


Fig. 19.5

4. **Circle passing through origin**
   The circle passing through (0, 0) must lie on the circle or (0, 0) satisfies the equation of the circle.
   ⇒ $c = 0$
   ⇒ $x^2 + y^2 + 2gx + 2fy = 0$

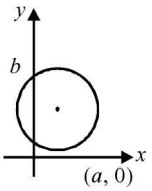


Fig. 19.6

Note A circle $x^2 + y^2 - ax - by = 0$ passes through the origin and cuts the *x*-axis at $(a, 0)$ and the *y*-axis at $(0, b)$.

5. **Circle centred at *x*-axis and touching *y*-axis at origin**
   $x^2 + y^2 + 2gx = 0$ is the circle.
   ⇒ (0, 0) and $(-2g, 0)$ are the end points of the diameter.

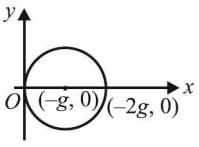


Fig. 19.7

6. **Circle centred at *y*-axis and touching *x*-axis at origin**
   $x^2 + y^2 + 2fy = 0$ is the circle.
   ⇒ (0, 0) and $(0, -2f)$ are the end points of the diameter.

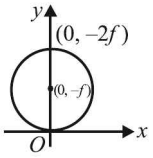


Fig. 19.8

7. **Circle touching both the axes**
   The centre of the circle of radius *r* touching both the axes *OX* and *OY* ≡ $(\pm r, \pm r)$, where ± can be selected according to the quadrant in which the circle lies.
   When the circle touches both (in the first quadrant), the equation is $(x - r)^2 + (y - r)^2 = r^2$.

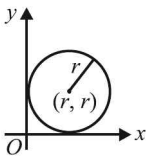


Fig. 19.9

When the circle touches both (in the second quadrant), the equation is $(x + r)^2 + (y - r)^2 = r^2$.
When the circle touches both (in the third quadrant), the equation is $(x + r)^2 + (y + r)^2 = r^2$.
When the circle touches both (in the fourth quadrant), the equation is $(x - r)^2 + (y + r)^2 = r^2$.

8. **Equation of a circle in diametric form**
   $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$ is the equation of the circle, where (x_1, y_1) and (x_2, y_2) are the ends.

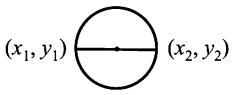


Fig. 19.10

9. **Parametric form of a circle**
   Consider the circle $(x - h)^2 + (y - k)^2 = r^2$ centred at $A \equiv (h, k)$ and radius *r*. Let $P \equiv (x, y)$.
   The coordinates of *P* can be expressed as
   $x = h + r\cos\theta$ and $y = k + r\sin\theta$
   These equations represent the coordinates of any point on the circle in terms of the parameter θ.
   For a circle centred at origin, the parametric forms of the equation are $x = r\cos\theta$ and $y = r\sin\theta$.

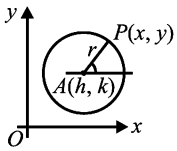


Fig. 19.11

INTERCEPTS MADE ON THE AXES BY A CIRCLE

$x^2 + y^2 + 2gx + 2fy + c = 0$ (i)
The length of intercepts on the *x*-axis is $|AB| = |x_2 - x_1|$
and on the *y*-axis it is $|CD| = |y_2 - y_1|$

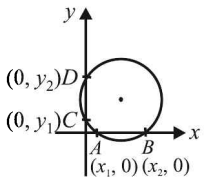


Fig. 19.12

1. **The circle intersects the *x*-axis, when $y = 0$**
   Now $x^2 + 2gx + c = 0$
   Since the circle intersects the *x*-axis at $A(x_1, 0)$ and $B(x_2, 0)$, then $x_1 + x_2 = -2g$, $x_1x_2 = c$
   $\therefore\ |AB| = |x_2 - x_1| = \sqrt{(x_2 + x_1)^2 - 4x_1x_2} = 2\sqrt{(g^2 - c)}$

2. **The circle intersects the *y*-axis, when $x = 0$**
   Now $y^2 + 2fy + c = 0$
   Since the circle intersects the *y*-axis at $C(0, y_1)$ and $D(0, y_2)$, then
   $|CD| = |y_2 - y_1| = \sqrt{(y_2 + y_1)^2 - 4y_2y_1} = 2\sqrt{(f^2 - c)}$

Note

- Intercepts are always positive.
- If the circle touches the *x*-axis, then $|AB| = 0$, so $c = g^2$ and if the circle touches the *y*-axis, then $|CD| = 0$, so $c = f^2$.
- If the circle touches both the axes, then $|AB| = 0 = |CD|$, so $c = g^2 = f^2$.

POSITION OF A POINT WITH RESPECT TO A CIRCLE

Let the equation of the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ having centre at $C \equiv (-g, -f)$ and radius $= r = \sqrt{g^2 + f^2 - c}$.

Then $\sqrt{g^2 + f^2 - c} < 0 \Rightarrow$ inside a circle (imaginary circle)

$\sqrt{g^2 + f^2 - c} > 0 \Rightarrow$ outside the circle

$\sqrt{g^2 + f^2 - c} = 0 \Rightarrow$ point of the circle

INTERSECTION OF A LINE AND A CIRCLE

Consider the circle $x^2 + y^2 = r^2$ and the line $y = mx + c$.

1. If $r > \left|\frac{c}{\sqrt{1+m^2}}\right|$, then the line intersects the circle at two distinct points.

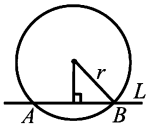


Fig. 19.13

2. If $r = \left|\frac{c}{\sqrt{1+m^2}}\right|$, then the line intersects the circle at two coincident points. It is called a *tangent*.
3. If $r < \left|\frac{c}{\sqrt{1+m^2}}\right|$, then the line does not intersect the circle.
4. A line $y = mx + c$ touches the circle $x^2 + y^2 = r^2$ if the perpendicular distance of the line from the centre of the circle is equal to the radius of the circle.

Thus, $\frac{|c|}{\sqrt{1+m^2}} = r \Rightarrow c = \pm r\sqrt{1+m^2}$

TWO CIRCLES TOUCHING EACH OTHER

1. When two circles touch each other externally, the distance between their centres $C_1C_2 = r_1 + r_2$. In such a case, the point of contact P divides the line joining C_1 and C_2 internally in the ratio $r_1 : r_2$. Thus,

$$\frac{C_1P}{C_2P} = \frac{r_1}{r_2}$$

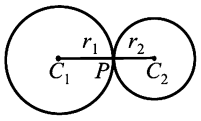


Fig. 19.14

2. When two circles touch each other internally, the distance between their centres $C_1C_2 = |r_1 - r_2|$. In such a case, the point of contact P divides the line joining C_1 and C_2 externally in the ratio $r_1 : r_2$. Thus,

$$\frac{C_1P}{C_2P} = \frac{r_1}{r_2}$$

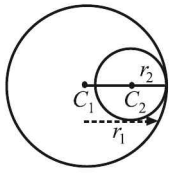


Fig. 19.15

3. The equation of the common tangent at the point of contact (P) of two circles S and S' is $S - S' = 0$.

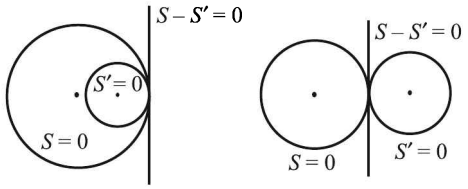


Fig. 19.16

EQUATION OF COMMON CHORD OF CIRCLES

The equation of common chord of two intersecting circles $S = 0$ and $S' = 0$ is given as $S - S' = 0$.

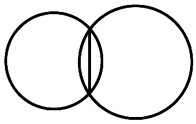


Fig. 19.17

TANGENT TO THE CIRCLE

1. **Equation of common chord at the point of contact** (x_1, y_1) to the circle $x^2 + y^2 = r^2$ (centred at origin) is given by $xx_1 + yy_1 = x_1^2 + y_1^2$
2. **Equation of the tangent at the point of contact** (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is given by $xx_1 + yy_1 + g(x + x_1) f(y + y_1) + c = 0$
3. **Condition for tangency:** A line L may be a tangent to any given circle S if the perpendicular distance of L from the centre of S is equal to the radius of the circle S.
   Let the line $y = mx + c$ is tangent to $x^2 + y^2 = r^2$ with radius r, then $c = \pm r\sqrt{1+m^2}$.
4. **Length of tangent:** The length of the tangent to a given circle from a given point (x_1, y_1) is given as

$$l = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$$

EQUATION OF NORMAL

The normal to a curve at any point P of a curve is the straight line which passes through P and perpendicular to the tangent at P. The equation of the normal to the circle $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$ at any point (x_1, y_1) is given by

$$y(x_1 + g) - x(y_1 + f) + fx_1 - gy_1 = 0$$

EQUATION OF CHORD OF CONTACT

A point $A(x_1, y_1)$ lies outside the circle $x^2 + y^2 = r^2$. Two tangents AP and AQ are drawn to the circle from A as shown in Fig. 19.18. The segment

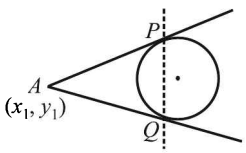


Fig. 19.18

PQ passing through the points of contact is called the *chord of contact* for A. The equation of the chord of contact (PQ) is

$$xx_1 + yy_1 = r^2.$$

The chord of contact for the general circle is given by

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

EQUATION OF CHORD WITH (x_1, y_1) AS THE MID-POINT

The equation of the chord of the circle $x^2 + y^2 = r^2$ with (x_1, y_1) as the mid-point of the chord is

$$xx_1 + yy_1 = x_1^2 + y_1^2.$$

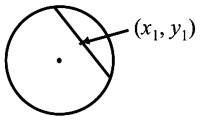


Fig. 19.19

If the chord lies in the general circle, its equation is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) = x_1^2 + y_1^2 + 2gx_1 + 2fy_1$$

POWER OF A POINT WITH RESPECT TO A CIRCLE

The power of a point $P(x_1, y_1)$ with respect to a given circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is defined as

$$\text{Power} = PA \times PB$$

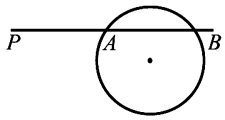


Fig. 19.20

where PAB is any line through P intersecting the circle at A and B. The power of a point P is also the square of the length of the tangent drawn from P to the circle.

POLE AND POLAR

If through a point $P(x_1, y_1)$ (lying inside or outside the circle) any straight line drawn intersects the circle in points Q and R, the locus of the point of intersection of the tangents at Q and R is called the **polar** of P and P is called the **pole** of the polar. The equation of the polar of the point (x_1, y_1) w.r.t. the circle $x^2 + y^2 = a^2$ is

$$xx_1 + yy_1 = a^2$$

The equation of the polar of the point (x_1, y_1) w.r.t. the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ is}$$

$$xx_1 + yy_1 + g(x + x_1) = f(y + y_1) + c = 0$$

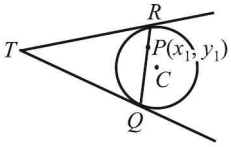


Fig. 19.21

FAMILY OF CIRCLES

1. The equation of the family of circles passing through the points of intersection of two given circles $S = 0$ and $S' = 0$ is given as

$$S + kS' = 0$$

where k is a parameter, $k \neq -1$.

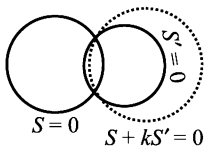


Fig. 19.22

2. The equation of the family of circles passing through the points of intersection of the circle $S = 0$ and the line $L = 0$ is given as

$$S + kL = 0$$

where k is a parameter.

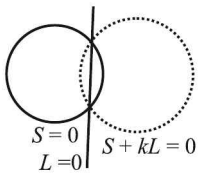


Fig. 19.23

3. The equation of the family of circles touching the circle $S = 0$ and the line $L = 0$ at their point of contact A is

$$S + kL = 0$$

where k is a parameter.

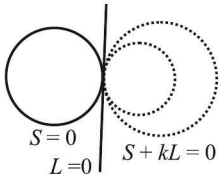


Fig. 19.24

4. The equation of the family of circles passing through two given points $A \equiv (x_1, y_1)$ and $B \equiv (x_2, y_2)$ is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + k = 0$$

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

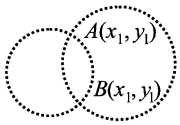


Fig. 19.25

5. The equation of the family of circles touching a given line $L = 0$ at a given point (x_1, y_1) on the line is

$$(x - x_1)^2 + (y - y_1)^2 + kL = 0$$

where k is a parameter.

RADICAL AXIS

The radical axis of two circles is a line passing through the points which satisfy the following property: *The lengths of the tangents drawn from any point (on the radical axis) to the two circles are equal.*

From the definition, $l_1 = l_2$

$$OT_1 = OT_2 \quad \sqrt{S} = \sqrt{S'}$$

$\Rightarrow$ $S - S' = 0$ is the equation of the radical axis.

Another definition: The radical axis of two circles $S = 0$ and $S' = 0$ is a line passing through all the points having the same power with respect to the two circles.

Note

- For circles intersecting in two points, the radical axis is the same as the common chord. Hence, $S - S' = 0$ is the equation of the common chord (or radical axis).

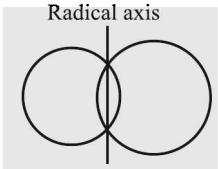


Fig. 19.26

- For circles touching each other, the radical axis is the same as the common tangent at the point of the contact. Hence, $S - S' = 0$ is the equation of the common tangent at the point of contact (or radical axis).

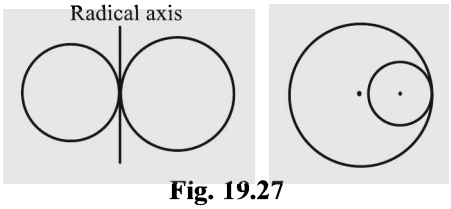


Fig. 19.27

- If $S = x^2 + y^2 + 2gx + 2fy + c = 0$ and $S' = x^2 + y^2 + 2g'x + 2f'y + c' = 0$ then the radical axis is $S - S' = 0$.

$\Rightarrow \quad 2(g - g')x + 2(f - f')y + c - c' = 0$

- The radical axis of two circles is always perpendicular to the line joining the centres of the circles.
- The radical axis of three circles, taken in pairs, meets in a point if the centres of the circles are non-collinear. The point of concurrency is known as the radical centre of the three circles.

ORTHOGONAL CIRCLES

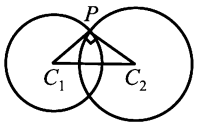


Fig. 19.28

Two circles cutting orthogonally are said to be orthogonal, if their tangents at a point of intersection are at right angles. If two circles are orthogonal, the square of the distance between their centres is equal to the sum of the squares of their radii. If C_1 and C_2 are the centres of two circles and P is the point of contact (intersection of circles) at which two tangents are drawn, then

$$[C_1C_2]^2 = [C_1P]^2 + [C_2P]^2$$

If $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ are two orthogonal circles, then

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

SOLVED PROBLEMS

1. The radius of the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ is

(a) 9 (b) 5

(c) -3 (d) -6

Ans. (b)

Solution: Equation of the circle is $x^2 + y^2 + 4x - 6y - 12 = 0$.

We know that the standard equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Comparing the given equation with the standard equalion, we get

$$g = 2, \; f = -3 \text{ and } c = -12.$$

We also know that the radius of the circle

$$= \sqrt{g^2 + f^2 - c} = \sqrt{(2)^2 + (-3)^2 + 12} = \sqrt{25} = 5.$$

2. The equation of the circle circumscribing a triangle formed by points (0, 0), (1, 0) and (0, 1) is

(a) $x^2 + y^2 + x + y = 0$ (b) $x^2 + y^2 + x - y + 2 = 0$

(c) $x^2 + y^2 - x - y = 0$ (d) $x^2 + y^2 + x - y - 2 = 0$

Ans. (c)

Solution: Coordinates of the vertices of the triangle are (0, 0), (1, 0) and (0, 1).

Let $x^2 + y^2 + 2gx + 2fy + c = 0$ be the equation of the circle which passes through the given vertices of the triangle.

Therefore, $c = 0$, $1 + 2g = 0$

$$\Rightarrow \; g = -\frac{1}{2} \text{ and } 1 + 2f = 0 \quad \Rightarrow \quad f = -\frac{1}{2}$$

Substituting the values of g, f and c in the standard equation, we get

$$x^2 + y^2 + 2\times\left(-\frac{1}{2}\right)x + 2\times\left(-\frac{1}{2}\right)y + c = 0$$

$$\Rightarrow \quad x^2 + y^2 - x - y = 0.$$

3. The area of a circle centred at (1, 2) and passing through (4, 6) is

(a) 5π sq. units (b) 15π sq. units

(c) 25π sq. units (d) 30π sq. units

Ans. (c)

Solution: Coordinates of the centre of the circle $(x_1, y_1) = (1, 2)$ and coordinates of the point through which it passes $(x_2, y_2) = (4, 6)$.

We know that the radius of the circle

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(4-1)^2 + (6-2)^2}$$

$$= \sqrt{9 + 16} = 5$$

Therefore, the area of the circle

$$\pi r^2 = \pi(5)^2 = 25\pi \text{ sq. units}$$

4. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have coordinates (3, 4) and (–4, 3), respectively, then the value of $\angle QPR$ is

(a) $\pi/2$ (b) $\pi/3$

(c) $\pi/4$ (d) $\pi/6$

Ans. (c)

Solution: Equation of the circle is $x^2 + y^2 = 25$;

ΔPQR is inscribed in the circle and co-ordinates of Q and R are (3, 4) and (–4, 3).

We know that the slope of OQ (m_1) = 4/3 and the slope of OR (m_2) = –3/4.

Since $m_1 \times m_2 = \frac{4}{3} \times \left(-\frac{3}{4}\right) = -1$,

therefore $\angle QOR = \frac{\pi}{2}$.

We also know that the angle subtended at the centre of a circle is twice the angle subtended in the alternate segment. Therefore, $\angle QPR = \frac{\pi}{4}$.

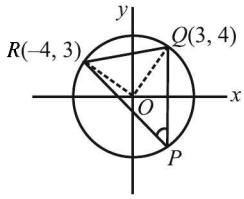


Fig. 19.29

5. Let AB be a chord of the circle $x^2 + y^2 = r^2$ subtending right angle at the centre. Then the locus of the centroid of the triangle PAB as P moves on the circle is

(a) a circle (b) a parabola

(c) an ellipse (d) a pair of straight lines

Ans. (a)

Solution: Equation of the circle is $x^2 + y^2 = r^2$.

Let co-ordinates of the point $P(x_1, y_1) = (r\cos\theta, r\sin\theta)$.

Since chord AB subtends right angle at the centre, therefore coordinates of point $A(x_2, y_2) = (r, 0)$ and point $B(x_3, y_3) = (0, r)$.

We know that coordinates of the centroid of the ΔPAB

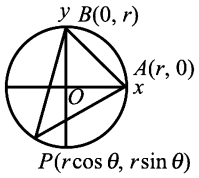


Fig. 19.30

$$(h,k) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

$$= \left(\frac{r\cos\theta + r + 0}{3}, \frac{r\sin\theta + 0 + r}{3}\right)$$

$$= \left(\frac{1}{3}(r + r\cos\theta), \frac{1}{3}(r + r\sin\theta)\right)$$

Therefore, $h = \frac{1}{3}(r + r\cos\theta)$

$$\Rightarrow \quad h - \frac{1}{3}r = \frac{1}{3}r\cos\theta \qquad \text{(i)}$$

and $k = \frac{1}{3}(r + r\sin\theta)$

$$\Rightarrow \quad k - \frac{1}{3}r = \frac{1}{3}r\sin\theta \qquad \text{(ii)}$$

Squaring and adding Eqs. (i) and (ii), we get

$$\left(h - \frac{1}{3}r\right)^2 + \left(k - \frac{1}{3}r\right)^2 = \frac{1}{9}r^2(\cos^2\theta + \sin^2\theta) = \frac{1}{9}r^2$$

Thus, the locus of the centroid of ΔPAB is

$\left(x - \frac{r}{3}\right)^2 + \left(y - \frac{r}{3}\right)^2 = \frac{r^2}{9}$, which is a circle.

6. The equation of the diameter of the circle $x^2 + y^2 + 2x - 4y - 11 = 0$, which bisects the chords intercepted on the line $2x - y + 3 = 0$, is

(a) $x + y - 7 = 0$ (b) $2x - y - 5 = 0$

(c) $x + 2y - 3 = 0$ (d) $2x + y - 3 = 0$

Ans. (c)

Solution: Equation of the circle is $x^2 + y^2 + 2x - 4y - 11 = 0$.

We know that the standard equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0.$$

Comparing the given equation with the standard equation, we get

$$g = 1, f = -2 \text{ and } c = -11$$

We also know that coordinates of the centre of the circle $(-g, -f) = (-1, 2)$.

Since the diameter perpendicularly bisects the chords intercepted on $2x - y + 3 = 0$,

therefore the equation of the diameter is $x + 2y + k = 0$.

We also know that the diameter passes through the centre $(-g, -f)$.

Therefore, $(-1) + 2\times 2 + k = 0 \quad \Rightarrow \quad k = -3$.

Thus, the equation of the diameter is $x + 2y - 3 = 0$.

7. The line $3x - 2y = k$ meets the circle $x^2 + y^2 = 4r^2$ at only one point, if k^2 is

(a) $52r^2$ (b) $20r^2$

(c) $\frac{20}{9}r^2$ (d) $\frac{52}{9}r^2$

Ans. (a)

Solution: Equation of the line is $3x - 2y = k$

$$\Rightarrow \quad y = \frac{3x}{2} - \frac{k}{2} \qquad \text{(i)}$$

and equation of the circle is $x^2 + y^2 = 4r^2$.

We know that the standard equation of the line is

$$y = mx + c$$

Comparing Eq. (i) with the standard equation, we get

$$m = \frac{3}{2} \text{ and } c = -\frac{k}{2}$$

We also know that the standard equation of the circle is

$$x^2 + y^2 = a^2 \qquad \text{(ii)}$$

Comparing the given equation of the circle with the standard Eq. (ii), we get

$$a^2 = 4r^2.$$

We also know that the condition for the line to meet at one point is $c^2 = a^2(1 + m^2)$.

Therefore, $\left(\frac{-k}{2}\right)^2 = 4r^2\left[1 + \left(\frac{3}{2}\right)^2\right]$

$$\Rightarrow \frac{k^2}{4} = 4r^2\left(1+\frac{9}{4}\right) \Rightarrow \frac{k^2}{4} = 4r^2\left(\frac{13}{4}\right)$$

$$\Rightarrow k^2 = 52r^2.$$

8. The line $x - y - 3 = 0$ touches the circle $x^2 + y^2 - 4x + 6y + 11 = 0$ at a point whose coordinates are

(a) (1, –2) (b) (1, 2)

(c) (–1, 2) (d) (–1, –2)

Ans. (a)

Solution: Equation of the line is $x - y - 3 = 0$

$$\Rightarrow x = y + 3 \quad \text{(i)}$$

and equation of the circle is

$$x^2 + y^2 - 4x + 6y + 11 = 0 \quad \text{(ii)}$$

Substituting the value of x from Eq. (i) in Eq. (ii), we get

$$(y+3)^2 + y^2 - 4(y+3) + 6y + 11 = 0$$

$$\Rightarrow y^2 + 6y + 9 + y^2 - 4y - 12 + 6y + 11 = 0$$

$$\Rightarrow 2y^2 + 8y + 8 = 0 \Rightarrow y^2 + 4y + 4 = 0$$

$$\Rightarrow (y+2)^2 = 0 \Rightarrow y = -2$$

Substituting this value of y in Eq. (i), we get $x = -2 + 3 = 1$.

Therefore, coordinates of the point = (1, –2).

9. The equation of the circle with centre (1, 2) and tangent $x + y - 5 = 0$ is

(a) $x^2 + y^2 + 2x - 4y + 6 = 0$

(b) $x^2 + y^2 - 2x - 4y + 3 = 0$

(c) $x^2 + y^2 - 2x - 4y + 8 = 0$

(d) $x^2 + y^2 - 2x - 4y + 8 = 0$

Ans. (b)

Solution: Coordinates of the centre of the circle (h, k) = (1, 2) and equation of the tangent is $x + y - 5 = 0$.

We know that the standard equation of the straight line is $ax + by + c = 0$.

Comparing the given equation with the standard equation, we get

$$a = 1,\ b = 1 \text{ and } c = -5.$$

We also know that the length of the perpendicular drawn from the centre (h, k) to the given line is equal to the radius of the circle, i.e.

$$(p) = \frac{|ah + bk + c|}{\sqrt{a^2 + b^2}} = \frac{|(1\times1) + (1\times2) - 5|}{\sqrt{(1)^2 + (1)^2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

We also know that the equation of the circle with (h, k) as coordinates of its centre and r as radius is

$$(x-h)^2 + (y-k)^2 = r^2$$

$$\Rightarrow (x-1)^2 + (y-2)^2 = (\sqrt{2})^2$$

$$\Rightarrow x^2 + 1 - 2x + y^2 + 4 - 4y = 2$$

$$\Rightarrow x^2 + y^2 - 2x - 4y + 3 = 0.$$

10. If the lines $3x - 4y + 10 = 0$ and $3x - 4y - 15 = 0$ are tangents to the circle, then its radius is

(a) 5/2 (b) 25/2

(c) 5 (d) 25

Ans. (a)

Solution: Equations of lines are

$$3x - 4y + 10 = 0 \quad \text{(i)}$$

and

$$3x - 4y - 15 = 0 \quad \text{(ii)}$$

We know that the given standard equation of the line is $ax + by + c = 0$.

Comparing Eqs. (i) and (ii) with the standard equation, we get

$$a = 3, b = -4, c_1 = 10 \text{ and } c_2 = -15.$$

Since tangents are parallel to each other, therefore distance between lines (i) and (ii)

$$= \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} = \frac{10 + 15}{\sqrt{9 + 16}} = \frac{25}{5} = 5$$

We also know that the distance between the lines is equal to the diameter of circle.

Therefore, radius of the circle

$$= \frac{1}{2} \text{ (distance between the lines)}$$

$$= \frac{1}{2} \times 5 = \frac{5}{2}.$$

11. The length of the tangent from (–3, 1) to the circle $3x^2 + 3y^2 - 5x - 6y - 12 = 0$ is

(a) 3 (b) $\sqrt{3}$

(c) 4 (d) $\sqrt{5}$

Ans. (a)

Solution: Coordinates of the point $P(x_1, y_1) = (-3, 1)$ and equation of the circle is $3x^2 + 3y^2 - 5x - 6y - 12 = 0$

The given equation may be written as

$$x^2 + y^2 - \frac{5}{3}x - 2y - 4 = 0 \quad \text{(i)}$$

We know that the standard equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0.$$

Comparing Eq. (i) with the standard equation, we get

$$g = -\frac{5}{6},\ f = -1 \text{ and } c = -4.$$

We also know that the length of the tangent from the given point to the circle

$$= \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c} = \sqrt{9 + 1 + 5 - 2 - 4}$$

$$= \sqrt{9} = 3.$$

12. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y - 2 = 0$ meets the straight line $5x - 2y + 6 = 0$ at point Q on the y-axis, then the length of PQ is

(a) 2 (b) 5

(c) $2\sqrt{5}$ (d) $3\sqrt{5}$

Ans. (b)

Solution: Equation of the circle is

$$x^2 + y^2 + 6x + 6y - 2 = 0 \quad \text{(i)}$$

and equation of the straight line is $5x - 2y + 6 = 0$

$$\Rightarrow \quad y = \frac{5}{2}x + 3 \quad \text{(ii)}$$

We know that the standard equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Comparing Eq. (i) with the standard equation, we get

$$g = 3, f = 3 \text{ and } c = -2.$$

We also know that coordinates of the centre of the circle

$$O(-g, -f) = (-3, -3)$$

and its radius $(r) = \sqrt{g^2 + f^2 - c} = \sqrt{9 + 9 + 2} = \sqrt{20}$

We also know that the standard equation of a straight line is $y = mx + c$.

Comparing Eq. (ii) with the standard equation, we get

$$m = \frac{5}{2} \text{ and } c = 3.$$

Thus, the straight line intersects the y-axis at $Q = (0, 3)$.

We also know that the distance between the point Q and centre O is

$$(QO) = \sqrt{(0+3)^2 + (3+3)^2} = \sqrt{45}$$

Therefore, the distance between points P and Q is

$$(PQ) = \sqrt{(QO)^2 - (r)^2} = \sqrt{45 - 20} = \sqrt{25} = 5$$

EXERCISE SET

1. The equation of the incircle of the triangle formed by the axes and the line $4x + 3y = 6$ is

(a) $x^2 + y^2 - 6x - 6y + 9 = 0$

(b) $4(x^2 + y^2 - x - y) + 1 = 0$

(c) $4(x^2 + y^2 + x + y) + 1 = 0$

(d) none of these

2. The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ will represent a real circle if

(a) $g^2 + f^2 - c < 0$ (b) $g^2 + f^2 - c \geq 0$

(c) always (d) none of these

3. The equation of the circle which passes through the centre of the circle $x^2 + y^2 + 8x + 10y - 7 = 0$ and is concentric with the circle $2x^2 + 2y^2 - 8x - 12y - 9 = 0$ is

(a) $x^2 + y^2 - 4x - 8y - 97 = 0$

(b) $x^2 + y^2 - 4x - 6y - 87 = 0$

(c) $x^2 + y^2 - 2x - 8y - 95 = 0$

(d) none of these

4. Consider four circles $(x \pm 1)^2 + (y \pm 1)^2 = 1$. Then the equation of the smaller circle touching these four circles is

(a) $x^2 + y^2 = 3 - \sqrt{2}$ (b) $x^2 + y^2 = 6 - 3\sqrt{2}$

(c) $x^2 + y^2 = 5 - 2\sqrt{2}$ (d) $x^2 + y^2 = 3 - 2\sqrt{2}$

5. If a circle passes through the point (1, 2) and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the equation of the locus of its centre is

(a) $x^2 + y^2 - 3x - 8y + 1 = 0$

(b) $x^2 + y^2 - 2x - 6y - 7 = 0$

(c) $2x + 4y - 9 = 0$

(d) $2x + 4y - 1 = 0$

6. The Cartesian equation of the curves $x = 7 + 4\cos\alpha$ and $y = -3 + 4\sin\alpha$ is

(a) $x^2 + y^2 - 14x + 6y + 42 = 0$

(b) $x^2 + y^2 - 6x + 14y + 21 = 0$

(c) $x^2 + y^2 - 10x + 12y + 28 = 0$

(d) none of these

7. The angle between tangents drawn from P to the circle $x^2 + y^2 + 4x - 6y + 9\sin^2\theta + 13\cos^2\theta = 0$ is 2θ. The locus of point P is

(a) $x^2 + y^2 + 4x - 6y - 9 = 0$

(b) $x^2 + y^2 + 4x - 6y - 4 = 0$

(c) $x^2 + y^2 + 4x - 6y + 9 = 0$

(d) $x^2 + y^2 + 4x - 6y + 4 = 0$

8. A line is drawn through a fixed point $P(\alpha, \beta)$ to cut the circle $x^2 + y^2 = r^2$ at A and B. Then $PA \cdot PB$ is equal to

(a) $(\alpha + \beta)^2 - r^2$ (b) $\alpha^2 + \beta^2 - r^2$

(c) $(\alpha - \beta)^2 + r^2$ (d) none of these

9. Parametric equations of the circle $2x^2 + 2y^2 - 5x - 7y - 3 = 0$ are

(a) $x=\frac{3}{4}+\frac{5}{4}\sqrt{2}\cos\theta, y=\frac{5}{4}+\frac{5}{4}\sqrt{2}\sin\theta$

(b) $x=\frac{5}{4}+\frac{7}{4}\sqrt{2}\cos\theta, y=\frac{7}{4}+\frac{7}{4}\sqrt{2}\sin\theta$

(c) $x=\frac{7}{4}+\frac{9}{4}\sqrt{2}\cos\theta, y=\frac{9}{4}+\frac{9}{4}\sqrt{2}\sin\theta$

(d) none of these

10. The condition that the straight line $cx-by+b^2=0$ may touch the circle $x^2+y^2=ax+by$ is

(a) $abc=1$ (b) $a=c$

(c) $b=ac$ (d) none of these

11. The circles $x^2+y^2+x+y=0$ and $x^2+y^2+x-y=0$ intersect at an angle of

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

12. The equation of the tangent lines to the circle $x^2+y^2=9$ which are parallel to the line $2x+y-3=0$ is

(a) $2x+y\pm3\sqrt{5}=0$ (b) $2x+y\pm7\sqrt{5}=0$

(c) $2x+y\pm2\sqrt{3}=0$ (d) none of these

13. If OA and OB are tangents from the origin to the circle $x^2+y^2+2gx+2fy+c=0$ and C is the centre of the circle, then the area of the quadrilateral $OACB$ is

(a) $\frac{1}{2}\sqrt{g^2+f^2-c}$ (b) $\sqrt{c(g^2+f^2-c)}$

(c) $c\sqrt{g^2+f^2-c}$ (d) $\sqrt{\frac{g^2+f^2-c^2}{c}}$

14. A circle of radius 5 unit touches both the axes and lies in the first quadrant. If the circle makes one complete roll on the x-axis along the positive direction of the x-axis, then its equation in the new position is

(a) $x^2+y^2+20\pi x-10y+100\pi^2=0$

(b) $x^2+y^2+20\pi x+10y+100\pi^2=0$

(c) $x^2+y^2-20\pi x-10y+100\pi^2=0$

(d) none of these

15. If $(\alpha, 3)$ and $(3, 5)$ are the extremities of the diameter of a circle with centre at $(2, \beta)$, then the values of α and β are

(a) $\alpha=1, \beta=4$ (b) $\alpha=4, \beta=2$

(c) $\alpha=4, \beta=1$ (d) none of these

16. The angle between the pair of tangents from the point $\left(1,\frac{1}{2}\right)$ to the circle $x^2+y^2+4x+2y-4=0$ is

(a) $\cos^{-1}\frac{4}{5}$ (b) $\sin^{-1}\frac{4}{5}$

(c) $\sin^{-1}\frac{3}{5}$ (d) none of these

17. If the distances from the the origin of the centres of three circles $x^2+y^2+2\lambda_k x-c^2=0$ $(k=1, 2, 3)$ are in G.P., then the lengths of the tangents drawn to them from any point of the circle $x^2+y^2=c^2$ are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

18. The radius of the circle passing through the points $(1, 2)$, $(5, 2)$ and $(5, -2)$ is

(a) $2\sqrt{5}$ (b) $3\sqrt{2}$

(c) $5\sqrt{2}$ (d) $2\sqrt{2}$

19. The locus of the point of intersection of tangents to the circles $x=a\cos\theta$ and $y=a\sin\theta$ at the points, whose parametric angles differ by $\pi/2$, is

(a) a straight line (b) a circle

(c) a pair of straight lines (d) none of these

20. The circle $x^2+y^2+4x-7y+12=0$ cuts an intercept on the y-axis equal to

(a) 7 (b) 4

(c) 3 (d) 1

21. The equation of a circle passing through $(3, -6)$ and touching both the axes is

(a) $x^2+y^2-6x+6y+9=0$

(b) $x^2+y^2+6x-6y+9=0$

(c) $x^2+y^2+30x-30y+225=0$

(d) none of these

22. The centres of a set of circles, each of radius 3, lie on the circle $x^2+y^2=25$. The locus of any point in the set is

(a) $4\le x^2+y^2\le64$ (b) $x^2+y^2\le25$

(c) $x^2+y^2\ge25$ (d) $3\le x^2+y^2\le9$

23. The equation of the line parallel to the line $3x+4y=0$ and touching the circle $x^2+y^2=9$ in the first quadrant is

(a) $3x+4y=9$ (b) $3x+4y=45$

(c) $3x+4y=15$ (d) none of these

24. The locus of the centre of a circle of diameter 4 which rolls on the outside of the circle $x^2+y^2+3x-6y-9=0$ is

(a) $x^2+y^2+3x-6y+5=0$

(b) $x^2+y^2+3x-6y-31=0$

(c) $x^2+y^2+3x-6y+\frac{29}{4}=0$

(d) none of these

25. Let a circle be $2x(x-a)+y(2y-b)=0$, $a\neq 0$, $b\neq 0$. Then the condition on a and b if two chords each bisected by the x-axis, can be drawn to the circle from $\left(a, \frac{b}{2}\right)$, is

(a) $a^2 > 2b^2$ (b) $a^2 < 2b^2$

(c) $a^2 = 2b^2$ (d) none of these

26. If the circle $x^2+y^2+2gx+2fy+c=0$ is touched by $y=x$ at P such that $OP = 6\sqrt{2}$, then the value of c is

(a) 36 (b) 144

(c) 72 (d) none of these

27. If one of the diameters of the circle $x^2+y^2-2x-6y+6=0$ is a chord to the circle with centre (2, 1), then the radius of the circle is

(a) $\sqrt{3}$ (b) $\sqrt{2}$

(c) 3 (d) 2

28. The length of the common chord of the two circles $(x-a)^2+(y-b)^2=c^2$ and $(x-b)^2+(y-a)^2=c^2$ is

(a) $\sqrt{4c^2+2(a-b)^2}$ (b) $\sqrt{4c^2-(a-b)^2}$

(c) $\sqrt{4c^2-2(a-b)^2}$ (d) $\sqrt{2c^2-2(a-b)^2}$

29. A triangle is formed by the lines whose combined equation is given by $(x+y-4)(xy-2x-y+2)=0$. The equation of its circumcircle is

(a) $x^2+y^2-5x-3y+8=0$

(b) $x^2+y^2-3x-5y+8=0$

(c) $x^2+y^2-3x-5y-8=0$

(d) none of these

30. The centre of a circle passing through points (0, 0), (1, 0) and touching the circle $x^2+y^2=9$ is

(a) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (b) $\left(\frac{1}{2}, \frac{3}{2}\right)$

(c) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (d) $\left(\frac{1}{2}, -2^{1/2}\right)$

Assertion-Reason Type Questions

These questions are Assertion-Reason type questions. Each of these questions contains two statements. Statement I (Assertion) and Statement II (Reason). Each of these questions has four alternative answers. Only one of them is the correct answer. Select the correct answer as given:

(a) Both Statement I and Statement II are true and Statement II is a correct explanation of Statement I.

(b) Both Statement I and Statement II are true but Statement II is not a correct explanation of Statement I.

(c) Statement I is true, Statement 2 is false.

(d) Statement I is false, Statement II is true.

31. Tangents are drawn from the point (17, 7) to the circle $x^2+y^2=169$.

Statement I: The tangents are perpendicular.

Statement II: The locus of points from which mutually perpendicular tangents can be drawn to the given circle is $x^2+y^2=338$.

32. **Statement I:** Locus of mid-points of chords of $x^2+y^2=a^2$ subtending constant angle β at the centre is

$$x^2+y^2=a^2\cos^2\frac{\beta}{2}.$$

Statement II: Such a locus must be a closed curve.

33. **Statement I:** The length of the tangent from the origin to the circle $x^2+y^2+4x+6y+\lambda=0$ is 4 if $\lambda=16$.

Statement II: Length of tangent is $\sqrt{5}$.

34. **Statement I:** The number of common tangents to the circles $x^2+y^2=4$ and $(x-5)^2+y^2=k^2$ will be 4 only if $k<1$.

Statement II: Two circles have four common tangents if the distance between the centres is greater than the sum of the radii.

Previous Years' Questions

35. If the two circles $(x-1)^2+(y-3)^2=r^2$ and $x^2+y^2-8x+2y+9=0$ intersect in two distinct points, then

(a) $2<r<8$ (b) $r<2$

(c) $r=2$ (d) $r>2$

[AIEEE 2004, AMU 2003]

36. The intercept on the line $y=x$ by the circle $x^2+y^2-2x=0$ is AB. The equation of the circle with AB as a diameter is

(a) $x^2+y^2-x-y=0$ (b) $x^2+y^2-x+y=0$

(c) $x^2+y^2+x+y=0$ (d) $x^2+y^2+x-y=0$

[AIEEE 2004]

37. If the lines $2x+3y+1=0$ and $3x-y-4=0$ lie along diameters of a circle of circumference 10π, then the equation of the circle is

(a) $x^2+y^2-2x+2y-23=0$

(b) $x^2+y^2-2x-2y-23=0$

(c) $x^2+y^2+2x+2y-23=0$

(d) $x^2+y^2+2x-2y-23=0$ **[AIEEE 2004]**

38. A circle touches the x-axis and also touches the circle with centre at (0, 3) and radius 2. The locus of the centre of the circle is

(a) a hyperbola (b) a parabola

(c) an ellipse (d) a circle

[AIEEE 2005]

39. If the circles $x^2 + y^2 + 2ax + cy + a = 0$ and $x^2 + y^2 - 3ax + dy - 1 = 0$ intersect in two distinct points P and Q, then the line $5x + by = 0$ passes through P and Q for
(a) exactly two values of a
(b) infinitely many values of a
(c) no value of a
(d) exactly one value of a **[AIEEE 2005]**

40. If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = p^2$ orthogonally, then the equation of the locus of its centre is
(a) $x^2 + y^2 - 2ax - 3by + (a^2 - b^2 - p^2) = 0$
(b) $2ax + 2by - (a^2 + b^2 - p^2) = 0$
(c) $x^2 + y^2 - ax - 4by + (a^2 + b^2 - p^2) = 0$
(d) $2ax + 2by - (a^2 - b^2 + p^2) = 0$ **[AIEEE 2005]**

41. Let C be the circle with centre (0, 0) and radius 3 units. The equation of the locus of the mid-points of the chords of the circle C that subtend an angle of $2\pi/3$ at its centre is
(a) $x^2 + y^2 = 1$ (b) $x^2 + y^2 = \frac{27}{4}$
(c) $x^2 + y^2 = \frac{9}{4}$ (d) $x^2 + y^2 = \frac{3}{2}$
[AIEEE 2006]

42. If the lines $3x - 4y - 7 = 0$ and $2x - 3y - 5 = 0$ are two diameters of a circle of area 49π sq. units, the equation of the circle is
(a) $x^2 + y^2 + 2x - 2y - 62 = 0$
(b) $x^2 + y^2 - 2x - 2y - 62 = 0$
(c) $x^2 + y^2 - 2x + 2y - 47 = 0$
(d) $x^2 + y^2 + 2x - 2y - 47 = 0$
[AIEEE 2006, 2003, DCE 2004]

43. Let $ABCD$ be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2CD$. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral $ABCD$ touching all the sides, then its radius is
(a) 3 (b) 2
(c) 3/2 (d) 1 **[AIEEE 2007]**

44. Consider a family of circles which are passing through the point (–1, 1) and are tangent to the x-axis. If (h, k) are the coordinates of the centre to the circles, then the set of values of k is given by the interval
(a) $0 < k < \frac{1}{2}$ (b) $k \geq \frac{1}{2}$
(c) $-\frac{1}{2} \leq k \leq \frac{1}{2}$ (d) $k \leq \frac{1}{2}$
[AIEEE 2007]

45. The point diametrically opposite to the point $P(1, 0)$ on the circle $x^2 + y^2 + 2x + 4y - 3 = 0$ is
(a) (–3, 4) (b) (–3, –4)
(c) (3, 4) (d) (3, –4) **[AIEEE 2008]**

46. If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$, then there is a circle passing through P, Q and (1, 1) and
(a) all values of p
(b) all except one value of p
(c) all except two values of p
(d) exactly one value of p **[AIEEE 2009]**

47. The circle $x^2 + y^2 = 4x + 8y + 5$ intersects the line $3x - 4y = m$ at two distinct points, if
(a) $-85 < m < -35$ (b) $-35 < m < 15$
(c) $15 < m < 65$ (d) $35 < m < 85$
[AIEEE 2010]

48. How many common tangents can be drawn to the following circles $x^2 + y^2 = 6x$ and $x^2 + y^2 + 6x + 2y + 1 = 0$?
(a) 4 (b) 3
(c) 2 (d) 1 **[DCE 2004]**

49. The length of the transversal common tangent to the circle $x^2 + y^2 = 1$ and $(x - t)^2 + y^2 = 1$ is $\sqrt{21}$, then t is equal to
(a) ± 2 (b) ± 5
(c) ± 3 (d) none of these
[DCE 2005]

50. If $2x - 4y = 9$ and $6x - 12y + 7 = 0$ are common tangents to the circle, then the radius of the circle is
(a) $\frac{\sqrt{3}}{5}$ (b) $\frac{17}{6\sqrt{5}}$
(c) $\frac{\sqrt{2}}{3}$ (d) $\frac{17}{3\sqrt{5}}$ **[DCE 2005]**

51. The equation of the circle concentric to the circle $2x^2 + 2y^2 - 3x + 6y + 2 = 0$ and having area double the area of this circle is
(a) $8x^2 + 8y^2 - 24x + 48y - 13 = 0$
(b) $16x^2 + 16y^2 + 24x - 48y - 13 = 0$
(c) $16x^2 + 16y^2 - 24x + 48y - 13 = 0$
(d) $8x^2 + 8y^2 + 24x - 48y - 13 = 0$ **[DCE 2006]**

52. The length of the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$ is
(a) $\frac{9}{2}$ (b) $2\sqrt{2}$
(c) $3\sqrt{2}$ (d) $\frac{3}{2}$ **[DCE 2007]**

53. If the line $y\cos\alpha = x\sin\alpha + a\cos\alpha$ is a tangent to the circle $x^2 + y^2 = a^2$, then

(a) $\sin^2\alpha = 1$ (b) $\cos^2\alpha = 1$

(c) $\sin^2\alpha = a^2$ (d) $\cos^2\alpha = a^2$

[DCE 2008]

54. The radius of the circle, which is touched by the line $y = x$ and has its centre on the positive direction of the x-axis and also cuts off a chord of length 2 unit along the line $\sqrt{3}y - x = 0$, is

(a) $\sqrt{5}$ (b) $\sqrt{3}$

(c) $\sqrt{2}$ (d) 1 **[DCE 2009]**

55. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have coordinates (3, 4) and (−4, 3), respectively, then $\angle OPR$ is equal to

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$ **[IIT-JEE 2000]**

56. Let AB be a chord of the circle $x^2 + y^2 = r^2$ subtending a right angle at the centre. Thus, the locus of the centroid of the triangle PAB as P moves on the circle is

(a) a parabola (b) a circle

(c) an ellipse (d) a pair of straight lines

[IIT-JEE 2001]

57. Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle, then $2r$ equals

(a) $\sqrt{PQ \cdot RS}$ (b) $\frac{PQ + RS}{2}$

(c) $\frac{2PQ \cdot RS}{PQ + RS}$ (d) $\sqrt{\frac{PQ^2 + RS^2}{2}}$

[IIT-JEE 2001]

58. If $a > 2b > 0$, then the positive value of m for which $y = mx - b\sqrt{1 + m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x - a)^2 + y^2 = b^2$ is

(a) $\frac{2b}{\sqrt{a^2 - 4b^2}}$ (b) $\frac{\sqrt{a^2 - 4b^2}}{2b}$

(c) $\frac{2b}{a - 2b}$ (d) $\frac{b}{a - 2b}$ **[IIT-JEE 2002]**

59. The centre of the circle inscribed in square formed by the lines $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$ is

(a) (4, 7) (b) (7, 4)

(c) (9, 4) (d) (4, 9) **[IIT-JEE 2003]**

60. A circle is given by $x^2 + (y - 1)^2 = 1$, another circle C touches it externally and also the x-axis, then the locus of its centre is

(a) $\{(x, y) : x^2 = 4y\} \cup \{(x, y) : y \le 0\}$

(b) $\{(x, y) : x^2 + (y - 1)^2 = 4\} \cup \{(x, y) : y \le 0\}$

(c) $\{(x, y) : x^2 = y\} \cup \{(0, y) : y \le 0\}$

(d) $\{(x, y) : x^2 = 4y\} \cup \{(0, y) : y \le 0\}$ **[IIT-JEE 2005]**

61. The intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$ is AB. The equation of the circle on AB as a diameter is

(a) $x^2 + y^2 - x - y = 0$ (b) $x^2 + y^2 - x + y = 0$

(c) $x^2 + y^2 + x + y = 0$ (d) $x^2 + y^2 + x - y = 0$

[Jamia Millia Islamia 2007]

62. The equation of the circle having $x - y - 2 = 0$ and $x - y + 2 = 0$ as two tangents and $x - y = 0$ as a diameter is

(a) $x^2 + y^2 + 2x - 2y + 1 = 0$

(b) $x^2 + y^2 - 2x + 2y - 1 = 0$

(c) $x^2 + y^2 = 2$

(d) $x^2 + y^2 = 1$ **[Jamia Millia Islamia 2008]**

63. The angle between the tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$

(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{8}$

[UPSEAT 1990, 2000]

64. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle of area 154 sq. units. Then the equation of this circle is

(a) $x^2 + y^2 - 2x - 2y - 35 = 0$

(b) $x^2 + y^2 - 2x + 2y - 47 = 0$

(c) $x^2 + y^2 + 2x - 2y + 77 = 0$

(d) $x^2 + y^2 + 2x + 2y + 84 = 0$

[MP PET 2009, UPSEE 2009, BITSAT 2008, AIEEE 2003, 2004, 2006, DCE 2004, IIT 1989]

65. The equation of the circle passing through (4, 6) and having centre (1, 2) is

(a) $x^2 + y^2 - 2x - 4y - 20 = 0$

(b) $x^2 + y^2 - 2x + 4y - 20 = 0$

(c) $x^2 + y^2 + 2x - 4y - 20 = 0$

(d) $x^2 + y^2 + 2x + 4y - 20 = 0$

[UPSEE 2005, BCECE 2004, MP PET 1999]

66. The lines $3x-4y+4=0$ and $6x-8y-7=0$ are tangents to the same circle. The radius of this circle is
(a) 3/5 (b) 5/9 (c) 3/4 (d) 5/6
[Orissa JEE 2008, AMU 2005, BCECE 2006, Kerala PET 2003, MH CET 1996, IIT 1984]

67. The length of the tangent to the circle $x^2 + y^2 - 2x - y - 7 = 0$ from $(-1, -3)$ is
(a) $2\sqrt{3}$ (b) 4 (c) $\sqrt{8}$ (d) 8
[EAMCET 2006, Kerala PET 2003, KCET 1984]

68. The number of common tangents to the circles $x^2+y^2-6x\ 2y+9=0$ and $x^2+y^2-14x-8y+61=0$ is
(a) 1 (b) 2 (c) 3 (d) 4
[EAMCET 2006, Kerala PET 2001, Orissa JEE 2005]

69. Radical axis of the circles $x^2+y^2+6x-2y-9=0$ and $x^2+y^2-2x+9y-11=0$ is given by
(a) $8x-11y+2=0$ (b) $8x-11y-2=0$
(c) $8x+11y+2=0$ (d) $8x+11y-2=0$
[Rajasthan PET 2003, 2004, KCET 1984, EAMCET 2001]

70. The equation of the circle passing through the points (1, 0) and (0, 1) and having the smallest radius is
(a) $x^2+y^2-2x-2y+1=0$
(b) $x^2+y^2-x-y=0$
(c) $x^2+y^2+2x+2y-7=0$
(d) $x^2+y^2+x+y-2=0$ **[AIEEE 2011]**

71. The length of the diameter of the circle which touches the x-axis at the point (1, 0) and passes through the point (2, 3) is
(a) 10/3 (b) 3/5 (c) 6/5 (d) 5/3
[AIEEE 2012]

ANSWERS

Exercise Set

1. (b)	**2.** (b)	**3.** (b)	**4.** (d)	**5.** (c)	**6.** (a)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (b)
11. (d)	**12.** (a)	**13.** (b)	**14.** (d)	**15.** (a)	**16.** (b)	**17.** (b)	**18.** (d)	**19.** (b)	**20.** (d)
21. (a)	**22.** (a)	**23.** (c)	**24.** (b)	**25.** (a)	**26.** (c)	**27.** (c)	**28.** (c)	**29.** (b)	**30.** (d)
31. (b)	**32.** (b)	**33.** (d)	**34.** (d)	**35.** (a)	**36.** (a)	**37.** (a)	**38.** (b)	**39.** (c)	**40.** (b)
41. (c)	**42.** (c)	**43.** (b)	**44.** (b)	**45.** (b)	**46.** (b)	**47.** (b)	**48.** (a)	**49.** (b)	**50.** (b)
51. (c)	**52.** (b)	**53.** (b)	**54.** (c)	**55.** (c)	**56.** (b)	**57.** (a)	**58.** (a)	**59.** (a)	**60.** (d)
61. (a)	**62.** (c)	**63.** (c)	**64.** (b)	**65.** (a)	**66.** (c)	**67.** (c)	**68.** (d)	**69.** (a)	**70.** (b)
71. (a)									

HINTS AND SOLUTIONS—EXERCISE SET

1. If the inradius is r, then the centre is (r, r) and its distance from the line $4x + 3y = 6$ is r.

So, $\left|\frac{4r+3r-6}{\sqrt{4^2+3^2}}\right| = r$

$\Rightarrow \quad |7r-6| = 5r$

$\Rightarrow \quad 7r-6 = \pm 5r$

$\Rightarrow \quad r = 3, \frac{1}{2}$

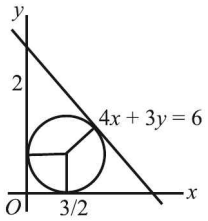


Fig. 19.31

Clearly from the figure, $r \neq 3$.

$\therefore$ The equation of the incircle is

$$\left(x-\frac{1}{2}\right)^2+\left(y-\frac{1}{2}\right)^2=\left(\frac{1}{2}\right)^2$$

$\Rightarrow \quad x^2+y^2-x-y+\frac{1}{4}=0.$

2. The given equation can be written as

$$[x-(-g)]^2+[y-(-f)]^2=(\sqrt{g^2+f^2-c})^2$$

So, the circle is real if

$$g^2-f^2-c \geq 0$$

3. The centre of the given circle is $(-4,-5)$ and the centre of the required circle is (2, 3).

$\Rightarrow$ The equation of the required circle is

$$(x-2)^2+(y-3)^2=100$$

$\Rightarrow \quad x^2+y^2-4x-6y-87=0.$

4. $A_1B_1 = \sqrt{4+4}$

$= 2\sqrt{2}$

$\Rightarrow \quad AB = 2\sqrt{2}-2$

$= 2(\sqrt{2}-1)$

Thus, the equation of the required circle is

$$x^2+y^2=(\sqrt{2}-1)^2$$

$$=3-2\sqrt{2}\ .$$

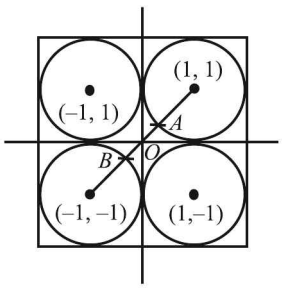


Fig. 19.32

5. Let the equation of the circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Since this passes through (1, 2) (given)

$\therefore \quad 5 + 2g + 4f + c = 0 \qquad$ (i)

Also, the circle $x^2 + y^2 = 4$ intersects the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ orthogonally.

$\therefore \quad 2(g \cdot 0 + f \cdot 0) = c - 4$

$\Rightarrow \quad c = 4$

$\Rightarrow$ Putting $c = 4$ in (i), we have

$$2g + 4f + 9 = 0$$

$\therefore$ The locus of the centre $(-g, -f)$ is $-2x - 4y + 9 = 0$

$\Rightarrow \quad 2x + 4y - 9 = 0$

6. As per the parametric equation of the circle, $h = 7$, $k = -3$, $r = 4$.

$\Rightarrow \quad (x-7)^2 + (y+3)^2 = 16$

$\Rightarrow \quad x^2 + 49 - 14x + y^2 + 9 + 6y = 16$

$\Rightarrow \quad x^2 + y^2 - 14x + 6y + 42 = 0$

7. Centre is (–2, 3) and radius is $2|\sin\theta|$, we have

$$CP = \frac{CA}{\sin\theta} = 2$$

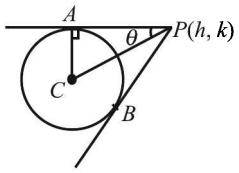


Fig. 19.33

Thus, $(h+2)^2 + (k-3)^2 = 4$

Hence, locus is $x^2 + y^2 + 4x - 6y + 9 = 0$.

8. The equation of any line through $P(\alpha, \beta)$ inclined at an angle θ with +ve x-axis is

$$\frac{x-\alpha}{\cos\theta} = \frac{y-\beta}{\sin\theta} = k$$

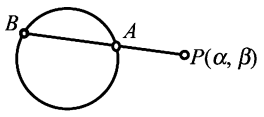


Fig. 19.34

Any point on this line is

$$(\alpha + k\cos\theta, \beta + k\sin\theta)$$

This point lies on the given circle if

$$(\alpha + k\cos\theta)^2 + (\beta + k\sin\theta)^2 = r^2$$

$\Rightarrow \quad k^2 + 2k(\alpha\cos\theta + \beta\sin\theta) + \alpha^2 + \beta^2 - r^2 = 0 \qquad$ (i)

which being quadratic in k, gives two values of k and hence the distance between two points A and B on the circle from the point P.

Let $PA = k_1, PB = k_2$ where k_1, k_2 are the roots of (i).

$\therefore \quad PA \cdot PB = k_1 k_2 = \alpha^2 + \beta^2 - r^2$

Alternative method

$PA \cdot PB$ is the power of the point $P(\alpha, \beta)$ with respect to the circle $x^2 + y^2 = r^2$

$\therefore \quad PA \cdot PB = \alpha^2 + \beta^2 - r^2$

9. The equation of the circle can be written as

$$x^2 + y^2 - \frac{5}{2}x - \frac{7}{2}y - \frac{3}{2} = 0$$

$$\Rightarrow \quad \left(x - \frac{5}{4}\right)^2 + \left(y - \frac{7}{4}\right)^2 = \frac{49}{8}$$

$$\Rightarrow \quad \left(x - \frac{5}{4}\right)^2 + \left(y - \frac{7}{4}\right)^2 = \left(\frac{7}{2\sqrt{2}}\right)^2$$

$$\Rightarrow \quad h = \frac{5}{4}, k = \frac{7}{4}, r = \frac{7\sqrt{2}}{4}$$

Therefore, the parametric equations of the given circle are

$$\Rightarrow \quad x = \frac{5}{4} + \frac{7\sqrt{2}}{4}\cos\theta, y = \frac{7}{4} + \frac{7}{4}\sqrt{2}\sin\theta.$$

10. Circle is given as $x^2 + y^2 = ax + by$

or $\quad x^2 + y^2 - ax - by = 0 \qquad$ (i)

The line is given as $cx - by + b^2 = 0$

or $\quad y = \left(\frac{c}{b}x + b\right) \qquad$ (ii)

On substituting the value of y from Eq. (ii) in Eq. (i), we get

$$x^2 + \left(\frac{c}{b}x + b\right)^2 - ax - b\left(\frac{c}{b}x + b\right) = 0$$

$$\Rightarrow \quad x^2 + \frac{c^2}{b^2}x^2 + 2cx + b^2 - ax - cx - b^2 = 0$$

$$\Rightarrow \quad x^2\left(1 + \frac{c^2}{b^2}\right) + x(c-a) = 0 \qquad \text{(iii)}$$

If it is a perfect square, then $c - a = 0$ or $c = a$.

11. The angle of intersection of two circles is given by

$$\cos\theta = \frac{r_1^2 + r_2^2 - C_1C_2^2}{2r_1r_2} \qquad \text{(i)}$$

where r_1 and r_2 are radii of two circles and C_1C_2 is the distance between their centres.

Here, $\quad r_1 = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{1}{2}} = r_2$ and $C_1C_2 = 1$

Substituting respective values in (i), we have

$\Rightarrow \cos\theta = 0 \qquad \therefore \ \theta = \frac{\pi}{2}$

12. Any line parallel to $2x+y-3=0$ is $2x+y+k=0$. This will be tangent to the circle $x^2+y^2=9$ if its distance from (0, 0) is 3.

Therefore, $\left|\frac{k}{\sqrt{5}}\right|=3 \Rightarrow |k|=3\sqrt{5} \Rightarrow k=\pm 3\sqrt{5}$

The equation of the tangent is $2x+y\pm 3\sqrt{5}=0$.

13. Since $OA = OB = \sqrt{c}$ (length of the tangent from the origin to the circle) and $CA = CB = \sqrt{g^2+f^2-c}$

Area of the quadrilateral $OACB$
= 2 (Area of the triangle OAC)

$$= 2\times\left(\frac{1}{2}\right)OA\times CA$$

$$= \sqrt{c}\sqrt{g^2+f^2-c}.$$

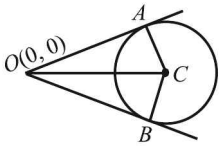


Fig. 19.35

14. The x-coordinate of the new position of the circle is 5 + (circumference of the circle)
i.e. $(5+10\pi)$.

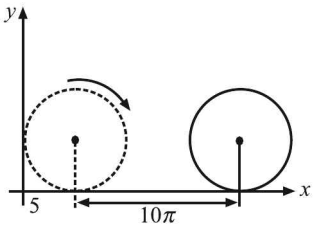


Fig. 19.36

The y-coordinate is 5 and the radius is also 5. Hence, the equation of the circle in the new position is

$$(x-5-10\pi)^2+(y-5)^2=5^2$$

15. As $(2, \beta)$ is the mid-point of the diameter, therefore,

$$\frac{\alpha+3}{2}=2, \frac{3+5}{2}=\beta$$

$$\Rightarrow \alpha=1, \beta=4$$

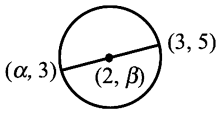


Fig. 19.37

16. The equation of the pair of tangents is $SS_1 = T^2$.

$$\Rightarrow (x^2+y^2+4x+2y-4)\left[1^2+\left(\frac{1}{2}\right)^2+4\cdot 1+2\cdot\frac{1}{2}-4\right]$$

$$=\left[x\cdot 1+y\cdot\frac{1}{2}+2(x+1)+\left(y+\frac{1}{2}\right)-4\right]^2$$

$$\Rightarrow \frac{9}{4}(x^2+y^2+4x+2y-4)=\frac{9}{4}(2x+y-1)^2$$

$$\Rightarrow 3x^2+4xy-8x-4y+5=0$$

$$\therefore \text{ The required angle } = \tan^{-1}\left|\frac{2\sqrt{(2)^2-3\times 0}}{3+0}\right|$$

$$= \tan^{-1}\frac{4}{3}=\sin^{-1}\frac{4}{5}.$$

17.

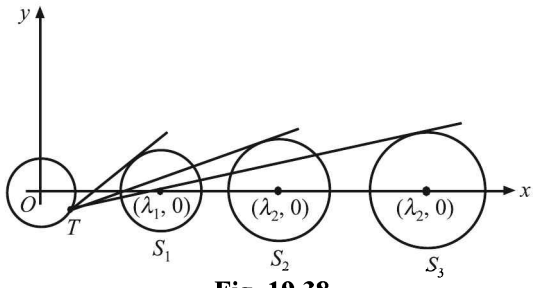


Fig. 19.38

Let (α, β) be any point on the locus of the circle

$$x^2+y^2=c^2$$

$$\therefore \quad \alpha^2+\beta^2=c^2 \qquad \text{(i)}$$

Now, $S_1: x^2+y^2+2\lambda_1 x-c^2=0$

also, $S_2: x^2+y^2+2\lambda_2 x-c^2=0$

$S_3: x^2+y^2+2\lambda_3 x-c^2=0$

Now, $|TS_1|=\sqrt{\alpha^2+\beta^2+2\lambda_1\alpha-c^2}$

$|TS_1|=\sqrt{2\lambda_1\alpha}$ [using (i)]

Similarly, $|TS_2|=\sqrt{2\lambda_2\alpha}$

$|TS_3|=\sqrt{2\lambda_3\alpha}$

Now, assume that $|TS_1|, |TS_2|, |TS_3|$ are in G.P.

$\Rightarrow \sqrt{2\lambda_1\alpha}, \sqrt{2\lambda_2\alpha}, \sqrt{2\lambda_3\alpha}$ are in G.P.

$\Rightarrow 2\lambda_1\alpha, 2\lambda_2\alpha, 2\lambda_3\alpha$ are in G.P.

$\Rightarrow \lambda_1, \lambda_2, \lambda_3$ are in G.P.

which is true.

Hence, our assumption was also true.

$\therefore$ Lengths of tangents drawn to the three circles from any point on $x^2+y^2=c^2$ are in G.P.

18. Now, $(x_1-5)^2+(y_1+2)^2=(x_1-5)^2+(y_1-2)^2$

$=(x_1-1)^2+(y_1-2)^2$

$\Rightarrow y_1=0$ and $x_1=3$

Therefore, the centre of the circle is (3, 0).

$\Rightarrow$ Radius = $\sqrt{4+4}=2\sqrt{2}$

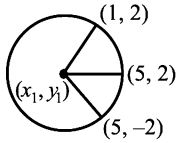


Fig. 19.39

19. Tangents at θ and $\left(\theta+\frac{\pi}{2}\right)$ are

$$x\cos\theta+y\sin\theta=a$$

and $$x\cos\left(\theta+\frac{\pi}{2}\right)+y\sin\left(\theta+\frac{\pi}{4}\right)=a$$

or $$x\cos\theta+y\sin\theta=a$$

and $$-x\cos\theta+y\sin\theta=a$$

Squaring and adding, we get $x^2+y^2=2a^2$.

20. The y-intercept can be obtained by substituting $x = 0$ in the equation.

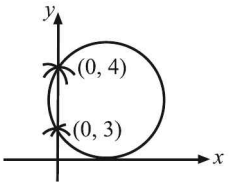


Fig. 19.40

$\Rightarrow \quad y^2 - 7y + 12 = 0$

$\Rightarrow \quad (y-4)(y-3) = 0$

$\Rightarrow$ The circle cuts the y-axis at two points (0, 3) and (0, 4), therefore, intercept = 1.

21.

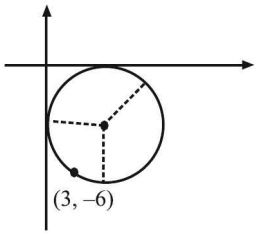


Fig. 19.41

The circle will touch the axes if the abscissa, ordinate of the centre of the circle and radius are equal. In (a), (b) and (c) options, the circle is touching the coordinate axes, but only equations of the circle represented by option (a) pass through (3, – 6).

22. Let (h, k) be any point in the set, then equation of circle is $(x-h)^2 + (y-k)^2 = 9$

But (h, k) lies on $x^2 + y^2 = 25$, then $h^2 + k^2 = 25$.

$\therefore \quad 2 \le$ distance between the centres of two circles ≤ 8

$\Rightarrow \quad 2 \le \sqrt{(h^2 + k^2)} \le 8 \quad \Rightarrow \quad 4 \le h^2 + k^2 \le 64$

$\therefore$ Locus of (h, k) is $4 \le x^2 + y^2 \le 64$.

23. Equation of the line parallel to $3x + 4y = 0$ is of the form $3x + 4y = k$. This line touches $x^2 + y^2 = 9$. Therefore,

$$x^2 + \left(\frac{k-3x}{4}\right)^2 = 9 \text{ or } 25x^2 - 6kx + (k^2 - 144) = 0$$

As the line is a tangent, we should have repeated roots of this equation.

$$\Rightarrow \quad k^2 = \frac{144 \times 25}{16} \text{ or } k = 15.$$

24. Let (h, k) be the coordinates of the centre of the circle which rolls on the outside of the circle $x^2 + y^2 + 3x - 6y - 9 = 0$. Then,

(Distance between centres) = (Sum of the radii)

$$\Rightarrow \quad \sqrt{\left(h + \frac{3}{2}\right) + (k-3)^2} = \frac{9}{2} + 2$$

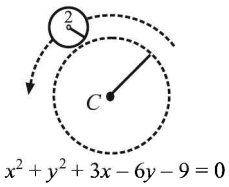


Fig. 19.42

$$\Rightarrow \quad h^2 + k^2 + 3h - 6k + \frac{9}{4} + 9 = \left(\frac{13}{2}\right)^2$$

$$\Rightarrow \quad h^2 + k^2 + 3h - 6k - 31 = 0$$

So, locus of (h, k) is $x^2 + y^2 + 3x - 6y - 31 = 0$.

25. The equation of the circle is $2x(x-a) + y(2y-b) = 0$

$$\Rightarrow \quad x^2 + y^2 - ax - \frac{b}{2}y = 0$$

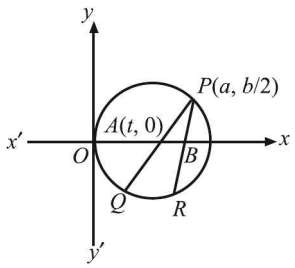


Fig. 19.43

Let PQ and PR be two chords drawn from $P\left(a, \frac{b}{2}\right)$ such that they are bisected by the x-axis.

Let $A(t, 0)$ be the mid-point of PQ. Then its equation is

$$tx + 0y - \frac{a}{2}(x+t) - \frac{b}{4}(y+0) = t^2 - at \qquad \text{(using T = } S'\text{)}$$

$$\Rightarrow \quad \left(t - \frac{a}{2}\right)x - \frac{b}{4}y = t^2 - \frac{a}{2}t$$

This passes through $P\left(a, \frac{b}{2}\right)$. Therefore,

$$\left(t - \frac{a}{2}\right)a - \frac{b^2}{8} = t^2 - \frac{a}{2}t$$

$$\Rightarrow \quad t^2 - \frac{3}{2}at + \left(\frac{a^2}{2} + \frac{b^2}{8}\right) = 0$$

This should give two distinct values of t for points A and B.

$$\therefore \quad \frac{9}{4}a^2 - 4\left(\frac{a^2}{2} + \frac{b^2}{8}\right) > 0$$

$$\Rightarrow \quad \frac{a^2}{4} - \frac{b^2}{2} > 0 \quad \Rightarrow \quad a^2 > 2b^2$$

26. The equation of the given line is $y = x$.

According to distance form, $\frac{x}{\cos\alpha} = \frac{y}{\sin\alpha} = r$,

where α is the angle made by the line $y = x$ with positive direction of the x-axis = $\pi/4$.

For point P, $r = 6\sqrt{2}$.

$\therefore$ Coordinates of P are given by

$$\frac{x}{\cos\pi/4} = \frac{y}{\sin\pi/4} = 6\sqrt{2}$$

$\Rightarrow x = 6, y = 6$

Since $P(6, 6)$ lies on $x^2 + y^2 + 2gx + 2fy + c = 0$

$\therefore \quad 72 + 12(g + f) + c = 0 \qquad$ (i)

Also, $y = x$ touches the given circle.

$\therefore$ The equation $2x^2 + 2x(g + f) + c = 0$ has equal roots

$\Rightarrow \quad 4(g + f)^2 = 8c \quad \Rightarrow \quad (g + f)^2 = 2c \qquad$ (ii)

Using (i), we have

$$[12(g + f)]^2 = [-(c + 72)]^2$$

$$\Rightarrow \quad 144(g + f)^2 = (c + 72)^2$$

$$\Rightarrow \quad 144(2c) = (c + 72)^2 \qquad \text{[using (ii)]}$$

$$\Rightarrow \quad (c - 72)^2 = 0$$

$$\therefore \quad c = 72$$

27. The given circle is $x^2 + y^2 - 2x - 6y + 6 = 0$ with centre $C(1, 3)$ and radius

$$= \sqrt{1 + 9 - 6} = 2.$$

Let AB be one of its diameters which is the chord of the other circle with centre at $C_1(2, 1)$.

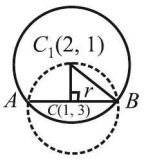


Fig. 19.44

Then in $\Delta C_1CB, C_1B^2 = CC_1^2 + CB^2$

$$\Rightarrow \quad r^2 = [(2 - 1)^2 + (1 - 3)^2] + 2^2$$

$$\Rightarrow \quad r^2 = 1 + 4 + 4 \quad \Rightarrow \quad r^2 = 9$$

$$\Rightarrow \quad r = 3$$

28. The equations of two circles are

$$S_1 \equiv (x - a)^2 + (y - b)^2 = c^2 \qquad \text{(i)}$$

and $$S_2 \equiv (x - b)^2 + (y - a)^2 = c^2 \qquad \text{(ii)}$$

The equation of the common chord of these circles is

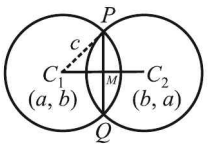


Fig. 19.45

$$S_1 - S_2 = 0$$

$$\Rightarrow \quad (x - a)^2 - (x - b)^2 + (y - b)^2 - (y - a)^2 = 0$$

$$\Rightarrow \quad (2x - a - b)(b - a) + (2y - b - a)(a - b) = 0$$

$$\Rightarrow \quad 2x - a - b - 2y + b + a = 0$$

$$\Rightarrow \quad x - y = 0$$

Let C_1 and C_2 be the centres of circles (i) and (ii), respectively. Then the coordinates of C_1 and C_2 are (a, b) and (b, a), respectively.

Now, $$C_1M = \left|\frac{a - b}{\sqrt{1 + 1}}\right| = \frac{|a - b|}{\sqrt{2}}$$

In the right triangle C_1PM, we have

$$PM = \sqrt{C_1P^2 - C_1M^2} = \sqrt{c^2 - \frac{(a - b)^2}{2}}$$

$$\therefore \quad PQ = 2PM = 2\sqrt{c^2 - \frac{(a - b)^2}{2}} = \sqrt{4c^2 - 2(a - b)^2}$$

29. The sides of the triangle are

$$x + y - 4 = 0, x - 1 = 0 \text{ and } y - 2 = 0.$$

So, the triangle is right-angled at (1, 2).

The hypotenuse is $x + y - 4 = 0$ whose ends are (1, 3) and (2, 2).

The circumcentre is $\left(\frac{1 + 2}{2}, \frac{3 + 2}{2}\right)$ and

circumradius is $\frac{1}{2}\sqrt{(1 - 2)^2 + (3 - 2)^2} = \frac{1}{\sqrt{2}}$

$\therefore$ The equation of the circle is

$$\left(x - \frac{3}{2}\right)^2 + \left(y - \frac{5}{2}\right)^2 = \frac{1}{2}$$

$$\therefore \quad x^2 + y^2 - 3x - 5y + 8 = 0$$

30. Let (h, k) be the centre of the circle passing through $(0, 0)$ and $(1, 0)$, then

$$(h - 0)^2 + (k - 0)^2 = (h - 1)^2 + (k - 0)^2$$

$$\Rightarrow \quad h = \frac{1}{2}$$

Since this circle touches $x^2 + y^2 = 9$ internally (only that is possible), we have distance between centres = difference of radii.

$$\Rightarrow \quad \sqrt{h^2 + k^2} = 3 - \sqrt{h^2 + k^2} \quad \Rightarrow \quad k = \pm\sqrt{2}.$$

Thus, the centre of such a circle should be either

$$\left(\frac{1}{2}, \sqrt{2}\right) \text{ or } \left(\frac{1}{2}, -\sqrt{2}\right).$$

31. Since (17, 7) lies on $x^2 + y^2 = 169$ and $x^2 + y^2 = 338$ is the director circle (director circle is the locus of point of inter-section of tangents) of the given circle, Statement I is true and follows from Statement II.

32.

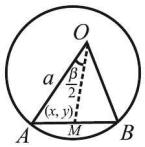


Fig. 19.46

$OM = \sqrt{x^2 + y^2}$ OM is also equal to $a\cos\frac{\beta}{2}$

$\Rightarrow \quad x^2 + y^2 = a^2\cos^2\frac{\beta}{2}$

which is satisfied by all possible mid-points. Statement II is correct.

Since such a locus must lie inside the given circle, so it does not explain Statement I.

33. Circle will be real if $g^2 + f^2 - c > 0, 4 + 9 - \lambda > 0$

$\Rightarrow \quad \lambda < 13$

Now, the length of the tangent from the origin is $\sqrt{\lambda}$.

$\Rightarrow \quad \sqrt{\lambda} = 4 \quad \Rightarrow \quad \lambda = 16$

For $\lambda = 16$, the circle is not real .

Thus, Statement I is wrong and Statement II is a standard result.

34. Centres are (0, 0), (5, 0). Applying $d > r_1 + r_2$, we get

$$5 > k + 2 \quad \Rightarrow \quad k < 3$$

$\Rightarrow$ Statement 2 is correct. But statement 1 is false.

35. The centres and radii of two circles are

$C_1(1,3), C_2(4,-1)$ and $r_1 = r, r_2 = \sqrt{16+1-8} = 3$

$r_1 - r_2 < C_1C_2 < r_1 + r_2$

Two circles intersect in two distinct points, then

$$r - 3 < \sqrt{(4-1)^2 + (-1-3)^2} < r + 3$$

$\Rightarrow \quad r - 3 < 5 < r + 3 \quad \Rightarrow \quad r < 8$ and $2 < r$

$\Rightarrow \quad 2 < r < 8$.

36. Solving $y = x$ and $x^2 + y^2 - 2x = 0$, we get the extremities of diameter of the required circles as (0, 0) and (1, 1).

37. The intersection point of two given lines is the centre of the circle, i.e. (1, –1).

Circumference of the circle = 10π

$\Rightarrow \quad 2\pi r = 10\pi \quad \Rightarrow \quad r = 5$

$\therefore$ Equation of the circle having centre (1, –1) and radius 5 is

$$(x-1)^2 + (y+1)^2 = 5^2$$

$\Rightarrow \quad x^2 + y^2 - 2x + 2y - 23 = 0.$

38. Circle with centre $C_1(0, 3)$ and radius 2 touches the circle with centre $C_2(\alpha, \beta)$ and radius $|\beta|$.

$\therefore$ Distance $C_1C_2 = 2 + |\beta|$

$\Rightarrow \quad \alpha^2 + (\beta - 3)^2 = 4 + \beta^2 + 4|\beta|$

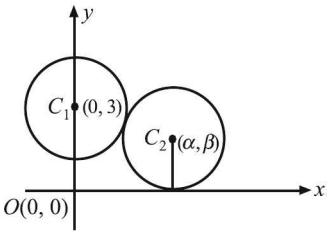


Fig. 19.47

Solving the above, we get

$$\alpha^2 = 10\beta - 5 \quad \Rightarrow \quad x^2 = Ay + A'$$

which is a parabola.

39. Chord through intersection points P and Q of the given circles is $S_1 - S_2 = 0$.

$\therefore \quad (x^2 + y^2 + 2ax + cy + a) - (x^2 + y^2 - 3ax + dy - 1) = 0$

$\Rightarrow \quad 5ax + (c - d)y + a + 1 = 0$

On comparing it with $5x + by - a = 0$, we get

$$\frac{5a}{5} = \frac{c-d}{b} = \frac{a+1}{-a}$$

$\Rightarrow \quad a(-a) = a + 1 \quad \Rightarrow \quad a^2 + a + 1 = 0$

which gives no real value of a.

Hence, the line passes through P and Q for no value of a.

40. Let $x^2 + y^2 = p^2$ and $x^2 + y^2 + 2gx + 2fy + c = 0$ intersect orthogonally. Locus of $(-g, -f)$ can be obtained by

$$a^2 + b^2 + 2ga + 2fb + p^2 = 0, p^2 = c$$

$$a^2 + b^2 + p^2 - 2ax - 2by = 0$$

41.

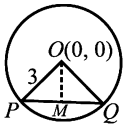


Fig. 19.48

Let PQ be the chord which subtends angle of $2\pi/3$ at the centre and its mid-point be $M(\alpha, \beta)$, then

$$\cos\frac{1}{2}\angle POQ = \frac{OM}{OP}$$

$\Rightarrow \quad \frac{1}{2} = \frac{\sqrt{\alpha^2 + \beta^2}}{3} \quad \Rightarrow \quad \alpha^2 + \beta^2 = \frac{9}{4}$

Locus of the mid-point of the chord = $x^2 + y^2 = \frac{9}{4}$.

42. Diameters intersect at (1, –1) and radius = $\sqrt{49} = 7$

$\therefore \quad (x-1)^2 + (y+1)^2 = 7^2$ is the required circle.

$\Rightarrow \quad x^2 + y^2 - 2x + 2y - 47 = 0.$

43. Here $\frac{1}{2} \times 3x_1 \times 2r = 18$

$\Rightarrow \quad x_1 \times r = 6 \qquad$ (i)

The equation of BC is $y=-\frac{2r}{x_1}(x-2x_1)$

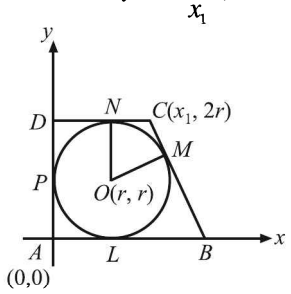


Fig. 19.49

BC is tangent to the circle

$$(x-r)^2+(y-r)^2=r^2$$

$\therefore$ Perpendicular distance of BC from centre = radius

$$\Rightarrow \frac{\left|r+\frac{2r}{x_1}(r-2x_1)\right|}{\sqrt{1+\frac{4r^2}{x_1^2}}}=r \Rightarrow \frac{2r^2}{x_1}-3r=r\sqrt{1+4\frac{r^2}{x_1^2}}$$

$$\Rightarrow (2r-3x_1)^2=x_1^2+4r^2$$

$$\Rightarrow r\cdot x_1=\frac{2}{3}x_1^2 \Rightarrow 3r=2x_1 \quad \text{(ii)}$$

From (i) and (ii), we get

$r=2$ units.

44. When the circle touches the x-axis, then

$(x-h)^2+(y-k)^2=k^2$ which passes through $(-1, 1)$

$\Rightarrow (-1-h^2)+(1-k)^2=k^2$

$\Rightarrow h^2+2h-2k+2=0 \Rightarrow (h+1)^2=2k-1$

$\therefore$ $2k-1\geq 0$ always.

$\Rightarrow k\geq\frac{1}{2}$.

45. Equation can be written as

$$(x+1)^2+(y+2)^2=(2\sqrt{2})^2$$

Mid-point of $A(1, 0)$ and $B(x_1, y_1)$ is centre $C(-1, -2)$.

$\Rightarrow x_1=-3, y_1=-4.$

46. Let $S_1+\lambda S_2=0$

Since it passes through (1, 1), then

$(1+1+3+7+2p-5)+\lambda(1+1+2+2-p^2)=0$

$$\Rightarrow \lambda=-\frac{7+2p}{6-p^2}$$

But $p^2\neq 6 \Rightarrow p\neq\pm\sqrt{6}$

But the other circle $x^2+y^2+2x+2y-6=0$ at $p=\pm\sqrt{6}$ also satisfies the point (1, 1).

So, $p=\pm\sqrt{6}$ is valid.

Now, $\lambda\neq -1 \Rightarrow \frac{7+2b}{6-p^2}\neq 1$

$\Rightarrow 7+2p\neq 6-p^2$

$\Rightarrow p^2+2p+1\neq 0 \Rightarrow p\neq -1$.

47. The centre of the circle is (2, 4) and radius is 5. The line will intersect the circle at two distinct points, if the distance of (2, 4) from $3x-4y=m$ is less than the radius of the circle.

$$\left|\frac{6-16-m}{5}\right|<5$$

$\Rightarrow -25<10+m<25 \Rightarrow -35<m<15$.

48. $x^2+y^2=6x$ (i)

and $x^2+y^2+6x+2y+1=0$ (ii)

From Eq. (i), $x^2-6x+y^2=0$

$\Rightarrow (x-3)^2+y^2=3^2$

$\therefore$ Centre (3, 0), $r=3$

From Eq. (ii)

$$x^2+6x+y^2+2y+1+3^2=3^2$$

$\Rightarrow (x+3)^2+(y+1)^2=3^2$

$\therefore$ Centre = $(-3, -1)$, radius = 3

Now, distance between centres

$$=\sqrt{(3+3)^2+1}=\sqrt{37}>r_1+r_2=6$$

$\therefore$ Circles do not cut each other.

$\Rightarrow$ 4 tangents (two direct and two transversal) are possible.

49. C_1 : (0, 0) with radius 1.

C_2 : $(t, 0)$ with radius 1.

$\therefore$ Required distance = $\sqrt{(t^2)-(1+1)^2}=\sqrt{21}$

$\Rightarrow t=\pm 5$.

50. The given common tangents are $2x-4y-9=0$ and $2x-4y+\frac{7}{3}=0$, which are parallel.

$\therefore$ Diameter = distance between tangents

= distance between parallel lines

$$=\frac{|c_1-c_2|}{\sqrt{a^2+b^2}}=\frac{\left|-9-\frac{7}{3}\right|}{\sqrt{2^2+(-4)^2}}$$

$$\Rightarrow d=\frac{34}{3\cdot 2\sqrt{5}}$$

$\therefore$ Radius $=\frac{17}{6\sqrt{5}}$.

51. The equation of the circle can be rewritten as

$$x^2 + y^2 - \frac{3}{2}x + 3y + 1 = 0$$

whose centre is $\left(\frac{3}{4}, -\frac{3}{2}\right)$

and radius, $r = \sqrt{\frac{9}{16} + \frac{9}{4} - 1} = \sqrt{\frac{29}{16}}$

$\therefore$ Area of the circle $= \pi r^2 = \frac{29\pi}{16}$

$\Rightarrow$ Area of the required circle $= 2 \times \frac{29\pi}{16} = \frac{29\pi}{8}$

Let R be the radius of the required circle.

$\therefore \quad R^2 = \frac{29}{8}$

Now, the equation of the circle is

$$\left(x - \frac{3}{4}\right)^2 + \left(y + \frac{3}{2}\right)^2 = \frac{29}{8}$$

$\Rightarrow \quad 16x^2 + 16y^2 - 24x + 48y - 13 = 0$.

52. The equation of the common chord PQ is

$2x + 1 = 0$ [i.e. $S_2 - S_1 = 0$]

Here, $C_1 = \left(-1, \frac{-3}{2}\right), r_1 = \frac{3}{2} = C_1P$

and $C_2 = \left(-2, \frac{-3}{2}\right), r_2 = \frac{\sqrt{17}}{2}$

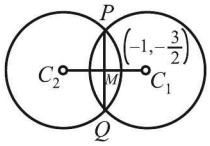


Fig. 19.50

C_1M = Perpendicular distance from C_1 to the common chord.

$\therefore \quad C_1M = \frac{|-2+1|}{\sqrt{2^2}} = \frac{1}{2}$

Now,

$$PQ = 2PM = 2\sqrt{(C_1P)^2 - (C_1M)^2} = 2\sqrt{\frac{9}{4} - \frac{1}{4}} = 2\sqrt{2}.$$

53. $y = x\tan\alpha + a$

Condition for tangency is $a^2 = a^2(1 + \tan^2\alpha)$

$[\because c^2 = a^2(1 + m^2)]$

$\Rightarrow \quad \sec^2\alpha = 1 \quad \Rightarrow \quad \cos^2\alpha = 1$

54. Since CA is perpendicular to the tangent,

$\therefore \quad r = \frac{|h - 0|}{\sqrt{1^2 + 1^2}} \quad \Rightarrow \quad h^2 = 2r^2$ (i)

Since CN is perpendicular to the chord of line, $y = \frac{x}{\sqrt{3}}$

$\therefore \quad CN = \frac{\left|\frac{h}{\sqrt{3}} - 0\right|}{\sqrt{\frac{1}{3} + 1}} = \frac{h}{2}$

In ΔBNC, $r^2 = 1^2 + \left(\frac{h}{2}\right)^2$

$r^2 = 1 + \frac{h^2}{4}$ (ii)

From Eqs. (i) and (ii), we get

$r^2 = 1 + \frac{2r^2}{4} \quad \Rightarrow \quad r = \sqrt{2}$.

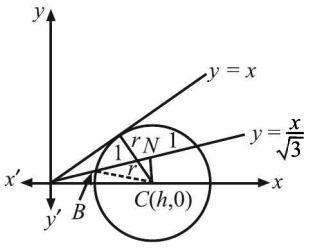


Fig. 19.51

55. Angle subtended by OR at centre O is 90°.

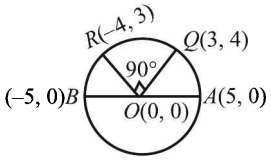


Fig. 19.52

$\left(\because m_1m_2 = \frac{4}{3} \cdot \frac{3}{-4} = -1\right)$

Hence, the angle at the circumference at P (any where) will be half of $\pi/2$, i.e. angle at P, i.e. $\angle OPR = \pi/4$.

56. Let (h, k) be the centroid of ΔPAB.

Then $h = \frac{r + 0 + r\cos\theta}{3}$, $k = \frac{0 + r + r\sin\theta}{3}$

$\Rightarrow \quad 3h = r(1 + \cos\theta)$

$P(r\cos\theta, r\sin\theta)$

$3k = r(1 + \sin\theta)$

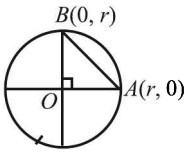


Fig. 19.53

$$\Rightarrow \quad \left(\frac{3h}{r} - 1\right)^2 + \left(\frac{3h}{r} - 1\right)^2 = \cos^2\theta + \sin^2\theta = 1$$

$\therefore$ Locus of (h, k) is $\left(\frac{3x}{r} - 1\right)^2 + \left(\frac{3y}{r} - 1\right)^2 = 1$

$\Rightarrow \quad (3x - r)^2 + (3y - r)^2 = r^2$

i.e. $\left(x - \frac{r}{3}\right)^2 + \left(y - \frac{r}{3}\right)^2 = \frac{r^2}{9}$

which is a circle.

57. $\tan\theta = \frac{PQ}{PR} = \frac{PQ}{2r}$

Also $\tan\left(\frac{\pi}{2} - \theta\right) = \frac{RS}{2r}$

i.e. $\cot\theta = \frac{RS}{2r}$

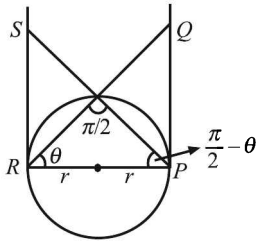


Fig. 19.54

$\therefore \quad \tan\theta \cdot \cot\theta = \frac{PQ \cdot RS}{4r^2}$

$\Rightarrow \quad 4r^2 = PQ \cdot RS \quad \Rightarrow \quad 2r = \sqrt{(PQ) \cdot (RS)}$.

58. Any tangent with slope m to the circle $x^2 + y^2 = b^2$ is
This is also the tangent to $(x-a)^2 + y^2 = b^2$

$$\therefore \quad \left[\frac{am - b\sqrt{1+m^2}}{\sqrt{1+m^2}}\right] = b$$

$$\Rightarrow \quad a^2m^2 + b^2(1+m^2) - 2mab\sqrt{1+m^2} = b^2(1+m^2)$$

$$\Rightarrow \quad a^2m^2 = 2amb\sqrt{1+m^2}$$

$$\Rightarrow \quad am = 2b\sqrt{1+m^2} \qquad \Rightarrow \quad a^2m^2 = 4b^2(1+m^2)$$

$$\Rightarrow \quad m^2(a^2 - 4b^2) = 4b^2$$

$$\Rightarrow \quad m = \frac{2b}{\sqrt{a^2 - 4b^2}}$$

59. $x^2 - 8x + 12 = 0 \quad \Rightarrow \quad (x-6)(x-2) = 0$

$y^2 - 14y + 45 = 0 \quad \Rightarrow \quad (y-5)(y-9) = 0$

Thus, the sides of the squares are

$$x = 2,\ x = 6,\ y = 5,\ y = 9$$

Then the centre of the circle inscribed in the square will be

$$\left(\frac{2+6}{2}, \frac{5+9}{2}\right) = (4, 7)$$

60. Let the centre of the circle C be (h, k). Then as this circle touches the x-axis its radius $= |k|$.

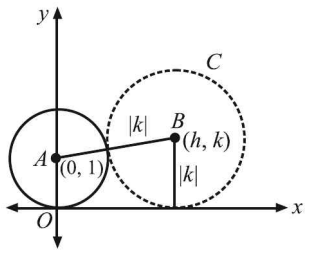


Fig. 19.55

Also it touches the given circle $x^2 + (y-1)^2 = 1$, centre (0, 1), radius 1, externally.

Therefore, the distance between centres = sum of radii

$$\Rightarrow \quad \sqrt{(h-0)^2 + (k-1)^2} = 1 + |k|$$

$$\Rightarrow \quad h^2 + k^2 - 2k + 1 = (1 + |k|)^2$$

$$\Rightarrow \quad h^2 + k^2 - 2k + 1 = 1 + 2|k| + k^2$$

$$\Rightarrow \quad h^2 = 2k + 2|k|$$

Therefore, locus of (h, k) is $x^2 = 2y + 2|y|$.

Now if $y > 0$, it becomes $x^2 = 4y$ and if $y \le 0$, it becomes $x = 0$.

Therefore, combining the two required locus, we get $\{(x, y) : x^2 = 4y\} \cup \{(0, y) : y \le 0\}$.

61. The intersection of the given line and circle is

$$x^2 + x^2 - 2x = 0$$

$$\Rightarrow \quad 2x(x-1) = 0 \quad \Rightarrow \quad x = 0,\ x = 1$$

and $y = 0,\ 1$

Let coordinates of A are (0, 0) and coordinates of B are (1, 1).

$\therefore$ The equation of the circle (AB as a diameter) is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

$$\Rightarrow \quad (x-0)(x-1) + (y-0)(y-1) = 0$$

$$\Rightarrow \quad x^2 + y^2 - x - y = 0$$

62. Since the equation of tangents $x - y - 2 = 0$ and $x - y + 2 = 0$ are parallel,

$\therefore$ Distance between them = Diameter of the circle

$$= \frac{2 - (-2)}{\sqrt{1^2 + 1^2}} \qquad \left(\because \frac{c_2 - c_1}{\sqrt{a^2 + b^2}}\right)$$

$$= \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$\therefore$ Radius $= \frac{1}{2}(2\sqrt{2}) = \sqrt{2}$

It is clear from the figure that the centre lies on the origin.

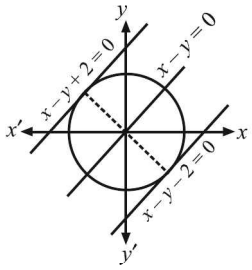


Fig. 19.56

$\therefore$ The equation of the circle is

$$(x-0)^2 - (y-0)^2 = (\sqrt{2})^2$$

$$\Rightarrow \quad x^2 + y^2 = 2$$

63. Any line through (0, 0) is $y = mx$. This touches the given circle.

$$\therefore \quad \frac{7m + 1}{\sqrt{m^2 + 1}} = 5$$

$$\Rightarrow \quad 49m^2 + 1 + 14m = 25m^2 + 25$$

$$\Rightarrow \quad 24m^2 + 14m - 24 = 0$$

$\Rightarrow$ $m_1 m_2 = -1$. Therefore, tangents are perpendicular.

64. Let the radius of the circle be r units.

Then, $\pi r^2 = 154$

$$\Rightarrow \quad r^2 = \frac{154}{\pi} = \frac{154 \times 7}{22} = 49$$

$$\Rightarrow \quad r = 7$$

Now, the centre of the circle is the point of intersection of any two diameters.

Solving the equations $2x - 3y = 5$ and $3x - 4y = 7$, we get

$$x = 1,\ y = -1$$

$\therefore$ The centre of the circle is (1, –1).

The equation of the circle with centre (1, –1) and radius 7 is

$$(x-1)^2+[y-(-1)]^2=7^2$$

i.e. $x^2+y^2-2x+2y-47=0$, which is the required equation of the circle.

65. The circle has centre $C(1, 2)$ and it passes through the point $P(4, 6)$.

$\therefore$ Radius = CP

$$=\sqrt{(4-1)^2+(6-2)^2}=\sqrt{9+16}=\sqrt{25}=5$$

Now, the equation of the circle with centre (1, 2) and radius 5 is

$$(x-1)^2+(y-2)^2=5^2$$

i.e. $(x^2-2x+1)+(y^2-4y+4)=25$

i.e. $x^2+y^2-2x-4y-20=0$

66. The given equations of the tangents are

$$3x-4y+4=0 \qquad \text{(i)}$$

and $6x-8y-7=0$, i.e., $3x-4y-\frac{7}{2}=0$ (ii)

Clearly, the two tangents are parallel to each other. Now, the distance between two parallel tangents of a circle is equal to the diameter of the circle.

$$\therefore \text{ Radius} = \frac{1}{2}\times\left|\frac{4-\left(-\frac{7}{2}\right)}{\sqrt{3^2+(-4)^2}}\right|=\frac{1}{2}\left|\frac{\frac{15}{2}}{\sqrt{25}}\right|=\frac{1}{2}\times\frac{3}{2}=\frac{3}{4}$$

67. The length of the tangent from a point $P(x_1, y_1)$ to a circle $x^2+y^2+2gx+2fy+c=0$ is

$$\sqrt{x_1^2+y_1^2+2gx_1+2fy_1+c}$$

$\therefore$ Length of tangent from (–1, –3) to the circle $x^2+y^2-2x-y-7=0$ is

$\sqrt{(1)^2+(-3)^2-2(-1)-(-3)-7}$, i.e. $\sqrt{8}$.

68. The first circle is

$$x^2+y^2-6x-2y+9=0 \qquad \text{(i)}$$

Its centre is $C_1(3, 1)$ and radius is

$$r_1=\sqrt{(-3)^2+(-1)^2-9}=1$$

The second circle is

$$x^2+y^2-14x-8y+61=0 \qquad \text{(ii)}$$

Its centre is $C_1(7, 4)$ and radius is

$$r_2=\sqrt{(-7)^2+(-4)^2-61}=\sqrt{4}=2.$$

Now, the distance between their centres C_1C_2

$$=\sqrt{(7-3)^2+(4-1)^2}=\sqrt{25}=5.$$

Sum of their radii = $|r_1+r_2|=3$.

$\therefore \quad C_1C_2>|r_1+r_2|$

So, the two circles do not intersect or touch each other. Hence, the number of common tangents is 4.

69. The given circles are

$$S_1\equiv x^2+y^2+6x-2y-9=0 \qquad \text{(i)}$$

$$S_2\equiv x^2+y^2-2x+9y-11=0 \qquad \text{(ii)}$$

Their radical axis is given by

$$S_1-S_2=0$$

i.e. $(x^2+y^2+2x-3y-6)-(x^2+y^2+x-8y-13)=0$

i.e. $x+5y+19=0$

Clearly, out of the given alternatives, only the point (1,–4) lies on line (0).

70. The equation of the circle whose diametric end points are (1, 0) and (0, 1) and has the smallest radius is

$$(x-1)(x-0)+(y-0)(y-1)=0$$

$$x^2+y^2-x-y=0$$

71. Now

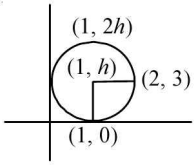
$h^2=(1-2)^2+(h-3)^2$

$0=1-6h+9$

$6h=10$

$h=5/3$

Now the diameter is $2h=10/3$



MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The radius of the circle passing through the points (1, 2), (5, 2) and (5, –2) is

(a) $5\sqrt{2}$ (b) $2\sqrt{5}$

(c) $3\sqrt{2}$ (d) $2\sqrt{2}$

2. The circle described on the line joining the points (0, 1) and (a, b) as the diameter cuts the x-axis in points whose abscissae are roots of the equation

(a) $x^2+ax+b=0$ (b) $x^2-ax+b=0$

(c) $x^2+ax-b=0$ (d) $x^2-ax-b=0$

3. The radius of the circle concentric with the circle $x^2+y^2-6x+12y+15=0$ and having four times its area is

(a) $3\sqrt{30}$ (b) $2\sqrt{30}$

(c) $\sqrt{30}$ (d) none of these

4. The straight line $y = mx + c$ cuts the circle $x^2 + y^2 = a^2$ at real points if

(a) $\sqrt{a^2(1+m^2)} \le |c|$ (b) $\sqrt{a^2(1-m^2)} \le |c|$

(c) $\sqrt{a^2(1+m^2)} \ge |c|$ (d) $\sqrt{a^2(1-m^2)} \ge |c|$

5. The locus of the centre of a circle of radius 2 which rolls on the outside of the circle $x^2 + y^2 + 3x - 6y - 9 = 0$ is

(a) $x^2 + y^2 + 3x - 6y - 5 = 0$

(b) $x^2 + y^2 + 3x - 6y - 31 = 0$

(c) $x^2 + y^2 + 3x - 6y + \frac{22}{4} = 0$

(d) none of these

6. The circle described on the line joining the points (0, 1), (a, b) as diameter cuts the x-axis at points whose abscissae are roots of the equation

(a) $x^2 + ax + b = 0$ (b) $x^2 - ax + b = 0$

(c) $x^2 + ax - b = 0$ (d) $x^2 - ax - b = 0$

7. If $(-3, 2)$ lies on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, which is concentric with the circle $x^2 + y^2 + 6x + 8y - 5 = 0$, then c is

(a) 11 (b) −11 (c) 24 (d) none of these

8. The centres of the circles $x^2 + y^2 - 6x - 8y - 7 = 0$ and $x^2 + y^2 - 4x - 10y - 3 = 0$ are the ends of the diameter of the circle

(a) $x^2 + y^2 - 5x - 9y + 26 = 0$

(b) $x^2 + y^2 + 5x - 9y + 14 = 0$

(c) $x^2 + y^2 + 5x - y - 14 = 0$

(d) $x^2 + y^2 + 5x + y + 14 = 0$

9. The intercepts made by the circle $x^2 + y^2 - 5x - 13y - 14 = 0$ on the x-axis and y-axis are respectively

(a) 9, 13 (b) 5, 13 (c) 9, 15 (d) none of these

10. Four distinct points $(2k, 3k)$, (1, 0), (0, 1) and (0, 0) lie on a circle for

(a) all integral values of k (b) $0 < k < 1$

(c) $k < 0$ (d) two values of k

11. A square is inscribed in the circle $x^2 + y^2 - 2x + 4y - 93 = 0$ with its sides parallel to the coordinate axes. The coordinates of its vertices are

(a) (−6, −9), (−6, 5), (8, −9), (8, 5)

(b) (−6, 9), (−6, −5), (8, −9), (8, 5)

(c) (−6, −9), (−6, 5), (8, 9), (8, 5)

(d) (−6, −9), (−6, 5), (8, −9), (8, −5)

12. A line is drawn through a fixed point $P(\alpha, \beta)$ to cut the circle $x^2 + y^2 = r^2$ at A and B. Then $PA \cdot PB$ is equal to

(a) $(a + b)^2 - r^2$ (b) $a^2 + b^2 - r^2$

(c) $(a - b)^2 + r^2$ (d) none of these

13. The two lines through (2, 3) from which the circle $x^2 + y^2 = 25$ intercepts chords of length 8 units have equations

(a) $2x + 3y = 13, x + 5y = 17$

(b) $y = 3, 12x + 5y = 39$

(c) $x = 2, 9x - 11y = 51$

(d) none of these

14. The radius of the circle with centre $(3, -1)$ and cutting a chord of length 6 on the line $2x - 5y + 18 = 0$ is

(a) 6 (b) 7 (c) $\sqrt{38}$ (d) $2\sqrt{5}$

15. The coordinates of the middle point of the chord cut off on $2x - 5y + 18 = 0$ by the circle $x^2 + y^2 - 6y + 2y - 54 = 0$ are

(a) (1, 4) (b) (2, 4) (c) (4, 1) (d) (1, 1)

16. The centre of the circle passing through the points (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$ is

(a) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (b) $\left(\frac{1}{2}, \frac{3}{2}\right)$

(c) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (d) $\left(\frac{1}{2}, -\sqrt{2}\right)$

17. One of the diameters of the circle $x^2 + y^2 - 12x + 4y + 6 = 0$ is given by

(a) $x + y = 0$ (b) $x + 3y = 0$

(c) $x = y$ (d) $3x + 2y = 0$

18. The length of the chord cut off by $y = 2x + 1$ from the circle $x^2 + y^2 = 2$ is

(a) 5/6 (b) 6/5 (c) $6/\sqrt{5}$ (d) $\sqrt{5}/6$

19. The circle $x^2 + y^2 - 6x - 10y + c = 0$ does not intersect or touch either axis and the point (1, 4) is inside the circle. Then the range of possible values of c is given by

(a) $c > 9$ (b) $c > 25$

(c) $c > 29$ (d) $25 < c < 29$

20. The position of the point (1, 2) w.r.t. the circle $x^2 + y^2 - 2x + 6y + 1 = 0$ is

(a) inside (b) outside

(c) on (d) none of these

21. The number of tangents that can be drawn from the point (8, 6) to the circle $x^2 + y^2 - 100 = 0$ is

(a) 0 (b) 1 (c) 2 (d) none of these

22. The coordinates of the middle point of the chord $2x - 5y + 18 = 0$ cut off by the circle $x^2 + y^2 - 6x + 2y - 54 = 0$ is

(a) (1, 4) (b) (2, 4) (c) (4, 1) (d) (1, 1)

23. If the circles $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, then k equals

(a) 2 or −(3/2) (b) −2 or (3/2)

(c) 2 or (3/2) (d) −2 or (3/2)

24. The equations of the circles which touch both the axes and the line $x = a$ are

(a) $x^2 + y^2 \pm ax \pm ay + \frac{a^2}{4} = 0$

(b) $x^2 + y^2 + ax \pm ay + \frac{a^2}{4} = 0$

(c) $x^2 + y^2 - ax \pm ay + \frac{a^2}{4} = 0$

(d) none of these

25. The tangent lines to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the line $4x + 3y + 5 = 0$ are given by

(a) $4x + 3y - 7 = 0, 4x + 3y + 15 = 0$

(b) $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$

(c) $4x + 3y - 17 = 0, 4x + 3y + 13 = 0$

(d) none of these

26. The line $3x + 5y + 9 = 0$ w.r.t. the circle $x^2 + y^2 - 4x + 6y + 5 = 0$ is

(a) a chord (b) a diameter

(c) a tangent (d) none of these

27. The condition so that the line $(x + g) \cos\theta + (y + f) \sin\theta = k$ is a tangent to $x^2 + y^2 + 2gx + 2fy + c = 0$ is

(a) $g^2 + f^2 = c + k^2$ (b) $g^2 + f^2 = c^2 + k$

(c) $g^2 + f^2 = c^2 + k^2$ (d) $g^2 + f^2 = c + k$

28. If the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ cut the coordinate axes in concyclic points, then

(a) $a_1a_2 = b_1b_2$ (b) $a_1b_1 = a_2b_2$

(c) $a_1b_2 = a_2b_2$ (d) none of these

29. If a circle passes through the points of intersection of the coordinate axes with the line $\lambda x - y + 1 = 0$ and $x - 2y + 3 = 0$, then the value of λ is

(a) 3 (b) 1/3

(c) 6 (d) none of these

30. The area of the triangle formed by the positive x-axis and the normal and tangent to the circle $x^2 + y^2 = 4$ at the point $(1, \sqrt{3})$ is

(a) $4\sqrt{3}$ (b) $2\sqrt{3}$

(c) $\sqrt{3}$ (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If $\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$ and $\left(d, \frac{1}{d}\right)$ are four distinct points on a circle of radius 4 units, then $abcd$ is equal to

(a) 4 (b) 16

(c) 1 (d) none of these

2. Form the point A (0, 3) on the circle $x^2 + 4x + (y - 3)^2$ a chord AB is drawn and extended to a point M such that $AM = 2AB$. The equation of the locus of M is

(a) $x^2 + 8x + y^2 = 0$ (b) $x^2 + 8x + (y - 3)^2 = 0$

(c) $(x - 3)^2 + 8x + y^2 = 0$ (d) $x^2 + 8x + 8y^2 = 0$

3. Two thin rods AB and CD of lengths $2a$ and $2b$ move along OX and OY, respectively, when O is the origin. The equation of the locus of the centre of the circle passing through the extremities of the two rods is

(a) $x^2 + y^2 = a^2 + b^2$ (b) $x^2 - y^2 = a^2 - b^2$

(c) $x^2 + y^2 = a^2 - b^2$ (d) $x^2 - y^2 = a^2 + b^2$

4. The condition that the chord $x \cos\alpha + y \sin\alpha - p = 0$ of $x^2 + y^2 - a^2 = 0$ may subtend a right angle at the centre of the circle is

(a) $a^2 = 2p^2$ (b) $p^2 = 2a^2$

(c) $a = 2p$ (d) $p = 2a$

5. If the distances from the origin of the centre of three circles $x^2 + y^2 + 2\lambda_i x - c^2$ ($i = 1, 2, 3$) are in G.P., then the lengths of the tangents drawn to them from any point on the circle $x^2 + y^2 = c^2$ are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

6. The coordinates of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$ which is farthest from the origin are

(a) (9, 3) (b) (8, 5)

(c) (12, 4) (d) none of these

7. The value of c for which the set $\{(x, y): x^2 + y^2 + 2x \le 1\} \cap \{(x, y): x - y + c \ge 0\}$ contains only one point in common is

(a) $(-\infty, -1] \cup [3, \infty)$ (b) $\{-1, 3\}$

(c) $\{-3\}$ (d) $\{-1\}$

8. Let x and y be the real numbers satisfying the equation $x^2 - 4x + y^2 + 3 = 0$. If the maximum and minimum values of $x^2 + y^2$ are M and m respectively, then the numerical value of $M - m$ is

(a) 2 (b) 8

(c) 15 (d) none of these

9. The area of the triangle formed by the tangents from the point (4, 3) to the circle $x^2 + y^2 = 9$ and the line joining their point of contact is

(a) 192/25 (b) 192

(c) 25 (d) 250

10. The circles $x^2 + y^2 + x + y = 0$ and $x^2 + y^2 + x - y = 0$ intersect at an angle of

(a) $\pi/6$ (b) $\pi/4$

(c) $\pi/3$ (d) $\pi/2$

11. The length of the tangent to the circle $x^2+y^2-2x-y=23$ from the point $(-1,-3)$ is

(a) 8 (b) $\sqrt{8}$

(c) $\sqrt{12}$ (d) none of these

12. AB is a diameter of a circle and C is any point on the circumference of the circle. Then

(a) the area of ΔABC is maximum when it is isosceles

(b) the area of ΔABC is minimum when it is equilateral

(c) the perimeter of ΔABC is maximum when it is right-angled

(d) none of these

13. The distance between the chords of contact of tangents to the circle $x^2+y^2+2gx+2fy+c=0$ from the origin and the point (g,f) is

(a) $\sqrt{g^2+f^2}$ (b) $\dfrac{\sqrt{g^2+f^2-c}}{2}$

(c) $\dfrac{g^2+f^2-c}{2\sqrt{g^2+f^2}}$ (d) $\dfrac{\sqrt{g^2+f^2+c}}{2\sqrt{g^2+f^2}}$

14. A circle passes through the point $\left(3,\sqrt{\dfrac{7}{2}}\right)$ and touches the line pair $x^2-y^2-2x+1=0$. The coordinates of the centre of the circle are

(a) (4 ,0) (b) (5, 0)

(c) (6, 0) (d) (0, 4)

15. If the tangent at $(1,2)$ to the circle $c_1: x^2+y^2=5$ intersects the circle c_2: $x^2+y^2=9$ at A and B and tangents at A and B to the second circle meet at point C, then the coordinates of C are

(a) (4, 5) (b) $\left(\dfrac{9}{15},\dfrac{18}{5}\right)$

(c) $(4,-5)$ (d) $\left(\dfrac{9}{5},\dfrac{18}{5}\right)$

16. If (2, 5) is an interior point of the circle $x^2+y^2-8x-12y+P=0$ and the circle neither cuts nor touches any one of the coordinate axes, then

(a) $P\in(36,47)$ (b) $P\in(16,47)$

(c) $P\in(16,36)$ (d) none of these

17. The centres of a set of circles, each of radius 3, lie on the circle $x^2+y^2=25$. The locus of any point in the set is

(a) $4\le x^2+y^2\le 64$ (b) $x^2+y^2\le 25$

(c) $x^2+y^2\ge 25$ (d) $3\le x^2+y^2\le 9$

18. The radius of the circle passing through the point (6, 2) and having $x+y=6$ as its normal and $x+2y=4$ as its diameter is

(a) 10 (b) $2\sqrt{5}$

(c) $5\sqrt{2}$ (d) $4\sqrt{5}$

19. A point $A(2, 1)$ is outside the circle $x^2+y^2+2gx+2fy+c=0$ and AP and AQ are tangents to the circle. The equation of the circle circumscribing the triangle APQ is

(a) $(x+g)(x-2)+(y+f)(y-1)=0$

(b) $(x+g)(x-2)-(y+f)(y-1)=0$

(c) $(x-g)(x+2)+(y-f)(y+1)=0$

(d) none of these

20. The locus of the mid-points of the chords of the circle $x^2+y^2+4x-6y-12=0$ which subtends an angle of $\pi/3$ radians at its circumference is

(a) $(x-2)^2+(y+3)^2=6.25$

(b) $(x+2)^2+(y-3)^2=6.25$

(c) $(x+2)^2+(y-3)^2=18.75$

(d) $(x+2)^2+(y+3)^2=18.75$

21. The locus of the centres of the circles such that the point (2, 3) is the mid-point of the chord $5x+2y=16$ is

(a) $2x-5y+11=0$ (b) $2x+5y-11=0$

(c) $2x+5y+11=0$ (d) none of these

22. Tangents OP and OQ are drawn from the origin O to the circle $x^2+y^2+2gx+2fy+c=0$. Then the equation of the circumcircle of the triangle OPQ is

(a) $x^2+y^2+2gx+2fy=0$ (b) $x^2+y^2+gx+fy=0$

(c) $x^2+y^2-gx-fy=0$ (d) $x^2+y^2-2gx-2fy=0$

23. The slope of the tangent at the point (h, h) of the circle $x^2+y^2=a^2$ is

(a) 0 (b) 1

(c) -1 (d) depends on h

24. The number of common tangents to the circles $x^2+y^2-x=0$ and $x^2+y^2+x=0$ is

(a) 2 (b) 1

(c) 4 (d) 3

25. If the circle $C_1: x^2+y^2=16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to 3/4, then the coordinates of the centre of C_2 are

(a) $\left(\pm\dfrac{9}{5},\pm\dfrac{12}{5}\right)$ (b) $\left(\pm\dfrac{9}{5},\mp\dfrac{12}{5}\right)$

(c) $\left(\pm\dfrac{12}{5},\pm\dfrac{9}{5}\right)$ (d) $\left(\pm\dfrac{12}{5},\mp\dfrac{9}{5}\right)$

26. If from any point P on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ tangents are drawn to the circle $x^2 + y^2 + 2gx + 2fy + c \sin^2 \alpha + (g^2 + f^2)\cos^2\alpha = 0$, then the angle between the tangents is

(a) α (b) 2α

(c) $\alpha/2$ (d) none of these

27. If the length of a common internal tangent to two circles is 7, and that of a common external tangent is 11, then the product of the radii of the two circles is

(a) 36 (b) 9

(c) 18 (d) 4

28. A circle touches the x-axis and also touches the circle with centre (0, 3) and radius 2. The locus of the centre of the circle is

(a) a circle (b) a parabola

(c) an ellipse (d) a hyperbola

29. A circle passes through the origin and has its centre on $y = x$. If it cuts $x^2 + y^2 - 4x - 6y + 10 = 0$ orthogonally, the equation of the circle is

(a) $x^2 + y^2 - x - y = 0$ (b) $x^2 + y^2 - 6x - 4y = 0$

(c) $x^2 + y^2 - 2x - 2y = 0$ (d) $x^2 + y^2 + 2x + 2y = 0$

30. If the two circles $x^2 + y^2 + 2g_1x + 2f_1y = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y = 0$ touch each other, then

(a) $f_1g_1 = f_2g_2$ (b) $\frac{f_1}{g_1} = \frac{f_2}{g_2}$

(c) $f_1f_2 = g_1g_2$ (d) none of these

ANSWERS

Level 1

1. (d)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (d)
11. (a)	**12.** (b)	**13.** (b)	**14.** (c)	**15.** (a)	**16.** (d)	**17.** (b)	**18.** (c)	**19.** (d)	**20.** (b)
21. (b)	**22.** (a)	**23.** (a)	**24.** (c)	**25.** (b)	**26.** (b)	**27.** (a)	**28.** (a)	**29.** (b)	**30.** (b)

Level 2

1. (c)	**2.** (b)	**3.** (b)	**4.** (a)	**5.** (b)	**6.** (a)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (d)
11. (d)	**12.** (a)	**13.** (c)	**14.** (a)	**15.** (d)	**16.** (a)	**17.** (a)	**18.** (b)	**19.** (a)	**20.** (b)
21. (a)	**22.** (b)	**23.** (c)	**24.** (d)	**25.** (b)	**26.** (b)	**27.** (c)	**28.** (b)	**29.** (c)	**30.** (b)

20

APOLLONIUS

Parabola

- ❑ Definition of Conic Section
- ❑ Important Rules
- ❑ Definition of Parabola
- ❑ General Form of Standard Parabola
- ❑ Properties of Parabola
- ❑ Standard Equation of Parabola
- ❑ Equation of Tangent
- ❑ Equation of a Tangent in Different Forms
- ❑ Equation of Normal
- ❑ Equation of a Normal in Different Forms
- ❑ Equation of Chord of Contact
- ❑ Equation of Chord of a Parabola Whose Mid-point is Given
- ❑ Equation of a Pair of Tangents
- ❑ Tangent to a Parabola
- ❑ Pole and Polar
- ❑ Equation of the Polar
- ❑ Properties of Pole and Polar
- ❑ Diameter
- ❑ Equation of the Diameter

☞ DEFINITION OF CONIC SECTION

It is the locus of the point (in plane) so that the ratio of its distance from a fixed point to its distance from a fixed line is constant. This ratio is known as **eccentricity** (denoted by e).

If $e = 1$, then the locus is a **parabola**.

If $e < 1$, then the locus is an **ellipse**.

If $e > 1$, then the locus is a **hyperbola**.

☞ IMPORTANT RULES

1. The fixed point is known as **focus** (denoted by S).
2. The fixed line is known as **directrix**.
3. A chord passing through the focus is known as **focal chord**.
4. The straight line through the focus and perpendicular to the directrix is known as **axis**.
5. The focal chord which is perpendicular to the axis is known as **latus rectum**.
6. The point(s) of intersection of the conic section and the axis is (are) called **vertex**.
7. A straight line drawn perpendicular to the axis and terminated at both the ends of the curve is known as **double ordinate**.

☞ DEFINITION OF PARABOLA

A **parabola** is the locus of a point which moves in a plane such that its distance from a fixed point (called the focus) is equal to its distance from a fixed straight line (called the directrix).

GENERAL FORM OF STANDARD PARABOLA

The general form of the standard parabola is $y^2 = 4ax$, where a is constant.

PROPERTIES

1. Vertex is at origin $O(0, 0)$.
2. Focus is at $S \equiv (a, 0)$.
3. Directrix is $x + a = 0$.
4. Axis is $y = 0$ (x-axis)
5. Length of latus rectum $= LL' = 4a$.
6. Ends of latus rectum are $L \equiv (a, 2a)$ and $L' \equiv (a, -2a)$.
7. The parametric equations are $x = at^2$ and $y = 2at$.

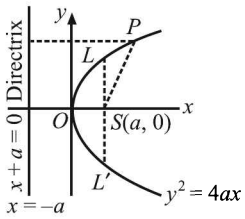
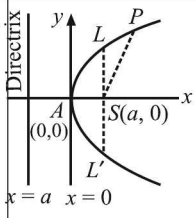
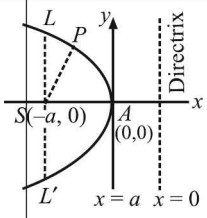
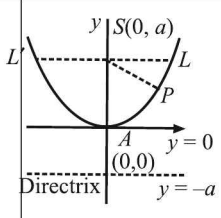
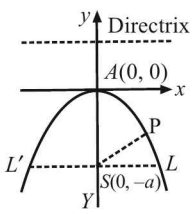


Fig. 20.1

Standard equation	$y^2 = 4ax (a > 0)$	$y^2 = -4ax (a > 0$	$x^2 = 4ay (a > 0)$	$x^2 = -4ay (a > 0)$
Shape of the parabola				
Vertex	$A(0, 0)$	$A(0, 0)$	$A(0, 0)$	$A(0, 0)$
Focus	$S(a, 0)$	$S(-a, 0)$	$S(0, a)$	$S = (0, -a)$
Equation of directrix	$x = -a$	$x = a$	$Y = -a$	$y = a$
Equation of axis	$Y = 0$	$y = 0$	$x = 0$	$x = 0$
Length of latus rectum	$4a$	$4a$	$4a$	$4a$
Equation of latus rectum	$X = a$	$x = -a$	$y = a$	$y = -a$
Equation of tangents at vertex	$x = 0$	$X = 0$	$Y = 0$	$y = 0$
Extremities of latus rectum	$(a, \pm 2a)$	$(-a, \pm 2a)$	$(\pm 2a, a)$	$(\pm 2a, -a)$
Focal distance of a point $P(x, y)$	$x + a$	$x - a$	$y + a$	$y - a$
Parametric coordinates	$(at^2, 2at))$	$(-at^2, 2at))$	$(2at, at^2)$	$(2at, -at^2)$
Eccentricity (e)	1	1	1	1

Note If the vertex of the parabola is (h, k), then the generalized form of the parabola is $(y-k)^2 = 4a(x-h)^2$. Its focus is at $(a+h, k)$. The equation of the directrix is $x+a-h=0$ and parametric equations are $x=h+at^2$ and $y=k+2at$.

EQUATION OF TANGENT

The various forms of the equation of tangents to the parabola $y^2 = 4ax$ are given as follows:

1. **Point form:** At the point of contact (x_1, y_1), the tangent is $yy_1 = 2a(x + x_1)$ with slope $= \dfrac{2a}{y_1}$.
2. **Parametric form:** At the point of contact $(at^2, 2at)$, the tangent is $ty = x + at^2$ with slope $= \dfrac{1}{t}$.
3. **Slope form:** If m is the slope of the tangent, then
   (i) tangent is $y = mx + \dfrac{a}{m}$ and
   (ii) the point of contact is $P \equiv \left(\dfrac{a}{m^2}, \dfrac{2a}{m}\right)$.

EQUATION OF A TANGENT IN DIFFERENT FORMS

1. The condition for $y = mx + c$ to be a tangent to the parabola $y^2 = 4ax$ is $c = \frac{a}{m}$.
2. The point of intersection of tangents drawn at two different points of contact $(at_1^2, 2at)$ and $(at_2^2, 2at_2)$ is $P \equiv [at_1t_2a(t_1 + t_2)]$.
3. The angle between tangents drawn at two different points of contact $(at_1^2, 2at)$ and $(at_2^2, 2at_2)$ is $\theta = \tan^{-1}\left|\frac{t_2 - t_1}{1 + t_1t_2}\right|$.

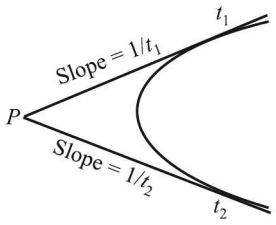


Fig. 20.2

EQUATION OF CHORD

1. The equation of the chord joining (x_1, y_1) and (x_2, y_2) is $y(y_1 + y_2) = 4ax + y_1y_2$.
2. The equation of the chord joining $(at_1^2, 2at)$ and $(at_2^2, 2at_2)$ is $y(t_1 + t_2) = 2(x + at_1t_2)$.

Note The chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ passes through the focus if $t_1t_2 = -1$.

EQUATION OF NORMAL

1. **Point form:** The equation of the normal drawn at a point (x_1, y_1) on the parabola is $2a(x_1, y_1) = y_1(x - x_1)$ with slope $-\frac{y_1}{2a}$.
2. **Parametric form:** The equation of the normal drawn at a point $(at^2, 2at)$ on the parabola is $tx + y = 2at + at^3$.
3. **Slope form:** The equation of the normal of slope m is $y = mx - 2am - am^3$ and the point of intersection with the parabola is $(am^2, -2am)$.

EQUATION OF NORMAL IN DIFFERENT FORMS

1. If the normal drawn at point $P(at_1^2, 2at_1)$ intersects the parabola again in $Q(at_1^2, 2at_1)$, then $t_1 = -t - \frac{2}{t}$.

Note PQ is normal to the curve at P and not at Q.

Fig. 20.3

2. The point of intersection of normals drawn at $P(t_1)$ and $Q(t_2)$ is $A \equiv [2a + a(t_1^2 + t_2^2 + t_1t_2) - at_1t_2(t_1 + t_2)]$.

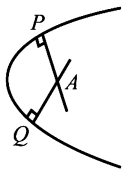


Fig. 20.4

3. If three normals are drawn through a point (h, k), their slopes are the roots of the cubic $k = mh - 2am\ am^3$ $\Rightarrow$ Sum of slopes $= m_1 + m_2 + m_3 = 0$ and sum of y coordinates of A, B, $C = 0$, i.e. $y_A + y_B + y_C = 0$.

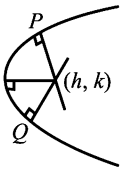


Fig. 20.5

EQUATION OF CHORD OF CONTACT

Let us consider a point R whose coordinates are (x_1, y_1) and which lies outside the parabola. From point R, two tangents are drawn to the parabola which meet the parabola at points P and Q. The line joining P and Q is known as chord of contact of the parabola and its equation is $yy_1 = 2a(x + x_1)$.

EQUATION OF CHORD OF A PARABOLA WHOSE MID-POINT IS GIVEN

Let us consider a chord PQ of the parabola whose mid-point is $R \equiv (x_1, y_1)$. The equation of this chord PQ in terms of coordinates of R is $S_1 = T$, where $S_1 = y_1^2 - 4ax_1$ and $T = yy_1 - 2a(x + x_1)$.

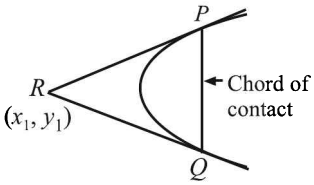


Fig. 20.6

EQUATION OF A PAIR OF TANGENTS

Let us consider a point R whose coordinates are (x_1, y_1) and which lies outside the parabola. From point R, two tangents are drawn to the parabola which meet the parabola in points P and Q. The equation of the pair of tangents RP and RQ is $SS_1 = T^2$, where $S = y^2 - 4ax$, $S_1 = y_1^2 - 4ax_1$ and $T = yy_1 - 2a(x + x_1)$.

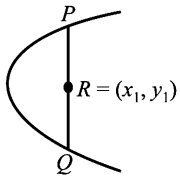


Fig. 20.7

TANGENT TO A PARABOLA

1. **Point form:** The equation of the tangent to the parabola $y^2 = 4ax$ at the point (x_1, y_1) is $yy_1 = 2a(x + x_1)$. The equations of tangents to all the standard forms of parabola at point (x_1, y_1) are given as follows:

Equation of parabola	Equation of tangent
$y^2 = 4ax$	$yy_1 = -2a(x + x_1)$
$x^2 = 4ay$	$xx_1 = 2a(y + y_1)$
$x^2 = -4ay$	$xx_1 = -2a(y + y_1)$

Note

- The equation of tangent at (x_1, y_1) to any second-degree curve can also be obtained by replacing x^2 by xx_1, y^2 by yy_1, x by $\frac{x+x_1}{2}$, y by $\frac{y+y_1}{2}$ and xy by $\frac{xy_1 + x_1y}{2}$ and without changing the constant (if any) in the equation of the curve.

2. **Slope form:** The equation of the tangent to the parabola $y^2 = 4ax$ in slope form is $y = mx + \frac{a}{m}$, where m is the slope of the tangent. The equation of tangent of slope m to the parabola $(y-k)^2 = 4a(x-h)$ is given by $y - k = m(x-h) + \frac{a}{m}$ and the coordinates of the point of contact are $\left(h + \frac{a}{m^2}, k + \frac{2a}{m}\right)$.

POLE AND POLAR

Let P be any point inside or outside a parabola. Suppose any straight line drawn through P intersects the parabola at Q and R. Then the locus of the point of intersection of the tangents to the parabola at Q and R is called the **polar** of the given point P with respect to the parabola and the point given called the **pole** of the polar.

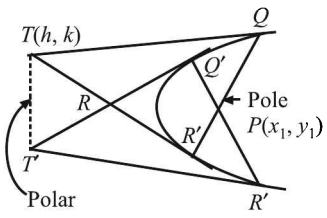


Fig. 20.8

EQUATION OF THE POLAR

The equation of the polar of a point (x_1, y_1) with respect to the parabola $y^2 = 4ax$ is $yy_1 = 2a(x + x_1)$.

PROPERTIES OF POLE AND POLAR

1. If the polar of $P(x_1, y_1)$ passes through $Q(x_2, y_2)$, then the polar of $Q(x_2, y_2)$ goes through (x_1, y_1) and such points are said to be **conjugate points**.
2. If the pole of a line $ax + by + c = 0$ lies on another line $a_1x + b_1y + c_1 = 0$, then the pole of the second line will lie on the first and such lines are said to be **conjugate lines**.

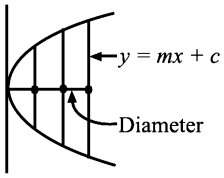


Fig. 20.9

DIAMETER

The locus of the middle points of a system of chords of a parabola is called a **diameter** and the chords are called its double ordinates.

EQUATION OF THE DIAMETER

Let (x_1, y_1) be the middle point of the chord $y = mx + c$ of the parabola $y^2 = 4ax$. Then $T = S_1$

$$\Rightarrow \quad yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$$

$$\text{Slope} = \frac{2a}{y_1} = m \Rightarrow k = \frac{2a}{m}$$

Hence, the locus of the mid-point is $y = \frac{2a}{m}$. It can be easily derived that the locus is a line parallel to the axis of the parabola.

SOLVED PROBLEMS

1. For the parabola $y^2 = 4x$, the point P whose focal distance is 17 is

(a) (16, 8) or (16, −8) (b) (8,8) or (8, −8)

(c) (4, 8) or (4, −8) (d) none of these

Ans. (a)

Solution: Let $P(x_1, y_1)$ be the point.

Since parabola is $y^2 = 4x$,

$\therefore \quad 4a = 4$

$\Rightarrow \quad a = 1$

$\therefore$ Focal distance = 17

$\therefore \quad x_1 + a = 17 \quad \Rightarrow \quad x_1 + 1 = 17$

$\Rightarrow \quad x_1 = 16$

$\therefore \quad y_1^2 = 4x_1 = 4(16) = 64$

$\therefore \quad y_1 = \pm 8$

$\therefore \quad P$ is (16, 8) or (16, −8)

2. If (0, 4) and (0, 2) are respectively the vertex and focus of a parabola, then its equation is

(a) $x^2 + 8y = 32$ (b) $y^2 + 8x = 32$

(c) $x^2 - 8y = 32$ (d) $y^2 - 8y = 32$

Ans. (a)

Solution: Since (0, 4) is the vertex and (0, 2) is the focus of the parabola, the parabola is a downward parabola.

Its equation is $(x - 0)^2 = 4a(y - 4)$

i.e. $X^2 = 4aY\ [X = x - 0, Y = y - 4]$

Its focus is given by $X = 0, Y = a$
i.e. $x = 0, y - 4 = a$ i.e. $y = 4 + a$
But the given focus is (0, 2).
$\therefore \quad 4 + a = 2 \quad \Rightarrow \quad a = -2$
$\therefore$ Required parabola is

$$(x-0)^2 = 4(-2)(y-4)$$

i.e. $$x^2 = -8(y-4) = -8y + 32$$

$\Rightarrow$ $$x^2 + 8y = 32$$

3. If the vertex and focus of a parabola are (3, 3) and (−3, 3) respectively, then its equation is

(a) $x^2 + 6x - 24y + 63 = 0$

(b) $x^2 - 6x + 24y - 63 = 0$

(c) $y^2 - 6y + 24x - 63 = 0$

(d) $y^2 + 6y - 24x + 63 = 0$

Ans. (c)

Solution: The equation of the line joining (3, 3) and (−3, 3), i.e. axis of the parabola is $y - 3 = 0$. Since the directrix is a line perpendicular to the axis, therefore its equation is $x + \lambda = 0$. The directrix intersects the axis at $(-\lambda, 3)$. Since the vertex is the mid-point of the line segment joining the focus and the point of intersection of the directrix and axis, therefore

$$\frac{-\lambda - 3}{2} = 3 \quad \Rightarrow \quad \lambda = -9$$

So, the equation of the directrix is $x - 9 = 0$.
Let $P(x, y)$ be any point on the parabola. Then by definition, $(x+3)^2 + (y-3)^2 = (x-9)^2$

$\Rightarrow$ $$y^2 - 6y + 24x - 63 = 0$$

4. The equation of the normal at the point of contact of a tangent $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ is

(a) $y = mx - 2am - am^3$

(b) $m^3 y = m^2 x - 2am^2 - a$

(c) $m^3 y = 2am^2 - m^2 x + a$

(d) none of these

Ans. (c)

Solution: The equation of the normal to $y^2 = 4ax$ at (x_1, y_2) is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1).$$

So, the equation of the normal at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ is

$$y - \frac{2a}{m} = -\frac{1}{m}\left(x - \frac{a}{m^2}\right)$$

$\Rightarrow$ $m^3 y - 2am^2 = -m^2 x + a \Rightarrow m^3 y = 2am^2 - m^2 x + a$

5. The equation of parabola is given by $y^2 + 8x - 12y + 20 = 0$. Tick the correct option given below:

(a) vertex (2, 6) (b) focus (1, 6)

(c) latus rectum 4 (d) none of these

Ans. (a)

Solution: $y^2 - 12y = -8x - 20$
Add 36 to both sides

$\therefore$ $(y-6)^2 = -8(x+2) \quad \Rightarrow \quad Y^2 = 4AX$

where $A = -2$
Axis is $Y = 0$ or $y - 6 = 0$ or $y = 6$
Vertex $X = 0, Y = 0,$ i.e. (2, 6)
Focus $x = A, Y = 0$ or $x - 2 = -2, y - 6 = 0$
or $x = 0, \; y = 6 \quad \therefore$ (0, 6)

Length of latus rectum $= 4A = -8 = 8$

6. The angle made by a double ordinate of length $8a$ at the vertex of the parabola $y^2 = 4ax$ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

Ans. (b)

Solution: $2y_1 = 8a \quad \Rightarrow \quad y_1 = 4a$
$\therefore \quad x_1 = 4a$
$\therefore$ Extremities of double ordinate PQ are $(4a, 4a)$ and $(4a, -4a)$ and vertex is (0, 0).
$m_1 m_2 =$ slope of $OP \times$ slope of $OQ = -1$
$\therefore \quad 90°$

7. The vertex of the parabola $x^2 + 2y = 8x - 7$ is

(a) $\left(4, \frac{7}{2}\right)$ (b) $\left(4, \frac{9}{2}\right)$

(c) $\left(\frac{9}{2}, 4\right)$ (d) (1, 0)

Ans. (b)

Solution: Since $x^4 - 8x = -2y - 7$

$$\Rightarrow \quad (x-4)^2 = -2\left(y - \frac{9}{2}\right)$$

$\therefore$ The vertex is at $\left(4, \frac{9}{2}\right)$.

8. The curve represented by $x = 3(\cot t + \sin t)$ and $y = 4(\cos t - \sin t)$ is

(a) an ellipse (b) a parabola

(c) a hyperbola (d) a circle

Ans. (a)

Solution: From the given relation, we have

$$\left(\frac{x}{3}\right)^2+\left(\frac{y}{4}\right)^2=(\cos t+\sin t)^2+(\cos t-\sin t)^2$$

$$\Rightarrow \frac{x^2}{9}+\frac{y^2}{16}=2 \quad \Rightarrow \quad \frac{x^2}{18}+\frac{y^2}{32}=1$$

which is an ellipse.

9. The equation of the directrix of the parabola $y^2+4y+4x+2=0$ is

(a) $x=-1$ (b) $x=1$

(c) $x=-\frac{3}{2}$ (d) $x=\frac{3}{2}$

Ans. (d)

Solution: The given equation can be written as

$$(y+2)^2=-4\left(x-\frac{1}{2}\right) \quad \Rightarrow \quad Y^2=-4x$$

Now the equation of the directrix is

$$X=-\left(-\frac{4}{4}\right)=1$$

$$\Rightarrow \quad x-\frac{1}{2}=1$$

$$\therefore \quad x=1+\frac{1}{2}=\frac{3}{2}$$

10. Three normals to the parabola $y^2 = x$ are drawn through a point $(c, 0)$, then

(a) $c=\frac{1}{4}$ (b) $c=\frac{1}{2}$

(c) $c>\frac{1}{2}$ (d) none of these

Ans. (c)

Solution: Equation of the normal to the parabola $y^2 = 4ax$ and having slope m is $y = mx - 2am - am^2$, where $4a = 1$ (in this case) and it passes through $(c, 0)$.

$\therefore \quad m(m^2-4c+2)=0$

$$\Rightarrow \quad m=\pm\sqrt{4c-2}$$

These will be real and distinct if $4c - 2 > 0$.

$$\therefore \quad c>\frac{1}{2}$$

11. Two tangents are drawn from the point $(-2, -1)$ to the parabola $y^2 = 4x$. If α is the angle between these tangents, then $\tan\alpha =$

(a) 3 (b) $\frac{1}{3}$

(c) 2 (d) $\frac{1}{2}$

Ans. (a)

Solution: Here, $a = 1$. Any tangent is $y=mx+\frac{1}{m}$.

It passes through $(-2, -1)$. Thus,

$$2m^2-m-1=0 \quad \Rightarrow \quad m=1,-\frac{1}{2}$$

$$\therefore \quad \tan\alpha=\frac{1+\frac{1}{2}}{1-\frac{1}{2}}=3$$

12. If the tangents at P and Q on a parabola meet in T, then SP, ST and SQ are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

Ans. (b)

Solution: If P and Q are t_1 and t_2, then T is

$$[at_1t_2,\ a(t_1+t_2)]$$

$$SP=a+x=a(1+t_1^2),\quad SQ=a(1+t_2^2)$$

$$ST^2=a^2(1+t_1t_2)^2+a^2(t_1+t_2)^2$$

$$ST^2=a^2(1+t_1^2+t_2^2+t_1^2t_2^2)$$

$$ST^2=a(1+t_1^2)\cdot a(1+t_2^2)=SP\cdot SQ$$

EXERCISE SET

1. The parabola $y = x^2 - 8x + 15$ cuts the x-axis at P and Q. A circle is drawn through P and Q so that the origin is outside it. The length of a tangent of the circle from O is

(a) 15 (b) $\sqrt{8}$

(c) $\sqrt{15}$ (d) 8

2. The tangents at the points $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$ are at right angles if

(a) $t_1t_2=-1$ (b) $t_1t_2=1$

(c) $t_1t_2=1$ (d) $t_1t_2=-2$

3. If y_1 and y_2 are the ordinates of two points P and Q on the parabola and y_3 is the ordinate of the points of intersection of tangents at P and Q, then

(a) y_1, y_2, y_3 are in A.P. (b) y_1, y_3, y_2 are in A.P.

(c) y_1, y_2, y_3 are in G.P. (d) y_1, y_3, y_2 are in G.P.

4. A tangent to a parabola $y^2 = 4ax$ is inclined at $\pi/3$ with the axis of the parabola. The point of contact is

(a) $\left(\frac{a}{3}, -\frac{2a}{\sqrt{3}}\right)$ (b) $(3a, -2\sqrt{3}a)$

(c) $(3a, 2\sqrt{3}a)$ (d) none of these

5. At any point P on the parabola $y^2-2y-4x+5=0$, a tangent is drawn which meets the directrix at Q. The locus of the points P which divides QP externally in the ratio $\frac{1}{2}:1$, is

(a) $(x+1)(1-y^2)+4=0$ (b) $x+1=0$

(c) $(1-y)^2-4=0$ (d) none of these

6. The point on $y^2=4ax$ nearest to the focus has its abcissae equal to

(a) $-a$ (b) a

(c) $a/2$ (d) 0

7. The tangents and normals at the ends of the latus rectum of a parabola form

(a) a cyclic quadrilateral (b) a rectangle

(c) a square (d) none of these

8. Find the lengths of the normals drawn from the point on the axis of the parabola $y^2 = 8x$ whose distance from the focus is 8.

(a) 10 (b) 8

(c) 9 (d) none of these

9. If t is the parameter for one end of a focal chord of the parabola $y^2 = 4ax$, then its length is

(a) $a\left(t+\frac{1}{t}\right)^2$ (b) $a\left(t-\frac{1}{t}\right)^2$

(c) $a\left(t+\frac{1}{t}\right)$ (d) $a\left(t-\frac{1}{t}\right)$

10. The vertex of the parabola $(y-2)^2=16(x-1)$ is

(a) (2, 1) (b) $(1,-2)$

(c) $(-1,2)$ (d) $(1,2)$

11. A ray of light moving parallel to the x-axis gets reflected from a parabolic mirror whose equation is $(y-2)^2 = 4(x+1)$. After reflection, the ray must pass through the point

(a) (2, 0) (b) $(0,-2)$

(c) (0, 2) (d) $(-1,-2)$

12. If two distinct chords of a parabola $y^2 = 4ax$ passing through $(a, 2a)$ are bisected on the line $x + y = 1$, then the length of the latus rectum can be

(a) 6 (b) -4

(c) 4 (d) none of these

13. The curve described parametrically by $x=t^2+t+1$ and $y=t^2-t+1$ represents

(a) a pair of lines (b) an ellipse

(c) a parabola (d) a hyperbola

14. The equation of the chord of the parabola $y^2 = 8x$ which is bisected at the point $(2, -3)$ is

(a) $4x+3y+1=0$ (b) $3x+4y-1=0$

(c) $4x-3y-1=0$ (d) $3x-4y+1=0$

15. The coordinates of the focus of the parabola $x^2 = 6y$ is

(a) (0, 0) (b) (3, 0)

(c) $\left(0,\frac{3}{2}\right)$ (d) (0, 5)

16. The locus of the foot of the perpendicular from the focus upon a tangent to the parabola $y^2 = 4ax$ is

(a) the directrix (b) tangents at the vertex

(c) $x = a$ (d) none of these

17. If the circle $x^2+y^2+2\lambda x=0$, $\lambda\in\mathbb{R}$, touches the parabola $y^2 = 4x$ externally, then

(a) $\lambda>0$ (b) $\lambda<0$

(c) $\lambda>1$ (d) none of these

18. The normal chord at a point t on the parabola $y^2 = 4ax$ subtends a right angle at the vertex. Then $t^2 =$

(a) 4 (b) 2

(c) 1 (d) 3

19. If points $(au^2, 2au)$ and $(av^2, 2av)$ are extremities of the focal chord of a parabola $y^2 = 4ax$, then

(a) $u+v=0$ (b) $u-v=0$

(c) $uv-1=0$ (d) $uv+1=0$

20. An equilateral triangle is inscribed in the parabola $y^2 = 4x$ one of whose vertex is at the vertex of the parabola. The length of each side of the triangle is

(a) $8\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{4\sqrt{3}}{2}$ (d) $\frac{8\sqrt{3}}{2}$

21. If $x+y=k$ is normal to $y^2=12x$, then k is

(a) 3 (b) 9

(c) -9 (d) -3

22. If the chord of contact of tangents from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, then the locus of P is

(a) a circle (b) a parabola

(c) a straight line (d) none of these

23. The equation of the x-axis of the parabola $2x^2 + 5y - 3x + 4 = 0$ is

(a) $x=\frac{3}{4}$ (b) $y=\frac{3}{4}$

(c) $x=-\frac{1}{2}$ (d) $x-3y=5$

24. If the tangent and normal at the extremities of a focal chord of a parabola intersect at (x_1, y_1) and (x_2, y_2), respectively, then

(a) $x_1 = x_2$ (b) $x_1 = y_2$

(c) $y_1 = y_2$ (d) $x_2 = y_1$

25. The length of the latus rectum of the parabola $9x^2 - 6x + 36y + 19 = 0$ is

(a) 4 (b) 6

(c) 9 (d) 36

26. AB and AC are tangents to a parabola $y^2 = 4ax$. If l_1, l_2 and l_3 are the lengths of perpendiculars from A, B, C on any tangent to the parabola, then

(a) l_1, l_2, l_3 are in G.P. (b) l_2, l_1, l_3 are in G.P.

(c) l_3, l_1, l_2 are in G.P. (d) l_3, l_2, l_1 are in G.P.

27. The equation of the parabola, whose vertex is (1, –2) and focus (1, –1), is

(a) $x^2 = 4(y+2)$ (b) $(x-1)^2 = 4(y+2)$

(c) $(x+1)^2 = 4(y-2)$ (d) $(x+1)^2 = 4(y+2)$

28. If $y + 3 = m_1(x+2)$ and $y + 3 = m_2(x + 2)$ are two tangents to the parabola $y^2 = 8x$, then

(a) $m_1 + m_2 = 0$ (b) $m_1 m_2 = -1$

(c) $m_1 m_2 = 1$ (d) none of these

29. The equation of the parabola with its axis on the y-axis and passing the through the origin and point (6, –3) is

(a) $x^2 = 12y$ (b) $x^2 = -12y$

(c) $y^2 = 12x + 6$ (d) $x^2 = -12x + 6$

30. If the common normal to the parabolas $y^2 = 4ax$ and $y^2 = 4c(x-b)$ is not horizontal, i.e. not parallel to x-axis, then

(a) $\frac{b}{a-c} > 2$ (b) $\frac{c}{b-a} > 2$

(c) $\frac{a}{c-b} > 2$ (d) none of these

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement I is false, Statement II is true.

(b) Statement I is true, Statement II is true, Statement II is a correct explanation for Statement I.

(c) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.

(d) Statement I is true, Statement II is false.

31. **Statement I:** Length of latus rectum of parabola $(3x - 7y + 4)^2 = 6(7x - 9y + 2)$ is 6.
Statement II: Length of latus rectum of parabola $y^2 = 4ax$ is $4a$.

32. **Statement I:** If the normals at the end points of a double ordinate at $x = 4$ of the parabola $y^2 = 4x$ meet the curve again at P and P', then $PP' = 12$ units.
Statement II: If the normal at t_1 of $y^2 = 4ax$ meets the parabola again at t_2 then $t_2 = t_1 - \frac{2}{t_1}$.

33. **Statement I:** If PQ and P_1Q_1 are two focal chords of the parabola $y^2 = 4ax$, then the locus of the point of intersection of chords PP_1 and QQ_1 is the directrix of the parabola.
Statement II: The locus of the point of intersection of perpendicular tangents of parabola is the directrix of the parabola.

34. **Statement I:** If the vertex of the parabola is at the point $(a, 0)$ and the directrix is the y-axis then the focus of the parabola is at the point $(2a, 0)$.
Statement II: Length of common chord of the parabola $y^2 = 12x$ and the circle $x^2 + y^2 = 9$ is equal to the length of latus rectum of the parabola.

Previous Years' Questions

35. If the normal at the point $(bt_1^2, 2bt_1)$ on a parabola $y^2 = 4bv$ meets the parabola again at the point $(bt_2^2, 2bt_2)$, then

(a) $t_2 = -t_1 - \frac{2}{t_1}$ (b) $t_2 = -t_1 + \frac{2}{t_1}$

(c) $t_2 = t_1 - \frac{2}{t_1}$ (d) $t_2 = t_1 + \frac{2}{t_1}$

[AIEEE 2003]

36. If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabolas $y^2 = 4ax$ and $x^2 = 4ay$, then

(a) $d^2 + (2b + 3c)^2 = 0$ (b) $d^2 + (3b + 2c)^2 = 0$

(c) $d^2 + (2b - 3c)^2 = 0$ (d) $d^2 + (3b - 2c)^2 = 0$

[AIEEE 2004]

37. Let P be a point (1, 0) and Q be a point on the locus of $y^2 = 8x$. The locus of the mid-point of PQ is

(a) $x^2 - 4y + 2 = 0$ (b) $x^2 + 4y + 2 = 0$

(c) $y^2 + 4x + 2 = 0$ (d) $y^2 - 4x + 2 = 0$

[AIEEE 2005]

38. The locus of the vertices of the family of parabolas $y=\frac{a^3x^2}{3}+\frac{a^2x}{2}-2a$ is

(a) $xy=\frac{64}{105}$ (b) $xy=\frac{105}{64}$

(c) $xy=\frac{3}{4}$ (d) $xy=\frac{35}{16}$

39. **Statement I:** The curve $y=-\frac{x^2}{2}+x+1$ is symmetric with respect to the line $x = 1$.

Statement II: A parabola is symmetric about its axis.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.

(c) Statement I is true, Statement II is false.

(d) Statement I is false, Statement II is true.

[AIEEE 2007]

40. The equation of the tangent to the parabola $y^2 = 8x$ is $y = x + 2$. The point on this line from which the other tangent to the parabola is perpendicular to the given tangent is

(a) (0, 2) (b) (2, 4)

(c) (–2, 0) (d) (–1, 1) **[AIEEE 2007]**

41. A parabola has the origin as its focus and the line $x = 2$ as the directrix. Then the vertex of the parabola is at

(a) (2, 0) (b) (0, 2)

(c) (1, 0) (d) (0, 1) **[AIEEE 2008]**

42. If two tangents drawn from a point P to the parabola $y^2 = 4x$ are at right angles, then the locus of P is

(a) $x = 1$ (b) $2x+1=0$

(c) $x=-1$ (d) $2x-1=0$

[AIEEE 2010]

43. If a focal chord of the parabola $y^2 = ax$ is $2x – y – 8 = 0$, then the equation of the directrix is

(a) $x + 4 = 0$ (b) $x – 4 = 0$

(c) $y – 4 = 0$ (d) $y + 4 = 0$

[MP PET 2005]

44. The equation of the parabola with vertex at the origin and directrix $y = 2$ is

(a) $y^2 = 8x$ (b) $y^2 = –8x$

(c) $y^2 = \sqrt{8}x$ (d) $x^2 = –8y$

[MP PET 2005]

45. The point of the parabola $y^2 = 18x$, for which the ordinate is three times the abscissa, is

(a) (6, 2) (b) (–2, –6)

(c) (3, 18) (d) (2, 6) **[MP PET 2003]**

46. The equation of the directrix of the parabola $x^2 + 8y – 2x = 7$ is

(a) $y = 3$ (b) $y = –3$

(c) $y = 2$ (d) $y = 0$ **[MP PET 2006]**

47. If $x^2+6x+20y-51=0$, then the axis of parabola is

(a) $x + 3 = 0$ (b) $x – 3 = 0$

(c) $x = 1$ (d) $x + 1 = 0$ **[OJEE 2004]**

48. The equation of the parabola which passes through the intersection of a straight line $x + y = 0$ and the circle $x^2 + y^2 + 4y = 0$ is

(a) $y^2 = 4x$ (b) $y^2 = x$

(c) $y^2 = 2x$ (d) none of these

[BISAT, OJEE 2005]

49. Let a focal chord of parabola $y^2 = 16x$ cuts it at points (f, g) and (h, k). Then, $f \cdot h$ is equal to

(a) 12 (b) 16

(c) 14 (d) none of these

[OJEE 2007]

50. Three normals to the parabola $y^2 = x$ pass through the point $(a, 0)$. Then

(a) $a=\frac{1}{2}$ (b) $a=\frac{1}{4}$

(c) $a>\frac{1}{2}$ (d) $a<\frac{1}{2}$ **[OJEE 2008]**

51. The locus of the middle points of the chords of the parabola $y^2 = 4ax$, which passes through the origin, is

(a) $y^2 = ax$ (b) $y^2 = 2ax$

(c) $y^2 = 4ax$ (d) $x^2 = 4ay$ **[OJEE 2008]**

52. The point of intersection of tangents at the ends of the latus rectum of the parabola $y^2 = 4x$ is equal to

(a) (1, 0) (b) (–1, 0)

(c) (0, 1) (d) (0, –1)

[Jamia Millia Islamia 2004]

53. The equation of the parabola with focus (0, 0) and directrix $x + y + 4$ is

(a) $x^2+y^2-2xy+8x+8y-16=0$

(b) $x^2+y^2-2xy+8x+8y=0$

(c) $x^2+y^2+8x+8y-16=0$

(d) $x^2-y^2+8x+8y-16=0$

[Jamia Millia Islamia 2006]

54. The equation of the line which is tangent to both the circle $x^2 + y^2 = 5$ and the parabola $y^2 = 40x$ is

(a) $2x-y\pm5=0$ (b) $2x-y+5=0$

(c) $2x-y-5=0$ (d) $2x+y+5=0$

[Jamia Millia Islamia 2008]

55. AB is a chord of the parabola $y^2 = 4ax$ with vertex at A. BC is drawn perpendicular to AB meeting the axes at C. The projection of BC on the axis of the parabola is
(a) 2 (b) $2a$
(c) $4a$ (d) $8a$
[Jamia Millia Islamia 2009]

56. The common tangent of the parabolas $y^2 = 4x$ and $x^2 = -8y$ is
(a) $y = x + 2$ (b) $y = x - 2$
(c) $y = 2x + 3$ (d) none of these
[DCE 2007]

57. If l denotes the semi-latus rectum of the parabola $y^2 = 4ax$ and SP and SQ denote the segments of any focal chord PQS being the focus, then SP, l and SQ are in the relation
(a) A.P. (b) G.P.
(c) H.P. (d) $l^2 = SP^2 + SQ^2$
[DCE 2009]

58. The focal chord of $y^2 = 16x$ is tangent to $(x-6)^2 + y^2 = 2$. Then the possible values of the slope of this chord are
(a) $1, -1$ (b) $-\frac{1}{2}, 2$
(c) $-2, \frac{1}{2}$ (d) $\frac{1}{2}, 1$ **[IIT-JEE 2001]**

59. The locus of the mid-point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix
(a) $x = -a$ (b) $x = -\frac{a}{2}$
(c) $x = 0$ (d) $x = \frac{a}{2}$
[IIT-JEE 2002]

60. The angle between the tangents drawn from the point (1, 4) to the parabola $y^2 = 4x$ is
(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$ **[IIT-JEE 2004]**

61. A tangent is drawn to the parabola $y = x^2 + 6$ at the point (1, 7) which also touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at
(a) $(-8, -6)$ (b) $(-7, -6)$
(c) $(-6, -7)$ (d) (6, 7) **[IIT-JEE 2005]**

62. The axis of a parabola is along the line $y = x$ and the distance of its vertex from the origin is $\sqrt{2}$ and that from its focus is $2\sqrt{2}$. If the vertex and focus both lie in the first quadrant, then the equation of the parabola is
(a) $(x + y)^2 = (x - y - 2)$ (b) $(x - y)^2 = (x + y - 2)$
(c) $(x - y)^2 = 4(x + y - 2)$ (d) $(x - y)^2 = 8(x + y - 2)$
[IIT-JEE 2006]

63. The equation of the common tangents to the parabola $y = x^2$ and $y = -(x-2)^2$ is
(a) $y = 4(x-1)$ (b) $y = 0$
(c) $y = -4(x-1)$ (d) $y = -30x - 50$
[IIT-JEE 2006]

64. The focus of the parabola $y^2 - x - 2y + 2 = 0$ is
(a) $\left(\frac{1}{4}, 0\right)$ (b) (1, 2)
(c) $\left(\frac{5}{4}, 1\right)$ (d) $\left(\frac{3}{4}, \frac{5}{2}\right)$
[UP SEE 2007]

65. If the tangent at extremities of a focal chord AB of the parabola $y^2 = 4ax$ intersects at a point C, then $\angle ACB$ is equal to
(a) $\pi/4$ (b) $\pi/3$
(c) $\pi/2$ (d) $\pi/6$ **[UP SEE 2009]**

ANSWERS

Exercise Set

1. (c)	**2.** (a)	**3.** (a)	**4.** (a)	**5.** (a)	**6.** (d)	**7.** (c)	**8.** (b)	**9.** (a)	**10.** (d)
11. (c)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (b)	**19.** (d)	**20.** (a)
21. (b)	**22.** (b)	**23.** (a)	**24.** (c)	**25.** (a)	**26.** (b)	**27.** (b)	**28.** (b)	**29.** (b)	**30.** (a)
31. (a)	**32.** (d)	**33.** (c)	**34.** (d)	**35.** (a)	**36.** (a)	**37.** (d)	**38.** (b)	**39.** (a)	**40.** (c)
41. (c)	**42.** (c)	**43.** (a)	**44.** (d)	**45.** (d)	**46.** (a)	**47.** (a)	**48.** (c)	**49.** (b)	**50.** (c)
51. (b)	**52.** (b)	**53.** (a)	**54.** (b) and (c)		**55.** (c)	**56.** (d)	**57.** (c)	**58.** (a)	**59.** (c)
60. (c)	**61.** (c)	**62.** (d)	**63.** (a) and (b)		**64.** (c)	**65.** (c)			

HINTS AND SOLUTIONS—EXERCISE SET

1. The points P and Q are (3, 0) and (5, 0).
Equation of the circle is $(x-3)(x-5) + y^2 + \lambda y = 0$.
The length of the tangent from $(0,0)$ is $\sqrt{S_{11}} = \sqrt{15}$.

2. Tangent at t_1 is $t_1 y = x + at_1^2$.
Tangent at t_2 is $t_2 y = x + at_2^2$.
Product of the slopes of the tangents

$$= \frac{1}{t_1} \cdot \frac{1}{t_2} = \frac{1}{t_1 t_2} = -1$$

$$\Rightarrow \quad t_1 t_2 = -1$$

3. Let the coordinates of P and Q be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ respectively. Then $y_1 = 2at_1$ and $y_2 = 2at_2$. The coordinates of the point of intersection of the tangents at P and Q are $(at_1t_2, a(t_1 + t_2))$.

$$\therefore \quad y_3 = a(t_1 + t_2) \quad \Rightarrow \quad y_3 = \frac{y_1 + y_2}{2}$$

$\Rightarrow$ y_1, y_2, y_3 are in A.P.

4. Equation of the tangent to $y^2 = 4ax$ is

$$y = mx + \frac{a}{m}$$

where $$m = \tan\frac{\pi}{3} = \sqrt{3}$$

$$\therefore \quad y = \sqrt{3}x + \frac{a}{\sqrt{3}}$$

Point of contact of the tangent with the parabola is

$$\left(\sqrt{3}x + \frac{a}{\sqrt{3}}\right)^2 = 4ax$$

$$\Rightarrow \quad 3x^2 + \frac{a^2}{3} + 2ax = 4ax \quad \Rightarrow \quad 9x^2 - 6ax + a^2 = 0$$

$$\Rightarrow \quad (3x - a)^2 = 0$$

$$\therefore \quad x = \frac{a}{3} \text{ and hence } y = \pm\frac{2a}{\sqrt{3}}$$

5. $(y-1)^2 = 4(x-1)$. P has coordinates $x = 1 + t^2, y = 1 + 2t$.
Tangent at P is $(x-1) - (y-1)t + t^2 = 0$
Directrix is $x = 0$

$$\therefore \quad Q = \left[0, t + 1 - \frac{1}{t}\right]$$

If $R(x, y)$ divides QP externally in the ratio 1 : 2, then

$$x = -(1 + t^2) \text{ and } y = 1 - \frac{2}{t}$$

$$\Rightarrow \quad t = \frac{2}{1-y}$$

$$\therefore \quad x + 1 + \frac{4}{(1-y)^2} = 0$$

or $(x+1)(1-y)^2 + 4 = 0$.

6. Focal distance of $P(x, y) = x + a$.
Point is nearest when $x + a$ is minimum which is so if $x = 0$.

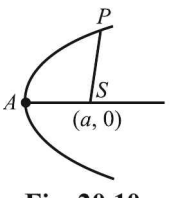


Fig. 20.10

7. The tangents at the ends of the latus rectum to $y^2 = 4ax$, i.e. at $L(a, 2a)$ and $L'(a, -2a)$ are $x - y + a = 0$ and $x + y + a = 0$ respectively.
Since they are perpendicular,
$\therefore$ their normals are also perpendicular.
Also, these tangents intersect at $P(-a, 0)$.

$$\therefore \quad PL = PL'$$

Hence, they form a square.

8. Here, $a = 2$, normal at t is $xt + y = 2t^3 + 4t$ and focus $= (2, 0)$. The point on the axis is $(10, 0)$.
The normal passes through $(10, 0)$.

$$\therefore \quad 10 = 2t^2 + 4 \quad \Rightarrow \quad t^2 = 3$$

The normal is at the point $(6, 4\sqrt{3})$.
The required length is $\sqrt{(10-6)^2 + (4\sqrt{3})^2} = 8$.

9. If t and t' are the ends of the focal chord of the parabola $y^2 = 4ax$, then its length $= a(t' - t)^2$
But for a focal chord $tt' = -1$

$$\therefore \quad t' = -\frac{1}{t}$$

$$\therefore \quad \text{Required length } = a\left(-\frac{1}{t} - t\right)^2 = a\left(t + \frac{1}{t}\right)^2.$$

10. Equation of the parabola is $(y-2)^2 = 16(x-1)$.
We know that the standard equation of parabola with (h, k) as coordinates of its vertex is

$$(y-k)^2 = 4a(x-h).$$

Comparing the given equation with the standard equation, we get $h = 1$, $k = 2$ and $a = 4$
We also know that coordinates of the vertex of the parabola $(h, k) = (1, 2)$

11. The given parabola is $(y-2)^2 = 4(x+1)$. Its axis is $y - 2 = 0$, which is parallel to the x-axis.
Thus, the ray is parallel to the axis of the parabola.
We know that any ray, which is parallel to the axis of the parabola, passes through the focus after the reflection.
Hence, the point is the focus $(0, 2)$.

12. Any point on the line $x + y = 1$ can be taken as $(t, 1 - t)$.
Equation of the chord with this as mid-point is

$$y(1-t) - 2a(x+t) = (1-t)^2 - 4at$$

It passes through $(a, 2a)$.
So, $$t^2 - 2t + 2a^2 - 2a + 1 = 0.$$
This should have 2 distinct real roots.
So, $$a^2 - a < 0 \quad \Rightarrow \quad 0 < a < 1$$
So, length of the latus rectum < 4.

13. $\frac{x+y}{2} = t^2 + 1, \frac{x-y}{2} = t$

$$\therefore \quad \frac{x+y}{2} = \frac{(x-y)^2}{4} + 1$$

$\Rightarrow (x-y)^2 = 2(x+y-2)$.

It represents a parabola.

14. Equation of the chord of the parabola $y^2 = 8x$ in terms of its mid-point is given by $T = S_1$

i.e. $yy_1 - 4(x+x_1) = y_1^2 - 8x_1$

$\Rightarrow \quad yy_1 - 4x = y_1^2 - 4x_1$

If mid-point is (2, –3), then $-3y - 4x = 9 - 8 = 1$

$\Rightarrow \quad 4x + 3y + 1 = 0$.

15. Equation of the parabola is $x^2 = 6y$.

We know that the standard equation of the parabola is $x^2 = 4ay$.

Comparing the given equation with the standard equation,

we get $4a = 6$ or $a = \frac{3}{2}$

We also know that coordinates of the focus of the parabola $(0, a) = \left(0, \frac{3}{2}\right)$.

16. Any line touching the parabola $y^2 = 4ax$ can be written as

$$y = mx + \frac{a}{m} \qquad \text{(i)}$$

Equation of a line passing through the focus $(a, 0)$ and perpendicular to (i) is

$$y = -\frac{1}{m}(x-a) \qquad \text{(ii)}$$

Let $P(h, k)$ be the point of intersection of (i) and (ii), then from (i) and (ii), we get $h = 0$.

So, the locus $P(h, k)$ is $x = 0$ which is a line tangent to the parabola $y^2 = 4ax$ at the vertex.

17. Equations are $x^2 + y^2 + 2\lambda x = 0$, $y^2 = 4x$.

Centre of circle is $(-\lambda, 0)$ and radius is $\sqrt{\lambda^2 + 0} = \lambda$.

Clearly the circle will touch the parabola externally if the centre of the circle lies on the –ve x-axis, i.e. $\lambda > 0$.

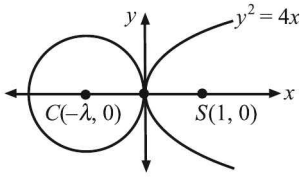


Fig. 20.11

18. Normal at t for parabola $y^2 = 4ax$ is

$$tx + y = 2at + at^3 \qquad \text{(i)}$$

Combined equation of the lines joining the vertex, i.e. origin to the points of intersection of the parabola and (i) is

$$y^2 = 4ax\left(\frac{tx+y}{2at+at^3}\right)$$

$\Rightarrow \quad (2t + t^3)y^2 = 4x(y + tx)$

Since (i) makes a right angle at the vertex,

$\therefore \quad 2t + t^3 - 4t = 0 \quad \Rightarrow \quad t^2 = 2$.

19. Coordinates of extremities of focal chord $(x_1, y_1) = (au^2, 2au)$ and $(x_2, y_2) = (av^2, 2av)$ and equation of parabola $y^2 = 4ax$.

We know that coordinates of the focus $(x, y) = (a, 0)$.

We know that the equation of the focal chord with (x_1, y_1) and (x_2, y_2) as extremity points is

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$\Rightarrow \quad (y - 2au) = \frac{2av - 2au}{av^2 - au^2}(x - au^2)$$

$$\Rightarrow \quad y - 2au = \frac{2}{v+u}(x - au^2)$$

Since the focal chord passes through the focus, therefore

$$0 - 2au = \frac{2}{v+u}(a - au^2)$$

$\Rightarrow \quad -uv - u^2 = 1 - u^2 \quad \Rightarrow \quad uv + 1 = 0$.

20. Let $AB = a$.

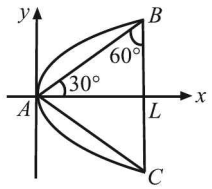


Fig. 20.12

Then $AL = a\cos 30° = \frac{\sqrt{3}a}{2}$

and $BL = a \sin 30° = \frac{a}{2}$. Thus, B is $\left(\frac{\sqrt{3}a}{2}, \frac{a}{2}\right)$.

Since P lies on the parabola $y^2 = 4x$,

$$\therefore \quad \left(\frac{a}{2}\right)^2 = 4\left(\frac{\sqrt{3}a}{2}\right)$$

$$\Rightarrow \quad \frac{a^2}{4} = 2\sqrt{3}a \quad \Rightarrow \quad a = 8\sqrt{3}. \qquad [\because a \neq 0]$$

21. $y = mx + c$ is a normal to $y^2 = 4ax$ if

$$c = -2am - am^3 \qquad \text{(i)}$$

Here $4a = 12$

$\Rightarrow \quad a = 3$

Also, $y = -x + k \quad \Rightarrow \quad m = -1, c = k$

$\therefore \quad k = -6(-1) - 3(-1) = 6 + 3$

$\therefore \quad k = 9$

22. Let $P(x_1, y_1)$ be a point.

$\therefore$ Chord of contact is $yy_1 = 2a(x + x_1)$

i.e. $x = \frac{y_1}{2a}y - x_1$

$x = my + c$, where $m = \frac{y_1}{2a} c = -x_1$

$\therefore$ Now, $x = mx + c$ touches

$$x^2 = 4by \text{ if } c = \frac{b}{m}$$

i.e. if $-x_1 = \frac{h}{y_1/2a} \Rightarrow -x_1 y_1 = 2ab$

$\therefore$ Locus of (x_1, y_1) is $-xy = 2ab$

i.e. $xy = -2ab$ which is a hyperbola.

23. Equation of the parabola is $2x^2 + 5y - 3x + 4 = 0$.

The given equation may be written as

$$x^2 - \frac{3}{2}x = -\frac{5}{2}y - 2$$

$$\Rightarrow \left(x - \frac{3}{4}\right)^2 - \frac{9}{16} = -\frac{5y}{2} - 2$$

$$\Rightarrow \left(x - \frac{3}{4}\right)^2 = -\frac{5}{2}y - \frac{23}{16}$$

$$\Rightarrow \left(x - \frac{3}{4}\right)^2 = -\frac{5}{2}\left(y + \frac{23}{40}\right) \quad \text{(i)}$$

We know that the standard equation of the parabola with (h, k) as coordinates of its vertex is

$$(x-h)^2 = 4a(y-k).$$

Comparing Eq. (i) with the standard equation, we get $h = \frac{3}{2}$.

We also know that the x-axis of the parabola is $(x - h) = 0$.

$$\Rightarrow x - \frac{3}{4} = 0 \Rightarrow x = \frac{3}{4}.$$

24. Let $P(at_1^2, 2at_1), Q(at_2^2, 2at_2)$ be a focal chord of the parabola $y^2 = 4ax$.

The tangents at P and Q meet at $(at_1t_2, a(t_1 + t_2))$.

$\therefore$ $x_1 = at_1t_2 = -a$ ($\because$ PQ is a focal chord $\therefore$ $t_1t_2 = -1$)

$y_1 = a(t_1 + t_2)$

Normals at P, Q meet at

$$(2a + a(t_1^2 + t_2^2 + t_1t_2), -at_1t_2(t_1 + t_2))$$

$\therefore$ $x_2 = 2a + a(t_1^2 + t_2^2 - 1)$

$\Rightarrow$ $y_2 = -a(-1)(t_1 + t_2) = a(t_1 + t_2)$

Clearly $y_1 = y_2$

25. Equation of the parabola is $9x^2 - 6x + 36y + 19 = 0$.

The given equation may be written as

$$9\left(x^2 - \frac{6}{9}x\right) = -36y - 19$$

$$\Rightarrow 9\left(x^2 - \frac{2}{3}x\right) = -36y - 19$$

$$\Rightarrow x^2 - \frac{2}{3}x + \frac{1}{9} = -4y - \frac{19}{9} + \frac{1}{9}$$

$$\Rightarrow \left(x - \frac{1}{3}\right) = -4y - 2$$

$$\Rightarrow \left(x - \frac{1}{3}\right)^2 = -4\left(y + \frac{1}{2}\right) \quad \text{(i)}$$

We know that the standard equation of parabola with (h, k) as coordinates of its vertex is

$$(x-h)^2 = -4a(y-k).$$

Comparing the Eq. (i) with the standard equation, we get $4a = 4$.

We also know that the length of the latus rectum of the parabola $(4a) = 4$.

26. Let $B(at_1^2, 2at_1)$ and $C(at_2^2, 2at_2)$ be points on $y^2 = 4ax$.

$\therefore$ A is $(at_1t_2, a(t_1 + t_2))$

Any tangent to $y^2 = 4ax$ is $ty = x + at^2$

$$\Rightarrow l_1 = \frac{at_1t_2 - a(t_1 + t_2) + at^2}{\sqrt{1 + t^2}},$$

$$l_2 = \frac{at_1^2 - 2at_1t + at_2}{\sqrt{1 + t^2}}$$

$$l_3 = \frac{at_2^2 - 2at_2t + at^2}{\sqrt{1 + t^2}}$$

Clearly $l_2l_3 = l_1^2$, therefore l_2, l_1, l_3 are in G.P.

27. Coordinates of the vertex parabola $(h, k) = (1, -2)$ and coordinates of its focus $(h, a + k) = (1, -1)$.

We know that $a = -1 - k = -1 + 2 = 1$.

We also know that the standard equation of the parabola with (h, k) as coordinates of its vertex is

$$(x-h)^2 = 4a(y-k)^2 \text{ or } (x-1)^2 = 4(y+2)$$

28. Clearly the two tangents, having slopes m_1 and m_2, meet the line $x + 2 = 0$, i.e. $x = -2$ which is the directrix of the parabola $y^2 = 8x$.

$\therefore$ The two tangents must be at right angles, i.e. $m_1m_2 = -1$.

29. Coordinates of the point $(x, y) = (6, -3)$.
We know that the standard equation of the parabola is

$$x^2 = 4ay \qquad \text{(i)}$$

Substituting $x = 6$ and $y = -3$ in Eq. (i), we get

$$(6)^2 = 4a(-3) \quad \Rightarrow \quad 36 = -12a$$

$$\Rightarrow \quad a = -3 .$$

Substituting the value of a in Eq. (i), we get the equation of the parabola $x^2 = -12y$.

30. For parabola $y^2 = 4ax$, the normal is

$$y = mx - 2am - am^3 \qquad \text{(i)}$$

For parabola $y^2 = 4c(x - b)$, the normal is

$$y = m(x-b) - 2cm - cm^3 \qquad \text{(ii)}$$

For common normal, (i) and (ii) must be identical.

$$\therefore \quad \frac{m}{m} = \frac{-2am - am^3}{-mb - 2cm - cm^3}$$

$$\Rightarrow \quad mb + 2cm + cm^3 = 2cm + am^3$$

$$\Rightarrow \quad b + 2c + cm^2 = 2a + am^2$$

$$\Rightarrow \quad m^2(c-a) = 2(a-c) - b$$

$$\Rightarrow \quad m = \pm\sqrt{\frac{2(a-c)-b}{c-a}}$$

[$\because$ common normal is other than the x-axis]

$$\therefore \quad m \neq 0 \quad \therefore \quad \frac{2(a-c)-b}{c-a} > 0$$

$$\therefore \quad -2 - \frac{b}{c-a} > 0 \quad \Rightarrow \quad \frac{b}{a-c} > 2 .$$

31. Since the lines $3x + 7y + 4 = 0$ and $7x - 9y + 2 = 0$ are not perpendicular, therefore latus rectum is not 6.

32. End points of double ordinate at $x = 4$ are $(4, \pm 4)$.

$$\therefore \quad t_1 = \pm 2$$

$$\therefore \quad t_2 = -t_1 - \frac{2}{t_1} = \pm 3$$

$\therefore$ $P(9, 6)$, $P'(9, -6)$ (Using $at^2, 2at$)(Here $a = 1$)

$\therefore$ $PP' = 12$ units.

33. Let coordinates of P and P_1 be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$.
Then coordinates of Q and Q_1 are

$$\left(\frac{a}{t_1^2}, \frac{-2a}{t_1}\right) \text{ and } \left(\frac{a}{t_2^2}, \frac{-2a}{t_2}\right)$$

$\therefore$ Equation of PQ is $y(t_1 + t_2) = 2(xat_1t_2)$ (i)

Replacing t_1 by $-\frac{1}{t_1}$ and t_2 by $-\frac{1}{t_2}$, the equation of P_1Q_1 becomes $y\left(-\frac{1}{t_1} - \frac{1}{t_2}\right) = 2\left[x + a\left(-\frac{1}{t_1}\right)\left(-\frac{1}{t_2}\right)\right]$

$$\Rightarrow \quad -y(t_1 + t_2) = 2(xt_1t_2 + a) \qquad \text{(ii)}$$

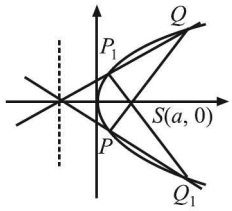


Fig. 20.13

$\therefore$ (i) + (ii) gives $0 = 2(1 + t_1t_2)(x + a)$

$\Rightarrow$ $x + a = 0$, which is a directrix.

34. In Statement I, focus is on the x-axis at a distance a from the vertex.
$\therefore$ Statement I is correct.
Statement II is false because length of the latus rectum or parabola is 12 which is greater than the diameter of the circle and the common chord is of the length less than the diameter.

35. Equation of the normal at point $(bt_1^2, 2bt_1)$ on the parabola is $y = -t_1x + 2bt_1 + bt_1^3$.
It also passes through $(bt_2^2, 2bt_2)$. Then

$$2bt_2 = -t_1.bt_2^2 + 2bt_1 + bt_1^3$$

$$\Rightarrow \quad 2t_2 - 2t_1 = -t_1(t_1^2 - t_1^2) \quad \Rightarrow \quad 2 = -t_1(t_2 + t_1)$$

$$\Rightarrow \quad t_2 = -t_1 - \frac{2}{t_1}$$

36. The common chord is the line $x - y = 0$. But the given line is $2bx + 3cy + 4d = 0$
$\therefore$ $d = 0$, $2b + 3c = 0$.

37. P is $(1, 0)$.

Let Q be the point $(2t^2, 4t)$.

[Any point on $y^2 = 4ax$ is $(at^2, 2at)$. Put $a = 2$]

Let mid-point of PQ be (h, k).

$$\therefore \quad 2h = 1 + 2t^2$$

$$2k = 4t \quad \Rightarrow \quad t = \frac{k}{2}$$

$$\therefore \quad 2h = 1 + 2.\frac{k^2}{4} = 1 + \frac{k^2}{2} \quad \Rightarrow \quad 4h = 2 + k^2$$

$\therefore$ Locus of (h, k) is $y^2 - 4x + 2 = 0$.

38. At a vertex of the parabola $y = ax^2 + bx + c$,

$$\frac{dy}{dx} = 0$$

We are given $y=\frac{1}{3}a^3x^2+\frac{1}{2}a^2x-2a$ (i)

$\Rightarrow \quad \frac{dy}{dx}=\frac{2}{3}a^3x+\frac{1}{2}a^2$

Now $\frac{dy}{dx}=0 \quad \Rightarrow \quad a=-\frac{3}{4x}$

Putting $a=-\frac{3}{4x}$ in (i), we get

$$y=\frac{1}{3}\cdot\left(-\frac{27}{64x^3}\right)\cdot x^2+\frac{1}{2}\cdot\frac{9}{16x^2}\cdot x-2\left(-\frac{3}{4x}\right)$$

$$=-\frac{9}{64x}+\frac{9}{32x}+\frac{3}{2x}$$

$$=\frac{-9+18+96}{64x}=\frac{105}{64x}$$

$\Rightarrow \quad xy=\frac{105}{64}$.

39. $y=-\frac{x^2}{2}+x+1 \quad \Rightarrow \quad y-\frac{3}{2}=-\frac{1}{2}(x-1)^2$

$\Rightarrow$ It is symmetric about $x = 1$.

40. Perpendicular tangents can be drawn only from a point on the directrix, $x = -2$.

$\therefore$ The point is $(-2, 0)$.

41. From the figure, it is clear that the vertex of the parabola is $(1, 0)$.

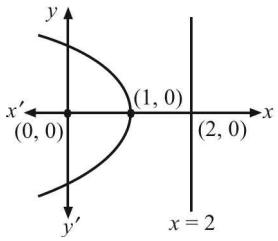


Fig. 20.14

42. We know that the locus of point P from which two perpendicular tangents are drawn to the parabola is the directrix of the parabola.

Hence, the required locus is $x = -1$.

43. Equation of the parabola is $y^2 = ax$

whose focus is $\left(\frac{a}{4}, 0\right)$.

Since the equation of the focal chord $2x-y-8=0$ passes through the focus $\left(\frac{a}{4}, 0\right)$, so

$$2\left(\frac{a}{4}\right)-0-8=0$$

$\Rightarrow \quad a = 16$

$\therefore$ Equation of the directrix is $x=-\frac{a}{4} \quad \Rightarrow \quad x=-4$.

44. Directrix of the parabola is $y = 2$

$\Rightarrow \quad a = -2$

$\therefore$ Required equation of the parabola is

$x^2=-4\cdot 2\cdot y \quad \Rightarrow \quad x^2=-8y$

45. $y^2 = 18x$ (i)

According to the given condition,

$y = 3x$ (ii)

From Eqs. (i) and (ii), we get

$(3x)^2=18x$ [from Eq. (i)]

$\Rightarrow \quad x^2=2x \quad \Rightarrow \quad x=0,2$

$\Rightarrow \quad y=0,\pm 6$

46. The equation can be rewritten as

$$(x-1)^2=-4\times(2)(y-1)$$

$\Rightarrow \quad X^2=-4aY$, where $X=x-1, Y=y-1$

So, equation of the directrix is $Y = a$

$\Rightarrow \quad y-1=2 \quad \Rightarrow \quad y=3$

47. $(x+3)^2=-20(y-3)$

This is of the form $X^2=-4aY$.

$\therefore$ Axis of such a parabola is given by

$X = 0$

$\Rightarrow x + 3 = 0$.

48. The intersection of the line and circle is $(0, 0)$ and $(2, -2)$.

Now, taking option (c),

i.e. $y^2 = 2x$

At point $(0, 0) \quad \Rightarrow \quad 0 = 0$

and at point $(2, -2) \quad \Rightarrow \quad (-2)^2 = 2(2)$

$\Rightarrow 4 = 4$.

49. Let (f, g) and (h, k) be $(4t_1^2, 8t_1)$ and $(4t_2^2, 8t_2)$ respectively.

Since they are end points of a focal chord,

$\therefore \quad t_1t_2=-1$

Now, $f\cdot h=4t_1^2\cdot 4t_2^2=16(t_1t_2)^2=16$

50. If three normals to the parabola $y^2 = 4ax$ pass through the point (h, k), then $h > 2a$.

Here, $h = a$ and $a=\frac{1}{4}$

$\therefore \quad a>2\cdot\frac{1}{4} \quad \Rightarrow \quad a>\frac{1}{2}$

51. Let the mid-point be (h, k).

$\therefore$ Equation of the chord is $T = S_1$

$$yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$$

Since it passes through the origin,

$\therefore \quad -2ax_1 = y_1^2 - 4ax_1 \quad \Rightarrow \quad y_1^2 = 2ax_1$

$\therefore$ Locus is $y^2 = 2ax$.

52. The coordinates at the ends of the latus rectum of the parabola $y^2 = 4x$ are $L(1, 2)$ and $L_1(1, -2)$.

Equations of tangents at L and L_1 are $2y = 2(x + 1)$ and $-2y = 2(x + 1)$, which give $x = -1$, $y = 0$.

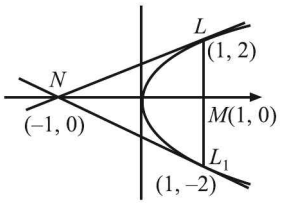


Fig. 20.15

53. Focus of the parabola is $S(0, 0)$ and equation of the directrix is $x + y = 4$.

Let $P(x, y)$ be any point on the parabola.

Then, $SP^2 = PM^2$

$$(x-0)^2 + (y-0)^2 = \left[\frac{x + y - 4}{\sqrt{1+1}}\right]^2$$

$$\Rightarrow \quad x^2 + y^2 = \frac{x^2 + y^2 + 16 + 2xy - 8y - 8x}{2}$$

$$\Rightarrow \quad x^2 + y^2 - 2xy + 8x + 8y - 16 = 0$$

54. Let the equation of the line be $y = mx + c$. Since this is the tangent to the circle $x^2 + y^2 = 5$,

$$\therefore \quad c = \pm a\sqrt{1 + m^2}$$

$$= \pm\sqrt{5}\sqrt{1 + m^2} \qquad \text{(i)}$$

Also, the above line is tangent to the parabola $y^2 = 40x$.

$$\therefore \quad c = \frac{a}{m} = \frac{10}{m} \qquad \text{(ii)}$$

From Eqs. (i) and (ii), we get

$$\frac{10}{m} = \pm\sqrt{5}\sqrt{1 + m^2}$$

$$\Rightarrow \quad m^4 + m^2 - 20 = 0 \quad \Rightarrow \quad (m^2 + 5)(m^2 - 4) = 0$$

$$\Rightarrow \quad m^2 = 4, m^2 \neq -5 \quad \Rightarrow \quad m = \pm 2$$

$$\Rightarrow \quad c = \pm 5$$

$$\therefore \quad y = \pm 2x \pm 5$$

55. In ΔABD, we have $\tan\theta = \dfrac{y}{x}$ (i)

In ΔBCD, we have

$$\tan(90° - \theta) = \frac{y}{CD}$$

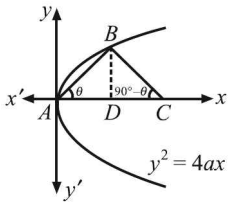


Fig. 20.16

$$\Rightarrow \quad CD = y\tan\theta = \frac{y^2}{x} \qquad \text{[using Eq. (i)]}$$

$$\Rightarrow \quad CD = \frac{4ax}{x} = 4a.$$

56. The equation of tangent of the parabola

$$y^2 = 4x \text{ is } y = mx + \frac{1}{m}$$

This is also the tangent to the parabola $x^2 = -8y$.

$$\therefore \quad x^2 = -\left(mx + \frac{1}{m}\right)$$

$$\Rightarrow \quad mx^2 + 8m^2x + 8 = 0 \text{ has equal roots}$$

$$\Rightarrow \quad 64m^4 = 32m \qquad [\because D = 0]$$

$$\Rightarrow \quad m = \frac{1}{\sqrt[3]{2}}$$

$\therefore$ Equation of the tangent is $y = \dfrac{1}{\sqrt[3]{2}}x + \sqrt[3]{2}$.

57. Let two points on the parabola be $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$.

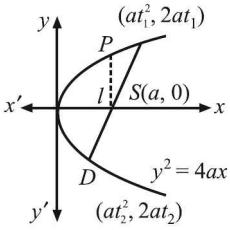


Fig. 20.17

Now, $SP = \sqrt{(a - at_1^2)^2 + (0 + 2at_1)^2}$

$$= a + at_1^2$$

$$SQ = \sqrt{(a - at_2^2)^2 + (0 - 2at_2)^2}$$

$$= a + at_2^2 = a + \frac{a}{t_1^2} \qquad (\because t_1t_2 = -1)$$

Now, $\dfrac{2 \times SP \times SQ}{SP + SQ} = \dfrac{2 \times a(1 + t_1^2) \times a\left(1 + \dfrac{1}{t_1^2}\right)}{(a + at_1^2) + \left(a + \dfrac{a}{t_1^2}\right)}$

$$= \frac{2a\left(2+\frac{1}{t_1^2}+t_1^2\right)}{\left(2+\frac{1}{t_1^2}+t_1^2\right)} = 2a = l$$

Hence, SP, l, SQ are in H.P.

58. Focal chord is $y = m(x-4)$ [$\because$ Focus is (4, 6)]

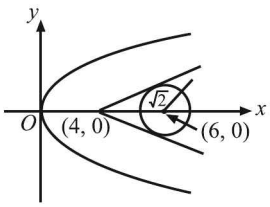


Fig. 20.18

This touches $(x-6)^2 + y^2 = 2$,

if $\frac{|m(6)-4m|}{\sqrt{1+m^2}} = \sqrt{2}$

$\Rightarrow \frac{4m^2}{1+m^2} = 2 \quad \Rightarrow \quad 4m^2 = 2+2m^2$

$\Rightarrow 2m^2 = 2 \quad \Rightarrow \quad m^2 = 1$

$\Rightarrow m = \pm 1$.

59. Let $P(at^2, 2at)$ be the moving point and $S(a, 0)$ the focus.

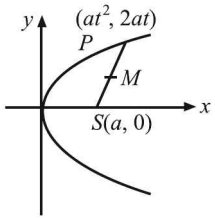


Fig. 20.19

If (x, y) is the mid-point of $[SP]$, then

$$x = \frac{at^2+a}{2}, y = \frac{2at+0}{2}$$

$\Rightarrow x - \frac{a}{2} = \frac{1}{2}at^2, y = at$

$\Rightarrow x - \frac{a}{2} = \frac{1}{2}a\left(\frac{y}{a}\right)^2 \quad \Rightarrow \quad y^2 = 2a\left(x-\frac{a}{2}\right)$

Shift the origin to the point $\left(\frac{a}{2}, 0\right)$ by putting

$$x - \frac{a}{2} = X \text{ and } y = Y$$

$\therefore \quad Y^2 = 2aX$ which is a parabola.

Its directrix is $X = -\frac{a}{2}$, i.e. $x - \frac{a}{2} = \frac{-a}{2}$, i.e. $x = 0$.

60. The tangent $y = mx + \frac{1}{m}$ [$\because 4a = 4 \Rightarrow a = 1$]

passes through (1, 4).

$\Rightarrow \quad 4 = m(1) + \frac{1}{m} \quad \Rightarrow \quad m^2 - 4m + 1 = 0$

$\Rightarrow \quad m = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$.

$\therefore \quad \tan\theta = \left|\frac{(2+\sqrt{3})-(2-\sqrt{3})}{1+(2+\sqrt{3})(2-\sqrt{3})}\right| = \left|\frac{2\sqrt{3}}{1+4-3}\right|$

$$= \left|\frac{2\sqrt{3}}{2}\right| = \left|\sqrt{3}\right| = \sqrt{3}.$$

Hence, $\theta = \frac{\pi}{3}$.

61. The given parabola is

$$y = x^2 + 6 \qquad \text{(i)}$$

and the given circle is

$$x^2 + y^2 + 16x + 12y + c = 0 \qquad \text{(ii)}$$

Tangent at (1, 7) to (i) is

$$y - 7 = 2(x-1) \quad \Rightarrow \quad 2x - y + 5 = 0 \qquad \text{(iii)}$$

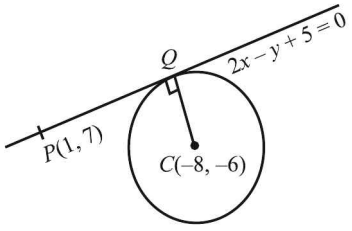


Fig. 20.20

Any line perpendicular to (iii) is

$$x + 2y + k = 0 \qquad \text{(iv)}$$

Since it passes through $C(-8,-6)$,

$\therefore \quad -8 - 12 + k = 0 \quad \Rightarrow \quad k = 20.$

Putting in (iv), equation of CQ is

$$x + 2y + 20 = 0 \qquad \text{(v)}$$

Equation of PQ is

$$2x - y + 5 = 0 \qquad \text{(vi)}$$

Solving (v) and (vi), we get Q as $(-6, -7)$.

62. Equation of the directrix is $y = -x$, i.e. $x + y = 0$.
Equation of the parabola is

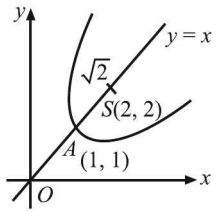


Fig. 20.21

$$(x-2)^2+(y-2)^2=\left(\frac{x+y}{\sqrt{2}}\right)^2$$

$\Rightarrow 2(x^2-4x+4+y^2-4y+4)=x^2+y^2+2xy$

$\Rightarrow x^2+y^2-2xy=8x+8y-16$

$\Rightarrow (x-y)^2=8(x+y-2)$.

63. Equation of the tangent to $x^2=y$ is

$$y=mx-\frac{1}{4}m^2 \qquad \text{(i)}$$

Equation of the tangent to $(x-2)^2=-y$ is

$$y=m(x-2)+\frac{1}{4}m^2 \qquad \text{(ii)}$$

(i) and (ii) are identical.

$\Rightarrow$ $m=0$ or 4

$\therefore$ Common tangents are $y=0$ and $y=4x-4$.

64. $(y-1)^2=x-1$

$\Rightarrow Y^2=X$, where $Y=y-1, X=x-1$

Here, $a=\frac{1}{4}$

$\therefore$ Focus is $(a, 0)$, i.e. $\left(\frac{1}{4}, 0\right)$

$\Rightarrow X=\frac{1}{4}, Y=0 \quad \Rightarrow \quad x-1=\frac{1}{4}, y-1=0$

$\Rightarrow x=\frac{5}{4}, y=1$

$\therefore$ Required focus is $\left(\frac{5}{4}, 1\right)$.

65. Let the extremities of focal chords be $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$.

The equations of the tangents at A and B are

$$t_1y=x+at_1^2 \text{ and } t_2y=x+at_2^2$$

which meet at the point C.

Slopes of these lines are $m_1=\frac{1}{t_1}, m_2=\frac{1}{t_2}$

Now, $m_1m_2=\frac{1}{t_1}\times\frac{1}{t_2}=\frac{1}{-1} \qquad [\because t_1t_2=-1]$

Hence, $\angle ACB=90°=\frac{\pi}{2}$.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. Which one of the following equations parametrically represents the equation to a parabolic profile?

(a) $x=3\cos t;\ y=4\sin t$

(b) $x^2-2=-2\cos t;\ y=4\cos^2\frac{t}{2}$

(c) $\sqrt{x}=\tan t;\sqrt{y}=\sec t$

(d) $x=\sqrt{1-\sin t};\ y=\sin\frac{t}{2}+\cos\frac{t}{2}$

2. If the line $y-\sqrt{3}x+3=0$ cuts the parabola $y^2=x+2$ at A and B, and if $P\equiv(\sqrt{3}, 0)$, then $PA\cdot PB$ is equal to

(a) $\frac{2(\sqrt{3}+2)}{3}$ (b) $\frac{4\sqrt{3}}{2}$

(c) $\frac{4(2-\sqrt{3})}{3}$ (d) $\frac{4(\sqrt{3}+2)}{3}$

3. A variable chord PQ of the parabola $y^2=4x$ is drawn parallel to the line $y=x$. If the parameters of the points P and Q on the parabola be p and q, respectively, then $p+q=$

(a) 1 (b) 1/2

(c) 2 (d) 4

4. Consider a focal chord PQ of the parabola $y^2=4ax$, $a\in R^+$. Tangent drawn at P and normal drawn at Q meet the axis of the parabola at T and N, respectively. Let P be $(at^2, 2at)$. If the angle between PT and QN is α and distance between PT and QN is d, then

(a) $d=\frac{a(t^2+1)^{3/2}}{t^2}$ (b) $0°<\alpha<90°$

(c) $d=0$ (d) none of these

5. If $y=2x-3$ is a tangent to the parabola $y^2=4a\left(x-\frac{1}{3}\right)$, then a is equal to

(a) 1 (b) -1

(c) 14/3 (d) $-14/3$

6. The equation of the parabola, whose vertex and focus lie on the positive side of the x-axis at distances a and a_1 form the origin respectively, is

(a) $y^2=4(a_1-a)x$ (b) $y^2=4(a_1-a)(x-a)$

(c) $y^2=4(a_1-a)(x-a_1)$ (d) none of these

7. The length of the focal chord of the parabola $y^2=4ax$ at a distance p from the vertex is

(a) $\frac{2a^2}{p}$ (b) $\frac{a^3}{p^2}$

(c) $\frac{4a^3}{p^2}$ (d) $\frac{p^2}{a}$

8. The vertex of a parabola is the point (a, b) and latus rectum is of length l. If the axis of the parabola is along the positive direction of the y-axis, then its equation is

(a) $(x+a)^2 = \frac{l}{2}(2y-2b)$ (b) $(x-a)^2 = \frac{l}{2}(2y-2b)$

(c) $(x+a)^2 = \frac{l}{4}(2y-2b)$ (d) $(x-a)^2 = \frac{l}{8}(2y-2b)$

9. The triangle PQR of area A is inscribed in the parabola $y^2 = 4ax$ such that the vertex P lies at the vertex of the parabola and the base QR is a focal chord. The modulus of the difference of the ordinates of the points Q and R is

(a) $A/2a$ (b) A/a
(c) $2A/a$ (d) $4A/a$

10. The length of the latus rectum of the parabola, whose focus is $\left(\frac{u^2}{2g}\sin 2\alpha, \frac{u^2}{2g}\cos 2\alpha\right)$ and directrix is $y = \frac{u^2}{2g}$, is

(a) $\frac{u^2}{g}\cos^2\alpha$ (b) $\frac{u^2}{g}\cos 2\alpha$

(c) $\frac{2u^2}{g}\cos 2\alpha$ (d) $\frac{2u^2}{g}\sin^2\alpha$

11. AB and AC are tangents to a parabola $y^2 = 4ax$. p_1, p_2 and p_3 are the lengths of the perpendiculars from A, B and C respectively on any tangent to the curve. Then p_2, p_1, p_3 are in

(a) A.P. (b) G.P.
(c) H.P. (d) none of these

12. If the line $x + y - 1 = 0$ touches the parabola $y^2 = kx$, then the value of k is

(a) 4 (b) -4
(c) 2 (d) -2

13. T is a point on the tangent to a parabola $y^2 = 4ax$ at a point P. TL and TN are the perpendiculars on the focal radius SP and the directrix of the parabola, respectively. Then

(a) $SL = 2(TN)$ (b) $3(SL) = 2(TN)$
(c) $SL = TN$ (d) $2(SL) = 3(TN)$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. The angle between tangents to the parabola $y^2 = 4ax$ at the points where it intersects the line $x - y - a = 0$ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

2. The mirror image of the parabola $y^2 = 4x$ in the tangents to the parabola at the point (1, 2) is

(a) $(x-1)^2 = 4(y+1)$ (b) $(x+1)^2 = 4(y+1)$
(c) $(x+1)^2 = 4(y-1)$ (d) $(x-1)^2 = 4(y-1)$

3. If by shifting the origin to (1, 2) the equation of the parabola $y^2 - 4y + 8x - 4 = 0$ becomes $y^2 = 4ax$, then a is equal to

(a) -2 (b) 2
(c) $\frac{1}{2}$ (d) none of these

4. The length of the chord of the parabola $x^2 = 4ay$ passing through the vertex and having slope $\tan\alpha$ is

(a) $4a\,\text{cosec}\,\alpha\cot\alpha$ (b) $4a\tan\alpha\sec\alpha$
(c) $4a\cos\alpha\cot\alpha$ (d) $4a\sin\alpha\tan\alpha$

5. If the normals from any point to the parabola $x^2 = 4x$ cut the line $y = 2$ at points whose abscissae are in A.P., then the slopes of the tangents at the 3 conormal points are in

(a) A.P. (b) G.P.
(c) H.P. (d) none of these

6. The normals at three points P, Q, R of the parabola $y^2 = 4ax$ meet at (h, k). The centroid of the triangles PQR lies on

(a) $x = 0$ (b) $y = 0$
(c) $x = -a$ (d) $y = a$

7. The length of the chord of the parabola $y^2 = 4ax$ passing through the vertex and making an angle θ with the axis is

(a) $4a\,\text{cosec}^2\theta$ (b) $4a\cos\theta\,\text{cosec}^2\theta$
(c) $4a\cot\theta\,\text{cosec}^2\theta$ (d) $2a\,\text{cosec}^2\theta$

8. The equation of the parabola whose focus is $(-8, -2)$ and the directrix is $2x - y - 9 = 0$ is

(a) $x^2 - 4xy + 4y^2 + 116x + 2y + 259 = 0$

(b) $x^2 + 4xy + y^2 + 116x + 2y + 259 = 0$

(c) $x^2 + 4xy + y^2 - 116x + 2y + 259 = 0$

(d) $x^2 - 4xy + 4y^2 + 116x - 2y + 259 = 0$

9. If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$, then one of the values of k is

(a) 4 (b) 1/4
(c) 8 (d) 1/8

10. A is a point on the parabola $y^2 = 4ax$. Then the normal at A cuts the parabola again at B. If AB subtends a right angle at the vertex, then the slope of AB is

(a) $\pm\sqrt{5}$ (b) ± 2

(c) $\pm\sqrt{3}$ (d) $\pm\sqrt{2}$

11. If tangents at A and B on the parabola $y^2 = 4ax$ intersect at T, then the ordinates of A, T and B are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

ANSWERS

Level 1

1. (b) **2.** (d) **3.** (c) **4.** (a) **5.** (d) **6.** (b) **7.** (c) **8.** (b) **9.** (c) **10.** (d)
11. (b) **12.** (b) **13.** (c)

Level 2

1. (d) **2.** (c) **3.** (a) **4.** (b) **5.** (b) **6.** (b) **7.** (b) **8.** (b) **9.** (a) **10.** (d)
11. (a)

21

APOLLONIUS

Ellipse

- Definition of Ellipse
- Standard Ellipse
- Important Properties
- Equation of Tangent
- Equation of a Tangent in Different Forms
- Equation of Normal
- Equation of Chord
- Auxiliary Circle
- Parametric Equation
- Equation of Chord of Contact
- Equation of a Pair of Tangents
- Equation of Chord of the Ellipse Whose Mid-point is Given
- Pole and Polar
- Equation of Polar
- Properties of Pole and Polar
- Diameter
- Equation of Diameter
- Conjugate Diameter
- Properties of Conjugate Diameter
- Director Circle

DEFINITION OF ELLIPSE

An **ellipse** is the locus of a point which moves so that its distance from a fixed point called *focus* is in a constant ratio to the distance from a fixed line called *directrix*. This ratio is called **eccentricity** and is denoted by e. For ellipse, **eccentricity is less than unity.**

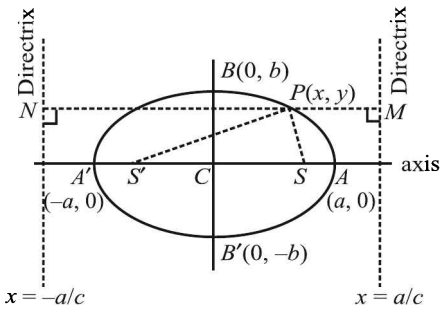


Fig. 21.1

STANDARD ELLIPSE

The general form of the standard ellipse is given by

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

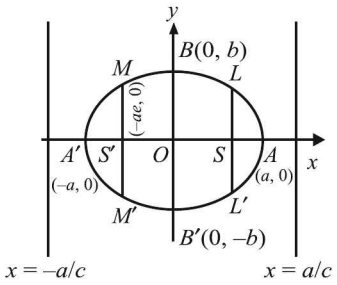
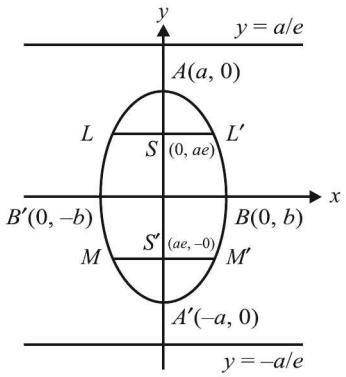
where a and b are constants.

IMPORTANT PROPERTIES

1. $AA' = 2a$ is the **major axis** of the ellipse. $BB' = 2b$ is the **minor axis** of the ellipse.
2. $S = (ae, 0)$ and $S' \equiv (-ae, 0)$ are foci.

3. Eccentricity $e = \frac{\sqrt{a^2 - b^2}}{a}$.

4. Latus rectum $LL' = \left(\frac{2b^2}{a}\right)$.

Standard equation	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \ (a > b)$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \ (a > b)$
Shape of the ellipse		
Centre	(0, 0)	(0, 0)
Equation of major axis	$y = 0$	$x = 0$
Equation of minor axis	$x = 0$	$y = 0$
Length of major axis	$2a$	$2a$
Length of minor axis	$2b$	$2b$
Foci	$(\pm ae, 0)$	$(0, \pm ae)$
Vertex	$(\pm a, 0)$	$(0, \pm a)$
Equation of directrix	$x = \pm \frac{a}{e}$	$y = \pm \frac{a}{e}$
Eccentricity (e)	$e = \sqrt{\frac{a^2 - b^2}{a^2}} = \sqrt{1 - \frac{b^2}{a^2}}$	$e = \sqrt{\frac{a^2 - b^2}{a^2}} = \sqrt{1 - \frac{b^2}{a^2}}$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Ends of latera recta	$\left(\pm ae, \pm \frac{b^2}{a}\right)$	$\left(\pm \frac{b^2}{a}, \pm ae\right)$
Parametric coordinates	$(a \cos \theta, b \sin \theta)$	$(a \cos \theta, b \sin \theta)$
Focal radii	$SP = a - ex_1$ and $S'P = a + ex_1$	$SP = a - ey_1$ and $S'P = a + ey_1$
Sum of focal radii $SP + S'P$	$2a$	$2a$
Distance between foci	$2ae$	$2ae$
Distance between directress	$\frac{2a}{e}$	$\frac{2a}{e}$
Tangents at the vertices	$x = \pm a$	$y = \pm a$

EQUATION OF TANGENT

1. **Point from:** The equation of the tangent drawn to the ellipse at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.
2. **Parametric form:** The equation of the tangent drawn to the ellipse at the point $(a\cos\theta, b\sin\theta)$ is
$$\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1.$$
3. **Slope form:** The equation of the tangent of the slope m can be taken as $y = mx + \sqrt{a^2m^2 + b^2}$, where the point of contact is $= \left[\frac{-a^2m}{\sqrt{a^2m^2+b^2}}, \frac{b^2}{\sqrt{a^2m^2+b^2}}\right]$.

EQUATION OF A TANGENT IN DIFFERENT FORMS

1. The equations of **two parallel tangents** of slope m that can be drawn to the ellipse are
$$y = mx \pm \sqrt{a^2m^2 + b^2}$$
2. The condition for $y = mx + c$ to be a tangent to the ellipse is $c^2 = a^2m^2 + b^2$ and the point of contact is $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$.
3. Tangents drawn to ellipse from a point (h, k): Let $y = mx + \sqrt{a^2m^2+b^2}$ be the tangent(s). The tangent passes through (h, k), so we get
$$\Rightarrow \quad k = mh + \sqrt{a^2m^2+b^2}$$
$$\Rightarrow \quad m^2(h^2 - a^2) - 2hk + k^2 - b^2 = 0$$
Since this is a quadratic equation, so two tangents can be drawn from a point (h, k) to an ellipse.

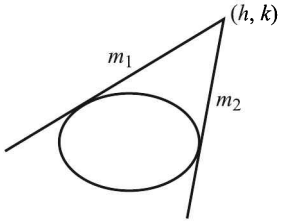


Fig. 21.2

4. Latus rectum: $LL' = \left(\frac{2b^2}{a}\right)$.

EQUATION OF NORMAL

1. **Point form:** The equation of the normal drawn to the ellipse at (x_1, y_1) is $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$.
2. **Parametric form:** The equation of the normal drawn to the ellipse at $(a\cos\theta, b\sin\theta)$ is $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$.
3. **Slope form:** The equation of the normal of slope m can be taken as $y = mx \pm \frac{m(a^2-b^2)}{\sqrt{(a^2+b^2m^2)}}$.

EQUATION OF CHORD

The equation of the chord joining the points $P(\theta)$ and $Q(\phi)$ is
$$\frac{x}{a}\cos\left(\frac{\theta+\phi}{2}\right) + \frac{y}{b}\sin\left(\frac{\theta+\phi}{2}\right) = \cos\left(\frac{\theta-\phi}{2}\right).$$

AUXILIARY CIRCLE

The circle drawn on the major axis AA' as diameter is known as the **auxiliary circle** and its equation is $x^2 + y^2 = a^2$. Let Q be a point on the auxiliary circle so that QM, perpendicular to the major axis, meets the ellipse at P. P and Q are called corresponding points on the ellipse and auxiliary circle, respectively. The angle θ is known as **eccentric angle** of the point P on the ellipse.

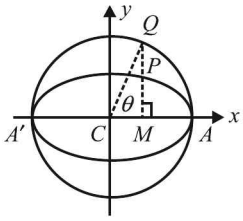


Fig. 21.3

Note CQ and not CP is inclined at θ with the x-axis.

PARAMETRIC EQUATION

In Fig. 2.3, $CM = CQ$ and $\cos\theta = a\cos\theta$.

$\Rightarrow$ X-coordinate of P and Q is $a\cos\theta$.

Putting $x = a\cos\theta$ in the equation of ellipse, we get $y = b\sin\theta$

$\Rightarrow$ $P \equiv (a\cos\theta, b\sin\theta)$.

Hence, $x = a\cos\theta$ and $y = b\sin\theta$ are the parametric equations of the ellipse where the parameter $\theta \in [0, 2\pi)$.

EQUATION OF CHORD OF CONTACT

Let us consider a point R whose coordinates are (x_1, y_1) and which lies outside the ellipse. From point R, two tangents are drawn to the ellipse which meet the ellipse at points P and Q. The line joining P and Q is known as *chord of contact* of the ellipse and its equation is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.

Note The above result is the same as the equation of the tangent drawn to the ellipse at (x_1, y_1).

EQUATION OF A PAIR OF TANGENTS

Let us consider a point R whose coordinates are (x_1, y_1) and which lies outside the ellipse. From point R, two tangents are drawn to the ellipse which meet the ellipse at points P and Q. The equation of the pair of tangents RP and RQ is $SS_1 = T2$,

where $S = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$, $S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$ and $T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$.

ANGLE BETWEEN A PAIR OF TANGENTS (DRAWN TO THE ELLIPSE $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ FROM A POINT)

This is given by $\tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b}$.

EQUATION OF CHORD OF THE ELLIPSE WHOSE MID-POINT IS GIVEN

Let us consider a chord PQ of the ellipse whose mid-point is $R \equiv (x_1, y_1)$. The equation of this chord PQ in terms of coordinates of R is $S_1 = T$, where $S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$ and $T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$.

POLE AND POLAR

Let P be any point inside or outside an ellipse. Suppose any straight line drawn through P intersects the ellipse at Q and R. Then the locus of the point of intersection of the tangents to the ellipse at Q and R is called the *polar* of the given point P with respect to the ellipse and the point P is called the *pole* of the polar.

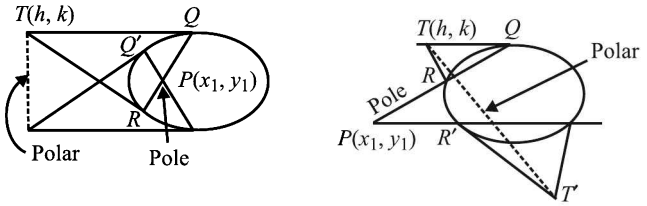


Fig. 21.4

EQUATION OF POLAR

The equation of the polar of the point (x_1, y_1) with respect to the ellipse is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.

PROPERTIES OF POLE AND POLAR

1. If the polar of $P(x_1, y_1)$ passes through $Q(x_2, y_2)$, then the polar of $Q(x_2, y_2)$ goes through $P(x_1, y_1)$ and such points are said to be **conjugate points**.
2. If the pole of a line $lx + my + n = 0$ lies on another line $l'x + m'y + n' = 0$, then the pole of the second line will lie on the first and such lines are said to be **conjugate lines**.

DIAMETER

The locus of the middle points of a system of parallel chords of an ellipse is called a *diameter*. It can be easily derived that the locus is a straight line which passes through the centre of the circle.

EQUATION OF DIAMETER

Let (x_1, y_1) be the middle point of the chord $y = mx + c$ of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Then $T = S_1$.

$$\Rightarrow \quad \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1}{a^2} + \frac{y_1}{b^2}$$

$$\text{Slope} = -\frac{b^2 x_1}{a^2 y_1} = m \quad \Rightarrow \quad y_1 = -\frac{b^2 x}{a^2 y}$$

Hence, the locus of the mid-point is $y = -\frac{b^2 x}{a^2 m}$.

It can also be defined as a chord which passes through the centre of the ellipse.

CONJUGATE DIAMETERS

Two diameters are said to be *conjugate* when each bisects the chords parallel to the other. If $y = mx$ and $y = m_1 x$ are two conjugate diameters of an ellipse, then $mm_1 = \frac{-b^2}{a^2}$.

PROPERTIES OF CONJUGATE DIAMETERS

1. The eccentric angles of the ends of a pair of conjugate diameters of an ellipse differ by a right angle.
2. The sum of the squares of any two conjugate semi-diameters of an ellipse is constant and equal to the sum of the square of the semi-axes of the ellipse, i.e. $CP^2 + CD^2 = a^2 + b^2$.
3. The product of the focal distances of a point on an ellipse is equal to the square of the semi-diameter which is conjugate to the diameter through the point.
4. The tangents at the extremities of a pair of conjugate diameters form a parallelogram whose area is constant and equal to the product of the axes.
5. The polar of any point with respect to the ellipse is parallel to the diameter to the one on which the point lies. Hence, obtain the equation of the chord whose mid-point is (x_1, y_1).

DIRECTOR CIRCLE

The locus of the point of intersection of the tangents to an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, which are perpendicular to each other, is called **director circle**.

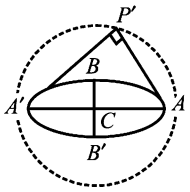


Fig. 21.5

Let any tangent in terms of slope to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ be $y=mx+\sqrt{(a^2m^2+b^2)}$.

Let it pass through (x_1, y_1).

$\therefore \quad y_1=mx_1+\sqrt{(a^2m^2+b^2)}$

$\Rightarrow \quad (y_1-mx_1)^2=a^2m^2+b^2$

$\Rightarrow \quad y_1^2-m^2x_1^2-2mx_1y_1=a^2m^2+b^2$

$\Rightarrow \quad m^2(y_1^2-a^2)-2mx_1y_1+y_1^2-b^2=0$

It is a quadratic equation in m. Let the slope of two tangents be m_1 and m_2.

$\Rightarrow \quad m_1m_2=\frac{y_1^2-b^2}{x_1^2-a^2}$

$-1=\frac{y_1^2-b^2}{x_1^2-a^2}$ ($\because$ tangents are perpendicular)

$\Rightarrow \quad -x_1^2+a^2=y_1^2-b^2 \Rightarrow x_1^2+y_1^2=a^2+b^2$

Hence, the locus of (x_1, y_1) is $x^2+y^2=a^2+b^2$.

So, the equation of the director circle is $x^2+y^2=a^2+b^2$.

SOLVED PROBLEMS

1. In an ellipse, the distance between its foci is 6 and its minor axis is 8. Then eccentricity is

(a) 3/5 (b) $1/\sqrt{52}$
(c) 1/2 (d) 4/5

Ans. (a)

Solution: $2ae=6 \Rightarrow a=3/e$

$2b=8 \Rightarrow b=4$

Also $b^2=a^2(1-e^2)$

$\Rightarrow \quad 16=\frac{9}{e^2}(1-e^2)=\frac{9}{e^2}-9$

$\Rightarrow \quad \frac{9}{e^2}=25 \Rightarrow e=\frac{3}{5}$

2. The eccentricity of ellipse $9x^2+5y^2-30y=0$ is

(a) 1/3 (b) 2/3
(c) 3/4 (d) none of these

Ans. (b)

Solution: $9x^2+5(y-3)^2=45$

$\Rightarrow \quad \frac{x^2}{5}+\frac{(y-3)^2}{9}=1 \Rightarrow a^2=5, b^2=9$

Using $a^2=b^2(1-e^2)$

$\Rightarrow \quad e=\frac{2}{3}$.

3. If the line $y=2x+c$ is tangent to the ellipse $\frac{x^2}{8}+\frac{y^2}{4}=1$, then c is

(a) ±4 (b) ±6
(c) ±1 (d) ±8

Ans. (b)

Solution. $c^2=a^2m^2+b^2=8\times4+4=36$

$\Rightarrow \quad c=\pm6$.

4. If the normal at any point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the axes at G and G', then $PG : PG' =$

(a) $a : b$ (b) $a^2 : b^2$
(c) $b^2 : a^2$ (d) $b : a$

Ans. (c)

Solution: Let $P(a\cos\theta, b\sin\theta)$ be a point on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

$\therefore$ Equation of normal is

$$(a\sec\theta)x-(b\operatorname{cosec}\theta)\,y=a^2-b^2$$

It meets the axes at

$$G\left(\frac{a^2-b^2}{a}\cos\theta, 0\right),\ G'\left(0, -\frac{(a^2-b^2)}{a}\sin\theta\right)$$

$\Rightarrow \quad \frac{PG}{PG'}=\frac{b^2}{a^2}$ (on calculation)

5. The angle between the pair of tangents drawn to the ellipse $3x^2+2y^2=5$ from the point (1, 2) is

(a) $\tan^{-1}\left(\frac{12}{5}\right)$ (b) $\tan^{-1}(6\sqrt{5})$
(c) $\tan^{-1}\left(\frac{12}{\sqrt{5}}\right)$ (d) $\tan^{-1}(12\sqrt{5})$

Ans. (a)

Solution: Equation of the pair of tangents to the ellipse from the point (1, 2) is $SS_1 = T_2$

$$(3x^2+2y^2-5)(3\cdot1^2+2\cdot2^2-5)=(3x\cdot1+2y\cdot2-5)^2$$

or $9x^2-4y^2-24xy+30x+40y-55=0$

$$\tan\theta=\frac{2\sqrt{h^2-ab}}{a+b}$$

where $a = 9$, $b = -4$, and $h = -12$.

6. If the length of the latus rectum is 5/2 and eccentricity is 1/2, then the equation of the ellipse in standard form is

(a) $\frac{x^2}{9}+\frac{y^2}{16}=1$ (b) $\frac{9x^2}{25}+\frac{12y^2}{25}=1$

(c) $\frac{9x^2}{25}+\frac{y^2}{25}=1$ (d) none of these

Ans. (b)

Solution: Let the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

Latus rectum $= \frac{2b^2}{a}=\frac{5}{2}$

$$\Rightarrow \frac{a^2(1-e^2)}{a}=\frac{5}{4} \Rightarrow a\left(1-\frac{1}{4}\right)=\frac{5}{4} \qquad \left(\because e=\frac{1}{2}\right)$$

$$\Rightarrow a=\frac{5}{3}$$

But $b^2=\frac{5a}{4}=\frac{5}{4}\cdot\frac{5}{3}=\frac{25}{12}$

$$\therefore \text{ Ellipse is } \frac{x^2}{\frac{25}{9}}+\frac{y^2}{\frac{25}{12}}=1, \text{ i.e. } \frac{9x^2}{25}+\frac{12y^2}{25}=1.$$

7. If the foci are at the points $S(2, 0)$ and $S'(-2, 0)$ and latus rectum is 6, then the equation of the ellipse in the standard form is

(a) $\frac{x^2}{16}+\frac{y^2}{12}=1$ (b) $\frac{x^2}{4}+\frac{y^2}{3}=1$

(c) $\frac{x^2}{4}+\frac{y^2}{12}=1$ (d) none of these

Ans. (a)

Solution: Clearly, $ae = 2$ and $\frac{2b^2}{a}=6$

$$\Rightarrow \frac{2a^2(1-e^2)}{a}=6 \Rightarrow 2a^2(1-e^2)=6a$$

$$\Rightarrow 2a^2-2a^2e^2=6a \Rightarrow 2a^2-2\cdot4=6a$$

$$\Rightarrow a^2-3a-4=0 \Rightarrow (a-4)(a+1)=0$$

$$\Rightarrow a=4,-1$$

When $a = 4$, $4e = 2$

$$\Rightarrow e=\frac{1}{2} \Rightarrow a=-1,-e=2$$

$$\Rightarrow e=-2$$

But $e > 0$ $\therefore \quad e=\frac{1}{2}$

$$\therefore \quad a=\frac{2}{e}=\frac{2}{1/2}=4$$

Now $b^2=a^2(1-e^2)$

$$=16\left(1-\frac{1}{4}\right)=\frac{16\times3}{4}=12$$

Hence, the ellipse is $\frac{x^2}{16}+\frac{y^2}{12}=1$.

8. The equation of the ellipse whose focus is (1, −1), directrix is $x-y-3=0$ and eccentricity is 1/2 is

(a) $7x^2+2xy+7y^2-10x+10y+7=0$

(b) $7x^2+2xy+7y^2+7=0$

(c) $7x^2+2xy+7y^2+10x-10y-7=0$

(d) none of these

Ans. (a)

Solution: Let $P(x, y)$ be any point on the ellipse.

$$\therefore \quad \frac{\sqrt{(x-1)^2+(y+1)^2}}{\frac{x-y-3}{\sqrt{2}}}=e=\frac{1}{2}$$

$$\Rightarrow x^2-2x+1+y^2+2y+1=\frac{(x-y-3)^2}{8}$$

$$\Rightarrow 8x^2+8y^2-16x+16y+16$$

$$= x^2+y^2+9-6x+6y-2xy$$

$$\Rightarrow 7x^2+7y^2+2xy-10x+10y+7=0$$

9. The eccentricity of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose latus rectum is half its minor axis, is

(a) $\sqrt{\frac{2}{3}}$ (b) $\frac{1}{\sqrt{2}}$

(c) $\frac{\sqrt{3}}{2}$ (d) none of these

Ans. (c)

Solution: $\frac{2b^2}{a}=\frac{1}{2}(2b)$

$$\frac{2b^2}{a}=b \Rightarrow 2b=a \qquad (\because b\neq0)$$

$\Rightarrow 4b^2 = a^2$ $\Rightarrow 4a^2(1-e^2) = a^2$

$\Rightarrow 4 - 4e^2 = 1$ $\Rightarrow 4e^2 = 3$

$\Rightarrow e^2 = \frac{3}{4}$ $\Rightarrow e = \frac{\sqrt{3}}{2}$ $(\because e > 0)$

10. The locus of the centre of the circle $x^2 + y^2 + 4x\cos\theta - 2y\sin\theta - 10 = 0$ is

(a) an ellipse (b) a circle

(c) a hyperbola (d) a parabola

Ans. (a)

Solution: If (α, β) is the centre, then $\alpha = -2\cos\theta$, $\beta = \sin\theta$

$\therefore \left(\frac{\alpha}{-2}\right)^2 + \beta = \cos^2\theta + \sin^2\theta = 1$

$\Rightarrow \frac{\alpha^2}{4} + \beta^2 = 1$

$\therefore$ Locus of (α, β) is $\frac{x^2}{4} + \frac{y^2}{1} = 1$, which is an ellipse.

11. If $a > 3$ and the ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ has eccentricity $\frac{\sqrt{3}}{2}$, then the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ is

(a) $\frac{7}{16}$ (b) $\frac{\sqrt{7}}{4}$

(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{2}{3}$

Ans. (b)

Solution: Here $b^2 = 4$, $e^2 = 3/4$.

Since $b^2 = a^2(1-e^2)$,

$\therefore 4 = a^2\left(1 - \frac{3}{4}\right) = a^2\left(\frac{1}{4}\right) \Rightarrow a^2 = 16$.

Now $a^2 = 16$, $b^2 = 4$

Since $b^2 = a^2(1-e^2)$, $\therefore 9 = 16(1-e^2)$

$\Rightarrow 1 - e^2 = \frac{9}{16} \Rightarrow e^2 = 1 - \frac{9}{16} = \frac{7}{16}$

$\Rightarrow e = \frac{\sqrt{7}}{4}$.

EXERCISE SET

1. The eccentricity of the ellipse $\frac{x^2}{255} + \frac{y^2}{289} = 1$ is

(a) $\frac{17}{8}$ (b) $\frac{8}{17}$

(c) $\frac{3}{4}$ (d) $\frac{4}{3}$

2. If the major axis of an ellipse is 3 times its minor axis, then its eccentricity is

(a) $\frac{2}{3}$ (b) $\frac{\sqrt{2}}{3}$

(c) $\frac{2\sqrt{2}}{3}$ (d) $\frac{4\sqrt{2}}{3}$

3. If the distance between the foci of an ellipse is equal to its minor axis, then eccentricity of the ellipse is

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{3}}$

(c) $\frac{1}{\sqrt{4}}$ (d) $\frac{1}{\sqrt{6}}$

4. The foci of the ellipse $25(x+1)^2 + 9(y+2)^2 = 255$ are

(a) (–1, 2) and (6, 1) (b) (–1, –2) and (1, 6)

(c) (1, –2) and (1, –6) (d) (–1, 2) and (–1, –6)

5. The length of the latus rectum of the ellipse $9x^2 + 4y^2 = 1$ is

(a) $\frac{3}{2}$ (b) $\frac{8}{3}$

(c) $\frac{4}{9}$ (d) $\frac{8}{9}$

6. The length of minor axes of the ellipse $16x^2 + 25y^2 = 400$ is

(a) 10 (b) 8

(c) 6 (d) 3

7. The equation of an ellipse whose vertices are (2, –2), (2, 4) and eccentricity 1/3 is

(a) $\frac{(x-2)^2}{9} + \frac{(y-1)^2}{8} = 1$ (b) $\frac{(x-2)^2}{8} + \frac{(y-1)^2}{9} = 1$

(c) $\frac{(x+2)^2}{8} + \frac{(y+1)^2}{9} = 1$ (d) $\frac{(x-2)^2}{9} + \frac{(y+1)^2}{8} = 1$

8. The equation of an ellipse whose latus rectum is 8 and eccentricity is $1/\sqrt{2}$ is

(a) $\frac{x^2}{18} + \frac{y^2}{32} = 1$ (b) $\frac{x^2}{8} + \frac{y^2}{9} = 1$

(c) $\frac{x^2}{64} + \frac{y^2}{32} = 1$ (d) $\frac{x^2}{16} + \frac{y^2}{24} = 1$

9. The equation $\frac{x^2}{2-r}+\frac{y^2}{r-5}+1=0$ represents an ellipse, if
(a) $r > 2$ (b) $r < 2$
(c) $r > 5$ (d) $2 < r < 5$

10. The equation of the normal at the point (0, 3) of the ellipse $9x^2+5y^2=45$ is
(a) $y - 3 = 0$ (b) $y + 3 = 0$
(c) x-axis (d) y-axis

11. The equations of the tangents to the ellipse $9x^2+16y^2 = 144$ from the point (2, 3) are
(a) $y = 3$ and $x = 5$ (b) $y = 3$ and $x = 2$
(c) $y = 2$ and $x = 3$ (d) $y = 3$ and $x + y = 5$

12. The locus of the middle point of the intercept of the tangent drawn from an external point to the ellipse $x^2+2y^2=2$, between the coordinate axes, is
(a) $\frac{1}{x^2}+\frac{1}{2y^2}=1$ (b) $\frac{1}{4x^2}+\frac{1}{2y^2}=1$
(c) $\frac{1}{2x^2}+\frac{1}{4y^2}=1$ (d) $\frac{1}{2x^2}+\frac{1}{y^2}=1$

13. If $\tan\theta_1\tan\theta_2=-\frac{a^2}{b^2}$, then the chord joining two points $P(\theta_1)$ and $Q(\theta_2)$ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ will subtend a right angle at
(a) focus (b) centre
(c) end of the major axis (d) end of the minor axis

14. If any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intercepts equal lengths l on the axes, then $l =$
(a) $\sqrt{a^2+b^2}$ (b) a^2+b^2
(c) $(a^2+b^2)^2$ (d) none of these

15. The equation of the ellipse with foci at (±3, 0) and vertices (±5, 0) is
(a) $\frac{x^2}{25}+\frac{y^2}{9}=1$ (b) $\frac{x^2}{25}+\frac{y^2}{16}=1$
(c) $\frac{x^2}{16}+\frac{y^2}{25}=1$ (d) none of these

16. The slope of a common tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and a concentric circle of radius is
(a) $\tan^{-1}\sqrt{\frac{r^2-b^2}{a^2-r^2}}$ (b) $\sqrt{\frac{r^2-b^2}{a^2-r^2}}$
(c) $\left(\frac{r^2-b^2}{a^2-r^2}\right)$ (d) $\sqrt{\frac{a^2-r^2}{r^2-b^2}}$

17. The eccentric angle of a point on the ellipse $\frac{x^2}{6}+\frac{y^2}{2}=1$ whose distance from the centre of the ellipse is 2 is
(a) $\frac{\pi}{4}$ (b) $\frac{3\pi}{2}$
(c) $\frac{5\pi}{3}$ (d) $\frac{7\pi}{6}$

18. Let P be a variable point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with foci S_1 and S_1. If A is the area of the triangle PS_1S_2, then the maximum value of A is
(a) ab (b) abe
(c) e (d) none of these

19. If $P(x, y)$, $F_1 = (3, 0)$, $F_2 = (-3, 0)$ and $16x^2+25y^2 = 400$, then $PF_1 + PF_2$ equals
(a) 6 (b) 8
(c) 10 (d) 12

20. P is a variable point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with AA' as the major axis. Then the maximum value of the area of the triangle APA' is
(a) ab (b) $2ab$
(c) $ab/2$ (d) none of these

21. The eccentricity of the ellipse whose pair of conjugate diameters are $y = x$ and $3y = -2x$ is
(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) none of these

22. The locus of the point of intersection of the perpendicular tangents to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is
(a) $x^2+y^2=9$ (b) $x^2+y^2=4$
(c) $x^2+y^2=13$ (d) $x^2+y^2=5$

23. If t is a variable, then the locus of the point of intersection of $\frac{x}{4}-\frac{y}{3}=t$ and $\frac{x}{4}+\frac{y}{3}=\frac{2}{t}$ is a/an
(a) parabola (b) ellipse
(c) hyperbola (d) pair of lines

24. An ellipse is described by using an endless string which is passed over two pins. If the axes are 6 cm and 4 cm, the necessary length of the string and the distance between the pins respectively in centimetres are
(a) $6, 2\sqrt{5}$ (b) $6, \sqrt{5}$
(c) $4, 2\sqrt{5}$ (d) $6+2\sqrt{5}, 2\sqrt{5}$

25. If the line joining foci subtends an angle of 90° at an extremity of the minor axis, then eccentricity e is

(a) $\frac{1}{\sqrt{6}}$ (b) $\frac{1}{\sqrt{3}}$

(c) $\frac{1}{\sqrt{2}}$ (d) none of these

26. The equation of the auxiliary circle of the ellipse $9x^2+4y^2-8y-32=0$ is

(a) $x^2+y^2-2y-8=0$ (b) $x^2+y^2-8y+7=0$

(c) $x^2+y^2-4x-5=0$ (d) none of these

27. The lengths of the axes of the conic $9x^2+4y^2-6x+4y+1=0$ are

(a) $\frac{1}{2}$, 9 (b) 3, $\frac{2}{5}$

(c) 1, $\frac{2}{3}$ (d) 3, 2

28. The eccentricity of the conic $2x^2+3y^2-4x+5y+4=0$ is

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$

(c) $\frac{1}{3}$ (d) $\frac{1}{\sqrt{3}}$

29. If S and T are the foci of an ellipse, B is an end of the minor axis and STB is an equilateral triangle, then the eccentricity of the ellipse is

(a) $\frac{1}{4}$ (b) $\frac{1}{3}$

(c) $\frac{1}{2}$ (d) $\frac{2}{3}$

30. A man running round a race course notes that the sum of the distances of two flag posts from him is always 10 m and the distance between the flag posts is 8 m. The area of the path he encloses in square metres is

(a) 8π (b) 12π

(c) 18π (d) 15π

31. If c is the centre and A and B are two points on the conic $4x^2+9y^2-8x-36y+4=0$ such that $\widehat{ACB}=\frac{\pi}{2}$, then $CA^{-2}+CB^{-2}$ is equal to

(a) $\frac{13}{36}$ (b) $\frac{36}{13}$

(c) $\frac{16}{33}$ (d) $\frac{33}{16}$

32. Let E be the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and C be the circle $x^2+y^2=9$. Let P and Q be the points (1, 2) and (2, 1), respectively. Then

(a) Q lies inside C but outside E

(b) Q lies outside both C and E

(c) P lies inside both C and E

(d) P lies inside C but outside E

33. The equation of the circle passing through the points of intersection of the ellipses $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$ is

(a) $x^2+y^2=a^2$ (b) $x^2+y^2=b^2$

(c) $x^2+y^2=\frac{a^2b^2}{a^2+b^2}$ (d) $x^2+y^2=\frac{2a^2b^2}{a^2+b^2}$

34. On the ellipse $4x^2+9y^2=1$, the points at which the tangents are parallel to the line $8x=9y$ are

(a) $\left(\frac{2}{5},\frac{1}{5}\right)$ (b) $\left(-\frac{2}{5},\frac{1}{5}\right)$

(c) $\left(-\frac{2}{5},-\frac{1}{5}\right)$ (d) none of these

35. The number of tangents from (7, 6) to the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ is

(a) 0 (b) 1

(c) 2 (d) none of these

36. The angle subtended by common tangents of two ellipses $4(x-4)^2+25y^2=100$ and $4(x+1)^2+y^2=4$ at the origin is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) none of these

37. The eccentricity of the ellipse which meets the straight line $\frac{x}{7}+\frac{y}{2}=1$ on the x-axis and the straight line $\frac{x}{3}-\frac{y}{5}=1$ on the y-axis and whose axis is the coordinate axis is

(a) $\frac{3\sqrt{2}}{7}$ (b) $\frac{2\sqrt{6}}{7}$

(c) $\frac{\sqrt{6}}{7}$ (d) none of these

38. A tangent at any point on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is cut by the tangents at the extremities of the major axis at T_1 and T_2. The circle with T_1T_2 as diameter passes through the point

(a) (2, 1) (b) $(\sqrt{5}, 0)$

(c) $(0, \sqrt{5})$ (d) $(0, -\sqrt{5})$

39. If the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is rotated 90° about the origin, the equation of the ellipse after rotation is

(a) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (b) $\frac{x^2}{a(a+b)}+\frac{y^2}{b(a+b)}=1$

(c) $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ (d) none of these

40. AB is a diameter of $x^2+9y^2=25$. The eccentric angle of A is $\pi/6$. Then the eccentric angle of B is

(a) $\frac{5\pi}{6}$ (b) $-\frac{5\pi}{6}$

(c) $-\frac{2\pi}{3}$ (d) none of these

41. The angle between the tangent drawn from the point (5, 4) to the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ is

(a) 60° (b) 90°

(c) 120° (d) 45°

Assertion-Reason Type Questions

These questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement 1 (Assertion) and Statement 2 (Reason). Each of these questions has four alternative answers. Only one of them is the correct answer. Select the correct answer.

(a) Both Statement 1 and Statement 2 are true and Statement 2 is a correct explanation of Statement 1.

(b) Both Statement 1 and Statement 2 are true but Statement 2 is not a correct explanation of Statement 1.

(c) Statement 1 is true, Statement 2 is false.

(d) Statement 1 is false, Statement 2 is true.

42. **Statement 1:** The foci of the ellipse $x^2+\lambda y^2=4$ will lie on the y-axis if $0<\lambda<1$.

Statement 2: Foci of any ellipse lie on the x-axis.

43. **Statement 1:** The distance of $P(1, 1)$ on the ellipse $4x^2+5y^2=9$ from the nearer focus is $\frac{3}{2}-\frac{1}{\sqrt{5}}$.

Statement 2: The focal distance of any point (x_1, y_1) on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a > b)$ is $a \pm ex_1$ where $e^2=\frac{a^2-b^2}{a^2}$.

44. **Statement 1:** If $A, C < 0$, then the equation $Ax^2+Bx+Cy^2+Dy=E$ represents an ellipse if $E+\frac{B^2}{4A}+\frac{D^2}{4C}<0$.

Statement 2: The equation $Px^2+Qx+R=0$ has real roots if $Q^2-4PR>0$.

45. **Statement 1:** The slopes of the tangents drawn through (3, 4) to the ellipse $x^2+4y^2=1$ are given by the equation $64m^2-96m+65=0$.

Statement 2: The line $y=mx\pm\sqrt{m^2+\frac{1}{4}}$ is a tangent to the ellipse $x^2+4y^2=1$ for all values of m.

Previous Years' Questions

46. The equation of the ellipse whose foci are (±2, 0) and eccentricity 1/2 is

(a) $\frac{x^2}{12}+\frac{y^2}{16}=1$ (b) $\frac{x^2}{16}+\frac{y^2}{12}=1$

(c) $\frac{x^2}{16}+\frac{y^2}{8}=1$ (d) none of these

[AIEEE 2002]

47. The eccentricity of an ellipse, with its centre at the origin is 1/2. If one of the directrices is $x = 4$, then the equation of the ellipse is

(a) $3x^2+4y^2=1$ (b) $3x^2+4y^2=12$

(c) $4x^2+3y^2=12$ (d) $4x^2+3y^2=1$

[AIEEE 2004, EAMCET 2008]

48. The area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

(a) ab (b) $2ab$ (c) a/b (d) $\sqrt{ab}$

[AIEEE 2005]

49. In an ellipse, the distance between its foci is 6 and minor axis is 8. Then its eccentricity is

(a) 1/2 (b) 4/5 (c) $1/\sqrt{5}$ (d) 3/5

[AIEEE 2006]

50. A focus of an ellipse is at the origin. The directrix is the line $x = 4$ and the eccentricity is 1/2, then the length of the semi major axis is

(a) 5/3 (b) 8/3 (c) 2/3 (d) 4/3

[AIEEE 2008]

51. The ellipse $x^2+4y^2=4$ is inscribed in a rectangle aligned with the coordinate axes, which in turn is inscribed in another ellipse that passes through the point (4, 0). Then the equation of the ellipse is

(a) $x^2+12y^2=16$ (b) $4x^2+48y^2=48$

(c) $4x^2+64y^2=48$ (d) $x^2+16y^2=16$

[AIEEE 2009]

52. The length of major axis of the ellipse $9x^2 + 7y^2 = 63$ is

(a) 3 (b) 9 (c) 6 (d) $2\sqrt{7}$

[DCE 2003]

53. How many real tangents can be drawn to the ellipse $5x^2 + 98y^2 = 32$ from the point (2, 3)?

(a) 2 (b) 1 (c) 0 (d) 3

[DCE 2004]

54. In an ellipse, the angle between the lines joining the foci with the positive end of the minor axis is a right angle. The eccentricity of the ellipse is

(a) $1/\sqrt{2}$ (b) $1/\sqrt{3}$ (c) $\sqrt{2}$ (d) $\sqrt{3}$

[DCE 2006]

55. The foci of an ellipse are $(0, \pm 4)$ and the equations for the directrices are $y = \pm 9$. The equation for the ellipse is

(a) $5x^2 + 9y^2 = 4$ (b) $2x^2 - 6y^2 = 28$

(c) $6x^2 + 3y^2 = 45$ (d) $9x^2 + 5y^2 = 180$

[UPSEE 2005, WBJEE 2006]

56. The number of maximum normals which can be drawn from a point to an ellipse is

(a) 4 (b) 2 (c) 1 (d) 3

[UPSEE 2006, WBJEE 2006]

57. If M_1 and M_2 are the feet of the perpendiculars from the foci S_1 and S_2 of the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$ on the tangent at any point P on the ellipse, then $(S_1M_1)(S_2M_2)$ is equal to

(a) 16 (b) 9 (c) 4 (d) 3

[AMU 2004]

58. For the ellipse $24x^2 + 9y^2 - 120x - 90y + 225 = 0$, the eccentricity is equal to

(a) 2/5 (b) 3/5 (c) $\sqrt{15/24}$ (d) 1/5

[AMU 2006]

59. **Statement 1:** An equation of a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$ is $y = 2x + 2\sqrt{3}$.

Statement 2: If the line $y = mx + (4\sqrt{3}/m)$, $(m \neq 0)$ is a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$, then m satisfies $m^4 + 2m^2 = 24$.

(a) Statement 1 is false, statement 2 is true.

(b) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1

(c) Statement 1 is true, statement 2 is true; statement 2 is a not a correct explanation for statement 1

(d) Statement 1 is true, statement 2 is false

60. An ellipse is drawn by taking a diameter of the circle $(x - 1)^2 + y^2 = 1$ as its semi-minor axis and a diameter of the circle $x^2 + (y - 2)^2 = 4$ as its semi-major axis. If the centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is

(a) $4x^2 + y^2 = 4$ (b) $x^2 + 4y^2 = 8$

(c) $4x^2 + y^2 = 8$ (d) $x^2 + 4y^2 = 16$

ANSWERS

Exercise Set

1. (b)	**2.** (c)	**3.** (a)	**4.** (d)	**5.** (a)	**6.** (b)	**7.** (b)	**8.** (c)	**9.** (d)	**10.** (d)
11. (d)	**12.** (c)	**13.** (b)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (b)	**19.** (c)	**20.** (a)
21. (a)	**22.** (c)	**23.** (c)	**24.** (d)	**25.** (c)	**26.** (a)	**27.** (c)	**28.** (d)	**29.** (c)	**30.** (d)
31. (a)	**32.** (d)	**33.** (d)	**34.** (b)	**35.** (c)	**36.** (c)	**37.** (b)	**38.** (b)	**39.** (a)	**40.** (b)
41. (b)	**42.** (c)	**43.** (a)	**44.** (a)	**45.** (a)	**46.** (b)	**47.** (b)	**48.** (b)	**49.** (d)	**50.** (b)
51. (a)	**52.** (c)	**53.** (c)	**54.** (a)	**55.** (d)	**56.** (a)	**57.** (b)	**58.** (c)	**59.** (b)	**60.** (d)

HINTS AND SOLUTIONS—EXERCISE SET

1. The equation of the ellipse is $\frac{x^2}{225} + \frac{y^2}{289} = 1$. We also know that the standard equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Comparing the given equation with the standard equation, we get $a^2 = 225$, $b^2 = 289$.

We also know that the relation for the eccentricity of an ellipse (e), when $b > a$, is

$$e^2 = \frac{b^2 - a^2}{b^2} = \frac{289 - 225}{289} = \frac{64}{289} \Rightarrow e = \frac{8}{17}.$$

2. The major axis of the ellipse $(2a) = 3 \times 2b$ (where $2b$ is the minor axis of the ellipse)

$\Rightarrow a = 3b$.

We know that the relation for the eccentricity of an ellipse (e), when $a > b$, is

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{(9b^2) - b^2}{9b^2} = \frac{8}{9}$$

$$\Rightarrow \quad e = \frac{2\sqrt{2}}{3}.$$

3. The minor axis of the ellipse $(2b) = 2ae$ (where $2ae$ is the distance between foci of the ellipse)

$\Rightarrow b = ae \Rightarrow b^2 = a^2e^2$.

We also know that the relation for the eccentricity of an ellipse (e) is

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{a^2 - a^2e^2}{a^2} = 1 - e^2$$

$$\Rightarrow 2e^2 = 1 \Rightarrow e^2 = \frac{1}{2}$$

$$\Rightarrow e = \frac{1}{\sqrt{2}}$$

4. The equation of the ellipse is $25(x+1)^2 + 9(y+2)^2 = 225$. The given equation may be written as

$$\frac{(x+1)^2}{9} + \frac{(y+2)^2}{25} = 1 \qquad \text{(i)}$$

We know that the standard equation of an ellipse with (h, k) as coordinates of its centre is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1.$$

Comparing Eq. (i) with the standard equation, we get $a^2 = 9, b^2 = 25, h = -1$ and $k = -2$.

We also know that the relation for the eccentricity of an ellipse (e), when $b > a$, is

$$e^2 = \frac{b^2 - a^2}{b^2} = \frac{25 - 9}{25} = \frac{16}{25}$$

$$\Rightarrow e = \frac{4}{5}.$$

We also know that coordinates of the foci of the given ellipse are

$$(h, k \pm be) = \left(-1, -2 \pm 5 \times \frac{4}{5}\right) = (-1, -2 \pm 4)$$

$= (-1, 2)$ and $(-1, -6)$.

5. The equation of the ellipse is $9x^2 + 4y^2 = 1$.

The given equation may be written as

$$\frac{x^2}{1/9} + \frac{y^2}{1/4} = 1 \qquad \text{(i)}$$

We know that the standard equation of an ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Comparing Eq. (i) with the standard equation, we get

$a^2 = \frac{1}{9}$ and $b^2 = \frac{1}{4}$.

We also know that the length of the latus rectum of an ellipse

$$= \frac{2b^2}{a} = \frac{2\left(\frac{1}{4}\right)}{\frac{1}{3}} = \frac{3}{2}.$$

6. The equation of the ellipse is $16x^2 + 25y^2 = 400$ or $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

We know that the standard equation of an ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Comparing the given equation with the standard equation, we get

$$a^2 = 25 \Rightarrow a = 5$$

and $$b^2 = 10 \Rightarrow b = 4$$

We know that the length of the minor axis = $2b$ = $2 \times 4 = 8$.

7. Coordinates of the vertices (A and A') of the ellipse = (2, –2) and (2, 4) and eccentricity (e) = 1/3.

We know that coordinates of the centre of the ellipse (h, k) = mid-point of the line joining the vertices A and

$$A' = \left(\frac{2+2}{2}, \frac{-2+4}{2}\right) = (2, 1).$$

Therefore, distance between vertices A and A' = $2b$

$$= \sqrt{(2-2)^2 + (-2-4)^2} = 6$$

$\Rightarrow b = 3$.

And $a^2 = b^2(1 - e^2) = 9\left(1 - \frac{1}{9}\right) = 8$.

We also know that the equation of an ellipse with (h, k) as coordinates of its centre is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\Rightarrow \frac{(x-2)^2}{8} + \frac{(y-1)^2}{9} = 1.$$

8. Latus rectum of the ellipse $\left(\frac{2b^2}{a}\right) = 8$

$$\Rightarrow b^2 = 4a \qquad \text{(i)}$$

and eccentricity of the ellipse (e) = $\frac{1}{\sqrt{2}}$.

We know that the relation for the eccentricity of an ellipse (e), when $a > b$, is $e^2 = \frac{a^2 - b^2}{a^2}$

$$\Rightarrow \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{a^2-4a}{a^2} \quad \Rightarrow \quad \frac{1}{2} = \frac{a(a-4)}{a^2} = \frac{a-4}{a}$$

$$\Rightarrow \ a = 8 \qquad \Rightarrow \quad a^2 = 64$$

Substituting the value of a in Eq. (i), we get

$$b^2 = 4a = 4\times 8 = 32.$$

We also know that the equation of the ellipse is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \Rightarrow \quad \frac{x^2}{64}+\frac{y^2}{32}=1.$$

9. The equation of the ellipse is $\frac{x^2}{2-r}+\frac{y^2}{r-5}+1=0.$

The given equation may be written as

$$\frac{x^2}{r-2}+\frac{y^2}{5-r}=1 \qquad \text{(i)}$$

We know that the standard equation of an ellipse is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Comparing Eq. (i) with the standard equation, we get $a^2 = r-2$ and $b^2 = 5-r$.

$\Rightarrow \quad r > 2$ and $r < 5 \quad \Rightarrow \quad 2 < r < 5.$

10. Coordinates of point $(x_1, y_1) = (0, 3)$ and equation of the ellipse is $9x^2+5y^2=45.$

$$\Rightarrow \qquad \frac{x^2}{5}+\frac{y^2}{9}=1 \qquad \text{(i)}$$

We know that the standard equation of an ellipse is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Comparing Eq. (i) with the standard equation, we get $a^2 = 5$ and $b^2 = 9$.

We also know that the equation of the normal at the point (x_1, y_1) to the ellipse is

$$\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2 \ \Rightarrow \ \frac{5x}{0}-\frac{9y}{3}=5-9$$

$\Rightarrow x = 0$. Therefore, answer is the y-axis.

11. The equation of the ellipse is $9x^2+16y^2=144.$

$$\Rightarrow \qquad \frac{x^2}{16}+\frac{y^2}{9}=1 \qquad \text{(i)}$$

We know that the standard equation of an ellipse is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1.$$

Comparing Eq. (i) with the standard equation, we get $a^2 = 16$ and $b^2 = 9$.

We also know that the equation of the tangent to the ellipse is

$$y = mx \pm \sqrt{a^2m^2+b^2} = mx \pm \sqrt{16m^2+9} \qquad \text{(ii)}$$

Since the tangents are passing through the point (2, 3), therefore $3 = 2m \pm \sqrt{16m^2+9}$

$$\Rightarrow \qquad 3-2m = \pm\sqrt{16m^2+9}.$$

Squaring both sides, we get $(3-2m)^2 = 16m^2+9$

$\Rightarrow \ 9+4m^2-12m = 16m^2+9$

$\Rightarrow \ -12m^2-12m = 0 \ \Rightarrow \ -12m(m+1) = 0$

$\Rightarrow m = 0$ and $m = -1$.

Substituting the value of m in Eq. (ii), we get $y = 3$ and $y = -x+5 \ \Rightarrow \ x + y = 5.$

12. The equation of the ellipse is $x^2+2y^2=2.$

The given equation may be written as $\frac{x^2}{2}+\frac{y^2}{1}=1.$

We know that the standard equation of an ellipse is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1.$$

Comparing the given equation with the standard equation, we get $a = \sqrt{2}$ and $b = 1$.

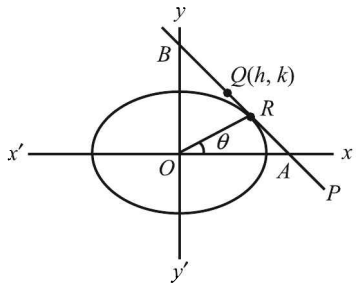


Fig. 21.6

Let R be the point of intercept of the tangent drawn from the point P on the ellipse. Let θ be the angle which the point R makes with the x-axis.

Therefore, coordinates of the point

$$R = (a\cos\theta, b\sin\theta) = (\sqrt{2}\cos\theta, \sin\theta)$$

We also know that the equation of the tangent at R is

$$\frac{x}{\sqrt{2}}\cos\theta + y\sin\theta = 1$$

Thus, coordinates of point $A = (\sqrt{2}\sec\theta, 0)$ and cordintantes of point $B = (0, \text{cosec}\,\theta)$.

Therefore, coordinates of the middle point Q of AB

$$= \left(\frac{\sqrt{2}\sec\theta+0}{2}, \frac{0+\text{cosec}\,\theta}{2}\right).$$

Thus, $h=\dfrac{\sec\theta}{\sqrt{2}} \Rightarrow \cos\theta=\dfrac{1}{h\sqrt{2}}$ (i)

and $k=\dfrac{\operatorname{cosec}\theta}{2} \Rightarrow \sin\theta=\dfrac{1}{2k}$ (ii)

Squaring and adding Eqs. (i) and (ii), we get

$$\frac{1}{2h^2}+\frac{1}{4k^2}=\cos^2\theta+\sin^2\theta=1.$$

Therefore, locus of the middle point of intercept of the tangent drawn from an external point P to the given ellipse is $\dfrac{1}{2x^2}+\dfrac{1}{4y^2}=1.$

13. $\tan\theta_1\tan\theta_1=-\dfrac{a^2}{b^2}$

Let $P(\theta_1)$ and $Q(\theta_2)$ be two points on the ellipse.

Slope of CP and CQ are $\dfrac{b\sin\theta_1}{a\cos\theta_1}$ and $\dfrac{b\sin\theta_2}{a\cos\theta_2}$

i.e. $\dfrac{b}{a}\tan\theta_1$ and $\dfrac{b}{a}\tan\theta_2$

Product of the slopes $=\dfrac{b^2}{a^2}\tan\theta_1\tan\theta_2$

$$=\frac{b^2}{a^2}\left(-\frac{a^2}{b^2}\right)=-1$$

$\therefore$ Chord will subtend a right angle at the centre.

14. The tangent at (x_1, y_1) is $\dfrac{xx_1}{a^2}+\dfrac{yy_1}{b^2}=1$

i.e. $\dfrac{x}{a^2/x_1}+\dfrac{y}{b^2/y_1}=1$

$\therefore \dfrac{a^2}{x_1}=l, \dfrac{b^2}{y_1}=l \quad \therefore x_1=\dfrac{a^2}{l}, y_1=\dfrac{b^2}{l}$

But $\dfrac{x_1^2}{a^2}+\dfrac{y_1^2}{b^2}=1 \Rightarrow \dfrac{a^4}{l^2a^2}+\dfrac{b^4}{b^2l^2}=1$

$\Rightarrow \dfrac{a^2+b^2}{l^2}=1 \Rightarrow l^2=a^2+b^2$

$\Rightarrow l=\sqrt{a^2+b^2}.$

15. Foci are $(\pm ae, 0)$ and vertices are $(\pm a, 0)$.

$\therefore$ $ae = 3$. Since $a = 5$,

$\therefore b^2=a^2(1-e^2)=25\left(1-\dfrac{9}{25}\right)=16$

$\therefore$ Ellipse is $\dfrac{a^2}{25}+\dfrac{y^2}{16}=1.$

16. Any tangent to the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ is

$$y=mx+\sqrt{a^2m^2+b^2}$$

This touches $x^2+y^2=r^2$ if $\dfrac{\sqrt{a^2m^2+b^2}}{\sqrt{1+m^2}}=r$

$\Rightarrow a^2m^2+b^2=r^2(1+m^2)$

$\Rightarrow m^2(a^2-r^2)=r^2-b^2 \Rightarrow m^2=\dfrac{r^2-b^2}{a^2-r^2}$

$\Rightarrow m=\sqrt{\dfrac{r^2-b^2}{a^2-r^2}}.$

17. Let θ be the eccentric angle.

$\therefore$ Point is $(\sqrt{6}\cos\theta, \sqrt{2}\sin\theta)$.

$\therefore (\sqrt{6}\cos\theta-0)^2+(\sqrt{2}\sin\theta-0)^2=2^2$

$\Rightarrow 6(1-\sin^2\theta)+2\sin^2\theta=4$

$\Rightarrow 4\sin^2\theta=2 \Rightarrow \sin^2\theta=\dfrac{1}{2}$

$\Rightarrow \sin\theta=\dfrac{1}{\sqrt{2}} \Rightarrow \theta=\dfrac{\pi}{4}$

18. Any point on the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ is

$$P(a\cos\theta, b\sin\theta).$$

Its foci are $S_1(ae,0)$ and $S_2(-ae,0)$.

$\therefore$ A, area of $\Delta PS_1S_2=\dfrac{1}{2}\begin{vmatrix}1 & a\cos\theta & b\sin\theta\\ 1 & ae & 0\\ 1 & -ae & 0\end{vmatrix}$

$$=\frac{1}{2}\left|-abe\sin\theta+0-0-abe\sin\theta\right|$$

$= abe\sin\theta$, which is max. when $\sin\theta = 1$.

$\therefore$ Max. value of $A = abe$.

19. $\dfrac{x^2}{25}+\dfrac{y^2}{16}=1$

$e^2=1-\dfrac{b^2}{a^2}=1-\dfrac{16}{25} \quad \therefore e=\dfrac{3}{5}$

$\therefore$ Foci are $(\pm ae, 0)$ or $(\pm 3, 0)$.

Hence, the given points F_1 and F_2 are foci of the ellipse.

$\therefore \quad PF_1+PF_2=2a=2\cdot 5=10$

In an ellipse, the sum of the focal distance of points is equal to the length of the major axis.

20. The maximum area corresponds to when P is at either end of the minor axis. Hence, the area for such a position of $P=\dfrac{1}{2}(2a)(b)=ab.$

21. Here $y = x$ and $y = -\frac{2}{3}x$ are conjugate diameters.

$$\therefore \quad mm' = -\frac{b^2}{a^2} \quad \Rightarrow \quad (1)\left(-\frac{2}{3}\right) = -\frac{b^2}{a^2}$$

$$\Rightarrow \quad \frac{2}{3} = \frac{b^2}{a^2} \quad \Rightarrow \quad 2a^2 = 3b^2$$

$$\Rightarrow \quad 2a^2 = 3a^2(1-e^2) \quad \Rightarrow \quad 2 = 3 - 3e^2$$

$$\Rightarrow \quad 3e^2 = 1 \quad \Rightarrow \quad e = \frac{1}{\sqrt{3}} \qquad [\because\ e > 0]$$

22. We know the locus of the point of intersection of two perpendicular tangents drawn on the ellipse is $x^2 + y^2 = a^2 + b^2$, which is called director circle.

The given equation of the ellipse is $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

Here, $a^2 = 9, b^2 = 4$

$\therefore$ Locus is $x^2 + y^2 = a^2 + b^2$

$$\Rightarrow \quad x^2 + y^2 = 9 + 4 \quad \Rightarrow \quad x^2 + y^2 = 13$$

23. Eliminating t, we have $\left(\frac{x}{4}\right)^2 - \left(\frac{y}{3}\right)^2 = t \times \frac{2}{t} = 2$ and this represents a hyperbola.

24. We know that

$$b^2 = a^2(1-e^2),\ a = 3,\ b = 2$$

$$\therefore \quad 4 = 9(1-e^2) \quad \text{or} \quad e = \frac{1}{3}\sqrt{5}$$

Hence, the distance between the pins is

$$2ae = 2 \cdot 3 \cdot \frac{1}{3} \cdot \sqrt{5} = 2\sqrt{5} \text{ cm}$$

Now the length of the endless string is

$$SPS'S = SP + S'P + SS'$$
$$= 2a + 2ae = 6 + 2\sqrt{5}$$

25. Let $S(ae, 0)$ and $S'(-ae, 0)$ be the foci and $B(0, b)$ be the end of the minor axis.

$$\therefore \quad SB \perp S'B$$

$\therefore$ $m_1 m_2 = -1$ where m_1 and m_2 are slopes of SB and $S'B$ respectively.

$$\therefore \quad \frac{b-0}{0-ae} \cdot \frac{b-0}{0+ae} = -1$$

$$\Rightarrow \quad \frac{b^2}{-a^2e^2} = -1 \quad \Rightarrow \quad \frac{b^2}{a^2} = e^2$$

$$\Rightarrow \quad \frac{a^2(1-e^2)}{a^2} = e^2 \quad \Rightarrow \quad 1 - e^2 = e^2$$

$$\Rightarrow \quad 2e^2 = 1 \quad \Rightarrow \quad e^2 = \frac{1}{2}$$

$$\Rightarrow \quad e = \frac{1}{\sqrt{2}}$$

26. The given ellipse is $9x^2 + 4y^2 - 8y - 32 = 0$

$$\Rightarrow \quad 9x^2 + 4(y^2 - 2y) - 32 = 0$$

$$\Rightarrow \quad 9x^2 + 4(y-1)^2 - 36 = 0$$

$$\Rightarrow \quad \frac{x^2}{4} + \frac{(y-1)^2}{9} = 1$$

The equation of the auxiliary circle is

$$x^2 + (y-1)^2 = 9$$

$$\Rightarrow \quad x^2 + y^2 - 2y - 8 = 0$$

27. The equation is $9x^2 + 4y^2 - 6x + 4y + 1 = 0$

$$\Rightarrow \quad 9\left(x^2 - \frac{2}{3}x + \frac{1}{3^2}\right) + 4\left(y^2 + y + \frac{1}{4}\right) + 1 - 1 - 1 = 0$$

$$\Rightarrow \quad \frac{\left(x - \frac{1}{3}\right)^2}{\left(\frac{1}{3}\right)^2} + \frac{\left(y + \frac{1}{2}\right)^2}{\left(\frac{1}{2}\right)^2} = 1 \qquad [\text{here, } a < b]$$

The length of the major axis = $2b = 2\left(\frac{1}{2}\right) = 1$

The length of the minor axis = $2a = 2\left(\frac{1}{3}\right) = \frac{2}{3}$.

28. The equation written in the standard form is

$$\frac{(x-1)^2}{\frac{1}{24}} + \frac{[y + (5/6)]^2}{\frac{1}{36}} = 1$$

which is an ellipse with eccentricity

$$= \left[\frac{(1/24) - (1/36)}{1/24}\right]^{1/2} = \frac{1}{\sqrt{3}}.$$

29. We have $S(ae, 0), T(-ae, 0)$ and $B(0, b)$. Since ΔSTB is an equilateral triangle,

$$\therefore \quad ST^2 = TB^2 \quad \Rightarrow \quad 4a^2e^2 = a^2e^2 + b^2$$

$$\Rightarrow \quad 3a^2e^2 = b^2 = a^2(1-e^2)$$

$$\Rightarrow \quad 3e^2 = 1 - e^2 \quad \Rightarrow \quad 4e^2 = 1$$

$$\Rightarrow \quad e = \frac{1}{2}.$$

30. Clearly the race course is an ellipse with flag posts as its foci. If a and b are the semi-major and minor axes of the ellipse, then $2a = 10$ and $2ae = 8$.

$\Rightarrow a = 5$ and $10e = 8$, i.e. $e = 4/5$.

Since $b^2 = a^2(1-e^2)$,

$$\therefore \quad b^2 = 25\left(1-\frac{16}{25}\right) = 9$$

$\Rightarrow b = 3$

$\therefore$ Area of the ellipse $= \pi ab = \pi \cdot 5 \cdot 3 = 15\pi$.

31. The equation can be written as $4(x-1)^2 + 9(y-2)^2 = 36$, which is an ellipse centered at (1, 2). If CA makes an angle θ with the major axis, then

$$A \equiv [1 + CA\cos\theta, 2 + CA\sin\theta]$$

$$B \equiv \left[1 + CB\cos\left(\frac{\pi}{2}+\theta\right), 2 + CB\sin\left(\frac{\pi}{2}+\theta\right)\right]$$

As A and B are points on the conic

$CA^2(4\cos^2\theta + 9\sin^2\theta) = 36$ and

$CB^2(4\sin^2\theta + 9\cos^2\theta) = 36$

giving $CA^{-2} + CB^{-2} = \dfrac{13}{36}$.

32. For $P(1, 2)$, $E' = +\text{ve}$, $C' = -\text{ve}$

$\therefore$ P lies outside the ellipse and inside the circle.

For $Q(2, 1)$, $E' = -\text{ve}$, $C' = -\text{ve}$

$\therefore$ Q lies inside both the ellipse and the circle.

33. The equation of any curve passing over the points of intersection of the curves $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ and $\dfrac{x^2}{b^2}+\dfrac{y^2}{a^2}=1$ will be

$$\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)+K\left(\frac{x^2}{b^2}+\frac{y^2}{a^2}-1\right)=0$$

where K is constant $\neq 0$.

$$\Rightarrow x^2\left(\frac{1}{a^2}+\frac{K}{b^2}\right)+y^2\left(\frac{1}{b^2}+\frac{K}{a^2}\right)-(K+1)=0 \qquad \text{(i)}$$

(i) is a circle if $\dfrac{1}{a^2}+\dfrac{K}{b^2}=\dfrac{1}{b^2}+\dfrac{K}{a^2}$

$\Rightarrow K = 1$

$\therefore$ The required circle is

$$x^2\left(\frac{1}{a^2}+\frac{1}{b^2}\right)+y^2\left(\frac{1}{b^2}+\frac{1}{a^2}\right)=2$$

$$\Rightarrow \quad x^2+y^2=\frac{2}{\frac{1}{a^2}+\frac{1}{b^2}}=\frac{2a^2b^2}{a^2+b^2}.$$

34. $\dfrac{dy}{dx}=-\dfrac{8x}{18y}=\dfrac{8}{9} \quad \Rightarrow \quad x=-2y.$

Solving, $16y^2+9y^2=1$

$$\therefore \quad y=\pm\frac{1}{5},\ x=\pm\frac{2}{5}$$

$\therefore$ Points are $\left(-\dfrac{2}{5}, \dfrac{1}{5}\right)$.

35. Let $S=\dfrac{x^2}{16}+\dfrac{y^2}{25}-1=0$

At point (7, 6), $S_1 > 0$. So two tangents can be drawn from this point.

36. $\dfrac{(x-4)^2}{25}+\dfrac{y^2}{4}=1$ and $(x-1)^2+\dfrac{y^2}{4}=1$

Slope of $OP \times$ slope of $OQ = \dfrac{2}{-1}\times\dfrac{2}{4}=-1$

$\therefore$ OP and OQ are perpendicular to each other.

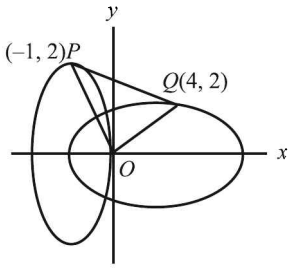


Fig. 21.7

37. Let the equation of the ellipse be $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$. It is given that it passes through (7, 0) and (0, –5).

$\therefore$ $a^2 = 49$ and $b^2 = 25$

Since $b^2 = a^2(1-e^2)$ $\quad\therefore\quad 25 = 49(1-e^2)$

$$\Rightarrow 1-e^2=\frac{25}{49} \quad \Rightarrow \quad e^2=1-\frac{25}{49}=\frac{24}{49}$$

$$\Rightarrow e=\frac{2\sqrt{6}}{7}$$

38. Any point on the given ellipse $\dfrac{x^2}{9}+\dfrac{y^2}{4}=1$ is (3 cos θ, 2 sin θ).

Tangent at P is $\dfrac{x}{3}\cos\theta+\dfrac{y}{2}\sin\theta=1$.

This meets $x = 3$ at $T\left[3, \dfrac{2(1-\cos\theta)}{\sin\theta}\right]$

and $x=-3$ at $T'\left[-3, \dfrac{2(1+\cos\theta)}{\sin\theta}\right]$.

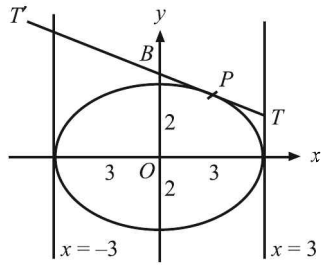


Fig. 21.8

Now the equation of the circle having TT' as diameter is

$$(x-3)(x+3)+\left(y-\frac{2(1-\cos\theta)}{\sin\theta}\right)\left(y-\frac{2(1+\cos\theta)}{\sin\theta}\right)=0$$

$\Rightarrow x^2+y^2-\frac{4}{\sin\theta}y-5=0$, which passes through $(\sqrt{5},0)$.

39. If the ellipse is rotated through an angle 90°, then the length of the major axis becomes $2b$ and the length of the minor axis becomes $2a$. So, the equation remains $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

40. Clearly, eccentric angle of $B = -\frac{5\pi}{6}$.

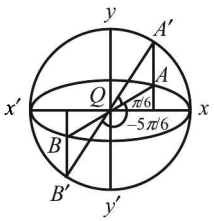


Fig. 21.9

41. The equation of the director circle of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ is $x^2+y^2=25+16$.

$\Rightarrow \quad x^2+y^2=41$

$\because$ The given point (5, 4) lies on the director circle. Therefore, the tangents drawn from this point to the ellipse make an angle 90°.

42. The equation of the ellipse is $\frac{x^2}{4}+\frac{y^2}{4/\lambda}=1$.

Foci will lie on the y-axis, if $\frac{4}{\lambda}>4$.

$\Rightarrow 0<\lambda<1$.

Thus, Statement 1 is true but Statement 2 is false.

43. From correct Statement 2, focal distance = $a - ex_1$.

Now $\frac{x^2}{9/4}+\frac{y^2}{9/5}=1$

$\Rightarrow e^2=\frac{(9/4)-(9/5)}{(9/4)}=\frac{1}{5} \quad \Rightarrow \quad e=\frac{1}{\sqrt{5}}$

Thus, focal distance = $\frac{3}{2}-\frac{1}{\sqrt{5}}$

$\Rightarrow$ Statement 1 is true and Statement 2 is a correct explanation.

44. The Statement 2 is obviously correct. Let us see whether or not it explains Statement 1 (or Statement 1 is wrong). The given equation can be written as

$$A\left(x^2+\frac{B}{A}x\right)+C\left(y^2+\frac{D}{C}\right)=E$$

or $A\left(x^2+\frac{B}{A}x+\frac{B^2}{4A^2}\right)+C\left(y^2+\frac{D}{C}+\frac{D^2}{4C^2}\right)$

$$=E+\frac{B^2}{4A}+\frac{D^2}{4C}$$

or $A\left(x+\frac{B}{2A}\right)^2+C\left(y+\frac{D}{2C}\right)^2=E+\frac{B^2}{4A}+\frac{D^2}{4C}$

Since A, $C > 0$, this represents an ellipse, if $E+\frac{B^2}{4A}+\frac{D^2}{4C}<0$

45. The ellipse is $x^2+\frac{y^2}{1/4}=1$.

$\Rightarrow$ Any tangent is

$$y=mx\pm\sqrt{a^2m^2+b^2}=mx\pm\sqrt{m^2+\frac{1}{4}}$$

If this goes through (3, 4), we have

$$4=3m\pm\sqrt{m^2+\frac{1}{4}}$$

or $(4-3m)^2=m^2+\frac{1}{4}$ or $8m^2-24m+\frac{63}{4}=0$

$\Rightarrow$ Both statements are correct and Statement 2 is a correct explanation.

46. Let the equation of the ellipse be

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \qquad \text{(i)}$$

By the question, $ae = 2$ and $e=\frac{1}{2}$.

$\therefore \quad a\left(\frac{1}{2}\right)=2 \quad \Rightarrow \quad a=4$

Now $b^2=a^2(1-e^2) \quad \Rightarrow b^2=16\left(1-\frac{1}{4}\right)=12$

$\therefore$ (i) becomes $\frac{x^2}{16}+\frac{y^2}{12}=1$

47. Eccentricity of an ellipse (e) = (1/2) with its centre at origin and equation of directrix $x = 4$. We know that the equation of the directrix of an ellipse is

$$x = \frac{a}{e} \qquad \Rightarrow \quad \frac{a}{e} = 4$$

$$\Rightarrow \frac{a}{4} = e \qquad \Rightarrow \quad \frac{a}{4} = \frac{1}{2}$$

$$\Rightarrow a = 2.$$

We also know that the equation for the eccentricity of an ellipse (e) when $a > b$ is $e^2 = \dfrac{a^2 - b^2}{a^2}$.

$$\Rightarrow \left(\frac{1}{2}\right)^2 = \frac{2^2 - b^2}{2^2} \qquad \Rightarrow \quad \frac{1}{4} = \frac{4 - b^2}{4}$$

$$\Rightarrow 1 = 4 - b^2 \qquad \Rightarrow \quad b^2 = 3$$

We also know that the equation of the ellipse with (0, 0) as coordinates of its centre is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \Rightarrow \quad \frac{x^2}{4} + \frac{y^2}{3} = 1$$

$$\Rightarrow 3x^2 + 4y^2 = 12.$$

48. Let $PQRS$ be a rectangle, where P is $(a\cos\theta, b\sin\theta)$.

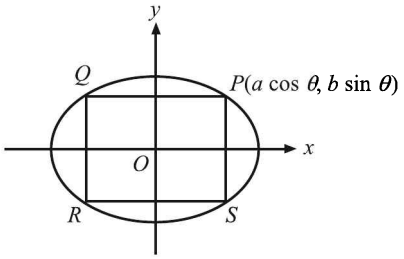


Fig. 21.10

$\therefore$ Area of the rectangle $= 4(a\cos\theta)(b\sin\theta)$

$$= 2ab\sin 2\theta$$

This is max. when $\sin 2\theta = 1$.

Hence, max. area $= 2ab(1) = 2ab$.

49. Here $2ae = 6$ and $2b = 4$ for $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

$$\Rightarrow e^2 = \frac{9}{16} \text{ as } 1 - e^2 = \frac{b^2}{a^2}$$

$$\Rightarrow e = \frac{3}{5}$$

50. Since $\dfrac{a}{e} - ae = 4$ and $e = \dfrac{1}{2}$,

$$\therefore \qquad 2a - \frac{a}{2} = 4$$

$$\Rightarrow \frac{3a}{2} = 4 \qquad \Rightarrow \quad a = \frac{8}{3}.$$

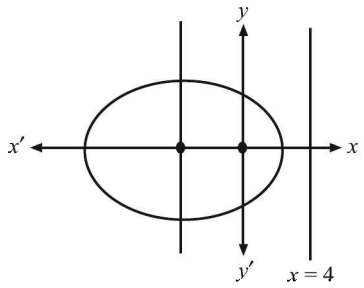


Fig. 21.11

51. Let the equation of the required ellipse be $\dfrac{x^2}{16} + \dfrac{y^2}{b^2} = 1$.

But the ellipse passes through the point (2, 1).

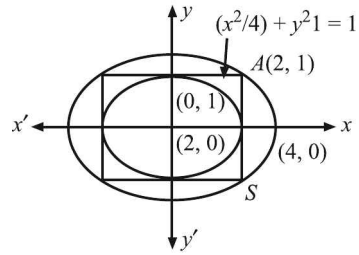


Fig. 21.12

$$\Rightarrow \frac{1}{4} + \frac{1}{b^2} = 1 \qquad \Rightarrow \quad \frac{1}{b^2} = \frac{3}{4}$$

$$\Rightarrow b^2 = \frac{4}{3}$$

Hence, the equation of the ellipse is $\dfrac{x^2}{16} + \dfrac{3y^2}{4} = 1$.

$$\Rightarrow \qquad x^2 + 12y^2 = 16$$

52. $\dfrac{x^2}{63/9} + \dfrac{y^2}{63/7} = 1$

$$\Rightarrow \qquad \frac{x^2}{(\sqrt{7})^2} + \frac{y^2}{(3)^2} = 1$$

$\therefore$ The length of the major axis $= 2 \times 3 = 6$.

53. Since (2, 3) lies inside the ellipse $5x^2 + 9y^2 = 32$, no tangent can be drawn.

54. Using the product of slopes of FP and $F'P$ as -1 and $b^2 = a^2(1 - e^2)$,

we get $e = \dfrac{1}{\sqrt{2}}$.

55. Here focus $be = 4$ and directrix $b/e = 4$

$\Rightarrow e = 2/3$ and $b^2 = 36$

$a^2 = 20$

$\therefore \quad \dfrac{x^2}{20} + \dfrac{y^2}{36} = 1$, i.e. $9x^2 + 5y^2 = 180$.

56. Let $ax\sec\theta - by\,\text{cosec}\,\theta = a^2 - b^2$ be any normal. If it passes through (h, k), $ah\sec\theta - bk\,\text{cosec}\,\theta = a^2 - b^2$.

$$\Rightarrow \frac{ah}{\cos\theta} - \frac{bk}{\sin\theta} = a^2 - b^2$$

$$\Rightarrow ah\cdot\frac{1+\tan^2(\theta/2)}{1-\tan^2(\theta/2)} - bk\cdot\frac{1+\tan^2(\theta/2)}{2\tan(\theta/2)} = a^2 - b^2$$

Since it is a fourth degree equation in $\tan(\theta/2)$, there may be at most four roots for $\tan(\theta/2)$.

$\Rightarrow$ There may be up to four normals to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ through (h, k).

57. We know the product of the perpendicular from the foci upon any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is b^2 always, when $a > b$ and a^2 always, when $a < b$.

Hence, the required answer is 9.

58. The given equation can be written as

$$24\left[x^2 - 2x\cdot\frac{5}{2} + \left(\frac{5}{2}\right)^2\right] + 9\left[y^2 - 10y + 25\right]$$

$$= 24\times\frac{25}{4}$$

$$\Rightarrow \frac{[x-(5/2)]^2}{(6\times25)/24} + \frac{(y-5)^2}{(6\times25)/9} = 1$$

$$\therefore \text{ Eccentricity} = \sqrt{1-\frac{(6\times25)/24}{(6\times25)/9}} = \sqrt{1-\frac{9}{24}}$$

$$= \sqrt{\frac{15}{24}}.$$

59. Equation of tangent to the ellipse $\frac{x^2}{2}+\frac{y^2}{4}=1$ is given by

$$y = mx \pm \sqrt{2m^2+4} \qquad \text{(i)}$$

Equation of tangent to the parabola $y^2 = 16\sqrt{3}x$ is given by

$$y = mx + \frac{4\sqrt{3}}{m} \qquad \text{(ii)}$$

On comparing Eqs. (i) and (ii), we get

$$\cancel{mx} \pm \sqrt{2m^2+4} = \cancel{mx} \pm \frac{4\sqrt{3}}{m}$$

$$\Rightarrow \frac{4\sqrt{3}}{m} = \pm\sqrt{2m^2+4}$$

$$\Rightarrow \frac{48}{m^2} = 2m^2+4 \quad \Rightarrow \quad 2m^4+4m^2-48=0$$

$$\Rightarrow m^4+2m^2-24=0 \quad \Rightarrow \quad (m^2+6)(m^2-4)=0$$

$$\Rightarrow m^2 = 4 \quad \Rightarrow \quad m = \pm2$$

$\Rightarrow$ Equations of the common tangent are $y = \pm2x + 2\sqrt{3}$

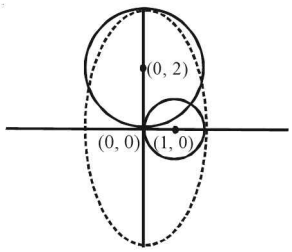
Statement 1 is true.

Statement 2 is obviously true.

60. Length of the semi-minor axis is 2, and length of the semi-major axis is 4, then the equation of the ellipse is

$$\frac{x^2}{16}+\frac{y^2}{4}=1$$

$$x^2 + 4y^2 = 16$$



MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. If the chord, joining two points whose eccentric angles are α and β, cuts the major axis of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at a distance d from the centre, then $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}$ is equal to

(a) 1 (b) d

(c) $\frac{d+a}{d-a}$ (d) none of these

2. If the normal at the point $P(\theta)$ to the ellipse $\frac{x^2}{14}+\frac{y^2}{5}=1$ intersects it again at the point $Q(2\theta)$, then $\cos\theta$ is equal to

(a) 2/3 (b) –(2/3)

(c) 3/4 (d) none of these

3. The length of the major axis of the ellipse $(5x-10)^2 + (5y+15)^2 = \frac{(3x-4y+7)^2}{4}$ is

(a) 10 (b) 20/3

(c) 20/7 (d) 4

4. The tangent and normal to the ellipse $x^2+4y^2=4$ at a point $P(\theta)$ on it meets the major axes at Q and R respectively. If $QR = 2$, then $\cos\theta$ is equal to

(a) –(2/3) (b) 2/5

(c) 1/3 (d) none of these

5. If $P \equiv (x, y)$, $F_1(3, 0)$, $F_2(-3, 0)$ and $16x^2 + 25y^2 = 400$, then $PF_1 + PF_2$ equals
(a) 8 (b) 10
(c) 6 (d) 12

6. A point moves in a plane so that its distances PA and PB from two fixed points A and B in the plane satisfy the relation $PA + PB = k$ ($k \neq 0$). Then the locus of P is
(a) a parabola (b) an ellipse
(c) a hyperbola (d) a branch of a hyperbola

7. Tangents are drawn from the points on the line $x - y - 5 = 0$ to $x^2 + 4y^2 = 4$. Then all the chords of contact pass through a fixed point, whose coordinates are
(a) $\left(\frac{4}{5}, -\frac{1}{5}\right)$ (b) $\left(\frac{1}{5}, -\frac{4}{5}\right)$
(c) $\left(\frac{4}{5}, \frac{1}{5}\right)$ (d) $\left(-\frac{4}{5}, \frac{1}{5}\right)$

8. If $(\sqrt{3})bx + ay = 2ab$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then the eccentric angle θ of the point of contact is
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

9. An ellipse has XY as its semi-minor axis. F_1 and F_2 are its foci and angle F_1YF_2 is a right angle. The eccentricity of the ellipse is
(a) $1/\sqrt{2}$ (b) 1/2
(c) $1/\sqrt{3}$ (d) $2/\sqrt{3}$

10. If the focal distance of an end of the minor axis of the ellipse (referred to its axes as the axes of x and y respectively) is k and the distance between its foci is $2h$, then its equation is
(a) $\frac{x^2}{k^2} + \frac{y^2}{h^2} = 1$ (b) $\frac{x^2}{k^2} + \frac{y^2}{k^2 - h^2} = 1$
(c) $\frac{x^2}{k^2} + \frac{y^2}{h^2 - k^2} = 1$ (d) $\frac{x^2}{k^2} + \frac{y^2}{k^2 + h^2} = 1$

11. A line segment of length $a + b$ moves in such a way that its ends are always on two fixed perpendicular straight lines. The locus of the point on this line which divides it into portions of lengths a and b is
(a) a parabola (b) a circle
(c) an ellipse (d) none of these

12. If α and β are the eccentric angles of the extremities of a focal chord of an ellipse, then the eccentricity of the ellipse is
(a) $\frac{\cos\alpha + \cos\beta}{\cos(\alpha - \beta)}$ (b) $\frac{\sin\alpha - \sin\beta}{\sin(\alpha - \beta)}$
(c) $\frac{\cos\alpha - \cos\beta}{\cos(\alpha - \beta)}$ (d) $\frac{\sin\alpha - \sin\beta}{\sin(\alpha - \beta)}$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose center is C, meets the major and the minor axes at P and Q respectively, then $\frac{a^2}{CP^2} + \frac{b^2}{CQ^2}$ is equal to
(a) 1 (b) –1
(c) $a + b$ (d) none of these

2. If the tangent to the ellipse $x^2 + 4y^2 = 16$ at the point $P'\phi'$ is a normal to the circle $x^2 + y^2 - 8x - 4y = 0$, then ϕ is equal to
(a) $\pi/2$ (b) $\pi/4$ (c) 1 (d) $-(\pi/4)$

3. The area of the rectangle formed by the perpendiculars from the centre of the ellipse to the tangent and normal at the point, whose eccentric angle is $\pi/4$, is
(a) $\left(\frac{a^2 - b^2}{a^2 + b^2}\right)ab$ (b) $\left(\frac{a^2 + b^2}{a^2 - b^2}\right)ab$
(c) $\frac{1}{ab}\left(\frac{a^2 - b^2}{a^2 + b^2}\right)ab$ (d) $\frac{1}{ab}\left(\frac{a^2 + b^2}{a^2 - b^2}\right)ab$

4. An ellipse having foci (3, 1) and (1, 1) passes through the point (1, 3) has the eccentricity
(a) $\sqrt{2} - 1$ (b) $\sqrt{3} - 1$
(c) $\frac{\sqrt{2} - 1}{2}$ (d) $\frac{\sqrt{3} - 1}{2}$

5. The equation $\frac{x^2}{10 - a} + \frac{y^2}{4 - a} = 1$ represents an ellipse if
(a) $a < 4$ (b) $a > 4$
(c) $4 < a < 10$ (d) $a > 10$

6. The equations of the tangents to the ellipse $4x^2 + 3y^2 = 5$, which are inclined at 60° to the axis of x, are
(a) $y = \sqrt{3}x \pm \sqrt{\frac{65}{12}}$ (b) $y = \sqrt{3}x \pm \sqrt{\frac{12}{65}}$
(c) $y = \frac{x}{\sqrt{3}} \pm \sqrt{\frac{65}{12}}$ (d) none of these

7. The line $lx+my+n=0$ is a normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ if

(a) $\frac{a^2}{m^2}+\frac{b^2}{l^2}=\frac{(a^2-b^2)^2}{n^2}$ (b) $\frac{a^2}{l^2}+\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$

(c) $\frac{a^2}{l^2}-\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$ (d) none of these

8. The equation of a diameter conjugate to a diameter $y=-\frac{b}{a}x$ of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

(a) $y=-\frac{b}{a}x$ (b) $y=-\frac{a}{b}x$

(c) $y=\frac{a}{b}x$ (d) none of these

9. If the foci are points (0, 1) and (0, –1) and the minor axis is of length 1, then the equation of the ellipse is

(a) $\frac{x^2}{1/4}+\frac{y^2}{5/4}=1$ (b) $\frac{x^2}{5/4}+\frac{y^2}{1/4}=1$

(c) $\frac{x^2}{3/4}+\frac{y^2}{1/4}=1$ (d) $\frac{x^2}{1/4}+\frac{y^2}{3/4}=1$

10. If α and β are eccentric angles of the ends of a focal chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$ is equal to

(a) $\frac{1-e}{1+e}$ (b) $\frac{e-1}{e+1}$

(c) $\frac{e+1}{e-1}$ (d) none of these

11. The equation $\sqrt{(x-3)^2+(y-1^2)}+\sqrt{(x+3)^2+(y-1)^2}$ = 6 represents

(a) an ellipse (b) a circle

(c) a pair of lines (d) a line

ANSWERS

Level 1

1. (c) **2.** (b) **3.** (b) **4.** (a) **5.** (b) **6.** (b) **7.** (a) **8.** (a) **9.** (a) **10.** (b)
11. (c) **12.** (d)

Level 2

1. (a) **2.** (a) **3.** (a) **4.** (a) **5.** (a) **6.** (a) **7.** (b) **8.** (a) **9.** (a) **10.** (b)
11. (d)

22

APOLLONIUS

Hyperbola

- Definition of Hyperbola
- Standard Hyperbola
- Important Properties
- Equation of Tangent
- Equation of Tangent in Different Forms
- Equation of Normal
- Auxiliary Circle
- Parametric Equations
- Equation of Chord of Contact
- Equation of a Pair of Tangents
- Equation of Chord of the Hyperbola Whose Mid-point is Given
- Pole and Polar
- Diameter
- Asymptotes of Hyperbola
- Conjugate Hyperbola
- Rectangular Hyperbola

DEFINITION OF HYPERBOLA

A hyperbola is the locus of a point which moves so that its distance from a fixed point called focus is in a constant ratio to its distance from a fixed straight line called directrix. This ratio is called *eccentricity* and is denoted by e. For a hyperbola, **eccentricity is greater than unity**.

STANDARD HYPERBOLA

The general form of the standard hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where a and b are constants.

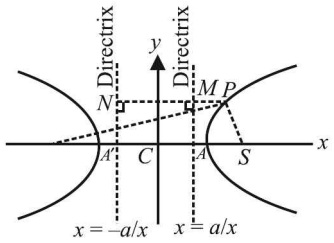
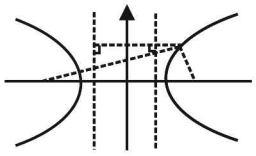
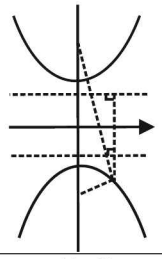


Fig. 22.1

IMPORTANT PROPERTIES

1. The centre of the hyperbola is $C \equiv (0, 0)$, vertices $A \equiv (a, 0)$ and $A \equiv (-a, 0)$.
2. AA' is known as the transverse axis.

3. $BB' = 2b$ is known as the conjugate axis.
4. Distance of P from foci is $S'P - SP = 2a$.
5. $SP = e(PM) \Rightarrow b^2 = a^2(e^2 - 1) \Rightarrow e^2 = 1 + \frac{b^2}{a^2}$
6. The chord LL' through the focus and perpendicular to the major axis is known as latus rectum and $LL' = \frac{2b^2}{a}$.

Standard equation	Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	Conjugate hyperbola $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 (a, b)$ or $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$
Shape of the hyperbola		
Centre	(0, 0)	(0, 0)
Equation of transverse axis	$y = 0$	$x = 0$
Equation of conjugate axis	$x = 0$	$y = 0$
Length of transverse axis	$2a$	$2b$
Length of conjugate axis	$2b$	$2a$
Foci	$(\pm ae, 0)$	$(0, \pm ae)$
Vertex	$(\pm a, 0)$	$(0, \pm b)$
Equation of directrix	$x = \pm\frac{a}{e}$	$y = \pm\frac{a}{e}$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{b}$
Eccentricity (e)	$e = \sqrt{\frac{a^2 + b^2}{a^2}}$	$e = \sqrt{\frac{a^2 + b^2}{b^2}}$
Parametric coordinates	$(a \sec\theta, b \tan\theta)$	$(a \sec\theta, b \tan\theta)$
Focal radii	$SP = ex_1 - a$ and $S'P = ex_1 + a$	$SP = ey_1 - b$ and $S'P = ey_1 + b$
Difference of focal radii ($S'P - SP$)	$2a$	$2b$
Tangents at the vertices	$x = \pm a$	$y = \pm b$

EQUATION OF TANGENT

1. **Point form:** The equation of a tangent drawn to the hyperbola at (x_1, y_2) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$.
2. **Parametric form:** The equation of a tangent drawn to the hyperbola at the point (a sec θ, b tan θ) is
$$\frac{x}{a}\sec\theta + \frac{y}{b}\tan\theta = 1.$$
3. **Slope form:** The equation of a tangent of slope m can be taken as $y = mx + \sqrt{a^2m^2 + b^2}$, where the point of contact is
$$\left[\frac{-a^2m}{\sqrt{a^2m^2 - b^2}}, \frac{-b^2}{\sqrt{a^2m^2 + b^2}}\right].$$

EQUATION OF A TANGENT IN DIFFERENT FORMS

1. The equations of **two parallel tangents** of slope m that can be drawn to the hyperbola is $y = mx \pm \sqrt{a^2m^2 - b^2}$.
2. The condition for $y = mx + c$ to be a tangent to the hyperbola is $c^2 = a^2m^2 + b^2$ and the point of contact is $\left(-\frac{a^2m}{c}, \frac{-b^2}{c}\right)$.
3. Tangents drawn to ellipse from a point (h, k): Let $y = mx + \sqrt{a^2m^2 - b^2}$ be the tangent(s).

   $\Rightarrow k = mh + \sqrt{a^2m^2 - b^2}$

   $\Rightarrow m^2(h^2 - a^2) - 2hk + k^2 + b^2 = 0$

   The solution of this quadratic equation will give the slopes m_1 and m_2 of the two tangents.

EQUATION OF NORMAL

1. **Point form:** The equation of a normal drawn to the hyperbola at (x_1, y_1) is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$.
2. **Parametric form:** The equation of a normal drawn to the hyperbola at $(a\sec\theta, b\tan\theta)$ is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$.
3. **Slope form:** The equation of a normal of slope m can be taken as $y = mx \mp \frac{m(a^2 + b^2)}{\sqrt{(a^2 - m^2b^2)}}$.

AUXILIARY CIRCLE

The circle drawn on the transverse axis AA' as diameter is known as the **auxiliary circle**. The equation of the auxiliary circle is $x^2 + y^2 = a^2$. Let P be a point on the hyperbola.

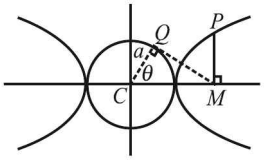


Fig. 22.2

Let PM be perpendicular to the x-axis. The tangent from M to the auxiliary circle touches at Q. P and Q are called corresponding points on the hyperbola and auxiliary circle respectively. The angle θ is known as **eccentric angle** of the point P on the hyperbola.

PARAMETRIC EQUATIONS

Let CQ make an angle θ with the x-axis. Then,

$$\Rightarrow \qquad CM = \frac{a}{\cos\theta} = a\sec\theta \quad \Rightarrow \quad x_p = a\sec\theta$$

Putting $x = a\sec\theta$ in the equation of the hyperbola, we get $y_p = b\tan\theta$

$$\Rightarrow \qquad x = a\sec\theta \text{ and } y = b\tan\theta$$

Hence, $x = a\sec\theta$ and $y = b\tan\theta$ are the parametric equations of the hyperbola, where the parameter $\theta \in [0, 2\pi)$.

EQUATION OF CHORD OF CONTACT

Let us consider a point R whose coordinates are (x_1, y_1) and which lies outside the hyperbola. From point R, two tangents are drawn to the hyperbola which meet the hyperbola at points P and Q. The line joining P and Q is known as *chord of contact* of the hyperbola and its equation is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$$

Note The above result is the same as the equation of the tangent drawn to the hyperbola at (x_1, y_1).

EQUATION OF A PAIR OF TANGENTS

Let us consider a point R whose coordinates are (x_1, y_1) and which lies outside the hyperbola. From point R, two tangents are drawn to the hyperbola which meet the hyperbola at points P and Q. The equation of the pair of the tangents RP and RQ is $SS_1 = T^2$,

where $S = S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$ and $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

EQUATION OF CHORD OF THE HYPERBOLA WHOSE MID-POINT IS GIVEN

Let us consider a chord PQ of the hyperbola whose mid-point is $R \equiv (x_1, y_1)$. The equation of this chord PQ in terms of coordinates of R is $S_1 = T$,

where $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$ and $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

POLE AND POLAR

Let P be any point inside or outside the hyperbola. Suppose any straight line drawn through P intersects the hyperbola at A and B. Then the locus of the point of intersection of the tangents to the hyperbola at A and B is called the **polar** of the given point P with respect to the hyperbola and the point P is called the **pole** of the polar.

EQUATION OF POLAR

The equation of the polar of the point (x_1, y_1) with respect to the hyperbola is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$.

DIAMETER

The locus of the middle points of a system of parallel chords of a hyperbola is called a **diameter**. It can be easily derived that the locus is a straight line which passes through the centre of the hyperbola.

EQUATION OF DIAMETER

Let (x_1, y_1) be the middle point of the chord $y = mx + c$ of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Then $T = S_1 \quad \Rightarrow \quad \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$

$\Rightarrow$ Slope $= \frac{x_1 b^2}{a^2 y_1} = m \quad \Rightarrow \quad y = \frac{b^2 x}{a^2 m}$

Hence, the locus of the mid-point of $y = \frac{b^2 x}{a^2 m}$. The diameter can also be defined as a chord which passes through the centre of the hyperbola.

ASYMPTOTES OF HYPERBOLA

The lines $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$, i.e. $y = \pm \frac{bx}{a}$ are called the **asymptotes** of the hyperbola. The curve comes close to these lines as $x \to \infty$ or $x \to -\infty$ but never meets them.

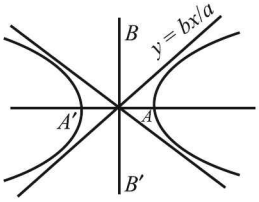


Fig. 22.4

1. Angle between asymptotes
$$= \tan^{-1} \frac{b}{a}$$
2. Asymptotes are the diagonals of the rectangle passing through A, B, A', B' with sides parallel to the axes.

CONJUGATE HYPERBOLA

Two hyperbolas such that the transverse and conjugate axes of one hyperbola are respectively the conjugate and transverse axes of the other are called **conjugate hyperbolas** of each other.

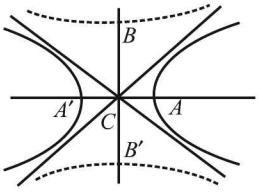


Fig. 22.5

Thus, the hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ are conjugate to each other. If e and e' are their eccentricities, then

$$b^2 = a^2(e^2 - 1) \qquad \text{(i)}$$

and
$$a^2 = b^2[(e')^2 - 1] \qquad \text{(ii)}$$

From (i) and (ii), we get $1/e^2 + 1/e'^2 = 1$.

RECTANGULAR HYPERBOLA

General Form

If asymptotes of the standard hyperbola are perpendicular to each other, then it is known as rectangular hyperbola. Thus,

$$2\tan^{-1}\frac{b}{a} = \frac{\pi}{2} \quad \Rightarrow \quad b = a$$

$\Rightarrow x^2 - y^2 = a^2$ is the general form of the equation of the rectangular hyperbola.

Properties of Rectangular Hyperbola

1. Eccentricity $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{2}$.
2. As the transverse and conjugate axes are equal, we also call it an equilateral hyperbola.

$xy = c^2$ HYPERBOLA

General Form

If we take the coordinate axes along asymptotes of a rectangular hyperbola, then its equation becomes $xy = c^2$. In parametric form, the equation is $x = ct$, $y = \frac{c}{t}$.

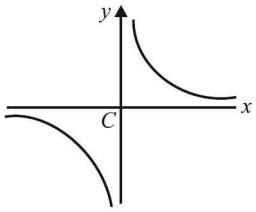


Fig. 22.6

Properties of $xy = c^2$ Hyperbola

1. The equation of the chord through $P(t_1)$ and $Q(t_2)$ is $x + y\,t_1 t_2 = c(t_1 + t_2)$.
2. The equation of the tangent at $P(t)$ is $x + yt^2 = 2ct$.
3. The equation of the normal at $P(t)$ is $xt^3 - yt - ct^4 + c = 0$.
4. The portion of the tangent between coordinate axes is bisected at the point of contact.
5. At (x_1, y_1), the tangent is $\frac{x}{x_1} + \frac{y}{y_1} = 2$.
6. At (x_1, y_1), the normal is $xx_1 - yy_1 = x_1^2 - y_1^2$.

SOLVED PROBLEMS

1. The eccentricity of the hyperbola $\sqrt{\frac{1999}{3}}(x^2 - y^2) = 1$ is

(a) $\sqrt{3}$ (b) $\sqrt{2}$

(c) 2 (d) $2\sqrt{2}$

Ans. (b)

Solution: Equation of the hyperbola is $\sqrt{\frac{1999}{3}}(x^2 - y^2) = 1$.

The given equation may be written as

$$\frac{x^2}{\left(\sqrt{\frac{3}{1999}}\right)^2} - \frac{y^2}{\left(\sqrt{\frac{3}{1999}}\right)^2} = 1 \qquad \text{(i)}$$

We know that the standard equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Comparing Eq. (i) with the standard equation, we get

$$a^2 = \left(\sqrt{\frac{3}{1999}}\right)^2 \text{ and } b^2 = \left(\sqrt{\frac{3}{1999}}\right)^2.$$

We also know that the relation for the eccentricity (e) of the hyperbola is

$$e^2 = \frac{a^2 + b^2}{a^2} = 1 + \frac{b^2}{a^2} = 1 + \frac{\sqrt{\frac{3}{1999}}}{\sqrt{\frac{3}{1999}}} = 1 + 1 = 2$$

or $e = \sqrt{2}$.

2. The length of the transverse axis of the hyperbola $3x^2 - 4y^2 = 32$ is

(a) $\frac{8\sqrt{2}}{\sqrt{3}}$ (b) $\frac{16\sqrt{2}}{\sqrt{3}}$

(c) $\frac{3}{32}$ (d) $\frac{64}{3}$

Ans. (a)

Solution: Equation of the hyperbola is $3x^2 - 4y^2 = 32$.

The given equation may be written as $\frac{x^2}{\frac{32}{3}} - \frac{y^2}{8} = 1$

$$\Rightarrow \quad \frac{x^2}{\left(\frac{4\sqrt{2}}{\sqrt{3}}\right)^2} - \frac{y^2}{\left(2\sqrt{2}\right)} = 1$$

Comparing the given equation with the standard equation, we get

$$a^2 = \left(\frac{4\sqrt{2}}{\sqrt{3}}\right)^2 \quad \text{or} \quad a = \frac{4\sqrt{2}}{\sqrt{3}}.$$

We also know that the length of the transverse axis of the hyperbola $= 2a = 2 \times \frac{4\sqrt{2}}{\sqrt{3}} = \frac{8\sqrt{2}}{\sqrt{3}}$.

3. The latus rectum of the hyperbola $16x^2 - 9y^2 = 144$ is

(a) 16/3 (b) 32/3

(c) 8/3 (d) 4/3

Ans. (b)

Solution: Equation of the hyperbola is $16x^2 - 9y^2 = 144$.

The given equation may be written as

$$\frac{x^2}{9} - \frac{y^2}{16} = 1 \qquad \text{(i)}$$

We know that the standard equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Comparing Eq. (i) with the standard equation, we get

$$a^2 = 9 \quad \text{or} \quad a = 3 \text{ and } b^2 = 16$$

We also know that the length of the latus rectum of the hyperbola

$$= \frac{2b^2}{a} = \frac{2 \times 16}{3} = \frac{32}{3}.$$

4. The latus rectum of the hyperbola $9x^2 - 16y^2 - 18x - 32y - 151 = 0$ is

(a) 9 (b) 3/2

(c) 9/4 (d) 9/2

Ans. (d)

Solution: Equation of the hyperbola is

$$9x^2 - 16y^2 - 18x - 32y - 151 = 0.$$

The given equation may be written as

$$9(x^2 - 2x + 1) - 16(y^2 + 2y + 1) = 151 + 9 - 16$$

$$\Rightarrow \quad 9(x-1)^2 - 16(y-1)^2 = 144$$

$$\Rightarrow \quad \frac{(x-1)^2}{16} - \frac{(y+1)^2}{9} = 1 \qquad \text{(i)}$$

We know that the standard equation of the hyperbola with (h, k) as coordinates of its centre is

$$\frac{(x-h)^2}{a^2} - \frac{(y+k)^2}{b^2} = 1.$$

Comparing Eq. (i) with the standard equation, we get

$$a^2 = 16 \quad \Rightarrow \quad a = 4 \text{ and } b^2 = 9.$$

We also know that the latus rectum of the hyperbola

$$= \frac{2b^2}{a} = \frac{2\times 9}{4} = \frac{9}{2}.$$

5. The equation of the hyperbola whose latus rectum is 8 and eccentricity is $\frac{3}{\sqrt{5}}$ is

(a) $x^2 - y^2 = 1$ (b) $\frac{x^2}{25} - \frac{y^2}{30} = 1$

(c) $\frac{x^2}{25} - \frac{y^2}{20} = 1$ (d) $\frac{x^2}{5} - \frac{y^2}{35} = 1$

Ans. (c)

Solution: Length of the latus rectum = 8 and eccentricity $(e) = \frac{3}{\sqrt{5}}$.

We know that the length of the latus rectum $\left(\frac{2b^2}{a}\right) = 8$

$$\Rightarrow \quad b^2 = 4a \qquad \text{(i)}$$

We also know that the relation for major axis (b) is

$$b^2 = a^2(e^2 - 1) \quad \Rightarrow \quad 4a = a^2(e^2 - 1)$$

$$\Rightarrow \quad a\left(\frac{9}{5} - 1\right) = 4 \quad \Rightarrow \quad a = 5.$$

Substituting the value of a in Eq. (i), we get

$$b^2 = 4\times 5 = 20.$$

We also know that the equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \Rightarrow \quad \frac{x^2}{25} - \frac{y^2}{20} = 1.$$

6. The equation of the hyperbola whose eccentricity is $\frac{\sqrt{5}}{2}$ and length of its minor axis is 1 is

(a) $x^2 - 4y^2 = 1$ (b) $4x^2 - 4y^2 = 1$

(c) $x^2 - y^2 = 1$ (d) $16x^2 - y^2 = 1$

Ans. (a)

Solution: Eccentricity of hyperbola $(e) = \frac{\sqrt{5}}{2}$ and length of its minor axis (a) = 1.

We know that the relation for the eccentricity (e) of the hyperbola is $e^2 = \frac{a^2 + b^2}{a^2}$

$$\Rightarrow \quad \frac{5}{4} = \frac{1 + b^2}{1} \quad \Rightarrow \quad \frac{5}{4} = 1 + b^2$$

$$\Rightarrow \quad \frac{5}{4} - 1 = b^2 \quad \Rightarrow \quad b^2 = \frac{1}{4}.$$

We also know that the equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \Rightarrow \quad \frac{x^2}{1} - \frac{y^2}{\left(\frac{1}{4}\right)} = 1$$

$$\Rightarrow \quad x^2 - 4y^2 = 1.$$

7. If the line $y = mx + 6$ is tangent to the hyperbola $\frac{x^2}{100} - \frac{y^2}{49} = 1$, then the value of m is

(a) $\sqrt{\frac{20}{17}}$ (b) $\sqrt{\frac{17}{20}}$

(c) $\sqrt{\frac{109}{49}}$ (d) $\sqrt{\frac{49}{100}}$

Ans. (b)

Solution: Equation of the line is $y = mx + 6$ and equation of the hyperbola is $\frac{x^2}{100} - \frac{y^2}{49} = 1$.

We know that the standard equation of the line is

$$y = mx + c.$$

Comparing the given equation of the line with the stand- and equation, we get $c = 6$.

We also know that the standard equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Comparing the given equation of the hyperbola with the standard equation, we get $a^2 = 100$ and $b^2 = 49$.

We also know that the condition for the line to be tangent to the hyperbola is

$$c^2 = a^2m^2 - b^2 \quad \Rightarrow \quad (6)^2 = 100m^2 - 49$$

$$\Rightarrow \quad 36 = 100m^2 - 49 \quad \Rightarrow \quad 100m^2 = 85$$

$$\Rightarrow \quad m^2 = \frac{85}{100} = \frac{17}{20} \quad \Rightarrow \quad m = \sqrt{\frac{17}{20}}.$$

8. The line $p = x\cos\alpha + y\sin\alpha$ becomes tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, if

(a) $p^2 = a^2\cos^2\alpha + b^2$

(b) $p^2 = a^2\cos\alpha - b^2\sin\alpha$

(c) $p = a\cos\alpha - b\sin\alpha$

(d) $p^2 = a^2\cos^2\alpha - b^2\sin^2\alpha$

Ans. (d)

Solution: Equation of the line is $p = x\cos\alpha + y\sin\alpha$ and equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

The given equation may be written as

$$y\sin\alpha = -x\cos\alpha + p$$

$$\Rightarrow \quad y = -x\cot\alpha + p\operatorname{cosec}\alpha$$

We know that the standard equation of the line is

$$y = mx + c$$

Comparing the given equation with the standard equation, we get

$$m = -\cot\alpha \text{ and } c = p\text{cosec}\alpha .$$

We also know that the condition for the given line to be tangent to the hyperbola is

$$c^2 = a^2m^2 - b^2 \quad \Rightarrow \quad p^2\text{cosec}^2\alpha = a^2\cot^2\alpha - b^2$$

$$\Rightarrow \quad p^2 = a^2\cos^2\alpha - b^2\sin^2\alpha .$$

9. The equation of the normal to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at the point $(8, 3\sqrt{3})$ is

(a) $x + y = 25$ (b) $y + 2x = 25$

(c) $2x + \sqrt{3}y = 25$ (d) $\sqrt{3}x + 2y = 25$

Ans. (c)

Solution: Equation of the hyperbola is $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (i)

and coordinates of the point $(x_1, y_1) = (8, 3\sqrt{3})$.

We know that the standard equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Comparing Eq. (i) with the standard equation, we get $a^2 = 16$ and $b^2 = 9$.

We also know that the equation of the normal to the given hyperbola at point (x_1, y_1) is

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2 \quad \Rightarrow \quad \frac{16x}{8} + \frac{9y}{3\sqrt{3}} = 4^2 + 3^2$$

$$\Rightarrow \quad \frac{16x}{8} + \frac{9y}{3\sqrt{3}} = 16 + 9 \quad \Rightarrow \quad 2x + \sqrt{3}y = 25.$$

10. The equation of the normal to the hyperbola $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point (a sec θ, b tan θ) is

(a) $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 - b^2$

(b) $\frac{ax}{\sec\theta} - \frac{by}{\tan\theta} = a - b$

(c) $\frac{ax}{\sec\theta} - \frac{by}{\tan\theta} = a^2 - b^2$

(d) $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$

Ans. (d)

Solution: Equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and coordinates of point $(x_1, y_1) = (a\sec\theta, b\sin\theta)$.

We know that the equation of the normal to the given hyperbola at (x_1, y_1) is

$$\frac{a^2(x - x_1)}{x_1} = \frac{-b^2(y - y_1)}{y_1}$$

$$\Rightarrow \quad \frac{a^2(x - a\sec\theta)}{a\sec\theta} = \frac{-b^2(y - b\tan\theta)}{b\tan\theta}$$

$$\Rightarrow \quad ax\tan\theta + by\sec\theta = a^2\tan\theta\sec\theta + b^2\sec\theta\tan\theta$$

$$\Rightarrow \quad \frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2.$$

11. The product of perpendiculars from any point on a hyperbola to its asymptotes is

(a) $\frac{a^2b^2}{a^2 + b^2}$ (b) $\frac{a^2 + b^2}{a^2b^2}$

(c) $\frac{ab}{\sqrt{a} + \sqrt{b}}$ (d) $\frac{ab}{a^2 + b^2}$

Ans. (a)

Solution: Equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0$. Let (x_1, y_1) be the coordinates of any point on the hyperbola.

Therefore, $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1 \quad \Rightarrow \quad b^2x_1^2 - a^2y_1^2 = a^2b^2$.

We also know that the relation for the asymptotes of the given hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$

$$\Rightarrow \quad b^2x^2 - a^2y^2 = 0.$$

Therefore, the product of perpendiculars from (x_1, y_1) to the asymptotes is given by

$$\frac{\left|Ax_1^2 + 2Hx_1y_1 + By_1^2\right|}{\sqrt{(A - B)^2 + 4H^2}} = \frac{b^2x_1^2 - a^2y_1^2}{\sqrt{(b^2 + a^2)^2}} = \frac{a^2b^2}{a^2 + b^2}$$

EXERCISE SET

1. The value of m, for which the line $y = mx + \frac{25\sqrt{3}}{3}$ is a normal to the conic $\frac{x^2}{16} - \frac{y^2}{9} = 1$, is

(a) $-\frac{2}{\sqrt{3}}$ (b) $\sqrt{3}$

(c) $-\frac{\sqrt{3}}{2}$ (d) none of these

2. Equations of the common tangents to the hyperbolas $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $\frac{y^2}{a^2}-\frac{x^2}{b^2}=1$ are

(a) $y=\pm x$ (b) $by=\pm ax$

(c) $y=\pm x\pm\sqrt{a^2-b^2}$ (d) none of these

3. If the line $x + 3y + 2 = 0$ and its perpendicular line are conjugate w.r.t. to $3x^2 - 5y^2 = 15$, then the equation of the conjugate line is

(a) $3x-y=15$ (b) $3x-y+12=0$

(c) $3x-y+10=0$ (d) $3x-y=4$

4. The foci of the hyperbola $4x^2 - 9y^2 - 36 = 0$ are

(a) $[\pm\sqrt{11},0]$ (b) $[\pm\sqrt{12},0]$

(c) $[\pm\sqrt{13},0]$ (d) $[0,\pm\sqrt{12}]$

5. The angle between the two asymptotes of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ is

(a) $\pi-2\tan^{-1}\frac{3}{4}$ (b) $\pi-2\tan^{-1}\frac{3}{2}$

(c) $2\tan^{-1}\frac{3}{4}$ (d) $\pi-2\tan^{-1}\frac{4}{3}$

6. The eccentricity of the conjugate hyperbola of the hyperbola $x^2 - 3y^2 = 1$ is

(a) 2 (b) $\frac{2}{\sqrt{3}}$

(c) 4 (d) $\frac{4}{3}$

7. The equation $16x^2-3y^2-32x+12y-44=0$ represents a hyperbola whose

(a) transverse axis is $4\sqrt{3}$ (b) conjugate axis is 4

(c) eccentricity is $\frac{\sqrt{19}}{\sqrt{3}}$ (d) centre is $(-1,2)$

8. The coordinates of a focus of the hyperbola $9x^2 - 16y^2 + 18x + 32y - 151 = 0$ are

(a) (– 1, 1) and (– 6, 1) (b) (4, 1) and (– 6, 1)

(c) (4, – 1) and (– 3, 1) (d) (– 1, 1) and (4, – 1)

9. A hyperbola has the asymptotes $x+2y=3$ and $x-y=0$ and passes through (2, 1). Its centre is

(a) (1, 2) (b) (2, 2

(c) (1, 1) (d) (2, 1)

10. The locus of the mid-points of the chords of the circle $x^2+y^2=16$ which are tangent to the hyperbola $9x^2 - 16y^2 = 144$ is

(a) $(x^2+y^2)^2=16x^2-9y^2$

(b) $(x^2+y^2)^2=9x^2-16y^2$

(c) $(x^2-y^2)^2=16x^2-9y^2$

(d) none of these

11. The eccentricity of the hyperbola $-\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is given by

(a) $e=+\sqrt{\frac{a^2+b^2}{a^2}}$ (b) $e=+\sqrt{\frac{a^2-b^2}{a^2}}$

(c) $e=+\sqrt{\frac{b^2-a^2}{a^2}}$ (d) $e=+\sqrt{\frac{a^2+b^2}{b^2}}$

12. If asymptotes of a hyperbola are at 90°, then

(a) eccentricity is $\sqrt{2}$ (b) eccentricity is 2

(c) eccentricity depends on the equation of the asymptotes

(d) none of these

13. If P is a point on the rectangular hyperbola $x^2 - y^2 = a^2$, C is its centre and S and S' are the two foci, then $SP \cdot S'P =$

(a) 2 (b) $(CP)^2$

(c) $(CS)^2$ (d) $(SS')^2$

14. The equation of the hyperbola passing through the origin and whose asymptotes are $3x + 4y = 5$ and $4x + 3y = 5$ is

(a) $x^2-y^2=1$

(b) $12x^2+12y^2+35xy-15x-15y=0$

(c) $12x^2+12y^2+25xy-35x-35y=0$

(d) $12x^2+12y^2+25xy-25x-25y=0$

15. The eccentricity of the conic $9x^2-16y^2=144$ is

(a) 4/5 (b) 5/4

(c) 4/3 (d) $\sqrt{7}$

16. The equation $\frac{x^2}{12-\lambda}+\frac{y^2}{8-\lambda}=1$ represents

(a) a hyperbola if $\lambda < 8$

(b) an ellipse if $\lambda > 8$

(c) a hyperbola if $8 < \lambda < 12$

(d) none of these

17. The locus of the poles with respect to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, of any tangent to the circle, whose diameter is the line joining the foci, is the ellipse $\frac{x^2}{a^4}+\frac{y^2}{b^4}$ equal to

(a) $\frac{1}{a^2-b^2}$ (b) $\frac{1}{a^2+b^2}$

(c) a^2+b^2 (d) none of these

18. If e and e' are the eccentricities of two conics S and S' and if $e^2+e'^2=3$, then both S and S' can be
(a) ellipses (b) parabolas
(c) hyperbolas (d) none of these

19. The equation of the hyperbola whose foci are (6, 5) and (−4, 5) and eccentricity is 5/4 is
(a) $\frac{(x-1)^2}{16}-\frac{(y-5)^2}{9}=1$
(b) $\frac{x^2}{16}-\frac{y^2}{9}=1$
(c) $\frac{(x-1)^2}{16}-\frac{(y-5)^2}{9}=-1$
(d) none of these

20. If the line $2x+\sqrt{6}y=2$ touches the hyperbola $x^2-2y^2=4$, then the point of contact is
(a) $(-2,\sqrt{6})$ (b) $(-5,2\sqrt{6})$
(c) $\left(\frac{1}{2},\frac{1}{\sqrt{6}}\right)$ (d) $(4,-\sqrt{6})$

21. PQ and RS are two perpendicular chords of the rectangular hyperbola $xy=c^2$. If C is the centre of the rectangular hyperbola, then the product of the slopes of CP, CQ, CR and CS is equal to
(a) −1 (b) 1
(c) 0 (d) none of these

22. Suppose a hyperbola has its transverse and conjugate axes coinciding with the major and minor axes of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ respectively. If the hyperbola passes through one of the foci of the ellipse and the product of the eccentricities of the hyperbola and ellipse is one, then
(a) the equation of the hyperbola is $\frac{x^2}{9}-\frac{y^2}{16}=1$
(b) the vertex of the hyperbola is $(5\sqrt{2},0)$
(c) the equation of the hyperbola is $\frac{x^2}{16}-\frac{y^2}{25}=1$
(d) none of these

23. The eccentricity of a hyperbola whose asymptotes are $3x-4y=7$ and $4x+3y=8$ is
(a) $\sqrt{2}$
(b) 2
(c) insufficient information
(d) none of these

24. The equation of the tangent to the curve $4x^2-9y^2=1$ which is parallel to $4y=5x+7$ is
(a) $24y-30x=17$ (b) $30y-24x=\pm\sqrt{161}$
(c) $24y-30x=\pm\sqrt{161}$ (d) none of these

25. A common tangent to $9x^2-16y^2=144$ and $x^2+y^2=9$ is
(a) $y=\frac{3}{\sqrt{7}}x+\frac{15}{\sqrt{7}}$ (b) $y=3\sqrt{\frac{2}{7}}x+\frac{15}{\sqrt{7}}$
(c) $y=2\sqrt{\frac{3}{7}}x+15\sqrt{7}$ (d) none of these

26. The point of intersection of two tangents to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, the product of whose slope is c^2, lies on the curve
(a) $y^2-b^2=c^2(x^2+a^2)$ (b) $y^2+a^2=c^2(x^2-b^2)$
(c) $y^2+b^2=c^2(x^2-a^2)$ (d) $y^2-a^2=c^2(x^2+b^2)$

27. The equation $16x^2-3y^2-+12y-44=0$ represents a hyperbola
(a) the length of whose transverse axis is $4\sqrt{3}$
(b) the length of whose conjugate axis is 4
(c) whose centre is (−1, 2)
(d) whose eccentricity is $\sqrt{\frac{19}{3}}$

28. If $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\,(a>b)$ and $x^2-y^2=c^2$ cuts at right angles, then
(a) $a^2+b^2=2c^2$ (b) $b^2-a^2=2c^2$
(c) $a^2-b^2=2c^2$ (d) $a^2b^2=2c^2$

29. One equation of the common tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=2$ is
(a) $2y=\sqrt{3}bx+ab$ (b) $y=2\sqrt{3}\frac{b}{a}x+2b$
(c) no common tangent (d) $ay=\sqrt{3}bx+2ab$

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) and (d) given below:
(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

30. **Statement I.** The equation of the director circle to the hyperbola $4x^2 - 3y^2 = 12$ is $x^2 + y^2 = 1$.
Statement II: Director circle is the locus of the point of intersection of perpendicular tangents to a hyperbola.

31. **Statement I:** The line $bx - ay = 0$ will not meet the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\,(a > b > 0)$.
Statement II: The line $y = mx + c$ does not meet the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, *if* $c^2 = a^2m^2 - b^2$.

32. **Statement I:** A hyperbola and its conjugate hyperbola have the same asymptotes.
Statement II: The difference between the second degree curve and pair of asymptotes is constant.

Previous Years' Questions

33. The eccentricity of the hyperbola $3x^2 - y^2 = 4$ is
(a) 1 (b) 2
(c) -2 (d) 5 **[AIEEE 2002]**

34. The equation of the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is
(a) $\frac{x}{x_1 + x_2} + \frac{y}{y_1 + y_2} = 1$ (b) $\frac{x}{x_1 - x_2} + \frac{y}{y_1 - y_2} = 1$
(c) $\frac{x}{y_1 + y_2} + \frac{y}{x_1 + x_2} = 1$ (d) $\frac{x}{y_1 - y_2} + \frac{y}{x_1 - x_2} = 1$
[AIEEE 2002]

35. If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide, then the value of b^2 is
(a) 1 (b) 5
(c) 7 (d) 9
[AIEEE 2003, M.N.R. 1992]

36. The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is
(a) a hyperbola (b) a parabola
(c) a circle (d) an ellipse
[AIEEE 2004]

37. A circle touches the x-axis and also touches the circle with centre (0, 3) and radius 2. The locus of the centre of the circle is
(a) a circle (b) an ellipse
(c) a parabola (d) a hyperbola
[AIEEE 2005]

38. For the hyperbola $\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$, which of the following remains constant when α varies?
(a) directrix (b) abscissa of vertices
(c) abscissa of foci (d) eccentricity
[AIEEE 2007]

39. The equation of the asymptotes of the hyperbola $2x^2 + 5xy + 2y^2 - 11x - 7y - 4 = 0$ is
(a) $2x^2 + 5xy - 2y^2 - 11x - 7y - 5 = 0$
(b) $2x^2 + 4xy + 2y^2 - 7x - 11y + 5 = 0$
(c) $2x^2 + 5xy + 2y^2 - 11x - 7y + 5 = 0$
(d) none of these **[DCE 2002]**

40. If $(a-2)x^2 + 9y^2 = 4$ represents a rectangular hyperbola, then a is equal to
(a) 0 (b) -7
(c) 1 (d) 3 **[DCE 2002]**

41. If e are e_1 are the eccentricities of hyperbolas $xy = c^2$ and $x^2 - y^2 = c^2$, then $e^2 + e_1^2$ is
(a) 1 (b) 4
(c) 6 (d) 8 **[DCE 2004]**

42. If e and e' are the eccentricities of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and its conjugate hyperbola, then the value of $\frac{1}{e^2} + \frac{1}{e'^2}$ is
(a) 0 (b) 1
(c) 2 (d) none of these
[DCE 2006]

43. The product of perpendiculars drawn from any point of a hyperbola to its asymptotes is
(a) $\frac{a^2b^2}{a^2 + b^2}$ (b) $\frac{a^2 + b^2}{a^2b^2}$
(c) $\frac{ab}{\sqrt{a} + \sqrt{b}}$ (d) $\frac{ab}{a^2 + b^2}$ **[DCE 2007]**

44. If the line $y = 2x + \lambda$ is a tangent to the hyperbola $36x^2 - 25y^2 = 3600$, then λ is equal to
(a) 16 (b) -16
(c) ± 16 (d) none of these
[DCE 2008]

45. The directrix of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is
(a) $x = \frac{9}{\sqrt{13}}$ (b) $y = \frac{9}{\sqrt{13}}$
(c) $x = \frac{6}{\sqrt{13}}$ (d) $y = \frac{6}{\sqrt{13}}$
[UP SEE 2003]

46. The eccentricity of the hyperbola conjugate to $x^2 - 3y^2 = 2x + 8$ is
(a) $2/\sqrt{3}$ (b) $\sqrt{3}$ (c) 2 (d) none of these
[UP SEE 2004]

47. The eccentricity of the hyperbola $9x^2 - 16y^2 - 18x - 64y - 199 = 0$ is
(a) 16/9 (b) 5/4 (c) 25/16 (d) 0
[MP Pet 2006, UP SEE 2005]

48. If the distance between the directrices of a rectangular hyperbola is 10, then the distance between its foci will be
(a) $10\sqrt{2}$ (b) 5 (c) $5\sqrt{2}$ (d) 20
[UP SEE 2009]

49. The equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is
(a) $3y = 9x + 2$ (b) $y = 2x + 1$
(c) $2y = x + 8$ (d) $y = x + 2$
[IIT-JEE 2002]

50. Which one of the following is independent of α in the hyperbola $\left(0 < \alpha < \frac{\pi}{2}\right)\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$?
(a) eccentricity (b) abscissa of foci
(c) directrix (d) vertex **[IIT-JEE 2003]**

51. If $2x + \sqrt{6}y = 2$ touches the hyperbola $x^2 - 2y^2 = 4$, then the point of contact is
(a) $(-2, \sqrt{6})$ (b) $(-5, 2\sqrt{6})$
(c) $\left(\frac{1}{2}, \frac{1}{\sqrt{6}}\right)$ (d) $(4, -\sqrt{6})$
[IIT-JEE 2004]

52. A hyperbola having the transversal axis of length 2 sin θ is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is
(a) $x^2\text{cosec}^2\theta - y^2\sec^2\theta = 1$
(b) $x^2\sec^2\theta - y^2\,\text{cosec}^2\theta = 1$
(c) $x^2\sec^2\theta - y^2\cos^2\theta = 1$
(d) $x^2\cos^2\theta - y^2\sin^2\theta = 1$ **[IIT-JEE 2007]**

53. If the chords of contact of tangents from two points (x_1, y_1) and (x_2, y_2) to the hyperbola $4x^2 - 9y^2 - 36$ are at right angels, then $\frac{x_1x_2}{y_1y_2}$ is equal to
(a) 9/4 (b) –9/4 (c) 81/16 (d) –81/16
[AMU 2004]

54. The equation of the director circle of the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$ is given by
(a) $x^2 + y^2 = 16$ (b) $x^2 + y^2 = 4$
(c) $x^2 + y^2 = 20$ (d) $x^2 + y^2 = 12$
[AMU 2006]

55. The equation of the conic with focus at (1, –1), directrix along $x - y + 1 = 0$ and with eccentricity $\sqrt{2}$ is
(a) $x^2 - y^2 = 1$ (b) $xy = 1$
(c) $2xy - 4x + 4y + 1 = 0$ (d) $2xy + 4x - 4y - 1 = 0$
[EAMCET 1994, Delhi CEE 1998]

56. If the distance between the foci and the distance between the directrices of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are in the ratio 3 : 2, then $a : b$ is
(a) $\sqrt{2}:1$ (b) $\sqrt{3}:\sqrt{2}$ (c) 1 : 2 (d) 2 : 1
[Jamia Millia Islamia 2008]

57. The ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ have in common
(a) centre only
(b) centre, foci and directrices
(c) centre, foci and vertices
(d) centre and vertices only **[KCET 2006)]**

58. Coordinates of the foci of a hyperbola are (–5, 3) and (7, 3) and eccentricity is 3/2. Then, length of its latus rectum is
(a) 20 (b) 10 (c) 40 (d) none of these
[OJEE 2007]

59. The length of the straight line $x - 3y = 1$ intercepted by the hyperbola $x^2 - 4y^2 = 1$ is
(a) $\frac{3}{5}\sqrt{10}$ (b) $\frac{6}{5}\sqrt{10}$ (c) $\frac{5}{3}\sqrt{10}$ (d) $\frac{5}{6}\sqrt{10}$
[OJEE 2008]

60. The eccentricity of the hyperbola which passes through (3, 0) and $(3\sqrt{2}, 2)$ is
(a) $\sqrt{13}$ (b) $\sqrt{13}/3$ (c) $\sqrt{13}/4$ (d) none of these
[WBJEE 2007]

61. The product of the perpendiculars drawn from any point on a hyperbola to its asymptotes is
(a) $\frac{ab}{\sqrt{a} + \sqrt{b}}$ (b) $\frac{ab}{a^2 + b^2}$
(c) $\frac{a^2b^2}{a^2 + b^2}$ (d) $\frac{a^2 + b^2}{a^2b^2}$
[Karnataka CEE 2000]

62. The equation of the hyperbola whose foci are (–2, 0) and (2, 0) and eccentricity is 2 is given by
(a) $x^2 - 3y^2 = 3$ (b) $3x^2 - y^2 = 3$
(c) $-x^2 + 3y^2 = 3$ (d) $-3x^2 + y^2 = 3$
[AIEEE 2011]

ANSWERS

Exercise Set

1. (a)	**2.** (c)	**3.** (b)	**4.** (c)	**5.** (c)	**6.** (a)	**7.** (c)	**8.** (b)	**9.** (c)	**10.** (a)
11. (d)	**12.** (a)	**13.** (b)	**14.** (c)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (c)	**19.** (a)	**20.** (d)
21. (b)	**22.** (a)	**23.** (a)	**24.** (c)	**25.** (b)	**26.** (c)	**27.** (d)	**28.** (c)	**29.** (d)	**30.** (d)
31. (c)	**32.** (b)	**33.** (b)	**34.** (a)	**35.** (c)	**36.** (a)	**37.** (c)	**38.** (c)	**39.** (c)	**40.** (b)
41. (b)	**42.** (b)	**43.** (a)	**44.** (c)	**45.** (a)	**46.** (c)	**47.** (b)	**48.** (d)	**49.** (d)	**50.** (b)
51. (d)	**52.** (a)	**53.** (d)	**54.** (d)	**55.** (c)	**56.** (a)	**57.** (d)	**58.** (b)	**59.** (b)	**60.** (b)
61. (c)	**62.** (b)								

HINTS AND SOLUTIONS—EXERCISE SET

1. $y = mx + \dfrac{25\sqrt{3}}{3}$ (i)

and $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$ (ii)

Here, Eq. (i) is normal to Eq. (ii). So

$$\frac{(a^2+b^2)^2}{c^2} = \frac{a^2}{m^2} - \frac{b^2}{1}$$

$$\Rightarrow \frac{(16+9)^2 \times 9}{625 \times 3} = \frac{16}{m^2} - \frac{9}{1} \quad \Rightarrow \quad \frac{16}{m^2} = 12$$

$$\Rightarrow m = \pm\frac{2}{\sqrt{3}}$$

2. The two hyperbolas are $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ and $\dfrac{x^2}{-b^2} - \dfrac{y^2}{-a^2} = 1$.

Equations of tangents are

$$y = mx \pm \sqrt{a^2m^2 - b^2} \quad \text{(i)}$$

and $$y = m_1x \pm \sqrt{-b^2m_1^2 - a^2} \quad \text{(ii)}$$

For common tangent, (i) and (ii) must be identical

i.e. $m = m_1$ and $a^2m^2 - b^2 = -b^2m^2 + a^2$

$\Rightarrow (a^2+b^2)(m^2-1) = 0 \quad \Rightarrow \quad m = \pm 1$

$\therefore$ Required equations are $y = \pm x \pm \sqrt{a^2 - b^2}$

3. A line perpendicular to $x + 3y + 2 = 0$ (i)

is $3x - y + k = 0$ (ii)

(i) and (ii) are conjugtate w.r.t. to

$$\frac{x^2}{5} - \frac{y^2}{3} = 1$$

$\Rightarrow 5(1)(3) - 3(3)(-1) = 2(k) \quad \Rightarrow \quad k = 12$

$\therefore$ Conjugate is $3x - y + 12 = 0$.

4. $a^2 = 9, b^2 = 4$

$\because \quad e^2 = 1 + \dfrac{b^2}{a^2} = \dfrac{13}{9}$

$\therefore$ Foci of hyperbola: $(\pm ae, 0)$

5. We know that the angle between two asymptotes of the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ is $2\tan^{-1}\left(\dfrac{b}{a}\right)$.

Equation of the given hyperbola is $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$.

Here, $a = 4$ and $b = 3$.

$\therefore$ Required angle $= 2\tan^{-1}\left(\dfrac{3}{4}\right)$.

6. The given hyperbola is $\dfrac{x^2}{1} - \dfrac{y^2}{\frac{1}{3}} = 1$.

Here $a^2 = 1$ and $b^2 = \dfrac{1}{3}$

Since $b^2 = a^2(e^2 - 1) \quad \Rightarrow \quad \dfrac{1}{3} = 1(e^2 - 1)$

$$\Rightarrow \quad e^2 = \frac{4}{3} \quad \Rightarrow \quad e = \frac{2}{\sqrt{3}}$$

If e' is the eccentricity of the conjugate hyperbola, then

$$\frac{1}{e^2} + \frac{1}{e'^2} = 1$$

$$\Rightarrow \frac{1}{e'^2} = 1 - \frac{1}{e^2} = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\Rightarrow e' = 2$$

7. The given equation can be written as

$$16(x^2 - 2x + 1 - 1) - 3(y^2 - 4y + 4 - 4) = 44$$

or $$16(x-1)^2 - 3(y-2)^2 = 44 + 16 - 12 = 48$$

$$\frac{(x-1)^2}{3} - \frac{(y-2)^2}{16} = 1$$

$$a^2 = 3, b^2 = 16 \quad \therefore \quad e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{16}{3} = \frac{19}{3}$$

$$\therefore \quad e = \sqrt{\frac{19}{3}}$$

8. The equation can be written as $\frac{(x+1)^2}{16}-\frac{(y-1)^2}{9}=1$.

$\therefore \quad e=\sqrt{1+\frac{9}{16}}=\frac{5}{4}$

$\therefore$ Its foci are given by $x+1=\pm\left(4\times\frac{5}{4}\right)$ and $y-1=0$

$\Rightarrow x=4$ or -6 and $y=1$.

9. Equations of asymptotes are $x+2y=3$ (i)

and $x-y=0$ (ii)

On solving Eqs. (i) and (ii), we get $x=1, y=1$

$\therefore$ The centre of the hyperbola is (1, 1) because asymptotes pass through the centre of the hyperbola.

10. The given hyperbola is $\frac{x^2}{16}-\frac{y^2}{9}=1$ (i)

Any tangent to (i) is

$$y=mx+\sqrt{16m^2-9} \qquad \text{(ii)}$$

Let (x_1, y_1) be the mid-point of the chord of the circle $x^2+y^2=16$.

Then the equation of the chord is $T=S_1$

i.e. $xx_1+yy_1-(x_1^2+y_1^2)=0$ (iii)

Since (ii) and (iii) represent the same line,

$$\therefore \quad \frac{m}{x_1}=\frac{-1}{y_1}=\frac{\sqrt{16m^2-9}}{-(x_1^2+y_1^2)}$$

$$\Rightarrow \quad m=-\frac{x_1}{y_1} \text{ and } (x_1^2+y_1^2)^2=y_1^2(16m^2-9)$$

$$\Rightarrow (x_1^2+y_1^2)^2=16.\frac{x_1^2}{y_1^2}y_1^2-9y_1^2=16.x_1^2-9y_1^2$$

$\therefore$ Locus of (x_1, y_1) is $(x^2+y^2)^2=16x^2-9y^2$

11. $\sqrt{1+\frac{a^2}{b^2}}=\sqrt{\frac{a^2+b^2}{b^2}}$ as the transverse axis is along the y-axis and the conjugate axis is along the x-axis, i.e. $a\to b$ and $b\to a$.

12. Since asymptotes are at 90°, it means it is a rectangular hyperbola.

$\therefore$ Eccentricity is $\sqrt{2}$.

13. Let P be (x, y). Centre C is (0, 0).

We know that $b^2=a^2(e^2-1)$.

$\therefore \quad a^2=a^2(e^2-1) \quad \Rightarrow \quad e=\pm\sqrt{2}$

$\therefore$ Foci are $S(a\sqrt{2},0)$ and $S'(-a\sqrt{2},0)$.

$\therefore$ Corresponding directrices are

$$x=\frac{a}{\sqrt{2}} \quad \text{and} \quad x=-\frac{a}{\sqrt{2}}$$

By the definition of hyperbola,

$$SP=e\left(\text{distance of } P \text{ from } x=\frac{a}{\sqrt{2}}\right)$$

$$=\sqrt{2}\left|x-\frac{a}{\sqrt{2}}\right|$$

Similarly, $S'P=\sqrt{2}\left|x+\frac{a}{\sqrt{2}}\right|$

$$\therefore \quad SP-S'P=2\left|x^2-\frac{a^2}{2}\right|$$

$$=2x^2-a^2=x^2+x^2-a^2$$

$$=x^2+y^2=(CP)^2$$

($\because P$ lies on the hyperbola $x^2-y^2=a^2$)

14. Given equations of asymptotes are $3x+4y=5$ and $4x+3y=5$.

$\therefore$ Equation of hyperbola is

$$(3x+4y-5)(4x+3y-5)=\lambda \qquad \text{(i)}$$

Since it passes through the origin,

$$\therefore \quad \lambda=25$$

From (i), we get $12x^2+12y^2+25xy-35x-35y=0$.

15. The hyperbola is $\frac{x^2}{16}-\frac{y^2}{9}=1$. Here $a^2=16$, $b^2=9$.

But $b^2=a^2(e^2-1) \quad \therefore \; 9=16(e^2-1)$

$$\therefore \quad e^2-1=\frac{9}{16} \quad \Rightarrow \quad e^2=\frac{9}{16}+1=\frac{25}{16}$$

$$\Rightarrow \quad e=\frac{5}{4}$$

16. $a^2=12-\lambda>0$ and $b^2=8-\lambda<0$

$\therefore \quad \lambda<12$ and $8<\lambda$

$\therefore \quad 8<\lambda<12$

17. Let $P(h, k)$ be the pole of any tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.

Then the equation of the polar is

$$\frac{hx}{a^2}-\frac{ky}{b^2}=1 \qquad \text{(i)}$$

$\therefore$ The equation of the circle described on the line joining $S(ae, 0)$ and $S'(-ae, 0)$ as diameter is

$$(x+ae)(x-ae)+y^2=0$$

$$\Rightarrow \quad x^2+y^2=a^2e^2$$

$$\Rightarrow \quad x^2+y^2=a^2+b^2 \quad [\because b^2=a^2(e^2-1)]$$

It is given that (i) touches the above circle.

Therefore, $\left|\frac{1}{\sqrt{\frac{h^2}{a^4}+\frac{k^2}{b^4}}}\right| = \sqrt{a^2+b^2}$

$\Rightarrow \quad \frac{h^2}{a^2}+\frac{k^2}{b^4} = \frac{1}{a^2+b^2}$

$\therefore$ The locus of the point is

$$\frac{x^2}{a^4}+\frac{y^2}{b^4} = \frac{1}{a^2+b^2}$$

18. For ellipse or parabola, $e^2 + e'^2 \le 2 \ne 3$.
For hyperbola, it is possible [$\because$ $e > 1$, e′ > 1]

19. Centre is mid-point of foci, i.e. (1, 5).
Distance between foci = $2ae = 10$

$\therefore \quad ae = 5 \quad \Rightarrow \quad a \cdot \frac{5}{4} = 5$

$\Rightarrow a = 4$

$\Rightarrow \quad b^2 = a^2(e^2-1) = 16\left(\frac{25}{16}-1\right) = 9$

$$\frac{(x-1)^2}{16} - \frac{(y-5)^2}{9} = 1.$$

20. As we know, the equation of the tangent at (x_1, y_1) is $xx_1 - 2yy_1 = 4$ which is the same as $2x + \sqrt{6}y = 2$.

$\therefore \quad \frac{x_1}{2} = \frac{-2y_1}{\sqrt{6}} = \frac{4}{2}$

$\Rightarrow \quad x_1 = 4$ and $y_1 = -\sqrt{6}$

Thus, the point of contact is $(4, -\sqrt{6})$.

21. Let t_1, t_1, t_3, t_4 be the parameters of the points P, Q, R, S respectively.

$\therefore \quad P$ is $\left(ct_1, \frac{c}{t_1}\right)$, Q is $\left(ct_2, \frac{c}{t_2}\right)$, R is $\left(ct_3, \frac{c}{t_3}\right)$,

S is $\left(ct_4, \frac{c}{t_4}\right)$

$$PQ \perp RS \quad \Rightarrow \quad \frac{\frac{c}{t_2}-\frac{c}{t_1}}{ct_2-ct_1} \cdot \frac{\frac{c}{t_4}-\frac{c}{t_3}}{ct_4-ct_3} = -1$$

$$\Rightarrow \quad -\frac{1}{t_1t_2} \cdot -\frac{1}{t_3t_4} = -1 \quad \Rightarrow \quad t_1t_2t_3t_4 = -1$$

Product of slopes of CP, CQ, CR, CS

$$= \frac{\frac{c}{t_1}}{ct_1} \cdot \frac{\frac{c}{t_2}}{ct_2} \cdot \frac{\frac{c}{t_3}}{ct_3} \cdot \frac{\frac{c}{t_4}}{ct_4} = \frac{1}{t_1^2t_2^2t_3^2t_4^2} = \frac{1}{(-1)^2} = 1$$

22. $e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{16}{25} = \frac{9}{25}$

$\therefore \quad e = \frac{3}{5}$

Hence, e' for the hyperbola $= \frac{5}{3}$ as $ee' = 1$.

Also foci of the ellipse are $(\pm ae, 0) = (\pm 3, 0)$.

The equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and it passes through $(\pm 3, 0)$.

$\therefore \quad a^2 = 9$

$e'^2 = 1 + \frac{b^2}{a^2}$ (for hyperbola)

$\therefore \quad \frac{25}{9} = 1 + \frac{b^2}{9} \quad \Rightarrow \quad b^2 = 16$

Hence, the equation of the hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

Its foci are at $(\pm ae', 0) = \left(\pm 3 \cdot \frac{5}{3}, 0\right) = (\pm 5, 0)$.

23. Since asymptotes $3x - 4y = 7$ and $4x + 3y = 8$ are perpendicular, therefore it is a rectangular hyperbola. So eccentricity is $\sqrt{2}$.

24. Let m be the slope of the tangent to $4x^2 - 9y^2 = 1$. Then

m = (slope of the line $4y = 5x + 7$) = 5/4

We have $\frac{x^2}{1/4} - \frac{y^2}{1/9} = 1$ or $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

where $a^2 = \frac{1}{4}, b^2 = \frac{1}{9}$

The equations of the tangents are

$$y = mx \pm \sqrt{a^2m^2 - b^2} \text{ or } y = \frac{5}{4}x \pm \sqrt{\frac{25}{64} - \frac{1}{9}}$$

or $30x - 24y \pm \sqrt{161} = 0$ or $24y - 30x = \pm\sqrt{161}$

25. Any tangent to the given hyperbola

$y = mx + \sqrt{a^2m^2 - b^2}$ is

$$mx - y + \sqrt{16m^2 - 9} = 0 \qquad \text{(i)}$$

If it touches the circle, then $p = r$ gives

$$\frac{\sqrt{16m^2-9}}{\sqrt{m^2+1}} = 3$$

$\therefore \quad 16m^2 - 9 = 9m^2 + 9 \quad \Rightarrow \quad 7m^2 = 18$

or $\quad m = \frac{3\sqrt{2}}{\sqrt{7}}$.

Putting in (i), we get option (b) as the answer.

26. Let the slopes of the two tangents to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}$ be cm and $\frac{c}{m}$.

The equations of tangents are

$$y=cmx+\sqrt{a^2c^2m^2-b^2} \qquad \text{(i)}$$

and $my-cx=\sqrt{a^2c^2-b^2m^2}$ (ii)

Squaring and subtracting Eq. (ii) from Eq. (i), we get

$$(y-cmx)^2-(my-cx)^2=a^2c^2m^2-b^2-a^2c^2+b^2m^2$$

$$\Rightarrow \quad (1-m^2)(y^2-c^2x^2)=-(1-m^2)(a^2c^2+b^2)$$

$$\Rightarrow \quad y^2+b^2=c^2(x^2-a^2).$$

27. We have $16(x^2-2x)-3(y^2-4y)=44$

$$\Rightarrow 16(x-1)^2-3(y-2y)^2=48$$

$$\Rightarrow \frac{(x-1)^2}{3}-\frac{(y-2)^2}{16}=1$$

This equation represents a hyperbola with eccentricity given by

$$e=\sqrt{1+\left(\frac{\text{Conjugate axis}}{\text{Transverse axis}}\right)}=\sqrt{1+\left(\frac{4}{\sqrt{3}}\right)^2}=\sqrt{\frac{19}{3}}$$

28. $\left(\frac{dy}{dx}\right)_i\left(\frac{dy}{dx}\right)_{ii}=-1$

$$\therefore \quad -\frac{b^2}{a^2}\cdot\frac{x^2}{y^2}=-1$$

$$\therefore \quad \frac{x^2}{a^2}=\frac{y^2}{b^2} \Rightarrow \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \Rightarrow x^2=\frac{a^2}{2},\ y^2=\frac{b^2}{2}$$

Putting in $x^2-y^2=c^2$, we get option (c) as the answer.

29. Equation of the tangent to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

$$y=mx+\sqrt{a^2m^2+b^2}$$

and equation of the tangent to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=2$ is

$$y=mx+\sqrt{2a^2m^2-2b^2}$$

For common tangent,

$$\Rightarrow \quad a^2m^2+b^2=2a^2m^2-2b^2$$

$$a^2m^2=3b^2 \qquad \Rightarrow \qquad m=\pm\frac{\sqrt{3}b}{a}$$

$\therefore$ Equation of the common tangent is $y=\frac{\sqrt{3}b}{a}x+2b.$

30. Now, $4x^2-3y^2=12$

$$\Rightarrow \frac{x^2}{3}-\frac{y^2}{4}=1$$

Then, the director circle is $x^2+y^2=3-4=-1$.

Does not exist, for existence $a>b$.

31. The intersection of the line $bx-ay=0$ and the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

$$\frac{x^2}{a^2}-\left(\frac{b}{a}\right)\frac{x^2}{b^2}=1 \quad \Rightarrow \quad 0=1$$

Hence, no solution exists. But Statement II is false.

32. Let the hyperbola be $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, then the pair of asymptotes is $\frac{x^2}{a^2}-\frac{y^2}{b^2}+\lambda=0.$

Then, $\Delta=0$

$\therefore \quad \lambda=0$

$\therefore$ The pair of asymptotes is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=0$ and the equation of conjugate hyperbola is $-\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

Then, the pair of asymptotes is $-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\mu=0$

Then, $\Delta=0$

$\therefore \quad \mu=0$

$\therefore$ The pair of asymptotes is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=0.$

33. $e^2=1+\frac{b^2}{a^2}=1+3=4$

$$\Rightarrow \quad e=2$$

$\because \quad a^2=\frac{4}{3}, b^2=\frac{4}{1}.$

34. The mid-point of the chord is $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$.

The equation of the chord in terms of its mid-point is $T=S_1$,

i.e. $x\left(\frac{y_1+y_2}{2}\right)+y\left(\frac{x_1+x_2}{2}\right)=2\left(\frac{x_1+x_2}{2}\right)\left(\frac{y_1+y_2}{2}\right)$

$$\Rightarrow \quad x(y_1+y_2)+y(x_1+x_2)=(x_1+x_2)(y_1+y_2)$$

$$\Rightarrow \quad \frac{x}{x_1+x_2}+\frac{y}{y_1+y_2}=1$$

35. For hyperbola, $e^2=1+\frac{b^2}{a^2}=1+\frac{81}{144}=\frac{225}{144}$

$$\therefore \quad e=\frac{15}{12}=\frac{5}{4}, \quad \text{i.e.} \qquad e>1.$$

Also $a^2 = \frac{144}{25}$

Hence, the foci are $(\pm ae, 0)$,

i.e. $\left(\pm\frac{12}{5}\cdot\frac{5}{4}, 0\right) = (\pm 3, 0)$

Now the foci coincide, therefore for ellipse,

$ae = 3$ or $a^2e^2 = 9$

$\Rightarrow a^2\left(1-\frac{b^2}{a^2}\right) = 9 \quad \Rightarrow \quad a^2 - b^2 = 9$

$\Rightarrow \quad 16 - 9 = b^2 \quad \Rightarrow \quad b^2 = 7.$

36. $y = mx + c$ touches $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

if $c^2 = a^2m^2 - b^2$

$\therefore \quad y = \alpha x + \beta$ touches if

$\beta^2 = a^2\alpha^2 - b^2$

$\therefore$ Locus of (α, β) is $y^2 = a^2x^2 - b^2$

$\Rightarrow a^2x^2 - y^2 = b^2$

$\Rightarrow \frac{x^2}{b^2/a^2} - \frac{y^2}{b^2} = 1$, which is a hyperbola.

$\therefore$ Locus of $P(\alpha, \beta)$ is a hyperbola.

37. Let (h, k) be the centre of the circle.

$\therefore \quad |C_1C_2| = r_1r_2$

$\Rightarrow \sqrt{(h-0)^2 + (k-3)^2} = |k \pm 2|$

Squaring, $h^2 = 5(2k-1) \quad \Rightarrow \quad h^2 = 10k - 5$

Hence, locus is $x^2 = 5(2y - 1)$, which is a parabola.

38. The given hyperbola is $\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1.$

$\therefore \quad a^2 = \cos^2\alpha$ and $b^2 = \sin^2\alpha$

Now $b^2 = a^2(e^2 - 1) \quad \Rightarrow \quad \sin^2\alpha = \cos^2\alpha(e^2 - 1)$

$\Rightarrow e^2 - 1 = \tan^2\alpha$

$\Rightarrow e^2 = 1 + \tan^2\alpha = \sec^2\alpha$

Foci are $(\pm ae, 0)$, i.e. $(\pm\cos\alpha\cdot\sec\alpha, 0)$

i.e. $(\pm 1, 0)$

Hence, abscissae of foci remain constant when α varies.

39. Since the equations of the pair of asymptotes of the hyperbola differ by the equations of hyperbolas by a constant only, we can suppose the required pair of asymptotes be

$$2x^2 + 5xy + 2y^2 - 11x - 7y + \lambda = 0 \qquad \text{(i)}$$

When compared to the general second-degree equation of x, y, we get

$$a = 2 = b = 2,\ h = \frac{5}{2},\ g = -\frac{11}{2},\ f = -\frac{7}{2},\ c = \lambda$$

To find λ, using $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$, we get $\lambda = 5$.

40. $a - 2$ must be equal to -9

$\Rightarrow a = -7.$

41. Both are rectangular hyperbolas and therefore will have $\sqrt{2}$ as eccentricity.

$\Rightarrow \qquad e^2 + e_1^2 = (\sqrt{2})^2 + (\sqrt{2})^2 = 4$

42. For hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, conjugate hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1.$$

$\therefore$ Eccentricities are e and e', such that

$$e^2 = \frac{a^2 + b^2}{a^2} \quad \text{and} \quad (e')^2 = \frac{a^2 + b^2}{b^2}$$

$$\therefore \quad \frac{1}{e} + \frac{1}{(e')^2} = \frac{a^2}{a^2 + b^2} + \frac{b^2}{a^2 + b^2}$$

$$= \frac{a^2 + b^2}{a^2 + b^2} = 1.$$

43. Let $(a\sec\theta, b\tan\theta)$ be any point to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The equations of assymtotes are

$$\frac{x}{a} - \frac{y}{b} = 0 \quad \text{and} \quad \frac{x}{a} + \frac{y}{b} = 0$$

Let p_1 = perpendicular length from

$$(a\sec\theta, b\tan\theta) \text{ on } \frac{x}{a} - \frac{y}{b} = 0$$

$$= \left|\frac{\sec\theta - \tan\theta}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}\right|$$

and p_2 = perpendicular on the other line $\frac{x}{a} + \frac{y}{b} = 0$

$$= \left|\frac{\sec\theta + \tan\theta}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}\right|$$

$$\therefore \quad p_1p_2 = \frac{\sec^2\theta - \tan^2\theta}{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{a^2b^2}{a^2 + b^2}.$$

44. If $y = 2x + \lambda$ is tangent to the given hyperbola,

then $\lambda = \pm\sqrt{a^2m^2 - b^2}$

$= \pm\sqrt{(100)(4) - 144} = \pm 16$ [$\because a^2 = 100, b^2 = 14$]

45. $x = \pm\dfrac{a}{e}$

Here, $e = \sqrt{1 + \dfrac{b^2}{a^2}} = \sqrt{1 + \dfrac{4}{9}} = \dfrac{\sqrt{13}}{3}$

$x = \pm\dfrac{9}{\sqrt{13}}$

46. $x^2 - 3y^2 = 2x + 8 \quad \Rightarrow \quad \dfrac{(x-1)^2}{9} - \dfrac{y^2}{3} = 1$

$\therefore \quad e = \sqrt{1 + \dfrac{3}{9}} = \dfrac{2}{\sqrt{3}}$

We know $\dfrac{1}{e^2} + \dfrac{1}{e'^2} = 1$

$\Rightarrow \quad \left(\dfrac{1}{e'}\right)^2 = 1 - \dfrac{3}{4} = \left(\dfrac{1}{2}\right)^2$

$\therefore \quad e' = 2$

47. $9x^2 - 16y^2 - 18x - 64y - 199 = 0$

$\Rightarrow \quad 9(x^2 - 2x + 1) - 16(y^2 + 4y + 4) = 144$

$\Rightarrow \quad \dfrac{(x-1)^2}{16} - \dfrac{(y+2)^2}{9} = 1$

$\therefore \quad e = \sqrt{1 + \dfrac{b^2}{a^2}} = \dfrac{5}{4}.$

48. Since the distance between directrices $\dfrac{2a}{e} = 10$

$\Rightarrow \quad a = \dfrac{10 \times \sqrt{2}}{2} = 5\sqrt{2}$

$\therefore$ Distance between foci, $2ae = 2 \times 5\sqrt{2} \times \sqrt{2}$

$= 20$

49. Any tangent to the parabola $y^2 = 8x$ is

$y = mx + \dfrac{2}{m}.$

Clearly options (a), (b) and (d) satisfy, where $m = 3, 2, 1$, respectively.

For $xy = -1$,

equations of tangents are $y - \sqrt{m} = m\left(x - \dfrac{1}{\sqrt{m}}\right)$

and $y + \sqrt{m} = m\left(x - \dfrac{1}{\sqrt{m}}\right).$

Clearly option (d) satisfies these equations.

50. The given hyperbola is $\dfrac{x^2}{\cos^2\alpha} - \dfrac{y^2}{\sin^2\alpha} = 1.$

Here $a^2 = \cos^2\alpha$, $b^2 = \sin^2\alpha$.

(i) $e^2 = 1 + \tan^2\alpha = \sec^2\alpha \quad \Rightarrow \quad e = \sec\alpha$

(ii) Foci $(\pm ae, 0)$, i.e. $(\pm\cos\alpha \cdot \sec\alpha, 0)$, i.e. $(\pm 1, 0)$

$\therefore$ Abscissa of foci $= \pm 1$

(iii) Directrix is $x = \pm\dfrac{c}{e} = \pm\dfrac{\cos\alpha}{\sec\alpha} = \pm\cos^2\alpha$

Hence, abscissa of foci is independent of α.

51. Let (x_1, y_1) be the point of contact.

Then $xx_1 - 2yy_1 = 4$ is the same as

$2x + \sqrt{6}y = 2.$

Comparing, $\dfrac{x_1}{2} = \dfrac{-2y_1}{\sqrt{6}} = \dfrac{4}{2}$

$\Rightarrow \quad x_1 = 4, y_1 = -\sqrt{6}.$

Hence, the point of contact is $(4, -\sqrt{6})$.

52. We have $3x^2 + 4y^2 = 12$, i.e. $\dfrac{x^2}{4} + \dfrac{y^2}{3} = 1.$

Here $a^2 = 4, b^2 = 3.$

Now $b^2 = a^2(1 - e^2) \quad \Rightarrow \quad 3 = 4(1 - e^2)$

$\Rightarrow \quad 1 - e^2 = \dfrac{3}{4} \quad \Rightarrow \quad e^2 = \dfrac{1}{4}$

$\Rightarrow \quad e = \dfrac{1}{2}$ [$\because e > 0$]

$\Rightarrow \quad ae = 2\left(\dfrac{1}{2}\right) = 1 \quad \Rightarrow \quad \sin\theta \cdot e = 1$ [$\because a = \sin\theta$]

$\Rightarrow \quad e = \operatorname{cosec}\theta$

$\therefore \quad b = a\sqrt{e^2 - 1} = \sin\theta\sqrt{\operatorname{cosec}^2\theta - 1}$

$= \sin\theta \cdot \cot\theta = \cos\theta.$

$\therefore$ The hyperbola is $\dfrac{x^2}{\sin^2\theta} - \dfrac{y^2}{\cos^2\theta} = 1$

$\Rightarrow \quad x^2\operatorname{cosec}^2\theta - y^2\sec^2\theta = 1.$

53. The equations of the chord of contact at $P(x_1, y_1)$ and $Q(x_2, y_2)$ are

$\dfrac{xx_1}{9} - \dfrac{yy_1}{4} = 1 \quad$ and $\quad \dfrac{xx_2}{9} - \dfrac{yy_2}{4} = 1$

with slopes $m_1 = \dfrac{\frac{x_1}{9}}{\frac{y_1}{4}} = \dfrac{4}{9}\dfrac{x_1}{y_1}$ and $m_2 = \dfrac{4}{9}\dfrac{x_2}{y_2}$

where $\quad m_1 m_2 = -1$

$\Rightarrow \quad \dfrac{x_1 x_2}{y_1 y_2} = -\dfrac{81}{16}.$

54. The equation of the director circle of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is given by $x^2+y^2=a^2-b^2$.

Hence, the required equation is

$$x^2+y^2=16-4=12$$

55. Let $P(x, y)$ be any point on the conic. Focus F is $(1, -1)$ and directrix is $x-y+1=0$.

Since $|PF|=e|PM|$, where PM is $\perp$ to $x-y+1=0$

$$\therefore\quad (x-1)^2+(y+1)^2=e^2\left(\frac{x-y+1}{\sqrt{2}}\right)^2$$

$$=2\frac{(x-y+1)^2}{2}\qquad[\because e=\sqrt{2}]$$

$$\Rightarrow\quad x^2-2x+1+y^2+2y+1=(x-y+1)^2$$

$$\Rightarrow\quad x^2+y^2-2x+2y+2=x^2+y^2+1-2xy+2x-2y$$

$$\Rightarrow\quad 2xy-4x+4y+1=0 \text{ is the required conic.}$$

56. Equation of the hyperbola is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.

Distance between the foci of the hyperbola = $2ae$ and the distance between the directrices = $2a/e$.

According to the question,

$$\frac{2ae}{2a/e}=\frac{3}{2}\quad\Rightarrow\quad e^2=\frac{3}{2}$$

Using $b^2=a^2(e^2-1)$, we get

$$\Rightarrow\qquad \frac{b^2}{a^2}=\frac{3}{2}-1\quad\Rightarrow\quad \frac{a}{b}=\frac{\sqrt{2}}{1}.$$

57. Let e and e' be the eccentricities of the ellipse and the hyperbola respectively.

$$\therefore\qquad e=\sqrt{\frac{a^2-b^2}{a^2}}=\sqrt{\frac{25-16}{25}}=\frac{3}{5}$$

and $$e'=\sqrt{\frac{a^2+b^2}{a^2}}=\sqrt{\frac{25+16}{25}}=\frac{\sqrt{41}}{5}$$

(i) Centre of the ellipse is (0, 0) and centre of the hyperbola is (0, 0).

(ii) Foci of the ellipse are $(\pm ae, 0)$ or $(\pm 3, 0)$ foci of the hyperbola are $(\pm ae', 0)$ or $(\pm\sqrt{41}, 0)$.

(iii) Directrices of the ellipse are $x=\pm\frac{a}{e}$

$$\Rightarrow\quad x=\pm\frac{25}{3}$$

Directrices of the hyperbola are $x=\pm\frac{a}{e}$

$$\Rightarrow\qquad x=\pm\frac{25}{\sqrt{41}}$$

(iv) Vertices of the ellipse are $(\pm a, 0)$ or $(\pm 5, 0)$.

Vertices of the hyperbola are $(\pm a, 0)$ or $(\pm 5, 0)$.

From the above discussions, they are common in centre and vertices.

58. Since the coordinates of foci of the hyperbola are (–5, 3) and (7, 3),

$$\therefore\quad 2ae=7-(-5)=12$$

$$\Rightarrow\qquad a=\frac{12\times 2}{3\times 2}=4\qquad[\because e=3/2]$$

$$e=\sqrt{1+\frac{b^2}{a^2}}$$

$$\Rightarrow\quad \frac{9}{4}-1=\frac{b^2}{16}$$

$$\Rightarrow\qquad b^2=20$$

Hence, the length of latus rectum $=\frac{2b^2}{a}=\frac{2\times 20}{4}=10$

59. The intersection points of the given curves are (1, 0) and $\left(-\frac{13}{5},-\frac{6}{5}\right)$.

$\therefore$ The distance between these two points

$$=\sqrt{\left(1+\frac{13}{5}\right)^2+\left(0+\frac{6}{5}\right)^2}$$

$$=\frac{1}{5}\sqrt{324+36}=\frac{6}{5}\sqrt{10}.$$

60. Any point of the hyperbola is $(a\sec\alpha, b\tan\alpha)$.

$\therefore\quad 3=a\sec\alpha$ and $0=b\tan\alpha$.

$\Rightarrow\quad a=3.$

Again $3\sqrt{2}=a\sec\beta$ and $2=b\tan\beta$

$\Rightarrow\quad \beta=45°$ and $b=2$

$\therefore$ Eccentricity $=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{1+\frac{4}{9}}=\frac{\sqrt{13}}{3}$

61. Product of the perpendiculars drawn from any point on a hyperbola to its asymptotes $=\frac{a^2b^2}{a^2+b^2}$.

62. $ae=2$

$e=2$

$$\therefore\quad a=1\quad\Rightarrow\quad b^2=1(4-1)$$

$$\Rightarrow\quad b^2=3\quad\Rightarrow\quad \frac{x^2}{1}-\frac{y^2}{3}=1$$

$$\Rightarrow\quad 3x^2-y^2=3$$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is

(a) 4/3 (b) $4/\sqrt{3}$

(c) $2/\sqrt{3}$ (d) none of these

2. The tangent at a point P on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ meets one of its directrices in F. If PF subtends an angle θ at the corresponding focus, then θ equals

(a) $\pi/4$ (b) $\pi/2$

(c) $3\pi/4$ (d) π

3. The tangents from $(1, 2\sqrt{2})$ to the hyperbola $16x^2 - 25y^2 = 400$ include between them an angle equal to

(a) $\pi/6$ (b) $\pi/4$

(c) $\pi/3$ (d) $\pi/2$

4. If a chord of a rectangular hyperbola, parallel to its conjugate axis, subtends angles θ_1 and θ_2 at its vertices, then

(a) $\theta_1+\theta_2=\frac{\pi}{2}$ (b) $\theta_1 + \theta_2 = \pi$

(c) $\theta_1+\theta_2=\frac{3\pi}{4}$ (d) none of these

5. If $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$ and $S(x_4, y_4)$ are four concyclic points on the rectangular hyperbola $xy = c^2$, the coordinates of the orthocentre of the ΔPQR are

(a) (x_4, y_4) (b) $(x_4, -y_4)$

(c) $(-x_4, -x_4)$ (d) $(-x_4, -y_4)$

6. If $(a\sec\theta, b\tan\theta)$ and $(a\sec\phi, b\tan\phi)$ are the ends of a focal chord of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, then $\tan\frac{\theta}{2}\tan\frac{\phi}{2}$ equals

(a) $\frac{e-1}{e+1}$ (b) $\frac{1-e}{1+e}$

(c) $\frac{1+e}{1-e}$ (d) $\frac{e+1}{e-1}$

7. The asymptotes of the hyperbola $xy = hx + ky$ are

(a) $x-k=0$ and $y-h=0$

(b) $x+h=0$ and $y+k=0$

(c) $x-k=0$ and $y+h=0$

(d) $x+k=0$ and $y-h=0$

8. The equation $16x^2-3y^2-32x-12y-44=0$ represents a hyperbola. Which one of the following is incorrect?

(a) the length of whose transverse axes is $2\sqrt{3}$

(b) the length of whose conjugate axes is 8

(c) whose centre is at $(1, -2)$

(d) whose eccentricity is $\sqrt{19}$

9. The combined equation of the asymptotes of the hyperbola $2x^2+5xy+2y^2+4x+5y=0$ is

(a) $2x^2+5xy+2y^2+4x+5y+2=0$

(b) $2x^2+5xy+2y^2+4x+5y-2=0$

(c) $2x^2+5xy+2y^2=0$

(d) none of these

10. The number of values of c such that the line $y = 2x + c$ meets the hyperbola $x^2 - y^2 = 1$ is

(a) 1 (b) 2

(c) 4 (d) infinite

11. P is a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. N is the foot of the perpendicular from P on the transverse axis. The tangent to the hyperbola at P meets the transverse axis at T. If O is the centre of the hyperbola, then $OT : ON$ is equal to

(a) e^2 (b) a^2

(c) b^2 (d) b^2/a^2

12. The value(s) of m for which $y = mx + 6$ is a tangent to the hyperbola $\frac{x^2}{100}-\frac{y^2}{49}=1$ is/are

(a) $\sqrt{\frac{17}{20}}$ (b) $\sqrt{\frac{20}{17}}$

(c) $-\sqrt{\frac{20}{17}}$ (d) $-\sqrt{\frac{17}{20}}$

13. If the chords of contact of the tangent from two points (x_1, y_1) and (x_2, y_2) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ are at right angles, then $\frac{x_1x_2}{y_1y_2}$ is equal to

(a) $-\frac{a^2}{b^2}$ (b) $-\frac{b^2}{a^2}$

(c) $-\frac{b^4}{a^4}$ (d) $-\frac{a^4}{b^4}$

14. The eccentricity of the hyperbola with latus rectum 12 and semi-conjugate axis $2\sqrt{3}$ is

(a) 2 (b) 3

(c) $\frac{\sqrt{3}}{2}$ (d) $2\sqrt{3}$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. The equation of the latus rectum of the hyperbola $(10x-5)^2+(10y-2)^2=9(3x+4y-7)^2$ is

(a) $y-1/5=-3/4(x-1/2)$

(b) $x-1/5=-3/4(y-1/2)$

(c) $y+1/5=-3/4(x+1/2)$

(d) $x+1/5=-3/4(y+1/2)$

2. The number of points from where a pair of perpendicular tangents can be drawn to the hyperbola $x^2\sec^2\alpha-y^2\operatorname{cosec}^2\alpha=1, \alpha\in(0,\pi/4)$ is

(a) 0 (b) 1

(c) 2 (d) infinite

3. A rectangular hyperbola whose centre is C is cut by any circle of radius r in four points P, Q, R and S. Then $CP^2+CQ+CR^2+CS^2=$

(a) r^2 (b) $2r^2$

(c) $3r^2$ (d) $4r^2$

4. If the angle between asymptotes of a hyperbola is α, then its eccentricity is equal to

(a) $\tan\frac{\alpha}{2}$ (b) $\cot\frac{\alpha}{2}$

(c) $\operatorname{cosec}\frac{\alpha}{2}$ (d) $\sec\frac{\alpha}{2}$

5. The angle between the tangents drawn from any point on the circle $x^2+y^2=3$ to the hyperbola $\frac{x^2}{4}-y^2=1$ is

(a) $\pi/3$ (b) $\pi/4$

(c) $\pi/2$ (d) $\pi/6$

6. The curve represented by $x=ae^{\theta}$, $y=be^{-\theta}$, $\theta\in\mathbb{R}$ is

(a) a hyperbola (b) an ellipse

(c) a parabola (d) a circle

7. The locus of the point of intersection of the straight lines $\frac{x}{a}+\frac{y}{b}=\lambda$ and $\frac{x}{a}-\frac{y}{b}=\frac{1}{\lambda}$ (λ is a variable) is

(a) a circle (b) a parabola

(c) an ellipse (d) a hyperbola

8. The line $lx+my+n=0$ will be a normal to the hyperbola $b^2x^2-a^2y^2=a^2b^2$ if

(a) $\frac{a^2}{l^2}+\frac{b^2}{m^2}=\frac{(a^2+b^2)^2}{n^2}$ (b) $\frac{a^2}{l^2}-\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$

(c) $\frac{a^2}{l^2}-\frac{b^2}{m^2}=\frac{(a^2+b^2)^2}{n^2}$ (d) none of these

9. A point moves in a plane so that its distances PA and PB from two fixed points A and B in the plane satisfy the relation $PA-PB=k$ ($k\neq 0$), then the locus of P is

(a) a parabola

(b) an ellipse

(c) a hyperbola

(d) a branch of a hyperbola

10. The curve represented by $x=a(\cosh\theta+\sinh\theta)$, $y=b(\cosh\theta-\sinh\theta)$ is

(a) a hyperbola (b) an ellipse

(c) a parabola (d) a circle

11. If the normal at $\left(ct,\frac{c}{t}\right)$ on the curve $xy=c^2$ meets the curve again at t, then

(a) $t'=-\frac{1}{t^3}$ (b) $t'=-\frac{1}{t}$

(c) $t'=\frac{1}{t^2}$ (d) $t'^2=-\frac{1}{t^2}$

12. If $x=9$ is the chord of contact of the hyperbola $x^2-y^2=9$, then the equation of the corresponding pair of tangents is

(a) $9x^2-8y^2+18x-9=0$

(b) $9x^2-8y^2-18x+9=0$

(c) $9x^2-8y^2-18x-9=0$

(d) $9x^2-8y^2+18x+9=0$

ANSWERS

Level 1

1. (c) **2.** (b) **3.** (d) **4.** (b) **5.** (d) **6.** (b) **7.** (a) **8.** (d) **9.** (a) **10.** (b)
11. (b) **12.** (a) and (b) **13.** (d) **14.** (a)

Level 2

1. (a) **2.** (d) **3.** (d) **4.** (c) and (d) **5.** (c) **6.** (a) **7.** (d) **8.** (c) **9.** (c)
10. (a) **11.** (a) **12.** (b)

23

W.R. HAMILTON

Vectors

- The Physical Quantities
- Representation of a Vector
- Types of Vectors
- Algebra of Vectors
- Properties of Multiplication of a Vector
- Linear Combination
- Components of a Vector
- Parallelogram Law of Vector Addition
- Triangle Law of Vector Addition
- Polygon Law of Vector Addition
- Properties of Vector Addition and Scalar Multiplication
- Position Vector of a Point
- Section (Ratio) Formula for Internal Division
- Section (Ratio) Formula for External Division
- Scalar Product (Dot Product)
- Properties of Scalar Product
- Projection of a Vector on a Line
- Properties of Cross Product
- Vector Triple Product
- Important Properties

☞ THE PHYSICAL QUANTITIES

The physical quantities may be divided into two groups: (i) scalars and (ii) vectors.

(i) **Scalar:** A scalar is a quantity having magnitude but no direction, for example, mass, length, time, temperature, volume, density, work, etc.

(ii) **Vector:** A vector is a quantity having both magnitude and direction, for example, force, velocity, acceleration, displacement, momentum, etc.

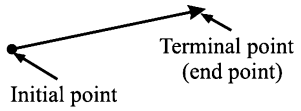


Fig. 23.1

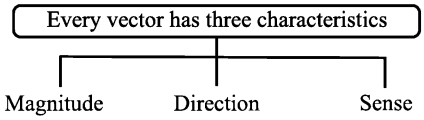


Fig. 23.2

☞ REPRESENTATION OF A VECTOR

A vector is represented by a directed line segment such that the length of the line segment is the magnitude of the vector and the direction of arrow marked at one end emphasizes the direction of the vector.

In Fig. 23.3, the directed line segment from A to B represents a vector and it is written as $\overrightarrow{AB}$. A is the initial point and B is the terminal point of the vector $\overrightarrow{AB}$. The modulus or module or magnitude of a vector $\overrightarrow{AB}$ is the positive number, which is the measure of its length and is denoted by $|\overrightarrow{AB}|$.

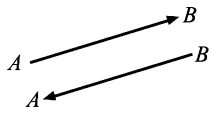


Fig. 23.3

1. Magnitude

The magnitude of $\overrightarrow{AB}$ is the measure of the length of vector $\overrightarrow{AB}$ and is denoted by $|\overrightarrow{AB}|$. So, $|\overrightarrow{AB}| = |\overrightarrow{BA}|$.

2. Direction

The direction of a vector $\overrightarrow{AB}$ is parallel to a given line.

3. Sense

Two parallel vectors $\overrightarrow{AB}$ and $\overrightarrow{BA}$ have different senses. The sense of vector $\overrightarrow{AB}$ is from A to B and that of $\overrightarrow{AB}$ is from B to A.

☞ TYPES OF VECTORS

1. Zero Vector or Null Vector

A vector whose initial and terminal points are coincident is called the **zero vector** or the **null vector**. The magnitude of the zero vector is zero and it can have any arbitrary direction and any line as its line of support.

2. Unit Vector

A vector whose modulus is unity is called a **unit vector**. The unit vector in the direction of a vector $\vec{a}$ is denoted by $\hat{a}$, read as "a cap". Thus, $|\hat{a}| = 1$.

3. Equal Vectors

Two vectors $\vec{a}$ and $\vec{b}$ which have equal magnitude, same direction (parallel) and same sense (arrow) are called **equal vectors**, which are written as $\vec{a} = \vec{b}$. They are

(i) of equal length
(ii) parallel vectors
(iii) of same sense

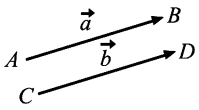


Fig. 23.4

4. Like Vectors

Vectors are said to be **like** when they have the same direction (parallel) and same sense (arrow), but their magnitudes may be different. They are

(i) parallel vectors
(ii) having same sense
(iii) vectors whose magnitudes may be different

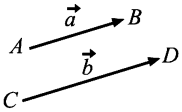


Fig. 23.5

5. Unlike Vectors

Vectors are said to be **unlike** when they have the same direction (parallel) and opposite sense (arrow), but their magnitudes may be different. They are

(i) parallel vectors
(ii) having opposite senses
(iii) vectors whose magnitudes may be different

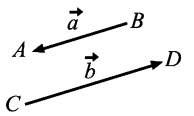


Fig. 23.6

6. Negative of Vector

The vector which has the same magnitude as the vector $\vec{a}$ but opposite direction is called the **negative** of $\vec{a}$ which is denoted by $-\vec{a}$.

Thus, if $\overrightarrow{AB} = \vec{a}$, then $\overrightarrow{BA} = -\vec{a}$.

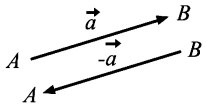


Fig. 23.7

7. Collinear Vectors

Vectors which are parallel to the same vector and the initial point of one vector coincides with the terminal point in common are called **collinear vectors**.

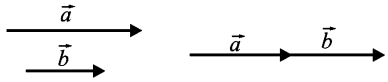


Fig. 23.8

8. Co-initial Vectors

Vectors having the same initial point are called **co-initial vectors**. As shown in Fig. 23.9, $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$ and $\overrightarrow{OD}$ are co-initial vectors.

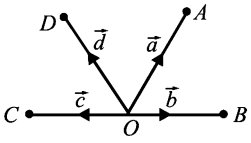


Fig. 23.9

9. Co-terminus Vectors

Co-terminus vectors are those vectors which have a common terminal point.

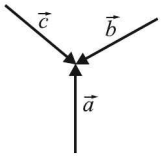


Fig. 23.10

10. Free Vectors

Vectors whose initial points are not specified are called **free vectors**.

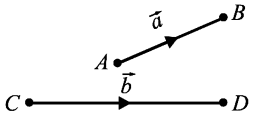


Fig. 23.11

11. Localized Vectors

A vector drawn parallel to a given vector, but through a specified point as initial point, is called **localized vector**.

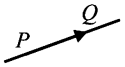


Fig. 23.12

☞ ALGEBRA OF VECTORS

1. Addition of Vectors

Let $\vec{a}$ and $\vec{b}$ be any two vectors. Then their **sum** or **resultant** is denoted by $\vec{a} + \vec{b}$.

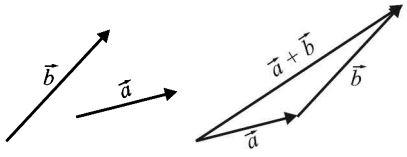


Fig. 23.13

2. Vector Subtraction

Vector subtraction is a special case of vector addition. Subtraction of a vector from another is performed by adding the corresponding negative vector, that is, if we seek $\vec{a} - \vec{b}$, we form $\vec{a} + (-\vec{b})$.

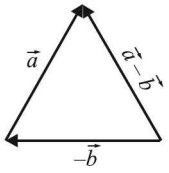


Fig. 23.14

Note If $\vec{a}$ and $\vec{b}$ are collinear vectors, then

- $\vec{a} + \vec{b}$ is a vector collinear with both the vectors $\vec{a}$ and $\vec{b}$, its magnitude $|\vec{a}+\vec{b}| = |\vec{a}|+|\vec{b}|$ if they have the same direction.
- If $\vec{a}$ and $\vec{b}$ have opposite directions, then the magnitude of $|\vec{a}+\vec{b}|$ is $|\vec{a}|-|\vec{b}|$ or $|\vec{b}|-|\vec{a}|$ depending upon $|\vec{a}|>|\vec{b}|$ or $|\vec{a}|<|\vec{b}|$.

MULTIPLICATION OF A VECTOR BY A SCALAR

Let n be a scalar and $\vec{a}$ be any vector. The product of a vector $\vec{a}$ and the number n ($n\vec{a}$) is a vector, whose magnitude is n times that of $\vec{a}$ and the direction is the same as $\vec{a}$. If n be a negative number, then the direction of $n\vec{a}$ is opposite to $\vec{a}$.

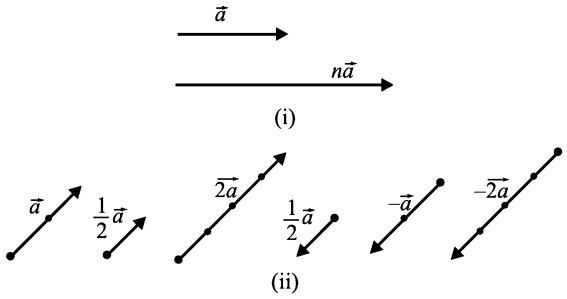


Fig. 23.15

PROPERTIES OF MULTIPLICATION OF A VECTOR BY A SCALAR

(i) $m(n\vec{a}) = (mn\vec{a})$

(ii) $(m + n)\vec{a} = m\vec{a} + n\vec{a}$

(iii) $m(\vec{a} - \vec{b}) = m\vec{a} - m\vec{b}$

(iv) $1 \cdot \vec{a} = \vec{a}$

If $\vec{a}$ is a vector, then $2\vec{a}$ is a vector whose magnitude is 2 times the magnitude of $\vec{a}$ and whose direction is the same as that of $\vec{a}$. $-5\vec{a}$ is a vector whose magnitude is 5 times the magnitude of $\vec{a}$ and whose direction is opposite to that of $\vec{a}$.

LINEAR COMBINATION

A vector r is called a linear combination of vectors $a, b, c, \ldots,$ if r can be expressed as the sum of the scalar multiple of these vectors as follows:

$\vec{r} = l\vec{a} + m\vec{b} + n\vec{c} + \cdots$, where l, m, n are scalars.

COMPONENTS OF A VECTOR (RECTANGULAR RESOLUTION OF A VECTOR)

Let Ox, Oy and Oz be the three rectangular axes. Let $\hat{i}$, $\hat{j}$ and $\hat{k}$ be three unit vectors and parallel to x, y and z axes, respectively. If $\overrightarrow{OP} = \vec{r}$ and the coordinates of P be (x, y, z), $\overrightarrow{OA} = x\hat{i}$, $\overrightarrow{OB} = y\hat{j}$, $\overrightarrow{OC} = z\hat{k}$.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}\vec{r} \qquad \text{(i)}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

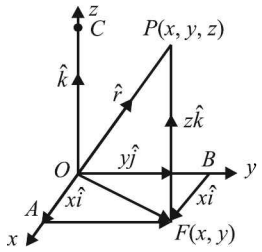


Fig. 23.16

1. **Direction ratio**: (x, y, z) are direction ratios of $\vec{r}$ if $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.
2. **Direction cosines** are $\cos\alpha = \frac{x}{|\vec{r}|}$, $\cos\beta = \frac{y}{|\vec{r}|}$ and $\cos\gamma = \frac{z}{|\vec{r}|}$, where $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$.

   $\therefore \quad \hat{r} = \cos\alpha\hat{i} + \cos\beta\hat{j} + \cos\gamma\hat{k}$

   $\therefore \quad \vec{r} = |\vec{r}|(\cos\alpha\hat{i} + \cos\beta\hat{j} + \cos\gamma\hat{k})$
3. **Unit vector** in a direction of r is

   $\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x}{|\vec{r}|}\hat{i} + \frac{y}{|\vec{r}|}\hat{j} + \frac{z}{|\vec{r}|}\hat{k}$, where $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$

PARALLELOGRAM LAW OF VECTOR ADDITION

The sum can also be obtained by bringing the initial points of $\vec{a}$ and $\vec{b}$ together and then completing the parallelogram $OABC$.

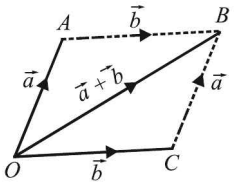


Fig. 23.17

TRIANGLE LAW OF VECTOR ADDITION

Given two vectors $\vec{a}$ and $\vec{b}$, their sum or resultant written as $(\vec{a} + \vec{b})$ is a vector obtained by joining the initial point of $\vec{a}$ to the terminal point of $\vec{b}$ giving a consistent direction by completing the triangle ΔOAB.

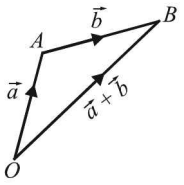


Fig. 23.18

POLYGON LAW OF VECTOR ADDITION

For adding more than two vectors, we have a polygon law of addition which is just an extension of the triangle law.

$\overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EF} = \overrightarrow{OF}$.

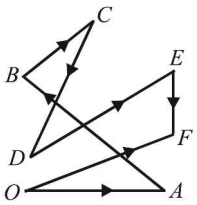


Fig. 23.19

Note If $(n - 1)$ sides of a polygon represent vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \ldots$, in consecutive order, the nth side represents their sum.

PROPERTIES OF VECTOR ADDITION AND SCALAR MULTIPLICATION

1. $\vec{a}+\vec{b}=\vec{b}+\vec{a}$ (commutativity)
2. $\vec{a}+(\vec{b}+\vec{c})=(\vec{a}+\vec{b})+\vec{c}$ (associativity)
3. $\vec{a}+\vec{0}=\vec{a}$ (additive identity)
4. $\vec{a}+(-\vec{a})=\vec{0}$ (additive inverse)
5. $(k_1+k_2)\vec{a}=k_1\vec{a}+k_2\vec{a}$ (multiplication by scalars)
6. $k(\vec{a}+\vec{b})=k\vec{a}+k\vec{b}$ (multiplication by scalars)
7. (i) $|\vec{a}+\vec{b}| \le |\vec{a}|+|\vec{b}|$ (ii) $|\vec{a}-\vec{b}| \ge |\vec{a}|-|\vec{b}|$

POSITION VECTOR OF A POINT

Consider a point $P(x, y, z)$ in space. $\overrightarrow{OP}$ is a vector with point at O and terminal point P. The directed line segment OP which fixes P uniquely is called the **position vector** of P in three-dimensional space.

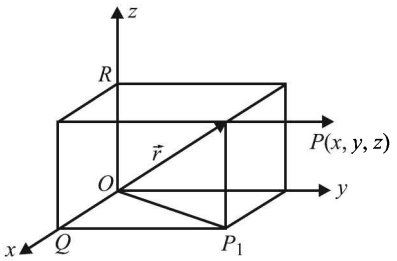


Fig. 23.20

(i) The vector from the origin to the point $P(x, y, z)$ is given by $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$. This is the position vector of the point P denoted as $P(\vec{r})$. $\overrightarrow{OP} = \sqrt{x^2 + y^2 + z^2} = |\vec{r}|$.

(ii) Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two points, then the length is given as $\overrightarrow{AB} = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$.

(iii) Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two points, then the position vector of AB = (p.v. of B) – (p.v. of A)

$\Rightarrow AB = (x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$

$\Rightarrow AB = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

Example: Vector with initial point $P(-4, 2)$ and terminal point $Q(0, -4)$ is $\overrightarrow{PQ} = (0+4)\hat{i} + (-4-2)\hat{j} = 4\hat{i} - 6\hat{j}$.

SECTION (RATIO) FORMULA FOR INTERNAL DIVISION

If A and B are two points with position vectors $\vec{a}$ and $\vec{b}$ respectively and let P be a point dividing AB internally in the ratio $m : n$. Then, the position vector of P is given by

$$\overrightarrow{OP} = \frac{m\vec{b} + n\vec{a}}{(m+n)}.$$

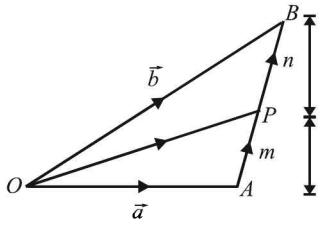


Fig. 23.21

Note If the point P is the mid-point, then it has position vector $\frac{\vec{a}+\vec{b}}{2}$.

SECTION (RATIO) FORMULA FOR EXTERNAL DIVISION

If A and B are two points with position vectors $\vec{a}$ and $\vec{b}$ respectively and let P be a point dividing AB externally in the ratio $m : n$. Then, the position vector of P is given by

$$\overrightarrow{OP} = \frac{m\vec{b} - n\vec{a}}{(m-n)}.$$

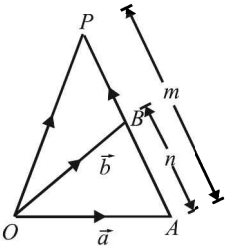


Fig. 23.22

SCALAR PRODUCT (DOT PRODUCT)

Consider two vectors $\vec{a}$ and $\vec{b}$ and let θ be the angle between these vectors. The scalar product of $\vec{a}$ and $\vec{b}$ is defined as

$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta=ab\cos\theta$$

θ between a and b is the angle between their directions as shown ($0\le\theta\le\pi$).

PROPERTIES OF SCALAR PRODUCT

Since $a\cdot b$ is a scalar quantity and hence it can be a positive (θ is acute) or a negative (θ is obtuse) real number.

1. If $\vec{a}=0$ or $\vec{b}=0$ or both $\vec{a}=0$ and $\vec{b}=0$, we define $\vec{a}\cdot\vec{b}=0$.
2. If $\vec{a}$ and $\vec{b}$ are orthogonal (perpendicular) vectors, we have $\theta=\frac{\pi}{2}$, $\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\frac{\pi}{2}=0$.
3. If $\vec{a}$ and $\vec{b}$ are like vectors, we have $\theta=0$,
$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos0^\circ=ab.$$
4. If $\vec{a}$ and $\vec{b}$ are unlike vectors, we have $\theta=\pi$,
$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\pi=-ab.$$
5. Scalar product is used to calculate the angle between vectors. Thus,
$$\cos\theta=\left|\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right| \quad\Rightarrow\quad \theta=\cos^{-1}\left|\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right|$$
6. $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{a}$ (commutative property)
$\vec{a}\cdot(\vec{b}+\vec{c})=\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}$ (distributive property)
$(l\vec{a})\cdot(m\vec{b})=lm(\vec{a}\cdot\vec{b})$ (where l, m are scalars)
7. $\vec{a}\cdot\vec{a}=|\vec{a}|^2=a^2=\vec{a}^2$
$(\vec{a}\pm\vec{b})^2=(\vec{a}\pm\vec{b})(\vec{a}\pm\vec{b})$
$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=a^2-b^2$
$(\vec{a}+\vec{b}+\vec{c})^2=a^2+b^2+c^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$
8. $\hat{i}\cdot\hat{i}=\hat{j}\cdot\hat{j}=\hat{k}\cdot\hat{k}=1$
$\hat{i}\cdot\hat{j}=\hat{j}\cdot\hat{k}=\hat{k}\cdot\hat{i}=0$
9. If $\vec{a}=a_x\hat{i}+a_y\hat{j}+a_z\hat{k}$ and $\vec{b}=b_x\hat{i}+b_y\hat{j}+b_z\hat{k}$
$\vec{a}\cdot\vec{b}=a_xb_x+a_yb_y+a_zb_z$, a_x: X_{comp} of $\vec{a}$ = projection of $\vec{a}$ on the x-axis.

Memory Tips

•	$\hat{i}$	$\hat{j}$	$\hat{k}$
$\hat{i}$	1	0	0
$\hat{j}$	0	1	0
$\hat{k}$	0	0	1

Fig. 23.23

PROJECTION OF A VECTOR ON ANOTHER VECTOR

Let two vectors be $\vec{a}$ and $\vec{b}$ and θ be the angle between them.

Projection of $\vec{a}$ on $\vec{b}$ $=\frac{(\vec{a}\cdot\vec{b})}{|\vec{b}|}$

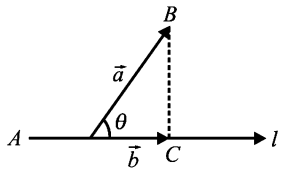


Fig. 23.24

Projection of $\vec{b}$ on $\vec{a}$ $=\frac{(\vec{a}\cdot\vec{b})}{|\vec{a}|}$

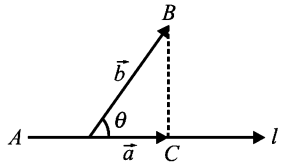


Fig. 23.25

Note If $\theta=\frac{\pi}{2}$ or $\theta=\frac{3\pi}{2}$, then the projection vector will be a zero vector.

VECTOR PRODUCT OF TWO VECTORS (CROSS PRODUCT)

The vector product of two vectors $\vec{a}$ and $\vec{b}$ is a vector quantity and is defined as $\vec{a}\times\vec{b}=|\vec{a}||\vec{b}|\sin\theta\hat{n}$, where θ is the angle between $\vec{a}$ and $\vec{b}$ and $\hat{n}$ is a unit vector denoting the direction of the axis.

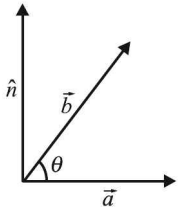


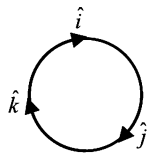
Fig. 23.26

If either $\vec{a}=\vec{0}$ or $\vec{b}=\vec{0}$, then θ is not defined and in this case $\vec{a}\times\vec{b}=0$.

PROPERTIES OF CROSS PRODUCT

1. $\vec{a}\times\vec{b}$ is a vector perpendicular to the plane containing $\vec{a}$ and $\vec{b}$.

2. If $\vec{a}$ and $\vec{b}$ are parallel, then the angle between them is π and $\vec{a}\times\vec{b}=|\vec{a}||\vec{b}|\sin\pi\hat{n}=\vec{0}$, therefore, $\vec{a}\times\vec{b}=\vec{0}$ ($\vec{a}$ and $\vec{b}$ are non-zero vectors) if $\vec{a}$ and $\vec{b}$ are parallel to each other, i.e. $\vec{a}\times\vec{b}=\vec{0} \Rightarrow \vec{a}\parallel\vec{b}$.
3. If $\theta=\frac{\pi}{2}$, then $\vec{a}\times\vec{b}=|\vec{a}||\vec{b}|\sin\frac{\pi}{2}\cdot\hat{n}=|\vec{a}||\vec{b}|\hat{n}$
4. $\hat{i}\times\hat{i}=\hat{j}\times\hat{j}=\hat{k}\times\hat{k}=\vec{0}$, $\hat{i}\times\hat{j}=\hat{k}$, $\hat{j}\times\hat{k}=\hat{i}, \hat{k}\times\hat{i}=\hat{j}$



Memory Tips

X	$\hat{i}$	$\hat{j}$	$\hat{k}$
$\hat{i}$	$\vec{0}$	$\hat{k}$	$-\hat{j}$
$\hat{j}$	$-\hat{k}$	$\vec{0}$	$\hat{i}$
$\hat{k}$	$\hat{j}$	$-\hat{i}$	$\vec{0}$

Fig. 23.27

5. $\vec{a}\times\vec{b}=-\vec{b}\times\vec{a} \Rightarrow \hat{j}\times\hat{i}=-\hat{k}, \hat{k}\times\hat{j}=-\hat{i}, \hat{i}\times\hat{k}=-\hat{j}$
6. Distributive: $\vec{a}\times(\vec{b}+\vec{c})=\vec{a}\times\vec{b}+\vec{a}\times\vec{c}$

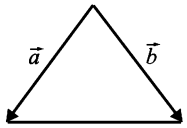


Fig. 23.28

7. If $\vec{a}$ and $\vec{b}$ represent the adjacent sides of a triangle, then the area of the triangle is given as $\frac{1}{2}|\vec{a}\times\vec{b}|$.
8. If $\vec{a}$ and $\vec{b}$ represent the adjacent sides of a parallelogram, then the area of the parallelogram is given as $|\vec{a}\times\vec{b}|$.

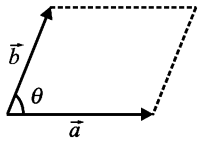


Fig. 23.29

9. If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$, then

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}$$

VECTOR TRIPLE PRODUCT

$$\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}$$

(a) Vector product is non-associative.

(b) $\vec{a}\times(\vec{b}\times\vec{c})$ is coplanar with $\vec{b}$ and $\vec{c}$ and perpendicular to $\vec{a}$.

SCALAR TRIPLE PRODUCT

If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$, $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$, then their scalar product is defined as

$$\vec{a}\cdot(\vec{b}\times\vec{c})=\begin{vmatrix}a_1&a_2&a_3\\b_1&b_2&b_3\\c_1&c_2&c_3\end{vmatrix}=[\vec{a}\ \vec{b}\ \vec{c}]$$

Geometrically this represents the volume of the parallelopiped whose edges are along $\vec{a}$, $\vec{b}$ and $\vec{c}$.

Properties of Scalar Triple Product

1. If any two vectors are same in $\vec{a}$, $\vec{b}$ and $\vec{c}$ or $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar, then $[\vec{a}\ \vec{b}\ \vec{c}]=0$.
2. $[\vec{a}\ \vec{b}\ \vec{c}]=[\vec{b}\ \vec{c}\ \vec{a}]=[\vec{c}\ \vec{a}\ \vec{b}]$
3. $\vec{a}\cdot(\vec{b}\times\vec{c})=(\vec{b}\times\vec{c})\cdot\vec{a}$
4. $\vec{a}\cdot(\vec{b}\times\vec{c})=-\vec{a}\cdot(\vec{c}\times\vec{b})$
5. $[\lambda\vec{a}\ \vec{b}\ \vec{c}]=\lambda[\vec{a}\ \vec{b}\ \vec{c}]$, where λ is a scalar.

RECIPROCAL SYSTEM OF VECTORS

If $\vec{a},\vec{b}$ and $\vec{c}$ are the non-zero, non-collinear and non-coplanar vectors, then the three vectors $\vec{a}',\vec{b}'$ and $\vec{c}'$ such that

$$\vec{a}'=\frac{\vec{b}\times\vec{c}}{[\vec{a}\vec{b}\vec{c}]},\ \vec{b}'=\frac{\vec{c}\times\vec{a}}{[\vec{a}\vec{b}\vec{c}]} \text{ and } \vec{c}'=\frac{\vec{a}\times\vec{b}}{[\vec{a}\vec{b}\vec{c}]}$$

are called **reciprocal system of vectors** for vectors $\vec{a},\vec{b}$ and $\vec{c}$.

Properties of Reciprocal System of Vectors

1. $\vec{a}\cdot\vec{a}'=\vec{b}\cdot\vec{b}'=\vec{c}\cdot\vec{c}'=1$
2. $\vec{a}\cdot\vec{b}'=\vec{a}\cdot\vec{c}'=\vec{b}\cdot\vec{a}'=\vec{b}\cdot\vec{c}'=\vec{c}\cdot\vec{a}'=\vec{c}\cdot\vec{b}'=0$
3. $[\vec{a}'\vec{b}'\vec{c}']=\frac{1}{[\vec{a}\vec{b}\vec{c}]}$
4. The vectors $\vec{a}',\vec{b}'$ and $\vec{c}'$ are non-zero, non-collinear and non-coplanar.
5. $\vec{r}=(\vec{r}\cdot\vec{a})\vec{a}'+(\vec{r}\cdot\vec{b})\vec{b}'+(\vec{r}\cdot\vec{c})\vec{c}'$
   $=(\vec{r}\cdot\vec{a}')\vec{a}+(\vec{r}\cdot\vec{b}')\vec{b}+(\vec{r}\cdot\vec{c}')\vec{c}$

LINEARLY DEPENDENT VECTORS

The set of coplanar vectors is referred to as the set of linearly dependent vectors.

LINEARLY INDEPENDENT VECTORS

The set of non-coplanar vectors is referred to as the set of linearly independent vectors.

SOLVED PROBLEMS

1. The values of x for which the angle between the vectors $\vec{a} = x\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = 2x\hat{i} + x\hat{j} - \hat{k}$ is acute, and the angle between the vectors $\vec{b}$ and the y-axis of ordinates is obtuse, are

(a) 1, 2 (b) –2, –3

(c) all $x < 0$ (d) all $x > 0$

Ans. (c)

Solution: Since the angle between the vectors $\vec{a}$ and $\vec{b}$ is acute and the angle between $\vec{b}$ and the y-axis is obtuse, therefore, $\vec{a}\cdot\vec{b} > 0$ and $\vec{b}\cdot\hat{j} < 0$,

$\Rightarrow 2x^2 - 3x + 1 > 0$ and $x < 0$

$\Rightarrow (2x-1)(x-1) > 0$ and $x < 0$

$\Rightarrow \left(x < \frac{1}{2} \text{ or } x > 1\right)$ and $x < 0 \Rightarrow$ all $x < 0$

2. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-zero vectors such that $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c}$, then

(a) $\vec{b} = \vec{c}$

(b) $\vec{a} \perp \vec{b}, \vec{c}$

(c) $\vec{a} \perp \vec{b} - \vec{c}$

(d) either $\vec{a} \perp \vec{b} - \vec{c}$ or $\vec{b} = \vec{c}$

Ans. (d)

Solution: Since $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c}$

$\Rightarrow \vec{a}(\vec{b} - \vec{c}) = 0$

either $\vec{a} \perp (\vec{b} - \vec{c})$ or $(\vec{b} - \vec{c}) = 0$

$\therefore$ $\vec{a} \perp (\vec{b} - \vec{c})$ or $\vec{b} = \vec{c}$

3. The values of x for which the angle between $\vec{a} = 2x^2\hat{i} + 4x\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + x\hat{k}$ is obtuse and the angle between $\vec{b}$ and the z-axis is acute and less than $\pi/6$, are

(a) $0 < x < \frac{1}{2}$ (b) $\frac{1}{2} < x < 15$

(c) $x > \frac{1}{2}$ or $x < 0$ (d) none of these

Ans. (d)

Solution: The angle between $\vec{a}$ and $\vec{b}$ is obtuse.

$\therefore$ $\vec{a}\cdot\vec{b} < 0 \Rightarrow 14x^2 - 8x + x < 0$

$\Rightarrow 7x(2x-1) < 0 \Rightarrow 0 < x < \frac{1}{2}$ (i)

$\because$ The angle between $\vec{b}$ and the z-axis is acute and less than $\pi/6$,

$$\frac{\vec{b}\cdot\vec{k}}{|\vec{b}||\vec{k}|} > \cos\frac{\pi}{6}$$

$$\left[\because \theta < \frac{\pi}{6} \Rightarrow \cos\theta > \cos\frac{\pi}{6} \Rightarrow \frac{\vec{b}\cdot\vec{k}}{|\vec{b}||\vec{k}|} > \cos\frac{\pi}{6}\right]$$

$$\Rightarrow \frac{x}{\sqrt{x^2+53}} > \frac{\sqrt{3}}{2} \Rightarrow 4x^2 > 3x^2 + 159$$

$\Rightarrow x^2 > 159 \Rightarrow x > \sqrt{159}$ or $x < -\sqrt{159}$ (ii)

Clearly, (i) and (ii) cannot hold together.

4. Let $\vec{p}, \vec{q}, \vec{r}$ be three mutually perpendicular vectors of the same magnitude. If a vector $\vec{x}$ satisfies the equation $\vec{p}\times\{\vec{x} - \vec{q}\times\vec{p}\} + \vec{q}\times\{\vec{x} - \vec{r}\times\vec{q}\} + \vec{r}\times\{\vec{x} - \vec{p}\times\vec{r}\} = \vec{0}$, then $\vec{x}$ is given by

(a) $\frac{1}{2}(\vec{p} + \vec{q} - 2\vec{r})$ (b) $\frac{1}{2}(\vec{p} + \vec{q} + \vec{r})$

(c) $\frac{1}{3}(\vec{p} + \vec{q} + \vec{r})$ (d) $\frac{1}{3}(2\vec{p} + \vec{q} - \vec{r})$

Ans. (b)

Solution: We have

$$\vec{p}\times\{\vec{x} - \vec{q}\times\vec{p}\} + \vec{q}\times\{\vec{x} - \vec{r}\times\vec{q}\} + \vec{r}\times\{\vec{x} - \vec{p}\times\vec{r}\} = \vec{0}$$

$$\Rightarrow (\vec{p}\cdot\vec{p})(\vec{x} - \vec{p}) - \{\vec{p}(\vec{x} - \vec{q})\}\vec{p} + (\vec{q}\cdot\vec{q})(\vec{x} - \vec{r})$$
$$- \{(\vec{q}\cdot(\vec{x} - \vec{r})\}\vec{q} + (\vec{r}\cdot\vec{r})(\vec{x} - \vec{p}) - \{\vec{r}\cdot(\vec{x} - \vec{p})\}\vec{r} = 0$$

$$\Rightarrow \left(|\vec{p}|^2 = |\vec{q}|^2 + |\vec{r}|^2\right)\vec{x} - \left(|\vec{p}|^2\vec{q} + |\vec{q}|^2\vec{r} + |\vec{r}|^2\vec{p}\right)$$
$$- [(\vec{p}\cdot\vec{x})\vec{p} + [(\vec{q}\cdot\vec{x})\vec{q} + (\vec{r}\cdot\vec{x})\vec{r}] = 0$$

$$\Rightarrow 3\lambda^2\vec{x} - \lambda^2(\vec{p} + \vec{q} + \vec{r}) - \{(\vec{p}\cdot\vec{x})\vec{p} + (\vec{q}\cdot\vec{x})\vec{q}$$
$$+ (\vec{r}\cdot\vec{x})\vec{r}\} = 0 \quad \text{(i)}$$

where $\lambda = |\vec{p}| = |\vec{q}| = |\vec{r}|$

Taking dot product with $\vec{p}$, we get

$$3\lambda^2(\vec{x}\cdot\vec{p}) - \lambda^2|\vec{p}|^2 - (\vec{p}\cdot\vec{x})|\vec{p}|^2 = 0 \quad [\because \vec{p}\cdot\vec{q} = \vec{p}\cdot\vec{r} = 0]$$

$$\Rightarrow 3\lambda^2(\vec{x}\cdot\vec{p}) - \lambda^4 - \lambda^2(\vec{p}\cdot\vec{x}) = 0$$

$$\Rightarrow \vec{p}\cdot\vec{x} = \frac{\lambda^2}{2}$$

Similarly, $\vec{q}\cdot\vec{x} = \frac{\lambda^2}{2}$ and $\vec{r}\cdot\vec{x} = \frac{\lambda^2}{2}$.

Substituting these values in (i), we get

$$3\lambda^2\vec{x} - \lambda^2(\vec{p} + \vec{q} + \vec{r}) - \frac{\lambda^2}{2}(\vec{p} + \vec{q} + \vec{r}) = \vec{0}$$

$$\Rightarrow \vec{x} = \frac{1}{2}(\vec{p} + \vec{q} + \vec{r})$$

5. The vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are of the same length and, taken pairwise, they form equal angles. If $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$, then $\vec{c}$ is equal to

(a) $\hat{i} + \hat{k}$ (b) $\hat{i} + 2\hat{j} + 3\hat{k}$

(c) $-\hat{i} + \hat{j} + 2\hat{k}$ (d) none of these

Ans. (a)

Solution: $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$. Then, $|\vec{a}| = |\vec{b}| = |\vec{c}|$

$$\Rightarrow \quad x^2 + y^2 + z^2 = 2 \qquad \text{(i)}$$

It is given that the angles between the vectors taken in pairs are equal, say θ. Therefore,

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{0+1+0}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$$

Similarly $\dfrac{\vec{a}\cdot\vec{c}}{|\vec{a}||\vec{c}|} = \dfrac{1}{2}$ and $\dfrac{\vec{b}\cdot\vec{c}}{|\vec{b}||\vec{c}|} = \dfrac{1}{2}$

$$\Rightarrow \frac{x+y}{\sqrt{2}\cdot\sqrt{2}} = \frac{1}{2} \text{ and } \frac{y+z}{\sqrt{2}\cdot\sqrt{2}} = \frac{1}{2}$$

$$\Rightarrow x + y = 1 \text{ and } y + z = 1$$

$$\Rightarrow y = 1 - x \text{ and } z = 1 - y = 1 - (1 - x) = x$$

$$\therefore \quad x^2 + y^2 + z^2 = 2$$

$$\Rightarrow 3x^2 - 2x - 1 = 0 \Rightarrow x = 1, -\frac{1}{3}$$

Now, $y = 1 - x \Rightarrow y = 0$ for $x = 1$ and $y = \dfrac{1}{4}$ for $x = -\dfrac{1}{3}$.

Hence, $\vec{c} = \hat{i} + 0\hat{j} + \hat{k}$ and $\vec{c} = -\dfrac{1}{3}\hat{i} + \dfrac{4}{3}\hat{j} - \dfrac{1}{3}\hat{k}$.

6. If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the vertices of an equilateral triangle whose orthocentre is at the origin, then

(a) $\vec{a} + \vec{b} + \vec{c} = 0$ (b) $\vec{a}^2 = \vec{b}^2 + \vec{c}^2$

(c) $\vec{a} + \vec{b} - \vec{c}$ (d) none of these

Ans. (a)

Solution: The position vector of the centroid of the triangle is $\dfrac{\vec{a} + \vec{b} + \vec{c}}{3}$.

Since the triangle is equilateral, therefore the orthocentre coincides with the centroid and hence $\dfrac{\vec{a} + \vec{b} + \vec{c}}{3} = 0$.

$$\therefore \quad \vec{a} + \vec{b} + \vec{c} = 0$$

7. The length of the longer diagonal of the parallelogram constructed on $5\vec{a} + 2\vec{b}$ and $\vec{a} - 3\vec{b}$, if it is given that $|\vec{a}| = 2\sqrt{2}$, $|\vec{b}| = 3$ and the angle between $\vec{a}$ and $\vec{b} = \dfrac{\pi}{4}$, is

(a) 15 (b) $\sqrt{113}$

(c) $\sqrt{593}$ (d) $\sqrt{369}$

Ans. (c)

Solution: The diagonals of the parallelogram are

$$\vec{\alpha} = 5\vec{a} + 2\vec{b} + \vec{a} - 3\vec{b} = 6\vec{a} - \vec{b}$$

$$\vec{\beta} = \pm(4\vec{a} + 5\vec{b})$$

Now, $|\vec{\alpha}| = |6\vec{a} - \vec{b}| = \sqrt{36|\vec{a}|^2 + |\vec{b}|^2 - 12(\vec{a}\cdot\vec{b})}$

$$= \sqrt{36\times 8 + 9 - 12\times 2\sqrt{2}\times 3\times\frac{1}{\sqrt{2}}} = 15$$

and $|\vec{\beta}| = |4\vec{a} + 5\vec{b}| = \sqrt{16|\vec{a}|^2 + 25|\vec{b}|^2 + 40(\vec{a}\cdot\vec{b})}$

$$= \sqrt{16\times 8 + 25\times 9 + 40\times 2\sqrt{2}\times 3\times\frac{1}{\sqrt{2}}} = \sqrt{593}.$$

8. If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors of equal magnitude, then the angle between $\vec{a} + \vec{b} + \vec{c}$ and $\vec{a}$ is

(a) $\cos^{-1}\left(\dfrac{1}{3}\right)$ (b) $\cos^{-1}\left(\dfrac{1}{\sqrt{3}}\right)$

(c) 0 (d) none of these

Ans. (b)

Solution: Let $\vec{a} = \hat{i}, \vec{b} = \hat{j}, \vec{c} = \hat{k}$

$$\therefore \quad (\vec{a} + \vec{b} + \vec{c})\cdot\vec{a} = |\vec{a} + \vec{b} + \vec{c}||\vec{a}|\cos\theta$$

$$\text{or,} \quad (\hat{i} + \hat{j} + \hat{k})\cdot\hat{i} = |\hat{i} + \hat{j} + \hat{k}||\hat{i}|\cos\theta$$

$$\text{or,} \quad 1 = \sqrt{3}\cos\theta$$

$$\therefore \quad \cos\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right).$$

9. The position vectors of three points are $2\vec{a} - \vec{b} + 3\vec{c}$, $\vec{a} - 2\vec{b} + \lambda\vec{c}$ and $\mu\vec{a} - 5\vec{b}$, where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors. The points are collinear when

(a) $\lambda = -2, \mu = \dfrac{9}{4}$ (b) $\lambda = -\dfrac{9}{4}, \mu = 2$

(c) $\lambda = \dfrac{9}{4}, \mu = -2$ (d) none of these

Ans. (c)

Solution: Let the points be A, B and C respectively, then

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$
$$= \vec{a} - 2\vec{b} + \lambda\vec{c} - (2\vec{a} - \vec{b} + 3\vec{c})$$
$$= -\vec{a} - \vec{b} + (\lambda - 3)\vec{c}$$
$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$$
$$= \mu\vec{a} - 5\vec{b} - (2\vec{a} - \vec{b} + 3\vec{c})$$
$$= (\mu - 2)\vec{a} - 4\vec{b} - 3\vec{c}$$

The points are collinear if $\overrightarrow{AB} = t\overrightarrow{AC}$

$\Rightarrow -1 = t(\mu - 2)$, $-1 = -4t$, $\lambda - 3 = -3t$

$\therefore \quad t = \frac{1}{4}$ and $-1 = \frac{1}{4}(\mu - 2)$, $\lambda - 3 = -\frac{3}{4}$

$\Rightarrow \mu = -2$, $\lambda = \frac{9}{4}$

10. The vector $\vec{a}$ is equal to

(a) $(\vec{a}\cdot\hat{i})\hat{i} + (\vec{a}\cdot\hat{j})\hat{j} + (\vec{a}\cdot\hat{k})\hat{k}$

(b) $(\vec{a}\cdot\hat{j})\hat{i} + (\vec{a}\cdot\hat{k})\hat{j} + (\vec{a}\cdot\hat{i})\hat{k}$

(c) $(\vec{a}\cdot\hat{k})\hat{i} + (\vec{a}\cdot\hat{i})\hat{j} + (\vec{a}\cdot\hat{j})\hat{k}$

(d) $(\vec{a}\cdot\vec{a})(\hat{i} + \hat{j} + \hat{k})$

Ans. (a)

Solution: Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$. Then,

$$\vec{a}\cdot\hat{i} = a_1, \vec{a}\cdot\hat{j} = a_2, \vec{a}\cdot\hat{k} = a_3$$

$\therefore \quad \vec{a} = (\vec{a}\cdot\hat{i})\hat{i} + (\vec{a}\cdot\hat{j})\hat{j} + (\vec{a}\cdot\hat{k})\hat{k}$.

11. A unit vector in the xy-plane that makes an angle of 45° with the vector $\hat{i} + \hat{j}$ and an angle of 60° with the vector $3\hat{i} + 4\hat{j}$ is

(a) $\hat{i}$ (b) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

(c) $\frac{\hat{i} - \hat{j}}{\sqrt{2}}$ (d) none of these

Ans. (d)

Solution: Let the unit vector be $\vec{p} = a\hat{i} + b\hat{j}$, so that $a^2 + b^2 = 1$.

Given, $(a\hat{i} + b\hat{j})\cdot(\hat{i} + \hat{j}) = |a\hat{i} + b\hat{j}||\hat{i} + \hat{j}|\cos 45°$

or, $a + b = \sqrt{a^2 + b^2}\sqrt{2}\cdot\frac{1}{\sqrt{2}} \quad (\because\ a^2 + b^2 = 1)$ (i)

and $(a\hat{i} + b\hat{j})\cdot(3\hat{i} + 4\hat{j}) = 1\times 5\times\cos 60°$

or, $3a - 4b = \frac{5}{2}$ (ii)

Solving (i) and (ii), we get $a = \frac{13}{14}, b = \frac{1}{14}$ but $a^2 + b^2 = 1$ is not satisfied.

12. If origin O is in the centre of the triangle ABC and a, b and c are the lengths of the sides, then the force $a\overrightarrow{OA} + b\overrightarrow{OB} + c\overrightarrow{OC} =$

(a) -1 (b) 2

(c) 0 (d) none of these

Ans. (c)

Solution:

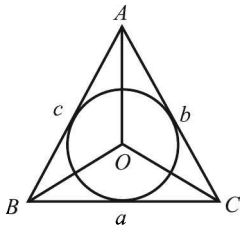


Fig. 23.30

If O is the origin, then the centre of ΔABC is given by

$$\frac{a\overrightarrow{OA} + b\overrightarrow{OB} + c\overrightarrow{OC}}{a + b + c} = 0$$

$\therefore \quad a\overrightarrow{OA} + b\overrightarrow{OB} + c\overrightarrow{OC} = 0$

13. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then $\left|\frac{\vec{a} - \vec{b}}{2}\right|$ is

(a) $\sin\frac{\theta}{2}$ (b) $\sin\theta$

(c) $2\sin\theta$ (d) $\sin 2\theta$

Ans. (a)

Solution: We have $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a}\cdot\vec{b}$

$\Rightarrow |\vec{a} - \vec{b}|^2 = 1 + 1 - 2|\vec{a}||\vec{b}|\cos\theta$

$\Rightarrow |\vec{a} - \vec{b}|^2 = 2 - 2\cos\theta$

$\Rightarrow |\vec{a} - \vec{b}|^2 = 4\sin^2\frac{\theta}{2} \Rightarrow \left|\frac{\vec{a} - \vec{b}}{2}\right|^2 = \sin^2\frac{\theta}{2}$

$\Rightarrow \left|\frac{\vec{a} - \vec{b}}{2}\right| = \sin\frac{\theta}{2}$.

14. A vector $\vec{a}$ has component $2p$ and 1 with respect to a rectangle Cartesian system. This system is rotated through a certain angle about the origin in the counter-clockwise sense. If with respect to the new system, $\vec{a}$ has components $p + 1$ and 1, then

(a) $p = 0$ (b) $p = 1$ or $p = -\frac{1}{3}$

(c) $p = -1$ or $p = \frac{1}{3}$ (d) $p = 1$ or $p = -1$

Ans. (b)

Solution: We have $\vec{a}=2p\hat{i}+\hat{j}$, after rotation $\vec{b}=(p+1)\hat{i}+\hat{j}$

Since rotation is made about the origin, $|\vec{a}|=|\vec{b}|$.

$$\Rightarrow\ 4p^2+1=(p+1)^2+1\ \Rightarrow\ p=1 \text{ or } -\frac{1}{3}$$

15. If the sides of a parallelogram are $2\hat{i}+4\hat{j}-5\hat{k}$ and $\hat{i}+2\hat{j}+3\hat{k}$, then the unit vector parallel to one of the diagonals is

(a) $\frac{1}{7}(3\hat{i}+6\hat{j}-2\hat{k})$ (b) $\frac{1}{7}(3\hat{i}-6\hat{j}-2\hat{k})$

(c) $\frac{1}{7}(-3\hat{i}+6\hat{j}-2\hat{k})$ (d) $\frac{1}{7}(3\hat{i}+6\hat{j}+2\hat{k})$

Ans. (a)

Solution: Let $\vec{a}=2\hat{i}+4\hat{j}-5\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}+3\hat{k}$.

Then the diagonals of the parallelogram are $\vec{p}=\vec{a}+\vec{b}$, $\vec{q}=\vec{b}-\vec{a}$, i.e. $\vec{p}=3\hat{i}+6\hat{j}-2\hat{k}$ and $\vec{q}=-\hat{i}-2\hat{j}+8\hat{k}$.

So, unit vectors along the diagonals are $\frac{1}{7}(3\hat{i}+6\hat{j}-2\hat{k})$ and $\frac{1}{\sqrt{69}}(-\hat{i}-2\hat{j}+8\hat{k})$.

16. The number of vectors of unit length perpendicular to the vectors $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=\hat{j}+\hat{k}$ is

(a) 1 (b) 2

(c) 4 (d) infinite

Ans. (b)

Solution: Required vectors are given by $\pm\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.

Hence, there are two vectors.

17. Let $\vec{a},\vec{b},\vec{c}$ be three non-coplanar vectors such that $\vec{r}_1=\vec{a}-\vec{b}+\vec{c}$, $\vec{r}_2=\vec{b}+\vec{c}-\vec{a}$, $\vec{r}_3=\vec{c}+\vec{a}+\vec{b}$, $\vec{r}=2\vec{a}-3\vec{b}+4\vec{c}$. If $\vec{r}=\lambda_1\vec{r}_1+\lambda_2\vec{r}_2+\lambda_3\vec{r}_3$, then

(a) $\lambda_1=7$ (b) $\lambda_1+\lambda_3=3$

(c) $\lambda_1+\lambda_2+\lambda_3=5$ (d) $\lambda_3+\lambda_2=2$

Ans. (b)

Solution:

$$\vec{r}=\lambda_1(\vec{a}-\vec{b}+\vec{c})+\lambda_2(\vec{b}+\vec{c}-\vec{a})+\lambda_3(\vec{c}+\vec{a}+\vec{b})$$

$$=\vec{a}(\lambda_1-\lambda_2+\lambda_3)+\vec{b}(-\lambda_1+\lambda_2+\lambda_3)+\vec{c}(\lambda_1+\lambda_2+\lambda_3)$$

also, $\vec{r}=2\vec{a}-3\vec{b}+4\vec{c}$

Comparing, we get

$$\lambda_1-\lambda_2+\lambda_3=2 \quad \text{(i)}$$

$$-\lambda_1+\lambda_2+\lambda_3=-3 \quad \text{(ii)}$$

$$\lambda_1+\lambda_2+\lambda_3=4 \quad \text{(iii)}$$

Adding (i) and (iii), we get $2(\lambda_1+\lambda_3)=6$

$\Rightarrow\ \lambda_1+\lambda_3=3$.

EXERCISE SET 1

1. If $\vec{u}=\vec{a}-\vec{b}$, $\vec{v}=\vec{a}+\vec{b}$ and $|\vec{a}|=|\vec{b}|=2$, then $|\vec{u}\times\vec{v}|$ is

(a) $2\sqrt{16-(\vec{a}\cdot\vec{b})^2}$ (b) $2\sqrt{4-(\vec{a}\cdot\vec{b})^2}$

(c) $\sqrt{16-(\vec{a}\cdot\vec{b})^2}$ (d) $\sqrt{4-(\vec{a}\cdot\vec{b})^2}$

2. The points with position vector $10\hat{i}+3\hat{j}$, $12\hat{i}-5\hat{j}$ and $a\hat{i}+11\hat{j}$ are collinear if a equals

(a) –8 (b) 4

(c) 8 (d) 12

3. Let $\vec{b}=4\hat{i}+3\hat{j}$ and $\vec{c}$ be two vectors perpendicular to each other in the xy-plane. Then a vector in the same plane having projections 1 and 2 along $\vec{b}$ and $\vec{c}$, respectively, is

(a) $\hat{i}+2\hat{j}$ (b) $2\hat{i}-\hat{j}$

(c) $2\hat{i}+\hat{j}$ (d) none of these

4. The number of vectors of unit length perpendicular to vectors $\vec{a}=(1,1,0)$ and $\vec{b}=(0,1,1)$ is

(a) one (b) two

(c) three (d) infinite

5. Two sides of a triangle are formed by the vectors $\vec{a}=3\hat{i}+6\hat{j}-2\hat{k}$ and $\vec{b}=4\hat{i}-\hat{j}+3\hat{k}$. The angles of the triangle are

(i) $\cos^{-1}\frac{7}{\sqrt{75}}$ (ii) $\cos^{-1}\sqrt{\frac{26}{75}}$

(iii) $\cos^{-1}\frac{3}{\sqrt{15}}$ (iv) $\cos^{-1}\frac{2}{3}$

(a) (i) and (iii) (b) (ii) and (iii)

(c) (iii) and (iv) (d) (i) and (ii)

6. If $\vec{a}=\hat{i}+\hat{j}$, $\vec{b}=\hat{j}+\hat{k}$ and $\vec{c}=\hat{k}+\hat{i}$, then the projection of vector $\vec{a}$ on vector $\vec{b}\times\vec{c}$ is

(a) $2\sqrt{3}$ (b) $2\sqrt{3}/3$

(c) $\sqrt{2}$ (d) none of these

7. If $\vec{a}$ and $\vec{b}$ are not perpendicular to each other and $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$, $\vec{r}\cdot\vec{c}=0$, then $\vec{r}$ is equal to

(a) $\vec{a}-\vec{c}$ (b) $\vec{b}+x\vec{a}$ for all scalars x

(c) $\vec{b}-\left(\frac{\vec{b}\cdot\vec{c}}{\vec{a}\cdot\vec{c}}\right)\cdot\vec{a}$ (d) none of these

8. The area of the triangle whose vertices are $\vec{a}, \vec{b}, \vec{c}$ is

(a) $\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|$ (b) $\frac{1}{2}\left|(\vec{b}-\vec{a})\times(\vec{c}-\vec{b})\right|$

(c) $2\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|$ (d) none of these

9. The area of the triangle whose two adjacent sides are $\vec{a} = 3\hat{i}+4\hat{j}$ and $\vec{b}=-5\hat{i}+7\hat{j}$ is

(a) 10 (b) 39

(c) 3 (d) 39/2

10. Let G be the centroid of ΔABC. If $\overrightarrow{AB}=\vec{a}$ and $\overrightarrow{AC}=\vec{b}$, then $\overrightarrow{AG}$, in terms of $\vec{a}$ and $\vec{b}$, is

(a) $\frac{2}{3}(\vec{a}+\vec{b})$ (b) $\frac{1}{6}(\vec{a}+\vec{b})$

(c) $\frac{1}{3}(\vec{a}+\vec{b})$ (d) $\frac{1}{2}(\vec{a}+\vec{b})$

11. If $\vec{a}, \vec{b}, \vec{c}$ are vectors such that $\vec{a}\cdot\vec{b}=0$ and $\vec{a}+\vec{b}=\vec{c}$, then

(a) $|\vec{a}|^2+|\vec{b}|^2=|\vec{c}|^2$ (b) $|\vec{a}|^2=|\vec{b}|^2+|\vec{c}|^2$

(c) $|\vec{b}|^2=|\vec{a}|^2+|\vec{c}|^2$ (d) none of these

12. If $\vec{a}\cdot\vec{b}=0$ and also $\vec{a}\times\vec{b}=0$, then

(a) $\vec{a}$ is parallel to $\vec{b}$

(b) $\vec{a}$ is perpendicular to $\vec{b}$

(c) either $\vec{a}$ or $\vec{b}$ is a non-zero vector

(d) none of these

13. If $\vec{r}=3\hat{i}+2\hat{j}-5\hat{k}$, $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$, $\vec{b}=\hat{i}+3\hat{j}-2\hat{k}$ and $\vec{c}=-2\hat{i}+\hat{j}-3\hat{k}$ such that $\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}$, then

(a) x, y, z are in A.P. (b) x, y, z are in G.P.

(c) x, y, z are in H.P. (d) $y, (x/2), z$ are in A.P.

14. The acute angle that the vector $2\hat{i}-2\hat{j}+\hat{k}$ makes with the plane contained by the two vectors $2\hat{i}+3\hat{j}-\hat{k}$ and $\hat{i}-\hat{j}+2\hat{k}$ is given by

(a) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

(c) $\tan^{-1}\left(\sqrt{2}\right)$ (d) $\cot^{-1}\left(\sqrt{2}\right)$

15. Let $\overrightarrow{AB}=3\hat{i}+\hat{j}-\hat{k}$ and $\overrightarrow{AC}=\hat{i}-\hat{j}+3\hat{k}$ and a point P on the line segment BC be equidistant from AB and AC, then $\overrightarrow{AP}$ is

(a) $2\hat{i}-\hat{k}$ (b) $\hat{i}-2\hat{k}$

(c) $2\hat{i}+\hat{k}$ (d) none of these

16. If $|\vec{a}|=4$, $|\vec{b}|=2$ and the angle between $\vec{a}$ and $\vec{b}$ is $\pi/6$, then $(\vec{a}\times\vec{b})^2$ is

(a) 48 (b) 32

(c) 16 (d) 8

17. If point A is $\vec{a}+2\vec{b}$, P is $\vec{a}$ and P divides AB in the ratio 2 : 3, then the position vector of B is

(a) $2\vec{a}-\vec{b}$ (b) $\vec{b}-2\vec{a}$

(c) $\vec{a}-3\vec{b}$ (d) $\vec{b}$

18. The vector $\vec{c}$ directed along the internal bisector of the angle between the vectors $\vec{a}=7\hat{i}-4\hat{j}-4\hat{k}$ and $\vec{b}=-2\hat{i}-\hat{j}+2\hat{k}$ with $|\vec{c}|=5\sqrt{6}$ is

(a) $\frac{5}{3}(\hat{i}-7\hat{j}+2\hat{k})$ (b) $\frac{5}{3}(5\hat{i}+5\hat{j}+2\hat{k})$

(c) $\frac{5}{3}(\hat{i}+7\hat{j}+2\hat{k})$ (d) $\frac{5}{3}(-5\hat{i}+5\hat{j}+2\hat{k})$

19. If $\hat{a}, \hat{b}, \hat{c}$ are unit vectors such that $\hat{a}+\hat{b}+\hat{c}=0$, then the value of $\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{c}+\hat{c}\cdot\hat{a}$ is

(a) 1 (b) 3

(c) $-(3/2)$ (d) none of these

20. If $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a}\times(\hat{b}\times\hat{c})=(1/2)\hat{b}$, then the angle between $\hat{a}$ and $\hat{c}$ is

(a) 30° (b) 45°

(c) 60° (d) 90°

21. The position vectors of the points A, B, C and D are $3\hat{i}-2\hat{j}-\hat{k}$, $2\hat{i}+3\hat{j}-4\hat{k}$, $-\hat{i}+\hat{j}+2\hat{k}$ and $4\hat{i}+5\hat{j}+\lambda\hat{k}$. It is known that these points are coplanar, then λ is equal to

(a) $-(146/17)$ (b) $-(137/17)$

(c) $-(154/17)$ (d) none of these

22. The unit vector perpendicular to each of the vectors $2\hat{i}-\hat{j}+\hat{k}$ and $3\hat{i}+4\hat{j}$ is

(a) $\frac{1}{\sqrt{146}}(4\hat{i}-3\hat{j}+11\hat{k})$ (b) $\frac{1}{\sqrt{146}}(-4\hat{i}+3\hat{j}+11\hat{k})$

(c) $\frac{1}{\sqrt{146}}(4\hat{i}+3\hat{j}+11\hat{k})$ (d) none of these

23. A unit vector perpendicular to the plane of $\vec{a}=2\hat{i}-6\hat{j}-3\hat{k}$, $\vec{b}=4\hat{i}+3\hat{j}-\hat{k}$ is

(a) $\frac{4\hat{i}+3\hat{j}-\hat{k}}{\sqrt{26}}$ (b) $\frac{2\hat{i}-6\hat{j}-3\hat{k}}{7}$

(c) $\frac{3\hat{i}-2\hat{j}+6\hat{k}}{7}$ (d) $\frac{2\hat{i}-3\hat{j}-6\hat{k}}{7}$

24. The line joining the points $6\vec{a}-4\vec{b}-5\vec{c}, -4\vec{c}$ and the line joining the points $-\vec{a}-2\vec{b}-3\vec{c}, \vec{a}+2\vec{b}-5\vec{c}$ intersect at

(a) $2\vec{c}$ (b) $-4\vec{c}$
(c) $8\vec{c}$ (d) none of these

25. If $\vec{r}\cdot\hat{i}=\vec{r}\cdot\hat{j}=\vec{r}\cdot\hat{k}$ and $|\vec{r}|=3$, then $\vec{r}=$

(a) $\pm 3(\hat{i}+\hat{j}+\hat{k})$ (b) $\pm\frac{1}{3}(\hat{i}+\hat{j}+\hat{k})$
(c) $\pm\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ (d) $\pm\sqrt{3}(\hat{i}+\hat{j}+\hat{k})$

26. Let $\vec{a},\vec{b},\vec{c}$ be three non-coplanar vectors and $\vec{p},\vec{q},\vec{r}$ be vectors defined by the relations

$$\vec{p}=\frac{\vec{b}\times\vec{c}}{[\vec{a}\vec{b}\vec{c}]},\quad \vec{q}=\frac{\vec{c}\times\vec{a}}{[\vec{a}\vec{b}\vec{c}]},\quad \vec{r}=\frac{\vec{a}\times\vec{b}}{[\vec{a}\vec{b}\vec{c}]}$$

then the value of the expression $(\vec{a}+\vec{b})\cdot\vec{p}+(\vec{b}+\vec{c})\cdot\vec{q}+(\vec{c}+\vec{a})\cdot\vec{r}$ is equal to

(a) 0 (b) 1
(c) 2 (d) 3

27. If the position vector of a point A is $\vec{a}+2\vec{b}$ and $\vec{a}$ divides $\overrightarrow{AB}$ in the ratio 2 : 3, then the position vector of B is

(a) $2\vec{a}-\vec{b}$ (b) $\vec{b}-2\vec{a}$
(c) $\vec{a}-3\vec{b}$ (d) $\vec{b}$

28. The position vectors of A and B are $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$. The length of the internal bisector of $\angle BOA$ of the triangle AOB is

(a) $\sqrt{\frac{136}{9}}$ (b) $\frac{\sqrt{139}}{9}$
(c) $\frac{20}{3}$ (d) $\sqrt{\frac{217}{9}}$

29. If $\vec{a},\vec{b},\vec{c}$ are mutually perpendicular vectors of equal magnitudes, then the angle between the vectors $\vec{a}$ and $\vec{a}+\vec{b}+\vec{c}$ is

(a) $\pi/3$ (b) $\pi/6$
(c) $\cos^{-1}(1/\sqrt{3})$ (d) $\pi/2$

30. For non-zero vectors $\vec{a},\vec{b},\vec{c}$, $(\vec{a}\times\vec{b})\cdot\vec{c}=|\vec{a}||\vec{b}||\vec{c}|$ holds if

(a) $\vec{a}\cdot\vec{b}=0,\ \vec{b}\cdot\vec{c}=0,\ \vec{c}\cdot\vec{a}\neq 0$
(b) $\vec{b}\cdot\vec{c}=0,\ \vec{c}\cdot\vec{a}=0,\ \vec{a}\cdot\vec{b}\neq 0$
(c) $\vec{c}\cdot\vec{a}=0,\ \vec{a}\cdot\vec{b}=0,\ \vec{b}\cdot\vec{c}\neq 0$
(d) $\vec{a}\cdot\vec{b}=0,\ \vec{b}\cdot\vec{c}=0,\ \vec{c}\cdot\vec{a}=0$

31. If $|\vec{a}|=|\vec{b}|$, then $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})$ is

(a) positive (b) negative
(c) zero (d) none of these

32. The area of the triangle whose vertices are $A(1,-1,2)$, $B(2,1,-1)$ and $C(3,-1,2)$ is

(a) 13 (b) $\sqrt{13}$
(c) 6 (d) $\sqrt{6}$

33. The sine of the angle between the vectors $\vec{a}=3\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=2\hat{i}-2\hat{j}+\hat{k}$ is

(a) $\sqrt{\frac{74}{99}}$ (b) $\sqrt{\frac{25}{99}}$
(c) $\sqrt{\frac{37}{99}}$ (d) $\frac{5}{\sqrt{41}}$

34. The area of the triangle having vertices as $\hat{i}-2\hat{j}+3\hat{k}$, $-2\hat{i}+3\hat{j}-\hat{k}, 4\hat{i}-7\hat{j}+7\hat{k}$ is

(a) 36 sq. units (b) 0 sq. unit
(c) 39 sq. units (d) 11 sq. units

35. The area of a parallelogram whose two adjacent sides are represented by the vectors $3\hat{i}-\hat{k}$ and $\hat{i}+2\hat{j}$ is

(a) $\frac{1}{2}\sqrt{17}$ (b) $\frac{1}{2}\sqrt{14}$
(c) $\sqrt{41}$ (d) $\frac{1}{2}\sqrt{7}$

36. A unit vector perpendicular to each of the vectors $-6\hat{i}+8\hat{k}$ and $8\hat{i}+6\hat{k}$ forming a right-handed system is

(a) $-\hat{j}$ (b) $\hat{j}$
(c) $\frac{6\hat{i}+8\hat{k}}{10}$ (d) $\frac{-6\hat{i}+8\hat{k}}{10}$

37. If $\hat{a},\hat{b},\hat{c}$ are mutually perpendicular unit vectors, then $|\hat{a}+\hat{b}+\hat{c}|$ is equal to

(a) $\sqrt{3}$ (b) 3
(c) 1 (d) 0

38. $\hat{i}\times(\vec{a}\times\hat{i})+\hat{j}\times(\vec{a}\times\hat{j})+\hat{k}\times(\vec{d}\times\hat{k})$ is always equal to

(a) $\vec{a}$ (b) $-\vec{a}$
(c) $2\vec{a}$ (d) $-2\vec{a}$

39. The area of the parallelogram whose diagonals are given by vectors $3\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-3\hat{j}+4\hat{k}$ is

(a) $10\sqrt{3}$ (b) $5\sqrt{3}$
(c) 8 (d) 4

40. The angle between the vectors $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$, when $\vec{a}=(1, 1, 4)$ and $\vec{b}=(1,-1, 4)$, is
(a) 90° (b) 45°
(c) 30° (d) 15°

41. If I is the centre of a circle inscribed in a triangle ABC, then $\left|\overrightarrow{BC}\right|\overrightarrow{IA}+\left|\overrightarrow{CA}\right|\overrightarrow{IB}+\left|\overrightarrow{AB}\right|\overrightarrow{IC}$ is
(a) $\vec{0}$ (b) $\overrightarrow{IA}+\overrightarrow{IB}+\overrightarrow{IC}$
(c) $\dfrac{\overrightarrow{IA}+\overrightarrow{IB}+\overrightarrow{IC}}{3}$ (d) none of these

42. $(\vec{a}\times\vec{b})^2$ is equal to
(a) $\vec{a}^2\vec{b}^2-(\vec{a}\cdot\vec{b})^2$ (b) $\vec{a}^2\vec{b}^2+(\vec{a}\cdot\vec{b})^2$
(c) $(\vec{a}\cdot\vec{b})^2$ (d) $\vec{a}^2\vec{b}^2$

43. The unit vector perpendicular to both $\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$ is
(a) $\hat{i}-\hat{j}+\hat{k}$ (b) $\hat{i}+\hat{j}+\hat{k}$
(c) $\dfrac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{3}}$ (d) $\dfrac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$

44. If θ is the angle between the unit vectors $\hat{a}$ and $\hat{b}$, then cos $(\theta/2)$ =
(a) $\dfrac{1}{2}\left|\hat{a}-\hat{b}\right|$ (b) $\dfrac{1}{2}\left|\hat{a}+\hat{b}\right|$
(c) $\dfrac{\left|\hat{a}-\hat{b}\right|}{\left|\hat{a}+\hat{b}\right|}$ (d) $\dfrac{\left|\hat{a}+\hat{b}\right|}{\left|\hat{a}-\hat{b}\right|}$

45. The vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}-3\hat{j}-5\hat{k}$ and $\sqrt{3}\hat{i}-4\hat{j}-4\hat{k}$ are the sides of a triangle, which is
(a) equilateral
(b) isosceles only
(c) right-angled only
(d) right-angled and isosceles

46. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is 60°. Then $\left|\hat{a}-\hat{b}\right|$ is equal to
(a) $\sqrt{5}$ (b) $\sqrt{3}$
(c) 0 (d) 1

47. If $|\vec{a}|=3, \left|\vec{b}\right|=4$ and the angle between $\vec{a}$ and $\vec{b}$ is 120°, then $\left|4\vec{a}+3\vec{b}\right|$ =
(a) 25 (b) 12
(c) 13 (d) 7

48. If $\vec{\alpha}=2\hat{i}+3\hat{j}-\hat{k}$, $\vec{\beta}=-\hat{i}+2\hat{j}-4\hat{k}$ and $\vec{\gamma}=\hat{i}+\hat{j}+\hat{k}$, then $(\vec{\alpha}\times\vec{\beta})\cdot(\vec{\alpha}\times\vec{\gamma})$ is
(a) 60 (b) 64
(c) 74 (d) –74

49. If $\vec{a}$ is perpendicular to $\vec{b}$ and $\vec{c}$, $|\vec{a}|=2, \left|\vec{b}\right|=3, |\vec{c}|=4$ and the angle between $\vec{b}$ and $\vec{c}$ is $2\pi/3$, then $[\vec{a}\vec{b}\vec{c}]$ is equal to
(a) $4\sqrt{3}$ (b) $6\sqrt{3}$
(c) $12\sqrt{3}$ (d) $18\sqrt{3}$

50. Let $\hat{a}$ and $\hat{b}$ be two unit vectors inclined at an angle θ, then sin $(\theta/2)$ is equal to
(a) $\dfrac{1}{2}\left|\hat{a}-\hat{b}\right|$ (b) $\dfrac{1}{2}\left|\hat{a}+\hat{b}\right|$
(c) $\left|\hat{a}-\hat{b}\right|$ (d) $\left|\hat{a}+\hat{b}\right|$

51. The scalar $\vec{A}\cdot(\vec{B}+\vec{C})\times(\vec{A}+\vec{B}+\vec{C})$ is
(a) 0 (b) $[\vec{A}\vec{B}\vec{C}]+[\vec{B}\vec{C}\vec{A}]$
(c) $2[\vec{A}\vec{B}\vec{C}]$ (d) $[\vec{A}\vec{B}\vec{C}]$

52. If $\vec{a}$ and $\vec{b}$ are two non-collinear vectors and $x\vec{a}+y\vec{b}$ = 0, then
(a) x = 0 but y is not necessarily zero
(b) y = 0 but x is not necessarily zero
(c) x = 0, y = 0
(d) none of these

53. Projection of $\hat{i}+2\hat{j}+3\hat{k}$ on $\hat{i}-2\hat{j}-2\hat{k}$ is equal to
(a) 3 (b) –3
(c) 9 (d) –9

54. If $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{a}\cdot\vec{b}$ = 0 and $\vec{a}\times\vec{b}=\vec{0}$, then
(a) $\vec{a}$ is parallel to $\vec{b}$
(b) $\vec{a}\perp\vec{b}$
(c) either $\vec{a}$ or $\vec{b}$ is a null vector
(d) none of these

55. The position vectors of the points A, B, C are $2\hat{i}+\hat{j}-\hat{k}$, $3\hat{i}-2\hat{j}+\hat{k}$ and $\hat{i}+4\hat{j}-3\hat{k}$, respectively. These points
(a) form an isosceles triangle
(b) form a right-angled triangle
(c) are collinear
(d) form a scalene triangle

EXERCISE SET 2

1. If three-unit vectors $\hat{a}, \hat{b}, \hat{c}$ satisfy $\hat{a}+\hat{b}+\hat{c}=\hat{0}$, then the angle between $\hat{a}$ and $\hat{b}$ is

(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$

(c) $\frac{\pi}{6}$ (d) $\frac{5\pi}{6}$

2. If $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ and $|\vec{a}|=\sqrt{37}, |\vec{b}|=3, |\vec{c}|=4$, then the angle between $\vec{b}$ and $\vec{c}$ is

(a) 30° (b) 45°

(c) 60° (d) 90°

3. If $\vec{a}=x\hat{i}+(x-1)\hat{j}+\hat{k}$ and $\vec{b}=(x+1)\hat{i}+\hat{j}+a\hat{k}$ always make an acute angle with each other for every value of $x \in \mathbb{R}$, then

(a) $a \in (-\infty, 2)$ (b) $a \in (2, \infty)$

(c) $a \in (-\infty, 1)$ (d) $a \in (1, \infty)$

4. If $\vec{e_1}=(1,1,1)$ and $\vec{e_2}=(1,1,-1)$ and $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{e_1}=2\vec{a}+\vec{b}$ and $\vec{e_2}=\vec{a}+2\vec{b}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(a) $\cos^{-1}\left(\frac{7}{9}\right)$ (b) $\cos^{-1}\left(\frac{7}{11}\right)$

(c) $\cos^{-1}\left(-\frac{7}{11}\right)$ (d) $\cos^{-1}\left(\frac{6\sqrt{2}}{11}\right)$

5. The angle between the line $\vec{r}=(-\hat{i}+3\hat{j}+3\hat{k})+t(2\hat{i}+3\hat{j}+6\hat{k})$ and the plane $\vec{r}\cdot(-\hat{i}+\hat{j}+\hat{k})$ is

(a) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(c) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (d) $\sin^{-1}\left(\frac{3}{\sqrt{2}}\right)$

6. If $\vec{a}$ and $\vec{b}$ are two vectors, then $(\vec{a}\times\vec{b})^2$ equals

(a) $\begin{vmatrix} \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{a} \\ \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{a} \end{vmatrix}$ (b) $\begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} \end{vmatrix}$

(c) $\begin{vmatrix} \vec{a}\cdot\vec{b} \\ \vec{b}\cdot\vec{a} \end{vmatrix}$ (d) none of these

7. If the angle between $\hat{i}+\hat{k}$ and $\hat{i}+\hat{j}+a\hat{k}$ is $\pi/3$, then the value of a is

(a) 0 or 2 (b) –4 or 0

(c) 0 or –2 (d) 2 or –2

8. The area of the parallelogram whose diagonals are $\vec{a}=3\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}-3\hat{j}+4\hat{k}$ is

(a) $10\sqrt{3}$ (b) $5\sqrt{3}$

(c) 8 (d) 4

9. For any three vectors $\vec{a}, \vec{b}$ and $\vec{c}$, $(\vec{a}-\vec{b})\cdot(\vec{b}+\vec{c})\times(\vec{c}+\vec{a})$ is equal to

(a) $2\vec{a}\cdot(\vec{b}\times\vec{c})$ (b) $[\vec{a}\vec{b}\vec{c}]$

(c) $[\vec{a}\vec{b}\vec{c}]^2$ (d) 0

10. If $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}\neq 0$, where $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar vectors, then for some scalar k

(a) $\vec{a}+\vec{c}=k\vec{b}$ (b) $\vec{a}+\vec{b}=k\vec{c}$

(c) $\vec{b}+\vec{c}=k\vec{a}$ (d) none of these

11. If the vector $\vec{b}=3\hat{i}+4\hat{k}$ is to be written as the sum of a vector $\vec{b_1}$ parallel to $\vec{a}=\hat{i}+\hat{j}$ and a vector $\vec{b_2}$ perpendicular to $\vec{a}$, then $\vec{b_1}$ is

(a) $\frac{3}{2}(\hat{i}+\hat{j})$ (b) $\frac{2}{3}(\hat{i}+\hat{j})$

(c) $\frac{1}{2}(\hat{i}+\hat{j})$ (d) $\frac{1}{3}(\hat{i}+\hat{j})$

12. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors, then $(\vec{a}+\vec{b}+\vec{c})\cdot[(\vec{a}+\vec{b})\times(\vec{a}+\vec{c})]$ is

(a) 0 (b) $2[\vec{a}\vec{b}\vec{c}]$

(c) $-[\vec{a}\vec{b}\vec{c}]$ (d) $[\vec{a}\vec{b}\vec{c}]$

13. If $\vec{u}=\vec{a}-\vec{b}$ and $\vec{v}=\vec{a}+\vec{b}$ and $|\vec{a}|=|\vec{b}|=2$, then $|\vec{u}\times\vec{v}|$ is equal to

(a) $2\sqrt{16-(\vec{a}\cdot\vec{b})^2}$ (b) $\sqrt{16+(\vec{a}\cdot\vec{b})^2}$

(c) $2\sqrt{4-(\vec{a}\cdot\vec{b})^2}$ (d) $\sqrt{4-(\vec{a}\cdot\vec{b})^2}$

14. If the vectors $a\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+5\hat{j}+a\hat{k}$ are perpendicular to each other, then a equals

(a) 6 (b) –6

(c) 5 (d) –5

15. A unit vector coplanar with $\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$, and perpendicular to $\hat{i}+\hat{j}+\hat{k}$ is

(a) $\left(\frac{\hat{j}-\hat{k}}{\sqrt{2}}\right)$ (b) $\left(\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}\right)$

(c) $\left(\frac{\hat{i}+\hat{j}+2\hat{k}}{\sqrt{6}}\right)$ (d) $\left(\frac{\hat{i}+2\hat{j}+\hat{k}}{\sqrt{6}}\right)$

16. If $\hat{a}$ and $\hat{b}$ are two unit vectors inclined to the x-axis at angles 30° and 120°, then $|\hat{a}+\hat{b}|$ equals

(a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{2}$

(c) $\sqrt{3}$ (d) 2

17. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{p}, \vec{q}, \vec{r}$ be the corresponding reciprocal system of vectors. Then the value of the expression $(\vec{a}+\vec{b}+\vec{c})\cdot(\vec{p}+\vec{q}+\vec{r})$ is
(a) 0 (b) 1
(c) 2 (d) 3

18. If $|\vec{a}|=|\vec{b}|=1$ and $|\vec{a}+\vec{b}|=\sqrt{3}$, then the value of $(3\vec{a}-4\vec{b})\cdot(2\vec{a}+5\vec{b})$ is
(a) –21 (b) –(21/2)
(c) 21 (d) 21/2

19. If $\vec{a}\times\vec{b}=\vec{c}, \vec{b}\times\vec{c}=\vec{a}$ and a, b, c be the moduli of the vectors $\vec{a}, \vec{b}, \vec{c}$ respectively, then
(a) $a = 1, b = 1$ (b) $c = 1, a = 1$
(c) $\vec{a}\cdot(\vec{b}\times\vec{c})=-1$ (d) $b = 1, c = a$

20. If $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=2\hat{i}-\hat{k}$ are two vectors, then the point of intersection of two lines $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$ and $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ is
(a) $\hat{i}+\hat{j}-\hat{k}$ (b) $\hat{i}-\hat{j}+\hat{k}$
(c) $3\hat{i}+\hat{j}-\hat{k}$ (d) $3\hat{i}-\hat{j}+\hat{k}$

21. If $\vec{a}=\hat{i}-2\hat{j}-3\hat{k}$, $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}+3\hat{j}-2\hat{k}$, then $(\vec{a}\times\vec{b})\times\vec{c}$ is
(a) $5(-\hat{i}+3\hat{j}+4\hat{k})$ (b) $4(-\hat{i}+3\hat{j}+4\hat{k})$
(c) $5(-\hat{i}-3\hat{j}-4\hat{k})$ (d) $4(\hat{i}+3\hat{j}+4\hat{k})$

22. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be vectors of magnitudes 3, 4 and 5, respectively. If $\vec{a}$ is perpendicular to $(\vec{b}+\vec{c})$, $\vec{b}$ is perpendicular to $(\vec{c}+\vec{a})$ and $\vec{c}$ is perpendicular to $(\vec{a}+\vec{b})$, then the magnitude of the vector $\vec{a}+\vec{b}+\vec{c}$ is
(a) 5 (b) $5\sqrt{2}$
(c) $5\sqrt{3}$ (d) 12

23. If D, E and F are respectively the mid-points of AB, AC and BC in ΔABC, then $\overrightarrow{BE}+\overrightarrow{AF}$ is equal to
(a) $\overrightarrow{DC}$ (b) $\frac{1}{2}\overrightarrow{BF}$
(c) $2\overrightarrow{BF}$ (d) $\frac{3}{2}\overrightarrow{BF}$

24. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors mutually $\perp$ to each other and $|\vec{a}|=1$, $|\vec{b}|=3$ and $|\vec{c}|=5$, then $[\vec{a}-2\vec{b}, \vec{b}-3\vec{c}, \vec{c}-4\vec{a}]$ is
(a) 0 (b) –24
(c) 3600 (d) –345

25. A unit vector in the xy-plane that makes an angle 45° with the vector $(\hat{i}+\hat{j})$ and an angle of 60° with the vector $(3\hat{i}-4\hat{j})$ is
(a) $\hat{i}$ (b) $\frac{1}{\sqrt{2}}(\hat{i}-\hat{j})$
(c) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$ (d) none of these

26. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{5\pi}{6}$ and the projection of $\vec{a}$ in the direction of $\vec{b}$ is $-\frac{6}{\sqrt{3}}$, then $|\vec{a}|$ =
(a) 6 (b) $\frac{\sqrt{3}}{2}$
(c) 12 (d) 4

27. If $\vec{a}$ and $\vec{b}$ are position vectors of the points A and B respectively, then the position vector of a point C in AB produced such that $AC = 3AB$ is
(a) $3\vec{a}-\vec{b}$ (b) $3\vec{a}-2\vec{b}$
(c) $\vec{a}-3\vec{b}$ (d) $3\vec{b}-2\vec{a}$

28. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a}\neq 0$ and $\vec{a}\times\vec{b}=2\vec{a}\times\vec{c}$, $|\vec{a}|=|\vec{c}|=1$, $|\vec{b}|=4$ and $|\vec{b}\times\vec{c}|=\sqrt{15}$. If $\vec{b}-2\vec{c}=\lambda\vec{a}$, then λ is equal to
(a) 1 (b) –4
(c) 3 (d) –2

29. For three vectors $\vec{u}, \vec{v}, \vec{w}$, which of the following expressions is not equal to any of the remaining three?
(a) $\vec{u}\cdot(\vec{v}\times\vec{w})$ (b) $(\vec{v}\times\vec{w})\cdot\vec{u}$
(c) $\vec{v}\cdot(\vec{u}\times\vec{w})$ (d) $(\vec{u}\times\vec{v})\cdot\vec{w}$

30. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors with magnitudes $|\vec{a}|=4$, $|\vec{b}|=4$ and $|\vec{c}|=2$ such that $\vec{a}$ is perpendicular to $(\vec{b}+\vec{c})$, $\vec{b}$ is perpendicular to $(\vec{c}+\vec{a})$ and $\vec{c}$ is perpendicular to $(\vec{a}+\vec{b})$, then it follows that $|\vec{a}+\vec{b}+\vec{c}|$ is equal to
(a) 9 (b) 6
(c) 5 (d) 4

31. If C is the mid-point of AB and P is any point outside AB, then
(a) $\overrightarrow{PA}+\overrightarrow{PB}=\overrightarrow{PC}$ (b) $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$
(c) $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=\vec{0}$ (d) $\overrightarrow{PA}+\overrightarrow{PB}+2\overrightarrow{PC}=\vec{0}$

32. Let $\vec{a}=\hat{i}-\hat{j}$, $\vec{b}=\hat{i}-\hat{k}$, $\vec{c}=\hat{k}-\hat{i}$. If $\hat{d}$ is a unit vector such that $\vec{a}\cdot\hat{d}=0=[\vec{b}\vec{c}\hat{d}]$, then $\hat{d}$ equals
(a) $\pm\frac{\hat{i}+\hat{j}-2\hat{k}}{\sqrt{6}}$ (b) $\pm\frac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{3}}$
(c) $\pm\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$ (d) $\pm\hat{k}$

33. The unit vector perpendicular to vectors $\hat{i}-\hat{j}$ and $\hat{i}+\hat{j}$ forming a right-handed system is

(a) $\hat{k}$ (b) $-\hat{k}$

(c) $\frac{1}{\sqrt{2}}(\hat{i}-\hat{j})$ (d) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) and (d) given below:

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

34. Statement I: For $a=-\frac{1}{\sqrt{3}}$, the volume of the parallelopiped formed by vectors $\hat{i}+a\hat{j}$, $a\hat{i}+\hat{j}+\hat{k}$ and $\hat{j}+a\hat{k}$ is maximum.

Statement II: The volume of the parallelopiped having three coterminous edges $\vec{a},\vec{b}$ and $\vec{c}$ is $\left|\vec{a}\vec{b}\vec{c}\right|$.

35. Statement I: If $\vec{a},\vec{b},\vec{c}$ are coplanar, then $\vec{a}\times\vec{b},\vec{b}\times\vec{c}$ and $\vec{c}\times\vec{a}$ are also coplanar.

Statement II: $[\vec{a}\times\vec{b}\ \vec{b}\times\vec{c}\ \vec{c}\times\vec{a}]=2[\vec{a}\vec{b}\vec{c}]^2$.

36. If $\vec{a}\cdot\vec{c}=\frac{3}{2}$, $\vec{b}\cdot\vec{d}=2$, $\vec{a}\cdot\vec{d}=3$ and $\vec{b}\cdot\vec{c}=\frac{1}{2}$

Statement I: $\vec{a}\times\vec{b}$, $\vec{c},\vec{d}$ are non-coplanar.

Statement II: $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d})=(\vec{b}\cdot\vec{c})(\vec{a}\cdot\vec{d})-(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d})$

37. Statement I: $|\vec{a}|=|\vec{b}|$ does not imply that $\vec{a}=\vec{b}$.

Statement II: If $\vec{a}=\vec{b}$, then $\vec{a}\cdot\vec{b}=|\vec{a}|^2=|\vec{b}|^2$.

Previous Years' Questions

38. If the vectors $\vec{c}$, $\vec{a}=x\hat{i}+y\hat{j}+z\hat{k}$ and $\vec{b}=\hat{j}$ are such that $\vec{a}\cdot\vec{c}$ and $\vec{b}$ form a right-handed system, then $\vec{c}$ is

(a) $z\hat{i}-x\hat{k}$ (b) $\vec{0}$

(c) $y\hat{j}$ (d) $-z\hat{i}+x\hat{k}$

[AIEEE 2002]

39. Consider points A, B, C and D with position vectors $7\hat{i}-4\hat{j}+7\hat{k}$, $\hat{i}-6\hat{j}+10\hat{k}$, $-\hat{i}-3\hat{j}+4\hat{k}$ and $5\hat{i}-\hat{j}+5\hat{k}$ respectively. Then, $ABCD$ is a

(a) square (b) rhombus

(c) rectangle (d) none of these

[AIEEE 2003]

40. A tetrahedron has vertices at $O(0, 0, 0)$, $A(1, 2, 1)$, $B(2, 1, 3)$ and $C(-1, 1, 2)$. Then, the angle between the faces OAB and ABC will be

(a) $\cos^{-1}\left(\frac{19}{35}\right)$ (b) $\cos^{-1}\left(\frac{17}{31}\right)$

(c) 30° (d) 90°

[AIEEE 2003]

41. If $\vec{u}$, $\vec{v}$ and $\vec{w}$ are three non-coplanar vectors, then $(\vec{u}+\vec{v}-\vec{w})\cdot[(\vec{u}-\vec{v})\times(\vec{v}-\vec{w})]$ equals

(a) 0 (b) $\vec{u}\cdot\vec{v}\times\vec{w}$

(c) $\vec{u}\cdot\vec{w}\times\vec{v}$ (d) $3\vec{u}\cdot\vec{v}\times\vec{w}$

[AIEEE 2003]

42. Let $\vec{a},\vec{b},\vec{c}$ are non-zero vectors such that $(\vec{a}\times\vec{b})\times\vec{c}=\frac{1}{3}|\vec{b}|\cdot|\vec{c}|\vec{a}$. If θ is an acute angle between the vectors $\vec{b}$ and $\vec{c}$, then sin θ is equal to

(a) $\frac{1}{3}$ (b) $\frac{\sqrt{2}}{3}$

(c) $\frac{2}{3}$ (d) $\frac{2\sqrt{2}}{3}$

[AIEEE 2004]

43. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of these are collinear. If the vector $\vec{a}+2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+3\vec{c}$ is collinear with $\vec{a}$ (λ being some non-zero scalar), then $\vec{a}+2\vec{b}+6\vec{c}$ is equal to

(a) $\lambda\vec{a}$ (b) $\lambda\vec{b}$

(c) $\lambda\vec{c}$ (d) $\vec{0}$

[AIEEE 2004]

44. Let $\vec{u},\vec{v},\vec{w}$ be such that $|\vec{u}|=1,|\vec{v}|=2,|\vec{w}|=3$. If the projection $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and $\vec{v}$ and $\vec{w}$ are perpendicular to each other, then $|\vec{u}-\vec{v}+\vec{w}|$ is equal to

(a) 2 (b) $\sqrt{7}$

(c) $\sqrt{14}$ (d) 14

[AIEEE 2004]

45. Let a, b and c be distinct non-negative numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k},\hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ lie in a plane, then c is

(a) the harmonic mean of a and b
(b) equal to zero
(c) the arithmetic mean of a and b
(d) the geometric mean of a and b

[AIEEE 2005]

46. The distance between the line $\vec{r} = 2\hat{i} - 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - \hat{j} + 4\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} + 5\hat{j} + \hat{k}) = 5$ is

(a) 10/3 (b) 3/10 (c) $10/3\sqrt{3}$ (d) 10/9

[AIEEE 2005]

47. Let $\vec{a} = \hat{i} - \hat{k}$, $\vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$ and $\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$. Then $[\vec{a}\vec{b}\vec{c}]$ depends on

(a) neither x nor y (b) both x and y

(c) only x (d) only y

[AIEEE 2006]

48. If $\hat{u}$ and $\hat{v}$ are unit vectors and θ is the acute angle between them, then $2\hat{u} \times 3\hat{v}$ is a unit vector for

(a) exactly two values of θ

(b) more than two values of θ

(c) no value of θ

(d) exactly one value of θ **[AIEEE 2007]**

49. The non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are related by $\vec{a} = 8\vec{b}$ and $\vec{c} = -7\vec{b}$. Then, the angle between $\vec{a}$ and $\vec{c}$ is

(a) π (b) 0 (c) $\pi/4$ (d) $\pi/2$

[AIEEE 2008]

50. The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then, which one of the following gives possible values of α and β?

(a) $\alpha = 1, \beta = 1$ (b) $\alpha = 2, \beta = 2$

(c) $\alpha = 1, \beta = 2$ (d) $\alpha = 2, \beta = 1$

[AIEEE 2009]

51. If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p and q are real numbers, then the equality $[3\vec{u}\ p\vec{v}\ p\vec{w}] - [p\vec{v}\ \vec{w}\ q\vec{u}] - [2\vec{w}\ q\vec{v}\ q\vec{u}] = 0$ holds for

(a) exactly two values of (p, q)

(b) more than two but not all values of (p, q)

(c) all values of (p, q)

(d) exactly one value of (p, q)

[AIEEE 2009]

52. If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then (λ, μ) is equal to

(a) (–3, 2) (b) (2, –3) (c) (–2, 3) (d) (3, –2)

[AIEEE 2010]

53. Let $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$. Then the vector $\vec{b}$ satisfying $\vec{a} \times \vec{b} + \vec{c} = 0$ and $\vec{a} \cdot \vec{b} = 3$ is

(a) $-\hat{i} + \hat{j} - 2\hat{k}$ (b) $2\hat{i} - \hat{j} + 2\hat{k}$

(c) $\hat{i} - \hat{j} - 2\hat{k}$ (d) $\hat{i} + \hat{j} - 2\hat{k}$

[AIEEE 2010]

54. Let P, Q, R and S be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}$, $3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral $PQRS$ must be a

(a) parallelogram, which is neither a rhombus nor a rectangle

(b) square

(c) rectangle, but not a square

(d) rhombus, but not a square

[IIT-JEE 2010]

55. If the scalar product of the vector $\hat{i} + \hat{j} + 2\hat{k}$ with the unit vector along $m\hat{i} + 2\hat{j} + 3\hat{k}$ is equal to 2, then one of the values of m is

(a) 3 (b) 4 (c) 5 (d) 6

[Kerala CEE 2010]

56. If $\vec{u}_1$ and $\vec{u}_2$ are vectors of unit length and θ is the angle between them, then $\frac{1}{2}|\vec{u}_2 - \vec{u}_1|$ is

(a) sin θ (b) sin $\theta/2$ (c) cos θ (d) cos $\theta/2$

[AMU 2009]

57. A constant force $\vec{F} = 2\hat{i} - 3\hat{j} + 2\hat{k}$ is acting on a particle such that the particle is displaced from the point (1, 2, 3) to the point (3, 4, 5). The work done by the force is

(a) 2 (b) 3 (c) 4 (d) 5

[OJEE 2009]

58. If the area of the parallelogram with $\vec{a}$ and $\vec{b}$ as two adjacent sides is 15 sq. units, then the area of the parallelogram having $3\vec{a} + 2\vec{b}$ and $\vec{a} + 3\vec{b}$ as two adjacent sides in sq. units is

(a) 120 (b) 105 (c) 75 (d) 45

[Jamia Millia Islamia 2008]

59. If $\vec{\alpha} = x(\vec{a} \times \vec{b}) + y(\vec{b} \times \vec{c}) + z(\vec{c} \times \vec{a})$ and $[\vec{a}\vec{b}\vec{c}] = \frac{1}{8}$, then $x + y + z$ is equal to

(a) $8\vec{\alpha} \cdot (\vec{a} + \vec{b} + \vec{c})$ (b) $\vec{\alpha} \cdot (\vec{a} + \vec{b} + \vec{c})$

(c) $8(\vec{a} + \vec{b} + \vec{c})$ (d) none of these

[Jamia Millia Islamia 2009]

60. If the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$, $-2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\lambda\hat{i} - \hat{j} + 2\hat{k}$ are linearly dependent, then the value of λ is equal to

(a) 0 (b) 1 (c) 2 (d) 3

[UP SEE 2009]

61. If the vectors $\vec{\alpha} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{\beta} = -\hat{i} + 2\hat{j} - 4\hat{k}$ and $\vec{\gamma} = \hat{i} + \hat{j} + \hat{k}$, then what is the value of $(\vec{\alpha} \times \vec{\beta}) \cdot (\vec{\alpha} \times \vec{\gamma})$?

(a) 47 (b) 74 (c) –74 (d) none of these

[BITSAT 2005]

62. The volume (in cubic unit) of the tetrahedron with edges $\hat{i}+\hat{j}+\hat{k}$, $\hat{i}-\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}-\hat{k}$ is
(a) 4 (b) 2/3 (c) 1/6 (d) 1/3
[EAMCET 2007]

63. The shortest distance between the straight lines through the points A_1 = (6, 2, 2) and A_2 = (–4, 0, –1), in the directions of (1, –2, 2) and (3, –2, –2), is
(a) 6 (b) 8 (c) 12 (d) 9
[VITEEE 2008]

64. If the vectors $p\hat{i}+\hat{j}+\hat{k}, \hat{i}+q\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+r\hat{k}$ $(p \neq q \neq r \neq 1)$ are coplanar, then the value of $pqr-(p+q+r)$ is
(a) 2 (b) 0 (c) –1 (d) –2
[AIEEE 2011]

65. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a}+3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+2\vec{c}$ is collinear with $\vec{a}$, then $\vec{a}+3\vec{b}+6\vec{c}$ is
(a) $\vec{a}$ (b) $\vec{c}$ (c) $\vec{0}$ (d) $\vec{a}+\vec{c}$
[AIEEE 2011]

66. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c}=\hat{a}+2\hat{b}$ and $\vec{d}=5\hat{a}-4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is
(a) $\pi/6$ (b) $\pi/2$ (c) $\pi/3$ (d) $\pi/4$
[AIEEE 2012]

67. Let $ABCD$ be a parallelogram such that $\overrightarrow{AB}=\vec{q}$, $\overrightarrow{AD}=\vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by
(a) $\vec{r}=3\vec{q}-\dfrac{3(\vec{p}\cdot\vec{q})}{(\vec{p}\cdot\vec{p})}\vec{p}$ (b) $\vec{r}=-\vec{q}+\left(\dfrac{\vec{p}\cdot\vec{q}}{\vec{p}\cdot\vec{p}}\right)\vec{p}$
(c) $\vec{r}=\vec{q}-\left(\dfrac{\vec{p}\cdot\vec{q}}{\vec{p}\cdot\vec{p}}\right)\vec{p}$ (d) $\vec{r}=-3\vec{q}+\dfrac{3(\vec{p}\cdot\vec{q})}{\vec{p}\cdot\vec{p}}\vec{p}$

ANSWERS

Exercise Set 1

1. (a)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (d)	**6.** (b)	**7.** (c)	**8.** (b)	**9.** (d)	**10.** (c)
11. (a)	**12.** (d)	**13.** (d)	**14.** (d)	**15.** (c)	**16.** (c)	**17.** (c)	**18.** (a)	**19.** (c)	**20.** (c)
21. (a)	**22.** (b)	**23.** (c)	**24.** (b)	**25.** (d)	**26.** (d)	**27.** (c)	**28.** (a)	**29.** (c)	**30.** (d)
31. (c)	**32.** (b)	**33.** (a)	**34.** (b)	**35.** (c)	**36.** (b)	**37.** (a)	**38.** (c)	**39.** (b)	**40.** (a)
41. (a)	**42.** (a)	**43.** (d)	**44.** (b)	**45.** (d)	**46.** (d)	**47.** (b)	**48.** (d)	**49.** (c)	**50.** (a)
51. (a)	**52.** (c)	**53.** (b)	**54.** (c)	**55.** (a)					

Exercise Set 2

1. (b)	**2.** (c)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (b)	**7.** (b)	**8.** (b)	**9.** (d)	**10.** (a)
11. (a)	**12.** (c)	**13.** (a)	**14.** (d)	**15.** (a)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (d)	**20.** (c)
21. (a)	**22.** (b)	**23.** (a)	**24.** (d)	**25.** (d)	**26.** (d)	**27.** (d)	**28.** (b)	**29.** (c)	**30.** (b)
31. (b)	**32.** (a)	**33.** (a)	**34.** (d)	**35.** (c)	**36.** (c)	**37.** (a)	**38.** (a)	**39.** (d)	**40.** (a)
41. (b)	**42.** (d)	**43.** (d)	**44.** (c)	**45.** (d)	**46.** (c)	**47.** (a)	**48.** (d)	**49.** (a)	**50.** (a)
51. (d)	**52.** (a)	**53.** (a)	**54.** (a)	**55.** (d)	**56.** (b)	**57.** (a)	**58.** (b)	**59.** (a)	**60.** (a)
61. (c)	**62.** (b)	**63.** (d)	**64.** (d)	**65.** (c)	**66.** (c)	**67.** (b)			

HINTS AND SOLUTIONS—EXERCISE SET 1

1. We have $\vec{u}=\vec{a}-\vec{b}$, $\vec{v}=\vec{a}+\vec{b}$

$\Rightarrow \vec{u}\times\vec{v}=(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})=2(\vec{a}\times\vec{b})$

$\Rightarrow |\vec{u}\times\vec{v}|=2|\vec{a}\times\vec{b}|=2\sqrt{|\vec{a}\times\vec{b}|^2}=2\sqrt{|\vec{a}|^2|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2}$

$=2\sqrt{16-(\vec{a}\cdot\vec{b})^2}$

2. $(12\hat{i}-5\hat{j})-(10\hat{i}+3\hat{j})=\lambda[(a-12)\hat{i}+16\hat{j}]$

Comparing, we get λ = 1/2, a = 8

3. Let $\vec{c}=x\hat{i}+y\hat{j}$.

Now, $\vec{b}\perp\vec{c}$ $\Rightarrow$ $\vec{b}\cdot\vec{c}=0$

$\Rightarrow 4x+3y=0$ $\Rightarrow \dfrac{x}{3}=\dfrac{y}{-4}=\lambda$

$\Rightarrow x=3\lambda, y=-4\lambda$

$\therefore \vec{c}=\lambda(3\hat{i}-4\hat{j})$

Let the required vector be $\alpha=p\hat{i}+q\hat{j}$. Then the projections of $\vec{\alpha}$ on $\vec{b}$ and $\vec{c}$ are $\vec{\alpha}\cdot\vec{b}/|\vec{b}|$ and $\vec{\alpha}\cdot\vec{c}/|\vec{c}|$ respectively.

$\therefore \quad \vec{\alpha}\cdot\dfrac{\vec{b}}{|\vec{b}|}=1$ and $\dfrac{\vec{\alpha}\cdot\vec{c}}{|\vec{c}|}=2$

$\Rightarrow 4p+3q=5$ and $3p-4q=10$

$\Rightarrow p=2, q=-1$

Hence, the required vector is $2\hat{i}-\hat{j}$.

4. Required vector $= \pm\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.

5. Since $\vec{a}\cdot\vec{b}=0$, the angle between the vectors $\vec{a}$ and $\vec{b}$ is $\pi/2$. Now, adding these two vectors, we get

$$\vec{c}=\vec{a}+\vec{b}=7\hat{i}+5\hat{j}+\hat{k}$$

So, the other two angles of the triangle are given by

$$\alpha=\cos^{-1}\frac{\vec{c}\cdot\vec{a}}{|\vec{c}||\vec{a}|}$$

$$\Rightarrow\ \alpha=\cos^{-1}\frac{(7)(3)+(5)(6)+(1)(-2)}{\sqrt{7^2+5^2+1^2}\sqrt{3^2+6^2+(-2)^2}}$$

$$\therefore\ \alpha=\cos^{-1}\frac{7}{\sqrt{75}}$$

and let $\beta=\cos^{-1}\dfrac{\vec{c}\cdot\vec{b}}{|\vec{c}||\vec{b}|}$

$$\Rightarrow\ \beta=\cos^{-1}\frac{(7)(4)+(5)(-1)+(1)(3)}{\sqrt{7^2+5^2+1^2}\sqrt{4^2+(-1)^2+3^2}}$$

$$\therefore\ \beta=\cos^{-1}\sqrt{\frac{26}{75}}$$

6. $\vec{b}\times\vec{c}=\hat{i}+\hat{j}-\hat{k}=\vec{d}$ (say)

$$\therefore\ \text{Projection of vector } \vec{a} \text{ on } \vec{d} = \frac{\vec{a}\cdot\vec{d}}{|\vec{d}|}=\frac{2}{\sqrt{3}}$$

7. Since $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$

$\Rightarrow (\vec{r}-\vec{b})\times\vec{a}=0 \quad \Rightarrow \vec{r}-\vec{b}$ is parallel to $\vec{a}$

$\therefore\ \vec{r}-\vec{b}=\lambda\vec{a}$ for some scalar λ

$\Rightarrow \vec{r}=\vec{b}+\lambda\vec{a}$ (i)

Now, $\vec{r}\perp\vec{c}$

$\Rightarrow \vec{r}\cdot\vec{c}=0 \quad \Rightarrow (\vec{b}+\lambda\vec{a})\cdot\vec{c}=0$

$\Rightarrow \vec{b}\cdot\vec{c}+\lambda(\vec{a}\cdot\vec{c})=0 \quad \Rightarrow \lambda=-\dfrac{\vec{b}\cdot\vec{c}}{\vec{a}\cdot\vec{c}}$

Putting the value of λ in (i), we get

$$\vec{r}=\vec{b}-\left(\frac{\vec{b}\cdot\vec{c}}{\vec{a}\cdot\vec{c}}\right)\vec{a}$$

8. Required area $= \dfrac{1}{2}|(\vec{b}-\vec{a})\times(\vec{c}-\vec{b})|$

9. Area $= \dfrac{1}{2}|\vec{a}\times\vec{b}|$

10. Let A be the origin. Then, $\overrightarrow{AB}=\vec{a}$, $\overrightarrow{AC}=\vec{b}$ implies that the position vectors of B and C are $\vec{b}$ and $\vec{c}$, respectively. Let AD be the median and G be the centroid. Then,

$$\text{P.V. of } D=\frac{\vec{a}+\vec{b}}{2},\ \text{P.V. of } G=\frac{\vec{a}+\vec{b}}{3}$$

$$\therefore\ \overrightarrow{AG}=\frac{\vec{a}+\vec{b}}{3}$$

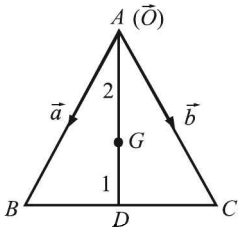


Fig. 23.31

11. Since $\vec{a}+\vec{b}=\vec{c}$, squaring, we get

$$\Rightarrow |\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b}=|\vec{c}|^2$$

$$\Rightarrow |\vec{a}|^2+|\vec{b}|^2+0=|\vec{c}|^2$$

$$\therefore\ |\vec{a}|^2+|\vec{b}|^2=|\vec{c}|^2$$

12. $\vec{a}$ and $\vec{b}$ cannot be parallel and perpendicular both at the same time. So in order to hold $\vec{a}\cdot\vec{b}=0$ and $\vec{a}\times\vec{b}=0$ simultaneously, either $\vec{a}$ or $\vec{b}$ must be zero.

13. We have $\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}$

$$\Rightarrow 3\hat{i}+2\hat{j}-5\hat{k}=x(2\hat{i}-\hat{j}+\hat{k})+y(\hat{i}+3\hat{j}-2\hat{k})+z(-2\hat{i}+\hat{j}-3\hat{k})$$

$$\Rightarrow 3=2x+y-2z, 2=-x+3y+z, -5=x-2y-3z$$

$\Rightarrow x=3,\ y=1,\ z=2$

Clearly, $x=y+z$

$\Rightarrow y, x/2, z$ are in A.P.

14. Let $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$, then the normal plane vector of the plane is

$$\vec{n}=\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&-1\\1&-1&2\end{vmatrix}=5\hat{i}-5\hat{j}-5\hat{k}$$

$\therefore$ Unit vector perpendicular to the plane of $\vec{a}$ and $\vec{b}$ is

$$\vec{n}=\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k}).$$

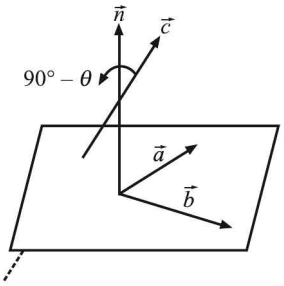


Fig. 23.32

If θ is the required angle between the plane and $\vec{c}=2\hat{i}-2\hat{j}+\hat{k}$, then $\left(\frac{\pi}{2}-\theta\right)$ is the angle between normal vector and $\vec{c}$. Thus,

$$\cos\left(\frac{\pi}{2}-\theta\right)=\frac{2\hat{i}-2\hat{j}+\hat{k}}{3}\cdot\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})=\frac{1}{\sqrt{3}}$$

$$\Rightarrow\ \theta=\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)=\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\therefore\ \theta=\cot^{-1}\left(\sqrt{2}\right)$$

15. Clearly, a point equidistant from AB and AC is on the bisector of the $\angle BAC$.
A vector along the bisector of $\angle BAC$ is

$$\frac{\overrightarrow{AB}}{\left|\overrightarrow{AB}\right|}+\frac{\overrightarrow{AC}}{\left|\overrightarrow{AC}\right|}=\frac{1}{\sqrt{11}}(4\hat{i}+2\hat{k})=\frac{2}{\sqrt{11}}(2\hat{i}+\hat{k})$$

Let $\overrightarrow{AP}=\lambda(2\hat{i}+\hat{k})$

$$\therefore\ \overrightarrow{BP}=\overrightarrow{AP}-\overrightarrow{AB}=\lambda(2\hat{i}+\hat{k})-(3\hat{i}+\hat{j}-\hat{k})$$

$$\Rightarrow\ \overrightarrow{BP}=(2\lambda-3)\hat{i}-\hat{j}+(\lambda+1)\hat{k}$$

Also, $\overrightarrow{BC}=\overrightarrow{AC}-\overrightarrow{AB}=-2\hat{i}-2\hat{j}+4\hat{k}$

Since $\overrightarrow{BP}\parallel\overrightarrow{BC}$. Therefore,

$$\overrightarrow{BP}=t\overrightarrow{BC}$$

$$\Rightarrow\ (2\lambda-3)\hat{i}-\hat{j}+(\lambda+1)\hat{k}=t(-2\hat{i}-2\hat{j}+4\hat{k})$$

$$\Rightarrow\ 2\lambda-3=-2t,\ -1=-2t,\ \lambda+1=4t$$

$$\Rightarrow\ \lambda=1,\ t=\frac{1}{2}$$

$$\therefore\ \overrightarrow{AP}=2\hat{i}+\hat{k}.$$

16. Since $(\vec{a}\times\vec{b})^2=\left|\vec{a}\times\vec{b}\right|^2$

$$\Rightarrow\ (\vec{a}\times\vec{b})^2=\left|\left|\vec{a}\right|\left|\vec{b}\right|\sin\frac{\pi}{6}\right|^2$$

$$\therefore\ (\vec{a}\times\vec{b})^2=16\times4\times\frac{1}{4}=16$$

17. If P.V. of B is $\vec{p}$, then

$$\vec{a}=\frac{2\vec{p}+3(\vec{a}+2\vec{b})}{2+3}$$

$$\Rightarrow\ \vec{p}=\vec{a}-3\vec{b}$$

18. The required vector $\vec{c}$ is given by

$$\Rightarrow\qquad \vec{c}=\lambda\left(\frac{\vec{a}}{\left|\vec{a}\right|}+\frac{\vec{b}}{\left|\vec{b}\right|}\right)$$

$$\Rightarrow\ \vec{c}=\lambda\left[\frac{1}{9}(7\hat{i}-4\hat{j}-4\hat{k})+\frac{1}{3}(-2\hat{i}-\hat{j}+2\hat{k})\right]$$

$$\Rightarrow\ \vec{c}=\frac{\lambda}{9}(\hat{i}-7\hat{j}+2\hat{k})$$

$$\Rightarrow\ \left|\vec{c}\right|=\pm\frac{\lambda}{9}\sqrt{1+49+4}=\pm\frac{\lambda}{9}\sqrt{54}$$

But, $\left|\vec{c}\right|=5\sqrt{6}$

$$\Rightarrow\ \pm\frac{\lambda}{9}\sqrt{54}=5\sqrt{6}\ \Rightarrow\ \lambda=\pm15$$

Hence, $\vec{c}=\pm\frac{15}{9}(\hat{i}-7\hat{j}+2\hat{k})=\pm\frac{5}{3}(\hat{i}-7\hat{j}+2\hat{k})$.

19. Since $(\vec{a}+\vec{b}+\vec{c})^2=0$,

$$\Rightarrow\ (\vec{a})^2+(\vec{b})^2+(\vec{c})^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0$$

$$\therefore\ \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-\frac{1}{2}(1+1+1)=-\frac{3}{2}$$

20. Since $\vec{a}\times(\vec{b}\times\vec{c})=\frac{1}{2}\vec{b}$

$$\Rightarrow\ (\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}=\frac{1}{2}\vec{b}$$

$$\Rightarrow\ \left[(\vec{a}\cdot\vec{c})-\frac{1}{2}\right]\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}=0$$

$$\Rightarrow\ (\vec{a}\cdot\vec{c})-\frac{1}{2}=0\ \text{and}\ \vec{a}\cdot\vec{b}=0$$

($\because$ $\vec{b}$ and $\vec{c}$ are non-collinear vectors)

$$\Rightarrow\ \cos\theta=\frac{1}{2},$$

where θ is the angle between $\vec{a}$ and $\vec{c}$.

$$\therefore\ \theta=\frac{\pi}{3}.$$

21. $\overrightarrow{AB}=-\hat{i}+5\hat{j}-3\hat{k}$, $\overrightarrow{AC}=-4\hat{i}+3\hat{j}+3\hat{k}$,

$\overrightarrow{AD}=\hat{i}+7\hat{j}+(\lambda+1)\hat{k}$

Vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ will be coplanar.

$$\Rightarrow \begin{vmatrix} -1 & 5 & -3 \\ -4 & 3 & 3 \\ 1 & 7 & \lambda+1 \end{vmatrix}=0 \quad \Rightarrow \lambda=-\frac{146}{17}.$$

22. $$(2\hat{i}-\hat{j}+\hat{k})\times(3\hat{i}+4\hat{j})=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & 4 & 0 \end{vmatrix}$$

$$= \hat{i}(-4)+\hat{j}(3)+\hat{k}(11)$$

$\therefore$ Required unit vector

$$=\frac{-4\hat{i}+3\hat{j}+11\hat{k}}{\sqrt{(-4)^2+3^2+11^2}}=\frac{-4\hat{i}+3\hat{j}+11\hat{k}}{\sqrt{146}}.$$

23. A unit vector perpendicular to the planes $\vec{a}$ and $\vec{b}$ = $\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$

Now, $$\vec{a}\times\vec{b}=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -6 & -3 \\ 4 & 3 & -1 \end{vmatrix}$$

$$= \hat{i}(6+9)-\hat{j}(-2+12)-\hat{k}(6+24)$$

$$= 15\hat{i}-10\hat{j}+30\hat{k}$$

and $|\vec{a}\times\vec{b}|=\sqrt{15^2+(-10)^2+(30)^2}=\sqrt{1225}=35$

$\therefore$ Required vector $=\dfrac{15\hat{i}-10\hat{j}+30\hat{k}}{35}=\dfrac{3\hat{i}-2\hat{j}+6\hat{k}}{7}$.

24. Equation of the line joining $6\vec{a}-4\vec{b}-5\vec{c}$ and $-4\vec{c}$ is

$$\vec{r}=(6\vec{a}-4\vec{b}-5\vec{c})+\lambda(-6\vec{a}+4\vec{b}+\vec{c})$$

$$\vec{r}=\vec{a}(6-6\lambda)+\vec{b}(-4+4\lambda)+\vec{c}(-5+\lambda) \qquad \text{(i)}$$

Also, equation of the line joining $-\vec{a}-2\vec{b}-3\vec{c}$ and $\vec{a}+2\vec{b}-5\vec{c}$ is

$$\vec{r}=(-\vec{a}-2\vec{b}-3\vec{c})+\mu(2\vec{a}+4\vec{b}-2\vec{c})$$

$$\vec{r}=\vec{a}(-1+2\mu)+\vec{b}(-2+4\mu)+\vec{c}(-3-2\mu) \qquad \text{(ii)}$$

Comparing (i) and (ii), we get

$$6-6\lambda=-1+2\mu,\ -4+4\lambda=-2+4\mu$$

and $$-5+\lambda=-3-2\mu$$

After solving, we get $\lambda=1$ and $\mu = 1/2$

Substituting the value of λ in (i), we get the point of intersection as $\vec{r}=-4\vec{c}$.

25. Let $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$. Since $\vec{r}\cdot\hat{i}=\vec{r}\cdot\hat{j}=\vec{r}\cdot\hat{k}$

$$\Rightarrow \qquad x=y=z \qquad \text{(i)}$$

Also $|\vec{r}|=\sqrt{x^2+y^2+z^2}=3 \quad \Rightarrow \quad x=\pm\sqrt{3}$

Hence, the required vector $\vec{r}=\pm\sqrt{3}(\hat{i}+\hat{j}+\hat{k})$.

26. $$(\vec{a}+\vec{b})\cdot\vec{p}=\frac{\vec{a}\cdot\vec{b}\times\vec{c}}{[\vec{a}\cdot\vec{b}\cdot\vec{c}]}+\frac{\vec{b}\cdot(\vec{b}\times\vec{c})}{[\vec{a}\cdot\vec{b}\cdot\vec{c}]}$$

$$=\frac{[\vec{a}\cdot\vec{b}\cdot\vec{c}]}{[\vec{a}\cdot\vec{b}\cdot\vec{c}]}+0=1.$$

Similarly $(\vec{b}+\vec{c})\cdot\vec{q}=1$ and $(\vec{c}+\vec{a})\cdot\vec{r}=1$.

$\therefore$ Total value = 3.

27. If $\vec{x}$ is the position vector of B, then $\vec{a}$ divides $\overrightarrow{AB}$ in the ratio 2 : 3.

$\therefore$ $$\vec{a}=\frac{2\vec{x}+3(\vec{a}+2\vec{b})}{2+3}$$

$\Rightarrow$ $5\vec{a}-3\vec{a}-6\vec{b}=2\vec{x} \quad \Rightarrow \quad \vec{x}=\vec{a}-3\vec{b}$

28. Here, OA = 3, OB = 6

$\therefore$ Internal bisector OD divides AB in the ratio 1 : 2.

$\Rightarrow$ Position vector of D is (2, 8/3, 2).

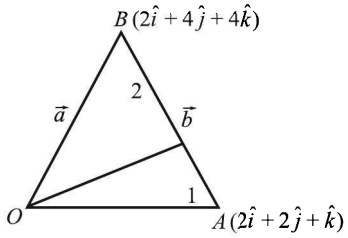


Fig. 23.33

$\therefore$ $$|OD|=\sqrt{4+\frac{64}{9}+4}=\sqrt{\frac{136}{9}}$$

29. Since $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular, so $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{c}=\vec{c}\cdot\vec{a}=0$.

Angle between $\vec{a}$ and $\vec{a}+\vec{b}+\vec{c}$ is

$$\cos\theta=\frac{\vec{a}(\vec{a}+\vec{b}+\vec{c})}{|\vec{a}||\vec{a}+\vec{b}+\vec{c}|} \qquad \text{(i)}$$

Now, $|\vec{a}|=|\vec{b}|=|\vec{c}|=a$

$$|\vec{a}+\vec{b}+\vec{c}|=\vec{a}^2+\vec{b}^2+\vec{c}^2+2\vec{a}\cdot\vec{b}+2\vec{b}\cdot\vec{c}+2\vec{c}\cdot\vec{a}$$

$$=\vec{a}^2+\vec{a}^2+\vec{a}^2+0+0+0$$

$\Rightarrow$ $|\vec{a}+\vec{b}+\vec{c}|^2=3a^2 \quad \Rightarrow \quad |\vec{a}+\vec{b}+\vec{c}|=\sqrt{3}a$

Putting this value in (i), we get $\theta=\cos^{-1}\dfrac{1}{\sqrt{3}}$.

30. $(\vec{a}\times\vec{b})\cdot\vec{c}$ represents the volume of the parallelopiped whose coterminous edges are $\vec{a}, \vec{b}, \vec{c}$ and this is $|\vec{a}||\vec{b}||\vec{c}|$ if it is a rectangular, i.e. if $\vec{a}\cdot\vec{b}=0, \vec{b}\cdot\vec{c}=0, \vec{c}\cdot\vec{a}=0$.

31. Given that $|\vec{a}|=|\vec{b}|$

Now, $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=\vec{a}\cdot\vec{a}+\vec{b}\cdot\vec{a}-\vec{b}\cdot\vec{a}-\vec{b}\vec{b}\ =0$

$(\because\ |\vec{a}|=|\vec{b}|)$

32. Here, $\overrightarrow{OA}=\hat{i}-\hat{j}+2\hat{k}$ and $\overrightarrow{OB}=2\hat{i}+\hat{j}-\hat{k}$ and $\overrightarrow{OC}=3\hat{i}-\hat{j}+2\hat{k}$.

This implies $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\hat{i}+2\hat{j}-3\hat{k}$ and $\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=2\hat{i}$

Hence, the required area is given by $=\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$

$$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&2&-3\\2&0&0\end{vmatrix}=-2(3\hat{j}+2\hat{k})$$

$\Rightarrow$ Area of the triangle $=\frac{1}{2}\times2|3\hat{j}+2\hat{k}|=\sqrt{13}$

33. $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&1&1\\2&-2&1\end{vmatrix}=\hat{i}(1+2)-\hat{j}(3-2)+\hat{k}(-6-2)$

$=3\hat{i}-\hat{j}-8\hat{k}$

$\therefore\ |\vec{a}\times\vec{b}|=|3\hat{i}-\hat{j}-8\hat{k}|$

$=\sqrt{9+1+64}=\sqrt{74}$

$\therefore\ \sin\theta=\frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}=\frac{\sqrt{74}}{\sqrt{9+1+1}\sqrt{4+4+1}}$

$=\frac{\sqrt{74}}{3\sqrt{11}}=\sqrt{\frac{74}{99}}$

34. Let $\vec{A}=\hat{i}-2\hat{j}+3\hat{k}$, $\vec{B}=-2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{C}=4\hat{i}-7\hat{j}+7\hat{k}$

$\therefore\ \overrightarrow{AB}=-2\hat{i}+3\hat{j}-\hat{k}-\hat{i}+2\hat{j}-3\hat{k}=-3\hat{i}+5\hat{j}-4\hat{k}$

and $\overrightarrow{AC}=4\hat{i}-7\hat{j}+7\hat{k}-\hat{i}+2\hat{j}-3\hat{k}=3\hat{i}-5\hat{j}+4\hat{k}$

$\therefore$ Area of $\Delta ABC=\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$

$$=\frac{1}{2}\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-3&5&-4\\3&-5&4\end{vmatrix}=-\frac{1}{2}\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&-5&4\\3&-5&4\end{vmatrix}$$

$=0$ [$\because$ two rows are identical]

35. The area of the parallelogram is given by $=|\overrightarrow{AB}\times\overrightarrow{AD}|$. Here, we are given adjacent sides and so

$$\overrightarrow{AB}\times\overrightarrow{AD}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&0&-1\\1&2&0\end{vmatrix}=2\hat{i}-\hat{j}+6\hat{k}$$

Hence, the required area is $=|2\hat{i}-\hat{j}+6\hat{k}|=\sqrt{41}$

36. $(-6\hat{i}+8\hat{k})\times(8\hat{i}+6\hat{k})\ =0\hat{i}+100\hat{j}+0\hat{k}$

$\therefore$ Required unit vector $=\frac{100\hat{j}}{100}=\hat{j}$.

37. Since $\hat{a},\hat{b},\hat{c}$ are mutually perpendicular unit vectors,

$\Rightarrow\ |\hat{a}|=|\hat{b}|=|\hat{c}|=1$ and $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=0$ (i)

Now, $|\hat{a}+\hat{b}+\hat{c}|^2=(\hat{a}+\hat{b}+\hat{c})\cdot(\hat{a}+\hat{b}+\hat{c}$

$=|\hat{a}|^2+|\hat{b}|^2+|\hat{c}|^2+2(\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{c}+\hat{c}\cdot\hat{a})$

$=1+1+1+0=3$

$\Rightarrow\ |\hat{a}+\hat{b}+\hat{c}|=\sqrt{3}$.

38. $\hat{i}\times(\vec{a}\times\hat{i})=(\hat{i}\cdot\hat{i})\vec{a}-(\vec{a}\cdot\hat{i})\hat{i}=\vec{a}-(\vec{a}\cdot\hat{i})\hat{i}$

Similarly, $\hat{j}\times(\vec{a}\times\hat{j})=\vec{a}-(\vec{a}\cdot\hat{j})\hat{j}$

and $\hat{k}\times(\vec{a}\times\hat{k})=\vec{a}-(\vec{a}\cdot\hat{k})\hat{k}$

$\Rightarrow\ \hat{i}\times(\vec{a}\times\hat{i})+\hat{j}\times(\vec{a}\times\hat{j})+\hat{k}\times(\vec{a}\times\hat{k})$

$=3\vec{a}-[(\vec{a}\cdot\hat{i})\hat{i}+(\vec{a}\cdot\hat{j})\hat{j}+(\vec{a}\cdot\hat{k})\hat{k}]=2\vec{a}$

39. Area $=\frac{1}{2}|(3\hat{i}+\hat{j}-2\hat{k})\times(\hat{i}-3\hat{j}+4\hat{k})|$

$=\frac{1}{2}|-2\hat{i}-14\hat{j}-10\hat{k}|$

$=\frac{\sqrt{4+196+100}}{2}=\frac{10\sqrt{3}}{2}=5\sqrt{3}$.

40. Given that $\vec{a}=(1,1,4)=\hat{i}+\hat{j}+4\hat{k}$

and $\vec{b}=(1,-1,4)=\hat{i}-\hat{j}+4\hat{k}$

$\therefore\ \vec{a}+\vec{b}=2\hat{i}+8\hat{k}\qquad\Rightarrow\ \vec{a}-\vec{b}=2\hat{j}$

Let θ be the angle between $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$, then

$\cos\theta=\frac{(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})}{|\vec{a}+\vec{b}||\vec{a}-\vec{b}|}$

$=\frac{(2\hat{i}+0\hat{j}+8\hat{k})\cdot(0\hat{i}+2\hat{j}+0\hat{k})}{\sqrt{2^2+0^2+8^2}\sqrt{0^2+2^2+0^2}}=\frac{0+0+0}{\sqrt{4+64}\sqrt{4}}=0$

$\Rightarrow\ \cos\theta=0\qquad\Rightarrow\ \theta=\frac{\pi}{2}=90°$.

41. We know that the position vector of the incentre of a ΔABC is given by

$$\frac{BC\cdot\vec{a}+CA\cdot\vec{b}+AB\cdot\vec{c}}{BC+CA+AB}$$

where $\vec{a},\vec{b},\vec{c}$ are the affixes of the vertices of the triangle.

Therefore, in the given triangle ABC, the affix of I is given by

$$\frac{|\overrightarrow{BC}|\overrightarrow{IA}+|\overrightarrow{CA}|\overrightarrow{IB}+|\overrightarrow{AB}|\overrightarrow{IC}}{|\overrightarrow{BC}|+|\overrightarrow{CA}|+|\overrightarrow{AB}|}$$

But it is the position vector of I with respect to I.

$$\therefore \quad \vec{0}=\frac{|\overrightarrow{BC}|\overrightarrow{IA}+|\overrightarrow{CA}|\overrightarrow{IB}+|\overrightarrow{AB}|\overrightarrow{IC}}{|\overrightarrow{BC}|+|\overrightarrow{CA}|+|\overrightarrow{AB}|}$$

$$\Rightarrow \quad |\overrightarrow{BC}|\overrightarrow{IA}+|\overrightarrow{CA}|\overrightarrow{IB}+|\overrightarrow{AB}|\overrightarrow{IC}=\vec{0}$$

42. $(\vec{a}\times\vec{b})^2=(\vec{a}\vec{b}\sin\theta)^2$

$$=\vec{a}^2\vec{b}^2(1-\cos^2\theta) = \vec{a}^2\vec{b}^2-\vec{a}^2\vec{b}^2\cos^2\theta$$
$$= \vec{a}^2\vec{b}^2-(\vec{a}\cdot\vec{b})^2.$$

43. Let $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=\hat{j}+\hat{k}$

Now, $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&0\\0&1&1\end{vmatrix}$

$$= \hat{i}(1)-\hat{j}(1)+\hat{k}(1)=\hat{i}-\hat{j}+\hat{k}$$

and $|\vec{a}\times\vec{b}|=\sqrt{(1)^2+(-1)^2+1^2}=\sqrt{3}$.

Unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{(\hat{i}+\hat{j})\times(\hat{j}+\hat{k})}{|(\hat{i}+\hat{j})\times(\hat{j}+\hat{k})|}=\frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$$

44. $(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})=|\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b}$

$$= 1 + 1 + 2(1)\,(1)\cos\theta = 2+2\cos\theta$$

or $|\vec{a}+\vec{b}|^2=2\cdot2\cos^2\frac{\theta}{2} \Rightarrow \cos\frac{\theta}{2}=\frac{1}{2}|\vec{a}+\vec{b}|$

45. $\overrightarrow{AB}=2\hat{i}-\hat{j}+\hat{k}$, $\overrightarrow{AB}\cdot\overrightarrow{DC}=2+3-5=0$

$\therefore$ ΔABC is right-angled.

$\overrightarrow{BC}=\hat{i}-3\hat{j}-5\hat{k}$, $\overrightarrow{CA}=\sqrt{3}\hat{i}-4\hat{j}-4\hat{k}$

$|\overrightarrow{CA}|=\sqrt{3+16+16}=\sqrt{35}$,

$|\overrightarrow{BC}|=\sqrt{1+9+25}=\sqrt{35}$

$\therefore$ $|\overrightarrow{BC}|=|\overrightarrow{CA}| \Rightarrow \Delta ABC$ is isosceles.

46. $|\vec{a}-\vec{b}|^2=|\vec{a}|^2+|\vec{b}|^2-2|\vec{a}||\vec{b}|\cos\theta$

$\Rightarrow |\vec{a}-\vec{b}|^2=1+1-2\cos60°=2-1$

$\Rightarrow |\vec{a}-\vec{b}|=1$

47. $|4\vec{a}+3\vec{b}|=\sqrt{(4\vec{a}+3\vec{b})\cdot(4\vec{a}+3\vec{b})}$

$$= \sqrt{16|\vec{a}|^2+9|\vec{b}|^2+24\vec{a}\cdot\vec{b}}$$

$$= \sqrt{144+144+24\times3\times4\times\left(\frac{-1}{2}\right)}=12$$

48. $\vec{\alpha}\times\vec{\beta}=-10\hat{i}+9\hat{j}+7\hat{k}$,

$\vec{\alpha}\times\vec{\gamma}=4\hat{i}-3\hat{j}-\hat{k}$.

$\therefore$ $(\vec{\alpha}\times\vec{\beta})\cdot(\vec{\alpha}\times\vec{\gamma})=-40-27-7=-74$

49. Given that $|\vec{a}|=2$, $|\vec{b}|=3$, $|\vec{c}|=4$

$\therefore$ $[\vec{a}\vec{b}\vec{c}]=\vec{a}\cdot(\vec{b}\times\vec{c})$

$$= \vec{a}\left(|\vec{b}||\vec{c}|\sin\frac{2\pi}{3}\hat{n}\right)$$

$$= |\vec{a}||\vec{b}||\vec{c}|\left(\sin\frac{2\pi}{3}\right)=2\times3\times4\times\frac{\sqrt{3}}{2}$$

$$= 12\sqrt{3}$$

50. $|\hat{a}-\hat{b}|=\sqrt{1^2+1^2-2\cdot1^2\cos\theta}=\sqrt{2(1-\cos\theta)}$

$$= \sqrt{2}\times\sqrt{2}\sin\left(\frac{\theta}{2}\right)=2\sin\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \sin\left(\frac{\theta}{2}\right)=\frac{|\hat{a}-\hat{b}|}{2}$$

51. $\vec{A}\cdot(\vec{B}+\vec{C})\times(\vec{A}+\vec{B}+\vec{C})$

$$= \vec{A}\cdot(\vec{B}\times\vec{A}+\vec{B}\times\vec{C}+\vec{C}\times\vec{A}+\vec{C}\times\vec{B})$$
$$= \vec{A}\cdot(\vec{B}\times\vec{A}+\vec{C}\times\vec{A})$$
$$= \vec{A}\cdot\vec{B}\times\vec{A}+\vec{A}\cdot\vec{C}\times\vec{A}$$
$$= [\vec{A}\vec{B}\vec{A}]+[\vec{A}\vec{C}\vec{A}]$$
$$= 0 + 0 = 0.$$

52. If $\vec{a}$ and $\vec{b}$ are two non-zero non-collinear vectors and x and y are two scalars such that $x\vec{a}+y\vec{b}=0$, then $x=0$, $y=0$. Otherwise one will be a scalar multiple of the other and hence collinear which is a contradiction.

53. Projection of $\vec{a}$ on $\vec{b}$ is $\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$.

$\Rightarrow$ Required projection $= \dfrac{(\hat{i}+2\hat{j}+2\hat{k}).(\hat{i}-2\hat{j}-2\hat{k})}{\sqrt{1+4+4}}$

$= -3$

54. $\vec{a}\cdot\vec{b}=0$ and $\vec{a}\times\vec{b}=0$ means either $\vec{a}$ or $\vec{b}$ is a null vector. ($\because$ both cannot be parallel and perpendicular)

55. Given that $\overrightarrow{OA}=2\hat{i}+\hat{j}-\hat{k}$, $\overrightarrow{OB}=3\hat{i}-2\hat{j}+\hat{k}$ and $\overrightarrow{OC}=\hat{i}+4\hat{j}-3\hat{k}$

$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=(3-2)\hat{i}+(-2-1)\hat{j}+(1+1)\hat{k}$

$=\hat{i}-3\hat{j}+2\hat{k}$

$|\overrightarrow{AB}|=\sqrt{1^2+(-3)^2+2^2}=\sqrt{1+9+4}=\sqrt{14}$

$\overrightarrow{BC}=\overrightarrow{OC}-\overrightarrow{OB}=(1-3)\hat{i}+(4+2)\hat{j}+(-3-1)\hat{k}$

$=-2\hat{i}+6\hat{j}-4\hat{k}$

$|\overrightarrow{BC}|=\sqrt{(-2)^2+6^2+(-4)^2}=\sqrt{4+36+16}=\sqrt{56}$

$\overrightarrow{CA}=\overrightarrow{OA}-\overrightarrow{OC}=(2-1)\hat{i}+(1-4)\hat{j}+(-1+3)\hat{k}$

$=\hat{i}-3\hat{j}+2\hat{k}$

$|\overrightarrow{CA}|=\sqrt{1^2+(-3)^2+(2)^2}=\sqrt{1+9+4}=\sqrt{14}$

It is clear that sides CA and AB of the triangle are equal.

$\therefore$ Points A, B, C form an isosceles triangle.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $\vec{a}+\vec{b}=-\vec{c}$

$\Rightarrow |\vec{a}+\vec{b}|^2=|\vec{c}|^2=1 \quad \Rightarrow |\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b}=1$

$\Rightarrow \vec{a}\cdot\vec{b}=-\dfrac{1}{2} \quad \Rightarrow |\vec{a}||\vec{b}|\cos\theta=-\dfrac{1}{2}$

$\Rightarrow \cos\theta=-\dfrac{1}{2} \quad \Rightarrow \theta=\dfrac{2\pi}{3}$

2. Given $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ and $|\vec{a}|=\sqrt{37}$, $|\vec{b}|=3$ and $|\vec{c}|=4$

Now, $\vec{a}+\vec{b}+\vec{c}=\vec{0} \quad \Rightarrow \quad \vec{a}=-(\vec{b}+\vec{c})$

$\Rightarrow |\vec{a}|^2=|-(\vec{b}+\vec{c})|^2$

$\Rightarrow |\vec{a}|^2=|\vec{b}|^2+|\vec{c}|^2+2|\vec{b}||\vec{c}|\cos\theta$

$\Rightarrow 9+16+24\cos\theta=37$

$\Rightarrow 37=25+24\cos\theta$

$\Rightarrow 24\cos\theta=12 \quad \Rightarrow \theta=60°$

3. $\vec{a}\cdot\vec{b}=[x\hat{i}+(x-1)\hat{j}+\hat{k}]\cdot[(x+1)\hat{i}+\hat{j}+a\hat{k}]$

$=x(x+1)+x-1+a=x^2+2x+a-1$

We must have $\vec{a}\cdot\vec{b}>0, \forall\, x\in\mathbb{R}$

$\Rightarrow x^2+2x+a-1>0, \forall\, x\in\mathbb{R}$

$\Rightarrow 4-4(a-1)<0 \quad \Rightarrow a>2$

4. We have $2\vec{a}+\vec{b}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{a}+2\vec{b}=\hat{i}+\hat{j}-\hat{k}$

Solving these two equations, we get

$\vec{a}=\dfrac{1}{3}(\hat{i}+\hat{j}+3\hat{k})$ and $\vec{b}=\dfrac{1}{3}(\hat{i}+\hat{j}-3\hat{k})$

$\therefore \cos\theta=\dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} \Rightarrow \cos\theta=-\dfrac{7}{11} \Rightarrow \theta=\cos^{-1}\left(-\dfrac{7}{11}\right)$

5. Angle between the line and the plane is given by

$$\sin\theta=\frac{(2\hat{i}+3\hat{j}+6\hat{k})\cdot(-\hat{i}+\hat{j}+\hat{k})}{\sqrt{4+9+36}\sqrt{1+1+1}}$$

$$=\frac{-2+3+6}{7\times\sqrt{3}}=\frac{7}{7\sqrt{3}}=\frac{1}{3}$$

$$\Rightarrow \quad \theta=\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

6. $(\vec{a}\times\vec{b})^2=|\vec{a}|^2\cdot|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2=\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b}\end{vmatrix}$

7. $\cos\dfrac{\pi}{3}=\dfrac{(\hat{i}+\hat{k})\cdot(\hat{i}+\hat{j}+a\hat{k})}{\sqrt{2}\sqrt{1+1+a^2}}$

$\Rightarrow \dfrac{1}{2}=\dfrac{1+a}{\sqrt{2}\sqrt{2+a^2}}$

$\Rightarrow \dfrac{1}{4}=\dfrac{(1+a)^2}{2(2+a^2)}$

$\Rightarrow 2+a^2=2(1+a^2+2a)$

$\Rightarrow a^2+4a=0 \quad \Rightarrow \quad a=0,-4$

8. $\Delta=\dfrac{1}{2}|\vec{a}\times\vec{b}|$

But $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 1 & -2\\ 1 & -3 & 4\end{vmatrix}=-2\hat{i}-14\hat{j}-10\hat{k}$

Hence, $\Delta=\dfrac{1}{2}|\vec{a}\times\vec{b}|=\dfrac{1}{2}\sqrt{4+196+100}=5\sqrt{3}$.

9. $(\vec{a}-\vec{b})\cdot(\vec{b}+\vec{c})\times(\vec{c}+\vec{a})$

$= (\vec{a}-\vec{b})\cdot[\vec{b}\times\vec{c}+\vec{b}\times\vec{a}+\vec{c}\times\vec{c}+\vec{c}\times\vec{a}]$

$= \vec{a}\cdot(\vec{b}\times\vec{c})+\vec{a}(\vec{b}\times\vec{a})+\vec{a}(\vec{c}\times\vec{a})$

$-\vec{b}(\vec{b}\times\vec{c})-\vec{b}(\vec{b}\times\vec{a})-\vec{b}(\vec{c}\times\vec{a})$

$= \vec{a}(\vec{b}\times\vec{c})-\vec{a}(\vec{b}\times\vec{c})$

$= [\vec{a}\vec{b}\vec{c}]-[\vec{a}\vec{b}\vec{c}]=0.$

10. Since $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}\neq 0 \Rightarrow \vec{a}\times\vec{b}-\vec{b}\times\vec{c}=0$

$\Rightarrow \vec{a}\times\vec{b}+\vec{c}\times\vec{b}=0 \Rightarrow (\vec{a}+\vec{c})\times\vec{b}=0$

$\Rightarrow \vec{a}+\vec{c}$ is parallel to $\vec{b}$

$\Rightarrow \vec{a}+\vec{c}=\hat{k}\vec{b}$.

11. $\vec{b}=\vec{b}_1+\vec{b}_2$, where $\vec{b}_1=x(\hat{i}+\hat{j})$ $[\because \vec{b}_1 \| \hat{i}+\hat{j}]$

$\vec{b}_2=y(\hat{i}-\hat{j}+\lambda\hat{k})$ $[\because \vec{b}_2\perp\hat{i}+\hat{j} \ \therefore \ \vec{b}_2\cdot(\hat{i}+\hat{j})=0]$

$\therefore \ \vec{b}=x(\hat{i}+\hat{j})+y(\hat{i}-\hat{j}+\lambda\hat{k})$

$\therefore \ 3\hat{i}+4\hat{k}=(x+y)\hat{i}+(x-y)\hat{j}+\lambda y\hat{k}$

$\therefore \ x+y=3, x-y=0, \lambda y=4$

$2x=3, \ x=\frac{3}{2}, \quad \therefore \ \vec{b}_1=\frac{3}{2}(\hat{i}+\hat{j})$.

12. Since $\vec{a},\vec{b},\vec{c}$ are three non-coplanar vectors, then

$(\vec{a}+\vec{b}+\vec{c})\cdot[(\vec{a}+\vec{b})\times(\vec{a}+\vec{c})]$

$= (\vec{a}+\vec{b}+\vec{c})\cdot[\vec{a}\times\vec{a}+\vec{a}\times\vec{c}+\vec{b}\times\vec{a}+\vec{b}\times\vec{c}]$

$= (\vec{a}+\vec{b}+\vec{c})\cdot[\vec{a}\times\vec{c}+\vec{b}\times\vec{a}+\vec{b}\times\vec{c}]$

$= [[\vec{a}\vec{a}\vec{c}]+[\vec{a}\vec{b}\vec{a}]+[\vec{a}\vec{b}\vec{c}]+[\vec{b}\vec{a}\vec{c}]+[\vec{b}\vec{b}\vec{c}]$

$+[\vec{c}\vec{a}\vec{c}]+[\vec{c}\vec{b}\vec{a}]+[\vec{c}\vec{b}\vec{c}]]$

$= -[\vec{a}\vec{b}\vec{c}]$.

13. $|\vec{u}\times\vec{v}| = |(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})|$

$= [2\vec{a}\times\vec{b}]=2|\vec{a}||\vec{b}|\sin\theta = 8\sin\theta$

$= 8\sqrt{1-\cos^2\theta}$

$= 8\sqrt{1-\left(\frac{\vec{a}\cdot\vec{b}}{4}\right)^2}$ $(\because \ \vec{a}\cdot\vec{b}=4\cos\theta)$

$= 2\sqrt{16-(\vec{a}\cdot\vec{b})^2}$

14. $a(-1)+2(5)+3(a)=0$

$\Rightarrow -a+10+3a=0 \Rightarrow 2a=-10$

$\Rightarrow a=-5$.

15. Let the unit vector be $a\hat{i}+b\hat{j}+c\hat{k}$.

$\because \ a\hat{i}+b\hat{j}+c\hat{k}$ is perpendicular to $\hat{i}+\hat{j}+\hat{k}$,

then $a+b+c=0$ (i)

and $a\hat{i}+b\hat{j}+c\hat{k}, (\hat{i}+\hat{j}+2\hat{k})$ and $(\hat{i}+2\hat{j}+\hat{k})$ are coplanar.

$\therefore \ \begin{vmatrix} a & b & c \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix}=0$

$\Rightarrow -3a+b+c=0$ (ii)

From Eqs. (i) and (ii), we get

$a = 0$ and $c = -b$

$\because \ a\hat{i}+b\hat{j}+c\hat{k}$ is a unit vector, then

$a^2+b^2+c^2=1 \Rightarrow 0+b^2+b^2=1$

$\Rightarrow b=\frac{1}{\sqrt{2}}$

$\therefore \ a\hat{i}+b\hat{j}+c\hat{k}=\frac{1}{\sqrt{2}}\hat{j}-\frac{1}{\sqrt{2}}\hat{k}=\frac{\hat{j}-\hat{k}}{\sqrt{2}}$

16. Angle between $\vec{a}$ and $\vec{b}$ $= 90°$

$\vec{a}\cdot\vec{b}=0$

Now, $(\vec{a}+\vec{b})^2=|\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b}$

$=|\vec{a}|^2+|\vec{b}|^2=1+1=2 \ \therefore \ |\vec{a}+\vec{b}|=\sqrt{2}$

17. $(\vec{a}+\vec{b}+\vec{c})\cdot(\vec{p}+\vec{q}+\vec{r})$

$= (\vec{a}\cdot\vec{p}+\vec{a}\cdot\vec{q}+\vec{a}\cdot\vec{r})+(\vec{b}\cdot\vec{p}+\vec{b}\cdot\vec{q}+\vec{b}\cdot\vec{r})$

$+(\vec{c}\cdot\vec{p}+\vec{c}\cdot\vec{q}+\vec{c}\cdot\vec{r})$

$[\because \ \vec{a}\cdot\vec{p}=1, \vec{b}\cdot\vec{q}=1, \vec{c}\cdot\vec{r}=1, \vec{a}\cdot\vec{q}=\vec{a}\cdot\vec{r}=0, \text{etc}]$

$= (1 + 0 + 0) + (0 + 1 + 0) + (0 + 0 + 1) = 3.$

18. Given that $|\vec{a}|=|\vec{b}|=1$ and $|\vec{a}+\vec{b}|=\sqrt{3}$

$\Rightarrow |\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b}=3 \Rightarrow \vec{a}\cdot\vec{b}=\frac{1}{2}$

$\therefore \ (3\vec{a}-4\vec{b})\cdot(2\vec{a}+5\vec{b})$

$= 6|\vec{a}|^2+15\vec{a}\cdot\vec{b}-8\vec{a}\cdot\vec{b}-20|\vec{b}|^2$

$= 6+7\vec{a}\cdot\vec{b}-20$ $[\because \ |\vec{a}|=|\vec{b}|=1]$

$= 6+\frac{7}{2}-20=\frac{7-28}{2}=-\frac{21}{2}$

19. Since $\vec{a}\times\vec{b}=\vec{c}$

$\therefore\ (\vec{b}\times\vec{c})\times\vec{b}=\vec{c} \quad\Rightarrow\ (\vec{b}\cdot\vec{b})\vec{c}-(\vec{b}\cdot\vec{c})\vec{b}=\vec{c}$

$\Rightarrow\ (\vec{b}\cdot\vec{b})=1,\ \vec{c}\cdot\vec{b}=0 \quad\Rightarrow\ |\vec{b}|^2=1$, i.e. $|\vec{b}|=1$

$\therefore\ \vec{b}$ is a unit vector. $\therefore\ |\vec{c}|=|\vec{a}|$, i.e. $c = a$.

20. Given that $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=2\hat{i}-\hat{k}$

Let $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$, then

$\vec{r}\times\vec{a}=\vec{b}\times\vec{a} \quad\Rightarrow\ (\vec{r}-\vec{b})\times\vec{a}=\vec{0}$

Now, $\vec{r}-\vec{b}=(x\hat{i}+y\hat{j}+z\hat{k})-(2\hat{i}-\hat{k})$

$= (x-2)\hat{i}+y\hat{j}+(z+1)\hat{k}$

$$\therefore\ (\vec{r}-\vec{b})\times\vec{a}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ x-2&y&z+1\\ 1&1&0\end{vmatrix}=\vec{0}$$

$\Rightarrow\ -(z+1)\hat{i}+(z+1)\hat{j}+(x-2-y)\hat{k}=\vec{0}$

Equating the coefficients of $\hat{i},\hat{j}$ and $\hat{k}$ we get

$$z=-1,\ x-y=2 \qquad \text{(i)}$$

Now, $\vec{r}\times\vec{b}=\vec{a}\times\vec{b} \quad\Rightarrow\ (\vec{r}-\vec{a})\times\vec{b}=\vec{0}$

and $\vec{r}-\vec{a}=(x-1)\hat{i}+(y-1)\hat{j}+z\hat{k}$

$$\therefore\quad (\vec{r}-\vec{a})\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ x-1&y-1&z\\ 2&0&-1\end{vmatrix}=\vec{0}$$

$\Rightarrow\ (-y+1)\hat{i}-\hat{j}(-x+1-2z)+(-2y+2)\hat{k}=\vec{0}$

$\Rightarrow\ y=1, x+2z=1 \qquad \text{(ii)}$

From Eqs. (i) and (ii), we get

$$x=3,\ y=1,\ z=-1$$

$\therefore\ \vec{r}=3\hat{i}+\hat{j}-\hat{k}$.

21. $(\vec{a}\times\vec{b})\times\vec{c}=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{b}\cdot\vec{c})\vec{a}$

$= (1-6+6)\vec{b}-(2+3+2)\vec{a}=\vec{b}-7\vec{a}$

$= 2\hat{i}+\hat{j}-\hat{k}-7\hat{i}+14\hat{j}+21\hat{k}$

$= -5\hat{i}+15\hat{j}+20\hat{k}$

$= 5(-\hat{i}+3\hat{j}+4\hat{k})$.

22. $(\vec{a}+\vec{b}+\vec{c})^2$

$= \vec{a}^2+\vec{b}^2+\vec{c}^2+2\vec{a}\cdot\vec{b}+2\vec{b}\cdot\vec{c}+2\vec{c}\cdot\vec{a}$

$= \vec{a}^2+\vec{b}^2+\vec{c}^2+\vec{a}\cdot(\vec{b}+\vec{c})+\vec{b}\cdot(\vec{c}+\vec{a})+\vec{c}(\vec{a}+\vec{b})$

$= 9+16+25+0+0+0=50$

$[\because\ \ \vec{a}\perp\vec{b}+\vec{c}, \vec{a}\cdot(\vec{b}+\vec{c})=0$; etc.$]$

$\therefore\ |\vec{a}+\vec{b}+\vec{c}|=5\sqrt{2}$.

23. Let O be the origin.

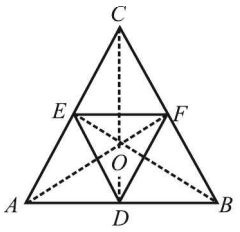


Fig. 23.34

$\therefore\ \overrightarrow{BE}+\overrightarrow{AF}=\overrightarrow{OE}-\overrightarrow{OB}+\overrightarrow{OF}-\overrightarrow{OA}$

$$=\frac{\overrightarrow{OA}+\overrightarrow{OC}}{2}-\overrightarrow{OB}+\frac{\overrightarrow{OB}+\overrightarrow{OC}}{2}-\overrightarrow{OA}$$

$$=\frac{\overrightarrow{OC}}{2}+\frac{\overrightarrow{OC}}{2}+\frac{\overrightarrow{OA}}{2}-\overrightarrow{OA}+\frac{\overrightarrow{OB}}{2}-\overrightarrow{OB}$$

$$=\overrightarrow{OC}-\frac{\overrightarrow{OA}+\overrightarrow{OB}}{2}=\overrightarrow{OC}-\overrightarrow{OD}=\overrightarrow{DC}$$

24. $[\vec{a}-2\vec{b}\quad \vec{b}-3\vec{c}\quad \vec{c}-4\vec{a}]$

$= (\vec{a}-2\vec{b})\times(\vec{b}-3\vec{c})\cdot(\vec{c}-4\vec{a})$

$= (\vec{a}\times\vec{b}-3\vec{a}\times\vec{c}+6\vec{b}\times\vec{c})\cdot(\vec{c}-4\vec{a})$

$= (\vec{a}\times\vec{b}\cdot\vec{c}-24\vec{b}\times\vec{c}\cdot\vec{a})$

$= -23(\vec{a}\times\vec{b})\cdot\vec{c}$

$= -69\times5=-345$

25. Let the unit vector in the xy-plane be $\hat{a}=x\hat{i}+y\hat{j}$.

$$\therefore\quad \cos45^\circ=\frac{(x\hat{i}+y\hat{j})\cdot(\hat{i}+\hat{j})}{\sqrt{x^2+y^2}\sqrt{1^2+1^2}}$$

$$\Rightarrow\ \frac{1}{\sqrt{2}}=\frac{x+y}{\sqrt{2}\sqrt{x^2+y^2}} \quad\Rightarrow\ 1=\frac{x+y}{\sqrt{x^2+y^2}}$$

$\Rightarrow\ x+y=\sqrt{x^2+y^2}$

Since $\hat{a}$ is a unit vector

$$\therefore\quad |\hat{a}|=\sqrt{x^2+y^2}=1 \quad\Rightarrow\quad x^2+y^2=1 \qquad \text{(i)}$$

Again $\cos60^\circ=\dfrac{(x\hat{i}+y\hat{j})\cdot(3\hat{i}-4\hat{j})}{\sqrt{x^2+y^2}\sqrt{3^2+4^2}}$

$$\Rightarrow\ \frac{1}{2}=\frac{3x-4y}{1(5)} \qquad\Rightarrow\ \frac{5}{2}=3x-4y$$

$$5=6x-8y \qquad \text{(ii)}$$

On solving Eqs. (i) and (ii), we get

$$x=\frac{13}{14},\ y=\frac{1}{14}$$

$\therefore \quad \hat{a} = \frac{1}{14}(13\hat{i} + \hat{j})$

No value in the given options satisfies the above relations. Thus, option (d) is correct.

26. $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} \quad \Rightarrow \cos\frac{5\pi}{6} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

Projection of $\vec{a}$ in the direction of $\vec{b}$

$= \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = -\frac{6}{\sqrt{3}}$

$\therefore \quad \cos\left(\frac{5\pi}{6}\right) = -\frac{6}{\sqrt{3}}\cdot\frac{1}{|\vec{a}|}$

$\Rightarrow -\frac{\sqrt{3}}{2} = -\frac{6}{\sqrt{3}}\cdot\frac{1}{|\vec{a}|} \quad \Rightarrow \quad |\vec{a}| = \frac{12}{3} = 4.$

27. Let $\vec{c}$ be the position vector of the point C.
Then as $AC = 3AB$,

$\Rightarrow \overrightarrow{AC} = 3\overrightarrow{AB}$

$\therefore \quad \vec{c} - \vec{a} = 3(\vec{b} - \vec{a}) = 3\vec{b} - 3\vec{a}$

$\Rightarrow \vec{c} = 3\vec{b} - 2\vec{a}.$

28. Given that $|\vec{a}| = |\vec{c}| = 1, \ |\vec{b}| = 4$

Let the angle between $\vec{b}$ and $\vec{c}$ be α, then $|\vec{b}\times\vec{c}| = \sqrt{15}$

$\Rightarrow |\vec{b}||\vec{c}|\sin\alpha = \sqrt{15} \quad \Rightarrow \sin\alpha = \frac{\sqrt{15}}{4\times1} = \frac{\sqrt{15}}{4}$

$\therefore \quad \cos\alpha = \sqrt{1-\sin^2\alpha} = \frac{1}{4}$

We have $\vec{b} - 2\vec{c} = \lambda\vec{a}$

On squaring both sides, we get $(\vec{b} - 2\vec{c})^2 = \lambda^2(\vec{a})^2$

$\Rightarrow |\vec{b}|^2 + 4|\vec{c}|^2 - 4\vec{b}\cdot\vec{c} = \lambda^2|\vec{a}|^2$

$\Rightarrow 16 + 4 - 4|\vec{b}||\vec{c}|\cos\alpha = \lambda^2$

$\Rightarrow 16 + 4 - 4\times4\times1\times\frac{1}{4} = \lambda^2$

$\Rightarrow \lambda^2 = 16 + 4 - 4 = 16$

$\Rightarrow \lambda = \pm4$

29. Since dot and cross can be interchanged without changing the cyclic order in a scalar triple product,

$\therefore \quad \vec{v}\cdot(\vec{u}\times\vec{w})$ is different from all the other three.

30. $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors with magnitudes $|\vec{a}| = 4, |\vec{b}| = 4$ and $|\vec{c}| = 2$. But given that $\vec{a}$ is perpendicular to $\vec{b} + \vec{c}$,

$\Rightarrow \vec{a}\cdot(\vec{b} + \vec{c}) = 0 \quad \Rightarrow \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} = 0 \qquad$ (i)

$\vec{b}$ is perpendicular to $(\vec{c} + \vec{a})$

$\Rightarrow \vec{b}\cdot(\vec{c} + \vec{a}) = 0 \quad \Rightarrow \vec{b}\cdot\vec{c} + \vec{b}\cdot\vec{a} = 0 \qquad$ (ii)

and $\vec{c}$ is perpendicular to $(\vec{a} + \vec{b})$

$\Rightarrow \vec{c}\cdot(\vec{a} + \vec{b}) = 0 \quad \Rightarrow \vec{c}\cdot\vec{a} + \vec{c}\cdot\vec{b} = 0 \qquad$ (iii)

On adding Eqs. (i), (ii) and (iii), we get

$$2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 0$$

Now, $|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

$= 4^2 + 4^2 + 2^2 = 36$

$\Rightarrow |\vec{a} + \vec{b} + \vec{c}| = 6.$

31. Since C is the mid-point of AB,

$\therefore \quad \overrightarrow{AC} = \overrightarrow{CB}$

$\Rightarrow \overrightarrow{PC} - \overrightarrow{PA} = \overrightarrow{PB} - \overrightarrow{PC} \quad \Rightarrow \overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$

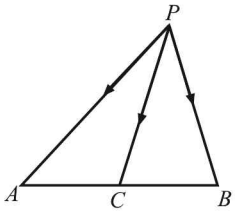


Fig. 23.35

32. Let $\hat{d} = x\hat{i} + y\hat{j} + z\hat{k}$

Since $\hat{d}$ is a unit vector,

where $\quad x^2 + y^2 + z^2 = 1 \qquad$ (i)

Also, $\vec{a}\cdot\hat{d} = 0 \quad \Rightarrow x - y = 0$

$\Rightarrow \quad x = y \qquad$ (ii)

$[\vec{b}\vec{c}\hat{d}] = 0$

$\Rightarrow \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ x & y & z \end{vmatrix} = 0 \quad \Rightarrow x + y + z = 0$

$\Rightarrow 2x + z = 0 \quad \Rightarrow z = -2x$

From (i), (ii) and (iii), we get

$x^2 + x^2 + 4x^2 = 1 \quad \Rightarrow x = \pm\frac{1}{\sqrt{6}}$

$\therefore \quad \hat{d} = \pm\left(\frac{1}{\sqrt{6}}\hat{i} + \frac{1}{\sqrt{6}}\hat{j} - \frac{2}{\sqrt{6}}\hat{k}\right) = \pm\left(\frac{\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{6}}\right)$

33. Required unit vector $= \frac{(\hat{i} - \hat{j})\times(\hat{i} + \hat{j})}{|(\hat{i} - \hat{j})\times(\hat{i} + \hat{j})|}$

$= \frac{2\hat{k}}{|2\hat{k}|} = \hat{k}.$

34. $V=\begin{vmatrix}1 & a & 0\\ a & 1 & 1\\ 0 & 1 & a\end{vmatrix}=a-1-a^3$

$\therefore \quad \dfrac{dV}{da}=1-3a^2=0$

$\therefore \quad a=\pm\dfrac{1}{\sqrt{3}} \quad \Rightarrow \quad \dfrac{d^2V}{da^2}=-6a$

$$\left(\frac{d^2V}{da^2}\right)_{(a=1/\sqrt{3})}=-\frac{6}{\sqrt{3}}(-ve)$$

$\therefore$ V is maximum at $a=\dfrac{1}{\sqrt{3}}$.

35. $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]=(\vec{a}\times\vec{b})\cdot[(\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})]$

$=(\vec{a}\times\vec{b})\cdot\{[\vec{b}\cdot(\vec{c}\times\vec{a})]\vec{c}-[\vec{c}\cdot(\vec{c}\times\vec{a})]\vec{b}\}$

$=(\vec{a}\times\vec{b})\cdot\{[\vec{b}\vec{c}\vec{a}]\vec{c}-0\}$

$=(\vec{a}\times\vec{b}\cdot\vec{c})\,[\vec{b}\vec{c}\vec{a}]=[\vec{a}\vec{b}\vec{c}]^2$

$\vec{a},\vec{b},\vec{c}$ are coplanar.

$\therefore$ $[\vec{a}\vec{b}\vec{c}]=0$ and then $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]=0$

36. Since $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d})=(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d})-(\vec{b}\cdot\vec{c})(\vec{a}\cdot\vec{d})$

$=\dfrac{3}{2}\times2-\dfrac{1}{2}\times3$

$=3-\dfrac{3}{2}=\dfrac{3}{2}$

37. If $\vec{a}=\vec{b}$, then $|\vec{a}|=|\vec{b}|$

Now, $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{a}=|\vec{a}|^2$ and $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{b}=|\vec{b}|^2$

$\therefore$ $\vec{a}\cdot\vec{b}=|\vec{a}|^2=|\vec{b}|^2$

But it is true that if $|\vec{a}|=|\vec{b}|$, it does not imply that $\vec{a}=\vec{b}$.

38. Here, $\vec{c}=\vec{b}\times\vec{a}$

$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 1 & 0\\ x & y & z\end{vmatrix}=z\hat{i}-x\hat{k}$

39. $|\overrightarrow{AB}|=\sqrt{(7-1)^2+(-4+6)^2+(7-10)^2}=7$

$|\overrightarrow{BC}|=\sqrt{(1+1)^2+(-6+3)^2+(10-4)^2}=7$

$|\overrightarrow{CD}|=\sqrt{(-1-5)^2+(-3+1)^2+(4-5)^2}=\sqrt{41}$

and $|\overrightarrow{DA}|=\sqrt{(5-7)^2+(-1+4)^2+(5-7)^2}=\sqrt{17}$

40. Vector perpendicular to face OAB is $\vec{n}_1$

$=\overrightarrow{OA}\times\overrightarrow{OB}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 1\\ 2 & 1 & 3\end{vmatrix}=5\hat{i}-\hat{j}-3\hat{k}$

Vector perpendicular to face ABC is $\vec{n}_2$

$=\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 2\\ -2 & -1 & 1\end{vmatrix}=\hat{i}-5\hat{j}-3\hat{k}$

$\therefore \quad \cos\theta=\dfrac{\vec{n}_1\cdot\vec{n}_2}{|\vec{n}_1|\cdot|\vec{n}_2|}$

$=\dfrac{5\times1+(-1)\times(-5)+(-3)(-3)}{\sqrt{5^3+(-1)^2+(-3)^2}\sqrt{1^2+(-5)^2+(-3)^2}}$

$=\dfrac{5+5+9}{\sqrt{35}\times\sqrt{35}}=\dfrac{19}{35}$

$\Rightarrow \quad \theta=\cos^{-1}\left(\dfrac{19}{35}\right)$

41. $(\vec{u}+\vec{v}-\vec{w})\cdot[(\vec{u}-\vec{v})\times(\vec{v}-\vec{w})]$

$=(\vec{u}+\vec{v}-\vec{w})\cdot[\vec{u}\times\vec{v}-\vec{u}\times\vec{w}+\vec{v}\times\vec{w}]$

$=\vec{u}\cdot\vec{v}\times\vec{w}-\vec{v}\cdot\vec{u}\times\vec{w}-\vec{w}\cdot\vec{u}\times\vec{v}$

$=\vec{u}\cdot\vec{v}\times\vec{w}+\vec{w}\cdot\vec{u}\times\vec{v}-\vec{w}\cdot\vec{u}\times\vec{v}$

$=\vec{u}\cdot\vec{v}\times\vec{w}$

42. $\dfrac{1}{3}|\vec{b}|\cdot|\vec{c}\,|\vec{a}=(\vec{a}\times\vec{b})\times\vec{c}$

$=(\vec{c}\cdot\vec{a})\vec{b}-(\vec{b}\cdot\vec{c})\,\vec{a}$

$\therefore \quad \vec{b}\cdot\vec{c}=-\dfrac{1}{3}|\vec{b}|\cdot|\vec{c}| \quad \Rightarrow \quad \cos\theta=-\dfrac{1}{3}$

$\Rightarrow \quad \sin\theta=\dfrac{2\sqrt{2}}{3}$

43. If $\vec{a}+2\vec{b}$ is collinear with $\vec{c}$, then

$$\vec{a}+2\vec{b}=t\vec{c} \qquad \text{(i)}$$

Also, if $\vec{b}+3\vec{c}$ is collinear with $\vec{a}$, then

$$\vec{b}+3\vec{c}=\lambda\vec{a} \quad \text{and} \quad \vec{b}=\lambda\vec{a}-3\vec{c} \qquad \text{(ii)}$$

On putting this value in Eq. (i), we get

$\vec{a}+2(\lambda\vec{a}-3\vec{c})=t\vec{c}$

$\vec{a}+2\lambda\vec{a}-6\vec{c}=t\vec{c}$

$(\vec{a}-6\vec{c})=t\vec{c}-2\lambda\vec{a}$

On comparing, we get

$1=-2\lambda \quad \Rightarrow \quad \lambda=-1/2$

and $-6=t \quad \Rightarrow \quad t=-6$

From Eq. (i), we get

$\vec{a}+2\vec{b}=-6\vec{c} \quad \Rightarrow \quad \vec{a}+2\vec{b}+6\vec{c}=\vec{0}$.

44. $|\vec{u}|=1, |\vec{v}|=2, |\vec{w}|=3$. By the given condition,

$\Rightarrow \dfrac{\vec{w}\cdot\vec{v}}{|\vec{u}|}=\dfrac{\vec{v}\cdot\vec{u}}{|\vec{u}|} \quad \Rightarrow \vec{v}\cdot\vec{u}=\vec{w}\cdot\vec{u}$

Also, $\vec{v}\cdot\vec{w}=0$

Now, $|\vec{u}-\vec{v}+\vec{w}|^2=\vec{u}^2+\vec{v}^2+\vec{w}^2-2\vec{u}\cdot\vec{v}-2\vec{v}\cdot\vec{w}+2\vec{w}\cdot\vec{u}$

$= 1+4+9-2\vec{v}\cdot\vec{w} \quad (\because \vec{v}\cdot\vec{u}=\vec{u}\cdot\vec{w})$

$= 14-2(0)=14 \quad \therefore |\vec{u}-\vec{v}+\vec{w}|=\sqrt{14}$

45. Since the given points lie in a plane,

$$\therefore \begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix}=0$$

Applying $C_1 \to C_1 - C_2$

$$\Rightarrow \begin{vmatrix} 0 & a & c \\ 1 & 0 & 1 \\ 0 & c & b \end{vmatrix}=0 \quad \Rightarrow -(ab-c^2)=0$$

$\Rightarrow c^2=ab$

Hence, c is G.M. of a and b.

46. The line is parallel to the plane as

$$(\hat{i}-\hat{j}+4\hat{k})\cdot(\hat{i}+5\hat{j}+\hat{k})=0$$

General point on the line is $(\lambda+2, -\lambda-2, 4\lambda+3)$ for $\lambda=0$, a point on this line is (2, –2, 3) and distance from $\vec{r}\cdot(\hat{i}+5\hat{j}+\hat{k})=5$.

or $x+5y+z=5$

$$\therefore \quad d=\frac{|2+5(-2)+3-5|}{\sqrt{1+25+1}} \quad \Rightarrow \quad d=\frac{|-10|}{3\sqrt{3}}=\frac{10}{3\sqrt{3}}$$

47. We have $\vec{a}=\hat{i}-\hat{k}$, $\vec{b}=x\hat{i}+\hat{j}+(1-x)\hat{k}$ and $\vec{c}=y\hat{i}+x\hat{j}+(1+x-y)\hat{k}$

$$[\vec{a}\vec{b}\vec{c}]=\begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix}$$

Applying $C_3 \to C_3 + C_1$

$$=\begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ y & x & 1+x \end{vmatrix}=1\begin{vmatrix} 1 & 1 \\ x & 1+x \end{vmatrix}$$

$= 1(1+x)-x=1$

Thus, $[\vec{a}\vec{b}\vec{c}]$ depends upon neither x nor y.

48. Let $(2\hat{u}\times 3\hat{v})$ is a unit vector.

$\therefore \quad |2\hat{u}\times 3\hat{v}|=1$

$\Rightarrow 6|\hat{u}||\hat{v}||\sin\theta|=1 \quad \Rightarrow \sin\theta=\dfrac{1}{6}$

Hence, there is exactly one value of θ for which $(2\hat{u}\times 3\hat{v})$ is a unit vector.

49. Since $\vec{a}=8\vec{b}$ and $\vec{c}=-7\vec{b}$

$\therefore$ $\vec{a}$ is parallel to $\vec{b}$ and $\vec{c}$ is anti-parallel to $\vec{b}$.

$\Rightarrow$ $\vec{a}$ and $\vec{c}$ are anti-parallel.

$\Rightarrow$ Angle between $\vec{a}$ and $\vec{c}$ is π.

50. Since $\vec{a}, \vec{b}, \vec{c}$ are coplanar,

$$\Rightarrow \begin{vmatrix} \alpha & 2 & \beta \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}=0$$

$\Rightarrow \alpha(1-0)-2(1-0)+\beta(1+0)=0$

$\Rightarrow \alpha+\beta=2$ which is possible for $\alpha=1, \beta=1$.

51. Since $[3\vec{u}\ p\vec{v}\ p\vec{w}]-[p\vec{v}\ \ \vec{w}\ q\vec{u}]-[2\vec{w}\ \ q\vec{v}\ \ q\vec{u}]=0$,

$\therefore \quad 3p^2[\vec{u}\cdot(\vec{v}\times\vec{w})]-pq[\vec{v}\cdot(\vec{w}\times\vec{u})]-2q^2[\vec{w}\cdot(\vec{v}\times\vec{u})]=0$

$\Rightarrow (3p^2-pq+2q^2)[\vec{u}\cdot(\vec{v}\times\vec{w})]=0$

But $[\vec{u}\vec{v}\vec{w}]\neq 0$

$\Rightarrow 3p^2-pq+2q^2=0$

$\Rightarrow p=q=0$.

52. Since the given vectors are mutually orthogonal, therefore

$$\vec{a}\cdot\vec{b}=2-4+2=0$$

$$\vec{a}\cdot\vec{c}=\lambda-1+2\mu=0 \qquad \text{(i)}$$

$$\vec{b}\cdot\vec{c}=2\lambda+4+\mu=0 \qquad \text{(ii)}$$

On solving Eqs. (i) and (ii), we get

$$\mu=2 \quad \text{and} \quad \lambda=-3$$

Hence, $(\lambda, m) = (-3, 2)$

53. We have $\vec{a}\times\vec{b}+\vec{c}=0$

$\Rightarrow \vec{a}\times(\vec{a}\times\vec{b})+\vec{a}\times\vec{c}=0$

$\Rightarrow (\vec{a}\cdot\vec{b})\vec{a}-(\vec{a}\cdot\vec{a})\vec{b}+\vec{a}\times\vec{c}=0$

$\Rightarrow 3\vec{a}-2\vec{b}+\vec{a}\times\vec{c}=0 \quad \Rightarrow \quad 2\vec{b}=3\vec{a}+\vec{a}\times\vec{c}$

$\Rightarrow 2\vec{b}=3\hat{j}-3\hat{k}-2\hat{i}-\hat{j}-\hat{k}=-2\hat{i}+2\hat{j}-4\hat{k}$

$\Rightarrow \vec{b}=-\hat{i}+\hat{j}-2\hat{k}$

54. $\overrightarrow{PQ}=6\hat{i}+\hat{j},\ \overrightarrow{QR}=-\hat{i}+3\hat{j},\ \overrightarrow{RS}=-6\hat{i}-\hat{j}$

$\overrightarrow{SP}=\hat{i}-3\hat{j},\ |\overrightarrow{PQ}|=\sqrt{37}=|\overrightarrow{RS}|$

$|\overrightarrow{QR}|=\sqrt{10}=|\overrightarrow{SP}|$

$\overrightarrow{PQ}\cdot\overrightarrow{QR}=-6+3=-3\neq 0$

$\overrightarrow{PQ}$ is anti-parallel to $\overrightarrow{RS}$ and their magnitudes are equal.

$\Rightarrow$ Quadrilateral $PQRS$ must be a parallelogram, which is neither a rhombus nor a rectangle.

55. $(\hat{i}+\hat{j}+2\hat{k})\cdot\left(\dfrac{m\hat{i}+2\hat{j}+3\hat{k}}{\sqrt{13+m^2}}\right)=2$

$\Rightarrow\ m+2+6=2\sqrt{13+m^2}$

$\Rightarrow\ (m+8)^2=4(13+m^3)$

$\Rightarrow\ m^2+16m+64=4m^2+52$

$\Rightarrow\ 3m^2-16m-12=0$

$\Rightarrow\ (3m+2)(m-6)=0$

$\Rightarrow\ m=6,-\dfrac{2}{3}$

56. $\left[\dfrac{1}{2}|\vec{u}_2-\vec{u}_1|\right]^2=\dfrac{1}{4}\left[|\vec{u}_2|^2+|\vec{u}_1|^2-2\vec{u}_2\cdot\vec{u}_1\right]$

$=\dfrac{1}{4}\left[1+1-2|\vec{u}_2||\vec{u}_1|\cos\theta\right]$

$=\dfrac{1}{4}\left[2-2\cos\theta\right]=\sin^2\dfrac{\theta}{2}$

$\Rightarrow\ \dfrac{1}{2}|\vec{u}_2-\vec{u}_1|=\sin\dfrac{\theta}{2}$

57. Let $\overrightarrow{OA}=\hat{i}+2\hat{j}+3\hat{k}$ and $\overrightarrow{OB}=3\hat{i}+4\hat{j}+5\hat{k}$

$\therefore\quad \overrightarrow{AB}=2\hat{i}+2\hat{j}+2\hat{k}$

$\therefore$ Work done, $W=\vec{F}\cdot\overrightarrow{AB}$

$=(2\hat{i}-3\hat{j}+2\hat{k})\cdot(2\hat{i}+2\hat{j}+2\hat{k})$

$=4-6+4=2$

58. Area $=|\vec{a}\times\vec{b}|=15$

If the sides are $(3\vec{a}+2\vec{b})$ and $(\vec{a}+3\vec{b})$, then the area of the parallelogram

$=|(3\vec{a}+2\vec{b})\times(\vec{a}+3\vec{b})|=7|\vec{a}\times\vec{b}|$

$=7\times15=105$ sq. units

59. We have $\vec{\alpha}=x(\vec{a}\times\vec{b})+y(\vec{b}\times\vec{c})+z(\vec{c}\times\vec{a})$

Taking dot product with $\vec{a},\vec{b},\vec{c}$ respectively, we get

$\vec{\alpha}\cdot\vec{a}=y[\vec{a}\vec{b}\vec{c}]\ \Rightarrow\ y=8(\vec{\alpha}\cdot\vec{a})$

$\vec{\alpha}\cdot\vec{b}=z[(\vec{c}\times\vec{a})\cdot\vec{b}]$

$\Rightarrow\ \vec{\alpha}\cdot\vec{b}=z[\vec{a}\vec{b}\vec{c}]\ \Rightarrow\ z=8(\vec{\alpha}\cdot\vec{b})$

and $\vec{\alpha}\cdot\vec{c}=x(\vec{a}\times\vec{b}\cdot\vec{c})$

$\Rightarrow\ \vec{\alpha}\cdot\vec{c}=x[\vec{a}\vec{b}\vec{c}]\ \Rightarrow\ x=8(\vec{\alpha}\cdot\vec{c})$

$\therefore\quad x+y+z=8\vec{\alpha}\cdot(\vec{a}+\vec{b}+\vec{c})$

60. Let $\vec{a}=\hat{i}-2\hat{j}+3\hat{k},\ \vec{b}=-2\hat{i}+3\hat{j}-4\hat{k}$ and $\vec{c}=\lambda\hat{i}-\hat{j}+2\hat{k}$

$\therefore\quad [\vec{a}\vec{b}\vec{c}]=0$

$\Rightarrow\ \begin{vmatrix}1&-2&3\\-2&3&-4\\\lambda&-1&2\end{vmatrix}=0$

$\Rightarrow\ 1(6-4)+2(-4+4\lambda)+3(2-3\lambda)=0$

$\Rightarrow\ \lambda=0$

61. $\because\quad \vec{\alpha}\times\vec{\beta}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&-1\\-1&2&-4\end{vmatrix}=-10\hat{i}+9\hat{j}+7\hat{k}$

$\therefore\quad (\vec{\alpha}\times\vec{\beta})\cdot(\vec{\alpha}\times\vec{\gamma})=\begin{vmatrix}-10&9&7\\2&3&-1\\1&1&1\end{vmatrix}$

$=-10(3+1)-9(2+1)+7(2-3)$

$=-74$

Alternate method

$(\vec{\alpha}\times\vec{\beta})\cdot(\vec{\alpha}\times\vec{\gamma})=\begin{vmatrix}\vec{\alpha}\cdot\vec{\alpha}&\vec{\alpha}\cdot\vec{\gamma}\\\vec{\alpha}\cdot\vec{\beta}&\vec{\beta}\cdot\vec{\gamma}\end{vmatrix}$

$=\begin{vmatrix}14&4\\8&-3\end{vmatrix}=-42-32$

$=-74$

62. Volume of tetrahedron $=\dfrac{1}{6}[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}]$

$=\dfrac{1}{6}\begin{vmatrix}1&1&1\\1&-1&1\\1&2&-1\end{vmatrix}$

$=\dfrac{1}{6}[-1+2+3]=\dfrac{2}{3}$ cu unit

63. Here, $\vec{a}_1=6\hat{i}+2\hat{j}+2\hat{k},\ \vec{a}_2=-4\hat{i}+0\hat{j}-\hat{k}$,

$\vec{b}_1=\hat{i}-2\hat{j}+2\hat{k}$ and $\vec{b}_2=3\hat{i}-2\hat{j}-2\hat{k}$

$\therefore$ Shortest distance $=\left|\dfrac{(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right|$

$=\left|\dfrac{(-10\hat{i}-2\hat{j}-3\hat{k})\cdot(8\hat{i}+8\hat{j}+4\hat{k})}{\sqrt{64+64+16}}\right|$

$=\left|\dfrac{-108}{12}\right|=9$

64. $\begin{vmatrix} p & 1 & 1 \\ 1 & q & 1 \\ 1 & 1 & r \end{vmatrix} = 0$

$\Rightarrow\ p(qr-1)+1(1-r)+1(1-q)=0$

$\Rightarrow\ pqr-p+1-r+1-q=0$

$\Rightarrow\ pqr-(p+q-r)=-2$

65. $\vec{a}+3\vec{b}=\lambda\vec{c}$ (i)

$\vec{b}+2\vec{c}=\mu\vec{a}$ (ii)

(i) – 3(ii) gives $(1+3\mu)\vec{a}-(\lambda+6)\vec{c}=0$

As $\vec{a}$ and $\vec{c}$ are non-collinear,

$\therefore\ 1+3\mu=0$ and $\lambda+6=0$

From (i), $\vec{a}+3\vec{b}+6\vec{c}=\vec{0}$

66. $\vec{c}\cdot\vec{d}=\vec{0}$

$\Rightarrow\ (\vec{a}+2\vec{b})\cdot(5\vec{a}-4\vec{b})=0$

$\Rightarrow\ 5|\vec{a}|^2+6\vec{a}\cdot\vec{b}-8|\vec{b}|^2=0$

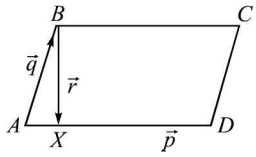
$\Rightarrow\ 5+6\vec{a}\cdot\vec{b}-8=0 \qquad [\because\ |\hat{a}|^2=1, |\hat{b}|=1]$

$\Rightarrow\ 6\vec{a}\cdot\vec{b}=3 \ \Rightarrow\ \vec{a}\cdot\vec{b}=1/2 \ \Rightarrow\ \theta=\pi/3$

67. $\overrightarrow{AX}=\dfrac{\vec{p}\cdot\vec{q}\vec{p}}{|\vec{p}||\vec{p}|}=\dfrac{\vec{p}\cdot\vec{q}}{|\vec{p}|^2}\vec{p}$

$\overrightarrow{BX}=\overrightarrow{BA}+\overrightarrow{AX}$

$=-\vec{q}+\dfrac{\vec{p}\cdot\vec{q}}{|\vec{p}|^2}\vec{p}$



MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. *OABCDE* is a regular hexagon of side 2 units in the *xy*-plane. *O* being the origin and *OA* taken along the *x*-axis. A point *P* is taken on a line parallel to the *z*-axis through the centre of the hexagon at a distance of 3 units from *O*. Then the vector $\overrightarrow{AP}$ is

(a) $-\hat{i}+3\hat{j}+\sqrt{5}\hat{k}$ (b) $\hat{i}-\sqrt{3}\hat{j}+5\hat{k}$

(c) $-\hat{i}+\sqrt{3}\hat{j}+\sqrt{5}\hat{k}$ (d) $\hat{i}+\sqrt{3}\hat{j}+\sqrt{5}\hat{k}$

2. Let $\vec{p}$ be the position vector of the orthocentre and $\vec{g}$ be the position vector of the centroid of the triangle *ABC*, where the circumcentre is the origin. If $\vec{p}=K\vec{g}$, then *K* is equal to

(a) 3 (b) 2 (c) 1/3 (d) 2/3

3. If $\hat{p}$ and $\hat{d}$ are two unit vectors and θ is the angle between them, then

(a) $\dfrac{1}{2}|\hat{p}-\hat{d}|=\sin\dfrac{\theta}{2}$ (b) $\hat{p}\times\hat{d}=\sin\theta$

(c) $\dfrac{1}{2}(\hat{p}-\hat{d})^2=1-\cos\theta$ (d) $\dfrac{1}{2}(\hat{p}-\hat{d})^2=1-\cos 2\theta$

4. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^2 \end{vmatrix}=0$ and the vectors $\vec{A}=(1,a,a^2)$, $\vec{B}=(1,b,b^2)$, $\vec{C}=(1,c,c^2)$ are non-coplanar, then the value of *abc* wil be

(a) –1 (b) 1 (c) 0 (d) none of these

5. *A*, *B*, *C* and *D* are four points in a plane with position vectors $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$, respectively such that

$(\vec{a}-\vec{d})\cdot(\vec{b}-\vec{c})=(\vec{b}-\vec{d})\cdot(\vec{c}-\vec{a})=0$.

Then for the triangle *ABC*, *D* is its

(a) incentre (b) circumcentre

(c) orthocentre (d) centroid

6. The unit vector perpendicular to the plane of the triangle *ABC* with position vectors $\vec{a},\vec{b},\vec{c}$ of the vertices *A*, *B*, *C* is

(a) $\dfrac{(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a})}{\Delta}$ (b) $\dfrac{(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a})}{2\Delta}$

(c) $\dfrac{(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a})}{4\Delta}$ (d) none of these

(where Δ is the area of the triangle *ABC*).

7. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+\hat{k}$, $\vec{c}=\hat{i}+2\hat{j}-\hat{k}$, then the value of $\begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$ is equal to

(a) 2 (b) 4 (c) 16 (d) 64

8. If vectors $\sec^2 A\hat{i}+\hat{j}+\hat{k}$, $\hat{i}+\sec^2 B\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+\sec^2 C\hat{k}$ are coplanar, then $\cot^2 A+\cot^2 B+\cot^2 C$ is equal to

(a) 1 (b) 2

(c) 0 (d) none of these

9. The value of $|a\times\hat{i}|^2+|a\times\hat{j}|^2+|a\times\hat{k}|^2$ is

(a) a^2 (b) $2a^2$

(c) $3a^2$ (d) none of these

10. $\vec{A},\vec{B},\vec{C}$ are three vectors respectively given by $2\hat{i}+\hat{k}$, $\hat{i}+\hat{j}+\hat{k}$ and $4\hat{i}-3\hat{j}+7\hat{k}$. Then vector $\vec{R}$, which satisfies the relation $\vec{R}\times\vec{B}=\vec{C}\times\vec{B}$ and $\vec{R}\cdot\vec{A}=0$, is

(a) $2\hat{i}-5\hat{j}+2\hat{k}$ (b) $-\hat{i}+4\hat{j}+2\hat{k}$

(c) $-\hat{i}-8\hat{j}+2\hat{k}$ (d) none of these

11. The points with position vectors $60\hat{i}+3\hat{j}$, $40\hat{i}-8\hat{j}$ and $a\hat{i}-52\hat{j}$ are collinear if
(a) $a=-40$ (b) $a=40$
(c) $a=20$ (d) none of these

12. Let O be the circumcentre, G be the centroid and O' be the orthocentre of a ΔABC. Three vectors are taken through O and are represented by $\vec{a}=\overrightarrow{OA}$, $\vec{b}=\overrightarrow{OB}$ and $\vec{c}=\overrightarrow{OC}$, then $\vec{a}+\vec{b}+\vec{c}$ is
(a) $\overrightarrow{OG}$ (b) $2\overrightarrow{OG}$
(c) $\overrightarrow{OO'}$ (d) none of these

13. Let $\vec{a}\equiv 2\hat{i}-\hat{j}+\hat{k}$, $\vec{b}\equiv\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}\equiv\hat{i}+\hat{j}-2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is of magnitude $\sqrt{2/3}$ is
(a) $2\hat{i}+3\hat{j}-3\hat{k}$ or $-2\hat{i}-\hat{j}+5\hat{k}$
(b) $2\hat{i}-3\hat{j}+3\hat{k}$ or $2\hat{i}-\hat{j}-5\hat{k}$
(c) $-2\hat{i}-\hat{j}+5\hat{k}$ or $2\hat{i}-3\hat{j}-3\hat{k}$
(d) $2\hat{i}+\hat{j}+5\hat{k}$ or $2\hat{i}-3\hat{j}-3\hat{k}$

14. Let $\overrightarrow{OA}=\hat{i}+3\hat{j}-2\hat{k}$ and $\overrightarrow{OB}=3\hat{i}+\hat{j}-2\hat{k}$. The vector $\overrightarrow{OC}$ bisecting the angle AOB and C being a point on the line AB is
(a) $4(\hat{i}+\hat{j}-\hat{k})$ (b) $2(\hat{i}+\hat{j}-\hat{k})$
(c) $\hat{i}+\hat{j}-\hat{k}$ (d) none of these

15. A parallelogram is constructed on $5\vec{a}+2\vec{b}$ and $\vec{a}-3\vec{b}$, where $|a|=2\sqrt{2}$ and $|b|=3$. If the angle between $\vec{a}$ and $\vec{b}$ is $\pi/4$, then the length of the longer diagonal is
(a) $\sqrt{473}$ (b) $\sqrt{593}$ (c) $\sqrt{474}$ (d) $\sqrt{594}$

16. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three non-coplanar vectors with $\vec{a}+\vec{b}+\vec{c}=m\vec{d}$ and $\vec{b}+\vec{c}+\vec{d}=n\vec{a}$, then
(a) $\vec{a}=\vec{b}+\vec{c}+\vec{d}$ (b) $\vec{d}=\vec{a}+\vec{b}+\vec{c}$
(c) $\vec{b}+\vec{c}=\vec{a}+\vec{d}$ (d) $\vec{a}+\vec{b}+\vec{c}+\vec{d}=\vec{0}$

17. Unit vectors $\hat{a}$ and $\hat{b}$ are perpendicular to each other and the unit vector $\hat{c}$ is inclined at an angle θ to both $\hat{a}$ and $\hat{b}$. If $\hat{c}=m(\hat{a}+\hat{b})+n(\hat{a}\times\hat{b})$, and m, n are real, then
(a) $\frac{\pi}{4}\le\theta\le\frac{3\pi}{4}$ (b) $\frac{\pi}{6}\le\theta\le\frac{2\pi}{3}$
(c) $0\le\theta\le\frac{\pi}{2}$ (d) none of these

18. If $\vec{a}+\vec{b}+\vec{c}=\alpha\vec{d}$, $\vec{b}+\vec{c}+\vec{d}=\beta\vec{a}$ and $\vec{a},\vec{b},\vec{c}$ are non-coplanar, then $\vec{a}+\vec{b}+\vec{c}+\vec{d}$ =
(a) $\alpha\vec{a}$ (b) $\beta\vec{b}$
(c) 0 (d) $(\alpha+\beta)\vec{c}$

19. If $\vec{a}\perp\vec{b}$, then $\vec{a}\times\{\vec{a}\times\{\vec{a}\times\{\vec{a}\times\{\vec{a}\times(\vec{a}\times\vec{b})\}\}\}\}$ is
(a) $-|\vec{a}|^2\vec{b}$ (b) $|\vec{a}|^6\vec{b}$
(c) $-|\vec{a}|^6\vec{b}$ (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. $\vec{a}$ and $\vec{b}$ are two given vectors. On these vectors as adjustment sides a parallelogram is constructed. The vector which is the altitude of the parallelogram and which is perpendicular to $\vec{a}$ is
(a) $\frac{(\vec{a}\cdot\vec{b})}{|\vec{a}|^2}\vec{a}-\vec{b}$ (b) $\frac{1}{|\vec{a}|^2}\left\{|\vec{a}|^2\vec{b}-(\vec{a}\cdot\vec{b})\vec{a}\right\}$
(c) $\frac{\vec{a}\times(\vec{a}\times\vec{b})}{|\vec{a}|^2}$ (d) $\frac{\vec{a}\times(\vec{b}\times\vec{a})}{|\vec{b}|^2}$

2. If vector $\vec{b}=\left(\tan\alpha,-1,2\sqrt{\sin\frac{\alpha}{2}}\right)$ and
$$\vec{c}=\left(\tan\alpha,\tan\alpha,-\frac{3}{\sqrt{\sin(\alpha/2)}}\right)$$
are orthogonal and vector $\vec{a}=(1,3,\sin 2\alpha)$ makes an obtuse angle with the z-axis, then
(a) $\alpha=\tan^{-1}(-2)$ (b) $\alpha=\tan^{-1}(-3)$
(c) $\alpha=\tan^{-1}(2)$ (d) $-2<\alpha<0$

3. The vectors $\hat{i}+2\hat{j}+3\hat{k}$, $2\hat{i}-\hat{j}+\hat{k}$ and $3\hat{i}+\hat{j}+4\hat{k}$ are so placed that the end point of one vector is the starting point of the next vector. Then the vectors are
(a) not coplanar
(b) coplanar but cannot form a triangle
(c) coplanar but can form a triangle
(d) coplanar and can form a right-angled triangle

4. Given are the vertices $A(2, 3, 1)$, $B(4, 1, -2)$, $C(6, 3, 7)$ and $D(-5, -4, 8)$ of a tetrahedron. The length of the altitude drawn from the vertex D is
(a) 7 (b) 9 (c) 11 (d) none of these

5. If three vectors $\vec{a},\vec{b},\vec{c}$ are such that $\vec{a}\neq 0$, $\vec{a}\times\vec{b}=2(\vec{a}\times\vec{c})$, $|\vec{a}|=|\vec{c}|=1$, $|\vec{b}|=4$ and the angle between $\vec{b}$ and $\vec{c}$ is $\cos^{-1}(1/4)$, then $\vec{b}-2\vec{c}=\lambda\vec{a}$, where λ is equal to
(a) 4 (b) –4 (c) 2 (d) –2

6. If $\hat{n}_1$ and $\hat{n}_2$ are two unit vectors and θ is the angle between them, then $\cos(\theta/2)$ is equal to

(a) $\frac{1}{2}|\hat{n}_1+\hat{n}_2|$ (b) $\frac{1}{2}|\hat{n}_1-\hat{n}_2|$

(c) $\frac{1}{2}|\hat{n}_1\cdot\hat{n}_2|$ (d) $\frac{1}{2}\frac{|\hat{n}_1\times\hat{n}_2|}{|\hat{n}_1||\hat{n}_2|}$

7. If a, b, c are pth, qth, rth terms of an H.P. and $\vec{u}=(q-r)\hat{i}+(r-p)\hat{j}+(p-q)\hat{k}$, $\vec{v}=\frac{\hat{i}}{a}+\frac{\hat{j}}{b}+\frac{\hat{k}}{c}$, then

(a) $\vec{u}$ and $\vec{v}$ are parallel vectors

(b) $\vec{u}$ and $\vec{v}$ are orthogonal vectors

(c) $\vec{u}$ and $\vec{v}$ are unit vectors

(d) $\vec{u}\times\vec{v}=\hat{i}+\hat{j}+\hat{k}$

8. If a non-zero vector $\vec{A}$ in the equations $\vec{A}\cdot\vec{B}=\vec{A}\cdot\vec{C}$ and $\vec{A}\times\vec{B}=\vec{A}\times\vec{C}$ holds simultaneously, then

(a) $\vec{A}$ is perpendicular to $\vec{B}-\vec{C}$

(b) $\vec{A}=\vec{B}$

(c) $\vec{B}=\vec{C}$

(d) $\vec{C}=\vec{A}$

9. If $\vec{\alpha}$, $\vec{\beta}$ and $\vec{\gamma}$ are the vertices of a Δ whose circumcentre is at the origin, then the orthocentre is given by

(a) $\frac{\vec{\alpha}+\vec{\beta}+\vec{\gamma}}{4}$ (b) $\frac{\vec{\alpha}+\vec{\beta}+\vec{\gamma}}{2}$

(c) $\vec{\alpha}+\vec{\beta}+\vec{\gamma}$ (d) $\frac{\vec{\alpha}+\vec{\beta}+\vec{\gamma}}{3}$

10. If $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=0$, then $(\vec{a}\times\vec{b})\times[(\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})]$ is equal to

(a) $\vec{0}$

(b) a vector $\perp$ plane of $\vec{a},\vec{b}$ and $\vec{c}$

(c) a scalar quantity

(d) none of these

11. Given that $(\vec{\alpha}+\vec{\beta})$ is perpendicular to $\vec{\beta}$ and $\vec{\alpha}$ is perpendicular to $(3\vec{\beta}+\vec{\alpha})$. This implies

(a) $|\vec{\alpha}|=\sqrt{2}|\vec{\beta}|$ (b) $|\vec{\alpha}|=\sqrt{3}|\vec{\beta}|$

(c) $|\vec{\alpha}|=|\vec{\beta}|$ (d) $|\vec{\alpha}|=1/|\vec{\beta}|$

12. Taken on side $\overrightarrow{AC}$ of a trianlge ABC, a point M such that $\overrightarrow{AM}=(1/3)\overrightarrow{AC}$. A point N is taken on the side $\overrightarrow{CB}$ such that $\overrightarrow{BN}=\overrightarrow{CB}$, then for the point of intersection X of $\overrightarrow{AB}$ and $\overrightarrow{MN}$, which of the following holds good?

(a) $\overrightarrow{XB}=(1/3)\overrightarrow{AB}$ (b) $\overrightarrow{AX}=(1/3)\overrightarrow{AB}$

(c) $\overrightarrow{XN}=(3/4)\overrightarrow{MN}$ (d) $\overrightarrow{XM}=3\overrightarrow{XN}$

13. The angles of a triangle, two of whose sides are represented by the vectors $\sqrt{3}(\hat{a}\times\vec{b})$ and $\vec{b}-(\hat{a}\cdot\vec{b})\hat{a}$, where $\vec{b}$ is a non-zero vector and $\hat{a}$ is a unit vector in the direction of $\hat{a}$ are

(a) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$; $\tan^{-1}\left(\frac{1}{2}\right)$; $\tan^{-1}\left(\frac{\sqrt{3}+2}{1-2\sqrt{3}}\right)$

(b) $\tan^{-1}(\sqrt{3})$; $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$; $\cot^{-1}(0)$

(c) $\tan^{-1}(\sqrt{3})$; $\tan^{-1}(2)$; $\tan^{-1}\left(\frac{\sqrt{3}+2}{2\sqrt{3}-1}\right)$

(d) $\tan^{-1}(\sqrt{3})$; $\tan^{-1}(\sqrt{2})$; $\tan^{-1}\left(\frac{\sqrt{2}+3}{3\sqrt{2}-1}\right)$

14. Find the vector $\vec{r}$ perpendicular to $\vec{a}=\hat{i}-2\hat{j}+5\hat{k}$ and $\vec{b}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{r}(2\hat{i}+\hat{j}+\hat{k})+8=0$.

(a) $-13\hat{i}+11\hat{j}+7\hat{k}$ (b) $13\hat{i}-11\hat{j}-7\hat{k}$

(c) $11\hat{i}+13\hat{j}+9\hat{k}$ (d) $7\hat{i}+13\hat{j}+11\hat{k}$

15. The diagonals of a parallelogram are given by the vectors $2\hat{i}+3\hat{j}-6\hat{k}$ and $3\hat{i}-4\hat{j}-\hat{k}$. The area of the parallelogram is

(a) $\frac{1}{2}\sqrt{1274}$ (b) $\sqrt{1274}$

(c) $\frac{1}{4}\sqrt{1274}$ (d) 1274

16. Image of the point P with position vector $7\hat{i}-\hat{j}+2\hat{k}$ in the line whose vector equation is $\vec{r}=(9\hat{i}+5\hat{j}+5\hat{k})+\lambda(\hat{i}+3\hat{j}+5\hat{k})$ has position vector

(a) $-9\hat{i}+5\hat{j}+2\hat{k}$ (b) $9\hat{i}+5\hat{j}+2\hat{k}$

(c) $9\hat{i}+5\hat{j}-2\hat{k}$ (d) $9\hat{i}-5\hat{j}-2\hat{k}$

17. If $\hat{a}$ and $\hat{b}$ are two unit vectors and θ is the angle between them, then the unit vector along their angular bisector is

(a) $\frac{\hat{a}-\hat{b}}{2\cos(\theta/2)}$ (b) $\frac{\hat{a}+\hat{b}}{2\cos(\theta/2)}$

(c) $\frac{\hat{a}-\hat{b}}{\cos(\theta/2)}$ (d) $\frac{\hat{a}+\hat{b}}{\cos(\theta/2)}$

18. If $\hat{a}$ and $\hat{b}$ are unit vectors such that the vector $\hat{a}+3\hat{b}$ is perpendicular to $7\hat{a}-5\hat{b}$ and $\hat{a}-4\hat{b}$ is perpendicular to $7\hat{a}-2\hat{b}$, then the angle between $\hat{a}$ and $\hat{b}$ is

(a) $\pi/6$ (b) $\pi/4$

(c) $\pi/3$ (d) $\pi/2$

19. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors and $\vec{p}, \vec{q}, \vec{r}$ are reciprocal vectors, then $(l\vec{a}+m\vec{b}+n\vec{c})(l\vec{p}+m\vec{q}+n\vec{r})=$

(a) $l+m+n$ (b) $l^3+m^3+n^3$

(c) $l^2+m^2+n^2$ (d) none of these

20. If $\vec{r}\cdot\vec{a}=\vec{r}\cdot\vec{b}=\vec{r}\cdot\vec{c}=1/2$ for some non-zero vector $\vec{r}$, then the area of the Δ formed by the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is

(a) $\left|\left[\vec{a}\vec{b}\vec{c}\right]\right|$ (b) $|\vec{r}|$

(c) $\left|\left[\vec{a}\vec{b}\vec{c}\right]\vec{r}\right|$ (d) none of these

ANSWERS

Level 1

1. (c) **2.** (a) **3.** (c) **4.** (a) **5.** (a) and (b) **6.** (a) **7.** (c) **8.** (d) **9.** (b)
10. (c) **11.** (a) **12.** (c) **13.** (a) **14.** (b) **15.** (b) **16.** (d) **17.** (a) **18.** (c) **19.** (c)

Level 2

1. (a), (b) and (c) **2.** (a) **3.** (b) **4.** (c) **5.** (d) **6.** (a) **7.** (b) **8.** (c) **9.** (c)
10. (a) **11.** (b) **12.** (c) **13.** (b) **14.** (a) **15.** (a) **16.** (c) **17.** (b) **18.** (c) **19.** (c)
20. (c)

24

LEONHARD EULER

Three-Dimensional Geometry

- ❑ Cartesian Coordinates in Three Dimensions
- ❑ Distance between Two Points
- ❑ Section Formulae
- ❑ Direction Cosines and Direction Ratios
- ❑ Angle between Two Lines
- ❑ Plane
- ❑ Distance between Two Parallel Planes
- ❑ Family of Planes
- ❑ Straight Line
- ❑ Skew Line
- ❑ Line and Plane
- ❑ Area of Triangle
- ❑ Sphere

☞ THREE-DIMENSIONAL COORDINATE SYSTEM

Let $x'Ox$, $y'Oy$ and $z'Oz$ be three mutually perpendicular lines intersecting at O such that two of them, viz. $y'Oy$ and $z'Oz$ lie in the plane of the paper and the third $x'Ox$ is perpendicular to the plane of the paper and is projecting out from the plane of the paper.

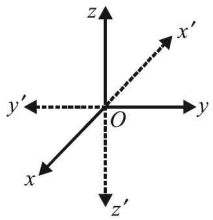


Fig. 24.1

☞ ORIGIN

The point O is called the *origin*, which is the point of intersection of three mutually perpendicular lines $x'Ox$, $y'Oy$ and $z'Oz$.

☞ COORDINATE AXES

The set of three mutually perpendicular lines $x'Ox$, $y'Oy$ and $z'Oz$ is called the **coordinate axes**. The coordinate axes $x'Ox$, $y'Oy$ and $z'Oz$ are called the x-axis, the y-axis and the z-axis respectively. The positive directions of the axes are $x'Ox$, $y'Oy$ and $z'Oz$ indicated by arrows.

☞ COORDINATE PLANES

1. The plane yOz containing the axes of y and z is called the yz-plane, i.e. $x = 0$.
2. The plane zOx containing the axes of z and x is called the xz-plane, i.e. $y = 0$.
3. The plane xOy containing the axes of y and x is called the xy-plane, i.e. $z = 0$.

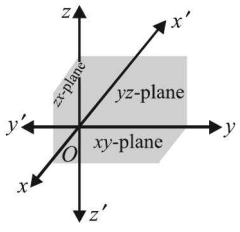


Fig. 24.2

SIGNS OF COORDINATES OF A POINT

The signs of the coordinates of a point determine the octant in which the point lies. The following table shows the signs of the coordinates in eight octants.

Octants / Coordinates	I	II	III	IV	V	VI	VII	VIII
x	+	−	−	+	+	−	−	+
y	+	+	−	−	+	+	−	−
z	+	+	+	+	−	−	−	−

DISTANCE BETWEEN TWO POINTS

The distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

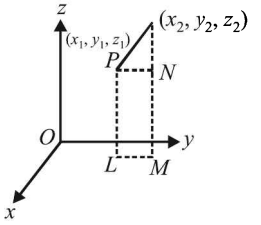


Fig. 24.3

Corollary: If one of the two points is taken as the origin $O(0, 0)$ and the other point as $P(x, y, z)$, then

$$OP = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2} = \sqrt{x^2 + y^2 + z^2}$$

Illustration 1. The direction ratios of the line segment joining from the origin to the point $(2, -3, 6)$ are

(a) (1, 0, 0) (b) (2, 3, 6)

(c) $(2, -3, 6)$ (d) $(0, -1, 6)$

Ans. (c)

Solution: Coordinates of the origin $(x_1, y_1, z_1) = (0, 0, 0)$. Coordinates of the second point $(x_2, y_2, z_2) = (2, -3, 6)$. We know that direction ratios of the line segment joining the origin (x_1, y_1, z_1) and the point (x_2, y_2, z_2) are

$$(a, b, c) = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$
$$= (2 - 0, -3 - 0, 6 - 0) = (2, -3, 6)$$

RATIO FORMULA

The coordinates of the point which divides the join of $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the ratio $m_1 : m_2$ are:

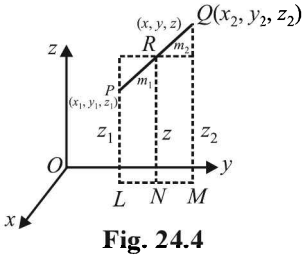


Fig. 24.4

Internally

$$\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}, \frac{m_1z_2 + m_2z_1}{m_1 + m_2}\right)$$

Externally

$$\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}, \frac{m_1z_2 - m_2z_1}{m_1 - m_2}\right)$$

The coordinates of the mid-point of the join of (x_1, y_1, z_1) and (x_2, y_2, z_2) are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$.

The coordinates of the point dividing the join of (x_1, y_1, z_1) and (x_2, y_2, z_2) in the ratio $k : 1$ internally are

$$\left(\frac{kx_2 + x_1}{k+1}, \frac{ky_2 + y_1}{k+1}, \frac{kz_2 + z_1}{k+1}\right).$$

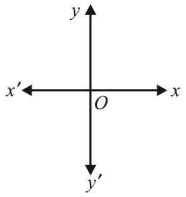
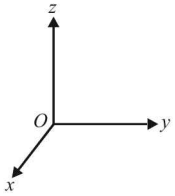
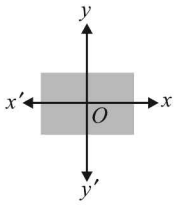
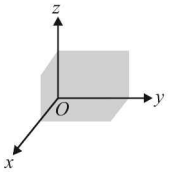
The centroid of a triangle with the vertices $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and C (x_3, y_3, z_3) is the point

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right).$$

The centroid of a tetrahedron with the vertices $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and C (x_3, y_3, z_3) and $D(x_4, y_4, z_4)$ is the point

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4}\right).$$

COMPARISON OF TWO-DIMENSIONAL AND THREE-DIMENSIONAL GEOMETRY

Two-dimensional geometry	Three dimensional geometry
Two axes: x-axis, y-axis 	Three axes: x-axis, y-axis, z-axis 
One plane: xy-plane 	Three planes: xy-plane, yz-plane, zy-plane 

Four quadrants					Eight octants			
	xOy'	$x'Oy$	$x'Oy'$	xOy'	$xOyz$ / $xOyz'$	$x'Oyz$ / $x'Oyz'$	$x'Oy'z$ / $x'Oy'z'$	$x'Oy'z$ / $x'Oy'z'$
Sign	+ +	− +	− −	+ −	+ + + / − − −	− + + / − + −	− − + / − − −	+ − + / + − −

TO FIND OUT THE POSITION OF THE POINT *P*, WHOSE COORDINATES ARE (*x*, *y*, *z*)

(i) Along Ox, measure $OA = x$, measure $AN = y$, parallel to Oy, then measure $NP = z$, parallel to Oz, P is the required point in space.

(ii) Along the axes Ox, Oy, Oz, measure $OA = x$, $OB = y$, $OC = z$. Through A, B, C, draw planes parallel to yz, zx, xy planes. The point of intersection of these planes is the required point P.

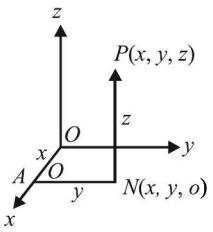


Fig. 24.5

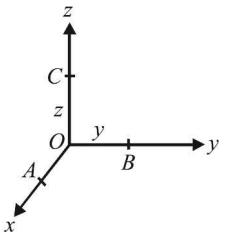


Fig. 24.6

DISTANCE BETWEEN TWO POINTS

The distance formula in Cartesian coordinates is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

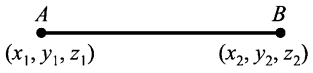


Fig. 24.7

$$AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$$

DIRECTION COSINES

If α, β and γ are the angles which a vector $\overrightarrow{OP}$ makes with the positive directions of the co-ordinate axes Ox, Oy and Oz respectively, then $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are known as the **direction cosines** of $\overrightarrow{OP}$ and are generally denoted by the letters l, m and n, respectively, i.e. $l = \cos\alpha$, $m = \cos\beta$ and $n = \cos\gamma$. The angles α, β and γ are known as the **direction angles** and satisfy the condition $0 \le \alpha, \beta, \gamma \le \pi$.

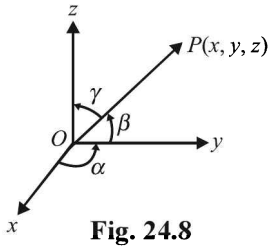


Fig. 24.8

In the above definition, the vector $\overrightarrow{OP}$ has its initial point at the origin. If a given vector does not have its initial point at the origin, then we can draw a parallel vector of the same magnitude having initial point at the origin. Clearly, $\overrightarrow{OP}$ makes angles $\pi - \alpha$, $\pi - \beta$, $\pi - \gamma$ with Ox, Oy, Oz, respectively. Therefore, the direction cosines of $\overrightarrow{OP}$ are $\cos(\pi - \alpha)$, $\cos(\pi - \beta)$, $\cos(\pi - \gamma)$ or $-l$, $-m$, $-n$.

DIRECTION COSINES OF A LINE

If α, β and γ are the angles which any line makes with the positive direction of the axis, then $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are called the direction cosines of the given line. The direction cosines are generally denoted as l, m, n.

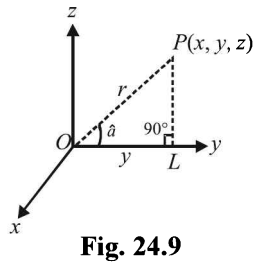


Fig. 24.9

Relation between Direction Cosines

If l, m and n are the line direction cosines of any line, then $l^2 + m^2 + n^2 = 1$, i.e. the sum of the square of the direction cosines of every line is unity.

USEFUL RELATION

If O is the origin and (x, y, z) are the coordinates of a point P, then

$$x = lr, y = mr, z = nr,$$

where l, m and n are the direction cosines of OP and r is the length of OP.

DIRECTION RATIOS

Let l, m and n be the direction cosines of a vector $\vec{r}$ and a, b and c be three numbers such that $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$.

Then, we say that the direction ratios of the direction numbers of vector $\vec{r}$ are proportional to a, b, c.

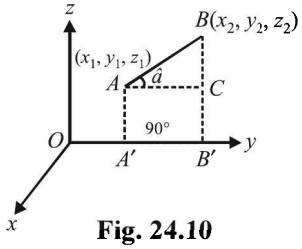


Fig. 24.10

DIRECTION RATIOS OF A LINE

Any three numbers a, b and c proportional to the direction cosines of a line are called **direction ratios** of a line. Thus,

$$l = \pm\frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm\frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \pm\frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

Direction ratios of the line joining two points (x_1, y_1, z_1) and (x_2, y_2, z_2) are proportional to $x_2 - x_1$, $y_2 - y_1$, $z_2 - z_1$. Thus,

$$l : m : n = \frac{x_2 - x_1}{r} : \frac{y_2 - y_1}{r} : \frac{z_2 - z_1}{r}$$
$$= (x_2 - x_1) : (y_2 - y_1) : (z_2 - z_1)$$

ANGLE BETWEEN TWO VECTORS WHOSE DIRECTION COSINES ARE l_1, m_1, n_1 AND l_2, m_2, n_2

This is given by

$$\cos^{-1}(l_1 l_2 + m_1 m_2 + n_1 n_2) = \cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

1. We know that $\sin^2\theta = 1 - \cos^2\theta$

$$\sin\theta = \sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}$$

2. The condition that the lines, whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2, are perpendicular if $l_1l_2 + m_1m_2 + n_1n_2 = 0$.
3. The condition that the lines, whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2, are parallel if $\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$.

ANGLE BETWEEN TWO VECTORS WHOSE DIRECTION RATIOS ARE a_1, b_1, c_1 AND a_2, b_2, c_2

This is given by

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Note The conditions that the lines whose direction ratios are a_1, b_1, c_1 and a_2, b_2, c_2 should be

(i) perpendicular, if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

(ii) parallel, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

CONDITION OF COLLINEARITY OF THREE POINTS

If $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ are three collinear points, then the direction cosines of AB and BC are equal. Thus,

$$\frac{x_2 - x_1}{x_3 - x_2} = \frac{y_2 - y_1}{y_3 - y_2} = \frac{z_2 - z_1}{z_3 - z_2}$$

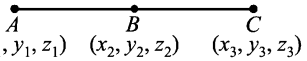


Fig. 24.11

COLLINEARITY OF THREE POINTS WHOSE POSITION VECTORS ARE GIVEN

Three points A, B and C with position vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively are collinear, if and only if there exist scalars μ_1, μ_2 and μ_3 not all zero, such that $\mu_1\vec{a} + \mu_2\vec{b} + \mu_3\vec{c} = 0$ and $\mu_1 + \mu_2 + \mu_3 = 0$.

CONDITION FOR THREE CONCURRENT LINES TO BE COPLANAR

Three lines AB, AC and AD with direction cosines l_1, m_1, n_1; l_2, m_2, n_2 and l_3, m_3, n_3 will be coplanar, if there exists a line AE with direction cosines l, m, n which is perpendicular to the three lines. Thus,

$$\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$$

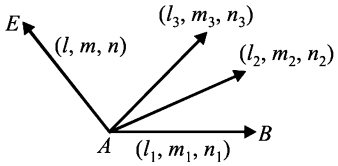


Fig. 24.12

PROJECTION OF THE JOIN OF TWO POINTS

The projection of the join of two points P and Q whose position vectors are $x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ and $x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$ on a vector $\vec{b} = l\hat{i} + m\hat{j} + n\hat{k}$ is $l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$.

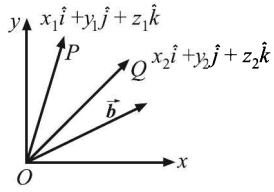


Fig. 24.13

Illustration 2. The projection of the line segment joining the points (–5, 1, 3) and (2, 3, 5) on a line with direction cosines $\frac{1}{3}, \frac{1}{3}, \frac{2}{3}$ is

(a) 1/3 (b) 7/3
(c) 11/3 (d) 13/3

Ans. (d)

Solution: Coordinates of the first point $(x_1, y_1, z_1) = (-5, 1, 3)$ and coordinates of the second point $(x_2, y_2, z_2) = (2, 3, 5)$;

direction cosines of the line $(l, m, n) = \left(\frac{1}{3}, \frac{1}{3}, \frac{2}{3}\right)$

We know that the projection of the line

$$(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n$$

$$= [2 - (-5)]\frac{1}{3} + (3 - 1)\frac{1}{3} + (5 - 3)\frac{2}{3} = \frac{7}{3} + \frac{2}{3} + \frac{4}{3} = \frac{13}{3}$$

EQUATION OF A LINE PASSING THROUGH A GIVEN POINT AND PARALLEL TO A GIVEN VECTOR

Vector form: The vector equation of a straight line passing through a fixed point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{b}$.

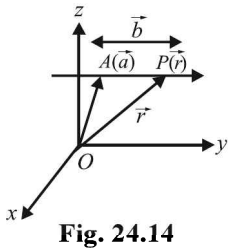


Fig. 24.14

Note

- In the above equation, $\vec{r}$ is the position vector of any point $P(x, y, z)$ on the line. Therefore, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.
- The position vector of any point on the line is taken as $\vec{a} + \lambda\vec{b}$.

Cartesian form: The Cartesian equation of a straight line passing through a fixed point (x_1, y_1, z_1) and having direction ratios proportional to a, b, c is given by

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}.$$

Corollary: The equation of a line having direction cosines l, m, n and passing through (x_1, y_1, z_1) is

$$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}.$$

☞ EQUATION OF A LINE PASSING THROUGH TWO GIVEN POINTS

Vector form: The vector equation of a line passing through two points with position vectors $\vec{a}$ and $\vec{b}$ is $\vec{r}=\vec{a}+\lambda\vec{b}$.

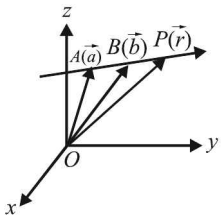


Fig. 24.15

Cartesian form: The Cartesian equation of a line passing through two given points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by

$$\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}.$$

☞ PERPENDICULAR DISTANCE OF A POINT (α, β, γ) FROM A LINE

Perpendicular distance of a point (α, β, γ) from a line

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$

is the equation of the given line. For perpendicular distance of a given point Q from the given line, follow the steps given below.

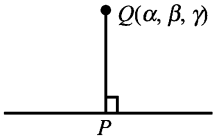


Fig. 24.16

Step 1: Obtain the general point on the given line in terms of r_1. Take this point P as a foot of the perpendicular drawn from Q on the given line.

Step 2: Obtain the direction ratio of PQ.

Step 3: Apply the condition of perpendicularity and get the value of r_1.

Step 4: Put this value of r_1 in the general point and obtain the coordinates of P.

Step 5: Using the distance formula, obtain PQ.

☞ ANGLE BETWEEN TWO LINES

Vector form: Let the vector equation of two given lines be

$$\vec{r}=\vec{a_1}+\lambda\vec{b_1} \quad \text{(i)}$$

and

$$\vec{r}=\vec{a_2}+\mu\vec{b_2} \quad \text{(ii)}$$

then the angle between two lines,

$$\cos\theta=\frac{\vec{b_1}\cdot\vec{b_2}}{|\vec{b_1}||\vec{b_2}|}$$

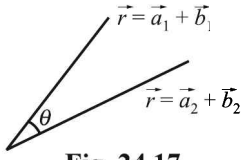


Fig. 24.17

1. **Condition of perpendicularity:** If the lines are perpendicular, then $\vec{b_1}$ and $\vec{b_2}$ are perpendicular. Thus, $\vec{b_1}\cdot\vec{b_2}=0$.
2. **Condition of parallelism:** If the lines are parallel, then $\vec{b_1}$ and $\vec{b_2}$ are parallel. Thus, $\vec{b_1}=\lambda\vec{b_2}$ for some scalar λ.

Cartesian form: Let the Cartesian equations of two given lines be

$$\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1} \text{ and } \frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$$

then, $\cos\theta=\dfrac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$

1. **Condition of perpendicularity:** If the lines are perpendicular, then $a_1a_2+b_1b_2+c_1c_2=0$.
2. **Condition of parallelism:** If the lines are parallel, then $\dfrac{a_1}{a_2}=\dfrac{b_1}{b_2}=\dfrac{c_1}{c_2}$.

Illustration 3. If lines $\dfrac{x-1}{-3}=\dfrac{y-2}{2k}=\dfrac{z-3}{2}$ and $\dfrac{x-1}{3k}=\dfrac{y-5}{1}=\dfrac{z-6}{-5}$ are mutually perpendicular, then k is equal to

(a) $-\dfrac{10}{7}$ (b) $-\dfrac{7}{10}$

(c) -10 (d) -7

Ans. (a)

Solution: Equations of lines are $\dfrac{x-1}{-3}=\dfrac{y-2}{2k}=\dfrac{z-3}{2}$ (i)

$$\frac{x-1}{3k}=\frac{y-5}{1}=\frac{z-6}{-5} \quad \text{(ii)}$$

These lines are perpendicular to each other.

$\therefore \quad -3(3k)+2k+2(-5)=0$

$\Rightarrow \quad -7k-10=0$

$\Rightarrow \quad k=-\dfrac{10}{7}$

☞ COPLANAR LINES

Two lines lying in the same plane are called **coplanar lines**.

☞ SKEW LINES

Two lines in space which are not coplanar are called **skew lines**. Skew lines are neither parallel nor intersecting.

LINE OF SHORTEST DISTANCE BETWEEN TWO SKEW LINES

Let two lines (1) and (2) be skew lines, then there is one unique line, which is perpendicular to both lines (1) and (2). This line SD is called the **line of shortest distance** between lines (1) and (2).

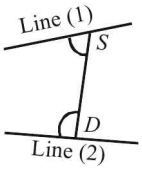


Fig. 24.18

SHORTEST DISTANCE BETWEEN TWO SKEW LINES

The length of the line segment SD intercepted by two lines (1) and (2) and perpendicular to both the lines is called the **shortest distance** between lines (1) and (2).

The method to find the shortest distance between two skew lines (vector form) is given by

$$\text{Shortest distance (S.D.)} = \left| \frac{(\vec{a_2} - \vec{a_1}) \cdot (\vec{b_1} \times \vec{b_2})}{|\vec{b_1} \times \vec{b_2}|} \right|$$

CONDITION FOR TWO GIVEN LINES TO INTERSECT

If the lines intersect, the shortest distance between them is zero. Thus, $(\vec{a_2} - \vec{a_1}) \cdot (\vec{b_1} \times \vec{b_2}) = 0$.

CONDITION TO BE INTERSECTING LINES (CARTESIAN FORM)

The condition to be intersecting lines is

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

SHORTEST DISTANCE BETWEEN TWO PARALLE LINES

Let two lines be

$$\vec{r} = \vec{a_1} + \lambda \vec{b} \qquad \text{(i)}$$

and

$$\vec{r} = \vec{a_2} + \mu \vec{b} \qquad \text{(ii)}$$

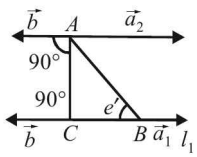


Fig. 24.19

Lines (i) and (ii) pass through the points $A(\vec{a_1})$ and $B(\vec{a_2})$ respectively and both are parallel to the vector $\vec{b}$. Let us draw a perpendicular AC from the point A to the line (ii).

Let θ be the angle between line (ii) and AB. Then,

$$\text{S.D.} = \frac{|(\vec{a_2} - \vec{a_1}) \times \vec{b}|}{|\vec{b}|}$$

PLANE

A surface is called a **plane** such that the line joining any two points on it lies wholly on this surface.

Fig. 24.20

EQUATION OF A PLANE, WHEN THE NORMAL TO THE PLANE AND THE DISTANCE OF THE PLANE FROM THE ORIGIN ARE GIVEN (NORMAL FORM)

Vector form: Let p be the distance of the given plane from the origin and let $\hat{n}$ be the unit vector perpendicular to the given plane. Let ON be the normal and P be any point on the given plane. Then

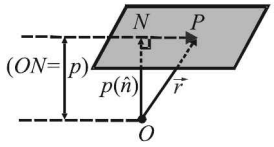


Fig. 24.21

$$\vec{r} \cdot \hat{n} = p$$

Cartesian form: Let $P(x, y, z)$ be any point on the given plane and $\hat{n}$ is the unit vector normal to the given plane. Then $lx + my + nz = p$ (l, m, n are direction cosines of the unit vector perpendicular to the plane).

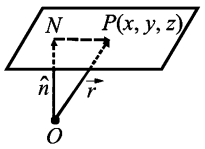


Fig. 24.22

REDUCING THE GENERAL EQUATION OF A PLANE INTO THE NORMAL FORM

Let the general equation of the plane be $ax + by + cz + d = 0$, then

$$\frac{-a}{\sqrt{a^2+b^2+c^2}}x + \frac{-b}{\sqrt{a^2+b^2+c^2}}y + \frac{-c}{\sqrt{a^2+b^2+c^2}}z = \frac{d}{\sqrt{a^2+b^2+c^2}}$$

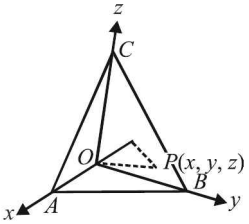


Fig. 24.23

EQUATION OF A PLANE PASSING THROUGH A GIVEN POINT AND PERPENDICULAR TO A GIVEN VECTOR

Vector form: Let the given plane pass through the point A, whose position vector is $\vec{a}$ and the plane is perpendicular to $\overrightarrow{ON}$. Let $\vec{p}$ be any point on the plane, with position vector $\vec{r}$, then

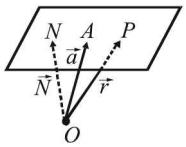


Fig. 24.24

$\overrightarrow{AP}$ = position vector of P – position vector of $A = \vec{r} - \vec{a}$

$\because$ $\overrightarrow{ON}$ is perpendicular to the given plane

$\Rightarrow$ It is also perpendicular to the line AP in this plane.

$\Rightarrow$ $\overrightarrow{AP} \cdot \overrightarrow{ON} = 0 \Rightarrow (\vec{r} - \vec{a}) \cdot \vec{N} = 0$

Cartesian form: Let $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\overrightarrow{ON} = l\hat{i} + m\hat{j} + n\hat{k}$, l, m, n being direction cosines, then $(\vec{r} - \vec{a}) \cdot \overrightarrow{ON} = 0$.

$\Rightarrow \quad (x - x_1)l + (y - y_1)m + (z - z_1)n = 0$

EQUATION OF A PLANE PASSING THROUGH A POINT AND PARALLEL TO TWO COPLANAR LINES

Vector form: Let the given plane be through a point A, whose position vector is $\vec{a}$. If this plane is parallel to two given lines represented by $\vec{b}$ and $\vec{c}$ and $\vec{r}$ is the position vector of any point P, on the plane, then $(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{c}) = 0$.

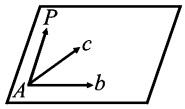


Fig. 24.25

Cartesian form: Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, then

$$\begin{vmatrix} x - a_1 & y - a_2 & z - a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

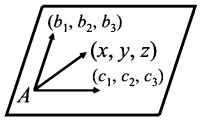


Fig. 24.26

Parametric form: $\vec{r} = \vec{a} + \lambda\vec{b} + k\vec{c}$, where λ and k are parameters.

CONDITION OF COPLANARITY OF TWO LINES

Vector form: Let the given lines be $\vec{r} = \overrightarrow{a_1} + \lambda\overrightarrow{b_1}$ and $\vec{r} = \overrightarrow{a_2} + \mu\overrightarrow{b_2}$, then the given lines are coplanar if $(\overrightarrow{a_2} - \overrightarrow{a_1}) \cdot (\overrightarrow{b_1} \times \overrightarrow{b_2}) = 0$.

Cartesian form: Let the two lines be $\dfrac{x - x_1}{a_1} = \dfrac{y - y_1}{b_1} = \dfrac{z - z_1}{c_1}$ and $\dfrac{x - x_2}{a_2} = \dfrac{y - y_2}{b_2} = \dfrac{z - z_2}{c_2}$, then the given lines are coplanar if

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

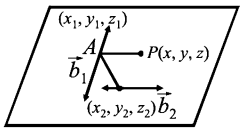


Fig. 24.27

Illustration 4. The lines $\dfrac{x-1}{k} = \dfrac{y-4}{1} = \dfrac{z-5}{1}$ and $\dfrac{x-2}{1} = \dfrac{y-3}{1} = \dfrac{4-z}{k}$ are coplanar if

(a) $k = 0$ or 4 (b) $k = 1$ or -1

(c) $k = 0$ or -1 (d) $k = -1$ or -2

Ans. (b)

Solution: For coplanarity of two lines, the condition is

$$\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$\Rightarrow \quad k^2 = 1 \quad \Rightarrow \quad k = \pm 1$

PLANE CONTAINING TWO LINES

Vector form: Let a plane contain two lines $\overrightarrow{b_1}$ and $\overrightarrow{b_2}$. $P(x, y, z)$ be a point on the plane. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\overrightarrow{a_1} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$, then

$(\vec{r} - \overrightarrow{a_1}) \cdot (\overrightarrow{b_1} \times \overrightarrow{b_2}) = 0$

Cartesian form: Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be the co-ordinates of the points A and B respectively. Then,

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

PLANE PASSING THROUGH THE INTERSECTION OF TWO GIVEN PLANES

Vector form: Let π_1 and π_2 be the two intersecting planes whose equations are given by

$\vec{r} \cdot \overrightarrow{n_1} = d_1$ and $\vec{r} \cdot \overrightarrow{n_2} = d_2$

For any scalar λ, we have

$\vec{r} \cdot \overrightarrow{n_1} + \lambda\vec{r} \cdot \overrightarrow{n_2} = d_1 + \lambda d_2$

$\vec{r}(\overrightarrow{n_1} + \lambda\overrightarrow{n_2}) = d_1 + \lambda d_2$

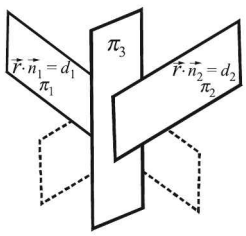


Fig. 24.28

Cartesian form: Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\overrightarrow{n_1} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$, $\overrightarrow{n_2} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$, then $\vec{r} \cdot (\overrightarrow{n_1} + \lambda\overrightarrow{n_2}) = d_1 + \lambda d_2$ changes to $(a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0$

PLANE PASSING THROUGH THREE GIVEN NON-COLLINEAR POINTS

If $A(\vec{a})$, $B(\vec{b})$ and $C(\vec{c})$ are given non-collinear points, then

Vector form:

$$(\vec{r} - \vec{a}) \cdot \left[(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})\right] = 0$$

Cartesian form:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

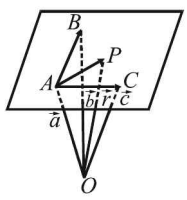


Fig. 24.29

☞ COPLANARITY OF FOUR GIVEN POINTS

Suppose four points are given and we have to find whether the given four points are coplanar or not. One of the simplest ways to find this is to first find the equation of the plane passing through any three of these given points and then to show that the remaining fourth point also lies on this plane.

Alternative method

Let the four points be A, B, C and D given. Four points will lie in a plane if three vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar.

$[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = 0.$

☞ INTERCEPT FORM

The equation of the plane in terms of the intercepts of a, b and c from the axes is

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1.$$

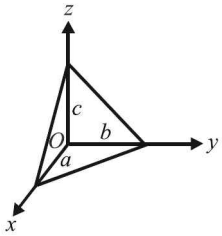


Fig. 24.30

☞ ANGLE BETWEEN TWO PLANES

The angle between two planes is equal to the angle between the normals to them from any point. Thus,

$$\vec{r}\cdot\hat{\eta}_1 = d_1 \qquad \text{(i)}$$

and

$$\vec{r}\cdot\hat{\eta}_2 = d_2 \qquad \text{(ii)}$$

$$\cos\theta = \frac{\hat{\eta}_1\cdot\hat{\eta}_2}{|\hat{\eta}_1||\hat{\eta}_2|}$$

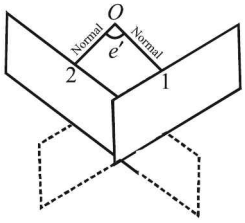


Fig. 24.31

Cartesian form: Let two planes be $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$, then

$$\cos\theta = \frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

☞ PARALLELISM AND PERPENDICULARITY OF TWO PLANES

Two planes are parallel or perpendicular according as the normals to them are parallel or perpendicular. Thus, the two planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ will be **parallel**, if $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$ and these planes will be **perpendicular**, if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

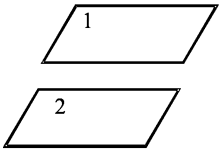


Fig. 24.32

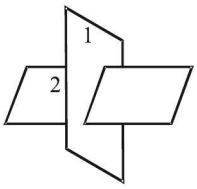


Fig. 24.33

Note The planes are

(i) **perpendicular** if $\hat{\eta}_1\cdot\hat{\eta}_2 = 0$ and (ii) **parallel** if $\hat{\eta}_1 = \pm\hat{\eta}_2$.

☞ LENGTH OF PERPENDICULAR OF A POINT FROM A PLANE BY BOTH VECTOR AND CARTESIAN METHODS

Vector form: The length of the perpendicular from the point $P(\vec{a})$ to the plane $\vec{r}\cdot\hat{\eta}=d$ is given by

$$P=\frac{|\vec{a}\cdot\vec{n}-d|}{|\hat{n}|}.$$

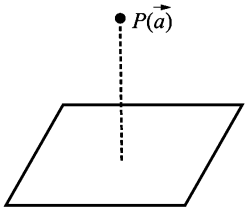


Fig. 24.34

Cartesian form: The length of the perpendicular from the point (x_1, y_1, z_1) to the plane $Ax + By + Cz + D = 0$ is given by

$$P=\left|\frac{Ax_1+By_1+Cz_1+D}{\sqrt{A^2+B^2+C^2}}\right|.$$

☞ EQUATION OF A PLANE BISECTING THE ANGLE BETWEEN TWO PLANES

Vector form: The equation of the plane bisecting the angle between planes $\vec{r}\cdot\vec{n}_1 = d_1$ and $\vec{r}\cdot\vec{n}_2 = d_2$ is

$$\left|\frac{\vec{r}\cdot\vec{n}_1-d_1}{\vec{n}_1}\right| = \left|\frac{\vec{r}\cdot\vec{n}_2-d_2}{\vec{n}_2}\right|.$$

Cartesian form: The equation of the plane bisecting the angle between the planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \pm\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}.$$

Note The plane bisecting the acute angle would make with either plane an angle less than 45°. Let θ be the angle between one of the given planes and one of the bisecting planes. Find cos θ and hence tan θ. If tan θ < I, the bisecting plane bisects the acute angle between the given planes. If tan θ> I, the bisecting plane bisects the obtuse angle between the two given planes.

☞ DISTANCE BETWEEN TWO PARALLEL PLANES

Vector form: The distance between two parallel planes $\vec{r}\cdot\vec{n} = d_1$ and $\vec{r}\cdot\vec{n} = d_2$ is given by

$$d=\frac{|d_1-d_2|}{|\vec{n}|}.$$

Cartesian form: The distance between two parallel planes $ax+by+cz+d_1=0$ and $ax+by+cz+d_2=0$ is given by

$$d=\left|\frac{d_2-d_1}{\sqrt{a^2+b^2+c^2}}\right|.$$

IMAGE OF A POINT IN A PLANE

If PQ is a line perpendicular to the given plane such that the mid-point of PQ lies on this plane, then the point Q is called the **image** of the point P in the plane. Let P be a point (x_1, y_1, z_1) outside the plane $ax + by + cz + d = 0$. Then the algorithms for finding the image of the point in the given plane are as follows:

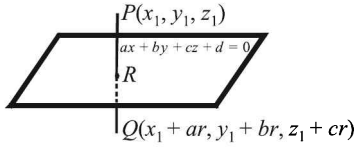


Fig. 24.35

Step 1: Write the equation of the line passing through P and the normal to the given plane as $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$.

Step 2: Write the coordinates of the image Q as (x_1+ar, y_1+br, z_1+cr).

Step 3: Find the coordinates of the mid-point R of PQ.

Step 4: Obtain the value of r by putting the coordinates of R in the equation of the plane.

Step 5: Put the value of r in the coordinates of Q.

Illustration 5. Image of the point (1, 2, 3) by the plane $x + y + z + 3 = 0$ is

(a) (5, 4, 3) (b) (7, 8, 9)

(c) (−5, −4, −3) (d) (−7, −8, −9)

Ans. (b)

Solution: Let the image of the point $P(1, 2, 3)$ be $Q(\alpha, \beta, \gamma)$. Thus, the equation of the line PQ will be

$\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-3}{1}=r$ (direction ratios of the line PQ are direction ratios of the normal to the plane)

$\therefore \quad (\alpha, \beta, \gamma) = (r + 1, r + 2, r + 3)$

also mid-point of $(1, 2, 3)$ and $r + 1, r + 2, r + 3$ lies on the plane, i.e. $\left(\frac{r+2}{2}, \frac{r+3}{2}, \frac{r+4}{2}\right)$ lies on the plane. Thus,

$x + y + z + 3 = 0$ or $x = -6$.

$\therefore$ Image is $(-5, -4, -3)$.

SPHERE

EQUATION OF SPHERE

1. **Central form:** Equation of the sphere, position vector of whose centre is $\vec{c}$ and radius is a, is $|\vec{r}-\vec{c}|=a$; $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$

   Cartesian form: $(x-c_1)^2+(y-c_2)^2+(z-c_3)^2=a^2$
2. **General form:** General equation of the sphere is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ with centre $(-u, -v, -w)$ and radius $\sqrt{u^2+v^2+w^2-d}$.
3. **Diameter form:** Equation of the sphere on the join of $A(\vec{a}), B(\vec{b})$ as diameter is $(\vec{r}-\vec{a})\cdot(\vec{r}-\vec{b})=0$; $\vec{a}=x_1\hat{i}+y_1\hat{j}+z_1\hat{k}, \vec{b}=x_2\hat{i}+y_2\hat{j}+z_2\hat{k}$.

   Cartesian form: $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)+(z-z_1)(z-z_2)=0$

RADIUS OF THE CIRCLE FORMED WHEN A PLANE CUTS A SPHERE

If a plane $Ax + By + Cz + d = 0$ intersects a sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$, then the radius (r) of the circle formed on the plane is

$$r=\sqrt{R^2-d^2};\ d=\frac{|-uA-vB-wC+d|}{\sqrt{A^2+B^2+C^2}}$$

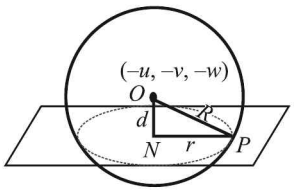


Fig. 24.36

SOLVED PROBLEMS

1. A point P lies on a line whose ends are $A(1, 2, 3)$ and $B(2, 10, 1)$. If z component of P is 7, then the coordinates of P are

   (a) (−1, −14, 7) (b) (1, −14, 7)

   (c) (−1, 14, 7) (d) (1, 14, 7)

Ans. (a)

Solution: The equation of a line passing through A and B is

$$\frac{x-1}{2-1}=\frac{y-2}{10-2}=\frac{z-3}{1-3}$$

$$\Rightarrow \quad \frac{x-1}{1}=\frac{y-2}{8}=\frac{z-3}{-2}=r$$

Putting $z = 7$, we have

$$\frac{x-1}{1}=\frac{y-2}{8}=\frac{7-3}{-2}=-2$$

$$\Rightarrow \quad x=1-2, y=2-16$$

$$\Rightarrow \quad x=-1, y=-14$$

$\therefore$ $P(-1, -14, 7)$

2. The equation of the plane which passes through the points (2, 3, –4) and (1, –1, 3) and is parallel to the x-axis is

(a) $7y-4z-5=0$ (b) $4y-7z-5=0$

(c) $4y+7z+5=0$ (d) $7y+4z-5=0$

Ans. (d)

Solution: Equation of the plane parallel to the x-axis is $by+cz+d=0$.

Since the plane passes through (2, 3, –4) and (1, –1, 3),

$$\therefore \quad 3b-4c+d=0$$
$$-b+3c+d=0$$
$$\Rightarrow \quad \frac{b}{-7}=\frac{c}{-4}=\frac{d}{5}=\lambda$$

$\therefore$ Equation of the plane is $7y + 4z - 5 = 0$.

3. If the direction cosines of two lines are $\langle l_1, m_1 n_1\rangle$ and $\langle l_2, m_2 n_3\rangle$, then the direction cosines of the line perpendicular to both the lines are

(a) $\langle m_1n_2-m_2n_1\ n_1l_2-n_2l_1\ l_1m_2-l_2m_1\rangle$

(b) $\langle l_1+l_2\ m_1+m_2\ n_1+n_2\rangle$

(c) $\langle l_1-l_2\ m_1-m_2\ n_1-n_2\rangle$

(d) none of these

Ans. (a)

Solution: Let $\langle a\ b\ c\rangle$ be the direction cosines of the line perpendicular to both the lines

$$al_1+bm_1+cn_1=0$$
$$al_2+bm_2+cn_2=0$$
$$\Rightarrow \quad \frac{a}{m_1n_2-m_2n_1}=\frac{b}{n_1l_2-n_2l_1}=\frac{c}{l_1m_2-l_2m_1}$$

4. The equation of the line passing through the point (1, 2, 3) and perpendicular to $x + 2y - 5z + 9 = 0$ is

(a) $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{-5}$ (b) $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z+5}{3}$

(c) $\frac{x+1}{1}=\frac{y+2}{2}=\frac{z+3}{-5}$ (d) $\frac{x+1}{1}=\frac{y+2}{2}=\frac{z-5}{3}$

Ans. (a)

Solution: Direction ratios of the normal to the plane

$$x+2y-5z+9=0 \text{ are } \langle 1, 2, -5\rangle$$

$\therefore$ Equation of the required line is

$$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{-5}$$

5. The vector equation of the line $\frac{x-2}{2}=\frac{2y-5}{-3}, z=-1$ is $\vec{r}=\left(2\hat{i}+\frac{5}{2}\hat{j}-\hat{k}\right)+\lambda\left(2\hat{i}-\frac{3}{2}\hat{j}+x\hat{k}\right)$, where x is equal to

(a) 0 (b) 1

(c) 2 (d) 3

Ans. (a)

Solution: The given line is $\frac{x-2}{2}=\frac{2y-5}{-3}, z=-1$

$$\Rightarrow \quad \frac{x-2}{2}=\frac{y-\frac{5}{2}}{-\frac{3}{2}}=\frac{z+1}{0}$$

This shows that the given line passes through the point $\left(2, \frac{5}{2}, -1\right)$ and has direction ratios $2, -\frac{3}{2}, 0$, so its direction cosines are $\frac{2}{\sqrt{2^2+(-3/2)^2}}, \frac{-3/2}{\sqrt{2^2+(-3/2)^2}}, \frac{0}{\sqrt{2^2+(-3/2)^2}}$ or $\left(\frac{4}{5}, -\frac{3}{5}, 0\right)$

Thus, the given line passes through the point having position vector $\vec{a}=2\hat{i}+\frac{5}{2}\hat{j}-\hat{k}$ and is parallel to the vector $\vec{b}=2\hat{i}-\frac{3}{2}\hat{j}+0\hat{k}$. So, its vector equation is

$$\vec{r}=\left(2\hat{i}+\frac{5}{2}\hat{j}-\hat{k}\right)+\lambda\left(2\hat{i}-\frac{3}{2}\hat{j}+0\hat{k}\right)$$

Hence, $x = 0$.

6. A line makes acute angles of α, β and γ with the coordinate axes such that $\cos\alpha+\cos\beta+\cos\gamma=\frac{2}{3}+\frac{1}{3}+\frac{2}{3}=\frac{5}{3}$ and $\cos\gamma\cos\alpha=\frac{4}{9}$, then $\cos\alpha+\cos\beta+\cos\gamma$ is equal to

(a) $\frac{25}{9}$ (b) $\frac{5}{9}$

(c) $\frac{5}{3}$ (d) $\frac{2}{3}$

Ans. (c)

Solution: $\cos\alpha\cos\beta=\cos\beta\cos\gamma=2/9$ and $\cos\gamma\cos\alpha=4/9$ then $\cos\alpha=2/3, \cos\beta=1/3$ and $\cos\gamma=2/3$

$$\therefore \quad \cos\alpha+\cos\beta+\cos\gamma=\frac{2}{3}+\frac{1}{3}+\frac{2}{3}=\frac{5}{3}$$

7. The equation of the plane that contains the point (1, –1, 2) and is perpendicular to each of the planes $2x + 3y - 2z = 5$ and $x + 2y - 3z = 8$ is

(a) $5x-4y-z=7$ (b) $x-4y-3y=7$

(c) $5x+4y-z=9$ (d) $x+4y-39=9$

Ans. (a)

Solution: Let $A(x-1)+B(y+1)+C(z-2)=0$ be the required plane which is perpendicular to $2x+3y-2z=5$ and $x+2y-3z=8$

$\Rightarrow 2A+3B-2C=0$ and $A+2B-3C=0$

$\Rightarrow \frac{A}{-5}=\frac{B}{4}=\frac{C}{1}$

8. Equation of the line of the shortest distance between the lines $\frac{x}{2}=\frac{y}{-3}=\frac{z}{1}$ and $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ is

(a) $3(x-21)=3y+92=3z-32$

(b) $\frac{x-(62/3)}{1/3}=\frac{y-31}{1/3}=\frac{z+(31/3)}{1/3}$

(c) $\frac{x-21}{1/3}=\frac{y-(92/3)}{1/3}=\frac{z+(32/3)}{1/3}$

(d) $\frac{x-2}{1/3}=\frac{y+3}{1/3}=\frac{z-1}{1/3}$

Ans. (a)

Solution: Let $P(2r_1-3r_1,r_1)$ and $Q(3r_2+2,-5r_2+1,2r_2-2)$ be the points on the given lines so that PQ is the line of the shortest distance between the given lines. Now direction ratios of PQ are

$$2r_1-3r_2-2,\ -3r_1+5r_2-1,\ r_1-2r_2+2$$

Since it is perpendicular to the given lines

$$2(2r_1-3r_2-2)-3(-3r_1+5r_2-1)+(r_1-2r_2+2)=0$$

and

$$3(2r_1-3r_2-2)-5(-3r_1+5r_2-1)+2(r_1-2r_2+2)=0$$

$\Rightarrow 14r_1-23r_2+1=0, 23r_1-38r_2+3=0$

$\Rightarrow r_1=31/3, r_2=19/3$

$\Rightarrow P$ is $(62/3,-31,31/3)$ and Q is $(21,-92/3,32/3)$ and direction ratios of PQ are 1/3, 1/3, 1/3.

Hence, equations of PQ as it passes through Q are

$$\frac{x-21}{1/3}=\frac{y+92/3}{1/3}=\frac{z-32/3}{1/3}$$

$$\Rightarrow \quad 3(x-21)=3y+92=3z-32$$

9. If a plane has intercepts a, b, c and is at a distance k units from the origin, then

(a) $\frac{3}{a^2}+\frac{3}{b^2}+\frac{3}{c^2}=\frac{1}{k^2}$ (b) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{k^2}$

(c) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{3}{k^2}$ (d) none of these

Ans. (b)

Solution: Let $\frac{x}{a}+\frac{x}{b}+\frac{z}{c}=1$ which has intercepts a, b, c and k units distance from the origin, then

$$k=\left|\frac{0\cdot\frac{1}{a}+0\cdot\frac{1}{b}+0\cdot\frac{1}{c}-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}\right|$$

$$\Rightarrow \frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{k^2}$$

10. The direction cosines of a line equally inclined to the axes are

(a) $\frac{1}{3},\frac{1}{3},\frac{1}{3}$ (b) $-\frac{1}{3},-\frac{1}{3},-\frac{1}{3}$

(c) $\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}$ (d) none of these

Ans. (c)

Solution: If a line makes angles α, β, γ with the axes, we have $\alpha=\beta=\gamma$

$\therefore \quad \cos\alpha=\cos\beta=\cos\gamma \Rightarrow l=m=n$

$\because \quad l^2+m^2+n^2=1$

$\Rightarrow \quad l^2+l^2+l^2 \Rightarrow 3l^2=1$

$\Rightarrow \quad l^2=\frac{1}{3} \Rightarrow l=\pm\frac{1}{\sqrt{3}}$

11. The angle between the lines whose direction ratios are a, b, c and $b-c$, $c-a$, $a-b$ is

(a) $\pi/4$ (b) $\pi/6$

(c) $\pi/2$ (d) $\pi/3$

Ans. (c)

Solution:

$$\cos\theta=\frac{a(b-c)+b(c-a)+a(a-b)}{\sqrt{a^2+b^2+c^2}\sqrt{(b-c)^2+(c-a)^2+(a-b)^2}}=0$$

$$\theta=\frac{\pi}{2}$$

12. The angle between the straight lines

$\frac{x+1}{2}=\frac{y-2}{5}=\frac{z+3}{4}$ and $\frac{x-1}{1}=\frac{y+2}{5}=\frac{z-3}{-3}$ is

(a) 45° (b) 30°

(c) 60° (d) 90°

Ans. (d)

Solution: The direction ratios of the lines are

$$a_1=2, b_1=5, c_1=4 \text{ and } a_2=1, b_2=2, c_2=-3$$

$$\therefore \quad \cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

$$=\frac{2\cdot1+5\cdot2+4(-3)}{\sqrt{2^2+5^2+4^2}\sqrt{1^2+2^2+(-3)^2}}$$

$$\Rightarrow \theta = \cos^{-1} = \frac{(2+10-12)}{\sqrt{4+25+16}\sqrt{1+4+9}} = \cos^{-1}(0)$$

$$\Rightarrow \theta = 90°$$

13. The acute angle between two lines whose direction ratios are 2, 3, 6 and 1, 2, 2 is

(a) $\cos^{-1}\left(\frac{20}{21}\right)$ (b) $\cos^{-1}\left(\frac{18}{21}\right)$

(c) $\cos^{-1}\left(\frac{17}{21}\right)$ (d) none of these

Ans. (a)

Solution: We have $a_1 = 2, b_1 = 3, c_1 = 6; a_2 = 1, b_2 = 2, c_2 = 2$

If θ is the angle between two lines whose direction ratios are given, then

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$= \frac{2\times1 + 3\times2 + 6\times2}{\sqrt{2^2+3^2+6^2}\sqrt{1^2+2^2+2^2}}$$

$$= \frac{2+6+12}{7\times3} = \frac{20}{21}$$

$$\theta = \cos^{-1}\frac{20}{21}$$

14. The angle between the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane $10x + 2y - 11z = 3$ is

(a) $\cos^{-1}\left(\frac{-8}{21}\right)$ (b) $\cos^{-1}\left(\frac{8}{21}\right)$

(c) $\sin^{-1}\left(\frac{-8}{21}\right)$ (d) $\sin^{-1}\left(\frac{8}{21}\right)$

Ans. (c)

Solution: $\sin\theta = \frac{2.10 + 3\times2 + 6(-11)}{\sqrt{2^2+3^2+6^2}\sqrt{10^2+2^2+(-11)^2}}$

$$= \frac{-40}{7\times15} = \frac{-8}{21}$$

15. The projection of the line segment joining the points (–1, 0, 3) and (2, 5, 1) on the line whose direction ratios are 6, 2, 3 is

(a) $\frac{22}{7}$ (b) $\frac{23}{7}$

(c) $\frac{24}{7}$ (d) $\frac{25}{7}$

Ans. (a)

Solution: The direction cosines l, m, n of the line are given by

$$\frac{l}{6} = \frac{m}{2} = \frac{n}{3} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{6^2+2^2+3^2}} = \frac{1}{\sqrt{49}} = \frac{1}{7}$$

$$\therefore \quad l = \frac{6}{7}, m = \frac{2}{7}, n = \frac{3}{7}$$

The required projection is given by

$$= l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

$$= \frac{6}{7}[2-(-1)] + \frac{2}{7}(5-0) + \frac{3}{7}(1-3)$$

$$= \frac{6}{7}\times3 + \frac{2}{7}\times5 + \frac{3}{7}\times(-2)$$

$$= \frac{18}{7} + \frac{10}{7} - \frac{6}{7} = \frac{18+10-6}{7} = \frac{22}{7}$$

16. The equation of the plane through the line of intersection of the planes $x + y + z = 1$ and $2x + 3y + 4z = 5$ which is perpendicular to the plane $x - y + z = 0$ is

(a) $x - z + 2 = 0$ (b) $x - y + 2 = 0$

(c) $y - z + 2 = 0$ (d) none of these

Ans. (a)

Solution: Here $x + y + z - 1 + \lambda(2x + 3y + 4z - 5) = 0$

$$\Rightarrow (1+2\lambda)x + (1+3\lambda)y + (1+4\lambda)z + (-1-5\lambda) = 0$$

which is perpendicular to $x - y + z = 0$

$$\Rightarrow (1+2\lambda) - (1+3\lambda) + (1+4\lambda) = 0$$

$$\Rightarrow 3\lambda = -1 \quad \Rightarrow \quad \lambda = -\frac{1}{3}$$

$$\therefore \frac{1}{3}x - \frac{1}{3}z + \left(-1 + \frac{5}{3}\right) = 0$$

$$\Rightarrow x - z + 2 = 0$$

17. The equation of the plane through the points (1, 1, 0), (–2, 2, –1) and (1, 2, 1) is

(a) $2x + 3y + 3z = 5$ (b) $2x - 3y + 3z = 5$

(c) $2x + 3y - 3z = 3$ (d) $2x + 3y - 3z = 5$

Ans. (d)

Solution: A plane through (1, 1, 0) is

$$a(x-1) + b(y-1) + cz = 0$$

It passes through (–2, 2, –1) and (1, 2, 1); hence, $-3a + b - c = 0$ and $0a + b + c = 0$

$$\Rightarrow \frac{a}{1+1} = \frac{b}{+3} = \frac{c}{-3}$$

$$\Rightarrow \frac{a}{2} = \frac{b}{3} = \frac{c}{-3}$$

Hence, the plane is $2(x-1)+3(y-1)-3z=0$

$\Rightarrow\ 2x+3y-3z=5$

18. The length of the perpendicular drawn from (1, 2, 3) to the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$ is

(a) 4 (b) 5

(c) 6 (d) 7

Ans. (d)

Solution: Direction cosines of the given line are

$$\frac{3}{\sqrt{17}},\frac{2}{\sqrt{17}},\frac{-2}{\sqrt{17}}$$

$$\therefore\ AM=\left|(6-1)\cdot\frac{3}{\sqrt{17}}+(7-2)\cdot\frac{2}{\sqrt{17}}+(7-3)\cdot\frac{-2}{\sqrt{17}}\right|=17$$

$$AP=\sqrt{(6-1)^2+(7-2)^2+(7-3)^2}$$

$$=\sqrt{25+25+16}=\sqrt{66}$$

$\therefore$ Length of the perpendicular

$$PM=\sqrt{AP^2-AM^2}$$

$$=\sqrt{66-17}=\sqrt{49}=7$$

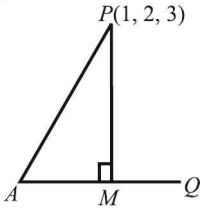


Fig. 24.37

EXERCISE SET 1

1. The shortest distance between lines $\vec{r}=6\hat{i}+2\hat{j}+2\hat{k}+\lambda(\hat{i}-2\hat{j}+2\hat{k})$ and $\vec{r}=-4\hat{i}-\hat{k}+\mu(3\hat{i}-2\hat{j}-2\hat{k})$ is

(a) 0 unit (b) +2 units

(c) +30 units (d) 9 units

2. The sine of the angle between the straight line $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ and the plane $2x-2y+z=5$ is

(a) $\frac{10}{6\sqrt{5}}$ (b) $\frac{4}{5\sqrt{2}}$

(c) $\frac{\sqrt{2}}{10}$ (d) $\frac{2\sqrt{3}}{5}$

3. Under what condition is a straight line $\frac{x-x_0}{l}=\frac{y-y_0}{m}=\frac{z-z_0}{n}$ parallel to the xy-plane?

(a) $l=0$ (b) $m=0$

(c) $n=0$ (d) $l=0, m=0$

4. If α, β and γ are the angles which a half ray makes with the positive directions of the axes, then $\sin^2\alpha+\sin^2\beta+\sin^2\gamma$ is equal to

(a) 1 (b) 2

(c) 0 (d) -1

5. The equations $|x| = p$, $|y| = p$, $|z| = p$ in the xyz space represent

(a) a cube (b) a rhombus

(c) a sphere (d) a point (p, p, p)

6. The ratio in which the line joining (2, 4, 5) and (3, 5, -4) is divided by the yz-plane is

(a) 2 : 3 (b) 3 : 2

(c) $-2:3$ (d) $4:-3$

7. The distance of the point $(-1, -5, -10)$ from the point of intersection of the line $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x-y+z=5$ is

(a) 14/5 (b) 11

(c) 13 (d) 15

8. The lines $\frac{x-1}{k}=\frac{y-4}{1}=\frac{z-5}{1}$ and $\frac{x-2}{1}=\frac{y-3}{1}=\frac{4-z}{k}$ are coplanar if

(a) $k = 0$ or 4 (b) $k = 1$ or -1

(c) $k = 0$ or -1 (d) $k = -1$ or -2

9. The angle between the lines $\vec{r}=(4\hat{i}-\hat{j})+s(2\hat{i}+\hat{j}-3\hat{k})$ and $\vec{r}=(\hat{i}-\hat{j}+2\hat{k})+t(\hat{i}-3\hat{j}+2\hat{k})$ is

(a) $3\pi/2$ (b) $\pi/3$

(c) $2\pi/3$ (d) $\pi/6$

10. A plane which is perpendicular to two planes $x-y+2z-4=0$ and $2x-2y+z=0$ passes through $(1,-2,1)$. The distance of the plane from the point (1, 2, , 2) is

(a) $2\sqrt{2}$ (b) 0

(c) 2 (d) $\sqrt{2}$

11. If lines $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-5}{1}=\frac{z-6}{-5}$ are mutually perpendicular, then k is equal to

(a) $-10/7$ (b) $-7/10$

(c) -10 (d) -7

12. The equation of the plane bisecting the obtuse angle between the planes $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$ is

(a) $23x-13y+32z+45=0$

(b) $6x-5y+17z+19=0$

(c) $3x+5y-11z-20=0$

(d) none of these

13. The equation of the plane through the points (1, 2, 3), (–1, 4, 2) and (3, 1, 1) is

(a) $5x + y + 12z - 23 = 0$ (b) $5x + 6y + 2z - 23 = 0$

(c) $x + 6y + 2z - 23 = 0$ (d) $x + y + z - 13 = 0$

14. If $\sin^{-1}\frac{1}{3}$ is the angle between the line $L_1 = \frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $P_1 = 2x - y + \sqrt{k}z + 4 = 0$, then the value of k is

(a) –7/3 (b) –3/7

(c) 5/3 (d) –5/3

15. The position vector of the point where the line $\vec{r} = \hat{i} - \hat{j} + \hat{k} + t(\hat{i} + \hat{j} - \hat{k})$ meets the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 5$ is

(a) $5\hat{i} + \hat{j} - \hat{k}$ (b) $5\hat{i} + 3\hat{j} - 3\hat{k}$

(c) $2\hat{i} + \hat{j} + 2\hat{k}$ (d) $5\hat{i} + \hat{j} + \hat{k}$

16. If the line makes angles α, β and γ with the x-axis, y-axis and z-axis, then the value of $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is

(a) –1 (b) 1

(c) –2 (d) 2

17. The equation of the plane which contains the lines $\vec{r} = \hat{i} + 2\hat{j} - \hat{k} + \lambda(\hat{i} + 2\hat{j} - \hat{k})$ and $\vec{r} = \hat{i} + 2\hat{j} - \hat{k} + \mu(\hat{i} + \hat{j} + 3\hat{k})$ must be

(a) $\vec{r} \cdot (7\hat{i} - 4\hat{j} - k) = 0$ (b) $\vec{r} \cdot (4\hat{i} - 7\hat{j} + k) = 5$

(c) $\vec{r} \cdot (\hat{i} + 2\hat{j} - k) = 0$ (d) $\vec{r} \cdot (-\hat{i} - 2\hat{j} + 3k) = 3$

18. The angle between the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and the plane $3x + 2y - 3z = 4$ is

(a) 45° (b) 0°

(c) $\cos^{-1}\left(\frac{24}{\sqrt{29}\sqrt{22}}\right)$ (d) 90°

19. The ratio in which the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 17$ divides the line joining the points $-2\hat{i} + 4\hat{j} + 7\hat{k}$ and $3\hat{i} - 5\hat{j} + 8\hat{k}$ is

(a) 1 : 10 (b) 3 : 10

(c) 3 : 5 (d) 1 : 5

20. The planes $ax + 4y + z = 0$, $2y + 3z - 1 = 0$ and $3x - 6z + 2 = 0$

(a) will meet on a line if $ab = 15$ and $a = 30$

(b) will have a common point if $ab = 15$ and $a = 30$

(c) will have no common point if $ab = 15$, $a \neq 30$

(d) will have no common point if $ab \neq 15$, $a \neq 30$

21. The equation of the plane in which the lines $\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$ and $\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$ lie is

(a) $17x - 47y - 24z + 172 = 0$

(b) $17x + 47y - 24z + 172 = 0$

(c) $17x + 47y + 24z + 172 = 0$

(d) $17x - 47y + 24z + 172 = 0$

22. The foot of the perpendicular from the origin to the plane $3x + 4y - 6z + 1 = 0$ is

(a) $-\frac{3}{61}, \frac{4}{61}, \frac{6}{61}$ (b) $\frac{-3}{61}, \frac{-4}{61}, \frac{6}{61}$

(c) $\frac{4}{61}, \frac{-3}{61}, \frac{5}{61}$ (d) none of these

23. A line makes angles of 45° and 60° with axes of x and y respectively. The angle made by the same line with the positive axis of z is

(a) 30° or 60° (b) 60° or 90°

(c) 90° or 120° (d) 60° or 120°

24. The radius of the sphere circumscribing the tetrahedron whose faces are $\frac{y}{b} + \frac{z}{c} = 0, \frac{z}{c} + \frac{x}{a} = 0, \frac{x}{a} + \frac{y}{b} = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ is

(a) $\frac{a^2 + b^2 + c^2}{2}$

(b) $\frac{2}{a^2 + b^2 + c^2}\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)$

(c) $\frac{a^2 + b^2 + c^2}{2}\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$

(d) $\frac{1}{2}\left(a^2 + b^2 + c^2\right)^{3/2}$

25. The image of the point $P(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$ is

(a) (3, 5, –2) (b) (–3, 5, 2)

(c) (3, –5, 2) (d) (–1, 4, 2)

26. The distance of the point (1, –2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is

(a) 5 (b) 3

(c) 1 (d) none of these

27. The direction ratios of the line segment joining from the origin to the point (2, –3, 6) is

(a) (1, 0, 0) (b) (2, 3, 6)

(c) (2, –3, 6) (d) (0, –1, 6)

28. The centre of the sphere which passes through four points (0, 0, 0), (0, 2, 0), (1, 0, 0) and (0, 0, 4) is

(a) $\left(\frac{1}{2}, 1, 2\right)$ (b) $\left(-\frac{1}{2}, 1, 2\right)$

(c) $\left(\frac{1}{2}, 1, -2\right)$ (d) $\left(1, \frac{1}{2}, 2\right)$

29. The line $\frac{x-3}{2} = \frac{y-4}{5} = \frac{z-6}{7}$

(a) lies in $3x + 2y + 4z - 6 = 0$

(b) is parallel to $2x - 5y + 3z = 9$

(c) is perpendicular to $2x - 5y + 3z = 0$

(d) passes through (1, 2, 3)

30. The direction ratios of the line segment joining the points (3, –2, 4) and (4, 1, 3) are

(a) (–1, 1, 2) (b) (3, 2, –3)

(c) (1, 3, –1) (d) (4, 2, 7)

31. The radius of the circle formed due to the intersection of the sphere $x^2 + y^2 + z^2 - 2y - 4z - 11 = 0$ and the plane $x + 2y + 2z - 15 = 0$ is

(a) $\sqrt{3}$ (b) $\sqrt{5}$

(c) $\sqrt{7}$ (d) 3

32. The point in which the line $\frac{x-1}{1} = \frac{y-1}{-1} = \frac{z+3}{1}$ cuts the surface $x^2 + y^2 + z^2 - 20 = 0$ is

(a) (0, 2, 4) (b) (0, 2, –4)

(c) (4, 2, 0) (d) (0, –2, –4)

33. The direction ratios of the segment joining the points $P\left(-1, \cos\frac{\pi}{2}, 2\right)$ and $Q\left(\sin\frac{\pi}{2}, 0, \tan\frac{\pi}{4}\right)$ are

(a) $(-1, -1, -1)$ (b) $(1, -3, 4)$

(c) $(2, 0, -1)$ (d) $(1, 3, 4)$

34. The angle between the line $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \mu(\hat{i} - \hat{j} + \hat{k})$ and the normal to the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 4$ is

(a) $\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ (b) $\cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

(c) $\tan^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ (d) $\cot^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

35. Using direction ratios, the points $A(1, 2, 3)$, $B(4, 0, 4)$ and $C(-2, 4, 2)$

(a) form an equilateral triangle

(b) are concyclic

(c) are non-collinear

(d) are collinear

36. The equation $3x^2 - 2y^2 - 4z^2 + xy + xz + 6yz = 0$ represents

(a) a pair of straight lines

(b) a pair of planes

(c) a sphere

(d) a pair of planes passing through the origin

37. The direction cosines of the line segment joining points $(-3, 1, 2)$ and $(1, 4, -10)$ are

(a) $\left(\frac{4}{13}, \frac{3}{13}, -\frac{12}{13}\right)$ (b) $\left(\frac{1}{13}, \frac{2}{13}, \frac{5}{13}\right)$

(c) $\left(-\frac{1}{2}, \frac{1}{3}, -\frac{1}{7}\right)$ (d) $\left(-\frac{5}{13}, \frac{4}{13}, \frac{2}{13}\right)$

38. If the projections of a directed line segment on the coordinate axes are 12, 4, 3, the direction cosines of the line are

(a) $\frac{-12}{13}, \frac{-4}{13}, \frac{-3}{13}$ (b) $\frac{12}{13}, \frac{4}{13}, \frac{3}{13}$

(c) $\frac{12}{13}, \frac{-4}{13}, \frac{3}{13}$ (d) $\frac{12}{13}, \frac{4}{13}, \frac{-3}{13}$

39. The projection of the line segment joining the points $(-5, 1, 3)$ and $(2, 3, 5)$ on a line with direction cosines $\frac{1}{3}, \frac{1}{3}, \frac{2}{3}$ is

(a) 1/3 (b) 7/3

(c) 11/3 (d) 13/3

40. The value of α for which the lines $\frac{x-1}{2} = \frac{2y-1}{3} = \frac{1-3z}{\alpha}$ and $\frac{x+1}{+2} = \frac{3y-5}{2} = \frac{z-4}{3}$ are perpendicular to each other is

(a) 3 (b) 2

(c) –3 (d) 5

41. The angle between any two diagonals of a cube is

(a) 45° (b) 60°

(c) 90° (d) $\tan^{-1}(2\sqrt{2})$

42. The equation of the plane through the line of intersection of $\vec{r} \cdot (\hat{i} - 2\hat{j} + 2\hat{k}) = 1$ and $\vec{r} \cdot (2\hat{i} - \hat{j}) + 2 = 0$ and perpendicular to $\vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) + 9 = 0$ is

(a) $\vec{r} \cdot (-5\hat{i} + \hat{j} + 2\hat{k}) = 7$ (b) $\vec{r} \cdot (5\hat{i} + \hat{j} + 2\hat{k}) = 7$

(c) $\vec{r} \cdot (5\hat{i} - \hat{j} - 2\hat{k}) = 7$ (d) none of these

43. If points $P(6, 3, 2)$, $Q(5, 1, 4)$ and $R(3, 3, 5)$ are the vertices of a triangle PQR, then the value of $\angle PQR$ is

(a) 0 (b) $\pi/6$

(c) $\pi/3$ (d) $\pi/2$

44. The lines whose direction cosines are given by the relations $a^2l + b^2m + c^2n = 0$ and $mn + nl + lm = 0$ are parallel if
(a) $(a^2 - b^2 + c^2)^2 = 4a^2c^2$ (b) $(a^2 + b^2 + c^2)2 = 4a^2c^2$
(c) $(a^2 - b^2 + c^2) = a^2c^2$ (d) none of these

45. The equation of the plane which bisects the line joining (2, 3, 4) and (6, 7, 8) is
(a) $x + y + z - 15 = 0$ (b) $x - y + z - 15 = 0$
(c) $x - y - z - 15 = 0$ (d) $x + y + z + 15 = 0$

46. The vectors of magnitudes a, $2a$, $3a$ meet at a point and their directions are along the diagonals of three adjacent faces of a cube. Then the magnitude of their resultant is
(a) $5a$ (b) $6a$
(c) $10a$ (d) $9a$

47. The distance of the point (–1, 2, 5) from the line which passes through (3, 4, 5) and whose direction cosines are proportional to (2, –3, 6) is
(a) $\frac{4\sqrt{61}}{7}$ (b) $\frac{2\sqrt{74}}{5}$
(c) $\frac{2\sqrt{37}}{5}$ (d) none of these

48. The equations of the line through the point (2, 3, –5) and equally inclined to the axis are
(a) $x - 2 = y - 3 = z + 5$ (b) $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z+1}{-5}$
(c) $\frac{x}{2} = \frac{y}{3} = \frac{z}{-5}$ (d) none of these

49. If the direction cosines of a line are $\left(\frac{1}{c}, \frac{1}{c}, \frac{1}{c}\right)$, then
(a) $0 < c < 1$ (b) $c > 2$
(c) $c = \pm\sqrt{2}$ (d) $c = \pm\sqrt{3}$

50. The points (5, 0, 2), (2, –6, 0), (4, –9, 6) and (7, –3, 8) are vertices of a
(a) square (b) rhombus
(c) rectangle (d) parallelogram

51. The perimeter of the triangle with vertices at (1, 0, 0), (0, 1, 0) and (0, 0, 1) is
(a) 3 (b) 2
(c) $2\sqrt{2}$ (d) $3\sqrt{2}$

52. The line $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{0}$
(a) is parallel to the x-axis
(b) is parallel to the y-axis
(c) is parallel to the z-axis
(d) lies in a plane parallel to the xy-plane

53. The xy-plane divides the line joining the points (–1, 3, 4) and (2, –5, 6)
(a) internally in the ratio 2 : 3
(b) externally in the ratio 2 : 3
(c) internally in the ratio 3 : 2
(d) externally in the ratio 3 : 2

54. The equation of the plane passing through the point (–2, –2, 2) and containing the line joining the points (1, 1, 1) and (1, –1, 2) is
(a) $x - 3y - 6z + 8 = 0$ (b) $x - 3y - 6z - 8 = 0$
(c) $2x + 3y + 6z + 8 = 0$ (d) none of these

EXERCISE SET 2

1. If $A(6, 3, 2)$, $B(5, 1, 4)$, $C(3, -4, 7)$ and $D(0, 2, 5)$ are four points, the projection of segment CD on the line AB is
(a) –3/13 (b) –13/7
(c) –13/3 (d) –7/13

2. A point on the xOz-plane divides the join of (5, –3, –2) and (1, 2, –2) at
(a) $\left(\frac{13}{5}, 0, -2\right)$ (b) $\left(\frac{13}{5}, 0, 2\right)$
(c) (5, 0, 2) (d) (5, 0, –2)

3. For what value of k, are the points (–6, 3, 2), (3, –2, 4), (5, 7, 3) and (–13, k, –1) coplanar?
(a) 13 (b) 15
(c) 0 (d) 17

4. Three lines with direction cosines $<1, 1, 2>$, $<\sqrt{3}-1, -\sqrt{3}-1, 4>$ and $<-\sqrt{3}-1, -\sqrt{3}-1, 4>$ enclose
(a) an equilateral triangle
(b) a right-angled triangle
(c) an isosceles triangle
(d) a right-angled isosceles triangle

5. The distance between the points (1, 4, 5) and (2, 2, 3) is
(a) 5 (b) 4
(c) 3 (d) 2

6. The distance between the parallel planes $2x - 2y + z + 3 = 0$ and $4x - 4y + 2z - 7 = 0$ is
(a) 13/12 (b) 1/6
(c) 13/6 (d) 1/12

7. The equation of the plane passing through the origin and containing the lines whose direction cosines are proportional to 1, –2, 2 and 2, 3, –1 is
(a) $x - 2y + 2z = 0$ (b) $2x + 3y - z = 0$
(c) $x + 5y - 3z = 0$ (d) $4x - 5y - 7z = 0$

8. A line makes an obtuse angle with the positive x-axis and angles $\pi/4$ and $\pi/3$ with the positive y-axis and z-axis respectively. Its direction cosines are

(a) $-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

(c) $-\frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}$ (d) $\frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}$

9. The area of the triangle whose vertices are (3, 4, –1), (2, 2, 1) and (3, –4, 3) is

(a) $\sqrt{29}$ (b) $\sqrt{32}$

(c) 6 (d) 7

10. The line $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is parallel to the plane

(a) $2x + y - 2z = 0$ (b) $3x + 4y + 5z = 7$

(c) $x + y + z = 2$ (d) $2x + 3y + 4z = 0$

11. If direction cosines of two lines are proportional to (2, 3, –6) and (3, –4, 5), then the acute angle between them is

(a) $\cos^{-1}\left(\frac{49}{36}\right)$ (b) $\cos^{-1}\left(\frac{18\sqrt{2}}{35}\right)$

(c) 96° (d) $\cos^{-1}\left(\frac{18}{35}\right)$

12. The equation of the plane passing through the point of intersection of planes $3x + 4y + 6z - 3 = 0$ and $2x - 4y + z + 6$ and passing through the origin is

(a) $2x + 3y + 4z = 0$ (b) $2x + y + 3z = 0$

(c) $8x + 4y + 13z = 0$ (d) none of these

13. The value of K so that $\frac{x-1}{-3} = \frac{y-2}{2K} = \frac{z-3}{2}$ and $\frac{x-1}{3K} = \frac{y-1}{1} = \frac{z-6}{-5}$ may be perpendicular is given by

(a) –10 (b) 10/7

(c) –10/7 (d) –7/10

14. If a line makes angles α, β, γ and δ with four diagonals of a cube, then the value of $\sin^2\alpha + \sin^2\beta + \sin^2\gamma + \sin^2\delta$ is

(a) 4/3 (b) 8/3

(c) 7/3 (d) 1

15. The equation of the plane containing the lines $\frac{x-2}{4} = \frac{y+1}{3} = \frac{z-3}{5}$ and $x + 2y + 3z - 9 = 0 = 2x - y + 2z - 11$ is

(a) $7x - 11y + z = 28$ (b) $7x - 11y + z - 28 = 0$

(c) $7x + 11y + z = 28$ (d) $7x - 11y + z + 83 = 0$

16. The angle between the two planes $3x - 4y + 5z = 0$ and $2x - y - 2z = 5$ is

(a) $\pi/3$ (b) $\pi/2$

(c) $\pi/6$ (d) $\pi/4$

17. The angle between the lines whose direction cosines are $\left(\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{\sqrt{3}}{2}\right)$ and $\left(\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{-\sqrt{3}}{2}\right)$ is

(a) π (b) $\pi/2$

(c) $\pi/3$ (d) $\pi/4$

18. If $A(5, 3, 2)$, $B(-1, 0, -4)$ and $C(1, 1, -2)$ are collinear, then the ratio in which B divides AC is

(a) 1 : 3 (b) 2 : 3

(c) 3 : –1 (d) 1 : 2

19. If the direction cosines of two lines are such that $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$, then the angle between them is

(a) π (b) $\pi/3$

(c) $\pi/4$ (d) $\pi/6$

20. The image of the point (1,2,3) by the plane $x + y + z + 3 = 0$ is

(a) (5, 4, 3) (b) (7, 8, 9)

(c) (–5, –4, –3) (d) (–7, –8, –9)

21. The equation of the plane passing through the points (2, 1, –1), (1, 1, 1) and (3, 3, 0) is

(a) $4x - 3y + 2z - 3 = 0$ (b) $4x - 4z + 1 = 0$

(c) $4x + 3y - 2z + 8 = 0$ (d) $4x - 3y + 5z - 16 = 0$

22. If a line makes angles α, β, γ with the coordinate axes, then

(a) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma - 1 = 0$

(b) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma - 2 = 0$

(c) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + 1 = 0$

(d) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + 1 = 0$

23. The point of intersection of the lines $\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ and $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ is

(a) (6, –9, 13) (b) (4, –6, 5)

(c) (5, –7, 6) (d) they do not intersect

24. The perpendicular distance of the point (6, 5, 8) from the y-axis is

(a) 5 units (b) 6 units

(c) 8 units (d) 10 units

25. The line of intersection of the planes $\vec{r} \cdot (3\vec{i} - \vec{j} + \vec{k}) = 1$ and $\vec{r} \cdot (\vec{i} + 4\vec{j} - 2\vec{k}) = 2$ is parallel to the vector

(a) $-2\vec{i} + 7\vec{j} + 13\vec{k}$ (b) $2\vec{i} + 7\vec{j} - 13\vec{k}$

(c) $-2\vec{i} - 7\vec{j} + 13\vec{k}$ (d) $2\vec{i} + 7\vec{j} + 13\vec{k}$

26. The ratio in which the yz-plane divides the line segment joining $(-3, 4, -2)$ and $(2, 1, 3)$ is
(a) $-4 : 1$ (b) $3 : 2$
(c) $-2 : 3$ (d) $1 : 4$

27. The points of intersection of the line $\frac{x+2}{4} = \frac{y+9}{3} = \frac{z-8}{-5}$ and the sphere $x^2 + y^2 + z^2 = 49$ are
(a) $(5, 2, 3), (6, 3, 2)$ (b) $(5, 2, 3), (6, -3, -2)$
(c) $(2, 6, 3), (6, 3, 2)$ (d) $(2, -6, 3)\ (6, -3, -2)$

Assertion-Reason Type Questions

These questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions has four alternative answers. Only one of them is the correct answer. Select the correct answer given as follows:
(a) Both Statement I and Statement II are true and Statement II is a correct explanation of Statement I.
(b) Both Statement I and Statement II are true but Statement II is not a correct explanation of Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

28. **Statement I:** The equations $x = 2$ and $y = 3$ represent a line in three dimensions.
Statement II: $x = 2$ and $y = 3$ are satisfied by infinite points in three dimensions.

29. **Statement I:** If l, m, n are direction cosines of a line, then l, $-m$, n can also be direction cosines.
Statement II: If a, b, c are direction cosines, then $a^2 + b^2 + c^2 = 1$.

30. **Statement I:** The plane $x + 2y + 3z - 1 = 0$ meets the line $x = y = z$ at $\left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$.
Statement II: The plane and the line in Statement 1 are unparallel.

31. **Statement I:** If the points $A(0, 0, 0)$, $B(2, 0, 0)$, $C(0, 4, 0)$ and $D(0, 0, 6)$ are at the same distance r from a point P, then $r = 12$.
Statement II: The system of equations

$$x^2 + y^2 + z^2 = r^2$$
$$(x-2)^2 + y^2 + z^2 = r^2$$
$$x^2 + (y-4)^2 + z^2 = r^2$$
$$x^2 + y^2 + (z-6)^2 = r^2$$

yields correct value of r.

32. **Statement I:** Three concurrent lines with direction cosines $l_1, m_1, n_1; l_2, m_2, n_2; l_3, m_3, n_3$ are coplanar if

$$\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0.$$

Statement II: Lines

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1}, \quad \frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2}$$

are coplanar if

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0.$$

Previous Years' Questions

33. The equation of the plane contaning the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, where
(a) $ax_1 + by_1 + cz_1 = 0$ (b) $al + bm + cn = 0$
(c) $\frac{a}{l} = \frac{b}{m} = \frac{c}{m}$ (d) $lx_1 + my_1 + nz_1 = 0$
[AIEEE 2002]

34. The direction ratios of the normal to the plane passing through $(1, 0, 0)$ and $(0, 1, 0)$ which makes an angle $\pi/4$ with the plane $x + y = 3$ are
(a) $1, \sqrt{2}, 1$ (b) $1, 1, \sqrt{2}$
(c) $1, 1, 2$ (d) $\sqrt{2}, 1, 1$
[AIEEE 2002]

35. The centre of the circle given by $\vec{r} \cdot (\hat{i} + 2\hat{j} + 2\hat{k}) = 15$ and $|\vec{r} - (\hat{j} + 2\hat{k})| = 4$ is
(a) $(0, 1, 2)$ (b) $(1, 3, 4)$
(c) $(-1, 3, 4)$ (d) none of these
[AIEEE 2002]

36. The tetrahedron has vertices $O(0, 0, 0)$, $A(1, 2, 1)$ $B(2, 1, 3)$ and $C(-1, 1, 2)$. Then the angle between the faces OAB and ABC will be
(a) $\cos^{-1}\left(\frac{17}{31}\right)$ (b) $30°$
(c) $90°$ (d) $\cos^{-1}\left(\frac{19}{35}\right)$
[AIEEE 2003]

37. Two systems of rectangular axes have the same origin. If a plane cuts them at distances a, b, c and a', b', c' from the origin, then

(a) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}+\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}=0$

(b) $\frac{1}{a^2}+\frac{1}{b^2}-\frac{1}{c^2}+\frac{1}{a'^2}+\frac{1}{b'^2}-\frac{1}{c'^2}=0$

(c) $\frac{1}{a^2}-\frac{1}{b^2}-\frac{1}{c^2}+\frac{1}{a'^2}-\frac{1}{b'^2}-\frac{1}{c'^2}=0$

(d) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}-\frac{1}{a'^2}-\frac{1}{b'^2}-\frac{1}{c'^2}=0$ **[AIEEE 2003]**

38. The lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar if

(a) $k=0$ or -1 (b) $k=1$ or -1

(c) $k=0$ or (d) $k=3$ or -3 **[AIEEE 2003]**

39. The radius of the circle in which the sphere $x^2+y^2+z^2+2x-2y-4z-19=0$ is cut by the plane $x+2y+2z+7=0$ is

(a) 1 (b) 2

(c) 3 (d) 4 **[AIEEE 2003]**

40. The shortest distance between the plane $12x+4y+3z=327$ and the sphere $x^2+y^2+z^2+4x-2y-6z=155$ is

(a) $11\frac{3}{4}$ (b) 13

(c) 39 (d) 26 **[AIEEE 2003]**

41. A line makes the same angle θ with each of the x and z axes. If the angle β, which it makes with the y-axis, is such that $\sin^2\beta=3\sin^2\theta$, then $\cos^2\theta$ equals

(a) 3/5 (b) 1/5

(c) 2/3 (d) 2/5 **[AIEEE 2004]**

42. The distance between two parallel planes $2x+y+2z=8$ and $4x+2y+4z+5=0$ is

(a) 3/2 (b) 5/2

(c) 7/2 (d) 9/2 **[AIEEE 2004]**

43. The intersection of the spheres $x^2+y^2+z^2+7x-2y-z=13$ and $x^2+y^2+z^2-3x+3y+4z=18$ is the same as the intersection of one of the spheres and the plane

(a) $x-y-z=1$ (b) $x-2y-z=1$

(c) $x-y-2z=1$ (d) $2x-y-z=1$ **[AIEEE 2004]**

44. If the straight lines $x=1+s, y=3-\lambda s, z=1+\lambda s$ and $x=\frac{t}{2}, y=1+t, z=2-t$, with parameters s and t respectively, are coplanars, then λ equals

(a) -2 (b) -1

(c) $-1/2$ (d) 0 **[AIEEE 2004]**

45. The plane $x+2y-z=4$ cuts the sphere $x^2+y^2+z^2-x+z-2=0$ in a circle of radius

(a) $\sqrt{2}$ (b) 2

(c) 1 (d) 3 **[AIEEE 2005]**

46. The angle between the lines $2x=3y=-z$ and $6x=-y=-4z$ is

(a) 30° (b) 45°

(c) 90° (d) 0° **[AIEEE 2005]**

47. If the plane $2ax-3ay+4az+6=0$ passes through the mid-point of the line joining the centres of the spheres $x^2+y^2+z^2+6x-8y-2z=13$ and $x^2+y^2+z^2-10x+4y-2z=8$, then a equals

(a) 1 (b) -1

(c) 2 (d) -2 **[AIEEE 2005]**

48. The two lines $x=ay+b, z=cy+d$ and $x=a'y+b', z=c'y+d$ are perpendicular to each other if

(a) $aa'+cc'=1$ (b) $\frac{a}{a'}+\frac{c}{c'}=-1$

(c) $\frac{a}{a'}+\frac{c}{c'}=1$ (d) $aa'+cc'=-1$ **[AIEEE 2006]**

49. The image of the point $(-1,3,4)$ in the plane $x-2y=0$ is

(a) $\left(-\frac{17}{3},-\frac{19}{3},4\right)$ (b) (15, 11, 4)

(c) $\left(-\frac{17}{3},-\frac{19}{3},1\right)$ (d) $\left(\frac{9}{5},-\frac{13}{5},4\right)$ **[AIEEE 2006]**

50. If a line makes an angle of $\pi/4$ with the positive direction of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is

(a) $\pi/3$ (b) $\pi/4$

(c) $\pi/2$ (d) $\pi/6$ **[AIEEE 2007]**

51. Let L be the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$. If L makes an angle α with the positive x-axis, then $\cos\alpha$ equals

(a) $1/\sqrt{3}$ (b) 1/2

(c) 1 (d) $1/\sqrt{2}$ **[AIEEE 2007]**

52. If (2, 3, 5) is one end of a diameter of the sphere $x^2+y^2+z^2-6x-12y-2z+20=0$, then the coordinates of the other end of the diameter are

(a) (4, 9, –3) (b) (4, –3, 3)

(c) (4, 3, 5) (d) (4, 3, –3) **[AIEEE 2007]**

53. The line passing through the points (5, 1, a) and (3, b, 1) crosses the yz-plane at the point $\left(0, \frac{17}{2}, -\frac{13}{2}\right)$. Then
(a) $a = 8, b = 2$ (b) $a = 2, b = 8$
(c) $a = 4, b = 6$ (d) $a = 6, b = 4$
[AIEEE 2008]

54. If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersect at a point, then the integer k is equal to
(a) -2 (b) -5
(c) 5 (d) 2 **[AIEEE 2008]**

55. Let the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lie in the plane $x + 3y - \alpha z + b = 0$. Then (α, β) equals
(a) $(6, -17)$ (b) $(-6, 7)$
(c) $(5, -15)$ (d) $(-5, 15)$ **[AIEEE 2010]**

56. A line AB in three-dimensional space makes angles 45° and 120° with the positive x-axis and the positive y-axis, respectively. If AB makes an acute angle θ with the positive z-axis, then θ equals
(a) 30° (b) 45°
(c) 60° (d) 75° **[AIEEE 2010]**

57. **Statement I:** The point $A(3, 1, 6)$ is the mirror image of the point $P(1, 3, 4)$ in the plane $x - y + z = 5$.
Statement II: The plane $x - y + z = 5$ bisects the line segment joining $A(3, 1, 6)$ and $B(1, 3, 4)$.
(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.
[AIEEE 2010]

58. The distance of the point (1, –5, 9) from the plane $x - y + z = 5$ measured along a straight line $x = y = z$ is
(a) $10\sqrt{3}$ (b) $5\sqrt{3}$ (c) $3\sqrt{10}$ (d) $3\sqrt{5}$
[AIEEE 2011]

59. The length of the perpendicular drawn from the point (3, –1, 11) to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is
(a) $\sqrt{29}$ (b) $\sqrt{33}$ (c) $\sqrt{53}$ (d) $\sqrt{66}$
[AIEEE 2011]

60. An equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is
(a) $x - 2y + 2z - 3 = 0$ (b) $x - 2y + 2z + 1 = 0$
(c) $x - 2y + 2z - 1 = 0$ (d) $x - 2y + 2z + 5 = 0$
[AIEEE 2012]

61. If the lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then k is equal to
(a) –1 (b) 2/9 (c) 9/2 (d) 0
[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (d)	**2.** (c)	**3.** (c)	**4.** (b)	**5.** (a)	**6.** (c)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (a)
11. (a)	**12.** (a)	**13.** (b)	**14.** (c)	**15.** (b)	**16.** (a)	**17.** (a)	**18.** (b)	**19.** (b)	**20.** (a)
21. (a)	**22.** (b)	**23.** (d)	**24.** (c)	**25.** (b)	**26.** (c)	**27.** (c)	**28.** (a)	**29.** (b)	**30.** (c)
31. (c)	**32.** (b)	**33.** (c)	**34.** (b)	**35.** (d)	**36.** (b) and (d)		**37.** (a)	**38.** (a) and (b)	
39. (d)	**40.** (d)	**41.** (d)	**42.** (a)	**43.** (d)	**44.** (a)	**45.** (a)	**46.** (a)	**47.** (a)	**48.** (a)
49. (d)	**50.** (a)	**51.** (d)	**52.** (d)	**53.** (b)	**54.** (a)				

Exercise Set 2

1. (c)	**2.** (a)	**3.** (d)	**4.** (a)	**5.** (c)	**6.** (c)	**7.** (d)	**8.** (c)	**9.** (c)	**10.** (a)
11. (b)	**12.** (c)	**13.** (c)	**14.** (b)	**15.** (a)	**16.** (b)	**17.** (c)	**18.** (c)	**19.** (b)	**20.** (c)
21. (a)	**22.** (c)	**23.** (c)	**24.** (d)	**25.** (a)	**26.** (b)	**27.** (d)	**28.** (b)	**29.** (a)	**30.** (b)
31. (d)	**32.** (b)	**33.** (b)	**34.** (b)	**35.** (b)	**36.** (d)	**37.** (d)	**38.** (c)	**39.** (c)	**40.** (b)
41. (a)	**42.** (c)	**43.** (d)	**44.** (a)	**45.** (c)	**46.** (c)	**47.** (d)	**48.** (d)	**49.** (d)	**50.** (c)
51. (a)	**52.** (a)	**53.** (d)	**54.** (b)	**55.** (b)	**56.** (c)	**57.** (b)	**58.** (a)	**59.** (c)	**60.** (a)
61. (c)									

HINTS AND SOLUTIONS—EXERCISE SET 1

1. $\vec{a_1} = 6\hat{i} + 2\hat{j} + 2\hat{k}, \quad \vec{b_1} = \hat{i} - 2\hat{j} + 2\hat{k}$

$\vec{a_2} = -4\hat{i} - \hat{k}, \quad \vec{b_2} = 3\hat{i} - 2\hat{j} - 2\hat{k}$

$$\text{Shortest distance} = \left|\frac{(a_2 - a_1)\cdot(\vec{b_1}\times\vec{b_2})}{|\vec{b_1}\times\vec{b_2}|}\right|$$

$$\text{Numerator} = \begin{vmatrix} -10 & -2 & -3 \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix} = -108$$

Denominator = 12

Shortest distance = 9 units

2. Let θ be the angle between the straight line and the plane.

$\therefore$ $90° - \theta$ is the angle between the normal to the plane and the line. Thus,

$$\cos(90° - \theta) = \frac{(3)(2) + 4(-2) + 5(1)}{\sqrt{9+16+25}\sqrt{4+4+1}}$$

$$\Rightarrow \qquad \sin\theta = \frac{3}{5\sqrt{2}(3)} = \frac{1}{5\sqrt{2}} = \frac{\sqrt{2}}{10}$$

3. Since the given line is parallel to the xy-plane, it means that the normal line is perpendicular to the z-axis.

$\therefore$ Direction ratios of z coordinate are zero,

i.e. $n = 0$

4. $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 3 - (\cos^2\alpha + \cos^2\beta + \cos^2\gamma)$

$= 3 - 1 = 2$

5. Since we are given the equal intercept of the coordinate axes, i.e. $|x| = p, |y| = p, |z| = p$.

Therefore, it makes a cube.

6. Let $A(2, 4, 5)$ and $B(3, 5, -4)$ be two points and C be the point in which the line AB meets the yz-plane. Let C divide AB in the ratio $k : 1$, then

$$C = \left(\frac{3k+2}{k+1}, \frac{5k+4}{k+1}, \frac{-4k+5}{k+1}\right)$$

But C lies on the yz-plane, therefore

$$\frac{3k+2}{k+1} = 0$$

$$\Rightarrow \qquad k = -\frac{2}{3}$$

7. Clerly the point $(2, -1, 2)$ lies on the line as well as on the plane

$\therefore$ Required distance of the point $(-1, -5, -10)$

$$= \sqrt{(-1-2)^2 + (-5+1)^2 + (-10-2)^2}$$

$$= \sqrt{9+16+144}$$

$$= \sqrt{169} = 13$$

8. For coplanarity of two lines, the condition is

$$\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$\Rightarrow \ k^2 = 1, \quad \Rightarrow \ k = \pm 1$$

9. Equation of the first line is $\frac{x-4}{2} = \frac{y+1}{1} = \frac{z}{-3}$ and that of the second line is $\frac{x-1}{1} = \frac{y+1}{-3} = \frac{z-2}{2}$. Angle between the lines

$$\theta = \cos^{-1}\left(\left|\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|\right)$$

$$= \cos^{-1}\left(\left|\frac{2-3-6}{\sqrt{14}\sqrt{14}}\right|\right) = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

10. $A(x-1) + B(y+2) + C(z-1) = 0$ is the plane passing through $(1, -2, 1)$ and parallel to the given planes which implies $A - B + 2C = 0, 2A - 2B + C = 0$

$$\frac{A}{1} = \frac{B}{1} = \frac{C}{0}$$

Therefore, $x + y + 1 = 0$ is the required plane.

Perpendicular distance from $(1, 2, 2)$ is

$$\frac{1+2+1}{\sqrt{1^2+1^2+0^2}} = 2\sqrt{2} \text{ units}$$

11. Equations of lines are

$$\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-2}{2} \qquad \text{(i)}$$

and $$\frac{x-1}{3k} = \frac{y-5}{1} = \frac{z-6}{-5} \qquad \text{(ii)}$$

These lines are perpendicular to each other.

$\therefore$ $-3(3k) + 2k + 2(-5) = 0$

$\Rightarrow$ $-7k - 10 = 0$

$$\Rightarrow \ k = -\frac{10}{7}$$

12. $$\frac{2x - y + 2z + 3}{\sqrt{2^2 + (-1)^2 + 2^2}} = \pm\frac{3x - 2y + 6z + 8}{\sqrt{3^2 + (-2)^2 + 6^2}}$$

$\Rightarrow$ $5x - y - 4z - 3 = 0$ and $23x - 13y + 32z + 45 = 0$ are the two planes.

θ_1 is the angle between

$2x - y + 2z + 3 = 0$ and $5x - y - 4z - 3 = 0$

$$\theta_1 = \cos^{-1}\frac{2\times5+(-1)(-1)+2(-4)}{\sqrt{42}\sqrt{9}}$$

$$= \cos^{-1}\frac{1}{\sqrt{42}} < \cos^{-1}\frac{1}{\sqrt{2}}$$

$\Rightarrow \quad \theta_1 < 45°$

$$\theta_2 = \cos^{-1}\frac{2(23)+(-1)(-13)+2(32)}{\sqrt{9}\sqrt{23^2+(-13)^2+(32)^2}}$$

$$= \cos^{-1}\frac{121}{3\times\sqrt{1722}} > \cos^{-1}\frac{1}{\sqrt{2}}$$

$\theta_2 > 45°$

13. Equation of the plane through (1, 2, 3) is

$a(x-1)+b(y-2)+c(z-3)=0$ (i)

It passes through (–1, 4, 2) and (3, 1, 1), so

$-2a+2b-c=0$ (ii)

and $2a-b-2c=0$ (iii)

From Eqs. (ii) and (iii), we have

$$\frac{a}{-5}=\frac{b}{-6}=\frac{c}{-2}$$

$\therefore$ Equation of the plane is

$-5x-6y-2z+5+12+6=0$

$\Rightarrow \quad 5x+6y+2z-23=0$

14. $\sin\theta = \dfrac{1\times2+2(-1)+2\sqrt{k}}{\sqrt{1^2+2^2+2^2}\sqrt{2^2+1+k}} = \dfrac{1}{3}$

$\Rightarrow \quad k = 5/3$

15. $\vec{r} = (\hat{i}-\hat{j}+\hat{k})+t(\hat{i}+\hat{j}-\hat{k})$

$= (1+t)\hat{i}-(1-t)\hat{j}+(1-t)\hat{k}$

Also, $\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})=5$

$\Rightarrow \quad (1+t)-(1-t)+(1-t)=5$

$\Rightarrow \quad 1+t=5 \Rightarrow t=4$

$\therefore \quad \vec{r} = (1+4)\hat{i}-(1-4)\hat{j}+(1-4)\hat{k} = 5\hat{i}+3\hat{j}-3\hat{k}$

16. Angles made with the x, y and z axes = α, β and γ

We know that $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

We also know that $\cos2\theta = 2\cos^2\theta-1$

Therefore, $\cos2\alpha+\cos2\beta+\cos2\gamma$

$= (2\cos^2\alpha-1)+(2\cos^2\beta-1)+(2\cos^2\gamma-1)$

$= 2(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)-3 = (2\times1)-3 = -1$

17. Required plane is $r\cdot\vec{n}=d$

where $\vec{n} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -1\\ 1 & 1 & 3\end{vmatrix} = 7\hat{i}-4\hat{j}-\hat{k}$

$d = \vec{n}\cdot\vec{a} = (7\hat{i}-4\hat{j}-\hat{k})(\hat{i}+2\hat{j}-\hat{k}) = 0$

18. Angle between the plane and the line is given by

$$\sin\theta = \frac{aa'+bb'+cc'}{\sqrt{a^2+b^2+c^2}\sqrt{a'^2+b'^2+c'^2}}$$

$$\therefore \quad \sin\theta = \frac{2\times(3/4)+3\times(2/4)-4\times(3/4)}{\sqrt{2^2+3^2+4^2}\sqrt{(3/4)^2+(2/4)^2+(-3/4)^2}}$$

$$= \frac{(6/4)+(6/4)-(12/4)}{\sqrt{4+9+16}\sqrt{(9/16)+(4/16)+(9/16)}} = 0$$

$\therefore \quad \sin\theta = \sin 0°$

$\Rightarrow \quad \theta = 0°$

19. Let the plane $\vec{r}\cdot(\hat{i}-2\hat{j}+3\hat{k})=17$ divide the line joining the given points (–2, 4, 7), $R(x, y, z)$, (3, –5, 8) in the ratio λ : 1.

$$\therefore \quad R(x,y,z) = \frac{3\lambda-2}{\lambda+1}, \frac{4-5\lambda}{\lambda+1}, \frac{8\lambda+7}{\lambda+1}$$

Using in the equation of the plane, we get

$(3\lambda-2)1+(4-5\lambda)(-2)+3(8\lambda+7)-17(\lambda+1)=0$

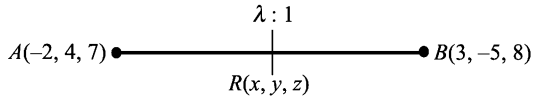


Fig. 24.38

$$\lambda = \frac{3}{10}$$

$$\therefore \quad \lambda : 1 = \frac{3}{10} : 1$$

Required ratio is 3 : 10.

20. $D = \begin{vmatrix}a & 4 & 1\\ 0 & 2 & 3\\ 3 & 0 & -b\end{vmatrix} = 30-2ab$

$$Dx = \begin{vmatrix}0 & 4 & 1\\ 1 & 2 & 3\\ -2 & 0 & -b\end{vmatrix} = 4b-20$$

and $$Dy = \begin{vmatrix}a & 0 & 1\\ 0 & 1 & 3\\ 3 & -2 & -b\end{vmatrix} = -ab+6a-3$$

and $Dz = -4a+12$

Now for $ab = 15$, $Dx = (60/a) - 20$, $Dy = 6a - 18$ and $Dz = -4a + 12$ will vanish for $a = 3$.

$\Rightarrow$ infinite points

$\Rightarrow$ meet on a line

For $ab = 15$ and for $a = 30$, we have no common point. So (b) is incorrect. Similarly (c) and (d) are incorrect. Hence, (a) is correct.

21. The equation of the plane in which the line $\frac{x-5}{4}=\frac{y-7}{4}=\frac{z+3}{-5}$ lies is

$$A(x-5)+B(y-7)+C(z+3)=0 \quad \text{(i)}$$

where A, B and C are the direction ratios of the plane. Since the first line lies on the plane,

$\therefore$ direction ratios of the normal to the plane is perpendicular to the line, so

$$4A+4B-5C=0 \quad \text{(ii)}$$

Since the line $\frac{x-8}{7}=\frac{y-4}{1}=\frac{z-5}{3}$ lies in this plane, the normal to the plane is perpendicular to this line.

$$7A+B+3C=0 \quad \text{(iii)}$$

From Eqs. (ii) and (iii), we get

$$\frac{A}{17}=\frac{B}{-47}=\frac{C}{-24}$$

$\therefore$ The required equation of the plane is

$$17(x-5)-47(y-7)+(-24)(z+3)=0$$

$$\Rightarrow \quad 17x-47y-24z+172=0$$

22. Equation of the plane is

$$3x+4y-6z+1=0 \quad \text{(i)}$$

The direction ratios of the normal to the plane (i) are 3, 4, –6. So the equations of the line through the origin and perpendicular to the plane (i) are

$$\frac{x}{3}=\frac{y}{4}=\frac{z}{-6}=\lambda \text{ (say)} \quad \text{(ii)}$$

$\therefore$ The coordinates of P on (ii) are $(3\lambda, 4\lambda, -6\lambda)$. Using this point in the equation of the plane, we get

$3(3\lambda)+4(4\lambda)-6(-6\lambda)+1=0 \quad \therefore \quad \lambda=-1/61$

$\therefore$ Required foot of the perpendicular is $\left(\frac{-3}{61}, \frac{-4}{61}, \frac{6}{61}\right)$.

23. Given: Angle made by the line with the x-axis (α) = 45° and angle made by the line with the y-axis (β) = 60°. Let γ be the angle which the line makes with the z-axis.

We know that $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

or $\cos^2 45°+\cos^2 60°+\cos^2\gamma=1$

or $\left(\frac{1}{\sqrt{2}}\right)^2+\left(\frac{1}{2}\right)^2+\cos^2\gamma=1$

or $\cos^2\gamma=1-\frac{1}{2}-\frac{1}{4}=\frac{4-2-1}{4}=\frac{1}{4}$

or $\cos\gamma=\pm(1/2)$ or $\gamma=60°$ or $120°$

24. Equation of the sphere is the same as the equation of the sphere passing through the points

$O(0, 0, 0)$, $A(-a, b, c)$, $B(a, -b, c)$, $C(a, b, -c)$

(on solving three equations taken at a time)

Hence, for the equation

$$x^2+y^2+z^2+2ux+2vy+2wz=0$$

$$u=-\frac{1}{a}(a^2+b^2+c^2),\ v=-\frac{1}{b}(a^2+b^2+c^2)$$

and $$w=-\frac{1}{c}(a^2+b^2+c^2)$$

$\therefore$ Radius $=\sqrt{u^2+v^2+w^2}$

$$=\frac{a^2+b^2+c^2}{2}\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}$$

25. The image of the point (x_1, y_1, z_1) in the plane $ax + by + cz + d = 0$ is

$$\frac{x-x_2}{a}=\frac{x-x_1}{b}=\frac{z-z_1}{c}$$

$$=\frac{-2(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$$

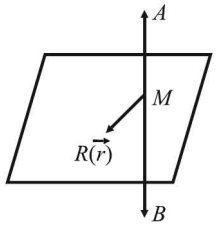


Fig. 24.39

$\therefore$ $\frac{x-1}{2}=\frac{y-3}{-1}=\frac{z-4}{1}$

$$=-2\left(\frac{2-3+4+3}{6}\right)$$

$\Rightarrow x=-3$, $y=5$ and $z=2$

Therefore, the image of the point is (–3, 5, 2).

26. Point of intersection of the line passing through x_1, y_1, z_1 and parallel to $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}$ and the plane $Ax + By + Kz + D = 0$ is given by the relation

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$

$$=\frac{Ax_1+By_1+Kz_1+D}{\sqrt{a^2+b^2+c^2}}$$

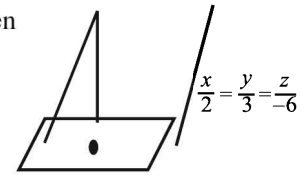


Fig. 24.40

$$\Rightarrow \quad \frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-6}=\frac{1\cdot1+(-1)(-2)+1(3)-5}{\sqrt{2^2+3^2+6^2}}=\frac{1}{7}$$

$\therefore$ $x-1=2/7$, $y+2=3/7$, $z-3=-(6/7)$

$x=9/7$, $y=-(11/7)$, $z=15/7$

$\therefore$ Point of intersection of the line and the plane is

$$(x, y, z)=\left(\frac{9}{7}-\frac{11}{7}, \frac{15}{7}\right)$$

Now, required distance = Distance from the given point to the point of intersection of the line and the plane

= Distance between (1, –2, 3) and $\left(\frac{9}{7}, \frac{-11}{7}, \frac{15}{7}\right)$

$$=\sqrt{\left(1-\frac{9}{7}\right)^2+\left(-2+\frac{11}{7}\right)^2+\left(3-\frac{15}{7}\right)^2}=1 \text{ unit}$$

27. Coordinates of the origin $(x_1, y_1, z_1) = (0, 0, 0)$;
coordinates of the second point $(x_2, y_2, z_2) = (2, -3, 6)$.
We know that the direction ratios of the line segment joining the origin (x_1, y_1, z_1) and the point (x_2, y_2, z_2) are
$(a, b, c) = (x_1 - x_1, y_2 - y_1, z_2 - z_1)$
$= [(2, 0), (-3) - 0, (6 - 0)] = (2, -3, 6)$

28. Let the equation of the sphere passing through the origin be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0.$$

It passes through (0, 2, 0).

$\therefore \quad 4 + 4v = 0 \Rightarrow v = -1$

Also, it passes through (1, 0, 0).

$\therefore \quad 1 + 2u = 0$

$\Rightarrow \quad u = \dfrac{-1}{2}$

Again it passes through (0, 0, 4).

$\therefore \quad 16 + 8w \Rightarrow w = -2$

$\therefore$ Centre of the sphere is $(-u, -v, -w) = \left(\dfrac{1}{2}, 1, 2\right)$

29. Line is $\dfrac{x-3}{2} = \dfrac{y-4}{5} = \dfrac{z-6}{7}$

$\therefore$ Direction ratios of $(2, 5, 7) = (a_1, b_1, c_1)$
Given plane is $2x - 5y + 3z = 9$
$\therefore$ Direction ratios of the normal of the plane are $(2, -5, 3)$ (say a_2, b_2, c_2)
Now $a_1a_2 + b_1b_2 + c_1c_2 = 4 - 25 + 21 = 0$
$\therefore$ The line is parallel to the plane.

30. Given: Coordinates of the first point $(x_1, y_1, z_1) = (3, -2, 4)$;
coordinates of the second point $(x_2, y_2, z_2) = (4, 1, 3)$. We know that the direction ratios of the line segment joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are

$$(a, b, c) = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$
$$= [(4 - 3), 1 - (-2), (3 - 4)] = (1, 3, -1)$$

31. Since the centre of the sphere
$x^2 + y^2 + z^2 - 2y - 4z - 11 = 0$ is $(0, 1, 2)$ and radius is 4,
distance of a plane $x + 2y + 2z - 15 = 0$ from (0, 1, 2)

$$= \frac{|0 + 2 + 4 - 15|}{\sqrt{1 + 4 + 4}} = \frac{9}{3} = 3$$

Now, $NP = \sqrt{OP^2 - ON^2}$

$= \sqrt{4^2 - 3^2} = \sqrt{16 - 9} = \sqrt{7}$

$\therefore$ Radius of the circle $= \sqrt{7}$

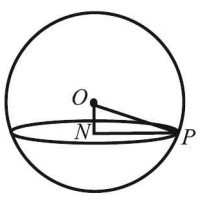


Fig. 24.41

32. Given $\dfrac{x-1}{1} = \dfrac{y-1}{-1} = \dfrac{z+3}{1}$

$\therefore \quad x = \lambda + 1$
$y = -\lambda + 1$
$z = \lambda - 3$

Now $x^2 + y^2 + z^2 - 20 = 0$

$\Rightarrow \quad (\lambda + 1)^2 + (-\lambda + 1)^2 + (\lambda - 3)^2 - 20 = 0$

$\Rightarrow \quad 3\lambda^2 - 6\lambda + 11 - 20 = 0$

$\Rightarrow \quad 3(\lambda^2 - 2\lambda - 3) = 0$

$\Rightarrow \quad \lambda^2 - 2\lambda - 3 = 0 \qquad \therefore \quad \lambda = -13$

$\therefore$ Required points are (0, 2, –4) and (4, –2, 0).

33. Coordinates of the point

$$P(x_1, y_1, z_1) = \left(-1, \cos\frac{\pi}{2}, 2\right) \text{ and point}$$

$$Q(x_2, y_2, z_2) = \left(\sin\frac{\pi}{2}, 0, \tan\frac{\pi}{4}\right)$$

We know that the direction ratios of the line segment joining the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are

$$(a, b, c) = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$
$$= \left(\sin\frac{\pi}{2} - (-1), 0 - \cos\frac{\pi}{2}, \tan\frac{\pi}{4} - 2\right)$$
$$= [(1 + 1), (0), (1 - 2)] = (2, 0, -1)$$

34. Direction ratios of the normal to the plane are (2, –1, 1).
Direction ratios of the line are (1, –1, 1).
Let θ be the angle between the line and the normal to the plane, then

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}} = = \frac{2\sqrt{2}}{3}$$

$$\Rightarrow \quad \theta = \cos^{-1}\frac{2\sqrt{2}}{3}$$

35. Coordinates of points $A(x_1, y_1, z_1) = (1, 2, 3)$, $B(x_2, y_2, z_2) = (4, 0, 4)$ and $C(x_3, y_3, z_3) = (-2, 4, 2)$
We know that the direction ratios of line AB are

$$(a_1, b_1, c_1) = (x_2 - x_1), (y_2 - y_1), (z_2 - z_1)$$
$$= [(4 - 1), (0 - 2), (4 - 3)] = (3, -2, 1)$$

or $a_1 = 3$, $b_1 = -2$ and $c_1 = 1$
We also know that the direction ratios of the line BC are

$$(a_2, b_2, c_2) = (x_3 - x_2), (y_3 - y_2), (z_3 - z_2)$$
$$= (-2 - 4), (4 - 0), (2 - 4) = (-6, 4, -2)$$

or $a_2 = -6$, $b_2 = 4$ and $c_2 = -2$

Thus, $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$. We also know that AB and BC are parallel lines.
Since both the lines pass through the common point B, therefore the three given points A, B and C are collinear.

36. The equation $3x^2-2y^2-4z^2+xy+xz+6yz=0$

$\Rightarrow\ 3x(x+y-z)-2y(x+y-z)+4z(x+y-z)=0$

$\Rightarrow\ (3x-2y+4z)(x+y-z)=0$

$3x-2y+4z=0$ and $x+y-z=0$

which represents planes, passing through the origin.

37. Coordinates of points $A(x_1, y_1, z_1) = (-3, 1, 2)$ and $B(x_2, y_2, z_2) = (1, 4, -10)$.
We know that the direction ratios of the line segment joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are

$$(a,b,c)=(x_2-x_1),(y_2-y_1),(z_2-z_1)$$
$$=(1+3,4-1,-10-2)=(4,3,-12)$$

We also know that the direction cosines of the line with direction ratios a, b, c are

$$(l,m,n)=\frac{a}{\sqrt{a^2+b^2+c^2}},\frac{b}{\sqrt{a^2+b^2+c^2}},\frac{c}{\sqrt{a^2+b^2+c^2}}$$
$$=\frac{4}{\sqrt{4^2+3^2+(-12)^2}},\frac{3}{\sqrt{4^2+3^2+(-12)^2}},$$
$$\frac{-12}{\sqrt{4^2+3^2+(-12)^2}}$$
$$=\left(\frac{4}{\sqrt{16+9+144}},\frac{3}{\sqrt{16+9+144}},\frac{-12}{\sqrt{16+9+144}}\right)$$
$$=\left(\frac{4}{13},\frac{3}{13},\frac{-12}{13}\right)$$

38. Let $CD = r$ be the given line and the direction cosines of CD be l, m, n.
Now projection of CD on the x-axis $= lr = 12$,
projection of $CD = r$ on the y-axis $= mr = 4$,
projection of $CD = r$ on the z-axis $= nr = 3$

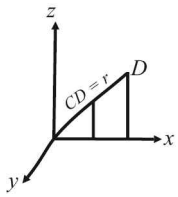


Fig. 24.42

$\therefore\ (l^2+m^2+n^2)(r)^2=12^2+4^2+3^2$

$r=\pm 13$

39. Coordinates of the first point $(x_1, y_1, z_1) = (-5, 1, 3)$ and coordinates of the second point $(x_2, y_2, z_2) = (2, 3, 5)$;

direction cosines of line $(l, m, n)=\left(\frac{1}{3},\frac{1}{3},\frac{2}{3}\right)$

We know that projection of the line

$(x_2-x_1)l+(y_2-y_1)m+(z_2-z_1)n$

$=[2-(-5)]\frac{1}{3}+(3-1)\frac{1}{3}+(5-3)\frac{2}{3}=\frac{7}{3}+\frac{2}{3}+\frac{4}{3}=\frac{13}{3}$

40. Lines are $\frac{x-1}{2}=\frac{y-\frac{1}{2}}{\frac{3}{2}}=\frac{\left(z-\frac{1}{3}\right)}{-\frac{\alpha}{3}}$ and

$$\frac{x+1}{2}=\frac{y-\frac{5}{3}}{\frac{2}{3}}=\frac{z-4}{3}$$

$\because$ as lines are perpendicular,

$\therefore\ l_1l_2+m_1m_2+n_1n_2=0$ or $a_1a_2+b_1b_2+c_1c_2=0$

$\therefore\ 2\times 2+\frac{2}{3}\times\frac{3}{2}-\alpha=0\ \Rightarrow\ \alpha=5$

41. The number of diagonals is 2.
We know that the direction ratios of the first diagonal of the cube $(a_1, b_1, c_1) = (1, 1, 1)$ and the direction ratios of the second diagonal of the cube $(a_2, b_2, c_2) = (1, 1, -1)$.
We also know that the relation for the angle between the two diagonals of a cube (θ) is

$$\cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\left[\sqrt{a_1^2+b_1^2+c_1^2}\right]\left[\sqrt{a_2^2+b_2^2+c_2^2}\right]}$$
$$=\frac{(1\times1)+(1\times1)+(1)\times(-1)}{\left[\sqrt{(1)^2+(1)^2+(1)^2}\right]\left[\sqrt{(1)^2+(1)^2+(-1)^2}\right]}$$
$$=\frac{2-1}{\left[\sqrt{3}\right]\left[\sqrt{3}\right]}=\frac{1}{3}\ \text{or}\ \sec\theta=3\ \text{ or }\ \sec^2\theta=9$$

We also know that $1+\tan^2\theta=\sec^2\theta$ or $1+\tan^2\theta=9$ or $\tan^2\theta=8$ or $\tan\theta=2\sqrt{2}$ or $\theta=\tan^{-1}(2\sqrt{2})$

42. Equation of any plane through the intersection of the given planes is

$$\left[\vec{r}\cdot(\hat{i}-2\hat{j}+2\hat{k})-1\right]+\lambda\left[\vec{r}\cdot(2\hat{i}-\hat{j})+2\right]=0$$

$$\Rightarrow\ \vec{r}\left[(1+2\lambda)\hat{i}+\hat{j}(-2-\lambda)+2\hat{k}\right]=1-2\lambda \qquad \text{(i)}$$

Now (i) will be perpendicular to $\vec{r}\cdot(\hat{i}+\hat{j}+2\hat{k})$ if

$$1(1+2\lambda)+(-2-\lambda)(1)+2\cdot2=0$$
$$(2\lambda-\lambda)+1-2+4=0\quad\therefore\ \lambda=-3$$

$\therefore$ Required equation of the plane is

$$\vec{r}\cdot(-5\hat{i}+\hat{j}+2\hat{k})=7$$

43. Coordiantes of points $P(x_1, y_1, z_1) = (6, 3, 2)$; coordinates of point $Q(x_2, y_2, z_2) = (5, 1, 4)$ and coordinates of point $R(x_3, y_3, z_3) = (3, 3, 5)$

We know that the direction ratios of QP are

$$(a_1, b_1, c_1) = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$
$$= (5-6, 1-3, 4-2) = (-1, -2, 2)$$

Similarly, the direction ratios of QR are

$$(a_2, b_2, c_2) = (x_3 - x_2, y_3 - y_2, z_3 - z_2)$$
$$= (3-5, 3-1, 5-4) = (-2, 2, 1)$$

Let $\angle PQR = \theta$

Therefore, $\cos\theta = \dfrac{a_1a_2 + b_1b_2 + c_1c_2}{\left[\sqrt{a_1^2 + b_1^2 + c_1^2}\right]\left[\sqrt{a_2^2 + b_2^2 + c_2^2}\right]}$

$$= \frac{-1\times(-2) + (-2\times 2) + (2)\times(1)}{\sqrt{1^2 + 2^2 + (-2)^2}\sqrt{(-2)^2 + 2^2 + (-1)^2}} = \frac{2 - 4 + 2}{\sqrt{9}\sqrt{9}} = 0$$

or $\theta = \dfrac{\pi}{2}$

44. Given relation

$$a^2l + b^2m + c^2n = 0 \quad \text{(i)}$$
$$mn + nl + lm = 0 \quad \text{(ii)}$$

Eliminating m from (i) and (ii), we have

$$-\frac{1}{b^2}(a^2l + c^2n)n + nl - \frac{1}{b^2}(a^2l + c^2n)l = 0$$

On simplification, we get

$$a^2\left(\frac{l}{n}\right)^2 + (a^2 - b^2 + c^2)\frac{l}{n} + c^2 = 0 \quad \text{(iii)}$$

Lines are parallel if roots of (iii) are equal. Thus,

$$(a^2 - b^2 + c^2)^2 - 4a^2c^2 = 0$$

45. Mid-point of (2, 3, 4) and (6, 7, 8) is

$$\left(\frac{2+6}{2}, \frac{3+7}{2}, \frac{4+8}{2}\right), \text{ i.e. } (4, 5, 6)$$

This lies on $x + y + z - 15 = 0$.

Hence, this is the required equation.

46. Let the sides of the cube be along the axes, then diagonals have direction cosines as

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \text{and} \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$\therefore$ The resultant vector is

$$\frac{a}{\sqrt{2}}(\hat{i} + \hat{j}) + \frac{2a}{\sqrt{2}}(\hat{i} + \hat{k}) + \frac{3a}{\sqrt{2}}(\hat{j} + \hat{k})$$
$$= \frac{a}{\sqrt{2}}(3\hat{i} + 4\hat{j} + 5\hat{k})$$

$\Rightarrow$ Magnitude of the resultant vector

$$= \frac{a}{\sqrt{2}}\sqrt{9 + 16 + 25} = \frac{a}{\sqrt{2}} \cdot \sqrt{50} = 5a$$

47. Direction cosines of the line AN are $\dfrac{2}{7}, \dfrac{-3}{7}$ and $\dfrac{6}{7}$. From Fig. 24.43,

$$AP = \sqrt{(3+1)^2 + (4-2)^2 + (5-5)^2} = \sqrt{20}$$

AN = projection of AP on the line

$$= \frac{2}{7}(3+1) - \frac{3}{7}(4-2) + \frac{6}{7}(5-5)$$
$$= 2/7$$

$\therefore$ Required distance

$$PN = \sqrt{AP^2 - PN^2}$$
$$= \sqrt{20 - 4/49} = \frac{4\sqrt{61}}{7}$$

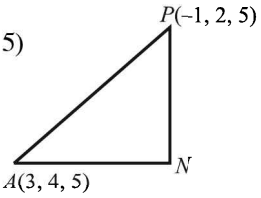


Fig. 24.43

48. Since the direction ratios of the line equally inclined to the axes are < 1, 1, 1>

$\therefore$ Equation of the line is

$$\frac{x-2}{1} = \frac{y-3}{1} = \frac{z+5}{1}$$

49. Since the direction ratios of a line are $\left(\dfrac{1}{c}, \dfrac{1}{c}, \dfrac{1}{c}\right)$

$$\therefore \left(\frac{1}{c}\right)^2 + \left(\frac{1}{c}\right)^2 + \left(\frac{1}{c}\right)^2 = 1$$

$\Rightarrow c^2 = 3$

$\Rightarrow c = \pm\sqrt{3}$

50. Let $A(5, 0, 2)$, $B(2, -6, 0)$, $C(4, -9, 6)$ and $D(7, -3, 8)$ be the vertices.

$$AB = \sqrt{(5-2)^2 + (0+6)^2 + (2-0)^2} = \sqrt{9 + 36 + 4} = 7$$

$$BC = \sqrt{(4-2)^2 + (-9+6)^2 + (6-0)^2} = \sqrt{4 + 9 + 36} = 7$$

$$CD = \sqrt{(7-4)^2 + (-3+9)^2 + (8-6)^2} = \sqrt{9 + 36 + 4} = 7$$

$$DA = \sqrt{(7-5)^2 + (-3+0)^2 + (8-2)^2} = \sqrt{4 + 9 + 36} = 7$$

$AB = BA = CD = DA$

Again $AC = \sqrt{(5-4)^2 + (0+9)^2 + (2-6)^2}$

$$= \sqrt{1 + 81 = 16} = \sqrt{98} = 7\sqrt{2}$$

$\therefore$ $AC^2 = AB^2 + BC^2$ $\therefore$ $\angle B = 90°$

$\therefore$ $ABCD$ is a square.

51. Let $A = (1, 0, 0)$, $B = (0, 1, 0)$ and $C = (0, 0, 1)$

Now, $AB = \sqrt{(0-1)^2 + (1-0)^2 + 0^2} = \sqrt{2}$

$BC = \sqrt{0^2 + (0-1)^2 + (1-0)^2} = \sqrt{2}$

and $CA = \sqrt{(1-0)^2 + 0^2 + (0-1)^2} = \sqrt{2}$

$\therefore$ Perimeter of the triangle $= AB + BC + CA$

$= \sqrt{2} + \sqrt{2} + \sqrt{2} = 3\sqrt{2}$

52. Since the direction cosines of the normal to the xy-plane are 0, 0, 1 and the direction ratios of the given line are 1, 2, 0

$\because \quad 1 \cdot 0 + 0 \cdot 2 + 1 \cdot 0 = 0$

$\therefore$ The normal of the xy-plane is perpendicular to the line. Hence, the given line lies in a plane parallel to the xy-plane.

53. Suppose the xy-plane divides at the joining of the given points in the ratio $\lambda : 1$. The coordinates of the points of division are $\left[\frac{2\lambda-1}{\lambda+1}, \frac{-5\lambda+3}{\lambda+1}, \frac{6\lambda+4}{\lambda+1}\right]$. Since the points lie on the xy-plane,

$\therefore \quad \frac{6\lambda+4}{\lambda+1} = 0 \Rightarrow \lambda = \frac{-2}{3}$

54. Equation of any plane passing through $(-2, -2, 2)$ is

$a(x+2) + b(y+2) + c(z-2) = 0 \quad$ (i)

Since it contains the line joining the points $(1, 1, 1)$ and $(1, -1, 2)$, we get

$3a + 3b - c = 0$ and $3a + b + 0c = 0$

Solving by cross multiplication, we get

$$\frac{a}{0+1} = \frac{b}{-3+0} = \frac{c}{3-9}$$

Putting $a = 1$, $b = -3$, $c = -6$ in (i), we get the required equation.

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $\overrightarrow{CD} = (0-3, 2+4, 5-7) = -3\hat{i} + 6\hat{j} - 2\hat{k}$

$\overrightarrow{AB} = (5-6, 1-3, 4-2) = -\hat{i} - 2\hat{j} + 2\hat{k}$

$\left|\overrightarrow{AB}\right| = \sqrt{1+4+4} = 3$

$\therefore$ A unit vector along $AB = \frac{1}{3}(-\hat{i} - 2\hat{j} + 2\hat{k})$

$\therefore$ Projection of $\overrightarrow{CD}$ on $\overrightarrow{AB}$

$= (-3\hat{i} + 6\hat{j} - 2\hat{k}) \cdot \frac{1}{3}(-\hat{i} - 2\hat{j} + 2\hat{k})$

$= \frac{1}{3}(3 - 12 - 4) = -\frac{13}{3}$

2. Let the point $P(x, y, z)$ divide the line joining the points A and B in the ratio $m : 1$

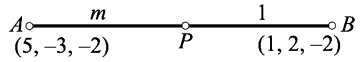


Fig. 24.44

Since point P is on the xOz plane,

$\therefore$ y coordinate $= 0$

$\Rightarrow \quad \frac{2m-3}{m+1} = 0 \Rightarrow m = \frac{3}{2}$

Now, $x = \frac{3 + 2 \times 5}{3+2} = \frac{13}{5}$

and $z = \frac{3 \times (-2) + 2 \times (-2)}{5} = -2$

$\therefore$ Required point is $\left(\frac{13}{5}, 0, -2\right)$.

3. Four given points are coplanar if

$$\begin{vmatrix} -6 & 3 & 2 & 1 \\ 3 & -2 & 4 & 1 \\ 5 & 7 & 3 & 1 \\ -13 & k & -1 & 1 \end{vmatrix} = 0 \qquad (R_1 \to R_1 - R_2)$$

$$\begin{vmatrix} -9 & 5 & -2 & 0 \\ 3 & -2 & 4 & 1 \\ 5 & 7 & 3 & 1 \\ -13 & k & -1 & 1 \end{vmatrix} = 0 \qquad (R_2 \to R_2 - R_3)$$

$$\begin{vmatrix} -9 & 5 & -2 & 0 \\ -2 & -9 & 1 & 0 \\ 5 & 7 & 3 & 1 \\ -13 & k & -1 & 1 \end{vmatrix} = 0 \qquad (R_3 \to R_3 - R_4)$$

$$\begin{vmatrix} -9 & 5 & -2 & 0 \\ -2 & -9 & 1 & 0 \\ 18 & 7-k & 4 & 0 \\ -13 & k & -1 & 1 \end{vmatrix} = 0$$

Expanding using C_4 and $R_3 \to R_3 + 2R_1$

$$\begin{vmatrix} -9 & 5 & -2 \\ -2 & -9 & 1 \\ 0 & 17-k & 0 \end{vmatrix} = 0 \qquad \text{or } -(17-k)(-13) = 0$$

or $k = 17$

4. If θ is the angle between the first and second lines, then

$$\cos\theta = \frac{1(\sqrt{3}-1)+(-\sqrt{3}-1)+2(4)}{\sqrt{1+1+4}\,\sqrt{(\sqrt{3}-1)^2+(-\sqrt{3}-1)^2+4^2}}$$

$\Rightarrow \theta = 60°$

Similarly, the angle between the first and third lines is also 60°.

$\therefore$ The triangle is equilateral.

5. The distance between the given points

$$= \sqrt{(2-1)^2+(2-4)^2+(3-5)^2}$$

$$= \sqrt{1+4+4} = 3$$

6. The given planes are $4x - 4y + 2z + 6 = 0$ and $4x - 4y + 2z - 7 = 0$

$$\therefore \text{ Required distance } = \frac{6-(-7)}{\sqrt{4^2+4^2+2^2}} = \frac{13}{6}$$

7. Any plane through the origin is $a(x-0)+b(y-0)+c(z-0)=0$. The plane contains the lines whose direction cosines are proportional to 1, –2, 2 and 2, 3, –1. So the normals to the plane are perpendicular to these lines.

$\therefore \quad a - 2b + 2c = 0$ and $2a + 3b - c = 0$

$$\therefore \quad \frac{a}{2-6} = \frac{b}{4+1} = \frac{c}{3+4}$$

$$\Rightarrow \quad \frac{a}{-4} = \frac{b}{5} = \frac{c}{7}$$

$\therefore$ Required plane is

$$-4(x-0)+5(y-0)+7(z-0)=0$$

$$\Rightarrow \quad -4x+5y+7z=0$$

i.e. $$4x-5y-7z=0$$

8. Let $m = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ and $n = \cos\frac{\pi}{3} = \frac{1}{2}$

$\because \quad l^2 + m^2 + n^2 = 1$

$$\Rightarrow \quad l = \sqrt{1-(m^2+n^2)}$$

$$= \sqrt{1-\left(\frac{1}{2}+\frac{1}{4}\right)}$$

$$= \sqrt{1-\frac{3}{4}}$$

$$\Rightarrow \quad l = \pm\frac{1}{2}$$

Since the line makes an obtuse angle, so we take $l = -\frac{1}{2}$

$\therefore$ Direction cosines are $-\frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}$.

9. The vertices of the projection of the triangle on the xy-plane are (3, 4, 0), (2, 2, 0), (3, –4, 0)

$$\therefore \quad \Delta xy = \frac{1}{2}\begin{vmatrix} 3 & 4 & 1 \\ 2 & 3 & 1 \\ 3 & -4 & 1 \end{vmatrix} = \frac{1}{2}\times 8 = 4$$

Similarly, $$\Delta yz = \frac{1}{2}\begin{vmatrix} 4 & -1 & 1 \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix} = \frac{1}{2}\times 8 = 4$$

and $$\Delta xz = \frac{1}{2}\begin{vmatrix} 3 & -1 & 1 \\ 2 & 1 & 1 \\ 3 & 3 & 1 \end{vmatrix} = \frac{1}{2}|-4| = 2$$

$$\therefore \text{ Required area } = \sqrt{\Delta^2 xy + \Delta^2 yz + \Delta^2 zx}$$

$$= \sqrt{4^2+4^2+2^2} = 6$$

10. Since $3(2)+4(1)+5(-2)=0$,

$\therefore$ the line is parallel to $2x + y - 2z = 0$.

11. Since direction cosines of two lines are proportional to (2, 3, –6) and (3, –4, 5),

$$\therefore \quad \cos\theta = \frac{|2\times 3+3\times(-4)-6\times 5|}{\sqrt{2^2+3^2+(-6)^2}\,\sqrt{3^2+(-4)^2+5^2}}$$

$$= \frac{|6-12-30|}{\sqrt{49}\,\sqrt{50}} \quad \Rightarrow \quad \theta = \cos^{-1}\left(\frac{18\sqrt{2}}{35}\right)$$

12. Any plane passing through the point of intersection of the planes $3x + 4y + 6z - 3 = 0$ and $(2x - 4y + z + 6) = 0$ is

$$(2x-4y+z+6)+\lambda(3x+4y+6z-3)=0 \qquad \text{(i)}$$

Since it passes through the origin,

$$6 - 3\lambda = 0$$

Putting $\lambda = 2$ in (i), we get the required equation as $8x + 4y + 13z = 0$.

13. Since the lines are perpendicular,

$\therefore \quad -9K + 2K - 10 = 0$

$$\therefore \quad 7K = -10 \quad \therefore \quad K = -\frac{10}{7}$$

14. We know that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$

where α, β, γ and δ are the angles with diagonals of the cube.

$\therefore \quad 1-\sin^2\alpha+1-\sin^2\beta+1-\sin^2\gamma+1-\sin^2\delta=\frac{4}{3}$

$\Rightarrow \quad \sin^2\alpha+\sin^2\beta+\sin^2\gamma+\sin^2\delta=\frac{8}{3}$

15. Let the equation of the given plane be

$$(x+2y+3z-9)+\lambda(2x-y+2z-11)=0 \qquad \text{(i)}$$

Its direction ratios are $1+2\lambda, 2-\lambda, 3+2\lambda$

Since plane (i) contains the line

$$\frac{x-2}{4}=\frac{y+1}{3}=\frac{z-3}{5},$$

$\therefore \quad 4(1+2\lambda)+3(2-\lambda)+5(3+2\lambda)=0$

or $\quad 15\lambda=-25 \quad \Rightarrow \quad \lambda=-5/3$

Putting $\lambda=-5/3$ in (i), we get the required equation.

16. $\cos\theta=\frac{3(2)+(-4)(-1)+5(-2)}{\sqrt{9+16+25}\sqrt{4+1+4}}=0$

$\therefore \quad \theta=\frac{\pi}{2}$

17. $\cos\theta=\left|l_1l_2+m_1m_2+n_1n_2\right|$

$$=\left|\frac{\sqrt{3}}{4}\times\frac{\sqrt{3}}{4}+\frac{1}{4}\times\frac{1}{4}+\frac{\sqrt{3}}{2}\times\left(\frac{-\sqrt{3}}{2}\right)\right|$$

$$=\left|\frac{3}{16}+\frac{1}{6}-\frac{3}{4}\right|=\left|-\frac{2}{4}\right|=\frac{1}{2}$$

$\Rightarrow \quad \theta=\frac{\pi}{3}$

18. Required point $=-1-5:1+1$

$$=-6:2=-3:1=3:-1$$

19. Given $l+m+n=0$ (i)

and $l^2+m^2-n^2=0$ (ii)

$\therefore \quad l^2+m^2-(-l-m)^2=0$

$\Rightarrow \quad 2lm=0$

$\Rightarrow \quad l=0$ or $m=0$

If $l=0$, then $n=-m$

$\Rightarrow \quad l:m:n=0:1:-1$

and if $m=0$, then $n=-l$

$\Rightarrow \quad l:m:n=1:0:-1$

$\therefore \quad \cos\theta=\frac{0+0+1}{\sqrt{0+1+1}\sqrt{0+1+1}}=\frac{1}{2}$

$\Rightarrow \quad \theta=\frac{\pi}{3}$

20. Let the image of the point $P(1, 2, 3)$ be $Q(\alpha, \beta, \gamma)$. Thus, the equation of the line PQ will be

$$\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-3}{1}=k$$

(the direction ratios of the line PQ are the direction ratios of the normal to the plane)

$\therefore \quad (\alpha, \beta, \gamma)=(k+1, k+2, k+3)$

also mid-point of $(1, 2, 3)$ and $k+1, k+2, k+3$ lies on the plane, i.e. $\left(\frac{k+2}{2}, \frac{k+4}{2}, \frac{k+6}{2}\right)$ lies on the plane

$$x+y+z+3=0 \quad \Rightarrow \quad k=-6$$

$\therefore$ The image is $(-5, -4, -3)$.

21. Clearly all the points satisfy the option (a),

i.e. $\quad 4x-3y+2z-3=0$

$[\because (2, 1, -1), (1, 1, 1), (3, 3, 0)$ do not satisfy (b), (c), (d)$]$

22. Since $2\cos^2\alpha+2\cos^2\beta+2\cos^2\gamma=2$

$[\because l^2+m^2+n^2=1]$

$\Rightarrow \quad 1+\cos 2\alpha+1+\cos 2\beta+1+\cos 2\gamma=2$

$\Rightarrow \quad \cos 2\alpha+\cos 2\beta+\cos 2\gamma+1=0$

23. Let $\frac{x-4}{1}=\frac{y+3}{-4}=\frac{z+1}{7}=\lambda$

$\Rightarrow \quad (x, y, z)=(\lambda+4, -4\lambda-3, 7\lambda-1)$

and $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}=\mu$

$\Rightarrow \quad (x, y, z)=(2\mu+1, -3\mu-1, 8\mu-10)$

$\therefore \quad \lambda+4=2\mu+1$

$-4\lambda-3=-3\mu-1$

$7\lambda-1=8\mu-10$

Solving, we get $\lambda=1$, $\mu=2$

$\therefore$ Point of intersection is $(5, -7, 6)$.

24. Perpendicular distance of the point $(6, 5, 8)$ from the y-axis $=\sqrt{6^2+8^2}=10$ units

25. The line of interesection of the planes $\vec{r}\cdot(3\vec{i}-\vec{j}+\vec{k})=1$ and $\vec{r}\cdot(\vec{i}+4\vec{j}-2\vec{k})=2$ is perpendicular to each of the normal vectors $\vec{n_1}=3\vec{i}-\vec{j}+\vec{k}$ and $\vec{n_2}=\vec{i}+4\vec{j}-2\vec{k}$.

$\therefore$ It is parallel to the vector

$$\vec{n_1}\times\vec{n_2}=(3\vec{i}-\vec{j}+\vec{k})\times(\vec{i}+4\vec{j}-2\vec{k})$$

$$=-2\vec{i}+7\vec{j}+13\vec{k}$$

26. The ratio in which the yz-plane divides the line segment
$= -x_1 : x_2 = -(-3) : 2 = 3 : 2$

27. Let $\frac{x+2}{4} = \frac{y+9}{3} = \frac{z-8}{-5} = k$ be a line.
Any point on the above line is

$$(4k-2, 3k-9, 8-5k) \qquad \text{(i)}$$

It lies on the sphere if

$$(4k-2)^2 + (3k-9)^2 + (8-5k)^2 = 49$$

On simplification, $k = 1, 2$
Putting these values in (i), we get the points of intersection as

$$(2, -6, 3), (6, -3, -2)$$

28. Statement I is true since $x = 2, y = 3$ are two planes whose line of intersection is represented by $x = 2$, $y = 3$.
Statement II is also true since $(2, 3, \beta)$, β arbitary, is the set of all points for which $x = 2$, $y = 3$. But it is not a correct explanation. Since it does not show that the set of points $(2, 3, \beta)$ is on a line.

29. Since $l^2 + m^2 + n^2 = 1$

$\Rightarrow\ l^2 + (-m^2) + (-n)^2 = 1$

Hence, (a) is correct.

30. Both statements are obviously correct but points of intersection is found by substituting $x = y = z = r$ in the equation of the plane to get $r = 1/6$

$\Rightarrow$ Intersection point is $\left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$

31. Statement II is definitely correct if P is the point (x, y, z) but as it yields $r = \sqrt{14}$, Statement I is false.

32. Statement I and Statement II are correct but Statement II does not explain Statement I. Let us observe this.

If $\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0,$

then the system of equations

$$ll_1 + mm_1 + nn_1 = 0$$
$$ll_2 + mm_2 + nn_2 = 0$$
$$ll_3 + mm_3 + nn_3 = 0$$

has non-zero solutions for l, m, n.
$\Rightarrow$ A line with direction cosines l, m, n is perpendicular to three lines, which is possible only when the three lines lie in a single plane.

33. When the given plane contains the given line, then the normal to the plane must be perpendicular to the line and the condition is $al + bm + cn = 0$.

34. Let the direction ratios of the normal to the plane be a, b, c.

$\therefore\ a(x-1) + b(y) + c(z) = 0 \qquad \text{(i)}$

$\Rightarrow\ a(0-1) + b(1) + c(0) = 0$ [by using (0, 1, 0) in (i)]

$\Rightarrow\ a - b + 0c = 0 \ \Rightarrow\ a = b$

Also the angle between (i) and $x + y + 0c = 3$ is $\pi/4$.

$$\therefore\ \cos\frac{\pi}{4} = \frac{a+a}{\sqrt{1^2+1^2}\sqrt{a^2+b^2+c^2}} = \frac{2a}{\sqrt{2}\sqrt{2a^2+c^2}}$$

$\Rightarrow\ 2a^2 + c^2 = 4a^2 \Rightarrow c = \pm\sqrt{2}a$

$\therefore$ Direction ratios are a, b, c, i.e. $a, a, \pm\sqrt{2}a$

$\therefore$ Required direction ratios are $1, 1, \sqrt{2}$ or $1, 1, -\sqrt{2}$.

Hence, direction ratios are $1, 1, \sqrt{2}$.

35. The equation of the line through the centre $\hat{j} + 2\hat{k}$ and normal to the given plane is

$$\vec{r} = \hat{j} + 2\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k}) \qquad \text{(i)}$$

This meets the plane for which

$$[\hat{j} + 2\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})] \cdot (\hat{i} + 2\hat{j} + 2\hat{k}) = 15$$

$\Rightarrow\ 6 + 9\lambda = 15 \Rightarrow \lambda = 1$

Putting in (i), we get

$$\vec{r} = \hat{j} + 2\hat{k} + (\hat{i} + 2\hat{j} + 2\hat{k}) = \hat{i} + 3\hat{j} + 4\hat{k}$$

Hence, the centre is (1, 3, 4).

36. The angle between the faces is equal to the angle between their normals.

$\therefore$ Vector perpendicular to the face OAB is

$$\overrightarrow{OA} \times \overrightarrow{OB} = 5\hat{i} - \hat{j} - 3\hat{k}$$

and vector perpendicular to the face ABC is

$$\overrightarrow{AB} \times \overrightarrow{AC} = \hat{i} - 5\hat{j} - 3\hat{k}$$

$\therefore$ Let θ be the angle between the faces OAB and ABC.

$$\therefore\ \cos\theta = \frac{(5\hat{i} - \hat{j} - 3\hat{k}).(\hat{i} - 5\hat{j} - 3\hat{k})}{\left|5\hat{i} - \hat{j} - 3\hat{k}\right|\left|\hat{i} - 5\hat{j} - 3\hat{k}\right|}$$

$\Rightarrow\ \cos\theta = \frac{19}{35}$

37. Consider Ox, Oy, Oz and Ox', Oy', Oz' as two systems of rectangular axes. Let their corresponding equations of plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \qquad \text{(i)}$$

and

$$\frac{x'}{a'} + \frac{y'}{b'} + \frac{z'}{c'} = 1 \qquad \text{(ii)}$$

Let the lengths of perpendiculars from the origin to Eqs. (i) and (ii) be the same.

$$\therefore \quad \frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}} = \frac{1}{\sqrt{\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}}}$$

$$\Rightarrow \quad \frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}-\frac{1}{a'^2}-\frac{1}{b'^2}-\frac{1}{c'^2}=0$$

38. If two lines are coplanar, then

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$\therefore \quad \begin{vmatrix} -1 & 1 & 1 \\ 1 & 1 & -k \\ k & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \quad -1(1+2k)-1(1+k^2)+1(2-k)=0$$

$$\Rightarrow \quad -k^2-3k=0$$

$$\Rightarrow \quad k=0 \text{ or } -3$$

39. The centre and radius of the the given sphere are

$C(-1, 1, 2)$ and $R=\sqrt{(-1)^2+(1)^2+(2)^2+19}=5$

Length of the perpendicular from the centre C on the plane

$$d=\frac{-1\times1+1\times2+2\times2+7}{\sqrt{1^2+2^2+2^2}}=4$$

$\therefore$ Radius of the circle $=\sqrt{R^2-d^2}=\sqrt{25-16}=3$

40. In order to determine the shortest distance between the plane and the sphere, we find the distance from the centre of the sphere to the plane – radius of the sphere

$\therefore$ Centre of the sphere is (–2, 1, 3).

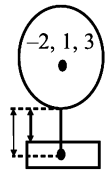


Fig. 24.45

$\therefore$ Required distance is

$$\frac{|-24+4+9-327|}{\sqrt{12^2+4^2+3^2}}-\sqrt{(2)^2+1^2+3^2+155}$$

$$=26-13=13 \text{ units}$$

41. If a line makes the angless α, β, γ with the x, y, z axes respectively, then

$$l^2+m^2+n^2=1$$

$$\Rightarrow \quad 2l^2+m^2=1 \text{ or } 2n^2+m^2=1$$

$$\Rightarrow \quad 2\cos^2\theta=1-\cos^2\beta(\alpha=\gamma=\theta)$$

$$2\cos^2\theta=\sin^2\beta$$

$$\Rightarrow \quad 2\cos^2\theta=3\sin^2\theta$$

$$\Rightarrow \quad 5\cos^2\theta=3$$

42. Distance between the given planes

$2x+y+2z-8=0$ and $2x+y+2z+\frac{5}{2}=0$ is

$$\frac{\left|\frac{5}{2}+8\right|}{\sqrt{4+1+4}}=\frac{21/2}{3}=\frac{7}{2}$$

43. The given spheres are

$$x^2+y^2+z^2+7x-2y-z-13=0 \qquad \text{(i)}$$

and $$x^2+y^2+z^2-3x+3y+4z-8=0 \qquad \text{(ii)}$$

Subtracting (ii) from (i), we get $10x-5y-5z-5=0$

$$\Rightarrow \quad 2x-y-z=1$$

44. The given lines $\frac{x-1}{1}=\frac{y-3}{-\lambda}=\frac{z-1}{\lambda}(=s)$ and $\frac{x}{1/2}=\frac{y-1}{1}=\frac{y-2}{-1}(=t)$ are coplanar if

$$\begin{vmatrix} 0+1 & 1-3 & 2-1 \\ 1 & -\lambda & \lambda \\ 1/2 & 1 & -1 \end{vmatrix} = 0$$

i.e. $$\begin{vmatrix} 1 & -2 & 1 \\ 1 & -\lambda & \lambda \\ 1/2 & 1 & -1 \end{vmatrix} = 0$$

if $$1(\lambda-\lambda)+2\left(-1-\frac{\lambda}{2}\right)+1\left(1+\frac{\lambda}{2}\right)=0$$

if $$-2-\lambda+1+\frac{\lambda}{2}=0 \text{ if } \frac{\lambda}{2}=-1$$

if $$\lambda=-2$$

45. The centre of the sphere is $\left(\frac{1}{2}, 0, -\frac{1}{2}\right)$ and

radius $=\sqrt{\frac{1}{4}+\frac{1}{4}+2}=\frac{\sqrt{10}}{2}$

Distance from the centre of the sphere to the given plane

$$=\left|\frac{\frac{1}{2}+\frac{1}{2}-4}{\sqrt{1+4+1}}\right|=\frac{3}{\sqrt{6}}$$

So, the radius of the circle $=\sqrt{\frac{10}{4}-\frac{9}{6}}=1$

46. The given lines can be rewritten as

$$\frac{x}{3}=\frac{y}{2}=\frac{z}{-6} \text{ and } \frac{x}{2}=\frac{y}{-12}=\frac{z}{-3}$$

$\therefore$ Angle between the lines is

$$\theta = \cos^{-1}\left(\frac{3\times 2+2(-12)-6(3)}{\sqrt{3^2+2^2+(-6)^2}\sqrt{(2)^2+(-12)^2+(-3)^2}}\right)$$

$$= 0$$

$\Rightarrow \quad \theta = 90°$

47. Centres of the given spheres are

$$C_1(-3, 4, 1) \text{ and } C_2(5, -2, 1)$$

Mid-point of $[C_1C_2]$ is $(1, 1, 1)$

The plane $2ax - 3ay + 4az + 6 = 0$ passes through $(1, 1, 1)$

$\Rightarrow \quad 2a(1) - 3a(1) + 4a(1) + 6 = 0$

$\Rightarrow \quad 3a = -6 \quad \Rightarrow \quad a = -2$

48. The given lines can be rewritten as

$$\frac{x-b}{a} = \frac{y-0}{1} = \frac{z-d}{c} \text{ and } \frac{x-b'}{a'} = \frac{y-0}{1} = \frac{z-d'}{c'}$$

These lines will be perpendicular if

$$aa' + 1 + cc' = 0.$$

49. Let (α, β, γ) be the image.

Then $\frac{\alpha-1}{2} - 2\left(\frac{\beta+3}{2}\right) = 0$

$$\Rightarrow \quad \alpha - 1 - 2\beta - 6 = 0 \Rightarrow \alpha - 3\beta = 7 \qquad \text{(i)}$$

and $$\frac{\alpha+1}{1} = \frac{\beta-3}{-2} = \frac{\gamma-4}{0} \qquad \text{(ii)}$$

Solving (i) and (ii), we get $\alpha = \frac{9}{5}, \beta = -\frac{13}{5}, \gamma = 4$

$\therefore$ Image is $\left(\frac{9}{5}, -\frac{13}{5}, 4\right)$.

50. Here $l = \cos\frac{\pi}{4}, m' = \cos\frac{\pi}{4}$

Let the line make an angle γ with the z-axis.

$\therefore \quad l^2 + m^2 + n^2 = 1$

$$\Rightarrow \quad \cos^2\frac{\pi}{4} + \cos^2\frac{\pi}{4} + \cos^2\gamma = 1$$

$$\Rightarrow \quad \frac{1}{2} + \frac{1}{2} + \cos^2\gamma = 1$$

$$\Rightarrow \quad \cos^2\gamma = 0 \quad \Rightarrow \quad \cos\gamma = 0 \quad \Rightarrow \quad \gamma = \frac{\pi}{2}$$

51. Let the direction cosines of the line L be l, m, n. Since the line is the line of intersection of the given planes, then the normal to the planes are perpendicular to the line L.

$\therefore \qquad 2l + 3m + n = 0 \qquad \text{(i)}$

and $\qquad l + 3m + 2n = 0 \qquad \text{(ii)}$

From Eqs. (i) and (ii), we get

$$\frac{l}{3} = \frac{m}{-3} = \frac{n}{3} = k \qquad \text{[say]}$$

We know $l^2 + m^2 + n^2 = 1$

$\therefore \quad (3k)^2 + (-3k)^2 + (3k)^2 = 1$

$$\Rightarrow \quad k = \frac{1}{3\sqrt{3}}$$

$$\therefore \quad l = \frac{1}{\sqrt{3}} \Rightarrow \cos\alpha = \frac{1}{\sqrt{3}}$$

52. Centre of a given sphere is $(3, 6, 1)$. Since one end of the diameter is $(2, 3, 5)$, let the other end of the diameter be (α, β, γ), then

$$\frac{\alpha+2}{2} = 3, \frac{\beta+3}{2} = 6, \frac{\gamma+5}{2} = 1$$

$\Rightarrow \quad \alpha = 4, \beta = 9$ and $\gamma = -3$

Hence, the coordinates of the other point are $(4, 9, -3)$.

53. Equation of the line passing through $(5, 1, \alpha)$ and $(3, \beta, 1)$ is

$$\frac{x-3}{5-3} = \frac{y-b}{1-b} = \frac{z-1}{a-1} \qquad \text{(i)}$$

Also, point $\left(0, \frac{17}{2}, -\frac{13}{2}\right)$ satisfies Eq. (i). So we get

$$-\frac{3}{2} = \frac{\frac{17}{2}-b}{1-b} = \frac{-\frac{13}{2}-1}{a-1}$$

From the 1st and 3rd terms, $a - 1 = \frac{\left(-\frac{15}{2}\right)}{\left(-\frac{3}{2}\right)} \Rightarrow a = 6$

From the 1st and 2nd terms, $-3(1-b) = 2\left(\frac{17}{2} - b\right) \Rightarrow b = 4$

54. Since two lines intersect at a point, so the shortest distance between them is zero.

$$\therefore \quad \begin{vmatrix} k & 2 & 3 \\ 3 & k & 2 \\ 1 & 1 & -2 \end{vmatrix} = 0$$

$$\left[\because \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \end{vmatrix} = 0\right]$$

$$\Rightarrow \quad k(-2k-2) - 2(-6-2) + 3(3-k) = 0$$

$$\Rightarrow \quad 2k^2 + 5k - 25 = 0$$

$\Rightarrow (2k-5)(k+5)=0$

$\Rightarrow k=\frac{5}{2},-5$

Hence, the integer value of k is -5.

55. Direction ratios of the given line are $(3,-5,2)$.
Direction ratios of the normal to the plane are $(1,3,-\alpha)$.
$\therefore$ The line is perpendicular to the normal.

$\Rightarrow 3(1)-5(3)+2(-\alpha)=0$

$\Rightarrow 3-15-2\alpha=0$

$\Rightarrow 2\alpha=-12$

$\Rightarrow \alpha=-6$

Also, point $(2, 1, -2)$ lies on the plane. So, $2 + 3 + 6(-2) + \beta = 0$

$\Rightarrow \beta=7$

$\therefore (\alpha,\beta)=(-6,7)$

56. $\cos^2 45^\circ+\cos^2 120^\circ+\cos^2\theta=1$

$\Rightarrow \frac{1}{2}+\frac{1}{4}+\cos^2\theta=1$

$\Rightarrow \cos^2\theta=1-\frac{3}{4}=\frac{1}{4}$

$\Rightarrow \cos\theta=\frac{1}{2}$ ($\because \theta$ is acute)

$\Rightarrow \theta=60^\circ$

57. The image of the point $(3, 1, 6)$ with respect to the plane $x-y+z=5$ is

$$\frac{x-3}{1}=\frac{y-1}{-1}=\frac{z-6}{1}=\frac{-2(3-1+6-5)}{1+1+1}=-2$$

$\Rightarrow x=3-2=1$

$y=1+2=3$

$z=6-2=4$

which shows that Statement I is true.
We observe that the line segment joining the points $A(3, 1, 6)$ and $B(1, 3, 4)$ is bisected by the plane $x - y + z = 5$. Hence, Statement II is true.
Thus, Statements I and II are true and Statement II is a correct explanation of Statement I.

58. The line through $P(1, -5, 9)$ parallel to $x = y = z$ is

$$\frac{x-1}{1}=\frac{y+5}{1}=\frac{z-9}{1}=\lambda$$

$Q(x=1+\lambda, y=-5+\lambda, z=9+\lambda)$

The given plane is $x - y + z = 5$.

$\therefore 1+\lambda+5-\lambda+9+\lambda=5$

$\Rightarrow \lambda=-10$

$\therefore Q(-9,-15,-1)$

$\therefore PQ=\sqrt{(1+9)^2+(15-5)^2+(9+1)^2}=\sqrt{300}=10\sqrt{3}$.

59. Let the feet of the perpendicular be $(2\alpha, 3\alpha+2, 4\alpha+3)$

$\Rightarrow D'$ ratio of the $\perp$ line $<2\alpha-3, 3\alpha+3, 4\alpha-8>$
and D' ratio of the line $< 2, 3, 4 >$

$\Rightarrow 2(2\alpha-3)+3(3\alpha+3)+4(4\alpha-8)=0$

$\Rightarrow 29\alpha-29=0 \Rightarrow \alpha=1$

$\Rightarrow$ feet of $\perp$ is $(2, 5, 7)$

$\Rightarrow$ length $\perp$ is $\sqrt{1^2+6^2+4^2}=\sqrt{53}$

60. Equation of a plane parallel to $x - 2y + 2z - 5 = 0$ is

$$x-2y+2z+d=0 \qquad \text{(i)}$$

Perpendicular distance from $O(0, 0, 0)$ to (i) is 1

$$\frac{|d|}{\sqrt{1+4+4}}=1 \Rightarrow |d|=3 \Rightarrow d=\pm 3$$

$\therefore x-2y+2z-3=0$

61. $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z}{1}$

$$\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$$

$\vec{a}(1,-1,1)$; $\vec{r}=\vec{a}+\lambda\vec{b}$

$\vec{b}(2,3,4)$

$\vec{c}(3,k,0)$; $\vec{r}=\vec{c}+\mu\vec{d}$

$\vec{d}(1,2,1)$

These lines will intersect if lines are coplaner.
$\vec{a}-\vec{c}, \vec{b}$ and $\vec{d}$ are coplaner.

$\therefore [\vec{a}-\vec{c},\vec{b},\vec{d}]=0$

$$\begin{vmatrix} 2 & k+1 & -1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix}=0$$

$\Rightarrow 2(-5)-(k+1)(-2)-1(1)=0$

$\Rightarrow 2(k+1)=11$

$\Rightarrow k=9/2$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. A plane mirror is placed at the origin so that the direction ratios of its normal are 1, –1, 1. A ray of light, coming along the positive direction of the x-axis, strikes the mirror. The direction cosines of the reflected ray are

(a) $\frac{1}{3},\frac{2}{3},\frac{2}{3}$ (b) $-\frac{1}{3},\frac{2}{3},\frac{2}{3}$

(c) $-\frac{1}{3},-\frac{2}{3},-\frac{2}{3}$ (d) $-\frac{1}{3},-\frac{2}{3},\frac{2}{3}$

2. A plane meets the coordinate axes at A, B, C and (α, β, γ) is the centroid of the triangle ABC. Then the equation of the plane is

(a) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$ (b) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=1$

(c) $\frac{3x}{\alpha}+\frac{3y}{\beta}+\frac{3z}{\gamma}=1$ (d) $\alpha x+\beta y+\gamma z=1$

3. The equations of the straight line through the origin and parallel to the line $(b+c)x+(c+a)y+(a+b)z = k=(b-c)x+(c-a)y+(a-b)z$ are

(a) $\frac{x}{b^2-c^2}=\frac{y}{c^2-a^2}=\frac{z}{a^2-b^2}$

(b) $\frac{x}{b}=\frac{y}{c}=\frac{z}{a}$

(c) $\frac{x}{a^2-bc}=\frac{y}{b^2-ca}=\frac{z}{c^2-ab}$

(d) none of these

4. A point moves so that the sum of the squares of its distances from the six faces of a cube given by $x=\pm1$, $y=\pm1$, $z=\pm1$ is 10 units. The locus of the point is

(a) $x^2+y^2+z^2=1$ (b) $x^2+y^2+z^2=2$

(c) $x+y+z=1$ (d) $x+y+z=2$

5. Which one of the following is the best condition for the plane $ax+by+cz+d=0$ to intersect the x-axis and the y-axis at equal angles?

(a) $|a|=|b|$ (b) $a=-b$

(c) $a=b$ (d) $a^2+b^2=1$

6. The equations of a straight line parallel to the x-axis are

(a) $\frac{x-a}{1}=\frac{y-b}{1}=\frac{z-c}{1}$ (b) $\frac{x-a}{0}=\frac{y-b}{1}=\frac{z-c}{1}$

(c) $\frac{x-a}{0}=\frac{y-b}{0}=\frac{z-c}{1}$ (d) $\frac{x-a}{1}=\frac{y-b}{0}=\frac{z-c}{0}$

7. The non-zero value of a for which the lines $2x-y+3z+4=0=ax+y-z+2$ and $x-3y+z=0=x+2y+z+1$ are coplaner is

(a) –2 (b) 4 (c) 6 (d) 0

8. A perpendicular is drawn from the point (0, 3, 4) to the plane $2x-2y+z=10$. The coordinates of the foot of the perpendicular are

(a) $\left(-\frac{8}{3},\frac{1}{3},\frac{16}{3}\right)$ (b) $\left(\frac{8}{3},\frac{1}{3},\frac{16}{3}\right)$

(c) $\left(\frac{8}{3},-\frac{1}{3},\frac{16}{3}\right)$ (d) $\left(\frac{8}{3},\frac{1}{3},-\frac{16}{3}\right)$

9. If the lines $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$, $\frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4}$ and $\frac{x+k}{3}=\frac{y-1}{2}=\frac{z-2}{h}$ are concurrent, then

(a) $h=-2, k=-6$ (b) $h=1/2, k=2$

(c) $h=6, k=2$ (d) $h=2, k=1/2$

10. In symmetric form, the equations of the lines $x+y-z=1$ and $2x-3y+z=2$ are

(a) $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ (b) $\frac{x}{2}=\frac{y}{3}=\frac{z-1}{5}$

(c) $\frac{x}{2}=\frac{y-1}{3}=\frac{z}{5}$ (d) $\frac{x-1}{2}=\frac{y}{3}=\frac{z}{5}$

11. Let the points $A(a, b, c)$ and $B(a', b', c')$ be at distance r and r' from the origin. The line AB passes through the origin when

(a) $\frac{a'}{a}=\frac{b'}{b}=\frac{c'}{c}$

(b) $aa'+bb'+cc'=rr'$

(c) $aa'+bb'+cc'=r^2+r'^2$

(d) none of these

12. The condition that the line $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ lies in the plane $ax+by+cz+d=0$ is

(a) $ax_1+by_1+cz_1+d=0$ and $al+bm+cn\neq0$

(b) $al+bm+cn=0$ and $ax_1+by_1+cz_1+d\neq0$

(c) $ax_1+by_1+cz_1+d=0$ and $al+bm+cn=0$

(d) $ax_1+by_1+cz_1=0$ and $al+bm+cn=0$

13. In the adjacent figure, P is any arbitary interior point of the triangle ABC such that the lines AA_1, BB_1 and CC_1 are concurrent at P. The value of $\frac{PA_1}{AA_1}+\frac{PB_1}{BB_1}+\frac{PC_1}{CC_1}$ is always equal to

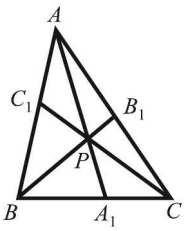


Fig. 24.46

(a) 1 (b) 2

(c) 3 (d) none of these

14. The locus of a point which is equidistant from the x-axis and the y-axis is
(a) the plane $x = y$
(b) the plane $x = -y$
(c) the pair of planes $x = \pm y$
(d) none of these

15. If the edges of a rectangular parallelopiped are 3, 2, 1, then the angle between a pair of diagonals is given by
(a) $\cos^{-1}(6/7)$ (b) $\cos^{-1}(3/7)$
(c) $\cos^{-1}(2/7)$ (d) none of these

16. If (p_1, q_1, r_1) is the image of (p, q, r) in the plane $ax + by + cz + d = 0$, then
(a) $\dfrac{p_1 - p}{a} = \dfrac{q_1 - q}{b} = \dfrac{r_1 - r}{c}$
(b) $a(p + p_1) + b(q + q_1) + c(r + r_1) + 2d = 0$
(c) both (a) and (b)
(d) none of these

17. The direction cosines of the lines bisecting the angle between the lines whose dircetion cosines are l_1, m_1, n_1 and l_2, m_2, n_2 and the angle between these lines is θ, are
(a) $\dfrac{l_1 + l_2}{\cos(\theta/2)}, \dfrac{m_1 + m_2}{\cos(\theta/2)}, \dfrac{n_1 + n_2}{\cos(\theta/2)}$
(b) $\dfrac{l_1 + l_2}{2\cos(\theta/2)}, \dfrac{m_1 + m_2}{2\cos(\theta/2)}, \dfrac{n_1 + n_2}{2\cos(\theta/2)}$
(c) $\dfrac{l_1 + l_2}{\sin(\theta/2)}, \dfrac{m_1 + m_2}{\sin(\theta/2)}, \dfrac{n_1 + n_2}{\sin(\theta/2)}$
(d) $\dfrac{l_1 - l_2}{2\sin(\theta/2)}, \dfrac{m_1 - m_2}{2\sin(\theta/2)}, \dfrac{n_1 - n_2}{2\sin(\theta/2)}$

18. The mirror images of the point (3, 4, 5) in the coordinate planes are A, B and C. The centroid of the triangle ABC is
(a) (0, 0, 0) (b) (–3, –4, –5)
(c) [(3/3), (4/3), (5/3)] (d) (1, 1, 1)

19. A variable plane at a distance of 1 unit from the origin cuts the coordinate axes at A, B, and C. If the centroid $D(x, y, z)$ of the triangle ABC satisfies the relation $\dfrac{1}{x^2} + \dfrac{1}{y^2} + \dfrac{1}{z^2} = k$, then the value of k is
(a) 9 (b) 3 (c) 1 (d) 1/3

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. Which of the following does not represent a straight line?
(a) $ax + by + cz + d = 0, a'x + by + cz + d = 0\ (a \neq a')$
(b) $ax + by + cz + d = 0, ax + b'y + cz + d = 0\ (b \neq b')$
(c) $ax + by + cz + d = 0, ax + by + c'z + d = 0\ (c \neq c')$
(d) $ax + by + cz + d = 0, ax + by + cz + d' = 0\ (d \neq d')$

2. For every point $P(x, y, z)$ on the xy-plane,
(a) $x = 0$ (b) $y = 0$
(c) $z = 0$ (d) none of these

3. A plane makes intercepts OA, OB, OC whose measurements are a, b, c on the axes Ox, Oy, Oz. The area of the ΔABC is
(a) $\dfrac{1}{2}(ab + bc + ca)$ (b) $\dfrac{1}{2}(a^2b^2 + b^2c^2 + c^2a^2)^{1/2}$
(c) $\dfrac{1}{2}abc(a + b + c)$ (d) $\dfrac{1}{2}(a + b + c)(ab + bc + ca)$

4. The foot of the perpendicular drawn from a point with position vector $\hat{i} - 4\hat{k}$ on the line joining the points having position vectors as $-11\hat{j} + 3\hat{k}$ and $2\hat{i} - 3\hat{j} + \hat{k}$ has the position vector
(a) $4\hat{i} + 5\hat{j} + 5\hat{k}$ (b) $4\hat{i} + 5\hat{j} - 5\hat{k}$
(c) $5\hat{i} + 4\hat{j} - 5\hat{k}$ (d) $4\hat{i} - 5\hat{j} + 5\hat{k}$

5. The straight lines, whose direction cosines are l_i, m_i, n_i which are the roots of $al + bm + cn = 0$, $fl^2 + gm^2 + hn^2 = 0$, are parallel if
(a) $\dfrac{f}{a} + \dfrac{g}{b} + \dfrac{h}{c} = 0$ (b) $\dfrac{a}{f} + \dfrac{b}{g} + \dfrac{c}{h} = 0$
(c) $\dfrac{f}{a^2} + \dfrac{g}{b^2} + \dfrac{h}{c^2} = 0$ (d) $\dfrac{a^2}{f} + \dfrac{b^2}{g} + \dfrac{c^2}{h} = 0$

6. The vector equation of the plane passing through the origin and the line of intersection of the plane $\vec{r} \cdot \vec{a} = \lambda$ and $\vec{r} \cdot \vec{b} = \mu$ is
(a) $\vec{r} \cdot (\lambda\vec{a} - \mu\vec{b}) = 0$ (b) $\vec{r} \cdot (\lambda\vec{b} - \mu\vec{a}) = 0$
(c) $\vec{r} \cdot (\lambda\vec{a} + \mu\vec{b}) = 0$ (d) $\vec{r} \cdot (\lambda\vec{b} + \mu\vec{a}) = 0$

7. The distance of the point (–2, –5, 7) from the point of intersection of the line $\dfrac{x}{2} = \dfrac{y-2}{3} = \dfrac{z-3}{4}$ and the plane $2x + y - z = 2$ is
(a) $4\sqrt{29}$ (b) $\sqrt{78}$
(c) $2\sqrt{29}$ (d) $2\sqrt{78}$

8. The one end of a diameter of the sphere $x^2 + y^2 + z^2 - 2x - 2y - 2z + 2 = 0$ is
(a) $\left(1 - \dfrac{1}{\sqrt{2}}, \dfrac{1}{2}, \dfrac{1}{2}\right)$ (b) $\left(-1 - \dfrac{1}{\sqrt{2}}, -\dfrac{3}{2}, -\dfrac{3}{2}\right)$
(c) $\left(1 - \dfrac{1}{\sqrt{2}}, -\dfrac{1}{2}, -\dfrac{1}{2}\right)$ (d) none of these

9. The length of the perpendicular from the origin to the plane passing through the point $\vec{a}$ and containing the line $\vec{r} = \vec{b} + \lambda\vec{c}$ is

(a) $\dfrac{[\vec{a}\vec{b}\vec{c}]}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}$ (b) $\dfrac{[\vec{a}\vec{b}\vec{c}]}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}|}$

(c) $\dfrac{[\vec{a}\vec{b}\vec{c}]}{|\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}$ (d) $\dfrac{[\vec{a}\vec{b}\vec{c}]}{|\vec{c}\times\vec{a}+\vec{a}\times\vec{b}|}$

10. The image of the point (1, 3, 4) in the plane $x + 2y - z + 3 = 0$ is

(a) (1, 1, –6) (b) (–1, –1, 6)
(c) (–1, 1, –6) (d) none of these

11. The reflection of the plane $2x - 3y + 4z - 3 = 0$ in the plane $x - y + z - 3 = 0$ is the plane

(a) $4x - 3y + 2z - 15 = 0$ (b) $x - 3y + 2z - 15 = 0$
(c) $4x + 3y - 2z + 15 = 0$ (d) none of these

12. The radius of the circle $x^2 + y^2 + z^2 = 25$, $x + y + z - 3\sqrt{3} = 0$, is

(a) 5 (b) 3
(c) 4 (d) none of these

13. The ratio of the distances from (–1, 1, –3) and (3, 2, 1) to the plane $2x + 5y - 7z + 9 = 0$ is

(a) 2 : 1 (b) 1 : 3
(c) 1 : 1 (d) 1 : 2

14. The foot of the perpendicular of the point $P(1, 2, 3)$ on the line $\vec{r} = \vec{a} + \lambda\vec{b} = (6\hat{i} + 7\hat{j} + 7\hat{k}) + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$ is

(a) $3\hat{i} - 5\hat{j} + 9\hat{k}$ (b) $3\hat{i} + 5\hat{j} - 9\hat{k}$
(c) $3\hat{i} + 5\hat{j} + 9\hat{k}$ (d) none of these

15. Chord CD is a diameter of the sphere $|\vec{r} - 2\hat{i} - \hat{j} + 6\hat{k}| = \sqrt{18}$ with the coordinate A as (3, 2, –2), then the co-ordinates of B equal

(a) (1, 0, 10) (b) (–1, 0, –10)
(c) (–3, –2, –2) (d) (1, 0, –10)

16. If a sphere of constant radius k passes through $O(0, 0, 0)$ and meets the axes in A, B, C and the centroid of ΔABC lies on $x^2 + y^2 + z^2 = \lambda k^2$, then the value of λ equals

(a) $\dfrac{9}{4}$ (b) $\dfrac{4}{9}$
(c) $\dfrac{2}{3}$ (d) $\dfrac{3}{2}$

17. The S.D. between the lines
$\vec{r} = (5\hat{i} + 7\hat{j} + 3\hat{k}) + \lambda(5\hat{i} - 16\hat{j} + 7\hat{k})$ and
$\vec{r} = (9\hat{i} + 13\hat{j} + 15\hat{k}) + \lambda(3\hat{i} + 8\hat{j} - 5\hat{k})$ is

(a) 8 units (b) 12 units
(c) 10 units (d) none of these

18. The equation of a line passing through (–1, 2, 3) and perpendicular to the plane $2x + 3y + z + 5 = 0$ is

(a) $\dfrac{x-1}{-1} = \dfrac{y+2}{1} = \dfrac{z-3}{-1}$ (b) $\dfrac{x+1}{-1} = \dfrac{y-2}{1} = \dfrac{z+3}{1}$

(c) $\dfrac{x+1}{2} = \dfrac{y-2}{3} = \dfrac{z+3}{1}$ (d) none of these

19. The equation $|\vec{r}|^2 - \vec{r}\cdot(2\hat{i} + 4\hat{j} - 2\hat{k}) - 10 = 0$ represents a

(a) circle (b) plane
(c) sphere of radius 4 (d) none of these

20. A mirror and a source of light are situated at the origin O and at a point on Ox respectively. A ray of light from the source strikes the mirror and is reflected. If the direction ratios of the normal to the plane are 1, –1, 1, then the direction ratios of the reflected ray are

(a) $\dfrac{1}{3}, \dfrac{2}{3}, \dfrac{2}{3}$ (b) $-\dfrac{1}{3}, \dfrac{2}{3}, \dfrac{2}{3}$

(c) $-\dfrac{1}{3}, -\dfrac{2}{3}, -\dfrac{2}{3}$ (d) $-\dfrac{1}{3}, -\dfrac{2}{3}, \dfrac{2}{3}$

ANSWERS

Level 1

1. (d) **2.** (a) **3.** (c) **4.** (b) **5.** (a) **6.** (d) **7.** (a) **8.** (b) **9.** (d) **10.** (d)
11. (a) **12.** (c) **13.** (a) **14.** (c) **15.** (a), (b) and (c) **16.** (c) **17.** (b) and (d) **18.** (c)
19. (a)

Level 2

1. (d) **2.** (c) **3.** (b) **4.** (b) **5.** (d) **6.** (b) **7.** (c) **8.** (a) **9.** (c) **10.** (b)
11. (a) **12.** (c) **13.** (d) **14.** (c) **15.** (d) **16.** (b) **17.** (d) **18.** (c) **19.** (b) **20.** (d)

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KARL PEARSON

Statistics

- Measures of Central Tendency
- Arithmetic Mean
- Median
- Quartiles, Deciles and Percentiles
- Mode
- Geometric Mean
- Harmonic Mean
- Dispersion
- Standard Deviation
- Combined Standard Deviation
- More Properties

MEASURES OF CENTRAL TENDENCY

Generally, average of a distribution is a measure of the middle or expected value of the distribution. Such values are known as **measures of central tendency**. These are as follows:

(a) Mathematical Average such as (i) Arithmetic Mean, (ii) Geometric Mean and (iii) Harmonic Mean

(b) Positional Average such as (i) Median, (ii) Mode, (iii) Quartiles, (iv) Deciles and (v) Percentiles

ARITHEMETIC MEAN

Individual Observations. $\bar{x} = \dfrac{\Sigma x}{N}$, Σx = sum of items, N = number of observations

Short-cut Method. $\bar{x} = A + \dfrac{\Sigma d}{N}$, $d = x - A$

Discrete Series. $\bar{x} = \dfrac{\Sigma f(x)}{N}$, f = frequency, N = total frequency.

Short-cut Method. $\bar{x} = A + \dfrac{\Sigma fd}{N}$, $d = x - A$, N = total frequency.

Continuous Series. *Direct Method:*

$\bar{x} = \dfrac{\Sigma fm}{N}$, m = mid-values of various classes, N = total frequency.

Short-cut Method. $\bar{x} = A + \dfrac{\Sigma fd}{N}$, A = assumed mean $d = m - A$, N = Total frequency.

Step Deviation Method. $\bar{x} = A + \dfrac{\Sigma fd'}{N} \times c$, A = assumed mean, $d' = \dfrac{m - A}{c}$, c = class size, m = class marks.

Combined Mean. If a series of N observations consists of two components with means $\bar{x}_1$ and $\bar{x}_2$ and number of items N_1 and N_2, then the combined mean is:

$$\bar{x}_{12} = \frac{N_1\bar{x}_1 + N_2\bar{x}_2}{N_1 + N_2}$$

MEDIAN

Firstly, we arrange the data in increasing or decreasing order.

Individual Observation. If N is odd, then Med = size of $[(N + 1)/2]$th item.

If N is even, Med = average of $(N/2)$th and $[(N/2) + 1]$th items.

Discrete Series. Firstly, arrange the data in ascending or descending order, find cumulative frequency, then

$$\text{Med} = \text{size of } \frac{N+1}{2} \text{ th item}$$

Continuous Series. Firstly, find the median class by using Med = size of $\dfrac{N+1}{2}$ th item, then apply

$$\text{Med} = l + \frac{(N/2) - c.f.}{f} \times i,$$

where

l = lower limit of the median class

$c.f.$ = cumulative frequency of the class preceding the median class

f = frequency of the median class

i = class interval of the median class

QUARTILES, DECILES AND PERCENTILES

The technique for determining quartiles, deciles and percentiles is the same as that of median.

In the case of individual observation and discrete series:

$$Q_2 = \text{size of } \frac{N+1}{4} \text{ th item}$$

$$Q_3 = \text{size of } \frac{3(N+1)}{4} \text{ th item}$$

$$D_1 = \text{size of } \frac{N+1}{10} \text{ th item}$$

$$D_7 = \text{size of } \frac{7(N+1)}{10} \text{ th item}$$

$$P_1 = \text{size of } \frac{N+1}{100} \text{ th item}$$

$$P_{89} = \text{size of } \frac{89(N+1)}{100} \text{ th item}$$

In the case of continuous series:

Firstly, find the class in which quartiles, deciles or percentiles lie by using

$$Q_1 = \text{size of } \frac{N}{4} \text{ th item}$$

$$Q_3 = \text{size of } \frac{3N}{4} \text{ th item}$$

$$D_1 = \text{size of } \frac{N}{10} \text{ th item}$$

$$D_7 = \text{size of } \frac{7N}{10} \text{ th item}$$

$$P_1 = \text{size of } \frac{N}{100} \text{ th item}$$

$$P_{89} = \text{size of } \frac{89N}{100} \text{ th item}$$

Then applying the formula for measuring the magnitude and interpolating the measurement, we get

$$Q_1 = l + \frac{(N/4) - c.f.}{f} \times i, \quad Q_3 = l + \frac{(3N/4) - c.f.}{f} \times i;$$

$$D_1 = l + \frac{(N/10) - c.f.}{f} \times i, \quad D_7 = l + \frac{(7N/10) - c.f.}{f} \times i;$$

$$P_1 = l + \frac{(N/100) - c.f.}{f} \times i, \quad P_{89} = l + \frac{(89N/100) - c.f.}{f} \times i$$

where

l = lower limit of the class in which Q_1, Q_3, D_1, D_7, P_1, P_{89} lie

$c.f.$ = cumulative frequency below the class in which these lie

f = frequency of these classes

i = class interval

MODE

It is that value of the variable which occurs itself the greatest number of times. Firstly, determine the modal class by grouping method or by inspection. Then we calculate mode by applying the formula

$$\text{Mode} = l_1 + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i$$

where

i = class interval of the modal class

l_1 = lower limit of the modal class

Δ_1 = difference between the frequency of the modal class the frequency of the pre-modal class (ignoring signs)

Δ_2 = difference between the frequency of the modal class and the frequency of the post-modal class (ignoring sings)

$$\therefore \quad M = l_1 + \frac{|f_1 - f_0|}{|f_1 - f_0| + |f_1 - f_2|} \times i$$

When mode is ill defined,

Mode = 3 Median – 2 Mean

GEOMETRIC MEAN

$$\text{G.M.} = (x_1, x_2, \ldots, x_n)^{1/n}$$

$$= \text{antilog}\left(\frac{\Sigma \log x}{N}\right)$$

Discrete Series: $\text{G.M.} = \text{antilog}\left(\frac{\Sigma f \log x}{N}\right)$

Continuous Series: $\text{G.M.} = \text{antilog}\ \frac{\Sigma f \log m}{N}$, where m is the mid-value of various classes, N = total frequency.

HARMONIC MEAN

$$\text{H.M.} = \frac{N}{\Sigma \frac{1}{x_i}} = \frac{N}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}, \quad x_i \neq 0$$

Discrete Series: $\text{H.M.} = \frac{N}{\Sigma (f/\text{m})}$, N = total frequency.

Continuous Series: $\text{H.M.} = \frac{N}{\Sigma (f/\text{m})}$, where m is the mid-point of various classes, N = Total frequency.

DISPERSION

Dispersion may be defined as the extent of the scatteredness of items around a measure of central tendency.

Methods of Measuring Dispersion

The following are the methods of measuring dispersion:

(i) Range

(ii) Semi-interquartile Range or Quartile Deviation

(iii) Mean Deviation

(iv) Standard Deviation

(i) *Range*

It is the difference between the highest and the lowest values in the series, i.e. Range = $x_h - x_l$

where x_h is the highest value and x_l is the lowest value.

(ii) *Quartile Deviation*

$$\text{Q.D.} = \frac{1}{2}(Q_3 - Q_1)$$

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

(iii) *Mean Deviation*

It is the mean of the absolute value of the deviations from some measure of central tendency.

$\text{M.D.} = \frac{\Sigma |D|}{N}$, where $|D|$ stands for the deviations from mean or median.

$\text{Coefficient of M.D.} = \frac{\text{Mean Deviation}}{\text{Mean}}$ (when deviations are taken from the mean)

$\text{Coefficient of M.D.} = \frac{\text{Mean Deviation}}{\text{Median}}$ (when deviations are taken from the median)

Discrete Series: $\text{M.D.} = \frac{\Sigma f |D|}{N}$, where $|D|$ denotes the deviations of the items from the median (or mean) ignoring signs.

Continuous Series: $\text{M.D.} = \frac{\Sigma f |D|}{N}$, where $|D|$ denotes the deviations of the mid-values of the classes from the mean (or median) ignoring signs.

STANDARD DEVIATION

The arithmetic mean of the square of deviations of the variable value from its actual arithmetic mean is known as **variance** and its square root is known as **standard deviation**.

Individual Observation. *Direct Method I:*

$$\sigma = \sqrt{\frac{\Sigma x^2}{N}}, \quad x = x - \bar{x}$$

Direct Method II: $\sigma = \sqrt{\frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2}$

Discrete Series. *Direct Method I:*

$$\sigma = \sqrt{\frac{\Sigma f x^2}{N}}, \quad x = x - \bar{x}$$

Direct Method II: $\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$

Short-cut Method: $\sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2}$

where

d = deviations of items from assumed mean

$N = \Sigma f$

Step Deviation Method.

$$\sigma = \sqrt{\frac{\Sigma fd'^2}{N} - \left(\frac{\Sigma fd'}{N}\right)^2} \times C$$

where $d' = \frac{X - A}{C}$, i.e. deviations of the items from the assumed mean divided by the common factor C.

Continuous Series. $\sigma = \sqrt{\frac{\Sigma fd'^2}{N} - \left(\frac{\Sigma fd'}{N}\right)^2}$, $d' = \frac{m - A}{C}$

Variance = σ^2

Coefficient of variation $= \frac{\sigma}{\text{Mean}} \times 100$

☞ COMBINED STANDARD DEVIATION

If there are two sets of observations containing n_1 and n_2 items with respective mean $\bar{x}_1$ and $\bar{x}_2$ and standards deviations σ_1 and σ_2, then the mean $\bar{x}$ and the standard deviations of the $n_1 + n_2$ observations, taken together, are

$$\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

$$\sigma^2 = \frac{1}{n_1 + n_2}[n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)]$$

$$d_1 = \bar{x}_{12} - \bar{x}_1,\ d_2 = \bar{x}_{12} - \bar{x}_2$$

☞ MORE PROPERTIES

(1) For a symmetrical distribution

Mean ± 0.6745 covers 50% of items
Mean ± 1σ covers 68.27% of items
Mean ± 2σ covers 95.45% of items
Mean ± 3σ covers 99.93% of items

(2) The sum of squares of deviations of items in the series from their arithmetic mean is minimum.

(3) $\text{Q.D.} = \frac{2}{3}\sigma$

$\text{M.D.} = \frac{4}{5}\sigma$, $4\sigma = 5\text{M.D.} = 6\text{Q.D.}$

$\text{Q.D.} = \frac{5}{6}\text{M.D.}$

SOLVED QUESTIONS

1. The mean of the series $1, 2, 4, 8, 16, \ldots, 2^n$ is

(a) $\frac{2^n - 1}{n}$ (b) $\frac{2^n - 1}{n+1}$

(c) $\frac{2^{n+1} - 1}{n+1}$ (d) $\frac{2^n + 1}{n}$

Ans. (c)

Solution: Given: series $1, 2, 4, 8, 16, \ldots, 2^n$.

We know that the mean of the given series

$$= \frac{1 + 2 + 4 + 8 + 16 + \cdots + 2^n}{n+1}$$

$$= \frac{1 + 2^1 + 2^2 + 2^3 + \cdots + 2^n}{n+1}$$

$$= \frac{2^{n+1} - 1}{(n+1)(2-1)} = \frac{2^{n+1} - 1}{n+1}$$

2. The average of n numbers $x_1, x_2, x_3, \ldots, x_n$ is M. If x_1 is replaced by x, then the new average is

(a) $M - x_1 + x$ (b) $\frac{(n-1)M - x_1 + x}{n}$

(c) $\frac{(n-1)M - x_1 + x}{n}$ (d) $\frac{nM - x_1 + x}{n}$

Ans. (d)

Solution: Given: Numbers $x_1, x_2, x_3, \ldots, x_n$ and their average = M. We know that the average of n numbers

$$M = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

or $$M + \frac{x}{n} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

or $$\frac{nM + x}{n} = \frac{x_1 + x_2 + x_3 + \cdots + x_n + x}{n}$$

or $$\frac{nM + x}{n} = \frac{x_1}{n} + \frac{x_2 + x_3 + \cdots + x_n + x}{n}$$

or $\dfrac{nM + x - x_1}{n} = \dfrac{x_2 + x_3 + \cdots + x_n + x}{n}$

or new average

$$= \left(\frac{x_2 + x_3 + \cdots + x_n + x}{n} \right) = \frac{nM - x_1 + x}{n}$$

3. The arithmetic mean for the following frequency distribution:

Class	0–8	8–16	16–24	24–32	32–40	40–48
Frequency	8	7	16	24	15	7

is

(a) 24.40 (b) 25.40

(c) 26.20 (d) 28.20

Ans. (b)

Solution: Given data

Class	0–8	8–16	16–24	24–32	32–40	40–48
Frequency	8	7	16	24	15	7

We know that mid-values of class (x) = 4, 12, 20, 28, 36, 44

We also know that $\Sigma f = 8 + 7 + 16 + 24 + 15 + 7 = 77$

Let the assumed mean be (A) = 28 and h = 8

Therefore, $\left(u = \dfrac{x - A}{h} \right) = \dfrac{4-28}{8}, \dfrac{12-28}{8}, \dfrac{20-28}{8},$

$\dfrac{28-28}{8}, \dfrac{36-28}{8}, \dfrac{44-28}{8} = -3, -2, -1, 0, 1, 2$

and $\Sigma fu = 8 \times (-3) + 7(-2) + 16 \times (-1) + 24 \times (-0) + (15 \times 1) + (7 \times 2)$

$= -24 - 14 - 16 + 0 + 15 + 14 = -25$

Therefore, the arithmetic mean of the data

$$= A + h\frac{\Sigma fu}{\Sigma f} = 28 + 8\left(-\frac{25}{77} \right) = 28 - \frac{200}{77} = 25.40$$

4. In the frequency distribution of the discrete data given below, the frequency x against value 0 is missing.

Variable x:	0	1	2	3	4	5
Frequency f:	x	20	40	40	20	4

If the mean is 2.5, then the missing frequency x will be

(a) 0 (b) 1

(c) 3 (d) 4

Ans. (d)

Solution: Given: Mean $(\bar{x}) = 2.5$ and discrete data

Variable x:	0	1	2	3	4	5
Frequency f:	x	20	40	40	20	4

We know that

$$\Sigma f = x + 20 + 40 + 40 + 20 + 4 = 124 + x$$

We also know that

$$\Sigma fx = (0 \times x) + (1 \times 20) + (2 \times 40) + (3 \times 40) + (4 \times 20) + (5 \times 4)$$

$$= 0 + 20 + 80 + 20 = 320$$

Therefore, mean $(\bar{x}) = \dfrac{\Sigma fx}{\Sigma f} = \dfrac{320}{124 + x}$ or $2.5 = \dfrac{320}{124 + x}$

or $124 + x = \dfrac{320}{2.5} = 128$ or $x = 128 - 124 = 4$

5. If a variate takes values $a, ar, ar^2, \ldots, ar^{n-1}$, which of the relations between means hold?

(a) $AH = G^2$ (b) $\dfrac{A+H}{2} = G$

(c) $A > G > H$ (d) $A = G = H$

Ans. (a)

Solution: $A = \dfrac{a(1 + r + r^2 + \cdots + r^{n-1})}{n} = \dfrac{a}{n}\dfrac{1 - r^n}{1 - r}$

$$G = (a \cdot ar \cdot ar^2 \ldots ar^{n-1})^{1/n}$$

$$= a \cdot r^{\frac{1+2+\cdots(n-1)}{n}} = ar^{\frac{n-1}{2}}$$

$$\frac{1}{H} = \frac{1}{na}\left(1 + \frac{1}{r} + \frac{1}{r^2} + \cdots + \frac{1}{r^{n-1}} \right) = \frac{1}{na} \cdot \frac{1 - (1/r^n)}{1 - (1/r)}$$

$$\Rightarrow H = \frac{na(1-r)r^{n-1}}{1 - r^n}$$

$$\therefore \quad AH = a^2 r^{n-1} = G^2$$

6. The mean weight of 150 persons in a group is 60 kg. The mean weight of men in the group is 70 kg and that of the women is 55 kg. The number of men and women are

(a) 50 and 100 (b) 60 and 90

(c) 100 and 50 (d) 75 and 75

Ans. (a)

Solution: Number of persons = 150; their mean weight = 60 kg; mean weight of men $(\bar{x}_1) = 70$ kg and mean weight of women $(\bar{x}_2) = 55$ kg .

Let n_1 and n_2 be the number of men and number of women respectively.

We know that the total number of persons $(n_1 + n_2) = 150$ or $n_2 = 150 - n_1$.

We also know that the mean weight of all persons

$$(\bar{x}) = \frac{(n_1\bar{x}_1 + n_2\bar{x}_2)}{n_1 + n_2}$$

or $$60 = \frac{70n_1 + 55n_2}{150}$$

or $70n_1 + 55n_2 = (60 \times 150) = 9000$

or $14n_1 + 11n_2 = 1800$ or $14n_1 + 11(150 - n_1) = 1800$

or $14n_1 + 1650 - 11n_1 = 1800$

or $3n_1 = (1800 - 1650) = 150$

or $n_1 = 50$ and $n_2 = 100$

7. The median of the observations 12, 17, 3, 14, 5, 8, 7, 15 is

(a) 12 (b) 14

(c) 8 (d) 10

Ans. (d)

Solution: Given observations are 12, 17, 3, 14, 5, 8, 7, 15.

We know that the number of observations (n) = 8.

Arranging the given observations in the ascending order, we get 3, 5, 7, 8, 12, 14, 15 and 17.

Since the number of observations is even, therefore their median

= Arithmetic mean of $\left(\frac{n}{2}\right)$th and $\left(\frac{n}{2}+1\right)$th observations

= Arithmetic mean of $\left(\frac{8}{2}\right)$th and $\left(\frac{8}{2}+1\right)$th observations

= Arithmetic mean of 4th and 5th observations

$= \frac{(8+12)}{2} = \frac{20}{2} = 10$

8. The marks obtained by 10 students in an examination are 22, 26, 14, 30, 18, 11, 35, 41, 12, 32. What is the median mark?

(a) 14 (b) 22

(c) 20 (d) 24

Ans. (d)

Solution: Number of students (n) = 10 and marks obtained by them = 22, 26, 14, 30, 18, 11, 35, 41, 12, 32

Arranging the given marks in the ascending order, we get 11, 12, 14, 18, 22, 26, 30, 32, 35, 41.

Since the number of students is even, therefore median of their marks

= Arithmetic mean of $\left(\frac{n}{2}\right)$th and $\left(\frac{n+2}{2}\right)$th marks

= Arithmetic mean of $\left(\frac{10}{2}\right)$th and $\left(\frac{10+2}{2}\right)$th marks

= Arithmetic mean of 5th and 6th marks

$= \frac{22+26}{2} = 24$

9. In a moderately skewed distribution, the values of mean and median are 5 and 6 respectively. Then the value of mode is

(a) 8 (b) 11

(c) 16 (d) 21

Ans. (a)

Solution: Given mean = 5 and median = 6

We know that in moderately skewed distribution, mode = (3 median – 2 mean)

$= (3 \times 6) - (2 \times 5) = 18 - 10 = 8$

10. If the difference between mean and mode is 63, then the difference between mean and median is

(a) 21 (b) 31.5

(c) 48.5 (d) 50.9

Ans. (a)

Solution: Given: Mean – mode = 63

We know that mode = 3 median – 2 mean

Let mean > median > mode

Therefore, mean – mode = 3 (mean – median)

or 63 = 3 (mean – median) or mean – median = 21

Therefore, difference between mean and median = 21.

11. Which of the following mode formulae is applicable for the continuous series?

(a) mode $= l + \left(\frac{f_1}{f - f_1 - f_2}\right) \times h$

(b) mode $= l + \left(\frac{f - f_1}{f - f_1 - f_2}\right) \times h$

(c) mode $= l + \left(\frac{f - f_1}{2f - f_1 - f_2}\right) \times h$

(d) mode $= l + \left(\frac{2f - f_1}{f - f_1 - f_2}\right) \times h$

Ans. (c)

Solution: We know that the mode value of a distribution is that value of the variable for which the frequency is maximum.

We also know that in the case of a discrete frequency distribution, the value of mode is determined by the method of a continuous series.

We also know that for a continuous series,

$$\text{mode} = l + \left(\frac{f - f_1}{2f - f_1 - f_2}\right) \times h$$

where l = lower limit of modal class, h = width of the modal class, f_1 = frequency of the class preceding the modal class, f_2 = frequency of the class following the modal class and f = frequency of the modal class.

12. The weighted A.M. of the first n natural numbers whose weights are equal to the corresponding numbers is equal to

(a) $2n + 1$ (b) $\frac{1}{2}(2n + 1)$

(c) $\frac{1}{3}(2n + 1)$ (d) $\frac{2n + 1}{6}$

Ans. (c)

Solution: The required weighted mean is

$$\bar{x} = \frac{1\cdot 1 + 2\cdot 2 + 3\cdot 3 + \cdots + n\cdot n}{1+2+3+\cdots+n}$$

$$= \frac{[n(n+1)(2n+1)]/6}{[n(n+1)]/2} = \frac{2n+1}{3}$$

13. The A.M. of n numbers of a series is $\bar{x}$. If the sum of the first $(n-1)$ terms is k, then the nth number is

(a) $\bar{x} - k$ (b) $n\bar{x} - k$

(c) $\bar{x} - nk$ (d) $n\bar{x} - nk$

Ans. (b)

Solution: Let the n numbers be $x_1, x_2, \ldots, x_n$. Then

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\Rightarrow \quad \bar{x} = \frac{x_1 + x_2 + \cdots + x_{n-1} + x_n}{n}$$

$$\Rightarrow \quad \bar{x} = \frac{k + x_n}{n} \qquad [\because \; x_1 + x_2 + \cdots + x_{n-1} = k]$$

$$\Rightarrow \quad x_n = n\bar{x} - k$$

14. The mean of a set of numbers is $\bar{x}$. If each number is divided by 3, then the new mean is

(a) $\bar{x}$ (b) $\bar{x} + 3$

(c) $3\bar{x}$ (d) $\dfrac{\bar{x}}{3}$

Ans. (d)

Solution: Let $x_1, x_2, \ldots, x_n$ be n numbers. Then $\bar{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$.

If each number is divided by 3, then the new mean $\bar{y}$ is given by

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i}{3}\right) = \frac{1}{3}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{\bar{x}}{3}$$

15. The mean and S.D. of 63 children on an arithmetic test are respectively 27.6 and 7.1. To them are added a new group of 26 children who had less training and whose mean is 19.2 and S.D. is 6.2. The values of the combined groupsdiffer from the original as to (i) the mean and (ii) the S.D. by

(a) 2.5, 0.7 (b) 2.3, 0.8

(c) 1.5, 0.9 (d) none of these

Ans. (a)

Solution: Mean of S.D. of the combined group is

$$m = \frac{63\times 27.6 + 26\times 19.2}{63+26} = 25.1$$

Thus, A.M. is decreased by $27.6 - 25.1 = 2.5$

$$\sigma^2 = \frac{63\times(7.1)^2 + 26\times(6.2)^2}{89} + \frac{63(25.1-27.6)^2 + 26(25.1-19.2)^2}{89}$$

$\Rightarrow \sigma = 7.8$ approx.

EXERCISE SET

1. The mean of n items is $\bar{x}$. If the first item is increased by 1, second by 2 and so on, then the new mean is

(a) $\bar{x} + n$ (b) $\bar{x} + \dfrac{n}{2}$

(c) $\bar{x} + \dfrac{n+1}{2}$ (d) none of these

2. If g_1 and g_2 are the geometric means of two series of n_1 and n_2 items. Then the G.M. of the series obtained on combining them is

(a) $[(g_1)^{n_1}\cdot(g_2)^{n_2}]^{\frac{1}{n_1+n_2}}$ (b) $(g_1\, g_2)^{\frac{n_1}{n_1+n_2}}$

(c) $(g_1\, g_2)^{\frac{n_2}{n_1+n_2}}$ (d) $(g_1\, g_2)^{\frac{n_1 n_2}{n_1+n_2}}$

3. If a variable takes discrete values $x+4$, $x-\dfrac{7}{2}$, $x-\dfrac{5}{2}$, $x-3$, $x-2, x+\dfrac{1}{2}$; $x-\dfrac{1}{2}$, $x+5 (x>0)$, then the median is

(a) $x - \dfrac{5}{4}$ (b) $x - \dfrac{1}{2}$

(c) $x - 2$ (d) $x + \dfrac{5}{4}$

4. If the coefficient of correlations between x and y is 0.28, covariance between x and y is 7.6 and the variance of x is 9, then the S.D. of y series is

(a) 9.8 (b) 10.1

(c) 9.05 (d) 10.05

5. If $\bar{x}$ is the mean of $x_1, x_2, x_3, \ldots, x_n$, then the mean of $(x_1 + a), (x_2 + a), \ldots, (x_n + a)$ is

(a) $\bar{x} + 2a$ (b) $\bar{x} + a$

(c) $\bar{x}$ (d) $2\bar{x} + a$

6. The median from the data:

Class	*Frequency*
110–120	6
120–130	25
130–140	48
140–150	72
150–160	116
160–170	60
170–180	38
180–190	22
190–200	3

is

(a) 153.79 (b) 153.91
(c) 165.18 (d) 165.93

7. If mean, mode and S.D. of a frequency distribution are 45, 49 and 4, respectively, then Karl Pearson's coefficient of skewness is

(a) –1 (b) 1
(c) 0 (d) –(1/2)

8. The mean and S.D. of 1, 2, 3, 4, 5, 6 is

(a) $\frac{7}{2}, \sqrt{\frac{35}{12}}$ (b) $\frac{7}{2}, \sqrt{3}$
(c) 3, 3 (d) $3, \frac{35}{12}$

9. The mean of 5 numbers is 18. If one number is excluded, their mean becomes 16, then the excluded number is

(a) 18 (b) 25
(c) 26 (d) 30

10. If the mean of the distribution

x	1	2	3	4	5
f	4	5	k	1	2

is 2.6, then the value of k is

(a) 8 (b) 10
(c) 12 (d) 18

11. A group of 10 items has mean 6. If the mean of 4 of these items is 7.5, then the mean of the remaining items is

(a) 6.5 (b) 5.5
(c) 4.5 (d) 5.0

12. The regression lines of x on y and y on x are $x = 4y + 5$ and $y = kx + 4$ respectively. The limits of k are given by

(a) $0 < k < \frac{1}{4}$ (b) $-1 < k < \frac{1}{2}$
(c) $-1 < k < 1$ (d) none of these

13. If a variable takes values 0, 1, 2,..., n with frequencies proportional to $q^n, nC_1pq^{n-1}, nC_2p^2q^{n-2},..., p^n$, where $p+q=1$, then the mean is

(a) np (b) nq
(c) npq (d) np^2

14. The mean age of a combined group of men and women is 25 years. If the mean age of the group of men is 26 and that of the group of women is 21, then the percentages of men and women in the group are

(a) 46, 60 (b) 80, 20
(c) 20, 80 (d) 60, 40

15. The means of two samples of size 200 and 300 were found to be 25 and 10, respectively. Their standard deviations were 3 and 4, respectively. The variance of the combined sample of size 500 is

(a) 64 (b) 65.2
(c) 67.2 (d) 64.2

16. The mean of 9 observations is 100 and that of 6 observations is 80, then the combined mean of all 15 observations is

(a) 100 (b) 92
(c) 90 (d) 80

17. One set containing 5 numbers has mean 8 and variance 18 and the second set containing 3 numbers has mean 8 and variance 24. Then the variance of the combined set of numbers is

(a) 42 (b) 20.25
(c) 18 (d) none of these

18. A person purchases 1 kg of tomatoes from each of the 4 places at the rate of 1 kg, 2 kg, 3 kg and 4 kg per rupee, respectively. On the average, if he has purchased x kg of tomatoes per rupee, then the value of x is

(a) 2.5 (b) 1.92
(c) 2 (d) none of these

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) and (d) given below:

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true; Statement II is false.
(d) Statement I is false; Statement II is true.

19. Suppose two groups of scores A and B are such that $A = (x, x+2, x+4)$ and $B = (x-2, x+2, x+6)$.

Statement I: Group B has more variability than group A.
Statement II: The value of mean for group B is more than that of group A.

20. Statement I: The algebraic sum of the deviations of 20 observations measured from 30 is 2. The mean value of the observation is 30.
Statement II: The sum of deviations from their mean is zero.

21. Statement I: The mean of the squares of the first n natural numbers is $\frac{(n+1)(2n+1)}{6}$.
Statement II: $\Sigma n = \frac{n(n+1)}{2}$.

22. Statement I: The standard deviation of the variable $\frac{ax+b}{c}$ is $\left|\frac{a}{c}\right|\alpha$.
Statement II: The standard deviation of a linear equation is $\sigma \times |\text{coefficient of } x|$.

Previous Years' Questions

23. In a class of 100 students, there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class are 72, then what is the average marks of the girls?
(a) 73 (b) 65
(c) 68 (d) 74 **[AIEEE 2002]**

24. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set
(a) is decreased by 2
(b) is two times the original median
(c) remains the same as that of the original set
(d) is increased by 2 **[AIEEE 2003]**

25. In an experiment with 15 observations on x, the following results were available $\Sigma x^2 = 2830$ and $\Sigma x = 170$. One observation 20 was replaced by 30 which was correct. The corrected variance is
(a) 80.33 (b) 78.00
(c) 188.66 (d) 177.33
[AIEEE 2003]

26. Consider the following statements:
(i) Mode can be computed from histogram
(ii) Median is not independent of change of scale
(iii) Variance is independent of change of origin and scale
Which of these is/are correct?
(a) Only (i) (b) Only (ii)
(c) Only (i) and (ii) (d) Only (i), (ii) and (iii)
[AIEEE 2004]

27. In a series of $2n$ observations, half of them equal a and the remaining half equal $-a$. If the standard deviation is 2, then $|a|$ equals
(a) $1/n$ (b) $\sqrt{2}$
(c) 2 (d) $\sqrt{2}/n$
[AIEEE 2004]

28. Let $x_1, x_2, \ldots, x_x$ be n observations such that $\Sigma x_i^2 = 400$ and $\Sigma x_i = 80$. Then, a possible value of n among the following is
(a) 12 (b) 9
(c) 18 (d) 15
[AIEEE 2005]

29. If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately
(a) 24.0 (b) 25.5
(c) 20.5 (d) 22.0
[AIEEE 2005]

30. Suppose a population A has 100 observations 101, 102, ..., 200 and another population B has 100 observations 151, 152,..., 250. If V_A and V_B represent the variances of the two populations respectively, then V_A/V_B is
(a) 9/4 (b) 4/9
(c) 2/3 (d) 1 **[AIEEE 2006]**

31. The average mark of boys in a class is 52 and that of girls is 42. The average mark of boys and girls combined is 50. The percentage of boys in the class is
(a) 40% (b) 20%
(c) 80% (d) 60%
[AIEEE 2007]

32. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then which one of the following gives the possible values of a and b?
(a) $a = 3, b = 4$ (b) $a = 0, b = 7$
(c) $a = 5, b = 2$ (d) $a = 1, b = 6$
[AIEEE 2008]

33. If the mean deviation of numbers $1, 1+d, 1+2d, \ldots, 1+100d$ from their mean is 255, then d is equal to
(a) 10.0 (b) 20.0
(c) 10.1 (d) 20.2
[AIEEE 2009]

34. Statement I: The variance of the first n even natural numbers is $(n^2 - 1)/4$.
Statement II: The sum of the first n natural numbers is $n(n + 1)/2$ and the sum of the squares of the first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$
(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

[AIEEE 2009]

35. For two data sets, each of size 5, the variances are given to be 4 and 5 and the corresponding means are given to be 2 and 4, respectively. The variance of the combined data set is
(a) 5/2 (b) 11/2
(c) 6 (d) 13/2

[AIEEE 2010]

36. The number of observations in a group is 40. If the average of the first 10 is 4.5 and that of the remaining 30 is 3.5, then the average of the whole group is
(a) 1/5 (b) 15/4
(c) 4 (d) 8

[AMU 1992, Delhi CEE 1996]

37. The mean of the values 0, 1, 2,..., n having corresponding weights ${}^nc_0, {}^nc_1, {}^nc_2, \ldots, {}^nc_n$ respectively is
(a) $\frac{2^n}{n+1}$ (b) $\frac{2^{n+1}}{n(n+1)}$
(c) $\frac{n+1}{2}$ (d) $\frac{n}{2}$

[AMU 1990, Ph. CET 1998]

38. The mean deviation of the numbers 3, 4, 5, 6, 7 is
(a) 0 (b) 1.2
(c) 5 (d) 25

[AMU 1992]

39. If the regression coefficients are 0.8 and 0.2, then the value of the coefficient of correlation is
(a) 0.16 (b) 0.4
(c) 0.04 (d) 0.164 **[DCE 2003]**

40. The coefficient of correlation (r) and the two regression coefficients b_{yx} and b_{xy} are related as
(a) $r = \frac{b_{xy}}{b_{yx}}$ (b) $r = b_{xy} \times b_{yx}$
(c) $r = b_{xy} + b_{yx}$ (d) $r = (\text{sign } b_{yx})\sqrt{b_{xy} b_{yx}}$

[DCE 2003]

41. If the variable takes the values 0, 1, 2,..., n with frequencies proportional to the binomial coefficients $C(n, 0)$, $C(n, 1)$, $C(n, 2)$,..., $C(n, n)$ respectively, then the variance of the distribution is
(a) n (b) $\sqrt{n}/2$
(c) $n/2$ (d) $n/4$ **[DCE 2004]**

42. The angle between two lines of regression is given by
(a) $\tan^{-1}\left[\frac{b_{xy} - (1/b_{yx})}{1 - (b_{xy}/b_{yx})}\right]$ (b) $\tan^{-1}\left(\frac{b_{yx} \cdot b_{xy} - 1}{b_{yx} + b_{xy}}\right)$
(c) $\tan^{-1}\left[\frac{b_{xy} - (1/b_{yx})}{1 + (b_{xy}/b_{yx})}\right]$ (d) $\tan^{-1}\left(\frac{b_{yx} - b_{xy}}{1 + b_{yx}.b_{xy}}\right)$

[DCE 2005]

43. If two lines of regression are $3\bar{x} - 2\bar{y} + 1 = 0$ and $2\bar{x} - \bar{y} - 2 = 0$, then $(\bar{x}, \bar{y})$ is
(a) (8, 5) (b) (5, 8)
(c) (5, 5) (d) (8, 8) **[DCE 2006]**

44. The two lines of regression are given by $3x + 2y = 26$ and $6x + y = 31$. The coefficient of correlation between x and y is
(a) –1/3 (b) 1/3
(c) –1/2 (d) 1/2 **[AMU 2003]**

45. If $\Sigma x = 15$, $\Sigma y = 36$, $\Sigma xy = 110$, $n = 5$, then cov(x, y) equals
(a) 1/5 (b) –1/5
(c) 2/5 (d) –2/5 **[AMU 2004]**

46. The mean deviation and coefficient of mean deviation from the data
Weight (in kg) 54, 50, 40, 42, 51, 45, 47, 55, 57 is
(a) 0.0900 (b) 0.0956
(c) 0.0056 (d) 0.0946 **[AMU 2005]**

47. If the standard deviation of a variable x is σ, then the standard deviation of another variable $(ax + b)/c$ is
(a) $(\sigma a + b)/c$ (b) $\sigma a/c$
(c) σ (d) none of these

48. The mode of the series 3, 4, 2, 6, 1, 7, 6, 7, 6, 8, 9, 5 is
(a) 5 (b) 6
(c) 7 (d) 8 **[AMU 2007]**

49. The median from the table

Value	7	8	10	9	11	12	1
Frequency	2	1	4	5	6	1	

is
(a) 100 (b) 10
(c) 110 (d) 1110 **[2009 AMU]**

50. For the arithmetic progression a, $(a+d)$, $(a+2d)$, $(a+3d)$,..., $(a+2nd)$, the mean deviation from the mean is
(a) $\frac{n(n+1)d}{2n-1}$ (b) $\frac{n(n+1)d}{2n+1}$
(c) $\frac{n(n-1)d}{2n+1}$ (d) $\frac{(n+1)d}{2}$

[Kerala CEE 2006]

51. The intersecting point of two regression lines is
(a) $(\bar{x}, 0)$ (b) $(0, \bar{y})$ (c) $(0, 0)$ (d) $(\bar{x}, \bar{y})$
[Kerala CEE 2004]

52. If the two lines of regression are $4x+3y+7=0$ and $3x+4y+8=0$, then the means of x and y are
(a) $\frac{-4}{7}, \frac{-11}{7}$ (b) $\frac{-4}{7}, \frac{11}{7}$ (c) $\frac{4}{7}, \frac{-11}{7}$ (d) 4, 7
[BCECE 2005]

53. For the given data, the calculation corresponding to all value of variates (x, y) is the following $\Sigma(x-\bar{x})^2=36$, $\Sigma(y-\bar{y})^2=25$, $\Sigma(x-\bar{x})(y-\bar{y})=20$.
The Kerl Pearson's correlation coefficient is
(a) 0.2 (b) 0.5 (c) 0.66 (d) 0.33
[Jamia Millia Islamia 2006]

54. The corelation coefficient between x and y from the following data $\Sigma x=40$, $\Sigma y=50$, $\Sigma xy=220$, $\Sigma x^2=200$ $\Sigma y^2=262$, $n=10$ is
(a) 0.89 (b) 0.76 (c) 0.91 (d) 0.98
[MP PET 2006]

55. In a bivariate data, $\Sigma x=30$, $\Sigma y=400$, $\Sigma x^2=196$, $\Sigma xy=850$, $n=10$. The regression coefficient of y on x is
(a) –3.1 (b) –3.2 (c) –3.3 (d) –3.4
[MP PET 2007]

56. If the lines of regression are $3x+12y=19$ and $3y+9x=46$, then r_{xy} will be
(a) 0.289 (b) –0.289 (c) 0.209 (d) none of these
[MP PET 2008]

57. If $x_1, x_2, \ldots, x_{18}$ are the observations such that $\sum_{j=1}^{18}(x_j-8)=9$ and $\sum_{j=1}^{18}(x_j-8)^2=45$, then the standard deviation of these observations is
(a) $\sqrt{81/34}$ (b) 5 (c) $\sqrt{5}$ (d) 3/2
[J & K CET 2004]

58. A scientist is weighing each of 30 fishes. Their mean weight worked out to be 30 g and the standard deviation to be 2 g. Later, it was found that the measuring scale was misaligned and always underreported every fish weight by 2 g. The correct mean and standard deviation (in gram) of fishes are respectively
(a) 32, 2 (b) 32, 4 (c) 28, 2 (d) 28, 4
[AIEEE 2011]

59. Let $x_1, x_2, \ldots, x_n$ be n observations and let $\bar{x}$ be their arithematic mean and σ^2 be their variance.
Statement 1: Variance of $2x_1, 2x_2, \ldots, 2x_n$ is $4\sigma^2$.
Statement 2: Arithmetic mean of $2x_1, 2x_2, \ldots, 2x_n$ is $4\bar{x}$.
(a) Statement 1 is false, statement 2 is true
(b) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1
(c) Satement 1 is true, statement 2 is true, statement 2 is not a correct explanation for statement 1
(d) Satement 1 is true, statement 2 is false
[AIEEE 2012]

ANSWERS

Exercise Set

1. (c)	**2.** (a)	**3.** (a)	**4.** (c)	**5.** (b)	**6.** (a)	**7.** (a)	**8.** (a)	**9.** (c)	**10.** (a)
11. (d)	**12.** (a)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (b)	**17.** (b)	**18.** (b)	**19.** (c)	**20.** (d)
21. (c)	**22.** (a)	**23.** (b)	**24.** (c)	**25.** (b)	**26.** (c)	**27.** (c)	**28.** (c)	**29.** (a)	**30.** (d)
31. (c)	**32.** (a)	**33.** (c)	**34.** (d)	**35.** (b)	**36.** (b)	**37.** (d)	**38.** (b)	**39.** (b)	**40.** (d)
41. (d)	**42.** (b)	**43.** (b)	**44.** (c)	**45.** (c)	**46.** (b)	**47.** (b)	**48.** (b)	**49.** (b)	**50.** (b)
51. (d)	**52.** (a)	**53.** (c)	**54.** (c)	**55.** (c)	**56.** (b)	**57.** (d)	**58.** (a)	**59.** (d)	

HINTS AND SOLUTIONS—EXERCISE SET

1. Let $x_1, x_2, \ldots, x_n$ be n items.
Then $\bar{x}=\frac{1}{n}\Sigma x_i$
Let $y_1=x_1+1$, $y_2=x_2+2$, $y_3+3, \ldots, y_n=x_n+n$. Then the mean of the new series is

$$\frac{1}{n}\sum_{n=1}^{n} y_i=\frac{1}{n}\sum_{n=1}^{n}(x_i+i)=\frac{1}{n}\sum_{n=1}^{n}x_i+\frac{1}{n}(1+2+3+\cdots+n)$$

$$\bar{y}=\bar{x}+\frac{1}{n}\cdot\frac{n(n+1)}{2}=\bar{x}+\frac{n+1}{2}$$

2. Let $g_1=(x_1x_2\ldots x_{n_1})^{1/n_1}$ and $g_2=(y_1y_2\ldots y_{n_2})^{1/n_2}$. If g is the G.M. of the combined series, then

$$g=[(x_1x_2\ldots x_{n_1})\times(y_1y_2\ldots y_{n_2})]^{1/(n_1+n_2)}$$
$$=(g_1^{n_1}\cdot g_2^{n_2})^{1/(n_1+n_2)}$$

3. Arranging the given values in ascending order of magnitude, we get

$$x-\frac{7}{2}, x-3, x-\frac{5}{2}, x-2, x-\frac{1}{2}, x+\frac{1}{2}, x+4, x+5$$

There are 8 observations in this series. Therefore,
Median = A.M. of 4th and 5th observations
= A.M. of $(x-2)$ and $(x-1/2)$

$$= \frac{x-2+x-(1/2)}{2} = x-\frac{5}{4}$$

4. $\rho = \dfrac{\text{cov}(x, y)}{\sigma_x \sigma_y}$

$$\Rightarrow \quad 0.28 = \frac{7.6}{\sigma_y \times 3}$$

$$\Rightarrow \quad \sigma_y = 9.05$$

5. Given mean of numbers $(x_1, x_2, x_3, \ldots, x_n) = \bar{x}$

We know that mean $(\bar{x}) = \dfrac{x_1 + x_2 + \cdots + x_n}{n}$

Let the mean of $(x_1 + a)$, $(x_2 + a), \ldots, (x_n + a) = \bar{x}$
Therefore, mean

$$(\bar{x}) = \frac{(x_1 + a) + (x_2 + a) + \cdots + (x_n + a)}{n}$$

$$= \frac{(x_1 + x_2 + \cdots + x_n) + na}{n}$$

$$= \frac{x_1 + x_2 + \cdots + x_n}{n} + a = (\bar{x}) + a$$

6.

Class interval	f	$c.f.$
110–120	6	6
120–130	25	31
130–140	48	79
140–150	72	151
150–160	116	267
160–170	60	327
170–180	38	365
180–190	22	387
190–200	3	390

$$N = \Sigma f = 390$$

Median = size of $(N/2)$th item = size of $(390/2) = 195$th item.

$\therefore$ Median class = 150 – 160

$l = 150,\ h = 10,\ f = 116,\ (N/2) = 195,\ C = 151$

$$\text{Median} = l + \frac{(N/2) - C}{f} \times h = 150 + \frac{195 - 151}{116} \times 10$$

$$= 150 + \frac{440}{116} = 150 + \frac{110}{29} = 150 + 3.79$$

$$= 153.79$$

7. Coefficient of skewness $= \dfrac{\text{Mean} - \text{Mode}}{\sigma}$

$$= \frac{45 - 49}{4} = \frac{-4}{4} = -1$$

8. Mean $\bar{x} = \dfrac{1+2+3+4+5+6}{6} = \dfrac{21}{6} = \dfrac{7}{2}$

$$\therefore \quad \text{S.D.} = \sigma = \sqrt{\frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2}$$

$$= \sqrt{\frac{1}{6}(1+4+9+16+25+36) - \frac{49}{4}}$$

$$= \sqrt{\frac{91}{6} - \frac{49}{4}} = \sqrt{\frac{182-147}{12}} = \sqrt{\frac{35}{12}}$$

9. Initial numbers = 5; initial mean = 18; final numbers = 4 and final mean = 16.
We know that the sum of the intial numbers = $(5 \times 18) = 90$
Similarly, the sum of the final numbers $= (4 \times 16) = 64$
Therefore, excluded number $= (90 - 64) = 26$

10. $\Sigma fx = 4 + 10 + 3k + 4 + 10 = 3k + 28$

$\Sigma f = 12 + k$

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\Rightarrow \quad 2.6 = \frac{3k + 28}{12 + k} \qquad \Rightarrow \quad \frac{13}{5} = \frac{3k + 28}{12 + k}$$

$$\Rightarrow \quad 156 + 13k = 15k + 140$$

$$\Rightarrow \quad 2k = 16 \qquad \Rightarrow \quad k = 8$$

11. Total of 10 items $= 10 \times 6 = 60$
Total of 4 items $= 4 \times 7.5 = 30$

$$\text{Mean of the remaining 6 items} = \frac{60 - 30}{6} = 5 = 5.0$$

12. We have $b_{yx} = k$ and $b_{xy} = 4$

$$\Rightarrow \qquad r^2 = b_{yx} \cdot b_{xy} = 4k$$

Now since r^2 is the square of the real quantity, it should be non-negative. Also since $0 < r^2 < 1$, we have $0 < 4k < 1$ or $0 < k < (1/4)$.

13. Mean $= \dfrac{(0 \cdot q^n + 1 \cdot {}^nC_1 pq^{n-1} + 2 {}^nC_2 p^2 q^{n-2} + \cdots + n \cdot p^n)k}{k(q^n + {}^nC_1 pq^{n-1} + {}^nC_2 p^2 q^{n-2} + \cdots + p^n)}$

$$= \frac{{}^nC_1 pq^{n-1} + 2 \cdot {}^nC_2 p^2 q^{n-2} + \cdots + n \cdot p^4}{(q+p)^n}$$

$$= npq^{n-1} + 2 \cdot \frac{n(n-1)}{2} p^2 q^{n-2} + \cdots + n \cdot p^n$$

$$(\because \quad q + p = 1)$$

$$= np[q^{n-1} + (n-1)pq^{n-2} + \cdots + p^{n-1}]$$

$$= np \cdot (q+q)^{n-1} = np(1)^{n-1} = np$$

$\therefore$ Mean = np

14. Let n_1 and n_2 be the number of men and women respectively, in a group. According to the given condition,

$$\frac{n_1 \times 26 + n_2 \times 21}{n_1 + n_2} = 25$$

$$\Rightarrow \quad 26n_1 + 21n_2 = 25n_1 + 25n_2$$

$$\Rightarrow \quad n_1 = 4n_2 \qquad \Rightarrow \quad \frac{n_1}{n_2} = \frac{4}{1}$$

$$\Rightarrow \quad \frac{n_1}{n_2} = \frac{80}{20}$$

15. Combined mean $\bar{x} = \dfrac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$$= \frac{200 \times 25 + 300 \times 10}{500} = 16$$

Let $d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$

$d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$

We know that $\sigma^2 = \dfrac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$

$$= \frac{200(9 + 81) + 300(16 + 36)}{500}$$

$$= \frac{33{,}600}{500} = 67.2$$

16. Given: Number of first observations $(n_1) = 9$ and its mean $(\bar{x}_1) = 100$;

Number of second observations $(n_2) = 6$ and its mean $(\bar{x}_2) = 80$.

We know that combined mean

$$(\bar{x}) = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{(9 \times 100) + (6 \times 80)}{(9 + 6)}$$

$$= \frac{900 + 480}{15} = \frac{1380}{15} = 92$$

17. Here $n_1 = 5,\ \bar{x}_1 = 8,\ \sigma_1^2 = 18$

$n_2 = 3,\ \bar{x}_2 = 8,\ \sigma_2^2 = 24$

$\bar{x}_{12}$ = combined mean $= \dfrac{5 \times 8 + 3 \times 8}{5 + 3} = \dfrac{64}{8} = 8$

Combined variance $= \dfrac{n_1(\sigma_1^2 + D_1^2) + n_2(\sigma_2^2 + D_2^2)}{n_1 + n_2}$

where $D_1 = \bar{x}_1 - \bar{x}_{12},\ D_2 = \bar{x}_2 - \bar{x}_{12}$

$\therefore \quad D_1 = 8 - 8;\ D_2 = 8 - 8 = 0$

Combined variance $= \dfrac{5(18) + 3(24)}{5 + 3}$

$$= \frac{90 + 72}{8} = \frac{162}{8} = 20.25$$

18. Since we are given the rate per rupees, harmonic mean willgive the correct answer

$$\text{H.M.} = \frac{4}{\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{4 \times 12}{12 + 6 + 4 + 3}$$

$$= \frac{48}{25} = 1.92 \text{ kg per rupee}$$

19. Since $A = (x,\ x+2,\ x+4)$ and $B = (x-2,\ x+2,\ x+6)$

$\therefore$ Mean of $A = \dfrac{x + x + 2 + x + 4}{3} = x + 2$

and mean of $B = \dfrac{x - 2 + x + 2 + x + 6}{3} = x + 2$

Hence, group B has more variability than group A.

20. Since deviations $= \Sigma(x_i - A)$

$\Rightarrow \quad 2 = \Sigma x_i - 20 \times 30$

$\Rightarrow \quad \Sigma x_i = 602$

$\therefore$ Mean $= \dfrac{602}{20} = 30.1$

21. Required mean $= \dfrac{\Sigma n^2}{n} = \dfrac{n(n+1)(2n+1)}{6n}$

$$= \frac{(n+1)(2n+1)}{6}$$

Statement II is also a true statement.

Hence, option (c) is correct.

22. Since $\sigma\left(\dfrac{ax+b}{c}\right) = \left|\dfrac{a}{c}\right|\sigma$

Hence, both statements are true and Statement II is a correct explanation for Statement I.

23. Since the total number of students = 100 and the number of boys = 70,

$\therefore$ Number of girls $= (100 - 70) = 30.$

Now, the total marks of 100 students $= 100 \times 72 = 7200$

And total marks of 70 boys $= 70 \times 75 = 5250$

$\Rightarrow$ Total marks of 30 girls $= 7200 - 5250 = 1950$

$\therefore$ Average marks of 30 girls $= \dfrac{1950}{30} = 65.$

24. Since the median is the 5th term and the increase is made in the last four terms, hence the median remains unchanged.

25. Given: $N = 15$

$$\Sigma x^2 = 2830,\ \Sigma x = 170$$

One observation 20 was replaced by 30, then

$$\Sigma x^2 = 2830 - 400 + 900 = 3330$$

$$\Sigma x = 170 - 20 + 30 = 180$$

$\therefore$ Variance, $\sigma^2 = \frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2$

$$= \frac{3330}{15} - \left(\frac{180}{15}\right)^2 = 222 - 144 = 78$$

26. It is true that mode can be computed from histogram and median is not independent of change of scale. But variance is independent of change of origin and not of scale.

27. In the $2n$ observations, half of them are equal to a and the remaining half are equal to $-a$. Then, the mean of the total $2n$ observations is equal to zero.

$\therefore$ S.D. $= \sqrt{\frac{\Sigma(x-\bar{x})^2}{N}}$

$\Rightarrow 2 = \sqrt{\frac{\Sigma x^2}{2n}}$

$\Rightarrow 4 = \frac{\Sigma x^2}{2n} = \frac{2na^2}{2n}$

$\Rightarrow a^2 = 4$

$\Rightarrow |a| = 2$

28. Given $\Sigma x_i^2 = 400$ and $\Sigma x_i = 80$ since $\sigma^2 \geq 0$

$\Rightarrow \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2 \geq 0$

$\Rightarrow \frac{400}{n} - \frac{6400}{n^2} \geq 0$

$\Rightarrow n \geq 16$

$\therefore n = 18$

29. Given that mean = 21 and median = 22

Using the relations

Mode = 3 Median – 2 Mean

$\therefore$ Mode $= 3(22) - 2(21)$

$= 66 - 42 = 24$

30. Since variance is independent of change of origin, hence, the variance of the observations 101, 102,..., 200 is the same as the variance of the observations 151, 152,..., 250.

$\therefore \quad V_A = V_B$

$\Rightarrow \quad \frac{V_A}{V_B} = 1$

31. Let the number of boys and girls be x and y.

$\therefore \quad 52x + 42y = 50(x + y)$

$\Rightarrow 52x + 42y = 50x + 50y$

$\Rightarrow 2x = 8y \qquad \Rightarrow x = 4y$

$\therefore$ Total number of students in the class

$= x + y = 4y + y = 5y$

$\therefore$ Required percentage of boys

$= \frac{4y}{5y} \times 100\% = 80\%$

32. According to the given condition,

$$6.80 = \frac{\left[\begin{array}{l}(6-a)^2 + (6-b)^2 + (6-8)^2 \\ \quad + (6-5)^2 + (6-10)^2\end{array}\right]}{5}$$

$\Rightarrow 34 = (6-a)^2 + (6-b)^2 + 4 + 1 + 16$

$\Rightarrow (6-a)^2 + (6-b)^2 = 13 = 9 + 4 = 3^2 + 2^2$

$\Rightarrow a = 3,\ b = 4$

33. $\bar{x} = \frac{\text{Sum of quantities}}{n} = \frac{n/2(a+l)}{n}$

$$= \frac{1}{2}(1 + 1 + 100d) = 1 + 50d$$

$$\text{M.D.} = \frac{1}{n}\Sigma|x_i - \bar{x}|$$

$\Rightarrow 255 = \frac{1}{101}(50d + 49d + \cdots + d + 0 + d + \cdots + 50d]$

$= \frac{2d}{101}\left(\frac{50 \times 51}{2}\right)$

$\Rightarrow d = \frac{255 \times 101}{50 \times 51} = 10.1$

34. Statement II is true.

Statement I: Sum of n even natural numbers = n (n + 1)

$$\text{Mean } (\bar{x}) = \frac{n(n+1)}{n} = n + 1$$

$$\text{Variance} = \left[\frac{1}{n}\Sigma(x_i)^2\right] - (\bar{x})^2$$

$$= \frac{1}{n}[2^2 + 4^2 + \cdots + (2n)^2] - (n+1)^2$$

$$= \frac{1}{n}2^2[1^2 + 2^2 + \cdots + n^2] - (n+1)^2$$

$$= \frac{4}{n}\frac{n(n+1)(2n+1)}{6} - (n+1)^2$$

$$= \frac{(n+1)[2(2n+1) - 3(n+1)]}{3}$$

$$= \frac{(n+1)(n-1)}{3} = \frac{n^2 - 1}{3}$$

$\therefore$ Statement I is false.

35. $\sigma_x^2 = 4$ and $\sigma_y^2 = 5$

Also $\bar{x} = 2$ and $\bar{y} = 4$

Now, $\frac{\Sigma x_i}{5} = 2 \Rightarrow \Sigma x_i = 10$

$$\frac{\Sigma y_i}{5} = 4 \Rightarrow \Sigma y_i = 20$$

Since $\sigma_x^2 = \frac{1}{5}(\Sigma x_i^2) - (\bar{x})^2$

$\Rightarrow \Sigma x_i^2 = 40.$

Similarly $\Sigma y_i^2 = 105$

$$\therefore \quad \sigma_z^2 = \frac{1}{10}(\Sigma x_i^2 + \Sigma y_i^2) - \left(\frac{\bar{x}+\bar{y}}{2}\right)^2$$

$$= \frac{1}{10}(40+105) - 9$$

$$= \frac{55}{10} = \frac{11}{2}$$

36. $\frac{x_1 + x_2 + \cdots + x_{10}}{10} = 4.5$

$\Rightarrow x_1 + x_2 + \cdots + x_{10} = 45$

and $\frac{x_{11} + x_{12} + \cdots + x_{40}}{30} = 3.5$

$\Rightarrow x_{11} + x_{12} + \cdots + x_{40} = 105$

$\therefore \quad x_1 + x_2 + \cdots + x_{40} = 105$

$$\therefore \quad \frac{x_1 + x_2 + \cdots + x_{40}}{40} = \frac{150}{40} = \frac{15}{4}$$

37. Required mean

$$= \frac{0 \cdot {}^nC_0 + 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + \cdots + n \cdot {}^nC_n}{{}^nC_0 + {}^nC_1 + \cdots + {}^nC_n}$$

$$= \frac{n \cdot 2^{n-1}}{2^n} = \frac{n}{2}$$

$$[\because \ 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + \cdots + n\,{}^nC_n = n \cdot 2^{n-1}]$$

38. Mean $\bar{x} = \frac{3+4+5+6+7}{5} = 5, \ N = 5$

x	$\lvert x - \bar{x} \rvert$
3	2
4	1
5	0
6	1
7	2

$$\Sigma \lvert x - \bar{x} \rvert = 6$$

$\therefore$ Mean deviation from the mean

$$= \frac{\Sigma \lvert x - \bar{x} \rvert}{N} = \frac{6}{5} = 1.2$$

39. $b_{yx} = 0.8, \ b_{xy} = 0.2$

Then $r = \sqrt{b_{yx} b_{yx}} = \sqrt{(0.8)(0.2)} = \sqrt{0.16}$

$\Rightarrow r = 0.4$

40. We know that

$$r = \pm\sqrt{b_{yx} \cdot b_{xy}}$$

Also we know that signs of r, $b_{xy} \cdot b_{yx}$ are all the same.

$\therefore \quad r = (\text{sign of } b_{yx})\sqrt{b_{yx} \cdot b_{xy}}$

41. Now, $\mu_1' = \frac{\sum_{r=0}^{n} r \cdot {}^nC_r}{\sum_{r=0}^{n} {}^nC_r} = \frac{n \cdot 2^{n-1}}{2^n} = \frac{n}{2}$

$$\mu_2' = \frac{\sum_{r=0}^{n} r^2 \cdot {}^nC_r}{\sum_{r=0}^{n} {}^nC_r} = \frac{n(n-1)}{2^n} \cdot 2^{n-2} + \frac{n}{2}$$

$$= \frac{n(n-1)}{4} + \frac{n}{2}$$

$\therefore$ Variance $\mu_2 = (\mu_2') - (\mu_1')^2 = \frac{n(n-1)}{4} + \frac{n}{2} - \frac{n^2}{4} = \frac{n}{4}$

42. The slopes of the lines of regression of y on x and x on y are $m_1 = b_{yx}$ and $m_2 = 1/b_{xy}$, respectively. Therefore, the angle between them is given by

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{b_{yx} - (1/b_{xy})}{1 + (b_{yx}/b_{xy})}$$

$$\Rightarrow \qquad \theta = \tan^{-1}\left(\frac{b_{yx} \times b_{xy} - 1}{b_{yx} \times b_{xy}}\right)$$

43. Given lines are $3\bar{x} - 2\bar{y} + 1 = 0$ (i)

and $2\bar{x} - \bar{y} - 2 = 0$ (ii)

On solving Eqs. (i) and (ii), we get $\bar{x} = 5, \ \bar{y} = 8$.

44. Given $3x + 2y = 26$

$$\Rightarrow \quad y = -\frac{3}{2}x + 13$$

and $6x + y = 31$

$$\Rightarrow \quad x = -\frac{1}{6}y + \frac{13}{6}$$

$$\therefore \quad r = -\sqrt{\left(\frac{-3}{2}\right)\left(\frac{-1}{6}\right)}$$

$$\Rightarrow \quad r = -\frac{1}{2}$$

45. $\text{cov}(x, y) = \frac{1}{n}\Sigma xy - \bar{x}\bar{y}$

$$= \frac{1}{5}(110) - \left(\frac{15}{5}\right)\left(\frac{36}{5}\right) = \frac{2}{5}$$

46. On arranging the terms in increasing order of magnitude, we have 40, 42, 45, 47, 50, 51, 54, 55, 57

Number of terms, $N = 9$

$\therefore$ Median $= \left(\frac{9+1}{2}\right)$th term = 5th term = 50 kg

Weight (kg)	*Deviation from median* (d)	$\lvert d \rvert$
40	−10	10
42	−8	8
45	−5	5
47	−3	3
50	0	0
51	1	1
54	4	4
55	5	5
57	7	7
		$\lvert d \rvert = 43$

$$\text{M.D. from median} = \frac{43}{9} = 4.78 \text{ kg}$$

$\therefore$ Coefficient of M.D. from median

$$= \frac{\text{M.D.}}{\text{median}}$$

$$= \frac{4.78}{50} = 0.0956$$

47. Standard deviation does not depend on origin but it depends on scale, so

$$\frac{ax+b}{c} = \frac{ax}{c} + \frac{b}{c}$$

$\Rightarrow$ Standard deviation of $\frac{ax+b}{c}$ is $\frac{a\sigma}{c}$.

48. In the given distribution, 6 occurs most of the times. Hence, mode of the series = 6.

49. Arranging the terms in increasing order, we have

Value (x)	*Frequency* (f)	*Cumulative frequency* (*c.f.*)
7	2	2
8	1	3
9	5	8
10	4	12
11	6	18
12	1	19
13	3	22

$\because \quad N = 22$

$\therefore$ Median number $= \frac{N+1}{2} = 11.5$

which comes under the cumulative frequency 12. The corresponding value of x will be the median, i.e. median = 10.

50. The mean of the series $a, a+d, \ldots, +a+2d$ is

$$\bar{x} = \frac{1}{2n+1}(a + a + d + a + 2d + \cdots + a + 2nd)$$

$$= \frac{1}{2n+1}\left[\frac{2n+1}{2}(a + a + 2nd)\right] = a + nd$$

$\therefore$ Mean deviation from mean

$$= \frac{1}{2n+1}\sum_{r=0}^{2n}\left|(a + rd) - (a + nd)\right|$$

$$= \frac{1}{2n+1}\sum_{r=0}^{2n}(r - n)d$$

$$= \frac{1}{2n+1}2d(1 + 2 + \cdots + n)$$

$$= \frac{n(n+1)}{2n+1}d$$

51. The intersecting point of two regression lines is on the mean, i.e. $(\bar{x}, \bar{y})$.

52. $4\bar{x} + 3\bar{y} + 7 = 0$ (i)

and $3\bar{x} + 4\bar{y} + 8 = 0$ (ii)

On solving Eqs. (i) and (ii), we get

$$\bar{x} = -\frac{4}{7} \quad \text{and} \quad \bar{y} = -\frac{11}{7}$$

53. $r_{xy} = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\sqrt{\Sigma(x - \bar{x})^2 \Sigma(y - \bar{y})^2}}$

$$= \frac{20}{\sqrt{36 \times 25}} = \frac{2}{3} = 0.66$$

54. Correlation coefficient

$$r=\frac{n\Sigma xy-\Sigma x\Sigma y}{\sqrt{[n\Sigma x^2-(\Sigma x^2)]}\sqrt{[n\Sigma y^2-(\Sigma y^2)]}}$$

$$=\frac{10(220)-40\times 50}{\sqrt{10(200)-(40)^2}\sqrt{10(262)-(50)^2}}$$

$$=\frac{200}{20\times 10.954}=\frac{200}{219.08}=0.91$$

55. $\text{cov}(x,y)=\frac{\Sigma xy}{n}-\frac{\Sigma x}{n}\cdot\frac{\Sigma y}{n}=\frac{1}{10}(850)-\left(\frac{30}{10}\right)\left(\frac{400}{10}\right)$

$=85-120=-35$

and $\text{var}(x)=\sigma_x^2=\frac{1}{n}\Sigma x^2-\left(\frac{\Sigma x}{n}\right)^2$

$=\frac{196}{10}-\left(\frac{30}{10}\right)^2=10.6$

$\therefore\quad b_{yx}=\frac{\text{cov}(x,y)}{\text{var}(x)}=\frac{-35}{10.6}=-3.3$

56. Let us assume that the line of regression y on x be $3x+12y=19$ and x on y be $3y+9x=46$.

$\therefore\quad b_{yx}=-\frac{3}{12}$ and $b_{xy}=-\frac{3}{9}=-\frac{1}{3}$

$\therefore\quad r_{xy}=-\sqrt{b_{yx}\times b_{xy}}=-\sqrt{\left(\frac{3}{12}\right)\times\left(\frac{1}{3}\right)}$

$=-\sqrt{\frac{1}{12}}=-\sqrt{0.083}$

$=-0.289$

57. Standard deviation $=\sqrt{\frac{\sum_{j=1}^{18}(x_j-8)^2}{n}-\left(\frac{\sum_{j=1}^{18}(x_j-8)}{n}\right)^2}$

$=\sqrt{\frac{45}{18}-\left(\frac{9}{18}\right)^2}=\sqrt{\frac{45}{18}-\frac{1}{4}}=\sqrt{\frac{81}{36}}$

$=\frac{9}{6}=\frac{3}{2}$

58. There is no change in the mean deviation if each observation is increased by a constant number while the mean is increased by that constant number.
Hence, A.M. = 30 + 2 + 32, M.D. = 2

59. $\sigma^2=\Sigma\frac{x_i^2}{n}-\left(\Sigma\frac{x_i}{n}\right)^2$

Variance of $2x_1, 2x_2, \ldots, 2x_n$

$=\Sigma\frac{(2x_i)^2}{n}-\left(\Sigma\frac{2x_i}{n}\right)^2=4\left[\Sigma\frac{x_i^2}{n}-\left(\Sigma\frac{x_i}{n}\right)^2\right]$

$=4\sigma^2$

Statement 1 is true.

A.M. of $2x_1, 2x_2, \ldots, 2x_n=\frac{2x_1+2x_2+\cdots+2x_n}{n}$

$=2\left(\frac{x_1+x_2+\cdots+x_n}{n}\right)=2\bar{x}$

Statement 2 is false.

Alternative method

Statement 2 is false because the arithmetic mean of $\frac{2x_1+2x_2+2x_3+\cdots+2x_n}{n}$ is $2\bar{x}$.

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. Mode of the data 3, 2, 5, 2, 3, 5, 6, 6, 3, 5, 2, 5 is
(a) 6 (b) 4 (c) 5 (d) 3

2. The median of 10, 14, 11, 9, 8, 12, 6 is
(a) 10 (b) 12 (c) 14 (d) 11

3. Mode of the distribution

[redacted]	4	5	6	7	8
	3	5	10	6	1

is
(a) 6 (b) 10
(c) 8 (d) none of these

4. The quartile deviation of the daily wages (in ₹) of 7 persons given as 12, 7, 15, 10, 17, 19, 25 is
(a) 145 (b) 5 (c) 9 (d) 4.5

5. The median of the following data 11, 29, 17, 21, 13, 31, 39, 19 is
(a) (21 + 13)/12 (b) 19
(c) 21 (d) 20

6. If the mode of a distribution is 18 and the mean is 24, then the median is
(a) 18 (b) 24 (c) 22 (d) 21

7. The mode of the following distribution

[redacted]	40	43	46	49	52	55
	5	8	16	9	7	3

is
(a) 40 (b) 46
(c) 55 (d) none of these

8. The median value of the following frequency distribution

	153–159	160–166	167–173	174–180	181–187
	2	4	6	4	4

is equal to
(a) 169 (b) 171 (c) 173 (d) 175

9. If in a moderately skewed distribution, the values of mode and mean are 6λ and 9λ respectively, then the value of the median is
(a) 8λ (b) 7λ (c) 6λ (d) 5λ

10. The mode and median of the data 82, 98, 73, 71, 43, 82, 90 are
(a) 80, 90 (b) 82, 82 (c) 77, 98 (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. Which of the following is not a measure of central tendency?
(a) mean (b) median
(c) mode (d) range

2. The weighted mean of the first n natural numbers whose weights are equal to the squares of the corresponding numbers is

(a) $\frac{n+1}{2}$ (b) $\frac{3n(n+1)}{2(2n+1)}$

(c) $\frac{(n+1)(2n+1)}{6}$ (d) $\frac{n(n+1)}{2}$

3. The mean of the following frequency table is 50.

Class	*Frequency*
0–20	17
20–40	f_1
40–60	32
60–80	f_2
80–100	19
Total	120

The missing frequencies are
(a) 28, 24 (b) 24, 36
(c) 38, 28 (d) none of these

4. The geometric mean of 1, 2, 2^2, 2^3,..., 2^n is
(a) $2^{2/n}$ (b) $2^{n/2}$ (c) $2^{(n-1)/2}$ (d) $2^{(n+1)/2}$

5. If $\bar{x}$ is the mean of a distribution, then $\sum f_i(x_i - \bar{x}) =$
(a) 0 (b) M.D.
(c) S.D. (d) none of these

6. The variance of the first n natural numbers is

(a) $\frac{n^2-1}{12}$ (b) $\frac{n^2-1}{6}$

(c) $\frac{n^2+1}{6}$ (d) $\frac{n^2+1}{12}$

7. If $\bar{x}_1$ and $\bar{x}_2$ are means of two distributions such that $\bar{x}_1 < \bar{x}_2$ and $\bar{x}$ is the mean of the combined distribution, then
(a) $\bar{x} < \bar{x}_1$ (b) $\bar{x} > \bar{x}_2$
(c) $\bar{x} = \frac{\bar{x}_1 + \bar{x}_2}{2}$ (d) $\bar{x}_1 < \bar{x} < \bar{x}_2$

8. The A.M. of n observations is $\bar{x}$. If the sum of $n - 5$ observations is a, then the mean of the remaining 5 observations is

(a) $\frac{n\bar{x} + a}{5}$ (b) $\frac{n\bar{x} - a}{5}$

(c) $n\bar{x} + a$ (d) none of these

9. A car completes the first half of its journey with a velocity v_1, and the rest half with velocity v_2. Then, the average velocity of the car for the whole journey is

(a) $\frac{v_1 + v_2}{2}$ (b) $\sqrt{v_1 v_2}$

(c) $\frac{2v_1 v_2}{v_1 + v_2}$ (d) none of these

10. Which one of the following measures is the most suitable one of central location for computing intelligence of students?
(a) mode (b) A.M.
(c) G.M. (d) median

ANSWERS

Level 1

1. (c) **2.** (a) **3.** (a) **4.** (d) **5.** (d) **6.** (c) **7.** (b) **8.** (b) **9.** (a) **10.** (b)

Level 2

1. (d) **2.** (b) **3.** (a) **4.** (b) **5.** (a) **6.** (a) **7.** (d) **8.** (b) **9.** (c) **10.** (d)

26 Probability

PIERRE DE FERMAT

- Experiment
- Random Experiment
- Sample Space
- Sample Point/Event Point
- Discrete Sample Space
- Types of Events
- Occurrence of an Event
- Probability of Occurrence of an Event
- Odds in Favour and Odds against an Event
- Algebra of Events
- Theorem of Total Probability
- Conditional Probability
- Multiplication Theorem of Probability
- Binomial Theorem on Probability
- Probability Distribution
- Bayes' Theorem

☞ EXPERIMENT

An operation which results in some well-defined outcomes is called an **experiment**.

☞ RANDOM EXPERIMENT

An experiment whose outcome cannot be predicted with certainty is called a **random experiment**. In other words, if an experiment is performed many times under similar conditions and the outcome of each time is not the same, then this experiment is called a random experiment.

Examples:

(a) Tossing of a fair coin
(b) Throwing of an unbiased die
(c) Drawing of a card from a well-shuffled pack of 52 playing cards

☞ SAMPLE SPACE

The set of all possible outcomes of a random experiment is called the **sample space** for that experiment. It is usually denoted by S.

Examples:

(a) When a die is thrown, any one of the numbers 1, 2, 3, 4, 5, 6 can come up. Therefore, sample space $S = \{1, 2, 3, 4, 5, 6\}$.
b) When a coin is tossed, either a head or a tail will come up, then the sample space w.r.t. the tossing of the coin is $S = \{H, T\}$.
(c) When two coins are tossed, then the sample space is $S = \{(H, H)\ (H, T)\ (T, H)\ (T, T)\}$.

☞ SAMPLE POINT/EVENT POINT

Each element of the sample space is called a **sample point** or an **event point**.
Example: When a die is thrown, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$, where 1, 2, 3, 4, 5 and 6 are the sample points.

☞ DISCRETE SAMPLE SPACE

A sample space S is called a **discrete sample space** if S is a finite set.

☞ EVENT

A subset of the sample space is called an **event**.

Note

- Sample space S plays the same role as universal set for all problems related to the particular experiment.
- ϕ is also the subset of S and is an impossible event.
- S is also a subset of S which is called a sure event or a certain event.

☞ TYPES OF EVENTS

A. Simple Event/Elementary Event

An event is called a **simple event** if it is a singleton subset of the sample space S.
Examples:

(a) When a coin is tossed, the sample space is $S = \{H, T\}$, then $A = \{H\}$ is the occurrence of head and $B = \{T\}$ is the occurrence of tail which are called simple events.

(b) When two coins are tossed, the sample space is $S = \{(H, H); (H, T); (T, H); (T, T)\}$, then $A = \{H, T\}$ is the occurrence of head on 1st and tail on 2nd which is called a simple event.

B. Mixed Event or Compound Event or Composite Event

A subset of the sample space S which contains more than one element is called a **mixed event** or when two or more events occur together, their joint occurrence is called a **compound event**.
Example: When a die is thrown, the sample space is

$$S = \{1, 2, 3, 4, 5, 6\}$$

then let $A = \{2, 4, 6\}$ is the event of the occurrence of even numbers and $B = \{1, 2, 4\}$ is the event of the occurrence of exponent of 2 which are mixed events.

Compound events are of two types:

Independent and dependent events: Two or more events are said to be independent if occurrence or non-occurrence of any of them does not affect the probability of occurrence or non-occurrence of the other event and those which affect the above probability are said to be dependent events.

C. Equally Likely Events

Outcomes are said to be equally likely when we have no reason to believe that one is more likely to occur than the other.
Example: When an unbiased die is thrown, all the six faces 1, 2, 3, 4, 5, 6 are equally likely to come up.

D. Exhaustive Events

A set of events is said to be **exhaustive** if one of them must necessarily happen every time the experiment is performed.
Example: When a die is thrown, events 1, 2, 3, 4, 5, 6 form an exhaustive set of events.

Note We can say that the total number of elementary events of a random experiment is called the exhaustive number of cases.

☞ OCCURRENCE OF AN EVENT

For a random experiment, let E be an event.

Let $E = \{a, b, c\}$. If the outcome of the experiment is either a or b or c, then we say the event has occurred.
Sample space: The outcomes of any type
Event: The outcomes of a particular type

☞ PROBABILITY OF OCCURRENCE OF AN EVENT

Let S be the sample space, then the probability of occurrence of an event E is denoted by $P(E)$ and is defined as

$$P(E) = \frac{n(E)}{n(S)} = \frac{\text{Number of elements in } E}{\text{Number of elements in } S}$$

$$P(E) = \frac{\text{Number of favourable/particular cases}}{\text{Total number of cases}}$$

Examples:

(a) When a coin is tossed, the sample space is $S = \{H, T\}$. Let E be the event of occurrence of a head
$\Rightarrow E = \{H\}$
$\therefore$ $P(E)$ = Probability of occurrence of head
$= \frac{n(E)}{n(S)} = \frac{1}{2}$

(b) When a die is tossed, the sample space $S = \{1, 2, 3, 4, 5, 6\}$. Let A be an event of occurrence of an odd number and B be an event of occurrence of a number greater than 4.
$\Rightarrow A = \{1, 3, 5\}$ and $B = \{5, 6\}$
$\therefore$ $P(A)$ = Probability of occurrence of an odd number
$= \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$ and
$P(B)$ = Probability of occurrence of a number greater than $4 = \frac{n(B)}{n(S)} = \frac{2}{6} = \frac{1}{3}$

Note

- Probability of occurrence of an event is a number lying between 0 and 1.
  Reasoning: Let S be the sample space and E be an event,
  $0 \le n(E) \le n(S)$

$$\frac{0}{n(S)} \le \frac{n(E)}{n(S)} \le \frac{n(S)}{n(S)}$$

- If ϕ is an impossible event, then the number of elements in ϕ is zero and conversely if $n(\phi) = 0$ where ϕ is an impossible event,
  if $P(E) = 0$
  $\Rightarrow E = \phi$ (impossible event)
  if $P(E) = 1$
  $\Rightarrow E = S$ (sample space event)

Statistical or empirical definition of probability: Let a random experiment be repeated n times under similar conditions, i.e. let n trails of a random experiment be made, where n is very large and the event E occurs r times in n trials, then

$$P(E) = \lim_{n\to\infty}\left(\frac{r}{n}\right)$$

ODDS IN FAVOUR AND ODDS AGAINST AN EVENT

Let S be the sample space and E be an event. Let E' denote the complement of the event E, then

A. Odds in Favour of Event *E*

$$\frac{n(E)}{n(E')} = \frac{\text{Number of cases favourable to } E}{\text{Number of cases against } E}$$

$$\frac{n(E)}{n(E')} = \frac{\frac{n(E)}{n(S)}}{\frac{n(E')}{n(S)}} = \frac{P(E)}{P(E')}$$

B. Odds against Event 'E'

$$\frac{n(E')}{n(E)} = \frac{\text{Number of cases against } E}{\text{Number of cases favourable to event } E}$$

$$\frac{n(E')}{n(E)} = \frac{\frac{n(E')}{n(S)}}{\frac{n(E)}{n(S)}}$$

$$\frac{n(E')}{n(E)} = \frac{P(E')}{P(E)}$$

ALGEBRA OF EVENTS

If A, B and C are any three events, then

(a) $A \cap B$ denotes the event of simultaneous occurrence of both the events A and B.

(b) $A \cup B$ denotes the event of occurrence of at least one of the events A and B.

(c) $A - B$ or $A \cap \bar{B}$ denotes the occurrence of event A but not B.

(d) $\bar{A} \cap \bar{B}$ denotes the occurrence of neither A nor B.

(e) $A \cup B \cup C$ denotes the occurrence of at least one event A, B or C.

(f) $A \cap B \cap C$ denotes the occurrence of all three events A, B and C.

(g) $(A \cap \bar{B}) \cup (\bar{A} \cap B)$ denotes the occurrence of exactly one of A and B.

(h) $(A \cap B) \cup (B \cap C) \cup (A \cap C) - (A \cap B \cap C)$ denotes that not only one occurs.

(i) $(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)$ denotes that one and only one occurs.

(j) $(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C)$ denotes that two and no more occur.

(k) $(A \cap B) \cup (B \cap C) \cup (A \cap C)$ denotes that at least two of A, B and C occur.

THEOREM OF TOTAL PROBABILITY/ ADDITION THEOREM OF PROBABILITY

If A and B are any two events in a sample space S, then the probability of occurrence of at least one of the events A and B is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A, B and C are any three events in sample space S, then the probability of occurrence of at least one of the events A and B is given by

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

In general,

$$P(A_1 \cup A_2 \cup \ldots \cup A_n) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n-1} P(A_1 \cap A_2 \cap \ldots \cap A_n)$$

Note

(a) If A and B are mutually exclusive events, then
$A \cap B = \phi$
i.e. $P(A \cap B) = 0$
$\therefore\ P(A \cap B) = P(A) + P(B)$

(b) If A, B and C are mutually exclusive events, then
(i) $A \cap B = \phi$
(ii) $B \cap C = \phi$
(iii) $C \cap A = \phi$
(iv) $A \cap B \cap C = \phi$
$\therefore \quad P(A \cup B \cup C) = P(A) + P(B) + P(C)$

(c) If $P(A \cup B) = P(A) + P(B)$, then it can be said that events A and B are mutually exclusive.

(d) $P(\overline{A} \cap \overline{B}) = 1 - P(A \cup B)$

(e) $P(\overline{A} \cup \overline{B}) = 1 - P(A \cap B)$

(f) $P(A) = P(A \cap B) + P(A \cap \overline{B})$

(g) $P(B) = P(B \cap A) + P(B \cap \overline{A})$

CONDITIONAL PROBABILITY

Let A and B be any two events such that $B \neq \phi$ or $n(B) \neq 0$ or $P(B) \neq 0$, then $P\left(\frac{A}{B}\right)$ denotes the conditional probability of occurrence of event A when B has already occurred.

Examples:

(a) Let a bag contain 2 red balls and 3 black balls. One ball is drawn from the bag and the ball is not replaced in the bag. Then a second ball is drawn from the bag. Let B be the occurrence of a red ball in the first draw and A denote the event of occurrence of a black ball in the second draw.
When a red ball has been drawn in the first draw, the number of balls left is 4 (1 red + 3 black).
$\therefore \quad P\left(\frac{A}{B}\right)$ = Probability of occurrence of a black ball in the second draw when a red ball has already been drawn= 3/4.

(b) When a die is thrown, the sample space is
$S = \{1, 2, 3, 4, 5, 6\}$
Let A = Occurrence of a number greater than 4, i.e.$\{5, 6\}$
Let B = Occurrence of an odd number

Case I: Probability of occurrence of A when B has already occurred. Now, in this case we need to find the probability of a number greater than 4, when an odd number has already been occurred.

Since the odd number has occurred, i.e. $\{1, 3, 5\}$, now, out of these three only 5 is greater than 4.

Hence, $P\left(\frac{A}{B}\right) = \frac{1}{3} = \frac{n(A \cap B)}{n(B)}$

Case II: Probability of occurrence of B when A has already occurred. In this case, we need to find the probability of an odd number when a number greater than 4 has already occurred. Since a number greater than 4 has occurred, i.e. $\{5, 6\}$.

Hence, $P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{n(A)} = \frac{1}{2}$

MULTIPLICATION THEOREM OF PROBABILITY

If A and B are any two events, then

$$P(A \cap B) = P(B) \cdot P\left(\frac{A}{B}\right) \quad [\because B = \phi] \qquad \text{(i)}$$

where $P\left(\frac{A}{B}\right)$ is the probability of A when B event has already occurred.

Now, if A and B are independent events, then the probability of occurrence of A is not affected by the occurrence or non-occurrence of event B.

$$\therefore \quad P\left(\frac{A}{B}\right) = P(A)$$

Hence, $P(A \cap B) = P(B) \cdot P\left(\frac{A}{B}\right)$ [Using (i)]

$$P(A \cap B) = P(B) \cdot P(A)$$

Also, if A and B are independent events, then

$$P\left(\frac{B}{A}\right) = P(B)$$

$$\Rightarrow \quad P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow \quad P(A \cap B) = P(A) \cdot P\left(\frac{B}{A}\right)$$

$$\therefore \quad P(A \cap B) = P(B) \cdot P(A) = P(AB)$$

Note

(a) The events A and ϕ are always independent.
$\therefore \quad P(A \cap \phi) = P(A) \cdot P(\phi) = 0$
i.e. an event A and an impossible event (i.e. null event) can never occur together.

(b) The events A and S are always independent.
$\therefore \quad P(A \cap S) = P(A) \cdot P(S) = P(A)$
i.e. occurrence of an event A along with the sample space event is the event A itself.

(c) Mutually exclusive events (none of which is an impossible event) are not independent.
Reasoning: Let A and B be two non-impossible mutually exclusive events.
$\therefore \quad P(A \cup B) = P(A) + P(B)$ (i)

Now, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

[Addition Theorem]

$\Rightarrow \quad P(A \cap B) = 0$ [Using (i)]

But since A and B are non-impossible events,

$\therefore \quad P(A)$ and $P(B) \neq 0$

$\Rightarrow \quad P(A) \cdot P(B) > 0$

[$\because$ Probability of an event is never negative]

Hence, $P(A \cap B) \neq P(A) \cdot P(B)$

$\therefore$ Events A and B are not independent.

Hence, mutually exclusive events (none of which is an impossible event) are not independent.

(d) Non-impossible independent events are not matually exclusive.

Reasoning: Let A and B be non-impossible independent events, then

$$P(A \cap B) = P(A) \cdot P(B) \quad \text{(i)}$$

Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

[Addition Theorem]

$\Rightarrow \quad P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$ (ii)

Since A and B are non-impossible events,

$\therefore \quad P(A)$ and $P(B) \neq 0$

$\Rightarrow \quad P(A) \cdot P(B)$ exists finitely

Hence, $P(A \cup B) \neq P(A) + P(B)$ [from (ii)]

$\Rightarrow$ the events are not mutually exclusive.

(e) If A and B are independent events, then A and $\overline{B}$ as well as $\overline{A}$ and B are independent events.

Reasoning: Since A and B are independent events.

$\therefore \quad P(A \cap B) = P(A) \cdot P(B)$

Now, $A = (A \cap \overline{B}) \cup (A \cap B)$

$\Rightarrow \quad P(A) = P(A \cap \overline{B}) \cup (A \cap B)$

As it can be seen in Fig. 26.1 that there is no interference between the elements of $(A \cap \overline{B})$ and $(A \cap B)$, i.e. the occurrence of $(A \cap \overline{B})$ does not affect the occurrence of $(A \cap B)$,

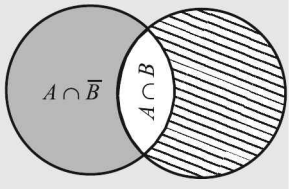


Fig. 26.1

i.e. $(A \cap \overline{B})$ and $(A \cap B)$ are mutually exclusive events.

$\therefore \quad P(A) = P(A \cap \overline{B}) + P(A \cap B)$

$\Rightarrow \quad P(A) = P(A \cap \overline{B}) + P(A) \cdot P(B)$

[$\because$ A and B are independent events]

$\Rightarrow \quad P(A \cap \overline{B}) = P(A) - P(A) \cdot P(B)$

$\Rightarrow \quad P(A \cap \overline{B}) = P(A)[1 - P(B)] = P(A)P(\overline{B})$

Hence, A and $\overline{B}$ are independent events.

Similarly, $\overline{A}$ and B are independent events.

(f) If A and B are independent events, then $\overline{A}$ and $\overline{B}$ are independent events.

Resoning: Since A and B are independent events,

$\therefore \quad P(A \cap B) = P(A) \cdot P(B)$ (i)

Now $P(\overline{A} \cap \overline{B}) = P(\overline{A \cup B})$

$\Rightarrow \quad P(\overline{A} \cap \overline{B}) = 1 - P(A \cup B)$

$\Rightarrow \quad P(\overline{A} \cap \overline{B}) = 1 - \{P(A) + P(B) - P(A \cap B)\}$

[Addition Theorem]

$\Rightarrow \quad P(\overline{A} \cap \overline{B}) = 1 - P(A) - P(B) + P(A) \cdot P(B)$

[Using (i)]

$\Rightarrow \quad P(\overline{A} \cap \overline{B}) = (1 - P(A))(1 - P(B)) = P(\overline{A}) \cdot P(\overline{B})$

Hence, if A and B are independent events, then $\overline{A}$ and $\overline{B}$ are also independent events.

BINOMIAL THEOREM ON PROBABILITY

If the experiment is repeated n times under similar conditions, we say that n trials of the experiment have been made. Let E be an event.

Let p = the probability of occurrence of event E in one trial and $q = 1 - p$ = probability of non-occurrence of event E in one trial such that $p + q = 1$.

Let X = number of successes, i.e. the number of times event E occurs in N trials.

Then the probability of occurrence of the event r times in n trials is denoted by

$P(X = r)$ or $P(r)$ and is given by

$P(X = r)$ or $P(r) = {}^nC_r p^r q^{n-r} = (r + 1)$th term in the expansion of $(q + p)^n$.

PROBABILITY DISTRIBUTION

A. Random Variable

A random variable is a real-valued function defined over the sample space of an experiment. In order words, a random variable is a real-valued function whose domain is the sample space of a random experiment. A random variable is usually denoted by the capital letters, X, Y, Z, etc.

B. Discrete Random Variable

A random variable which can take only a finite or countably infinite number of values is called a **discrete random variable**.

C. Continuous Random Variable

A random variable which can take any value between two given limits is called a **continuous random variable**.

D. Probability Distribution of the Random Variable

If the values of a random variable together with the corresponding probabilities are given, then this description is called a **probability distribution of the random varible**.

Example:

Probability distribution when two coins are tossed:

Let X denote the number of heads occurred, then

$P(X = 0)$ = probability of occurrence of zero head

$$= P(TT) = \frac{1}{2}\frac{1}{2} = \frac{1}{4}$$

$P(X = 1)$ = probability of occurrence of one head

$$= P(HT) + P(TH) = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}$$

$P(X = 2)$ = probability of occurrence of two heads

$$= P(HH) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

Thus, the probability distribution when two coins are tossed is as given below:

X	0	1	2
$P(X)$	1/4	1/2	1/4

☞ BINOMIAL DISTRIBUTION

Let E be an event.

Let P = probability of occurrence of event E in one trial and $q = 1 - p$ = probability of non-occurrence of event E in one trial. Clearly $p + q = 1$.

Let the occurrence of the event E be called a success.

Let X denote the number of successes in n trials.

Then $P(X = r)$ probability of r successes $= {}^nC_r p^r q^{n-r}$.

Therefore, the probability distribution of the random variable X is as given below:

X	0	1	2	⋯	r	⋯	n
$P(X)$	q^n	${}^nC_1 pq^{n-1}$	${}^nC_2 p^2 q^{n-2}$	⋯	${}^nC_r p^r q^{n-r}$	⋯	p^n

☞ BAYES' THEOREM

If $E_1, E_2, E_3, \ldots, E_n$ are n mutually exclusive and exhaustive events and A is an event which occurs together (in conjunction) with either of E_i, i.e. if $E_1, E_2, E_3, \ldots, E_n$ forms a partition of the sample space S and A is any event, then

$$P\left(\frac{E_k}{A}\right) = \frac{P(E_k)\cdot P\left(\frac{A}{E_k}\right)}{P(E_1)\cdot P\left(\frac{A}{E_1}\right) + P(E_2)\cdot P\left(\frac{A}{E_2}\right) + \cdots + P(E_n)\cdot P\left(\frac{A}{E_n}\right)}$$

☞ POISSON DISTRIBUTION

$$P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!} \qquad (\lambda = \text{mean})$$

Note

(a) If $E_1, E_2, \ldots, E_n$ forms a partition of S and A is any event, then

$$P(A) = P(E_1)\cdot P\left(\frac{A}{E_1}\right) + P(E_2)\cdot P\left(\frac{A}{E_2}\right) + \cdots + P(E_n)\cdot P\left(\frac{A}{E_n}\right)$$

$$\left\{\because\ P(E_i \cap A) = P(E_i)\cdot P\left(\frac{A}{E_i}\right)\right\}$$

(b) If $P(E_1) = P(E_2) = - \cdots = P(E_n)$, then by Bayes' theorem,

$$P\left(\frac{E_k}{A}\right) = \frac{P(E_k)\cdot P\left(\frac{A}{E_k}\right)}{P\left(\frac{A}{E_1}\right) + P\left(\frac{A}{E_2}\right) + \cdots + P\left(\frac{A}{E_n}\right)}$$

(c) The probabilities $P(E_1), P(E_2), \ldots, P(E_n)$ which are known before the experiment takes place are called priori probabilities and $P\left(\frac{E_i}{A}\right)$ are called posteriori probabilities.

SOLVED PROBLEMS

1. From 4 children, 2 women and 4 men, 4 persons are selected. The probability that there are exactly 2 children among the selected is

(a) 7/3 (b) 3/7

(c) 2/7 (d) 2/9

Ans. (b)

Solution: Total number of selection $= {}^{10}C_4 = 210$

2 children can be selected in three combinations

(i) 2 children and 2 women $= {}^4C_2 \times {}^2C_2 = 6$

(ii) 2 children and 2 men $= {}^4C_2 \times {}^4C_2 = 36$

(iii) 2 children, 1 man and 1 woman $= {}^4C_2 \times {}^4C_1 \times {}^2C_1 = 48$

Total favourable cases = 48 + 36 + 6 = 90

$$\therefore \text{ Required probability} = \frac{90}{210} = \frac{9}{21} = \frac{3}{7}$$

2. A bag contains 8 red and 5 white balls. Three balls are drawn at random. Find the probability that one ball is red and two balls are white.

(a) 40/143 (b) 40/147

(c) 41/141 (d) 39/147

Ans. (a)

Solution: $n(S)$ = Number of ways of selecting 3 balls out of 13.

$$= {}^{13}C_3 = \frac{13 \cdot 12 \cdot 11}{1 \cdot 2 \cdot 3} = 286$$

$n(E)$ = Number of ways of selecting 1 red ball out of 8 and 2 white balls out of 5

$= {}^{8}C_1 \times {}^{5}C_2 = 8 \times 10 = 80$

$\therefore$ Required probability $= \frac{n(E)}{n(S)} = \frac{80}{286} = \frac{40}{143}$

3. Four cards are drawn at random from a pack of 52 playing cards. The probability of getting all the four cards of the same suit is

(a) 44/4165 (b) 22/4165

(c) 11/4165 (d) none of these

Ans. (a)

Solution: Total number of equally likely cases $= {}^{52}C_4$

There are four suits, i.e. club, spade, heart and diamond, each of 13 cards. All the four cards are of the same suit means that the four cards drawn are either club cards or spade cards or heart cards or diamond cards.

$\therefore$ Total number of ways of getting all the four cards of the same suit

$$= {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4$$

$$= 4 \cdot {}^{13}C_4$$

Hence, the required probability $= 4 \cdot \frac{{}^{13}C_4}{{}^{52}C_4}$

$$= 4 \cdot \frac{\frac{13 \times 12 \times 11 \times 10}{4 \times 3 \times 2 \times 1}}{\frac{52 \times 51 \times 50 \times 49}{4 \times 3 \times 2 \times 1}} = 4 \cdot \frac{13 \times 12 \times 11 \times 10}{52 \times 51 \times 50 \times 49}$$

$$= \frac{12 \times 11 \times 10}{51 \times 50 \times 49} = \frac{12 \times 11}{51 \times 5 \times 49} = \frac{4 \times 11}{17 \times 5 \times 49} = \frac{44}{4165}$$

4. A bag contains 4 red and 3 black balls. A second bag contains 2 red and 4 black balls. One bag is selected at random. From the selected bag, one ball is drawn. The probability that the ball drawn to be red is

(a) 13/42 (b) 14/42

(c) 19/42 (d) none of these

Ans. (c)

Solution: A red ball can be drawn in two mutually exclusive ways:

(i) Selecting bag I and then drawing a red ball from it

(ii) Selecting bag II and then drawing a red ball from it

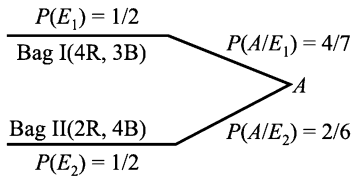


Fig. 26.2

Let E_1, E_2 and A denote the events defined as follows:

E_1 = Selecting bag I

E_2 = Selecting bag II

$\Rightarrow A$ = Drawing a red ball

Since one of the two bags is selected randomly,

$$\therefore \quad P(E_1) = \frac{1}{2} \text{ and } P(E_2) = \frac{1}{2}$$

Now, $P(A/E_1)$ = Probability of drawing a red ball when the first bag has been chosen = 4/7

[$\because$ The first bag contains 4 red and 3 black balls]

and $P(A/E_2)$ = Probability of drawing a red ball when the second bag has been selected =2/6

[$\because$ The second bag contains 2 red and 4 black balls]

Using the law of total probability, we have P (red ball)

$$= P(A)$$

$$= P(E_1)P(A/E_1) + P(E_2)P(A/E_2)$$

$$= \frac{1}{2} \times \frac{4}{7} + \frac{1}{2} \times \frac{2}{6} = \frac{19}{42}$$

5. A mapping is selected at random from set $A = \{1, 2, \ldots, 10\}$ into itself. The probability that the mapping selected is an injective is

(a) $10/10^9$ (b) $9!/10^9$

(c) 9/10! (d) none of these

Ans. (b)

Solutions: $n(A) = 10$ [$n(A)$ represents number of elements of A]

$\therefore$ Number of functions/mappings from A to $A = 10^{10}$. Out of 10^{10}, the number of injective functions is 10!

$$\therefore \quad P(A) = \frac{10!}{10^{10}} = \frac{10 \times 9!}{10.10^9} = \frac{9!}{10^9 !}$$

6. A card is drawn at random from a pack of 100 cards numbered 1 to 100. The probability of drawing a number which is a square is

(a) 1/5 (b) 2/5

(c) 1/10 (d) none of these

Ans. (c)

Solutions: Total ways = 100

Squares of the following numbers lie between 1 and 100: 1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2, 10^2, which are 10 in number. So, the required probability $= \frac{10}{100} = \frac{1}{10}$

7. An urn contains 7 green and 5 yellow balls. Two balls are drawn at a time. The probability that both balls are of the same colour is

(a) 1/33 (b) 5/33

(c) 7/22 (d) 31/66

Ans. (d)

Solution: Required probability $= \frac{{}^{7}C_2}{{}^{12}C_2} + \frac{{}^{5}C_2}{{}^{12}C_2} = \frac{21+10}{60} = \frac{31}{66}$

8. The probability that in a random arrangement of the letters of the word FAVOURABLE, the two A's do not come together is

(a) 1/5 (b) 1/10

(c) 9/10 (d) 4/5

Ans. (d)

Solution: Total number of arrangements of the letters of the word FAVOURABLE is $\frac{10!}{2!}$

Now say 2A's as one letter.

$\therefore$ Total number of arrangements if 'AA' is considered as one letter is 9!.

$\therefore$ Number of ways in which 'AA' never comes together is $\frac{10!}{2!} - 9!$.

$\therefore$ Required probability $= \frac{\frac{10!}{2!} - 9!}{\frac{10!}{2!}} = \frac{4}{5}$

9. Three factories A, B and C produce the same product. The factory A produces twice as many as B produces, while the factories B and C produce the same quantity. It is known that 2% of the products of A as well as C are defective, while 4% of the products of B are defective. All the products of the three factories are stocked together. If a product is selected at random from the stock, what is the probability that the product is defective?

(a) 1/40 (b) 1/30

(c) 1/50 (d) 1/20

Ans. (a)

Solution: Let A, B and C produce $2n$, n and n products.

Let E_1 = event of a product being produced at A

E_2 = event of a product being produced at B

E_3 = event of a product being produced at C

$$P(E_1) = \frac{2n}{2n+n+n} = \frac{1}{2}$$

$$P(E_2) = \frac{n}{2n+n+n} = \frac{1}{4}$$

$$P(E_3) = \frac{1}{4}$$

Also, $P(D/E_1)$ = probability of a product being defective when produced at A, $P(D/E_1) = \frac{2}{100} = \frac{1}{50}$

Similarly, $P(D/E_2) = \frac{4}{100} = \frac{1}{25}$

and $P(D/E_3) = \frac{2}{100} = \frac{1}{50}$

As $P(E_1) + P(E_2) + P(E_3) = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$,

events E_1, E_2, E_3 are exhaustive.

Also, they are mutually exclusive. The probability that the product is defective

$$= P(E_1)\cdot P(D/E_1) + P(E_2)\cdot P(D/E_2) + P(E_3)\cdot P(D/E_3)$$

$$= \frac{1}{2}\cdot\frac{1}{50} + \frac{1}{4}\cdot\frac{1}{25} + \frac{1}{4}\cdot\frac{1}{50}$$

$$= \frac{1}{100} + \frac{1}{100} + \frac{1}{200} = \frac{1}{40}$$

10. Two events A and B have probabilities 0.25 and 0.50 respectively. The probability that both A and B occur simultaneously is 0.14. Then the probability that neither A nor B occurs is

(a) 0.25 (b) 0.39

(c) 0.11 (d) none of these

Ans. (b)

Solution: $P(A) = 0.25$, $P(B) = 0.50$

$$P(A \cap B) = 0.14$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.25 + 0.50 - 0.14$$

$$= 0.75 - 0.14 = 0.61$$

$$P(\overline{A} \cap \overline{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - 0.61 = 0.39$$

11. Three dice are thrown. The probability that the sum of the numbers appearing on the dice is greater than fifteen is

(a) 5/216 (b) 5/108

(c) 15/216 (d) none of these

Ans. (b)

Solution: As the sum is greater than 15,

so we need the cases in which the sum is 16, 17, 18.

Now the cases when the sum is 16 are (6, 5, 5), (5, 6, 5), (5, 5, 6), (6, 6, 4), (6, 4, 6) and (4, 6, 6).

Cases when the sum is 17 are (6, 6, 5), (6, 5, 6) and (5, 6, 6).

Cases when the sum is 18 is only one (6, 6, 6).

$\therefore$ Total number of favourable cases = 10

Total number of cases = Number of sample points of the sample space = 6^3

$\therefore$ Required probability $= \frac{10}{216} = \frac{5}{108}$

12. In a test, an examinee either guesses or copies or knows the answer to a multiple-choice question with four choices, only one answer being correct. The probability that he makes a guess is 1/3 and the probability that he copies the answer is 1/6. The probability that his answer is correct, given he copies it, is 1/8. Find the probability that he knew the answer to the question, given that he correctly answers it.

(a) $\frac{26}{29}$ (b) $\frac{22}{29}$

(c) $\frac{24}{29}$ (d) $\frac{25}{29}$

Ans. (c)

Solution: Let G = the event of answering by guess work
C = the event of answering by copying
K = the event of answering by knowing
A = the event of answering correctly

From the question, $P(G)=\frac{1}{3}, P(C)=\frac{1}{6}$

As the question is answered by either guessing or copying or knowing

$$P(K)=1-\frac{1}{3}-\frac{1}{6}=\frac{1}{2}$$

Now, $P(A/G)$ = probability of answering correctly by guessing =1/4

[$\because$ there are 4 choices and only one of them is correct]

$P(A/C)$ = probability of answering correctly by copying = 1/8

$P(A/K)$ = probability of answering correctly by knowing = 1

[$\because$ he knows he certainly answers correctly]

We have to find $P(K/A)$, i.e. the probability that he knew when he answered correctly.

By Bayes' theorem,

$$P(K/A)=\frac{P(K)\cdot P(A/K)}{P(G)\cdot P(A/G)+P(C)\cdot P(A/C)+P(K)\cdot P(A/K)}$$

$$=\frac{\frac{1}{2}\cdot 1}{\frac{1}{3}\cdot\frac{1}{4}+\frac{1}{6}\cdot\frac{1}{8}+\frac{1}{2}\cdot 1}$$

$$=\frac{\frac{1}{2}}{\frac{1}{12}+\frac{1}{48}+\frac{1}{2}}=\frac{\frac{1}{2}}{\frac{29}{48}}=\frac{24}{29}$$

13. Five men in a company of 20 are graduates. If three men are picked at random, then the probability that there is at least one graduate is

(a) $1-\left(\frac{{}^{15}C_3}{{}^{20}C_3}\right)$ (b) $\frac{{}^{15}C_3}{{}^{20}C_3}$

(c) $\left(\frac{{}^{15}C_3}{{}^{20}C_3}\right)^2$ (d) none of these

Ans. (a)

Solution: Probability that none is graduate = $\left(\frac{{}^{15}C_3}{{}^{20}C_3}\right)$

$\therefore$ Required probability $=1-\frac{{}^{15}C_3}{{}^{20}C_3}$

14. The probability that a man can hit a target is 3/4. He tries 5 times. The probability that he will hit the target at most one time is

(a) $\left(\frac{1}{4}\right)^3$ (b) $\left(\frac{3}{4}\right)^5$

(c) $\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3$ (d) none of these

Ans. (a)

Solution: $P(A)$ = 3/4

$P(\bar{A})=1/4 \quad n=5$

Let X represent the number of times he hits the targets.

$\therefore \quad P(X=r)={}^5C_r\left(\frac{3}{4}\right)^r\left(\frac{1}{4}\right)^{5-r}$

$\therefore \quad P(X=r\le 1)=P(0)+P(1)$

$$=\left(\frac{1}{4}\right)^5+5\left(\frac{3}{4}\right)^1\left(\frac{1}{4}\right)^4=\left(\frac{1}{4}\right)^4\left[\frac{1}{4}+\frac{15}{4}\right]$$

$$=\left(\frac{1}{4}\right)^3$$

15. Seven chits are numbered 1 to 7. Three are drawn one by one with replacements. The probability that the least number on any selected chit is 5 is

(a) $1-(2/7)^4$ (b) $4\cdot(2/7)^4$

(c) $(3/7)^3$ (d) none of these

Ans. (c)

Solution: Total ways = 7.

In a single try, 5 or 6 or 7 can come in 3 ways. So the probability in a single try is 3/7 when three chits are drawn with replacement.

The required probability = $\left(\frac{3}{7}\right)^3$

since cards are drawn with replacement.

16. If 3 distinct numbers are chosen randomly from {1, 2, ..., 100}, then the probability that all are divisible by both 2 and 3 is

(a) 4/25 (b) 4/35
(c) 4/33 (d) 4/1155

Ans. (d)

Solution: The number of numbers divisible by 6 from the numbers 1 to 100 are 6, 12, ..., 96

or $96 = 6 + (n-1)6$

$\therefore \quad n = 16$

$\therefore$ Probability $= 16\dfrac{^{16}C_3}{^{100}C_3} = \dfrac{4}{1155}$

17. In a box containing 100 bulbs, 10 are defective. What is the probability that out of a sample of 5 bulbs, none is defective (the bulbs are selected with replacement)?

(a) 10^{-5} (b) $\left(\dfrac{1}{2}\right)^5$

(c) $\left(\dfrac{9}{10}\right)^5$ (d) $\dfrac{9}{10}$

Ans. (c)

Solution: Probability for a defective bulb $= \dfrac{10}{100} = \dfrac{1}{10}$

$\therefore$ Probability for a non-defective bulb $= 1 - \dfrac{1}{10} = \dfrac{9}{10}$

$\therefore$ Required probability $= \dfrac{9}{10} \times \dfrac{9}{10} \times \dfrac{9}{10} \times \dfrac{9}{10} \times \dfrac{9}{10} = \left(\dfrac{9}{10}\right)^5$

18. What is the chance of getting multiple of 2 on one and multiple of 3 on the other in a single throw of two dice?

(a) 1/3 (b) 7/36
(c) 11/36 (d) none of these

Ans. (c)

Solution: Total ways = 36

Favourable ways:

(i) Multiple of two on the first die are (2, 4, 6) and multiple of three on the second die are (3, 6), so there are $3 \times 2 = 6$ ways.

(ii) Multiple of three on the first die and multiple of two on the second die. There are 3 × 2 = 6 ways. But one outcome (6, 6) is common in both. So, the total number of favourable ways is 6 + 6 – 1 = 11

$\therefore$ Required probability = 11/36.

EXERCISE SET 1

1. An urn contains 9 red, 7 white and 4 black balls. Out of them, one ball is drawn at random. The probability that the ball drawn will be a red or black is

(a) 9/20 (b) 7/20
(c) 11/20 (d) 13/20

2. A speaks truth 3 times out of 4 while B, 7 times out of 10. A ball is drawn at random from a bag containing one black ball and five other balls of different colours. Both A and B report that a black ball has been drawn from the bag. Find the probability of their assertion being true.

(a) 7/12 (b) 5/12
(c) 3/12 (d) 9/12

3. In a purse there are 9 five-paisa coins and 1 one-rupee coin. In another purse there are all 10 five-paise coins. Nine coins are taken from the former and put into the second and then 9 coins are taken from the latter and put into the first. What is the chance that the one-rupee coin is still in the first purse?

(a) 1/9 (b) 10/19
(c) 5/19 (d) none of these

4. An urn contains 6 white and 4 black balls. A fair die is rolled and that number of balls are chosen from the urn. The probability that the balls selected are white is

(a) 1/5 (b) 1/6
(c) 1/7 (d) 1/8

5. Two numbers a and b are chosen at random from the set of the first 30 natural numbers. The probability that $a^2 - b^2$ is divisible by 3 is

(a) 9/87 (b) 12/87
(c) 15/87 (d) 47/87

6. The probabilities of solving a problem by three students A, B and C are 1/2, 1/3 and 1/4 respectively. The probability that the problem will be solved is

(a) 1/4 (b) 1/2
(c) 3/4 (d) 1/3

7. The probability of India winning a test match against West Indies is 1/2. Assuming independence from match to match, the probability that in a five-match series, India's second win occurs at the third test is

(a) 1/8 (b) 1/4
(c) 1/2 (d) 2/3

8. A lot contains 20 articles. The probability that the lot contains exactly 2 defective articles is 0.4 and the probability that the lot contains exactly 3 defective articles is 0.6. Articles are drawn from the lot at random one by one without replacement and tested till all the defective articles are found. What is the probability that the testing procedure ends at the 12th testing?

(a) 89/1900 (b) 99/1900
(c) 78/1900 (d) 59/1900

9. The probability that a student is not a swimmer is 1/5. Then the probability that out of five students, four are swimmers is

(a) ${}^5C_4\left(\frac{4}{5}\right)^4\frac{1}{5}$ (b) $\left(\frac{4}{5}\right)^4\frac{1}{5}$

(c) ${}^5C_1\frac{1}{5}\left(\frac{4}{5}\right)^4$ (d) none of these

10. One of 10 keys opens the door. If we try the keys one after another, then which of the following is wrong?

(a) The probability that the door is opened on the first attempt is 1/10

(b) The probability that the door is opened on the second attempt is 1/10

(c) The probability that the door is opened on the tenth attempt is 1/10

(d) none of these

11. The probability that when 12 balls are distributed among three boxes, the first will contain three balls is

(a) $\frac{2^9}{3^{12}}$ (b) $\frac{{}^{12}C_3\cdot 2^9}{3^{12}}$

(c) $\frac{{}^{12}C_3\cdot 2^{12}}{3^{12}}$ (d) none of these

12. The probability that certain electronic component fails when used for the first time is 0.10. If it does not fail immediately, the probability that it lasts for one year is 0.99. The probability that a new component will last for one year is

(a) 0.9 (b) 0.1

(c) 0.119 (d) 0.891

13. A die and two coins are tossed. The probability that both the coins show heads and the die shows 1 or 2 is

(a) 7/12 (b) 4/7

(c) 1/12 (d) none of these

14. A fair coin is tossed 100 times. The probability of getting tails 1, 3, ..., 49 times is

(a) 1/2 (b) 1/4

(c) 1/8 (d) 1/16

15. There are two groups of bags. The first group has four bags each containing 6 white and 3 black balls and the second group has three bags each containing 2 white and 4 black balls. From these groups, one bag is selected at random and a ball is drawn from it which is found to be black. The probability of it coming from a bag belonging to the first group is

(a) 1/5 (b) 2/5

(c) 3/5 (d) 2/3

16. In a college, 25% of the boys and 10% of the girls are offered mathematics. The girls constitute 60% of the total number of students. If a student is selected at random and is found to be studying mathematics, the probability that the student is a girl is

(a) 1/6 (b) 3/8

(c) 5/8 (d) 5/6

17. Three ships A, B and C sail from England to India. If the ratio of their arriving safely are 2 : 5, 3 : 7 and 6 : 11 respectively, then the probability of all the ships for arriving safely is

(a) 18/595 (b) 6/17

(c) 3/10 (d) 2/7

18. A student appears for tests I, II and III. The student is successful if he passes in either tests I and II or tests I and III. The probabilities of the student passing in tests I, II and III are respectively, p, q and 1/2. If the probability that the student is successful is 1/2, then $p(1 + q) =$

(a) 1/2 (b) 1

(c) 3/2 (d) 3/4

19. A symmetrical coin is tossed three times. Let A and B be the two events such that

A: First throw is a head

B: Last throw is a tail. Then

(a) A and B are mutually exclusive and independent

(b) A and B are neither mutually exclusive nor independent

(c) A and B are mutually exclusive but not independent

(d) A and B are independent but not mutually exclusive

20. There are 5 address envelopes. Letters are written to corresponding persons but the letters are placed at random in envelopes. The probability that none of the letters enters into the correct envelope is

(a) 1/5! (b) 11/120

(c) 11/30 (d) none of these

21. If birth to a male child and birth to a female child are equal-probable, then what is the probability that at least one of the three children born to a couple is a male?

(a) 4/5 (b) 7/8

(c) 8/7 (d) 1/2

22. A box contains 10 good articles and 6 with defects. If one article is taken out at random, what is the probability that it is either good or has a defect?

(a) 64/64 (b) 48/64

(c) 36/64 (d) 24/64

23. If two events A and B are such that $P(A') = 0.3$, $P(B) = 0.4$, $P(A \cap B') = 0.5$, then $P(B/A \cap B') =$

(a) 0.20 (b) 0.25

(c) 0.30 (d) 0.35

24. One hundred cards are numbered from 1 to 100. The probability that a randomly chosen card has a digit 5 is

(a) 1/100 (b) 9/100

(c) 19/100 (d) none of these

25. Three dice are thrown simultaneously. The probability of getting a sum of 15 is
(a) 5/108 (b) 1/72
(c) 5/72 (d) 5/36

26. A bag contains 5 white and 3 black balls and 4 balls are successively drawn out and not replaced. The probability that they are alternately of different colours is
(a) 1/196 (b) 2/7
(c) 13/56 (d) 1/7

27. Of the cigarette smoking population, 70% are men and 30% are women. 10% of the men and 20% of the women smoke Wills, then the probability that a person smoking a Wills to be a male is
(a) 1/5 (b) 7/13
(c) 5/13 (d) 7/10

28. A and B are two independent witnesses (i.e. there is no collusion between them) in a case. The probability that A will speak the truth is x and the probability that B will speak the truth is y. A and B agree in a certain statement. The probability that the statement is true is
(a) $\frac{x-y}{x+y}$ (b) $\frac{xy}{1+x+y+xy}$
(c) $\frac{x-y}{1-x-y+2xy}$ (d) $\frac{xy}{1-x-y+2xy}$

29. A car is parked by a driver amongst 25 cars in a row, not at either end. When he returns he finds that 10 places are empty. The probability that both the neighbouring places of the driver's car are vacant is
(a) 9/92 (b) 15/92
(c) 21/92 (d) 27/92

30. The probability that out of 5 persons, all having birthday in March, at least two have the same birthday is
(a) $\frac{{}^{31}C_5}{(31)^5}$ (b) $\frac{(31)^5 - {}^{31}C_5}{(31)^5}$
(c) $\frac{(31)^3 \times {}^{31}C_2}{(31)^5}$ (d) none of these

31. What is the probability that when one die is thrown, the number appearing on top is even?
(a) 1/6 (b) 1/3
(c) 1/2 (d) none of these

32. A bag contains 7 red and 2 black balls. Another bag contains 5 red and 3 black balls. If one ball is drawn from each bag, then the probability that the balls are red is
(a) 24/35 (b) 35/72
(c) 45/72 (d) 50/72

33. A card is drawn from a pack of 52 cards. A gambler bets that it is a space or an ace. What are odds against his winning this bet?
(a) 17 : 52 (b) 52 : 17
(c) 9 : 4 (d) 4 : 9

34. The probability of getting 10 in a single throw of three fair dice is
(a) 1/6 (b) 1/8
(c) 1/9 (d) none of these

35. A seven-digit number is formed by using 0, 1, 2, 3, 4, 8, 9 without repetition. Then the probability that it is divisible by 4 is
(a) $\frac{53\times 4!}{5\times 6!}$ (b) $\frac{53}{180}$
(c) $\frac{53}{6\times 6!}$ (d) none of these

36. Among 15 players, 8 are batsmen and 7 are bowlers. The probability that a team is chosen of 6 batsmen and 5 bowlers is
(a) $\frac{{}^8C_6 \times {}^7C_5}{{}^{15}C_{11}}$ (b) $\frac{{}^8C_6 + {}^7C_5}{{}^{15}C_{11}}$
(c) $\frac{15}{28}$ (d) none of these

37. The probability of getting at least one tail in 4 throws of a coin is
(a) 15/16 (b) 1/16
(c) 1/4 (d) 1/8

38. Two cards are drawn from a pack of 52 cards. What is the probability that one of them is a queen and the other is an ace?
(a) 2/663 (b) 2/13
(c) 4/663 (d) 8/663

39. Odds in favour of an event A are 2 to 1 and odds in favour of $A \cup B$ are 3 to 1. Consistent with this information, the smallest and largest values for the probability of event B are given by
(a) $\frac{1}{6} \le P(B) \le \frac{1}{3}$ (b) $\frac{1}{3} \le P(B) \le \frac{1}{2}$
(c) $\frac{1}{12} \le P(B) \le \frac{3}{4}$ (d) none of these

40. Among the divisors of 504, two are chosen randomly, then the probability that the divisors chosen are even is
(a) 3/4 (b) 1/24
(c) 1/18 (d) 51/92

41. A die is rolled three times. The probability of getting a larger number than the previous number each time is
(a) $\frac{15}{216}$ (b) $\frac{5}{54}$
(c) $\frac{13}{216}$ (d) $\frac{1}{18}$

42. Two dice are thrown. The probability that the sum of the points on the two dice will be 7 is
(a) 5/36 (b) 1/6
(c) 3/36 (d) 2/9

43. Two dice are thrown. The number of sample points in the sample space when six does not appear on any one side is
(a) 11 (b) 30
(c) 18 (d) 25

44. A coin is tossed 10 times. The probability that on the 10th throw it is observed 5th head is
(a) ${}^9C_5\frac{1}{2^5}$ (b) $\frac{{}^9C_4}{2^{10}}$
(c) $\frac{{}^9C_5}{2^{10}}$ (d) none of these

45. Twenty-five coins are tossed simultaneously. The probability that the fifth coin will fall with head upwards is
(a) $\frac{5}{25}$ (b) $\frac{5}{2^{25}}$
(c) $\frac{1}{2}$ (d) none of these

46. The probability that Krishna will be alive 10 years hence is 7/15 and the probability that Hari will be alive after 10 years is 7/10. The probability that both Krishna and Hari will be alive 10 years hence is
(a) 21/150 (b) 24/150
(c) 49/150 (d) 56/150

47. Two dice are thrown. What is the probability that the sum of the numbers appearing on the two dice is 11, if 5 appears on the first?
(a) 1/36 (b) 1/6
(c) 5/6 (d) none of these

48. The probability that a teacher will give an unannounced test during any class meeting is 1/5. If a student is absent twice, the probability that he will miss at least one test is
(a) 7/25 (b) 9/25
(c) 16/25 (d) 24/25

49. In a bag there are six balls of unknown colours; three balls are drawn at random and found to be all black. Find the probability that no black ball is left in the bag (or find the probability that the bag contains exactly three black balls).
(a) 1/35 (b) 1/34
(c) 1/40 (d) 1/38

50. The probability that at least one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, then $P(\bar{A})+P(\bar{B})$ is
(a) 0.4 (b) 0.8
(c) 1.2 (d) 1.4

51. In a lottery there were 90 tickets numbered 1 to 90. Five tickets were drawn at random. The probability that two of the tickets drawn were numbered 15 and 89 is
(a) 2/801 (b) 2/623
(c) 1/267 (d) 1/623

52. A six-faced die is so biased that it is twice as likely to show an even number as an odd number when thrown. It is thrown twice. The probability that the sum of two numbers thrown is even is
(a) 1/12 (b) 1/6
(c) 1/3 (d) 5/9

53. The probability that a 50-year-old man will be alive at 60 is 0.83 and the probability that a 45-year-old woman will be alive at 55 is 0.87.
(1) The probability that both will be alive is 0.7221.
(2) At least one of them will be alive is 0.9779.
(a) Only (1) is true
(b) Only (2) is true
(c) Both (1) and (2) are true
(d) Both (1) and (2) are false

54. If two squares are chosen at random on a chess board, the probability that they have a side in common is
(a) 1/9 (b) 1/18
(c) 2/7 (d) none of these

55. An unbiased die with faces marked 1, 2, 3, 4, 5 and 6 is rolled four times. Out of four face values obtained, the probability that the minimum face value is not less than 2 and the maximum face value is not greater than 5 is
(a) 16/81 (b) 1/81
(c) 80/81 (d) 65/81

EXERCISE SET 2

1. If A and B are two independent events, then the value of $P\left(\frac{A}{B}\right)$ is
(a) 0 (b) 1
(c) $P(A)$ (d) $P(B)$

2. Given two events A and B. If odds against A are as 2 : 1 and those in favour of $A \cup B$ are as 3 : 1, then
(a) $\frac{1}{2} \le P(B) \le \frac{3}{4}$ (b) $\frac{5}{12} \le P(B) \le \frac{3}{4}$
(c) $\frac{1}{4} \le P(B) \le \frac{3}{5}$ (d) none of these

3. If $P(A \cap B) = \frac{1}{2}$, $P(\bar{A} \cap \bar{B}) = \frac{1}{3}$, $P(A) = p$, $P(B) = 2p$, then the value of p is given by
(a) 1/3 (b) 7/18
(c) 4/9 (d) 1/2

4. If one card is drawn randomly from a pack of 52 cards, then the probability that it is a king or a spade is
(a) 1/26 (b) 3/26
(c) 4/13 (d) 3/13

5. If A speaks the truth in 75% cases and B in 80% cases, then the probability that they contradict each other, in stating the same statement, is
(a) 7/20 (b) 13/20
(c) 12/20 (d) 2/5

6. There are 20 cards. 10 of these cards have the letter I printed on them and the other 10 have the letter T printed on them. If three cards are picked up at random and kept in the same order, the probability of making the word IIT is
(a) 4/27 (b) 5/38
(c) 1/8 (d) 9/80

7. A coin is tossed 4 times. The probability that at least one head turns up is
(a) 1/16 (b) 2/16
(c) 14/16 (d) 15/16

8. Two dice are thrown twice. Then the probability that at least one of the two throws, one die comes up with the number 4 is
(a) 11/36 (b) 11/42
(c) 13/58 (d) 14/62

9. A mapping is selected at random from the set of all the mappings of the set $A = \{1, 2, \ldots, n\}$ into itself. The probability that the mapping selected is an injection is
(a) $\frac{1}{n^n}$ (b) $\frac{1}{n!}$
(c) $\frac{(n-1)!}{n^{n-1}}$ (d) $\frac{n!}{n^{n-1}}$

10. Two athletes A and B participate in a race along with other athletes. If the chance of A winning the race is 1/6 and that of B winning the same race is 1/8, then the chance that neither wins the race is
(a) 1/4 (b) 7/24
(c) 17/24 (d) 35/48

11. A box contains 24 identical balls of which 12 are white and 12 are black. The balls are drawn at random from the box one at a time with replacement. The probability that a white ball is drawn for the 4th time on the 7th draw is
(a) 5/64 (b) 27/32
(c) 5/32 (d) 1/2

12. A drawer contains 50 bolts and 150 nuts. Half of the bolts and half of the nuts are rusted. If one item is chosen at random, then the probability that it is rusted or a bolt is
(a) 2/5 (b) 3/8
(c) 5/8 (d) 6/13

13. Suppose n (≥ 3) persons are sitting in a row. Two of them are selected at random. The probability that they are not together is
(a) $1 - \frac{2}{n}$ (b) $\frac{2}{n-1}$
(c) $1 - \frac{1}{n}$ (d) none of these

14. An urn contains 9 white and 1 black balls and another urn contains 10 white balls. Nine balls are transferred from the first urn to the second urn and then 9 balls are replaced to the first urn. The probability that the black ball is still in the first urn is
(a) $\frac{{}^{18}C_9 + 9 \times {}^{18}C_8}{{}^{10}C_9 \times {}^{19}C_9}$ (b) $\frac{{}^{19}C_9 + 9 \times {}^{18}C_8}{{}^{10}C_9 \times {}^{19}C_9}$
(c) $\frac{{}^{19}C_9 + 8 \times {}^{19}C_8}{{}^{10}C_9 \times {}^{19}C_9}$ (d) $\frac{{}^{19}C_9 + 8 \times {}^{18}C_8}{{}^{10}C_9 \times {}^{19}C_9}$

15. A bag contains 7 red and 2 white balls and another bag contains 5 red and 4 white balls. Two balls are drawn, one from each bag. The probability that both the balls are white is
(a) 2/9 (b) 2/3
(c) 8/81 (d) 35/81

16. The probability of choosing at random a number that is divisible by 6 or 8 from among 1 to 90 is
(a) 1/6 (b) 1/30
(c) 11/80 (d) 23/90

17. A box contains tickets numbered 1 to 20. Three tickets are drawn from the box with replacement. The probability that the largest number on the tickets is 7 is
(a) $\frac{2}{19}$ (b) $\frac{7}{20}$
(c) $1 - \left(\frac{7}{20}\right)^3$ (d) none of these

18. A biased die is tossed and the respective probabilities for various faces to turn up are

Face	1	2	3	4	5	6
Probability	0.1	0.24	0.19	0.18	0.15	0.14

If an even face has turned up, then the probability that it is face 2 or face 4 is
(a) 0.25 (b) 0.42
(c) 0.75 (d) 0.9

19. A number is chosen at random among the first 120 natural numbers. The probability of the number chosen being a multiple of 5 or 15 is
(a) 1/8 (b) 1/5
(c) 1/24 (d) 1/6

20. Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selects the winning horse is
(a) 4/5 (b) 3/5
(c) 1/5 (d) 2/5

21. A father has 3 children with at least 1 boy. The probability that he has 2 boys and 1 girl is
(a) 1/4 (b) 1/3
(c) 2/3 (d) none of these

22. The probability that out of 10 persons, all born in April, at least two have the same birthday is
(a) $\frac{{}^{30}C_{10}}{(30)^{10}}$ (b) $1-\frac{{}^{30}C_{10}}{30!}$
(c) $\frac{(30)^{10}-{}^{30}C_{10}}{(30)^{10}}$ (d) none of these

23. A four-digit number is formed by the digits 1, 2, 3, 4 with no repetition. The probability that the number is odd is
(a) 0 (b) 1/3
(c) 1/4 (d) none of these

24. If x is a random variable with probability function $P(x) = {}^3C_x\left(\frac{1}{2}\right)^3$, $x = 0, 1, 2, 3$, then the value of $E(x^2 + 2)$ is
(a) 7 (b) 17/4
(c) 5 (d) 4

25. Three cards are drawn from a bag containing n cards marked $1, 2, 3, \ldots, n$. The probability that the three cards form a sequence is
(a) $\frac{6}{n(n-1)}$ (b) $\frac{3}{n(n-1)}$
(c) $\frac{6}{n(n-2)}$ (d) $\frac{3}{n(n-2)}$

26. The probability of throwing 16 in one throw with three dice is
(a) 1/36 (b) 1/18
(c) 1/72 (d) 1/9

27. In a class of 125 students, 70 passed in mathematics, 55 in statistics and 30 in both. The probability that a student selected at random from the class has passed in only one subject is
(a) 13/25 (b) 3/25
(c) 17/25 (d) 8/25

28. If three vertices of a regular hexagon are chosen at random, then the chance that they form an equilateral triangle is
(a) 1/3 (b) 1/5
(c) 1/10 (d) 1/2

29. A fair coin is tossed 100 times. The probability of getting tails on odd number of times is
(a) 1/2 (b) 1/8
(c) 3/8 (d) none of these

30. A bag contains 5 white, 7 black and 4 white balls. Three balls are drawn from the bag at random, then the probability that all the three balls are white is
(a) 3/16 (b) 3/5
(c) 1/60 (d) 1/56

31. Ten apples are distributed at random among 6 persons. The probability that at least one of them will receive none is
(a) $\frac{6}{143}$ (b) $\frac{{}^{14}C_4}{{}^{15}C_5}$
(c) $\frac{137}{143}$ (d) none of these

32. In a box, there are 2 red, 3 black and 4 white balls. Out of these, three balls are drawn together, then the probability of the balls drawn are of the same colour is
(a) 1/84 (b) 1/21
(c) 5/84 (d) 1/27

33. The probability of getting a sum of 12 in four throws of an ordinary die is
(a) $\frac{1}{6}\left(\frac{5}{6}\right)^3$ (b) $\left(\frac{5}{6}\right)^4$
(c) $\frac{1}{36}\left(\frac{5}{6}\right)^2$ (d) none of these

34. A bag contains 5 brown and 4 white socks. A man pulls out two socks. The probability that both the socks are of the same colour is
(a) 9/108 (b) 18/108
(c) 36/108 (d) 48/108

35. Two dice are rolled one after another. The probability that the number on the first is less than or equal to the number on the second is
(a) $\frac{5}{12}$ (b) $\frac{7}{12}$
(c) $\frac{5}{18}$ (d) $\frac{13}{18}$

36. The probability that the 13th day of a randomly chosen month is a Friday is
(a) 1/12 (b) 1/7
(c) 1/84 (d) none of these

37. Out of 40 consecutive natural numbers, two are chosen at random. Then the probability that the sum of these numbers is odd is
(a) 1/2 (b) 14/29
(c) 20/39 (d) 29/31

38. If the letters of the word PROBABILITY are written at random in a row, the probability that the two B's occur consecutively is
(a) $\frac{2}{11}$ (b) $\frac{1}{11}$
(c) $\frac{4}{11}$ (d) $\frac{1}{\underline{11}}$

39. Three identical dice are rolled. The probability that the same number will appear on each of them is
(a) 1/6 (b) 1/36
(c) 1/18 (d) 3/28

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:
(a) Statement I is false, Statement II is true.
(b) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(c) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement II.
(d) Statement I is true, Statement II is false.

40. **Statement I:** If A, B and C are 3 events such that $P(A)=\frac{1}{4}$, $P(B)=\frac{2}{3}$, $P(C)=\frac{1}{3}$, then A, B and C are mutually exclusive events.
Statement II: If $P(A\cup B\cup C)=P(A)+P(B)+P(C)$, then A, B and C are mutually exclusive events.

41. If A and B are two independent events such that
Statement I: $P(A)=\frac{1}{5}$ and $P(B)=\frac{1}{2}$, then $P(B/A)=\frac{1}{2}$.
Statement II: $P(B/A) = P(B)$.

42. Two persons A and B speak the truth with probabilities p and $2p$ respectively.
Statement I: If A and B contradict each other with probability 1/2, then there are two values of p.
Statement II: Every quadratic equation with real coefficients has two real roots.

43. **Statement I:** If 10 coins are thrown simultaneously, then the probability of getting 4 heads is equal to the probability of getting 6 heads.
Statement II: ${}^nC_r = {}^nC_{n-r}$ and $P(H)=P(T)$.

44. **Statement I:** If in sample space two events A and B are such that $P(A) = .5$ and $P(B) = .5$, then $P(A\cap\overline{B})$ can be determined.
Statement II: $P(A\cap\overline{B})=P(A)-P(A\cap B)$.

Previous Years' Questions

45. A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both of them win a prize. The probability that they will not win a prize in a single trial is
(a) 1/25 (b) 24/25
(c) 2/25 (d) none of these
[AIEEE 2002]

46. A problem in mathematics is given to three students A, B and C and their respective probabilities for solving the problem are $\frac{1}{2}, \frac{1}{3}$ and $\frac{1}{4}$. The probability that the problem is solved is
(a) $\frac{3}{4}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$ **[AIEEE 2002]**

47. If A and B are two mutually exclusive events, then
(a) $P(A)<P(\overline{B})$ (b) $P(A)>P(B)$
(c) $P(A)<P(B)$ (d) none of these
[AIEEE 2002]

48. A fair die is tossed eight times. The probability that a third six is observed on the eight throw is
(a) $\frac{{}^7C_2\times 5^5}{6^7}$ (b) $\frac{{}^7C_2\times 5^5}{6^8}$
(c) $\frac{{}^7C_2\times 5^5}{6^6}$ (d) none of these
[AIEEE 2002]

49. In a class of 100 students, there are 70 boys whose average mark in a subject is 75. If the average mark of the whole class is 72, then what is the average mark of the girls?
(a) 73 (b) 65
(c) 68 (d) 79 **[AIEEE 2002]**

50. If A and B are events such that $P(A \cup B) = 3/4$, $P(A \cap B) = 1/4$, $P(\bar{A}) = 2/3$, then $P(\bar{A} \cap B)$ is
(a) 1/4 (b) 5/8
(c) 3/8 (d) 5/12 **[AIEEE 2002]**

51. If events A, B and C are mutually exclusive such that $P(A) = \frac{3x+1}{3}$, $P(B) = \frac{1-x}{4}$, $P(C) = \frac{1-2x}{2}$, then the set of possible values of x is in the interval
(a) $\left[\frac{1}{2}, \frac{2}{3}\right]$ (b) $\left[\frac{1}{3}, \frac{13}{3}\right]$
(c) [0, 1] (d) $\left[\frac{1}{3}, \frac{1}{2}\right]$ **[AIEEE 2003]**

52. If the mean and variance of a random variable X having a binomial distribution are 4 and 2 respectively, then $P(X = 1)$ is
(a) 1/4 (b) 1/16
(c) 1/32 (d) 1/8 **[AIEEE 2003]**

53. If $(1 + 3P)/3$, $(1 - P)/4$ and $(1 - 2P)/2$ are the probabilities of three mutually exclusive events, then the set of all values is
(a) $\frac{1}{3} \le P \le \frac{1}{2}$ (b) $\frac{1}{2} < P < \frac{2}{3}$
(c) $\frac{1}{2} \le P \le \frac{2}{3}$ (d) $\frac{1}{3} < P < \frac{1}{2}$
[AIEEE 2003]

54. If the mean and variance of a B.D. are 4 and 2 respectively, then the probability of 2 successes is
(a) $\frac{128}{256}$ (b) $\frac{219}{256}$
(c) $\frac{37}{256}$ (d) $\frac{28}{256}$ **[AIEEE 2004]**

55. A random variable X has the probability distribution

X	1	2	3	4	5	6	7	8
$P(X)$	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

for the event $E = \{X \text{ is a prime number}\}$ and $F = \{X < 4\}$, then the probability $P(E \cup F)$ is
(a) 0.35 (b) 0.77
(c) 0.87 (d) 0.50 **[AIEEE 2004]**

56. The probability that A speaks the truth is 4/5 and B speaks the truth is 3/4. The probability that they contradict each other is
(a) $\frac{7}{20}$ (b) $\frac{1}{5}$
(c) $\frac{3}{20}$ (d) $\frac{4}{5}$ **[AIEEE 2004]**

57. Let $x_1, x_2, \ldots, x_n$ be n distributions, such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then a possible value of n among the following is
(a) 18 (b) 15
(c) 12 (d) 9 **[AIEEE 2005]**

58. Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is
(a) 1/9 (b) 2/9
(c) 7/9 (d) 8/9 **[AIEEE 2005]**

59. A random variable X has Poisson distribution with mean 2. The $P(X > 1.5)$ equals
(a) 0 (b) $2/e^2$
(c) $3/e^2$ (d) $1 - \frac{3}{e^2}$ **[AIEEE 2005]**

60. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$, where $\bar{A}$ stands for the complement of event A. Then events A and B are
(a) equally likely but not independent
(b) equally likely and mutually exclusive
(c) mutually exclusive and independent
(d) independent but not equally likely **[AIEEE 2005]**

61. At a telephone enquiry system, the number of phone calls regarding relevent enquiry follows the Poisson distribution with average of 5 phone calls during 10-minute time intervals. The probability that there is at the most one phone call during a 10-minute time period is
(a) $\frac{6}{5^e}$ (b) $\frac{5}{6}$
(c) $\frac{6}{55}$ (d) $\frac{6}{e^2}$ **[AIEEE 2006]**

62. The average mark of boys in a class is 52 and that of girls is 42. The average mark of boys and girls combined is 50. The percentage of boys in the class is
(a) 80 (b) 60
(c) 40 (d) 20 **[AIEEE 2007]**

63. Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2 respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane is
(a) 0.14 (b) 0.2
(c) 0.7 (d) 0.06 **[AIEEE 2007]**

64. A pair of two dice is thrown independently three times. The probability of getting a score of exactly 9 twice is
(a) 8/9 (b) 8/729
(c) 8/243 (d) 1/729 **[AIEEE 2007]**

65. It is given that events A and B are such that $P(A) = 1/4$, $P\left(\frac{A}{B}\right)=\frac{1}{2}$, $P\left(\frac{B}{A}\right)=\frac{2}{3}$, then $P(B) =$

(a) 1/3 (b) 2/3
(c) 1/2 (d) 1/6 **[AIEEE 2008]**

66. A die is thrown. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is

(a) 2/5 (b) 3/5
(c) 0 (d) 1 **[AIEEE 2008]**

67. In a binomial distribution B [n, p = (1/4)], if the proba-bility of at least one success is greater than or equal to 9/10, then n is greater than

(a) $\frac{1}{\log_{10}4+\log_{10}3}$ (b) $\frac{9}{\log_{10}4-\log_{10}3}$
(c) $\frac{4}{\log_{10}4-\log_{10}3}$ (d) $\frac{1}{\log_{10}4-\log_{10}3}$

[AIEEE 2009]

68. One ticket is selected at random from 50 tickets numbered 00, 01, 02, ..., 49. Then the probability that the sum of the digits on the selected tickets is 8, given that the product of these digits is zero, equals

(a) 1/7 (b) 5/14
(c) 1/50 (d) 1/14 **[AIEEE 2009]**

69. Three numbers are chosen at random without replacement from {1, 2, 3,..., 8}. The probability that their minimum is 3, given that their maximum is 6, is

(a) 3/8 (b) 1/5
(c) 1/4 (d) 2/5 **[AIEEE 2012]**

ANSWERS

Exercise Set 1

1. (d)	**2.** (a)	**3.** (b)	**4.** (a)	**5.** (d)	**6.** (c)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (d)
11. (b)	**12.** (d)	**13.** (c)	**14.** (b)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (b)	**19.** (d)	**20.** (c)
21. (b)	**22.** (a)	**23.** (b)	**24.** (c)	**25.** (a)	**26.** (d)	**27.** (b)	**28.** (d)	**29.** (b)	**30.** (b)
31. (c)	**32.** (b)	**33.** (c)	**34.** (b)	**35.** (b)	**36.** (a)	**37.** (a)	**38.** (d)	**39.** (c)	**40.** (d)
41. (b)	**42.** (b)	**43.** (d)	**44.** (b)	**45.** (c)	**46.** (c)	**47.** (b)	**48.** (b)	**49.** (a)	**50.** (c)
51. (a)	**52.** (d)	**53.** (c)	**54.** (b)	**55.** (a)					

Exercise Set 2

1. (c)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (b)	**7.** (d)	**8.** (a)	**9.** (c)	**10.** (c)
11. (c)	**12.** (c)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (d)	**17.** (d)	**18.** (c)	**19.** (b)	**20.** (d)
21. (b)	**22.** (c)	**23.** (d)	**24.** (c)	**25.** (a)	**26.** (a)	**27.** (a)	**28.** (c)	**29.** (a)	**30.** (d)
31. (c)	**32.** (c)	**33.** (a)	**34.** (d)	**35.** (b)	**36.** (c)	**37.** (c)	**38.** (a)	**39.** (b)	**40.** (a)
41. (b)	**42.** (d)	**43.** (b)	**44.** (a)	**45.** (b)	**46.** (a)	**47.** (d)	**48.** (b)	**49.** (b)	**50.** (d)
51. (d)	**52.** (c)	**53.** (a)	**54.** (d)	**55.** (b)	**56.** (a)	**57.** (a)	**58.** (a)	**59.** (d)	**60.** (d)
61. (d)	**62.** (a)	**63.** (a)	**64.** (c)	**65.** (a)	**66.** (d)	**67.** (d)	**68.** (d)	**69.** (b)	

HINTS AND SOLUTIONS—EXERCISE SET 1

1. P(Red and Black) = P(Red) + P(Black)

$$=\frac{9}{20}+\frac{4}{20}=\frac{13}{20}$$

2. Let B = the event of a black ball drawn from the bag
T = the event of both A and B speaking the truth
L = the event of both A and B speaking lies

Clearly, $P(B)=\frac{1}{6}$ and $P(\overline{B})=1-\frac{1}{6}=\frac{5}{6}$

Obviously, B and $\overline{B}$ are exhaustive and mutually exclusive.

Now, $P(T/B)$ = probability of both telling the truth when a black ball is drawn

$$=\frac{3}{4}\times\frac{7}{10}$$

[$\because$ probability of A telling the truth = 3/4, probability of B telling the truth = 7/10]

$P(L/\overline{B})$ = probability of both telling lies when a black ball is not drawn

$$=\left(1-\frac{3}{4}\right)\left(1-\frac{7}{10}\right)=\frac{1}{4}\cdot\frac{3}{10}$$

We have to find $P(B/T)$, i.e. the probability that a black ball was drawn when they spoke the truth.

By Bayes' theorem,

$$P(B/T)=\frac{P(B)\cdot P(T/B)}{P(B)\cdot P(T/B)+P(\bar{B})\cdot P(L/\bar{B})}$$

$$=\frac{\frac{1}{6}\cdot\frac{3}{4}\cdot\frac{7}{10}}{\frac{1}{6}\frac{3}{4}\cdot\frac{7}{10}+\frac{5}{6}\cdot\frac{1}{4}\cdot\frac{3}{10}}$$

$$=\frac{21}{21+15}=\frac{21}{36}=\frac{7}{12}$$

3. There are two ways in which the one-rupee coin is still in the first bag

(1) It did not go to the 2nd bag at all.

(2) It went to the 2nd bag and then was put back.

$$P(1)=\frac{1}{{}^{10}C_1}\times\frac{{}^{19}C_9}{{}^{19}C_9}=\frac{1}{10}$$

$$P(2)=\frac{{}^{9}C_8\times{}^{1}C_1}{{}^{10}C_9}\times\frac{{}^{18}C_8\times{}^{1}C_1}{{}^{10}C_9}=\frac{81}{190}$$

$\therefore$ Required probability $=\frac{1}{10}+\frac{81}{190}=\frac{10}{19}$

4. Let E_i denote the event that i appears on the die and let A denote the event that only white balls are drawn.

We have $P(E_i)=\frac{1}{6}$ for i = 1, 2, 3, 4, ..., 6.

Also, $P(A/E_i)=\frac{{}^{6}C_i}{{}^{10}C_i}$

Thus, $P(A)=\sum_{i=1}^{6}P(E_i)P(A/E_i)$

$$=\frac{1}{6}\left[\frac{6}{10}+\frac{15}{45}+\frac{20}{120}+\frac{15}{210}+\frac{6}{252}+\frac{1}{210}\right]=\frac{1}{5}$$

5. The total number of ways of choosing two numbers out of 1, 2, 3, ..., 30 is ${}^{30}C_2=435$

So, exhaustive number of cases = 435

Since $a^2 - b^2$ is divisible by 3, either a or b or both are divisible by 3 or none of a and b is divisible by 3. Thus, the favourable number of cases

$$={}^{10}C_2+{}^{20}C_2=235$$

Hence, the required probability $=\frac{235}{435}=\frac{47}{87}$.

6. $P(A)=\frac{1}{2}P(\bar{A})=\frac{1}{2}$

$P(B)=\frac{1}{3}P(\bar{B})=\frac{2}{3}$

$P(C)=\frac{1}{4}P(\bar{C})=\frac{3}{4}$

Problem will be solved even if one of them solves it; so frist we calculate the probability that it is not solved.

$$P(\bar{A})\times P(\bar{B})\times P(\bar{C})=\frac{1}{2}\times\frac{2}{3}\times\frac{3}{4}=\frac{1}{4}$$

$\therefore$ The required probability $=1-\frac{1}{4}=\frac{3}{4}$.

7. P(India) = $\frac{1}{2}$. Since India's second win occurs at the third test,

$\therefore$ the first win will occur in the first or the second test.

$\therefore$ possible cases are WLW or LWW (W for win, L for lose)

$\therefore$ Out of 3 tests, India will win twice.

$\therefore$ Required probability = 2/3.

8. Let E_2 = event of containing exactly 2 defective articles

E_3 = event of containing exactly 3 defective articles

E = event of getting the last defective at the 12th testing

From the question, $P(E_2)$ = 0.4 and $P(E_3)$ = 0.6

$\therefore$ $P(E_2)+P(E_3)=0.4+0.6=1$

So, E_2 and E_3 are exhaustive.

Also, E_2 and E_3 are mutually exclusive.

$\therefore$ The required probability

$$P(E)=P(E_2)\cdot P(E/E_2)+P(E_3)\cdot P(E/E_3) \quad \text{(i)}$$

Now, $P(E/E_2)$ = probability of getting one defective and 10 non-defective in the first 11 draws and the last defective in the 12th draw

$$=\frac{{}^{2}C_1\times{}^{18}C_{10}}{{}^{20}C_{11}}\times\frac{1}{9}=2\times\frac{18!}{10!8!}\times\frac{11!9!}{20!}\times\frac{1}{9}$$

$$=2\times\frac{11\times9}{20\times19}\times\frac{1}{9}=\frac{11}{190}$$

$P(E/E_3)$ probability of getting two defective and 9 non-defective in the first 11 draws and the last defective in the 12th draw

$$=\frac{{}^{3}C_2\times{}^{17}C_9}{{}^{20}C_{11}}\times\frac{1}{9}$$

$$=3\times\frac{17!}{9!8!}\times\frac{11!9!}{20!}\times\frac{1}{9}=\frac{3\times11\times10\times9}{20\times19\times18}\times\frac{1}{9}=\frac{11}{228}$$

$\therefore$ By (i), the required probability

$$=\frac{2}{5}\times\frac{11}{190}+\frac{3}{5}\times\frac{11}{228}$$

$$=\frac{11}{5}\left(\frac{1}{95}+\frac{1}{76}\right)=\frac{11}{5}\times\frac{45}{1900}=\frac{99}{1900}$$

9. $P(\text{Student is a swimmer}) = \dfrac{4}{5}$

$P(\text{Student is not a swimmer}) = \dfrac{1}{5}$

So the required probability $= {}^5C_4\left(\dfrac{4}{5}\right)^4\dfrac{1}{5}$.

10. Since out of 10 keys only one can open the door, so the door will open in the first attempt with probability 1/10. Now, if he fails in the first attempt which has the probability 9/10, then he will attempt next time with different keys. So, the probability that the door will open in the second attempt is $\dfrac{9}{10}\times\dfrac{1}{9}=\dfrac{1}{10}$.

The probability that the door will open only in the tenth attempt is equal to the probability that the door not opened in the first nine attempts which is equal to

$$\frac{9}{10}\times\frac{8}{9}\times\frac{7}{8}\times\cdots\times\frac{2}{3}\times\frac{1}{2}=\frac{1}{10}$$

(As each time he tries with a new key and keeps away the key which does not open the door.)

11. Generally each ball has 3 ways of placement in any one of the three boxes and there are 12 balls. Once 3 balls are in the first box, there are only 2 boxes for the remaining 9 balls.

$\therefore$ Required probability $= \dfrac{{}^{12}C_3\times 2^9}{3^{12}}$

12. $P(\text{the electronic component fails when first used}) = P(F) = 0.10$

$\therefore \quad P(F') = 1 - P(F) = 0.90$

Let E be the event that a new component will last for one year, then

$$P(F) = P(F)\cdot P\left(\frac{E}{F}\right) + P(E')P\left(\frac{E}{F'}\right)$$

[total probability theorem]

$$= 0.10\times 0 + 0.90\times 0.99 = 0.891$$

13. Probability of getting head on the coin $= \dfrac{1}{2}$

Probability of getting 1 or 2 on die $= \dfrac{2}{6}$

$\therefore$ Probability of getting heads on both coins and 1 or 2 on the die is $= \dfrac{1}{2}\times\dfrac{1}{2}\times\dfrac{2}{6}=\dfrac{1}{12}$

14. Let p = probability of getting a tail in a single trial = 1/2,
n = number of trials = 100
and X = number of trials in 100 trials

We have $P(X=r) = {}^{100}C_r p^r q^{n-r} = {}^{100}C_r\left(\dfrac{1}{2}\right)^r\left(\dfrac{1}{2}\right)^{100-r}$

$$= {}^{100}C_r\left(\frac{1}{2}\right)^{100}$$

Now, $P(X=1)+P(X=3)+\cdots+P(X=49)$

$$= {}^{100}C_1\left(\frac{1}{2}\right)^{100} + {}^{100}C_3\left(\frac{1}{2}\right)^{100} + \cdots + {}^{100}C_{49}\left(\frac{1}{2}\right)^{100}$$

$$= ({}^{100}C_1 + {}^{100}C_3 + \cdots + {}^{100}C_{49})\left(\frac{1}{2}\right)^{100}$$

But ${}^{100}C_1 + {}^{100}C_3 + \cdots + {}^{100}C_{99} = 2^{99}$

Also, ${}^{100}C_{99} = {}^{100}C_1 = {}^{100}C_{97} = {}^{100}C_3, \ldots, {}^{100}C_{51} = {}^{100}C_{49}$

Thus, $2({}^{100}C_1 + {}^{100}C_3 + \cdots + {}^{100}C_{49}) = 2^{99}$

$\Rightarrow \quad {}^{100}C_1 + {}^{100}C_3 + \cdots + {}^{100}C_{49} = 2^{98}$

Therefore, probability of the required event $= \dfrac{2^{98}}{2^{100}} = \dfrac{1}{4}$

15. B_1 = a bag is selected from the first group
B_2 = a bag is selected from the second group
A = a black ball is drawn

$\therefore \quad P(B_1)=\dfrac{4}{7}, P(B_2)=\dfrac{3}{7}$

Also, $P\left(\dfrac{A}{B_1}\right)=\dfrac{3}{9}=\dfrac{1}{3}, P\left(\dfrac{A}{B_2}\right)=\dfrac{4}{6}=\dfrac{2}{3}$

$\therefore$ Required probability $= P\left(\dfrac{B_1}{A}\right)$

$$\Rightarrow \quad P(E) = \frac{P(B_1)P\left(\dfrac{A}{B_1}\right)}{P(B_1)\cdot P\left(\dfrac{A}{B_1}\right) + P(B_2)\cdot P\left(\dfrac{A}{B_2}\right)}$$

$$\therefore \quad P(E) = \frac{\dfrac{4}{7}\times\dfrac{1}{3}}{\dfrac{4}{7}\times\dfrac{1}{3}+\dfrac{3}{7}\times\dfrac{2}{3}} = \frac{4}{10}=\frac{2}{5}$$

16. Let the total number of students be 100 in which 60% are girls and 40% are boys.
Number of boys = 40, number of girls = 60
25% of boys are offered mathematics

$$= \frac{25}{100}\times 40 = 10 \text{ boys}$$

10% of girls are offered mathematics

$$= \frac{10}{100}\times 60 = 6 \text{ girls}$$

It means, 16 students are offered mathematics.

$\therefore$ Required probability $= \frac{6}{16} = \frac{3}{8}$

17. Ratios of the ships A, B and C for arriving safely = 2 : 5, 3 : 7 and 6 : 11.

We know that probability of ship A for arriving safely

$$= \frac{2}{2+5} = \frac{2}{7}$$

Similarly, probability of ship B for arriving safely

$$= \frac{3}{3+7} = \frac{3}{10}$$

and probability of ship C for arriving safely

$$= \frac{6}{6+11} = \frac{6}{17}$$

Therefore, probability of all the ships for arriving safely

$$= \frac{2}{7} \times \frac{3}{10} \times \frac{6}{17} = \frac{18}{595}$$

18. If A, B and C represent events that the student is successful in tests I, II, III, respectively, then P (the student is successful) =

$$P[(A \cap B \cap C') \cup (A \cap B' \cap C) \cup (A \cap B \cap C)]$$

$$= P(A \cap B \cap C') + P(A \cap B' \cap C) + P(A \cap B \cap C)$$

$$= P(A)P(B)P(C') + P(A)P(B')P(C) + P(A)P(B)P(C)$$

[$\therefore$ A, B, C are independent events]

$$= pq\left(1 - \frac{1}{2}\right) + p(1-q)\frac{1}{2} + pq\frac{1}{2} = pq + \frac{1}{2}p - \frac{1}{2}pq$$

$$= \frac{1}{2}(pq + p)$$

$$\therefore \quad \frac{1}{2}p(1+q) = \frac{1}{2} \quad \Rightarrow \quad p(1+q) = 1$$

19. The relevant sample space is

$A = \{HHT, HTH, HTT, HHH\}$ and

$B = \{HHT, HTT, THT, TTT\}$

Now $(A \cap B) = \{HHT, HTT\}$

$$\therefore \quad P(A) = \frac{1}{2}, P(B) = \frac{1}{2}, P(A \cap B) = \frac{1}{4}$$

$$\therefore \quad P(A \cap B) = P(A) \cdot P(B)$$

$\Rightarrow$ A and B are independent

Also, since $(A \cap B)$ is not a null set,

$\therefore$ A and B are non-mutually exclusive events.

20. Number of ways of addressing the n letters = $n!$

Number of ways no one letter will go to the right envelope is

$$n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right]$$

Required probability

$$= n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right] \times \frac{1}{n!} \quad [\text{putting } n = 5]$$

$$\therefore \quad \text{Required probability} = \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120}$$

$$= \frac{1}{120}(60 - 20 + 5 - 1) = \frac{44}{120} = \frac{11}{30}$$

21. $S = \{BBB, BBG, BGB, GBB, GBG, BGG, GGG\}$

and $E = \{BBB, BBG, BGB, GBB, GGB, GBG, BGG\}$

$n(E) = 7$ and $n(S) = 8$

$$P(E) = \frac{n(E)}{n(S)} = \frac{7}{8}$$

22. Number of good articles $n(G)$ = 10 and number of defected articles $n(D)$ = 6. Therefore, total number of articles $n(T) = 10 + 6 = 16$

We know that the probability that the taken out article is good, $P(G) = \frac{n(G)}{n(T)} = \frac{10}{16}$

Similarly, the probability that the taken out article is defective, $P(D) = \frac{n(D)}{n(T)} = \frac{6}{16}$

We also know that the probability of the article being either good or defective, $(P) = P(G) + P(D)$

$$= \frac{10}{16} + \frac{6}{16} = \frac{16}{16} = \frac{64}{64}$$

23. We have $P(A \cap B') = P(A) - P(A \cap B)$

$$\Rightarrow \quad P(A \cap B) = P(A) - P(A \cap B') = 0.7 - 0.5 = 0.2$$

Now, $P(A \cap B') = P(A) + P(B') - P(A \cap B')$

$$= 0.7 + 0.6 - 0.5 = 0.8$$

We have

$$P[B \cap (A \cup B')] = P[(B \cap A) \cup (B \cap B')] = P(A \cap B)$$

$$\Rightarrow \quad P(A \cup B')P(B/A \cup B') = P(A \cap B)$$

$$\Rightarrow \quad 0.8P(B/A \cup B')$$

$$= 0.2 \Rightarrow P(B/A \cup B') = \frac{0.2}{0.8} = 0.25$$

24. There are 10 numbers from 50 to 59 such that each has a digit 5 and there are 9 other numbers, 5, 15, 25, 35, 45, 65, 75, 85, 95 each containing a digit 5. Thus, the favourable number of cases is 19. The exhaustive number of cases is 100.

Hence, the required probability = 19/100.

25. $n(S) = 6^3 = 216$

$n(E)$ = sum of three numbers = $x + y + z = 15$

where $1 \le x \le 6, 1 \le y \le 6, 1 \le z \le 6$

$n(E)$ = coefficient of x^{15} in $(x + x^2 + \cdots + x^6)^3$

= coefficient of x^{12} in $(1 + x + \cdots + x^5)^3$

= coefficient of x^{12} in $\left(\frac{1-x^6}{1-x}\right)^3$

= coefficient of x^{12} in $(1 - x^6)^3 (1 - x)^{-3}$

= coefficient of x^{12} $(1 - 3x^6 + 3x^{12} - x^{18})$

$({}^2C_0 x^0 + {}^3C_1 x^1 + {}^4C_2 x^2 + \cdots + {}^8C_6 x^6 + \cdots + {}^{14}C_{12} x^{12})$

$= {}^2C_0 \times 3 + {}^8C_6 \times (-3) + {}^{14}C_{12}$

$= 3 + \frac{8}{2} \times \frac{7}{1} \times (-3) + \frac{14}{2} \times \frac{13}{1}$

$= 3 - 84 + 91$

$= 10$

$\therefore$ Required probability $= \frac{10}{216} = \frac{5}{108}$

26. Required probability $= P(WBWB) + P(BWBW)$

$$= \left(\frac{{}^5C_1 \times {}^3C_1 \times {}^4C_1 \times {}^2C_1}{{}^8C_1 \times {}^7C_1 \times {}^6C_1 \times {}^5C_1}\right) + \left(\frac{{}^5C_1 \times {}^3C_1 \times {}^4C_1 \times {}^2C_1}{{}^8C_1 \times {}^7C_1 \times {}^6C_1 \times {}^5C_1}\right)$$

$$= \frac{1}{14} + \frac{1}{14} = \frac{2}{14} = \frac{1}{7}$$

27. Number of persons = 100

Number of men smoking cigarette = 70

Number of women smoking cigarette = 30 and 10% of men out of 70 are smoking Wills. And 20% of women out of 30 are smoking Wills.

We know that the number of men smoking Wills $= \frac{70 \times 10}{100} = 7$

And the number of women smoking Wills $= \frac{30 \times 20}{100} = 6$

We also know that the number of persons smoking Wills $= 7 + 6 = 13$

Therefore, the probability that a person smoking a Wills to be a male is $= \frac{7}{13}$.

28. A and B will agree in a certain statement if both speak the truth or both tell a lie. We define the following events:

E_1 = A and B both speak the truth $\Rightarrow P(E_1) = xy$

E_2 = A and B both tell a lie $\Rightarrow P(E_2) = (1 - x)(1 - y)$

E = A and B agree in a certain statement

Clearly, $P(E/E_1) = 1$ and $P(E/E_2) = 1$

The required probability is $P(E_1/E)$. Using Bayes' theorem,

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2)}$$

$$= \frac{xy \times 1}{xy \times 1 + (1 - x)(1 - y)1} = \frac{xy}{1 - x - y + 2xy}$$

29. Out of 24 places (except driver's car), 10 are vacant. 10 vacant places out of 24 can be selected in ${}^{24}C_{10}$ ways. If the neighbouring places on either side of the driver's car are vacant, then out of the remaining 22 places, 8 are vacant which can be selected in ${}^{22}C_8$ ways.

$\therefore$ Required probability $= \frac{{}^{22}C_8}{{}^{22}C_{10}} = \frac{15}{92}$

30. Number of days in the month of March = 31

$\therefore$ $n(S) = (31)^5$

$n(A)$ = at least two have the same birthday

= $n(S)$ – number of ways in which all candidates have the different birthday

$= (31)^5 - {}^{31}C_5$

$\therefore$ $P(A) = \frac{(31)^5 - {}^{31}C_5}{(31)^5}$

31. Number of favourable cases = 3

Required probability $= \frac{3}{6} = \frac{1}{2}$

32. In the first bag, the number of red balls = 7 and the number of black balls = 2 and in the second bag, the number of red balls = 5 and the number of black balls = 3

We know that the total balls in the first bag = 7 + 2 = 9

And the total balls in the second bag = 5 + 3 = 8

We also know that the probability of drawing one red ball from the first bag = $\frac{7}{9}$

Similarly, the probability of drawing one red ball from the second bag = $\frac{5}{8}$

Therefore, the probability that both balls are red

$$= \frac{7}{9} \times \frac{5}{8} = \frac{35}{72}$$

33. Probability of the card being a space or an ace $= \frac{16}{52} = \frac{4}{13}$

Hence, odds in favour is 4 : 9,

so that odds against his winning is 9 : 4.

34. With three dice, the cases favourable for the event of getting 10 are

$$\left.\begin{matrix}1+3+6\\1+4+5\\2+3+5\end{matrix}\right\}\times 3!+\left.\begin{matrix}2+2+6\\2+4+4\\3+3+4\end{matrix}\right\}\times\frac{3!}{2!}$$

$\Rightarrow\ 3\times 3!+3\times\frac{3!}{2!}=18+9=27$ ways

Hence, the required probability $=\frac{27}{6^3}=\frac{1}{8}$

35. Two or more than two-digit numbers will be divisible by 4 if its last two digits are divisible by 4, so we need the numbers of the form 04, 08, 20, 24, 28, 32, 40, 80, 84, 92. There are twelve cases out of which 5 cases are like 04, 08, 20, 40, 80. Total digits are $=5\times 5!$

and numbers in other 7 cases are $=4\times 4!\times 7$

Number of favourable cases are $=5\times 5!+4\times 7\times 4!$

$=(25+28)4!$

$=53\times 4!$

Total number of 7-digit numbers $=7!-6!$

$=6\cdot 6!$

$\therefore$ Required probability $=\frac{53\times 4!}{6\times 6\times 5\times 4!}=\frac{53}{180}$

36. Total number of ways of selecting 11 players $={}^{15}C_{11}$

Favourable cases $={}^{8}C_{6}\times{}^{7}C_{5}$

$\therefore$ Required probability $=\frac{{}^{8}C_{6}\times{}^{7}C_{5}}{{}^{15}C_{11}}$

37. Number of die = 1

We know that the probability of getting head (not tail) on coin $=\frac{1}{2}$

We also know that the probability of getting head in 4 throws $=\left(\frac{1}{2}\right)^4=\frac{1}{16}$

Therefore, the probability of getting at least one tail in 4 throws $=1-\frac{1}{16}=\frac{15}{16}$

38. Required probability $=\frac{4\cdot 4}{52\cdot 51}\times 2=\frac{8}{663}$ (As it can be, the first card is queen and the 2nd card is ace or the 1st card is ace and the 2nd card is queen)

39. Since $P(A)=\frac{2}{3}, P(A\cup B)=\frac{3}{4}$ (given)

$\therefore\ P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$\Rightarrow\ \frac{3}{4}=\frac{2}{3}+P(B)-P(A\cap B)\ \Rightarrow\ P(A\cap B)=P(B)-\frac{1}{12}$

Now, $A\cap B\subset A$

$\Rightarrow\ P(A\cap B)\le P(A)\ \Rightarrow\ P(B)-\frac{1}{12}\le\frac{2}{3}$

$\Rightarrow\ P(B)\le\frac{3}{4}$

Now, $P(A\cap B)\ge 0$

$\Rightarrow\ P(B)-\frac{1}{12}\ge 0\ \Rightarrow\ P(B)\ge\frac{1}{12}$

Hence, $\frac{1}{12}\le P(B)\le\frac{3}{4}$

40. $504=2^3 3^2 7^1$

$\therefore$ Total number of divisors $=(3+1)(2+1)(1+1)$

$=4\times 3\times 2$

$=24$

$\therefore\ n(S)={}^{24}C_2$

Let A denote even divisors which are

$=3\ (2+1)\ (1+1)$

$=3\times 3\times 2$

$=18$

$\therefore\ n(A)={}^{18}C_2$

$\therefore\ P(A)=\frac{{}^{18}C_2}{{}^{24}C_2}=\frac{18}{24}\times\frac{17}{23}=\frac{51}{92}$

41. The total number of ways $=6^{3=216}$

If the second number is i (>1), then the total number of favourable ways

$$=\sum_{i=1}^{5}(i-1)(6-i)=20$$

$\therefore$ Required probability $=\frac{20}{216}=\frac{5}{54}$

42. Number of dice = 2

We know that the total the number of all possible outcomes $=6\times 6=36$

And favourable outcomes are (1, 6) (2, 5) (3, 4) (4, 3) (5, 2) (6, 1) or the number of favourable outcomes = 6.

Therefore, the probability of getting a sum of 7 in both dice $(P)=\frac{6}{36}=\frac{1}{6}$

43. The number of sample points $=5\times 5=25$

44. $P(H)=1/2=P(T)$

Here in the 10th throw, 5th head appears which means in nine throws 4 times head appear.

$\therefore$ Required probability $={}^{9}C_4\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^1$

$={}^{9}C_4\frac{1}{2^{10}}=\frac{{}^{9}C_4}{2^{10}}$

45. There is no condition on other coins. Only the 5th coin will fall head upwards. So, probability = 1/2.

46. The probability that both Krishna and Hari will be alive after 10 years is

$$\frac{7}{15}\times\frac{7}{10}=\frac{49}{150}$$

47. Since we are given that 5 appears on the first die, so to get the sum 11, six must be on the second and hence, the required probability = 1/6.

48. **D I** Probability of student (if he misses no test)

$$P(S)=\left(1-\frac{1}{5}\right)\times\left(1-\frac{1}{5}\right)=\frac{16}{25}$$

Hence, probability that he will miss at least one test

$$=1-\frac{16}{25}=\frac{9}{25}$$

D II The probability that one test is held

$$=2\times\frac{1}{5}\times\frac{4}{5}=\frac{8}{25}$$

Probability that the test is held on both days $=\frac{1}{5}\times\frac{1}{5}=\frac{1}{25}$

Thus, the probability that the student misses at least one test $=\frac{8}{25}+\frac{1}{25}=\frac{9}{25}$

49. Let E_3 = the event of the bag containing 3 black balls and 3 balls of other colours

E_4 = the event of the bag containing 4 black balls and 2 balls of other colours

E_5 = the event of the bag containing 5 black balls and 1 ball of another colour

E_6 = the event of the bag containing 6 black balls

Clearly, they are equally probable and no other possibilities exist as 3 balls drawn at random are black which means the bag has at least 3 black balls.

$$\therefore\quad P(E_1)=P(E_2)=P(E_3)=\frac{1}{4}\qquad[\because \text{their sum} = 1]$$

Let B = the event of drawing 3 black balls.

Now, $P(B/E_3)$ = probability of drawing 3 black balls when E_3 happens

$$=\frac{{}^3C_3}{{}^6C_3}=\frac{1}{20}$$

Similarly, $P(B/E_4)=\frac{{}^4C_3}{{}^6C_3}=\frac{1}{5}$

$$P(B/E_5)=\frac{{}^5C_3}{{}^6C_3}=\frac{1}{2}$$

$$P(B/E_6)=\frac{{}^6C_3}{{}^6C_3}=1$$

The bag will contain no black ball after drawing 3 black balls if B took place in conjunction with E_3.

So, we have to find $P(E_3/B)$.

By Bayes' theorem for equally probable events,

$$P(E_3/B)=\frac{P(B/E_3)}{P(B/E_3)+P(B/E_4)+P(B/E_5)+P(B/E_6)}$$

$$=\frac{\frac{1}{20}}{\frac{1}{20}+\frac{1}{5}+\frac{1}{2}+1}=\frac{\frac{1}{20}}{\frac{35}{20}}=\frac{1}{35}$$

50. It is given that $P(A\cup B)=0.6$ and $P(A\cap B)=0.2$

Therefore, $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$\Rightarrow\quad 0.6=P(A)+P(B)-0.2\Rightarrow P(A)+P(B)=0.8$

$\Rightarrow\quad 1-P(\overline{A})+1-P(\overline{B})=0.8\Rightarrow P(\overline{A})+P(\overline{B})=1.2$

51. Out of 90 tickets, two tickets have already been considered, instead of selecting 5 tickets we have to select only 3 tickets out of 88 tickets.

$\therefore$ Required probability $=\frac{{}^{88}C_3}{{}^{90}C_5}$

$$=\frac{\frac{88\times87\times86}{3\times2\times1}}{\frac{90\times89\times88\times87\times86}{5\times4\times3\times2\times1}}$$

$$=\frac{5\times4}{90\times89}=\frac{2}{801}$$

52. Let the probability for appearance of odd number be p.

$\therefore$ Probability for appearance of even number is $2p$.

$$\therefore\qquad p+2p=1\Rightarrow p=\frac{1}{3}$$

Sum of two numbers is even means both numbers are either even or odd

$\therefore$ Required probability $=\frac{1}{3}\times\frac{1}{3}+\frac{2}{3}\times\frac{2}{3}=\frac{5}{9}$

53. (1) The probability that both will be alive 10 years hence = The probability that the man and his wife both will be alive 10 years. Hence

$$= 0.83 \times 0.87 = 0.7221$$

(2) The probability that at least one of them will be alive $= 1 - p$ (That none of them remains alive 10 years hence)

$$=1-(1-0.83)(1-0.87)=1-0.17\times0.13$$

$$=0.9779$$

54. The total number of cases = 64×63

2	3	3	3	3	3	3	2
3	4	4	4	4	4	4	3
3	4	4	4	4	4	4	3
3	4	4	4	4	4	4	3
3	4	4	4	4	4	4	3
3	4	4	4	4	4	4	3
3	4	4	4	4	4	4	3
2	3	3	3	3	3	3	2

Fig. 26.4

For each of the four corner squares, the favourable number of cases is 2.

For each of the 24 non-corner squares on either side of the chess board, the favourable number of cases is 3.

For each of the 36 remaining squares, the favourable number of cases is 4.

Thus, the total number of favourable cases

$$= 4\times 2 + 24\times 3 + 36\times 4 = 224$$

Hence, the required probability $= \dfrac{224}{64\times 63} = \dfrac{1}{18}$

55. Let A = getting not less than 2 and not greater than 5.

$\Rightarrow$ $A = \{2, 3, 4, 5\}$

$\Rightarrow$ $P(A) = \dfrac{4}{6}$

But the die is rolled four times, therefore the probability in getting four throws

$$= \left(\frac{4}{6}\right)\left(\frac{4}{6}\right)\left(\frac{4}{6}\right)\left(\frac{4}{6}\right) = \frac{16}{81}$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. Two events = A and B

Since A and B are independent events, therefore $P(A \cap B) = P(A) \cdot P(B)$

We know that

$$P\left(\frac{A}{B}\right) = \frac{P(A\cap B)}{P(B)} = \frac{P(A)\cdot P(B)}{P(B)} = P(A)$$

2. $P(A) = \dfrac{1}{3}, P(A \cup B) = \dfrac{3}{4}$

Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B) \le P(A) + P(B)$

$\Rightarrow$ $\dfrac{3}{4} \le \dfrac{1}{3} + P(B) \Rightarrow \dfrac{5}{12} \le P(B)$

Also, $B \subseteq A \cup B \Rightarrow P(B) \le P(A \cup B) = \dfrac{3}{4}$

$\therefore$ $\dfrac{5}{12} \le P(B) \le \dfrac{3}{4}$

3. Since $P(\bar{A} \cap \bar{B}) = \dfrac{1}{3}$,

$\Rightarrow$ $P(\overline{A \cup B}) = \dfrac{1}{3}$ $\Rightarrow$ $1 - P(A \cup B) = \dfrac{1}{3}$

$\Rightarrow$ $P(A \cup B) = \dfrac{2}{3}$ $\Rightarrow$ $P(A) + P(B) - P(A \cap B) = \dfrac{2}{3}$

$\Rightarrow$ $p + 2p - \dfrac{1}{2} = \dfrac{2}{3}$ $\Rightarrow$ $p = \dfrac{7}{18}$

4. Probability of getting a king $= \dfrac{4}{52} = \dfrac{1}{13}$

Probability of getting a spade $= \dfrac{13}{52} = \dfrac{1}{4}$

Probability of getting a king and a spade $= \dfrac{1}{52}$

$\therefore$ P(king or spade) = P(king) + P(spade) $-P$(king and spade)

$$= \frac{1}{13} + \frac{1}{4} - \frac{1}{52} = \frac{4+13-1}{52} = \frac{16}{52} = \frac{4}{13}$$

5. A speaking the truth = 75% and

B speaking the truth = 80%

We know that probability of A speaking the truth

$$P(A) = \frac{75}{100}$$

and probability of A speaking the false

$$P(\bar{A}) = 1 - \frac{75}{100} = \frac{25}{100}$$

Similarly, probability of B speaking the truth $P(B) = \dfrac{80}{100}$

And probability of B speaking the false

$$P(\bar{B}) = 1 - \frac{80}{100} = \frac{20}{100}$$

We also know that the probability of contradiction (A speaking the truth and B speaking the false or B speaking the truth and A speaking the false)

$$= P(A)\times P(\bar{B}) + P(B)\times P(\bar{A})$$

$$= \left(\frac{75}{100}\times\frac{20}{100}\right) + \left(\frac{80}{100}\times\frac{25}{100}\right) = \frac{35}{100} = \frac{7}{20}$$

6. Consider the following events:

$A \rightarrow$ getting a card with mark I in the first draw

$B \rightarrow$ getting a card with mark I in the second draw

$C \rightarrow$ getting a card with mark T in the third draw

Then, the required probability $= P(A \cap B \cap C)$

$$\Rightarrow \quad P(E) = P(A)P\left(\frac{B}{A}\right)P\left(\frac{C}{A \cap B}\right)$$

$$\therefore \quad P(E) = \frac{10}{20} \times \frac{9}{19} \times \frac{10}{18} = \frac{5}{38}$$

7. The probability of getting head and tail in one toss is $\frac{1}{2}$.

P(at least one H) $= 1 - P$(no head in four toss)

$= 1 - P$(four tails)

$$= 1 - \left(\frac{1}{2}\right)^4 = 1 - \frac{1}{16} = \frac{15}{16}$$

8. Given: Number of dice = 2 and number of throws = 2

We know that total events in dice throw $n(S) = 36$

We also know that events in which the first thrown die shows 4,

$(A) = \{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\}$

or $n(A) = 6$

And events in which the second thrown die shows 4,

$(B) = \{1, 4\}, (2, 4), (3, 4), (4, 4), (5, 4), (6, 4)$.

Similarly, getting 4 in each throw

$$(A \cap B) = \{(4, 4)\} \quad \text{or} \quad n(A \cap B) = 1$$

Probability of the first throw showing 4, $P(A) = \frac{6}{36}$

And probability of the second throw showing 4,

$$P(B) = \frac{6}{36} \quad \text{and} \quad P(A \cap B) = \frac{1}{36}$$

We also know that from the addition theorem of probability that at least one of the two throws shows 4

$$P(A \cup B) = P(A) + P(B) - (A \cap B)$$

$$= \frac{6}{36} + \frac{6}{36} - \frac{1}{36} = \frac{11}{36}$$

9. Total number of functions from A to itself $= n^n$

Out of these functions, $n!$ functions are injective mappings (one-one and onto)

So, the required probability $= \frac{n!}{n^n} = \frac{(n-1)!}{n^{n-1}}$

10. Chance of A winning the race $= P(A) = \frac{1}{6}$ and

chance of B winning the race $= P(B) = \frac{1}{8}$

$\therefore$ Chance of winning the race $= \frac{1}{6} + \frac{1}{8} = \frac{14}{48} = \frac{7}{24}$

Hence, chance that neither wins the race $= 1 - \frac{7}{24} = \frac{17}{24}$

11. Probability of getting a white ball at any draw is

$$P = \frac{12}{24} = \frac{1}{2}$$

The probability of getting a white ball 4th time in the 7th draw

= P(getting 3 white balls in 6 draws)

$\times P$(white ball at the 7th draw)

$$= {}^6C_3\left(\frac{1}{2}\right)^6 \cdot \frac{1}{2} = \frac{20}{2^7} = \frac{5}{32}$$

12. Given: Number of bolts = 50 and number of nuts = 150

Therefore, total number of items in drawer = 50 + 150 = 200. Since half the bolts and nuts are rusted, therefore number of rusted bolts = 25.

And number of rusted nuts = 75.

Therefore, total number of rusted items = 25 + 75 = 100

We know that the probability that the chosen item is rusted $P(A) = \frac{100}{200}$

Similarly, the probability that the chosen item is bolt $P(B) = \frac{50}{200}$

Therefore, the probability that the chosen item is rusted and a bolt $P(A \cap B) = \frac{25}{200}$

We also know from the addition theorem of probability that the chosen item is rusted or a bolt

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{100}{200} + \frac{50}{200} - \frac{25}{200} = \frac{125}{200} = \frac{5}{8}$$

13. The total number of ways of choosing 2 persons out of n is ${}^nC_2$. After selecting two persons when the remaining $n-2$ persons sit in a row, $n-1$ places are created between them in which 2 persons can be seated in ${}^{n-1}P_2$ ways. So,

the required probability $= \frac{{}^{n-1}P_2}{{}^nC_2} = \frac{n-2}{n} = 1 - \frac{2}{n}$

14. The black ball will remain in the first urn in the following ways:

(1) 9 white balls are transferred from the 1st urn to the second and then 9 white balls are replaced

$$\therefore \text{ Probability } = \frac{{}^9C_9}{{}^{10}C_9} \times \frac{{}^{19}C_9}{{}^{19}C_9}$$

(2) 8 white balls and 1 black ball is transferred from the 1st urn and then the black ball is replaced

$$\therefore \text{ Probability } = \frac{{}^{9}C_8 \times {}^{1}C_1}{{}^{10}C_9} \times \frac{{}^{18}C_8 \times {}^{1}C_1}{{}^{19}C_9}$$

$$= \frac{9 \times {}^{18}C_8}{{}^{10}C_9 \times {}^{19}C_9}$$

$$\therefore \text{ Required probability } = \frac{{}^{19}C_9 + 9 \times {}^{18}C_8}{{}^{10}C_9 \times {}^{19}C_9}$$

15. Required probability $= \frac{{}^{2}C_1}{{}^{9}C_1} \times \frac{{}^{4}C_1}{{}^{9}C_1} = \frac{8}{81}$

16. Series of numbers = 1 to 90

We know that numbers divisible by 6 in the series from 1 to 90 = 15

Similarly, numbers divisible by 8 in the series from 1 to 90 = 11

We also know that numbers divisible by both 6 and 8 (multiple of 24) in series from 1 to 90 = 3

Therefore, the probability of choosing numbers divisible by 6 or 8 $(P) = \frac{15+11-3}{90} = \frac{23}{90}$

17. Let X denote the largest number on the 3 tickets drawn. Then

$$P(X \le 7) = \left(\frac{7}{20}\right)^3 \text{ and } P(X \le 6) = \left(\frac{6}{20}\right)^3$$

Then, the required probability

$$= P(X = 7) = \left(\frac{7}{20}\right)^3 - \left(\frac{6}{20}\right)^3$$

18. Let A be the event that an even face turns up and B be the event that it is 2 or 4.

Then, $P(A) = 0.24 + 0.18 + 0.14 = 0.56$

and $P(B) = 0.24 + 0.18 = 0.42$

The required probability

$$P(E) = P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)}{P(A)} \qquad [\because B \subset A]$$

$$\therefore \quad P(E) = \frac{0.42}{0.56} = \frac{3}{4} = 0.75$$

19. In the first 120 natural numbers, total number of multiple of 5, $n(A)$ = 24 and total number of multiple of 15, $n(B)$ = 8 and $n(A \cap B) = 8$

$$\therefore \quad n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$= 24 + 8 - 8 = 24$$

$$\therefore \text{ Required probability } = \frac{24}{120} = \frac{1}{5}$$

20. Given: Total number of horses = 5 and number of selected horses = 2

We know that the number of ways to select 2 horses out of 5 horses

$$= {}^{5}C_2 = \frac{5!}{2!(5-2)} = \frac{5 \times 4 \times 3!}{2!3!} = 10$$

Similarly, the number of ways to select the winning horse

$$= {}^{2}C_1 = \frac{2!}{1!(2-1)} = \frac{2!}{1!1!} = 2$$

We also know that the probability that the selected horse is the winning horse $= \frac{2}{10} + \frac{2}{10} = \frac{4}{10} = \frac{2}{5}$

21. Consider the following events:

A: father has at least 1 boy; B: father has 2 boys and 1 girl; then, A = 1 boy and 2 girls; 2 boys and 1 girl; 3 boys and no girl

$A \cap B$ = 2 boys and 1 girl

Now, the required probability

$$= P(A/B) = \frac{P(A \cap B)}{P(A)} = \frac{1}{3}$$

22. There are 30 days in April. So, each person can have birthday in 30 days.

$$\therefore \quad n(S) = 30 \times 30 \times \cdots \times 30 \text{ (10 times)} = 30^{10}$$

$$n(E) = n(S) - \begin{pmatrix} \text{The number of ways in which 10} \\ \text{persons can have different birthdays} \end{pmatrix}$$

$$\therefore \quad n(E) = 30^{10} - {}^{30}C_{10}$$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{30^{10} - {}^{30}C_{10}}{30^{10}}$$

23. Numbers are 1, 2, 3, 4

Possibilities for unit place digit (either 1 or 3) = 2

Possibilities for ten place digit = 3

Possibilities for hundred place digit = 2

Possibilities for thousand place digit = 1

$\therefore$ Number of favourable outcomes $= 2 \times 3 \times 2 \times 1 = 12$

Number of numbers formed by 1, 2, 3, 4 (without repetitions) = 4!

$$\therefore \text{ Required probability } = \frac{12}{4 \times 3 \times 2} = \frac{1}{2}$$

24. $E(x^2 + 2) = E(x^2) + 2$

$$= x_1^2 P_1 + x_2^2 P_2 + x_3^2 P_3 + x_4^2 P_4 + 2$$

$$= 0 + 1^2 \times \frac{3}{8} + 2^2 \times \frac{3}{8} + 9 \times \frac{1}{8} + 2 = 5$$

25. The sequences can be 1, 2, 3, 4, $\ldots, (n-2), (n-1), n$ which are $(n-2)$ in number.

Hence, the required probability $= \dfrac{n-2}{{}^{n}C_3} = \dfrac{6}{n(n-1)}$

26. The total number of possible ways = 216
Let A, B, C be three dices, then the favourable number ways in which the sum of 16 occurs are

A	B	C	Sum
6	6	4	16
6	4	6	16
4	6	6	16
6	5	5	16
5	5	6	16
5	6	5	16

$\therefore$ Number of favourable ways = 6

$\therefore$ Required probability $= \dfrac{6}{216} = \dfrac{1}{36}$

27. Consider the following events:
A = A student has passed in mathematics
B = A student has passed in statistics

Then, $P(A) = \dfrac{70}{125}, P(B) = \dfrac{55}{125}, P(A \cap B) = \dfrac{30}{125}$

Required probability is

$$P(A \cap \overline{B}) + P(\overline{A} \cap B) = P(A) + P(B) - 2P(A \cap B)$$

$$= \frac{70}{125} + \frac{55}{125} - \frac{60}{125} = \frac{65}{125} = \frac{13}{25}$$

28.

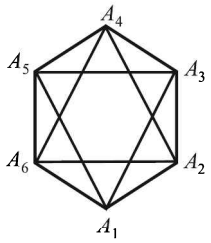


Fig. 26.5

The only equilateral triangles possible are $A_1A_3A_5$ and $A_2A_4A_6$

$\therefore$ Required probability $= \dfrac{2}{{}^{6}C_3} = \dfrac{2}{20} = \dfrac{1}{10}$

29. Total number of cases = 100
Number of favourable cases

$$= {}^{100}C_1 + {}^{100}C_3 + \cdots + {}^{100}C_{99}$$

$$= 2^{100-1} = 2^{99}$$

$\therefore$ Required probability $= \dfrac{2^{99}}{2^{100}} = \dfrac{1}{2}$

30. Number of white balls = 5; number of black balls = 7 and number of red balls = 4
Therefore, total number of balls = 5 + 7 + 4 = 16.
We know that the number of ways to draw 3 balls out of 16 balls $= {}^{16}C_3$

$$= \frac{16!}{3!(16-3)!} = \frac{16 \times 15 \times 14 \times 13!}{3!13!} = \frac{16 \times 15 \times 14}{3!}$$

Similarly, the number of ways to draw 3 white balls out of 5 white balls $= {}^{5}C_3$

$$= \frac{5!}{3!(5-3)!} = \frac{5 \times 4 \times 3 \times 2!}{3!2!} = \frac{5 \times 4 \times 3}{3!}$$

Therefore, probability that all three balls are white

$$(P) = \frac{{}^{5}C_3}{{}^{16}C_3} = \frac{5 \times 4 \times 3}{16 \times 15 \times 14} = \frac{1}{56}$$

31. Here,

$$\begin{pmatrix}\text{Probability that at}\\ \text{least one of them}\\ \text{will receive none}\end{pmatrix} = 1 - \begin{pmatrix}\text{Probability of each}\\ \text{receiving at least one}\end{pmatrix}$$

Required probability $= 1 - \dfrac{n(E)}{n(S)}$

Now, the number of integral solutions of

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 10$$

such that $x_1 \geq 1, x_2 \geq 1, \ldots, x_6 \geq 1$ gives $n(E)$ and the number of integral solutions of $x_1 + x_2 + \cdots + x_5 + x_6 = 10$ such that $x_1 \geq 0, x_2 \geq 0, \ldots, x_6 \geq 0$ gives $n(S)$

$\therefore$ The required probability $= 1 - \dfrac{{}^{10-1}C_{6-1}}{{}^{10+6-1}C_{6-1}}$

$\Rightarrow P(E) = 1 - \dfrac{{}^{9}C_5}{{}^{15}C_5} = \dfrac{137}{143}$

32. Number of red balls = 2; number of black balls = 3 and number of white balls = 4.
Therefore, total number of balls = 2 + 3 + 4 = 9
We know that the number of ways to draw 3 balls out of 9 balls $= {}^{9}C_3$

$$= \frac{9!}{3!(9-3)!} = \frac{9 \times 8 \times 7 \times 6!}{3!6!} = \frac{9 \times 8 \times 7}{3!} = 84$$

Since the red balls are only 2, therefore it is not possible to draw 3 red balls.
We also know that the number of ways to draw 3 black balls out of 3 black balls $= {}^{3}C_3$

$$= \frac{3!}{3!(3-3)!} = \frac{3!}{3!0!} = 1$$

Similarly, the number of ways to draw 3 white balls out of 4 white balls $= {}^4C_3 = \frac{4!}{3!(4-3)!} = \frac{4\times 3!}{3!1!} = 4$

Therefore, probability of 3 balls are of the same colour

$$(P) = \frac{{}^3C_3 + {}^4C_3}{{}^9C_3} = \frac{1+4}{84} = \frac{5}{84}$$

33. $n(S) = 6\times 6\times 6\times 6$

$n(E) =$ the number of integral solutions of

$$x_1 + x_2 + x_3 + x_4 = 12$$

where $1 \le x_1 \le 6, \ldots, 1 \le x_4 \le 6$

$n(E) =$ coefficient of x^{12} in $(x + x^2 + \cdots + x^6)^4$

$\Rightarrow$ $n(E) =$ coefficient of x^8 in $\left(\frac{1-x^6}{1-x}\right)^4$

$\Rightarrow$ $n(E) =$ coefficient of x^8 in

$$(1-x^6)^4\ ({}^3C_0 + {}^4C_1 x + {}^5C_2 x^2 + \cdots)$$

$\Rightarrow$ $n(E) = {}^{11}C_8 - 4\cdot {}^5C_2 = 125$

$\therefore$ $P(E) = \frac{125}{6\times 6\times 6\times 6}$

34. Number of brown socks = 5 and number of white socks = 4

Therefore, the total number of socks = 5 + 4 = 9

We know that the total number of ways to pull 2 socks out of 9 socks $= {}^9C_2 = \frac{9!}{2!(9-2)!} = \frac{9\times 8\times 7!}{2!7!} = \frac{9\times 8}{2!}$

And the number ways to pull 2 brown socks out of 5 brown socks $= {}^5C_2 = \frac{5!}{2!(5-2)!} = \frac{5\times 4\times 3!}{2!3!} = \frac{5\times 4}{2!}$

Similarly, the number of ways to pull 2 white socks out of 4 white socks $= {}^4C_2 = \frac{4!}{2!(4-2)!} = \frac{4\times 3\times 2!}{2!2!} = \frac{4\times 3}{2!}$

Thus, the probability that both the socks are of the same colour

$$P(E) = \frac{{}^5C_2 + {}^4C_2}{{}^9C_2} = \frac{(5\times 4)+(4\times 3)}{9\times 8} = \frac{32}{72} = \frac{4}{9} = \frac{48}{108}$$

35. Let A_i be the event that the number i is found on the first die and B_s be the event that the number s is found on the second.

$\therefore$ The probability that

$$i \le s = \sum_{s=i}^{6} P(A_i \cap B_s),\ i = 1, 2, \ldots, 6$$

$$= \sum_{s=i}^{6} P(A_i)P(B_s) \quad (\because A_i \text{ and } B_s \text{ are independent})$$

$$= \sum_{s=i}^{6} \frac{1}{6}\cdot P(B_s)$$

$$= \frac{1}{6}\left[\sum_{s=1}^{6} P(B_s) + \sum_{s=2}^{6} P(B_s) + \cdots + \sum_{s=6}^{6} P(B_s)\right]$$

$$= \frac{1}{6}\left[\frac{6}{6} + \frac{5}{6} + \frac{4}{6} + \frac{3}{6} + \frac{2}{6} + \frac{1}{6}\right] = \frac{21}{36} = \frac{7}{12}$$

36. Any month out of 12 months can be chosen with probability = 1/12

There are 7 possible ways in which the month can start and it will be a Friday on 13th day if the first day of the month is Sunday, whose probability is 1/7

Hence, the required probability $= \frac{1}{12}\times\frac{1}{7} = \frac{1}{84}$

37. Number of consecutive natural numbers = 40

We know that to get the sum of two numbers as odd, one number must be even and the other must be odd. Therefore, the total number of ways to get 2 numbers out of 40 numbers $= {}^{40}C_2$

$$= \frac{40!}{2!(40-2)!} = \frac{40\times 39\times 38!}{2!38!} = \frac{40\times 39}{2\times 1}$$

Similarly, the number of ways to get 1 even number out of 20 even numbers $= {}^{20}C_1$

$$= \frac{20!}{1!(20-1)!} = \frac{20\times 19!}{1!19!} = \frac{20}{1!}$$

And the number of ways to get 1 odd number out of 20 odd numbers $= {}^{20}C_1$

$$= \frac{20!}{1!(20-1)!} = \frac{20\times 19!}{1!19!} = \frac{20}{1!}$$

Thus, the probability that the sum of two number is odd

$P(E)$ = Probability of 1 even number and 1 odd number

$$= \frac{{}^{20}C_1 \times {}^{20}C_1}{{}^{40}C_2} = \frac{20\times 20\times 2\times 1}{40\times 39} = \frac{20}{39}$$

38. The word PROBABILITY contains one P, one R, one O, two B's, one A, two I's, one L, one T and one Y, on the whole 11 letters.

Hence, the total number of arrangements $= \frac{|11}{|2|2}$

If the two B's should come consecutively, the number of ways of arrangements $= \frac{|10}{|2}$ (as two B's are considered as one unit and so there are 10 units with repetition of two I's)

Hence, the required probability $= \frac{|10}{|2}\times\frac{|2|2}{|11} = \frac{2}{11}$

39. Let A_i denote an event of getting number i ($i = 1, 2, \ldots, 6$,) on each die

Then $A_i = i = 1, 2, \ldots, 6$ are mutually exclusive events.

$\therefore$ Required probability is

$$P(E) = P(A_1) + P(A_2) + \cdots + P(A_6) \qquad \text{(i)}$$

Now, $P(A_i)$ = Probability of getting number i on each die

$$\Rightarrow\ P(A_i) = \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216}$$

So, the required probability $= \dfrac{6}{216} = \dfrac{1}{36}$ [From (i)]

40. For mutually exclusive events,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$= \frac{1}{4} + \frac{2}{3} + \frac{1}{3} = \frac{5}{4} > 1, \text{ not possible}$$

41. $P(A \cap B) = P(A) \cdot P(B/A)$ (i)

Since A and B are independent,

$\therefore\ P(A \cap B) = P(A)P(B)$ (ii)

$\therefore$ From (i) and (ii), $P(B/A) = P(B) = \dfrac{1}{2}$

42. Statement II is false. For example, $x^2 + x + 1 = 0$ has no real roots

$P(A$ and B contradict each other)

$$= p(1-2p) + 2p(1-p) = \frac{1}{2} \text{ (given)}$$

$$\Rightarrow\ 8p^2 - 6p + 1 = 0 \quad \Rightarrow \quad p = \frac{1}{2}, \frac{1}{4}$$

43. In $\left(\dfrac{1}{2} + \dfrac{1}{2}\right)^{10}$,

probability of getting 4 heads

$$= {}^{10}C_4 \left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^6$$

$$= {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4$$

= probability of getting 6 heads $\quad (\because\ {}^{10}C_4 = {}^{10}C_6)$

44. $P(A \cap \overline{B}) = P(A) - P(A \cap B)$

$$= 0.5 - P(A \cap B)$$

Hence, $P(A \cap \overline{B})$ cannot be determined.

45. The total number of ways in which numbers can be chosen is $25 \times 25 = 625$

The number of ways in which both players can choose the same number is 25.

$\therefore$ Probability that they win a prize $= \dfrac{25}{625} = \dfrac{1}{25}$

Thus, the probability that they will not win a prize

$$= 1 - \frac{1}{25} = \frac{24}{25}$$

46. $P(A) = \dfrac{1}{2}$, $P(B) = \dfrac{1}{3}$ and $P(C) = \dfrac{1}{4}$

$\therefore\quad P(\overline{A}) = 1 - \dfrac{1}{2} = \dfrac{1}{2}$, $P(\overline{B}) = 1 - \dfrac{1}{3} = \dfrac{2}{3}$

and $\quad P(\overline{C}) = 1 - \dfrac{1}{4} = \dfrac{3}{4}$

Hence, the required probability

$$= 1 - P(\overline{A})P(\overline{B})P(\overline{C})$$

$$= 1 - \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = 1 - \frac{1}{4} = \frac{3}{4}$$

47. Since A and B are mutually exclusive events,

$\therefore\ A \cap B = \phi$

$\Rightarrow\ A \subseteq \overline{B}$ and $B \subseteq \overline{A}$

$\Rightarrow\ P(A) \le P(\overline{B})$ and $P(B) \le P(\overline{A})$

48. Required probability $= {}^7C_2 \left(\dfrac{1}{6}\right)^2 \left(\dfrac{5}{6}\right)^5 \times \dfrac{1}{6} = \dfrac{{}^7C_2 \times 5^5}{6^8}$

49. Using $\bar{x} = \dfrac{(x_1 + x_2 + \cdots + x_{100})}{100} = 72$

$\therefore\qquad x_1 + \cdots + x_{100} = 7200 \qquad \text{(i)}$

Again, $\quad \dfrac{x_1 + x_2 + \cdots + x_{70}}{70} = 75$

$\therefore\quad x_1 + \cdots + x_{70} = 75 \times 70 = 5250 \qquad \text{(ii)}$

$\therefore$ Average mark of 30 girls $= \dfrac{7200 - 5250}{30} = 65$

50. $P(A \cup B) = 3/4, P(A \cap B) = 1/4,$

$P(\overline{A}) = 2/3 \quad \Rightarrow \quad P(A) = 1/3$

$\therefore\ P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$$\frac{1}{4} = \frac{1}{3} + P(B) - \frac{3}{4} \quad \Rightarrow \quad P(B) = \frac{2}{3}$$

$$P(\overline{A} \cap B) = P(B) - P(A \cap B) = \frac{2}{3} - \frac{1}{4} = \frac{8-3}{12} = \frac{5}{12}$$

51. A, B and C are mutually exclusive.

$\therefore\ 0 \le P(A) + P(B) + P(C) \le 1 \qquad \text{(i)}$

$0 \le P(A) \cdot P(B) \cdot P(C) \le 1 \qquad \text{(ii)}$

Now on solving (i) and (ii), we get $\dfrac{1}{3} \le x \le \dfrac{1}{2}$

52. $\left.\begin{matrix} np=4 \\ npq=2 \end{matrix}\right\} \Rightarrow q=\frac{1}{2}, p=\frac{1}{2}, n=8$

$$P(X=1)={}^{8}C_1\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^7=8\cdot\frac{1}{2^8}=\frac{1}{2^5}=\frac{1}{32}$$

53. Since $(1+3P)/3$, $(1-P)/4$ and $(1-2P)/2$ are the probabilities of the three events, we must have

$$0\le\frac{1+3P}{3}\le 1, 0\le\frac{1-P}{4}\le 1 \text{ and } 0\le\frac{1-2P}{2}\le 1$$

$$\Rightarrow -\frac{1}{3}\le P\le\frac{2}{3}, -3\le P\le 1 \text{ and } -\frac{1}{2}\le P\le\frac{1}{2}$$

Also $\frac{1+3P}{3}, \frac{1-P}{4}$ and $\frac{1-2P}{2}$ are the probabilities of three mutually exclusive events

$$\Rightarrow \quad 0\le\frac{1+3P}{3}+\frac{1-P}{4}+\frac{1-2P}{2}\le 1$$

$$\Rightarrow \quad 0\le 4+12P+3-3P+6-12P\le 12$$

$$\Rightarrow \quad \frac{1}{3}\le P\le\frac{13}{3}$$

Thus, the required values of P are such that

$$\text{max.}\left\{-\frac{1}{3}, -3, \frac{1}{2}, \frac{1}{3}\right\}\le P\le\text{min.}\left\{\frac{2}{3}, 1, \frac{1}{2}, \frac{13}{3}\right\}$$

$$\Rightarrow \frac{1}{3}\le P\le\frac{1}{2}$$

54. Given $np = 4$ and $npq = 2$

$$q=\frac{npq}{np}=\frac{2}{4}=\frac{1}{2} \text{ so } p=1-\left(\frac{1}{2}\right)=\frac{1}{2}$$

Now $npq = 2 \quad \therefore n = 8$

$\therefore$ B.D. is given by $P(X=r)={}^{8}C_r p^r q^{n-r}$

$$\therefore \quad P(X=r=2)={}^{8}C_r\left(\frac{1}{2}\right)^8=\frac{28}{256}$$

55. From the given table, prime numbers are 2, 3, 5, 7. Let E denote prime numbers and F denote the numbers <4.

$\therefore \quad P(E) = P(2 \text{ or } 3 \text{ or } 5 \text{ or } 7)$ (events 2, 3, 5, 7 are E)

$$= P(2)+P(3)+(5)+P(7)=0.62$$

$$P(F) = P(1 \text{ or } 2 \text{ or } 3) \text{ (events 1, 2, 3, are } E)$$

$$= P(1)+P(2)+P(3)=0.50$$

$$P(E\cap F)=P(2 \text{ or } 3)=P(2)+P(3)=0.35$$

$$P(E\cup F)=P(E)+P(F)-P(E\cap F)$$

$$=0.62+0.50-0.35=0.77$$

56. $P(A)=\frac{4}{5} \quad \therefore \quad P(\bar{A})=\frac{1}{5}$

$$P(B)=\frac{3}{4} \quad \therefore \quad P(\bar{B})=\frac{1}{4}$$

Now we need $P(A)P(\bar{B})+P(B)P(\bar{A})$

$$=\frac{4}{5}\times\frac{1}{4}+\frac{3}{4}\times\frac{1}{5}=\frac{4}{20}+\frac{3}{20}=\frac{7}{20}$$

57. Using the well-known fact that the root mean square of numbers $\ge$ A.M. of the numbers

$$\Rightarrow \sqrt{\frac{400}{n}}\ge\frac{80}{n} \quad \Rightarrow \quad \frac{20}{\sqrt{n}}\ge\frac{80}{n}$$

$$\Rightarrow \sqrt{n}\ge 4 \quad \Rightarrow \quad n\ge 16$$

$\Rightarrow n = 17$ but not given in choice

$\therefore n = 18$ is the correct number.

58. Number of houses = 3 = Number of favourable cases
Number of applicants = 3 $\therefore$ Total number of events = 3^3 (because each candidate can apply in 3 ways)

$$\therefore \text{ Required probability } = \frac{3}{3^3}=\frac{1}{9}$$

59. $P(X=r)=\frac{e^{-\lambda}\lambda^r}{r!}$ (λ = mean)

$$\therefore \quad P(X=r>1.5)=P(2)+P(3)+\cdots+\infty$$

$$=1-[P(0)+P(1)]$$

$$=1-\left[e^{-2}+\frac{e^{-2}\times 2^2}{2}\right]=1-\frac{3}{e^2}$$

60. $P(\overline{A\cup B})=1-P(A\cup B)$

$$=1-P(A)-P(B)+P(A\cap B)$$

$$\frac{1}{6}=1-\frac{3}{4}-P(B)+\frac{1}{4}$$

$$P(B)=\frac{1}{2}-\frac{1}{6}\Rightarrow P(B)=\frac{4}{12}=\frac{1}{3}$$

Now, $P(A\cap B)=\frac{1}{4}=\frac{1}{3}\times\frac{3}{4}=P(A)P(B)$ so events are independent but not equally likely as $P(A)\ne P(B)$.

61. According to the Poisson distribution,

$$P(X=r)=\frac{e^{-m}m^r}{r!}$$

$$\therefore \quad P(X\le 1)=P(X=0)+P(X=1)$$

$$=e^{-m}+\frac{e^{-m}\cdot m}{1!}$$

Given m = mean = 5

$$\therefore \quad P(x\le 1)=e^{-5}+5\times e^{-5}=e^{-5}(1+5)=\frac{6}{e^5}$$

62. Let x and y be the number of boys and girls in a class, respectively.

$$\frac{52x+42y}{x+y}=50 \Rightarrow x=4y \Rightarrow \frac{x}{y}=\frac{4}{1} \text{ and } \frac{x}{x+y}=\frac{4}{5}$$

$\therefore$ Required percentage of boys

$$=\frac{x}{x+y}\times 100=\frac{4}{5}\times 100=80\%$$

63. Required probability $= P(\bar{\text{I}} \cap \text{II}) = P(\bar{\text{I}})P(\text{II})$

$$= [1-P(\text{I})]\,P(\text{II})$$
$$= (1-0.3)(0.2)=(0.7)(0.2)$$
$$= 0.14$$

64. Possibilities of getting 9 are (5, 4), (4, 5), (6, 3), (3, 6)

$\therefore$ $p=\frac{4}{36}=\frac{1}{9}$ and $q=1-\frac{1}{9}=\frac{8}{9}$

$\therefore$ Required probability $= {}^3C_2 q^1 p^2$

$$=(3)\left(\frac{8}{9}\right)\left(\frac{1}{9}\right)^2=\frac{8}{243}$$

65. We know that

$$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}=\frac{P(A\cap B)}{P(A)}\times\frac{P(A)}{P(B)}$$

$$P\left(\frac{A}{B}\right)=P\left(\frac{B}{A}\right)\cdot\frac{P(A)}{P(B)}$$

$\therefore$ $P(B)=\frac{2/3\times 1/4}{1/2}=\frac{1}{3}$

$\left[\text{By using } P\left(\frac{B}{A}\right)=\frac{2}{3}, P(A)=\frac{1}{4} \text{ and } P\left(\frac{A}{B}\right)=\frac{1}{2}\right]$

66. $A = \{4, 5, 6\}$

Also $B = \{1, 2, 3, 4,\}$

We have $A \cup B = \{1, 2, 3, 4, 5, 6\} = S$

where S is the sample space of the experiment of throwing a die. $P(S) = 1$, since it is a sure event.

Hence, $P(A \cup B) = 1$

67. Probability of at least one success = 1 – No success

$$=1-{}^nC_n q^n$$

where $q = 1 - p = 3/4$

We want $1-\left(\frac{3}{4}\right)^n \geq \frac{9}{10}$

$$\Rightarrow \frac{1}{10}\geq\left(\frac{3}{4}\right)^n \Rightarrow \left(\frac{3}{4}\right)^n \leq \frac{1}{10}$$

Taking logarithm on base 10, we have

$$n\log_{10}(3/4)\leq \log_{10}10^{-1}$$

$$\Rightarrow n\geq\frac{1}{\log_{10}4-\log_{10}3}$$

68. Any number in the set

$S = \{00, 01, 02, \ldots, 49\}$ is of the form ab, where $a \in \{0, 1, 2, 3, 4\}$ and $b \in \{0, 1, 2, \ldots, 9\}$. For the product of digits to be zero, the number must be either of the form 0 which are 5 in number, because

$$x \in \{1, 2, 3, 4\}$$

or of the form $0x$ which are 10 in numbers because

$$x \in \{0, 1, 2, \ldots, 9\}$$

The only number common to both = 00

Thus, the number of numbers in S, the product of whose digit is zero = 10 + 5 – 1 = 14

Of these, the number whose sum of digits is 8 is just one, i.e. 08.

$\therefore$ The required probability = 1/14

69. Let the event be (Given: {1, 2, 3,…, 8})

A: Maximum of three numbers is 6.

B: Minimum of three numbers is 3.

$P(A) = \frac{{}^5C_2}{{}^8C_3}$ (the number < 6 are 5)

$P(B) = \frac{{}^5C_2}{{}^8C_3}$ (the number > 3 are 5)

$P(B \cap A) = \frac{{}^2C_1}{{}^8C_3}$

$$P\left(\frac{B}{A}\right)=\frac{P(B\cap A)}{P(A)}=\frac{{}^2C_1}{{}^5C_2}=\frac{2}{10}=\frac{1}{5}$$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. An experiment results in four possible outcomes S_1, S_2, S_3 and S_4 with probabilities P_1, P_2, P_3 and P_4 respectively. Which one of the following probability assignments is possible?

[Assume that S_1, S_2, S_3, S_4 are mutually exclusive]

(a) $P_1 = 0.25, P_2 = 0.35, P_3 = 0.10, P_4 = 0.05$

(b) $P_1 = 0.40, P_2 = 0.20, P_3 = 0.60, P_4 = 0.20$

(c) $P_1 = 0.30, P_2 = 0.60, P_3 = 0.10, P_4 = 0.10$

(d) $P_1 = 0.20, P_2 = 0.30, P_3 = 0.40, P_4 = 0.10$

2. An integer x is chosen from the first 100 positive integers. The probability that $x + (100/x) > 50$ is

(a) 1/10 (b) 11/20

(c) 1/2 (d) none of these

3. A and B having equal skill are playing a game of best of 5 points. After A has won two points and B has won one point, the probability that A will win the game is
(a) 1/2 (b) 2/3
(c) 3/4 (d) none of these

4. Five boys and three girls are seated at random in a row. The probability that no boy sits between two girls is
(a) 1/56 (b) 1/8
(c) 3/28 (d) none of these

5. In a convex hexagon, two diagonals are drawn at random. The probability that the diagonals intersect at an interior point of the hexagon is
(a) 5/12 (b) 7/12
(c) 2/5 (d) none of these

6. A natural number x is chosen at random from 150 natural numbers. The probability that $\frac{x^2-70x+1200}{x-35}<0$ is
(a) 23/30 (b) 1/50
(c) 11/50 (d) none of these

7. Two cards are randomly selected represented by x_1, x_2. The probability that x_1, x_2 satisfies $4 \le x_1, x_2 \le 9$ is
(a) 23/221 (b) 1/221
(c) 17/221 (d) none of these

8. Four five-rupee coins, 3 two-rupee coins and 2 one-rupee coins are stacked together in a column at random. The probability that the coins of the same denomination are consecutive is
(a) 13/9! (b) 1/210
(c) 1/35 (d) none of these

9. Two cards are drawn at random from a pack of 52 cards. The probability of getting at least a spade and an ace is
(a) 1/34 (b) 8/221
(c) 1/26 (d) 2/51

10. A point is selected randomly from the inside of the circle. The probability that the point is closer to the centre than the boundary of the circle is
(a) 4/5 (b) 3/4
(c) 1/4 (d) none of these

11. Four gentlemen and 4 ladies take seats at random round a table. The probability that they are sitting alternately is
(a) 4/35 (b) 1/70
(c) 2/35 (d) 1/35

12. If $\frac{1+3P}{3}, \frac{1-P}{4}$ and $\frac{1-3P}{3}$ are the probabilities of three M.E. and exhaustive events, then the set of values of P is
(a) $\left[-\frac{1}{9}, \frac{1}{3}\right]$ (b) $\left[-\frac{1}{3}, \frac{1}{3}\right]$
(c) $\left[\frac{1}{9}, \frac{1}{3}\right]$ (d) $\left[\frac{1}{3}, 1\right]$

13. If the integers a and b are chosen randomly between 1 and 100, then the probability that a number of the form $7^a + 7^b$ is divisible by 5 is
(a) 1/7 (b) 1/5
(c) 1/48 (d) 1/4

14. A man has a bunch of n keys, only one of which exactly opens the lock. The man tries to open the lock by using the keys randomly. The probability that he opens the lock at the jth attempt by assuming that the rejected key has already been tried is
(a) $\frac{j}{n}$ (b) $\frac{1}{n}$
(c) $\left(1-\frac{1}{n}\right)^{j-1} \times \frac{1}{n}$ (d) none of these

15. If $a \in [-10, 0]$, then the probability that the graph of the function $y = x^2 + 2(a+3)x - (2a+b)$ is strictly above the x-axis is
(a) 3/5 (b) 2/5
(c) 1/5 (d) none of these

16. $x_1, x_2, x_3, \ldots, x_{50}$ are fifty real numbers such that $x_r < x_r + 1$ for $r = 1, 2, 3, \ldots, 49$. Five numbers out of these are picked up at random. The probability that the five numbers have x_{20} as the middle number is
(a) $\frac{{}^{20}C_2 \times {}^{30}C_2}{{}^{50}C_5}$ (b) $\frac{{}^{30}C_2 \times {}^{19}C_2}{{}^{50}C_5}$
(c) $\frac{{}^{19}C_2 \times {}^{31}C_3}{{}^{50}C_5}$ (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. A bag contains 7 tickets marked with the numbers 0, 1, 2, 3, 4, 5, 6, respectively. A ticket is drawn and replaced. Then the chance that after 4 drawings, the sum of the numbers drawn is 8 is
(a) 165/2401 (b) 149/2401
(c) 3/49 (d) none of these

2. A biased coin with probability p, $0 < p < 1$, of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is 2/5, then p equals
(a) 1/3 (b) 2/3
(c) 2/5 (d) 3/5

3. Numbers 1, 2, 3, …, 100 are written down on each of the cards A, B and C. One number is selected at random from each of the cards. The probability that the numbers so selected can be the measures (in cm) of three sides of a right-angled triangle, no two of which are similar, is

(a) $\frac{4}{100^3}$ (b) $\frac{3}{50^3}$

(c) $\frac{3!}{100^3}$ (d) none of these

4. Three numbers are chosen at random without replacement from the set $A=\{x|1\le x\le 10, x\in N\}$. The probability that the minimum of the chosen numbers is 3 and maximum is 7 is

(a) 1/12 (b) 1/15

(c) 1/40 (d) none of these

5. In a purse, there are 10 coins, all 5 naya paisa except one which is a rupee. In another purse, there are 10 coins all 5 naya paisa. Nine coins are taken out from the former purse and put into the latter and then 9 coins are taken out from the latter and put into the former. Then the chance that the rupee is still in the first purse is

(a) 9/19 (b) 10/19

(c) 4/9 (d) none of these

6. A letter is known to have come from either KRISHNAGIRI or DHARMAPURI. On the post mark only the two consecutive letters "RI" are visible. Then the chance that it came from KRISHNAGIRI is

(a) 3/5 (b) 2/3

(c) 9/14 (d) none of these

7. If $\frac{(1+3p)}{3}, \frac{(1-p)}{4}$ and $\frac{(1-2p)}{2}$ are the probabilities of three mutually exclusive events, then the set of all values of p is

(a) $\left[\frac{1}{2},\frac{2}{3}\right]$ (b) $\left[\frac{1}{3},\frac{1}{2}\right]$

(c) $\left[\frac{1}{4},\frac{1}{2}\right]$ (d) $\left[\frac{1}{3},\frac{2}{3}\right]$

8. Three natural numbers are taken at random from the set $A=\{x:1\le x\le 100, x\in N\}$. The probability that the A.M. of the numbers taken is 25 is

(a) $\frac{^{77}C_2}{^{100}C_3}$ (b) $\frac{^{25}C_2}{^{100}C_3}$

(c) $\frac{^{74}C_{72}}{^{100}C_{97}}$ (d) none of these

9. From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category is

(a) 1/2 (b) 1/3

(c) 2/3 (d) none of these

10. The probability that out of 10 persons, all born in April, at least two have the same birthday is

(a) $\frac{^{30}C_{10}}{(30)^{10}}$ (b) $1-\frac{^{30}C_{10}}{30!}$

(c) $\frac{(30)^{10}-{}^{30}C_{10}}{(30)^{10}}$ (d) none of these

11. Let p be the probability that a man aged x years will die in a year time. The probability that out of n men $A_1, A_2, A_3, \ldots, A_n$ each aged x years, A_1 will die and will be the first to die is

(a) $\frac{1-p^n}{n}$ (b) $\frac{p}{n}$

(c) $\frac{p(1-p)^{n-1}}{n}$ (d) $\frac{1-(1-p)^n}{n}$

12. The contents of urns I and II are as follows:

Urn I: 4 white and 5 black balls

Urn II: 3 white and 6 black balls

One urn is chosen at random and a ball is drawn and its colour is noted and replaced back to the urn. Again a ball is drawn from the same urn, colour is noted and replaced. The process is repeated 4 times and as a result, one ball of white colour and 3 of black colour are noted. Find the probability that the chosen urn was I.

(a) $\frac{125}{287}$ (b) $\frac{64}{127}$

(c) $\frac{25}{287}$ (d) $\frac{79}{192}$

13. A die is weighted so that the probability of different faces to turn up is as given

Number	1	2	3	4	5	6
Probability	0.2	0.1	0.1	0.3	0.1	0.2

If $P\left(\frac{A}{B}\right)=P_1$ and $P\left(\frac{B}{A}\right)=P_2$ and $P\left(\frac{C}{A}\right)=P_3$, then the values of P_1, P_2, P_3 respectively are [Take the events A, B and C as $A=\{1,2,3\}$, $B=\{2,3,5\}$ and $C=\{2,4,6\}$]

(a) $\frac{2}{3},\frac{1}{3},\frac{1}{4}$ (b) $\frac{1}{3},\frac{1}{3},\frac{1}{6}$

(c) $\frac{1}{4},\frac{1}{3},\frac{1}{6}$ (d) $\frac{2}{3},\frac{1}{6},\frac{1}{4}$

14. If one ball is drawn at random from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, and 1 white and 3 black balls, then the probability that 2 white and 1 black balls be drawn is

(a) $\frac{13}{32}$ (b) $\frac{1}{4}$

(c) $\frac{1}{32}$ (d) $\frac{3}{16}$

15. A and B draw two cards each, one after another, from a well-shuffled pack of 52 cards. The probability that all the four cards drawn are the same suit is

(a) $\frac{44}{85\times 49}$ (b) $\frac{11}{85\times 49}$

(c) $\frac{13\times 24}{17\times 25\times 49}$ (d) none of these

16. Fifteen coupons are numbered 1, 2, 3, ..., 15 respectively. Seven coupons are selected at random one at a time with replacement. The probability that the largest number appearing on a selected coupon is 9 is

(a) $\left(\frac{9}{16}\right)^6$ (b) $\left(\frac{8}{15}\right)^7$

(c) $\left(\frac{3}{5}\right)^7$ (d) $\frac{9^7-8^7}{15^7}$

17. Let A and B be two independent events such that $P(A)=\frac{1}{5}$, $P(A\cup B)=\frac{7}{10}$. Then $P(\bar{B})$ is equal to

(a) 3/8 (b) 2/7

(c) 7/9 (d) none of these

18. The probability that at least one of the events A and B occurs is 3/5. If A and B occur simultaneously with probability 1/5, then $P(A') + P(B')$ is

(a) 2/5 (b) 4/5

(c) 6/5 (d) 7/5

19. A coin is tossed 7 times. Each time a man calls a head, the probability that he wins the toss on more occasions is

(a) 1/4 (b) 5/8

(c) 1/2 (d) none of these

ANSWERS

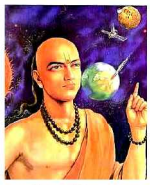
Level 1

1. (d)	**2.** (b)	**3.** (c)	**4.** (c)	**5.** (a)	**6.** (c)	**7.** (d)	**8.** (b)	**9.** (c)	**10.** (c)
11. (d)	**12.** (c)	**13.** (b)	**14.** (b)	**15.** (c)	**16.** (b)				

Level 2

1. (b)	**2.** (a)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (c)	**9.** (c)	**10.** (c)
11. (d)	**12.** (a)	**13.** (d)	**14.** (a)	**15.** (a)	**16.** (d)	**17.** (a)	**18.** (c)	**19.** (c)	

27



ARYA BHATT

Trigonometric Ratios, Identities and Equations

- ❑ Measurement of Angles
- ❑ Relation between Radians and Degrees
- ❑ Reciprocal, Quotient and Square Relations
- ❑ Values of Trigonometric Ratios
- ❑ Formulae for Any Two Angles A and B
- ❑ Transformation of Products and Sums
- ❑ C and D Formula
- ❑ Trigonometrical Ratios of Angles $2A$, $3A$ and Half Angle Formulae
- ❑ Trigonometrical Equations
- ❑ T-ratio of Some Special Angles
- ❑ Some Useful Identities
- ❑ Greatest and Least Values of the Expression $\alpha \sin\theta + \beta \cos\theta$

☞ MEASUREMENT OF ANGLES

1. Degree Measure (Sexagesimal System)

In this system, an angle is measured in degrees, minutes and seconds. A complete rotation describes 360°.

1 right angle = 90° (90 degrees)

1° = 60′ (60 minutes)

1′ = 60″ (60 seconds)

2. Radian Measure (Circular System)

The unit of angle in this system is radian. One radian, written as 1^c, is the measure of an angle subtended at the centre of a circle of radius r by an arc of length r.

2π radians = 360° or π radians = 180°

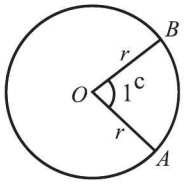


Fig. 27.1

3. Grade Measure (Centesimal System)

In this system, each right angle is divided into 100 equal parts called **grades**.

1 right angle = 100^g (100 grades)

1^g = 100′ (100 minutes)

1′ = 100″ (100 seconds)

π radians = 180° = 200^g

Note Sum of the interior angles of a polygon of n sides = $[(2n-4)\times 90]°$

RELATION BETWEEN RADIANS AND DEGREES

1. π radians = 180° $\Rightarrow 1^c = \left(\frac{180}{\pi}\right)^\circ \approx 57°17'44.8''$
2. 180° = π radians $\Rightarrow 1° = \left(\frac{\pi}{180}\right)^c \approx 0.01746^c$

RELATION BETWEEN THE LENGTH OF AN ARC AND THE ANGLE SUBTENDED BY THE ARC AT THE CENTRE OF THE CIRCLE

$l = r\theta$, where l = length of the arc AB
r = radius of the circle
θ = angle (in radian) subtended at the centre

Also, $l = \frac{2\pi r\theta}{360}$, where θ = angle (in degrees) subtended at the centre.

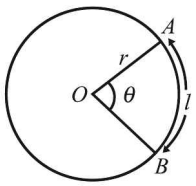


Fig. 27.2

RELATION BETWEEN THE AREA OF SECTOR AND THE ANGLE SUBTENDED BY THE ARC AT THE CENTRE OF THE CIRCLE

$S = \frac{1}{2}r^2\theta$, where S = area of sector AOB.

r = radius of the circle
θ = angle subtended by arc AB at the centre

Also, $S = \frac{\pi r^2\theta}{360}$, where θ = angle (in degrees) subtended by arc AB at the centre.

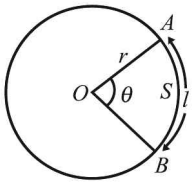


Fig. 27.3

RECIPROCAL RELATIONS

1. $\sin\theta \cdot \operatorname{cosec}\theta = 1$
2. $\cos\theta \cdot \sec\theta = 1$
3. $\tan\theta \cdot \cot\theta = 1$

QUOTIENT RELATIONS

1. $\tan\theta = \frac{\sin\theta}{\cos\theta}$
2. $\cot\theta = \frac{\cos\theta}{\sin\theta}$

SQUARE RELATIONS

1. $\sin^2\theta + \cos^2\theta = 1$
2. $\sin^2\theta = 1 - \cos^2\theta$
3. $\cos^2\theta = 1 - \sin^2\theta$
4. $1 + \tan^2\theta = \sec^2\theta$
5. $\tan^2\theta = \sec^2\theta - 1$
6. $\sec^2\theta - \tan^2\theta = 1$
7. $1 + \cot^2\theta = \operatorname{cosec}^2\theta$
8. $\cot^2\theta = \operatorname{cosec}^2\theta - 1$
9. $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

II	Sin θ and cosec θ (+) Remaining 4(–)	All (+)	I
	(90° + θ) and (180° – θ)	(90° – θ) and (360° + θ)	
III	tan θ and cot θ (+) Remaining 4(–)	cos θ and sec θ (+) Remaining 4(–)	IV
	(180° + θ) and (270° – θ)	(270° + θ) and (360° – θ)	

SIMPLE RULES TO REMEMBER

After - School - To - College
↓ ↓ ↓ ↓
A S T C

(90° ± θ) and (270° ± θ)	(180° ± θ) and (360° ± θ)	
Change	No change	
sin $\theta \leftrightarrow$ cos θ	sin $\theta \rightarrow$ sin θ	cosec $\theta \rightarrow$ cosec θ
tan $\theta \leftrightarrow$ cot θ	cos $\theta \rightarrow$ cos θ	sec $\theta \rightarrow$ sec θ
cosec $\theta \leftrightarrow$ sec θ	tan $\theta \rightarrow$ tan θ	cot $\theta \rightarrow$ cot θ

VALUES OF TRIGONOMETRIC RATIOS OF SOME STANDARD ANGLES

Angles	Degrees	0°	30°	45°	60°	90°	180°	270°	360°
	Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
sin θ		0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	–1	0
cos θ		1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	–1	0	1
tan θ		0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	0	∞	0
cosec θ		∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	∞	–1	∞
sec θ		1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞	–1	∞	1
cot θ		∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	∞	0	∞

∞ = Undefined

Note

$\sin(-x) = -\sin x$ $\quad$ $\operatorname{cosec}(-x) = -\operatorname{cosec} x$

$\cot(-x) = -\cot x$ $\quad$ $\cos(-x) = +\cos x$

$\tan(-x) = -\tan x$

$\sec(-x) = +\sec x$

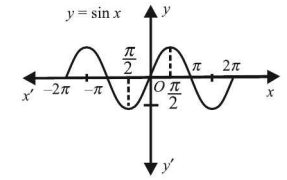


Fig. 27.4

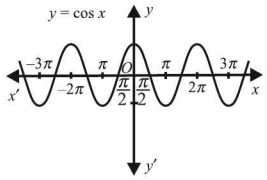


Fig. 27.5

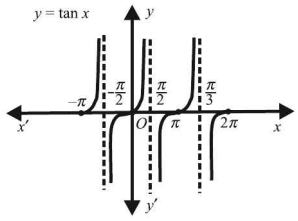


Fig. 27.6

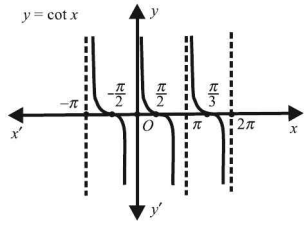


Fig. 27.7

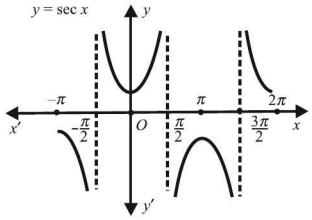


Fig. 27.8

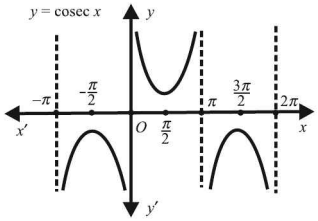


Fig. 27.9

FORMULAE FOR ANY TWO ANGLES *A* AND *B*

1. $\sin(A+B) = \sin A\cos B + \cos A\sin B$
2. $\sin(A-B) = \sin A\cos B - \cos A\sin B$
3. $\cos(A+B) = \cos A\cos B - \sin A\sin B$
4. $\cos(A-B) = \cos A\cos B + \sin A\sin B$
5. $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A\tan B}$

   $\Rightarrow \tan\left(\dfrac{\pi}{4} + A\right) = \dfrac{1+\tan A}{1-\tan A}$
6. $\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A\tan B}$

   $\Rightarrow \tan\left(\dfrac{\pi}{4} - A\right) = \dfrac{1-\tan A}{1+\tan A}$
7. $\cot(A+B) = \dfrac{\cot A\cot B - 1}{\cot B + \cot A}$
8. $\cot(A-B) = \dfrac{\cot A\cot B + 1}{\cot B - \cot A}$
9. $\sin(A+B)\sin(A-B) = (\sin^2 A - \sin^2 B)$

   or $\quad (\cos^2 B - \cos^2 A)$
10. $\cos(A+B)\cos(A-B) = (\cos^2 A - \sin^2 B)$

    or $\quad (\cos^2 B - \sin^2 A)$
11. $\sin(A+B+C) = \cos A\cos B\cos C(\tan A + \tan B + \tan C - \tan A\tan B\tan C)$
12. $\cos(A+B+C) = \cos A\cos B\cos C\,(1 - \tan A\tan B - \tan B\tan C - \tan C\tan A)$
13. $\tan(A+B+C) = \dfrac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - \tan A\tan B - \tan B\tan C - \tan C\tan A}$

TRANSFORMATION OF PRODUCTS AND SUMS

1. $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
2. $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
3. $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
4. $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

C AND *D* FORMULA

1. $\sin C + \sin D = 2\sin\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$
2. $\sin C - \sin D = 2\cos\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$
3. $\cos C + \cos D = 2\cos\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$
4. $\cos C - \cos D = -2\sin\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$

Add to Memory

1. $S + S = 2SC$ $\quad$ 2. $S - S = 2CS$
3. $C + C = 2CC$ $\quad$ 4. $C - C = -2SS$

where S = sin and C = cos

TRIGONOMETRICAL RATIOS OF ANGLE 2*A*

1. $\sin 2A = 2\sin A\cos A \Rightarrow \sin 2A = \dfrac{2\tan A}{1+\tan^2 A}$
2. $\cos 2A = (\cos^2 A - \sin^2 A)$ or $(1 - 2\sin^2 A)$

   or $\quad (2\cos^2 A - 1) = \dfrac{1-\tan^2 A}{1+\tan^2 A}$
3. $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

4. $\cot 2A = \dfrac{\cot^2 A - 1}{2\cot A}$

5. $\sin 2A = \dfrac{2\tan A}{1 + \tan^2 A}$

TRIGONOMETRICAL RATIOS OF ANGLE 3A

1. $\sin 3A = 3\sin A - 4\sin^3 A$
2. $\cos 3A = 4\cos^3 A - 3\cos A$
3. $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

TRIGONOMETRICAL RATIOS OF HALF ANGLE FORMULAE

1. $\sin A = 2\sin\dfrac{A}{2}\cos\dfrac{A}{2}$
2. $\cos A = \left(\cos^2\dfrac{A}{2} - \sin^2\dfrac{A}{2}\right)$ or $\left(1 - 2\sin^2\dfrac{A}{2}\right)$ or $\left(2\cos^2\dfrac{A}{2} - 1\right)$
3. $1 + \cos A = 2\cos^2\dfrac{A}{2}$
4. $1 - \cos A = 2\sin^2\dfrac{A}{2}$
5. $\tan A = \dfrac{2\tan(A/2)}{1 - \tan^2(A/2)}$
6. $\cot A = \dfrac{\cot^2(A/2) - 1}{2\cot(A/2)}$
7. $\sin A = \dfrac{2\tan(A/2)}{1 + \tan^2(A/2)}$
8. $\cos A = \dfrac{1 - \tan^2(A/2)}{1 + \tan^2(A/2)}$
9. $\sin\dfrac{A}{2} = \sqrt{\dfrac{1 - \cos A}{2}}$
10. $\cos\dfrac{A}{2} = \sqrt{\dfrac{1 + \cos A}{2}}$
11. $\tan\dfrac{A}{2} = \sqrt{\dfrac{1 - \cos A}{1 + \cos A}}$

TRIGONOMETRICAL EQUATIONS

1. If $\sin x = 0$ or $\tan x = 0$, then $x = n\pi$
2. If $\cos x = 0$ or $\cot x = 0$, then $x = (2n + 1)\dfrac{\pi}{2}$
3. If $\sin x = \sin y$, then $x = n\pi + (-1)^n y$
4. If $\cos x = \cos y$, then $x = 2n\pi \pm y$
5. If $\tan x = \tan y$, then $x = n\pi + y$

T-RATIO OF SOME SPECIAL ANGLES

1. $\sin 7\dfrac{1^\circ}{2} = \dfrac{\sqrt{4 - \sqrt{6} - \sqrt{2}}}{2\sqrt{2}}$
2. $\cos 7\dfrac{1^\circ}{2} = \dfrac{\sqrt{4 + \sqrt{6} + \sqrt{2}}}{2\sqrt{2}}$
3. $\sin 9^\circ = \dfrac{\sqrt{3 + \sqrt{5}} - \sqrt{5 - \sqrt{5}}}{4}$
4. $\cos 9^\circ = \dfrac{\sqrt{3 + \sqrt{5}} + \sqrt{5 - \sqrt{5}}}{4}$
5. $\sin 15^\circ = \dfrac{\sqrt{3} - 1}{2\sqrt{2}}$
6. $\cos 15^\circ = \dfrac{\sqrt{3} + 1}{2\sqrt{2}}$
7. $\tan 15^\circ = 2 - \sqrt{3}$
8. $\tan 18^\circ = \dfrac{\sqrt{5} - 1}{4}$
9. $\cos 18^\circ = \dfrac{\sqrt{10 + 2\sqrt{5}}}{4}$
10. $\tan 18^\circ = \dfrac{\sqrt{25 - 10\sqrt{5}}}{5}$
11. $\sin 22\dfrac{1^\circ}{2} = \dfrac{\sqrt{2 - \sqrt{2}}}{2}$
12. $\sin 22\dfrac{1^\circ}{2} = \dfrac{\sqrt{2 + \sqrt{2}}}{2}$
13. $\tan 22\dfrac{1^\circ}{2} = \sqrt{2} - 1$
14. $\sin 36^\circ = \dfrac{\sqrt{10 - 2\sqrt{5}}}{4}$
15. $\cos 36^\circ = \dfrac{\sqrt{5} + 1}{4}$
16. $\tan 36^\circ = \sqrt{5 - 2\sqrt{5}}$
17. $\sin 54^\circ = \dfrac{\sqrt{5} + 1}{4}$
18. $\cos 54^\circ = \dfrac{\sqrt{10 - 2\sqrt{5}}}{4}$
19. $\sin 72^\circ = \dfrac{\sqrt{10 + 2\sqrt{5}}}{4}$
20. $\cos 72^\circ = \dfrac{\sqrt{5} - 1}{4}$

SOME USEFUL IDENTITIES

1. $\sin\alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \cdots + \sin[a + (n-1)\beta]$

$$= \frac{\sin\left[\alpha + (n-1)\dfrac{\beta}{2}\right]\sin\left(\dfrac{n\beta}{2}\right)}{\sin\left(\dfrac{\beta}{2}\right)}$$

$$= \frac{\sin\left(\dfrac{\text{1st angle} + \text{last angle}}{2}\right)\cdot\sin\left(\dfrac{n}{2} \times \begin{matrix}\text{difference of two}\\ \text{consecutive angles}\end{matrix}\right)}{\sin\left(\dfrac{1}{2} \times \text{difference of two consecutive angles}\right)}$$

2. $\cos\alpha+\cos(\alpha+\beta)+\cos(\alpha+2\beta)+\cdots+\cos[\alpha+(n-1)\beta]$
$$=\frac{\cos\left[\alpha+(n-1)\frac{\beta}{2}\right]\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$$

3. $\cos A\cos 2A\cos 4A\ldots\cos 2^{n-1}A=\frac{1}{2^n\sin A}\cdot\sin(2^nA)$

4. $\cos n\theta=\cos^n\theta-{}^nC_2\cos^{n-2}\sin^2\theta+{}^nC_4\cos^{n-4}\theta\sin^4-\cdots$

5. $\sin n\theta={}^nC_1\cos^{n-1}\theta\sin\theta-{}^nC_3\cos^{n-3}\sin^3\theta+\cdots$

6. $\tan n\theta=\frac{{}^nC_1\tan\theta-{}^nC_3\tan^3+\cdots}{1+{}^nC_2\tan^2\theta+{}^nC_4\tan^4\theta-\cdots}$

7. $\sin\theta\cdot\sin\left(\frac{\pi}{3}-\theta\right)\cdot\sin\left(\frac{\pi}{3}+\theta\right)=\frac{1}{4}\sin 3\theta$

8. $\cos\theta\cdot\cos\left(\frac{\pi}{3}-\theta\right)\cdot\cos\left(\frac{\pi}{3}+\theta\right)=\frac{1}{4}\cos 3\theta$

9. $\tan\theta\cdot\tan\left(\frac{\pi}{3}-\theta\right)\cdot\tan\left(\frac{\pi}{3}+\theta\right)=\tan 3\theta$

GREATEST AND LEAST VALUES OF THE EXPRESSION $a\sin\theta+b\cos\theta$

Given expression: $a\sin\theta+b\cos\theta$

Let $a=r\cos\alpha$ and $b=r\sin\alpha$, then

$r=\sqrt{a^2+b^2}$ and $a\sin\theta+b\cos\theta=r\cos\alpha\sin\theta+r\sin\alpha\cos\theta$

$=r\sin(\theta+\alpha)$

Now, since $-1\le\sin(\theta+\alpha)\le 1$,

$\therefore\quad -r\le r\sin(\theta+\alpha)\le r$,

i.e. $-\sqrt{a^2+b^2}\le a\sin\theta+b\cos\theta\le\sqrt{a^2+b^2}$

$\therefore$ The greatest and the least values of $a\sin\theta+b\cos\theta$ are $\sqrt{a^2+b^2}$ and $-\sqrt{a^2+b^2}$, respectively.

SOLVED PROBLEMS

1. If $\tan\theta+\sec\theta=\sqrt{3}$, $0<\theta<\pi$, then θ is equal to or the least positive value of θ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$

(c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$

Ans. (b)

Solution: $\sec\theta+\tan\theta=\sqrt{3}$, $\sec^2\theta-\tan^2\theta=1$

$\therefore\quad \sec\theta-\tan\theta=\frac{1}{\sqrt{3}}$

Subtracting, $2\tan\theta=\sqrt{3}-\frac{1}{\sqrt{3}}=\frac{3-1}{\sqrt{3}}$

$\Rightarrow\quad \tan\theta=\frac{1}{\sqrt{3}}$

$\therefore\quad \theta=30^\circ=\frac{1}{\sqrt{3}}=\frac{\pi}{6}$ [$\because\ 0<\theta<\pi$]

2. $\cos 2\theta+2\cos\theta$ is always

(a) greater than –(3/2)

(b) less than or equal to 3/2

(c) greater than or equal to –(3/2)

(d) none of these

Ans. (a)

Solution: We have

$\cos 2\theta+2\cos\theta=2\cos^2\theta-1+2\cos\theta$

$=2\left(\cos\theta+\frac{1}{2}\right)^2-\frac{3}{2}$

Now, $2\left(\cos\theta+\frac{1}{2}\right)^2\ge 0$ for all θ

$\Rightarrow\quad 2\left(\cos\theta+\frac{1}{2}\right)^2-\frac{3}{2}\ge\frac{-3}{2}$ for all θ

$\Rightarrow\quad \cos 2\theta+2\cos\theta\ge\frac{-3}{2}$ for all θ

3. If A and B are positive acute angles satisfying the equations $3\cos^2A+2\cos^2B=4$ and $\frac{3\sin A}{\sin B}=\frac{2\cos B}{\cos A}$, then $A+2B$ is

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

Ans. (a)

Solution: $\frac{3\sin A}{\sin B}=\frac{2\cos B}{\cos A}\Rightarrow\frac{3\sin A}{\cos A}=\frac{2\cos B}{\cos A}\cdot\frac{\sin B}{\cos A}$

$$\Rightarrow \tan A = \frac{\sin 2B}{3\cos^2 A} = \frac{\sin 2B}{\cos 2B}\cdot\frac{\cos 2B}{3\cos^2 A}$$

$$= \frac{\tan 2B(2\cos^2 B - 1)}{3\cos^2 A}$$

$$= \frac{\tan 2B(4 - 3\cos^2 A - 1)}{3\cos^2 A}$$

$$[\because \quad 2\cos^2 B = 4 - 3\cos^2 A]$$

$$= \tan 2B \cdot \tan^2 A$$

$$\Rightarrow \tan A \cdot \tan 2B = 1$$

i.e. $\sin A \sin 2B = \cos A \cos 2B$

$$\Rightarrow \cos(A + 2B) = 0 \Rightarrow A + 2B = \frac{\pi}{2}$$

4. $\sin\frac{\pi}{n} + \sin\frac{3\pi}{n} + \sin\frac{5\pi}{n} + \cdots$ up to n terms is equal to

(a) 1 (b) 2

(c) 3 (d) 0

Ans. (d)

Solution: $\sin\frac{\pi}{n} + \sin\frac{3\pi}{n} + \sin\frac{5\pi}{n} + \cdots$ up to n terms

$$= \frac{\sin(n\cdot 2\pi)/2n}{\sin(2\pi/2n)}\cdot\sin\left[\frac{2\cdot(\pi/n) + (n-1)(2\pi/n)}{2}\right]$$

$$= \frac{\sin\pi}{\sin(\pi/n)}.\sin\left(\frac{2\pi + 2n\pi - 2\pi}{2n}\right)$$

$$= \frac{\sin^2\pi}{\sin(\pi/n)} = 0$$

$\therefore$ $= \sin\frac{\pi}{n} + \sin\frac{3\pi}{n} + \sin\frac{5\pi}{n} + \cdots$ up to n terms $= 0$

5. If $\sin\theta + \cos\theta = a$, then the value of $\sin^4\theta + \cos^4\theta$ is

(a) $1 - \frac{1}{2}(a^2-1)^2$ (b) $1 + \frac{1}{2}(a^2-1)^2$

(c) $\frac{1}{2}(a^2-1)^2$ (d) none of these

Ans. (a)

Solution: $\sin^4\theta + \cos^4\theta = (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta$

$$= 1 - \frac{1}{2}(\sin 2\theta)^2 = 1 - \frac{1}{2}(a^2-1)^2$$

$$[\because \quad \sin\theta + \cos\theta = a \text{ ; squaring}$$

$$1 + \sin 2\theta = a^2 \Rightarrow \sin 2\theta = a^2 - 1]$$

6. If θ lies in the second quadrant, then the value of $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}} + \sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$ is equal to

(a) $2\sec\theta$ (b) $-2\sec\theta$

(c) $2\text{cosec}\,\theta$ (d) none of these

Ans. (b)

Solution: $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}} + \sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \frac{1-\sin\theta+1+\sin\theta}{\sqrt{1-\sin^2\theta}}$

$$= \frac{2}{\sqrt{\cos^2\theta}} = \frac{2}{|\cos\theta|}$$

$$= -\frac{2}{\cos\theta} = -2\sec\theta \qquad \left(\because \quad \frac{\pi}{2} < \theta < \pi\right)$$

7. The value of $\left(\frac{\cos A + \cos B}{\sin A - \sin B}\right)^n + \left(\frac{\sin A + \sin B}{\cos A - \cos B}\right)^n$ (where n is even) is

(a) $2\tan^n\left(\frac{A-B}{2}\right)$ (b) $2\cot^n\left(\frac{A-B}{2}\right)$

(c) 0 (d) none of these

Ans. (b)

Solution: $\left(\frac{\cos A + \cos B}{\sin A - \sin B}\right)^n + \left(\frac{\sin A + \sin B}{\cos A - \cos B}\right)^n$

$$= \left[\frac{2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)}\right]^n$$

$$+ \left[\frac{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{B-A}{2}\right)}\right]^n$$

$$= \cot^n\left(\frac{A-B}{2}\right) + \cot^n\left(\frac{B-A}{2}\right)$$

$$= \cot^n\left(\frac{A-B}{2}\right) + (-1)^n\cot\left(\frac{A-B}{2}\right)$$

$$= 2\cot^n\left(\frac{A-B}{2}\right) \qquad (\because \quad n \text{ is given})$$

8. If $x + y + z = xyz$, then the value of

$x(1-y^2)(1-z^2) + y(1-z^2)(1-x^2) + z(1-x^2)(1-y^2)$ is

(a) xyz (b) $2xyz$

(c) $4xyz$ (d) none of these

Ans. (c)

Solution: Let $x = \tan A$, $y = \tan B$ and $z = \tan C$

Given $x + y + z = xyz$

$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$

$\Rightarrow \tan A + \tan B = -\tan C + \tan A \tan B \tan C$

$= -\tan C(1 - \tan A \tan B)$

$\Rightarrow \dfrac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$

$\Rightarrow \tan(A + B) = \tan(-C)$

$\Rightarrow A + B = n\pi - C$

$\Rightarrow A + B + C = n\pi$

$\Rightarrow 2A + 2B + 2C = 2n\pi$

$\Rightarrow \tan(2A + 2B + 2C) = \tan 2n\pi = 0$

$\Rightarrow \dfrac{[\tan 2A + \tan 2B + \tan 2C - \tan 2A \tan 2B \tan 2C]}{[1 - \tan 2A \tan 2B - \tan 2B \tan 2C - \tan 2C \tan 2A]} = 0$

$\Rightarrow \tan 2A + \tan 2B + \tan 2C = \tan 2A \tan 2B \tan 2C$

$\Rightarrow \dfrac{2\tan A}{1 - \tan^2 A} + \dfrac{2\tan B}{1 - \tan^2 B} + \dfrac{2\tan C}{1 - \tan^2 C}$

$= \dfrac{2\tan A}{1 - \tan^2 A} \cdot \dfrac{2\tan B}{1 - \tan^2 B} \cdot \dfrac{2\tan C}{1 - \tan^2 C}$

Hence, on putting $\tan A = x$, $\tan B = y$, $\tan C = z$, we get

$$\frac{2x}{1-x^2} + \frac{2y}{1-y^2} + \frac{2z}{1-z^2} = \frac{2x}{1-x^2} \cdot \frac{2y}{1-y^2} \cdot \frac{2z}{1-z^2}$$

On multiplying both sides by $\dfrac{1}{2}(1-x^2)(1-y^2)(1-z^2)$, we get

$x(1-y^2)(1-z^2) + y(1-z^2)(1-x^2) + z(1-x^2)(1-y^2)$

$= 4xyz$

9. If $\sin A = (4/5)$ and $\cos B = -(12/13)$, where A and B lie in the first and third quadrants respectively, then $\cos(A + B)$ is equal to

(a) 56/65 (b) $-(56/65)$

(c) 16/65 (d) $-(16/65)$

Ans. (d)

Solution: We have $\sin A = \dfrac{4}{5}$ and $\cos B = -\dfrac{12}{13}$

Now, $\cos(A + B) = \cos A \cos B - \sin A \sin B$

$= \sqrt{1 - \dfrac{16}{25}}\left(-\dfrac{12}{13}\right) - \dfrac{4}{5}\sqrt{1 - \dfrac{144}{169}}$

$= -\dfrac{3}{5} \times \dfrac{12}{13} - \dfrac{4}{5}\left(\dfrac{5}{-13}\right)$

$= -\dfrac{36}{65} + \dfrac{20}{65} = -\dfrac{16}{25}$

10. If $\cos 5\theta = a\cos\theta + b\cos^3\theta + c\cos^5\theta + d$, then

(a) $a + b + c = 2$ (b) $a + b + c + d = 1$

(c) $b = -20$ (d) none of these

Ans. (b)

Solution: $\cos 5\theta = \cos(4\theta + \theta) = \cos 4\theta \cos\theta - \sin 4\theta \sin\theta$

$= (2\cos^2 2\theta - 1)\cos\theta - 2\sin 2\theta \cos 2\theta \sin\theta$

$= [2(2\cos^2\theta - 1)^2 - 1]\cos\theta - 4\sin\theta\cos\theta$

$(2\cos^2\theta - 1)\sin\theta$

$= [2(4\cos^4\theta - 4\cos^2\theta + 1) - 1]\cos\theta - 4\sin^2\theta$

$(2\cos^3\theta - \cos\theta)$

$= 8\cos^5\theta - 8\cos^3\theta + \cos\theta - 4(1 - \cos^2\theta)$

$(2\cos^3\theta - \cos\theta)$

$= 8\cos^5\theta - 8\cos^3\theta + \cos\theta - 4(2\cos^3\theta$

$-2\cos^5\theta - \cos\theta + \cos^3\theta)$

$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

Hence $a = 5$, $b = -20$, $c = 16$, $d = 0$

$\therefore \quad a + b + c + d = 1$

11. If $0° < \theta < 180°$, then

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2(1 + \cos\theta)}}}}$$

there being n number of 2's, is equal to

(a) $2\cos\dfrac{\theta}{2^n}$ (b) $2\cos\dfrac{\theta}{2^{n-1}}$

(c) $2\cos\dfrac{\theta}{2^{n+1}}$ (d) none of these

Ans. (a)

Solution: $\sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2(1 + \cos\theta)}}}}$, there being n numbers of 2's

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2\left(2\cos^2\frac{\theta}{2}\right)}}}}$$

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2\left(1 + \cos\frac{\theta}{2}\right)}}}}$$

there being $(n - 1)$ number of 2's

$= \ldots = \sqrt{2\left(1 + \cos\dfrac{\theta}{2^{n-1}}\right)} = 2\cos\dfrac{\theta}{2^n}$

12. A circular wire of diameter 10 cm is cut and placed along the circumference of a circle of diameter 1 m. The angle subtended by the wire at the centre of the circle is equal to

(a) $\frac{\pi}{4}$ rad (b) $\frac{\pi}{3}$ rad

(c) $\frac{\pi}{5}$ rad (d) $\frac{\pi}{10}$ rad

Ans. (c)

Solution: Diameter of the circular wire = 10 cm

$\therefore$ Length of the wire = 10π cm

Hence, the required angle $= \frac{\text{length of the arc}}{\text{radius of the big circle}}$

$= \frac{10\pi}{50} = \frac{\pi}{5}$ rad

13. The angles of a triangle are in A.P. The number of degrees in the least is to be the number of radians in the greatest as 60 : π. Then the greatest angle is

(a) 120° (b) 90°

(c) 135° (d) 105°

Ans. (b)

Solution: Let the angles of the triangle be

$$(a-d)^\circ,\ a^\circ,\ (a+d)^\circ$$

Their sum $= a-d+a+a+d = 180$

$\Rightarrow\ 3a = 180\ \Rightarrow\ a = 60$

Again by the given condition,

$$\frac{a-d}{(a+d)\frac{\pi}{180}} = \frac{60}{\pi}\ \Rightarrow\ \frac{3(a-d)}{a+d} = 1$$

$\Rightarrow\ 3a - 3d = a + d$

$\Rightarrow\ 2a = 4d \quad \Rightarrow\ a = 2d$

$\therefore\ 60 = 2d \quad \therefore\ d = 30$

Hence, the greatest angle = 60 + 30 = 90°.

14. The difference between two acute angles of a right-angled triangle is $3\pi/10$ radians. The angles in degrees are

(a) 60°, 30° (b) 70°, 20°

(c) 72°, 18° (d) none of these

Ans. (c)

Solution: Let the angles in degrees be x and $y (x > y)$

$\therefore\ x + y = 90^\circ$

Given $x - y = \frac{3\pi}{10} = 54^\circ$

$\Rightarrow\ 2x = 144^\circ$, i.e. $x = 72^\circ$

$2y = 36$, i.e. $y = 18^\circ$

15. A cow is tied to a post with a rope. If the cow moves along the circular path always keeping the rope tight and describes 44 m, when it has traced out 72° at the centre, the length of the rope is

(a) 35 m (b) 22 m

(c) 56 m (d) 45 m

Ans. (a)

Solution: $\theta = 72^\circ$

Let r = length of the rope

Since $\theta = \frac{l}{r}$

$$\therefore\ r = \frac{l}{\theta} = \frac{44\times 5}{2\pi} = \frac{44\times 5\times 7}{2\times 22} = 35 \text{ m}$$

16. If $\tan A = 2\tan B + \cot B$, then $2\tan(A-B)$ is equal to

(a) $\tan B$ (b) $2\tan B$

(c) $\cot B$ (d) $2\cot B$

Ans. (c)

Solution: $\tan A = 2\tan B + \cot B$ (i)

Now, $2\tan(A-B) = 2\left(\frac{\tan A - \tan B}{1+\tan A\tan B}\right)$

$= 2\frac{(2\tan B - \cot B - \tan B)}{1+(2\tan B+\cot B)\tan B}$ [from Eq. (i)]

$= 2\frac{\tan B + \cot B}{2(1+\tan^2 B)} = \frac{\cot B(\tan^2 B+1)}{(1+\tan^2 B)} = \cot B$

17. The minute hand of a clock is 10 cm long. How far does the tip of the hand move in 20 min?

(a) $\frac{3}{20}$ cm (b) $\frac{3\pi}{20}$ cm

(c) $\frac{20}{3}$ cm (d) $\frac{20\pi}{3}$ cm

Ans. (d)

Solution: Angle moved in 20 min $= 120^\circ = \frac{2\pi}{3}$

Here r = 10 cm

Using $\theta = \frac{l}{r}$, we get

$$l = r\theta = 10\left(\frac{2\pi}{3}\right) = \frac{20\pi}{3} \text{ cm}$$

18. If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then $\frac{\tan x}{\tan y}$ is equal to

(a) $\frac{b}{a}$ (b) $\frac{a}{b}$

(c) ab (d) none of these

Ans. (b)

Solution: $\frac{\sin(x+y)}{\sin(x-y)}=\frac{a+b}{a-b}$

$\Rightarrow \frac{\sin(x+y)+\sin(x-y)}{\sin(x-y)-\sin(x-y)}=\frac{(a+b)+(a-b)}{(a+b)-(a-b)}$

$\Rightarrow \frac{2\sin x\cos y}{2\cos x\sin y}=\frac{2a}{2b} \quad \Rightarrow \quad \frac{\tan x}{\tan y}=\frac{a}{b}$

19. If $\cos A = m\cos B$ and $\cot\frac{A+B}{2}=\lambda\tan\frac{B-A}{2}$, then λ is

(a) $\frac{m}{m-1}$ (b) $\frac{m+1}{m}$

(c) $\frac{m+1}{m-1}$ (d) none of these

Ans. (c)

Solution: We have $\cos A = m\cos B$

$\Rightarrow \frac{\cos A}{\cos B}=\frac{m}{1}$

$\Rightarrow \frac{\cos A+\cos B}{\cos A-\cos B}=\frac{m+1}{m-1}$

$\Rightarrow \frac{2\cos[(A+B)/2]\{\cos[(B-A)/2]\}}{2\sin[(A+B)/2]\{\sin[(B-A)/2]\}}=\frac{m+1}{m-1}$

$\Rightarrow \cot\frac{A+B}{2}=\left(\frac{m+1}{m-1}\right)\tan\frac{B-A}{2}$

But $\cot\frac{A+B}{2}=\lambda\tan\frac{B-A}{2}$

$\therefore \quad \lambda=\frac{m+1}{m-1}$

EXERCISE SET 1

1. The value of $\sin^2\frac{2\pi}{3}+\cos^2\frac{5\pi}{6}-\tan^2\frac{3\pi}{4}$ is

(a) 1/2 (b) 1/4

(c) 1/3 (d) none of these

2. A circular wire of radius 3 cm is cut and bent so as to lie along the circumference at a loop whose radius is 48 cm. The angle in grades which is subtended at the centre of the loop is

(a) 50 grades (b) 20 grades

(c) 25 grades (d) none of these

3. The value of tan 20° tan 40° tan 60° tan 80° is equal to

(a) 1 (b) 2

(c) 3 (d) $\sqrt{3/2}$

4. If $\operatorname{cosec}\theta-\sin\theta=a^3$ and $\sec\theta-\cos\theta=b^3$, then $a^2b^2(a^2+b^2)=$

(a) 2 (b) $\sin^2\theta$

(c) $\cos^2\theta$ (d) 1

5. If cos 20° = k and $\cos x=2k^2-1$, then the possible values of x between 0° and 360° are

(a) 140° and 180° (b) 40° and 140°

(c) 50° and 130° (d) 40° and 320°

6. If $\operatorname{cosec}\theta-\cot\theta=p$, then the value of $\operatorname{cosec}\theta$ is

(a) $\frac{1}{2}\left(p+\frac{1}{p}\right)$ (b) $\frac{1}{2}\left(1-\frac{1}{p}\right)$

(c) $\left(1+\frac{1}{p}\right)$ (d) $\left(1-\frac{1}{p}\right)$

7. The value of the expression $3\cos\theta+4\sin\theta$ lies between

(a) –7 and 7 (b) –25 and 25

(c) –1 and 1 (d) –5 and 5

8. If $\frac{x}{\cos\theta}=\frac{y}{\cos[\theta-(2\pi/3)]}=\frac{z}{\cos[\theta+(2\pi/3)]}$, then $x + y + z$ is equal to

(a) 1 (b) 0

(c) –1 (d) none of these

9. If $\sin\alpha=\frac{2pq}{p^2+q^2}$, then $\sec\alpha-\tan\alpha=$

(a) $\frac{p-q}{p+q}$ (b) $\frac{pq}{p^2+q^2}$

(c) $\frac{p+q}{p-q}$ (d) none of these

10. $\tan 3x-\tan 2x-\tan x$ is equal to

(a) $\tan x\tan 2x\tan 3x$

(b) $-\tan x\tan 2x\tan 3x$

(c) $\tan x\tan 2x-\tan x\tan 3x-\tan 2x\tan 3x$

(d) none of these

11. If $a\cos\theta-b\sin\theta=c$, then $a\sin\theta+b\cos\theta=$

(a) $\pm\sqrt{a^2+b^2-c^2}$ (b) $\pm\sqrt{a^2-b^2+c^2}$

(c) $\pm\sqrt{b^2+c^2-a^2}$ (d) $\pm\sqrt{a^2+b^2+c^2}$

12. The value of cos 15° – sin 15° is

(a) $\frac{1}{\sqrt{2}}$ (b) $-\frac{1}{\sqrt{2}}$

(c) $\frac{1}{2\sqrt{2}}$ (d) 0

13. The value of $6(\sin^6\theta+\cos^6\theta)-9(\sin^4\theta+\cos^4\theta)+4$ is
(a) –3 (b) 0
(c) 1 (d) 3

14. The value of cos 20° cos 40° cos 60° cos 80° is equal to
(a) $\frac{3}{8}$ (b) $\frac{1}{8}$
(c) $\frac{1}{16}$ (d) $\frac{3}{16}$

15. If $A+B+C=\frac{3\pi}{2}$, then $\cos 2A+\cos 2B+\cos 2C$ is equal to
(a) $1-4\cos A\cos B\cos C$ (b) $4\sin A\sin B\sin C$
(c) $1+2\cos A\cos B\cos C$ (d) $1-4\sin A\sin B\sin C$

16. If $\sec\theta+\tan\theta=p$, then the value of sin θ is
(a) $\frac{p^2+1}{p^2-1}$ (b) $\frac{p^2-1}{p^2+1}$
(c) $\frac{p^3+1}{p^3-1}$ (d) $\frac{p^3-1}{p^3+1}$

17. Let $\sin x+\sin y=a$ and $\cos x+\cos y=b$. Then $\sin(x+y)$ is equal to
(a) $\frac{ab}{a+b}$ (b) $\frac{ab}{a^2+b^2}$
(c) $\frac{2ab}{a^2+b^2}$ (d) $\frac{ab}{2(a^2+b^2)}$

18. If $A+B+C=\pi$ and $\cos A=\cos B\cos C$, then tan $B\cdot$ tan C is equal to
(a) 1/2 (b) 2
(c) 1 (d) –(1/2)

19. If $\sin x+\sin^2 x=1$, then $\cos^8 x+2\cos^6 x+\cos^4 x=$
(a) 0 (b) –1
(c) 2 (d) 1

20. The value of $\frac{\cot 54^\circ}{\tan 36^\circ}+\frac{\tan 20^\circ}{\cot 70^\circ}=$
(a) 3 (b) 2
(c) 0 (d) 1

21. If $A=\sin^2\theta+\cos^4\theta$, then for all real values of θ,
(a) $1\le A\le 2$ (b) $\frac{3}{4}\le A\le 1$
(c) $\frac{13}{16}\le A\le 1$ (d) $\frac{3}{4}\le A\le\frac{13}{16}$

22. $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}}+\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$ is equal to
(a) $2\sin\theta$ (b) $2\cos\theta$
(c) $\frac{2}{|\cos\theta|}$ (d) $\frac{2}{|\sin\theta|}$

23. The value of tan 15° + cot 15° is
(a) $\sqrt{3}$ (b) $2\sqrt{3}$
(c) 4 (d) –4

24. If $\tan x+\cot x=2$, then $\sin^{2n}x+\cos^{2n}x$ is equal to
(a) 2^n (b) –(1/2)
(c) 1/2 (d) none of these

25. If $\sin\theta+\cos\theta=a$, then the value of $|\sin\theta-\cos\theta|$ is
(a) $\sqrt{2-a^2}$ (b) $\sqrt{2+a^2}$
(c) $\sqrt{a^2-2}$ (d) none of these

26. $\frac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta}=$
(a) $\frac{1}{2}\tan\theta$ (b) $\frac{1}{2}\cot\theta$
(c) tan θ (d) cot θ

27. The maximum value of $4\sin^2 x+3\cos^2 x+\sin\frac{x}{2}+\cos\frac{x}{2}$ is
(a) $4+\sqrt{2}$ (b) $3+\sqrt{2}$
(c) 9 (d) 4

28. If $\sin 2\theta=\cos 3\theta$ and θ is an acute angle, then sin θ equals
(a) $\frac{\sqrt{5}-1}{4}$ (b) $-\left(\frac{\sqrt{5}-1}{4}\right)$
(c) $\frac{\sqrt{5}+1}{4}$ (d) $\frac{-\sqrt{5}-1}{4}$

29. If $\frac{3\pi}{4}<\alpha<\pi$, then $\sqrt{2\cot\alpha+\frac{1}{\sin^2\alpha}}$ is equal to
(a) $1+\cot\alpha$ (b) $-1-\cot\alpha$
(c) $1-\cot\alpha$ (d) $-1+\cot\alpha$

30. If $\sec A+\tan A=\frac{3}{2}$, then
(a) $\sin A=\frac{5}{13}$ (b) $\sin 2A=\frac{5}{13}$
(c) $\sin A=\frac{12}{13}$ (d) $\sin 2A=\frac{12}{13}$

31. If $\cos\theta = \frac{1}{2}\left(x + \frac{1}{x}\right)$, then $\frac{1}{2}\left(x^2 + \frac{1}{x^2}\right)$ is equal to

(a) $\sin 2\theta$ (b) $\cos 2\theta$

(c) $\tan 2\theta$ (d) none of these

32. $\cos^2 48° - \sin^2 12° =$

(a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{8}$

(c) $\frac{\sqrt{3}-1}{4}$ (d) $\frac{\sqrt{3}+1}{2\sqrt{2}}$

33. If $1 + \sin x + \sin^2 x + \sin^3 x + \cdots + \infty$ is equal to $4 + 2\sqrt{3}$, $0 < x < \pi$, then $x =$

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ or $\frac{\pi}{6}$ (d) $\frac{\pi}{3}$ or $\frac{2\pi}{3}$

34. The value of $\sin^2 5° + \sin^2 10° + \sin^2 15° + \cdots + \sin^2 90°$ is

(a) $7\frac{1}{2}$ (b) $8\frac{1}{2}$

(c) $1\frac{1}{2}$ (d) $9\frac{1}{2}$

35. If for real values of x, $\cos\theta = x + \frac{1}{x}$, then

(a) θ is an acute angle
(b) θ is a right angle
(c) θ is an obtuse angle
(d) no value of θ is possible

36. If $m = \sin^6 x + \cos^6 x$, then m belongs to the interval

(a) $\left[\frac{7}{4}, \frac{5}{4}\right]$ (b) $\left[\frac{1}{2}, \frac{5}{8}\right]$

(c) $\left[\frac{1}{4}, 1\right]$ (d) none of these

37. If $\sin(A + B + C) = 1$, $\tan(A - B) = \frac{1}{\sqrt{3}}$ and $\sec(A + C) = 2$, then

(a) $A = 120°$, $B = 60°$, $C = 0°$
(b) $A = 60°$, $B = 30°$, $C = 0°$
(c) $A = 90°$, $B = 60°$, $C = 30°$
(d) none of these

38. The value of $\frac{\cot^2\theta + 1}{\cot^2\theta - 1}$ is equal to

(a) $\sin\theta$ (b) $\cos 2\theta$

(c) $\operatorname{cosec} 2\theta$ (d) $\sec 2\theta$

39. Total number of solutions of $\sin^4 x + \cos^4 x = \sin x \cdot \cos x$ $(0 \le x \le 2\pi)$ is equal to

(a) 2 (b) 4

(c) 6 (d) none of these

40. If $\tan^2\theta = 2\tan^2\phi + 1$, then $\cos 2\theta + \sin^2\phi$ equals

(a) -1 (b) 0

(c) 1 (d) none of these

41. If $\sin^6\theta + \cos^6\theta + K\cos^2\theta = 1$, then K is equal to

(a) $\frac{1}{2}\tan^2 2\theta$ (b) $\frac{1}{4}\tan^2 2\theta$

(c) $4\cot^2 2\theta$ (d) $\frac{3}{4}\tan^2 2\theta$

42. If $a\sec\alpha - c\tan\alpha = d$ and $b\sec\alpha + d\tan\alpha = c$, then

(a) $a^2 + c^2 = b^2 + d^2$ (b) $a^2 + d^2 = b^2 + c^2$

(c) $a^2 + b^2 = c^2 + d^2$ (d) $ab = cd$

43. The sum of all the solutions of the equation $\cot\theta = \sin 2\theta$ $(\theta \ne n\pi, n \text{ integer}, 0 \le \theta \le \pi)$ is

(a) $3\pi/2$ (b) π

(c) $3\pi/4$ (d) 2π

44. If $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$, then $\cos\theta_1 + \cos\theta_2 + \cos\theta_3 =$

(a) 3 (b) 2

(c) 1 (d) 0

45. If $A = \sin^9\theta + \cos^{14}\theta$, then for all values of θ,

(a) $A \ge 1$ (b) $0 < A \le 1$

(c) $1 < 2A \le 3$ (d) none of these

46. If θ is an acute angle and $\tan\theta = 1/\sqrt{7}$, then the value of $\frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta}$ is

(a) 3/4 (b) 1/2

(c) 2 (d) 5/4

47. If $0 \le x \le 2\pi$, then the number of solutions of the equation $3(\sin x + \cos x) - 2(\sin^3 x + \cos^3 x) = 8$ is

(a) 0 (b) 1

(c) 2 (d) 4

48. The value of $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$ is

(a) 1/8 (b) $-(1/8)$

(c) 1 (d) 0

49. The value of $\cos 9° - \sin 9°$ is

(a) $-\frac{\sqrt{5-\sqrt{5}}}{2}$ (b) $\frac{5+\sqrt{5}}{4}$

(c) $\frac{1}{2}\sqrt{5-\sqrt{5}}$ (d) none of these

50. The general solution of $x-5\sin x\cos x-6\cos^2 x=0$ is
(a) $x=n\pi-\pi/4,\ n\in Z$ only
(b) $n\pi+\tan^{-1}6,\ n\in Z$ only
(c) both (a) and (b)
(d) none of these

51. The value of $\sin\frac{\pi}{7}+\sin\frac{2\pi}{7}+\sin\frac{3\pi}{7}$ is
(a) $\cot\frac{\pi}{14}$ (b) $\frac{1}{2}\cot\frac{\pi}{14}$
(c) $\tan\frac{\pi}{14}$ (d) $\frac{1}{2}\tan\frac{\pi}{14}$

52. If $\tan^2\theta=1-e^2$, then $\sec\theta+\tan^3\theta\,\text{cosec}\,\theta=$
(a) $(1-e^2)^{3/2}$ (b) $(2-e^2)^{1/2}$
(c) $(2-e^2)^{3/2}$ (d) none of these

53. If $2\cos\theta=a+\left(\frac{1}{a}\right)$, then $a^2+\left(\frac{1}{a^2}\right)$ is equal to
(a) $\cos 2\theta$ (b) $\sin 2\theta$
(c) $2\cos 2\theta$ (d) $2\sin 2\theta$

54. If $x,\ y\in[0,\ 2\pi]$ and $\sin x+\sin y=2$, then the value of $x+y$ is
(a) π (b) $\pi/2$
(c) 3π (d) none of these

55. If $b\sin\beta=a\sin(2\alpha+\beta)$, then $\frac{\cot(\alpha+\beta)}{\cot\alpha}=$
(a) $\frac{b-a}{b+a}$ (b) $\frac{b+a}{b-a}$
(c) $\frac{a-b}{a+b}$ (d) none of these

56. The minimum value of $9\tan^2\theta+4\cot^2\theta$ is
(a) 13 (b) 9
(c) 6 (d) 12

57. If $\frac{\cos^4 x}{\cos^2 y}+\frac{\sin^4 x}{\sin^2 y}=1$, then $\frac{\cos^4 y}{\cos^2 x}+\frac{\sin^4 y}{\sin^2 x}=$
(a) -1 (b) 1
(c) 0 (d) 2

58. The equation $3^{\sin 2x+2\cos^2 x}+3^{1-\sin 2x+2\sin^2 x}=28$ is satisfied for the values of x, given by
(a) $\cos x=0$ (b) $\tan x=-1$
(c) $\tan x=1$ (d) none of these

59. The tangents of two acute angles are 3 and 2. The sine of twice of their difference is
(a) 7/24 (b) 7/25
(c) 7/50 (d) 7/48

60. The general values of x for which cos $2x$, 1/2 and sin $2x$ are in A.P. are given by
(a) $n\pi,\ n\pi+\frac{\pi}{2}$ (b) $n\pi,\ n\pi+\frac{\pi}{4}$
(c) $n\pi+\frac{\pi}{4}$ (d) $n\pi$

EXERCISE SET 2

1. If $\tan x=\frac{2b}{a-c},\ a\neq c$
$y=a\cos^2 x+2b\sin x\cos x+c\sin^2 x,$
$z=a\sin^2 x-2b\sin x\cos x+c\cos^2 x$, then
(a) $y=z$ (b) $y-z=a+c$
(c) $y-z=a-c$ (d) $y-z=(a-c)^2+4b^2$

2. The most general value of θ, satisfying the two equations $\cos\theta=-\frac{1}{\sqrt{2}},\ \tan\theta=1$, is
(a) $2n\pi\pm\frac{5\pi}{4}$ (b) $2n\pi+\frac{\pi}{4}$
(c) $n\pi+\frac{5\pi}{4}$ (d) $(2n+1)\pi+\frac{\pi}{4}$

3. $\sin\theta+\sqrt{3}\cos\theta=6x-x^2-11,\ 0\le\theta\le 4\pi,\ x\in\mathbb{R}$ holds for
(a) no value of x and θ
(b) one value of x and two values of θ
(c) two values of x and two values of θ
(d) three pairs of values of $(x,\ \theta)$

4. If $\log_{\cos x}\sin x=0.5$, when $0<x<90°$, then the value of $\cos x=$
(a) $\frac{\sqrt{5}+1}{4}$ (b) $\frac{\sqrt{5}-1}{4}$
(c) $\frac{\sqrt{5}-1}{2}$ (d) $\frac{\sqrt{5}+1}{2}$

5. If $x=X\cos\theta-Y\sin\theta,\ y=X\sin\theta+Y\cos\theta$ and $x^2+4xy+y^2=AX^2+BY^2,\ 0\le\theta\le\frac{\pi}{2},\ n\in Z$, then
(a) $\theta=\frac{\pi}{6}$ (b) $\theta=\frac{\pi}{4}$
(c) $A=3$ (d) $B=1$

6. If $\frac{\cos^4\alpha}{\cos^2\beta}+\frac{\sin^4\alpha}{\sin^2\beta}=1$, then the value of $\frac{\cos^4\beta}{\cos^2\alpha}+\frac{\sin^4\beta}{\sin^2\alpha}$ is equal to
(a) 0 (b) 1
(c) 2 (d) 4

7. If $\tan\theta+\sec\theta=e^x$, then $\cos\theta$ equals
(a) $\frac{e^x+e^{-x}}{2}$ (b) $\frac{2}{e^x+e^{-x}}$
(c) $\frac{e^x-e^{-x}}{2}$ (d) $\frac{e^x-e^{-x}}{e^x+e^{-x}}$

8. If $\tan a\theta-\tan b\theta=0$, then the values of θ form a series in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

9. The general solution of $\sin x-3\sin 2x+\sin 3x=\cos x-3\cos 2x+\cos 3x$ is
(a) $n\pi+\frac{\pi}{8}$ (b) $\frac{n\pi}{2}+\frac{\pi}{8}$
(c) $(-1)^n\left(\frac{n\pi}{2}\right)+\left(\frac{\pi}{8}\right)$ (d) $2n\pi+\cos^{-1}\left(\frac{3}{2}\right)$

10. If $\cos x=\tan y$, $\cos y=\tan z$, $\cos z=\tan x$, then the value of $\sin x$ is
(a) sin 18° (b) cos 18°
(c) 2 sin 18° (d) 2 cos 18°

11. The value of the expression $1-\frac{\sin^2 x}{1+\cos x}+\frac{1+\cos x}{\sin x}-\frac{\sin x}{1-\cos x}$ is equal to
(a) 1 (b) 0
(c) $\sin x$ (d) $\cos x$

12. The maximum value of $\sin^8\theta+\cos^{16}\theta$ is
(a) 1 (b) 3/2
(c) 3/4 (d) none of these

13. The equation $\sin^4 x-2\cos^2 x+a^2=0$ is solvable if
(a) $-\sqrt{3}\le a\le\sqrt{3}$ (b) $-\sqrt{2}\le a\le\sqrt{2}$
(c) $-1\le a\le 1$ (d) none of these

14. The value of $\left(1+\cos\frac{\pi}{8}\right)\cdot\left(1+\cos\frac{3\pi}{8}\right)\cdot\left(1+\cos\frac{5\pi}{8}\right)\cdot\left(1+\cos\frac{7\pi}{8}\right)$ is equal to
(a) $\frac{1}{8}$ (b) $\frac{1}{2}$
(c) $\frac{1+\sqrt{2}}{2\sqrt{2}}$ (d) $-\frac{1}{8}$

15. If α and β are different values of x satisfying $a\cos x+b\sin x=c$, then $\tan[(\alpha+\beta)/2]=$
(a) $a+b$ (b) $a-b$
(c) b/a (d) a/b

16. The expression $\frac{\cos 6x+6\cos 4x+15\cos 2x+10}{\cos 5x+5\cos 3x+10\cos x}$ is equal to
(a) $\cos 2x$ (b) $2\cos x$
(c) $\cos^2 x$ (d) $\cos x$

17. If $D=\begin{vmatrix}1 & \cos\alpha & 1\\ -\sin\alpha & 1 & -\cos\alpha\\ -1 & \sin\alpha & 1\end{vmatrix}$, then D lies in the interval
(a) [–2, 2] (b) [2, 4]
(c) [0, 4] (d) $[2-\sqrt{2},\ 2+\sqrt{2}]$

18. It the equation $\sec\theta+\operatorname{cosec}\theta=c$ has two real roots between 0 and 2π, then
(a) $c^2<8$ (b) $c^2>8$
(c) $c^2=8$ (d) none of these

19. If $\frac{2\sin\theta}{1+\cos\theta+\sin\theta}=y$, then $\frac{1-\cos\theta+\sin\theta}{1+\sin\theta}$ is equal to
(a) $1-y$ (b) $1+y$
(c) $1/y$ (d) y

20. If $\sin^4 x+\cos^4 y+2=4\sin x\cos y$, $0\le x,\ y\le\pi/2$, then $\sin x+\cos y$ equals
(a) 0 (b) –2
(c) 2 (d) none of these

21. The number of real roots of the equation $\sec\theta+\operatorname{cosec}\theta=\sqrt{15}$ lying between 0 and π is
(a) 0 (b) 2
(c) 4 (d) 8

22. The solutions of the equation $(\sqrt{3}-1)\sin\theta+(\sqrt{3}+1)\cos\theta=2$ are
(a) $2n\pi\pm\frac{\pi}{4}+\frac{\pi}{12}$ (b) $n\pi+(-1)^n\frac{\pi}{4}-\frac{\pi}{12}$
(c) $2n\pi\pm\frac{\pi}{4}-\frac{\pi}{12}$ (d) $n\pi$

23. $\sum_{r=1}^{n-1}\cos^2\frac{r\pi}{n}$ is equal to
(a) $\frac{n}{2}-1$ (b) $\frac{n}{2}-\frac{1}{2}$
(c) $\frac{n}{2}$ (d) none of these

24. If $5\cos 2\theta + 2\cos^2\frac{\theta}{2} + 1 = 0, -\pi < \theta < \pi$, then $\theta =$

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{3}, \cos^{-1}\left(\frac{3}{5}\right)$

(c) $\cos^{-1}\left(\frac{3}{5}\right)$ (d) $\pm\frac{\pi}{3}, \pi - \cos^{-1}\left(\frac{3}{5}\right)$

25. The value of $\sin^6 x + \cos^6 x$ is

(a) π (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

26. $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}$ is equal to

(a) an integer
(b) a negative rational number
(c) a positive rational number
(d) an inrrational number

27. If $\sin^2 x + \cos^4 x = \frac{13}{16}$, then the general solution for x is

(a) $\frac{n\pi}{2} \pm \frac{\pi}{6}$ (b) $n\pi \pm \frac{\pi}{6}$

(c) $n\pi + \frac{\pi}{6}$ (d) $2n\pi \pm \frac{\pi}{6}$

28. The general solution for $|\sin x| = \cos x$ is

(a) $n\pi + \frac{\pi}{4}$ (b) $2n\pi + \frac{\pi}{4}$

(c) $2n\pi \pm \frac{\pi}{4}$ (d) none of these

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(b) Statement I is true, Statement II is false.
(c) Statement I is false, Statement II is a true.
(d) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

29. Statement I: If $\sec\theta + \cos\theta = 2$, then $\sec^n\theta + \cos^n\theta = 2^n$

Statement II: If $a + b = 2$, $ab = 1$, then $a = b = 1$

30. Statement I: $\sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right)$

$= 3\sin x \sin\left(x + \frac{2\pi}{3}\right) \cdot \sin\left(x + \frac{4\pi}{3}\right)$

Statement II: If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

31. Statement I: $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7} = -\frac{1}{8}$

Statement II: $\cos\theta\cos 2\theta\cos 4\theta \ldots \cos 2^{n-1}\theta = -\frac{1}{2^n}$ if $\theta = \frac{\pi}{2^n - 1}$

32. Statement I: $\cos\theta = x + \frac{1}{x}$ is not possible $[x \in \mathbb{R} - (0)]$

Statement II: A.M. $\geq$ G.M.

33. Statement I: $\tan 3A - \tan 2A - \tan A$

$= \tan 3A \tan 2A \tan A$

Statement II: If $x = y + z$, then

$\tan x - \tan y - \tan z = \tan x \tan y \tan z$

Previous Years' Questions

34. The value of $\frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ}$ is

(a) 1 (b) $\sqrt{3}$

(c) $\sqrt{3}/2$ (d) 2

[AIEEE 2002]

35. If $y = \sec^2\theta + \cos^2$, $\theta \neq 0$, then

(a) $y = 0$ (b) $y \leq 2$

(c) $y \geq -2$ (d) $y > 2$

[AIEEE 2002]

36. The number of integral values of k for which the equation $7\cos x + 5\sin x = 2k + 1$ has a solution is

(a) 4 (b) 8
(c) 10 (d) 12

[AIEEE 2002]

37. The number of distinct real roots of

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

in the interval $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$ is

(a) 0 (b) 2
(c) 1 (d) 3

[AIEEE 2002]

38. If $\tan\theta = -(4/3)$, then $\sin\theta$ is

(a) $-\frac{4}{5}$ but not $\frac{4}{5}$ (b) $-\frac{4}{5}$ or $\frac{4}{5}$

(c) $\frac{4}{5}$ but not $-\frac{4}{5}$ (d) none of these

[AIEEE 2002]

39. If α is a root of $25\cos^2\theta + 5\cos\theta - 12 = 0$, $\frac{\pi}{2} < \alpha < \pi$, then $\sin 2\alpha$ is equal to

(a) 24/25 (b) –(24/25)

(c) 13/18 (d) –(13/18)

[AIEEE 2002]

40. The number of solutions of the equation $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$ is

(a) 5 (b) 0

(c) 2 (d) none of these

[AIEEE 2002]

41. If $\sin(\alpha+\beta) = 1$ and $\sin(\alpha-\beta) = 1/2$, then $\tan(\alpha+2\beta)\cdot\tan(2\alpha+\beta)$ is equal to

(a) 1 (b) –1

(c) zero (d) none of these

[AIEEE 2002]

42. If $\tan\alpha = \frac{m}{m+1}$ and $\tan\beta = \frac{1}{2m+1}$, then $\alpha+\beta =$

(a) $\pi/3$ (b) $\pi/4$

(c) $\pi/6$ (d) none of these

[AIEEE 2002]

43. If $3\sin x + 4\cos x = 5$, then the value of $4\sin x - 3\cos x$ is

(a) 0 (b) 1 (c) 25 (d) 0

[AIEEE 2002]

44. $3\cos x + 4\sin x + 5$ has a minimum value

(a) 5 (b) 7 (c) –7 (d) 0

[AIEEE 2002]

45. If $u = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta} + \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$, then the difference between the maximum and minimum values of u^2 is given by

(a) $2(a^2+b^2)$ (b) $2\sqrt{a^2+b^2}$

(c) $(a+b)^2$ (d) $(a-b)^2$

[AIEEE 2004]

46. Let α and β be such that $\pi < \alpha < -\beta < 3\pi$. If $\sin\alpha + \sin\beta = -(21/65)$ and $\cos\alpha + \cos\beta = -(17/65)$, then the value of $\cos(\alpha-\beta)/2$ is

(a) $-\frac{3}{\sqrt{130}}$ (b) $\frac{3}{\sqrt{130}}$ (c) $\frac{6}{65}$ (d) $-\frac{6}{65}$

[AIEEE 2004]

47. In a triangle PQR, $\angle R = (\pi/2)$ if $\tan(P/2)$ and $\tan(Q/2)$ are the roots of $ax^2 + bx + c = 0$, $a \neq 0$, then

(a) $c = a+b$ (b) $a = b+c$

(c) $b = a+c$ (d) $b = c$

[AIEEE 2005]

48. The number of values of x in the interval $[0, 3\pi]$ satisfying the equation $2\sin^2 x + 5\sin x - 3 = 0$ is

(a) 6 (b) 1 (c) 2 (d) 4

[AIEEE 2006]

49. If $0 < x < \pi$ and $\cos x + \sin x = 1/2$, then $\tan x$ is

(a) $\frac{(4-\sqrt{7})}{3}$ (b) $-\frac{(4+\sqrt{7})}{3}$

(c) $\frac{(1+\sqrt{7})}{4}$ (d) $\frac{(1-\sqrt{7})}{4}$

[AIEEE 2006]

50. Let A and B denote the statements

A: $\cos\alpha + \cos\beta + \cos\gamma = 0$

B: $\sin\alpha + \sin\beta + \sin\gamma = 0$

If $\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) = -(3/2)$, then

(a) A is true and B is false

(b) A is false and B is true

(c) Both A and B are true

(d) Both A and B are false **[AIEEE 2009]**

51. The value of $\tan 1° \tan 2° \tan 3° \ldots \tan 89°$ is equal to

(a) –1 (b) 2

(c) $\pi/2$ (d) 1 **[DCE 2003]**

52. If $\tan\theta + \tan\left(\theta+\frac{\pi}{3}\right) + \tan\left(\theta+\frac{2\pi}{3}\right) = 3$, then

(a) $\tan 2\theta = 1$ (b) $\tan 3\theta = 1$

(c) $\tan^2\theta = 1$ (d) $\tan^3\theta = 1$

[DCE 2004]

53. The value of the expression $\cos 1° \cdot \cos 2° \ldots \cos 179°$ equals

(a) 0 (b) 1

(c) $1/\sqrt{2}$ (d) –1 **[DCE 2005]**

54. If $\sin^3 x \sin 3x = \sum_{m=0}^{n} C_m \cos mx$ is an identity in x, where $C_0, C_1, \ldots, C_n$ are constants and $C_n \neq 0$, then the value of n equals

(a) 2 (b) 4 (c) 6 (d) 8

[DCE 2005]

55. If $\alpha + \beta = (\pi/2)$ and $\beta + \gamma = \alpha$, then the value of $\tan\alpha$ equals

(a) $\tan\beta + \tan\gamma$ (b) $2(\tan\beta + \tan\gamma)$

(c) $\tan\beta + 2\tan\gamma$ (d) $2\tan\beta + \tan\gamma$

[DCE 2005]

56. The expression $\tan^2\alpha + \cot^2\alpha$ is
(a) ≥ 2 (b) ≤ 2
(c) ≥ -2 (d) none of these
[DCE 2007]

57. The maximum value of $\cos^2\left(\frac{\pi}{3}-x\right)-\cos^2\left(\frac{\pi}{3}+x\right)$ is
(a) $-(\sqrt{3}/2)$ (b) 1/2 (c) $\sqrt{3}/2$ (d) 3/2
[DCE 2008]

58. If $\tan x = b/a$, then the value of $a\cos 2x + b\sin 2x$ is
(a) a (b) $a-b$ (c) $a+b$ (d) b
[AMU 2002]

59. The general value of θ obtained from the equation $\cos 2\theta = \sin\alpha$ is
(a) $\theta = 2n\pi \pm \left(\frac{\pi}{2}-\alpha\right)$ (b) $\theta = \frac{n\pi + (-1)^n\alpha}{2}$
(c) $\theta = n\pi \pm \left(\frac{\pi}{4}-\frac{\alpha}{2}\right)$ (d) $2\theta = \frac{\pi}{4}-\alpha$
[AMU 2002]

60. If $A = \sin^2\theta + \cos^4\theta$, then for all real values of θ,
(a) $1 \leq A \leq 2$ (b) $(3/4) \leq A \leq 1$
(c) $(13/16) \leq A \leq 1$ (d) $(3/4) \leq A \leq (13/16)$
[AMU 2003]

61. If $5\cos 2\theta + 2\cos^2(\theta/2) + 1 = 0$, $-\pi < \theta < \pi$, then θ is equal to
(a) $\pi/3$ (b) $\pi/3$, $\cos^{-1}(3/5)$
(c) $\cos^{-1}(3/5)$ (d) $\pi/3$, $\pi - \cos^{-1}(3/5)$
[AMU 2006]

62. If $\frac{\tan 3x - \tan 2x}{1+\tan 3x\tan 2x} = 1$, then $x =$
(a) ϕ (b) $\pi/4$
(c) $[n\pi + (\pi/4),\ n = 1, 2, 3\ldots]$
(d) $[2n\pi + (\pi/4),\ n = 1, 2, 3\ldots]$ **[AMU 2007]**

63. The value of $\frac{\sin 55^\circ - \cos 55^\circ}{\sin 10^\circ}$ is
(a) $1/\sqrt{2}$ (b) 2 (c) 1 (d) $\sqrt{2}$
[VITEEE 2007]

64. The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is
(a) 0 (b) 5 (c) 6 (d) 10
[Jamia Millia Islamia 2006]

65. The possible values of $\theta \in (0, \pi)$ such that $\sin\theta + \sin 4\theta + \sin 7\theta = 0$ are
(a) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$
(b) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$
(c) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$
(d) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$ **[AIEEE 2011]**

66. In a ΔPQR, if $3\sin P + 4\cos Q = 6$ and $4\sin Q + 3\cos P = 1$, then the angle R is equal to
(a) $5\pi/6$ (b) $\pi/6$ (c) $\pi/4$ (d) $3\pi/4$
[AIEEE 2012]

ANSWERS

Exercise Set 1

1. (a)	**2.** (c)	**3.** (c)	**4.** (d)	**5.** (d)	**6.** (a)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (d)
11. (a)	**12.** (a)	**13.** (c)	**14.** (c)	**15.** (d)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (d)	**20.** (b)
21. (b)	**22.** (d)	**23.** (c)	**24.** (d)	**25.** (a)	**26.** (c)	**27.** (a)	**28.** (a)	**29.** (b)	**30.** (a)
31. (b)	**32.** (b)	**33.** (d)	**34.** (d)	**35.** (d)	**36.** (c)	**37.** (b)	**38.** (d)	**39.** (a)	**40.** (b)
41. (d)	**42.** (c)	**43.** (a)	**44.** (d)	**45.** (b)	**46.** (a)	**47.** (c)	**48.** (a)	**49.** (c)	**50.** (c)
51. (b)	**52.** (c)	**53.** (c)	**54.** (a)	**55.** (a)	**56.** (d)	**57.** (b)	**58.** (a) and (b)		**59.** (b)
60. (b)									

Exercise Set 2

1. (c)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (b) and (c)		**6.** (b)	**7.** (b)	**8.** (a)	**9.** (b)
10. (c)	**11.** (d)	**12.** (a)	**13.** (b)	**14.** (a)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (d)
20. (c)	**21.** (c)	**22.** (a)	**23.** (a)	**24.** (d)	**25.** (d)	**26.** (b)	**27.** (b)	**28.** (c)	**29.** (c)
30. (d)	**31.** (d)	**32.** (d)	**33.** (d)	**34.** (c)	**35.** (d)	**36.** (b)	**37.** (c)	**38.** (b)	**39.** (b)
40. (c)	**41.** (a)	**42.** (b)	**43.** (a)	**44.** (d)	**45.** (d)	**46.** (a)	**47.** (a)	**48.** (d)	**49.** (b)
50. (c)	**51.** (d)	**52.** (b)	**53.** (a)	**54.** (c)	**55.** (c)	**56.** (a)	**57.** (c)	**58.** (a)	**59.** (c)
60. (b)	**61.** (d)	**62.** (a)	**63.** (d)	**64.** (c)	**65.** (d)	**66.** (b)			

HINTS AND SOLUTIONS—EXERCISE SET 1

1. $E=\sin^2\frac{2\pi}{3}+\cos^2\frac{5\pi}{6}-\tan^2\frac{3\pi}{4}$

$\Rightarrow E=\left(\frac{\sqrt{3}}{2}\right)^2+\left(-\frac{\sqrt{3}}{2}\right)^2-(-1)^2$

$\Rightarrow E=\frac{3}{4}+\frac{3}{4}-1=\frac{6-4}{4}$

$E=\frac{1}{2}$

2. Circumference of a circle of radius 3 cm = $2\pi(3)$ = 6π cm

$\therefore$ Length of the wire (l) = 6π cm

Radius (r) = 48 cm

Since $\theta=\frac{l}{r}\Rightarrow\frac{6\pi}{48}=\frac{\pi}{8}$ radians

$=\frac{200}{8}$ grades = 25 grades

3. We have $\tan 20^\circ\tan 40^\circ\tan 60^\circ\tan 80^\circ$

$=\frac{\sin 20^\circ\sin 40^\circ\sin 80^\circ}{\cos 20^\circ\cos 40^\circ\cos 80^\circ}\tan 60^\circ$

Here, numerator $=(\sin 20^\circ\sin 40^\circ\sin 80^\circ)$

$=\frac{\sin 20^\circ}{2}(2\sin 40^\circ\sin 80^\circ)$

$=\frac{\sin 20^\circ}{2}(\cos 40^\circ-\cos 120^\circ)$

$=\frac{1}{2}\sin 20^\circ\left(1-2\sin^2 20^\circ+\frac{1}{2}\right)$

$=\frac{1}{2}\sin 20^\circ\left(\frac{3}{2}-2\sin^2 20^\circ\right)$

$=\frac{1}{4}[3\sin 20^\circ-4\sin^3 20^\circ]$

$=\frac{\sin 60^\circ}{4}=\frac{\sqrt{3}}{8}$

Now, denominator $=\cos 20^\circ\cos 40^\circ\cos 80^\circ$

$=\frac{\sin 2^3 20^\circ}{2^3\sin 20^\circ}=\frac{\sin 160^\circ}{8\sin 20^\circ}$

$=\frac{\sin 20^\circ}{8\sin 20^\circ}=\frac{1}{8}$

Hence, $\tan 20^\circ\tan 40^\circ\tan 80^\circ=\frac{\sqrt{3}/8}{1/8}=\sqrt{3}$

4. $\operatorname{cosec}\theta-\sin\theta=a^3$ and $\sec\theta-\cos\theta=b^3$

$\Rightarrow\frac{1-\sin^2\theta}{\sin\theta}=a^3\qquad\Rightarrow\frac{1-\cos^2\theta}{\cos\theta}=b^3$

$\Rightarrow\frac{\cos^2\theta}{\sin\theta}=a^3\qquad\Rightarrow\frac{\sin^2\theta}{\cos\theta}=b^3$

Now, $a^6b^3=\frac{\cos^4\theta}{\cos\theta}=\cos^3\theta$

Also, $a^3b^6=\frac{\sin^4\theta}{\sin\theta}=\sin^3\theta$

$\Rightarrow\cos\theta=a^2b$ and $\sin\theta=ab^2$

Squaring and adding, we have

$(a^2b)^2+(ab^2)^2=1$

$\Rightarrow\quad a^4b^2+a^2b^4=1$

$\therefore\quad a^2b^2(a^2+b^2)=1$

5. $\cos x=2k^2-1=2\cos^2 20^\circ-1=\cos 40^\circ$

[$\because\ \cos 2\theta=2\cos^2\theta-1$]

$\cos(360^\circ-40^\circ)=\cos 320^\circ$

$\therefore\ x=40^\circ$ and 320°

6. Since $\operatorname{cosec}\theta-\cot\theta=p$ (i)

$\Rightarrow\quad\operatorname{cosec}\theta+\cot\theta=\frac{1}{p}$ (ii)

Adding (i) and (ii), we have

$\operatorname{cosec}\theta=\frac{1}{2}\left(p+\frac{1}{p}\right)$

7. Put $\left.\begin{matrix}3=r\cos\alpha\\4=r\sin\alpha\end{matrix}\right]\Rightarrow r^2=9+16=25$

$\Rightarrow r=5$

$\therefore\ 3\cos\theta+4\sin\theta=r(\cos\theta\cos\alpha+\sin\theta\sin\alpha)$

$=r\cos(\theta-\alpha)=5\cos(\theta-\alpha)$

Its maximum value = 5 (1) = 5

and minimum value = –5

8. We have $\frac{x}{\cos\theta}=\frac{y}{\cos[\theta-(2\pi/3)]}=\frac{z}{\cos[\theta+(2\pi/3)]}$

Therefore, each ratio is equal to

$\frac{x+y+z}{\cos\theta+\cos[\theta-(2\pi/3)]+\cos[[\theta+(2\pi/3)]]}$

$=\frac{x+y+z}{\cos\theta+2\cos\theta\cos(2\pi/3)}$

$=\frac{x+y+z}{0}$

$\Rightarrow\ x+y+z=0$

9. Since $\sin\alpha = \dfrac{2pq}{p^2+q^2}$ (given)

$$\cos\alpha = \frac{p^2-q^2}{p^2+q^2}$$

$$\Rightarrow \tan\alpha = \frac{2pq}{p^2-q^2} \text{ and } \sec\alpha = \frac{p^2+q^2}{p^2-q^2}$$

$$\Rightarrow \sec\alpha - \tan\alpha = \frac{p^2+q^2-2pq}{p^2-q^2}$$

$$\Rightarrow \sec\alpha - \tan\alpha = \frac{(p-q)^2}{(p+q)(p-q)}$$

$$\therefore \sec\alpha - \tan\alpha = \frac{p-q}{p+q}$$

10. $\tan 3x = \tan(2x+x) = \dfrac{\tan 2x + \tan x}{1-\tan 2x \tan x}$

$\therefore \tan 3x - \tan 3x \tan 2x \tan x = \tan 2x + \tan x$

$\therefore \tan 3x - \tan 2x - \tan x = \tan 3x \tan 2x \tan x$

11. Since $a\cos\theta - b\sin\theta = c$ (i)

Let $a\sin\theta + b\cos\theta = E$ (ii)

Squaring and adding (i) and (ii), we have

$\Rightarrow c^2 + E^2 = a^2 + b^2$

$\Rightarrow E = \pm\sqrt{a^2+b^2-c^2}$

12. $\cos 15° = \cos(45° - 30°)$

$$= \cos 45°\cos 30° + \sin 45°\sin 30°$$

$$= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}\cdot\frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$= \sin 15° = \sin(45° - 30°)$$

$$= \sin 45°\cos 30° - \cos 45°\sin 30°$$

$$= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\cdot\frac{1}{2} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$\therefore \cos 15° - \sin 15° = \dfrac{2}{2\sqrt{2}} = \dfrac{1}{\sqrt{2}}$

13. $6(\sin^6\theta + \cos^6\theta) - 9(\sin^4\theta + \cos^4\theta) + 4$

$$= 6[(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)]$$

$$- 9[(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta] + 4$$

$$= 6[1 - 3\sin^2\theta\cos^2\theta] - 9(1 - 2\sin^2\theta\cos^2\theta) + 4$$

$$= 6 - 9 + 4 = 1$$

14. $\cos 20°\cos 40°\cos 60°\cos 80°$

$$= \frac{1}{2}\cdot\frac{1}{2}(2\cos 20°\cos 80°)\cos 40°$$

$$= \frac{1}{4}(\cos 100° + \cos 60°)\cos 40°$$

$$= \frac{1}{8}(2\cos 100°\cos 40°) + \frac{1}{8}\cos 40°$$

$$= \frac{1}{8}(\cos 140° + \cos 60°) + \frac{1}{8}\cos 40°$$

$$= \frac{1}{8}[\cos(180° - 40°)] + \frac{1}{8}\times\frac{1}{2} + \frac{1}{8}\cos 40°$$

$$= -\frac{1}{8}\cos 40° + \frac{1}{16} + \frac{1}{8}\cos 40° = \frac{1}{16}$$

15. Since $A + B + C = \dfrac{3\pi}{2}$

$\therefore \cos 2A + \cos 2B + \cos 2C$

$$= 2\cos(A+B)\cos(A-B) + \cos 2C$$

$$= 2\cos\left(\frac{3\pi}{2} - C\right)\cos(A-B) + 1 - 2\sin^2 C$$

$$= 1 - 2\sin C\left[\cos(A-B) + \sin\left(\frac{3\pi}{2} - (A+B)\right)\right]$$

$$= 1 - 2\sin C[\cos(A-B) - \cos(A+B)]$$

$$= 1 - 4\sin A\sin B\sin C$$

16. $\sec\theta + \tan\theta = p$ (given)

$$\Rightarrow \sec\theta - \tan\theta = \frac{1}{p}$$

Adding, we have

$$2\sec\theta = p + \frac{1}{p} \quad \text{(i)}$$

Subtracting, we have

$$2\tan\theta = p - \frac{1}{p} \quad \text{(ii)}$$

Dividing (ii) by (i), we have

$$\sin\theta = \frac{p^2-1}{p^2+1}$$

17. Squaring and adding, we get

$$1 + 1 + 2(\cos x\cos y + \sin x\sin y) = a^2 + b^2$$

i.e. $a^2 + b^2 = 2 + 2\cos(x-y)$

i.e. $a^2 + b^2 = 2[1 + \cos(x-y)]$

Again

$ab = \sin x\cos x + \sin y\cos y + \sin x\cos y + \cos x\sin y$

$= \frac{1}{2}[\sin 2x + \sin 2y] + \sin(x+y)$

$= \frac{1}{2}\cdot 2\sin(x+y)\cos(x-y) + \sin(x+y)$

$ab = \sin(x+y)[1+\cos(x-y)]$

$\therefore \quad \frac{ab}{a^2+b^2} = \frac{\sin(x+y)}{2}$

$\therefore \quad \sin(x+y) = \frac{2ab}{a^2+b^2}$

18. Since $A+B+C=\pi$

$\Rightarrow A = \pi-(B+C)$

We have $\cos A = \cos B\cos C$

$\Rightarrow \cos[\pi-(B+C)] = \cos B\cos C$

$\Rightarrow -\cos(B+C) = \cos B\cos C$

$\Rightarrow -(\cos B\cos C - \sin B\sin C) = \cos B\cos C$

$\Rightarrow \sin B\sin C = 2\cos B\cos C$

$\Rightarrow \tan B\tan C = 2$

19. Let $\sin x + \sin^2 x = 1$

$\Rightarrow \sin x = 1-\sin^2 x = \cos^2 x$

$\therefore \quad \sin x = \cos^2 x$

Now, $E = \cos^8 x + 2\cos^6 x + \cos^4 x$

$\Rightarrow E = \sin^4 x + 2\sin^3 x + \sin^2 x$

$\Rightarrow E = (\sin^2 x + \sin x)^2 = 1$

20. $\frac{\cot 54^\circ}{\tan 36^\circ} + \frac{\tan 20^\circ}{\cot 70^\circ}$

$= \frac{\cot(90^\circ-36^\circ)}{\tan 36^\circ} + \frac{\tan(90^\circ-70^\circ)}{\cot 70^\circ}$

$= \frac{\tan 36^\circ}{\tan 36^\circ} + \frac{\cot 70^\circ}{\cot 70^\circ} = 1+1 = 2$

21. We have $A = \sin^2\theta + \cos^4\theta$

$= \sin^2\theta + \cos^2\theta\cos^2\theta$

$\le \sin^2\theta + \cos^2\theta \qquad (\text{since } \cos^2\theta \le 1)$

$\Rightarrow \qquad \sin^2\theta + \cos^4\theta \le 1 \Rightarrow A \le 1$

Again $\sin^2\theta + \cos^4\theta = 1-\cos^2\theta + \cos^4\theta$

$= \cos^4\theta - \cos^2\theta + 1$

$= \left(\cos^2\theta - \frac{1}{2}\right)^2 + \frac{3}{4} \ge \frac{3}{4}$

Hence, $\frac{3}{4} \le A \le 1$

22. Let $E = \sqrt{\frac{1+\cos\theta}{1-\cos\theta}} + \sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$

$\Rightarrow E = \frac{1+\cos\theta+1-\cos\theta}{\sqrt{1-\cos^2\theta}} = \frac{2}{\sqrt{\sin^2\theta}}$

$\therefore \quad E = \frac{2}{|\sin\theta|}$

23. $\tan 15^\circ + \cot 15^\circ = \tan 15^\circ + \frac{1}{\tan 15^\circ}$

$= \frac{1+\tan^2 15^\circ}{\tan 15^\circ}$

$= 2\left[\frac{1}{\frac{2\tan 15^\circ}{1+\tan^2 15^\circ}}\right] = \frac{2}{\sin 30^\circ} = \frac{2}{\frac{1}{2}} = 4$

24. Since $\tan x + \frac{1}{\tan x} = 2$

$\Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}$

$\therefore \quad \sin x = \frac{1}{\sqrt{2}}$ and $\cos x = \frac{1}{\sqrt{2}}$

Hence, $\sin^{2n} x + \cos^{2n} x = \frac{1}{2^n} + \frac{1}{2^n} = \frac{1}{2^{n-1}}$

25. $|\sin\theta - \cos\theta|^2 = (\sin\theta - \cos\theta)^2$

$= 1 - \sin 2\theta$

and $\sin\theta - \cos\theta = a$

$\Rightarrow 1 + \sin 2\theta = a^2 \Rightarrow \sin 2\theta = a^2 - 1$

$\Rightarrow 1 - \sin 2\theta = 2 - a^2$

$\therefore \quad |\sin\theta - \cos\theta| = \sqrt{2-a^2}$

26. $\frac{\sin\theta + \sin 2\theta}{1+\cos\theta+\cos 2\theta}$

$= \frac{\sin\theta + 2\sin\theta\cos\theta}{2\cos^2\theta + \cos\theta}$

$= \frac{\sin\theta(1+2\cos\theta)}{\cos\theta(1+2\cos\theta)} = \tan\theta$

27. Maximum value of $4\sin^2 x + 3\cos^2 x$, i.e. $\sin^2 x + 3$ is 4 and that of $\sin\frac{x}{2} + \cos\frac{x}{2}$ is $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$, both attained at $x = \frac{\pi}{2}$. Hence, the given function has maximum value $4+\sqrt{2}$.

28. $\sin 2\theta = \cos 3\theta$

$\Rightarrow \quad 2\sin\theta\cos\theta = 4\cos^3\theta - 3\cos\theta$

$\Rightarrow \quad 2\sin\theta = 4(1-\sin^2\theta) - 3$

$\Rightarrow \quad 4\sin^2\theta + 2\sin\theta - 1 = 0 \quad \Rightarrow \quad \sin\theta = \dfrac{\sqrt{5}-1}{4}$

29. $\sqrt{2\cot\alpha + \dfrac{1}{\sin^2\alpha}} = \sqrt{2\cot\alpha + \operatorname{cosec}^2\alpha}$

$= \sqrt{2\cot\alpha + 1 + \cot^2\alpha} = |1+\cot\alpha|$

$= -1 - \cot\alpha$

[since $\cot\alpha < -1$ where $\dfrac{3\pi}{4} < \alpha < \pi$, $|1+\cot\alpha|$

$= -1 - \cot\alpha$]

30. $\sec^2 A - \tan^2 A = 1$, $\sec A + \tan A = \dfrac{3}{2}$

$\therefore \quad \sec A - \tan A = \dfrac{2}{3}$

$\therefore \quad 2\sec A = \dfrac{3}{2} + \dfrac{2}{3} = \dfrac{13}{6}$

$\sec A = \dfrac{13}{12}$ or $\cos A = \dfrac{12}{13}$

$\therefore \quad \sin A = \dfrac{5}{13}$

31. $\cos\theta = \dfrac{1}{2}\left(x + \dfrac{1}{x}\right)$

$\Rightarrow \quad x + \dfrac{1}{x} = 2\cos\theta$ (i)

We know that $x^2 + \dfrac{1}{x^2} = \left(x + \dfrac{1}{x}\right)^2 - 2$

$= (2\cos\theta)^2 - 2 = 4\cos^2\theta - 2$

$= 2\cos 2\theta$ [From Eq. (i)]

$\therefore \quad \dfrac{1}{2}\left(x^2 + \dfrac{1}{x^2}\right) = \dfrac{1}{2} \times 2\cos 2\theta = \cos 2\theta$

32. Since

$\cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)$

$\Rightarrow \quad \cos^2 48° - \sin^2 12° = \cos(48°+12°)\cos(48°-12°)$

$\therefore \quad \cos^2 48° - \sin^2 12° = \dfrac{1}{2}\cos 36° = \dfrac{\sqrt{5}+1}{8}$

33. Since $0 < x < \pi$, therefore, $\sin x > 0$

We have $1 + \sin x + \sin^2 x + \sin^3 x + \cdots + \infty = 4 + 2\sqrt{3}$

$\Rightarrow \quad \dfrac{1}{1-\sin x} = 4 + 2\sqrt{3} \quad \Rightarrow \quad \sin x = 1 - \dfrac{1}{4+2\sqrt{3}}$

$\Rightarrow \quad \sin x = \dfrac{3+2\sqrt{3}}{4+2\sqrt{3}} = \dfrac{\sqrt{3}}{2}$

$\Rightarrow \quad \dfrac{\pi}{3} \text{ or } \dfrac{2\pi}{3}$

34. $\sin^2 5° + \sin^2 10° + \sin^2 15° + \cdots + \sin^2 85° + \sin^2 90°$

$= (\sin^2 5° + \sin^2 85°) + (\sin^2 10° + \sin^2 80°)$

$+ (\sin^2 15° + \sin^2 75°) + \cdots + (\sin^2 40° + \sin^2 50°)$

$+ \sin^2 45° + \sin^2 90°$

$= [\sin^2 5° + \sin^2(90°-5°)] + [\sin^2 10° + \sin^2(90°-10°)]$

$+ [\sin^2 15° + \sin^2(90°-15°)] + \cdots + (1)^2$

$= (\sin^2 5° + \cos^2 5°) + (\sin^2 10° + \cos^2 10°) +$

$+ (\sin^2 15° + \cos^2 15°) + \cdots + (\sin^2 40° + \cos^2 40°) + \dfrac{1}{2} + 1$

$= \dfrac{(1)+(1)+(1)+\cdots+(1)}{8\,\text{times}} + \dfrac{1}{2} + 1$

$= 8 + \dfrac{1}{2} + 1 = 9\dfrac{1}{2}$

35. The quadratic equation is $x^2 - x\cos\theta + 1 = 0$

Since x is real, therefore, discriminant ≥ 0

$\Rightarrow \quad B^2 - 4AC \geq 0 \Rightarrow \cos^2\theta \geq 4(1)(1) \quad \Rightarrow \quad \cos^2\theta \geq 4$

which is impossible because $\cos^2\theta$ is not greater than 1.

36. Since $m = (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x\cos^2 x \cdot 1$

$$m = 1 - \frac{3}{4}\sin^2 2x$$

Now, $0 \leq \dfrac{3}{4}\sin^2 2x \leq \dfrac{3}{4} \forall x$

Multiplying by (–) sign and thus reversing the inequality, we have

$$-\frac{3}{4} \leq -\frac{3}{4}\sin^2 2x \leq 0$$

Adding 1 throughout, we have

$$1 - \frac{3}{4} \leq 1 - \frac{3}{4}\sin^2 2x \leq 1$$

$\therefore \quad E \in \left[\dfrac{1}{4}, 1\right]$

37. $\left.\begin{array}{l} A+B+C=90^\circ \\ A-B=30^\circ \\ A+C=60^\circ \end{array}\right] \Rightarrow B=30^\circ,\ A=60^\circ,\ C=0^\circ$

38. $\dfrac{\cot^2\theta+1}{\cot^2\theta-1}=\dfrac{1+\tan^2\theta}{1-\tan^2\theta}=\dfrac{1}{\cos^2\theta-\sin^2\theta}$

$$=\frac{1}{\cos 2\theta}=\sec 2\theta$$

39. $\sin^4 x+\cos^4 x=\sin x\cdot\cos x$

$\Rightarrow (\sin^2 x+\cos^2 x)^2-2\sin^2 x\cdot\cos^2 x=\sin x\cdot\cos x$

$\Rightarrow 1-\dfrac{\sin^2 2x}{2}=\dfrac{\sin 2x}{2}$

$\Rightarrow \sin^2 2x+\sin 2x-2=0$

$\Rightarrow (\sin 2x+2)(\sin 2x-1)=0$

$\Rightarrow \sin 2x=1$

$\Rightarrow 2x=(4n+1)\dfrac{\pi}{2} \Rightarrow x=(4n+1)\dfrac{\pi}{4},\ n\in Z$

$\Rightarrow x=\dfrac{\pi}{4},\dfrac{5\pi}{4}$, thus there are two solutions.

40. $\tan^2\theta=2\tan^2\phi+1$

$\Rightarrow 1+\tan^2\theta=2(1+\tan^2\phi)$

$\Rightarrow \sec^2\theta=2\sec^2\phi$

$\Rightarrow \cos^2\phi=2\cos^2\theta$

$\Rightarrow \cos^2\phi=1+\cos 2\theta$

$\Rightarrow \cos 2\theta=\cos^2\phi-1$

$\Rightarrow \cos 2\theta=-\sin^2\phi$

$\therefore \sin^2\phi+\cos\theta=0$

41. $\sin^6\theta+\cos^6\theta+K\cos^2 2\theta=1$

$\sin^6\theta+\cos^6\theta=(\sin^2\theta+\cos^2\theta)$

$$(\sin^4\theta+\cos^4\theta-\sin^2\theta\cos^2\theta)$$

$=(1)\,[(\sin^2\theta+\cos^2\theta)^2-3\sin^2\theta\cos^2\theta]$

$=1-3\sin^2\theta\cos^2\theta$

$=1-3\left[\dfrac{2\sin\theta\cos\theta}{2}\right]^2$

$=1-\dfrac{3}{4}\sin^2 2\theta=1-\dfrac{3}{4}[1-\cos^2 2\theta]$

$=\dfrac{1}{4}+\dfrac{3}{4}\cos^2 2\theta$

$\therefore \dfrac{1}{4}+\dfrac{3}{4}\cos^2 2\theta+K\cos^2 2\theta=1$

$\Rightarrow K\cos^2 2\theta=\dfrac{3}{4}-\dfrac{3}{4}\cos^2 2\theta=\dfrac{3}{4}\sin^2 2\theta$

$\Rightarrow K=\dfrac{3}{4}\tan^2 2\theta$

42. Given, $a\sec\alpha=d+c\tan\alpha$ (i)

and $b\sec\alpha=c-d\tan\alpha$ (ii)

On squaring and adding Eqs. (i) and (ii), we get

$$(a^2+b^2)\sec^2\alpha=d^2+c^2\tan^2\alpha+2dc\tan\alpha$$
$$+c^2+d^2\tan^2\alpha-2dc\tan\alpha$$

$\Rightarrow (a^2+b^2)\sec^2\alpha=c^2(\tan^2\alpha+1)+d^2(1+\tan^2\alpha)$

$$=(c^2+d^2)\sec^2\alpha$$

$\therefore a^2+b^2=c^2+d^2$

43. From the given relation

$\cos\theta=(2\sin\theta\cos\theta)\sin\theta,\ \sin\theta\neq 0$

$\Rightarrow \sin\theta=\pm\dfrac{1}{\sqrt{2}}$ or $\cos\theta=0$

$\Rightarrow \theta=\dfrac{\pi}{4},\dfrac{3\pi}{4},\dfrac{\pi}{2}$

Then the sum of roots is $3\pi/2$.

44. Since $\sin\theta\in[-1,1]$

and $\sin\theta_1+\sin\theta_2+\sin\theta_3=3$ (given)

$\Rightarrow$ each $\sin\theta=1$

$\therefore$ each $\cos\theta=0$

Hence, $\cos\theta_1+\cos\theta_2+\cos\theta_3=0$

45. $A=\sin^8\theta+\cos^{14}\theta$

Since $\sin^8\theta\geq 0$ and $\cos^{14}\theta\geq 0$ $\therefore$ $A\geq 0$

But $\sin^8\theta+\cos^{14}\theta=0$ is not possible

$[\because \sin\theta=0 \Rightarrow \cos\theta\neq 0]$

$\therefore$ $A>0$ $[\because \cos\theta=0 \Rightarrow \sin\theta\neq 0]$

Again $\sin^2\theta<1 \Rightarrow (\sin^2\theta)^4\leq\sin^2\theta$

$\Rightarrow \sin^8\theta\leq\sin^2\theta$

$\cos^2\theta<1$

$\Rightarrow (\cos^2\theta)^7\leq\cos^2\theta$

$\Rightarrow \cos^{14}\theta\leq\cos^2\theta$

$\therefore A=\sin^8\theta+\cos^{14}\theta\leq\sin^2\theta+\cos^2\theta=1$

$\therefore A\leq 1$. Thus $0<A\leq 1$

46. $\tan\theta = \dfrac{1}{\sqrt{7}}$

$$\therefore \quad \frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta} = \frac{(1+\cot^2\theta)-(1+\tan^2\theta)}{(1+\cot^2\theta)+(1+\tan^2\theta)}$$

$$= \frac{(\cot^2\theta - \tan^2\theta)}{2+\tan^2\theta+\cot^2\theta}$$

$$= \frac{7-(1/7)}{2+(1/7)+7} = \frac{48}{64} = \frac{3}{4}$$

47. The given equation is $3(\sin x + \cos x) - 2(\sin x + \cos x)(1 - \sin x \cos x) = 8$

$\Rightarrow (\sin x + \cos x)(3 - 2 + 2\sin x \cos x) = 8$

$\Rightarrow (\sin x + \cos x)(\sin^2 x + \cos^2 x + 2\sin x \cos x) = 8$

$\Rightarrow (\sin x + \cos x)^3 = 8$

$\Rightarrow \sin x + \cos x = 2$

The above is not possible. Hence, the given equation has no solution.

48. Let $E = \cos\dfrac{\pi}{7}\cos\dfrac{2\pi}{7}\cos\dfrac{3\pi}{7}$

$$\Rightarrow E = \cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\left(\pi - \frac{4\pi}{7}\right)$$

$$\Rightarrow E = -\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}$$

$$E = -\left\{\frac{\sin[2^3\cdot(\pi/7)]}{2^3\sin(\pi/7)}\right\} = -\frac{\sin(8\pi/7)}{8\sin(\pi/7)} = \frac{1}{8}$$

49. $(\cos 9^\circ - \sin 9^\circ)^2 = 1 - \sin 18^\circ$

$$= 1 - \frac{\sqrt{5}-1}{4} = \frac{5-\sqrt{5}}{4}$$

$$\therefore \quad \cos 9^\circ - \sin 9^\circ = \frac{\sqrt{5-\sqrt{5}}}{2} \qquad (\cos 9^\circ > \sin 9^\circ)$$

50. Dividing the given equation by $\cos^2 x$, as $\cos x = 0$ does not satisfy the equation,

we have $\tan^2 x - 5\tan x - 6 = 0$

$\Rightarrow (\tan x + 1)(\tan x - 6) = 0$

$\Rightarrow \tan x = -1$ or $\tan x = 6$

If $\tan x = -1 = \tan(-\pi/4)$, then $x = n\pi - \pi/4,\ \forall n \in Z$

and if $\tan x = 6 = \tan\alpha$ (say) $\Rightarrow \alpha = \tan^{-1}6$, then

$+\alpha = n\pi + \tan^{-1}6,\ n \in I$

Hence, $x = n\pi - \pi/4,\ n\pi + \tan^{-1}6,\ \forall n \in Z$

51. Let $E = \sin\dfrac{\pi}{7} + \sin\dfrac{2\pi}{7} + \sin\dfrac{3\pi}{7}$

$$\Rightarrow E = \frac{1}{2\sin(\pi/7)}\left(2\sin^2\frac{\pi}{7} + 2\sin\frac{\pi}{7}\sin\frac{2\pi}{7} + 2\sin\frac{\pi}{7}\sin\frac{4\pi}{7}\right)$$

$$\Rightarrow E = \frac{1}{2\sin(\pi/7)}\left(1 - \cos\frac{2\pi}{7} + \cos\frac{\pi}{7} - \cos\frac{3\pi}{7} + \cos\frac{2\pi}{7} - \cos\frac{4\pi}{7}\right)$$

$$\Rightarrow E = \frac{1}{2\sin(\pi/7)}\left(1 + \cos\frac{\pi}{7}\right) = \frac{2\cos^2(\pi/14)}{4\sin(\pi/14)\cos(\pi/14)}$$

$$\therefore \quad E = \frac{1}{2}\cot\frac{\pi}{14}$$

52. $\sec\theta + \tan^3\theta\operatorname{cosec}\theta$

$$= \sec\theta\left(1 + \tan^3\theta\cdot\frac{\operatorname{cosec}\theta}{\sec\theta}\right)$$

$$= \sec\theta(1+\tan^2\theta)$$

$$= \sec\theta\cdot\sec^2\theta = \sec^3\theta = (2-e^2)^{3/2}$$

53. Since $a + \dfrac{1}{a} = 2\cos\theta$ (given)

Squaring both sides, we have

$$a^2 + \frac{1}{a^2} + 2 = 4\cos^2\theta$$

$$\Rightarrow a^2 + \frac{1}{a^2} = 2(2\cos^2\theta - 1) = 2\cos 2\theta$$

54. Since $x \in [0, 2\pi]$ and $y \in [0, 2\pi]$ and $\sin x + \sin y = 2$, it is possible only when $\sin x = 1$ and $\sin y = 1$.

$$\Rightarrow \qquad x = \frac{\pi}{2} \text{ and } y = \frac{\pi}{2}$$

Hence, $x + y = \pi$

55. $\dfrac{b}{a} = \dfrac{\sin(2\alpha+\beta)}{\sin\beta}$

$$\frac{b-a}{b+a} = \frac{\sin(2\alpha+\beta)-\sin\beta}{\sin(2\alpha+\beta)+\sin\beta}$$

$$= \frac{2\cos(\alpha+\beta)\sin\alpha}{2\sin(\alpha+\beta)\cos\alpha}$$

$$= \frac{\tan\alpha}{\tan(a+\beta)} = \frac{\cot(\alpha+\beta)}{\cot\alpha}$$

56. Since A.M. ≥ G.M.

$$\Rightarrow \frac{9\tan^2\theta+4\cot^2\theta}{2} \geq \sqrt{4\cot^2\theta\cdot 9\tan^2\theta}$$

$$\Rightarrow 9\tan^2\theta+4\cot^2\theta \geq 12$$

$\therefore$ The minimum value of $9\tan^2\theta+4\cot^2\theta$ is 12.

57. $\dfrac{\cos^4 x}{\cos^2 y}+\dfrac{\sin^4 x}{\sin^2 y}=1$

$$\Rightarrow \frac{\cos^4 x\sin^2 y+\sin^4 x\cos^2 y}{\cos^2\sin^2 y}=1$$

$$\frac{\frac{1}{4}(1+\cos 2x)^2\frac{1}{2}(1-\cos 2y)+\frac{1}{4}(1-\cos 2x)^2}{\frac{1}{2}(1+\cos 2y)\frac{1}{2}(1-\cos 2y)}$$

$$= \frac{\frac{1}{2}(1+\cos 2y)}{\frac{1}{2}(1+\cos 2y)\frac{1}{2}(1-\cos 2y)}$$

$$= 1$$

$$\Rightarrow \frac{1}{8}[(1+\cos 2x)^2(1-\cos 2y)+(1-\cos 2x)^2(1+\cos 2y)]$$

$$=\frac{1}{4}[1-\cos^2 2y]$$

$$\Rightarrow [(1+\cos 2x)^2+(1-\cos 2x)^2]-\cos 2y[(1+\cos 2x)^2$$

$$-(1-\cos 2x)^2]=2[1-\cos^2 2y]$$

$$\Rightarrow 2[1+\cos^2 2x]-\cos 2y[4\cos 2x]=2-2\cos^2 2y$$

$$\Rightarrow 2\cos^2 2x-4\cos 2x\cos 2y=-2\cos^2 2y$$

$$\Rightarrow \cos^2 2x-2\cos 2x\cos 2y=-\cos^2 2y$$

$$\Rightarrow \cos^2 2x+\cos^2 2y-2\cos 2x\cos 2y=0$$

$$\Rightarrow (\cos 2x-\cos 2y)^2=0$$

$$\Rightarrow \cos 2x=\cos 2y$$

Now $\dfrac{\cos^4 y}{\cos^2 x}+\dfrac{\sin^4 y}{\sin^2 x}$

$$=\frac{\frac{1}{4}(1+\cos 2y)^2}{\frac{1}{2}(1+\cos 2x)}+\frac{\frac{1}{4}(1-\cos 2y)^2}{\frac{1}{2}(1-\cos 2x)}$$

$$=\frac{\frac{1}{4}(1+\cos 2x)^2}{\frac{1}{2}(1+\cos 2y)}+\frac{\frac{1}{4}(1-\cos 2x)^2}{\frac{1}{2}(1-\cos 2y)}$$

$$=\frac{\cos^4 x}{\cos^2 y}+\frac{\sin^4 x}{\sin^2 y}=1$$

58. $3^{\sin 2x+2\cos^2 x}+3^{1-\sin 2x+2\sin^2 x}=28$

$$\Rightarrow 3^{\sin 2x+2\cos^2 x}+3^{3-(\sin 2x+2\cos^2 x)}=28$$

Let $y=3^{\sin 2x+2\cos^2 x}$

$$\therefore \quad y+\frac{27}{y}=28$$

$$\Rightarrow y^2-28y+27=0$$

$$\Rightarrow y=27 \text{ or } 1$$

If $y=27$, then $3^{\sin 2x+2\cos^2 x}=3^3$

$$\Rightarrow \sin 2x+(2\cos^2 x-1)=2$$

$$\Rightarrow \sin 2x+\cos 2x=2(1)$$

Dividing both sides by $\sqrt{2}$, we have

$$\sin\left(2x+\frac{\pi}{4}\right)=\sqrt{2}$$

which is not possible

Now, if $y = 1$

$$\Rightarrow 3^{\sin 2x+2\cos^2 x}=1=3^0$$

$$\Rightarrow \sin 2x+2\cos^2 x=0$$

$$\Rightarrow 2\cos x(\sin x+\cos x)=0$$

$$\therefore \quad \cos x=0 \text{ or } \tan x=-1$$

59. If $\tan\alpha=3$, $\tan\beta=2$

So, $$\tan(\alpha-\beta)=\frac{1}{7}$$

$$\sin 2(\alpha-\beta)=\frac{2\tan(\alpha-\beta)}{1+\tan^2(\alpha-\beta)}=\frac{7}{25}$$

60. Since $\cos x$, $\dfrac{1}{2}$, $\sin 2x$ are in A.P.,

$$\Rightarrow \cos 2x+\sin 2x=2\left(\frac{1}{2}\right)$$

$$\Rightarrow \cos 2x+\sin 2x=1$$

$$\Rightarrow \sin 2x=1-\cos 2x=2\sin^2 x$$

$$\Rightarrow 2\sin x\cos x=2\sin^2 x$$

$$\Rightarrow \sin x(\cos x-\sin x)=0$$

either $\sin x=0$ or $\cos x-\sin x=0$

$x=n\pi$ or $\tan x=1$

$\therefore \quad x=n\pi$ or $x=n\pi+\dfrac{\pi}{4}$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $y-z=(a-c)\cos 2x+2b\sin 2x$

$$=(a-c)\frac{1-t^2}{1+t^2}+2b\frac{2t}{1+t^2}, \text{ where } t=\tan x$$

$$=\frac{(a-c)[(a-c)^2-4b^2]+8b^2(a-c)}{(a-c)^2+4b^2}=a-c$$

2. Since the value of θ lying between 0 and 2π and satisfying the given equations is $\theta = (5\pi/4)$,

$\therefore$ the general value of θ satisfying the given equations is

$$\theta=2n\pi+\frac{5\pi}{4}=(2n+1)\pi+\frac{\pi}{4}$$

Since $\cos\theta=-\frac{1}{\sqrt{2}}$ (–ve) and $\tan\theta=1$ (+ve)

$\therefore$ The angle θ lies in the III quadrant.

3. $\sin\theta+\sqrt{3}\cos\theta=6x-x^2-11=-2-(x-3)^2$

$\Rightarrow \sin\theta+\sqrt{3}\sin\theta\le -2$

But the minimum value of $\sin\theta+\sqrt{3}\cos\theta$

$=-\sqrt{1+3}=-\sqrt{4}=-2$

$\therefore \sin\theta+\sqrt{3}\cos\theta=-2$ and then $x = 3$

$\therefore$ $x = 3$ and $\cos\left(\theta-\frac{\pi}{6}\right)=-1$

i.e. $\theta-\frac{\pi}{6}=\pi, 3\pi$

4. $\log_{\cos x}\sin x=0.5$

$\sin x=\sqrt{\cos x}$

$\sin^2 x=\cos x$

$1-\cos^2 x=\cos x$

$\cos^2 x+\cos x-1=0$

$\cos x=\frac{-1\pm\sqrt{1+4}}{2}=\frac{\sqrt{5}-1}{2}$ (taking +ve given)

5. Let $E=x^2+4xy+y^2$

Substituting respective values, we have

$$E=(X\cos\theta-Y\sin\theta)^2+4(X\cos\theta-Y\sin\theta)$$
$$(X\sin\theta+Y\cos\theta)+(X\sin\theta+Y\cos\theta)^2$$

$$E=(1+4\sin\theta\cos\theta)X^2+4(\cos^2\theta-\sin^2\theta)XY$$
$$+(1-4\sin\theta\cos\theta)Y^2$$

Now, $x^2+4xy+y^2=AX^2+BY^2$ (given)

$\Rightarrow (1+2\sin 2\theta)X^2+4\cos 2\theta XY+(1-2\sin 2\theta)Y^2$

$=AX^2+BY^2$

Comparing respective coefficients, we have

$\Rightarrow \cos 2\theta=0,\ A=1+2\sin 2\theta,\ B=1-2\sin 2\theta$

$\Rightarrow \theta=\frac{\pi}{4}$ and $A=1+2=3,\ B=1-2=-1$

6. Putting $\alpha=\beta$ in the L.H.S. of the given expression, we get one which is equal to R.H.S. It means $\alpha=\beta$ is true.

So, $\frac{\cos^4\alpha}{\cos^2\alpha}+\frac{\sin^4\alpha}{\sin^2\alpha}=\cos^2\alpha+\sin^2\alpha=1$

7. Given $\tan\theta+\sec\theta=e^x$ (i)

$\therefore$ $\sec\theta-\tan\theta=e^{-x}$ (ii)

On adding Eqs. (i) and (ii), we get

$\therefore$ $2\sec\theta=e^x+e^{-x}$

$\Rightarrow$ $\cos\theta=\frac{2}{e^x+e^{-x}}$

8. $\tan a\theta=\tan b\theta$

$\Rightarrow a\theta=n\pi+b\theta,\ n\in I$

$\Rightarrow (a-b)\theta=n\pi$

$\Rightarrow \theta=\frac{n\pi}{(a-b)}, n\in I$

Putting $n=\cdots,-3,-2,-1,0,1,2,3,\ldots$, we get

$$\theta=\ldots,-\frac{3\pi}{(a-b)},-\frac{2\pi}{(a-b)},-\frac{\pi}{(a-b)},0,\frac{\pi}{(a-b)},$$
$$\frac{2\pi}{(a-b)},\frac{3\pi}{(a-b)},\cdots$$

Thus, the values of θ form an A.P. with common difference $\pi/(a - b)$.

9. The given equations can be written as

$\sin 3x+\sin x-3\sin 2x=\cos 3x+\cos x-3\cos 2x$

$\Rightarrow 2\sin 2x\cos x-3\sin 2x=2\cos 2x\cos x-3\cos x$

$\Rightarrow \sin 2x(2\cos x-3)=\cos 2x(2\cos x-3)=0$

$\therefore \sin 2x=\cos 2x$ as $2\cos x-3\neq 0$

$\Rightarrow \tan 2x=1=\tan\frac{\pi}{4}$

$\Rightarrow 2x=n\pi+\frac{\pi}{4}$

$\therefore$ $x=\frac{n\pi}{2}+\frac{\pi}{8}$

10. $\cos x = \tan y \Rightarrow \tan^2 y = \cos^2 x$

$\Rightarrow \sec^2 y - 1 = \cos^2 x \Rightarrow \sec^2 y = 1 + \cos^2 x$

$\Rightarrow \sec y = \sqrt{1+\cos^2 x}$

$\Rightarrow \cos y = \dfrac{1}{\sqrt{1+\cos^2 x}} = \tan z$

$\Rightarrow \tan^2 x = \dfrac{1}{1+\cos^2 x} \Rightarrow \sec^2 z + 1 = \dfrac{1}{1+\cos^2 x}$

$\Rightarrow \sec^2 z = 1 + \dfrac{1}{1+\cos^2 x} \Rightarrow \sec^2 z = \dfrac{2+\cos^2 x}{1+\cos^2 x}$

$\Rightarrow \sec z = \sqrt{\dfrac{2+\cos^2 x}{1+\cos^2 x}}$

$\Rightarrow \cos z = \sqrt{\dfrac{1+\cos^2 x}{2+\cos^2 x}} = \tan x$

$\Rightarrow \dfrac{1+\cos^2 x}{2+\cos^2 x} = \tan^2 x \Rightarrow \dfrac{1+\cos^2 x}{2+\cos^2 x} = \dfrac{1-\cos^2 x}{\cos^2 x}$

$\Rightarrow \cos^2 x + \cos^4 x = 2 - 2\cos^2 x + \cos^2 x - \cos^4 x$

$\Rightarrow 2\cos^4 x + 2\cos^2 x - 2 = 0$

$\Rightarrow \cos^4 x + \cos^2 x - 1 = 0$

$\Rightarrow \cos^2 x = \left(\dfrac{-1 \pm \sqrt{5}}{2}\right) \Rightarrow \cos^2 x = \dfrac{-1+\sqrt{5}}{2}$

$[\because \cos^2 x > 0]$

$\Rightarrow \sin^2 x = 1 - \left(\dfrac{-1 \pm \sqrt{5}}{2}\right) = \dfrac{2+1-\sqrt{5}}{2} = \dfrac{3-\sqrt{5}}{2}$

$= \dfrac{6-\sqrt{5}}{4} = \left(\dfrac{\sqrt{5}-1}{2}\right)^2$

$\Rightarrow \sin x = \dfrac{\sqrt{5}-1}{2} \Rightarrow \sin x = 2\sin 18°$

11. $\left(\dfrac{1+\cos x - \sin^2 x}{1+\cos x}\right) + \left(\dfrac{1-\cos^2 x - \sin^2 x}{\sin x(1-\cos x)}\right)$

(tanking L.C.M.)

$= \cos x$

12. Since $\sin^8 \theta \le \sin^2 \theta$ and $\cos^{16} \theta \le \cos^2 \theta$

$\therefore \sin^8 \theta + \cos^{16} \theta \le \sin^2 \theta + \cos^2 \theta = 1$

$\therefore$ The maximum value of $\sin^8 \theta + \cos^{16} \theta = 1$

13. $\sin^4 x - 2\sin^2 x + a^2 - 2 = 0$

If $\sin^2 x = y$, then the equation reduces to

$$y^2 + 2y + a^2 - 2 = 0$$

Now, $\Delta = 4 - 4(a^2 - 2) \ge 0$

$\Rightarrow 3 - a^2 \ge 0$

$\Rightarrow a^2 \le 3$ (i)

$\Rightarrow y = \sin^2 x = -1 \pm \sqrt{3-a^2}$

Now, $y = \sin^2 x$ is positive

$\therefore 0 \le y \le 1$

$\Rightarrow 0 \le -1 \pm \sqrt{3-a^2} \le 1$

$\Rightarrow 1 \le \pm\sqrt{3-a^2} \le 2$

$\Rightarrow 1 \le 3 - a^2 \le 4$

$\Rightarrow -2 \le -a^2 \le 1$

$\Rightarrow a^2 - 2 \le 0$ and $a^2 + 1 \ge 0$ (ii)

Using (i) and (ii), we have

$$a^2 - 2 \le 0$$

or $$(a+\sqrt{2})(a-\sqrt{2}) \le 0$$

$\therefore$ $$-\sqrt{2} \le a \le \sqrt{2}$$

14. $\left(1 - \cos^2 \dfrac{\pi}{8}\right) \cdot \left(1 - \cos^2 \dfrac{3\pi}{8}\right)$

$$\left[\begin{array}{l} \cos\dfrac{5\pi}{8} = \cos\left(\pi - \dfrac{3\pi}{8}\right) = -\cos\dfrac{3\pi}{8} \\ \cos\dfrac{7\pi}{8} = \cos\left(\pi - \dfrac{\pi}{8}\right) = -\cos\dfrac{\pi}{8} \end{array}\right]$$

$= \dfrac{1}{4}\left(2\sin\dfrac{\pi}{8}\sin\dfrac{3\pi}{8}\right)^2$

$= \dfrac{1}{4}\left(\cos\dfrac{\pi}{4} - \cos\dfrac{\pi}{2}\right)^2$

$= \dfrac{1}{4}\left(\dfrac{1}{\sqrt{2}}\right)^2 = \dfrac{1}{8}$

15. $a\cos x + b\sin x = c$

$\Rightarrow a\left[\dfrac{1-\tan^2(x/2)}{1+\tan^2(x/2)}\right] + b\left[\dfrac{2\tan(x/2)}{1+\tan^2(x/2)}\right] = c$

$\Rightarrow (a+c)\tan^2\dfrac{x}{2} - 2b\tan\dfrac{x}{2} + (c-a) = 0$

Suppose this equation has roots $\tan\dfrac{\alpha}{2}$ and $\tan\dfrac{\beta}{2}$.

$\therefore \quad \tan\frac{\alpha}{2}+\tan\frac{\beta}{2}=\frac{2b}{a+c}$

and $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{c-a}{a+c}$

Now, $\tan\left(\frac{\alpha}{2}+\frac{\beta}{2}\right)=\frac{\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}}{1-\tan\frac{\alpha}{2}\tan\frac{\beta}{2}}=\frac{\frac{2b}{a+c}}{1-\frac{c-a}{a+c}}$

$\therefore \quad \tan\left(\frac{\alpha+\beta}{2}\right)=\frac{b}{a}$

16. Putting $x = 0°$ in the given expression, we get value 2 . Putting $x = 0°$ in the options also, we get option (b) as correct.

17. $D=\begin{vmatrix} 1 & \cos\alpha & 1 \\ -\sin\alpha & 1 & -\cos\alpha \\ -1 & \sin\alpha & 1 \end{vmatrix}$

$=1\cdot(1+\sin\alpha\cos\alpha)-\cos\alpha(-\sin\alpha-\cos\alpha)$

$+1\cdot(-\sin^2\alpha+1)$

[Expanding by R_1]

$=1+\sin\alpha\cos\alpha+\sin\alpha\cos\alpha+\cos^2\alpha-\sin^2\alpha+1$

$=2+2\sin\alpha\cos\alpha+(\cos^2\alpha-\sin^2\alpha)$

$=2+\sin 2\alpha+\cos 2\alpha$

$=2+\sqrt{2}\left(\cos 2\alpha\cdot\frac{1}{\sqrt{2}}+\sin 2\alpha\frac{1}{\sqrt{2}}\right)$

$=2+\sqrt{2}\cos\left(2\alpha-\frac{\pi}{4}\right)$

Since $-1\le\cos\left(2\alpha-\frac{\pi}{4}\right)\le 1$,

$\therefore \quad -\sqrt{2}\le\sqrt{2}\cos\left(2\alpha-\frac{\pi}{4}\right)\le\sqrt{2}$

$\Rightarrow \quad 2-\sqrt{2}\le 2+\sqrt{2}\cos\left(2\alpha-\frac{\pi}{4}\right)\le 2+\sqrt{2}$

$\Rightarrow \quad 2-\sqrt{2}\le D\le 2+\sqrt{2}$

$\Rightarrow \quad D\in[2-\sqrt{2},2+\sqrt{2}]$

18. Since $\sec\theta+\operatorname{cosec}\theta=c$,

$\Rightarrow \quad \sqrt{1+\tan^2\theta}+\sqrt{1+\cot^2\theta}=c$

$\Rightarrow \quad \sqrt{1+t^2}+\sqrt{1+\frac{1}{t^2}}=c$, where $\tan\theta=t$

$\Rightarrow \quad \sqrt{1+t^2}\left(\frac{t+1}{t}\right)=c$

Squaring, we have

$(t^2+1)(t^2+2t+1)=c^2t^2$

$\Rightarrow \quad (t^2+t+1)^2=(c^2+1)t^2$

$\Rightarrow \quad t^2+t+1\mp t\sqrt{c^2+1}=0$

$\Rightarrow \quad t^2+t+1+t\sqrt{c^2+1}=0$

or $\quad t^2+t+1-t\sqrt{c^2+1}=0$

Now, discriminant of

$t^2+t\left(1-\sqrt{c^2+1}\right)+1=0$ is

$\Delta_1=[1-\sqrt{c^2+1}]^2-4$

Now, discriminant of

$t^2+t\left(1+\sqrt{c^2+1}\right)+1=0$ is

$\Delta_2=[1+\sqrt{c^2+1}]^2-4$

We have $\Delta_2=2\left[\sqrt{c^2+1}-1\right]>0$ and $\Delta_1>0$

$\Rightarrow \quad \left(1-\sqrt{c^2+1}\right)^2>4$

$\Rightarrow \quad 1+c^2+1-2\sqrt{c^2+1}>4$

$\Rightarrow \quad c^2-2>2\sqrt{c^2+1}$

$\Rightarrow \quad c^4-4c^2+4>4c^2+4$

$\Rightarrow \quad c^4-8c^2>0$

$\Rightarrow \quad c^2>8$

19. $\frac{2\sin\theta}{1+\cos\theta+\sin\theta}=y$

$=\frac{4\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}}{1+\left(2\cos^2\frac{\theta}{2}-1\right)+2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}}$

$=\frac{2\sin\frac{\theta}{2}}{\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)}$

and rationalizing, we get $\frac{1-\cos\theta+\sin\theta}{1+\sin\theta}$

Alternative method: Putting $\theta = 0°$ in y, we get zero and putting $\theta = 0°$ in the required expression, we also get the value zero.

20. We have $\sin^4 x+\cos^4 y+2-4\sin x\cos y=0$

$\Rightarrow (\sin^2 x-1)^2+(\cos^2 y-1)^2+2\sin^2 x+2\cos^2 y$
$-4\sin x\cos y=0$

$\Rightarrow (\sin^2 x-1)^2+(\cos^2 y-1)^2+2(\sin x-\cos y)^2=0$

$\Rightarrow \sin^2 x-1=0, \cos^2 y-1=0, \sin x-\cos y=0$

$\Rightarrow \sin x+\cos y=2 \qquad \left[\because\ 0\le x,\ y\le\frac{\pi}{2}\right]$

21. $\sec\theta+\operatorname{cosec}\theta=\sqrt{15}$

$\Rightarrow \sin\theta+\cos\theta=\sqrt{15}\sin\theta\cos\theta$

Squaring, we have

$\Rightarrow 1+\sin 2\theta=15\sin^2\theta\cos^2\theta$

$\Rightarrow 1+\sin 2\theta=\frac{15}{4}\sin^2 2\theta$

$\Rightarrow 15\sin^2 2\theta-4\sin 2\theta-4=0$

$\Rightarrow \sin 2\theta=\frac{2}{3}, -\frac{2}{5}$

Since $0<\theta<\pi$

$\therefore \qquad 0<2\theta<2\pi$

So, $\sin 2\theta = (2/5)$ will give two values of 2θ between 0 and 2π.

Also, $\sin 2\theta = -(2/5)$ will give two values of 2θ between 0 and 2π.

Hence, 4 values/solutions of θ between 0 and π are possible.

22. $\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)\sin\theta+\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)\cos\theta=\frac{1}{\sqrt{2}}$

$$\cos\left(\theta-\frac{\pi}{12}\right)=\cos\frac{\pi}{4}$$

$$\theta-\frac{\pi}{12}=2n\pi\pm\frac{\pi}{4}$$

23. $\sum_{r=1}^{n-1}\cos^2\frac{r\pi}{n}=\frac{1}{2}\sum_{n=1}^{n-1}\left(1+\cos\frac{2r\pi}{n}\right)$

$$=\frac{1}{2}(n-1)+\frac{1}{2}\left[\cos\frac{2\pi}{n}+\cos\frac{4\pi}{n}+\cdots+\cos\frac{(2n-2)\pi}{n}\right]$$

$$=\frac{1}{2}(n-1)+\frac{1}{2}\frac{\sin(n-1)\frac{\pi}{n}}{\sin\frac{\pi}{n}}\cos\left(\frac{2\pi}{n}+(n-2)\frac{\pi}{n}\right)$$

$$=\frac{1}{2}(n-1)-\frac{1}{2}=\frac{n}{2}-1$$

24. Now $5(2\cos^2\theta-1)+(1+\cos\theta)+1=0$

$\Rightarrow 10\cos^2\theta+\cos\theta-3=0$

$\Rightarrow (5\cos\theta+3)(2\cos\theta-1)=0$

$\therefore \quad \theta=\frac{\pi}{3}, -\frac{\pi}{3}$

and $\cos^{-1}\left(-\frac{3}{5}\right)$, i.e. $\pi-\cos^{-1}\left(\frac{3}{5}\right)$

25. $(\sin^2 x+\cos^2 x)^3-3\sin^2 x\cdot\cos^2 x(\sin^2 x+\cos^2 x)$

$$=1-\frac{3}{4}(2\sin x\cdot\cos x)^2=1-\frac{3}{4}\sin^2 2x$$

$$=1-\frac{3}{8}(2\sin^2 2x)=1-\frac{3}{8}(1-\cos 4x)$$

$$=1-\frac{3}{8}+\frac{3}{8}\cos 4x=\frac{5}{8}+\frac{3}{8}\cos 4x$$

Period of $\frac{2\pi}{4}=\frac{\pi}{2}$

26. $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$

$$=\frac{\sin 3(\pi/7)}{\sin(\pi/7)}\cdot\cos\frac{4\pi}{7}$$

$$=\frac{2\sin(3\pi/7)\cos(4\pi/7)}{2\sin(\pi/7)}=\frac{\sin\pi+\sin[-(\pi/7)]}{2\sin[-(\pi/7)]}$$

$$=0-\frac{\sin(\pi/7)}{2\sin(\pi/7)}=-\frac{1}{2}$$

which is a –ve rational number.

27. $1-\cos^2 x+\cos^4 x-\frac{13}{16}=0$

So, $\cos^2 x=\frac{1}{4}$ and $\cos^2 x=\frac{3}{4}$

$$\cos^2 x=\left(\frac{1}{2}\right)^2=\cos^2\left(\frac{\pi}{3}\right)$$

$$x=n\pi\pm\frac{\pi}{3}$$

and $\cos^2 x=\left(\frac{\sqrt{3}}{2}\right)^2=\cos^2\left(\frac{\pi}{6}\right)$

$$x=n\pi\pm\frac{\pi}{6}$$

28. $|\sin x| = \cos x$

$\Rightarrow$ R.H.S. $\geq 0 \Rightarrow x$ lies in I or IV quadrant.

These are equal for $x = \frac{\pi}{4}$ or $\frac{7\pi}{4}$

Hence, the general solution is $2n\pi \pm \frac{\pi}{4}$.

29. $\sec\theta + \cos\theta = 2$

$\Rightarrow \frac{\sec\theta + \cos\theta}{2} = 1$ and $\sqrt{\sec\theta \cos\theta} = 1$

$\Rightarrow$ A.M. = G.M. $\Rightarrow \sec\theta = \cos\theta = 1$

Then $\sec^n\theta + \cos^n\theta = 1^n + 1^n = 2$

$\therefore$ Statement I is false and Statement II is true.

30. $\sin x + \sin\left(x + \frac{2\pi}{3}\right) + \sin\left(x + \frac{4\pi}{3}\right)$

$= \sin x + 2\sin(x+\pi)\cdot\cos\left(-\frac{\pi}{3}\right)$

$= \sin x - 2\cos 60 \sin x = 0$

$\therefore \sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right)$

$= 3\sin x \sin\left(x + \frac{2\pi}{3}\right)\sin\left(x + \frac{4\pi}{3}\right)$

31. $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}$

$= \frac{1}{2\sin\frac{\pi}{7}}\left(2\sin\frac{\pi}{7}\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\right)$

$= \frac{1}{8}\frac{\sin\frac{8\pi}{7}}{\sin\frac{\pi}{7}} = \frac{1}{8}\frac{\sin\left(\pi + \frac{\pi}{7}\right)}{\sin\frac{\pi}{7}} = -\frac{1}{8}$

$\therefore$ Statement I is true.

Again as $\theta = \frac{\pi}{2^n - 1} \Rightarrow 2^n\theta = \pi + \theta$

$\therefore \sin(2^n\theta) = -\sin\theta$

Now $\cos\theta\cos 2\theta \ldots \cos 2^{n-1}\theta$

$= \frac{1}{2}\frac{(\sin\theta\cos\theta\cos 2\theta \ldots \cos 2^{n-1}\theta)}{\sin\theta} = \frac{1}{2^n}\frac{\sin(2^n\theta)}{\sin\theta}$

$= -\frac{1}{2^n}\frac{\sin\theta}{\sin\theta} = -\frac{1}{2^n}$

$\therefore$ Statement II is true.

32. A.M. $\geq$ G.M.

$\Rightarrow \frac{x + (1/x)}{2} \geq 1 \Rightarrow x + \frac{1}{x} \geq 2$

$\Rightarrow \cos\theta \geq 2$, which is not possible.

33. $\because 3A = 2A + A$

$\tan 3A = \frac{\tan 2A + \tan A}{1 - \tan 2A \tan A}$

$= \tan 3A - \tan 2A - \tan A = \tan 3A \tan 2A \tan A$

34. $\frac{1 - \tan^2 15°}{1 + \tan^2 15°} = \cos 30° = \frac{\sqrt{3}}{2}$

35. $y = \sec^2\theta + \cos^2\theta$

$= (\sec\theta - \cos\theta)^2 + 2$

≥ 2

But $y = 2$ when $\theta = 0$

$\therefore \quad y > 2$

36. $7\cos x + 5\sin x = 2k + 1$

$-\sqrt{74} \leq (7\cos x + 5\sin x) \leq \sqrt{74}$

So, $-\sqrt{74} \leq 2k + 1 \leq \sqrt{74}$

$-8.6 - 1 \leq 2k \leq 8.6 - 1$

$\frac{-9.6}{2} \leq k \leq \frac{7.6}{2}$

$-4.8 \leq k \leq 3.8$

$k = -4, -3, -2, -1, 0, 1, 2, 3.$

Thus, 8 solutions are possible.

37. $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$

If two rows or two columns are identical, then the value of the determinant is zero.

So, $\sin x = \cos x$

$\tan x = 1$

$x = \frac{\pi}{4}$

38. Here $\tan\theta < 0 \Rightarrow \theta$ lies in II or IV quadrant

$\Rightarrow \sin\theta > 0$ or $\sin\theta < 0$

Now, $\tan\theta = -\frac{4}{3} \Rightarrow \sin\theta = \pm\frac{4}{5}$

Hence, $\sin\theta = \frac{-4}{5}$ or $\frac{4}{5}$

39. Since α is a root of $25\cos^2\theta+5\cos\theta-12=0$,

$\therefore\quad 25\cos^2\alpha+5\cos\alpha-12=0$

$$\Rightarrow\ \cos\alpha=\frac{-5\pm\sqrt{25+1200}}{50}=-\frac{4}{5}\qquad\left[\because\frac{\pi}{2}<\alpha<\pi\right]$$

and $\sin\alpha=\sqrt{1-\frac{16}{25}}=\frac{3}{5}$

$$\therefore\quad \sin 2\alpha=2\sin\alpha\cos\alpha=2\left(\frac{3}{5}\right)\left(\frac{-4}{5}\right)=\frac{-24}{25}$$

40. $\tan x+\sec x=2\cos x$

$$\frac{\sin x}{\cos x}+\frac{1}{\cos x}=2\cos x$$

$$\sin x+1=2\cos^2 x$$

$$\sin x+1=2(1-\sin^2 x)$$

$$2\sin^2 x+\sin x-1=0$$

$$\sin x=\frac{-1\pm\sqrt{1+8}}{4}$$

$$\sin x=\frac{-1\pm 3}{4}$$

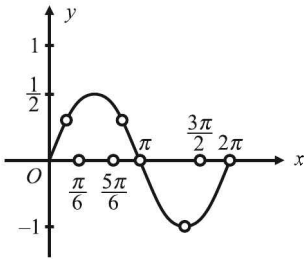


Fig. 27.10

$$\sin x=-1\ \text{ and }\ \sin x=\frac{1}{2}$$

So, $x=\frac{\pi}{6},\frac{5\pi}{6},\frac{3\pi}{2}$

But $x = (3\pi/2)$ is not a solution of the given equation.
So, two solutions are $(\pi/6)$ and $(5\pi/6)$.

41. $\sin(\alpha+\beta)=1\ \Rightarrow\ \alpha+\beta=\frac{\pi}{2}$

$$\sin(\alpha-\beta)=\frac{1}{2}\Rightarrow\alpha-\beta=\frac{\pi}{6}$$

Solving, we get

$$\alpha=\frac{\pi}{3}\ \text{ and }\ \beta=\frac{\pi}{6}$$

Now $\tan(\alpha+2\beta)\tan(2\alpha+\beta)$

$$=\tan\left(\frac{\pi}{3}+\frac{\pi}{3}\right)\tan\left(\frac{2\pi}{3}+\frac{\pi}{6}\right)$$

$$=\tan\frac{2\pi}{3}\tan\frac{5\pi}{6}$$

$$=\left(-\cot\frac{\pi}{3}\right)\left(-\cot\frac{\pi}{6}\right)$$

$$=\left(-\frac{1}{\sqrt{3}}\right)(-\sqrt{3})=1$$

42. $\tan\alpha=\frac{m}{m+1},\ \tan\beta=\frac{1}{2m+1}$

$$\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$$

$$=\frac{[m/(m+1)]+[1/(2m+1)]}{1-[m/(m+1)]\cdot[1/(2m+1)]}$$

$$=1$$

$$\tan(\alpha+\beta)=\tan\frac{\pi}{4}$$

$$\alpha+\beta=\frac{\pi}{4}$$

43. If $3\sin x+4\cos x=5$

Squaring both the sides, we get

$$9\sin^2 x+16\cos^2 x+24\sin x\cos x=25$$

$$9(1-\cos^2 x)+16(1-\sin^2 x)+24\sin x\cos x=25$$

$$9\cos^2 x+16\sin^2 x-24\sin x\cos x=0$$

$$(4\sin x-3\cos x)^2=0$$

$\therefore\quad (4\sin x-3\cos x)=0$

44. Given: $3\cos x+4\sin x=-5$

So, the minimum value of $(3\cos x+4\sin x+5)$ is 0.

45. $u^2=(a^2\cos^2\theta+b^2\sin^2\theta)+(a^2\sin^2\theta+b^2\sin^2\theta)$

$$+2\sqrt{a^2\cos^2\theta+b^2\sin^2\theta}\sqrt{a^2\sin^2\theta+b^2\cos^2\theta}$$

$$=a^2+b^2+2$$

$$\sqrt{\sin^2\theta\cos^2\theta(a^4+b^4)+a^2b^2(\sin^4\theta+\cos^4\theta)}$$

$$=(a^2+b^2)+2$$

$$\sqrt{a^2b^2(1-2\sin^2\theta\cos^2\theta)+(a^4+b^4)\sin^2\theta\cos^2\theta}$$

$$=(a^2+b^2)+2\sqrt{a^2b^2+(a^2-b^2)^2\sin^2\theta\cos^2\theta}$$

$$=(a^2+b^2)+2\sqrt{a^2b^2+\frac{(a^2-b^2)^2}{4}\sin^2 2\theta}$$

Maximum $u^2=(a^2+b^2)+2\sqrt{a^2b^2+\frac{(a^2-b^2)^2}{4}}$

Minimum $u^2=(a^2+b^2)+2ab$

$$\therefore\ \text{Difference}=2\sqrt{a^2b^2+\frac{(a^2-b^2)^2}{4}}-2ab$$

$$=\sqrt{4a^2b^2+a^4+b^4-2a^2b^2}-2ab$$

$$=\sqrt{(a^2+b^2)^2}-2ab$$

$$=a^2+b^2-2ab=(a-b)^2$$

46. We have $\sin\alpha+\sin\beta=-\frac{21}{65}$ (i)

$$\cos\alpha+\cos\beta=-\frac{27}{65} \quad \text{(ii)}$$

Squaring (i), we get

$$\sin^2\alpha+\sin^2\beta+2\sin\alpha\sin\beta=\left(\frac{21}{65}\right)^2$$

Squaring (ii), we get

$$\cos^2\alpha+\cos^2\beta+2\cos\alpha\cos\beta=\left(\frac{27}{65}\right)^2$$

Adding (i) and (ii), we have

$$2+2\cos(\alpha-\beta)=\frac{1}{(65)^2}((27)^2+(21)^2)$$

$$=\frac{1}{(65)^2}[729+441]$$

$$\Rightarrow\ 2+2\cos(\alpha-\beta)=\frac{1}{(65)^2}(1170)=\frac{18}{65}$$

$$\Rightarrow\ 1+\cos(\alpha-\beta)=\frac{9}{65}$$

$$\Rightarrow\ 2\cos^2\frac{\alpha-\beta}{2}=\frac{9}{65}$$

$$\Rightarrow\ \cos\frac{\alpha-\beta}{2}=-\frac{3}{\sqrt{130}}$$

$$\left[\because \pi<\alpha-\beta<3\pi\Rightarrow\frac{\pi}{2}<\frac{\alpha-\beta}{2}<\frac{3\pi}{2}\Rightarrow\cos\left(\frac{\alpha-\beta}{2}\right)<0\right]$$

47. Since $\tan\frac{P}{2}$ and $\tan\frac{Q}{2}$ are roots of $ax^2+bx+c=0$

$$\therefore \quad \tan\frac{P}{2}+\tan\frac{Q}{2}=-\frac{b}{a} \quad \text{(i)}$$

and $$\tan\frac{P}{2}\cdot\tan\frac{Q}{2}=\frac{c}{a} \quad \text{(ii)}$$

Now, $$\tan\left(\frac{P}{2}+\frac{Q}{2}\right)=\frac{\tan\frac{P}{2}+\tan\frac{Q}{2}}{1-\tan\frac{P}{2}\cdot\tan\frac{Q}{2}}$$

$$\Rightarrow\ \tan 45^\circ=\frac{-(b/a)}{1-(c/a)}$$

$$\Rightarrow\ 1=\frac{-b}{a-c} \quad\Rightarrow\quad a-c=-b$$

$$\Rightarrow\ c=a+b$$

48. $2\sin^2 x+5\sin x-3=0$

$\Rightarrow\ (2\sin x-1)(\sin x+3)=0$

$\Rightarrow\ (2\sin x-1)(\sin x+3)=0$

$\Rightarrow\ \sin x=(1/2)$ $[\because \sin x\neq -3]$

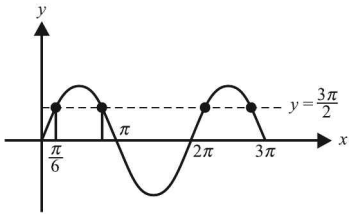


Fig. 27.11

It is clear from the figure that the curve intersects the line at four points in the given interval.
Hence, the number of solutions is 4.

49. $\cos x+\sin x=\frac{1}{2}$

$$\Rightarrow \quad \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}+\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}=\frac{1}{2}$$

Let $\tan\frac{x}{2}=t$, then

$$\frac{1-t^2}{1+t^2}+\frac{2t}{1+t^2}=\frac{1}{2}$$

$$\Rightarrow\ 3t^2-4t-1=0\ \Rightarrow\ t=\frac{2\pm\sqrt{7}}{3}$$

$$\Rightarrow\ t=\tan\frac{x}{2}=\frac{2+\sqrt{7}}{3}$$

$$\left[\because\ 0<\frac{x}{2}<\frac{\pi}{2},\ \tan\frac{x}{2}\text{ is positive}\right]$$

Now, $$\tan x=\frac{2\tan\frac{x}{2}}{1-\tan^2\frac{x}{2}}$$

$$=\frac{2\left(\frac{2+\sqrt{7}}{3}\right)}{1-\left(\frac{2+\sqrt{7}}{3}\right)^2}=-\frac{3(2+\sqrt{7})}{1+2\sqrt{7}}\times\frac{1-2\sqrt{7}}{1-2\sqrt{7}}$$

$$\Rightarrow\ \tan x=-\left(\frac{4+\sqrt{7}}{3}\right)$$

50. $\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)=-\frac{3}{2}$

$\Rightarrow 2[\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)]+3=0$

$\Rightarrow 2[\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)]$

$+\sin^2\alpha+\cos^2\alpha+\sin^2\beta+\cos^2\beta+\sin^2\gamma+\cos^2\gamma=0$

$\Rightarrow (\sin\alpha+\sin\beta+\sin\gamma)^2+(\cos\alpha+\cos\beta+\cos\gamma)^2=0$

51. $\tan1^\circ\tan2^\circ\tan3^\circ\ldots\tan89^\circ$

$=\tan1^\circ\tan2^\circ\tan3^\circ\ldots\tan45^\circ\cot44^\circ\ldots\cot2^\circ\cot1^\circ$

$=(\tan1^\circ\cot1^\circ)(\tan2^\circ\cot2^\circ)\ldots\tan45^\circ=1\cdot1\ldots1=1$

52. We know that $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$

Given, $\tan\theta+\tan\left(\theta+\frac{\pi}{3}\right)+\tan\left(\theta+\frac{2\pi}{3}\right)=3$

$$\Rightarrow \tan\theta+\left(\frac{\tan\theta+\tan\frac{\pi}{3}}{1-\tan\theta\tan\frac{\pi}{3}}\right)+\left(\frac{\tan\theta+\tan\frac{2\pi}{3}}{1-\tan\theta\tan\frac{2\pi}{3}}\right)=3$$

$$\Rightarrow \tan\theta+\frac{\tan\theta+\sqrt{3}}{1-\sqrt{3}\tan\theta}+\frac{\tan\theta-\sqrt{3}}{1+\sqrt{3}\tan\theta}=3$$

$\Rightarrow \tan\theta(1-3\tan^2\theta)+(\tan\theta+\sqrt{3})(1+\sqrt{3}\tan\theta)$

$+(\tan\theta-\sqrt{3})(1-\sqrt{3}\tan\theta)$

$=3(1-3\tan^2\theta)$

$\Rightarrow \tan\theta-3\tan^3\theta+\tan\theta+\sqrt{3}\tan^2\theta+\sqrt{3}$

$+3\tan\theta+\tan\theta-\sqrt{3}\tan^2\theta-\sqrt{3}+3\tan\theta$

$=3-9\tan^2\theta$

$\Rightarrow 3(3\tan\theta-\tan^3\theta)=3(1-3\tan^2\theta)$

$$\Rightarrow \frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}=1 \Rightarrow \tan3\theta=1$$

53. $\cos1^\circ\cdot\cos2^\circ\ldots\cos179^\circ$

$=\cos1^\circ\cdot\cos2^\circ\ldots\cos90^\circ\ldots\cos179^\circ$

$=\cos1^\circ\cdot\cos2^\circ\ldots0^\circ\ldots\cos179^\circ=0$

54. Given, $\sin^3x\sin3x=\sum_{m=0}^{n}C_m\cos mx$

$$\Rightarrow \left(\frac{3\sin x-\sin3x}{4}\right)\sin3x=\sum_{m=0}^{n}C_m\cos mx$$

$$\Rightarrow \frac{3}{8}(2\sin3x\sin x)-\frac{1}{8}(2\sin^3 3x)=\sum_{m=0}^{n}C_m\cos mx$$

$$\Rightarrow \frac{3}{8}(\cos2x-\cos4x)-\frac{1}{8}(1-\cos6x)=\sum_{m=0}^{n}C_m\cos mx$$

$$\Rightarrow \frac{1}{8}\cos6x+\frac{3}{8}(\cos2x-\cos4x)-\frac{1}{8}$$

$=C_0\cos0+C_1\cos x+C_2\cos2x+\cdots+C_n\cos nx$

$\therefore \quad n=6$

55. Given, $\alpha+\beta=\frac{\pi}{2},\ \beta+\gamma=\alpha$

$$\Rightarrow \beta=\frac{\pi}{2}-\alpha,\ \beta+\gamma=\alpha$$

$\Rightarrow \tan\beta=\tan\left(\frac{\pi}{2}-\alpha\right)$ and $\tan(\beta+\gamma)=\tan\alpha$

$\Rightarrow \tan\beta=\cot\alpha$ (i)

and $\frac{\tan\beta+\tan\gamma}{1-\tan\beta\tan\gamma}=\tan\alpha$

$\Rightarrow \frac{\tan\beta+\tan\gamma}{1-\cot\alpha\tan\gamma}=\frac{\tan\alpha}{1}$ [from Eq. (i)]

$\Rightarrow \tan\beta+\tan\gamma=\tan\alpha-\tan\gamma$

$\Rightarrow \tan\alpha=\tan\beta+2\tan\gamma$

56. $\because\ (\tan\alpha-\cot\alpha)^2\geq0$

$\Rightarrow \tan^2\alpha+\cot^2\alpha-2\geq0$

$\Rightarrow \tan^2\alpha+\cot^2\alpha\geq2$

57. $\cos^2\left(\frac{\pi}{3}-x\right)-\cos^2\left(\frac{\pi}{3}+x\right)$

$=\sin\left(\frac{\pi}{3}-x+\frac{\pi}{3}+x\right)\sin\left(\frac{\pi}{3}+x-\frac{\pi}{3}+x\right)$

$=\sin\frac{2\pi}{3}\sin x=\frac{\sqrt{3}}{2}\sin2x$

[since the maximum value of $\sin 2x$ is 1]

Its maximum value is $\frac{\sqrt{3}}{2}$.

58. $a\cos2x+b\sin2x=a\left(\frac{1-\tan^2x}{1+\tan^2x}\right)+b\left(\frac{2\tan x}{1+\tan^2x}\right)$

Put $\tan x=\frac{b}{a}$

$\Rightarrow$ which gives option (a) as the answer.

59. $\cos2\theta=\sin\alpha$

$$\cos2\theta=\cos\left(\frac{\pi}{2}-\alpha\right)$$

$$2\theta=2n\pi\pm\left(\frac{\pi}{2}-\alpha\right)$$

$$\theta=n\pi\pm\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)$$

60. $A = 1 - \cos^2\theta + \cos^4\theta$

$$A = \left(\cos^2\theta - \frac{1}{2}\right)^2 + \frac{3}{4}$$

$A \geq 3/4$

Put $\cos^2\theta = 1$ for maximum

$$A = \left(1 - \frac{1}{2}\right)^2 + \frac{3}{4} = 1$$

Thus, $3/4 \leq A \leq 1$

61. $5(2\cos^2\theta - 1) + (1 + \cos\theta) + 1 = 0$

$(5\cos\theta + 3)(2\cos\theta - 1) = 0$

$\cos\theta = 1/2$, $\cos\theta = -3/5$

$$\theta = \frac{\pi}{3},\ \pi - \cos^{-1}\left(\frac{3}{5}\right)$$

62. $\tan(3x - 2x) = \tan\frac{\pi}{4}$

$\tan x = \tan\frac{\pi}{4}$

$x = n\pi + \frac{\pi}{4}$

Put $n = 0$; $x = \frac{\pi}{4}$

Put $n = 1$; $x = \frac{5\pi}{4}$

These values of x do not satisfy the given equation. So, there is no solution. Solution set is ϕ.

63. $\dfrac{\sin 55° - \cos 55°}{\sin 10°} = \dfrac{\sin 55° - \sin 35°}{\sin 10°}$

$$= \frac{2\cos 45° \cdot \sin 10°}{\sin 10°} = \sqrt{2}$$

64. $3\sin^2 x - 7\sin x + 2 = 0$

$(3\sin x - 1)(\sin x - 2) = 0$

$\Rightarrow$ sin x = 1/3 or 2

$\Rightarrow$ sin x = 1/3 $[\because \sin x \neq 2]$

Let $\sin^{-1}\frac{1}{3} = \alpha$, $0 < \alpha < \frac{\pi}{2}$, then $\alpha, \pi - \alpha, 2\pi + \alpha, 3\pi - \alpha, 4\pi + \alpha, 5\pi - \alpha$ are the solutions in $[0, 5\pi]$.

Hence, the number of the required solutions is 6.

65. $\sin 4\theta + 2\sin 4\theta\cos 3\theta = 0$ $[\because \theta, \in (0, \pi)]$

$\sin 4\theta(1 + 2\cos 3\theta) = 0$

$\sin 4\theta = 0$ or $\cos 3\theta = -1/2$

$4\theta = n\pi, n \in I$ or $3\theta = 2n\pi \pm \frac{2\pi}{3}, n \in I$

$\theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$ or $\theta = \frac{2\pi}{9}, \frac{8\pi}{9}, \frac{4\pi}{9}$

66. $3\sin P + 4\cos Q = 6$ (i)

$4\sin Q + 3\cos P = 1$ (ii)

From Eqs. (i) and (ii) $\angle P$ is obtuse. $[\because P + Q + R = \pi]$

$(3\sin P + 4\cos Q)^2 + (4\sin Q + 3\cos P)^2 = 37$

$\Rightarrow$ $9 + 16 + 24(\sin P\cos Q + \cos P\sin Q) = 37$

$\Rightarrow$ $24\sin(P + Q) = 12$

$\Rightarrow$ $\sin(P + Q) = 1/2$ $\Rightarrow$ $P + Q = \pi/6$ or $5\pi/6$

$\Rightarrow$ $R = 5\pi/6$ or $\pi/6$

If $R = 5\pi/6$ then $0 < P, Q < \pi/6$

$\Rightarrow$ $\cos Q < 1$ and $\sin P < 1/2$

$\Rightarrow$ $3\sin P + 4\cos Q < 11/2$

So, $R = \pi/6$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The number of solutions of the equation tan x + sec x = 2 cos x lying in the interval $[0, 2\pi]$ is

(a) 0 (b) 1

(c) 2 (d) 3

2. The values of θ lying between 0 and $\pi/2$ and satisfying the equation $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+\sin^4\theta \end{vmatrix} = 0$ are

(a) $\frac{7\pi}{24}$ and $\frac{11\pi}{24}$ (b) $\frac{7\pi}{24}$ and $\frac{5\pi}{24}$

(c) $\frac{5\pi}{24}$ and $\frac{\pi}{24}$ (d) none of these

3. $\dfrac{\sin 3\alpha}{\cos 2\alpha} < 0$ if α lies in

(a) $(13\pi/48, 14\pi/48)$ (b) $(14\pi/48, 18\pi/48)$

(c) $(18\pi/48, 23\pi/48)$ (d) any of these intervals

4. If $\dfrac{\sin(x+y)}{\sin(x-y)} = \dfrac{a+b}{a-b}$, then $\dfrac{\tan x}{\tan y}$ is equal to

(a) b/a (b) a/b

(c) ab (d) none of these

5. If $2\sin^2[(\pi/2)\cos^2 x] = 1 - \cos(\pi\sin 2x)$ $x \neq (2n+1)\pi/2$, $n \in I$, then cos $2x$ is equal to

(a) 1/5 (b) 3/5

(c) 4/5 (d) 1

6. If $\sin 2\theta = \cos 3\theta$ and θ is an acute angle, then $\sin\theta$ equals

(a) $\frac{\sqrt{5}-1}{4}$ (b) $-\frac{\sqrt{5}-1}{4}$

(c) $\frac{\sqrt{5}+1}{4}$ (d) $\frac{-\sqrt{5}-1}{4}$

7. The values of x between 0 and 2π which satisfy the equation $\sin x\sqrt{8\cos^2 x} = 1$ are in A.P. The common difference of the A.P. is

(a) $\pi/8$ (b) $\pi/4$ (c) $3\pi/8$ (d) $5\pi/8$

8. If $\alpha+\beta+\gamma = 2\pi$, then

(a) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(b) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\beta}{2}\tan\frac{\gamma}{2}+\tan\frac{\gamma}{2}\tan\frac{\alpha}{2} = 1$

(c) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2} = -\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(d) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}+\tan\frac{\beta}{2}\tan\frac{\gamma}{2}+\tan\frac{\gamma}{2}\tan\frac{\alpha}{2} = 0$

9. A solution of the equation $(1-\tan\theta)(1+\tan\theta)\sec^2\theta + 2^{\tan^2\theta} = 0$, where θ lies in the interval $(-\pi/2, \pi/2)$, is given by

(a) $\theta = 0$ (b) $\theta = \pi/3$

(c) $\theta = -\pi/3$ (d) $\theta = \pi/6$

10. The value of $\sin 10° + \sin 20° + \sin 30° + \cdots + \sin 360°$ is

(a) 1 (b) 0

(c) –1 (d) none of these

11. Let k = 1°, then $2\sin 2k + 4\sin 4k + 6\sin 6k + \cdots + 180\sin 180k$ is equal to

(a) 90 cos k (b) 90 tan 89°

(c) 90 tan k (d) 90 cot 89°

12. The value of the expression $3(\sin\theta-\cos\theta)^4 + 6(\sin\theta+\cos\theta)^2 + 4(\sin^6\theta+\cos^6\theta)$ is

(a) 1 (b) –1 (c) 13 (d) 0

13. The least positive solution of $\cot\left(\frac{\pi}{3\sqrt{3}}\sin 2x\right) = \sqrt{3}$ lies in

(a) $\left(0, \frac{\pi}{6}\right]$ (b) $\left(\frac{\pi}{9}, \frac{\pi}{6}\right]$ (c) $\left(\frac{\pi}{9}, \frac{\pi}{3}\right]$ (d) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

14. The value of $e^{\log_{10}\tan 1° + \log_{10}\tan 2° + \log_{10}\tan 3° + \cdots + \log_{10}\tan 89°}$ is

(a) 0 (b) e

(c) $1/e$ (d) none of these

15. The number of solutions of $\theta \in [0, 2\pi]$ satisfying the equation $(\log_{\sqrt{3}}\tan\theta)\left[\sqrt{(\log_{\tan\theta}3 + \log_{\sqrt{3}}3\sqrt{3})}\right] = -1$ is

(a) 0 (b) 1 (c) 2 (d) >2

16. If α is an acute angle and $\sin\frac{\alpha}{2} = \sqrt{\frac{x-1}{2x}}$, then $\tan\alpha$ is

(a) $\sqrt{\frac{x-1}{x+1}}$ (b) $\frac{\sqrt{x-1}}{x+1}$

(c) $\sqrt{x^2-1}$ (d) $\sqrt{x^2+1}$

17. If $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$, then which of the following is/are correct?

(a) $\cos^{-1}\cos x = |x|$ (b) $\sin^{-1}\sin x = |x|$

(c) $\cos^{-1}\cos|x| = |x|$ (d) $\sin^{-1}\sin|x| = |x|$

18. The value of $\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}+\cos\frac{7\pi}{7}$ is

(a) 1 (b) –1

(c) 0 (d) none of these

19. The minimum value of $27^{\cos 2x}\cdot 81^{\sin 2x}$ is

(a) 1 (b) 1/9

(c) 1/81 (d) 1/243

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If $\sin A = \frac{336}{625}$, where $450° < A < 540°$, then $\sin\frac{A}{4} =$

(a) 3/5 (b) –(3/5)

(c) 4/5 (d) –(4/5)

2. If $x, y > 0$, then the range of

$$\sin^{-1}\left(\frac{x}{1+x^2}\right)+\sin^{-1}\left(\frac{2y}{1+y^2}\right) \text{ is}$$

(a) $[0, \pi]$ (b) $(-\pi, \pi]$

(c) $\left[0, \frac{2\pi}{3}\right]$ (d) $\left(0, \frac{2\pi}{3}\right]$

3. If $\cos A + \cos B = m$ and $\sin A + \sin B = n$, where m, $n \neq 0$, then $\sin(A+B)$ is equal to

(a) $\frac{mn}{m^2+n^2}$ (b) $\frac{2mn}{m^2+n^2}$

(c) $\frac{m^2+n^2}{2mn}$ (d) $\frac{mn}{m+n}$

4. If $y=\dfrac{\sec^2\theta-\tan\theta}{\sec^2\theta+\tan\theta}$, then

(a) $\dfrac{1}{3}<y<3$ (b) $y\notin\left[\dfrac{1}{3},3\right]$

(c) $-3<y<-\dfrac{1}{3}$ (d) none of these

5. The number of solutions of the equation $\tan^{-1}\left(\dfrac{x}{1-x^2}\right)+\tan^{-1}\left(\dfrac{1}{x^3}\right)=\dfrac{3\pi}{4}$, belonging to the interval (0, 1), is

(a) 0 (b) 1 (c) 2 (d) infinite

6. The number of values of x between 0 and 2π that satisfy the equation $\sin x+\sin 2x+\sin 3x=\cos x+\cos 2x$ is

(a) 3 (b) 4 (c) 5 (d) 6

7. $\cos(540°-\theta)-\sin(630°-\theta)$ is equal to

(a) 0 (b) 2 cos θ

(c) 2 sin θ (d) $\sin\theta-\cos\theta$

8. The number of solutions of $\dfrac{\tan 3x-\tan 2x}{1+\tan 3x\tan 2x}=1$ in $[-2\pi, 2\pi]$ is

(a) 2 (b) 4

(c) 8 (d) none of these

9. $2\sec^2\alpha-\sec^4\alpha-2\text{cosec}^2\alpha+\text{cosec}^4\alpha=15/4$ if $\tan\alpha$ is equal to

(a) $1/\sqrt{2}$ (b) $1/2$ (c) $1/2\sqrt{2}$ (d) $1/4$

10. If $\cos p\theta+\cos q\theta=0$, then the different values of θ are in A.P. whose common difference is

(a) $\dfrac{\pi}{p+q}$ (b) $\dfrac{\pi}{p-q}$ (c) $\dfrac{2\pi}{p\pm q}$ (d) $\dfrac{3\pi}{p\pm q}$

11. If $\dfrac{2\sin\alpha}{1+\cos\alpha+\sin\alpha}=x$, then $\dfrac{\cos\alpha}{1+\sin\alpha}$ is equal to

(a) $1/x$ (b) x (c) $1+x$ (d) $1-x$

12. The number of solutions of the equation $(2+\sqrt{3})\cos x=1-\sin x$ in the interval $[-\pi, \pi]$ is

(a) 2 (b) 3 (c) 4 (d) none of these

13. If tan 25° = x, then $\dfrac{\tan 155°-\tan 115°}{1+\tan 155°\tan 115°}$ is equal to

(a) $\dfrac{1-x^2}{2x}$ (b) $\dfrac{1+x^2}{2x}$

(c) $\dfrac{1+x^2}{1-x^2}$ (d) $\dfrac{1-x^2}{1+x^2}$

14. The number of soultions of the equation $\text{cosec}\,\theta-\cot\theta=1$ in $[0, 2\pi]$ is

(a) 4 (b) 3 (c) 2 (d) 1

15. If A = 130° and $x=\sin A+\cos A$, then

(a) $x>0$ (b) $x<0$ (c) $x=0$ (d) $x\geq 0$

16. The number of all possible triplets (a_1, a_2, a_3) such that $a_1+a_2\cos 2x+a_3\sin^2 x=0$ for all x is

(a) 0 (b) 1 (c) 3 (d) infinite

17. If $\dfrac{\sin x}{a}=\dfrac{\cos x}{b}=\dfrac{\tan x}{c}=k$, then $bc+\dfrac{1}{ck}+\dfrac{ak}{1+bk}$ is

(a) $k\left(a+\dfrac{1}{a}\right)$ (b) $\dfrac{1}{k}\left(a+\dfrac{1}{a}\right)$

(c) $\dfrac{1}{k^2}$ (d) $\dfrac{a}{k}$

18. The general value of θ satisfying both the equations $\cos\theta=\dfrac{-1}{\sqrt{2}}$ and tan θ = 1 is

(a) $n\pi+(\pi/4)$ (b) $n\pi+(5\pi/4)$

(c) $2n\pi+(\pi/4)$ (d) $2n\pi+(5\pi/4)$

19. The minimum value of $27\tan^2\theta+3\cot^2\theta$ is

(a) 9 (b) 18 (c) 27 (d) 30

ANSWERS

Level 1

1. (c) **2.** (a) **3.** (a) **4.** (b) **5.** (b) **6.** (a) **7.** (b) **8.** (a) **9.** (b) and (c)
10. (b) **11.** (b) **12.** (c) **13.** (a) **14.** (d) **15.** (c) **16.** (c) **17.** (a), (c) and (d) **18.** (b)
19. (d)

Level 2

1. (c) **2.** (d) **3.** (b) **4.** (a) **5.** (a) **6.** (a) **7.** (a) **8.** (d) **9.** (a) **10.** (c)
11. (d) **12.** (a) **13.** (a) **14.** (d) **15.** (a) **16.** (d) **17.** (b) **18.** (d) **19.** (b)

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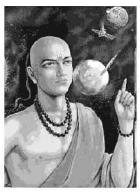

ARYA BHATT

Properties of Triangles

- Some Conditional Identities
- Trigonometrical Relations between Sides and Angles
- Napier's Analogies (Laws of Tangent)
- Area of a Triangle
- Important Concepts
- Ratio Formula
- Circumradius, Inradius and Exradii
- Regular Polygon

SOME CONDITIONAL IDENTITIES

If $A + B + C = \pi$

$\Rightarrow A + B = \pi - C, B + C = \pi - A$

$\Rightarrow \sin(A + B) = \sin(\pi - C) = \sin C$

$\Rightarrow \tan(A + C) = \tan(\pi - B) = -\tan B$, etc.

where A, B, C are all positive, then

1. $\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$
2. $\sin 2A - \sin 2B + \sin 2C = 4\cos A \sin B \cos C$
3. $\cos 2A + \cos 2B + \cos 2C = 1 - 4 \cos A \cos B \cos C$
4. $\cos 2A + \cos 2B + \cos 2C = 1 - 4 \sin A \sin B \cos C$
5. $\sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C = \sin A \sin B \sin C$
6. $\cos A \sin B \cos C + \sin A \cos B \sin C + \sin A \sin B \cos C = 1 + \cos A \cos B \cos C$
7. $\sin^2 A + \sin^2 B + \sin^2 C = 2(1 + \cos A \cos B \cos C)$
8. $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2 \cos A \cos B \cos C$
9. $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$
10. $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$
11. $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$
12. $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

13. $\dfrac{\cot A+\cot B}{\tan A+\tan B}+\dfrac{\cot B+\cot C}{\tan B+\tan C}+\dfrac{\cot C+\cot A}{\tan C+\tan A}=1$

14. $\cot\dfrac{A}{2}+\cot\dfrac{B}{2}+\cot\dfrac{C}{2}=\cot\dfrac{A}{2}\cot\dfrac{B}{2}\cot\dfrac{C}{2}$

15. $\tan\dfrac{A}{2}\tan\dfrac{B}{2}+\tan\dfrac{B}{2}\tan\dfrac{C}{2}+\tan\dfrac{C}{2}\tan\dfrac{A}{2}=1$

TRIGONOMETRICAL RELATIONS BETWEEN SIDES AND ANGLES FOR ANY ΔABC

1. $\dfrac{a}{\sin A}=\dfrac{b}{\sin B}=\dfrac{c}{\sin C}=2R,$

   where R = circumradius (sine rule)

2. $\cos A=\dfrac{b^2+c^2-a^2}{2bc}$ 3. $\cos B=\dfrac{c^2+a^2-b^2}{2ca}$

4. $\cos C=\dfrac{a^2+b^2-c^2}{2ab}$ 5. $a\sin B+b\cos A=c$, etc.

HALF ANGLE FORMULAE

In any ΔABC, if $2s=a+b+c$, then we have

1. $\sin\dfrac{A}{2}=\sqrt{\dfrac{(s-b)(s-c)}{bc}}$

2. $\sin\dfrac{B}{2}=\sqrt{\dfrac{(s-c)(s-a)}{ca}}$

3. $\sin\dfrac{C}{2}=\sqrt{\dfrac{(s-a)(s-b)}{ab}}$

4. $\cos\dfrac{A}{2}=\sqrt{\dfrac{s(s-a)}{bc}}$

5. $\cos\dfrac{B}{2}=\sqrt{\dfrac{s(s-b)}{ca}}$

6. $\cos\dfrac{C}{2}=\sqrt{\dfrac{s(s-c)}{ab}}$

7. $\tan\dfrac{A}{2}=\sqrt{\dfrac{(s-b)(s-c)}{s(s-a)}}=\dfrac{\Delta}{s(s-a)}$

8. $\tan\dfrac{B}{2}=\sqrt{\dfrac{(s-c)(s-a)}{s(s-b)}}=\dfrac{\Delta}{s(s-b)}$

9. $\tan\dfrac{C}{2}=\sqrt{\dfrac{(s-a)(s-b)}{s(s-c)}}=\dfrac{\Delta}{s(s-c)}$

NAPIER'S ANALOGIES (LAWS OF TANGENT)

In any ΔABC, we have

1. $\tan\left(\dfrac{B-C}{2}\right)=\dfrac{(b-c)}{(b+c)}\cot\dfrac{A}{2}$

2. $\tan\left(\dfrac{C-A}{2}\right)=\dfrac{(c-a)}{(c+a)}\cot\dfrac{B}{2}$

3. $\tan\left(\dfrac{A-B}{2}\right)=\dfrac{(a-b)}{(a+b)}\cot\dfrac{C}{2}$

AREA OF A TRIANGLE

The area of any triangle ABC is given by the **Heron's formula** as

$$\Delta=\sqrt{s(s-a)(s-b)(s-c)},\quad \text{where } s=\frac{1}{2}(a+b+c)$$

$$\Delta=\frac{1}{2}ab\sin C=\frac{1}{2}bc\sin A=\frac{1}{2}ca\sin B$$

FORMULAE FOR AREA OF A TRIANGLE

1. $\Delta=\dfrac{1}{2}ab\sin C$ and so $\sin C=\dfrac{2\Delta}{ab}$

2. $\Delta=\dfrac{1}{2}bc\sin A$ and so $\sin A=\dfrac{2\Delta}{bc}$

3. $\Delta=\dfrac{1}{2}ca\sin B$ and so $\sin B=\dfrac{2\Delta}{ca}$

IMPORTANT DEDUCTIONS FROM NAPIER'S ANALOGIES

1. $\tan\dfrac{A}{2}\tan\dfrac{B}{2}=\dfrac{s-c}{s}$ and $\cot\dfrac{A}{2}\cot\dfrac{B}{2}=\dfrac{s}{s-c}$

2. $\tan\dfrac{B}{2}\tan\dfrac{C}{2}=\dfrac{s-a}{s}$ and $\cot\dfrac{B}{2}\cot\dfrac{C}{2}=\dfrac{s}{s-a}$

3. $\tan\dfrac{C}{2}\tan\dfrac{A}{2}=\dfrac{s-b}{s}$ and $\cot\dfrac{C}{2}\cot\dfrac{A}{2}=\dfrac{s}{s-b}$

4. $\tan\dfrac{A}{2}+\tan\dfrac{B}{2}=\dfrac{c}{s}\cot\dfrac{C}{2}$ or $\dfrac{c(s-c)}{\Delta}$

5. $\tan\dfrac{B}{2}+\tan\dfrac{C}{2}=\dfrac{a}{s}\cot\dfrac{A}{2}$ or $\dfrac{c(s-a)}{\Delta}$

6. $\tan\dfrac{C}{2}+\tan\dfrac{A}{2}=\dfrac{b}{s}\cot\dfrac{B}{2}$ or $\dfrac{b(s-b)}{\Delta}$

7. $\tan\frac{A}{2}-\tan\frac{B}{2}=\frac{(a-b)(s-c)}{\Delta}$

8. $\tan\frac{B}{2}-\tan\frac{C}{2}=\frac{(b-c)(s-a)}{\Delta}$

9. $\tan\frac{C}{2}-\tan\frac{A}{2}=\frac{(c-a)(s-b)}{\Delta}$

IMPORTANT CONCEPTS

The lengths of the medians AL, BM and CN of a ΔABC are given by

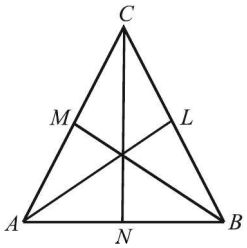

Fig. 28.1

$$AL=\frac{1}{2}\sqrt{b^2+c^2-2bc\cos A}=\frac{1}{2}\sqrt{2b^2+2c^2-a^2}$$

$$BM=\frac{1}{2}\sqrt{c^2+a^2-2ca\cos B}=\frac{1}{2}\sqrt{2c^2+2a^2-b^2}$$

$$CN=\frac{1}{2}\sqrt{a^2+b^2-2ab\cos C}=\frac{1}{2}\sqrt{2a^2+2b^2-c^2}$$

RATIO FORMULA

In the ΔABC, AD divides BC in the ratio $m:n$ at D and $\angle BAC$ in two parts as $\angle BAD=\alpha$, $\angle CAD=\beta$. If $\angle ADB=\theta$, then

1. $(m+n)\cot\theta=n\cot\beta-m\cot\alpha$
2. $(m+n)\cot\theta=m\cot C-n\cot B$

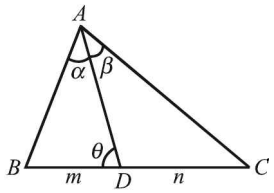

Fig. 28.2

CIRCUMRADIUS, INRADIUS AND EXRADII

In the ΔABC, let the circumradius be R, inradius be r and the three exradii corresponding to the vertices A, B and C be r_1, r_2 and r_3, respectively. Then

1. $R=\frac{a}{2\sin A}=\frac{abc}{4\Delta}$
$=\frac{b}{2\sin B}=\frac{c}{2\sin C}$

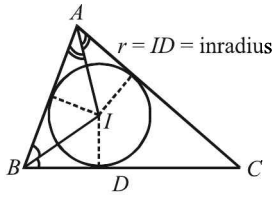

Fig. 28.3

2. $r=\frac{\Delta}{s}=(s-a)\tan\frac{A}{2}=(s-b)\tan\frac{B}{2}=(s-c)\tan\frac{C}{2}$
$=4R\sin\frac{A}{2}\cdot\sin\frac{B}{2}\cdot\sin\frac{C}{2}$

3. $r_1=\frac{\Delta}{s-a}=s\tan\frac{A}{2}=4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

4. $r_2=\frac{\Delta}{s-b}=s\tan\frac{B}{2}=4R\sin\frac{B}{2}\cos\frac{C}{2}\cos\frac{A}{2}$

5. $r_3=\frac{\Delta}{s-b}=s\tan\frac{C}{2}=4R\sin\frac{C}{2}\cos\frac{A}{2}\cos\frac{B}{2}$

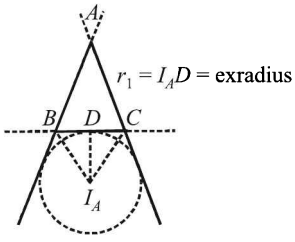

Fig. 28.4

REGULAR POLYGON

1. A regular polygon of n sides will have its vertices on a circle. Let O be the centre, r be the radius of the circle, and a be the length of each side, then clearly, in the $\angle OA_1A_2$, $\angle A_1OA_2=\frac{2\pi}{n}$ because ΔOA_1A_2 will be one of the n equal triangles with a vertex at O.

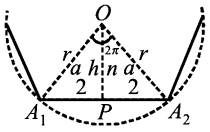

Fig. 28.5

2. $\frac{a/2}{h}=\tan\frac{1}{2}\left(\frac{2\pi}{n}\right)$, $\frac{a/2}{r}=\sin\frac{1}{2}\left(\frac{2\pi}{n}\right)$, etc.

3. Each interior angle of the regular polygon
$=\frac{2(n-2)}{n}\times 90°$.
Each exterior angle of the regular polygon $=\frac{2\pi}{n}$.

4. Sum of internal angles $=\pi(n-2)$

5. R(radius of circumcircle) $=\frac{a}{2}\operatorname{cosec}\frac{\pi}{n}$

6. r(radius of incircle) $=\frac{a}{2}\cot\frac{\pi}{n}$

SOLVED PROBLEMS

1. In a ΔABC, if $a = 2b$ and $\angle A = 3\angle B$, then angle A is

(a) 30° (b) 45°

(c) 60° (d) 90°

Ans. (d)

Solution: By sine formula, we have

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\Rightarrow \frac{a}{b} = \frac{\sin A}{\sin B} \quad \Rightarrow \quad 2 = \frac{\sin 3B}{\sin B},$$

$$\left[\because a = 2b \Rightarrow \frac{a}{b} = 2 \text{ and } \angle A = 3\angle B\right]$$

$$\Rightarrow 2\sin B = 3\sin B - 4\sin^3 B$$

$$\Rightarrow 4\sin^3 B - \sin B = 0$$

$$\Rightarrow \sin B(4\sin^2 B - 1) = 0$$

$$\Rightarrow \sin B = 0 \quad \text{or} \quad \sin^2 B = \frac{1}{4}$$

$$\Rightarrow \sin B = \frac{1}{2} \qquad [\because \sin B \text{ cannot be zero or negative}]$$

$$\Rightarrow \angle B = 30°$$

$$\therefore \angle A = 3\angle B = 3 \times 30° = 90°$$

2. If $a = \sqrt{3}+1, b = 2, c = \sqrt{2}$, then $\angle B$ is equal to

(a) 15° (b) 45°

(c) 60° (d) 90°

Ans. (b)

Solution: By cosine formula, we have

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca} = \frac{(\sqrt{2})^2 + (\sqrt{3}+1)^2 - 2^2}{2\times(\sqrt{2})\times(\sqrt{3}+1)}$$

$$= \frac{2 + (4 + 2\sqrt{3}) - 4}{2\sqrt{2}(\sqrt{3}+1)} = \frac{2 + 2\sqrt{3}}{2\sqrt{2}(\sqrt{3}+1)}$$

$$= \frac{2(\sqrt{3}+1)}{2\sqrt{2}(\sqrt{3}+1)} = \frac{1}{\sqrt{2}}$$

$$= \cos 45°$$

$$\Rightarrow \quad B = 45°$$

3. The area of the equilateral triangle which contains three coins of unity radius is

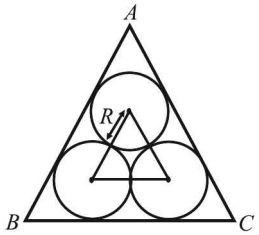

Fig. 28.6

(a) $6 + 4\sqrt{3}$ sq. units (b) $8 + \sqrt{3}$ sq. units

(c) $4 + \frac{7\sqrt{3}}{2}$ sq. units (d) $12 + 2\sqrt{3}$ sq. units

Ans. (a)

Solution:

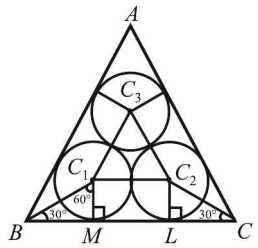

Fig. 28.7

We know that the area of an equilateral triangle of side a is $\frac{1}{2}\cdot a\cdot a\cdot \sin 60° = \frac{\sqrt{3}}{4}a^2$. Hence, we have only to calculate the sides of the triangle

$$C_1M = 1 = C_2L$$

$$\therefore \quad BM = C_1M \cot 30° = \sqrt{3} = BL$$

Also $C_1C_2 = ML = 1 + 1 = 2$

$$\therefore \quad BC = BM + ML + LC = 2\sqrt{3} + 2 = 2(\sqrt{3}+1) = 0$$

$$\therefore \quad \text{Area} = \frac{1}{4}\sqrt{3}\cdot 4(\sqrt{3}+1)^2 = \sqrt{3}\cdot(4 + 2\sqrt{3})$$

$$= 6 + 4\sqrt{3} \text{ sq. units}$$

4. If the roots of the equation $6x^2 - 5x + 1 = 0$ are $\tan\frac{A}{2}$ and $\tan\frac{B}{2}$, where A, B, C are the angles of the triangle and a, b, c are the sides of the triangle, then

(a) $a^2 + b^2 > c^2$ (b) $a^2 + b^2 = c^2$

(c) $a^2 - b^2 = c^2$ (d) none of these

Ans. (b)

Solution: $$\tan\left(\frac{A}{2} + \frac{B}{2}\right) = \frac{\tan\frac{A}{2} + \tan\frac{B}{2}}{1 - \tan\frac{A}{2}\tan\frac{B}{2}} = \frac{\frac{5}{6}}{1 - \frac{1}{6}} = 1$$

$$\therefore \quad \frac{A}{2} + \frac{B}{2} = \frac{\pi}{4} \text{ or } A + B = \frac{\pi}{2}$$

$$\therefore \quad \angle C = \frac{\pi}{2}$$

$\therefore$ $a^2 + b^2 = c^2$ is a right-angled triangle.

5. If the angles of a triangle are in the ratio of 2 : 3 : 7, then the sides are in the ratio of

(a) $\sqrt{2}:2:(\sqrt{3}+1)$ (b) $\sqrt{2}:(\sqrt{2}+1):3$

(c) $\sqrt{2}:\sqrt{3}:(\sqrt{3}+1)$ (d) $\sqrt{3}:(\sqrt{3}+1):(\sqrt{3}+\sqrt{2})$

Ans. (a)

Solution: Let the angles of the triangle be $2m$, $3m$ and $7m$ $= 180°$

$\Rightarrow 12m = 180° \Rightarrow m = 15°$

$\therefore$ The three angles are $\angle A = 30°$, $\angle B = 45°$ and $\angle C = 105°$.

Now, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = k$

$\Rightarrow \dfrac{a}{\sin 30°} = \dfrac{b}{\sin 45°} = \dfrac{c}{\sin 105°} = k$

$\Rightarrow \dfrac{a}{\left(\frac{1}{2}\right)} = \dfrac{b}{\left(\frac{1}{\sqrt{2}}\right)} = \dfrac{c}{\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)} = k$

$\Rightarrow a = \dfrac{k}{2}, b = \dfrac{k}{\sqrt{2}}, c = \left(\dfrac{\sqrt{3}+1}{2\sqrt{2}}\right)k$

$\therefore a:b:c = \dfrac{k}{2}:\dfrac{k}{\sqrt{2}}:\left(\dfrac{\sqrt{3}+1}{2\sqrt{2}}\right)k = \dfrac{\sqrt{2}k}{2\sqrt{2}}:\dfrac{2k}{2\sqrt{2}}:\left(\dfrac{\sqrt{3}+1}{2\sqrt{2}}\right)k$

$= \sqrt{2}:2:(\sqrt{3}+1)$

6. If p_1, p_2, p_3 are the altitudes of a triangle ABC from the vertices A, B, C, and Δ is the area of the triangle, then $p_1^{-2} + p_2^{-2} + p_3^{-2}$ is equal to

(a) $\dfrac{a+b+c}{2\Delta^2}$ (b) $\dfrac{a+b+c}{4\Delta}$

(c) $\dfrac{a^2+b^2+c^2}{2\Delta}$ (d) $\dfrac{a^2+b^2+c^2}{4\Delta^2}$

Ans. (d)

Solution: Let AD, BE and CF be the altitudes of ΔABC and form the vertices A, B and C on the sides a, b and c, respectively.

Then, $AD = p_1$, $BE = p_2$, $CF = p_3$

$$\Delta = \frac{1}{2}\times BC\times AD = \frac{1}{2}\times a\times p_1 \Rightarrow p_1 = \frac{2\Delta}{a};$$

$$\Delta = \frac{1}{2}\times AC\times BE = \frac{1}{2}\times b\times p_2 \Rightarrow p_2 = \frac{2\Delta}{b};$$

$$\Delta = \frac{1}{2}\times AB\times CF = \frac{1}{2}\times c\times p_3 \Rightarrow p_3 = \frac{2\Delta}{c}.$$

$\therefore p_1^{-2} + p_2^{-2} + P_3^{-3}$

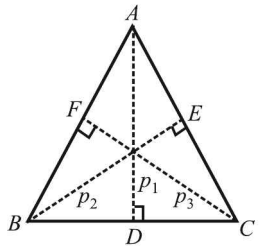

Fig. 28.8

$$= \left(\frac{2\Delta}{a}\right)^{-2} + \left(\frac{2\Delta}{b}\right)^{-2} + \left(\frac{2\Delta}{c}\right)^{-2}$$

$$= \frac{a^2}{4\Delta^2} + \frac{b^2}{4\Delta^2} + \frac{c^2}{4\Delta^2} = \frac{a^2+b^2+c^2}{4\Delta^2}$$

7. The expression

$\dfrac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ is equal to

(a) $\cos^2 A$ (b) $\sin^2 A$

(c) $\cos A \cos B \cos C$ (d) none of these

Ans. (b)

Solution: Given expression $= \dfrac{2s(2s-2a)(2s-2b)(2s-2c)}{4b^2c^2}$

$$= \frac{4s(s-a)(s-b)(s-c)}{b^2c^2} = 4\left(\frac{\Delta}{bc}\right)^2$$

$$= 4\left(\frac{abc}{4Rbc}\right)^2 = \left(\frac{a}{2R}\right)^2 = \sin^2 A$$

8. In a triangle ABC, if $\cos A = \dfrac{\sin B}{2\sin C}$, then the triangle is

(a) equilateral (b) isosceles

(c) right-angled (d) none of these

Ans. (b)

Solution: $\cos A = \dfrac{\sin B}{2\sin C}$

$\Rightarrow \dfrac{b^2+c^2-a^2}{2bc} = \dfrac{b}{2c} \Rightarrow b^2 + c^2 - a^2 - b^2 = 0$

$\Rightarrow c^2 = a^2$

Thus, the triangle is isosceles.

9. If in a triangle ABC, $\cos A \sin B = \sin C$, then the value of $\tan\dfrac{A}{2}$, when $3b - 5c = 0$, is

(a) $\dfrac{1}{2}$ (b) $\dfrac{3}{4}$

(c) $\dfrac{1}{\sqrt{3}}$ (d) 0.33

Ans. (a)

Solution: $\cos A = \dfrac{\sin C}{\sin B} = \dfrac{c}{b} = \dfrac{3}{5}$

$$\therefore \quad \frac{1-\tan^2\frac{A}{2}}{1+\tan^2\frac{A}{2}} = \frac{3}{5} \quad \text{or} \quad 2 = 8\tan^2\frac{A}{2}$$

$$\therefore \quad \tan^2\frac{A}{2} = \frac{1}{4} \quad \text{or} \quad \tan\frac{A}{2} = \frac{1}{2}$$

10. In a triangle ABC, if $\left(1+\dfrac{a}{b}+\dfrac{c}{b}\right)\left(1+\dfrac{b}{c}-\dfrac{a}{c}\right) = k$, then k lies in the interval

(a) $[-1, 1]$ (b) $[0, 1]$

(c) $[-4, 4]$ (d) $[0, 4]$

Ans. (d)

Solution: Given $\left(\dfrac{b+a+c}{b}\right)\left(\dfrac{c+b-a}{c}\right) - k$

or $(b+c)^2 - a^2 = kbc$

or $b^2 + c^2 - a^2 = (k-2)bc$

or $\dfrac{b^2+c^2-a^2}{2bc} = \dfrac{k-2}{2}$ or $\cos A = \dfrac{k-2}{2}$

$\because \quad -1 \le \cos A \le 1$

$$\therefore \quad -1 \le \frac{k-2}{2} \le 1 \Rightarrow k \in [0,4]$$

11. In a triangle ABC, $a : b : c = 4 : 5 : 6$. The ratio of the radius of the circumcircle to that of the incircle is

(a) 16/9 (b) 16/7

(c) 11/7 (d) 7/16

Ans. (b)

Solution: We have $R = \dfrac{abc}{4\Delta}$ and $r = \dfrac{\Delta}{s}$

$$\frac{R}{r} = \frac{abc}{4\Delta}\cdot\frac{s}{\Delta}$$

$$= \frac{abc}{4(s-a)(s-b)(s-c)}$$

Since $a:b:c = 4:5:6$

$$\Rightarrow \quad \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k \quad \text{(say)}$$

Thus, $$\frac{R}{r} = \frac{(4k)(5k)(6k)}{4\left(\frac{15k}{2}-4k\right)\left(\frac{15k}{2}-5k\right)\left(\frac{15k}{2}-6k\right)}$$

$$= \frac{120k^3\cdot 2}{k^3\cdot 7\cdot 5\cdot 3} = \frac{16}{7}$$

12. If CF is the internal bisector of the angle C of the ΔABC, then CF is equal to

(a) $\dfrac{2ab}{a+b}\cos\dfrac{C}{2}$ (b) $\dfrac{a+b}{2ab}\cos\dfrac{C}{2}$

(c) $\dfrac{b\sin A}{\sin\left(B+\dfrac{C}{2}\right)}$ (d) none of these

Ans. (a) and (c)

Solution:

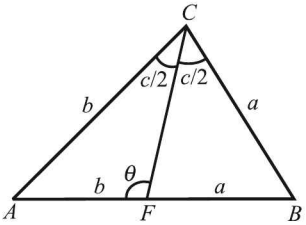

Fig. 28.9

$$\Delta = \Delta_1 + \Delta_2$$

$$\frac{1}{2}ab\sin C = \frac{1}{2}b(CF)\sin\frac{C}{2} + \frac{1}{2}a(CF)\sin\frac{C}{2}$$

or $$CF = \frac{ab\sin C}{(a+b)\sin\frac{C}{2}} = \frac{2ab\sin\frac{C}{2}}{a+b} \Rightarrow \text{option (a)}$$

Again in ΔCFB by sine rule,

$$\frac{CF}{\sin B} = \frac{a}{\sin(\pi-\theta)} = \frac{a}{\sin\theta} = \frac{a}{\sin\left(B+\frac{C}{2}\right)}$$

$\because \quad \theta + B + \dfrac{C}{2} = \pi$

$$\therefore \quad CF = \frac{a\sin B}{\sin\left(B+\frac{C}{2}\right)} = \frac{b\sin A}{\sin\left(B+\frac{C}{2}\right)} \Rightarrow \text{option (c)}$$

13. In a triangle ABC,

$$\frac{a^2\sin(B-C)}{\sin B+\sin C} + \frac{b^2\sin(C-A)}{\sin C+\sin A} + \frac{c^2\sin(A-B)}{\sin A+\sin B} =$$

(a) abc (b) $a+b+c$

(c) 0 (d) 1

Ans. (c)

Solution: $$\frac{a^2\sin(B-C)}{\sin B+\sin C} = \frac{a2R\sin A\sin(B-C)}{\sin B+\sin C}$$

$$= \frac{a2R\sin(B+C)\sin(B-C)}{\sin B+\sin C}$$

$$= \frac{a2R(\sin^2 B - \sin^2 C)}{\sin B + \sin C}$$

$$= (2R\sin A)2R(\sin B - \sin C)$$

$$= 4R^2(\sin A\sin B - \sin A\sin C)$$

By symmetry, the required sum = 0

14. Two adjacent sides of a cyclic and quadrilateral are 2 and 5 and the angle between them is 60°. If the third side is 3, then the remaining fourth side is

(a) 2 (b) 3
(c) 4 (d) 5

Ans. (a)

Solution: In ΔABD

$$\cos 60° = \frac{2^2 + 5^2 - BD^2}{2(2)(5)}$$

$\Rightarrow BD^2 = 19$

Now in ΔBCD,

$$\cos 120° = \frac{CD^2 + 9 - 19}{(2)(3)(CD)}$$

$\Rightarrow CD^2 + 3CD - 10 = 0$

$\Rightarrow CD = -5, 2$

$\Rightarrow CD = 2$ $(\because CD \neq -5)$

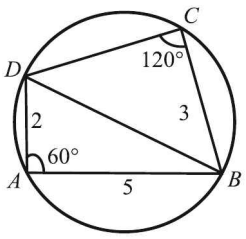

Fig. 28.10

EXERCISE SET 1

1. In a triangle ABC, if $a = 5$, $b = 4$ and $\cos(A - B) = \frac{31}{32}$, then the third side is equal to

(a) 36 (b) $6\sqrt{6}$
(c) 6 (d) $(216)^{1/3}$

2. In a ΔABC, $\operatorname{cosec} A(\sin B\cos C + \cos B\sin C)$ is equal to

(a) $\frac{c}{a}$ (b) $\frac{a}{c}$
(c) 1 (d) $\frac{c}{ab}$

3. In a ΔABC, $bc\cos^2\frac{A}{2} + ca\cos^2\frac{B}{2} + ab\cos^2\frac{C}{2} =$

(a) $(s-a)^2$ (b) $(s-b)^2$
(c) $(s-c)^2$ (d) s^2

4. In a triangle ABC, $\frac{1+\cos(A-B)\cos C}{1+\cos(A-C)\cos B}$ is equal to

(a) $\frac{a-b}{a-c}$ (b) $\frac{a+b}{a+c}$
(c) $\frac{a^2-b^2}{a^2-c^2}$ (d) $\frac{a^2+b^2}{a^2+c^2}$

5. In a triangle ABC, if $\cot\frac{A}{2} = \frac{b+c}{a}$, then the triangle ABC is

(a) isosceles (b) equilateral
(c) right-angled (d) none of these

6. In a ΔABC, if $\cot A$, $\cot B$, $\cot C$ are in A.P., then a^2, b^2, c^2 are in

(a) A.P. (b) G.P.
(c) H.P. (d) none of these

7. If the sides of a triangle are in A.P., then the cotangent of half the angles will be in

(a) H.P. (b) G.P.
(c) A.P. (d) no particular order

8. In a triangle PQR, if $\sin P$, $\sin Q$, $\sin R$ are in A.P., then

(a) the altitudes are in A.P.
(b) the altitudes are in H.P.
(c) the medians are in A.P.
(d) the medians are in H.P.

9. If the sides a, b, c of ΔABC are in A.P., then $\cos A\cot\frac{1}{2}A$, $\cos B\cot\frac{1}{2}B$, $\cos C\cot\frac{1}{2}C$ are in

(a) A.P. (b) G.P.
(c) H.P. (d) none of these

10. In a ΔABC, if $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$ and the side $a = 2$, then the area of the triangle is

(a) 1 sq. unit (b) 2 sq. units
(c) $\frac{\sqrt{3}}{2}$ sq. unit (d) $\sqrt{3}$ sq. units

11. In a ΔABC, $\angle C = 90°$, $a = 3$ and $b = 4$. D is a point on AB, so that $\angle BCD = 30°$, then $CD =$

(a) $\frac{24}{4+4\sqrt{3}}$ (b) $\frac{24}{3+4\sqrt{3}}$
(c) $\frac{24}{4+3\sqrt{3}}$ (d) $\frac{24}{3+3\sqrt{3}}$

12. $a^3\cos(B-C) + b^3\cos(C-A) + c^3\cos(A-B) =$

(a) $3abc$ (b) $3(a+b+c)$
(c) $abc(a+b+c)$ (d) 0

13. If x, y, z are perpendiculars drawn from the vertices of the triangle having sides a, b and c, then the value of $\frac{bx}{c}+\frac{cy}{a}+\frac{az}{a}$ will be

(a) $\frac{a^2+b^2+c^2}{2R}$ (b) $\frac{a^2+b^2+c^2}{R}$

(c) $\frac{a^2+b^2+c^2}{4R}$ (d) $\frac{2(a^2+b^2+c^2)}{R}$

14. If the hypotenuse of a right-angled triangle is four times the length of the perpendicular drawn from the opposite vertex to it, then the difference of two acute angles will be

(a) 75° (b) 15°
(c) 60° (d) 30°

15. In any triangle $ABC, \prod\left[\frac{\sin^2 A+\sin A+1}{\sin A}\right]$ is always greater than

(a) 9 (b) 3
(c) 27 (d) 10

16. In a ΔABC, if AD is the median and $\angle A = 60°$, then $4AD^2$ is equal to

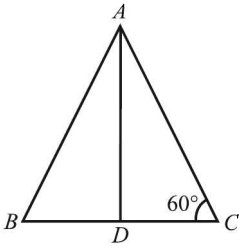

Fig. 28.11

(a) b^2+c^2-bc (b) $2b^2+c^2-2bc$
(c) b^2+c^2+2bc (d) none of these

17. If r_1, r_2, r_3 are the exradii of a triangle ABC, then $\frac{b-c}{r_1}+\frac{c-a}{r_2}+\frac{a-b}{r_3}$ is equal to

(a) 0 (b) $a+b+c$

(c) $\frac{a+b+c}{2}$ (d) $3r$

18. If in a triangle $ABC, \frac{a^2-b^2}{a^2+b^2}=\frac{\sin(A-B)}{\sin(A+B)}$, then the triangle is

(a) right-angled or isosceles
(b) right-angled and isosceles
(c) equilateral
(d) none of these

19. In a triangle ABC, $2ac\ \sin\frac{1}{2}(A-B+C)$ is equal to

(a) $a^2+b^2+c^2$ (b) $c^2+a^2-b^2$
(c) $b^2-a^2-a^2$ (d) $c^2-a^2-b^2$

20. In a triangle ABC, if $b=2, c=\sqrt{3}$ and $\angle A = 30°$, then the inradius of the triangle is

(a) $\frac{\sqrt{3}+1}{2}$ (b) $\sqrt{3}+1$

(c) $\frac{\sqrt{3}-1}{2}$ (d) $\sqrt{3}-1$

21. $(b+c)\cos A+(c+a)\cos B+(a+b)\cos C =$

(a) $b+c-a$ (b) $c+a-b$
(c) $a+b-c$ (d) $a+b+c$

22. A circle of radius 4 unit is inscribed in an equilateral triangle ABC. The sides AB and AC of the triangle are produced and a circle is drawn touching the side BC and produced parts of sides AB and AC. The area of the circle drawn is

(a) 16π sq. units (b) 64π sq. units
(c) 144π sq. units (d) 256π sq. units

23. If the sides of a triangle are a, b and $\sqrt{a^2+ab+b^2}$, then the largest angle is

(a) 105° (b) 120°
(c) 150° (d) 135°

24. If the sides a, b, c of a triangle ABC are the roots of $x^3-11x^2+38x-40=0$, then $\Sigma\frac{\cos A}{a} =$

(a) $\frac{3}{4}$ (b) 1

(c) $\frac{9}{16}$ (d) none of these

25. In a triangle, if $\left(1-\frac{r_1}{r_2}\right)\left(1-\frac{r_1}{r_3}\right)=2$, then the triangle is

(a) right-angled (b) equilateral
(c) isosceles (d) none of these

26. If $\frac{1}{a+c}+\frac{1}{b+c}=\frac{3}{a+b+c}$, then $\angle C =$

(a) 90° (b) 60°
(c) 45° (d) 30°

27. In a triangle ABC, $\frac{b+c}{a}$ is equal to

(a) $\frac{\cos\frac{1}{2}(B-C)}{\sin\frac{1}{2}A}$ (b) $\frac{\sin\frac{1}{2}(B-C)}{\cos\frac{1}{2}A}$

(c) $\frac{\cos\frac{1}{2}(B+C)}{\sin\frac{1}{2}A}$ (d) $\frac{\cos\frac{1}{2}(B+C)}{\cos\frac{1}{2}A}$

28. If $A = 60°$, then $\frac{b}{c+a}+\frac{c}{a+b}=$

(a) 0 (b) 1

(c) s (d) S

29. If in a triangle ABC, the sides AB and AC are perpendicular, then the true equation is

(a) $\tan A+\tan B=0$ (b) $\tan B+\tan C=0$

(c) $\tan A+2\tan C=0$ (d) $\tan B\cdot\tan C=1$

30. If the angles A, B, C of a triangle are in A.P. and the sides a, b, c are in G.P., then a^2, b^2, c^2 are in

(a) A.P. (b) H.P.

(c) G.P. (d) none of these

31. In a triangle, the lengths of the two larger sides are 10 cm and 9 cm, respectively. If the angles of the triangle are in A.P., then the length of the third side in centimetres can be

(a) $5-\sqrt{6}$ only

(b) $5+\sqrt{6}$ only

(c) $5-\sqrt{6}$ or $5+\sqrt{6}$

(d) neither $5-\sqrt{6}$ nor $5+\sqrt{6}$

32. In a triangle ABC, if $2a+4b^2+c^2=4ab+2ac$, then the numerical value of $\cos B$ is equal to

(a) 0 (b) $\frac{1}{8}$

(c) $\frac{3}{8}$ (d) $\frac{7}{8}$

33. If in a triangle ABC,

$$\frac{(2\cos A)}{a}+\frac{(\cos B)}{b}+\frac{(2\cos C)}{c}=\frac{a}{bc}+\frac{b}{ca},$$

then the value of the angle A in degrees is

(a) 90° (b) 135°

(c) 45° (d) none of these

34. If in a triangle ABC, the sides a, b, c are in A.P., then

(a) $B>60°$ (b) $B<60°$

(c) $B\le 60°$ (d) $B=|A-C|$

35. If in a ΔABC, $\sin A$, $\sin B$ and $\sin C$ are in A.P., then

(a) the altitudes are in A.P.

(b) the altitudes are in H.P.

(c) the medians are in G.P.

(d) the medians are in A.P.

36. A circle is inscribed in an equilateral triangle of side a. The area of any square inscribed in this circle is

(a) a^2 (b) $a^2/4$

(c) $a^2/3$ (d) $a^2/6$

37. Let PQ and RS be the tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle, then $2r$ equals

(a) $\sqrt{PQ\cdot RS}$ (b) $\frac{PQ+RS}{2}$

(c) $\frac{2PQ\cdot RS}{PQ+RS}$ (d) $\sqrt{\frac{PQ^2+RS^2}{2}}$

38. If $\cos A=\frac{\sin B}{2\sin C}$, then in ΔABC,

(a) $c=a$ (b) $b=c$

(c) $\cos B=\cos 2C$ (d) $\sin B=\sin 2C$

39. If $b+c=3a$, then the value of $\cot\frac{B}{2}\cot\frac{C}{2}=$

(a) 1 (b) 2

(c) $\sqrt{3}$ (d) $\sqrt{2}$

40. If Δ denotes the area of any triangle with semi-perimeters, then

(a) $\Delta<\frac{s^2}{2}$ (b) $\Delta>\frac{s^2}{4}$

(c) $\Delta<\frac{s^2}{4}$ (d) $\Delta<s^2$

41. If in a triangle ABC, $\frac{\cos A+2\cos C}{\cos A+2\cos B}=\frac{\sin B}{\sin C}$, then the triangle can be

(a) equilateral (b) isosceles

(c) right-angled (d) obtuse-angled

42. $\frac{b-c}{a}\cos^2\frac{1}{2}A+\frac{c-a}{b}\cos^2\frac{1}{2}B+\frac{a-b}{c}\cos^2\frac{1}{2}C=$

(a) $(s-a)(s-b)$ (b) $(s-b)(s-c)$

(c) $(s-c)(s-a)$ (d) 0

43. If D is the mid-point of the side BC of a triangle ABC and AD is perpendicular to AC, then

(a) $b^2=a^2-c^2$ (b) $a^2+b^2=5c^2$

(c) $3b^2=a^2-c^2$ (d) $3a^2=b^2-3c^2$

44. If the sides a, b and c of a triangle are in A.P., then $\tan\frac{A}{2}+\tan\frac{C}{2}=$

(a) $\frac{2}{3}\cot\frac{A}{2}$ (b) $\frac{2}{3}\cot\frac{B}{2}$

(c) $\frac{2}{3}\cot\frac{C}{2}$ (d) none of these

45. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to the vertices A, B and C respectively. Suppose $a = 6$, $b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to

(a) 2 (b) 4

(c) 3 (d) 6

46. If the cotangents of half the angles of a triangle are in A.P., then the sides are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

47. In a triangle ABC, if the angle $B = 60°$, then

(a) $(a-b)^2+ab=c^2$ (b) $(b-c)^2+bc=a^2$

(c) $(c-a)^2+ca=b^2$ (d) $a^2+b^2+c^2=2b^2+ac$

48. If in a triangle ABC, $S=a^2-(b-c)^2$, then $\tan A$ is equal to

(a) 1/2 (b) 8/17

(c) 8/15 (d) 15/16

49. In a triangle ABC, if $\angle B = 60°$, then the expression $(a+b+c)(a-b+c)$ is equal to

(a) $3ab$ (b) $3bc$

(c) $3ca$ (d) $3abc$

50. In a triangle ABC, if $a^4+b^4+c^4=2(a^2+c^2)b^2$, then the angle B is

(a) 45°, 135° (b) 45°, 45°

(c) 45°, 90° (d) none of these

51. The expression $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ is equal to

(a) $\cos^2 A$ (b) $\sin^2 A$

(c) $\cos 2A$ (d) $1-\cos A$

52. If the angles A, B, C of the triangle ABC are in A.P., then $\frac{a+c}{\sqrt{(a^2-ac+c^2)}}=$

(a) $2\cos\frac{A+C}{2}$ (b) $2\sin\frac{A+C}{2}$

(c) $2\cos\frac{A-C}{2}$ (d) $2\sin\frac{A-C}{2}$

53. The perimeter of a triangle is 6 times the A.M. of the sum of its angles. If the side $a = 1$, then the angle A is

(a) $\pi/6$ (b) $\pi/4$

(c) $\pi/3$ (d) $\pi/2$

EXERCISE SET 2

1. If the radius of a circumcircle of an isosceles triangle PQR is equal to $PQ(=PR)$, then $\angle P =$

(a) $\pi/6$ (b) $\pi/3$

(c) $\pi/2$ (d) $2\pi/3$

2. Two sides of a triangle are $\sqrt{3}+1$ and $\sqrt{3}-1$ and the included angle is 60°. The difference of the remaining angles is

(a) 90° (b) 60°

(c) 45° (d) 30°

3. In a ΔABC, $\frac{\sin(A-B)}{\sin(A+B)}=$

(a) $\frac{a^2-b^2}{c^2}$ (b) $\frac{a^2+b^2}{c^2}$

(c) $\frac{c^2}{a^2-b^2}$ (d) $\frac{c^2}{a^2+b^2}$

4. If the sides of a triangle are 3 cm, 2 cm and 4 cm, then the cosine of the greatest angle is equal to

(a) 7/8 (b) 11/16

(c) –1/4 (d) 1/4

5. If the lengths of the sides of a triangle are 6, 10 and 14 cm, then the triangle is

(a) obtuse-angled (b) acute-angled

(c) right-angled (d) isosceles

6. In any ΔABC, $\sum a^3\sin(B-C)$ is equal to

(a) $ab+bc+ca$ (b) $3(a+b+c)$

(c) $3abc$ (d) 0

7. In a ΔABC, $a(b\cos C - c\cos B)$ is equal to

(a) 0 (b) b^2-c^2

(c) a^2 (d) none of these

8. If $c^2=a^2+b^2$, then $4s(s-a)(s-b)(s-c)=$

(a) s^4 (b) b^2c^2

(c) c^2a^2 (d) a^2b^2

9. If in a ΔABC, $\sin^3 A+\sin^3 B+\sin^3 C = 3\sin A\sin B\sin C$, then

$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ equals

(a) 0

(b) $(a+b+c)(ab+bc+ca)$

(c) $(a+b+c)^3$

(d) none of these

10. If the angles of a triangle are 30° and 45° and the included side is $(\sqrt{3}+1)$ cm, then the area of the triangle is

(a) $\frac{1}{\sqrt{3}-1}$ (b) $\sqrt{3}+1$

(c) $\frac{1}{\sqrt{3}+1}$ (d) none of these

11. In Fig. 28.12, ABC is a triangle in which $\angle C = 90°$ and $AB = 5$ cm. D is a point on AB such that $AD = 3$ cm and $\angle ACD = 60°$. Then the length of AC is

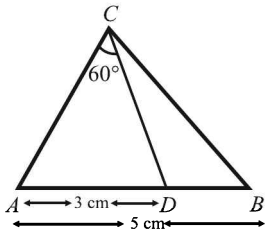

Fig. 28.12

(a) $\frac{3}{\sqrt{7}}$ cm (b) $\sqrt{\frac{7}{3}}$ cm

(c) $5\sqrt{\frac{3}{7}}$ cm (d) none of these

12. In a triangle ABC, the maximum value of $\cos A + \cos B + \cos C$ is

(a) 1/2 (b) 3/2

(c) 2/3 (d) 1/3

13. In a ΔABC, $\frac{a\cos A + b\cos B + c\cos C}{a+b+c}$ equals

(a) $\frac{R}{r}$ (b) $\frac{r}{R}$

(c) $\frac{R}{2r}$ (d) $\frac{2r}{R}$

where R and r are circumradius and inradius, respectively.

14. If in any triangle, $a\cos^2\frac{C}{2} + c\cos^2\frac{A}{2} = \frac{3b}{2}$, then its sides are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

15. In a triangle, if $\sin A : \sin C = \sin(A - B) : \sin(B - C)$, then $a^2 : b^2 : c^2$ are in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

Assertion-Reason Type Questions

In this section, each question consists of two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions has the following four choices (a), (b), (c) and (d), only one of which is the correct answer:

(a) Both Statement I is true and Statement II is true; Statement II is a correct explanation of Statement I.

(b) Both Statement I is true and Statement II is true; Statement II is not a correct explanation of Statement II.

(c) Statement I is true but Statement II is false.

(d) Statement I is false but Statement II is true.

16. **Statement I:** In a triangle ABC, if $\sin^2 A + \sin^2 B + \sin^2 C = 2$, then one of the angles must be 90°.

Statement II: In any triangle ABC,

$$\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A\cos B\cos C$$

17. **Statement I:** In any ΔABC, if A is obtuse, then $\tan B \tan C < 1$.

Statement II: In any ΔABC, we have $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

18. **Statement I:** If the measures of two angles of a triangle are 45° and 60°, then the ratio of the smallest and the greatest sides is $(\sqrt{3}-1):1$.

Statement II: The greatest side of a triangle is opposite to its greatest angle.

19. **Statement I:** If the lengths of two sides of a triangle are the roots of the equation $x^2 - 12x + 35 = 0$ and the angle opposite to the third side is obtuse, then the square of the length of the third side is greater than 74.

Statement II: In a ΔABC, $\cos C = \frac{a^2+b^2-c^2}{2ab}$.

20. **Statement I:** In a ΔABC, if $\tan A : \tan B : \tan C = 1 : 2 : 3$, then $A = 45°$.

Statement II: If $p : q : r = 1 : 2 : 3$, then $p = 1$.

Previous Years' Questions

21. In a triangle ABC, $2ca\sin\frac{A-B+C}{2} =$

(a) $a^2 + b^2 - c^2$ (b) $c^2 + a^2 - b^2$

(c) $b^2 - c^2 - a^2$ (d) $c^2 - a^2 - b^2$

[AIEEE 2002]

22. In a ΔABC, $\tan\frac{A}{2} = \frac{5}{6}$, $\tan\frac{C}{2} = \frac{2}{5}$, then

(a) a, c, b are in A.P. (b) a, b, c are in A.P.

(c) b, a, c are in A.P. (d) a, b, c are in G.P.

[AIEEE 2002]

23. In a ΔABC, $a = 4$, $b = 3$, $\angle A = 60°$, then c is the root of the equation

(a) $c^2 - 3c - 7 = 0$ (b) $c^2 + 3c + 7 = 0$

(c) $c^2 - 3c + 7 = 0$ (d) $c^2 + 3c - 7 = 0$

[AIEEE 2002]

24. The sides of a triangle are $3x + 4y$, $4x + 3y$ and $5x + 5y$, where $x, y > 0$, then the triangle is
(a) right-angled (b) obtuse-angled
(c) equilateral (d) none of these
[AIEEE 2002]

25. In a triangle with sides a, b, c, $r_1 > r_2 > r_3$ (which are exradii), then
(a) $a > b > c$ (b) $a < b < c$
(c) $a > b$ and $b < c$ (d) $a < b$ and $b > c$
[AIEEE 2002]

26. In a ΔABC, medians AD and BE are drawn. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$, then the area of the ΔABC is
(a) 8/3 (b) 16/3
(c) $32/3\sqrt{3}$ (d) 64/3 **[AIEEE 2003]**

27. The sum of the radii of the inscribed and circumscribed circles for an n-sided regular polygon of side a is
(a) $a\cot\left(\frac{\pi}{n}\right)$ (b) $\frac{a}{2}\cot\left(\frac{\pi}{2n}\right)$
(c) $a\cot\left(\frac{\pi}{2n}\right)$ (d) $\frac{a}{4}\cot\left(\frac{\pi}{2n}\right)$
[AIEEE 2003]

28. If in a ΔABC, $a\cos^2\left(\frac{C}{2}\right) + c\cos^2\left(\frac{A}{2}\right) = \frac{3b}{2}$, then the sides a, b and c
(a) are in A.P. (b) are in G.P.
(c) are in H.P. (d) satisfy $a + b = c$
[AIEEE 2003]

29. The sides of a triangle are $\sin\alpha$, $\cos\alpha$ and $\sqrt{1+\sin\alpha\cos\alpha}$ for some $0 < \alpha < \frac{\pi}{2}$. Then the greatest angle of the triangle is
(a) 60° (b) 90°
(c) 120° (d) 150° **[AIEEE 2004]**

30. In a ΔABC, let ΔABC $\angle C = \frac{\pi}{2}$, if r is the inradius and R is the circumradius of the ΔABC, then $2(r + R)$ equals
(a) $c + a$ (b) $a + b + c$
(c) $a + b$ (d) $b + c$ **[AIEEE 2005]**

31. If in a ΔABC, the altitudes from the vertices A, B, C on opposite sides are in H.P., then $\sin A$, $\sin B$, $\sin C$ are in
(a) H.P.
(b) arithmetico-geometric progression
(c) A.P.
(d) G.P. **[AIEEE 2005]**

32. In a ΔPQR, $\angle R = \frac{\pi}{2}$, if $\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of $ax^2 + bx + c = 0$, $a \neq 0$, then
(a) $b = a + c$ (b) $b = c$
(c) $c = a + b$ (d) $a = b + c$
[AIEEE 2005]

33. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of the same length x. The maximum area enclosed by the park is
(a) $\sqrt{\frac{x^3}{8}}$ (b) $\frac{1}{2}x^2$
(c) πx^2 (d) $\frac{3}{2}x^2$ **[AIEEE 2006]**

34. For a regular polygon, let r and R be the radii of the inscribed and the circumscribed circles. A false statement among the following is
(a) there is a regular polygon with $\frac{r}{R} = \frac{1}{2}$
(b) there is a regular polygon with $\frac{r}{R} = \frac{1}{\sqrt{2}}$
(c) there is a regular polygon with $\frac{r}{R} = \frac{2}{3}$
(d) there is a regular polygon with $\frac{r}{R} = \frac{\sqrt{3}}{2}$
[AIEEE 2010]

35. The sides of a triangle are in the ratio $1:\sqrt{3}:2$, then the angles of the triangle are in the ratio
(a) 1 : 3 : 5 (b) 2 : 3 : 2
(c) 3 : 2 : 1 (d) 1 : 2 : 3
[IIT-JEE 2004]

36. In a ΔABC, among the following which one is true?
(a) $(b + c)\cos\frac{A}{2} = a\sin\left(\frac{B + C}{2}\right)$
(b) $(b + c)\cos\left(\frac{B + C}{2}\right) = a\sin\frac{A}{2}$
(c) $(b - c)\cos\left(\frac{B - C}{2}\right) = a\cos\left(\frac{A}{2}\right)$
(d) $(b - c)\cos\frac{A}{2} = a\sin\left(\frac{B - C}{2}\right)$ **[IIT-JEE 2005]**

37. If the inradius of a circle which is inscribed in an isosceles triangle one of whose angle is $2\pi/3$ is $\sqrt{3}$, then the area of the triangle is

(a) $4\sqrt{3}$ (b) $12-7\sqrt{3}$

(c) $12+7\sqrt{3}$ (d) none of these

[IIT-JEE 2006]

38. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C, respectively, then the value of the expression $\frac{a}{c}\sin 2C+\frac{c}{a}\sin 2A$ is

(a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$

(c) 1 (d) $\sqrt{3}$ **[IIT-JEE 2010]**

39. Let ABC be a triangle such that $\angle ACB=\frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C, respectively. The value (s) of x for which $a=x^2+x+1$, $b=x^2-1$ and $c=2x+1$ is (are)

(a) $-(2+\sqrt{3})$ (b) $1+\sqrt{3}$

(c) $2+\sqrt{3}$ (d) $4\sqrt{3}$ **[IIT-JEE 2010]**

40. If the angles A, B and C of a triangle are in A.P. with common difference 15°, then the angle A is equal to

(a) 45° (b) 60°

(c) 75° (d) 30° **[DCE 2003]**

41. In a triangle, vertex angles A, B, C and side BC are given. The area of the ΔABC is

(a) $\frac{s(s-a)(s-b)(s-c)}{2}$ (b) $\frac{b^2\sin C\sin A}{\sin B}$

(c) $ab\sin C$ (d) $\frac{1}{2}\cdot\frac{a^2\sin B\sin C}{\sin A}$

[DCE 2003]

42. In any ΔABC, $\sum a^3\cos(B-C)$ is equal to

(a) $3abc$ (b) $ab+bc+ca$

(c) $3(a+b+c)$ (d) 0 **[DCE 2006]**

43. In a ΔABC, a, c, A are given and b_1, b_2 are two values, if the third side b is such that $b_2=2b_1$, then $\sin A$ is equal to

(a) $\frac{\sqrt{9a^2-c^2}}{8a^2}$ (b) $\frac{\sqrt{9a^2-c^2}}{8c^2}$

(c) $\sqrt{\frac{9a^2+c^2}{8a^2}}$ (d) none of these

[DCE 2006]

44. In a ΔABC, if $b=2$, $\angle C=60°$, $c=\sqrt{6}$, then a is equal to

(a) $\sqrt{3}-1$ (b) $\sqrt{3}$

(c) $\sqrt{3}+1$ (d) none of these

[DCE 2006]

45. In a ΔABC, if $\angle A=\frac{\pi}{2}$, $b=4$, $c=3$, then the value of $\frac{R}{r}$ is equal to

(a) $\frac{5}{2}$ (b) $\frac{7}{2}$

(c) $\frac{9}{2}$ (d) $\frac{35}{24}$ **[DCE 2007]**

46. In any ΔABC, $\frac{\tan\frac{A}{2}-\tan\frac{B}{2}}{\tan\frac{A}{2}+\tan\frac{B}{2}}$ is equal to

(a) $\frac{a-b}{a+b}$ (b) $\frac{a-b}{c}$

(c) $\frac{a-b}{a+b+c}$ (d) $\frac{c}{a+b}$ **[DCE 2008]**

47. In a ΔABC, if $\sin^2\frac{A}{2}, \sin^2\frac{B}{2}, \sin^2\frac{C}{2}$ are in H.P., then a, b, c will be in

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

[DCE 2008]

48. The sides BC, CA and AB of a ΔABC are of lengths a, b and c, respectively. If D is the mid-point of BC and AD is perpendicular to AC, then the value of $\cos A\cos C$ is

(a) $\frac{3(a^2-c^2)}{2ac}$ (b) $\frac{2(a^2-c^2)}{3bc}$

(c) $\frac{(a^2-c^2)}{3ac}$ (d) $\frac{2(c^2-a^2)}{3ac}$

[DCE 2009]

49. In a ΔABC, if $AB=1$, $AC=2$ and $\angle A=6°$, then its largest angle is equal to

(a) 75° (b) 90°

(c) 120° (d) 135° **[UP SEE 2009]**

50. If $\Delta=a^2-(b-c)^2$, where Δ is the area of ΔABC, then $\tan A$ is equal to

(a) 15/16 (b) 8/17

(c) 8/15 (d) 1/2

[Kerala CEE 2005]

51. In a triangle ABC, the value of $\dfrac{\cot\frac{A}{2}\cot\frac{B}{2}-1}{\cot\frac{A}{2}\cot\frac{B}{2}}$ is

(a) $\dfrac{a}{a+b+c}$ (b) $\dfrac{c}{a+b+c}$

(c) $\dfrac{2a}{a+b+c}$ (d) $\dfrac{2b}{a+b+c}$

[Kerala CEE 2009]

52. In a ΔABC, $2ac\sin\dfrac{A-B+C}{2}$ is equal to

(a) $a^2+b^2-c^2$ (b) $c^2+a^2-b^2$

(c) $b^2-a^2-c^2$ (d) $c^2-a^2-b^2$

[WB JEE 2010]

53. If in a ΔABC, $4\sin A = 4\sin B = 3\sin C$, then $\cos C$ is equal to

(a) 1/3 (b) 1/9

(c) 1/27 (d) 1/18

[J & K CET 2008]

ANSWERS

Exercise Set 1

1. (c)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (a)	**7.** (c)	**8.** (b)	**9.** (a)	**10.** (d)
11. (b)	**12.** (a)	**13.** (a)	**14.** (c)	**15.** (c)	**16.** (d)	**17.** (a)	**18.** (a)	**19.** (b)	**20.** (c)
21. (d)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (a)	**26.** (b)	**27.** (a)	**28.** (b)	**29.** (d)	**30.** (a)
31. (a)	**32.** (d)	**33.** (a)	**34.** (c)	**35.** (b)	**36.** (d)	**37.** (a)	**38.** (a) and (d)		**39.** (b)
40. (c)	**41.** (b) and (c)		**42.** (d)	**43.** (c)	**44.** (b)	**45.** (c)	**46.** (a)	**47.** (c)	**48.** (c)
49. (c)	**50.** (a)	**51.** (b)	**52.** (c)	**53.** (a)					

Exercise Set 2

1. (d)	**2.** (a)	**3.** (a)	**4.** (c)	**5.** (a)	**6.** (d)	**7.** (b)	**8.** (d)	**9.** (a)	**10.** (a)
11. (c)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (a)	**16.** (a)	**17.** (a)	**18.** (a)	**19.** (a)	**20.** (c)
21. (b)	**22.** (b)	**23.** (a)	**24.** (b)	**25.** (a)	**26.** (c)	**27.** (b)	**28.** (a)	**29.** (c)	**30.** (c)
31. (c)	**32.** (c)	**33.** (b)	**34.** (c)	**35.** (d)	**36.** (d)	**37.** (c)	**38.** (d)	**39.** (b)	**40.** (b)
41. (d)	**42.** (a)	**43.** (b)	**44.** (c)	**45.** (a)	**46.** (b)	**47.** (c)	**48.** (d)	**49.** (b)	**50.** (c)
51. (b)	**52.** (b)	**53.** (b)							

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Using $\tan^2\dfrac{A-B}{2}=\dfrac{1-\cos(A-B)}{1+\cos(A-B)}$,

we get

$$\tan\frac{A-B}{2}=\frac{1}{\sqrt{63}}=\frac{1}{3\sqrt{7}}$$

also $\tan\left(\dfrac{A-B}{2}\right)=\dfrac{a-b}{a+b}\cot C/2$

or $\dfrac{1}{3\sqrt{7}}=\dfrac{5-4}{5+4}\cot C/2$ or $\tan C/2=\dfrac{\sqrt{7}}{3}$

$$\cos C=\frac{1-\tan^2 C/2}{1+\tan^2 C/2}=\frac{1-7/9}{1+7/9}=\frac{1}{8}$$

or $\dfrac{a^2+b^2-c^2}{2ab}=\dfrac{1}{8}$ or $\dfrac{25+16-c^2}{2\times5\times4}=\dfrac{1}{8}$

or $c=6$

2. We have $\operatorname{cosec}A(\sin B\cos C+\cos B\sin C)$

$$=\left(\frac{\sin B}{\sin C}\cos C+\frac{\sin C}{\sin A}\cos B\right)$$

$$=\left(\frac{b}{a}\cos C+\frac{c}{a}\cos B\right)=1$$

3. $bc\cos^2\dfrac{A}{2}+ca\cos^2\dfrac{B}{2}+ab\cos^2\dfrac{C}{2}$

$$=bc\left[\frac{s(s-a)}{bc}\right]+ca\left[\frac{s(s-b)}{ac}\right]+\frac{ab[s(s-c)]}{ab}$$

$$\left[\because\cos^2\frac{A}{2}=\frac{s(s-a)}{bc},\cos^2\frac{B}{2}=\frac{s(s-b)}{ac},\ \cos^2\frac{c}{2}=\frac{s(s-c)}{ab}\right]$$

$=s(s-a)+s(s-b)+s(s-c)$

$=s[s-a+s-b+s-c]$

$= s[3s - (a + b + c)]$

$= s[3s - 2s]$ $[\because 2s = a + b + c]$

$= s(s) = s^2$

So, correct option is (d).

4. $\dfrac{1+\cos C\cos(A-B)}{1+\cos(A-C)\cos B} = \dfrac{1-\cos(A+B)\cos(A-B)}{1-\cos(A-C)\cos(A+C)}$

$[\because A + B + C = \pi]$

$\Rightarrow \dfrac{1-\cos^2 A+\sin^2 B}{1-\cos^2 A+\sin^2 C} = \dfrac{\sin^2 A+\sin^2 B}{\sin^2 A+\sin^2 C}$

$= \dfrac{a^2+b^2}{a^2+c^2}$

5. $\dfrac{\cot\frac{A}{2}}{\sin\frac{A}{2}} = \dfrac{\sin B+\sin C}{\sin A} = \dfrac{2\sin\frac{B+C}{2}\cos\frac{B-C}{2}}{2\sin\frac{A}{2}\cos\frac{A}{2}}$

or $\cos\dfrac{A}{2} = \cos\dfrac{B-C}{2}$ or $A = B - C$

or $\pi - B - C = B - C$ $\therefore B = \pi/2$

6. $\cot B - \cot A = \cot C - \cot B$

$\dfrac{\sin(A-B)}{\sin A} = \dfrac{\sin(B-C)}{\sin C}$

or $\sin(A-B)\sin(A+B) = \sin(B-C)\sin(B+C)$

or $\sin^2 A - \sin^2 B = \sin^2 B - \sin^2 C$

or $a^2 - b^2 = b^2 - c^2$ $\therefore$ a^2, b^2, c^2 are in A.P.

7. Let $\cot\dfrac{A}{2}, \cot\dfrac{B}{2}$ and $\cot\dfrac{C}{2}$ be in A.P.

Then $2\cot\dfrac{B}{2} = \cot\dfrac{C}{2} + \cot\dfrac{A}{2}$

$\Rightarrow 2\sqrt{\dfrac{s(s-b)}{(s-a)(s-c)}} = \sqrt{\dfrac{s(s-c)}{(s-a)(s-b)}} + \sqrt{\dfrac{s(s-a)}{(s-b)(s-c)}}$

R.H.S. $= \sqrt{\dfrac{s}{(s-b)}}\left(\sqrt{\left(\dfrac{s-c}{s-a}\right)} + \sqrt{\dfrac{s-a}{s-c}}\right)$

$= \sqrt{\dfrac{s}{(s-b)}}\left(\dfrac{s-c+s-a}{\sqrt{(s-a)(s-c)}}\right)$

$= \sqrt{\dfrac{s}{(s-b)}}\left(\dfrac{2s-a-c}{\sqrt{(s-a)(s-c)}}\right)$

$= 2\sqrt{\dfrac{s}{(s-b)}}\sqrt{\left(\dfrac{(s-b)^2}{(s-a)(s-c)}\right)}$

Since sides are in A.P.,

$\therefore$ $a + c = 2b$ and $a + b + c = 2s$

$\Rightarrow 2(s-b) = 2s - a - c$

$= 2\sqrt{\dfrac{s(s-b)}{(s-a)(s-c)}} =$ L.H.S.

Therefore, our assumption is true.

8. If sides of the Δ are p, q, r, then $\sin P$, $\sin Q$, $\sin R$ are in A.P. which means p, q, r are in A.P. The area of

$\Delta = \dfrac{1}{2}\times PL\times p$

$= \dfrac{1}{2}\times QM\times q = \dfrac{1}{2}\times RN\times r$

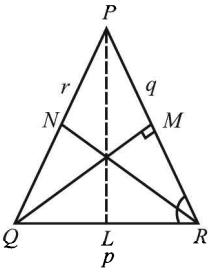

Fig. 28.13

$\therefore$ Altitudes are: $PL = \dfrac{2\Delta}{p}, QM = \dfrac{2\Delta}{q}, RN = \dfrac{2\Delta}{r}$

$\because$ p, q, r are in A.P., so the altitudes PL, QM, RN are in H.P.

9. $\cot\dfrac{A}{2} - \sin A = \dfrac{\cos(A/2)}{\sin(A/2)} - 2\sin\dfrac{A}{2}\cos\dfrac{A}{2}$

$= \cos\dfrac{A}{2}\cdot\dfrac{[1-2\sin^2(A/2)]}{\sin(A/2)} = \cot\dfrac{A}{2}\cos A$ (i)

Since each of $\cot\dfrac{A}{2}, \cot\dfrac{B}{2}, \cot\dfrac{C}{2}$ is in A.P. and $\sin A$, $\sin B$, $\sin C$ are in A.P., thier differences are also in A.P.

Hence from (i), $\cot\dfrac{A}{2}\cos A, \cot\dfrac{B}{2}\cos B, \cot\dfrac{C}{2}\cos C$ are also in A.P.

10. We have $\dfrac{\cos A}{a} = \dfrac{\cos B}{b} = \dfrac{\cos C}{c}$

$\Rightarrow \dfrac{\cos A}{k\sin A} = \dfrac{\cos B}{k\sin B} = \dfrac{\cos C}{k\sin C}$

$\Rightarrow \cot A = \cot B = \cot C$

$\Rightarrow A = B = C = 60°$

So, ΔABC is an equilateral triangle.

$\therefore$ $\Delta = \dfrac{\sqrt{3}}{4}a^2 = \dfrac{\sqrt{3}}{4}(2)^2$ $[\because a = 2$ (given)]

$= \sqrt{3}$

11. Let $CD = x$, clearly area of $(\Delta ACD + \Delta BCD)$ = area of ΔABC

$\Rightarrow \dfrac{1}{2}4x\cdot\sin 60° + \dfrac{1}{2}3x\cdot\sin 30°$

$\Rightarrow \dfrac{1}{2}\cdot\dfrac{4\sqrt{3}}{2}\cdot x\cdot\dfrac{1}{2}\cdot\dfrac{3x}{2} = \dfrac{1}{2}\cdot 12$

$\Rightarrow x = \dfrac{24}{3+4\sqrt{3}}$

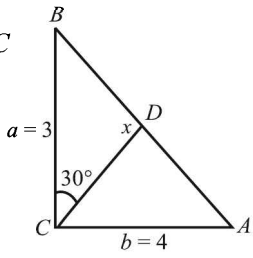

Fig. 28.14

12. By sine rule,

$$T_1 = k^3 \sin^3 A \cos(B-C)$$

$$= k^3 \sin^2 A \sin(B+C)\cos(B-C)$$

$$= \frac{1}{2}k^3[\sin^2 A(\sin 2B + \sin 2C)]$$

$$= \frac{1}{2}k^3[2\sin^2 A \sin B \cos B + 2\sin^2 A \sin C \cos C]$$

$\therefore$ $\sum a^3 \cos(B-C) =$ sum of the terms

$$= \sum k^3 \sin A \sin B(\sin A \cos B + \cos A \sin B)$$

$$= \sum k^3 \sin A \sin B \sin(A+B)\text{, three pairs}$$

$$= \sum k^3 \sin A \sin B \sin C = \sum abc = 3abc$$

13. Let the area of the triangle be Δ, then according to the question,

$$\Delta = \frac{1}{2}ax = \frac{1}{2}by = \frac{1}{2}cz$$

$$\therefore \quad \frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = \frac{b}{c}\left(\frac{2\Delta}{a}\right) + \frac{c}{a}\left(\frac{2\Delta}{b}\right) + \frac{a}{b}\left(\frac{2\Delta}{c}\right)$$

$$= \frac{2\Delta(b^2+c^2+a^2)}{abc} = \frac{2(a^2+b^2+c^2)}{abc} \cdot \frac{abc}{4R}$$

$$\left[\because \Delta = \frac{abc}{4R}\right]$$

$$= \frac{a^2+b^2+c^2}{2R}$$

14. Given $BC = 4AD$

$\Rightarrow$ $a = 4c \sin B$

$\Rightarrow$ $\sin A = 4 \sin C \sin B$

$\Rightarrow$ $1 = 2[\cos(C-B) - \cos(C+B)]$

$$\frac{1}{2} = \cos(C-B)$$

$[\because \cos(B+C) = \cos 90° = 0]$

$\therefore$ $C - B = 60°$

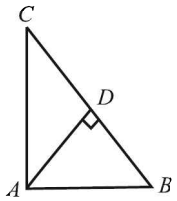

Fig. 28.15

15. $\sin A + \frac{1}{\sin A} + 1 > 2 + 1 = 3$ by A.M. $\geq$ G.M.

Each term is greater than 3.

$\therefore$ $E > 3 \cdot 3 \cdot 3 = 27$

16. $\because \angle A = 60° \quad \Rightarrow \quad \cos 60° = \frac{b^2+c^2-a^2}{2bc}$

$\Rightarrow$ $b^2 + c^2 - a^2 = bc$

Now, $AB^2 + AC^2 = 2(AD^2 + BD^2)$

$$\Rightarrow \quad c^2 + b^2 = 2AD^2 + 2\left(\frac{a}{2}\right)^2$$

$$\Rightarrow \quad 2b^2 + 2c^2 - a^2 = 4AD^2$$

$$\Rightarrow \quad b^2 + c^2 + bc = 4AD^2 \qquad (\because \ b^2 + c^2 - a^2 = bc)$$

17. We know that $r_1 = \frac{\Delta}{s-a}$, etc.

Given expression

$$= \frac{1}{\Delta}[(s-a)(b-c) + (s-b)(c-a) + (s-a)(a-b)] = 0$$

18. $\frac{\sin(A-B)}{\sin(A+B)} = \frac{k^2(\sin^2 A - \sin^2 B)}{k^2(\sin^2 A + \sin^2 B)}$ [by sine formula]

or $\frac{\sin(A-B)}{\sin C} = \frac{\sin(A-B)\sin(A+B)}{\sin^2 A + \sin^2 B}$

or $\sin(A-B)\left[\frac{1}{\sin C} - \frac{\sin C}{\sin^2 A + \sin^2 B}\right] = 0$

$\therefore$ Either $\sin(A-B) = 0$ $\therefore$ $A = B$, i.e. Δ is isosceles

or $\sin^2 A + \sin^2 B = \sin^2 C$ or $a^2 + b^2 = c^2$

$\therefore$ Δ is right-angled.

19. $\because$ $A + B + C = 180°$

$\Rightarrow$ $A - B + C = 180° - 2B$

$$\therefore \quad 2ac \sin\left(\frac{A-B+C}{2}\right) = 2ac \sin(90° - B)$$

$$= 2ac \cos B = c^2 + a^2 - b^2$$

20. $\cos 30° = \frac{b^2+c^2-a^2}{2bc}$ or $\frac{\sqrt{3}}{2} = \frac{4+3-a^2}{2 \times 2 \times \sqrt{3}}$

$$6 = 7 - a^2 \quad \Rightarrow \quad a = 1$$

$$s = \frac{a+b+c}{2} = \frac{3+\sqrt{3}}{2}$$

and area $= \Delta = \frac{1}{2}bc \sin A = \frac{\sqrt{3}}{2}$

$$\therefore \quad r = \frac{\Delta}{s} = \frac{\sqrt{3}}{2} \times \frac{2}{\sqrt{3}(\sqrt{3}+1)} = \frac{1}{\sqrt{3}+1} = \frac{\sqrt{3}-1}{2}$$

21. Rewriting the left-hand side, we get

$\sum(b \cos C + c \cos B) = a + b + c$

22. Circle drawn is the excircle opposite to A

$$\therefore \quad \frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{3}{r_1} \quad \text{(For equilateral } \Delta, r_1 = r_2 = r_3\text{)}$$

$\therefore$ $r_1 = 3 \times 4 = 12$

$\therefore$ Area $= \pi r_1^2 = 144\pi$ sq. units

23. The largest side is $\sqrt{a^2+ab+b^2}$. If the largest angle is θ, then $\cos\theta = \dfrac{a^2+b^2-(a^2+ab+b^2)}{2ab}$

$= -\dfrac{1}{2} = \cos 120° \Rightarrow \theta = 120°$

24. $\Sigma \dfrac{\cos A}{a} = \dfrac{\Sigma(b^2+c^2-a^2)}{2abc}$

$= \dfrac{a^2+b^2+c^2}{2abc} = \dfrac{(\Sigma a)^2 - 2\Sigma ab}{2abc}$

$= \dfrac{(11)^2 - 2(38)}{2\times 40} = \dfrac{45}{80} = \dfrac{9}{16}$

25. Since $\left(1-\dfrac{r_1}{r_2}\right)\left(1-\dfrac{r_1}{r_3}\right) = 2$,

$\therefore \left(1-\dfrac{s-b}{s-a}\right)\left(1-\dfrac{s-c}{s-a}\right) = 2 \Rightarrow \dfrac{(b-a)(c-a)}{(s-a)^2} = 2$

$\Rightarrow 2bc - 2ab - 2ac + 2a^2 = b^2 + c^2 + a^2 + 2bc - 2ab - 2ac$

$\Rightarrow a^2 = b^2 + c^2$

So, the triangle is right-angled.

26. $c^2 = a^2 + b^2 - 2ab\cos C$

or $a^2 + b^2 - c^2 = 2ab\cos C$ (i)

Given $\dfrac{a+b+2c}{(a+c)(b+c)} = \dfrac{3}{(a+b+c)}$

or $(a+b+2c)(a+b+c) = 3(c+a)(c+b)$

or $(a+b)^2 + 2c^2 + 3c(a+b) = 3[c^2 + c(a+b) + ab]$

i.e. if $a^2 + b^2 + 2ab + 2c^2 = 3c^2 + 3ab$

i.e. if $a^2 + b^2 - c^2 = ab$

or $2ab\cos C = ab$ [by (i)]

$\therefore \cos C = \dfrac{1}{2}$ or $C = 60°$

27. $\because \dfrac{\sin A}{a} = \dfrac{\sin B}{b} = \dfrac{\sin C}{c}$

$\therefore \dfrac{b+c}{a} = \dfrac{\sin B + \sin C}{\sin A}$

$= \dfrac{\sin\left(\dfrac{B+C}{2}\right)\cos\left(\dfrac{B-C}{2}\right)}{\sin\dfrac{A}{2}\cos\dfrac{A}{2}}$

$= \dfrac{\cos\left(\dfrac{B-C}{2}\right)}{\sin\dfrac{A}{2}}$

$[\because A+B+C = p \Rightarrow A = p-(B+C)]$

28. $\cos A = \dfrac{1}{2}$

$\Rightarrow b^2 + c^2 = a^2 + bc$ (i)

L.H.S. $= \dfrac{b^2+c^2+a(b+c)}{a^2+bc+a(b+c)} = 1$ [by (i)]

29. We have $A + B + C = 180°$

$\Rightarrow A = 180° - (B + C)$

$\Rightarrow \tan A = \tan(180° - B - C)$

$\Rightarrow \tan 90° = -\tan(B + C)$

$\Rightarrow \infty = -\dfrac{\tan B + \tan C}{1 - \tan B \tan C}$

$\Rightarrow 1 - \tan B \tan C = 0$

$\Rightarrow \tan B \tan C = 0$

30. $2B = A + C \Rightarrow 3B = \pi$

$\therefore B = \pi/3$ (i)

Also, a, b, c are in G.P. $\Rightarrow b^2 = ac$ (ii)

$\cos B = \cos 60° = \dfrac{1}{2} = \dfrac{c^2+a^2-b^2}{2ca}$

$\therefore ca = c^2 + a^2 - b^2$ or $2b^2 = c^2 + a^2$ [by (ii)]

$\therefore a^2, b^2, c^2$ are in A.P.

31. We know that in a triangle, the larger side has a larger angle opposite to it. Since angles $\angle A$, $\angle B$ and $\angle C$ are in A.P.,

$\Rightarrow 2B = A + C$

$\because A + B + C = \pi$

$\Rightarrow B = 60°$

$\therefore \cos B = \dfrac{a^2+c^2-b^2}{2ac}$

$\Rightarrow \cos 60° = \dfrac{1}{2} = \dfrac{100+a^2-81}{20a}$

$\Rightarrow a^2 + 19 = 10a$

$\Rightarrow a^2 - 10a + 19 = 0$

$\therefore a = \dfrac{10 \pm \sqrt{100-76}}{2} = 5 \pm \sqrt{6}$

32. $2a^2 + 4b^2 + c^2 = 4ab + 2ac$

$\Rightarrow (a-2b)^2 + (a-c)^2 = 0$

$\Rightarrow a = 2b$ and $a = c$

$\Rightarrow a = 2b = c$ or $\frac{a}{1} = \frac{b}{1/2} = \frac{c}{1}$

so that $\cos B = \frac{c^2 + a^2 - b^2}{2ca} = \frac{1+1-(1/4)}{2\times1\times1} = \frac{7}{8}$

33. Combine first and third and put the value of cos B.

$\therefore \quad \frac{2}{ac}\cdot(b) + \frac{1}{b}\frac{c^2 + a^2 - b^2}{2ca} = \frac{a^2 + b^2}{abc}$

or $4b^2 + c^2 + a^2 - b^2 = 2a^2 + 2b^2$

$\therefore \quad b^2 + c^2 = a^2 \quad \therefore \quad \angle A = 90°$

34. Since a, b, c are in A.P., $2b = a + c$

$\Rightarrow \qquad 2\sin B = \sin A + \sin C$

$\Rightarrow 4\sin(B/2)\cos(B/2) = 2\sin[(A+C)/2]\cos[(A-C)/2]$

$= 2\cos(B/2)\cos((A-C)/2)$

$\Rightarrow \sin(B/2) = \frac{1}{2}\cos[(A-C)/2] \le 1/2$

$\therefore \quad B \le 60°$

35. Let p_1, p_2, p_3 be the altitudes of ΔABC. Then,

$\Delta = \frac{1}{2}ap_1 = \frac{1}{2}bp_2 = \frac{1}{2}cp_3$

$\Rightarrow \quad a = \frac{2\Delta}{p_1}, b = \frac{2\Delta}{p_2}, c = \frac{2\Delta}{p_3}$

Now, $\sin A, \sin B, \sin C$ are in A.P.

$\Rightarrow$ a, b, c are in A.P.

$\Rightarrow \quad \frac{2\Delta}{p_1}, \frac{2\Delta}{p_2}, \frac{2\Delta}{p_3}$ are in A.P.

$\Rightarrow \quad \frac{1}{p_1}, \frac{1}{p_2}, \frac{1}{p_3}$ are in A.P.

$\Rightarrow$ p_1, p_2, p_3 are in H.P.

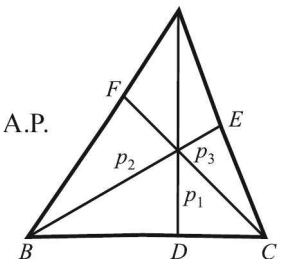

Fig. 28.17

36. Let a be the length of each side of the equilateral triangle ABC (Fig. 28.18). Then r, the radius of the incircle = (1/3) AP (the altitude, median and the angle bisector of angle A)

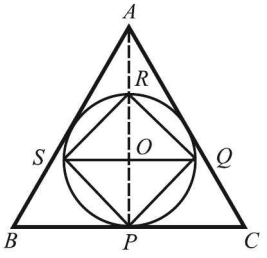

Fig. 28.18

$\Rightarrow r = \frac{1}{3}\sqrt{a^2 - \frac{a^2}{4}} = \frac{a}{2\sqrt{3}}$

Area of the square $PQRS$ inscribed in this circle

$= PQ^2 = OP^2 + OQ^2 = 2r^2 = 2\times\frac{a^2}{4\times3} = \frac{a^2}{6}$

37. In triangles PQR and PRS, we have $\tan\left(\frac{\pi}{2} - \theta\right) = \frac{PQ}{PR}$

and $\tan\theta = \frac{RS}{PR}$

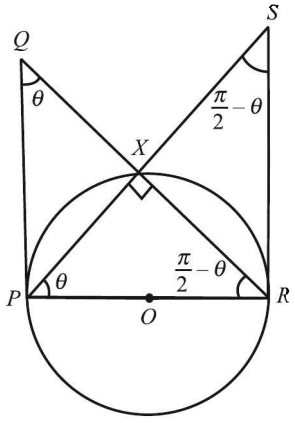

Fig. 28.19

$\Rightarrow \quad \cot\theta \times \tan\theta = \frac{PQ}{PR}\times\frac{RS}{PR}$

$\Rightarrow \quad PR^2 = PQ\cdot RS \Rightarrow PR = \sqrt{PQ\cdot RS} \Rightarrow 2r = \sqrt{PQ\cdot RS}$

38. From the given relation, we have

$\frac{b^2 + c^2 - a^2}{2bc} = \frac{b}{2c}$

$\Rightarrow \quad a^2 - c^2 = 0 \quad \Rightarrow \quad a = c$

So, option (a) is correct and $a = c \Rightarrow A = C$ and the given relation gives $\sin B = \sin 2C$

39. $b + c = 3a$ (i)

$\cot\frac{B}{2}\cot\frac{C}{2} = \frac{s}{s-a} = \frac{2s}{2s-2a}$

$\frac{a+b+c}{b+c-a} = \frac{4a}{2a} = 2$ [by (i)]

40. Let a, b, c be the lengths of the sides of the triangle. Now, A.M. > G.M.

$\Rightarrow \quad \frac{s+s-a+s-b+s-c}{4} > [s(s-a)(s-b)(s-c)]^{1/4}$

$\Rightarrow \quad \frac{s}{2} > \Delta^{2/4} \quad \Rightarrow \quad \frac{s^2}{4} > \Delta \quad \Rightarrow \quad \Delta < \frac{s^2}{4}$

41. The given relation can be written as

$\cos A(\sin B - \sin C) + \sin 2B - \sin 2C = 0$

or $\cos A(\sin B-\sin C)+2\cos(B+C)\sin(B-C)=0$

or $\cos A(\sin B-\sin C)+2\cos(180°-A)\ \sin(B-C)=0$

or $\cos A[\sin B-\sin C-2\sin(B-C)]=0$

from which it follows that either $\cos A=0$, so that $A=\pi/2$ and ΔABC is right-angled, or

$$\sin B-\sin C-2\sin(B-C)=0$$

$\Rightarrow$ $(b-c)-2(b\cos C-c\cos B)=0$ [by the law of sines]

$$\Rightarrow\ (b-c)-2\left(\frac{a^2+b^2-c^2}{2a}-\frac{c^2+a^2-b^2}{2a}\right)=0$$

$$\Rightarrow\ a(b-c)-2(b^2-c^2)=0$$

$$\Rightarrow\ (b-c)[a-2(b+c)]=0$$

$\Rightarrow$ $b-c=0$

$\Rightarrow$ triangle is isosceles $\quad(\because\ b+c>a)$

42. $T_1=\dfrac{b-c}{a}\cos^2\dfrac{1}{2}A$

$$=\frac{b-c}{a}\cdot\frac{s(s-a)}{bc}=\frac{s(s-a)(b-c)}{abc}$$

$\therefore$ L.H.S. $=\dfrac{s}{abc}[\Sigma(s-a)(b-c)]$

$$=\frac{s}{abc}[s\Sigma(b-c)-\Sigma a(b-c)]=0$$

43. In the ΔCAD, we have

$$\cos C=\frac{AC}{CD}\Rightarrow\frac{a^2+b^2-c^2}{2bc}=\frac{b}{a/2}\Rightarrow a^2-c^2=3b^2$$

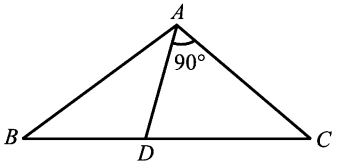

Fig. 28.20

44. $2b=a+c\quad\therefore\ 3b=2s$

$$\tan\frac{A}{2}+\tan\frac{C}{2}=\frac{b}{s}\cot\frac{B}{2}=\frac{2b}{2s}\cot\frac{B}{2}$$

$$=\frac{2b}{3b}\cot\frac{B}{2}=\frac{2}{3}\cot\frac{B}{2}$$

45. We have

$$\text{area}=\frac{1}{2}ab\sin C\quad\Rightarrow\quad 5\sqrt{3}=\frac{1}{2}\times6\times10\times\sin C$$

$$\Rightarrow\ \sin C=\frac{\sqrt{3}}{2}\quad\Rightarrow\quad C=120°$$

$\therefore$ $c^2=a^2+b^2-2ab\cos C$

$$\Rightarrow\ c^2=36+100-120\times-\frac{1}{2}=196\Rightarrow c=14$$

So, $2s=a+b+c\quad\Rightarrow\quad 2s=30\quad\Rightarrow\quad s=15$

$$\therefore\quad r=\frac{\Delta}{s}=\frac{15\sqrt{3}}{15}=\sqrt{3}\quad\Rightarrow\quad r^2=3$$

46. $\cot\dfrac{A}{2}=\sqrt{\dfrac{s(s-a)}{(s-b)(s-c)}}=\dfrac{s(s-a)}{S}$

Cancel s/S and we have $s-a, s-b, s-c$ which are in A.P.

$\therefore$ a, b, c are in A.P.

47. Since $\cos B=\dfrac{c^2+a^2-b^2}{2ac}$

$$\therefore\quad\cos=60°=\frac{c^2+a^2-b^2}{2ca}\quad\Rightarrow\quad\frac{1}{2}=\frac{c^2+a^2-b^2}{2ca}$$

$\Rightarrow$ $ca=c^2+a^2-b^2$

$\Rightarrow$ $(c-a)^2+ca=b^2$

48. $S=(a+b-c)(a-b+c)$

or $S^2=[2(s-b).2(s-c)]^2$

$$s(s-a)(s-b)(s-c)=16(s-b)^2(s-c)^2$$

$$\therefore\quad\frac{(s-b)(s-c)}{s(s-a)}=\frac{1}{16}$$

or $\tan^2\dfrac{A}{2}=\dfrac{1}{16}$ or $\tan\dfrac{A}{2}=\dfrac{1}{4}$

$$\therefore\quad\tan A=\frac{2\tan(A/2)}{1-\tan^2(A/2)}=\frac{2\cdot(1/4)}{1-(1/16)}=\frac{8}{15}$$

49. Since $B=60°,\quad\therefore\cos B=\cos 60°=\dfrac{1}{2}$

$$\Rightarrow\quad\frac{c^2+a^2-b^2}{2ca}=\frac{1}{2}\Rightarrow c^2+a^2-b^2=ca$$

$$\Rightarrow\ (c+a)^2-b^2=3ca$$

$$\Rightarrow\ (c+a+b)(c+a-b)=3ca$$

50. $\cos B=\left(\dfrac{c^2+a^2-b^2}{2ca}\right)$

or $\cos^2 B=\dfrac{a^4+b^4+c^4-2b^2c^2-2a^2b^2+2a^2c^2}{4a^2c^2}$

or $\cos^2 B=\dfrac{1}{2}$ by the given relation

$$\therefore\quad\cos B=\pm\frac{1}{\sqrt{2}}\quad\therefore B=45°\text{ or }135°$$

51. Given expression

$$= \frac{1}{4b^2c^2}[(b+c)^2 - a^2][a^2 - (b-c)^2]$$

$$= \frac{1}{4b^2c^2}[b^2 + c^2 - a^2 + 2bc][a^2 - (b^2 + c^2 - 2bc)]$$

$$= \frac{1}{4b^2c^2}[2bc\cos A + 2bc][2bc - (b^2 + c^2 - a^2)]$$

$$= \frac{2bc}{4b^2c^2}[\cos A + 1](2bc - 2bc\cos A)$$

$$= (1+\cos A)(1-\cos A)$$

$$= 1 - \cos^2 A = \sin^2 A$$

52. $2B = A + C \Rightarrow 3B = 180° \quad \therefore B = 60°$

and $A + C = 120°$ (i)

Now $b^2 = c^2 + a^2 - 2ca\cos B = c^2 + a^2 - ca$ (ii)

Hence, $\dfrac{a+c}{\sqrt{(a^2 - ac + c^2)}} = \dfrac{a+c}{b} = \dfrac{\sin A + \sin C}{\sin B}$ [by (ii)]

$$= \frac{2\sin[(A+C)/2]\cos[(A-C)/2]}{\sin B}$$

$$= \frac{2\sin 60° \cos\dfrac{A-C}{2}}{\sin 60°} = 2\cos\left(\frac{A-C}{2}\right) \quad \text{[by (i)]}$$

53. $a + b + c = 6\dfrac{(\sin A + \sin B + \sin C)}{3}$

Since $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$,

$\therefore \quad a + b + c = k(\sin A + \sin B + \sin C)$

$\therefore \quad k(\sin A + \sin B + \sin C) = 2(\sin A + \sin B + \sin C)$

$\Rightarrow \quad k = 2$

But $\dfrac{a}{\sin A} = k = 2 \quad \therefore \dfrac{1}{\sin A} = 2 \qquad (\because a = 1)$

$\Rightarrow \sin A = \dfrac{1}{2} \quad \Rightarrow \quad A = \dfrac{\pi}{6}$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. In ΔPQR, the circumradius $PQ = PR = R_1$

$$\therefore \quad \frac{PQ}{2R_1\sin R} = \frac{QR}{2R_1\sin P} = \frac{PR}{2R_1\sin Q}$$

$$\Rightarrow \quad \sin R = \sin Q = \frac{1}{2}$$

$$\Rightarrow \quad \angle R = \angle Q \geq \frac{\pi}{6}$$

$$\therefore \quad \angle P = \frac{2\pi}{3}$$

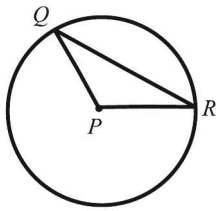

Fig. 28.21

2. $a = \sqrt{3} + 1, b = \sqrt{3} - 1, C = 60°$

By Napier's analogy,

$$\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2}$$

$$= \frac{2}{2\sqrt{3}}\cot 30° = \frac{1}{\sqrt{3}}\cdot\sqrt{3} = 1$$

$$\therefore \quad \frac{A-B}{2} = 45° \quad \Rightarrow \quad A - B = 90°$$

3. Let $E = \dfrac{\sin(A-B)}{\sin(A+B)} = \dfrac{\sin A\cos B - \cos A\sin B}{\sin C}$

$$\Rightarrow \quad E = \frac{a}{c}\cos B - \frac{b}{c}\cos A$$

$$\Rightarrow \quad E = \frac{a}{c}\frac{(a^2 + c^2 - b^2)}{2ac} - \frac{b}{c}\frac{(b^2 + c^2 - a^2)}{2bc}$$

$$\Rightarrow \quad E = \frac{1}{2c^2}[a^2 + c^2 - b^2 - b^2 - c^2 + a^2]$$

$$\therefore \quad E = \frac{a^2 - b^2}{c^2}$$

4. Cosine of the greatest angle

$$= \frac{3^2 + 2^2 - 4^2}{2\cdot 3\cdot 2} = -\frac{3}{12} = -\frac{1}{4}$$

5. The angle opposite to the side of length 14 cm

$$\cos\theta = \frac{36 + 100 - 196}{2\cdot 6\cdot 10} = -\frac{1}{2}$$

$\Rightarrow \quad \theta = 120°$

$\therefore \quad \Delta$ is an obtuse-angled Δ.

6. $a^3\sin(B-C) = ka^2\sin(B+C)\sin(B-C)$

$[\because a = k\sin A = k\sin(180° - \overline{B+C} = k\sin(B+C)]$

$= k^3\sin^2 A(\sin^2 B - \sin^2 C)$

$\therefore \quad \Sigma a^3\sin(B-C) = 0$

7. $a(b\cos C - c\cos B)$

$$= a\left(b\cdot\frac{a^2 + b^2 - c^2}{2ab} - c\cdot\frac{c^2 + a^2 - b^2}{2ca}\right)$$

$$= a\left(\frac{a^2+b^2-c^2}{2a} - \frac{c^2+a^2-b^2}{2a}\right)$$

$$= \frac{1}{2}(a^2+b^2-c^2-c^2-a^2+b^2) = b^2-c^2$$

8. Since $c^2 = a^2 + b^2$,

$\Rightarrow$ the triangle is a right-angled triangle, then $\Delta = \frac{1}{2}ab$

$\Rightarrow \Delta^2 = \frac{a^2b^2}{4}$

Now $4s(s-a)(s-b)(s-c) = 4\Delta^2$

$\Rightarrow 4s(s-a)(s-b)(s-c) = 4\cdot\frac{a^2b^2}{4} = a^2b^2$

9. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a+b+c & b & c \\ b+c+a & c & a \\ c+a+b & a & b \end{vmatrix}$

[Operating $C_1 \to C_1 + C_2 + C_3$]

$$= (a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$$

$$= (a+b+c)\begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-c & b-a \end{vmatrix}$$

[Operating $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_2$]

$= (a+b+c)[(1)(c-b)(b-a) - (a-c)^2]$

$= (a+b+c)(bc - ac - b^2 + ab - a^2 - c^2 + 2ac)$

$= (a+b+c)(bc + ca + ab - a^2 - b^2 - c^2)$

$= -(a^3 + b^3 + c^3 - 3abc)$

$= -[(2R\sin A)^3 + (2R\sin B)^3 + (2R\sin C)^3$

$- 3(2R\sin A)(2R\sin B)(2R\sin C)]$

$= -8R^3(\sin^3 A + \sin^3 B + \sin^3 C - 3\sin A\sin B\sin C)$

$= -8R^3(0) = 0$ [by the given condition]

10. Using sine formula, we have

$$\frac{\sqrt{3}+1}{\sin 105^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 45^\circ}$$

$\Rightarrow 2\sqrt{2} = 2b = \sqrt{2}c$

$\Rightarrow b = \sqrt{2}, c = 2$

Area of $\Delta ABC\ (\Delta) = \frac{1}{2}bc\sin A$

$\Rightarrow \Delta = \frac{1}{2}2\sqrt{2}\sin 105^\circ$

$\therefore \Delta = \frac{\sqrt{3}+1}{2} = \frac{1}{\sqrt{3}-1}$

11. By (m, n) theorem, $(3 + 2)\cot\angle CDA$
$= 2\cot 30^\circ - 3\cot 60^\circ$

$\Rightarrow \cot\angle CDA = \frac{1}{5}\left(2\sqrt{3} - \frac{3}{\sqrt{3}}\right)$

$= \frac{1}{5}\left(2\sqrt{3} - \sqrt{3}\right) = \frac{\sqrt{3}}{5}$

By sine formula, $\frac{AC}{\sin\angle CDA} = \frac{AD}{\sin 60^\circ}$

$\Rightarrow AC = \frac{3\left(\frac{5}{2/7}\right)}{\sqrt{3}/2} = \frac{15}{\sqrt{21}} = \frac{5\sqrt{3}}{\sqrt{7}} = 5\sqrt{\frac{3}{7}}$ cm

12. Let $E = \cos A + \cos B + \cos C$

$\Rightarrow E = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$\Rightarrow E = 1 + 4\sqrt{\frac{(s-b)(s-c)}{bc}}\sqrt{\frac{(s-a)(s-c)}{ac}}$

$\sqrt{\frac{(s-a)(s-b)}{ab}}$

Since A.M. $\geq$ G.M.,

$\Rightarrow \frac{s-b+s-c}{2} \geq \sqrt{(s-b)(s-c)}$

$\Rightarrow \frac{s-a+s-c}{2} \geq \sqrt{(s-a)(s-c)}$

$\Rightarrow \frac{s-a+s-b}{2} \geq \sqrt{(s-a)(s-b)}$

$\therefore E \leq 1 + \frac{4}{abc}\left(\frac{s-b+s-c}{2}\right)\left(\frac{s-a+s-c}{2}\right)$

$\left(\frac{s-a+s-b}{2}\right)$

$\Rightarrow E \leq 1 + \frac{4}{8abc}(abc) = 1 + \frac{1}{2} = \frac{3}{2}$

13. $\frac{a\cos A + b\cos B + c\cos C}{a+b+c}$

$= \frac{2R\sin A\cos A + 2R\sin B\cos B + 2R\sin C\cos C}{2s}$

$= \frac{R}{2s}(\sin 2A + \sin 2B + \sin 2D)$

$$= \frac{R}{2s}(4\sin A\sin B + \sin C)$$

$$= \frac{2R}{s}\cdot\frac{abc}{8R^3} = \frac{abc}{4sR^2}$$

But $R = \dfrac{abc}{4A}$ and $r = \dfrac{\Delta}{s}$

$\therefore$ Given expression $= \dfrac{4\Delta R}{4\cdot\frac{\Delta}{r}\cdot R^2} = \dfrac{r}{R}$

14. Since $a\cos^2\dfrac{C}{2} + c\cos^2\dfrac{A}{2} = \dfrac{3b}{2}$,

$\Rightarrow a\dfrac{s(s-c)}{ab} + c\dfrac{s(s-a)}{bc} = \dfrac{3b}{2}$

$\Rightarrow s(s-c+s-a) = \dfrac{3}{2}b^2$

$\Rightarrow s[2s-(a+c)] = \dfrac{3}{2}b^2$

$\Rightarrow s\cdot b = \dfrac{3}{2}b^2$ (i)

$2s = 3b$

Now, $2s = a+b+c$

$\Rightarrow a+b+c-3b = 0$ [using (i)]

$\Rightarrow a+c = 2b$

$\therefore$ a, b, c are in A.P.

15. Given,

$$\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$$

$$\Rightarrow \frac{\sin A}{\sin C} = \frac{\sin A\cos B - \cos A\sin B}{\sin B\cos C - \cos B\sin C}$$

$$\Rightarrow \frac{a}{c} = \frac{a\cos B - b\cos A}{b\cos C - c\cos B}$$ [using sine rule]

$\Rightarrow ab\cos C - ac\cos B = ac\cos B - bc\cos A$

$\Rightarrow ab\cos C + bc\cos A = 2ac\cos B$

$\Rightarrow \dfrac{a^2+b^2-c^2}{2} + \dfrac{b^2+c^2-a^2}{2} = a^2+c^2-b^2$

$\Rightarrow a^2+c^2 = 2b^2$

$\therefore$ a^2, b^2, c^2 are in A.P.

16. In a triangle ABC, we have $A + B + C = \pi$

$\therefore$ $\cos 2A + \cos 2B + \cos 2C$

$= 2\cos(A+B)\cos(A-B) + 2\cos^2 C - 1$

$= -2\cos C\cos(A-B) + 2\cos^2 C - 1$

$= -2\cos C[\cos(A-B) - \cos C] - 1$

$= -2\cos C[\cos(A-B) + \cos(A+B)] - 1$

$= -2\cos C[2\cos A\cos B] - 1$

$= -1 - 4\cos A\cos B\cos C$

So, Statement II is true.

And $\sin^2 A + \sin^2 B + \sin^2 C = 2$

$\Rightarrow 1-\cos 2A + 1 - \cos 2B + 1 - \cos 2C = 4$

$\Rightarrow \cos 2A + \cos 2B + \cos 2C = -1$

$\Rightarrow 4\cos A\cos B\cos C = 0$ [using Statement II]

$\Rightarrow \cos A = 0$ or $\cos B = 0$ or $\cos C = 0$

$\Rightarrow A = \dfrac{\pi}{2}$ or $B = \dfrac{\pi}{2}$ or $C = \dfrac{\pi}{2}$

So, Statement I is true and Statement II is a correct explanation for Statement I.

17. In any ΔABC, we have

$A + B + C = \pi$

$\Rightarrow \tan(A + B + C) = \tan\pi$

$$\Rightarrow \frac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - \tan A\tan B - \tan B\tan C - \tan C\tan A} = 0$$

$\Rightarrow \tan A + \tan B + \tan C = \tan A\tan B\tan C$

So, Statement II is true.

Now, $\tan A + \tan B + \tan C = \tan A\tan B\tan C$

$$\Rightarrow \tan A = -\frac{\tan B + \tan C}{1 - \tan B\tan C}$$

$$\Rightarrow \frac{\tan B + \tan C}{1 - \tan B\tan C} > 0$$ [$\because A$ is obtuse $\therefore \tan A < 0$]

$\Rightarrow \tan B\tan C < 1$

So, Statement I is true and Statement II is a correct explanation for Statement I.

18. Clearly, Statement II is true.

If the measures of two angles of a triangle are 45° and 60°, then the measure of its third angle is 75°. Therefore, the ratio of the smallest and greatest sides is

$$\sin 45° : \sin 75° = \frac{1}{\sqrt{2}} : \frac{\sqrt{3}+1}{2\sqrt{2}} = 2:\sqrt{3}+1 = \sqrt{3}-1:1$$

Thus, Statement I is true. Also Statement II is a correct explanation for Statement I.

19. Clearly, Statement II is true.

Clearly, $x = 7, 5$ are the roots of the given equation. So, let $a = 7$, $b = 5$ be the lengths of two sides of ΔABC. It is given that the angle C is obtuse. Thus, $\cos C < 0$

$$\Rightarrow \frac{a^2+b^2-c^2}{2ab} < 0 \quad \Rightarrow \quad c^2 > a^2 + b^2$$

$\Rightarrow\ c^2 > 49 + 25 \ \Rightarrow\ c^2 > 74$

So, Statement II is also true and it is a correct explanation for Statement I.

20. In a ΔABC, we have

$$\tan A : \tan B : \tan C = 1 : 2 : 3$$

So, let $\tan A = x$, $\tan B = 2x$, and $\tan C = 3x$

We know that

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$\Rightarrow\ x + 2x + 3x = 6x^3$

$\Rightarrow\ x = 1 \qquad [\because\ x \neq -1, 0]$

$\therefore\ \tan A = 1 \ \Rightarrow\ A = 45°$

So, Statement I is true.

Clearly, Statement II is not true, because $p : q : r = 1 : 2 : 3$ for

$$p = 3,\ q = 6 \text{ and } r = 9, \text{ but } p \neq 1$$

21. $A + B + C = p \Rightarrow A + C = p - B$

$$\Rightarrow\ \frac{A - B + C}{2} = \frac{\pi - 2B}{2} = \frac{\pi}{2} - B$$

$$\therefore\ 2ca\sin\frac{A - B + C}{2} = 2ca\sin\left(\frac{\pi}{2} - B\right)$$

$$= 2ca\cos B = 2ca\frac{c^2 + a^2 - b^2}{2ca}$$

$$= c^2 + a^2 - b^2$$

22. Since $\tan\frac{A}{2} = \frac{5}{6}, \tan\frac{C}{2} = \frac{2}{5}$

Now, $\tan\frac{A}{2}\tan\frac{C}{2} = \frac{5}{6} \times \frac{2}{5}$

$$\Rightarrow\ \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \cdot \sqrt{\frac{(s-a)(s-b)}{s(s-c)}} = \frac{1}{3}$$

$$\Rightarrow\ \frac{s-b}{s} = \frac{1}{3} \ \Rightarrow\ 3s - 3b = s$$

$$\Rightarrow\ 2s = 3b \ \Rightarrow\ a + b + c = 3b$$

$\Rightarrow\ a + c = 2b$

Hence, a, b, c are in A.P.

23. Given $a = 4$, $b = 3$ and $\angle A = 60°$

$$\therefore\ \cos 60° = \frac{c^2 + 9 - 16}{2 \times 3 \times c} \ \Rightarrow\ \frac{1}{2} = \frac{c^2 + 7}{2 \times 3c}$$

$$\Rightarrow\ c^2 - 7 = 3c \ \Rightarrow\ c^2 - 3c - 7 = 0$$

Thus, c is the root of the above equation.

24. Let $a = 3x + 4y$, $b = 4x + 3y$, $c = 5x + 5y$

Cleraly, C is the greatest side.

$\therefore$ The greatest angle C is given by

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$= \frac{(3x + 4y)^2 + (4x + 3y)^2 - (5x + 5y)^2}{2(3x + 4y)(4x + 3y)}$$

$$= \frac{-2xy}{2(12x^2 + 25xy + 12y^2)} < 0$$

$\therefore\ C > 90°$

Hence, the triangle is obtuse-angled.

25. Here $r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c}$

Now, $r_1 > r_2 > r_3$

$$\Rightarrow\ \frac{\Delta}{s-a} > \frac{\Delta}{s-b} > \frac{\Delta}{s-c} \Rightarrow \frac{1}{s-a} > \frac{1}{s-b} > \frac{1}{s-c}$$

$$\Rightarrow\ s - a < s - b < s - c \Rightarrow -a < -b < -c$$

$\Rightarrow\ a > b > c$

26. Given, $AD = 4$ and $BD = DC$

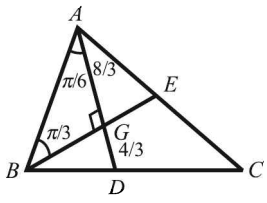

Fig. 28.22

In the ΔABG, $\tan\frac{\pi}{3} = \frac{AG}{BG}$

$$\Rightarrow\ BG = AG\cot\frac{\pi}{3}$$

$$\Rightarrow\ BG = \frac{8}{3} \times \frac{1}{\sqrt{3}} = \frac{8}{3\sqrt{3}}$$

Area of $\Delta ADB = \frac{1}{2} \times BG \times AD$

$$= \frac{1}{2} \times \frac{8}{3\sqrt{3}} \times 4 = \frac{16}{3\sqrt{3}}$$

Since the median divides a triangle into two triangles of equal area, therefore,

$$\text{area of } \Delta ABC = 2 \times \text{area of } \Delta ADB$$

$$= 2 \times \frac{16}{3\sqrt{3}} = \frac{32}{3\sqrt{3}}$$

27. $AB = a$

$ON \perp AB$ and $AN = BN$

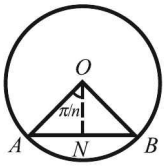

Fig. 28.23

In the ΔAON, $\tan\frac{\pi}{n} = \frac{AN}{ON}$

$$\Rightarrow\ \cos C = -\frac{1}{2} \ \Rightarrow\ \angle C = 120°$$

and $\sin\frac{\pi}{n} = \frac{AN}{OA}$

$$\Rightarrow\ OA = AN\ \text{cosec}\frac{\pi}{n} = \frac{a}{2}\text{cosec}\frac{\pi}{n}$$

Sum of the radii = $ON + OA$

$$= \frac{a}{2}\cot\frac{\pi}{n} + \frac{a}{2}\operatorname{cosec}\frac{\pi}{n} = \frac{a}{2}\left[\frac{\cos\frac{\pi}{n}}{\sin\frac{\pi}{n}} + \frac{1}{\sin\frac{\pi}{n}}\right]$$

$$= \frac{a}{2}\left[\frac{1+\cos\frac{\pi}{n}}{\sin\frac{\pi}{n}}\right] = \frac{a}{2}\left[\frac{1+2\cos^2\frac{\pi}{2n}-1}{2\sin\frac{\pi}{2n}\cos\frac{\pi}{2n}}\right]$$

$$= \frac{a}{2}\cot\frac{\pi}{2n}$$

28. Given, $a\cos^2\left(\frac{C}{2}\right) + c\cos^2\left(\frac{A}{2}\right) = \frac{3b}{2}$

$$\Rightarrow a\left[\frac{s(s-c)}{ab}\right] + c\left[\frac{s(s-a)}{bc}\right] = \frac{3b}{2}$$

$$\Rightarrow \frac{s(s-c+s-a)}{b} = \frac{3b}{2} \quad \Rightarrow 2s(2s-c-a) = 3b^2$$

$$\Rightarrow (a+b+c)b = 3b^2 \quad \Rightarrow 2b = a+c$$

29. Let $a = \sin\alpha, b = \cos\alpha, c = \sqrt{1+\sin\alpha\cos\alpha}$

Here, we see that the greatest side is c.

$$\therefore \cos C = \frac{a^2+b^2-c^2}{2ab}$$

$$\Rightarrow \cos C = \frac{\sin^2\alpha + \cos^2\alpha - 1 - \sin\alpha\cos\alpha}{2\sin\alpha\cos\alpha}$$

$$\Rightarrow \cos C = -\frac{\sin\alpha\cos\alpha}{2\sin\alpha\cos\alpha}$$

$$\Rightarrow \cos C = -\frac{1}{2} \Rightarrow \angle C = 120°$$

30. We know that $\frac{c}{\sin C} = 2R$

$$\Rightarrow c = 2R \qquad \text{(i)}$$

and $\tan\frac{C}{2} = \frac{r}{s-c} \Rightarrow \tan\frac{\pi}{4} = \frac{r}{s-c}$ $(\because \angle C = 90°)$

$$\therefore r = s-c$$

$$\Rightarrow r = \frac{a+b+c}{2} - c$$

$$\Rightarrow 2r = a+b-c \qquad \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2(r+R) = a+b$$

31. In the ΔBAD,

$$\cos(90°-B) = \frac{AD}{c} \Rightarrow AD = c\sin B$$

Similarly, $BE = a\sin C$ and $CF = b\sin A$

Since AD, BE, CF are in H.P.,

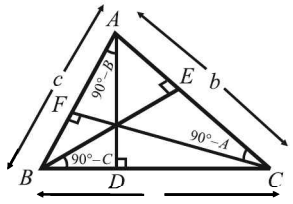

Fig. 28.24

$\therefore$ $c\sin B, a\sin C, b\sin A$ are in H.P.

$$\Rightarrow \frac{1}{\sin C\sin B}, \frac{1}{\sin A\sin C}, \frac{1}{\sin B\sin A} \text{ are in A.P.}$$

$\Rightarrow$ $\sin A, \sin B, \sin C$ are in A.P.

32. Since $\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of the equation

$ax^2 + bx + c = 0$,

$$\therefore \tan\frac{P}{2} + \tan\frac{Q}{2} = -\frac{b}{a} \text{ and } \tan\frac{P}{2}\tan\frac{Q}{2} = \frac{c}{a}$$

Also, $\frac{P}{2} + \frac{Q}{2} + \frac{R}{2} = \frac{\pi}{2}$

(as P, Q, R are the angles of the triangle)

$$\Rightarrow \frac{P+Q}{2} = \frac{\pi}{2} - \frac{R}{2} \Rightarrow \frac{P+Q}{2} = \frac{\pi}{4}$$

Now, $\tan\left(\frac{P}{2}+\frac{Q}{2}\right) = 1 \Rightarrow \frac{\tan\frac{P}{2}+\tan\frac{Q}{2}}{1-\tan\frac{P}{2}\tan\frac{Q}{2}} = 1$

$$\Rightarrow \frac{-\frac{b}{a}}{1-\frac{c}{a}} = 1 \Rightarrow -\frac{b}{a} = 1-\frac{c}{a}$$

$$\Rightarrow -b = a-c \Rightarrow c = a+b$$

33. Area $= \frac{1}{2} \times \text{base} \times \text{altitude}$

$$= \frac{1}{2} \times (2x\cos\theta) \times (x\sin\theta)$$

$$= \frac{1}{2}x^2\sin 2\theta$$

(Since the maximum value of $\sin 2\theta$ is 1)

$\therefore$ Maximum area $= \frac{1}{2}x^2$

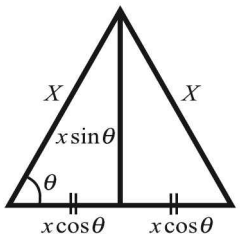

Fig. 28.25

34.

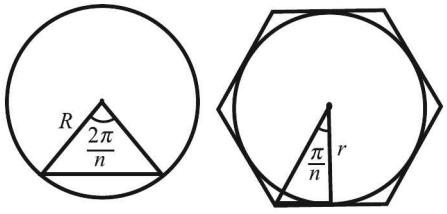

Fig. 28.26

$\because \quad \frac{a}{2R} = \sin\frac{\pi}{n}$

and $\frac{a}{2r} = \tan\frac{\pi}{n} \quad \therefore \frac{r}{R} = \cos\frac{\pi}{n}$

For $n = 3$, we get $\frac{r}{R} = \frac{1}{2}$

For $n = 4$, we get $\frac{r}{R} = \frac{1}{\sqrt{2}}$

For $n = 6$, we get $\frac{r}{R} = \frac{\sqrt{3}}{2}$

35. Let $a : b : c = 1 : \sqrt{3} : 2 \Rightarrow c^2 = a^2 + b^2$

$\therefore$ The triangle is right-angled at C

or $\angle C = 90°$ and $\frac{a}{b} = \frac{1}{\sqrt{3}}$

In the ΔBAC, $\tan A = \frac{a}{b} = \frac{1}{\sqrt{3}}$

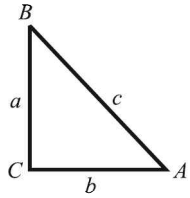

Fig. 28.27

$\Rightarrow \quad \angle A = 30°$

and $\angle B = 60°$ $(\because \angle A + \angle B = 90°)$

$\therefore$ Ratio of angles, $\angle A : \angle B : \angle C = 30° : 60° : 90°$

$\Rightarrow A : B : C = 1 : 2 : 3$

36. Let a, b, c are the sides of ΔABC.

Now,

$$\frac{b+c}{a} = \frac{k(\sin B + \sin C)}{k \sin A}$$

$$= \frac{2\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}}$$

$$\Rightarrow \frac{b+c}{a} = \frac{\cos\left(\frac{B-C}{2}\right)}{\sin\frac{A}{2}}. \text{ Also, } \frac{b-c}{a} = \frac{\sin\left(\frac{B-C}{2}\right)}{\cos\frac{A}{2}}$$

Therefore, option (d) is the answer.

37. Let $AB = AC = a$ and $\angle A = 120°$

$\therefore$ Area of triangle $= \frac{1}{2}a^2 \sin 120°$

where $a = AD + BD$

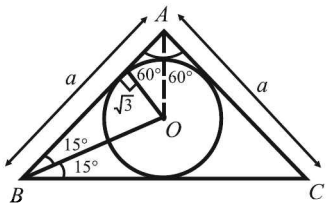

Fig. 28.28

$$= \sqrt{3}\tan 30° + \sqrt{3}\cot 15°$$

$$= 1 + \frac{\sqrt{3}}{\tan(45° - 15°)}$$

$$= 1 + \sqrt{3}\left(\frac{1 + \tan 45° \tan 30°}{\tan 45° - \tan 30°}\right)$$

$$= 1 + \sqrt{3}\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$$

$\therefore \quad a = 4 + 2\sqrt{3} \quad \Rightarrow$ Area of the triangle

$$= \frac{1}{2}(4 + 2\sqrt{3})^2\left(\frac{\sqrt{3}}{2}\right) = 12 + 7\sqrt{3}$$

38. Since A, B and C are in A.P.,

$\Rightarrow \quad 2B = A + C \Rightarrow \angle B = 60°$

$$\therefore \quad \frac{a}{c}(2\sin C \cos C) + \frac{c}{a}(2\sin A \cos A)$$

$$= 2k(a\cos C + c\cos A) = 2k(b)$$

$$= 2\sin B = \sqrt{3} \qquad [\text{using } b = a\cos C + c\cos A]$$

39. Using $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$, we get

$$\frac{\sqrt{3}}{2} = \frac{(x^2 + x + 1)^2 + (x^2 - 1) + (2x + 1)^2}{2(x^2 + x + 1)(x^2 + 1)}$$

$$\Rightarrow (x+2)(x+1)(x-1)x + (x^2-1)^2$$

$$= \sqrt{3}(x^2 + x + 1)(x^2 - 1)$$

$$\Rightarrow x^2 + 2x + (x^2 - 1) = \sqrt{3}(x^2 + x + 1)$$

$$\Rightarrow (2 - \sqrt{3})x^2 + (2 - \sqrt{3})x - (\sqrt{3} + 1) = 0$$

$$\Rightarrow x = -(2 + \sqrt{3}) \text{ and } x = 1 + \sqrt{3}$$

But, $x = -(2 + \sqrt{3}) \Rightarrow c$ is negative

$\therefore \quad x = 1 + \sqrt{3}$ is the only solution.

40. Given, angles A, B, C of ΔABC are in A.P. with d (common difference) 15°

$\therefore \quad \angle B = \angle A + 15°$ and $\angle C = \angle A + 30°$

Also, $\angle A + \angle B + \angle C = 180°$

$\Rightarrow \quad \angle A + \angle A + 15° + \angle A + 30° = 180°$

$\Rightarrow \quad \angle A = 45°$

$\therefore \quad \angle A = 45° + 15° = 60°$

41. Given, the vertex angles are A, B, C and side is BC.

$\therefore$ Area $= \frac{1}{2}a(k \sin C)\sin B$

$$= \frac{1}{2} \cdot \frac{a^2 k \sin C \sin B}{a}$$

$$= \frac{1}{2} \cdot \frac{a^2 k \sin C \sin B}{k \sin A}$$

$$= \frac{1}{2} a^2 \cdot \frac{\sin B \sin C}{\sin A}$$

42. $\sum a^3 \cos(B-C)$

$= \sum k^3 \sin^3 A \cos(B-C)$ $(\because\ a = k \sin A)$

$= k^3 \sum \sin^2 A \sin(B+C)\cos(B-C)$

$= \frac{k^3}{2} \sum \sin^2 A(\sin 2B + \sin 2C)$

$= \frac{k^3}{2}[\sin^2 A(\sin 2B + \sin 2C)$

$+ \sin^2 B(\sin 2C + \sin 2A) + \sin^2 C(\sin 2A + \sin 2B)]$

$= k^3 \sum[\sin^2 A(\sin B \cos B) + \sin^2 B(\sin A \cos A)]$

$= k^3 \sum \sin A \sin B \sin(A+B)$

$= k^3[\sin A \sin B \sin C + \sin B \sin C \sin A$

$+ \sin C \sin A \sin B]$

$= 3(k \sin A)(k \sin B)(k \sin C) = 3abc$

43. Since $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$,

$\Rightarrow\ b^2 - 2bc \cos A + (c^2 - a^2) = 0$

It is given that b_1 and b_2 are the roots of this equation.

Therefore, $b_1 + b_2 = 2c \cos A$ and $b_1 b_2 = c^2 - a^2$

$\Rightarrow\ 3b_1 = 2c \cos A$ and $2b_1^2 = c^2 - a^2$ $(\because\ b_2 = 2b_2)$

$\Rightarrow\ 2\left(\frac{2c}{3}\cos A\right)^2 = c^2 - a^2$

$\Rightarrow\ 8c^2(1 - \sin^2 A) = 9c^2 - 9a^2 \Rightarrow \sin A = \sqrt{\frac{9a^2 - c^2}{8c^2}}$

44. We have $\frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow \sin B = \frac{b \sin C}{c}$

$\Rightarrow\ \sin B = \frac{2 \sin 60°}{\sqrt{6}}$ $(\because\ b = 2, c = \sqrt{6}, \angle C = 60°)$

$= \frac{2}{\sqrt{6}} \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{2}}$

$\Rightarrow\ \angle B = 45°$ $(\because\ \angle B \neq 135°)$

$\Rightarrow\ A = 180° - \angle(B + C) = 75°$

Now, $\frac{\sin A}{a} = \frac{\sin B}{b}$

$\Rightarrow\ a = \frac{b \sin A}{\sin B} = \frac{2 \sin 75°}{\sin 45°} = \sqrt{3} + 1$

45. Given, $\angle A = \frac{\pi}{2}$, then $a^2 = b^2 + c^2 = 4^2 + 3^2 = 25$ and

$\frac{a}{\sin A} = 2R \Rightarrow a = 2R$ and $a = 5$

Also, $r = \frac{\Delta}{s} = \frac{bc}{a+b+c}$ $\left(\because \Delta = \frac{bc}{2}\right)$

$\therefore\ \frac{R}{r} = \frac{a(a+b+c)}{2bc} = \frac{5 \times 12}{2 \times 4 \times 3} = \frac{5}{2}$

46. $\frac{\tan\frac{A}{2} - \tan\frac{B}{2}}{\tan\frac{A}{2} + \tan\frac{B}{2}} = \frac{\sqrt{\frac{(s-b)(s-c)}{s(s-a)}} - \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}}{\sqrt{\frac{(s-b)(s-c)}{s(s-a)}} + \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}}$

$= \frac{(s-b)\sqrt{s(s-c)} - (s-a)\sqrt{s(s-c)}}{(s-b)\sqrt{s(s-c)} + (s-a)\sqrt{s(s-c)}}$

$= \frac{\sqrt{s(s-c)}(s-b-s+a)}{\sqrt{s(s-c)}(s-b+s-a)} = \frac{a-b}{c}$

47. Here, $\frac{1}{\sin^2\frac{A}{2}}, \frac{1}{\sin^2\frac{B}{2}}, \frac{1}{\sin^2\frac{C}{2}}$ are in A.P.

$\Rightarrow\ \frac{1}{\sin^2\frac{C}{2}} - \frac{1}{\sin^2\frac{B}{2}} = \frac{1}{\sin^2\frac{B}{2}} - \frac{1}{\sin^2\frac{A}{2}}$

$\Rightarrow\ \frac{ab}{(s-a)(s-b)} - \frac{ac}{(s-a)(s-c)}$

$= \frac{ac}{(s-a)(s-c)} - \frac{bc}{(s-b)(s-c)}$

$\Rightarrow\ \left(\frac{a}{s-a}\right)\left(\frac{b(s-c) - c(s-b)}{(s-a)(s-c)}\right)$

$= \left(\frac{c}{s-c}\right)\left(\frac{a(s-b) - b(s-a)}{(s-a)(s-b)}\right)$

$\Rightarrow\ ab + bc = 2ac \Rightarrow \frac{1}{c} + \frac{1}{a} = \frac{2}{b}$

Hence, a, b, c are in H.P.

48. Using sine rule in the ΔACD, we have

$$\frac{\sin(90^\circ - C)}{b} = \frac{\sin 90^\circ}{\frac{a}{2}}$$

$$\Rightarrow \quad \cos C = \frac{2b}{a}$$

and in the ΔABD, we get

$$\frac{\sin(A - 90^\circ)}{\frac{a}{2}} = \frac{(\sin 90^\circ + C)}{c}$$

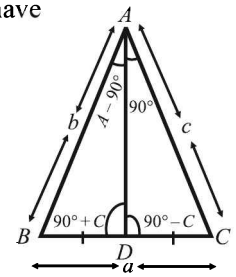

Fig. 28.29

$$\Rightarrow \quad -\frac{\cos A}{\frac{a}{2}} = \frac{\cos C}{c} \quad \Rightarrow \quad \cos A = -\frac{b}{c}$$

$$\Rightarrow \quad \frac{b^2 + c^2 - a^2}{2bc} = -\frac{b}{c} \quad \Rightarrow \quad c^2 - a^2 = 3b^2$$

$$\therefore \quad \cos A \cos C = -\frac{2b^2}{ac} = \frac{2}{3ac}(c^2 - a^2)$$

49. Given that $A = \angle 60^\circ$, $AB = c = 1$, $AC = b = 2$

$$\therefore \quad \frac{\sin 60^\circ}{a} = \frac{\sin B}{2} = \frac{\sin C}{1}$$

The last two terms will be equal when $\angle B = 90^\circ$ and $\angle C = 30^\circ$, which is also satisfied by the fact that the sum of the three angles of a triangle is 180°.
Hence, the largest angle is 90°.

50. $\Delta = 2bc - (b^2 + c^2 - a^2) = 2bc(1 - \cos A)$

$$= 2bc \cdot 2\sin^2 \frac{A}{2} \qquad \text{(i)}$$

But $\Delta = \frac{1}{2}bc \sin A = \frac{1}{2}bc \cdot 2\sin\frac{A}{2}\cos\frac{A}{2}$

$$\Rightarrow \quad \Delta = bc \sin\frac{A}{2}\cos\frac{A}{2} \qquad \text{(ii)}$$

On dividing Eq. (ii) by Eq. (i), we get

$$\tan\frac{A}{2} = \frac{1}{4}$$

$$\therefore \quad \tan A = \frac{2\tan\frac{A}{2}}{1 - \tan^2\frac{A}{2}} = \frac{\frac{1}{2}}{1 - \frac{1}{16}} = \frac{8}{15}$$

51. $$\frac{\cot\frac{A}{2}\cot\frac{B}{2} - 1}{\cot\frac{A}{2}\cot\frac{B}{2}} = 1 - \tan\frac{A}{2}\tan\frac{B}{2}$$

$$= 1 - \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}\sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$$

$$= 1 - \frac{s-c}{s} = \frac{c}{s} = \frac{c}{a+b+c}$$

52. $$2ac\sin\left(\frac{A - B + C}{2}\right)$$

$$= 2ac\sin\left(\frac{180^\circ - 2B}{2}\right)$$

$$= 2ac\sin(90^\circ - B) = 2ac\cos B = a^2 + c^2 - b^2$$

53. Given, $\frac{\sin A}{1/4} = \frac{\sin B}{1/4} = \frac{\cos C}{1/3}$

$$\therefore \quad \frac{\sin A}{3} = \frac{\sin B}{3} = \frac{\cos C}{4}$$

Here, $a = 3k$, $b = 3k$ and $c = 4k$, where k is a proportionality constant.

$$\therefore \quad \cos C = \frac{9k^2 + 9k^2 - 16k^2}{2 \times 3k \times 3k} = \frac{1}{9}$$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. In a ΔABC, $\frac{c+b}{c-b}$, $\tan\frac{A}{2}$ is equal to

(a) $\tan\left(\frac{A}{2} + B\right)$ (b) $\cot\left(\frac{A}{2} + B\right)$

(c) $\tan\left(A + \frac{B}{2}\right)$ (d) none of these

2. If the area of a ΔABC is λ, then $a^2 \sin B + b^2 \sin 2A$ is equal to

(a) 2λ (b) λ

(c) 4λ (d) none of these

3. If k is the perimeter of the ΔABC, then $b\cos^2\frac{C}{2} + c\cos^2\frac{B}{2}$ is equal to

(a) k (b) $2k$

(c) $k/2$ (d) none of these

4. If R denotes the circumradius, then in a ΔABC, $\frac{b^2 - c^2}{2aR}$ is equal to

(a) $\cos(B - C)$ (b) $\sin(B - C)$

(c) $\cos B - \cos C$ (d) none of these

5. In a ΔABC, $A : B : C = 3 : 5 : 4$. Then $a+b+c\sqrt{2}$ is equal to
(a) $2b$ (b) $2c$
(c) $3b$ (d) $3a$

6. If in a ΔABC, $a^2 \cos^2 A = b^2 + c^2$, then
(a) $A < \frac{\pi}{4}$ (b) $\frac{\pi}{4} < A < \frac{\pi}{2}$
(c) $A > \frac{\pi}{2}$ (d) $A = \frac{\pi}{2}$

7. In a ΔABC, $\tan\frac{A}{2}$ and $\tan\frac{B}{2}$ satisfy $6x^2 - 5x + 1 = 0$. Then
(a) $a^2 + b^2 > c^2$ (b) $a^2 - b^2 = c^2$
(c) $a^2 + b^2 = c^2$ (d) none of these

8. In a ΔABC, if $\tan\frac{A}{2} = \frac{5}{6}$ and $\tan\frac{B}{2} = \frac{20}{37}$, then
(a) $2a = b + c$ (b) $a > b > c$
(c) $2c = a + b$ (d) none of these

9. In a ΔABC, $a = 2b$ and $|A - B| = \frac{\pi}{3}$. The measure of $\angle C$ is
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{6}$ (d) none of these

10. If in a ΔABC, $AC = 12$, $BC = 13$ and $AB = 5$, then the distance of A from BC is
(a) $\frac{25}{13}$ (b) $\frac{60}{13}$
(c) $\frac{65}{12}$ (d) none of these

11. In a ΔABC, $\cos A = \frac{3}{5}$ and $\cos B = \frac{5}{13}$. The value of $\cos C$ can be
(a) $\frac{7}{13}$ (b) $\frac{12}{13}$
(c) $\frac{33}{65}$ (d) none of these

12. Two angles of a triangle are $\pi/6$ and $\pi/4$, and the length of the included side is $(\sqrt{3}+1)$ cm. The area of the triangle is
(a) $\frac{\sqrt{3}-1}{2}$ cm^2 (b) $\frac{\sqrt{3}}{2}$ cm^2
(c) $\frac{\sqrt{3}+1}{2}$ cm^2 (d) none of these

13. If in a ΔABC, $\sin^3 A + \sin^3 B + \sin^3 C = 3\sin A \cdot \sin B \cdot \sin C$, then the value of the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is
(a) 0
(b) $(a + b + c)^3$
(c) $(a + b + c)(ab + bc + ca)$
(d) none of these

14. If in a ΔABC, $A = 90°$, then $\tan^{-1}\frac{b}{a+c} + \tan^{-1}\frac{c}{a+b}$ is equal to
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) $\tan^{-1}\frac{a}{b+c}$ (d) none of these

15. If in a ΔABC, the values of $\cot A$, $\cot B$, $\cot C$ are in A.P., then
(a) a, b, c are in A.P.
(b) a^2, b^2, c^2 are in A.P.
(c) $\cos A, \cos B, \cos C$ are in A.P.
(d) none of these

16. If in a ΔABC, $\frac{a}{\cos A} = \frac{b}{\cos B}$, then
(a) $2\sin A \sin B \sin C = 1$
(b) $\sin^2 A + \sin^2 B = \sin^2 C$
(c) $2\sin A \cos B = \sin C$
(d) none of these

17. If in a ΔABC, $2\cos A \sin C = \sin B$, then the triangle is
(a) equilateral (b) isosceles
(c) right-angled (d) none of these

18. If in a ΔABC, $3a = b + c$, then $\tan\frac{B}{2} \cdot \tan\frac{C}{2}$ is equal to
(a) $\tan\frac{A}{2}$ (b) 1
(c) 2 (d) none of these

19. If in a ΔABC, $c\cos^2\frac{A}{2} + a\cos^2\frac{C}{2} = \frac{3b}{2}$, then a, b, c are in
(a) G.P. (b) H.P.
(c) A.P. (d) none of these

20. The sides of a triangle are in A.P. and its area is $\frac{3}{5}\times$ (area of an equilateral triangle of the same perimeter). Then the ratio of the sides is
(a) 1 : 2 : 3 (b) 3 : 5 : 7
(c) 1 : 3 : 5 (d) none of these

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. If the base angles of a triangle are $22\frac{1}{2}^\circ$ and $112\frac{1}{2}^\circ$, then the altitude of the triangle is equal to

(a) the base (b) $\frac{1}{3}$rd of the base

(c) $\frac{1}{2}$ of the base (d) $\frac{1}{4}$th of the base

2. If the sides a, b, c of a triangle ABC are the roots of $x^3 - 11x^2 + 38x - 40 = 0$, then $\Sigma\frac{\cos A}{a} =$

(a) 3/4 (b) 1

(c) 9/16 (d) none of these

3. If $x, y > 0$, then the triangle whose sides are given by $3x + 4y$, $4x + 3y$ and $5x + 5y$ units is

(a) right-angled (b) obtuse-angled

(c) acute-angled (d) none of these

4. If in a triangle ABC, $\frac{a\sin B + b\sin A}{\sqrt{\sin A \sin B}} = 4$, $\angle C = \frac{\pi}{3}$, then $a^2 + b^2 - c^2 =$

(a) 4 (b) 6

(c) 8 (d) 10

5. If $A = 60°$, $\frac{b}{c+a} + \frac{c}{a+b} =$

(a) 0 (b) 1

(c) s (d) S

6. In a triangle, $\cot A : \cot B : \cot C = 30 : 19 : 6$, then $a : b : c$

(a) 5 : 6 : 7 (b) 6 : 7 : 5

(c) 7 : 6 : 5 (d) none of these

7. If $(a-b)(s-c) = (b-c)(s-a)$, then r_1, r_2, r_3 are in

(a) H.P. (b) G.P.

(c) A.P. (d) none of these

8. If twice the square of the diameter of a circle is equal to the sum of the squares of the sides of the inscribed triangle ABC, then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to

(a) 2 (b) 3

(c) 4 (d) 1

9. In a triangle ABC, if $\frac{a}{1} = \frac{b}{\sqrt{3}} = \frac{c}{2}$, then

(a) $A + B - C = 90°$

(b) the triangle is acute-angled

(c) A, B, C are in A.P.

(d) the triangle is abtuse-angled

10. The expression $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ is equal to

(a) $\cos^2 A$ (b) $\sin^2 A$

(c) $\cos A \cos B \cos C$ (d) none of these

11. In a triangle ABC, $(a+b+c)(b+c-a) = \lambda\, bc$ if

(a) $\lambda > d$ (b) $\lambda > 0$

(c) $0 < \lambda < 4$ (d) $\lambda > 4$

12. In a triangle ABC, $2ac \sin\frac{(A-B+C)}{2}$ is equal to

(a) $a^2 + b^2 - c^2$ (b) $c^2 - a^2 - b^2$

(c) $a^2 - a^2 - a^2$ (d) $c^2 + a^2 - b^2$

13. In any triangle ABC, $b^2 \sin 2C + c^2 \sin 2C + c^2 \sin 2B$ is equal to

(a) Δ (b) 2Δ

(c) 3Δ (d) 4Δ

14. If a, b, A are given in a triangle and c_1 and c_2 are two possible values of the third side such that $c_1^2 + c_1c_2 + c_2^2 = a^2$, then A is equal to

(a) 30° (b) 60°

(c) 90° (d) 145°

15. In a triangle ABC, if $s - a$, $s - b$, $s - c$ are in G.P., then $\frac{\sin^2 A + \sin^2 C}{\sin A + \sin C} =$

(a) $\sin B$ (b) $\cos B$

(c) $\sin[(a + C)/2]$ (d) $\sin[(A - C)/2]$

16. If $\cos A = \frac{\sin B}{2\sin C}$, then ΔABC is

(a) equilateral (b) isosceles

(c) right-angled (d) none of these

17. If in a triangle ABC, $\frac{\cos A}{a} = \frac{\tan C}{c}$, then $\sin(B + C)$ is equal to

(a) $\cos B \cos C$ (b) $\cos A \cos C$

(c) $\cos A \cos B$ (d) $\sin B \sin C$

18. In a triangle ABC, $1 - \tan(A/2)\tan(B/2)$ is equal to

(a) $\frac{2a}{b+c-a}$ (b) $\frac{2b}{c+a-b}$

(c) $\frac{2c}{a+b-c}$ (d) $\frac{2c}{a+b+c}$

19. In a triangle ABC, if $\tan(A/2) = p$ and $\tan(B/2) = q$, then $\dfrac{2(p+q)(1-pq)}{(1+p^2)(1+q^2)}$ is equal to

(a) $\sin A$ (b) $\sin B$

(c) $\sin C$ (d) $\sin A + \sin B$

20. If R is the circumradius of a triangle ABC, then the area of its pedal triangle is

(a) $(1/2)R^2 \sin A \sin B \sin C$

(b) $(1/2)R^2 \sin 2A \sin 2B \sin 2C$

(c) $(1/2)R^2 \cos A \cos B \cos C$

(d) $(1/2)R^2 \cos 2A \cos 2B \cos 2C$

ANSWERS

Level 1

1. (a)	**2.** (c)	**3.** (c)	**4.** (b)	**5.** (c)	**6.** (c)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (b)
11. (c)	**12.** (c)	**13.** (a)	**14.** (a)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (c)	**20.** (b)

Level 2

1. (c)	**2.** (c)	**3.** (b)	**4.** (a)	**5.** (b)	**6.** (a)	**7.** (c)	**8.** (a)	**9.** (c)	**10.** (b)
11. (c)	**12.** (d)	**13.** (d)	**14.** (b)	**15.** (a)	**16.** (b)	**17.** (b)	**18.** (d)	**19.** (c)	**20.** (b)

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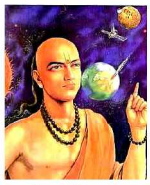


ARYA BHATT

Inverse Trigonometric Functions

- Inverse Function
- Inverse Trigonometric Function
- Domain, Range and Principal Values
- Graph of Inverse Circular Functions
- Properties of Inverse Circular Functions
- Some Other Special Properties

☞ INVERSE FUNCTION

Let $f: [a, b] \to [c, d]$ be a bijection [one-one onto function]. Then, the inverse function f^{-1} exists as $f^{-1}\ [c, d] \to [a, b]$. If $x \in [a, b]$ and $y \in [c, d]$ such that $y = f(x)$, then $x = f^{-1}(y)$.

☞ INVERSE TRIGONOMETRIC FUNCTION

If $\sin\theta = x$, then $\theta = \sin^{-1} x$ (read as sine inverse x). Thus, we see that '$\sin^{-1} x$' is the symbol which denotes an angle or a number. Thus,

$$\sin\frac{\pi}{6} = \frac{1}{2} \quad \Rightarrow \quad \frac{\pi}{6} = \sin^{-1}\left(\frac{1}{2}\right)$$

The expressions $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, etc. are called inverse trigonometric functions.

1. The symbol 'arc sin x' is also used for $\sin^{-1} x$.
2. $\sin^{-1} x$ and $(\sin x)^{-1}$ have different meanings, $(\sin x)^{-1} = \dfrac{1}{\sin x}$.

☞ DOMAIN, RANGE AND PRINCIPAL VALUES OF INVERSE CIRCULAR FUNCTIONS

Function	*Domain*	*Range*	*Principal value branch*
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$-\frac{\pi}{2} \le y \le \frac{\pi}{2}$, where $y = \sin^{-1} x$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$	$0 \le y \le \pi$, where $y = \cos^{-1} x$

$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$-\frac{\pi}{2} < y < \frac{\pi}{2}$, where $y = \tan^{-1} x$
$\text{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$	$-\frac{\pi}{2} \le y \le \frac{\pi}{2}$, $y \ne 0$, where $y = \text{cosec}^{-1} x$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$	$0 \le y \le \pi$, $y \ne \left\{\frac{\pi}{2}\right\}$, where $y = \sec^{-1} x$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$	$0 < y < \pi$, where $y = \cot^{-1} x$

☞ GRAPH OF INVERSE CIRCULAR FUNCTIONS

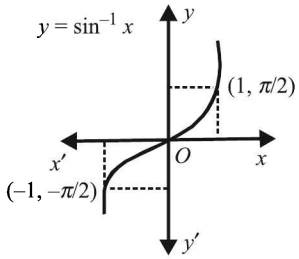


Fig. 29.1

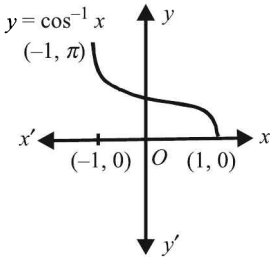


Fig. 29.2

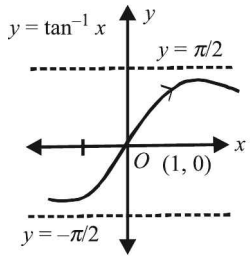


Fig. 29.3

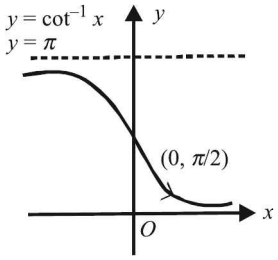


Fig. 29.4

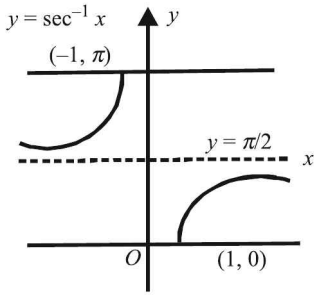


Fig. 29.5

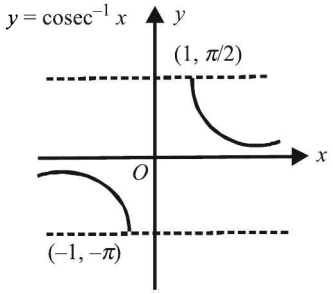


Fig. 29.6

☞ PROPERTIES OF INVERSE CIRCULAR FUNCTIONS

1. $\sin^{-1}(\sin\theta) = \theta$, $\quad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$
2. $\cos^{-1}(\cos\theta) = \theta$, $\quad 0 \le \theta \le \pi$
3. $\tan^{-1}(\tan\theta) = \theta$, $\quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$
4. $\cot^{-1}(\cot\theta) = \theta$, $\quad 0 < \theta < \pi$
5. $\sec^{-1}(\sec\theta) = \theta$, $\quad 0 \le \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \le \pi$
6. $\text{cosec}^{-1}(\text{cosec}\,\theta) = \theta$, $\quad -\frac{\pi}{2} \le \theta < 0$ or $0 < \theta \le \frac{\pi}{2}$
7. $\sin(\sin^{-1} x) = x$, $\quad -1 \le x \le 1$
8. $\cos(\cos^{-1} x) = x$, $\quad -1 \le x \le 1$
9. $\tan(\tan^{-1} x) = x$, $\quad -\infty < x < \infty$
10. $\cot(\cot^{-1} x) = x$, $\quad -\infty < x < \infty$
11. $\sec(\sec^{-1} x) = x$, $\quad -\infty < x \le -1$ or $1 \le x < \infty$
12. $\text{cosec}(\text{cosec}^{-1} x) = x$, $\quad -\infty < x \le -1$ or $1 \le x < \infty$
13. $\sin^{-1}(-x) = -\sin^{-1} x$
14. $\cos^{-1}(-x) = \pi - \cos^{-1} x$
15. $\tan^{-1}(-x) = -\tan^{-1} x$
16. $\cot^{-1}(-x) = \pi - \cot^{-1} x$
17. $\sec^{-1}(-x) = \pi - \sec^{-1} x$
18. $\text{cosec}^{-1}(-x) = -\text{cosec}^{-1} x$
19. $\sin^{-1}\frac{1}{x} = \text{cosec}^{-1} x$
20. $\cos^{-1}\frac{1}{x} = \sec^{-1} x$
21. $\tan^{-1}\frac{1}{x} = \cot^{-1} x$
22. $\text{cosec}^{-1}\frac{1}{x} = \sin^{-1} x$
23. $\sec^{-1}\frac{1}{x} = \cos^{-1} x$

24. $\cot^{-1}\frac{1}{x}=\tan^{-1}x$

25. $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$, for all $x\in[-1,1]$

26. $\tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}$, for all $x\in\mathbb{R}$

27. $\sec^{-1}x+\text{cosec}^{-1}x=\frac{\pi}{2}$, for all $x\in(-\infty,-1]\cup[1,\infty)$

28. $\sin^{-1}x+\sin^{-1}y=\sin^{-1}\left[x\sqrt{1-y^2}+y\sqrt{1-x^2}\right]$
if $-1\le x,y\le 1$

29. $\sin^{-1}x-\sin^{-1}y=\sin^{-1}\left[x\sqrt{1-y^2}-y\sqrt{1-x^2}\right]$
if $-1\le x,y\le 1$

30. $\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left[xy-\sqrt{1-x^2}\sqrt{1-y^2}\right]$
if $x+y\ge 0$

31. $\cos^{-1}x-\cos^{-1}y=\cos^{-1}\left[xy+\sqrt{1-x^2}\sqrt{1-y^2}\right]$
if $x-y\ge 0$

32. $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1-xy}\right)$
if $x>0,\ y>0,\ xy<1$

33. $\tan^{-1}x+\tan^{-1}y=\pi+\tan^{-1}\left(\frac{x+y}{1-xy}\right)$
if $x>0,\ y>0,\ xy>1$

34. $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1-xy}\right)-\pi$
if $x<0,\ y<0,\ xy>1$

35. $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\left(\frac{x-y}{1+xy}\right)$
if $x>0,\ y<0,\ xy>-1$

36. $\tan^{-1}x-\tan^{-1}y=\pi+\tan^{-1}\left(\frac{x-y}{1+xy}\right)$
if $x>0,\ y<0,\ xy<-1$

37. $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\left(\frac{x-y}{1+xy}\right)-\pi$
if $x<0,\ y>0,\ xy<-1$

38. $2\tan^{-1}x=\tan^{-1}\left(\frac{2x}{1-x^2}\right)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

39. $2\sin^{-1}x=\sin^{-1}(2x\sqrt{1-x^2})$

40. $2\cos^{-1}x=\cos^{-1}(2x^2-1)$

41. $3\sin^{-1}x=\sin^{-1}(3x-4x^3)$

42. $3\cos^{-1}x=\cos^{-1}(4x^3-3x)$

43. $3\tan^{-1}x=\tan^{-1}\left(\frac{3x-x^2}{1-3x^2}\right)$

☞ SOME OTHER SPECIAL PROPERTIES

1. $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z$
$=\tan^{-1}\left[\frac{x+y+z-xyz}{1-(xy+xz+yz)}\right]$
$=\tan^{-1}\left(\frac{S_1-S_3}{1-S_2}\right)$
where $S_1=\sum x$, $S_2=\sum xy$, $S_3=\sum xyz$

2. If $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\frac{\pi}{2}$, then
$xy+yz+zx=1$

3. If $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\pi$, then
$x+y+z=xyz$

4. If $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\frac{\pi}{2}$, then
$x^2+y^2+z^2+2xyz=1$

5. If $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\pi$, then
$x\sqrt{1-x^2}+y\sqrt{1-y^2}+z\sqrt{1-z^2}=2xyz$

6. If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=3\pi$, then $xy+yz+zx=3$

7. If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$, then
$x^2+y^2+z^2+2xyz=1$

8. If $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=3\pi/2$, then
$xy+yz+zx=3$

9. If $\sin^{-1}x+\sin^{-1}y=\theta$, then $\cos^{-1}x+\cos^{-1}y=\pi-\theta$

10. If $\cos^{-1}x+\cos^{-1}y=\theta$, then $\sin^{-1}x+\sin^{-1}y=\pi-\theta$

SOLVED PROBLEMS

1. The principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ is

(a) $-\frac{2\pi}{3}$ (b) $-\frac{\pi}{3}$

(c) $\frac{4\pi}{3}$ (d) $\frac{5\pi}{3}$

Ans. (b)

Solution: $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\left[\because \sin^{-1}(-x) = -\sin^{-1}x\right]$

$= -\frac{\pi}{3} \qquad \left[\because \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}\right]$

2. $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$ is equal to

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{2\pi}{3}$

Ans. (d)

Solution: $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} + 2\left(\frac{\pi}{6}\right)$

$= \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$

3. The simplified expression of $\sin(\tan^{-1}x)$, for any real number x, is given by

(a) $-\frac{1}{\sqrt{1+x^2}}$ (b) $-\frac{x}{\sqrt{1+x^2}}$

(c) $\frac{1}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{1+x^2}}$

Ans. (d)

Solution: $\sin(\tan^{-1}x) = \sin\left[\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)\right] = \frac{x}{\sqrt{1+x^2}}$

Another method:

Let $x = \tan\theta$. Then $\tan^{-1}x = \theta$

$\sin(\tan^{-1}x) = \sin\theta$

$= \frac{\tan\theta}{\sec\theta} = \frac{\tan\theta}{\sqrt{1+\tan^2\theta}} = \frac{x}{\sqrt{1+x^2}}$

4. The value of $\sin\left[2\cos^{-1}\frac{\sqrt{5}}{3}\right]$ is

(a) $\frac{2\sqrt{5}}{9}$ (b) $\frac{\sqrt{5}}{9}$

(c) $\frac{4\sqrt{5}}{9}$ (d) $\frac{2\sqrt{5}}{3}$

Ans. (c)

Solution: $\sin\left[2\cos^{-1}\frac{\sqrt{5}}{3}\right]$

$= \sin\left[2\sin^{-1}\sqrt{1-\frac{5}{9}}\right]$

$\left[\because \cos^{-1}x = \sin^{-1}\sqrt{1-x^2}\right]$

$= \sin\left[2\sin^{-1}\left(\frac{2}{3}\right)\right]$

$= \sin\left\{\sin^{-1}\left[2\cdot\left(\frac{2}{3}\right)\sqrt{1-\frac{4}{9}}\right]\right\}$

$\left[\because 2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})\right]$

$= \frac{4}{3}\cdot\sqrt{\frac{5}{9}} = \frac{4\sqrt{5}}{9}$

5. If $\tan^{-1}\left(\frac{a}{x}\right) + \tan^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$, then x is equal to

(a) $\sqrt{ab}$ (b) $2\sqrt{ab}$

(c) $\sqrt{2ab}$ (d) $\sqrt{\frac{ab}{2}}$

Ans. (a)

Solution: $\tan^{-1}\left(\frac{a}{x}\right) + \tan^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$

$\Rightarrow \tan^{-1}\left(\frac{(a/x)+(b/x)}{1-(a/x)\cdot(b/x)}\right) = \frac{\pi}{2}$

$\Rightarrow \frac{(a+b)/x}{1-(ab/x^2)} = \tan\left(\frac{\pi}{2}\right) = \infty$

$\Rightarrow 1 - \frac{ab}{x^2} = 0 \Rightarrow \frac{ab}{x^2} = 1 \Rightarrow x^2 = ab \Rightarrow x = \sqrt{ab}$

6. $\tan^{-1}\left(\frac{m}{n}\right)-\tan^{-1}\left(\frac{m-n}{m+n}\right)$ is equal to

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$

Ans. (a)

Solution: $\tan^{-1}\left(\frac{m}{n}\right)-\tan^{-1}\left(\frac{m-n}{m+n}\right)$

$$=\tan^{-1}\left[\frac{\frac{m}{n}-\frac{m-n}{m+n}}{1+\frac{m}{n}\left(\frac{m-n}{m+n}\right)}\right]$$

$$=\tan^{-1}\left[\frac{\left(\frac{m^2+n^2}{n(m+n)}\right)}{\left(\frac{n^2+m^2}{n(m+n)}\right)}\right]=\tan^{-1}(1)=\frac{\pi}{4}$$

7. The value of $\cos\left(2\sin^{-1}\frac{\sqrt{3}}{2}\right)+\sin\left[2\cos^{-1}\left(-\frac{1}{2}\right)\right]$ is

(a) $-\frac{\sqrt{3}+1}{2}$ (b) -1

(c) 1 (d) $\frac{\sqrt{3}+1}{2}$

Ans. (a)

Solution: $\cos\left(2\sin^{-1}\frac{\sqrt{3}}{2}\right)+\sin\left[2\cos^{-1}\left(-\frac{1}{2}\right)\right]$

$$=\cos\left[2\left(\frac{\pi}{3}\right)\right]+\sin\left[2\cos^{-1}\left(-\cos\frac{\pi}{3}\right)\right]$$

$$=\cos\left(\frac{2\pi}{3}\right)+\sin\left\{2\left[\pi-\cos^{-1}\left(\cos\frac{\pi}{3}\right)\right]\right\}$$

$$\left[\because \cos^{-1}(-\alpha)=\pi-\cos^{-1}\alpha\right]$$

$$=\cos\left(\frac{2\pi}{3}\right)+\sin\left[2\pi-2\left(\frac{\pi}{3}\right)\right]$$

$$=\cos\left(\pi-\frac{\pi}{3}\right)+\sin\left(\frac{4\pi}{3}\right)=\cos\left(\pi-\frac{\pi}{3}\right)+\sin\left(\pi+\frac{\pi}{3}\right)$$

$$=-\cos\frac{\pi}{3}-\sin\frac{\pi}{3}=-\frac{\sqrt{3}}{2}-\frac{1}{2}=-\frac{\sqrt{3}+1}{2}$$

8. If $\sin^{-1}\left(\frac{3}{x}\right)+\sin^{-1}\left(\frac{4}{x}\right)=\frac{\pi}{2}$, then x is equal to

(a) 3 (b) 5

(c) 7 (d) 11

Ans. (b)

Solution: $\sin^{-1}\left(\frac{3}{x}\right)+\sin^{-1}\left(\frac{4}{x}\right)=\frac{\pi}{2}$

$$\Rightarrow \sin^{-1}\frac{4}{x}=\frac{\pi}{2}-\sin^{-1}\frac{3}{x}$$

$$\Rightarrow \sin^{-1}\frac{4}{x}=\cos^{-1}\frac{3}{x} \qquad \left[\because \sin^{-1}\alpha+\cos^{-1}\alpha=\frac{\pi}{2}\right]$$

$$\Rightarrow \sin^{-1}\frac{4}{x}=\sin^{-1}\sqrt{1-\frac{9}{x^2}} \quad \left[\because \cos^{-1}\alpha=\sin^{-1}\sqrt{1-\alpha^2}\right]$$

$$\Rightarrow \frac{4}{x}=\sqrt{1-\frac{9}{x^2}}$$

$$\Rightarrow \frac{16}{x^2}=1-\frac{9}{x^2} \Rightarrow \frac{25}{x^2}=1 \Rightarrow x^2=25 \Rightarrow x=5$$

9. If $5\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)+7\sin^{-1}\left(\frac{2x}{1+x^2}\right)-4\tan^{-1}\left(\frac{2x}{1-x^2}\right)-\tan^{-1}x=5\pi$, then x is equal to

(a) 0 (b) $1/\sqrt{3}$

(c) 1 (d) $\sqrt{3}$

Ans. (d)

Solution: Let $x=\tan\theta$.

Then $5\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)+7\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

$$-4\tan^{-1}\left(\frac{2x}{1-x^2}\right)-\tan^{-1}x=5\pi$$

$$\Rightarrow 5\cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)+7\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

$$-4\tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)-\tan^{-1}(\tan\theta)=5\pi$$

$$\Rightarrow 5\cos^{-1}(\cos 2\theta)+7\sin^{-1}(\sin 2\theta)-4\tan^{-1}(\tan 2\theta)$$

$$-\tan^{-1}(\tan\theta)=5\pi$$

$$\Rightarrow 5\cdot(2\theta)+7\cdot(2\theta)-4\cdot(2\theta)-\theta=5\pi$$

$$\Rightarrow 10\theta+14\theta-8\theta-\theta=5\pi$$

$$\Rightarrow 15\theta=5\pi$$

$$\Rightarrow \theta=\frac{\pi}{3} \Rightarrow \tan\theta=\tan\frac{\pi}{3} \Rightarrow x=\sqrt{3}$$

10. Evaluate $\cot^{-1}\left(\frac{xy+1}{x-y}\right)+\cot^{-1}\left(\frac{yz+1}{y-z}\right)+\cot^{-1}\left(\frac{zx+1}{z-x}\right)$

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

Ans. (a)

Solution: $\cot^{-1}\left(\frac{xy+1}{x-y}\right)+\cot^{-1}\left(\frac{yz+1}{y-z}\right)+\cot^{-1}\left(\frac{zx+1}{z-x}\right)$

$= \tan^{-1}\left(\frac{x-y}{1+xy}\right)+\tan^{-1}\left(\frac{y-z}{1+yz}\right)+\tan^{-1}\left(\frac{z-x}{1+zx}\right)$

$= (\tan^{-1}x-\tan^{-1}y)+(\tan^{-1}y-\tan^{-1}z)$

$+(\tan^{-1}z-\tan^{-1}x)=0$

11. If $\cos^{-1}\sqrt{p}+\cos^{-1}\sqrt{1-p}+\cos^{-1}\sqrt{1-q}=\frac{3\pi}{4}$, the value of q is

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{3}$

(c) $\frac{1}{2}$ (d) 1

Ans. (c)

Solution: Let

$\cos^{-1}\sqrt{p}=A,\ \cos^{-1}\sqrt{1-p}=B,\ \cos^{-1}\sqrt{1-q}=C$

Then $\sqrt{p}=\cos A, \sqrt{1-p}=\cos B$ and $\sqrt{1-q}=\cos C$

$\therefore\ \cos^{-1}\sqrt{p}+\cos^{-1}\sqrt{1-p}+\cos^{-1}\sqrt{1-q}=\frac{3\pi}{4}$

$\Rightarrow\ A+B+C=\frac{3\pi}{4}$

$\Rightarrow\ A+B=\frac{3\pi}{4}-C$

$\Rightarrow\ \cos(A+B)=\cos\left(\frac{3\pi}{4}-C\right)=\cos\left[\pi-\left(C+\frac{\pi}{4}\right)\right]$

$\Rightarrow\ \cos(A+B)=-\cos\left(C+\frac{\pi}{4}\right)$

$\Rightarrow\ \cos A\cos B-\sin A\sin B$

$=-\left[\cos C\cos\frac{\pi}{4}-\sin C\sin\frac{\pi}{4}\right]$

$\Rightarrow\ \sqrt{p}\sqrt{1-p}-\sqrt{1-p}\sqrt{p}=-\frac{1}{\sqrt{2}}(\sqrt{1-q}-\sqrt{q})$

$\left[\because \cos A=\sqrt{p}\Rightarrow\sin A=\sqrt{1-\left(\sqrt{p}\right)^2}=\sqrt{1-p}\right.$

$\cos B=\sqrt{1-p}\Rightarrow\sin B=\sqrt{1-\left(\sqrt{1-p}\right)^2}=\sqrt{p}$

$\left.\cos C=\sqrt{1-q}\Rightarrow\sin C=\sqrt{1-\left(\sqrt{1-q}\right)^2}=\sqrt{q}\right]$

$\Rightarrow\ 0=-\frac{1}{2}\left(\sqrt{1-q}-\sqrt{q}\right)$

$\Rightarrow\ \sqrt{q}=\sqrt{1-q}$

$\Rightarrow\ q=1-q\ \Rightarrow\ 2q=1\ \Rightarrow\ q=\frac{1}{2}$

12. If $\tan^{-1}(1-x)$, $\tan^{-1}x$ and $\tan^{-1}(1+x)$ are in A.P., then

(a) $x^3+x^2+x-1=0$ (b) $x^3+x^2-1=0$

(c) $x^3-x^2+1=0$ (d) $x^3-x+1=0$

Ans. (b)

Solution: $\tan^{-1}(1-x), \tan^{-1}x, \tan^{-1}(1+x)$ are in A.P.

$\Rightarrow\ 2\tan^{-1}x=\tan^{-1}(1-x)+\tan^{-1}(1+x)$

$\Rightarrow\ \tan^{-1}\left(\frac{2x}{1-x^2}\right)=\tan^{-1}\left[\frac{(1-x)+(1+x)}{1-(1-x)(1+x)}\right]$

$\Rightarrow\ \tan^{-1}\left(\frac{2x}{1-x^2}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$

$\Rightarrow\ \frac{2x}{1-x^2}=\frac{2}{x^2}\ \Rightarrow\ \frac{x}{1-x^2}=\frac{1}{x^2}\ \Rightarrow\ x^3+x^2-1=0$

13. The sum of n terms of the series $\tan^{-1}\frac{1}{1+1+1^2}$ $+\tan^{-1}\frac{1}{1+2+2^2}+\tan^{-1}\frac{1}{1+3+3^2}+\cdots+\infty$ is

(a) $\frac{\pi}{4}$ (b) $\tan^{-1}\left(\frac{n}{n+1}\right)$

(c) $\tan^{-1}\left(\frac{n}{n+2}\right)$ (d) $\tan^{-1}\left(\frac{n+1}{n+2}\right)$

Ans. (c)

Solution: The nth term of the given series is

$t_n=\tan^{-1}\frac{1}{1+n+n^2}=\tan^{-1}\left[\frac{(n+1)-n}{1+(n+1)n}\right]$

$=\tan^{-1}(n+1)-\tan^{-1}n$

$\therefore \quad t_1 = \tan^{-1}2 - \tan^{-1}1,\ t_2 = \tan^{-1}3 - \tan^{-1}2, \ldots,$

$$t_n = \tan^{-1}(n+1) - \tan^{-1}n$$

and so, $S_n = t_1 + t_2 + \cdots + t_n = \tan^{-1}(n+1) - \tan^{-1}1$

$$= \tan^{-1}\left[\frac{(n+1)-1}{1+(n+1)\cdot 1}\right]$$

$$= \tan^{-1}\left(\frac{n}{n+2}\right)$$

14. The sum of the series $\tan^{-1}\frac{1}{1+1+1^2} + \tan^{-1}\frac{1}{1+2+2^2} + \tan^{-1}\frac{1}{1+3+3^2} + \cdots + \infty$ is equal to

(a) $\pi/6$ (b) $\pi/4$

(c) $\pi/3$ (d) $\pi/2$

Ans. (b)

Solution: The nth term of the given series is

$$t_n = \tan^{-1}\frac{1}{1+n+n^2} = \tan^{-1}\left[\frac{(n+1)-n}{1+n(n+1)}\right]$$

$$= \tan^{-1}(n+1) - \tan^{-1}n$$

$\therefore \quad t_1 = \tan^{-1}2 - \tan^{-1}1,\ t_2 = \tan^{-1}3 - \tan^{-1}2, \ldots,$

$$t_n = \tan^{-1}(n+1) - \tan^{-1}n$$

$\therefore \quad S_n = t_1 + t_2 + \cdots + t_n = \tan^{-1}(n+1) - \tan^{-1}1$

and so, $S_\infty = \lim_{n\to\infty}\left[\tan^{-1}(n+1) - \tan^{-1}1\right]$

$$= \tan^{-1}\infty - \tan^{-1}1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

EXERCISE SET 1

1. $\sin^{-1}(3x - 2 - x^2) + \cos^{-1}(x^2 - 4x + 3) = \frac{\pi}{4}$ can have a solution for $x \in$

(a) [1, 2] (b) $\left(\frac{3+\sqrt{5}}{2},\ 2+\sqrt{2}\right)$

(c) $\left(\frac{3-\sqrt{5}}{2},\ 2-\sqrt{2}\right)$ (d) none of these

2. If $a\sin^{-1}x - b\cos^{-1}x = c$, then $a\sin^{-1}x + b\cos^{-1}x$ is equal to

(a) 0 (b) $\frac{\pi ab + c(b-a)}{a+b}$

(c) $\pi/2$ (d) $\frac{\pi ab + c(a-b)}{a+b}$

3. The value of $\cot^{-1}\frac{3}{4} + \sin^{-1}\frac{5}{13}$ is

(a) $\sin^{-1}\frac{63}{65}$ (b) $\sin^{-1}\frac{12}{13}$

(c) $\sin^{-1}\frac{65}{68}$ (d) $\sin^{-1}\frac{5}{12}$

4. $\tan^{-1}\left[\cos\left(2\tan^{-1}\frac{3}{4}\right) + \sin\left(2\cot^{-1}\frac{1}{2}\right)\right]$ is

(a) not real (b) equal to $\pi/4$

(c) greater than $\pi/4$ (d) less than $\pi/4$

5. If $\frac{1}{2}\sin^{-1}\left[\frac{3\sin 2\theta}{5+4\cos 2\theta}\right] = \tan^{-1}x$, then $x =$

(a) $\tan 3\theta$ (b) $3\tan\theta$

(c) $(1/3)\tan\theta$ (d) $3\cot\theta$

6. If $\theta = \tan^{-1}a$, $\phi = \tan^{-1}b$ and $ab = -1$, then $(\theta - \phi)$ is equal to

(a) 0 (b) $\pi/4$

(c) $\pi/2$ (d) none of these

7. The value of $\tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$ is

(a) $\pi/4$ (b) $\pi/2$

(c) π (d) 0

8. If $\tan^{-1}\frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} = \alpha$, then x^2 is equal to

(a) $\sin\alpha$ (b) $\cos 2\alpha$

(c) $\cos\alpha$ (d) $\sin 2\alpha$

9. The value of $\cot^{-1}\frac{xy+1}{x-y} + \cot^{-1}\frac{yz+1}{y-z} + \cot^{-1}\frac{zx+1}{z-x}$ is

(a) 0

(b) 1

(c) $\cot^{-1}x + \cot^{-1}y + \cot^{-1}z$

(d) none of these

10. If $\cot^{-1}\sqrt{\cos\alpha} - \tan^{-1}\sqrt{\cos\alpha} = u$, then $\sin u =$

(a) $\tan^2\alpha$ (b) $\tan 2\alpha$

(c) $\tan^2(\alpha/2)$ (d) $\cot^2(\alpha/2)$

11. If $\tan^{-1}\frac{a}{x} + \tan^{-1}\frac{b}{x} + \tan^{-1}\frac{c}{x} + \tan^{-1}\frac{d}{x} = \frac{\pi}{2}$, then $x^4 - x^2\sum ab + abcd$ is equal to

(a) -1 (b) 0

(c) 1 (d) 2

12. If $x^2+y^2+z^2=r^2$, then $\tan^{-1}\left(\frac{xy}{zr}\right)+\tan^{-1}\left(\frac{yz}{xr}\right)+\tan^{-1}\left(\frac{xz}{yr}\right)$ is equal to

(a) π (b) $\pi/2$
(c) 0 (d) none of these

13. If $\alpha=\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{1}{3}$ and $\beta=\cos^{-1}\frac{4}{5}+\cos^{-1}\frac{1}{3}$, then

(a) $\alpha<\beta$ (b) $\alpha=\beta$
(c) $\alpha>\beta$ (d) none of these

14. $\sin^{-1}x+\sin^{-1}\frac{1}{x}+\cos^{-1}x+\cos^{-1}\frac{1}{x}$ is equal to

(a) π (b) $\pi/2$
(c) $3\pi/2$ (d) none of these

15. The number of the positive integral solutions of $\tan^{-1}x+\cos^{-1}\frac{y}{\sqrt{1+y^2}}=\sin\frac{3}{\sqrt{10}}$ is

(a) 1 (b) 2
(c) 3 (d) 4

16. $\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$ is equal to

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$
(c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{4}$

17. Let $f:\left[\frac{1}{2},1\right]\to[-1,1]$ be given by $f(x)=4x^3-3x$, then $f^{-1}(x)$ is given by

(a) $\cos\left(\frac{1}{3}\cos^{-1}x\right)$ (b) $3\cos(\sin^{-1}x)$
(c) $3\sin^{-1}(\cos x)$ (d) none of these

18. If $\sin^{-1}\alpha+\sin^{-1}\beta+\sin^{-1}\gamma=\frac{3\pi}{2}$, then $\alpha\beta+a\gamma+\beta\gamma$ is equal to

(a) 1 (b) 0
(c) 3 (d) –3

19. The value of $\sin(\cot^{-1}x)$ is

(a) $(1+x^2)^{3/2}$ (b) $(1+x^2)^{-3/2}$
(c) $(1+x^2)^{1/2}$ (d) $(1+x^2)^{-1/2}$

20. $\cos^{-1}\{\cos[2\cot^{-1}(\sqrt{2}-1)]\}$ is equal to

(a) $\sqrt{2}-1$ (b) $1-\sqrt{2}$
(c) $\pi/4$ (d) $3\pi/4$

21. $\cos\tan^{-1}\sin\cot^{-1}x$ is equal to

(a) x (b) 1
(c) $\frac{x^2+1}{x^2+2}$ (d) $\sqrt{\frac{x^2+1}{x^2+2}}$

22. If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$, then

(a) $x^2+y^2=z^2$ (b) $x^2+y^2+z^2=0$
(c) $x^2+y^2+z^2=1-2xyz$ (d) none of these

23. The value of $3\tan^{-1}a$ is

(a) $\tan^{-1}\left(\frac{3a+a^3}{1+3a^2}\right)$ (b) $\tan^{-1}\left(\frac{3a-a^3}{1+3a^2}\right)$
(c) $\tan^{-1}\left(\frac{3a+a^3}{1-3a^2}\right)$ (d) $\tan^{-1}\left(\frac{3a-a^3}{1-3a^2}\right)$

24. If $\cos^{-1}\frac{x}{2}+\cos^{-1}\frac{y}{3}=\theta$, then $9x^2-12xy\cos\theta+4y^2=$

(a) 36 (b) $-36\sin^2\theta$
(c) $36\sin^2\theta$ (d) $36\cos^2\theta$

25. If $2\tan^{-1}x=\sin^{-1}\frac{2a}{1+a^2}+\cos^{-1}\frac{1-b^2}{1+b^2}$, then x is equal to

(a) $\frac{a+b}{1+ab}$ (b) $\frac{a+b}{1-ab}$
(c) $\frac{a-b}{1+ab}$ (d) $\frac{a^2-b^2}{1+a^2b^2}$

26. The value of $\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\cdots+\tan^{-1}\left(\frac{1}{n^2+n+1}\right)$ is

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
(c) $\frac{2\pi}{3}$ (d) 0

27. If $\left(\tan^{-1}x\right)^2+\left(\cot^{-1}x\right)^2=\frac{5\pi^2}{8}$, then x is equal to

(a) –1 (b) 1
(c) 0 (d) none of these

28. If $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\frac{\pi}{2}$, then the value of $xy+yz+xz$ is

(a) 0 (b) 1
(c) xyz (d) $x+y+z$

29. The number of real solutions of the equation $\tan^{-1}\sqrt{x^2-3x+2}+\cos^{-1}\sqrt{4x-x^2-3}=\pi$ is
(a) one (b) two
(c) zero (d) infinite

30. If $\sin^{-1}(x)+\sin^{-1}(y)+\sin^{-1}(z)=\frac{3\pi}{2}$, then the value of $(x^{100}+y^{100}+z^{100})-(x^{101}+y^{101}+z^{101})$ is
(a) 0 (b) 1
(c) 2 (d) 4

31. $\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x\right)$, $x\neq 0$, is equal to
(a) x (b) $2x$
(c) $2/x$ (d) none of these

32. If $3\sin^{-1}\left(\frac{2x}{1+x^2}\right)-4\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)+2\tan^{-1}\left(\frac{2x}{1-x^2}\right)=\frac{\pi}{3}$, then the value of x is
(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{2}}$
(c) 1 (d) 2

33. If $\sin^{-1}\left(\frac{1}{3}\right)+\sin^{-1}\left(\frac{2}{3}\right)=\sin^{-1}x$, then x is equal to
(a) $\frac{4+\sqrt{5}}{9}$ (b) $\frac{4\sqrt{2}+\sqrt{5}}{9}$
(c) $\frac{\sqrt{3}+1}{6}$ (d) 1

34. If $\sin[\cos^{-1}(x+1)]=\cos(\tan^{-1}x)$, then the value of x is
(a) $-(1/2)$ (b) 1/2
(c) 0 (d) 9/4

35. In $a\Delta ABC$, if $A=\tan^{-1}2$ and $B=\tan^{-1}3$, then C is equal to
(a) $\pi/3$ (b) $\pi/4$
(c) $\pi/6$ (d) none of these

36. If $\sin^{-1}x+\sin^{-1}y=\frac{2\pi}{3}$, then $\cos^{-1}x+\cos^{-1}y=$
(a) $2\pi/3$ (b) $\pi/3$
(c) $\pi/6$ (d) π

37. If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=3\pi$, then the value of $xy+yz+zx$ is
(a) 1 (b) 0
(c) –3 (d) 3

38. If α,β,γ are the roots of the equation $x^3+px^2+2x+p=0$, then the general value of $\tan^{-1}\alpha+\tan^{-1}\beta+\tan^{-1}\gamma$ is
(a) $n\pi$
(b) $\frac{n\pi}{2}$
(c) $(2n+1)\frac{\pi}{2}$
(d) dependent upon the value of p

39. $6\sin^{-1}(x^2-6x+8.5)=\pi$, if
(a) $x=1$ (b) $x=2$
(c) $x=3$ (d) $x=4$

40. If $\sin^{-1}x-\cos^{-1}x=\frac{\pi}{6}$, then $x=$
(a) 1/2 (b) $-(1/2)$
(c) $\sqrt{3}/2$ (d) none of these

41. If $\cos^{-1}p+\cos^{-1}q+\cos^{-1}r=\pi$, then the value of $p^2+q^2+r^2=2pqr$ is
(a) 3 (b) 1
(c) 2 (d) –1

42. If $\sin^{-1}\sqrt{x^2+2x+1}+\sec^{-1}\sqrt{x^2+2x+1}=\frac{\pi}{2}$, $x\neq 0$, then the value of $2\sec^{-1}\frac{x}{2}+\sin^{-1}\frac{x}{2}=$
(a) $-\frac{x}{2}$ (b) $-\frac{3\pi}{2},\frac{\pi}{2}$
(c) $\frac{3\pi}{2}$ (d) $-\frac{3\pi}{2}$

43. The sum of infinite terms of the series $\cot^{-1}\left(1^2+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+\cdots$ is
(a) $\pi/4$ (b) $\tan^{-1}2$
(c) $\tan^{-1}3$ (d) none of these

44. If $\cos^{-1}\left(\frac{x}{a}\right)+\cos^{-1}\left(\frac{y}{b}\right)=\alpha$, then the value of $\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}$ is
(a) $\sin^2\alpha$ (b) $\cos^2\alpha$
(c) $\tan^2\alpha$ (d) $\cot^2\alpha$

45. If $\alpha \le \sin^{-1} x + \cos^{-1} x + \tan^{-1} x \le \beta$, then

(a) $\alpha = 0$ (b) $\beta = \frac{\pi}{2}$

(c) $\alpha = \frac{\pi}{2}$ (d) $\beta = \pi$

46. The value of $\tan\left[\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right]$ is

(a) $\frac{6}{17}$ (b) $\frac{17}{6}$

(c) $\frac{7}{6}$ (d) $\frac{16}{7}$

EXERCISE SET 2

1. If $x + \frac{1}{x} = 2$, the principle value of $\sin^{-1} x$ is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) π (d) $\frac{3\pi}{2}$

2. If $0 < x < 1$, then $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{1+x}\right)$ is equal to

(a) $\frac{1}{2}\cos^{-1} x$ (b) $\cos^{-1}\frac{\sqrt{1+x}}{2}$

(c) $\sin^{-1}\frac{\sqrt{1-x}}{2}$ (d) all of the above

3. The value of $\cos\left[\tan^{-1}\left(\tan\frac{15\pi}{4}\right)\right]$ is

(a) $\frac{1}{\sqrt{2}}$ (b) $-\frac{1}{\sqrt{2}}$

(c) 1 (d) none of these

4. If $\sin^{-1} x = \theta + \beta$ and $\sin^{-1} y = \theta - \beta$, then $1 + xy =$

(a) $\sin^2\theta + \sin^2\beta$ (b) $\sin^2\theta + \cos^2\beta$

(c) $\cos^2\theta + \cos^2\theta$ (d) $\cos^2\theta + \sin^2\beta$

5. The set of values of λ for which $x^2 - \lambda x + \sin^{-1}(\sin 4) > 0$ for all $x \in \mathbb{R}$ is

(a) ϕ (b) (–2, 2)

(c) $\mathbb{R}$ (d) none of these

6. The value of $\sin\{\cot^{-1}[\cos(\tan^{-1} x)]\}$ is

(a) $\sqrt{\frac{x^2+2}{x^2+1}}$ (b) $\sqrt{\frac{x^2+1}{x^2+2}}$

(c) $\frac{x}{\sqrt{x^2+2}}$ (d) $\frac{1}{\sqrt{x^2+2}}$

7. Exhaustive set of values of parameter a so that $\sin^{-1} x - \tan^{-1} x = a$ has a solution

(a) $[-\pi/6, \pi/6]$ (b) $[-\pi/4, \pi/4]$

(c) $[-\pi/2, \pi/2]$ (d) none of these

8. If $\tan^{-1}\frac{a+x}{a} + \tan^{-1}\frac{a-x}{a} = \frac{\pi}{6}$, then $x^2 =$

(a) $2\sqrt{3}a$ (b) $\sqrt{3}a$

(c) $2\sqrt{3}a^2$ (d) none of these

9. The value of $\cot^{-1}\left\{\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}\right\}$, where $0 < x < 2\pi$, is

(a) $\pi - \frac{x}{2}$ (b) $2\pi - x$

(c) $\frac{x}{2}$ (d) $2\pi - \frac{x}{2}$

10. The value of $\cos^{-1}\sqrt{\frac{2}{3}} - \cos^{-1}\frac{\sqrt{6}+1}{2\sqrt{3}}$ is equal to

(a) $\pi/3$ (b) $\pi/4$

(c) $\pi/2$ (d) $\pi/6$

11. If $A = \tan^{-1}\left(\frac{x\sqrt{3}}{2k-x}\right)$ and $B = \tan^{-1}\left(\frac{2x-k}{k\sqrt{3}}\right)$, then the value of $A - B$ is

(a) 0° (b) 45°

(c) 60° (d) 30°

12. If $0 \le x \le 1$, then $\tan\left(\frac{1}{2}\sin^{-1}\frac{2x}{1+x^2} + \frac{1}{2}\cos^{-1}\frac{1-x^2}{1+x^2}\right)$ is equal to

(a) $\frac{2x}{1-x^2}$ (b) 0

(c) $\frac{2x}{1+x^2}$ (d) x

13. The smallest and the largest values of $\tan^{-1}\left(\frac{1-x}{1+x}\right)$, $0 \le x \le 1$, are

(a) 0, π (b) 0, $\frac{\pi}{4}$

(c) $-\frac{\pi}{4}, \frac{\pi}{4}$ (d) $\frac{\pi}{4}, \frac{\pi}{2}$

14. If $(\sin^{-1}x)^2+(\cos^{-1}x)^2=\frac{5\pi^2}{8}$, then $x=$

(a) ±1 (b) $\pm\frac{1}{2}$

(c) $-\frac{1}{\sqrt{2}}$ (d) none of these

15. The range of $f(x)=\sin^{-1}x+\tan^{-1}x+\sec^{-1}x$ is

(a) $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$ (b) $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$

(c) $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$ (d) none of these

16. The number of solutions of the equation $2\sin^{-1}\sqrt{x^2-x+1}+\cos^{-1}\sqrt{x^2-x}=\frac{3\pi}{2}$ is

(a) 1 (b) 0

(c) 2 (d) ∞

17. If $\cos^{-1}\left(\frac{p}{a}\right)+\cos^{-1}\left(\frac{q}{b}\right)=\alpha$, then $\frac{p^2}{a^2}+k\cos\alpha+\frac{q^2}{b^2}=\sin^2\alpha$, where k is equal to

(a) $\frac{2pq}{ab}$ (b) $-\frac{2pq}{ab}$

(c) $\frac{pq}{ab}$ (d) $-\frac{pq}{ab}$

18. If x satisfies the equation $t^2-t-2>0$, then there exists a value for

(a) $\sin^{-1}x$ (b) $\cos^{-1}x$

(c) $\sec^{-1}x$ (d) none of these

19. Statement I: If $-1\le x\le1$, then $\sin^{-1}(-x)=-\sin^{-1}x$ and $\cos^{-1}(-x)=\pi-\cos^{-1}x$.

Statement II: If $-1\le x\le1$, then $\cos^{-1}x=2\sin^{-1}\sqrt{\frac{1-x}{2}}=2\cos^{-1}\sqrt{\frac{1+x}{2}}$

(a) Statement I is false, Statement II is true.

(b) Statement I is true, Statement II is true; Statement II is not a correct explanation of Statement I.

(c) Statement I is true, Statement II is true; Statement II is a correct explanation of Statement I.

(d) Statement I is true, Statement II is false.

20. If $3\tan^{-1}\left(\frac{1}{2+\sqrt{3}}\right)-\tan^{-1}\frac{1}{x}=\tan^{-1}\frac{1}{3}$, then x is equal to

(a) 1 (b) 2

(c) 3 (d) $\sqrt{2}$

21. If $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\pi$, then $x^4+y^4+z^4+4x^2y^2z^2=K(x^2y^2+y^2z^2+z^2x^2)$, where $K=$

(a) 1 (b) 2

(c) 4 (d) none of these

22. $\sec^2(\tan^{-1}2)+\text{cosec}^2(\cot^{-1}3)$ is equal to

(a) 5 (b) 10

(c) 15 (d) 20

23. The value of $\cos(2\cos^{-1}x+\sin^{-1}x)$ at $x=(1/5)$ is

(a) $\sqrt{6}/5$ (b) $-2(\sqrt{6}/5)$

(c) 2/5 (d) $2\sqrt{6}$

24. The number of real solutions of $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{(x^2+x+1)}=\frac{\pi}{2}$ is

(a) zero (b) one

(c) two (d) infinite

Assertion-Reason Type Questions

The following questions are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice given as follows:

(a) Statement I is false, Statement II is true.

(b) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

(c) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.

(d) Statement I is true, Statement II is false.

25. Statement I: If $\sum_{i=1}^{2n}\sin^{-1}x_i=n\pi,\ n\in N$, then

$$\sum_{i=1}^{n}x_i=\sum_{i=1}^{n}x_i^2=\sum_{i=1}^{n}x_i^3$$

Statement II: $-\frac{\pi}{2}\le\sin^{-1}x\le\frac{\pi}{2},\ \forall x\in[-1,1]$

26. Statement I: If $x<0$, then $\tan^{-1}x+\tan^{-1}\left(\frac{1}{x}\right)=\frac{\pi}{2}$

Statement-II: $\tan^{-1}x+\cot^{-1}x=\frac{\pi}{2},\forall x\in\mathbb{R}$

Previous Years' Questions

27. $\tan^{-1}\left(\frac{1}{4}\right)+\tan^{-1}\left(\frac{2}{9}\right)$ is equal to

(a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ (b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$

(c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ (d) $\tan^{-1}\left(\frac{1}{2}\right)$

[AIEEE 2002]

28. The maximum value of $(\sec^{-1}x)^2+(\operatorname{cosec}^{-1}x)^2$ is equal to

(a) $\frac{\pi^2}{2}$ (b) $\frac{5\pi^2}{4}$

(c) π^2 (d) none of these

[AIEEE 2002]

29. The value of $\tan^{-1}\sqrt{\frac{a\lambda}{bc}}+\tan^{-1}\sqrt{\frac{b\lambda}{ca}}+\tan^{-1}\sqrt{\frac{c\lambda}{ab}}$, where $a,b,c\in\mathbb{R}^+$ and $\lambda=\frac{abc}{ab+bc+ca}$, is equal to

(a) $\pi/4$ (b) π

(c) $\pi/2$ (d) none of these

[AIEEE 2002]

30. $\tan\left(2\cos^{-1}\frac{3}{5}\right)$ is equal to

(a) $\frac{8}{3}$ (b) $\frac{24}{25}$

(c) $\frac{7}{25}$ (d) $-\frac{24}{7}$ **[AIEEE 2002]**

31. The trigonometric equation $\sin^{-1}x=2\sin^{-1}a$ has a solution for

(a) $\frac{1}{2}<|a|<\frac{1}{\sqrt{2}}$ (b) all real values of a

(c) $|a|\le\frac{1}{\sqrt{2}}$ (d) $|a|\ge\frac{1}{\sqrt{2}}$

[AIEEE 2003]

32. If $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$, then $4x^2-4xy\cos\alpha+y^2$ is equal to

(a) $4\sin^2\alpha$ (b) $-2\cos^2\alpha$

(c) $2\sin 2\alpha$ (d) $-4\cos 2\alpha$

[AIEEE 2005]

33. If $\sin^{-1}\left(\frac{x}{5}\right)+\operatorname{cosec}^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$, then the value of x is

(a) 3 (b) 4

(c) 5 (d) 1 **[AIEEE 2007]**

34. The value of $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$ is

(a) $\frac{5}{17}$ (b) $\frac{6}{17}$

(c) $\frac{3}{17}$ (d) $\frac{4}{17}$ **[AIEEE 2008]**

35. $\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right)+\tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right)+\tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right)+\cdots+\tan^{-1}\left(\frac{1}{c_n}\right)$ is equal to

(a) $-\tan^{-1}\left(\frac{x}{y}\right)$ (b) $-\tan^{-1}\left(\frac{y}{x}\right)$

(c) $\tan^{-1}\left(\frac{x}{y}\right)$ (d) $\tan^{-1}\left(\frac{y}{x}\right)$

[DCE 2002]

36. The value of $\sin^{-1}(\sin 10)$ is

(a) 10 (b) $\pi-10$

(c) $2\pi-10$ (d) $3\pi-10$ **[DCE 2002]**

37. The range of $\tan^{-1}x$ is

(a) $(-\pi,\pi)$ (b) $(0,\pi)$

(c) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ (d) $\left(\frac{\pi}{2},\pi\right)$ **[DCE 2002]**

38. The value of $\sin^{-1}x+\cos^{-1}x$ is

(a) $-\pi$ (b) $-\frac{\pi}{2}$

(c) $\frac{\pi}{2}$ (d) π **[DCE 2002]**

39. If $\tan^{-1}\frac{1-x}{1+x}=\frac{1}{2}\tan^{-1}x$, then the value of x is

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{3}$ (d) 2 **[DCE 2006]**

40. The equation $\sin^{-1}x-\cos^{-1}x=\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ has

(a) no solution
(b) unique solution
(c) infinite number of solutions
(d) none of these **[DCE 2006]**

41. If $\sqrt{3}+i=(a+ib)(c+id)$, then $\tan^{-1}\left(\frac{b}{a}\right)+\tan^{-1}\left(\frac{d}{c}\right)$ has the value

(a) $\frac{\pi}{3}+2n\pi,\ n\in I$ (b) $n\pi+\frac{\pi}{6},\ n\in I$

(c) $n\pi-\frac{\pi}{3},\ n\in I$ (d) $2n\pi-\frac{\pi}{3},\ n\in I$

[DCE 2007]

42. If $\sin^{-1}\frac{2a}{1+a^2}-\cos^{-1}\frac{1-b^2}{1+b^2}=\tan^{-1}\frac{2x}{1-x^2}$, then the value of x is

(a) a (b) b

(c) $\frac{a+b}{1-ab}$ (d) $\frac{a-b}{1+ab}$ **[DCE 2007]**

43. Let $\cos(2\tan^{-1}x)=\frac{1}{2}$, then the value of x is

(a) $\frac{1}{\sqrt{3}}$ (b) $1-\sqrt{3}$

(c) $1-\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$ **[DCE 2007]**

44. The value of $\sin\left(\sin^{-1}\frac{1}{3}+\sec^{-1}3\right)+\cos\left(\tan^{-1}\frac{1}{2}+\tan^{-1}2\right)$ is

(a) 1 (b) 2

(c) 3 (d) 4 **[UP SEE 2005]**

45. The value of $\cot^{-1}9+\text{cosec}^{-1}\frac{\sqrt{41}}{4}$ is

(a) 0 (b) $\frac{\pi}{6}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$ **[UP SEE 2005]**

46. $3\tan^{-1}a$ is equal to

(a) $\tan^{-1}\frac{3a-a^3}{1+3a^2}$ (b) $\tan^{-1}\frac{3a+a^3}{1+3a^2}$

(c) $\tan^{-1}\frac{3a-a^3}{1-3a^2}$ (d) $\tan^{-1}\frac{3a+a^3}{1-3a^2}$

[UP SEE 2007]

47. The value of $2\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}$ is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$ (d) π **[UP SEE 2007]**

48. If $\tan^{-1}(x-1)+\tan^{-1}x+\tan^{-1}(x+1)=\tan^{-1}3x$, then x is equal to

(a) $\pm\frac{1}{2}$ (b) $0,\frac{1}{2}$

(c) $0,-\frac{1}{2}$ (d) $0,\pm\frac{1}{2}$

[UP SEE 2008]

49. If $\cot(\cos^{-1}x)=\sec\left(\tan^{-1}\frac{a}{\sqrt{b^2-a^2}}\right)$, then x is equal to

(a) $\frac{b}{\sqrt{2b^2-a^2}}$ (b) $\frac{a}{\sqrt{2b^2-a^2}}$

(c) $\frac{\sqrt{2b^2-a^2}}{a}$ (d) $\frac{\sqrt{2b^2-a^2}}{b}$

[UP SEE 2009]

50. The value of $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{7}{8}$ is

(a) $\tan^{-1}\frac{7}{8}$ (b) $\cot^{-1}15$

(c) $\tan^{-1}15$ (d) $\tan^{-1}\frac{25}{24}$

[UP SEE 2009]

51. The relation $\tan^{-1}\left(\frac{1+x}{1-x}\right)=\frac{\pi}{4}+\tan^{-1}x$ holds true for all

(a) $x\in\mathbb{R}$ (b) $x\in(-\infty,1)$

(c) $x\in(-1,\infty)$ (d) $x\in(-\infty,-1)$

[J&K CET 2003]

52. The value of $\cos^{-1}(\cos 12)-\sin^{-1}(\sin 14)$ is

(a) -2 (b) $4\pi+2$

(c) $8\pi-26$ (d) $6\pi-14$

[J&K CET 2005]

53. $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)-\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ is equal to

(a) 0 (b) 1

(c) $\tan^{-1}x$ (d) $\tan^{-1}2x$

[J&K CET 2006]

54. The value of $\cos^{-1}\left(-\frac{1}{2}\right)$, among the following, is

(a) $\frac{9\pi}{3}$ (b) $\frac{8\pi}{3}$

(c) $\frac{5\pi}{3}$ (d) $\frac{11\pi}{3}$

[J&K CET 2007]

55. $\tan^{-1}\frac{x}{\sqrt{a^2-x^2}}$ is equal to

(a) $\sin^{-1}\frac{x}{a}$ (b) $\sin\frac{2x}{a}$

(c) $2\sin^{-1}\frac{x}{a}$ (d) $\frac{1}{2}\sin^{-1}\frac{x}{a}$

[J&K CET 2008]

56. The value of $\cos^{-1}[\sin(7\pi/6)]$ is
(a) $-(\pi/3)$ (b) $\pi/6$
(c) $\pi/3$ (d) $2\pi/3$
[J&K CET 2009]

57. The value of $\sin^{-1}[\cos(4095°)]$ is equal to
(a) $-(\pi/3)$ (b) $\pi/6$
(c) $-(\pi/4)$ (d) $\pi/4$
[Kerala CEE 2010]

58. If $[\sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1}x]=1$, where [.] denotes the greatest integer function, then x belongs to the interval
(a) [tan sin cos 1, tan sin cos sin 1]
(b) [tan sin cos 1, tan sin cos sin 1]
(c) [1, −1]
(d) [sin cos tan 1, sin cos sin tan 1]
[Kerala CEE 2008]

ANSWERS

Exercise Set 1

1. (d)	**2.** (d)	**3.** (a)	**4.** (c)	**5.** (c)	**6.** (c)	**7.** (d)	**8.** (d)	**9.** (a)	**10.** (c)
11. (b)	**12.** (b)	**13.** (a)	**14.** (a)	**15.** (b)	**16.** (c)	**17.** (a)	**18.** (c)	**19.** (d)	**20.** (d)
21. (d)	**22.** (c)	**23.** (d)	**24.** (c)	**25.** (b)	**26.** (b)	**27.** (a)	**28.** (b)	**29.** (c)	**30.** (a)
31. (c)	**32.** (a)	**33.** (b)	**34.** (a)	**35.** (b)	**36.** (b)	**37.** (d)	**38.** (a)	**39.** (b) and (d)	
40. (c)	**41.** (b)	**42.** (c)	**43.** (b)	**44.** (a)	**45.** (a) and (d)		**46.** (b)		

Exercise Set 2

1. (b)	**2.** (d)	**3.** (a)	**4.** (b)	**5.** (a)	**6.** (b)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (d)
11. (d)	**12.** (a)	**13.** (b)	**14.** (c)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (c)	**19.** (b)	**20.** (b)
21. (b)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (b)	**26.** (a)	**27.** (d)	**28.** (b)	**29.** (c)	**30.** (d)
31. (c)	**32.** (a)	**33.** (a)	**34.** (b)	**35.** (c)	**36.** (d)	**37.** (c)	**38.** (c)	**39.** (b)	**40.** (b)
41. (b)	**42.** (d)	**43.** (a)	**44.** (a)	**45.** (c)	**46.** (c)	**47.** (a)	**48.** (d)	**49.** (a)	**50.** (c)
51. (b)	**52.** (c)	**53.** (c)	**54.** (b)	**55.** (a)	**56.** (d)	**57.** (c)	**58.** (a)		

HINTS AND SOLUTIONS—EXERCISE SET 1

1. Since the given equation is

$$\sin^{-1}(3x-2-x^2)+\cos^{-1}(x^2-4x+3)=\frac{\pi}{4},$$

$$\Rightarrow \sin^{-1}[-(x-1)(x-2)]+\cos^{-1}(x-3)(x-1)=\frac{\pi}{4}$$

for $x\in[1,2]$, we have

$$\sin^{-1}[-(x-1)(x-2)]\in\left[0,\frac{\pi}{2}\right)$$

and $\cos^{-1}(x-3)(x-1)\in\left[\frac{\pi}{2},\pi\right]$

Hence, in this range, the equation possesses two solutions.

Also, $-1\le-(x-1)(x-2)\le1$

and $-1\le x^2-4x+3\le1$

$$\Rightarrow \left(2-\sqrt{2}\le x\le\frac{3+\sqrt{5}}{2}\right)$$

2. We have $b\sin^{-1}x+b\cos^{-1}x=\frac{b\pi}{2}$ and

$$a\sin^{-1}x-b\cos^{-1}x=c \qquad \text{(given)}$$

$$\Rightarrow (a+b)\sin^{-1}x=\frac{b\pi}{2}+c$$

$$\Rightarrow \sin^{-1}x=\frac{(b\pi)/2+c}{a+b}$$

Similarly, $\cos^{-1}x=\frac{(a\pi)/2-c}{a+b}$

So that $a\sin^{-1}x+b\cos^{-1}x=\frac{\pi ab+c(a-b)}{a+b}$

3. Let $\cot^{-1}\frac{3}{4}=\theta \Rightarrow \cot\theta=\frac{3}{4}$

and $\sin\theta=\frac{1}{\sqrt{1+\cot^2\theta}}=\frac{1}{\sqrt{1+\left(\frac{9}{16}\right)}}=\frac{4}{5}$

$$\therefore \quad \cot^{-1}\frac{3}{4}+\sin^{-1}\frac{5}{13}=\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{5}{13}$$

$$=\sin^{-1}\left[\frac{4}{5}\sqrt{1-\frac{25}{169}}+\frac{5}{13}\sqrt{1-\frac{16}{25}}\right]$$

$$=\sin^{-1}\left[\frac{4}{5}\cdot\frac{12}{13}+\frac{5}{13}\cdot\frac{3}{5}\right]$$

$$=\sin^{-1}\left[\frac{48+15}{65}\right]=\sin^{-1}\frac{63}{65}$$

4. Let $E=\tan^{-1}\left\{\underbrace{\cos\left[2\tan^{-1}\left(\frac{3}{4}\right)\right]}_{E_1}+\underbrace{\sin\left[2\cot^{-1}\left(\frac{1}{2}\right)\right]}_{E_2}\right\}$

Now, $E_1=\cos\left\{\cos^{-1}\left[\frac{1-(3/4)^2}{1+(3/4)^2}\right]\right\}$

$\Rightarrow \quad E_1=\cos\cos^{-1}\left(\frac{7}{25}\right)=\frac{7}{25}$

Also $E_2=\sin\left[2\cot^{-1}\left(\frac{1}{2}\right)\right]=\sin\left(2\tan^{-1}2\right)$

$\Rightarrow \quad E_2=\sin\left\{\sin^{-1}\left[\frac{2\cdot 2}{1+(2)^2}\right]\right\}$

$\Rightarrow \quad E_2=\sin\sin^{-1}\left(\frac{4}{5}\right)=\frac{4}{5}$

$\therefore \quad E=\tan^{-1}\left(\frac{7}{25}+\frac{4}{5}\right)$

$\Rightarrow \quad E=\tan^{-1}\left(\frac{27}{25}\right)=\tan^{-1}\left(1+\frac{2}{25}\right)>\frac{\pi}{4}$

[$\because$ tangent function is an increasing function]

5. $\frac{3\sin 2\theta}{5+4\cos 2\theta}=\frac{6\tan\theta}{9+\tan^2\theta}=\frac{2\tan\phi}{1+\tan\phi}=\sin 2\phi$

where $\tan\theta=3\tan\phi$

$\Rightarrow \quad \left(\frac{1}{2}\right)\sin^{-1}\left[\frac{3\sin 2\theta}{5+4\cos 2\theta}\right]=\phi=\tan^{-1}\left[\left(\frac{1}{3}\right)\tan\theta\right]$

so that $x=(1/3)\tan\theta$

6. Given that, $\theta=\tan^{-1}a$ and $\phi=\tan^{-1}b$ and $ab=-1$

$\therefore \quad \tan\theta\tan\phi=ab=-1$

$\Rightarrow \quad \tan\theta=-\cot\phi$

$\Rightarrow \quad \tan\theta=\tan\left(\frac{\pi}{2}+\phi\right)$

$\Rightarrow \quad \theta-\phi=\frac{\pi}{2}$

7. The given expression can be written as

$$E=\tan^{-1}\left(a\sqrt{\frac{a+b+c}{abc}}\right)+\tan^{-1}\left(b\sqrt{\frac{a+b+c}{abc}}\right)+\tan^{-1}\left(c\sqrt{\frac{a+b+c}{abc}}\right)$$

$\Rightarrow \quad E=\tan^{-1}(ay)+\tan^{-1}(by)+\tan^{-1}(cy)$

where $y=\sqrt{\frac{a+b+c}{abc}}$

$\therefore \quad y=\tan^{-1}\left(\frac{ay+by+cy-abc\,y^3}{1-aby^2-bcy^2-ac\,y^2}\right)$

$\Rightarrow \quad y=\tan^{-1}\left[y\left(\frac{a+b+c-abcy^2}{1-y^2(ab+bc+ca)}\right)\right]$

$\therefore \quad y=\tan^{-1}0=0$ or π

8. From the given relation, we have

$$\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}=\tan\alpha$$

$\Rightarrow \quad \frac{2\sqrt{1+x^2}}{2\sqrt{1-x^2}}=\frac{1+\tan\alpha}{1-\tan\alpha}$

(by componendo and dividendo)

$\Rightarrow \quad \frac{1+x^2}{1-x^2}=\left(\frac{\sin\alpha+\cos\alpha}{\cos\alpha-\sin\alpha}\right)^2=\frac{1+\sin 2\alpha}{1-\sin 2\alpha}$

$\Rightarrow \quad x^2=\sin 2\alpha$

9. $\cot^{-1}\frac{xy+1}{x-y}+\cot^{-1}\frac{yz+1}{y-z}+\cot^{-1}\frac{zx+1}{z-x}$

$=\cot^{-1}y-\cot^{-1}x+\cot^{-1}z-\cot^{-1}y+\cot^{-1}x-\cot^{-1}z$

$=0$

10. Since $\cot^{-1}\sqrt{\cos\alpha}-\tan^{-1}\sqrt{\cos\alpha}=u$,

$\Rightarrow \quad \frac{\pi}{2}-\tan^{-1}\sqrt{\cos\alpha}-\tan^{-1}\sqrt{\cos\alpha}=u$

$\Rightarrow \quad \frac{\pi}{2}-2\tan^{-1}\sqrt{\cos\alpha}=u$

$\Rightarrow \quad \cos^{-1}\left(\frac{1-\cos\alpha}{1+\cos\alpha}\right)=\frac{\pi}{2}-u$

$\therefore \quad \tan^2\frac{\alpha}{2}=\cos\left(\frac{\pi}{2}-u\right)=\sin u$

11. We have from the given equation,

$$\tan^{-1}\frac{(a+b)x}{x^2-ab}=\frac{\pi}{2}-\tan^{-1}\frac{(c+d)x}{x^2-cd}$$

$$\Rightarrow \tan^{-1}\frac{(a+b)x}{x^2-ab}=\cot^{-1}\frac{(c+d)x}{x^2-cd}$$

$$=\tan^{-1}\frac{x^2-cd}{(c+d)x}$$

$$\Rightarrow (x^2-ab)(x^2-cd)=(a+b)(c+d)x^2$$

$$\Rightarrow x^4-x^2\sum ab+abcd=0$$

12. $x^2+y^2+z^2=r^2$

Now, $\tan^{-1}\left(\frac{xy}{zr}\right)+\tan^{-1}\left(\frac{yz}{xr}\right)+\tan^{-1}\left(\frac{xz}{yr}\right)$

$$=\tan^{-1}\left[\frac{\frac{xy}{zr}+\frac{yz}{xr}+\frac{xz}{yr}-\frac{xyz}{r^3}}{1-\left(\frac{x^2+y^2+z^2}{r^2}\right)}\right]$$

$$=\tan^{-1}\left[\frac{\frac{xy}{zr}+\frac{yz}{xr}+\frac{xz}{yr}-\frac{xyz}{r^3}}{1-\frac{r^2}{r^2}}\right]$$

$$=\tan^2\infty=\frac{\pi}{2}$$

13. $\alpha=\sin^{-1}\left[\frac{4}{5}\sqrt{1-\frac{1}{9}}+\frac{1}{3}\sqrt{1-\frac{16}{25}}\right]$

$$\Rightarrow \alpha=\sin^{-1}\left[\frac{8\sqrt{2}}{15}+\frac{3}{15}\right]=\sin^{-1}\left(\frac{8\sqrt{2}+3}{15}\right)$$

Since $\frac{8\sqrt{2}+3}{15}<1$,

$$\therefore \quad a<\frac{\pi}{2} \qquad \text{(i)}$$

Now, $\beta=\left(\frac{\pi}{2}-\sin^{-1}\frac{4}{5}+\frac{\pi}{2}-\sin^{-1}\frac{1}{3}\right)=\pi-\alpha>\frac{\pi}{2}$

$$\therefore \quad \alpha<\beta \qquad \text{[using (i)]}$$

14. $\sin^{-1}x+\sin^{-1}\frac{1}{x}+\cos^{-1}x+\cos^{-1}\frac{1}{x}$

$$=(\sin^{-1}x+\cos^{-1}x)+\left[\sin^{-1}\left(\frac{1}{x}\right)+\cos^{-1}\left(\frac{1}{x}\right)\right]$$

$$=\frac{\pi}{2}+\frac{\pi}{2}=\pi$$

15. The given equation can be written as

$$\tan^{-1}x+\tan^{-1}\frac{1}{y}=\tan^{-1}3$$

$$\Rightarrow \tan^{-1}\left[\frac{x+(1/y)}{1-x\cdot(1/y)}\right]=\tan^{-1}3$$

$$\Rightarrow xy+1=3(y-x)$$

$$\therefore \quad y=\frac{3x+1}{3-x}$$

There is only one equation in two variables, so we choose $x = k$

$$\therefore \quad y=\frac{3k+1}{3-k}$$

Further, we have to find only +ve and integral solutions. This is possible only when $k = 1$ or 2. Hence, (1, 2) and (2, 7) are only two +ve integral solutions.

16. $\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$

$$=\frac{\pi}{3}+2\cdot\frac{\pi}{6}=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$$

17. Let $g(x)$ be the inverse of f, such that

$$f[g(x)]=x \qquad \text{(i)}$$

Given that $f(x)=4x^3-3x$

$$\therefore \quad f[g(x)]=4[g(x)]^3-3[g(x)]$$

Let $g(x)=\cos\theta$ (ii)

$$\Rightarrow f[g(x)]=4\cos^3\theta-3\cos\theta=\cos 3\theta$$

$$\Rightarrow x=\cos 3\theta \qquad \text{[using (i)]}$$

$$\Rightarrow \theta=\frac{1}{3}\cos^{-1}(x) \qquad \text{(iii)}$$

Hence, the inverse of the function is

$$g(x)=\cos\left[\frac{1}{3}\cos^{-1}(x)\right] \qquad \text{[using (ii) and (iii)]}$$

18. Since $-\frac{\pi}{2}\le\sin^{-1}x\le\frac{\pi}{2}$,

$$\therefore \quad \sin^{-1}\alpha=\frac{\pi}{2},\ \sin^{-1}\beta=\frac{\pi}{2} \text{ and } \sin^{-1}\gamma=\frac{\pi}{2}$$

$$\therefore \quad \alpha=\beta=\gamma=1$$

Thus, $\alpha\beta+\alpha\gamma+\gamma\beta=3$

19. Inverse function is $\sin(\cot^{-1}x)$.

Let $\cot^{-1}x=\theta$ or $\cot\theta=x$

Therefore, $\sin\theta=\frac{1}{\sqrt{1+x^2}}$ or $\theta=\sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

Therefore, $\sin(\cot^{-1}x)=\sin\theta=\sin\left(\sin^{-1}\dfrac{1}{\sqrt{1+x^2}}\right)$

$=\dfrac{1}{\sqrt{1+x^2}}=(1+x^2)^{-1/2}$

20. Let $\cot^{-1}(\sqrt{2}-1)=\theta$

$\Rightarrow \cot\theta=\sqrt{2}-1$

$\therefore \cos^{-1}(\cos 2\theta)=\cos^{-1}\left(\dfrac{1-\tan^2\theta}{1+\tan^2\theta}\right)$

$\Rightarrow \cos^{-1}(\cos 2\theta)=\cos^{-1}\left(\dfrac{\cot^2\theta-1}{\cot^2\theta+1}\right)$

$\Rightarrow \cos^{-1}(\cos 2\theta)=\cos^{-1}\left(\dfrac{2-2\sqrt{2}}{4-2\sqrt{2}}\right)$

$\therefore \cos^{-1}(\cos 2\theta)=\cos^{-1}\left(-\dfrac{1}{\sqrt{2}}\right)=\dfrac{3\pi}{4}$

21. Let $\cot^{-1}x=\alpha$

$\therefore \cot\alpha=x \Rightarrow \sin\alpha=\dfrac{1}{\sqrt{1+x^2}}$

Thus, the given expression $=\cos\tan^{-1}\dfrac{1}{\sqrt{1+x^2}}$

Let $\tan^{-1}\dfrac{1}{\sqrt{1+x^2}}=\theta \Rightarrow \tan\theta=\dfrac{1}{\sqrt{1+x^2}}$

or $\cos\theta=\sqrt{\dfrac{x^2+1}{x^2+2}}$

22. $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$

$\Rightarrow \cos^{-1}[xy-\sqrt{1-y^2}\sqrt{1-x^2}]=\pi-\cos^{-1}z$

$\Rightarrow xy-\sqrt{1-x^2}\sqrt{1-y^2}=\cos(\pi-\cos^{-1}z)$

$\Rightarrow xy-\sqrt{1-x^2}\sqrt{1-y^2}=-z$

$\Rightarrow xy+z=\sqrt{1-x^2}\sqrt{1+y^2}$

On squaring both sides, we get

$x^2y^2+z^2+2xyz=1-x^2-y^2+x^2y^2$

$\Rightarrow x^2+y^2+z^2=1-2xyz$

23. Inverse function is $3\tan^{-1}a$.

We know that $\tan 3A=\dfrac{3\tan A-\tan^3 A}{1-3\tan^2 A}$

or $3A=\tan^{-1}\left[\dfrac{3\tan A-\tan^3 A}{1-3\tan^2 A}\right]$ (i)

Substituting $\tan A=a$ and $A=\tan^{-1}(a)$ in Eq. (i), we get

$3\tan^{-1}a=\tan^{-1}\left(\dfrac{3a-a^3}{1-3a^2}\right)$

24. $\theta=\cos^{-1}\left(\dfrac{x}{2}\cdot\dfrac{y}{3}-\sqrt{1-\dfrac{x^2}{4}}\sqrt{1-\dfrac{y^2}{9}}\right)$

$\Rightarrow \cos\theta=\dfrac{xy}{6}-\dfrac{\sqrt{4-x^2}\sqrt{9-y^2}}{6}$

$\Rightarrow (xy-6\cos\theta)^2=(4-x^2)(9-y^2)$

$=36-9x^2-4y^2+x^2y^2$

$\Rightarrow 36\cos^2\theta-12xy\cos\theta+4y^2+9x^2=36$

$\therefore 9x^2-12xy\cos\theta+4y^2=36\sin^2\theta$

25. $2\tan^{-1}x=2\tan^{-1}a+2\tan^{-1}b$

or $\tan^{-1}x=\tan^{-1}\dfrac{a+b}{1-ab}$

26. Series is $\tan^{-1}\left(\dfrac{1}{3}\right)+\tan^{-1}\left(\dfrac{1}{7}\right)+\tan^{-1}\left(\dfrac{1}{13}\right)+\cdots+\tan^{-1}\left(\dfrac{1}{n^2+n+1}\right)$

We know that the nth term of the series

$=T_n=\tan^{-1}\left(\dfrac{1}{1+n+n^2}\right)=\tan^{-1}\left[\dfrac{(n+1)-n}{1+(n+1)n}\right]$

$=\tan^{-1}(n+1)-\tan^{-1}n$. Substituting n = 1, 2, 3,..., n and adding, we get the sum of the series

$(S_n)=\tan^{-1}(n+1)-\tan^{-1}1$

Substituting $n=\infty$, we get

$(S_\infty)=\tan^{-1}\infty-\dfrac{\pi}{4}=\dfrac{\pi}{2}-\dfrac{\pi}{4}=\dfrac{\pi}{4}$

27. Since $\left(\tan^{-1}x\right)^2+\left(\cot^{-1}x\right)^2=\dfrac{5\pi^2}{8}$,

$\Rightarrow \left(\tan^{-1}x+\cot^{-1}x\right)^2-2\tan^{-1}x\left(\dfrac{\pi}{2}-\tan^{-1}x\right)=\dfrac{5\pi^2}{8}$

$\Rightarrow \dfrac{\pi^2}{4}-2\times\dfrac{\pi}{2}\tan^{-1}x+2\left(\tan^{-1}x\right)^2=\dfrac{5\pi^2}{8}$

$\Rightarrow 2\left(\tan^{-1}x\right)^2-\pi\tan^{-1}x-\dfrac{3\pi^2}{8}=0$

$\Rightarrow \tan^{-1} x = -\frac{\pi}{4}, \frac{3\pi}{4}$

Since $3\pi/4$ does not lie in the principal value of the argument,

$\therefore \quad \tan^{-1} x = -\frac{\pi}{4} \quad \Rightarrow \quad x = -1$

28. Inverse function is $\tan^{-1}(x) + \tan^{-1}(y) + \tan^{-1}(z) = \frac{\pi}{2}$.

The given function may be written as

$$\tan^{-1}\left(\frac{x+y}{1-xy}\right) + \tan^{-1} z = \frac{\pi}{2}$$

or $$\tan^{-1}\left\{\frac{\left[\frac{(x+y)}{(1-xy)}\right]+z}{1-\left[\frac{(x+y)}{(1-xy)}\right]\cdot z}\right\} = \tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right] = \frac{\pi}{2}$$

or $$\frac{x+y+z-xyz}{1-xy-yz-zx} = \tan\left(\frac{\pi}{2}\right) = \infty = \frac{1}{0}$$

or $1 - xy - yz - zx = 0$

or $xy + yz + zx = 1 \qquad \left[\because \quad \tan\left(\frac{\pi}{2}\right) = \infty\right]$

29. Since $\sqrt{x^2 - 3x + 2} \geq 0$,

$\Rightarrow 0 \leq \tan^{-1}\sqrt{x^2 - 3x + 2} < \frac{\pi}{2}$ (i)

Also, $\sqrt{4x - x^2 - 3} \geq 0$

$\Rightarrow 0 \leq \cos^{-1}\sqrt{4x - x^2 - 3} \leq \frac{\pi}{2}$ (ii)

Adding (i) and (ii), we have

$0 \leq \tan^{-1}\sqrt{x^2 - 3x + 2} + \cos^{-1}\sqrt{4x - x^2 - 3} < \pi$

$\therefore$ The given equation possesses no solution.

30. Inverse function is

$$\sin^{-1}(x) + \sin^{-1}(y) + \sin^{-1}(z) = \frac{3\pi}{2}$$

Since $\left|\sin^{-1} x\right| \leq \frac{\pi}{2}$, therefore

$$\sin^{-1} x = \sin^{-1} y = \sin^{-1} z = \frac{\pi}{2}$$

or $\sin^{-1} x = \frac{\pi}{2}$ or $x = \sin\frac{\pi}{2} = 1$

Similarly, $y = 1$ and $z = 1$. Therefore,

$(x^{100} + y^{100} + z^{100}) - (x^{101} + y^{101} + z^{101})$

$= [(1)^{100} + (1)^{100} + (1)^{100}] - [(1)^{101} + (1)^{101} + (1)^{101}]$

$= 3 - 3 = 0$

31. Let $\cos^{-1} x = \theta$

Also, $E = \frac{1 + \tan\theta/2}{1 - \tan\theta/2} + \frac{1 - \tan\theta/2}{1 + \tan\theta/2}$

$$E = \frac{2(1 + \tan^2\theta/2)}{1 - \tan^2\theta/2}$$

$\therefore \quad E = \frac{2}{\cos x} = \frac{2}{x}$

32. Inverse function is $3\sin^{-1}\left(\frac{2x}{1+x^2}\right) - 4\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

$+ 2\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{\pi}{3}$

Substituting $x = \tan\theta$, we get

$$3\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) - 4\cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right) + 2\tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)$$

$$= \frac{\pi}{3}$$

or $3\sin^{-1}(\sin 2\theta) - 4\cos^{-1}(\cos 2\theta) + 2\tan^{-1}(\tan 2\theta) = \frac{\pi}{3}$

$$\left[\because \frac{2\tan\theta}{1+\tan^2\theta} = \sin 2\theta \quad \frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos 2\theta\right]$$

or $3(2\theta) - 4(2\theta) + 2(2\theta) = \frac{\pi}{3}$

or $2\theta = \frac{\pi}{3}$ or $\theta = \frac{\pi}{6}$

$\left[\because \sin^{-1}(\sin\theta) = \theta, \cos^{-1}(\cos\theta) = \theta \text{ and } \tan^{-1}(\tan\theta) = \theta\right]$

Therefore, $x = \tan\theta = \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$

33. Since $\sin^{-1}\left(\frac{1}{3}\right) + \sin^{-1}\left(\frac{2}{3}\right) = \sin^{-1} x$

$$\Rightarrow \sin^{-1}\left(\frac{1}{3}\sqrt{1-\frac{4}{9}} + \sqrt{1-\frac{1}{9}}\frac{2}{3}\right) = \sin^{-1} x$$

$$\Rightarrow \frac{\sqrt{5}}{9} + \frac{4\sqrt{2}}{9} = x$$

$$\therefore \quad x = \frac{4\sqrt{2} + \sqrt{5}}{9}$$

34. Inverse function is $\sin[\cot^{-1}(x+1)] = \cos(\tan^{-1} x)$

We know that

$$\cot^{-1}(x+1) = \sin^{-1}\left(\frac{1}{\sqrt{(1+x)^2 + 1}}\right)$$

$$= \sin^{-1}\left(\frac{1}{\sqrt{(x^2 + 2x + 2)}}\right)$$

Therefore,

$$\sin[\cot^{-1}(x+1)] = \sin\left\{\sin^{-1}\left[\frac{1}{\sqrt{(x^2+2x+2)}}\right]\right\}$$

$$= \frac{1}{\sqrt{(x^2+2x+2)}}$$

We also know that $\tan^{-1} x = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

Therefore,

$$\cos(\tan^{-1} x) = \cos\left\{\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right\} = \frac{1}{\sqrt{1+x^2}}$$

$$\left[\because \cos(\cos^{-1}\theta) = \theta\right]$$

Thus, $\frac{1}{\sqrt{x^2+2x+2}} = \frac{1}{\sqrt{1+x^2}}$ or $x^2+2x+2 = x^2+1$

or $x = -\frac{1}{2}$

35. Since A, B, C are angles of a triangle, $A+B+C = \pi$

$$\Rightarrow \quad C = \pi - (A+B) \qquad \text{(i)}$$

Now, $A+B = \tan^{-1} 2 + \tan^{-1} 3$

$$= \pi + \tan^{-1}\left(\frac{2+3}{1-2\cdot 3}\right)$$

$$\left[\because \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)\right.$$

$$\left.\text{for } x>0, y>0 \text{ and } xy>1\right]$$

$$= \pi + \tan^{-1}(-1) = \pi - \tan^{-1} 1$$

$$= \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$\therefore$ From (i), we have $C = \pi - \frac{3\pi}{4} = \frac{\pi}{4}$

36. $\left(\frac{\pi}{2} - \cos^{-1} x\right) + \left(\frac{\pi}{2} - \cos^{-1} y\right) = \frac{2\pi}{3}$

$\therefore \quad \cos^{-1} x + \cos^{-1} y = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$

37. Inverse function is

$$\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$$

We know that $0 \le \cos^{-1} x \le \pi$. Therefore,

$$\cos^{-1} x = \cos^{-1} y = \cos^{-1} z = \pi$$

or $x = \cos\pi = -1$, $y = \cos\pi = -1$, $z = \cos\pi = -1$

Thus,

$$xy + yz + zx = (-1)\times(-1) + (-1)\times(-1) + (-1\times -1) = 3$$

38. Since $\alpha, \beta, \gamma \in x^3 + px^2 + 2x + p = 0$

$\therefore$

$$\alpha+\beta+\gamma = -p$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 2$$

$$\alpha\beta\gamma = -p$$

Hence, the required general value is

$$E = n\pi + \tan^{-1}\frac{\alpha+\beta+\gamma-a\beta\gamma}{1-\alpha\beta-\beta\gamma-\gamma\alpha}$$

$$\Rightarrow \quad E = n\pi + \tan^{-1}\frac{-p+p}{1-2} = n\pi + \tan^{-1}(0) = n\pi$$

39. Since $6\sin^{-1}(x^2-6x+8.5) = \pi$,

$$\Rightarrow \quad \sin^{-1}(x^2-6x+8.5) = \frac{\pi}{6}$$

$$\Rightarrow \quad x^2-6x+8.5 = \sin\left(\frac{\pi}{6}\right)$$

$$\Rightarrow \quad x^2-6x+8.5 = 0.5$$

$$\Rightarrow \quad x^2-6x+8 = 0$$

$$\Rightarrow \quad (x-4)(x-2) = 0$$

$\therefore \quad x = 2, 4$

40. We know that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

Adding, we get

$$2\sin^{-1} x = 30^\circ + 90^\circ = 120^\circ$$

$\therefore \quad \sin^{-1} x = 60^\circ \qquad \therefore \quad x = \sin 60^\circ = \frac{\sqrt{3}}{2}$

41. The equation is $\cos^{-1} p + \cos^{-1} q + \cos^{-1} r = \pi$.

The given equation may be written as

$$\cos^{-1} p + \cos^{-1} q = \pi - \cos^{-1} r$$

or $\cos^{-1}(pq + \sqrt{1-p^2}\sqrt{1-q^2}) = -\cos^{-1} r = \cos^{-1}(-r)$

$$\left[\because \cos^{-1}(-\theta) = \pi - \cos^{-1}\theta\right]$$

or $pq + \sqrt{1-p^2}\sqrt{1-q^2} = (-r)$

or $pq + r = -\sqrt{1-p^2}\sqrt{1-q^2}$

Squaring both sides of this equation, we get

$$= p^2q^2 + 2pqr + r^2 = (1-p^2)(1-q^2)$$

$$= 1 - q^2 - p^2 + p^2q^2$$

or $p^2 + q^2 + r^2 + 2pqr = 1$

42. $\sec^{-1} t = \cos^{-1}\frac{1}{t}$ and $\sin^{-1} z + \cos^{-1} z = \frac{\pi}{2}$

$\therefore \quad \sin^{-1}\sqrt{x^2+2x+1} + \cos^{-1}\frac{1}{\sqrt{x^2+2x+1}} = \frac{\pi}{2}$

The above equation implies that

$$\sqrt{x^2+2x+1}=\frac{1}{\sqrt{x^2+2x+1}}$$

or $x^2+2x+1=1$ or $x(x+2)=0$

$\therefore \quad x=0,-2 \quad \therefore \quad x=-2$ as $x\neq 0$

$$2\sec^{-1}\frac{x}{2}+\sin^{-1}\frac{x}{2}$$

$$=2\cos^{-1}(-1)+\sin^{-1}(-1)$$

$$=\cos^{-1}(-1)+\cos^{-1}(-1)+\sin^{-1}(-1)$$

$$=\pi+\frac{\pi}{2}=\frac{3\pi}{2}$$

43. Let $T_n=\cot^{-1}\left(n^2+\frac{3}{4}\right)$

$$\Rightarrow T_n=\cot^{-1}\left(\frac{4n^2+3}{4}\right)$$

$$\Rightarrow T_n=\tan^{-1}\left(\frac{4}{4n^2+3}\right)=\tan^{-1}\left[\frac{1}{n^2+(3/4)}\right]$$

$$\Rightarrow T_n=\tan^{-1}\left\{\frac{1}{1+[n^2-(1/4)]}\right\}$$

$$\Rightarrow T_n=\tan^{-1}\left\{\frac{1}{1+[n-(1/2)][n+(1/2)]}\right\}$$

$$\Rightarrow T_n=\tan^{-1}\left\{\frac{[n+(1/2)]-[n-(1/2)]}{1+[n+(1/2)][n-(1/2)]}\right\}$$

$$\Rightarrow T_n=\tan^{-1}\left(n+\frac{1}{2}\right)-\tan^{-1}\left(n-\frac{1}{2}\right)$$

Now, $S_n=T_1+T_2+T_3+\cdots+T_n$

$$\Rightarrow S_n=\tan^{-1}\left(\frac{3}{2}\right)-\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{5}{2}\right)-\tan^{-1}\left(\frac{3}{2}\right)$$

$$+\tan^{-1}\left(\frac{7}{2}\right)-\tan^{-1}\left(\frac{5}{2}\right)+\cdots+\tan^{-1}\left(n+\frac{1}{2}\right)-\tan^{-1}\left(n-\frac{1}{2}\right)$$

$$\therefore \quad S_\infty=\tan^{-1}\infty-\tan^{-1}\left(\frac{1}{2}\right)$$

$$\Rightarrow S_\infty=\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right)$$

$$\Rightarrow S_\infty=\cot^{-1}\left(\frac{1}{2}\right)$$

$$\therefore \quad S_\infty=\tan^{-1}2$$

44. Given: Inverse equation is $\cos^{-1}\left(\frac{x}{a}\right)+\cos^{-1}\left(\frac{y}{b}\right)=\alpha$

We know that

$$\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left[xy-\sqrt{1-x^2}\sqrt{1-y^2}\right]$$

Therefore, $\cos^{-1}\left(\frac{x}{a}\right)+\cos^{-1}\left(\frac{y}{b}\right)$

$$=\cos^{-1}\left[\frac{x}{a}\cdot\frac{y}{b}-\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}\right]=\alpha$$

or $\frac{xy}{ab}-\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}=\cos\alpha$

or $\frac{xy}{ab}-\cos\alpha=\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}$

Squaring both sides, we get

$$\left(\frac{xy}{ab}-\cos\alpha\right)^2=\left(1-\frac{x^2}{a^2}\right)\left(1-\frac{y^2}{b^2}\right)$$

or $\frac{x^2y^2}{a^2b^2}+\cos^2\alpha-\frac{2xy}{ab}\cos\alpha=1-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{x^2y^2}{a^2b^2}$

or $\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}=1-\cos^2\alpha=\sin^2\alpha$

45. Since $\sin^{-1}x+\cos^{-1}x+\tan^{-1}x=\frac{\pi}{2}+\tan^{-1}x$

Now, $-\frac{\pi}{2}\le\tan^{-1}x\le\frac{\pi}{2}$

$$\Rightarrow 0\le\frac{\pi}{2}+\tan^{-1}x\le\pi$$

$\therefore \quad \alpha=0,\beta=\pi$

46. Give: Inverse function is $\tan\left[\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right]$

Let $\cos^{-1}\left(\frac{4}{5}\right)=\theta$ or $\cos\theta=\frac{4}{5}$

Therefore $\tan\theta=\frac{3}{4}$ or $\theta=\tan^{-1}\left(\frac{3}{4}\right)$

Substituting $\cos^{-1}\left(\frac{4}{5}\right)=\theta=\tan^{-1}\left(\frac{3}{4}\right)$ in the given function, we get

$$\tan\left[\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right]=\tan\left[\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right]$$

$$\left[\because\ \cos^{-1}\left(\frac{4}{5}\right)=\tan^{-1}\left(\frac{3}{4}\right)\right]$$

We know $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1+xy}\right)$. Thus,

$$\tan\left[\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right]=\tan\left[\tan^{-1}\left\{\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{3}{4}.\frac{2}{3}}\right\}\right]$$

$$=\tan\left[\tan^{-1}\frac{(17/12)}{(6/12)}\right]=\tan\left[\tan^{-1}\left(\frac{17}{6}\right)\right]=\frac{17}{6}$$

$$\left(\because\ \tan^{-1}(\tan\theta)=\theta\ \text{ when }-\frac{\pi}{2}<\theta<\frac{\pi}{2}\right)$$

HINTS AND SOLUTIONS—EXERCISE SET 2

1. $x+\frac{1}{x}=2$

$\Rightarrow\ x=1$

So, the principal value of $\sin^{-1}x$ is $\pi/2$.

2. Let $x=\cos\theta$. Then,

$$0<x<1\ \Rightarrow\ 0<\cos\theta<1\ \Rightarrow\ 0<\theta<\frac{\pi}{2}$$

Now, $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{1+x}\right)=\tan^{-1}\sqrt{\frac{1-x}{1+x}}$

$$=\tan^{-1}\left(\tan\frac{\theta}{2}\right)=\frac{\theta}{2}=\frac{1}{2}\cos^{-1}x$$

Thus, option (a) is true.

$$\cos^{-1}\sqrt{\frac{1+x}{2}}=\cos^{-1}\left(\cos\frac{\theta}{2}\right)=\frac{1}{2}\theta=\frac{1}{2}\cos^{-1}x$$

So, option (b) is true.

$$\sin^{-1}\sqrt{\frac{1-x}{2}}=\sin^{-1}\left(\sin\frac{\theta}{2}\right)=\frac{\theta}{2}=\frac{1}{2}\cos^{-1}x$$

So, option (c) is true.

3. We have $\cos\left\{\tan^{-1}\left[\tan\left(\frac{15\pi}{4}\right)\right]\right\}$

$$=\cos\left\{\tan^{-1}\left[\tan\left(4\pi-\frac{\pi}{4}\right)\right]\right\}$$

$$=\cos\left\{\tan\left[\tan^{-1}\left(-\frac{\pi}{4}\right)\right]\right\}=\cos\left(-\frac{\pi}{4}\right)=\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}$$

4. Obviously, $x=\sin(\theta+\beta)$ and $y=\sin(\theta-\beta)$

$\Rightarrow\ 1+xy=1+\sin(\theta+\beta)\sin(\theta-\beta)$

$$=1+\sin^2\theta-\sin^2\beta=\sin^2\theta+\cos^2\beta$$

5. We have $\sin^{-1}(\sin 4)=\sin^{-1}[\sin(\pi-4)]=\pi-4$

$\therefore\ x^2-\lambda x+\sin^{-1}(\sin 4)>0$ for all $x\in\mathbb{R}$

$\Rightarrow\ x^2-\lambda x+(\pi-4)>0$ for all $x\in\mathbb{R}$

$\Rightarrow\ \lambda^2-4(\pi-4)<0\ \Rightarrow\ \lambda^2+16-4\pi<0$

But $\lambda^2+16-4\pi>0$ for all $\lambda\in\mathbb{R}$

So, there is no value of λ for which the given inequation holds true.

6. We have $\sin\{\cot^{-1}[\cos(\tan^{-1}x)]\}$

$$\sin\left\{\cot^{-1}\left[\frac{1}{\sqrt{1+\tan^2(\tan^{-1}x)}}\right]\right\}=\sin\left[\cot^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right]$$

$$=\frac{1}{\sqrt{1+\cot^2\left\{\cot^{-1}\frac{1}{\sqrt{1+x^2}}\right\}}}$$

$$=\frac{1}{\sqrt{1+[1/(1+x^2)]}}=\sqrt{\frac{1+x^2}{2+x^2}}$$

7. Let $f(x)=\sin^{-1}x-\tan^{-1}x$

Clearly, $f(x)$ is defined for all $x\in[-1,1]$.

Also, $f'(x)=\frac{1}{\sqrt{1-x^2}}-\frac{1}{1+x^2}>0$ for all $x\in(-1,1)$

$\Rightarrow\ f(x)$ is increasing on $(-1, 1)$.

$\Rightarrow\ f(-1)\le f(x)\le f(1)$

$$\Rightarrow\ -\frac{\pi}{2}+\frac{\pi}{4}<\sin^{-1}x-\tan^{-1}x\le\frac{\pi}{2}-\frac{\pi}{4}$$

$$\Rightarrow\ -\frac{\pi}{4}\le a\le\frac{\pi}{4}\ \Rightarrow\ a\in\left[-\frac{\pi}{4},\frac{\pi}{4}\right]$$

8. The equation is $\tan^{-1}\frac{a+x}{a}+\tan^{-1}\frac{a-x}{a}=\frac{\pi}{6}$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{a+x}{a}+\frac{a-x}{a}}{1-\frac{a+x}{a}\cdot\frac{a-x}{a}}\right)=\frac{\pi}{6}$$

$$\Rightarrow \frac{2a^2}{x^2}=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}} \Rightarrow x^2=2\sqrt{3}a^2$$

9. Since $\sqrt{1\pm\sin x}=\cos\frac{x}{2}\pm\sin\frac{x}{2}$

$$\Rightarrow \text{L.H.S.}=\cot^{-1}\left[\frac{2\cos(x/2)}{-2\sin(x/2)}\right]=\cot^{-1}\left(-\cot\frac{x}{2}\right)$$

$$\Rightarrow \text{L.H.S.}=\cot^{-1}\left[\cot\left(\pi-\frac{x}{2}\right)\right]=\pi-\frac{x}{2}$$

10. $\cos^{-1}\sqrt{\frac{2}{3}}-\cos^{-1}\left(\frac{\sqrt{6}+1}{2\sqrt{3}}\right)$

$$=\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)-\tan^{-1}\left(\frac{\sqrt{3}-\sqrt{2}}{1+\sqrt{6}}\right)$$

$$=\tan^{-1}\left(\frac{\frac{1}{\sqrt{2}}-\frac{\sqrt{3}-\sqrt{2}}{1+\sqrt{6}}}{1+\frac{\sqrt{3}-\sqrt{2}}{\sqrt{2}(1+\sqrt{6})}}\right)$$

$$=\tan^{-1}\left(\frac{1+\sqrt{6}-\sqrt{6}+2}{\sqrt{2}+2\sqrt{3}+\sqrt{3}-\sqrt{2}}\right)$$

$$=\tan^{-1}\left(\frac{3}{3\sqrt{3}}\right)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{6}$$

11. $\tan(A-B)=\frac{\tan A-\tan B}{1+\tan A\tan B}=\frac{\frac{x\sqrt{3}}{2K-x}-\frac{2x-K}{K\sqrt{3}}}{1+\frac{x\sqrt{3}}{2K-x}\cdot\frac{2x-K}{K\sqrt{3}}}=\frac{1}{\sqrt{3}}$

$$\Rightarrow A-B=30^\circ$$

12. We know that $\sin^{-1}\left(\frac{2x}{1+x^2}\right)=2\tan^{-1}x$, if $-1\le x\le 1$

and $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)=2\tan^{-1}x$, if $0\le x<\infty$

$$\therefore \tan\left(\frac{1}{2}\sin^{-1}\frac{2x}{1+x^2}+\frac{1}{2}\cos^{-1}\frac{1-x^2}{1+x^2}\right)$$

$$=\tan(2\tan^{-1}x) \quad \text{for } 0\le x\le 1$$

$$=\tan\left(\tan^{-1}\frac{2x}{1-x^2}\right)=\frac{2x}{1-x^2}$$

13. We have $\tan^{-1}\left(\frac{1-x}{1+x}\right)=\tan^{-1}1-\tan^{-1}x=\frac{\pi}{4}-\tan^{-1}x$

Since $0\le x\le 1$, therefore, $0\le\tan^{-1}x\le\frac{\pi}{4}$

$$\Rightarrow 0\ge-\tan^{-1}x\ge\frac{-\pi}{4} \Rightarrow \frac{\pi}{4}\ge\frac{\pi}{4}-\tan^{-1}x\ge 0$$

$$\Rightarrow \frac{\pi}{4}\ge\tan^{-1}\left(\frac{1-x}{1+x}\right)\ge 0$$

14. We have $(\sin^{-1}x)^2+(\cos^{-1}x)^2=\frac{5\pi^2}{8}$,

$$\Rightarrow (\sin^{-1}x)^2+\left(\frac{\pi}{2}-\sin^{-1}x\right)^2=\frac{5\pi^2}{8}$$

$$\Rightarrow 2(\sin^{-1}x)^2-\pi\sin^{-1}x-\frac{3\pi^2}{8}=0$$

$$\Rightarrow \sin^{-1}x=\frac{\pi\pm 2\pi}{4} \Rightarrow \sin^{-1}x=-\frac{\pi}{4}\Rightarrow x=-\frac{1}{\sqrt{2}}$$

15. $f(x)=\sin^{-1}x+\tan^{-1}x+\sec^{-1}x$. Clearly domain of $f(x)$ is $x=\pm 1$. Thus, the range is $[f(1), f(-1)]$, i.e. $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$.

16. The given equation is meaningful if

$$0\le\sqrt{x^2-x+1}\le 1,\ x^2-x\ge 0 \text{ and } 0\le\sqrt{x^2-x}\le 1$$

$$\Rightarrow x^2-x+1\le 1,\ x^2-x\ge 0 \text{ and } x^2-x\le 1$$

$$\Rightarrow x^2-x\le 0,\ x^2-x\ge 0 \text{ and } x^2-x\le 1$$

$$\Rightarrow x^2-x=0 \Rightarrow x=0, 1$$

Clearly, these values of x satisfy the given equation.

17. $\cos^{-1}\left[\frac{p}{a}\cdot\frac{q}{b}-\sqrt{\left(1-\frac{p^2}{a^2}\right)}\sqrt{\left(1-\frac{q^2}{b^2}\right)}\right]=\alpha$

$$\Rightarrow \frac{pq}{ab}-\sqrt{\left(1-\frac{p^2}{a^2}\right)}\sqrt{\left(1-\frac{q^2}{b^2}\right)}=\cos\alpha$$

$$\Rightarrow \left(\frac{pq}{ab}-\cos\alpha\right)^2=1-\frac{p^2}{a^2}-\frac{q^2}{b^2}+\frac{p^2q^2}{a^2b^2}$$

$$\Rightarrow \frac{p^2q^2}{a^2b^2}+\cos^2\alpha-\frac{2pq}{ab}\cos\alpha=1-\frac{p^2}{a^2}-\frac{q^2}{b^2}+\frac{p^2q^2}{a^2b^2}$$

$$\Rightarrow \frac{p^2}{a^2}-\frac{2pq}{ab}\cos\alpha+\frac{q^2}{b^2}=1-\cos^2\alpha=\sin^2\alpha$$

$$\therefore k=-\frac{2pq}{ab}$$

18. $(t+1)(t-2)>0 \rightarrow t<-1$ or $t>2$

$\therefore \quad x<-1$ or $x>2$ (i)

Now both $\sin^{-1}x$ and $\cos^{-1}x$ are defined for $-1\le x\le 1$ but $\sec^{-1}x$ is defined for $x\le -1$ or $x\ge 1$. Hence, because of (i) only $\sec^{-1}x$ is defined.

19. Clearly, Statement I is true.

Putting $x=\cos\theta$, we get

$$2\sin^{-1}\sqrt{\frac{1-x}{2}}=2\cos^{-1}\sqrt{\frac{1+x}{2}}=\theta=\cos^{-1}x$$

So, Statement II is also true.

20. The given equation can be written as

$$3\tan^{-1}\left(2-\sqrt{3}\right)=\tan^{-1}\left(\frac{1}{x}\right)+\tan^{-1}\left(\frac{1}{3}\right)$$

$$\Rightarrow \quad 3\tan^{-1}\left(2-\sqrt{3}\right)=\tan^{-1}\left(\frac{1}{x}\right)+\tan^{-1}\left(\frac{1}{3}\right)$$

$$\Rightarrow \quad 3(15^\circ)=\tan^{-1}\frac{(1/x)+(1/3)}{1-(1/x)\times(1/3)}$$

$$\Rightarrow \quad 1=\frac{3+x}{3x-1} \quad \Rightarrow \quad x=2$$

21. Since $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\pi$,

$$\therefore \quad \sin^{-1}x+\sin^{-1}y=\pi-\sin^{-1}z$$

$$\Rightarrow \quad \sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)=\pi-\sin^{-1}(z)$$

$$\Rightarrow \quad x\sqrt{1-y^2}+y\sqrt{1-x^2}=\sin\left[\pi-\sin^{-1}(z)\right]$$

$$\Rightarrow \quad \sin(\sin^{-1}z)=z$$

$$\Rightarrow \quad x^2(1-y^2)=z^2+y^2(1-x^2)-2zy\sqrt{1-x^2}$$

$$\Rightarrow \quad (x^2-y^2-z^2)^2=4y^2z^2(1-x^2)$$

$$\Rightarrow \quad x^4+y^4+z^4-2x^2y^2-2x^2z^2+2y^2z^2=4y^2z^2-4x^2y^2z^2$$

$$\Rightarrow \quad x^4+y^4+z^4+4x^2y^2z^2=2(x^2y^2+y^2z^2+z^2x^2)$$

$$\Rightarrow \quad K=2$$

22. $\sec^2(\tan^{-1}2)+\operatorname{cosec}^2(\cot^{-1}3)$

$$=\sec^2\left(\cos^{-1}\frac{1}{\sqrt{1+2^2}}\right)+\operatorname{cosec}^2\left(\tan^{-1}\frac{1}{3}\right)$$

$$\left[\because \quad \tan^{-1}x=\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) \quad \text{and} \quad \cot^{-1}x=\tan^{-1}\left(\frac{1}{x}\right)\right]$$

$$=\sec^2(\sec^{-1}\sqrt{5})+\operatorname{cosec}^2\left(\sin^{-1}\frac{1/3}{\sqrt{1+(1/3)}}\right)$$

$$\left[\because \tan^{-1}x=\sin^{-1}\frac{x}{\sqrt{1+x^2}}\right]$$

$$=\sec^2(\sec^{-1}\sqrt{5})+\operatorname{cosec}^2(\operatorname{cosec}^{-1}\sqrt{10})$$

$$=[\sec(\sec^{-1}\sqrt{5})]^2+[\operatorname{cosec}(\operatorname{cosec}^{-1}\sqrt{10})]^2$$

$$=(\sqrt{5})^2+(\sqrt{10})^2=5+10=15$$

Another method:

$$\sec^2(\tan^{-1}2)+\operatorname{cosec}^2(\cot^{-1}3)$$

$$=1+\tan^2(\tan^{-1}2)+1+\cot^2(\cot^{-1}3)$$

$$=1+2^2+1+3^2=1+4+1+9=15$$

23. $\cos(2\cos^{-1}x+\sin^{-1}x)$

$$=\cos\left(\cos^{-1}x+\frac{\pi}{2}\right)$$

$$=-\sin(\cos^{-1}x)$$

Let $\cos^{-1}x=\theta$

$\therefore \quad x=\cos\theta$

$\therefore \quad \sqrt{1-x^2}=\sin\theta$

$\therefore \quad \cos(2\cos^{-1}x+\sin^{-1}x)=-\sin(\theta)=-\sqrt{1-x^2}$

At $x=\frac{1}{5}$, value $=-\sqrt{1-\frac{1}{25}}=-\sqrt{\frac{24}{25}}=-\frac{2\sqrt{6}}{5}$

24. $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{(x^2+x+1)}=\frac{\pi}{2}$

$$=\tan^{-1}\sqrt{x^2+x}=\frac{\pi}{2}-\sin^{-1}\sqrt{x^2+x+1}$$

$$=\cos^{-1}\sqrt{x^2+x+1}$$

$$\Rightarrow \quad \tan^{-1}\sqrt{x^2+x}=\tan^{-1}\left(\frac{\sqrt{1-\left(\sqrt{x^2+x+1}\right)}}{\sqrt{x^2+x+1}}\right)$$

$$\Rightarrow \quad \sqrt{x^2+x}=\frac{\sqrt{-x^2-x}}{\sqrt{x^2+x+1}}$$

The only real solutions of this equation are $x=0$ and $x=-1$, i.e. there are only two real solutions.

25. Since the maximum value of $\sin^{-1}x_i$ is $\pi/2$ and it occurs when $x_i=1$,

$\therefore \quad \sum_{i=1}^{2n}\sin^{-1}x_i=n\pi$ is possible only

when $x_1=x_2=\cdots=x_{2n}=1$

$$\therefore \quad \sum_{i=1}^{n} x_i = n, \sum_{i=1}^{n} x_i^2 = \underbrace{1^2+1^2+\cdots+1^2}_{n \text{ times}} = n$$

$$\sum_{i=1}^{n} x_i^3 = \underbrace{1^3+1^3+\cdots+1^3}_{n \text{ times}} = n$$

26. For $x<0$, $\tan^{-1}\frac{1}{x} = -\pi + \cot^{-1} x$

$$\therefore \quad \tan^{-1} x + \tan^{-1}\frac{1}{x} = \tan^{-1} - \pi + \cot^{-1} x$$

$$= \frac{\pi}{2} - \pi \quad \left(\because \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}\right)$$

$$= -\frac{\pi}{2} \text{ and not } \frac{\pi}{2}$$

27. $\because \quad \tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$,

$$\therefore \quad \tan^{-1}\frac{1}{4} + \tan^{-1}\frac{2}{9} = \tan^{-1}\left(\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\times\frac{2}{9}}\right)$$

$$= \tan^{-1}\left(\frac{1}{2}\right)$$

28. $I = (\sec^{-1} x)^2 + (\operatorname{cosec}^{-1} x)^2$

$$= (\sec^{-1} x + \operatorname{cosec}^{-1} x)^2 - 2\sec^{-1} x \cdot \operatorname{cosec}^{-1} x$$

$$= \frac{\pi^2}{4} - 2\sec^{-1} x\left(\frac{\pi}{2} - \sec^{-1} x\right)$$

$$= \frac{\pi^2}{4} + 2\left[(\sec^{-1} x)^2 - \frac{\pi}{2}\cdot(\sec^{-1} x) + \frac{\pi^2}{16} - \frac{\pi^2}{16}\right]$$

$$\Rightarrow \quad \frac{\pi^2}{8} + 2\left[\left(\sec^{-1} x - \frac{\pi}{4}\right)^2\right] \quad \Rightarrow \quad I_{\max} = \frac{\pi^2}{8} + 2\left[\frac{9\pi^2}{16}\right]$$

$$\Rightarrow \quad \frac{\pi^2}{8} + \frac{9\pi^2}{8} = I_{\max} \quad \Rightarrow \quad I_{\max} = \frac{5\pi^2}{4}$$

29. Since $\sum\sqrt{\frac{a\lambda}{bc}}\cdot\sqrt{\frac{b\lambda}{ca}} = \sum\frac{\lambda}{c} = 1$

$$\Rightarrow \quad \tan^{-1}\sqrt{\frac{a\lambda}{bc}} + \tan^{-1}\sqrt{\frac{b\lambda}{ca}} + \tan^{-1}\sqrt{\frac{c\lambda}{ab}} = \frac{\pi}{2}$$

30. $\tan\left(2\cos^{-1}\frac{3}{5}\right) = \tan\left[2\tan^{-1}\left(\frac{\sqrt{1-(3/5)^2}}{3/5}\right)\right]$

$$\left[\because \quad \cos^{-1} x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)\right]$$

$$= \tan\left[2\tan^{-1}\left(\frac{4}{3}\right)\right]$$

$$\left[\textit{Note: } \frac{4}{3} > 1 \Rightarrow \tan^{-1}\left(\frac{4}{3}\right) > \tan^{-1}(1) \Rightarrow \tan^{-1}\left(\frac{4}{3}\right) > \frac{\pi}{4}\right.$$

$$\Rightarrow 2\tan^{-1}\left(\frac{4}{3}\right) > \frac{\pi}{2} \text{ since } \tan^{-1} x \text{ is increasing in } \left[0, \frac{\pi}{2}\right].$$

$$\left.\text{So, } \tan\left\{2\tan^{-1}\left(\frac{4}{3}\right)\right\} \text{ is } -\text{ve}\right]$$

$$= \tan\left\{\tan^{-1}\left[\frac{2\times(4/3)}{1-(4/3)^2}\right]\right\}$$

$$= \tan\left\{\tan^{-1}\left[\frac{8}{3}\times\frac{9}{(-7)}\right]\right\} = \frac{-24}{7}$$

31. $\sin^{-1} x = 2\sin^{-1} a$

Since $-\frac{\pi}{2} \le \sin^{-1} x \le \frac{\pi}{2}$,

$$\Rightarrow \quad -\frac{\pi}{2} \le 2\sin^{-1} a \le \frac{\pi}{2}$$

$$\Rightarrow \quad -\frac{\pi}{4} \le \sin^{-1} a \le \frac{\pi}{4}$$

$$\Rightarrow \quad \sin\left(-\frac{\pi}{4}\right) \le a \le \sin\frac{\pi}{4}$$

$$\Rightarrow \quad -\frac{1}{\sqrt{2}} \le a \le \frac{1}{\sqrt{2}}$$

$$\therefore \quad |a| \le \frac{1}{\sqrt{2}}$$

32. $\cos^{-1} x - \cos^{-1}\frac{y}{2} = \alpha$

$$\Rightarrow \quad \cos^{-1}\left[x\frac{y}{2} - \sqrt{(1-x^2)\left(1-\frac{y^2}{4}\right)}\right] = \alpha$$

$$\left[\because \quad \cos^{-1}\alpha + \cos^{-1}\beta \right.$$

$$\left. = \cos^{-1}(\alpha\beta - \sqrt{(1-\alpha^2)(1-\beta)^2})\right]$$

$$\Rightarrow \quad \frac{1}{2}[xy - \sqrt{(1-x^2)(4-y^2)}] = \cos\alpha$$

$$\Rightarrow \quad xy - 2\cos\alpha = \sqrt{(1-x^2)(4-y^2)}$$

$$\Rightarrow \quad (xy - 2\cos\alpha)^2 = (1-x^2)(4-y^2)$$

$$\Rightarrow \quad x^2y^2 - 4xy\cos\alpha + 4\cos^2\alpha = 4 - 4x^2 - y^2 + x^2y^2$$

$\Rightarrow\ 4x^2 - 4xy\cos\alpha + y^2 = 4 - 4\cos^2\alpha$

$\Rightarrow\ 4x^2 - 4xy\cos\alpha + y^2 = 4\sin^2\alpha$

33. $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$

$\Rightarrow\ \sin^{-1}\left(\frac{x}{5}\right) + \sin^{-1}\left(\frac{4}{5}\right) = \frac{\pi}{2}$

$\Rightarrow\ \sin^{-1}\left(\frac{x}{5}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{4}{5}\right) = \cos^{-1}\left(\frac{4}{5}\right) = \sin^{-1}\left(\frac{3}{5}\right)$

34. $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3}\right) = \cot\left(\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3}\right)$

$= \cot\left\{\tan^{-1}\left[\frac{(3/4)+(2/3)}{1-(1/2)}\right]\right\}$

$= \cot\left[\tan^{-1}\left(\frac{17}{6}\right)\right] = \frac{6}{17}$

35. $\tan^{-1}\left(\frac{c_1x - y}{c_1y + x}\right) + \tan^{-1}\left(\frac{c_2 - c_1}{1 + c_2c_1}\right) + \tan^{-1}\left(\frac{c_3 - c_2}{1 + c_3c_2}\right)$

$+ \cdots + \tan^{-1}\left(\frac{1}{c_n}\right)$

$\tan^{-1}\left(\frac{\frac{x}{y} - \frac{1}{c_1}}{1 + \frac{x}{y}\cdot\frac{1}{c_1}}\right) + \tan^{-1}\left(\frac{\frac{1}{c_1} - \frac{1}{c_2}}{1 + \frac{1}{c_1}\cdot\frac{1}{c_2}}\right) + \tan^{-1}\left(\frac{\frac{1}{c_2} - \frac{1}{c_3}}{1 + \frac{1}{c_2}\cdot\frac{1}{c_3}}\right)$

$+ \cdots + \tan^{-1}\left(\frac{1}{c_n}\right)$

$= \left[\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{1}{c_1}\right)\right] + \left[\tan^{-1}\left(\frac{1}{c_1}\right) - \tan^{-1}\left(\frac{1}{c_2}\right)\right]$

$+ \left[\tan^{-1}\left(\frac{1}{c_2}\right) - \tan^{-1}\left(\frac{1}{c_3}\right)\right]$

$+ \cdots + \left[\tan^{-1}\left(\frac{1}{c_{n-1}}\right) - \tan^{-1}\left(\frac{1}{c_n}\right)\right] + \tan^{-1}\left(\frac{1}{c_n}\right)$

$= \tan^{-1}\left(\frac{x}{y}\right) + \left[-\tan^{-1}\left(\frac{1}{c_1}\right) + \tan^{-1}\left(\frac{1}{c_1}\right)\right] + \left[\tan^{-1}\left(\frac{1}{c_2}\right)\right.$

$\left. + \tan^{-1}\left(\frac{1}{c_2}\right)\right] + \cdots + \left[-\tan^{-1}\left(\frac{1}{c_n}\right) + \tan^{-1}\left(\frac{1}{c_n}\right)\right]$

$= \tan^{-1}\left(\frac{x}{y}\right)$

36. Let $\sin^{-1}(\sin 10) = \theta$, where $\frac{-\pi}{2} \le \theta \frac{\pi}{2}$

Then, $\sin\theta = \sin 10$ (i)

Now, we know that $3\pi < 10 < 3\pi + \frac{\pi}{2}$

Let $10 = 3\pi + \alpha$ (ii)

where $0 < \alpha < \frac{\pi}{2}$

From (i) and (ii), we get

$\sin\theta = \sin(3\pi + \alpha) = -\sin\alpha = \sin(-\alpha)$

$\Rightarrow\ \theta = -\alpha \qquad \left[\because -\frac{x}{2} < -\alpha < 0\right]$

$\Rightarrow\ \theta = 3\pi - 10$ [using (i)]

$\Rightarrow\ \sin^{-1}(\sin 10) = 3\pi - 10 \qquad [\because \sin^{-1}(\sin 10) = \theta]$

37. The range of $\tan^{-1}x$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

38. Let $\sin^{-1}x = \theta$, then

$\sin^{-1}x = \theta \ \Rightarrow\ x = \sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$

$\Rightarrow\ \cos^{-1}x = \frac{\pi}{2} - \theta$

$\Rightarrow\ \cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$

$\Rightarrow\ \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

39. $\tan^{-1}\frac{1-x}{1+x} = \frac{1}{2}\tan^{-1}x$

Let $x = \tan\theta$

$\therefore\ \tan^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right) = \frac{1}{2}\tan^{-1}(\tan\theta)$

$\Rightarrow\ \tan^{-1}\left\{\tan\left(\frac{\pi}{4} - \theta\right)\right\} = \frac{1}{2}\tan^{-1}(\tan\theta)$

$\Rightarrow\ \frac{\pi}{4} - \theta = \frac{\theta}{2} \Rightarrow \theta = \frac{\pi}{6}$

$\therefore\ x = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$

40. $\sin^{-1}x - \cos^{-1}x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\Rightarrow\ \sin^{-1}x - \cos^{-1}x = \frac{\pi}{6}$ (i)

But $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ (ii)

On solving Eqs. (i) and (ii), we get

$$\sin^{-1} x = \frac{\pi}{3} \text{ and } \cos^{-1} x = \frac{\pi}{6}$$

$\Rightarrow$ $x = \frac{\sqrt{3}}{2}$ is the unique solution.

41. $(\sqrt{3}+i) = (a+ib)(c+id)$

$$= (ac-bd) + i(ad+bc)$$

On comparing the real and imaginary parts on both sides, we get

$$ac - bd = \sqrt{3} \text{ and } ad + bc = 1$$

Now, $\tan^{-1}\left(\frac{b}{a}\right) + \tan^{-1}\left(\frac{d}{c}\right)$

$$= \tan^{-1}\left(\frac{bc+ad}{ac-bd}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= n\pi + \frac{\pi}{6},\ n \in I$$

42. $\sin^{-1}\frac{2a}{1+a^2} - \cos^{-1}\frac{1-b^2}{1+b^2} = \tan^{-1}\frac{2x}{1-x^2}$

$\Rightarrow$ $2\tan^{-1} a - 2\tan^{-1} b = 2\tan^{-1} x$

$\Rightarrow$ $\tan^{-1}\left(\frac{a-b}{1+ab}\right) = \tan^{-1} x$

$\Rightarrow$ $x = \frac{a-b}{1+ab}$

43. $\cos(2\tan^{-1} x) = \frac{1}{2}$

$\Rightarrow$ $\cos\left[\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right] = \frac{1}{2}$

$$\left[\because\ 2\tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right]$$

$\Rightarrow$ $\frac{1-x^2}{1+x^2} = \frac{1}{2}$ $\Rightarrow$ $2 - 2x^2 = 1 + x^2$

$\Rightarrow$ $3x^2 = 1$ $\Rightarrow$ $x = \frac{1}{\sqrt{3}}$

44. $\sin\left(\sin^{-1}\frac{1}{3} + \sec^{-1} 3\right) + \cos\left(\tan^{-1}\frac{1}{2} + \tan^{-1} 2\right)$

$$= \sin\left(\sin^{-1}\frac{1}{3} + \cos^{-1}\frac{1}{3}\right) + \cos\left(\tan^{-1}\frac{1}{2} + \cot^{-1}\frac{1}{2}\right)$$

$$\left[\because\ \sec^{-1} x = \cos^{-1}\frac{1}{x} \text{ and } \tan^{-1} x = \cot^{-1}\frac{1}{x}\right]$$

$$= \sin\frac{\pi}{2} + \cos\frac{\pi}{2}$$

$$\left[\because \sin^{-1}\alpha + \cos^{-1}\alpha = \frac{\pi}{2} \text{ and } \tan^{-1}\beta + \cot^{-1}\beta = \frac{\pi}{2}\right]$$

$$= 1 + 0 = 1$$

45. $\cot^{-1} 9 = \tan^{-1}\left(\frac{1}{9}\right)$ $\left[\because \cot^{-1} x = \tan^{-1}\left(\frac{1}{x}\right)\right]$

Let $\text{cosec}^{-1}\frac{\sqrt{41}}{4} = \theta$. Then $\text{cosec}^{-1}\theta = \frac{\sqrt{41}}{4}$

$\therefore$ $\cot\theta = \sqrt{\text{cosec}^2\theta - 1} = \sqrt{\frac{41}{16} - 1} = \sqrt{\frac{25}{16}} = \frac{5}{4}$

$\Rightarrow$ $\tan\theta = \frac{1}{\cot\theta} = \frac{4}{5}$ $\Rightarrow$ $\theta = \tan^{-1}\left(\frac{4}{5}\right)$

i.e. $\text{cosec}^{-1}\frac{\sqrt{41}}{4} = \tan^{-1}\frac{4}{5}$

$\therefore$ $\cot^{-1} 9 + \text{cosec}^{-1}\frac{\sqrt{41}}{4} = \tan^{-1}\left(\frac{1}{9}\right) + \tan^{-1}\left(\frac{4}{5}\right)$

$$= \tan^{-1}\left\{\frac{(1/9)+(4/5)}{1-[(1/9)\times(4/5)]}\right\} = \tan^{-1}\left[\frac{(41/45)}{(41/45)}\right]$$

$$= \tan^{-1}(1) = \frac{\pi}{4}$$

46. $3\tan^{-1} a = 2\tan^{-1} a + \tan^{-1} a$

$$= \tan^{-1}\left(\frac{2a}{1-a^2}\right) + \tan^{-1} a$$

$$= \tan^{-1}\left\{\frac{[2a/(1-a^2)]+a}{1-[2a^2/(1-a^2)]}\right\}$$

$$= \tan^{-1}\left\{\frac{3a-a^3}{1-3a^2}\right\}$$

47. $2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} = \tan^{-1}\left[\frac{2/3}{1-(1/9)}\right] + \tan^{-1}\frac{1}{7}$

$$\left[\because 2\tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)\right]$$

$$= \tan^{-1}\left(\frac{6}{8}\right) + \tan^{-1}\left(\frac{1}{7}\right)$$

$$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{7}\right)$$

$$= \tan^{-1}\left[\frac{(3/4)+(1/7)}{1-[(3/4)\times(1/7)]}\right]$$

$$= \tan^{-1}\left(\frac{25}{25}\right) = \tan^{-1} 1 = \frac{\pi}{4}$$

48. $\tan^{-1}(x-1)+\tan^{-1}x=\tan^{-1}3x-\tan^{-1}(x+1)$

$\Rightarrow \tan^{-1}\left[\dfrac{(x-1)+x}{1-(x-1)x}\right]=\tan^{-1}\left[\dfrac{3x-(x+1)}{1+3x(x+1)}\right]$

$\Rightarrow (1+3x^2+3x)(2x-1)=(1-x^2+x)(2x-1)$

$\Rightarrow (2x-1)(4x^2+2x)=0$

$\Rightarrow x=0,\pm\dfrac{1}{2}$

49. $\cot(\cos^{-1}x)=\sec\left(\tan^{-1}\dfrac{a}{\sqrt{b^2-a^2}}\right)$

$\therefore \cot\left[\cot^{-1}\left(\dfrac{x}{\sqrt{1-x^2}}\right)\right]=\sec\left(\sec^{-1}\dfrac{b}{\sqrt{b^2-a^2}}\right)$

$\Rightarrow \dfrac{x}{\sqrt{1-x^2}}=\dfrac{b}{\sqrt{b^2-a^2}}$

$\Rightarrow x^2(b^2-a^2)=b^2-b^2x^2$

$\Rightarrow x^2(2b^2-a^2)=b^2$

$\Rightarrow x=\dfrac{b}{\sqrt{2b^2-a^2}}$

50. Now, $\tan^{-1}\left(\dfrac{1}{2}\right)+\tan^{-1}\left(\dfrac{1}{3}\right)+\tan^{-1}\dfrac{7}{8}$

$=\tan^{-1}\left(\dfrac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\cdot\frac{1}{3}}\right)+\tan^{-1}\dfrac{7}{8}=\tan^{-1}\left(\dfrac{\frac{5}{6}}{\frac{5}{6}}\right)+\tan^{-1}\dfrac{7}{8}$

$=\tan^{-1}(1)+\tan^{-1}\dfrac{7}{8}=\tan^{-1}\left[\dfrac{1+(7/8)}{1-(7/8)}\right]=\tan^{-1}(15)$

51. $\tan^{-1}\left(\dfrac{1+x}{1-x}\right)=\dfrac{\pi}{4}+\tan^{-1}x$

R.H.S. $=\dfrac{\pi}{4}+\tan^{-1}x=\tan^{-1}1+\tan^{-1}x$

$=\tan^{-1}\left(\dfrac{1+x}{1-x}\right)$, if $x<1$ $\quad\therefore\quad x\in(-\infty,1)$

52. We have $\left(3\pi+\dfrac{\pi}{2}\right)<12<4\pi$

$\left[\because\ 3\pi+\dfrac{\pi}{2}\approx 3(3.142)+\dfrac{3.142}{2}=10.997\right.$

$\left.\text{and } 4\pi\approx 4(3.142)=12.568\right]$

Also, $4\pi<14<4\pi+\dfrac{\pi}{2}$

$\left[\because\ 4\pi\approx 4(3.142)=12.568\right.$

$\left.4\pi+\dfrac{\pi}{2}\approx 12.568+\dfrac{3.142}{2}=14.139\right]$

We put $12=4\pi-\alpha$, where $0<\alpha<\dfrac{\pi}{2}$ and $14=4\pi+\beta$, where $0<\beta<\dfrac{\pi}{2}$.

Then, $\cos^{-1}(\cos 12)-\sin^{-1}(\sin 14)$

$=\cos^{-1}[\cos(4\pi-\alpha)]-\sin^{-1}[\sin(4\pi+\beta)]$

$=\cos^{-1}(\cos\alpha)-\sin^{-1}(\sin\beta)$

$\left[\because\ \cos(4\pi-x)=\cos x\right.$

$\left.\text{and } \sin(4\pi+x)=\sin x\right]$

$=\alpha-\beta=(4\pi-12)-(14-4\pi)=8\pi-26$

53. Let $x=\tan\theta$. Then

$\tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right)-\tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$

$=\tan^{-1}\left(\dfrac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}\right)-\tan^{-1}\left(\dfrac{2\tan\theta}{1-\tan^2\theta}\right)$

$=\tan^{-1}(\tan 3\theta)-\tan^{-1}(\tan 2\theta)=3\theta-2\theta=\theta=\tan^{-1}x$

54. Let $\theta=\cos^{-1}\left(-\dfrac{1}{2}\right)$

$\Rightarrow \cos\theta=-\dfrac{1}{2}=-\cos\left(\dfrac{\pi}{3}\right)=\cos\left(\pi-\dfrac{\pi}{3}\right)=\cos\left(\dfrac{2\pi}{3}\right)$

$\Rightarrow \theta=2n\pi\pm\dfrac{2\pi}{3}\ \Rightarrow\ \theta=\dfrac{4\pi}{3},\dfrac{8\pi}{3},\dfrac{10\pi}{3},\dfrac{14\pi}{3},\cdots$

55. Let $x=a\sin\theta$

Then, $\tan^{-1}\dfrac{x}{\sqrt{a^2-x^2}}=\tan^{-1}\left(\dfrac{a\sin\theta}{\sqrt{a^2-a^2\sin^2\theta}}\right)$

$=\tan^{-1}\left(\dfrac{a\sin\theta}{a\cos\theta}\right)=\tan^{-1}(\tan\theta)=\theta=\sin^{-1}\dfrac{x}{a}$

56. $\cos^{-1}\left[\sin\left(\dfrac{7\pi}{6}\right)\right]=\cos^{-1}\left[\sin\left(\pi+\dfrac{\pi}{6}\right)\right]$

$=\cos^{-1}\left[-\sin\dfrac{\pi}{6}\right]=\pi-\cos^{-1}\left[\sin\dfrac{\pi}{6}\right]$

$\left[\because \cos^{-1}(-\alpha)=\pi-\cos^{-1}\alpha\right]$

$=\pi-\cos^{-1}\left(\dfrac{1}{2}\right)=\pi-\dfrac{\pi}{3}=\dfrac{2\pi}{3}$

57. $\cos(4095°) = \cos(45\times 90° + 45°)$

$$= -\sin 45° = -\sin\frac{\pi}{4} = \sin\left(-\frac{\pi}{4}\right)$$

$$\therefore \quad \sin^{-1}[\cos(4095°)] = \sin^{-1}\sin\left(-\frac{\pi}{4}\right) = -\frac{\pi}{4}$$

58. We have $[\sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1}x] = 1$

$$\Rightarrow \quad 1 \le \sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1}x \le \frac{\pi}{2}$$

$$\Rightarrow \quad \sin 1 \le \cos^{-1}\sin^{-1}\tan^{-1}x \le 1$$

$$\Rightarrow \quad \cos\sin 1 \ge \sin^{-1}\tan^{-1}x \ge \cos 1$$

$$\Rightarrow \quad \sin\cos\sin 1 \ge \tan^{-1}x \ge \sin\cos 1$$

$$\Rightarrow \quad \tan\sin\cos\sin 1 \ge x \ge \tan\sin\cos 1$$

Hence, $x \in [\tan\sin\cos 1, \tan\sin\cos\sin 1]$

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 1

1. The equations $2\sin^{-1}x + 3\sin^{-1}y = \frac{5\pi}{2}$ and $y = kx - 5$ hold simultaneously when k equals

(a) 2 (b) 4
(c) 6 (d) no such k exists.

2. If a, b, c are positive real numbers and

$$\theta = \tan^{-1}\sqrt{\frac{ak}{cb}} + \tan^{-1}\sqrt{\frac{bk}{ca}} + \tan^{-1}\sqrt{\frac{ck}{ab}},$$

where $k = a + b + c$, then θ equals

(a) $\pi/2$ (b) $\pi/4$
(c) π (d) none of these

3. $\cos^{-1}\left[\cos(2\cot^{-1}(\sqrt{2}-1)\right]$ is equal to

(a) $\sqrt{2}-1$ (b) $1-\sqrt{2}$
(c) $\pi/4$ (d) $3\pi/4$

4. A solution of the equation $\tan^{-1}(1+x) + \tan^{-1}(1-x) = \frac{\pi}{2}$ is

(a) $x = 1$ (b) $x = -1$
(c) $x = 0$ (d) $x = \pi$

5. If $x_1, x_2, x_3, \ldots, x_n$ are positive numbers in decreasing order, then $\cot^{-1}\left(\frac{1+x_1x_2}{x_1-x_2}\right) + \cot^{-1}\left(\frac{1+x_2x_3}{x_2-x_3}\right) + \cdots +$

$\cot^{-1}\left(\frac{1+x_{n-1}x_n}{x_{n-1}-x_n}\right) + \cot^{-1}\left(\frac{1+x_nx_1}{x_n-x_1}\right)$ is equal to

(a) $n\pi$ (b) π
(c) $\pi/2$ (d) none of these

6. The value of $\sin(2\sin^{-1}0.8)$ is

(a) 0.48 (b) 0.96
(c) 0.6 (d) none of these

7. The equation $\sin^{-1}x - \cos^{-1}x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ has

(a) no solution
(b) unique solution
(c) infinite number of solutions
(d) none of these

8. The equation $\sin^{-1}x = |x-a|$ will have at least one solution if

(a) $a \in [-1, 1]$ (b) $a \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(c) $a \in \left[1, 1+\frac{\pi}{2}\right]$ (d) $a \in \left(1-\frac{\pi}{2}, 1+\frac{\pi}{2}\right)$

9. If $\tan^{-1}\frac{1}{1+(1)\cdot(2)} + \tan^{-1}\frac{1}{1+(2)\cdot(3)} + \tan^{-1}\frac{1}{1+(3)\cdot(4)}$

$+ \cdots + \tan^{-1}\frac{1}{1+n(n+1)} = \tan^{-1}\theta$, then θ equals

(a) $\frac{n}{n+1}$ (b) $\frac{n+1}{n+2}$
(c) $\frac{n}{n+2}$ (d) $\frac{n-1}{n+2}$

10. If $\tan(x + y) = 33$ and $x = \tan^{-1}3$, then y will be

(a) 0.3 (b) $\tan^{-1}(1.3)$
(c) $\tan^{-1}(0.3)$ (d) $\tan^{-1}(1/18)$

11. In a triangle ABC, $\angle C = \frac{\pi}{2}$ and $\sin^{-1}(x) = \sin^{-1}\left(\frac{ax}{c}\right)$ $+\sin^{-1}\left(\frac{bx}{c}\right)$, where a, b, c are the sides of the triangle, then the total number of different values of x is

(a) 2 (b) 3
(c) 1 (d) none of these

12. If $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-\cdots\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-\cdots\right)=\frac{\pi}{2}$ for $0<|x|<\sqrt{2}$, then x equals

(a) 1/2 (b) 1
(c) $-(1/2)$ (d) -1

13. The sum of the infinite series $\cot^{-1}2+\cot^{-1}8+\cot^{-1}18+\cot^{-1}32+\cdots$ is equal to

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{8}$

14. If x_1, x_2, x_3, x_4 are the roots of the equation $x^4-x^3\sin 2\beta+x^2\cos^2\beta-x\cos\beta-\sin\beta=0$, then $\sum_{i=1}^{4}\tan^{-1}x_i$ is equal to

(a) $\pi-\beta$ (b) $\pi-2\beta$
(c) $(\pi/2)-\beta$ (d) $(\pi/2)-2\beta$

15. The value of $2\arcsin\frac{1}{2}+3\arctan(-1)+2\arccos\left(-\frac{1}{2}\right)$ is

(a) $\frac{11\pi}{12}$ (b) $\frac{13\pi}{12}$
(c) $\frac{2\pi}{7}$ (d) π

MULTIPLE CHOICE TYPE QUESTIONS—LEVEL 2

1. $\cos^{-1}\left(\frac{1}{2}x^2+\sqrt{1-x^2}\cdot\sqrt{1-\frac{x^2}{4}}\right)=\cos^{-1}\frac{x}{2}-\cos^{-1}x$ holds for

(a) $|x|\le 1$ (b) $x\in\mathbb{R}$
(c) $0\le x\le 1$ (d) $-1\le x\le 0$

2. A solution to the equation $\tan^{-1}(1+x)+\tan^{-1}(1-x)=\frac{\pi}{2}$ is

(a) $x=1$ (b) $x=-1$
(c) $x=0$ (d) $x=\pi$

3. The set of values of a for which $x^2+ax+\sin^{-1}(x^2-4x+5)+\cos^{-1}(x^2-4x+5)=0$ has at least one solution is

(a) $(-\infty,-\sqrt{2\pi}]\cup[\sqrt{2\pi},\infty)]$
(b) $(-\infty,-\sqrt{2\pi})\cup(\sqrt{2\pi},\infty)$
(c) $\mathbb{R}$
(d) none of these

4. If $\cos^{-1}x>\sin^{-1}x$, then

(a) $x<0$ (b) $-1<x<10$
(c) $0\le x<\frac{1}{\sqrt{2}}$ (d) $-1\le x<\frac{1}{\sqrt{2}}$

5. All possible values of p and q for which

$$\cos^{-1}\sqrt{p}+\cos^{-1}\sqrt{1-p}+\cos^{-1}\sqrt{1-q}=\frac{3\pi}{4}$$

holds are

(a) $p=1, q=\frac{1}{2}$ (b) $q>1, p=\frac{1}{2}$
(c) $0\le p\le 1, q=\frac{1}{2}$ (d) none of these

6. Which one of the following is correct?

(a) $\tan 1>\tan^{-1}1$ (b) $\tan 1<\tan^{-1}1$
(c) $\tan 1=\tan^{-1}1$ (d) none of these

7. The number of solutions of the equation $\sin^{-1}(1-x)-2\sin^{-1}x=\frac{\pi}{2}$ is

(a) 0 (b) 1
(c) 2 (d) more than two

8. The value of $\cos\left(\frac{1}{2}\cos^{-1}\frac{1}{8}\right)$ is equal to

(a) $\frac{3}{4}$ (b) $-\frac{3}{4}$
(c) $\frac{1}{16}$ (d) $\frac{1}{4}$

9. The value of $\tan^{-1}\left(\frac{a}{b+c}\right)+\tan^{-1}\left(\frac{b}{c+a}\right)$, if $\angle C=90°$ in a triangle ABC, is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{2}$ (d) π

10. The number of solutions of the equation $\tan^{-1}\left(\frac{1}{2x+1}\right)+\tan^{-1}\left(\frac{1}{4x+1}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$ is

(a) 0 (b) 1
(c) 2 (d) 3

11. If $u = \cot^{-1}\sqrt{\tan\alpha} - \tan^{-1}\sqrt{\tan\alpha}$, then $\tan\left(\frac{\pi}{4} - \frac{u}{2}\right)$ is equal to

(a) $\sqrt{\tan\alpha}$ (b) $\sqrt{\cot\alpha}$

(c) $\tan\alpha$ (d) $\cot\alpha$

12. The value of $\cot^{-1}3 + \text{cosec}^{-1}\sqrt{5}$ is

(a) $\pi/3$ (b) $\pi/2$

(c) $\pi/4$ (d) $\pi/6$

13. If $2\sin^{-1}x - 3\cos^{-1}x = 4$, then $2\sin^{-1}x + 3\cos^{-1}x$ equals

(a) $\frac{6\pi - 4}{5}$ (b) $\frac{4 - 6\pi}{5}$

(c) $3\pi/2$ (d) 0

14. $\sum_{r=1}^{\infty}\cot^{-1}\left(r^2 + \frac{3}{4}\right)$ equals

(a) $\frac{\pi}{2}$ (b) $\cot^{-1}2$

(c) $\frac{\pi}{6}$ (d) $\tan^{-1}2$

15. Given $\pi < \theta < \frac{3\pi}{2}$, then the value of the expression $\sqrt{4\sin^4\theta + \sin^2 2\theta} + 4\cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right)$ is

(a) 2 (b) $2 + 4\sin\theta$

(c) $2 - 4\sin\theta$ (d) none of these

ANSWERS

Level 1

1. (c)	**2.** (c)	**3.** (d)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (c)
11. (b)	**12.** (b)	**13.** (b)	**14.** (b)	**15.** (b)					

Level 2

1. (c)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (a)	**7.** (b)	**8.** (a)	**9.** (a)	**10.** (b)
11. (a)	**12.** (c)	**13.** (a)	**14.** (c)	**15.** (a)					

30

GEORGE BOOLE

Heights and Distances

- ❑ Angles of Elevation and Depression
- ❑ Geometrical Properties and Formulae for a Triangle
- ❑ Trigonometrical Results
- ❑ $m - n$ Theorem of Trigonometry
- ❑ To Express Sides of a Triangle

☞ ANGLES OF ELEVATION AND DEPRESSION

(i) Angle of elevation: Let P be an object and OX be a horizontal line; an observer at O to perceive the object P has to elevate his eye from the direction OX to the direction OP. As such, we define $\angle XOP$ as the **angle of elevation** of P at O.

(ii) Angle of depression: Let P be an object and OX be a horizontal line; an observer at O to perceive the object P has to depress his eye from the direction OX to the direction OP. As such, we define $\angle XOP$ as the **angle of depression** of P at O.

(iii) Angular elevation: If PM is perpendicular to OX, then the angle XOP is also called the altitude or the **angular elevation** of MP at O.

(iv) Bearing of a point: Let NS and EW stand for lines in the north-south and east-west directions respectively. Then the acute angle which the line OA makes with NS is called the **bearing** of the point A from O.

The bearing of A may be indicated, precisely, by giving the size of the angle and specifying whether it is measured from ON (or OS) and whether to east (or west). For example, in Fig. 30.2 , OA is in the direction 30° east of north. Thus, the bearing is written as $N\ 30°\ E$.

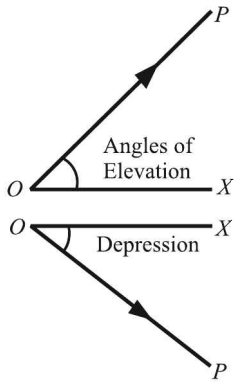


Fig. 30.1

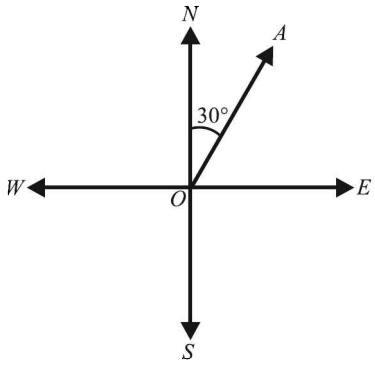


Fig. 30.2

Note

(i) Angles of elevation and depression are always acute angles.

(ii) Unless stated to the contrary, it is assumed that the height of the observer is neglected, and that the angles of elevation and depression are measured from the group.

(iii) Any perpendicular line to a plane is perpendicular to all lines lying in the plane.

GEOMETRICAL PROPERTIES AND FORMULAE FOR A TRIANGLE

(a) In a triangle, the internal bisector of an angle divides the opposite side in the ratio of the arms of the angle. Thus,

$$\frac{BD}{DC} = \frac{c}{b}$$

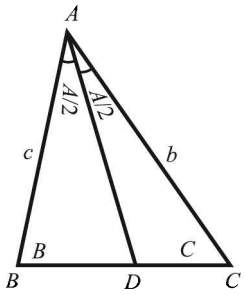


Fig. 30.3

(b) In an isosceles triangle, the median is perpendicular to the base.

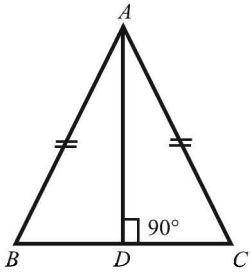


Fig. 30.4

(c) In similar triangles, the corresponding sides are proportional.

(d) The exterior angle is equal to the sum of the interior opposite angles.

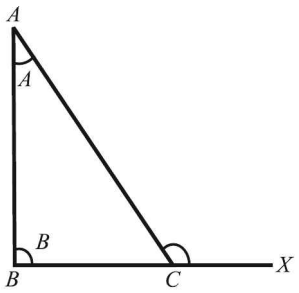


Fig. 30.5

(e) **Sine formula:** In any triangle ABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

where R is the radius of the circumcircle of ΔABC.

Also $R = \dfrac{abc}{4\Delta}$

where Δ is the area of ΔABC.

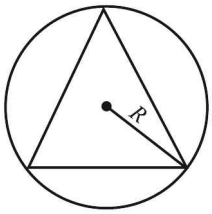


Fig. 30.6

(f) **Cosine formula**

$a^2 = b^2 + c^2 - 2bc \cos A$

$$\Rightarrow \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

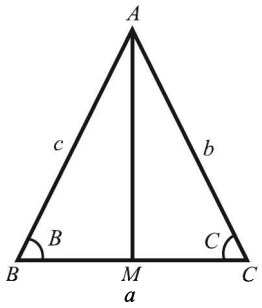


Fig. 30.7

(g) **Projection formula**

$a = b \cos C + c \cos B$

(h) If in ΔABC, D is the mid-point of BC, then

$AB^2 + AC^2 = 2(AD^2 + DB^2)$

(i) In a right-angled triangle ABC, the mid-point D of the hypotenuse AC is equidistant from its vertices A, B and C. That is,

$AD = BD = CD$

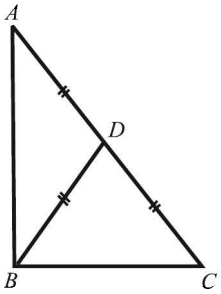


Fig. 30.8

TRIGONOMETRICAL RESULTS

(a) $\tan A \pm \tan B = \dfrac{\sin(A \pm B)}{\cos A \cos B}$

(b) $\cot A \pm \cot B = \dfrac{\sin(B \pm A)}{\sin A \sin B}$

(c) $\sin(A+B)\sin(A-B)$

$= \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

(d) $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B$

m – n THEOREM OF TRIGONOMETRY

(a) $(m+n)\cot\theta = n\cot B - m\cot C$

(b) $(m+n)\cot\theta = m\cot\alpha - n\cot B$

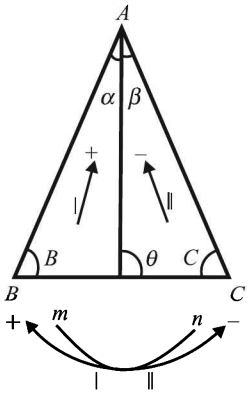


Fig. 30.9

☞ TO EXPRESS ONE SIDE OF A RIGHT-ANGLED TRIANGLE IN TERMS OF OTHER SIDES

Let $AB = H$ (hypotenuse) and let $\angle ABC = \theta$. The side opposite to θ is a perpendicular p and the remaining side is base b.

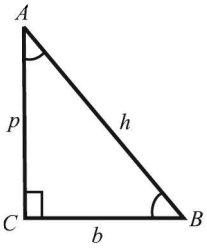


Fig. 30.10

$b = p(\cot\theta)$

$p = b(\tan\theta)$

$p = h(\sin\theta)$

$b = h(\cos\theta)$

(a) $a = h(\cot\alpha - \cot\beta)$

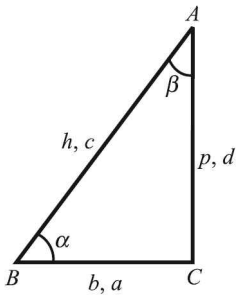


Fig. 30.11

(b) If $AD = BE$, then $x = y\tan\left(\frac{\alpha+\beta}{2}\right)$

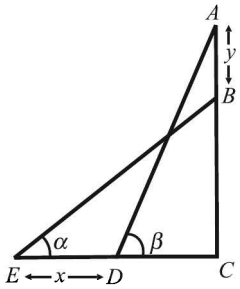


Fig. 30.12

(c) $h = \frac{a\sin(\alpha+\beta)}{\sin(\beta-\alpha)}$

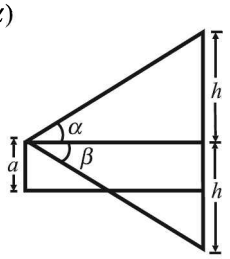


Fig. 30.13

(d) $h = \frac{H\sin(\beta-\alpha)}{\cos\alpha\cos\beta}$ and $H = \frac{\cot\alpha}{\cos\alpha - \cos\beta}$

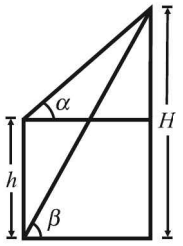


Fig. 30.14

(e) $h = \frac{a}{\sqrt{\cot^2\alpha + \cot^2\beta}}$

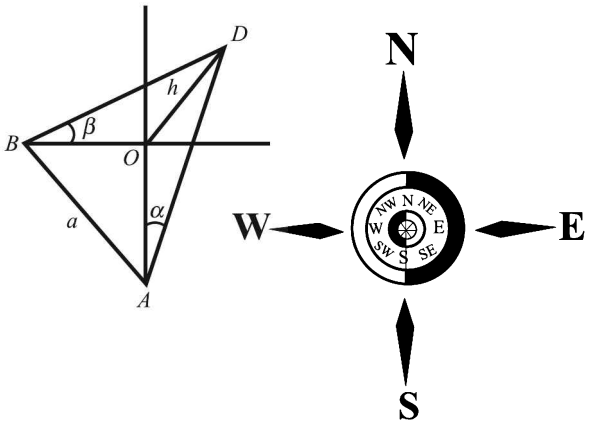


Fig. 30.15

SOLVED PROBLEMS

1. If a flagstaff 6 m high placed on the top of a tower throws a shadow of $2\sqrt{3}$ m on the ground, then the angle that the sun makes with the ground is

(a) 15° (b) 30°

(c) 45° (d) 60°

Ans. (d)

Solution: Shadow of tower (OP) = OA and shadow of flag-staff (PQ) = $AB = 2\sqrt{3}$.

Let α be the angle made by the sun with the ground.

Let the distance between O and A (OA) = x, height of the tower (OP) = h, distance between O and B (OB) = $x + 2\sqrt{3}$ and (OQ) = $h + 6$

We know that in ΔOAP,

$$\tan\alpha = \frac{OP}{OA} = \frac{h}{x} \quad \text{(i)}$$

We also know that in ΔOBQ,

$$\tan\alpha = \frac{OQ}{OB} = \frac{h+6}{x+2\sqrt{3}} \quad \text{(ii)}$$

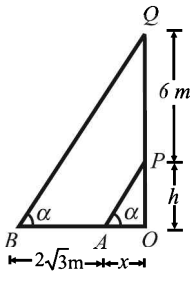


Fig. 30.16

Solving Eqs. (i) and (ii), we get

$$\frac{h}{x}=\frac{h+6}{x+2\sqrt{3}}$$

or $h(x+2\sqrt{3})=x(h+6)$ or $hx+2\sqrt{3}h=hx+6x$

or $2\sqrt{3}h=6x$ or $x=\frac{\sqrt{3}h}{3}$ or $x=\frac{h}{\sqrt{3}}$

or $\frac{h}{\tan\alpha}=\frac{h}{\sqrt{3}}$ or $\tan\alpha=\sqrt{3}$ or $\alpha=60°$

2. From the top of a light house 60 m high with its base at the sea level, the angle of depression of a boat is 15°. The distance of the boat from the foot of the light house is

(a) $\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)60$ m (b) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)60$ m

(c) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ m (d) none of these

Ans. (b)

Solution: Let P be the top of the light house at a height of 60 m.

If A is the position of the boat, then

$\angle OAP=\angle LPA=15°$

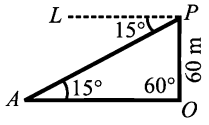


Fig. 30.17

Now, $\tan 15°=\frac{OP}{OA}=\frac{60}{OA}$

$\therefore$ $OA=60\cot 15°=60\cot(45°-30°)$

$=60\cdot\frac{\cot 45°\cot 30°+1}{\cot 30°-\cot 45°}=\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)60$ m

3. A pole subtends an angle of 30° at the foot of a tower of height H. And at a distance d from the tower, the angle of depression of the foot of the pole from the top of the tower is 60°. The height of the pole is

(a) $\frac{H}{3}$ (b) $\frac{H}{d}$

(c) $\frac{H}{3d}$ (d) $\frac{3H}{d}$

Ans. (a)

Solution: Angle subtended by the pole at the foot of the tower = 30°; height of tower = H; distance between tower and pole = d and angle of depression at the foot of the pole = 60°

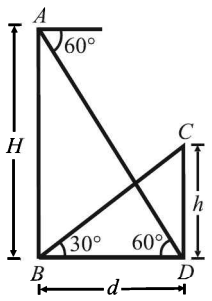


Fig. 30.18

We know that in ΔCBD,

$$\tan 30°=\frac{h}{d} \quad \text{(i)}$$

Similarly, in ΔABD,

$$\tan 60°=\frac{h}{d} \quad \text{(ii)}$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{\tan 30°}{\tan 60°}=\frac{\frac{h}{d}}{\frac{H}{d}}=\frac{h}{H} \quad \text{or} \quad \frac{\frac{1}{\sqrt{3}}}{\sqrt{3}}=\frac{h}{H}$$

or $\frac{1}{3}=\frac{h}{H}$ or $h=\frac{H}{3}$

4. The angle of elevation of the top of the tower at any point of the ground is 30° and if moving 20 m towards the tower it becomes 60°, then the height of the tower is

(a) 10 m (b) $10\sqrt{3}$ m

(c) $\frac{10}{\sqrt{3}}$ (d) 30

Ans. (b)

Solution: Angle of elevation of the top of a tower at D = 30°; distance between C and D (CD) = 20 m and angle of elevation of the top of the tower at C = 60°

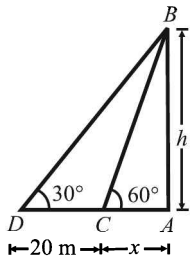


Fig. 30.19

Let AB be the tower of height h and x be the distance between A and C.

We know that in ΔACB,

$$\tan 60°=\frac{h}{x} \quad \text{or} \quad \sqrt{3}=\frac{h}{x}$$

or $h=\sqrt{3}x$ (i)

We also know that in ΔABD,

$$\tan 30°=\frac{h}{x+20} \quad \text{or} \quad \frac{1}{\sqrt{3}}=\frac{h}{x+20}$$

or $\sqrt{3}h=x+20$ or $\sqrt{3}h-20=x$ (ii)

Substituting the value of x in Eq. (i), we get

$h=\sqrt{3}(\sqrt{3}h-20)$ or $h=3h-20\sqrt{3}$

or $-2h=-20\sqrt{3}$

or $h=10\sqrt{3}$

5. AB is a vertical pole. The end A is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion BC subtends angle β at P. If $AP=nAB$, then $\tan\beta=$

(a) $\frac{n}{2n^2+1}$ (b) $\frac{n}{n^2-1}$

(c) $\frac{n}{n^2+1}$ (d) none of these

Ans. (a)

Solution: C is the middle point of the pole AB and P is a point on the horizontal plane. subtended $\angle BPC = \beta$. Let $\angle APC = \alpha$.

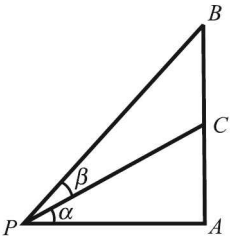


Fig. 30.20

$$\therefore \quad \tan\alpha = \frac{AC}{AP} = \frac{AB}{2AP} = \frac{1}{2n}$$

$$\left(\because AP = nAB\right)$$

In ΔAPB, $\tan(\alpha+\beta) = \dfrac{AB}{AP} = \dfrac{1}{n}$

Now $\tan\beta = \tan[(\alpha+\beta)-\alpha]$

$$= \frac{\tan(\alpha+\beta)-\tan\alpha}{1+\tan(\alpha+\beta)\tan\alpha} = \frac{\frac{1}{n}-\frac{1}{2n}}{1+\frac{1}{n}\times\frac{1}{2n}} = \frac{n}{2n^2+1}$$

$$\left\{\because \tan x - \tan y = \frac{\tan x - \tan y}{1+\tan x \tan y}\right\}$$

6. An aeroplane (P) flying at a height of 300 m above the ground passes vertically above another plane (Q) at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45°, respectively. The height of the lower plane from the ground is

(a) 50 m (b) 100 m

(c) $100\sqrt{3}$ m (d) $150(\sqrt{3}+1)$ m

Ans. (c)

Solution: Height of the aeroplane from the ground (OP) = 300 m; angle of elevation of the plane $P(\alpha) = 60°$ and angle of elevation of the aeroplane $Q(\beta) = 45°$

Let x be the height of the lower aeroplane OQ. We know that in ΔPAO,

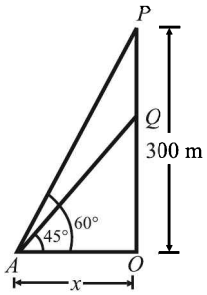


Fig. 30.21

$$\tan 60° = \frac{PO}{OA} \quad \text{or} \quad \sqrt{3} = \frac{300}{x}$$

$$\text{or} \quad x = \frac{300}{\sqrt{3}} = 100\sqrt{3} \text{ m}$$

Since the $\angle QAO$ is equal to 45°, therefore the height of the lower aeroplane from the ground (x) is $100\sqrt{3}$ m.

7. A tower of height b subtends an angle at a point O on the level of the foot of the tower and at a distance a from the foot of the tower. If a pole mounted on the tower also subtends an equal angle at O, then the height of the pole is

(a) $b\left(\dfrac{a^2-b^2}{a^2+b^2}\right)$ (b) $b\left(\dfrac{a^2+b^2}{a^2-b^2}\right)$

(c) $a\left(\dfrac{a^2-b^2}{a^2+b^2}\right)$ (d) $a\left(\dfrac{a^2+b^2}{a^2-b^2}\right)$

Ans. (b)

Solution: Height of tower = b and distance between point O and foot of tower = a. Let h be the height of the pole AC and α be the angles subtended by AB at O and angle subtended by AC at O. We know that in ΔABO,

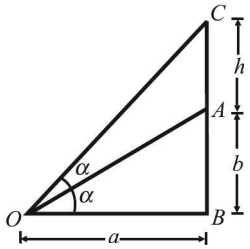


Fig. 30.22

$$\tan\alpha = \frac{AB}{OB} = \frac{b}{a} \qquad \text{(i)}$$

Similarly, in ΔCBO,

$$\tan 2\alpha = \frac{BC}{OB} = \frac{b+h}{a}$$

$$\text{or} \quad \frac{2\tan\alpha}{1-\tan^2\alpha} = \frac{b+h}{a} \qquad \text{(ii)}$$

Substituting the value of $\tan\alpha$ from Eq. (i) in Eq. (ii), we get

$$\frac{b+h}{a} = \frac{2\cdot\left(\frac{b}{a}\right)}{1-\frac{b^2}{a^2}} \text{ or } b+h = \frac{2b\cdot a^2}{(a^2-b^2)} \text{ or } h = \frac{2a^2b}{a^2-b^2} - b$$

$$= \frac{2a^2b-a^2b+b^3}{a^2-b^2} = \frac{a^2b+b^3}{(a^2-b^2)} = \frac{b(a^2+b^2)}{a^2-b^2}$$

$$= b\left(\frac{a^2+b^2}{a^2-b^2}\right)$$

8. The shadow of a tower standing on a level plane is found to be 60 m longer than when the sun's altitude is 30° that of the sun's altitude is 45°, then the height of the tower is

(a) $10\sqrt{3}$ m (b) $\dfrac{30}{\sqrt{3}+1}$ m

(c) $30(\sqrt{3}+1)$ m (d) $50(\sqrt{3}+1)$ m

Ans. (c)

Solution: Angle of elevation of the shadow of a tower at $A = 30°$. Distance between points B and $A(BA) = 60$ m and angle of elevation = 45° at B

Let the height of the tower be h, and the distance between O and B (OB) = x.

We know that in ΔOAP,

$$\tan 30° = \frac{OP}{AO}$$

or $\frac{1}{\sqrt{3}} = \frac{h}{x+60}$

or $\sqrt{3}h = x + 60$ or

$$\sqrt{3}h - 60 = x \qquad \text{(i)}$$

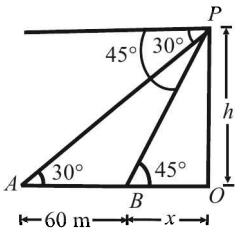


Fig. 30.23

We know that in ΔOBP, $\tan 45° = \frac{OP}{OB}$

or $1 = \frac{h}{x}$ or $h = x$ (ii)

Substituting the value of x in Eq. (i), we get

$$\sqrt{3}h - 60 = h \quad \text{or} \quad \sqrt{3}h - h = 60$$

or $h(\sqrt{3}-1) = 60$ or $h = \frac{60}{\sqrt{3}-1}$

or $h = \frac{60(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = 30(\sqrt{3}+1)$ m

Therefore, the height of tower = $30(\sqrt{3}+1)$ m

EXERCISE SET

1. The height of a chimney when it is found that, on walking towards it 50 m in a horizontal line through its base, the angle of elevation of its top changes from 30° to 45° is

(a) 69.04 m (b) 59.04 m
(c) 49.04 m (d) none of these

2. A tower stands at the top of a hill whose height is 3 times the height of the tower. The tower is found to subtend at a point 3 km away on the horizontal through the foot of the hill, an angle θ, where $\tan\theta = \frac{1}{9}$. The height of the tower is

(a) 12 (b) 3
(c) $\frac{9 \pm \sqrt{33}}{8}$ (d) none of these

3. The upper part of a tree broken over by the wind makes an angle of 30° with the ground and the distance from the root to the point where the top of the tree touches the ground is 10 m. The height of the tree is

(a) 17.3 m (b) 18.3 m
(c) 16.3 m (d) none of these

4. An observer on the top of a tree finds the angle of depression of a car moving towards the tree to be 30°. After 3 min, this angle becomes 60°. After how much more time, will the car reach the tree?

(a) 4 min (b) 4.5 min
(c) 1.5 min (d) 2 min

5. The angle of elevation of the top of a tower from point A due south of the tower is α and from point B due east of the tower is β. If $AB = d$, the height of the tower is

(a) $\frac{d}{\sqrt{\tan^2\alpha - \tan^2\beta}}$ (b) $\frac{d}{\sqrt{\tan^2\alpha + \tan^2\beta}}$
(c) $\frac{d}{\sqrt{\cot^2\alpha - \cot^2\beta}}$ (d) $\frac{d}{\sqrt{\cot^2\alpha + \cot^2\beta}}$

6. Two vertical lamp posts of equal heights stand on either side of a road 50 m wide. At a point on the road, between the two lamp posts, the angles of elevation of the tops of the lamp posts are 60°and 30°. Then the height of each post and the position of the point from one pole are

(a) 21.65 m, 12.5 m (b) 21.65 m, 20 m
(c) 23 m, 12.5 m (d) none of these

7. From the top of a cliff of height a, the angle of depression of the foot of a certain tower is found to be double the angle of elevation of the top of the tower of height h. If θ is the angle of elevation, then its value is

(a) $\cos^{-1}\sqrt{\frac{2h}{a}}$ (b) $\sin^{-1}\sqrt{\frac{2h}{a}}$
(c) $\sin^{-1}\sqrt{\frac{a}{2-h}}$ (d) $\tan^{-1}\sqrt{3-\frac{2h}{a}}$

8. The angles of depression of the top and the foot of a chimney as seen from the top of a second chimney, which is 150 m high and standing on the same level as the first are θ and ϕ respectively, then the distance between their tops when $\tan\theta = \frac{4}{3}$ and $\tan\phi = \frac{5}{2}$ is

(a) $\frac{150}{\sqrt{3}}$ m (b) $100\sqrt{3}$ m
(c) 150 m (d) 100 m

9. The base of a rock is circular. The angles of elevation of the peak of the rock at the end points of a diameter are 30° and 60°. If the height of the peak is 500 m, then the diameter of the rock is

(a) $1000\sqrt{3}$ m (b) $\frac{2000}{\sqrt{3}}$ m
(c) $\frac{1000}{\sqrt{3}}$ m (d) $2000\sqrt{3}$ m

10. A man in a boat rowed away from a cliff 150 m high takes 2 min to change the angle of elevation of the top of the cliff from 60° to 45°. The speed of the boat is

(a) $\frac{9-3\sqrt{3}}{2}$ km/h (b) $\frac{9+3\sqrt{3}}{2}$ km/h

(c) $\frac{9\sqrt{3}}{2}$ km/h (d) none of these

11. The top of a hill observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. The height of the hill is

(a) $\frac{h\cot q}{\cot q-\cot p}$ (b) $\frac{h\cot p}{\cot p-\cot q}$

(c) $\frac{h\tan p}{\tan p-\tan q}$ (d) none of these

12. From the ship, the angle of elevation of point A at the top of a cliff is 19°. After the ship has sailed 800 m directly towards the cliff, the angle of elevation of A is 44°. Then the height of the cliff
(given tan 19° = 0.3443, tan 44° = 0.9657 from tables) is

(a) 418.15 m (b) 438.15 m
(c) 428.03 m (d) none of these

13. A flagstaff 5 m high is placed on a building 25 m high. If the flag and building both subtend equal angles on the observer at a height 30 m, the distance between the observer and the top of the flag is

(a) $\frac{5\sqrt{3}}{2}$ (b) $5\sqrt{\frac{3}{2}}$

(c) $5\sqrt{\frac{2}{3}}$ (d) $\frac{5\sqrt{2}}{3}$

14. If the angle of elevation of a cloud from a foot h metres above a lake is β and the angle of depression of its reflection in the lake is α, then the height is

(a) $\frac{h\sin(\alpha+\beta)}{\sin(\alpha-\beta)}$ (b) $\frac{h\sin(\alpha+\beta)}{\sin(\beta-\alpha)}$

(c) $\frac{h\sin(\beta-\alpha)}{\sin(\alpha+\beta)}$ (d) none of these

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct option from the codes (a), (b), (c) and (d) given below:

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true; Statement II is false.
(d) Statement I is false; Statement II is true.

15. **Statement I:** A flagstaff of length 100 m stands on tower of height h. If at a point on the ground, the angles of elevation of the tower and top of the flagstaff are 30° and 45°, then $h = 50(\sqrt{3}+1)$ m.

Statement II: A flagstaff of length d stands on tower of height h. If at a point on the ground, the angles of elevation of the tower and top of the flagstaff are α and β, then $h = \frac{d\cot\beta}{\cot\alpha-\cot\beta}$.

16. **Statement I:** Over a tower AB of height 10 m, there is a flagstaff BC. AB and BC are making equal angles at a point distant 15 m from the foot A of the tower. The height of the flagstaff is 30 m.

Statement II: Over a tower AB of height h metres, there is a flagstaff BC. AB and BC are making equal angles at a point distant d metre from the foot A of the tower. The height of the flagstaff is $\frac{h(d^2+h^2)}{(d^2-h^2)}$ metres.

17. **Statement I:** The angle of elevation of a cloud from a point 100 m above the surface of a lake is 30° and the angle of depression of its reflection in the lake is 60°. The height of the cloud is 200 m.

Statement II: The angle of elevation of a cloud from a point h metres above the surface of a lake is θ and the angle of depression of its reflection in the lake is ϕ. The height of the cloud is $\frac{h\sin(\theta+\phi)}{\sin(\phi-\theta)}$ metres.

Previous Years' Questions

18. The upper $\left(\frac{3}{4}\right)$th portion of a vertical pole subtends an angle $\tan^{-1}\left(\frac{3}{5}\right)$ at a point in the horizontal plane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is

(a) 40 m (b) 60 m
(c) 80 m (d) 20 m **[AIEEE 2003]**

19. A person standing on the bank of a river observes that the angle of elevation of the top of the tree on the opposite bank of the river is 60° and when he retires 40 m away from the tree, the angle of elevation becomes 30°. The breadth of the river is

(a) 20 m (b) 30 m
(c) 40 m (d) 60 m **[AIEEE 2004]**

20. A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that AB $(= a)$ subtends an angle of 60° at the foot of the tower and the angle of elevation of the top of the tower from A or B is 30°. The height of the tower is

(a) $\frac{2a}{\sqrt{3}}$ (b) $2a\sqrt{3}$

(c) $\frac{a}{\sqrt{3}}$ (d) $\sqrt{3}$ **[AIEEE 2007]**

21. AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is 60°. He moves away from the pole along the line BC to a point D such that $CD = 7$ m. From D, the angle of elevation of the point A is 45°. Then the height of the pole is

(a) $\frac{7\sqrt{3}}{2}\left(-\frac{1}{\sqrt{3}+1}\right)$ m (b) $\frac{7\sqrt{3}}{2}\left(\frac{1}{\sqrt{3}-1}\right)$ m

(c) $\frac{7\sqrt{3}}{2}(\sqrt{3}+1)$ m (d) $\frac{7\sqrt{3}}{2}(\sqrt{3}-1)$ m

[AIEEE 2008]

22. A person observes the angle of elevation of a building as 30°. The person proceeds towards the building with a speed of $25(\sqrt{3}-1)$ m/h. After two hours, he observes the angle of elevation as 45°. The height of the building (in metres) is

(a) $50(\sqrt{3}-1)$ (b) $50(\sqrt{3}+1)$

(c) 50 (d) 100 **[UP SEE 2003]**

23. Two pillars of equal height stand on either side of a roadway which is 60 m wide. At a point in the roadway between the pillars, the elevation angles of the top of pillars are 60° and 30°. The height of the pillars is

(a) $15\sqrt{3}$ m (b) $\frac{15}{\sqrt{3}}$ m

(c) 15 m (d) 20 m **[UP SEE 2004]**

24. A person standing on the bank of a river observes that the angles subtended by a tree on the opposite bank is 60°. When he retreats 20 ft from the bank, he finds the angle to be 30°. The breadth of the river in feet is

(a) 15 (b) $15\sqrt{3}$

(c) $10\sqrt{3}$ (d) 10

[UP SEE 2005, WBJEE 2006]

25. A flag is standing vertically on a tower of height b. On a point at a distance a from the foot of the tower, the flag and the tower subtend equal angles. The height of the flag is

(a) $b\cdot\frac{a^2+b^2}{a^2-b^2}$ (b) $a\cdot\frac{a^2-b^2}{a^2+b^2}$

(c) $b\cdot\frac{a^2-b^2}{a^2+b^2}$ (d) $a\cdot\frac{a^2+b^2}{a^2-b^2}$

[UP SEE 2006]

26. The angle of elevation of the sun if the length of the shadow of a tower is $\sqrt{3}$ times the height of the tower is

(a) 150° (b) 30°

(c) 60° (d) 45° **[UP SEE 2007]**

27. From the top of a cliff 50 m high, the angles of depression of the top and bottom of a tower are observed to be 30° and 45°. The height of the tower is

(a) 50 m (b) $50\sqrt{3}$ m

(c) $50(\sqrt{3}-1)$ m (d) $50\left(1-\frac{\sqrt{3}}{3}\right)$ m

[UP SEE 2009]

28. When the length of the shadow of the pole is equal to the height of the pole, then the elevation of the source of light is

(a) 75° (b) 60°

(c) 45° (d) 30° **[DCE 2006]**

29. The angle of elevation of the top of a tower from a point on the ground is 30° and it is 60° when it is viewed from a point located 40 m away from the initial point towards the tower. The height of the tower is

(a) $-20\sqrt{3}$ m (b) $\frac{\sqrt{3}}{20}$ m

(c) $-\frac{\sqrt{3}}{20}$ m (d) $20\sqrt{3}$ m **[DCE 2007]**

30. A house subtends a right angle at the window of an opposite house and the angle of elevation of the window from the bottom of the first house is 60°. If the distance between the two houses is 6 m, then the height of the first house is

(a) $8\sqrt{3}$ m (b) $6\sqrt{3}$ m

(c) $4\sqrt{3}$ m (d) none of these

[DCE 2008]

31. The angle of elevation of the top of a hill from a point is α. After walking b metres towards the top of a slope inclined at an angle β to the horizon, the angle of elevation of the top becomes γ. Then, the height of the hill is

(a) $\frac{b\sin\alpha\sin(\gamma-\beta)}{\sin(\gamma-\alpha)}$ (b) $\frac{b\sin\alpha\sin(\gamma-\alpha)}{\sin(\gamma-\beta)}$

(c) $\frac{b\sin(\gamma-\beta)}{\sin(\gamma-\alpha)}$ (d) $\frac{\sin(\gamma-\beta)}{b\sin\alpha\sin(\gamma-\alpha)}$

[DCE 2009]

32. From a point on the level ground, the angle of elevation of the top of a pole is 30°. On moving 20 m nearer, the angle of elevation is 45°. Then the height of the pole (in metres) is

(a) $10(\sqrt{3}-1)$ (b) $10(\sqrt{3}+1)$

(c) 15 (d) 20 **[EAMCET 2002]**

33. A tower subtends angles α, 2α and 3α respectively at points A, B and C, all lying on a horizontal line through the foot of the tower. Then AB/BC is equal to

(a) $\frac{\sin 3\alpha}{\sin 2\alpha}$ (b) $1+2\cos 2\alpha$

(c) $2\cos 2\alpha$ (d) $\frac{\sin 2\alpha}{\sin\alpha}$

[EAMCET 2003]

34. An aeroplane flying with uniform speed horizontally 1 km above the ground is observed at an elevation of 60°. After 10 s if the elevation is observed to be 30°, then the speed of the plane (in km/h) is

(a) $\frac{240}{\sqrt{3}}$ (b) $200\sqrt{3}$

(c) $240\sqrt{3}$ (d) $\frac{120}{\sqrt{3}}$

[EAMCET 2004]

35. A tower of x metres high has a flagstaff at its top. The tower and the flagstaff subtend equal angles at a point distant y metres from the foot of the tower. Then the length of the flagstaff (in metres) is

(a) $\frac{y(x^2-y^2)}{(x^2+y^2)}$ (b) $\frac{x(y^2+x^2)}{(y^2-x^2)}$

(c) $\frac{x(x^2+y^2)}{(x^2-y^2)}$ (d) $\frac{x(x^2-y^2)}{(x^2+y^2)}$

[EAMCET 2005]

36. From the top of a hill h metres high, the angles of depressions of the top and the bottom of a pillar are α and β, respectively. The height (in metres) of the pillar is

(a) $\frac{h(\tan\beta-\tan\alpha)}{\tan\beta}$ (b) $\frac{h(\tan\alpha-\tan\beta)}{\tan\alpha}$

(c) $\frac{h(\tan\beta+\tan\alpha)}{\tan\beta}$ (d) $\frac{h(\tan\beta+\tan\alpha)}{\tan\alpha}$

[EAMCET 2008]

37. P is a point on the segment joining the feet of two vertical poles of heights a and b. The angles of elevation of the tops of the poles from P are 45° each. Then, the square of the distance between the tops of the poles is

(a) $\frac{a^2+b^2}{2}$ (b) a^2+b^2

(c) $2(a^2+b^2)$ (d) $4(a^2+b^2)$

[EAMCET 2009]

38. $ABCD$ is a rectangular field. A vertical lamp post of height 12 m stands at the corner A. If the angle of elevation of its top from B is 60° and from C is 45°, then the area of the field is

(a) $48\sqrt{2}$ sq. m (b) $48\sqrt{3}$ sq. m

(c) 48 sq. m (d) $12\sqrt{2}$ sq. m

[Kerala CEE 2005]

39. A tower subtends an angle α at a point A in the plane of its base and the angle of depression of the foot of the tower at a point b feet just above A is β. Then the height of the tower is

(a) $b\tan\alpha\cot\beta$ (b) $b\cot\alpha\tan\beta$

(c) $b\cot\alpha\cot\beta$ (d) $b\tan\alpha\tan\beta$

[Kerala CEE 2006]

40. A flagpole stands on a building of height 450 ft and an observer on a level ground is 300 ft from the base of the building. The angle of elevation of the bottom of the flagpole is 30° and the height of the flagpole is 50 ft. If θ is the angle of elevation of the top of the flagpole, then $\tan\theta$ is equal to

(a) $\frac{4}{3\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{9}{2}$ (d) $\frac{\sqrt{3}}{5}$

[Kerala CEE 2008]

41. From the top of a tower, the angle of depression of a point on the ground is 60°. It the distance of this point from the tower is $\frac{1}{\sqrt{3}+1}$ m, then the height of the tower is

(a) $\frac{4\sqrt{3}}{2}$ m (b) $\frac{\sqrt{3}+3}{2}$ m

(c) $\frac{3-\sqrt{3}}{2}$ m (d) $\frac{\sqrt{3}}{2}$ m

[Kerala CEE 2009]

42. A house of height 100 m subtends a right angle at the window of an opposite house. If the height of the window is 64 m, then the distance between the two houses is
(a) 48 m (b) 36 m
(c) 54 m (d) 72 m
[WBJEE 2007]

43. On one bank of a river, there is a tree. On another bank, an observer makes an angle of elevation of 60° at the top of the tree. The angle of elevation of the top of the tree at a distance 20 m away from the bank is 30°. The width of the river is
(a) 20 m (b) 10 m
(c) 5 m (d) 1 m **[BCECE 2007]**

44. The elevation of an object on a hill is observed from a certain point in the horizontal plane through its base, to be 30°. After walking 120 m towards it on the level ground, the elevation is found to be 60° . Then the height of the object (in metres) is
(a) 120 (b) $60\sqrt{3}$
(c) $120\sqrt{3}$ (d) 60 **[BCECE 2008]**

45. ABC is a triangular park with $AB = AC = 100$ m. A clock tower is situated at the point of BC. The angles of elevation at the top of the tower at A and B are $\cot^{-1} 3.2$ and $\text{cosec}^{-1} 2.6$ respectively. The height of the tower is
(a) 16 m (b) 25 m
(c) 50 m (d) none of these
[Jamia Millia Islamia 2009]

ANSWERS

Exercise Set

1. (a)	**2.** (c)	**3.** (a)	**4.** (c)	**5.** (d)	**6.** (a)	**7.** (d)	**8.** (d)	**9.** (b)	**10.** (a)
11. (b)	**12.** (d)	**13.** (b)	**14.** (a)	**15.** (d)	**16.** (d)	**17.** (a)	**18.** (a)	**19.** (a)	**20.** (c)
21. (c)	**22.** (c)	**23.** (a)	**24.** (d)	**25.** (a)	**26.** (b)	**27.** (d)	**28.** (c)	**29.** (d)	**30.** (a)
31. (a)	**32.** (b)	**33.** (b)	**34.** (c)	**35.** (b)	**36.** (a)	**37.** (c)	**38.** (a)	**39.** (a)	**40.** (a)
41. (c)	**42.** (a)	**43.** (b)	**44.** (b)	**45.** (b)					

HINTS AND SOLUTIONS—EXERCISE SET

1. $\tan 30° = \dfrac{x}{50+y}$

$\Rightarrow (0.58)(50+y) = x$ (i)

$\tan 45° = \dfrac{x}{y}$

$\Rightarrow x = y$ [using (i)]

$\therefore 29 + 0.58x = x$

$\Rightarrow x = 69.04$

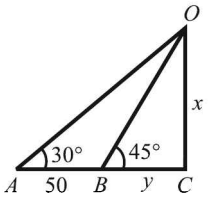


Fig. 30.24

2. Given $\tan\theta = \dfrac{1}{9}$

In ΔABC, $\tan\phi = \dfrac{3h}{3} = h$

In ΔABD, $\tan(\theta+\phi) = \dfrac{4h}{3}$

$\Rightarrow \dfrac{\tan\theta + \tan\phi}{1 - \tan\phi\tan\theta} = \dfrac{4h}{3}$

$\Rightarrow \dfrac{\frac{1}{9} + h}{1 - \frac{h}{9}} = \dfrac{4h}{3}$ $\Rightarrow \dfrac{1+9h}{9-h} = \dfrac{4h}{3}$

$\Rightarrow 3 + 27h = 36h - 4h^2$ $\Rightarrow 4h^2 - 9h + 3 = 0$

$\Rightarrow h = \dfrac{9 \pm \sqrt{81-48}}{2\times 4} = \dfrac{9 \pm \sqrt{33}}{8}$

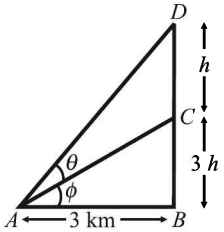


Fig. 30.25

3. Let x be the height of the tree.

Then $\tan 30° = \dfrac{10}{x}$

$\Rightarrow x = 10\sqrt{3} = 17.3$ m

4. In ΔABC, $\tan 60° = \dfrac{h}{BC}$

$\Rightarrow BC = h\cot 60°$ (i)

In ΔABD, $\tan 30° = \dfrac{h}{BD}$

$\Rightarrow BD = h\cot 30°$

$\Rightarrow BC + CD = h\cot 30°$

$\Rightarrow CD = h\cot 30° - BC$

$\Rightarrow d = h\cot 30° - h\cot 60°$ [from Eq. (i)]

$\therefore$ Speed of the car $= \dfrac{\text{distance from } D \text{ to } C}{\text{time taken}}$

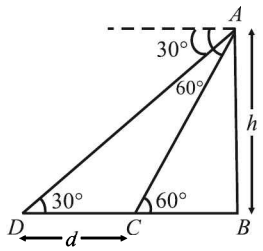


Fig. 30.26

$$= \frac{d}{3} = \frac{h\cot 30° - h\cot 60°}{3}$$

$\therefore$ Time taken from C to $B = \dfrac{\text{distance}}{\text{speed}}$

$$= \frac{h\cot 60°}{\frac{h(\cot 30° - \cot 60°)}{3}} = \frac{\frac{1}{\sqrt{3}} \times 3}{\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)}$$

$$= \frac{3}{2} = 1.5 \text{ min}$$

5. Let h be the height of the tower.

$\therefore$ In ΔOBT, right angled at O,

$OB = h\cot\beta$

Also, in triangle OAT right angled at O, we have

$OA = h\cot\alpha$

$$d = \sqrt{OB^2 + OA^2}$$

$$d = h\sqrt{\cot^2\beta + \cot^2\alpha}$$

$$\therefore \quad h = \frac{d}{\sqrt{\cot^2\alpha + \cot^2\beta}}$$

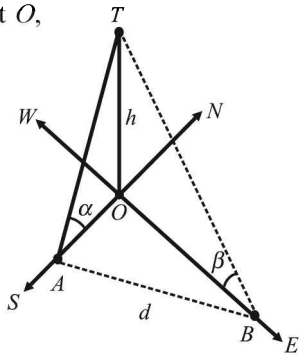


Fig. 30.27

6. Let BD be the road.

$PD = x$, $BP = 50 - x$

Let AB and CD be lamp posts of height y each.

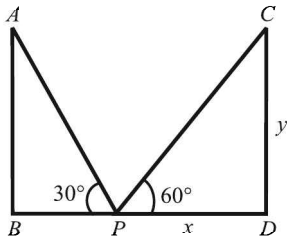


Fig. 30.28

$$\therefore \quad \tan 60° = \frac{y}{x} \quad \Rightarrow \quad y = x\sqrt{3}$$

$$\tan 30° = \frac{y}{50 - x} \quad \Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{x\sqrt{3}}{50 - x}$$

$\Rightarrow 50 - x = 3x$

$\Rightarrow 4x = 50 \Rightarrow x = 12.5$ m

$\therefore \quad y = (12.5)\sqrt{3} = 21.65$ m

7. In ΔDEC, $\tan\theta = \dfrac{DE}{CE}$

$$\Rightarrow \quad \tan\theta = \frac{h - a}{CE}$$

$$\Rightarrow \quad CE = \frac{h - a}{\tan\theta} \qquad \text{(i)}$$

In ΔABC,

$$\tan 2\theta \frac{a}{AB} = \frac{a}{CE} \quad (\because CE = AB)$$

$$\Rightarrow \quad \frac{2\tan\theta}{1 - \tan^2\theta} = \frac{a}{\frac{h - a}{\tan\theta}}$$

[from Eq. (i)]

$$\Rightarrow \quad a\tan\theta = (h - a) \times \frac{2\tan\theta}{1 - \tan^2\theta}$$

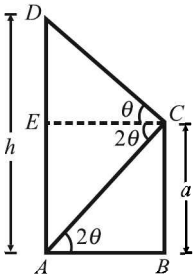


Fig. 30.29

$$\Rightarrow \quad a\tan\theta(1 - \tan^2\theta) - 2(h - a)\tan\theta = 0$$

$$\Rightarrow \quad \tan\theta(a - a\tan^2\theta - 2h + 2a) = 0$$

$\because \quad \tan\theta \neq 0$

$$\therefore \quad \tan^2\theta = \frac{-2h + 3a}{a}$$

$$\Rightarrow \quad \theta = \tan^{-1}\sqrt{\frac{-2h + 3a}{a}} = \tan^{-1}\sqrt{3 - \frac{2h}{a}}$$

8. $\tan\theta = \dfrac{4}{3}$ and $\tan\phi = \dfrac{5}{2}$ (i)

In ΔABE, $\tan\phi = \dfrac{150}{d}$

$\Rightarrow \quad d = 150\cot\phi$

$$= 150 \times \frac{2}{5} = 60 \text{ m} \qquad \text{(ii)}$$

In ΔDCE, $\tan\theta = \dfrac{h}{d}$

$$\Rightarrow \quad \frac{4}{3} = \frac{h}{d} \qquad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad h = \frac{4}{3}(60) \qquad \text{[from Eq. (ii)]}$$

$\Rightarrow \quad h = 80$ m

Now, in ΔDCE, $DE^2 = DC^2 + CE^2$

$\Rightarrow x^2 = 60^2 + 80^2 = 10000$

$\Rightarrow x = 100$ m

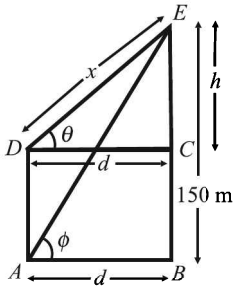


Fig. 30.30

9. Let OP be the peak and A and B be the extremities of diameter of the base of the hill.

$\therefore$ From the figure, we have

In ΔAOP, we have

$$d_1 = \frac{500}{\tan 60°}$$

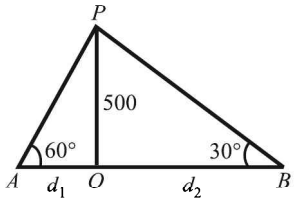


Fig. 30.31

In ΔBOP, we have

$$d_2 = 500\cot 30°$$

$\therefore$ Diameter of the Hill $D = d_1 + d_2$

$\Rightarrow$ $D = 500(\cot 60° + \cot 30°)$

$$D = 500\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = \frac{2000}{\sqrt{3}} \text{ m}$$

10. Let OP be the cliff of height 150 m and let A and B be two positions of the boat at an interval of 2 min.

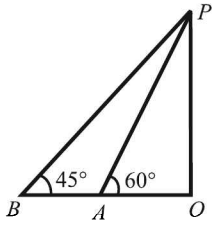


Fig. 30.32

$\angle OAP = 60°$ and $\angle OBP = 45°$

In ΔOAP and ΔOBP,

$$OA = OP\cot 60° = \frac{150}{\sqrt{3}} = 50\sqrt{3} \text{ m}$$

and $OB = OP \cot 45° = 150$ m

$\therefore$ Speed of the boat

$$= \frac{\text{distance } AB}{\text{time}} = \frac{50(3-\sqrt{3})}{2} \text{ m/min}$$

$$= 25(3-\sqrt{3})\frac{60}{1000} \text{ km/h}$$

$$= \frac{9-3\sqrt{3}}{2} \text{ km/h}$$

11. Let $AD = h$ be the height of the building and BP be the height of the hill. $BP = h + x$.

Then $\tan q = \dfrac{h+x}{y}$

and $\tan p = \dfrac{x}{y}$

$\Rightarrow$ $\tan q = \dfrac{h+x}{x\cot p}$

$\Rightarrow$ $x\cot p = (h+x)\cot q$

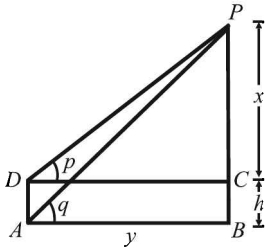


Fig. 30.33

$\Rightarrow$ $x = \dfrac{h\cot q}{\cot p - \cot q}$

$\therefore$ $h + x = \dfrac{h\cot p}{\cot p - \cot q} + h \Rightarrow h + x = \dfrac{h\cot q}{\cot p - \cot q}$

12. Let OA be the cliff and the ship be at point C, $\angle OCA = 44°$.

When ship has sailed 800 m, it is at B.

$\angle CBA = 19°$

Let $OC = x$, $OA = h$

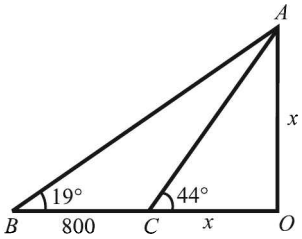


Fig. 30.34

$\therefore$ $\tan 44° = \dfrac{h}{x}$, $\tan 19° = \dfrac{h}{800+x}$

$\Rightarrow$ $x = \dfrac{h}{\tan 44°} = \dfrac{h}{0.9657} = (1.0355)h$

$\Rightarrow$ $(800+x)\tan 19° = h \Rightarrow 800\tan 19° + x\tan 19° = h$

$\Rightarrow$ $800(0.3443) + h(1.0355)(0.3443) = h$

$\Rightarrow$ $h(0.6435) = 275.62 \Rightarrow h = 428.07$

13. In ΔOBC, we have

$$\tan\alpha = \frac{5}{x} \quad \text{(i)}$$

Also, $\tan 2\alpha = \dfrac{30}{x}$ (ii)

Dividing (ii) by (i), we have

$$\tan 2\alpha = \frac{30}{5}\tan\alpha$$

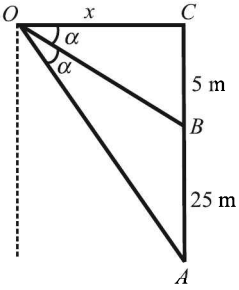


Fig. 30.35

$\Rightarrow$ $\dfrac{2\tan\alpha}{1-\tan^2\alpha} = 6\tan\alpha$

$\Rightarrow$ $\tan^2\alpha = \dfrac{2}{3}$ $\Rightarrow$ $\tan\alpha = \sqrt{\dfrac{2}{3}}$

$\therefore$ $x = 5\cot\alpha = 5\cdot\sqrt{\dfrac{3}{2}}$

14. Let AM be the lake, P the cloud and P' its reflection in the lake.

Let the height of cloud $MP = x$, then $MP' = x$

In the right-angled ΔPBC,

$$\frac{BC}{CP} = \cot\beta$$

$\therefore$ $BC = (x-h)\cot\beta$ (i)

In the right-angled $\Delta P'BC$,

$$\frac{BC}{CP'} = \cot\alpha$$

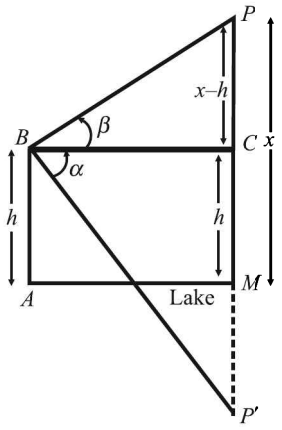


Fig. 30.36

$\Rightarrow BC = CP'\cot\alpha = (CM + MP')\cos\alpha$

$\Rightarrow BC = (x+h)\cot\alpha$ (ii)

From (i) and (ii), we get

$$(x-h)\cot\beta = (x+h)\cot\alpha$$

$\Rightarrow (x-h)\tan\alpha = (x+h)\tan\beta$

$\Rightarrow x(\tan\alpha - \tan\beta) = h(\tan\alpha + \tan\beta)$

$$\Rightarrow x\left(\frac{\sin\alpha}{\cos\alpha} - \frac{\sin\beta}{\cos\beta}\right) = h\left(\frac{\sin\alpha}{\cos\alpha} + \frac{\sin\beta}{\cos\beta}\right)$$

$\Rightarrow x(\sin\alpha\cos\beta - \cos\alpha\sin\beta)$

$= h(\sin\alpha\cos\beta + \cos\alpha\sin\beta)$

$\Rightarrow x\sin(\alpha-\beta) = h\sin(\alpha+\beta)$

$$\Rightarrow \text{Hence, } x = \frac{h\sin(\alpha+\beta)}{\sin(\alpha-\beta)}$$

15. Using the relation,

$$h = \frac{d\cot\beta}{\cot\alpha - \cot\beta} = \frac{100\cot 45°}{\cot 30° - \cot 45°}$$

$$= \frac{100\times 1}{\sqrt{3}-1}\times\frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{100\sqrt{3}}{2} = 50\sqrt{3}$$

16. Using the relation, height of flagstaff $= \dfrac{h(d^2+h^2)}{d^2-h^2}$

$$= \frac{10(225+100)}{225-100}$$

$$= \frac{3250}{125} = 26 \text{ m}$$

17. The height of the cloud

$$= h\left(\frac{\sin(\theta+\phi)}{\sin(\phi-\theta)}\right) = \frac{100\sin(30°+60°)}{\sin(60°-30°)}$$

$$= \frac{100\times 1}{1/2} = 200 \text{ m}$$

18. Let h be the height of the pole. The position PQ subtends an angle β at O such that

$$\beta = \tan^{-1}\frac{3}{5}$$

i.e. $\tan\beta = \dfrac{3}{5}$

Let $\theta = \alpha + \beta$

Then $\beta = \theta - \alpha$

$$\Rightarrow \tan\beta = \frac{\tan\theta - \tan\alpha}{1+\tan\theta\tan\alpha}$$

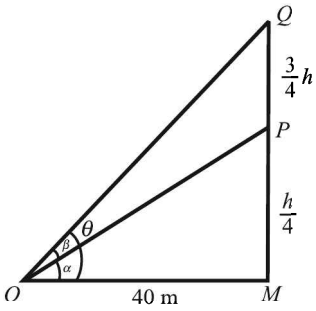


Fig. 30.37

$$\Rightarrow \frac{3}{5} = \frac{\frac{h}{40} - \frac{h/4}{40}}{1 + \frac{h}{40}\cdot\frac{h/4}{40}}$$

$$\Rightarrow \frac{3}{5} = \frac{\frac{h}{40} - \frac{h}{160}}{1 + \frac{h^2}{40\times 160}} \quad \Rightarrow \frac{3}{5} = \frac{3h}{160}\times\frac{40\times 160}{40\times 160 + h^2}$$

$$\Rightarrow \frac{3}{5} = \frac{120h}{6400+h^2} \quad \Rightarrow 3\times 6400 + 3h^2 = 600h$$

$\Rightarrow 6400 + h^2 = 200h \quad \Rightarrow h^2 - 200h + 6400 = 0$

$\Rightarrow (h-40)(h-160) = 0 \Rightarrow 40, 160$

Hence, h = 40 m

19. Let CD (= h) be the height of the tree and BC (= x) be the width of the river.

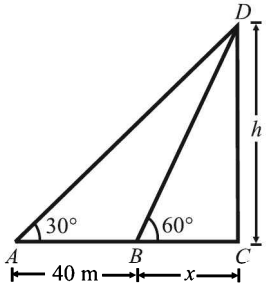


Fig. 30.38

In ΔBCD, $\tan 60° = \dfrac{CD}{BC}$

$$\Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow h = x\sqrt{3} \quad \text{(i)}$$

In ΔACD, $\tan 30° = \dfrac{CD}{AC}$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{40+x} \quad \Rightarrow \quad h\sqrt{3} = 40 + x$$

$\Rightarrow \quad 3x = 40 + x$ [from Eq. (i)]

$\Rightarrow \quad x = 20$ m

Hence, breadth of the river is 20 m.

20. Let h be the height of a tower. Since $\angle AOB = 60°$,
$\therefore$ ΔOAB is an equilateral triangle.

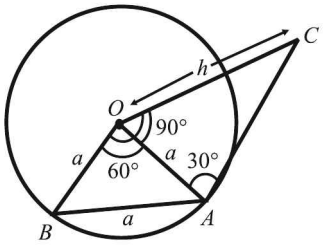


Fig. 30.39

$\therefore \quad OA = OB = AB = a$

In ΔACD, $\tan 30° = \dfrac{CD}{AC}$

$\Rightarrow \quad h = \dfrac{a}{\sqrt{3}}$

21. In ΔABC, $BC = h \cot 60°$ and in ΔABD,
$BD = h \cot 45°$
Since $BD - BC = DC$,

$\Rightarrow \quad h\cot 45° - h\cot 60° = 7$

$\Rightarrow \quad h = \dfrac{7}{\cot 45° - \cot 60°}$

$= \dfrac{7}{\left(1 - \dfrac{1}{\sqrt{3}}\right)}$

$= \dfrac{7\sqrt{3}(\sqrt{3}+1)}{2}$ m

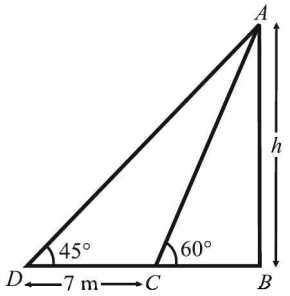


Fig. 30.40

22. Let DC be the height of the building.
$\therefore$ AB = speed × time

$= 25(\sqrt{3}-1) \cdot 2$

$= 50(\sqrt{3}-1)$

In ΔDBC,

$\tan 45° = \dfrac{DC}{BC} \Rightarrow Bc = h$

In ΔDAC,

$\tan 30° = \dfrac{h}{50(\sqrt{3}-1)+h}$

$\Rightarrow \quad 50(\sqrt{3}-1) + h = \sqrt{3}h \quad \Rightarrow \quad h = 50$ m

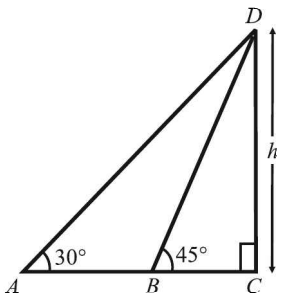


Fig. 30.41

23. Let AB and ED be two pillars and BD be the road.
Given, $AB = ED = h$
Let $BC = x \Rightarrow CD = 60 - x$

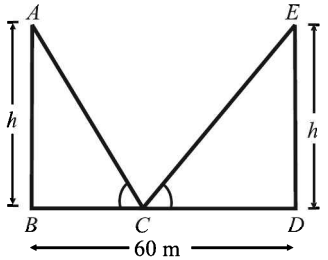


Fig. 30.42

From ΔABC,

$h = x \tan 60°$ (i)

and from ΔEDC,

$h = (60 - x)\tan 30°$ (ii)

From Eqs. (i) and (ii), we get

$x \tan 60° = (60 - x)\tan 30°$

$\Rightarrow \quad x\sqrt{3} = (60 - x)\dfrac{1}{\sqrt{3}}$

$\Rightarrow \quad 3x = 60 - x \quad \Rightarrow \quad 4x = 60$

$\Rightarrow \quad x = 15$

From Eq. (i), we get $h = 15\sqrt{3}$ m

$\therefore$ Height of the pillar is $15\sqrt{3}$ m.

24. Let AB be the tree of height h feet and breadth of the river be x feet.

In ΔABC, $\tan 60° = \dfrac{h}{x}$

$\Rightarrow \quad h = \sqrt{3}x$ (i)

and in ΔABD, $\tan 30° = \dfrac{h}{x+20}$

$\Rightarrow \quad \dfrac{1}{\sqrt{3}} = \dfrac{h}{x+20}$

$\Rightarrow \quad x + 20 = \sqrt{3}h$

$\Rightarrow \quad x + 20 = 3x$ [from Eq. (i)]

$\Rightarrow \quad 2x = 20 \quad \Rightarrow \quad x = 10$

$\therefore$ Breadth of the river is 10 ft.

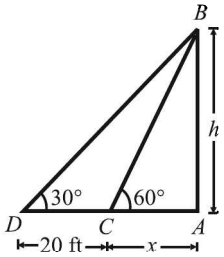


Fig. 30.43

25. Let the height of the flag be h.
In ΔARQ,

$\tan\alpha = \dfrac{b}{a}$ (i)

and in ΔPRQ,

$\tan 2\alpha = \dfrac{h+b}{a}$ (ii)

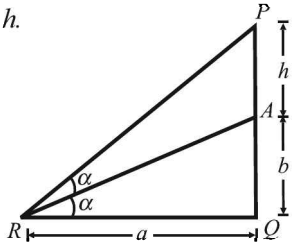


Fig. 30.44

$$\Rightarrow \quad \frac{2\tan\alpha}{1-\tan^2\alpha}=\frac{h+b}{a}$$

$$\Rightarrow \quad \frac{2\times\frac{b}{a}}{1-\frac{b^2}{a^2}}=\frac{h+b}{a} \qquad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad \frac{2ab}{a^2-b^2}=\frac{h+b}{a}\Rightarrow h=\frac{b(a^2+b^2)}{(a^2-b^2)}$$

26. Let the height of the tower be $BC = h$.

Then shadow of the tower $AB=\sqrt{3}h$

In ΔABC,

$$\tan\alpha=\frac{BC}{AB}$$

$$\Rightarrow \quad \tan\alpha=\frac{h}{\sqrt{3}h}$$

$$\Rightarrow \quad \tan\alpha=\frac{1}{\sqrt{3}}=\tan 30^\circ$$

$$\Rightarrow \quad \alpha=30^\circ$$

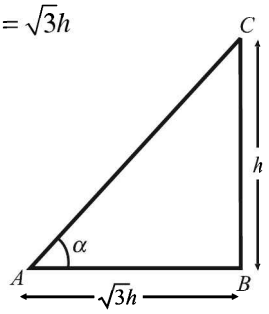


Fig. 30.45

27. Let the height of the cliff be BD = 50 m and height of the tower be $AE = h$ metres.

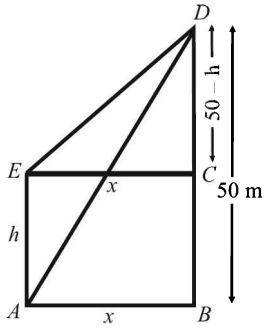


Fig. 30.46

In ΔDEC, $\tan 30^\circ=\dfrac{50-h}{x}$

$$\Rightarrow \quad x=\frac{50-h}{\frac{1}{\sqrt{3}}}=\sqrt{3}(50-h) \qquad \text{(i)}$$

and in ΔBAD, $\tan 45^\circ=\dfrac{50}{x} \quad\Rightarrow\quad x=50$ m

From Eq. (i), $50=\sqrt{3}(50-h)$

$$\Rightarrow \quad h=\frac{50(\sqrt{3}-1)}{\sqrt{3}}\text{ m} \quad\Rightarrow\quad h=50\left(1-\frac{\sqrt{3}}{3}\right)\text{m}$$

28. Let AB be a pole of height h and $BC = x$ be the shadow of pole and θ be the angle of elevation of the source of light.

Given, length of shadow of pole = height of pole

$$\Rightarrow \quad x=h \qquad \text{(i)}$$

In ΔABC, $\tan\theta=\dfrac{h}{x}$

$\Rightarrow \tan\theta = 1$

$\Rightarrow \theta = 45^\circ$

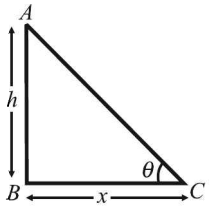


Fig. 30.47

29. In ΔBCD, $\tan 60^\circ=\dfrac{CD}{BC}=\dfrac{h}{x}$

$$\Rightarrow \quad h=\sqrt{3}x \qquad \text{(i)}$$

and in ΔACD,

$$\tan 30^\circ=\frac{CD}{AC}=\frac{h}{x+40}$$

$$\Rightarrow \quad \frac{1}{\sqrt{3}}=\frac{h}{40+x}$$

$$40+x=h\sqrt{3}$$

$$\Rightarrow \quad 40+x=3x$$

$$\Rightarrow \quad x=20 \qquad \text{[using Eq. (i)]}$$

On putting this value in Eq. (i), we get $h=20\sqrt{3}$ m

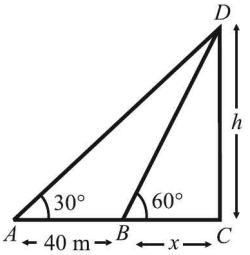


Fig. 30.48

30. Let D be the position of window whose height is H metres and BC be the position of the house.

In ΔABD, $\tan 60^\circ=\dfrac{AD}{AB}$

$$\Rightarrow \quad \sqrt{3}=\frac{H}{6} \Rightarrow h=6\sqrt{3}$$

In ΔDAB, $DB^2=(6\sqrt{3})^2+(6)^2$

$\Rightarrow DB=\sqrt{144}-12$ m

In ΔDCB, $\cos 30^\circ=\dfrac{DB}{BC}=\dfrac{12}{h}$

$$\Rightarrow \quad h=\frac{12}{\sqrt{3}/2}=8\sqrt{3}\text{ m}$$

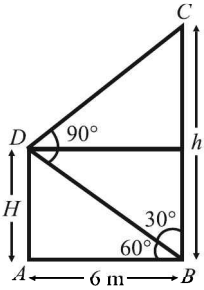


Fig. 30.49

31. In ΔABE, $\sin\beta=\dfrac{BE}{b}$

$\Rightarrow BE = h_1 = b\sin\beta$

Using sine rule in

ΔAED, $\dfrac{\sin(\alpha-\beta)}{ED}=\dfrac{\sin(\gamma-\alpha)}{b}$

$$\Rightarrow \quad ED=\frac{b\sin(\alpha-\beta)}{\sin(\gamma-\alpha)}$$

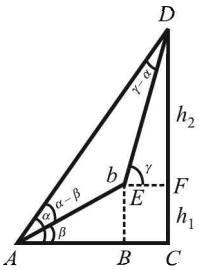


Fig. 30.50

Now, in ΔFED, $\sin\gamma = \dfrac{h_2}{ED}$

$\Rightarrow\ h_2 = \dfrac{b\sin(\alpha-\beta)\sin\gamma}{\sin(\gamma-\alpha)}$

$\therefore$ Total height, CD

$$= h_1 + h_2 = b\sin\beta + \frac{b\sin(\alpha-\beta)\sin\gamma}{\sin(\gamma-\alpha)}$$

$$= \frac{b[\sin\beta\sin(\gamma-\alpha)+\sin(\alpha-\beta)\sin\gamma]}{\sin(\gamma-\alpha)}$$

$$= \frac{\begin{matrix}b[\sin\beta(\sin\gamma\cos\alpha-\cos\gamma\sin\alpha)\\+\sin\gamma(\sin\alpha\cos\beta-\sin\beta\cos\alpha)]\end{matrix}}{\sin(\gamma-\alpha)}$$

$$= \frac{\begin{matrix}b[\sin\beta\sin\gamma\cos\alpha-\sin\beta\cos\gamma\sin\alpha\\+\sin\gamma\sin\alpha\sin\beta-\sin\gamma\sin\beta\cos\alpha]\end{matrix}}{\sin(\gamma-\alpha)}$$

$$= \frac{b\sin\alpha\sin(\gamma-\beta)}{\sin(\gamma-\alpha)}$$

32. Let $AC = x$ metres, $AB = h$ and $DC = 20$ m

In ΔABC, $\tan 45° = \dfrac{h}{x}$

$\because\ h = x$

In ΔABD, $\tan 30° = \dfrac{h}{20+x}$

$$\frac{1}{\sqrt{3}} = \frac{h}{20+h}$$

$$20 + h = \sqrt{3}h$$

$\because\ \dfrac{20}{(\sqrt{3}-1)} = h \quad \therefore\ h = 10(\sqrt{3}+1)$

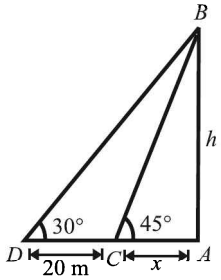


Fig. 30.51

33. In ΔECD, $\tan 3\alpha = \dfrac{h}{CD}$

$\Rightarrow\ CD = h\cot 3\alpha$ (i)

In ΔEBD, $\tan 2\alpha = \dfrac{h}{BD}$

$\Rightarrow\ BD = h\cot 2\alpha$ (ii)

In ΔEAD, $\tan\alpha = \dfrac{h}{AD}$

$\Rightarrow\ AD = h\cot\alpha$ (iii)

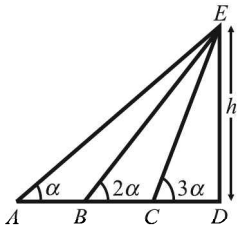


Fig. 30.52

From Eqs. (ii) and (iii), we have

$$AD - BD = h\cot\alpha - h\cot 2\alpha$$

$$AB = h(\cot\alpha - \cot 2\alpha) \qquad \text{(iv)}$$

From Eqs. (i) and (iv), we have

$$BD - CD = h\cot 2\alpha - h\cot 3\alpha$$

$\Rightarrow\ BC = h(\cot 2\alpha - \cot 3\alpha)$ (v)

From Eqs. (iv) and (v), we have

$$\frac{AB}{BC} = \frac{h(\cot\alpha - \cot 2\alpha)}{h(\cot 2\alpha - \cot 3\alpha)}$$

$$= \frac{\dfrac{\cos\alpha}{\sin\alpha} - \dfrac{\cos 2\alpha}{\sin 2\alpha}}{\dfrac{\cos 2\alpha}{\sin 2\alpha} - \dfrac{\cos 3\alpha}{\sin 3\alpha}} = \frac{\dfrac{\sin(2\alpha-\alpha)}{\sin\alpha\sin 2\alpha}}{\dfrac{\sin(3\alpha-2\alpha)}{\sin 2\alpha\sin 3\alpha}}$$

$$= \frac{\sin 3\alpha}{\sin\alpha} = 3 - 4\sin^2\alpha = 3 - 2(1-\cos 2\alpha)$$

$$= 1 + 2\cos 2\alpha$$

34. In ΔDAP, $\tan 60° = \dfrac{DP}{AP}$

$$AP = \frac{1}{\sqrt{3}} \quad [\because\ EQ = DP = 1]$$

In ΔEAQ, $\tan 30° = \dfrac{EQ}{AP + PQ}$

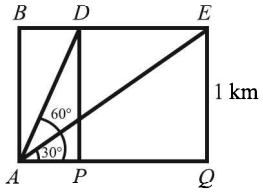


Fig. 30.53

$\Rightarrow$

$$\frac{1}{\sqrt{3}} = \frac{1}{1/\sqrt{3} + PQ}$$

$$PQ = \frac{2}{\sqrt{3}}\ \text{km}$$

$$\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{2/\sqrt{3}}{\dfrac{10}{60\times 60}} = 240\sqrt{3}\ \text{km/h}$$

35. Let BC be the height of the tower and CD be the height of the flagstaff.

In ΔBAC, $\tan\theta = \dfrac{x}{y}$ (i)

In ΔDAB, $\tan 2\theta = \dfrac{x+h}{y}$

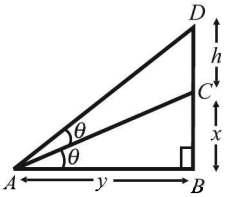


Fig. 30.54

$$\Rightarrow\ \frac{2\tan\theta}{1-\tan^2\theta} = \frac{x+h}{y}$$

$$\Rightarrow\ \frac{2\left(\dfrac{x}{y}\right)}{1-\dfrac{x^2}{y^2}} = \frac{x+h}{y} \qquad \text{[from Eq. (i)]}$$

$$\Rightarrow\ 2xy^2 - xy^2 + x^3 = (y^2 - x^2)h$$

$$\Rightarrow\ h = \frac{x(x^2+y^2)}{(y^2-x^2)}$$

36. Let h be the height of the hill and height of the pillar be H.

So, $AC = h$ and $BC = h - H$

Let $AD = a$

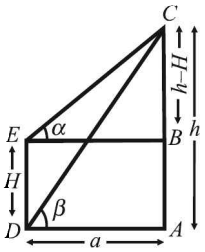


Fig. 30.55

In a ΔADC, $\tan\beta = \dfrac{h}{a}$

$\because \quad a = h\cot\beta \qquad$ (i)

In ΔBCE, $\tan\alpha = \dfrac{h-H}{a}$

$a\tan\alpha = h - H$

$H = h - a\tan\alpha$

$H = h - h\tan\alpha\cot\beta \qquad$ [by Eq. (i)]

$$H = h\left(1 - \frac{\tan\alpha}{\tan\beta}\right) = h\left(\frac{\tan\beta - \tan\alpha}{\tan\beta}\right)$$

37. In ΔAPD, $\tan 45° = \dfrac{a}{AP}$

$\Rightarrow \quad AP = a$

and in ΔBPC, $\tan 45° = \dfrac{b}{PB}$

$\Rightarrow \quad PB = b$

$\therefore \quad DE = a + b$

and $CE = b - a$

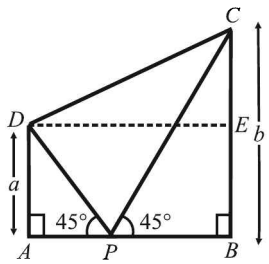


Fig. 30.56

In ΔDEC,

$DC^2 = DE^2 + EC^2$

$= (a+b)^2 + (b-a)^2 = 2(a^2 + b^2)$

38. In ΔABE, $\tan 60° = \dfrac{12}{AB}$

$\Rightarrow \quad AB = 4\sqrt{3}$ m

and in ΔACE, $\tan 45° = \dfrac{12}{AC}$

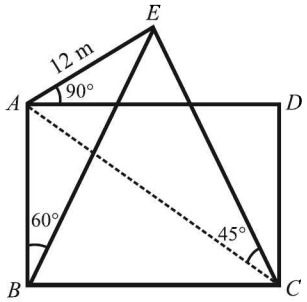


Fig. 30.57

$\Rightarrow \quad AC = 12$ m

In ΔABC, $BC = \sqrt{AC^2 - AB^2} = \sqrt{144 - 48} = 4\sqrt{6}$ m

$\therefore$ Area of the rectangular field = $AB \times BC$

$= 4\sqrt{3} \times 4\sqrt{6} = 48\sqrt{2}$ sq. m

39. Let CD be the tower.

In ΔACM, $\tan\beta = \dfrac{b}{AC}$

$\Rightarrow \quad AC = b\cot\beta$

and in ΔADC,

$\tan\alpha = \dfrac{CD}{AC}$

$\Rightarrow \quad CD = b\cot\beta\tan\alpha$

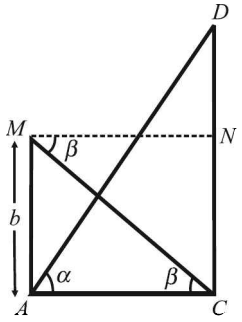


Fig. 30.58

40. In ΔDCE, $\tan 30° = \dfrac{150}{CD}$

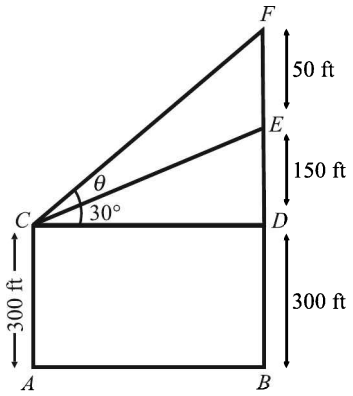


Fig. 30.59

$\Rightarrow \quad CD = \sqrt{3} \times 150$

Now, in ΔDCE, $\tan\theta = \dfrac{DF}{CD} = \dfrac{200}{\sqrt{3}\cdot 150} = \dfrac{4}{3\sqrt{3}}$

41. Let h be the height of the tower.

In ΔABC, $\tan 60° = \dfrac{h}{\frac{1}{(\sqrt{3}+1)}}$

$\Rightarrow \quad \dfrac{\sqrt{3}}{\sqrt{3}+1} = \dfrac{h}{1}$

$\Rightarrow \quad h = \dfrac{\sqrt{3}(\sqrt{3}-1)}{3-1} = \dfrac{3-\sqrt{3}}{2}$ m

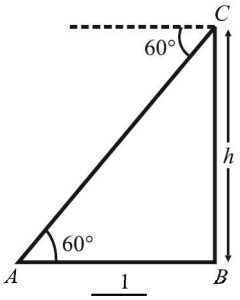


Fig. 30.60

42. In ΔADE, $\tan\theta = \dfrac{64}{x} \qquad$ (i)

In ΔBCE, $\tan(90° - \theta) = \dfrac{36}{x}$

$\cot\theta = \dfrac{36}{x} \qquad$ (ii)

By (i) and (ii),

$\Rightarrow \quad 1 = \left(\dfrac{64}{x}\right)\left(\dfrac{36}{x}\right) \quad x = 48$ m

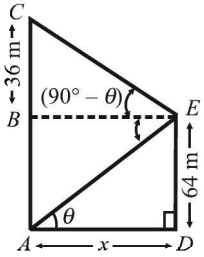


Fig. 30.61

43. Let h metres be the height of the tree CD and x metres be the width of the river.

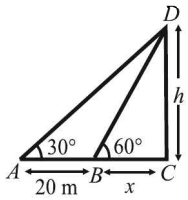


Fig. 30.62

In ΔBCD, $\tan 60° = \frac{h}{x}$

$\Rightarrow h = \sqrt{3}x$ (i)

and in ΔACD,

$\tan 30° = \frac{h}{x+20}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+20} \Rightarrow 3x = x+20$ [from Eq. (i)]

$\Rightarrow x = 10$ m

44. In ΔCAD, $\tan 30° = \frac{CD}{AC}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{120+x}$

$\Rightarrow \sqrt{3}h = 120 + x$ (i)

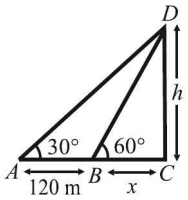


Fig. 30.63

and in ΔCBD, $\tan 60° = \frac{CD}{BC}$

$\Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow h = \sqrt{3}x$ (ii)

From Eqs. (i) and (ii), we get $x = 60$ m

On putting $x = 60$ in Eq. (i), we get

$h = 60\sqrt{3}$ m

45. Let OP be the clock tower standing at the mid-point O of the side BC of ΔABC. Let $\alpha = \angle PAO = \cot^{-1}3.2$ and $\beta = \angle PBO = \text{cosec}^{-1}2.6$

Then, $\cot\alpha = 3.2$

and $\text{cosec}\beta = 2.6$

$\cot\beta = \sqrt{\text{cosec}^2\beta - 1} = \sqrt{(2.6)^2 - 1} = 2.4$

In ΔPAO and ΔPBO, we have

$AO = h\cot\alpha = 3.2h$

and $BO = h\cot B = 2.4h$

In ΔABO,

$AB^2 = OA^2 + OB^2$

$\Rightarrow 100^2 = (3.2h)^2 + (2.4h)^2$

$\Rightarrow 100^2 = 16h^2$

$\Rightarrow h^2 = 625 \Rightarrow h = 25$ m

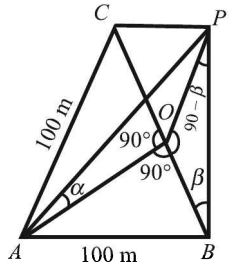


Fig. 30.64

MULTIPLE CHOICE TYPE QUESTIONS

1. A pole stands vertically inside a triangular park ABC. If the angle of elevation of the top of the pole from each corner of the park is the same, then in ΔABC, the foot of the pole is at the

(a) centroid (b) circumcentre
(c) incentre (d) orthocentre

2. The length of the shadows of a vertical pole of height h, thrown by the sun's rays at three different moments are h, $2h$ and $3h$. The sum of the angles of elevation of the rays at these three moments is equal to

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

3. Three vertical poles of heights h_1, h_2 and h_3 at the vertices A, B and C of ΔABC subtend angles α, β and γ respectively at the circumcentre of the triangle. If $\cot\alpha, \cot\beta$ and $\cot\gamma$ are in A.P., then h_1, h_2, h_3 are in

(a) A.P. (b) G.P.
(c) H.P. (d) none of these

4. A circular ring of radius 3 cm is suspended horizontally from a point 4 cm vertically above the centre by 4 strings attached at equal intervals to its circumference. If the angle between two consecutive strings be θ, then $\cos\theta$ =

(a) 4/5 (b) 4/25
(c) 16/25 (d) none of these

5. A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that AB $(= a)$ subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30°. The height of the tower is

(a) $\frac{a}{\sqrt{3}}$ (b) $\sqrt{3}a$
(c) $\frac{2a}{\sqrt{3}}$ (d) $2\sqrt{3}a$

6. The angle of elevation of a ladder leaning against a wall is

(a) 16.12 m (b) 17.26 m
(c) 18.11 m (d) none of these

7. A spherical balloon whose radius is r cm subtends at an observer's eye an angle α, when an angular deviation of the centre is β. The height of the centre of the balloon is

(a) $r\,\text{cosec}\frac{\alpha}{2}$ (b) $r\,\text{cosec}\frac{\alpha}{2}\sin\beta$
(c) $r\,\text{cosec}\alpha\,\text{cosec}\beta$ (d) none of these

8. The angles of elevation of the top of a T.V. tower from three points A, B, C in a straight line through the foot of the tower are α, 2α, 3α, respectively. If $AB = a$, the height of the tower is

(a) $a\tan\alpha$ (b) $a\sin\alpha$

(c) $a\sin 2\alpha$ (d) $a\sin 3\alpha$

9. From the bottom of a pole of height h, the angle of elevation of the top of a tower is α. The pole subtends an angle β at the top of a tower. The height of the tower is

(a) $\dfrac{h\sin\alpha\cos(\alpha+\beta)}{\cos\beta}$ (b) $\dfrac{h\sin\alpha\cos(\alpha-\beta)}{\sin\beta}$

(c) $\dfrac{h\sin\alpha\sin(\alpha+\beta)}{\sin\beta}$ (d) $\dfrac{h\sin\alpha\sin(\alpha-\beta)}{\sin\beta}$

10. ABC is a triangular park with $AB = AC = 100$ m. A clock tower is situated at the mid-point of BC. The angles of elevation of the top of the tower at A and B are $\cot^{-1}3.2$ and $\text{cosec}^{-1}2.6$, respectively. The height of the tower is

(a) 16 m (b) 25 m

(c) 50 m (d) none of these

11. The angle of elevation of a cloud from a point h metres above is $\theta°$ and the angle of depression of its reflection in the lake is ϕ. Then the height is

(a) $\dfrac{h\sin(\phi-\theta)}{\sin(\phi+\theta)}$ (b) $\dfrac{h\sin(\phi+\theta)}{\sin(\phi-\theta)}$

(c) $\dfrac{h\sin(\theta+\phi)}{\sin(\theta-\phi)}$ (d) none of these

12. Each side of a square subtends an angle of 60° at a tower h metres high standing in the centre of the square. If a is the length of each side of the square, then

(a) $2a^2 = h^2$ (b) $2h^2 = a^2$

(c) $3a^2 = 2h^2$ (d) $2h^2 = 3a^2$

ANSWERS

1. (b) **2.** (a) **3.** (a) **4.** (c) **5.** (a) **6.** (c) **7.** (b) **8.** (c) **9.** (b) **10.** (b)
11. (b) **12.** (b)

31

GEORGE BOOLE

Mathematical Reasoning

- Mathematical Logic
- Statement
- Types of Sentences
- Truth Value and Truth Table
- Elementary Operation of Logic
- Negation
- Conditional Sentence
- Biconditional Statement
- Some Useful Properties
- Valid Argument
- Methods of Testing the Validity of Arguments
- Duality
- Duality Symbol
- Algebra of Statements

MATHEMATICAL LOGIC

Logic is the subject which deals with the principles of reasoning. Logic is sometimes defined as the *science of proof.*

Mathematics and other science subjects deals with the reasoning and arguments and so every student of mathematics and other science subjects should know the principles of logic.

STATEMENT

We communicate our ideas or thoughts with the help of sentences in a particular language.

Note

1. A true statement is known as a valid statement.
2. A false statement is known as an invalid statement.
3. A statement cannot be both true and false at the same time.
4. Mathematical identities are considered to be statements.

TYPES OF SENTENCES

Following types of sentences are generally used in our daily normal life.

1. Assertive Sentence

A sentence that makes an assertion is called as assertive sentence or a declarative sentence.

Examples:
(i) The sun is a star.
(ii) Delhi is the capital of India.

2. Imperative Sentence

A sentence that expresses a request or a command is called an imperative sentence.
Examples:
(i) Give me a glass of water.
(ii) Go to school.

3. Exclamatory Sentence

A sentence that expresses some strong feeling is called an exclamatory sentence.
Examples:
(i) May I help you!
(ii) Hurrah! we have won the match.

4. Interrogative Sentence

A sentence that asks some question is called an interrogative sentence.
Examples:
(i) How young are you?
(ii) Where is your bag?

5. Optative Sentence

A sentence that expresses a wish is called an optative sentence.
Examples:
(i) God bless you.
(ii) Wish you best of luck.

6. Logical Statement or Proposition

A logical statement is any statement which is either true or false or equivalently valid or invalid. The falseness or truth of a statement is called its truth value. An open sentence is not a statement.
Examples:
(i) 'Every rectangle is a square' is a false statement.
(ii) 'Good morning to all' is not a statement.

7. Open Statement

A sentence which contains one or more variables such that when certain values are given to the variable it becomes a statement, is called an open statement.

8. Compound Statement

If two or more simple statements are combined by the use of words such as 'and', 'or', 'not', 'if', 'then', and 'if and only if', then the resulting statement is called a compound statement.

☞ TRUTH VALUE AND TRUTH TABLE

Truth Values

A statement can be either 'true' or 'false' which are called truth values of a statement and they are represented by the symbols T, F respectively.

Truth Table

A truth table is a summary of truth values of the resulting statements for all possible assignment of values to the variables appearing in a compound statement. The number of rows depends on the number of statements.

1. Truth table for single statement p:
   Number of rows = $2^1 = 2$.

p
T
F

2. Truth table for two statements p and q:
   Number of rows = $2^2 = 4$

p	q
T	T
T	F
F	T
F	F

☞ ELEMENTARY OPERATION OF LOGIC

Concavities: The words 'and', 'but', 'not', 'either … or', 'neither … nor', 'if … then', '… if and only if …' etc. are called concavities.

1. Conjuction

A compound sentence formed by two simple sentences p and q using connective 'and' is called the conjunction of p and q and it is represented by $p \wedge q$.

Example: Let p and q be two statements.
p: 5 is an integer.
q : The sun is shining.
$p \wedge q$: 5 is an integer and sun is shining.
The truth values of the conjunction $p \wedge q$:

1. The statement $p \wedge q$ has the truth value T whenever both p and q have the truth values T.
2. The statement $p \wedge q$ has the truth value F whenever either p or q or both have the truth value F.

Truth Table

q	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

2. Disjunction

A compound sentence formed by two simple sentences p and q using concavities 'or' is called the disjunction of p and q and it is represented by $p \vee q$.

Example: Let p and q are two statements.

p: Three plus two is five.

q: Two is a even number.

$p \vee q$: Three plus two is five or two is a even number.

Truth Table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

3. Negation

A statement which is formed by changing the truth values of a given statement by using the word like 'no', 'not', etc. is called negation of the given statement. If p is a statement, then negation of p is denoted by $\sim p$.

Example: p: Raju is fat, $\sim p$: Raju is not fat.

Note If p is true, then $\sim p$ is false. If p is false, then $\sim p$ is true.

Truth Table

p	$\sim p$
T	F
F	T

4. Conditional Sentence

When two simple sentences p and q are connected by the phase 'if and then', the new sentence is called the conditional sentence of p and q and it is denoted by $p \Rightarrow q$.

Note $p \Rightarrow q$ is true in all cases except when p is true and q is false. $p \Rightarrow q$ also means:

(i) p is sufficient for q (ii) q is necessary for p

(iii) p leads to q (iv) q if p

Truth Table

p	q	$p \Rightarrow q$	$q \Rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

5. Biconditional Statement

When two simple sentences are connected by the phrase 'if and only if' the new sentence is called the biconditional statement. It is denoted by the symbol '$\Leftrightarrow$'.

Note

(i) $p \Leftrightarrow q$ is true if both p and q have the same truth value, i.e. if either both p and q are true or both are false.

(ii) $p \Leftrightarrow q$ is false if p and q have opposite truth values.

$p \Leftrightarrow q$ means:

(i) p is a necessary and sufficient condition for q.

(ii) q is a necessary and sufficient condition for p.

(iii) if p then q and if q then p.

(iv) q if and only if p.

Truth Table

p	q	$p \Leftrightarrow q$
T	T	T
F	T	F
T	F	F
F	F	T

6. Tautology

The compound statements which are true for every value of their components are called tautology.

Example: Consider $p \wedge \sim p$.

Truth Table

p	$\sim p$	$p \wedge \sim q$
T	F	T
F	T	T

From the truth table, we observe that last columns is always true.

7. Contradiction (Fallacy)

The compound statements which are false for every value of their components are called contradiction (fallacy).

Example: Consider $p \vee \sim p$

Truth Table

p	$\sim p$	$p \wedge \sim q$
T	F	F
F	T	F

From the truth table, we observe that the last columns is always false.

SOME USEFUL PROPERTIES

1. $\sim(\sim p) = p$
2. $\sim(p \wedge q) = \sim p \vee \sim q$
3. $\sim(p \vee q) = \sim p \vee \sim q$
4. $(p \wedge q) \wedge r = p \wedge (q \wedge r)$
5. $p \Rightarrow q = (\sim q) \Rightarrow (\sim p)$
6. $p \Rightarrow q = \sim p \vee q$
7. $\sim(p \Rightarrow q) = p \wedge \sim q$
8. $(p \Leftrightarrow q) \Leftrightarrow r = p \Leftrightarrow (q \Leftrightarrow r)$

VALID ARGUMENT

An argument is said to be a valid argument if the conclusion p is true whenever all the hypotheses $p_1, p_2, p_3, \ldots, p_n$ are true or equivalently argument is valid when it is a tatuology, otherwise the argument is called an invalid argument.

METHODS OF TESTING THE VALIDITY OF ARGUMENTS

1. Construct the truth table for conditional statements $p_1 \wedge p_2 \wedge p_3 \wedge \ldots \wedge p_n \rightarrow p$.
2. Check the last column of the truth table. If the column contains T only, then the given argument is valid. Otherwise, it is an invalid argument.

DUALITY

Two compound statements S_1 and S_2 are said to be duals of each other, if one can be obtained from the other by replacing $\wedge$ by $\vee$ and $\vee$ by $\wedge$. The conectives $\wedge$ and $\vee$ are also called duals of each other.

DUALITY SYMBOL

Let $S(p, q) = p \wedge q$ be a compound statement. Then $S^*(p, q) = p \wedge q$, where $S^*(p, q)$ is the dual statement of $S(p, q)$.

ALGEBRA OF STATEMENTS

1. (a) $p \vee p \Leftrightarrow p$
   (b) $p \wedge p \Leftrightarrow p$ (Idempotent law)
2. (a) $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$
   (b) $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$ (Associative law)
3. (a) $p \vee q \Leftrightarrow q \vee p$
   (b) $p \wedge q \Leftrightarrow q \wedge p$ (Commutative law)
4. (a) $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
   (b) $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$(Distributive law)
5. (a) $p \vee \mathrm{F} \Leftrightarrow p$
   (b) $p \wedge \mathrm{T} \Leftrightarrow \mathrm{T}$
   (c) $p \wedge \mathrm{F} \Leftrightarrow \mathrm{F}$
6. (a) $p \vee \sim p \Leftrightarrow \mathrm{T}$
   (b) $p \wedge \mathrm{T} \Leftrightarrow p$
   (c) $p \wedge \sim p \Leftrightarrow \mathrm{F}$
7. (a) $p \vee (p \wedge q) \Leftrightarrow p$
   (b) $p \wedge (p \vee q) \Leftrightarrow p$ (Absorption law)
8. (a) $\sim(p \vee q) \Leftrightarrow \sim p \wedge \sim q$
   (b) $\sim(p \wedge q) \Leftrightarrow \sim p \vee \sim q$ (De morgan's law)

Note

(i) The number of rows of a table depends on the number of statements.
   (a) If p is false, then $\sim p$ is true.
   (b) If p is true, then $\sim p$ is false.

(ii) (a) The converse of $p \Rightarrow q$ is $q \Rightarrow p$.
   (b) The inverse of $p \Rightarrow q$ is $\sim p \Rightarrow \sim q$.

(iii) The contrapositive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$.

(iv) A statement which is neither a tautology nor a contradiction is a contingency.

SOLVED PROBLEMS

1. Which of the following is a statement?
   (a) May you live long!
   (b) May God bless you!
   (c) The sun is a star.
   (d) Hurrah! we have won the match.

Ans. (b)

Solution: "The sun is a star" is a statement.

2. If p: It rains today, q: I go to the school, r: I shall meet my friends and s: I shall go for a movie, then which of the following is the proposition?
If it does not rain or if I do not go to school, then I shall meet my friends and go for a movie.
   (a) $\sim(p \wedge q) \Rightarrow (r \wedge s)$ (b) $(\sim p \wedge \sim q) \Rightarrow (r \wedge s)$
   (c) $\sim(p \vee q) \Rightarrow (r \vee s)$ (d) none of these

Ans. (a)

Solution: The correct result is

$$(\sim p \vee \sim q) \Rightarrow (r \wedge s) = \sim(p \wedge q) \Rightarrow (r \wedge s)$$

3. Which of the following is not a statement?
   (a) Every set is a finite set.
   (b) 8 is less than 6.
   (c) Where are you going?
   (d) The sum of interior angles of a triangle is 180 degree.

Ans. (c)

Solution: "Where are you going" is not a statement.

4. The logically equivalent proposition of $p \Leftrightarrow q$ is
   (a) $(p \wedge q) \vee (p \wedge q)$ (b) $(p \Rightarrow q) \wedge (q \Rightarrow p)$
   (c) $(p \wedge q) \vee (q \Rightarrow p)$ (d) $(p \wedge q) \Rightarrow (p \vee q)$

Ans. (b)

Solution: $(p \Rightarrow q) \wedge (q \Rightarrow p)$ means $p \Leftrightarrow q$.

5. When does the current flow through the following circuit?
   (a) a, b, c should be open
   (b) a, b, c should be closed
   (c) always
   (d) none of these

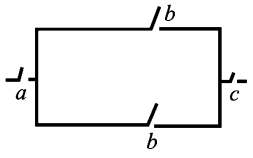


Fig. 31.1

Ans. (b)

Solution: a, b, c should be closed for the current to flow.

6. Which of the following is not a statement?
   (a) Give me a glass of water.
   (b) Asia is a continent.
   (c) The earth revolved round the sun.
   (c) The number 6 has two prime factors 2 and 3.

Ans. (a)

Solution: "Give me a glass of water" is not a statement.

7. Which of the following is true for any two statements p and q?

(a) $\sim[p \vee (\sim q)] \equiv (\sim p) \wedge q$

(b) $(p \vee q) \vee (\sim q)$ is a tautology

(c) $(p \wedge q) \wedge (\sim q)$ is a contradiction

(d) $\sim[p \wedge (\sim q)]$ is a tautology

Ans. (a)

Solution:

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim(p \wedge \sim q)$	$\sim p \wedge q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	F	T	T
F	F	T	T	T	F	F

$\therefore \quad \sim[p \vee (\sim q)] \equiv (\sim p) \wedge q$

8. Which Venn diagram represents the truth of the statement 'No policeman is a thief'?

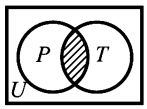
(a) 

Fig. 31.2

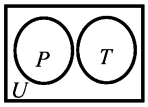
(b) 

Fig. 31.3

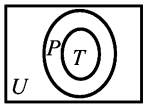
(c) 

Fig. 31.4

(d) none of these

Ans. (b)

Solution: No policeman is a thief means $P \cap T = \phi$, i.e. there is no common element (area) between P and T.

9. Negation of "$2 + 3 = 5$ of $8 < 10$" is

(a) $2+3 \neq 5$ and < 10 (b) $2 + 3 = 5$ and $8 \not< 10$

(c) $2+3 \neq 5$ or $8 \not< 10$ (d) none of these

Ans. (c)

Solution: Let p: $2 + 3 = 5$, q : $8 < 10$

Given proposition is $p \wedge q$.

Its negation is $\sim(p \wedge q) = \sim p \vee \sim q$

$\therefore$ We have $2+3 \neq 5$ or $8 \neq 10$.

10. The contrapositive of the inverse of $p \Rightarrow \sim q$ is

(a) $\sim q \Rightarrow p$ (b) $p \Rightarrow q$

(c) $\sim q \Rightarrow \sim p$ (d) $\sim p \Rightarrow \sim q$

Ans. (a)

Solution: The inverse of $p \Rightarrow \sim q$ is $\sim p \Rightarrow q$

The contrapositive of $\sim p \Rightarrow q$ is $\sim q \Rightarrow p$

[contrapositive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$]

EXERCISE SET

1. Which of the following is the inverse of the proposition 'If a number is a prime, then it is odd'?

(a) If a number is not a prime, then it is odd

(b) If a number is not odd, then it is prime

(c) If a number is prime, then it is odd

(d) If a number is not odd, then it is not a prime

2. $\sim(p \wedge q)$ is equal to

(a) $\sim p \vee \sim q$ (b) $\sim p \wedge \sim q$

(c) $\sim p \vee q$ (d) $p \vee \sim q$

3. If $p \Rightarrow (q \vee r)$ is false, then the truth values of p, q, r are respectively

(a) F, T, T (b) T, T, F

(c) T, F, F (d) F, F, F

4. If p: Ram works hard

q: Ram gets good grade

Then the verbal form for $(\sim p \rightarrow q)$ is

(a) If Ram works hard, then he gets good grade

(b) If Ram does not work hard, then he gets good grade

(c) Ram works hard if and only if he gets good grade

(d) If Ram does not work hard, then he does not get good grade

5. $p \Rightarrow q$ can also be written as

(a) $p \Rightarrow \sim q$ (b) $\sim p \vee q$

(c) $\sim q \Rightarrow \sim p$ (d) none of these

6. A compound sentence formed by two simple statements p and q using connective 'and' is called

(a) conjunction (b) disjunction

(c) implication (d) none of these

7. If p: Ajay is tall

q: Ajay is intelligent

then the symbolic statement $\sim p \vee q$ means

(a) Ajay is not tall or he is intelligent

(b) Ajay is tall or he is intelligent

(c) Ajay is not tall and he is intelligent

(d) Ajay is not tall then he is intelligent

8. If p and q are simple propositions, then $p \Rightarrow q$ is false when

(a) p is true and q is true

(b) p is false and q is true

(c) p is true and q is false

(d) both p and q are false

9. If $p = \Delta ABC$ is an equilateral and q = each angle is 60°, then the symbolic form of the statement "ΔABC is an equilateral iff each of its angle is of 60°" is
(a) $p \vee q$ (b) $p \wedge q$ (c) $p \Rightarrow q$ (d) $p \Leftrightarrow q$

10. Which of the following logically equivalent to $\sim(\sim p \Rightarrow q)$?
(a) $p \wedge q$ (b) $\sim p \wedge q$ (c) $\sim p \wedge \sim q$ (d) $p \wedge \sim q$

11. $\sim(p \Rightarrow q) \Leftrightarrow \sim p \vee \sim q$ is
(a) a tautology
(b) a contradiction
(c) neither a tautology nor a contradiction
(d) cannot come to any conclusion

12. $\sim[(\sim p) \wedge q]$ is equal to
(a) $p \wedge (\sim q)$ (b) $p \vee q$ (c) $p \vee (\sim q)$ (d) $\sim p \wedge \sim q$

13. Which of the following is not logically equivalent to the proposition:
"A real number is either rational or irrational"?
(a) If a number is neither rational nor irrational, then it is not real
(b) If a number is not rational or not irrational, then it is not real
(c) If a number is not real, then it is neither rational nor irrational
(d) If a number is real, then it is rational or irrational

14. Which of the following is always true?
(a) $(p \Rightarrow q) \equiv \sim q \Rightarrow \sim p$ (b) $\sim(p \vee q) \equiv p \vee q$
(c) $\sim(p \Rightarrow q) \equiv p \wedge \sim q$ (d) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

15. Which Venn diagram represents the truth of the statement "No child is naughty" where
U = universal set of human beings
C = set of children
N = set of naughty persons

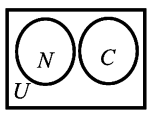
(a) 

Fig. 31.5

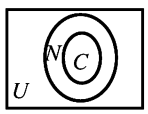
(b) 

Fig. 31.6

(c)

Fig. 31.7

(d) none of these

16. Negation of "New Delhi is in India and London is in England" is
(a) New Delhi is in India and London is not in England
(b) New Delhi is in India or London is not in England
(c) New Delhi is not in India or London is in England
(d) New Delhi is not in India or London is not in England

17. Which of the following is an open statement?
(a) x is a natural number
(b) Give me a glass of water
(c) Wish you best of luck
(d) Good morning to all.

Assertion-Reason Type Questions

Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) or (d) given below:
(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true; Statement II is false.
(d) Statement I is false; Statement II is true.

18. **Statement I:** If p = '11 is an integer', q = 'Apple is sour', the symbolic form of 'if apple is not sour, then 11 is an odd integer is $\sim q \Rightarrow p$.
Statement II: It is a biconditional statement.

19. **Statement I:** "Ram is rich or happy" is a statement.
Statement II: It is disjunction of the pair of statements.

20. **Statement I:** The dual of $p \vee (\sim q) \wedge (\sim p)$ is $p \wedge (\sim q) \vee (\sim p)$.
Statement II: In a dual statement, we interchange their operators $\wedge$ and $\vee$.

21. **Statement I:** The converse of $p \Rightarrow q$ is $q \Rightarrow p$.
Statement II: The converse of inverse of $p \Rightarrow q$ is $\sim p \Rightarrow \sim q$.

Previous Years' Questions

22. Let p be the statement "x is an irrational number", q be the statement "y is a transcendental number" and r be the statement "x is a rational number iff y is a transcendental number".
Statement I: r is equivalent to either q or p.
Statement II: r is equivalent to $\sim(p \leftrightarrow \sim q)$.
(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.
[AIEEE 2008]

23. The statement $p \rightarrow (q \rightarrow p)$ is equivalent to
(a) $p \rightarrow (p \leftrightarrow q)$ (b) $p \rightarrow (p \rightarrow q)$
(c) $p \rightarrow (p \vee q)$ (d) $p \rightarrow (p \wedge q)$
[AIEEE 2008]

24. **Statement I:** $\sim(p \leftrightarrow \sim q)$ is equivalent to $p \leftrightarrow q$.
Statement II: $\sim(p \leftrightarrow \sim q)$ is a tautology.
(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.
[AIEEE 2009]

25. Le S be a non-empty subset of R. Consider the following statement:
P: There is a rational number $x \in S$ such that $x > 0$.
Which of the following statements is the negation of the statement P?
(a) There is a rational number $x \in S$ such that $x \leq 0$
(b) There is no rational number $x \in S$ such that $x \leq 0$
(c) Every rational number $x \in S$ satisfies $x \leq 0$
(d) $x \in S$ and $x \leq 0 \Rightarrow x$ is not rational
[AIEEE 2010]

26. $\sim(p \vee q) \vee (\sim p \wedge q)$ is logically equivalent to
(a) $\sim p$ (b) p
(c) q (d) $\sim q$ **[DCE 2008]**

27. The negation of the statement "he is rich and happy" is given by
(a) he is not rich and not happy
(b) he is not rich or not happy
(c) he is rich and happy
(d) he is not rich and happy **[MHT CET 2006]**

28. If p: A man is happy
q: A man is rich
then the statement, "If a man is not happy, then he is not rich" is written as
(a) $\sim p \rightarrow \sim q$ (b) $\sim q \rightarrow p$
(c) $\sim q \rightarrow \sim p$ (d) $q \rightarrow \sim p$
[MHT CET 2004]

29. Dual of $(x' \vee y')' = x \wedge y$ is
(a) $(x' \vee y') = x \vee y$ (b) $(x' \wedge y')' = x \vee y$
(c) $(x' \wedge y')' = x \wedge y$ (d) none of the above
[MHT CET 2008]

30. Which of the following statements has the truth value 'F'?
(a) A quadratic equation has always a real root
(b) The number of ways of seating 2 persons in two chairs out of n persons is $P(n, 2)$
(c) The cube roots of unity are in G.P.
(d) none of these **[MHT CET 2007]**

31. If $S(p, q, r) = (\sim p) \vee [\sim(q \wedge r)]$ is a compound statement, then $S(\sim p, \sim q, \sim r)$ is
(a) $\sim S(p, q, r)$ (b) $S(p, q, r)$
(c) $p \vee (q \wedge r)$ (d) $p \vee (q \vee r)$
[Kerala CEE 2010]

32. Let p: 7 is not greater than 4 and q: Paris is in France be two statements.
Then, $\sim(p \vee q)$ is the statement of
(a) 7 is greater than 4 or Paris is not in France
(b) 7 is not greater than 4 and Paris is not in France
(c) 7 is greater than 4 and Paris is not in France
(d) 7 is greater than 4 and Paris is in France
[Kerala CEE 2010]

33. If p: 4 is an even prime number, q: 6 is a divisor of 12 and r: the HCF of 4 and 6 is 2, then which one of the following is true?
(a) $(p \wedge q)$ (b) $(p \vee q) \wedge \sim r$
(c) $\sim(q \wedge r) \vee \sim p$ (d) $\sim p \vee (q \wedge r)$
[Kerala CEE 2007]

34. Let p be the statement 'Ravi races' and let q be the statement 'Ravi wins'. Then the verbal translation of $\sim[p \vee (\sim q)]$ is
(a) Ravi does not race and Ravi does not win
(b) It is not true that Ravi races and that Ravi does not win
(c) Ravi does not race and Ravi wins
(d) It is not true that Ravi races or that Ravi does not win **[Kerala CEE 2008]**

35. If p, q and r are simple propositions with truth values T, F, T, then the truth value of $(\sim p \vee q) \wedge \sim q \rightarrow p$ is
(a) true (b) false
(c) true, if r is false (d) none of these
[BCECE 2008]

36. Let p and q be two statements, then $(p \vee q) \vee \sim p$ is
(a) a tautology (b) a contradiction
(c) both (a) and (b) (d) none of these
[BCECE 2003]

37. If $(p \wedge \sim r) \rightarrow (\sim p \vee p)$ is false, then the truth values of p, q and r are respectively
(a) T, F and F (b) F, F and T
(c) F, T and T (d) T, F and T
[KECT 2006]

38. Which of the following is not a proposition?
(a) 3 is prime
(b) $\sqrt{2}$ is irrational
(c) Mathematics is interesting
(d) 5 is an even integer **[KCET 2003]**

39. If $p \Rightarrow (\sim p \vee q)$ is false, the truth values of p and q are respectively
(a) F, T (b) F, F (c) T, T (d) T, F
[Karnataka C.E.T. 2002]

40. The negative of $q \vee \sim(p \wedge r)$ is
(a) $\sim q \wedge \sim(p \wedge r)$ (b) $\sim q \wedge (p \wedge r)$
(c) $\sim q \vee (p \wedge r)$ (d) $\sim q \vee (p \wedge r)$
[Karnataka C.E.T. 1997]

41. The false statement in the following is
(a) $p \wedge (\sim p)$ is a contradiction
(b) $(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$ is a contradiction
(c) $\sim(\sim p) \Leftrightarrow p$ is a tautology
(d) $p \vee (\sim p)$ is a tautology

42. $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is
(a) a tautology
(b) a contradiction
(c) both a tautology and a contradiction
(d) neither a tautology nor a contradiction
[Karnataka C.E.T. 2003]

43. The only statement among the following that is a tautology is
(a) $A \wedge (A \vee B)$ (b) $A \vee (A \wedge B)$
(c) $[A \wedge (A \to B)] \to B$ (d) $B \to [A \wedge (A \to B)]$
[AIEEE 2011]

44. The negation of the statement "If I become a teacher, then I will open a school" is
(a) I will become a teacher and I will not open a school
(b) Either I will not become a teacher or I will not open a school
(c) Neither I will become a teacher nor I will open a school
(d) I will not become a teacher or I will open a school
[AIEEE 2012]

ANSWERS

Exercise Set

1. (d)	**2.** (a)	**3.** (c)	**4.** (b)	**5.** (b)	**6.** (a)	**7.** (a)	**8.** (c)	**9.** (d)	**10.** (c)
11. (c)	**12.** (c)	**13.** (b)	**14.** (c)	**15.** (a)	**16.** (d)	**17.** (a)	**18.** (c)	**19.** (b)	**20.** (a)
21. (a)	**22.** (*)	**23.** (c)	**24.** (c)	**25.** (c)	**26.** (a)	**27.** (b)	**28.** (a)	**29.** (b)	**30.** (a)
31. (d)	**32.** (c)	**33.** (d)	**34.** (c)	**35.** (a)	**36.** (a)	**37.** (a)	**38.** (c)	**39.** (d)	**40.** (b)
41. (b)	**42.** (b)	**43.** (c)	**44.** (a)						

Note: (*) denotes that none of the given option is correct.

HINTS AND SOLUTIONS—EXERCISE SET

1. If *p: A* number is a prime
q: It is odd
We have $p \Rightarrow q$
The inverse of $p \Rightarrow q$ is $\sim p \Rightarrow \sim q$
i.e. if a number is not a prime, then it is not odd.

2. $\sim(p \wedge q) \equiv \sim p \vee \sim q$

3. $p \Rightarrow q$ is false only when p is true and q is false
$p \Rightarrow q$ is false when p is true and $q \vee r$ is false, and $q \vee r$ is false when both q and r are false.

4. $\sim p$: Ram does not work hard
Use '$\to$' symbol for then
$(\sim p \to q)$ means, if Ram does not work hard, then he gets good grade

5. $p \Rightarrow q \equiv \sim p \vee q$

6. A compound sentence formed by two simple statements p and q using connective 'and' is called conjunction.

7. $\sim p$: Ajay is not tall
Use '$\vee$' symbol for so or
$\sim p \vee q$ mean Ajay is not tall or he is intelligent.

8. $p \Rightarrow q$ is false, when p it true and q is false
Since q, r are false $\therefore$ $q \vee r$ is false
Since r is false $\therefore$ $\sim r$ is true
Since $p \wedge \sim r$ is true $\therefore$ p is true

9. The symbolic form of given statement is $p \Rightarrow q$.

10. Since $\sim(p \Rightarrow q) = p \wedge \sim q$, $\sim(\sim p \Rightarrow q) = \sim p \wedge \sim q$

11.

p	q	$p \Rightarrow q$	$\sim(p \Rightarrow q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$\sim(p \Rightarrow q) \Leftrightarrow \sim p \vee \sim q$
T	T	T	F	F	F	F	T
T	F	F	T	F	T	T	T
F	T	T	F	T	F	T	F
F	F	T	F	T	T	T	F

The last column shows that the result is neither a tautology nor a contradiction

12. $\sim[(\sim p) \wedge q] \equiv \sim(\sim p) \vee \sim q \equiv p \vee \sim q$

13. $\sqrt{3}$ is not rational but it is real

14. If $p \Rightarrow q \equiv \sim p \vee q$
$\sim(p \Rightarrow q) \equiv p \wedge \sim q$
It means $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$

15. "No child is naughty" means $C \cap N = \phi$
i.e. there is no common element between C and N.

16. Let p: New Delhi is in India
q: London is in England
We have $p \wedge q$
Its negation is $\sim(p \wedge q)$
We know that $\sim(p \wedge q) = \sim p \vee \sim q$
i.e. New Delhi is not in India or London is not in England.

17. "x is a rational number" is an open statement.

18. Statement I is a true statement but Statement II is not a true statement.

19. Statement I is a true statement.
Statement II is also a true statement but it is not a proper explanation for Statement I.

20. Both statement are true and Statement II is the correct explanation for Statement I.

21. Both statements are true and Statement II is the correct explanation for Statement I.

22. Here, p: x is an irrational number, q: y is a transcendental number, r: x is a rational number, iff y is a transcendental number.
$\Rightarrow$ $r: \sim p \leftrightarrow q$
$I: r \equiv q \vee p$
and $II: r \equiv \sim q(p \leftrightarrow \sim q)$

p	q	$\sim p$	$\sim q$	$\sim p \leftrightarrow q$	$q \vee p$	Statement I $(p \leftrightarrow \sim q)$	Statement II $\sim(p \leftrightarrow \sim q)$
T	T	F	F	F	T	F	T
T	F	F	T	T	T	T	F
F	T	T	F	T	T	T	F
F	F	T	T	F	F	F	T

It is clear from the table that r is not equivalent to either of the statements.
Hence, none of the given options is correct.

23.

q	p	$q \to p$	$p \to (q \to p)$	$p \vee q$	$p \to (p \vee q)$
T	T	F	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	F	T	T	F	T

$\therefore$ Statement $p \to (q \to p)$ is equivalent to $p \to (p \vee q)$.

24.

p	q	$p \leftrightarrow q$	$\sim q$	$(p \leftrightarrow \sim q)$	$\sim(p \leftrightarrow \sim q)$
T	T	T	F	F	T
T	F	F	T	T	F
F	T	F	F	T	F
F	F	T	T	F	T

25. P: There is rational number $x \in S$ such that $x > 0$
$\sim P$: Every rational number $x \in S$ satisfies $x \le 0$

26. $\sim(p \vee q) \vee (\sim p \wedge q) \equiv (\sim p \wedge \sim p) \vee (\sim p \wedge q)$
$\equiv \sim p \wedge (\sim q \vee q) \equiv \sim p \wedge t \equiv \sim p$

27. The negation of the given statment is "he is not rich or not happy".

28. 'If a man is not happy, then he is not rich' is written as $\sim p \to \sim q$.

29. Dual of $(x' \vee y')' = x \wedge y$ is $(x' \wedge y')' = x \vee y$

30. The roots of a quadratic equation can be imaginary

31. $S(p, q, r) = (\sim p) \vee [\sim(q \wedge r)]$
$= (\sim p) \vee [\sim q \vee \sim r]$
$\Rightarrow$ $S(\sim p, \sim q, \sim r) = p \vee (q \vee r)$

32. $\sim(p \vee q) \equiv \sim p \wedge \sim q$
$\therefore$ 7 is greater than 4 and Paris is not in France.

33. $\because$ p: 4 is an even prime number.
q: 6 is a divisor of 12 and r: the HCF of 4 and 6 is 2.
$\therefore$ $\sim p \vee (q \wedge r)$ is true.

34. p: Ravi races, q: Ravi wins
$\therefore$ The statement of given proposition $\sim[p \vee (\sim q)]$ is equivalent to $\sim p \wedge q$.
"It is not true that Ravi races or that Ravi does not win".

35.

p	q	r	$\sim p$	$\sim q$	$\sim p \vee q$	$(\sim p \vee q) \wedge \sim q$	$(\sim p \vee q) \wedge \sim q \to p$
T	F	T	F	T	F	F	T

36.

p	q	$p \vee q$	$\sim p$	$(p \vee q) \vee \sim p$
T	T	T	F	T
T	F	T	F	T
F	T	T	T	T
F	F	F	T	T

It is clear that $(p \vee q) \vee \sim p$ is a tautology.

37.

p	q	r	$\sim p$	$\sim r$	$p \wedge \sim r$	$(\sim p \vee q)$	$p \vee \sim r$ $(\sim p \vee q)$
T	T	T	F	F	F	T	T
T	T	F	F	T	T	T	T
T	F	T	F	F	F	F	T
T	F	F	F	T	T	F	F
F	T	T	T	F	F	T	T
F	T	F	T	T	F	T	T
F	F	T	T	F	F	T	T
F	F	F	T	T	F	T	T

Hence, $(p \wedge \sim r) \to (\sim p \vee q)$ is F
When $p = \text{T}$, $q = \text{F}$, $r = \text{F}$

38. Mathematics is interesting is not a proposition.

39. $p \Rightarrow (\sim p \vee q)$ is false means p is true and $\sim p \vee q$ is false.
$\Rightarrow$ p is true and both $\sim p$ and q are false
$\Rightarrow$ p is true and q is false

40. $\sim[q \vee \sim(p \wedge r)] = \sim q \wedge [\sim(p \wedge r)]$
$= \sim q \wedge (p \wedge r)$

41. $p \Rightarrow q$ is logically equivalent to $\sim q \Rightarrow \sim p$
$\therefore \quad (p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$
is a tautology but not a contradiction.

42. $(p \wedge \sim q) \wedge (\sim p \wedge q) = (p \wedge \sim p) \wedge (\sim p \wedge q)$
$= f \wedge f = f$
(By using associative and commutative laws)
$\therefore \quad (p \wedge \sim q) \wedge (\sim p \wedge q)$ is a contradiction.

43.

A	T	F	T	F
B	F	T	T	F
$A \vee B$	T	T	T	F
$A \wedge B$	F	F	T	F
$A \wedge (A \vee B)$	T	F	T	F
$A \vee (A \wedge B)$	T	F	T	F
$A \rightarrow B$	F	T	T	T
$A \wedge (A \rightarrow B)$	F	F	T	F
$[A \wedge (A \rightarrow B)] \rightarrow B$	T	T	T	T
$B \rightarrow [A \wedge (A \rightarrow B)]$	T	F	T	T

44. Let p: I become a teacher.
q: I will open a school.
Negation of $p \rightarrow q$ is $\sim(\sim p \vee q) = p \wedge \sim q$
i.e. I will become a teacher and I will not open a school.

MULTIPLE CHOICE TYPE QUESTIONS

1. If p and q are true and r and s are false then which of the following expressions has the truth value false?
(a) $p \vee (q \vee r)$
(b) $[p \wedge (q \wedge r)] \vee [(p \vee q) \wedge (r \vee s)]$
(c) $[\sim(p \wedge q) \vee \sim r] \vee \{[(\sim p \vee q) \vee \sim r] \wedge s\}$
(d) none of three

2. Which of the following is not a tautology?
(a) $(p \wedge q) \Rightarrow p$
(b) $(p \wedge q) \Rightarrow (p \Rightarrow q)$
(c) $\sim(p \wedge q) \vee [(\sim p) \wedge q] \vee p$
(d) $(q \Rightarrow p) \Rightarrow p$

3. All equivalent formula for $p \wedge (q \Leftrightarrow r)$ which is free from biconditionals as wall as conditionals is
(a) $p \vee [(\sim q \vee r) \wedge (\sim r \vee q)]$
(b) $p \wedge [(\sim q \vee r) \wedge (\sim r \vee q)]$
(c) $p \wedge [(\sim q \wedge r) \wedge (\sim r \vee q)]$
(d) $p \wedge [(\sim q \vee r) \wedge (\sim r \wedge q)]$

4. Choose the correct alternative form the following:
(i) $(p \wedge q) \rightarrow (p \vee q)$ is a theorem
(ii) $(p \wedge q) \rightarrow (p \vee q)$ is a fallacy
(iii) $\sim(p \vee q) \leftrightarrow (\sim p \vee \sim q)$ is a fallacy
(iv) $(p \wedge q) \rightarrow (p \vee q)$ is a theorem
(a) (i) only (b) (i) and (iv)
(c) (i) and (iii) (d) none of these

5. The dual of the statement $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ is
(a) $(p \vee q) \wedge r \equiv p \wedge (q \vee r)$
(b) $(p \vee q) \vee r \equiv p \wedge (q \vee r)$
(c) $(p \vee q) \vee r \equiv p \vee (q \vee r)$
(d) $(p \wedge q) \wedge r \equiv p \wedge (q \vee r)$

6. If p: 2 is an even number,
q: 2 is a prime number, and $r : 2+2=2^2$,
then the symbolic statement $p \rightarrow (q \vee r)$ means
(a) 2 is an even number and 2 is a prime number or $2+2=2^2$
(b) 2 is an even number then 2 is a prime number or $2+2=2^2$
(c) 2 is an even number or 2 is a prime number then $2+2=2^2$
(d) If 2 is not an even number, then 2 is a prime number or $2+2=2^2$

7. Let p : I am brave,
q : I will climb the mount Everest
The symbolic form of a statement,
'I am neither brave nor I will climb the mount Everest' is
(a) $p \wedge q$ (b) $\sim(p \wedge q)$ (c) $\sim p \wedge \sim q$ (d) $\sim p \wedge q$

8. Which of the following is not logically equivalent to the proposition: 'A real number is either rational or irrational'.
(a) If a number is neither rational nor irrational, then it is not real
(b) If a number is not a rational or not an irrational, then it is not real
(c) If a number is not real, then it is neither rational nor irrational
(d) If a number is real, then it is rational or irrational

9. The negation of the proposition: "If we control population growth, we prosper" is
(a) If we do not control population growth, we prosper
(b) If we control population growth, we do not prosper
(c) We control population but we do not prosper
(d) We do not control population, but we prosper

10. If p is any statement, t is tautology and c is a contradiction, then which of the following is not correct?
(a) $p \vee (\sim p) = c$ (b) $p \vee t = t$
(c) $p \wedge t = p$ (d) $p \wedge c = c$

11. Which of the following is a contradiction?
(a) $(p \wedge q) \wedge \sim(p \vee q)$ (b) $p \vee (-p \wedge q)$
(c) $(p \Rightarrow q) \Rightarrow p$ (d) none of these

12. Which of the following is the inverse of the proposition: 'If a number is a prime, then it is odd'?
(a) If a number is not prime, then it is odd
(b) If a number is not prime, then it is even
(c) If a number is not odd, then it is not a prime
(d) If a number is not odd, then it is prime

13. The inverse of the statement 'If x is zero, then we cannot divide by x' is
(a) If we cannot divide by x, then x is zero
(b) If we cannot divide by x, then x is not zero
(c) If x is not zero, then we divide by x
(d) none of three

14. The negation of the statement $(p \vee q) \wedge r$ is
(a) $(\sim p \vee \sim q) \vee \sim q$ (b) $(\sim p \vee \sim q) \vee \sim r$
(c) $\sim(p \vee q) \to r$ (d) $p \wedge q$

ANSWERS

1. (d) **2.** (d) **3.** (b) **4.** (a) **5.** (a) **6.** (b) **7.** (c) **8.** (b) **9.** (c) **10.** (a)
11. (a) **12.** (b) **13.** (c) **14.** (b)

Mock Test Set 1

1. Let $X = \{1, 2, 3, 4\}$, then one-one onto mappings $f: X \to X$ such that $f(1) = 1$, $f(2) \neq 2$ and $f(4) \neq 4$ are given by

(a) $\{(1, 1), (2, 4), (3, 2), (4, 3)\}$

(b) $\{(1, 2), (2, 4), (3, 3), (4, 2)\}$

(c) $\{(1, 2), (2, 4), (3, 3), (4, 2)\}$

(d) none of these

2. In the binomial expansion of $(a-b)^n$, $n \geq 5$, if the sum of the 5th and 6th terms is zero, then a/b equals

(a) $\frac{n-5}{6}$ (b) $\frac{n-4}{5}$

(c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$

3. $\int_0^{10\pi} |\sin x| dx$ is

(a) 20 (b) 8

(c) 10 (d) 18

4. If the sides of a triangle are in the ratio $1:\sqrt{3}:2$, then the angles of the triangle are in the ratio

(a) 1 : 3 : 5 (b) 2 : 3 : 4

(c) 3 : 2 : 1 (d) 1 : 2 : 3

5. The unit vector perpendicular to each of the vectors $2\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} + 4\hat{j}$ is

(a) $\frac{1}{\sqrt{146}}(4\hat{i} - 3\hat{j} + 11\hat{k})$

(b) $\frac{1}{\sqrt{146}}(-4\hat{i} + 3\hat{j} + 11\hat{k})$

(c) $\frac{1}{\sqrt{146}}(4\hat{i} + 3\hat{j} + 11\hat{k})$

(d) none of these

6. If $\begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ is a square root of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then α, β and γ satisfy

(a) $1 + \alpha^2 + \beta\gamma = 0$ (b) $1 - \alpha^2 + \beta\gamma = 0$

(c) $1 - \alpha^2 - \beta\gamma = 0$ (d) $1 + \alpha^2 - \beta\gamma = 0$

7. $2^{3n} - 7n - 1$ is divisible by

(a) 64 (b) 36

(c) 49 (d) 25

8. There are four machines and it is known that exactly two of them are faulty. They are tested, one by one, in a random order till both the faulty machines are identified. Then the probability that only two tests are needed is

(a) 1/3 (b) 1/6

(c) 1/2 (d) 1/4

9. If the statements are *p:* It is raining, *q*: It is pleasant, then the statement 'It is neither raining nor pleasant' can be expressed in symbols as

(a) $p \vee \sim q \wedge q$ (b) $\sim p \wedge \sim q$

(c) $p \to q$ (d) none of these

10. If $I(m, n) = \int_0^1 t^m (1+t)^n dt$; $m, n \in \mathbb{R}$, then $I(m, n)$ is

(a) $\frac{n}{1+m} I[(m+1), (n-1)]$

(b) $\frac{m}{n+1} I[(m+1), (n-1)]$

(c) $\frac{2^n}{1+m} - \frac{n}{1+m} I[(m+1, n-1)]$

(d) $\frac{2^n}{1+m} - \frac{m}{1+n} I[(m+1), (n-1)]$

11. If $\sin^{-1}\left(\frac{2a}{1+a^2}\right)+\sin^{-1}\left(\frac{2b}{1+b^2}\right)=2\tan^{-1}x$, then x is

(a) $\frac{a-b}{1+ab}$ (b) $\frac{b}{1+ab}$

(c) $\frac{b}{1-ab}$ (d) $\frac{a+b}{1-ab}$

12. The angle between the line $\frac{x+1}{3}=\frac{y-1}{2}=\frac{z-2}{4}$ and the plane $2x+y-3z+4=0$ is

(a) 30° (b) $\cos^{-1}\left(\frac{4}{\sqrt{406}}\right)$

(c) $\sin^{-1}\left(\frac{-4}{\sqrt{406}}\right)$ (d) none of these

13. The length of the latus rectum of the parabola $169[(x-1)^2+(y-3)^2]=(5x-12y+17)^2$ is

(a) 12/13 (b) 14/13

(c) 28/13 (d) 25/13

14. The unit vector which is orthogonal to the vector $\vec{a}=3\hat{i}+2\hat{j}+6\hat{k}$ and is coplanar with the vectors $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ is

(a) $\frac{2\hat{i}-6\hat{j}+\hat{k}}{\sqrt{41}}$ (b) $\frac{2\hat{i}-3\hat{j}}{\sqrt{13}}$

(c) $\frac{3\hat{j}-\hat{k}}{\sqrt{10}}$ (d) $\frac{4\hat{i}-3\hat{j}-3\hat{k}}{\sqrt{34}}$

15. The value of the integral $I=\int_0^1 x(1-x)^n\,dx$ is

(a) $\frac{1}{n+2}$ (b) $\frac{1}{n+1}-\frac{1}{n+2}$

(c) $\frac{1}{n+1}+\frac{1}{n+2}$ (d) $\frac{1}{n+1}$

16. The value of $f(0)$ in which the function $\frac{\log_e(1-ax)-\log_e(1-bx)}{x}$ is continuous at $x = 0$ is

(a) $b - a$ (b) $a + b$

(c) $-(a + b)$ (d) $a - b$

17. The domain of the function $f(x)=\sqrt{x-\sqrt{1-x^2}}$ is

(a) [–1, 1] (b) $\left[\frac{1}{\sqrt{2}},1\right]$

(c) $\left[-1,-\frac{1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},1\right]$

(d) $\left(-\infty,-\frac{1}{\sqrt{2}}\right)\cup\left[\frac{1}{\sqrt{2}},+\infty\right]$

18. $\int\frac{\cos x}{\cos(x-\alpha)}dx$ equals

(a) $x\cos\alpha+\sin\alpha\log\left|\cos(x-\alpha)\right|+c$

(b) $x\cos\alpha-\sin\alpha\log\left|\cos(x-\alpha)\right|+c$

(c) $\sin\alpha-\sin\alpha\log\left|\cos(x-\alpha)\right|+c$

(d) none of these

19. The length of the chord joining the points (4 cos θ, 4 sin θ) and [4 cos (θ+60°), 4sin (θ+60°)] of the circle $x^2+y^2=16$ is

(a) 16 (b) 2

(c) 8 (d) 4

20. The angle of elevation of a ladder leaning against a house is 58° and the foot of the ladder is 9.6 m from the house. Then the length of the ladder is

(a) 15.36 m (b) 16.36 m

(c) 14.36 m (d) none of these

21. The area of the triangle formed by the line $x+y=2$ and angle bisectors of the pair of straight lines $x^2-y^2+2y=1$ is

(a) 2 sq. units (b) 4 sq. units

(c) 6 sq. units (d) 8 sq. units

22. The term independent of x in the expansion of $(1+x)^n[1+(1/x)]^n$ is

(a) $C_0^2+2C_1^2+\cdots+(n+1)C_n^2$

(b) $(C_0+C_1+\cdots+C_n)^2$

(c) $C_0^2+C_1^2+\cdots+C_n^2$

(d) none of these

23. The equation of the circle whose radius is 5 and which touches the circle $x^2+y^2-2x-4y-20=0$ at the point (5, 5) is

(a) $x^2+y^2+18x+16y+120=0$

(b) $x^2+y^2-18x-16y+120=0$

(c) $x^2+y^2-18x+16y+120=0$

(d) $x^2+y^2+18x-16y+120=0$

24. If $y=\tan^{-1}\left[\frac{\sin x+\cos x}{\cos x-\sin x}\right]$, then $dy/dx=$

(a) 1/2 (b) π/4

(c) 0 (d) 1

25. Let [x] denotes the greatest integer less than or equal to x. If $f(x)=x\sin\pi x$, then $f(x)$ is

(a) not continuous at $x = 0$

(b) differentiable at $x = 1$

(c) continuous in (–1, 0)

(d) none of these

26. If the sides of a triangle are in A.P. and one angle is 90°, then the ratio of the sides of the triangle is
(a) 1 : 2 : 3 (b) 2 : 3 : 4
(c) 3 : 4 : 5 (d) 4 : 5 : 6

27. If $a^{1/x} = b^{1/y} = c^{1/z}$ and a, b, c are in G.P., then x, y and z are in
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

The following 3 questions numbered 28 to 30 contain Statement I (Assertion) and Statement II (Reason). Mark the correct choice out of the four given choices:

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

28. **Statement I:** If $\frac{a}{a_1}, \frac{b}{b_1}, \frac{c}{c_1}$ are in A.P., then a_1, b_1, c_1 are in G.P.

Statement II: If $ax^2 + bx + c = 0$ and $a_1x^2 + b_1x + c_1 = 0$ have a common root and $\frac{a}{a_1}, \frac{b}{b_1}, \frac{c}{c_1}$ are in A.P., then a_1, b_1, c_1 are in G.P.

29. **Statement I:** Suppose a 2×2 matrix A has determinant 2. If $B = 9A^2$, then the determinant of B^T is equal to 36.
Statement II: If A, B, C are three square matrices such that $C = AB$, then $|C| = |A||B|$.

30. **Statement I:** If A, B, C are the angles of a triangle such that angle A is obtuse, then $\tan B \tan C > 1$.
Statement II: In any triangle ABC,
$$\tan A = \frac{\tan B + \tan C}{\tan B \tan C - 1}.$$

ANSWERS

1. (a)	**2.** (b)	**3.** (a)	**4.** (d)	**5.** (b)	**6.** (c)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (c)
11. (d)	**12.** (c)	**13.** (c)	**14.** (c)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (a)	**19.** (d)	**20.** (a)
21. (a)	**22.** (c)	**23.** (b)	**24.** (d)	**25.** (c)	**26.** (a)	**27.** (d)	**28.** (d)	**29.** (a)	**30.** (d)

Mock Test Set 2

1. If $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$, then $A^n (n\in N)$ equals

(a) $\begin{bmatrix}n+2 & 5-n\\ n & -n\end{bmatrix}$ (b) $\begin{bmatrix}3n & -4n\\ n & -n\end{bmatrix}$

(c) $\begin{bmatrix}3^n & (-4)^n\\ 1 & (-1)^n\end{bmatrix}$ (d) none of these

2. The domain of the definition of the function $f(x)$ = $\sqrt{\sin^{-1}(2x)+\frac{\pi}{6}}$ for real value x is

(a) $\left[-\frac{1}{4},\frac{1}{2}\right]$ (b) $\left[-\frac{1}{2},\frac{1}{2}\right]$

(c) $\left[-\frac{1}{2},\frac{1}{9}\right]$ (d) $\left[-\frac{1}{4},\frac{1}{4}\right]$

3. If $\hat{a}$ and $\hat{b}$ are two unit vectors, then the vector $(\hat{a}+\hat{b})\times(\hat{a}\times\hat{b})$ is parallel to the vector

(a) $\hat{a}-\hat{b}$ (b) $\hat{a}+\hat{b}$

(c) $2\hat{a}-\hat{b}$ (d) $2\hat{a}+\hat{b}$

4. The mean of the values of $0, 1, 2,\ldots, n$ having corresponding weights ${}^nC_0, {}^nC_1, {}^nC_2,\ldots, {}^nC_n$ respectively is

(a) $\frac{2^n}{n+1}$ (b) $\frac{2^{n+1}}{n(n+1)}$

(c) $\frac{n+1}{2}$ (d) $\frac{n}{2}$

5. If the angles of a triangle are in the ratio 4 : 1 : 1, then the ratio of the largest side to the perimeter is

(a) $1:1+\sqrt{3}$ (b) 2 : 3

(c) $\sqrt{3}:2+\sqrt{3}$ (d) $1:2+\sqrt{3}$

6. If $x>1$, $y>1$, $z>1$ are in G.P., then $\frac{1}{1+\log x}, \frac{1}{1+\log y}, \frac{1}{1+\log z}$ are in

(a) A.P. (b) H.P.

(c) G.P. (d) none of these

7. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1-z_3}{z_2-z_3}=\frac{1-i\sqrt{3}}{2}$ are the vertices of a triangle which is

(a) of area zero (b) right-angled isosceles

(c) equilateral (d) obtuse-angled isosceles

8. The value of x for which the sixth term in the expansion of $\left[2^{\log_2(\sqrt{9^{x-1}+7})}+\frac{1}{2^{1/5\log_2(3^{x-1}+1)}}\right]^7$ is 84 is

(a) 4 (b) 3

(c) 2 (d) 1

9. $\lim\limits_{x\to 0}\frac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ is

(a) 2 (b) –2

(c) 1/2 (d) –1/2

10. If a is a parameter, then the equation of the family of lines having y-intercept 4 is

(a) $\frac{x}{a}-\frac{y}{4}=1$ (b) $\frac{x}{a}+\frac{y}{4}=1$

(c) $\frac{x}{a}-\frac{y}{4}-1=0$ (d) $\frac{x}{a}+\frac{y}{4}-1=0$

11. $\int e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)dx$ equals

(a) $\frac{e^{\tan^{-1}x}}{1+x^2}+c$ (b) $xe^{\tan^{-1}x}+c$

(c) $x\frac{\tan^{-1}x}{1+x^2}+c$ (d) $x^2\frac{\tan^{-1}x}{1+x^2}+c$

12. Let L_1 be a straight line passing through the origin and L_2 be the straight line $x+y=1$. If the intercepts made by the circle $x^2+y^2-x+3y=0$ on L_1 and L_1 are equal, then which of the following equations can represent L_1?

(a) $x+y=0$ (b) $x-y=0$

(c) $x+7y=7$ (d) $x-7y=0$

13. If the line $x-1=0$ is the directrix of the parabola $y^2-kx+8=0$, then one of the values of k is

(a) 1/8 (b) 8

(c) 4 (d) 1/4

14. If $\vec{a}=(\hat{i}+\hat{j}+\hat{k})$, $\vec{a}\cdot\vec{b}=1$ and $\vec{a}\times\vec{b}=\hat{j}-\hat{k}$, then $\vec{b}$ is

(a) $\hat{i}-\hat{j}+\hat{k}$ (b) $2\hat{j}-\hat{k}$

(c) $\hat{i}$ (d) $2\hat{i}$

15. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is

(a) 1 (b) 1/6

(c) 1/3 (d) 2/3

16. $\sim(p\Rightarrow q)=0$ is

(a) $(\sim p)\vee q$ (b) $\sim p\wedge(\sim q)$

(c) $p\wedge(\sim q)$ (d) none of these

17. A fair coin is tossed repeatedly. If head appears on the first four tosses, then the probability of head appearing on the fifth toss equals

(a) 1/2 (b) 1/32

(c) 31/32 (d) 1/5

18. The area between the hyperbola $xy = c^2$, the x-axis and the ordinates at a and b with $a > b$, is

(a) $c^2\log\left(\frac{a}{b}\right)$ (b) $c^2\log\left(\frac{b}{a}\right)$

(c) $c^2\log(ab)$ (d) none of these

19. The differential equation representing the family of curves $y^2=2c(x+\sqrt{c})$, where c is a positive parameter, is of

(a) order 1, degree 3 (b) order 2

(c) degree 3 (d) none of these

20. The centre of the smallest circle passing through the origin lies on the line $y=x+1$. The coordinates of the centre of this circle are

(a) $\left(-\frac{1}{2},\frac{1}{2}\right)$ (b) (0, 1)

(c) $\left(\frac{1}{2},-\frac{1}{2}\right)$ (d) none of these

21. The vectors $2\hat{i}-\hat{j}+\hat{k},\hat{i}-3\hat{j}-5\hat{k}$ and $\sqrt{3}\hat{i}-4\hat{j}-4\hat{k}$ are the sides of a triangle, which is

(a) equilateral

(b) isosceles only

(c) right-angled only

(d) right-angled and isosceles

22. A tower MPQ surmounted by a spiral QR subtends on a horizontal plane. At the extremity A of a line horizontal AM it is found that MP and QR subtend equal angles. If MP = 3 m, PQ = 28 m and QR = 5 m, then MA equals

(a) $\sqrt{36\times93}$ (b) $\sqrt{18\times93}$

(c) $\sqrt{34\times36}$ (d) $\sqrt{34\times93}$

23. If 3 letters are placed in the three addressed envelopes, then find the mean and variance of x, where x denotes the number of correct despatches.

(a) 1, 1 (b) 1, 2

(c) 2, 1 (d) 4, 2

24. The length of the common chord of the ellipse $\frac{(x-1)^2}{9}+\frac{(y-2)^2}{4}=1$ and the circle $(x-1)^2+(y-2)^2=1$ is

(a) 2 (b) $\sqrt{3}$

(c) 4 (d) 0

25. If $f(x)=\frac{x+x^2+\cdots+x^n-n}{x-1}$, $x\neq1$, then the value of $f(1)$ such that f is continuous is

(a) $\frac{n(n-1)}{2}$ (b) $\frac{n(n+1)}{2}$

(c) $\frac{n(n+3)}{2}$ (d) $\frac{n(n-3)}{2}$

26. The value of $\int\frac{\cos4x+1}{\cot x-\tan x}dx=1$ is

(a) $\frac{1}{8}\cos4x+C$ (b) $\frac{1}{4}\cos4x-C$

(c) $\frac{1}{4}\cos4x+C$ (d) $-\frac{1}{8}\cos4x+C$

27. If $x = a$ divides the area bounded by the x-axis, part of the curve $y=1+\frac{8}{x^2}$ and the ordinates $x = 2$, $x = 4$ into equal parts, then $a =$

(a) 2 (b) $\frac{2}{\sqrt{2}}$

(c) $\frac{1}{\sqrt{2}}$ (d) $2\sqrt{2}$

The following 3 questions numbered 28 to 30 contain Statement I (Assertion) and Statement II (Reason). Mark the correct choice out of the following four choices given.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation of Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation of Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

28. Statement I: If the number $\frac{z-1}{z+1}$ is purely imaginary, then $|z|=1$.

Statement II: If $|z|=1$, $z \neq -1$ and $\omega=\frac{z-1}{z+1}$, then the real part of ω is zero.

29. Statement I: If $a > 0$ and $b^2-4ac<0$, then the value of the integral $\int\frac{dx}{ax^2+bx+c}$ will be of the type $\mu\tan^{-1}\frac{x+A}{B}+C$, where A, B, C, μ are constants.

Statement II: If $a > 0$, $b^2-4ac<0$, then ax^2+bx+c can be written as the sum of two factors.

30. If $P(\overline{B})=(P(A\cup B)^2$, then

Statement I: $P(A\cup B)$ is at least $\frac{\sqrt{5}-1}{2}$.

Statement II: $A\cap(B\cap A)=A\cup(B\cup A)$.

ANSWERS

1. (d)	**2.** (a)	**3.** (a)	**4.** (d)	**5.** (c)	**6.** (b)	**7.** (c)	**8.** (c)	**9.** (c)	**10.** (b)
11. (b)	**12.** (b)	**13.** (c)	**14.** (c)	**15.** (b)	**16.** (c)	**17.** (a)	**18.** (b)	**19.** (a)	**20.** (a)
21. (d)	**22.** (b)	**23.** (a)	**24.** (d)	**25.** (c)	**26.** (d)	**27.** (d)	**28.** (b)	**29.** (a)	**30.** (b)

Mock Test Set 3

1. If $|z|=1$ and $w=\dfrac{z-1}{z+1}$ (where $z\neq -1$), then Re (w) is

(a) 0 (b) $-\dfrac{1}{|z+1|^2}$

(c) $\left|\dfrac{z}{z+1}\right|\dfrac{1}{|z+1|^2}$ (d) $\dfrac{\sqrt{2}}{|z+1|^2}$

2. An n-digit number is a positive number with exactly n digits. Nine hundred distinct n-digit numbers are to be formed using only the three digits 2, 5 and 7. The smallest values of n for which this is possible is

(a) 6 (b) 7
(c) 8 (d) 9

3. In the binomial expansion of $\left(\sqrt[4]{4}+\dfrac{1}{\sqrt[4]{3}}\right)^n$ the ratio of 9th term from beginning to the nth term from the end is 1728 : 1. Then n equals

(a) 12 (b) 16
(c) 18 (d) 28

4. If $(1+x)^n=\sum_{r=0}^{n}a_r x^r$ and $b_r=1+\dfrac{a_r}{a_r-1}$ and $\prod_{r=1}^{n}b_r=\dfrac{(101)^{100}}{100!}$, then n equals

(a) 99 (b) 100
(c) 101 (d) none of these

5. A six-faced die is so biased that it is twice as likely to show an even number as an odd number when thrown. It is thrown twice. The probability that the sum of two numbers thrown is even is

(a) 1/12 (b) 1/6
(c) 1/3 (d) 5/9

6. The maximum value of $f(x)=\dfrac{x}{1+4x+x^2}$ on [–1. 1] is

(a) –(1/4) (b) –(1/3)
(c) 1/6 (d) 1/5

7. $\int e^{\tan^{-1}x}\left(\dfrac{1+x+x^2}{1+x^2}\right)dx$ equals

(a) $\dfrac{1}{x}e^{\tan^{-1}x}+c$ (b) $xe^{\tan^{-1}x}+c$

(c) $x^2e^{\tan^{-1}x}+c$ (d) none of these

8. If $\int_0^x f(t)dt=x+\int_x^1 tf(1)dt$, then the value of $f(1)$ is

(a) 1/2 (b) 0
(c) 1 (d) –(1/2)

9. If a circle passes through the points of intersection of the coordinate axes with the lines $\lambda x-y+1=0$ and $x-2y+3=0$, then the value of λ is

(a) 2 (b) 1/4
(c) 6 (d) 3

10. Let $(a\sec\theta, b\tan\theta)$ and $(a\sec\phi, b\tan\phi)$ be the ends of a focal chord of $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$, then $\tan\dfrac{\theta}{2}\tan\dfrac{\phi}{2}$ is

(a) $\dfrac{1+e}{1-e}$ (b) $\dfrac{1-e}{1+e}$

(c) $\dfrac{e+1}{e-1}$ (d) none of these

11. A value of C for which the conclusion of the Mean Value Theorem holds for the function $f(x)=\log_e x$ on the interval [1, 3] is

(a) $(1/2)\log_e 3$ (b) $\log_3 e$
(c) $\log_e 3$ (d) $2\log_3 e$

12. The function $f(x)=\tan^{-1}(\sin x+\cos x)$ is an increasing function in

(a) $\left(-\dfrac{\pi}{2},\dfrac{\pi}{4}\right)$ (b) $\left(0,\dfrac{\pi}{2}\right)$

(c) $\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$ (d) $\left(\dfrac{\pi}{4},\dfrac{\pi}{2}\right)$

13. The function $f(x)=\begin{cases}\dfrac{\sin x^{1/3}\log(1+3x)}{(\tan^{-1}\sqrt{x})^2(e^{5x^{1/3}}-1)}; & x\neq 0\\ a; & x=0\end{cases}$ is continuous at $x = 0$ if

(a) $a = 0$ (b) $a = 2$
(c) $a = (5/3)$ (d) $a = (3/5)$

14. A square is formed by the following two pairs of straight lines: $y^2-14y+45=0$ and $x^2-8x+12=0$. A circle is inscribed in it. The centre of the circle is

(a) (7, 4) (b) (4, 7)
(c) (3, 7) (d) [(3/8), 4]

15. The angle between the lines whose direction cosines are given by $l^2+m^2-n^2=0$ and $l+m+n=0$ is

(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$

16. The number of real solutions of

$$\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$$ is

(a) zero (b) one
(c) two (d) infinite

17. The sum of all the 4-digit numbers that can be formed by using the digits 2, 4, 6, 8 (repetition of digits not allowed) is

(a) 133,320 (b) 123,330
(c) 113,230 (d) 323,430

18. If a, b, c, d are in G.P., then $\dfrac{(a^2+b^2+c^2)(b^2+c^2+d^2)}{(ab+bc+cd)^2}=$

(a) 1 (b) 2
(c) 3 (d) none of these

19. The length of the longest interval in which the function $3\sin x-4\sin^3 x$ is increasing is

(a) $\pi/3$ (b) $\pi/2$
(c) $3\pi/2$ (d) π

20. The parameter on which the value of the determinant

$$\begin{vmatrix}1 & a & a^2\\ \cos(p-d)x & \cos px & \cos(p+d)x\\ \sin(p-d)x & \sin px & \sin(p+d)x\end{vmatrix}$$ deos not depend upon is

(a) a (b) p
(c) d (d) x

21. The value of x for which $\sin[\cot^{-1}(1+x)]=\cos(\tan^{-1}x)$ is

(a) 1/2 (b) 1
(c) 0 (d) –1/2

22. Tangents are drawn to $3x^2-2y^2=6$ from a point P. If the product of the slopes of the tangents is 2, find the locus of P.

(a) $2x^2-y^2=7$ (b) $x^2+2y^2=7$
(c) $y^2-2x^2=7$ (d) $2x^2+y^2=7$

23. Let $I=\int\dfrac{e^x dx}{e^{4x}+e^{2x}+1}$, $J=\int\dfrac{e^{-x}dx}{e^{-4x}+e^{-2x}+1}$. Then for an arbitrary constant c, $J-I=$

(a) $\dfrac{1}{2}\ln\left(\dfrac{e^{4x}-e^{2x}+1}{e^{4x}+e^{2x}+1}\right)+c$

(b) $\dfrac{1}{2}\ln\left(\dfrac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right)+c$

(c) $\dfrac{1}{2}\ln\left(\dfrac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}\right)+c$

(d) $\dfrac{1}{2}\ln\left(\dfrac{e^{4x}+e^{2x}+1}{e^{4x}-e^{2x}+1}\right)+c$

24. If $a_n=\sum_{r=0}^{n}\dfrac{1}{{}^nC_r}$, then $\sum_{r=0}^{n}\dfrac{r}{{}^nC_r}$ equals

(a) $(n-1)a_n$ (b) na_n
(c) $\dfrac{n}{2}a_n$ (d) none of these

25. The integer k for which the inequality $x^2-2(4k-1)x+15k^2-2k-7>0$ is valid for any real x is

(a) 2 (b) 3
(c) 4 (d) none of these

26. Two tangents are drawn from the point (–2, –1) to the parabola $y^2 = 4x$. If α is the angle between these tangents, then $\tan\alpha =$

(a) 3 (b) 1/3
(c) 2 (d) 1/2

The following 4 questions numbered 27 to 30 are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the corrcet answer. You have to select the correct choice given as follows:

(a) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(b) Statement I is true, Statement II is false.
(c) Statement I is false, Statement II is true.
(d) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

27. Statement I: If the mean of a set of observations x_1, $x_2,\ldots,\ x_n$ is $\bar{x}$, then the mean of the observation $x_i + 2 (i = 1, 2,\ldots, n)$ is $\bar{x} + 2$.

Statement II: If each observation is increased by 2, then the mean is increased by 2.

28. Statement I: $D^{48}(\sin^4 x) = 2^{93}\cos 4x - 2^{47}\cos 2x$.

Statement II: If $y = \sin(ax + b)$, then

$D^n y = a^n \sin\left(\frac{n\pi}{2} + ax = b\right)$, where $D = \frac{d}{dx}$.

29. Statement I: $S_n = n^3 + 3n^2 + 5n + 3$ is divisible by 3.

Statement II: $t_n = 3\lambda$.

30. Statement I: The term independent of x in the expansion of $\left(x + \frac{1}{x} + 2\right)^{2m}$ is $\frac{(4m)!}{(2m)!(2m)!}$.

Statement II: The coefficient of x_a in the expansion of $(1 + x)^n$ is ${}^nC_a$.

ANSWERS

1. (a)	**2.** (b)	**3.** (d)	**4.** (b)	**5.** (d)	**6.** (c)	**7.** (b)	**8.** (a)	**9.** (a)	**10.** (d)
11. (d)	**12.** (d)	**13.** (d)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (a)	**18.** (a)	**19.** (a)	**20.** (b)
21. (d)	**22.** (a)	**23.** (c)	**24.** (c)	**25.** (b)	**26.** (a)	**27.** (d)	**28.** (c)	**29.** (d)	**30.** (d)

Sample Paper I

1. The shortest distance between the line $y - x = 1$ and the curve $x = y^2$ is

(a) $\frac{3\sqrt{2}}{8}$ (b) $\frac{2\sqrt{3}}{8}$

(c) $\frac{3\sqrt{2}}{5}$ (d) $\frac{\sqrt{3}}{4}$

2. $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval [–1, 1],

(a) $P(-1)$ is the minimum and $P(1)$ is the maximum of P

(b) $P(-1)$ is not the minimum but $P(1)$ is the maximum of P

(c) $P(-1)$ is the minimum and $P(1)$ is the maximum of P

(d) neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of P

3. The area of the region bounded by the parabola $(y-2)^2 = x-1$, the tangent to the parabola at the point (2, 3) and the x-axis is

(a) 3 (b) 6

(c) 9 (d) 12

4. Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $y'(1)$ equals

(a) –1 (b) 1

(c) log 2 (d) –log 2

5. One ticket is selected at random from 50 tickets numberd 00, 01, 02,..., 49. Then the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero, equals

(a) $\frac{1}{14}$ (b) $\frac{1}{7}$

(c) $\frac{5}{14}$ (d) $\frac{1}{50}$

6. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 and c_2 are arbitrary constants, is

(a) $y' = y^2$ (b) $y'' = y'y$

(c) $yy'' = y'$ (d) $yy'' = (y')^2$

7. The sum to the infinity of the series $1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \cdots$ is

(a) 2 (b) 3

(c) 4 (d) 6

8. The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the coordinate axes, which in turn is inscribed in another ellipse that passes through the point (4, 0). Then the equation of the ellipse is

(a) $x^2 + 16y^2 = 16$ (b) $x^2 + 12y^2 = 16$

(c) $4x^2 + 48y^2 = 48$ (d) $4x^2 + 64y^2 = 48$

9. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is

(a) 0 (b) 2

(c) 7 (d) 8

10. Three distinct points A, B and C are given in the two-dimensional coordinate plane such that the ratio of the distance of any one of them from the point (1, 0) to the distance from the point (–1, 0) is equal to 1/3. Then the circumcentre of the triangle ABC is at the point

(a) (0, 0) (b) [(5/4), 0]

(c) [(5/2), 0] (d) [(5/3), 0]

11. If $\left|z - \frac{4}{2}\right| = 2$, then the maximum value of $|z|$ is equal to

(a) $\sqrt{3}+1$ (b) $\sqrt{5}+1$

(c) 2 (d) $2+\sqrt{2}$

12. The projections of a vector on the three coordinate axes are 6, –3, 2, respectively. The direction cosines of the vector are

(a) 6, –3, 2 (b) $\frac{6}{5}, -\frac{3}{5}, \frac{2}{5}$

(c) $\frac{6}{7}, -\frac{3}{7}, \frac{2}{7}$ (d) $-\frac{6}{7}, -\frac{3}{7}, \frac{2}{7}$

13. If P and Q are the points of intersection of the circles $x^2+y^2+3x+7y+2p-5=0$ and $x^2+y^2+2x+2y-p^2=0$, then there is a circle passing through P, Q and (1, 1) for

(a) all values of p
(b) all except one value of p
(c) all except two values of p
(d) exactly one value of p

14. In a binomial distribution $B[n, p = (1/4)]$, if the probability of at least one success is greater than or equal to 9/10, then n is greater than

(a) $\frac{1}{\log_{10}4-\log_{10}3}$ (b) $\frac{1}{\log_{10}4+\log_{10}3}$

(c) $\frac{9}{\log_{10}4+\log_{10}3}$ (d) $\frac{4}{\log_{10}4+\log_{10}3}$

15. For real x, let $f(x)=x^3+5x+1$, then

(a) f is one-one but not onto $\mathbb{R}$
(b) f is onto $\mathbb{R}$ but not one-one
(c) f is one-one and onto $\mathbb{R}$
(d) f is neither one-one nor onto $\mathbb{R}$

16. $\int_0^{\pi}[\cot x]dx$, where [] denotes the greatest integer function, is equal to

(a) $\pi/2$ (b) 1
(c) –1 (d) $-\pi/2$

17. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on the shelf so that the dictionary is always in the middle. Then the number of such arrangements is

(a) less than 500
(b) at least 500 but less than 750
(c) at least 750 but less than 1000
(d) at least 1000

18. Let the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ lie in the plane $x+3y-\alpha z+\beta=0$. Then (α, β) equals

(a) (6, –17) (b) (–6, 7)
(c) (5, –15) (d) (–5, 5)

19. If $\vec{u},\vec{v},\vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality $[3\vec{u}, p\vec{v}, p\vec{w}]-[p\vec{v}, \vec{w}, q\vec{u}]-[2\vec{w}, q\vec{v}, q\vec{u}]=0$ holds for

(a) exactly one value of (p, q)
(b) exactly two values of (p, q)
(c) more than two but not all values of (p, q)
(d) all values of (p, q)

20. If A, B and C are three sets such that $A\cap B=A\cap C$ and $A\cup B=A\cup C$, then

(a) $A=B$ (b) $A=C$
(c) $B=C$ (d) $A\cap B=\phi$

21. The lines $p(p^2+1)x-y+q=0$ and $(p^2+1)^2x+(p^2+1)y+2q=0$ are perpendicular to a common line for

(a) no value of p
(b) exactly one value of p
(c) exactly two values of p
(d) more than two values of p

22. Let A and B denote the statements

A: $\cos\alpha+\cos\beta+\cos\gamma=0$

B: $\sin\alpha+\sin\beta+\sin\gamma=0$

If $\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)=-\frac{3}{2}$, then

(a) A is true and B is false
(b) A is false and B is true
(c) both A and B are true
(d) both A and B are fale

23. If the roots of the equation $bx^2+cx+a=0$ are imaginary, then for all real values of x, the expression $3b^2x^2+6bcx+2c^2$ is

(a) greater than $4ab$ (b) less than $4ab$
(c) greater than $-4ab$ (d) less than $-4ab$

24. If the mean deviation of the numbers 1, $1+d$, $1+2d$, …, $1+100d$ from their mean is 255, then d is equal to

(a) 10.0 (b) 20.0
(c) 10.1 (d) 20.2

25. Let a, b, c be such that $b(a+c)\neq 0$.

If $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix}+\begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix}=0$,

then the value of n is

(a) zero (b) any even integer
(c) any odd integer (d) any integer

Directions: Questions 26 to 30 are Assertion-Reason type questions. Each of these questions contains two statements: Statement I (Assertion) and Statement II (Reason). Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice.

26. Let $f(x)=(x+1)^2-1,\ x\geq -1$.

Statement I: The set $\{x: f(x)=f^{-1}(x)\}=\{0,-1\}$

Statement II: f is a bijection.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

27. Let $f(x)=x|x|$ and $g(x)=\sin x$.

Statement I: $g\circ f$ is twice differetiable at $x = 0$ and its derivative is continuous at that point.

Statement II: $g\circ f$ is twice differentiable at $x = 0$.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

28. Statement I: The variance of the first n even natural numbers is $\frac{n^2-1}{4}$.

Statement II: The sum of the first n natural numbers is $\frac{n(n+1)}{2}$ and the sum of the squares of the first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

29. Statement I: $\sim(p\leftrightarrow\sim q)$ is equivalent to $p\leftrightarrow q$.

Statement II: $\sim(p\leftrightarrow\sim q)$ is a tautology.

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

30. Let A be a 2×2 matrix.

Statement I: adj (adj$A$) = A

Statement II: $|\text{adj}\,A|=|A|$

(a) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.
(b) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

ANSWERS

1. (a)	**2.** (b)	**3.** (c)	**4.** (a)	**5.** (a)	**6.** (d)	**7.** (b)	**8.** (b)	**9.** (b)	**10.** (b)
11. (b)	**12.** (c)	**13.** (c)	**14.** (a)	**15.** (c)	**16.** (d)	**17.** (d)	**18.** (b)	**19.** (a)	**20.** (c)
21. (b)	**22.** (c)	**23.** (c)	**24.** (c)	**25.** (c)	**26.** (b)	**27.** (c)	**28.** (d)	**29.** (c)	**30.** (b)

HINTS AND SOLUTIONS—SAMPLE PAPER I

1. Given line is

$$x-y+1=0 \qquad \text{(i)}$$

Let $P(x,y)$ be any point on the parabola. Then perpendicular distance of P from (i)

$$P=\frac{x-y+1}{\sqrt{2}}=\frac{y^2-y+1}{\sqrt{2}} \qquad (\because x=y^2)$$

$$P=\frac{\left(y-\frac{1}{2}\right)^2+1-\frac{1}{4}}{\sqrt{2}}=\frac{\left(y-\frac{1}{2}\right)^2+\frac{3}{4}}{\sqrt{2}}$$

P is minimum when $y-\frac{1}{2}=0 \Rightarrow y=\frac{1}{2}$

$$P_{\min}=\frac{3}{4\sqrt{2}}=\frac{3\sqrt{2}}{8}.$$

2. $P'(x)=4x^3+3ax^2+2bx+c$

At $x = 0$, $P'(x)=0$

$\therefore\ P(x)=x^4+ax^3+bx^2+d$

Now $P(-1)<P(1)$

$$1-a+b+d<1+a+b+d$$
$$-2a<0 \quad\Rightarrow\quad a>0$$

Now, $P(x)$ being a polynomial is differentiable in $(-1, 1)$.

Therefore, critical points are $x = -1, 0, 1$

Since $P(-1) < P(1)$, the maximum of $P(x)$ can occur at $x = 0, 1$.

Now, $P'(x) = 4x^3 + 3a^2x + 2bx = x(4x^2 + 3ax + 2b)$

Since only root of $P'(x) = 0$ is $x = 0$, thus $4x^2 + 3ax + 2b = 0$ has no real roots.

$\therefore \quad 9a^2 - 32b < 0 \quad \Rightarrow \quad -32b < -9a^2$

or $\quad b > \frac{9}{32}a^2 \quad \Rightarrow \quad b > 0$

Thus, $a > 0, b > 0$

Now, $P(x)$ is increasing in $[0, 1]$.

$\therefore \quad P(1)$ is maximum.

Similarly, $P(x)$ is decreasing in $[-1, 0]$

$\therefore \quad P(-1)$ cannot be minimum.

3. Given parabola is $(y-2)^2 = x - 1$ (i)

Let $x - 1 = X, \; y - 2 = Y$

Equation (i) becomes

$$Y^2 = X \qquad \text{(ii)}$$

Axis of (ii) is $Y = 0 \Rightarrow y - 2 = 0$

$\Rightarrow y = 2$

Vertex of (ii) is $X = 0, Y = 0$

$\Rightarrow x - 1 = 0$ and $y - 2 = 0$; $A(1, 2)$

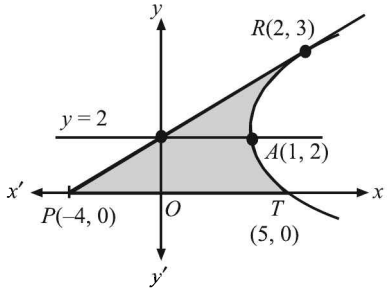


Fig. S.1

Parabola (i) meets the x-axis where $y = 0$, then $x = 5$, i.e. at $(5, 0)$.

From (i), $2(y-2)\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{2(y-2)}$

At $(2, 3)$, $\left(\frac{dy}{dx}\right)_{(2,3)} = \frac{1}{2}$

or $\qquad x - 2y + 4 = 0 \qquad \text{(iii)}$

This tangent meets the axis at $P(-4, 0)$ and $Q(0, 2)$.

The shaded area is the required area which is area TPRAT.

Required area $= \int_0^3 (x_{\text{parabola}} - x_{\text{tangent}})\,dy$

$= \int_0^3 \{[(y-2)^2 + 1] - (2y - 4)\}\,dy$

$= \int_0^3 (y^2 - 6y + 9)\,dy = \left[\frac{y^3}{3} - 3y^2 + 9y\right]_0^3$

$= 9 - 27 + 27 = 9$ sq. units

4. Given equation is

$$x^{2x} - 2^{x^x}\cot y - 1 = 0 \qquad \text{(i)}$$

We need the value of y' when $x = 1$

From (i), when $x = 1$,

$1 - 2\cot y - 1 = 0 \Rightarrow \cot y = 0$

$\Rightarrow y = \frac{\pi}{2}$

We know $\frac{d}{dx}(x^x) = x^x \log(1 + x)$

$\because \quad \frac{d}{dx}(x^{2x}) = 2x^{2x}\log(1 + x)$

Differentiating (i) w.r.t. x, we get

$$2x^{2x}(1 + \log x) - 2\left[x^x(1 + \log x)\cot y - x^x \operatorname{cosec}^2 y \frac{dy}{dx}\right] = 0 \qquad \text{(ii)}$$

At $x = 1, \; y = \frac{\pi}{2}$

Substituting in (ii), we get

$$2 - 2\left[\cot\frac{\pi}{2} - \operatorname{cosec}^2\frac{\pi}{2}\frac{dy}{dx}\right] = 0$$

$$\left[\cot\frac{\pi}{2} = 0, \operatorname{cosec}^2\left(\frac{\pi}{2}\right) = 1\right]$$

$$2\frac{dy}{dx} = -2 \Rightarrow \frac{dy}{dx} = -1 \quad \text{or} \quad y'(1) = -1$$

5. Tickets having product of digits as zero are 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 20, 30, 40

Number of such tickets = 14

There is only one ticket numbered as 08, sum of whose digits is 8.

$\therefore$ Total number of cases = 14

Favourable number of cases = 1.

6. $y = c_1 e^{c_2 x}$ (i)

$y' = c_1 c_2 e^{c_2 x} = c_2 y \qquad (\because c_1 e^{c_2 x} = y)$ (ii)

$y'' = c_2 y'$ (iii)

From (i) and (ii), we have

$$\frac{y'}{y} = c_2$$

Substituting the value of c_2 in (iii), the required differential equation is

$$y'' = \frac{y'}{y} \cdot y' \quad \text{or} \quad yy'' = (y')^2$$

7. Let $S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \cdots + \infty$

$$\frac{1}{3}S = 1 + \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} + \cdots + \infty$$

$$\frac{2}{3}S = 1 + \frac{1}{3} + 4\left(\frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \cdots\right)$$

(series in brackets is infinite G.P.)

$$= \frac{4}{3} + 4\left[\frac{1/3^2}{1-(1/3)}\right] \left(S = \frac{a}{1-r}\right)$$

$$\frac{2}{3}S = \frac{4}{3} + 4 \cdot \left(\frac{3}{2} \times \frac{1}{9}\right) = \frac{4}{3} + \frac{2}{3} = 2$$

or $S = 3$

8. Given ellipse is $\frac{x^2}{4} + \frac{y^2}{1} = 1$

Comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, we get $a = 2$, $b = 1$

Let the second ellipse be $\frac{x^2}{4^2} + \frac{y^2}{b^2} = 1$

The ellipse passes through $P(2, 1)$.

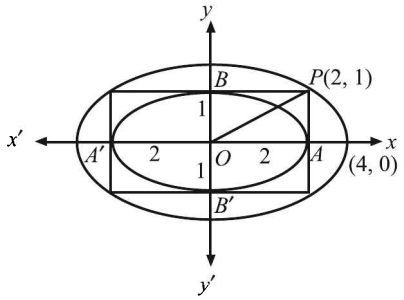


Fig. S.2

$$\therefore \quad \frac{4}{16} + \frac{1}{b^2} = 1$$

$$\frac{1}{b^2} = \frac{3}{4} \Rightarrow b^2 = \frac{4}{3}$$

The equation of the second ellipse is

$$\frac{x^2}{4^2} + \frac{y^2}{4/3} = 1 \Rightarrow \frac{x^2}{16} + \frac{3y^2}{4} = 1$$

or $$x^2 + 12y^2 = 16$$

9. $8^{2n} - (62)^{2n+1}$

$$= (9-1)^{2n} - (9 \times 7 - 1)^{2n+1}$$

$$= (9-1)^{2n} - (1-63)^{2n+1}$$

$$= \left[1 - {}^{2n}C_0(9) + {}^{2n}C_1(9)^2 + \cdots\right]$$

$$+ \left[1 - {}^{(2n+1)}C_1 \cdot 63 + {}^{(2n+1)}C_2(63)^2 + \cdots\right]$$

$$= 2 + [-{}^{2n}C_0 9 + {}^{2n}C_1(9)^2 - \cdots - {}^{2n+1}C_1 63$$

$$+ {}^{2n+1}C_2(63)^2 + \cdots]$$

The expression within bracket is divisible by 9, so the remainder is 2.

10. Let the coordinates of one of the points A, B, C be (x, y), say, $A(x, y)$.

$P(1, 0)$, $Q(-1, 0)$

$$PA = \sqrt{(x-1)^2 + y^2}$$

$$QA = \sqrt{(x+1)^2 + y^2}$$

$$\frac{PQ}{QA} = \frac{1}{3} \Rightarrow \frac{PA^2}{QA^2} = \frac{1}{9}$$

or $9PA^2 = QA^2$

$\Rightarrow 9[(x-1)^2 + y^2] = (x+1)^2 + y^2$

$\Rightarrow 8x^2 + 8y^2 - 20x + 8 = 0$

$\Rightarrow x^2 + y^2 - \frac{5}{2}x + 1 = 0$, which is a circle with centre at $\left(\frac{5}{4}, 0\right)$.

The centre of this circle is the circumcircle of the ΔABC.

$\therefore$ The circumcentre is at $\left(\frac{5}{4}, 0\right)$.

11. $\left|z - \frac{4}{z}\right| = 2$

$$|z| - 4\left|\frac{1}{z}\right| \le 2$$

$|z| - \frac{4}{|z|} \le 2$ or $|z|^2 - 2|z| \le 4$ $\quad \left(\begin{array}{c} |z_1 - z_2| \ge \big||z_1| - |z_2|\big| \\ \therefore \big||z_1| - |z_2|\big| \le |z_1 - z_2| \end{array}\right)$

Adding 1 to both sides, we get

$$|z|^2 - 2|z| + 1 \le 4 + 1 \quad \Rightarrow \quad (|z| - 1)^2 \le 5$$

or $-\sqrt{5} \le |z| - 1 \le \sqrt{5}$ $(x^2 \le a \Rightarrow -\sqrt{a} \le x \le \sqrt{a})$

or $1 - \sqrt{5} \le |z| \le \sqrt{5} + 1$

$\therefore$ $|z| \le \sqrt{5} + 1$. The maximum value of $|z|$ is $\sqrt{5} + 1$.

12. Let the direction cosines of the vector be l, m, n and its magnitude be r. Then projections of this vector on the coordinate axes are lr, mr, nr.

$\therefore$ $lr = 6,\ mr = -3,\ nr = 2$

Squarring and adding, we have

$$(l^2 + m^2 + n^2)r^2 = 6^2 + (-3)^2 + 2^2 = 49$$

$$r^2 = 49 \qquad (\because l^2 + m^2 + n^2 = 1)$$

$\Rightarrow$ $r = 7$

$$l = \frac{6}{r} = \frac{6}{7},\ m = \frac{-3}{r} = \frac{-3}{7},\ n = \frac{2}{r} = \frac{2}{7}$$

$\therefore$ $l = \frac{6}{7},\ m = \frac{-3}{7},\ n = \frac{2}{7}$.

13. Equation of a circle passing through the points of intersection P and Q of the given circles is

$$x^2 + y^2 + 3x + 7y + (2p - 5) + \lambda(x^2 + y^2 + 2x + 2y - p^2) = 0 \qquad \text{(i)}$$

$$(S_1 + \lambda S_2) = 0$$

Circle (i) passes through the point (1, 1)

$\therefore$ $1 + 1 + 3 + 7 + (2p - 5) + \lambda(1 + 1 + 2 + 2 - p^2) = 0$

$$\lambda = \frac{-(2p + 7)}{6 - p^2} = \frac{2p + 7}{p^2 - 6}$$

$\therefore$ For λ to exist, $p^2 - 6 \ne 0 \Rightarrow p \ne \pm\sqrt{6}$.

Hence, Eq. (i) can hold for all real values of p except for two values $p = \pm\sqrt{6}$.

14. $p = \frac{1}{4},\ q = \frac{3}{4},\ n = n$

$$p(r) = {}^nC_r \cdot q^{r-r} p^r \qquad \text{(i)}$$

P (at least one event) $= 1 - P(0) \ge \frac{9}{10}$ (ii)

Form (i), we have

$$P(0) = {}^nC_0 \cdot \left(\frac{3}{4}\right)^n \cdot \left(\frac{1}{4}\right)^0 = \left(\frac{3}{4}\right)^n$$

From (ii), we have

$$1 - \left(\frac{3}{4}\right)^n \ge \frac{9}{10} \Rightarrow -\left(\frac{3}{4}\right)^n \ge -\frac{1}{10}$$

$$\Rightarrow \qquad \left(\frac{3}{4}\right)^n \ge \frac{1}{10}$$

Taking logs to the base 10, we get

$$n \log_{10}\left(\frac{3}{4}\right) \le \log_{10}(10^{-1})$$

$$n(\log_{10} 3 - \log_{10} 4) \le -1 \Rightarrow n(\log_{10} 4 - \log_{10} 3) \ge 1$$

$$\Rightarrow n \ge \frac{1}{\log_{10} 4 - \log_{10} 3}.$$

15. $f(x) = x^3 + 5x + 1$

$f'(x) = 3x^2 + 5 > 0,\ \forall x \in \mathbb{R}$

Thus, $f(x)$ is strictly an increasing function on $\mathbb{R}$.

Therefore, $f(x)$ is one-one.

Now, $f(x) = x^3 + 5x + 1$, being a polynomial function, is continous in $\mathbb{R}$, so it takes all values in $(-\infty, \infty)$.

Therefore, $f(x)$ is onto.

16. If x is not an integer, then note

$$[x] + [-x] = -1 \qquad \text{(i)}$$

and in case x is an integer, then

$$[x] + [-x] = 0$$

Now, let $I = \int_0^{\pi} [\cot x]\,dx$ (ii)

$$I = \int_0^{\pi} [\cot(\pi - x)]\,dx$$

$$\Rightarrow I = \int_0^{\pi} [-\cot x]\,dx \qquad \text{(iii)}$$

Adding (ii) and (iii), we get

$$2I = \int_0^{\pi} ([\cot x] + [-\cot x])\,dx$$

$$= \int_0^{\pi} (-1)\,dx = -\pi$$

$$I = -\frac{\pi}{2}.$$

17. The number of ways to select 4 novels out of 6 novels $= {}^6C_4 = {}^6C_2 = 15$.

The number of ways to select 1 dictionary out of 3 dictionaries $= {}^3C_1 = 3$.

Thus, the total number of ways to select 4 novels and 1 dictionary $= 15 \times 3 = 45$.

These can be arranged in 4 places (because dictionary is in the middle) in 4! = 24 ways.

The total number of ways $= 45 \times 24 = 1080$.

18. A point on the line is $P(2, 1, -2)$ and direction ratios of the line are proportional to 3, –5, 2.
The direction ratios of the normal to the plane are proportional to 1, 3, $-\alpha$.
The line lies in the plane if the point $P(2, 1, -2)$ lies in the plane and the normal to the plane is perpendicular to the line.

$\therefore \quad 2+3+2\alpha+\beta=0$ (i)

Also, $3-15-2\alpha=0$ (ii)

From (ii), $\alpha=-6$

From (iii), $\beta=7$

Thus, $(\alpha, \beta)=(-6, 7)$.

19. We know $[\vec{a}\vec{b}\vec{c}]=\vec{a}\cdot(\vec{b}\times\vec{c})=(\vec{a}\times\vec{b})\cdot\vec{c}=\vec{c}\cdot(\vec{a}\times\vec{b})$

($\because$ Dot and cross are interchangeable)

$$[3\vec{u}\ p\vec{v}\ p\vec{w}]=3p^2\vec{u}\cdot(\vec{v}\times\vec{w})$$

$$[3\vec{u}\ p\vec{v}\ p\vec{w}]=3p^2[\vec{u}\ \vec{v}\ \vec{w}] \quad \text{(i)}$$

Now, $[p\vec{v}\ \vec{w}\ q\vec{u}]=pq\ \vec{v}(\vec{w}\times\vec{u})=pq\ (\vec{v}\times\vec{w})\cdot\vec{u}$

(Dot and cross interchanged)

$$= pq\ \vec{u}(\vec{v}\cdot\vec{w})$$

$$[p\vec{v}\ \vec{w}\ q\vec{u}]=pq\ [\vec{u}\ \vec{v}\ \vec{w}] \quad \text{(ii)}$$

Again, $[2\vec{w}q\ \vec{v}\ \ q\vec{u}]=2q^2\ \vec{w}\cdot(\vec{v}\times\vec{u})=2q^2\ (\vec{w}\times\vec{v})\cdot\vec{u}$

(Dot and cross interchanged)

$$= -2q^2\ (\vec{v}\times\vec{w})\cdot\vec{u}$$

$$[2\vec{w}\ q\vec{v}\ \ q\vec{u}]=-2q^2\ \vec{u}\cdot[\vec{v}\times\vec{w}]=-2q^2[\vec{u}\ \vec{v}\ \vec{w}] \quad \text{(iii)}$$

From (i), (ii) and (iii), we get

$$[3\vec{u}\ p\vec{v}\ p\vec{w}]-[p\vec{v}\ \vec{w}\ q\vec{u}]-[2\vec{w}\ q\vec{v}\ \ q\vec{u}]$$

$$=(3p^2-pq+2q^2)[\vec{u}\ \vec{v}\ \vec{w}]=0 \quad \text{(iv)}$$

Since $\vec{u}$, $\vec{v}$, $\vec{w}$ are non-coplanar, $[\vec{u}\ \vec{v}\ \vec{w}]\neq 0$

$\therefore \quad 3p^2-pq+2q^2=0$

which gives imaginary values of p and q, except when $p = 0$, $q = 0$.
Thus, the given equality is true for only one value of (p, q).

20. $A\cup B=A\cup C$

$$(A\cup B)\cap B=(A\cup C)\cap B$$

$$B=(A\cap B)\cup(C\cap B)$$

$$B=(A\cap B)\cup(B\cap C) \quad \text{(i)}$$

Again, $A\cup C=A\cup B$

$\therefore \quad (A\cup C)\cap C=(A\cup B)\cap C$

$$= (A\cap C)\cup(B\cap C)$$

$$C=(A\cap B)\cup(B\cap C) \quad \text{(ii)}$$

From (i) and (ii), we have

$$B = C$$

21. Slope of the first line $= \dfrac{-\text{coeff. of } x}{\text{coeff. of } y}=p(p^2+1)=m_1$

Slope of the second line

$$=\frac{-\text{coeff. of } x}{\text{coeff. of } y}=\frac{(p^2+1)^2}{p^2+1}=-(p^2+1)=m_2$$

Since these lines are perpendicular to the same line, these lines must be parallel.

$\therefore \quad m_1=m_2$

$$p(p^2+1)=-(p^2+1) \Rightarrow p=-1$$

Thus, the given lines are perpendicular to a common line for exactly one value of p.

22. A: $\cos\alpha+\cos\beta+\cos\gamma=0$ (i)

B: $\sin\alpha+\sin\beta+\sin\gamma=0$ (ii)

From (i) and (ii) on squaring and adding, we get

$$(\cos\alpha+\cos\beta+\cos\gamma)^2=0 \quad \text{(iii)}$$

$$(\sin\alpha+\sin\beta+\sin\gamma)^2=0 \quad \text{(iv)}$$

From (iii) and (iv),

or $(\cos^2\alpha+\sin^2\alpha)+(\cos^2\beta+\sin^2\beta)+(\cos^2\gamma+\sin^2\gamma)$

$+2[(\cos\alpha\cos\beta+\sin\alpha\sin\beta)+(\cos\beta\cos\gamma$

$+\sin\beta\sin\gamma)+(\cos\gamma\cos\alpha+\sin\gamma\sin\alpha)=0$

$1+1+1+2[\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha)]=0$

or $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha)=-\dfrac{3}{2}$

or $\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)=-\dfrac{3}{2}$

Thus, both A and B are true.

23. Root of $bx^2+cx+a=0$ has imaginary roots

$$c^2-4ab<0 \Rightarrow c^2<4ab \quad \text{(i)}$$

Let $y=3b^2x^2+6bcx+2c^2$ (ii)

Equation (ii) represents an upward parabola, whose vertex is $\left(\dfrac{-c}{b}, -c^2\right)$

The equation $y=ax^2+bx+c$ represents an upward parabola, if $a > 0$ with vertex $\left(\dfrac{-b}{2a}, \dfrac{4ac-b^2}{4a}\right)$. Thus, from (ii), the least value of y is $-c^2$.

$\therefore \quad -c^2>-4ab$ by (i).

24. Mean deviation $= \dfrac{\sum|x-\bar{x}|}{n}$ (i)

For given data with 101 terms (n = 101),

$$\bar{x} = \frac{1+1+d+1+2d+\cdots+1+100d}{101}$$

$$= \frac{101+d(1+2+\cdots+100)}{101}$$

$$= \frac{101}{101} + \frac{d[(100\times101)/2]}{101} = 1+50d$$

Now, $\sum|x-\bar{x}| = |1-(1+50d)| + |1+d-(1+5d)|$

$$+|1+2d-(1+50d)| + \cdots + |(1+100d)-(1+50d)|$$

$$\sum|(x-\bar{x})| = 50d+49d+48d+\cdots+0d+\cdots$$

$$+49d+50d$$

$$= 2(1+2+3+.....+50)d = 2\times\left(\frac{50\times51}{2}\right)d$$

$$= (50\times51)d$$

From (i),

$$\text{Mean deviation} = \frac{(50\times51)d}{101}$$

$$255 = \left(\frac{50\times51}{101}\right)d \quad \Rightarrow \quad d = 10.1$$

25. Let

$$\Delta = \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ a & -b & c \end{vmatrix} = 0$$

[Taking $(-1)^n$ common from the last row of the second determinant]

Interchanging rows and columns in the second determinant, we get

$$\Delta = \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & a-1 & a \\ b+1 & b-1 & -b \\ c-1 & c+1 & c \end{vmatrix}$$

Interchanging $C_2 \leftrightarrow C_3$ and $C_2 \leftrightarrow C_1$ successively,

$$\Delta = \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix}$$

Now, $\Delta = 0$, if n is any odd integer.

26. $f(x) = (x+1)^2 - 1;\ x \geq -1$ (i)

Let $y = f(x) = x^2 + 2x;\ x \geq -1$

$f(x)$ being a polynomial is continuous for $x \geq -1$, so $f(x)$ is an onto function. Also $f(x)$ is one-one. Thus, $f(x)$ is a bijection, which is evident from the graph of $f(x)$. From (i),

$$y = (x+1)^2 - 1 \quad \Rightarrow \quad y+1 = (x+1)^2$$

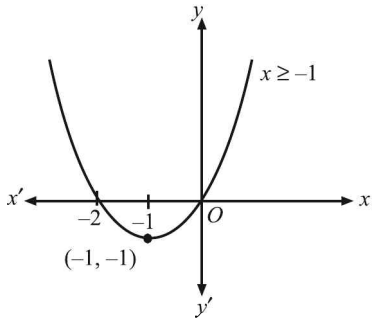


Fig. S.3

$$x+1 = \sqrt{y+1} \quad \Rightarrow \quad x = \sqrt{y+1} - 1$$

or $$f^{-1}(x) = \sqrt{x+1} - 1$$

Now given, $f^{-1}(x) = f(x)$

$\Rightarrow \sqrt{x+1} - 1 = (x+1)^2 - 1$

$\Rightarrow \sqrt{x+1} = (x+1)^2 \Rightarrow x+1 = (x+5)^4$

$\Rightarrow (x+1)^4 - (x+1) = 0 \Rightarrow (x+1)[(x+1)^3 - 1] = 0$

$\Rightarrow x+1 = 0$ or $(x+1)^3 = 1$

$x = 1, x = 0$

[other values from $(x+1)^3 = 1$ are imaginary]

Thus, the set $[x: f(x) = f^{-1}(x)] = (0, -1)$.

Thus, both Statement I and Statement II are true.

Statement II states that f is a bijection, but this is not the reason for $f^{-1}(x) = f(x)$.

Hence, Statement II is not a correct explanation for Statement I.

27. $f(x) = x|x|,\ g(x) = \sin x$

$f(x) = -x^2;\ x \leq 0$

$= x^2;\ x \geq 0$

$[\because |x| = -x, \text{ if } x \leq 0;\ |x| = x \text{ if } x \geq 0]$

$$g \circ f = g[f(x)] = \begin{cases} -\sin x^2; & x \leq 0 \\ \sin x^2; & x \geq 0 \end{cases}$$

$(g \circ f)' = -2x\cos x^2;\ x \leq 0 = 2x\cos x^2;\ x \geq 0$

At $x = 0$; for $(g \circ f)$; L.H.D. = R.H.D.

Since $g \circ f$ is differentiable at $x = 0$.

Now, $(g \circ f)' = -2x\cos x^2;\ x \le 0 = 2x\cos x^2;\ x \ge 0$ is also continuous at $x = 0$.
Statement I is true.

$$(g \circ f)'' = -2\cos x^2 + 4x^2 \sin x^2; \quad x \le 0$$
$$= 2\cos x^2 - 4x^2 \sin x^2; \quad x \ge 0$$

At $x = 0$,

L.H.D. = $(g \circ f)'(0 - h) = -2$,

R.H.D. = $(g \circ f)'(0 + h) = 2$

L.H.D. $\neq$ R.H.D.

$(g \circ f)''$ does not exist at $x = 0$. Hence, Statement II is false.

28. We know that

$$\text{variance} = \sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{(i)}$$

x: 2, 4, 6, 8,..., $2n$; $n \in N$

$$\sum x^2 = 2^2 + 4^2 + 6^2 + 8^2 + \cdots + (2n)^2$$

$$4(1^2 + 2^2 + 3^2 + 4^2 + \cdots + n^2)$$

$$= 4\left[\frac{n(n+1)(2n+1)}{6}\right] = \frac{2}{3}n(n+1)(2n+1)$$

$$\sum x = 2 + 4 + 6 + 8 + \cdots + 2n$$
$$= 2[1 + 2 + 3 + 4 + \cdots + n]$$
$$= 2 \cdot \frac{n(n+1)}{2} = n(n+1)$$

Now from (i), we have

$$\sigma^2 = \frac{2}{3}(n+1)(2n+1) - (n+1)^2$$

$$= (n+1)\left[\frac{2}{3}(2n+1) - (n+1)\right]$$

$$= (n+1)\left[\frac{4n+2-3n-3}{3}\right]$$

$$\sigma^2 = \frac{(n+1)(n-1)}{3} = \frac{n^2-1}{3}$$

Thus, Statement I is flase and Statement II is true.

29. Statement I: $\sim(p \leftrightarrow \sim q)$ is equivalent to $p \leftrightarrow q$.
Statement II: $\sim(p \leftrightarrow \sim q)$ is a tautology.

Notes: Before attempting this question, the student is required to remember the following table:

p	q	$p \vee q$	$p \wedge q$	$p \to q$	$p \leftrightarrow q$
T	T	T	T	T	F
T	F	T	F	F	T
F	T	T	F	T	T
F	F	F	F	T	F

(i) $p \to q$ is false (F) if and only if p is true (T) and q is false (F); in all other cases it is true (T).

(ii) $p \leftrightarrow q$ is true (T) if and only if both p and q are true (T) or p and q are both false (F).

(iii) Tautology: The compound statement involving p and q as component statements is called a tautology if this statement is true (T) for all values of p and q.

(iv) Fallacy: The compound statement involving p and q as component statements is called a fallacy if it is false (F) for all values of p and q.

p	q	$\sim q$	$p \leftrightarrow \sim q$	$\sim(p \leftrightarrow \sim q)$	$p \leftrightarrow q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T

From the last two columns $\sim(p \leftrightarrow \sim q) \equiv p \leftrightarrow q$
Thus, Statement I is correct. But $\sim(p \leftrightarrow \sim q)$ is not a tautology because all values are not true. Hence, Statement II is false.

30. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\text{adj}\, A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, $\text{adj}\,(\text{adj}\, A) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = A$.

Thus, for a second-order matrix,

$$\text{adj}\,(\text{adj}\, A) = A$$

Also, $\text{adj}(\text{adj}\, A) = |A|^{n-2} A = A \qquad (\because\ n = 2)$

Thus, Statement I is correct.
Also for a square matrix of order n,

$$|\text{adj}\, A| = |A|^{n-1}$$

For a second-order matrix A, i.e. when $n = 2$,

$$|\text{adj}\, A| = |A|^{2-1} = |A|$$

Statement II is correct.
But Statement II cannot be a correct explanation for Statement I.

Sample Paper II

1. Consider 5 independent Bernoulli's trials each with probability of success p. If the probability of at least one failure is greater than or equal to 31/32, then p lies in the interval

(a) $\left(\frac{11}{12}, 1\right]$ (b) $\left(\frac{1}{2}, \frac{3}{4}\right]$

(c) $\left(\frac{3}{4}, \frac{11}{12}\right]$ (d) $\left[0, \frac{1}{2}\right]$

2. The coefficient of x^7 in the expansion of $(1 - x - x^2 + x^3)^6$ is

(a) 132 (b) 144

(c) −132 (d) −144

3. $\lim_{x \to 2} \left\{ \frac{\sqrt{1 - \cos[2(x-2)]}}{x-2} \right\}$

(a) equals $\frac{1}{\sqrt{2}}$ (b) does not exist

(c) equals $\sqrt{2}$ (d) equals $-\sqrt{2}$

4. Let $\mathbb{R}$ be the set of real numbers.

Statement 1: $A = \{(x, y) \in R \times R: y - x \text{ is an integer}\}$ is an equivalence relation on $\mathbb{R}$.

Statement 2: $A = \{(x, y) \in R \times R: x = \alpha y$ for some rational number $\alpha\}$ is an equivalence relation on $\mathbb{R}$.

(a) Statement 1 is false, Statement 2 is true.

(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

(d) Statement 1 is true, Statement 2 is false.

5. Let α and β be real numbers and z be a complex number. If $z^2 + \alpha z - \beta = 0$ has two distinct roots on the line Re $z = 1$, then it is necessary that

(a) $\beta \in (1, \infty)$ (b) $\beta \in (0, 1)$

(c) $\beta \in (-1, 0)$ (d) $|\beta| = 1$

6. $\frac{d^2x}{dy^2}$ equals

(a) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$ (b) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

(c) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$ (d) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

7. The number of values of k for which the linear equations

$4x + ky + 2z = 0$

$kx + 4y + z = 0$

$2x + 2y + z = 0$

possess a non-zero solution is

(a) 0 (b) 3

(c) 2 (d) 1

8. **Statement 1:** The point $A(1, 0, 7)$ is the mirror image of the point $B(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$.

Statement 2: The line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ bisects the line segment joining $A(1, 0, 7)$ and $B(1, 6, 3)$.

(a) Statement 1 is false, Statement 2 is true.

(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

(d) Statement 1 is true, Statement 2 is false.

9. Consider the following statements

P: Suman is brilliant

Q: Suman is rich

R: Suman is honest

The negation of the statement "Suman is brilliant and dishonest if and only if Suman is rich" can be expressed as

(a) $\sim(P \wedge \sim R) \leftrightarrow Q$ (b) $\sim P \wedge (Q \leftrightarrow \sim R)$

(c) $\sim[Q \leftrightarrow (P \wedge \sim R)]$ (d) $\sim Q \leftrightarrow \sim P \wedge R$

10. The lines $L_1: y - x = 0$ and $L_2: 2x + y = 0$ intersect the line $L_3: y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

Statement 1: The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$.
Statement 2: In any triangle, the bisector of an angle divides the triangle into two similar triangles.

(a) Statement 1 is false, Statement 2 is true.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(d) Statement 1 is true, Statement 2 is false.

11. A man saves ₹ 200 in each of the first three months of his service. In each of the subsequent months, his saving increases by ₹ 40 more than the saving of the immediately previous month. His total saving from the start of the service will be ₹ 11,040 after

(a) 21 months (b) 18 months
(c) 19 months (d) 20 months

12. The equation of the ellipse whose axes are the axes of coordinates and which passes through the point (–3, 1) and has eccentricity $\sqrt{\frac{2}{5}}$ is

(a) $5x^2 + 3y^2 - 32 = 0$ (b) $3x^2 + 5y^2 - 32 = 0$
(c) $5x^2 + 3y^2 - 48 = 0$ (d) $3x^2 + 5y^2 - 15 = 0$

13. If $A = \sin^2 x + \cos^4 x$, then for all real x,

(a) $\frac{3}{4} \le A \le \frac{13}{16}$ (b) $\frac{3}{4} \le A \le 1$
(c) $\frac{13}{16} \le A \le 1$ (d) $1 \le A \le 2$

14. The value of $\int_0^1 \frac{8\log(1+x)}{1+x^2}\,dx$ is

(a) $\log 2$ (b) $\pi \log 2$
(c) $\frac{\pi}{8}\log 2$ (d) $\frac{\pi}{2}\log 2$

15. If the angle between the line $x = \frac{y-1}{2} = \frac{z-3}{\lambda}$ and the plane $x + 2y + 3z$ is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then λ equals

(a) 5/3 (b) 2/3
(c) 3/2 (d) 2/5

16. For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t}\sin t\,dt$. Then f has

(a) local maximum at π and local minimum at 2π
(b) local maximum at π and 2π
(c) local minimum at π and 2π
(d) local minimum at π and local maximum at 2π

17. The domain of the function $f(x) = \frac{1}{\sqrt{|x| - x}}$ is

(a) $(-\infty, \infty) - \{0,\}$ (b) $(-\infty, \infty)$
(c) $(0, \infty)$ (d) $(-\infty, 0)$

18. If the mean deviation about the median of the numbers $a, 2a, \ldots, 50a$ is 50, then $|a|$ equals

(a) 5 (b) 2
(c) 3 (d) 4

19. If $\vec{a} = \frac{1}{\sqrt{10}}(3\hat{i} + \hat{k})$ and $\vec{b} = \frac{1}{7}(2\hat{i} + 3\hat{j} - 6\hat{k})$, then the value of $(2\vec{a} - \vec{b}) \cdot \left[(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})\right]$ is

(a) 3 (b) –5
(c) –3 (d) 5

20. The values of p and q for which the function

$$f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x + x^2} - \sqrt{x}}{x^{\frac{3}{2}}}, & x > 0 \end{cases}$$

is continuous for all x in $\mathbb{R}$, are

(a) $p = \frac{1}{2}, q = \frac{3}{2}$ (b) $p = \frac{1}{2}, q = -\frac{3}{2}$
(c) $p = \frac{5}{2}, q = \frac{1}{2}$ (d) $p = -\frac{3}{2}, q = \frac{1}{2}$

21. The two circles $x^2 + y^2 = ax$ and $x^2 + y^2 = c^2 (c > 0)$ touch each other if

(a) $|a| = 2c$ (b) $2|a| = c$
(c) $|a| = c$ (d) $a = 2c$

22. Let I be the purchase value of a piece of equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$ depreciates at a rate given by the differential equation $\frac{dV(t)}{dt} = -k(T - t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is

(a) e^{-kT} (b) $T^2 - \frac{1}{k}$

(c) $I - \frac{kT^2}{2}$ (d) $I - \frac{k(T-t)^2}{2}$

23. If C and D are two events such that $C \subset D$ and $P(D) \neq 0$, then the correct statement among the following is

(a) $P(C:D) = \frac{P(D)}{P(C)}$ (b) $P(C:D) = P(C)$

(c) $P(C:D) \geq P(C)$ (d) $P(C:D) < P(C)$

24. Let A and B be two symmetric matrices of order 3.
Statement 1: $A(BA)$ and $(AB)A$ are symmetric matrices.
Statement 2: AB is a symmetric matrix if the matrix multiplication of A with B is commutative.

(a) Statement 1 is false, Statement 2 is true.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(d) Statement 1 is true, Statement 2 is false.

25. If ω ($\neq 1$) is a cube root of unity, and $(1+\omega)^7 = A + B\omega$, then (A, B) equals

(a) $(-1, 1)$ (b) $(0, 1)$
(c) $(1, 1)$ (d) $(1, 0)$

26. **Statement 1:** The number of ways of distributing 10 identical balls in 4 distinct boxes such that no box is empty is ${}^9C_3$.
Statement 2: The number of ways of choosing any 3 places from 9 different places is ${}^9C_3$.

(a) Statement 1 is false, Statement 2 is true.
(b) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(c) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(d) Statement 1 is true, Statement 2 is false.

27. The shortest distance between the line $y - x = 1$ and the curve $x = y^2$ is

(a) $\frac{4}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{4}$

(c) $\frac{3\sqrt{2}}{8}$ (d) $\frac{8}{3\sqrt{2}}$

28. The area of the region enclosed by the curves $y = x$, $x = e$, $y = \frac{1}{x}$ and the positive x-axis is

(a) $\frac{5}{2}$ sq. units (b) $\frac{1}{2}$ sq. units

(c) 1 sq. unit (d) $\frac{3}{2}$ sq. units

29. If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\log 2)$ is equal to

(a) -2 (b) 7
(c) 5 (d) 13

30. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 0$. Then the vector $\vec{d}$ is equal to

(a) $\vec{c} - \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{b}$ (b) $\vec{b} - \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{c}$

(c) $\vec{c} + \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{b}$ (d) $\vec{b} + \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{c}$

ANSWERS

1. (d)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (c)	**8.** (b)	**9.** (c)	**10.** (d)
11. (a)	**12.** (b)	**13.** (b)	**14.** (b)	**15.** (b)	**16.** (a)	**17.** (d)	**18.** (d)	**19.** (b)	**20.** (d)
21. (c)	**22.** (c)	**23.** (c)	**24.** (c)	**25.** (c)	**26.** (c)	**27.** (c)	**28.** (d)	**29.** (b)	**30.** (a)

HINTS AND SOLUTIONS—SAMPLE PAPER II

1. p(at least one failure) = $1 - p$(all success)
$= 1 - {}^5C_5 p^5$

Hence, $1 - {}^5C_5 p^5 \geq \frac{31}{32}$

$\Rightarrow p^5 \leq \frac{1}{32} \Rightarrow p \leq \frac{1}{2}$

Hence $p \in \left[0, \frac{1}{2}\right]$

2. $(1 - x - x^2 + x^3)^6 = (1 - x^2)^6 (1 - x)^6$
$= (C_0 - C_1 x^2 + C_2 x^4 + C_3 x^6 + \cdots + C_6 x^{12})$
$(C_0 - C_1 x + C_2 x^2 + \cdots + C_6 x^6)$

Coefficient of x^7 is
$(-C_1)(-C_5) + (C_2)(-C_3) + (-C_3)(-C_1)$
$= (-6) \times (-6) + 15 \times (-20) + (-20) \times (-6)$
$= 36 - 300 + 120 = -300 + 156 = -144$

3. $\lim\limits_{x\to 2}\left\{\dfrac{\sqrt{1-\cos[2(x-2)]}}{x-2}\right\}$

$$= \lim_{x\to 2}\frac{\sqrt{2}|\sin(x-2)|}{x-2}$$

$x-2=t,\ x\to 2,\ t\to 0$

$$= \lim_{t\to 0}\sqrt{2}\frac{|\sin t|}{t}$$

L.H.L. $= \lim\limits_{t\to 0}-\sqrt{2}\dfrac{\sin t}{t} = -\sqrt{2}$

R.H.L. $= \lim\limits_{t\to 0}\sqrt{2}\dfrac{\sin t}{t} = \sqrt{2}$

So L.H.L. $\neq$ R.H.L., hence does not exist.

4. Statement 1:

$A=\{(x,y)\in R\times R: y-x \text{ is an integer}\}$

$(x,x)\in R\times R$ as $x-x=0$ (integer)

$\Rightarrow$ reflexive

Also, $(x, y)\in R\times R$ and $(y, z)\in R\times R$

$\Rightarrow$ $x-y=$ integer and $y-z=$ integer

So, $x-z=(x-y)+(y-z)$

$=$ integer $-$ integer

$=$ integer

$\Rightarrow$ transitive

$\Rightarrow$ equivalence relation on $\mathbb{R}$.

Statement 2:

Obviously reflexive and symmetric

Let $(x,y)\in R\times R\Rightarrow x=\alpha y$

$(y,z)\in R\times R\Rightarrow y=\beta z$

So, $x=\alpha y=\alpha(\beta z)$

$(\alpha\beta)z$

$\Rightarrow$ transitive

$\Rightarrow$ equivalence relation

5. $z^2+\alpha z-\beta=0$

$$\Rightarrow\ z=\frac{-\alpha\pm\sqrt{\alpha^2+4\beta}}{2}=-\frac{\alpha}{2}\pm\frac{\sqrt{\alpha^2+4\beta}}{2}$$

Now, $-\dfrac{\alpha}{2}=1$

$\Rightarrow$ $\alpha=-2$

Now, $\alpha^2+4\beta<0$

$\Rightarrow$ $4-4\beta<0$ $\Rightarrow$ $-\beta<-1$

$\Rightarrow$ $\beta>1$

Hence, $\beta\in(1,\infty)$

6. $\dfrac{dy}{dx}=\dfrac{1}{\frac{dx}{dy}}$ $\Rightarrow$ $\dfrac{d^2y}{dx^2}=\dfrac{d}{dx}\left(\dfrac{dx}{dy}\right)^{-1}$

$$=\frac{d}{dy}\cdot\left(\frac{dx}{dy}\right)^{-1}\cdot\frac{dy}{dx}=-\left(\frac{dx}{dy}\right)^{-2}\cdot\frac{d^2x}{dy^2}\cdot\frac{dy}{dx}$$

$$=-\frac{1}{\left(\frac{dx}{dy}\right)^3}\cdot\frac{d^2x}{dy^2}$$

$$\Rightarrow\ -\frac{d^2x}{dy^2}\left(\frac{dx}{dy}\right)^3=\frac{d^2x}{dy^2}$$

$$\Rightarrow\ \frac{d^2x}{dy^2}=-\frac{d^2y}{dx^2}\left(\frac{dy}{dx}\right)^{-3}$$

7. $\begin{vmatrix}4 & k & 2\\ k & 4 & 1\\ 2 & 2 & 1\end{vmatrix}=0$

$$\xrightarrow{R_1-2R_3}\begin{vmatrix}0 & k-4 & 0\\ k & 4 & 1\\ 2 & 2 & 1\end{vmatrix}=0$$

$\Rightarrow$ $(k-4)(2-k)=0$ $\Rightarrow$ $k=2, 4.$

8. Direction of line AB $=(1-1, 6-0, 3-7)$

$=(0,6,-4)$

So if B is the mirror image of A $\Rightarrow$ AB and $\dfrac{x}{1}=\dfrac{y-1}{2}=\dfrac{z-2}{3}$ are perpendicular.

So, $a_1a_2+b_1b_2+c_1c_2$

$=0\times 1+6\times 2-4\times 3=0$

$\Rightarrow$ AB and the given line are perpendicular

Also, mid-point of AB, i.e. (1, 3, 5) has to satisfy the equation of the given line $\dfrac{x}{1}=\dfrac{y-1}{2}=\dfrac{z-2}{3}$,

$\Rightarrow$ $\dfrac{1}{1}=\dfrac{3-1}{2}=\dfrac{5-2}{3}$

So, Statement 1 is correct.

Statement 2:

By the definition of the mirror image, the line is perpendicular to the line joining $A(1, 0, 7)$ and $B(1, 6, 3)$.

So, Statement 2 is correct .

9. $\sim[P\wedge\sim(R\Leftrightarrow Q)]$

$=\sim[Q\Leftrightarrow(P\wedge\sim R)]$

10. The equation of the angle bisector

$$\frac{x-y}{\sqrt{2}}=-\frac{2x+y}{\sqrt{5}}$$

$(\sqrt{5}+2\sqrt{2})x+(\sqrt{2}-\sqrt{5})y=0$

$P\equiv(-2,-2), Q\equiv(1,-2), R\equiv\left(\frac{2(\sqrt{2}-\sqrt{5})}{\sqrt{5}+2\sqrt{2}},-2\right)$

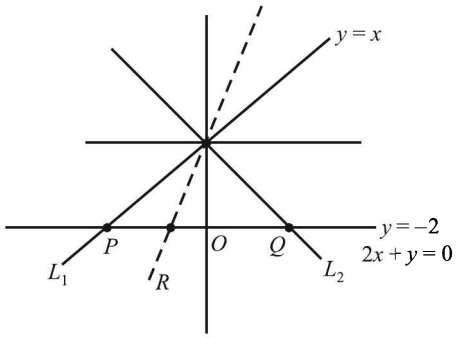


Fig. Q.1

$$PR=2-\frac{2(\sqrt{5}-\sqrt{2})}{\sqrt{5}+2\sqrt{2}}$$

$$=\frac{6\sqrt{2}}{\sqrt{5}+2\sqrt{2}}$$

$$RQ=1+\frac{(\sqrt{5}-\sqrt{2})2}{\sqrt{5}+2\sqrt{2}}=\frac{3\sqrt{5}}{\sqrt{5}+2\sqrt{2}}$$

$PR:RQ=2\sqrt{2}:\sqrt{5}$

Statement 1 is true.

Statement 2 is false.

11. Total saving in $(n+3)$ months

$$=200+200+200+240+280+\cdots+(200+40n)$$

$$=400+\frac{(200+200+40n)(n+1)}{2}$$

$$=400+(200+20n)(n+1)$$

Now, $400+(200+20n)(n+1)=11{,}040$

$\Rightarrow 20(10+n)(n+1)=11040-400=10{,}640$

$\Rightarrow 10n+n^2+10+n=532$

$\Rightarrow n^2+11n-522=0$

$\Rightarrow n^2+29n-18n-522=0$

$\Rightarrow n(n+29)-18(n+29)=0$

$\Rightarrow (n+29)(n-18)=0$

$\Rightarrow n=18$

Hence, the total months $(18+3)=21$.

12. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Given that it passes through (–3, 1)

$$\Rightarrow \frac{9}{a^2}+\frac{1}{b^2}=1 \qquad \text{(i)}$$

Also given $e^2=1-\frac{b^2}{a^2}$

$$\Rightarrow \frac{2}{5}=1-\frac{b^2}{a^2} \quad \Rightarrow \quad \frac{b^2}{a^2}=1-\frac{2}{5}=\frac{3}{5}$$

$$\Rightarrow b^2=\frac{3a^2}{5} \qquad \text{(ii)}$$

From (i) and (ii), we get

$$\frac{9}{a^2}+\frac{5}{3a^2}=1 \quad \Rightarrow \quad 9+\frac{5}{3}=a^2$$

$$\Rightarrow a^2=\frac{32}{3}$$

From (ii), we have

$$b^2=\frac{3}{5}\times\frac{32}{3}=\frac{32}{5}$$

So the ellipse is $\frac{x^2}{\frac{32}{3}}+\frac{y^2}{\frac{32}{5}}=1$

$\Rightarrow 3x^2+5y^2-32=0.$

13. If $A=\sin^2 x+\cos^2 x\cdot\cos^2 x\le\sin^2 x+\cos^2 x=1$ (i)

Also, $A=(1-\cos^2 x)+(\cos^4 x)$

$$=\cos^4 x-\cos^2 x+1$$

$$=\left(\cos^2 x-\frac{1}{2}\right)^2+1-\frac{1}{4}$$

$$=\left(\cos^2 x-\frac{1}{2}\right)^2+\frac{3}{4}\ge\frac{3}{4} \qquad \text{(ii)}$$

From (i) and (ii), we get

$$\frac{3}{4}\le A\le 1$$

14. $I=\int_0^1\frac{8\log(1+x)}{1+x^2}\,dx$

Putting $x=\tan\theta$, we obtain

$$I=8\int_0^{\frac{\pi}{4}}\log(1+\tan\theta)\,d\theta$$

$$I=8\int_0^{\frac{\pi}{4}}\log\left[1+\tan\left(\frac{\pi}{4}-\theta\right)\right]d\theta$$

$$I = 8\int_0^{\frac{\pi}{4}} \log\left(\frac{2}{1+\tan\theta}\right) d\theta$$

$$I = \pi \log 2$$

15. $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{1+4+3\lambda}{\sqrt{5+\lambda^2}\sqrt{14}}\right)$

$$\Rightarrow \sqrt{1-\frac{5}{14}} = \frac{5+3\lambda}{\sqrt{5+\lambda^2}\sqrt{14}}$$

$$\Rightarrow 3 = \frac{5+3\lambda}{\sqrt{5+\lambda^2}} \quad \Rightarrow \quad 45+9\lambda^2 = 25+9\lambda^2+30\lambda$$

$$\Rightarrow \lambda = \frac{2}{3}$$

16. $f'(x) = \sqrt{x}\sin x$

$f'(x)$ changes its sign from (+) to (−) at $x = \pi$ and from (−) to (+) at $x = 2\pi$.

$\therefore$ $f(x)$ has local maximum at π and local minimum at $x = 2\pi$.

17. $f(x)$ $\begin{cases} \dfrac{1}{\sqrt{-2x}} & \text{if } x < 0 \\ \dfrac{1}{\sqrt{0}} & \text{if } x > 0 \end{cases}$

$\therefore$ $f(x)$ is defined for $x \in (-\infty, 0)$.

18. Median $= \dfrac{25a+26a}{2} = 25.5a$

$\therefore$ Mean deviation

$$= \frac{|a-25.5a|+|2a-25.5a|+\cdots+|50a-25.5a|}{50} = 50$$

$$\Rightarrow 2\left(|0.5a|+|1.5a|+\cdots+|24.5a|\right) = 2500$$

$$\Rightarrow 2\times\frac{25}{2}\left(|a|(25)\right) = 2500$$

$$\Rightarrow |a| = 4$$

19. $|a| = 1$, $|b| = 1$,

$\vec{a}\cdot\vec{b} = 0$,

$$(2\vec{a}-\vec{b})\cdot\left[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})\right]$$

$$= (2\vec{a}-\vec{b})\cdot(\vec{b}-2\vec{a})$$

$$= -\left|2\vec{a}-\vec{b}\right|^2 = -(4|\vec{a}|^2+|\vec{b}|^2-4\vec{a}\cdot\vec{b})$$

$$= -5$$

20. L.H.L. $= \lim\limits_{x\to 0^-}(p+2) = p+2$

R.H.L. $= \lim\limits_{x\to 0^+}\dfrac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}}$

$$= \lim_{x\to 0^+}\frac{x^2}{x^{3/2}\left[\sqrt{x+x^2}+\sqrt{x}\right]}$$

$$= \lim_{x\to 0^+}\frac{1}{\sqrt{1+x}+1} = \frac{1}{2}$$

$f(0) = q$

$$p+2 = q = \frac{1}{2} \quad \Rightarrow \quad q = \frac{1}{2},\ p = \frac{-3}{2}$$

21. $x^2+y^2 = ax$

$x^2+y^2 = c^2$

$$\Rightarrow ax = c^2 \quad \Rightarrow \quad x = \frac{c^2}{a}$$

$\therefore$ $y^2 = c^2 - \dfrac{c^4}{a^2} = 0$ for one point of intersection

$$\Rightarrow a^2c^2 = c^4 \qquad \Rightarrow |ac| = |c^2|$$

$$\Rightarrow |a| = |c| \qquad \Rightarrow |a| = c \qquad (\because c > 0)$$

22. $\dfrac{dV(t)}{dt} = -k(T-t)$

$$V(t) = \frac{k(T-t)^2}{2} + c$$

at $t = 0$, $V(t) = I$

$$I = \frac{kT^2}{2} + c \quad \Rightarrow \quad c = I - \frac{kT^2}{2}$$

$$V(t) = \frac{k(T-t)^2}{2} + I - \frac{kT^2}{2}$$

$$V(T) = I - \frac{kT^2}{2}$$

23. $P\left(\dfrac{C}{D}\right) = \dfrac{P(C\cap D)}{P(D)} = \dfrac{P(C)}{P(D)}$ $(\because C \subset D)$

$$\therefore \quad P\left(\frac{C}{D}\right) = \frac{P(C)}{P(D)} \geq P(C)$$

24. Statement 1: $(A(BA))' = (BA)'A'$

$= A'B'A' = A(BA)$

$((AB)A)' = A'(AB)' = A'B'A'$

$= (AB)A$

Statement 1 is true.

Statement 2: $(AB)' = B'A' = BA = AB \quad (\because AB = BA)$
Statement 2 is correct but not a correct explanation for Statement 1.

25. $(1+\omega)^7 = A + B\omega$

$\Rightarrow (-\omega^2)^7 = A + B\omega \quad \Rightarrow \quad -\omega^2 = A + B\omega$

$\Rightarrow A + B\omega + \omega^2 = 0$ which is possible for $(A, B) \equiv (1, 1)$

26. Let the number of balls to be put in the four boxes be x_1, x_2, x_3, x_4 respectively such that $x_1, x_2, x_3, x_4 \neq 0$, then $x_1, x_2, x_3, x_4 = 10$
The number of ways of distributing 10 identical balls in 4 distinct boxes such that no box is empty is

$^{10-1}C_{4-1} = {}^9C_3$

So, Statement 1 is true and Statement 2 is obviously true.

27. For a point on the parabola $y^2 = x$ nearest to the line $y - x = 1$, the normal passing through it will be perpendicular to the line $y - x = 1$.

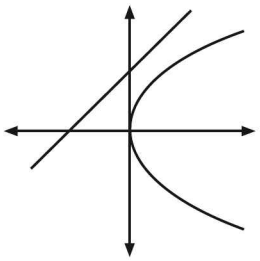


Fig. Q.2

$\therefore$ (Slope of normal) $\times$ 1 = –1

$\Rightarrow -t = -1 \quad \Rightarrow \quad t = 1$

Thus, the point is $\left(\frac{1}{4}, \frac{2}{4}\right) \equiv \left(\frac{1}{4}, \frac{1}{2}\right)$

$\therefore$ Shortest distance $= \left|\frac{\frac{1}{4} - \frac{1}{2} + 1}{\sqrt{2}}\right| = \frac{3}{4\sqrt{2}} = \frac{3\sqrt{2}}{8}$

28. $A = \int_0^1 x\,dx + \int_1^e \frac{1}{x}\,dx$

$= \frac{1}{2} + 1 = \frac{3}{2}$

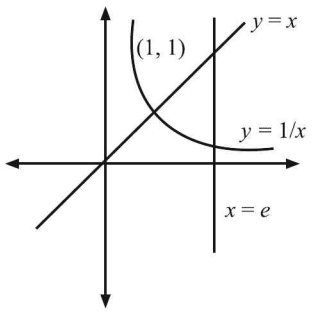


Fig. Q.3

29. $\frac{dy}{y+3} = dx \quad \Rightarrow \quad \log_e(y+3) = x + c$

$\Rightarrow y + 3 = e^{x+c} \quad \Rightarrow \quad y + 3 = e^x \cdot e^c$

$5 = e^c \qquad [\because y(0) = 2]$

$\therefore \quad y + 3 = 5e^x$

$\therefore \quad y(\log_e 2) = 10 - 3 = 7$

30. $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$

$(\vec{b} \times \vec{c}) \times \vec{a} = (\vec{b} \times \vec{d}) \times \vec{a}$

$(\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b} = (\vec{a} \cdot \vec{b})\vec{d} - (\vec{a} \cdot \vec{d})\vec{b}$

$\vec{d} = \vec{c} - \frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\vec{b}$

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